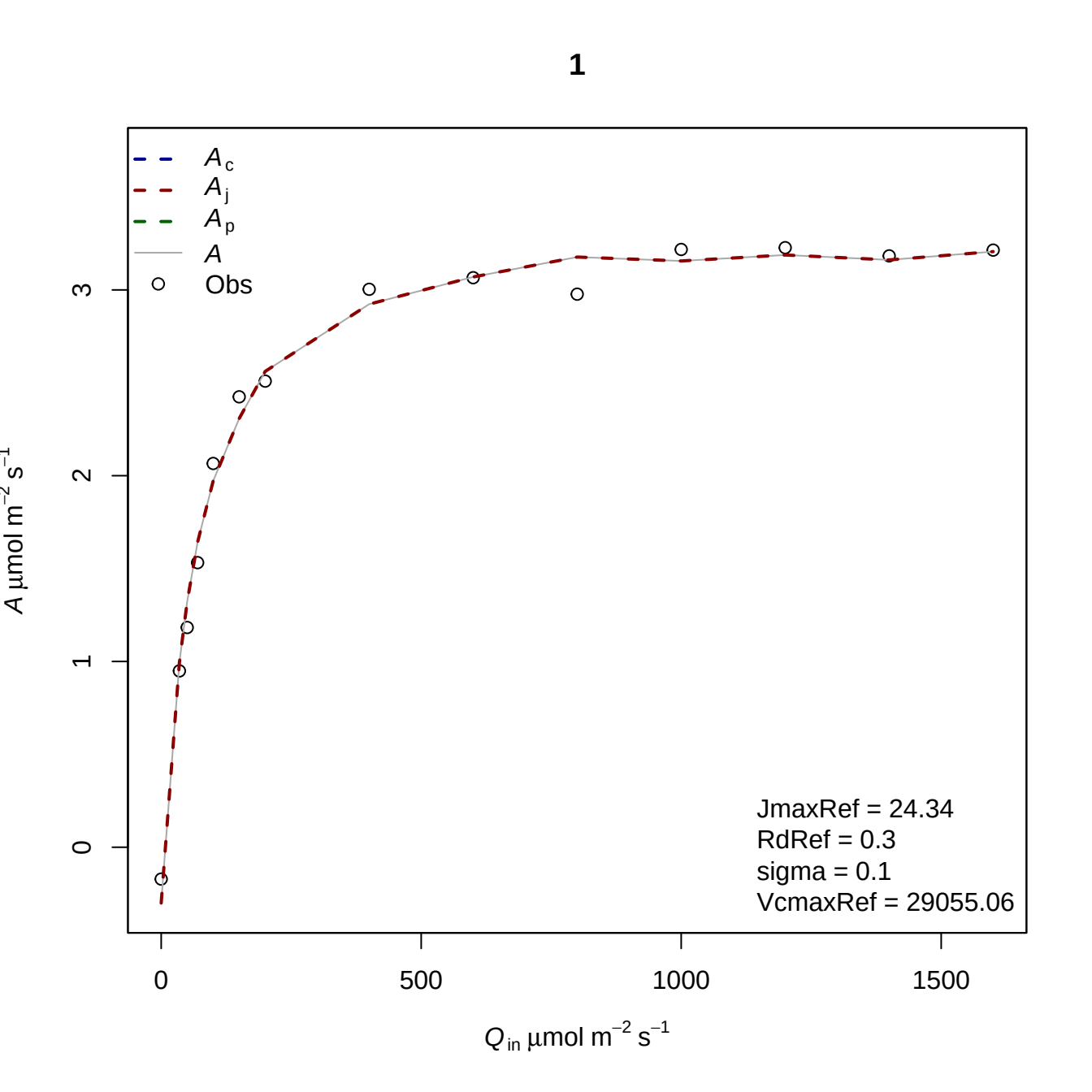
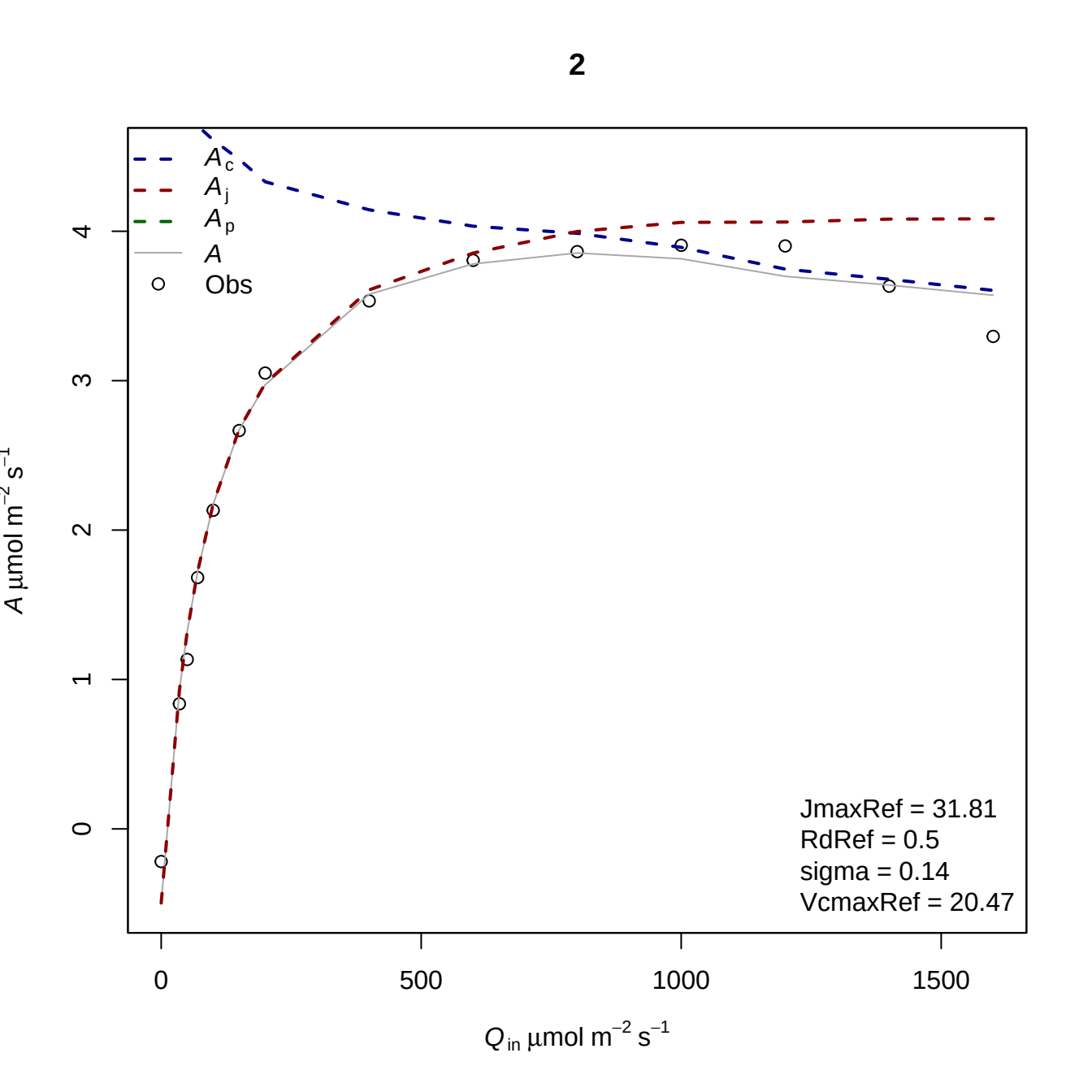
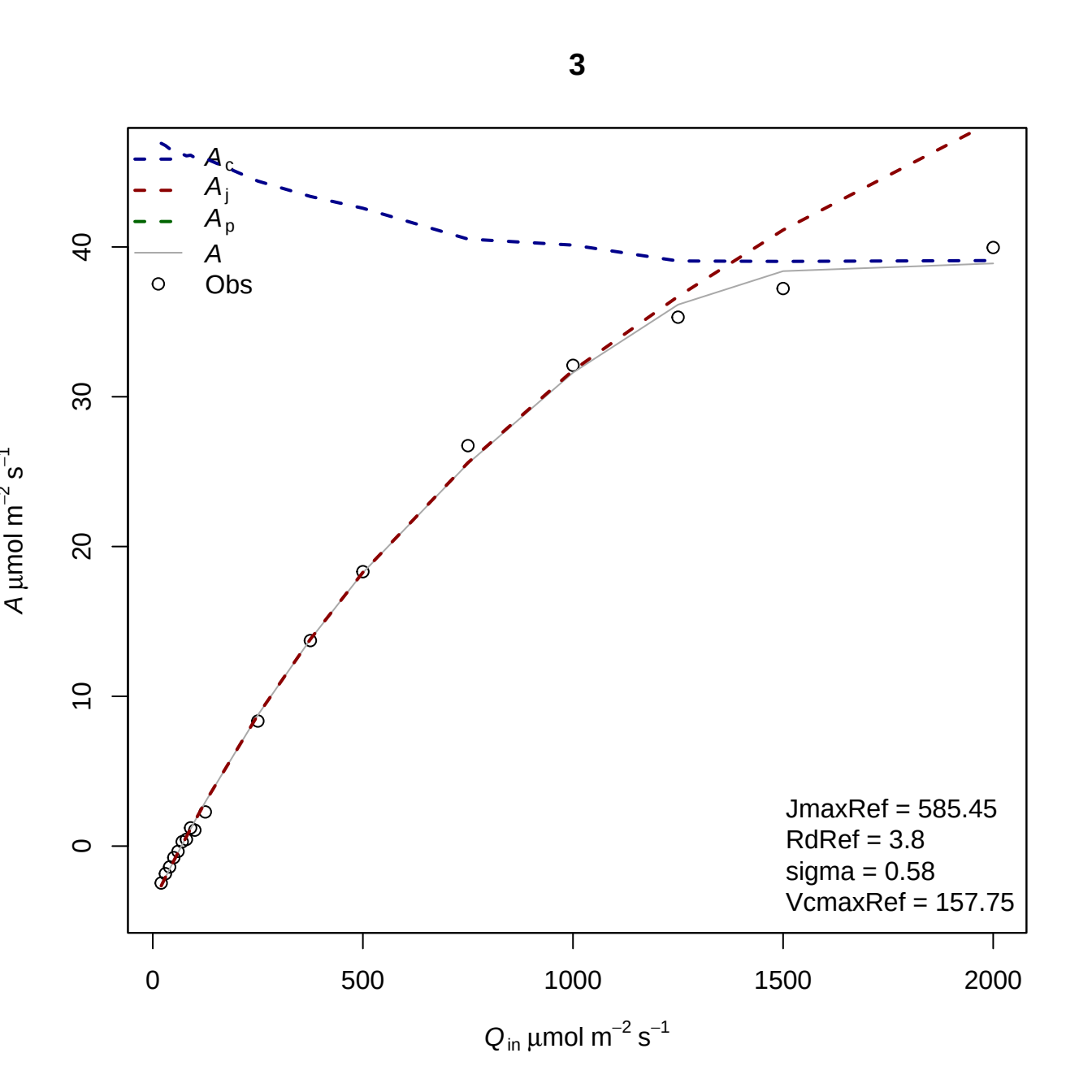
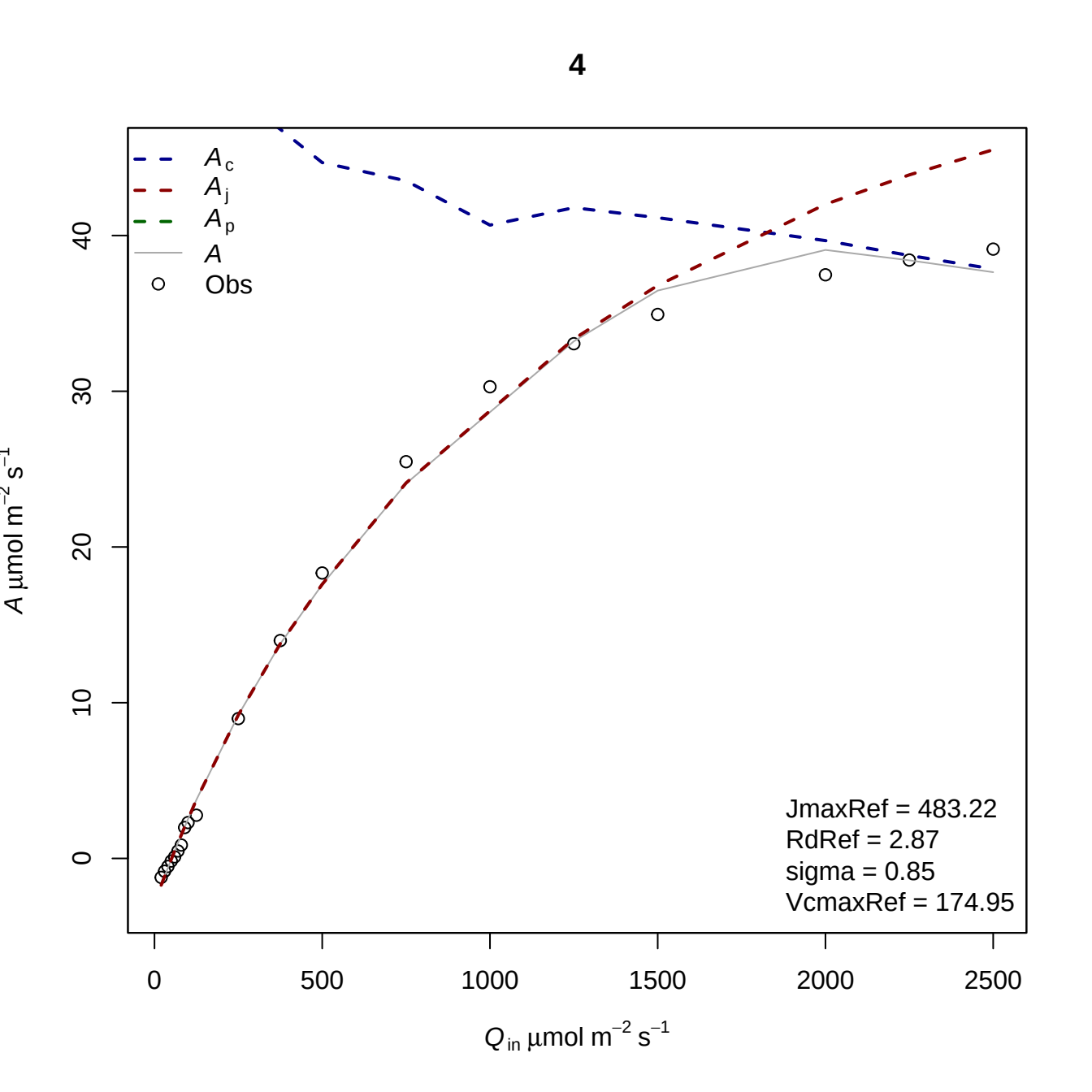
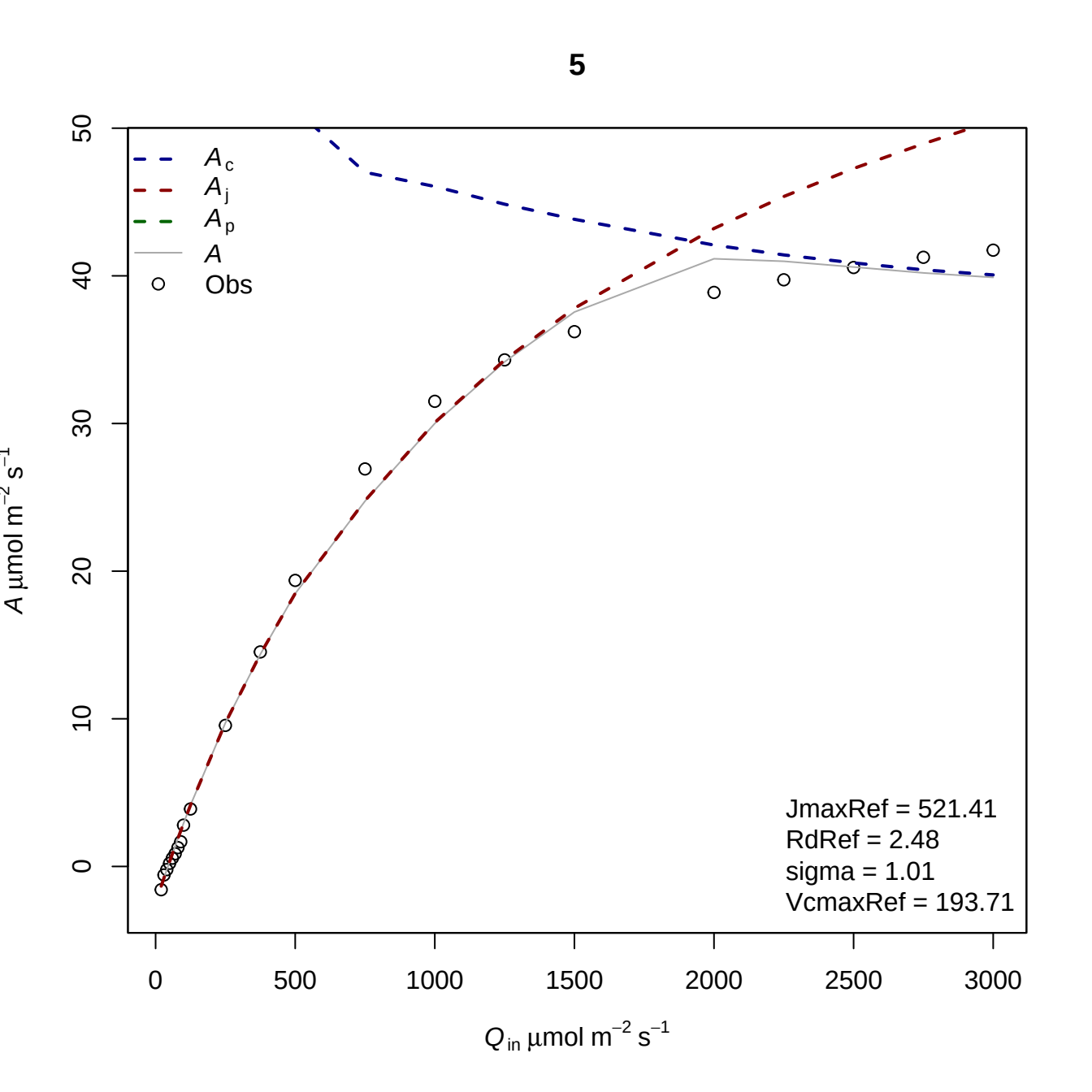
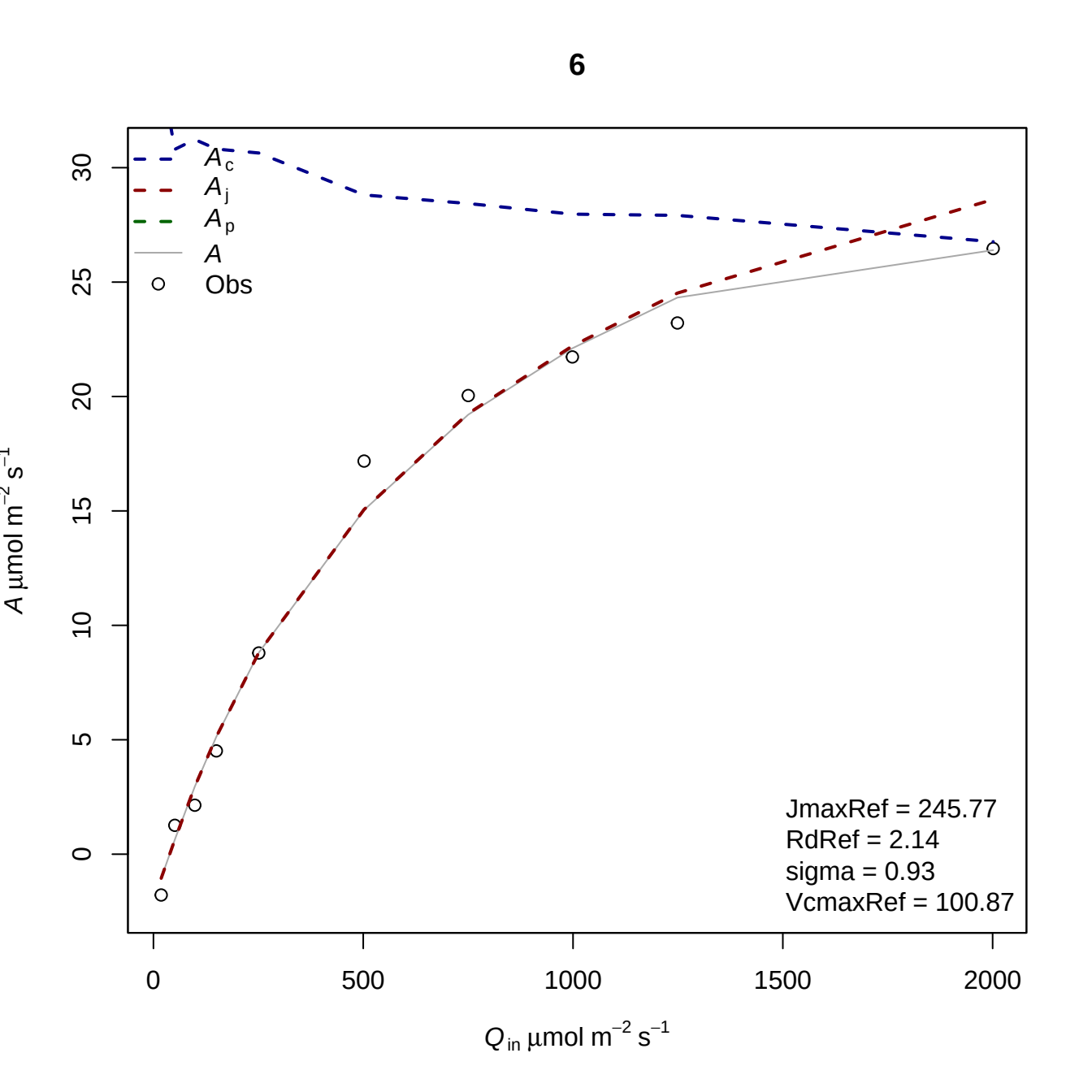

1
A_c
A_j
A_p
A
Obs
JmaxRef = 24.34
RdRef = 0.3
sigma = 0.1
VcmaxRef = 29055.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500

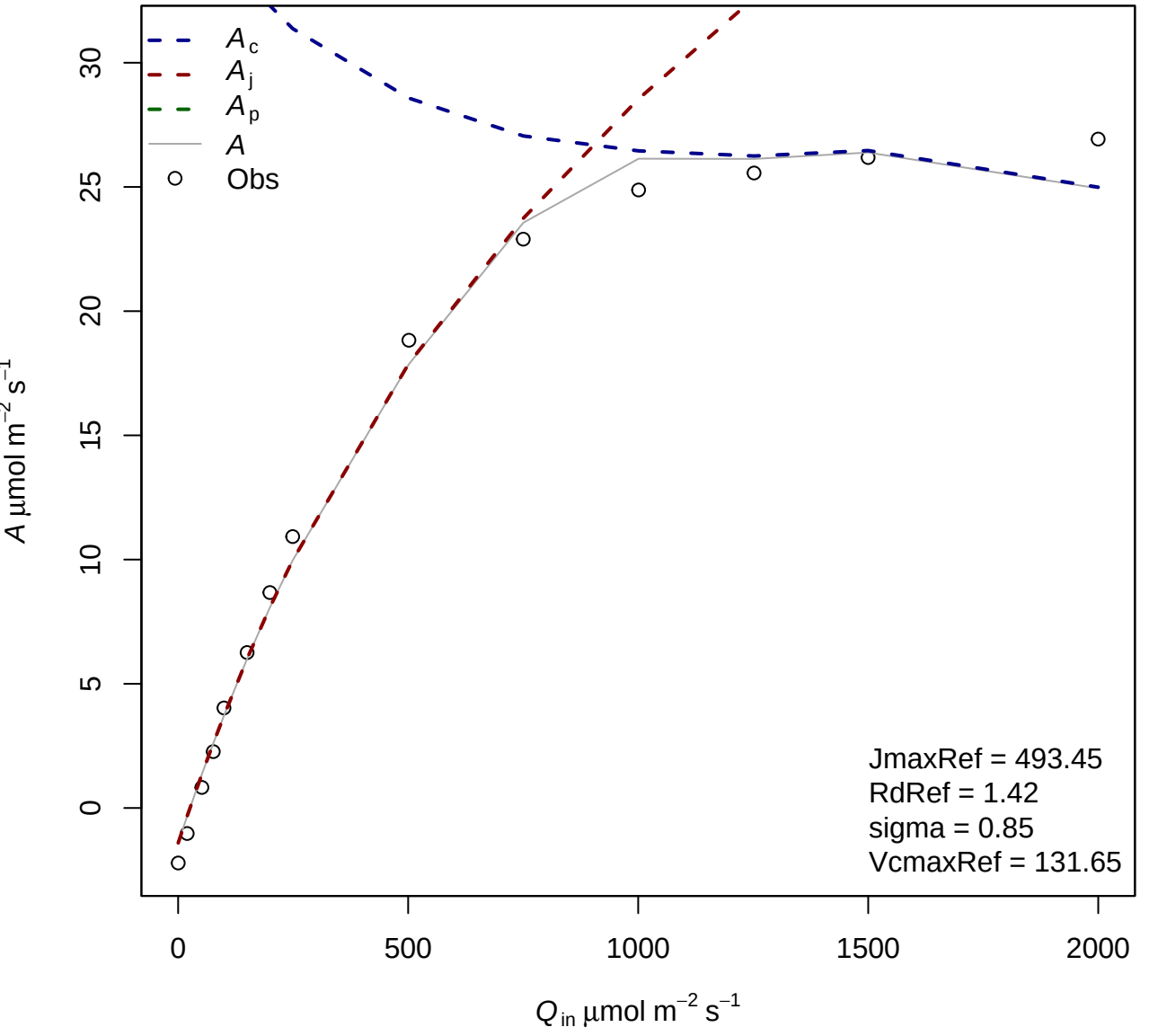

2
A_c
A_j
A_p
A
Obs
JmaxRef = 31.81
RdRef = 0.5
sigma = 0.14
VcmaxRef = 20.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

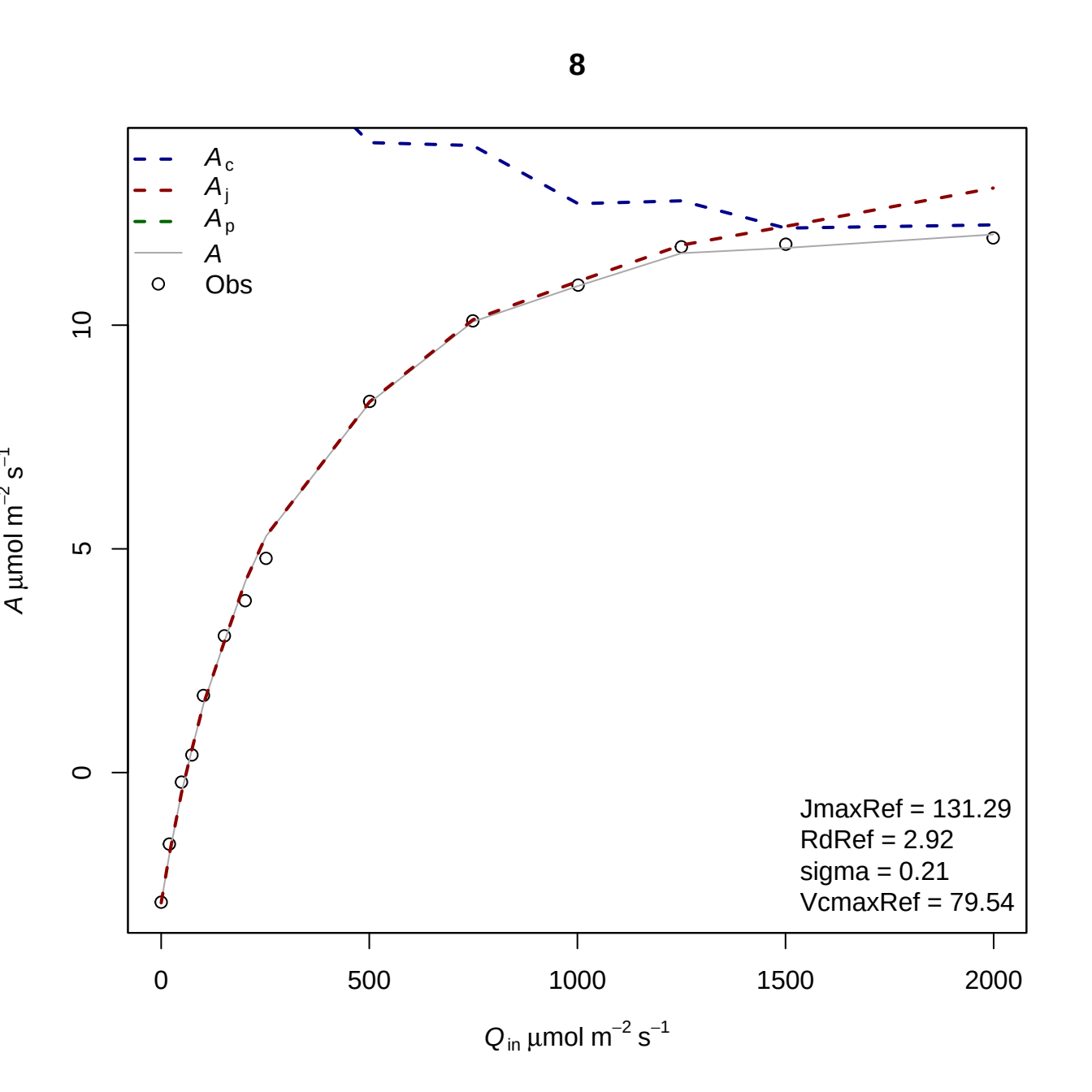

3
A_c
A_j
A_p
A
Obs
JmaxRef = 585.45
RdRef = 3.8
sigma = 0.58
VcmaxRef = 157.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000

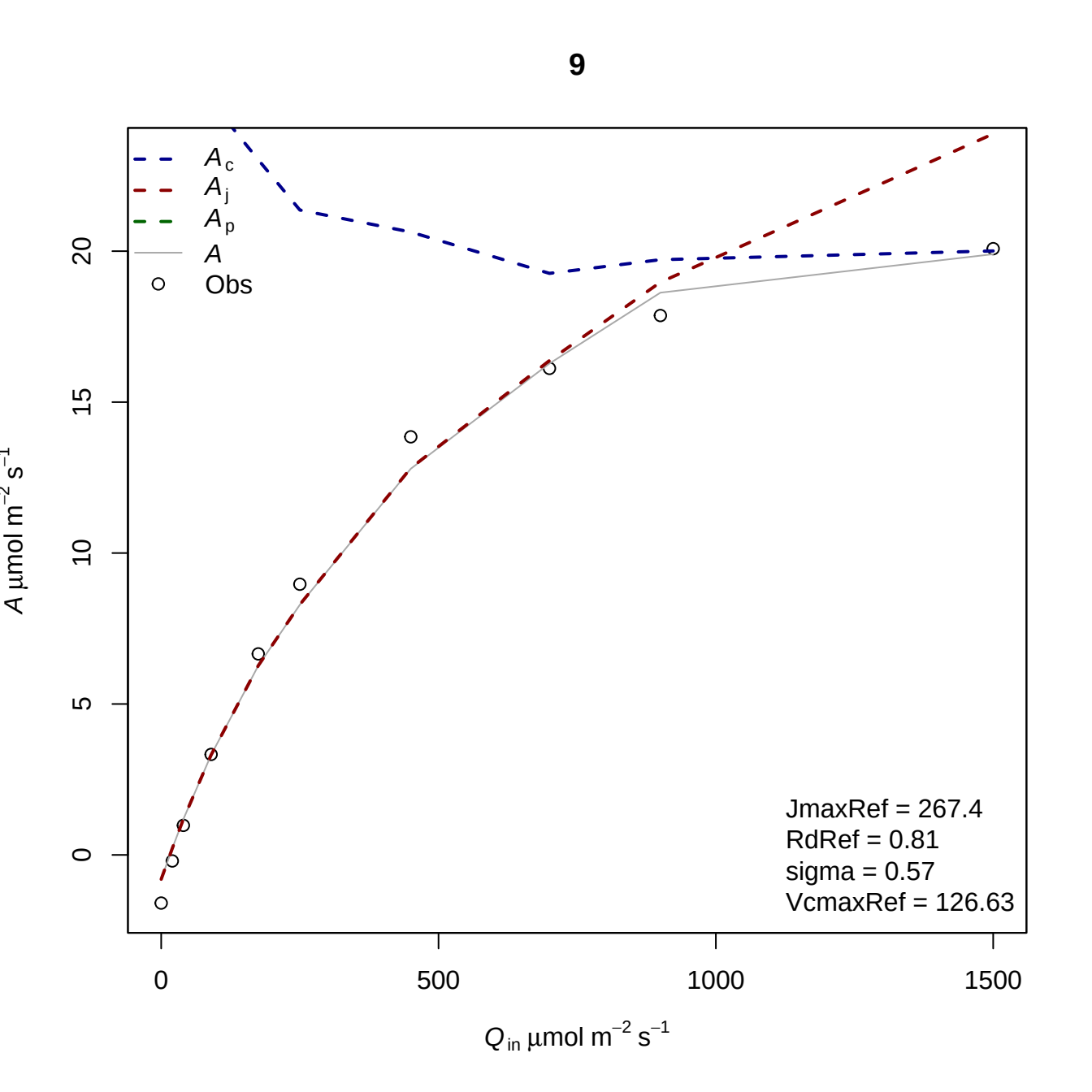

4
A_c
A_j
A_p
A
Obs
JmaxRef = 483.22
RdRef = 2.87
sigma = 0.85
VcmaxRef = 174.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000
2500

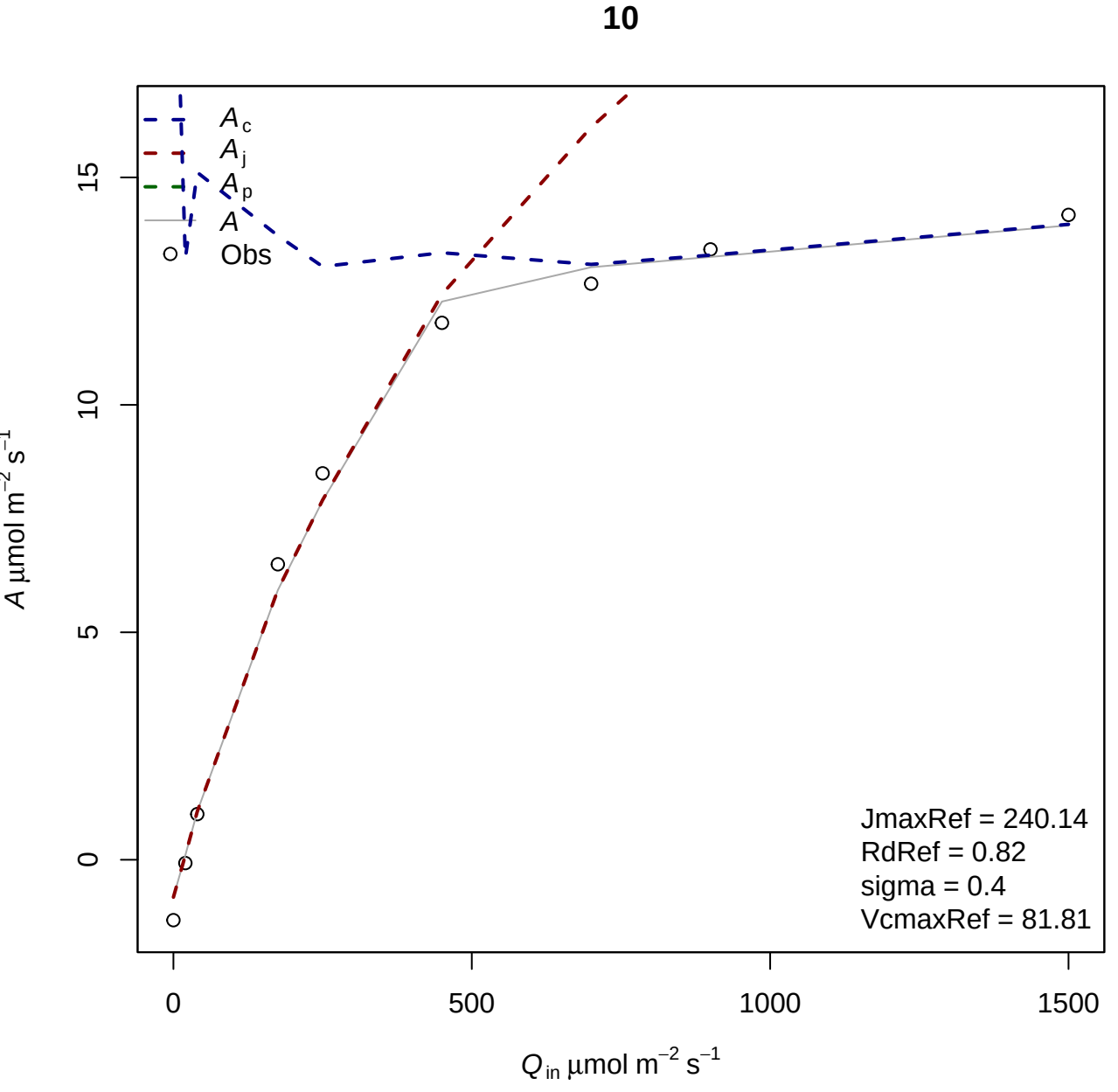

5
A_c
A_j
A_p
A
Obs
JmaxRef = 521.41
RdRef = 2.48
sigma = 1.01
VcmaxRef = 193.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
50
0
500
1000
1500
2000
2500
3000

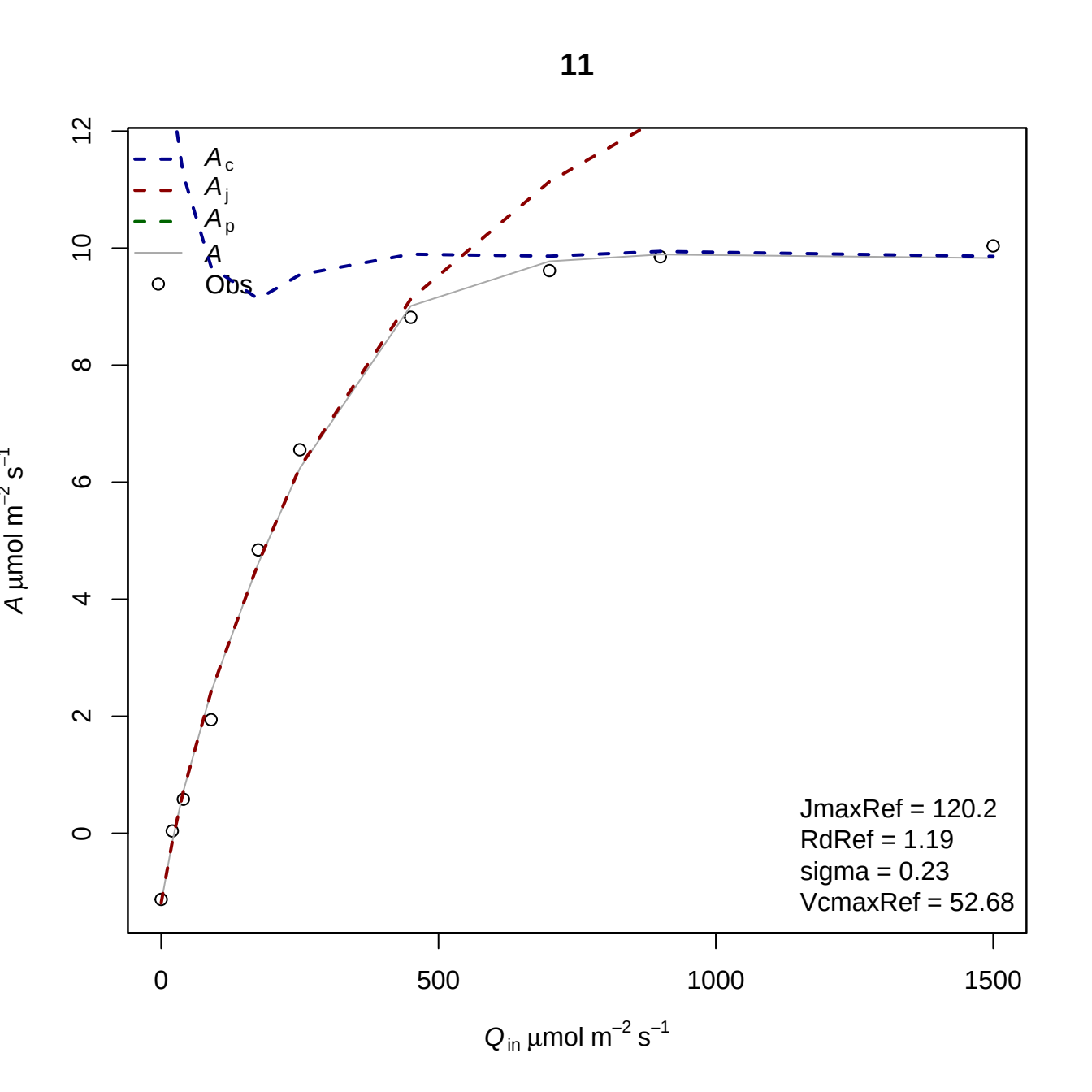

6
A_c
A_j
A_p
A
Obs
JmaxRef = 245.77
RdRef = 2.14
sigma = 0.93
VcmaxRef = 100.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

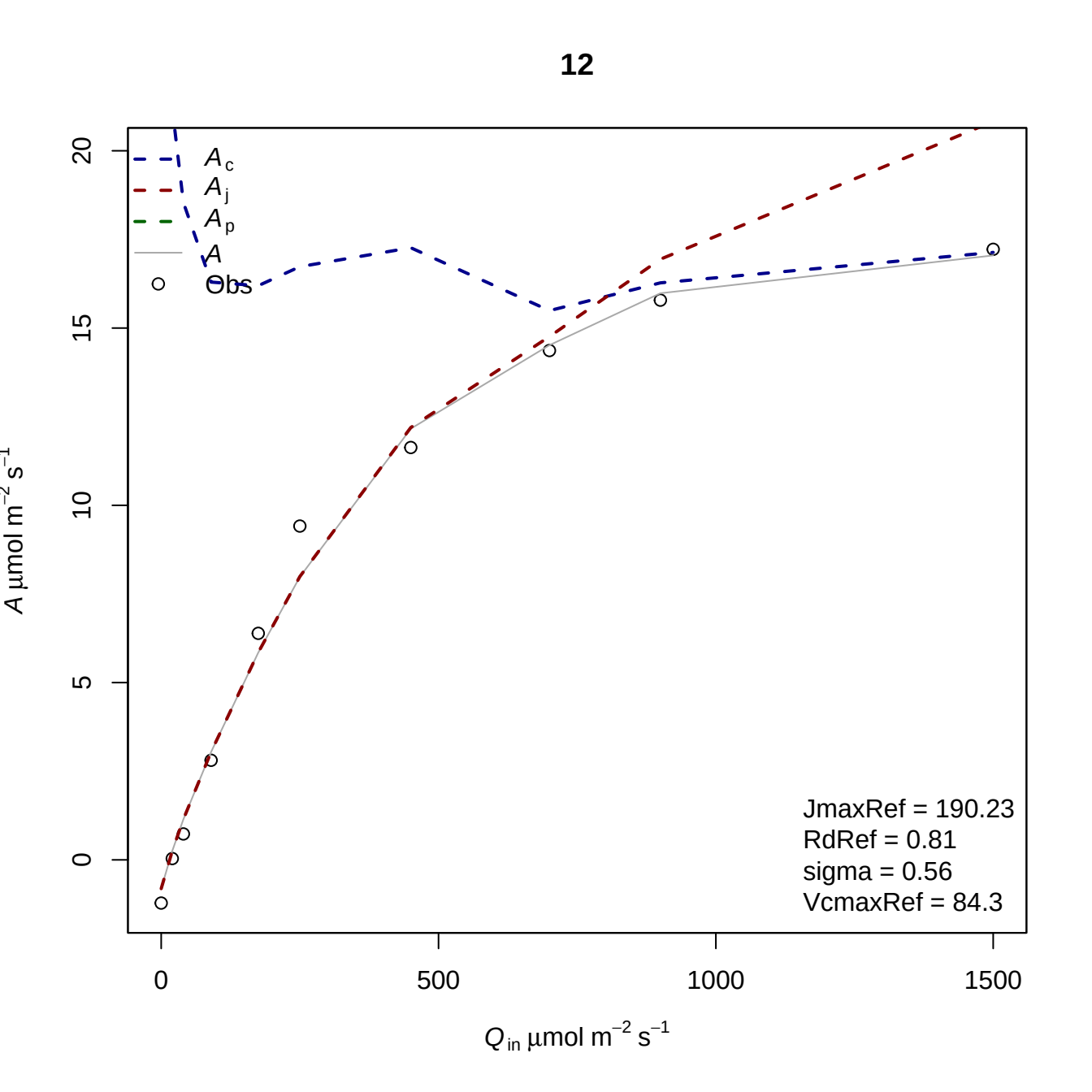

A_c
A_j
A_p
A
Obs
JmaxRef = 493.45
RdRef = 1.42
sigma = 0.85
VcmaxRef = 131.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

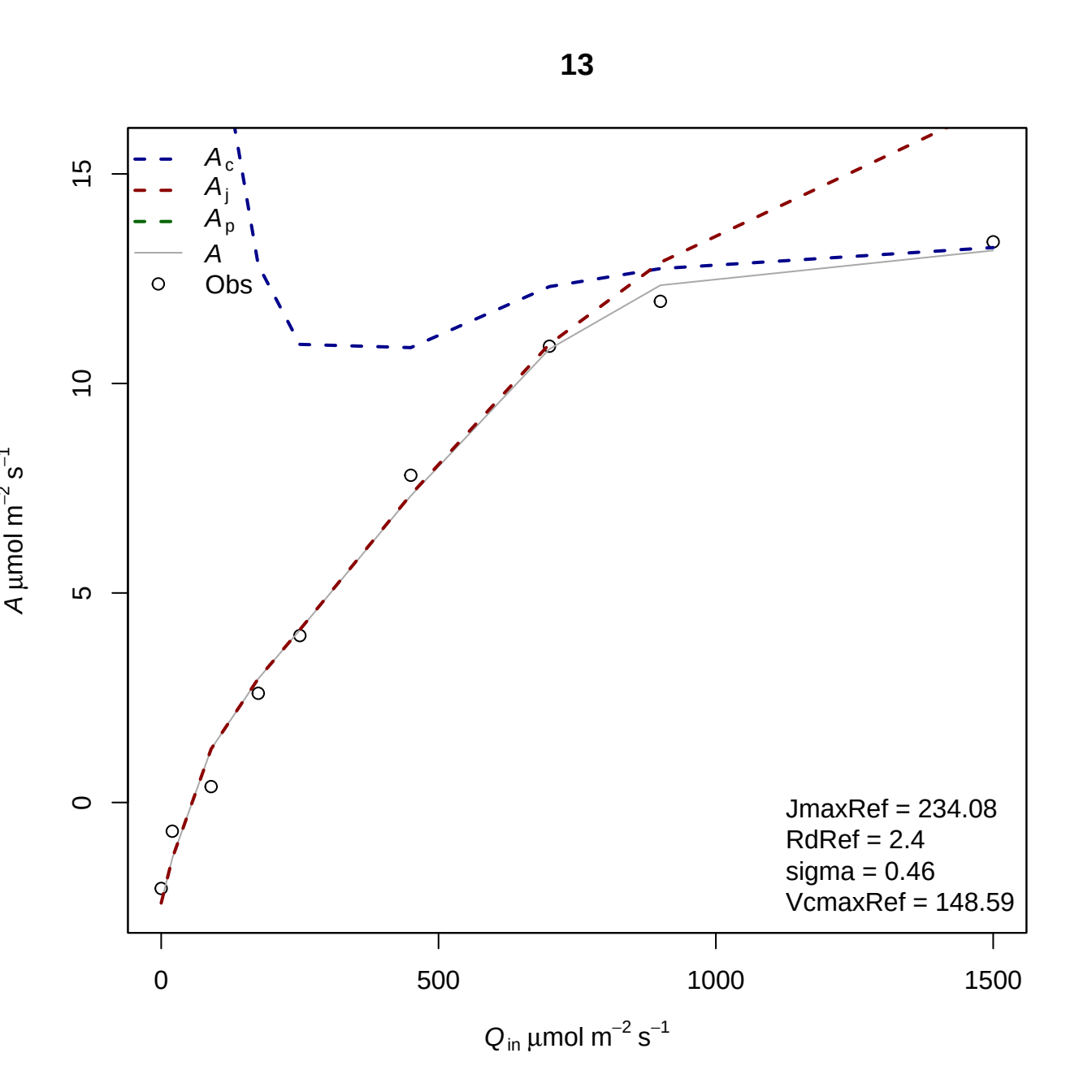

8
A_c
A_j
A_p
A
Obs
JmaxRef = 131.29
RdRef = 2.92
sigma = 0.21
VcmaxRef = 79.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

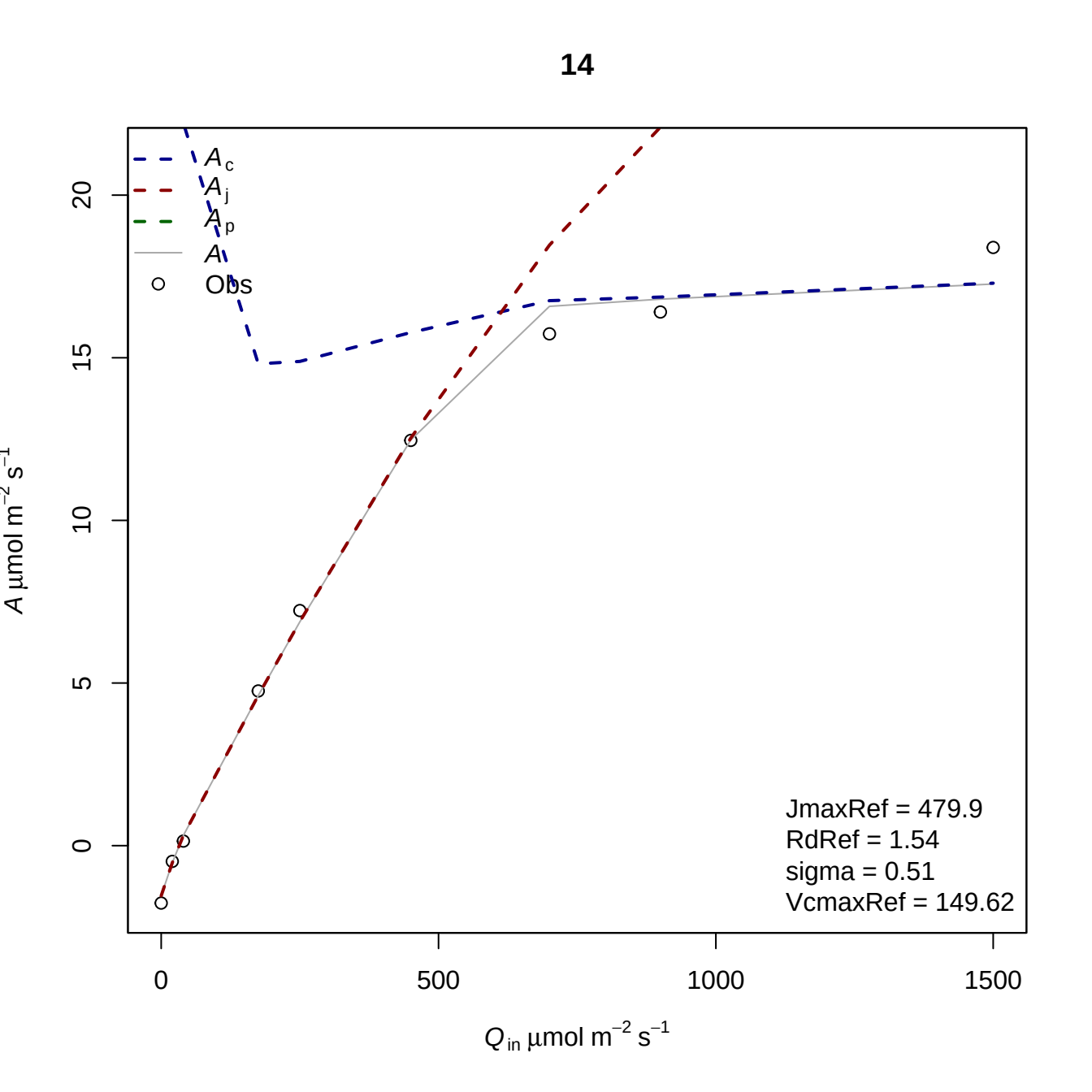

9
A_c
A_j
A_p
A
Obs
JmaxRef = 267.4
RdRef = 0.81
sigma = 0.57
VcmaxRef = 126.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

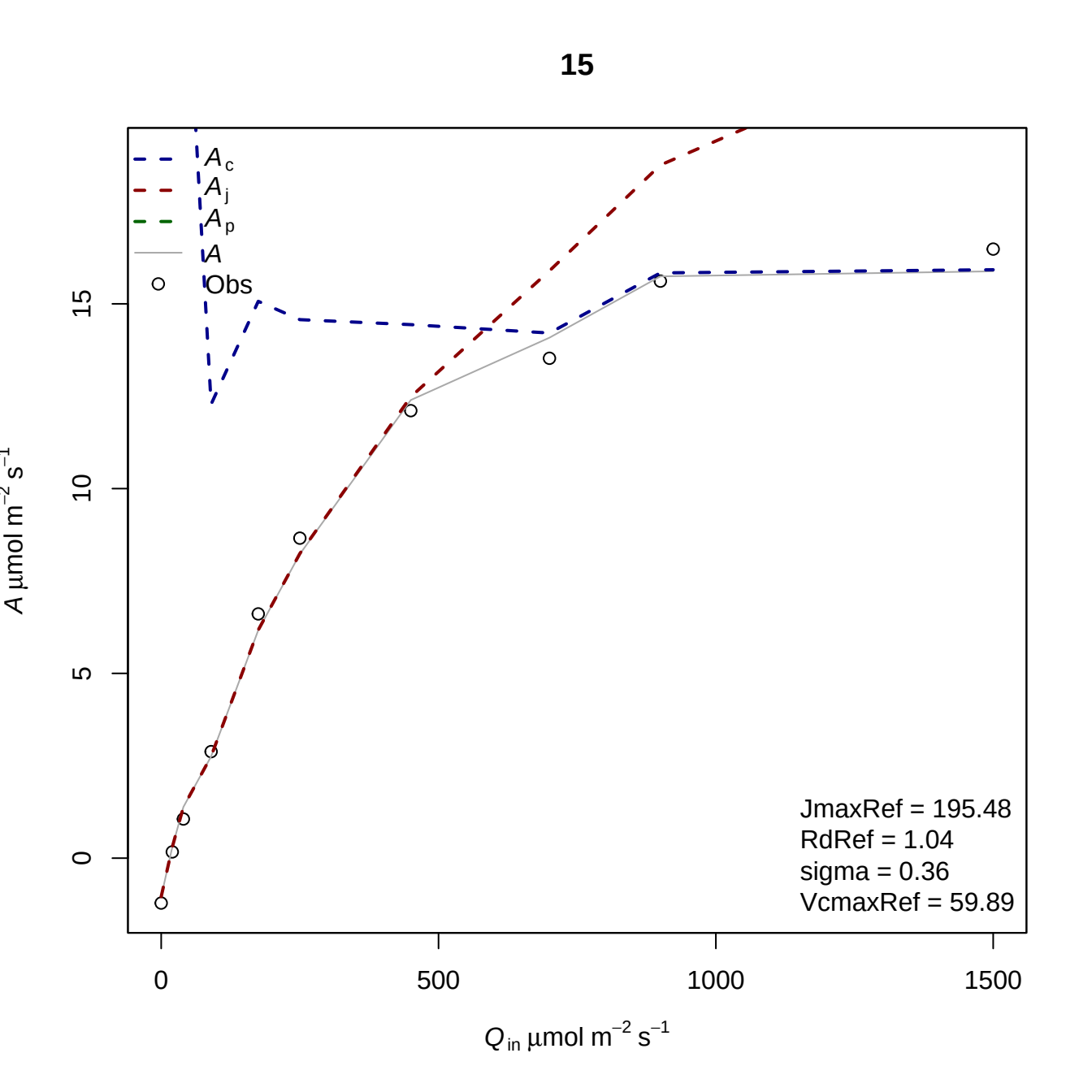

10
A_c
A_j
A_p
A
Obs
JmaxRef = 240.14
RdRef = 0.82
sigma = 0.4
VcmaxRef = 81.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

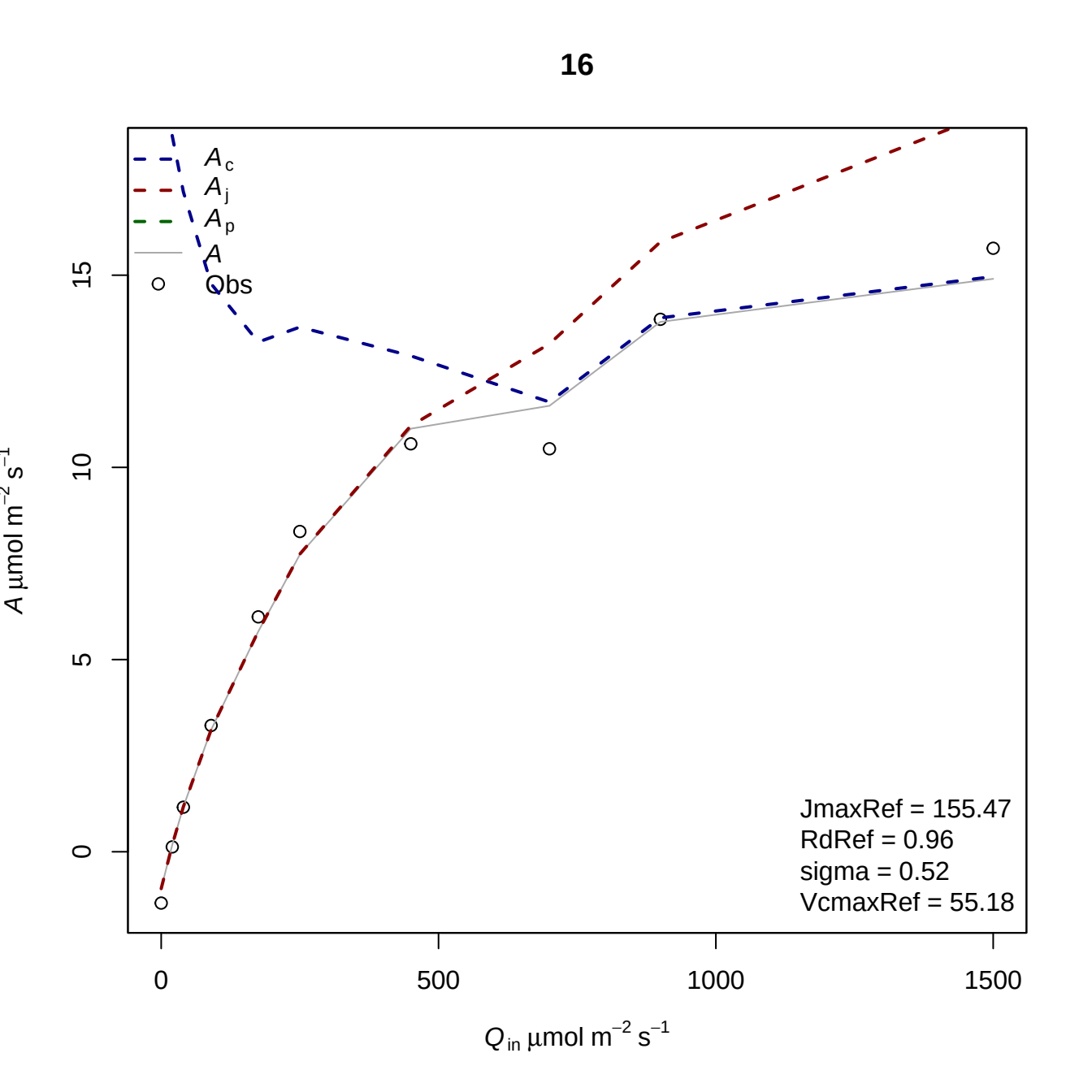

11
A_c
A_j
A_p
A
Obs
JmaxRef = 120.2
RdRef = 1.19
sigma = 0.23
VcmaxRef = 52.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

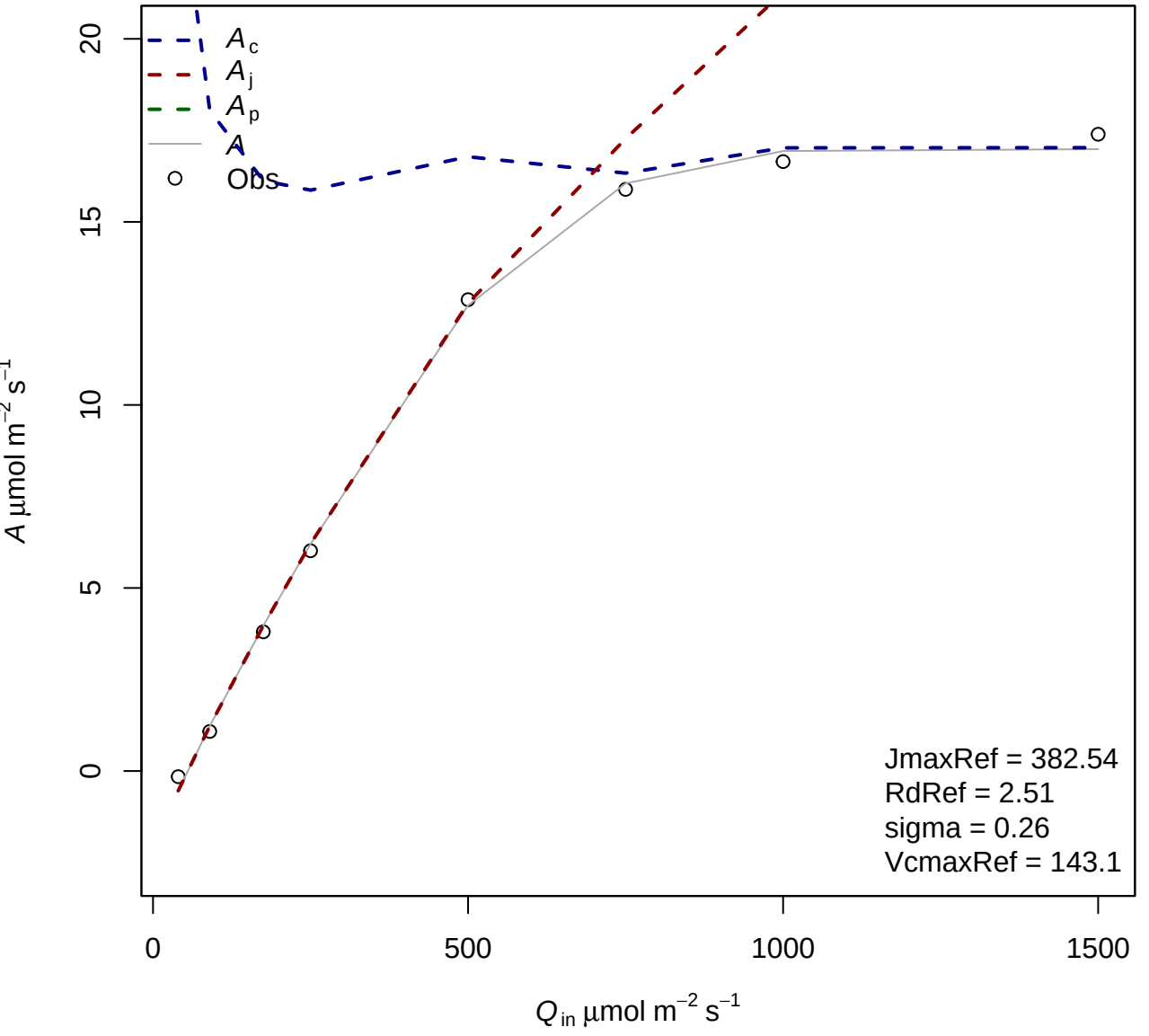

12
A_c
A_j
A_p
A
Obs
JmaxRef = 190.23
RdRef = 0.81
sigma = 0.56
VcmaxRef = 84.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

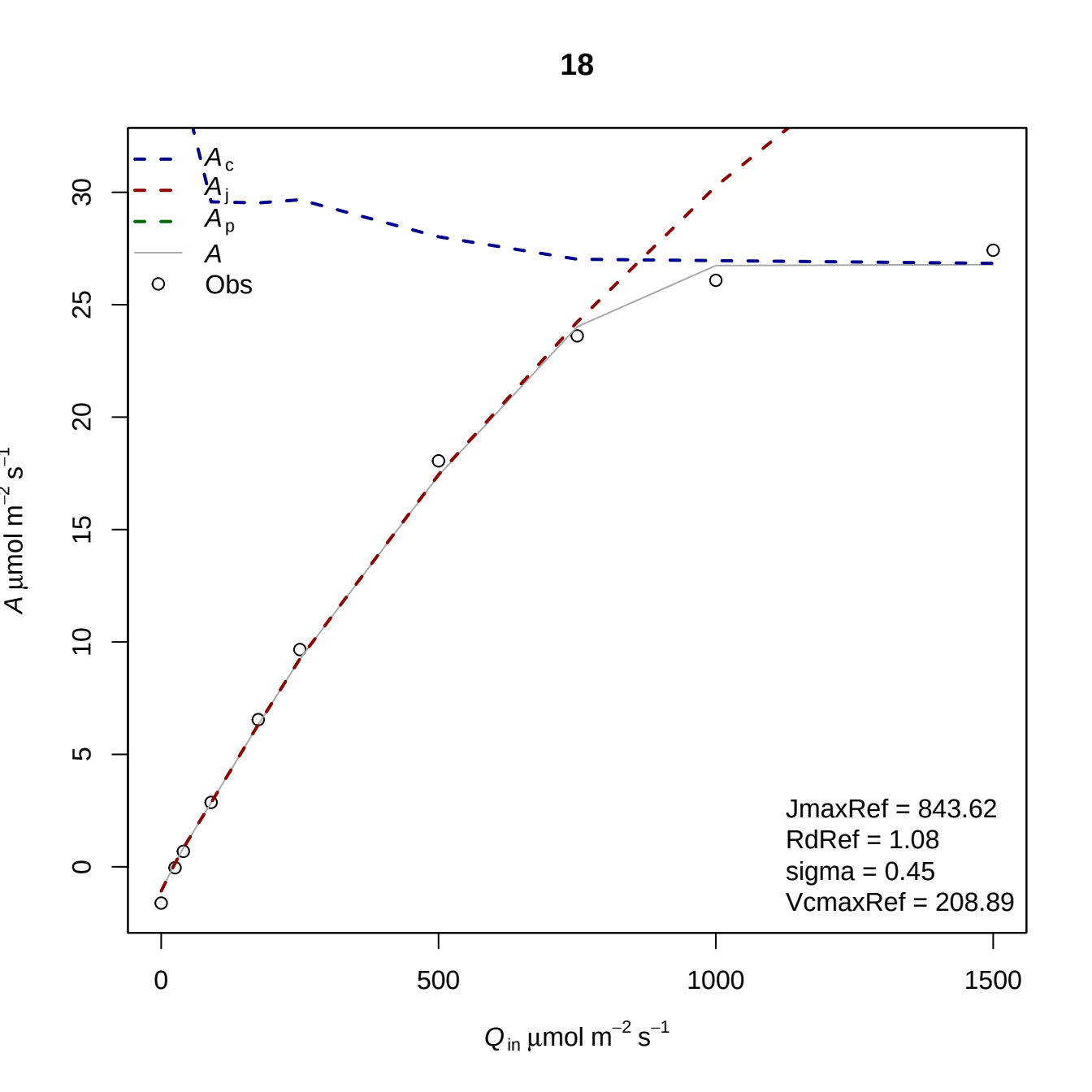

13
A_c
A_j
A_p
A
Obs
JmaxRef = 234.08
RdRef = 2.4
sigma = 0.46
VcmaxRef = 148.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

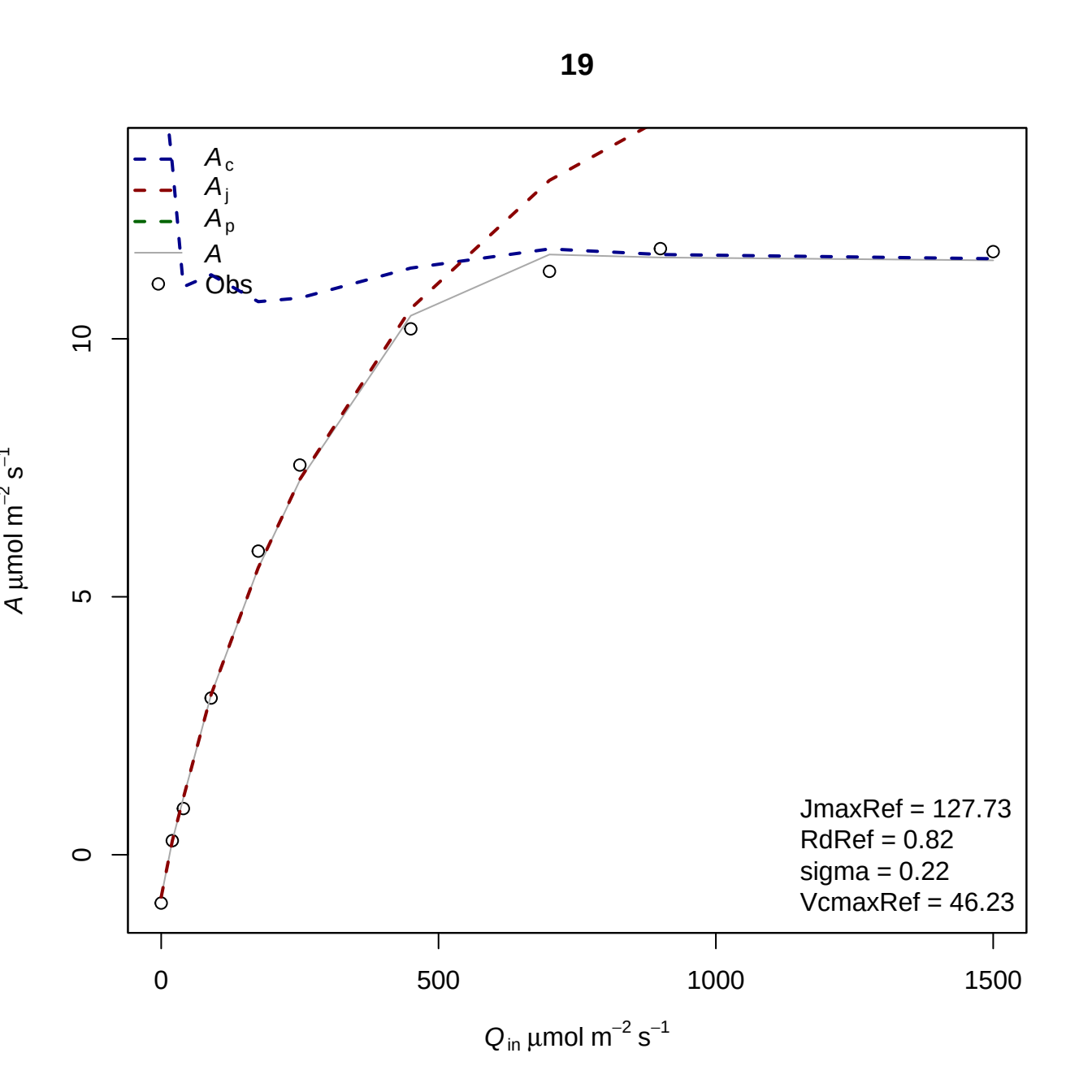

14
A_c
A_j
A_p
A
Obs
JmaxRef = 479.9
RdRef = 1.54
sigma = 0.51
VcmaxRef = 149.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

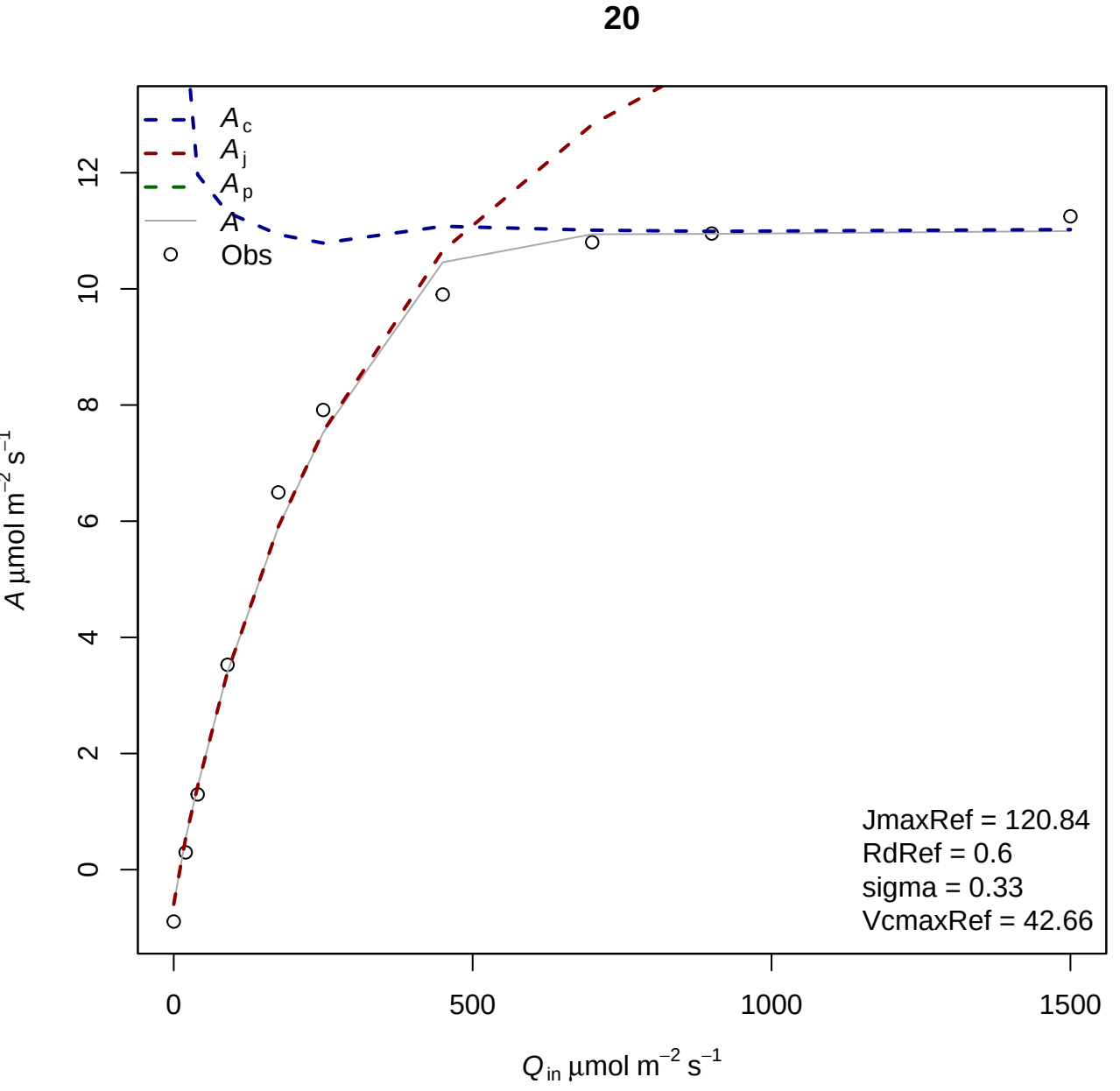

15
A_c
A_j
A_p
A
Obs
JmaxRef = 195.48
RdRef = 1.04
sigma = 0.36
VcmaxRef = 59.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

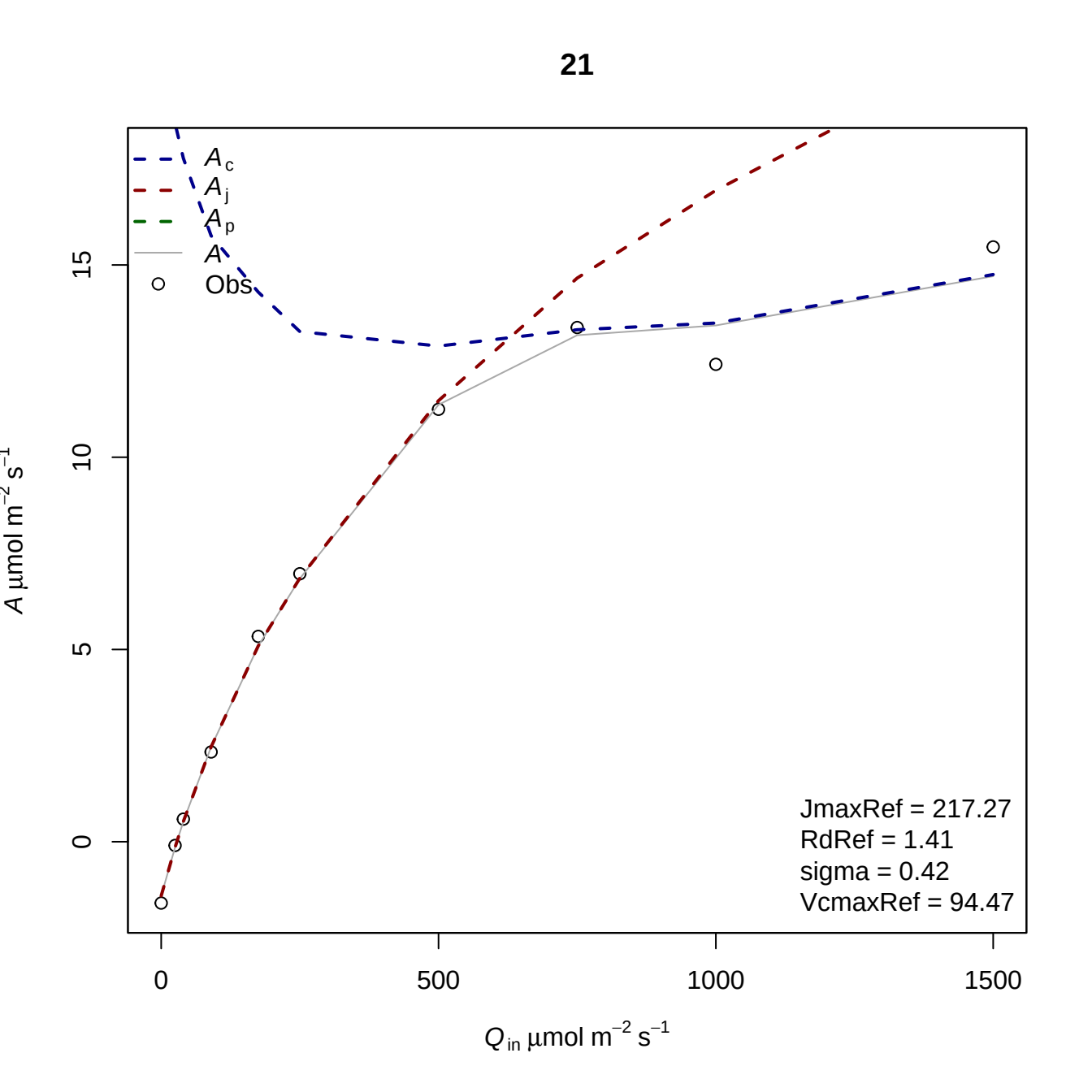

16
A_c
A_j
A_p
A
Obs
JmaxRef = 155.47
RdRef = 0.96
sigma = 0.52
VcmaxRef = 55.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

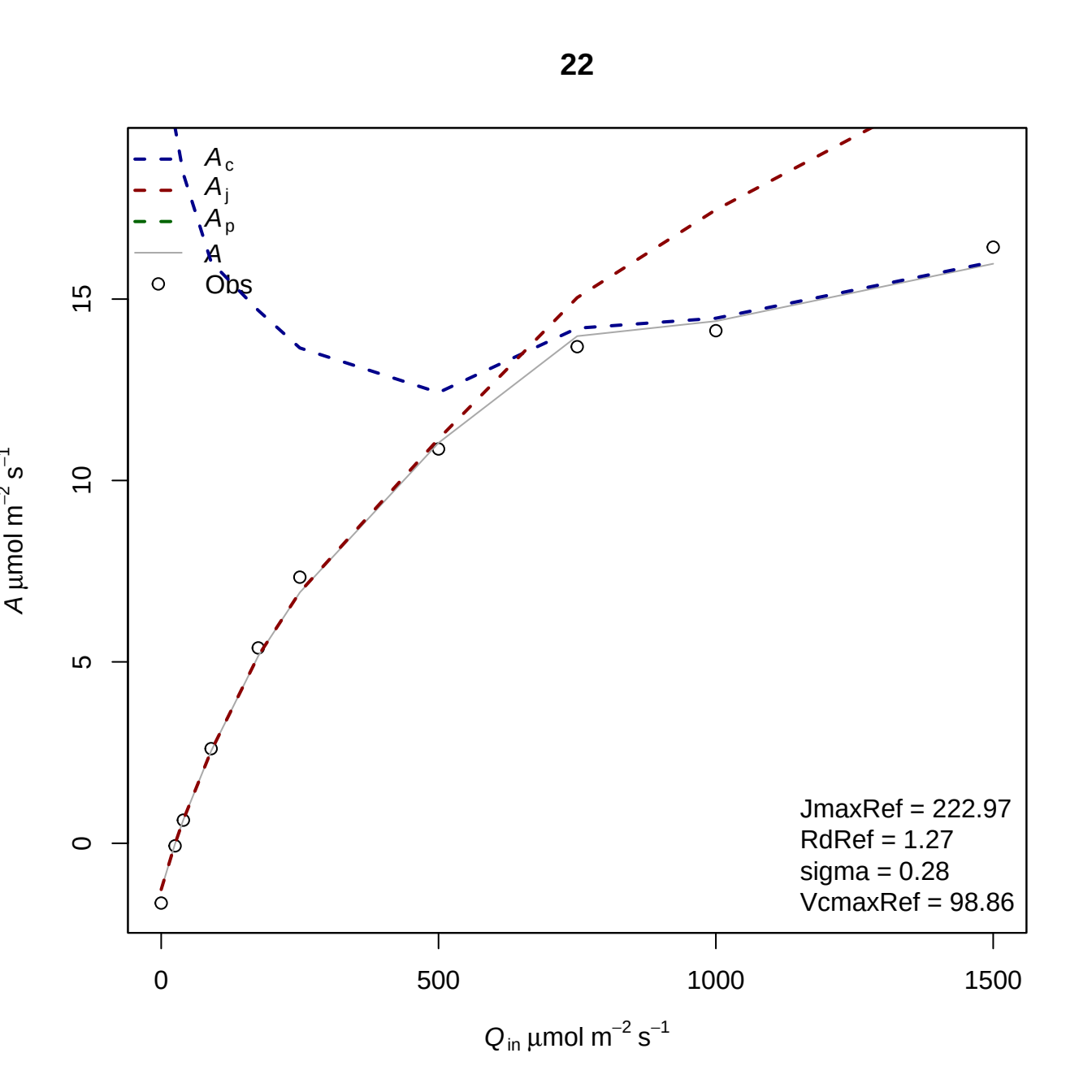

A_c
A_j
A_p
A
Obs
JmaxRef = 382.54
RdRef = 2.51
sigma = 0.26
VcmaxRef = 143.1
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

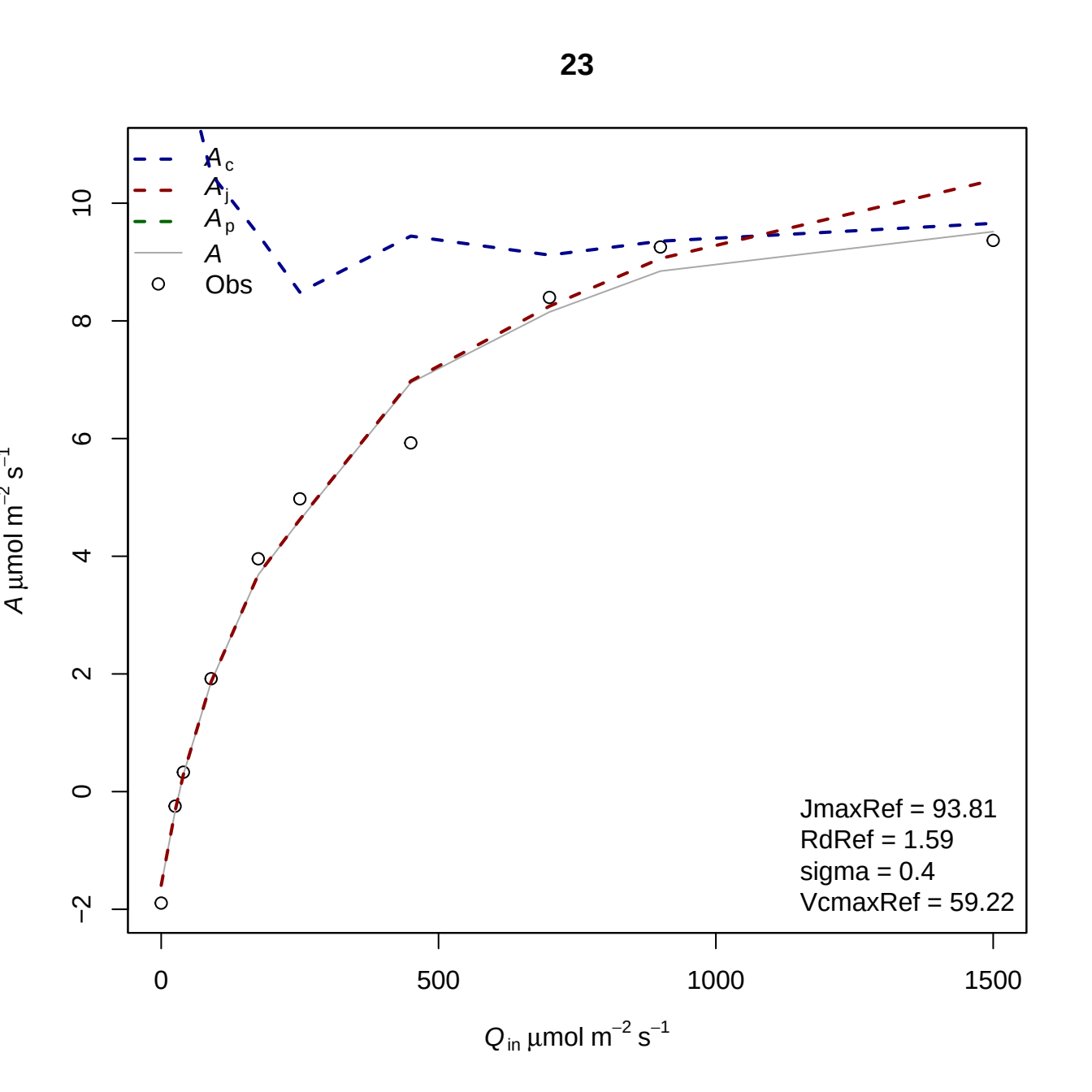

18
A_c
A_j
A_p
A
Obs
JmaxRef = 843.62
RdRef = 1.08
sigma = 0.45
VcmaxRef = 208.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500

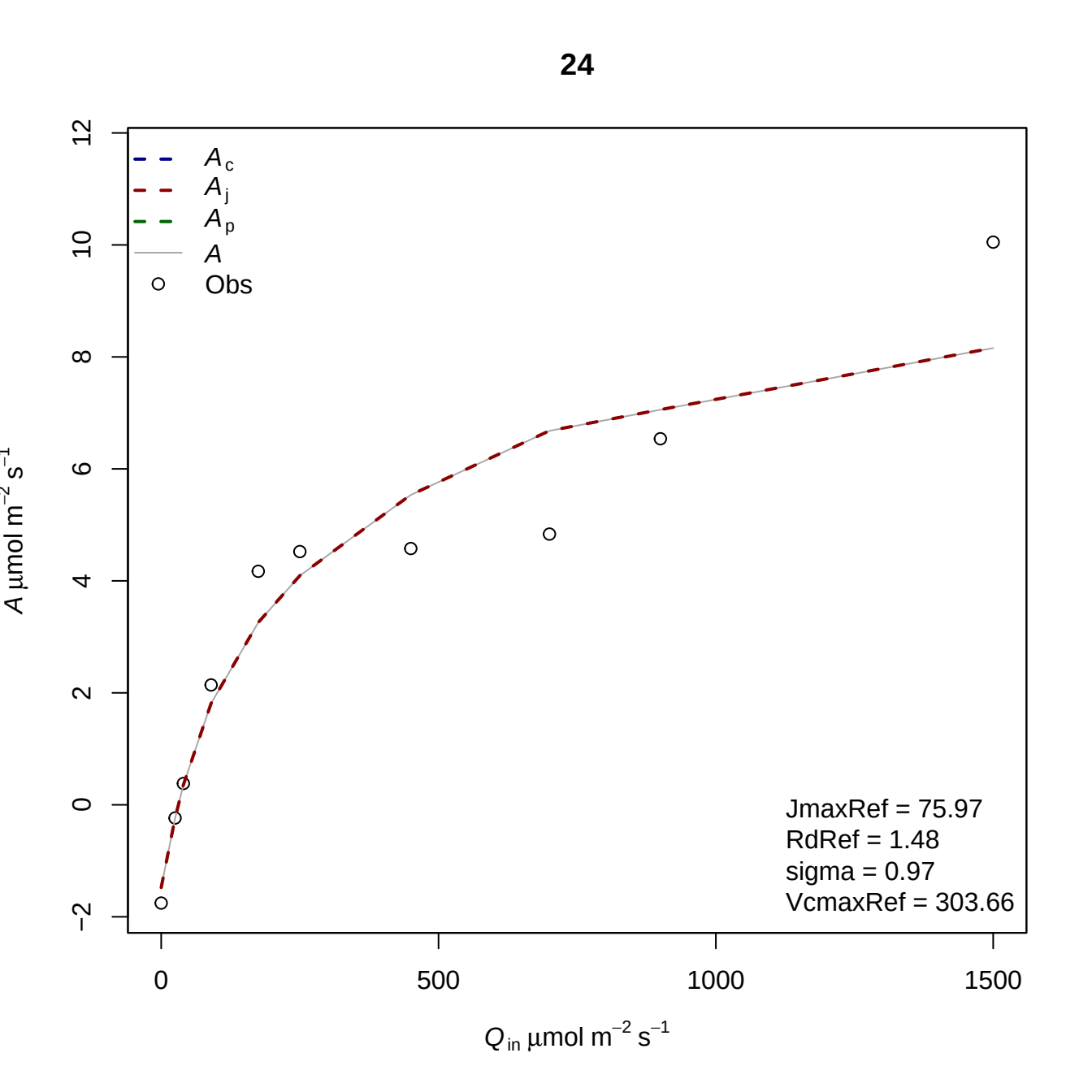

19
A_c
A_j
A_p
A
Obs
JmaxRef = 127.73
RdRef = 0.82
sigma = 0.22
VcmaxRef = 46.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

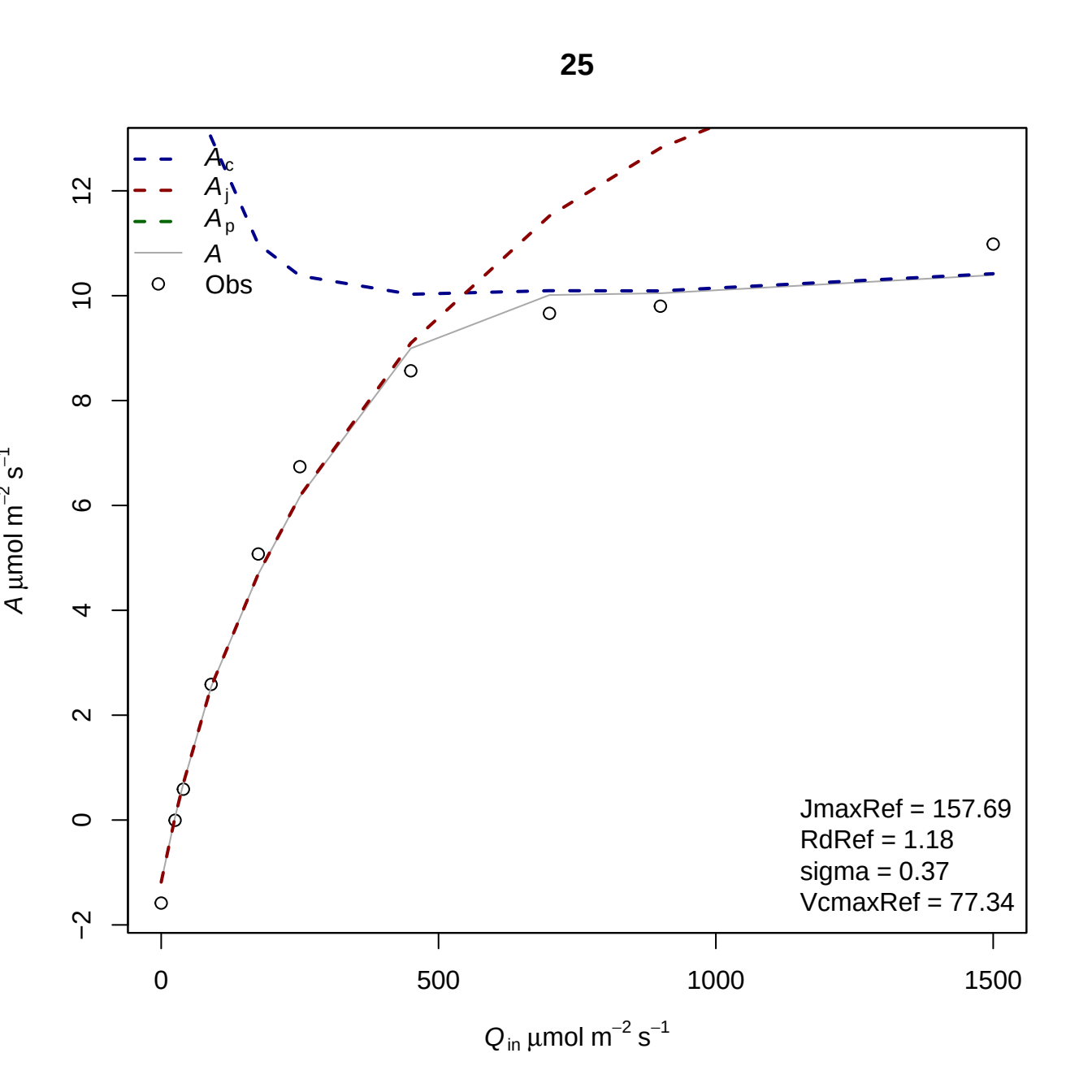

20
A_c
A_j
A_p
A
Obs
JmaxRef = 120.84
RdRef = 0.6
sigma = 0.33
VcmaxRef = 42.66
0
2
4
6
8
10
12
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

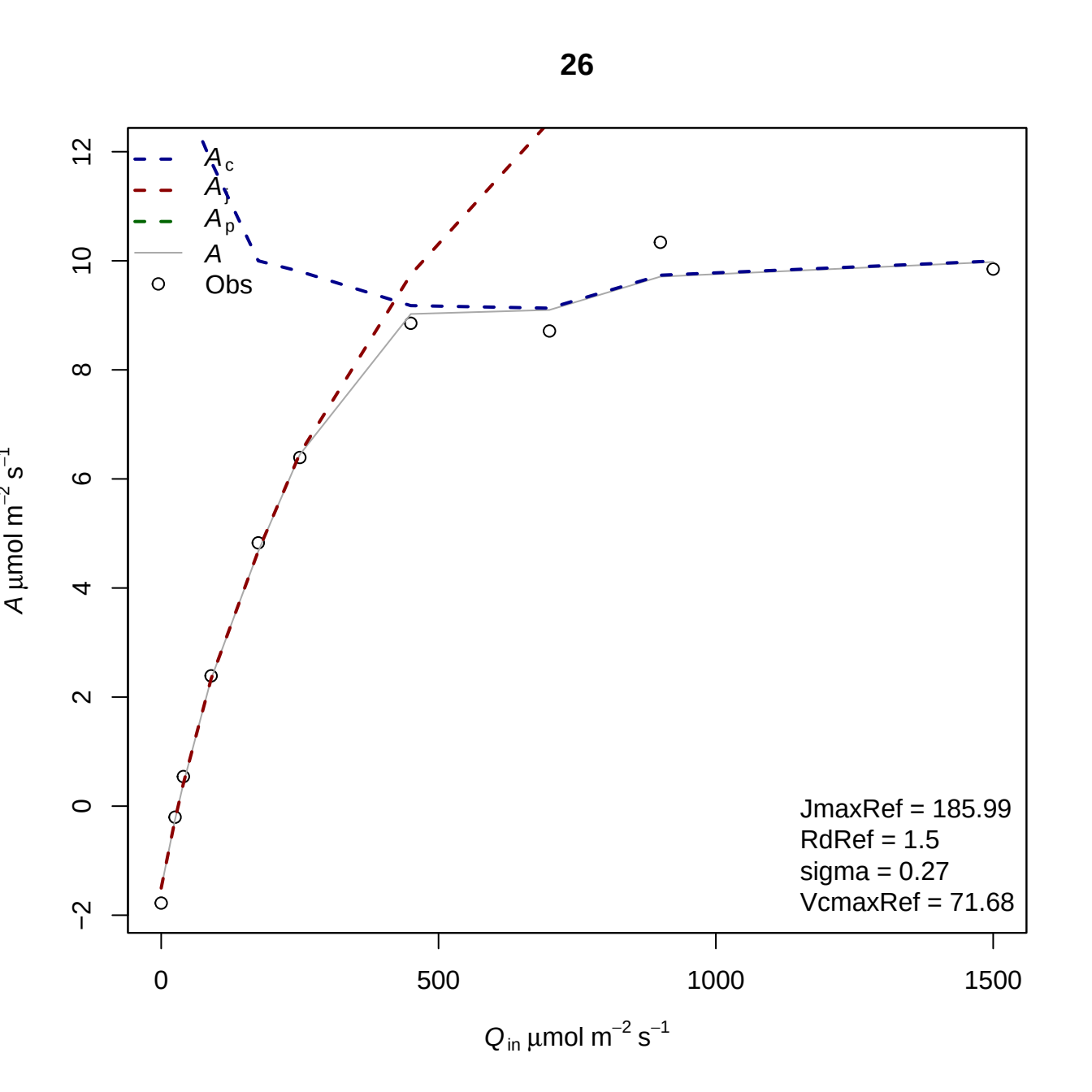

21
A_c
A_j
A_p
A
Obs
JmaxRef = 217.27
RdRef = 1.41
sigma = 0.42
VcmaxRef = 94.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

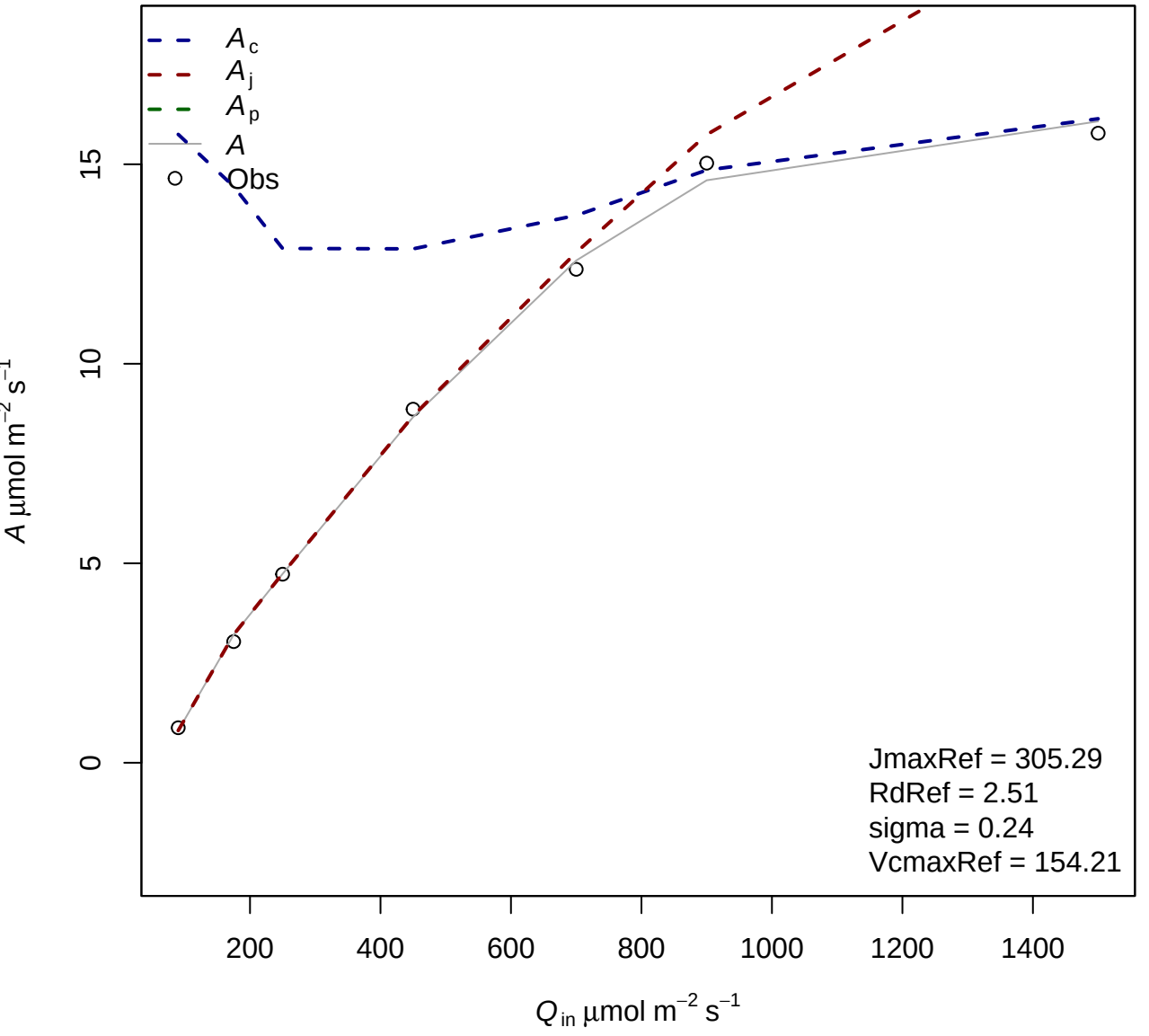

22
A_c
A_j
A_p
A
Obs
JmaxRef = 222.97
RdRef = 1.27
sigma = 0.28
VcmaxRef = 98.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

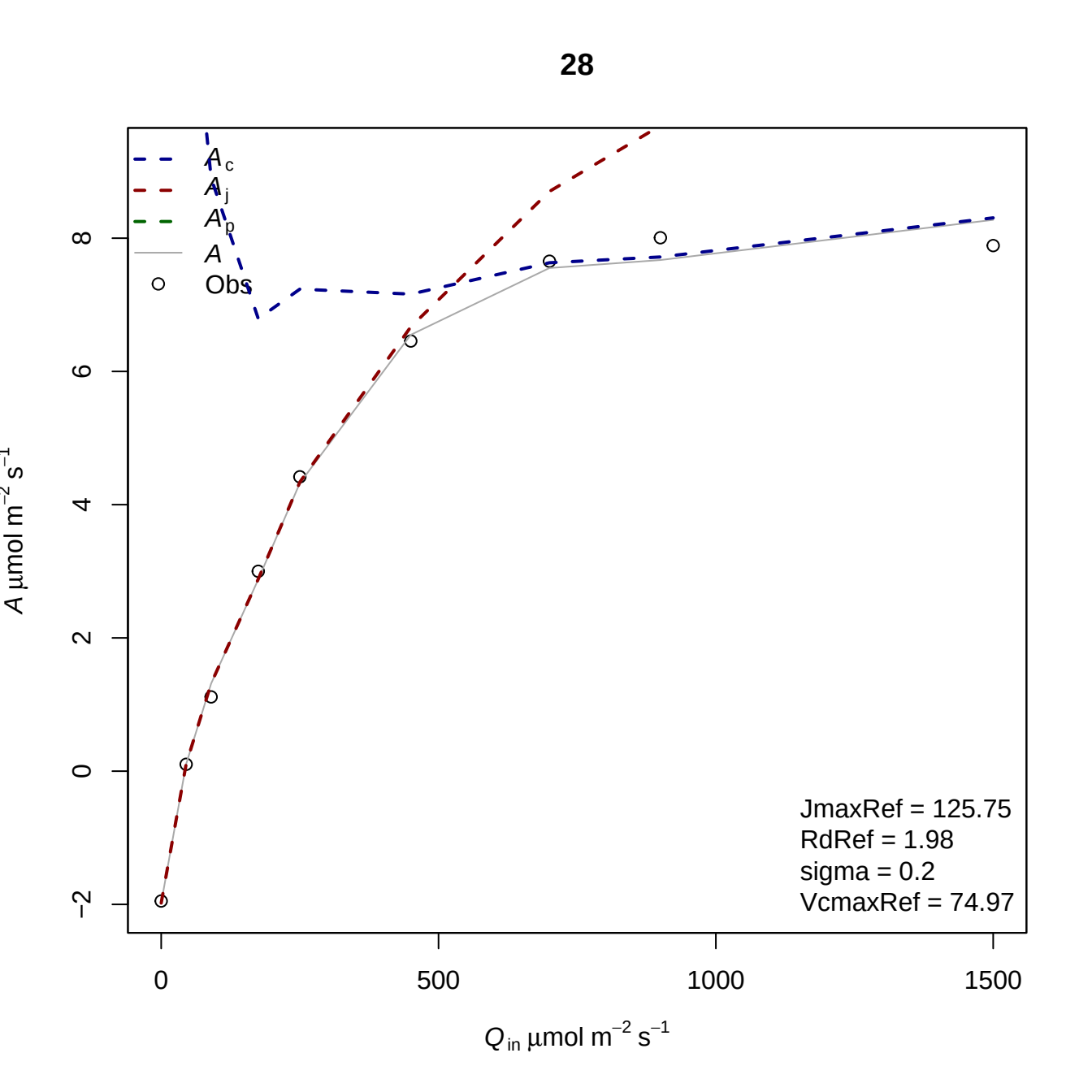

23
A_c
A_j
A_p
A
Obs
JmaxRef = 93.81
RdRef = 1.59
sigma = 0.4
VcmaxRef = 59.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

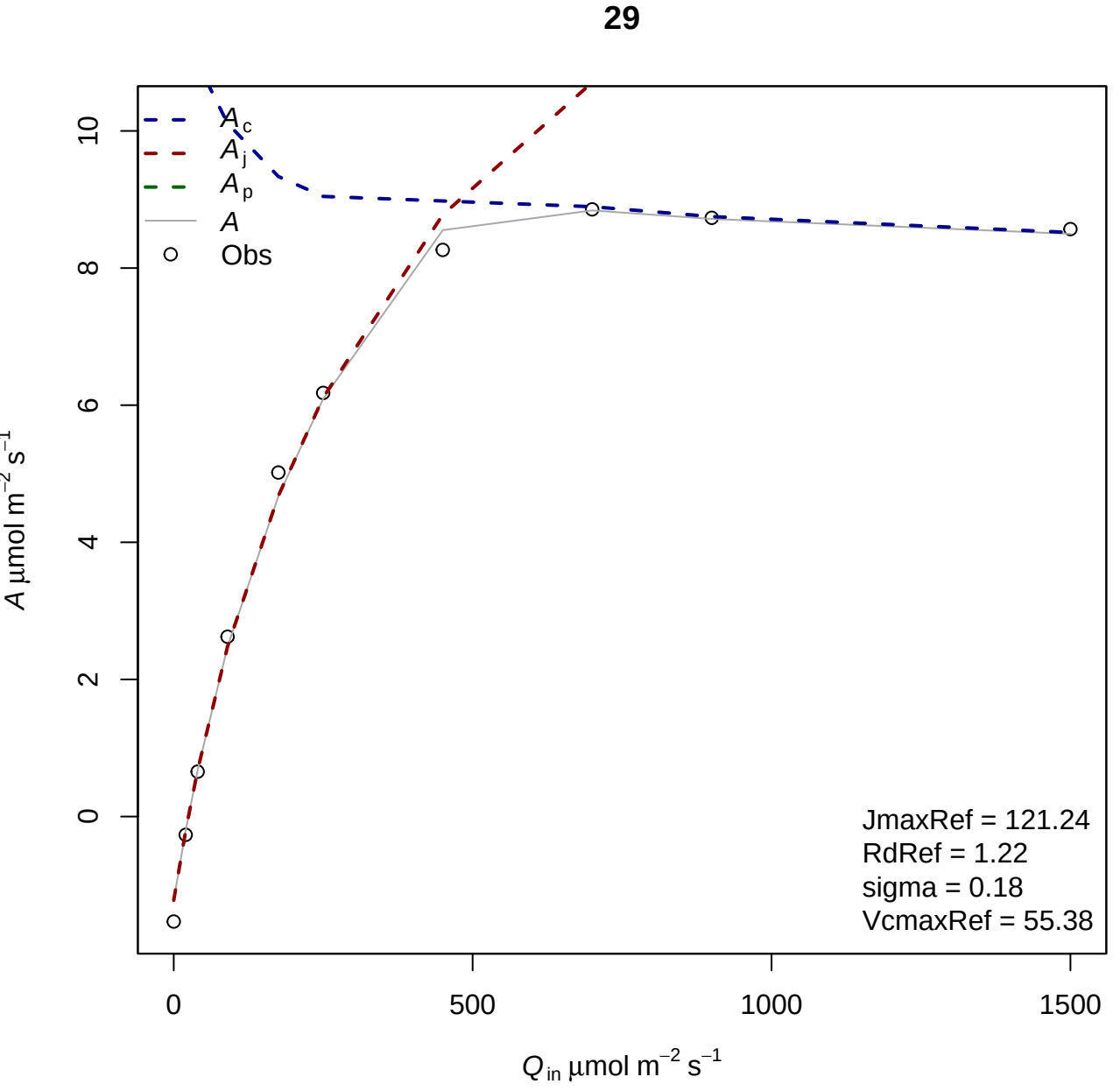

24
A_c
A_j
A_p
A
Obs
JmaxRef = 75.97
RdRef = 1.48
sigma = 0.97
VcmaxRef = 303.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

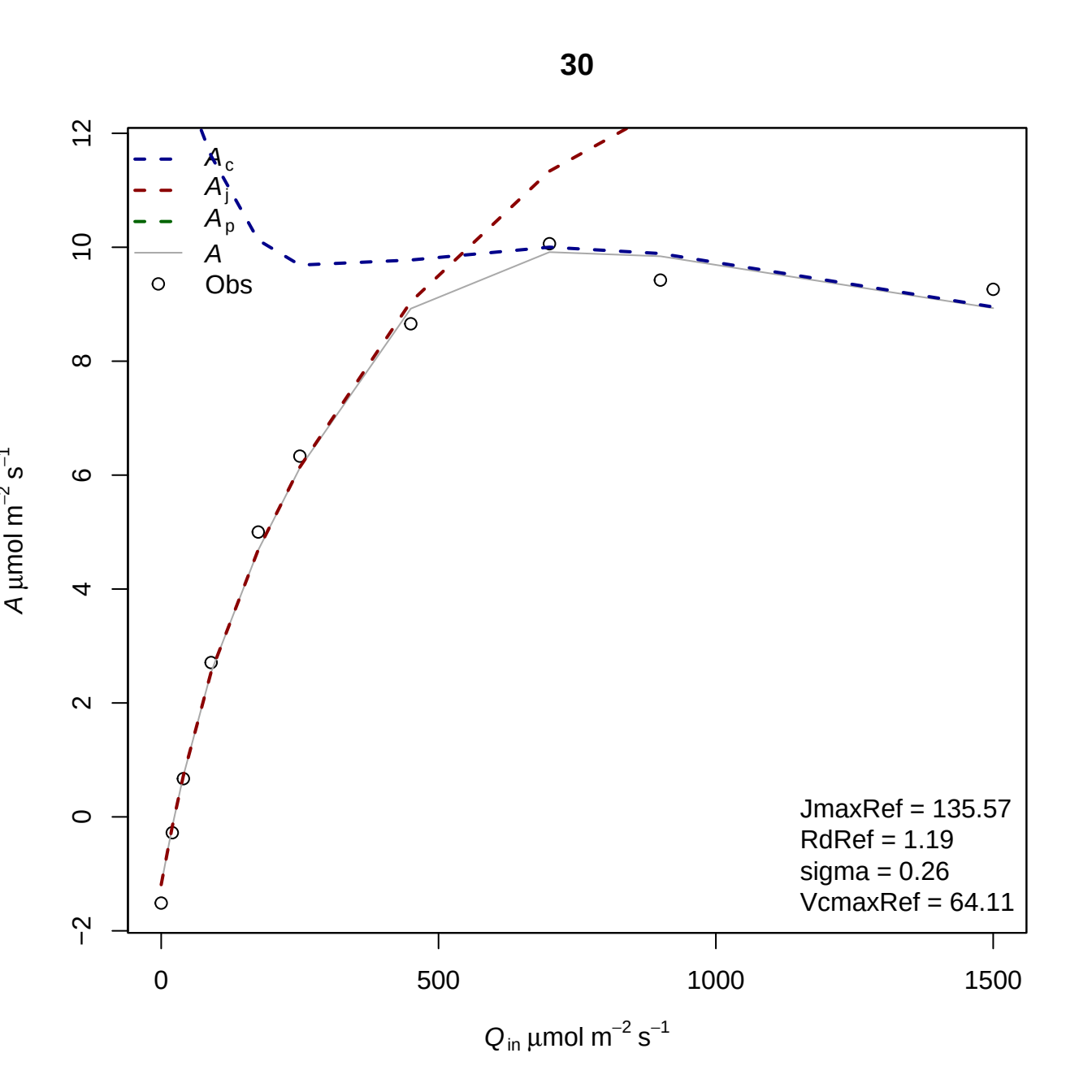

25
A_c
A_j
A_p
A
Obs
JmaxRef = 157.69
RdRef = 1.18
sigma = 0.37
VcmaxRef = 77.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

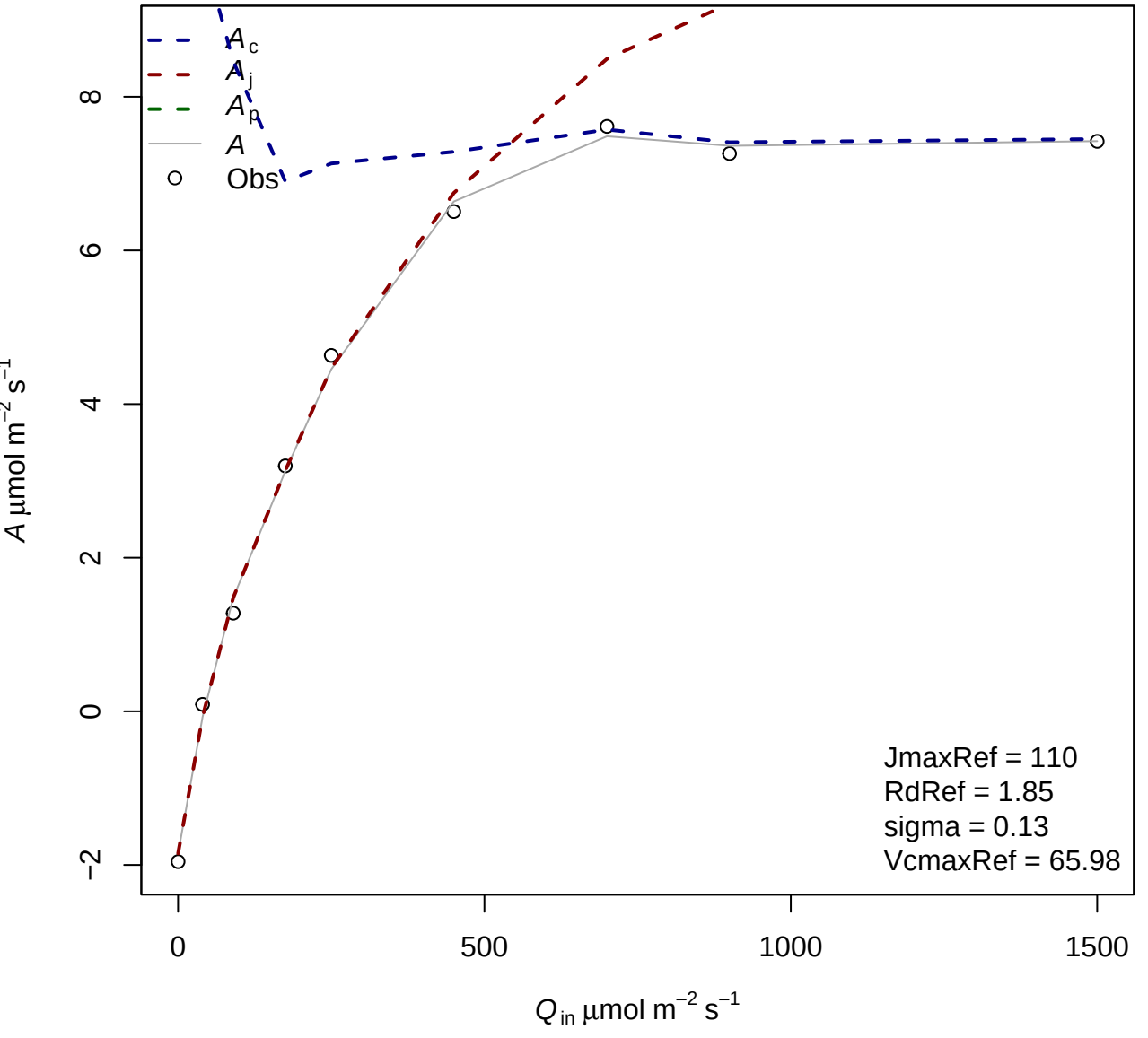

26
A_c
A_j
A_p
A
Obs
JmaxRef = 185.99
RdRef = 1.5
sigma = 0.27
VcmaxRef = 71.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

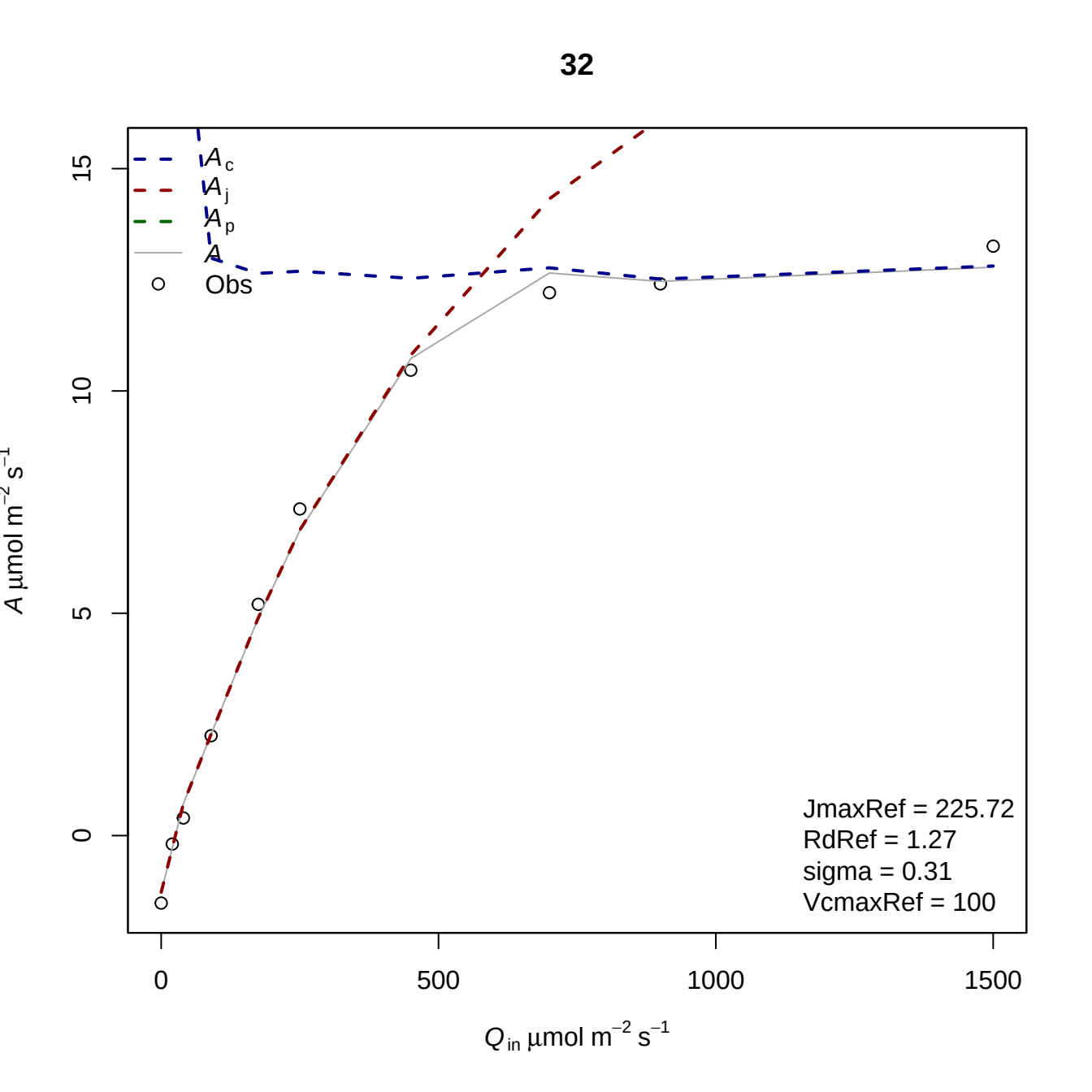

A_c
A_j
A_p
A
Obs
JmaxRef = 305.29
RdRef = 2.51
sigma = 0.24
VcmaxRef = 154.21
0
5
10
15
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

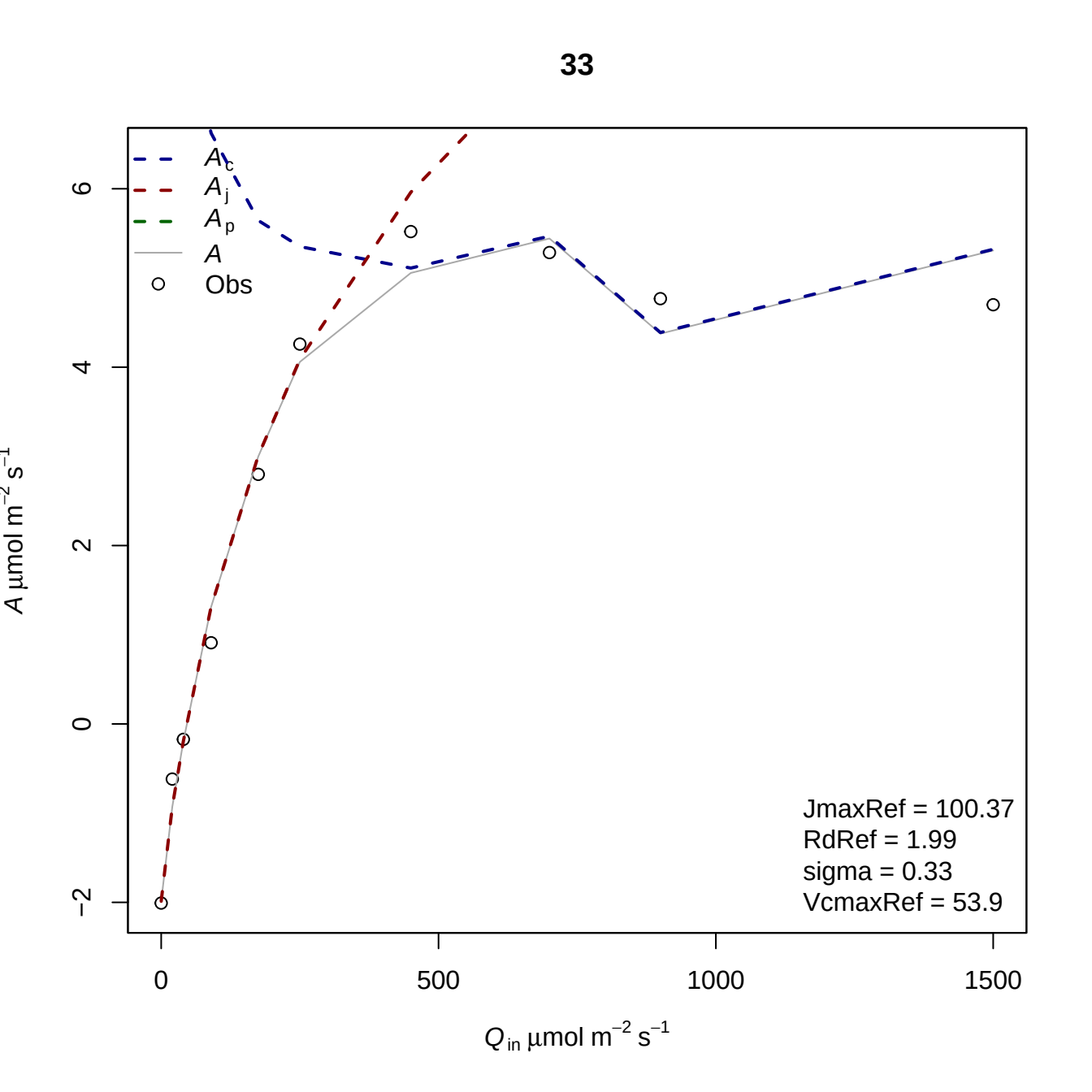

28
A_c
A_j
A_p
A
Obs
JmaxRef = 125.75
RdRef = 1.98
sigma = 0.2
VcmaxRef = 74.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

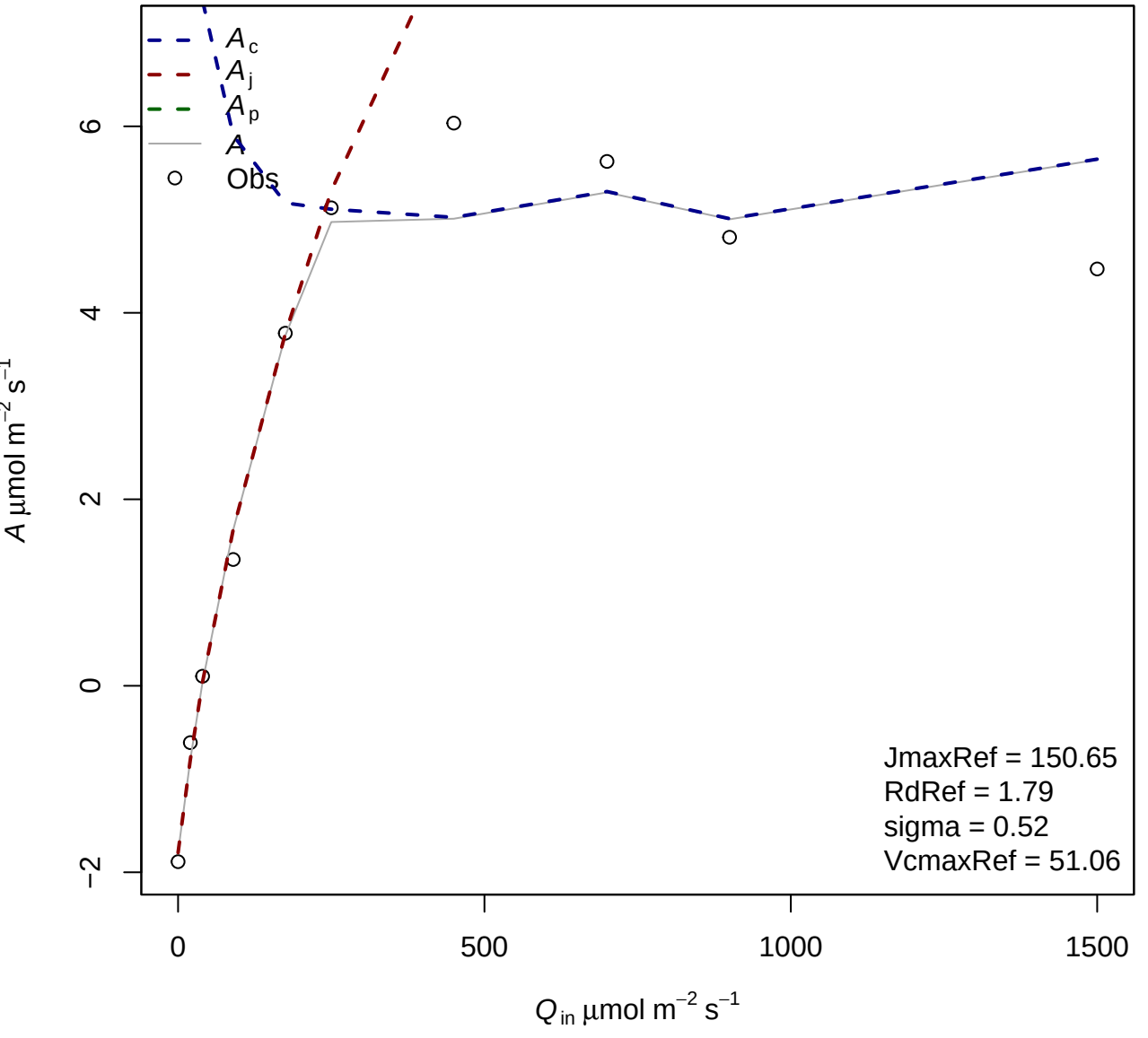

29
A_c
A_j
A_p
A
Obs
JmaxRef = 121.24
RdRef = 1.22
sigma = 0.18
VcmaxRef = 55.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

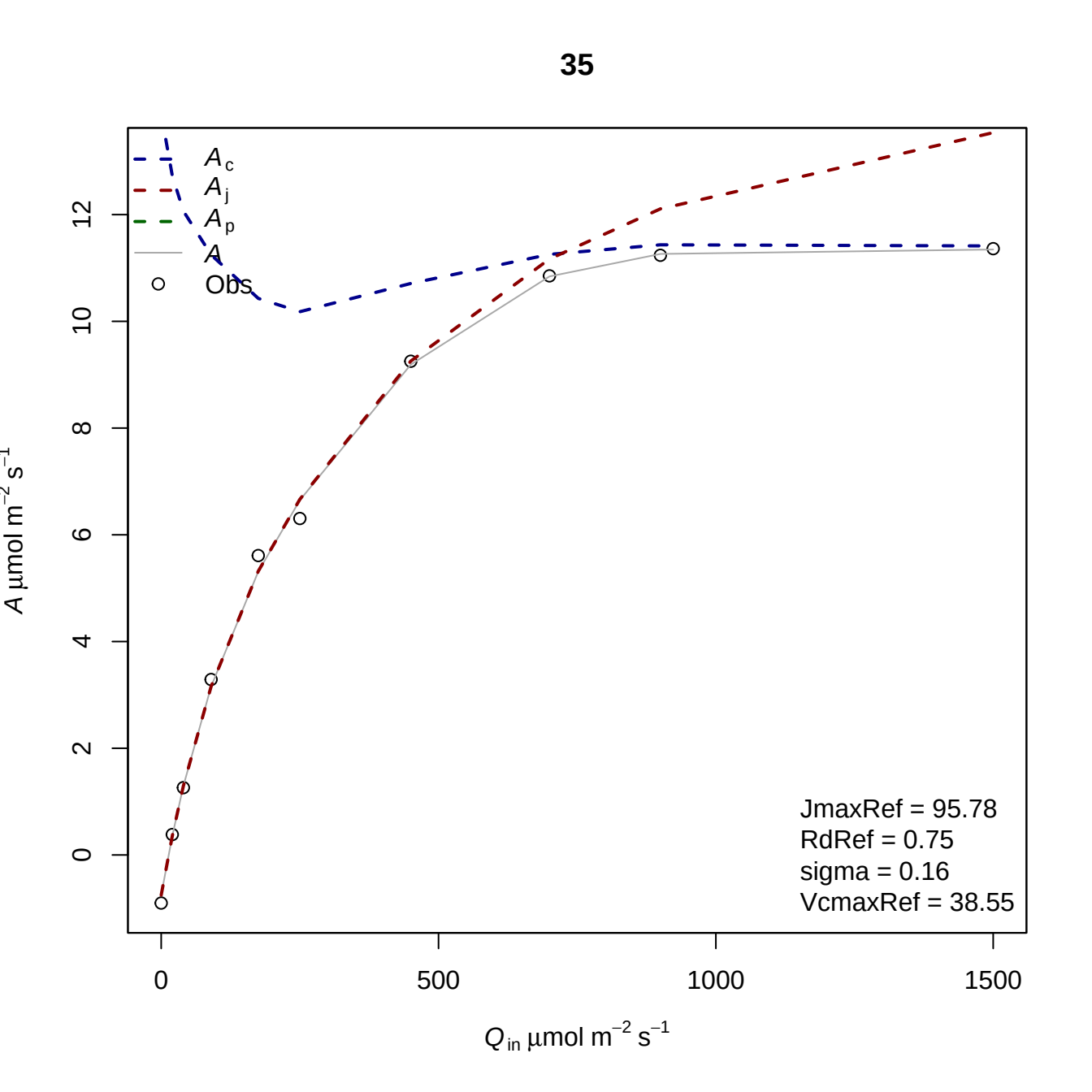

30
A_c
A_j
A_p
A
Obs
JmaxRef = 135.57
RdRef = 1.19
sigma = 0.26
VcmaxRef = 64.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

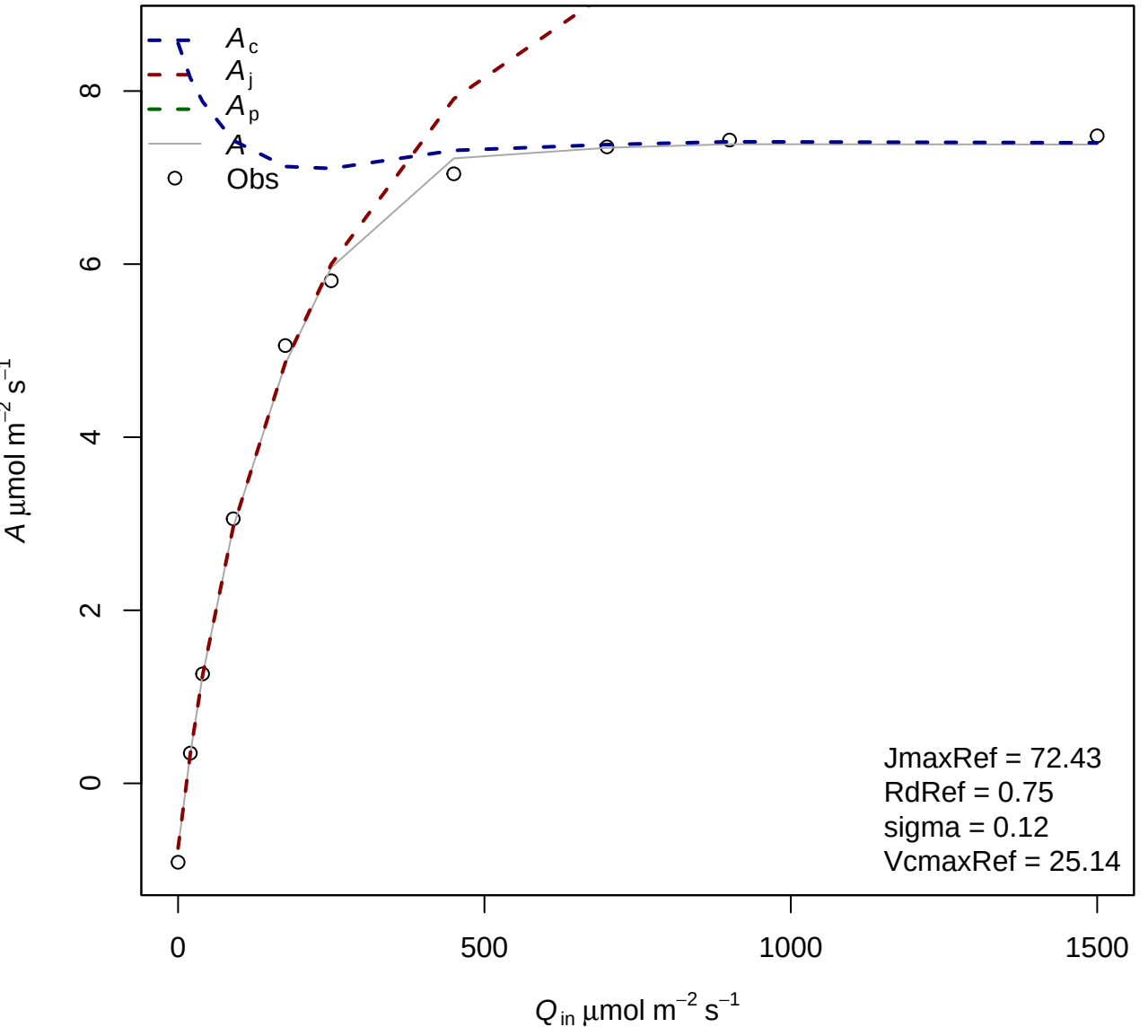

A_c
A_j
A_p
A
Obs
JmaxRef = 110
RdRef = 1.85
sigma = 0.13
VcmaxRef = 65.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

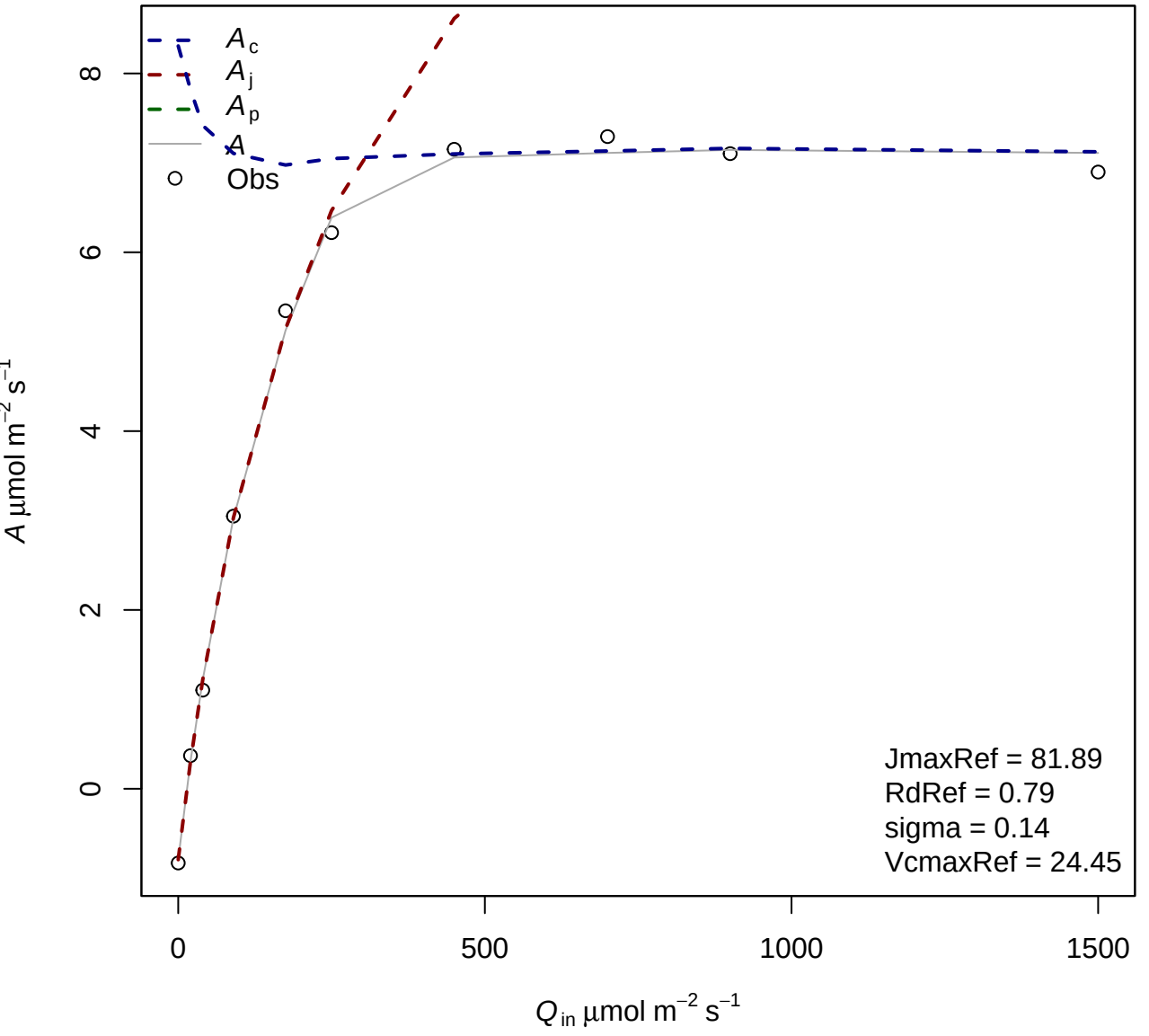

32
A_c
A_j
A_p
A
Obs
JmaxRef = 225.72
RdRef = 1.27
sigma = 0.31
VcmaxRef = 100
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

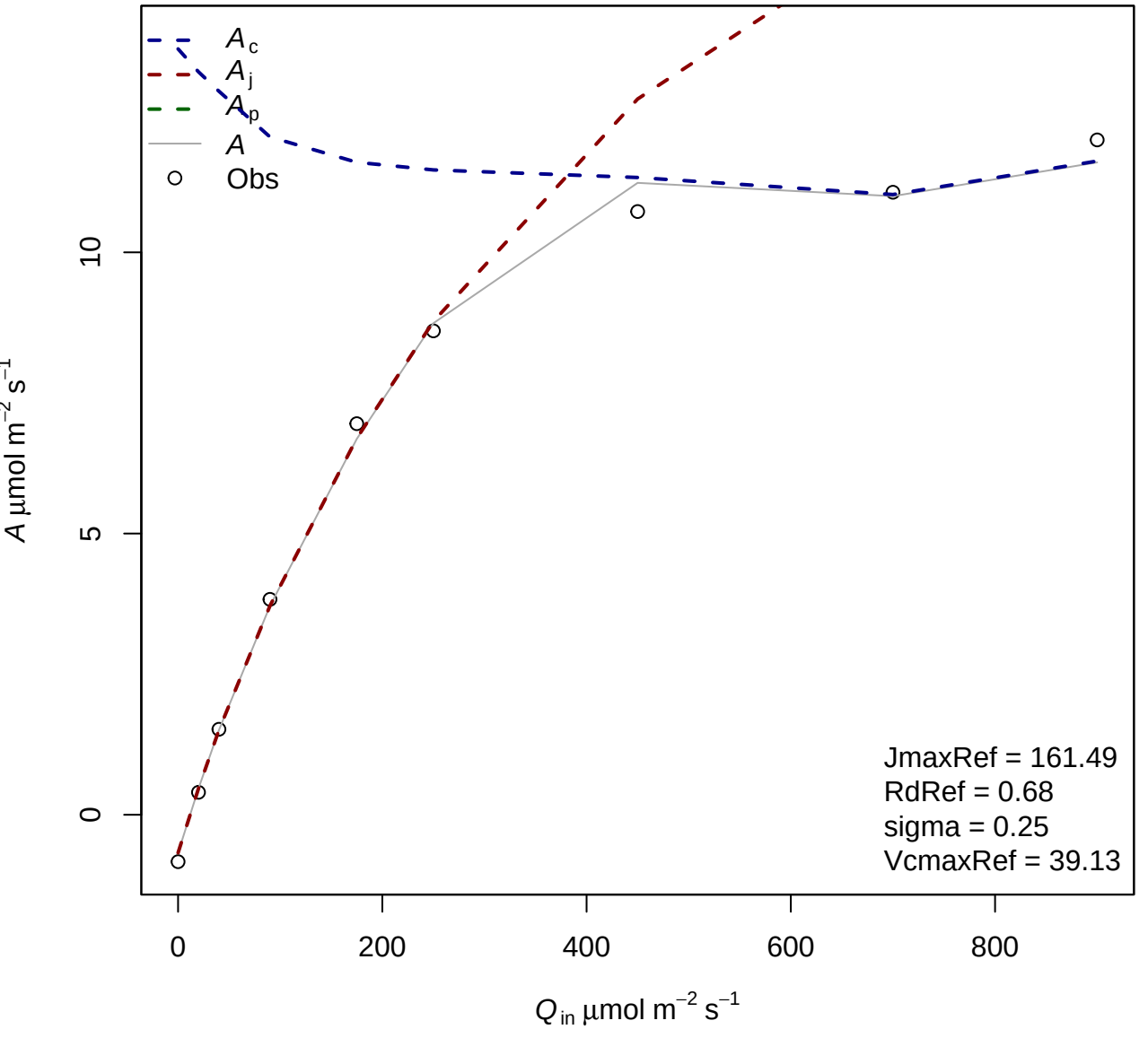

33
A_c
A_j
A_p
A
Obs
JmaxRef = 100.37
RdRef = 1.99
sigma = 0.33
VcmaxRef = 53.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

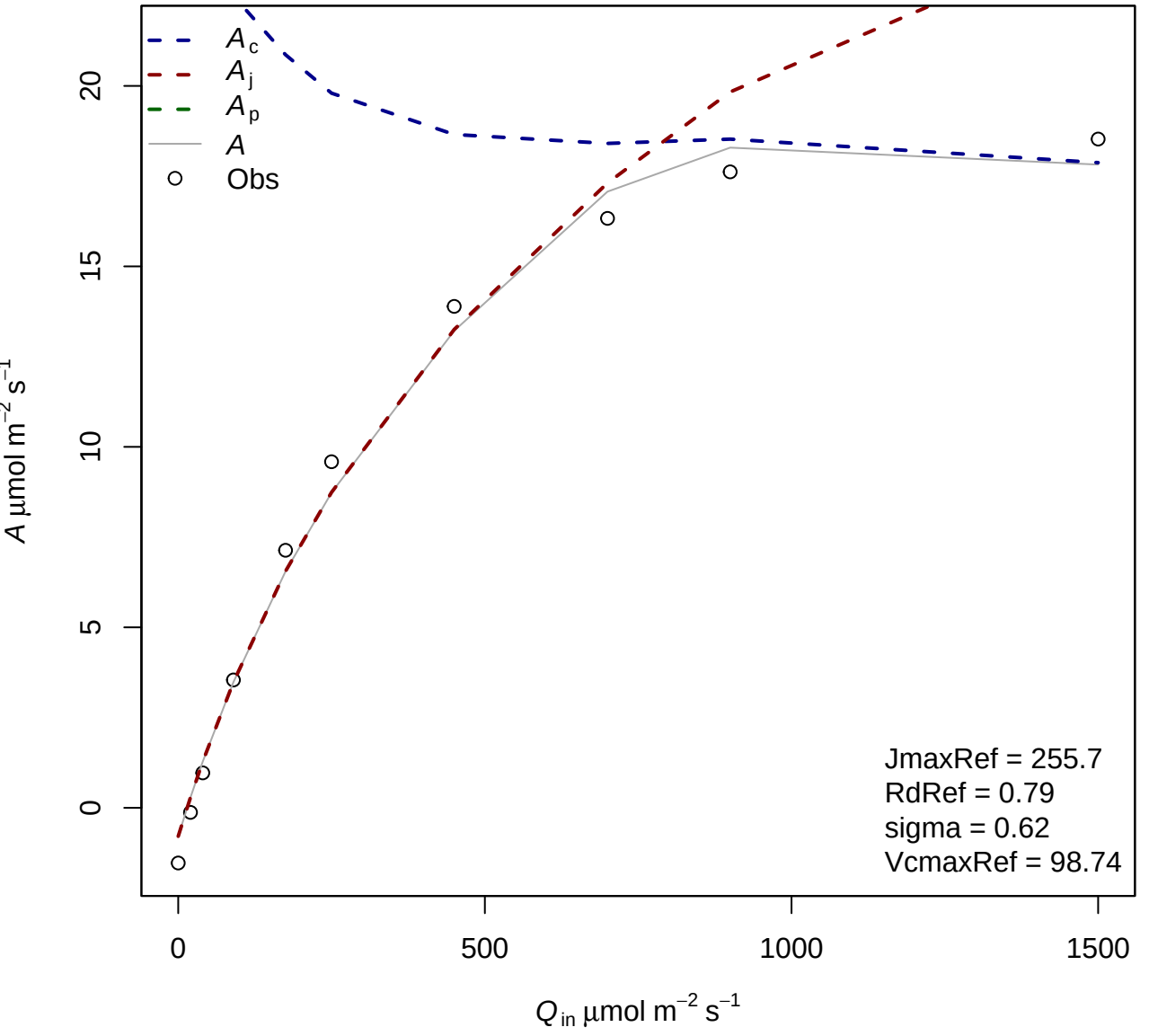

A_c
A_j
A_p
A
Obs
JmaxRef = 150.65
RdRef = 1.79
sigma = 0.52
VcmaxRef = 51.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

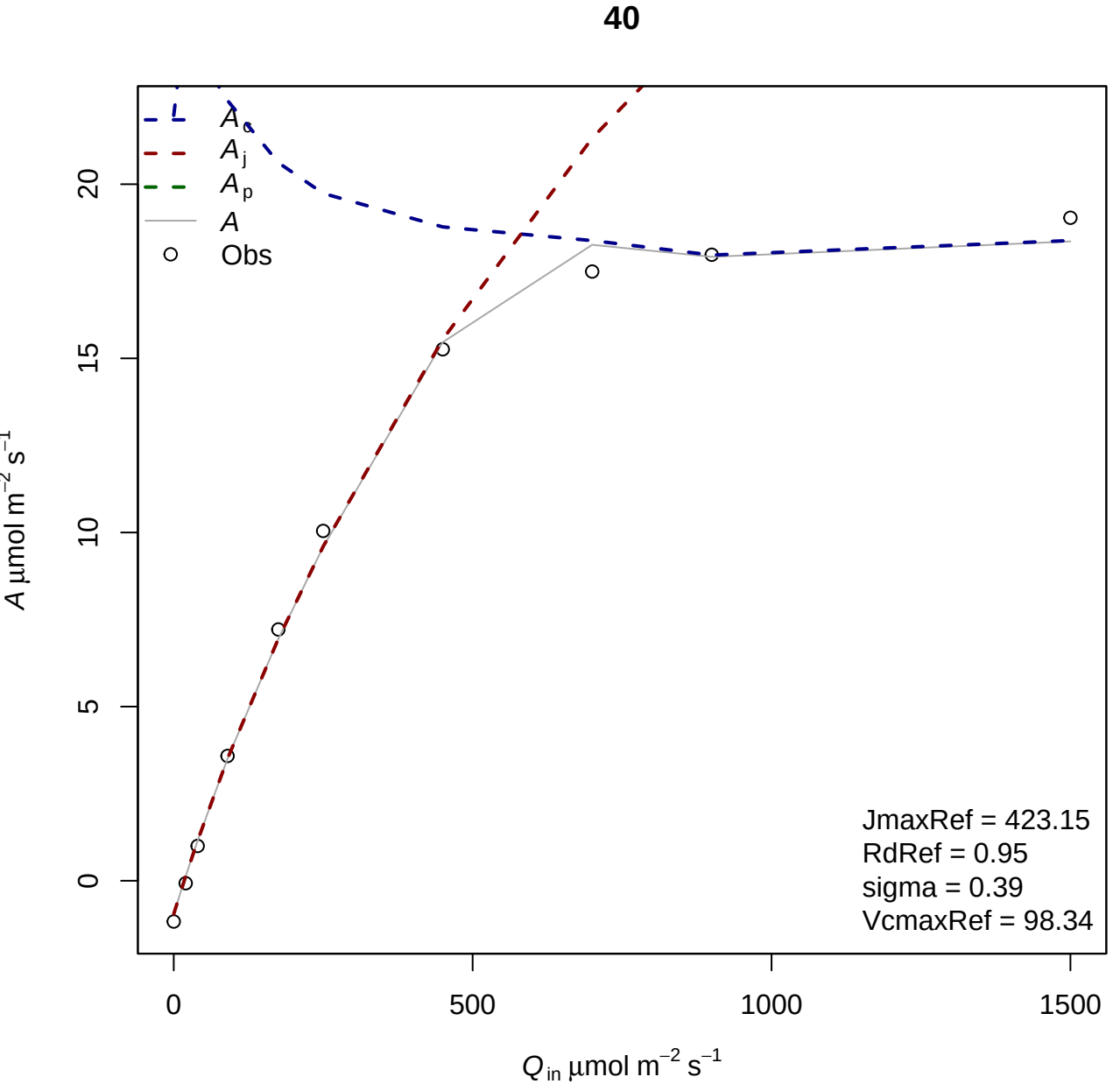

35
A_c
A_j
A_p
A
Obs
JmaxRef = 95.78
RdRef = 0.75
sigma = 0.16
VcmaxRef = 38.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

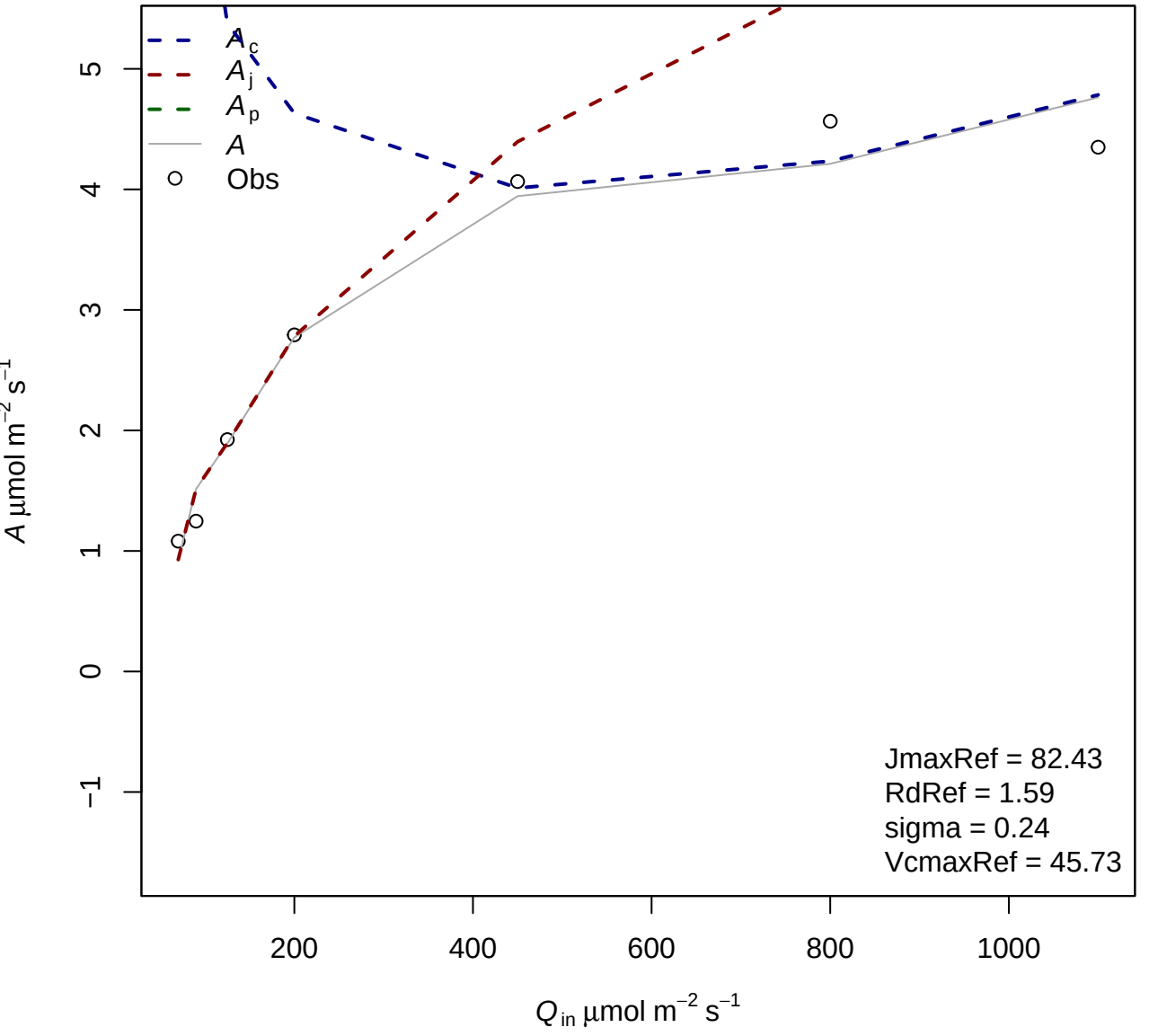

A_c
A_j
A_p
A
Obs
JmaxRef = 72.43
RdRef = 0.75
sigma = 0.12
VcmaxRef = 25.14
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

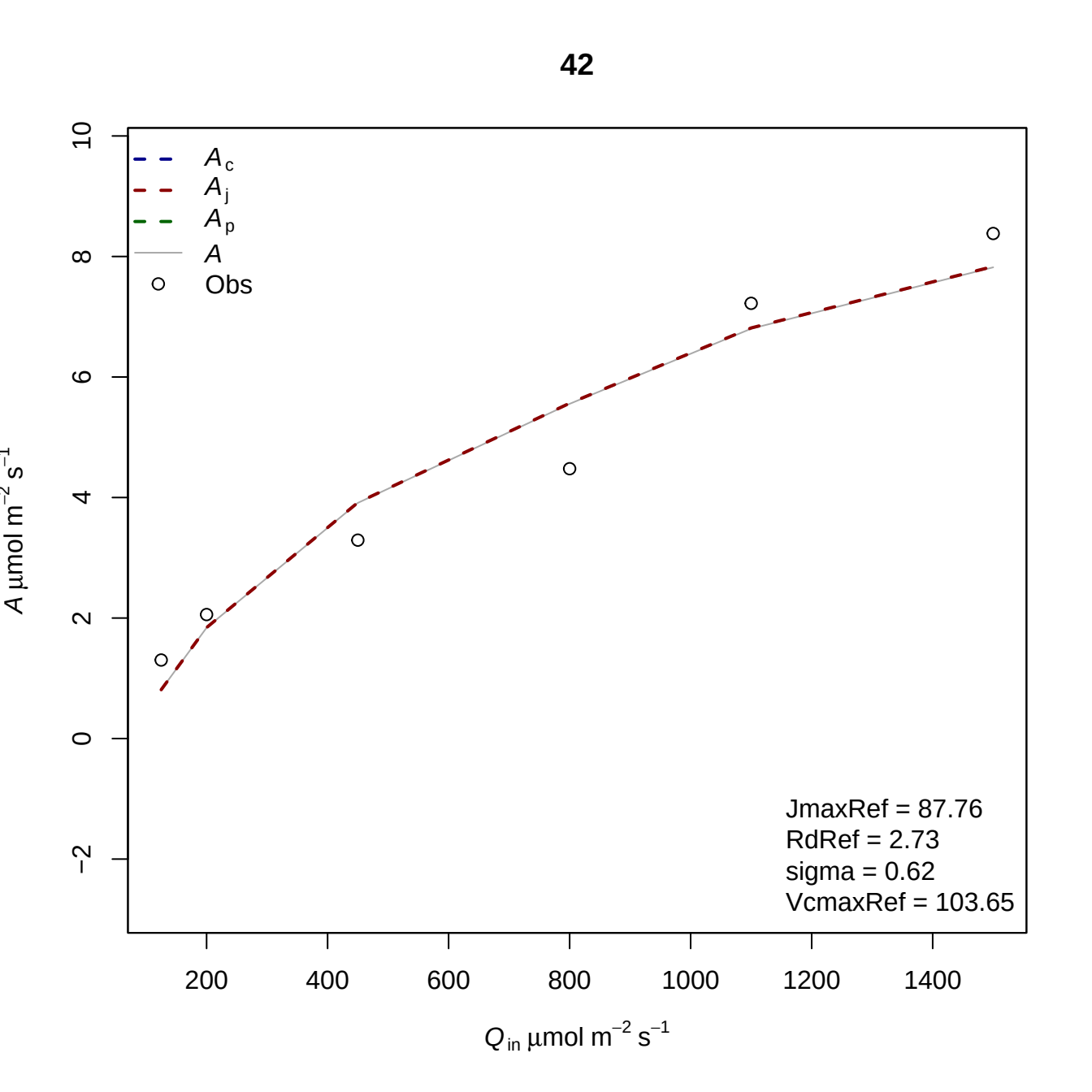

A_c
A_j
A_p
A
Obs
JmaxRef = 81.89
RdRef = 0.79
sigma = 0.14
VcmaxRef = 24.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

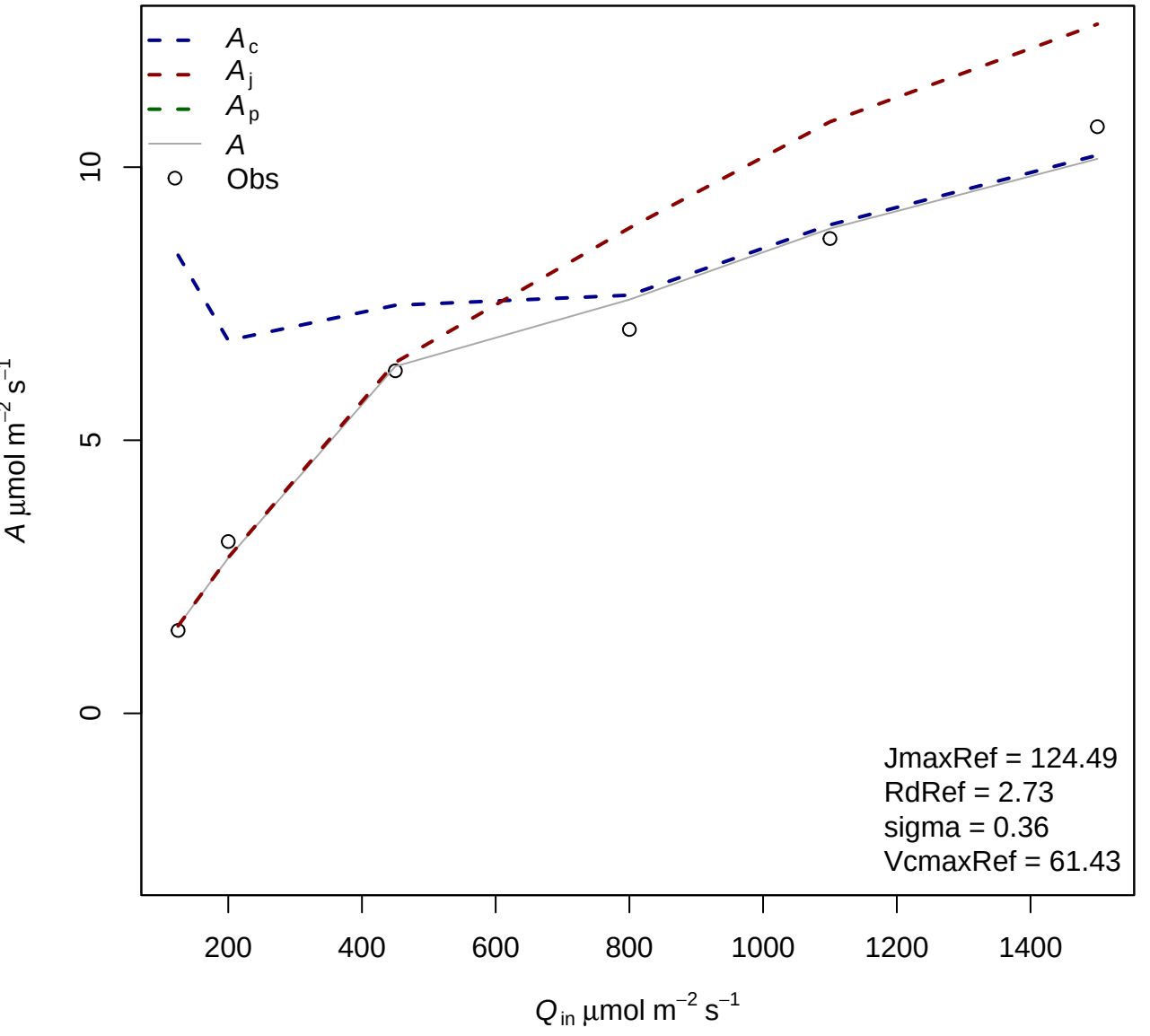

A_c
A_j
A_p
A
Obs
JmaxRef = 161.49
RdRef = 0.68
sigma = 0.25
VcmaxRef = 39.13
A µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
Q_{in} µmol m^{-2} s^{-1}

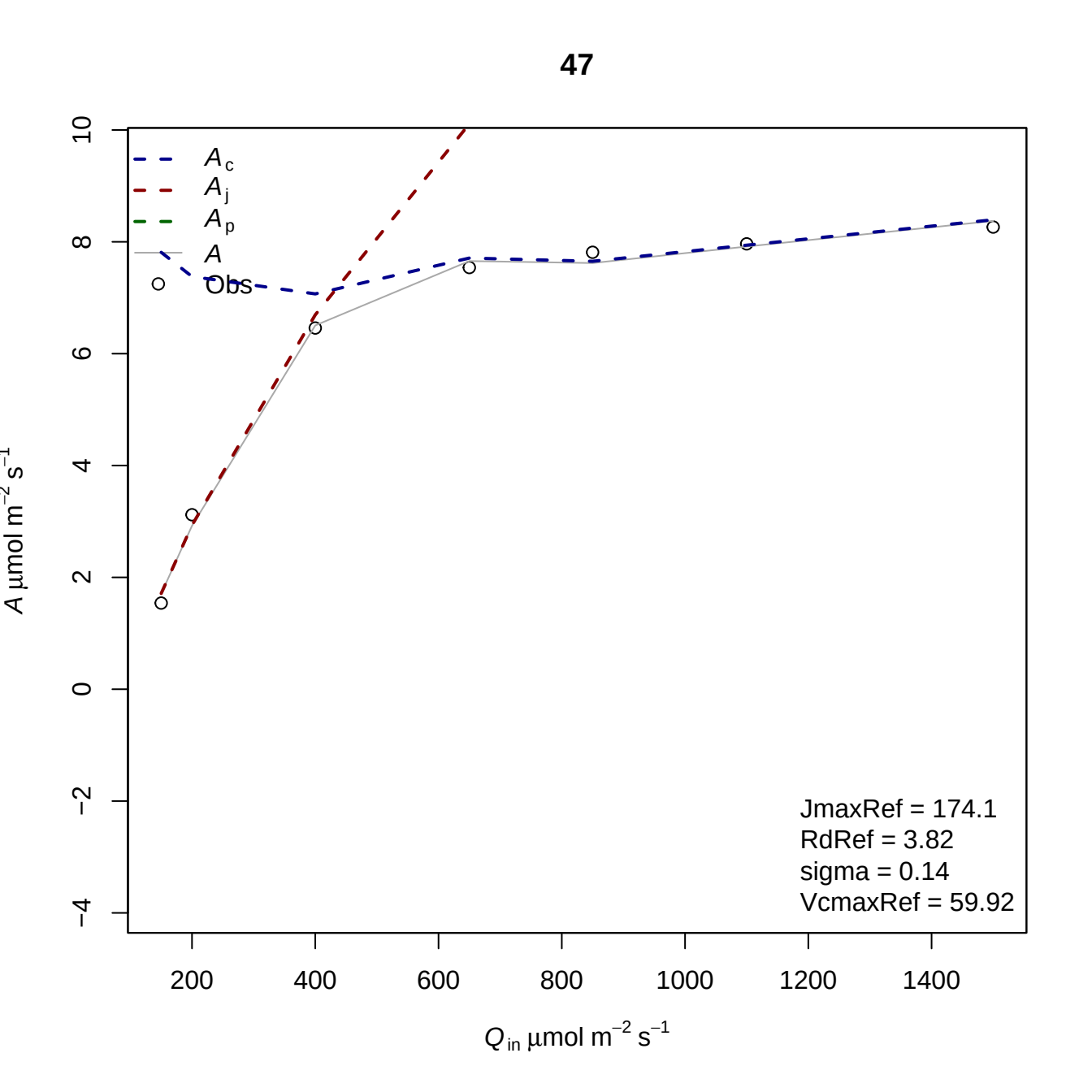

A_c
A_j
A_p
A
Obs
JmaxRef = 255.7
RdRef = 0.79
sigma = 0.62
VcmaxRef = 98.74
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

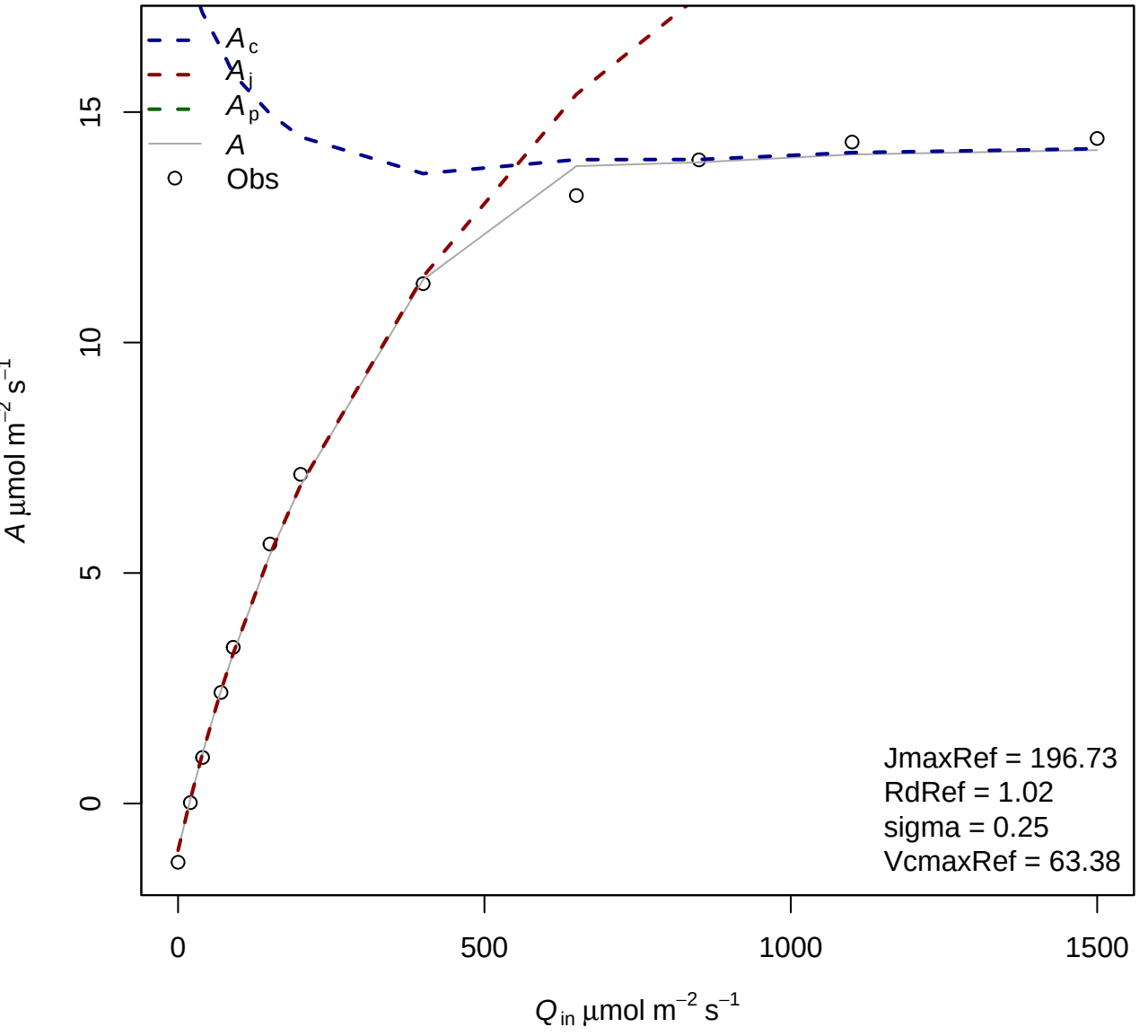

40
A_c
A_j
A_p
A
Obs
JmaxRef = 423.15
RdRef = 0.95
sigma = 0.39
VcmaxRef = 98.34
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

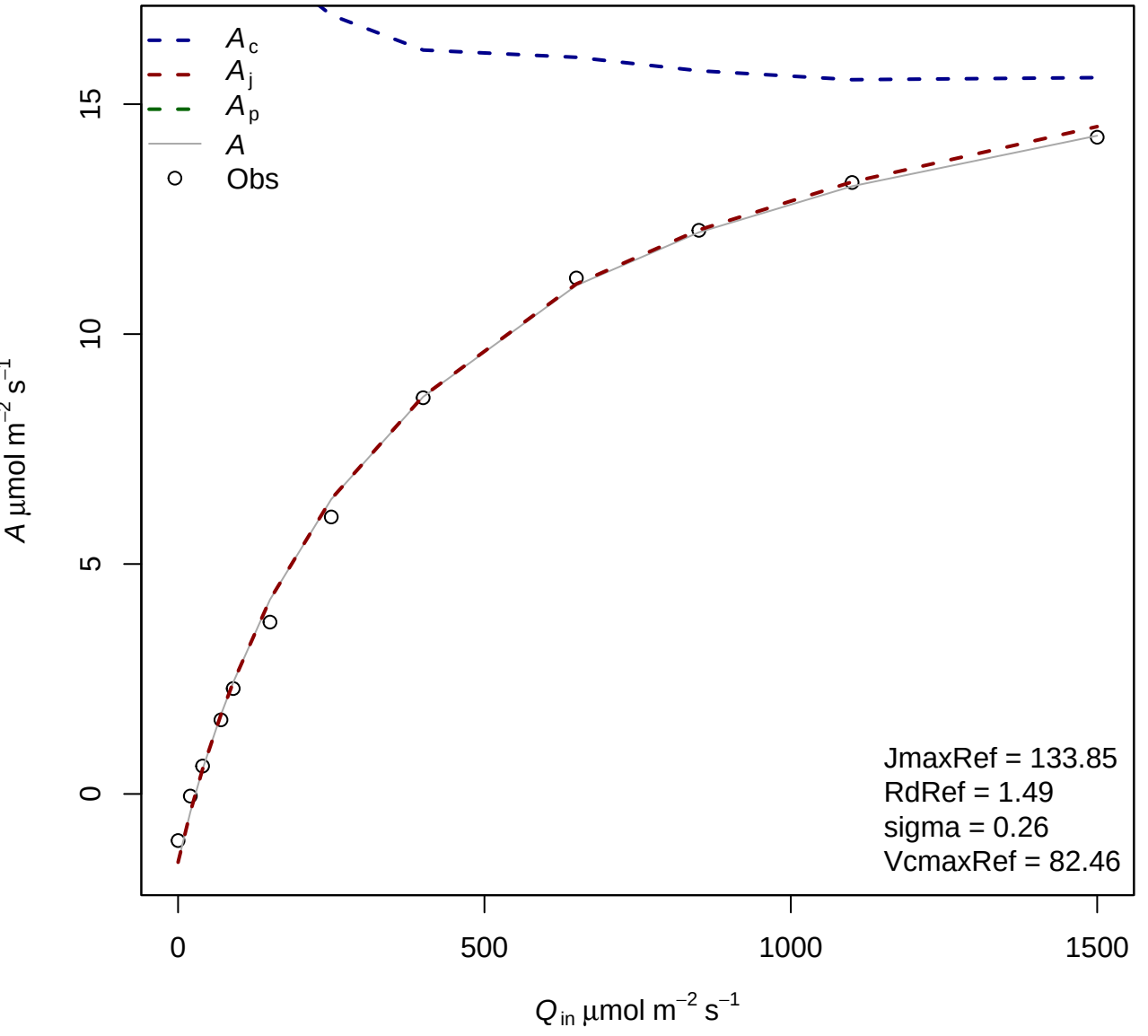

A_c
A_j
A_p
A
Obs
JmaxRef = 82.43
RdRef = 1.59
sigma = 0.24
VcmaxRef = 45.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
200
400
600
800
1000

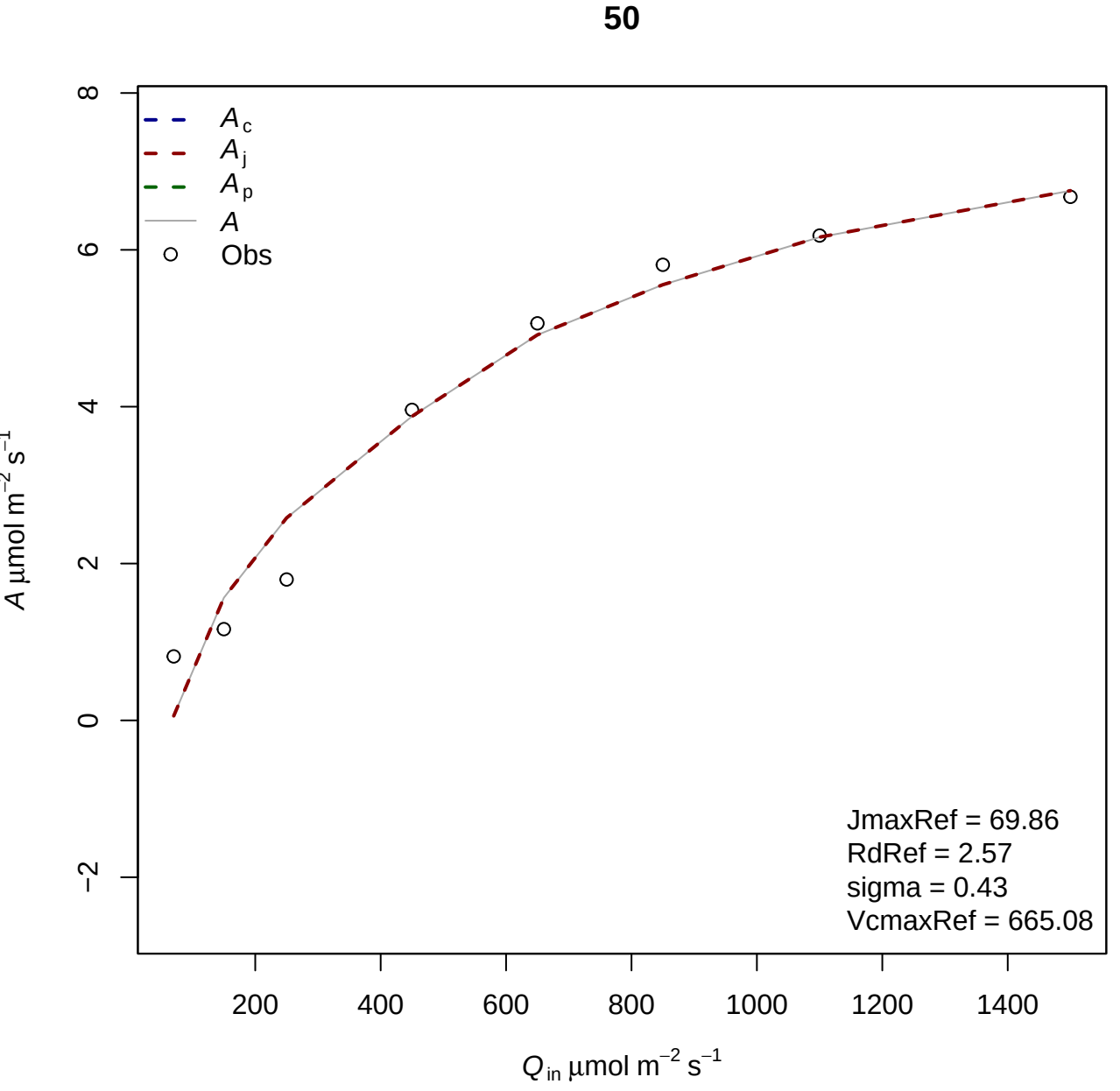

42
A_c
A_j
A_p
A
Obs
JmaxRef = 87.76
RdRef = 2.73
sigma = 0.62
VcmaxRef = 103.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
200
400
600
800
1000
1200
1400

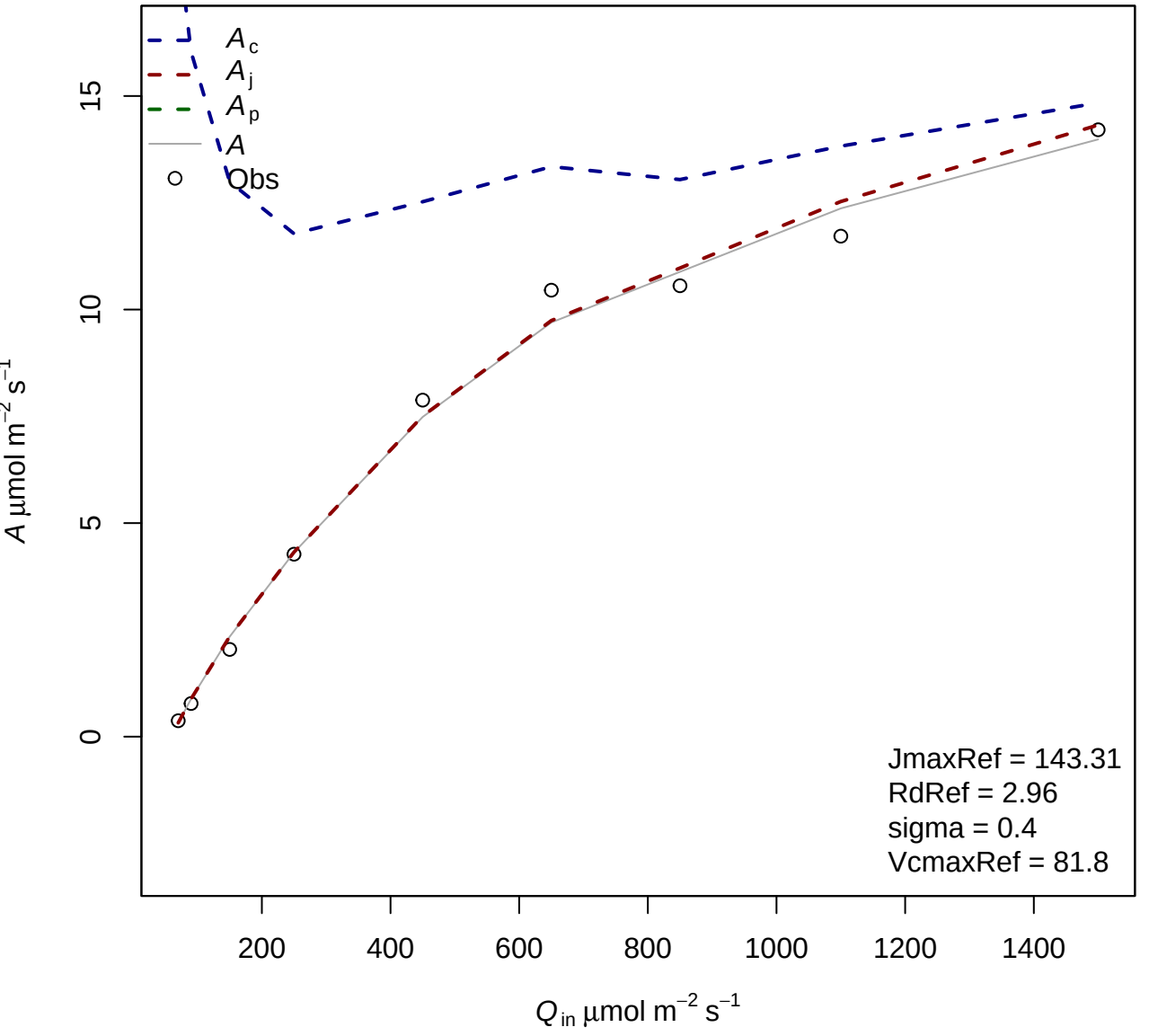

A_c
A_j
A_p
A
Obs
JmaxRef = 124.49
RdRef = 2.73
sigma = 0.36
VcmaxRef = 61.43
0
5
10
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

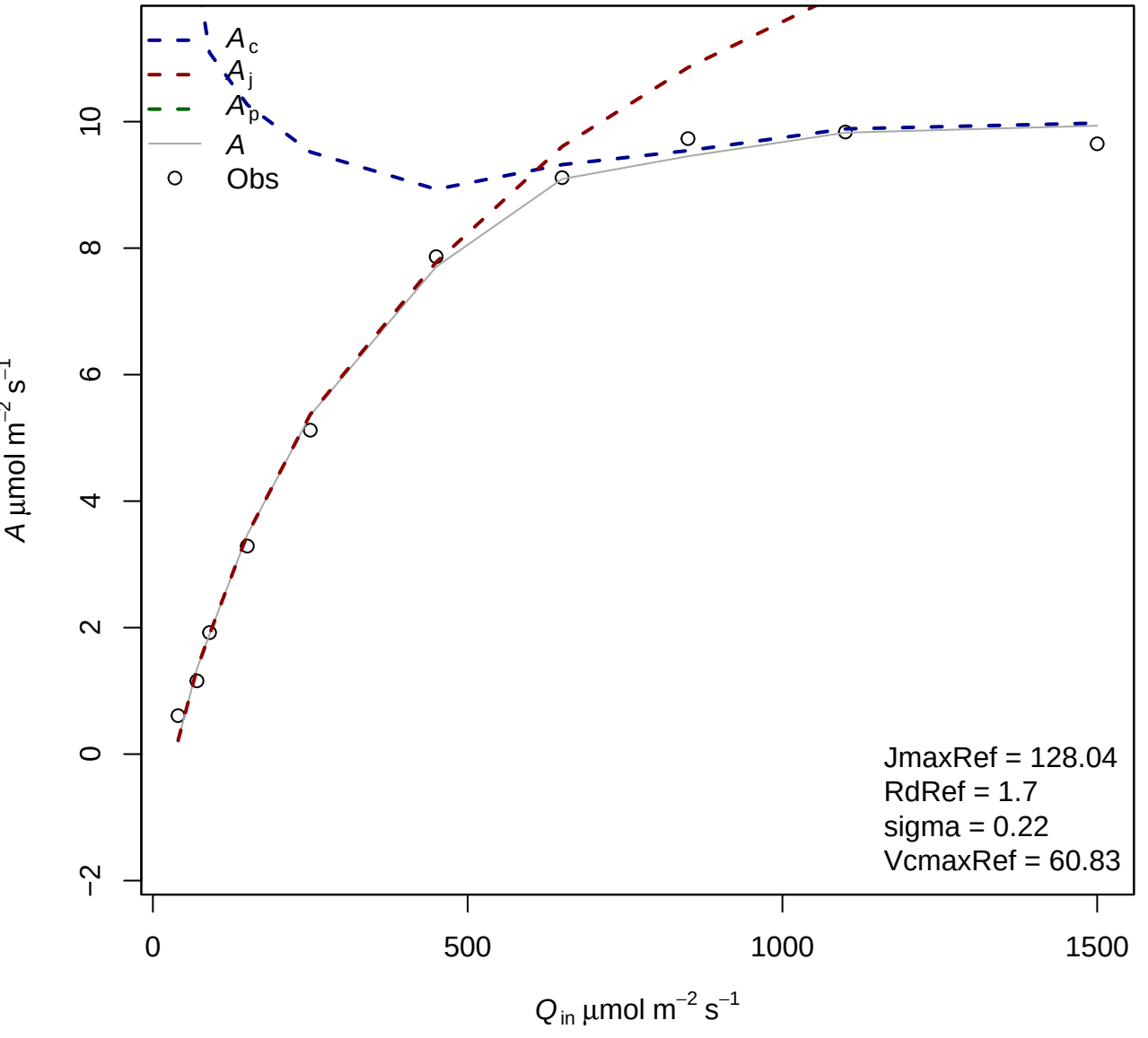

47
A_c
A_j
A_p
A
Obs
JmaxRef = 174.1
RdRef = 3.82
sigma = 0.14
VcmaxRef = 59.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
200
400
600
800
1000
1200
1400

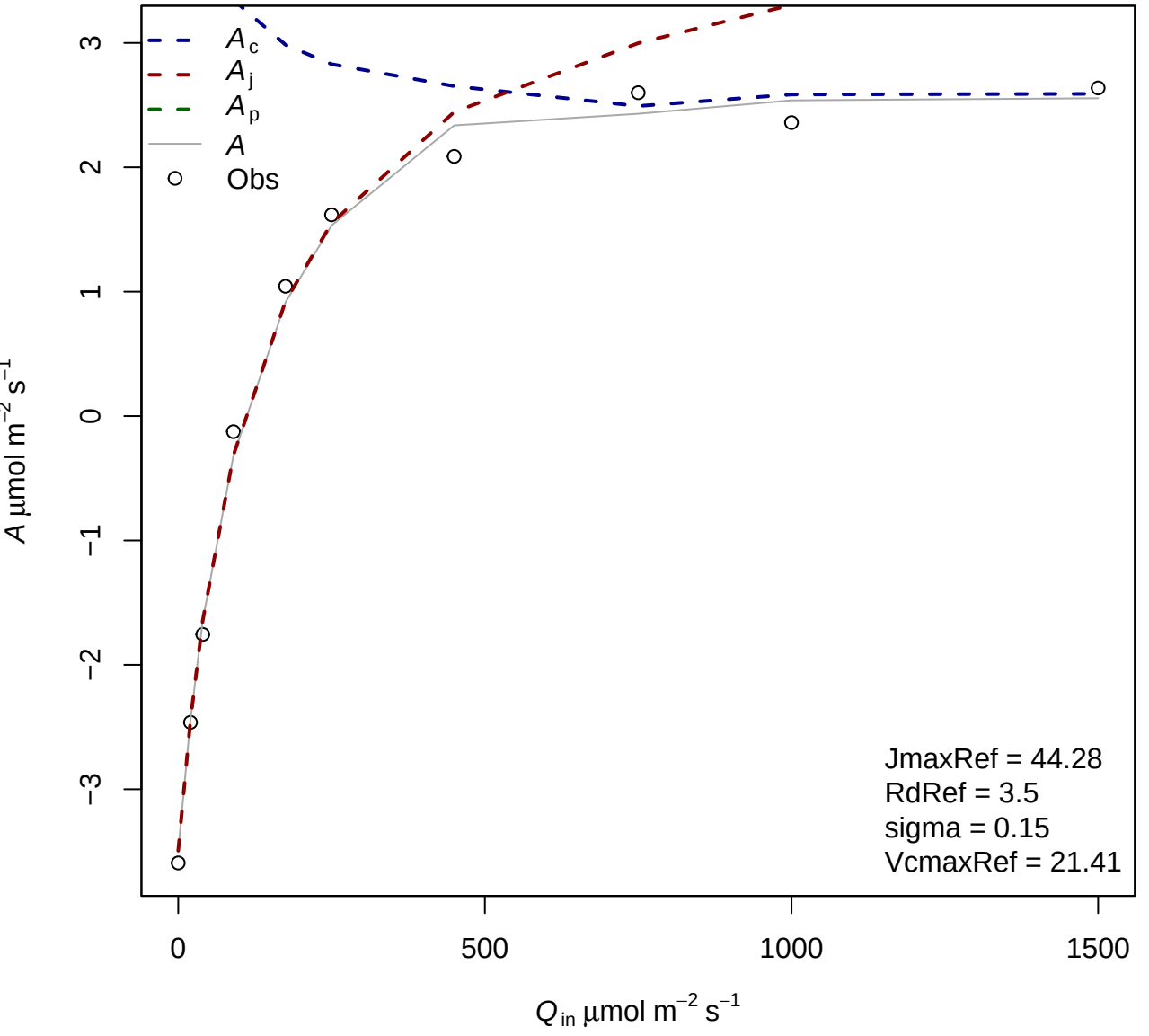

A_c
A_j
A_p
A
Obs
JmaxRef = 196.73
RdRef = 1.02
sigma = 0.25
VcmaxRef = 63.38
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

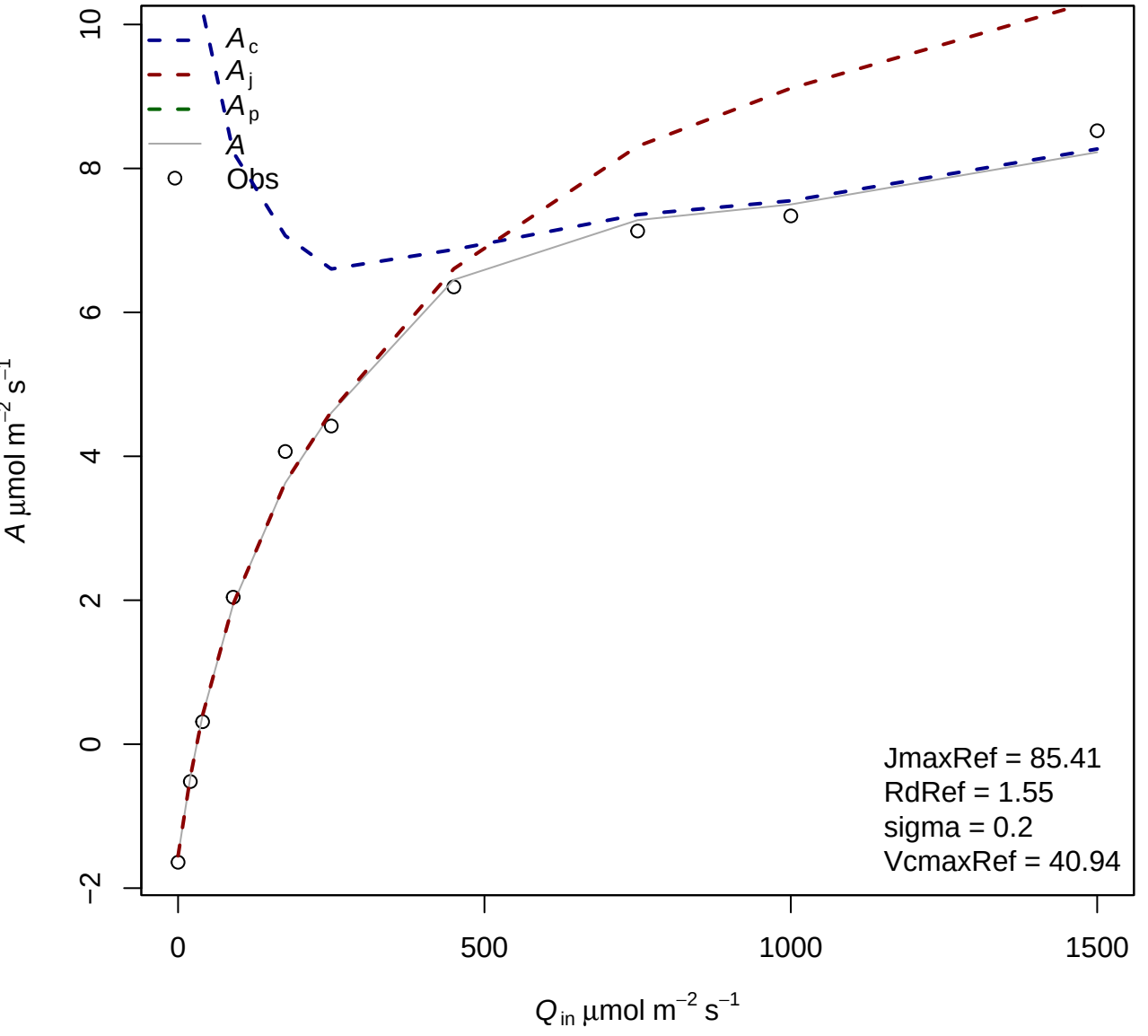

A_c
A_j
A_p
A
Obs
JmaxRef = 133.85
RdRef = 1.49
sigma = 0.26
VcmaxRef = 82.46
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

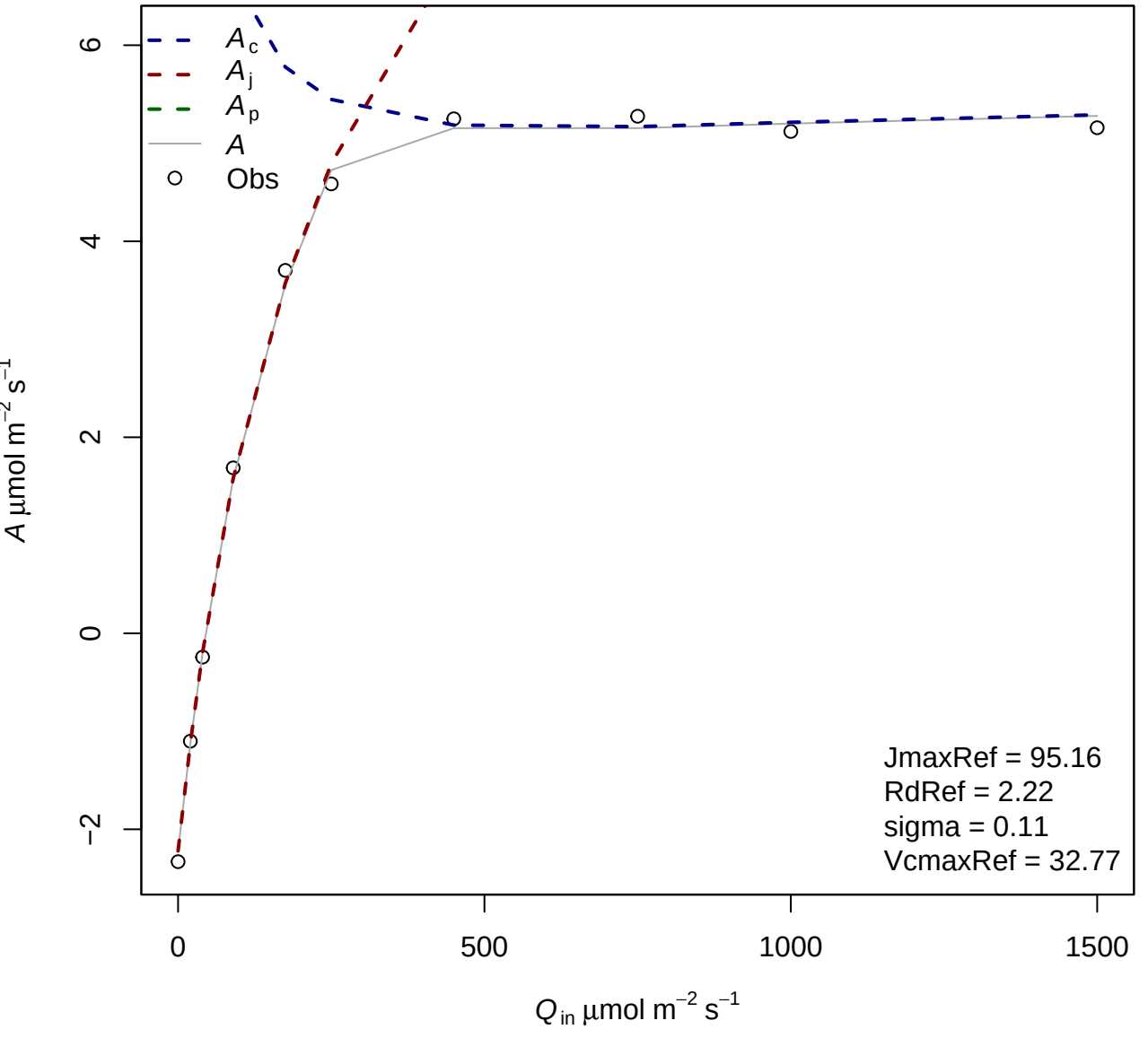

50
A_c
A_j
A_p
A
Obs
JmaxRef = 69.86
RdRef = 2.57
sigma = 0.43
VcmaxRef = 665.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
200
400
600
800
1000
1200
1400

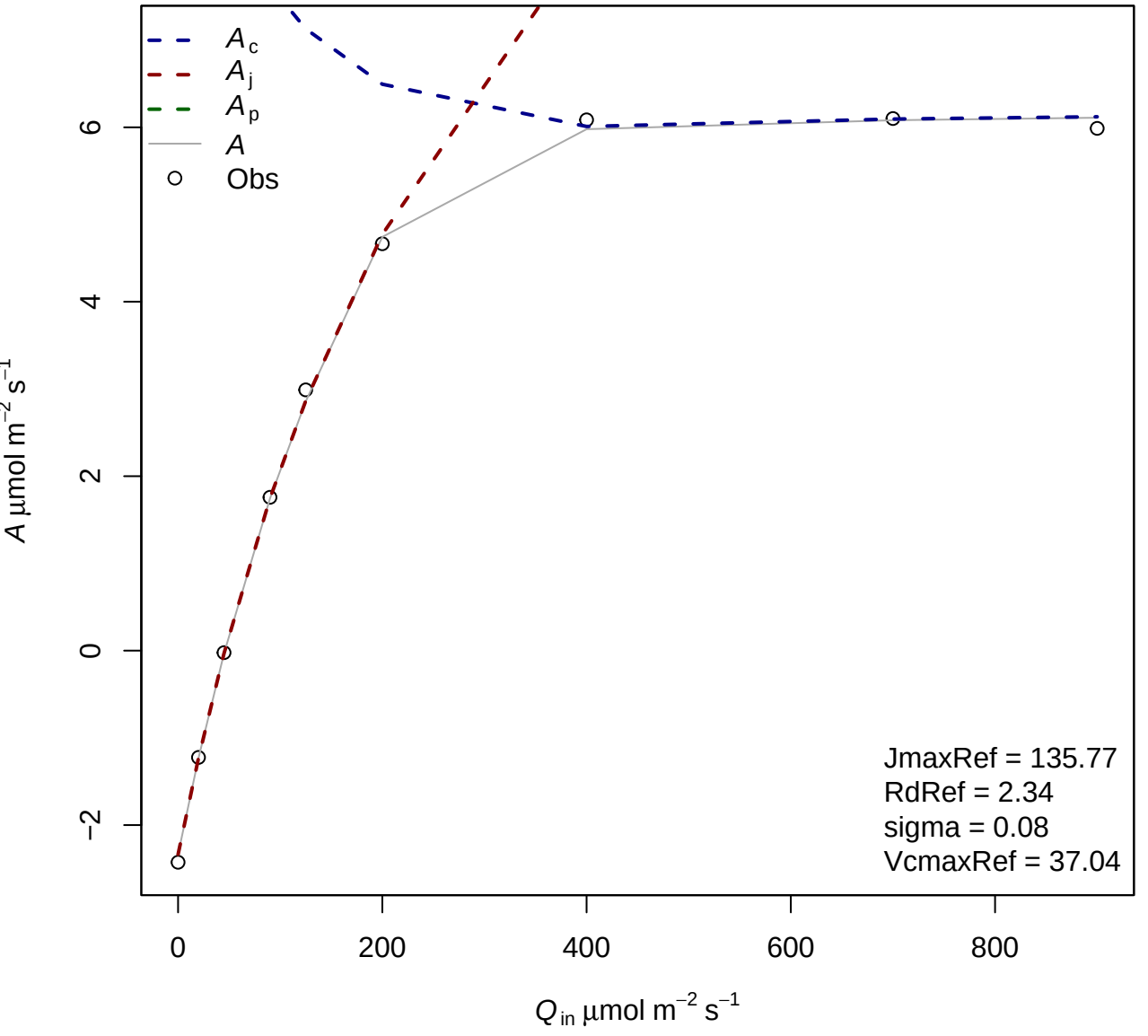

A_c
A_j
A_p
A
Obs
JmaxRef = 143.31
RdRef = 2.96
sigma = 0.4
VcmaxRef = 81.8
0
5
10
15
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

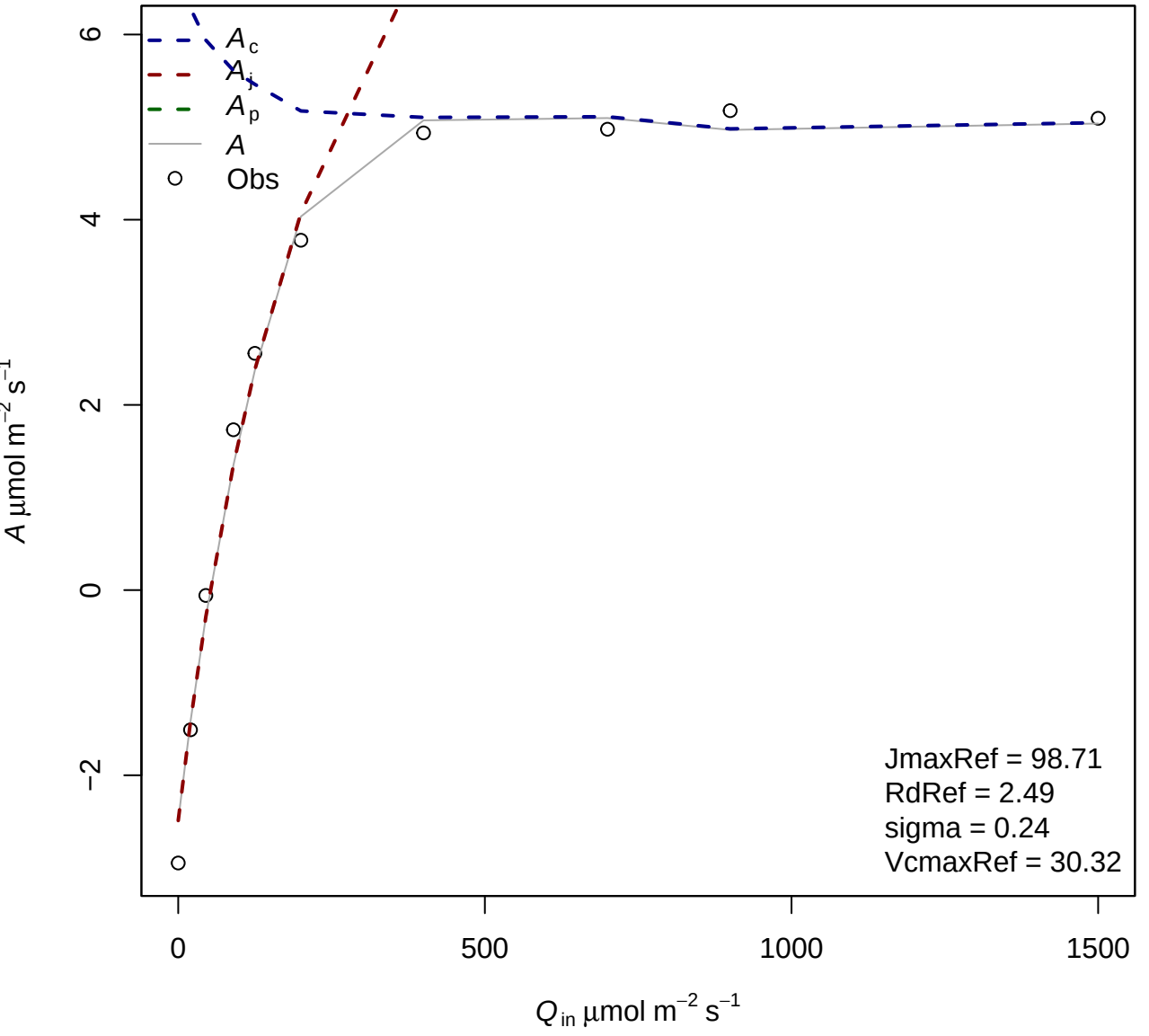

A_c
A_j
A_p
A
Obs
JmaxRef = 128.04
RdRef = 1.7
sigma = 0.22
VcmaxRef = 60.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

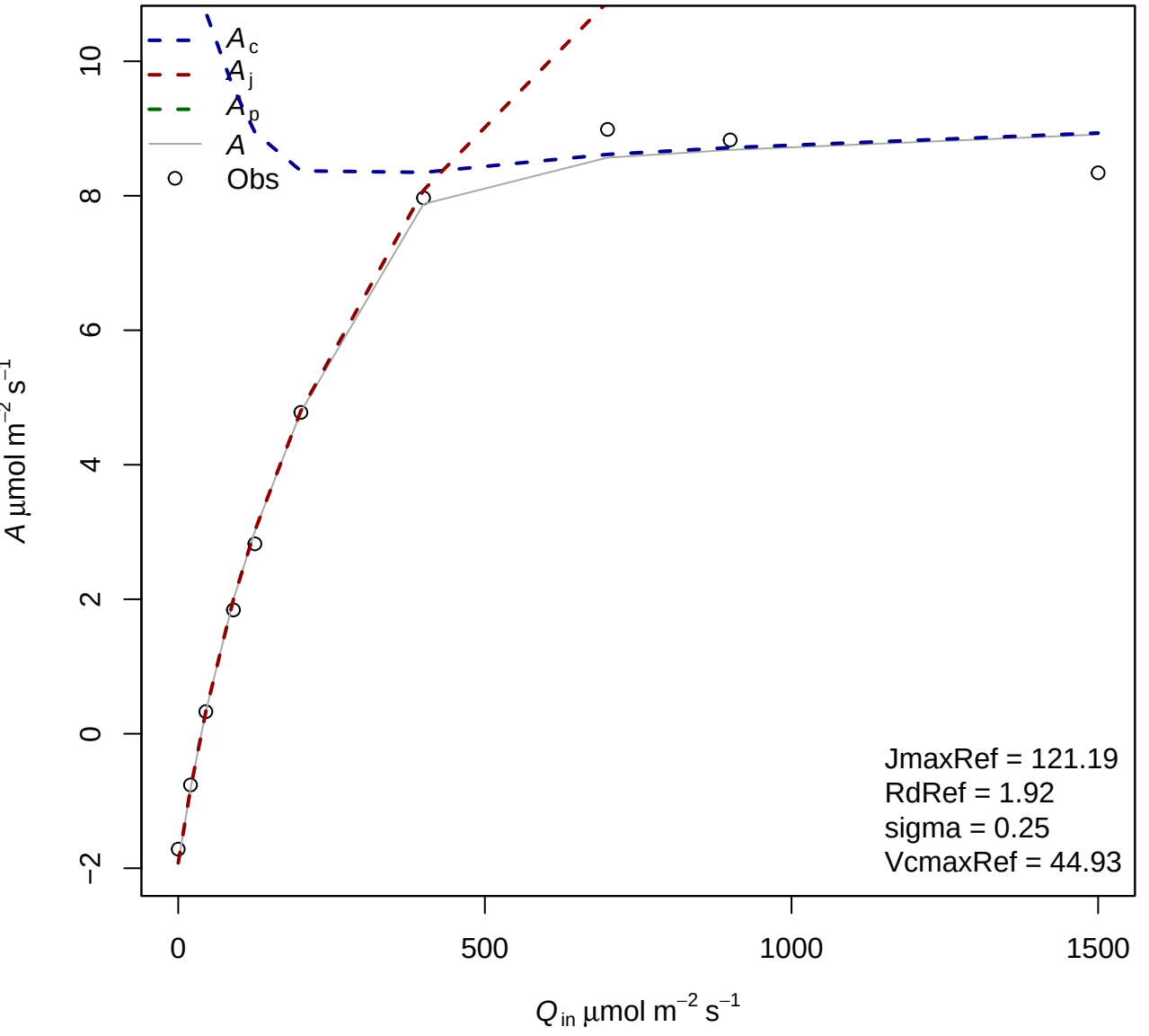

A_c
A_j
A_p
A
Obs
JmaxRef = 44.28
RdRef = 3.5
sigma = 0.15
VcmaxRef = 21.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
−2
−3
0
500
1000
1500

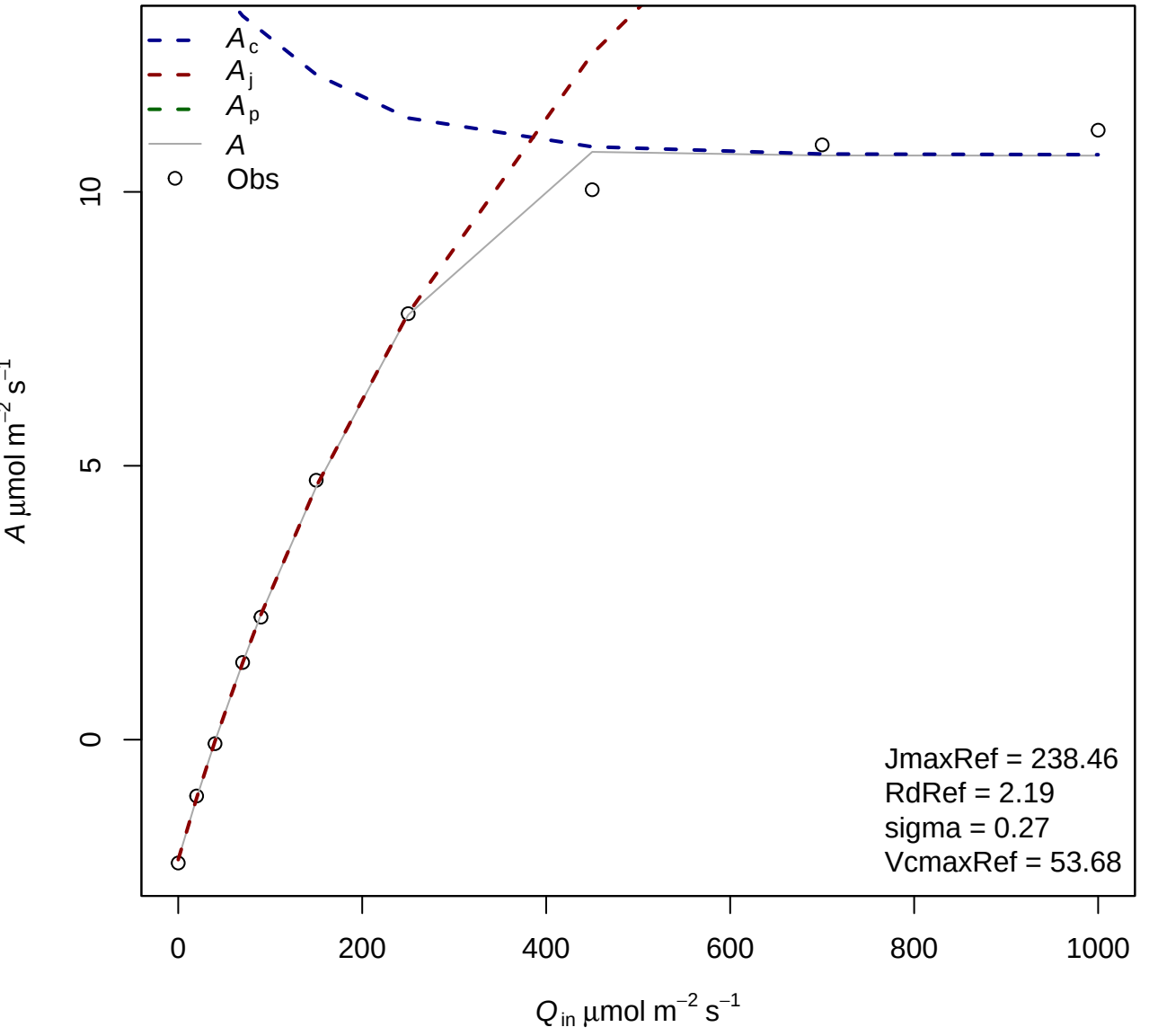

A_c
A_j
A_p
A
Obs
JmaxRef = 85.41
RdRef = 1.55
sigma = 0.2
VcmaxRef = 40.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

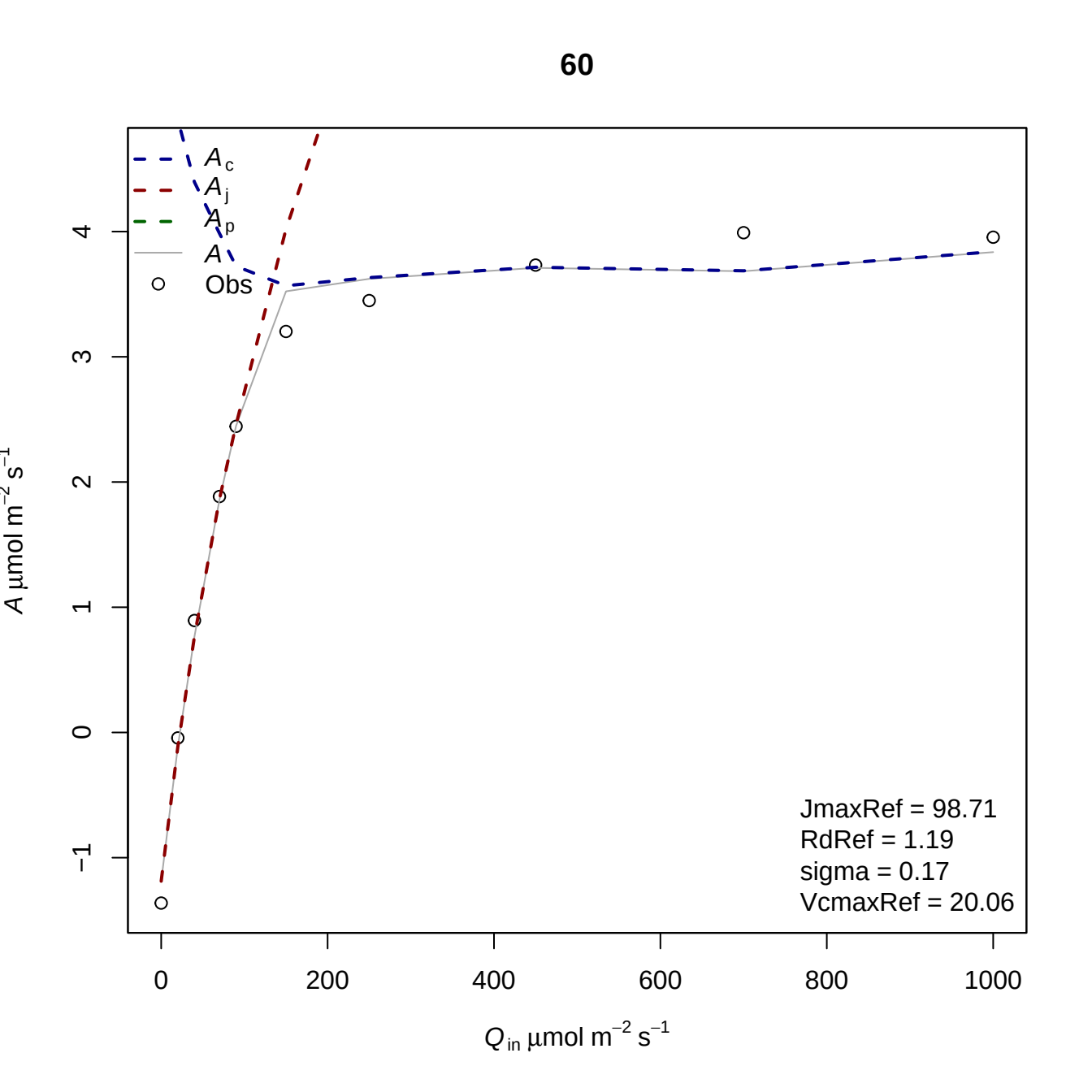

A_c
A_j
A_p
A
Obs
JmaxRef = 95.16
RdRef = 2.22
sigma = 0.11
VcmaxRef = 32.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

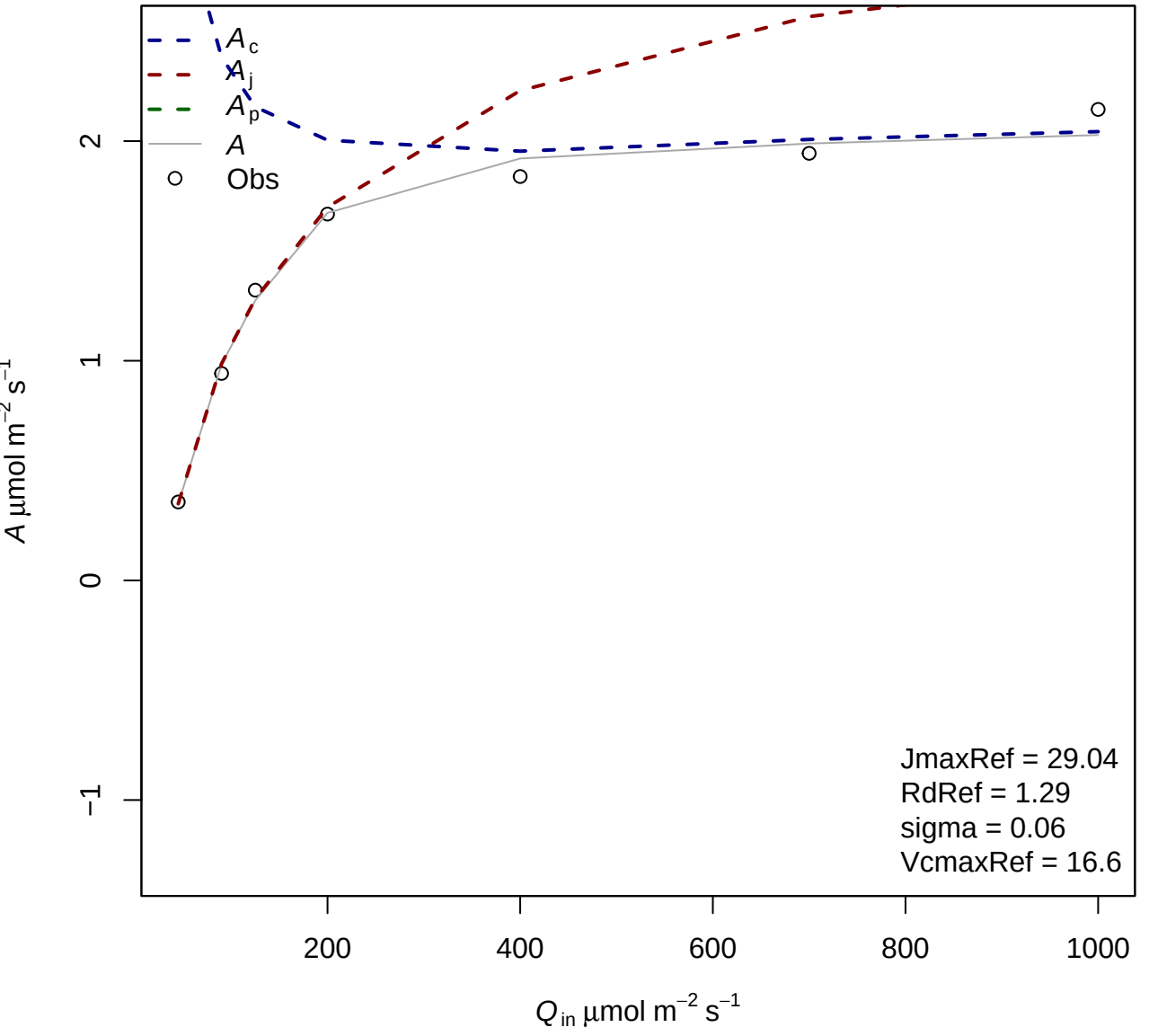

A_c
A_j
A_p
A
Obs
JmaxRef = 135.77
RdRef = 2.34
sigma = 0.08
VcmaxRef = 37.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800

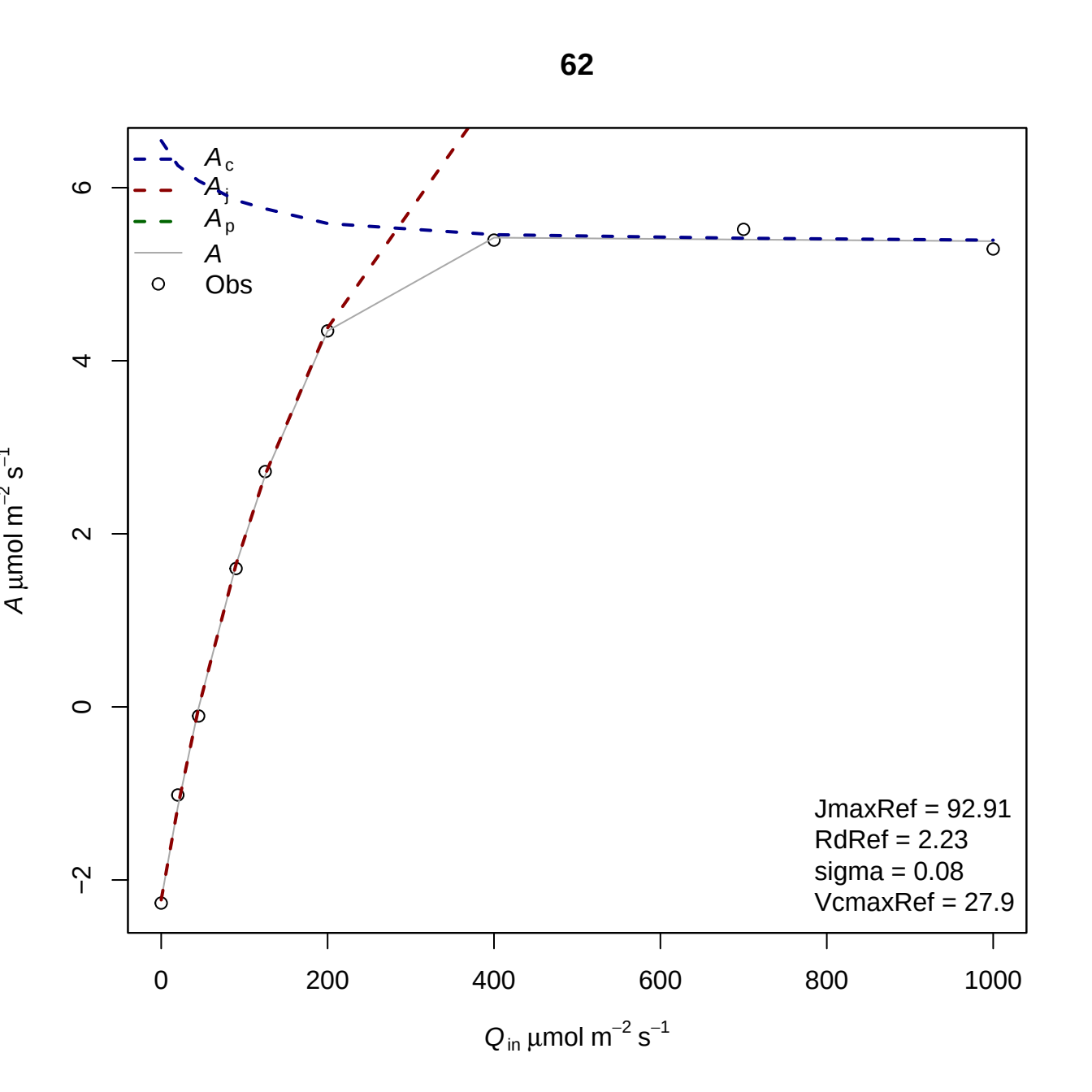

A_c
A_j
A_p
A
Obs
JmaxRef = 98.71
RdRef = 2.49
sigma = 0.24
VcmaxRef = 30.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

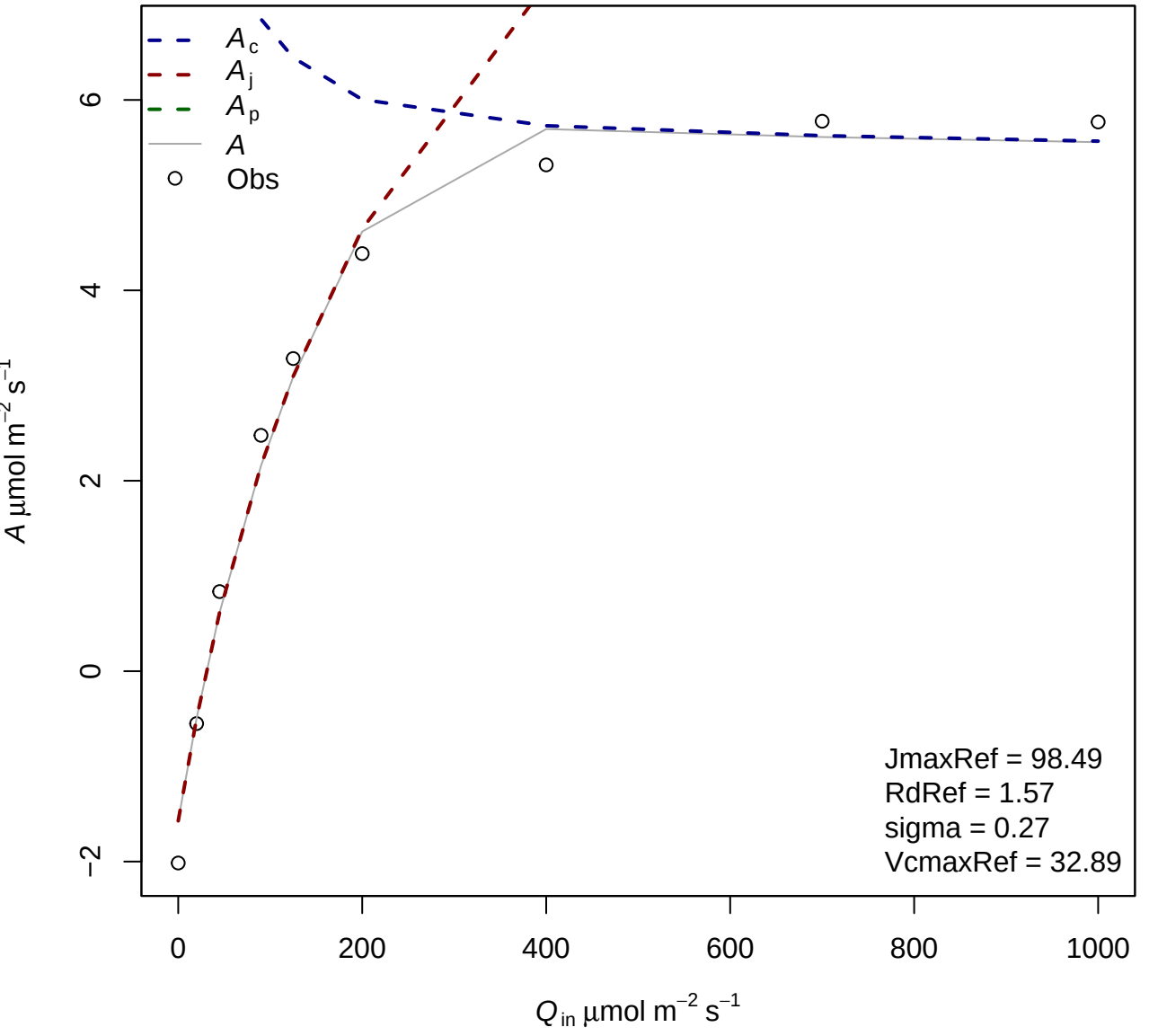

A_c
A_j
A_p
A
Obs
JmaxRef = 121.19
RdRef = 1.92
sigma = 0.25
VcmaxRef = 44.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

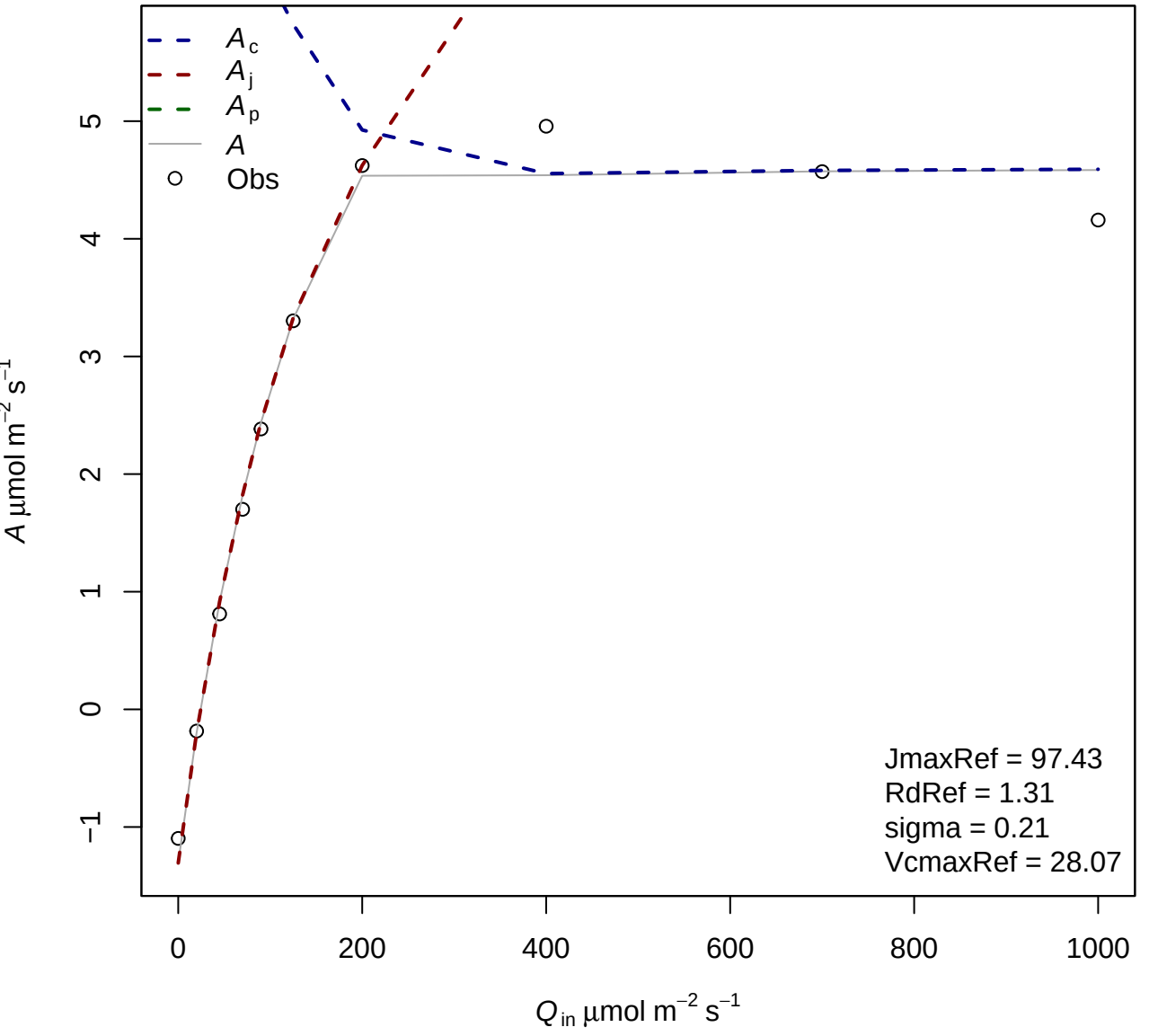

A_c
A_j
A_p
A
Obs
JmaxRef = 238.46
RdRef = 2.19
sigma = 0.27
VcmaxRef = 53.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000

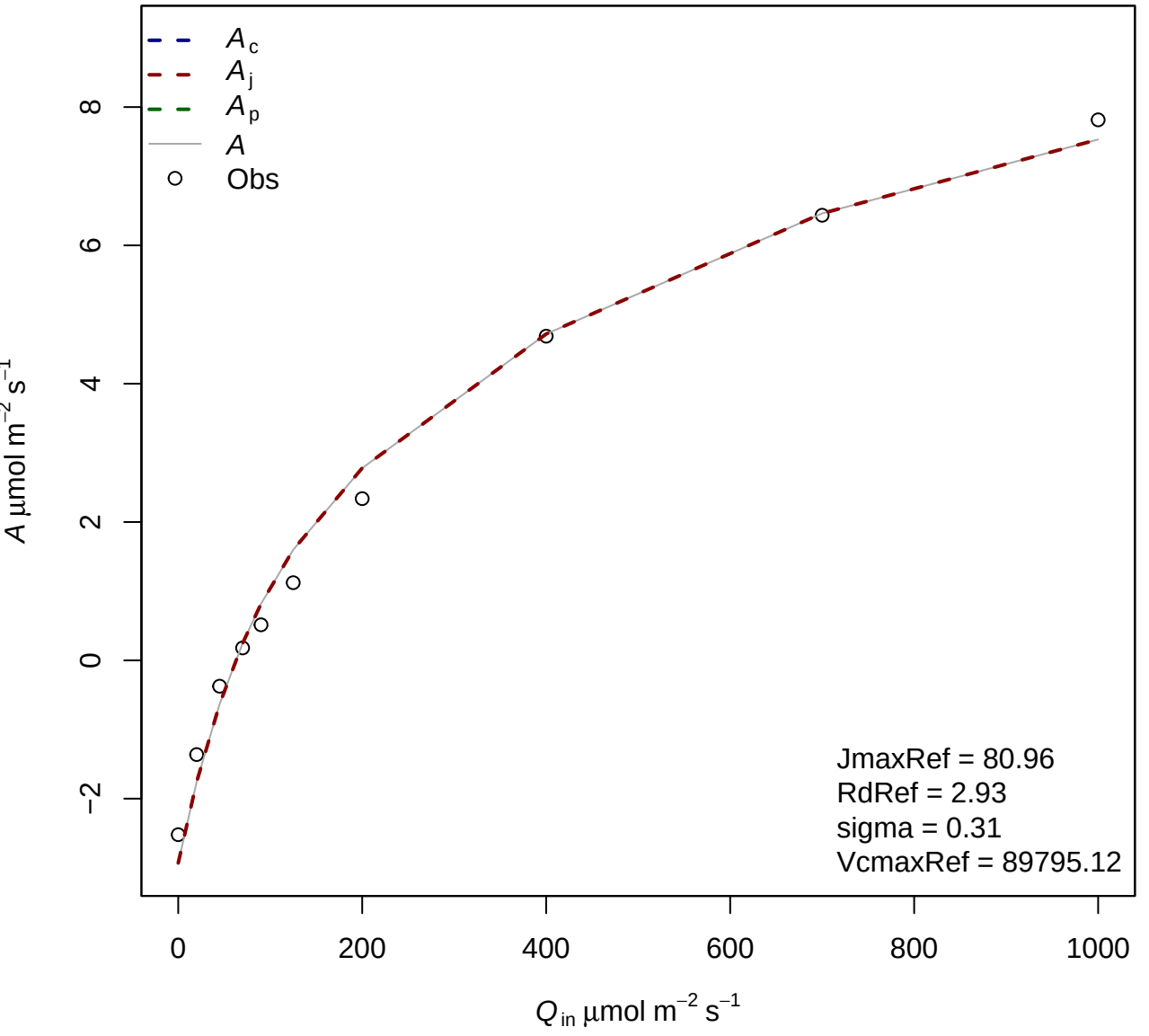

60
A_c
A_j
A_p
A
Obs
JmaxRef = 98.71
RdRef = 1.19
sigma = 0.17
VcmaxRef = 20.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

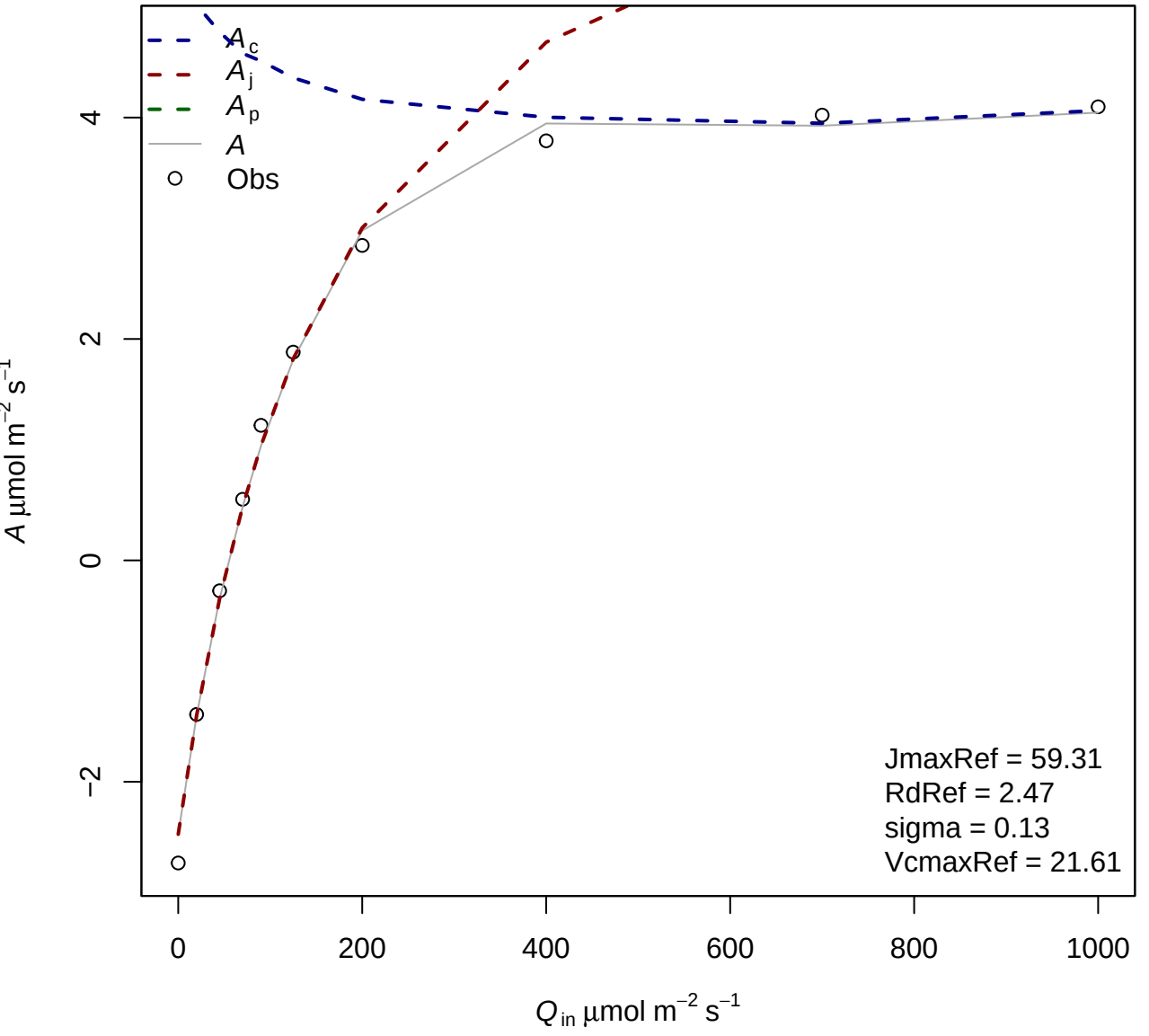

A_c
A_j
A_p
A
Obs
JmaxRef = 29.04
RdRef = 1.29
sigma = 0.06
VcmaxRef = 16.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
200
400
600
800
1000

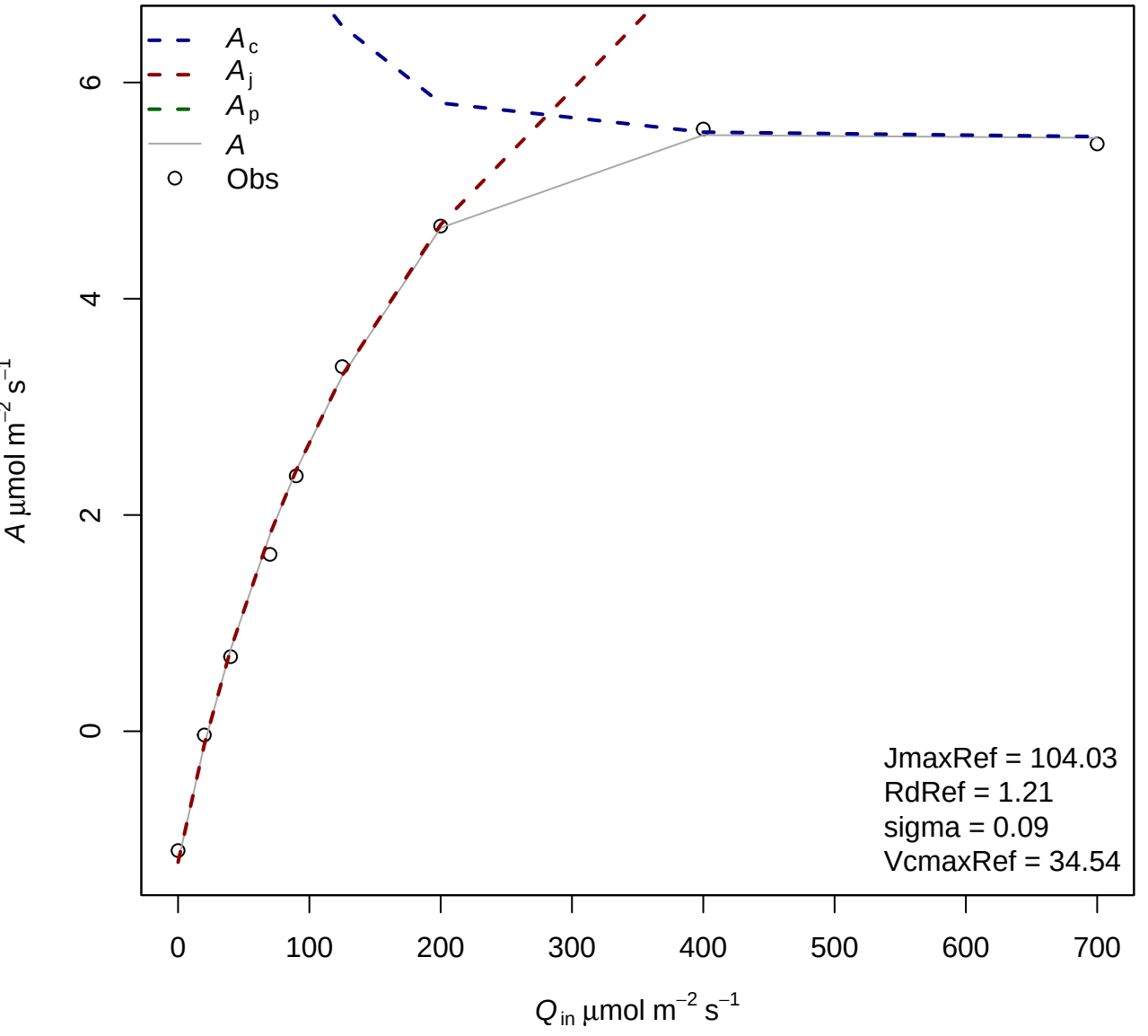

62
A_c
A_j
A_p
A
Obs
JmaxRef = 92.91
RdRef = 2.23
sigma = 0.08
VcmaxRef = 27.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000

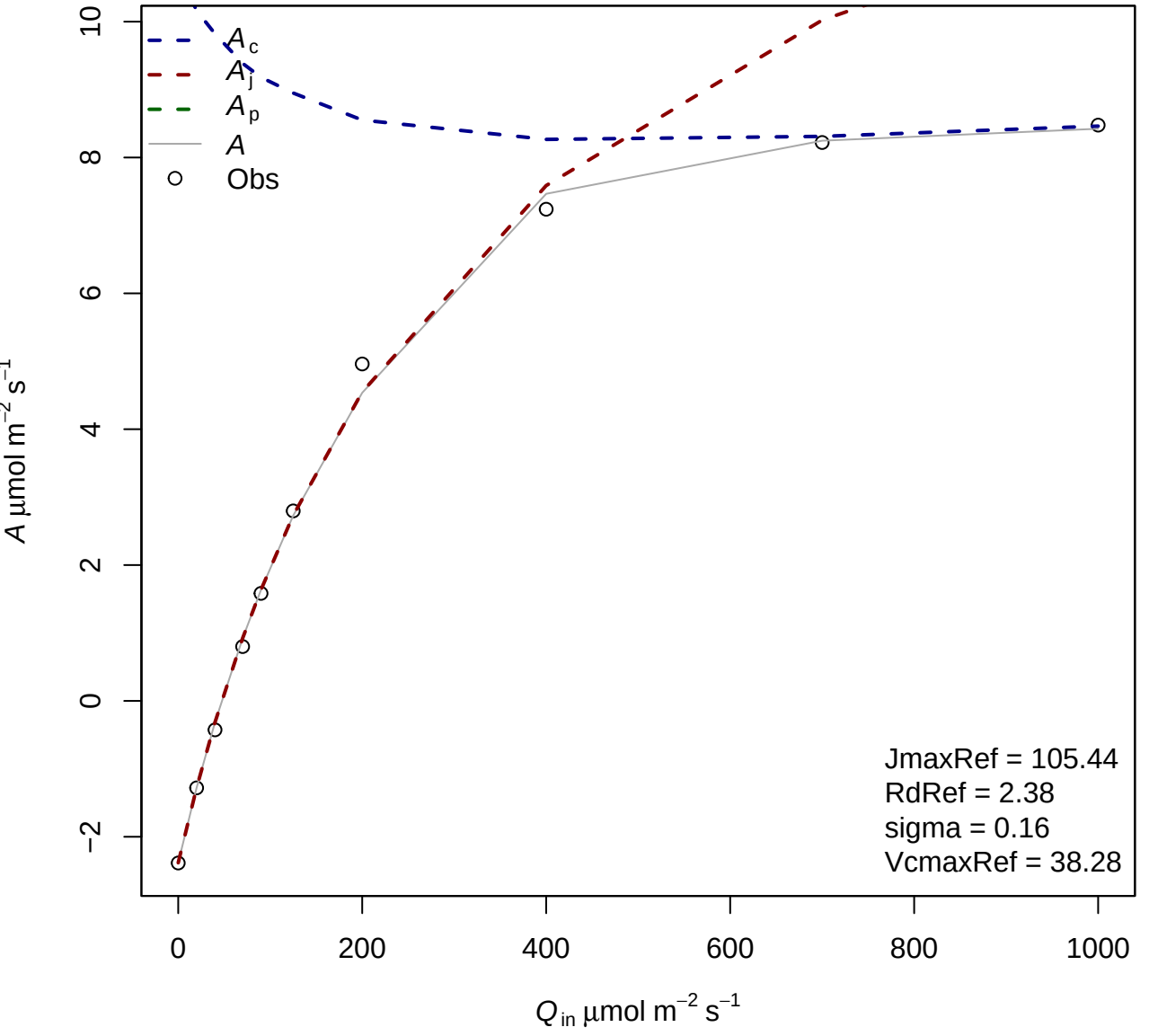

A_c
A_j
A_p
A
Obs
JmaxRef = 98.49
RdRef = 1.57
sigma = 0.27
VcmaxRef = 32.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000

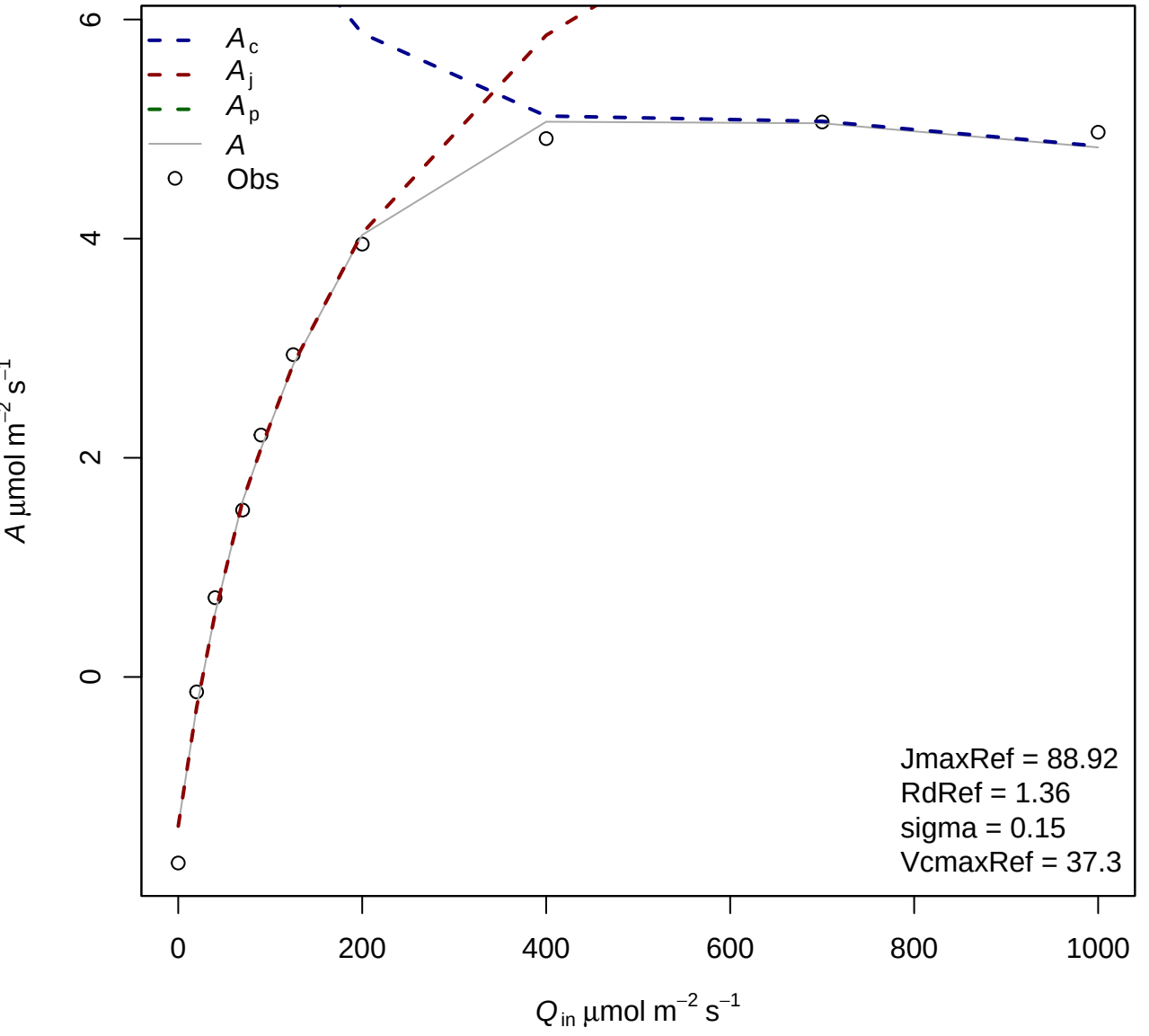

A_c
A_j
A_p
A
Obs
JmaxRef = 97.43
RdRef = 1.31
sigma = 0.21
VcmaxRef = 28.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
200
400
600
800
1000

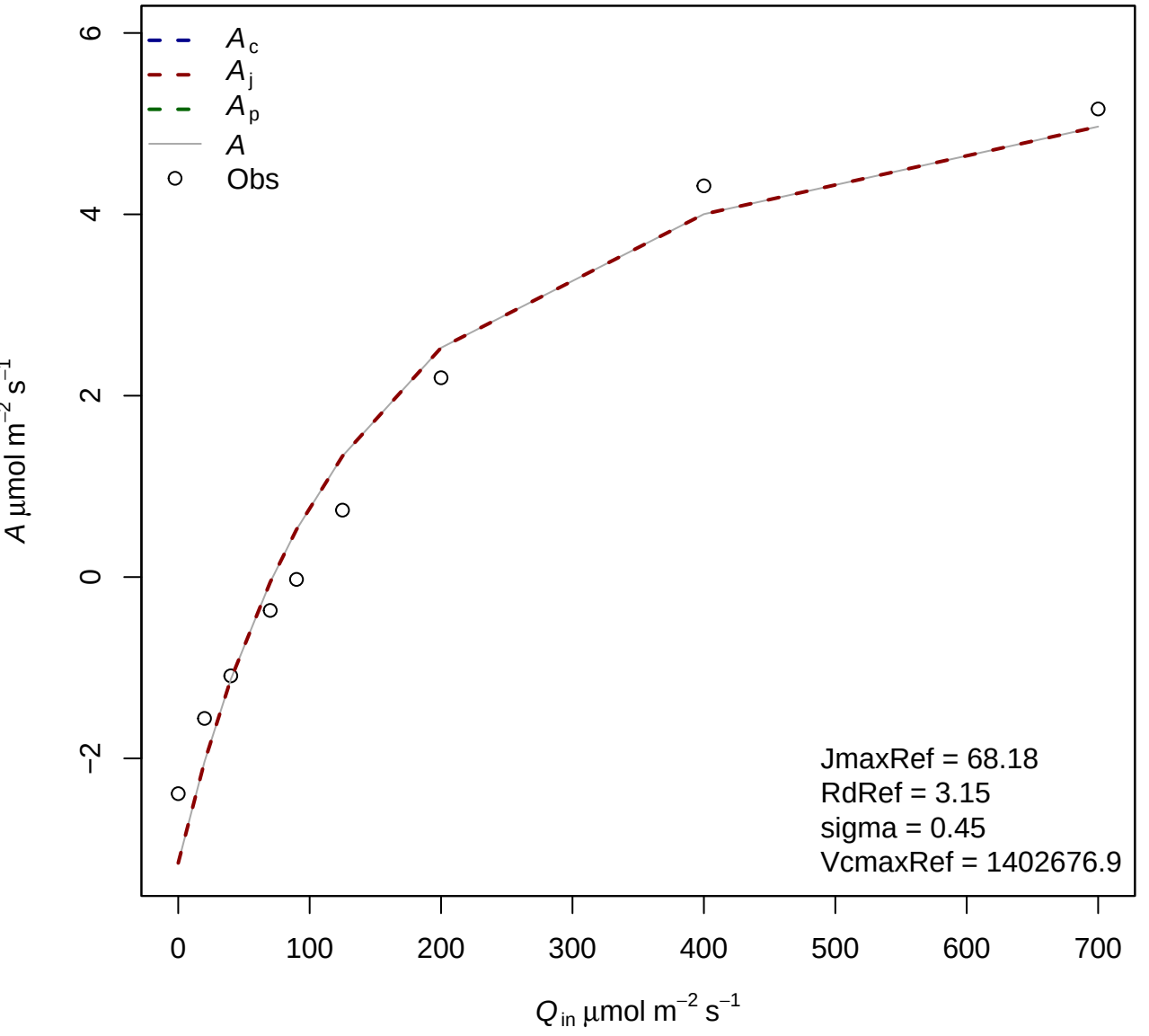

A_c
A_j
A_p
A
Obs
JmaxRef = 80.96
RdRef = 2.93
sigma = 0.31
VcmaxRef = 89795.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000

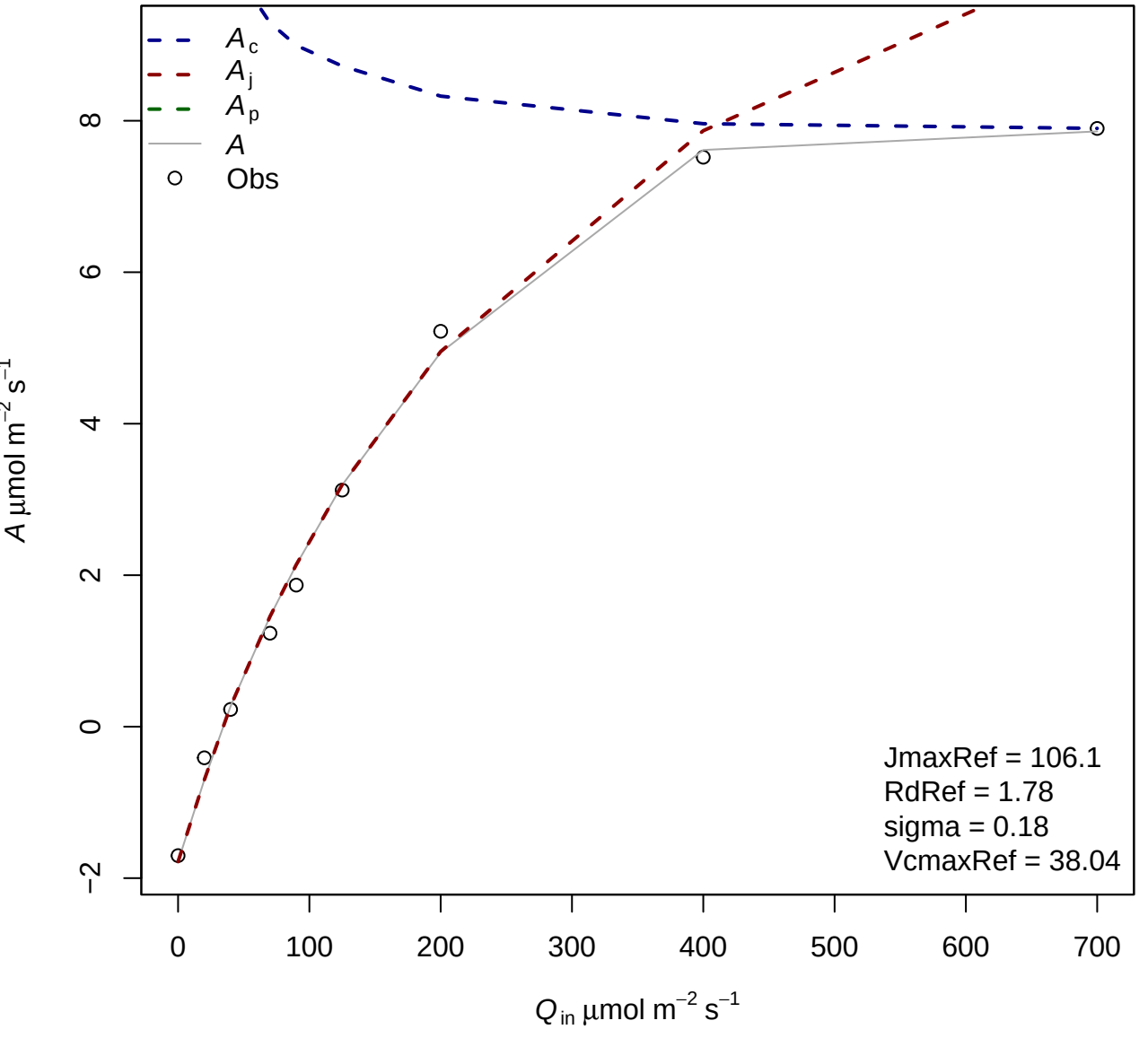

A_c
A_j
A_p
A
Obs
JmaxRef = 59.31
RdRef = 2.47
sigma = 0.13
VcmaxRef = 21.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000

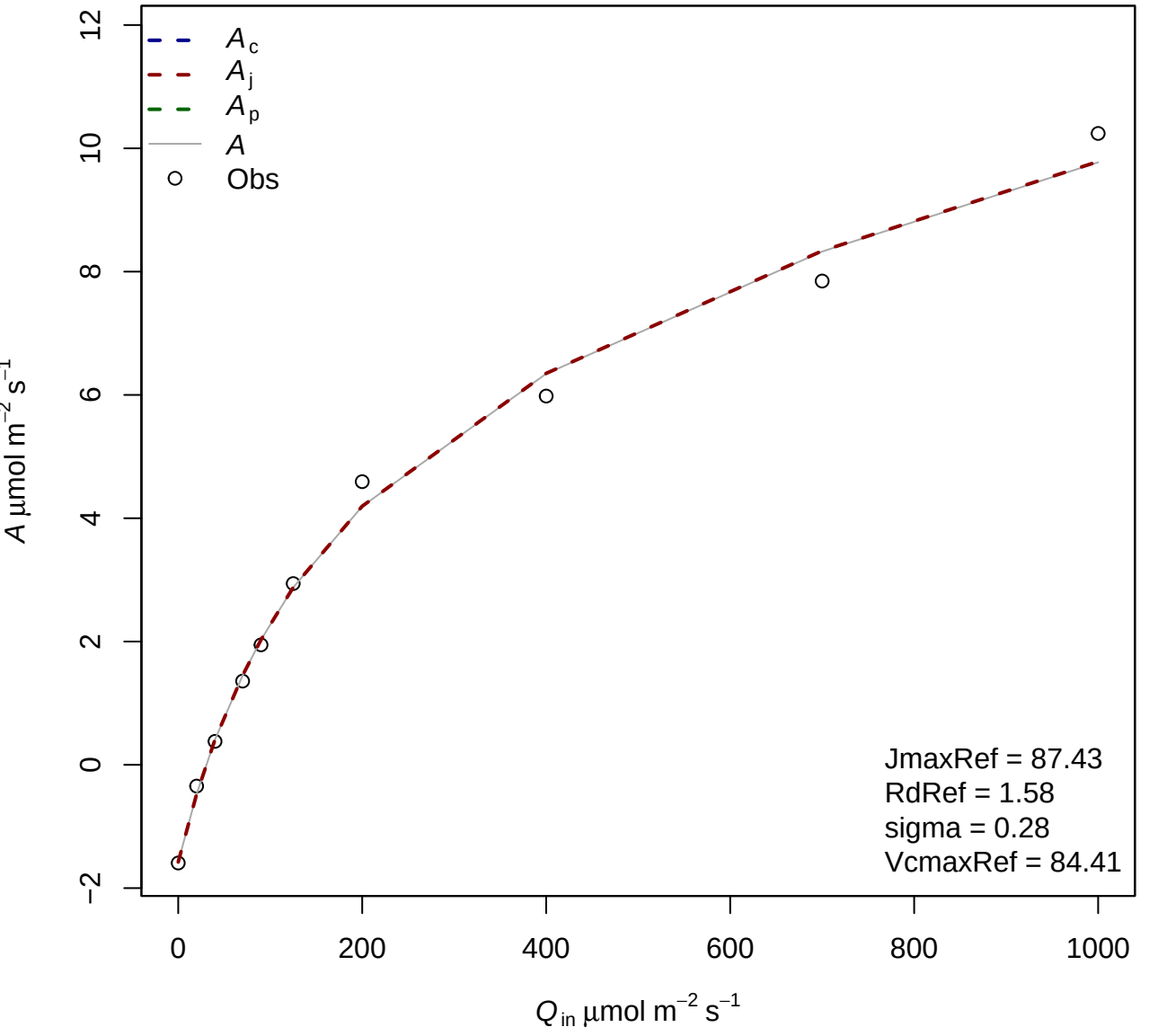

A_c
A_j
A_p
A
Obs
JmaxRef = 104.03
RdRef = 1.21
sigma = 0.09
VcmaxRef = 34.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
100
200
300
400
500
600
700

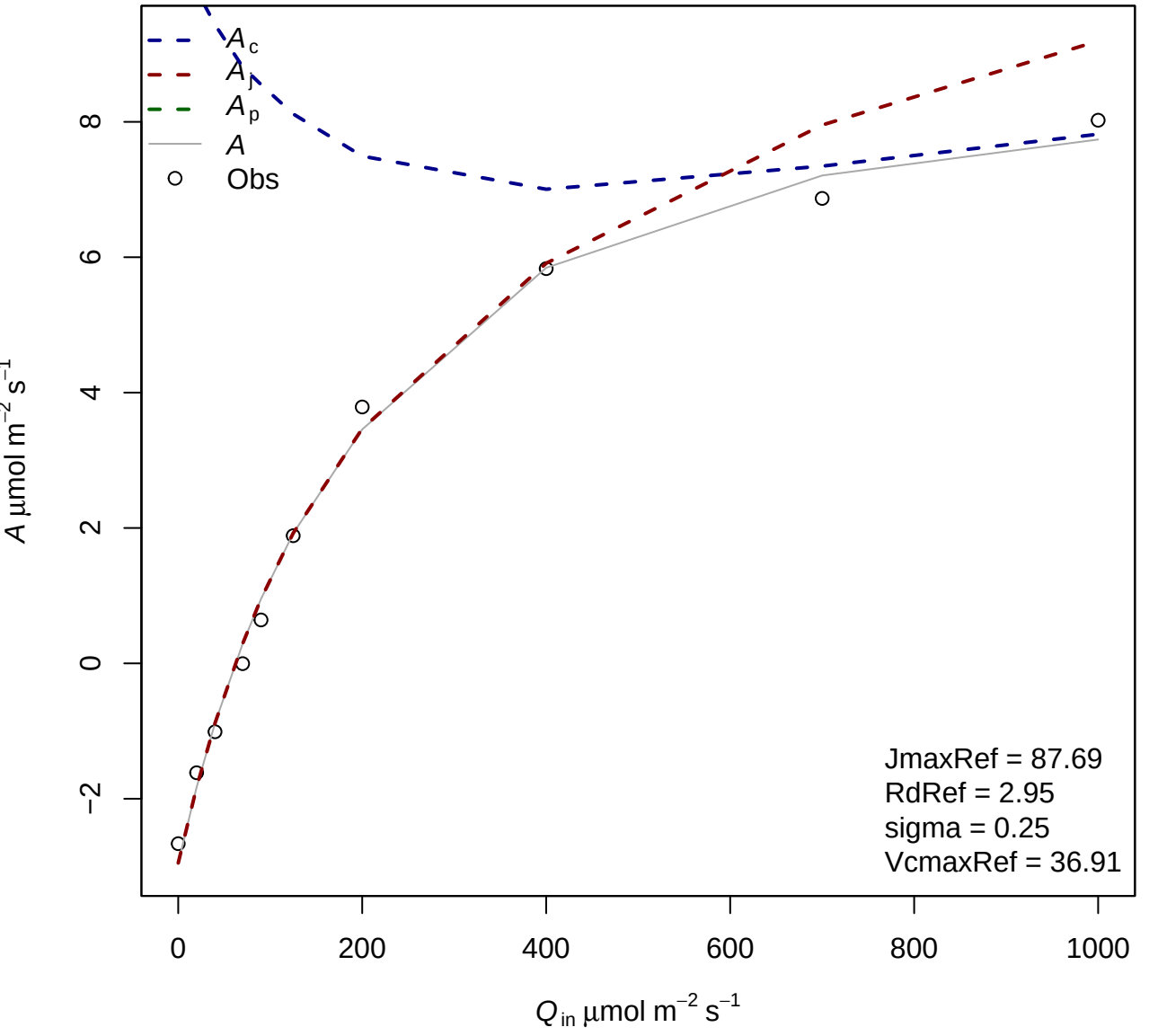

A_c
A_j
A_p
A
Obs
JmaxRef = 105.44
RdRef = 2.38
sigma = 0.16
VcmaxRef = 38.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
200
400
600
800
1000

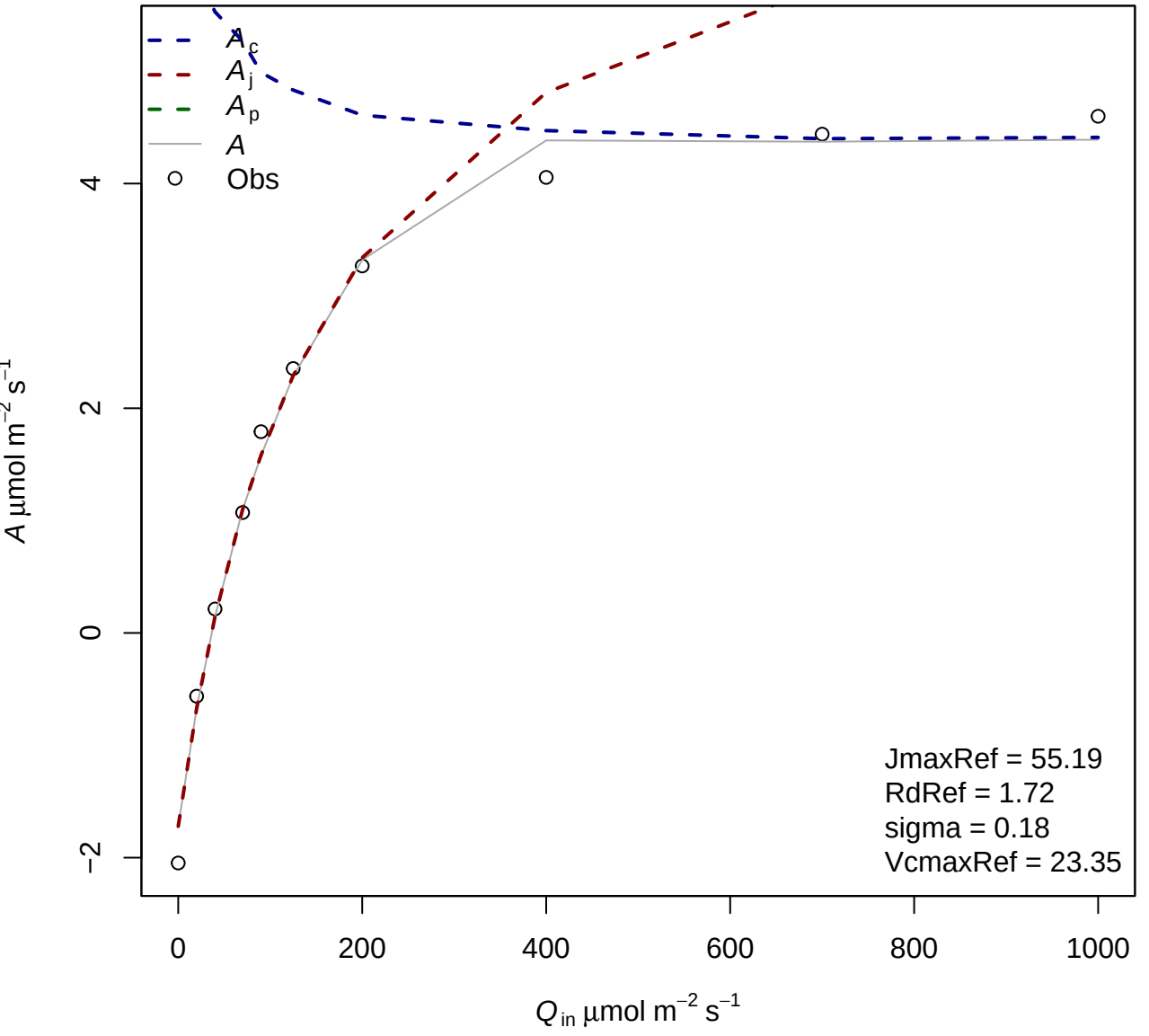

A_c
A_j
A_p
A
Obs
JmaxRef = 88.92
RdRef = 1.36
sigma = 0.15
VcmaxRef = 37.3
0
2
4
6
0
200
400
600
800
1000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

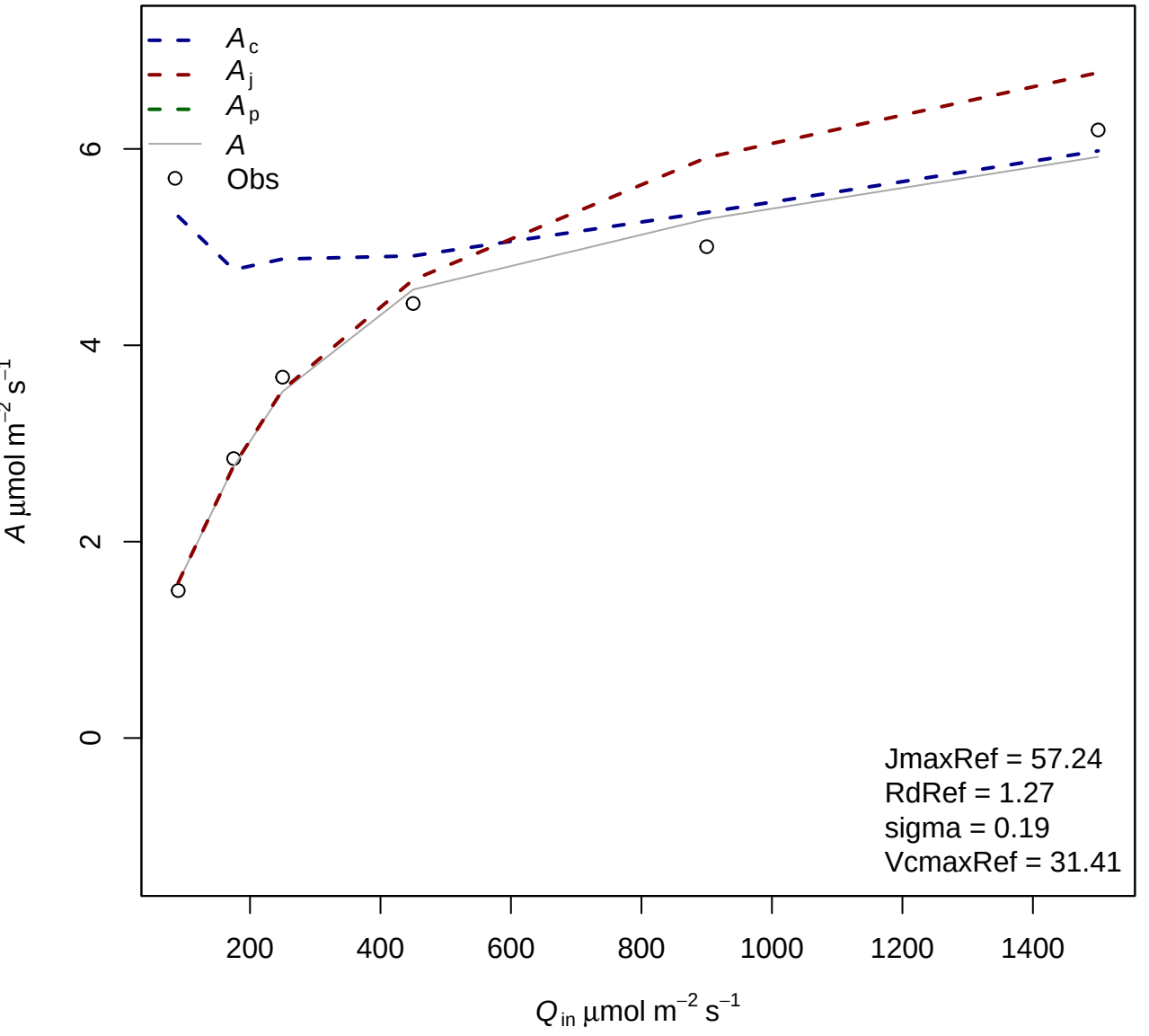

A_c
A_j
A_p
A
Obs
JmaxRef = 68.18
RdRef = 3.15
sigma = 0.45
VcmaxRef = 1402676.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
100
200
300
400
500
600
700

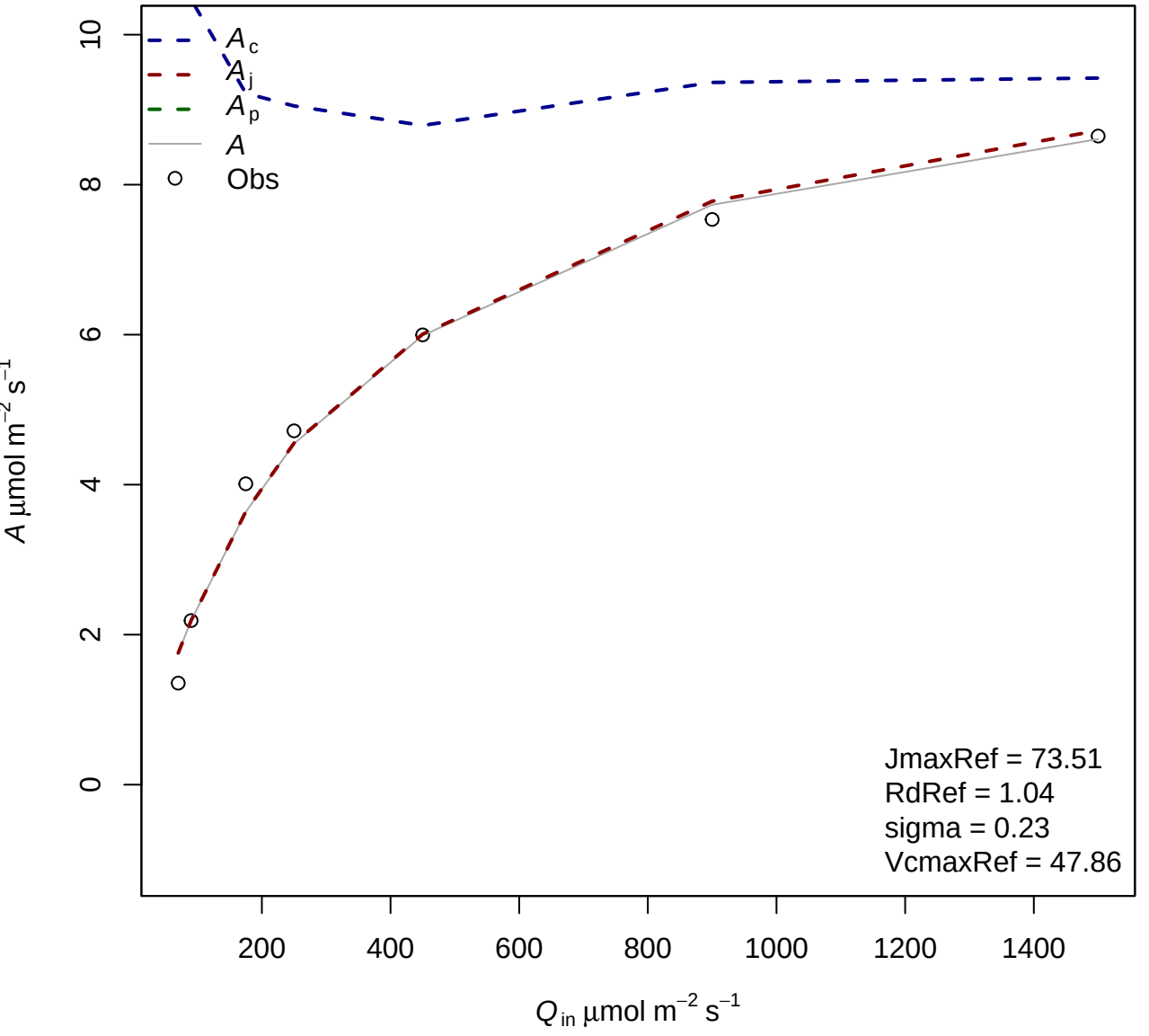

A_c
A_j
A_p
A
Obs
JmaxRef = 106.1
RdRef = 1.78
sigma = 0.18
VcmaxRef = 38.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
100
200
300
400
500
600
700

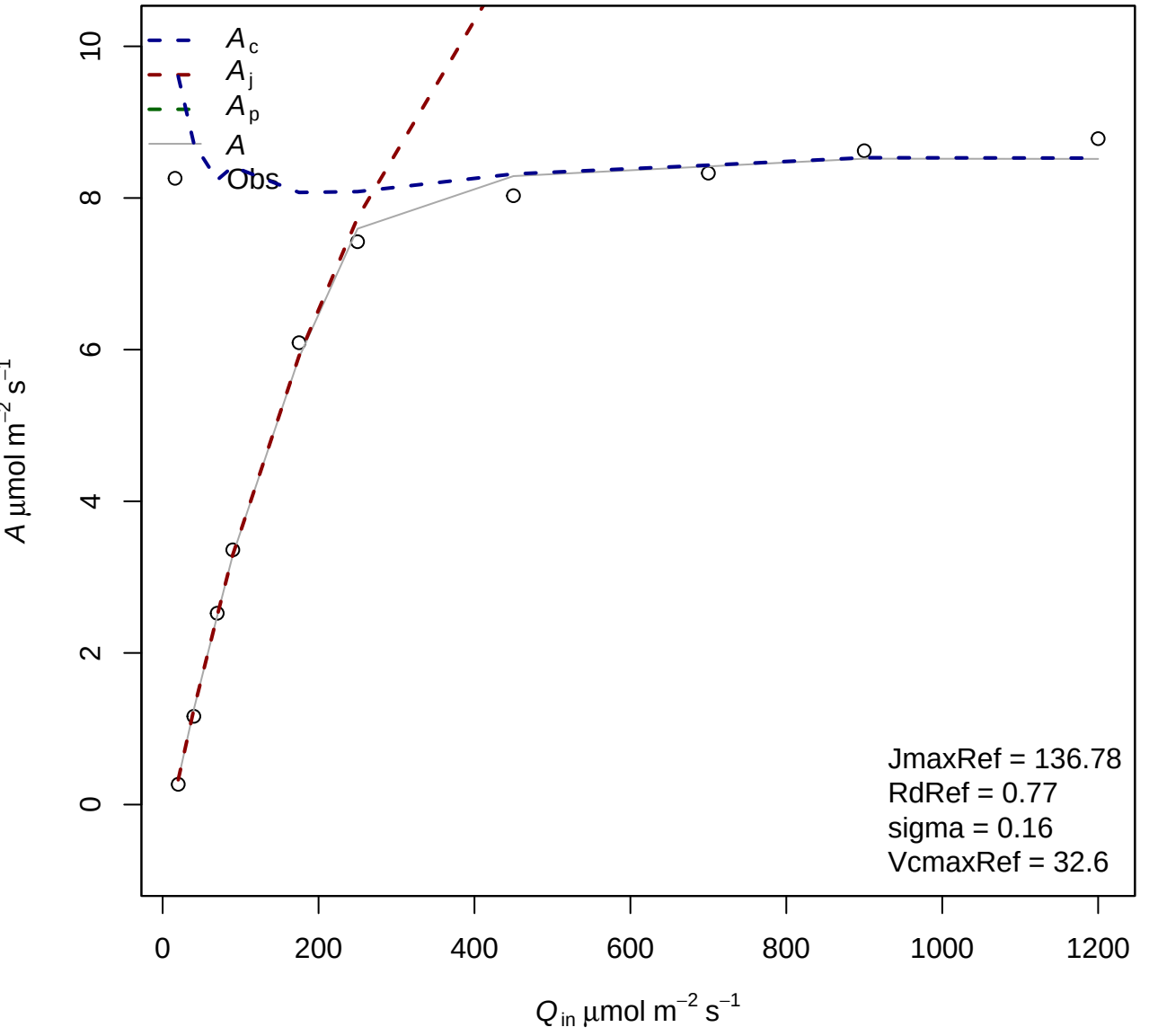

A_c
A_j
A_p
A
Obs
JmaxRef = 87.43
RdRef = 1.58
sigma = 0.28
VcmaxRef = 84.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
200
400
600
800
1000

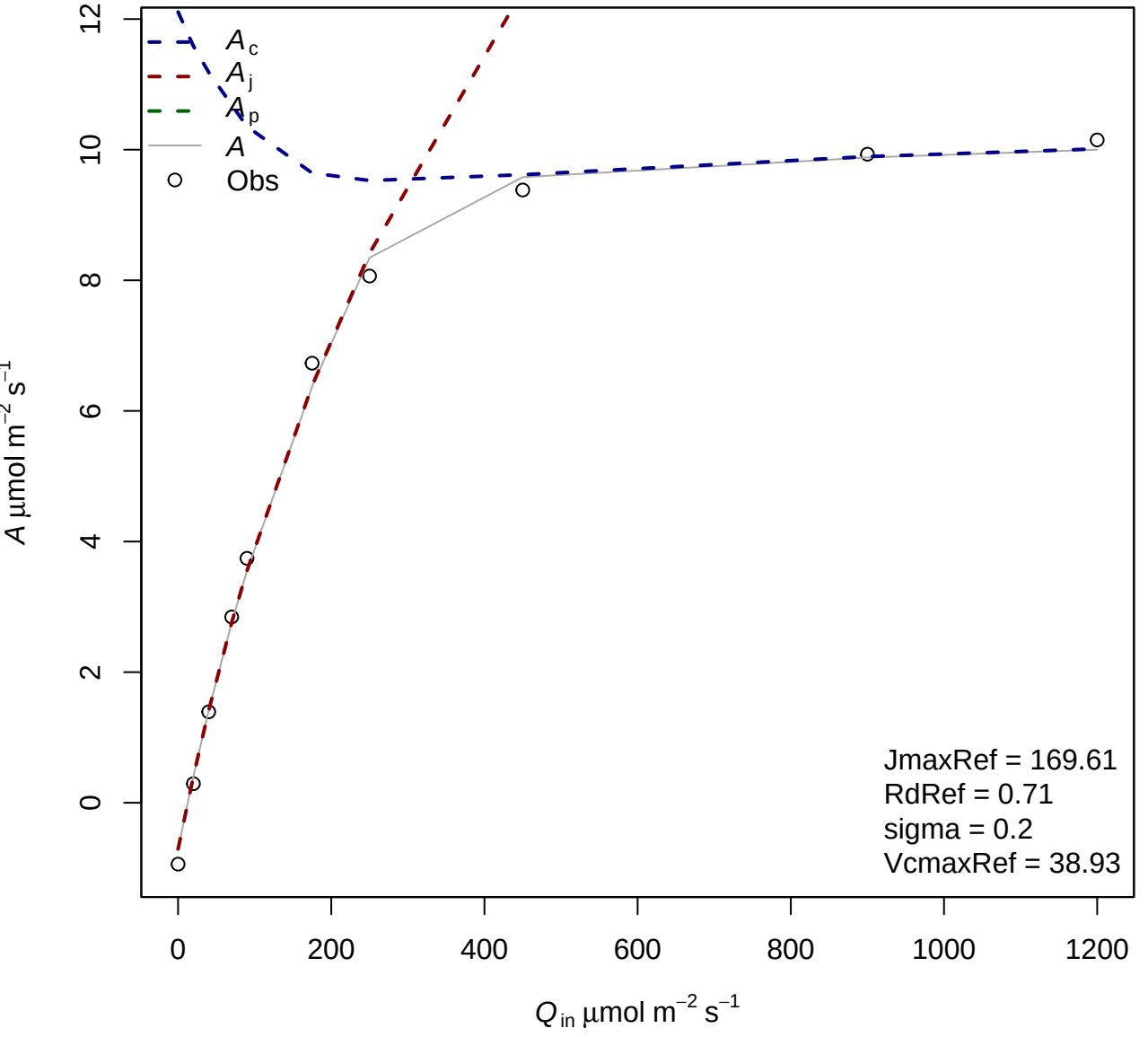

A_c
A_j
A_p
A
Obs
JmaxRef = 87.69
RdRef = 2.95
sigma = 0.25
VcmaxRef = 36.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000

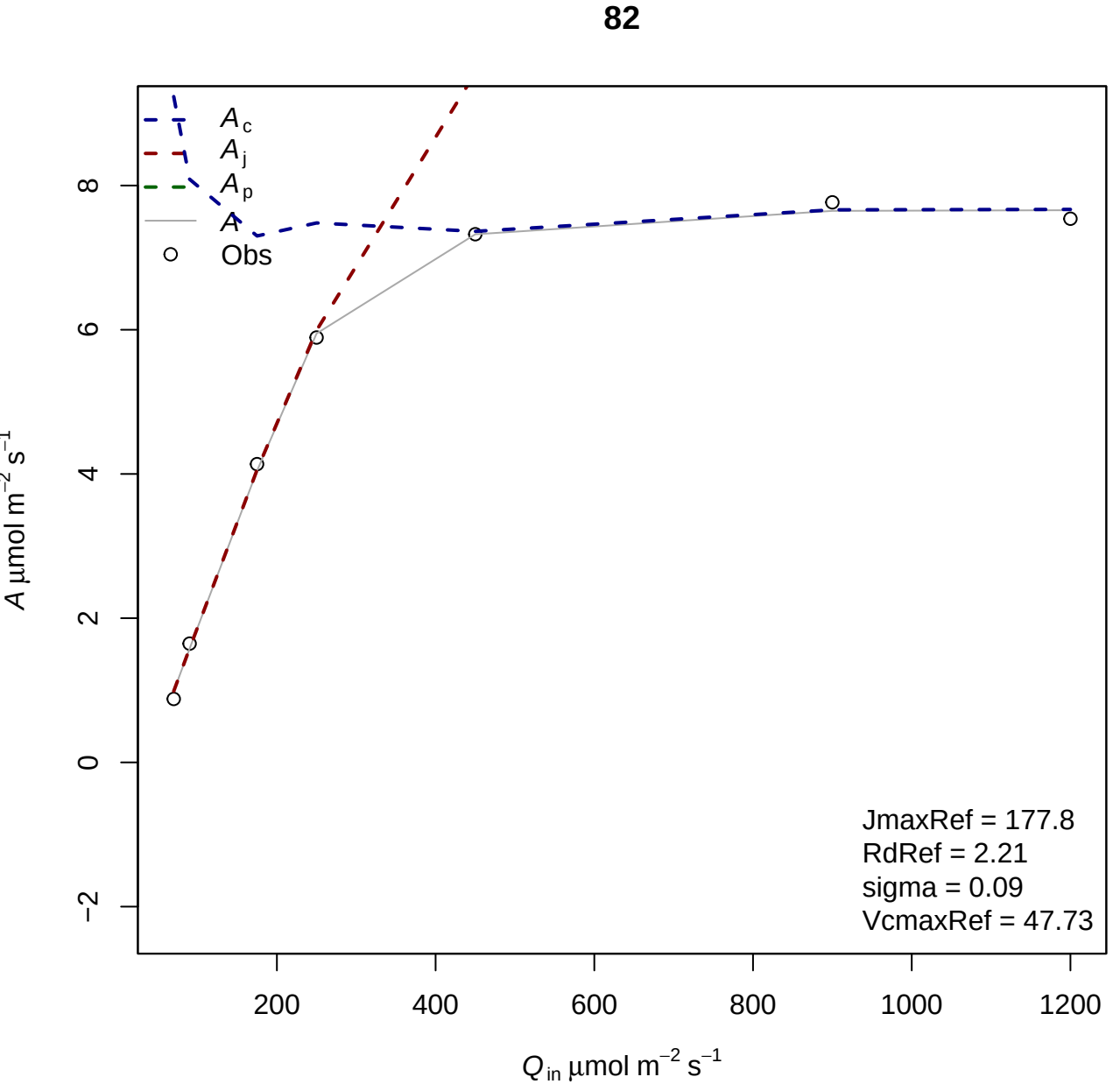

A_c
A_j
A_p
A
Obs
JmaxRef = 55.19
RdRef = 1.72
sigma = 0.18
VcmaxRef = 23.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000

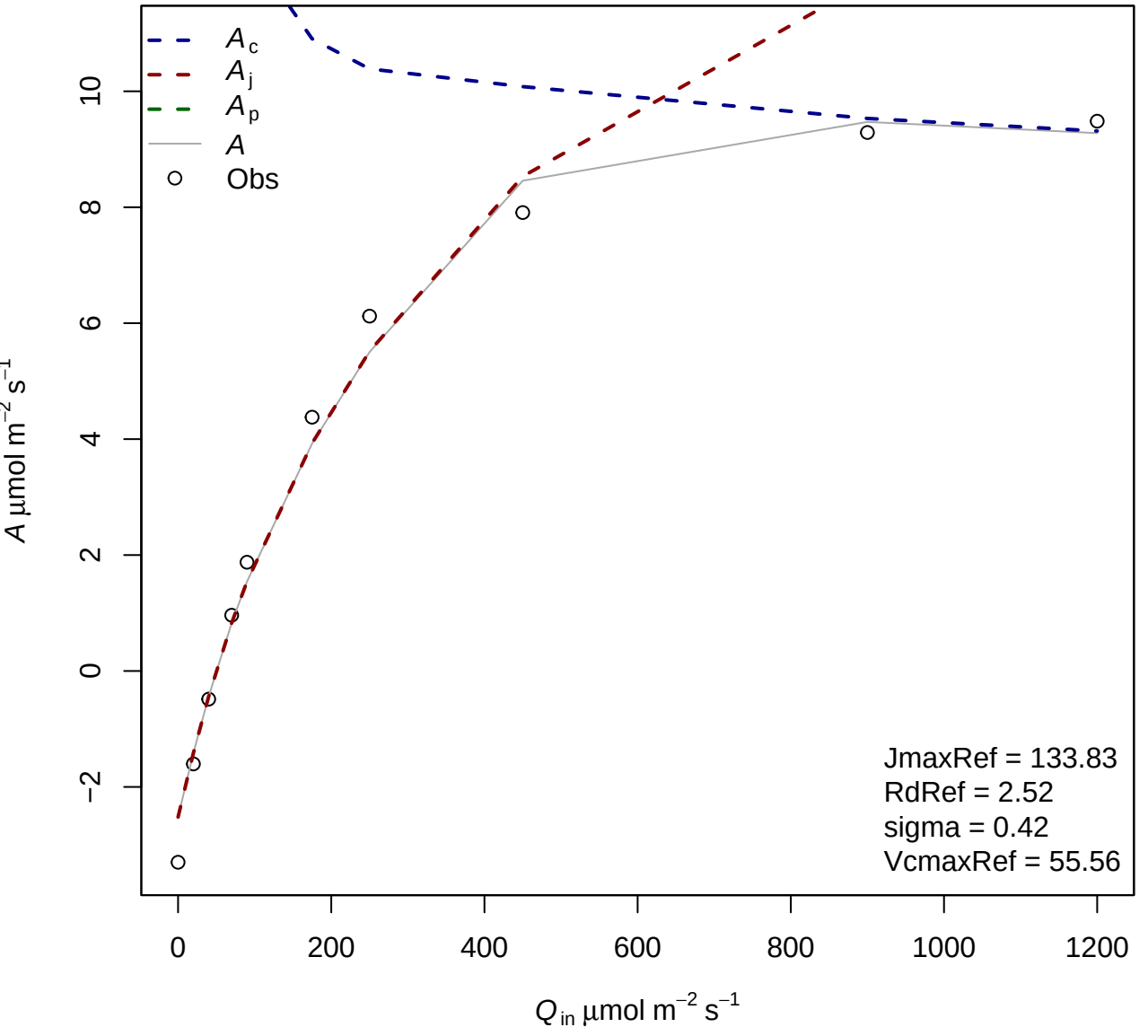

A_c
A_j
A_p
A
Obs
JmaxRef = 57.24
RdRef = 1.27
sigma = 0.19
VcmaxRef = 31.41
0
2
4
6
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

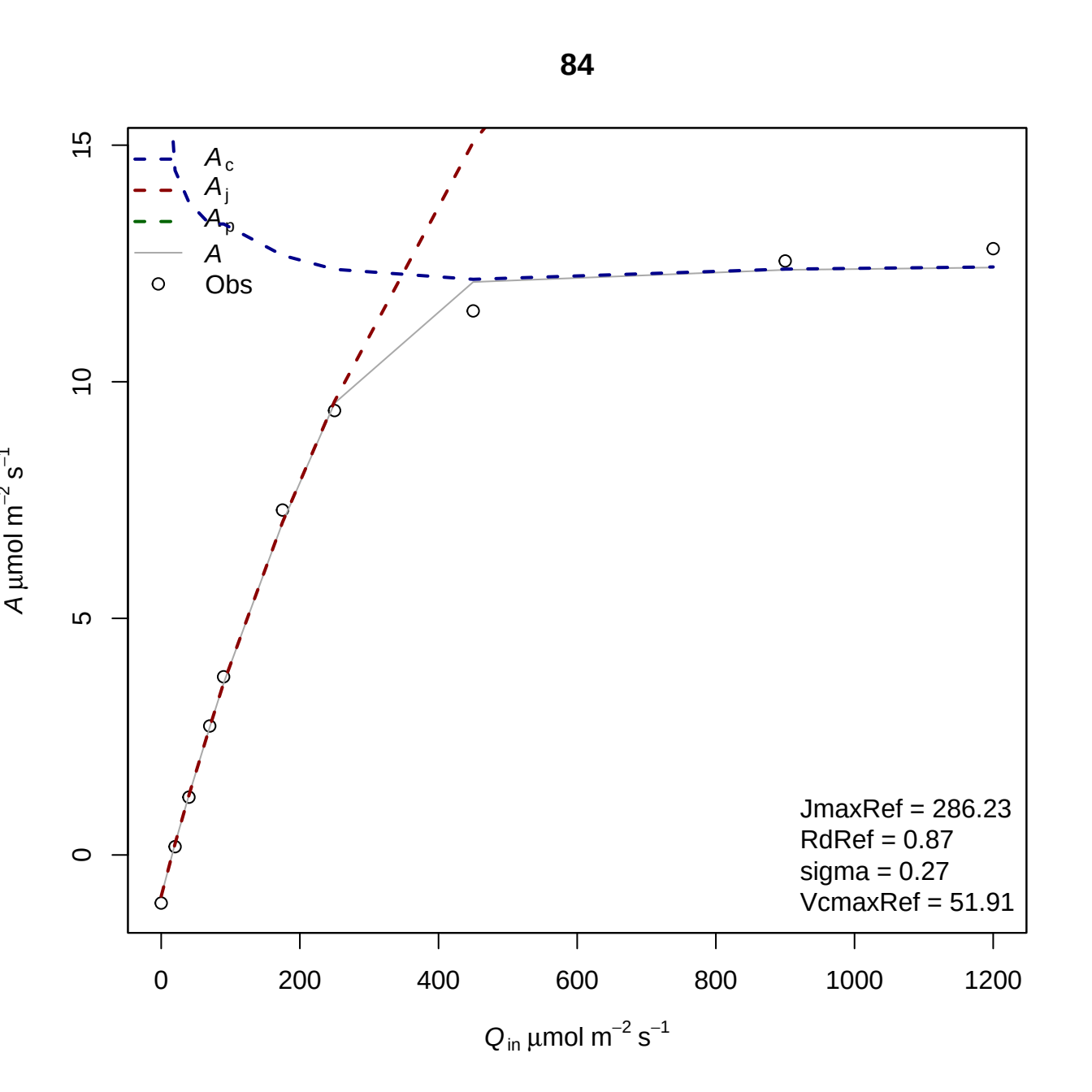

A_c
A_j
A_p
A
Obs
JmaxRef = 73.51
RdRef = 1.04
sigma = 0.23
VcmaxRef = 47.86
0
2
4
6
8
10
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

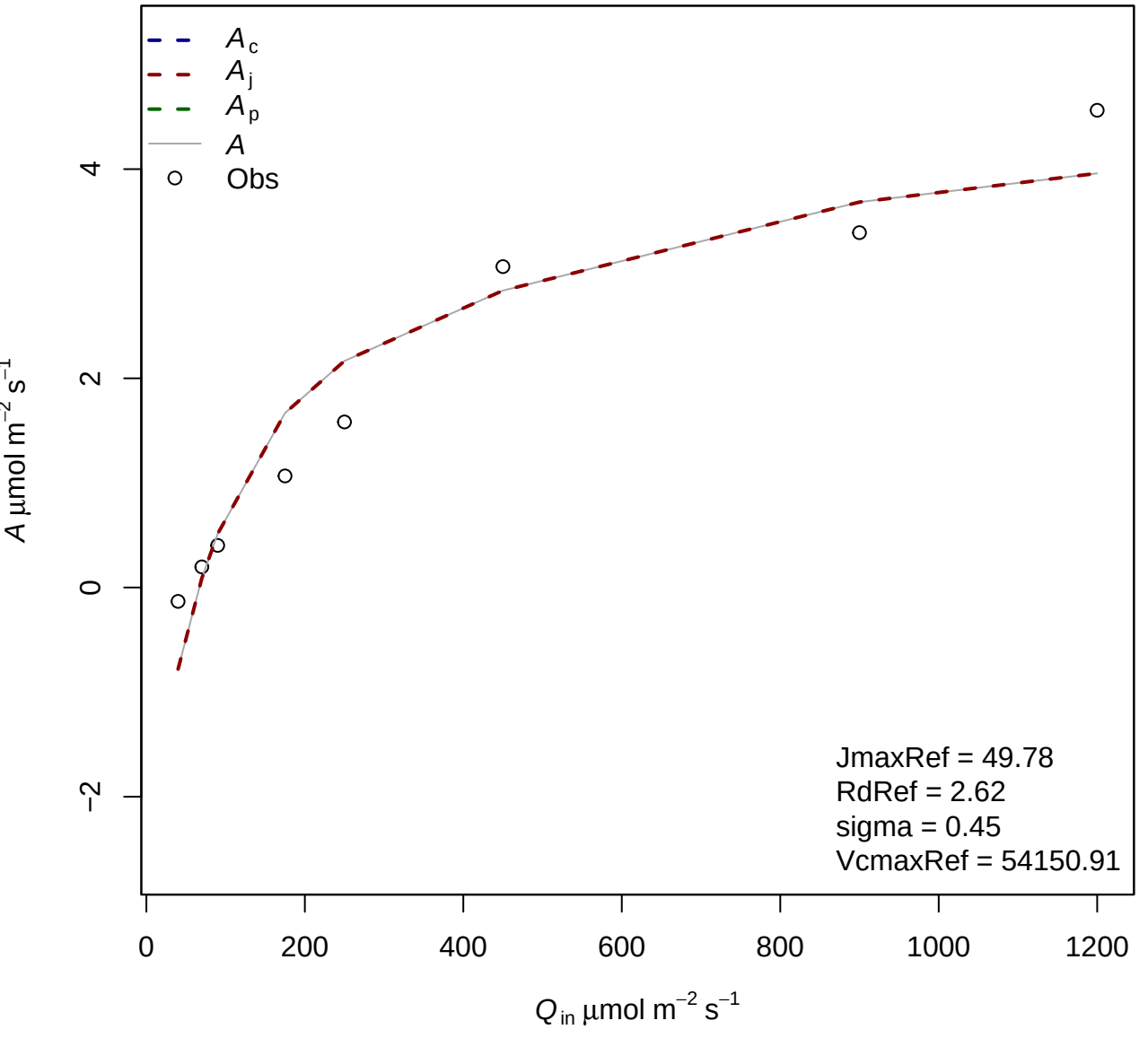

A_c
A_j
A_p
A
Obs
JmaxRef = 136.78
RdRef = 0.77
sigma = 0.16
VcmaxRef = 32.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
200
400
600
800
1000
1200

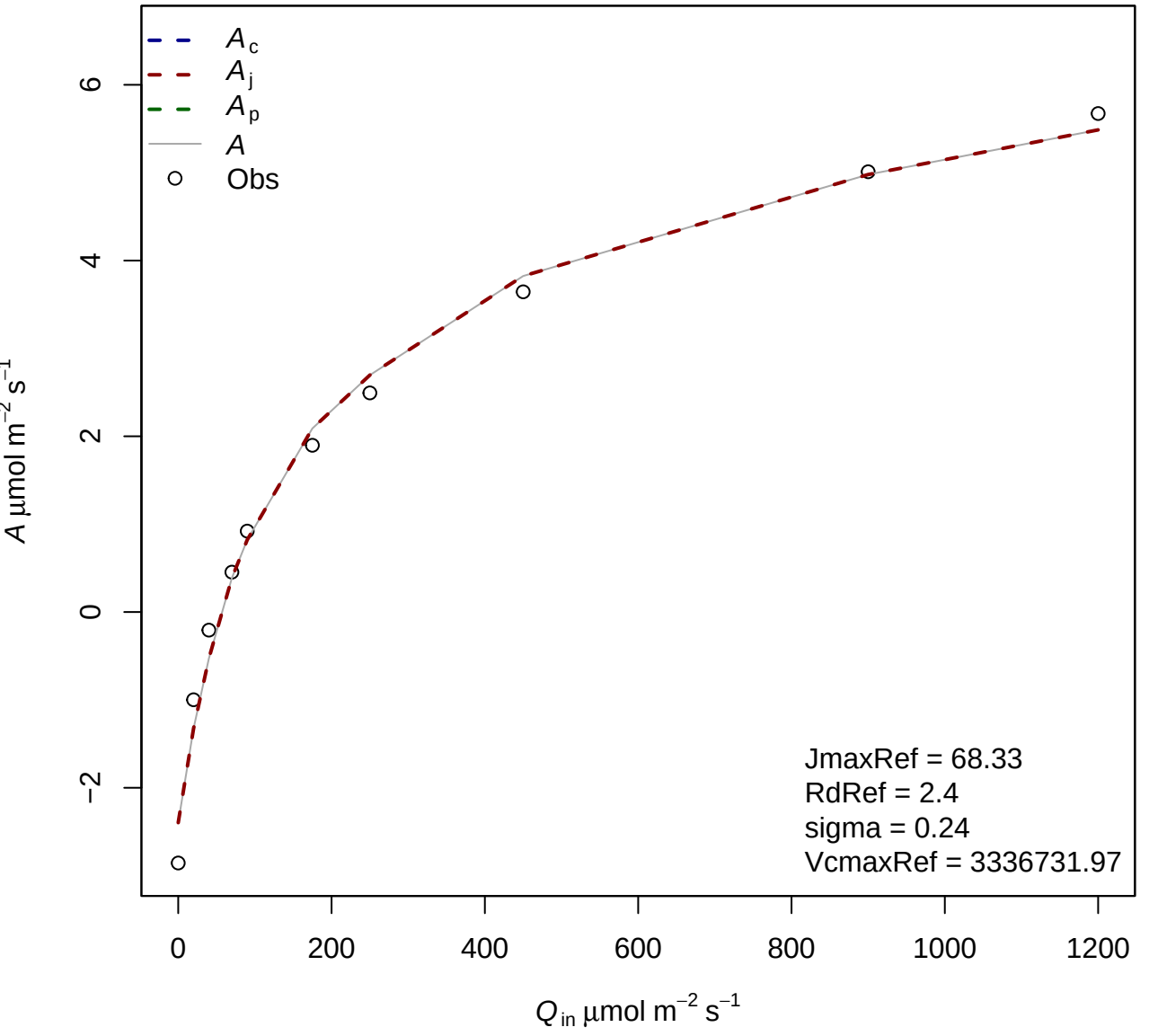

A_c
A_j
A_p
A
Obs
JmaxRef = 169.61
RdRef = 0.71
sigma = 0.2
VcmaxRef = 38.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

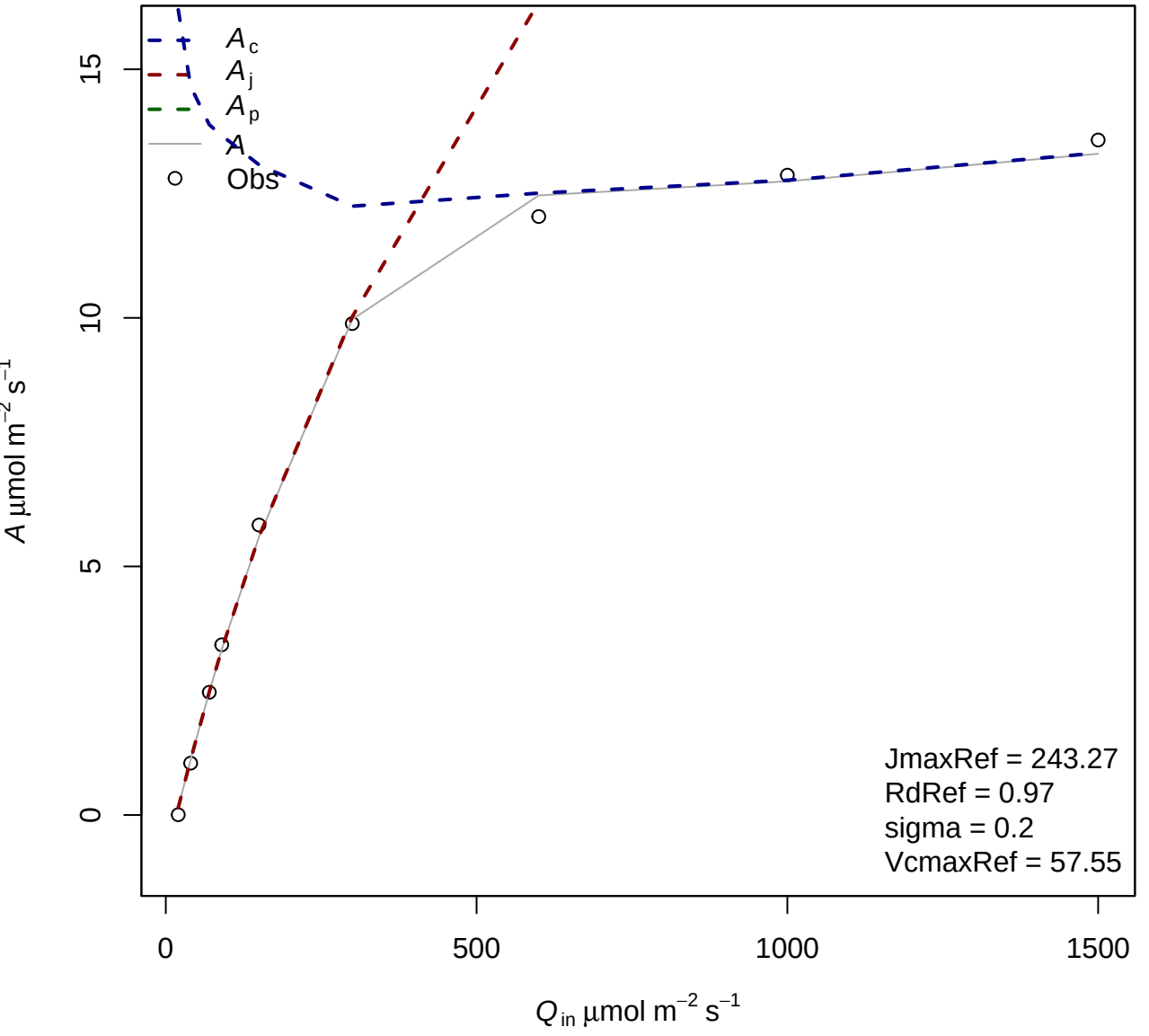

82
A_c
A_j
A_p
A
Obs
JmaxRef = 177.8
RdRef = 2.21
sigma = 0.09
VcmaxRef = 47.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
200
400
600
800
1000
1200

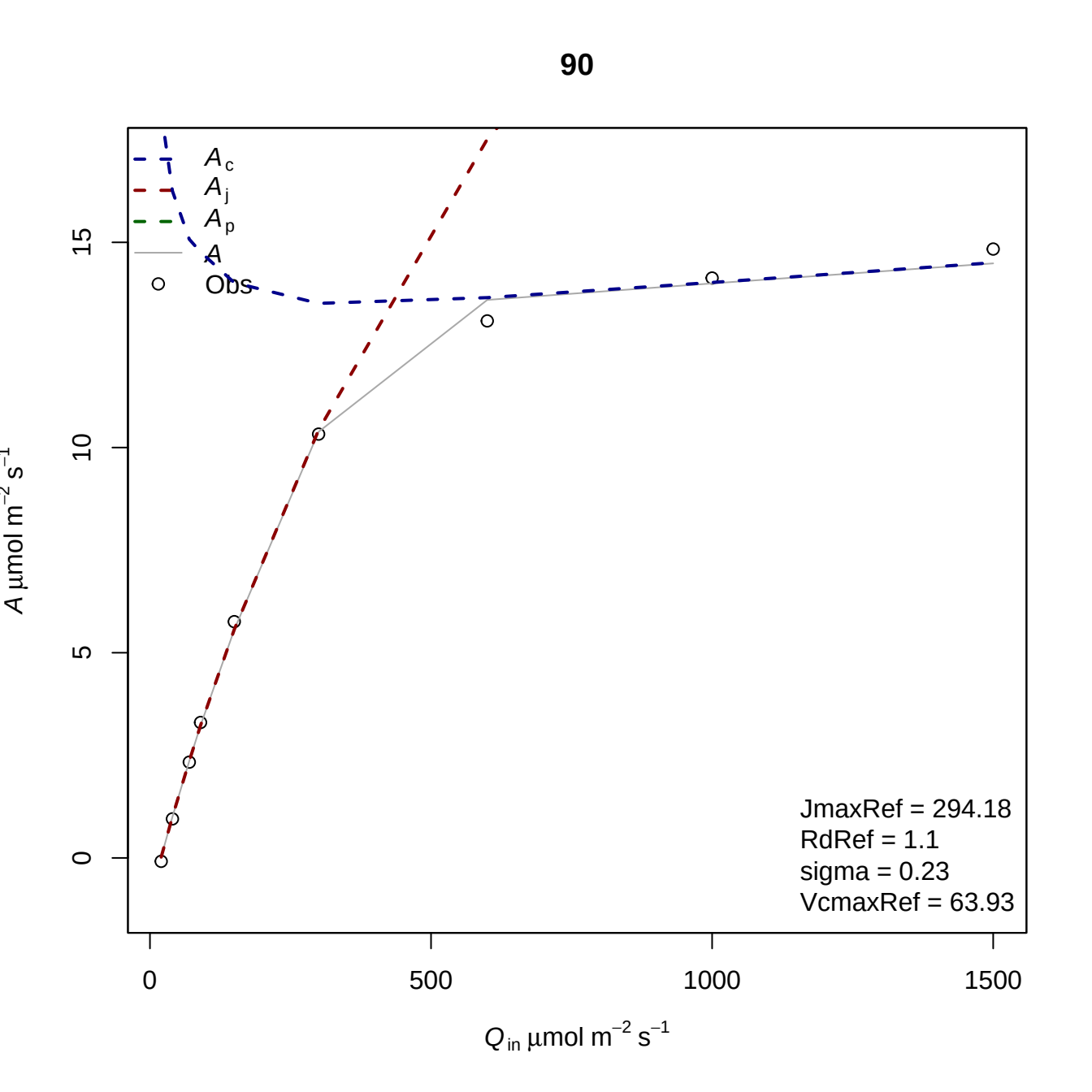

A_c
A_j
A_p
A
Obs
JmaxRef = 133.83
RdRef = 2.52
sigma = 0.42
VcmaxRef = 55.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200

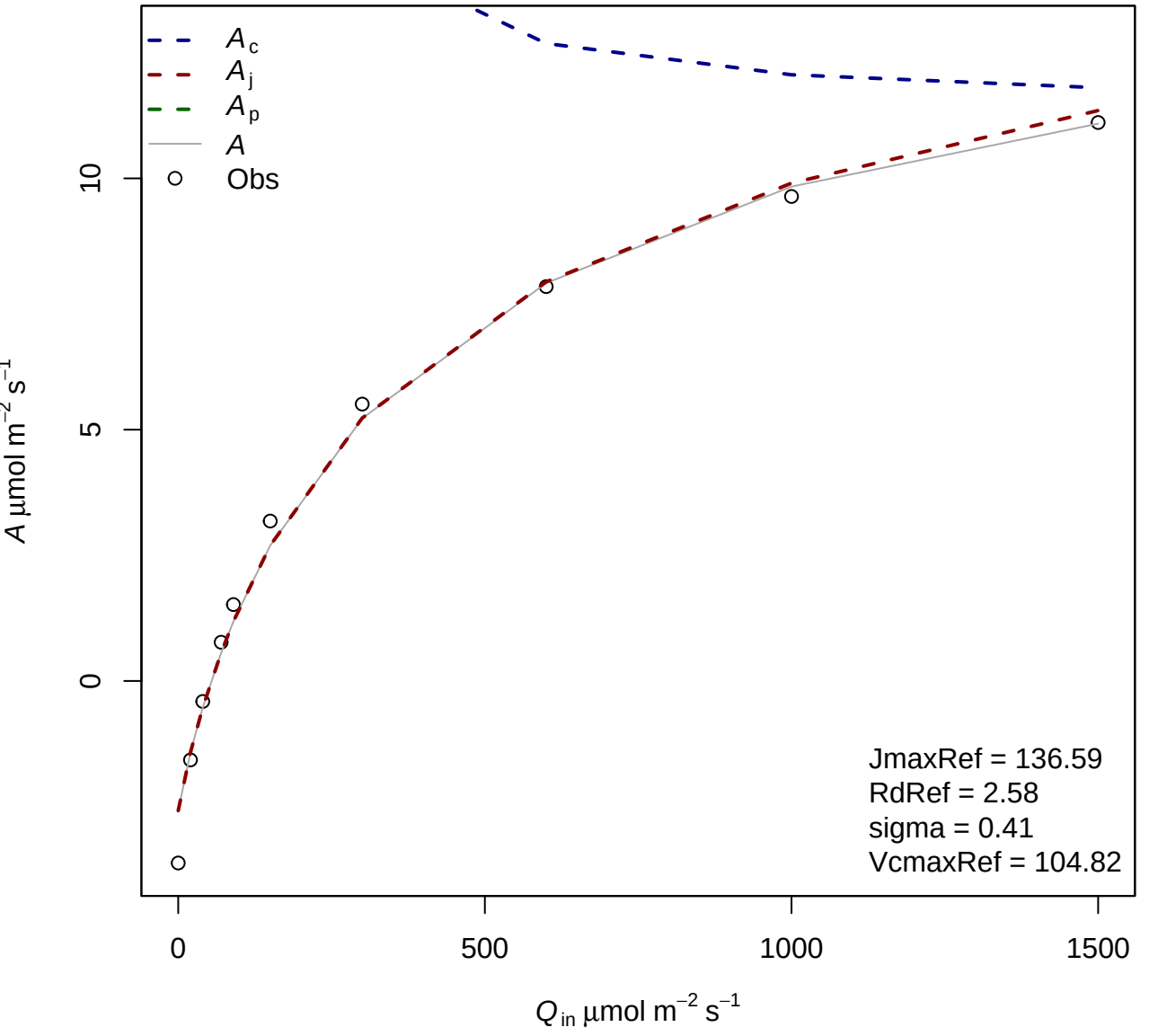

84
A_c
A_j
A_p
A
Obs
JmaxRef = 286.23
RdRef = 0.87
sigma = 0.27
VcmaxRef = 51.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200

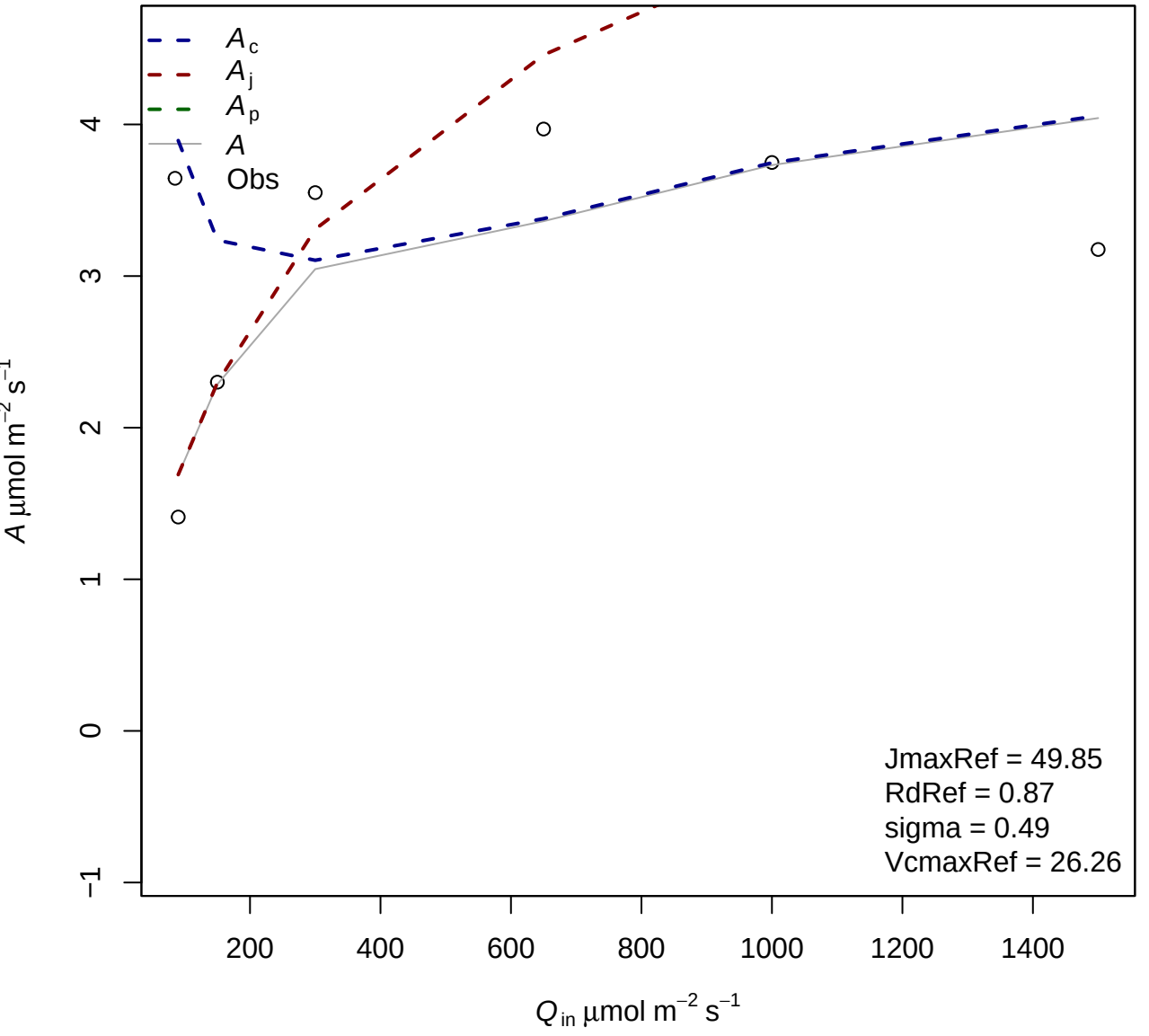

A_c
A_j
A_p
A
Obs
JmaxRef = 49.78
RdRef = 2.62
sigma = 0.45
VcmaxRef = 54150.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000
1200

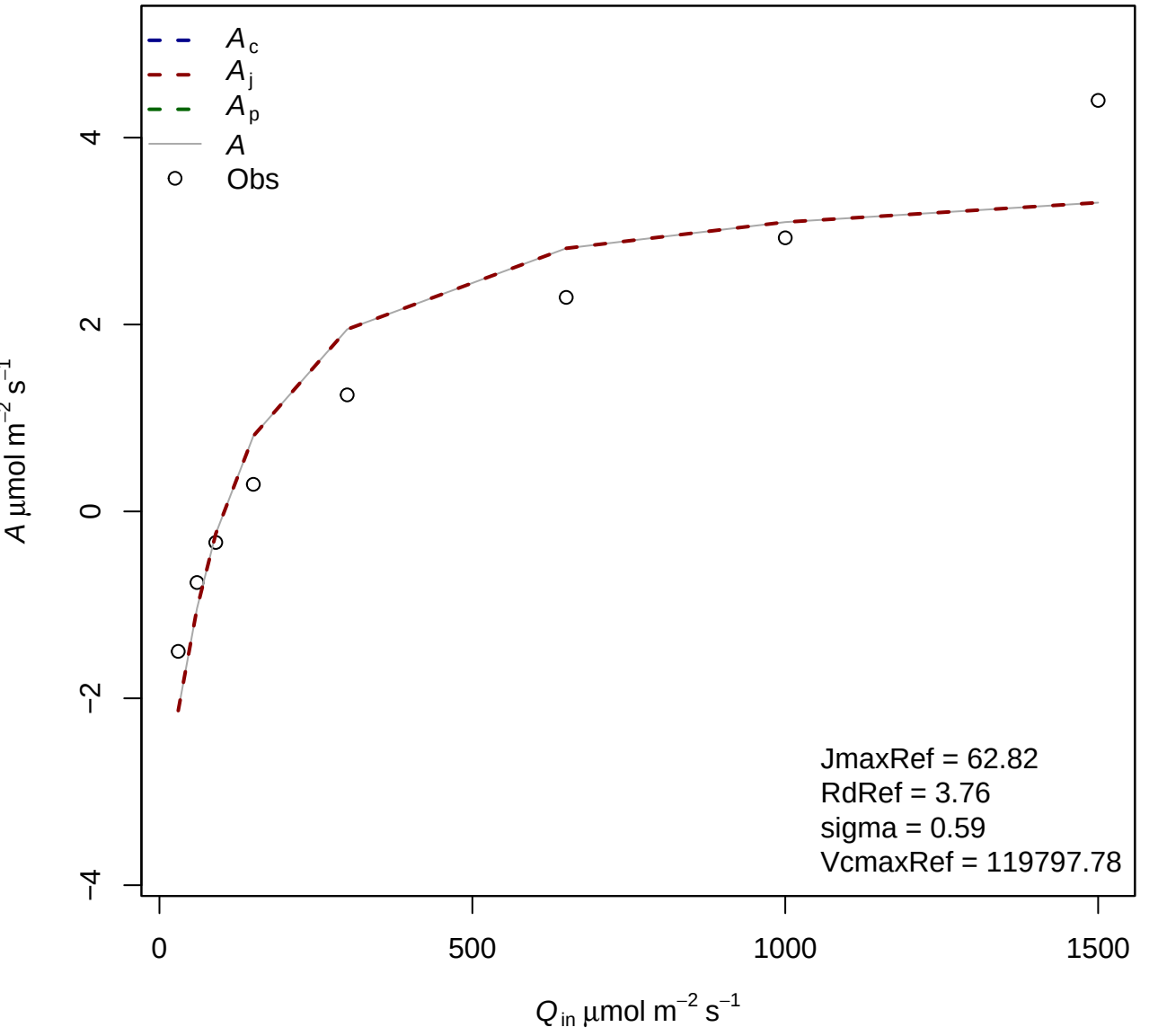

A_c
A_j
A_p
A
Obs
JmaxRef = 68.33
RdRef = 2.4
sigma = 0.24
VcmaxRef = 3336731.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200

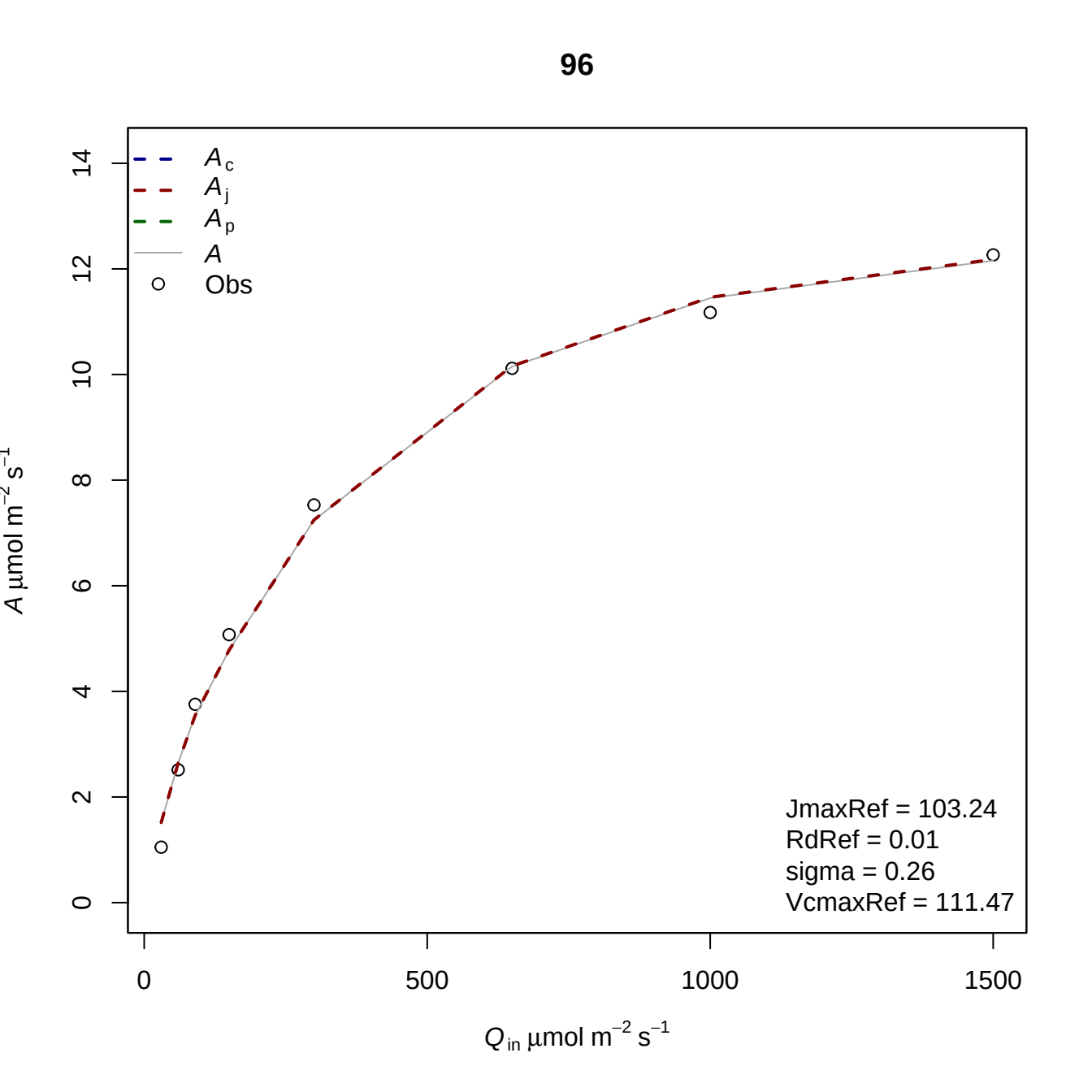

A_c
A_j
A_p
A
Obs
JmaxRef = 243.27
RdRef = 0.97
sigma = 0.2
VcmaxRef = 57.55
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

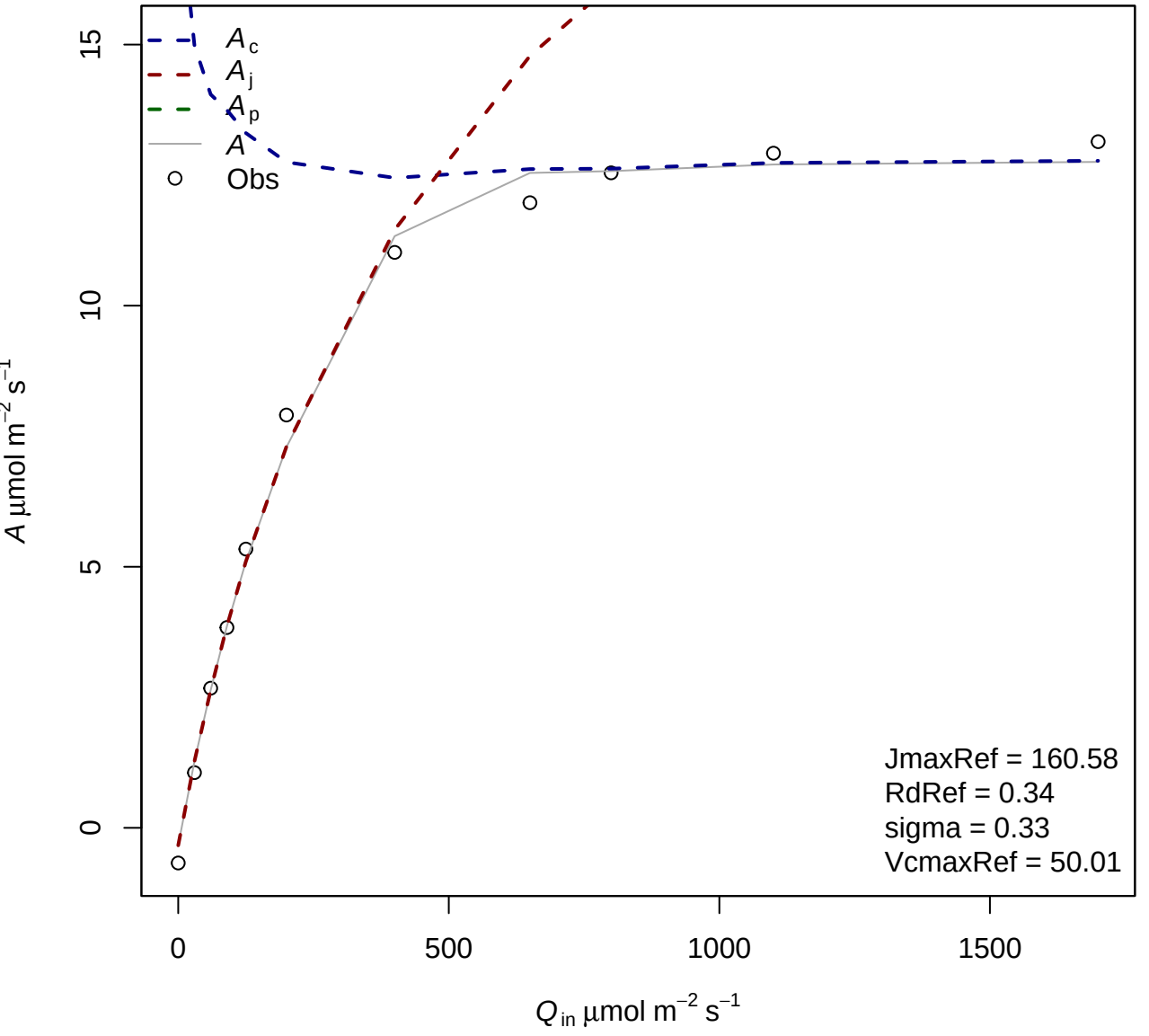

90
A_c
A_j
A_p
A
Obs
JmaxRef = 294.18
RdRef = 1.1
sigma = 0.23
VcmaxRef = 63.93
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

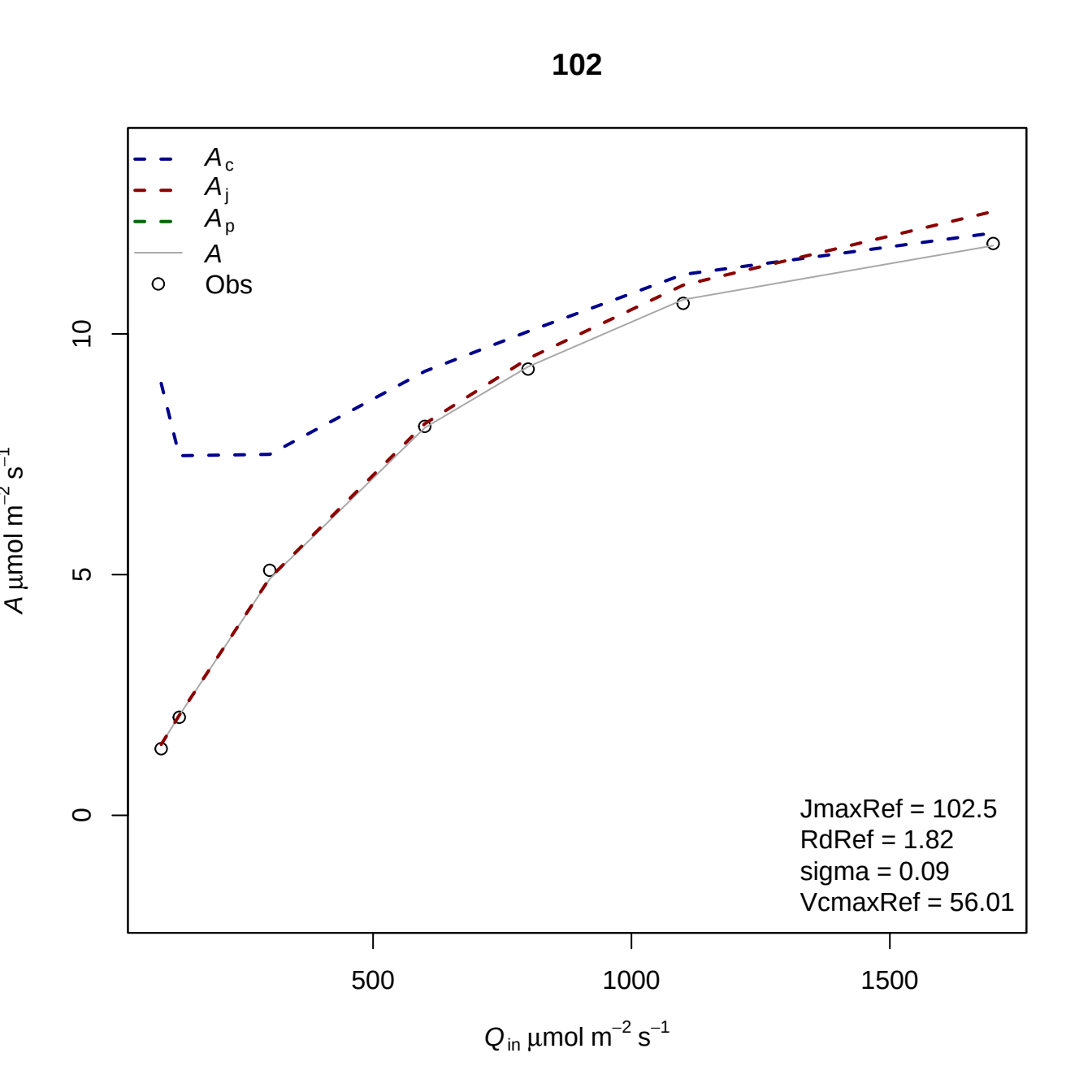

A_c
A_j
A_p
A
Obs
JmaxRef = 136.59
RdRef = 2.58
sigma = 0.41
VcmaxRef = 104.82
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

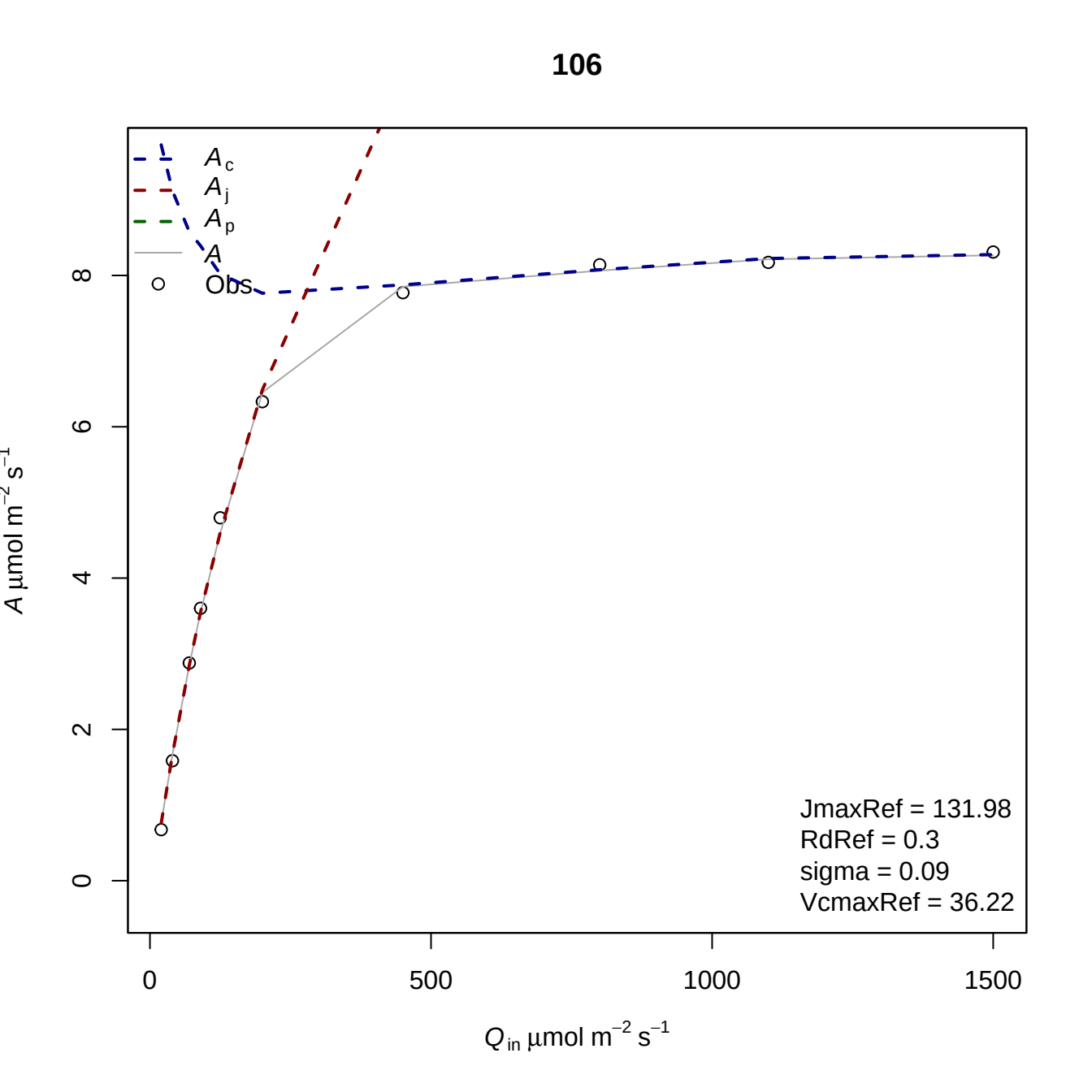

A_c
A_j
A_p
A
Obs
JmaxRef = 49.85
RdRef = 0.87
sigma = 0.49
VcmaxRef = 26.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
200
400
600
800
1000
1200
1400

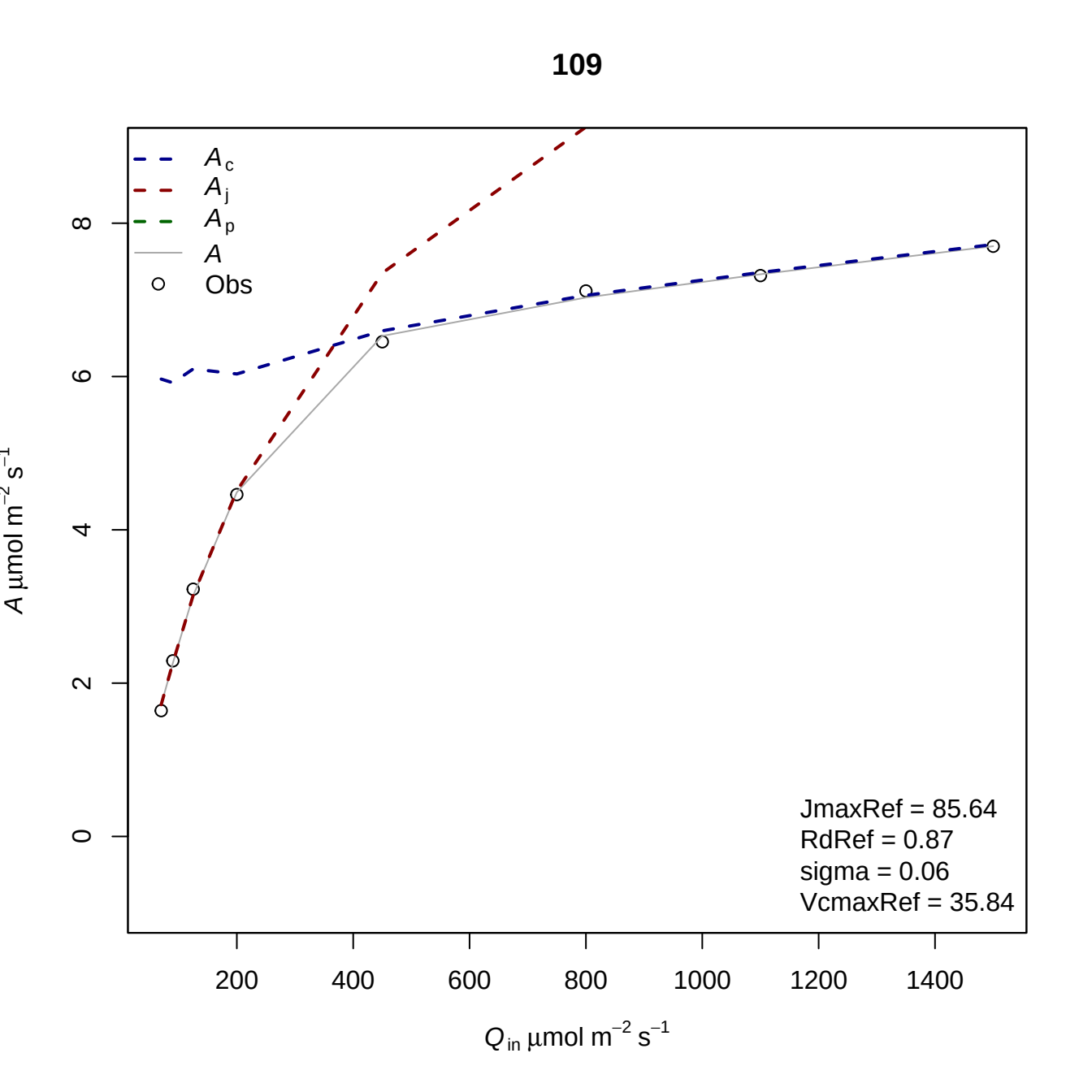

A_c
A_j
A_p
A
Obs
JmaxRef = 62.82
RdRef = 3.76
sigma = 0.59
VcmaxRef = 119797.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
500
1000
1500

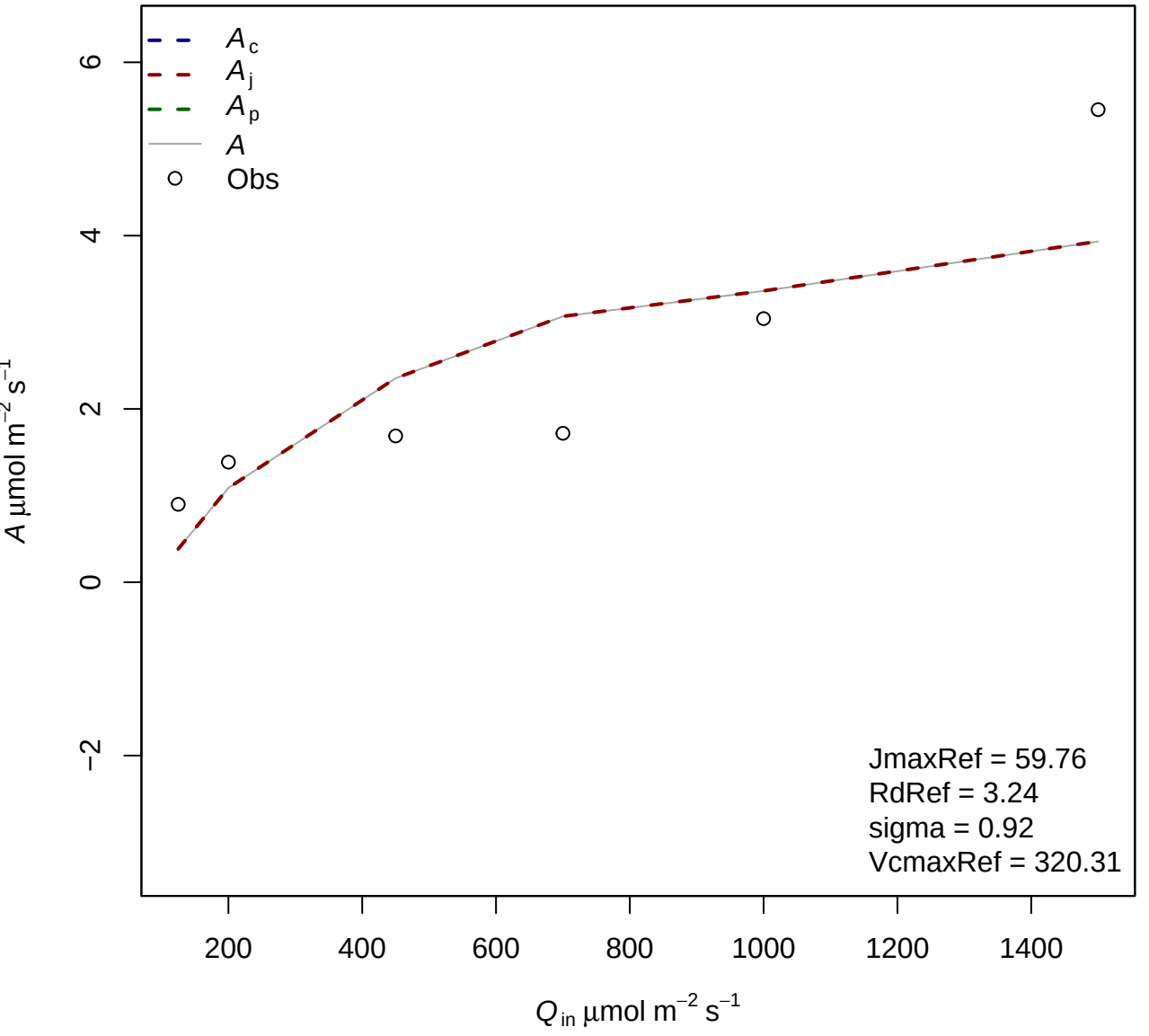

96
A_c
A_j
A_p
A
Obs
JmaxRef = 103.24
RdRef = 0.01
sigma = 0.26
VcmaxRef = 111.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

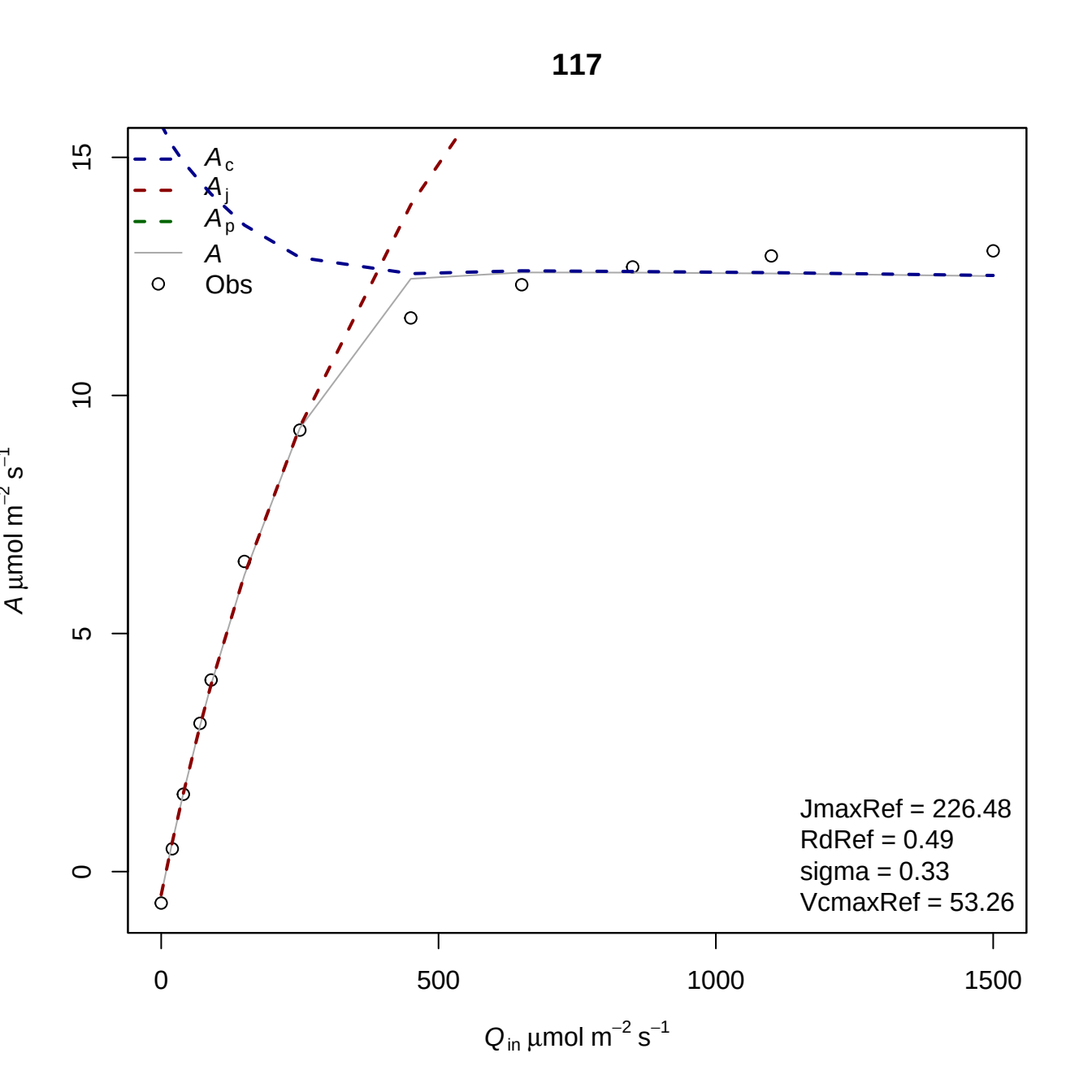

A_c
A_j
A_p
A
Obs
JmaxRef = 160.58
RdRef = 0.34
sigma = 0.33
VcmaxRef = 50.01
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

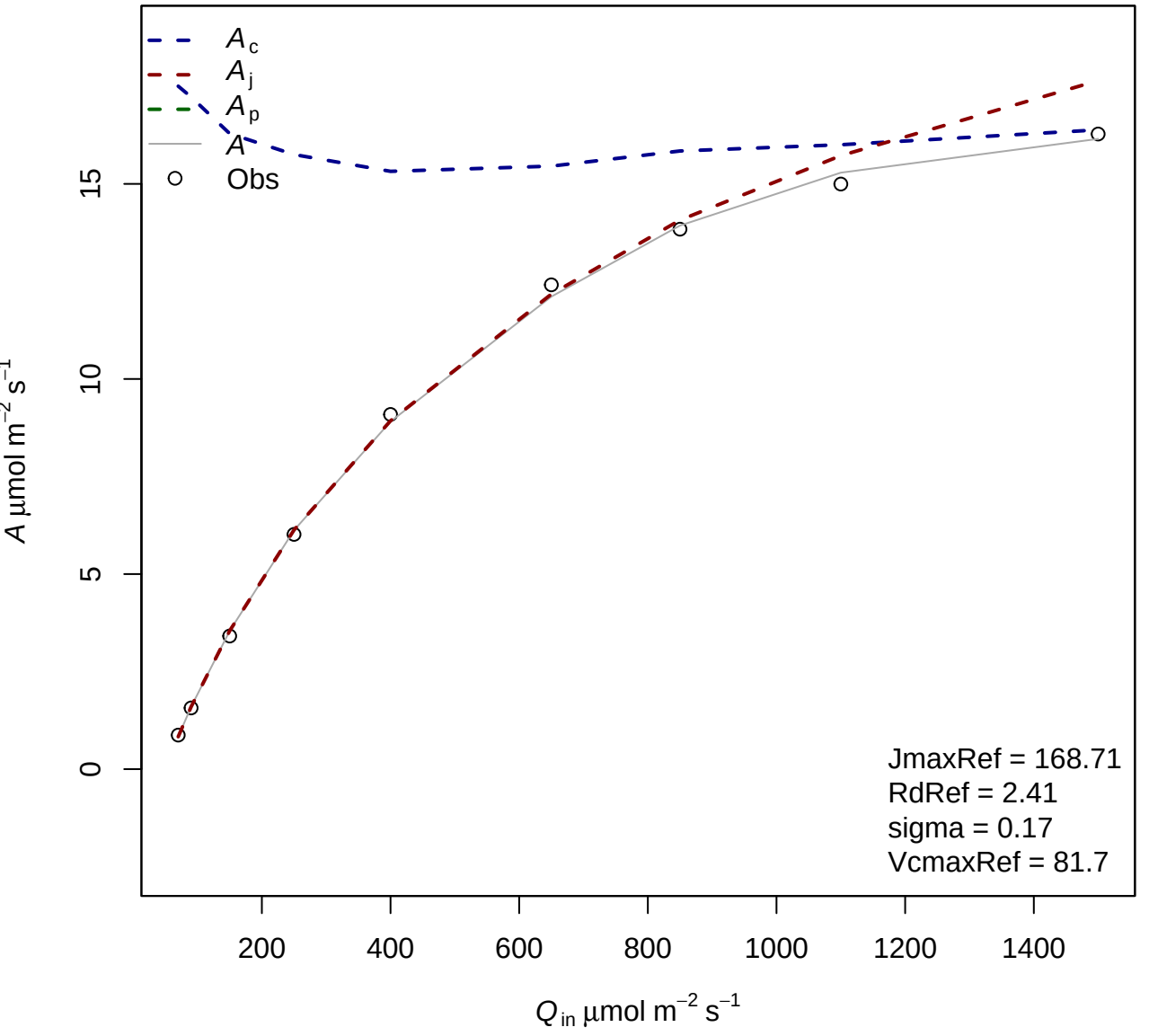

102
A_c
A_j
A_p
A
Obs
JmaxRef = 102.5
RdRef = 1.82
sigma = 0.09
VcmaxRef = 56.01
0
5
10
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

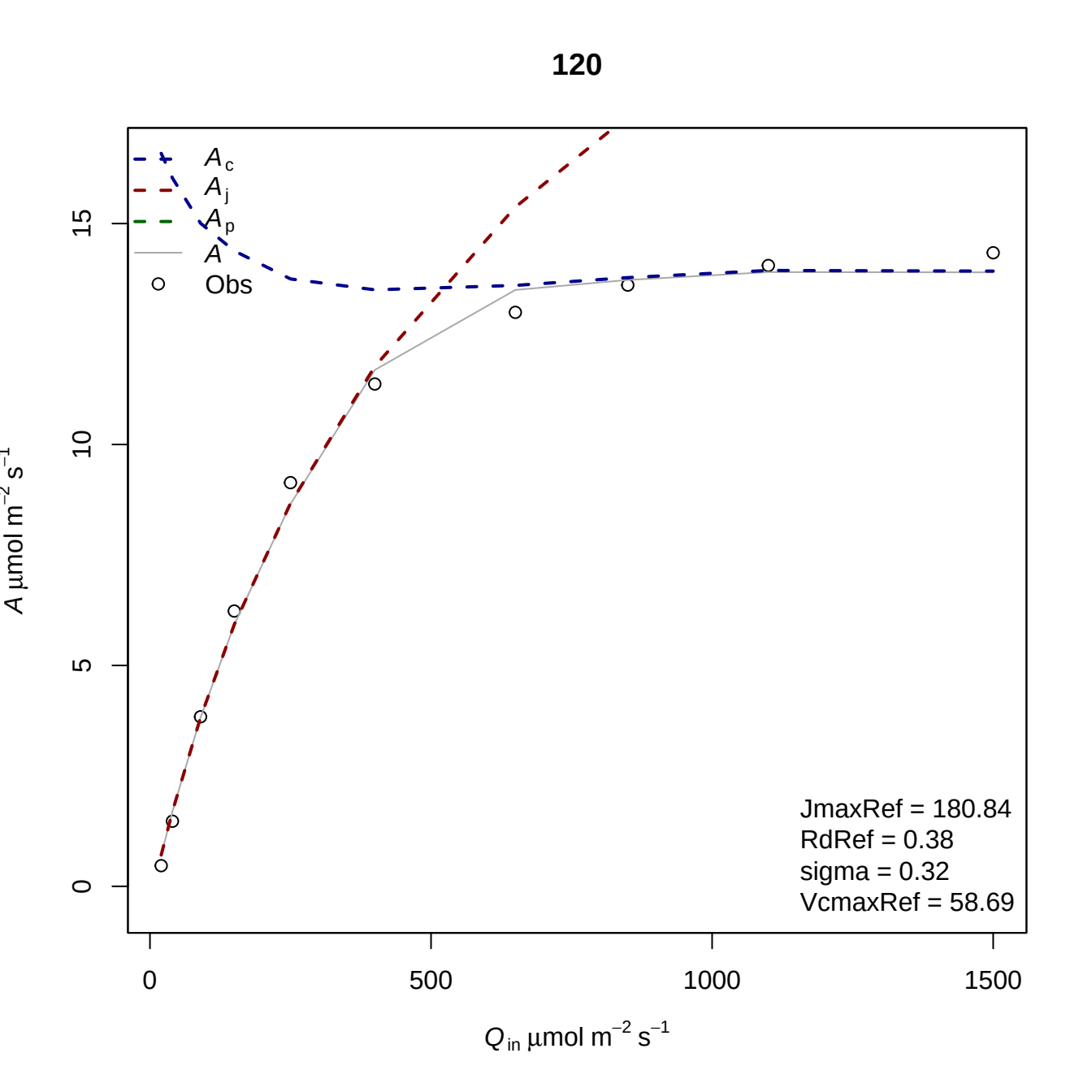

106
A_c
A_j
A_p
A
Obs
JmaxRef = 131.98
RdRef = 0.3
sigma = 0.09
VcmaxRef = 36.22
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

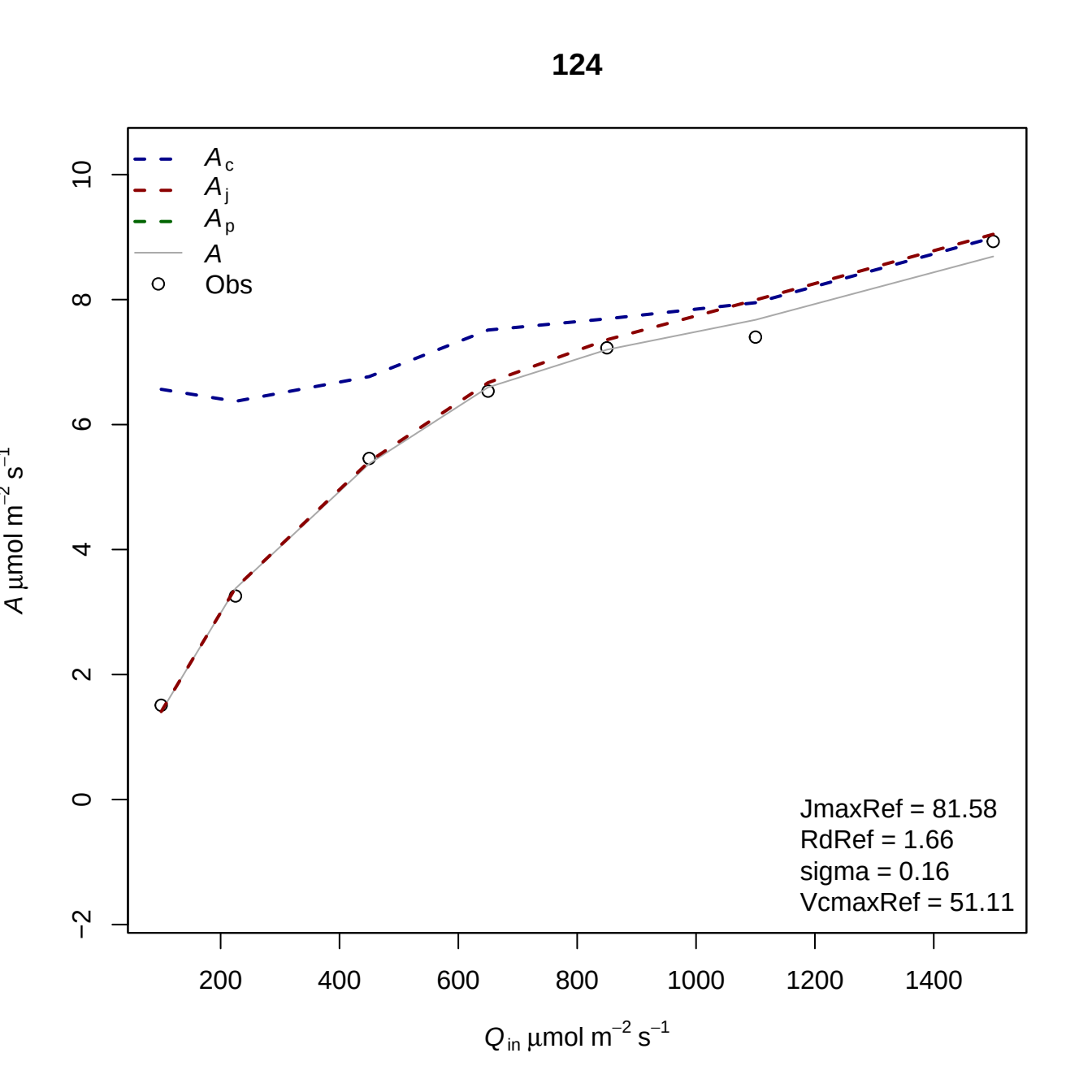

109
A_c
A_j
A_p
A
Obs
JmaxRef = 85.64
RdRef = 0.87
sigma = 0.06
VcmaxRef = 35.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
200
400
600
800
1000
1200
1400

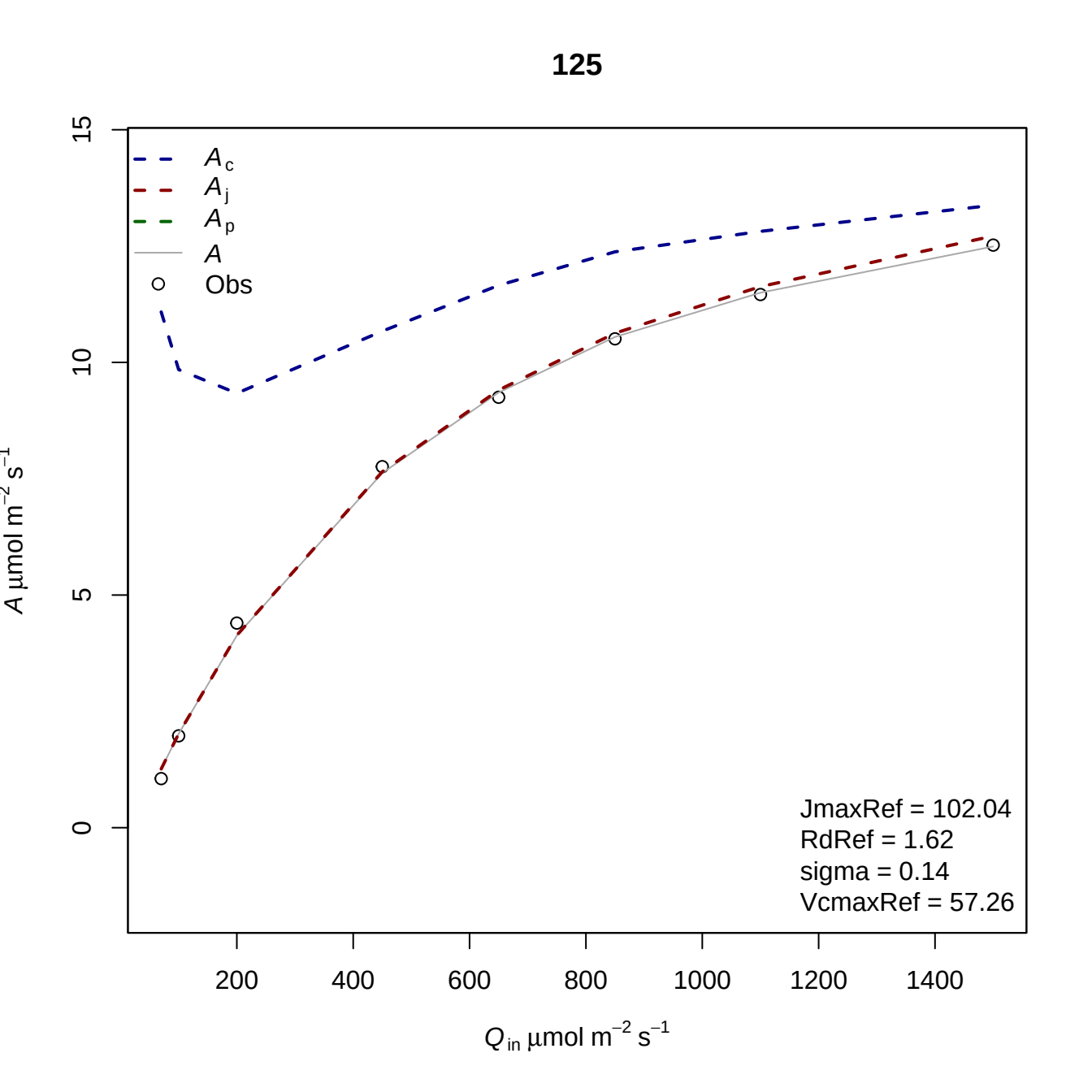

A_c
A_j
A_p
A
Obs
6
4
2
0
−2
A µmol m^{-2} s^{-1}
200
400
600
800
1000
1200
1400
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 59.76
RdRef = 3.24
sigma = 0.92
VcmaxRef = 320.31

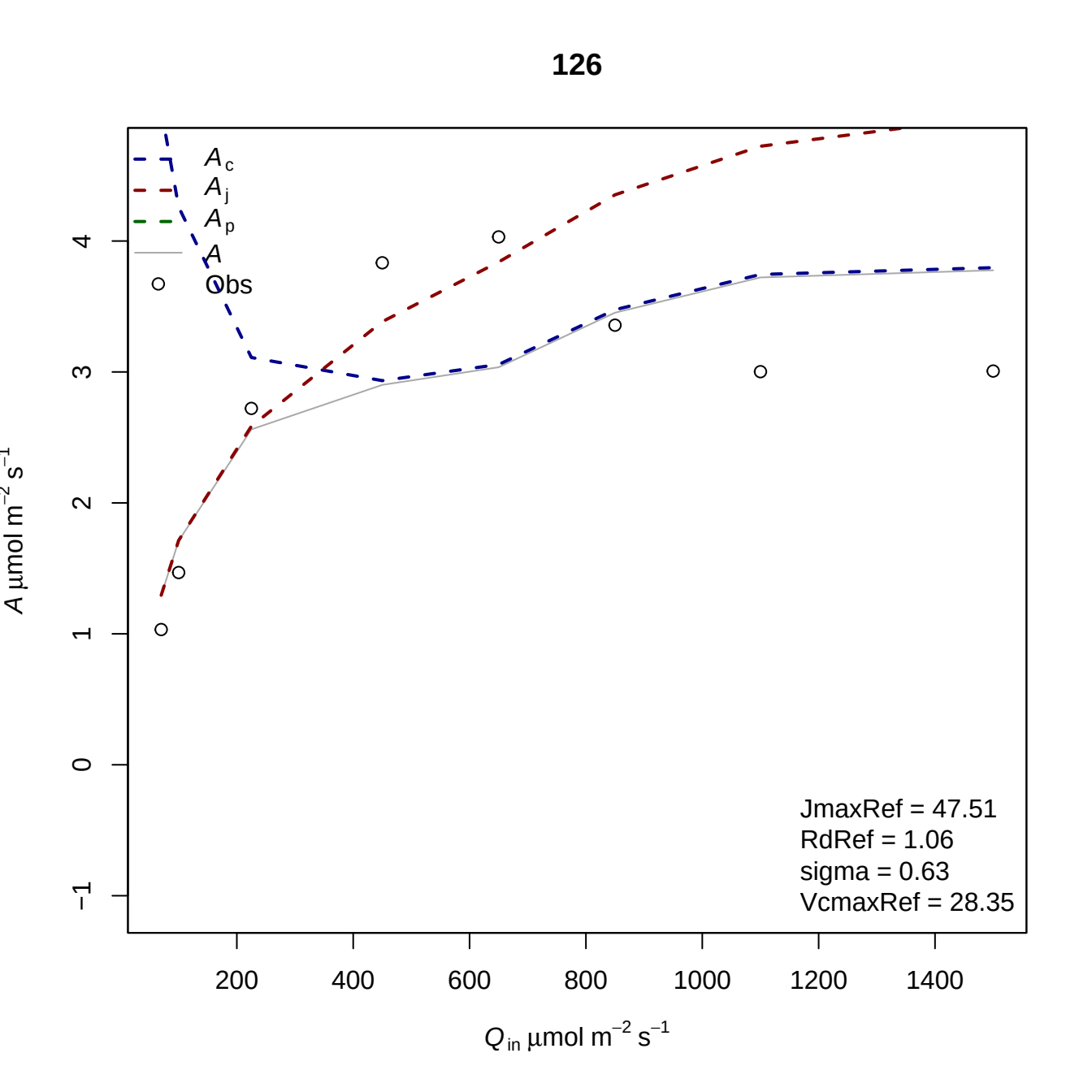

117
A_c
A_j
A_p
A
Obs
JmaxRef = 226.48
RdRef = 0.49
sigma = 0.33
VcmaxRef = 53.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

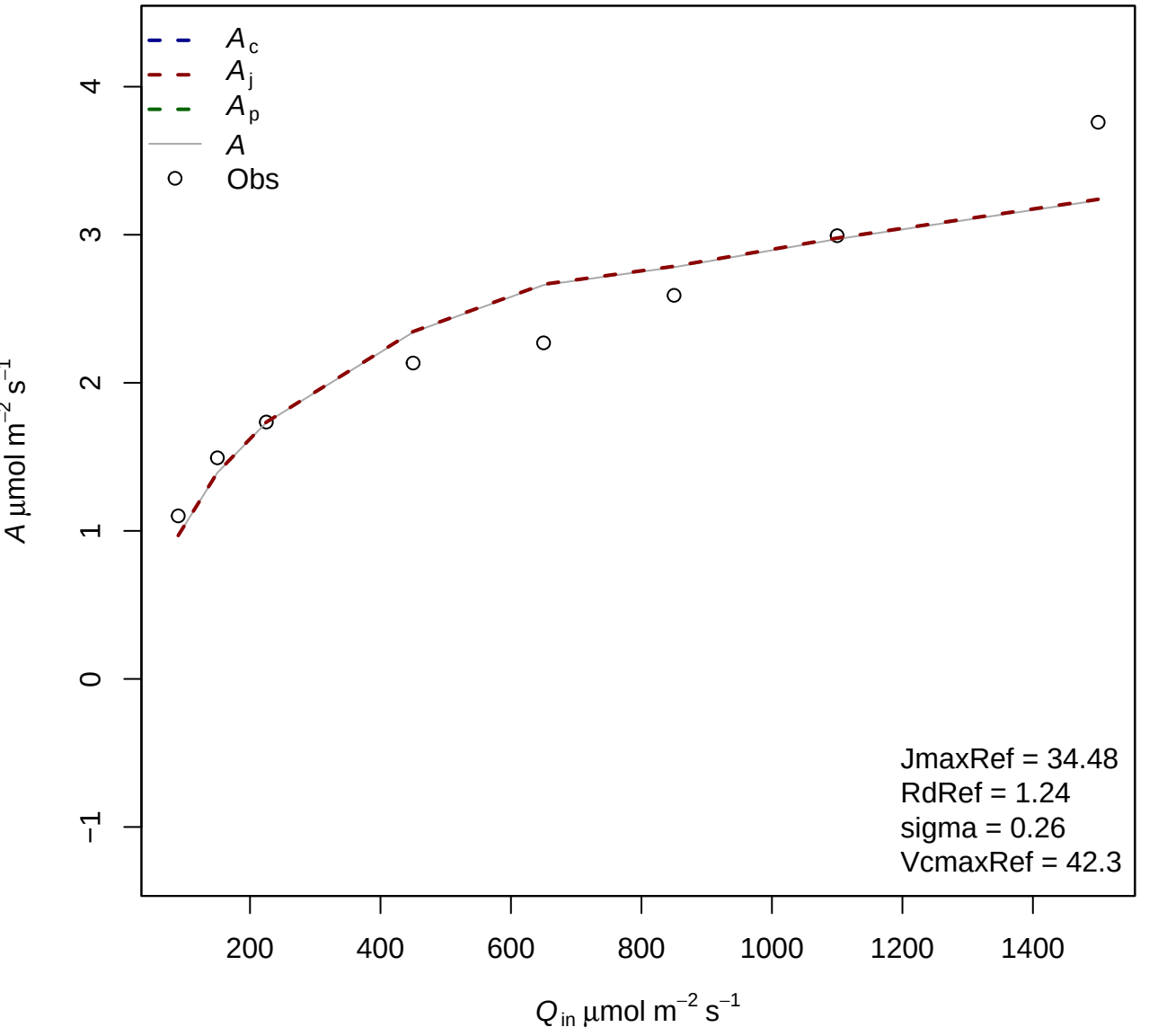

A_c
A_j
A_p
A
Obs
JmaxRef = 168.71
RdRef = 2.41
sigma = 0.17
VcmaxRef = 81.7
0
5
10
15
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

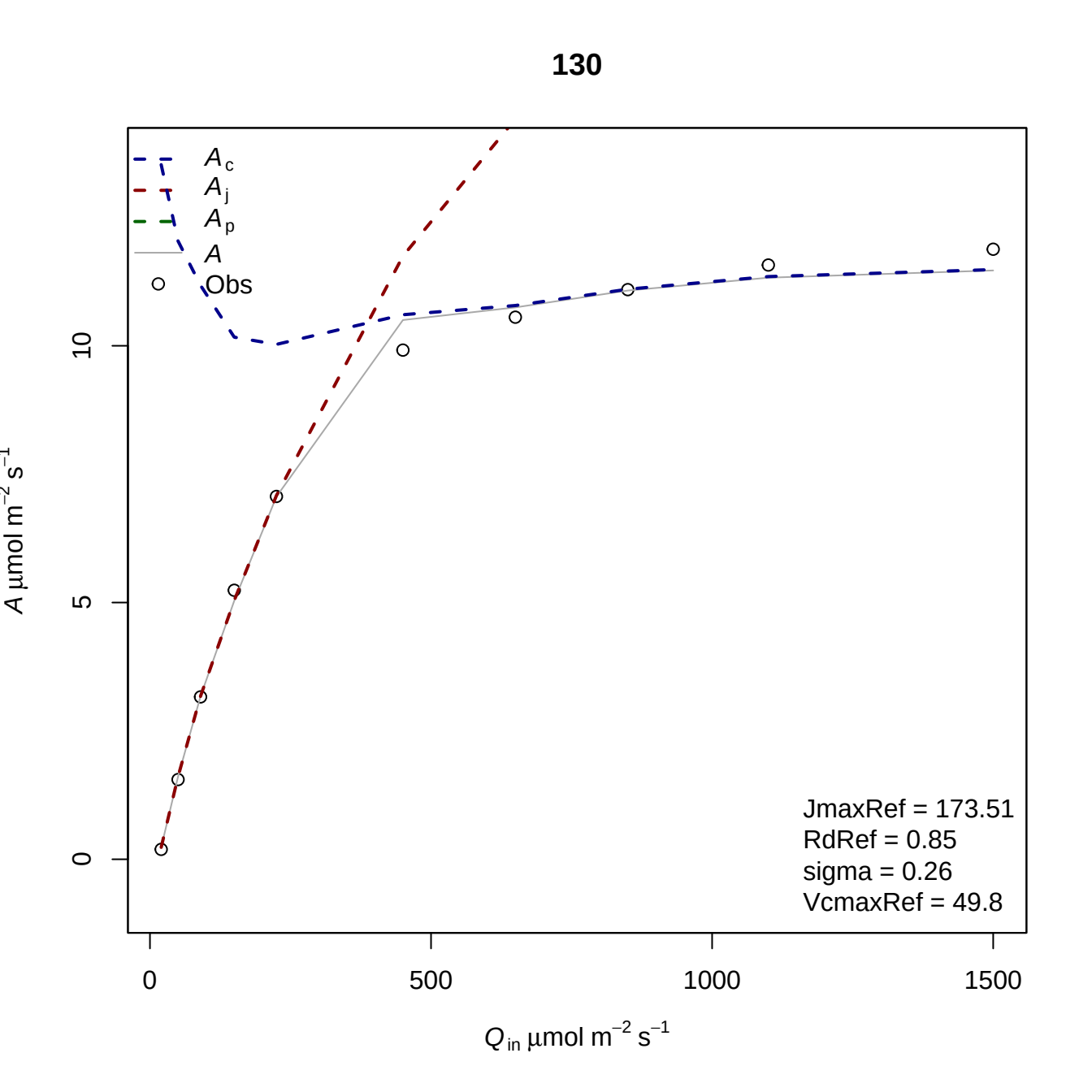

120
A_c
A_j
A_p
A
Obs
JmaxRef = 180.84
RdRef = 0.38
sigma = 0.32
VcmaxRef = 58.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

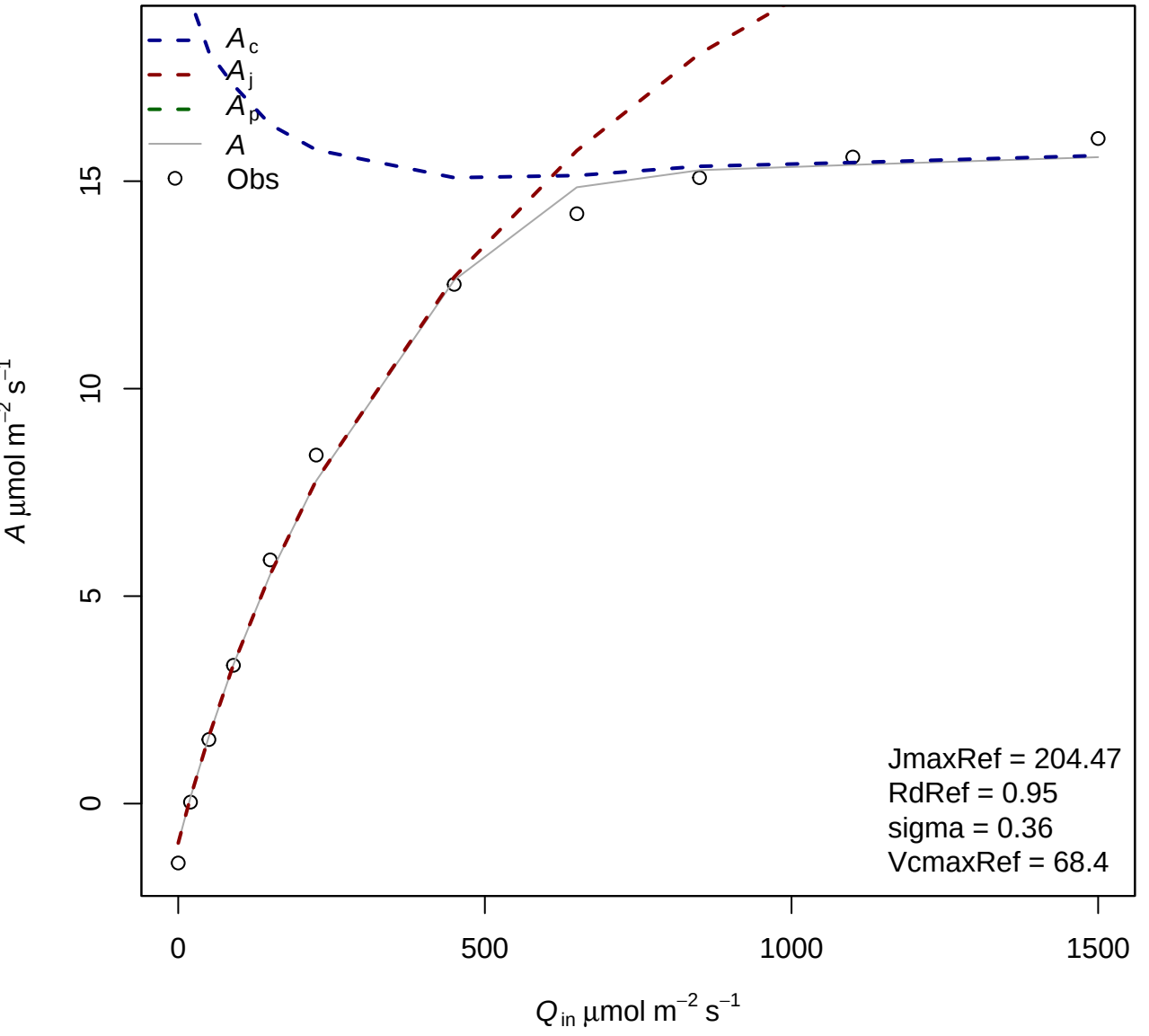

124
A_c
A_j
A_p
A
Obs
JmaxRef = 81.58
RdRef = 1.66
sigma = 0.16
VcmaxRef = 51.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
200
400
600
800
1000
1200
1400

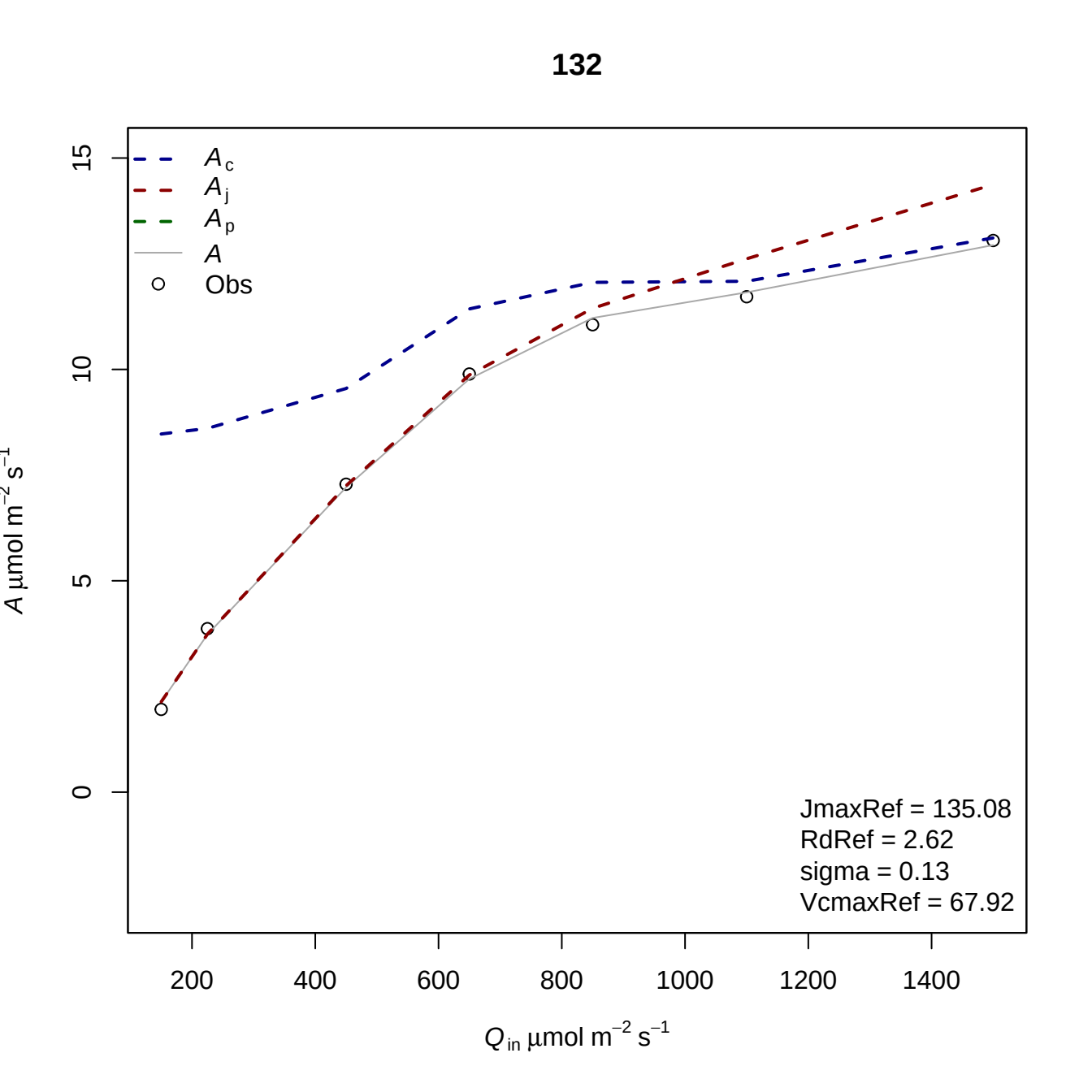

125
A_c
A_j
A_p
A
Obs
JmaxRef = 102.04
RdRef = 1.62
sigma = 0.14
VcmaxRef = 57.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
200
400
600
800
1000
1200
1400

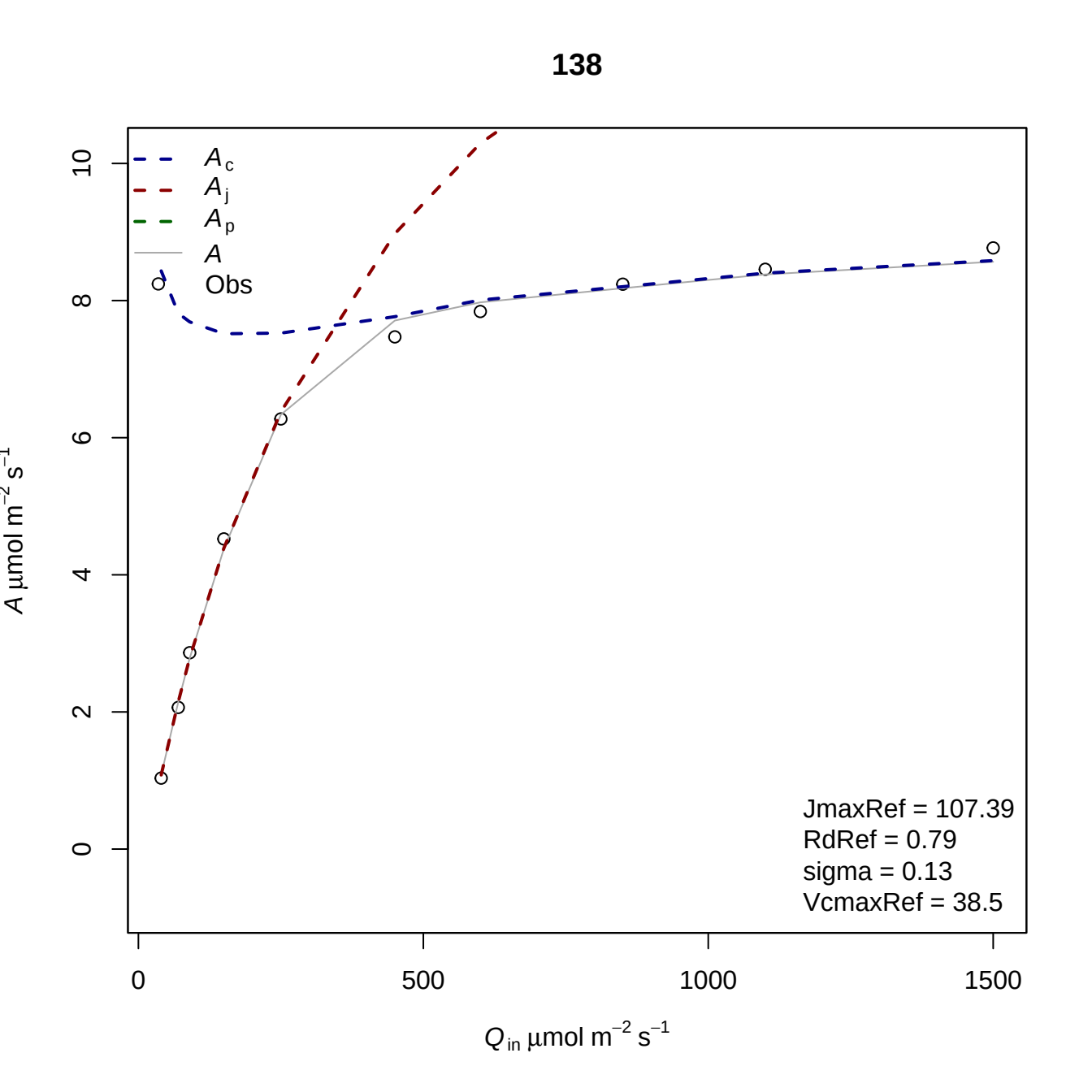

126
A_c
A_j
A_p
A
Obs
JmaxRef = 47.51
RdRef = 1.06
sigma = 0.63
VcmaxRef = 28.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
200
400
600
800
1000
1200
1400

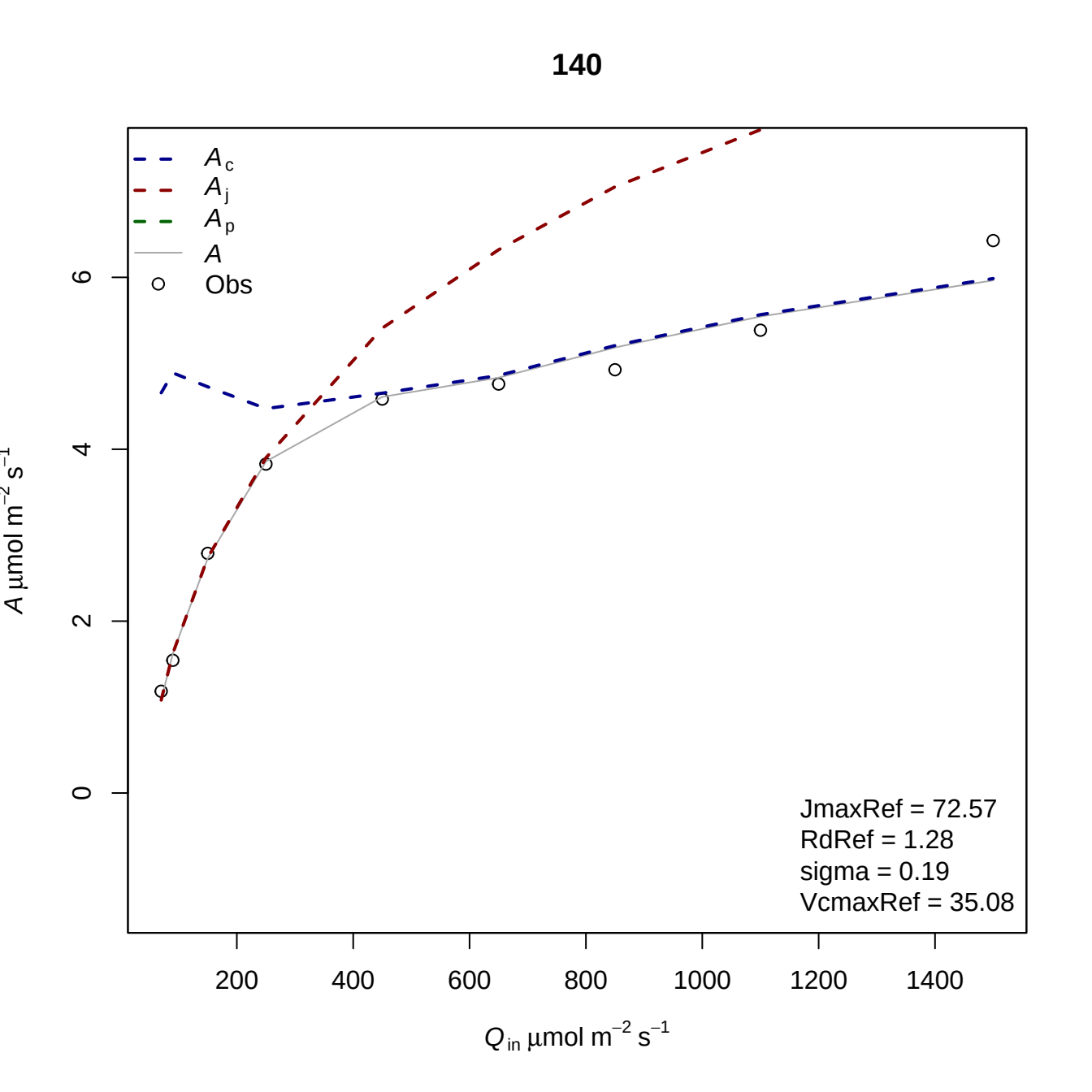

A_c
A_j
A_p
A
Obs
JmaxRef = 34.48
RdRef = 1.24
sigma = 0.26
VcmaxRef = 42.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
200
400
600
800
1000
1200
1400

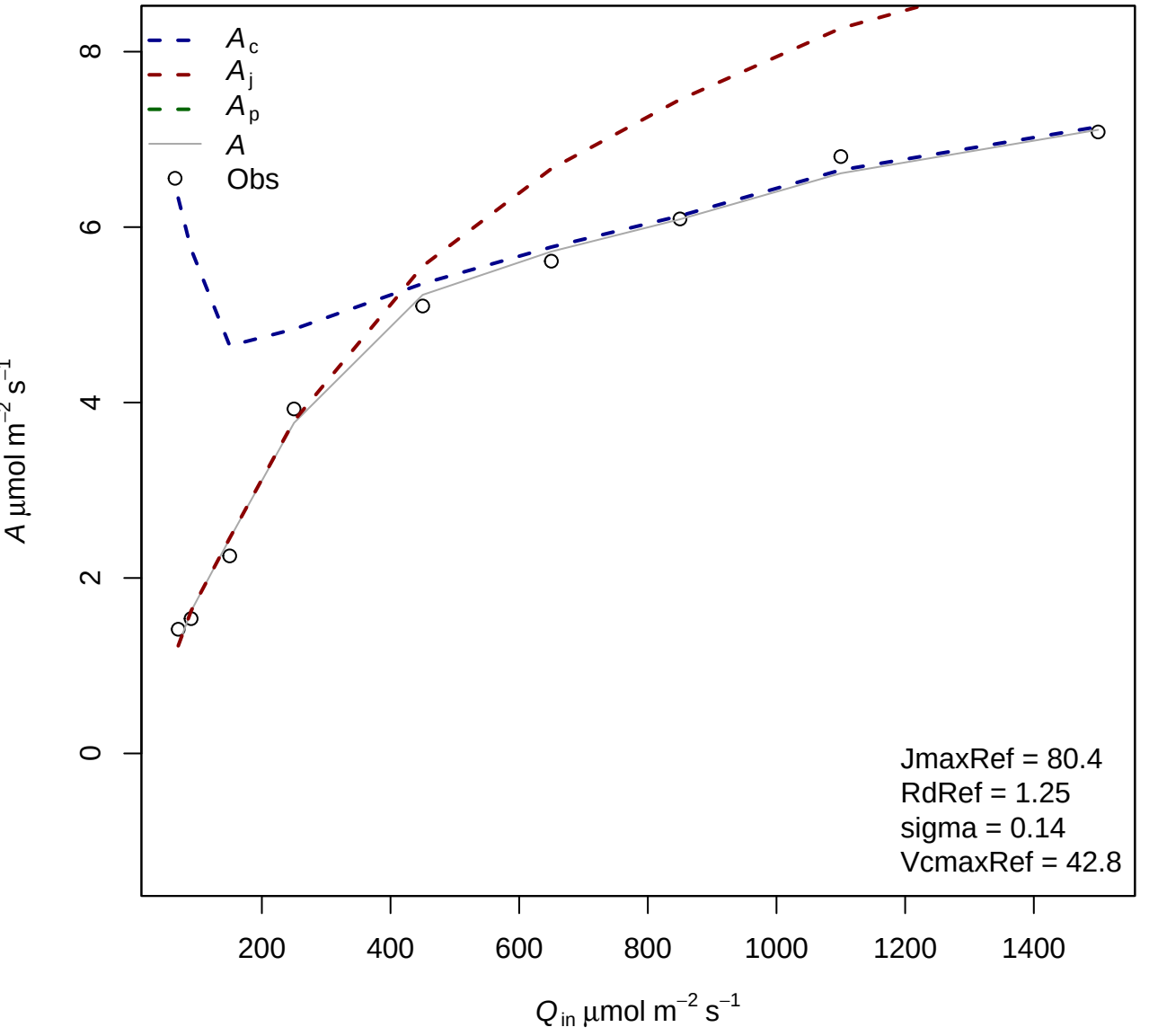

130
A_c
A_j
A_p
A
Obs
JmaxRef = 173.51
RdRef = 0.85
sigma = 0.26
VcmaxRef = 49.8
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

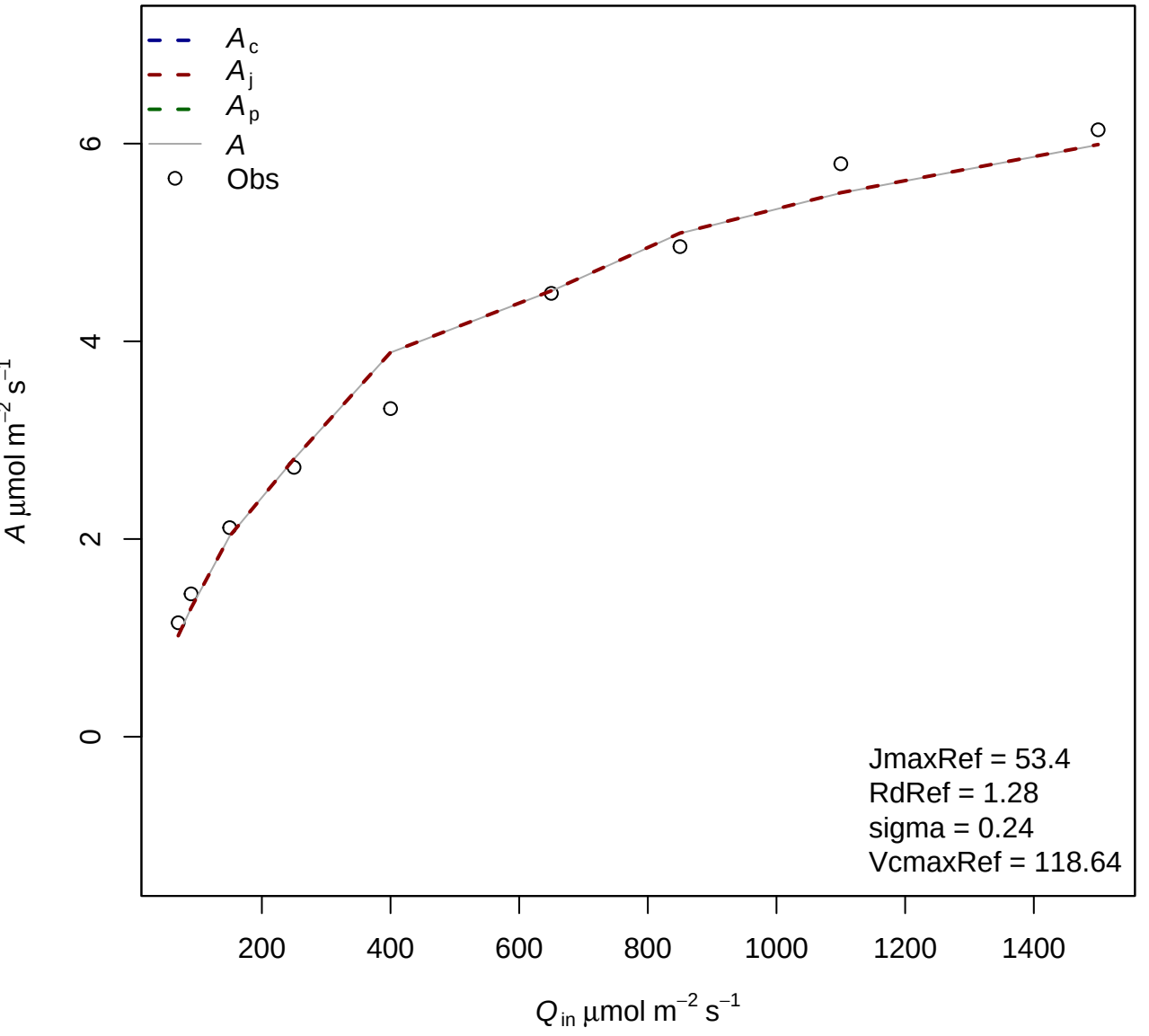

A_c
A_j
A_p
A
Obs
JmaxRef = 204.47
RdRef = 0.95
sigma = 0.36
VcmaxRef = 68.4
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

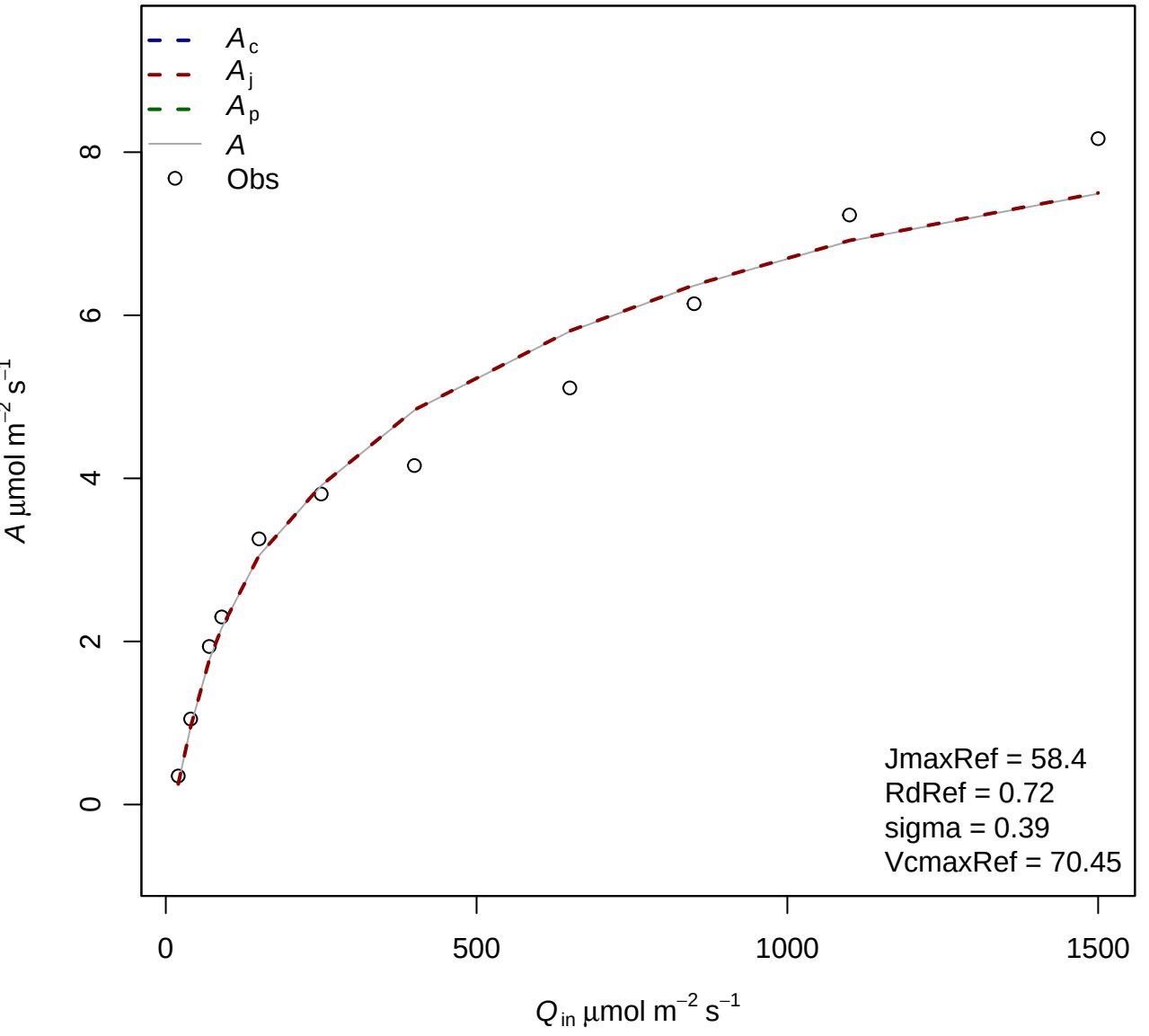

132
A_c
A_j
A_p
A
Obs
JmaxRef = 135.08
RdRef = 2.62
sigma = 0.13
VcmaxRef = 67.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
200
400
600
800
1000
1200
1400

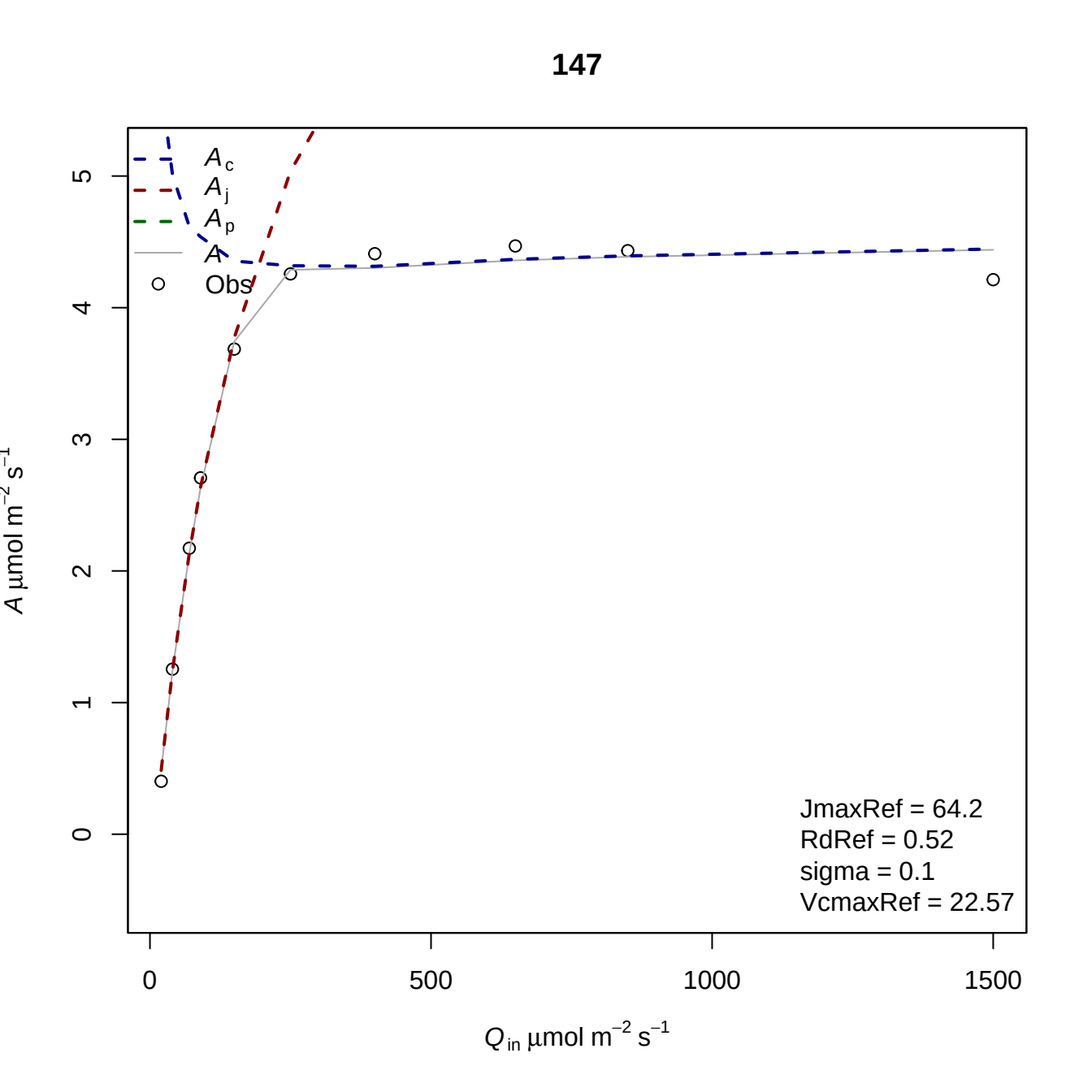

138
A_c
A_j
A_p
A
Obs
JmaxRef = 107.39
RdRef = 0.79
sigma = 0.13
VcmaxRef = 38.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

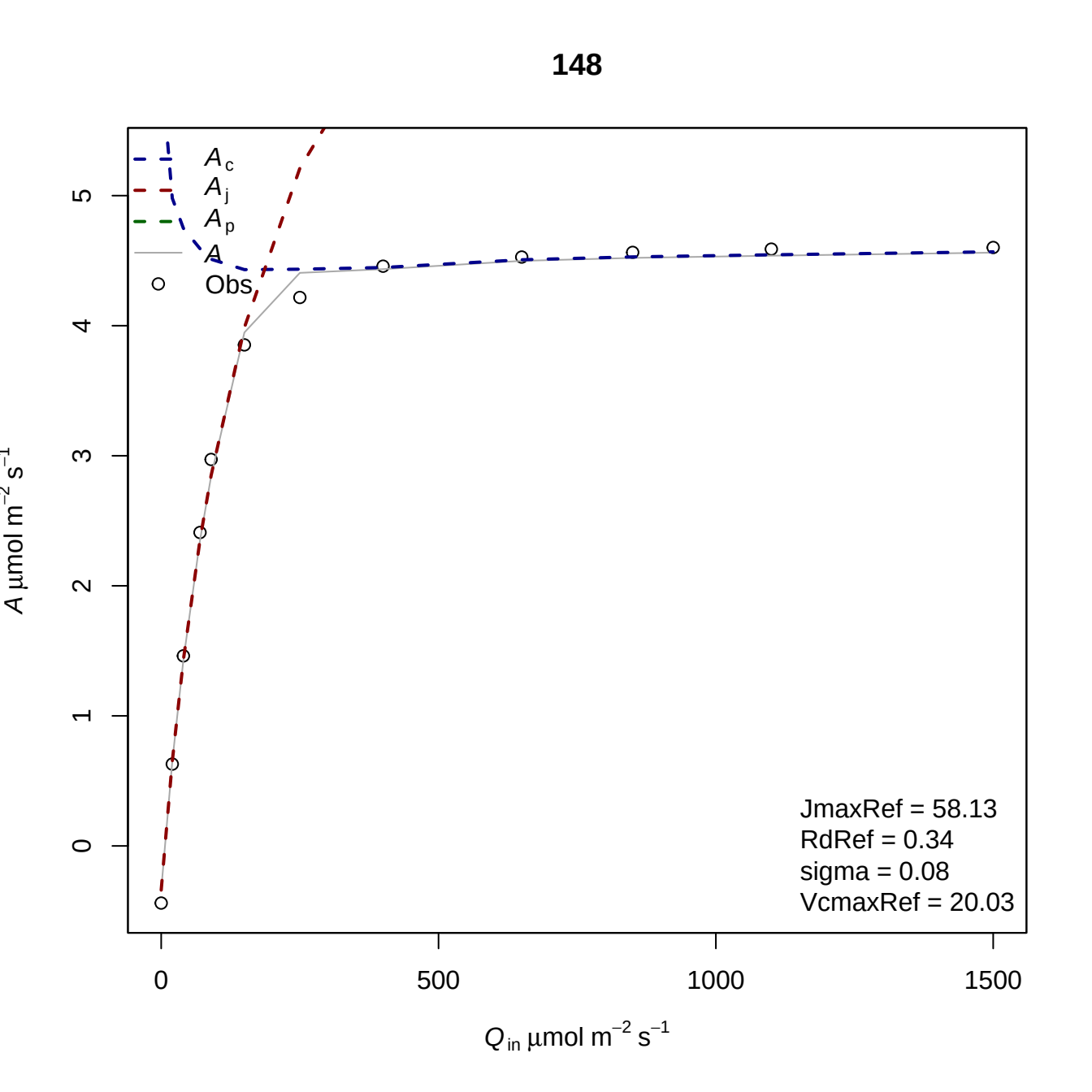

140
A_c
A_j
A_p
A
Obs
JmaxRef = 72.57
RdRef = 1.28
sigma = 0.19
VcmaxRef = 35.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
200
400
600
800
1000
1200
1400

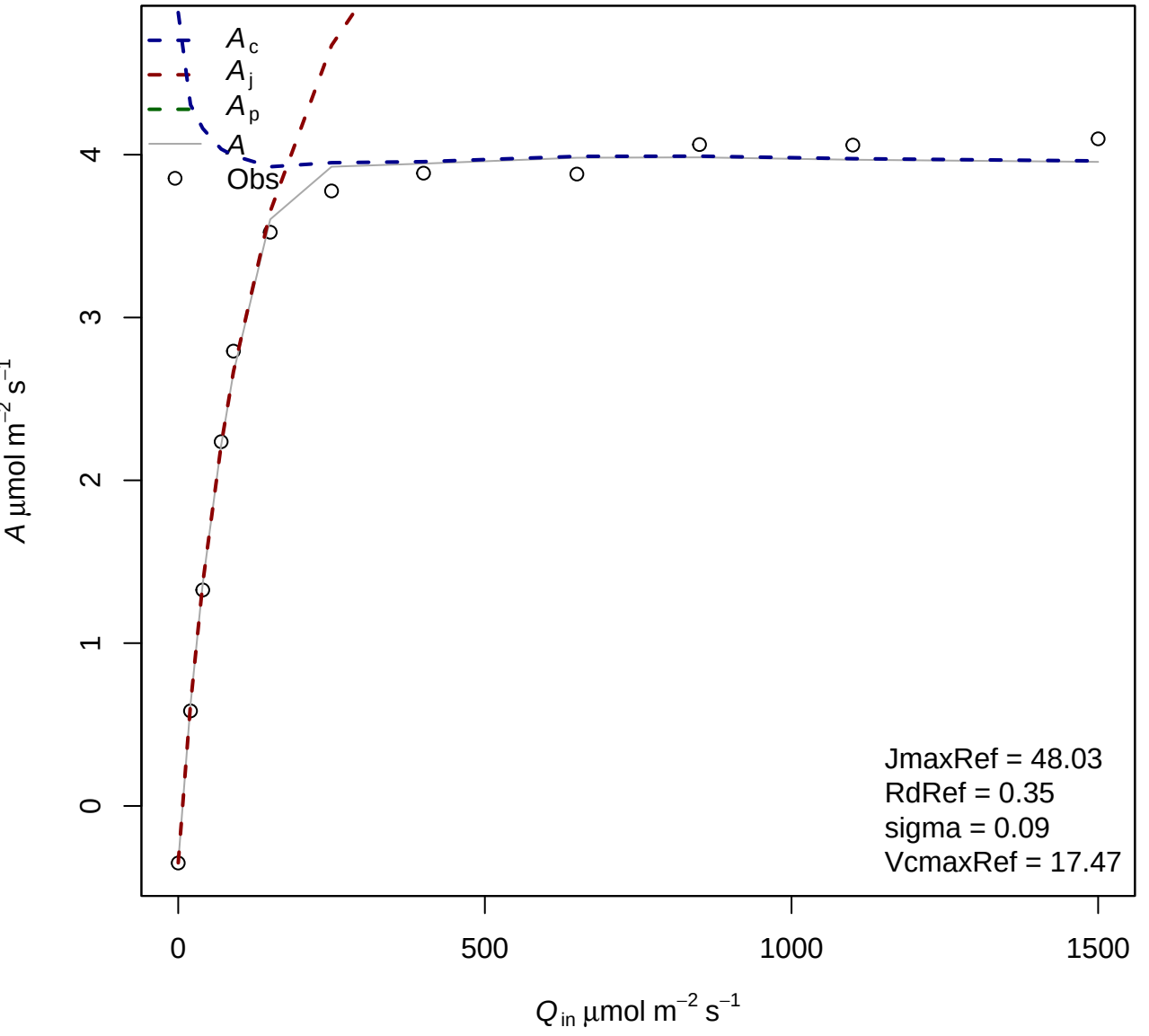

A_c
A_j
A_p
A
Obs
JmaxRef = 80.4
RdRef = 1.25
sigma = 0.14
VcmaxRef = 42.8
0
2
4
6
8
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

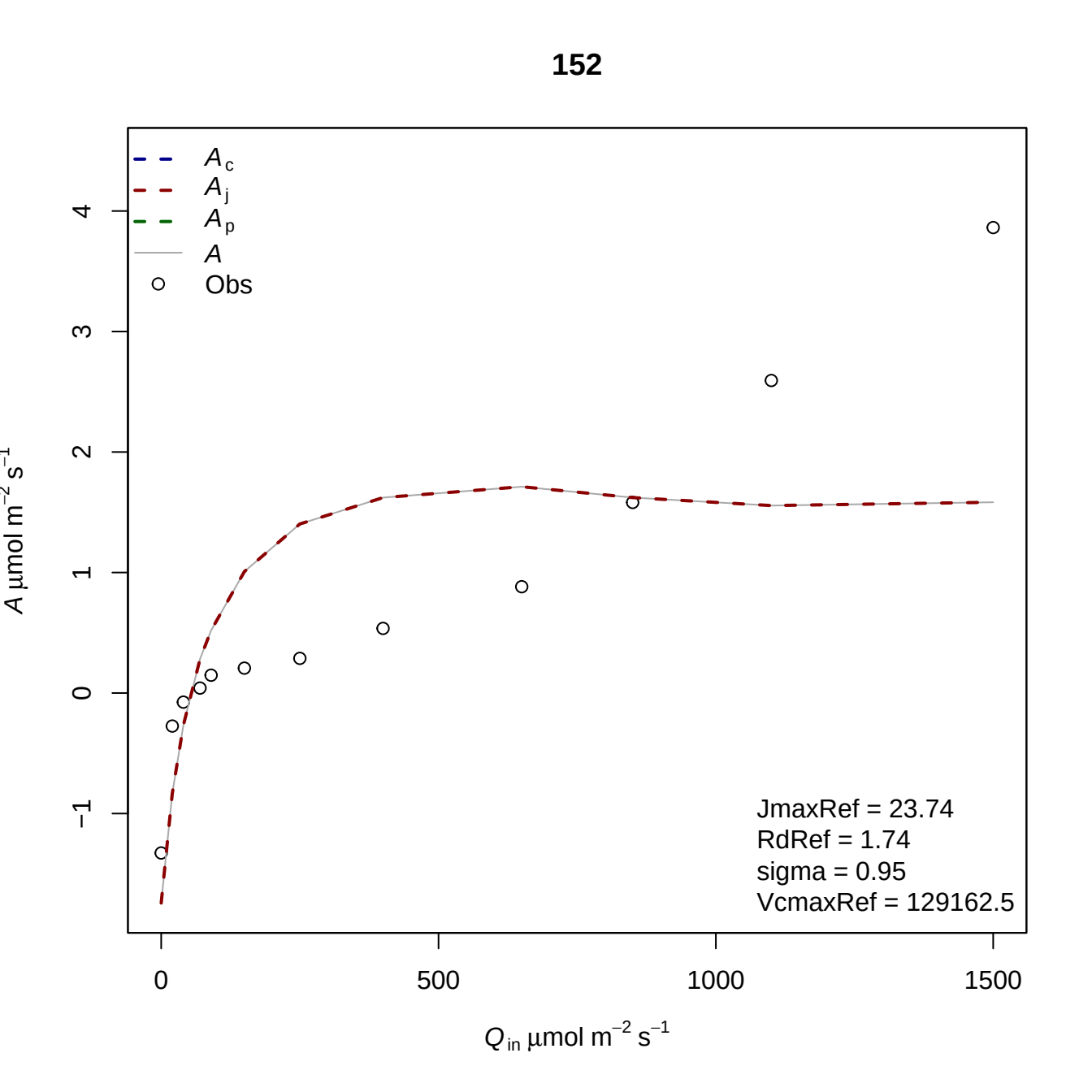

A_c
A_j
A_p
A
Obs
JmaxRef = 53.4
RdRef = 1.28
sigma = 0.24
VcmaxRef = 118.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
200
400
600
800
1000
1200
1400

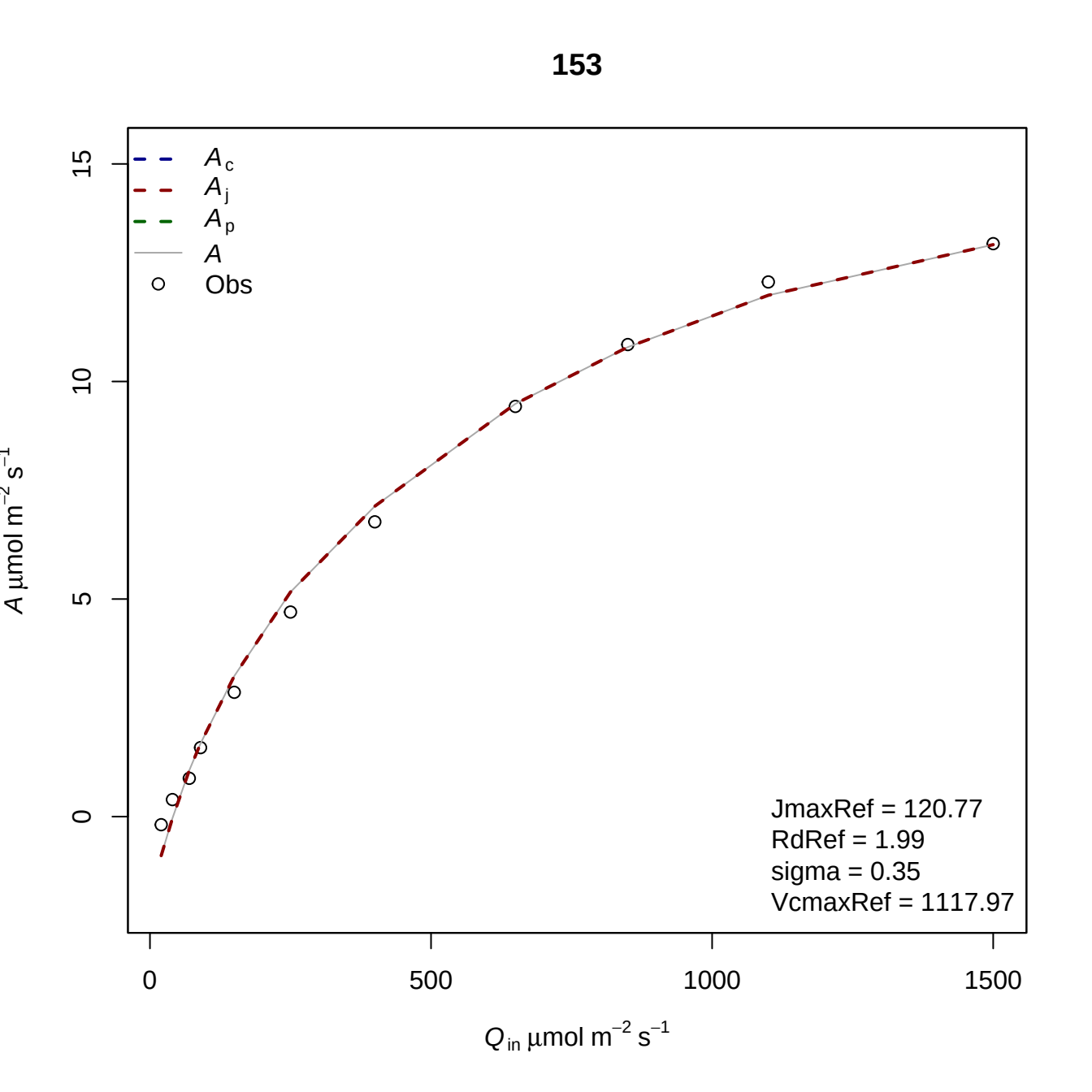

A_c
A_j
A_p
A
Obs
JmaxRef = 58.4
RdRef = 0.72
sigma = 0.39
VcmaxRef = 70.45
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

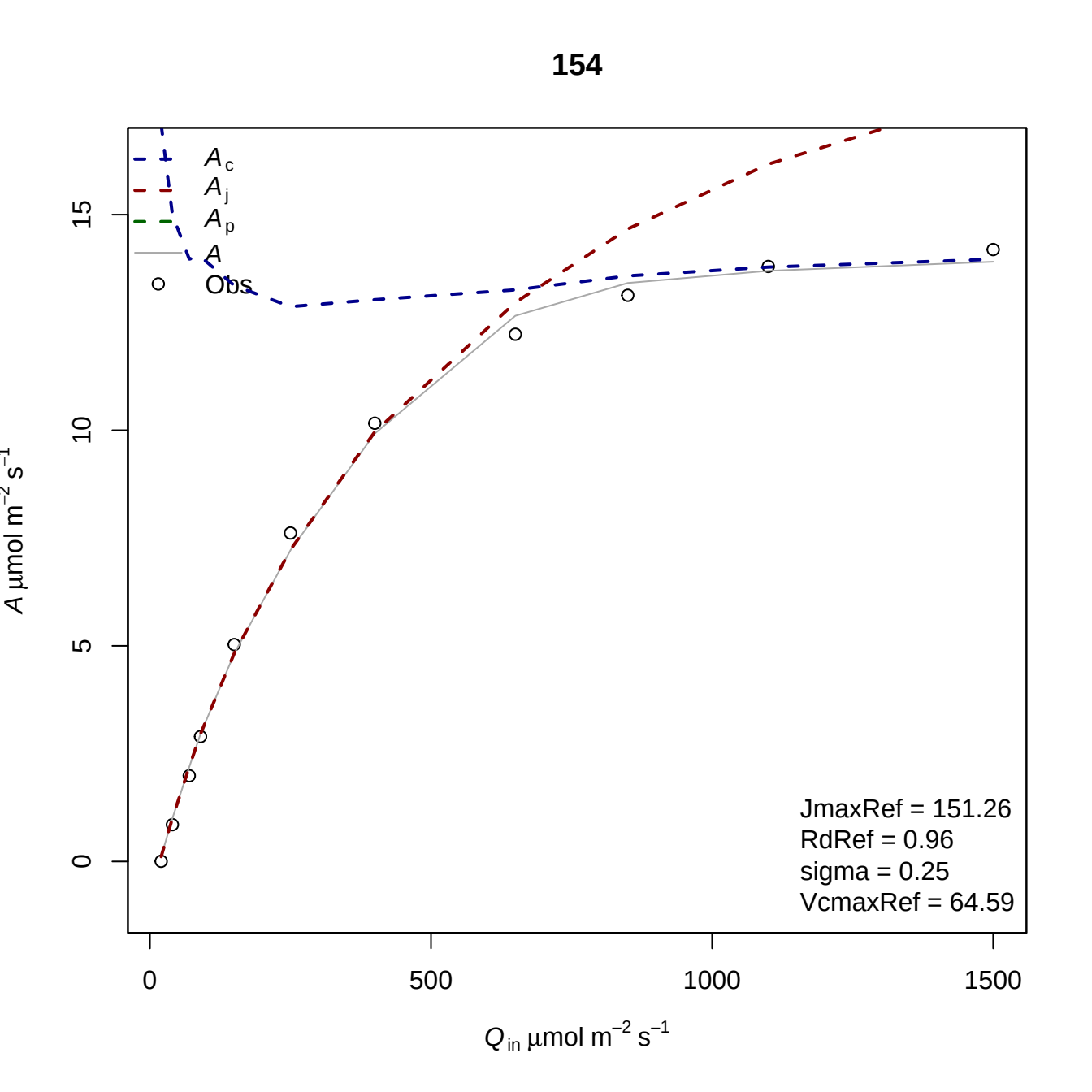

147
A_c
A_j
A_p
A
Obs
JmaxRef = 64.2
RdRef = 0.52
sigma = 0.1
VcmaxRef = 22.57
0
1
2
3
4
5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

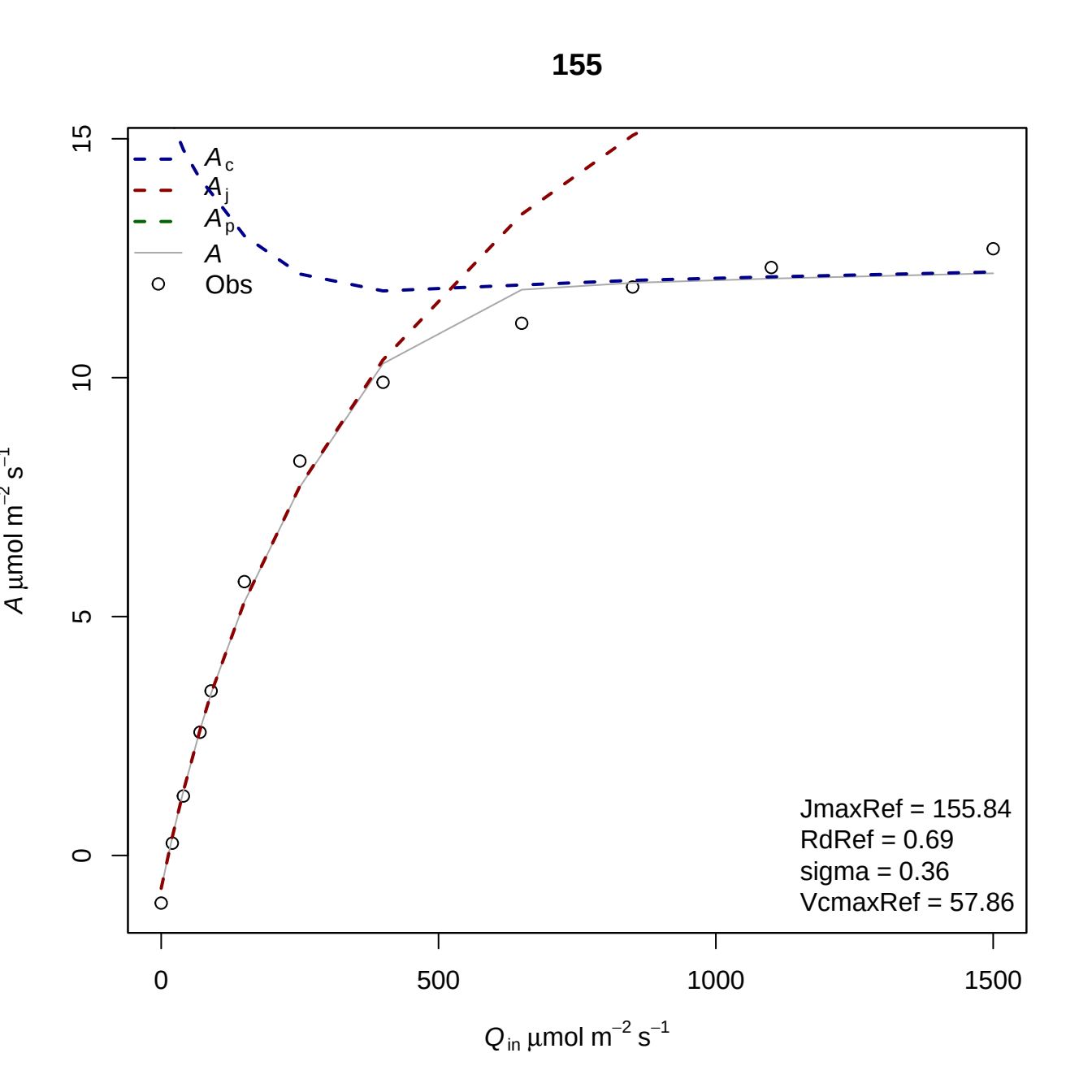

148
A_c
A_j
A_p
A
Obs
JmaxRef = 58.13
RdRef = 0.34
sigma = 0.08
VcmaxRef = 20.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

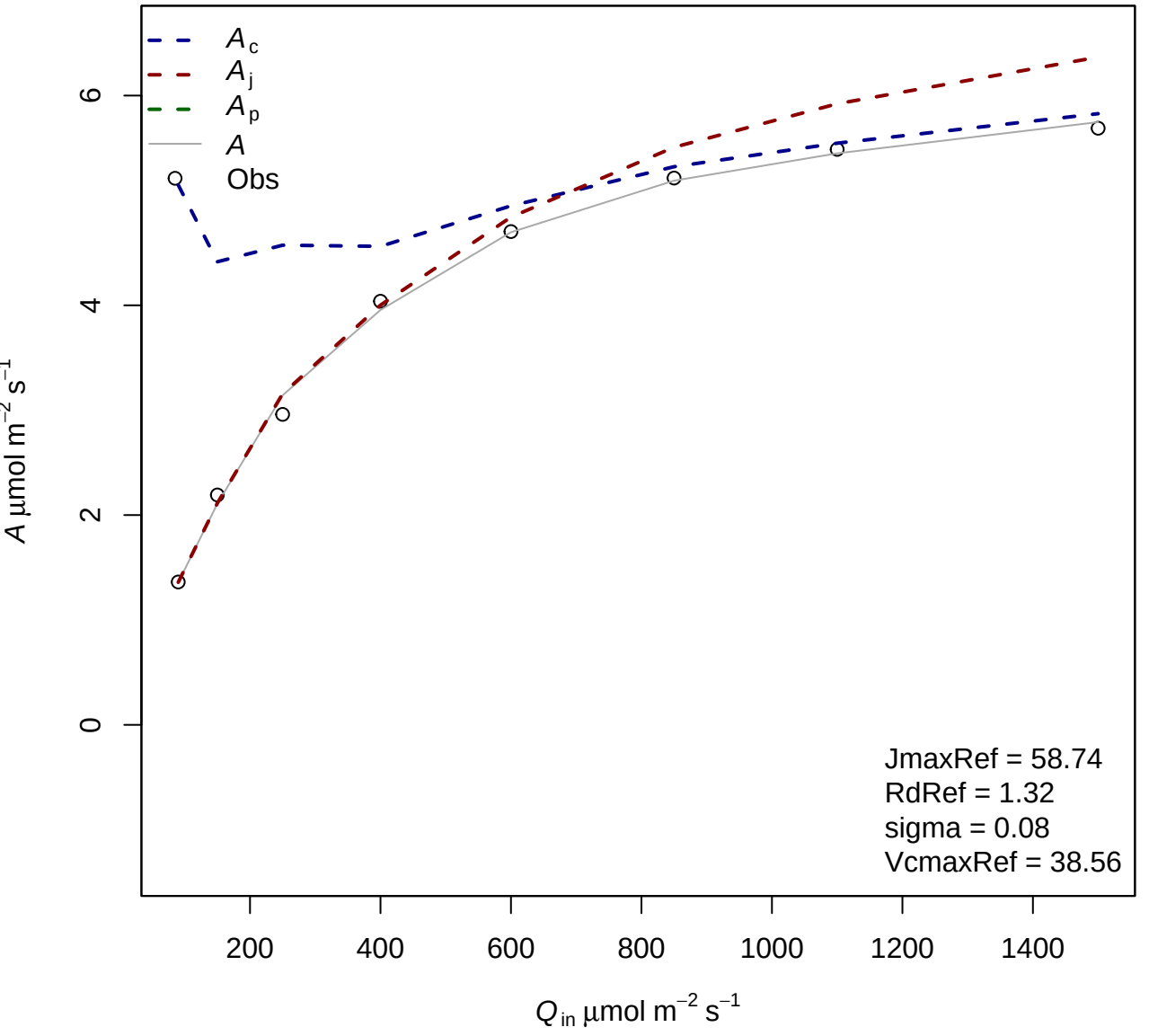

A_c
A_j
A_p
A
Obs
JmaxRef = 48.03
RdRef = 0.35
sigma = 0.09
VcmaxRef = 17.47
0
1
2
3
4
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

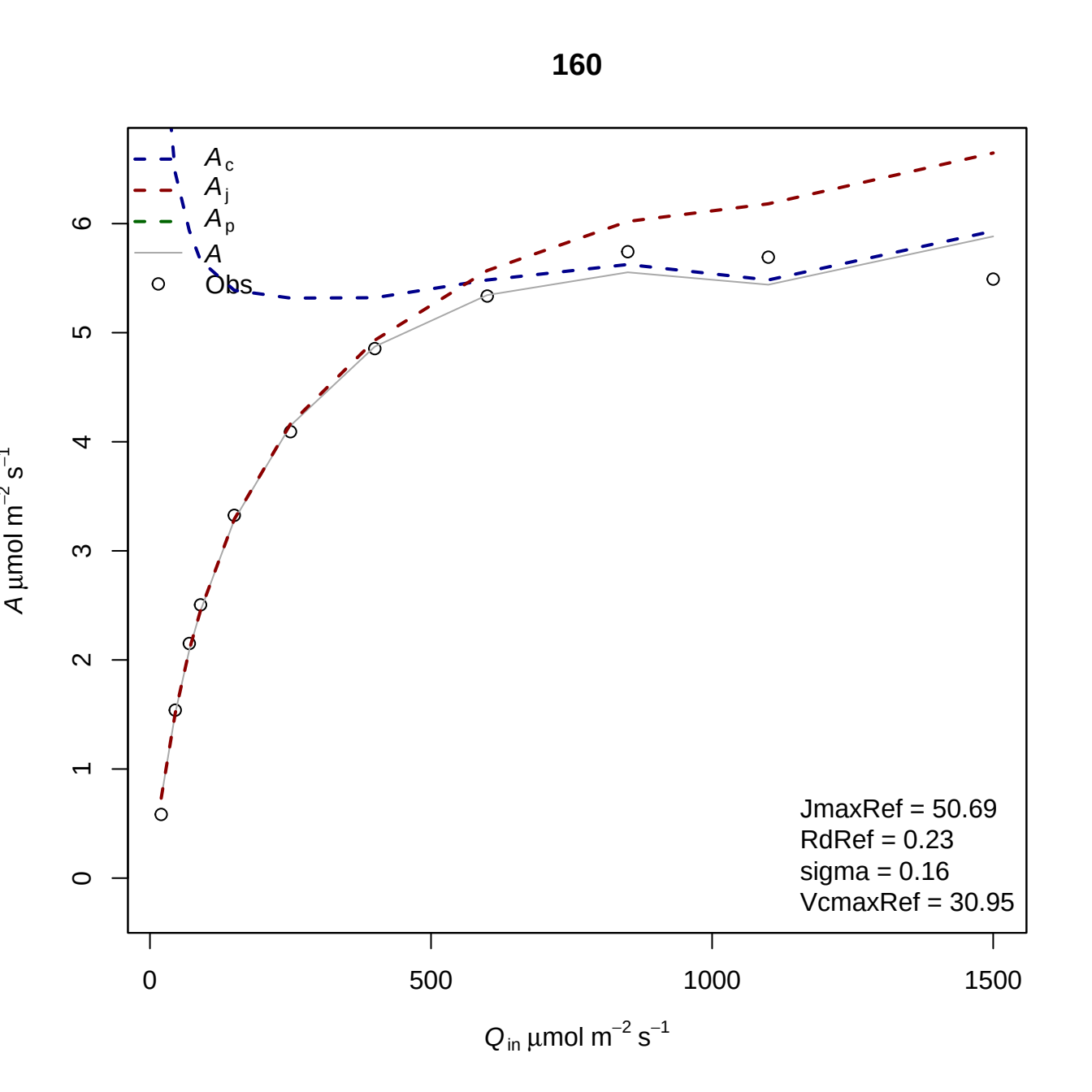

152
A_c
A_j
A_p
A
Obs
JmaxRef = 23.74
RdRef = 1.74
sigma = 0.95
VcmaxRef = 129162.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

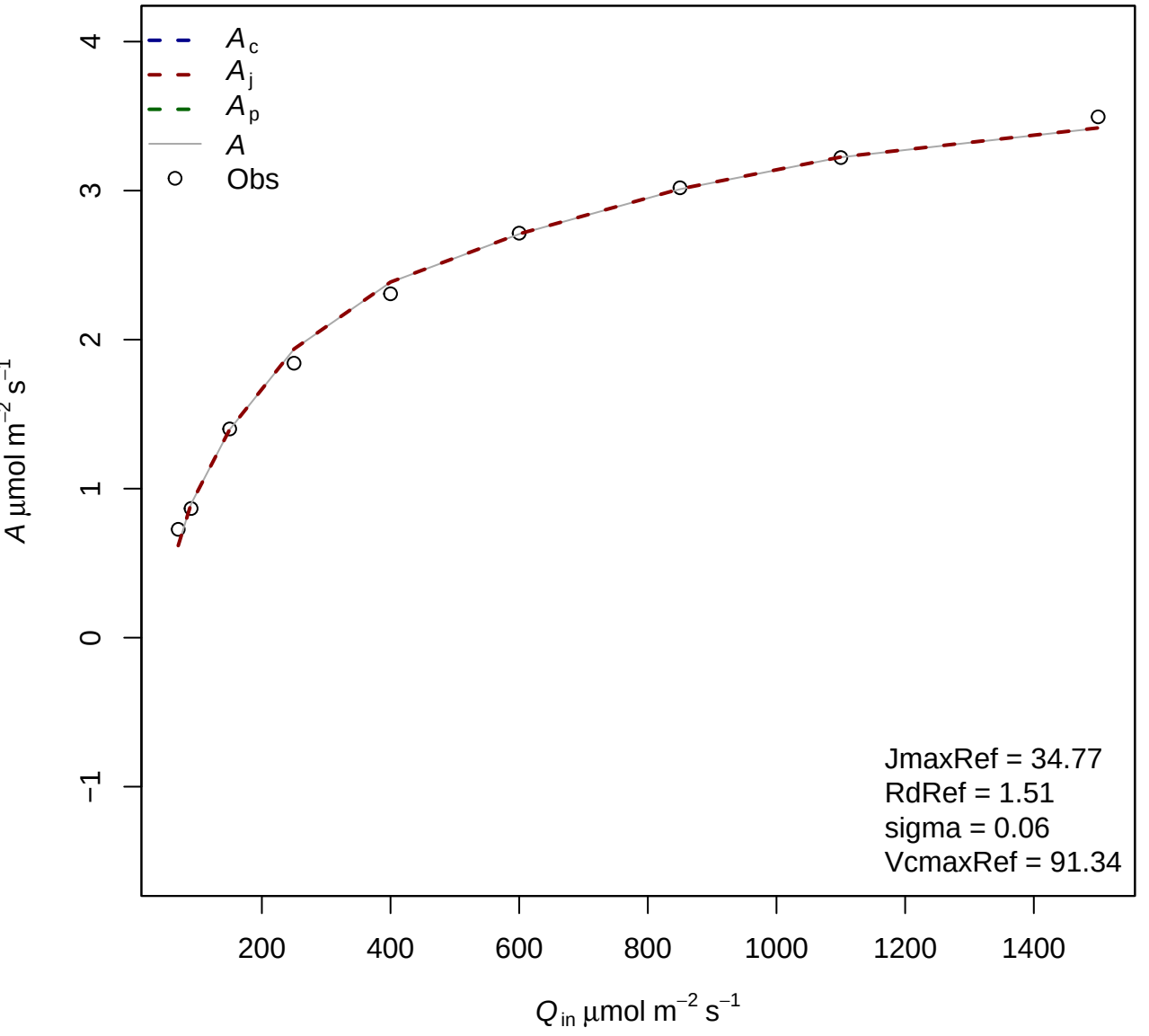

153
A_c
A_j
A_p
A
Obs
JmaxRef = 120.77
RdRef = 1.99
sigma = 0.35
VcmaxRef = 1117.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

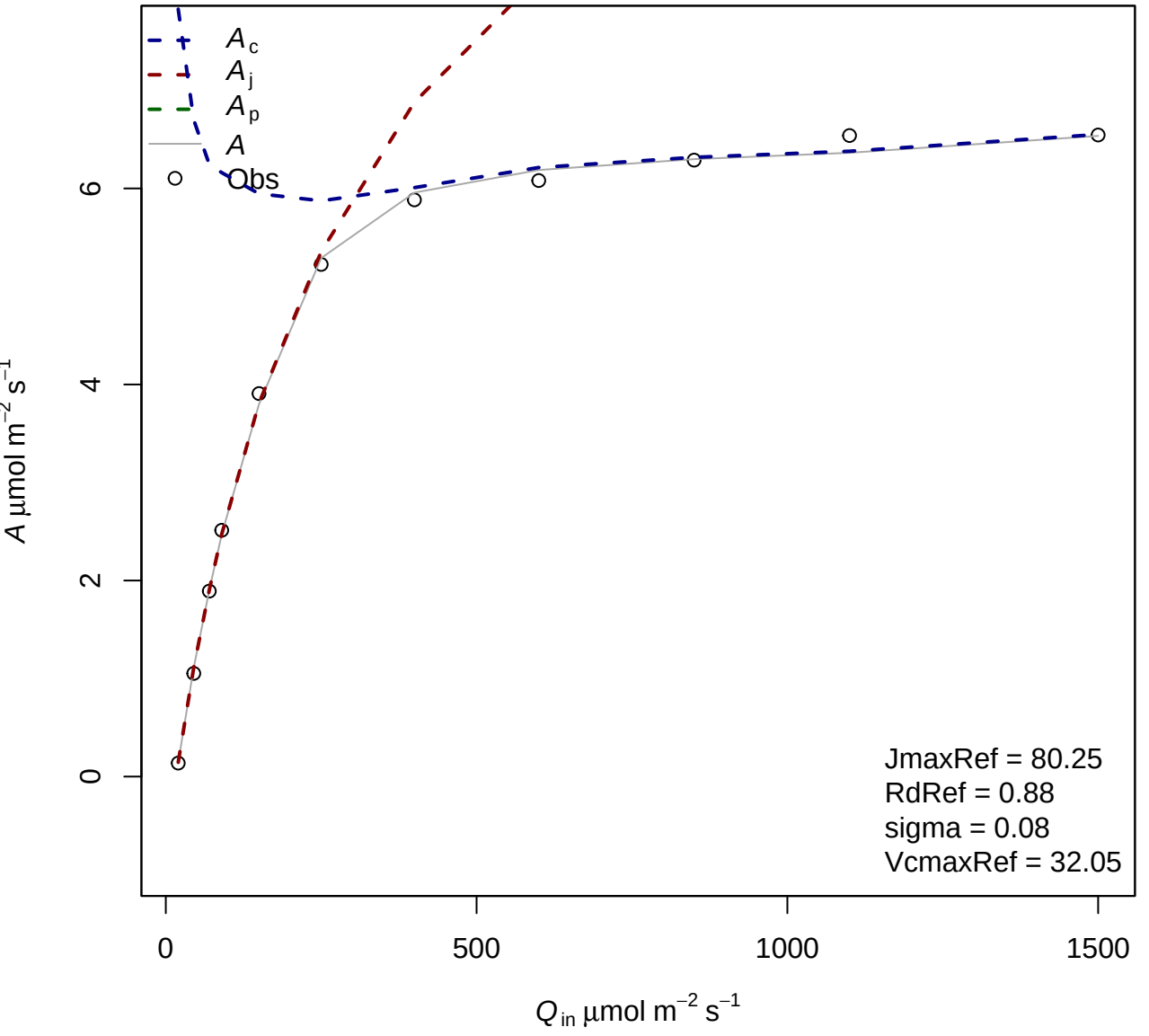

154
A_c
A_j
A_p
A
Obs
JmaxRef = 151.26
RdRef = 0.96
sigma = 0.25
VcmaxRef = 64.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

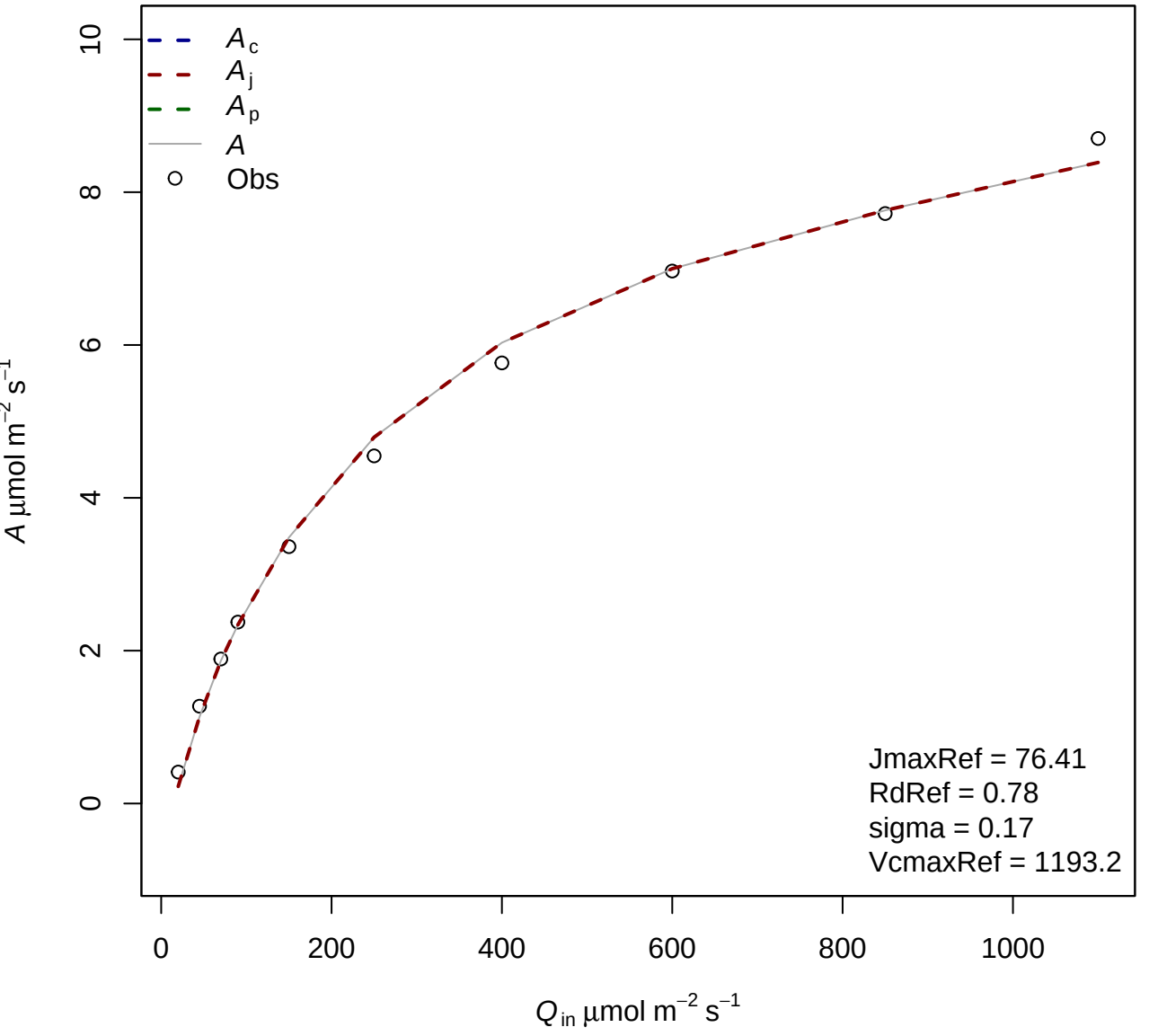

155
A_c
A_j
A_p
A
Obs
JmaxRef = 155.84
RdRef = 0.69
sigma = 0.36
VcmaxRef = 57.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

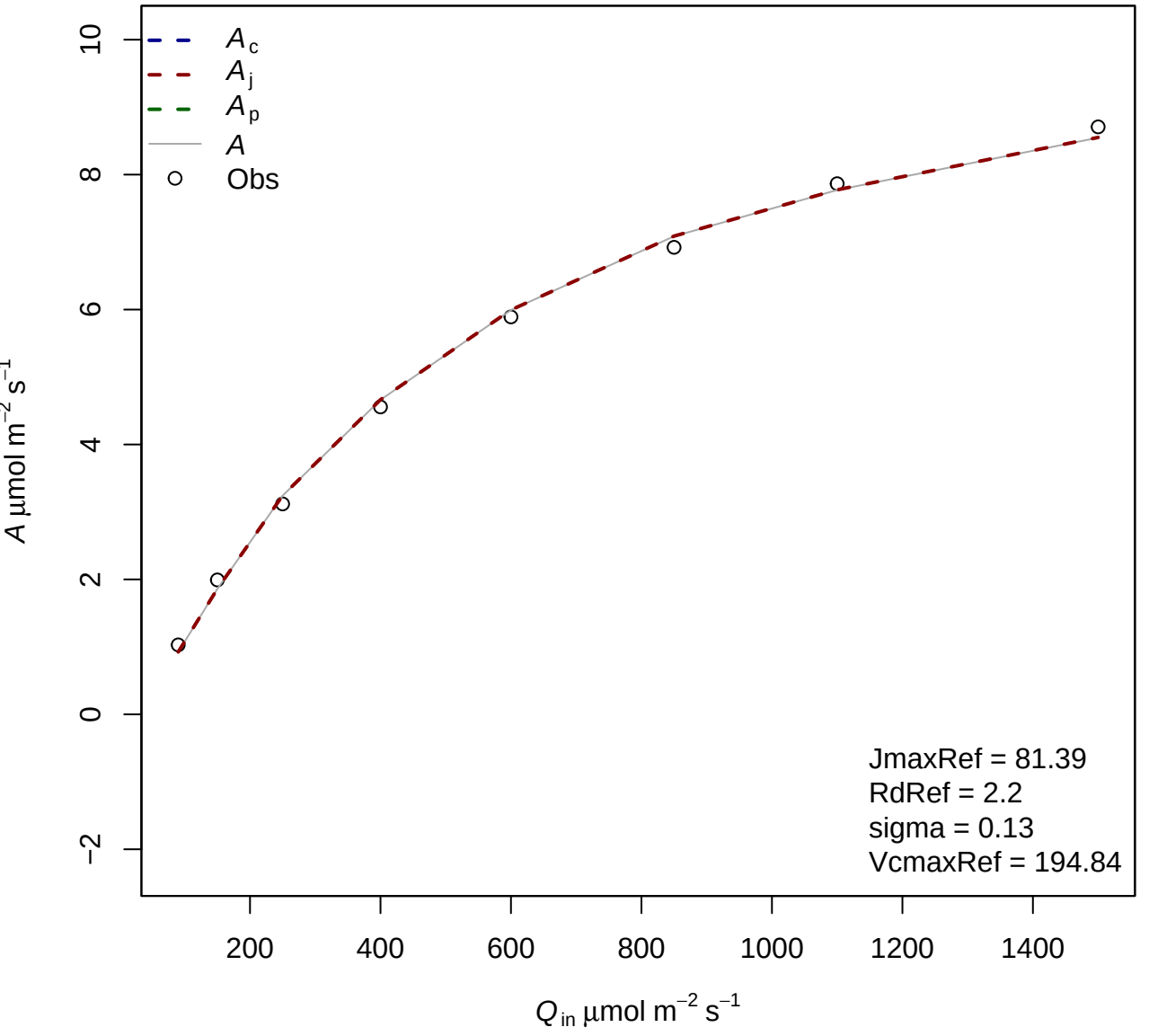

A_c
A_j
A_p
A
Obs
0
2
4
6
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 58.74
RdRef = 1.32
sigma = 0.08
VcmaxRef = 38.56

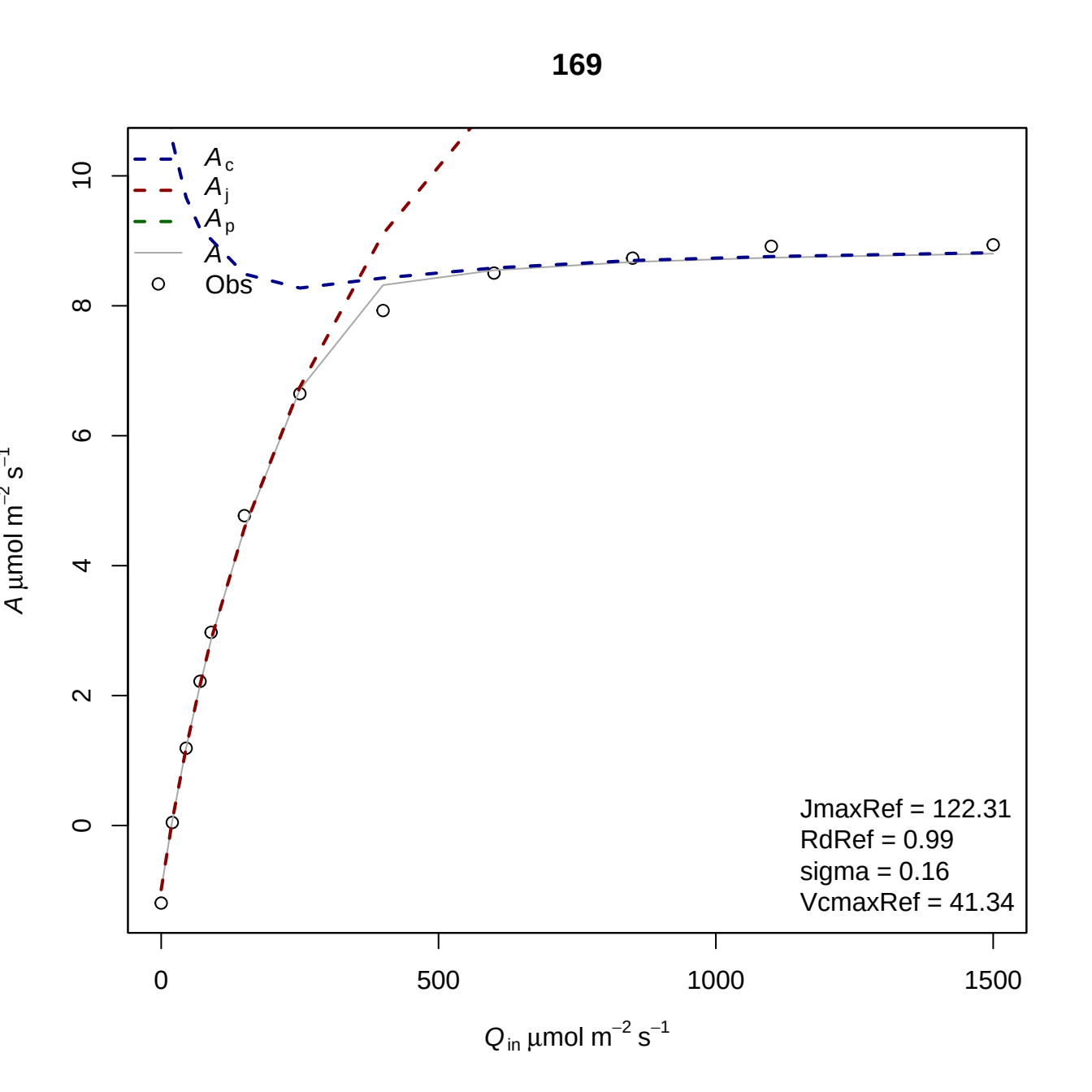

160
A_c
A_j
A_p
A
Obs
JmaxRef = 50.69
RdRef = 0.23
sigma = 0.16
VcmaxRef = 30.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

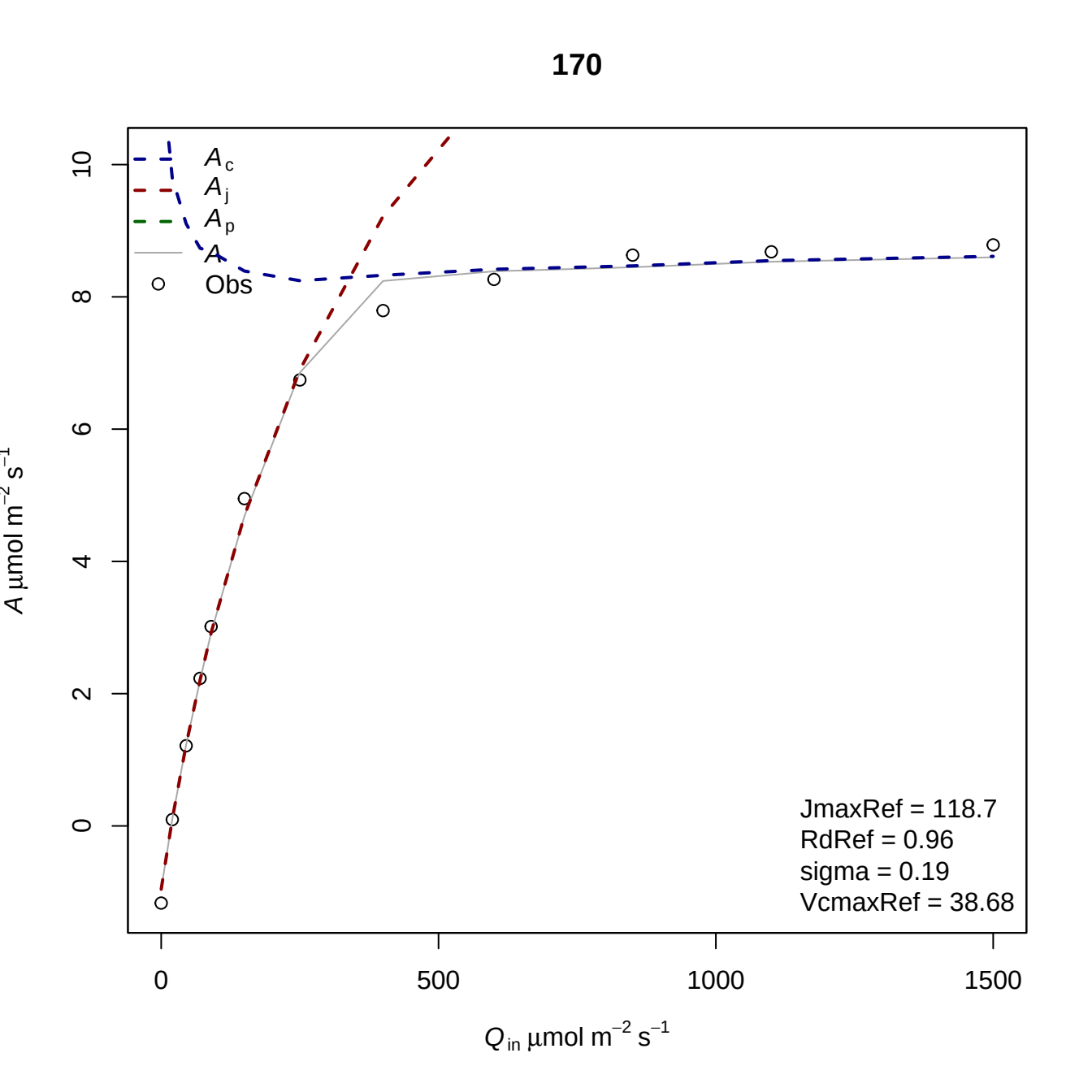

A_c
A_j
A_p
A
Obs
JmaxRef = 34.77
RdRef = 1.51
sigma = 0.06
VcmaxRef = 91.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
200
400
600
800
1000
1200
1400

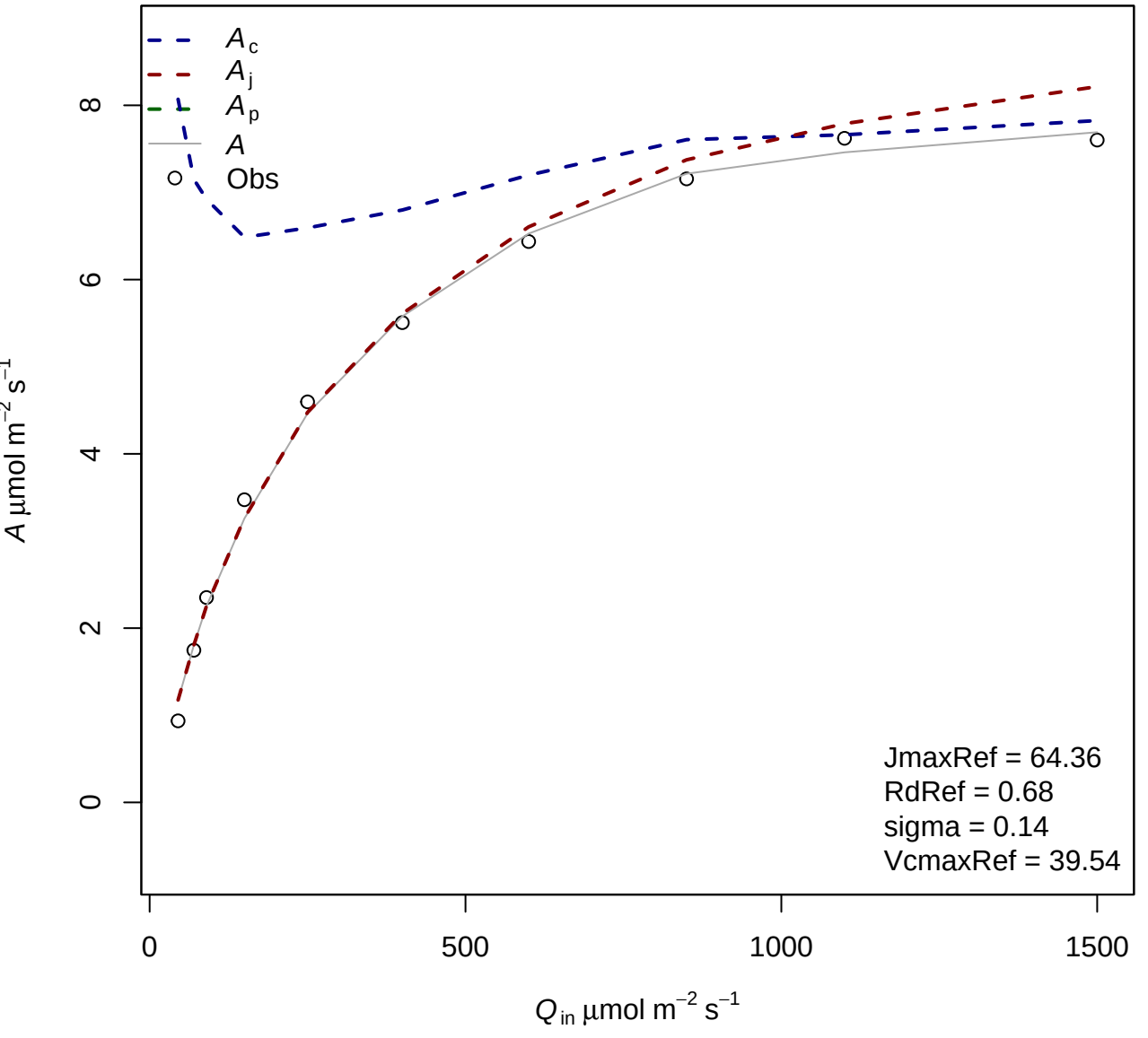

A_c
A_j
A_p
A
Obs
JmaxRef = 80.25
RdRef = 0.88
sigma = 0.08
VcmaxRef = 32.05
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

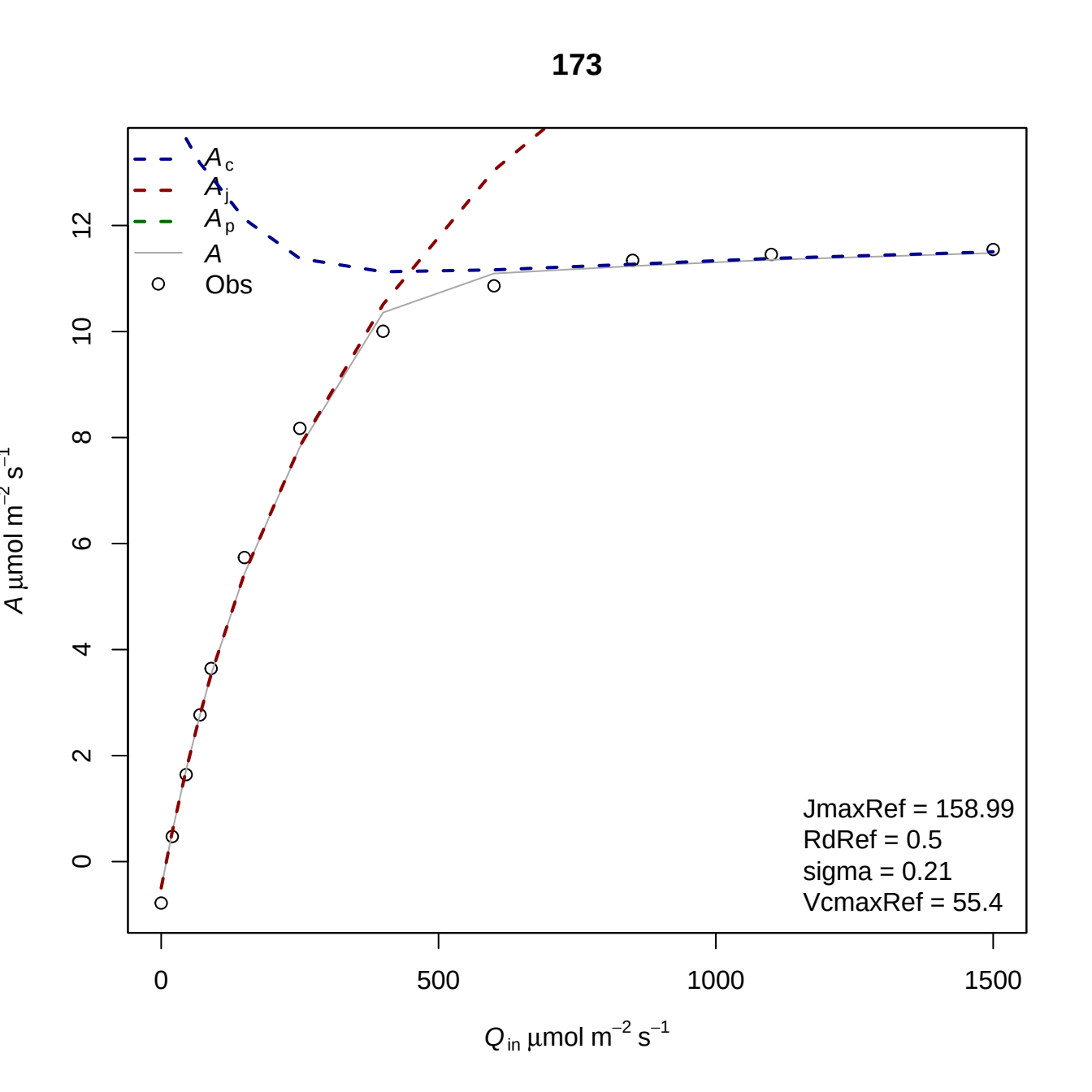

A_c
A_j
A_p
A
Obs
JmaxRef = 76.41
RdRef = 0.78
sigma = 0.17
VcmaxRef = 1193.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
200
400
600
800
1000

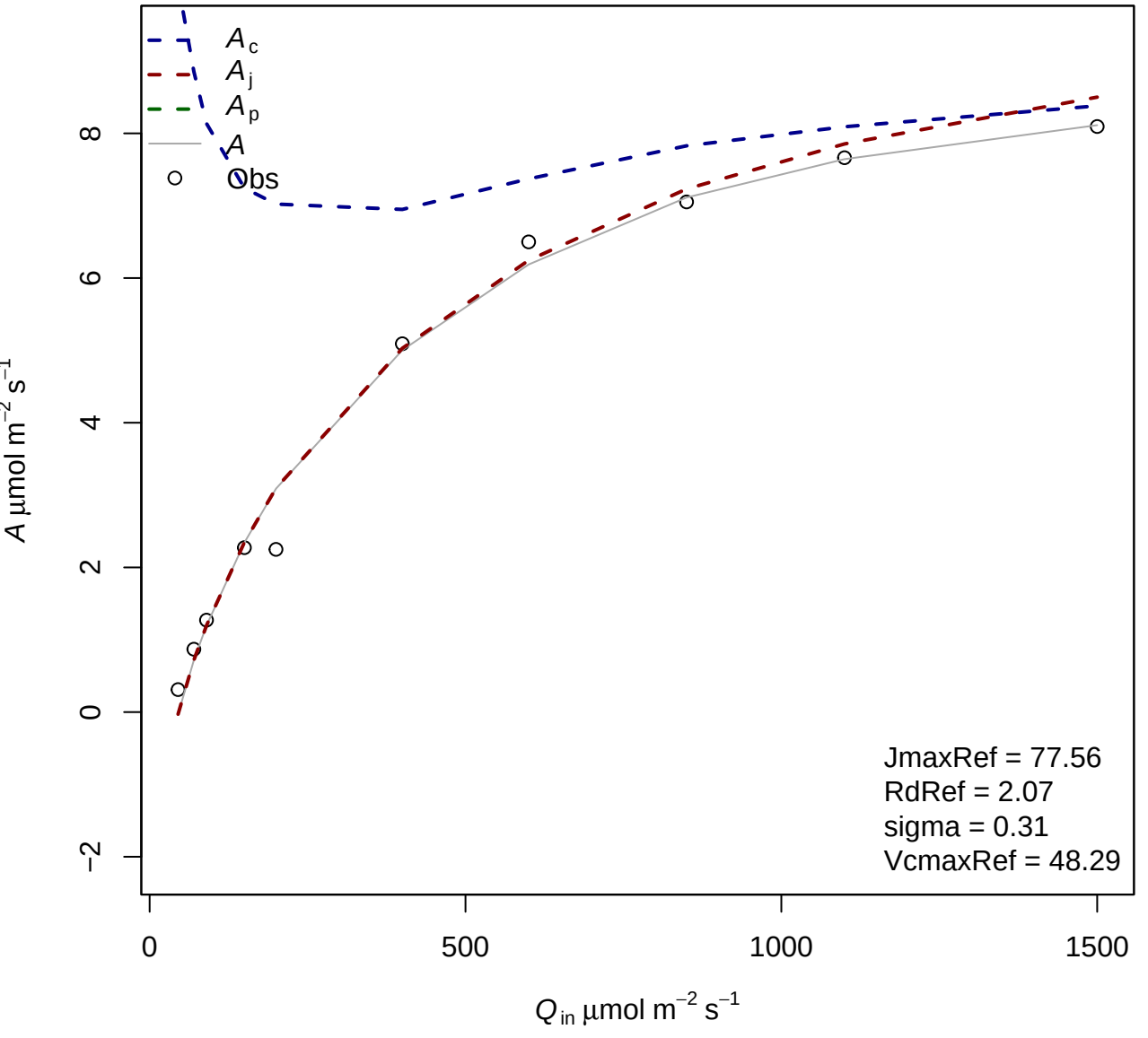

A_c
A_j
A_p
A
Obs
JmaxRef = 81.39
RdRef = 2.2
sigma = 0.13
VcmaxRef = 194.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
200
400
600
800
1000
1200
1400

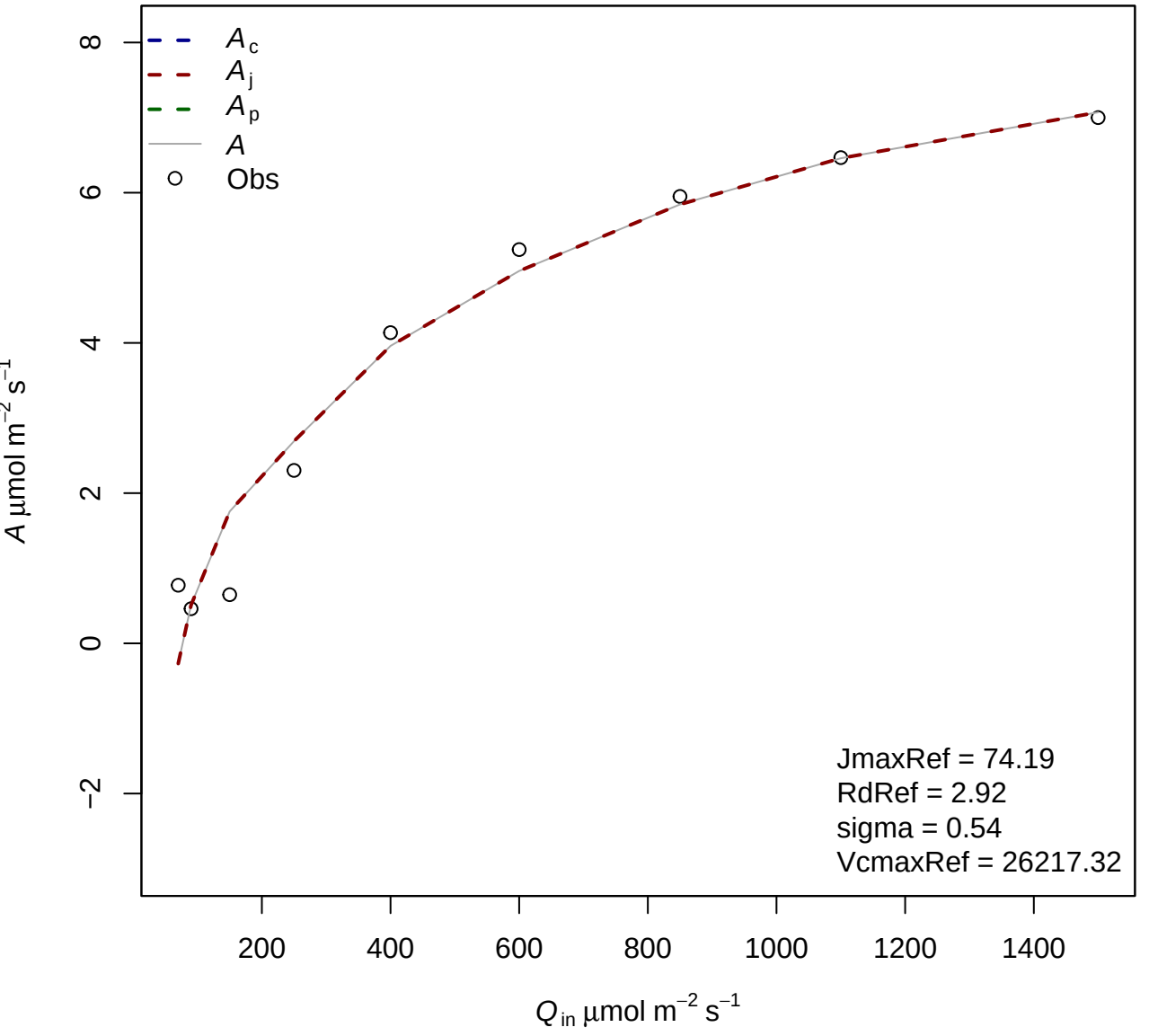

169
A_c
A_j
A_p
A
Obs
JmaxRef = 122.31
RdRef = 0.99
sigma = 0.16
VcmaxRef = 41.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

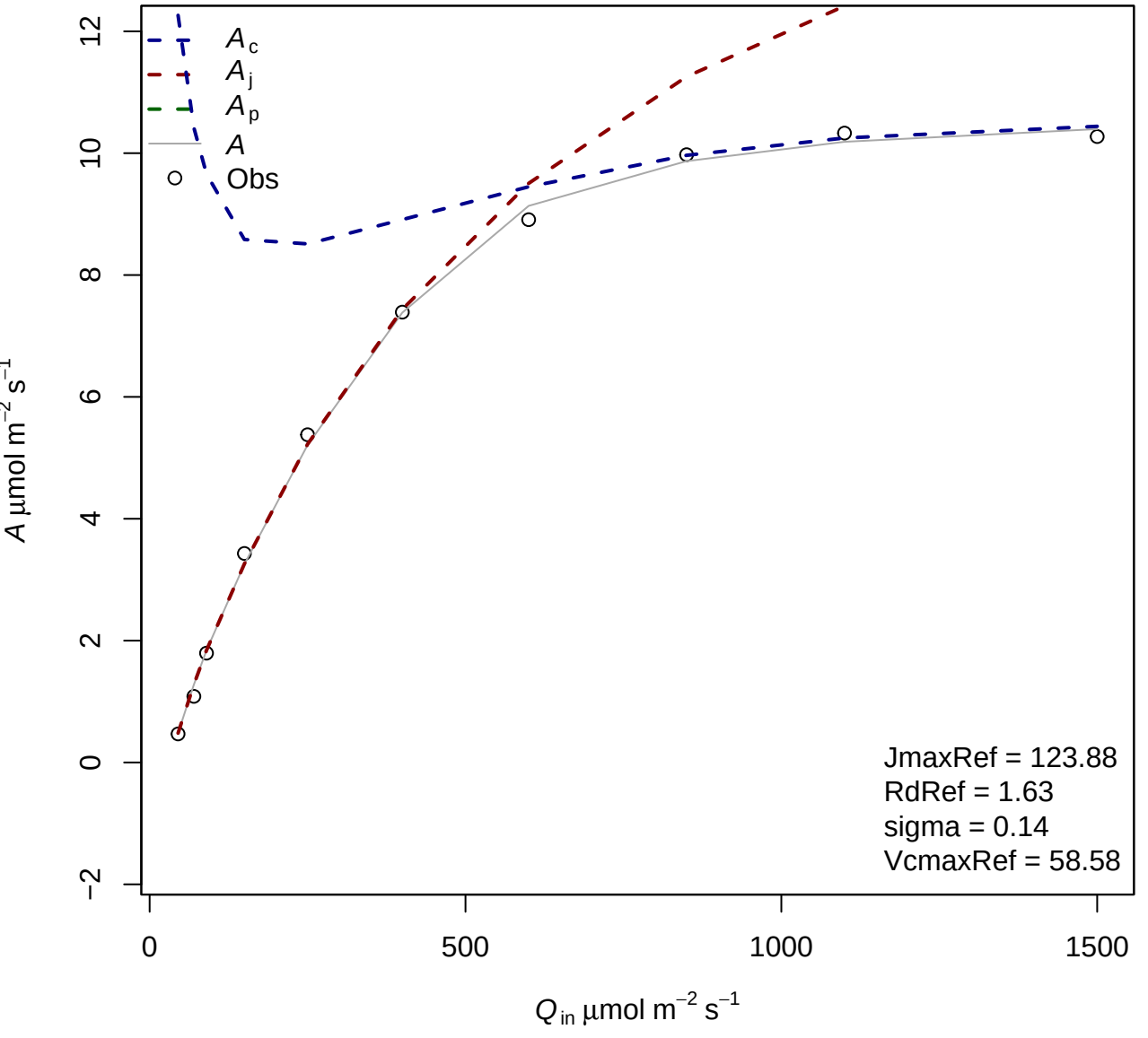

170
A_c
A_j
A_p
A
Obs
JmaxRef = 118.7
RdRef = 0.96
sigma = 0.19
VcmaxRef = 38.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

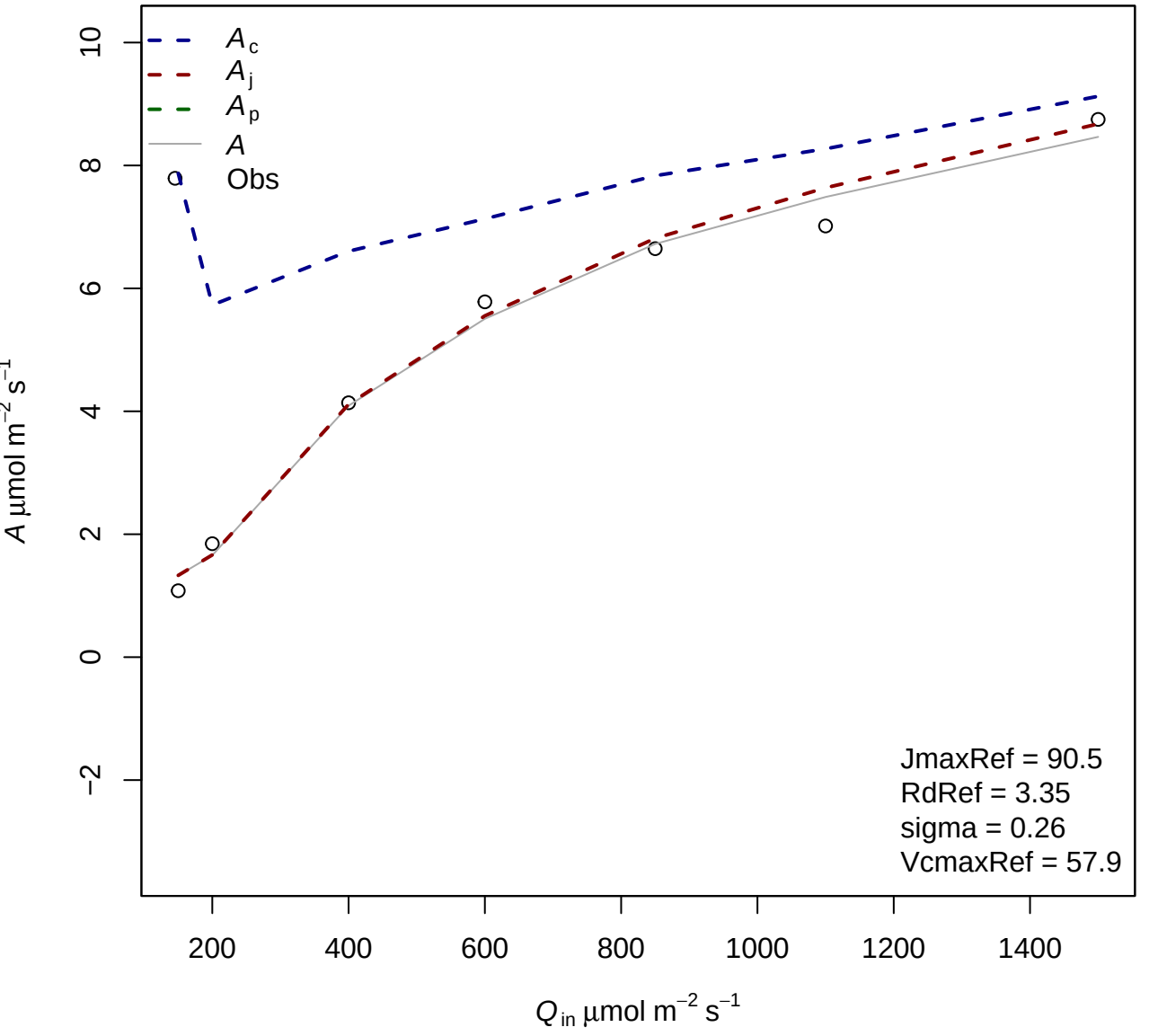

A_c
A_j
A_p
A
Obs
JmaxRef = 64.36
RdRef = 0.68
sigma = 0.14
VcmaxRef = 39.54
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

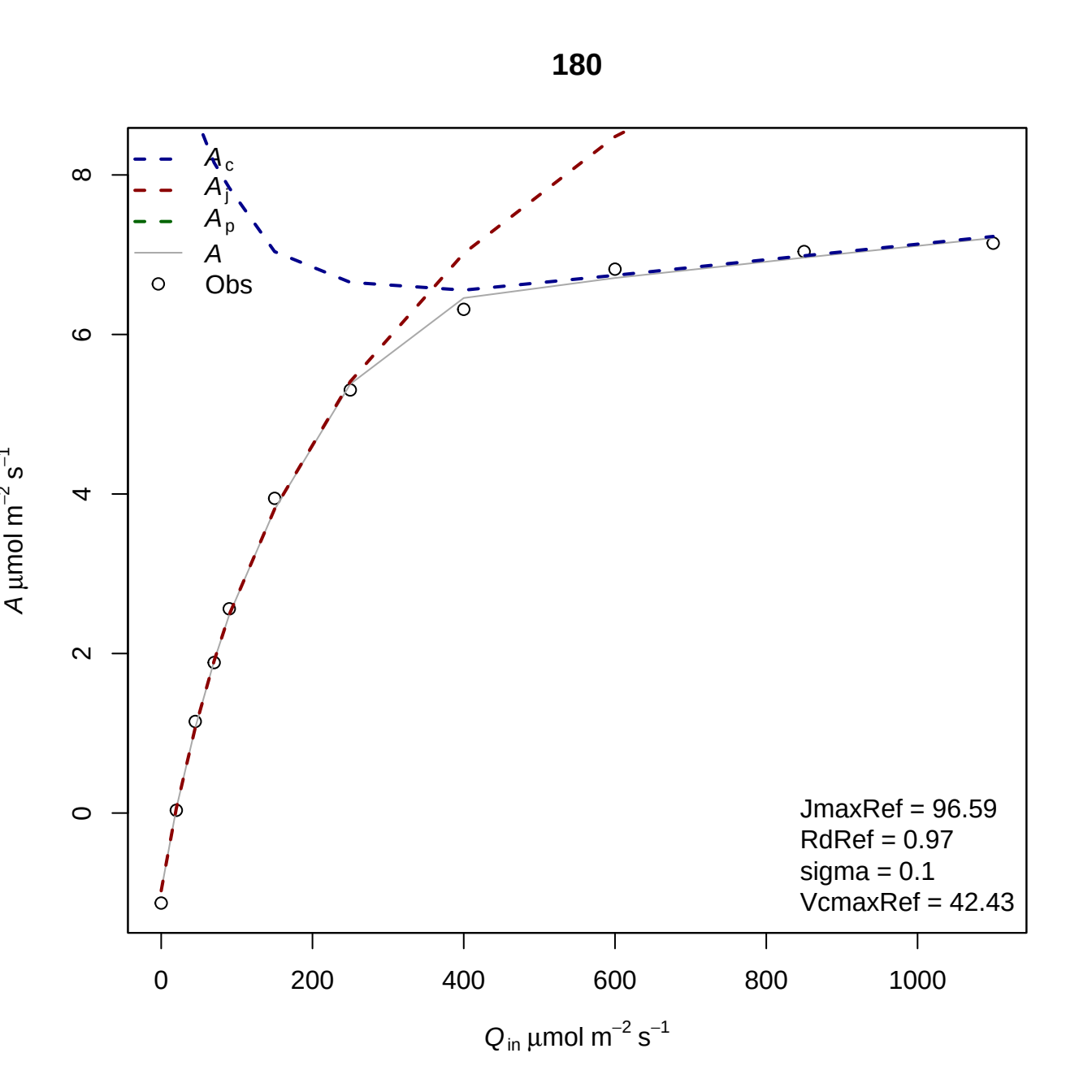

173
A_c
A_j
A_p
A
Obs
JmaxRef = 158.99
RdRef = 0.5
sigma = 0.21
VcmaxRef = 55.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

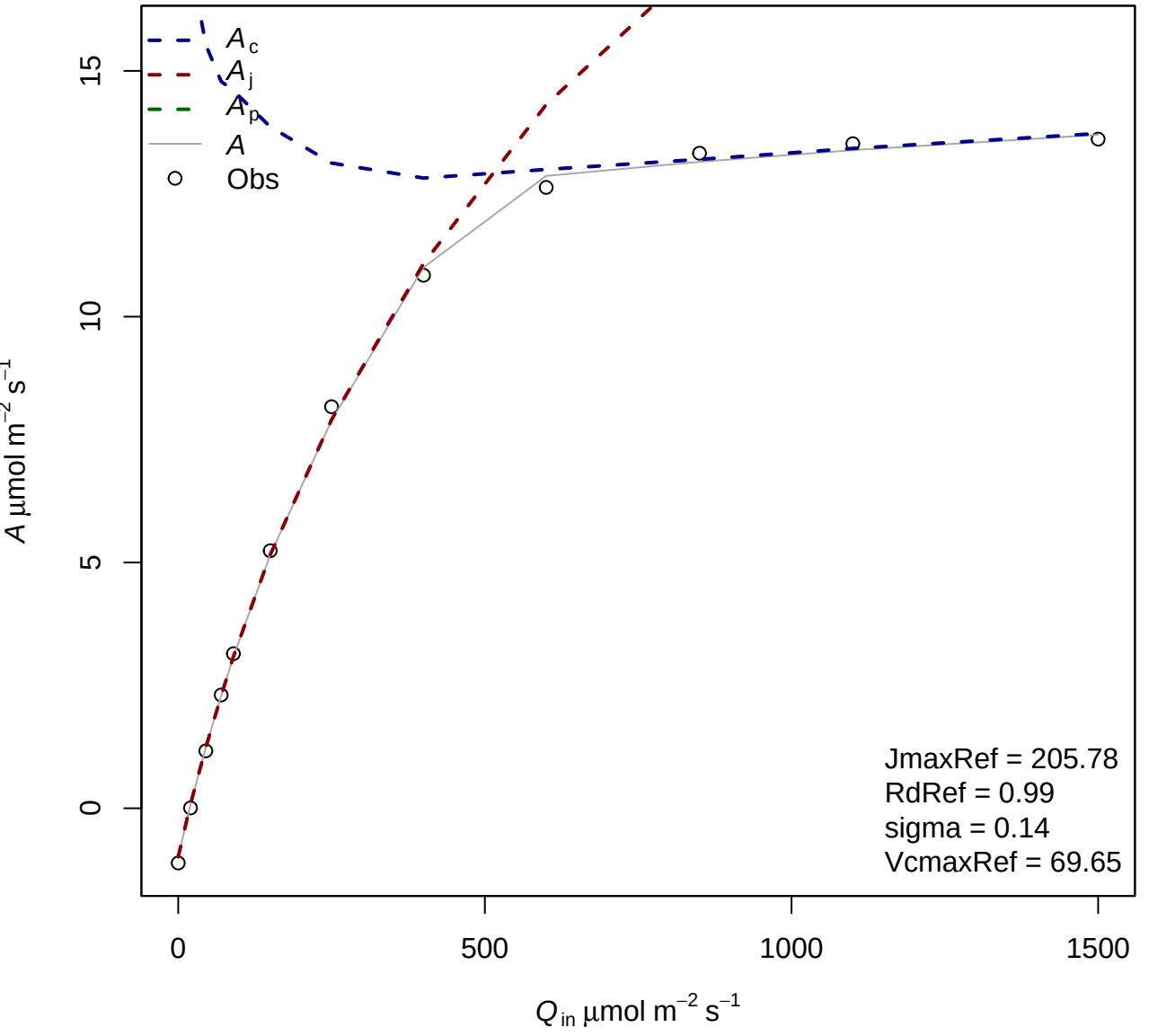

A_c
A_j
A_p
A
Obs
JmaxRef = 77.56
RdRef = 2.07
sigma = 0.31
VcmaxRef = 48.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

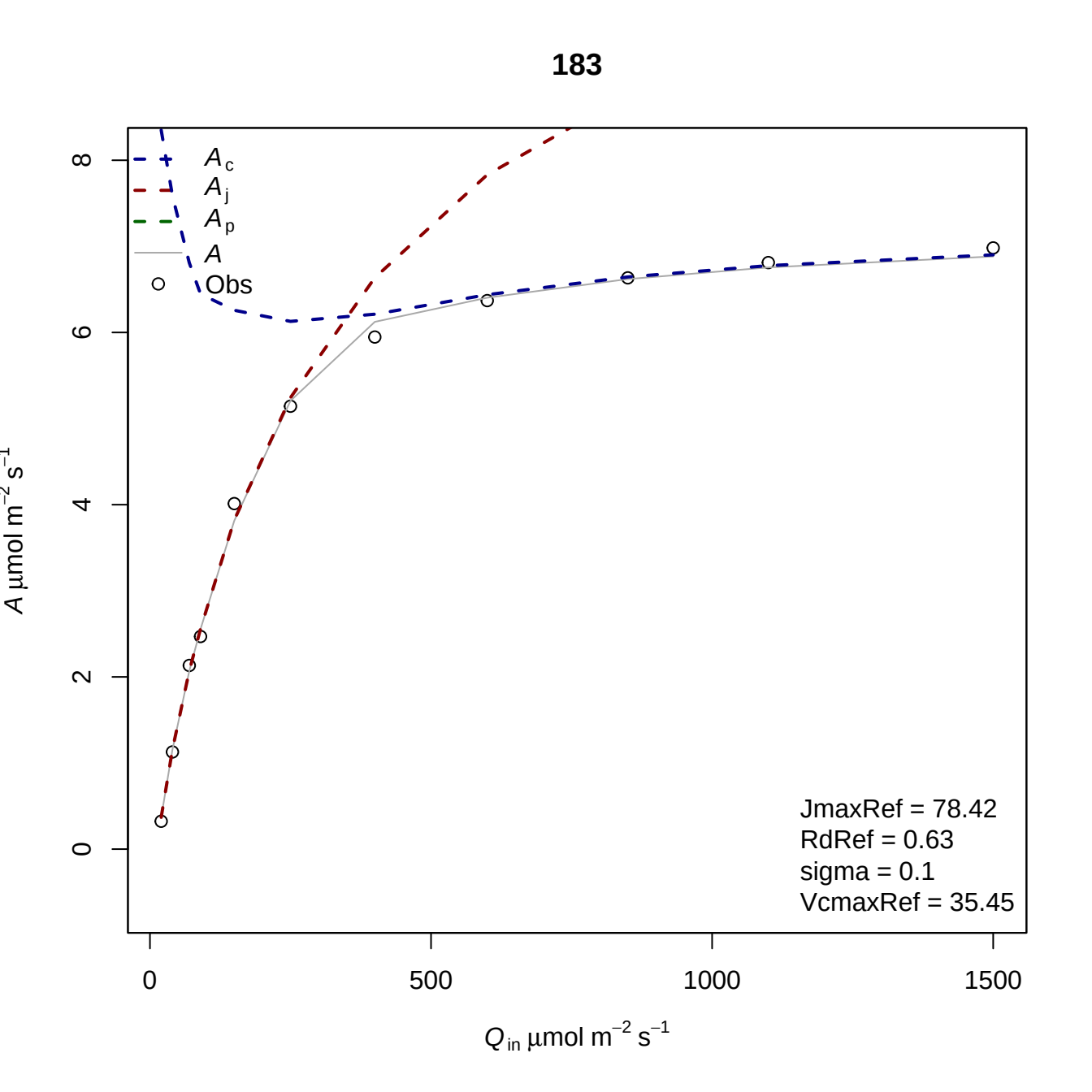

A_c
A_j
A_p
A
Obs
JmaxRef = 74.19
RdRef = 2.92
sigma = 0.54
VcmaxRef = 26217.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
200
400
600
800
1000
1200
1400

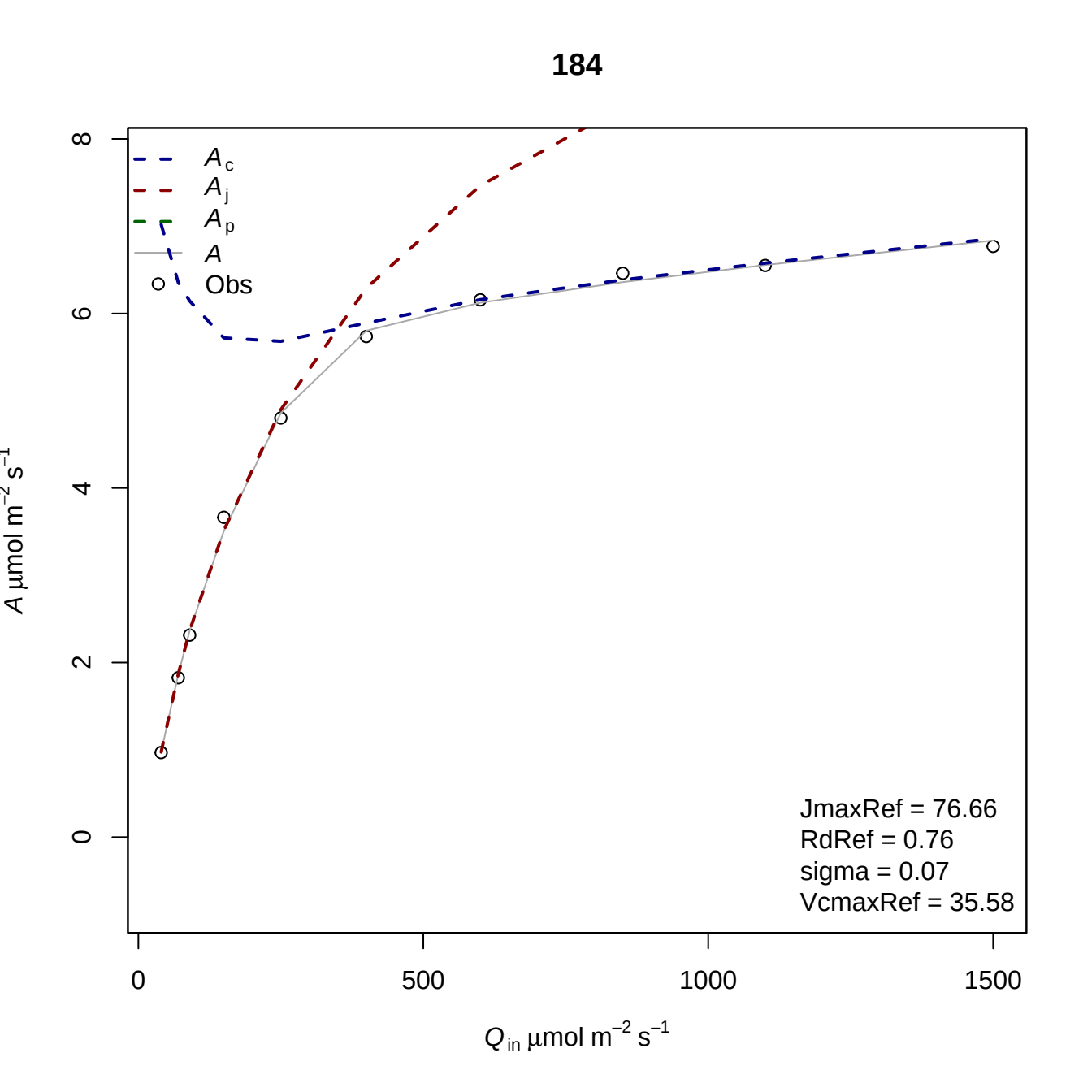

A_c
A_j
A_p
A
Obs
JmaxRef = 123.88
RdRef = 1.63
sigma = 0.14
VcmaxRef = 58.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

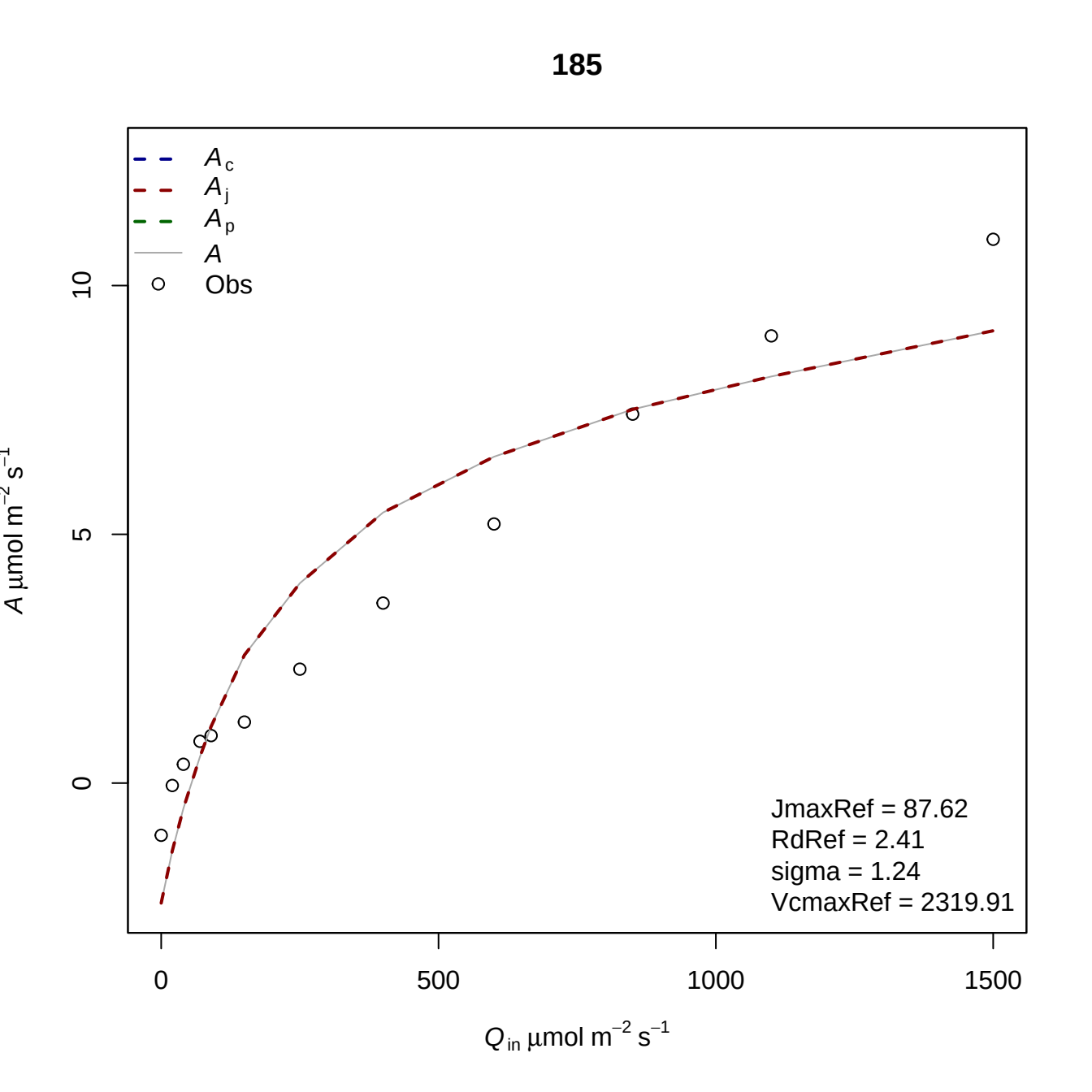

A_c
A_j
A_p
A
Obs
JmaxRef = 90.5
RdRef = 3.35
sigma = 0.26
VcmaxRef = 57.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
200
400
600
800
1000
1200
1400

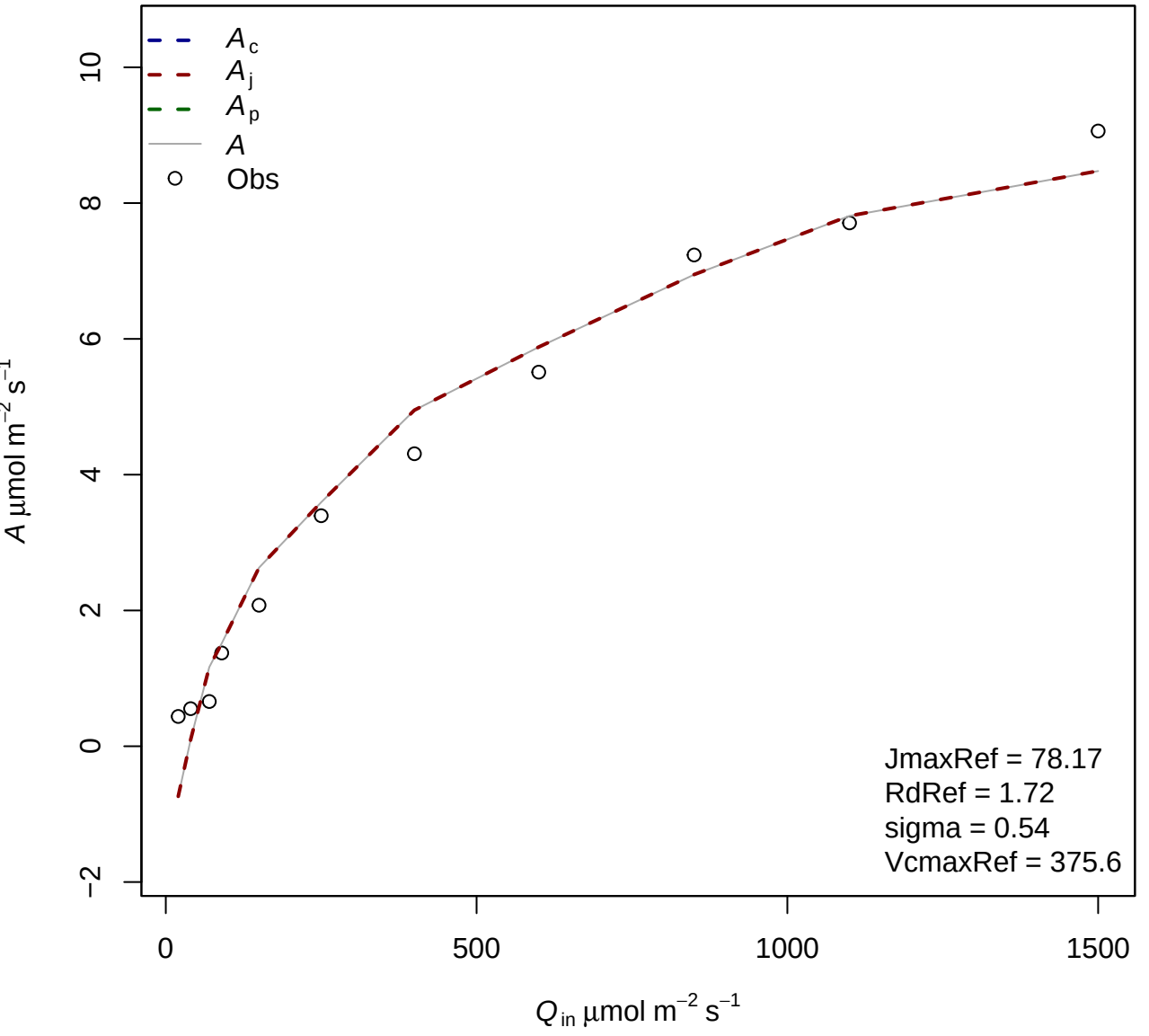

180
A_c
A_j
A_p
A
Obs
JmaxRef = 96.59
RdRef = 0.97
sigma = 0.1
VcmaxRef = 42.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
200
400
600
800
1000

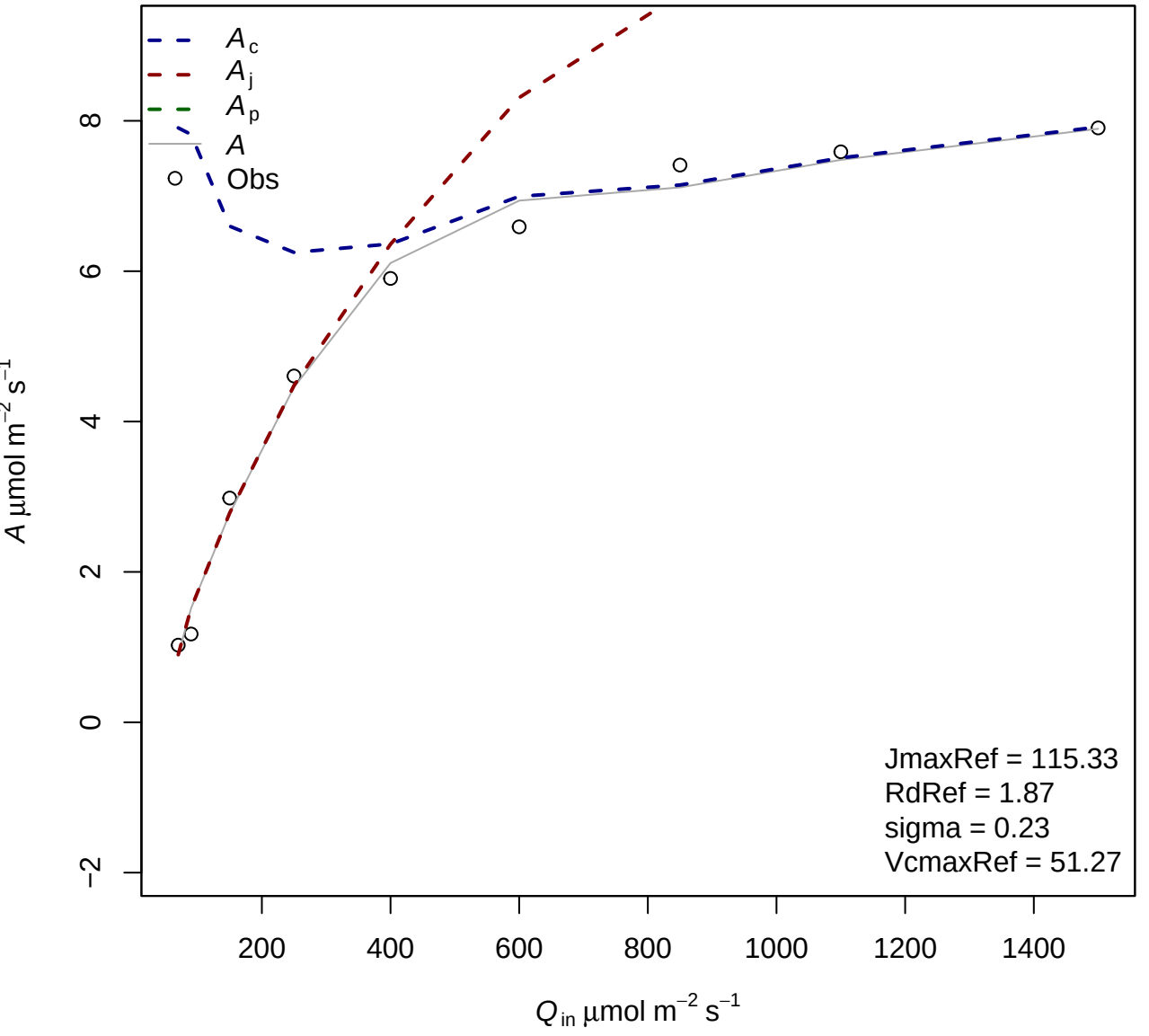

A_c
A_j
A_p
A
Obs
JmaxRef = 205.78
RdRef = 0.99
sigma = 0.14
VcmaxRef = 69.65
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

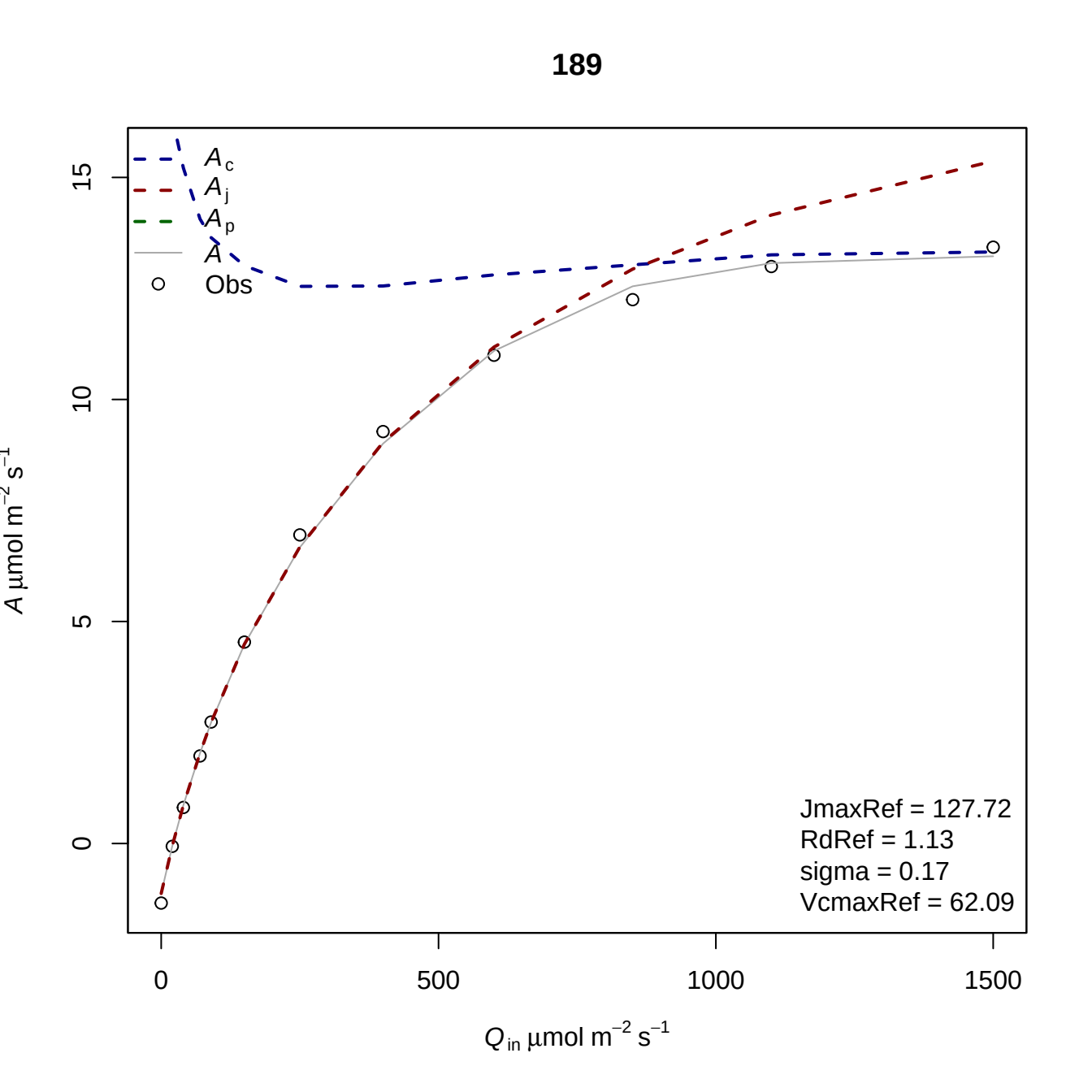

183
A_c
A_j
A_p
A
Obs
JmaxRef = 78.42
RdRef = 0.63
sigma = 0.1
VcmaxRef = 35.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

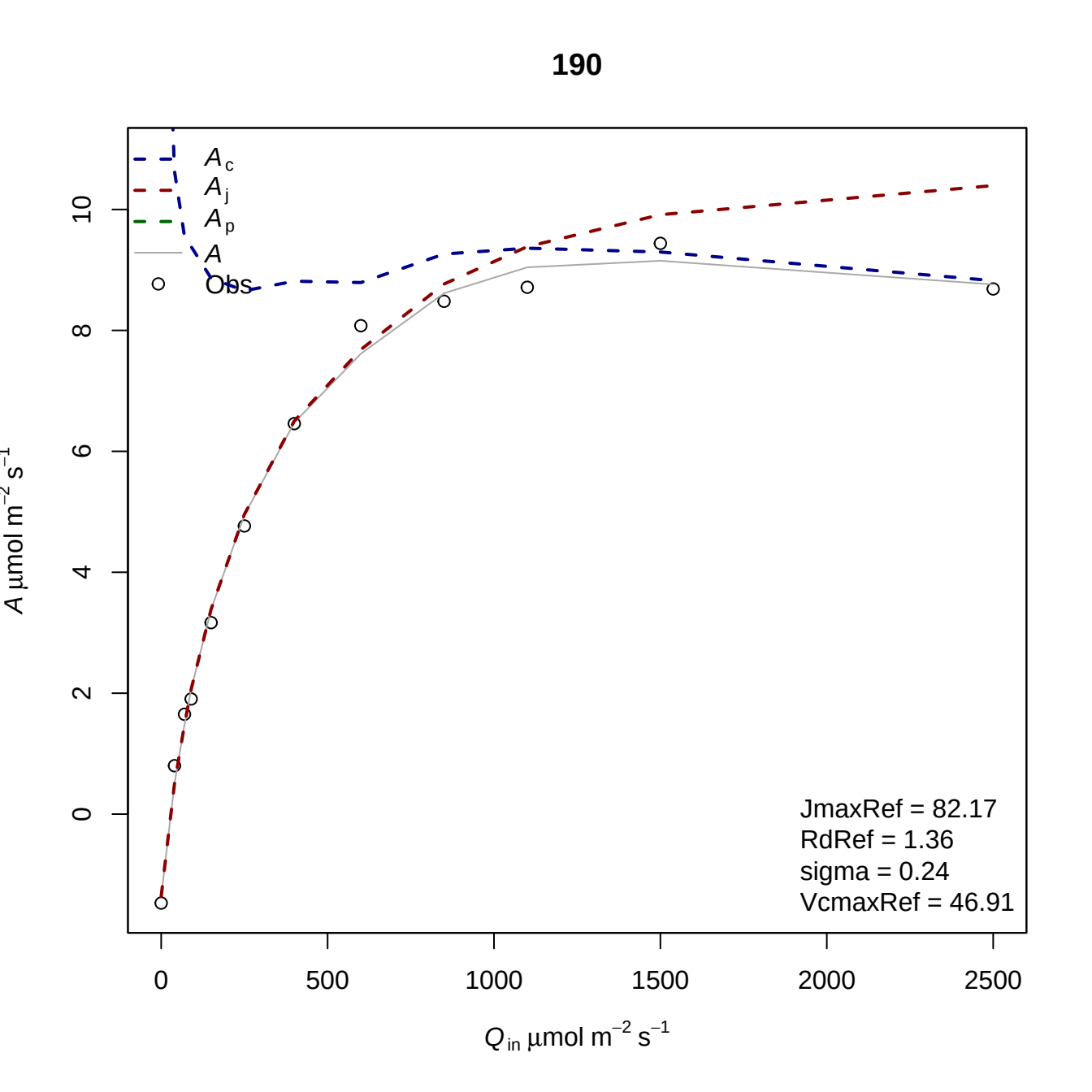

184
A_c
A_j
A_p
A
Obs
JmaxRef = 76.66
RdRef = 0.76
sigma = 0.07
VcmaxRef = 35.58
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

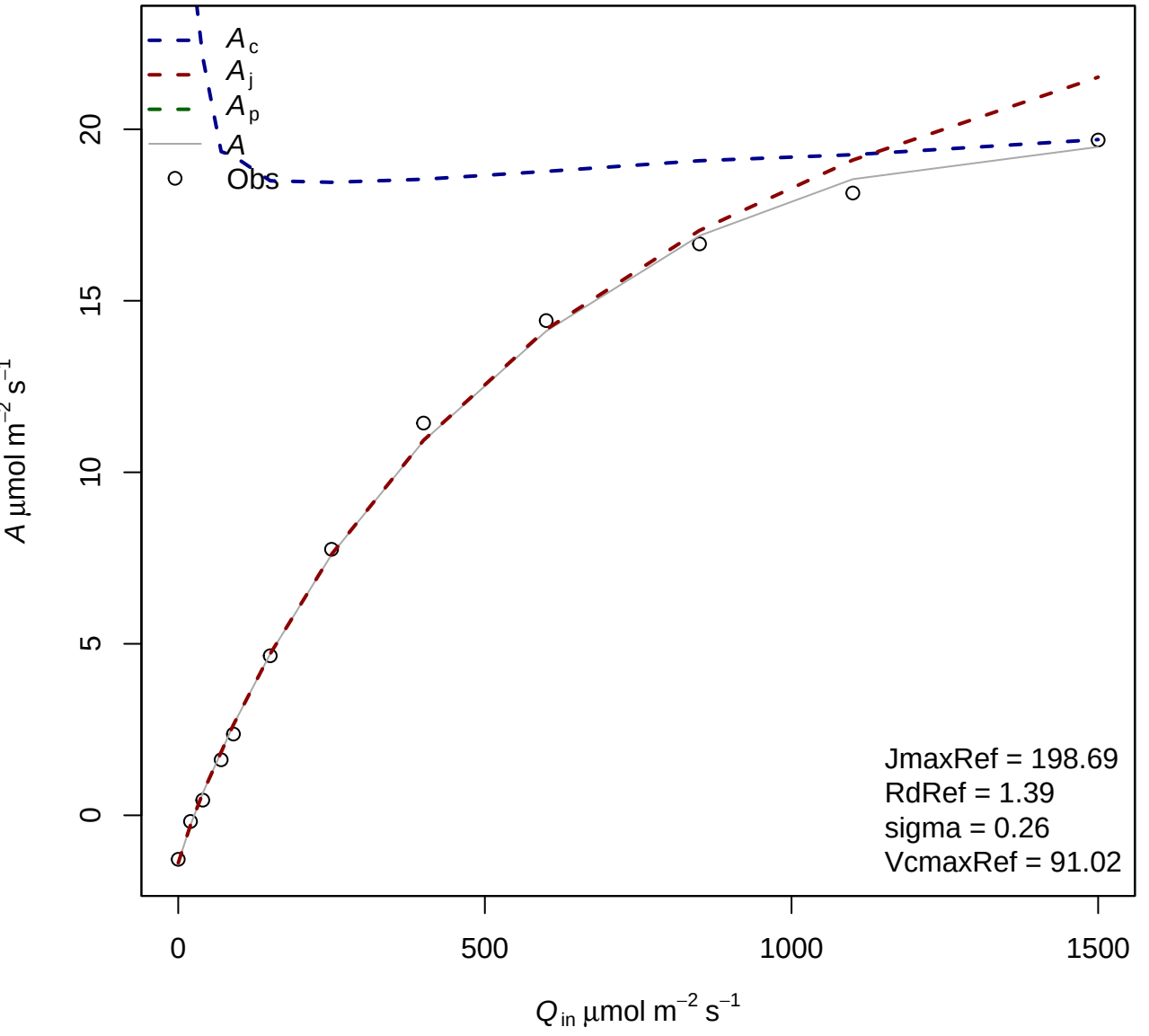

185
A_c
A_j
A_p
A
Obs
JmaxRef = 87.62
RdRef = 2.41
sigma = 1.24
VcmaxRef = 2319.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

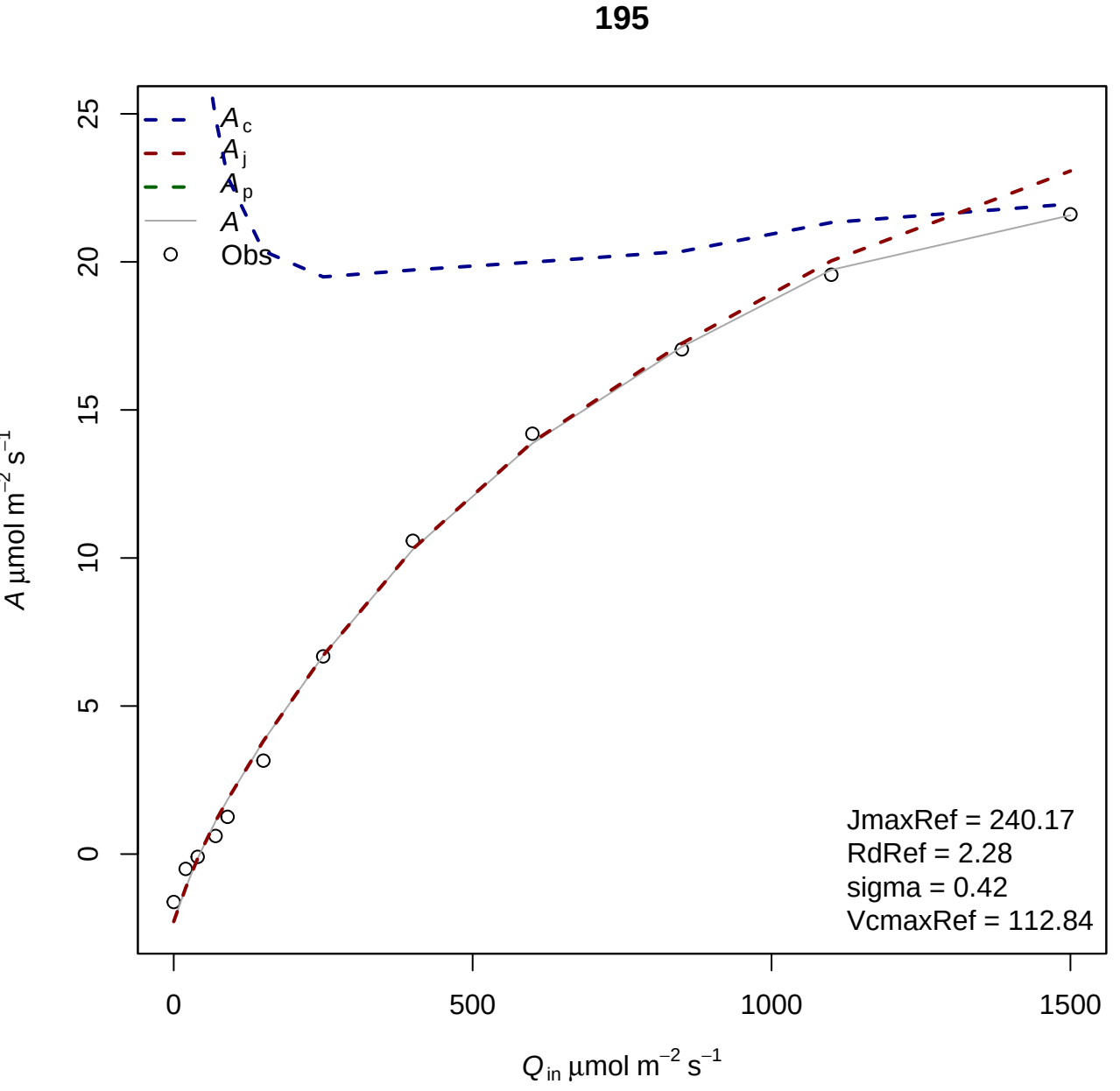

A_c
A_j
A_p
A
Obs
JmaxRef = 78.17
RdRef = 1.72
sigma = 0.54
VcmaxRef = 375.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

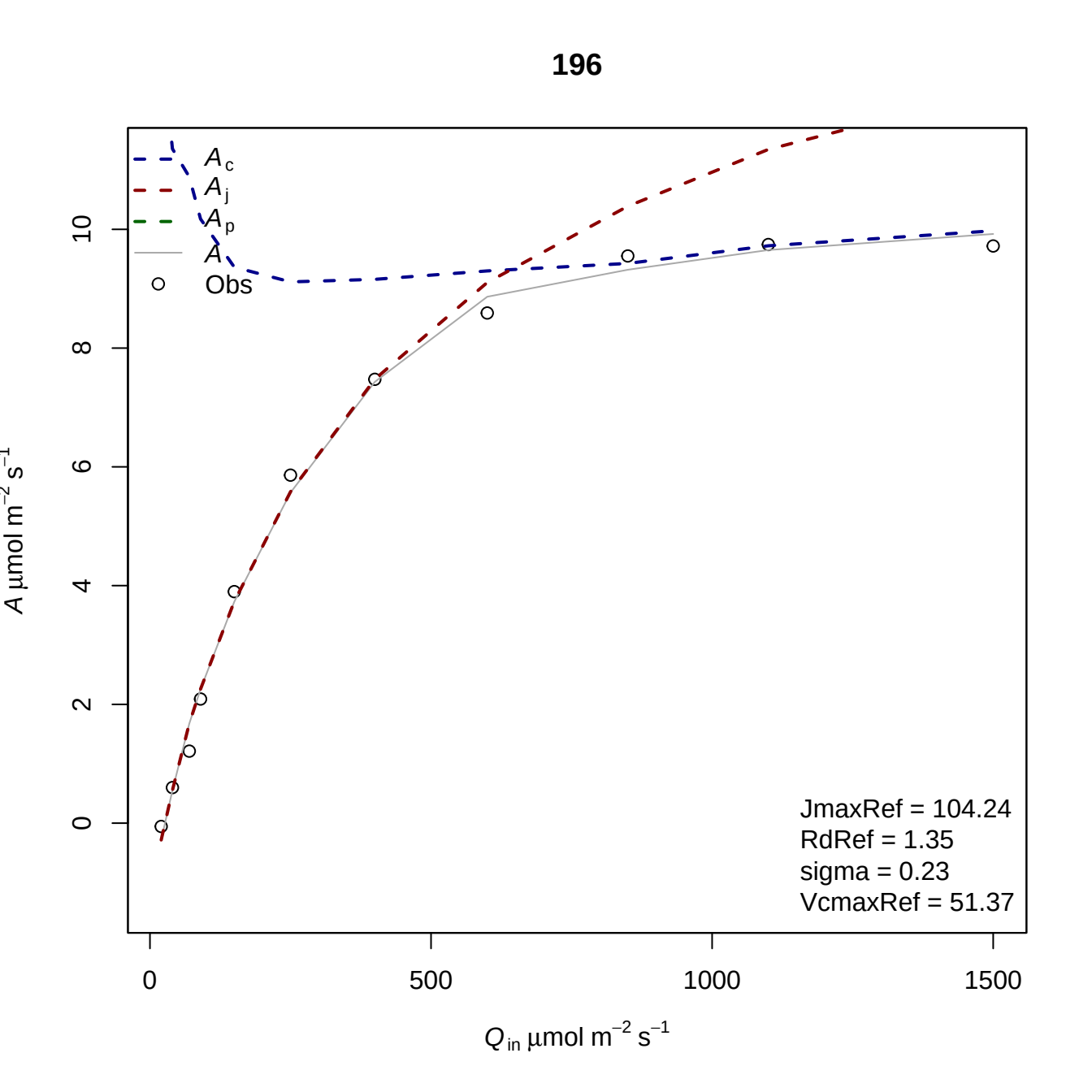

A_c
A_j
A_p
A
Obs
JmaxRef = 115.33
RdRef = 1.87
sigma = 0.23
VcmaxRef = 51.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
200
400
600
800
1000
1200
1400

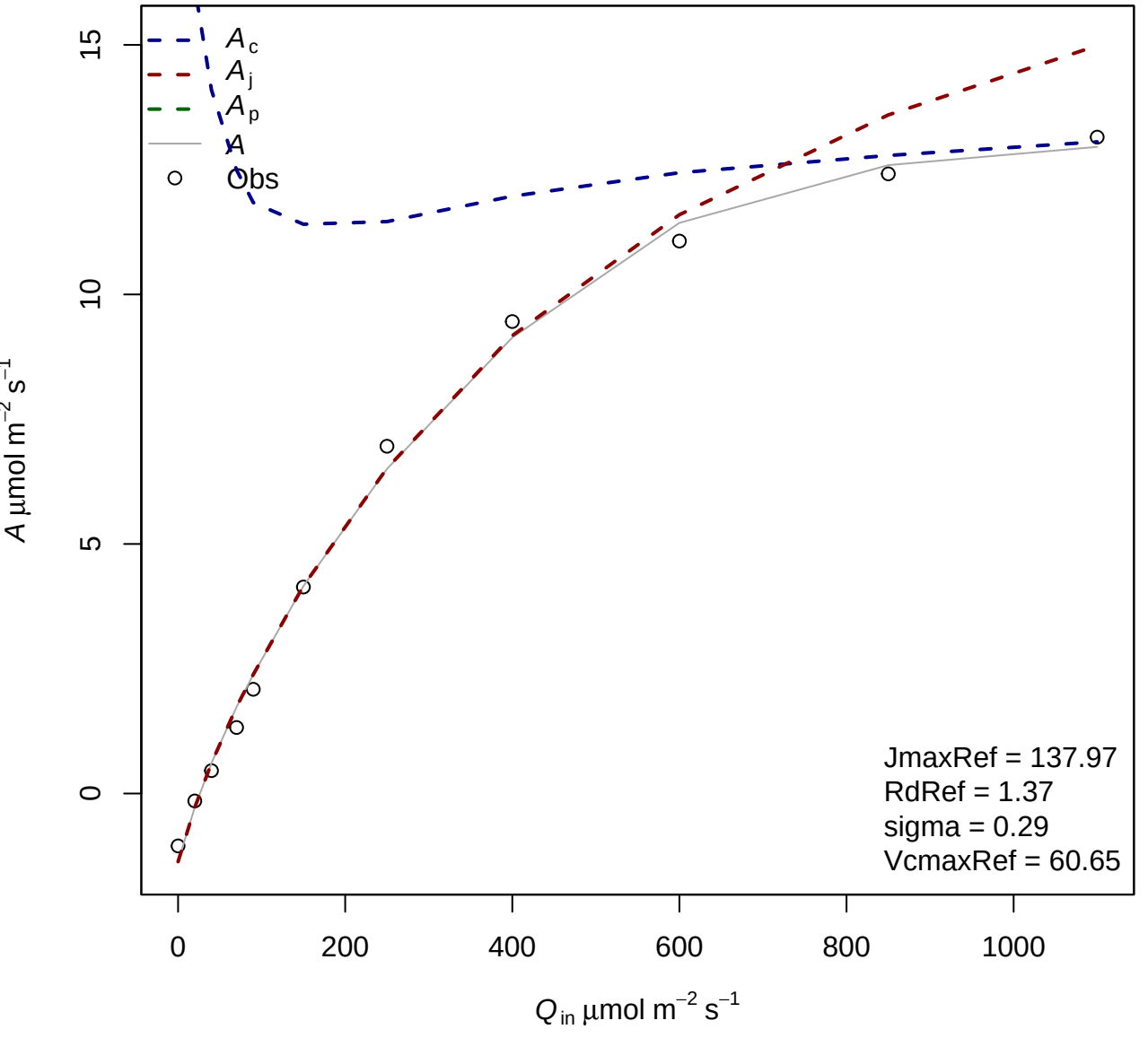

189
A_c
A_j
A_p
A
Obs
JmaxRef = 127.72
RdRef = 1.13
sigma = 0.17
VcmaxRef = 62.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

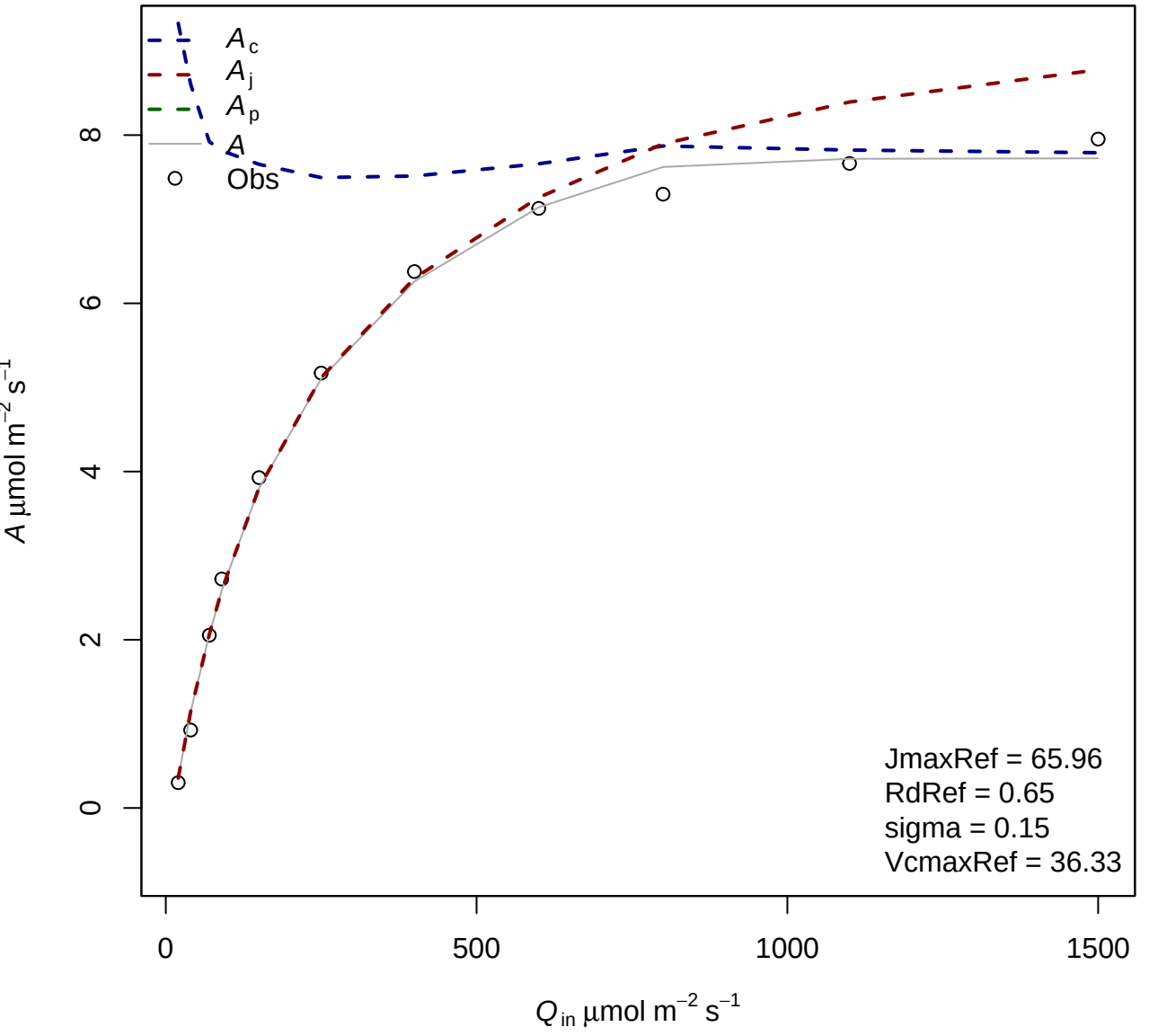

190
A_c
A_j
A_p
A
Obs
JmaxRef = 82.17
RdRef = 1.36
sigma = 0.24
VcmaxRef = 46.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000
2500

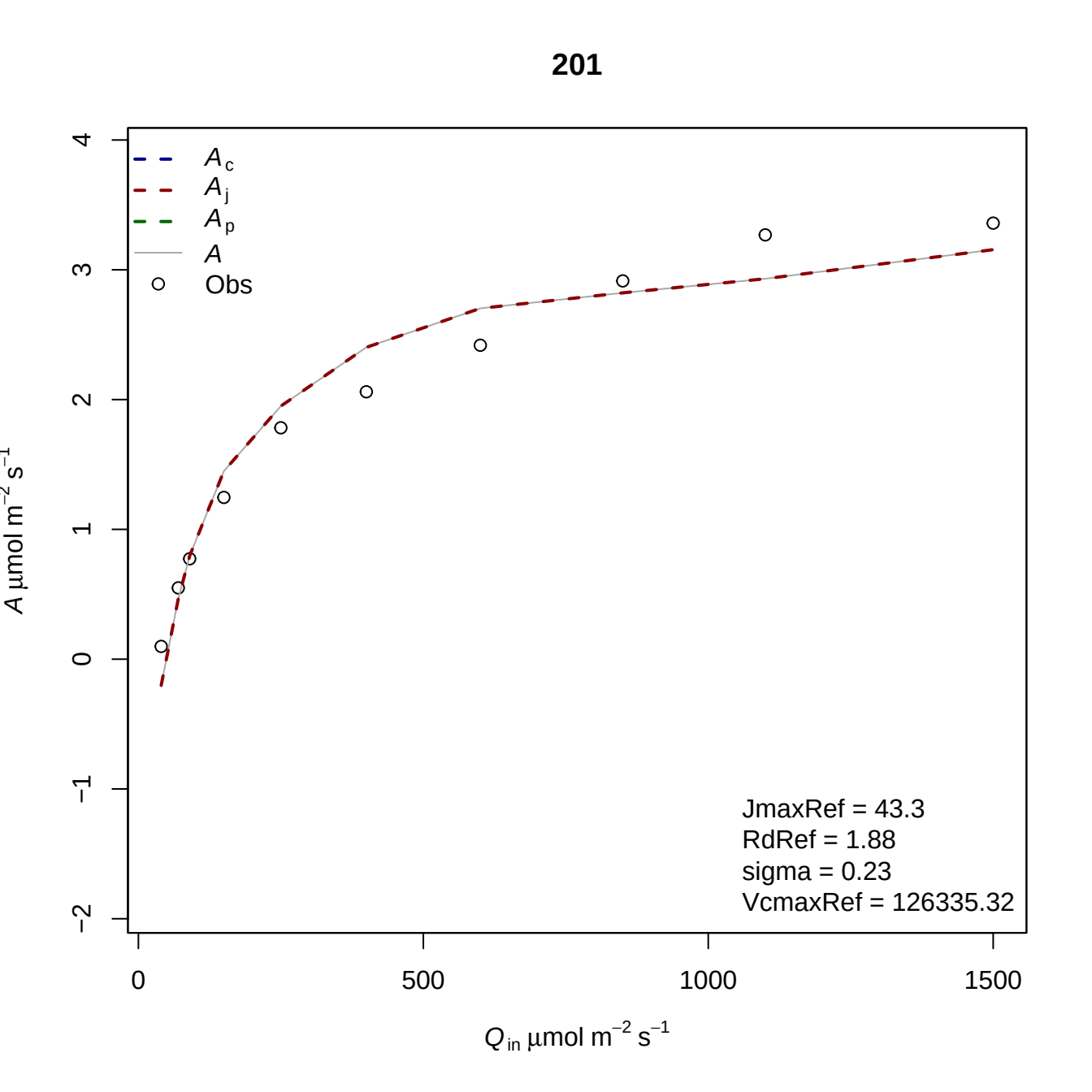

A_c
A_j
A_p
A
Obs
JmaxRef = 198.69
RdRef = 1.39
sigma = 0.26
VcmaxRef = 91.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

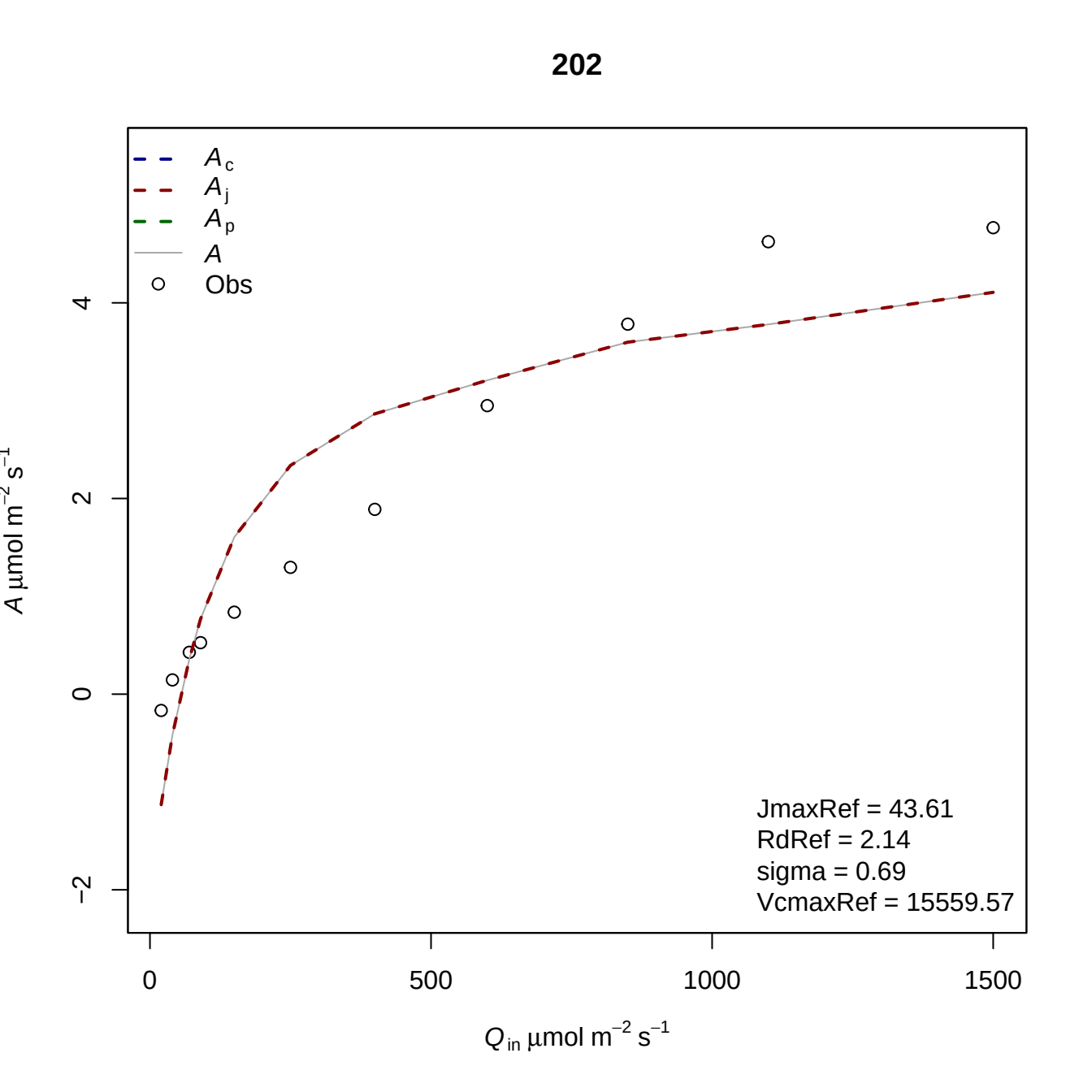

195
A_c
A_j
A_p
A
Obs
JmaxRef = 240.17
RdRef = 2.28
sigma = 0.42
VcmaxRef = 112.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

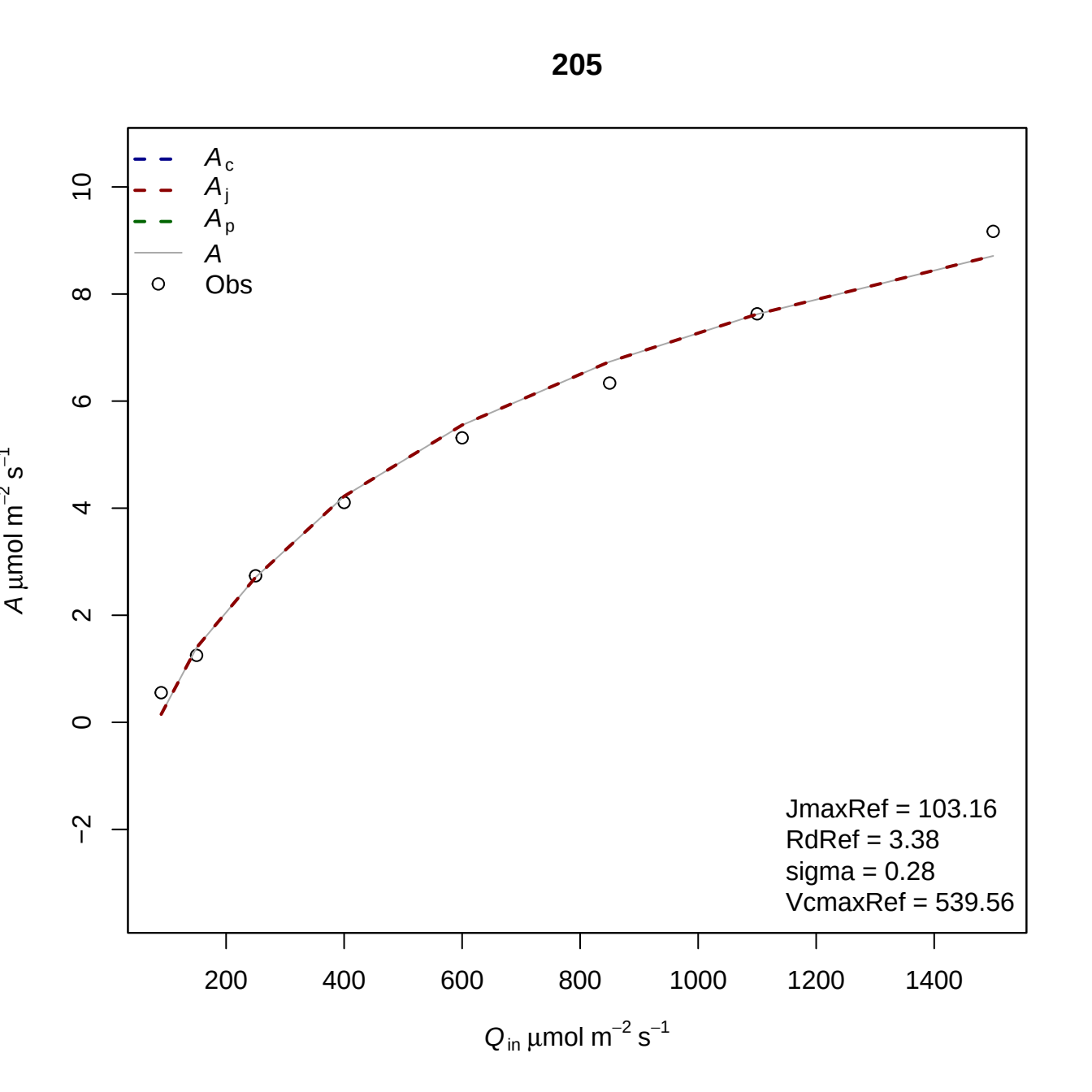

196
A_c
A_j
A_p
A
Obs
JmaxRef = 104.24
RdRef = 1.35
sigma = 0.23
VcmaxRef = 51.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

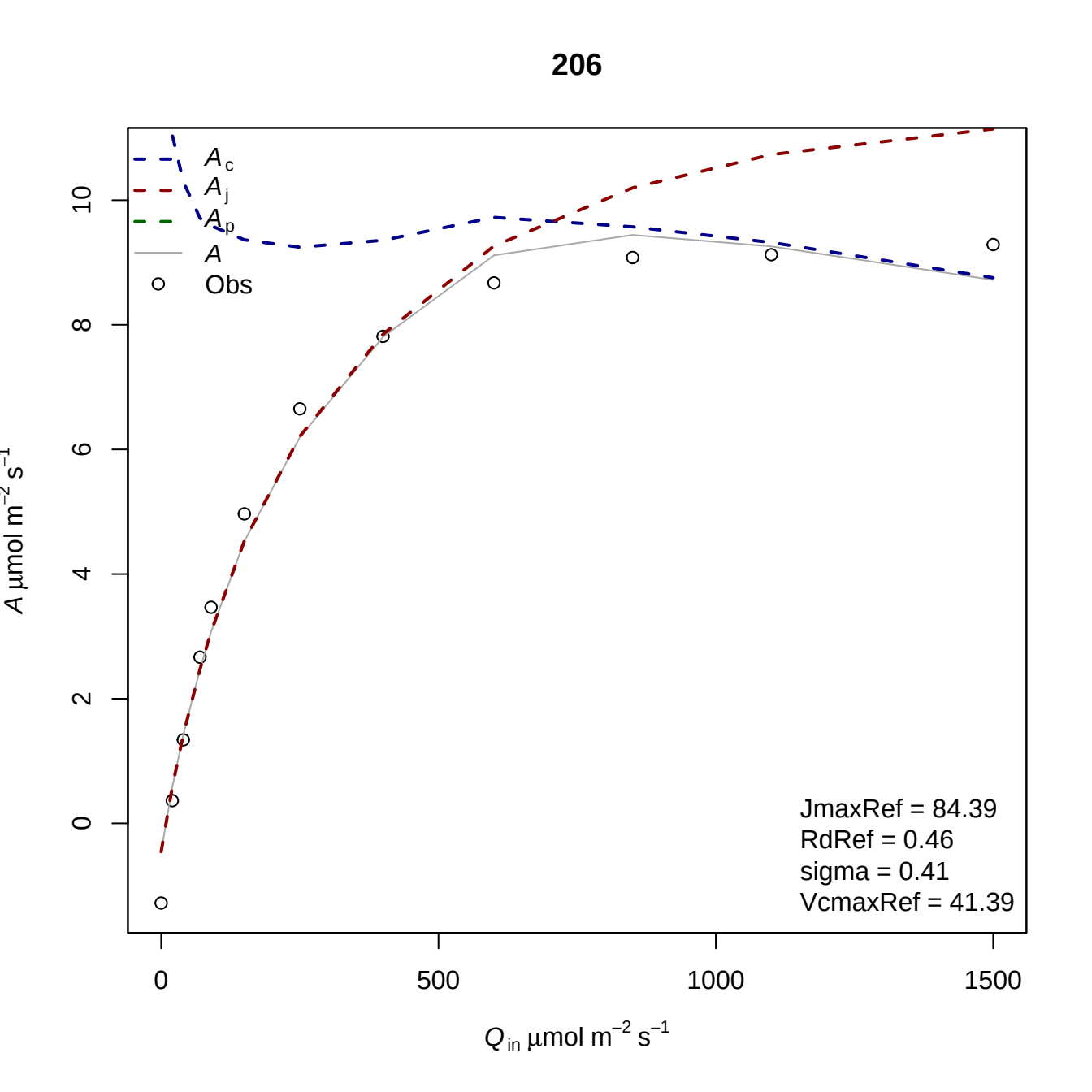

A_c
A_j
A_p
A
Obs
JmaxRef = 137.97
RdRef = 1.37
sigma = 0.29
VcmaxRef = 60.65
0
5
10
15
0
200
400
600
800
1000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

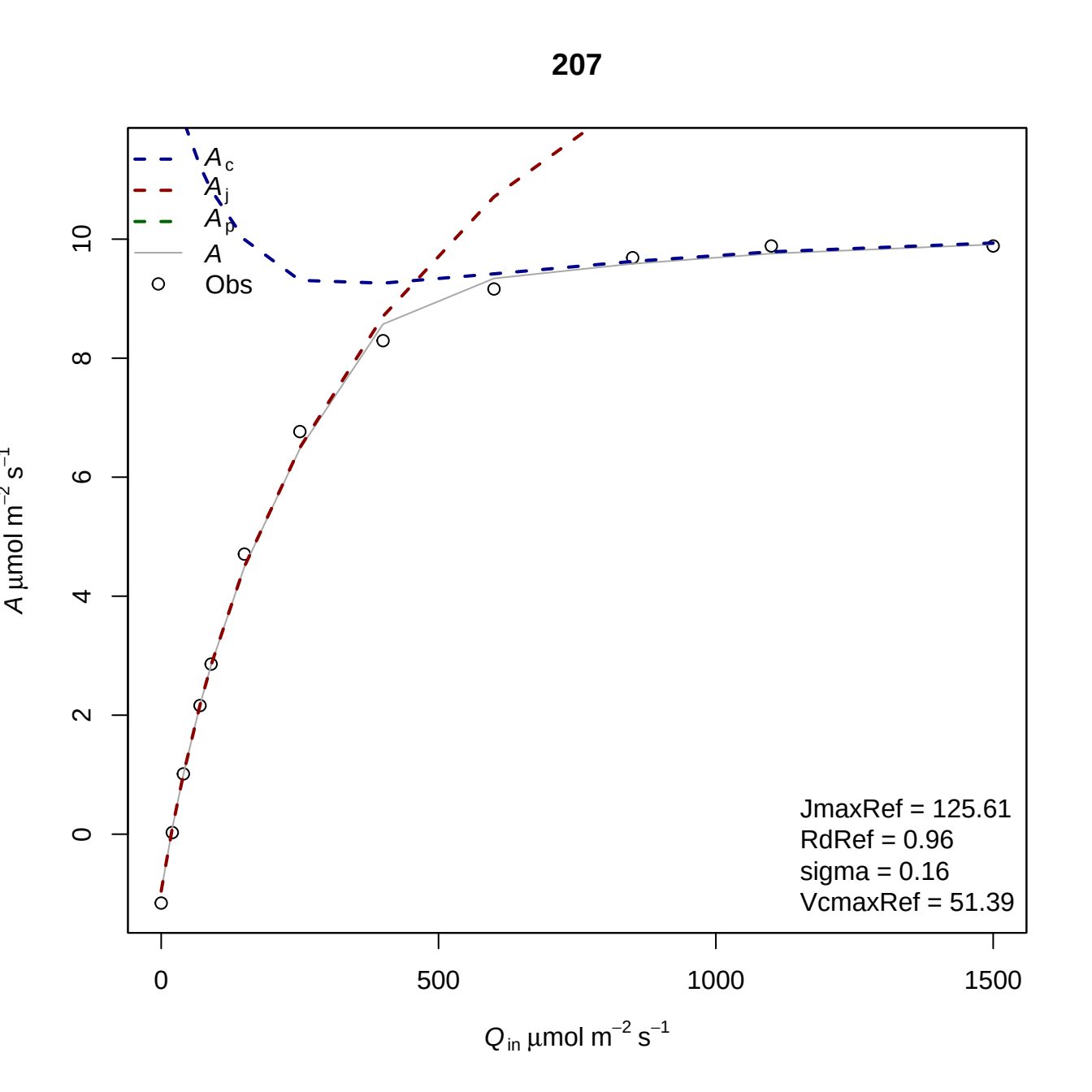

A_c
A_j
A_p
A
Obs
JmaxRef = 65.96
RdRef = 0.65
sigma = 0.15
VcmaxRef = 36.33
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

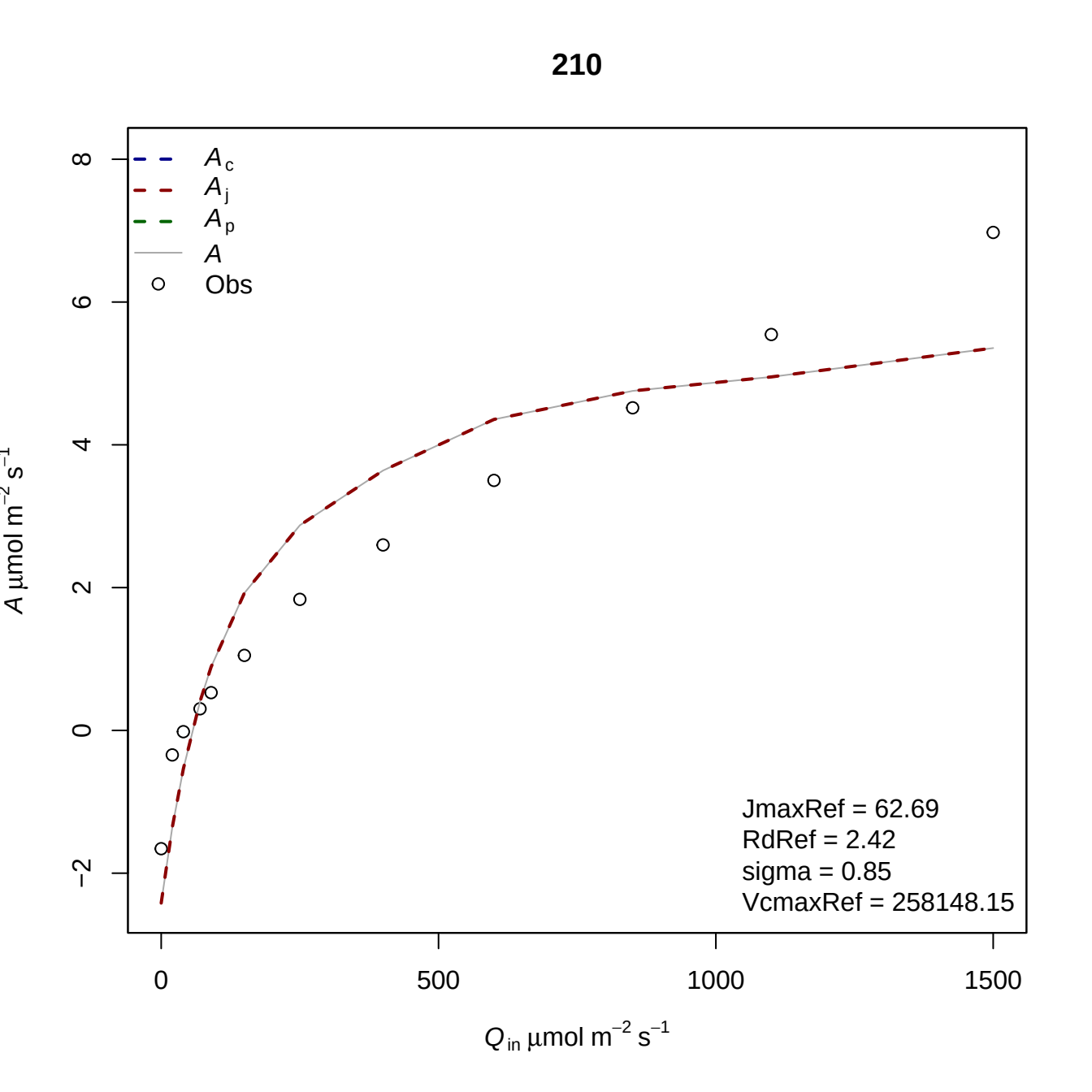

201
A_c
A_j
A_p
A
Obs
JmaxRef = 43.3
RdRef = 1.88
sigma = 0.23
VcmaxRef = 126335.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500

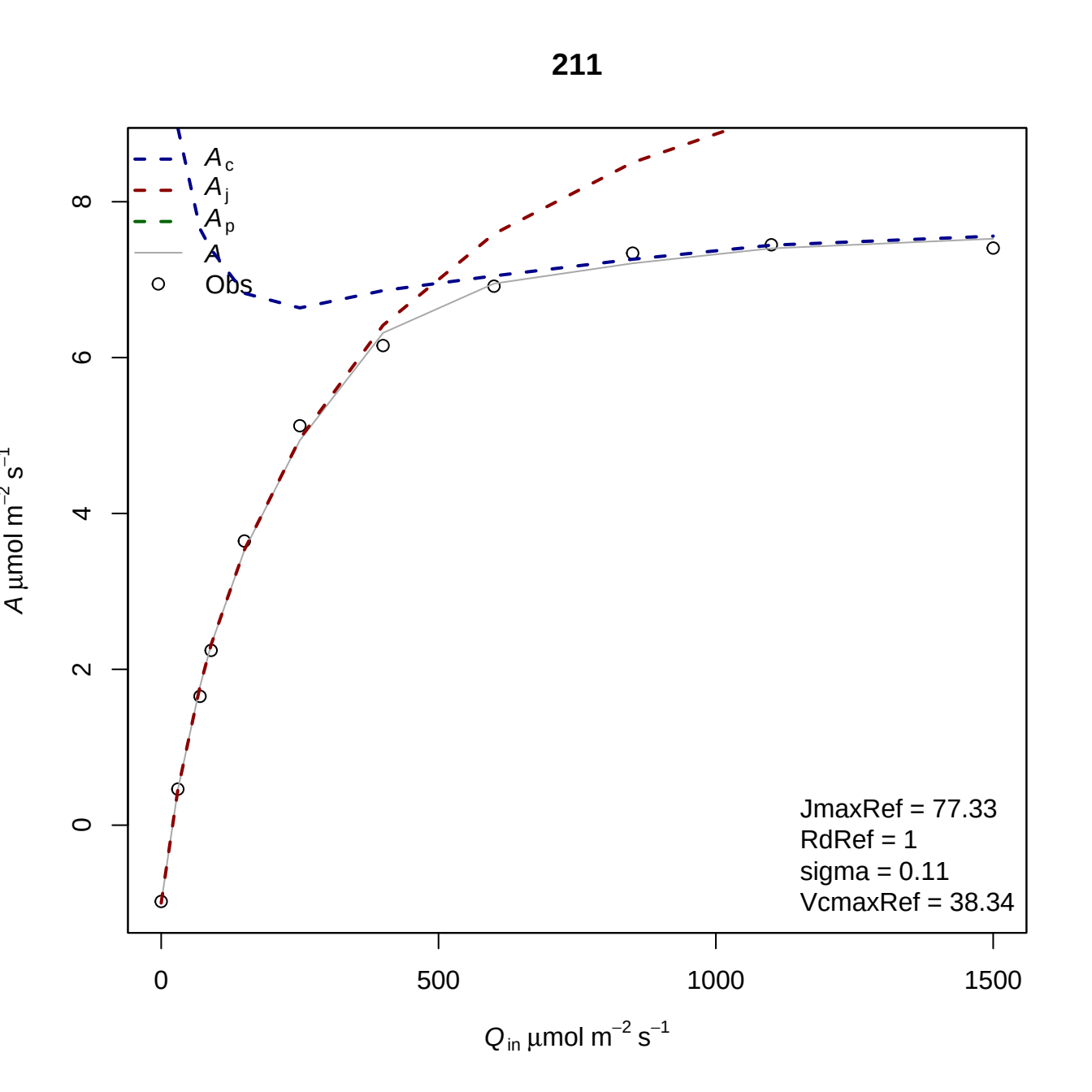

202
A_c
A_j
A_p
A
Obs
JmaxRef = 43.61
RdRef = 2.14
sigma = 0.69
VcmaxRef = 15559.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

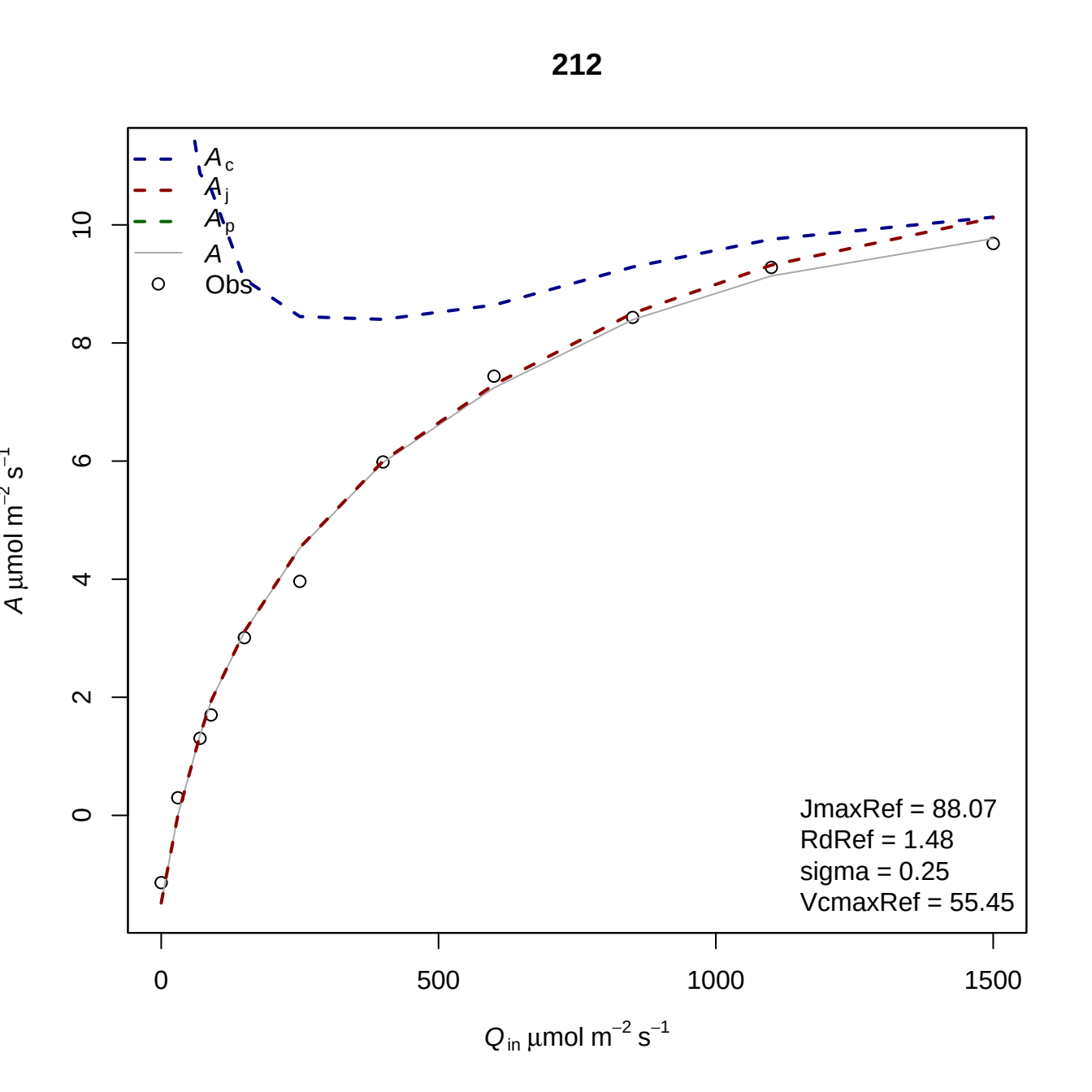

205
A_c
A_j
A_p
A
Obs
JmaxRef = 103.16
RdRef = 3.38
sigma = 0.28
VcmaxRef = 539.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
200
400
600
800
1000
1200
1400

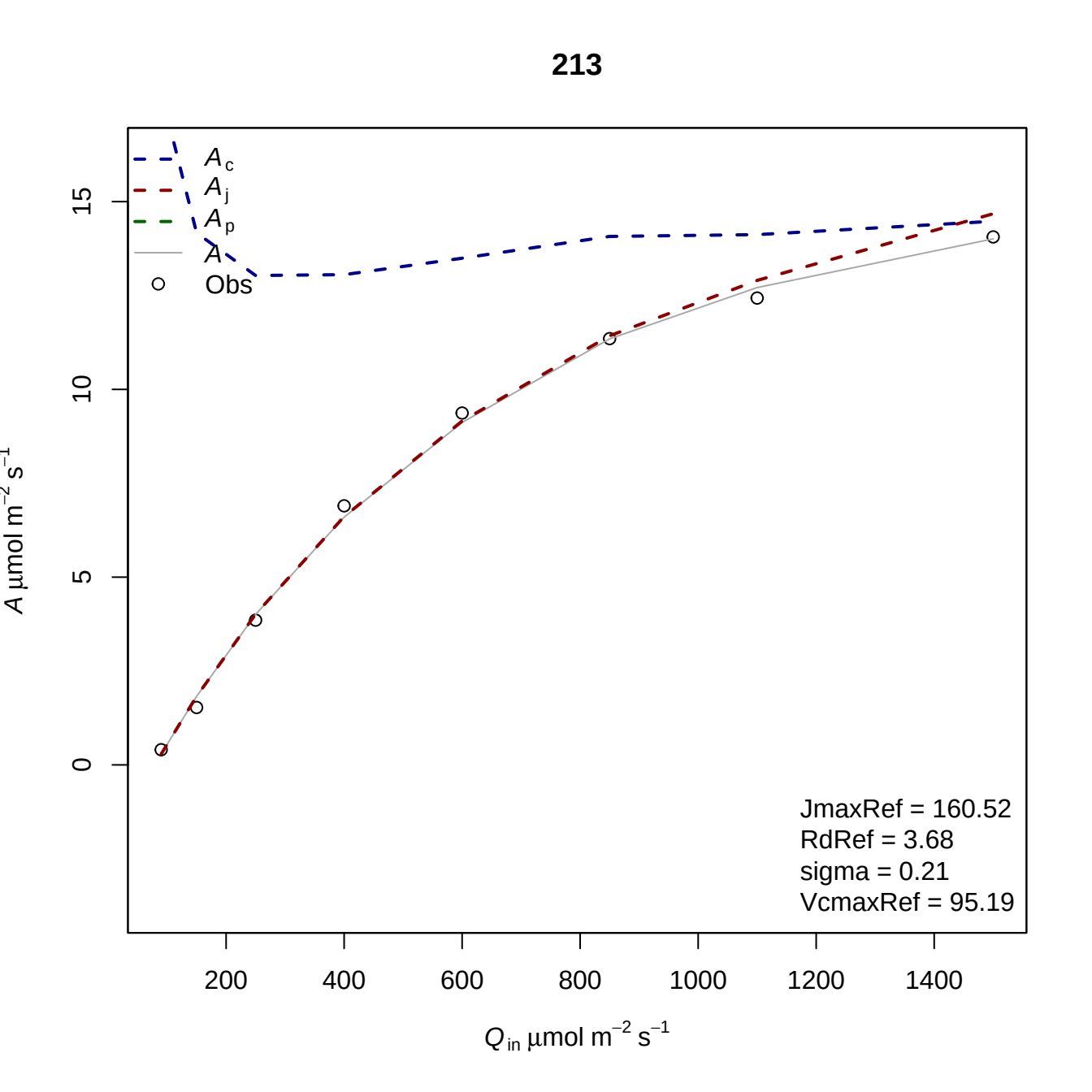

206
A_c
A_j
A_p
A
Obs
JmaxRef = 84.39
RdRef = 0.46
sigma = 0.41
VcmaxRef = 41.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

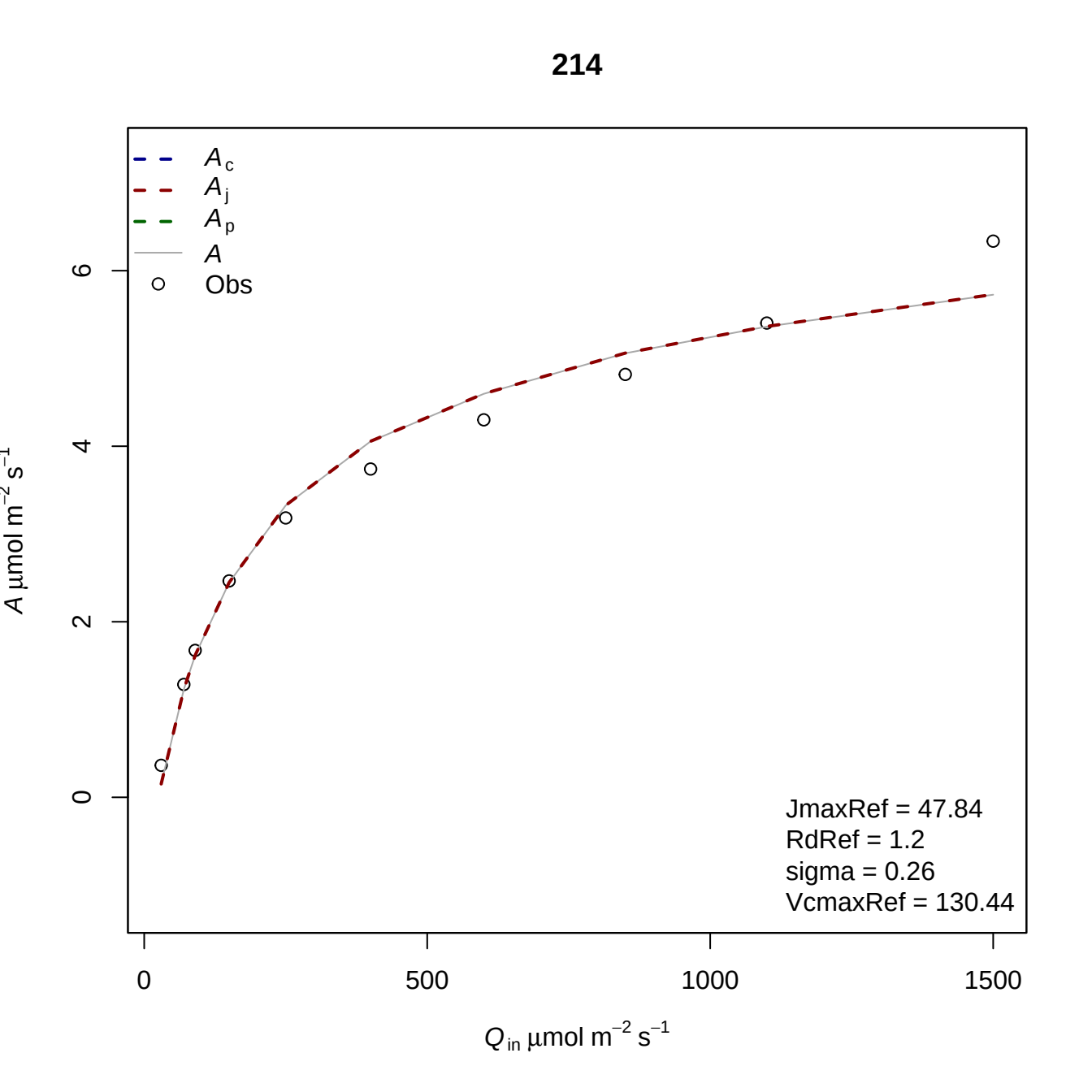

207
A_c
A_j
A_p
A
Obs
JmaxRef = 125.61
RdRef = 0.96
sigma = 0.16
VcmaxRef = 51.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

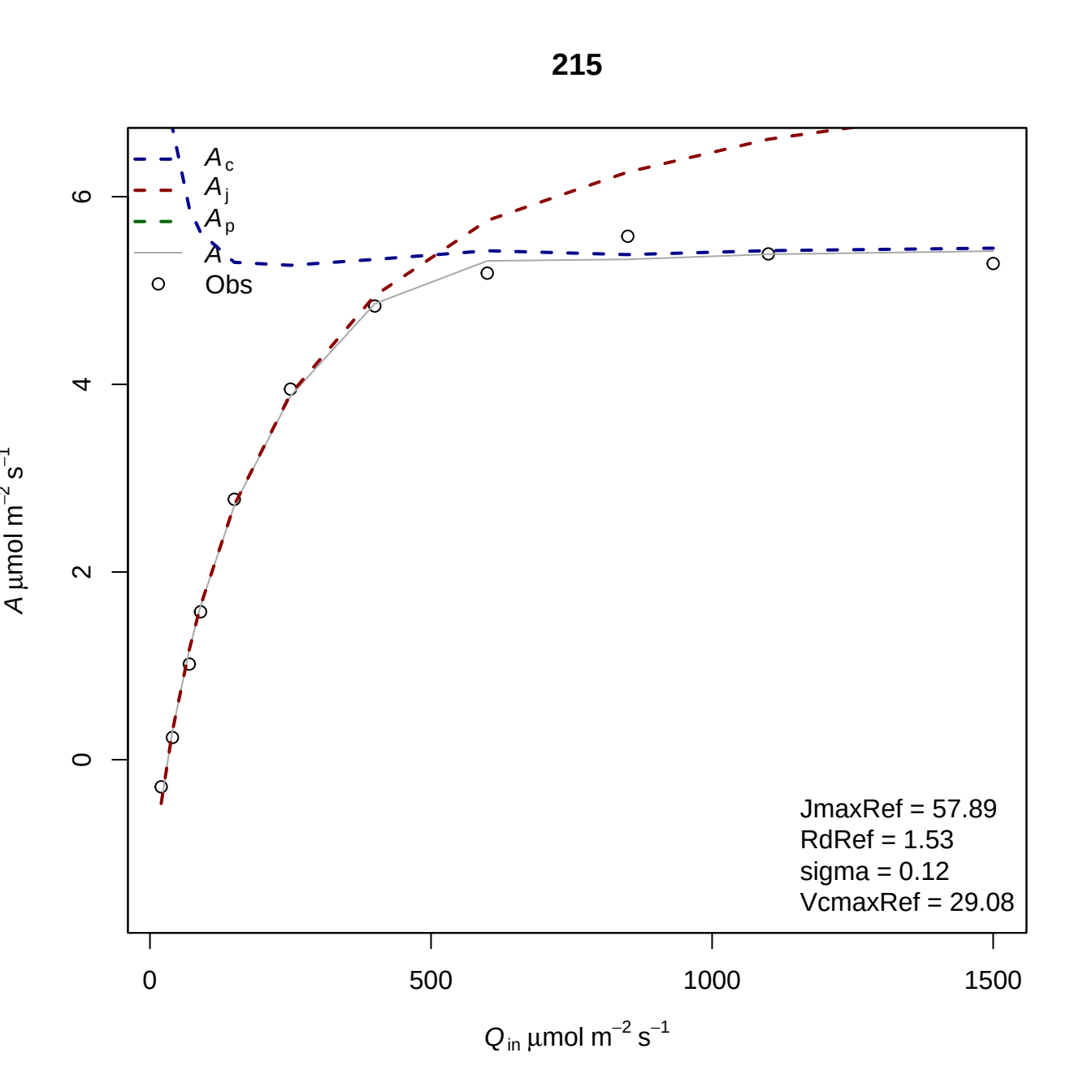

210
A_c
A_j
A_p
A
Obs
JmaxRef = 62.69
RdRef = 2.42
sigma = 0.85
VcmaxRef = 258148.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

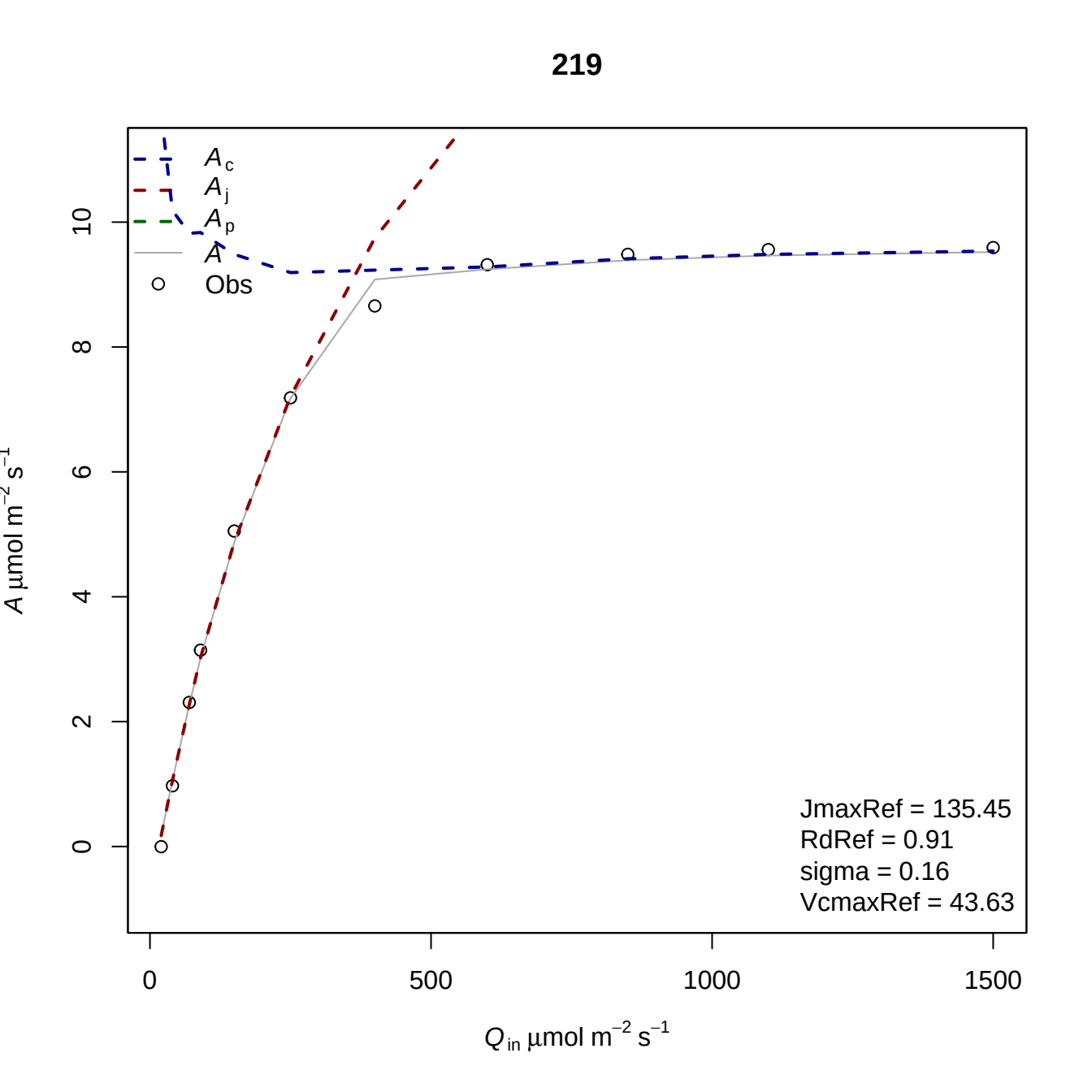

211
A_c
A_j
A_p
A
Obs
JmaxRef = 77.33
RdRef = 1
sigma = 0.11
VcmaxRef = 38.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

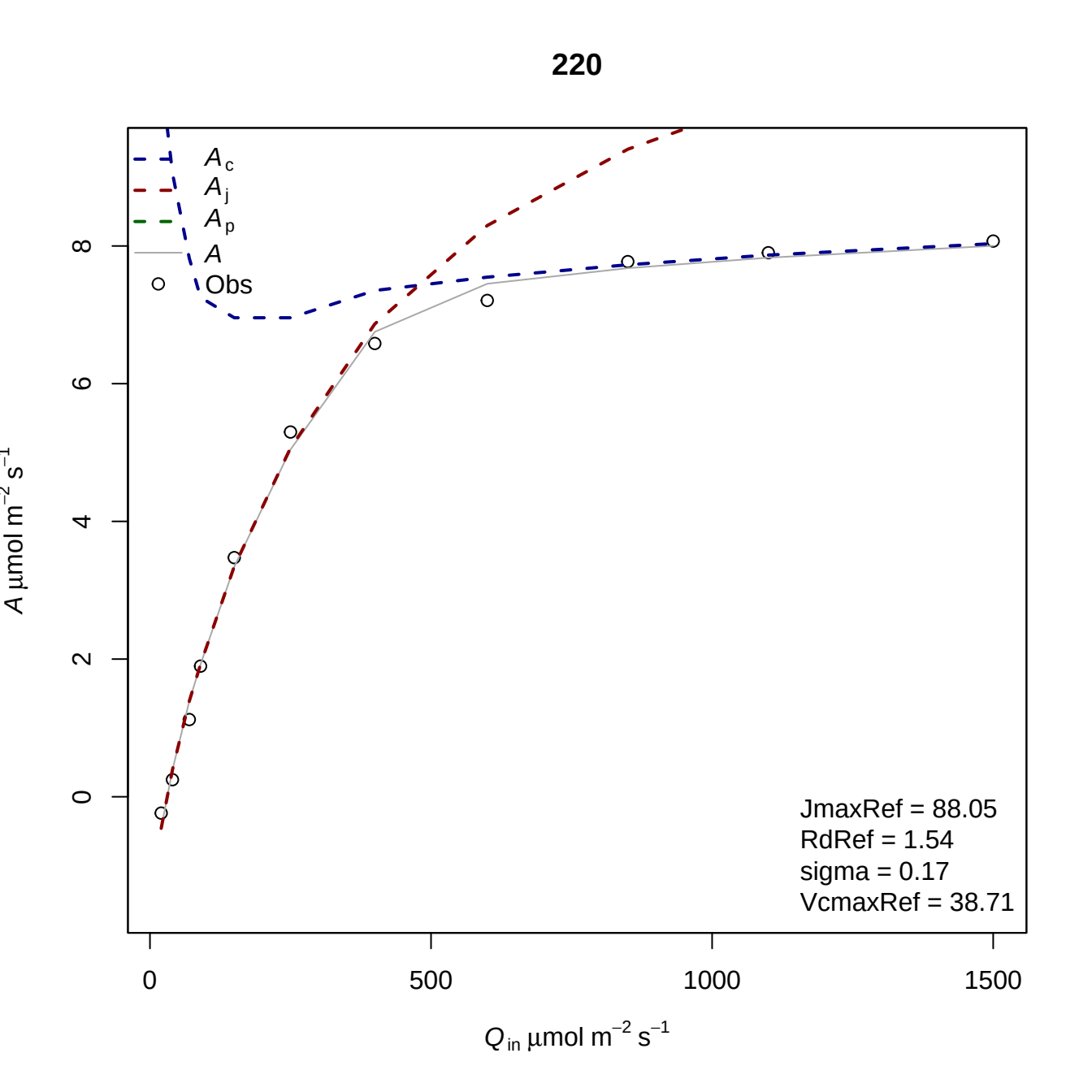

212
A_c
A_j
A_p
A
Obs
JmaxRef = 88.07
RdRef = 1.48
sigma = 0.25
VcmaxRef = 55.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

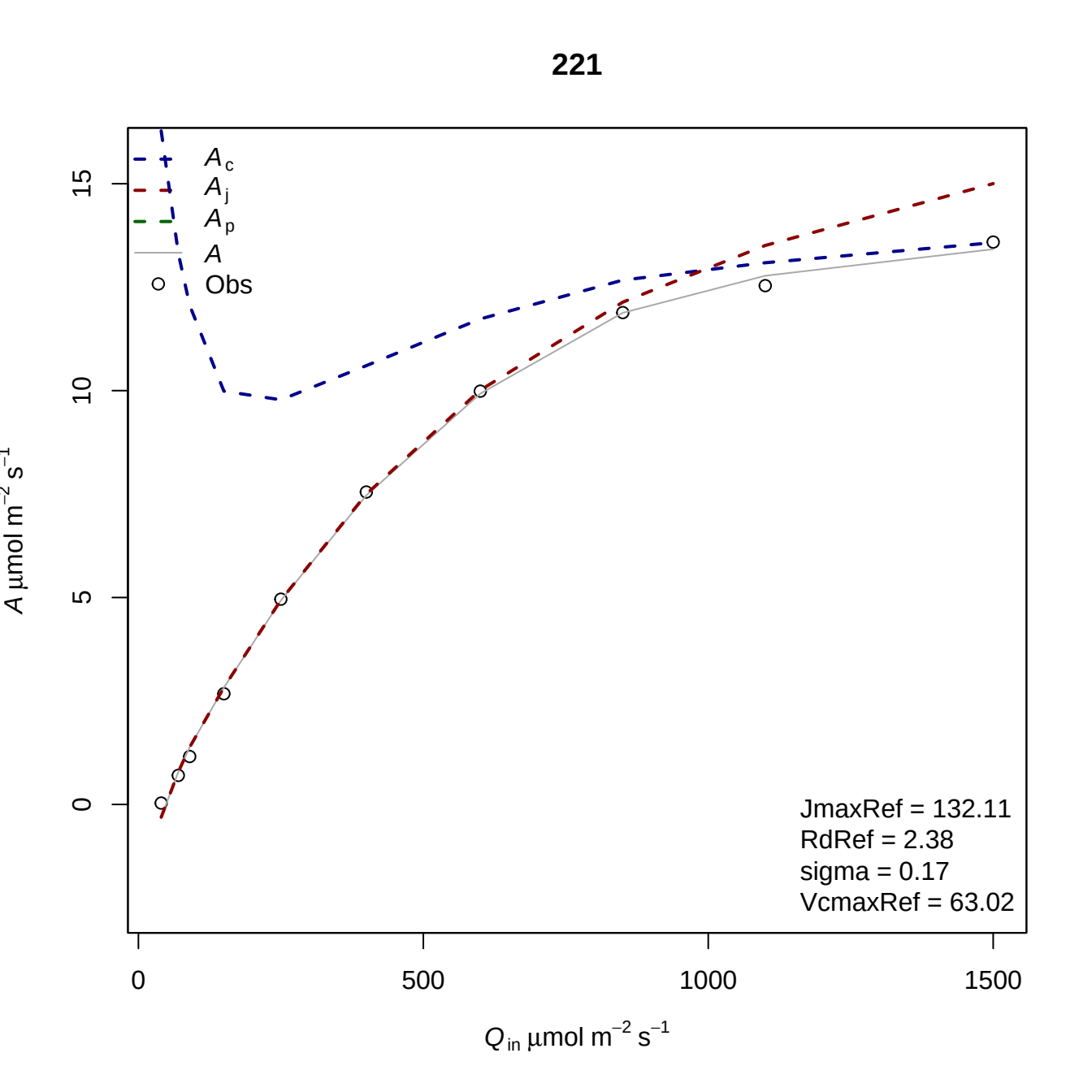

213
A_c
A_j
A_p
A
Obs
JmaxRef = 160.52
RdRef = 3.68
sigma = 0.21
VcmaxRef = 95.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
200
400
600
800
1000
1200
1400

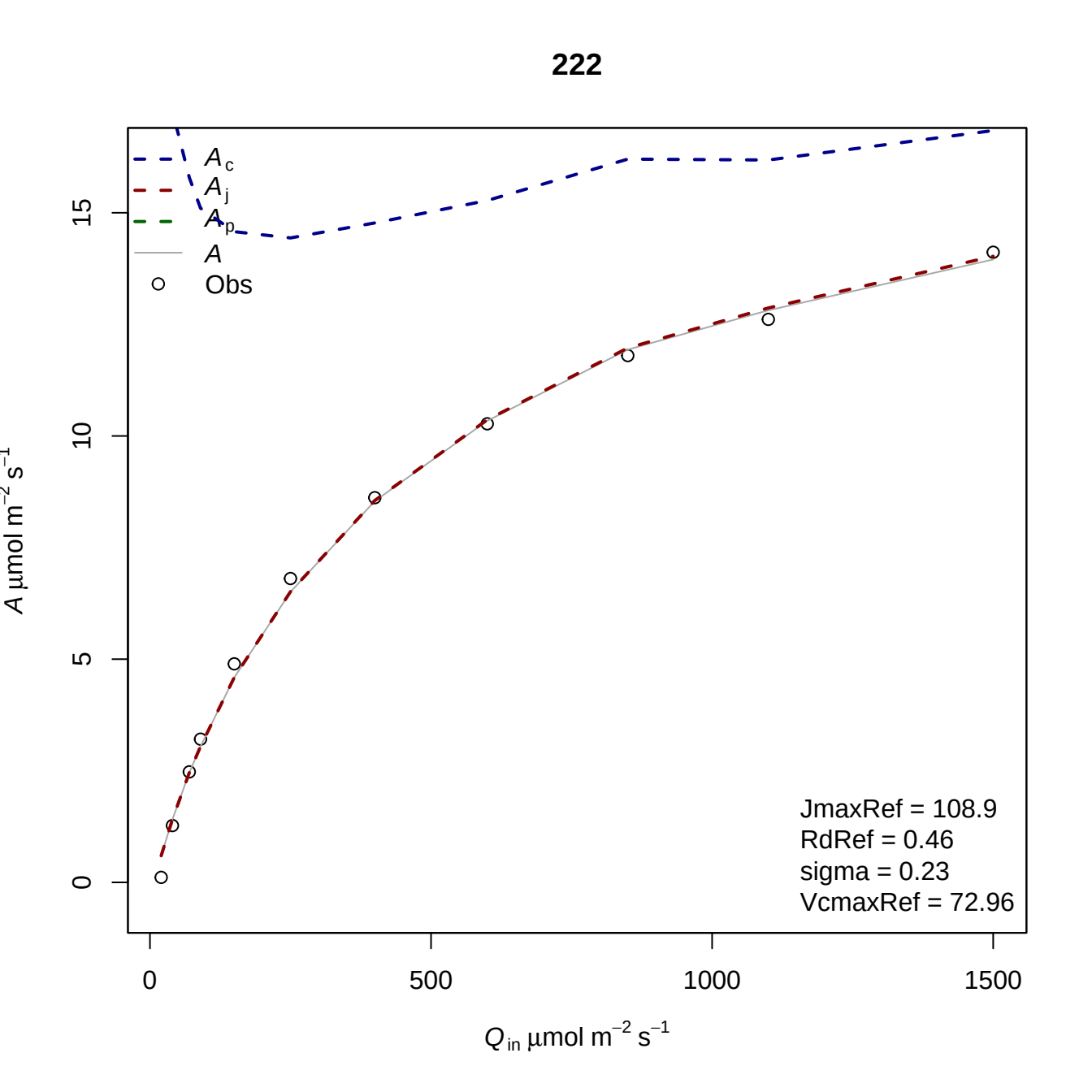

214
A_c
A_j
A_p
A
Obs
JmaxRef = 47.84
RdRef = 1.2
sigma = 0.26
VcmaxRef = 130.44
A µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

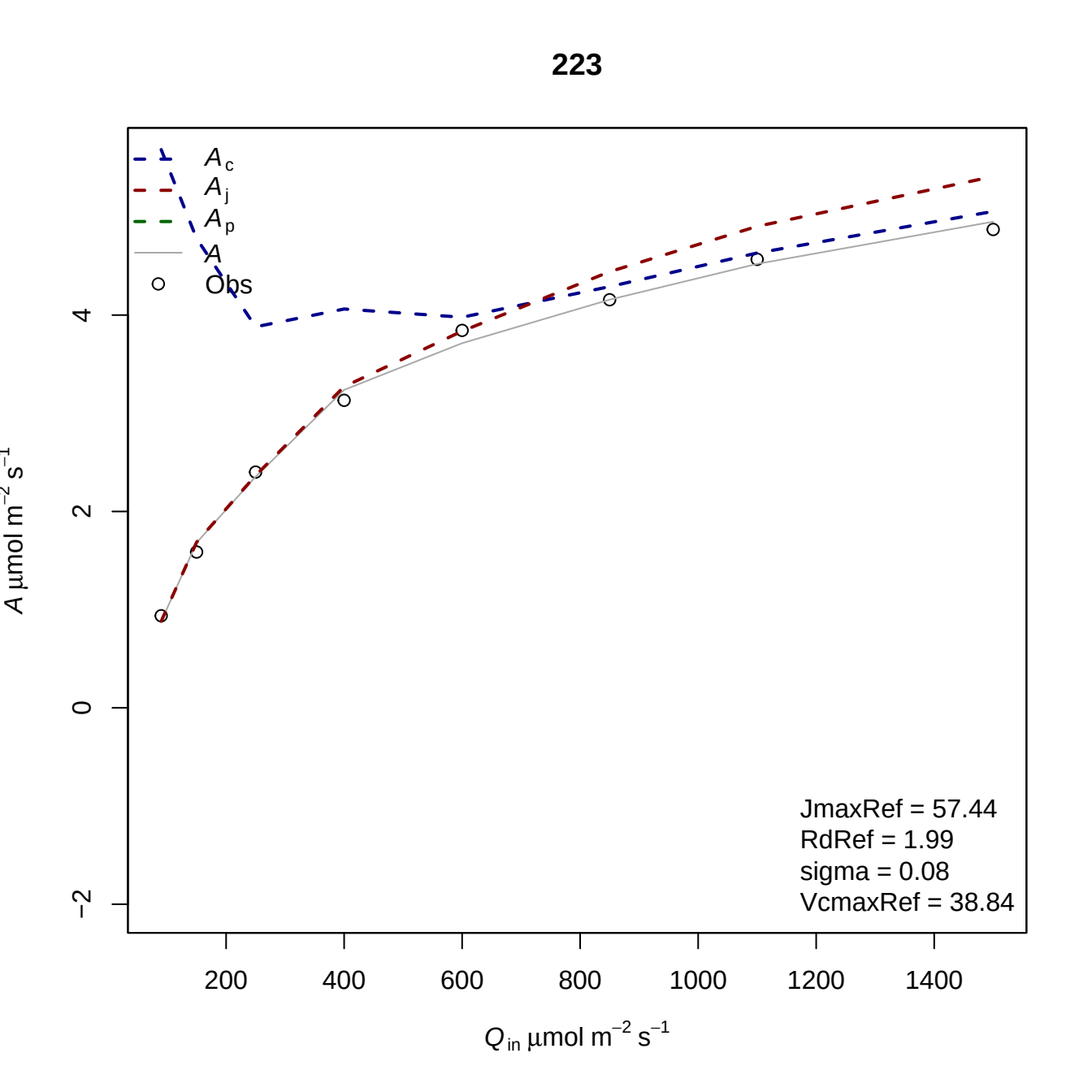

215
A_c
A_j
A_p
A
Obs
JmaxRef = 57.89
RdRef = 1.53
sigma = 0.12
VcmaxRef = 29.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

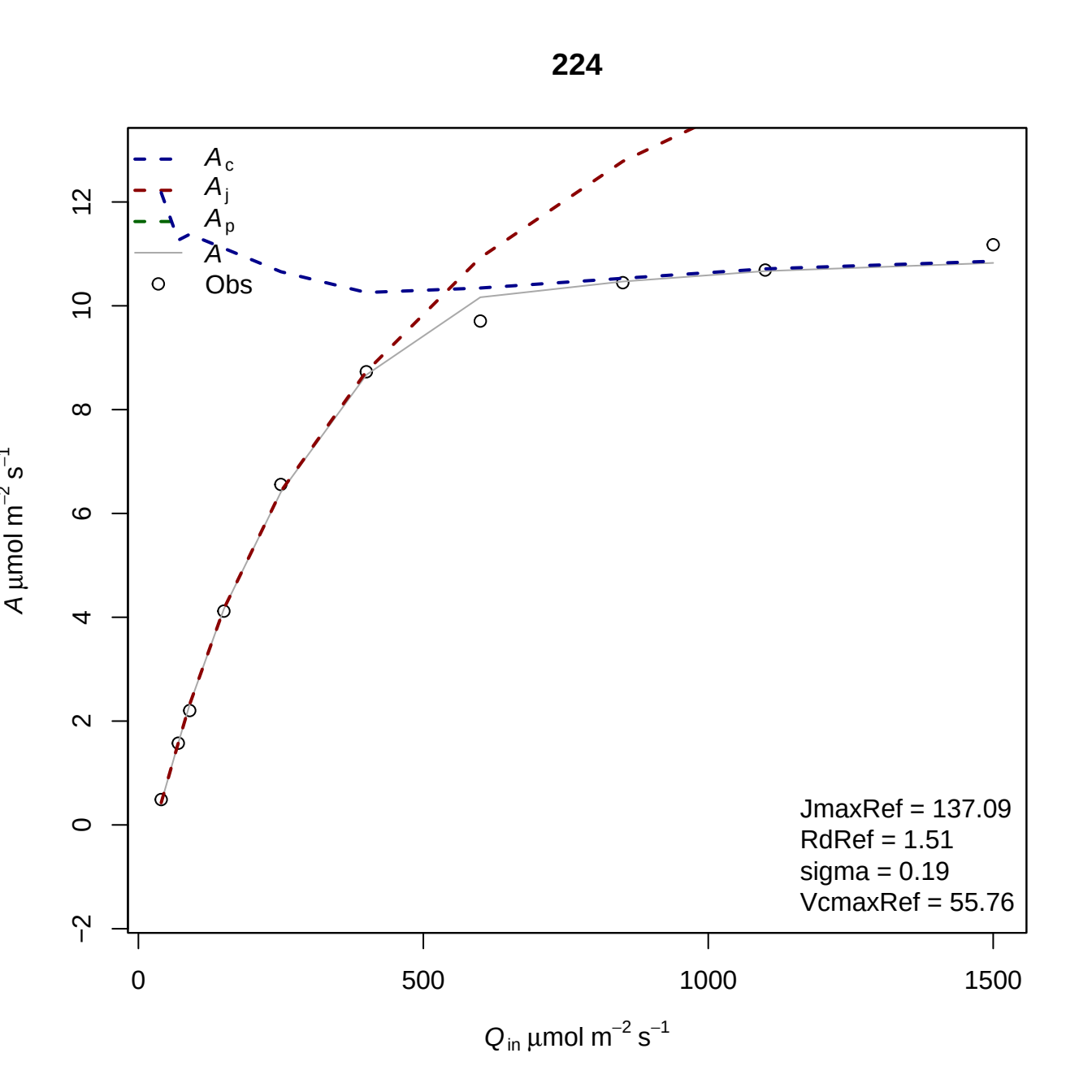

219
A_c
A_j
A_p
A
Obs
JmaxRef = 135.45
RdRef = 0.91
sigma = 0.16
VcmaxRef = 43.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

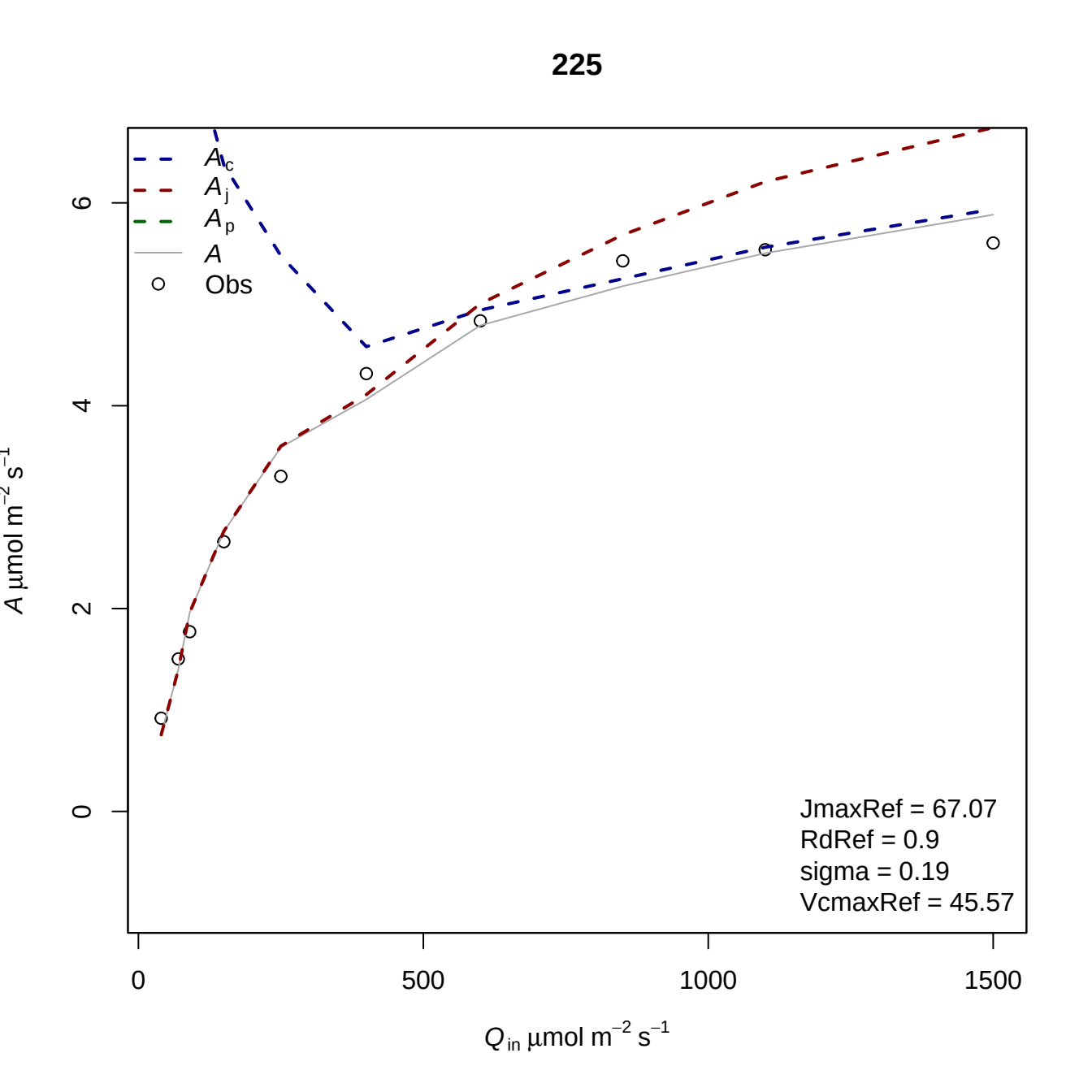

220
A_c
A_j
A_p
A
Obs
JmaxRef = 88.05
RdRef = 1.54
sigma = 0.17
VcmaxRef = 38.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

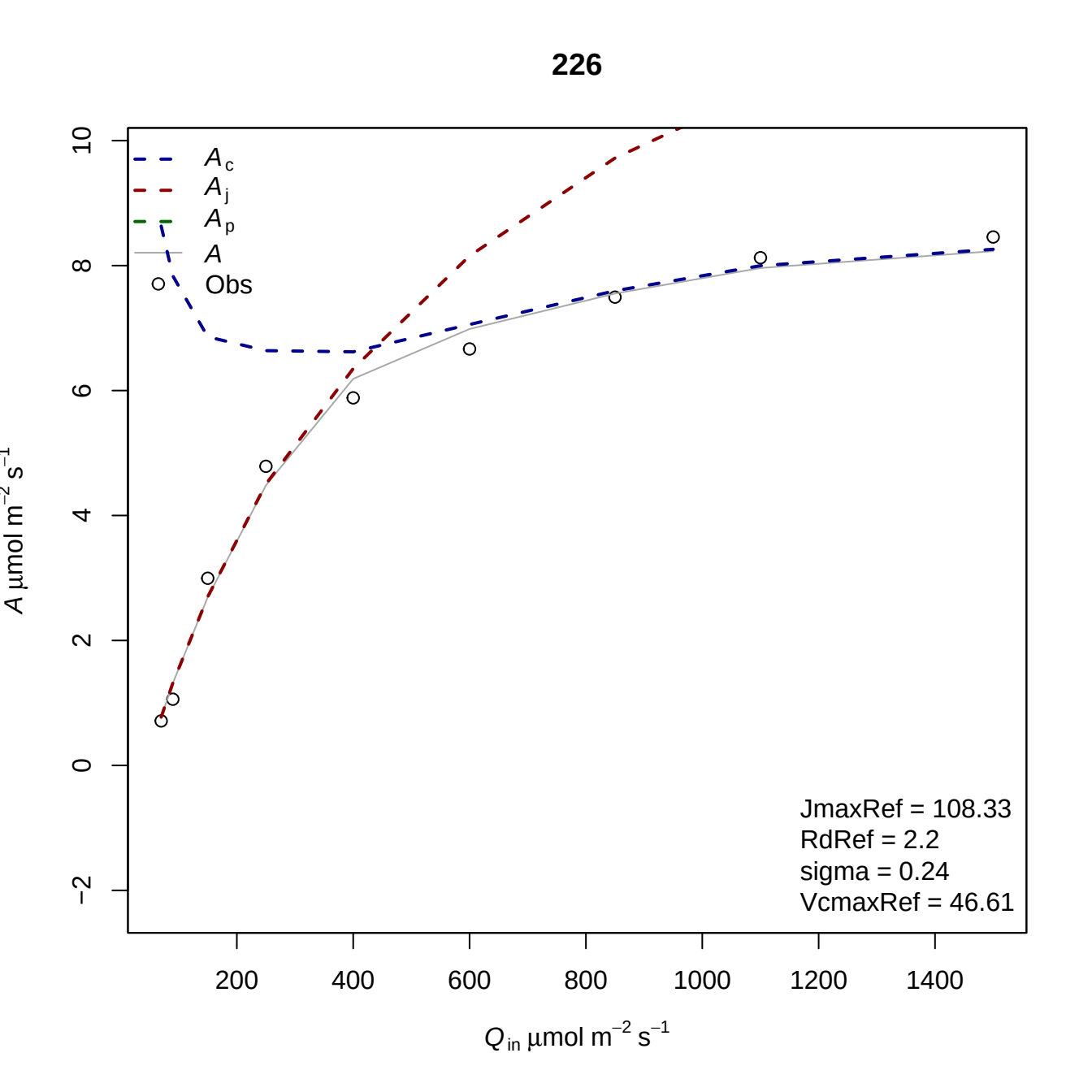

221
A_c
A_j
A_p
A
Obs
JmaxRef = 132.11
RdRef = 2.38
sigma = 0.17
VcmaxRef = 63.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

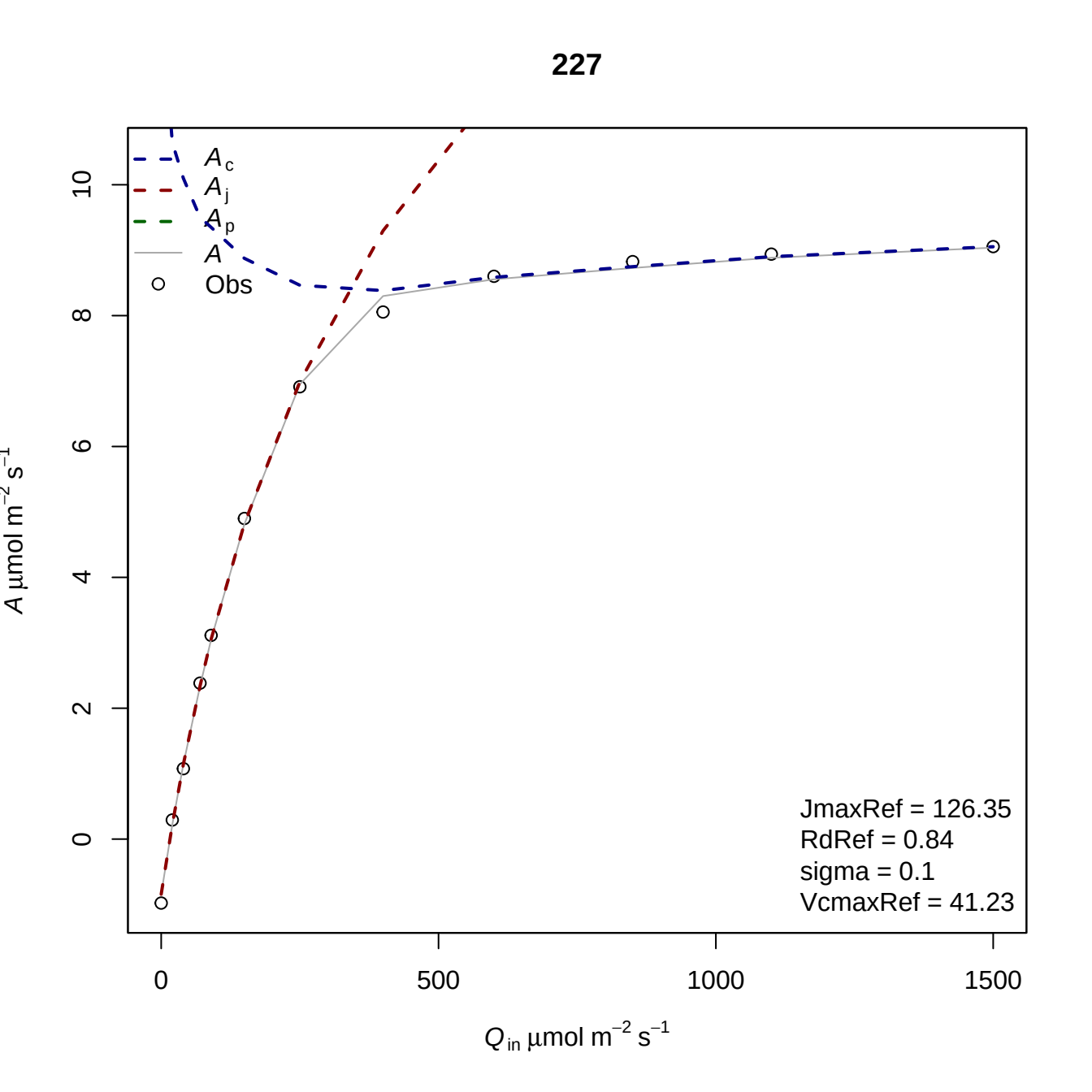

222
A_c
A_j
A_p
A
Obs
JmaxRef = 108.9
RdRef = 0.46
sigma = 0.23
VcmaxRef = 72.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

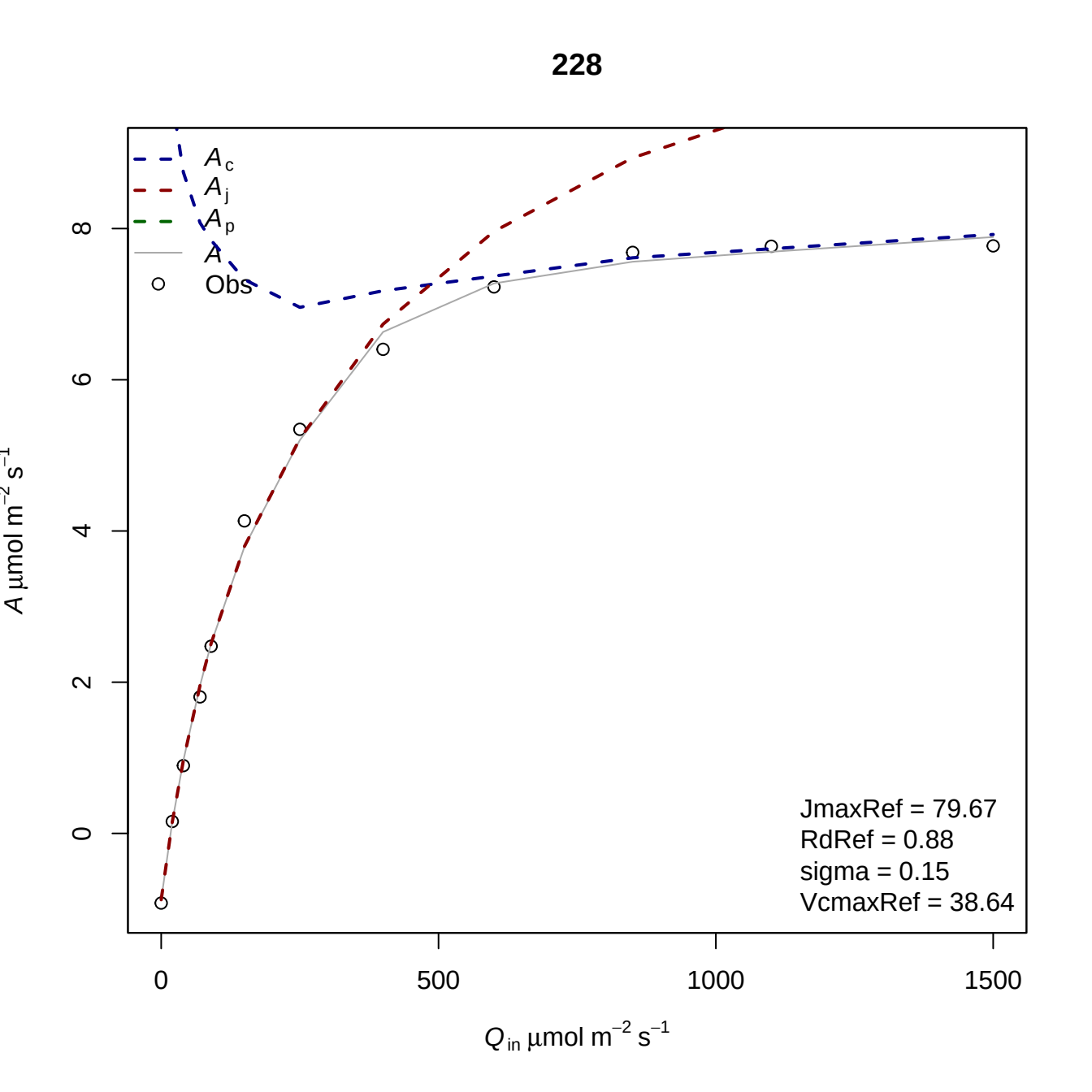

223
A_c
A_j
A_p
A
Obs
JmaxRef = 57.44
RdRef = 1.99
sigma = 0.08
VcmaxRef = 38.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
200
400
600
800
1000
1200
1400

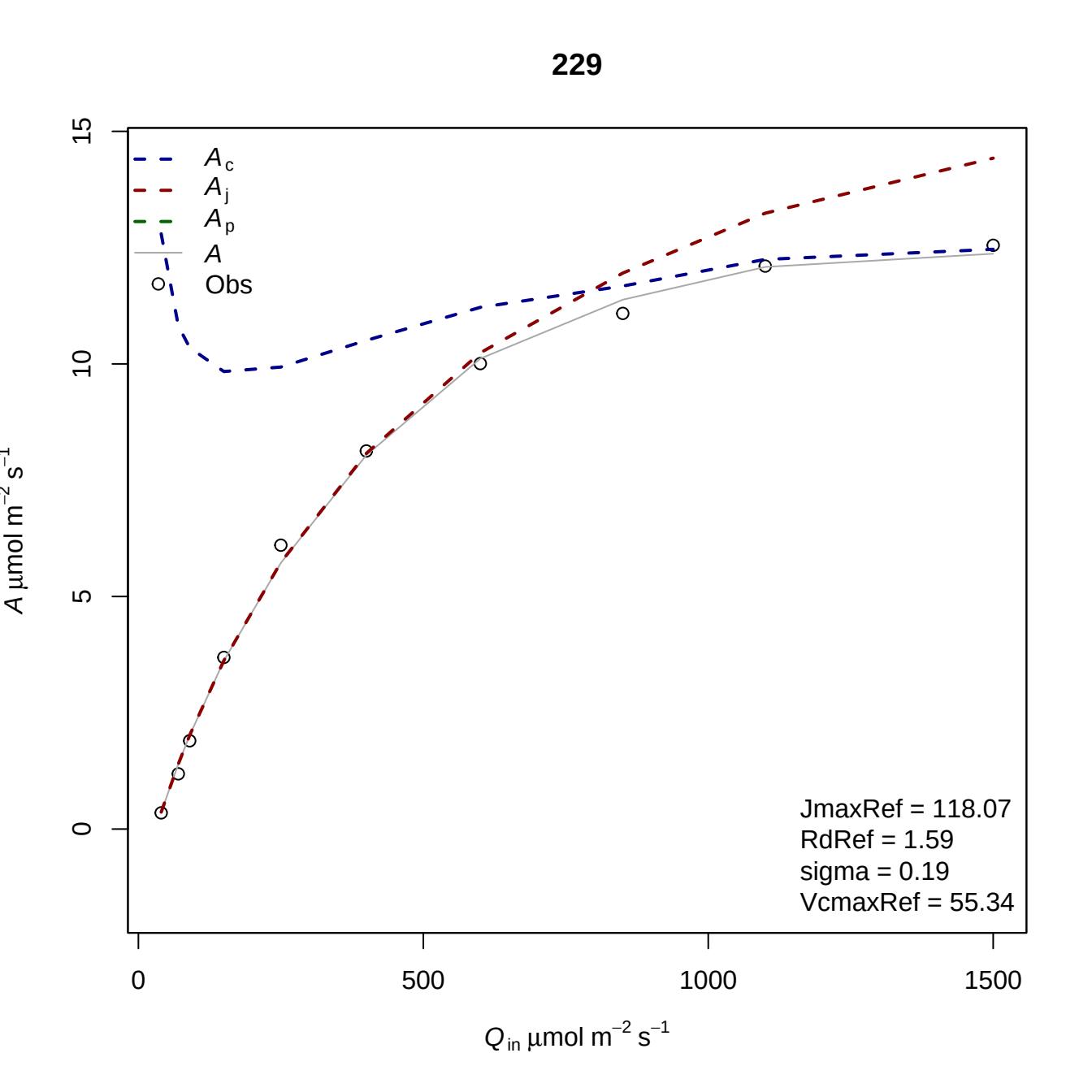

224
A_c
A_j
A_p
A
Obs
JmaxRef = 137.09
RdRef = 1.51
sigma = 0.19
VcmaxRef = 55.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

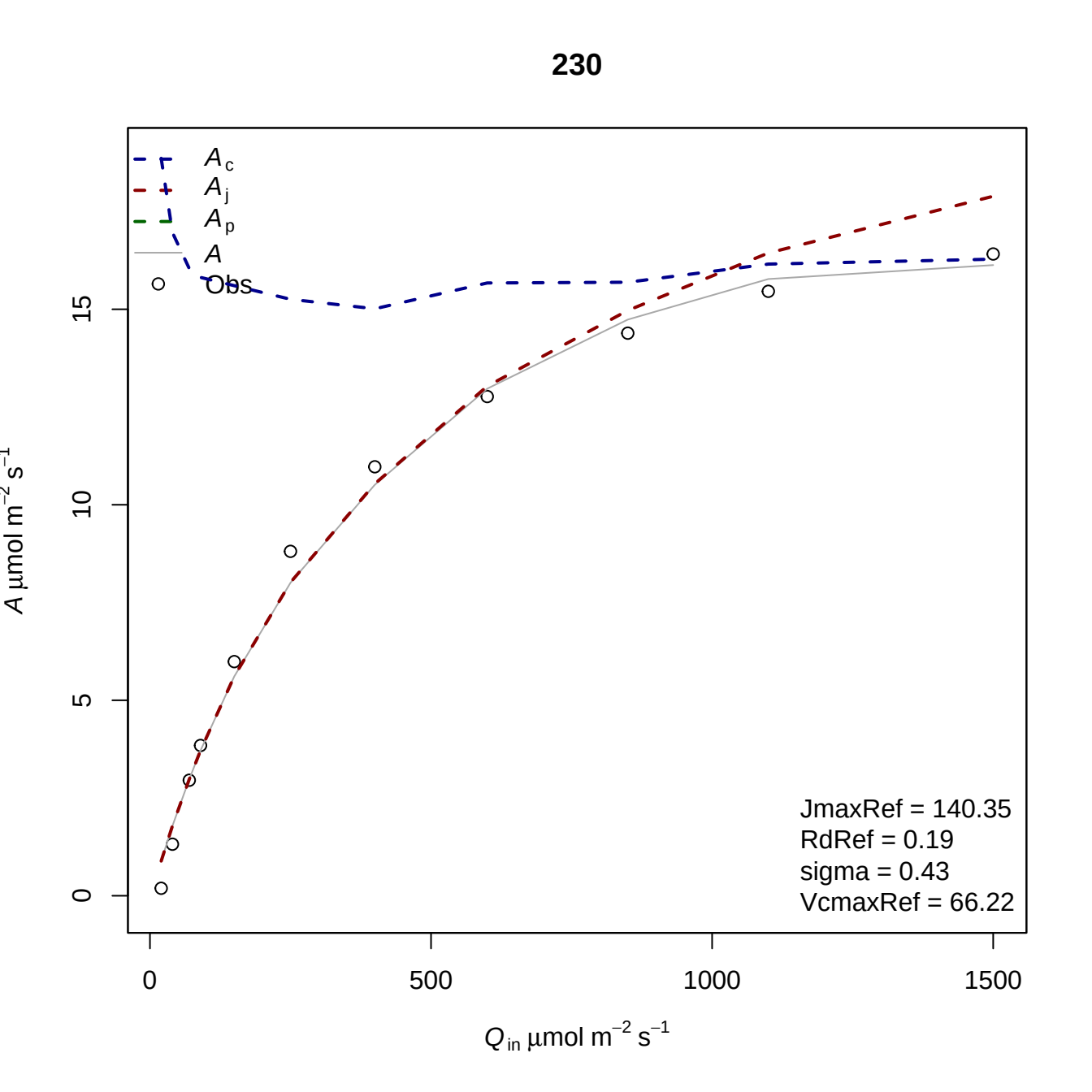

225
A_c
A_j
A_p
A
Obs
JmaxRef = 67.07
RdRef = 0.9
sigma = 0.19
VcmaxRef = 45.57
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

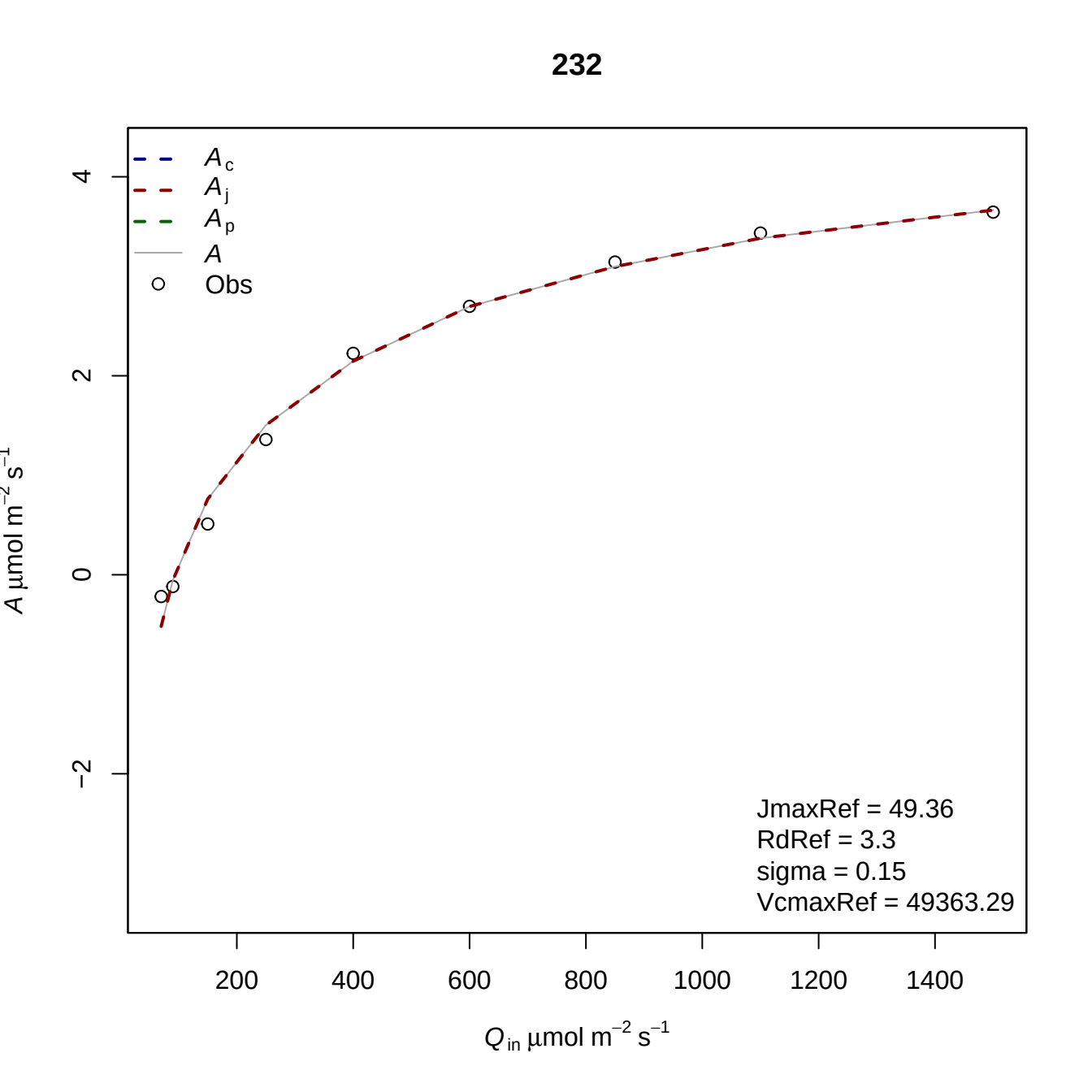

226
A_c
A_j
A_p
A
Obs
JmaxRef = 108.33
RdRef = 2.2
sigma = 0.24
VcmaxRef = 46.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
200
400
600
800
1000
1200
1400

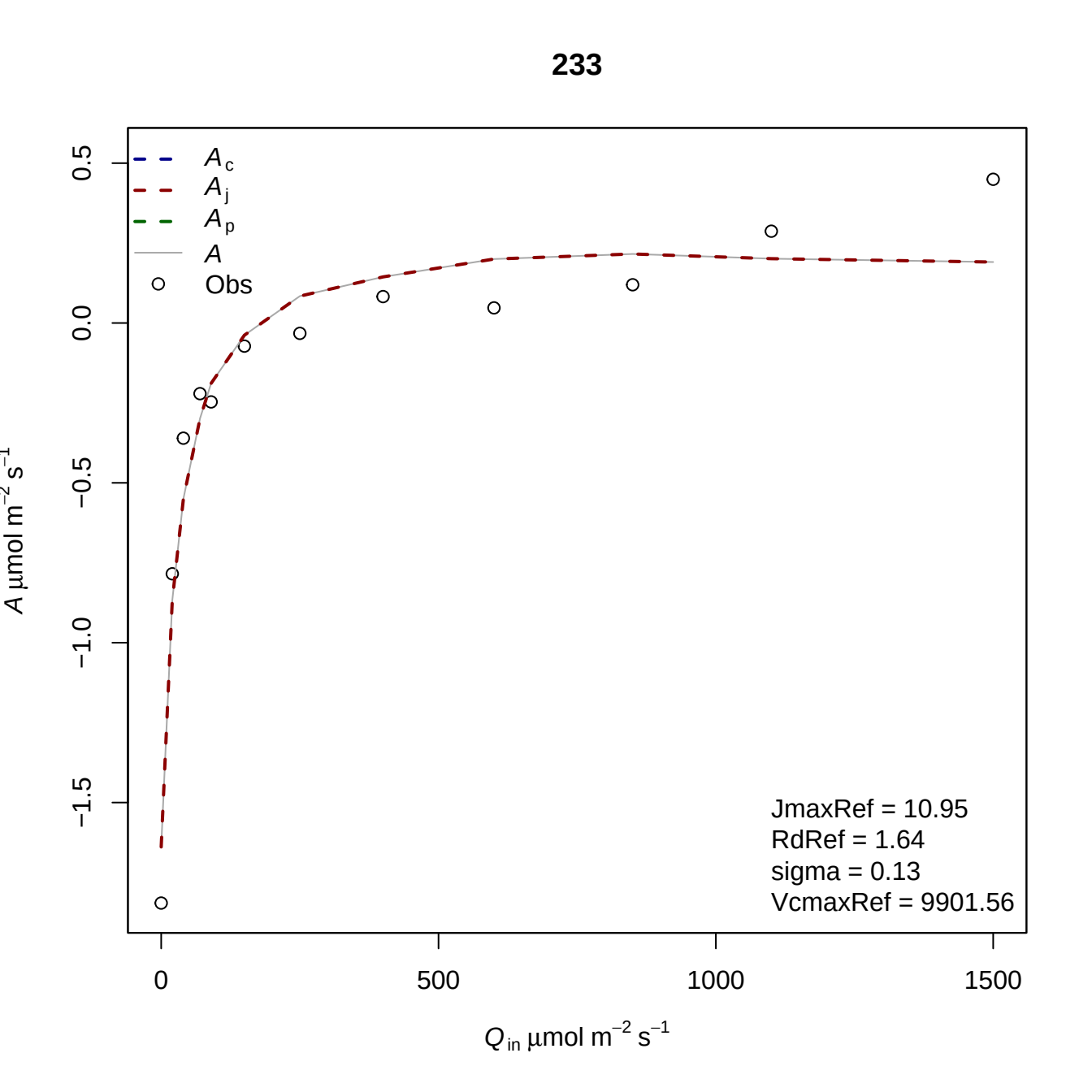

227
A_c
A_j
A_p
A
Obs
JmaxRef = 126.35
RdRef = 0.84
sigma = 0.1
VcmaxRef = 41.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

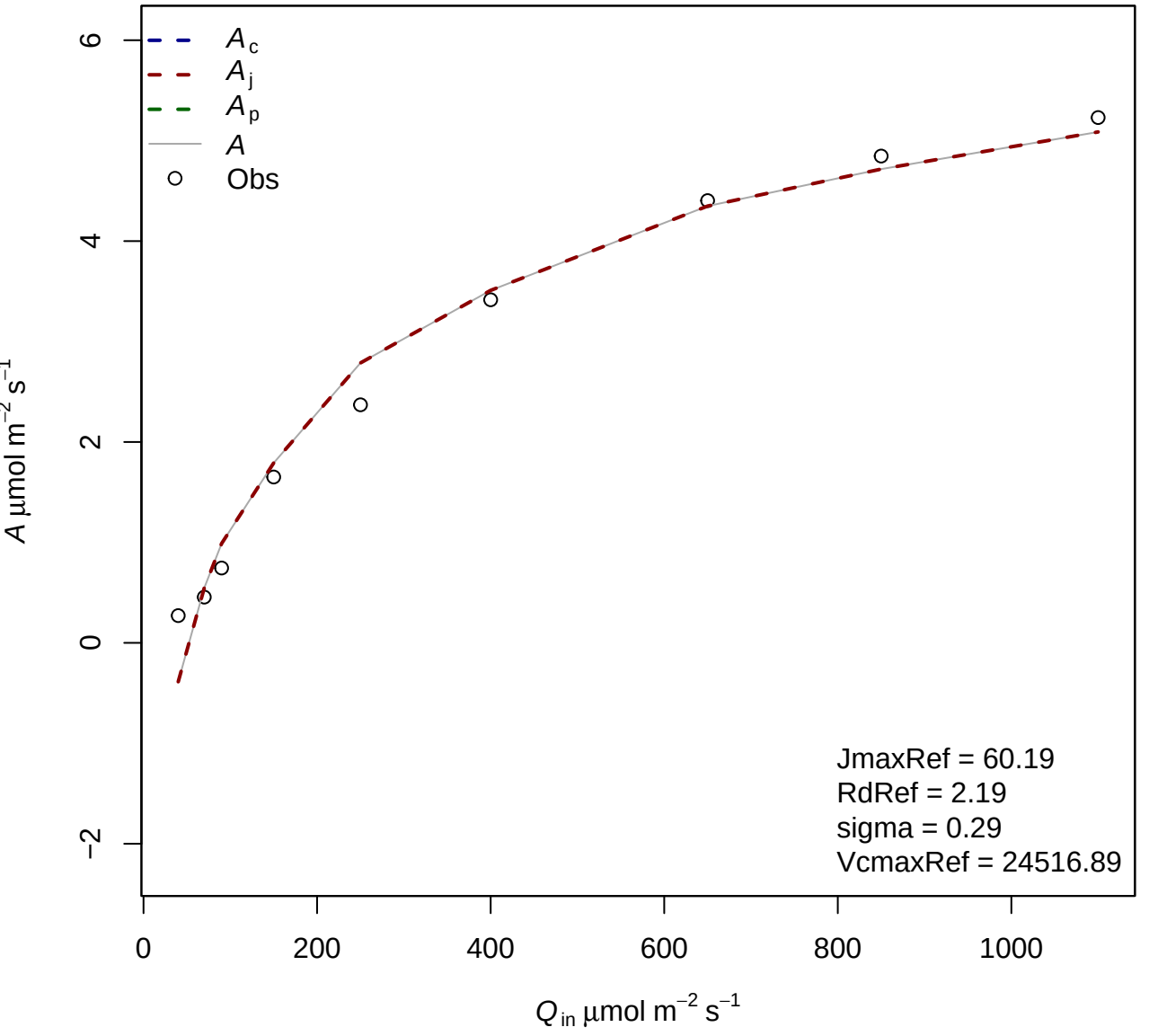

228
A_c
A_j
A_p
A
Obs
JmaxRef = 79.67
RdRef = 0.88
sigma = 0.15
VcmaxRef = 38.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

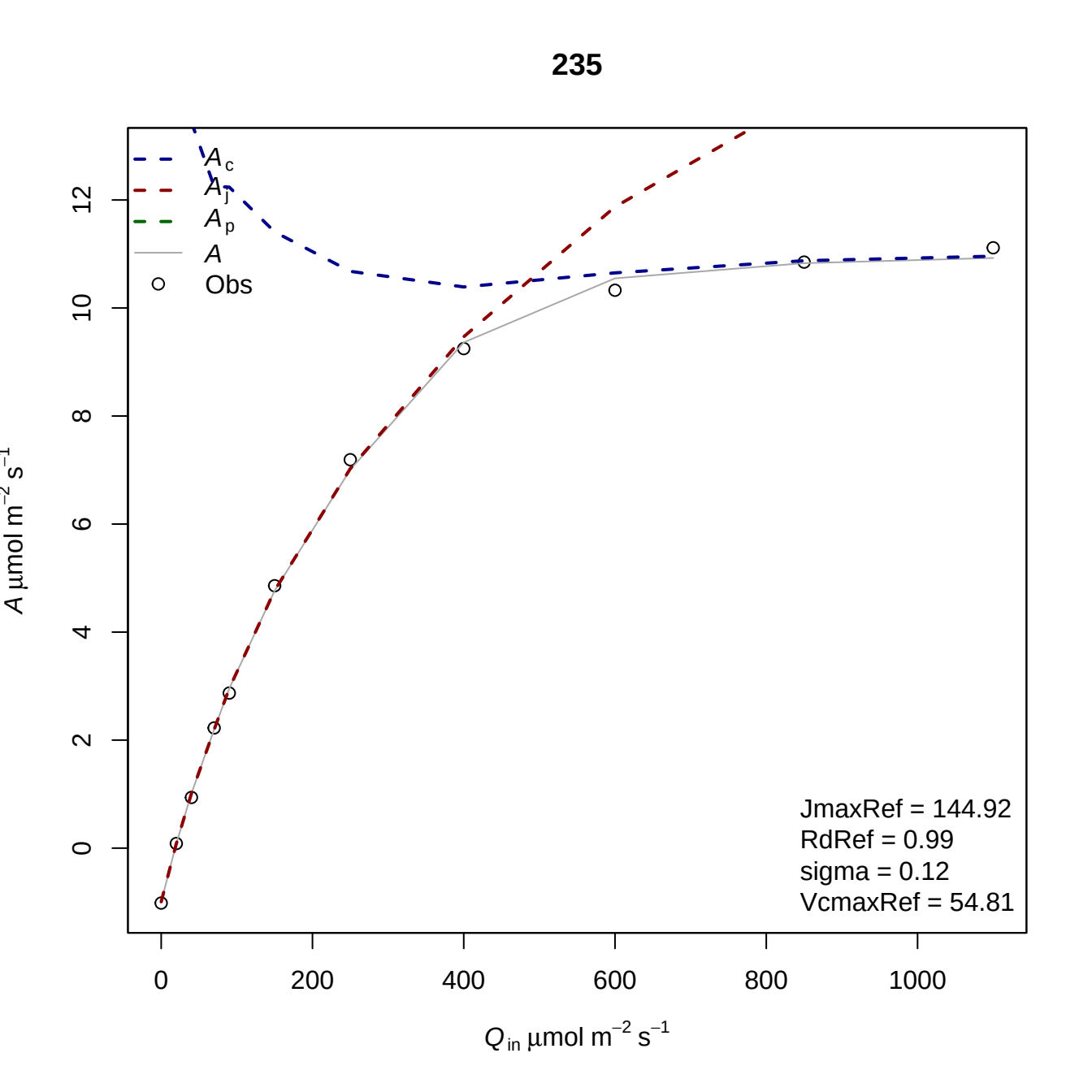

229
A_c
A_j
A_p
A
Obs
JmaxRef = 118.07
RdRef = 1.59
sigma = 0.19
VcmaxRef = 55.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

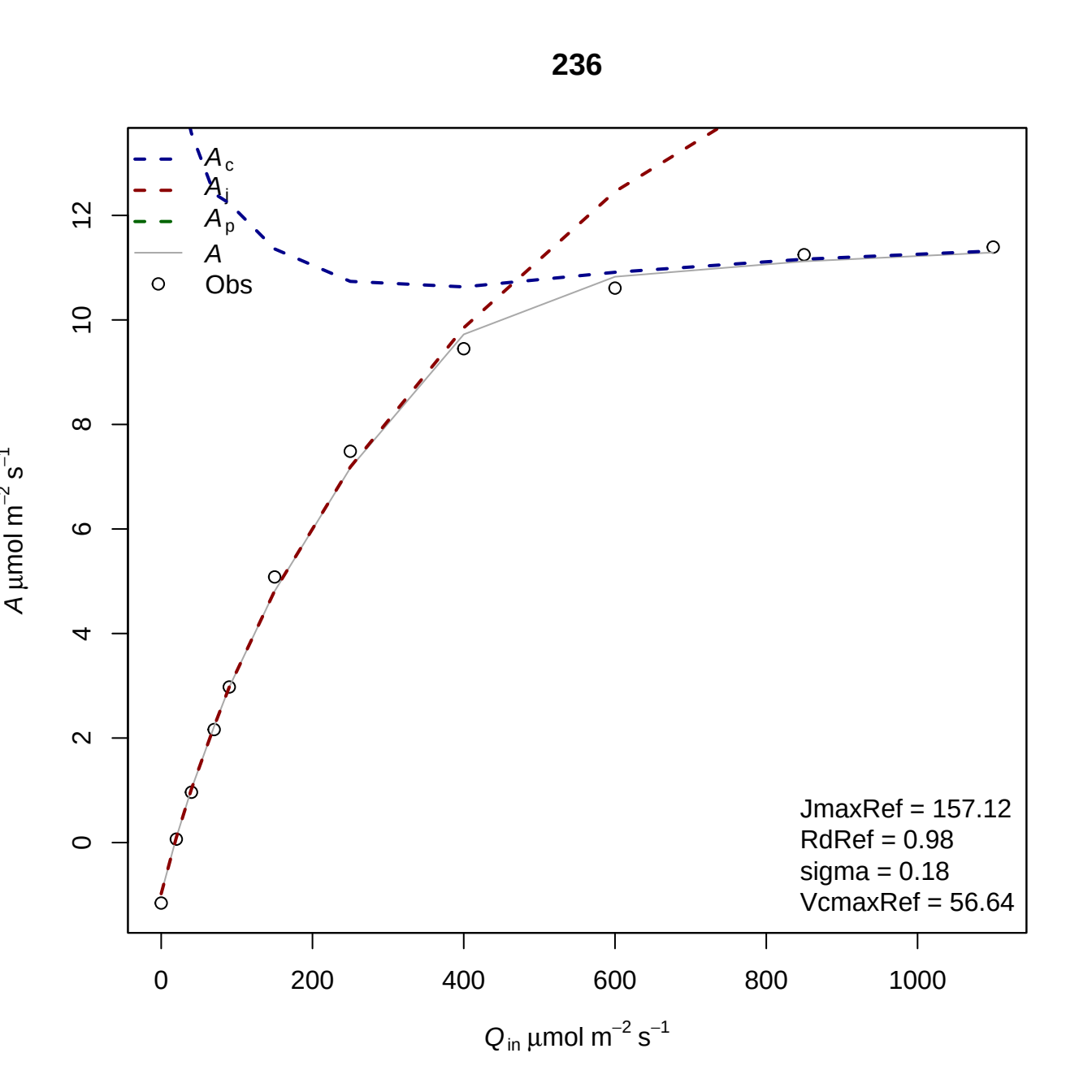

230
A_c
A_j
A_p
A
Obs
JmaxRef = 140.35
RdRef = 0.19
sigma = 0.43
VcmaxRef = 66.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

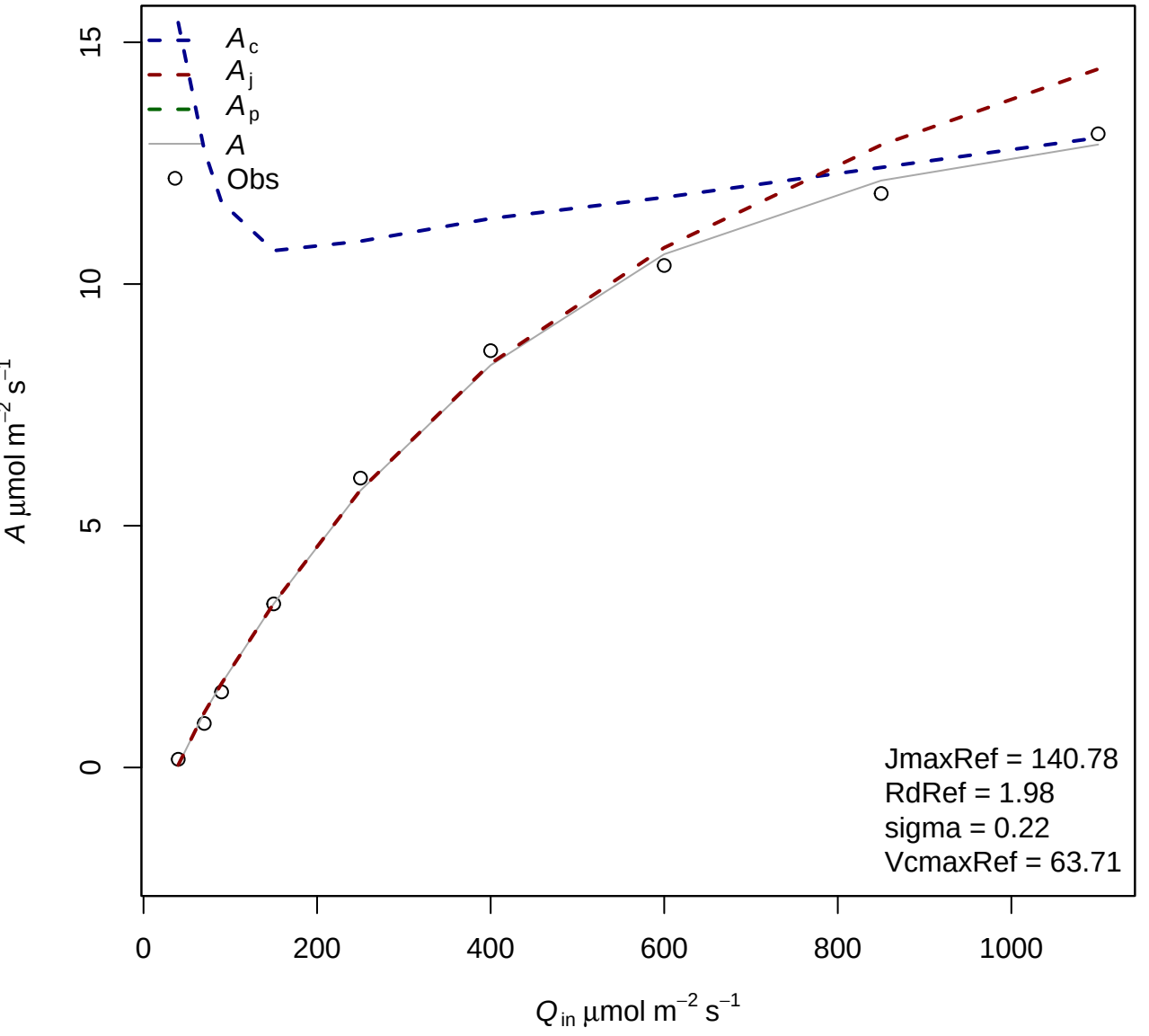

232
A_c
A_j
A_p
A
Obs
JmaxRef = 49.36
RdRef = 3.3
sigma = 0.15
VcmaxRef = 49363.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
200
400
600
800
1000
1200
1400

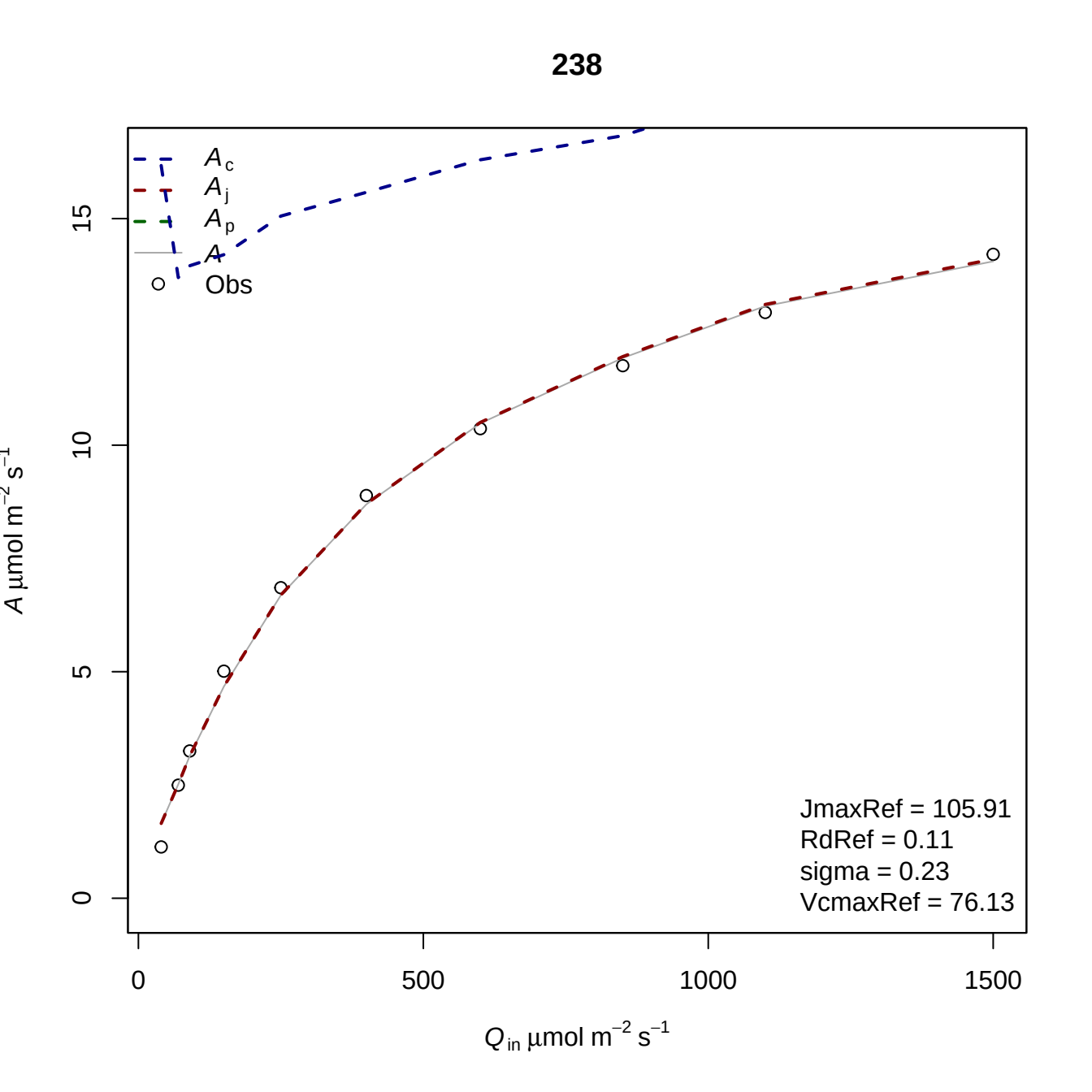

233
A_c
A_j
A_p
A
Obs
JmaxRef = 10.95
RdRef = 1.64
sigma = 0.13
VcmaxRef = 9901.56
0.5
0.0
−0.5
−1.0
−1.5
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

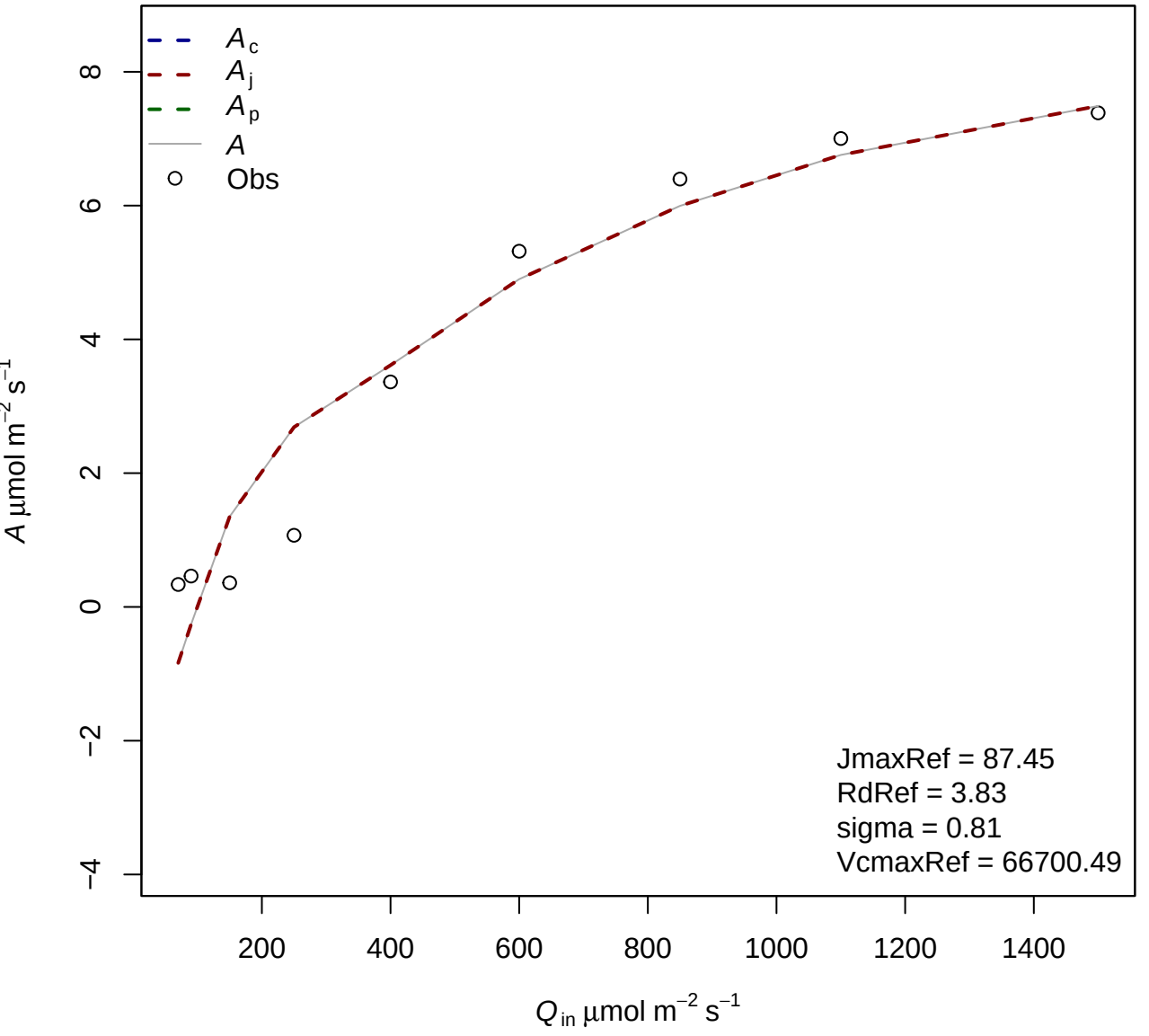

A_c
A_j
A_p
A
Obs
JmaxRef = 60.19
RdRef = 2.19
sigma = 0.29
VcmaxRef = 24516.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000

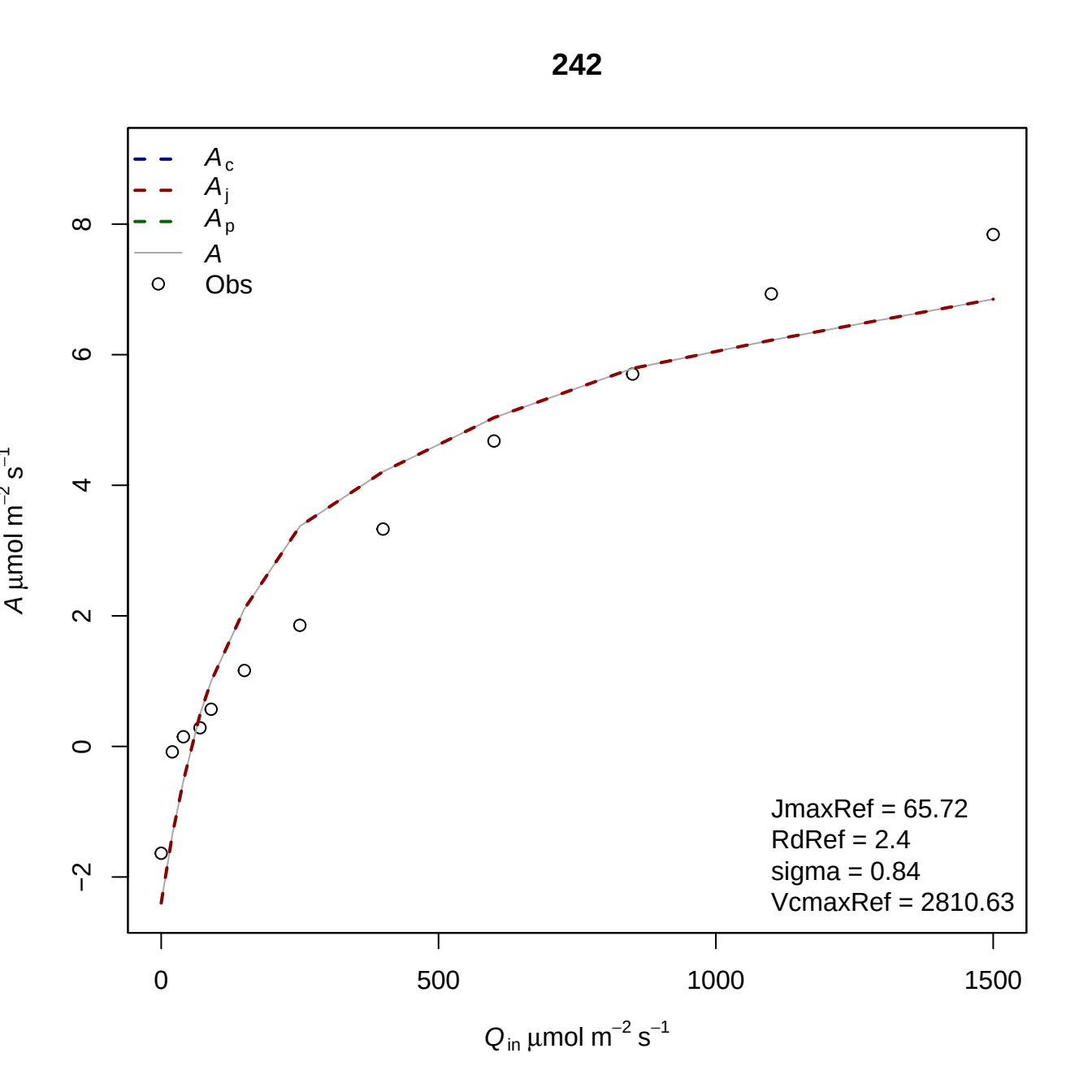

235
A_c
A_J
A_p
A
Obs
JmaxRef = 144.92
RdRef = 0.99
sigma = 0.12
VcmaxRef = 54.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000

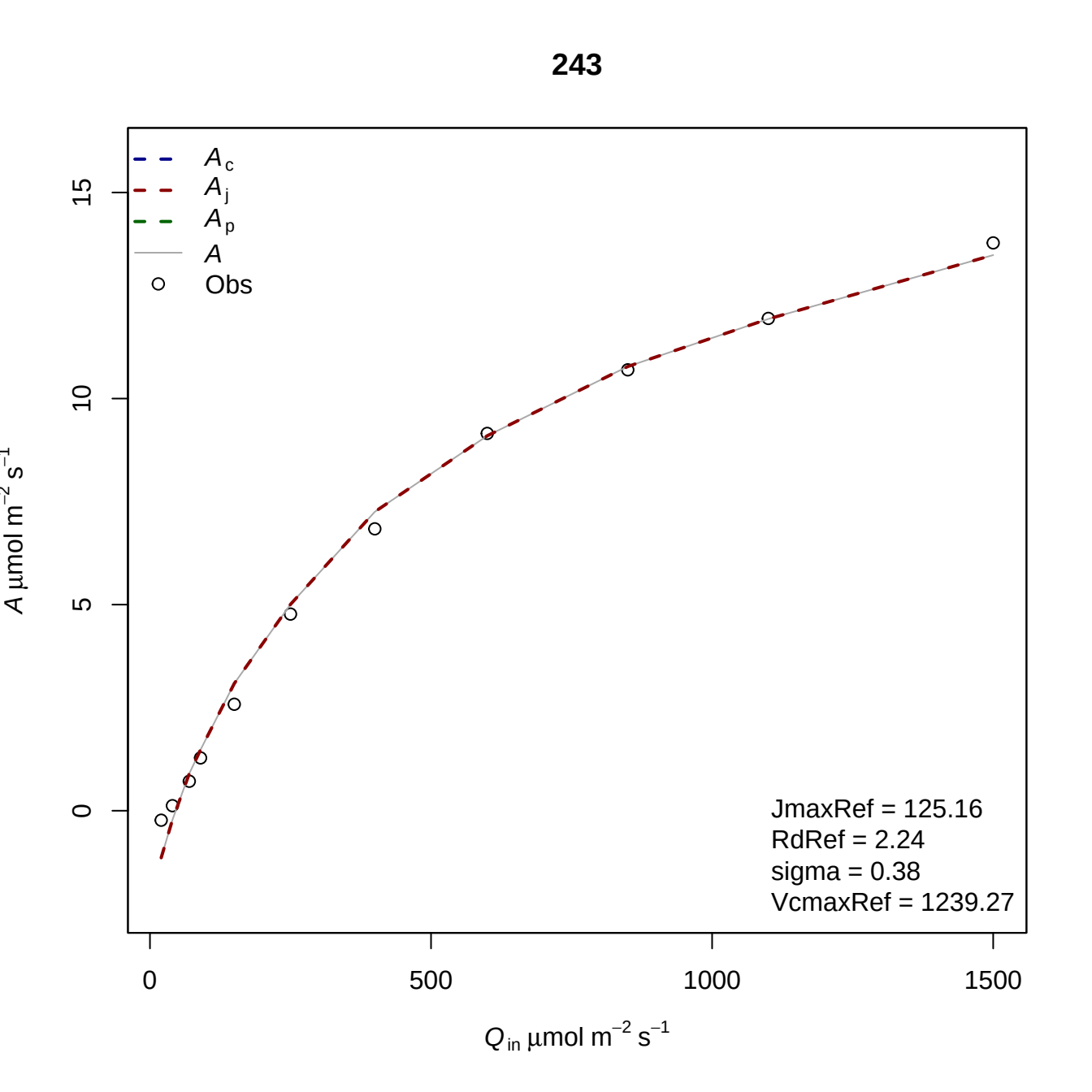

236
A_c
A_j
A_p
A
Obs
JmaxRef = 157.12
RdRef = 0.98
sigma = 0.18
VcmaxRef = 56.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000

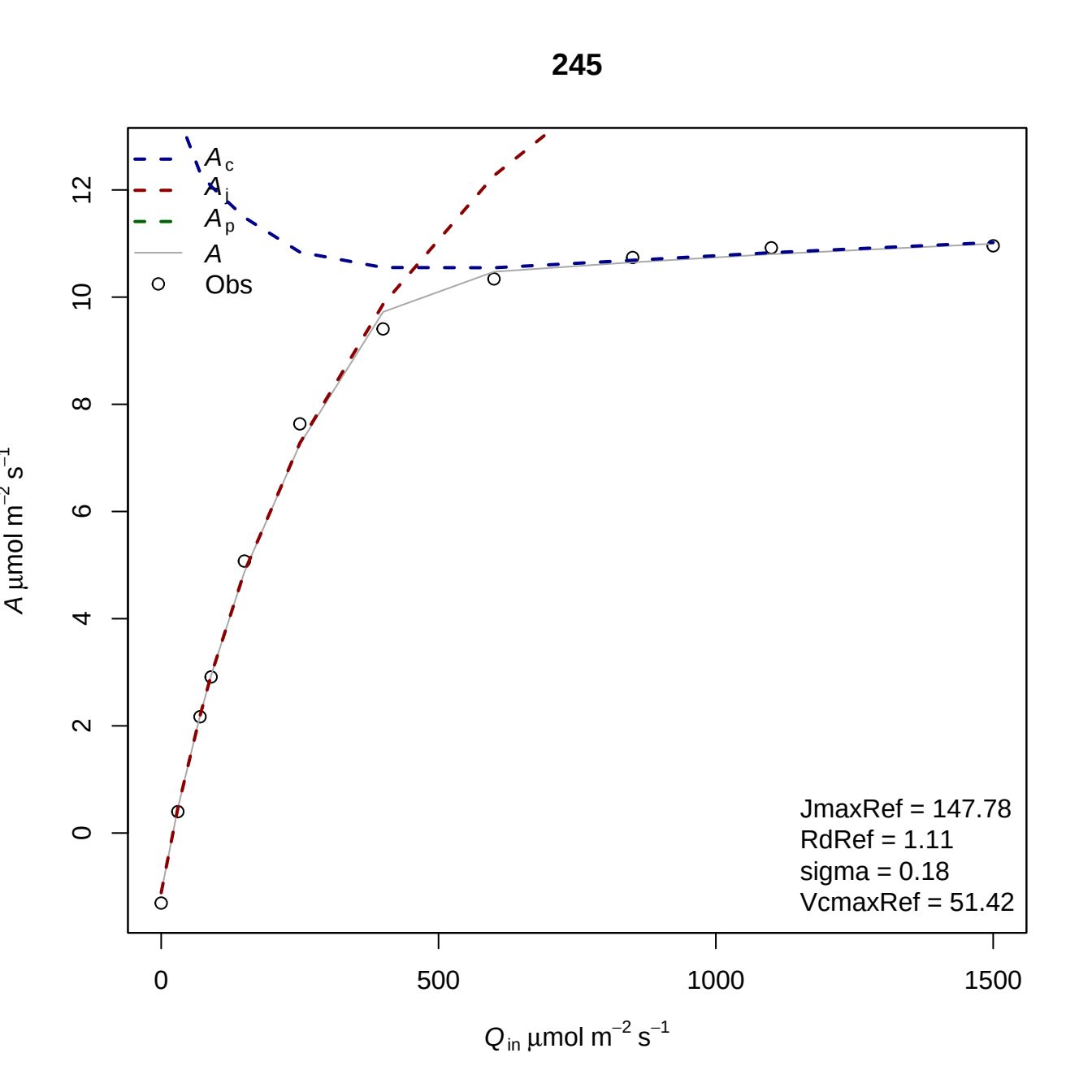

A_c
A_j
A_p
A
Obs
JmaxRef = 140.78
RdRef = 1.98
sigma = 0.22
VcmaxRef = 63.71
0
5
10
15
0
200
400
600
800
1000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

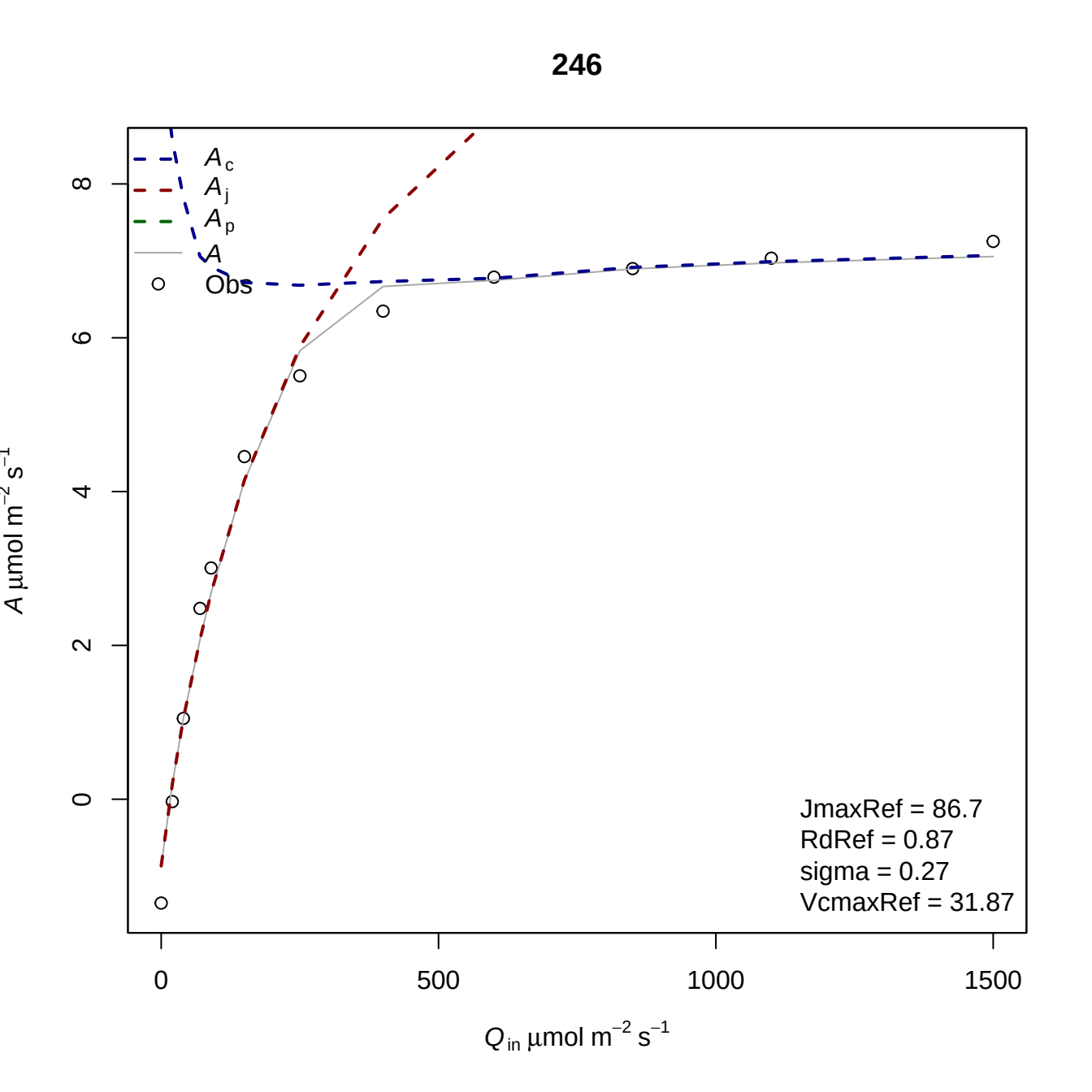

238
A_c
A_j
A_p
A
Obs
JmaxRef = 105.91
RdRef = 0.11
sigma = 0.23
VcmaxRef = 76.13
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

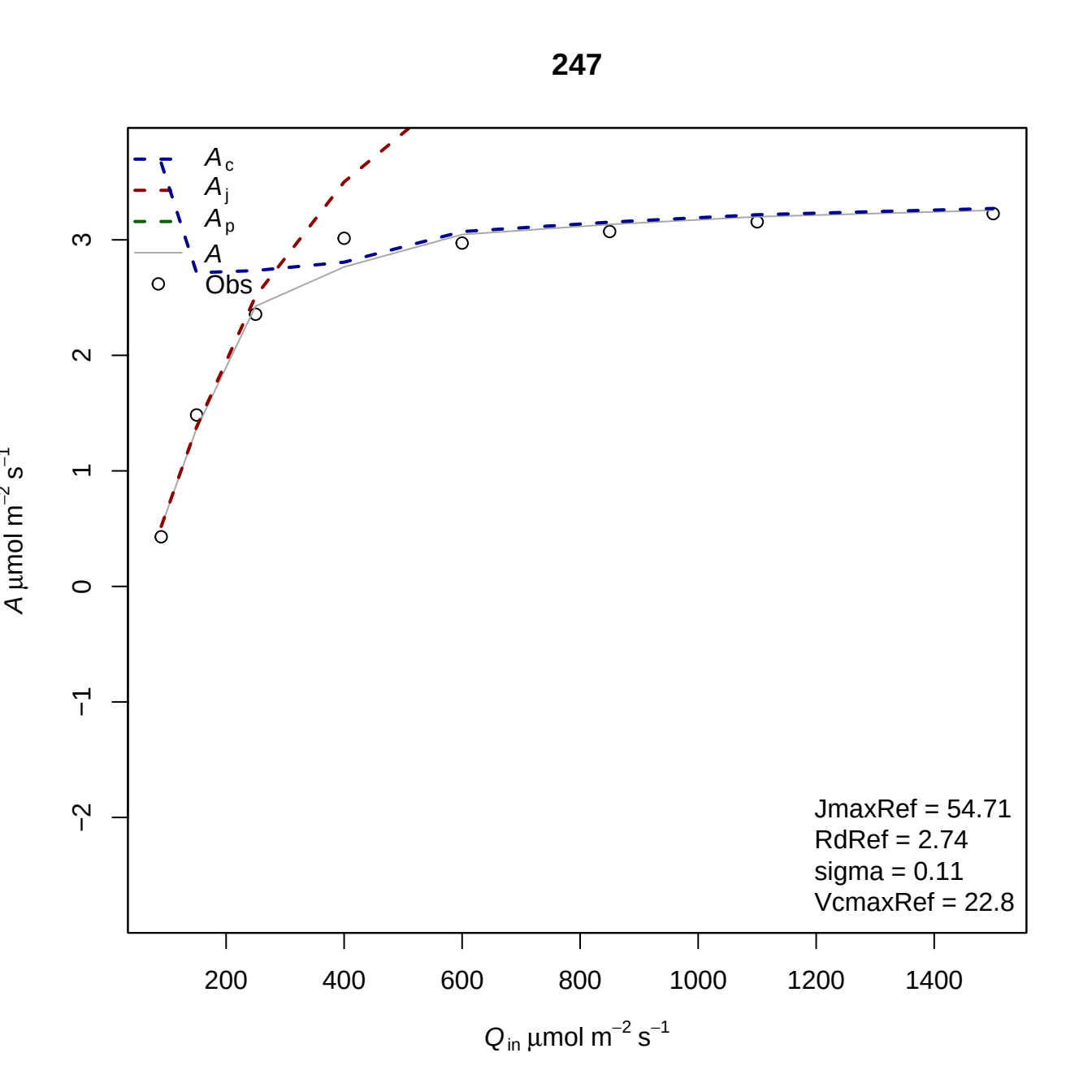

A_c
A_j
A_p
A
Obs
JmaxRef = 87.45
RdRef = 3.83
sigma = 0.81
VcmaxRef = 66700.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
200
400
600
800
1000
1200
1400

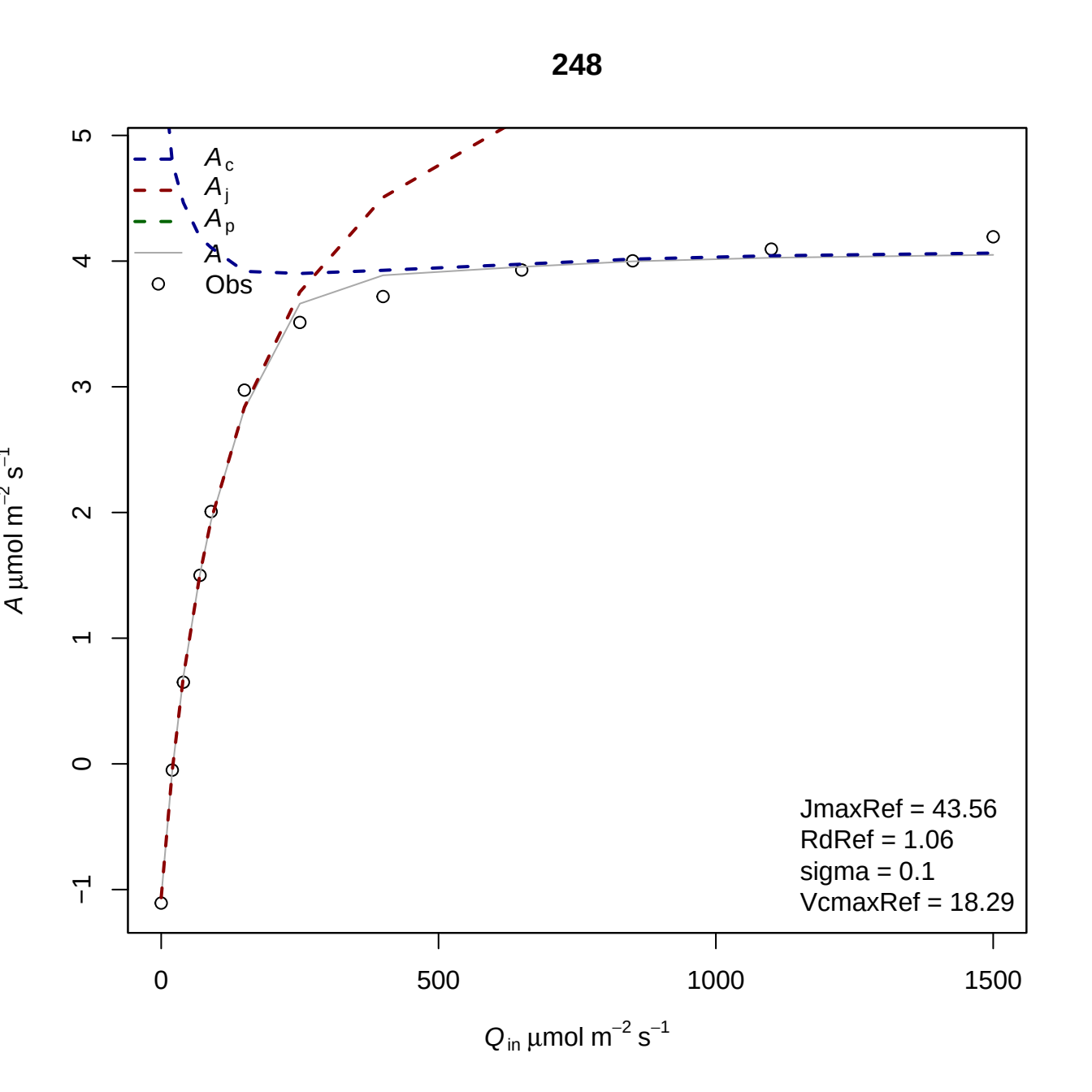

242
A_c
A_j
A_p
A
Obs
JmaxRef = 65.72
RdRef = 2.4
sigma = 0.84
VcmaxRef = 2810.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

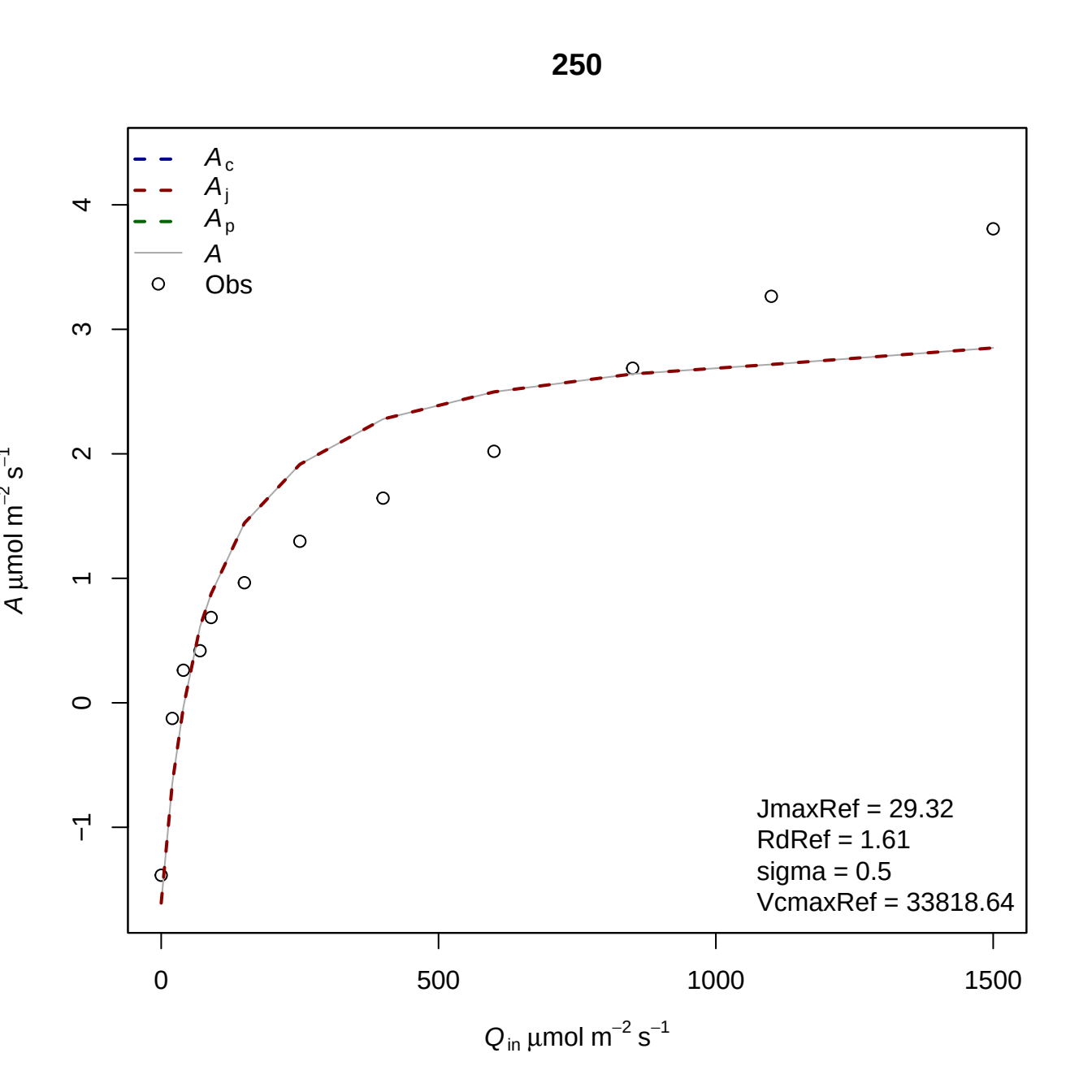

243
A_c
A_j
A_p
A
Obs
JmaxRef = 125.16
RdRef = 2.24
sigma = 0.38
VcmaxRef = 1239.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

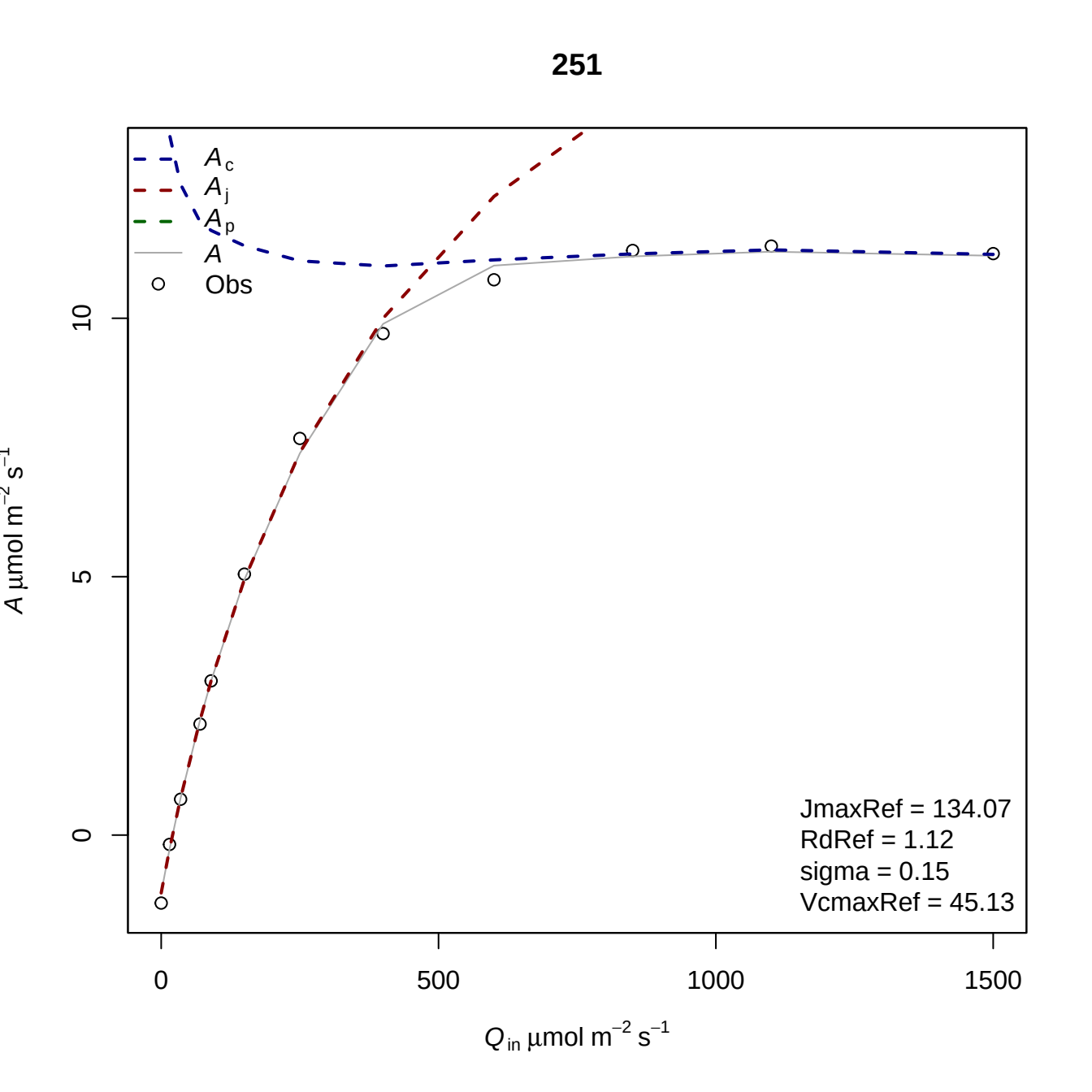

245
A_c
A_j
A_p
A
Obs
JmaxRef = 147.78
RdRef = 1.11
sigma = 0.18
VcmaxRef = 51.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

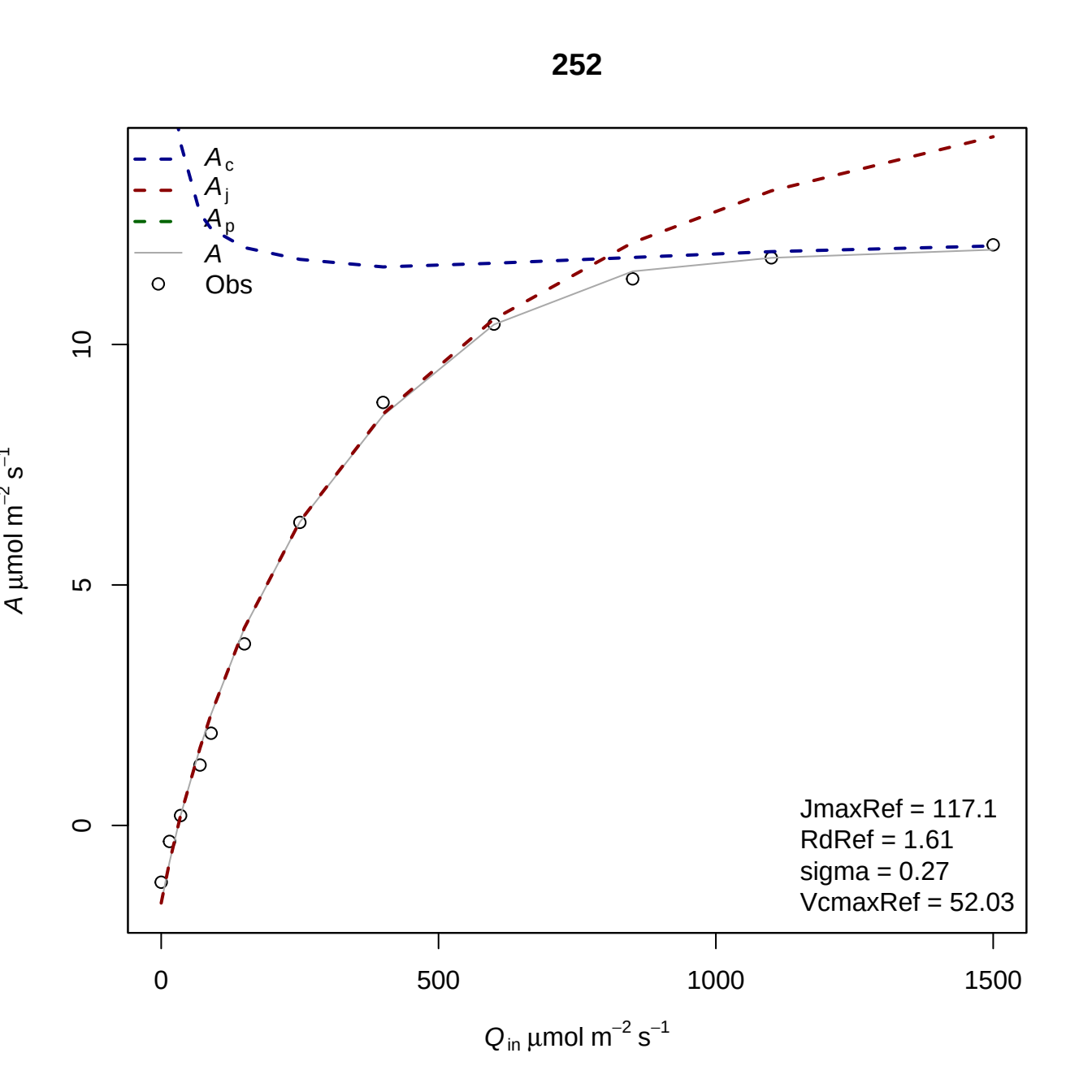

246
A_c
A_j
A_p
A
Obs
JmaxRef = 86.7
RdRef = 0.87
sigma = 0.27
VcmaxRef = 31.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

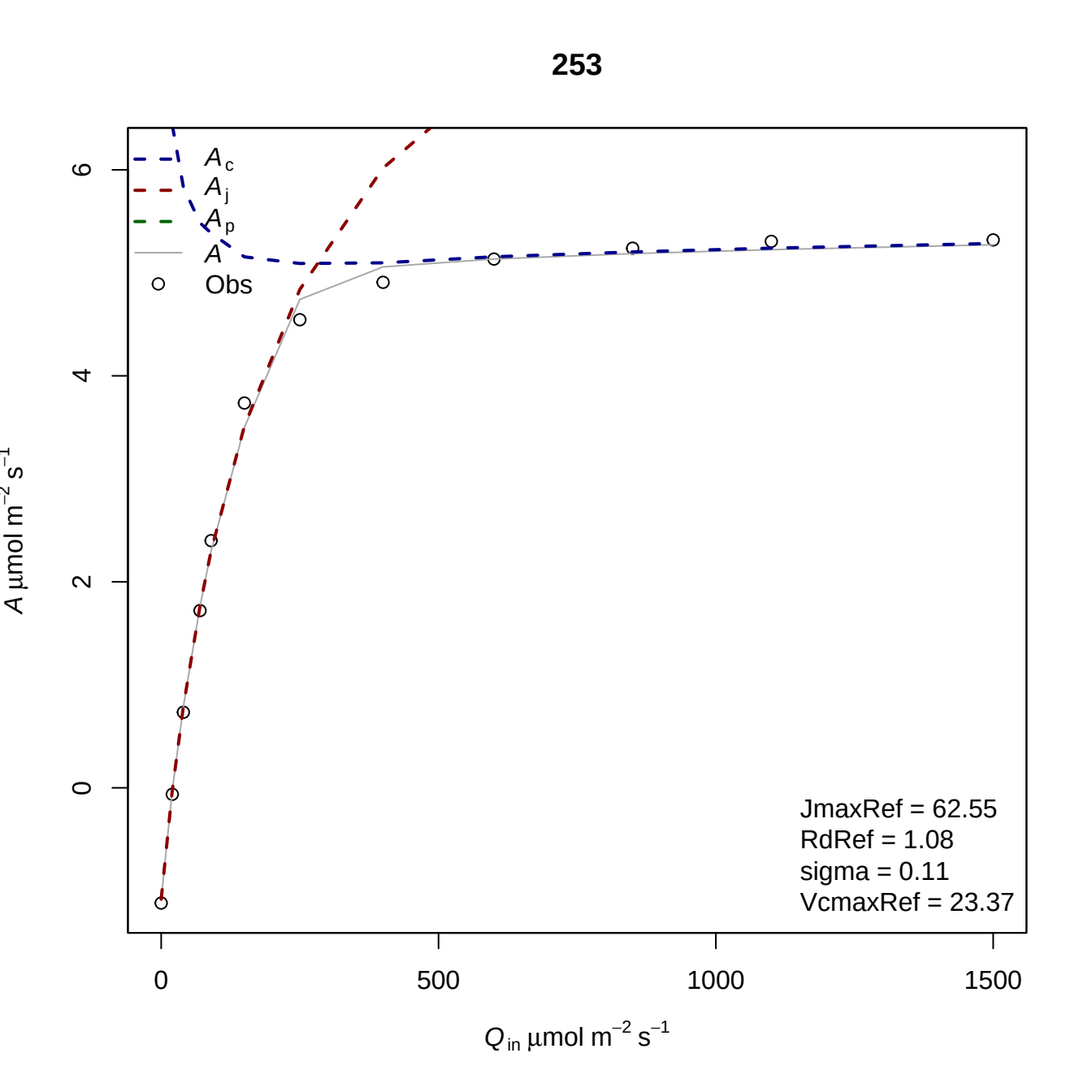

247
A_c
A_j
A_p
A
Obs
JmaxRef = 54.71
RdRef = 2.74
sigma = 0.11
VcmaxRef = 22.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
200
400
600
800
1000
1200
1400

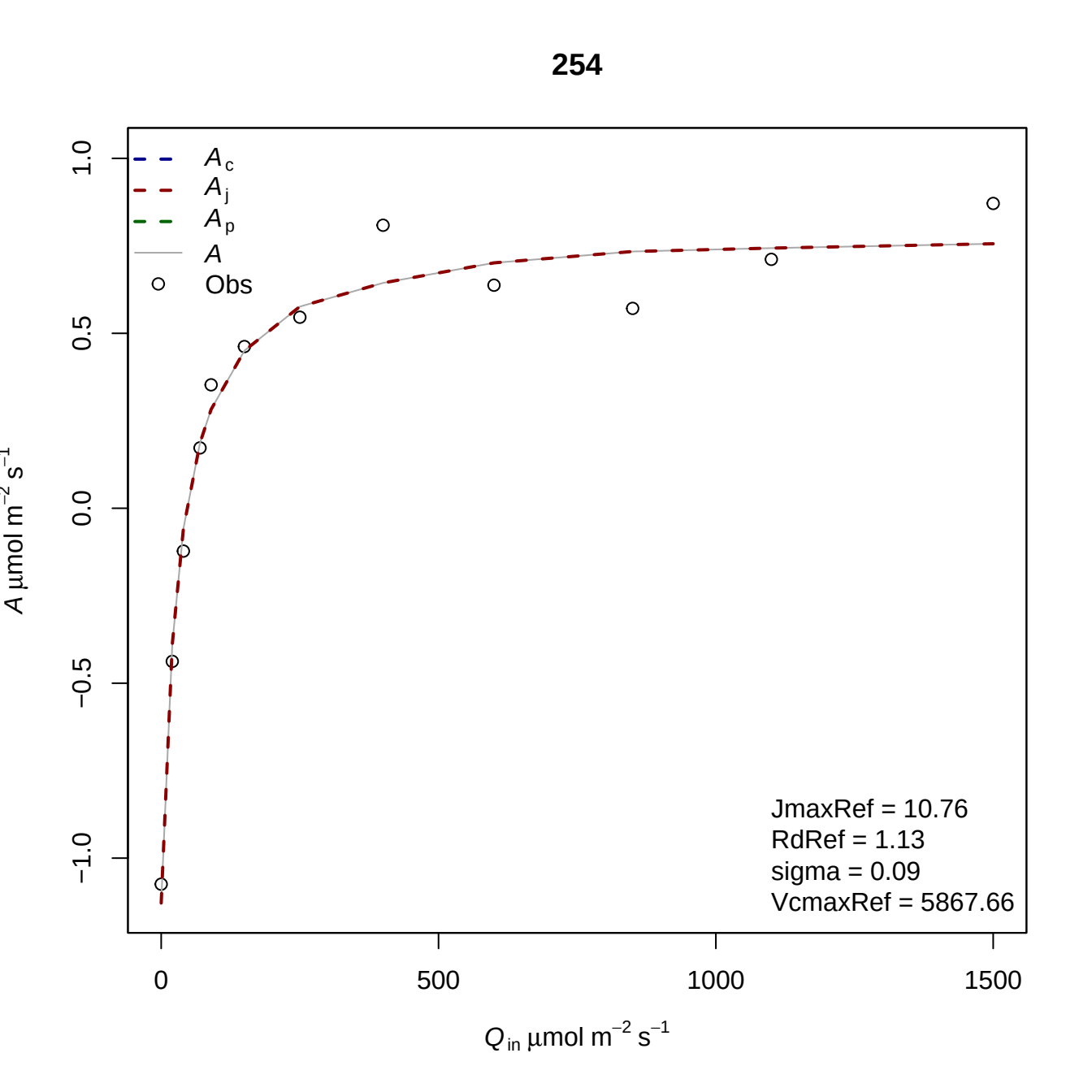

248
A_c
A_j
A_p
A
Obs
JmaxRef = 43.56
RdRef = 1.06
sigma = 0.1
VcmaxRef = 18.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

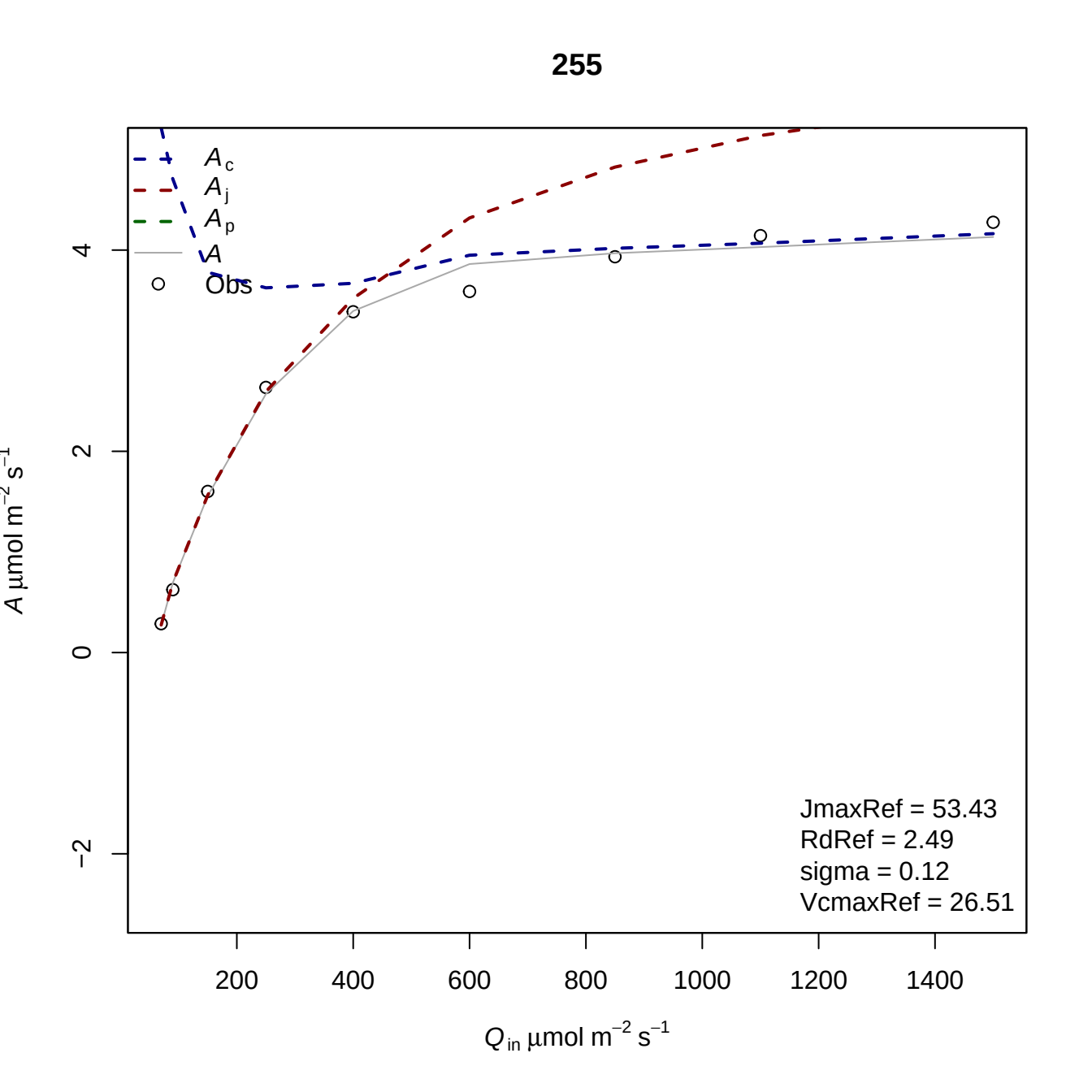

250
A_c
A_j
A_p
A
Obs
JmaxRef = 29.32
RdRef = 1.61
sigma = 0.5
VcmaxRef = 33818.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

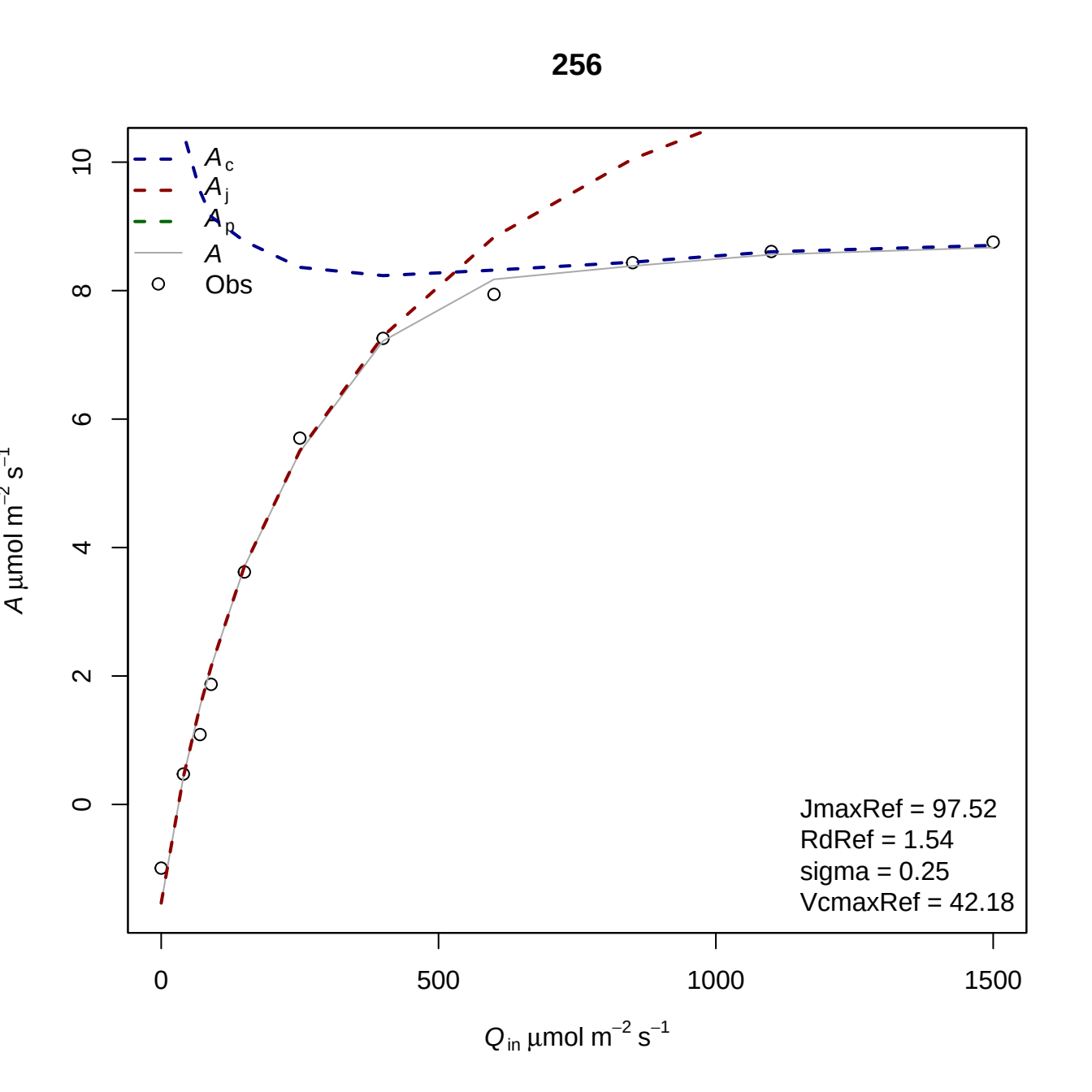

251
A_c
A_j
A_p
A
Obs
JmaxRef = 134.07
RdRef = 1.12
sigma = 0.15
VcmaxRef = 45.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

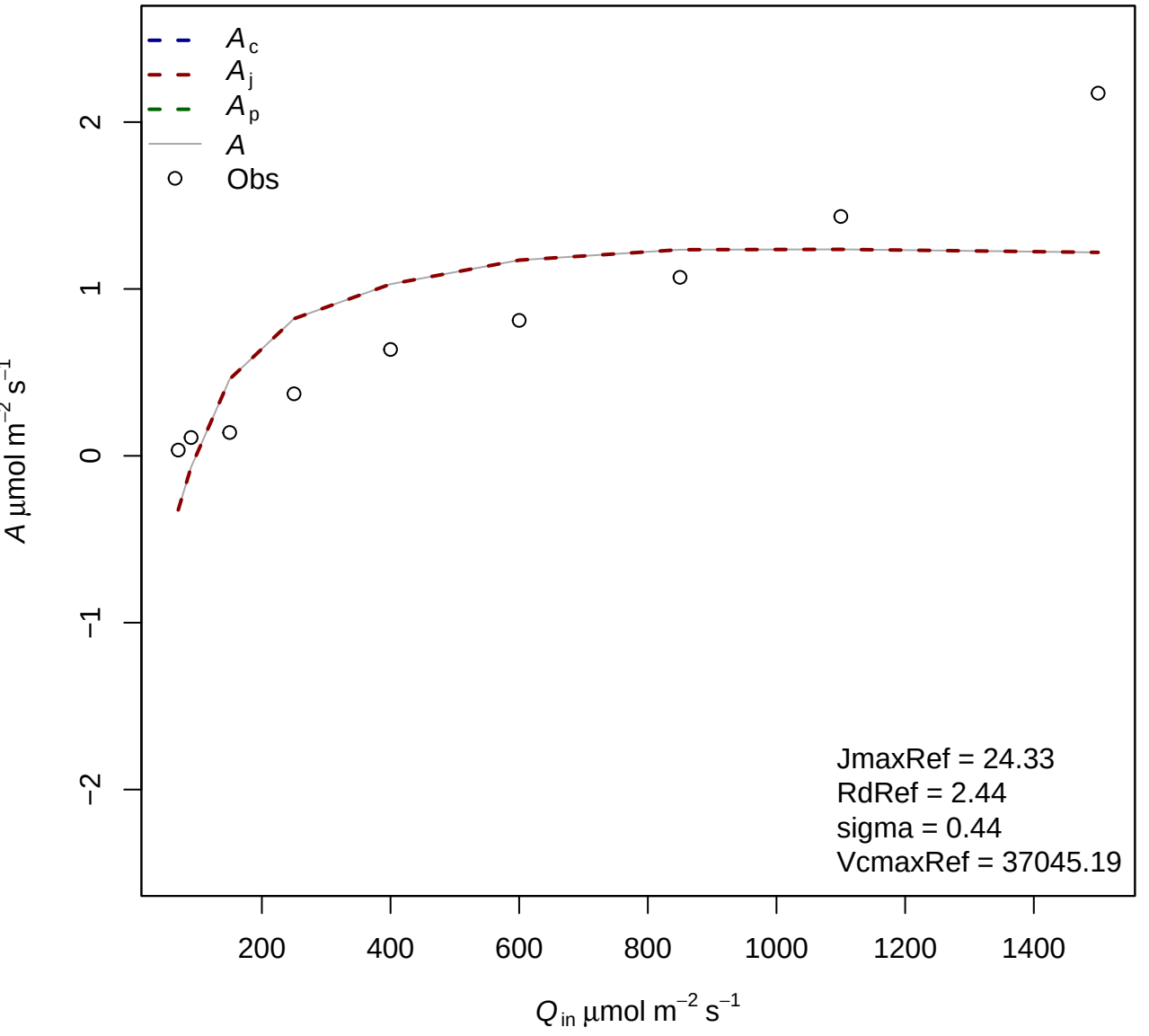

252
A_c
A_j
A_p
A
Obs
JmaxRef = 117.1
RdRef = 1.61
sigma = 0.27
VcmaxRef = 52.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

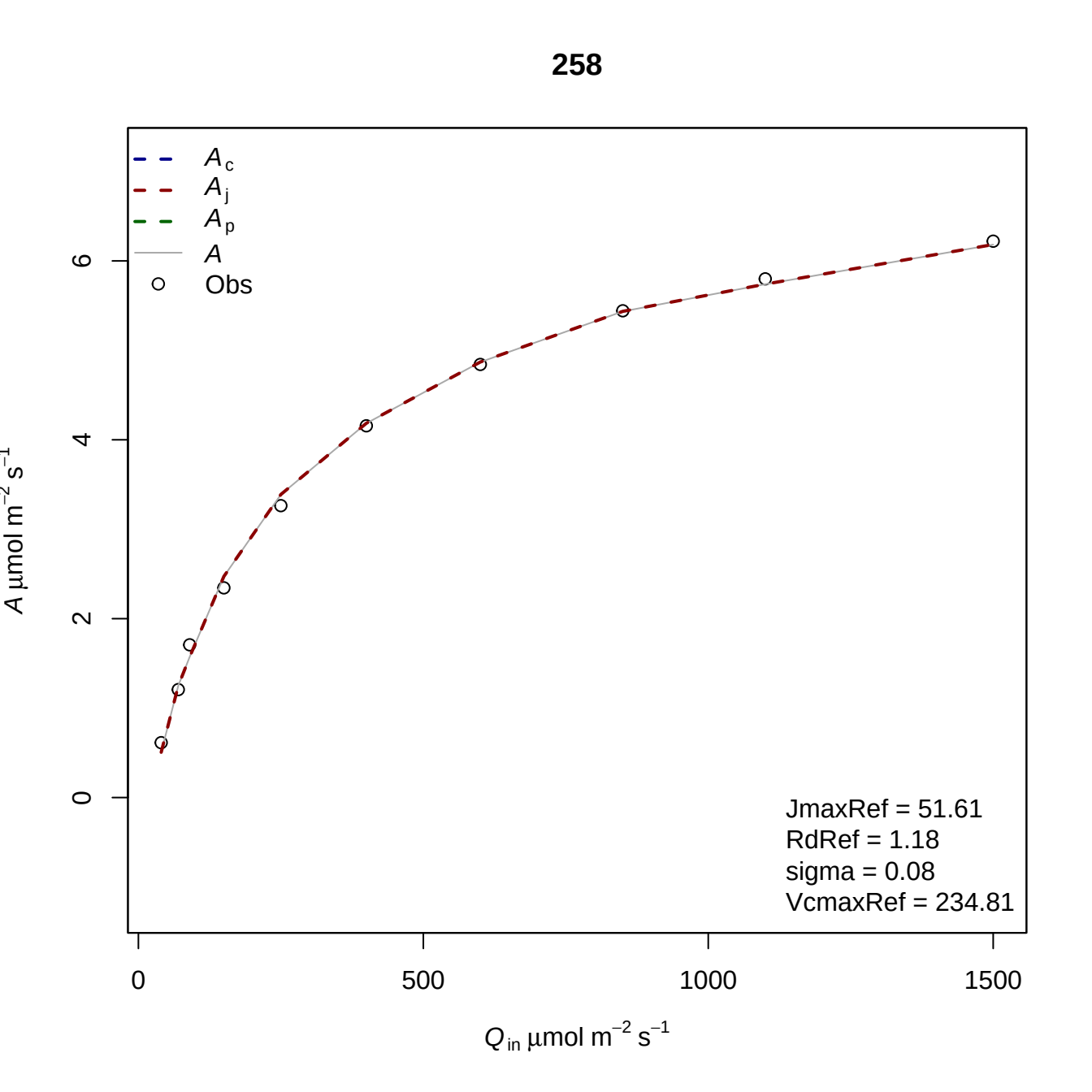

253
A_c
A_j
A_p
A
Obs
JmaxRef = 62.55
RdRef = 1.08
sigma = 0.11
VcmaxRef = 23.37
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

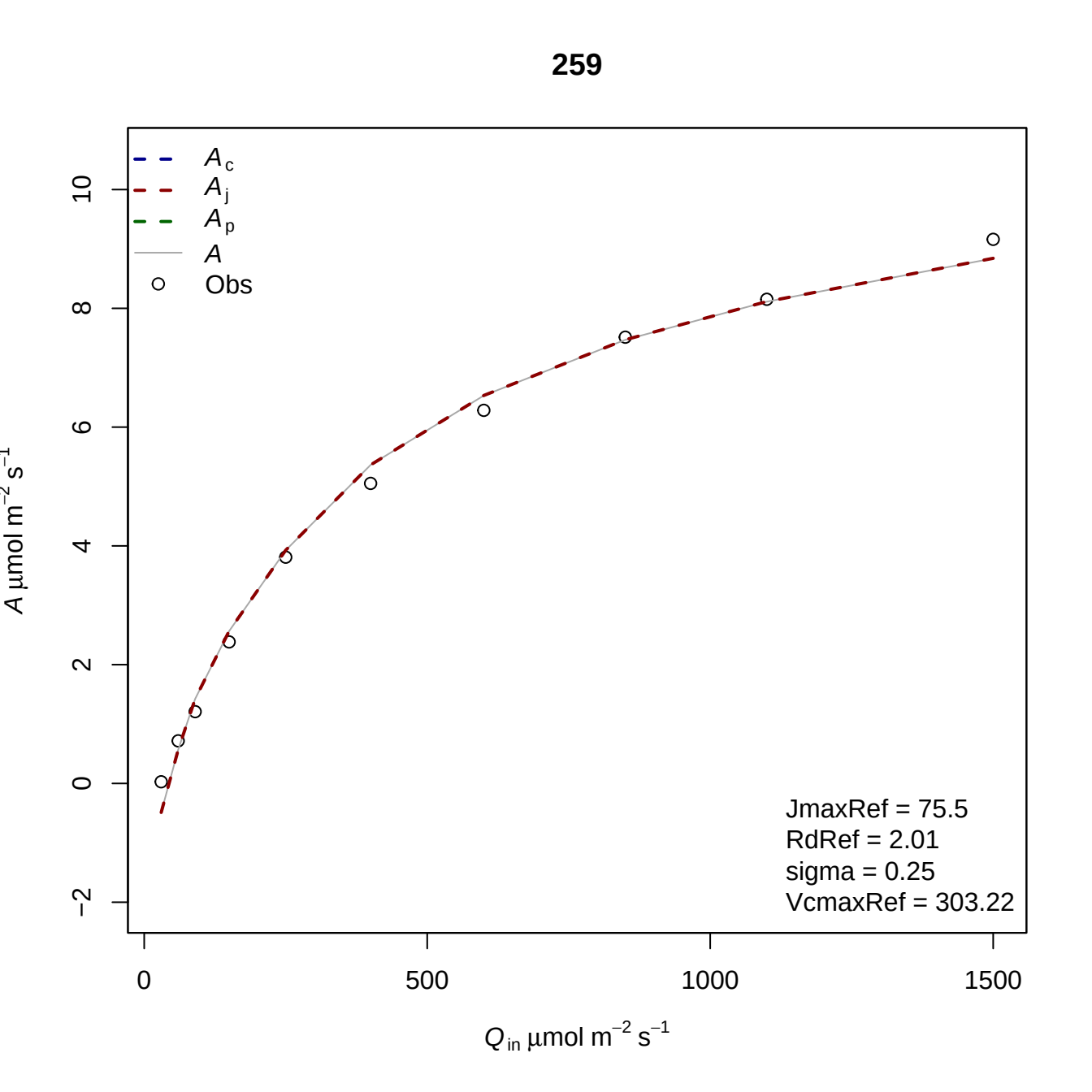

254
A_c
A_j
A_p
A
Obs
JmaxRef = 10.76
RdRef = 1.13
sigma = 0.09
VcmaxRef = 5867.66
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

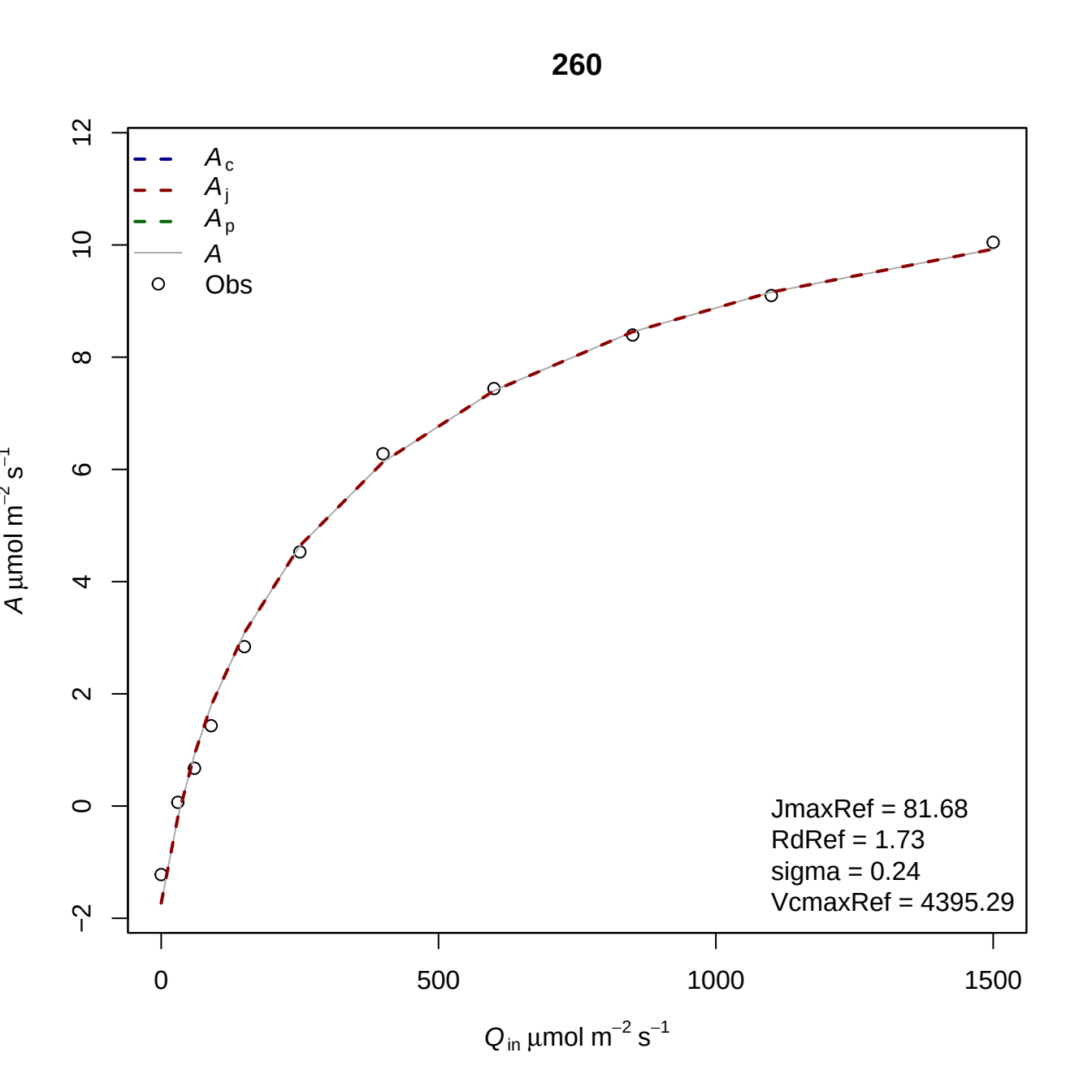

255
A_c
A_j
A_p
A
Obs
JmaxRef = 53.43
RdRef = 2.49
sigma = 0.12
VcmaxRef = 26.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
200
400
600
800
1000
1200
1400

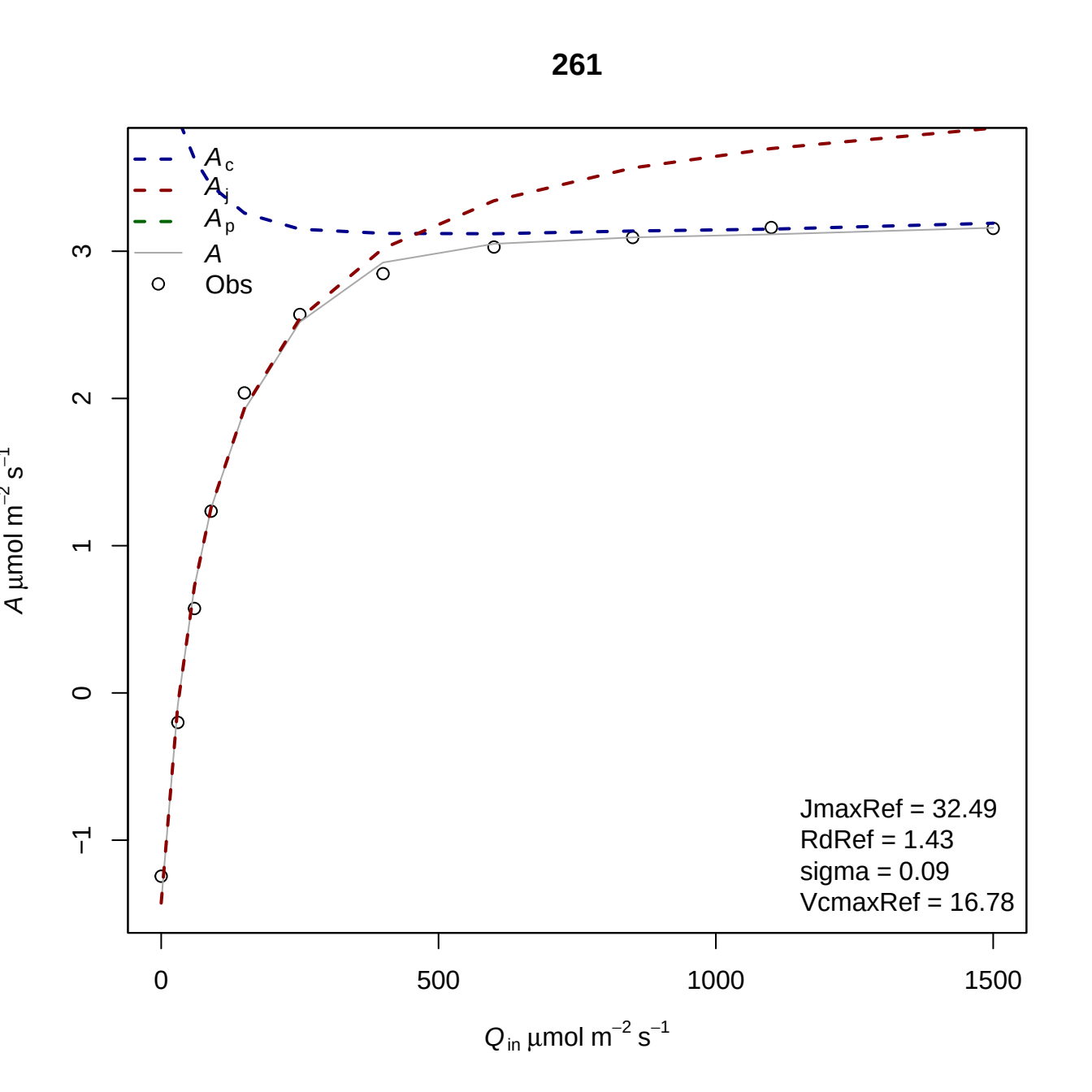

256
A_c
A_j
A_p
A
Obs
JmaxRef = 97.52
RdRef = 1.54
sigma = 0.25
VcmaxRef = 42.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

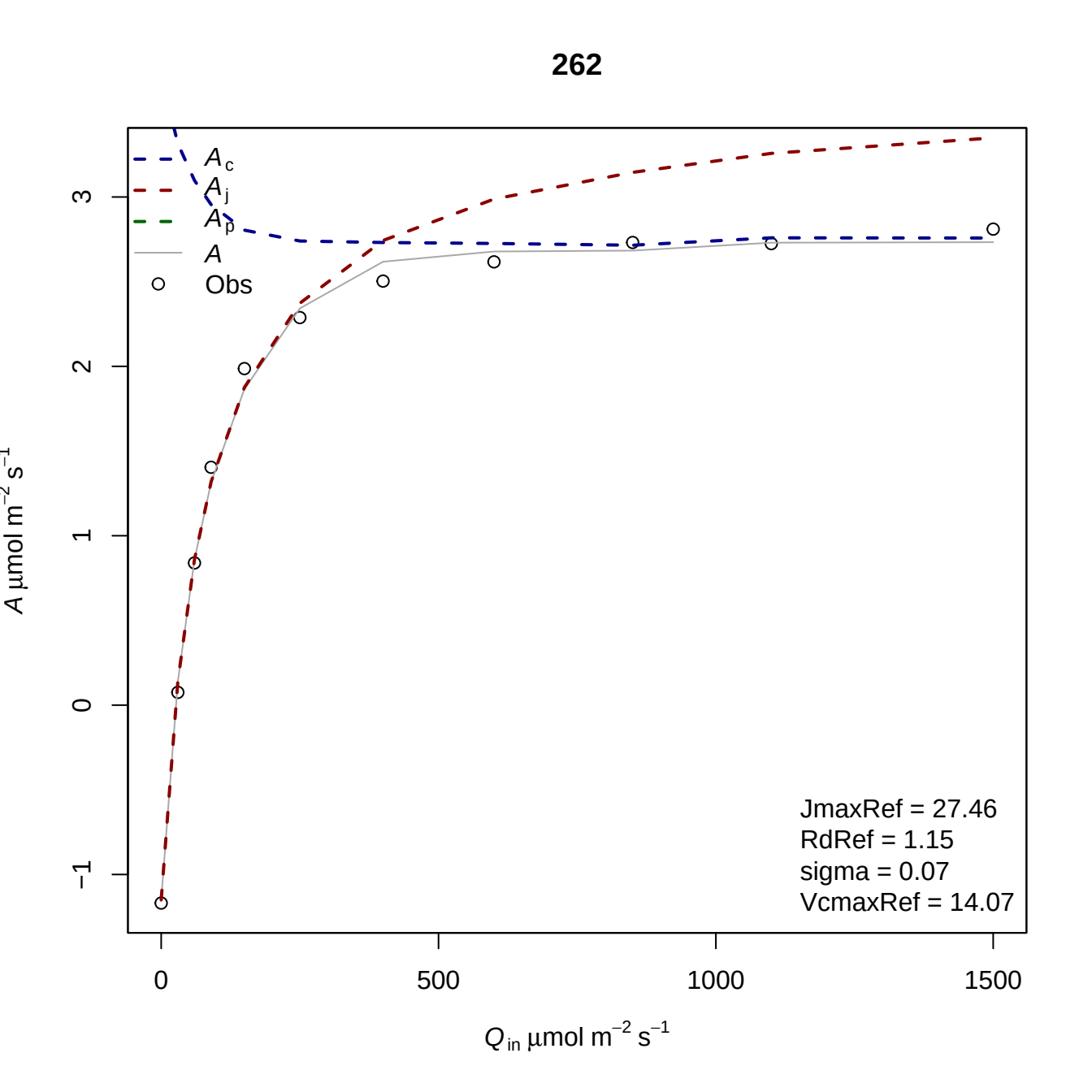

A_c
A_j
A_p
A
Obs
JmaxRef = 24.33
RdRef = 2.44
sigma = 0.44
VcmaxRef = 37045.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
200
400
600
800
1000
1200
1400

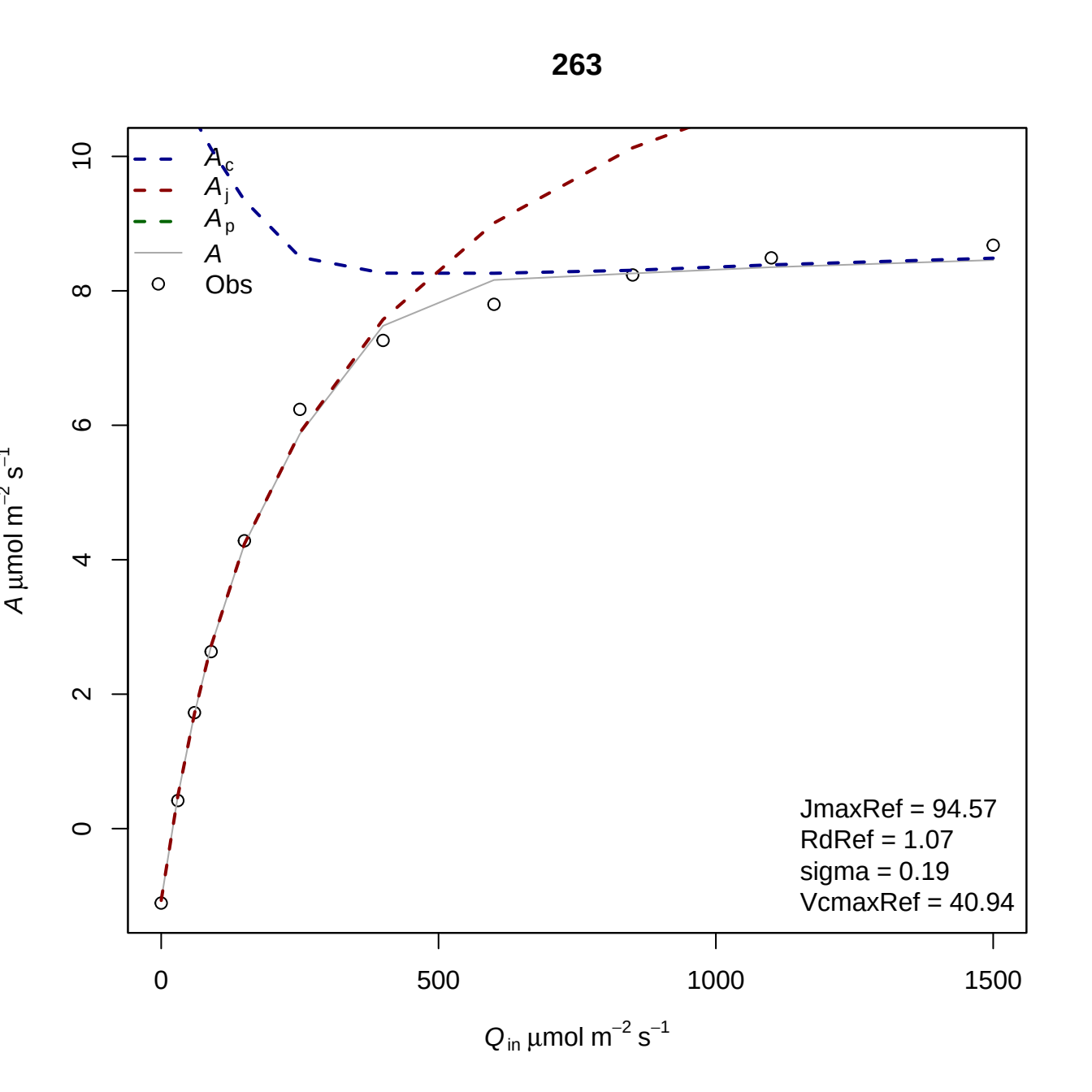

258
A_c
A_j
A_p
A
Obs
JmaxRef = 51.61
RdRef = 1.18
sigma = 0.08
VcmaxRef = 234.81
A µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

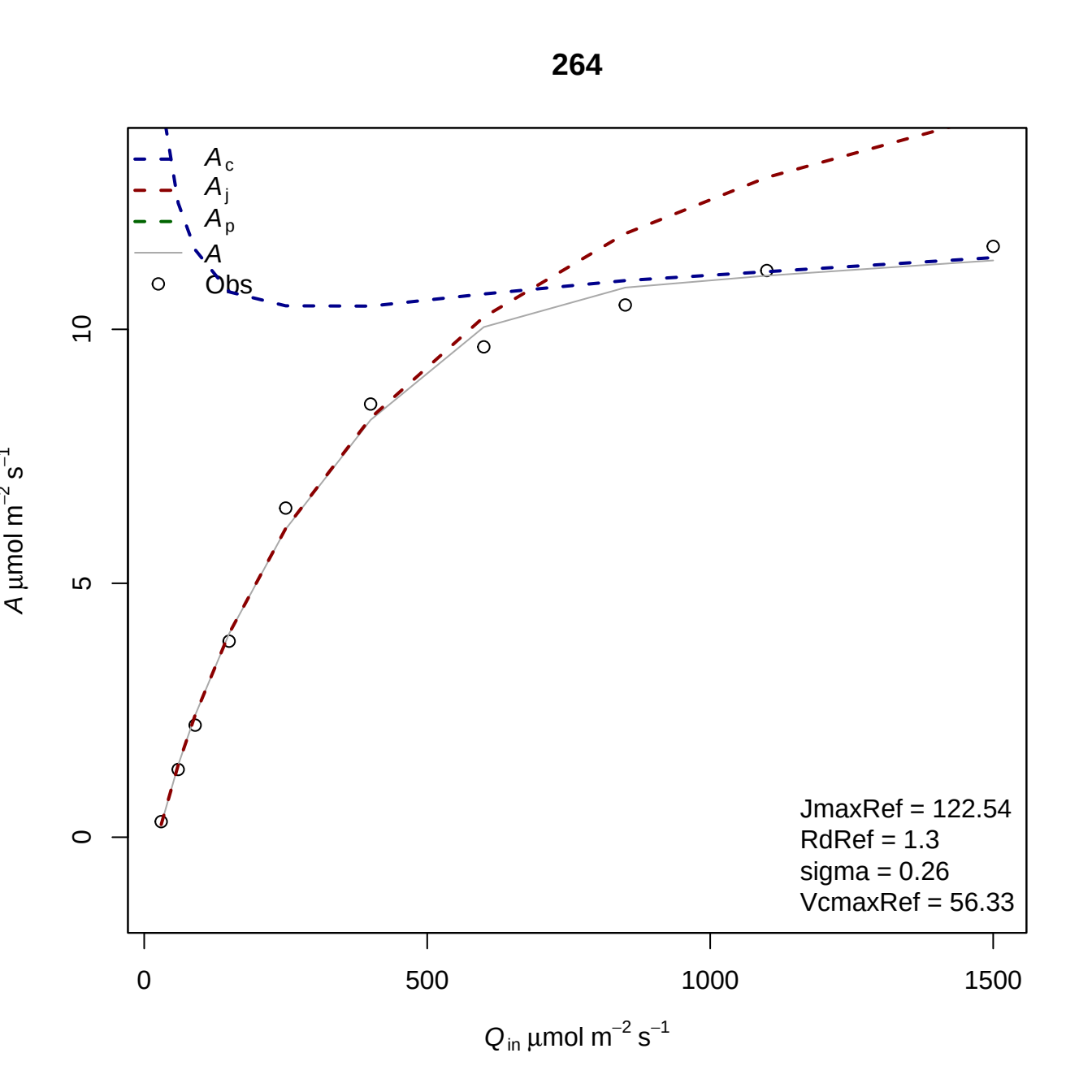

259
A_c
A_j
A_p
A
Obs
JmaxRef = 75.5
RdRef = 2.01
sigma = 0.25
VcmaxRef = 303.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

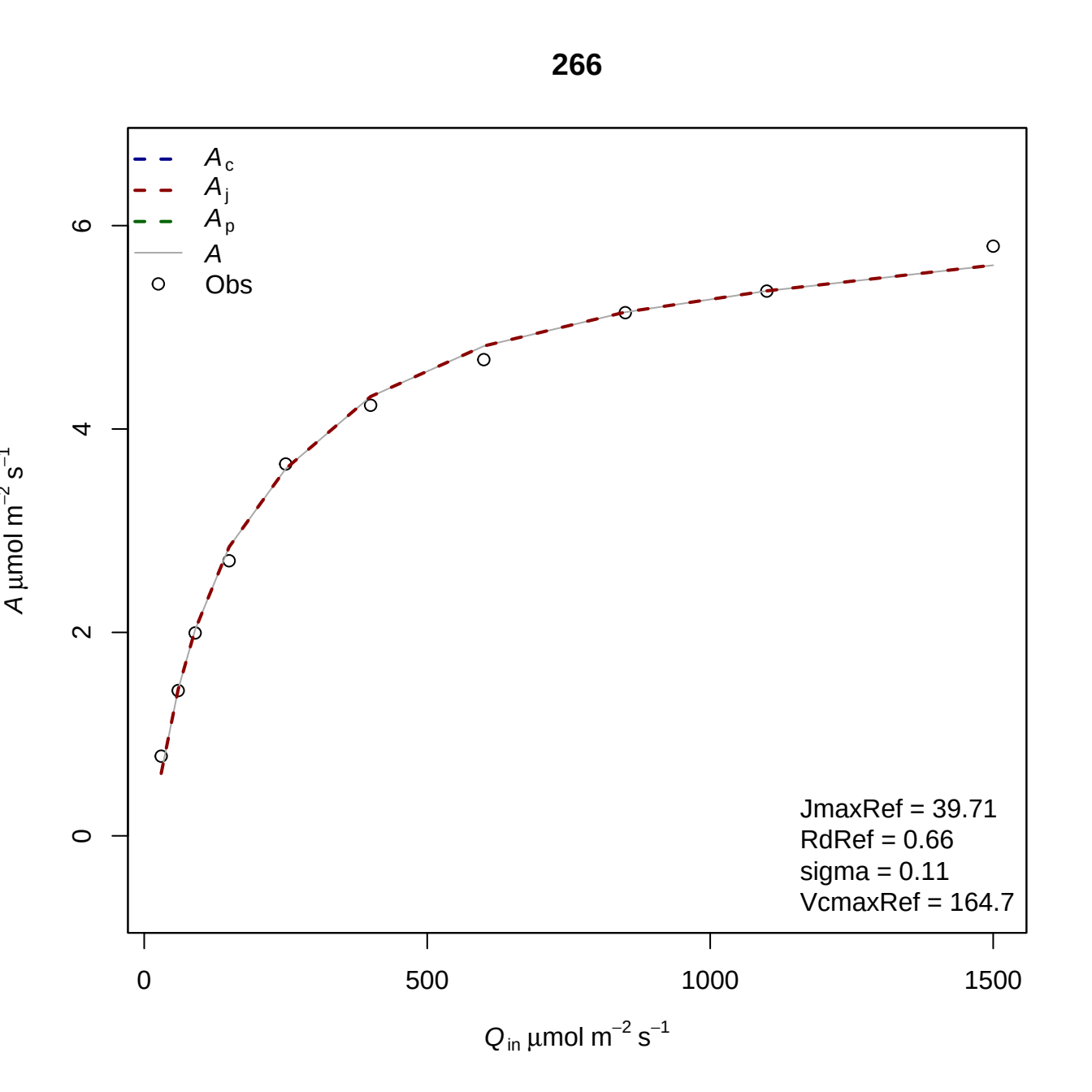

260
A_c
A_j
A_p
A
Obs
JmaxRef = 81.68
RdRef = 1.73
sigma = 0.24
VcmaxRef = 4395.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

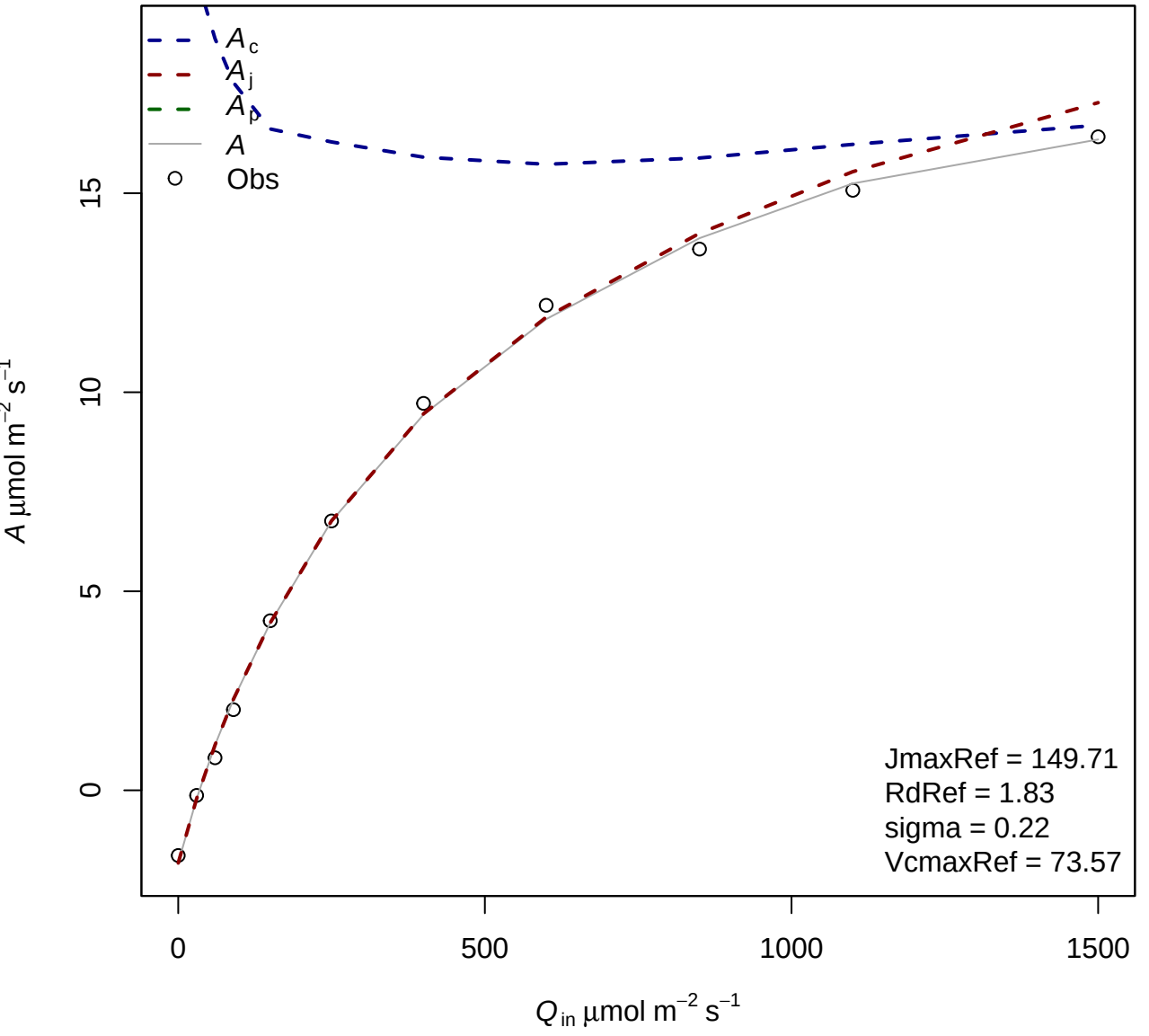

261
A_c
A_j
A_p
A
Obs
JmaxRef = 32.49
RdRef = 1.43
sigma = 0.09
VcmaxRef = 16.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

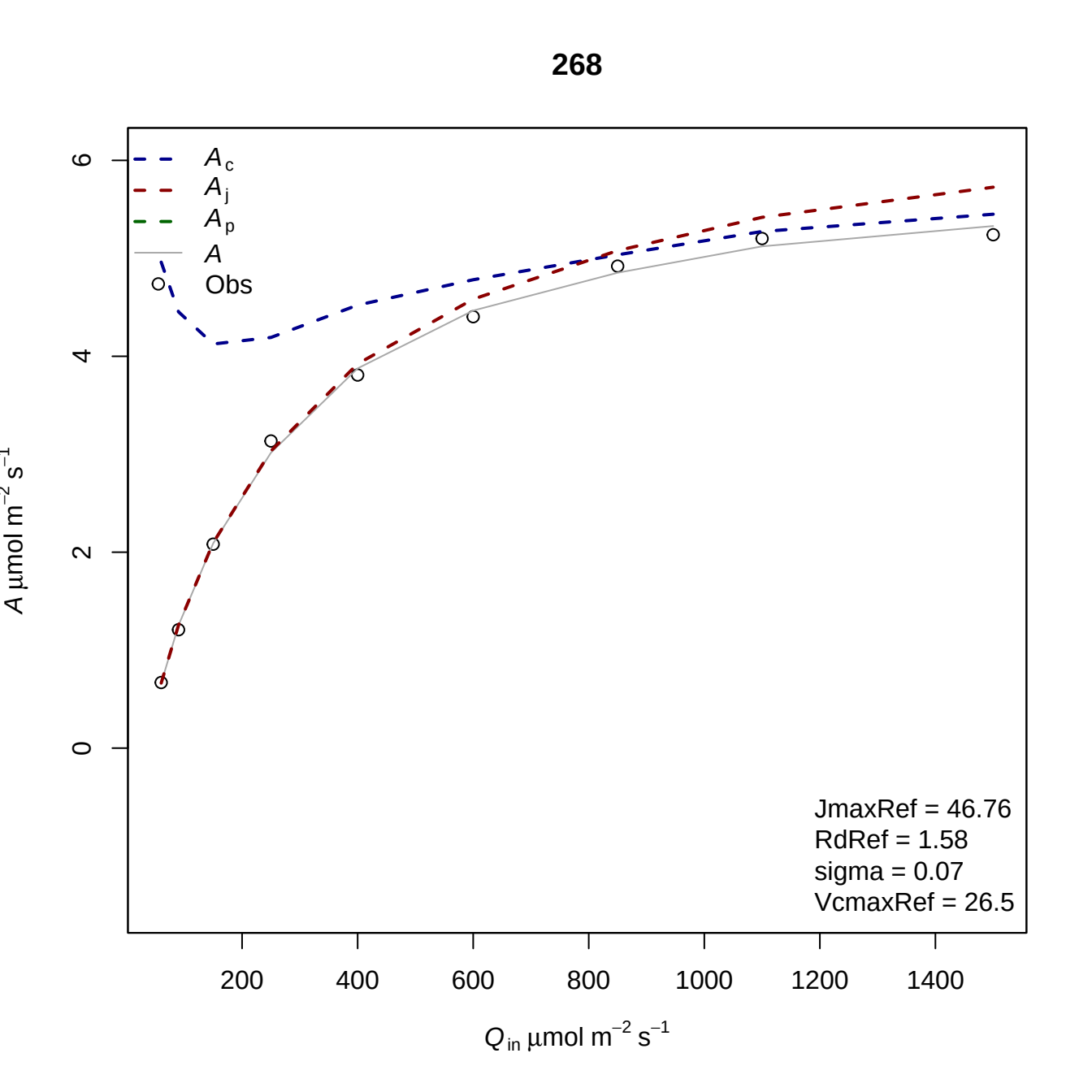

262
A_c
A_j
A_p
A
Obs
JmaxRef = 27.46
RdRef = 1.15
sigma = 0.07
VcmaxRef = 14.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

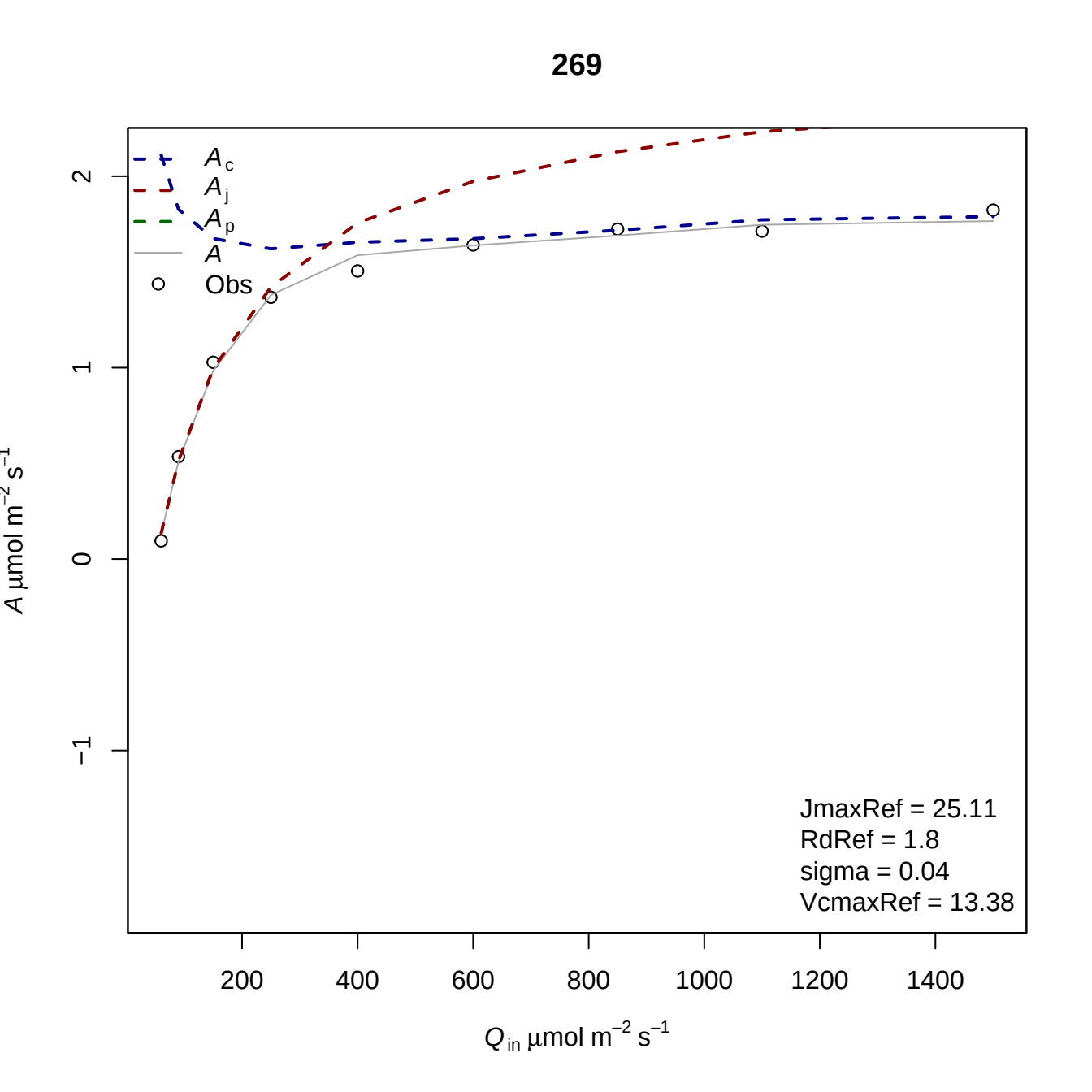

263
A_c
A_j
A_p
A
Obs
JmaxRef = 94.57
RdRef = 1.07
sigma = 0.19
VcmaxRef = 40.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

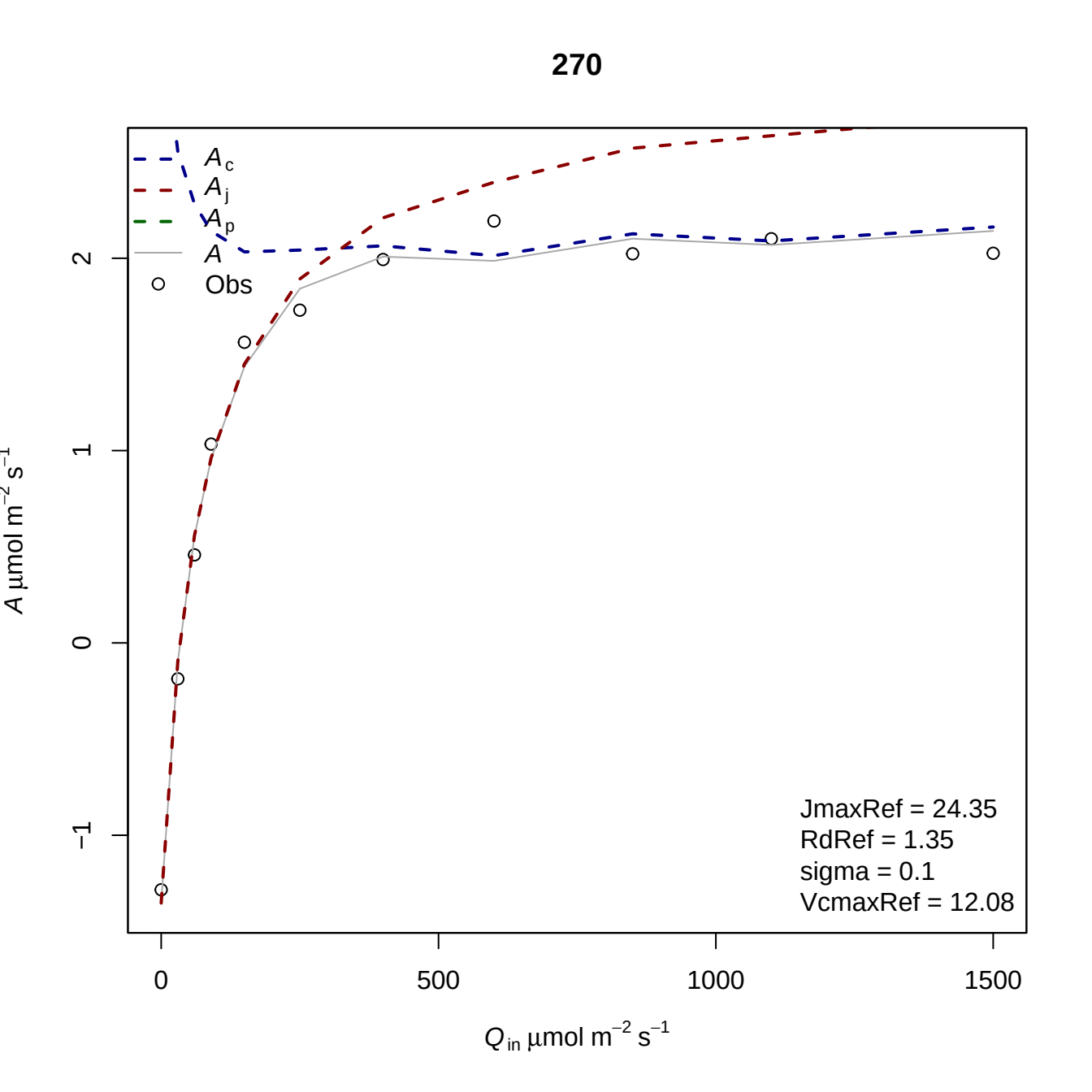

264
A_c
A_j
A_p
A
Obs
JmaxRef = 122.54
RdRef = 1.3
sigma = 0.26
VcmaxRef = 56.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

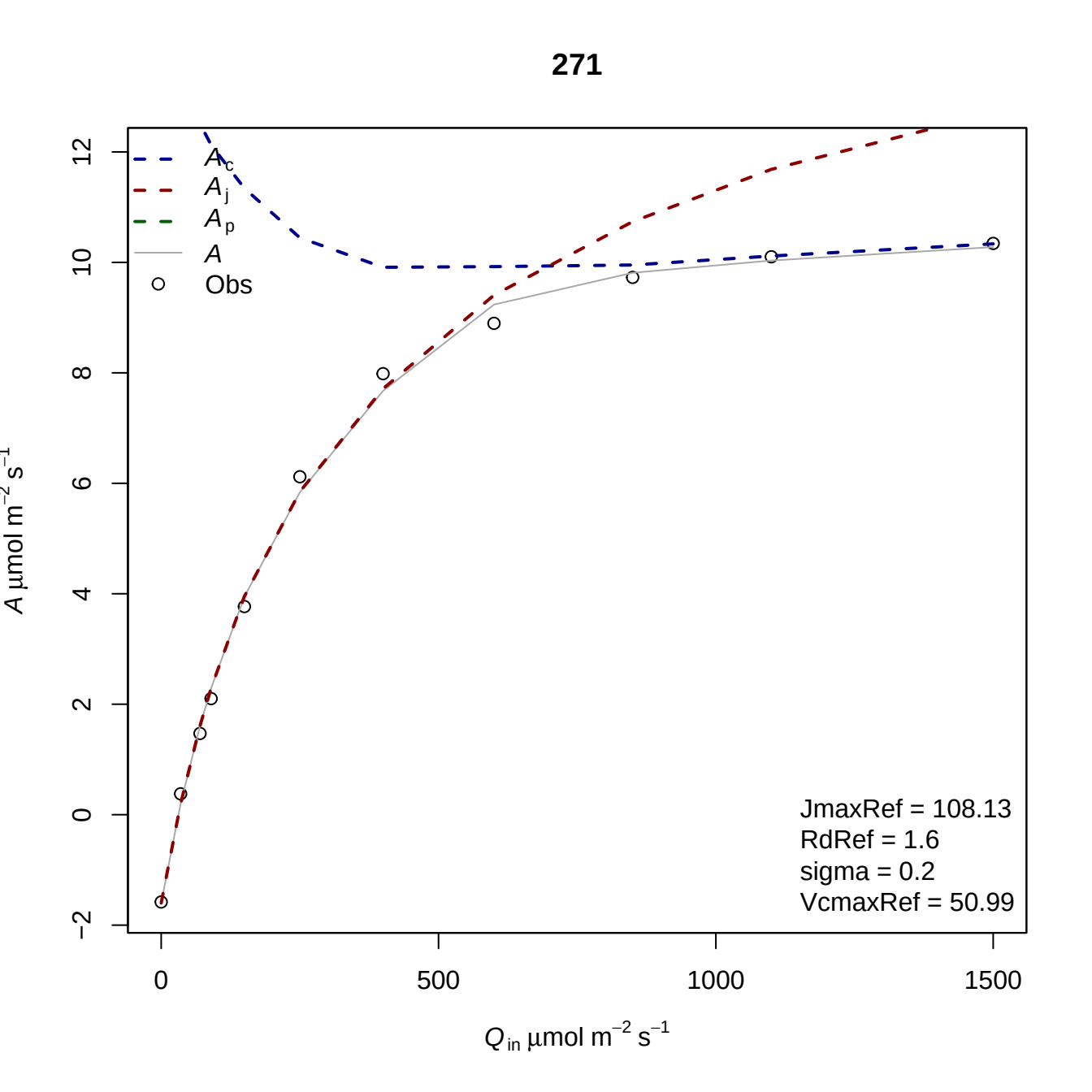

266
A_c
A_j
A_p
A
Obs
JmaxRef = 39.71
RdRef = 0.66
sigma = 0.11
VcmaxRef = 164.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

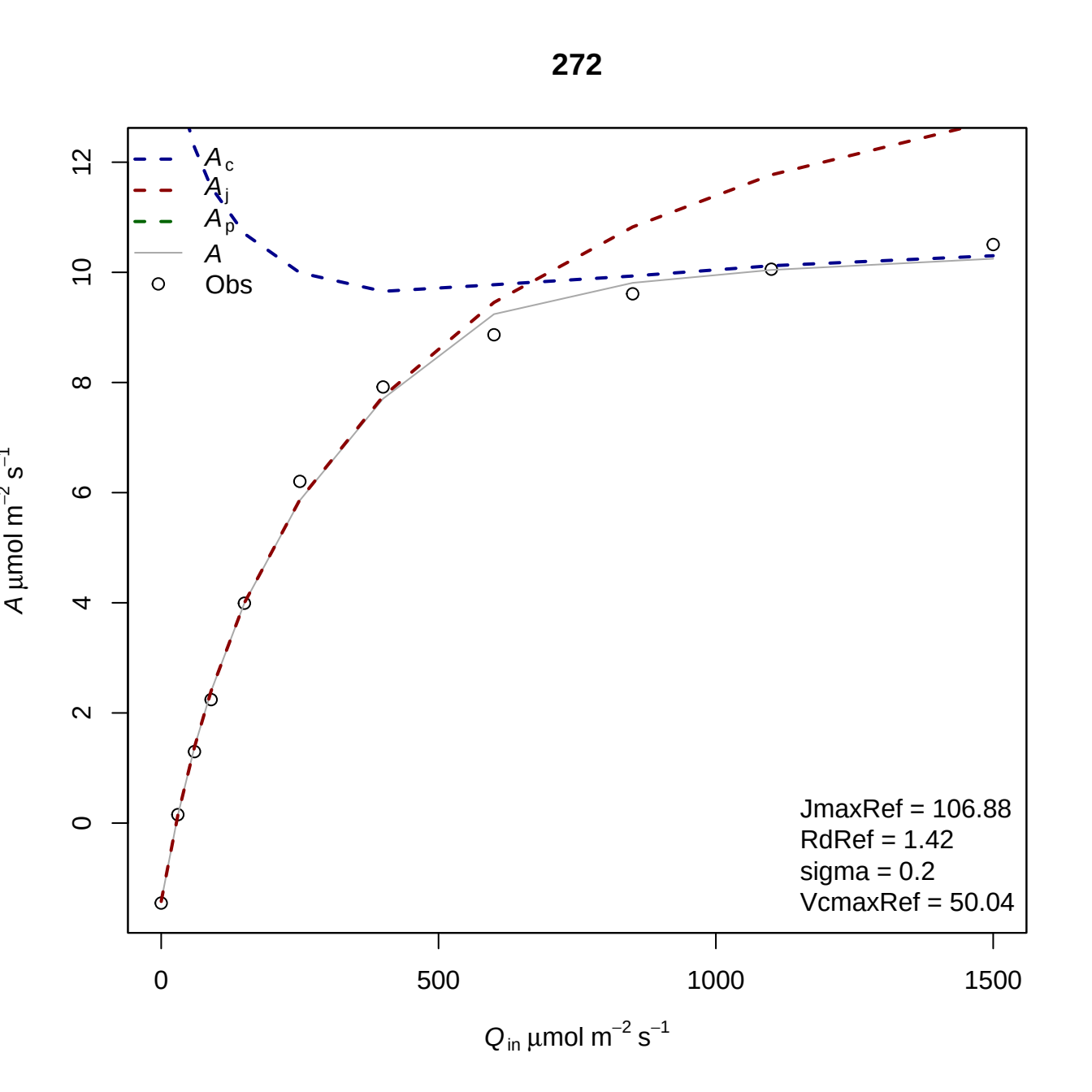

A_c
A_j
A_p
A
Obs
JmaxRef = 149.71
RdRef = 1.83
sigma = 0.22
VcmaxRef = 73.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

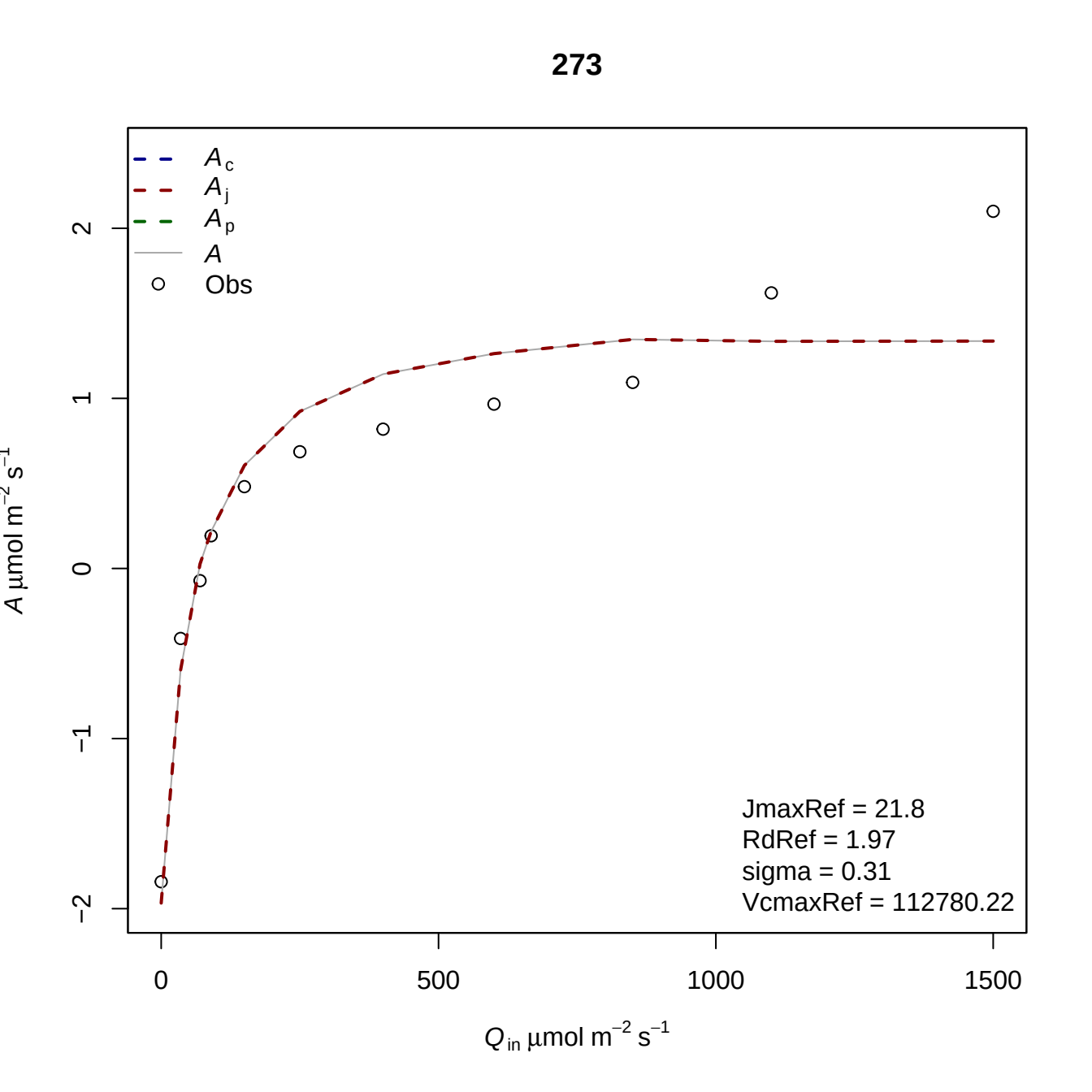

268
A_c
A_j
A_p
A
Obs
JmaxRef = 46.76
RdRef = 1.58
sigma = 0.07
VcmaxRef = 26.5
0
2
4
6
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

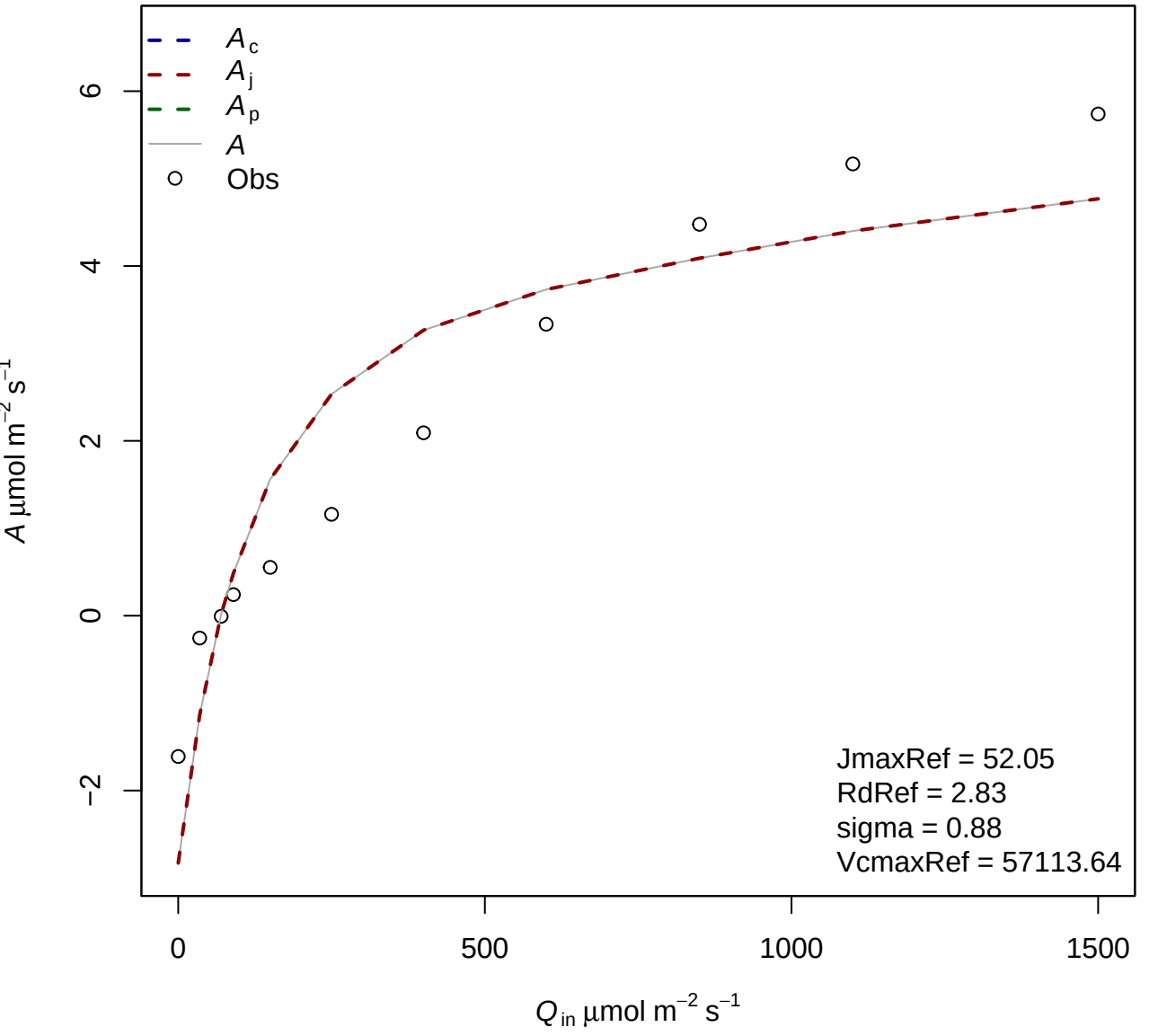

269
A_c
A_j
A_p
A
Obs
2
1
0
−1
A µmol m^{-2} s^{-1}
200
400
600
800
1000
1200
1400
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 25.11
RdRef = 1.8
sigma = 0.04
VcmaxRef = 13.38

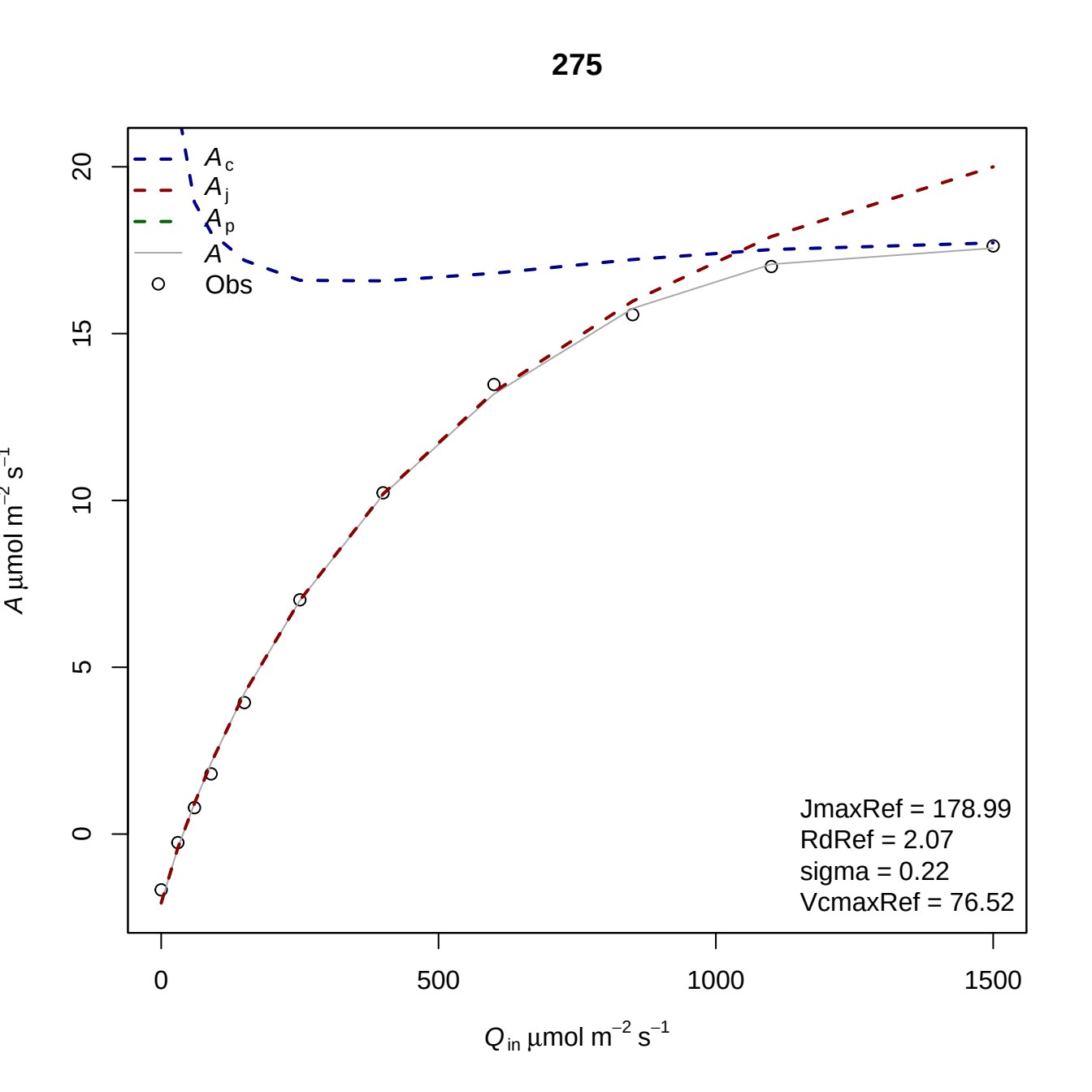

270
A_c
A_j
A_p
A
Obs
JmaxRef = 24.35
RdRef = 1.35
sigma = 0.1
VcmaxRef = 12.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500

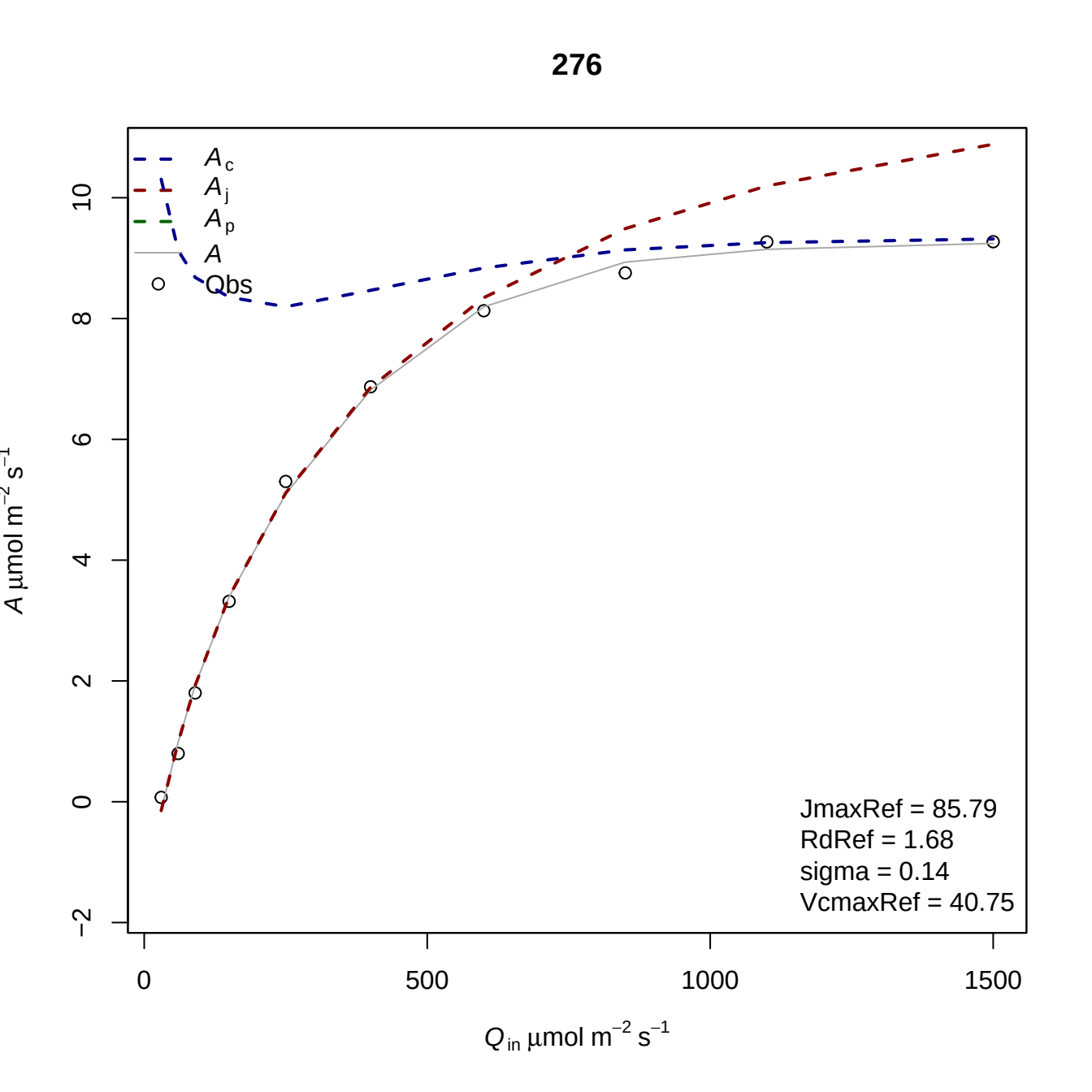

271
A_c
A_j
A_p
A
Obs
JmaxRef = 108.13
RdRef = 1.6
sigma = 0.2
VcmaxRef = 50.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

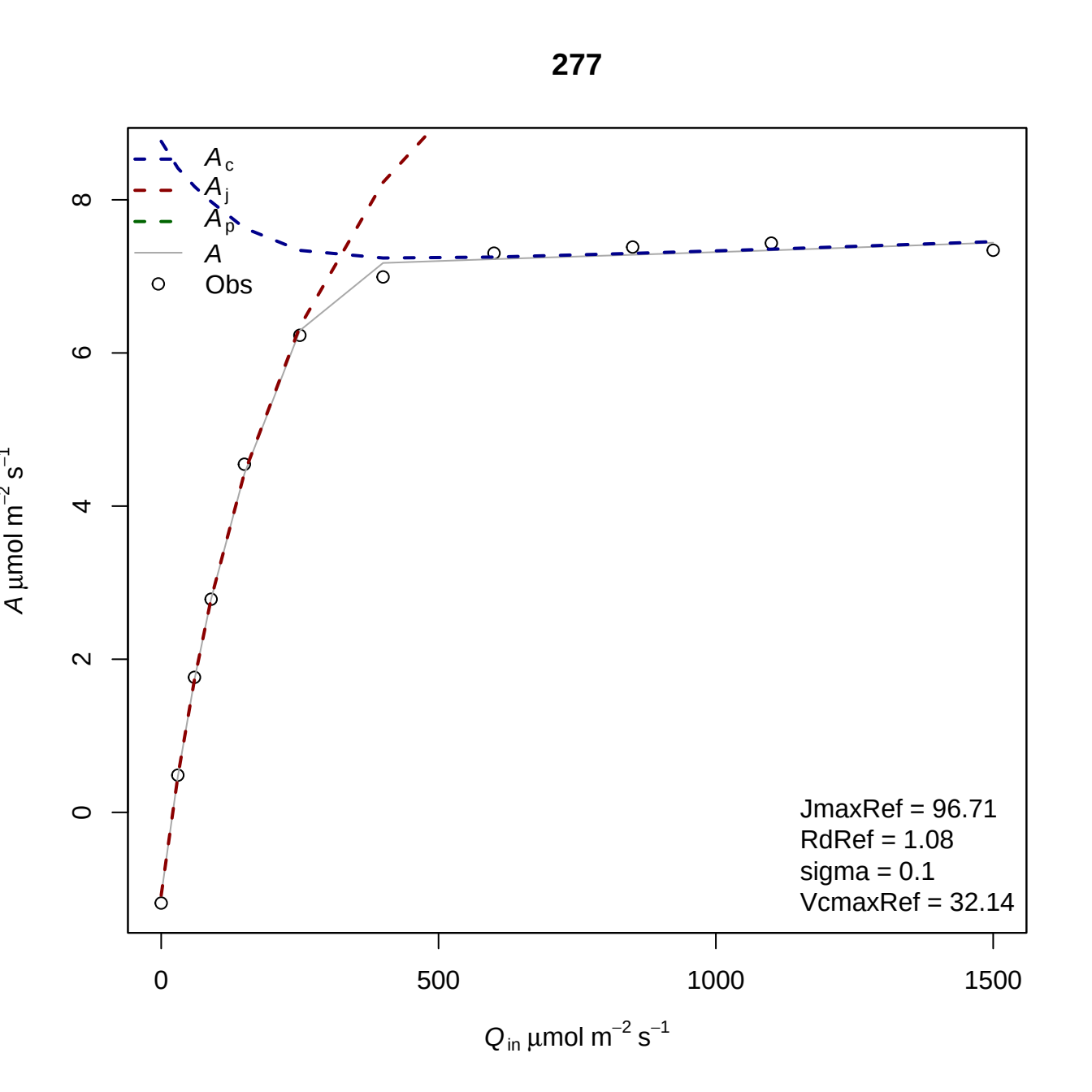

272
A_c
A_j
A_p
A
Obs
JmaxRef = 106.88
RdRef = 1.42
sigma = 0.2
VcmaxRef = 50.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

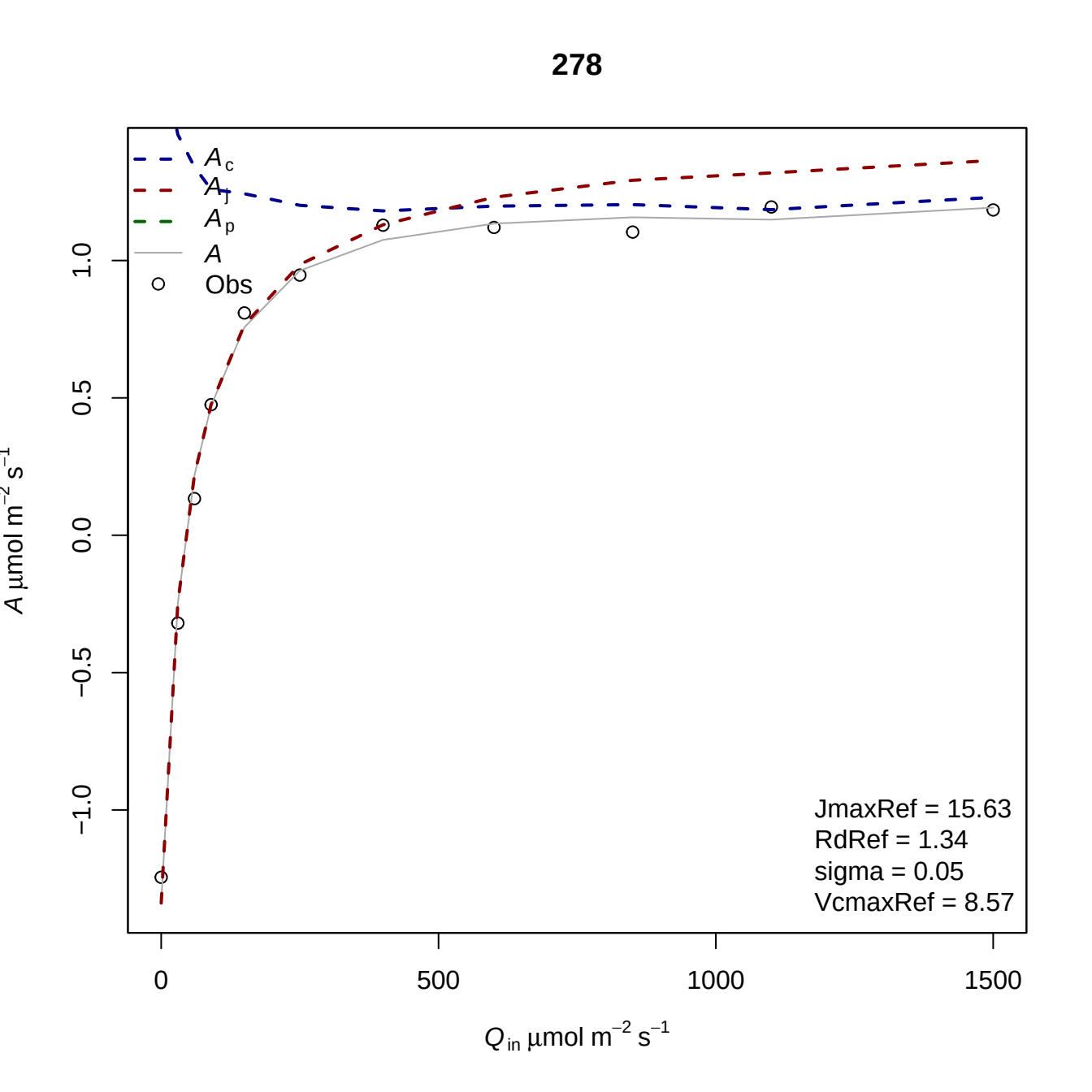

273
A_c
A_j
A_p
A
Obs
JmaxRef = 21.8
RdRef = 1.97
sigma = 0.31
VcmaxRef = 112780.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
−2
0
500
1000
1500

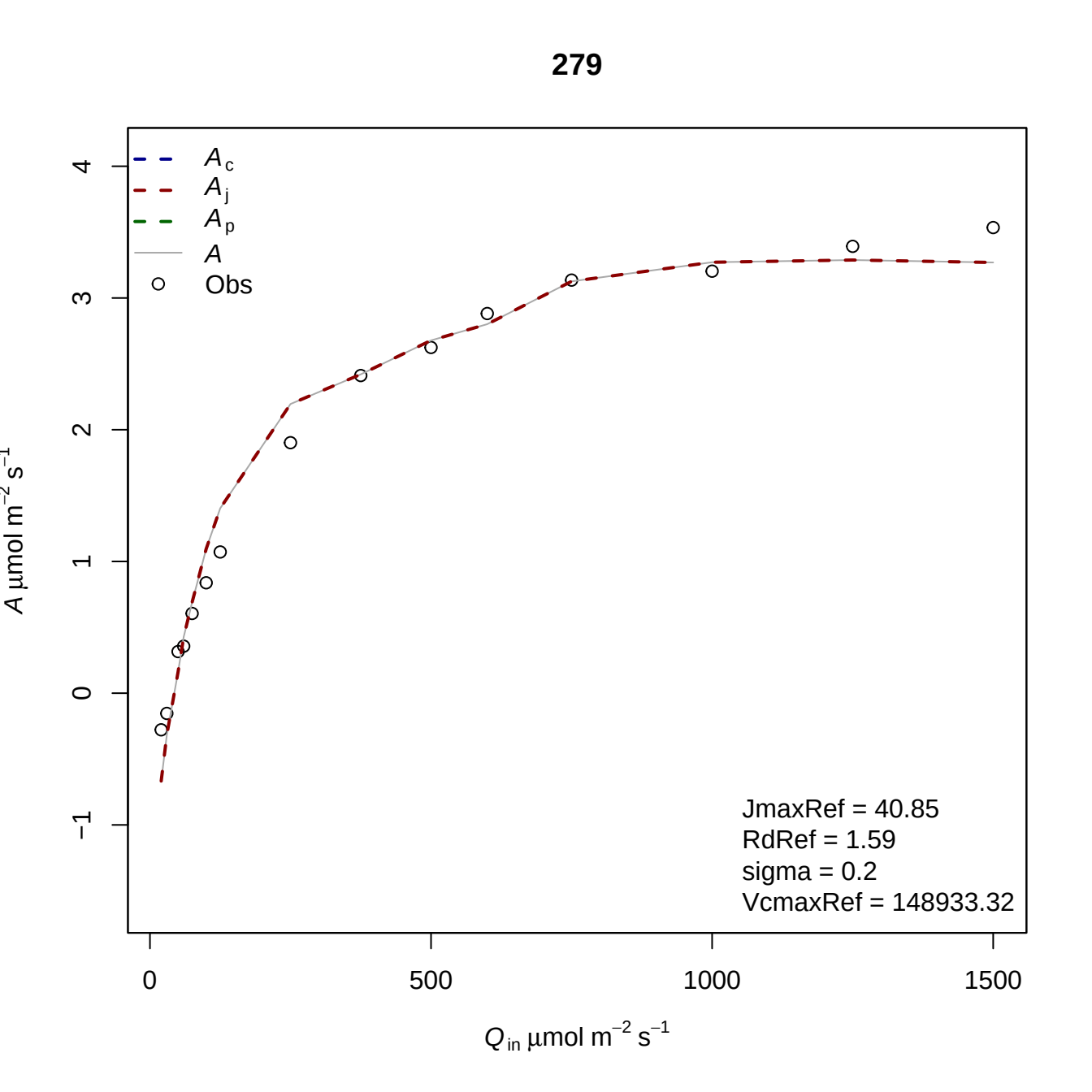

A_c
A_j
A_p
A
Obs
JmaxRef = 52.05
RdRef = 2.83
sigma = 0.88
VcmaxRef = 57113.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

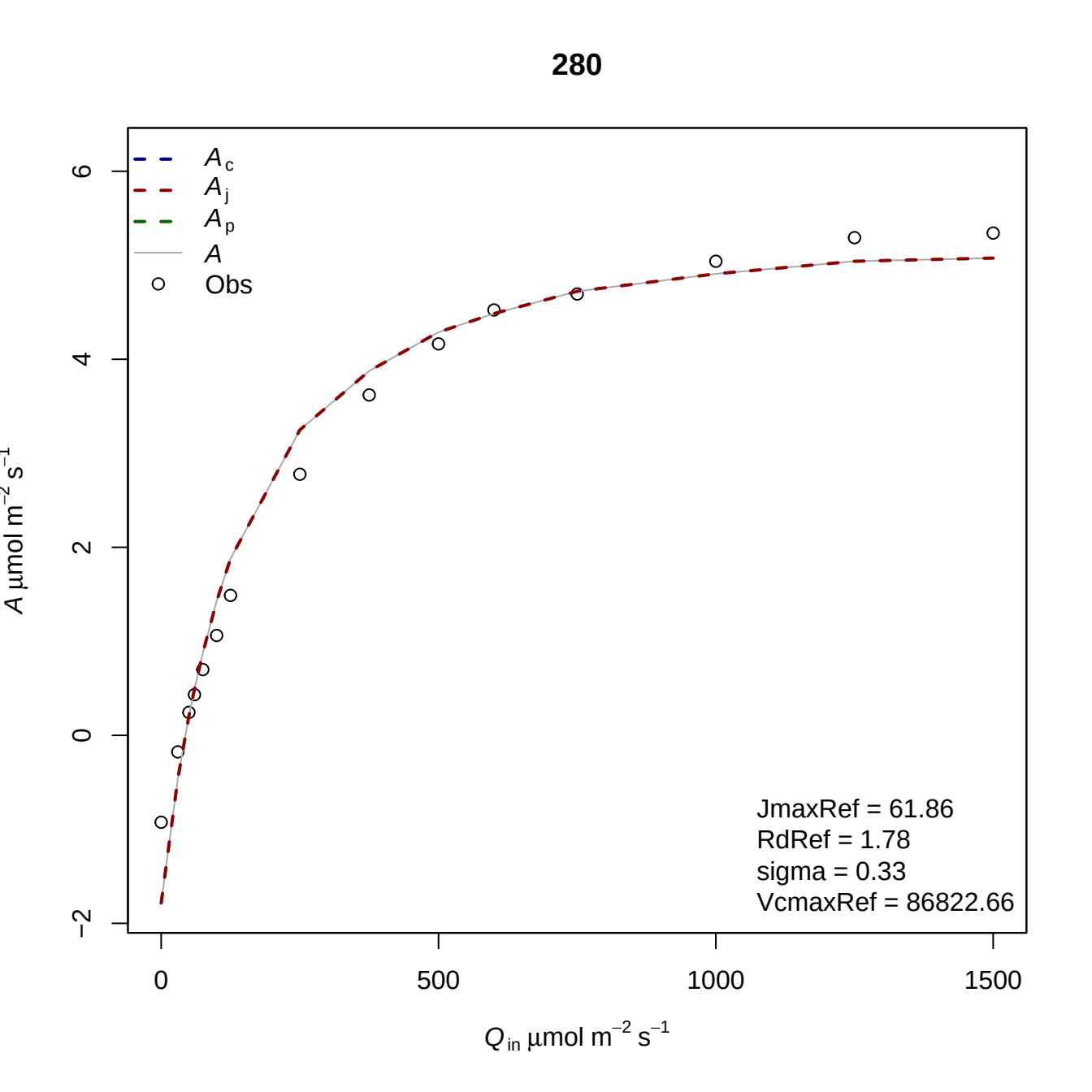

275
A_c
A_j
A_p
A
Obs
JmaxRef = 178.99
RdRef = 2.07
sigma = 0.22
VcmaxRef = 76.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

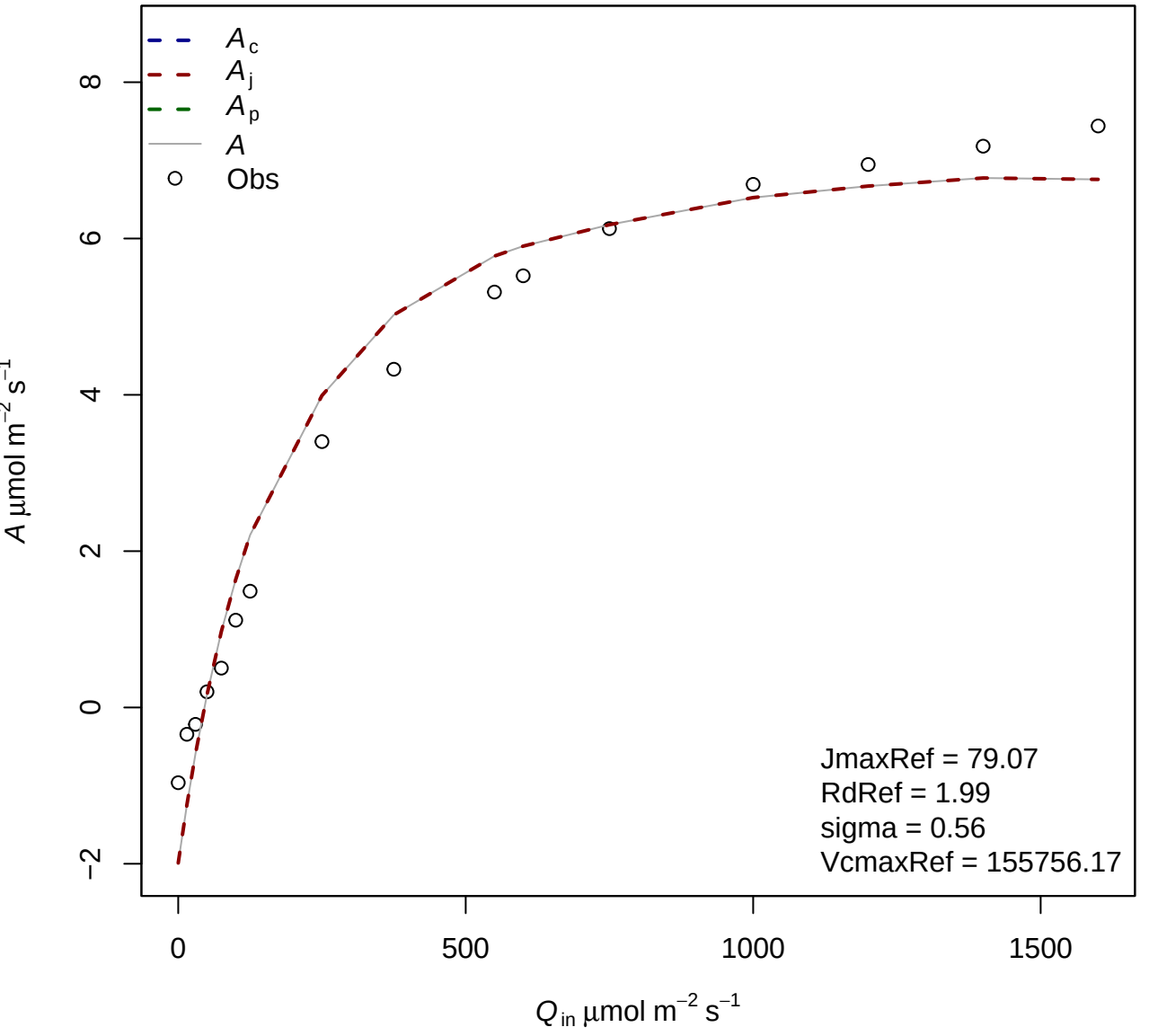

276
A_c
A_j
A_p
A
Qbs
JmaxRef = 85.79
RdRef = 1.68
sigma = 0.14
VcmaxRef = 40.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

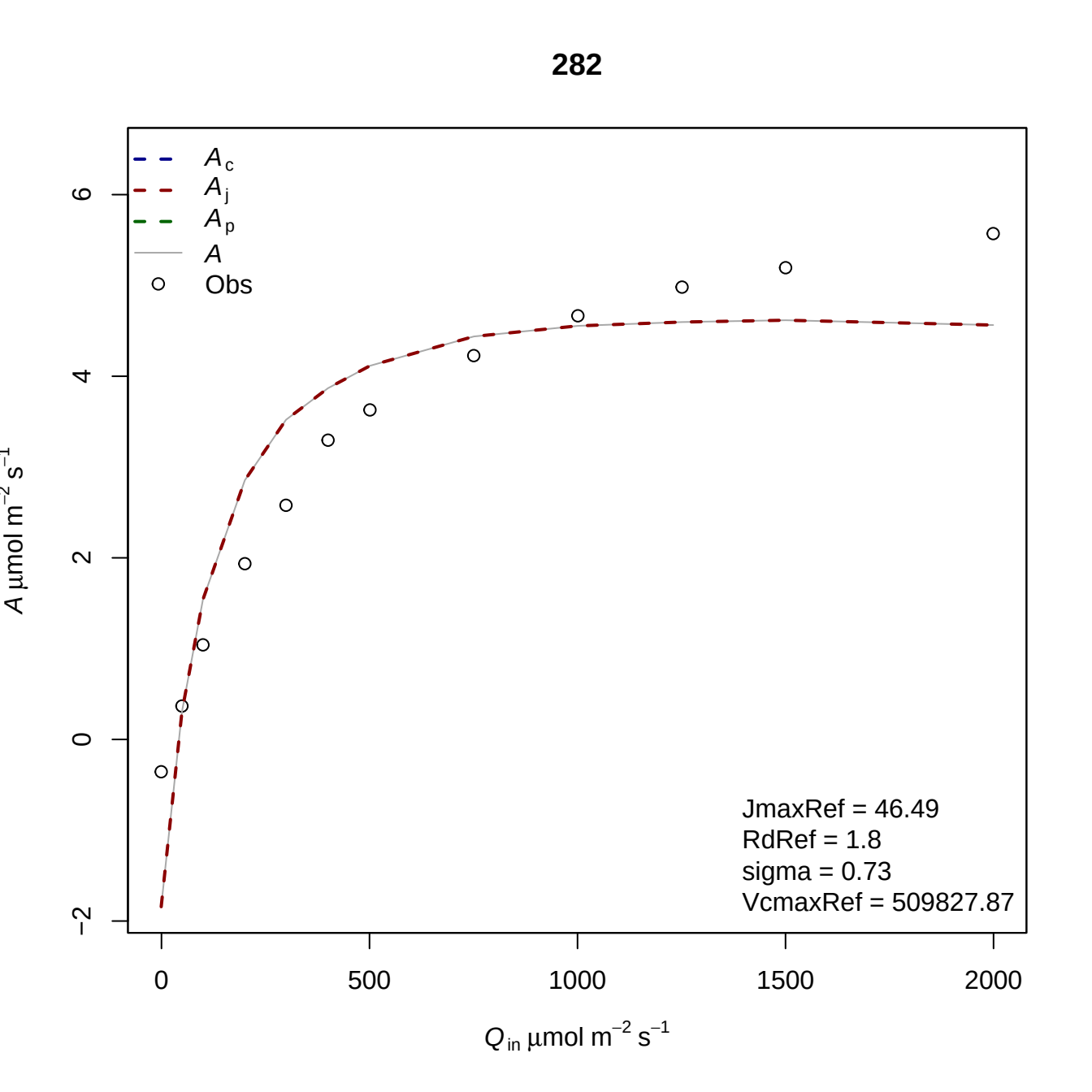

277
A_c
A_j
A_p
A
Obs
JmaxRef = 96.71
RdRef = 1.08
sigma = 0.1
VcmaxRef = 32.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

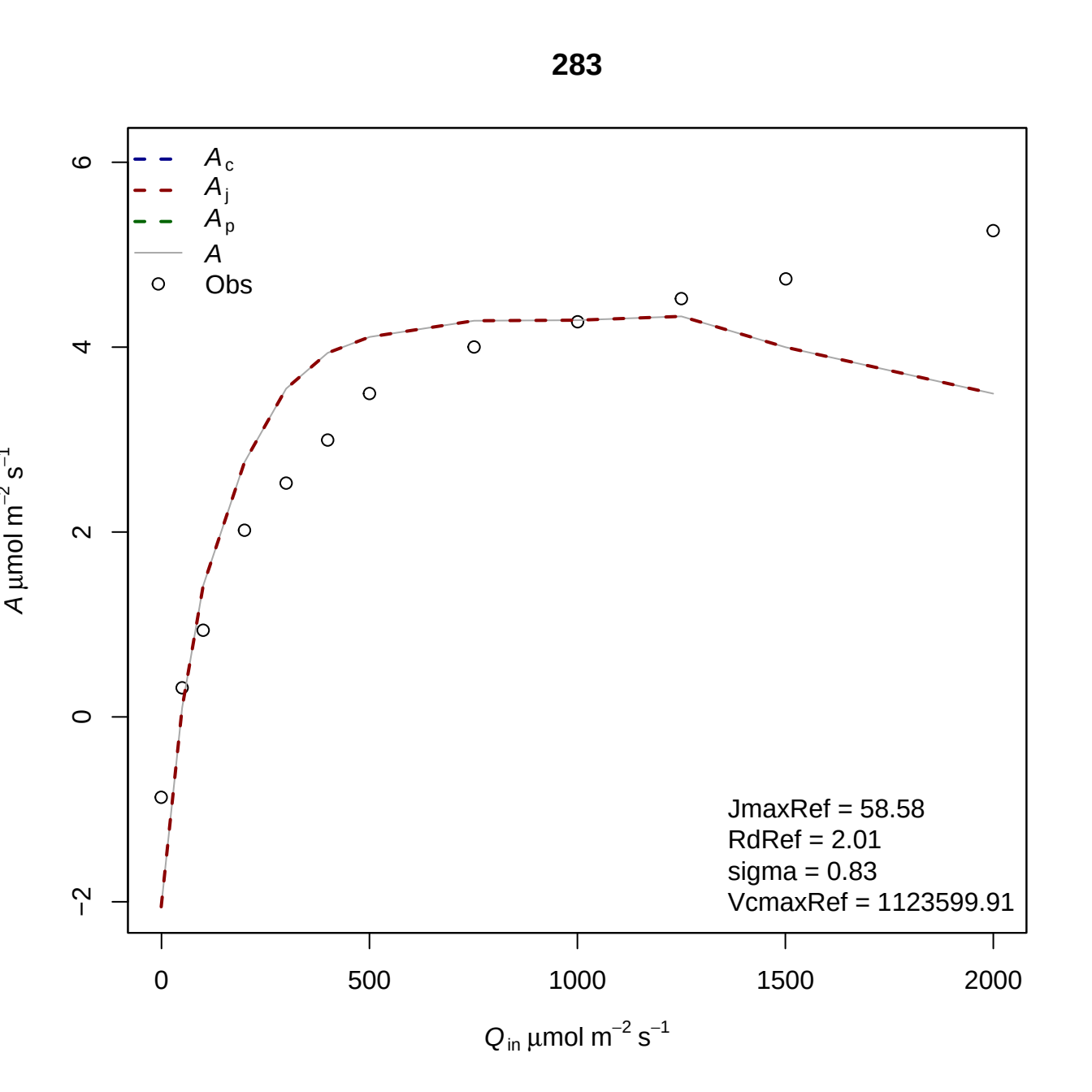

278
A_c
A_j
A_p
A
Obs
JmaxRef = 15.63
RdRef = 1.34
sigma = 0.05
VcmaxRef = 8.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1.0
−0.5
0.0
0.5
1.0
0
500
1000
1500

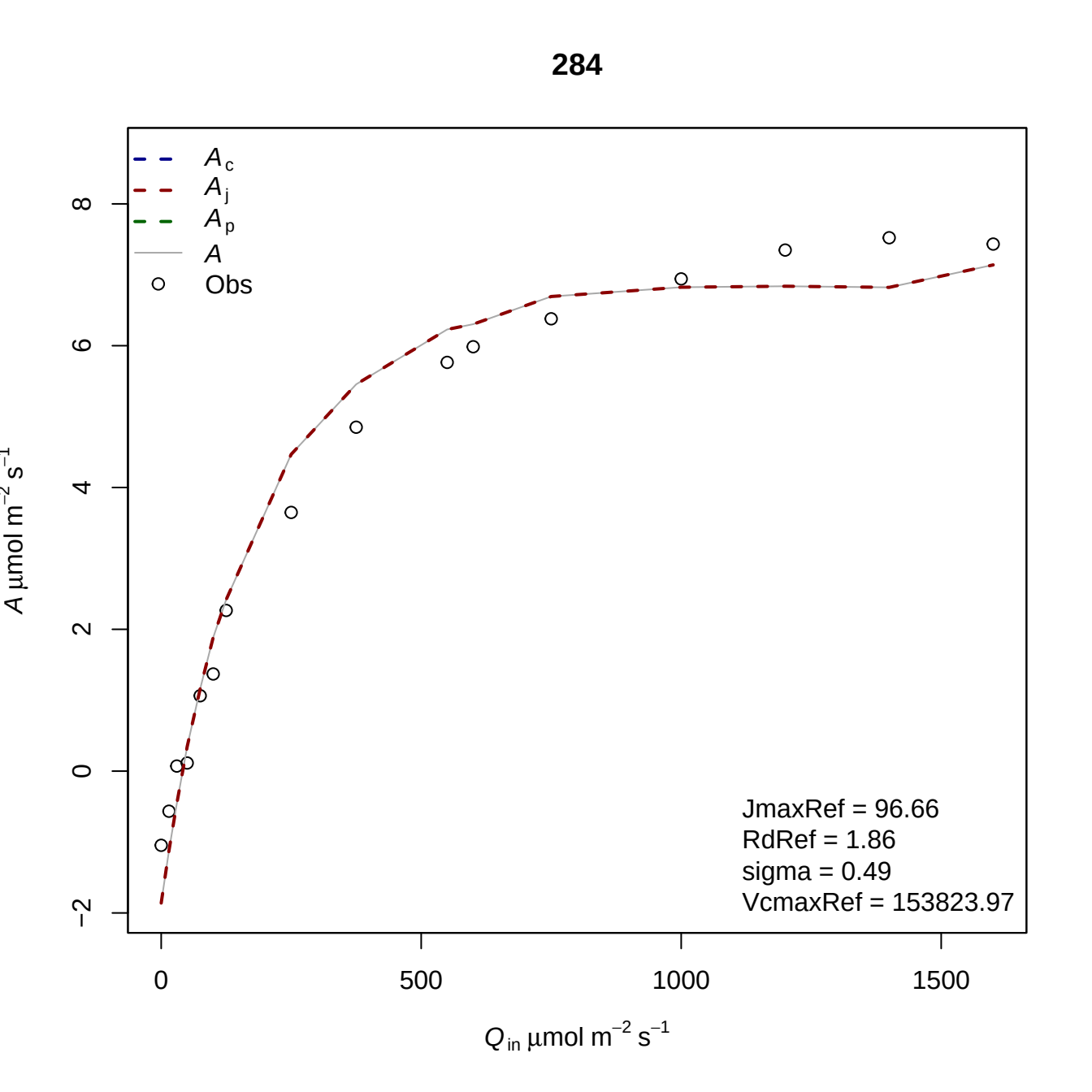

279
A_c
A_j
A_p
A
Obs
JmaxRef = 40.85
RdRef = 1.59
sigma = 0.2
VcmaxRef = 148933.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

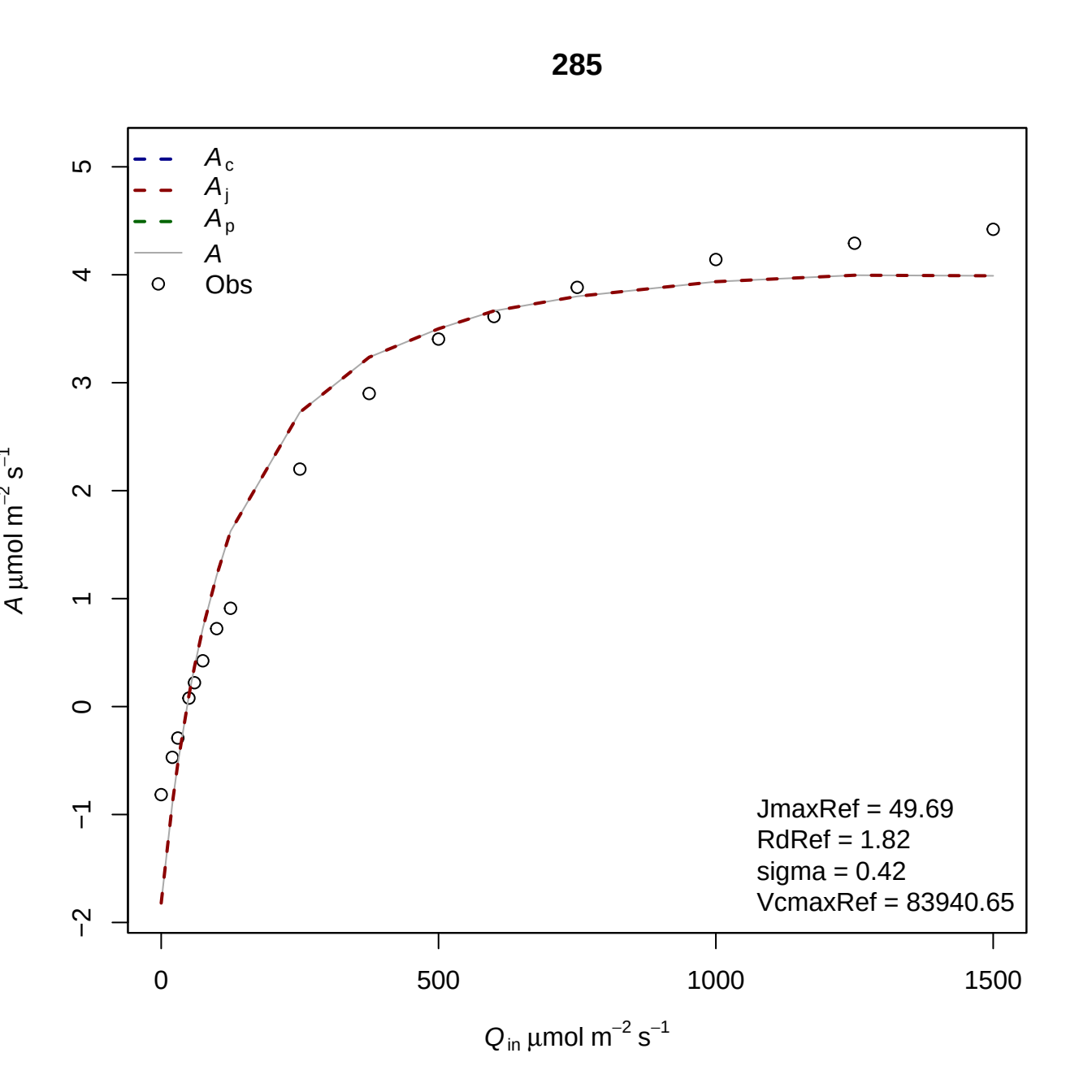

280
A_c
A_j
A_p
A
Obs
JmaxRef = 61.86
RdRef = 1.78
sigma = 0.33
VcmaxRef = 86822.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

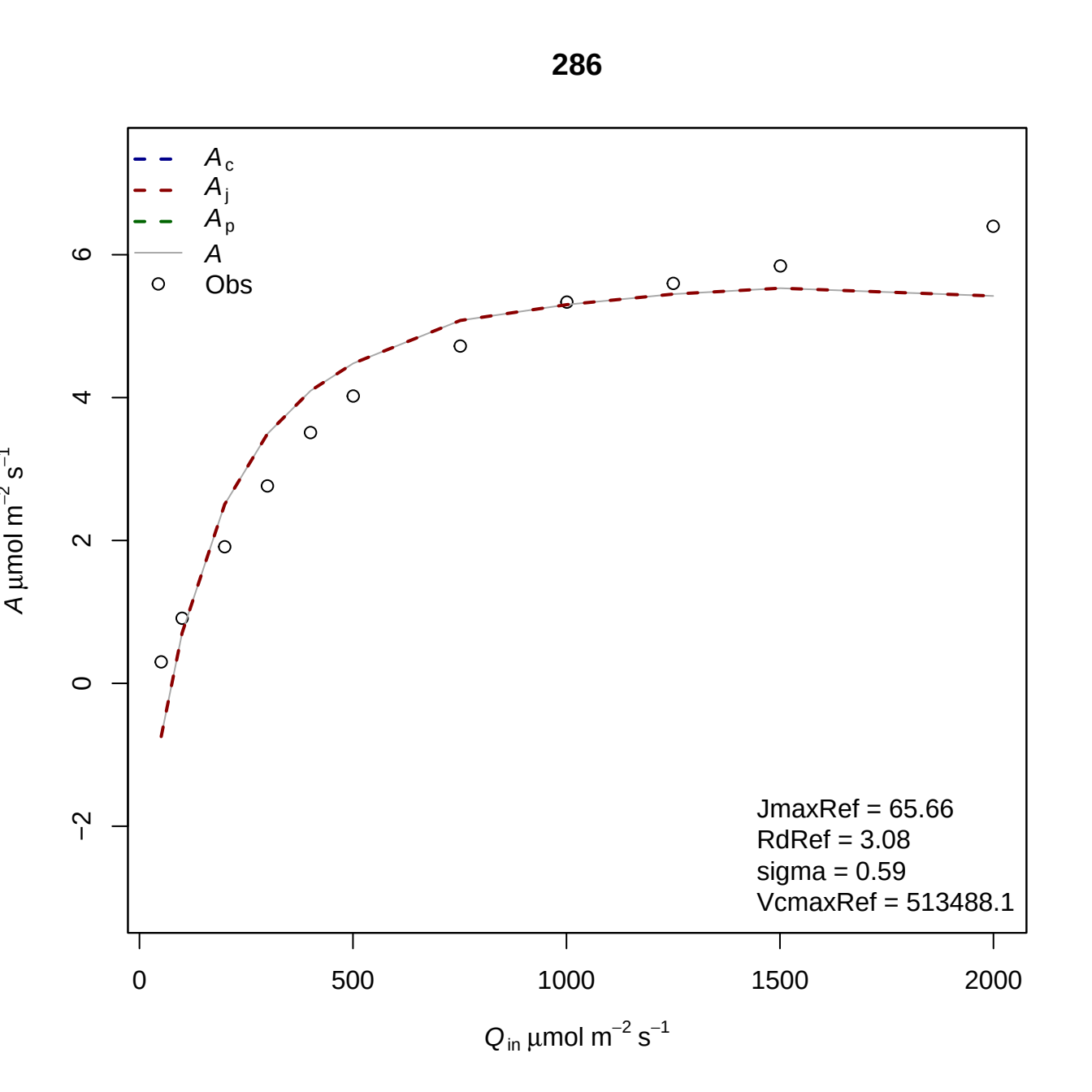

A_c
A_j
A_p
A
Obs
JmaxRef = 79.07
RdRef = 1.99
sigma = 0.56
VcmaxRef = 155756.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

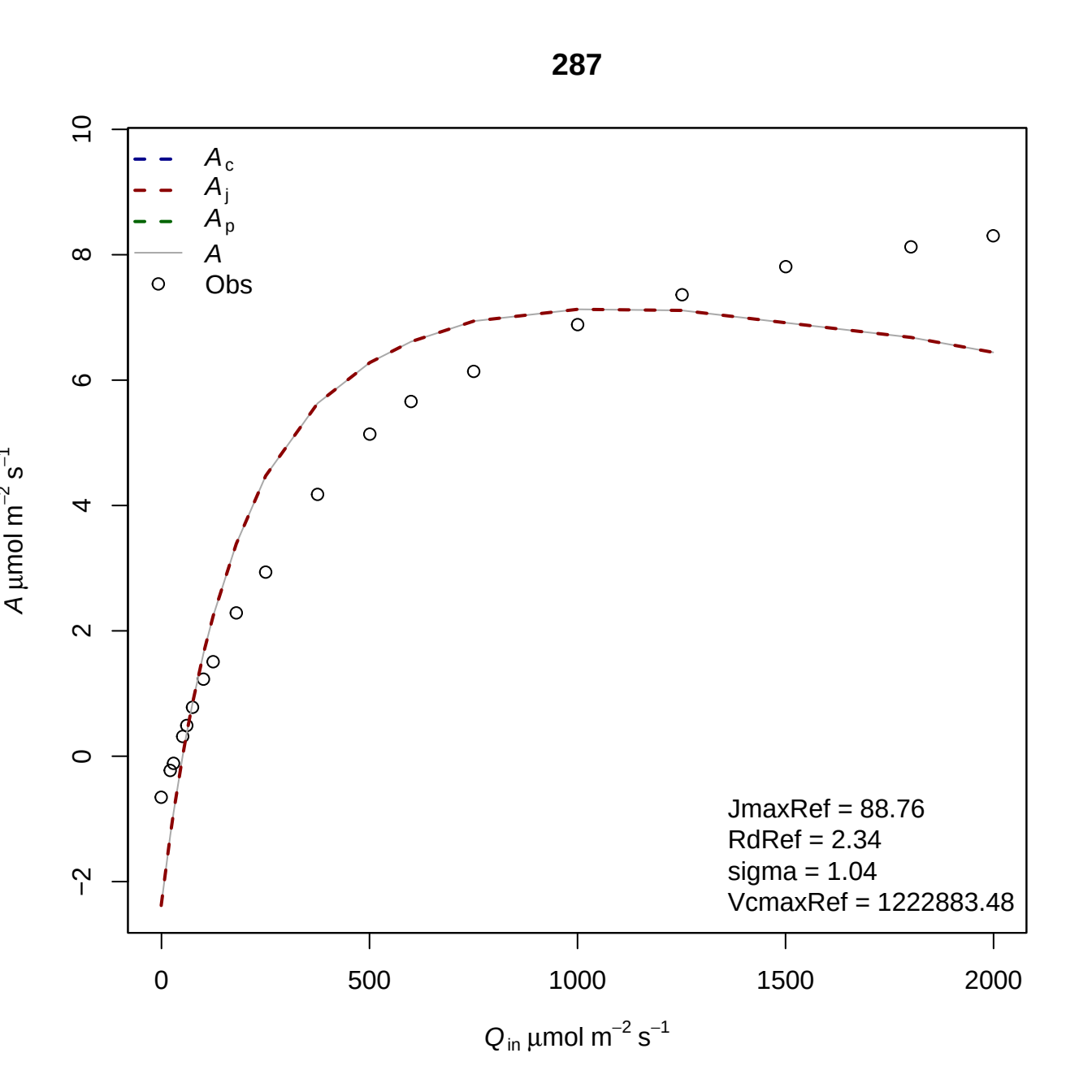

282
A_c
A_j
A_p
A
Obs
JmaxRef = 46.49
RdRef = 1.8
sigma = 0.73
VcmaxRef = 509827.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

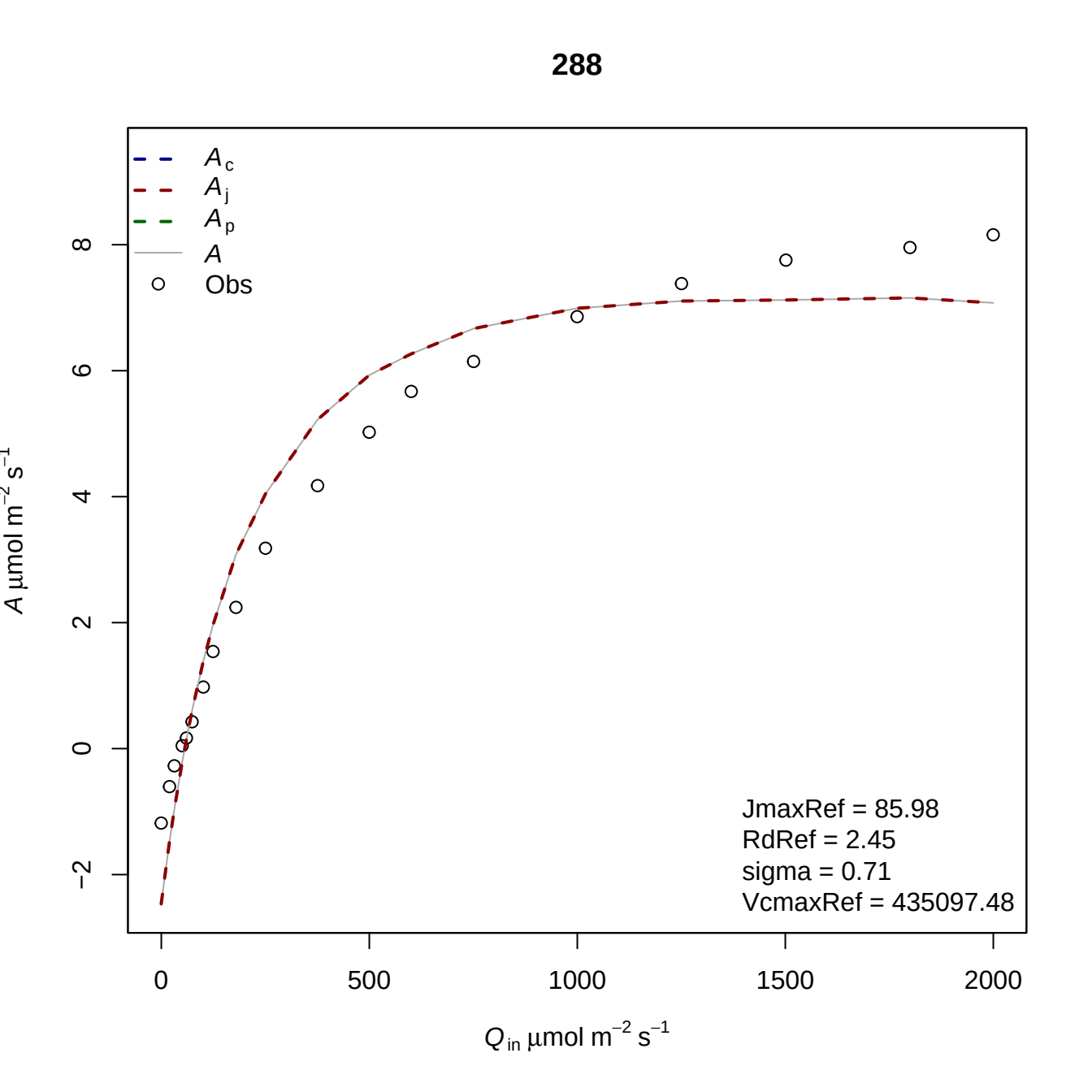

283
A_c
A_j
A_p
A
Obs
JmaxRef = 58.58
RdRef = 2.01
sigma = 0.83
VcmaxRef = 1123599.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

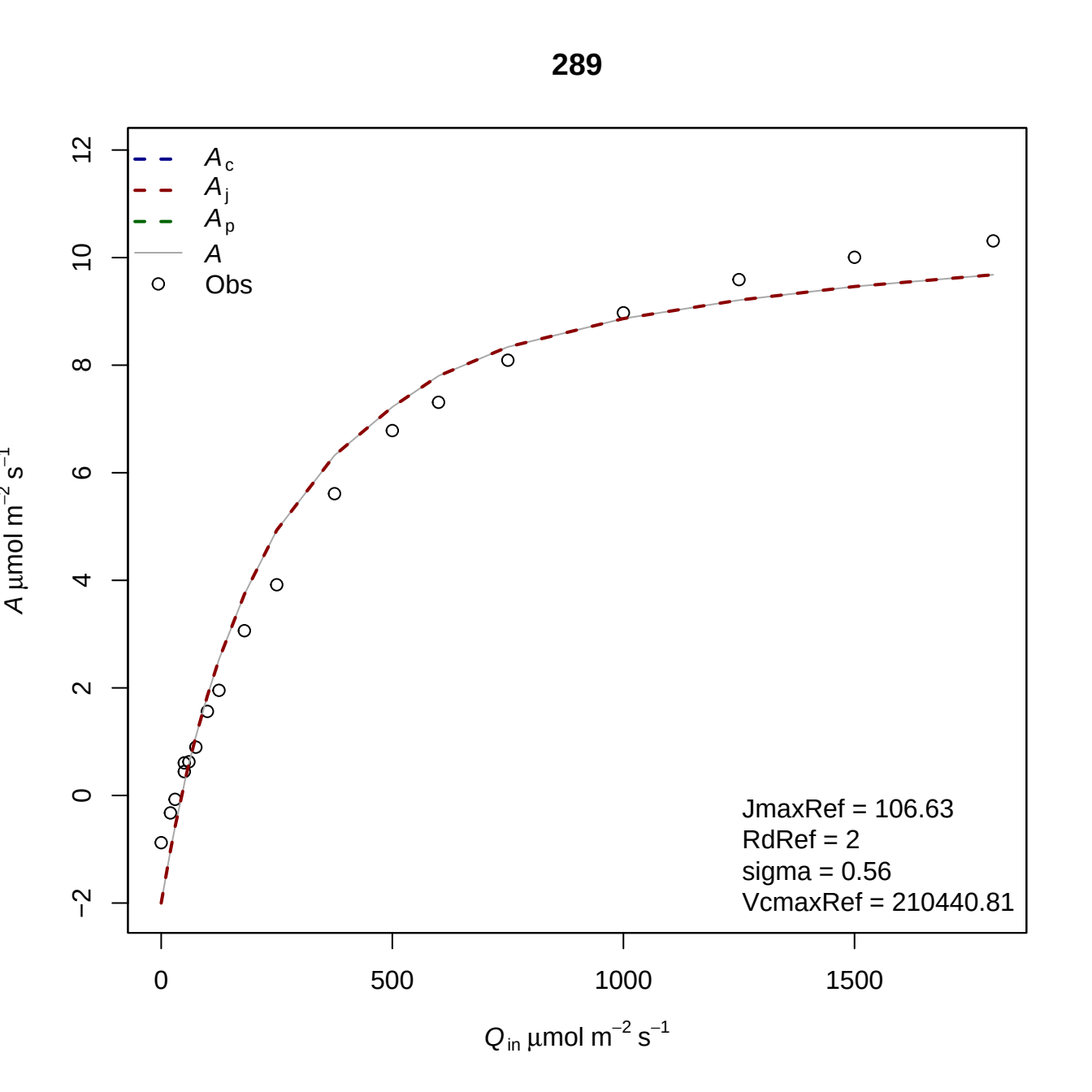

284
A_c
A_j
A_p
A
Obs
JmaxRef = 96.66
RdRef = 1.86
sigma = 0.49
VcmaxRef = 153823.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

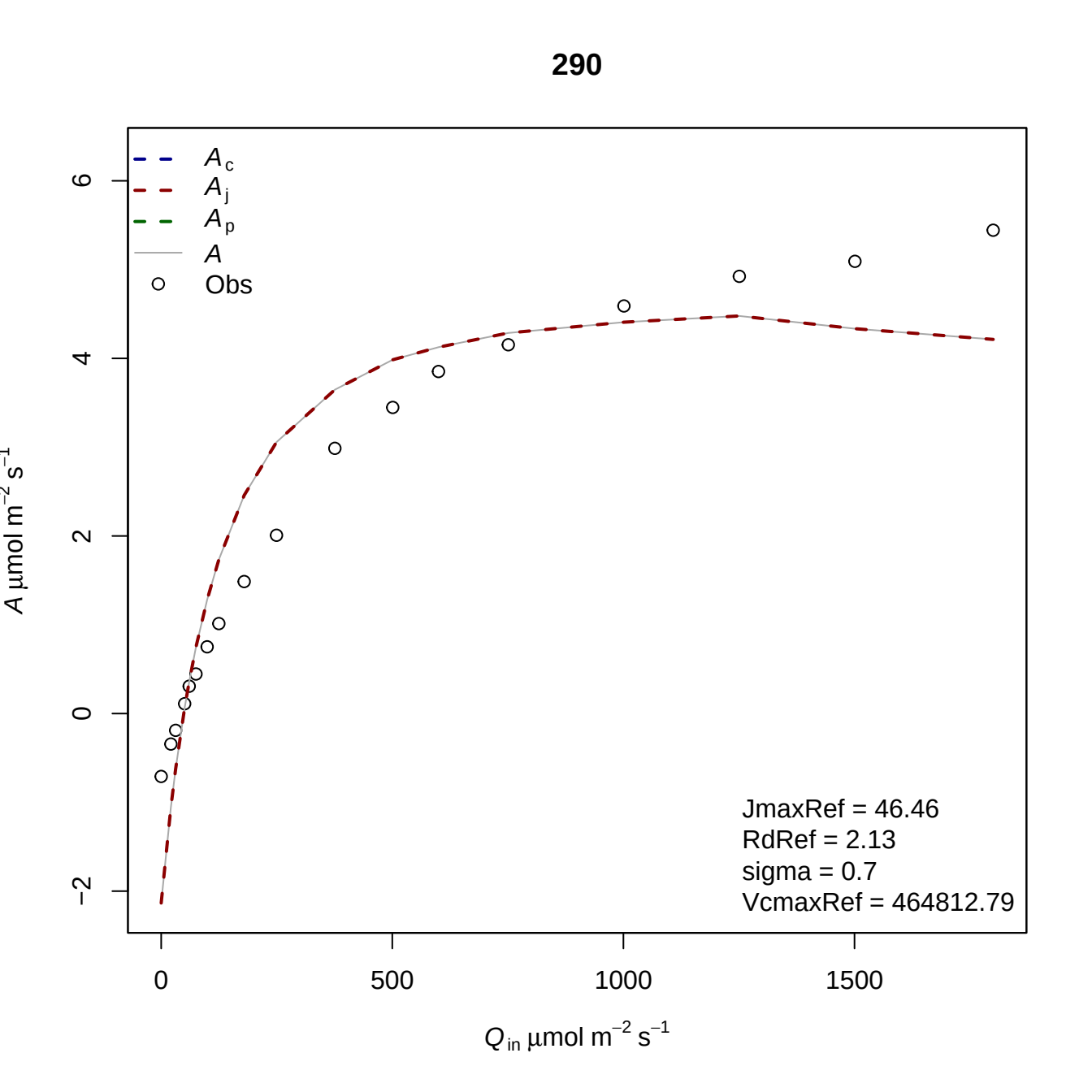

285
A_c
A_j
A_p
A
Obs
JmaxRef = 49.69
RdRef = 1.82
sigma = 0.42
VcmaxRef = 83940.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
5
4
3
2
1
0
−1
−2
0
500
1000
1500

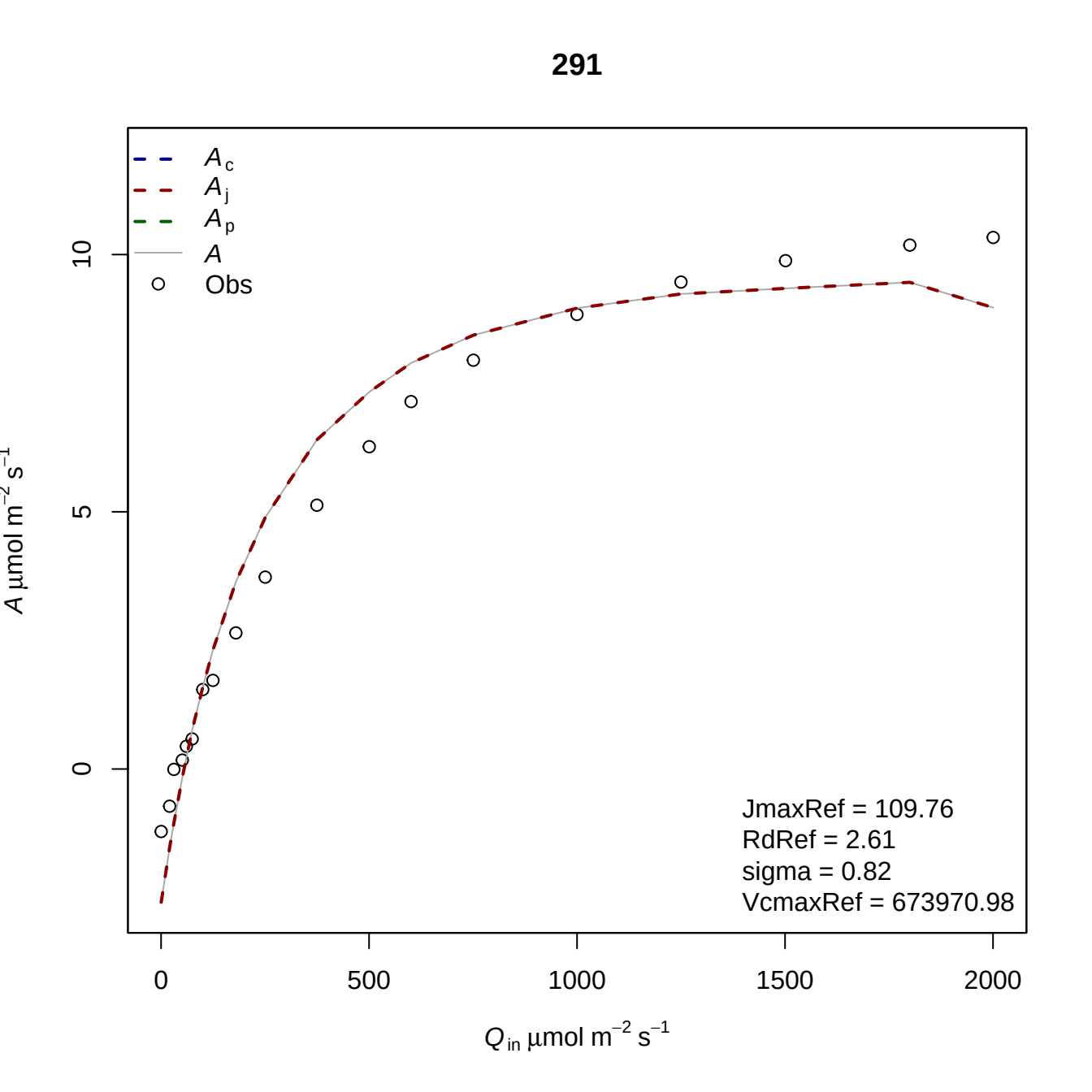

286
A_c
A_j
A_p
A
Obs
JmaxRef = 65.66
RdRef = 3.08
sigma = 0.59
VcmaxRef = 513488.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

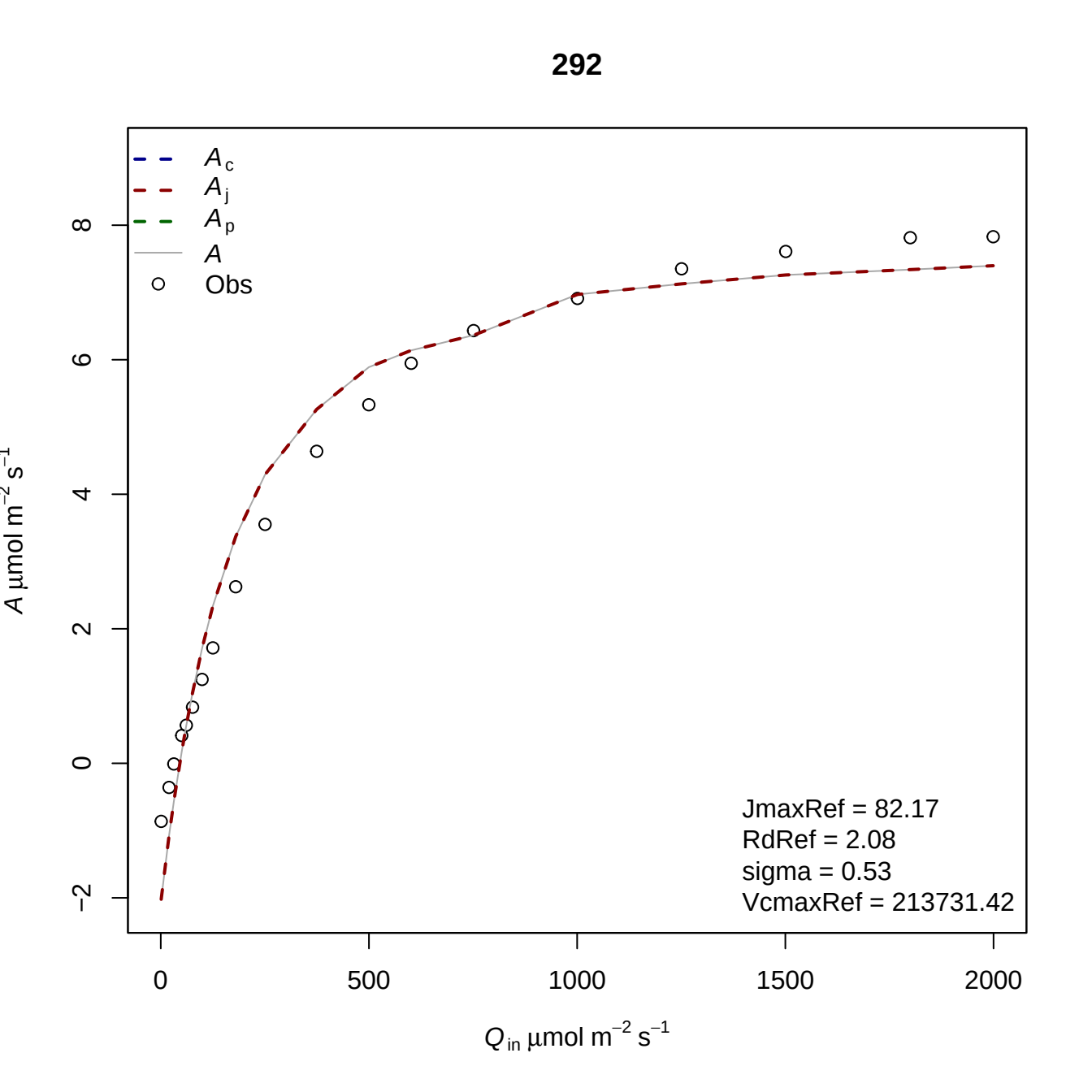

287
A_c
A_j
A_p
A
Obs
JmaxRef = 88.76
RdRef = 2.34
sigma = 1.04
VcmaxRef = 1222883.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

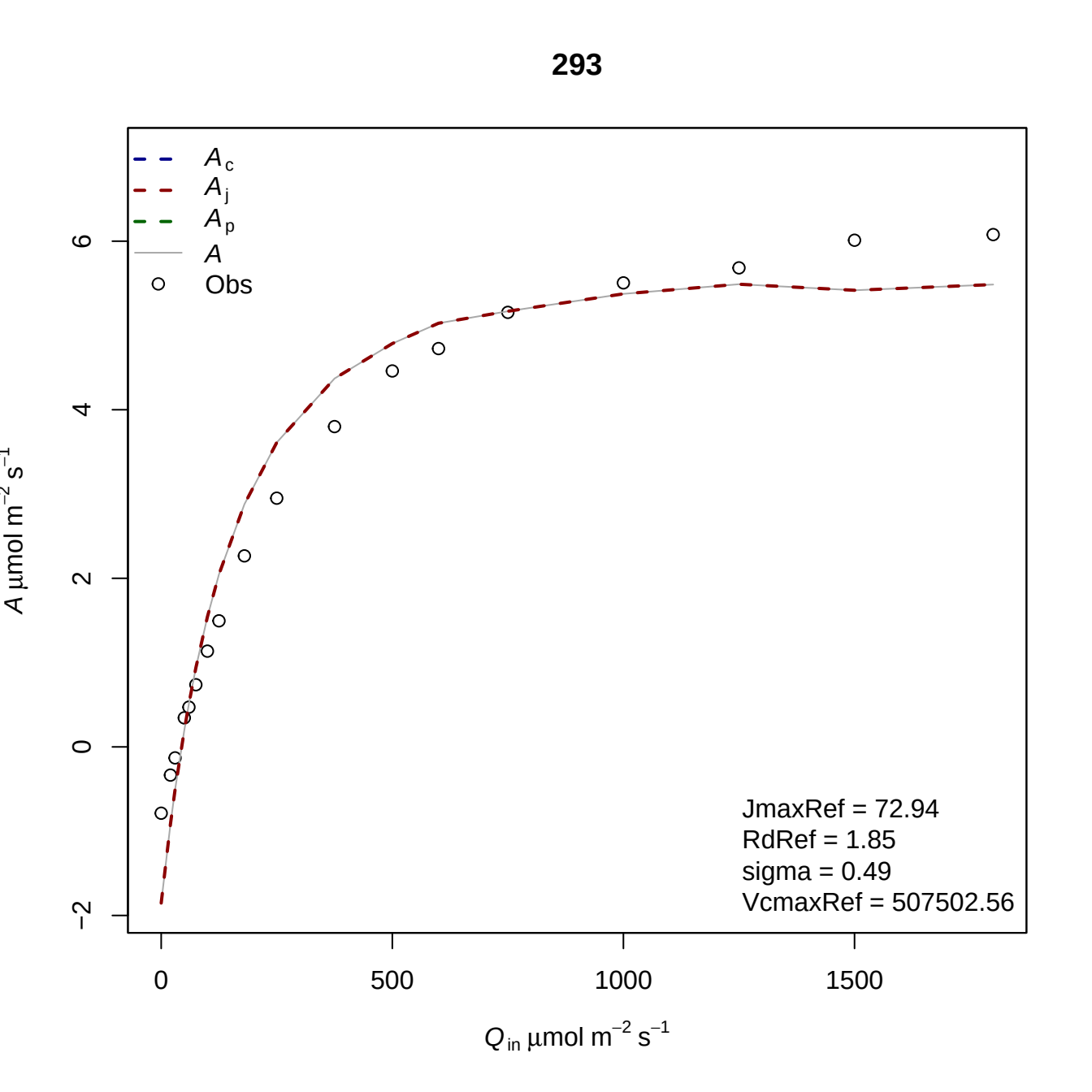

288
A_c
A_j
A_p
A
Obs
JmaxRef = 85.98
RdRef = 2.45
sigma = 0.71
VcmaxRef = 435097.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

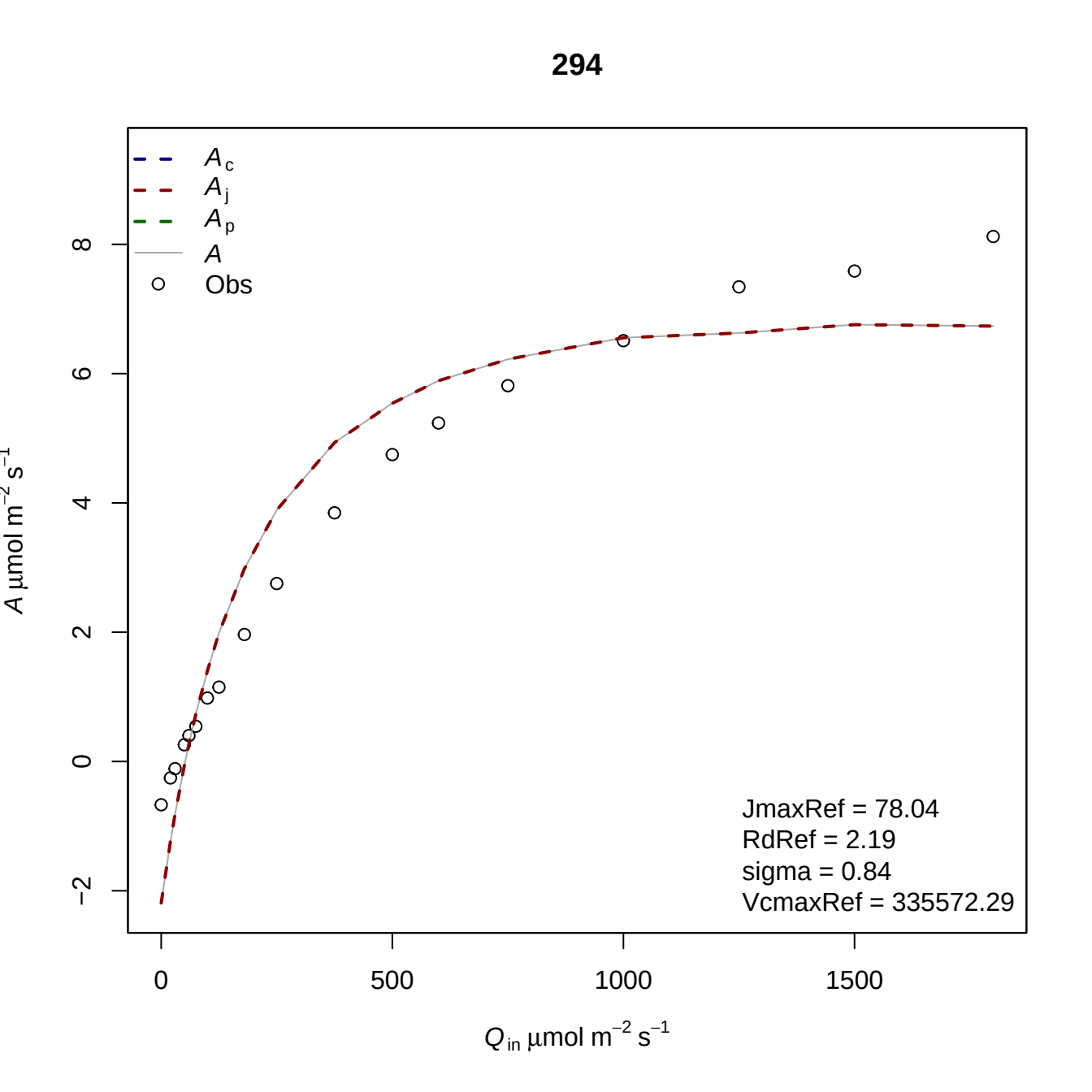

289
A_c
A_j
A_p
A
Obs
JmaxRef = 106.63
RdRef = 2
sigma = 0.56
VcmaxRef = 210440.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

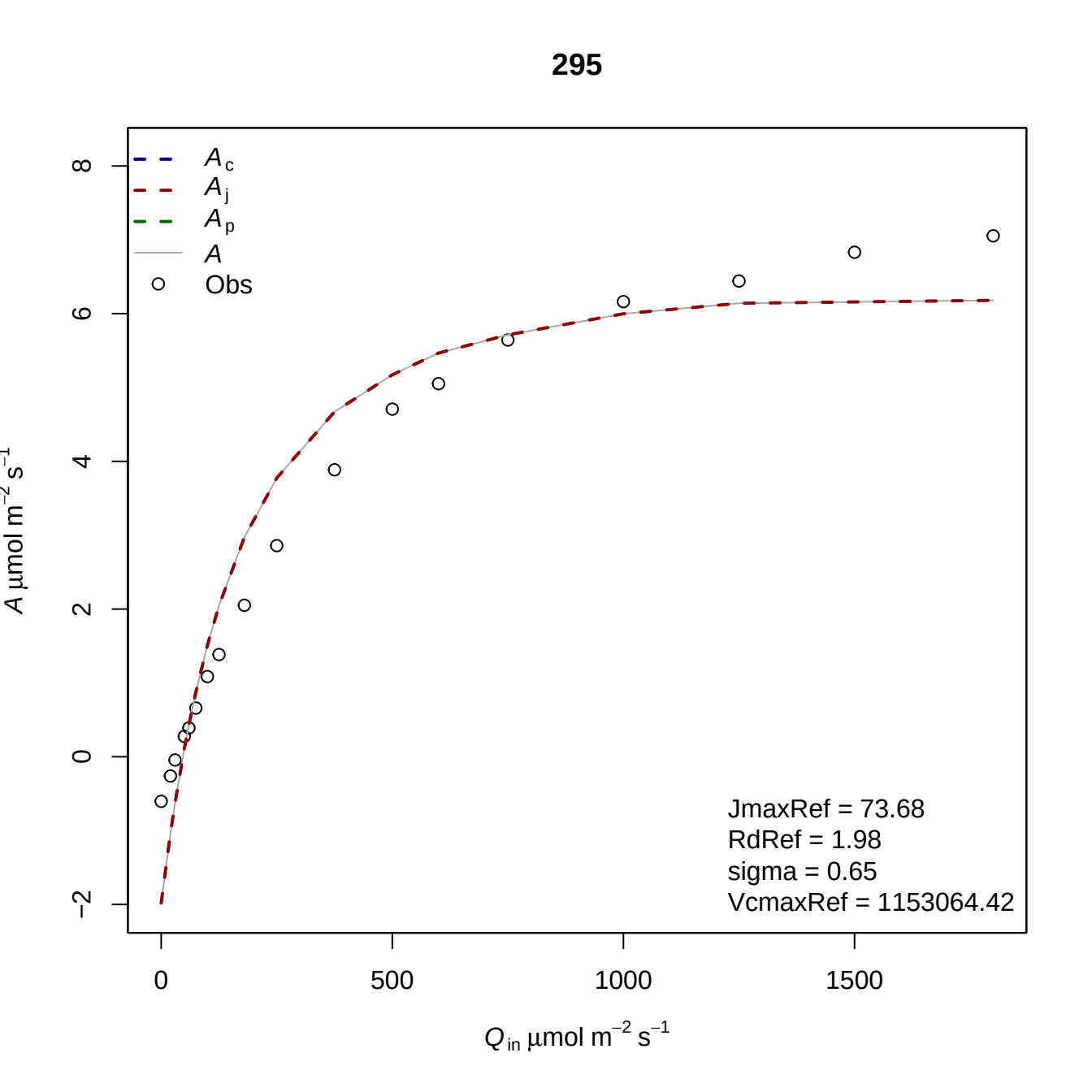

290
A_c
A_j
A_p
A
Obs
JmaxRef = 46.46
RdRef = 2.13
sigma = 0.7
VcmaxRef = 464812.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

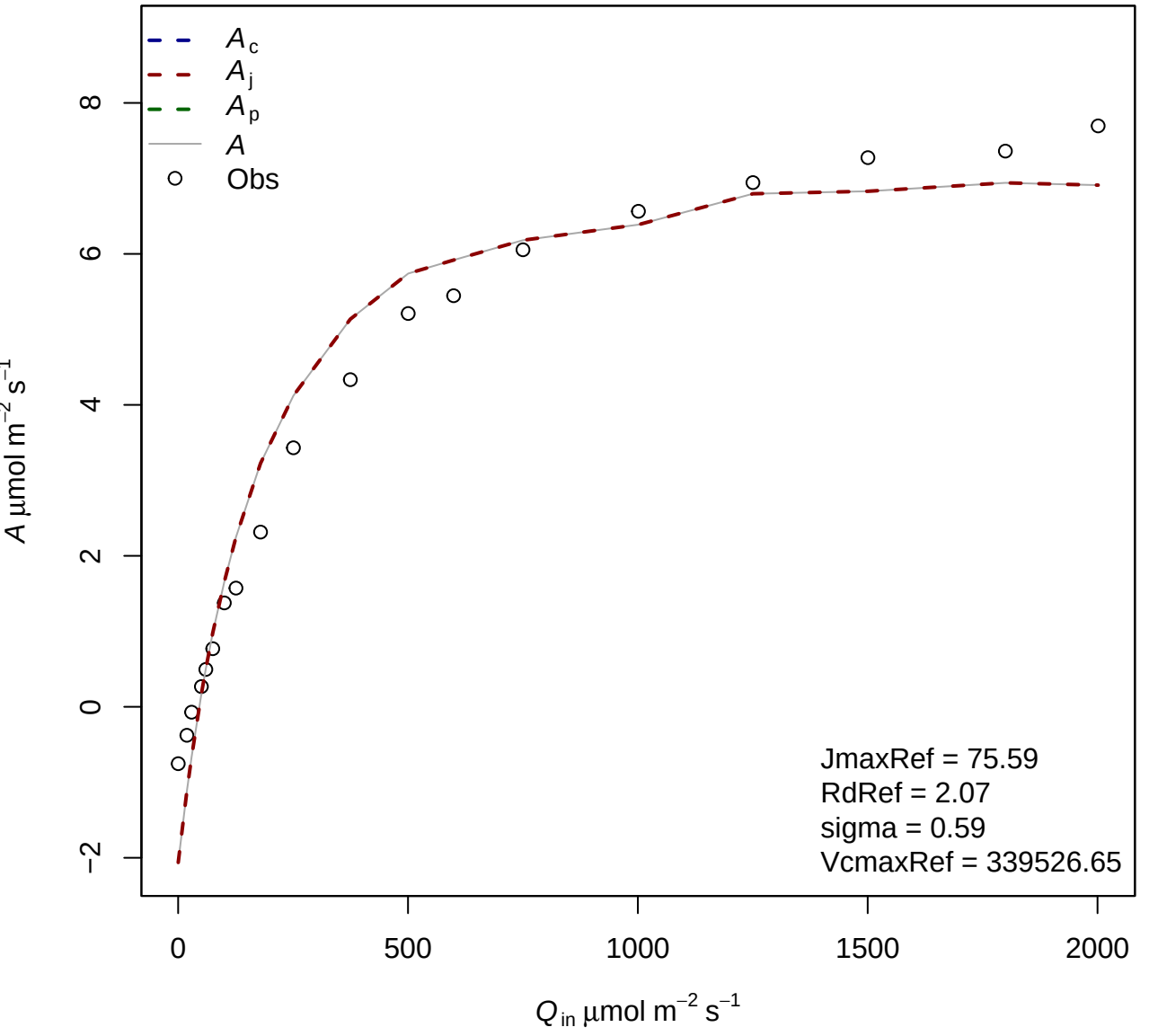

291
A_c
A_j
A_p
A
Obs
JmaxRef = 109.76
RdRef = 2.61
sigma = 0.82
VcmaxRef = 673970.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

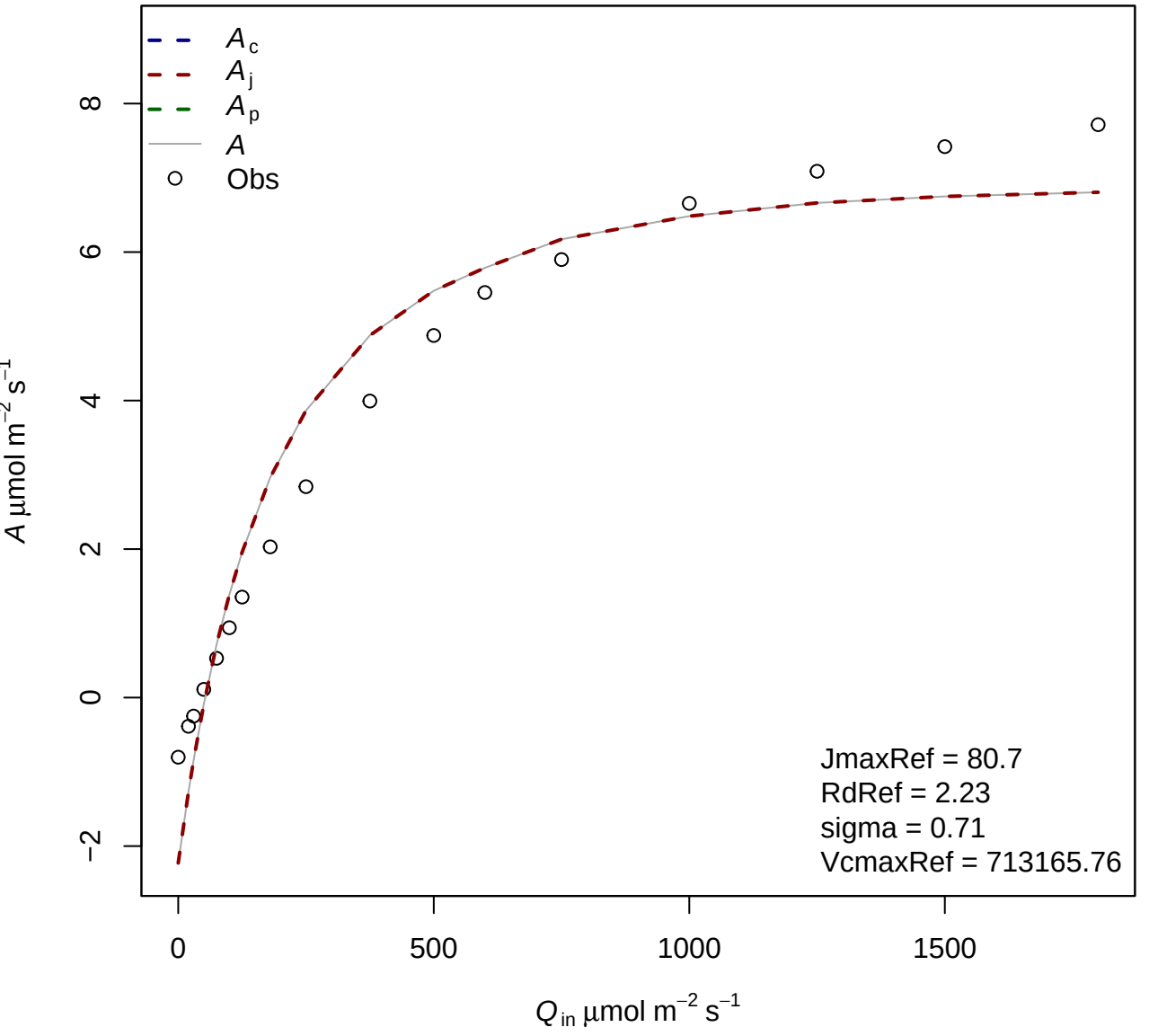

292
A_c
A_j
A_p
A
Obs
JmaxRef = 82.17
RdRef = 2.08
sigma = 0.53
VcmaxRef = 213731.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

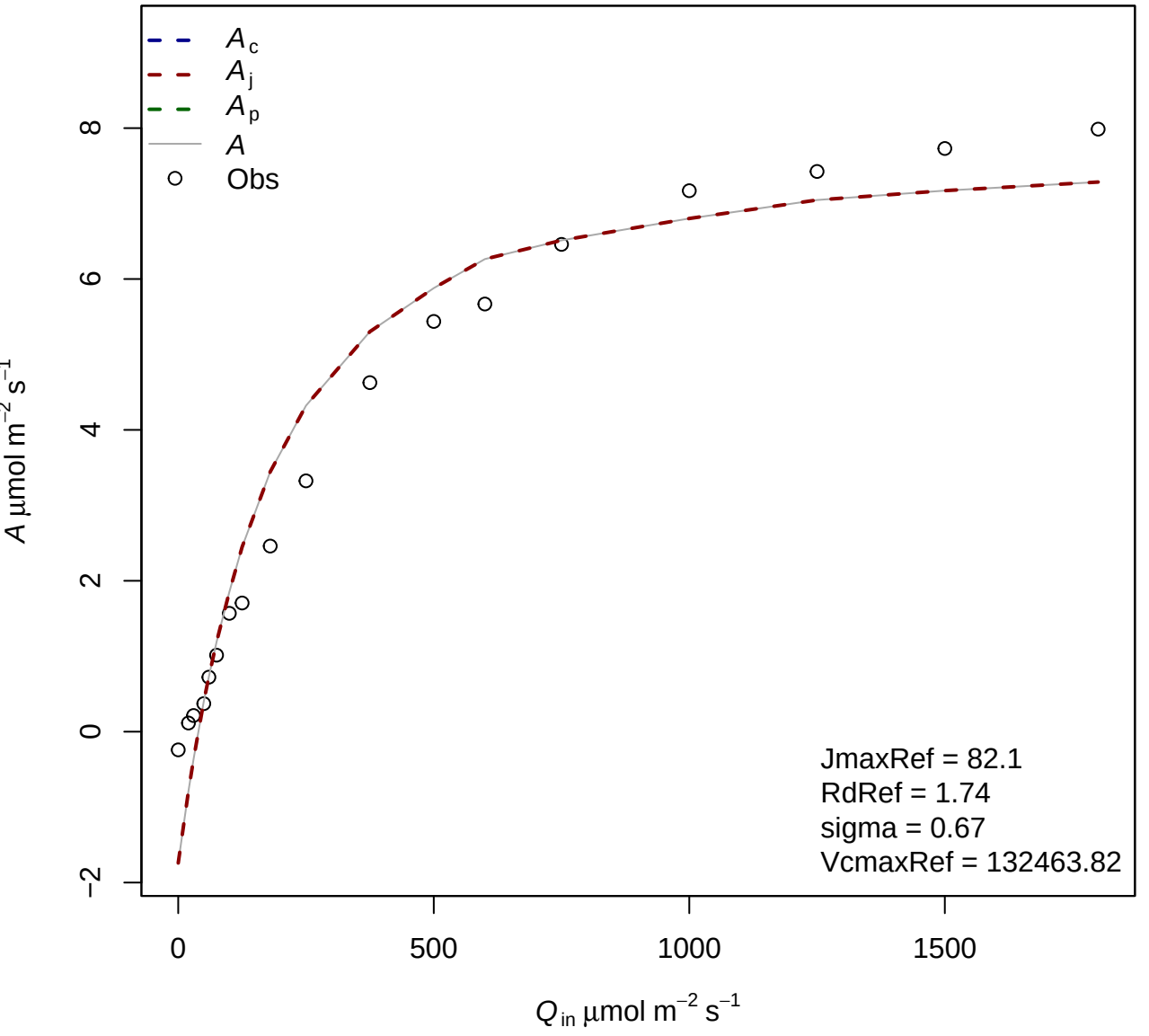

293
A_c
A_j
A_p
A
Obs
JmaxRef = 72.94
RdRef = 1.85
sigma = 0.49
VcmaxRef = 507502.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

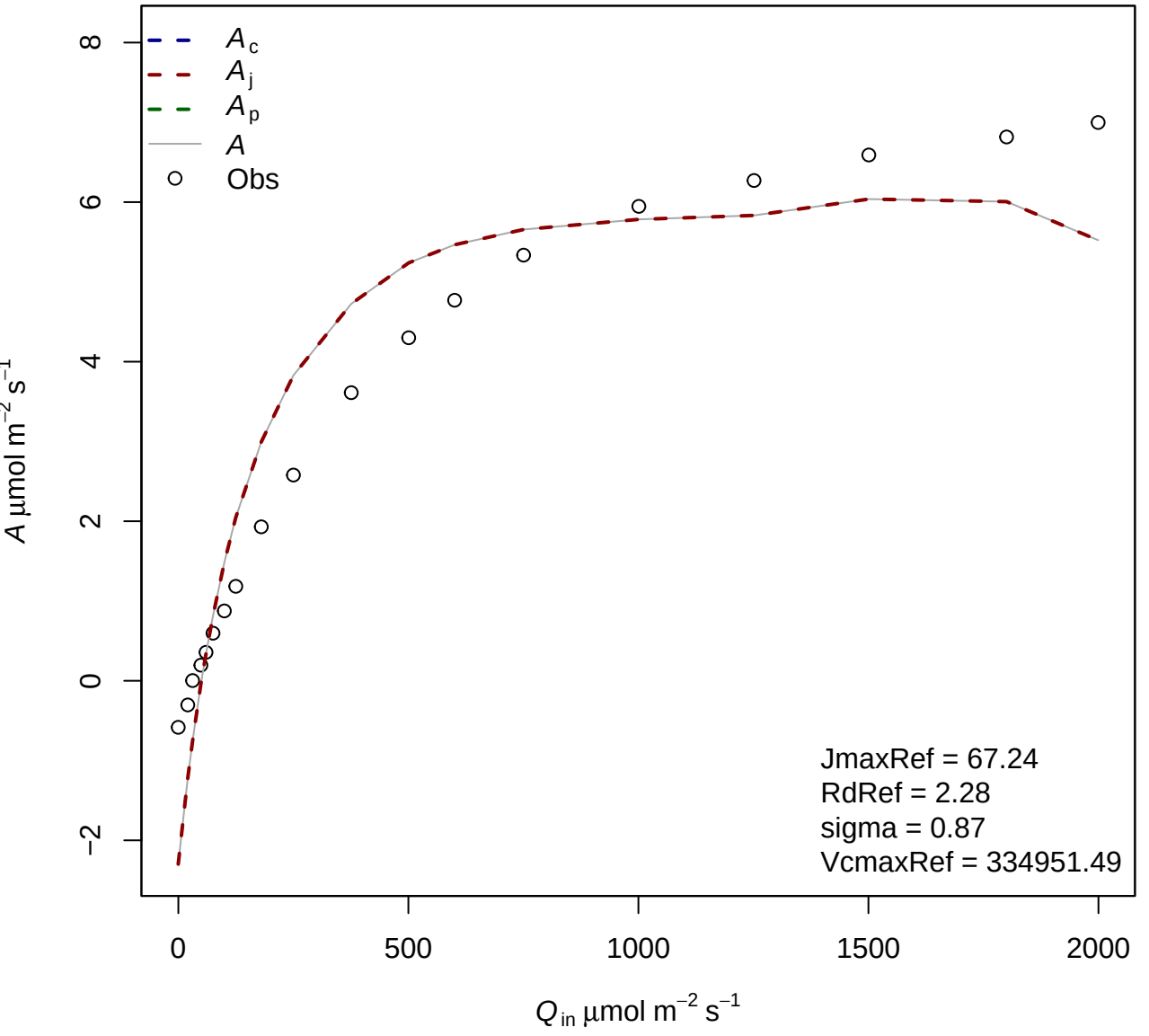

294
A_c
A_j
A_p
A
Obs
JmaxRef = 78.04
RdRef = 2.19
sigma = 0.84
VcmaxRef = 335572.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

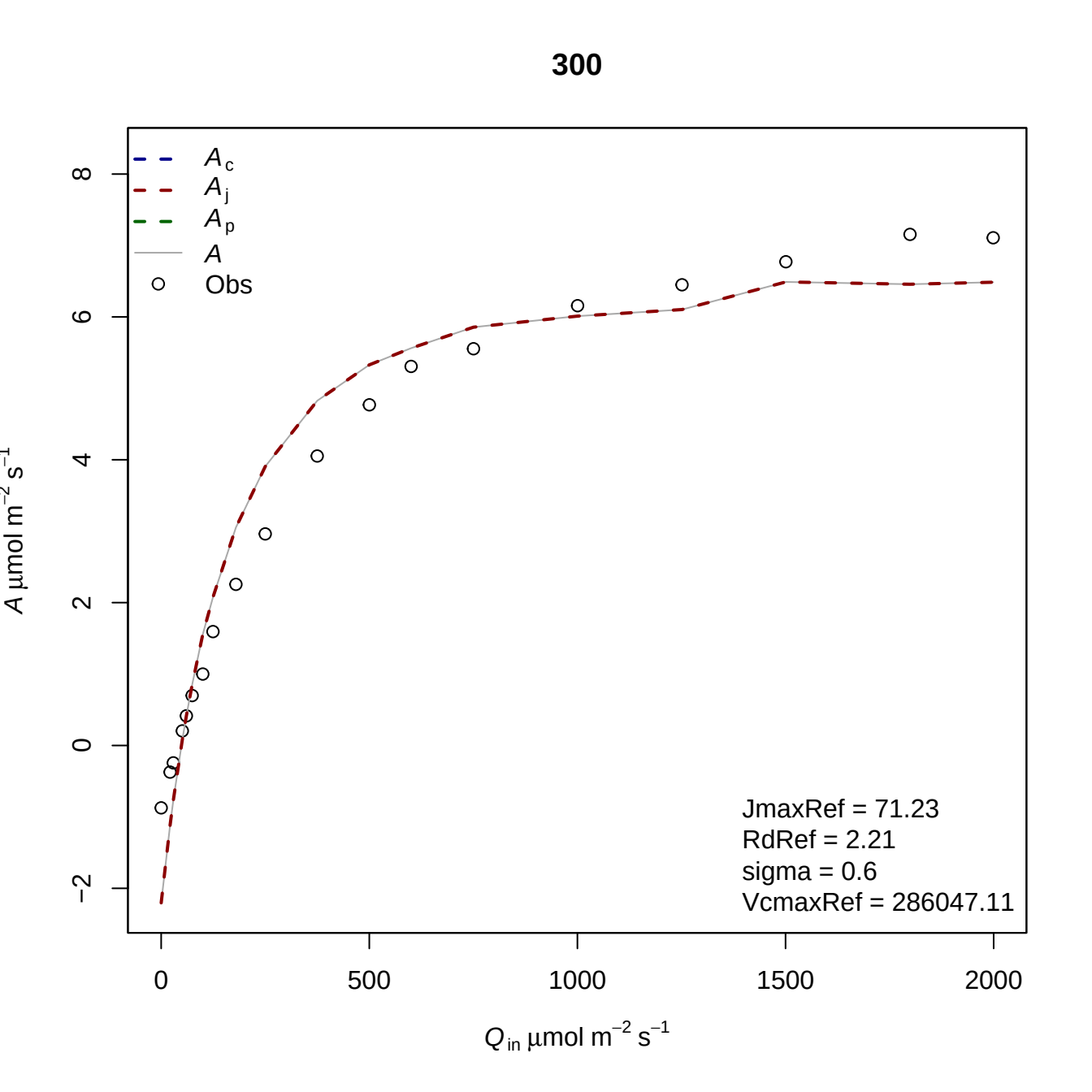

295
A_c
A_j
A_p
A
Obs
JmaxRef = 73.68
RdRef = 1.98
sigma = 0.65
VcmaxRef = 1153064.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

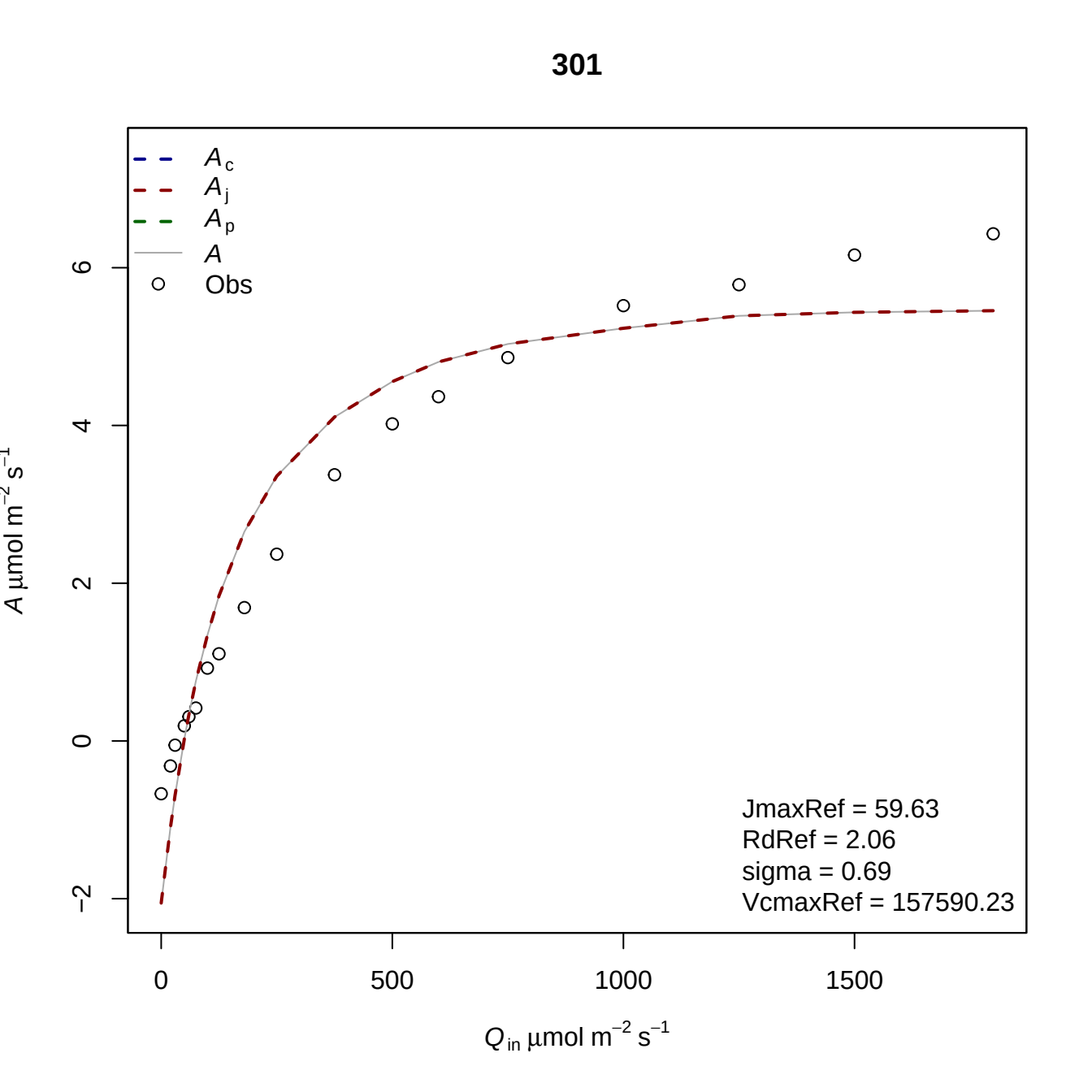

A_c
A_j
A_p
A
Obs
JmaxRef = 75.59
RdRef = 2.07
sigma = 0.59
VcmaxRef = 339526.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

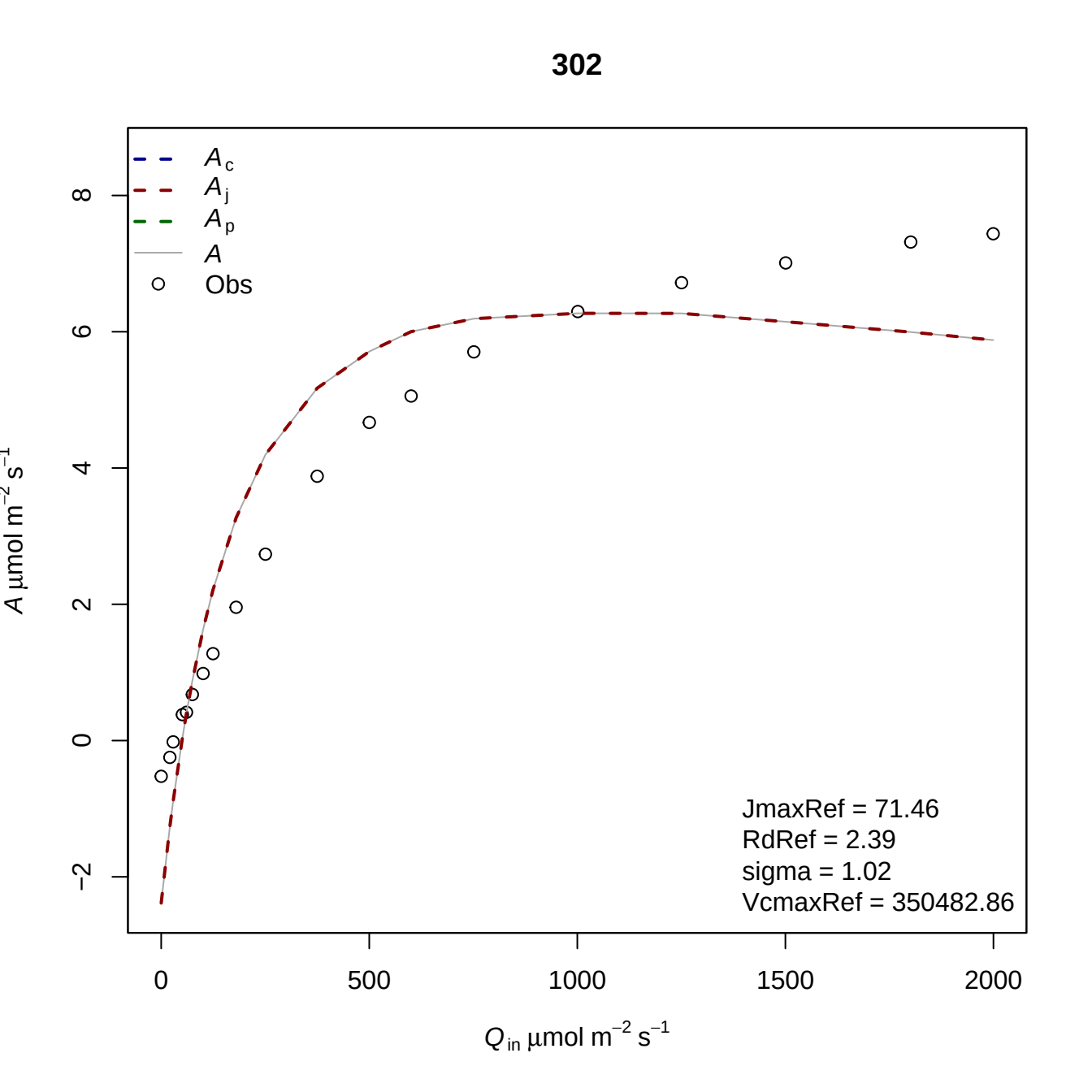

A_c
A_j
A_p
A
Obs
JmaxRef = 80.7
RdRef = 2.23
sigma = 0.71
VcmaxRef = 713165.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

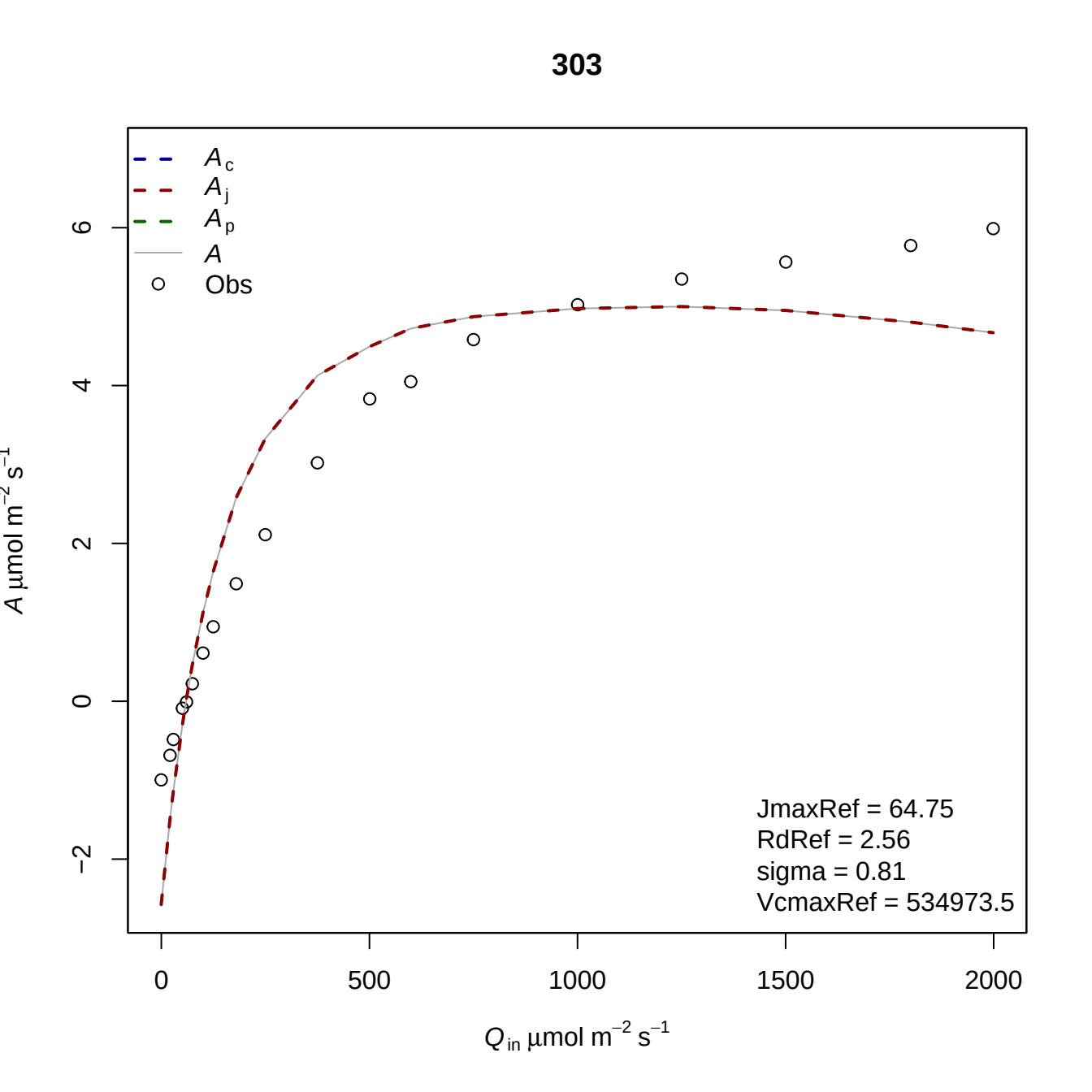

A_c
A_j
A_p
A
Obs
JmaxRef = 82.1
RdRef = 1.74
sigma = 0.67
VcmaxRef = 132463.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

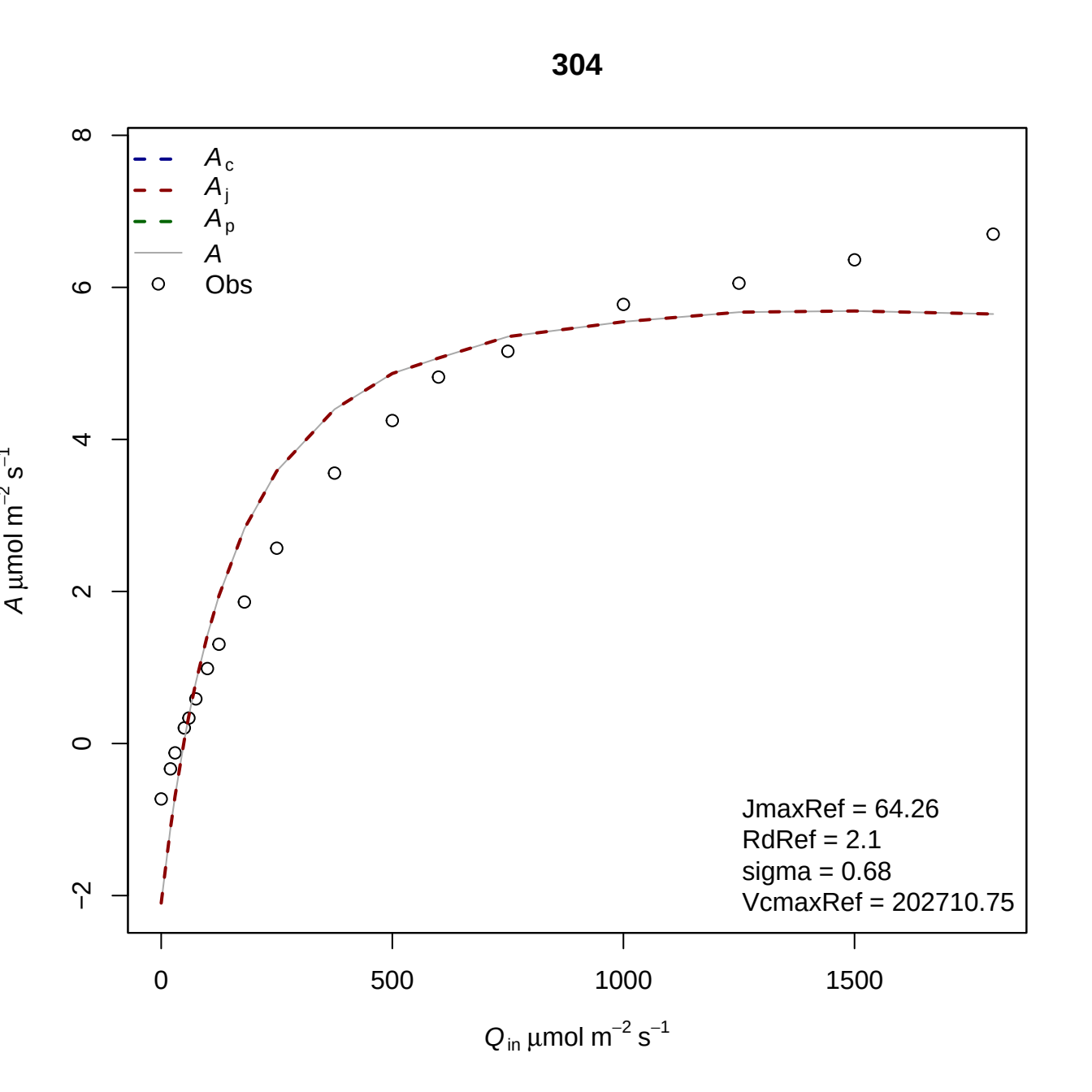

A_c
A_j
A_p
A
Obs
JmaxRef = 67.24
RdRef = 2.28
sigma = 0.87
VcmaxRef = 334951.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

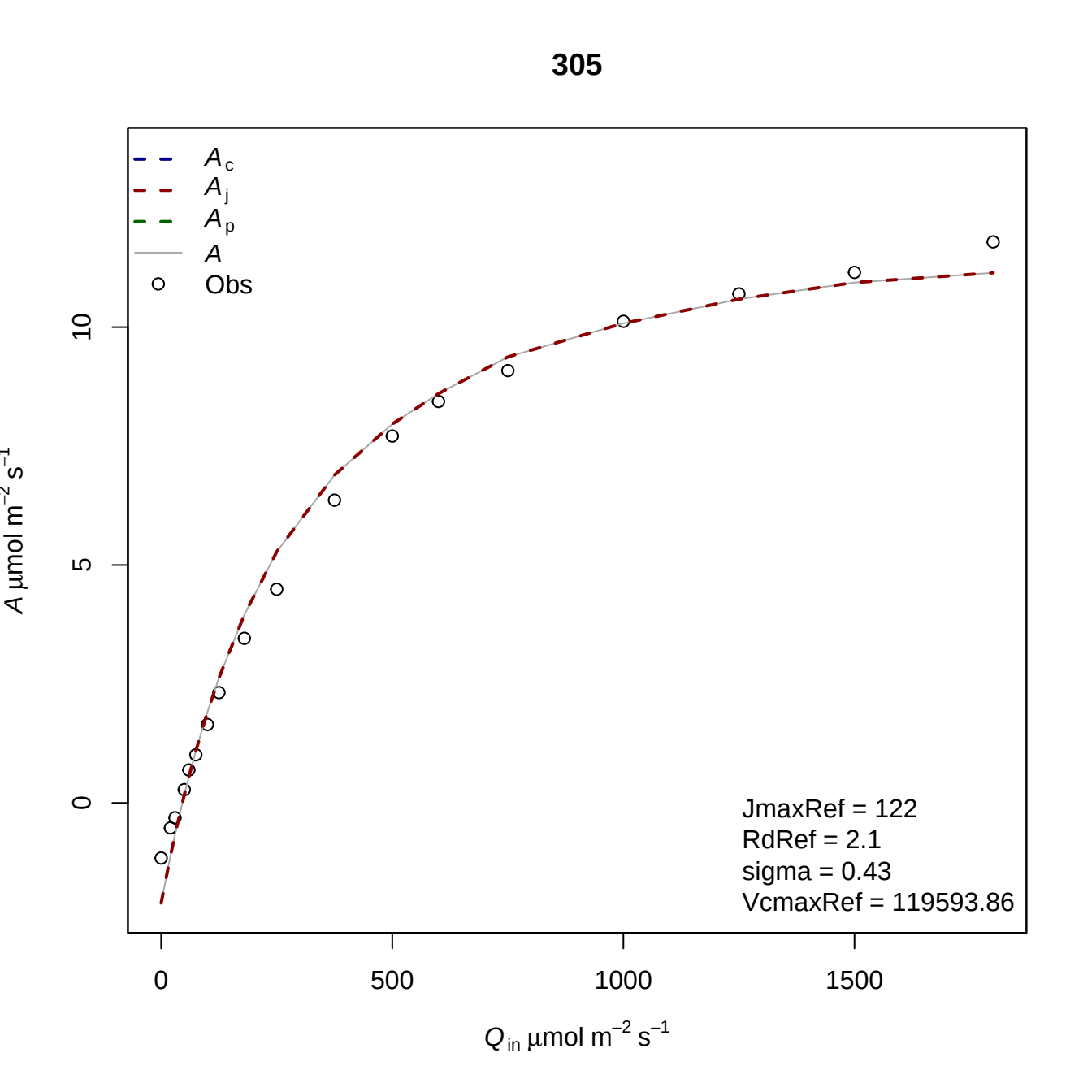

300
A_c
A_j
A_p
A
Obs
JmaxRef = 71.23
RdRef = 2.21
sigma = 0.6
VcmaxRef = 286047.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

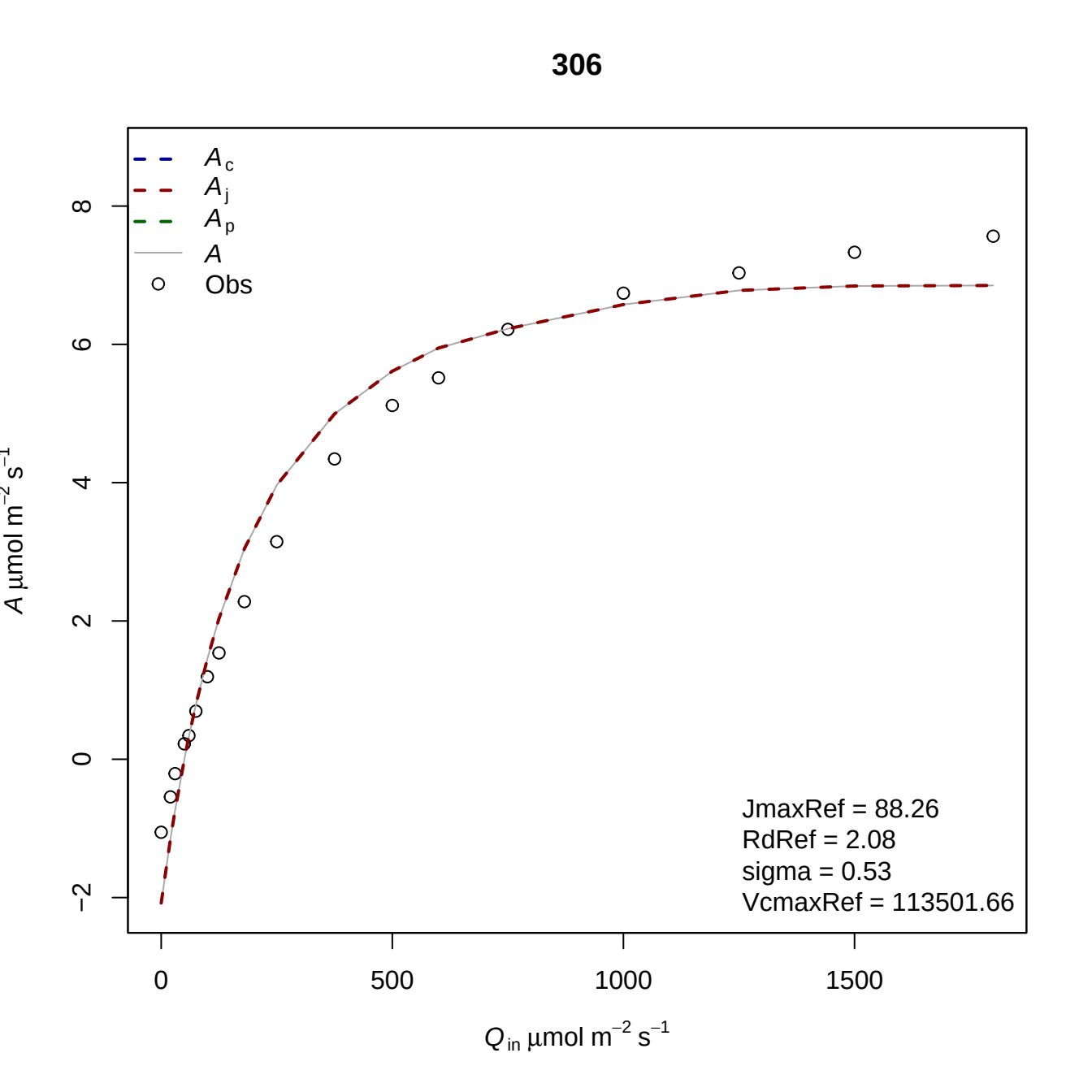

301
A_c
A_j
A_p
A
Obs
JmaxRef = 59.63
RdRef = 2.06
sigma = 0.69
VcmaxRef = 157590.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

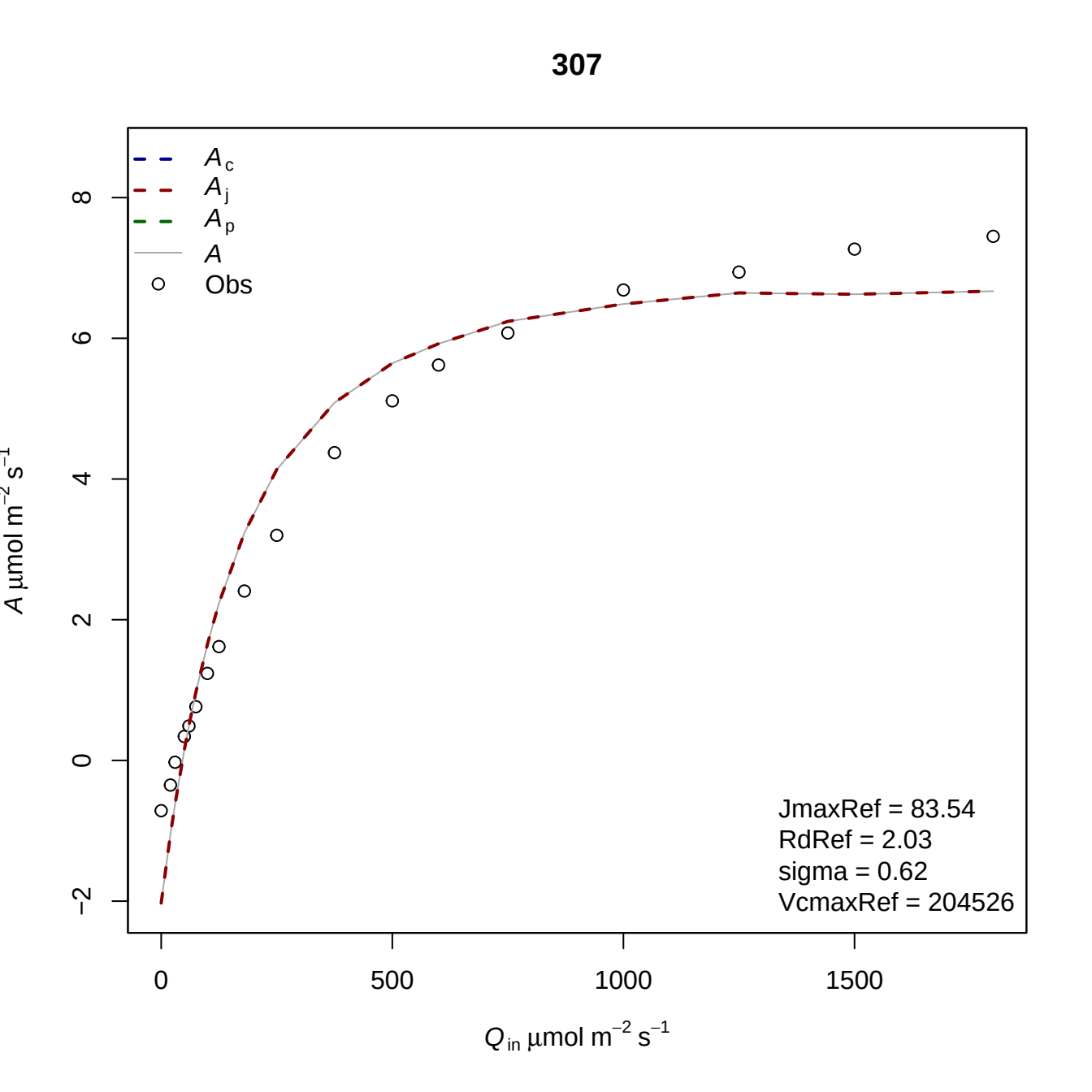

302
A_c
A_j
A_p
A
Obs
JmaxRef = 71.46
RdRef = 2.39
sigma = 1.02
VcmaxRef = 350482.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

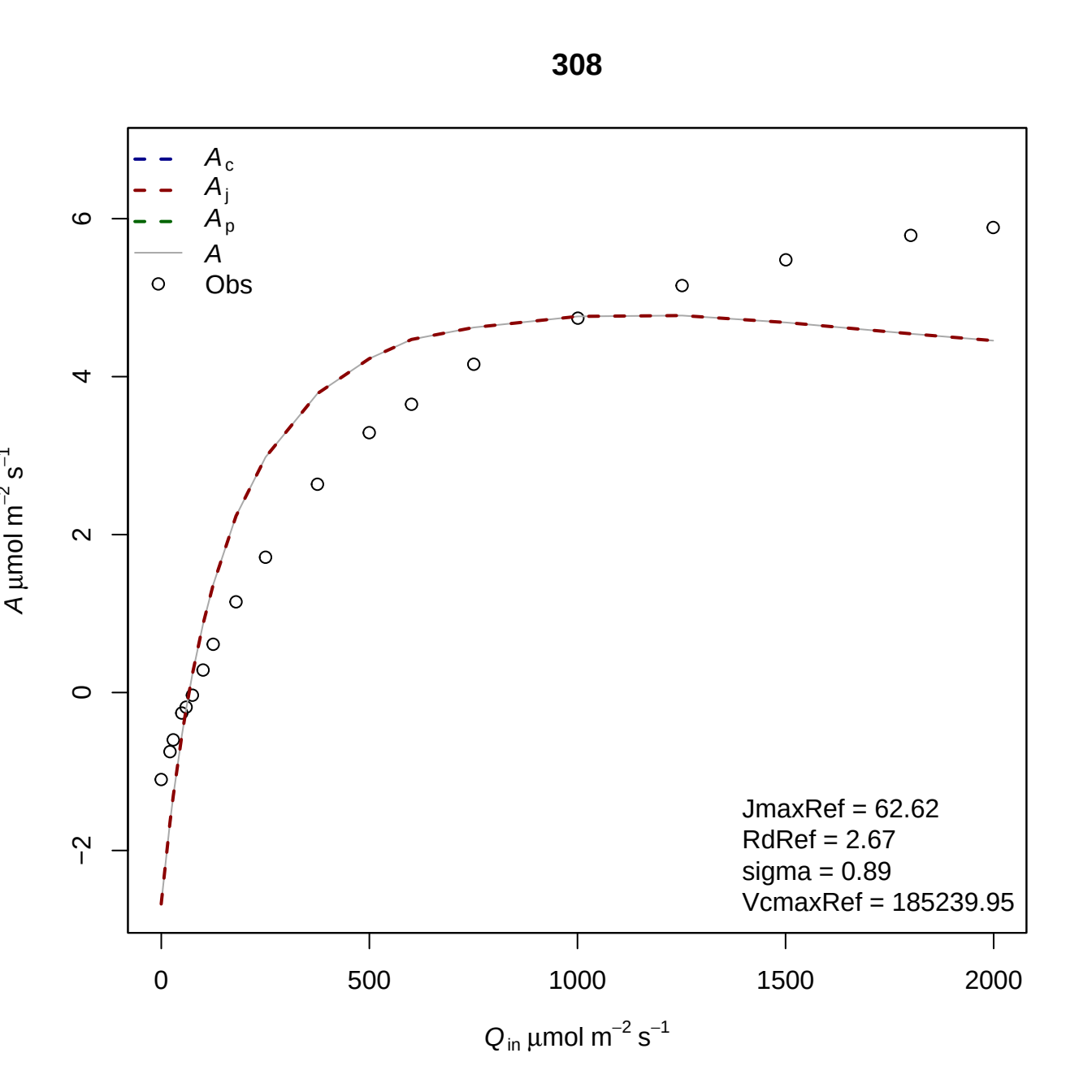

303
A_c
A_j
A_p
A
Obs
JmaxRef = 64.75
RdRef = 2.56
sigma = 0.81
VcmaxRef = 534973.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

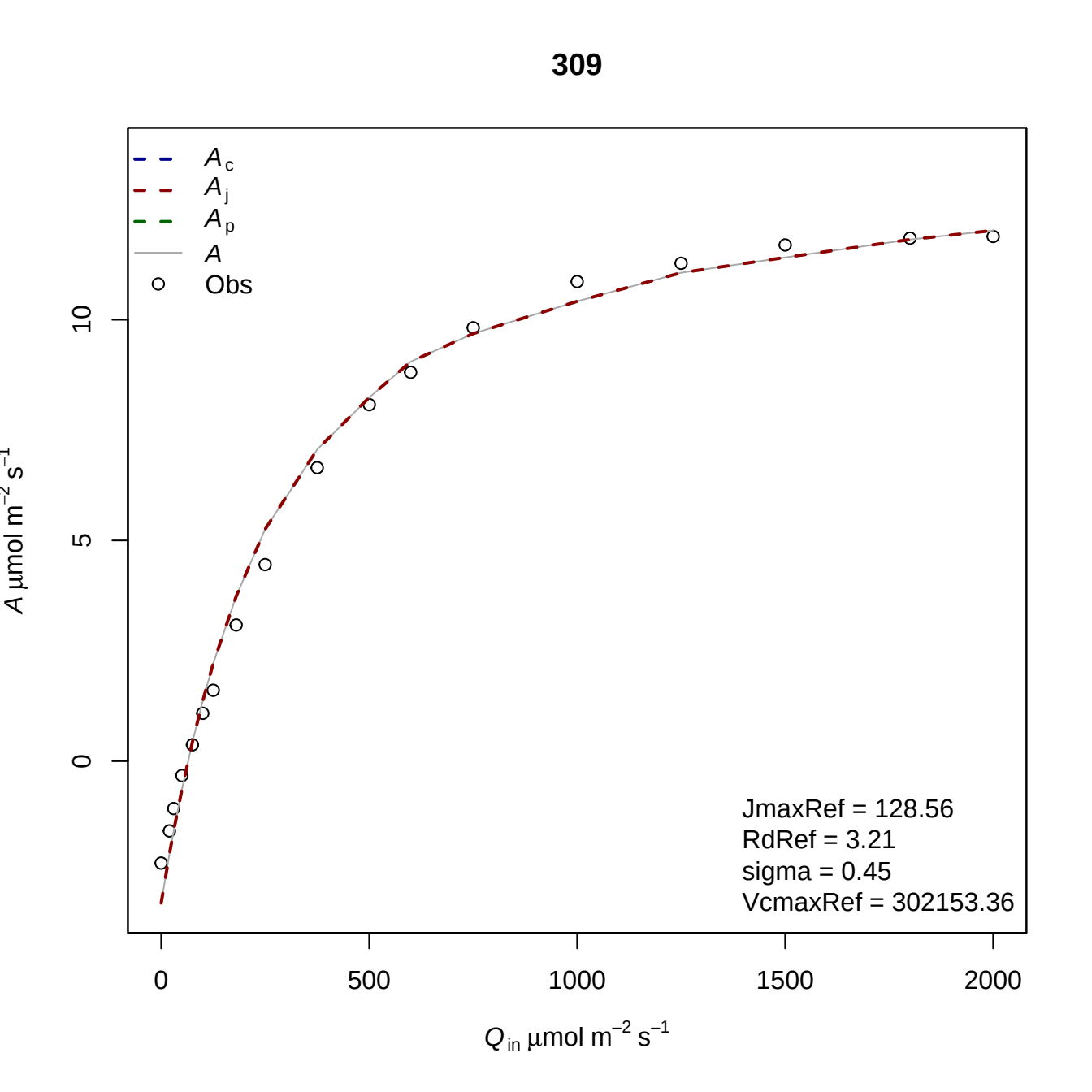

304
A_c
A_j
A_p
A
Obs
JmaxRef = 64.26
RdRef = 2.1
sigma = 0.68
VcmaxRef = 202710.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

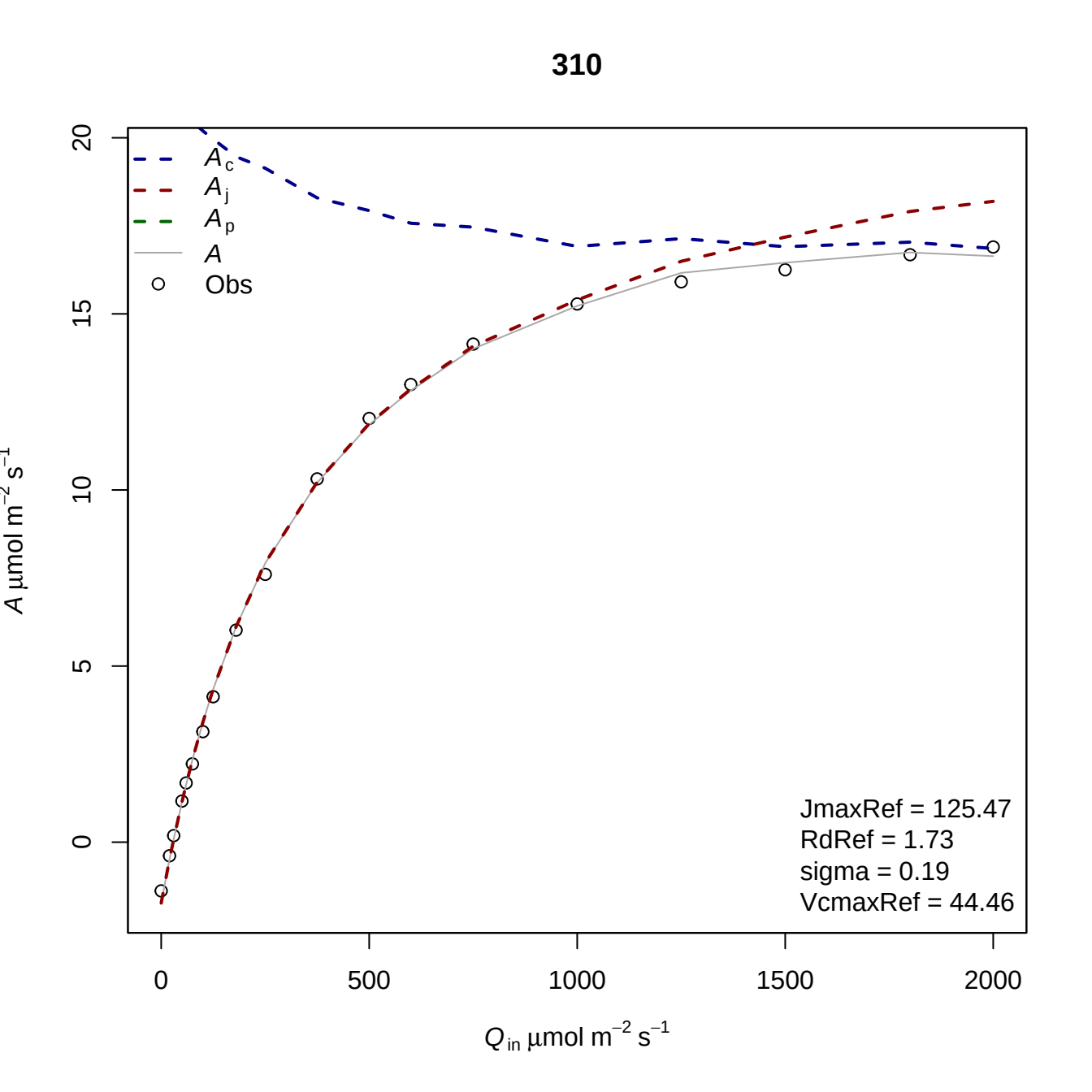

305
A_c
A_j
A_p
A
Obs
JmaxRef = 122
RdRef = 2.1
sigma = 0.43
VcmaxRef = 119593.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

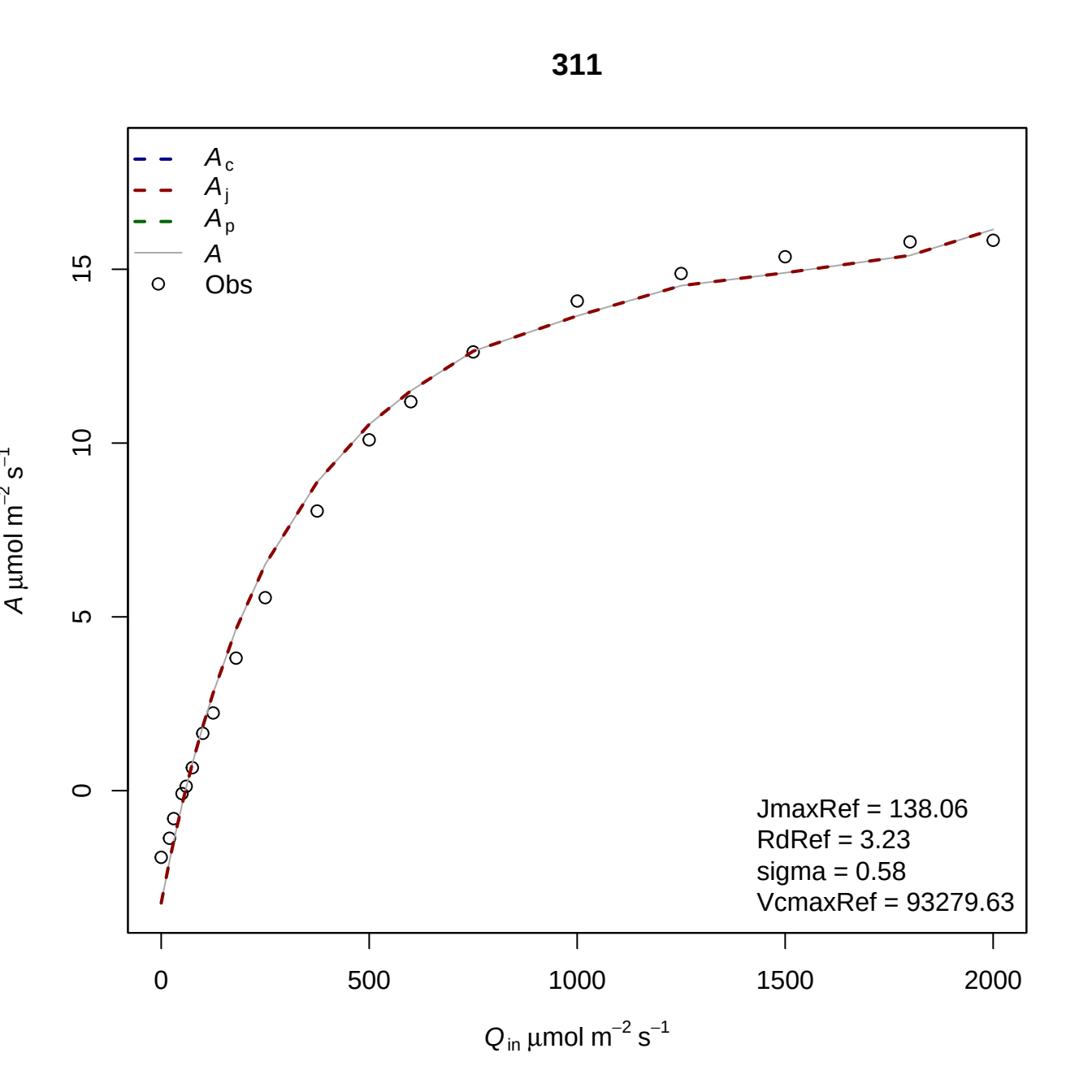

306
A_c
A_j
A_p
A
Obs
JmaxRef = 88.26
RdRef = 2.08
sigma = 0.53
VcmaxRef = 113501.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

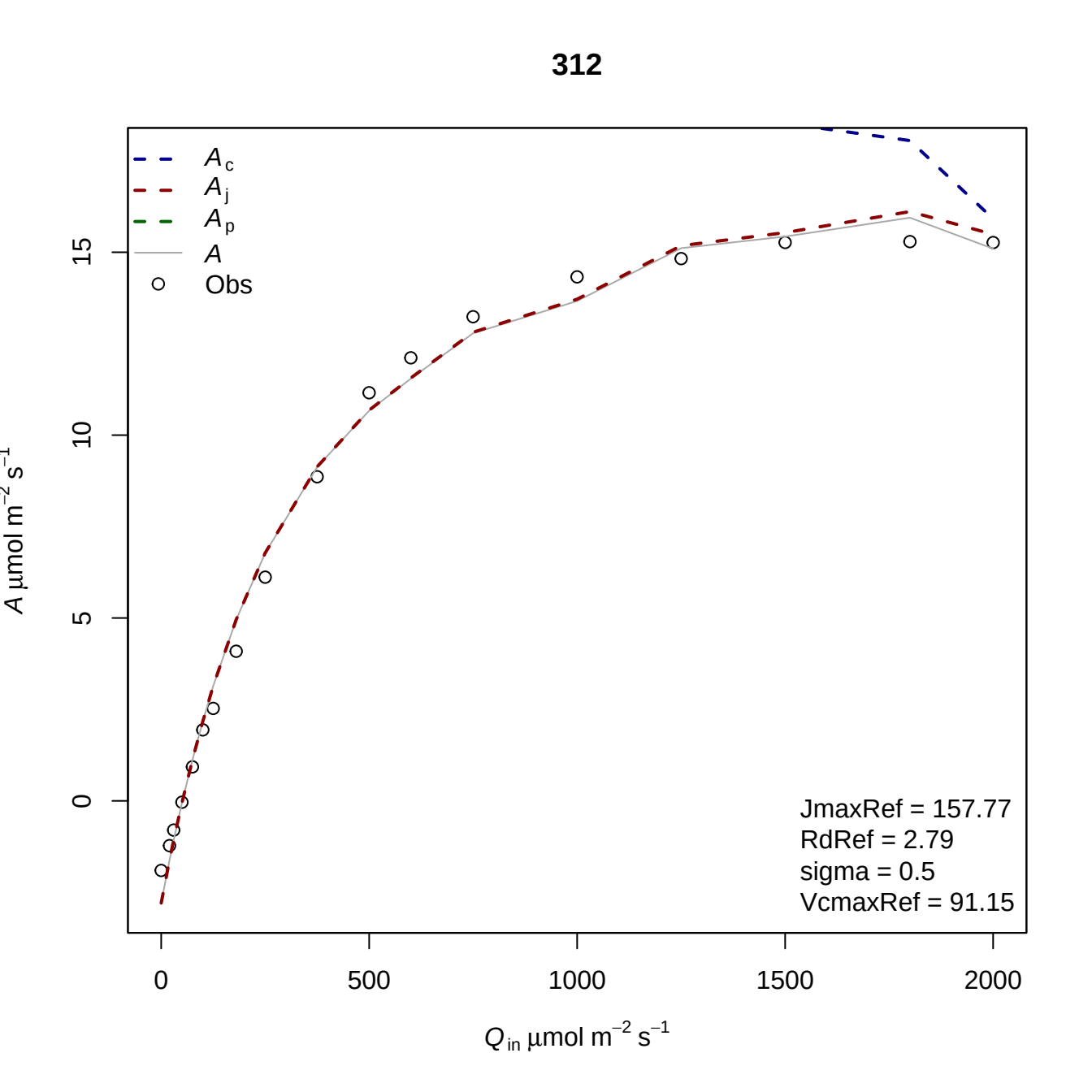

307
A_c
A_j
A_p
A
Obs
JmaxRef = 83.54
RdRef = 2.03
sigma = 0.62
VcmaxRef = 204526
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

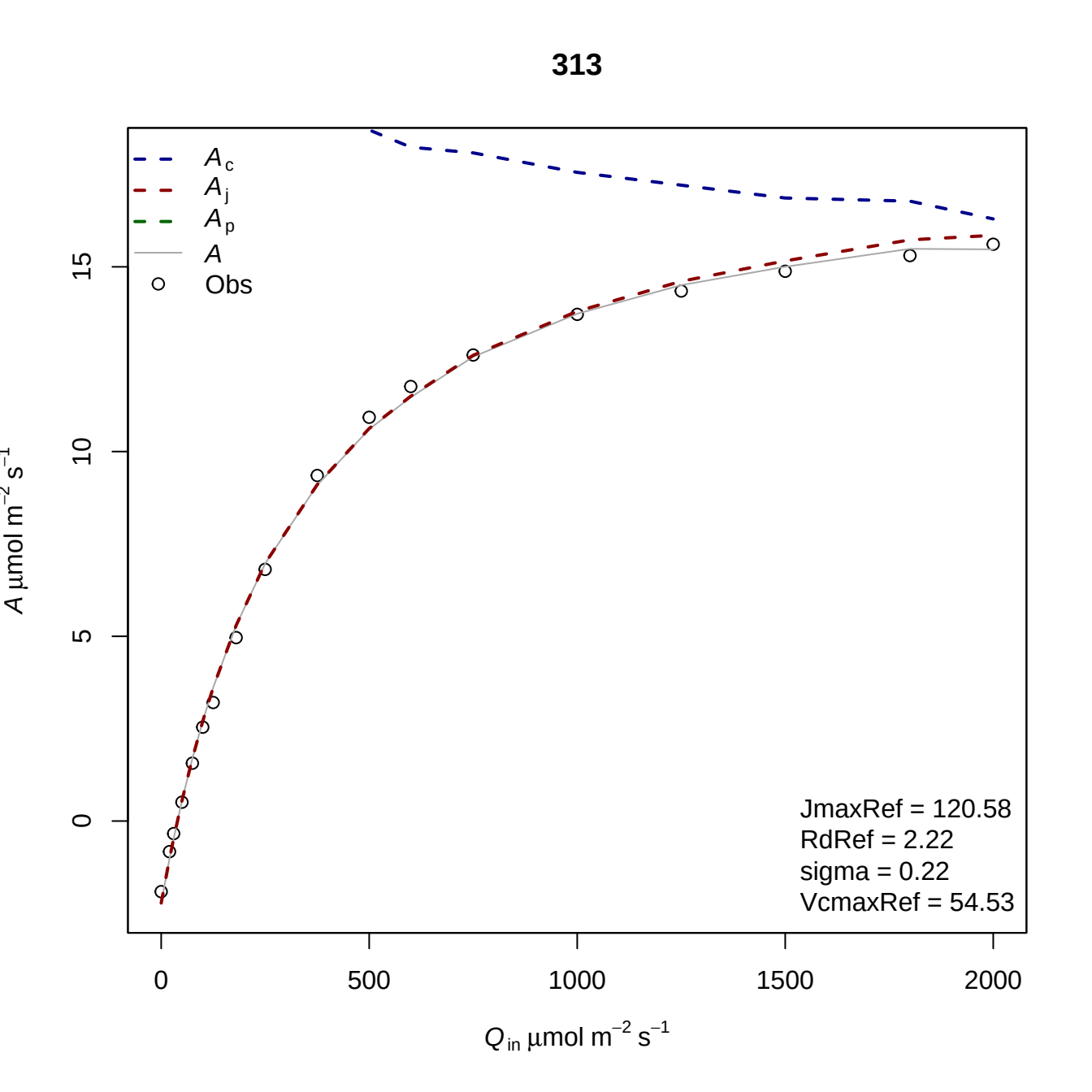

308
A_c
A_j
A_p
A
Obs
JmaxRef = 62.62
RdRef = 2.67
sigma = 0.89
VcmaxRef = 185239.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

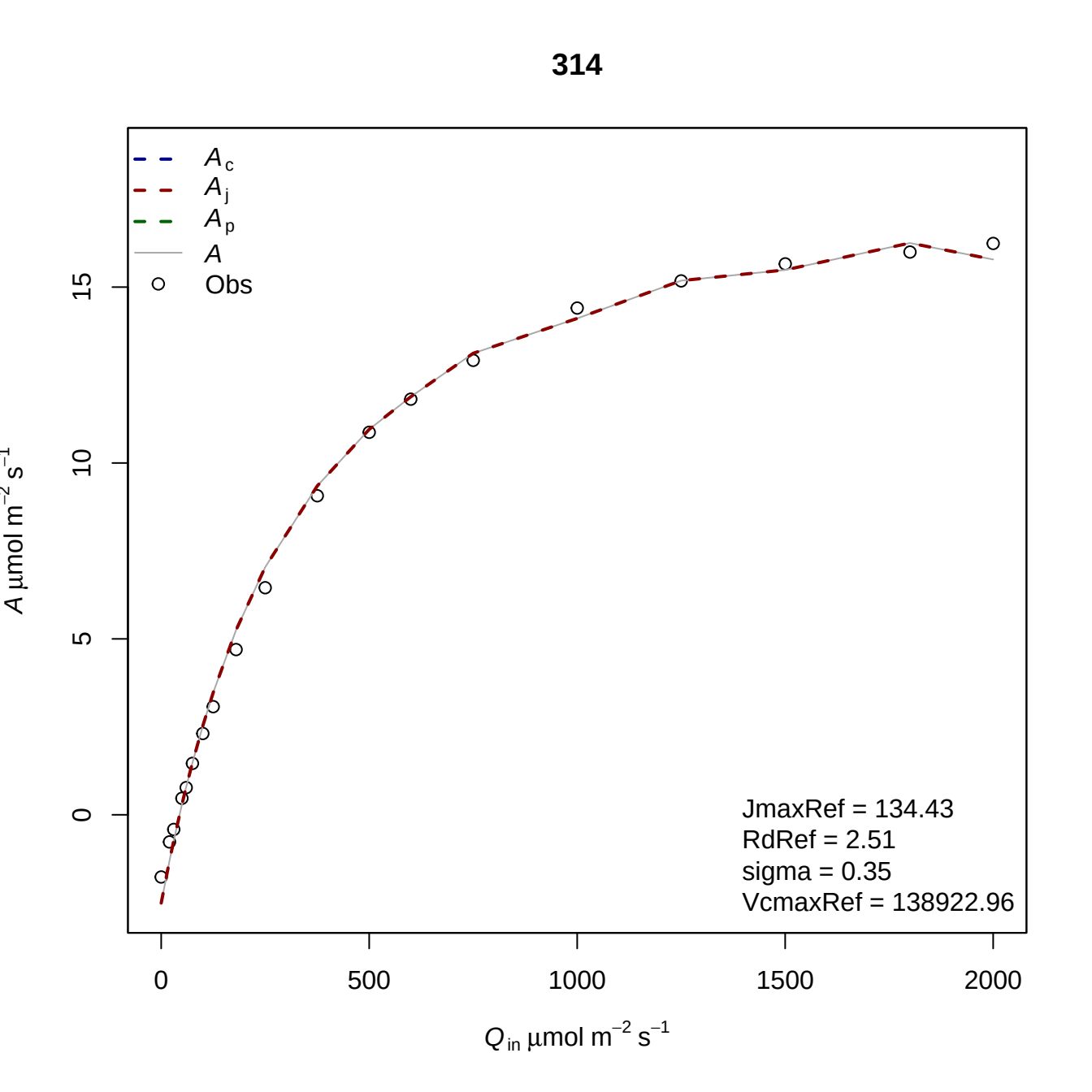

309
A_c
A_j
A_p
A
Obs
JmaxRef = 128.56
RdRef = 3.21
sigma = 0.45
VcmaxRef = 302153.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

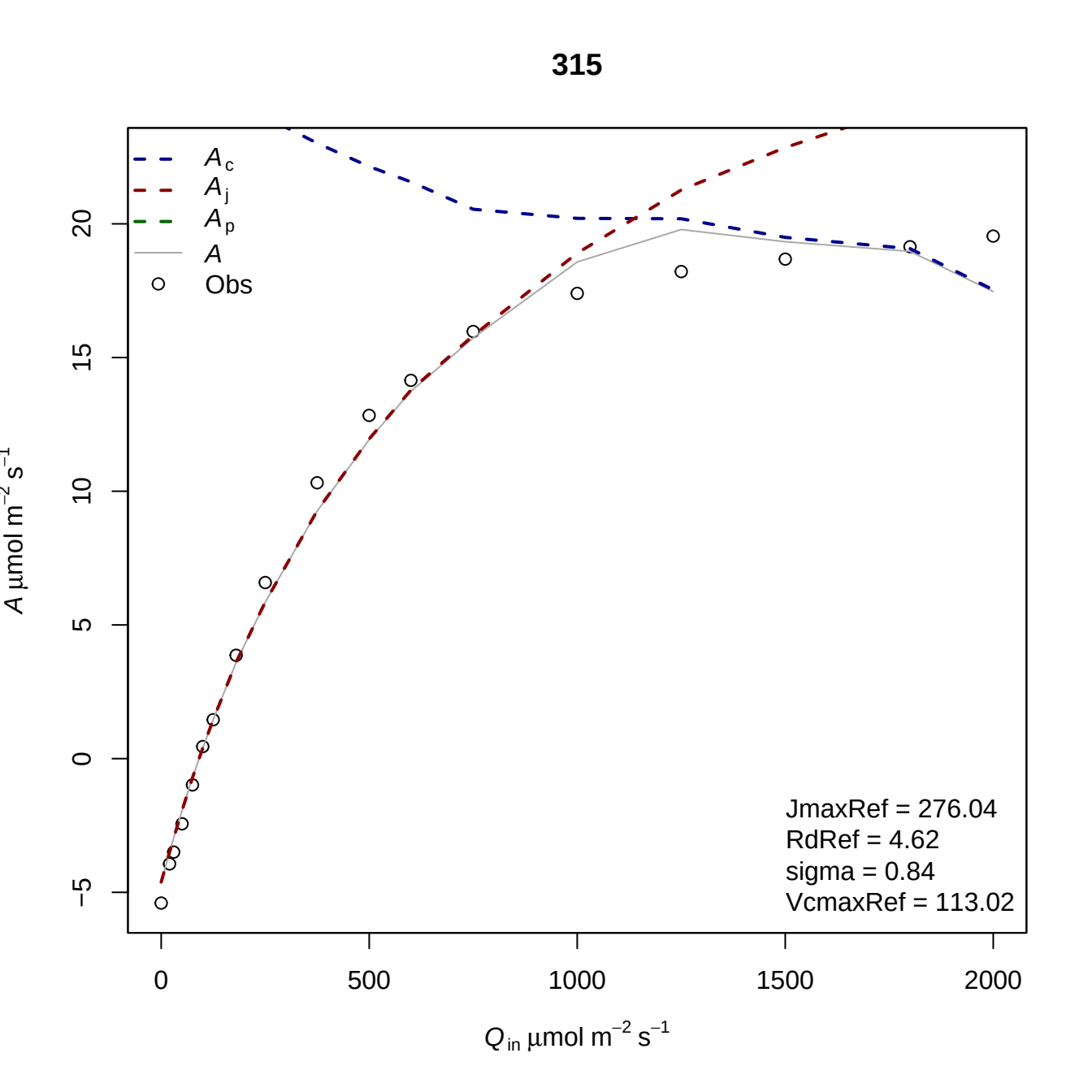

310
A_c
A_j
A_p
A
Obs
JmaxRef = 125.47
RdRef = 1.73
sigma = 0.19
VcmaxRef = 44.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

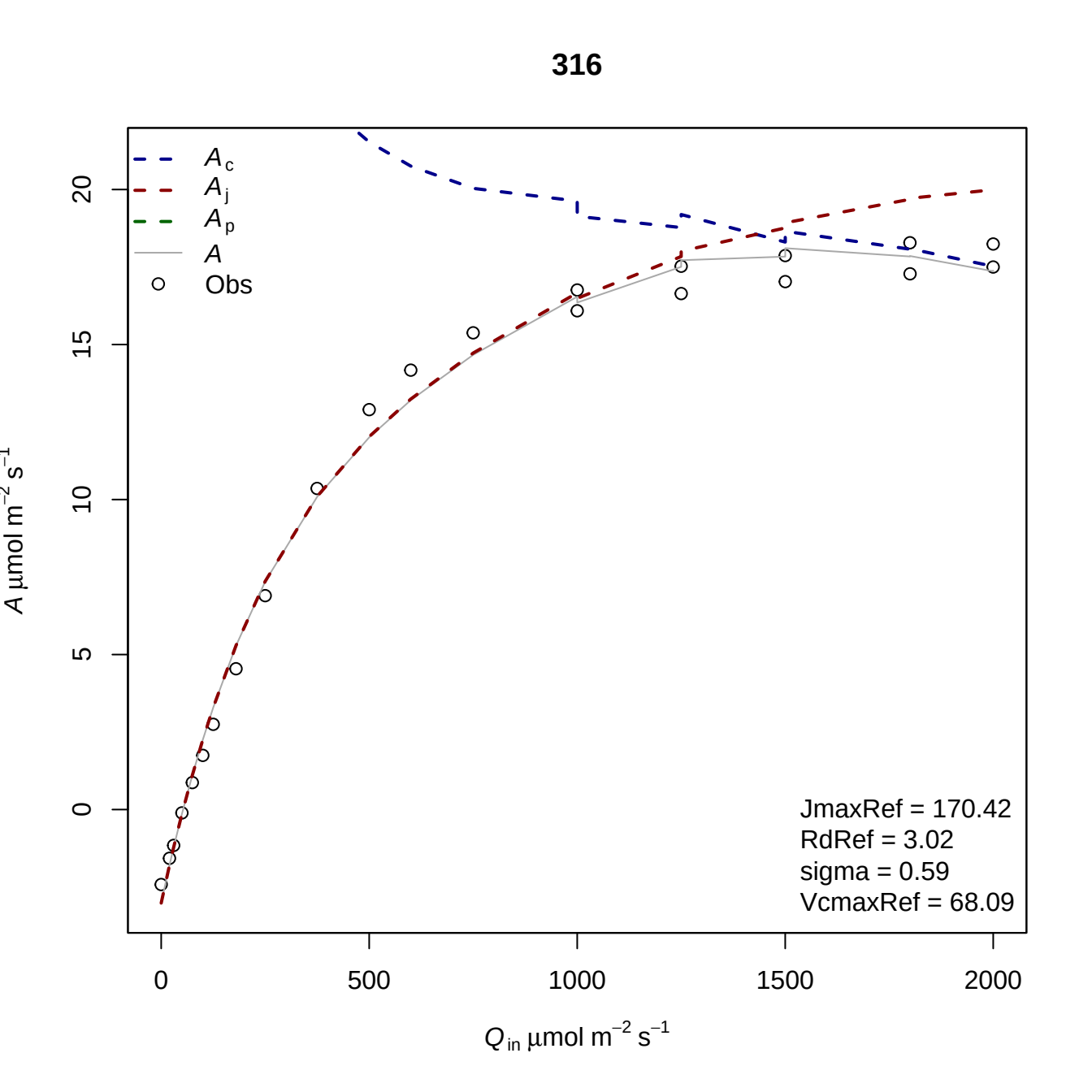

311
A_c
A_j
A_p
A
Obs
JmaxRef = 138.06
RdRef = 3.23
sigma = 0.58
VcmaxRef = 93279.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

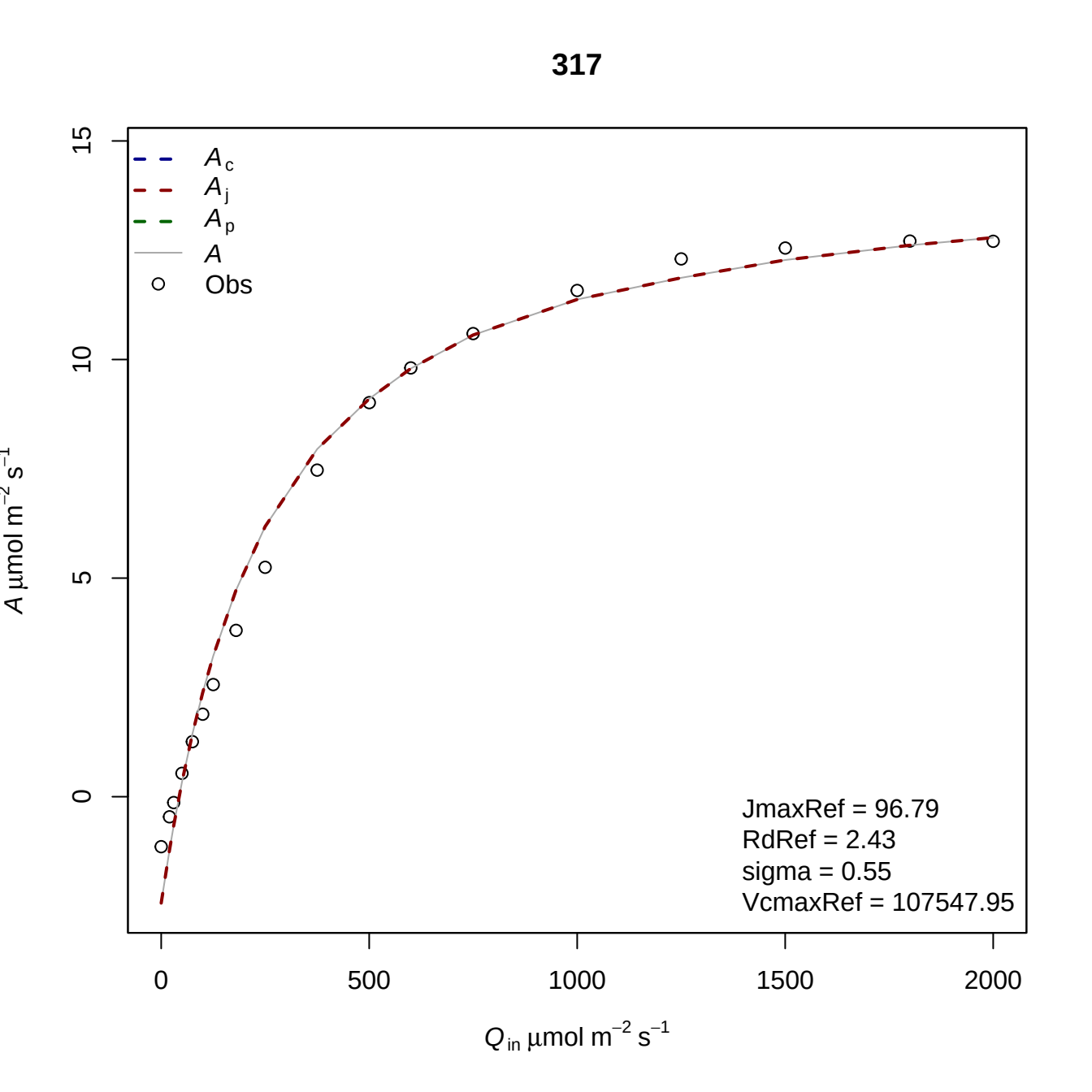

312
A_c
A_j
A_p
A
Obs
JmaxRef = 157.77
RdRef = 2.79
sigma = 0.5
VcmaxRef = 91.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

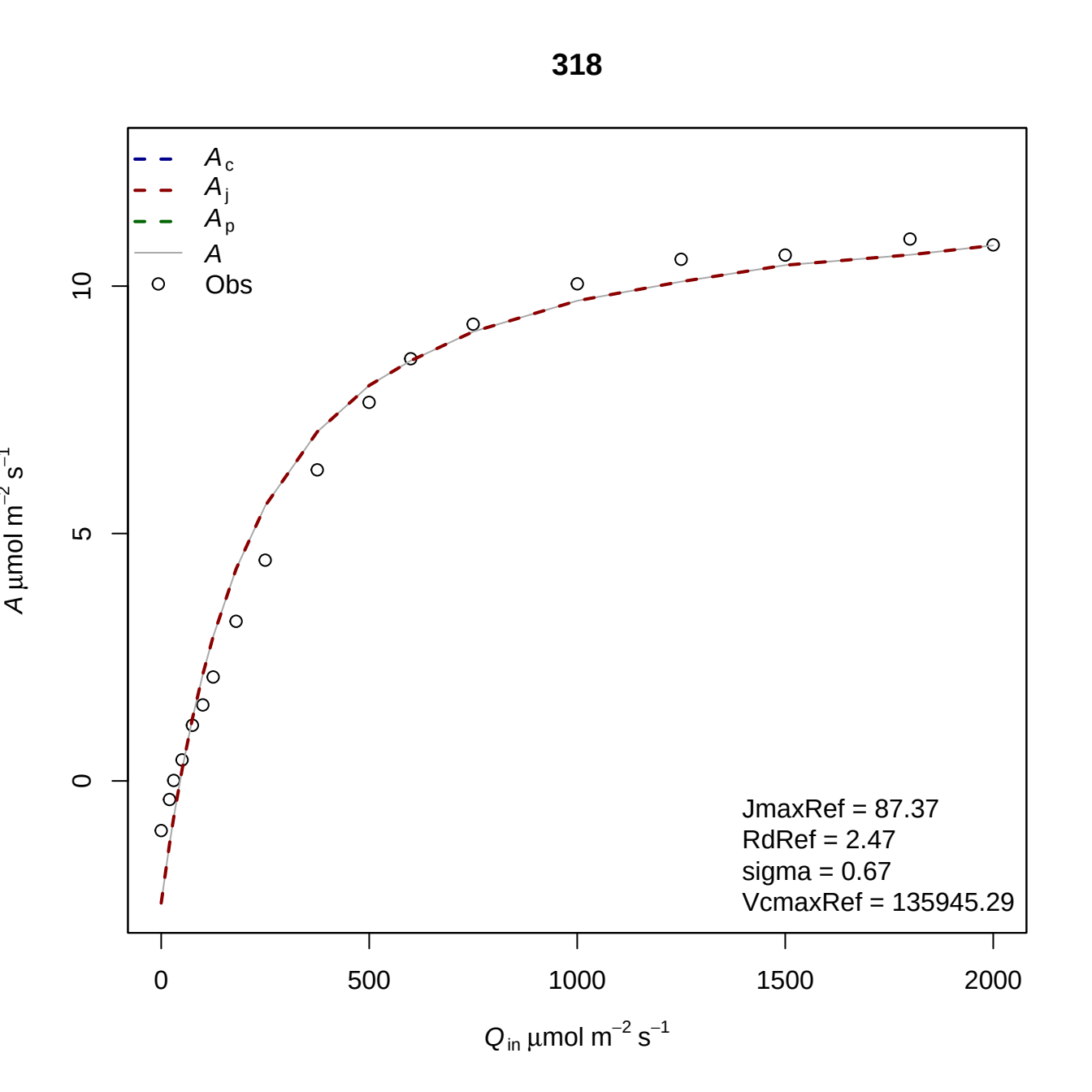

313
A_c
A_j
A_p
A
Obs
JmaxRef = 120.58
RdRef = 2.22
sigma = 0.22
VcmaxRef = 54.53
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

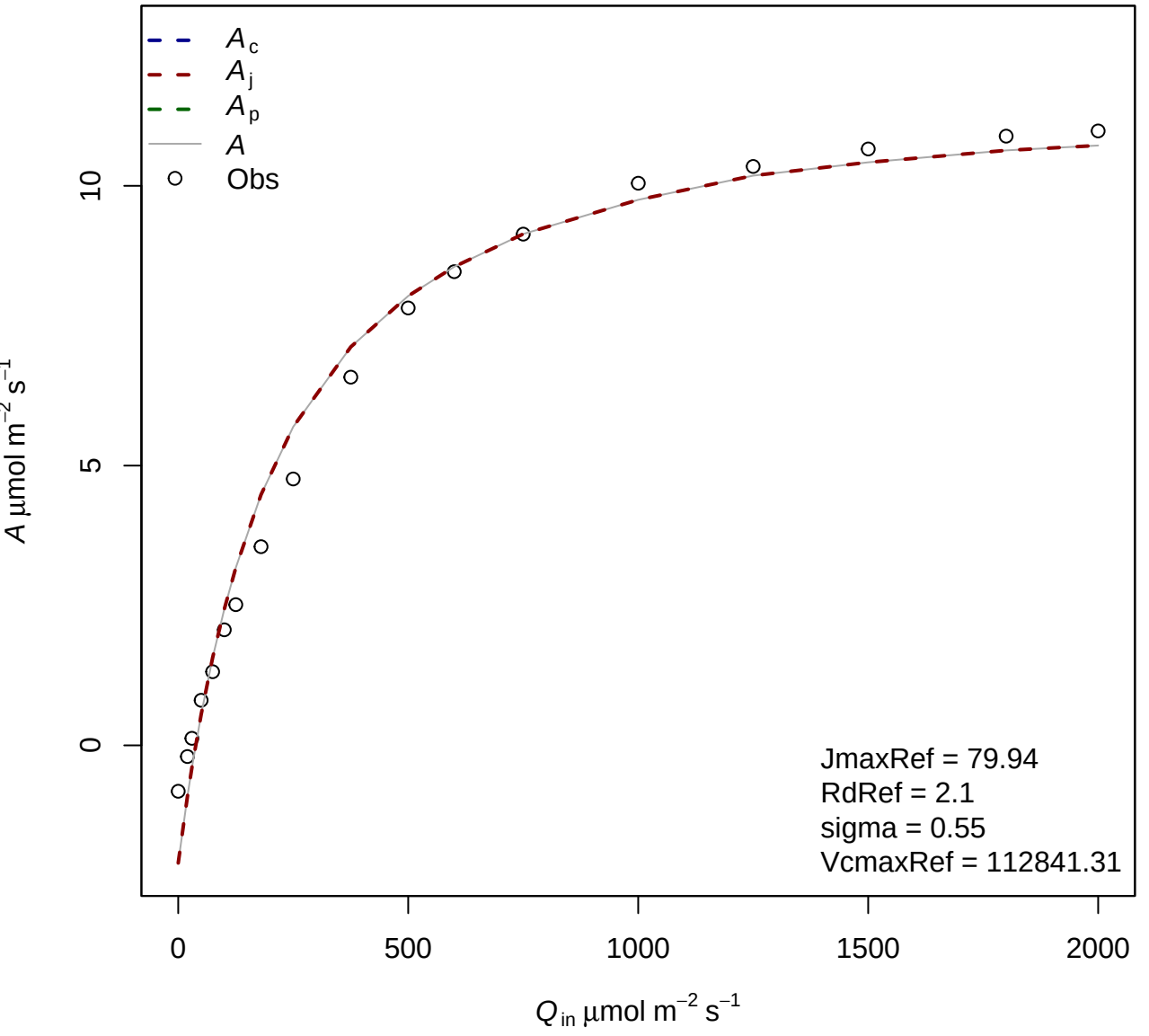

314
A_c
A_j
A_p
A
Obs
JmaxRef = 134.43
RdRef = 2.51
sigma = 0.35
VcmaxRef = 138922.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

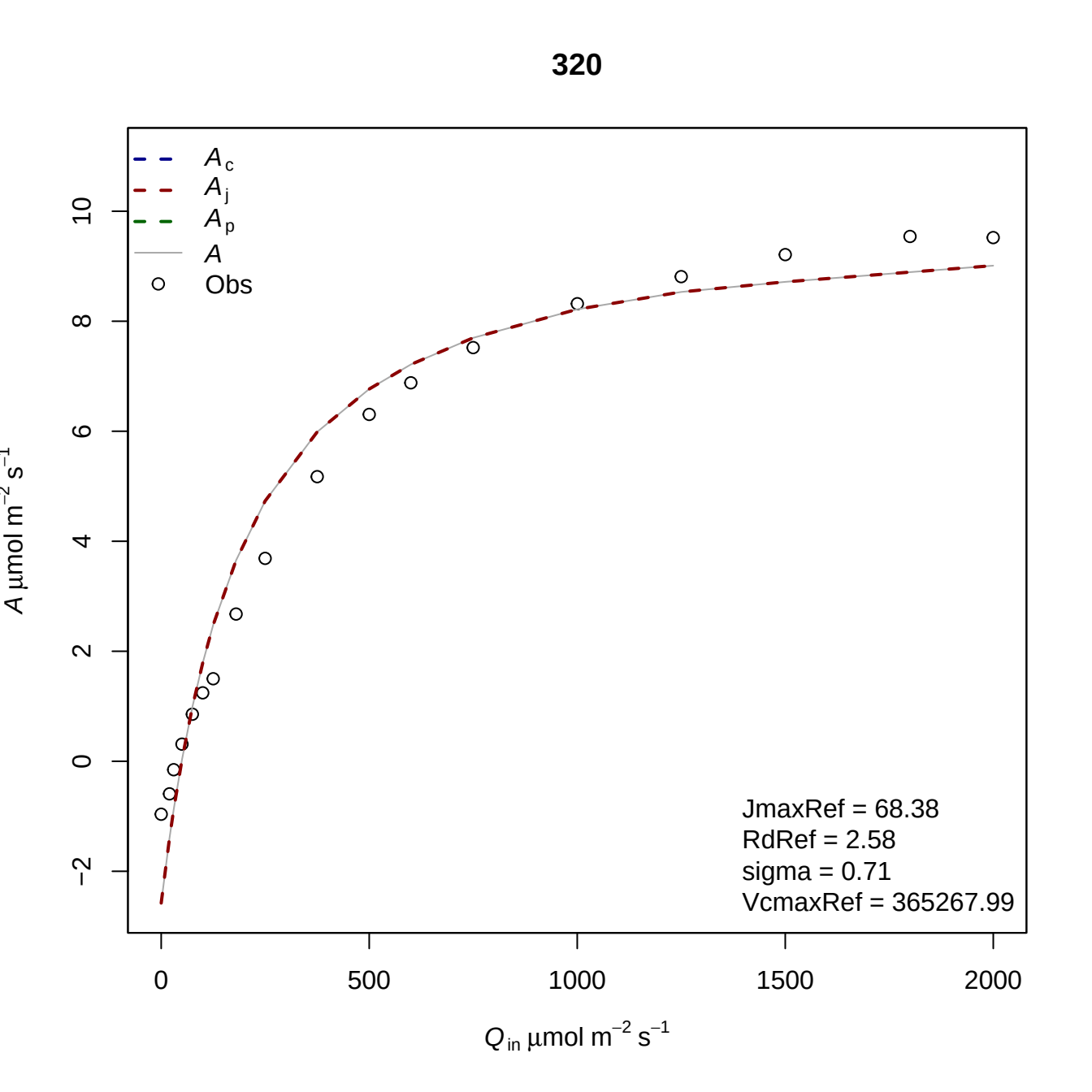

315
A_c
A_j
A_p
A
Obs
JmaxRef = 276.04
RdRef = 4.62
sigma = 0.84
VcmaxRef = 113.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

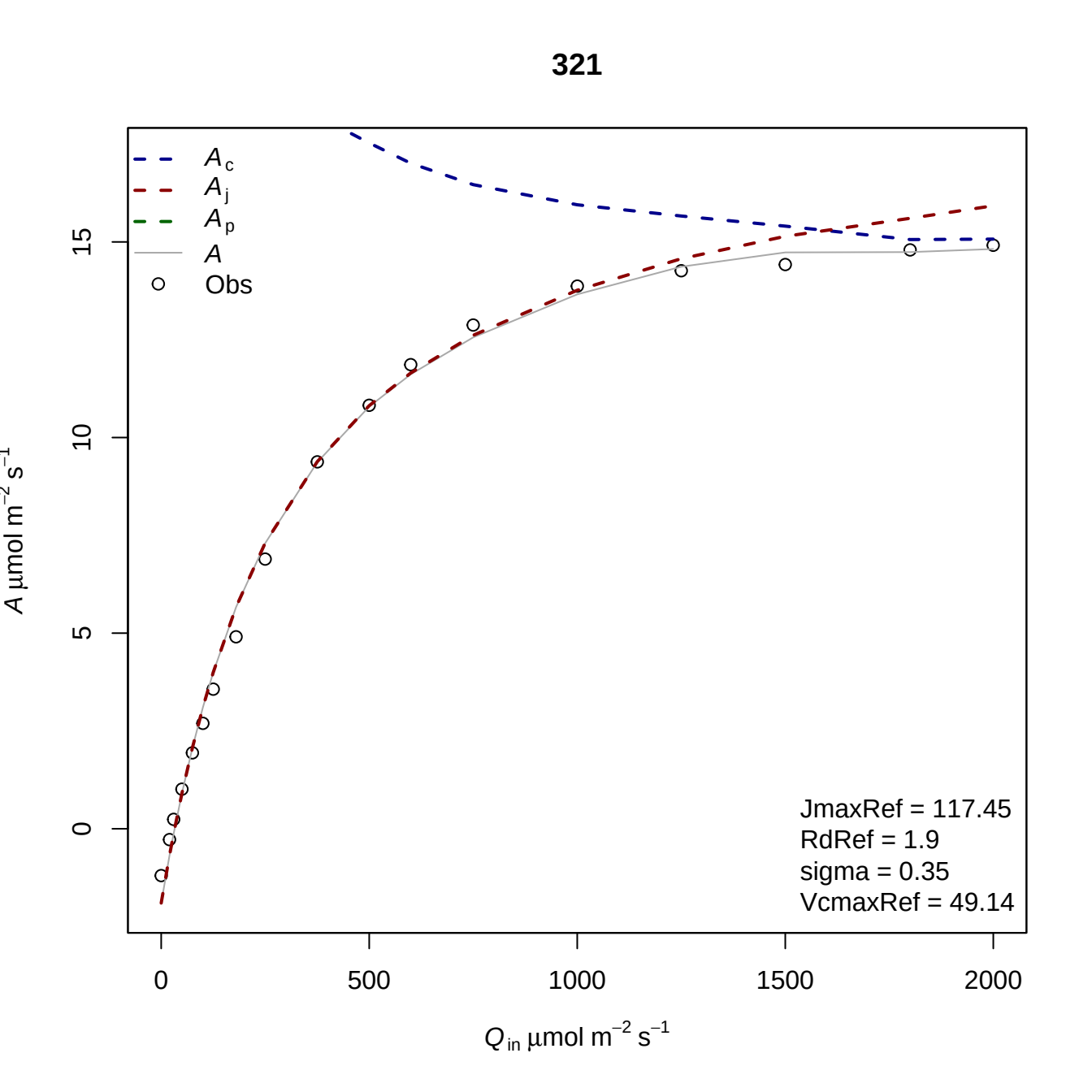

316
A_c
A_j
A_p
A
Obs
JmaxRef = 170.42
RdRef = 3.02
sigma = 0.59
VcmaxRef = 68.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

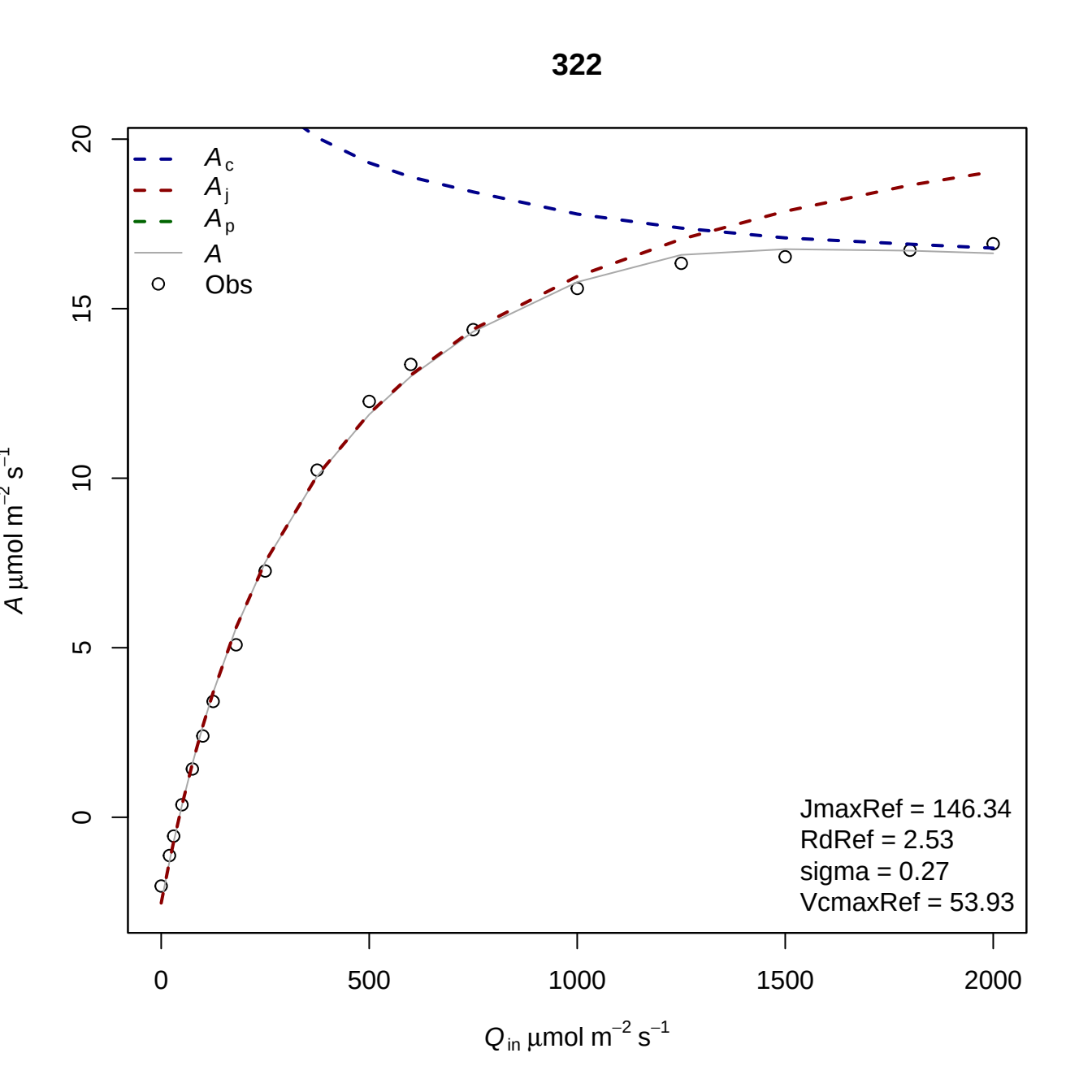

317
A_c
A_j
A_p
A
Obs
JmaxRef = 96.79
RdRef = 2.43
sigma = 0.55
VcmaxRef = 107547.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

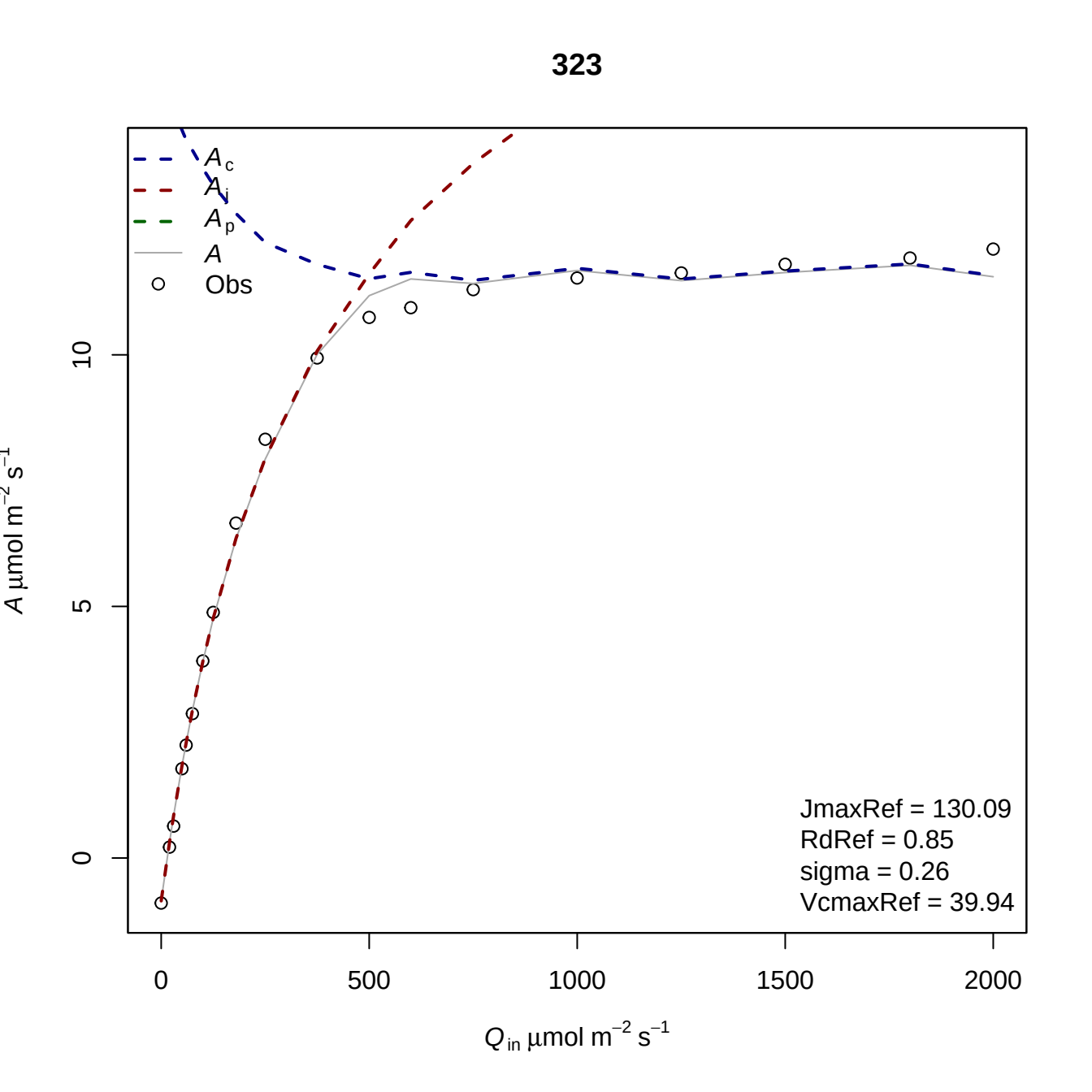

318
A_c
A_j
A_p
A
Obs
JmaxRef = 87.37
RdRef = 2.47
sigma = 0.67
VcmaxRef = 135945.29
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

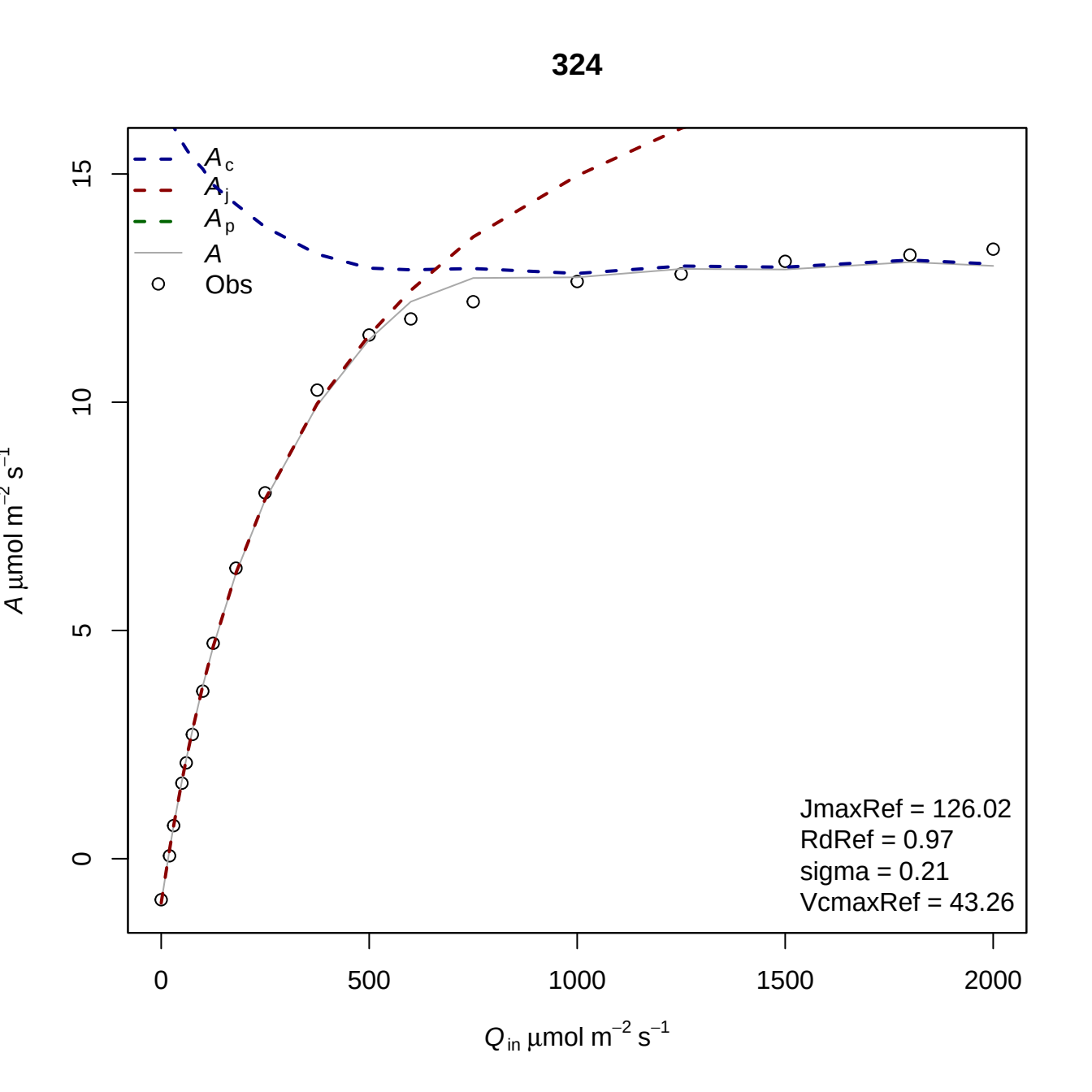

A_c
A_j
A_p
A
Obs
JmaxRef = 79.94
RdRef = 2.1
sigma = 0.55
VcmaxRef = 112841.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

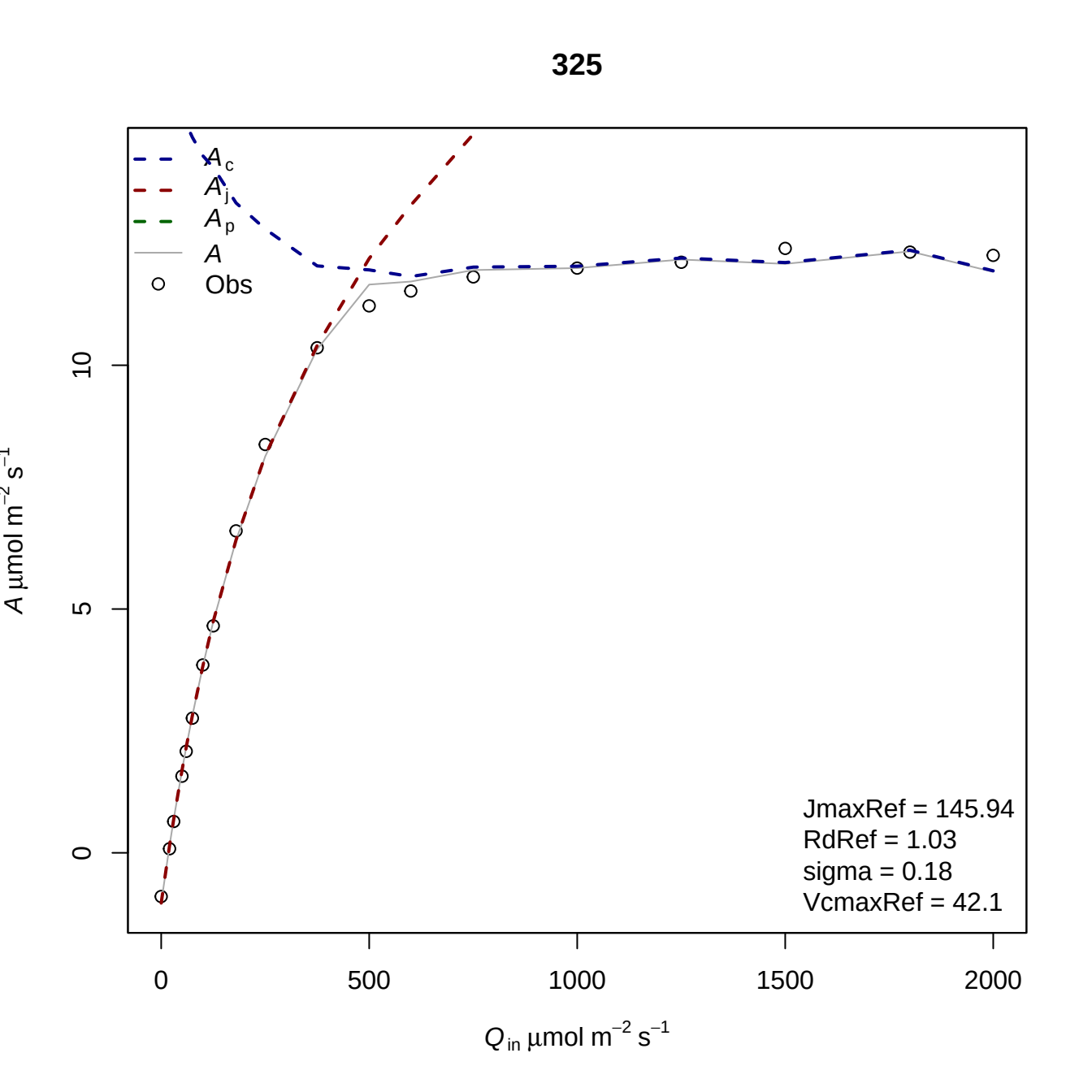

320
A_c
A_j
A_p
A
Obs
JmaxRef = 68.38
RdRef = 2.58
sigma = 0.71
VcmaxRef = 365267.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

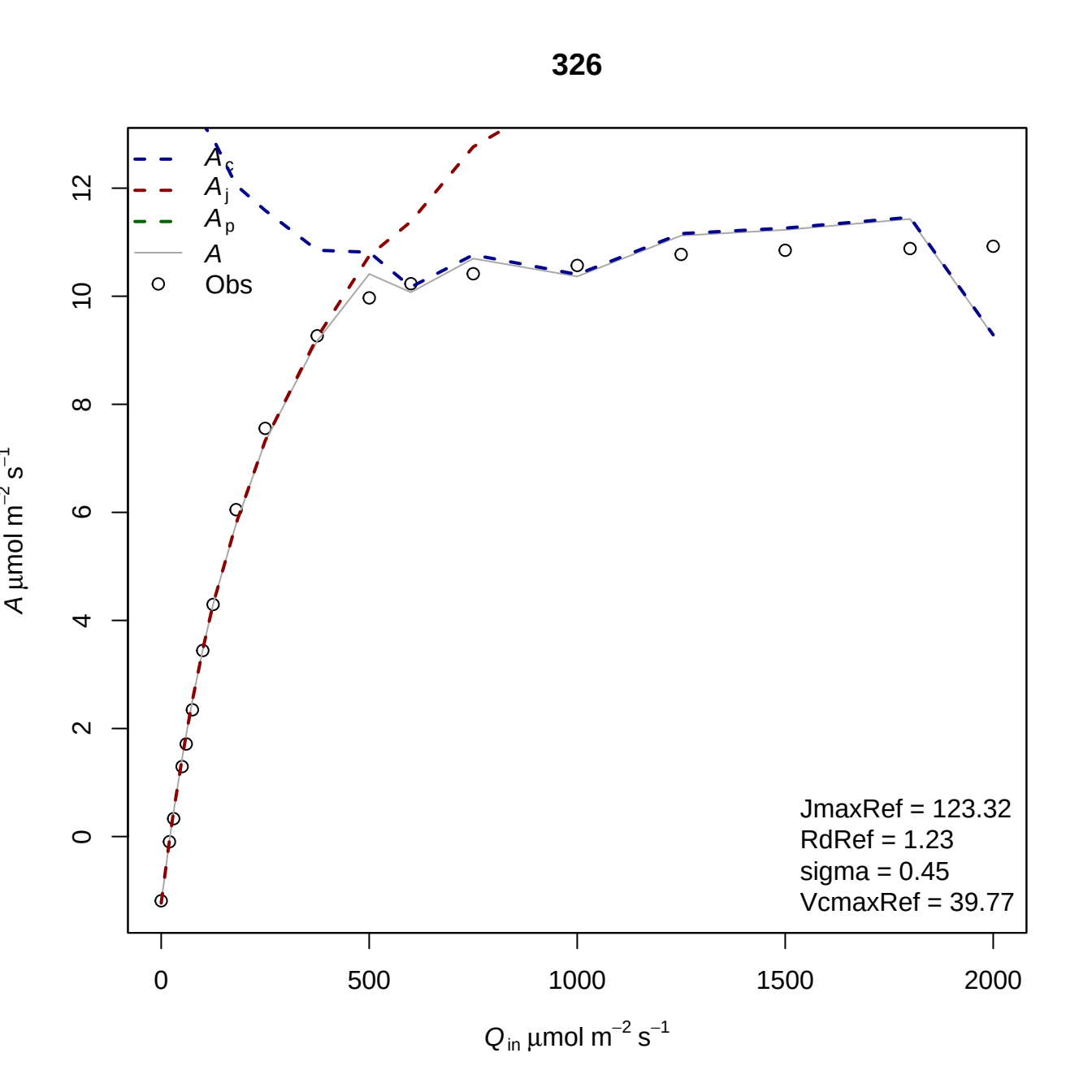

321
A_c
A_j
A_p
A
Obs
JmaxRef = 117.45
RdRef = 1.9
sigma = 0.35
VcmaxRef = 49.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

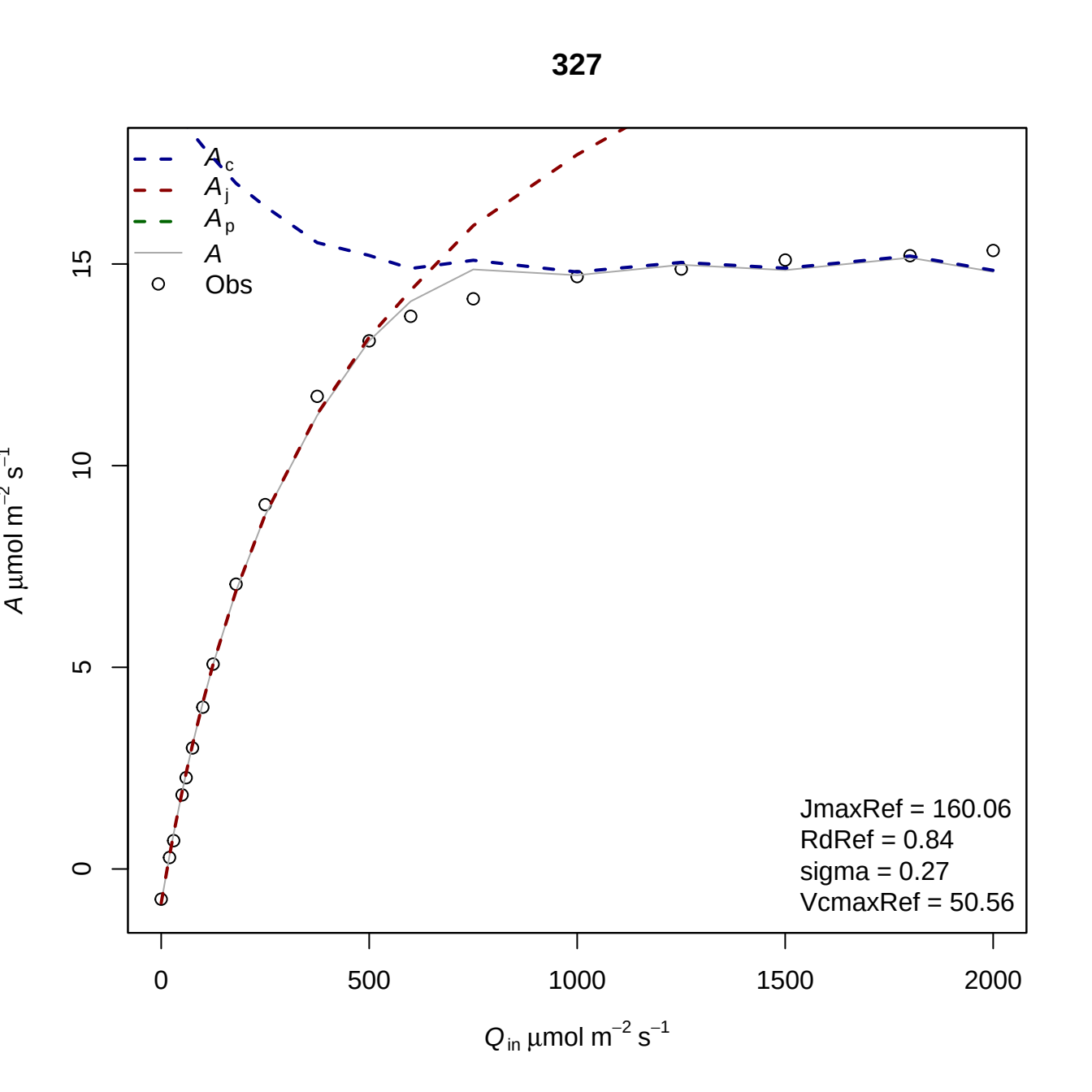

322
A_c
A_j
A_p
A
Obs
JmaxRef = 146.34
RdRef = 2.53
sigma = 0.27
VcmaxRef = 53.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

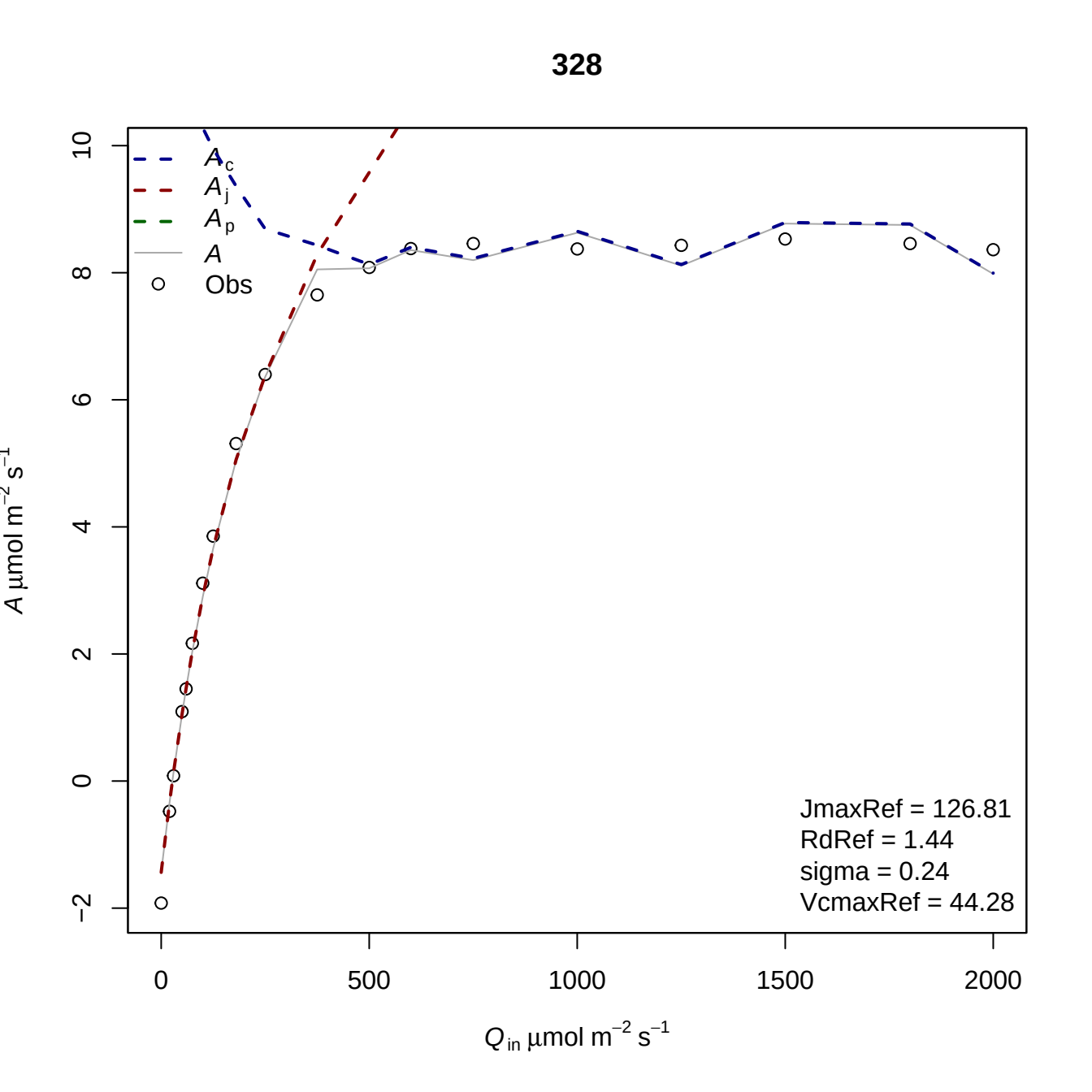

323
A_c
A_j
A_p
A
Obs
JmaxRef = 130.09
RdRef = 0.85
sigma = 0.26
VcmaxRef = 39.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

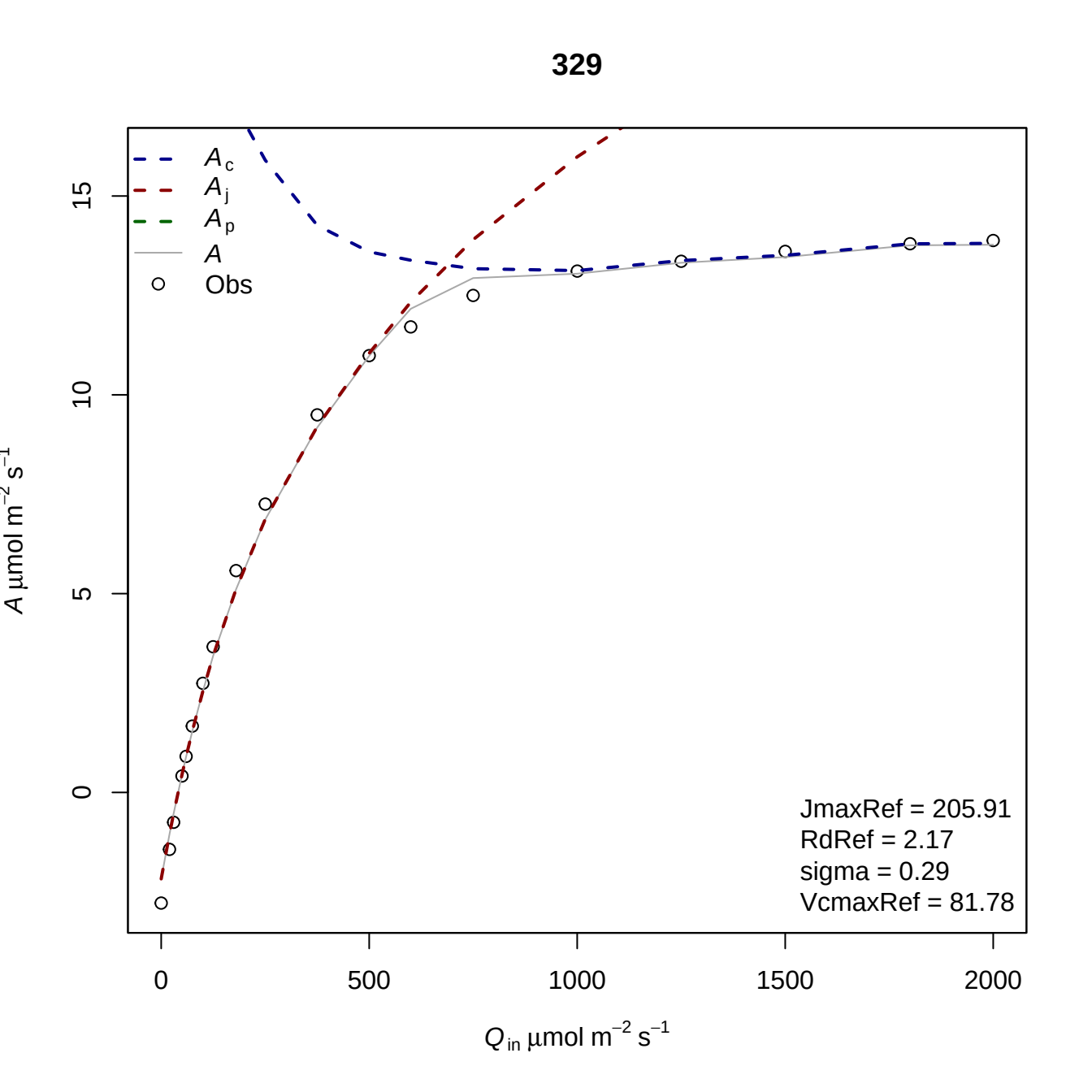

324
A_c
A_j
A_p
A
Obs
JmaxRef = 126.02
RdRef = 0.97
sigma = 0.21
VcmaxRef = 43.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

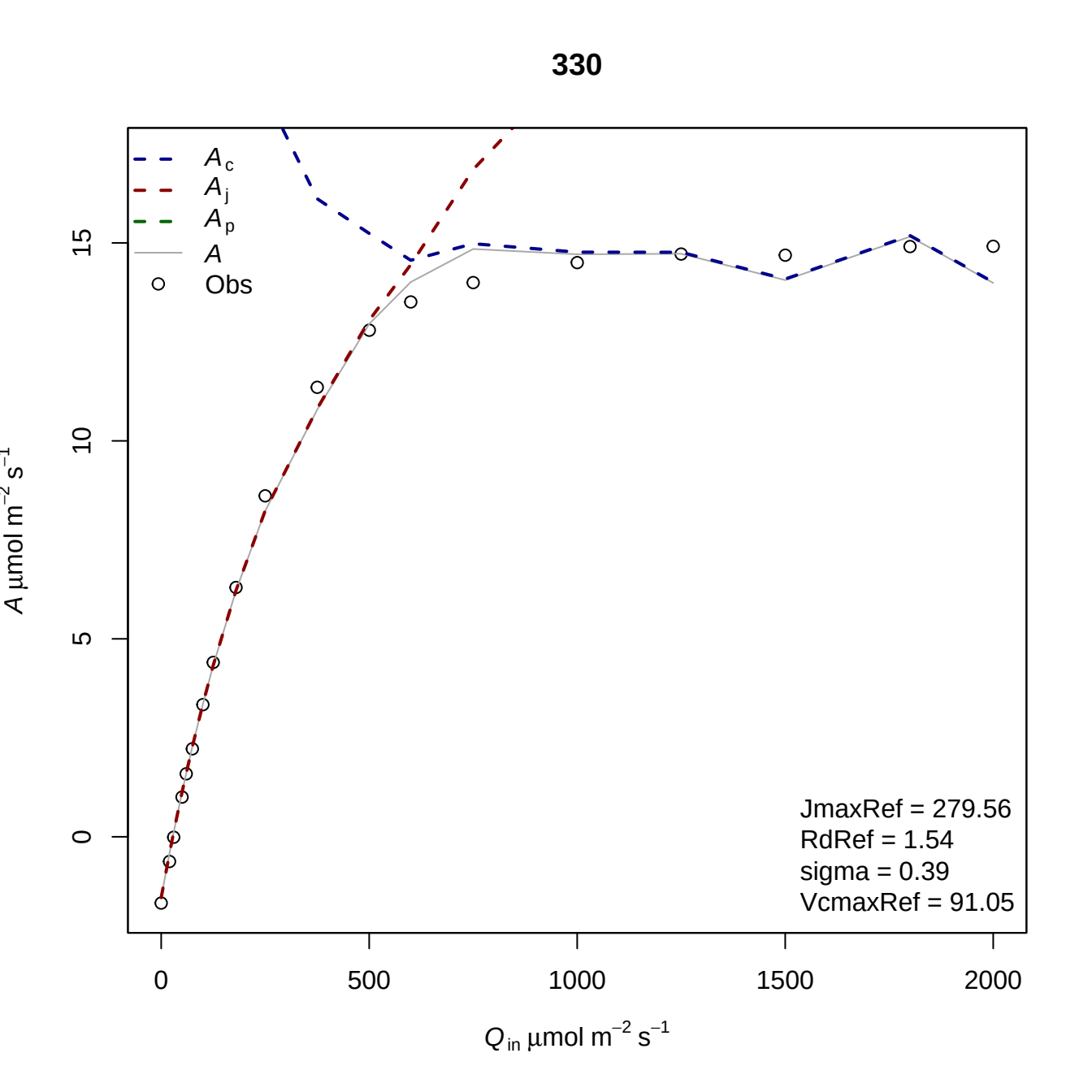

325
A_c
A_j
A_p
A
Obs
JmaxRef = 145.94
RdRef = 1.03
sigma = 0.18
VcmaxRef = 42.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

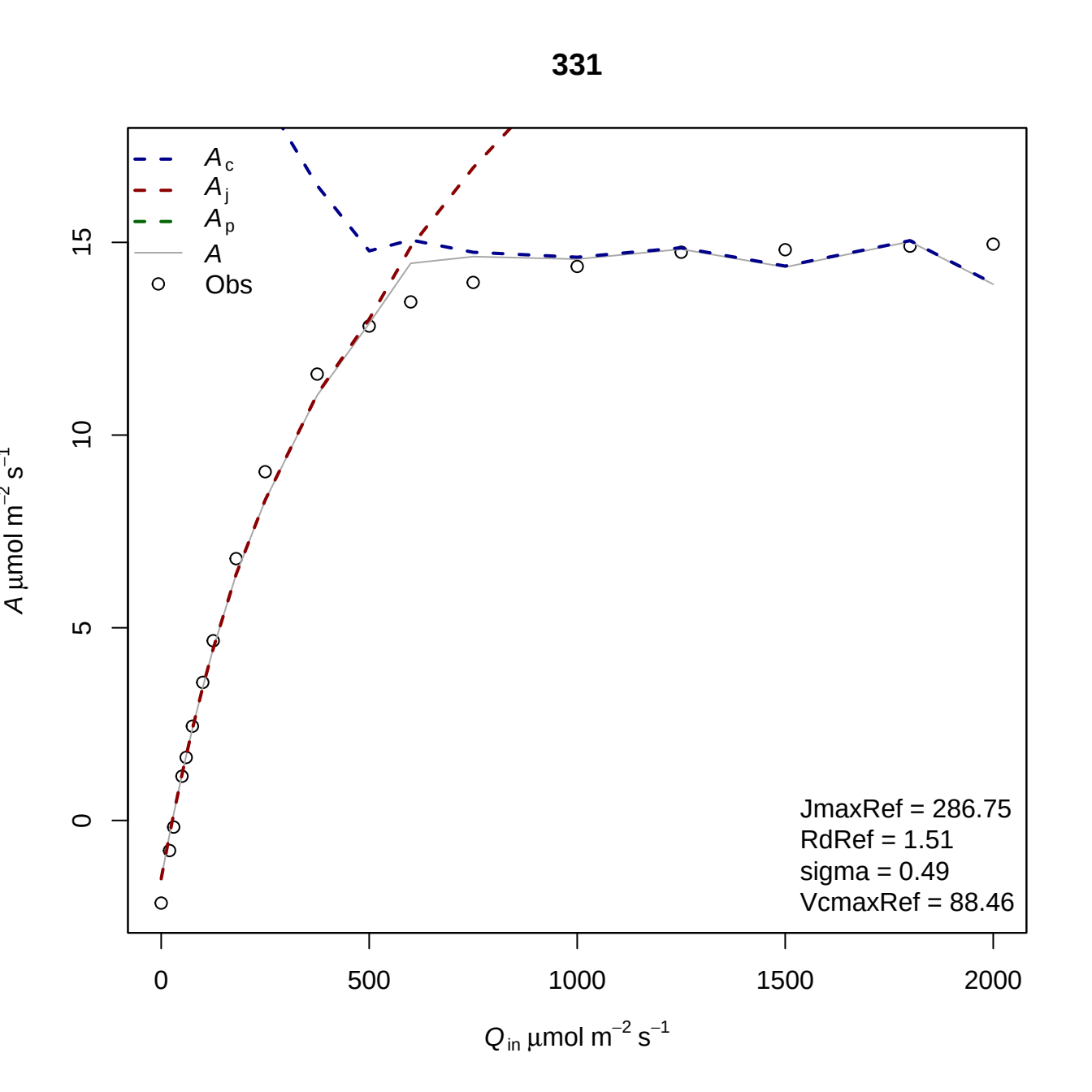

326
A_c
A_j
A_p
A
Obs
JmaxRef = 123.32
RdRef = 1.23
sigma = 0.45
VcmaxRef = 39.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

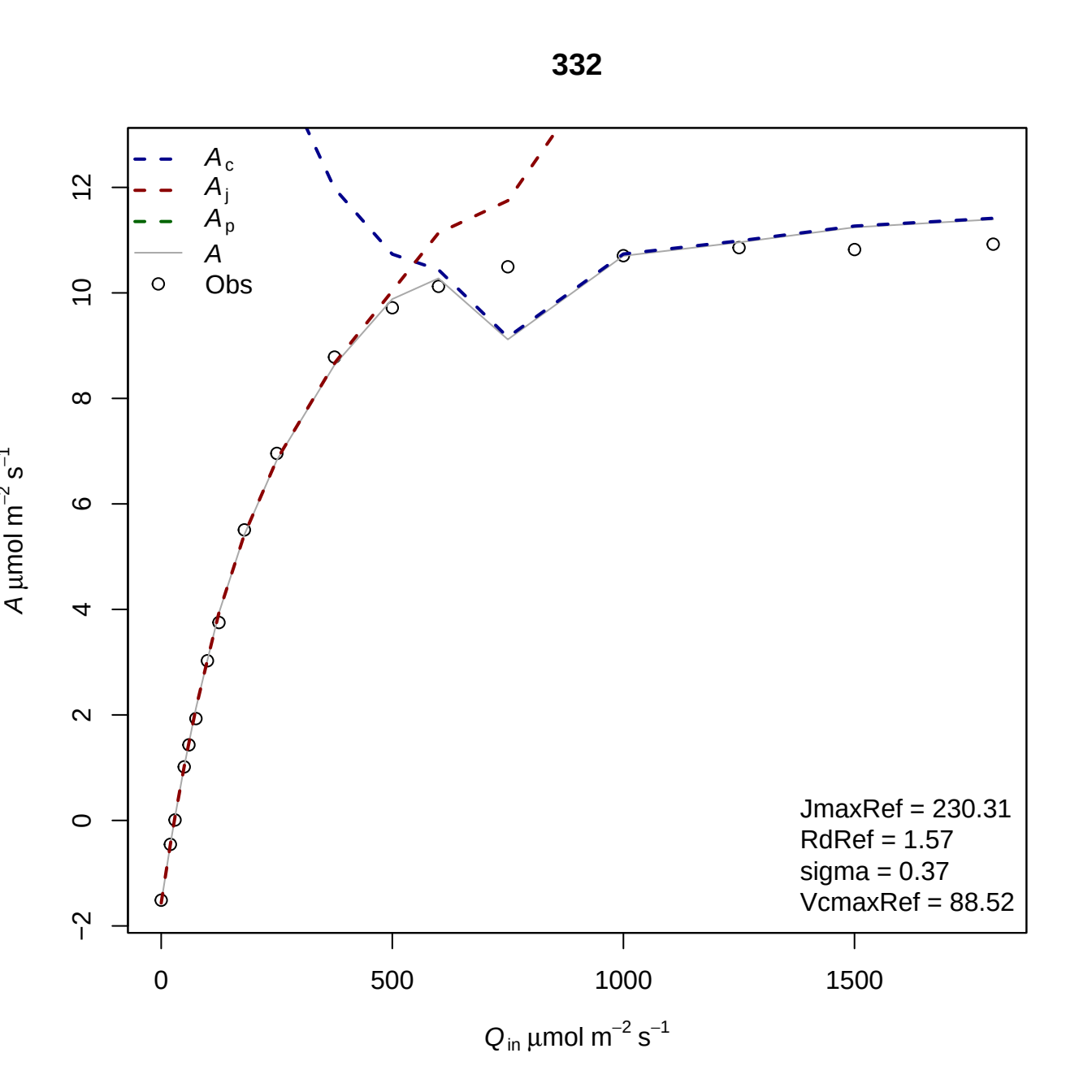

327
A_c
A_j
A_p
A
Obs
JmaxRef = 160.06
RdRef = 0.84
sigma = 0.27
VcmaxRef = 50.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

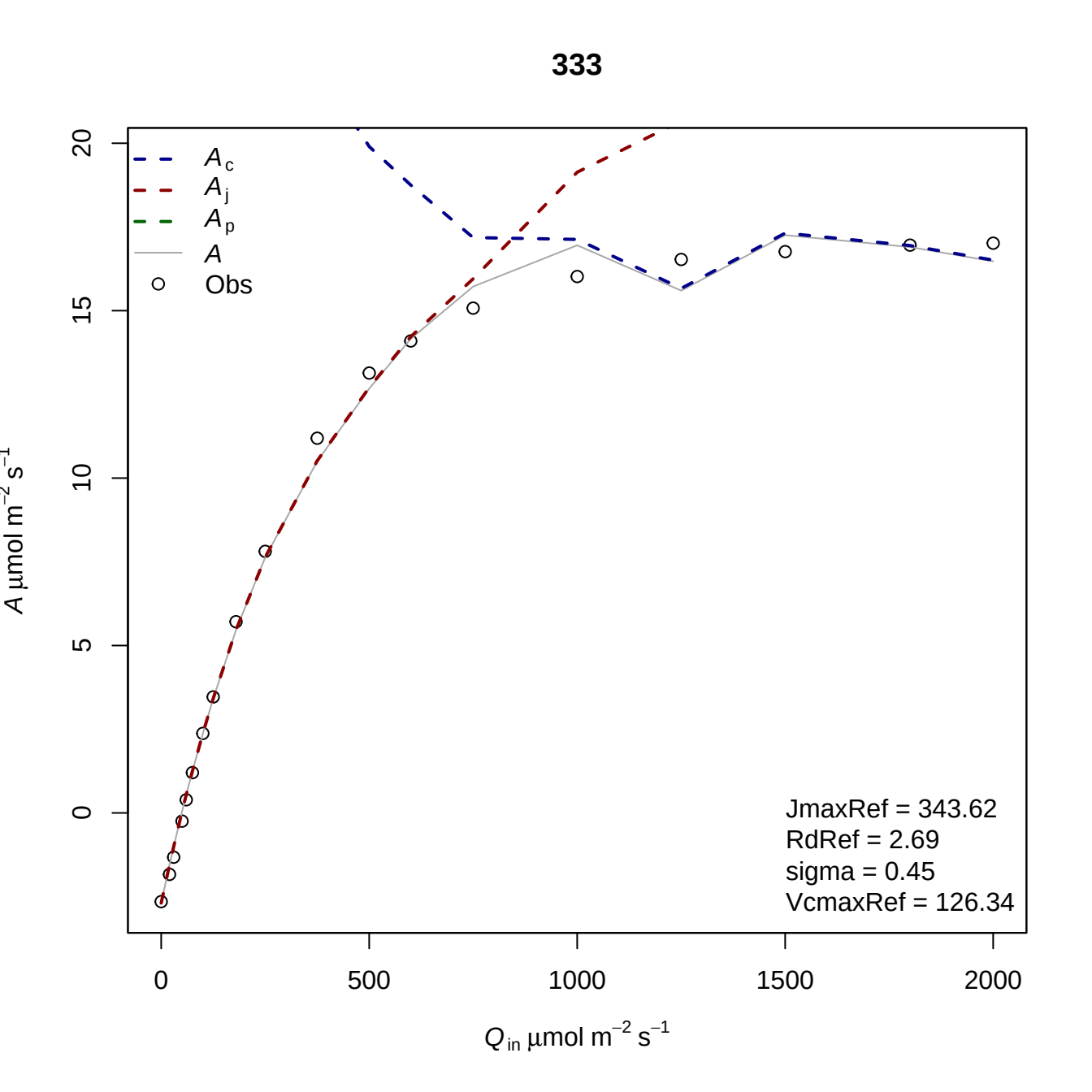

328
A_c
A_j
A_p
A
Obs
JmaxRef = 126.81
RdRef = 1.44
sigma = 0.24
VcmaxRef = 44.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

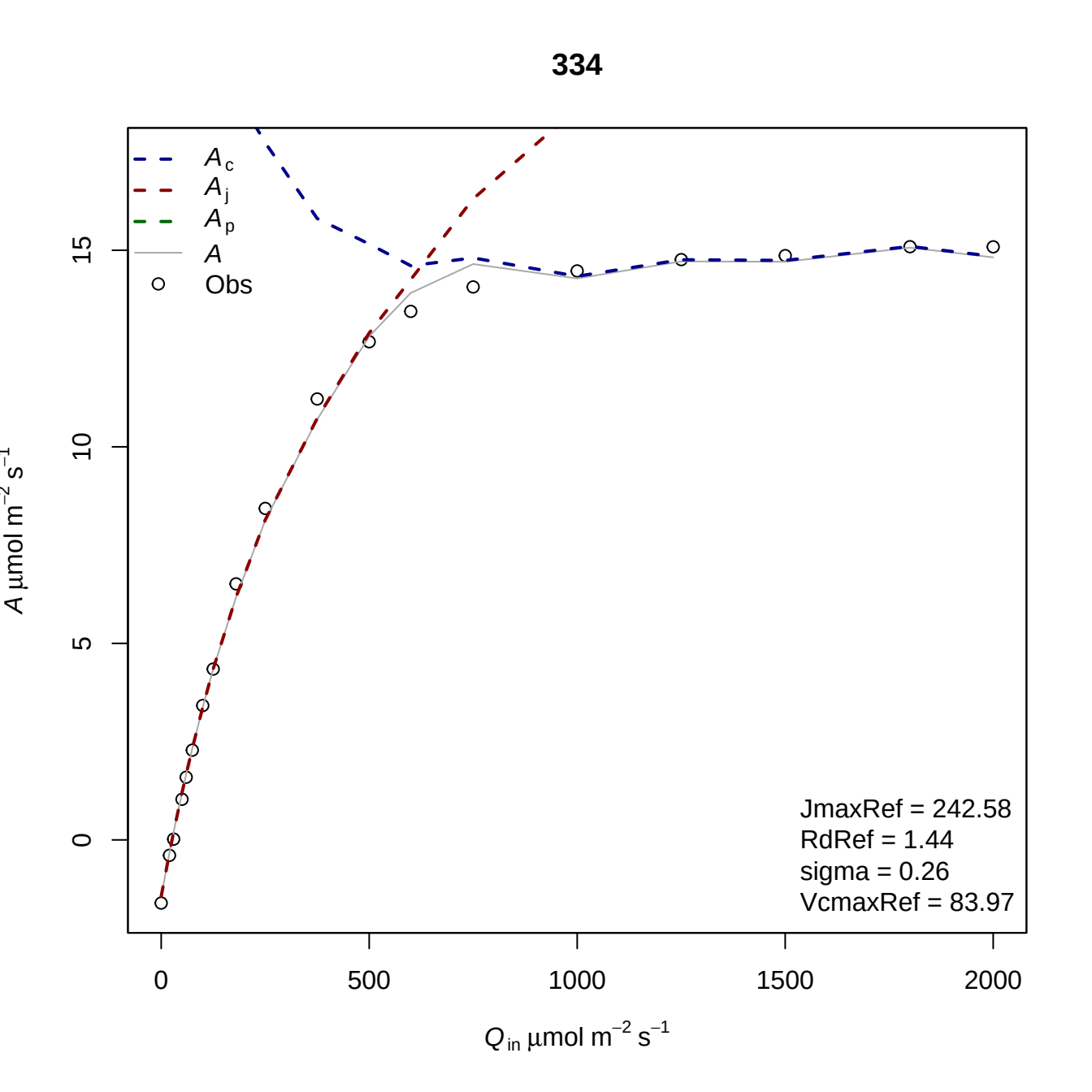

329
A_c
A_j
A_p
A
Obs
JmaxRef = 205.91
RdRef = 2.17
sigma = 0.29
VcmaxRef = 81.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

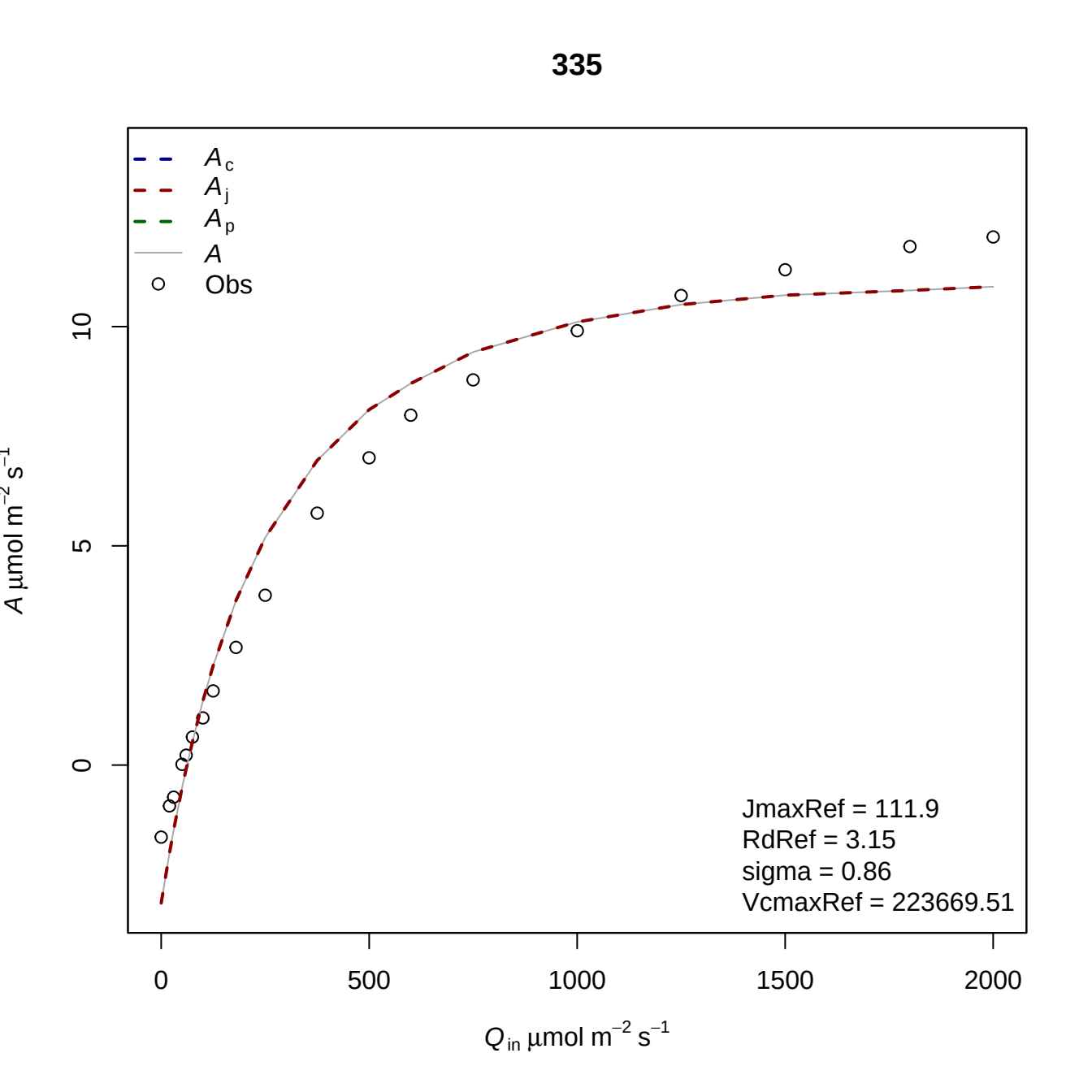

330
A_c
A_j
A_p
A
Obs
JmaxRef = 279.56
RdRef = 1.54
sigma = 0.39
VcmaxRef = 91.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

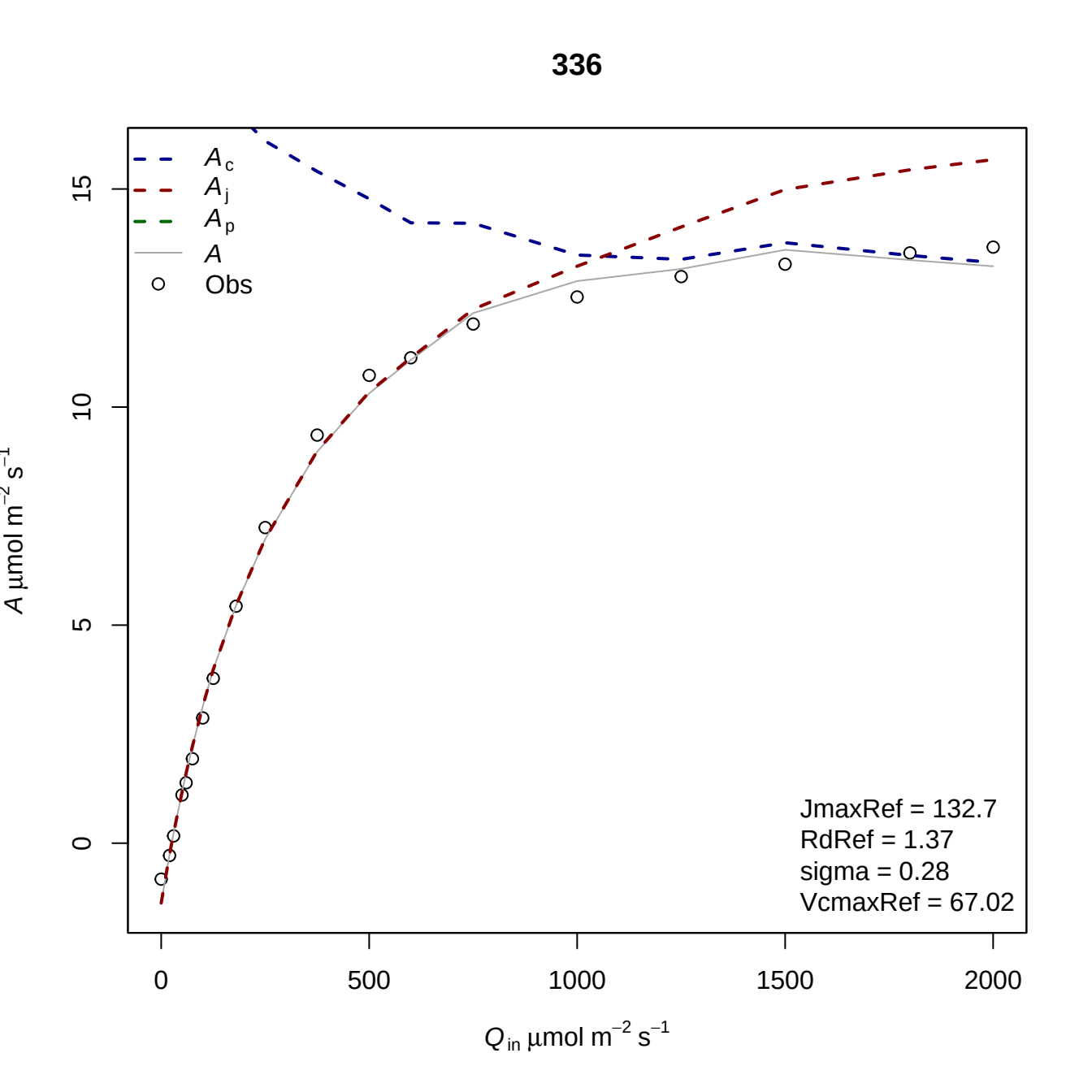

331
A_c
A_j
A_p
A
Obs
JmaxRef = 286.75
RdRef = 1.51
sigma = 0.49
VcmaxRef = 88.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

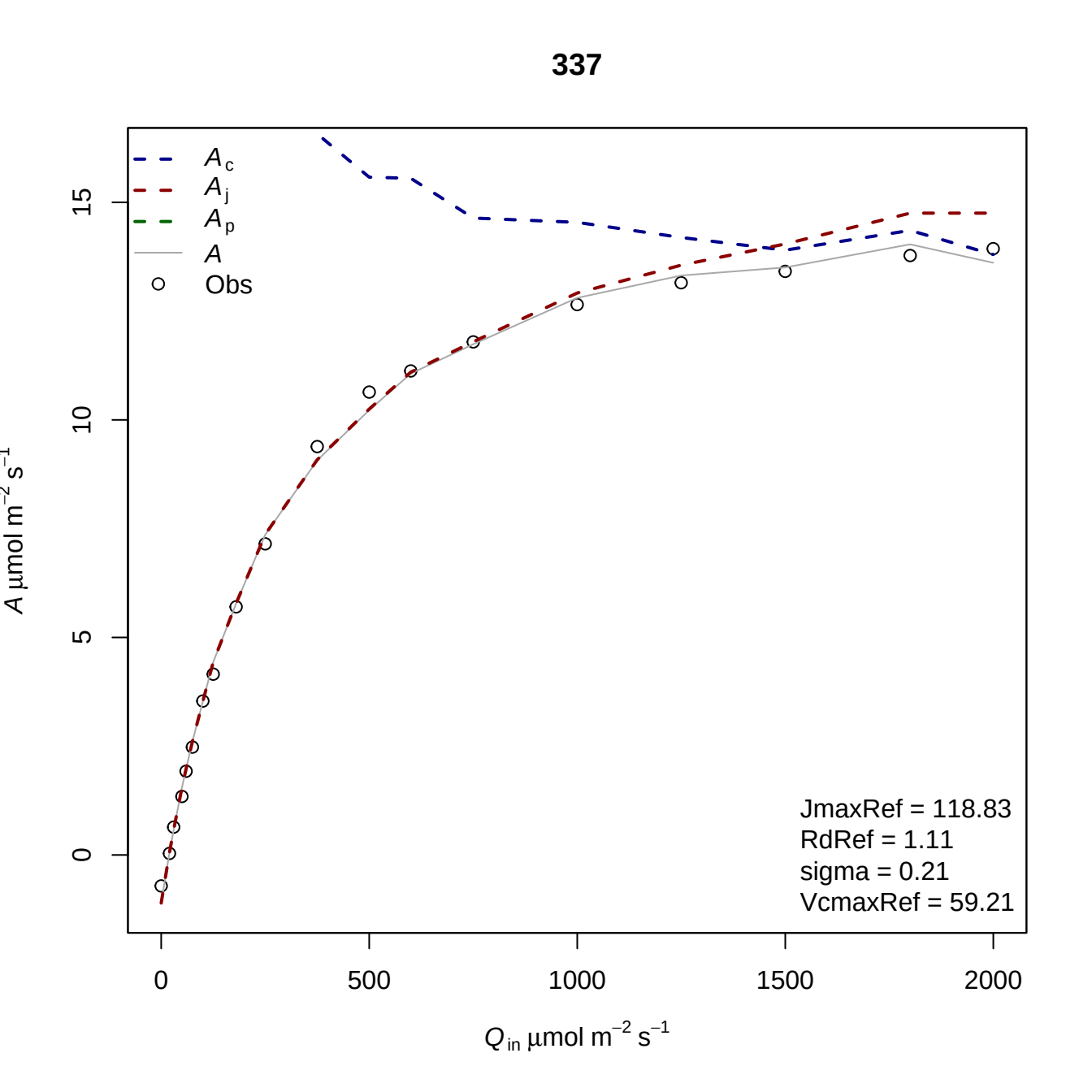

332
A_c
A_j
A_p
A
Obs
JmaxRef = 230.31
RdRef = 1.57
sigma = 0.37
VcmaxRef = 88.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

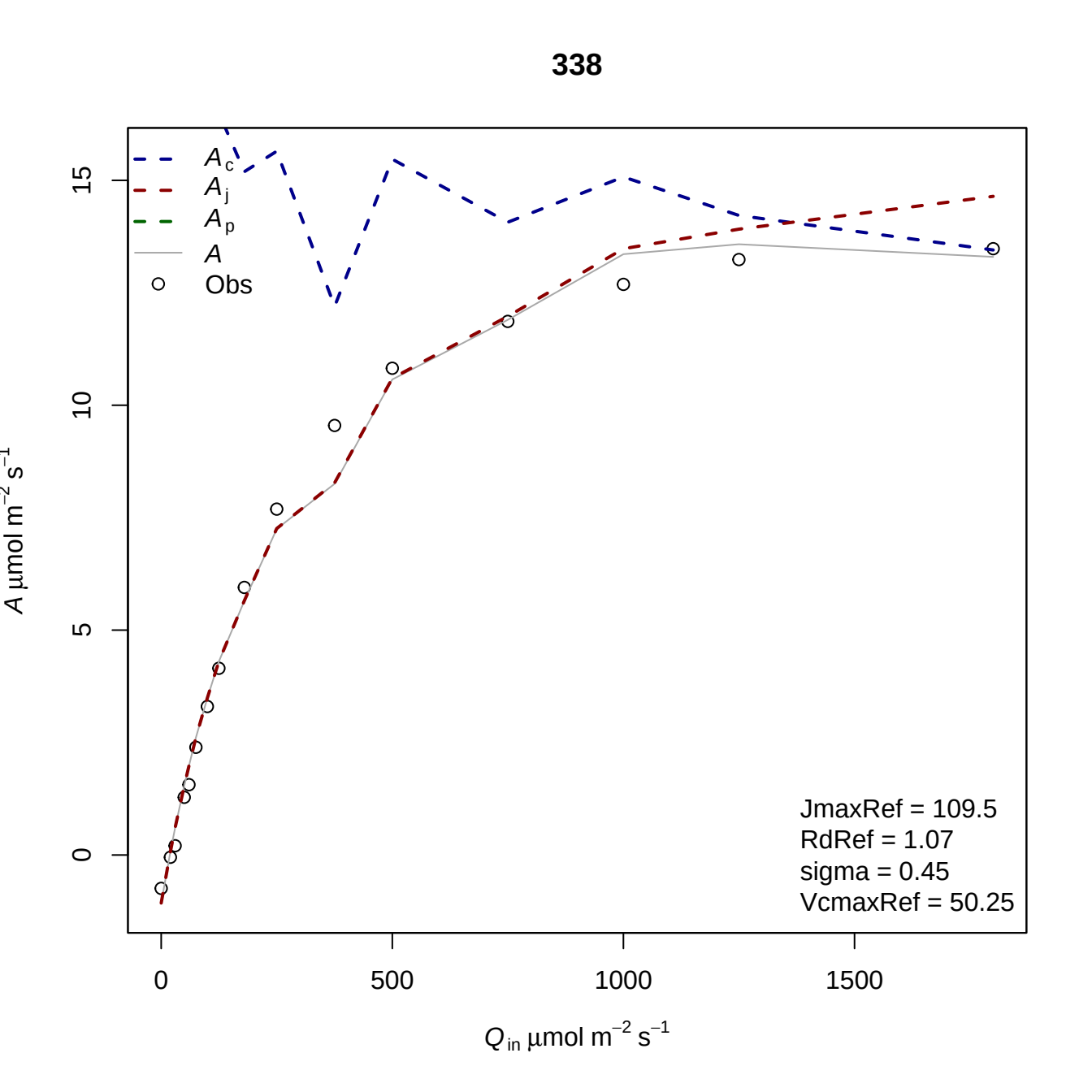

333
A_c
A_j
A_p
A
Obs
JmaxRef = 343.62
RdRef = 2.69
sigma = 0.45
VcmaxRef = 126.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

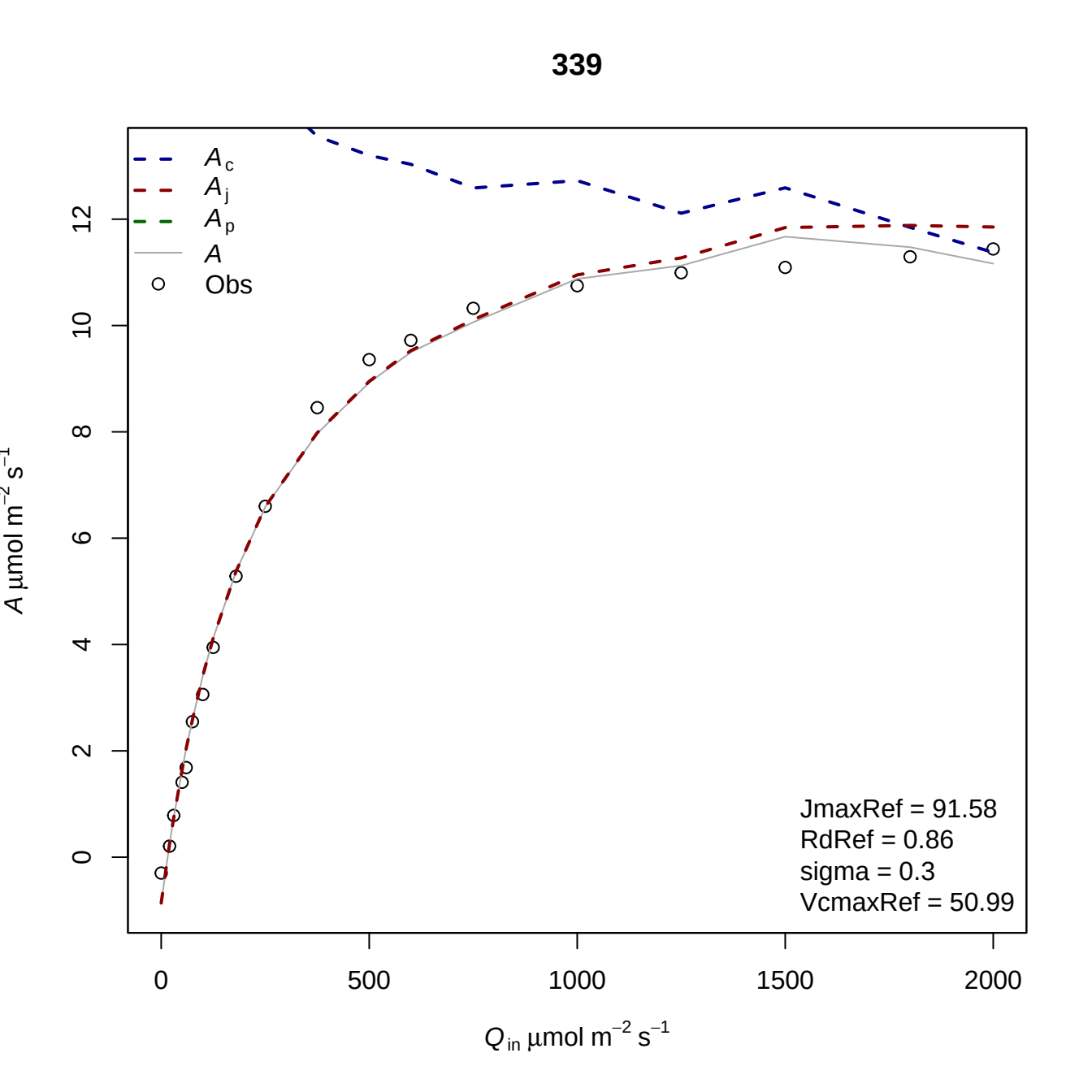

334
A_c
A_j
A_p
A
Obs
JmaxRef = 242.58
RdRef = 1.44
sigma = 0.26
VcmaxRef = 83.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

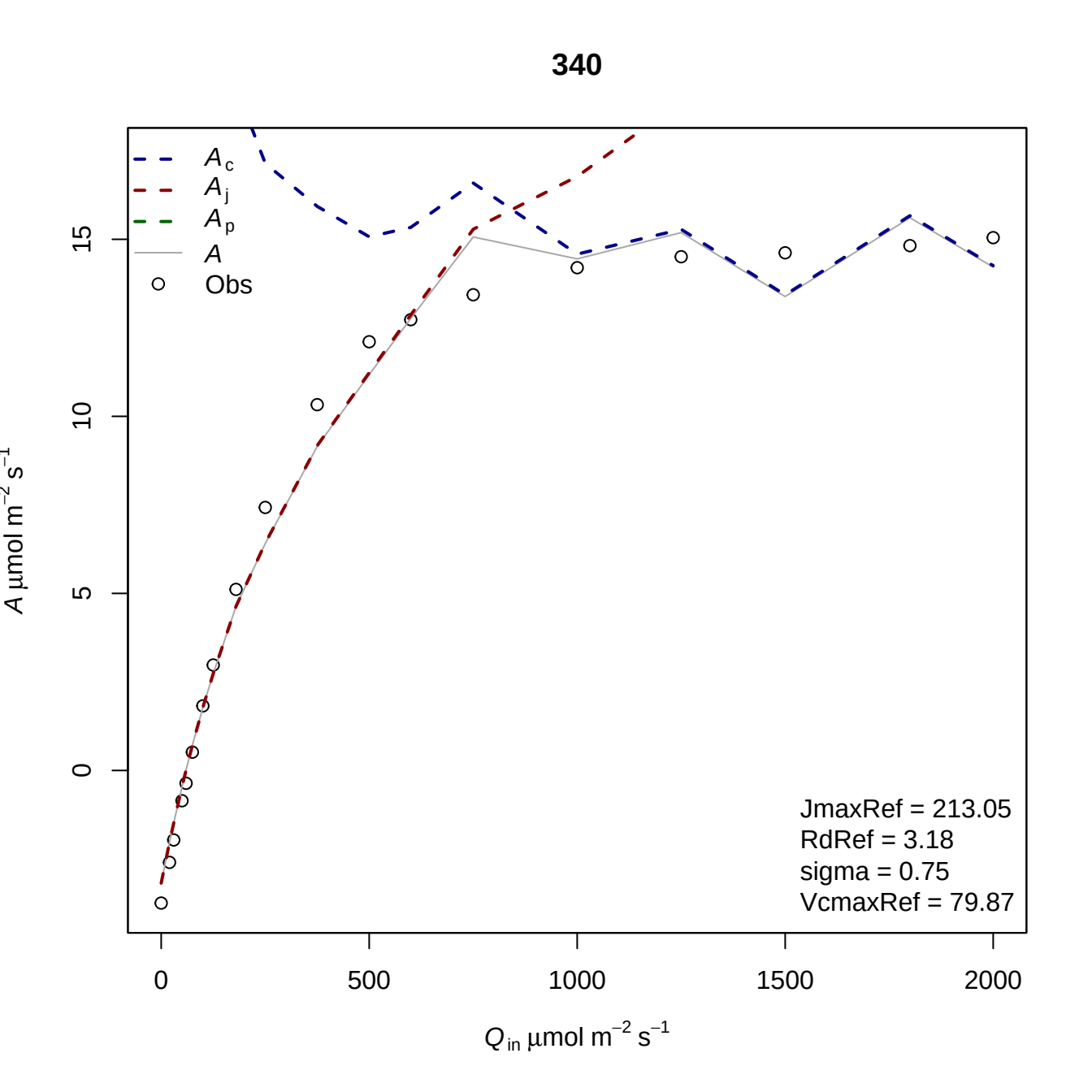

335
A_c
A_j
A_p
A
Obs
JmaxRef = 111.9
RdRef = 3.15
sigma = 0.86
VcmaxRef = 223669.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

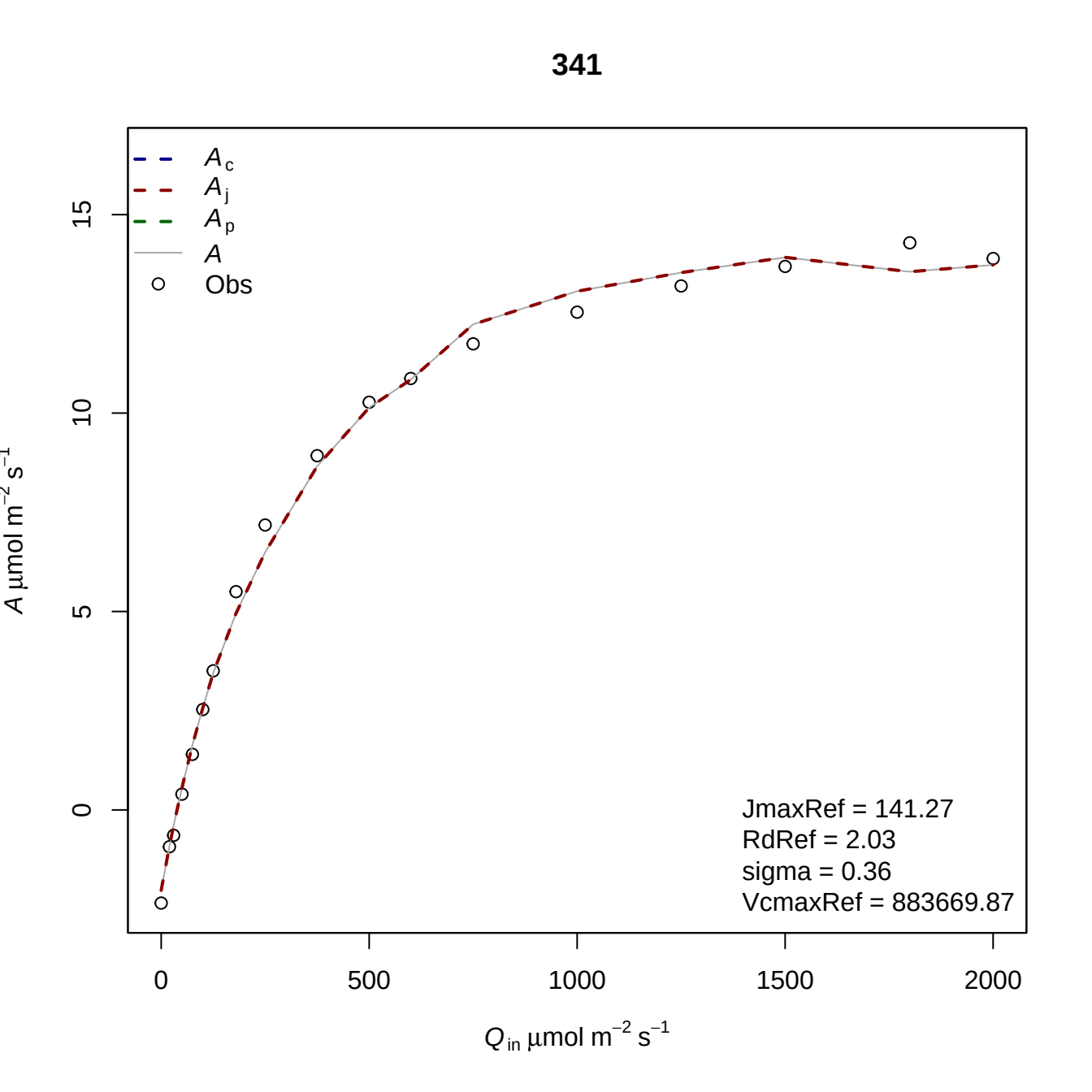

336
A_c
A_j
A_p
A
Obs
JmaxRef = 132.7
RdRef = 1.37
sigma = 0.28
VcmaxRef = 67.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

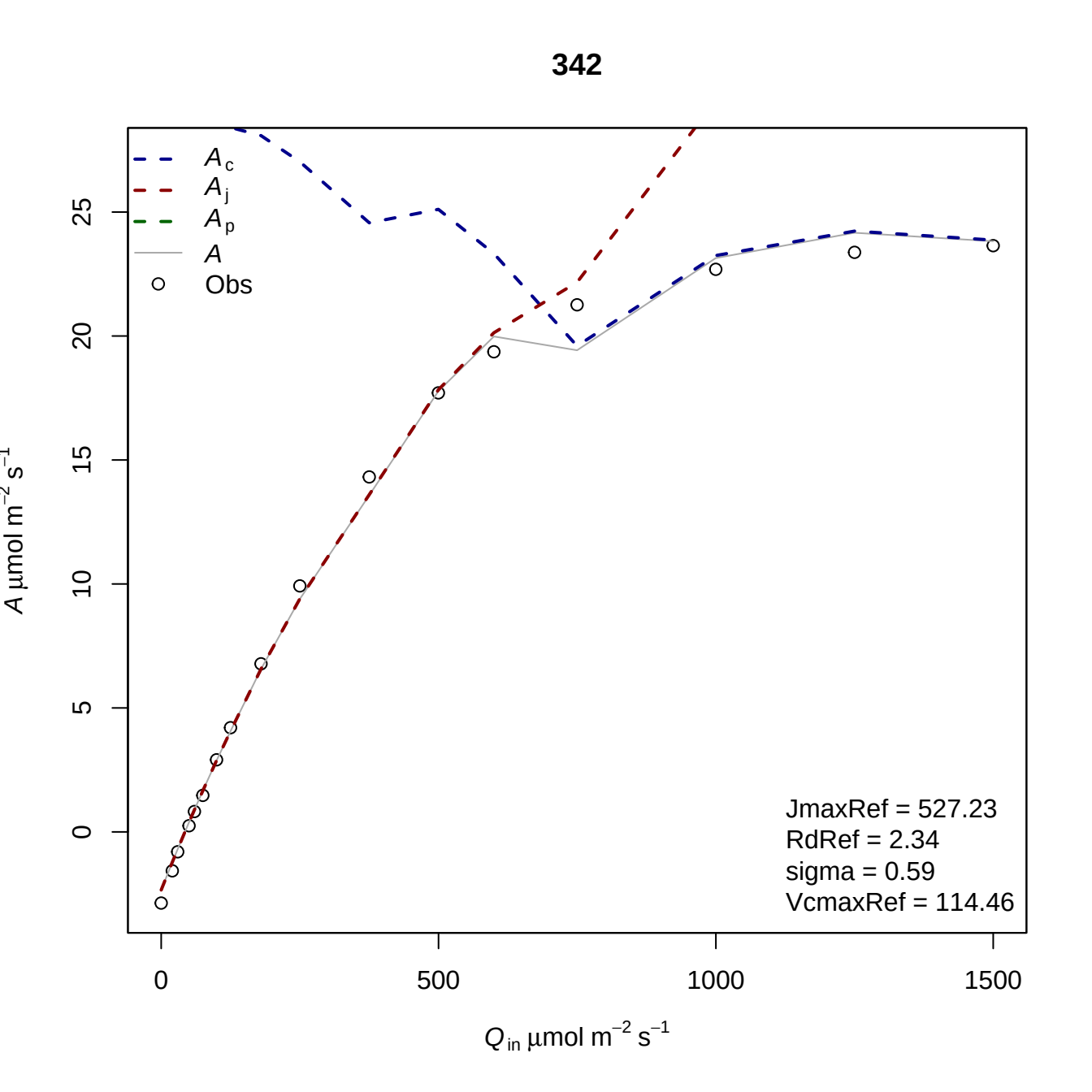

337
A_c
A_j
A_p
A
Obs
JmaxRef = 118.83
RdRef = 1.11
sigma = 0.21
VcmaxRef = 59.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

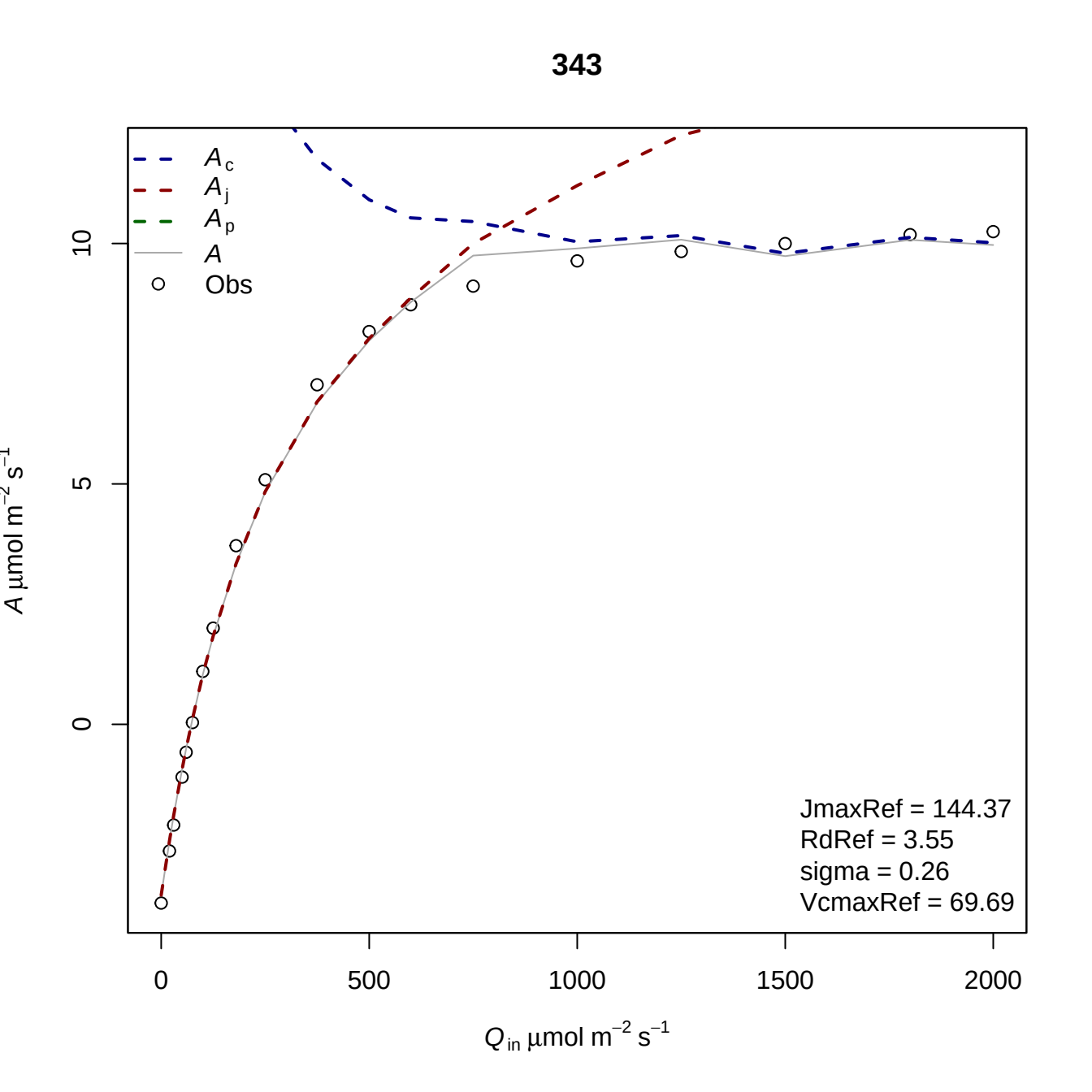

338
A_c
A_j
A_p
A
Obs
JmaxRef = 109.5
RdRef = 1.07
sigma = 0.45
VcmaxRef = 50.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

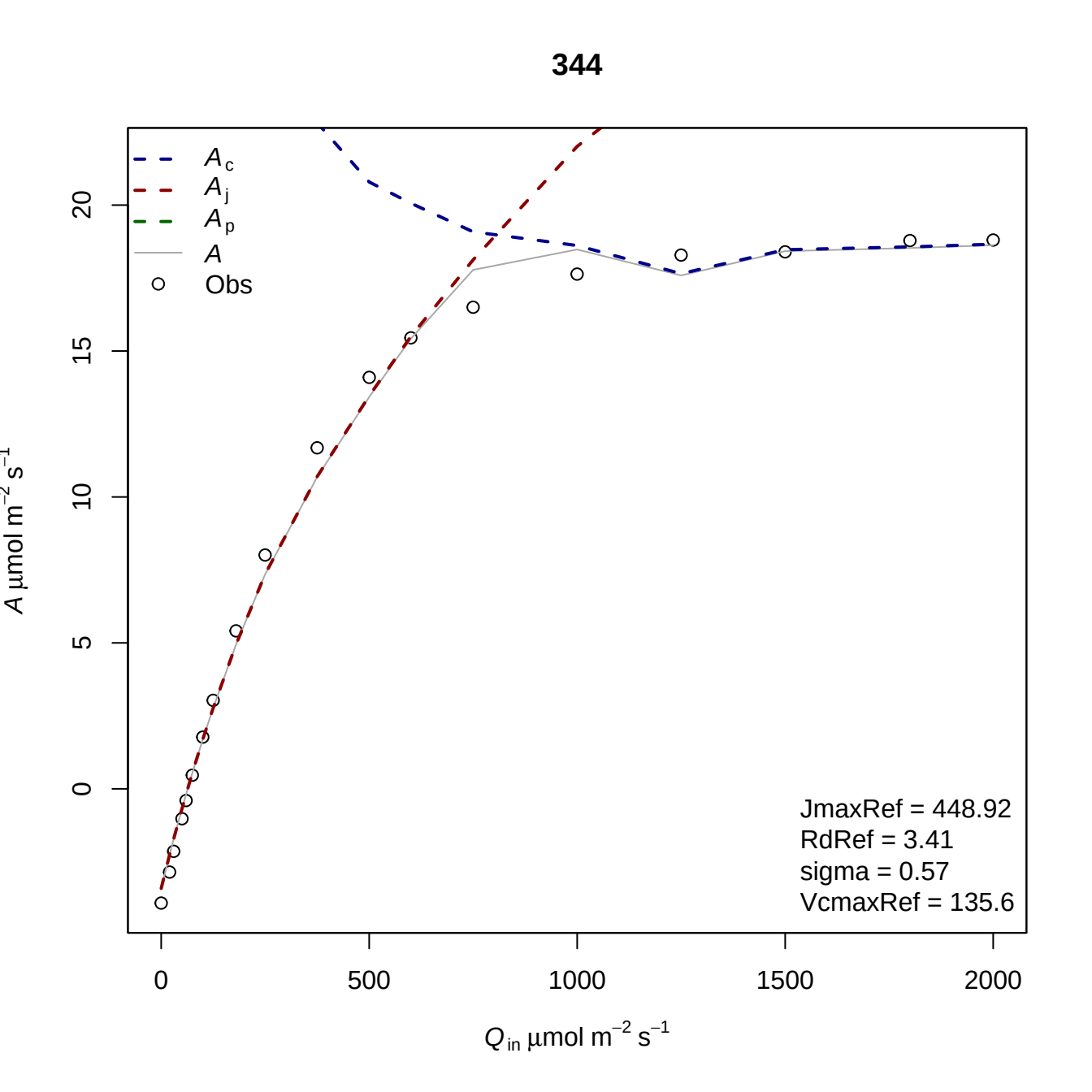

339
A_c
A_j
A_p
A
Obs
JmaxRef = 91.58
RdRef = 0.86
sigma = 0.3
VcmaxRef = 50.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

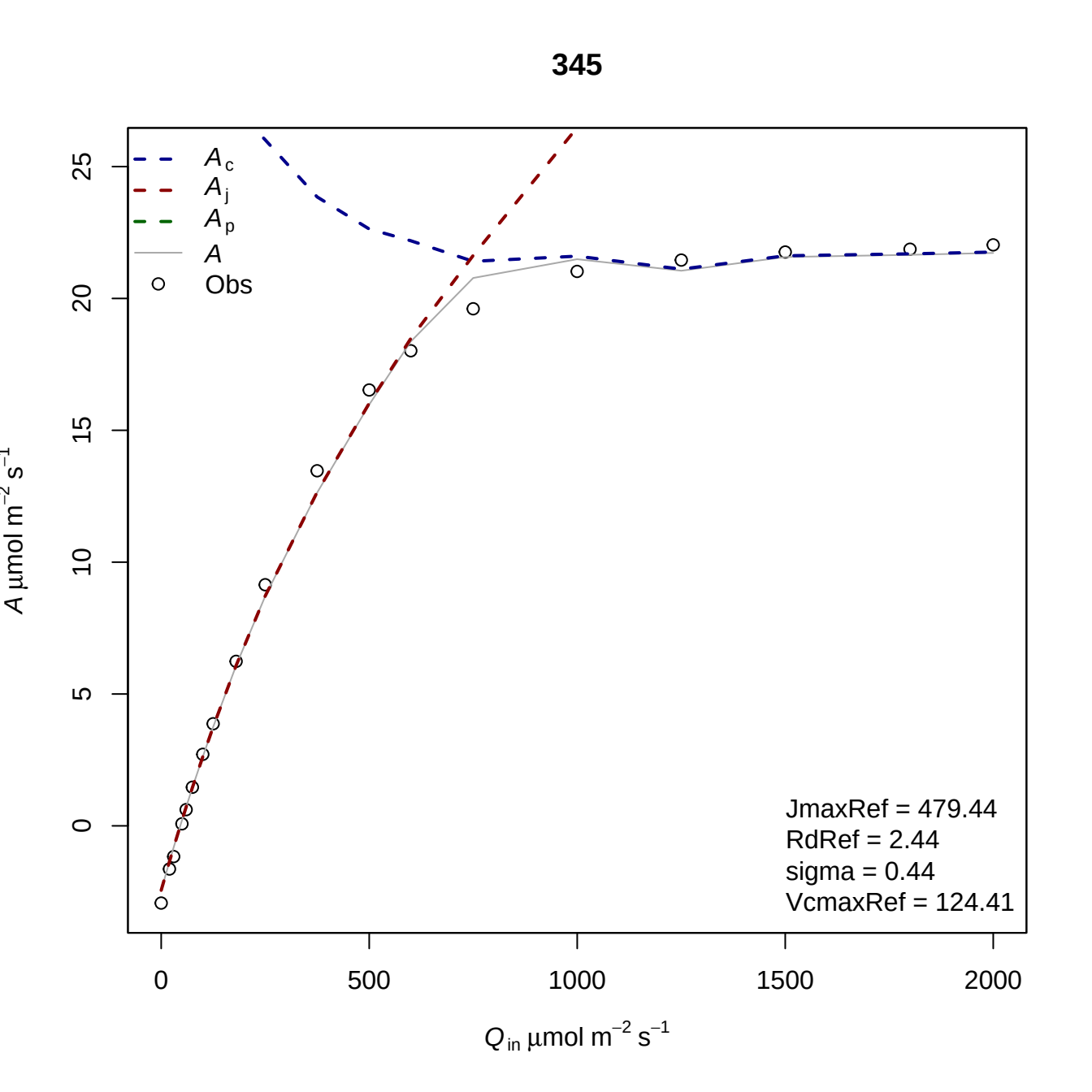

340
A_c
A_j
A_p
A
Obs
JmaxRef = 213.05
RdRef = 3.18
sigma = 0.75
VcmaxRef = 79.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

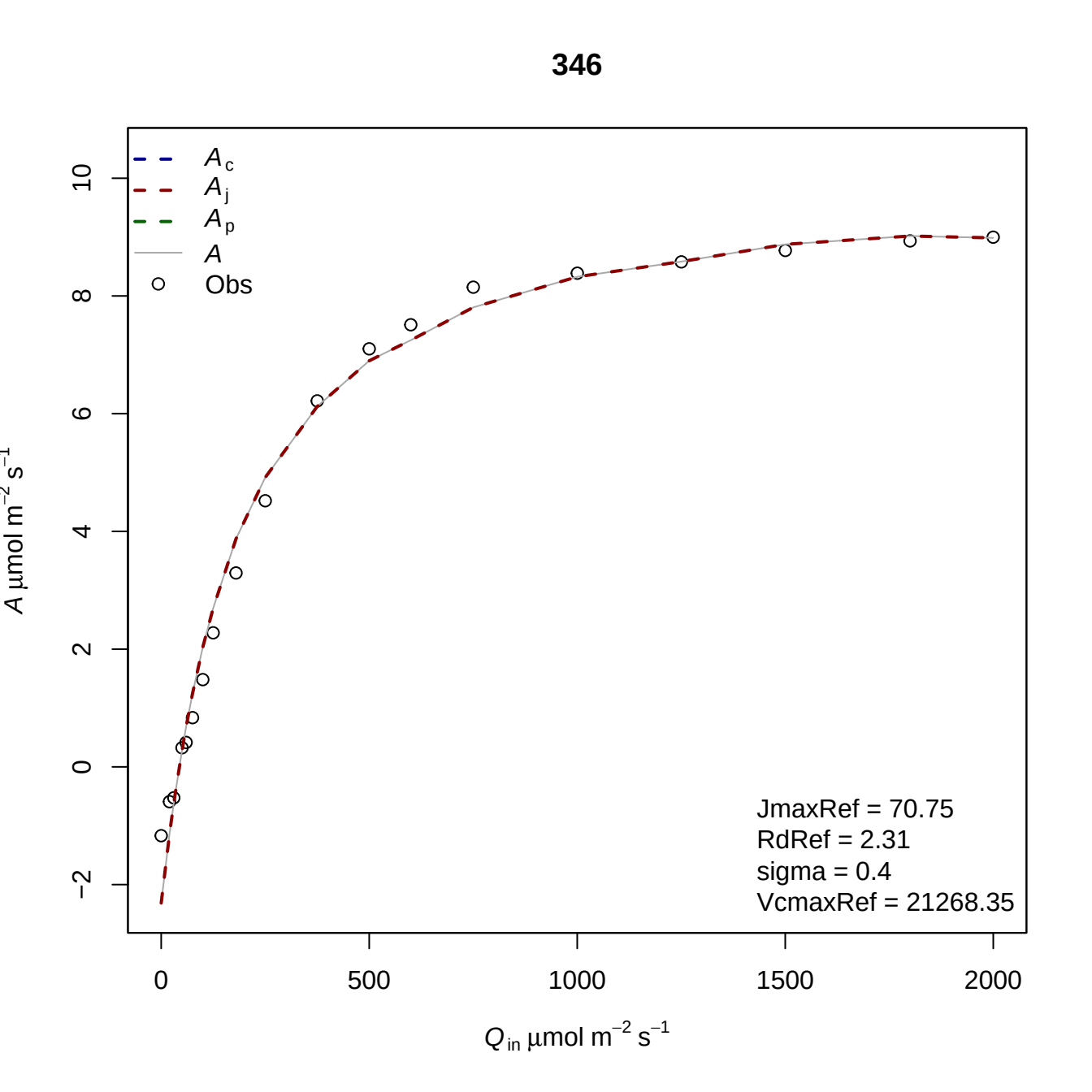

341
A_c
A_j
A_p
A
Obs
JmaxRef = 141.27
RdRef = 2.03
sigma = 0.36
VcmaxRef = 883669.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

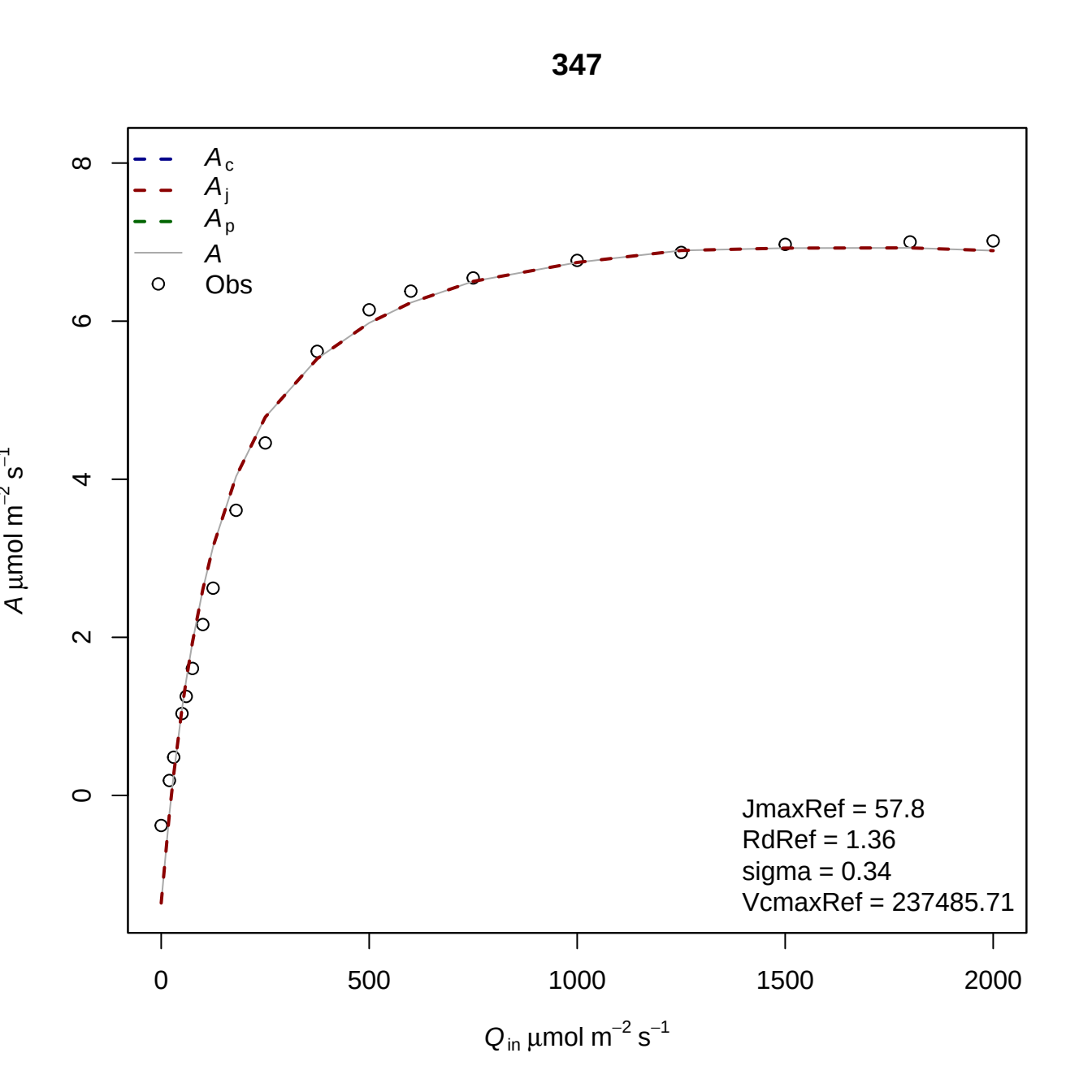

342
A_c
A_j
A_p
A
Obs
JmaxRef = 527.23
RdRef = 2.34
sigma = 0.59
VcmaxRef = 114.46
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

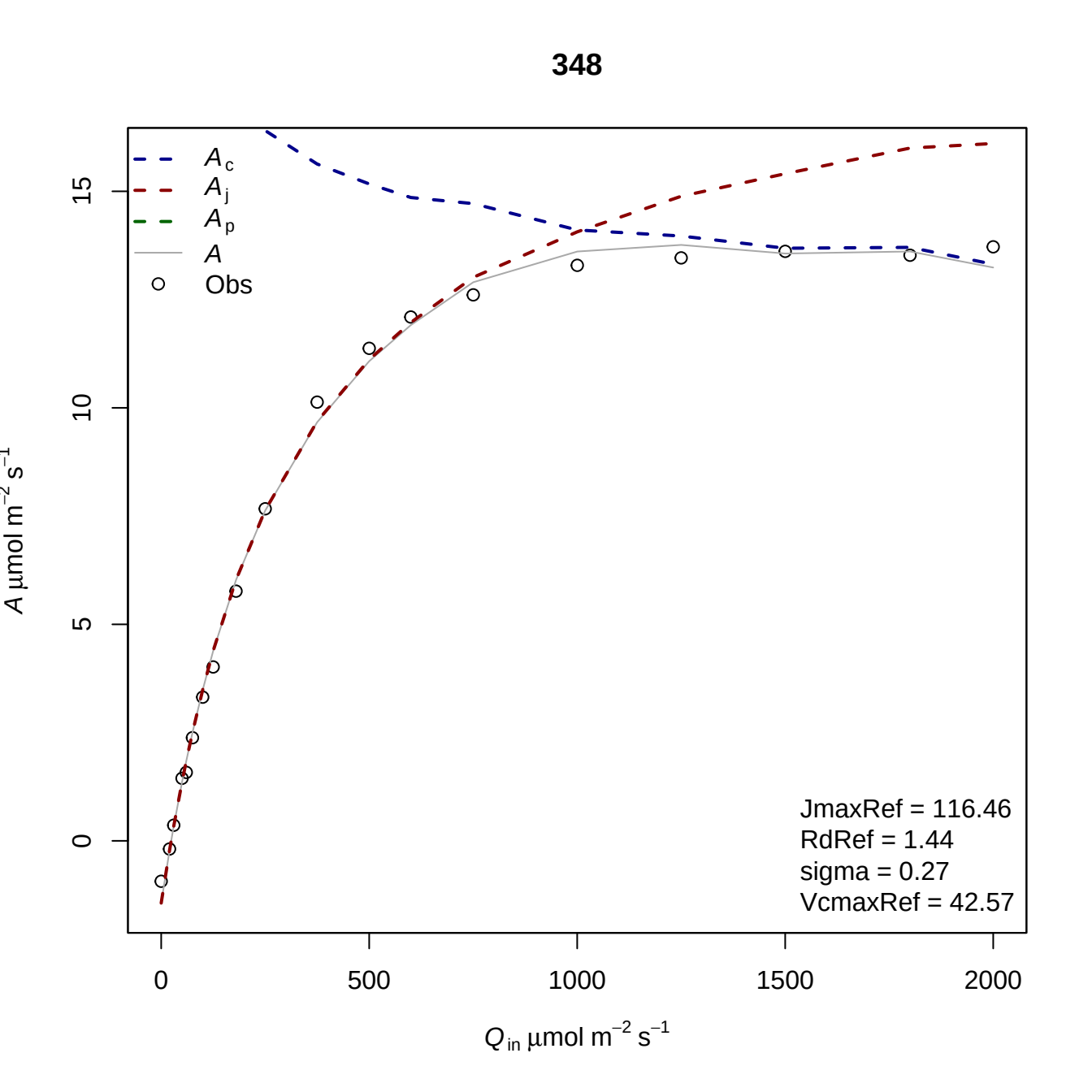

343
A_c
A_j
A_p
A
Obs
JmaxRef = 144.37
RdRef = 3.55
sigma = 0.26
VcmaxRef = 69.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

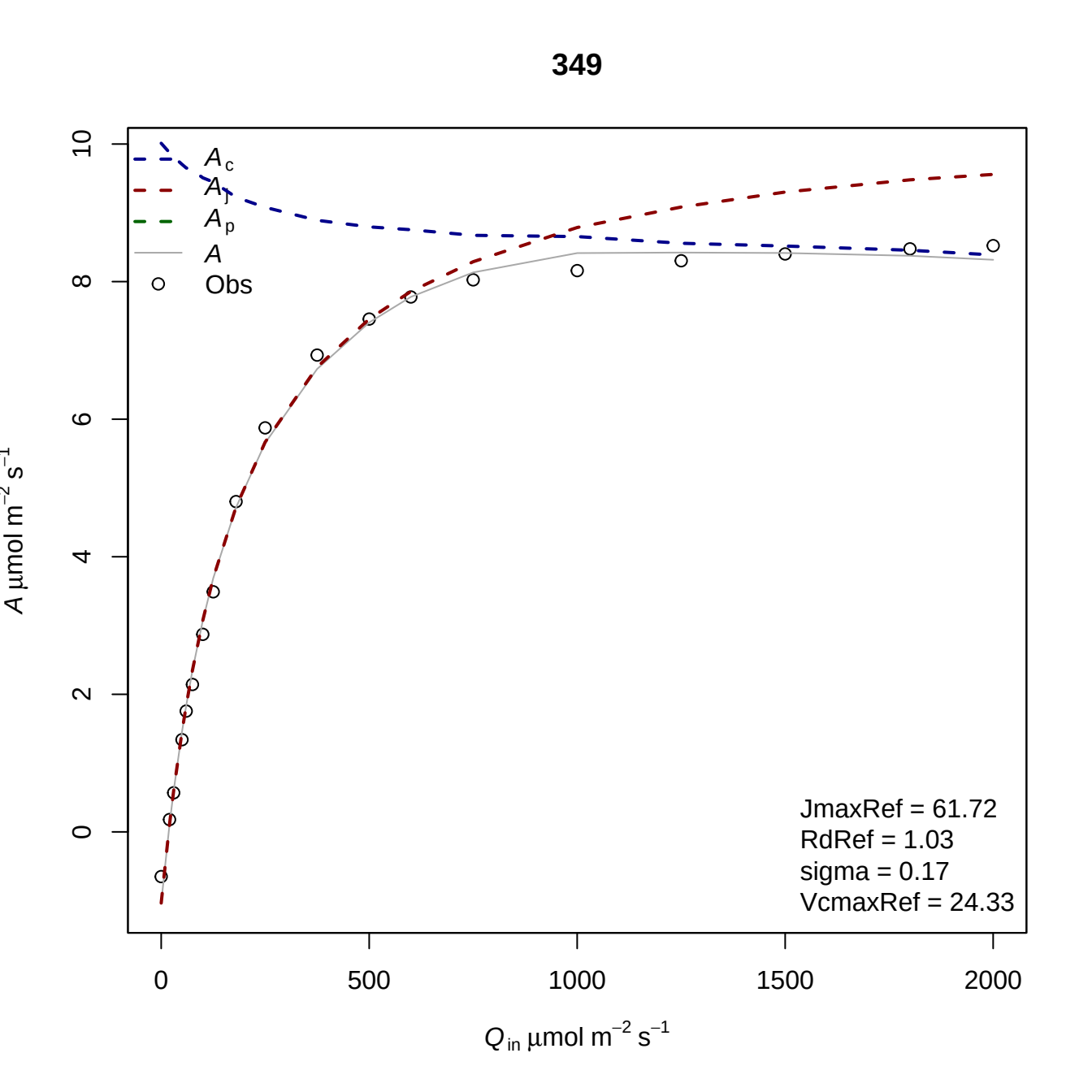

344
A_c
A_j
A_p
A
Obs
JmaxRef = 448.92
RdRef = 3.41
sigma = 0.57
VcmaxRef = 135.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

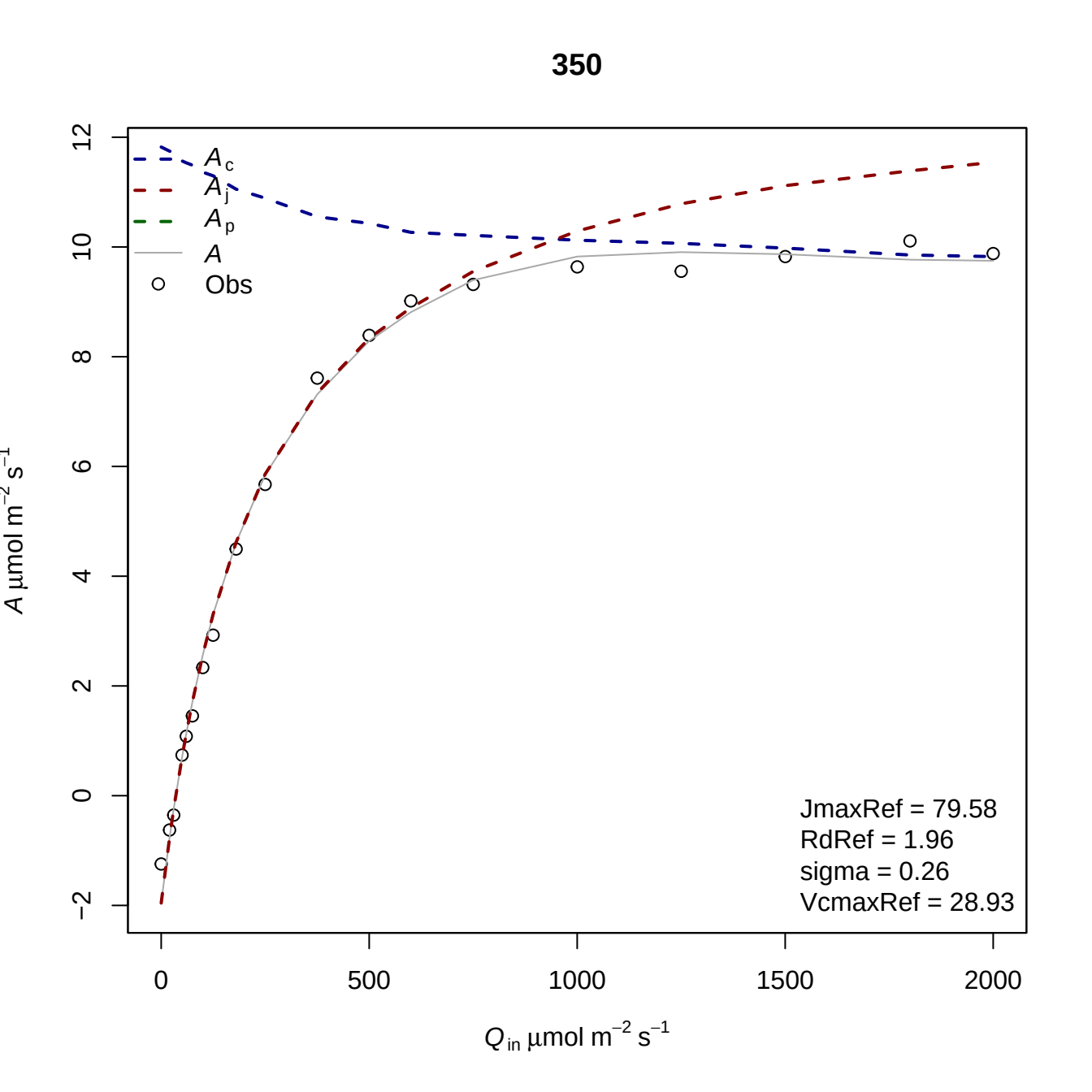

345
A_c
A_j
A_p
A
Obs
JmaxRef = 479.44
RdRef = 2.44
sigma = 0.44
VcmaxRef = 124.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

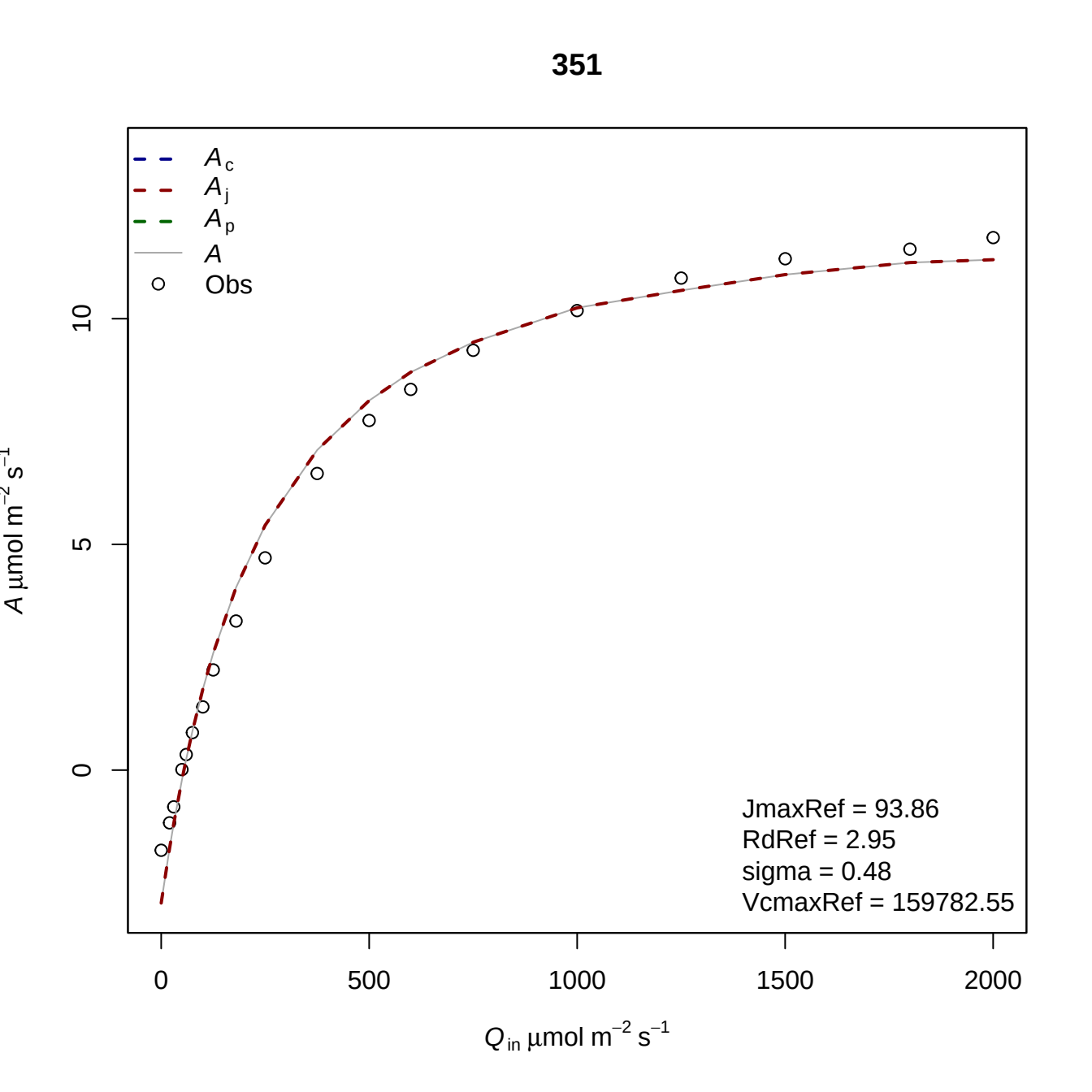

346
A_c
A_j
A_p
A
Obs
JmaxRef = 70.75
RdRef = 2.31
sigma = 0.4
VcmaxRef = 21268.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

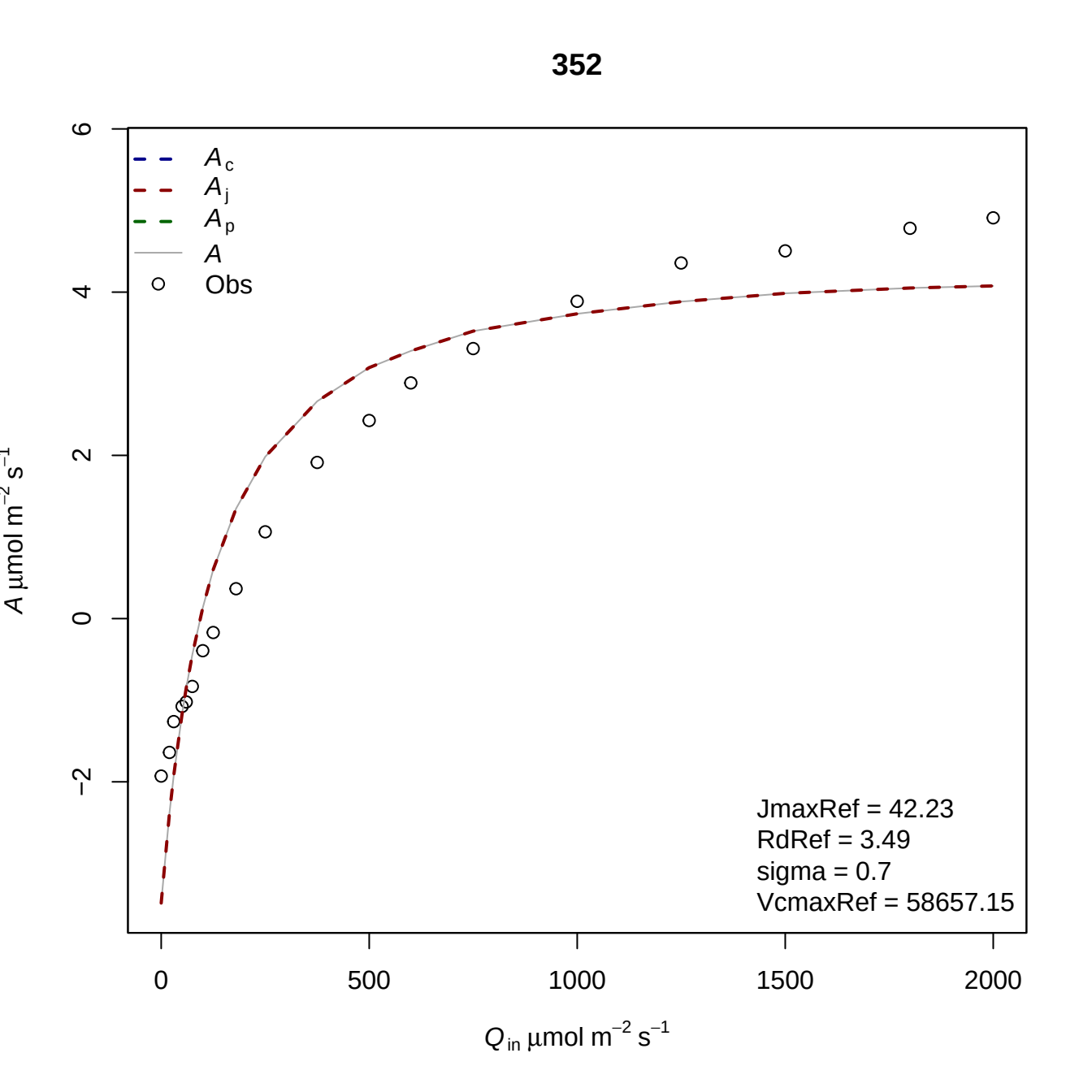

347
A_c
A_j
A_p
A
Obs
JmaxRef = 57.8
RdRef = 1.36
sigma = 0.34
VcmaxRef = 237485.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

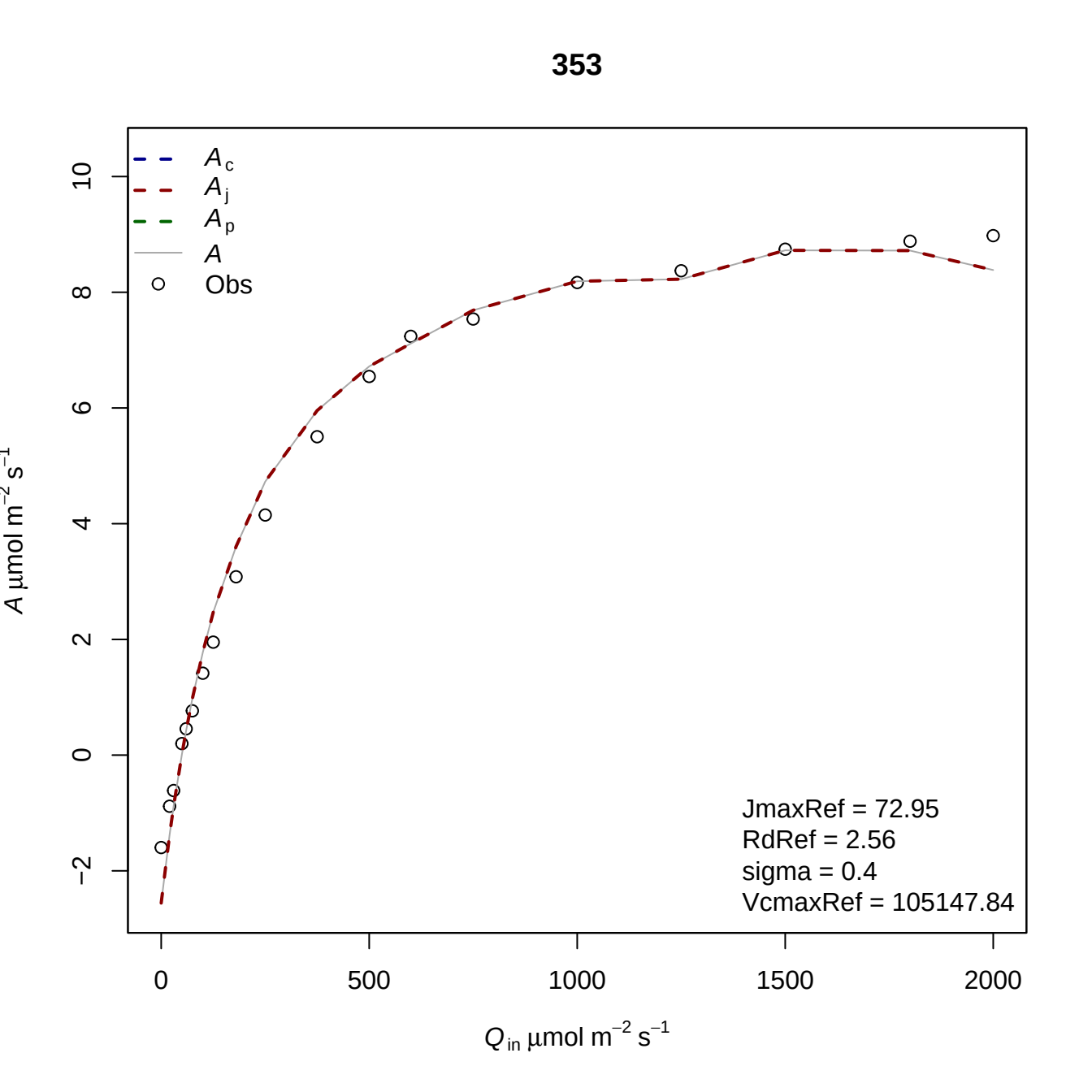

348
A_c
A_j
A_p
A
Obs
JmaxRef = 116.46
RdRef = 1.44
sigma = 0.27
VcmaxRef = 42.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

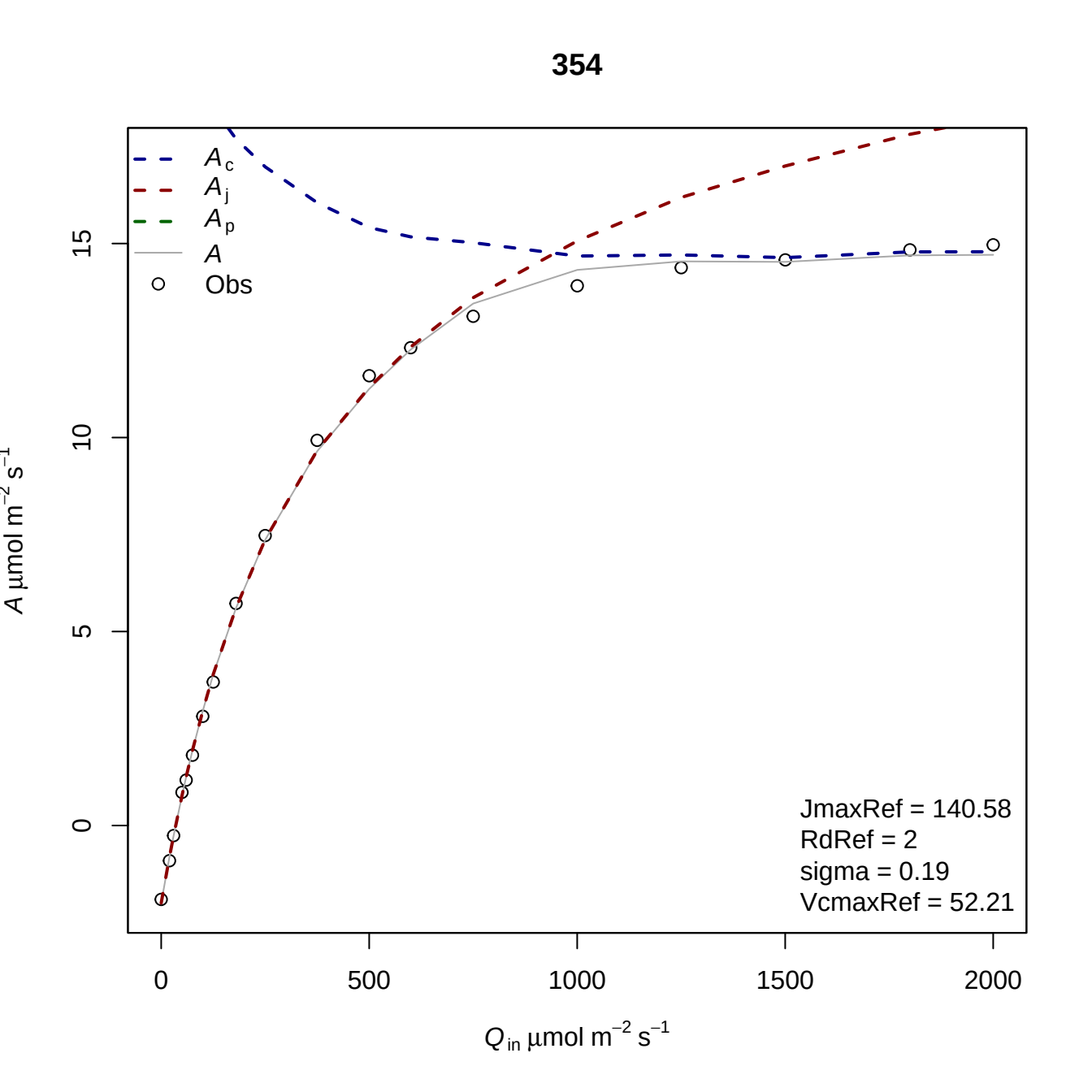

349
A_c
A_j
A_p
A
Obs
JmaxRef = 61.72
RdRef = 1.03
sigma = 0.17
VcmaxRef = 24.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

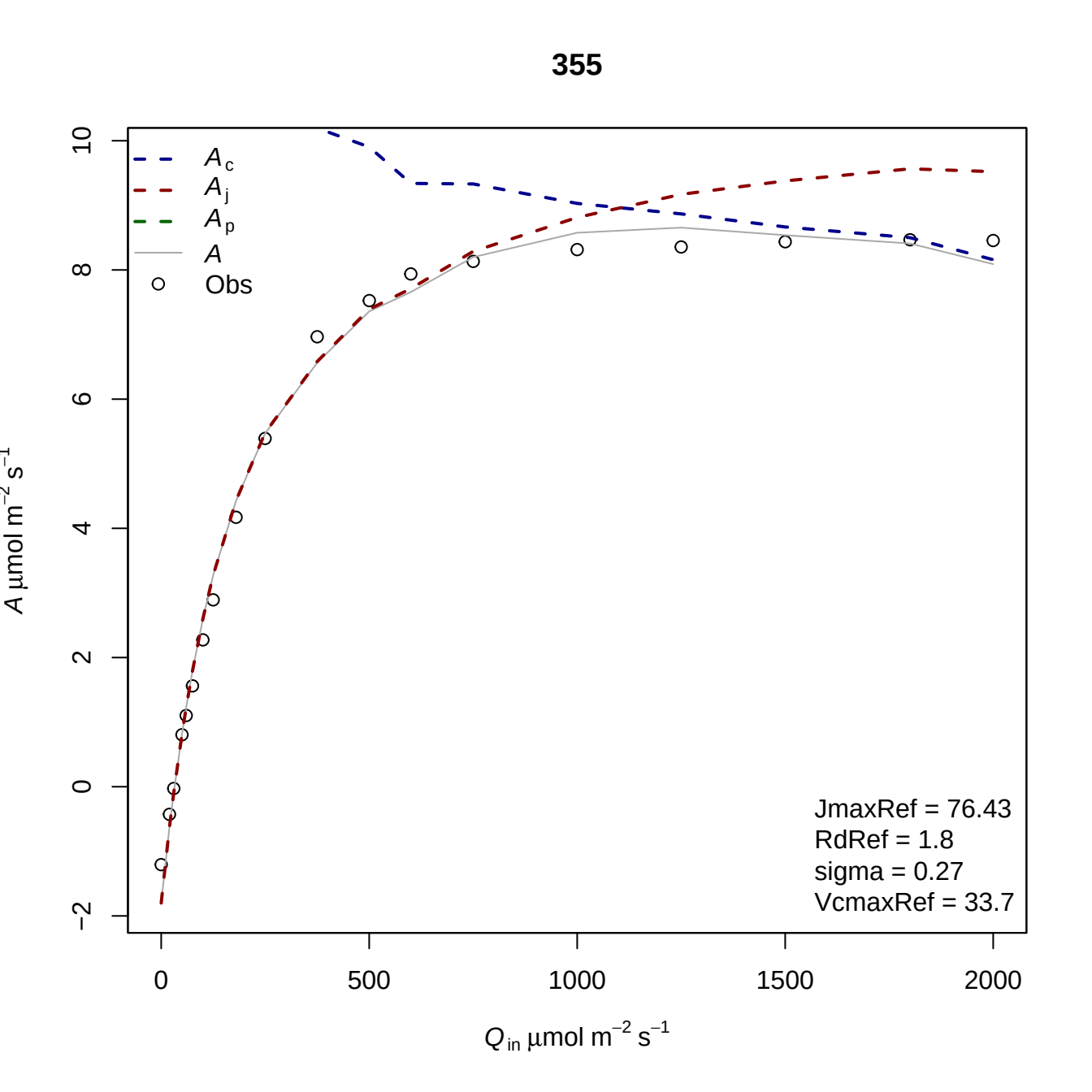

350
A_c
A_j
A_p
A
Obs
JmaxRef = 79.58
RdRef = 1.96
sigma = 0.26
VcmaxRef = 28.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

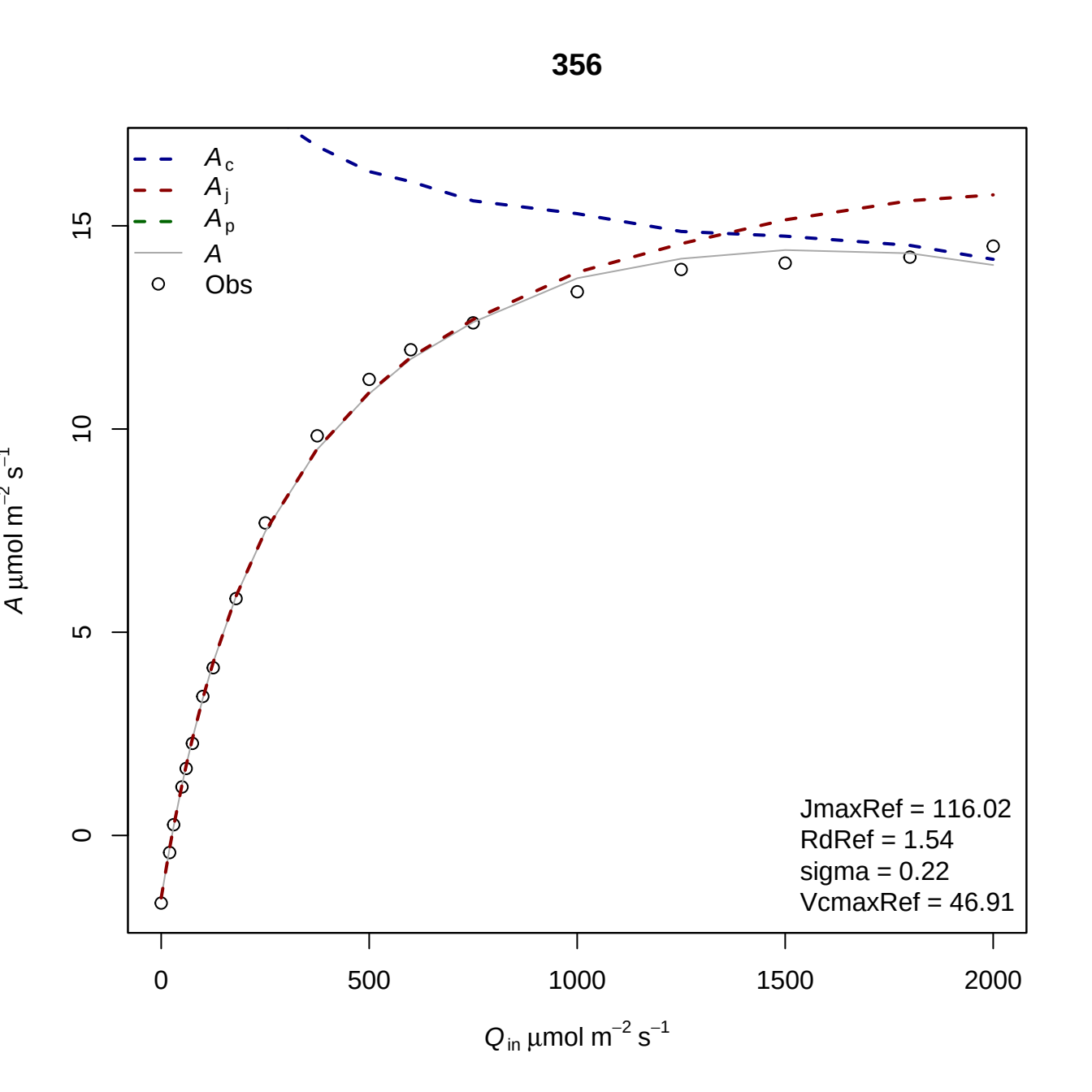

351
A_c
A_j
A_p
A
Obs
JmaxRef = 93.86
RdRef = 2.95
sigma = 0.48
VcmaxRef = 159782.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

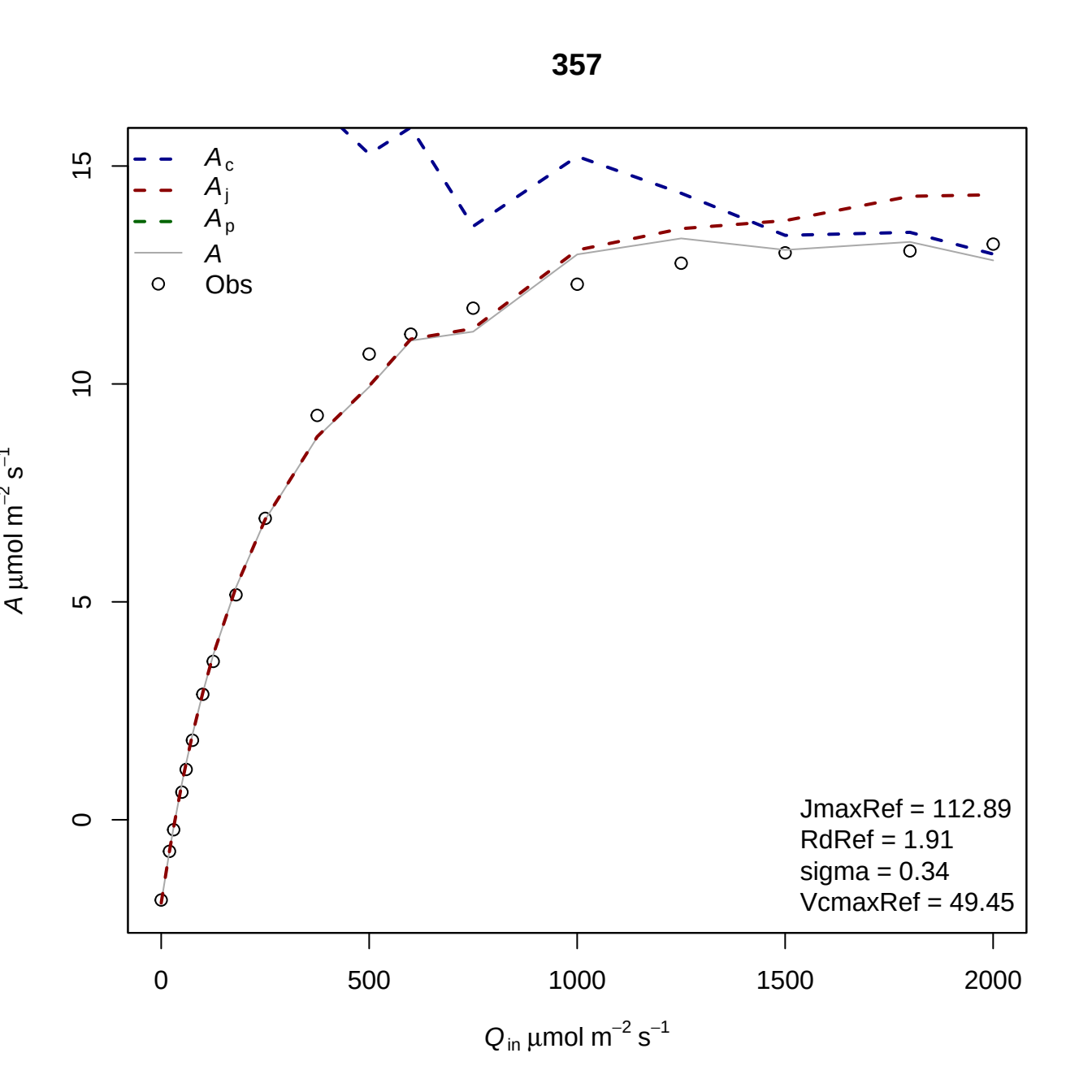

352
A_c
A_j
A_p
A
Obs
JmaxRef = 42.23
RdRef = 3.49
sigma = 0.7
VcmaxRef = 58657.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

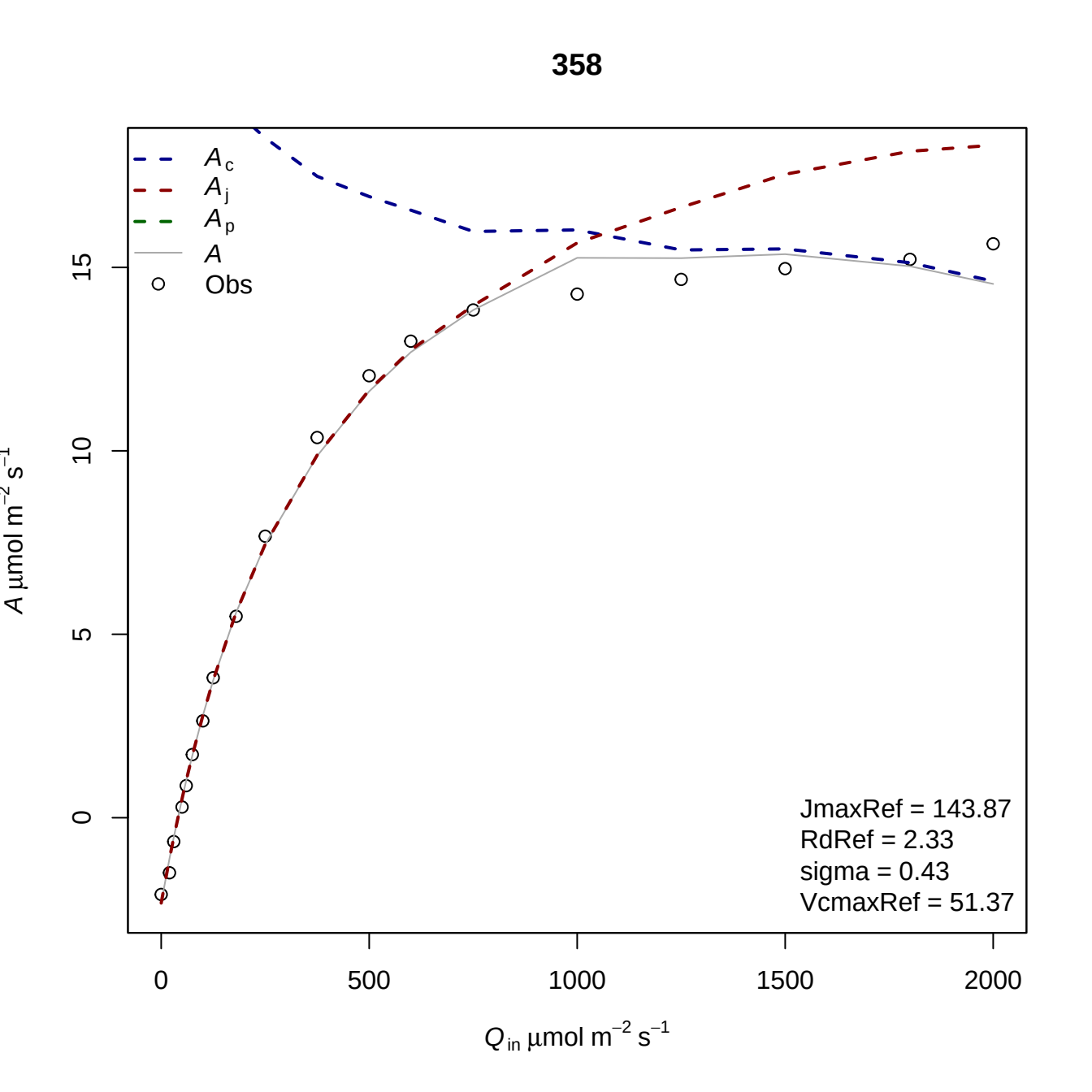

353
A_c
A_j
A_p
A
Obs
JmaxRef = 72.95
RdRef = 2.56
sigma = 0.4
VcmaxRef = 105147.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

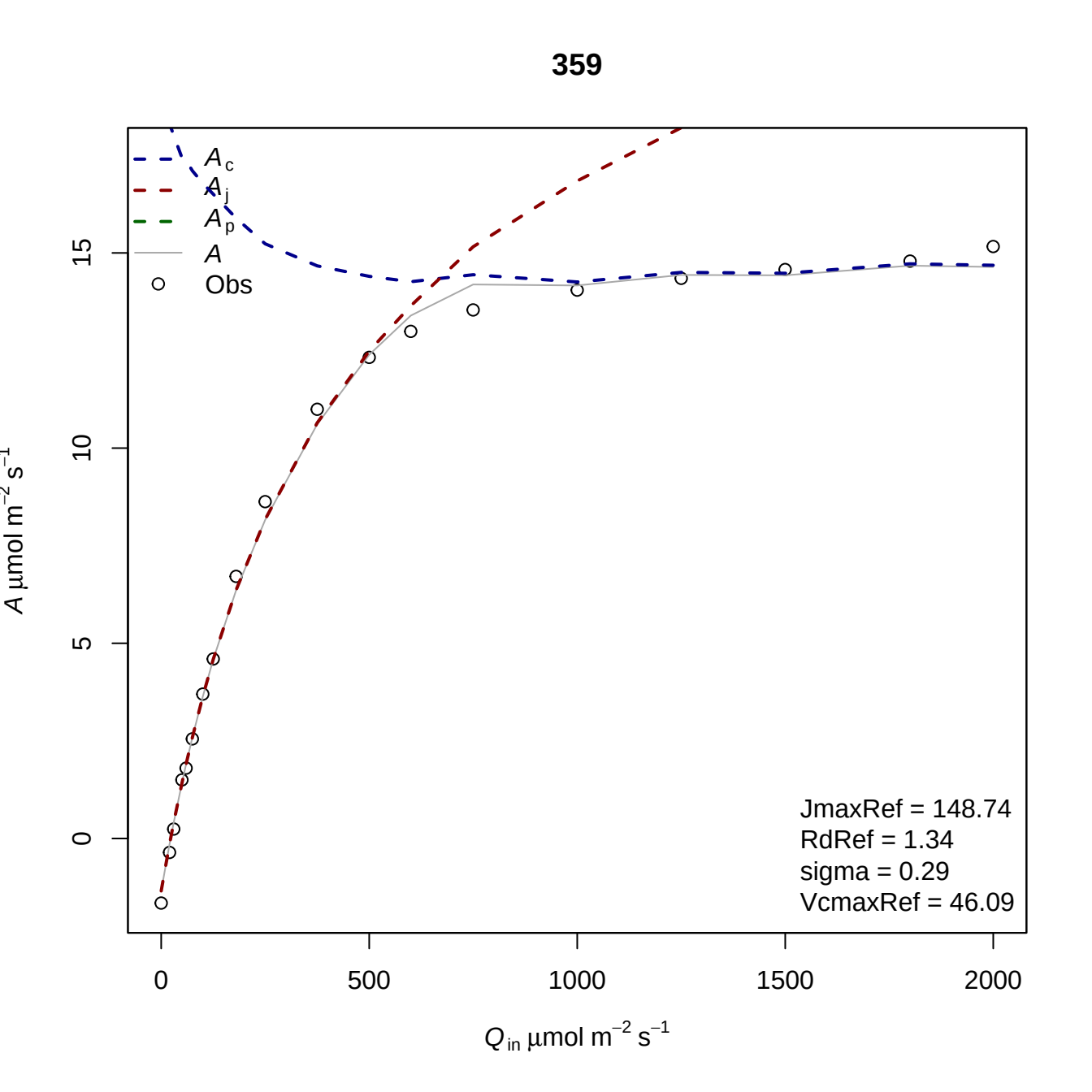

354
A_c
A_j
A_p
A
Obs
JmaxRef = 140.58
RdRef = 2
sigma = 0.19
VcmaxRef = 52.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

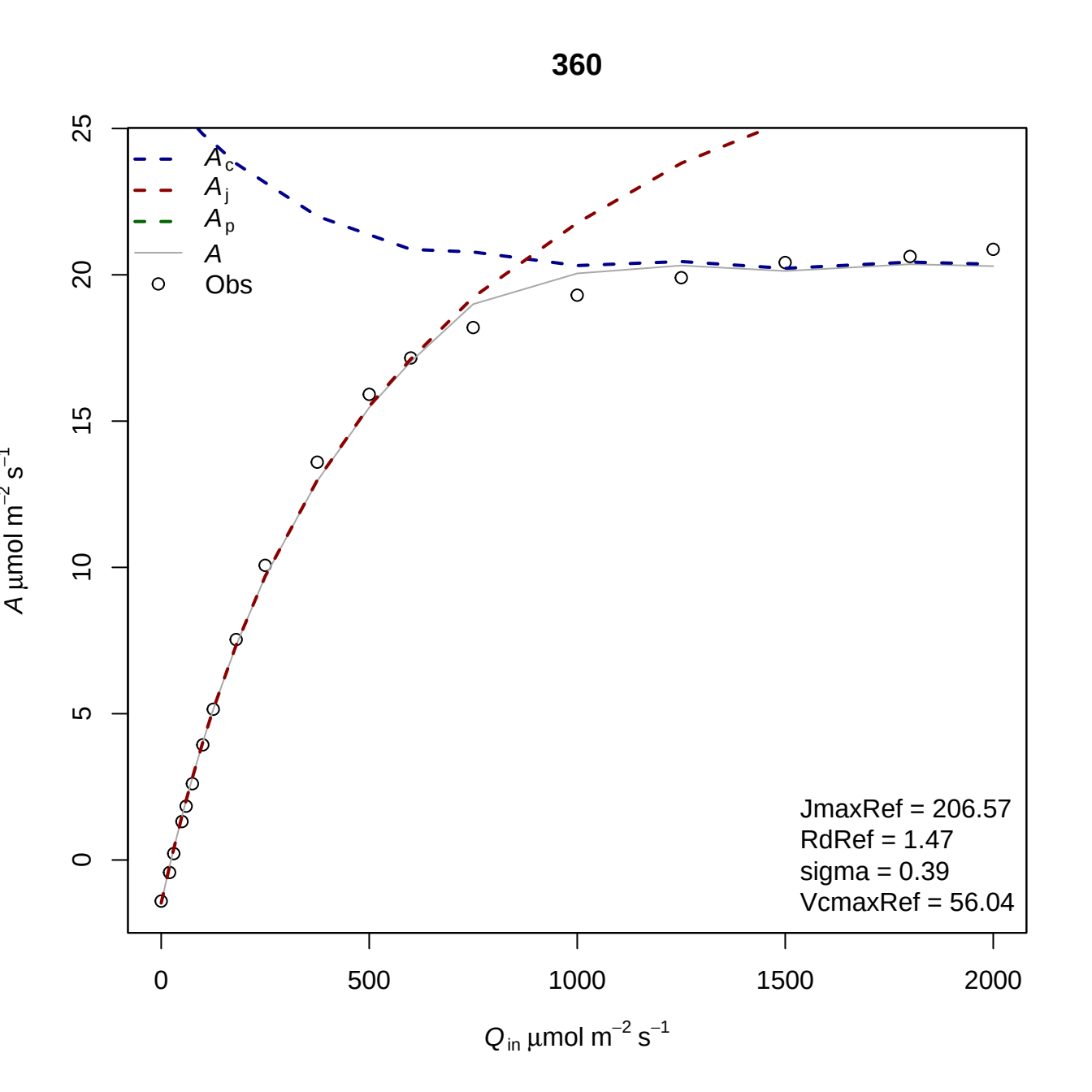

355
A_c
A_j
A_p
A
Obs
JmaxRef = 76.43
RdRef = 1.8
sigma = 0.27
VcmaxRef = 33.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

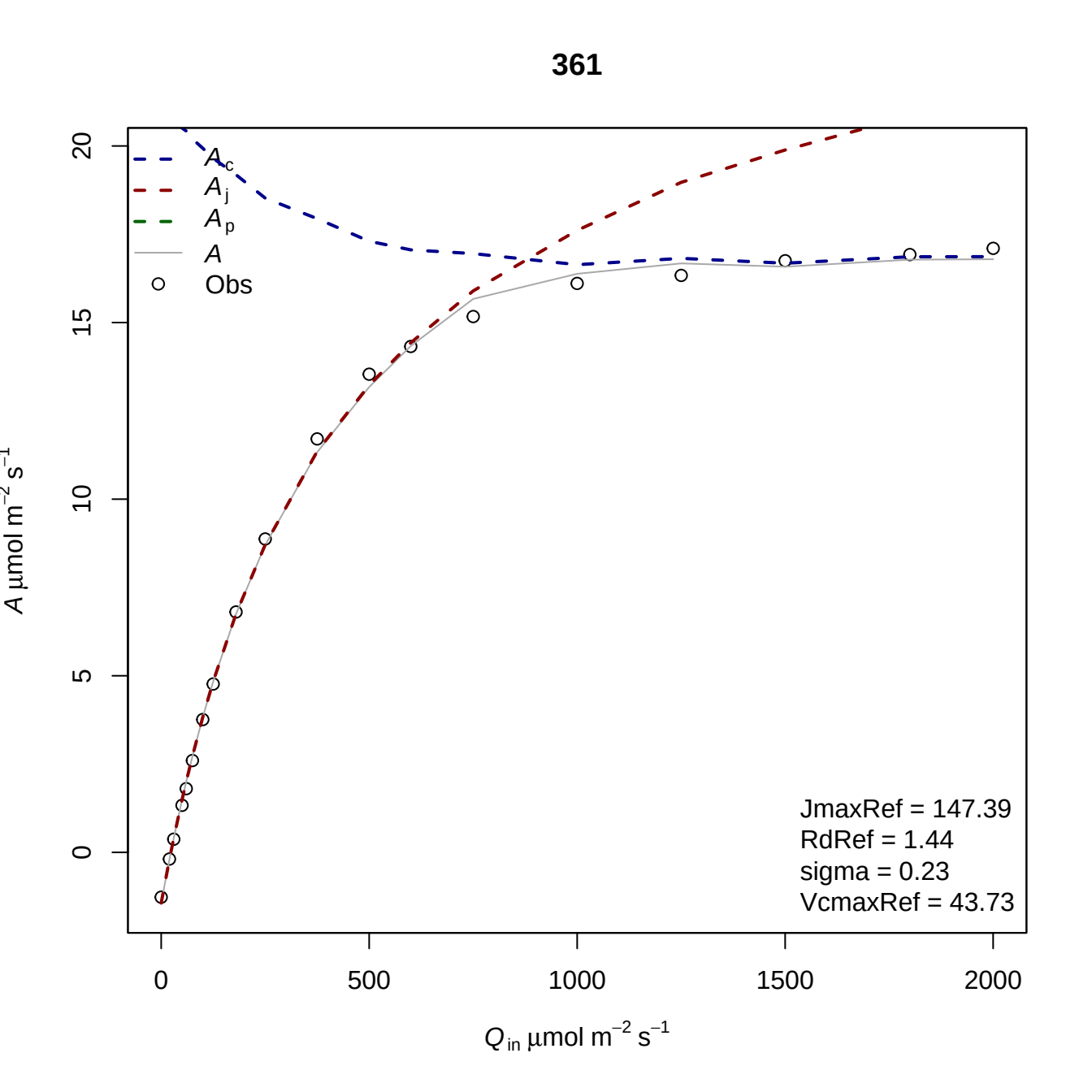

356
A_c
A_j
A_p
A
Obs
JmaxRef = 116.02
RdRef = 1.54
sigma = 0.22
VcmaxRef = 46.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

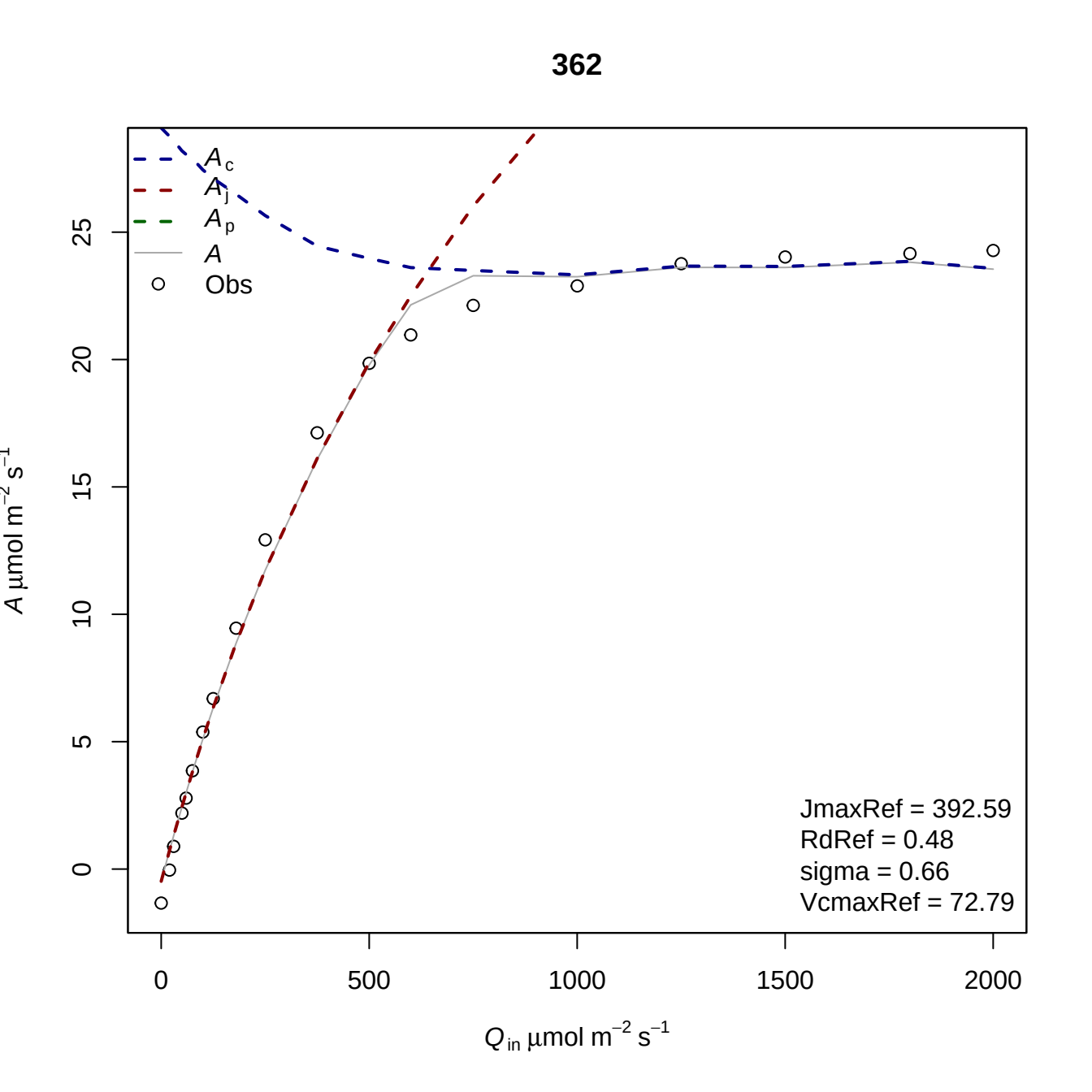

357
A_c
A_j
A_p
A
Obs
JmaxRef = 112.89
RdRef = 1.91
sigma = 0.34
VcmaxRef = 49.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

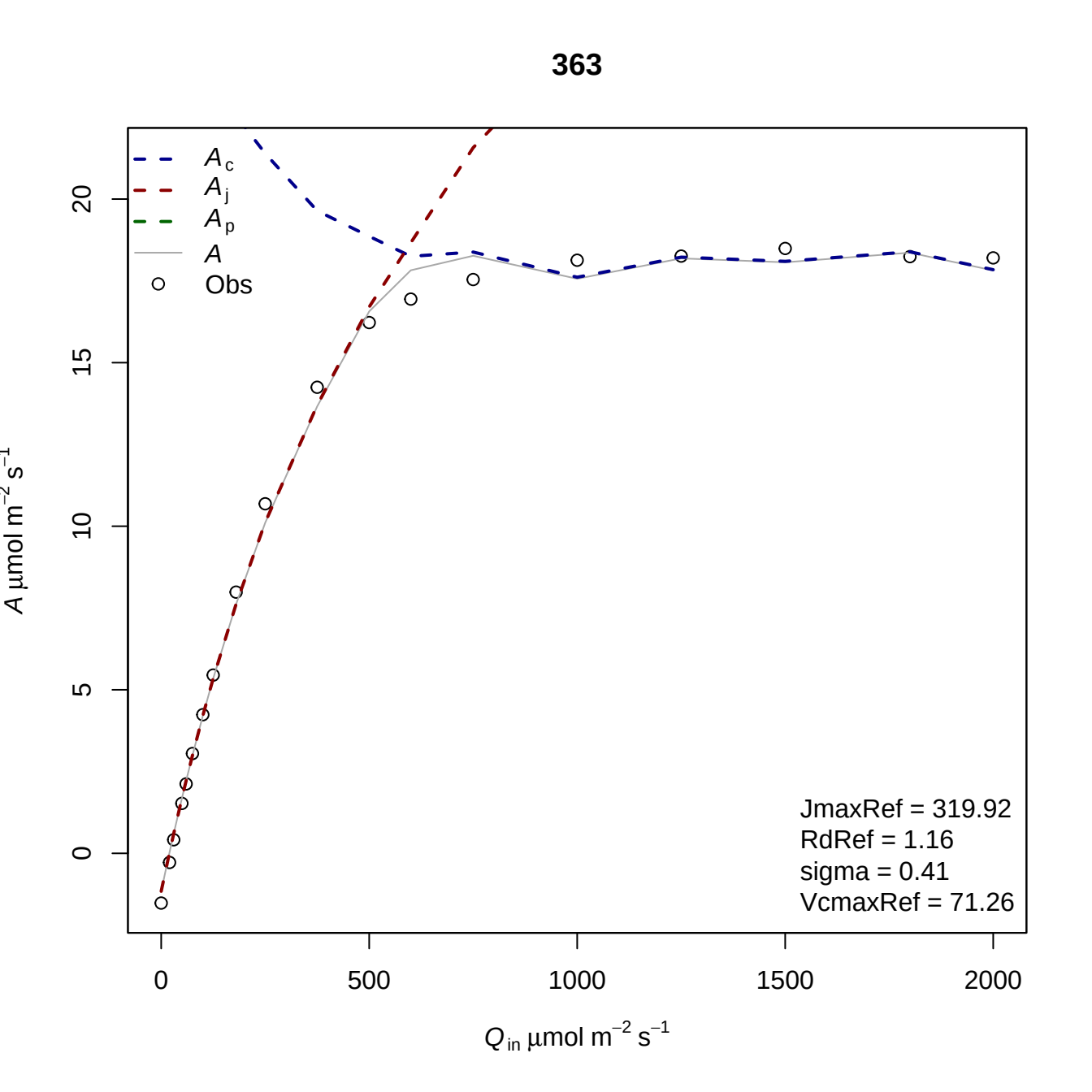

358
A_c
A_j
A_p
A
Obs
JmaxRef = 143.87
RdRef = 2.33
sigma = 0.43
VcmaxRef = 51.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

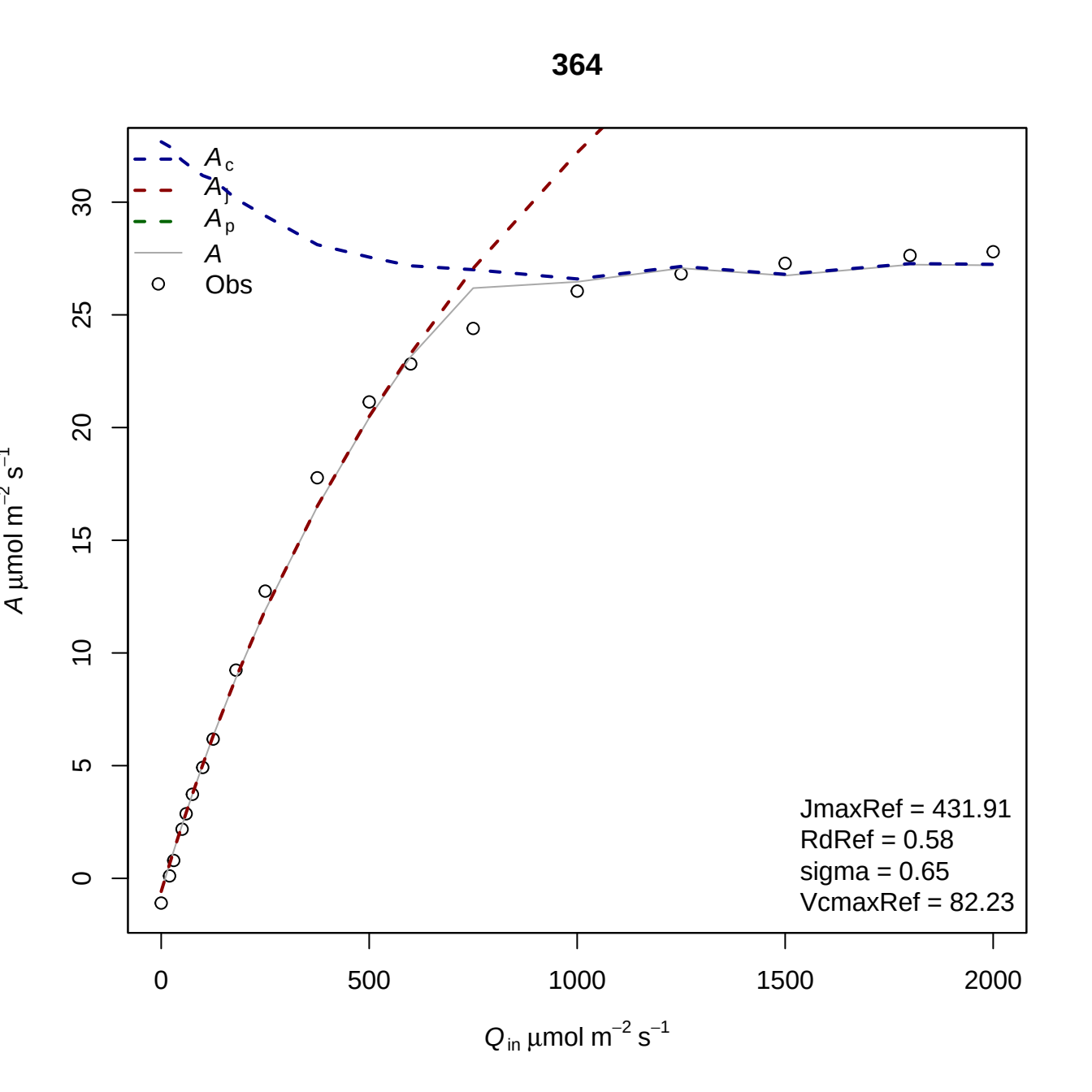

359
A_c
A_j
A_p
A
Obs
JmaxRef = 148.74
RdRef = 1.34
sigma = 0.29
VcmaxRef = 46.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

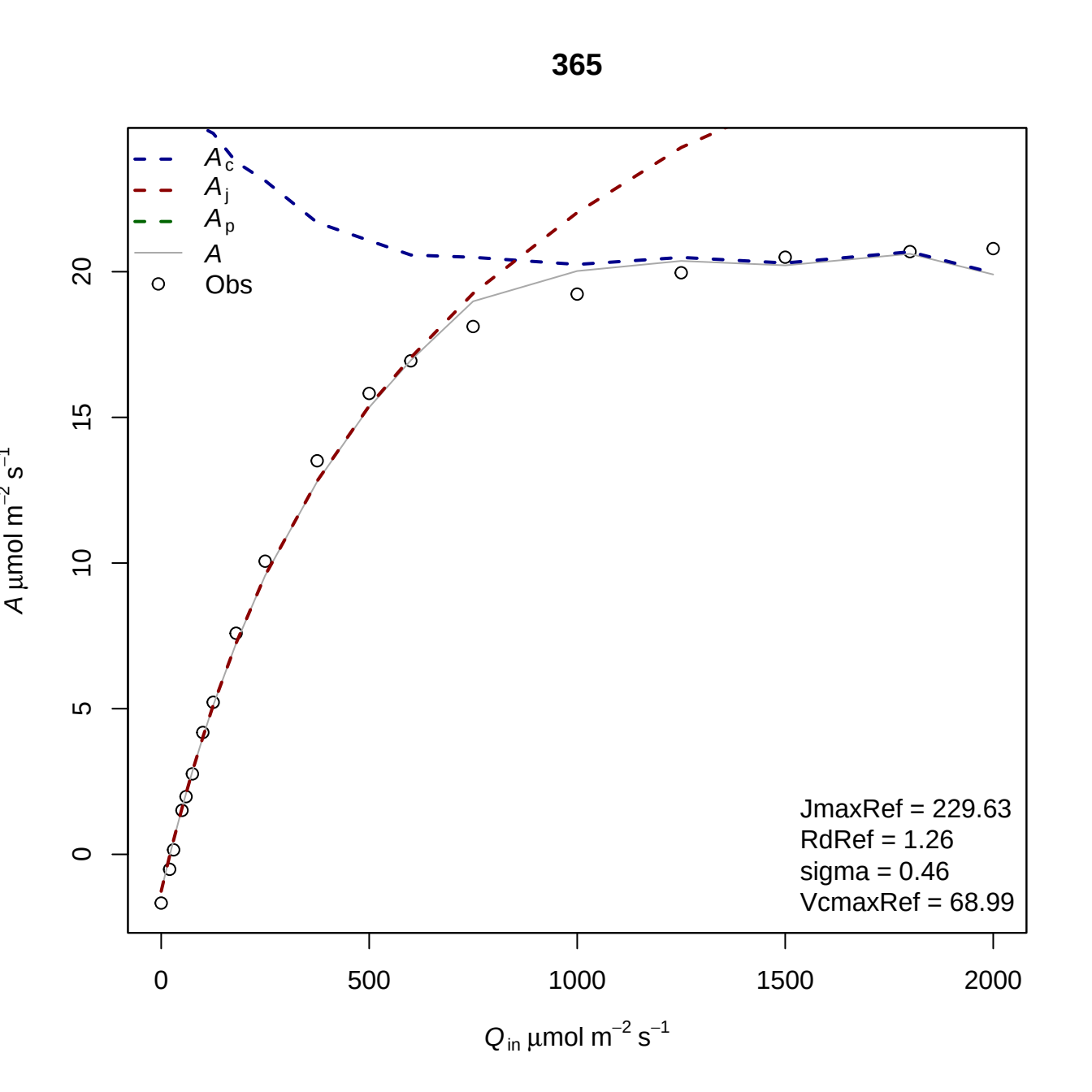

360
A_c
A_j
A_p
A
Obs
JmaxRef = 206.57
RdRef = 1.47
sigma = 0.39
VcmaxRef = 56.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

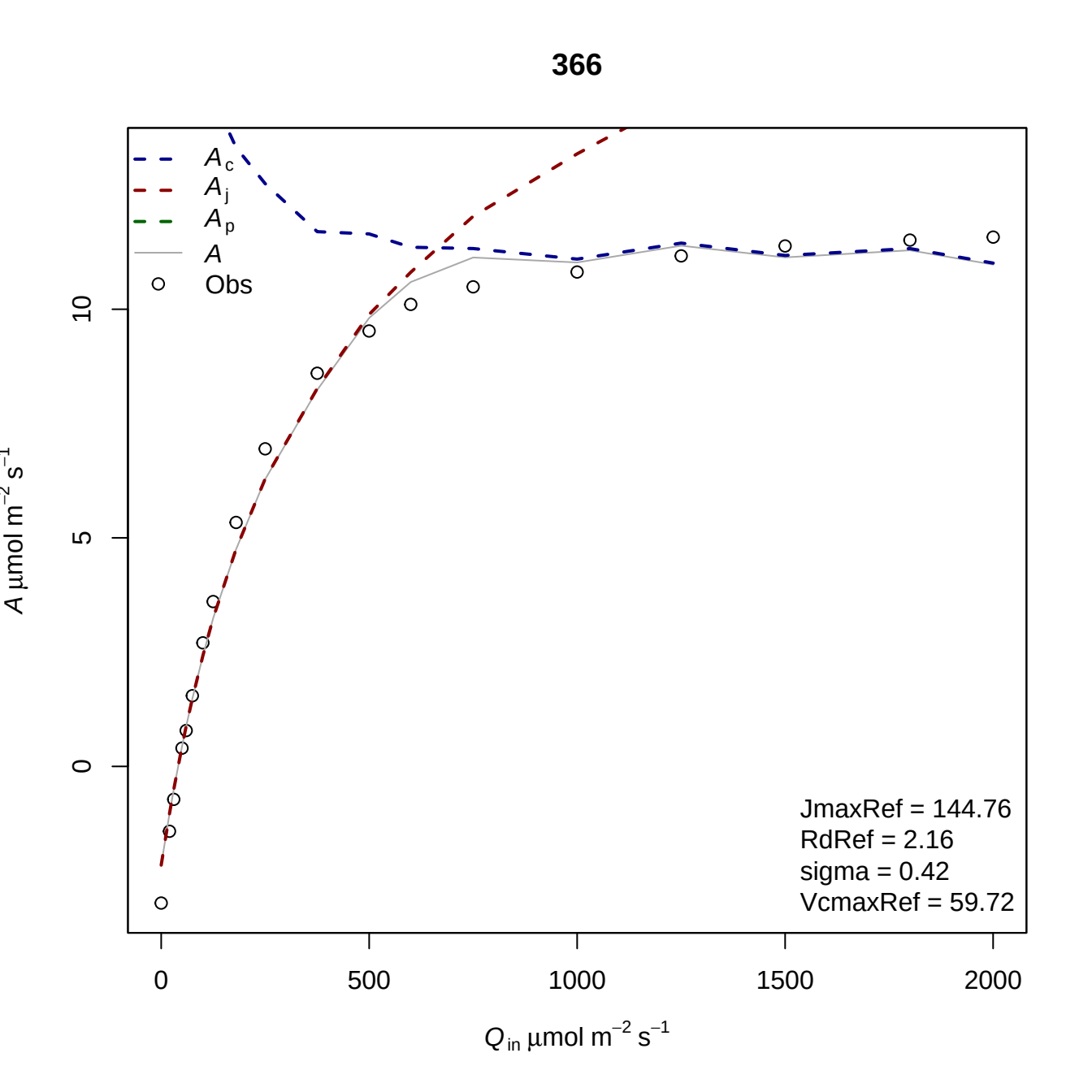

361
A_c
A_j
A_p
A
Obs
JmaxRef = 147.39
RdRef = 1.44
sigma = 0.23
VcmaxRef = 43.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

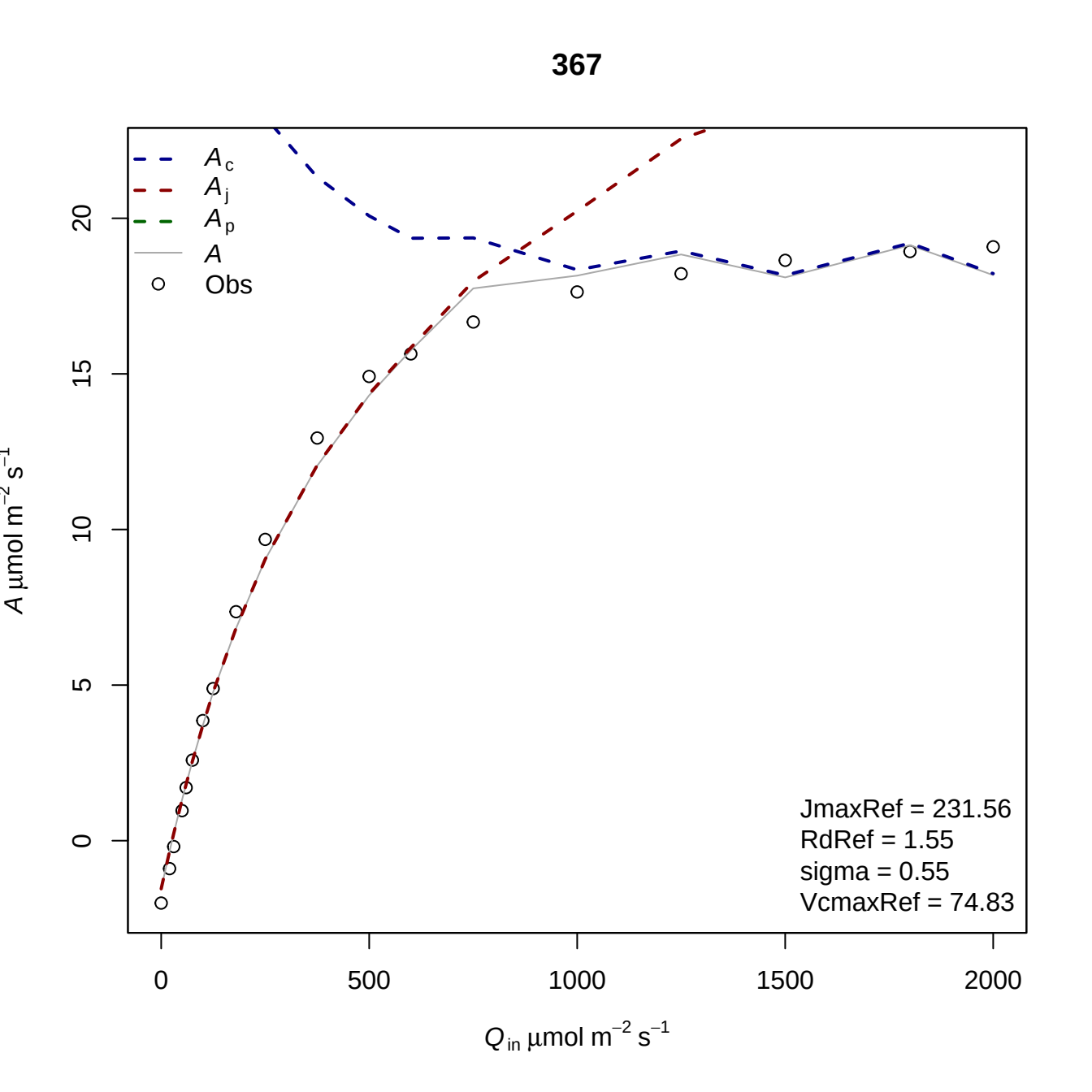

362
A_c
A_j
A_p
A
Obs
JmaxRef = 392.59
RdRef = 0.48
sigma = 0.66
VcmaxRef = 72.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

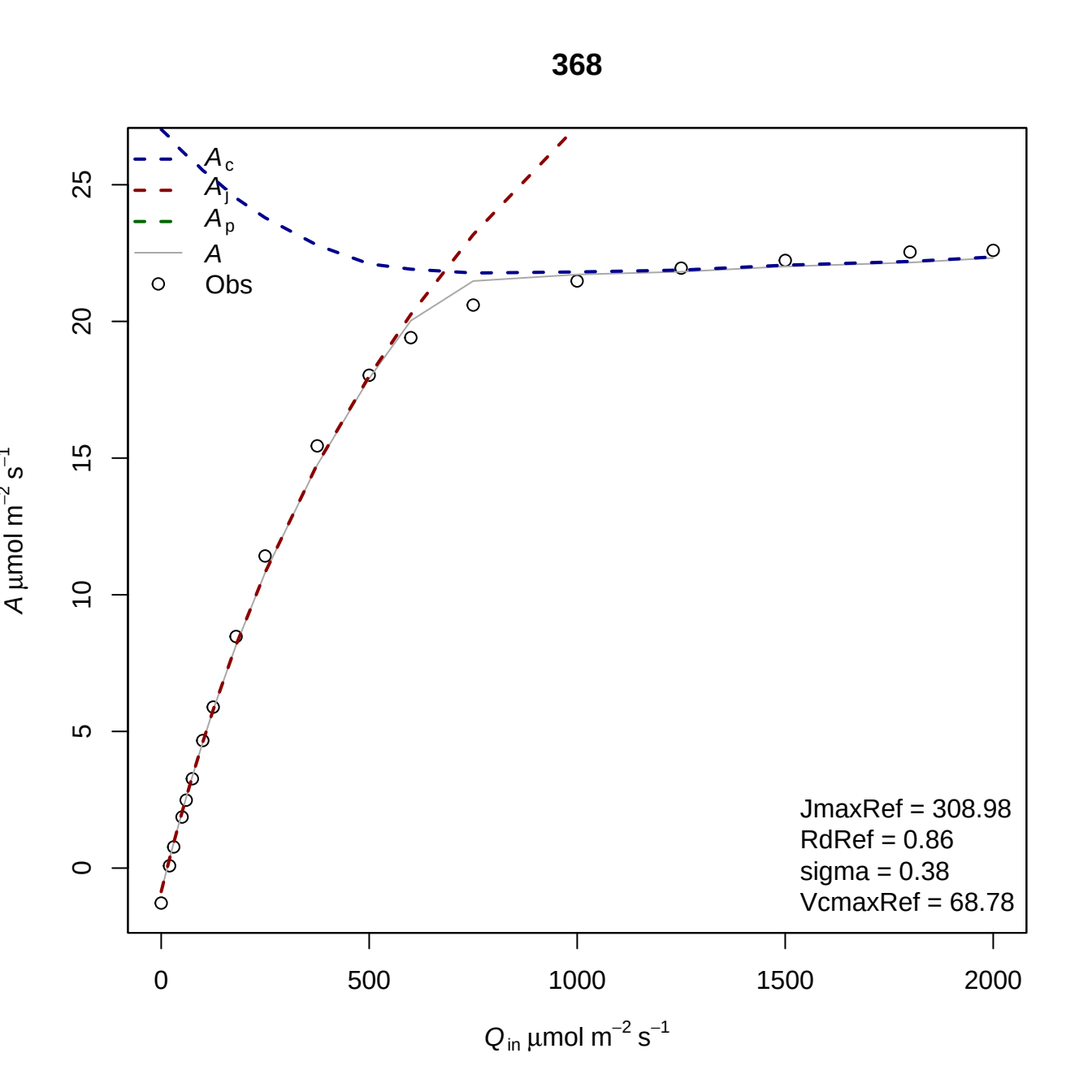

363
A_c
A_j
A_p
A
Obs
JmaxRef = 319.92
RdRef = 1.16
sigma = 0.41
VcmaxRef = 71.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

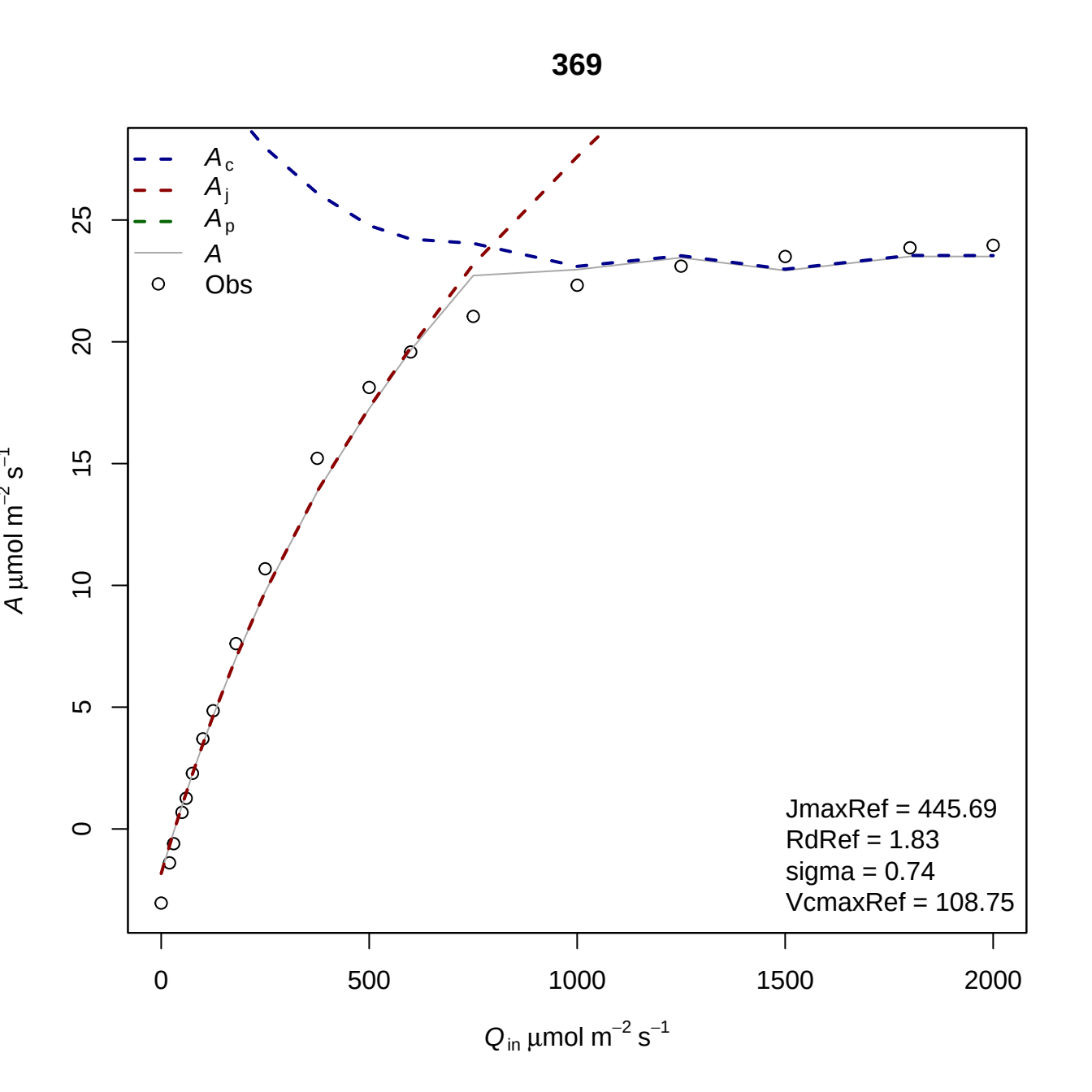

364
A_c
A_j
A_p
A
Obs
JmaxRef = 431.91
RdRef = 0.58
sigma = 0.65
VcmaxRef = 82.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

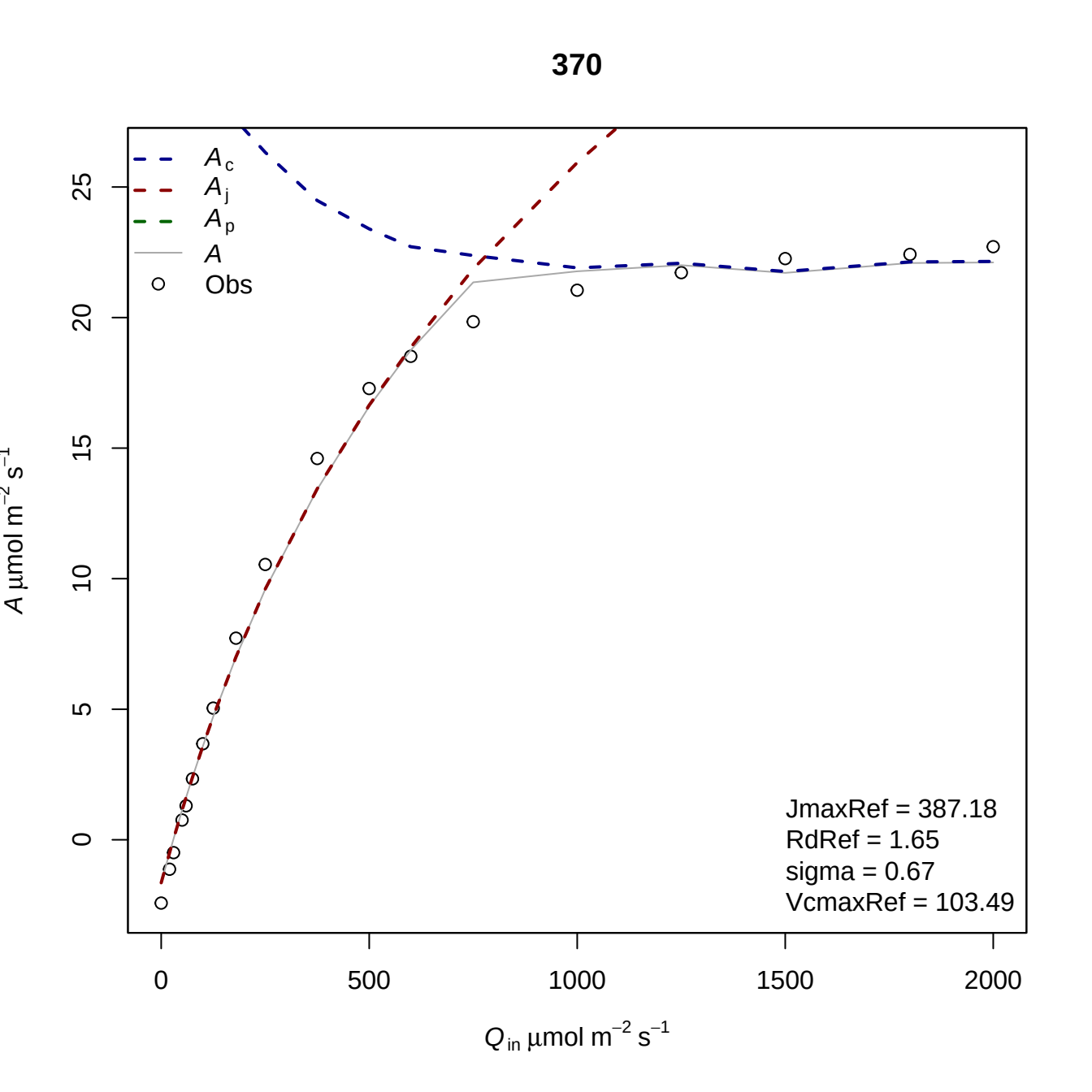

365
A_c
A_j
A_p
A
Obs
JmaxRef = 229.63
RdRef = 1.26
sigma = 0.46
VcmaxRef = 68.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

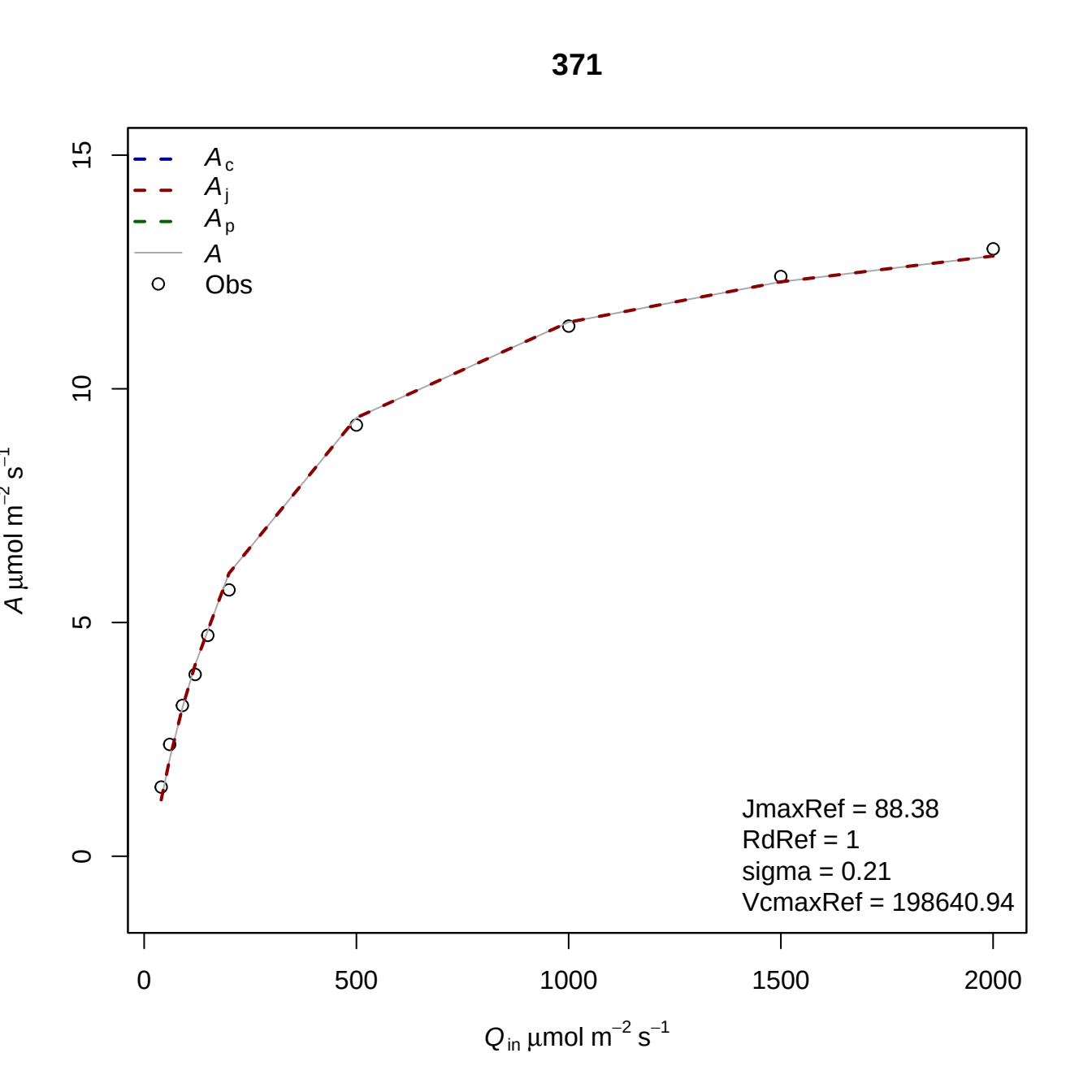

366
A_c
A_j
A_p
A
Obs
JmaxRef = 144.76
RdRef = 2.16
sigma = 0.42
VcmaxRef = 59.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

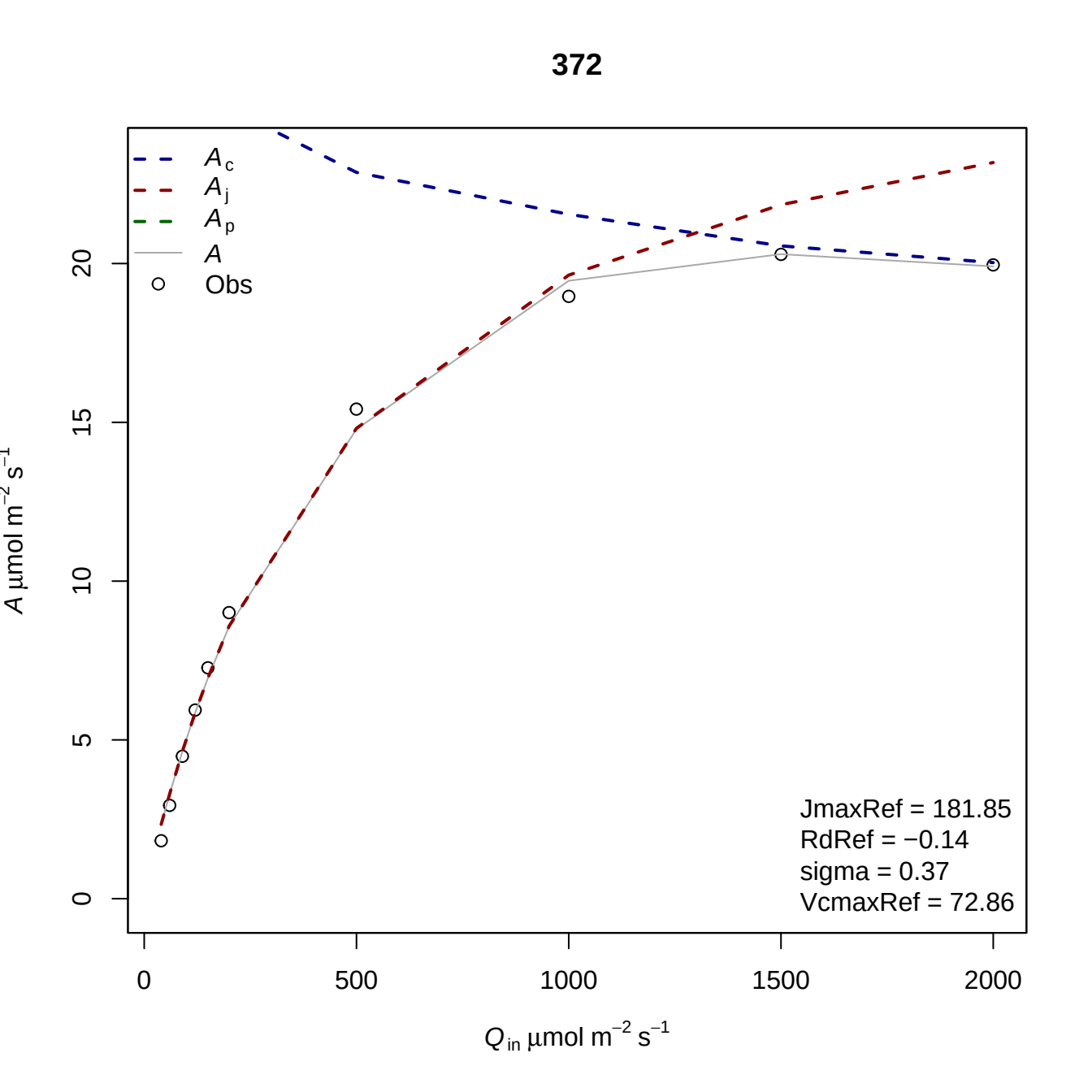

367
A_c
A_j
A_p
A
Obs
JmaxRef = 231.56
RdRef = 1.55
sigma = 0.55
VcmaxRef = 74.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

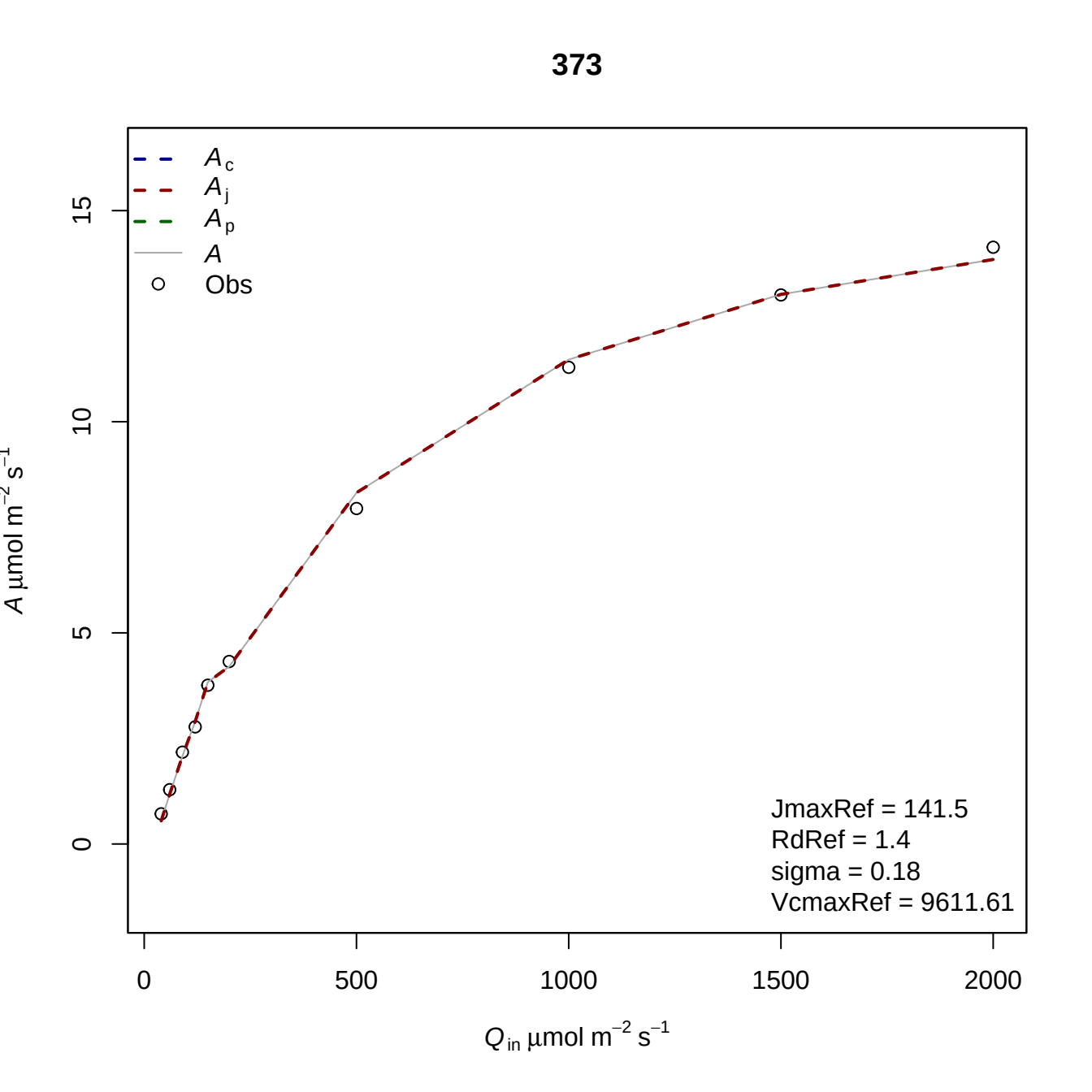

368
A_c
A_j
A_p
A
Obs
JmaxRef = 308.98
RdRef = 0.86
sigma = 0.38
VcmaxRef = 68.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

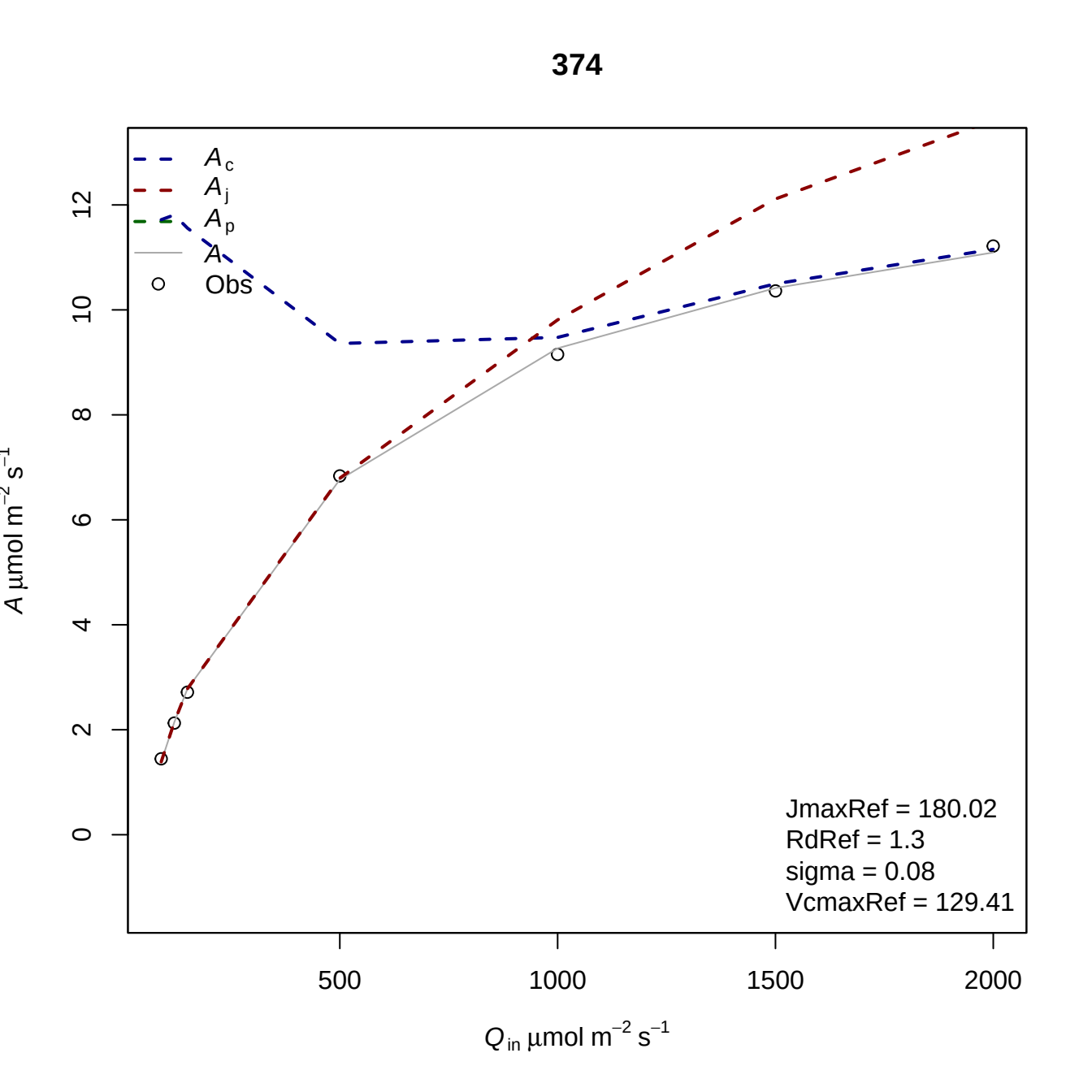

369
A_c
A_j
A_p
A
Obs
JmaxRef = 445.69
RdRef = 1.83
sigma = 0.74
VcmaxRef = 108.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

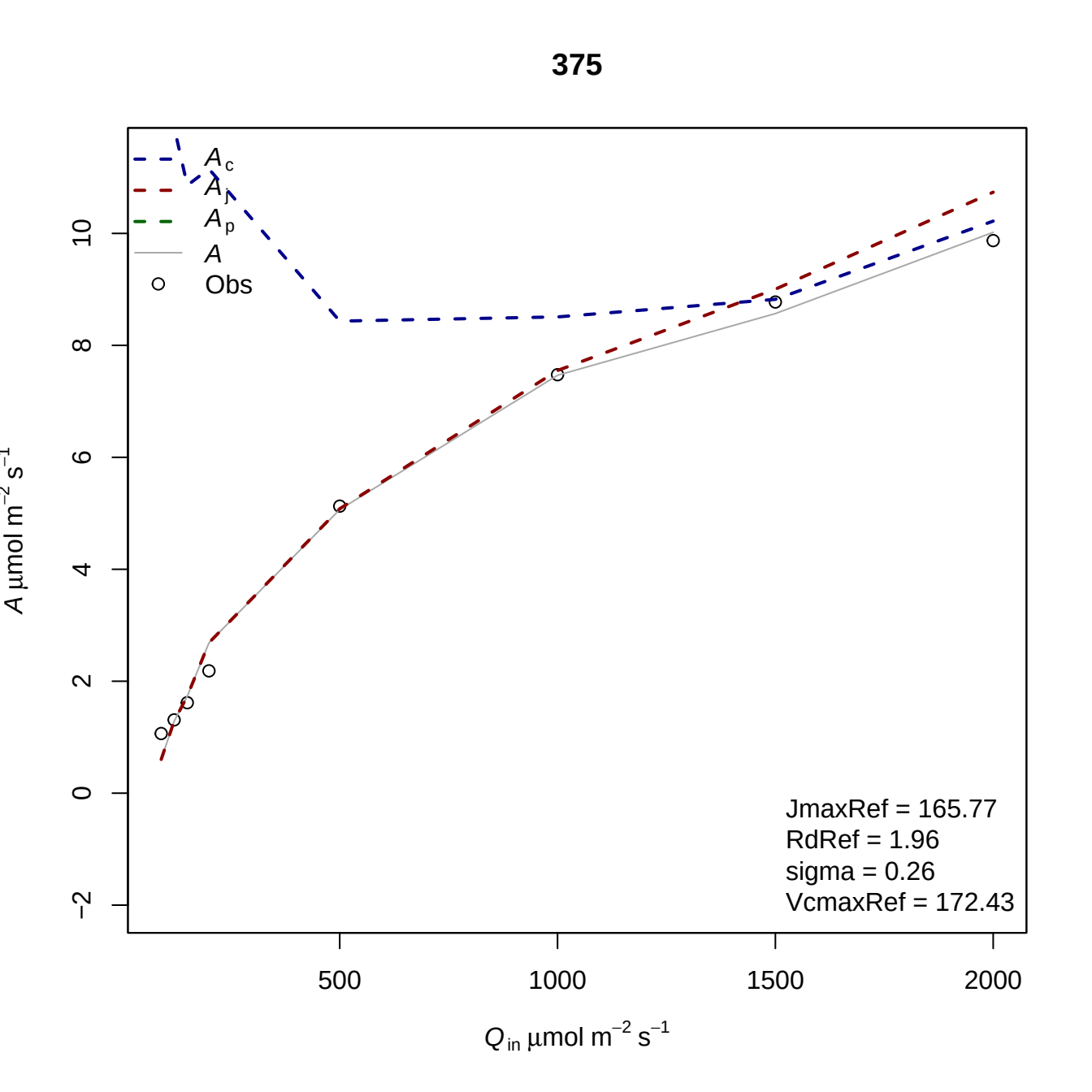

370
A_c
A_j
A_p
A
Obs
JmaxRef = 387.18
RdRef = 1.65
sigma = 0.67
VcmaxRef = 103.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

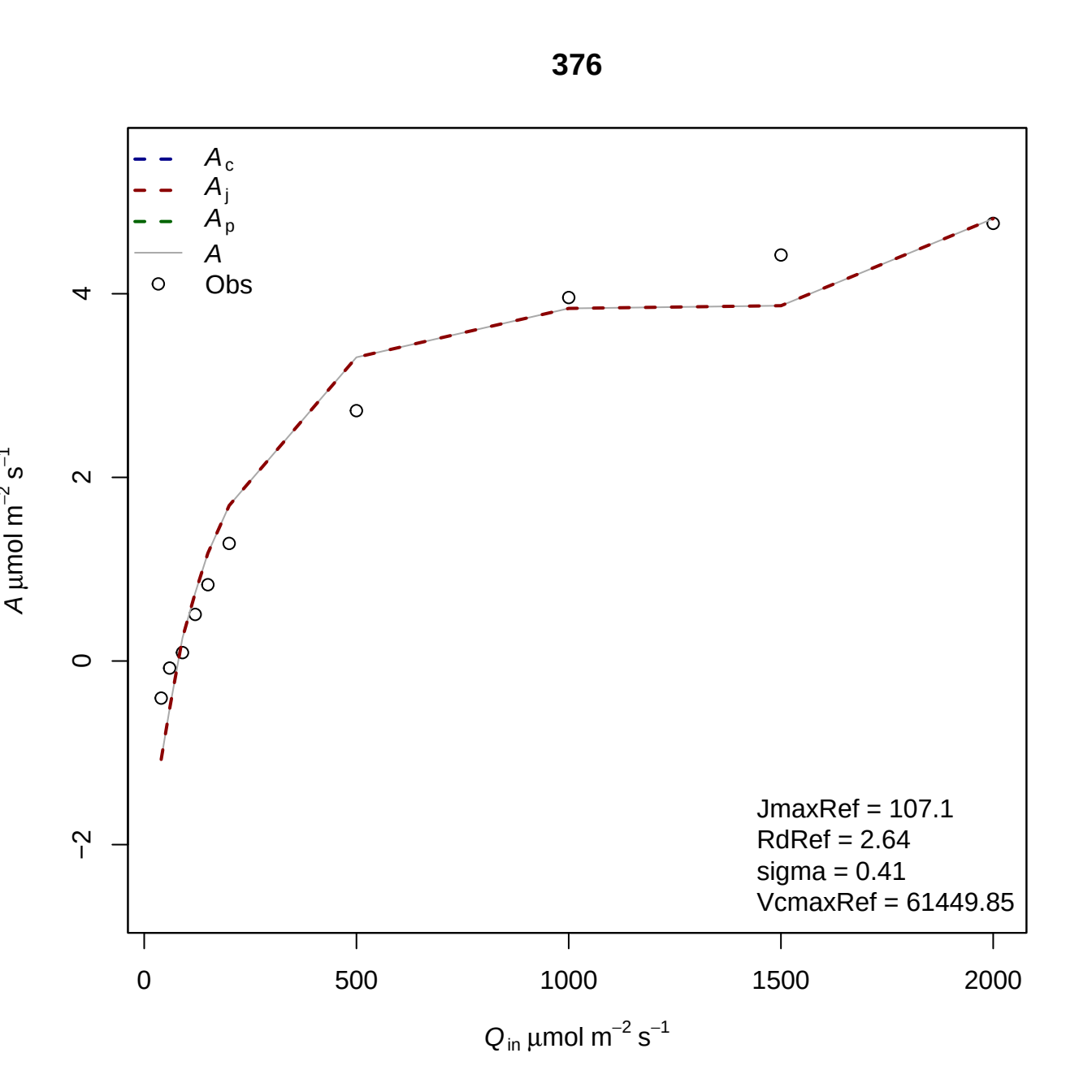

371
A_c
A_j
A_p
A
Obs
JmaxRef = 88.38
RdRef = 1
sigma = 0.21
VcmaxRef = 198640.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

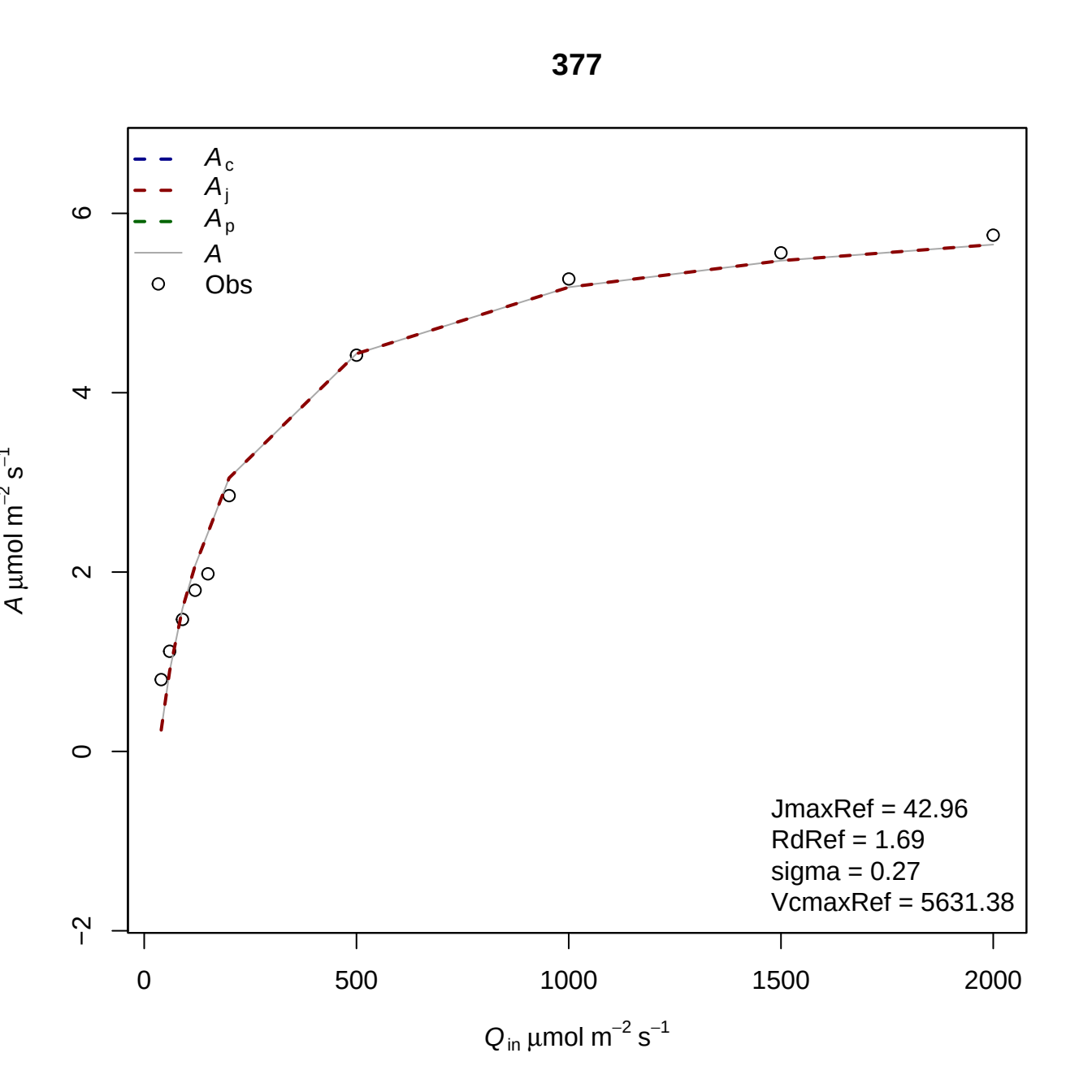

372
A_c
A_j
A_p
A
Obs
JmaxRef = 181.85
RdRef = −0.14
sigma = 0.37
VcmaxRef = 72.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

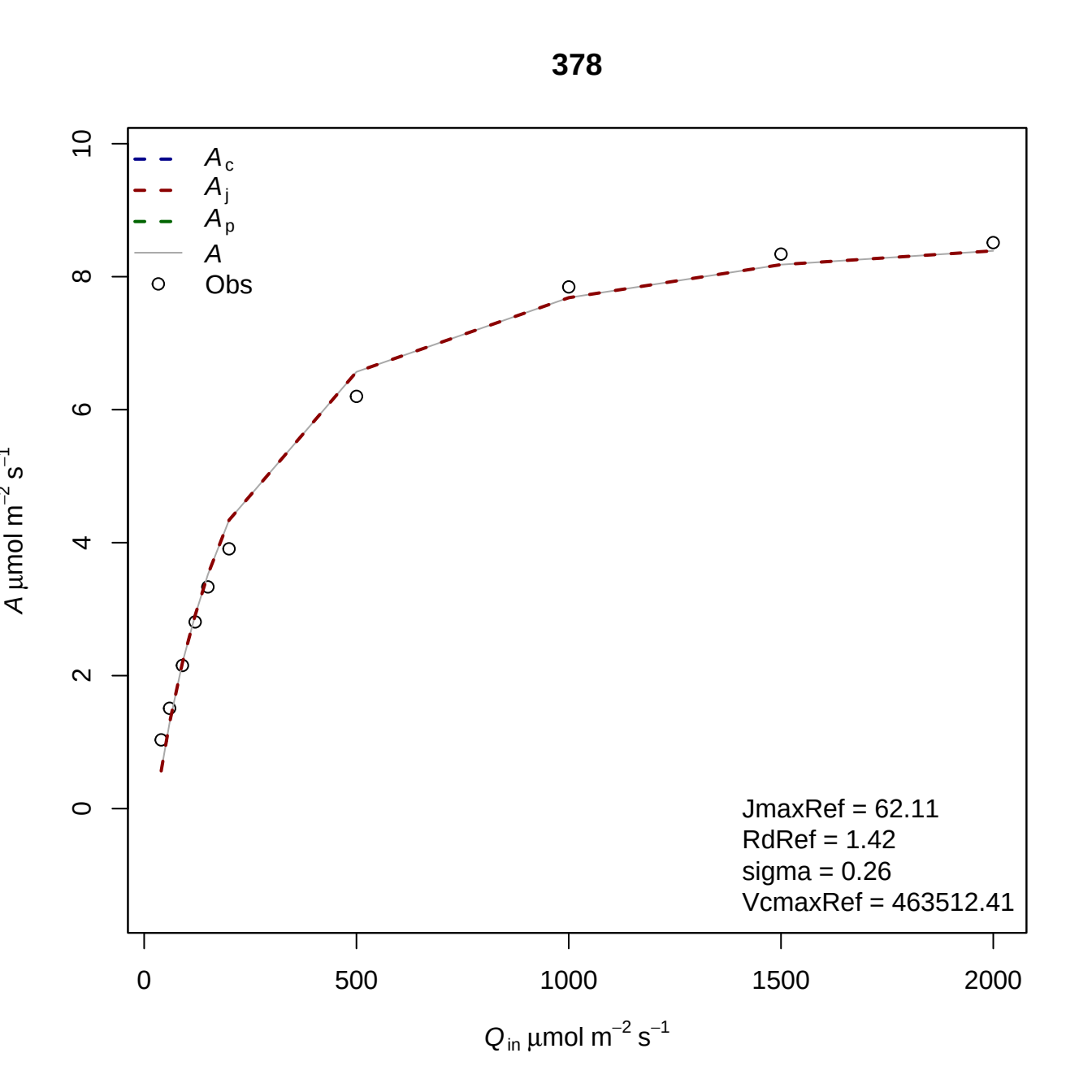

373
A_c
A_j
A_p
A
Obs
JmaxRef = 141.5
RdRef = 1.4
sigma = 0.18
VcmaxRef = 9611.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

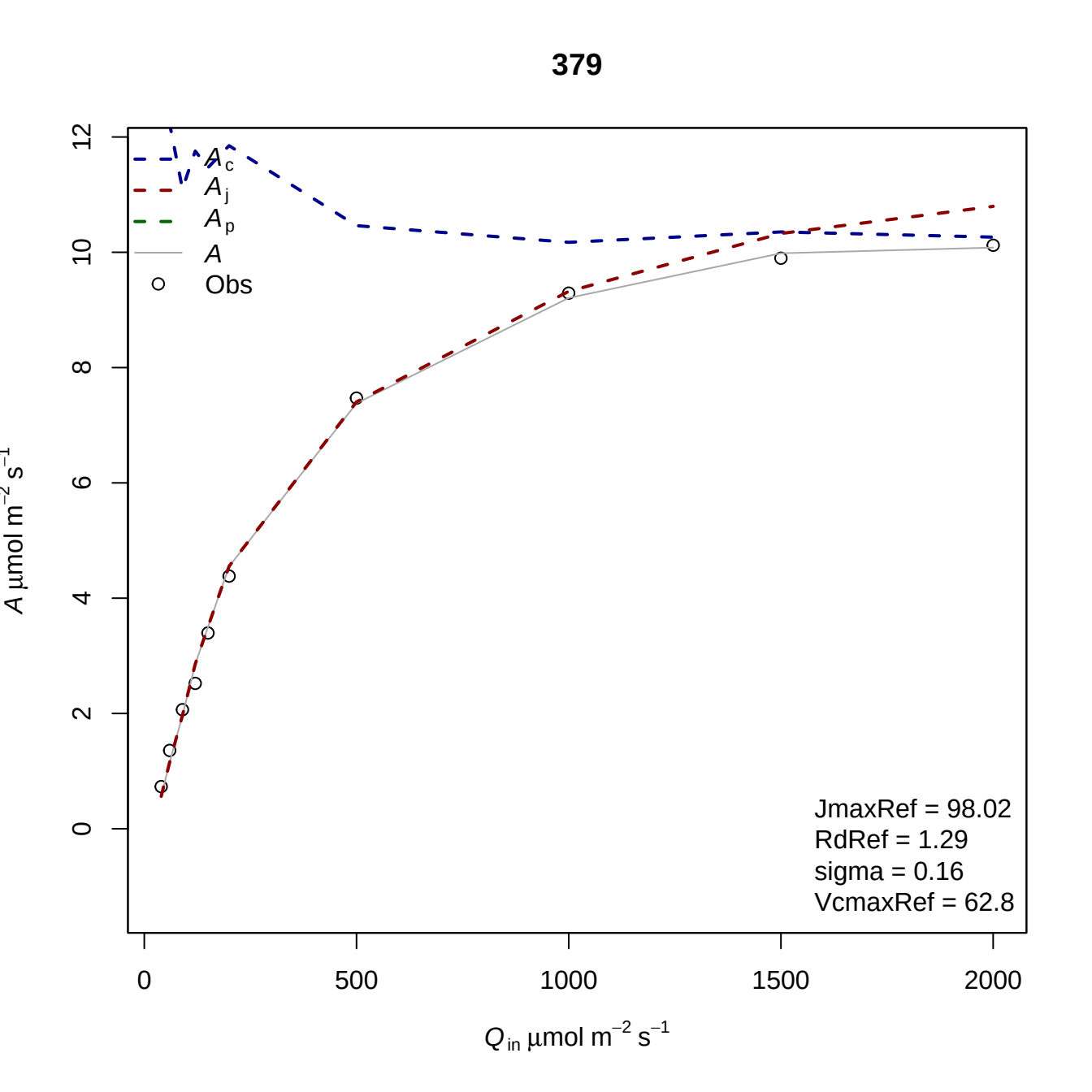

374
A_c
A_j
A_p
A
Obs
JmaxRef = 180.02
RdRef = 1.3
sigma = 0.08
VcmaxRef = 129.41
0
2
4
6
8
10
12
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

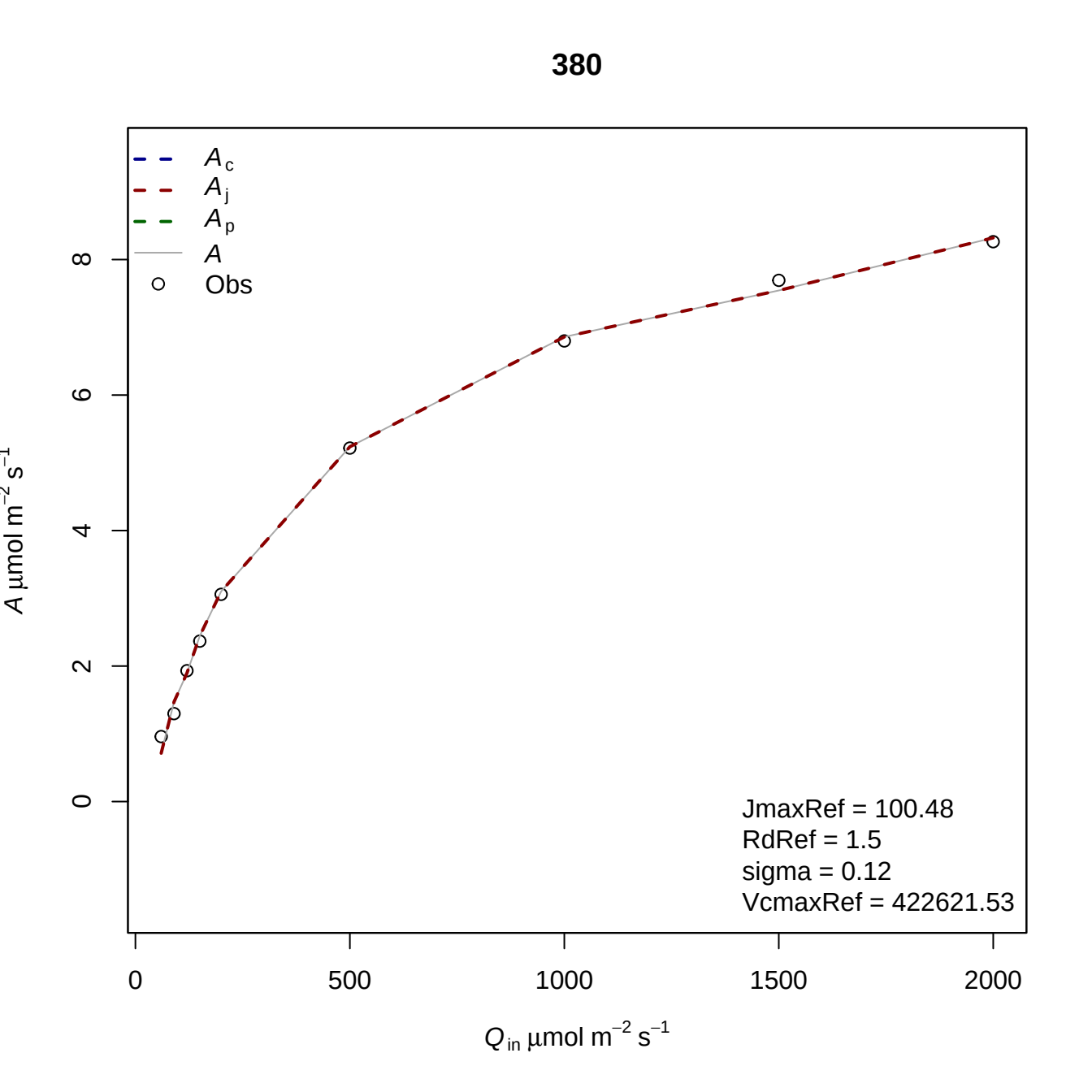

375
A_c
A_j
A_p
A
Obs
JmaxRef = 165.77
RdRef = 1.96
sigma = 0.26
VcmaxRef = 172.43
−2
0
2
4
6
8
10
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

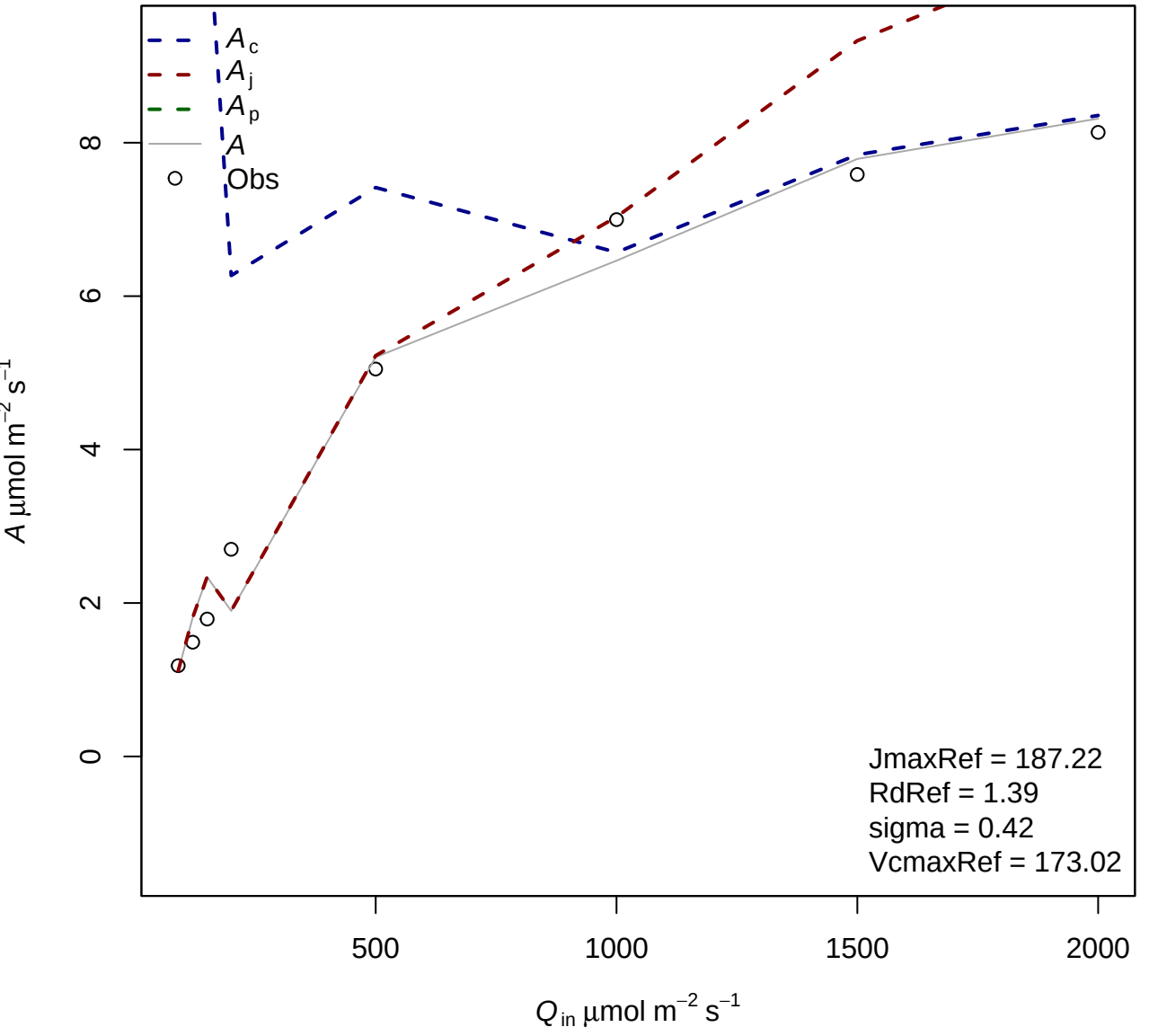

376
A_c
A_j
A_p
A
Obs
JmaxRef = 107.1
RdRef = 2.64
sigma = 0.41
VcmaxRef = 61449.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

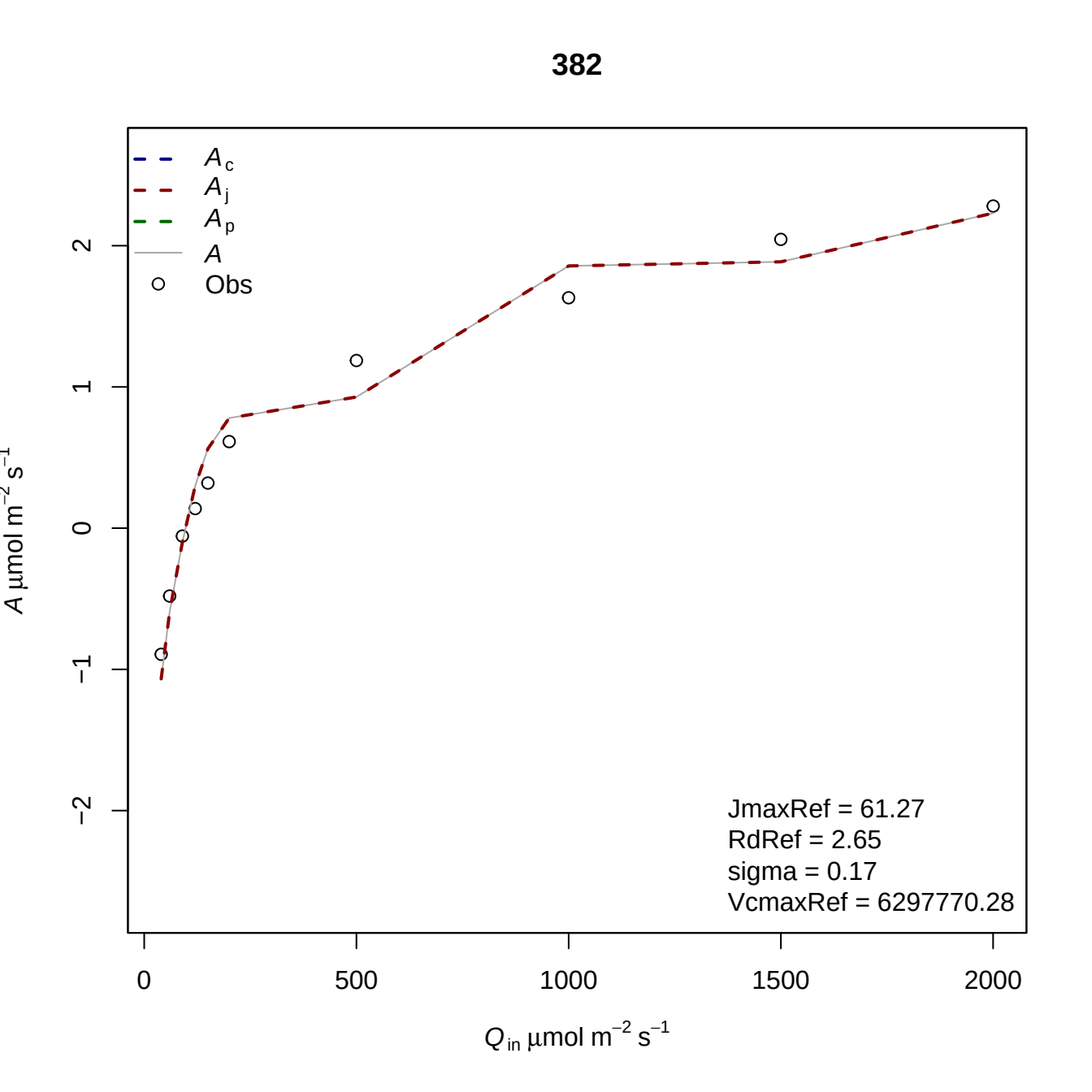

377
A_c
A_j
A_p
A
Obs
JmaxRef = 42.96
RdRef = 1.69
sigma = 0.27
VcmaxRef = 5631.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

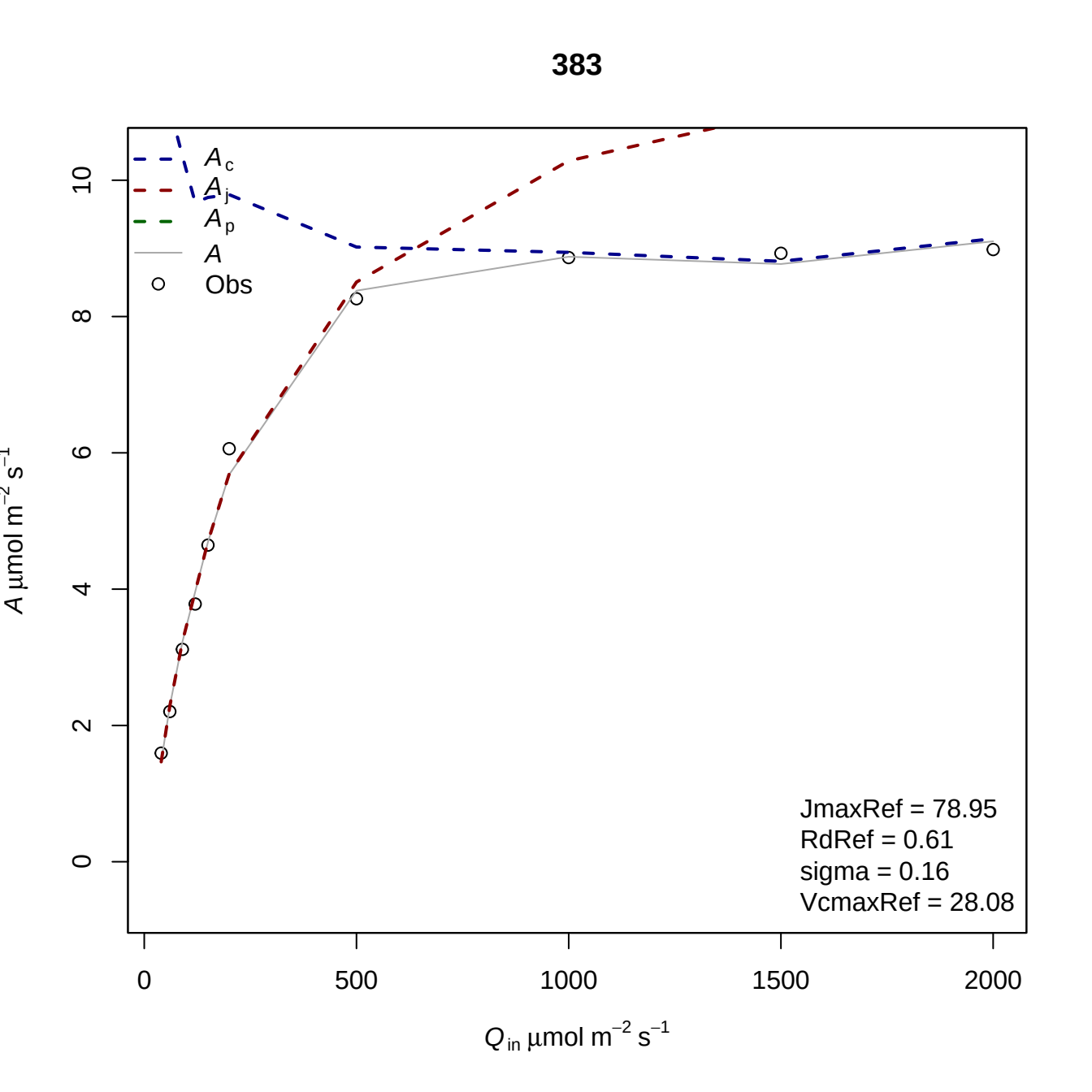

378
A_c
A_j
A_p
A
Obs
JmaxRef = 62.11
RdRef = 1.42
sigma = 0.26
VcmaxRef = 463512.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

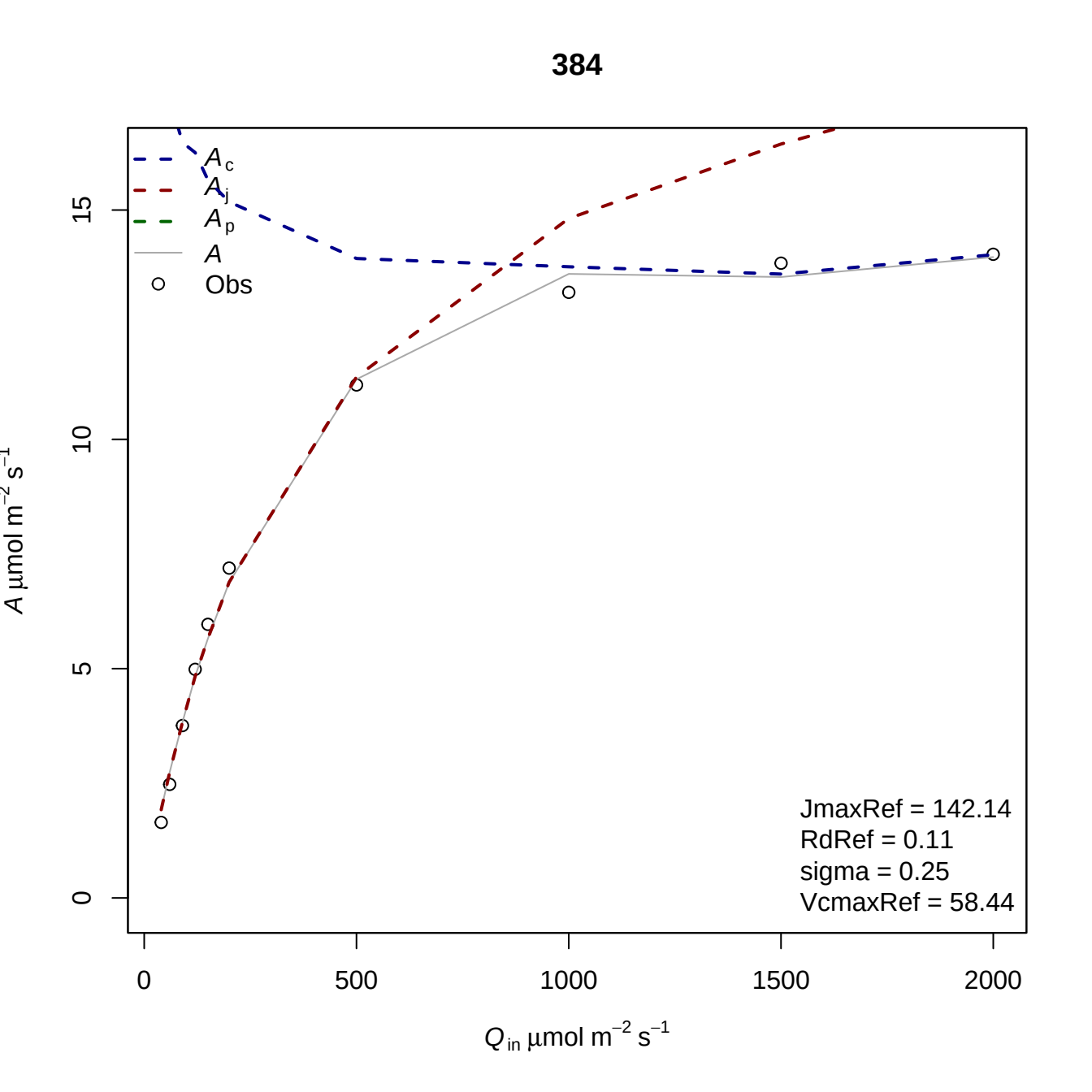

379
A_c
A_j
A_p
A
Obs
JmaxRef = 98.02
RdRef = 1.29
sigma = 0.16
VcmaxRef = 62.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

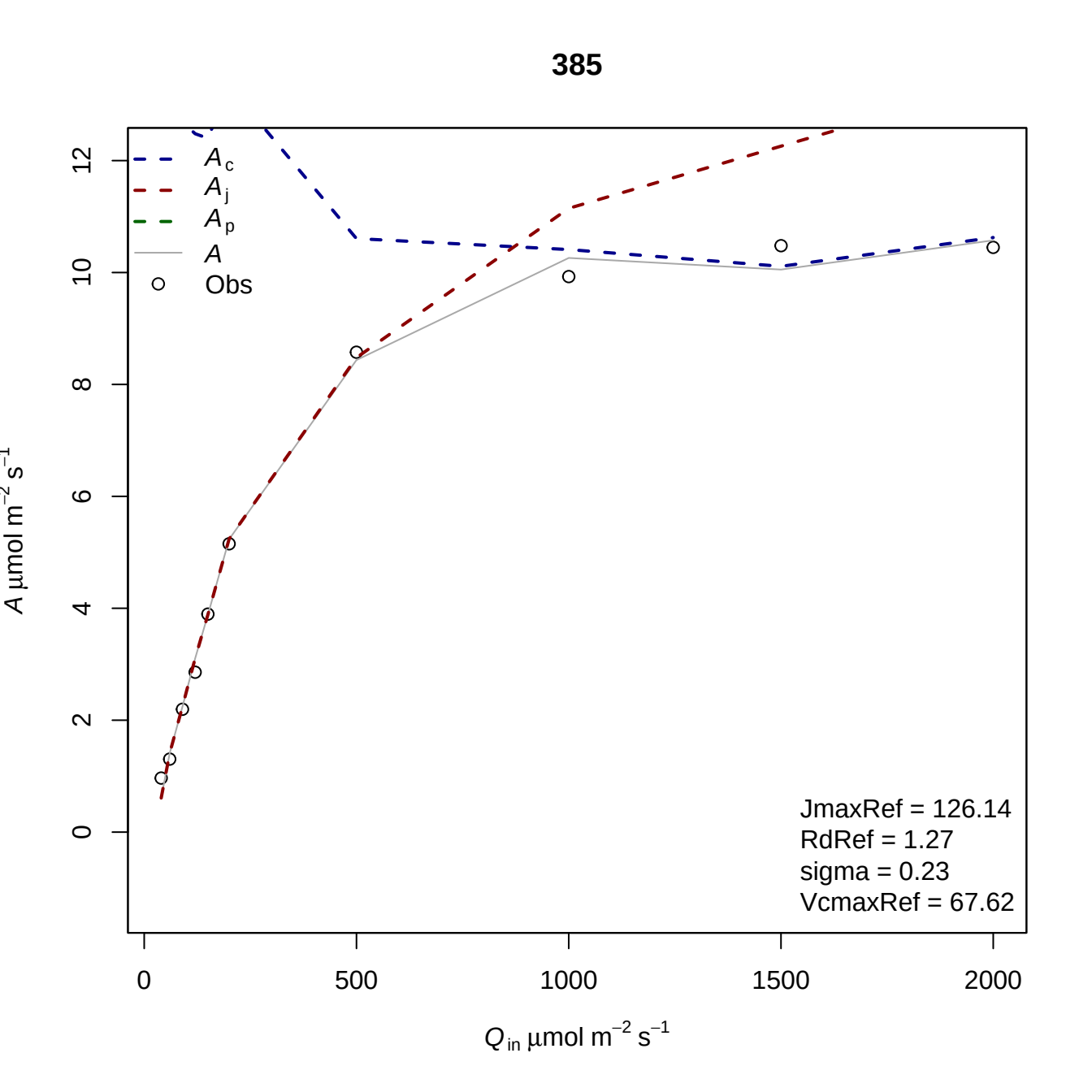

380
A_c
A_j
A_p
A
Obs
JmaxRef = 100.48
RdRef = 1.5
sigma = 0.12
VcmaxRef = 422621.53
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

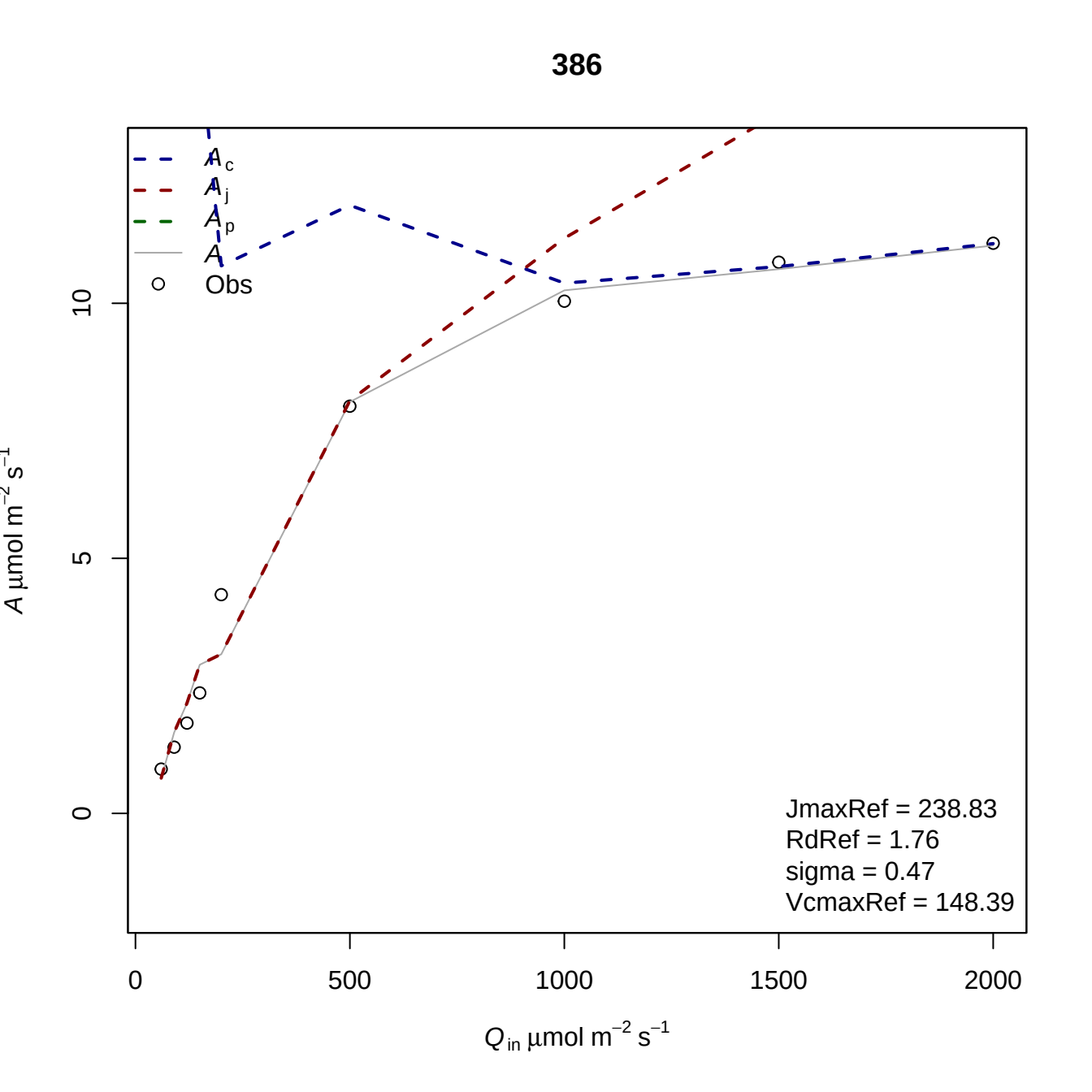

A_c
A_j
A_p
A
Obs
JmaxRef = 187.22
RdRef = 1.39
sigma = 0.42
VcmaxRef = 173.02
0
2
4
6
8
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

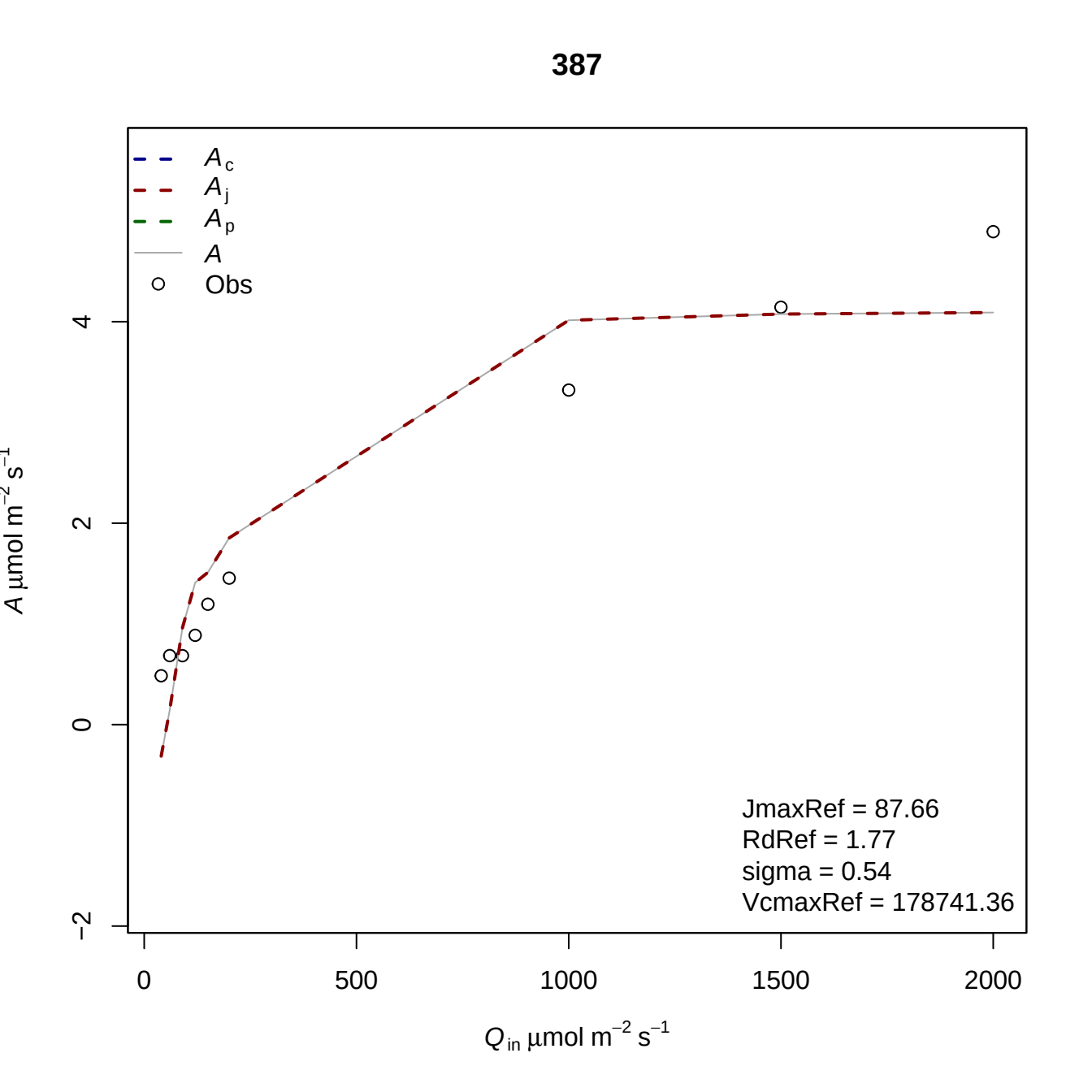

382
A_c
A_j
A_p
A
Obs
JmaxRef = 61.27
RdRef = 2.65
sigma = 0.17
VcmaxRef = 6297770.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
−2
0
500
1000
1500
2000

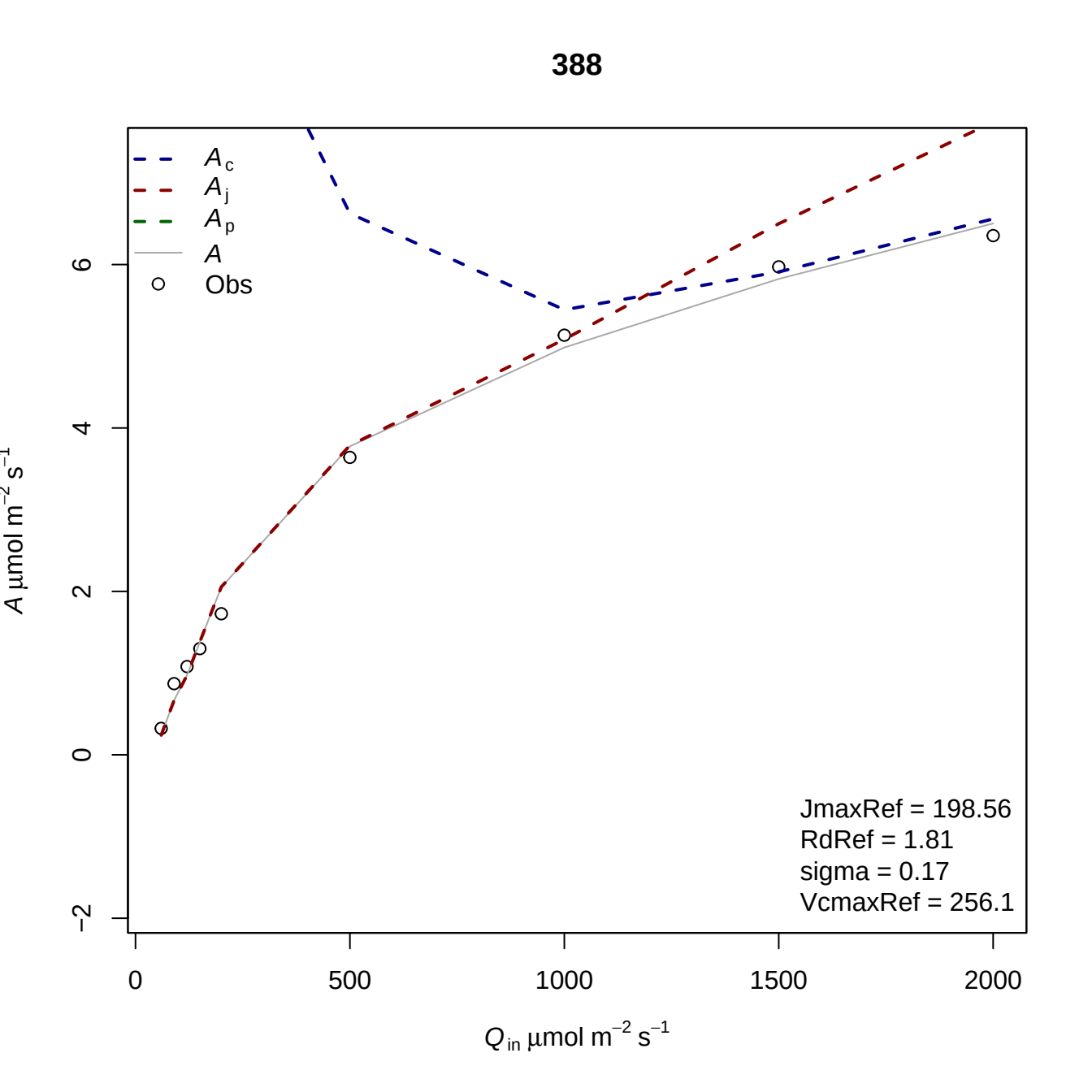

383
A_c
A_j
A_p
A
Obs
JmaxRef = 78.95
RdRef = 0.61
sigma = 0.16
VcmaxRef = 28.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

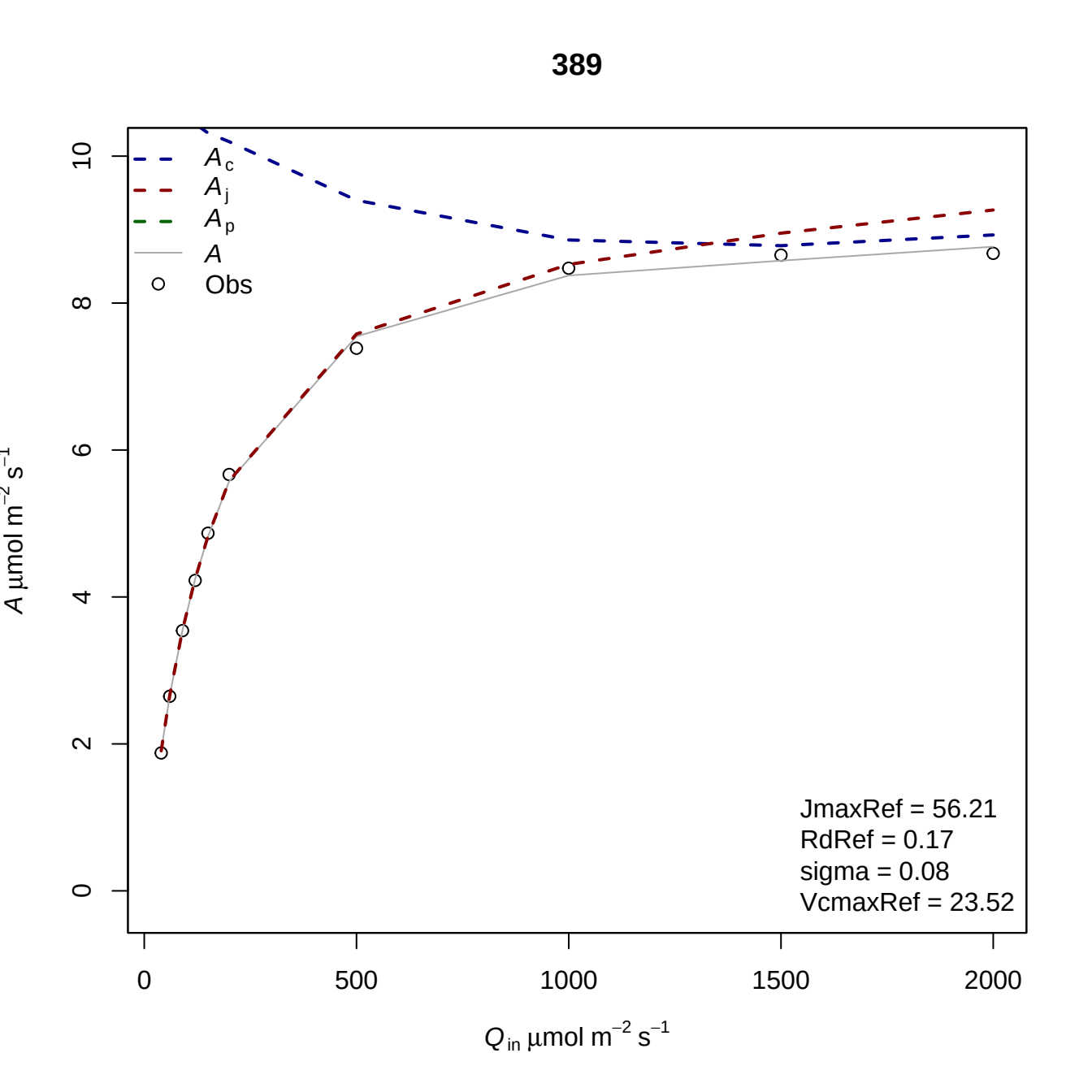

384
A_c
A_j
A_p
A
Obs
JmaxRef = 142.14
RdRef = 0.11
sigma = 0.25
VcmaxRef = 58.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

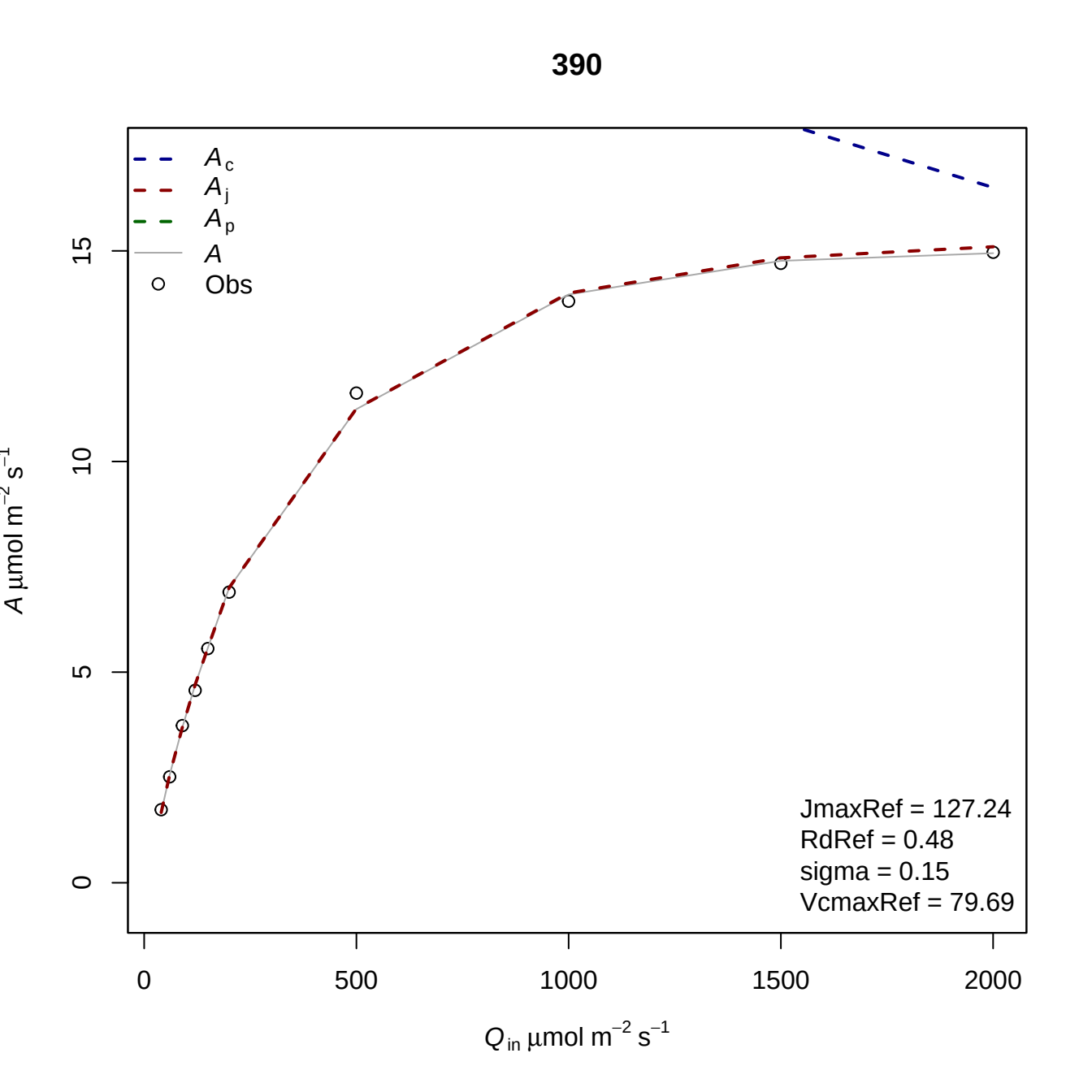

385
A_c
A_j
A_p
A
Obs
JmaxRef = 126.14
RdRef = 1.27
sigma = 0.23
VcmaxRef = 67.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

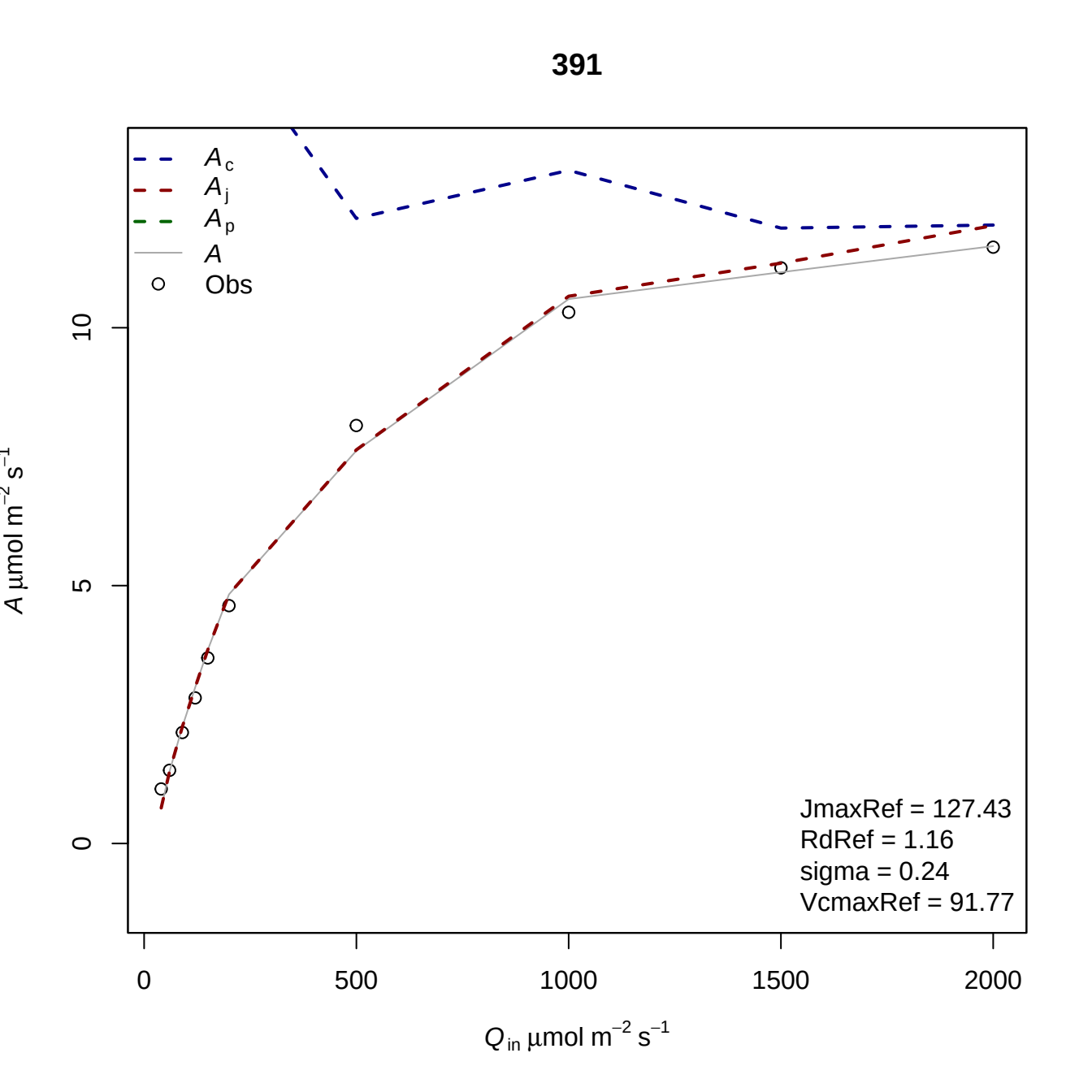

386
A_c
A_j
A_p
A
Obs
JmaxRef = 238.83
RdRef = 1.76
sigma = 0.47
VcmaxRef = 148.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

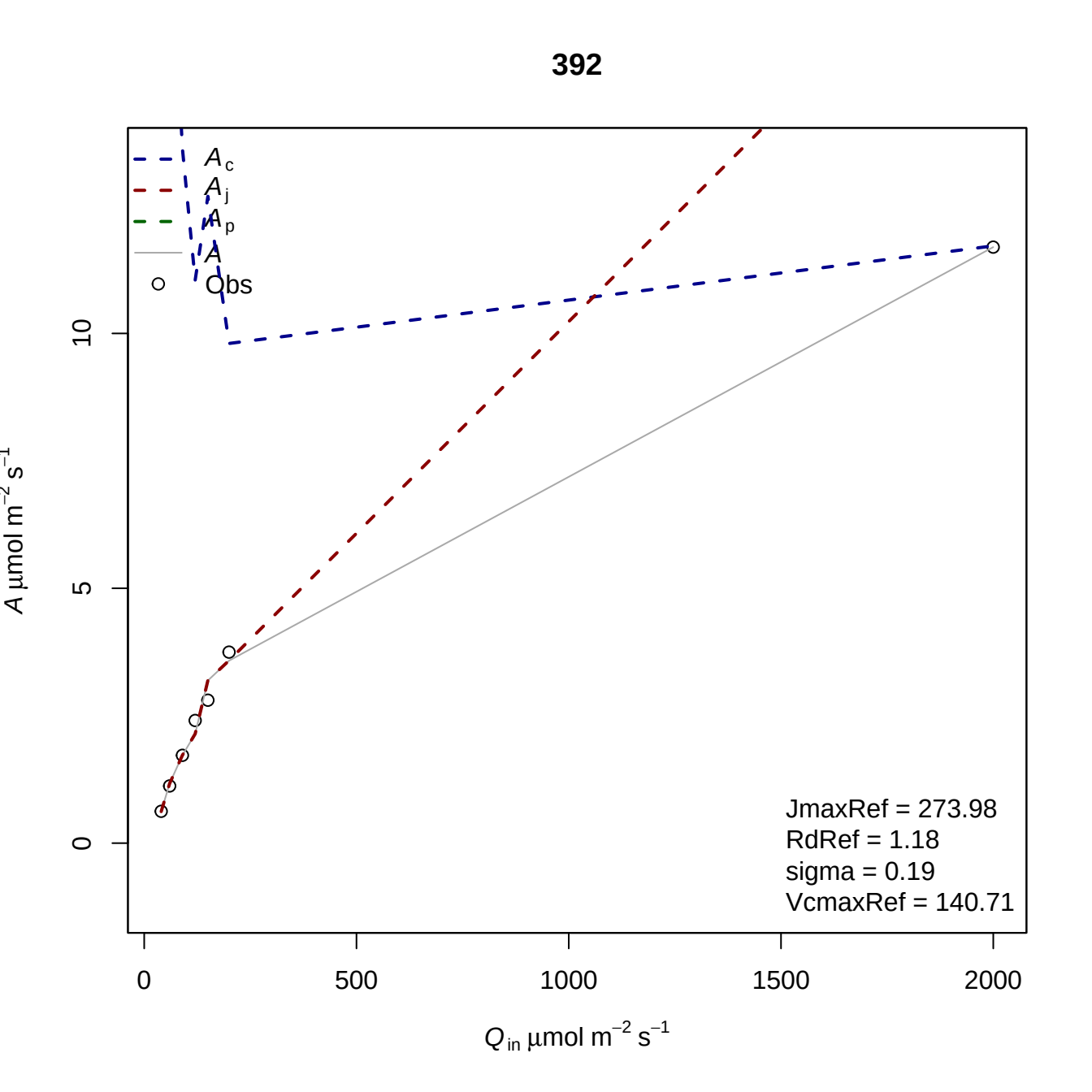

387
A_c
A_j
A_p
A
Obs
JmaxRef = 87.66
RdRef = 1.77
sigma = 0.54
VcmaxRef = 178741.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

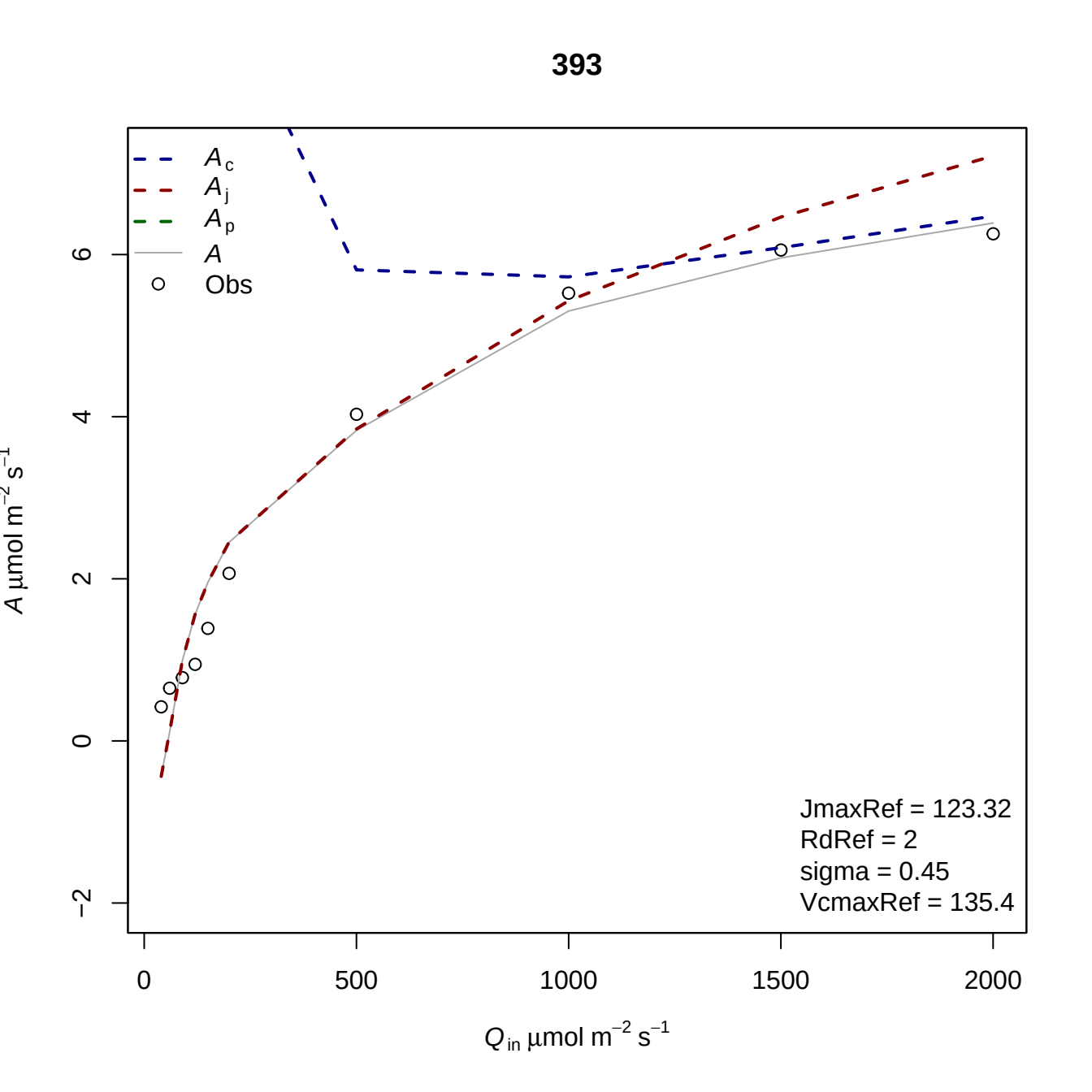

388
A_c
A_j
A_p
A
Obs
JmaxRef = 198.56
RdRef = 1.81
sigma = 0.17
VcmaxRef = 256.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

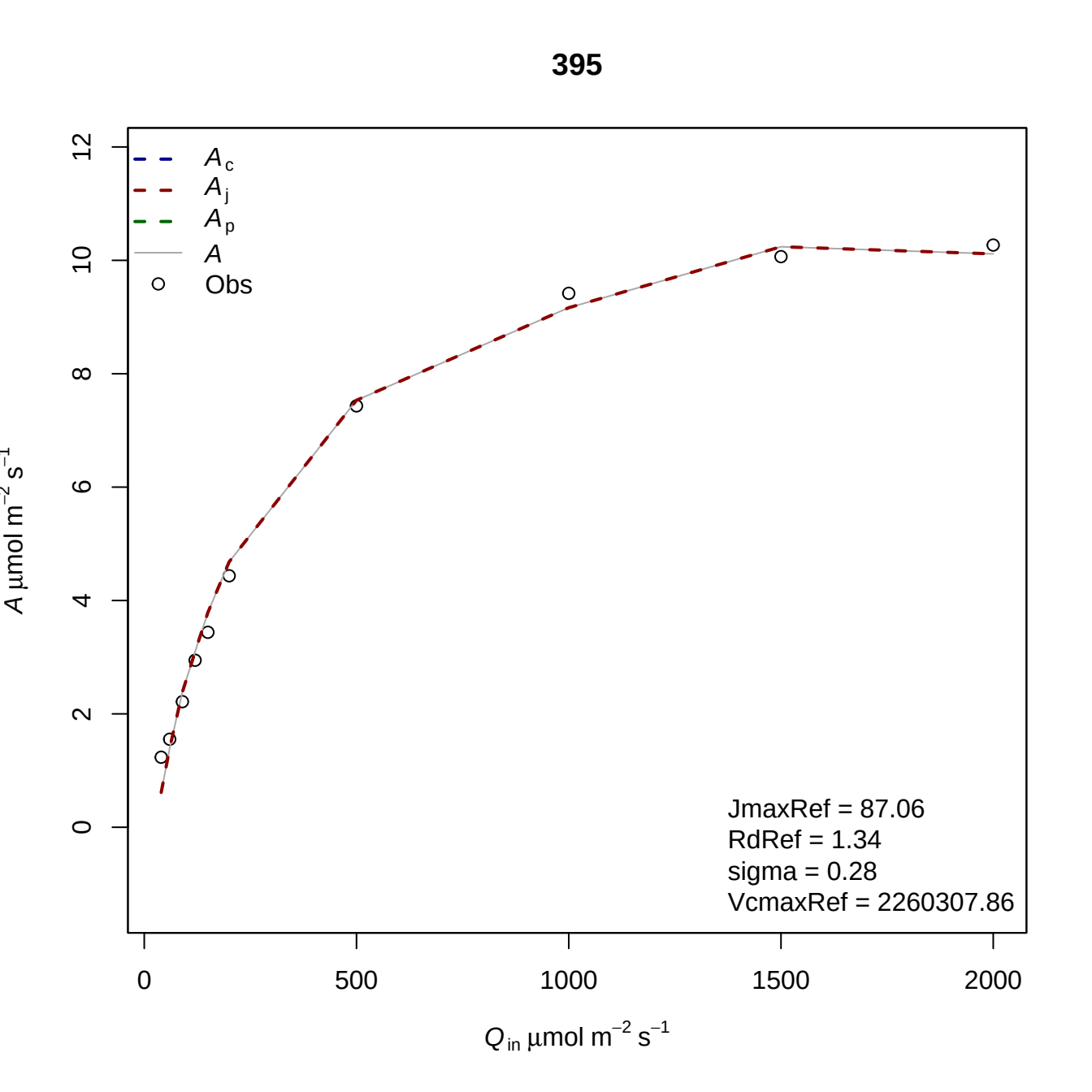

389
A_c
A_j
A_p
A
Obs
JmaxRef = 56.21
RdRef = 0.17
sigma = 0.08
VcmaxRef = 23.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

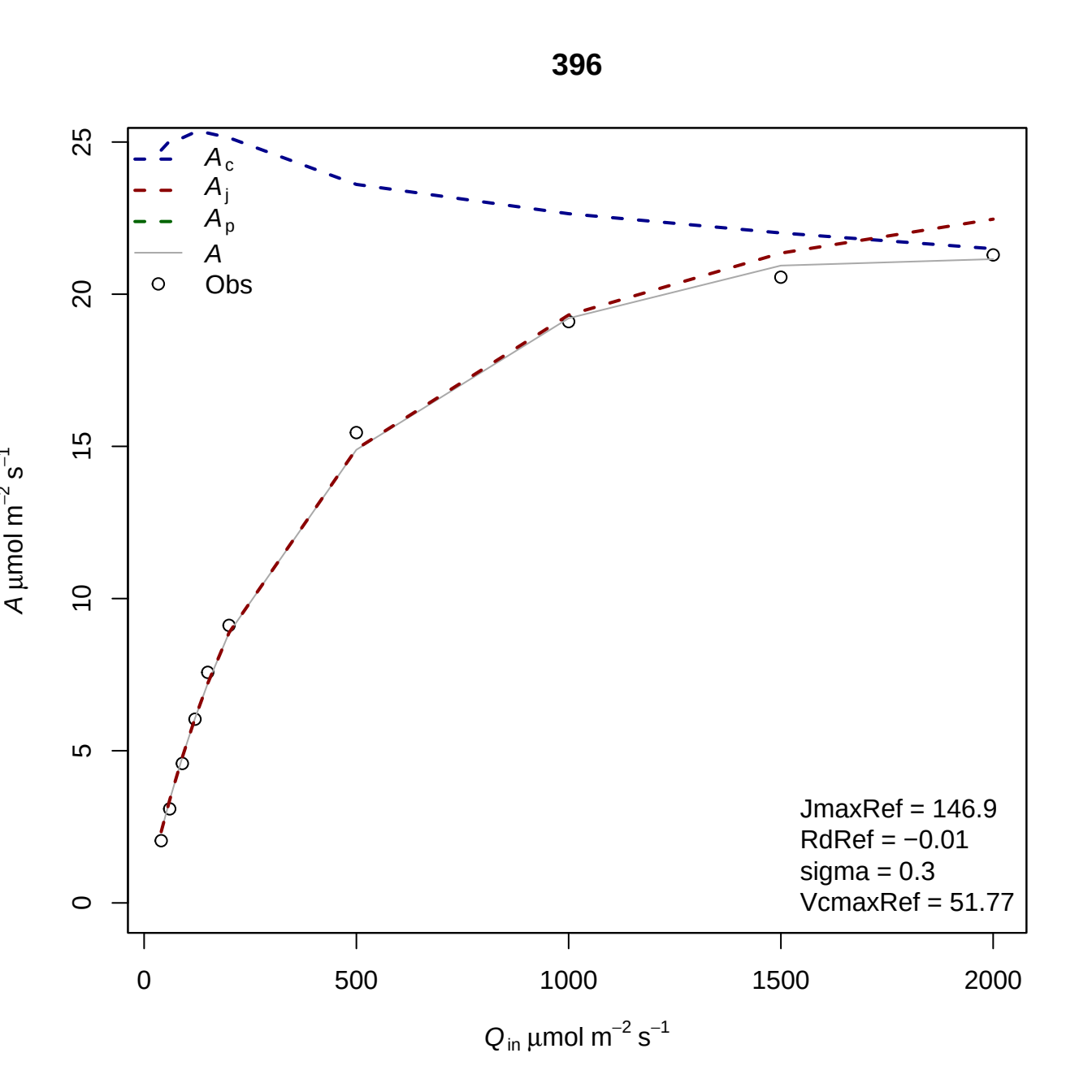

390
A_c
A_j
A_p
A
Obs
JmaxRef = 127.24
RdRef = 0.48
sigma = 0.15
VcmaxRef = 79.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

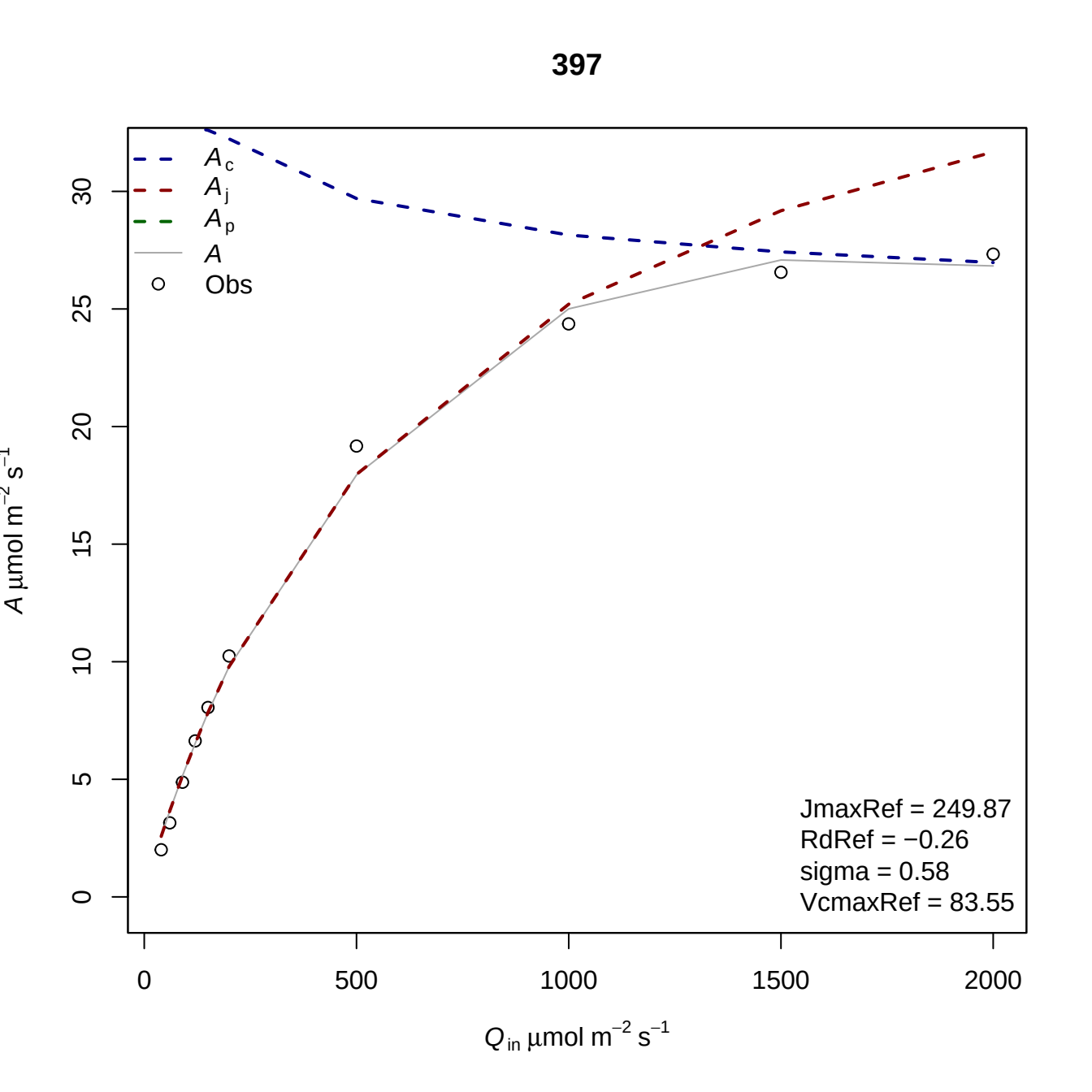

391
A_c
A_j
A_p
A
Obs
JmaxRef = 127.43
RdRef = 1.16
sigma = 0.24
VcmaxRef = 91.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

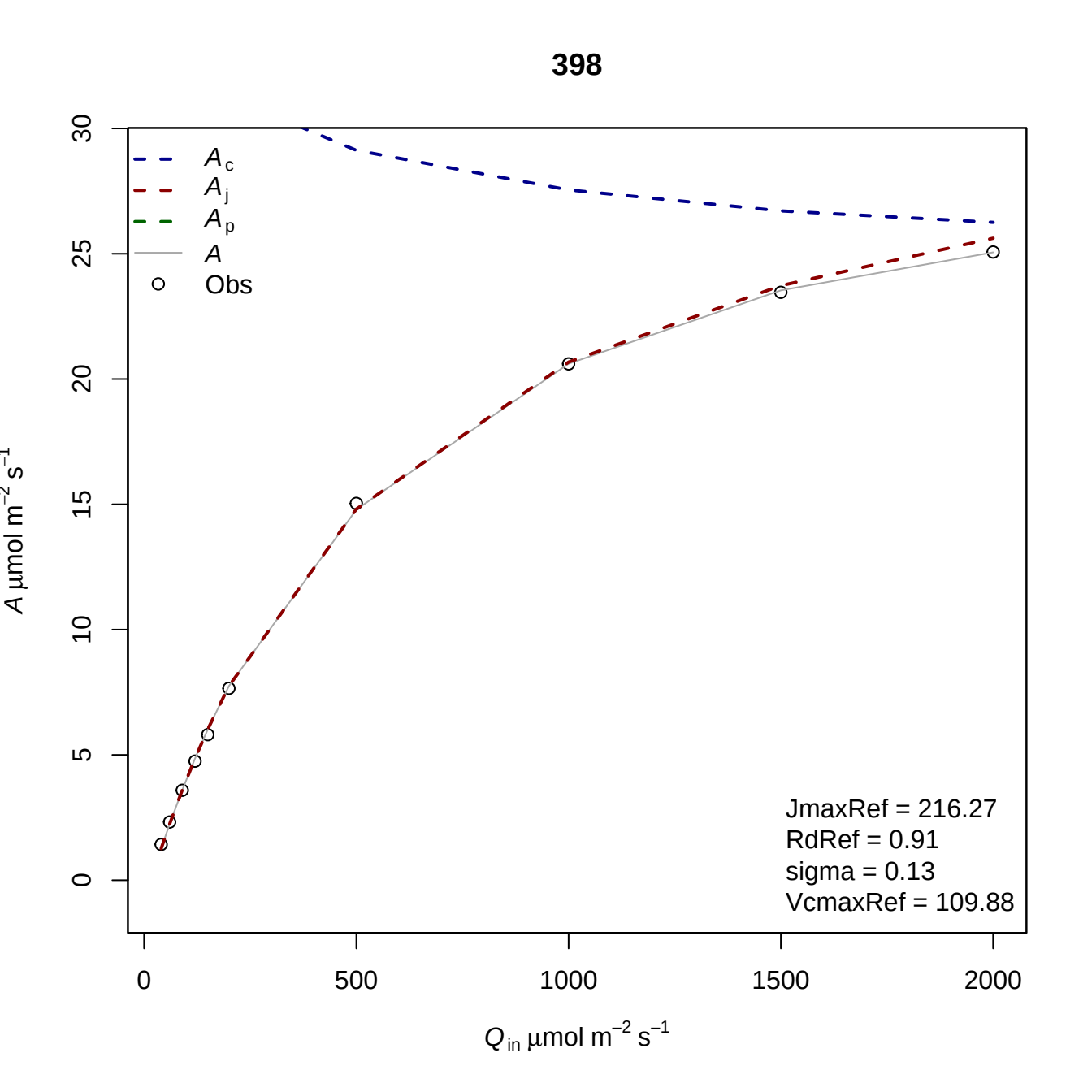

392
A_c
A_j
A_p
A
Obs
JmaxRef = 273.98
RdRef = 1.18
sigma = 0.19
VcmaxRef = 140.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

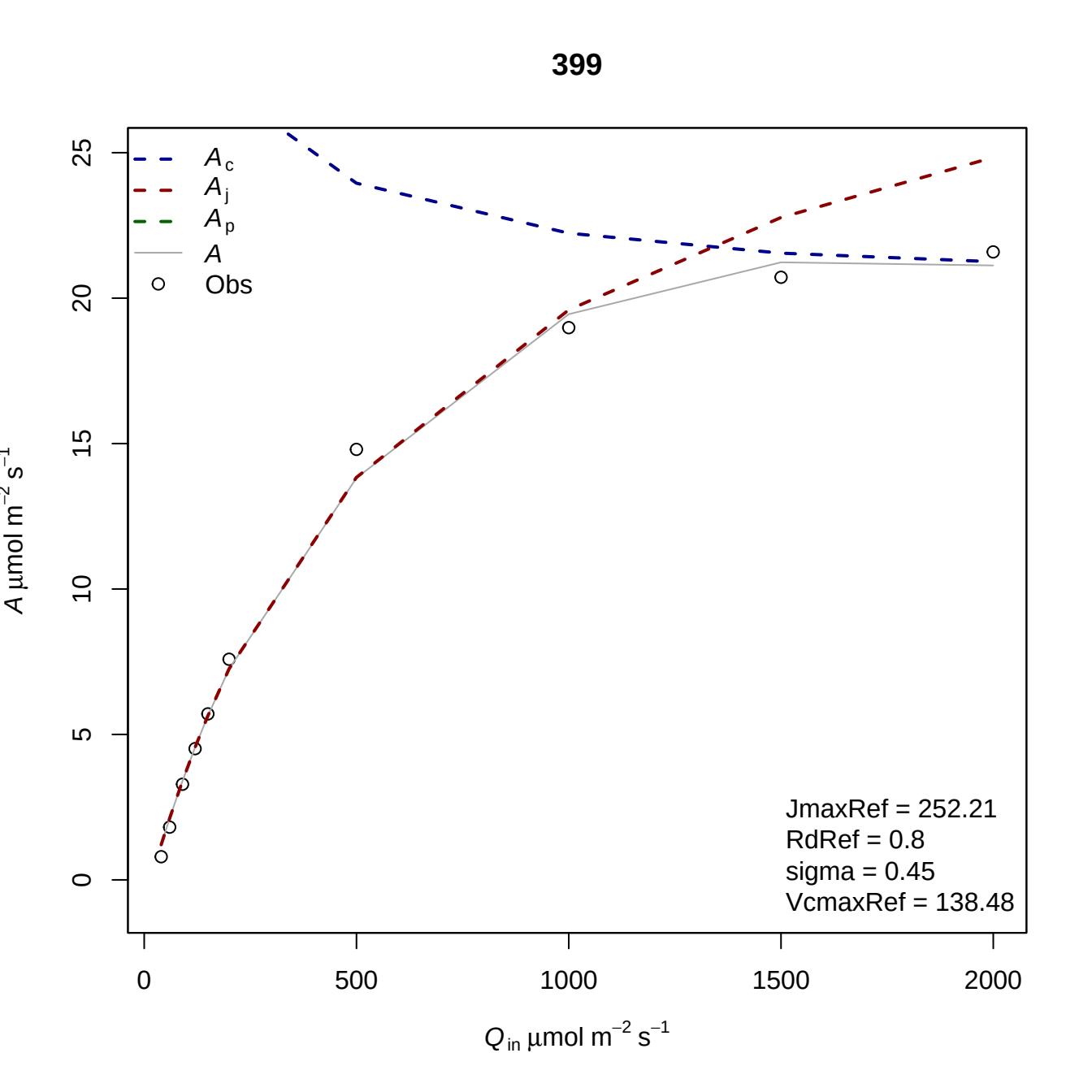

393
A_c
A_j
A_p
A
Obs
JmaxRef = 123.32
RdRef = 2
sigma = 0.45
VcmaxRef = 135.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

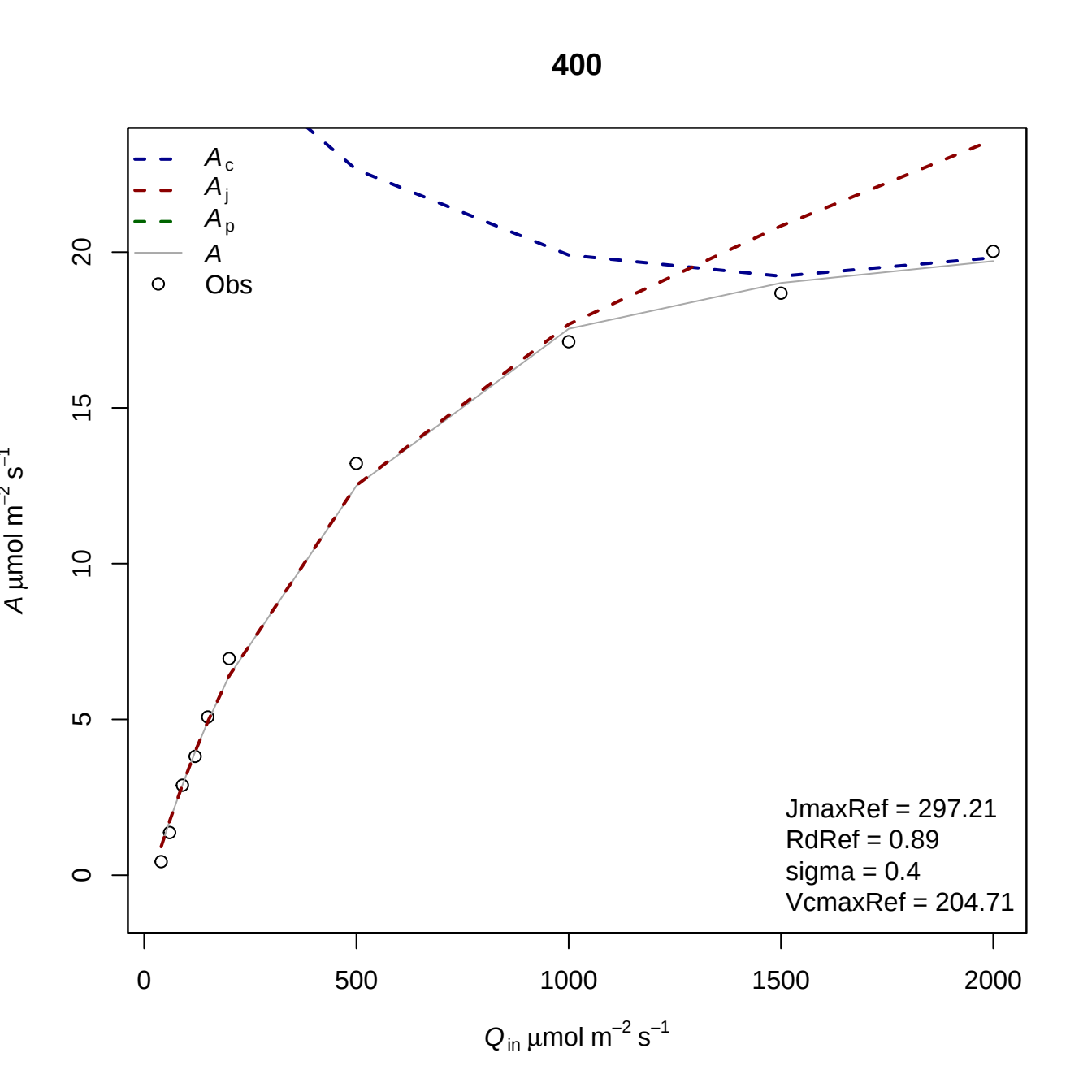

395
A_c
A_j
A_p
A
Obs
JmaxRef = 87.06
RdRef = 1.34
sigma = 0.28
VcmaxRef = 2260307.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

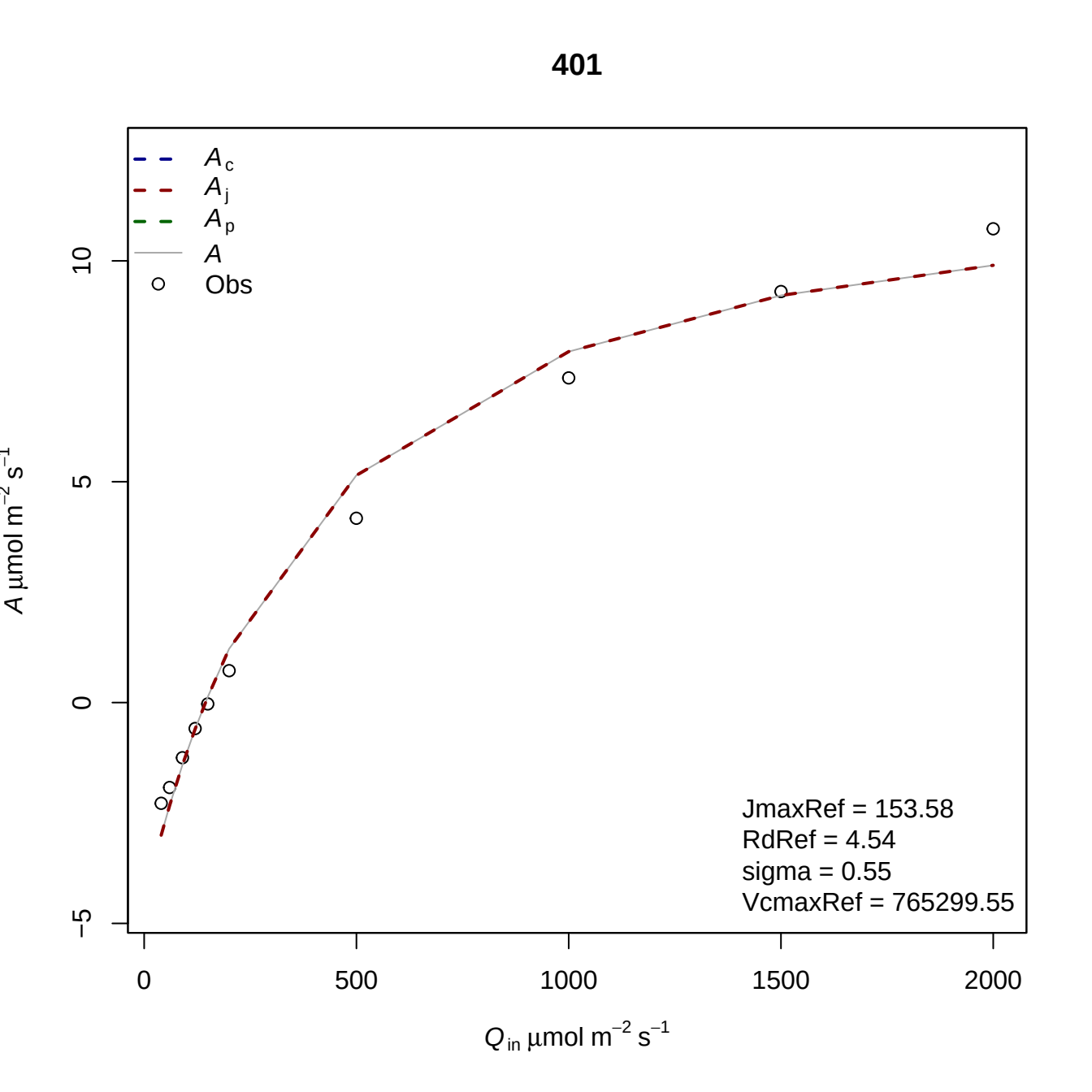

396
A_c
A_j
A_p
A
Obs
JmaxRef = 146.9
RdRef = −0.01
sigma = 0.3
VcmaxRef = 51.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

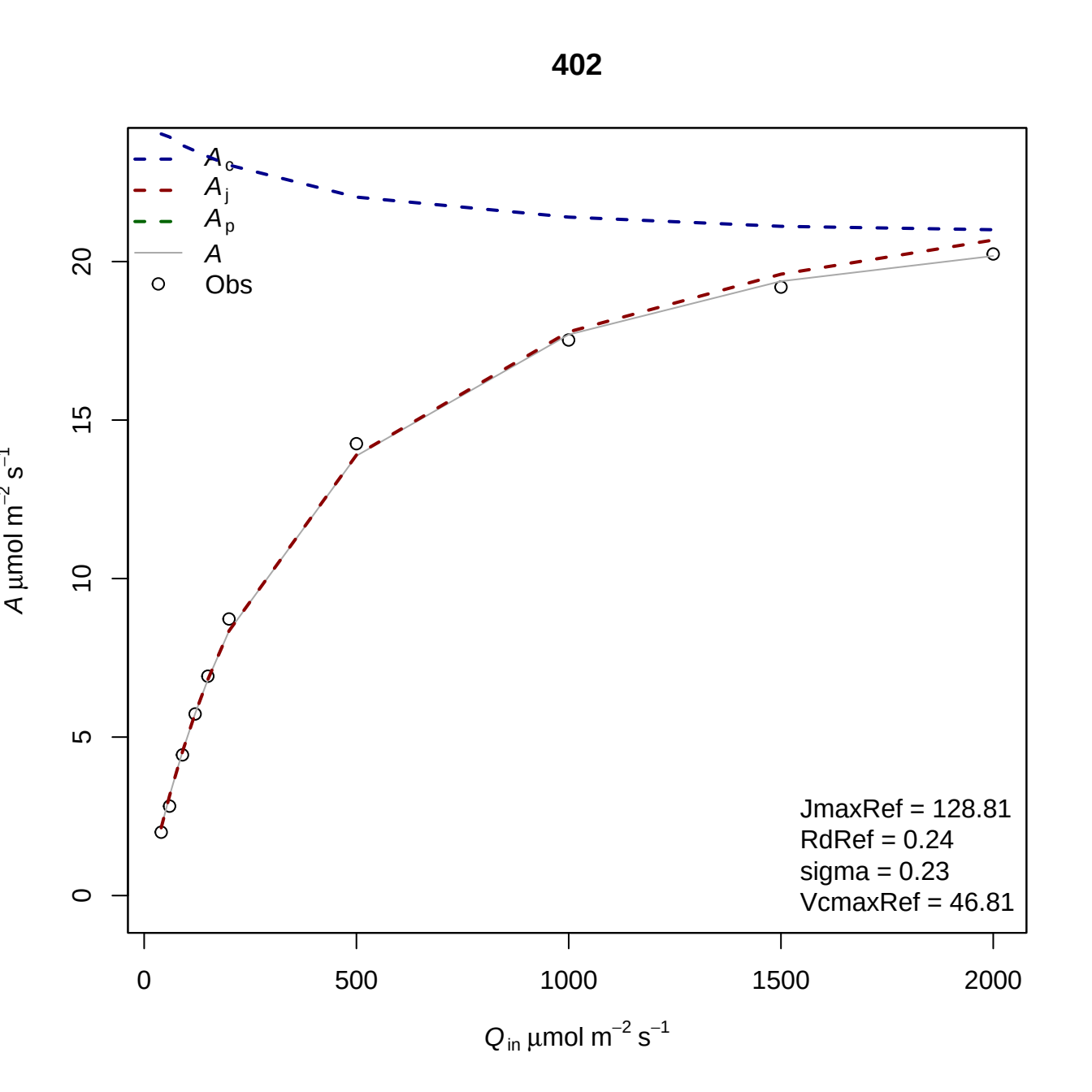

397
A_c
A_j
A_p
A
Obs
JmaxRef = 249.87
RdRef = −0.26
sigma = 0.58
VcmaxRef = 83.55
0
5
10
15
20
25
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

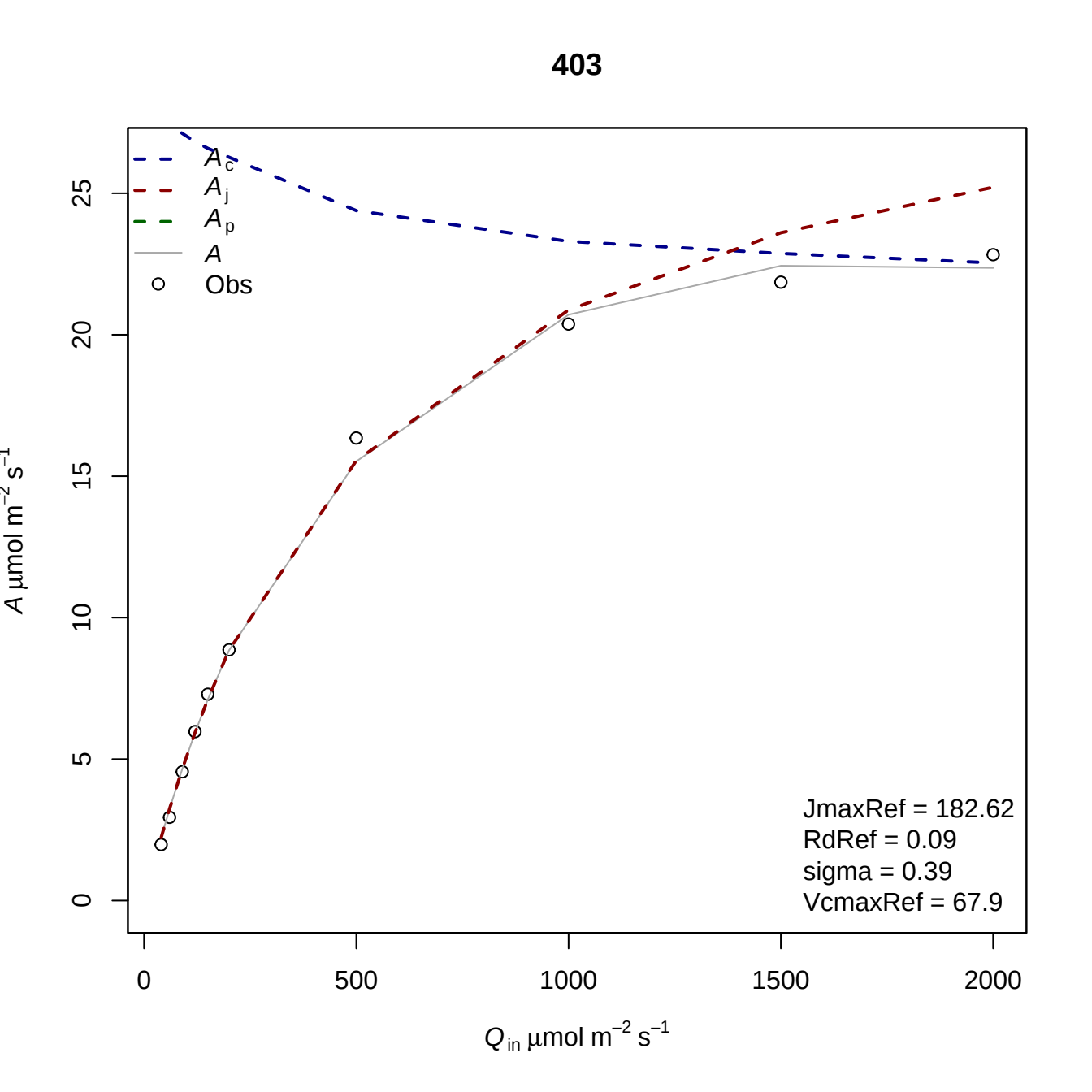

398
A_c
A_j
A_p
A
Obs
JmaxRef = 216.27
RdRef = 0.91
sigma = 0.13
VcmaxRef = 109.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

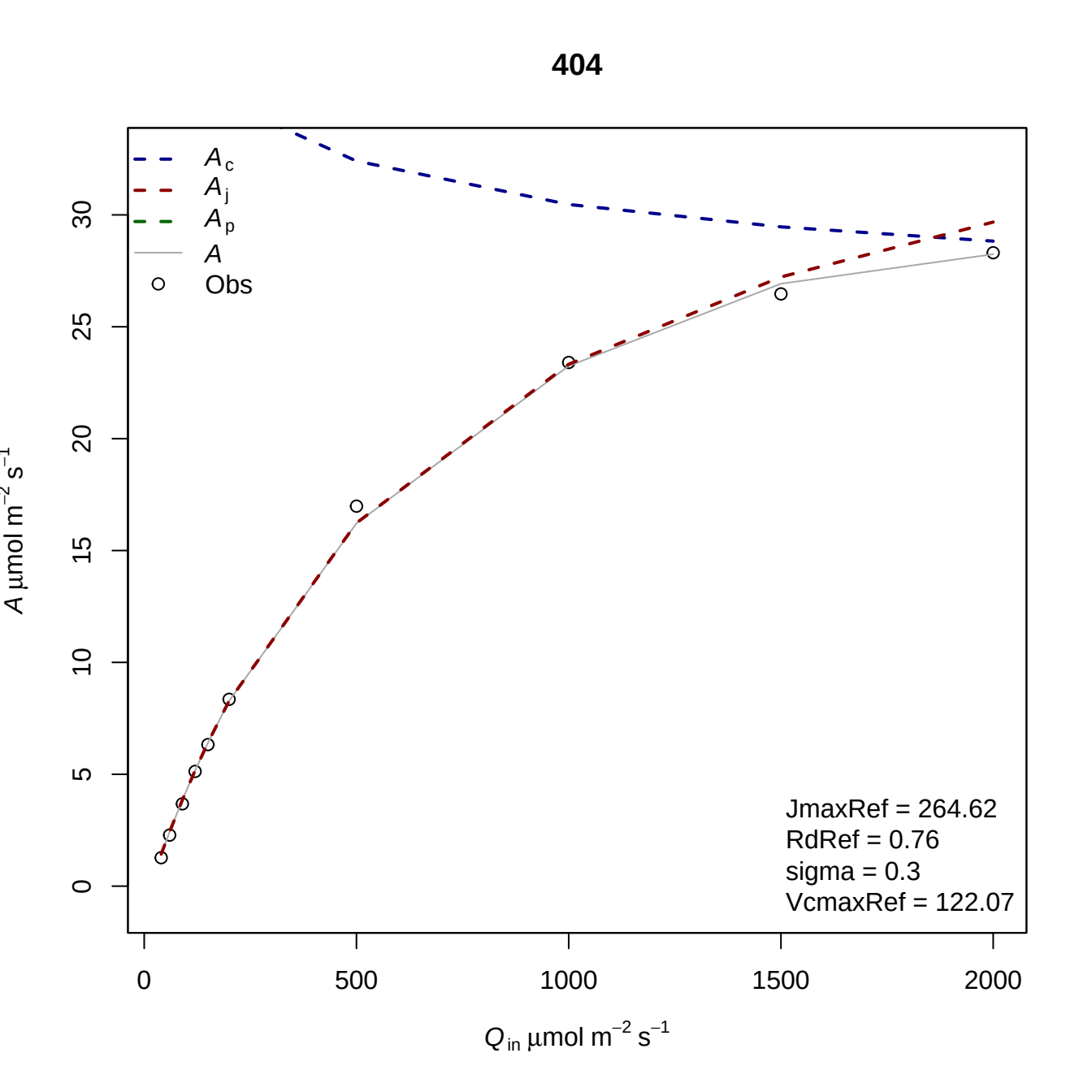

399
A_c
A_j
A_p
A
Obs
JmaxRef = 252.21
RdRef = 0.8
sigma = 0.45
VcmaxRef = 138.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

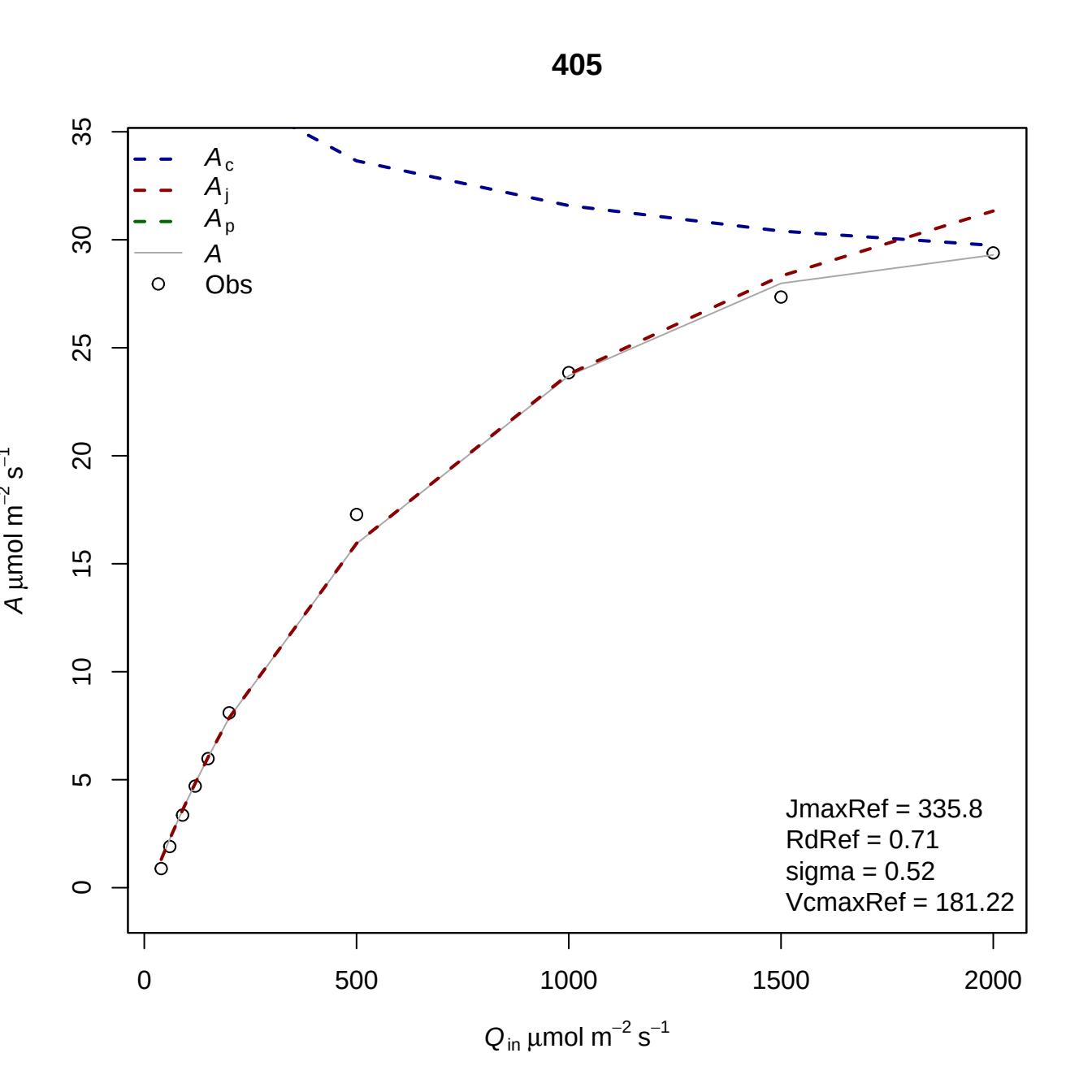

400
A_c
A_j
A_p
A
Obs
JmaxRef = 297.21
RdRef = 0.89
sigma = 0.4
VcmaxRef = 204.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

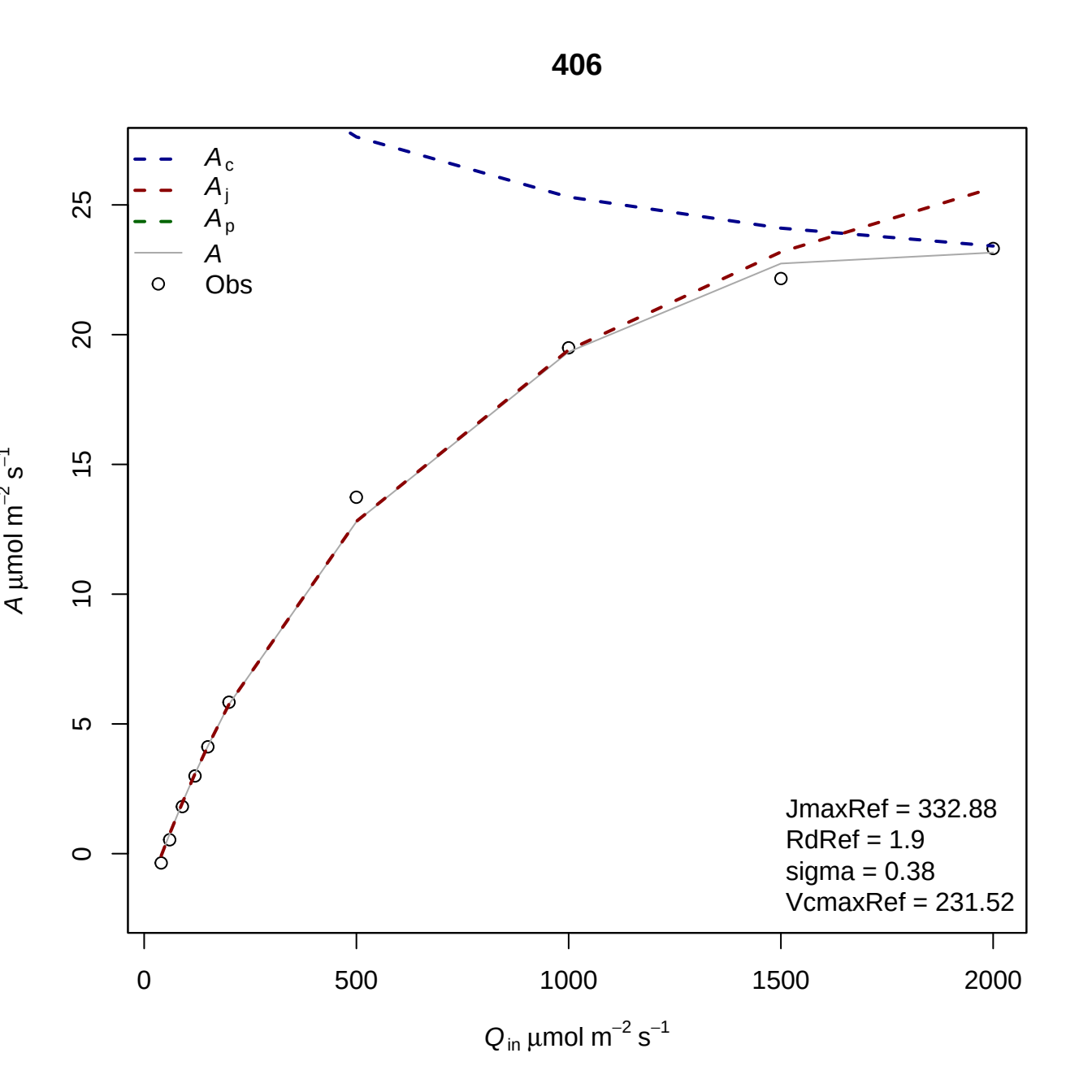

401
A_c
A_j
A_p
A
Obs
JmaxRef = 153.58
RdRef = 4.54
sigma = 0.55
VcmaxRef = 765299.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

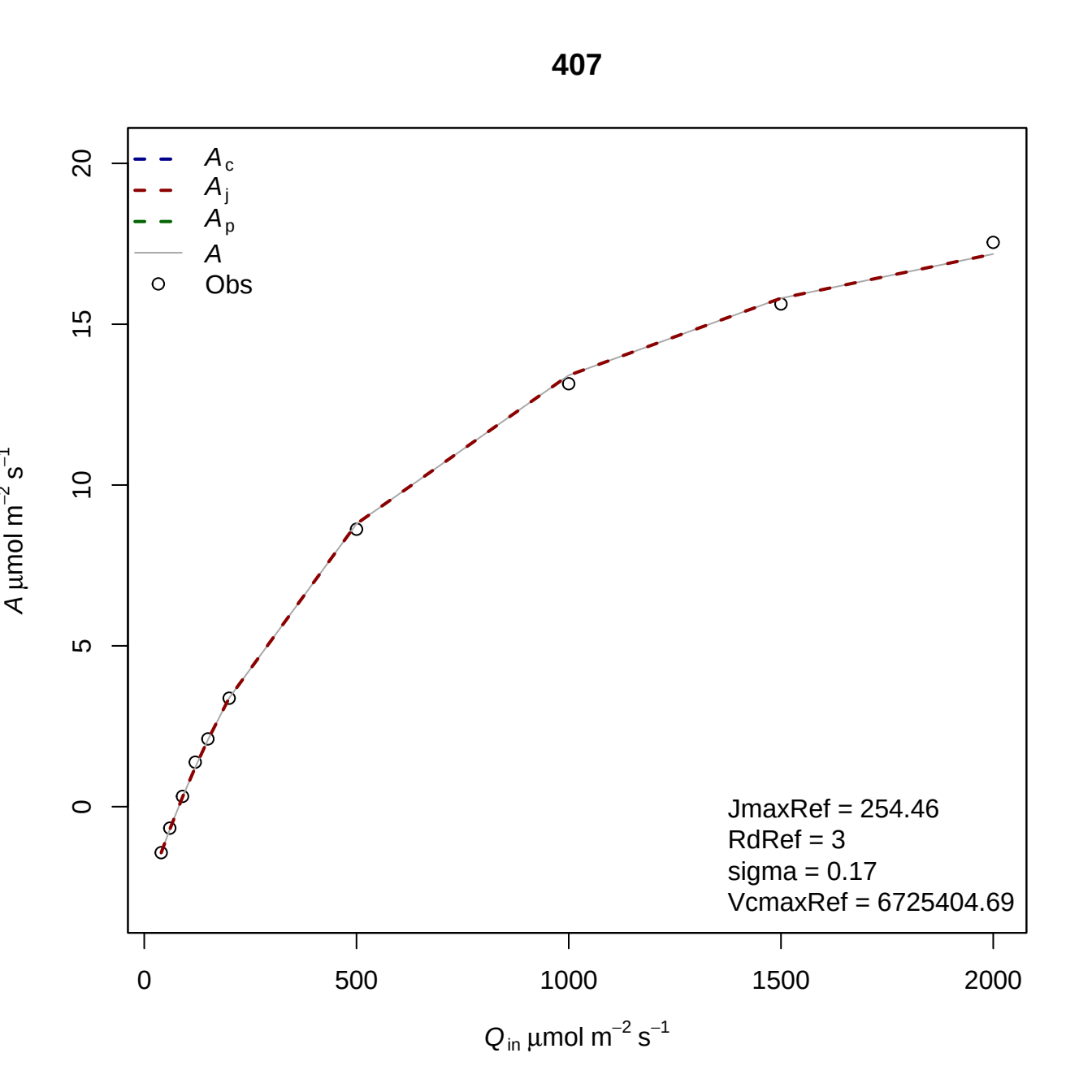

402
A_c
A_j
A_p
A
Obs
JmaxRef = 128.81
RdRef = 0.24
sigma = 0.23
VcmaxRef = 46.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

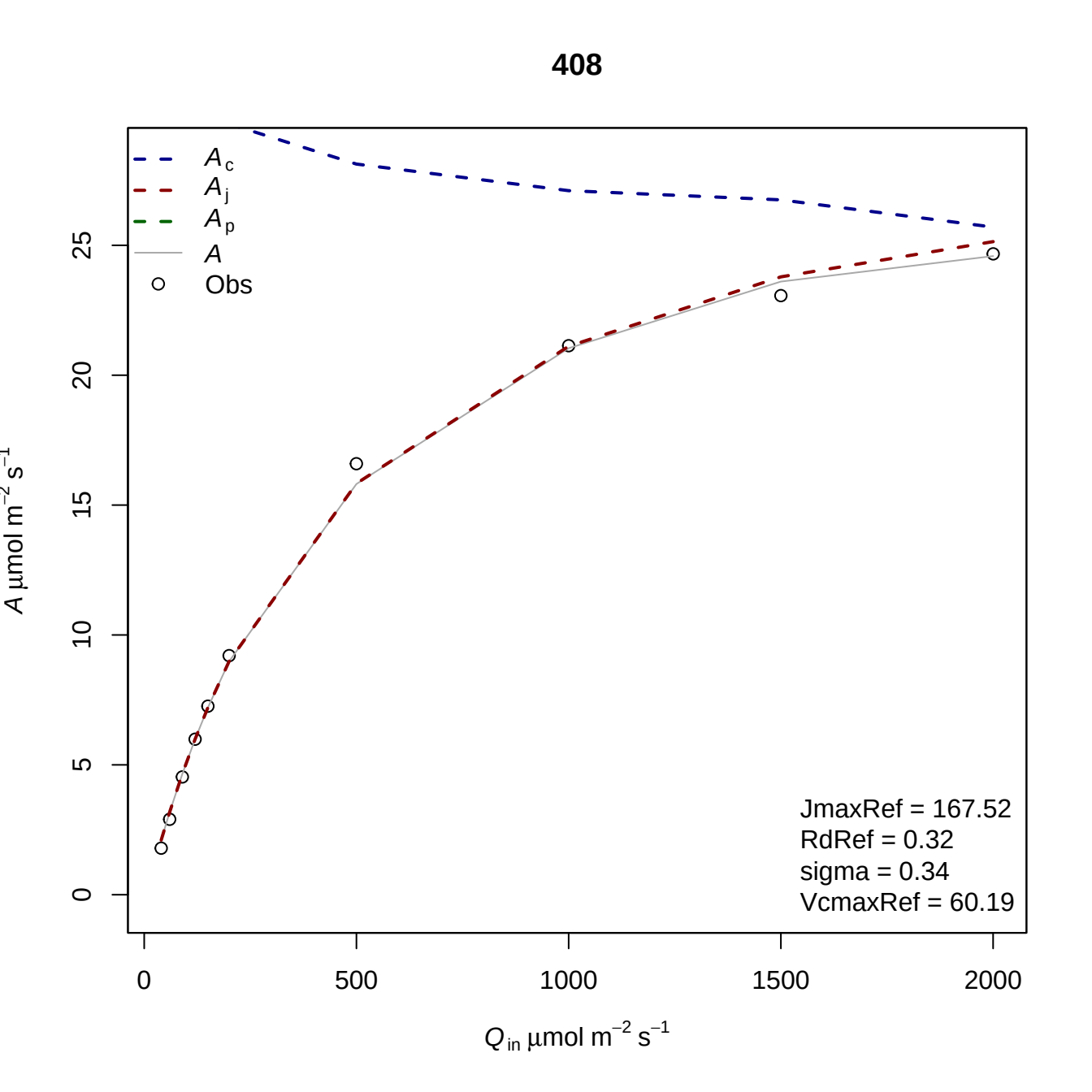

403
A_c
A_j
A_p
A
Obs
JmaxRef = 182.62
RdRef = 0.09
sigma = 0.39
VcmaxRef = 67.9
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

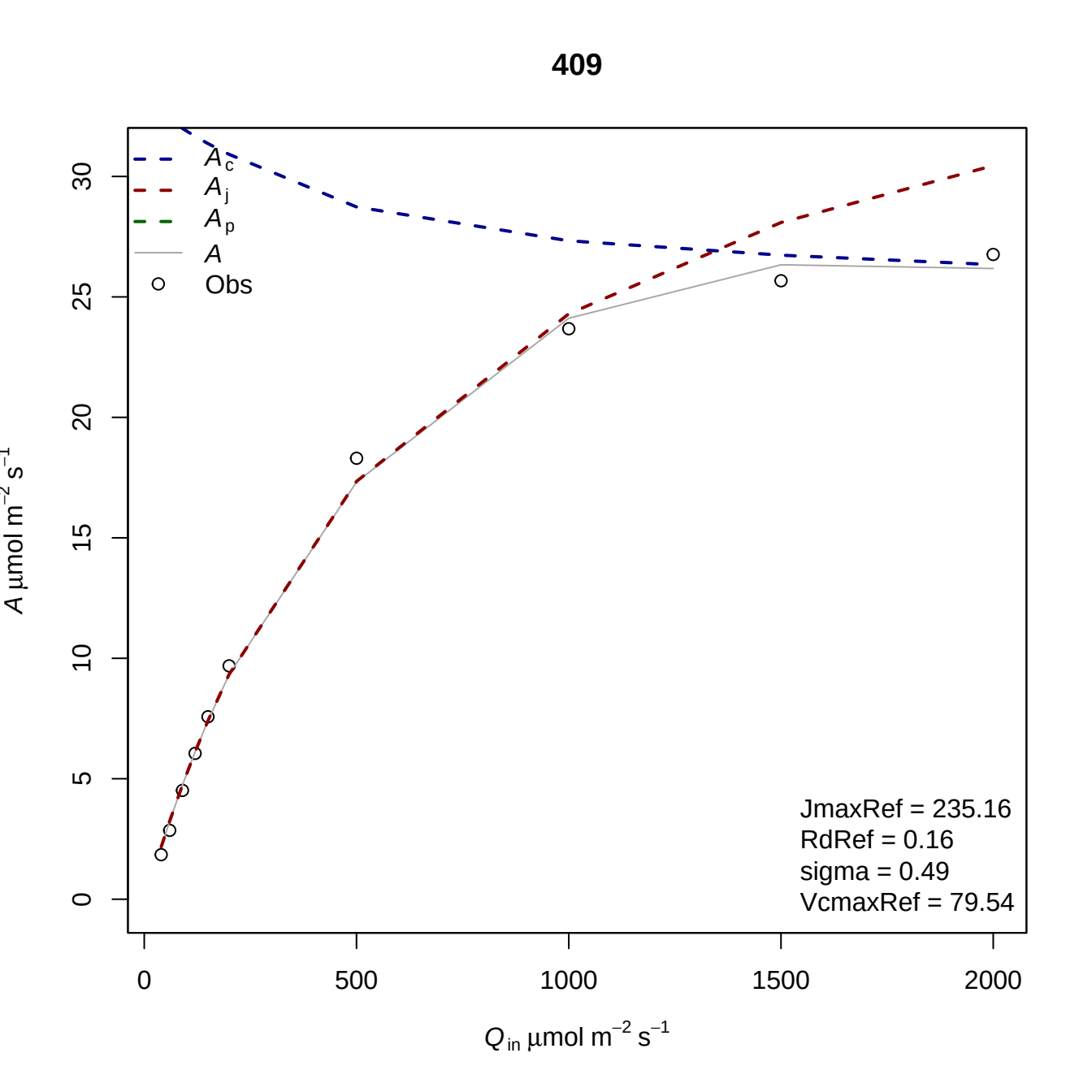

404
A_c
A_j
A_p
A
Obs
JmaxRef = 264.62
RdRef = 0.76
sigma = 0.3
VcmaxRef = 122.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

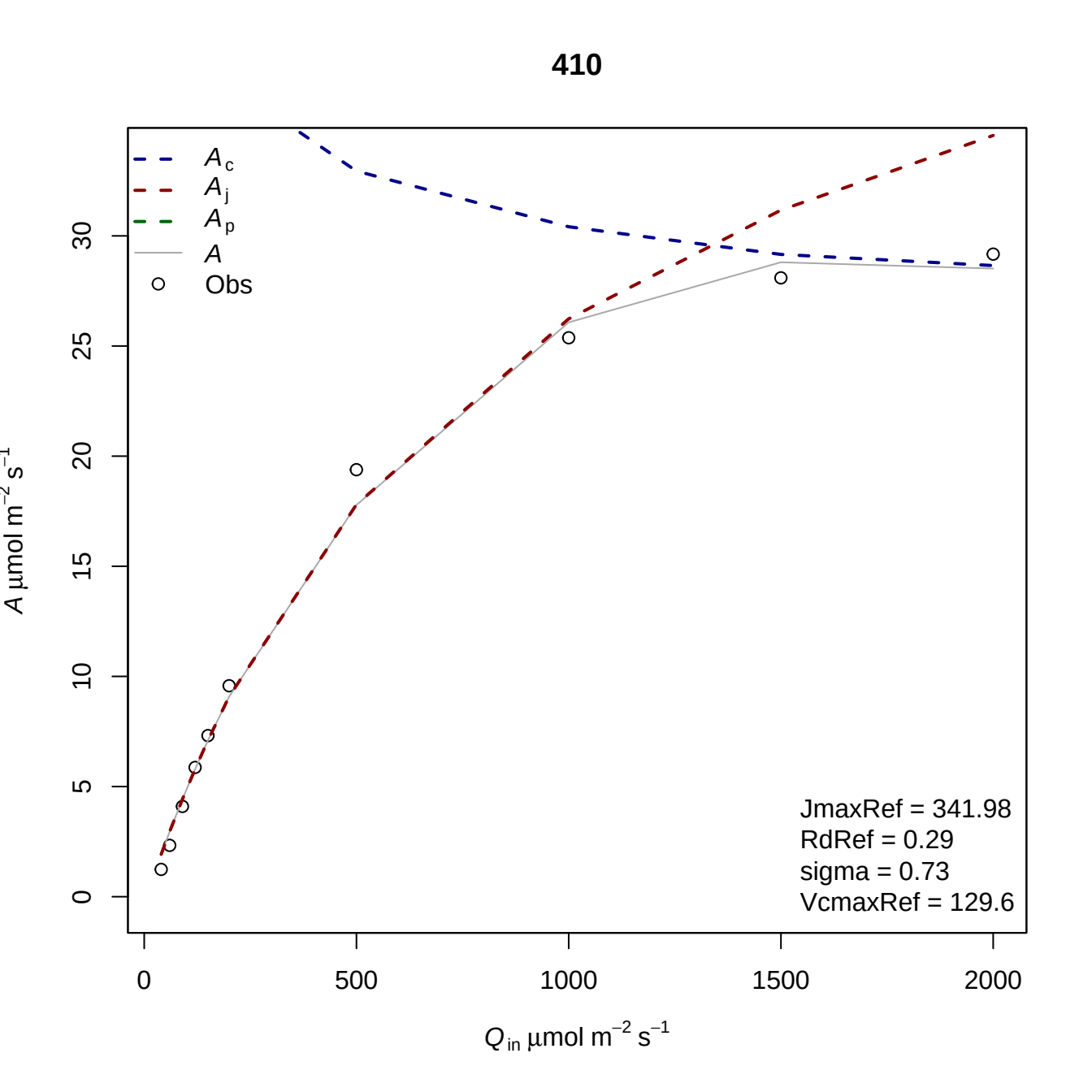

405
A_c
A_j
A_p
A
Obs
JmaxRef = 335.8
RdRef = 0.71
sigma = 0.52
VcmaxRef = 181.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

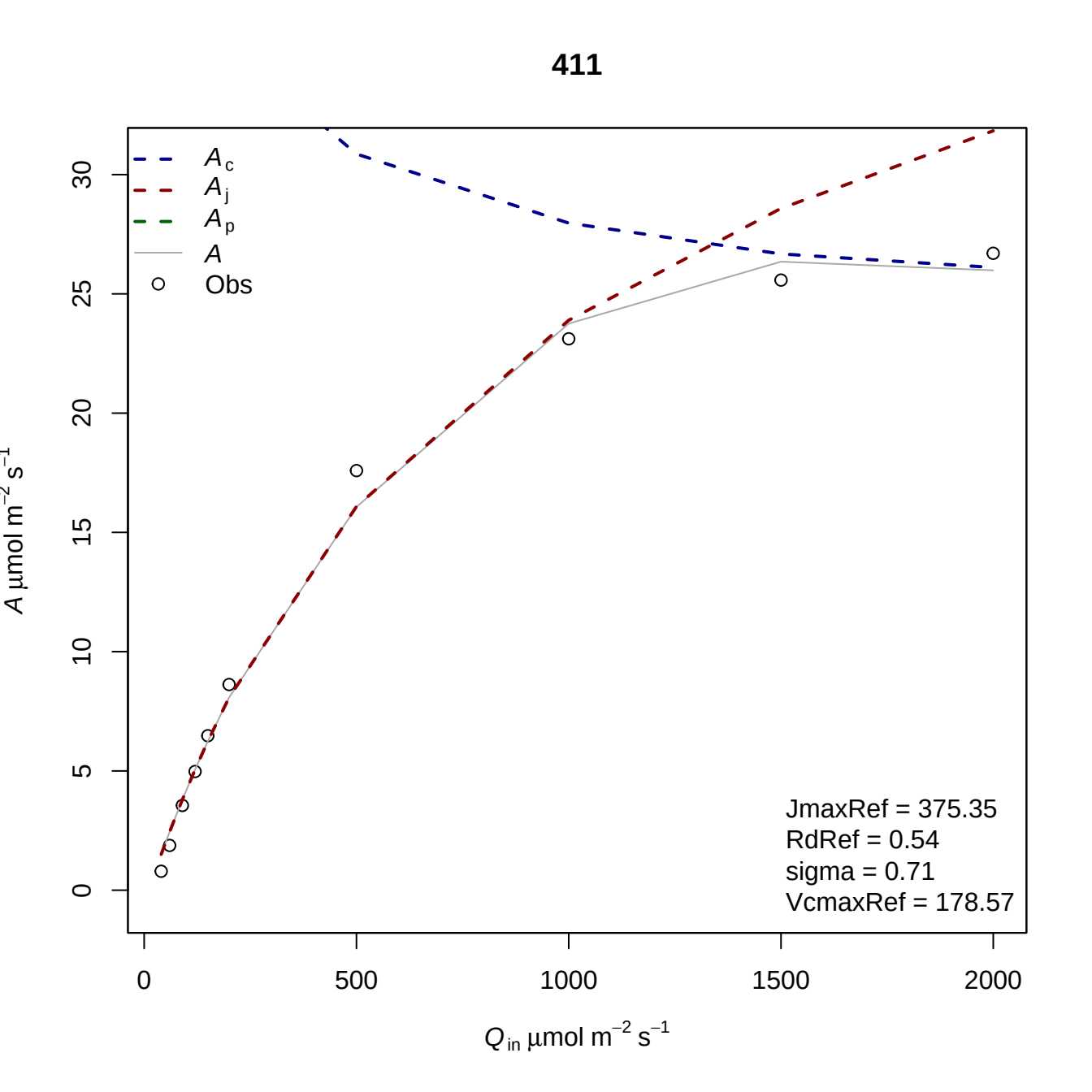

406
A_c
A_j
A_p
A
Obs
JmaxRef = 332.88
RdRef = 1.9
sigma = 0.38
VcmaxRef = 231.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

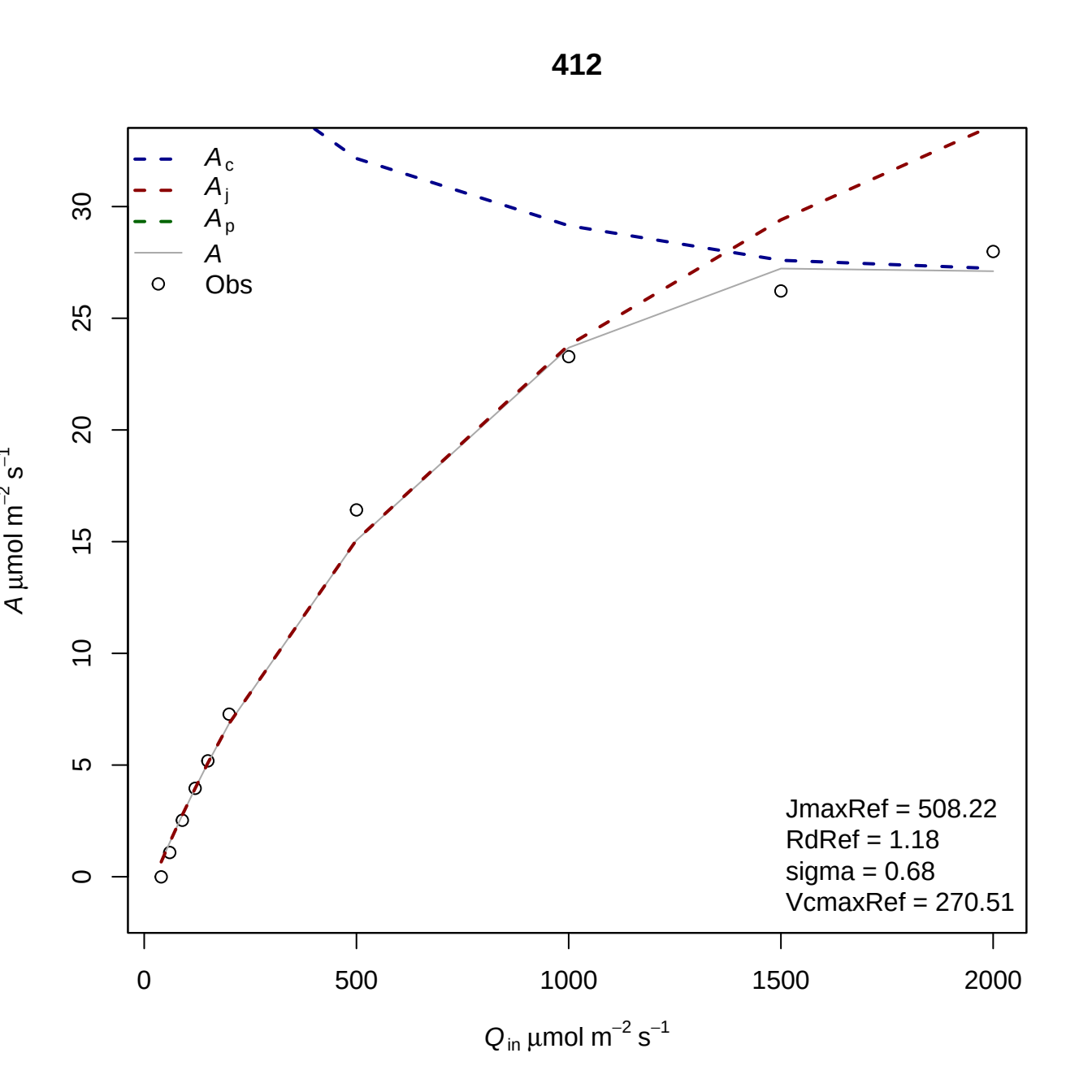

407
A_c
A_j
A_p
A
Obs
JmaxRef = 254.46
RdRef = 3
sigma = 0.17
VcmaxRef = 6725404.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

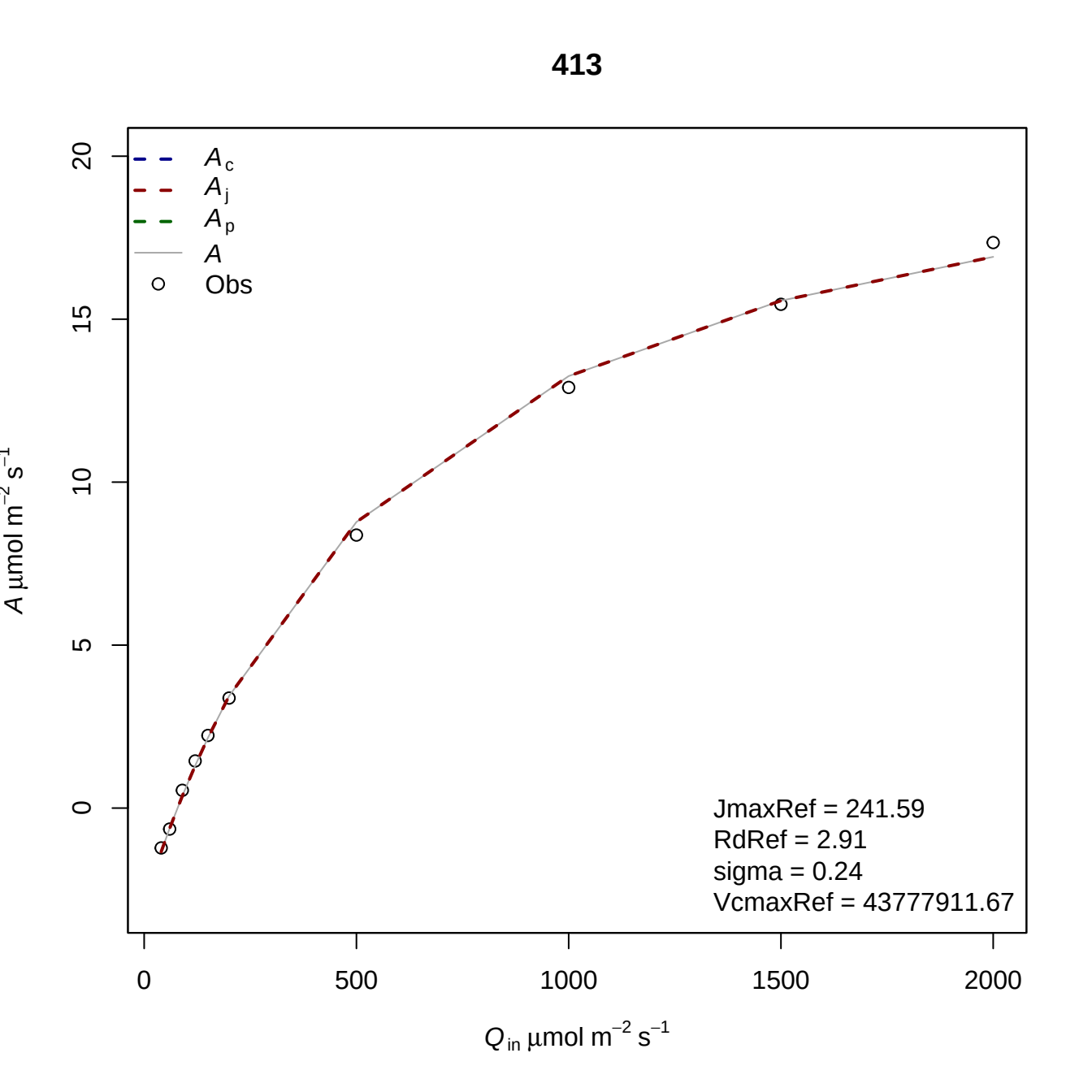

408
A_c
A_j
A_p
A
Obs
JmaxRef = 167.52
RdRef = 0.32
sigma = 0.34
VcmaxRef = 60.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

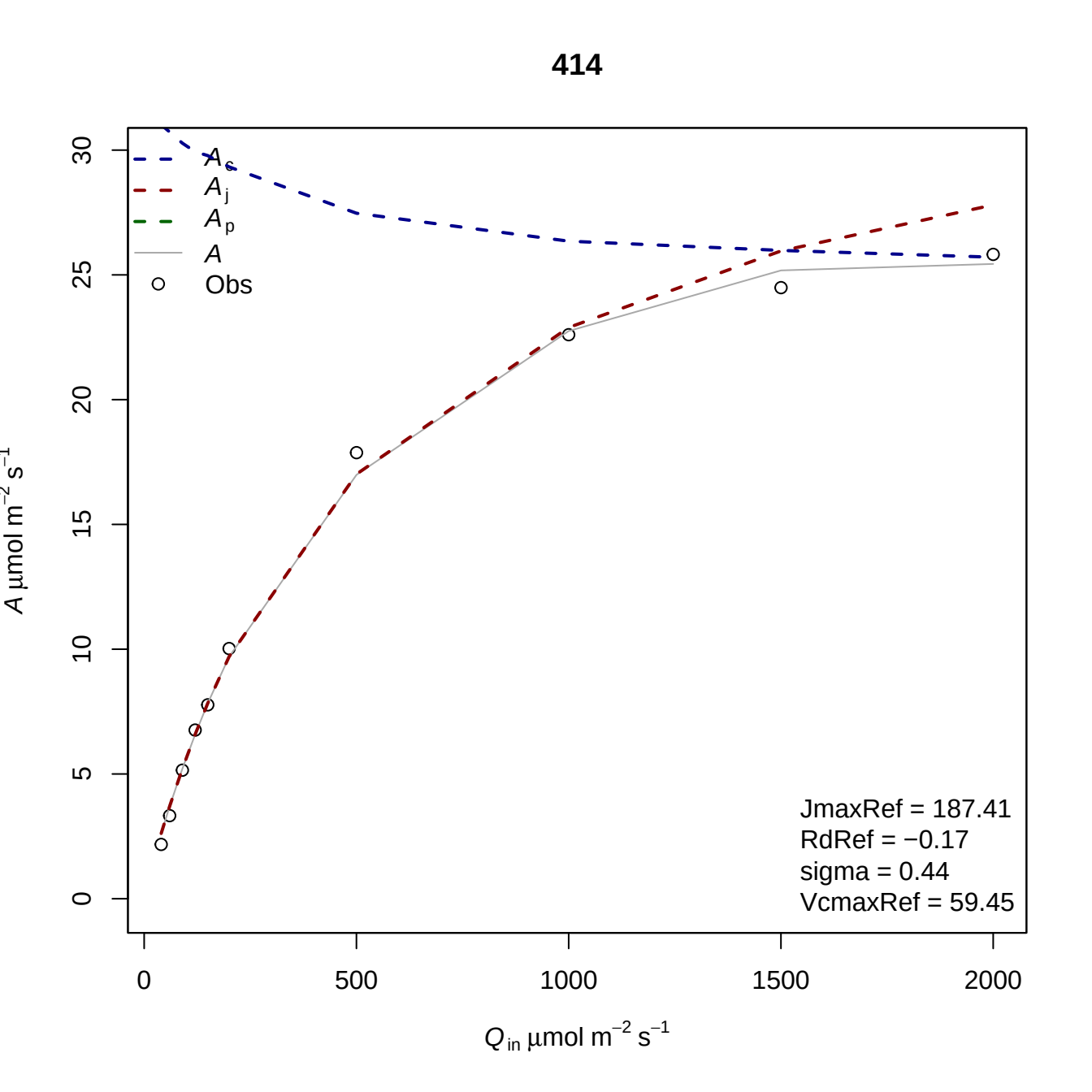

409
A_c
A_j
A_p
A
Obs
JmaxRef = 235.16
RdRef = 0.16
sigma = 0.49
VcmaxRef = 79.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

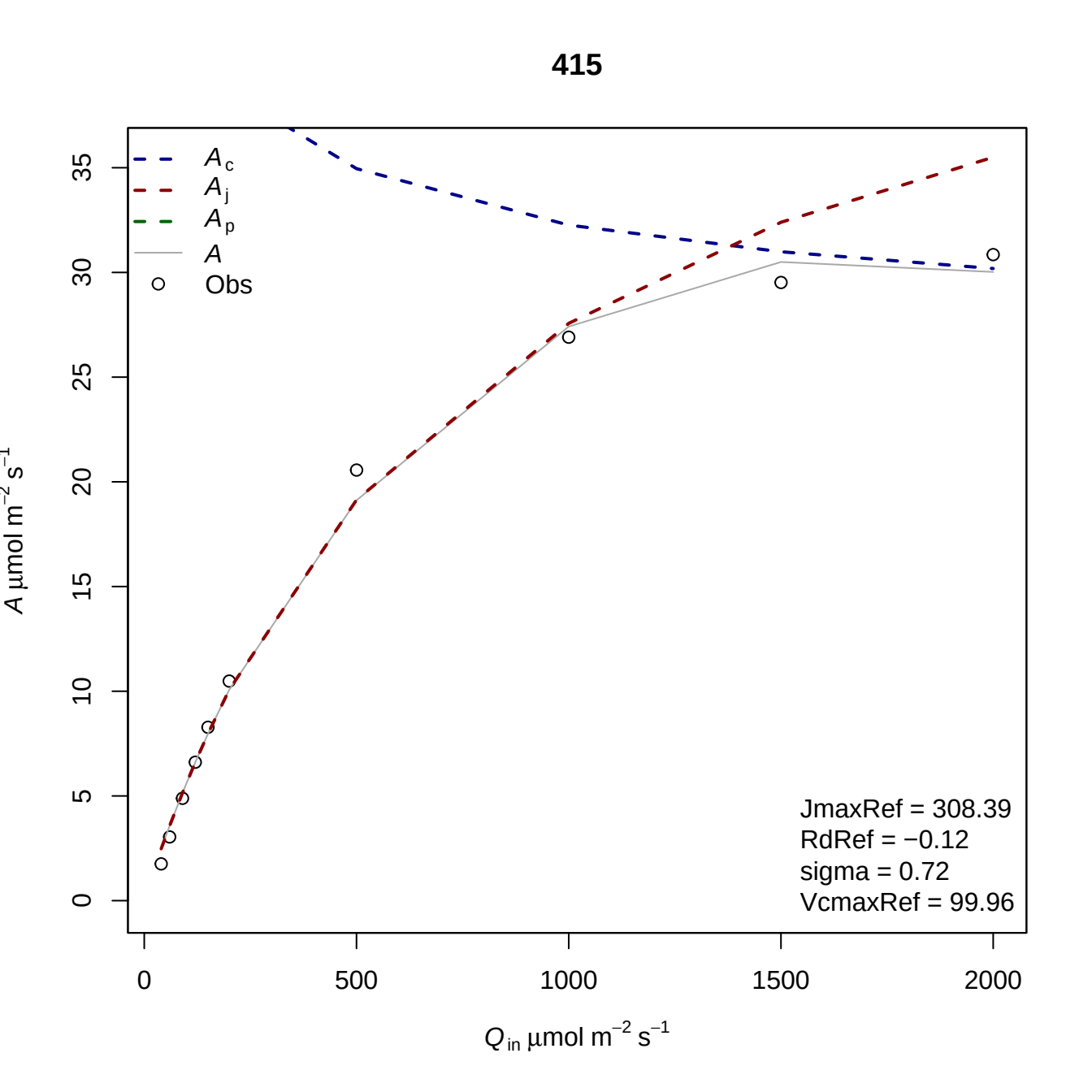

410
A_c
A_j
A_p
A
Obs
JmaxRef = 341.98
RdRef = 0.29
sigma = 0.73
VcmaxRef = 129.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

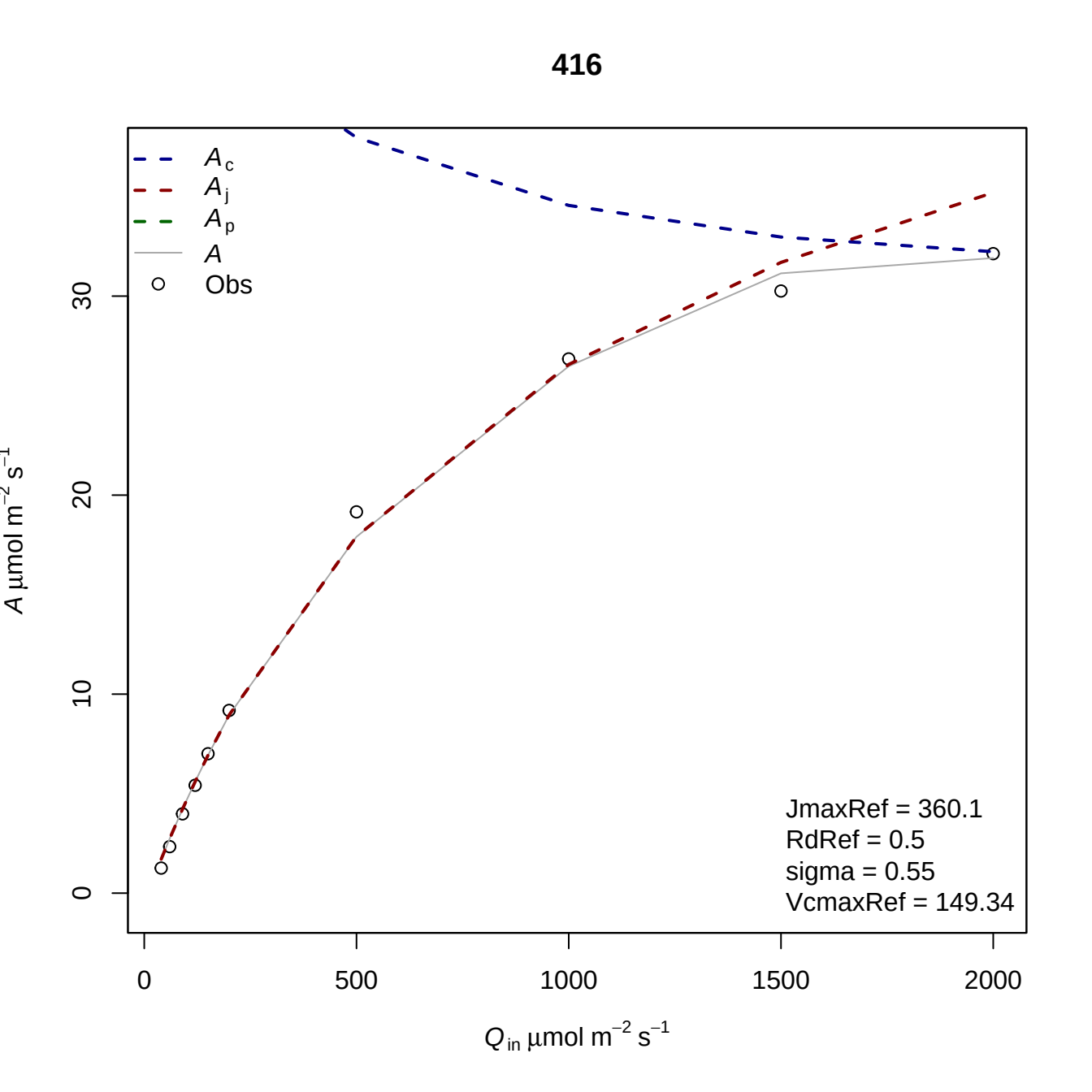

411
A_c
A_j
A_p
A
Obs
JmaxRef = 375.35
RdRef = 0.54
sigma = 0.71
VcmaxRef = 178.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

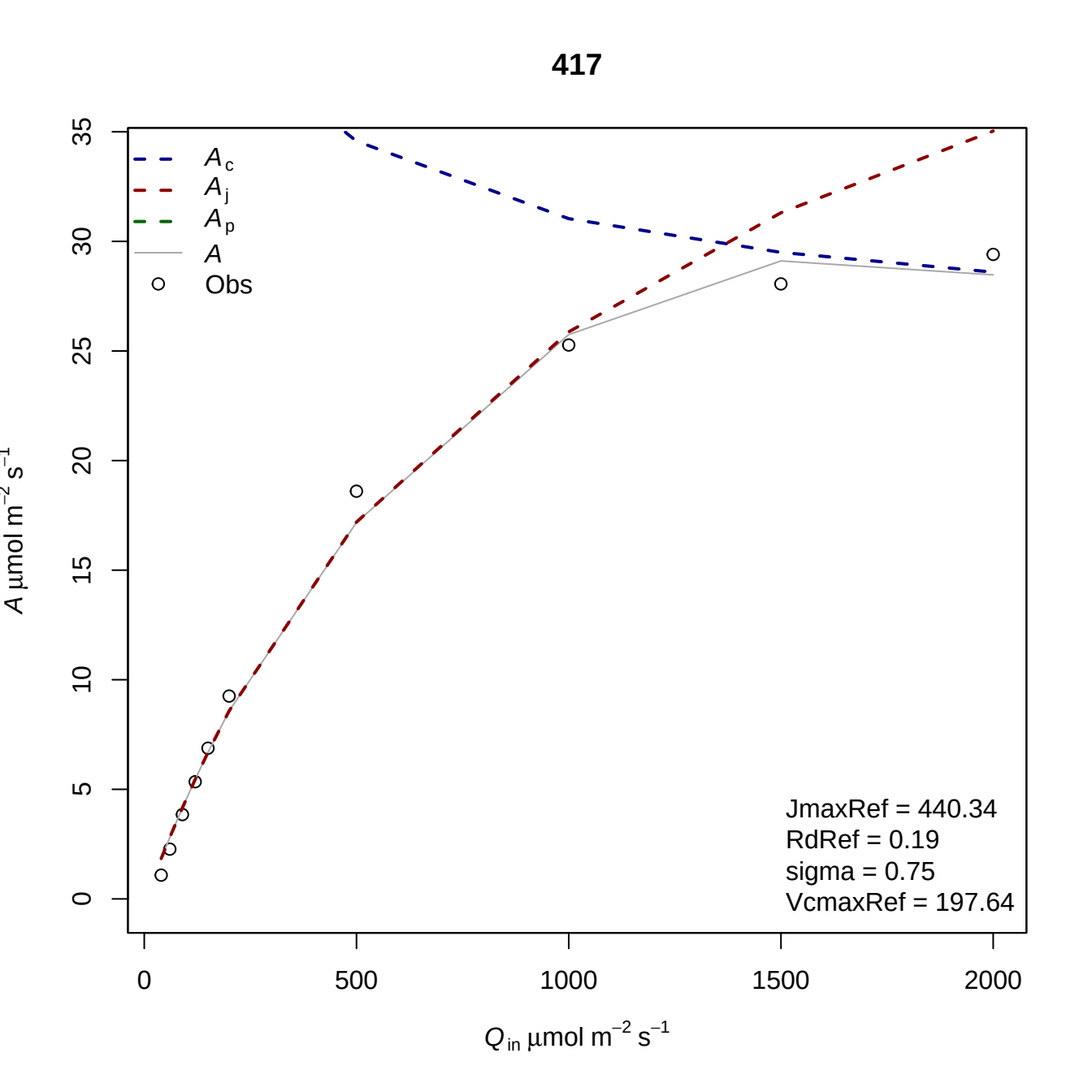

412
A_c
A_j
A_p
A
Obs
JmaxRef = 508.22
RdRef = 1.18
sigma = 0.68
VcmaxRef = 270.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

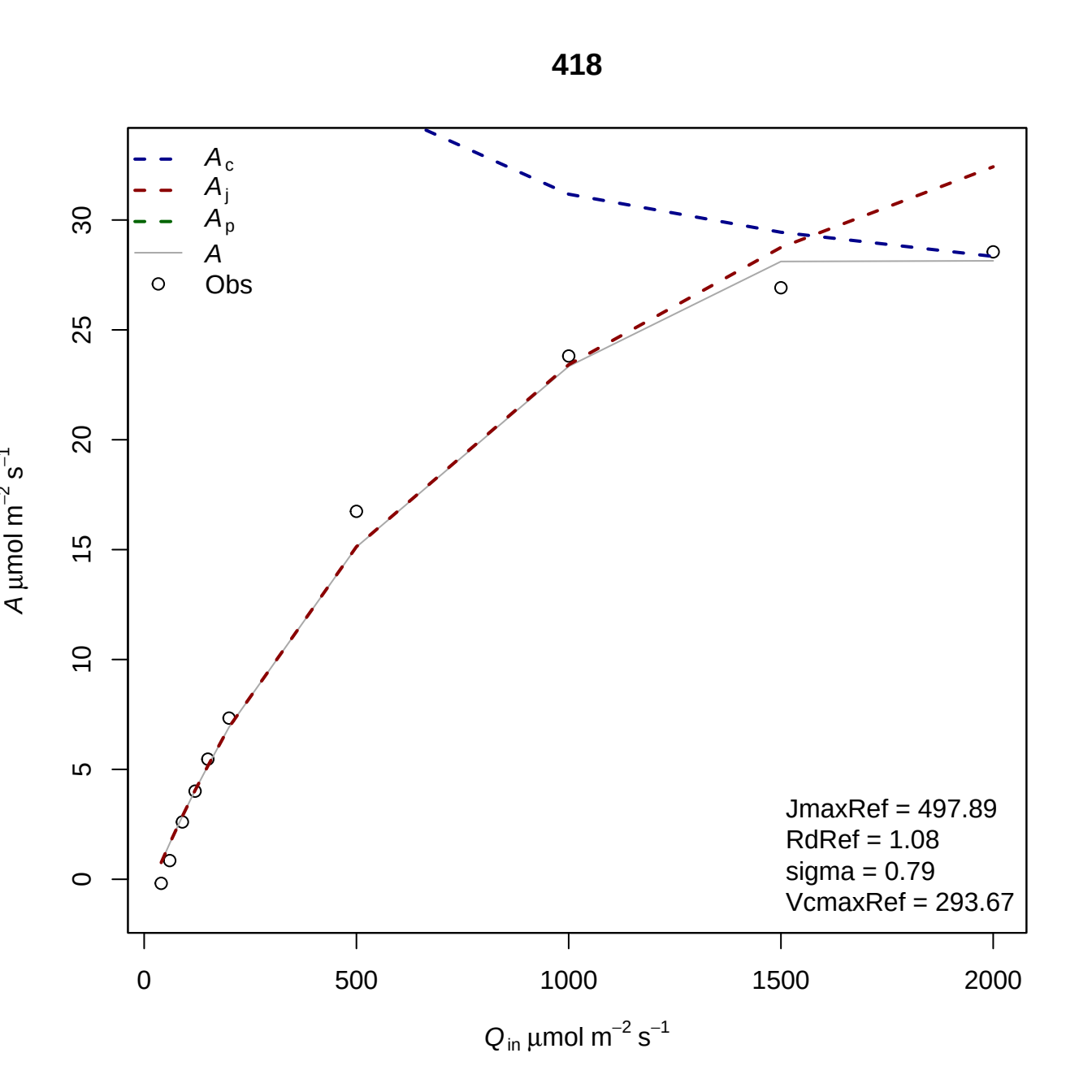

413
A_c
A_j
A_p
A
Obs
JmaxRef = 241.59
RdRef = 2.91
sigma = 0.24
VcmaxRef = 43777911.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

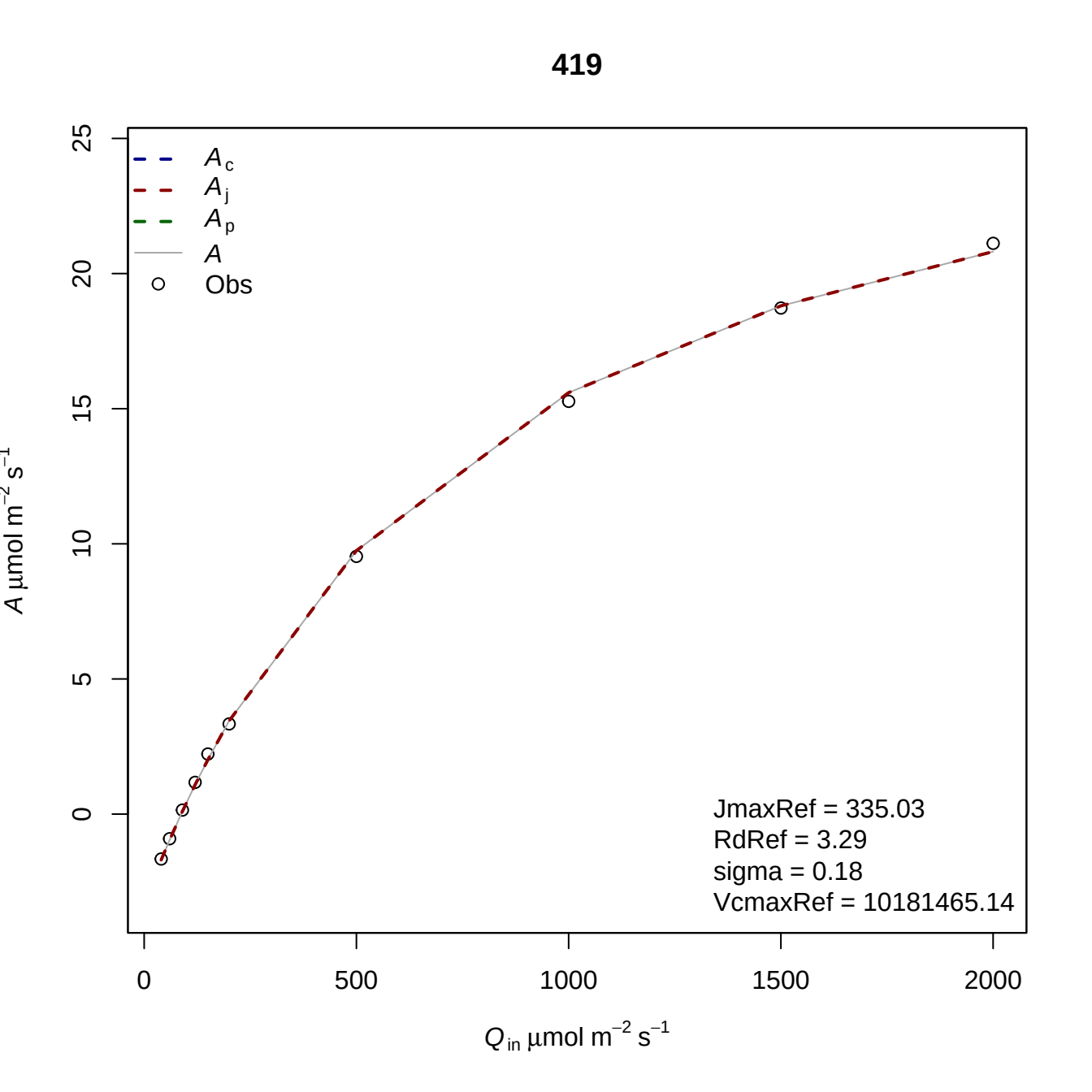

414
A_c
A_j
A_p
A
Obs
JmaxRef = 187.41
RdRef = −0.17
sigma = 0.44
VcmaxRef = 59.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

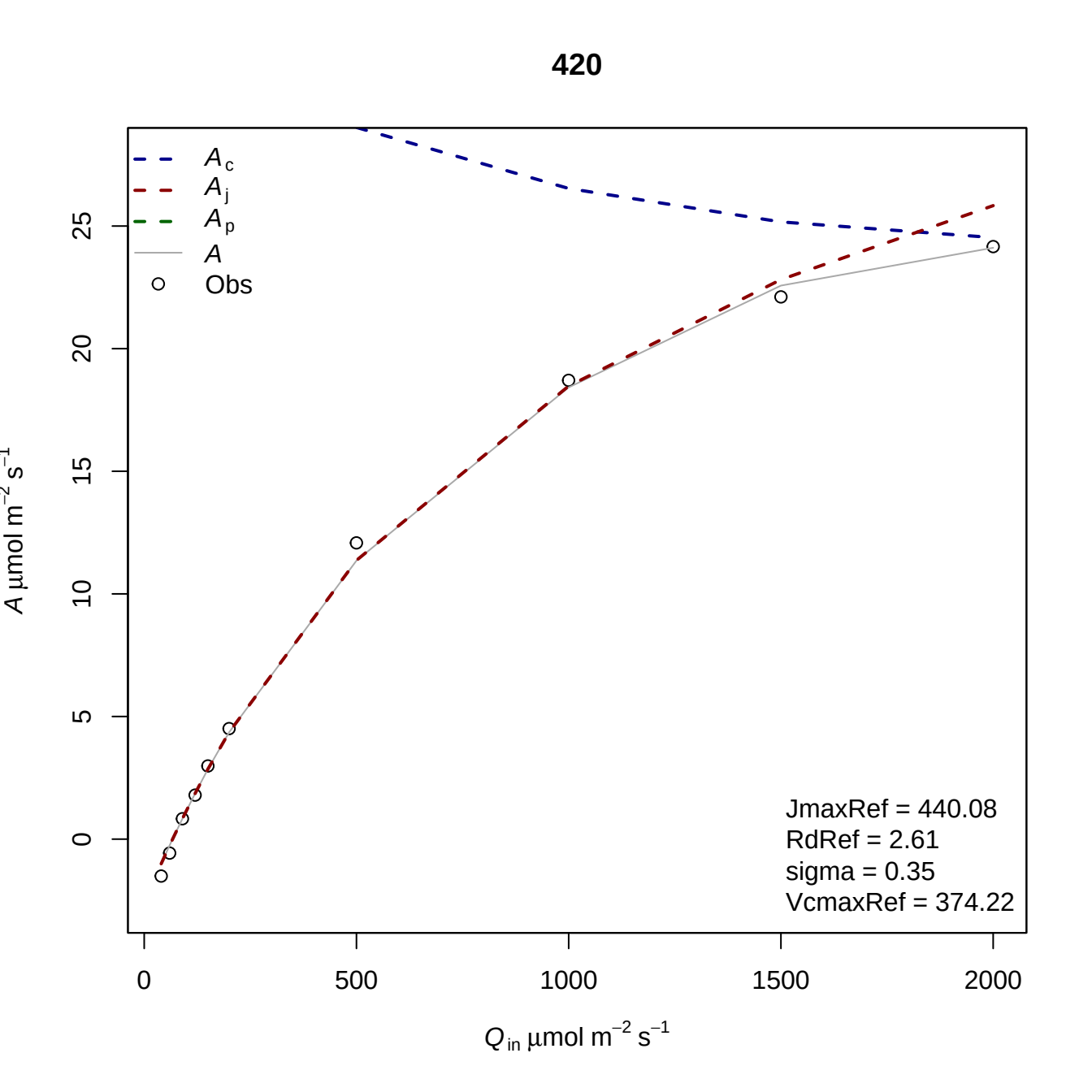

415
A_c
A_j
A_p
A
Obs
JmaxRef = 308.39
RdRef = −0.12
sigma = 0.72
VcmaxRef = 99.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

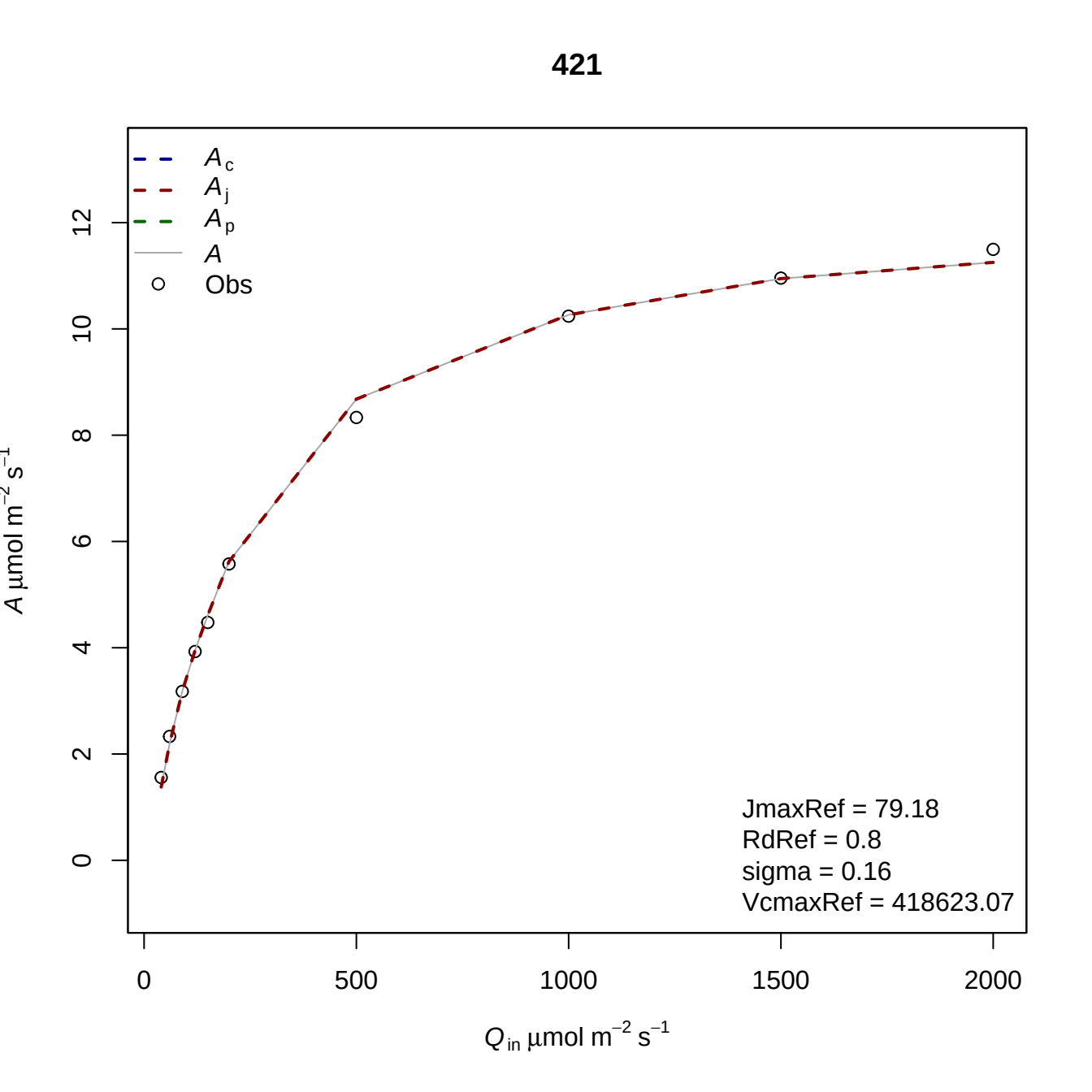

416
A_c
A_j
A_p
A
Obs
JmaxRef = 360.1
RdRef = 0.5
sigma = 0.55
VcmaxRef = 149.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

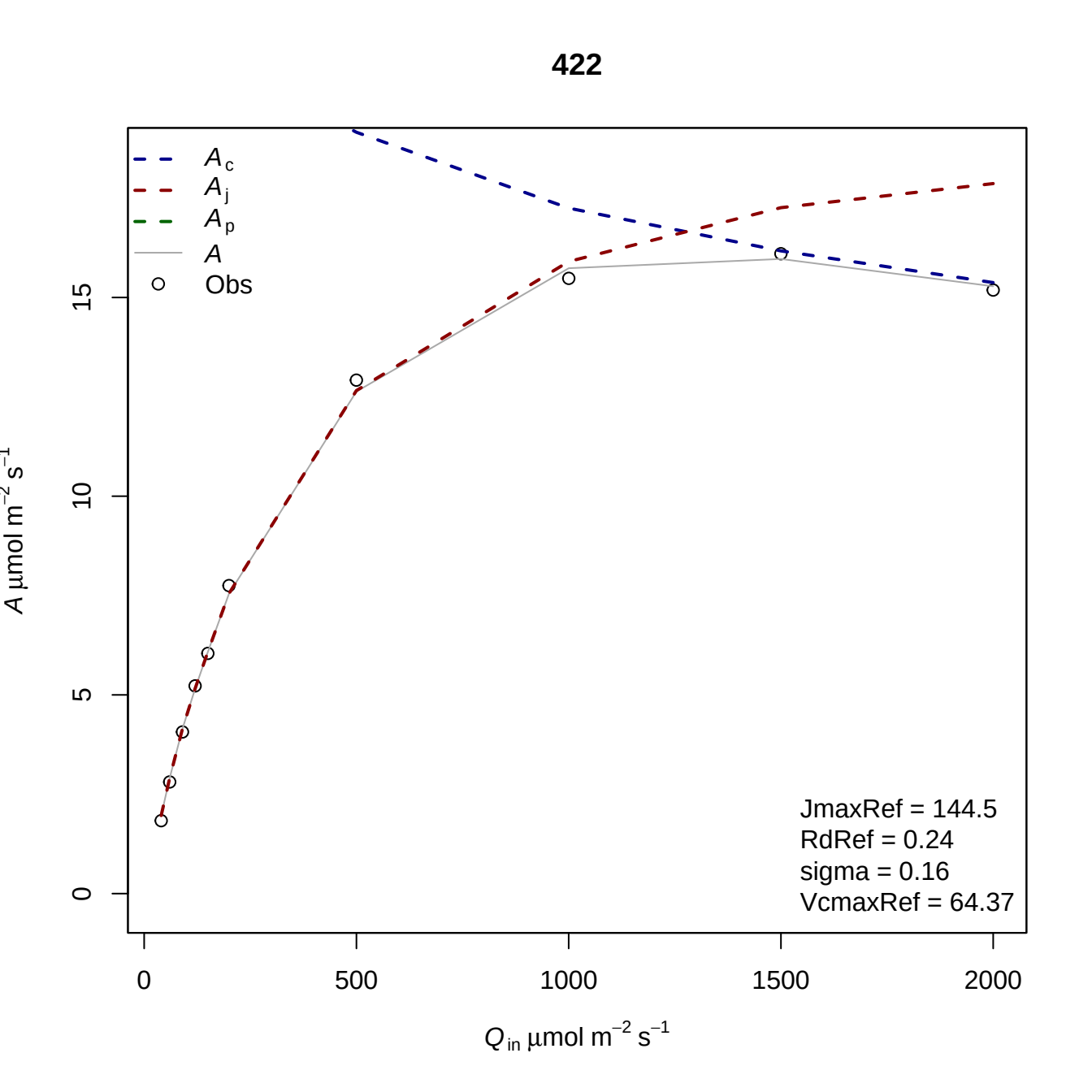

417
A_c
A_j
A_p
A
Obs
JmaxRef = 440.34
RdRef = 0.19
sigma = 0.75
VcmaxRef = 197.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

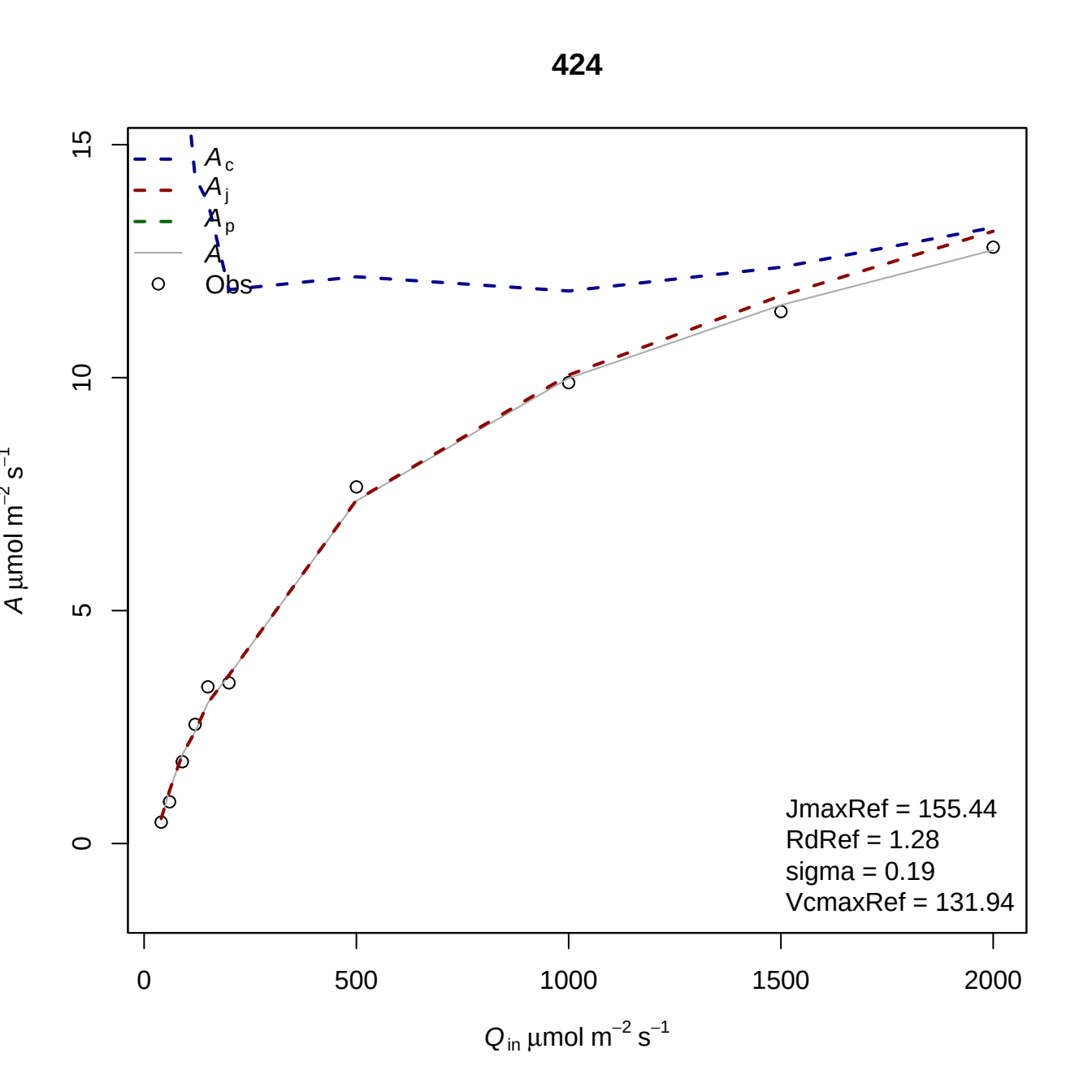

418
A_c
A_j
A_p
A
Obs
JmaxRef = 497.89
RdRef = 1.08
sigma = 0.79
VcmaxRef = 293.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

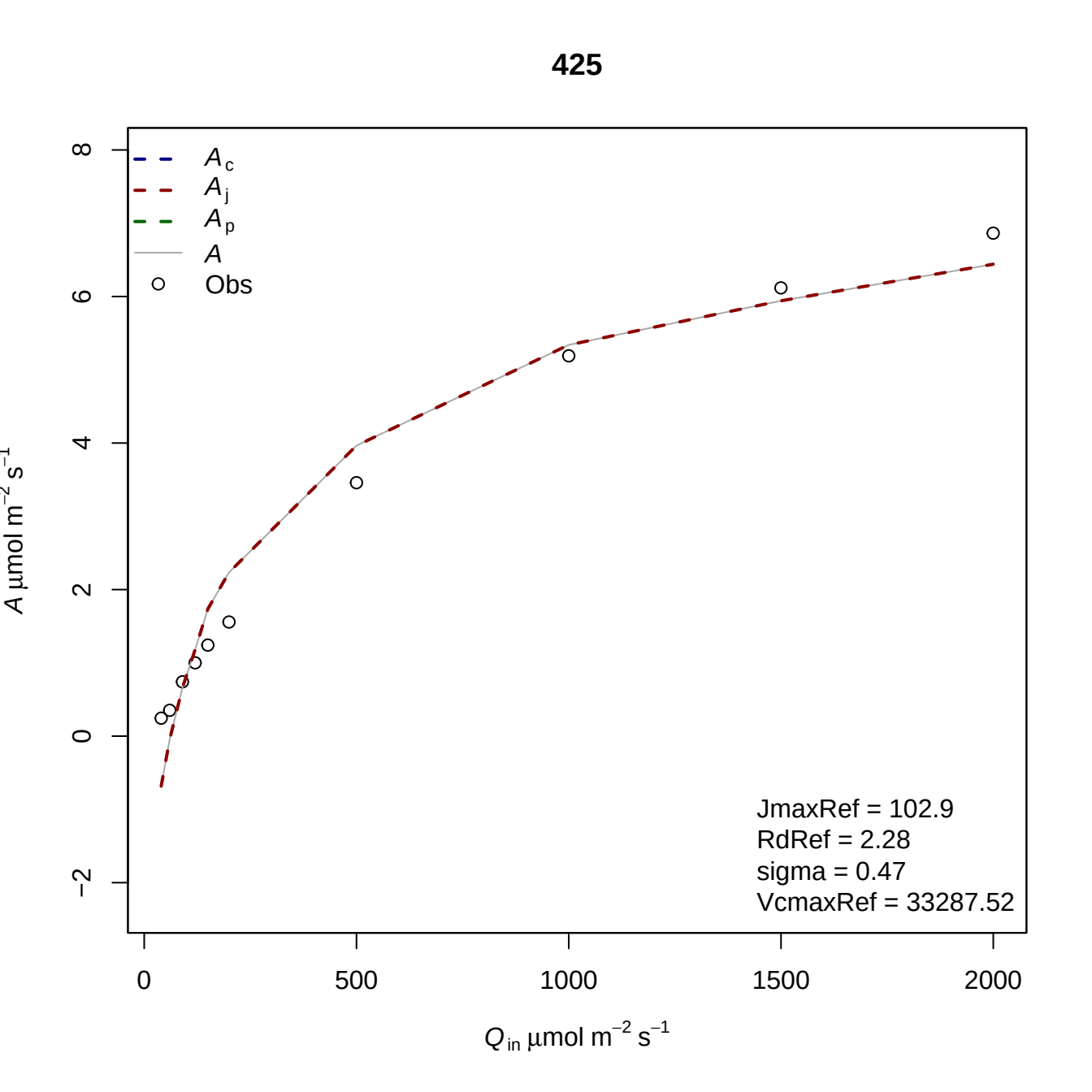

419
A_c
A_j
A_p
A
Obs
JmaxRef = 335.03
RdRef = 3.29
sigma = 0.18
VcmaxRef = 10181465.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

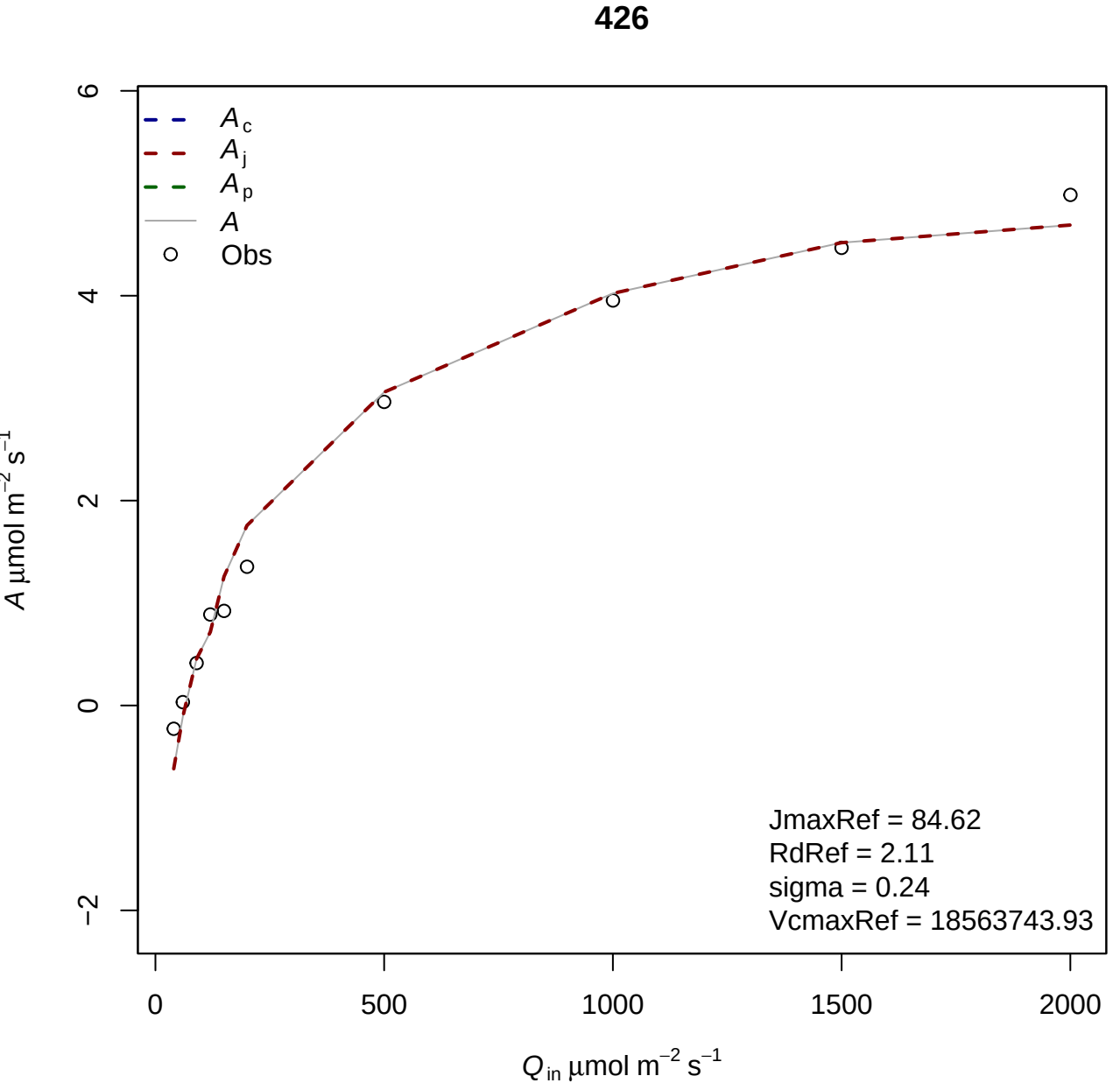

420
A_c
A_j
A_p
A
Obs
JmaxRef = 440.08
RdRef = 2.61
sigma = 0.35
VcmaxRef = 374.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

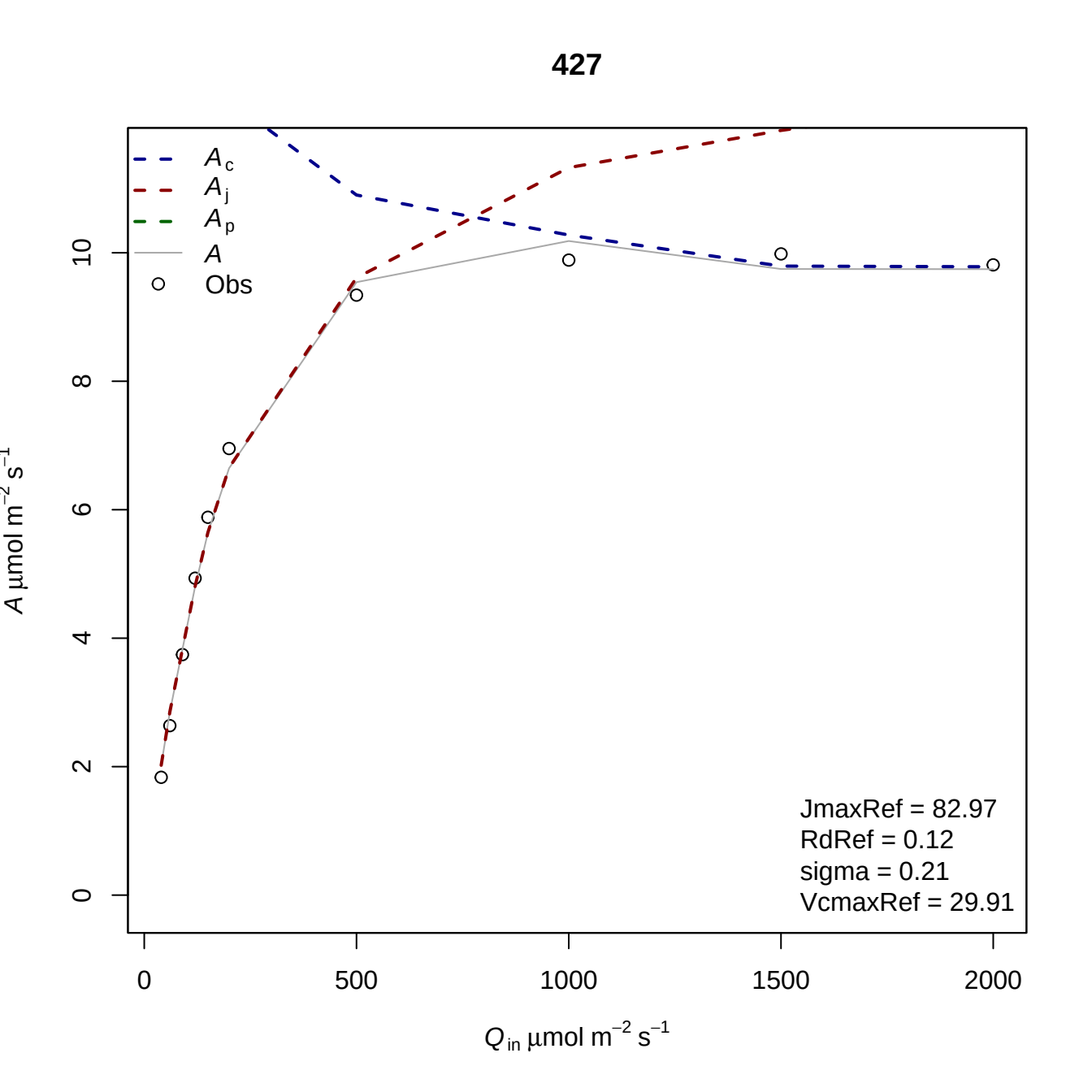

421
A_c
A_j
A_p
A
Obs
JmaxRef = 79.18
RdRef = 0.8
sigma = 0.16
VcmaxRef = 418623.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

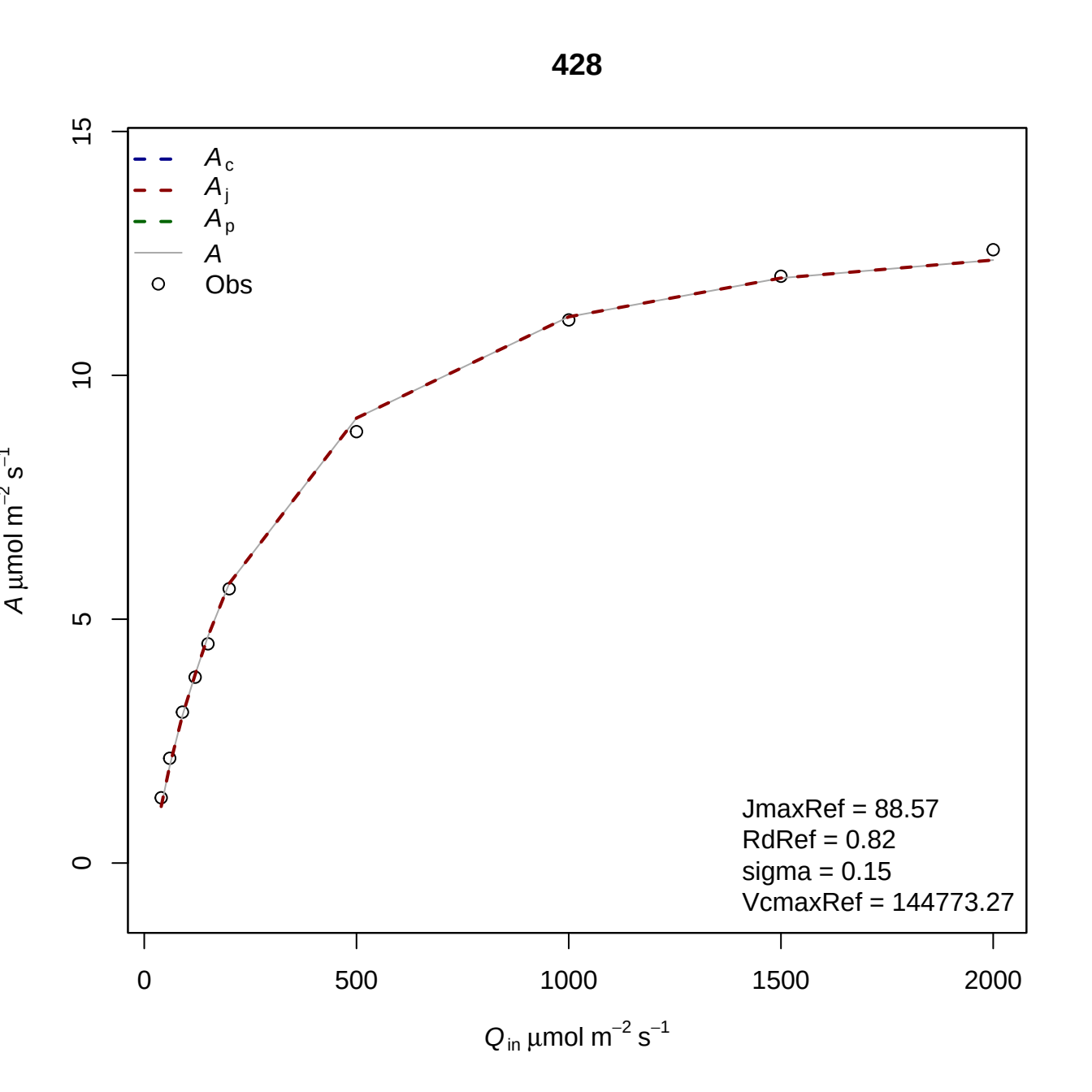

422
A_c
A_j
A_p
A
Obs
JmaxRef = 144.5
RdRef = 0.24
sigma = 0.16
VcmaxRef = 64.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

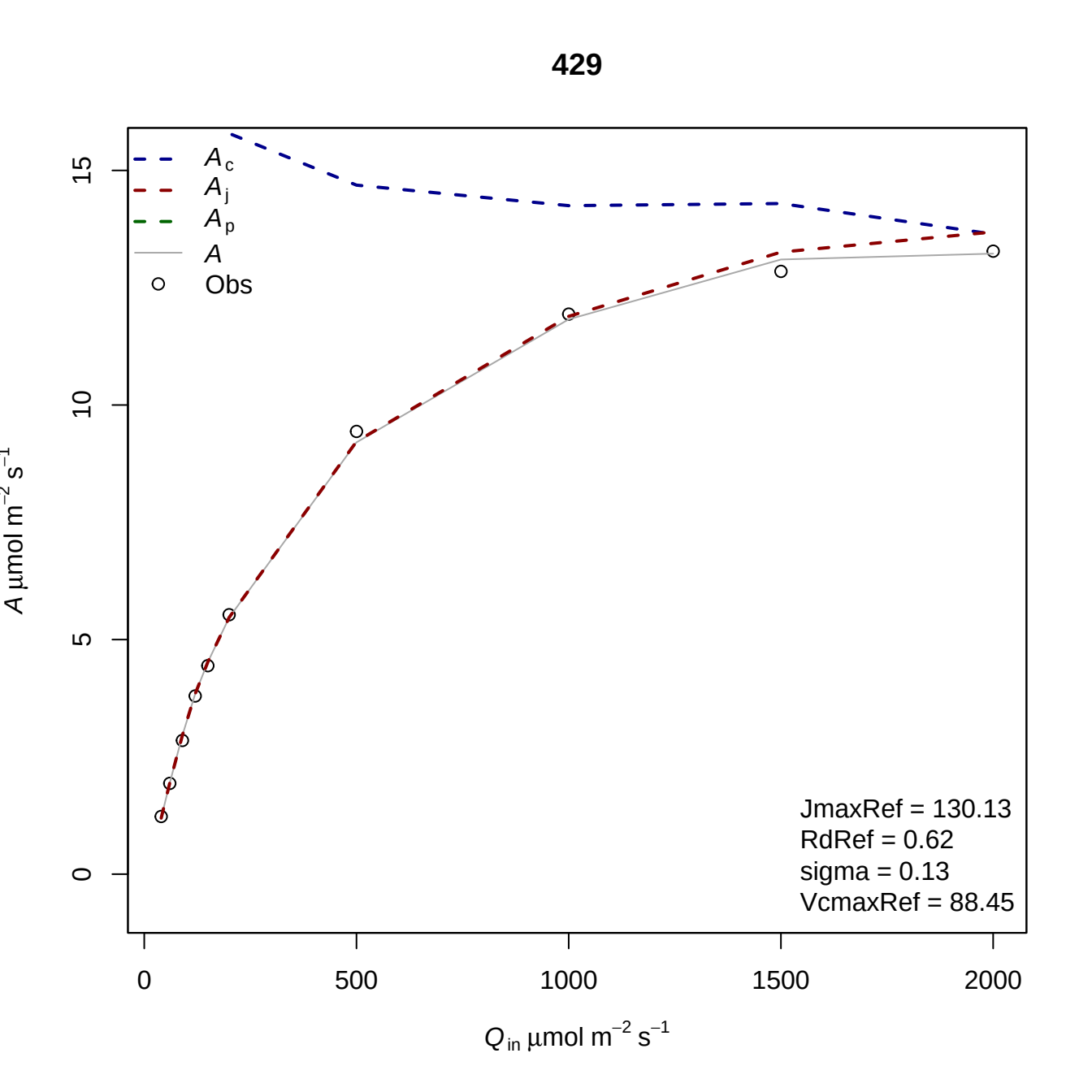

424
A_c
A_j
A_p
A
Obs
JmaxRef = 155.44
RdRef = 1.28
sigma = 0.19
VcmaxRef = 131.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

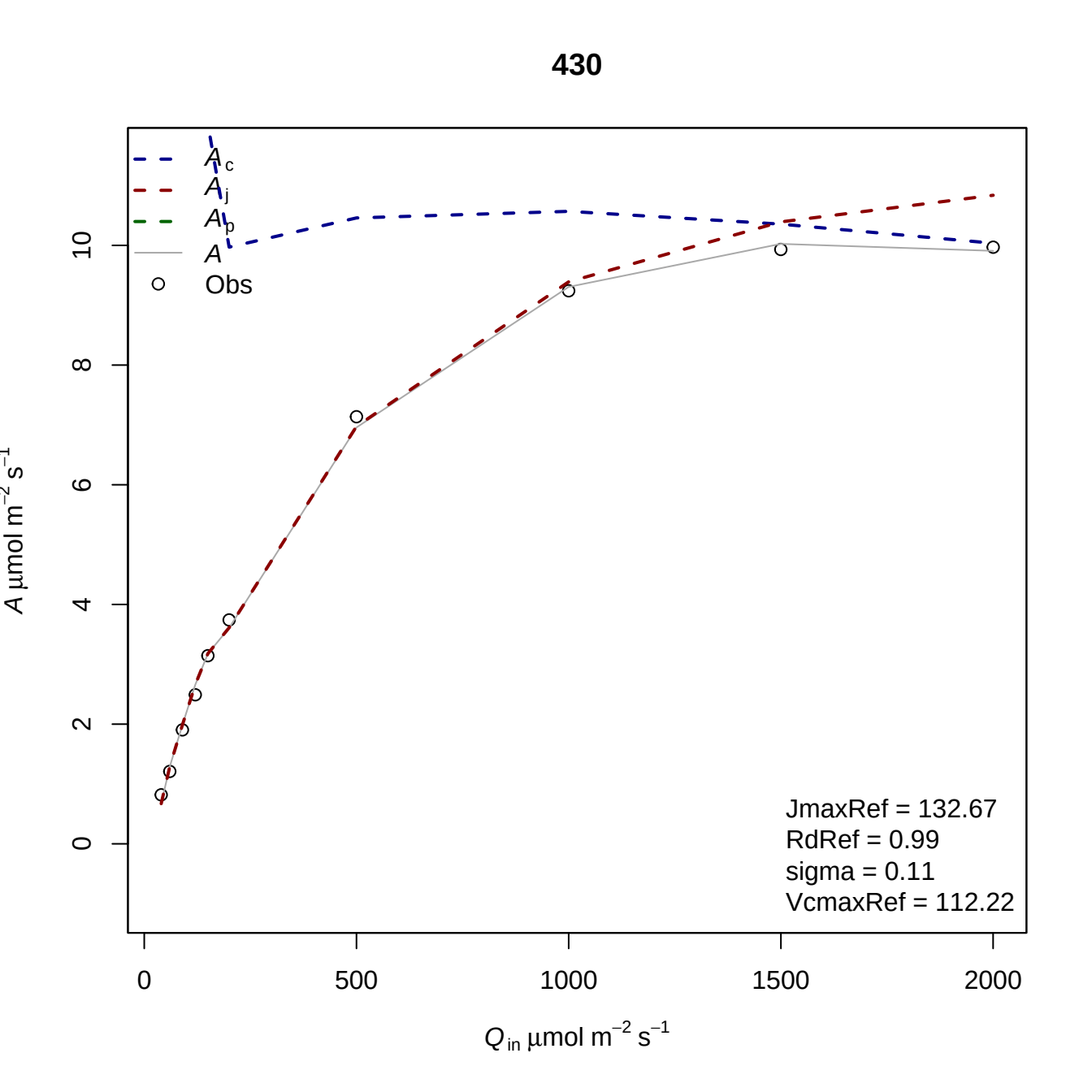

425
A_c
A_j
A_p
A
Obs
JmaxRef = 102.9
RdRef = 2.28
sigma = 0.47
VcmaxRef = 33287.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

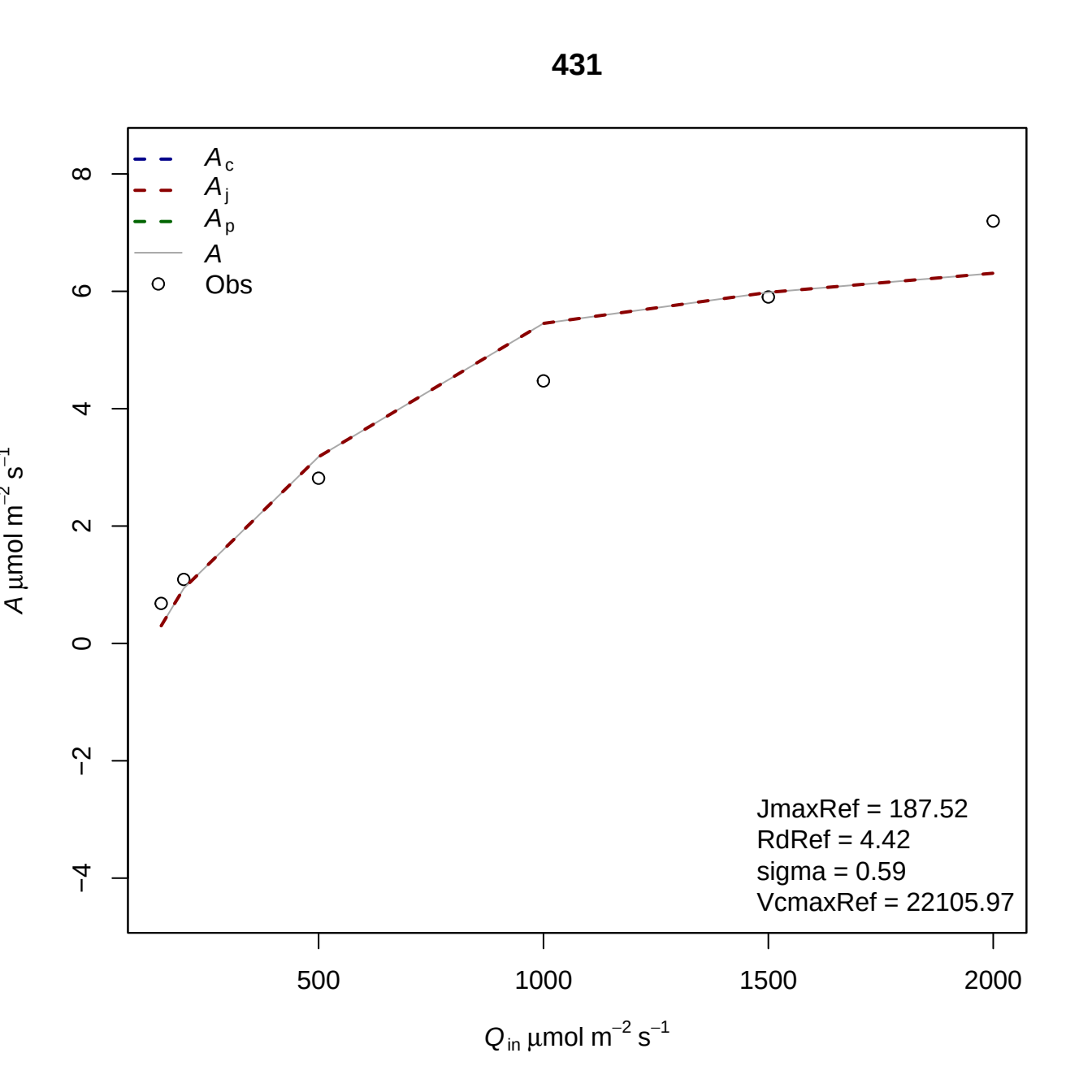

426
A_c
A_j
A_p
A
Obs
JmaxRef = 84.62
RdRef = 2.11
sigma = 0.24
VcmaxRef = 18563743.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

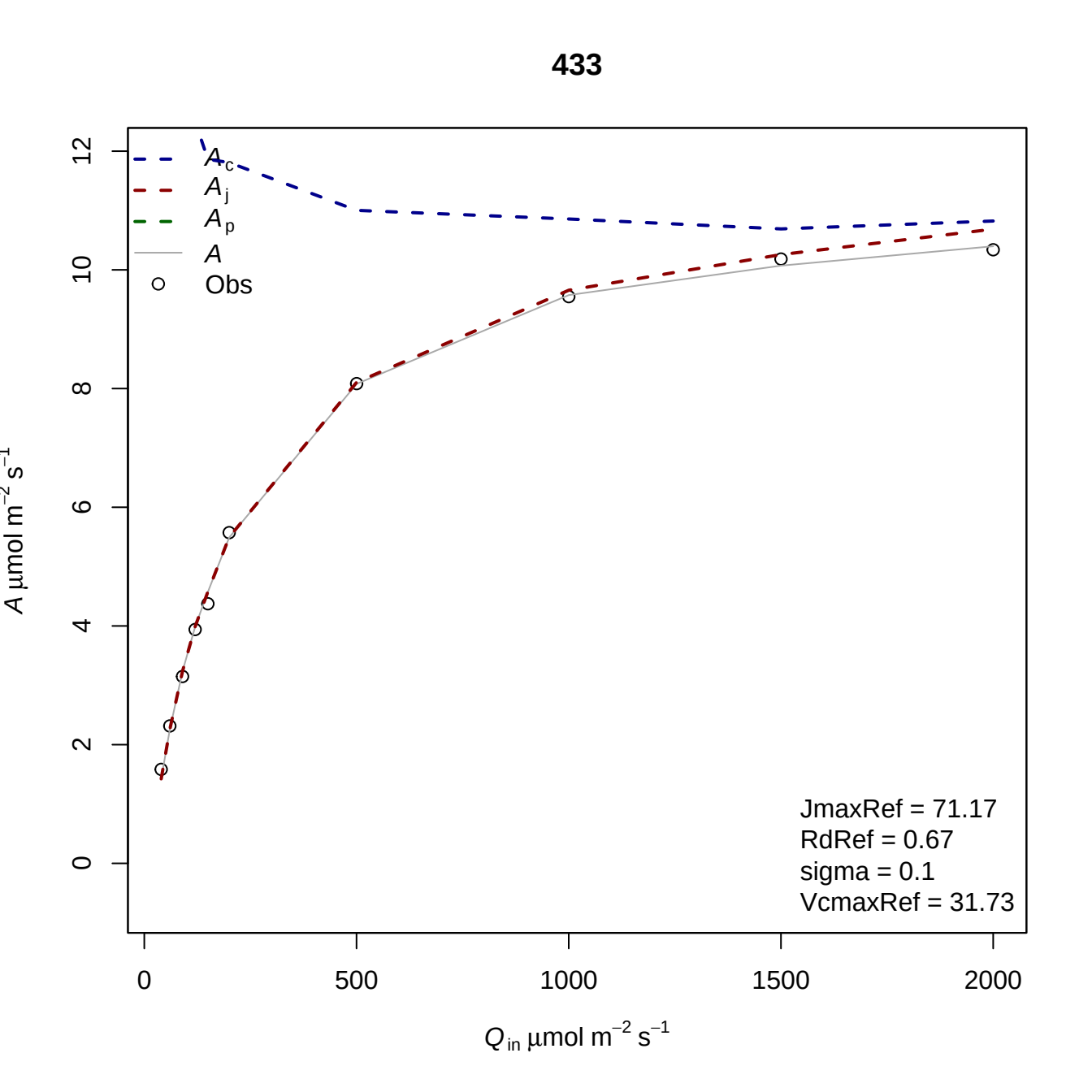

427
A_c
A_j
A_p
A
Obs
JmaxRef = 82.97
RdRef = 0.12
sigma = 0.21
VcmaxRef = 29.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

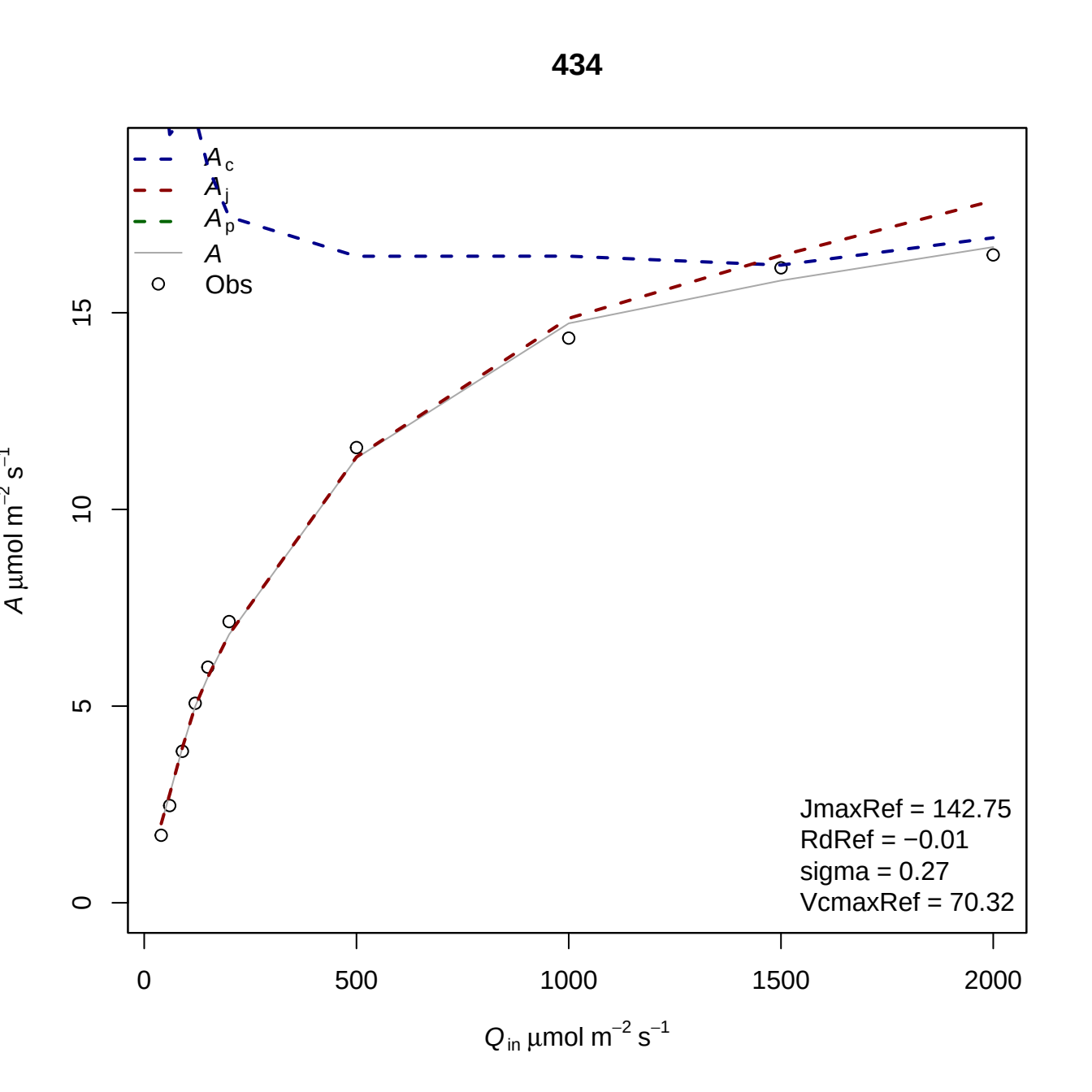

428
A_c
A_j
A_p
A
Obs
JmaxRef = 88.57
RdRef = 0.82
sigma = 0.15
VcmaxRef = 144773.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

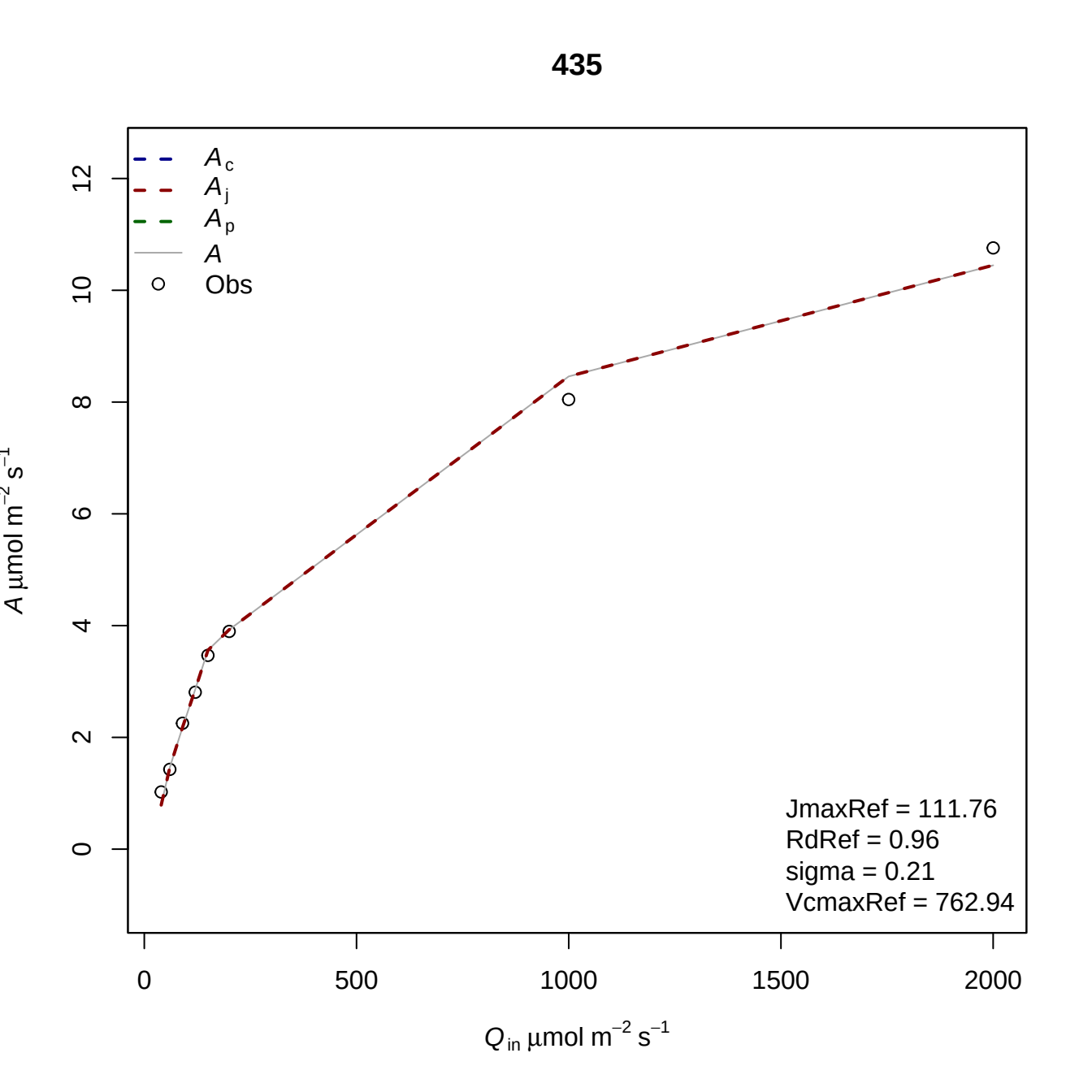

429
A_c
A_j
A_p
A
Obs
JmaxRef = 130.13
RdRef = 0.62
sigma = 0.13
VcmaxRef = 88.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

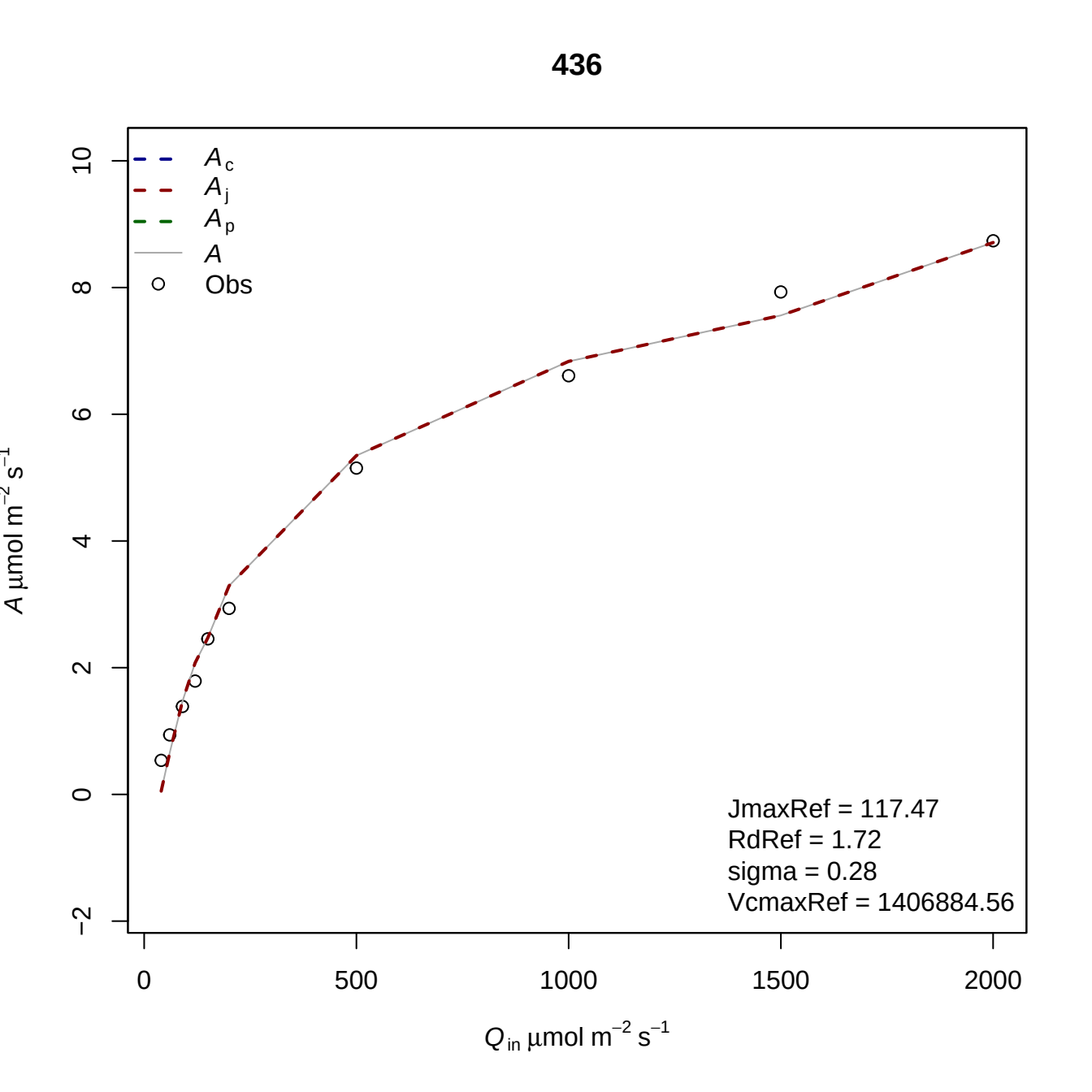

430
A_c
A_j
A_p
A
Obs
JmaxRef = 132.67
RdRef = 0.99
sigma = 0.11
VcmaxRef = 112.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

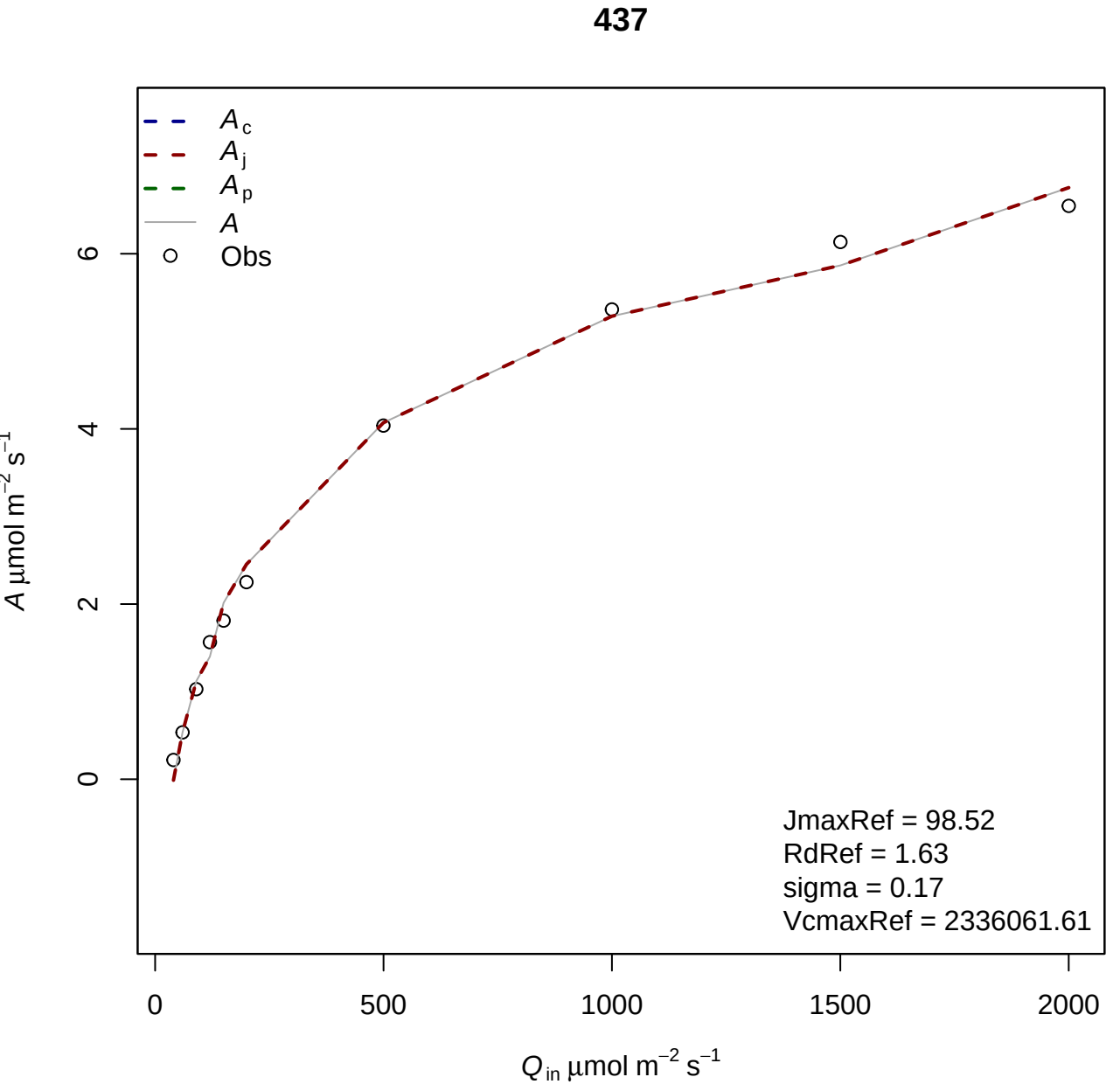

431
A_c
A_j
A_p
A
Obs
JmaxRef = 187.52
RdRef = 4.42
sigma = 0.59
VcmaxRef = 22105.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
500
1000
1500
2000

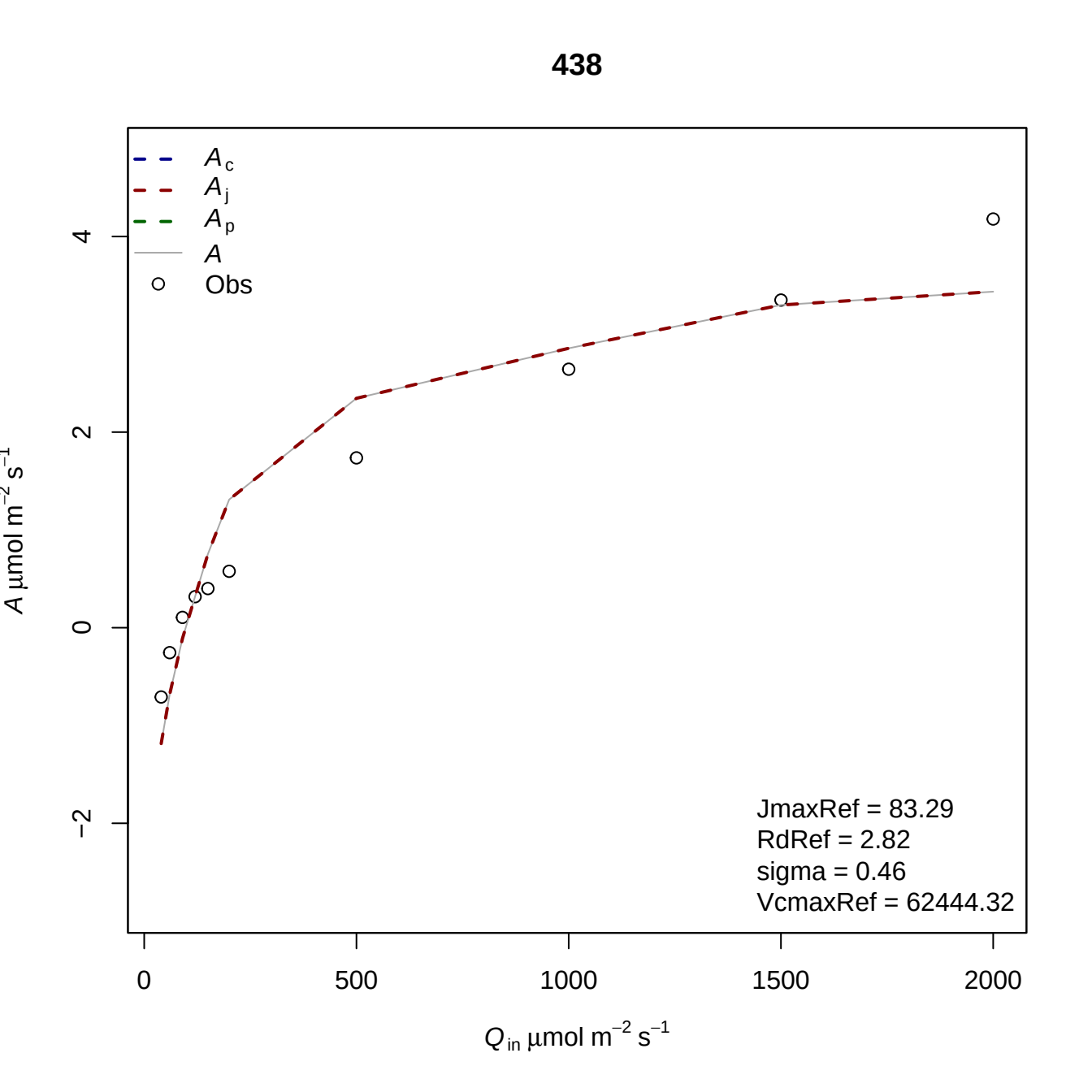

433
A_c
A_j
A_p
A
Obs
JmaxRef = 71.17
RdRef = 0.67
sigma = 0.1
VcmaxRef = 31.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

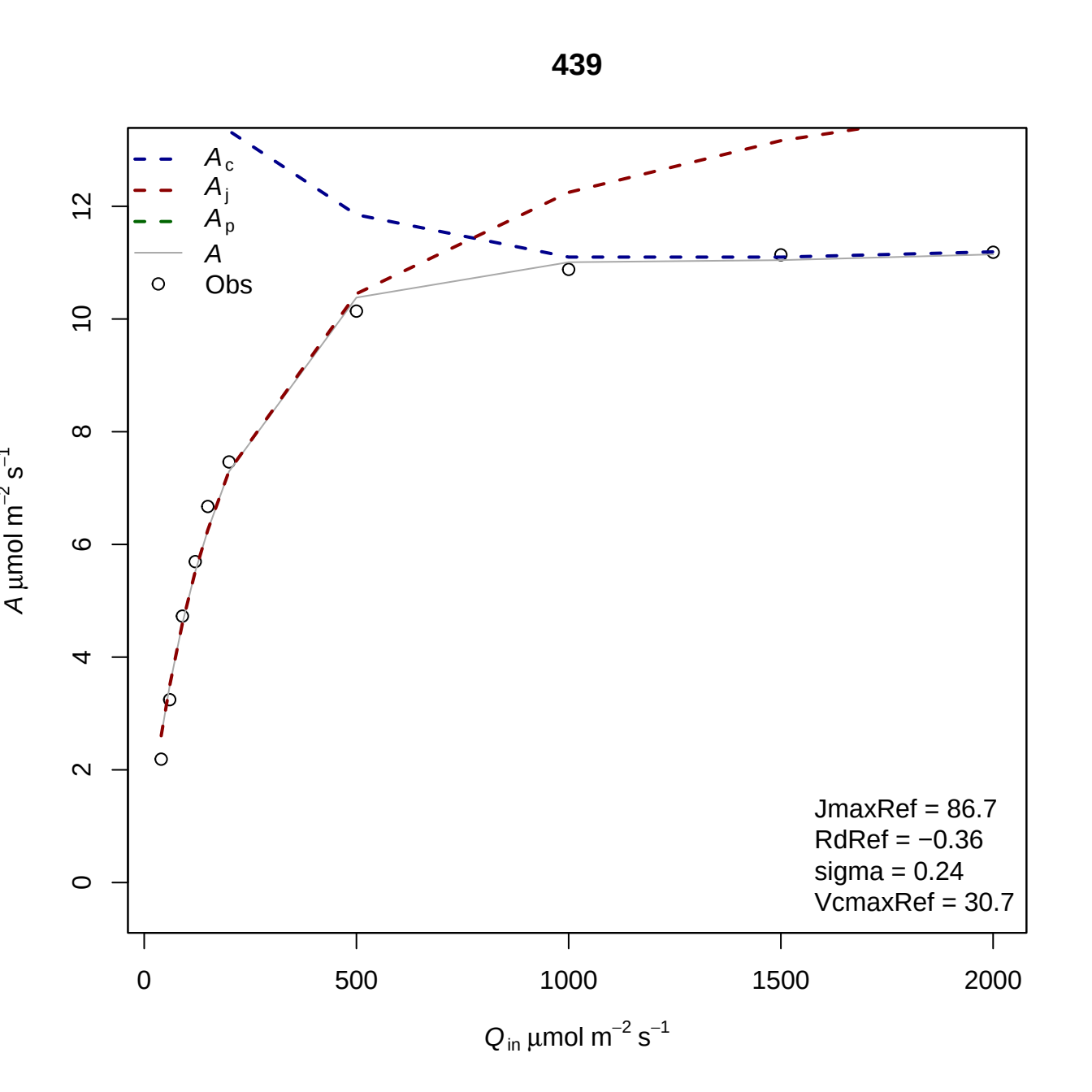

434
A_c
A_j
A_p
A
Obs
JmaxRef = 142.75
RdRef = −0.01
sigma = 0.27
VcmaxRef = 70.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

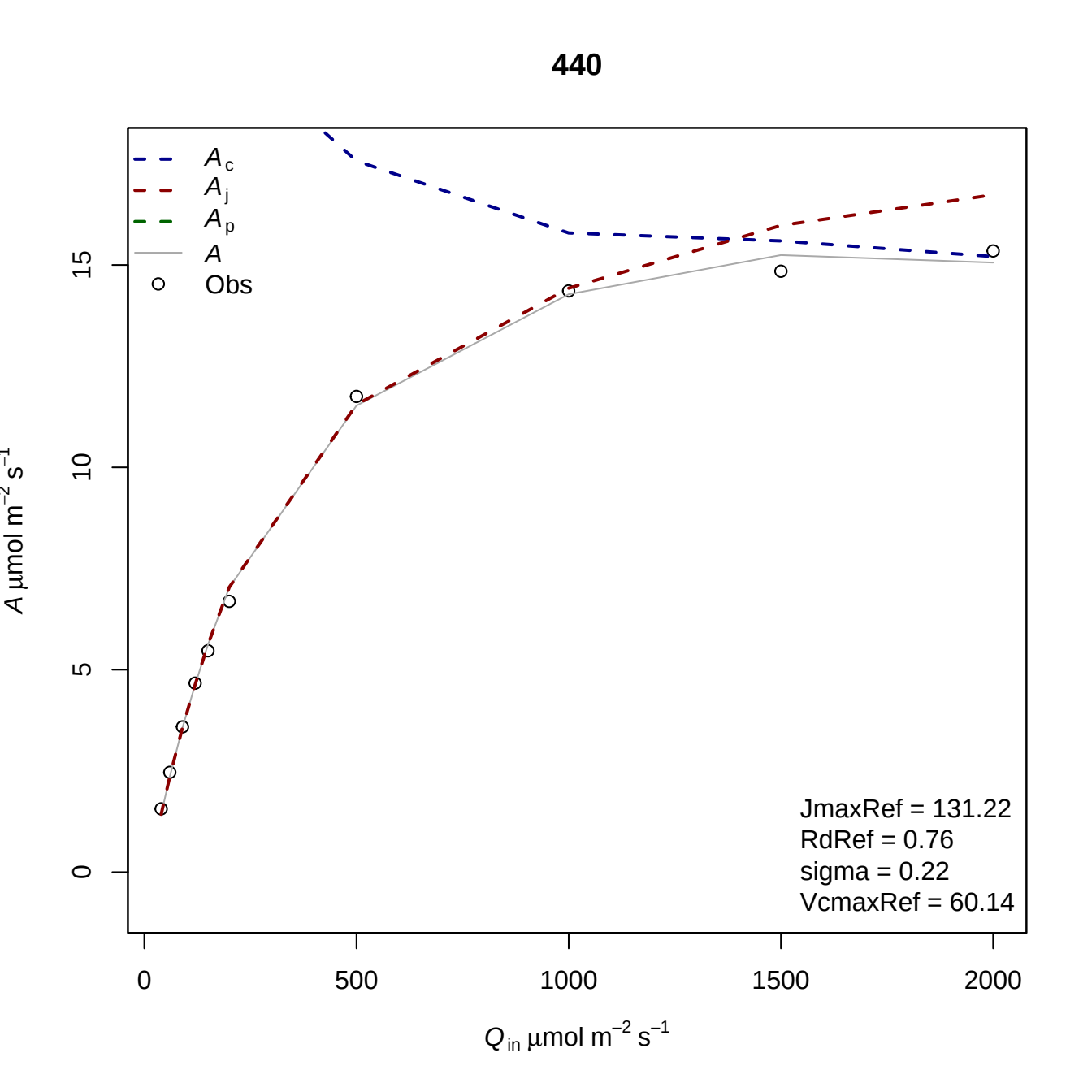

435
A_c
A_j
A_p
A
Obs
JmaxRef = 111.76
RdRef = 0.96
sigma = 0.21
VcmaxRef = 762.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

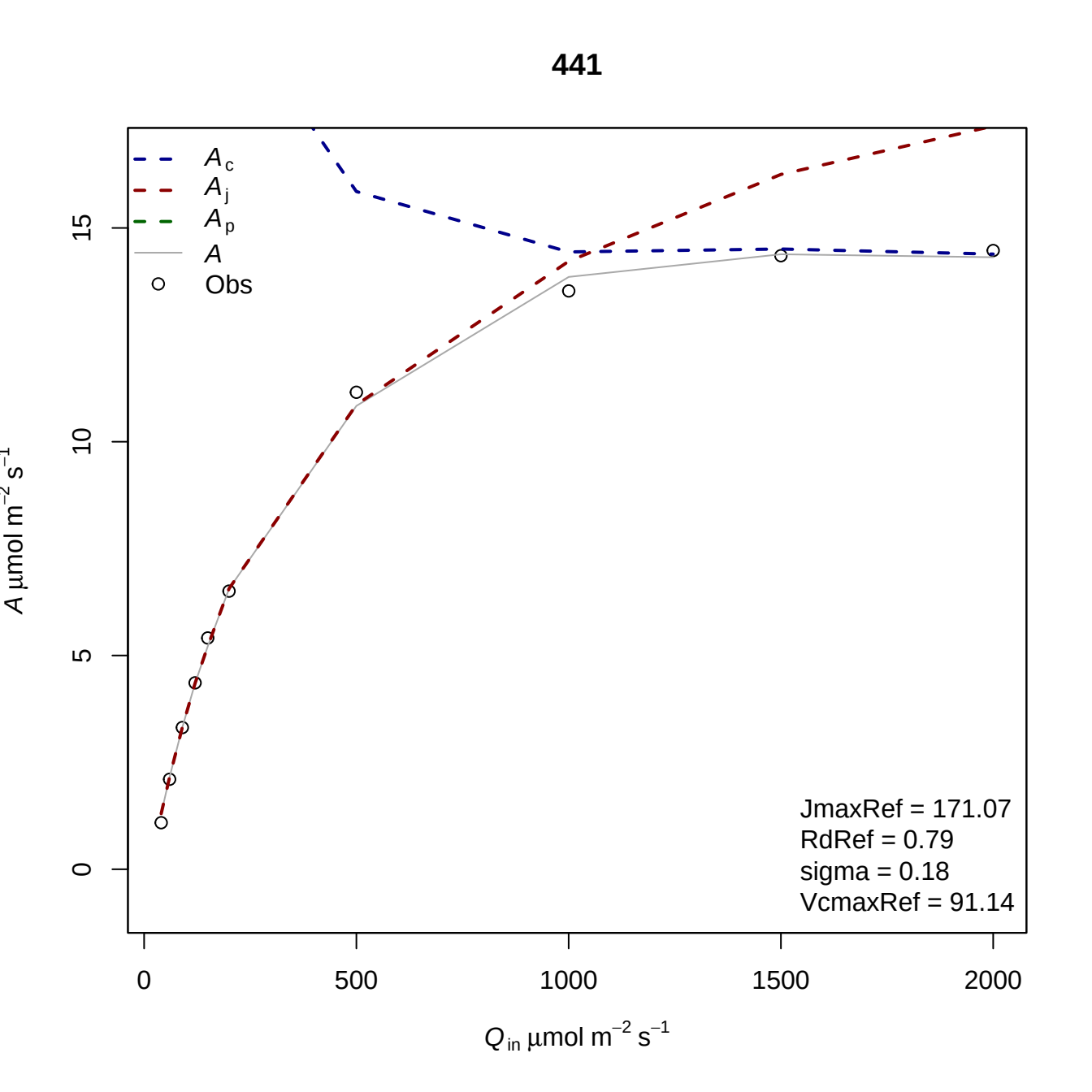

436
A_c
A_j
A_p
A
Obs
JmaxRef = 117.47
RdRef = 1.72
sigma = 0.28
VcmaxRef = 1406884.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

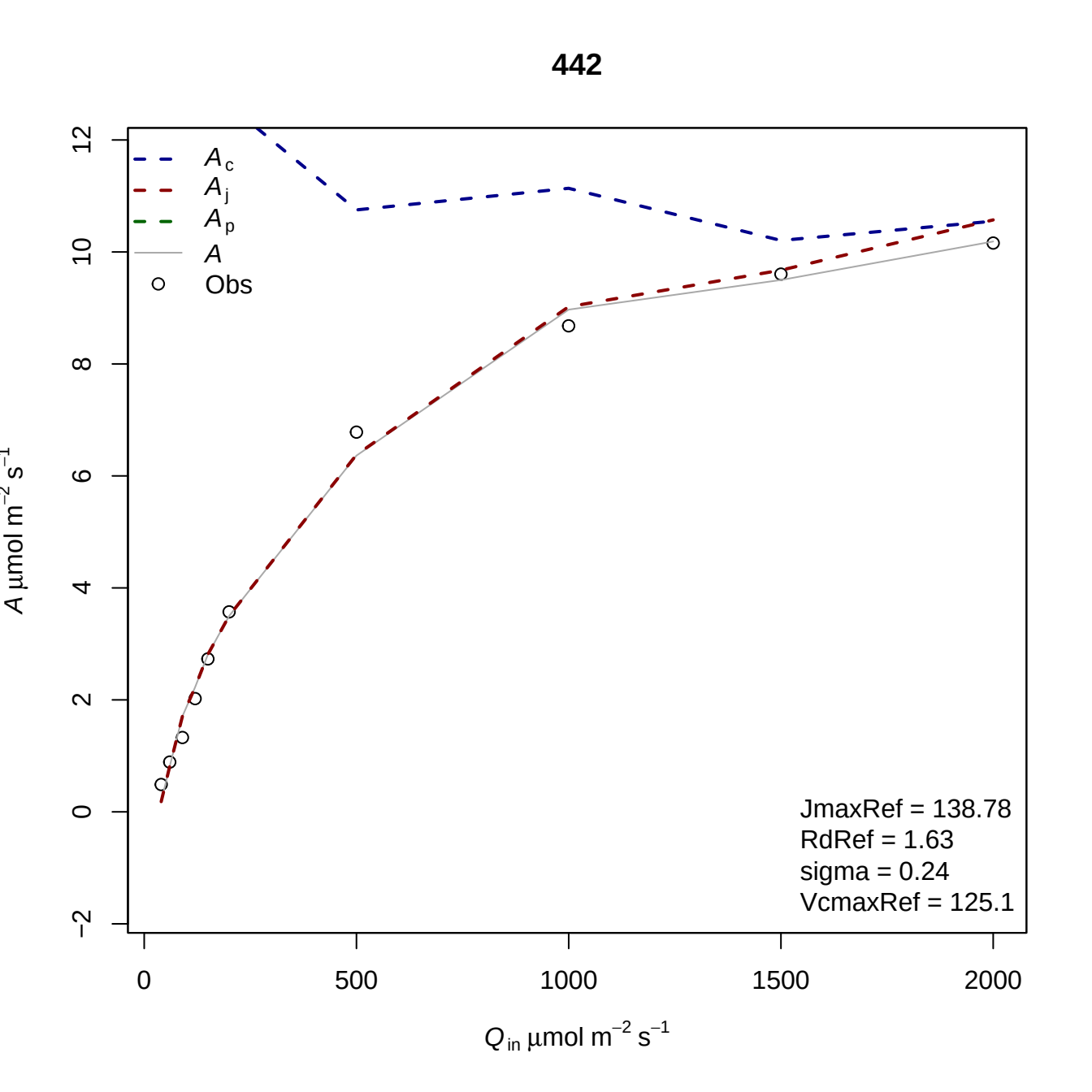

437
A_c
A_j
A_p
A
Obs
A µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 98.52
RdRef = 1.63
sigma = 0.17
VcmaxRef = 2336061.61

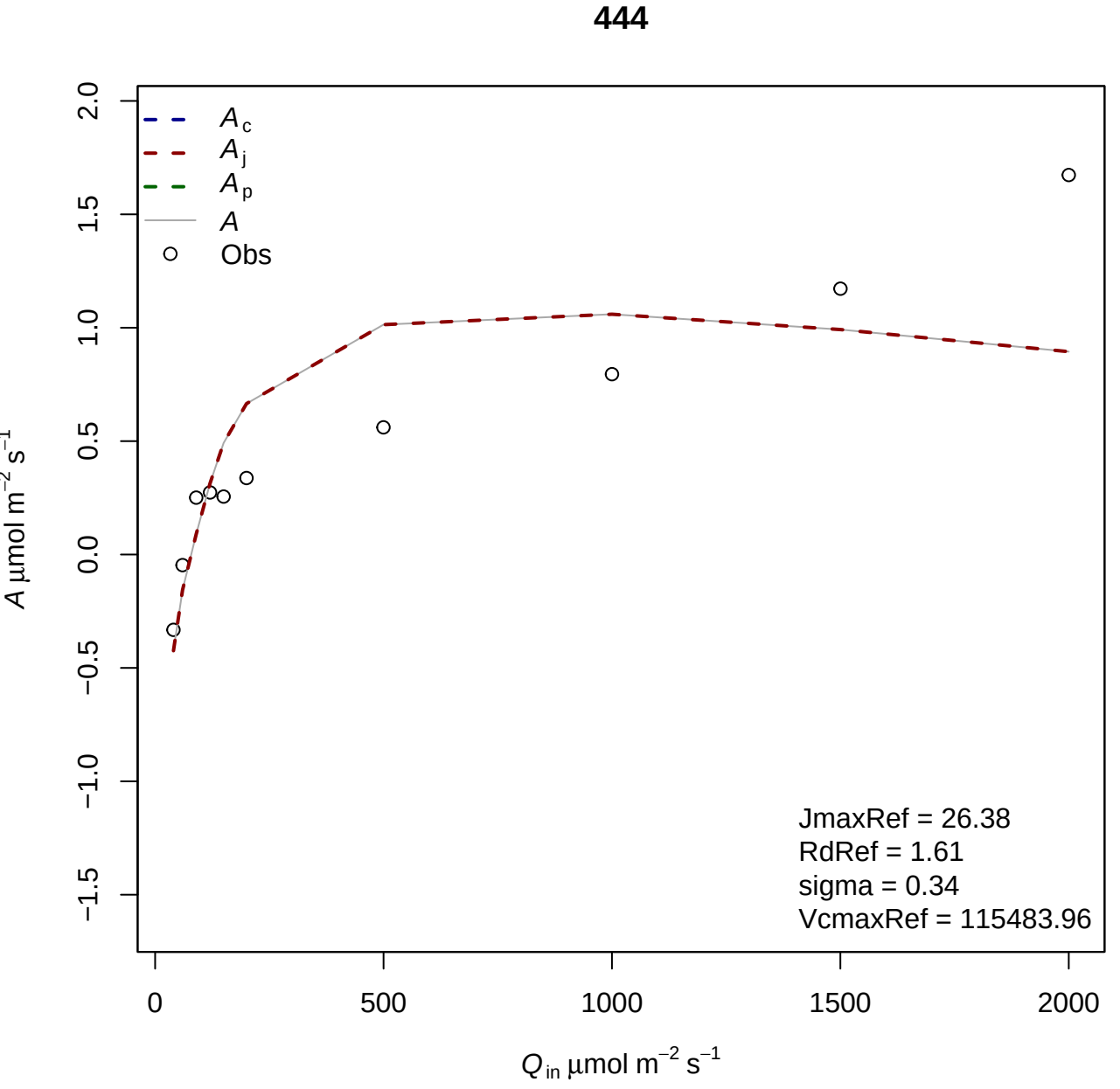

438
A_c
A_j
A_p
A
Obs
JmaxRef = 83.29
RdRef = 2.82
sigma = 0.46
VcmaxRef = 62444.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

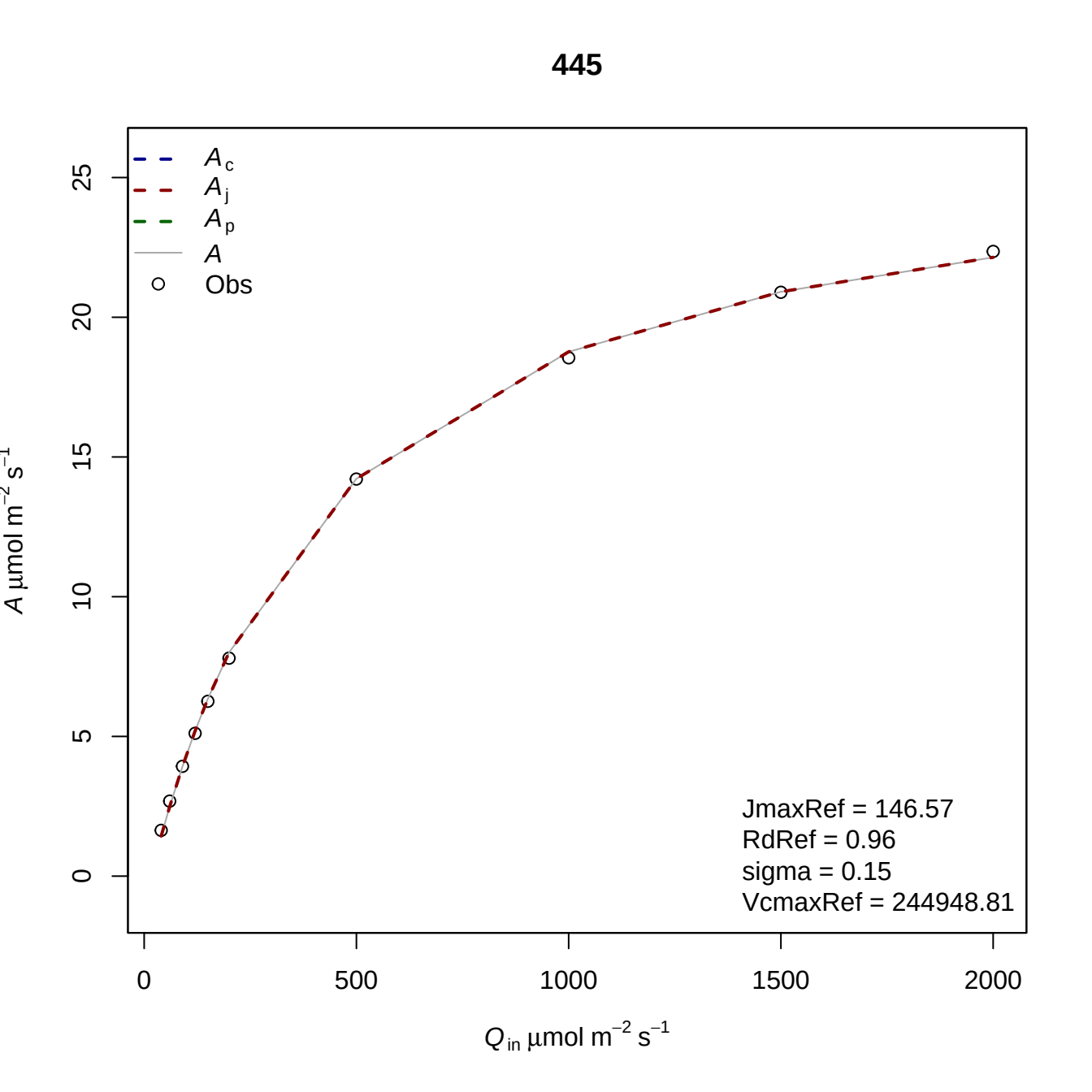

439
A_c
A_j
A_p
A
Obs
JmaxRef = 86.7
RdRef = −0.36
sigma = 0.24
VcmaxRef = 30.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

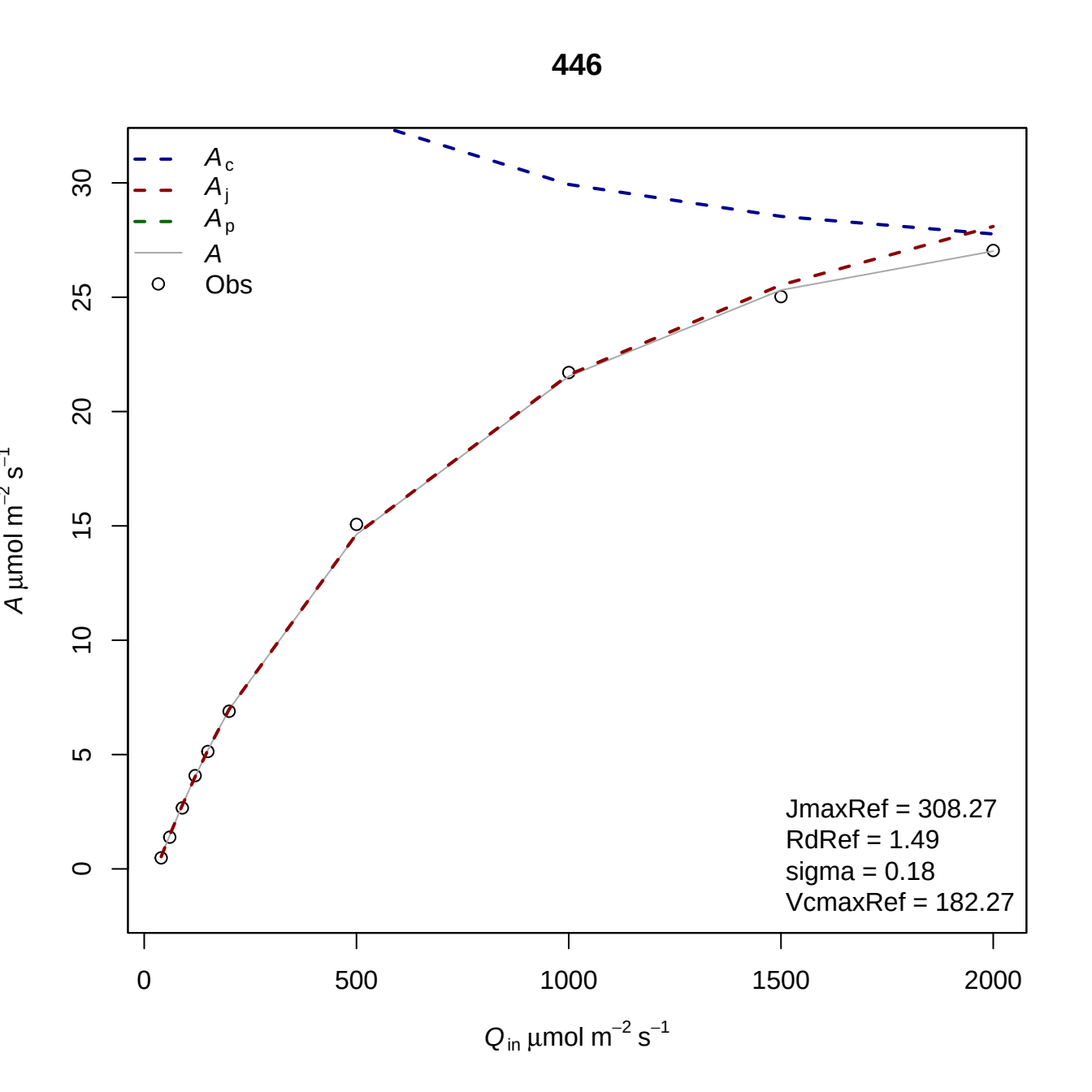

440
A_c
A_j
A_p
A
Obs
JmaxRef = 131.22
RdRef = 0.76
sigma = 0.22
VcmaxRef = 60.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

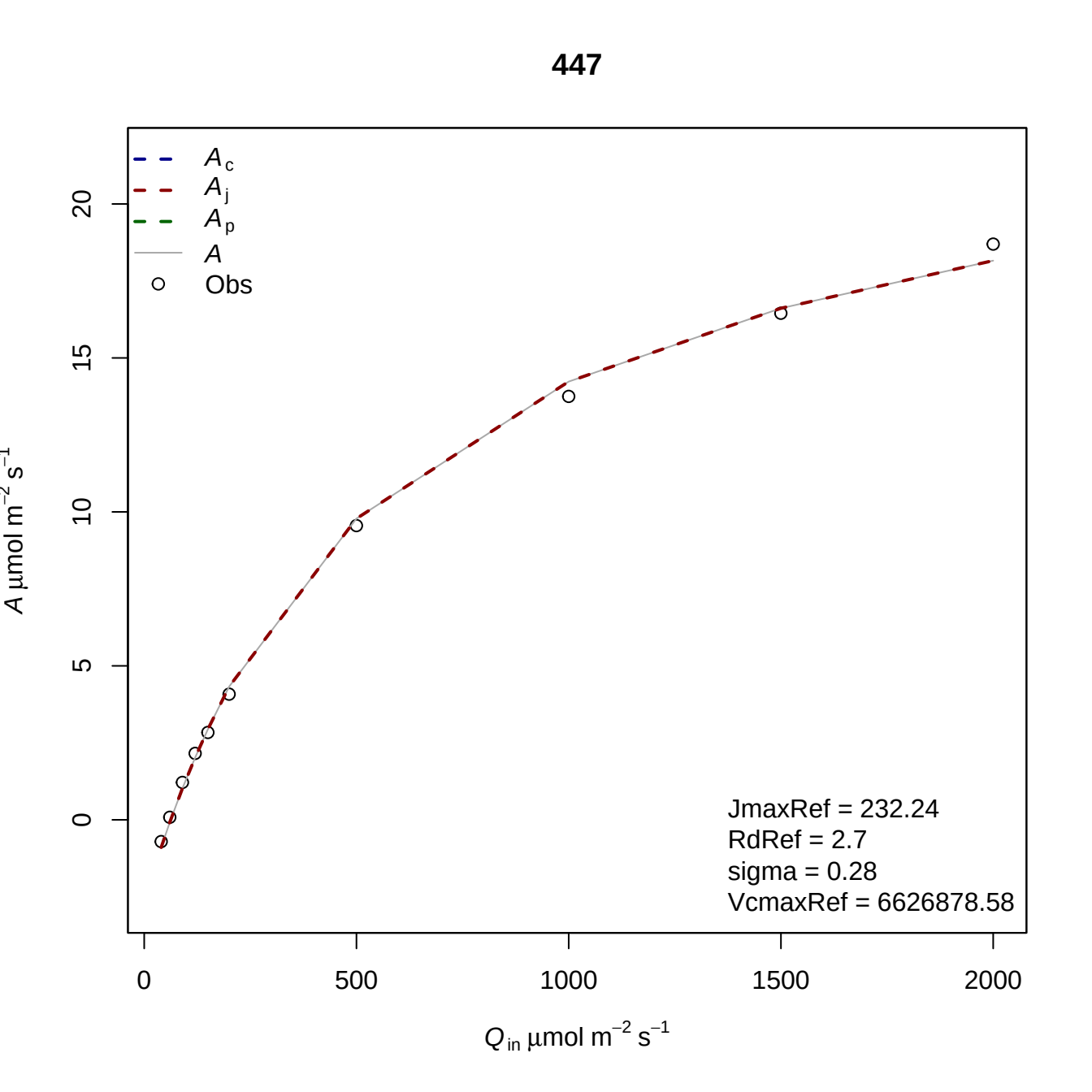

441
A_c
A_j
A_p
A
Obs
JmaxRef = 171.07
RdRef = 0.79
sigma = 0.18
VcmaxRef = 91.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

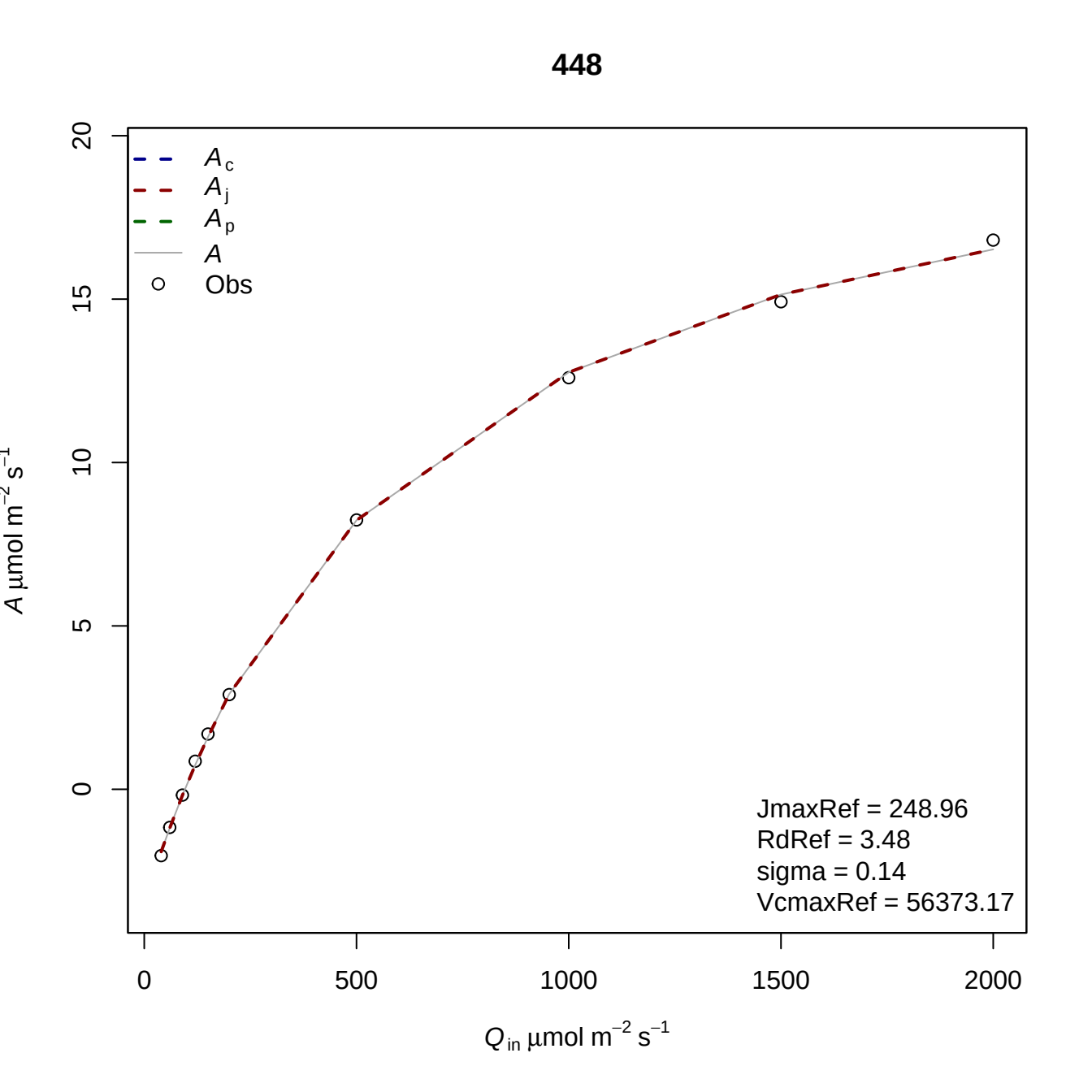

442
A_c
A_j
A_p
A
Obs
JmaxRef = 138.78
RdRef = 1.63
sigma = 0.24
VcmaxRef = 125.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

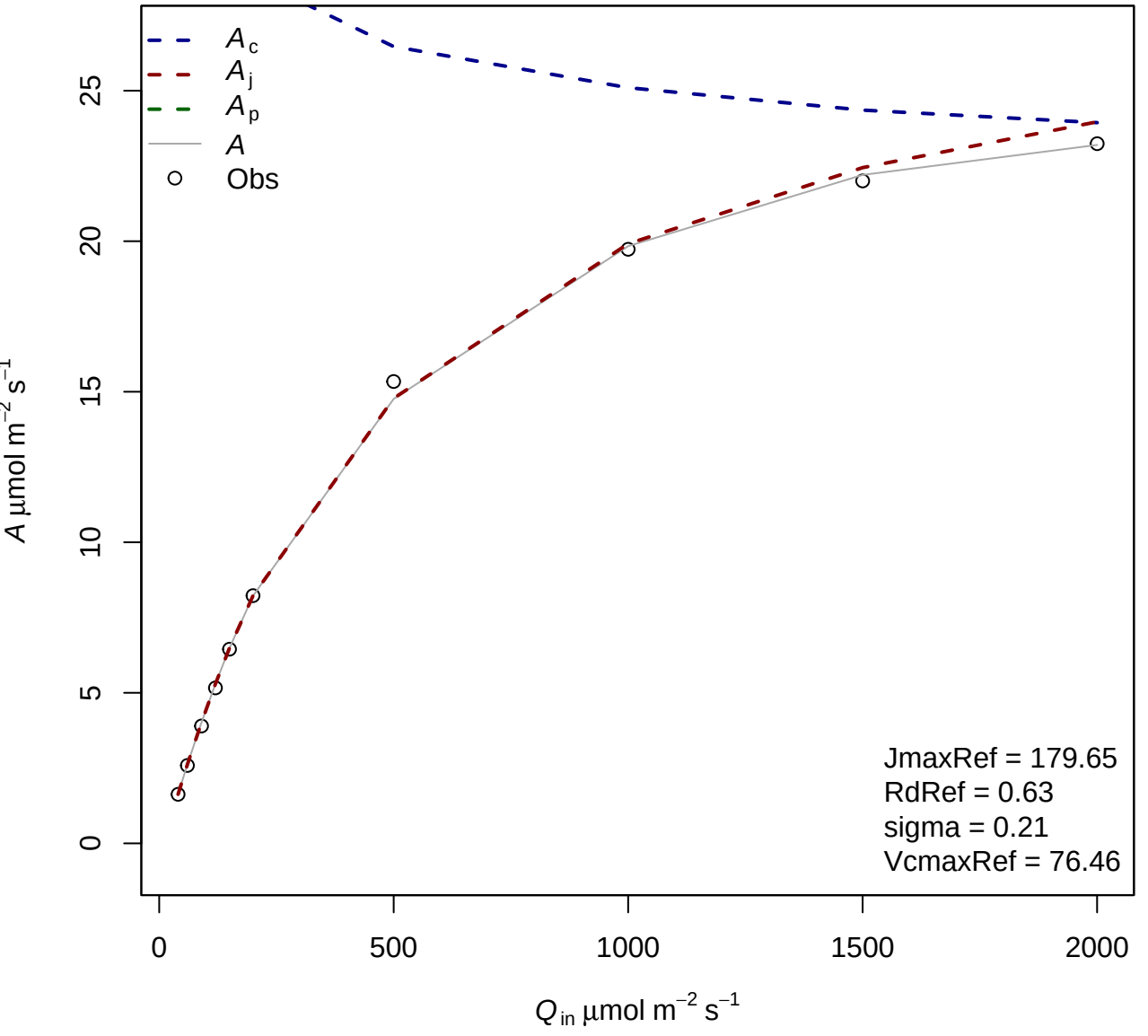

444
A_c
A_j
A_p
A
Obs
JmaxRef = 26.38
RdRef = 1.61
sigma = 0.34
VcmaxRef = 115483.96
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

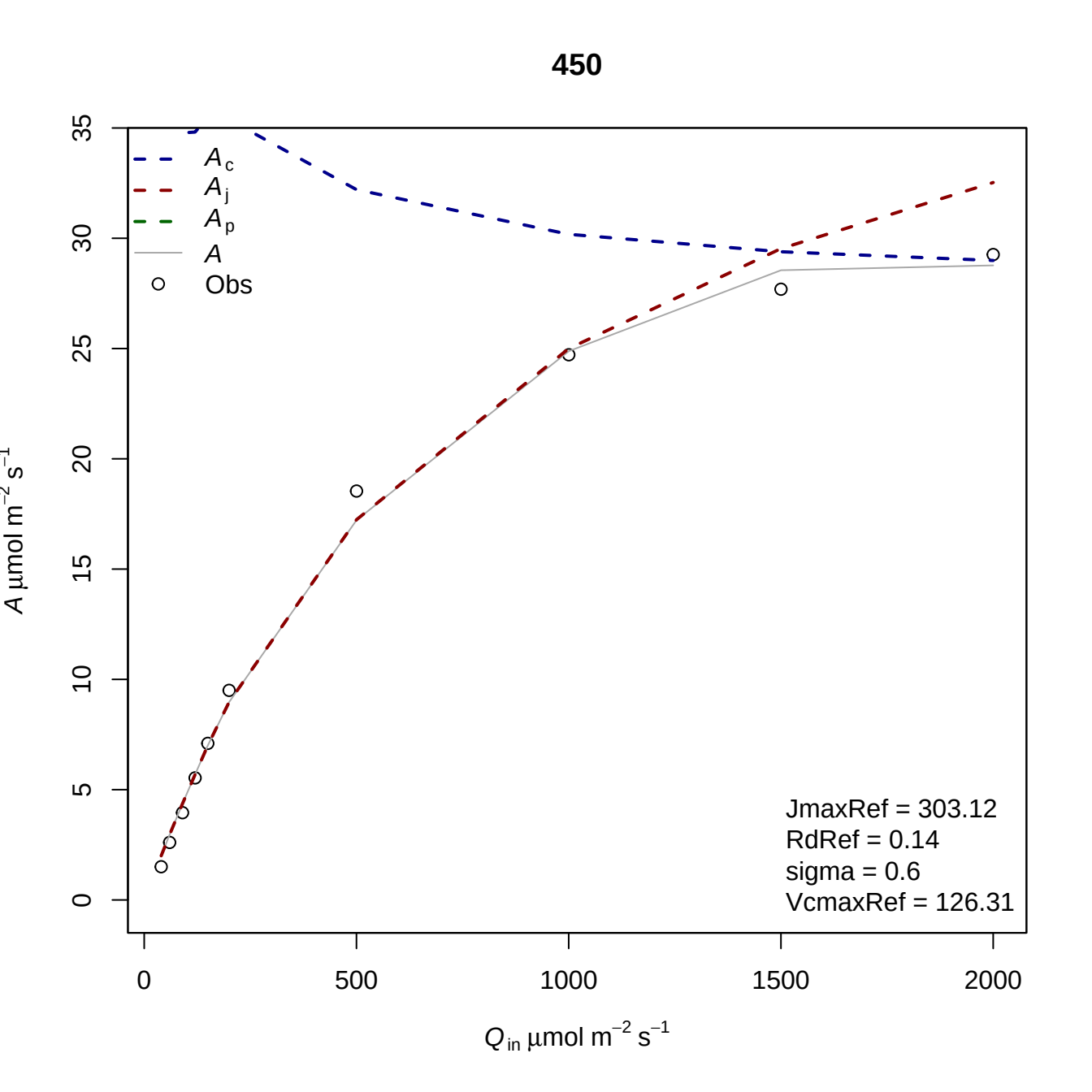

445
A_c
A_j
A_p
A
Obs
JmaxRef = 146.57
RdRef = 0.96
sigma = 0.15
VcmaxRef = 244948.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

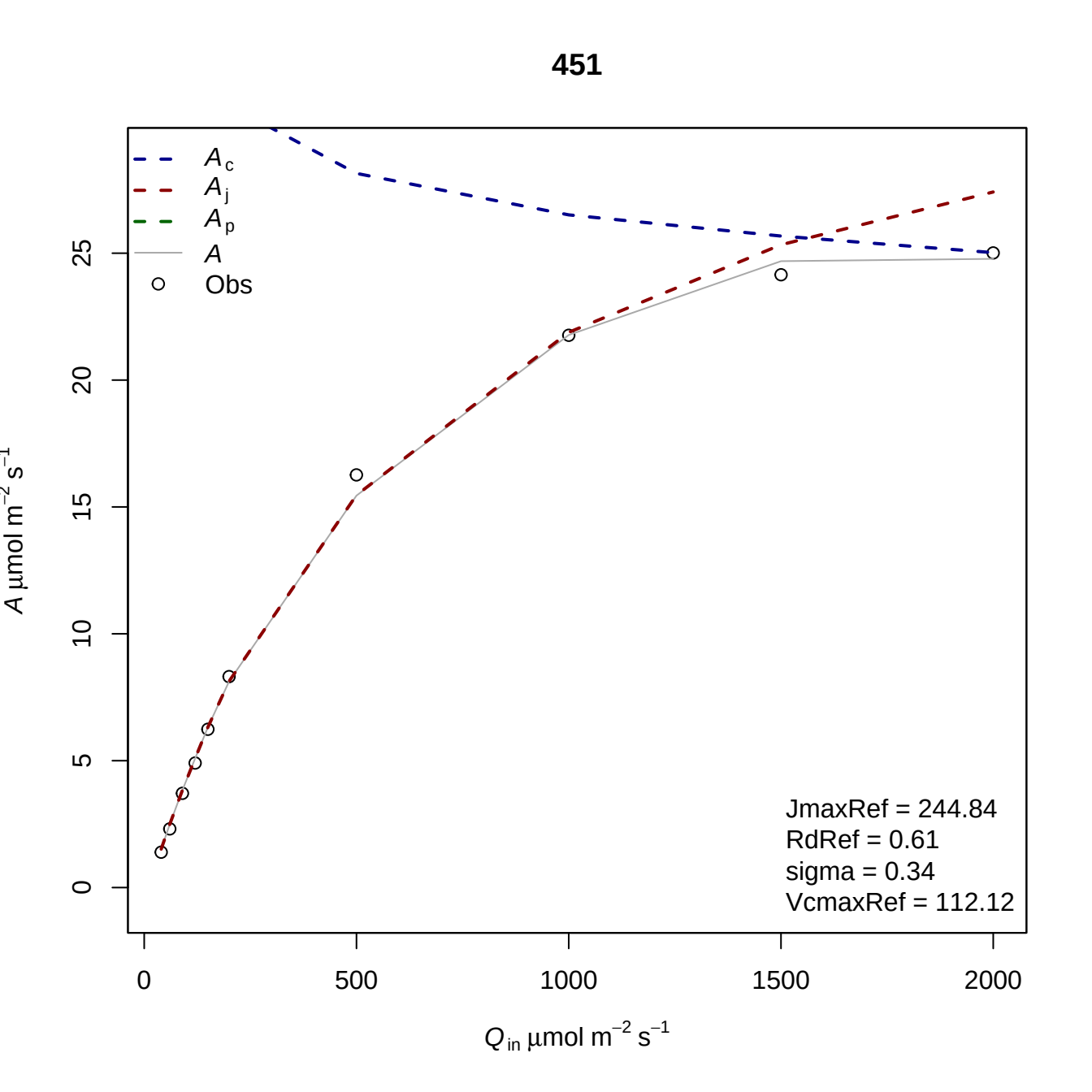

446
A_c
A_j
A_p
A
Obs
JmaxRef = 308.27
RdRef = 1.49
sigma = 0.18
VcmaxRef = 182.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

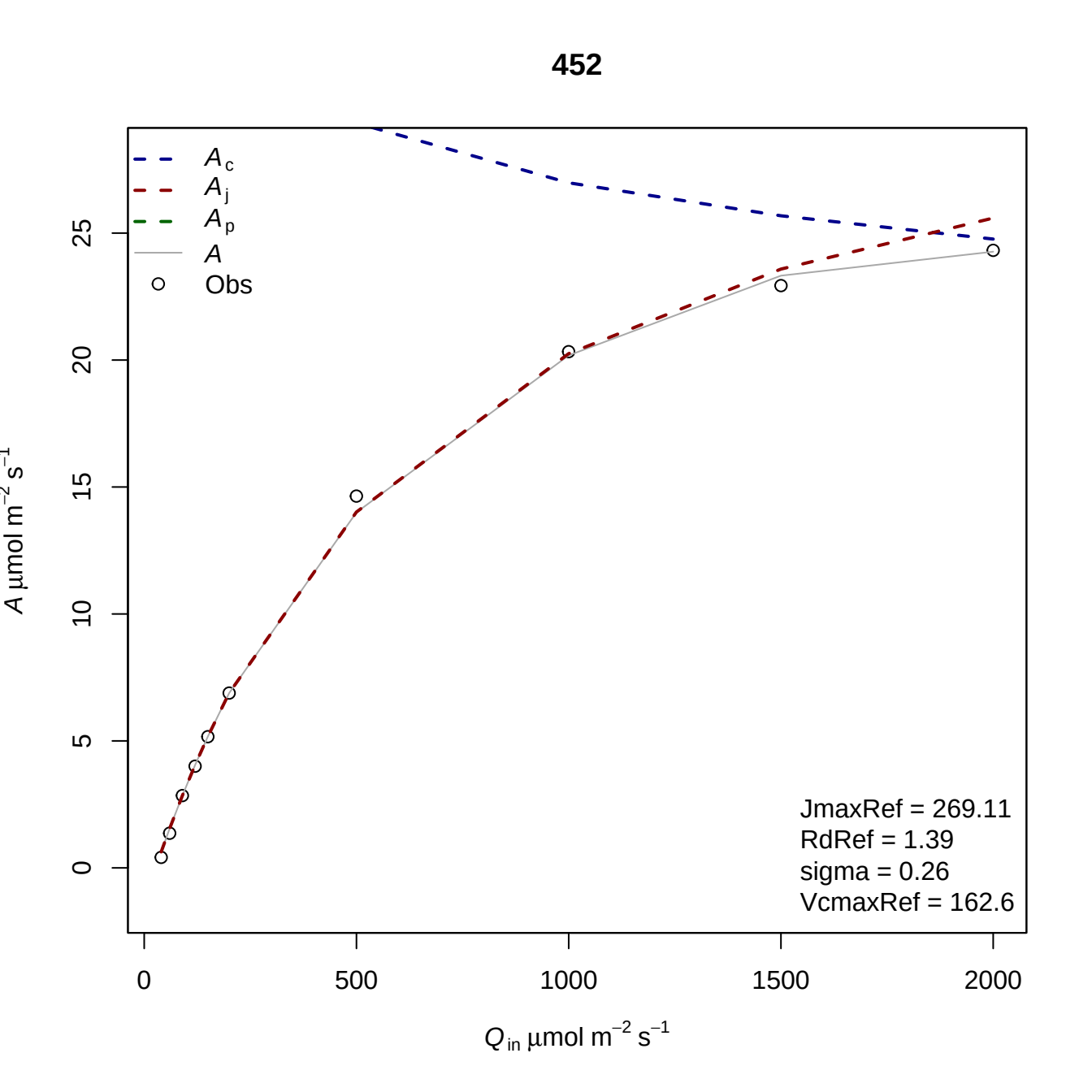

447
A_c
A_j
A_p
A
Obs
JmaxRef = 232.24
RdRef = 2.7
sigma = 0.28
VcmaxRef = 6626878.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

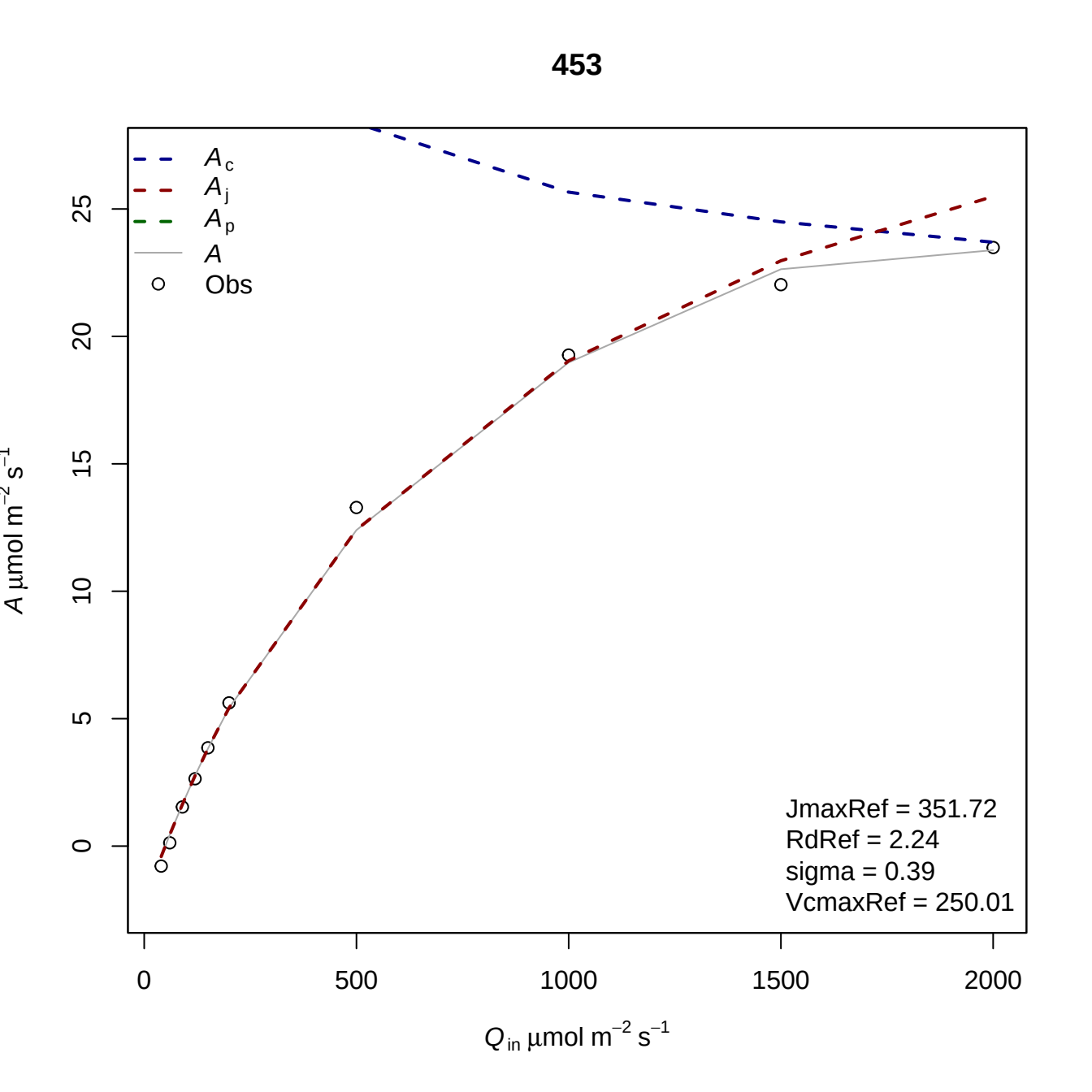

448
A_c
A_j
A_p
A
Obs
JmaxRef = 248.96
RdRef = 3.48
sigma = 0.14
VcmaxRef = 56373.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

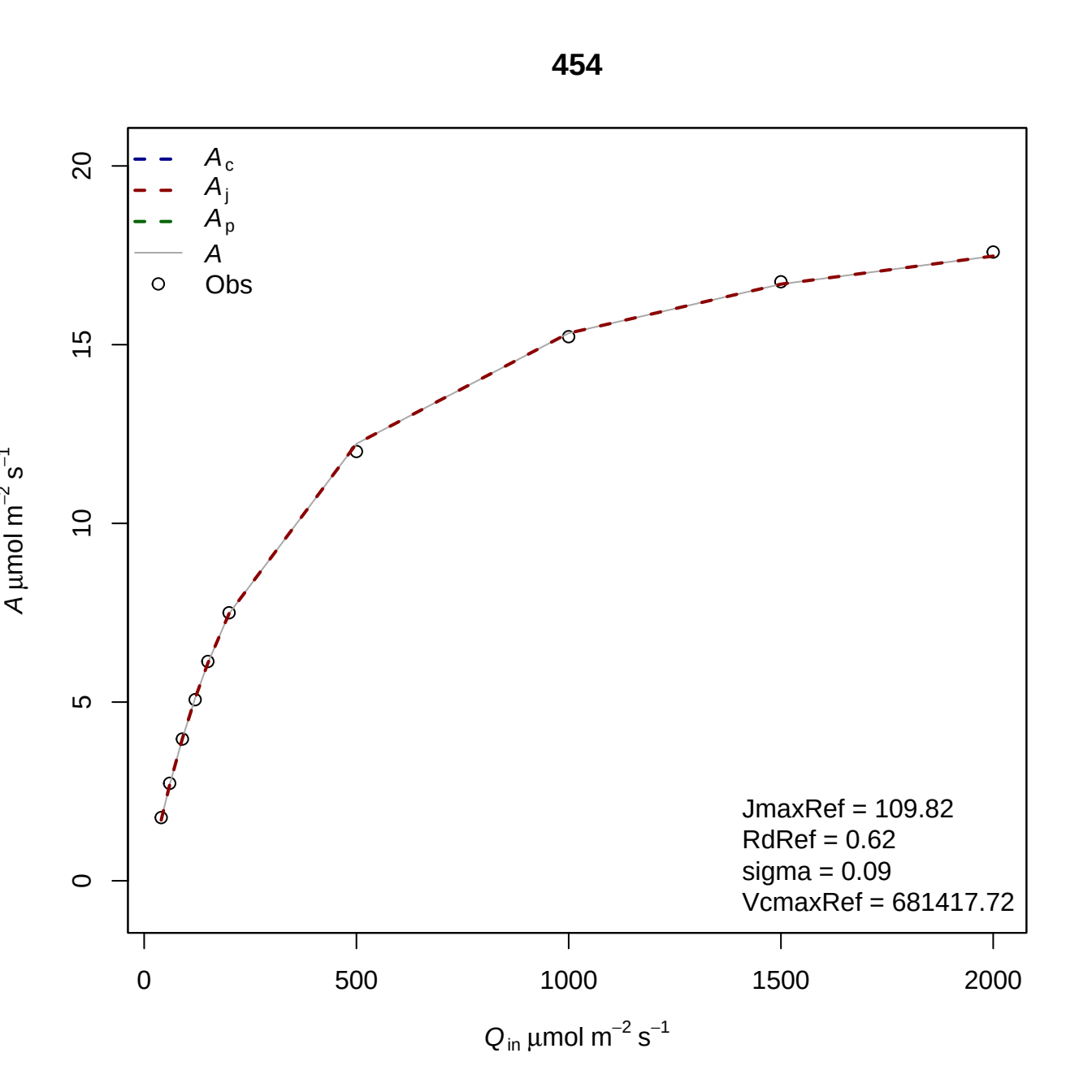

A_c
A_j
A_p
A
Obs
JmaxRef = 179.65
RdRef = 0.63
sigma = 0.21
VcmaxRef = 76.46
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

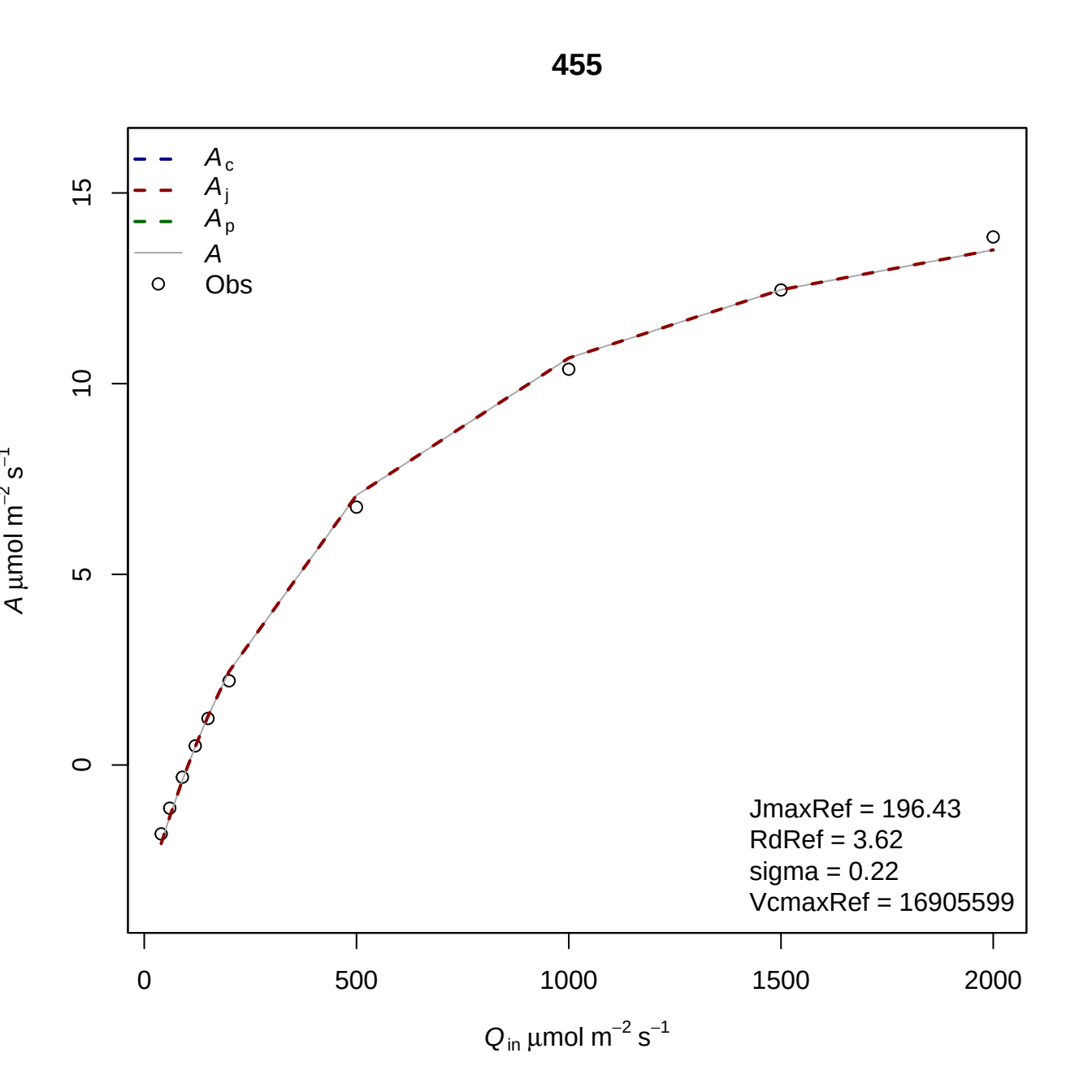

450
A_c
A_j
A_p
A
Obs
JmaxRef = 303.12
RdRef = 0.14
sigma = 0.6
VcmaxRef = 126.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

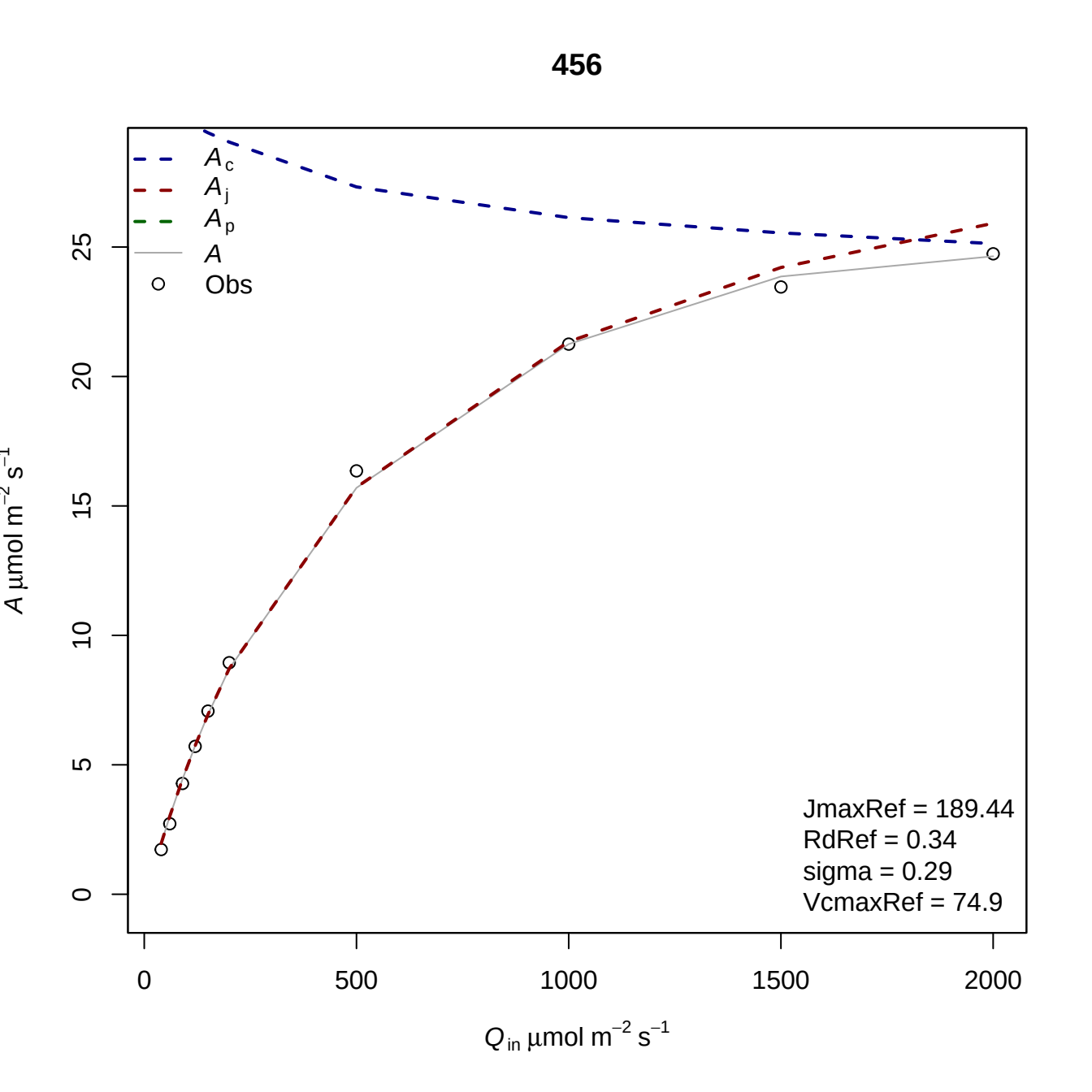

451
A_c
A_j
A_p
A
Obs
JmaxRef = 244.84
RdRef = 0.61
sigma = 0.34
VcmaxRef = 112.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

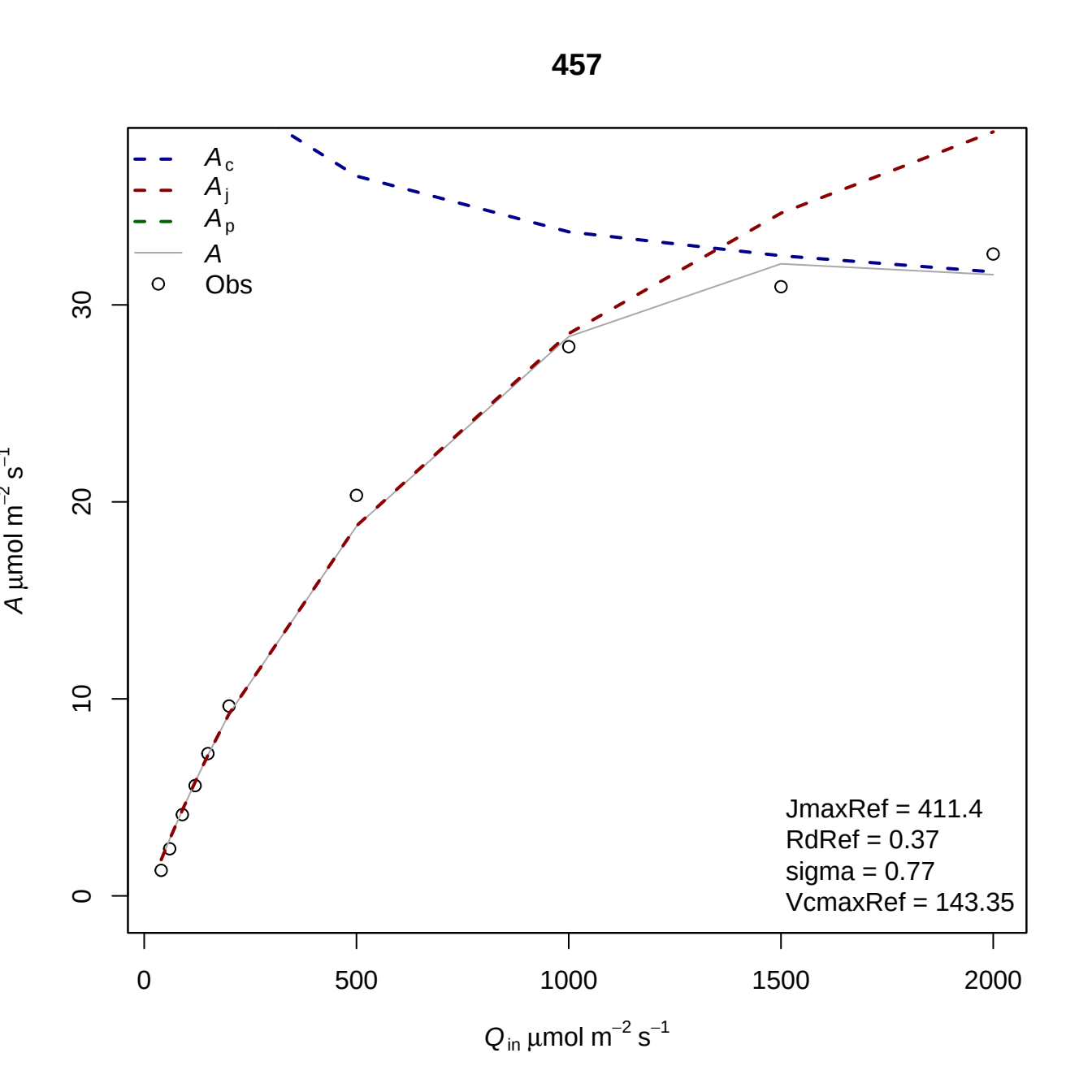

452
A_c
A_j
A_p
A
Obs
JmaxRef = 269.11
RdRef = 1.39
sigma = 0.26
VcmaxRef = 162.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

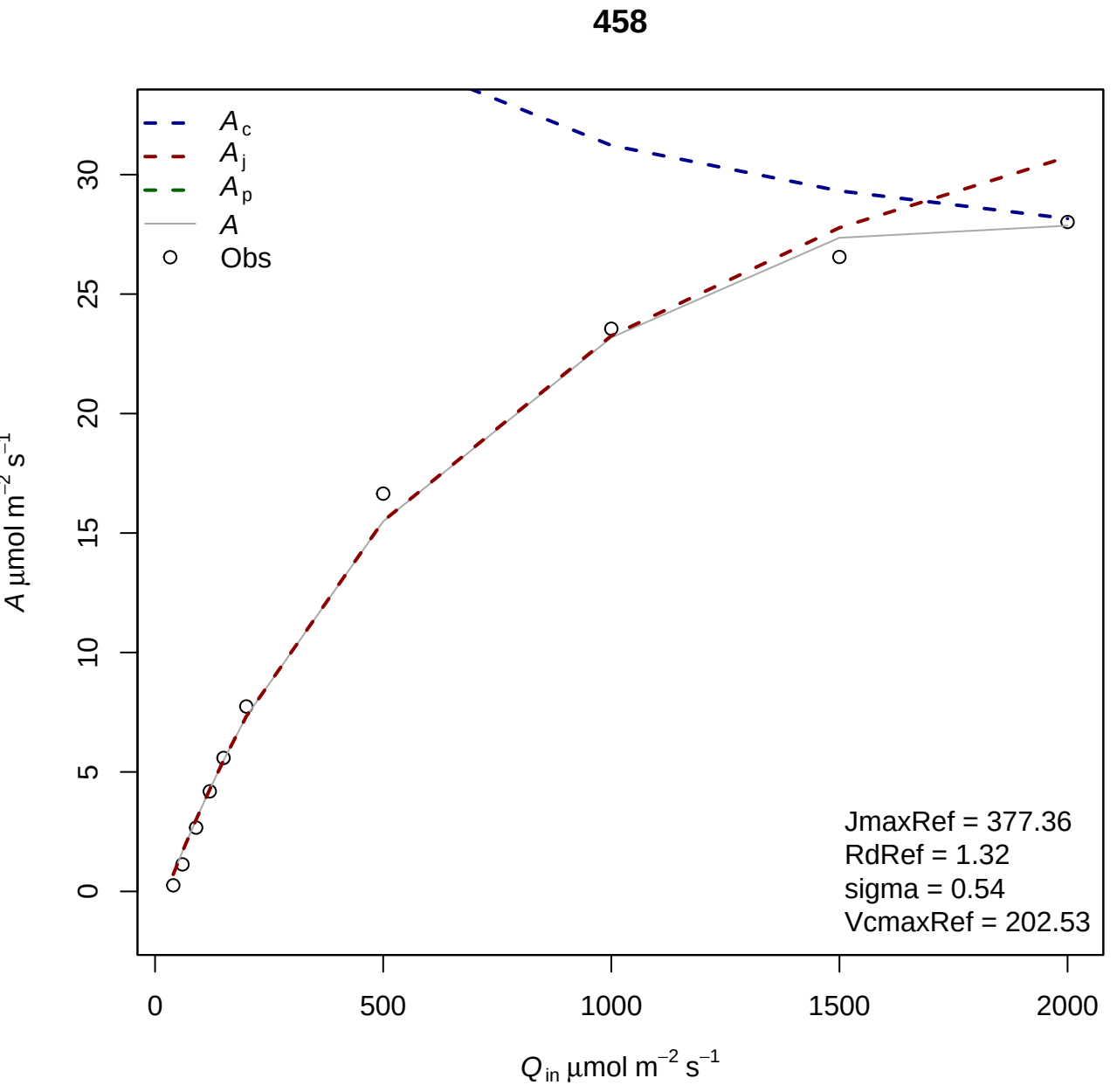

453
A_c
A_j
A_p
A
Obs
JmaxRef = 351.72
RdRef = 2.24
sigma = 0.39
VcmaxRef = 250.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

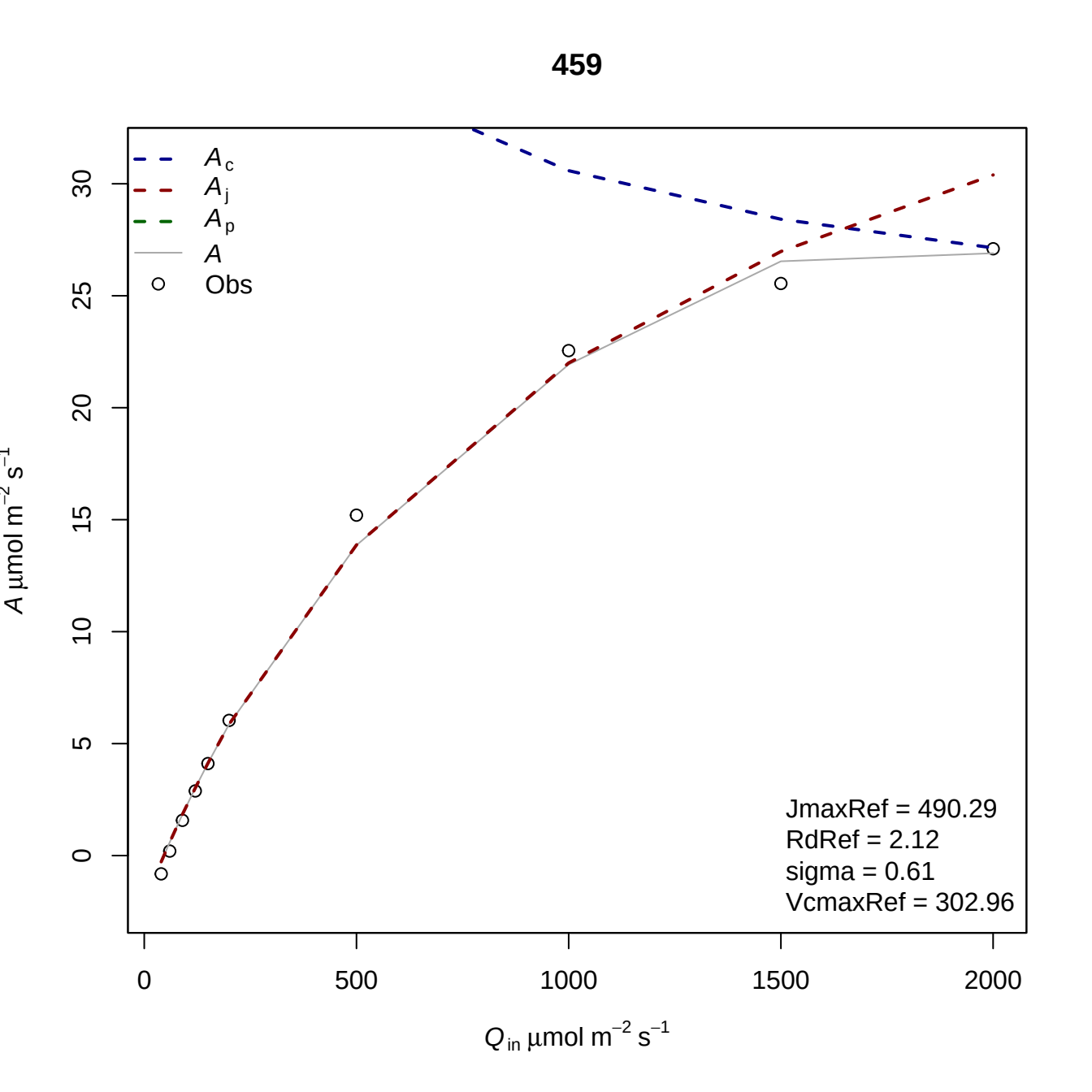

454
A_c
A_j
A_p
A
Obs
JmaxRef = 109.82
RdRef = 0.62
sigma = 0.09
VcmaxRef = 681417.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

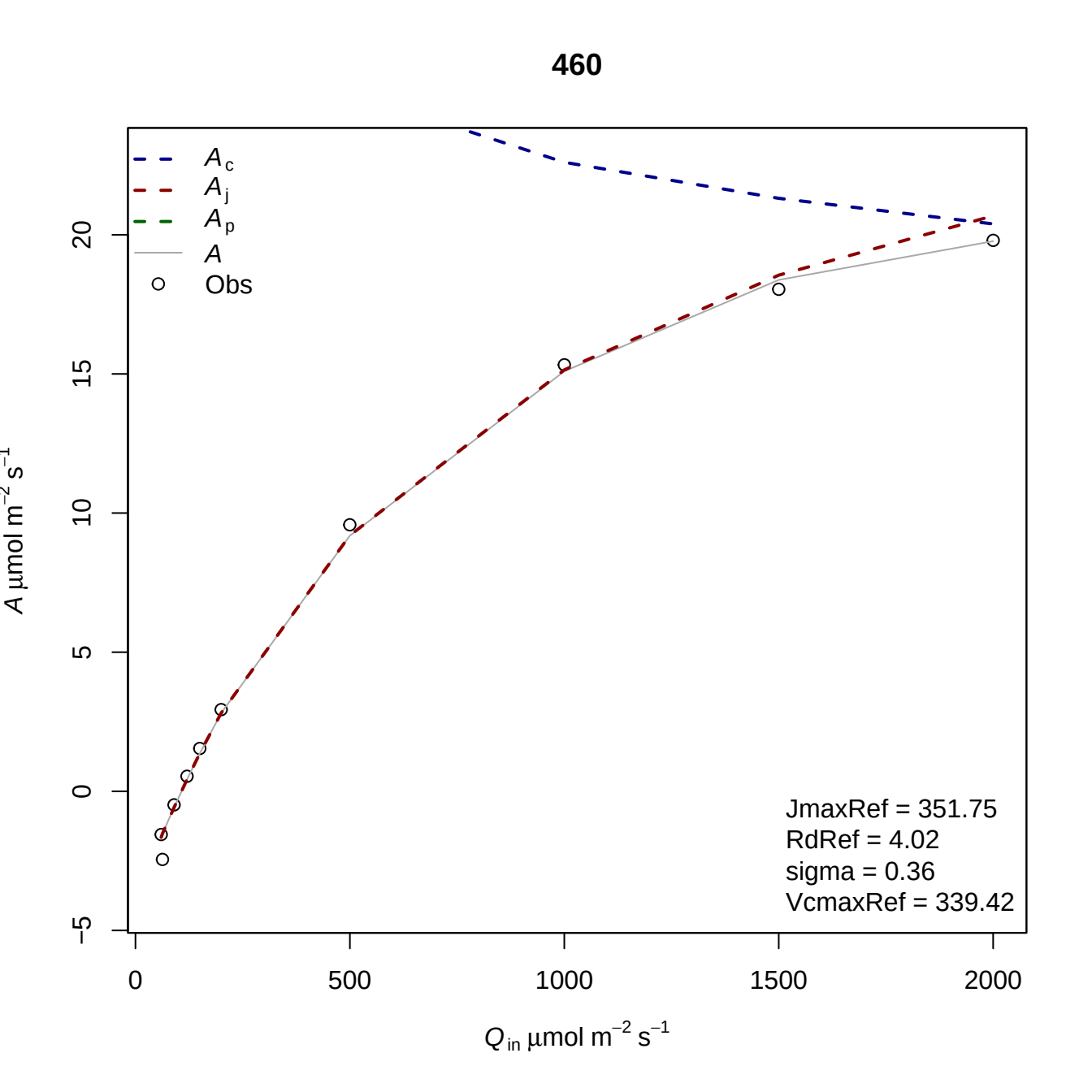

455
A_c
A_j
A_p
A
Obs
JmaxRef = 196.43
RdRef = 3.62
sigma = 0.22
VcmaxRef = 16905599
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

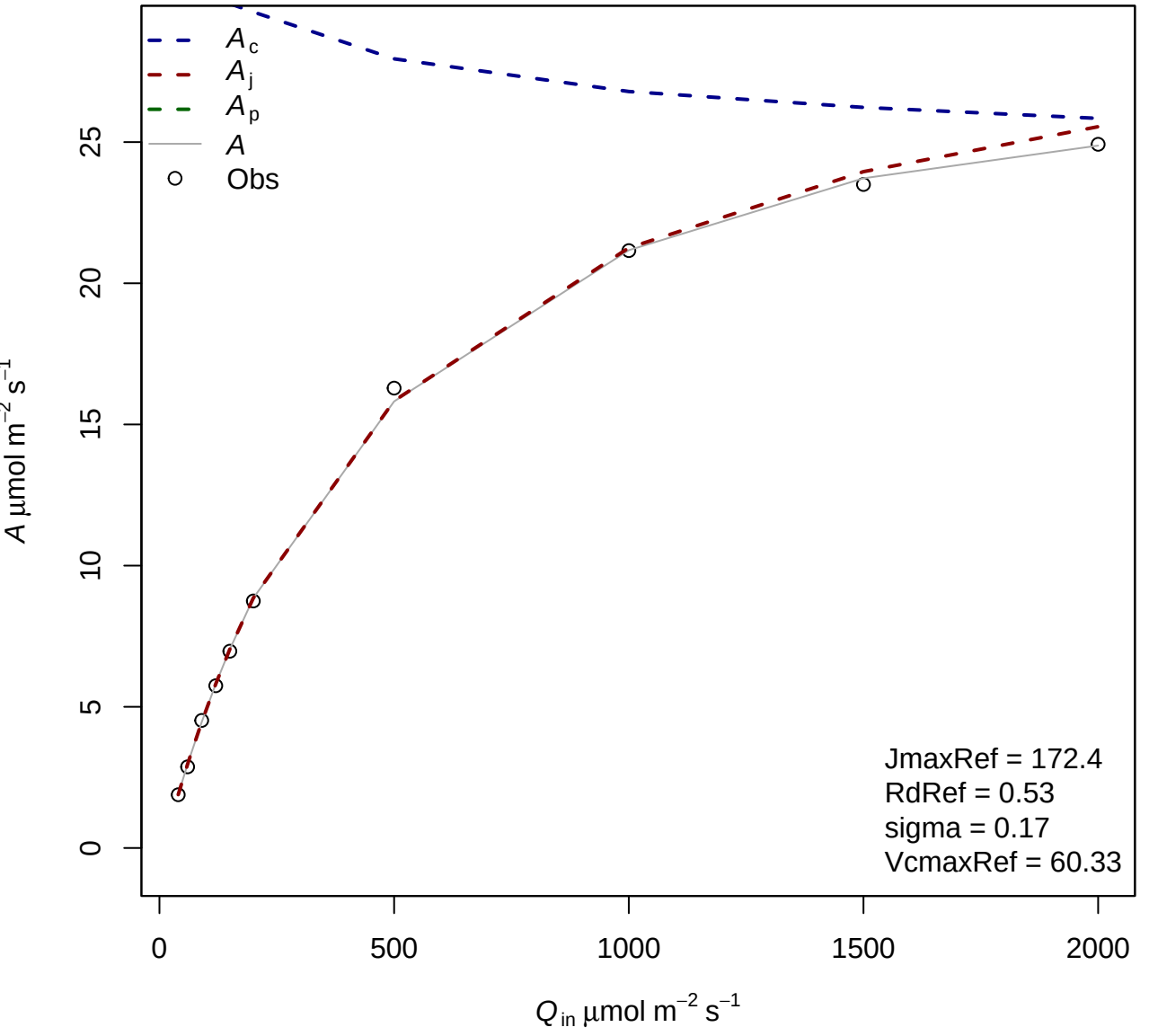

456
A_c
A_j
A_p
A
Obs
JmaxRef = 189.44
RdRef = 0.34
sigma = 0.29
VcmaxRef = 74.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

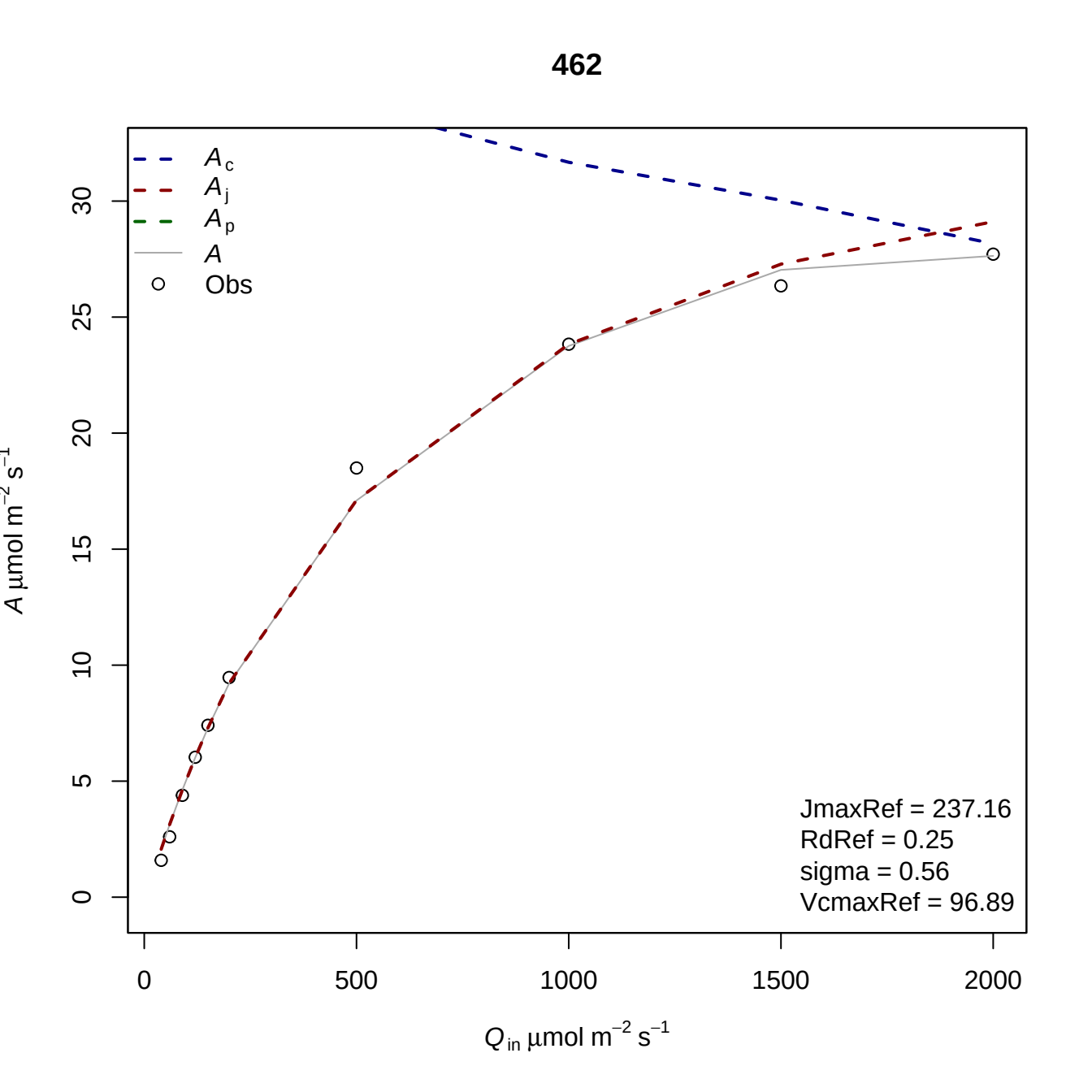

457
A_c
A_j
A_p
A
Obs
JmaxRef = 411.4
RdRef = 0.37
sigma = 0.77
VcmaxRef = 143.35
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

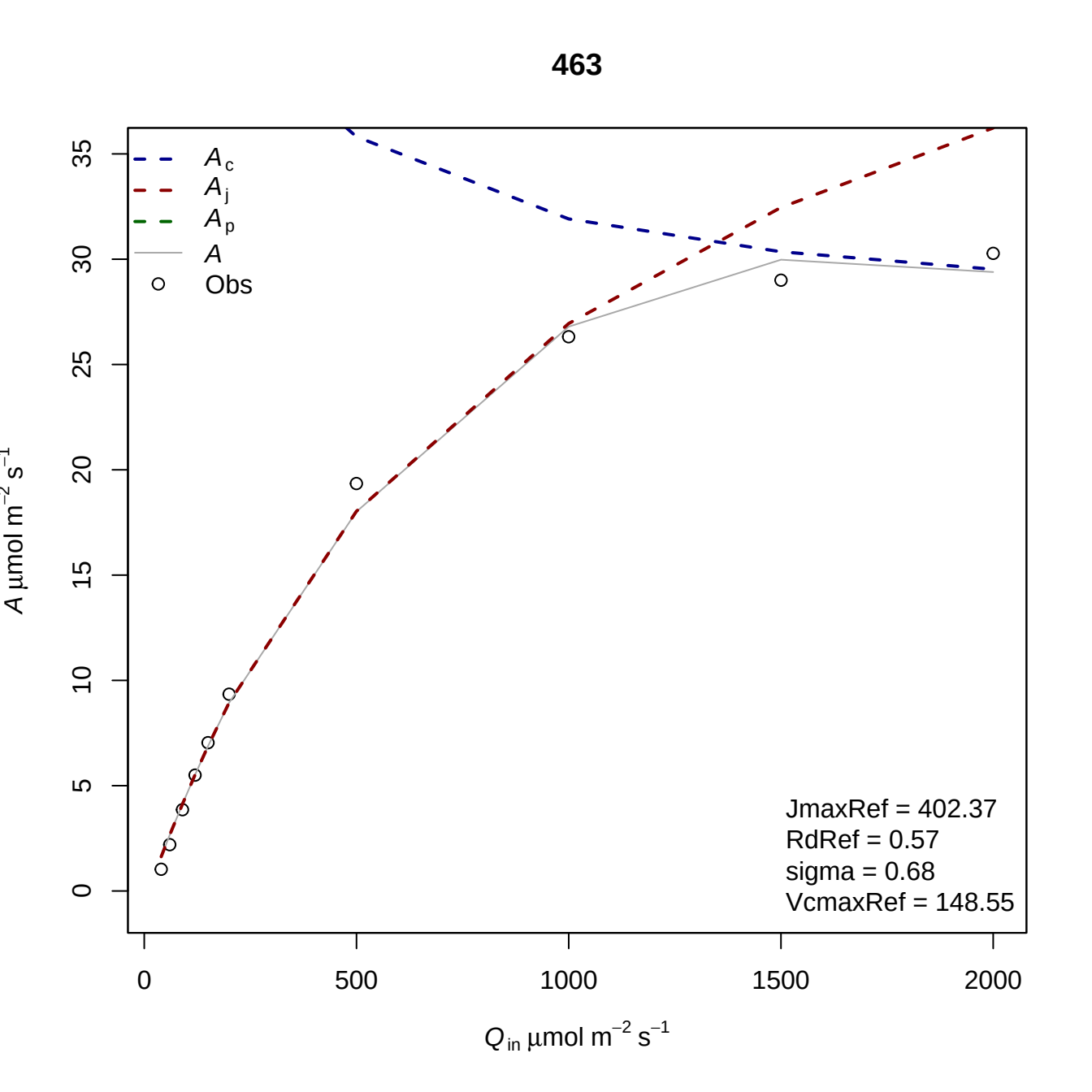

458
A_c
A_j
A_p
A
Obs
JmaxRef = 377.36
RdRef = 1.32
sigma = 0.54
VcmaxRef = 202.53
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

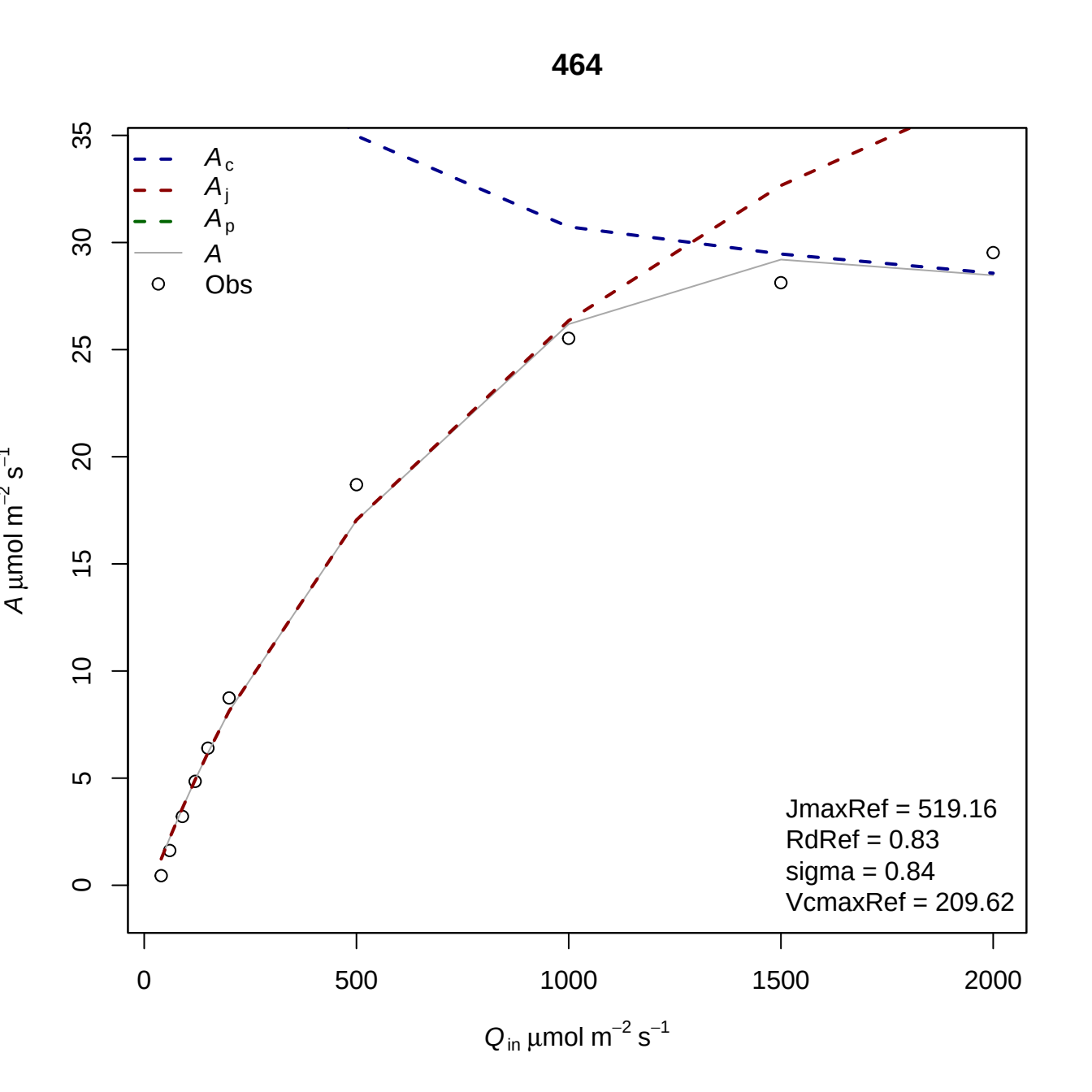

459
A_c
A_j
A_p
A
Obs
JmaxRef = 490.29
RdRef = 2.12
sigma = 0.61
VcmaxRef = 302.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

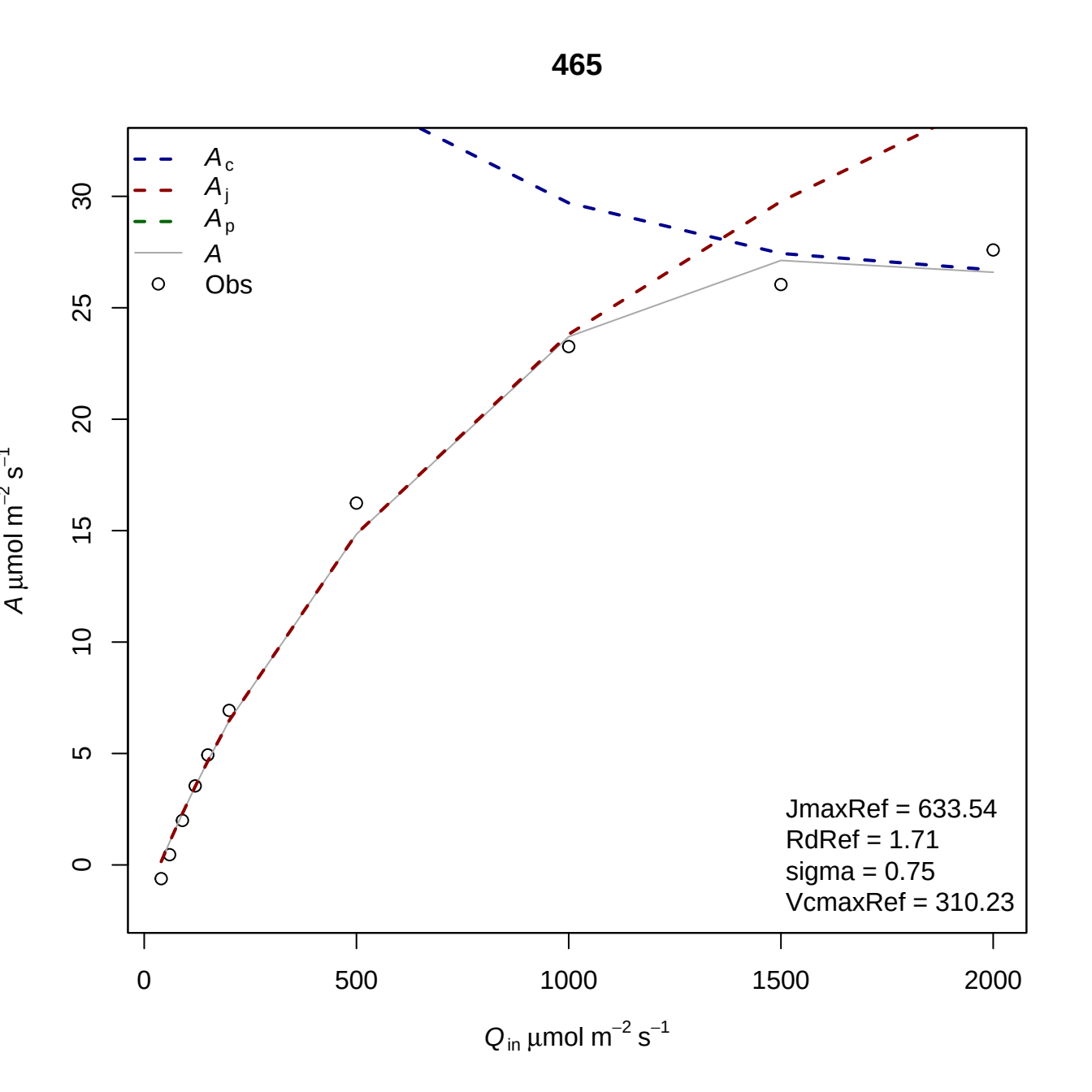

460
A_c
A_j
A_p
A
Obs
JmaxRef = 351.75
RdRef = 4.02
sigma = 0.36
VcmaxRef = 339.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

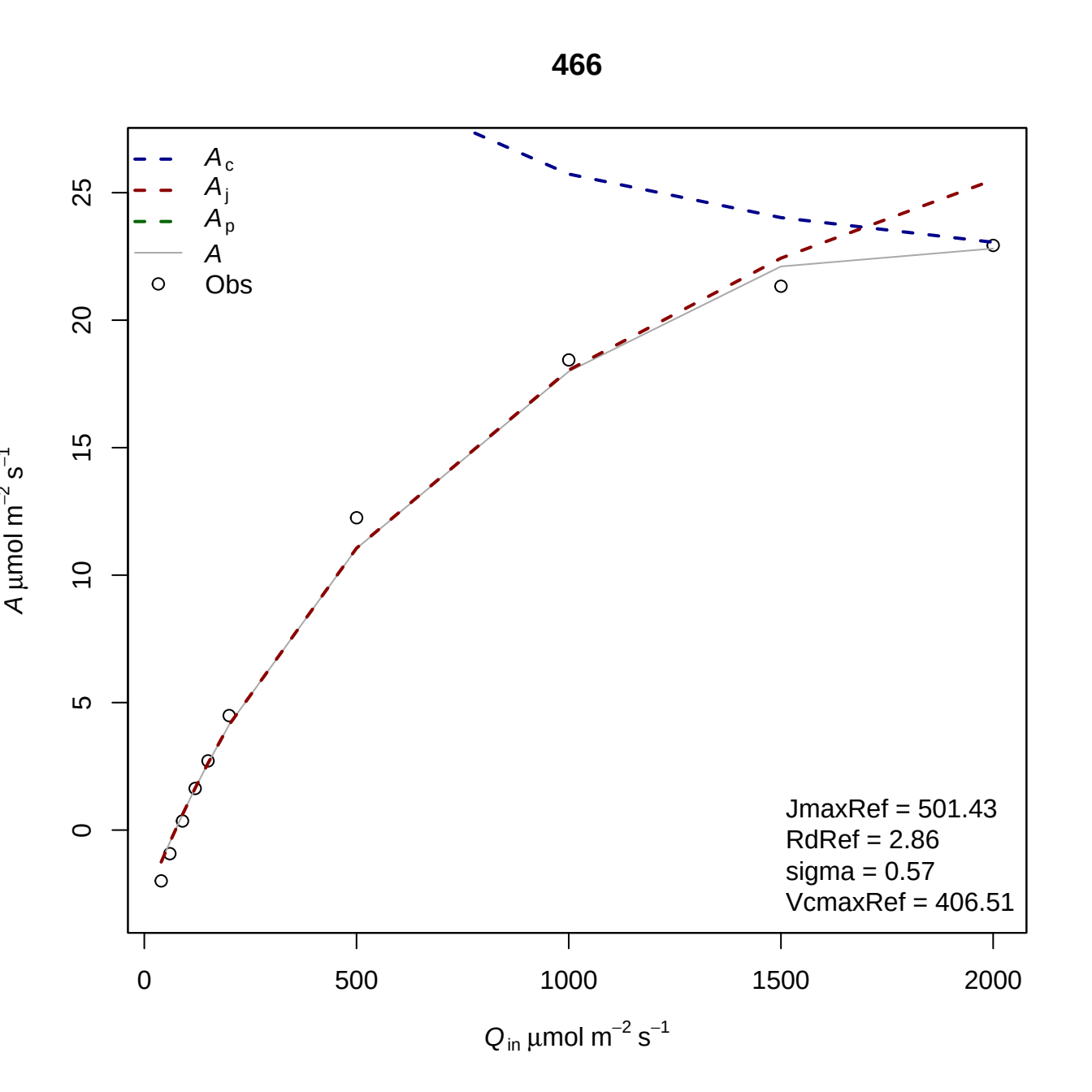

A_c
A_j
A_p
A
Obs
JmaxRef = 172.4
RdRef = 0.53
sigma = 0.17
VcmaxRef = 60.33
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

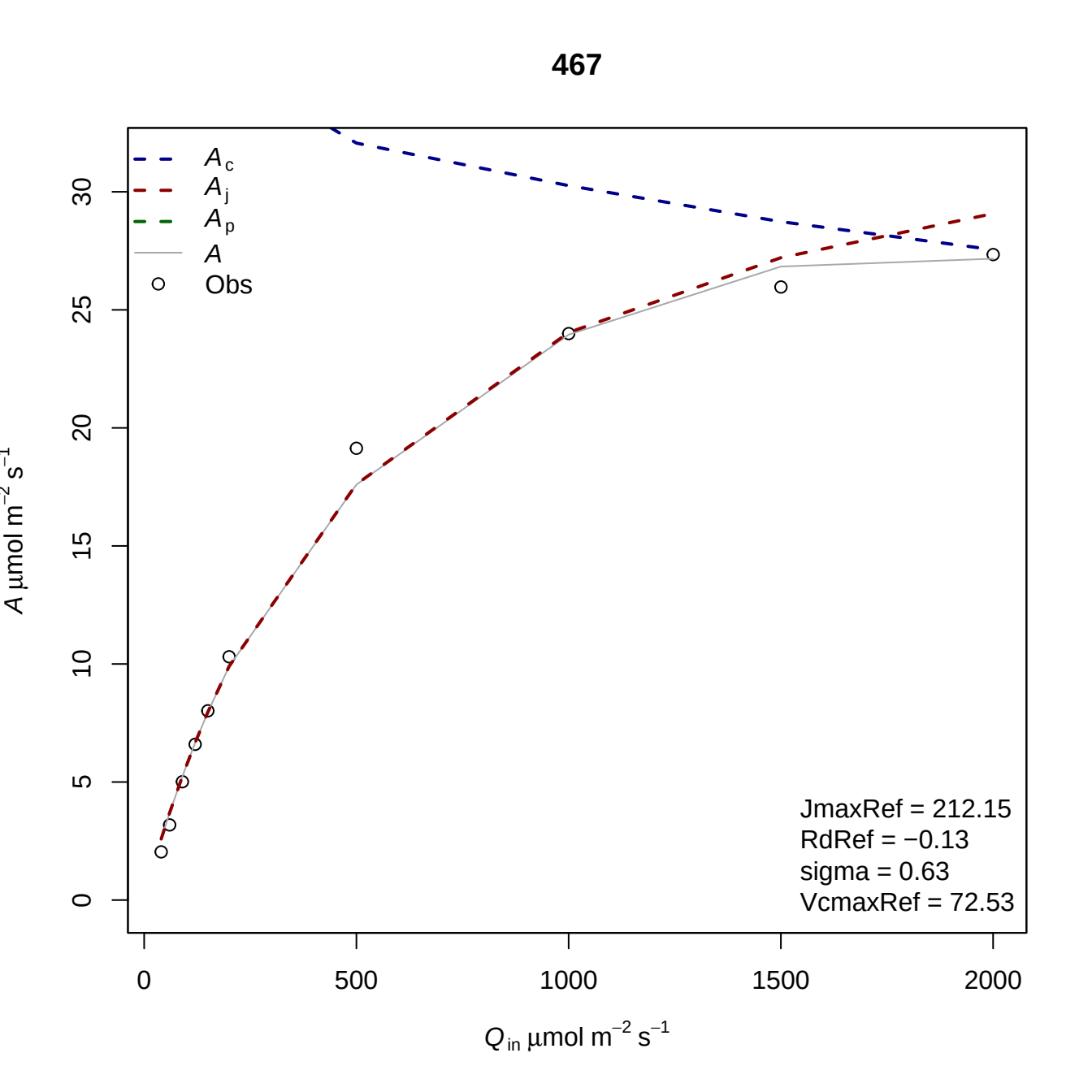

462
A_c
A_j
A_p
A
Obs
JmaxRef = 237.16
RdRef = 0.25
sigma = 0.56
VcmaxRef = 96.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

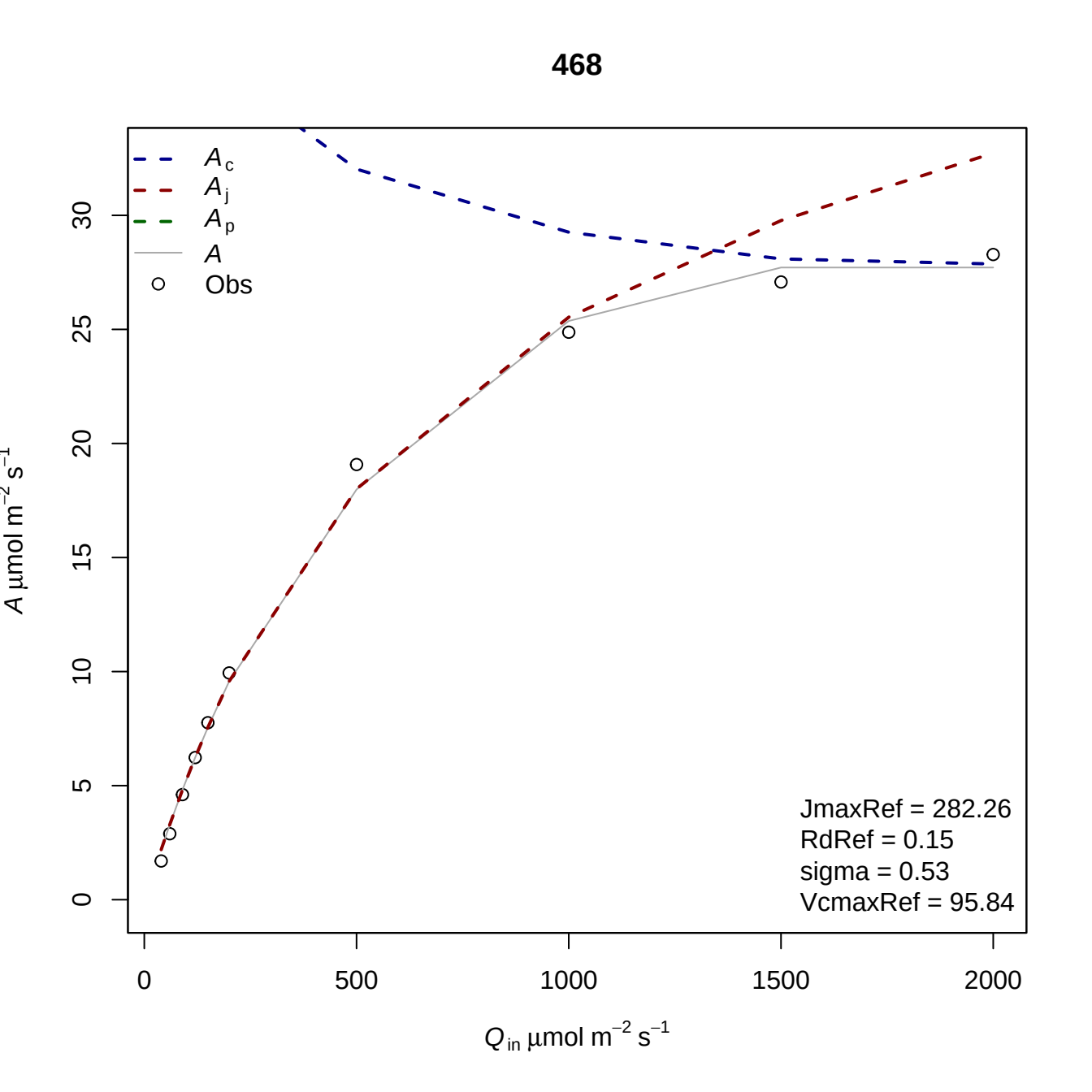

463
A_c
A_j
A_p
A
Obs
JmaxRef = 402.37
RdRef = 0.57
sigma = 0.68
VcmaxRef = 148.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

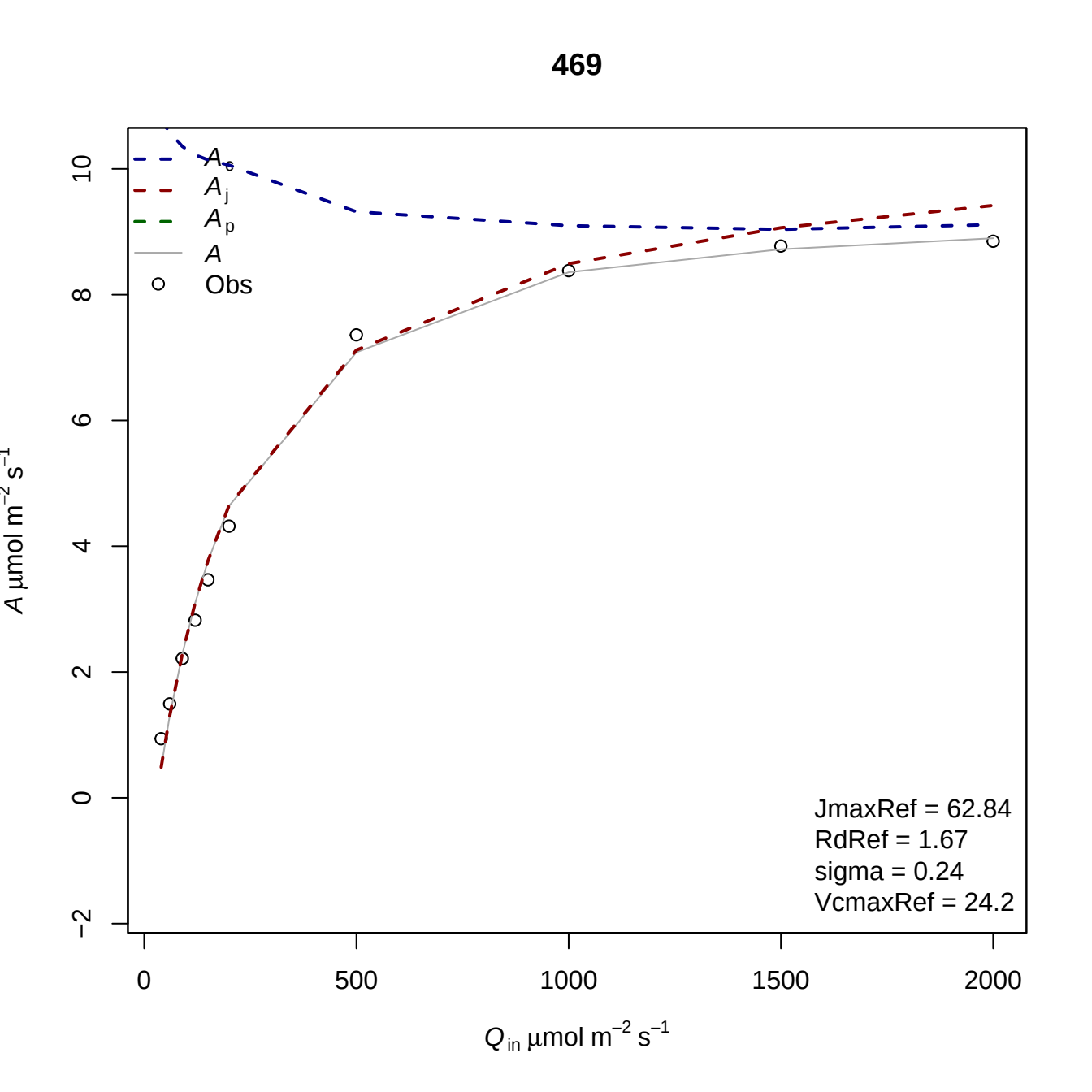

464
A_c
A_j
A_p
A
Obs
JmaxRef = 519.16
RdRef = 0.83
sigma = 0.84
VcmaxRef = 209.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

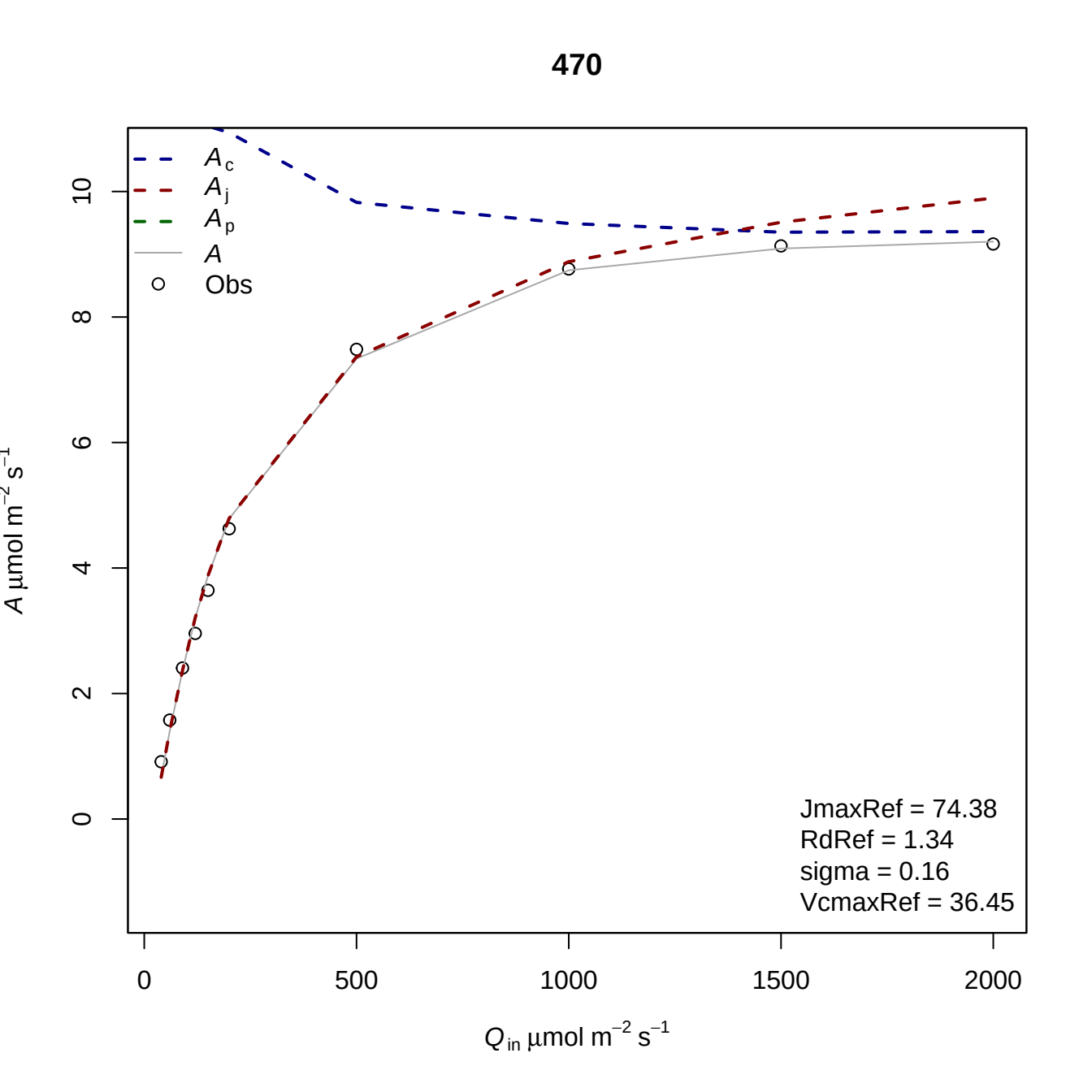

465
A_c
A_j
A_p
A
Obs
JmaxRef = 633.54
RdRef = 1.71
sigma = 0.75
VcmaxRef = 310.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

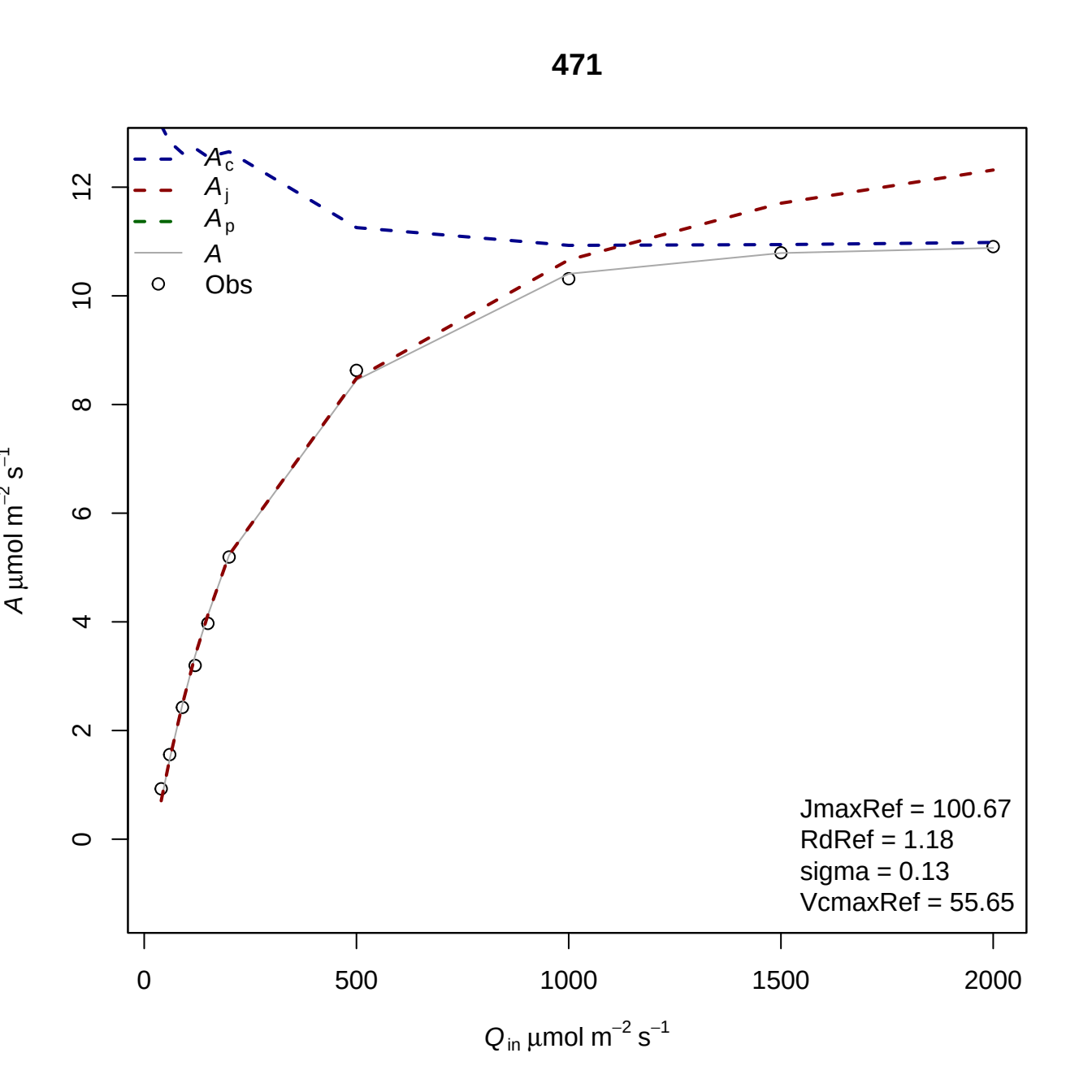

466
A_c
A_j
A_p
A
Obs
JmaxRef = 501.43
RdRef = 2.86
sigma = 0.57
VcmaxRef = 406.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

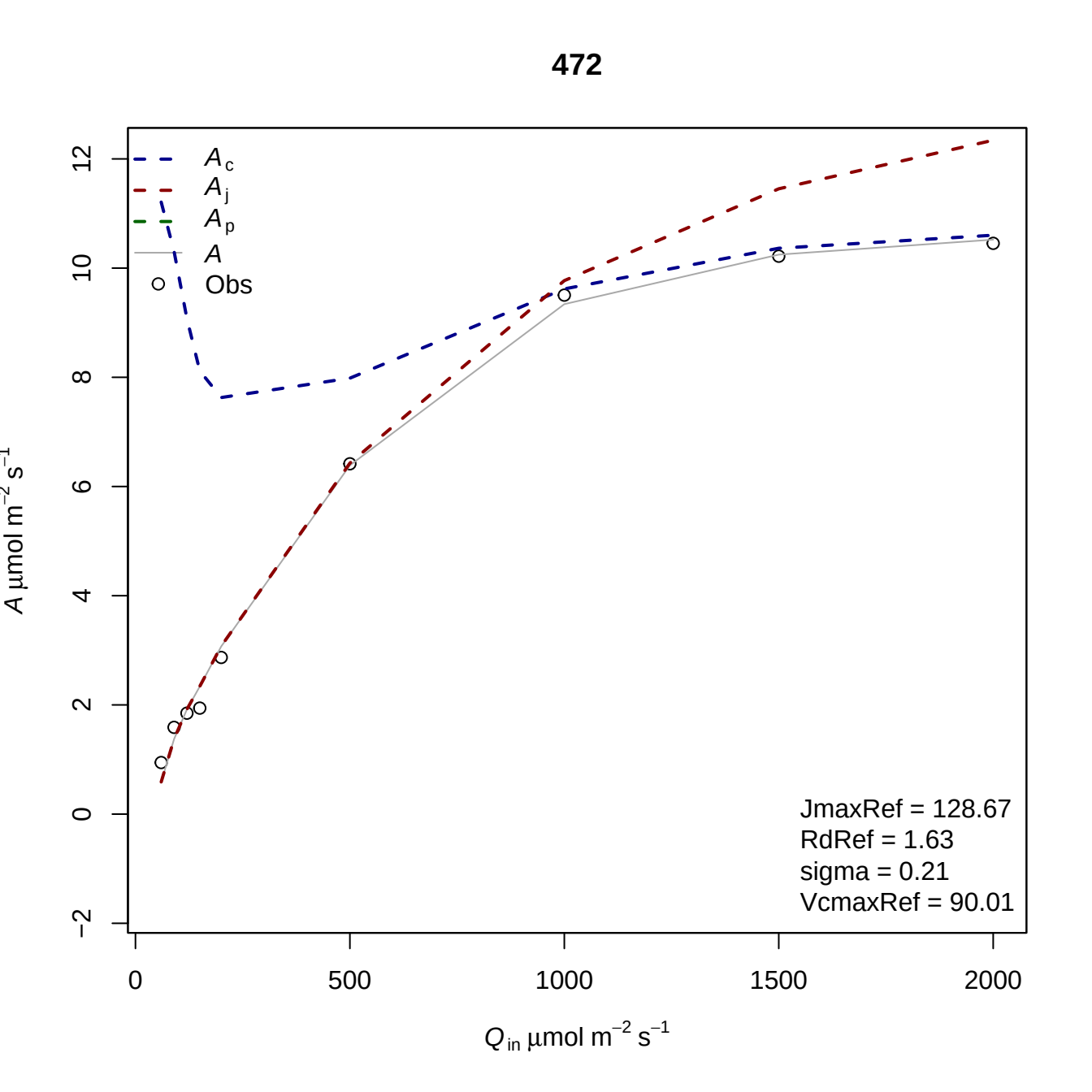

467
A_c
A_j
A_p
A
Obs
JmaxRef = 212.15
RdRef = −0.13
sigma = 0.63
VcmaxRef = 72.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

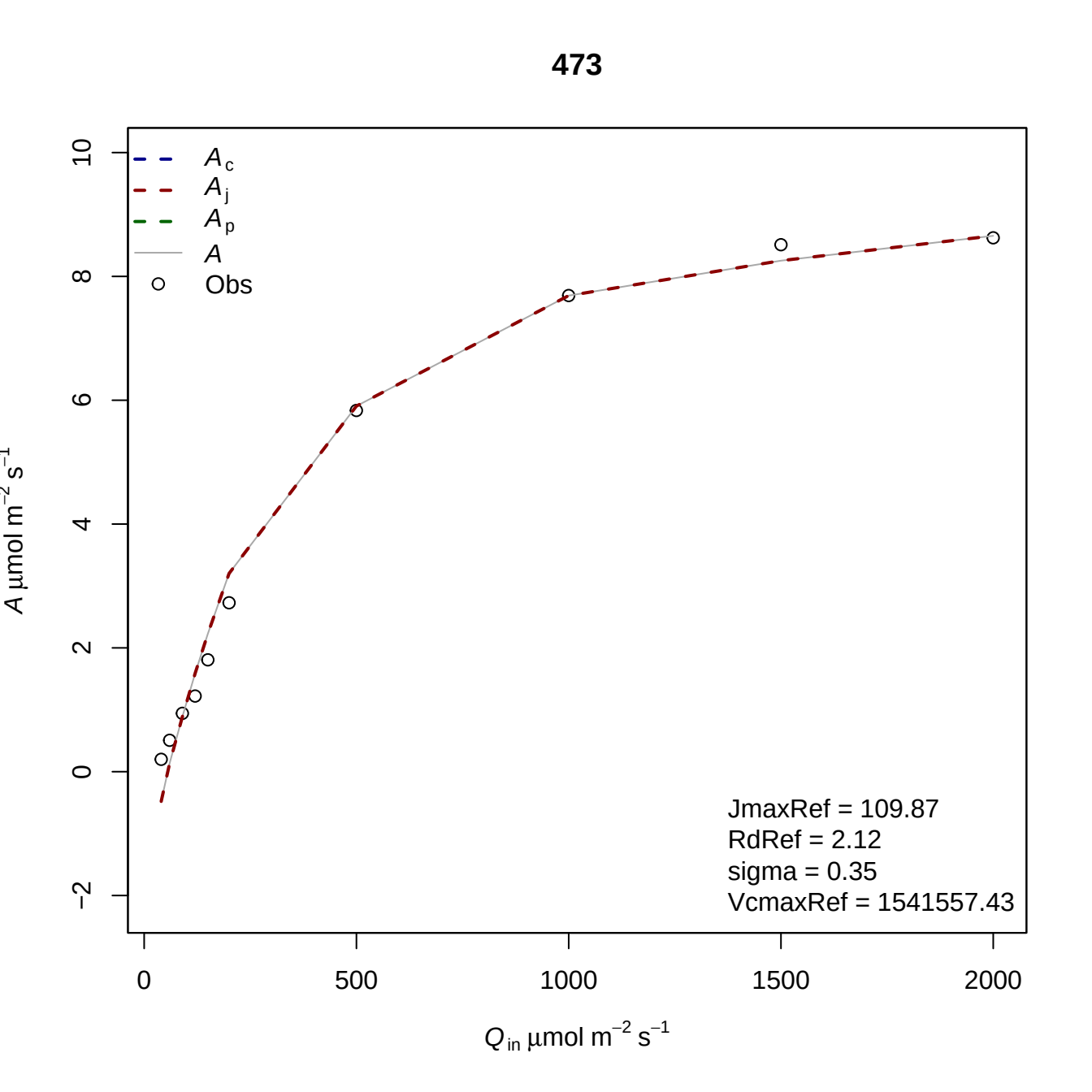

468
A_c
A_j
A_p
A
Obs
JmaxRef = 282.26
RdRef = 0.15
sigma = 0.53
VcmaxRef = 95.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

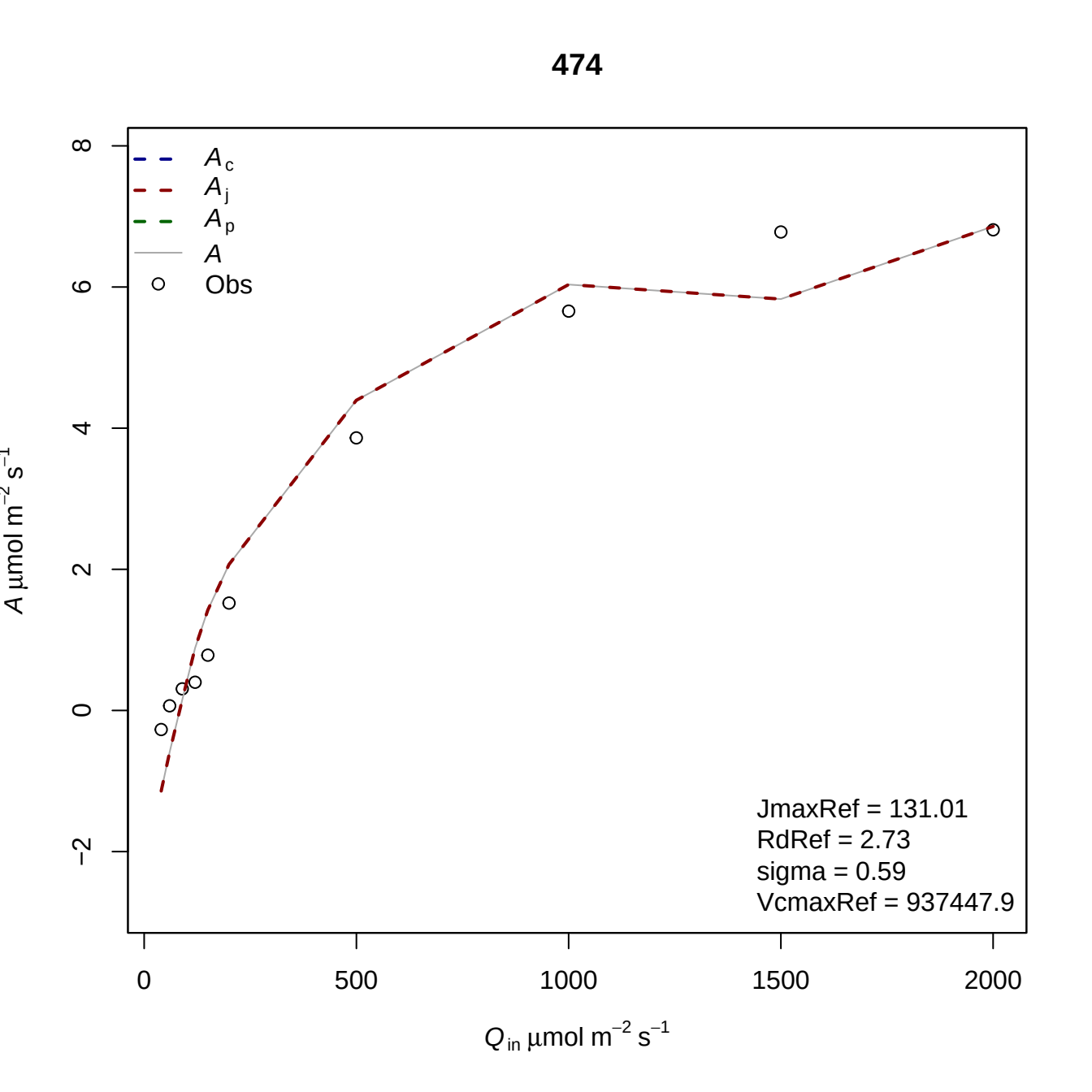

469
A_c
A_j
A_p
A
Obs
JmaxRef = 62.84
RdRef = 1.67
sigma = 0.24
VcmaxRef = 24.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

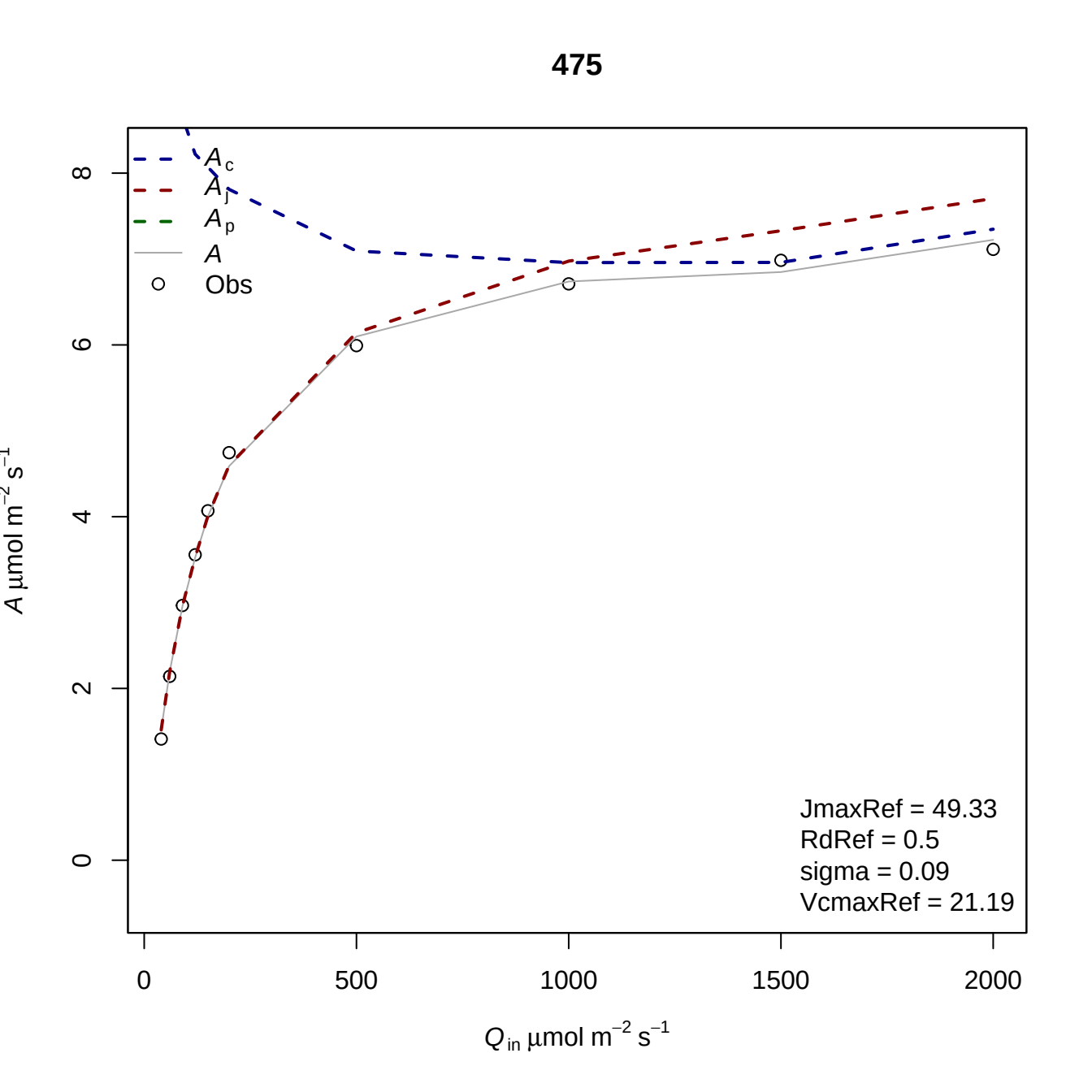

470
A_c
A_j
A_p
A
Obs
JmaxRef = 74.38
RdRef = 1.34
sigma = 0.16
VcmaxRef = 36.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

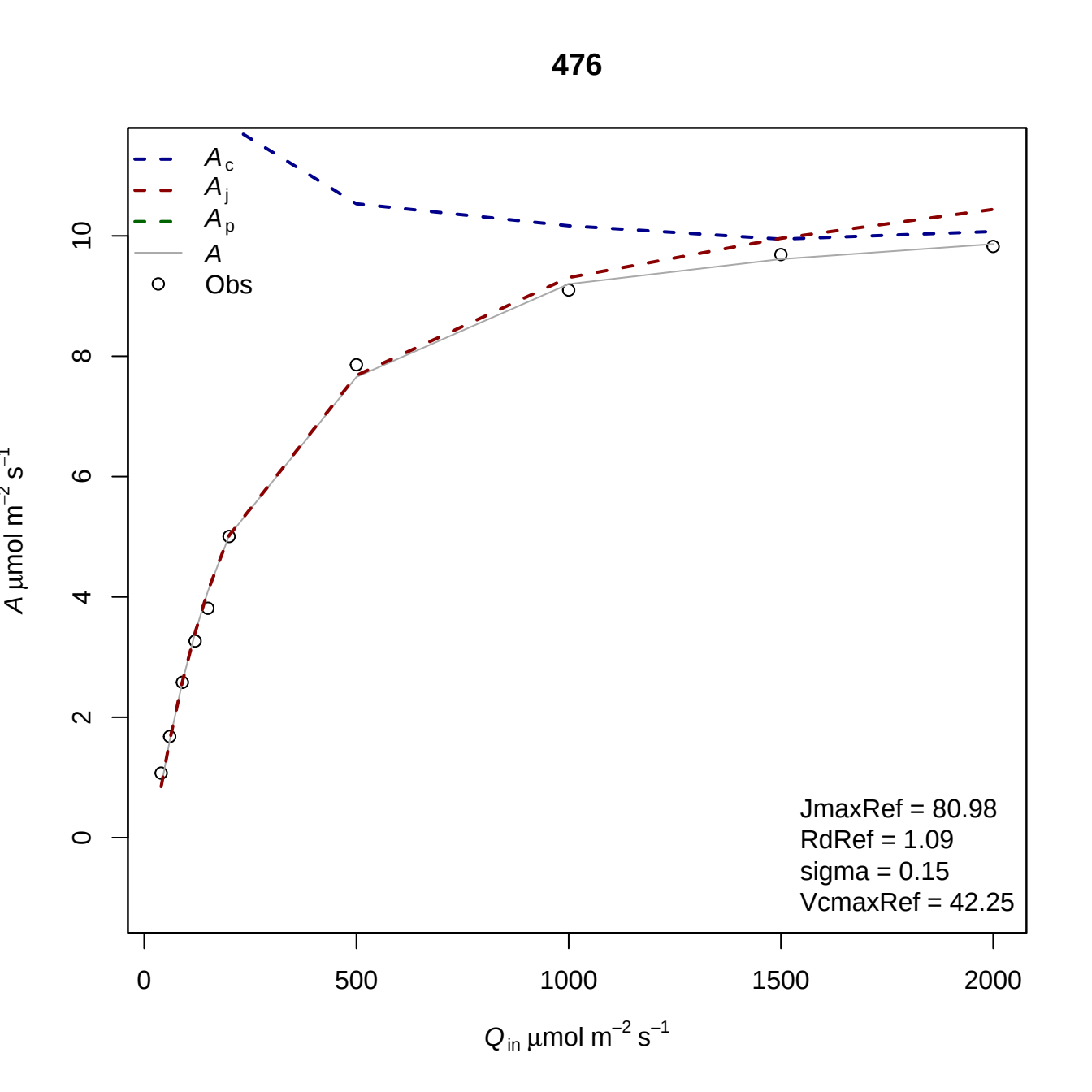

471
A_c
A_j
A_p
A
Obs
JmaxRef = 100.67
RdRef = 1.18
sigma = 0.13
VcmaxRef = 55.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

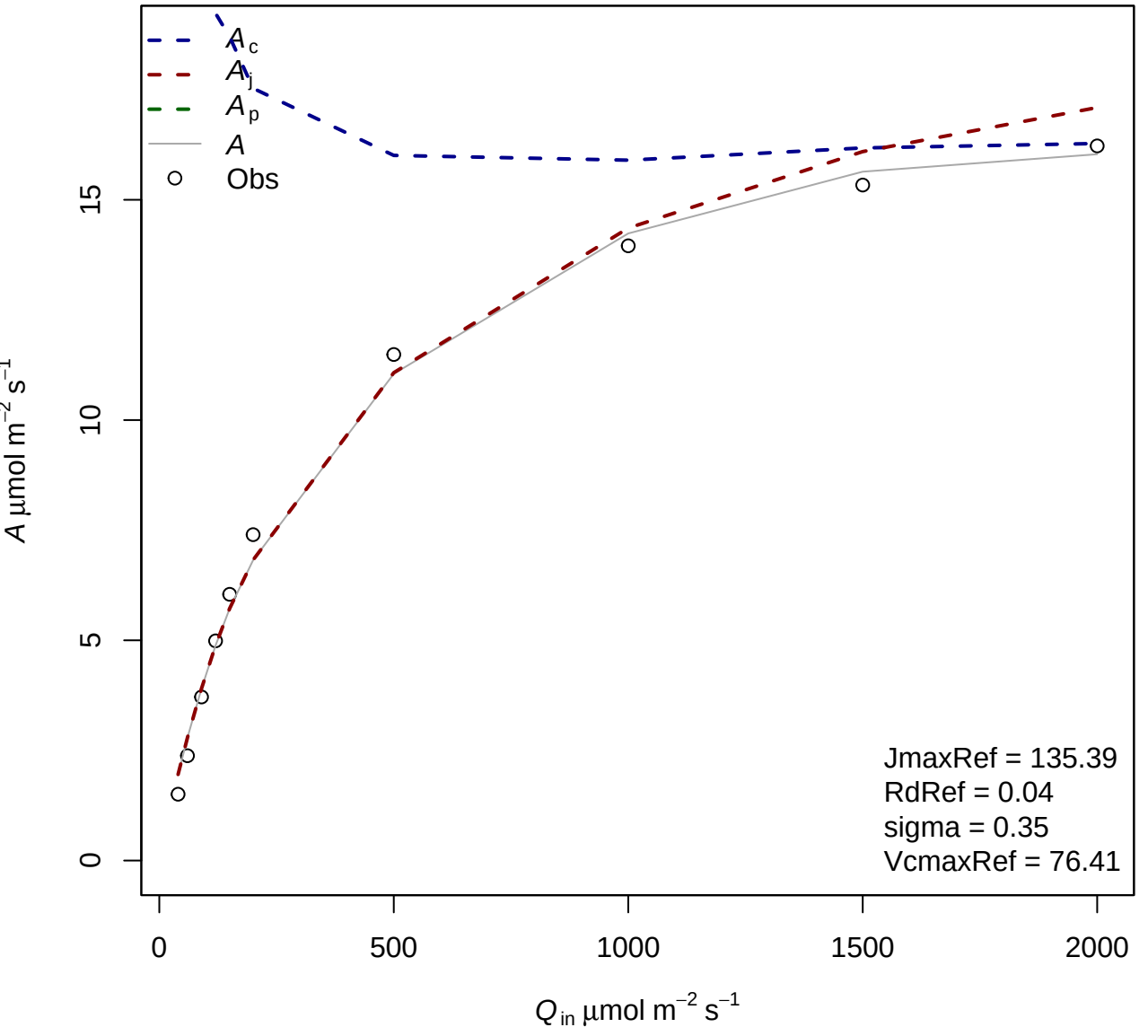

472
A_c
A_j
A_p
A
Obs
JmaxRef = 128.67
RdRef = 1.63
sigma = 0.21
VcmaxRef = 90.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

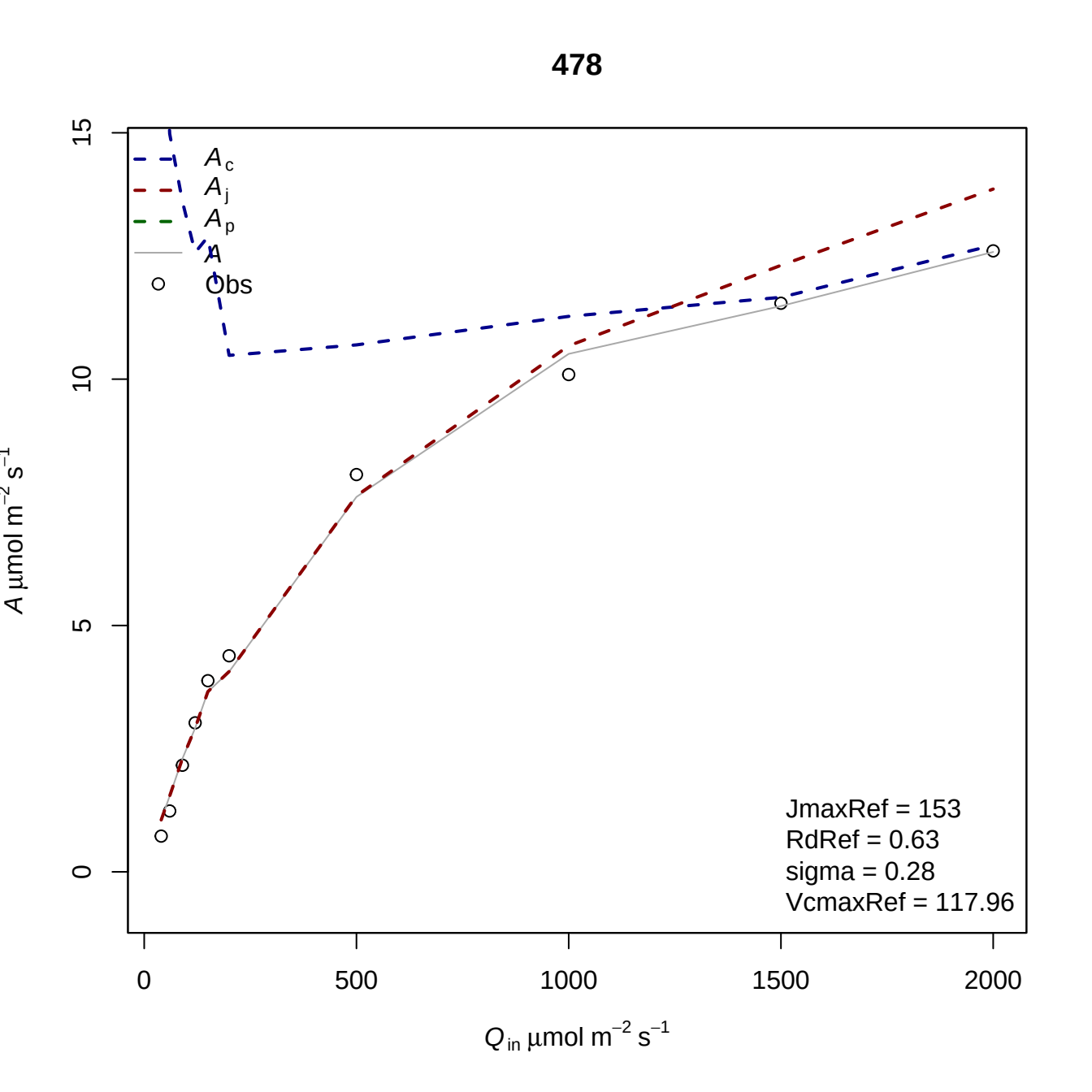

473
A_c
A_j
A_p
A
Obs
JmaxRef = 109.87
RdRef = 2.12
sigma = 0.35
VcmaxRef = 1541557.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

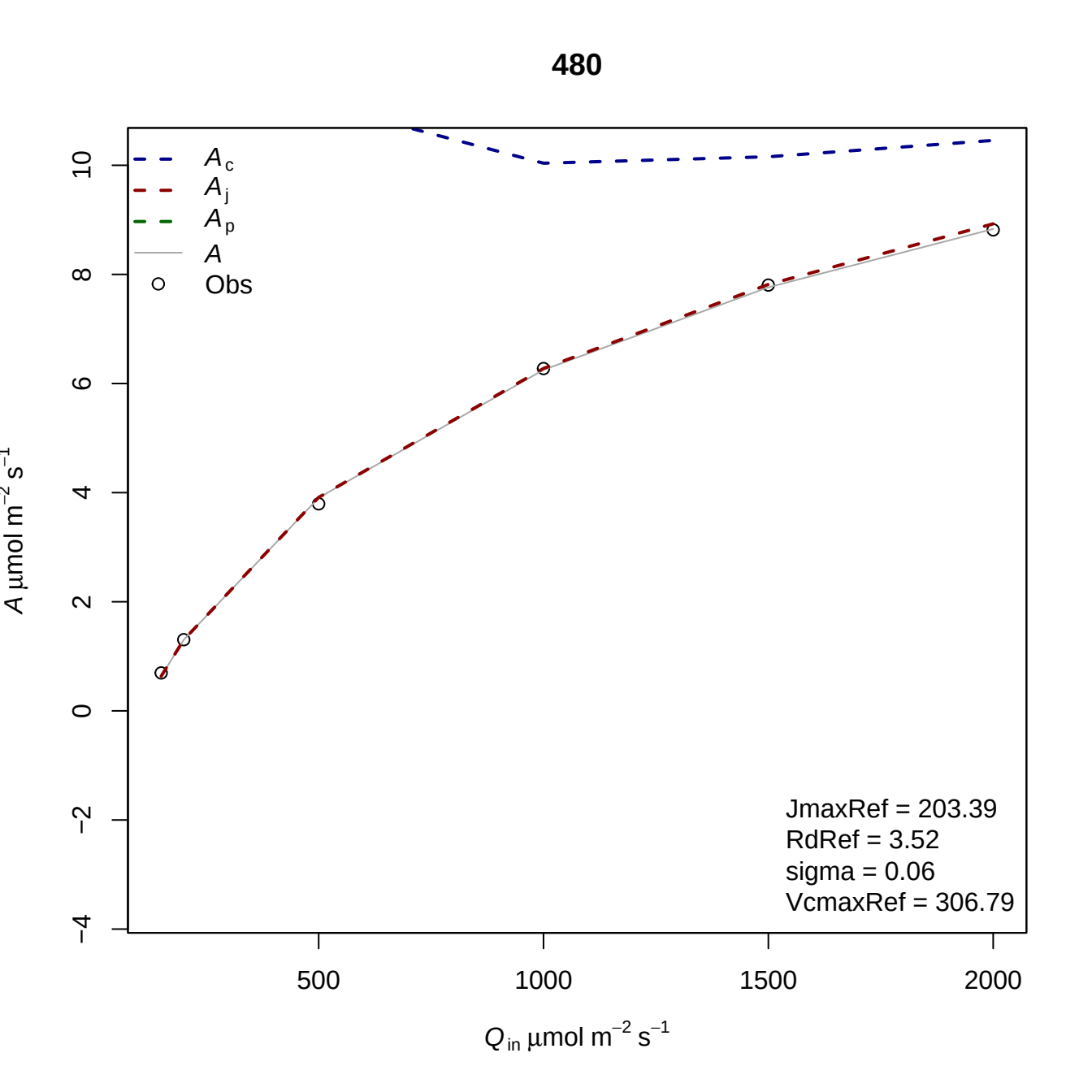

474
A_c
A_j
A_p
A
Obs
JmaxRef = 131.01
RdRef = 2.73
sigma = 0.59
VcmaxRef = 937447.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

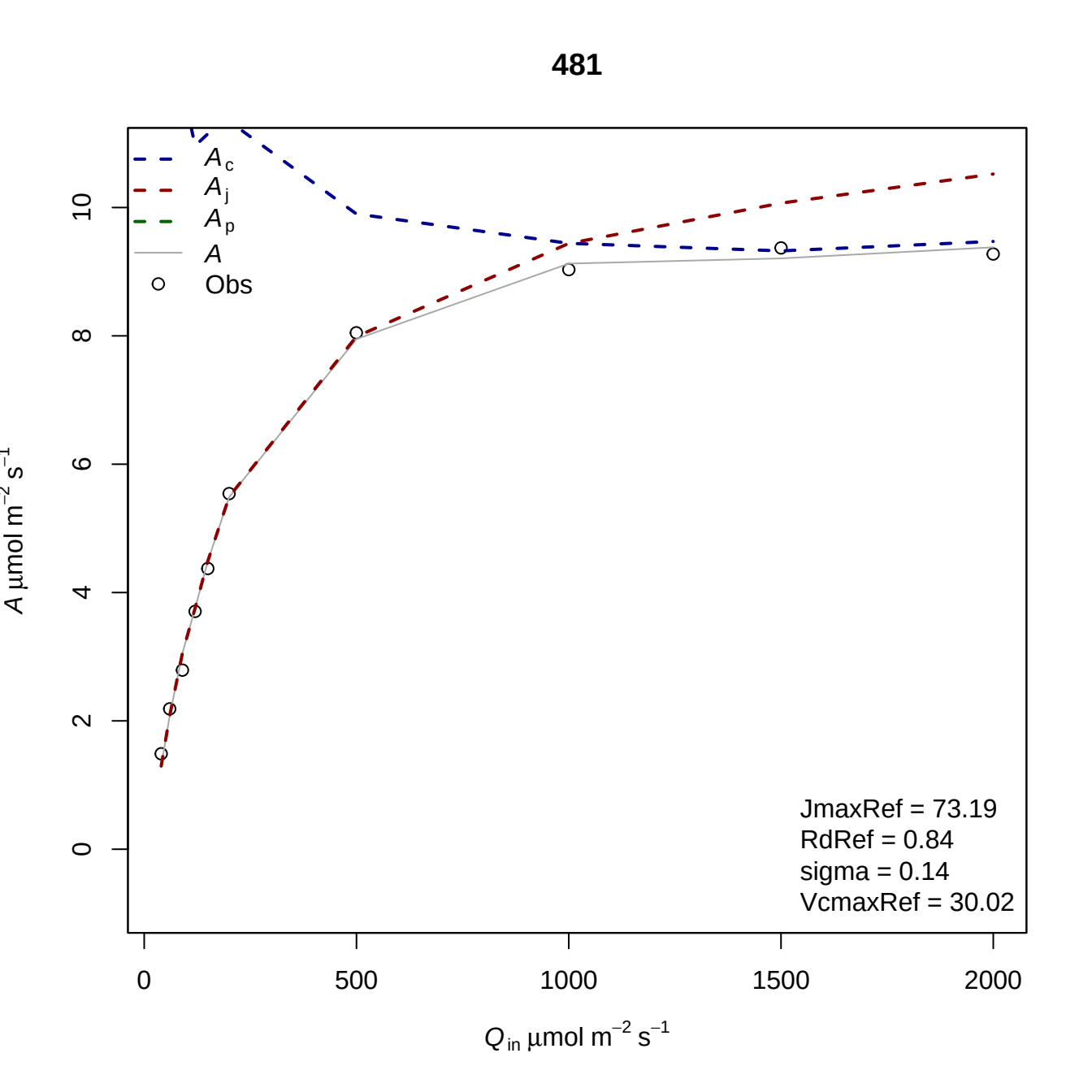

475
A_c
A_j
A_p
A
Obs
JmaxRef = 49.33
RdRef = 0.5
sigma = 0.09
VcmaxRef = 21.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

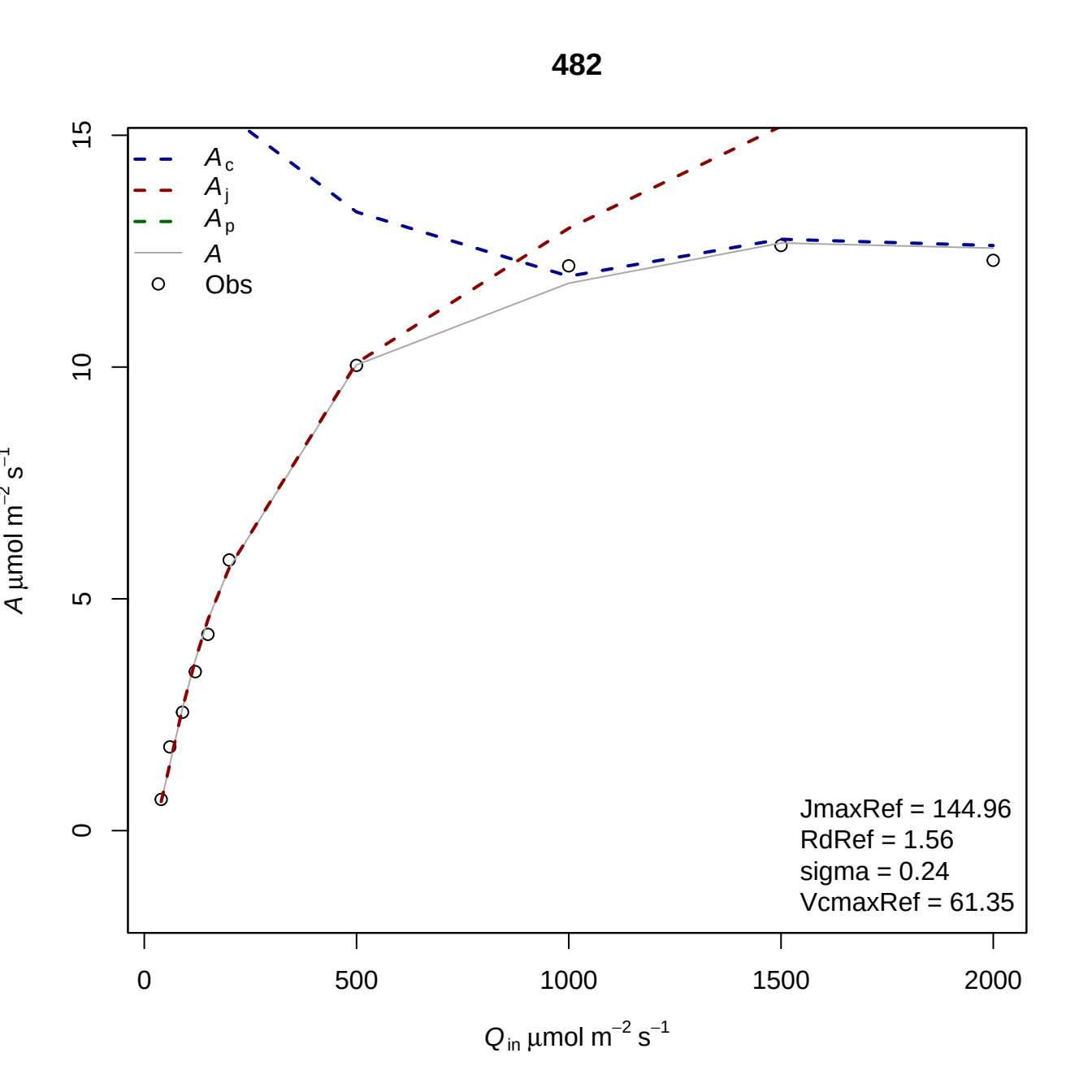

476
A_c
A_j
A_p
A
Obs
JmaxRef = 80.98
RdRef = 1.09
sigma = 0.15
VcmaxRef = 42.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

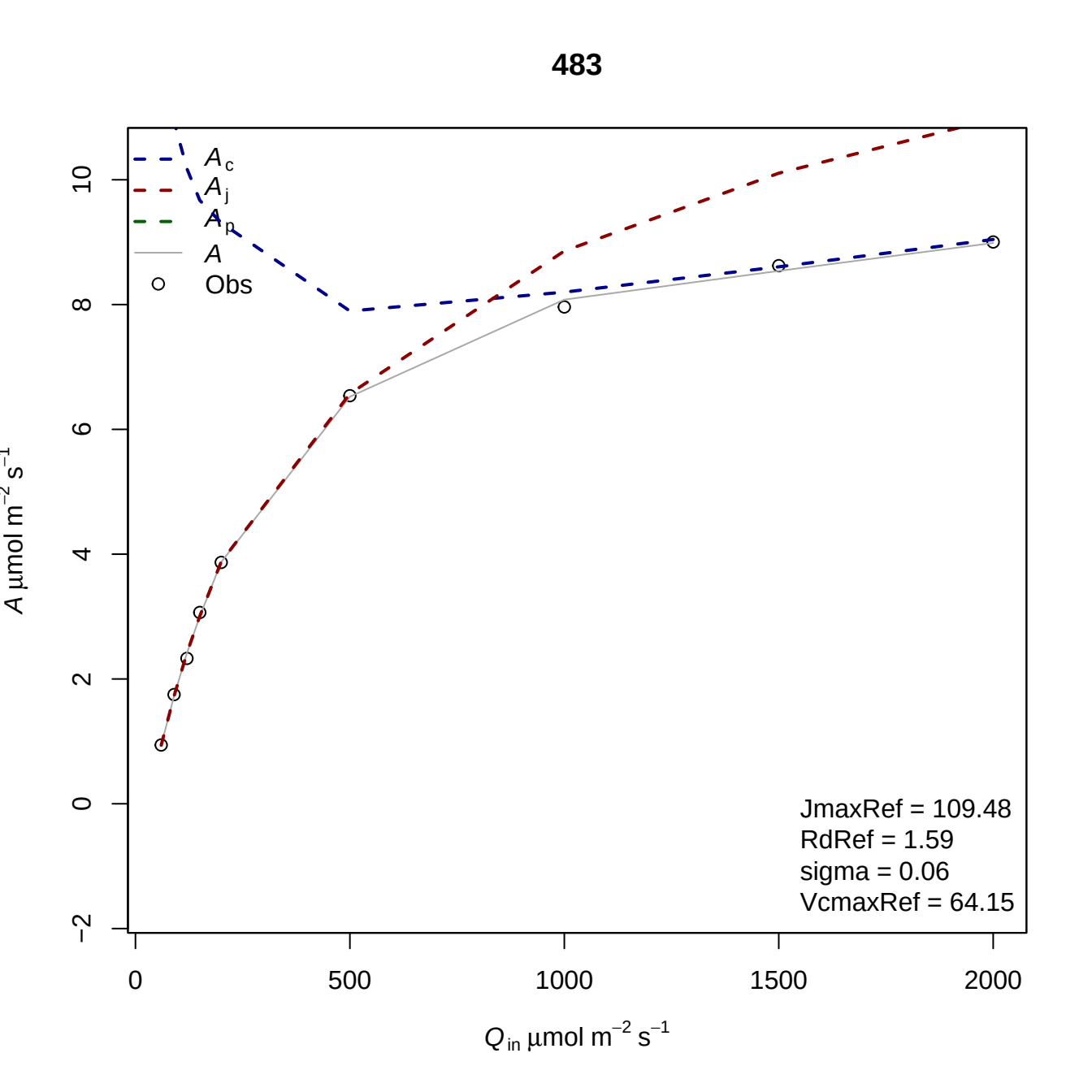

A_c
A_j
A_p
A
Obs
JmaxRef = 135.39
RdRef = 0.04
sigma = 0.35
VcmaxRef = 76.41
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

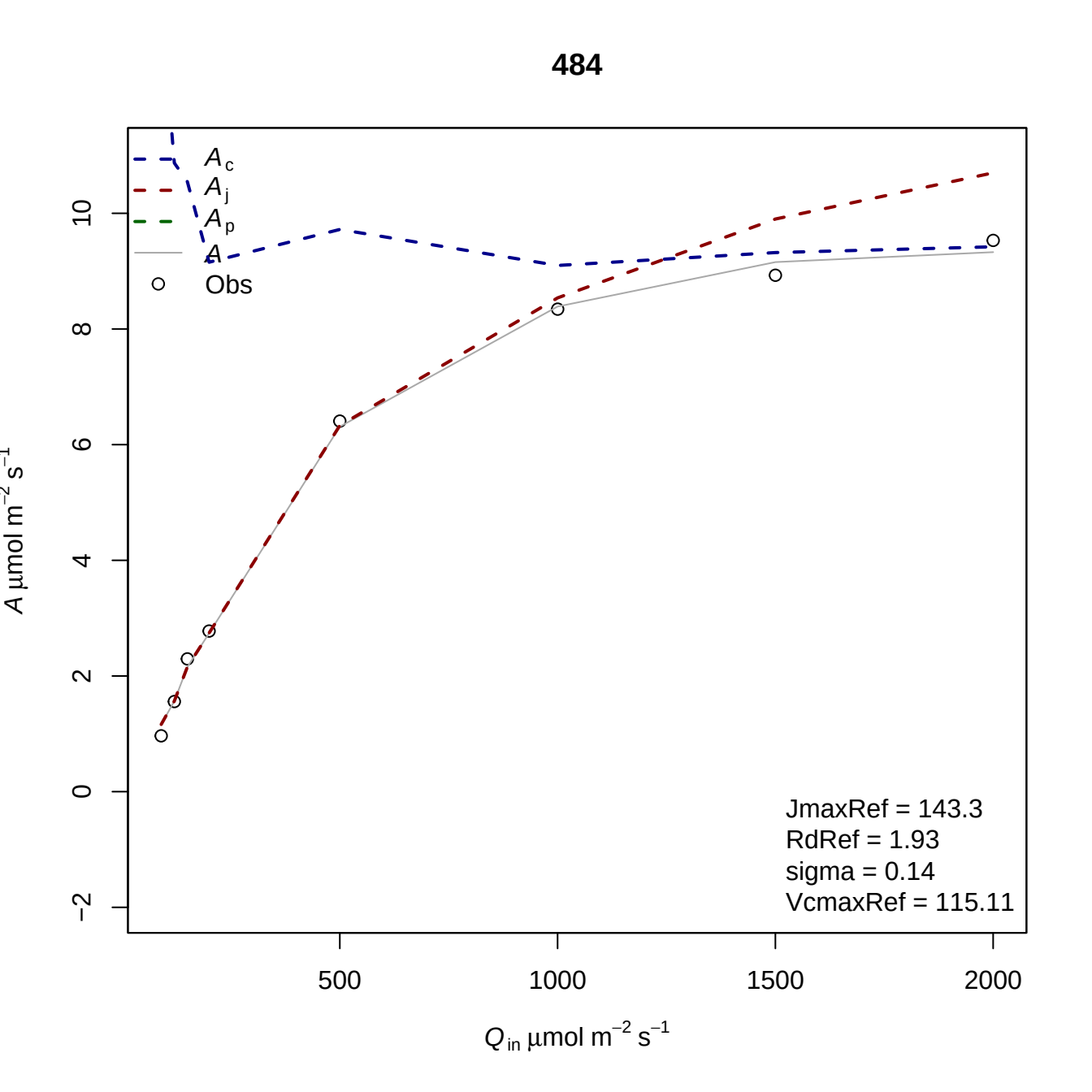

478
A_c
A_j
A_p
A
Obs
JmaxRef = 153
RdRef = 0.63
sigma = 0.28
VcmaxRef = 117.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

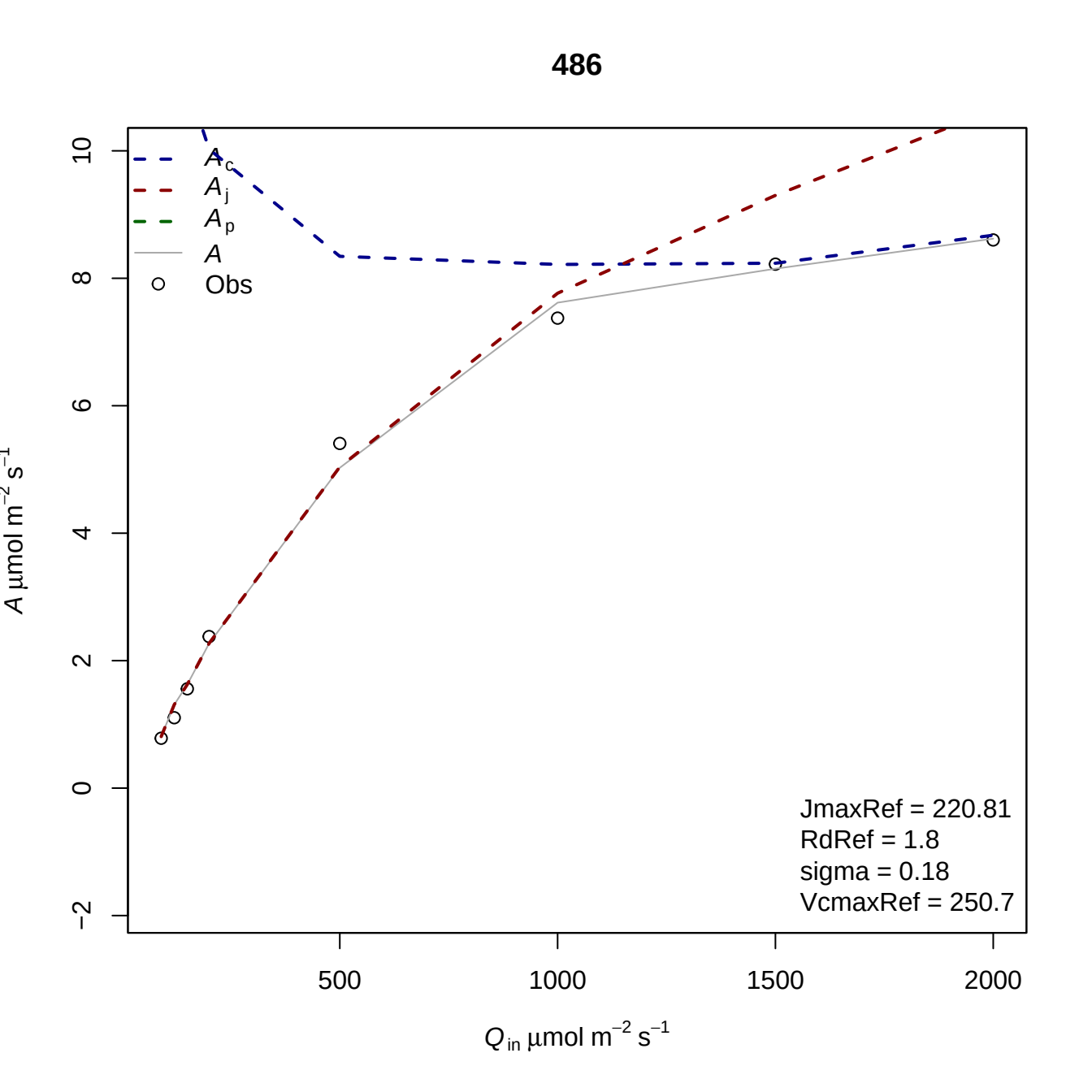

480
A_c
A_j
A_p
A
Obs
JmaxRef = 203.39
RdRef = 3.52
sigma = 0.06
VcmaxRef = 306.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
500
1000
1500
2000

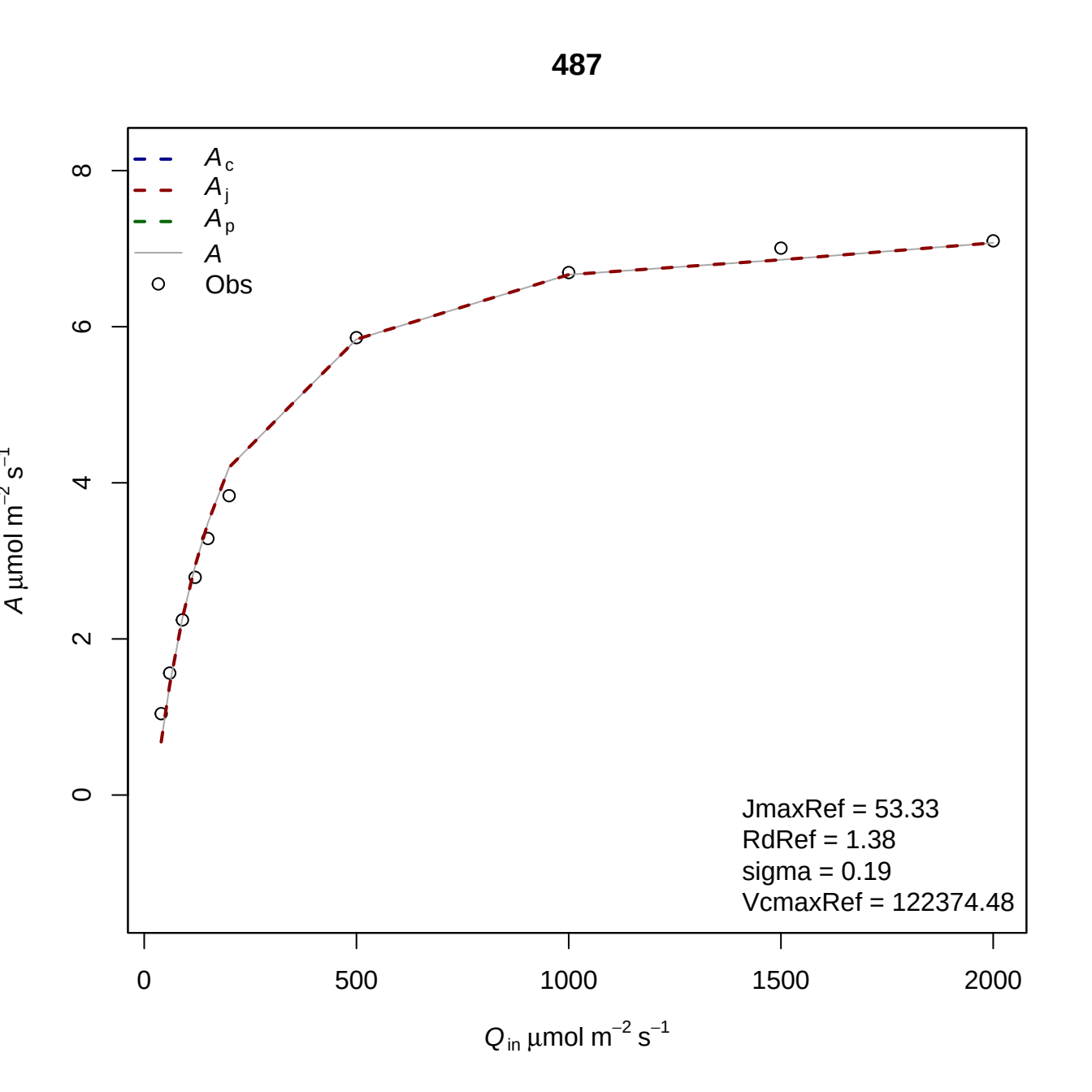

481
A_c
A_j
A_p
A
Obs
JmaxRef = 73.19
RdRef = 0.84
sigma = 0.14
VcmaxRef = 30.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

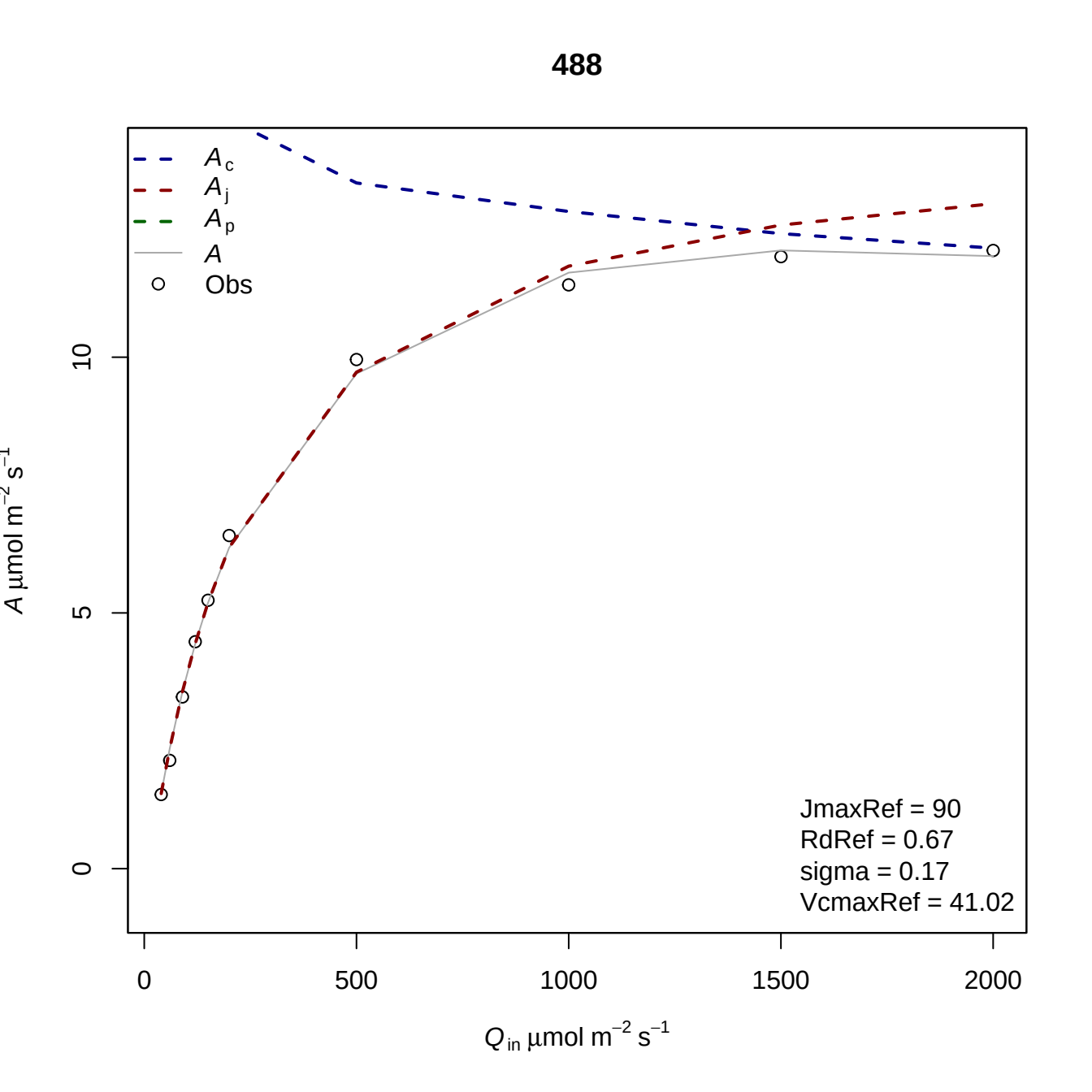

482
A_c
A_j
A_p
A
Obs
JmaxRef = 144.96
RdRef = 1.56
sigma = 0.24
VcmaxRef = 61.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

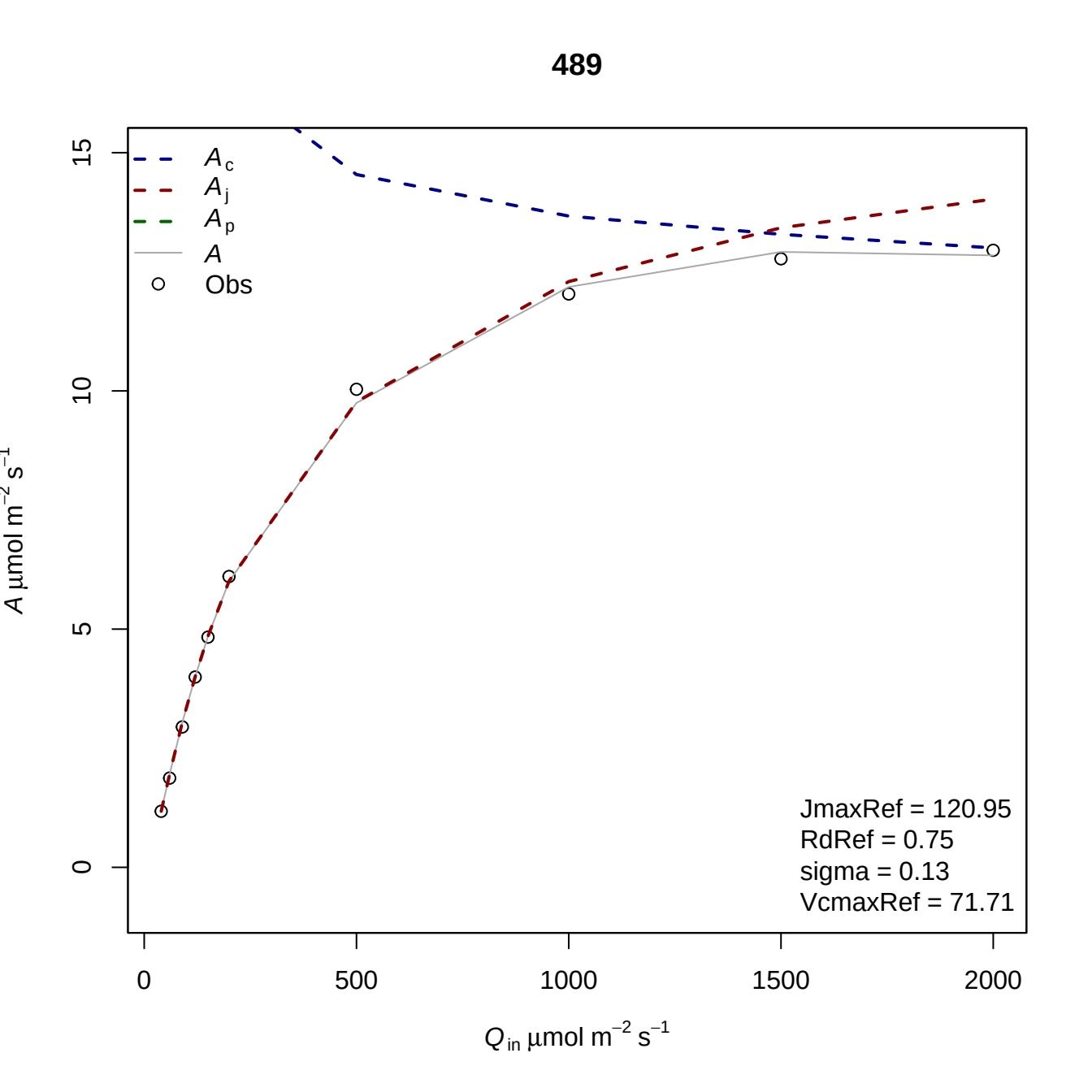

483
A_c
A_j
A_p
A
Obs
JmaxRef = 109.48
RdRef = 1.59
sigma = 0.06
VcmaxRef = 64.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

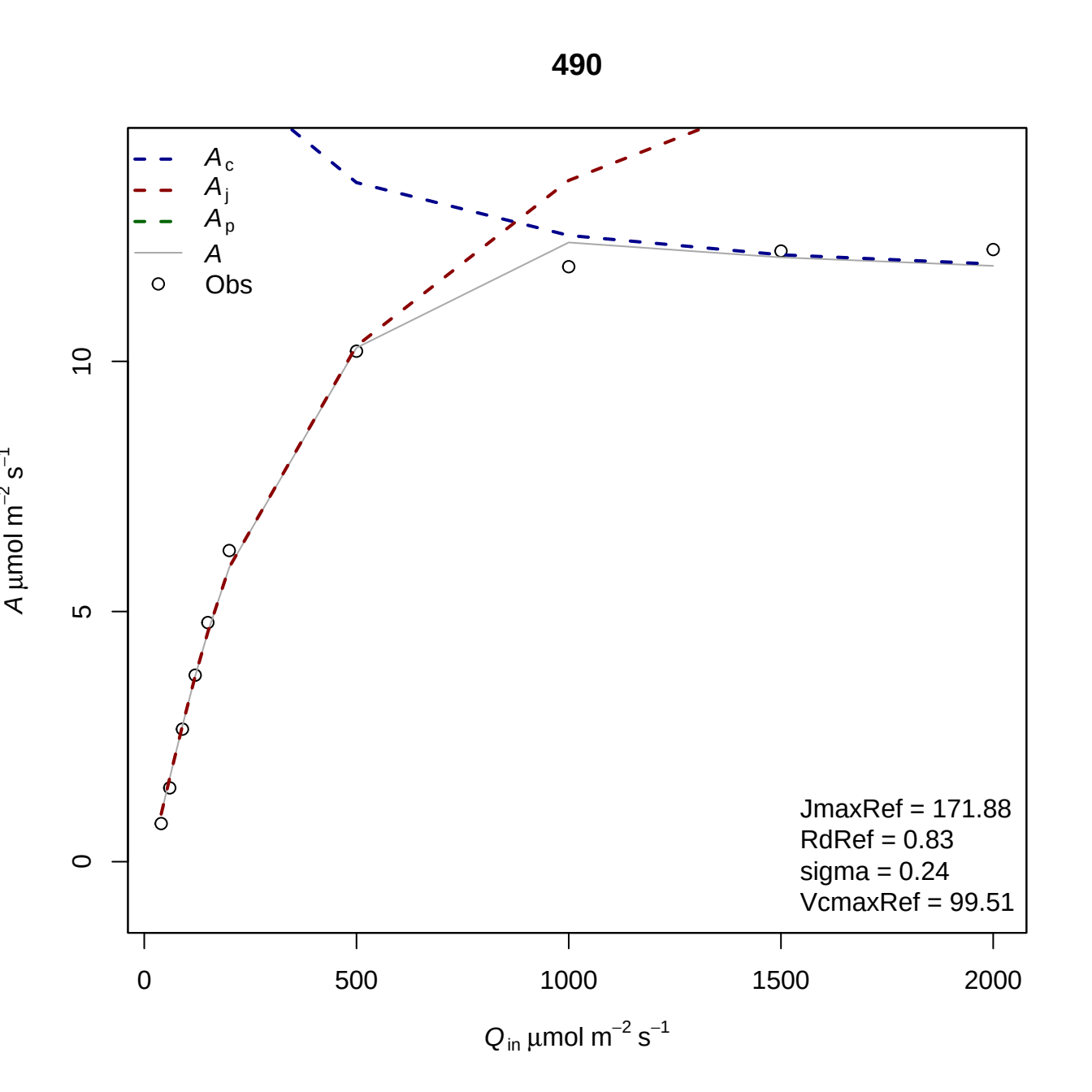

484
A_c
A_j
A_p
A
Obs
JmaxRef = 143.3
RdRef = 1.93
sigma = 0.14
VcmaxRef = 115.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
500
1000
1500
2000

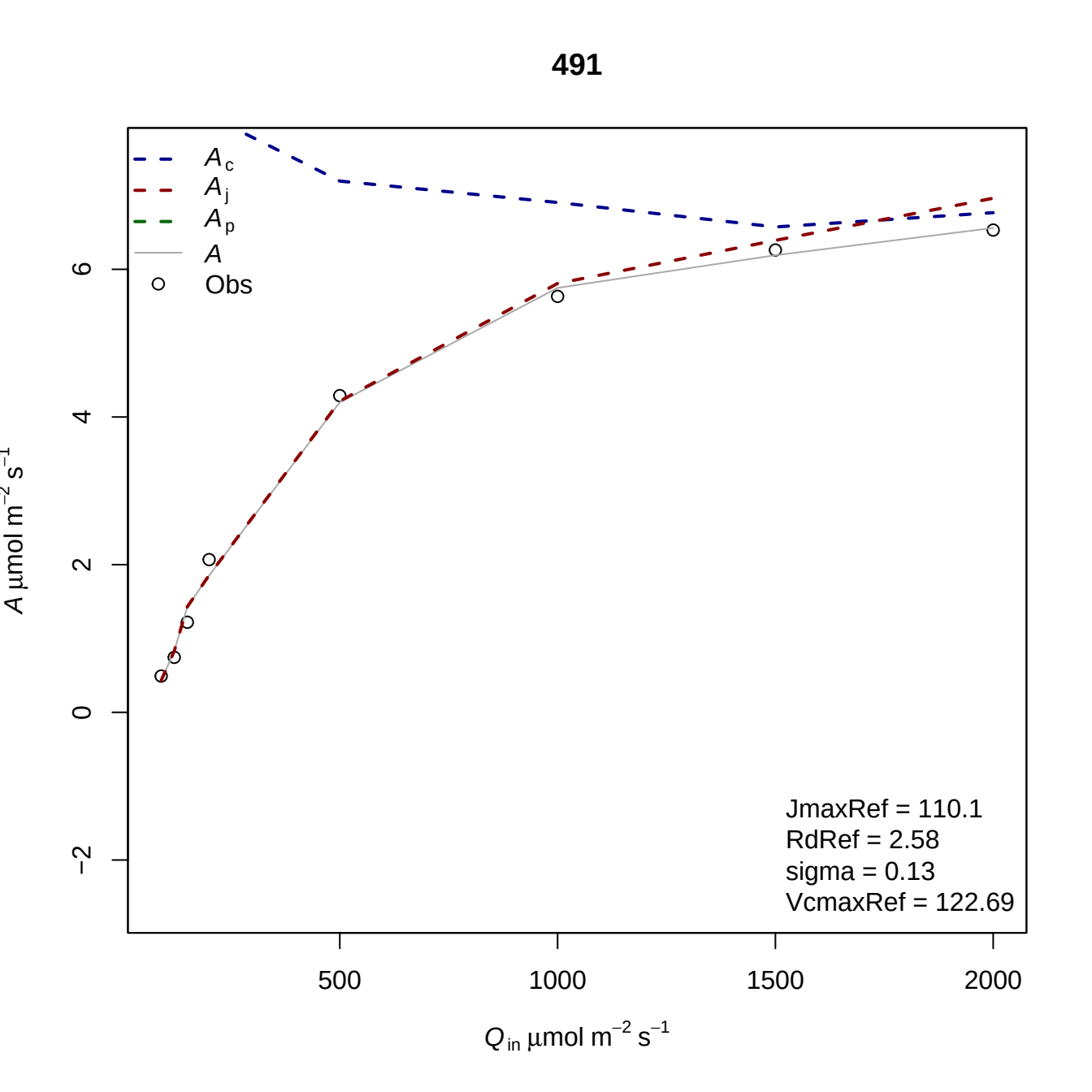

486
A_c
A_j
A_p
A
Obs
JmaxRef = 220.81
RdRef = 1.8
sigma = 0.18
VcmaxRef = 250.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
500
1000
1500
2000

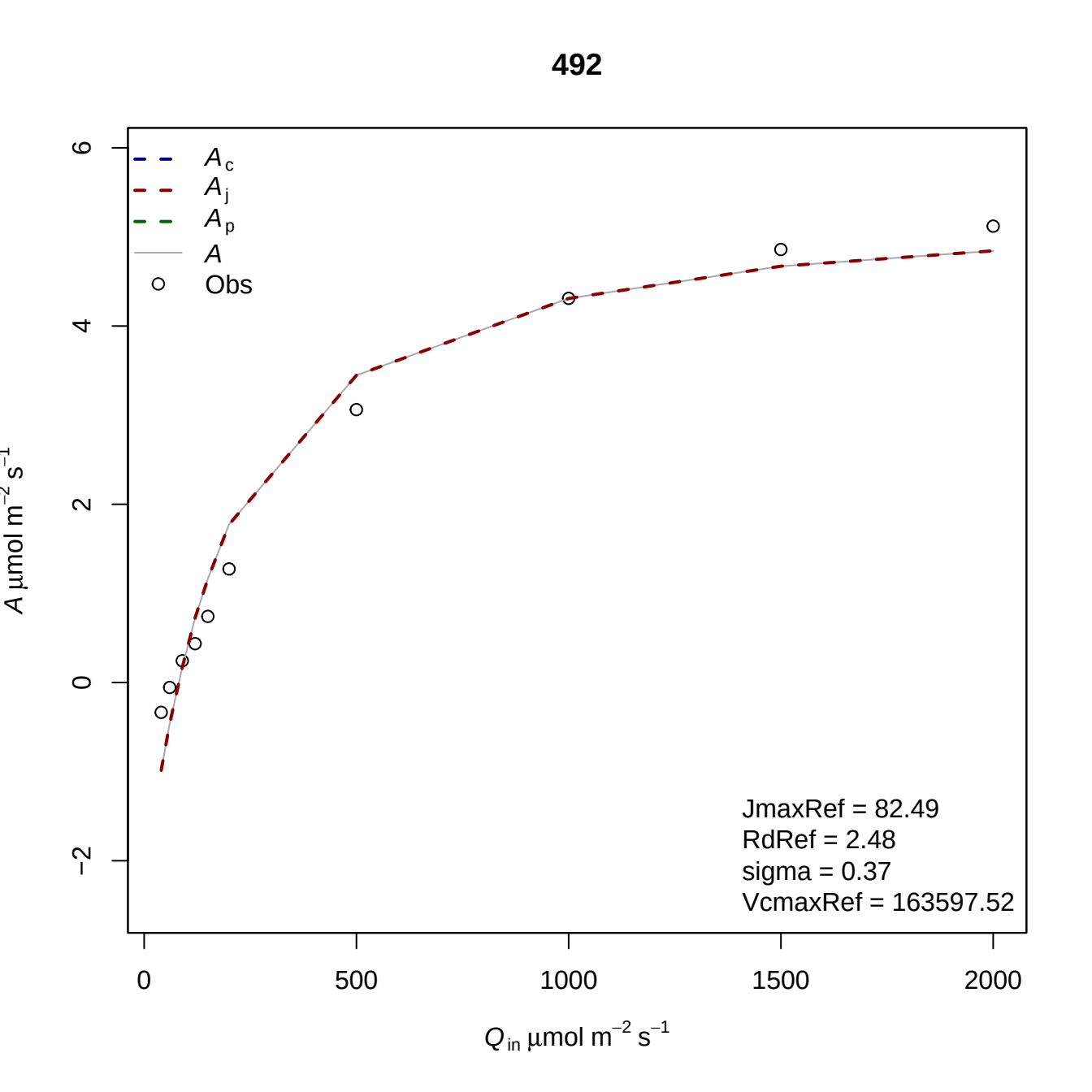

487
A_c
A_j
A_p
A
Obs
JmaxRef = 53.33
RdRef = 1.38
sigma = 0.19
VcmaxRef = 122374.48
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

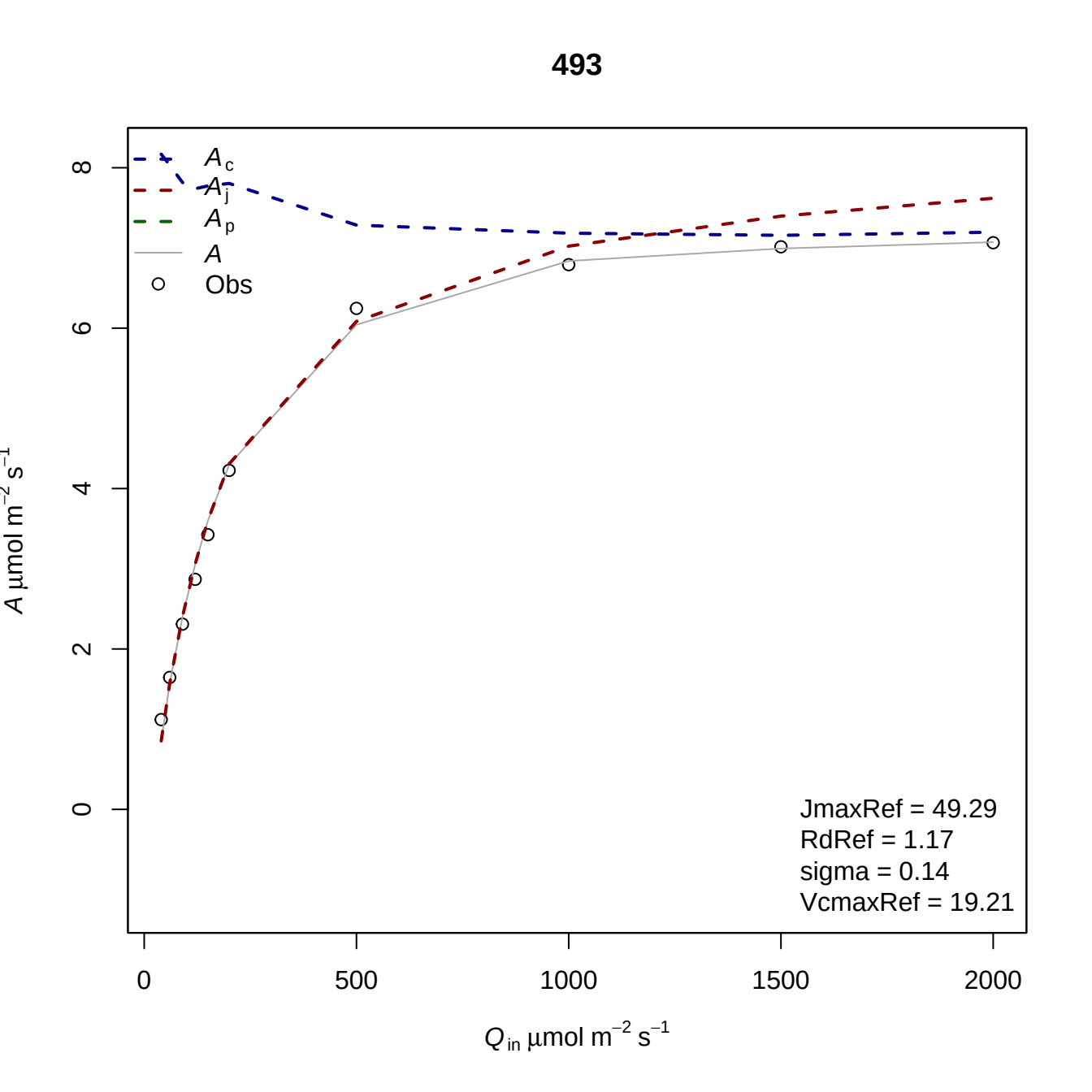

488
A_c
A_j
A_p
A
Obs
JmaxRef = 90
RdRef = 0.67
sigma = 0.17
VcmaxRef = 41.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

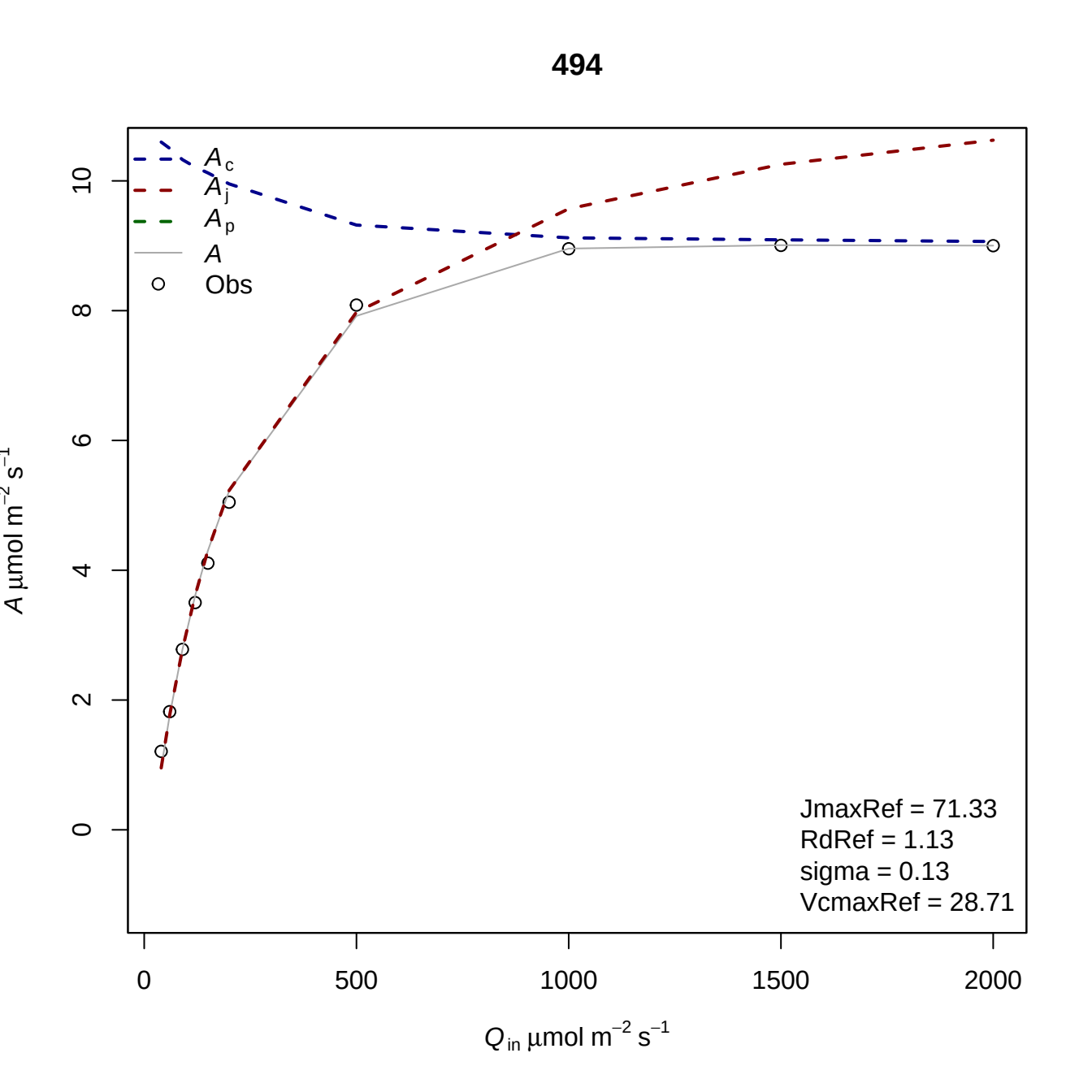

489
A_c
A_j
A_p
A
Obs
JmaxRef = 120.95
RdRef = 0.75
sigma = 0.13
VcmaxRef = 71.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

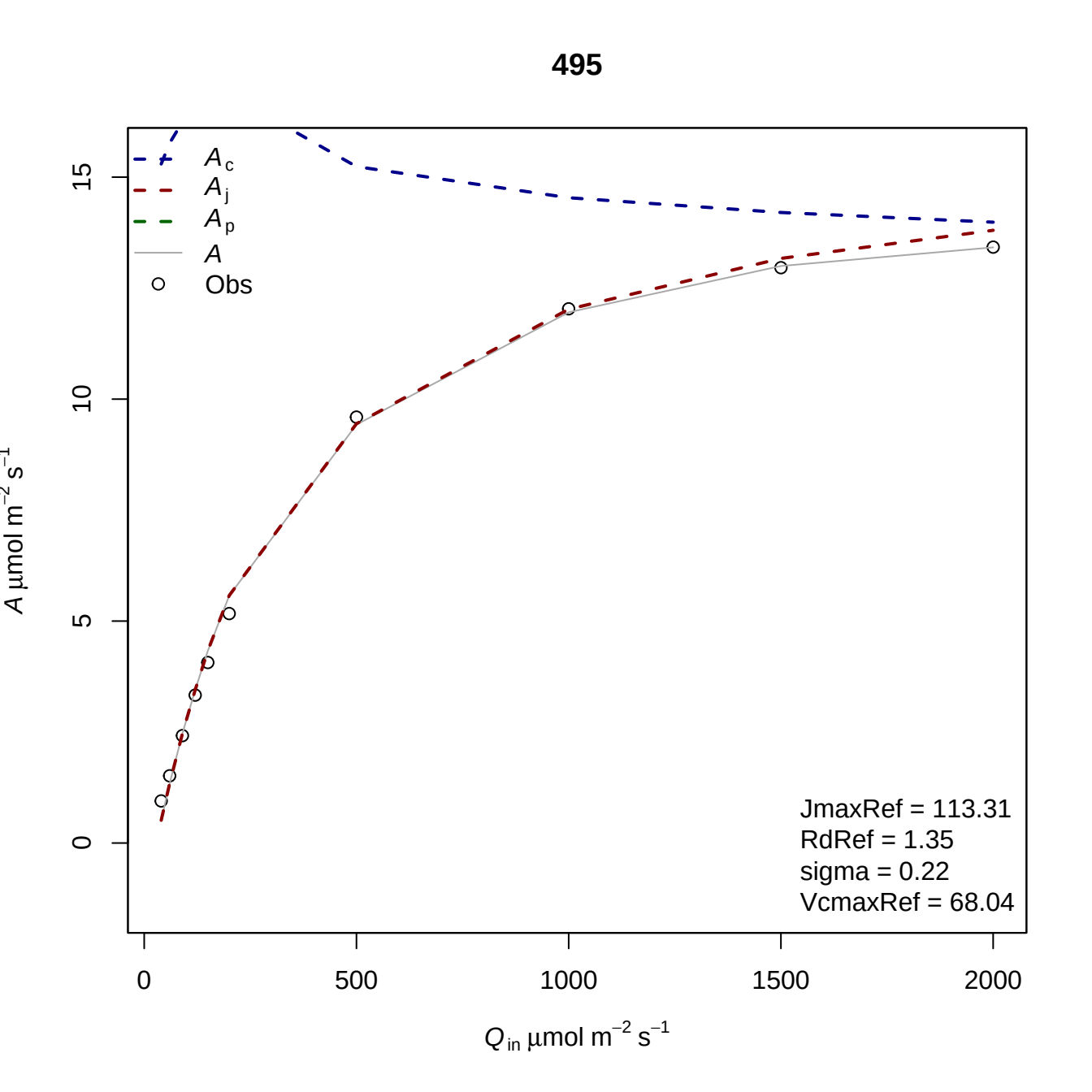

490
A_c
A_j
A_p
A
Obs
JmaxRef = 171.88
RdRef = 0.83
sigma = 0.24
VcmaxRef = 99.51
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

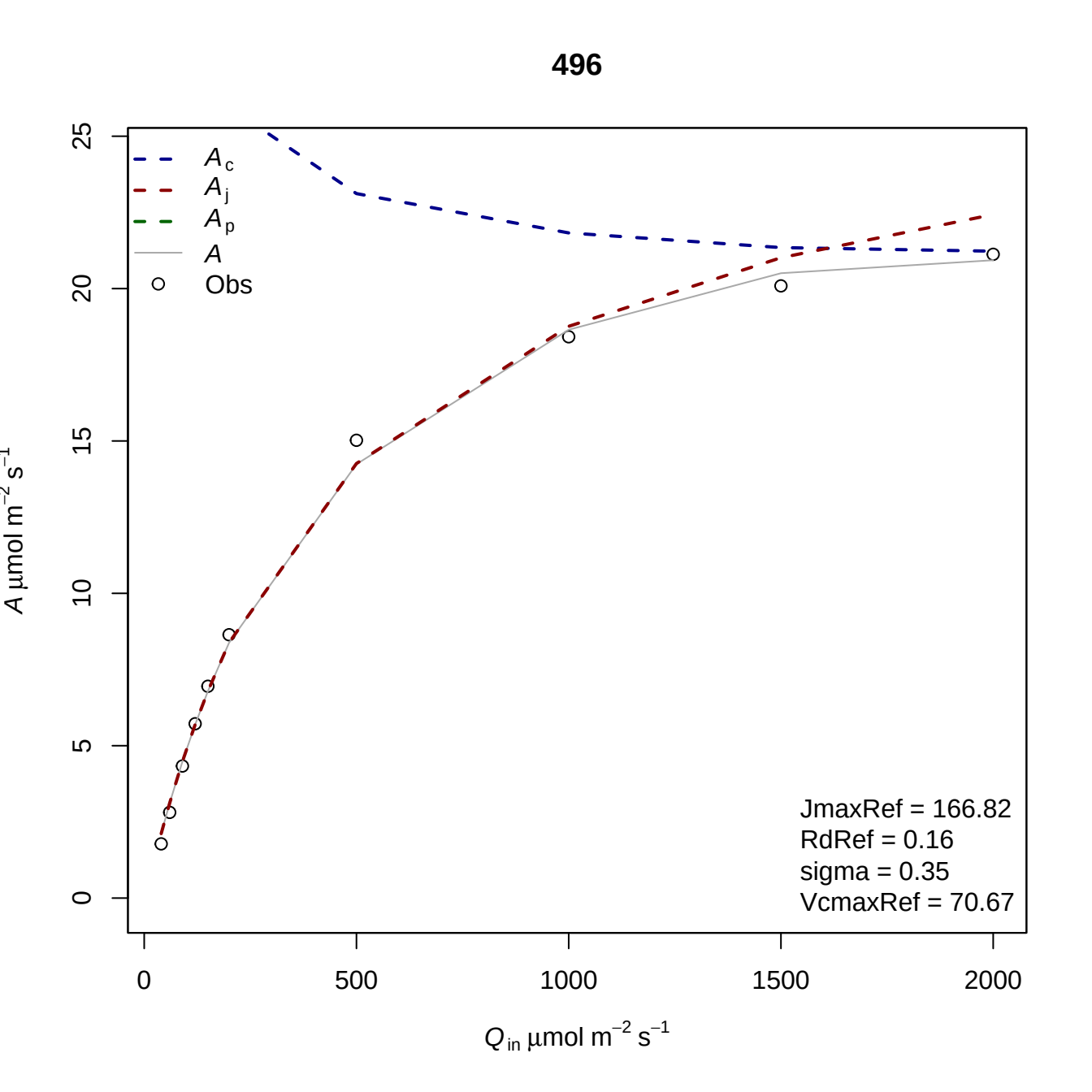

491
A_c
A_j
A_p
A
Obs
JmaxRef = 110.1
RdRef = 2.58
sigma = 0.13
VcmaxRef = 122.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
500
1000
1500
2000

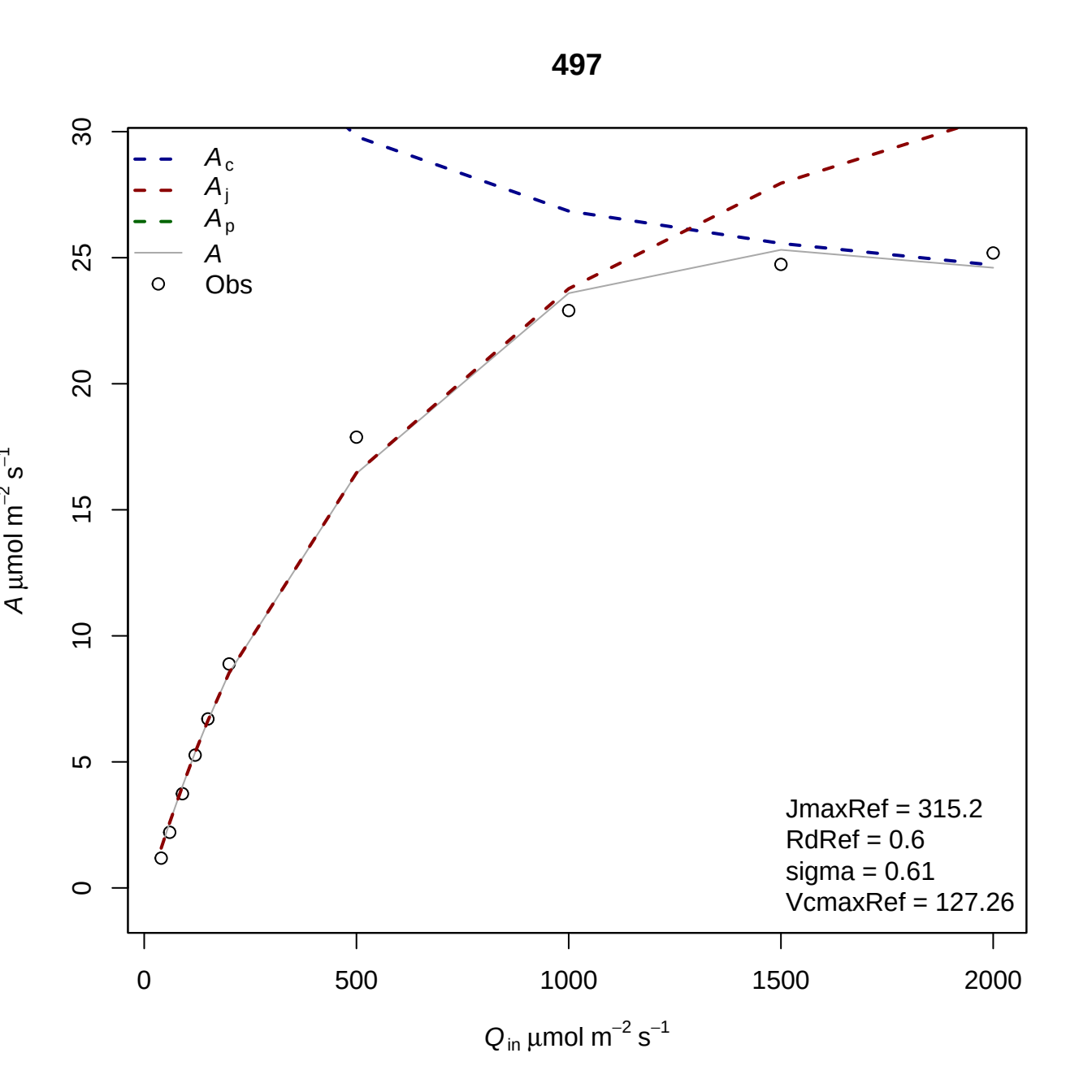

492
A_c
A_j
A_p
A
Obs
JmaxRef = 82.49
RdRef = 2.48
sigma = 0.37
VcmaxRef = 163597.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

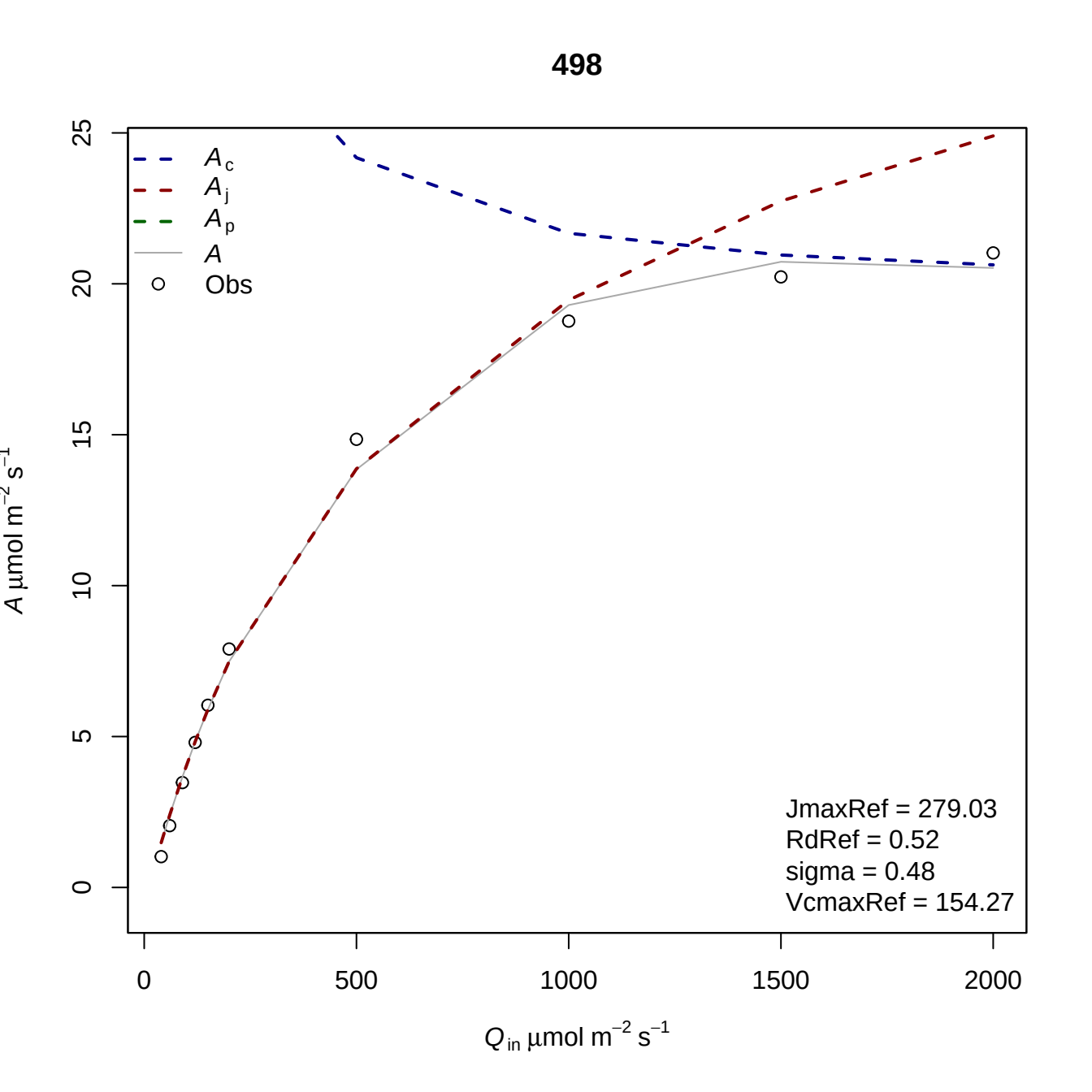

493
A_c
A_j
A_p
A
Obs
JmaxRef = 49.29
RdRef = 1.17
sigma = 0.14
VcmaxRef = 19.21
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

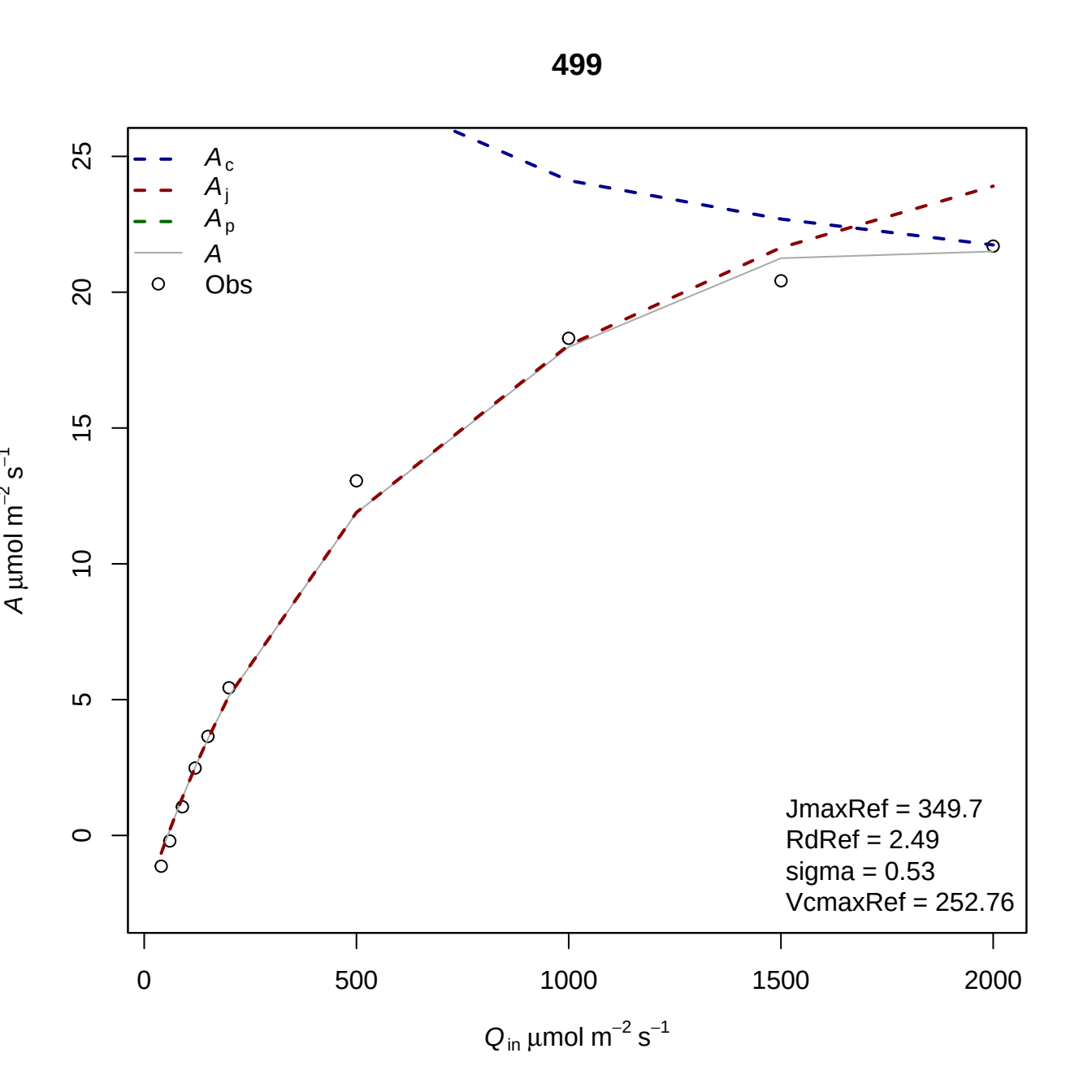

494
A_c
A_j
A_p
A
Obs
JmaxRef = 71.33
RdRef = 1.13
sigma = 0.13
VcmaxRef = 28.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

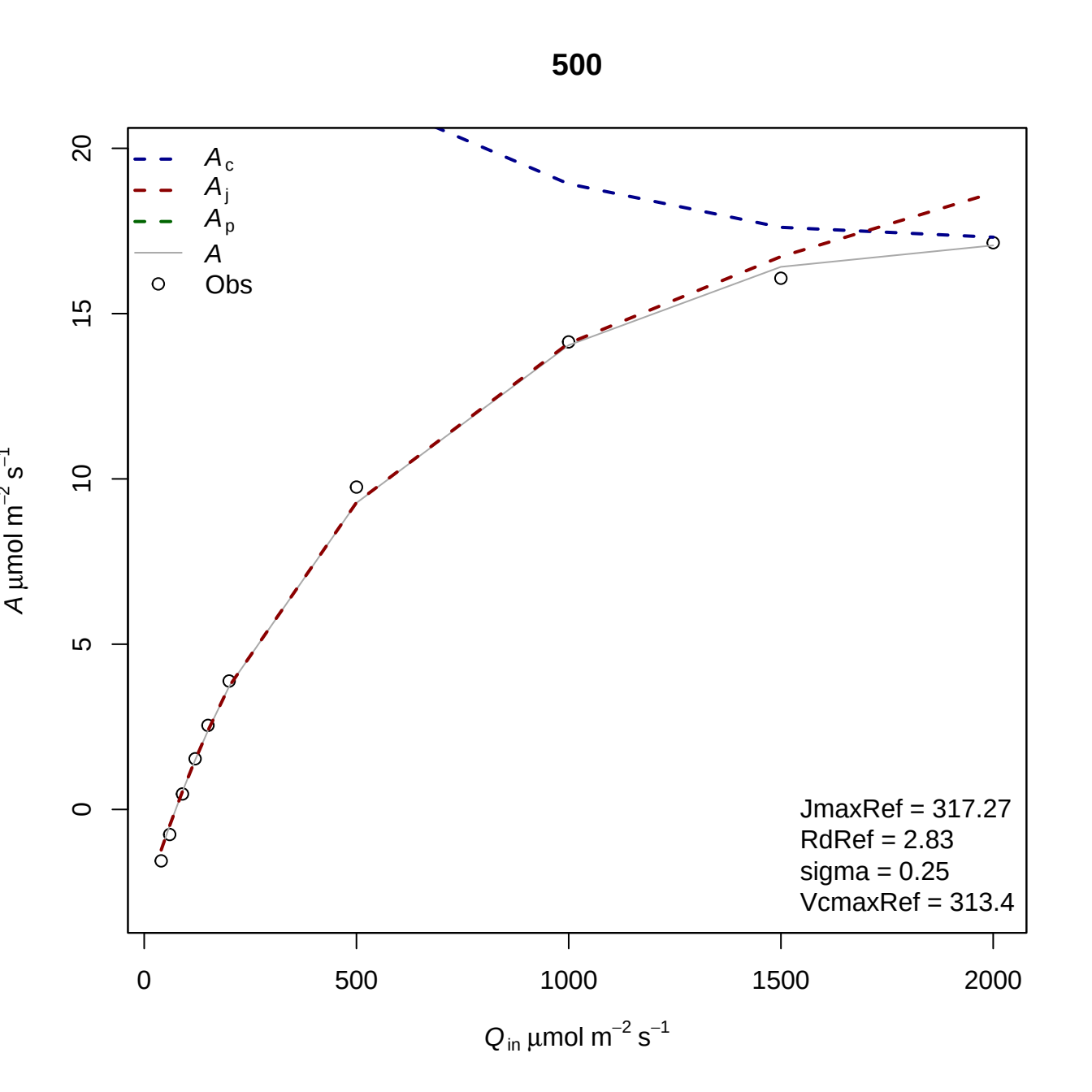

495
A_c
A_j
A_p
A
Obs
JmaxRef = 113.31
RdRef = 1.35
sigma = 0.22
VcmaxRef = 68.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

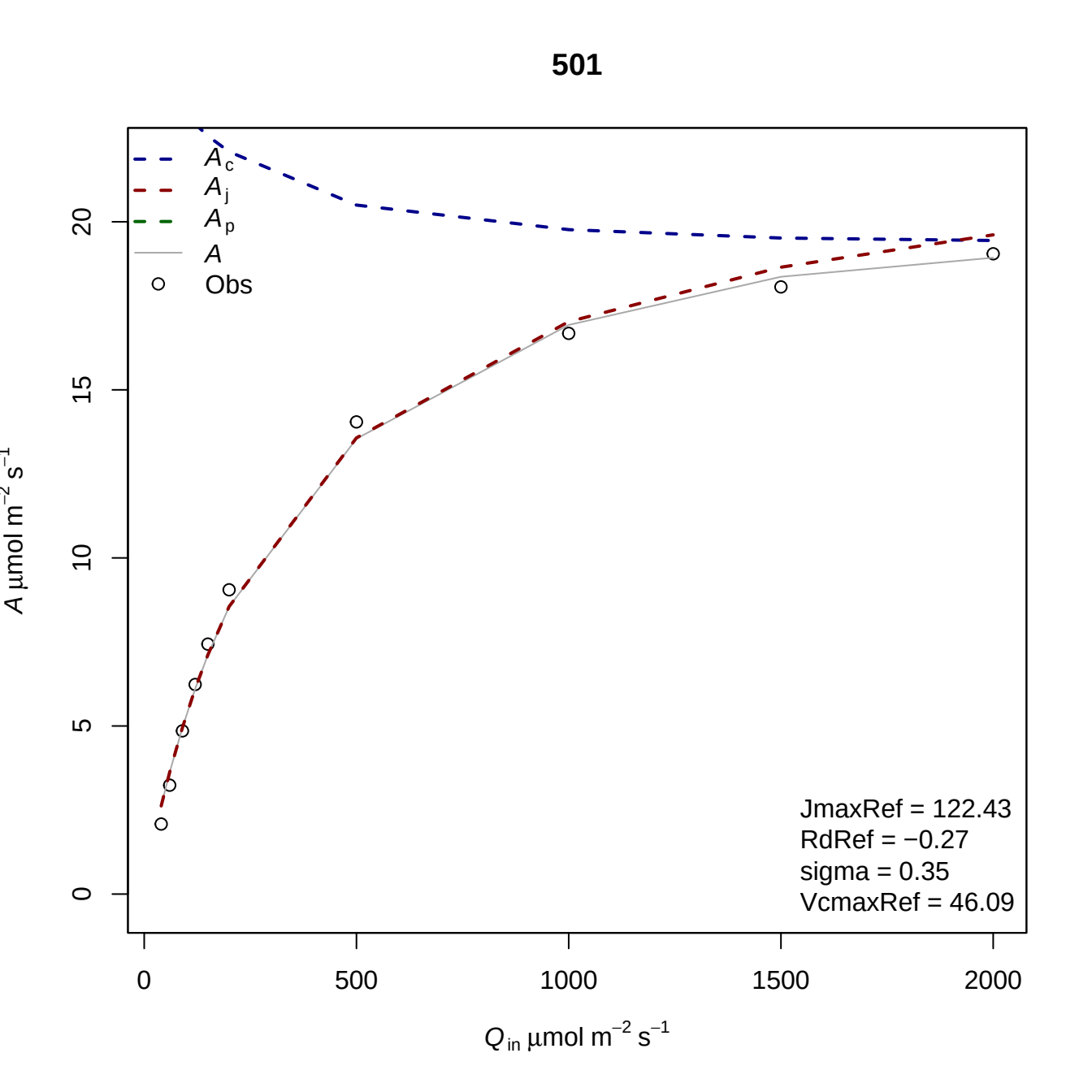

496
A_c
A_j
A_p
A
Obs
JmaxRef = 166.82
RdRef = 0.16
sigma = 0.35
VcmaxRef = 70.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

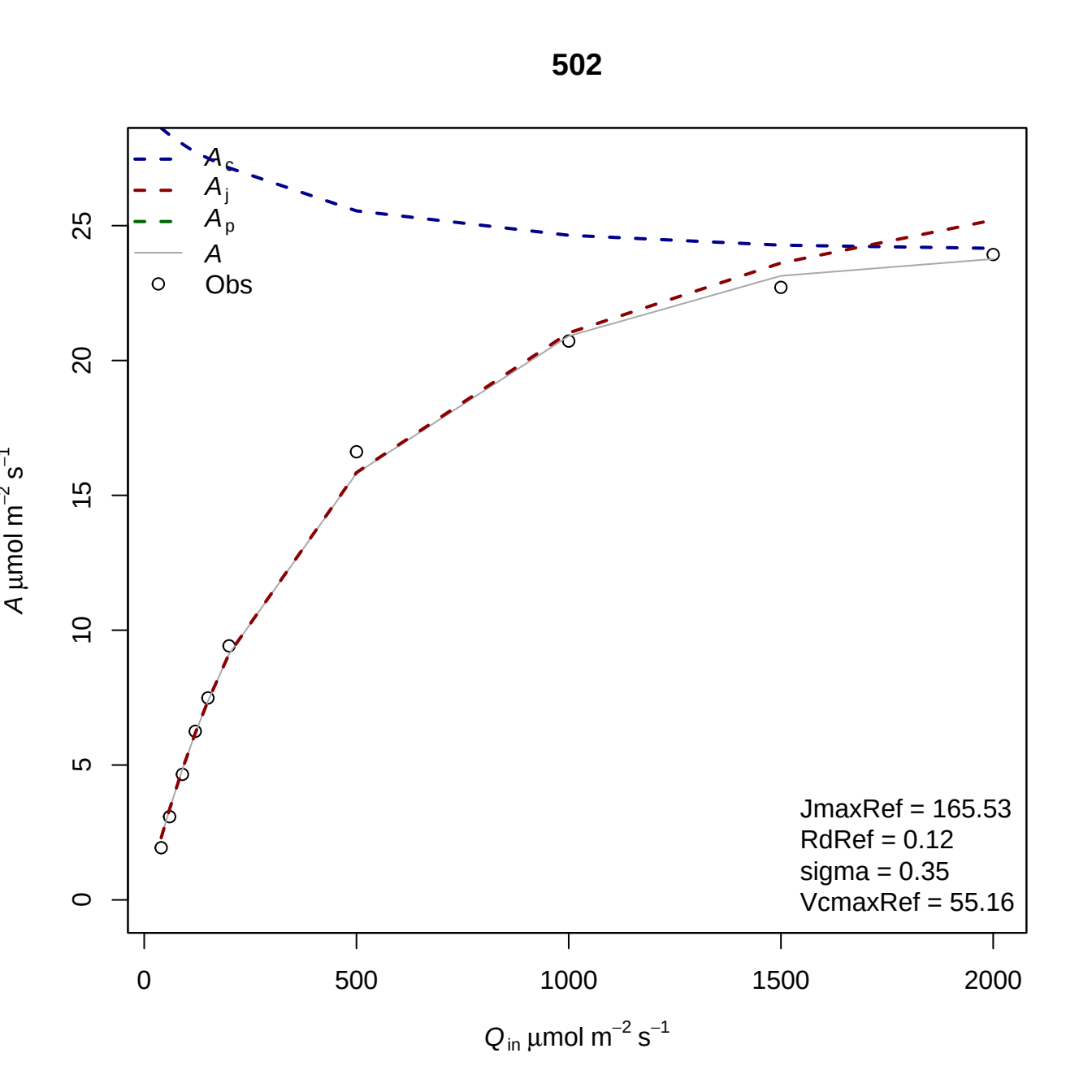

497
A_c
A_j
A_p
A
Obs
JmaxRef = 315.2
RdRef = 0.6
sigma = 0.61
VcmaxRef = 127.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

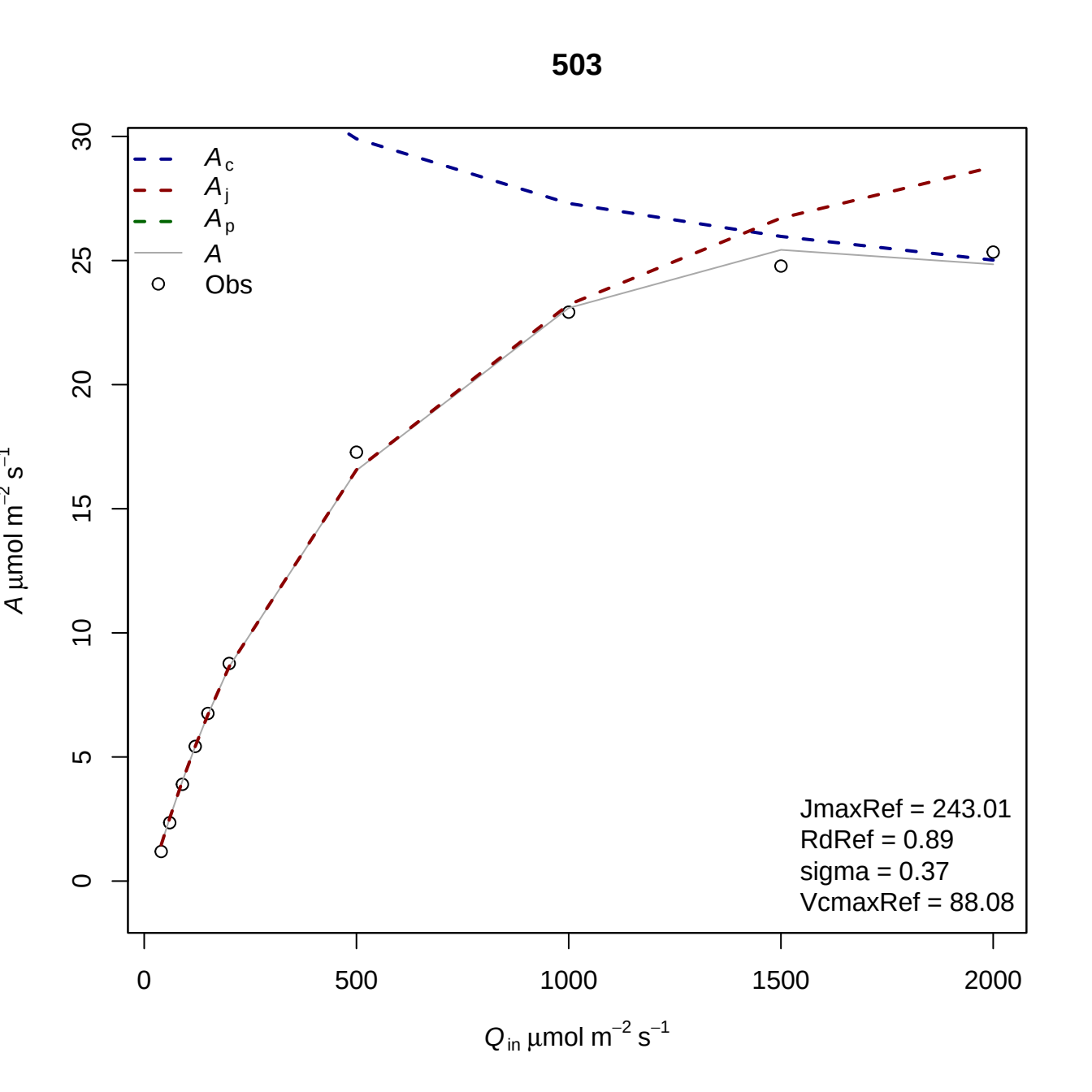

498
A_c
A_j
A_p
A
Obs
JmaxRef = 279.03
RdRef = 0.52
sigma = 0.48
VcmaxRef = 154.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

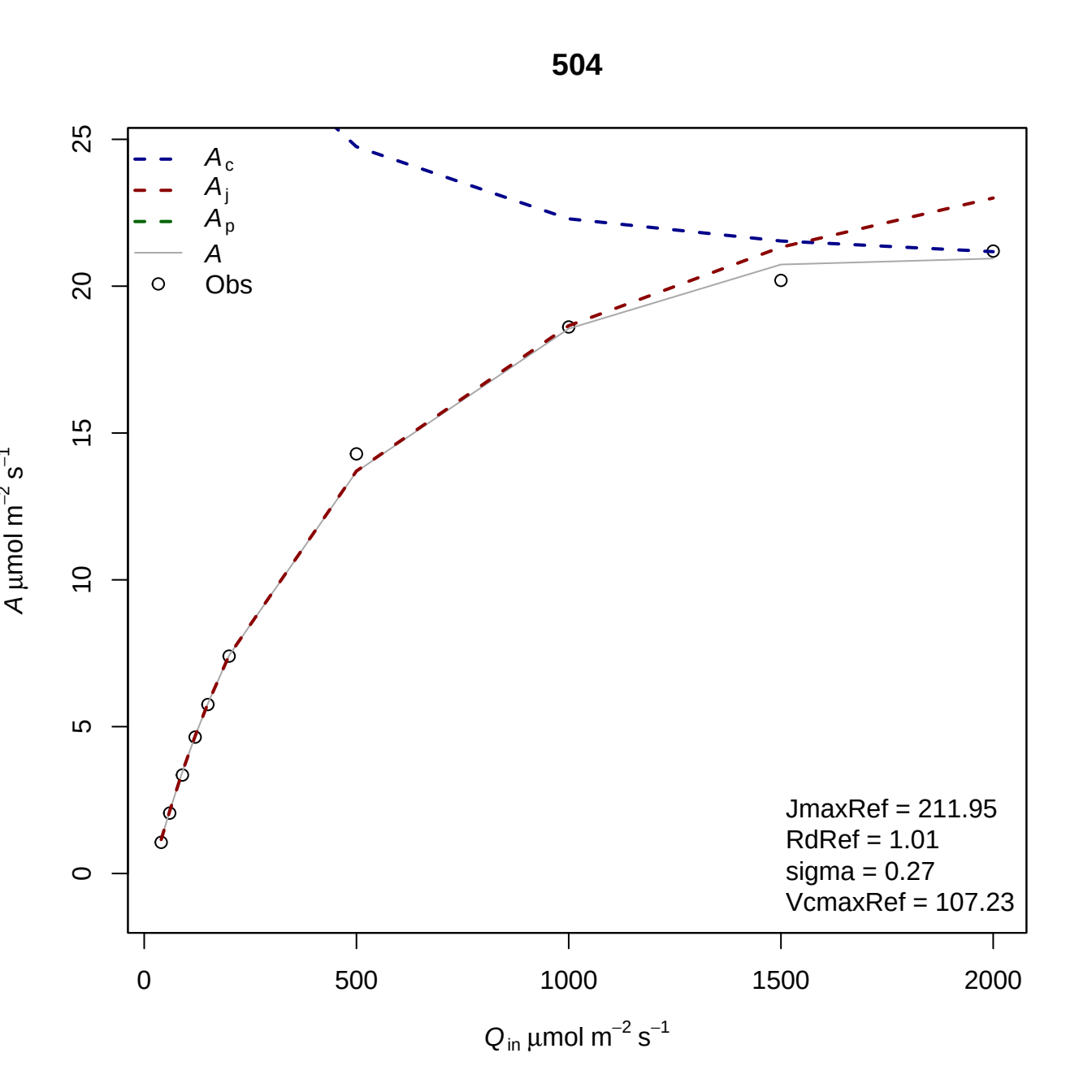

499
A_c
A_j
A_p
A
Obs
JmaxRef = 349.7
RdRef = 2.49
sigma = 0.53
VcmaxRef = 252.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

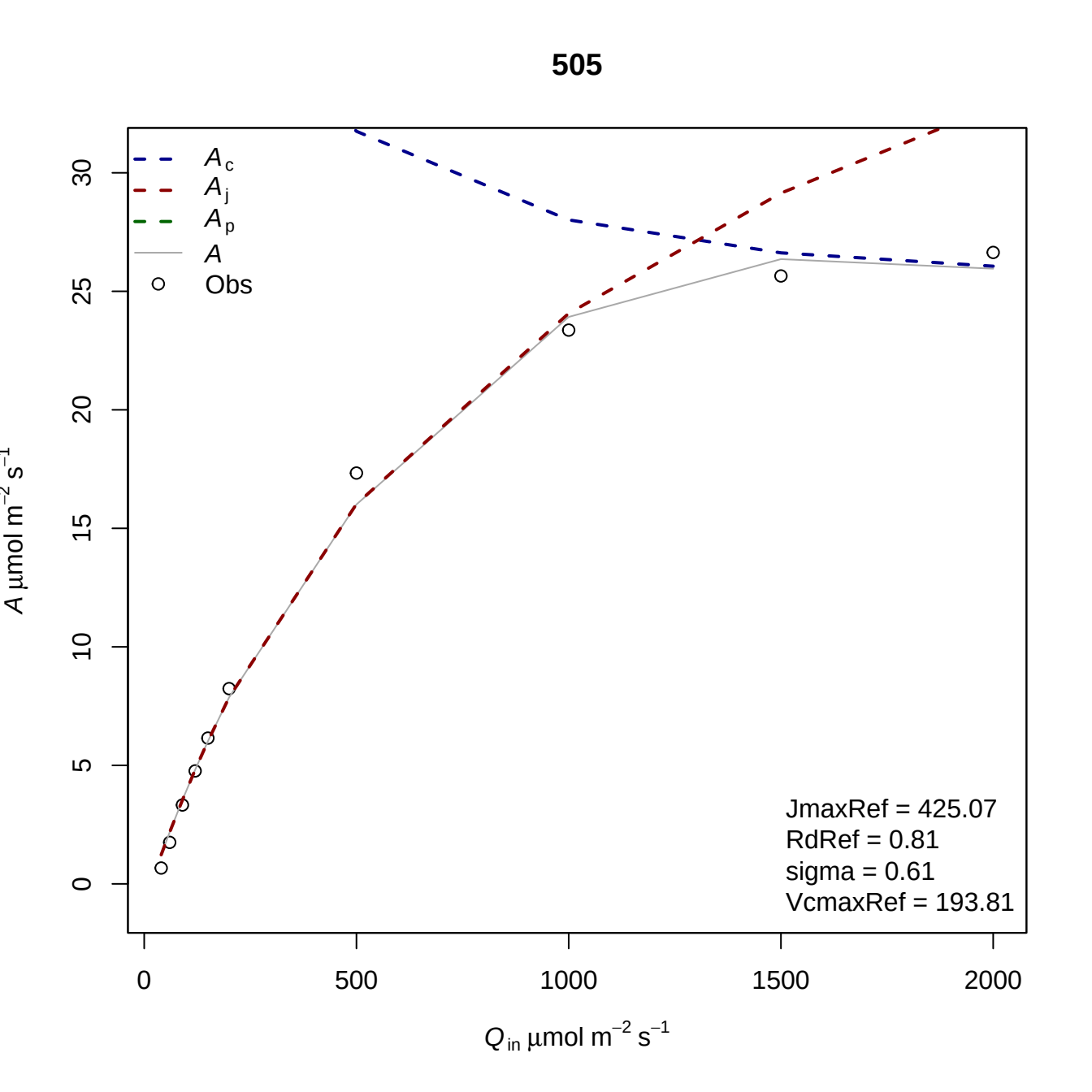

500
A_c
A_j
A_p
A
Obs
JmaxRef = 317.27
RdRef = 2.83
sigma = 0.25
VcmaxRef = 313.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

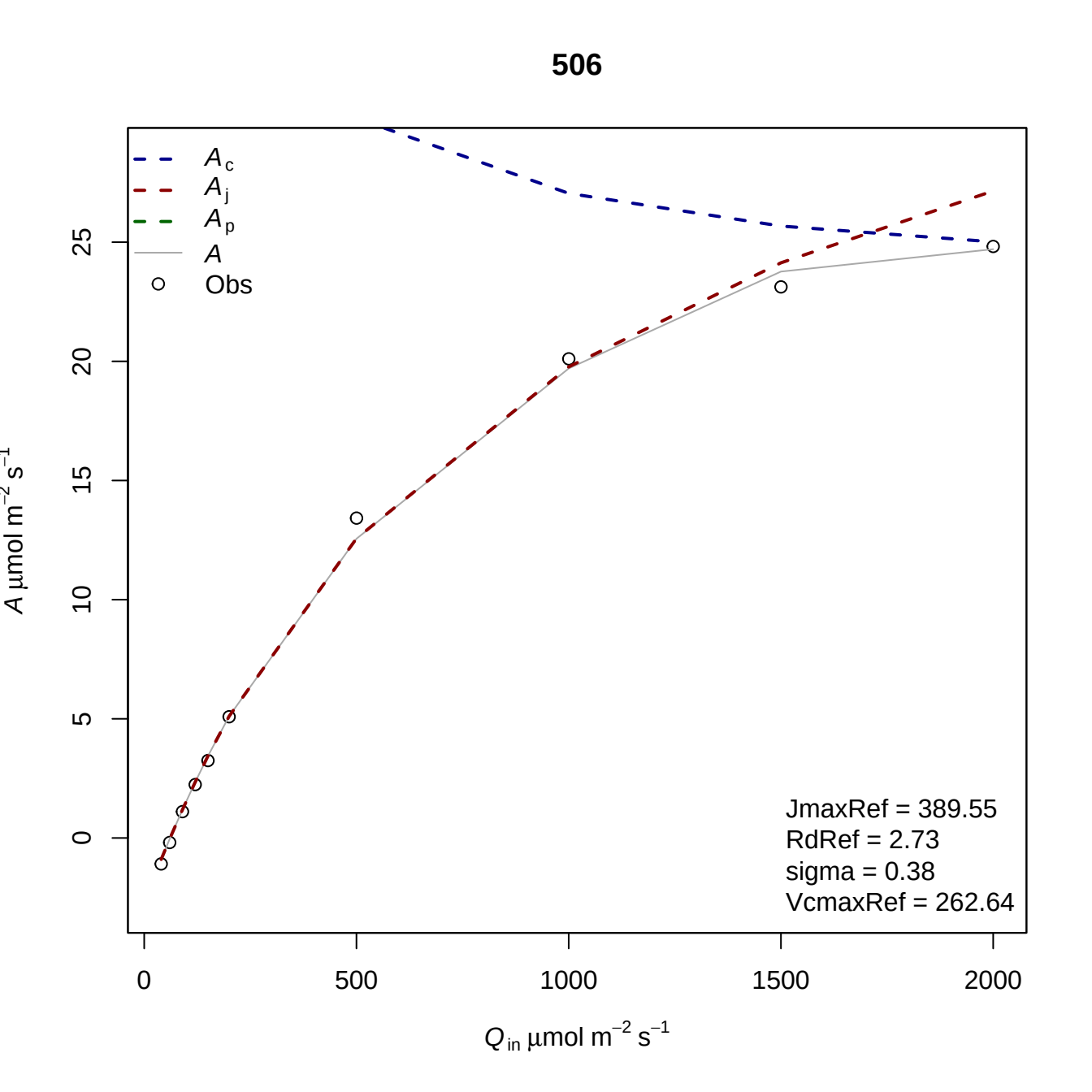

501
A_c
A_j
A_p
A
Obs
JmaxRef = 122.43
RdRef = −0.27
sigma = 0.35
VcmaxRef = 46.09
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

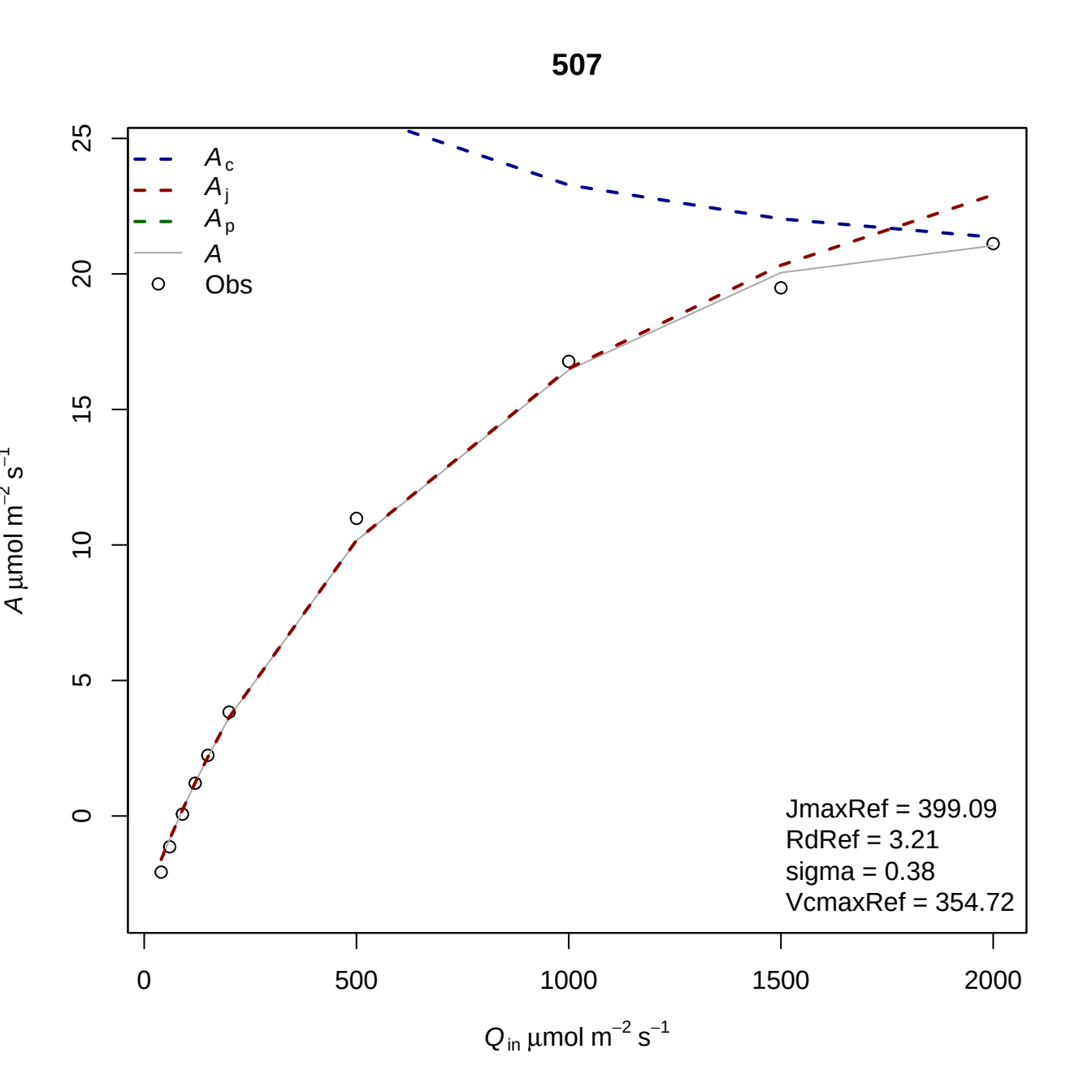

502
A_c
A_j
A_p
A
Obs
JmaxRef = 165.53
RdRef = 0.12
sigma = 0.35
VcmaxRef = 55.16
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

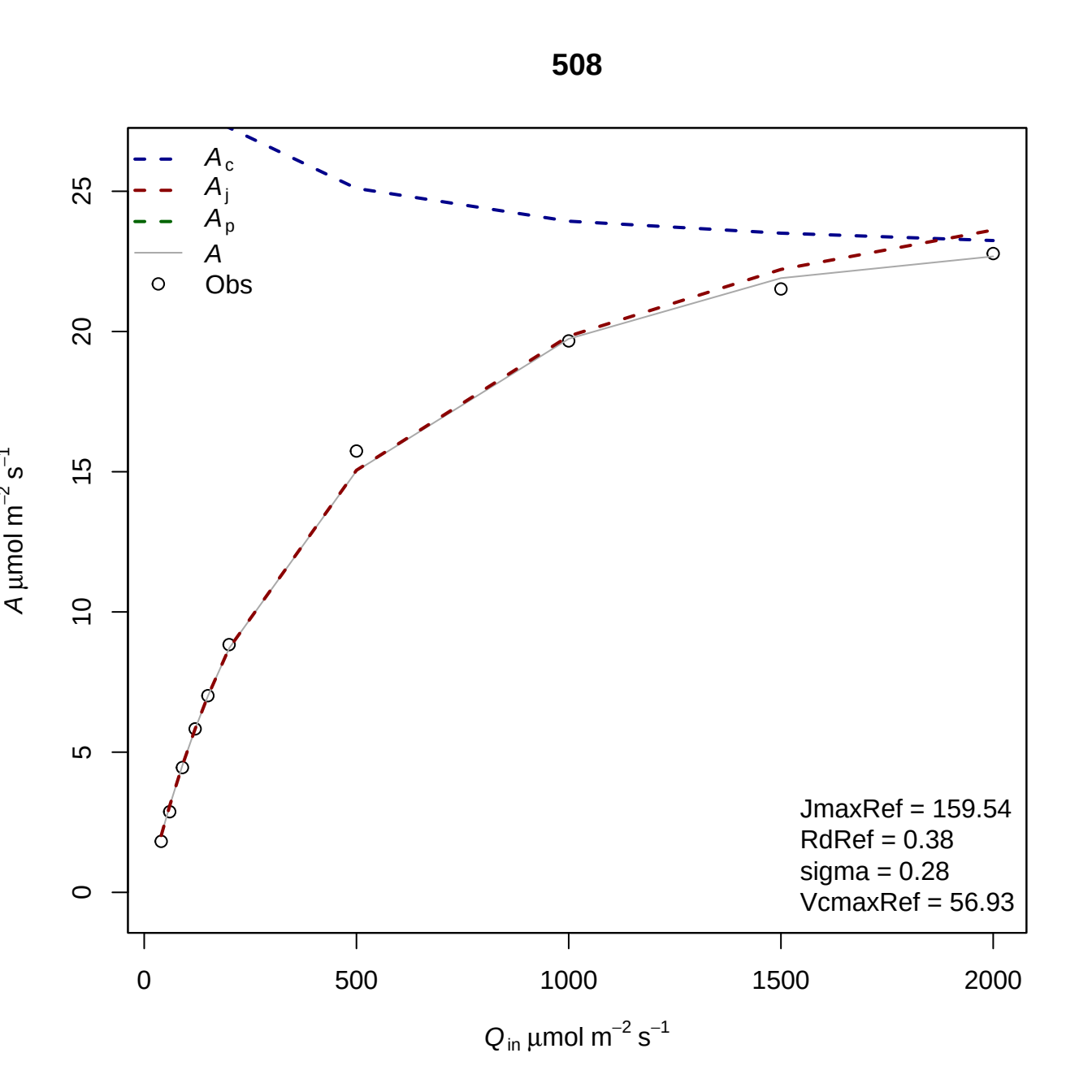

503
A_c
A_j
A_p
A
Obs
JmaxRef = 243.01
RdRef = 0.89
sigma = 0.37
VcmaxRef = 88.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

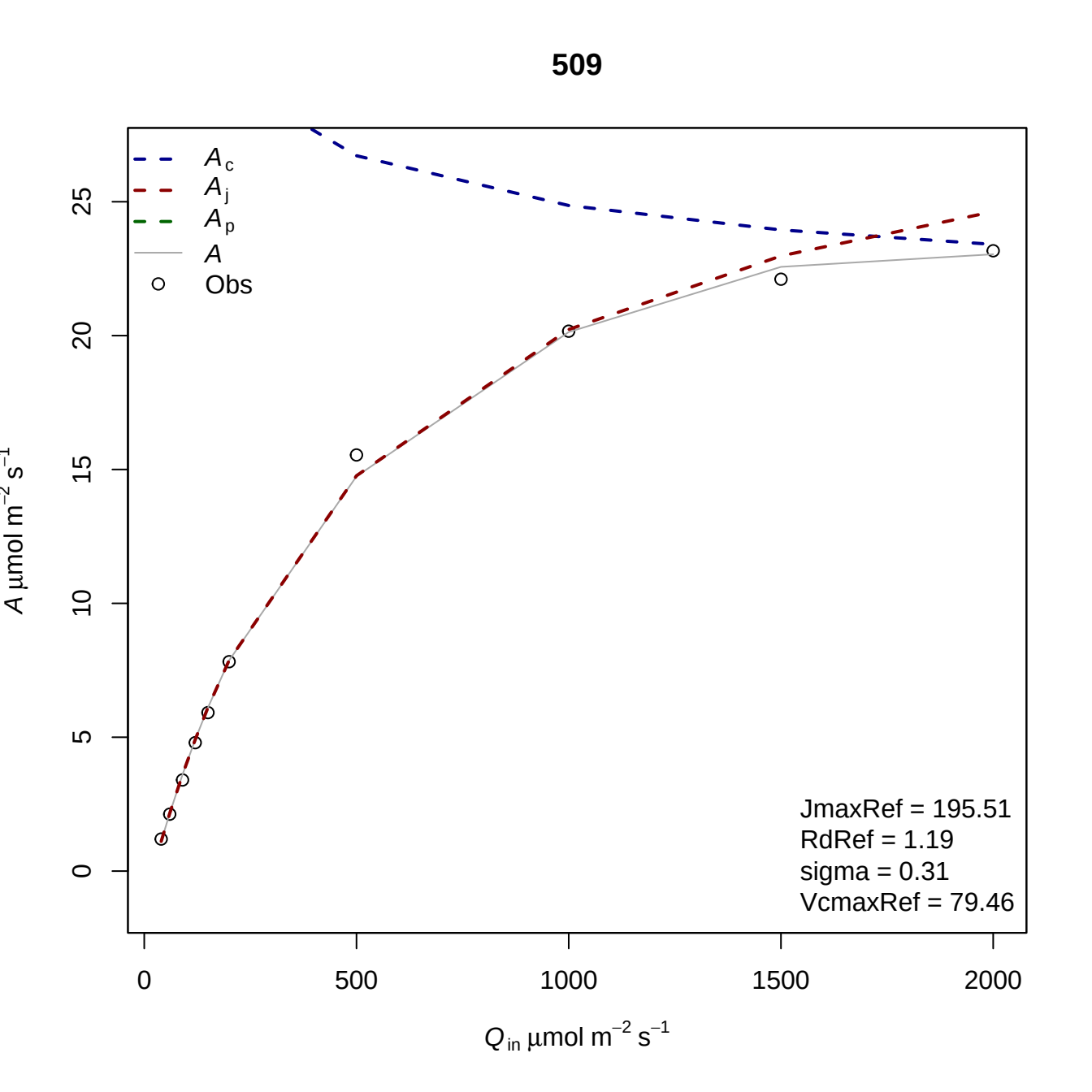

504
A_c
A_j
A_p
A
Obs
JmaxRef = 211.95
RdRef = 1.01
sigma = 0.27
VcmaxRef = 107.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

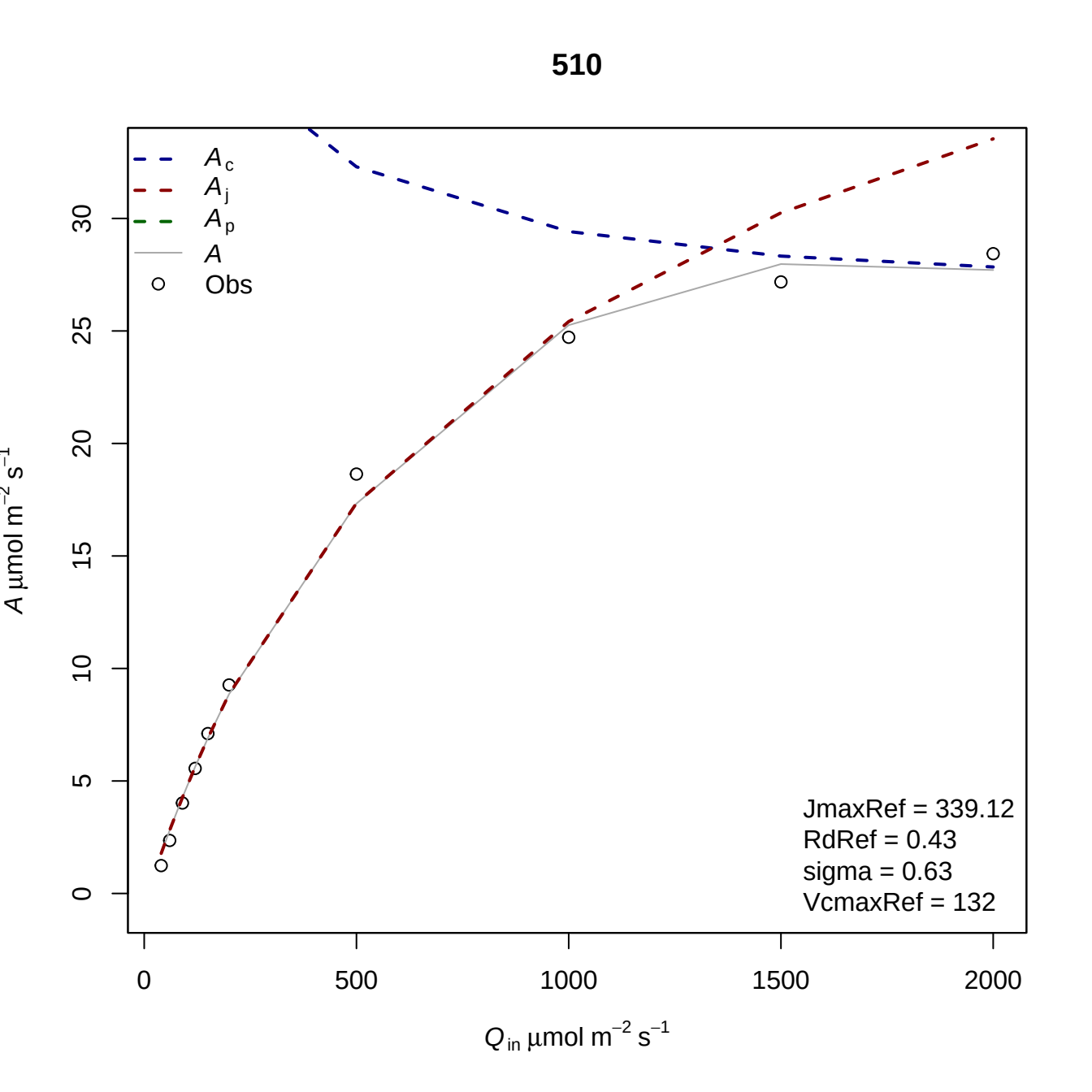

505
A_c
A_j
A_p
A
Obs
JmaxRef = 425.07
RdRef = 0.81
sigma = 0.61
VcmaxRef = 193.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

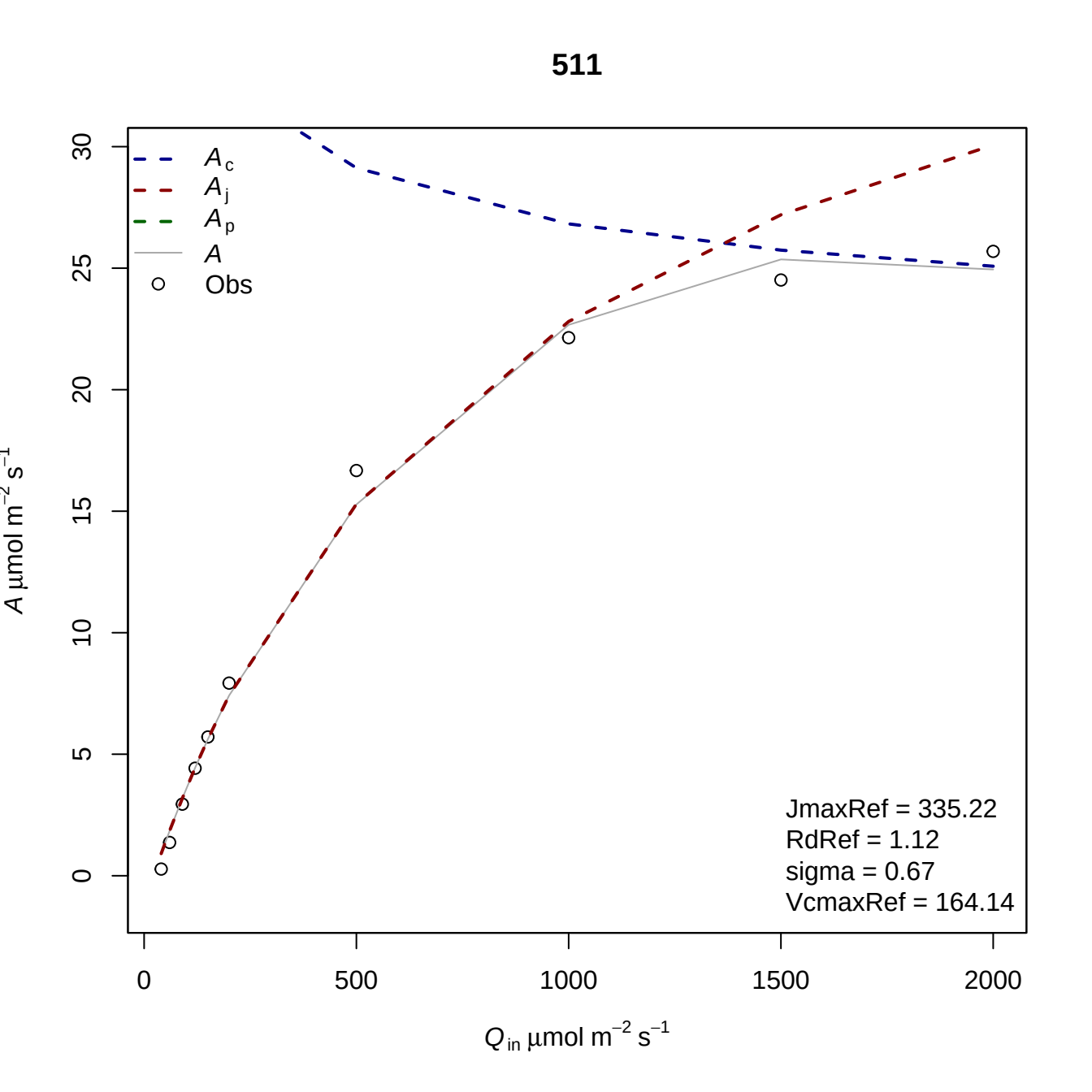

506
A_c
A_j
A_p
A
Obs
JmaxRef = 389.55
RdRef = 2.73
sigma = 0.38
VcmaxRef = 262.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

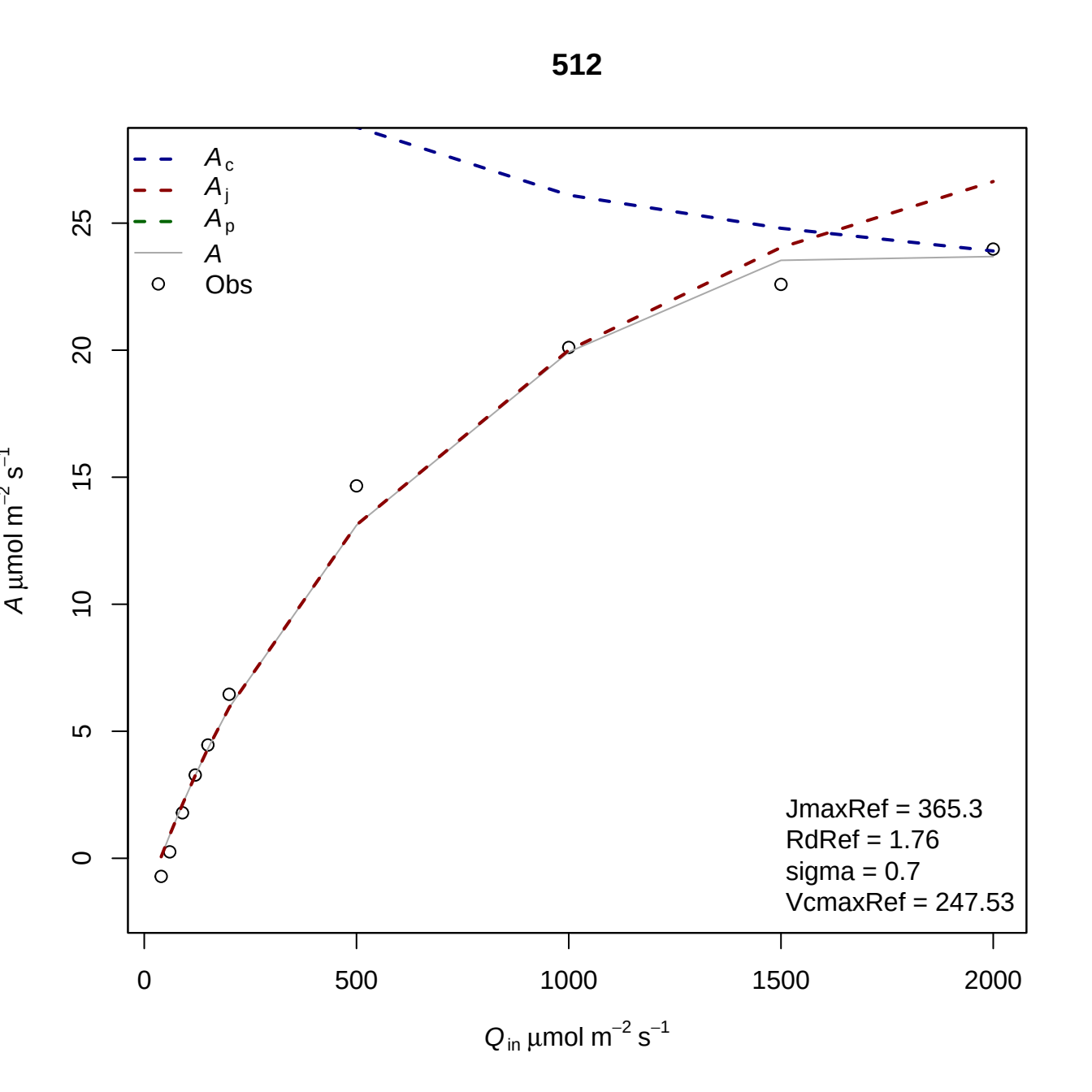

507
A_c
A_j
A_p
A
Obs
JmaxRef = 399.09
RdRef = 3.21
sigma = 0.38
VcmaxRef = 354.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

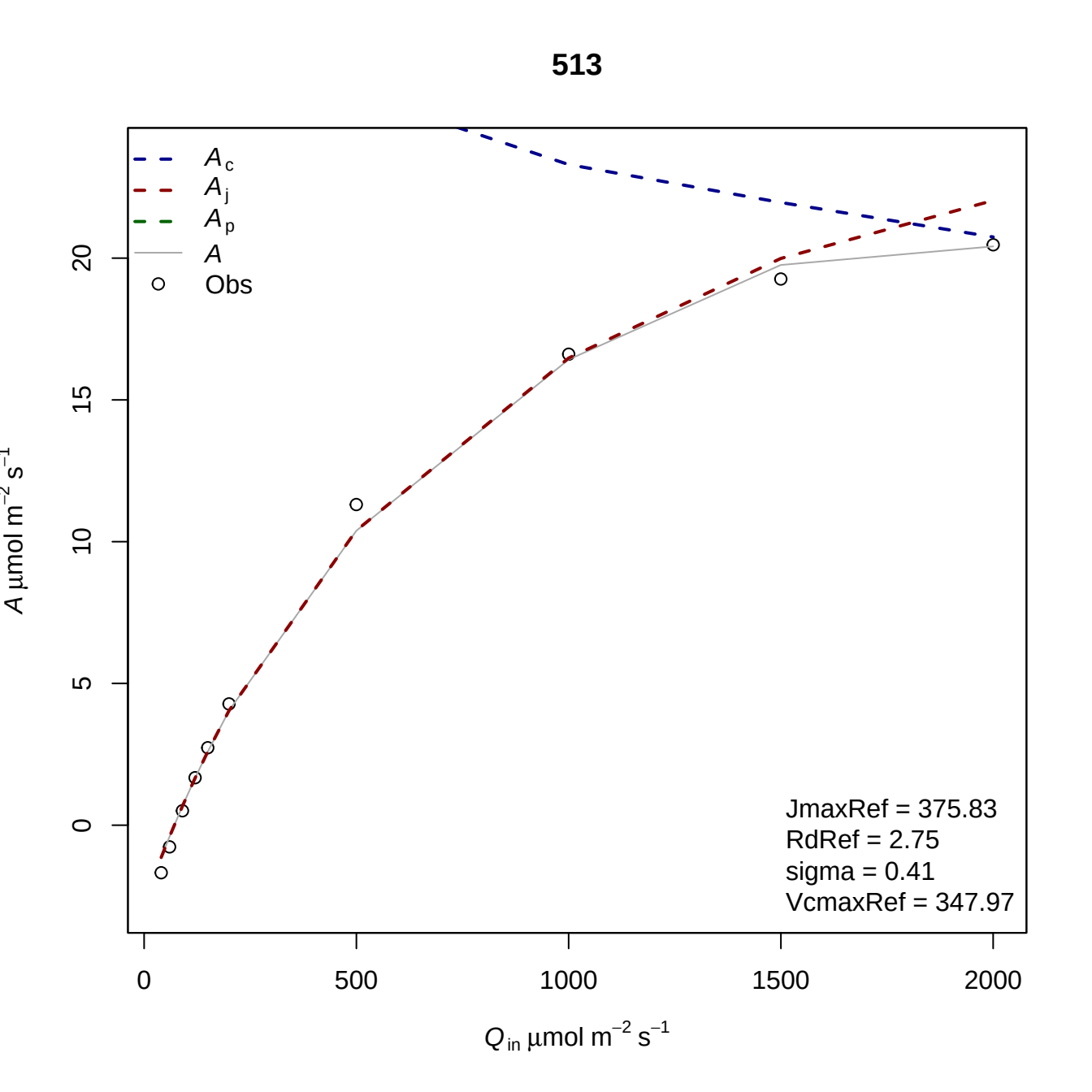

508
A_c
A_j
A_p
A
Obs
JmaxRef = 159.54
RdRef = 0.38
sigma = 0.28
VcmaxRef = 56.93
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

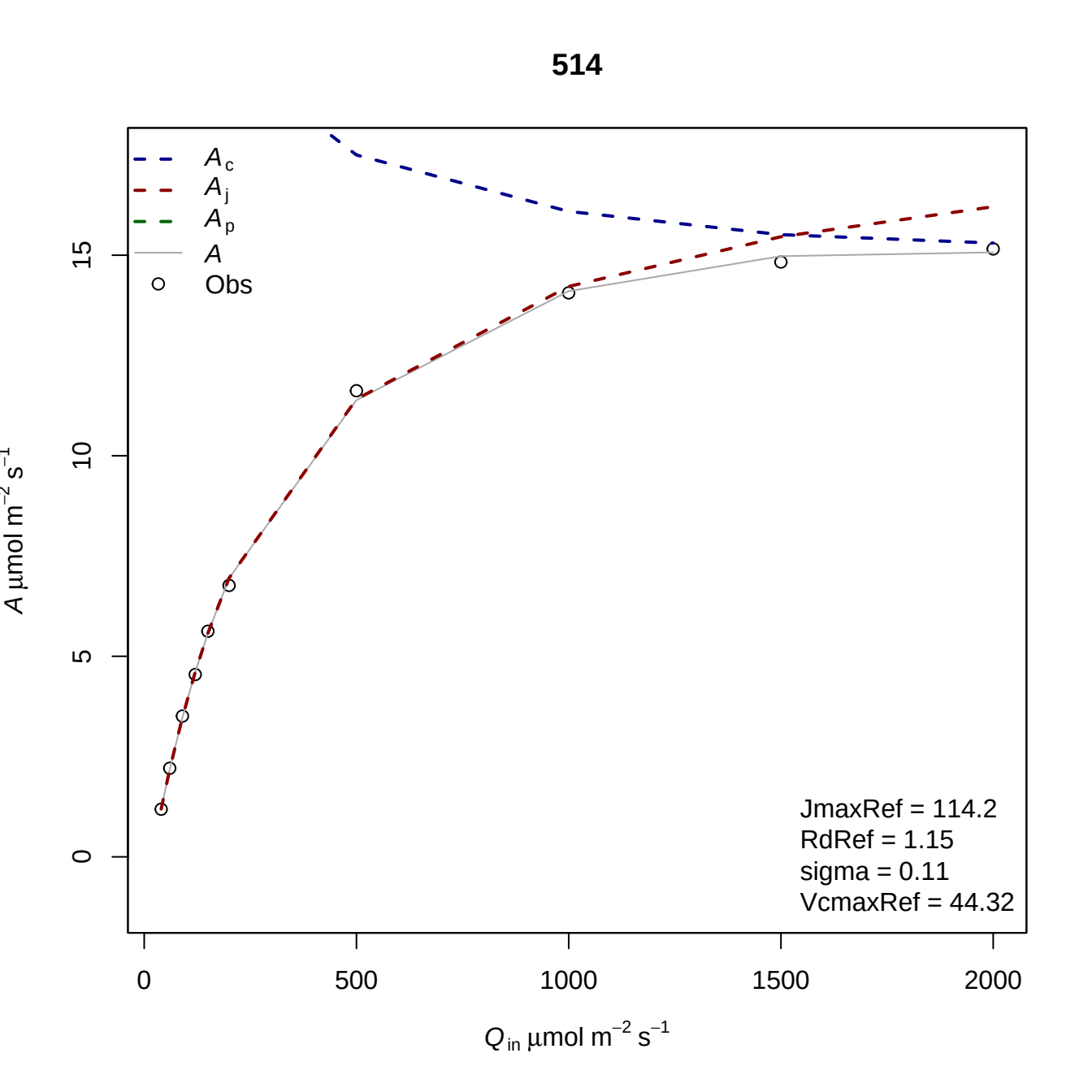

509
A_c
A_j
A_p
A
Obs
JmaxRef = 195.51
RdRef = 1.19
sigma = 0.31
VcmaxRef = 79.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

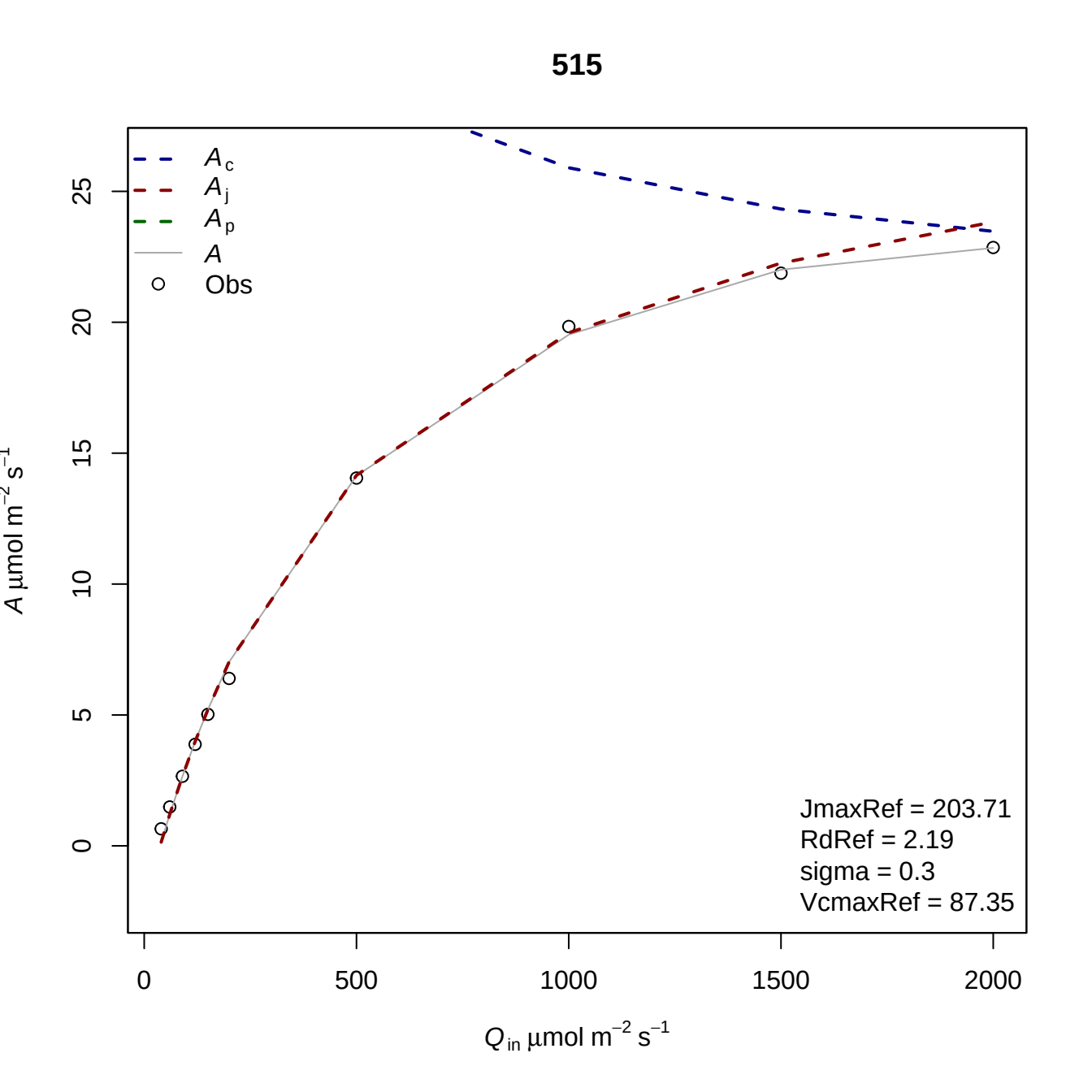

510
A_c
A_j
A_p
A
Obs
JmaxRef = 339.12
RdRef = 0.43
sigma = 0.63
VcmaxRef = 132
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

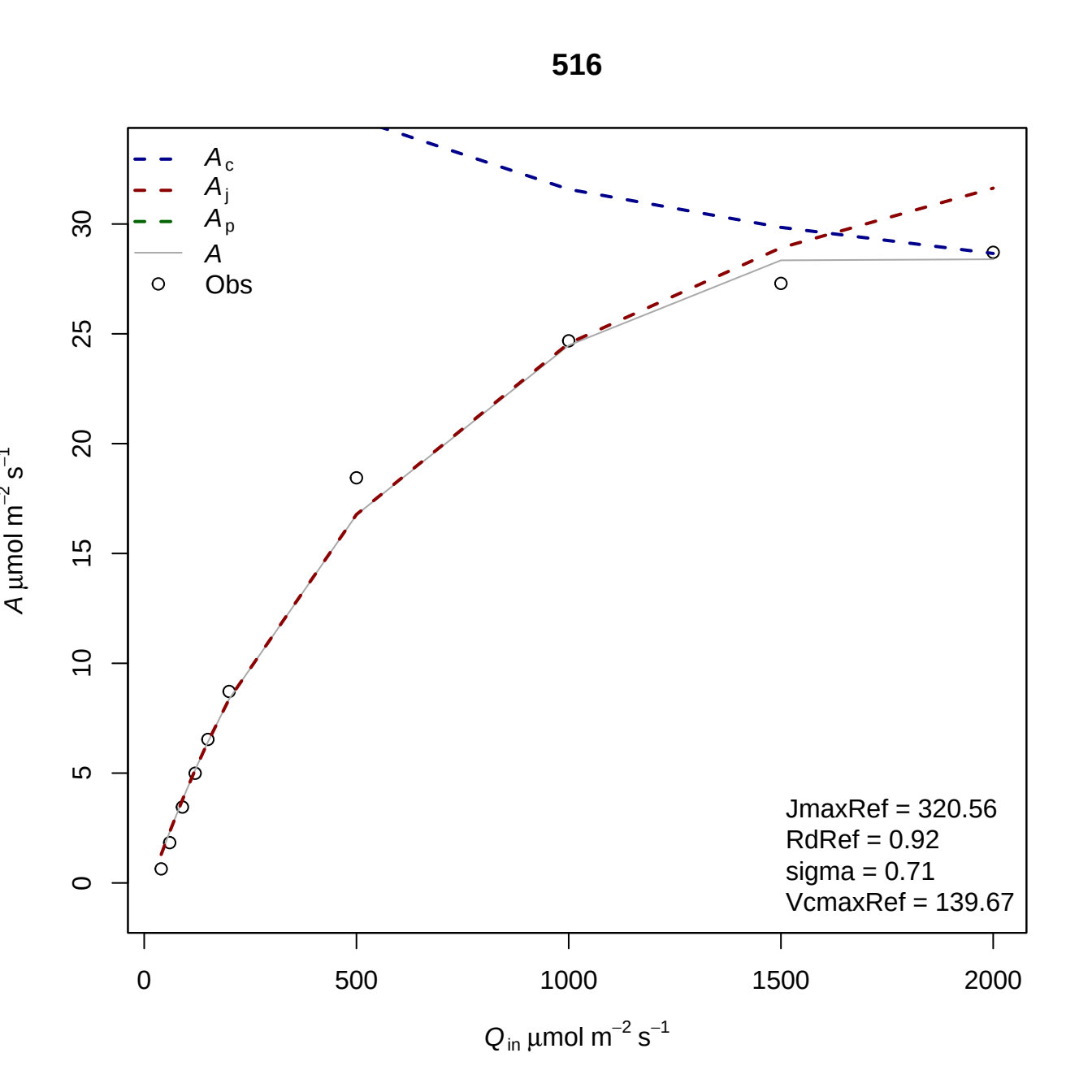

511
A_c
A_j
A_p
A
Obs
JmaxRef = 335.22
RdRef = 1.12
sigma = 0.67
VcmaxRef = 164.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

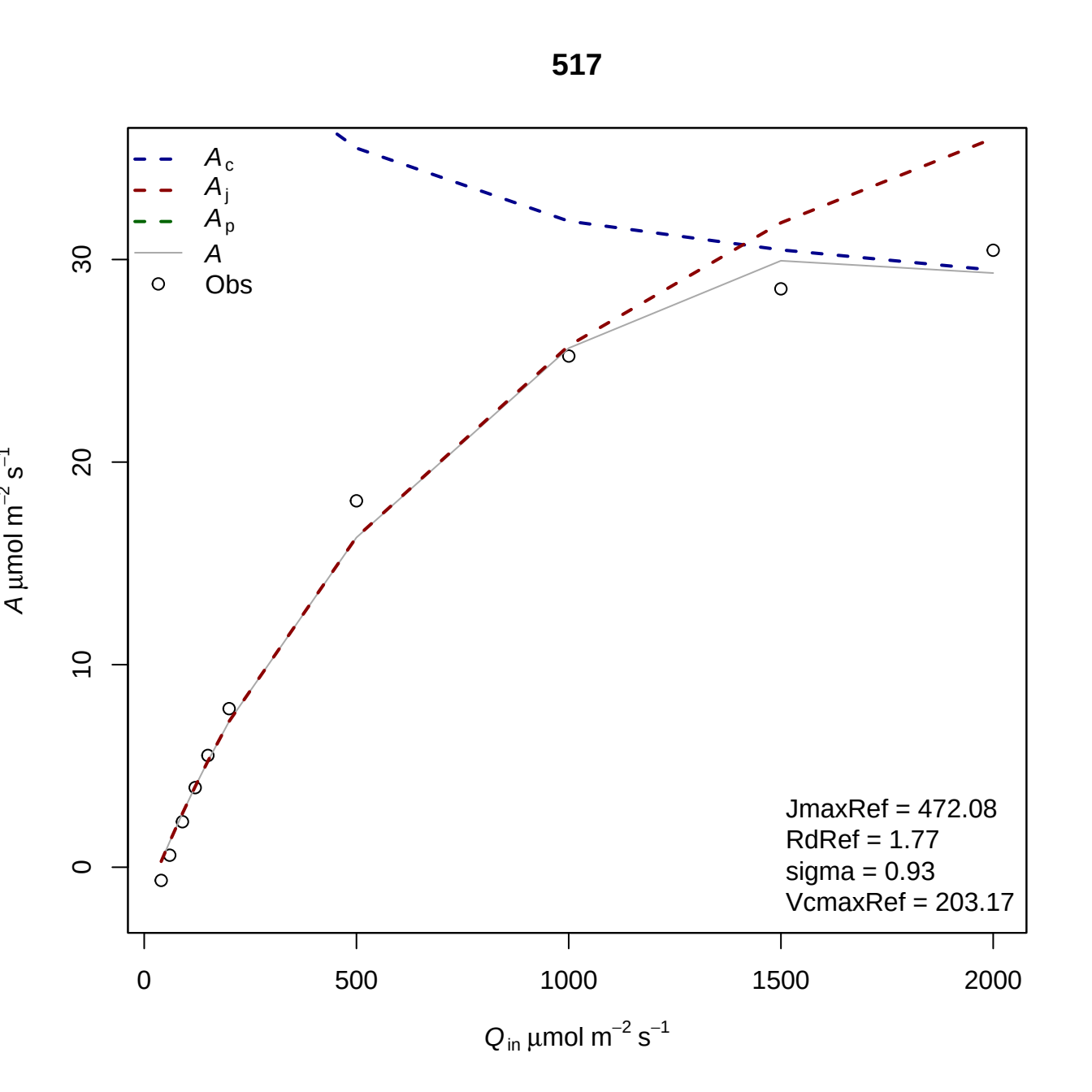

512
A_c
A_j
A_p
A
Obs
JmaxRef = 365.3
RdRef = 1.76
sigma = 0.7
VcmaxRef = 247.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

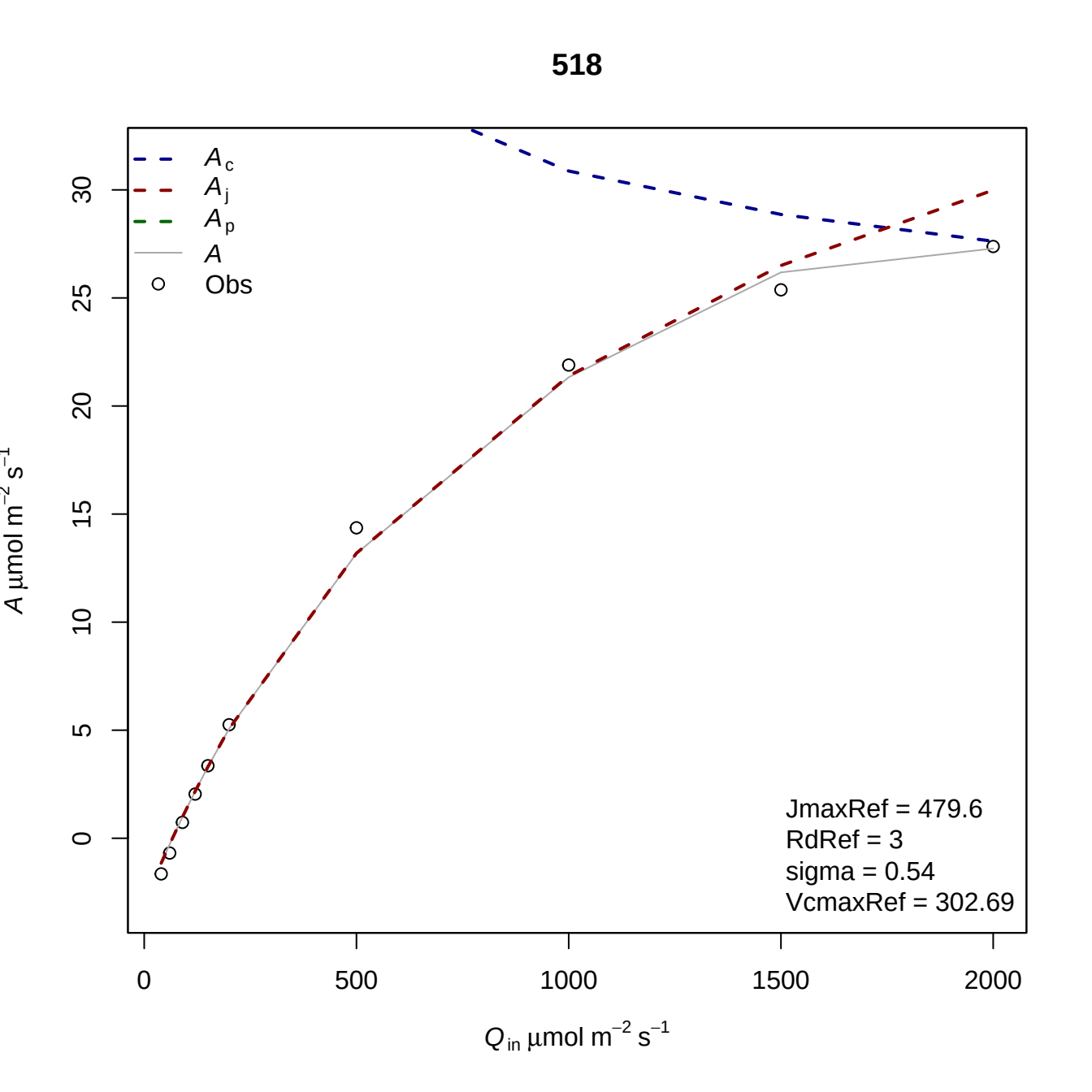

513
A_c
A_j
A_p
A
Obs
JmaxRef = 375.83
RdRef = 2.75
sigma = 0.41
VcmaxRef = 347.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

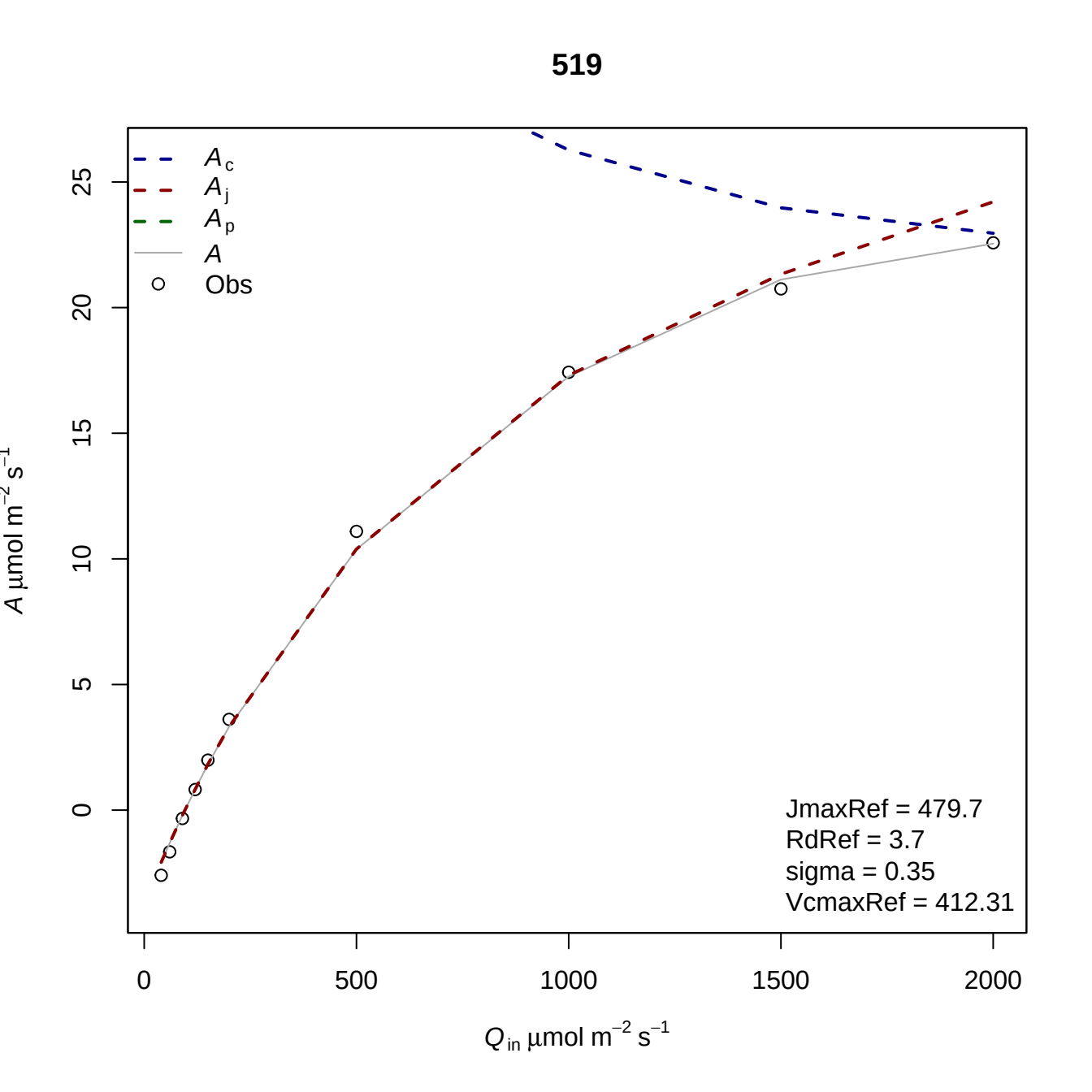

514
A_c
A_j
A_p
A
Obs
JmaxRef = 114.2
RdRef = 1.15
sigma = 0.11
VcmaxRef = 44.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

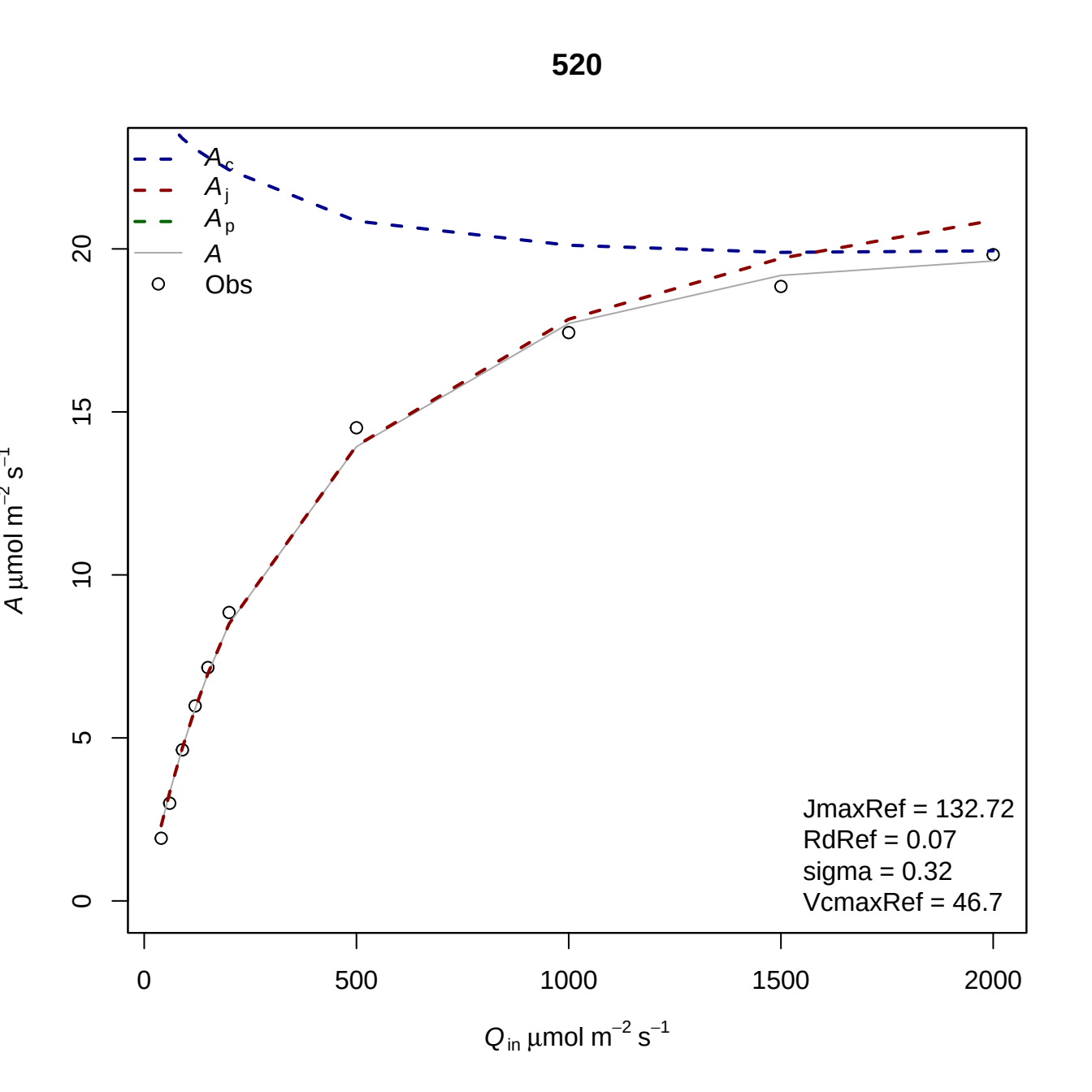

515
A_c
A_j
A_p
A
Obs
JmaxRef = 203.71
RdRef = 2.19
sigma = 0.3
VcmaxRef = 87.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

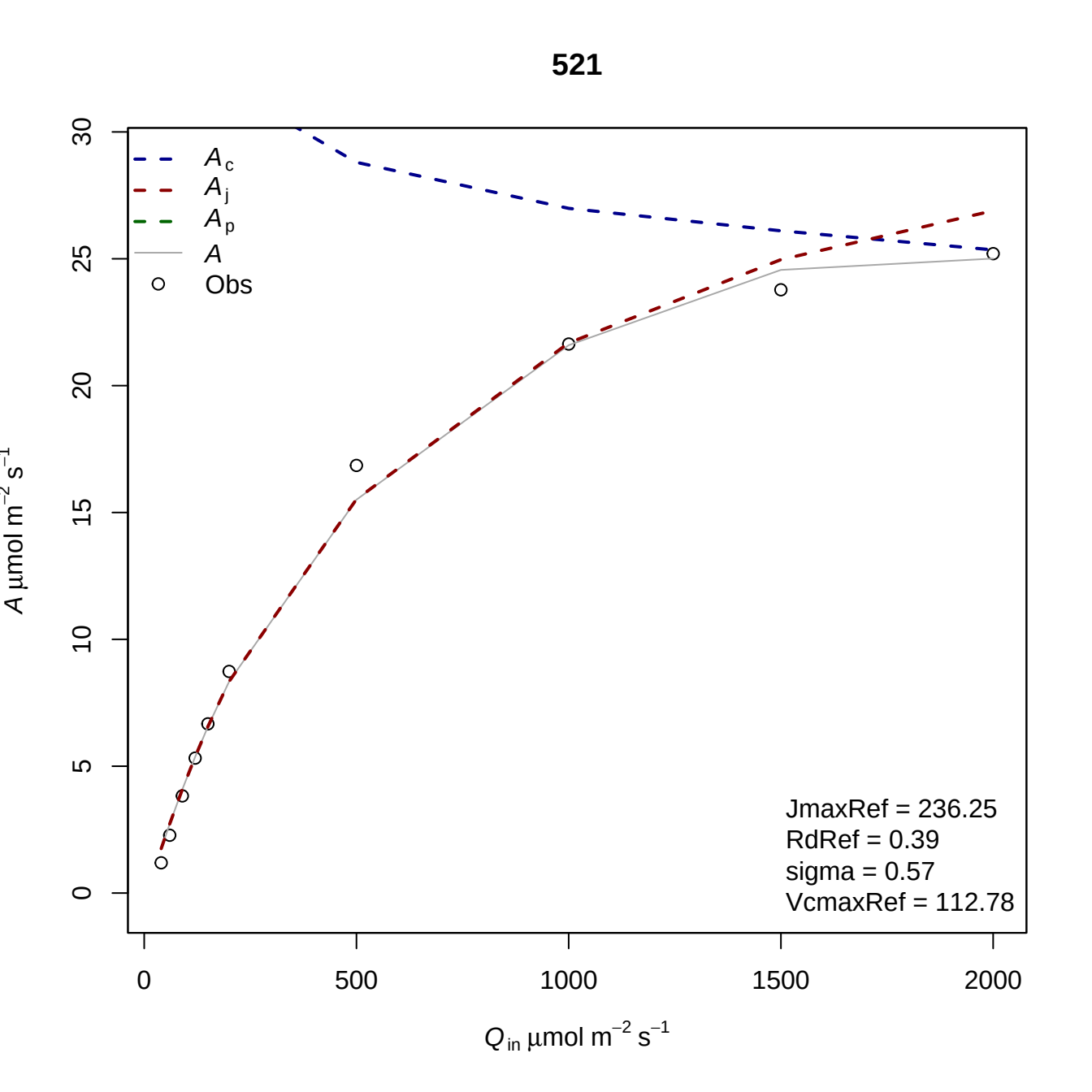

516
A_c
A_j
A_p
A
Obs
JmaxRef = 320.56
RdRef = 0.92
sigma = 0.71
VcmaxRef = 139.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

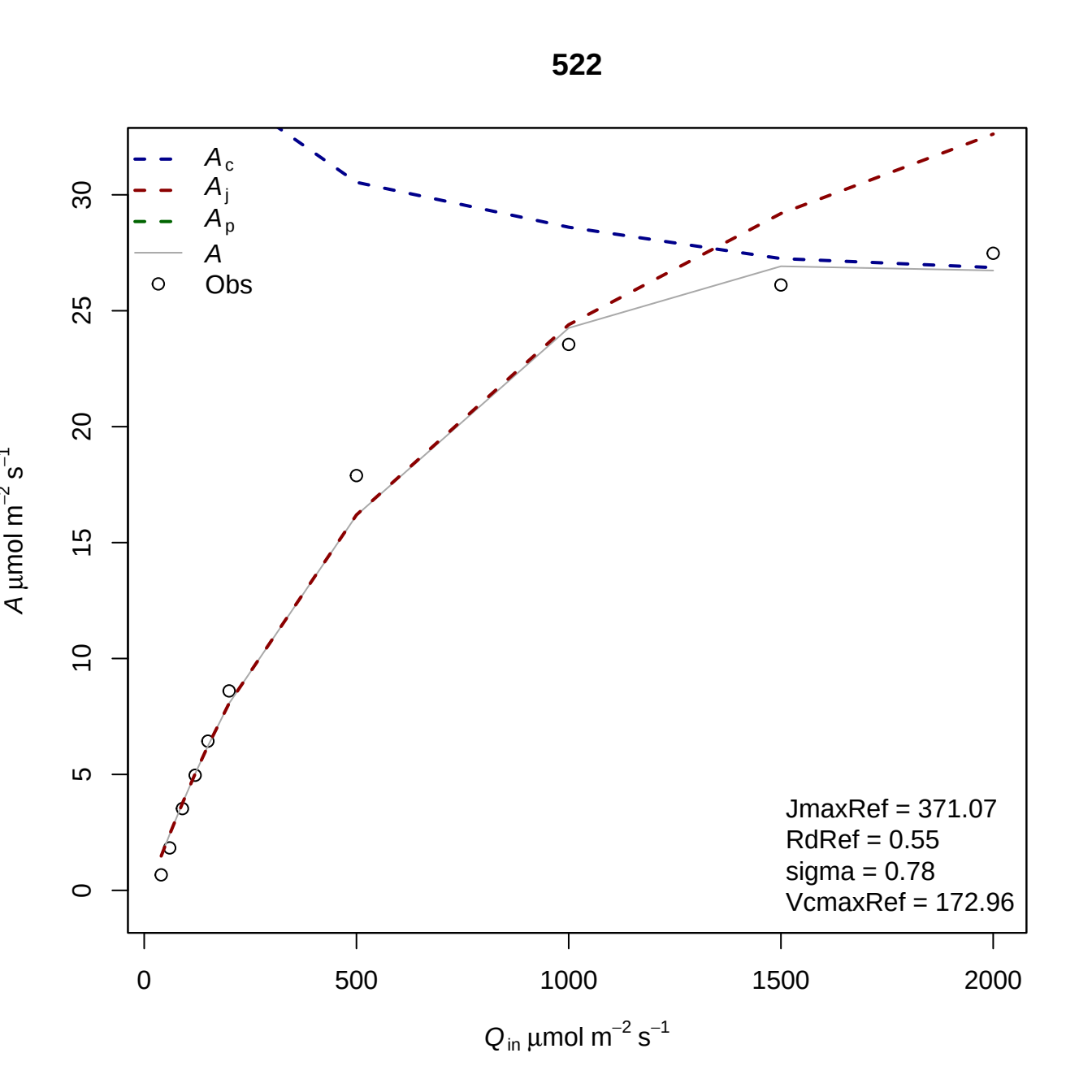

517
A_c
A_j
A_p
A
Obs
JmaxRef = 472.08
RdRef = 1.77
sigma = 0.93
VcmaxRef = 203.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

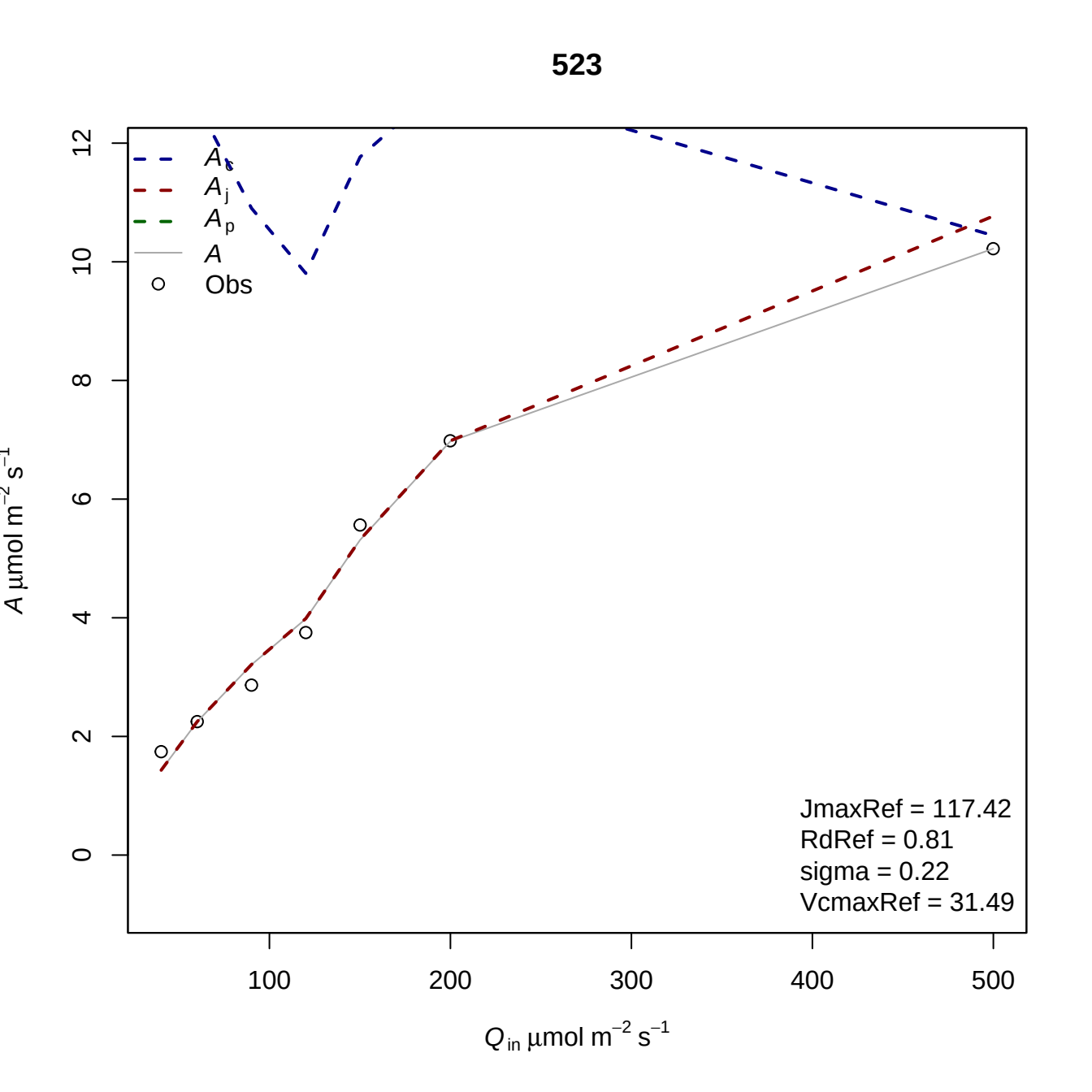

518
A_c
A_j
A_p
A
Obs
JmaxRef = 479.6
RdRef = 3
sigma = 0.54
VcmaxRef = 302.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

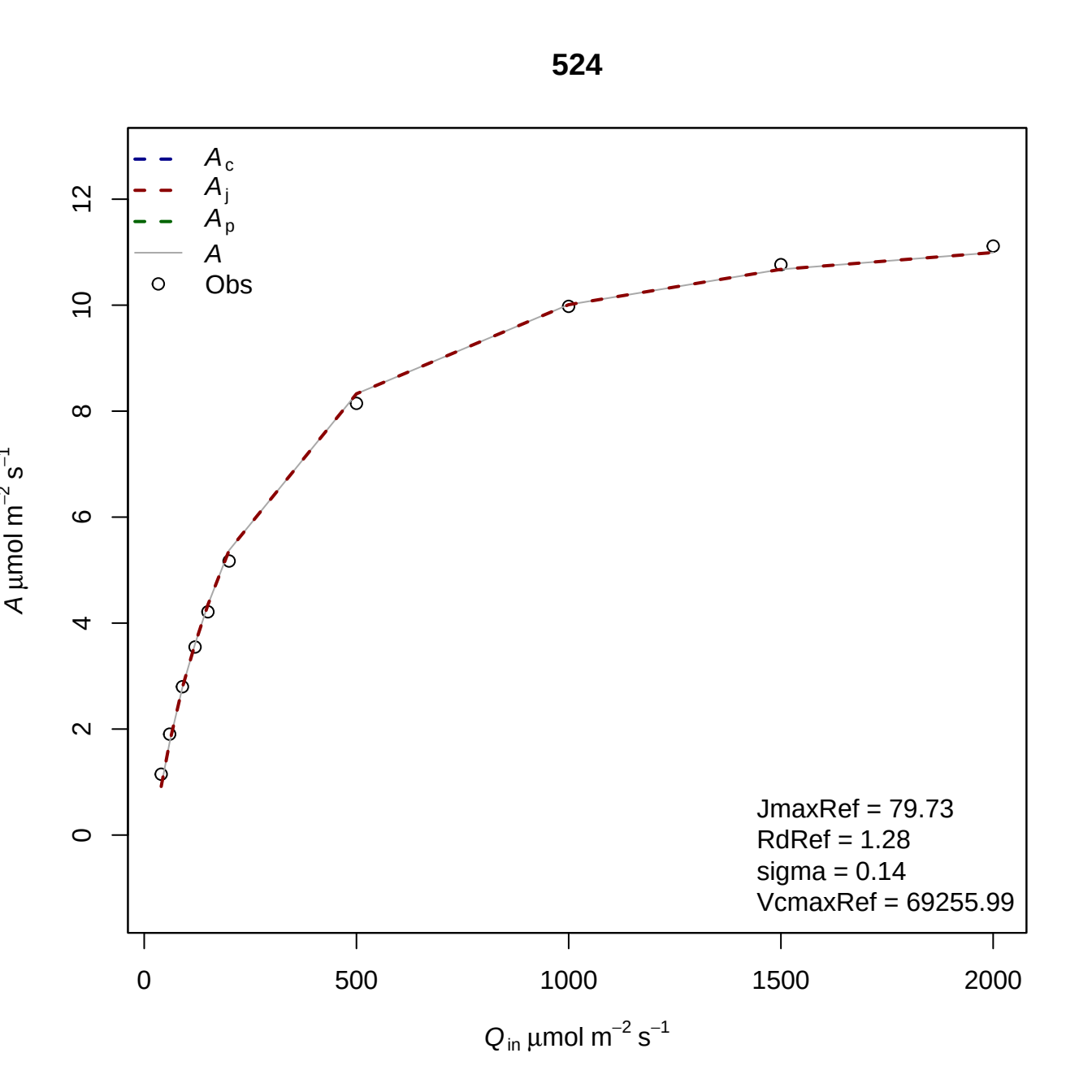

519
A_c
A_j
A_p
A
Obs
JmaxRef = 479.7
RdRef = 3.7
sigma = 0.35
VcmaxRef = 412.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

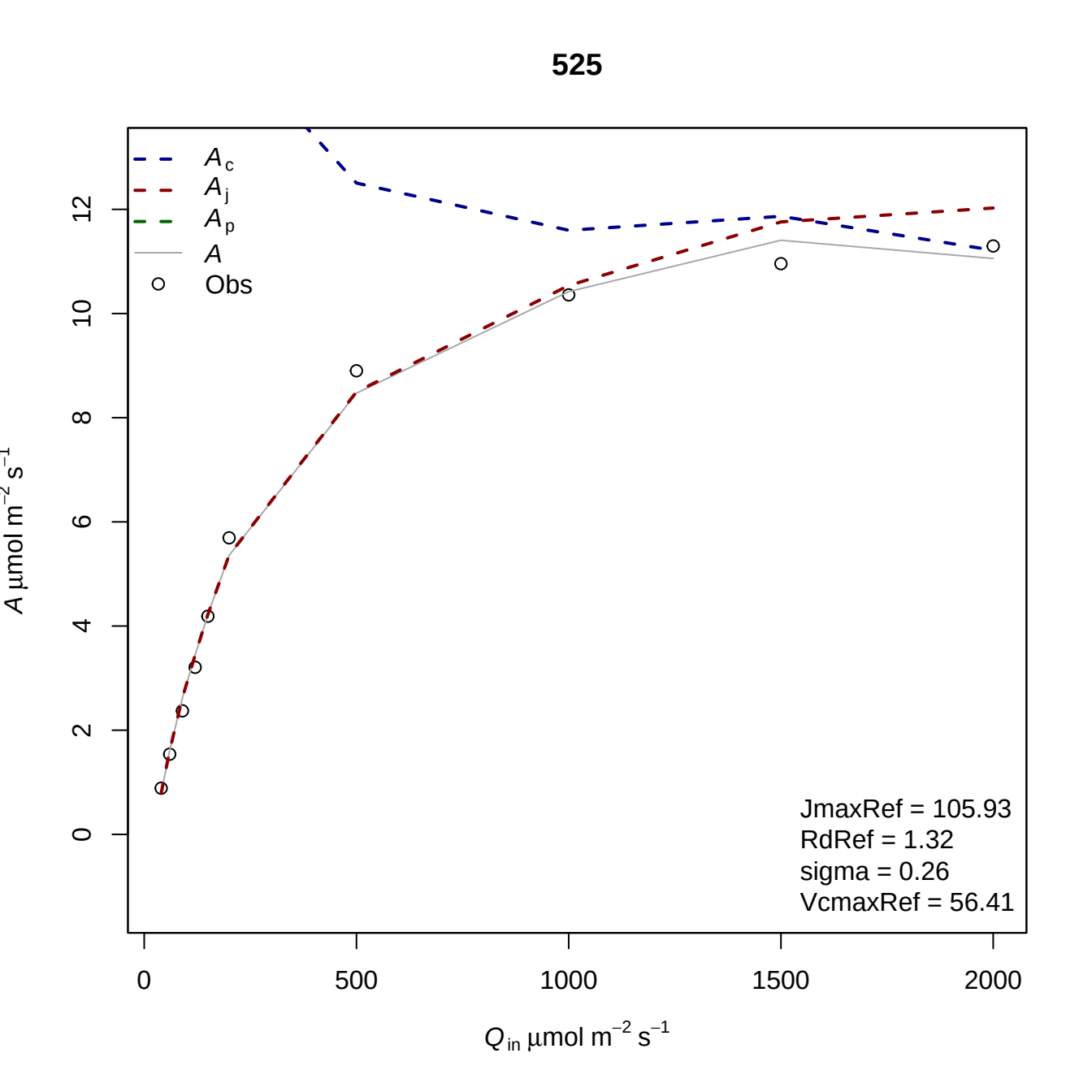

520
A_c
A_j
A_p
A
Obs
JmaxRef = 132.72
RdRef = 0.07
sigma = 0.32
VcmaxRef = 46.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

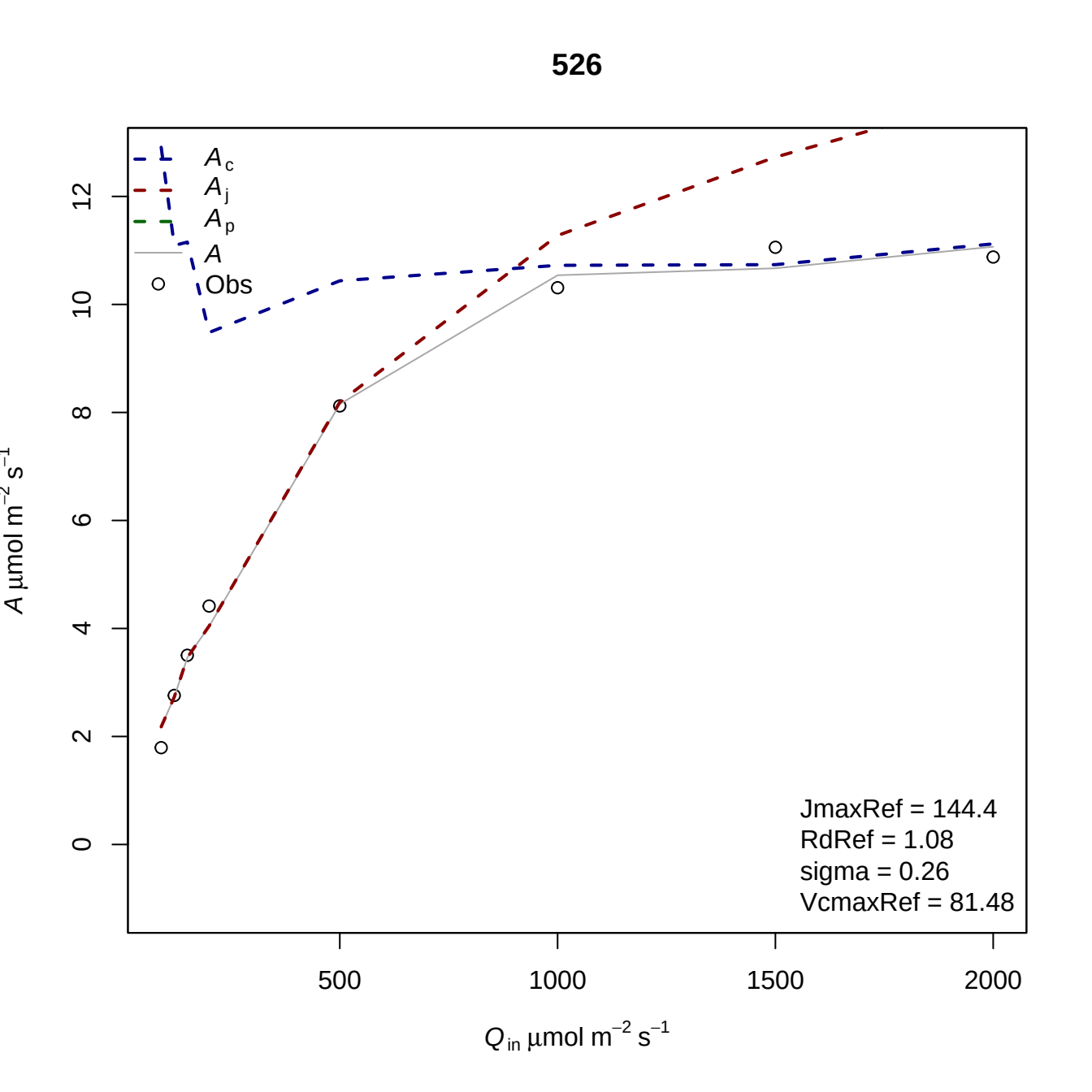

521
A_c
A_j
A_p
A
Obs
JmaxRef = 236.25
RdRef = 0.39
sigma = 0.57
VcmaxRef = 112.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

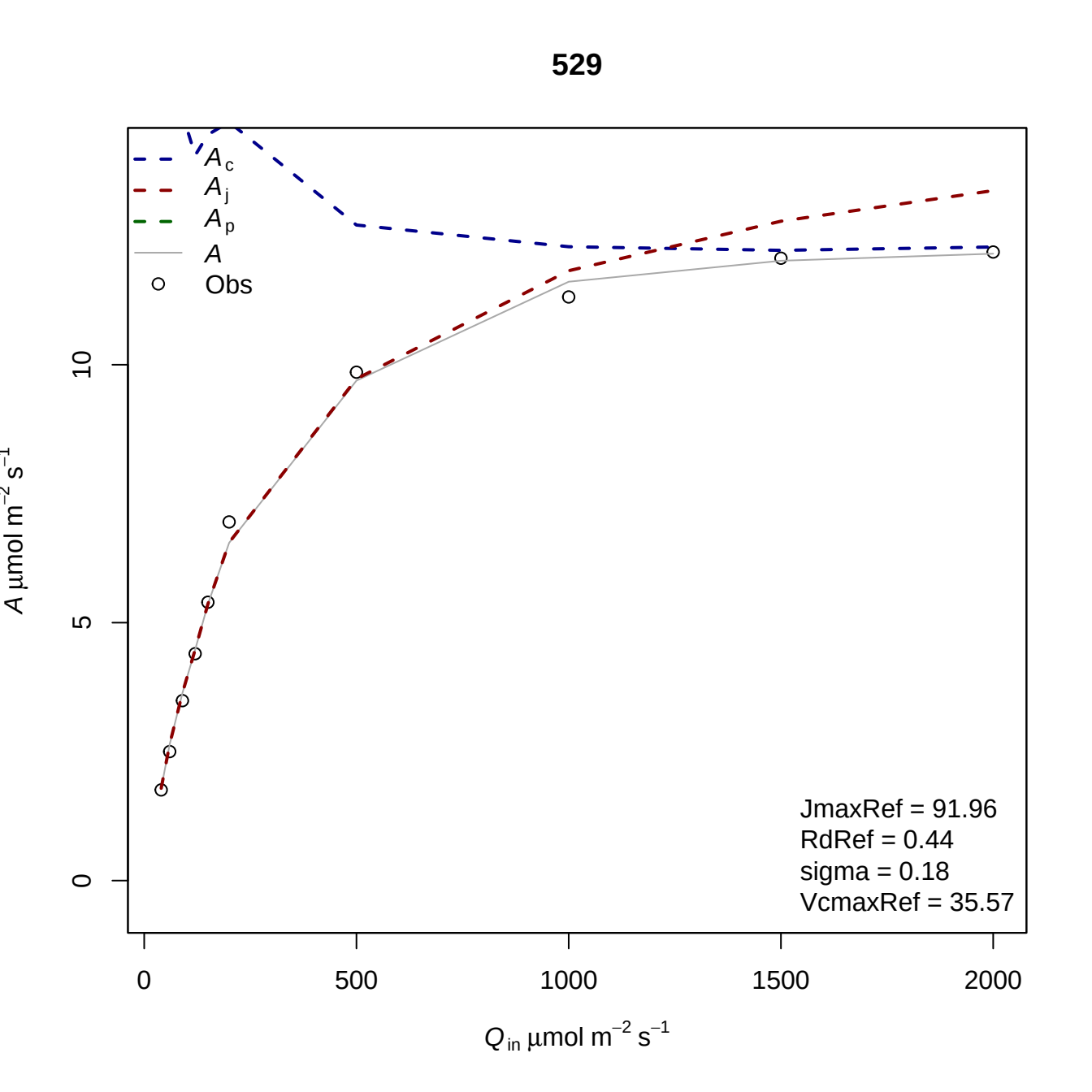

522
A_c
A_j
A_p
A
Obs
JmaxRef = 371.07
RdRef = 0.55
sigma = 0.78
VcmaxRef = 172.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

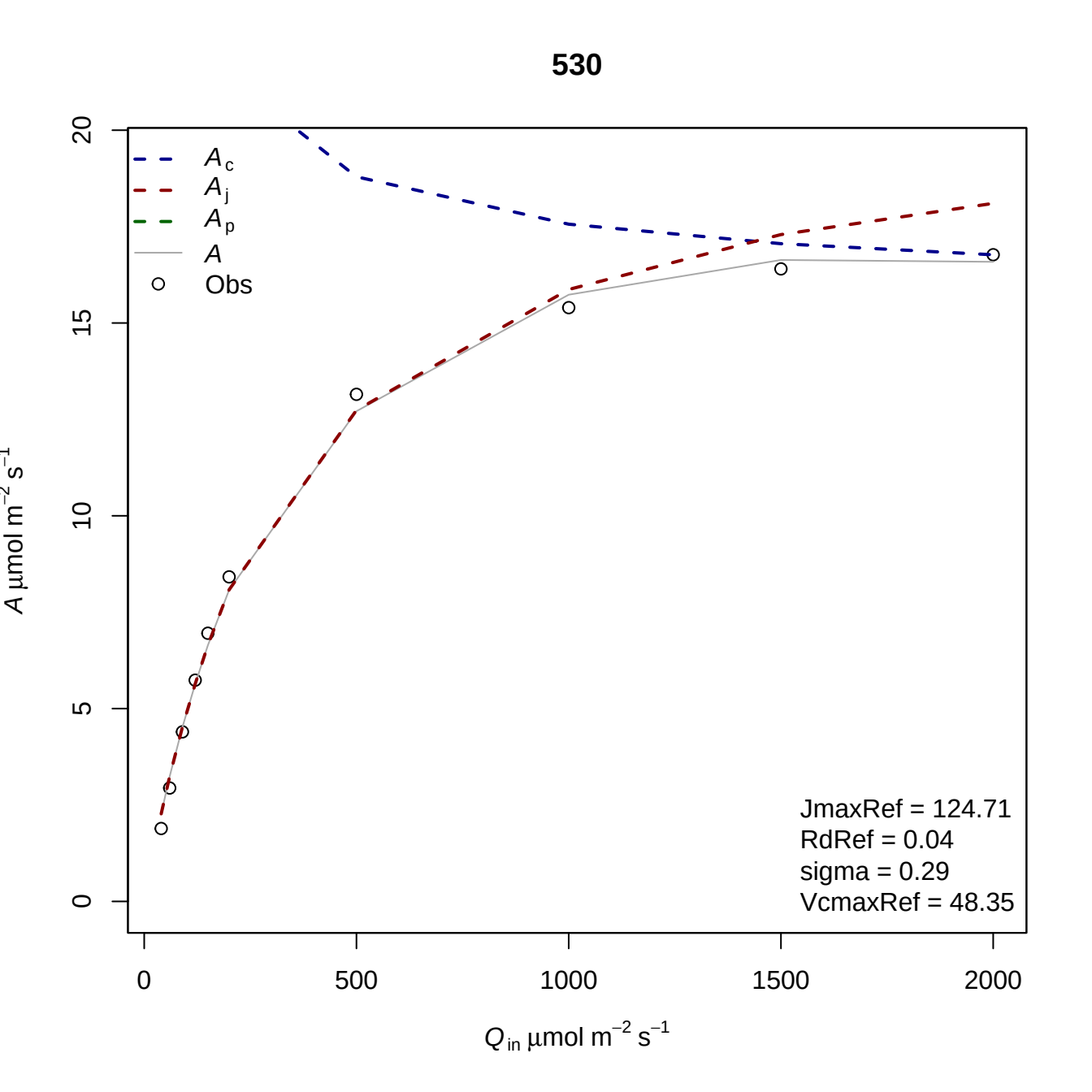

523
A_c
A_j
A_p
A
Obs
JmaxRef = 117.42
RdRef = 0.81
sigma = 0.22
VcmaxRef = 31.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
100
200
300
400
500

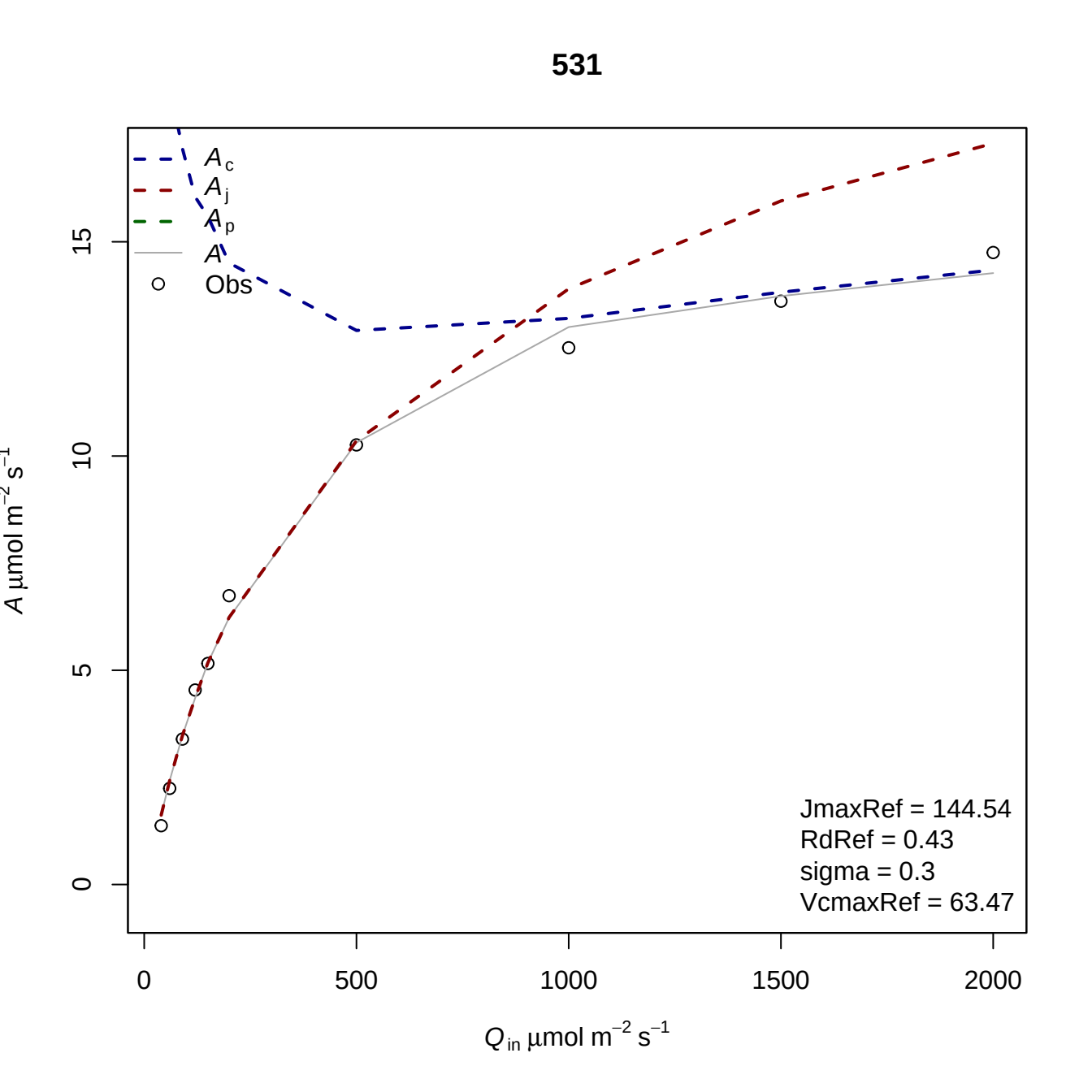

524
A_c
A_j
A_p
A
Obs
JmaxRef = 79.73
RdRef = 1.28
sigma = 0.14
VcmaxRef = 69255.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

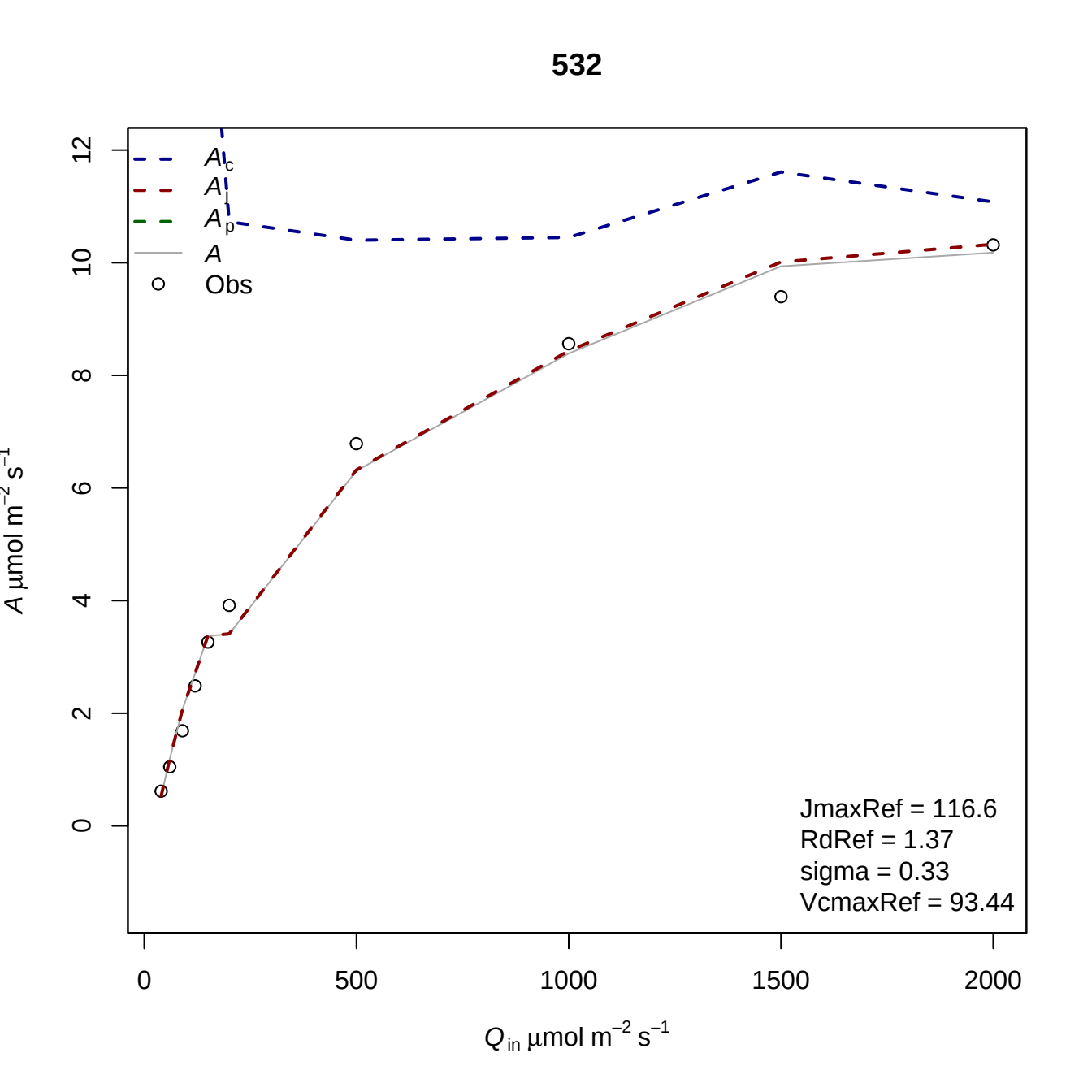

525
A_c
A_j
A_p
A
Obs
JmaxRef = 105.93
RdRef = 1.32
sigma = 0.26
VcmaxRef = 56.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

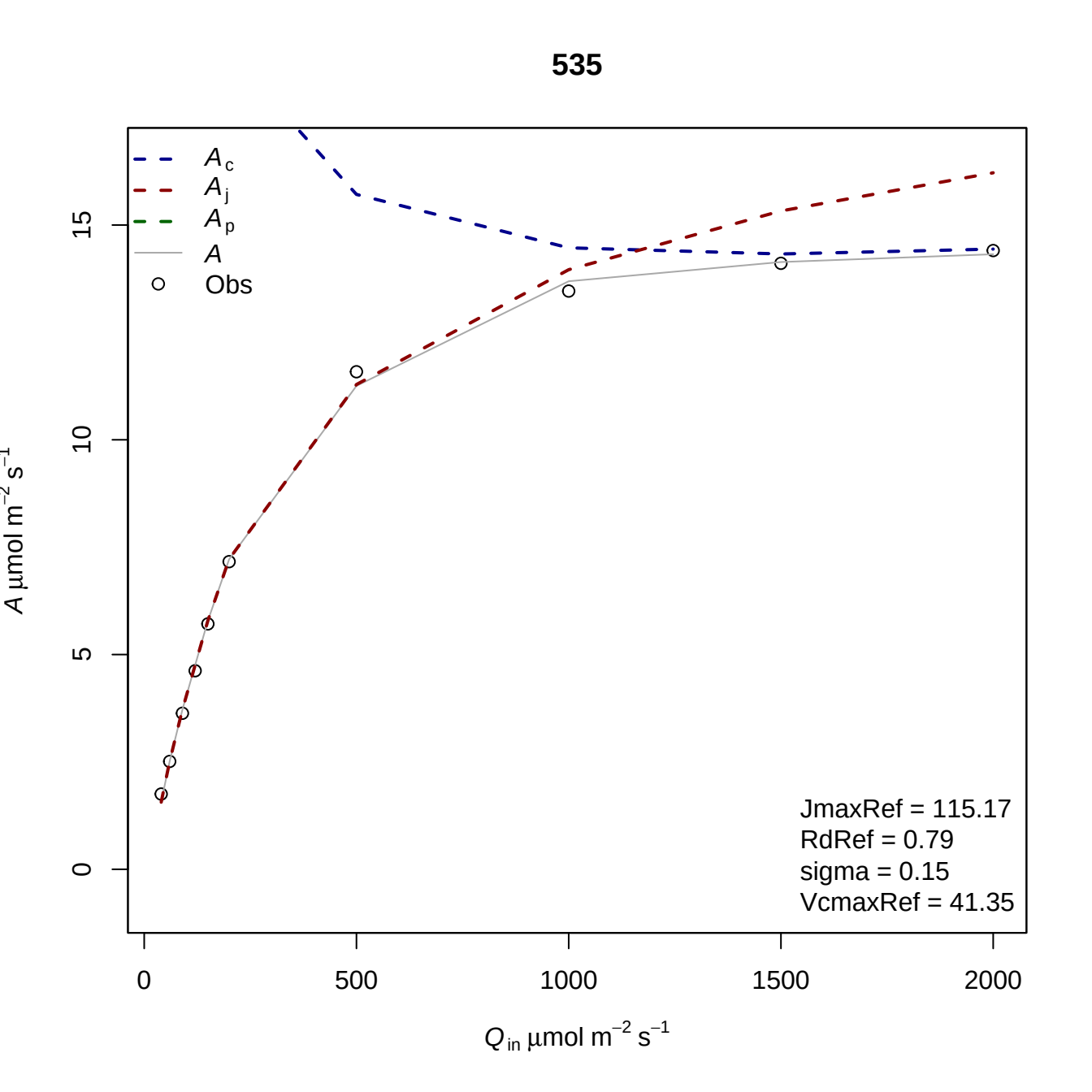

526
A_c
A_j
A_p
A
Obs
JmaxRef = 144.4
RdRef = 1.08
sigma = 0.26
VcmaxRef = 81.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
500
1000
1500
2000

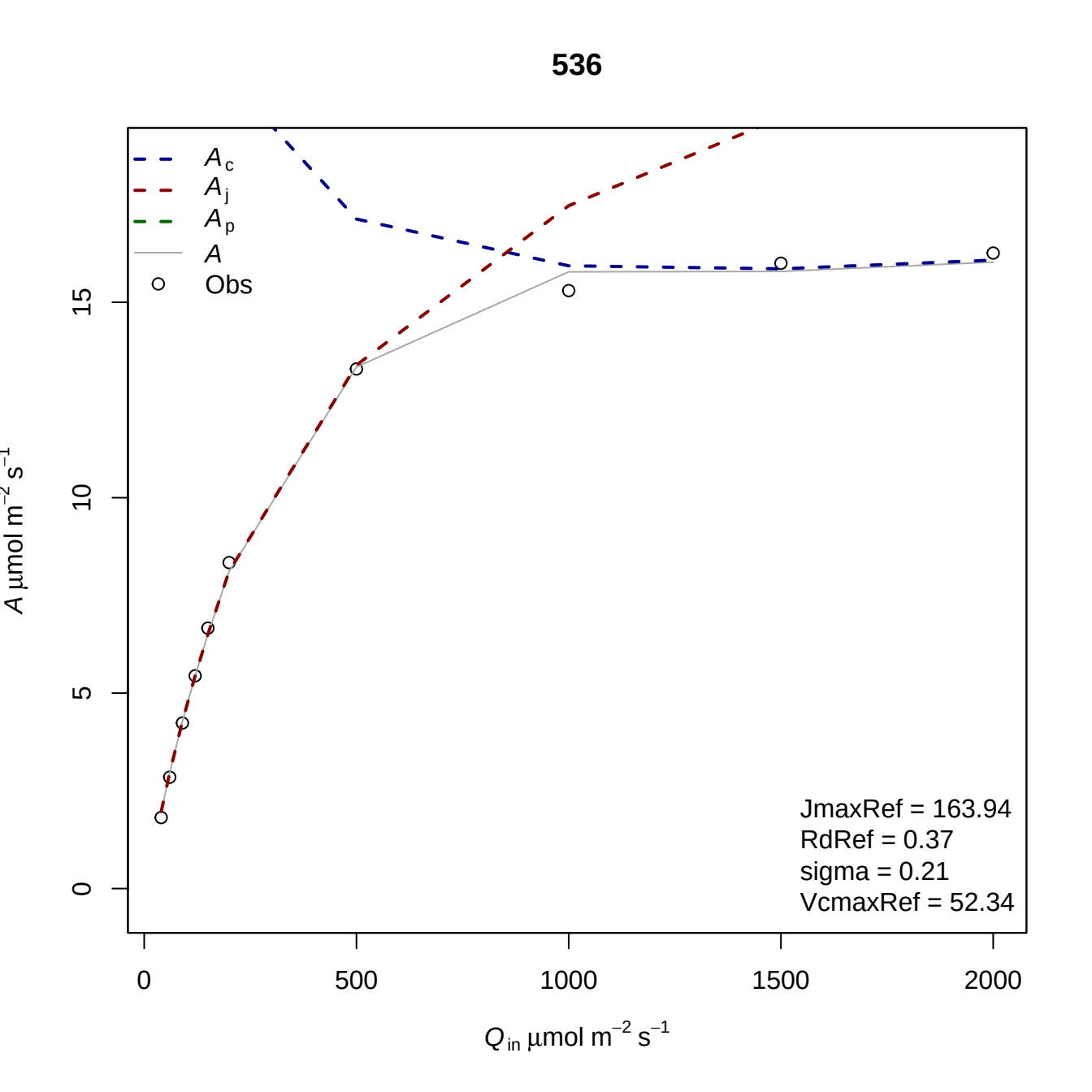

529
A_c
A_j
A_p
A
Obs
JmaxRef = 91.96
RdRef = 0.44
sigma = 0.18
VcmaxRef = 35.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

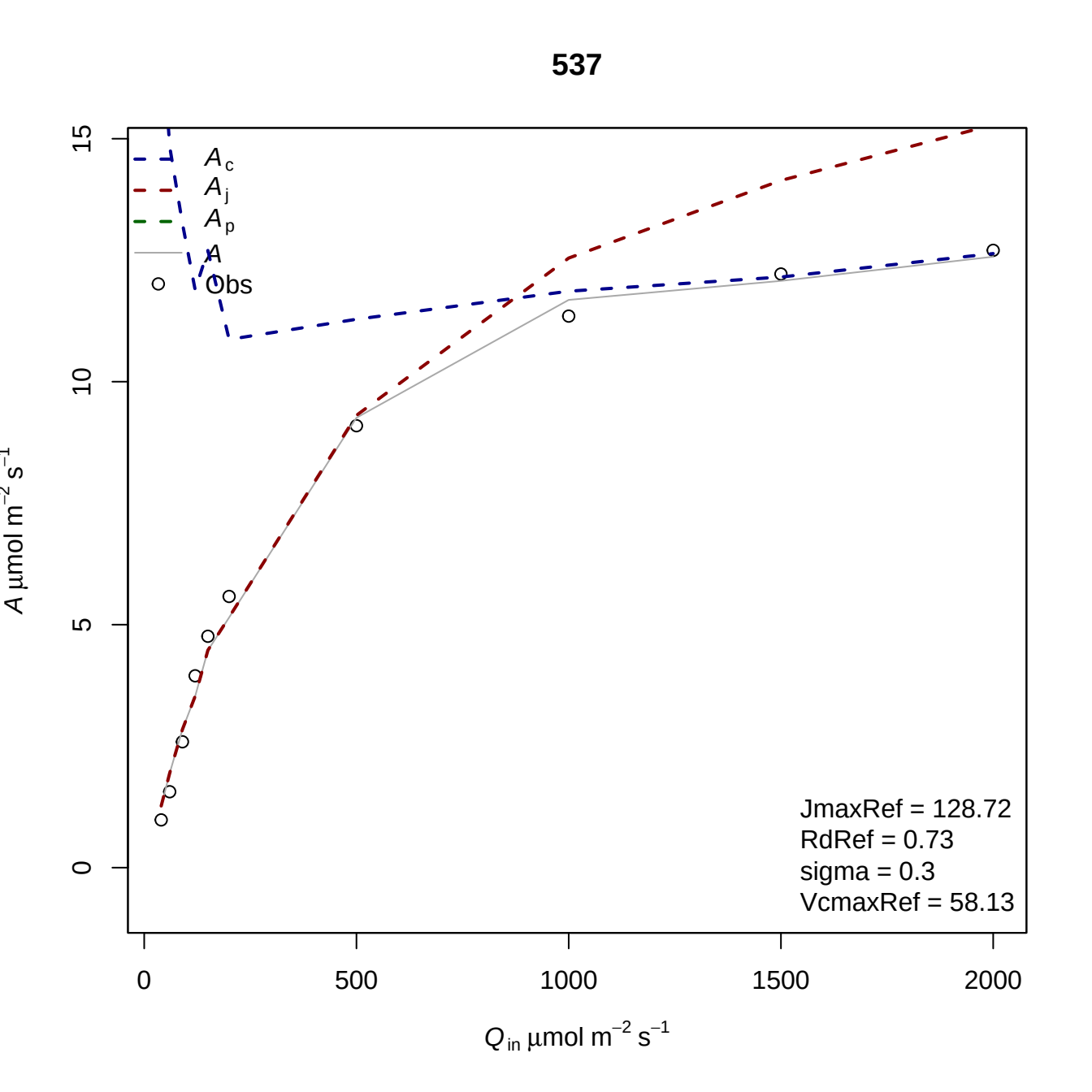

530
A_c
A_j
A_p
A
Obs
JmaxRef = 124.71
RdRef = 0.04
sigma = 0.29
VcmaxRef = 48.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

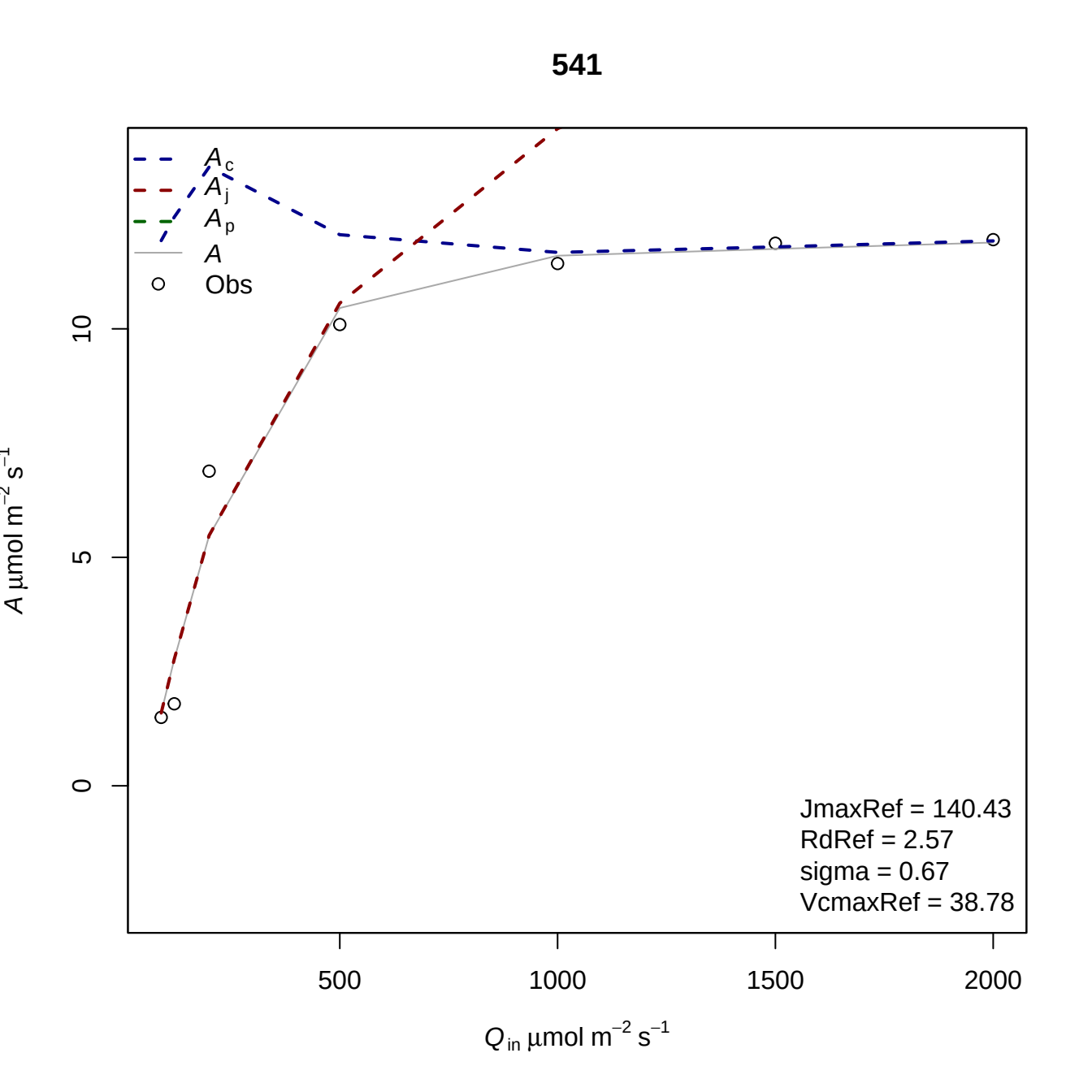

531
A_c
A_j
A_p
A
Obs
JmaxRef = 144.54
RdRef = 0.43
sigma = 0.3
VcmaxRef = 63.47
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

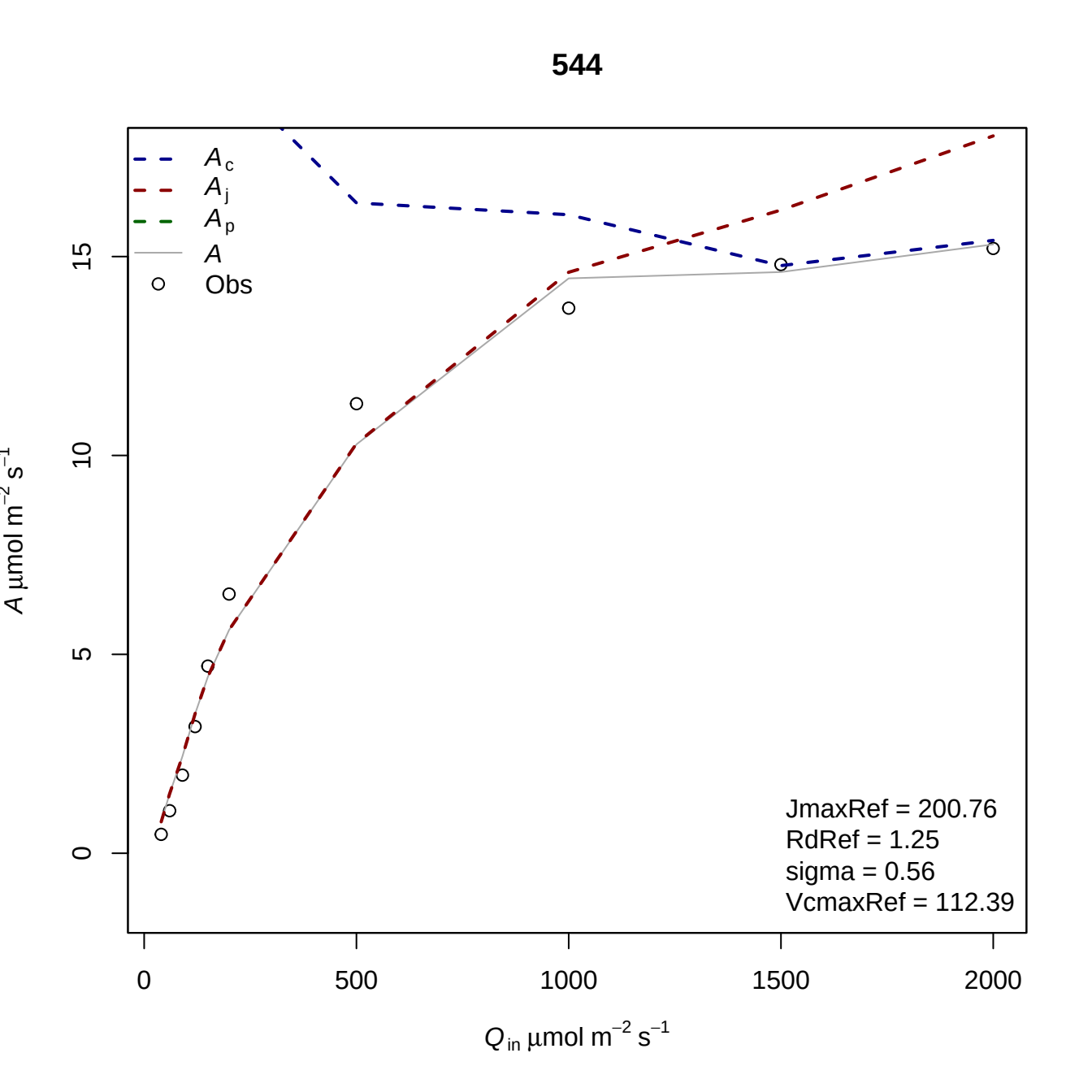

532
A_c
A_j
A_p
A
Obs
JmaxRef = 116.6
RdRef = 1.37
sigma = 0.33
VcmaxRef = 93.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

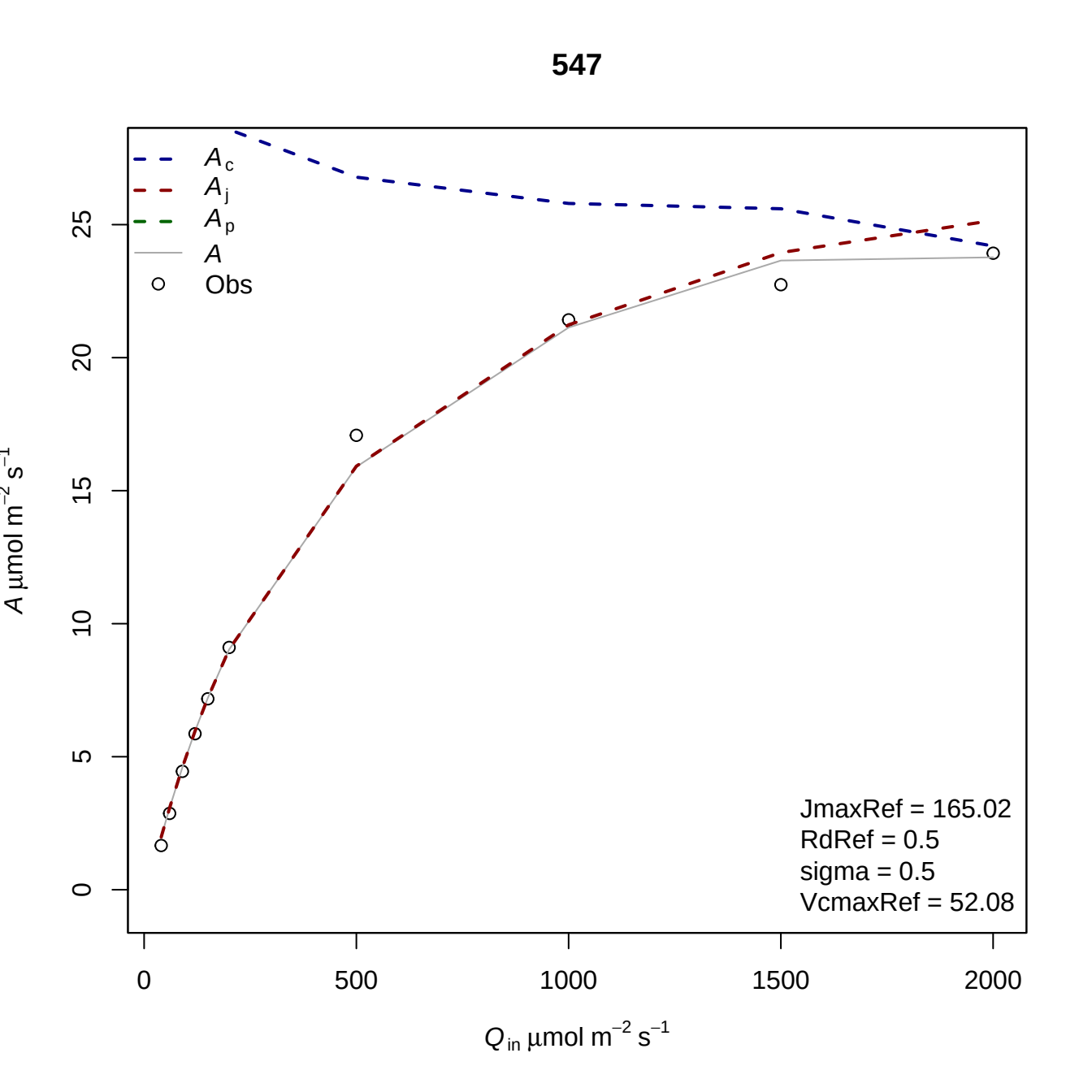

535
A_c
A_j
A_p
A
Obs
JmaxRef = 115.17
RdRef = 0.79
sigma = 0.15
VcmaxRef = 41.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

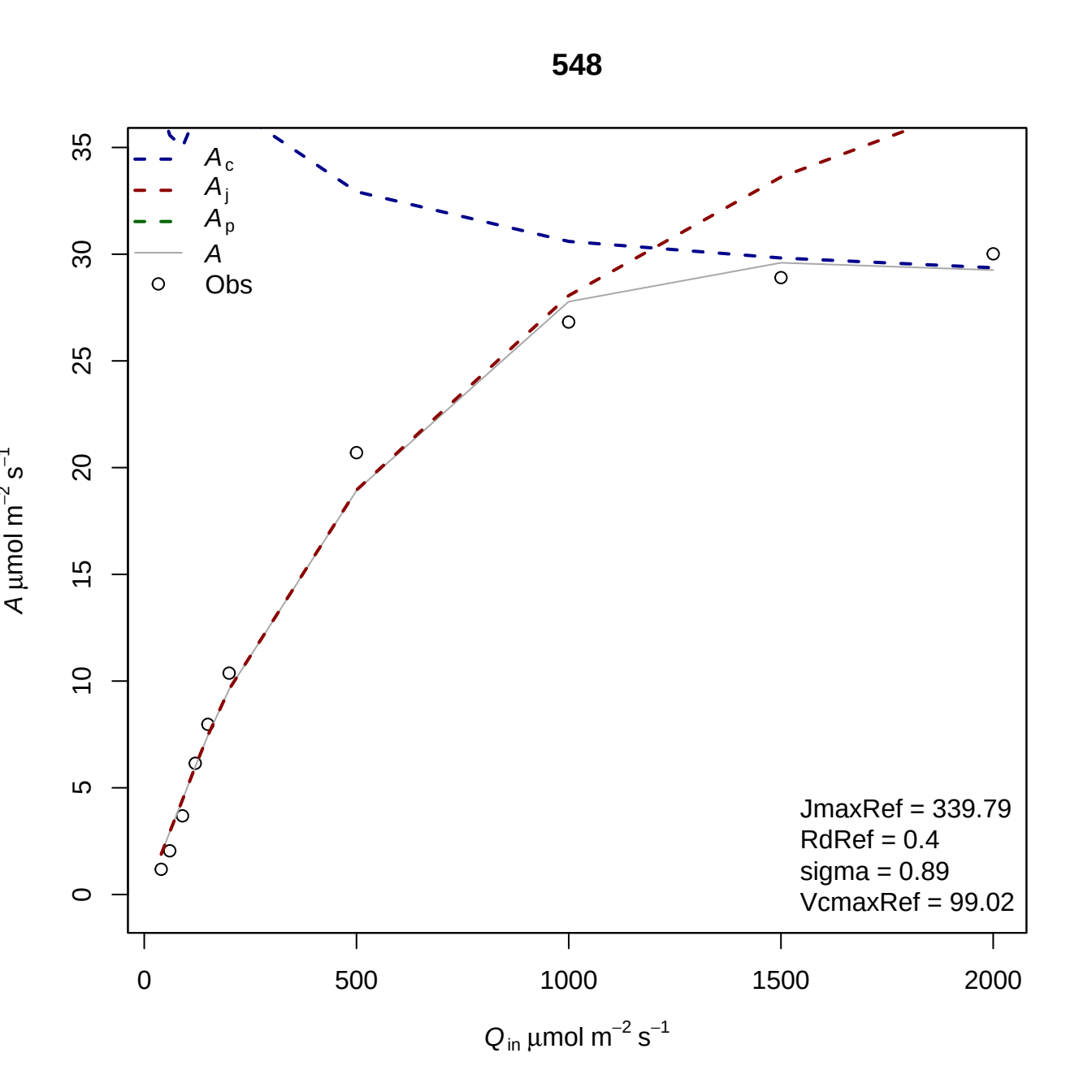

536
A_c
A_j
A_p
A
Obs
JmaxRef = 163.94
RdRef = 0.37
sigma = 0.21
VcmaxRef = 52.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

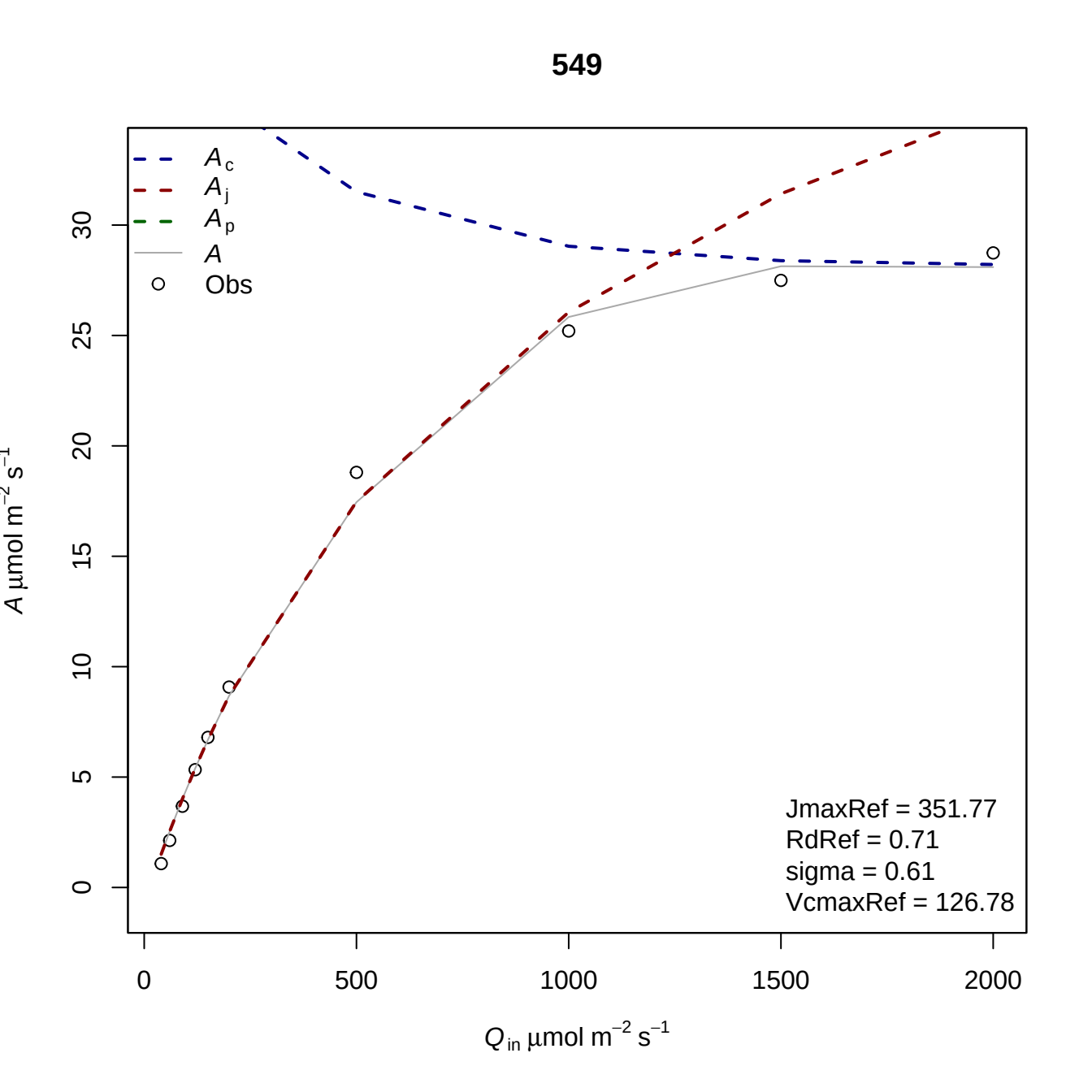

537
A_c
A_j
A_p
A
Obs
JmaxRef = 128.72
RdRef = 0.73
sigma = 0.3
VcmaxRef = 58.13
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

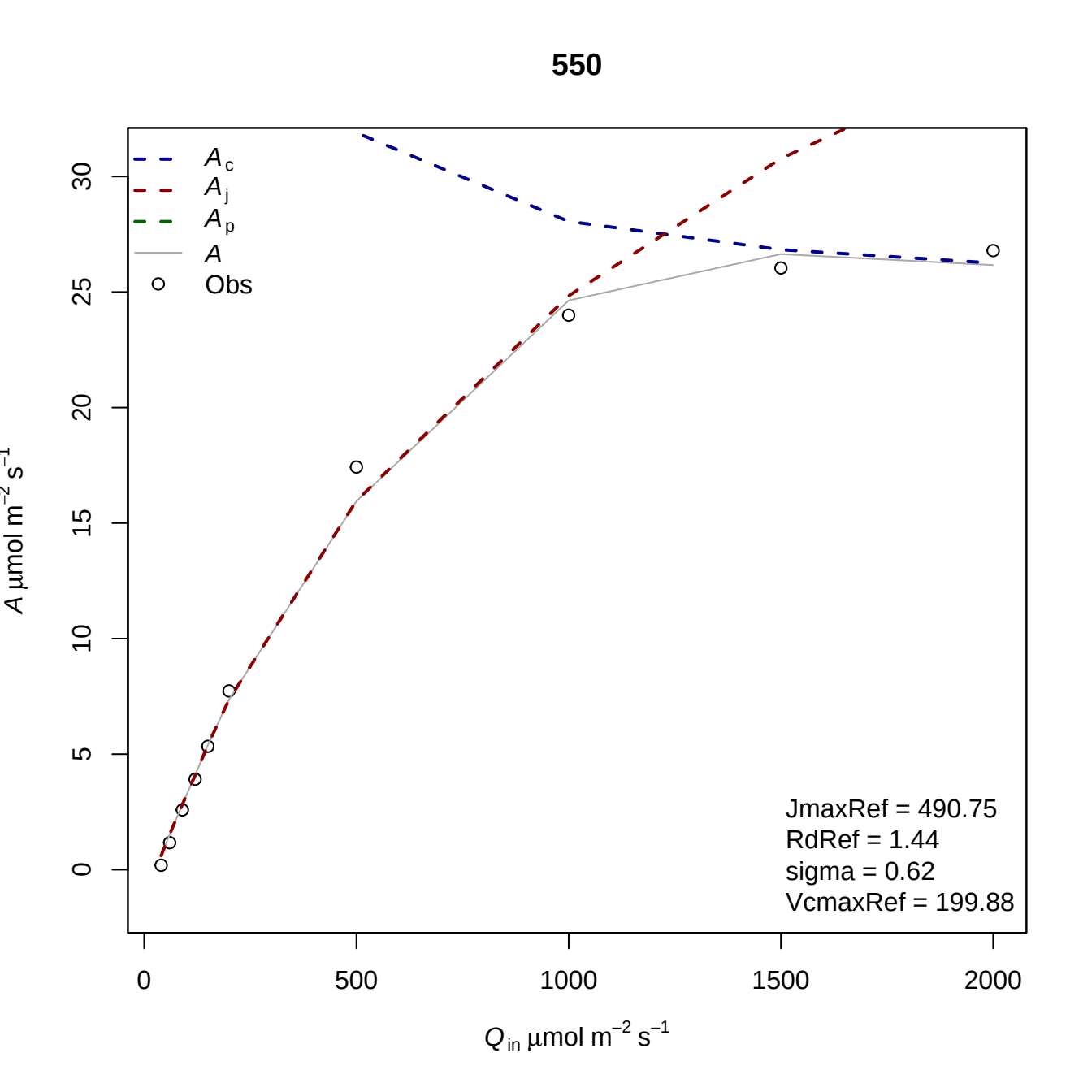

541
A_c
A_j
A_p
A
Obs
JmaxRef = 140.43
RdRef = 2.57
sigma = 0.67
VcmaxRef = 38.78
0
5
10
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

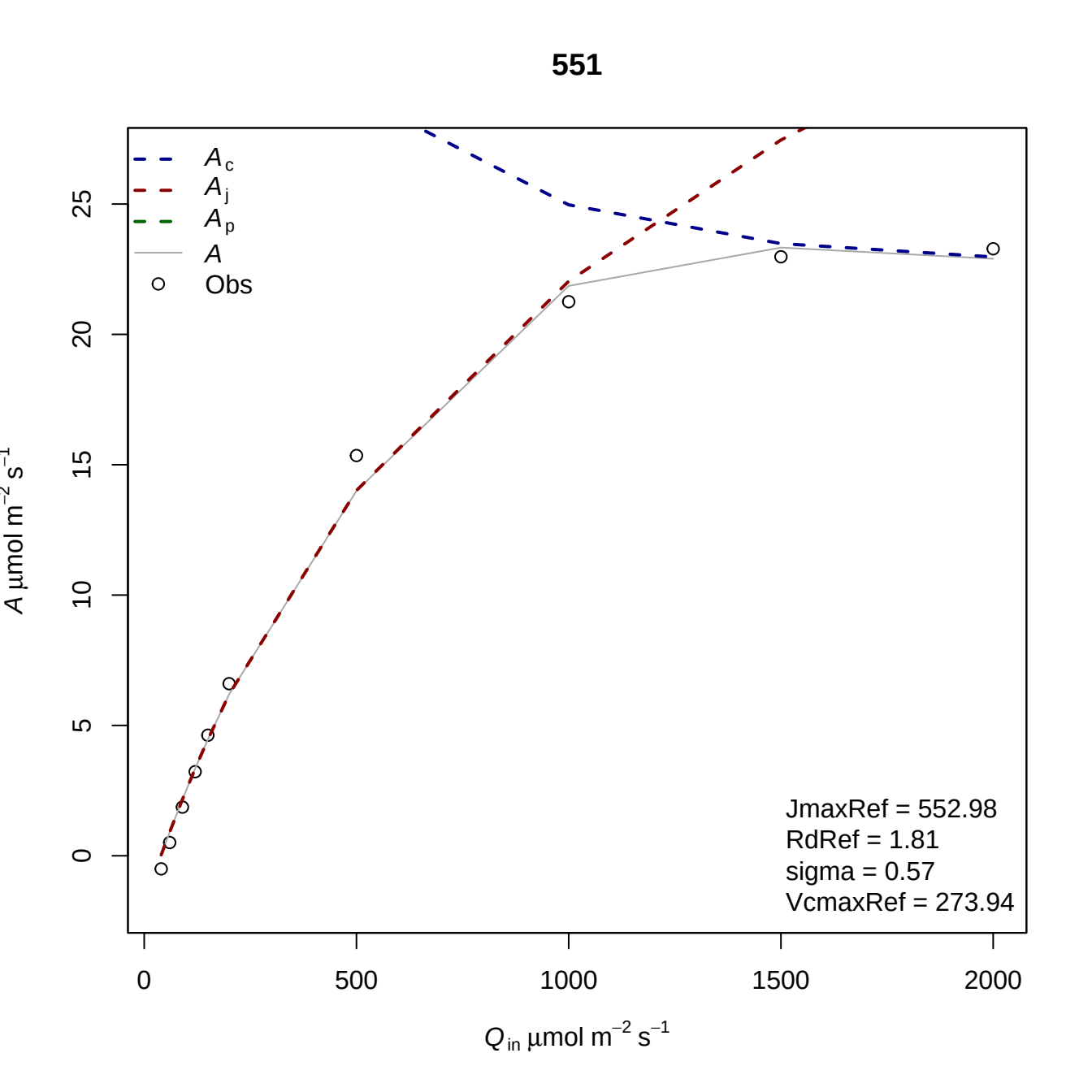

544
A_c
A_j
A_p
A
Obs
JmaxRef = 200.76
RdRef = 1.25
sigma = 0.56
VcmaxRef = 112.39
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

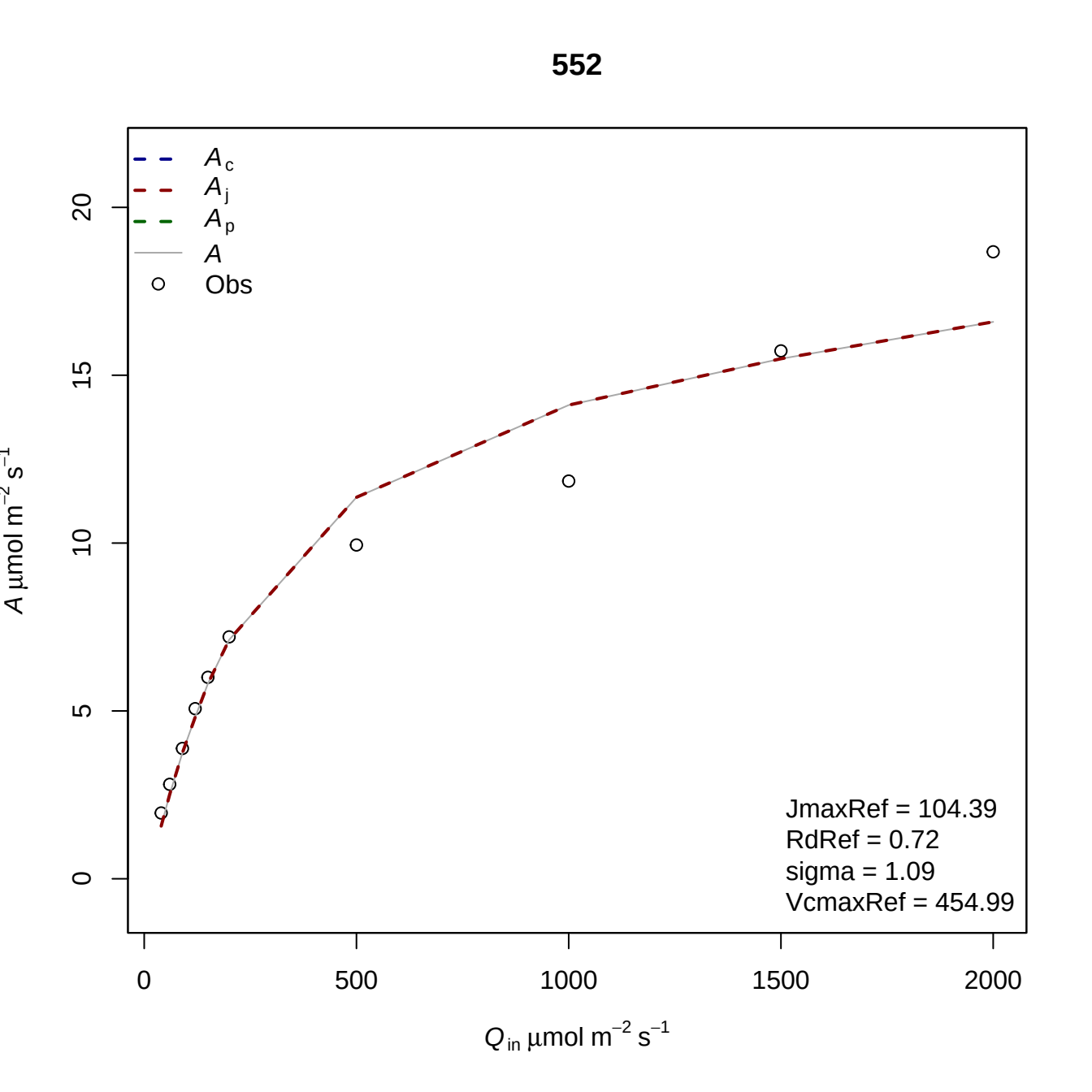

547
A_c
A_j
A_p
A
Obs
JmaxRef = 165.02
RdRef = 0.5
sigma = 0.5
VcmaxRef = 52.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

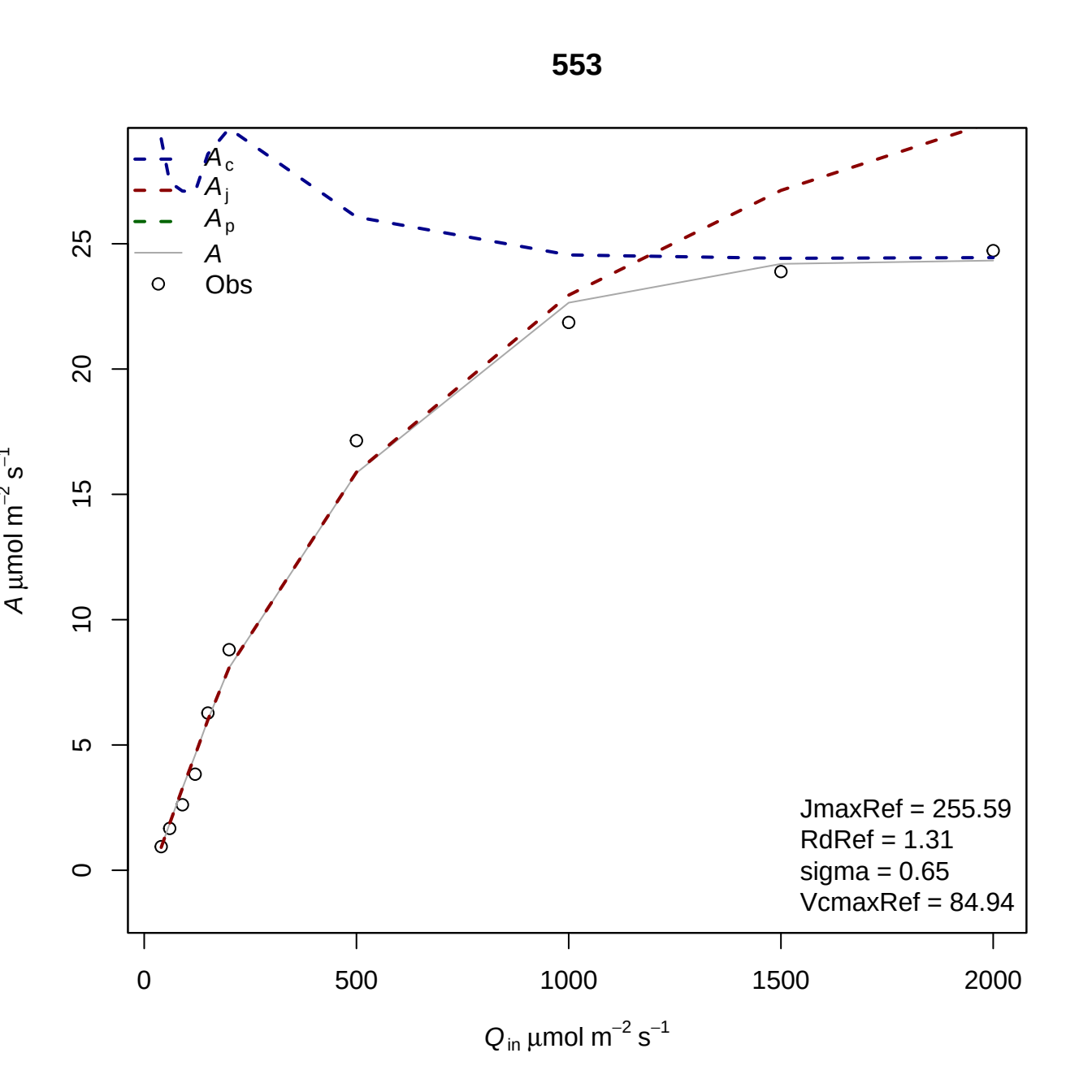

548
A_c
A_j
A_p
A
Obs
JmaxRef = 339.79
RdRef = 0.4
sigma = 0.89
VcmaxRef = 99.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

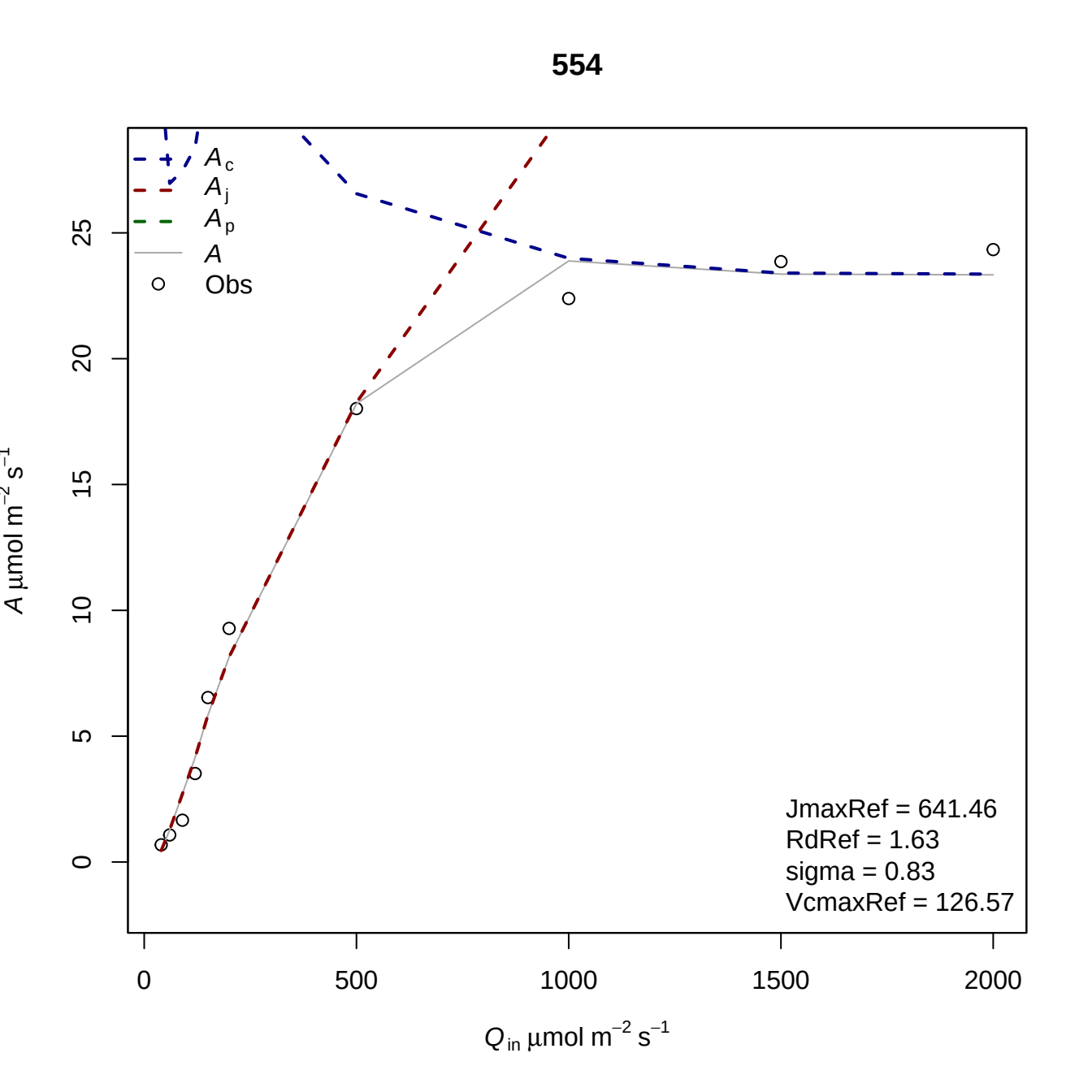

549
A_c
A_j
A_p
A
Obs
JmaxRef = 351.77
RdRef = 0.71
sigma = 0.61
VcmaxRef = 126.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

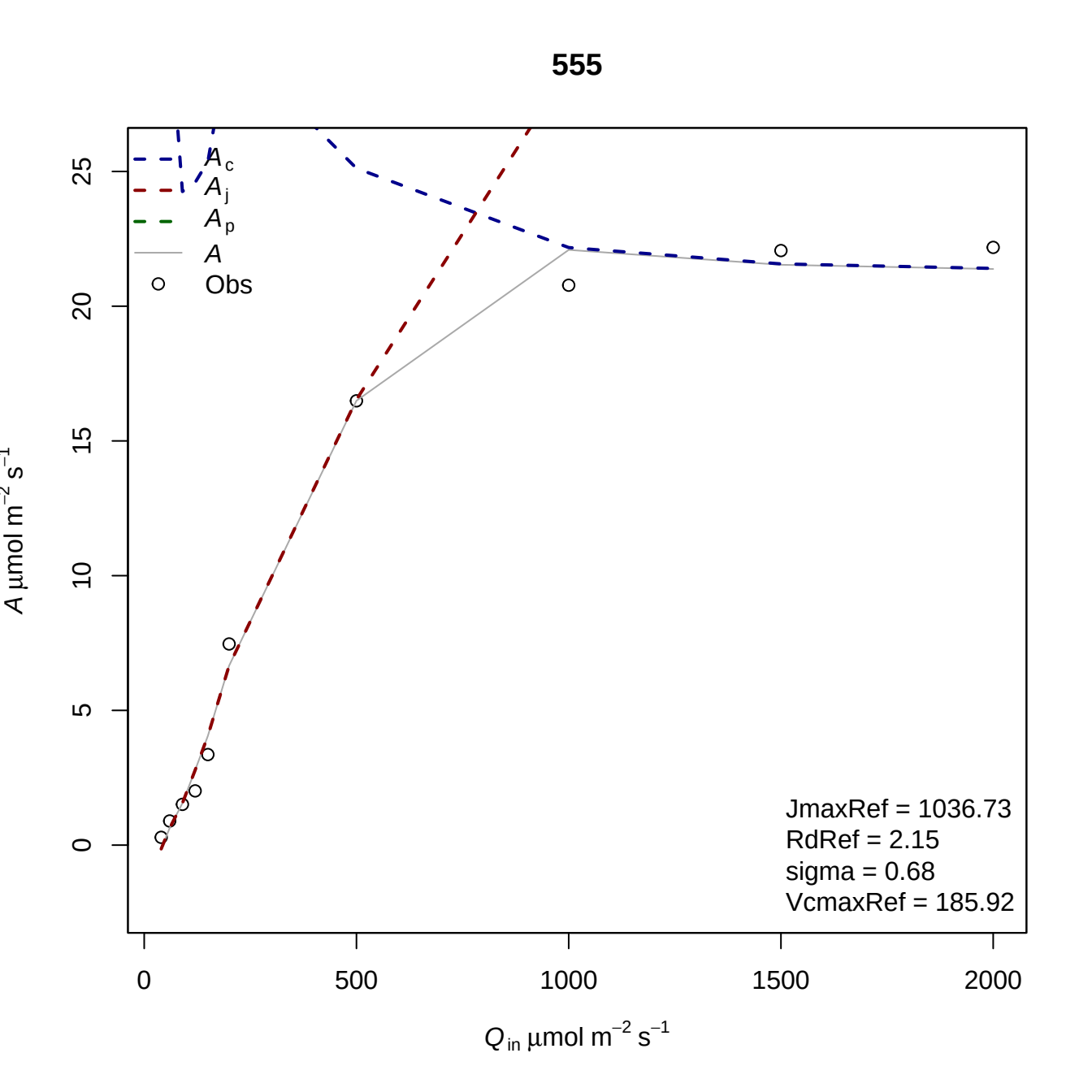

550
A_c
A_j
A_p
A
Obs
JmaxRef = 490.75
RdRef = 1.44
sigma = 0.62
VcmaxRef = 199.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

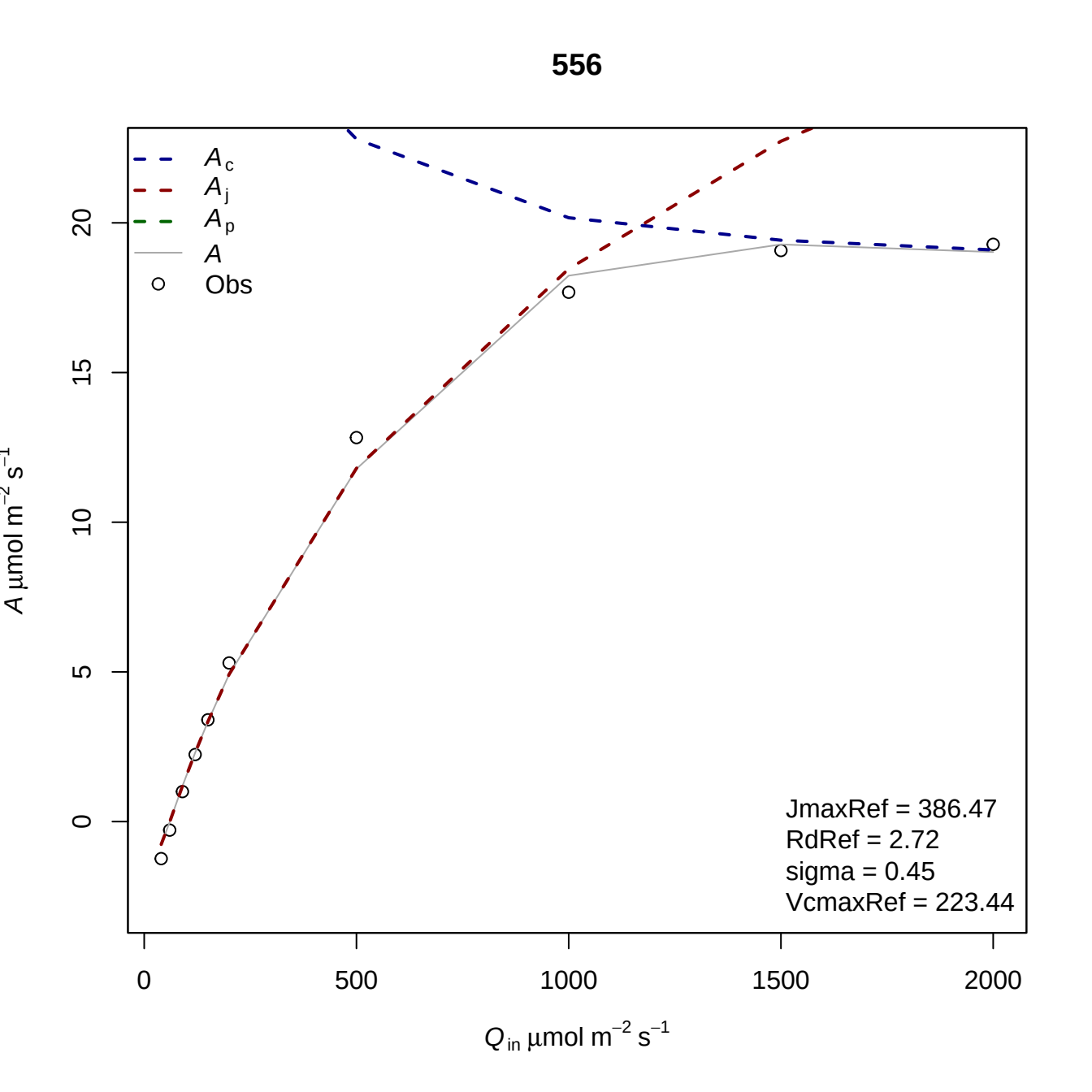

551
A_c
A_j
A_p
A
Obs
JmaxRef = 552.98
RdRef = 1.81
sigma = 0.57
VcmaxRef = 273.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

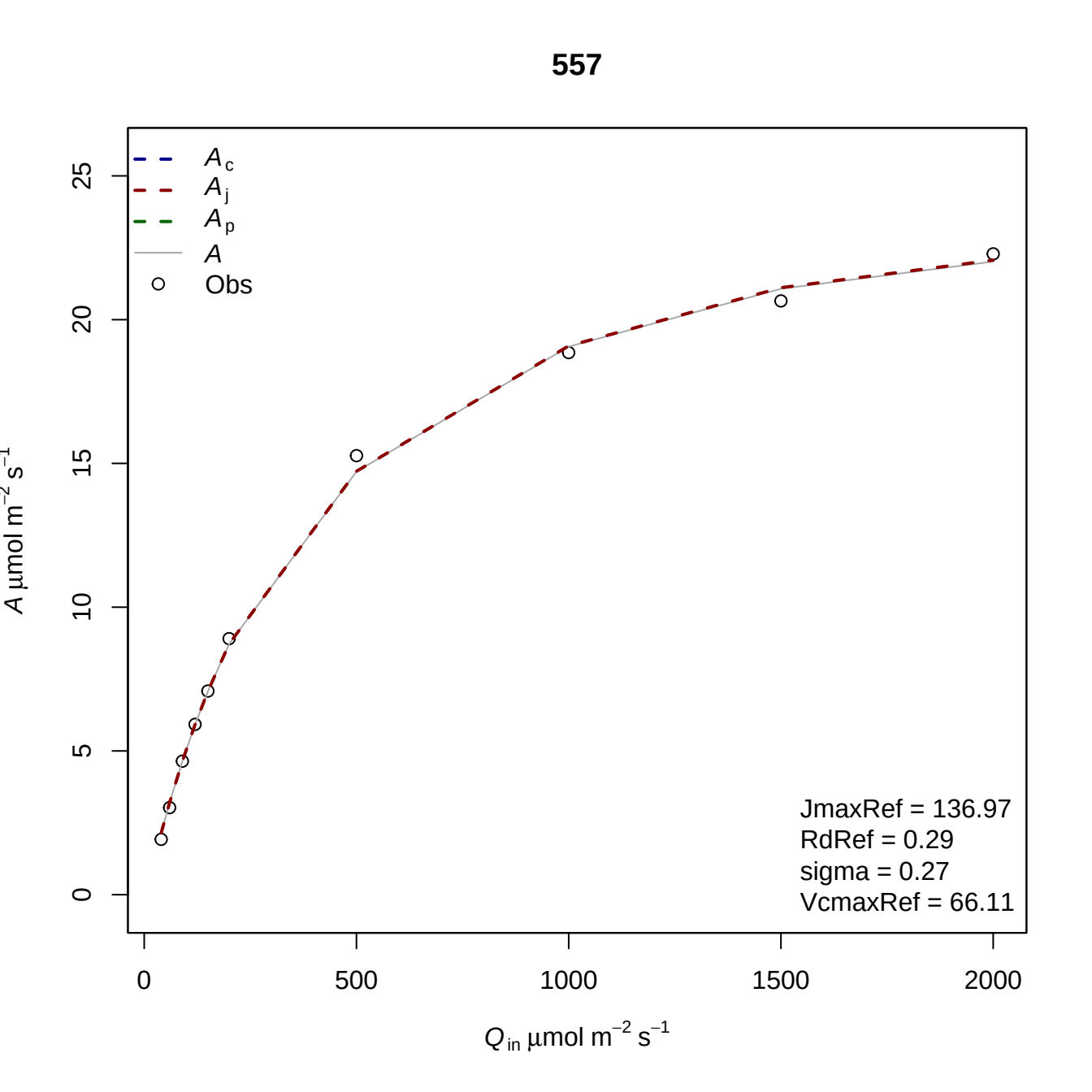

552
A_c
A_j
A_p
A
Obs
JmaxRef = 104.39
RdRef = 0.72
sigma = 1.09
VcmaxRef = 454.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

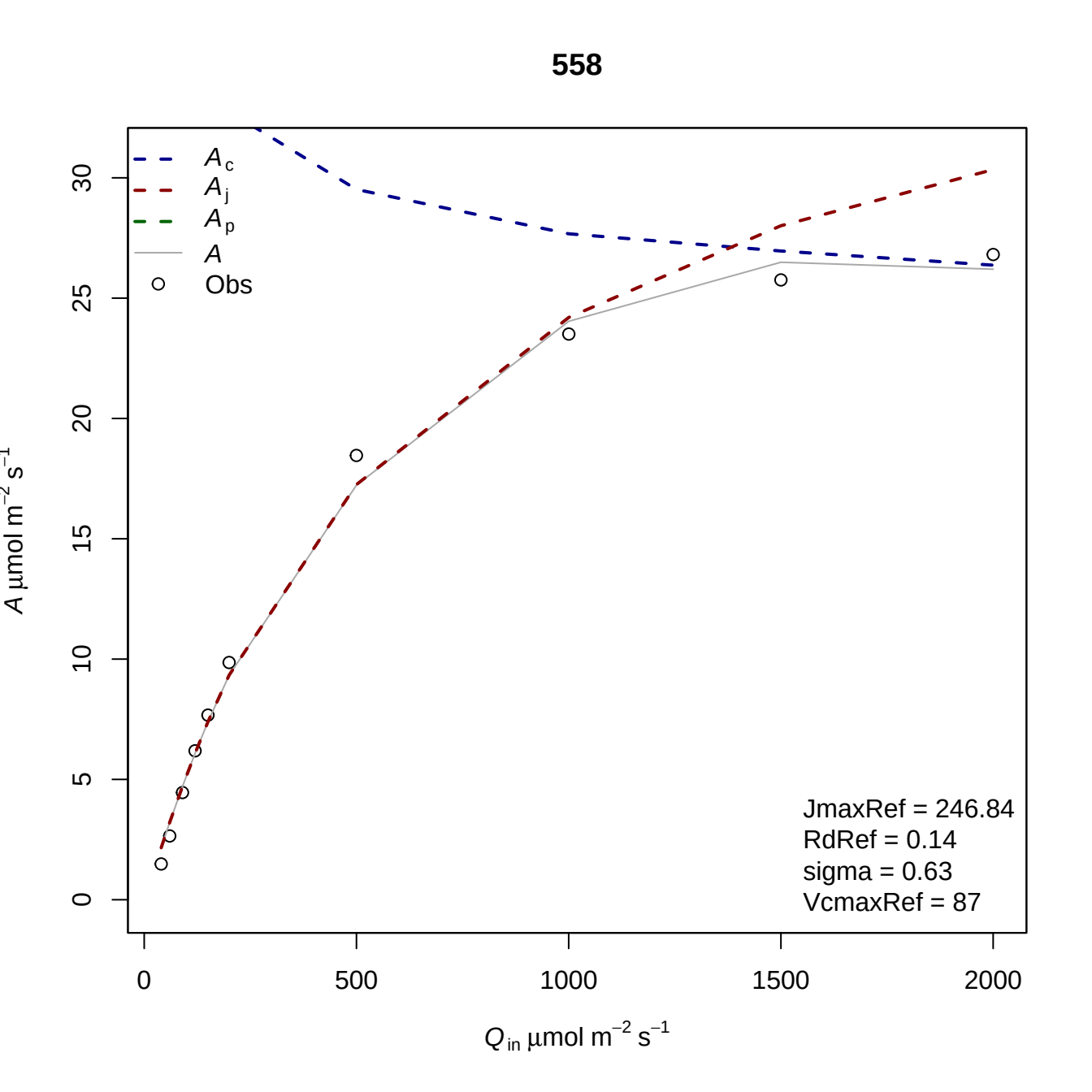

553
A_c
A_j
A_p
A
Obs
JmaxRef = 255.59
RdRef = 1.31
sigma = 0.65
VcmaxRef = 84.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

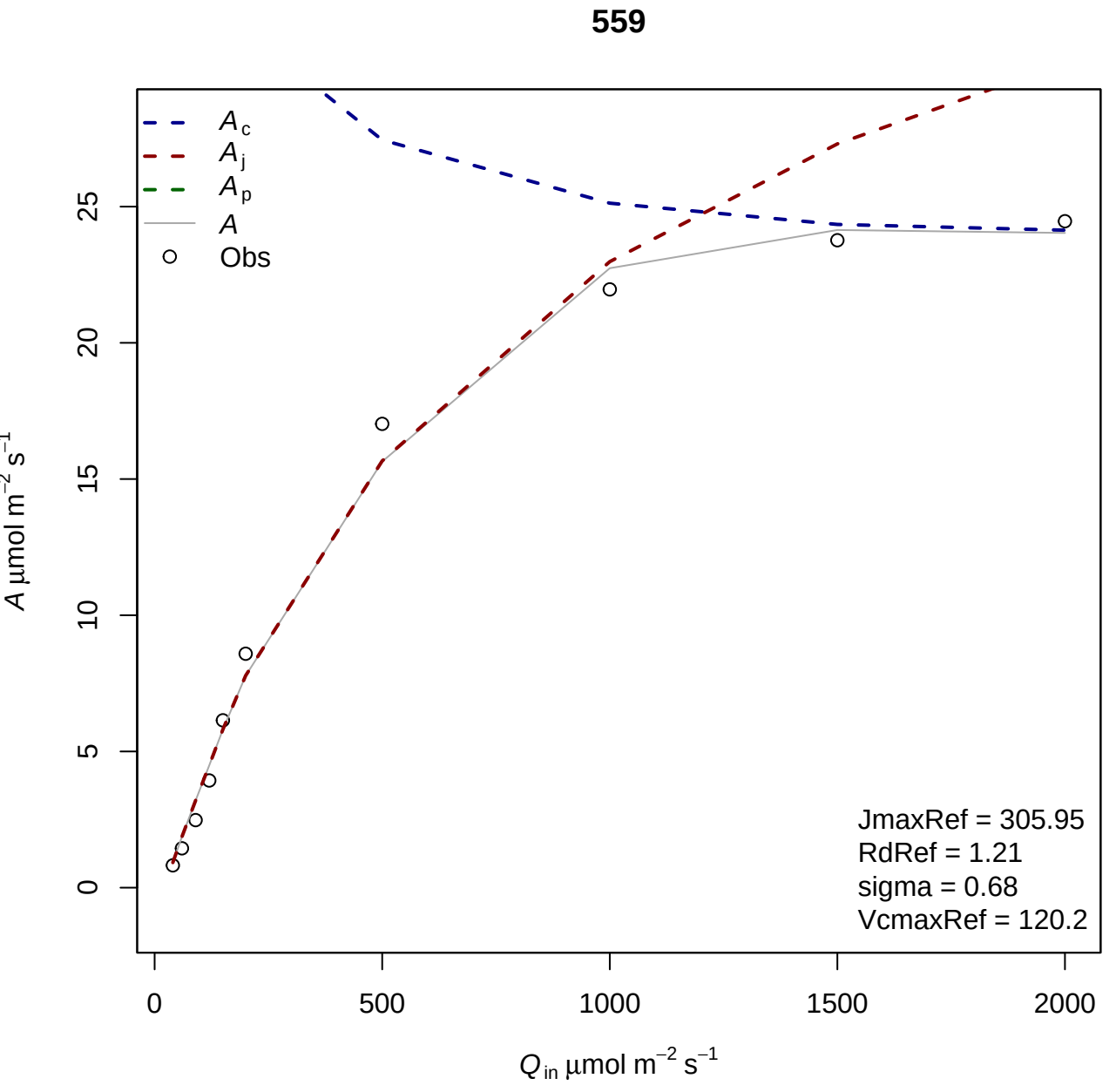

554
A_c
A_j
A_p
A
Obs
JmaxRef = 641.46
RdRef = 1.63
sigma = 0.83
VcmaxRef = 126.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

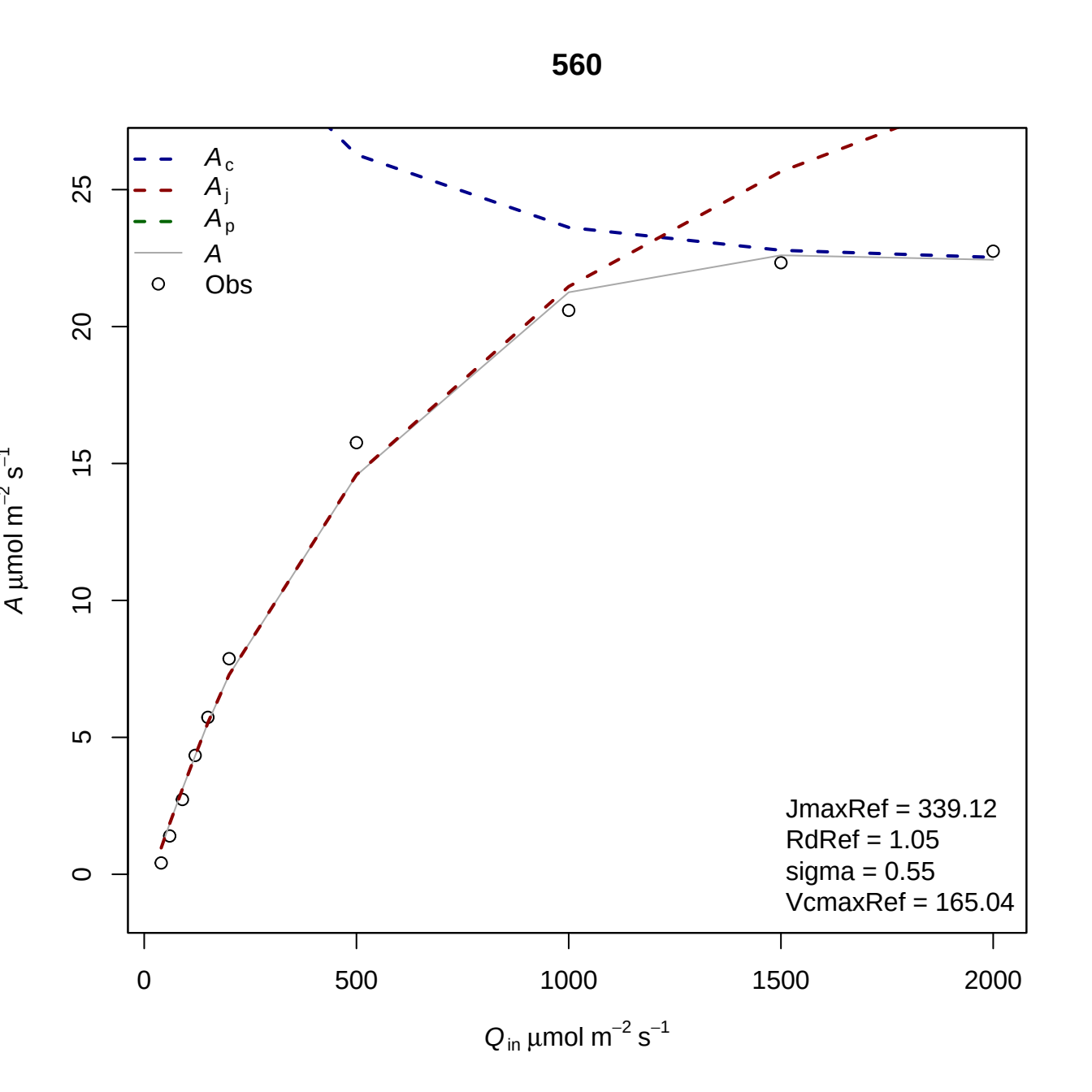

555
A_c
A_j
A_p
A
Obs
JmaxRef = 1036.73
RdRef = 2.15
sigma = 0.68
VcmaxRef = 185.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

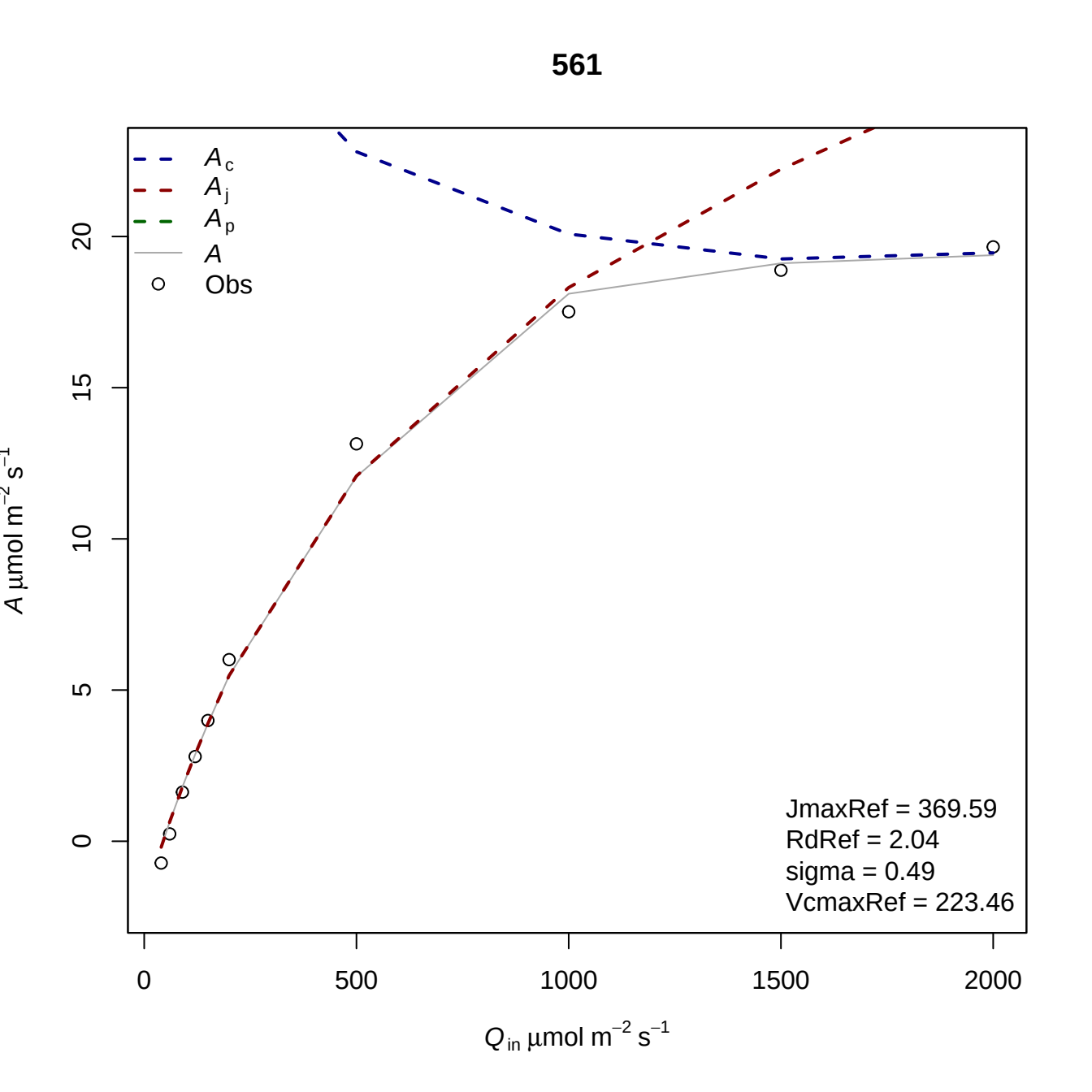

556
A_c
A_j
A_p
A
Obs
JmaxRef = 386.47
RdRef = 2.72
sigma = 0.45
VcmaxRef = 223.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

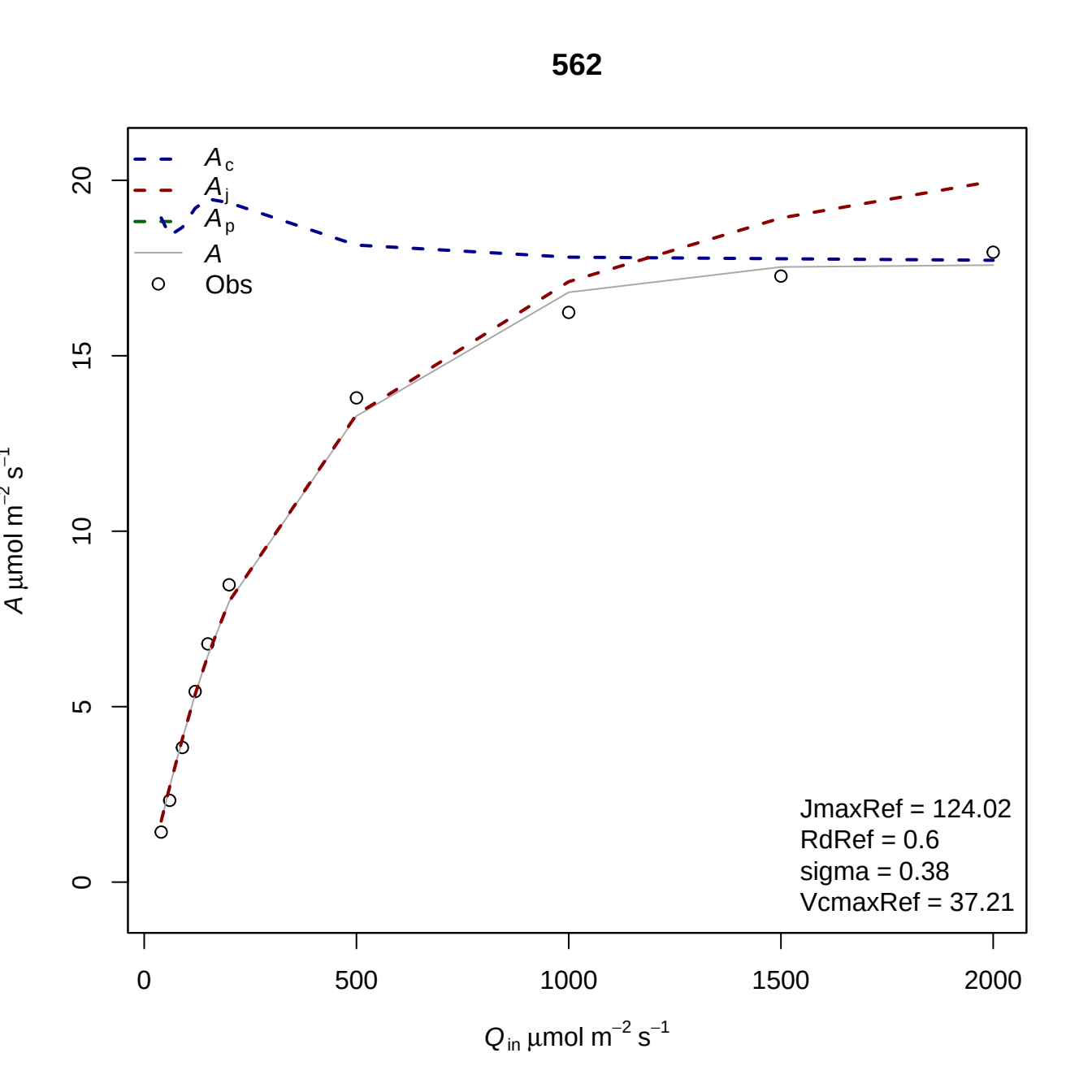

557
A_c
A_j
A_p
A
Obs
JmaxRef = 136.97
RdRef = 0.29
sigma = 0.27
VcmaxRef = 66.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

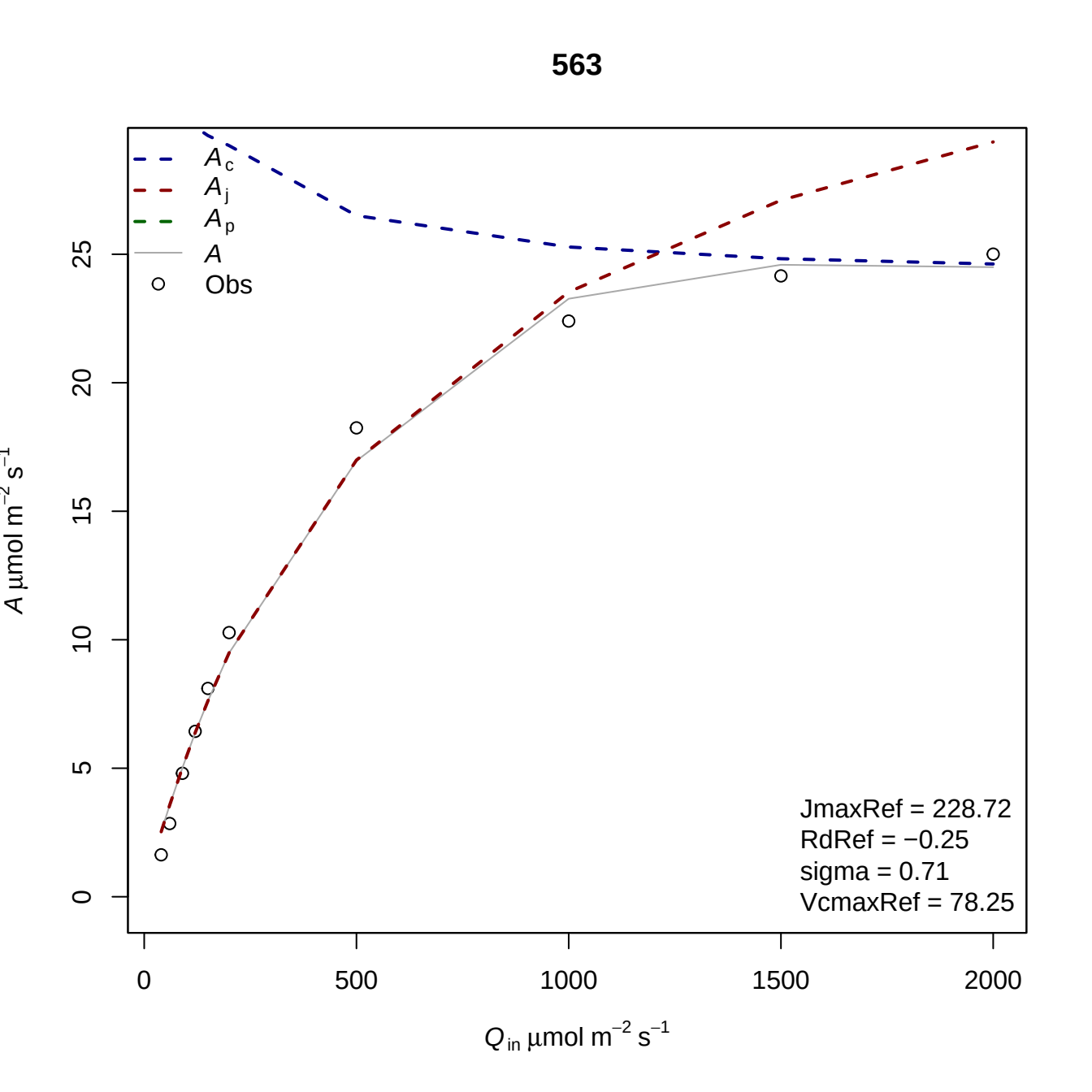

558
A_c
A_j
A_p
A
Obs
JmaxRef = 246.84
RdRef = 0.14
sigma = 0.63
VcmaxRef = 87
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

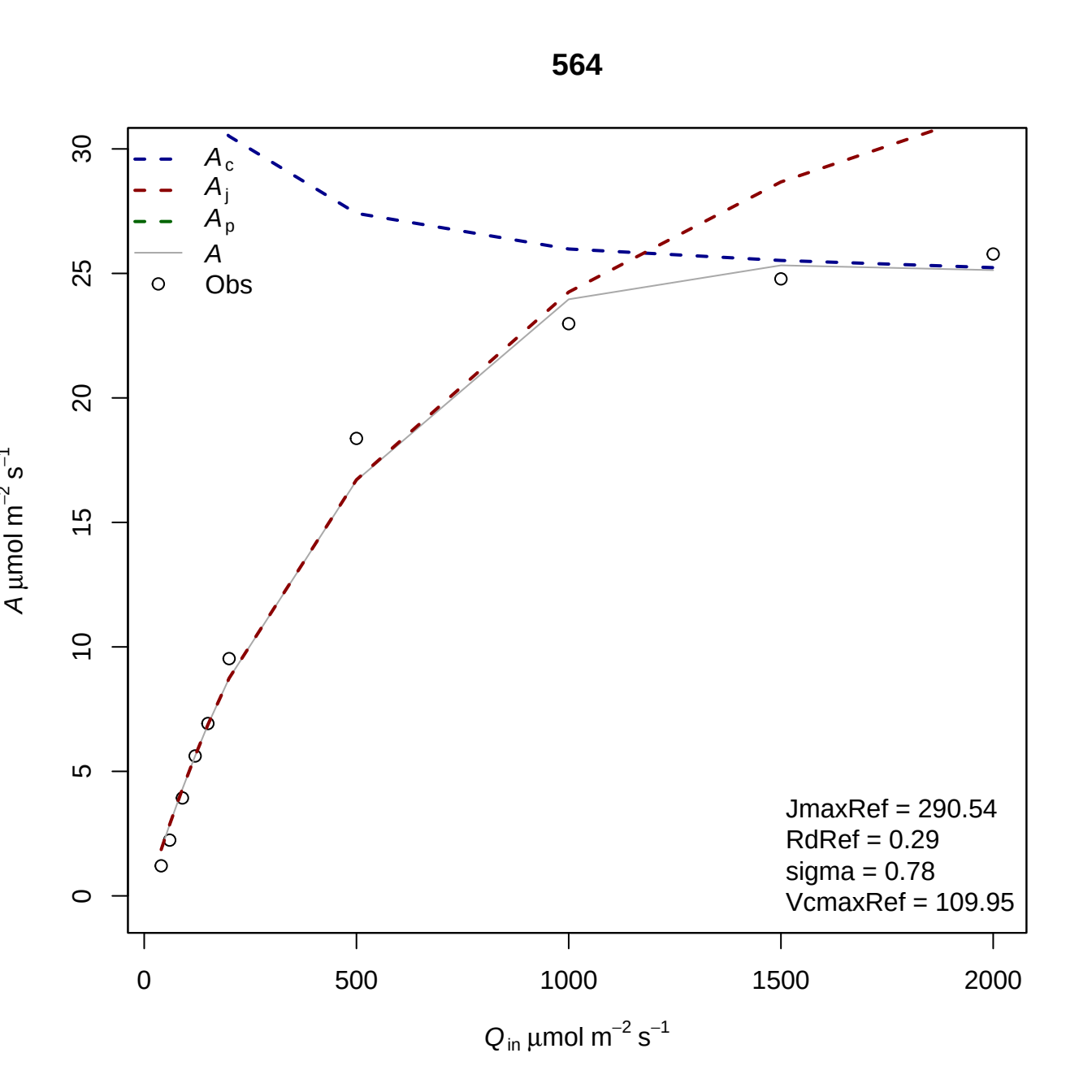

559
A_c
A_j
A_p
A
Obs
JmaxRef = 305.95
RdRef = 1.21
sigma = 0.68
VcmaxRef = 120.2
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

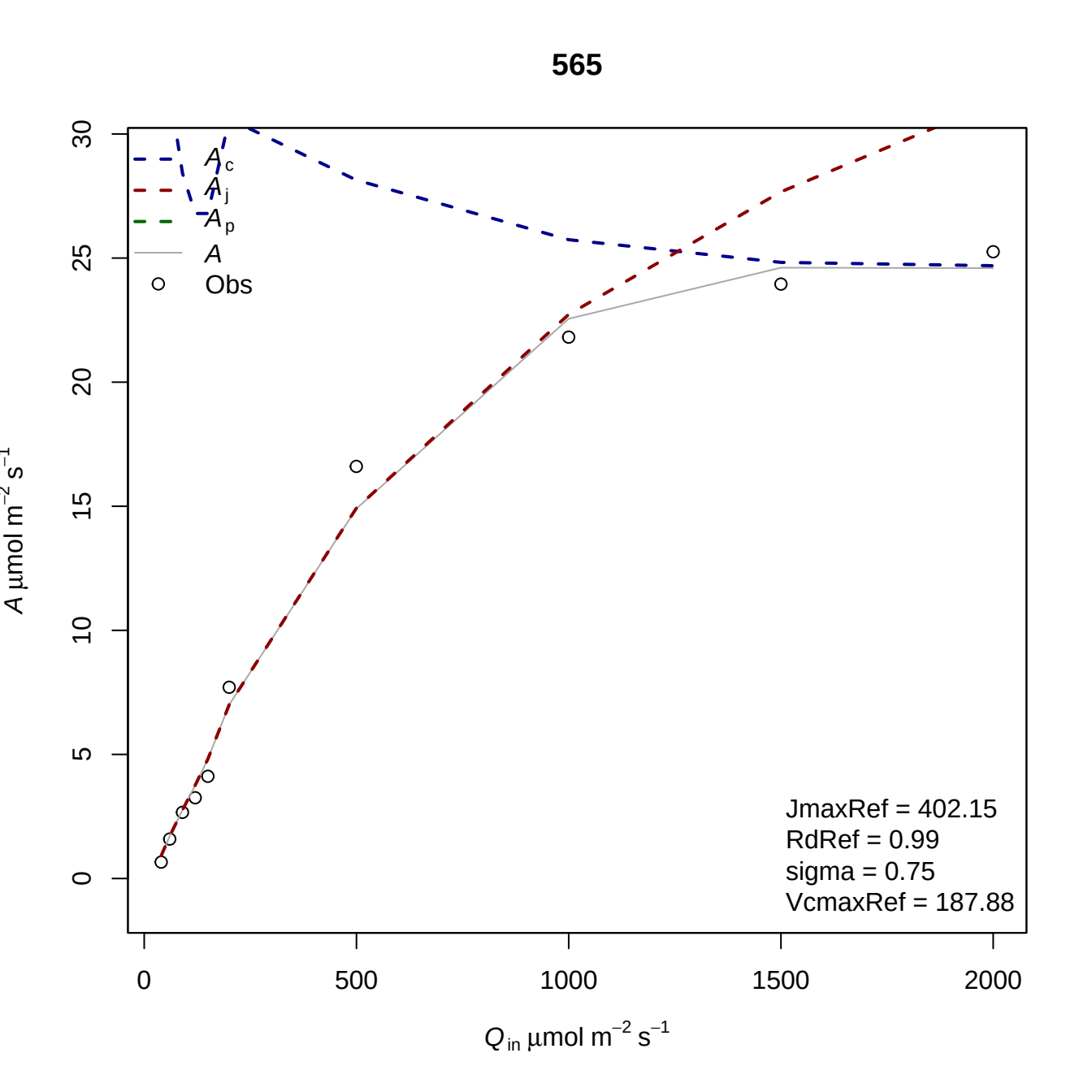

560
A_c
A_j
A_p
A
Obs
JmaxRef = 339.12
RdRef = 1.05
sigma = 0.55
VcmaxRef = 165.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

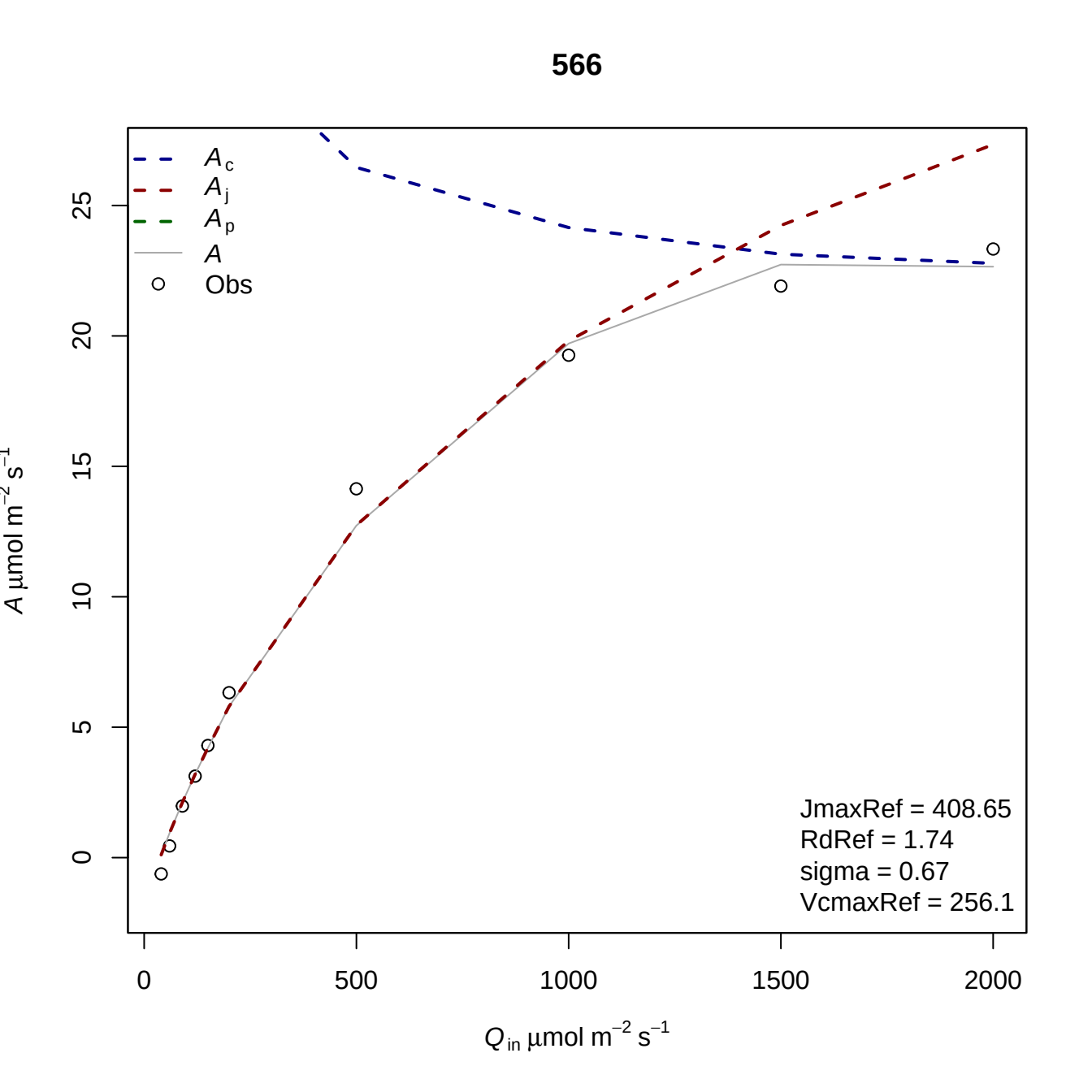

561
A_c
A_j
A_p
A
Obs
JmaxRef = 369.59
RdRef = 2.04
sigma = 0.49
VcmaxRef = 223.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

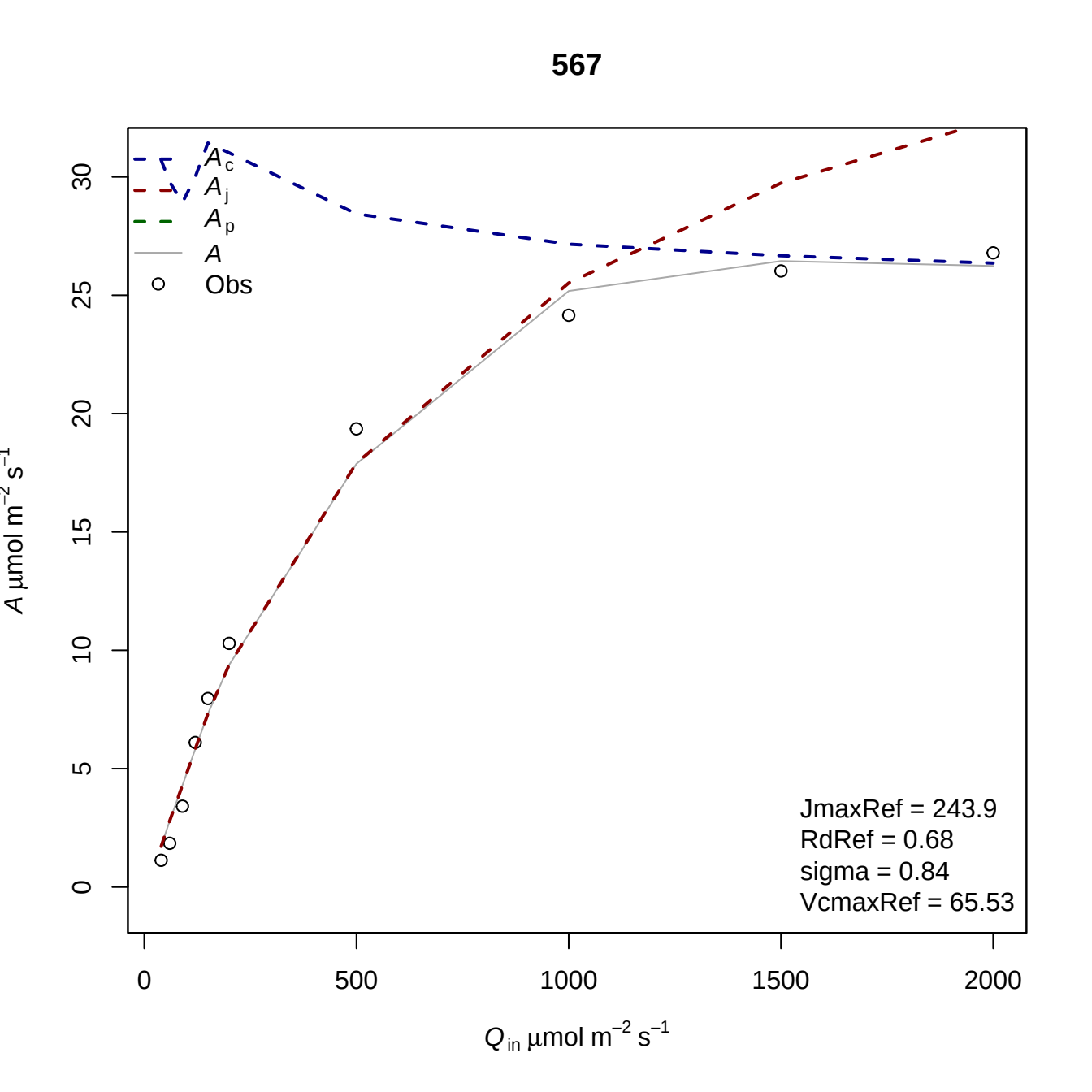

562
A_c
A_j
A_p
A
Obs
JmaxRef = 124.02
RdRef = 0.6
sigma = 0.38
VcmaxRef = 37.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

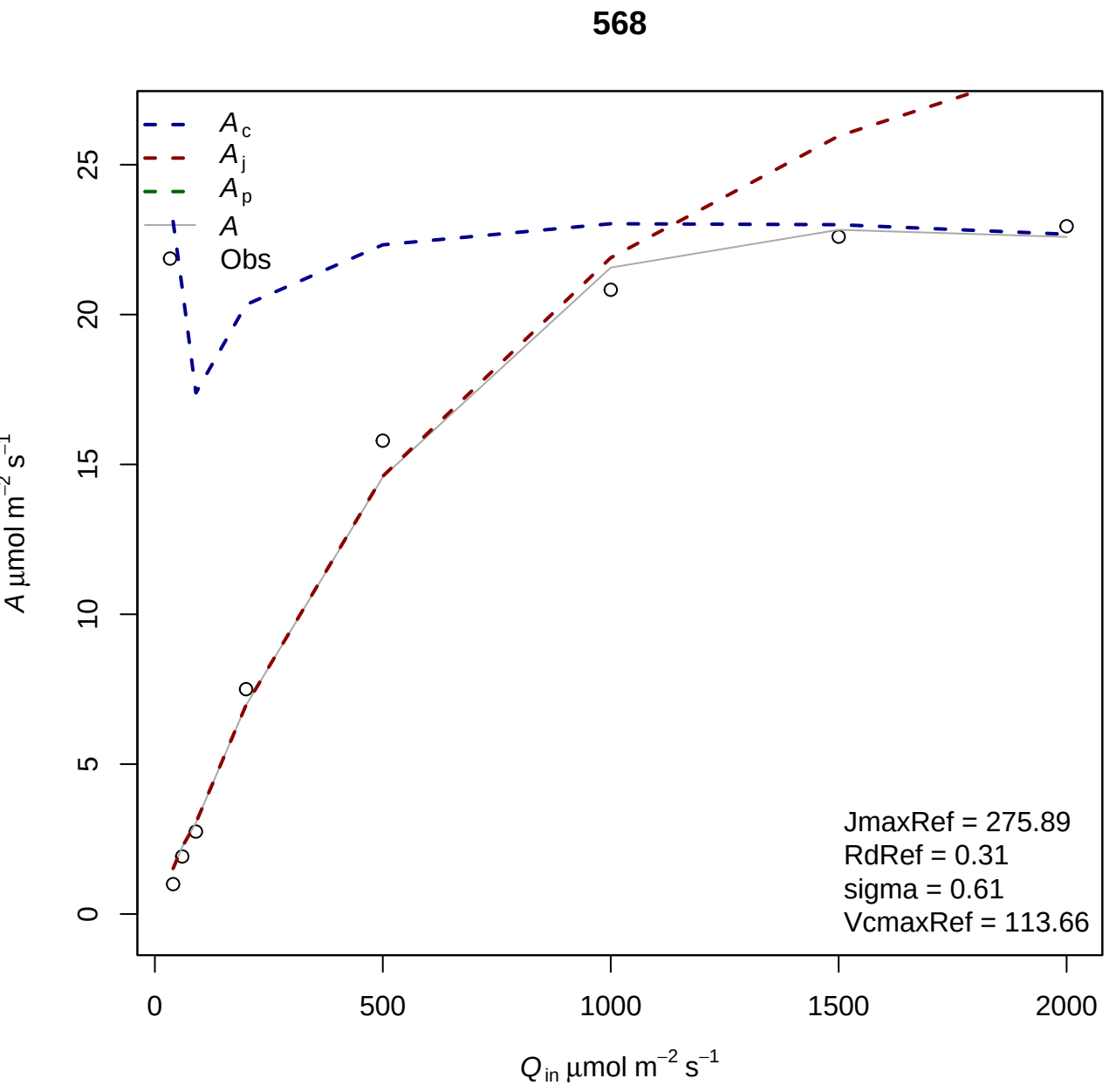

563
A_c
A_j
A_p
A
Obs
JmaxRef = 228.72
RdRef = −0.25
sigma = 0.71
VcmaxRef = 78.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

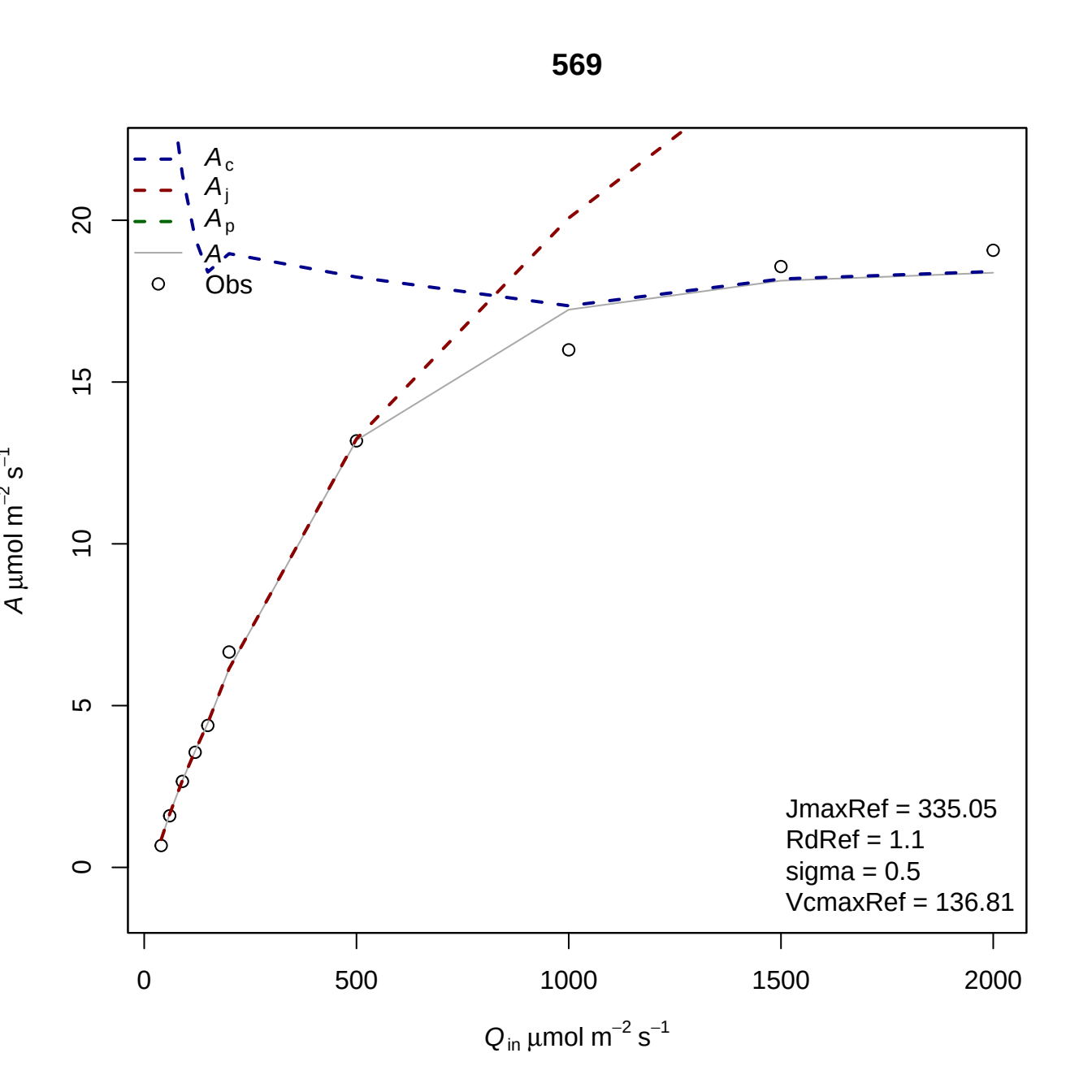

564
A_c
A_j
A_p
A
Obs
JmaxRef = 290.54
RdRef = 0.29
sigma = 0.78
VcmaxRef = 109.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

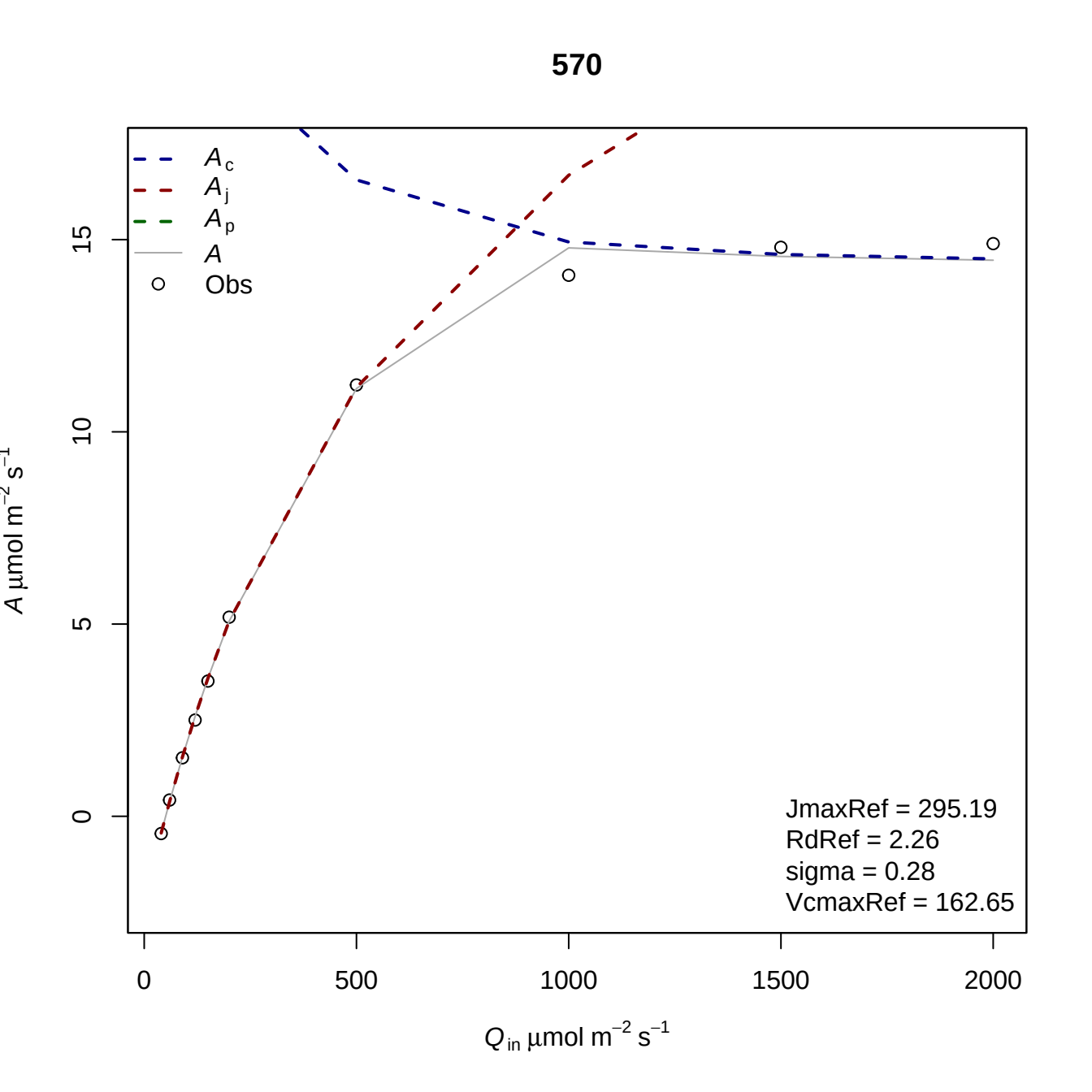

565
A_c
A_j
A_p
A
Obs
JmaxRef = 402.15
RdRef = 0.99
sigma = 0.75
VcmaxRef = 187.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

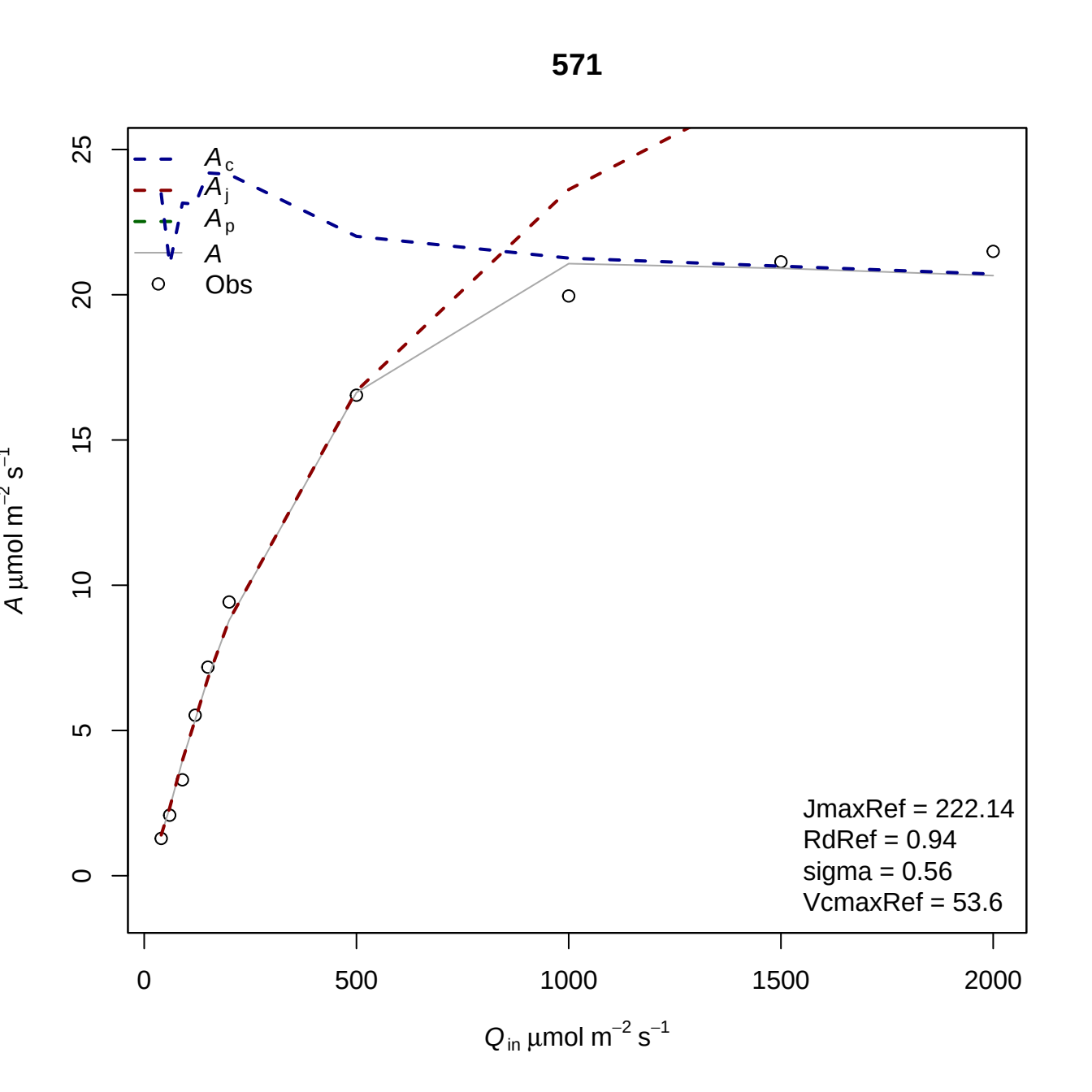

566
A_c
A_j
A_p
A
Obs
JmaxRef = 408.65
RdRef = 1.74
sigma = 0.67
VcmaxRef = 256.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

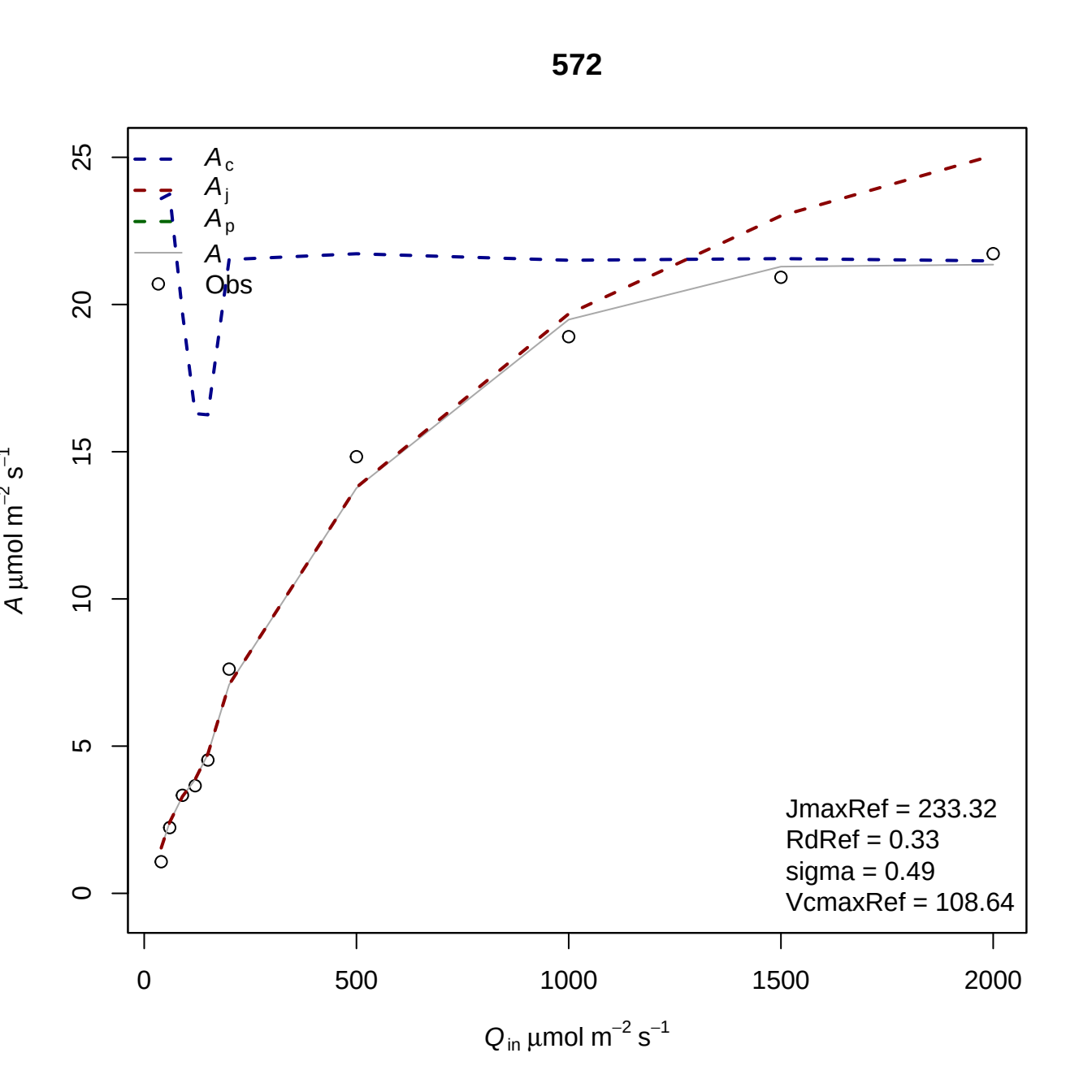

567
A_c
A_j
A_p
A
Obs
JmaxRef = 243.9
RdRef = 0.68
sigma = 0.84
VcmaxRef = 65.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

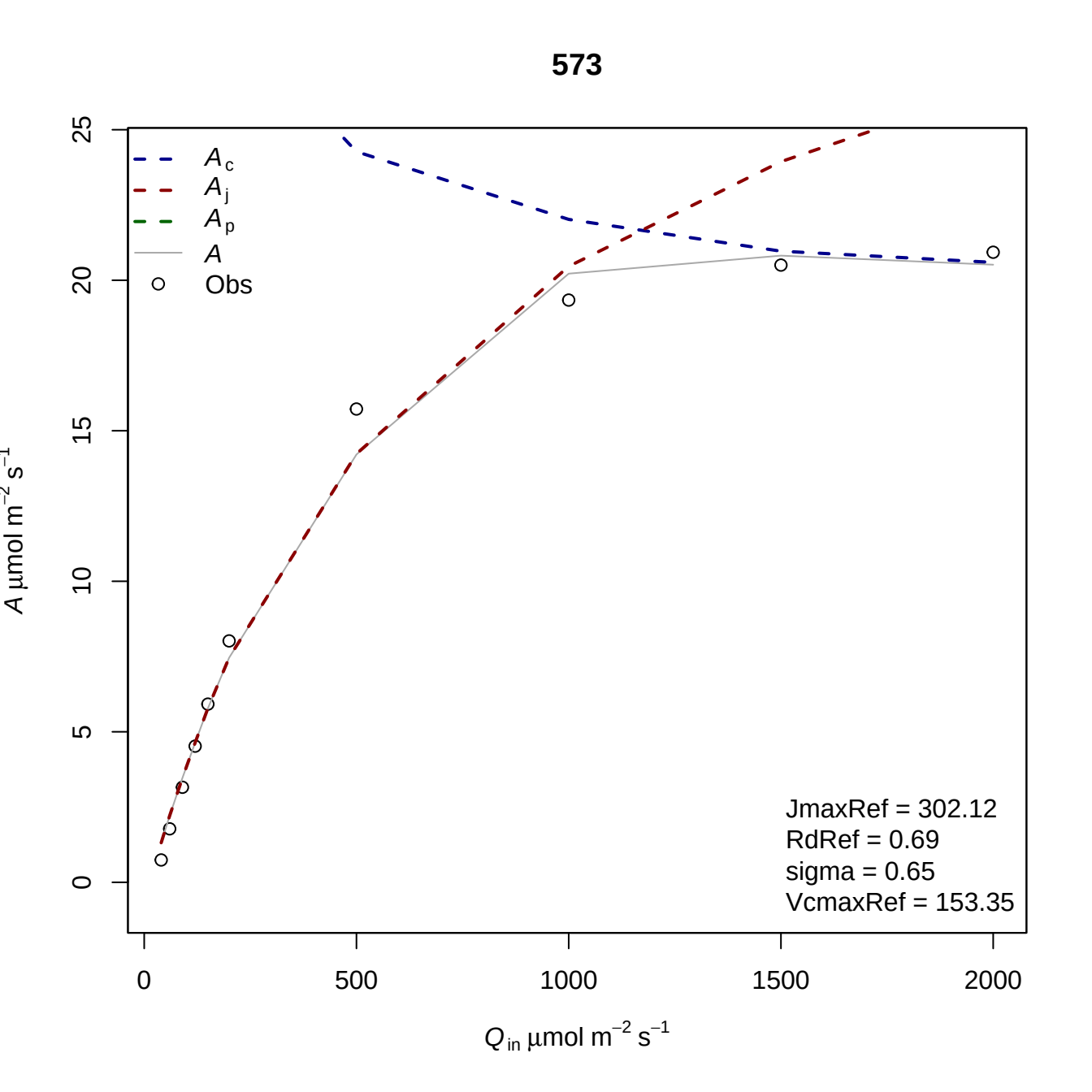

568
A_c
A_j
A_p
A
Obs
JmaxRef = 275.89
RdRef = 0.31
sigma = 0.61
VcmaxRef = 113.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

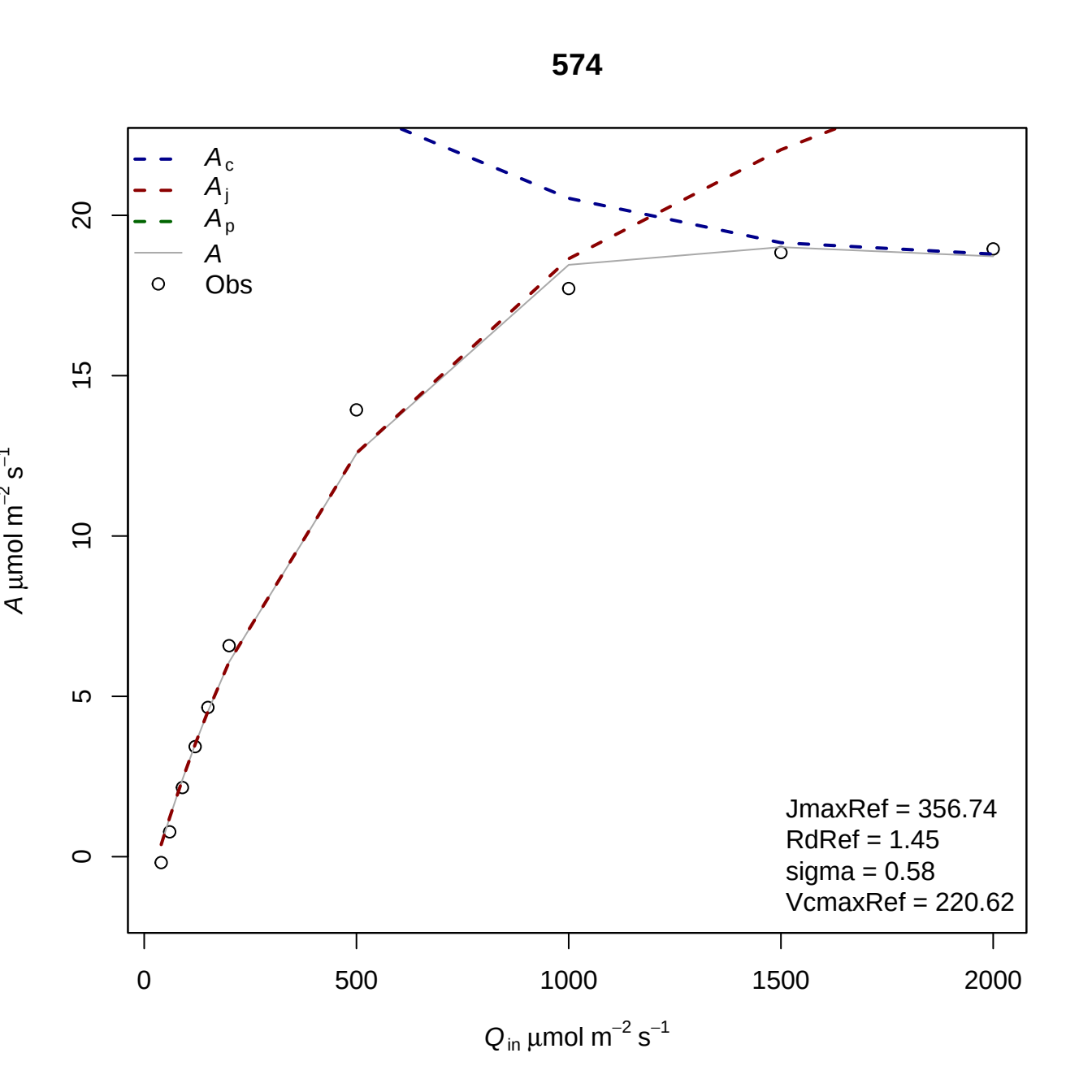

569
A_c
A_j
A_p
A
Obs
JmaxRef = 335.05
RdRef = 1.1
sigma = 0.5
VcmaxRef = 136.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

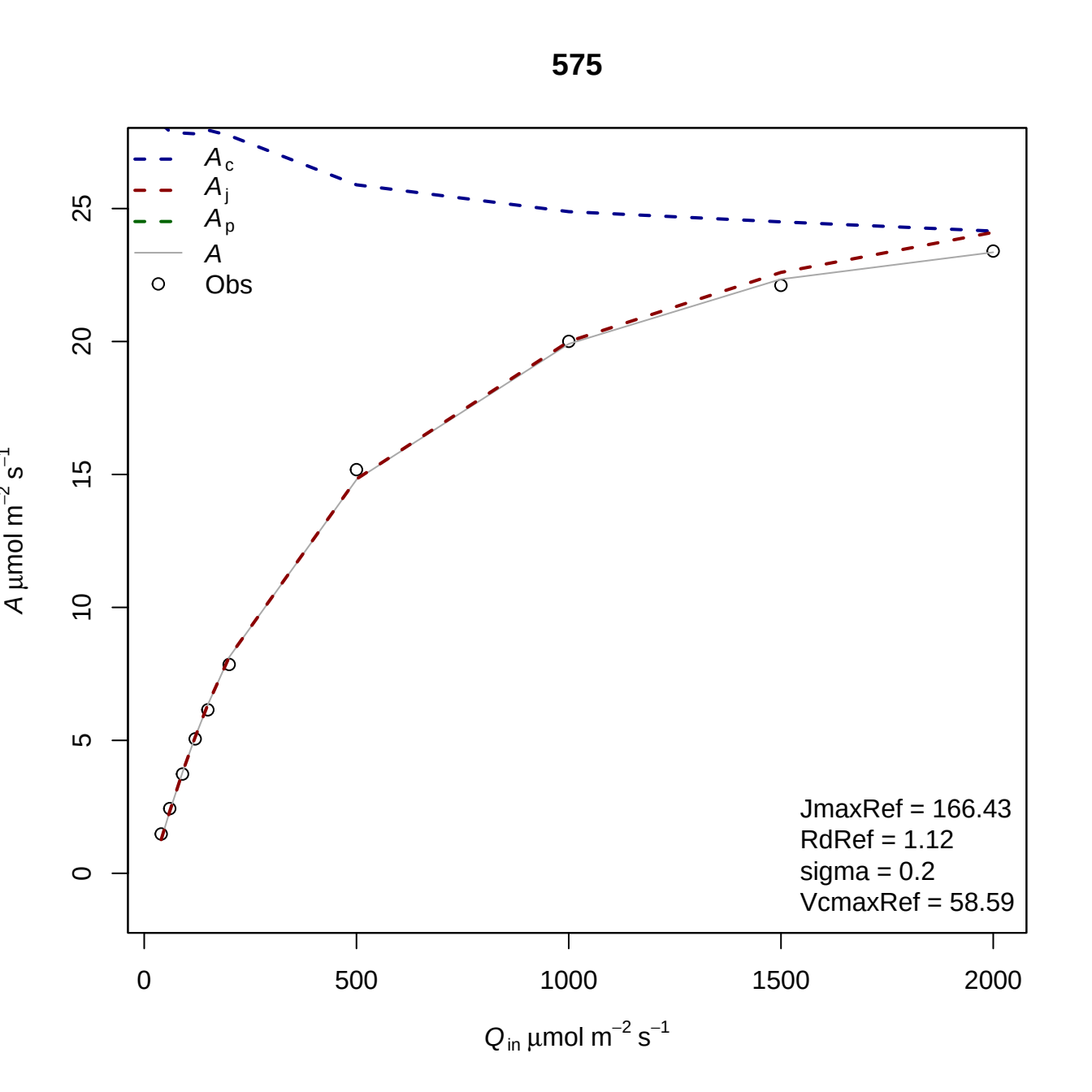

570
A_c
A_j
A_p
A
Obs
JmaxRef = 295.19
RdRef = 2.26
sigma = 0.28
VcmaxRef = 162.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

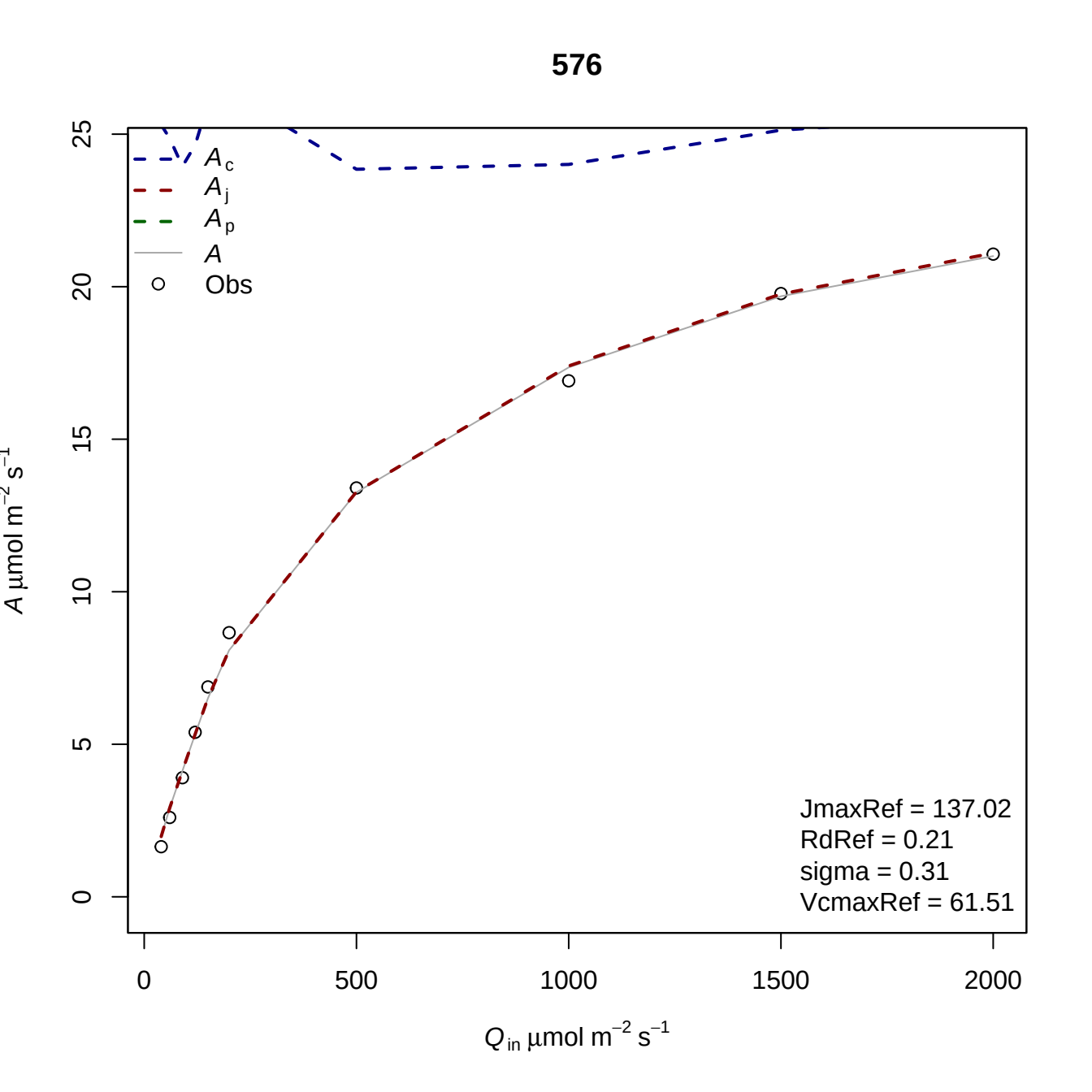

571
A_c
A_j
A_p
A
Obs
JmaxRef = 222.14
RdRef = 0.94
sigma = 0.56
VcmaxRef = 53.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

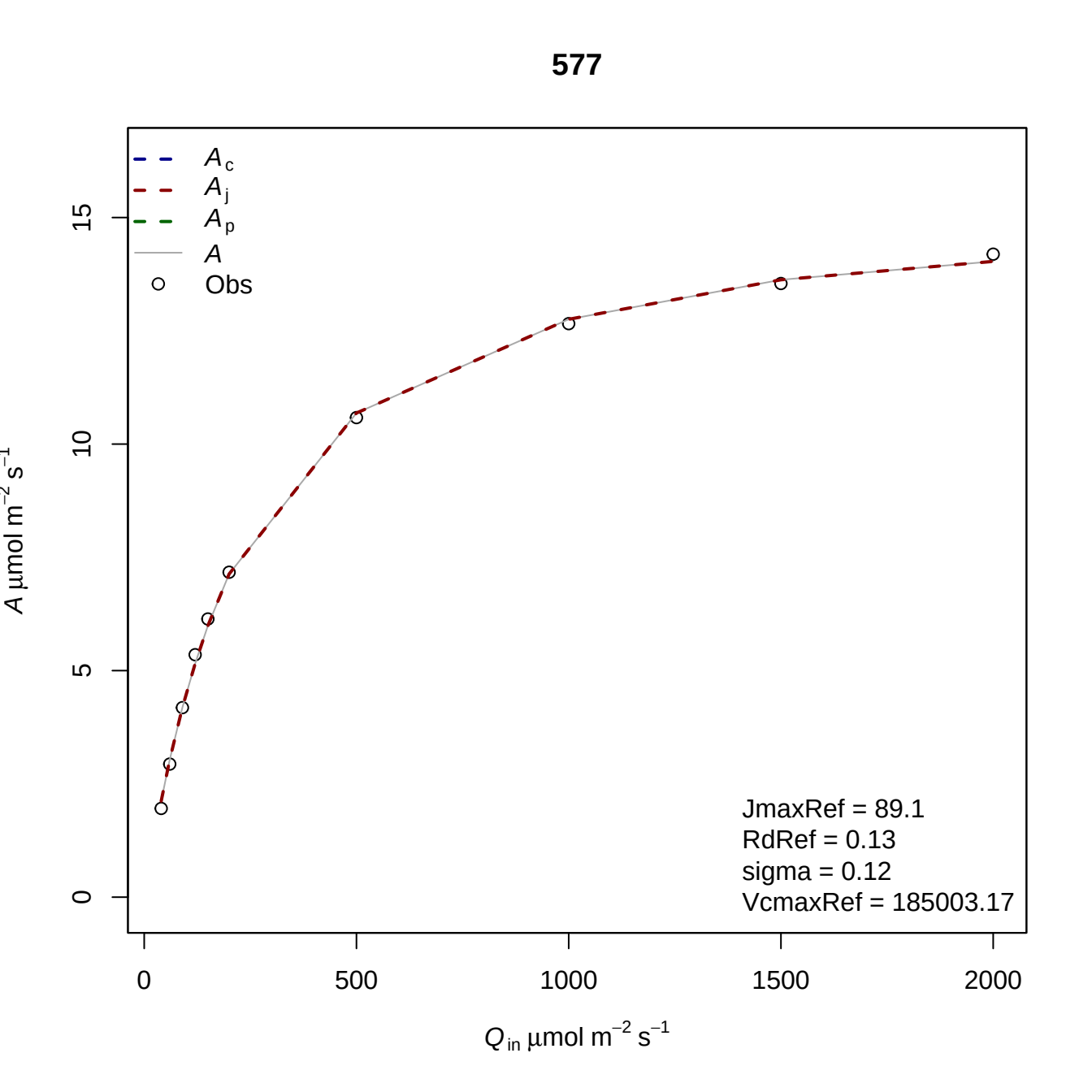

572
A_c
A_j
A_p
A
Obs
JmaxRef = 233.32
RdRef = 0.33
sigma = 0.49
VcmaxRef = 108.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

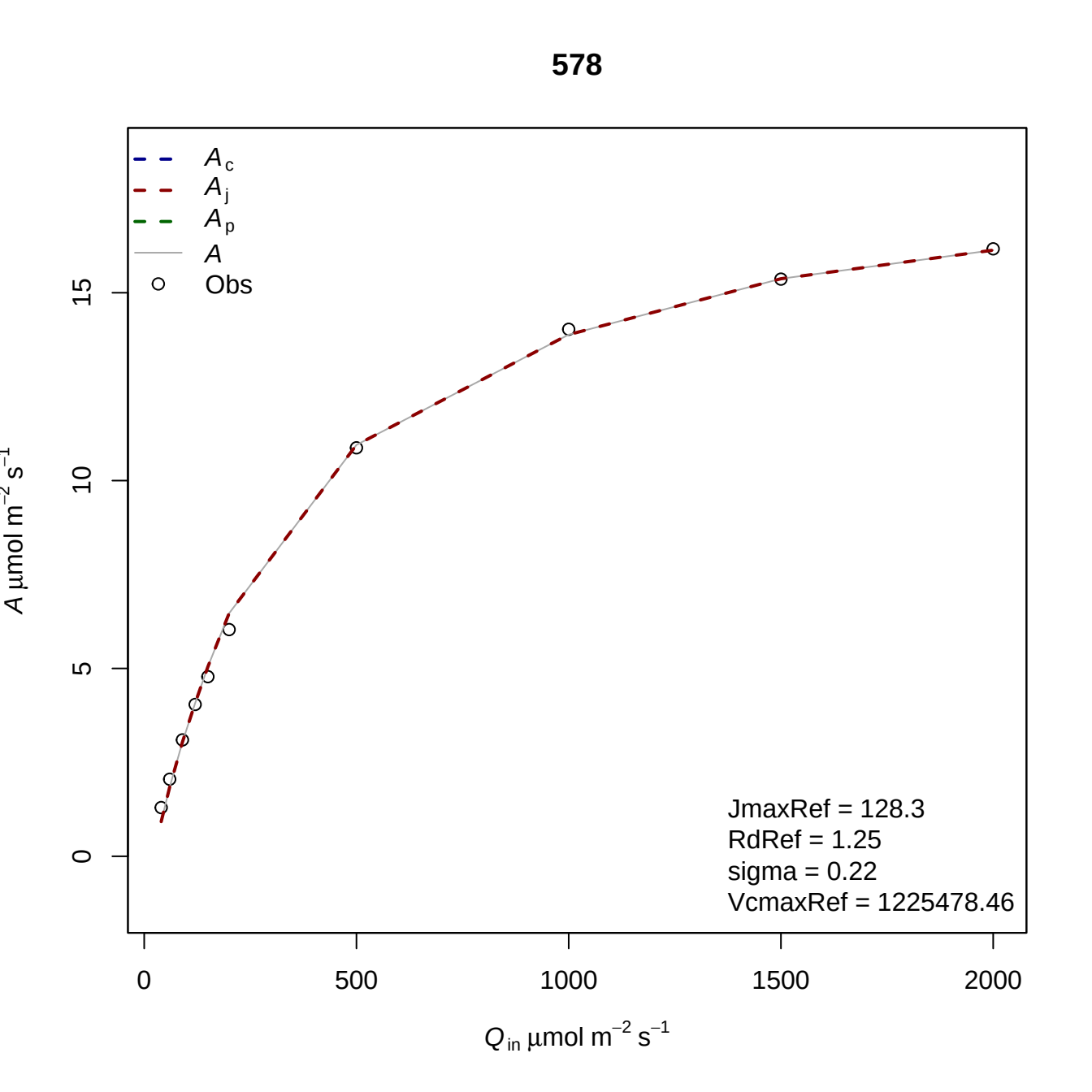

573
A_c
A_j
A_p
A
Obs
JmaxRef = 302.12
RdRef = 0.69
sigma = 0.65
VcmaxRef = 153.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

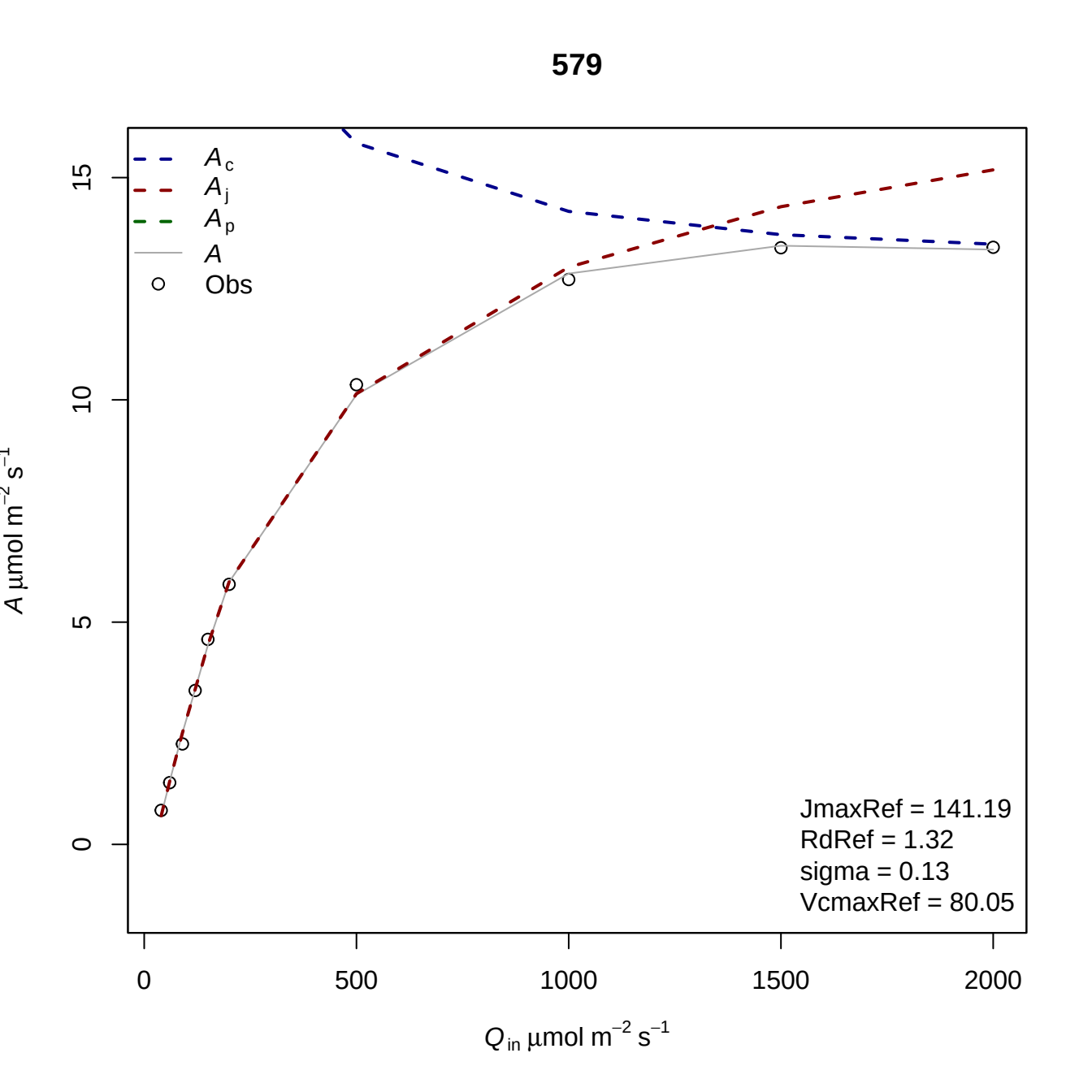

574
A_c
A_j
A_p
A
Obs
JmaxRef = 356.74
RdRef = 1.45
sigma = 0.58
VcmaxRef = 220.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

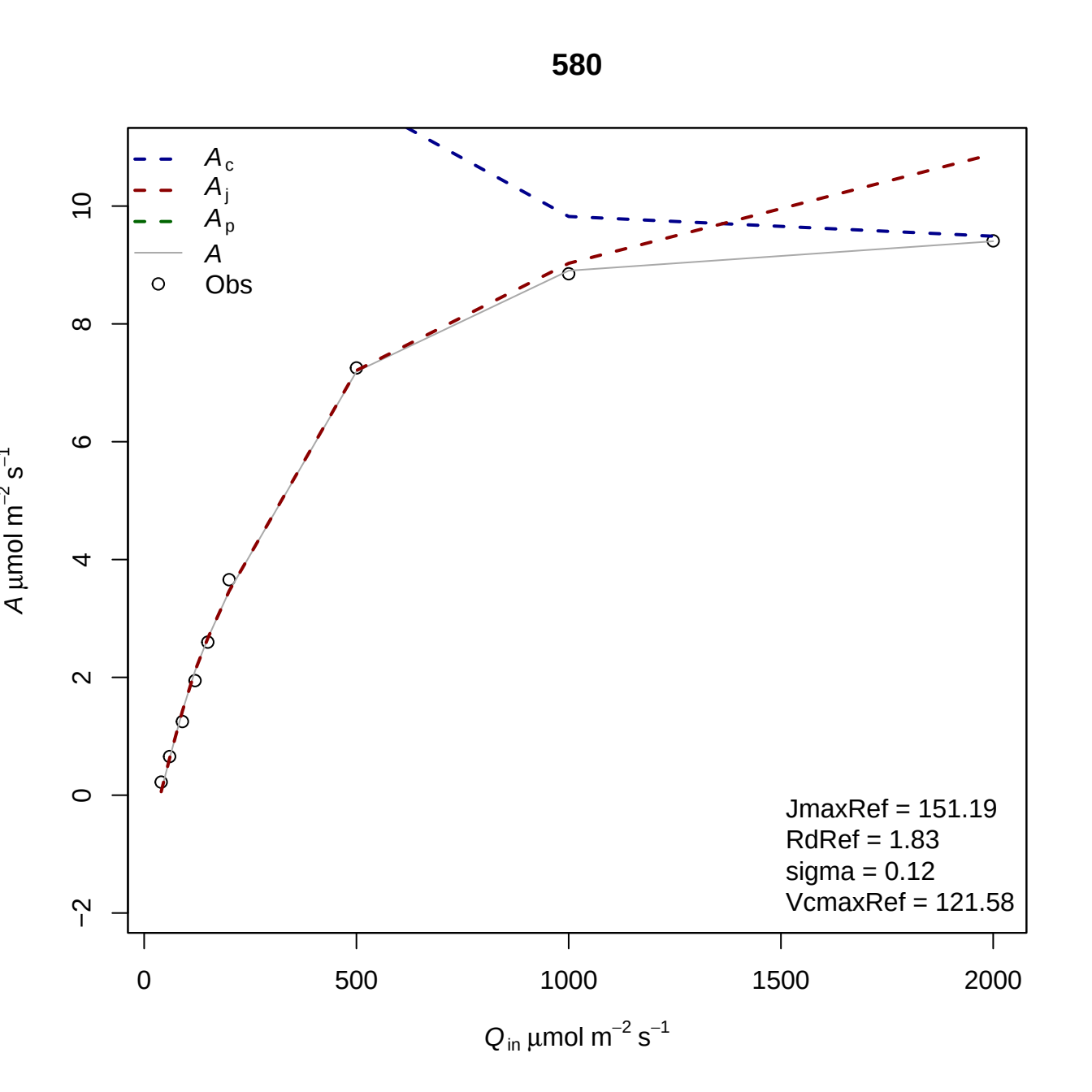

575
A_c
A_j
A_p
A
Obs
JmaxRef = 166.43
RdRef = 1.12
sigma = 0.2
VcmaxRef = 58.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

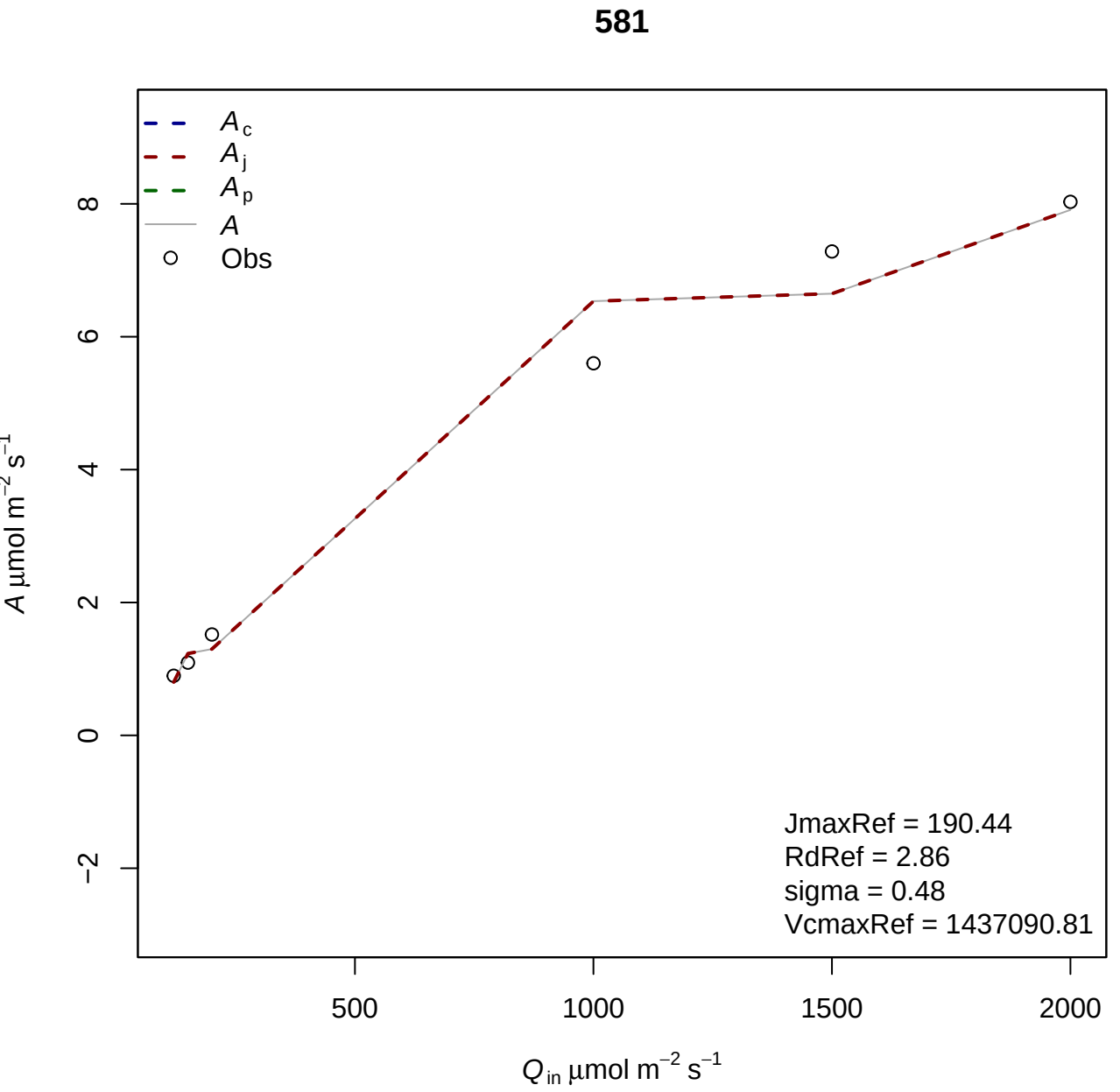

576
A_c
A_j
A_p
A
Obs
JmaxRef = 137.02
RdRef = 0.21
sigma = 0.31
VcmaxRef = 61.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

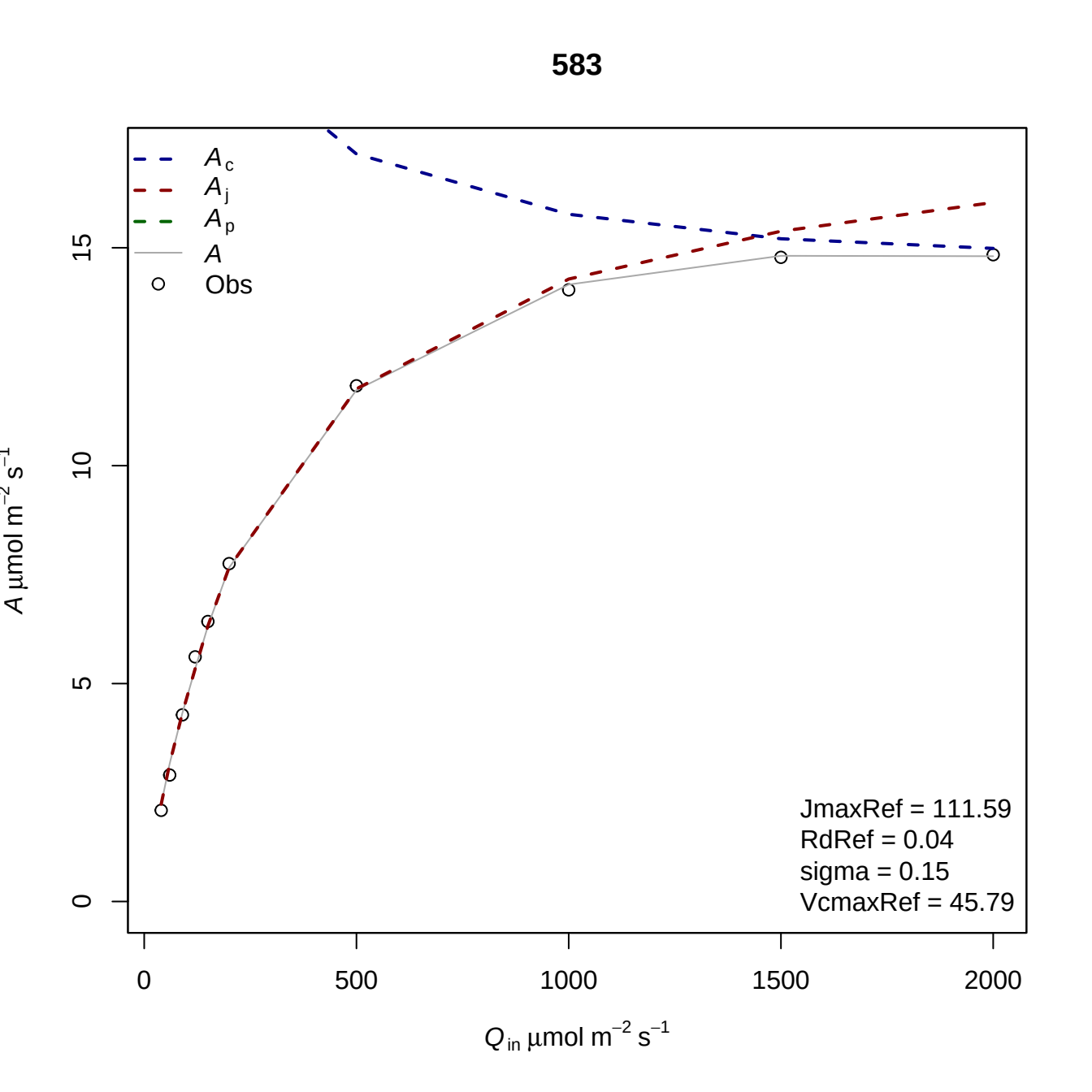

577
A_c
A_j
A_p
A
Obs
JmaxRef = 89.1
RdRef = 0.13
sigma = 0.12
VcmaxRef = 185003.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

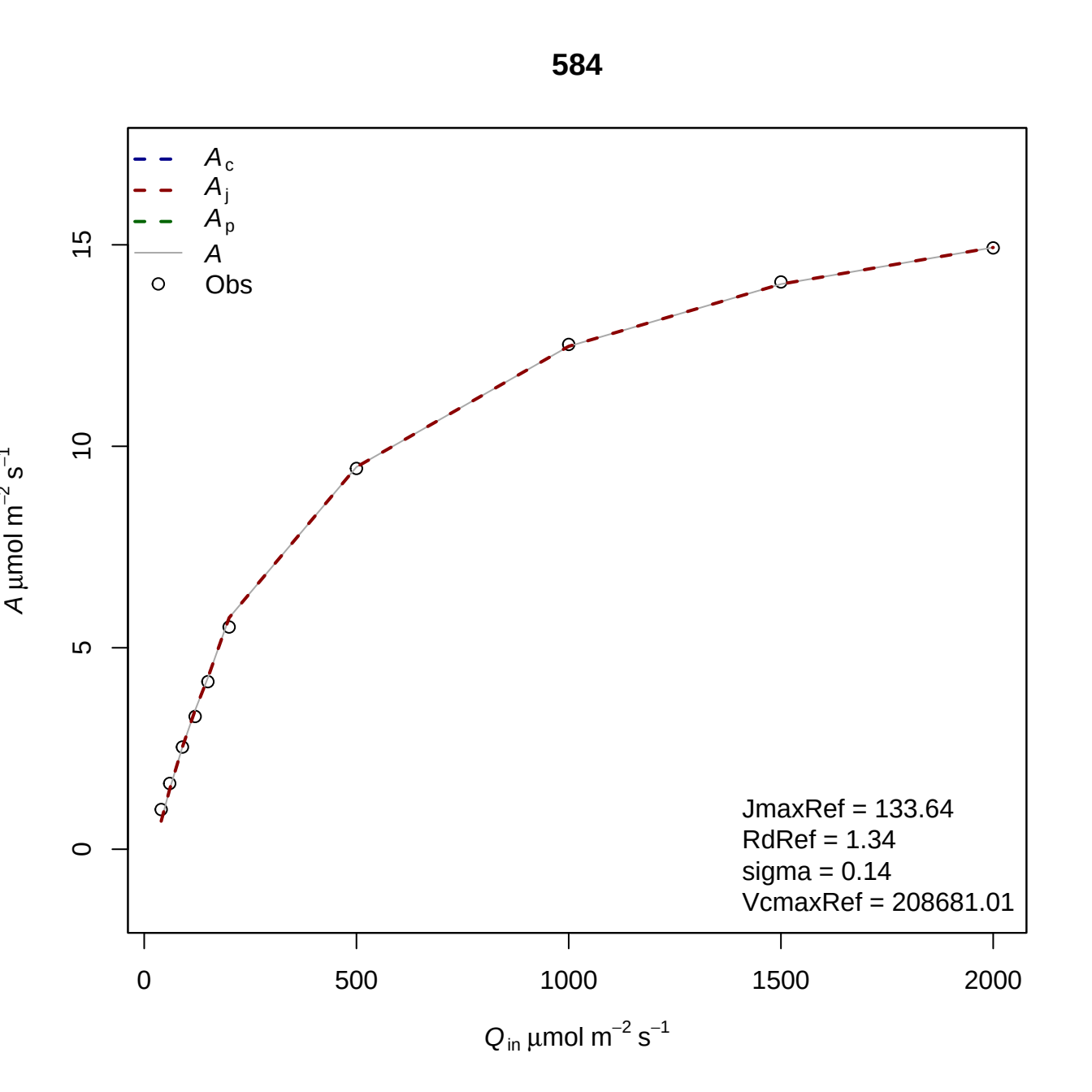

578
A_c
A_j
A_p
A
Obs
JmaxRef = 128.3
RdRef = 1.25
sigma = 0.22
VcmaxRef = 1225478.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

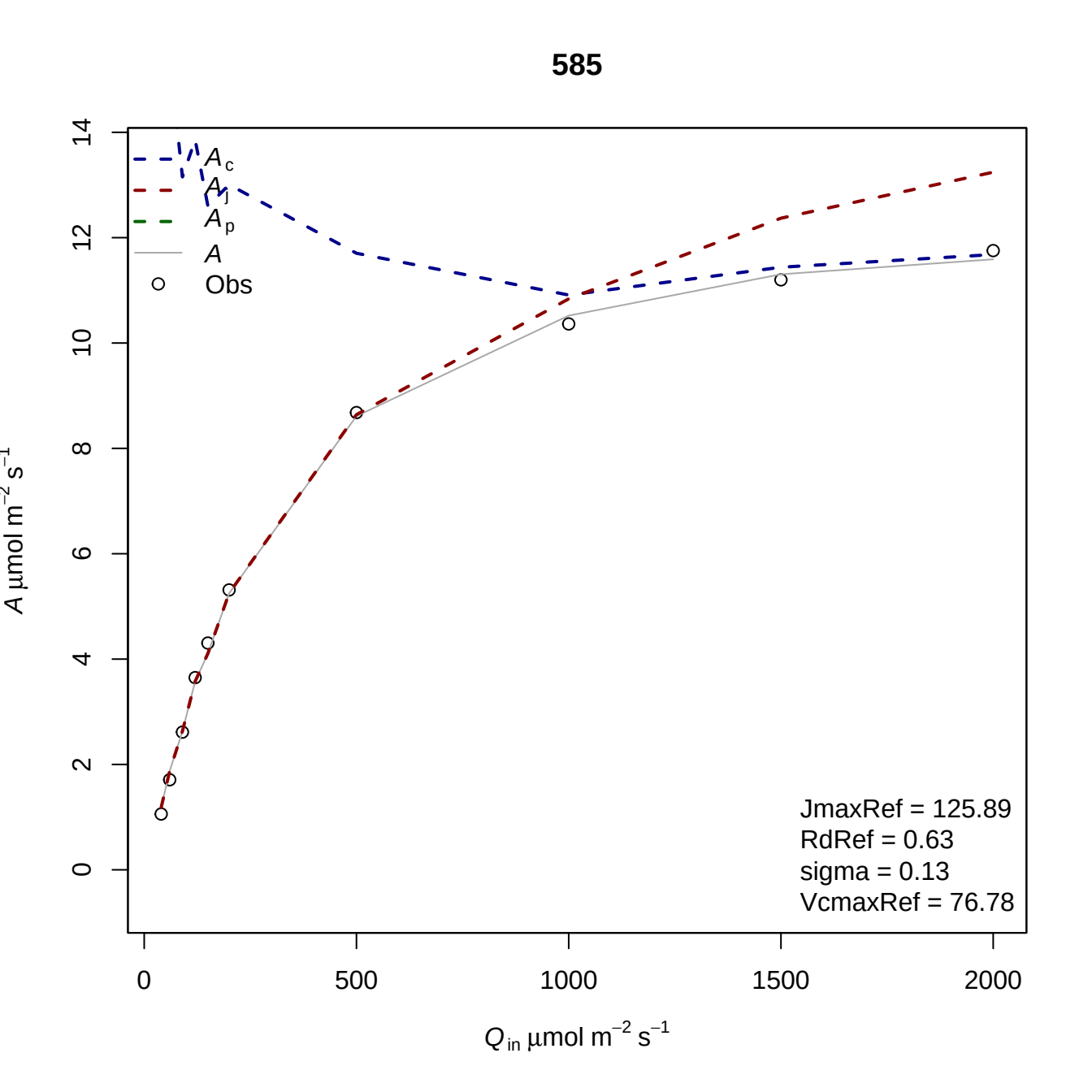

579
A_c
A_j
A_p
A
Obs
JmaxRef = 141.19
RdRef = 1.32
sigma = 0.13
VcmaxRef = 80.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

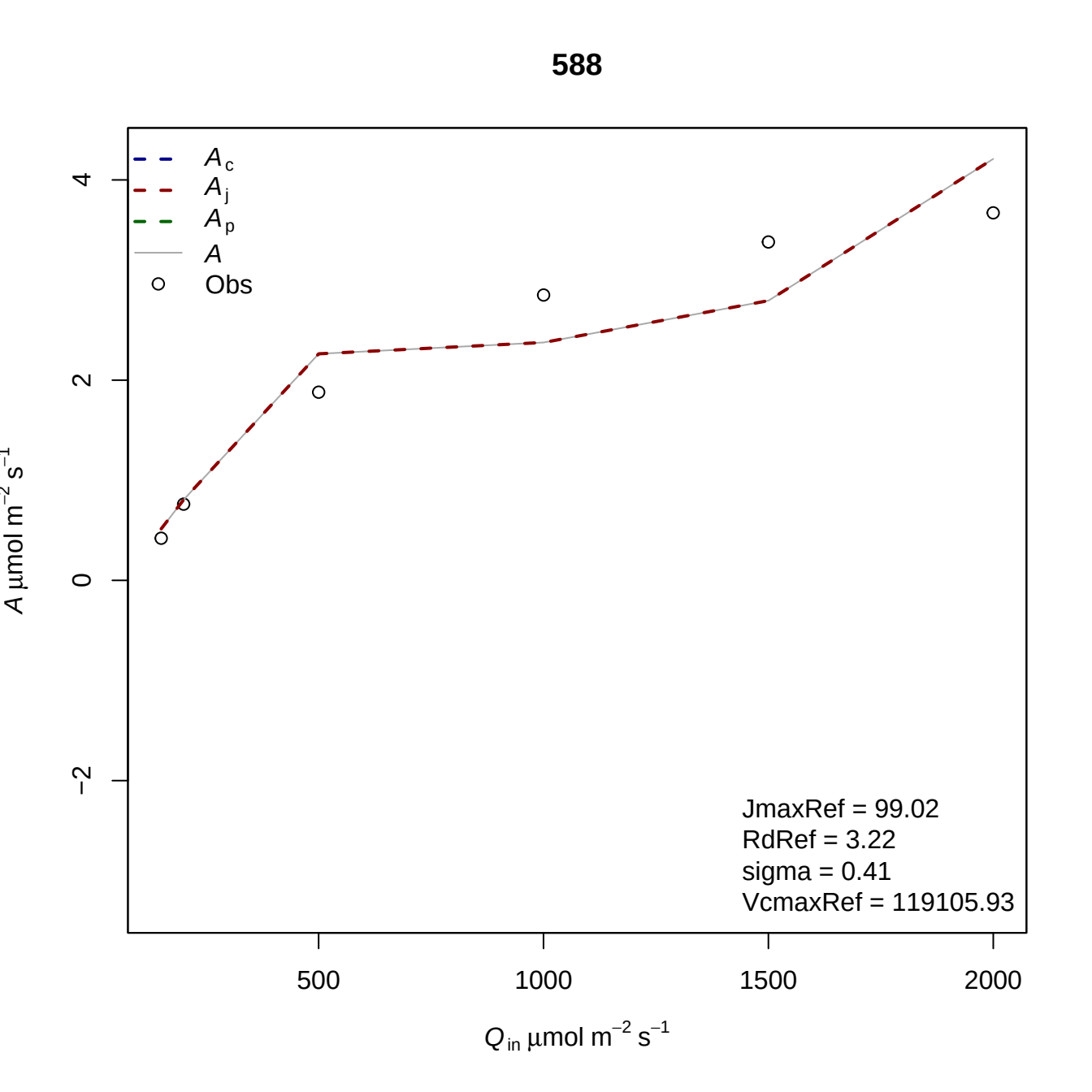

580
A_c
A_j
A_p
A
Obs
JmaxRef = 151.19
RdRef = 1.83
sigma = 0.12
VcmaxRef = 121.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

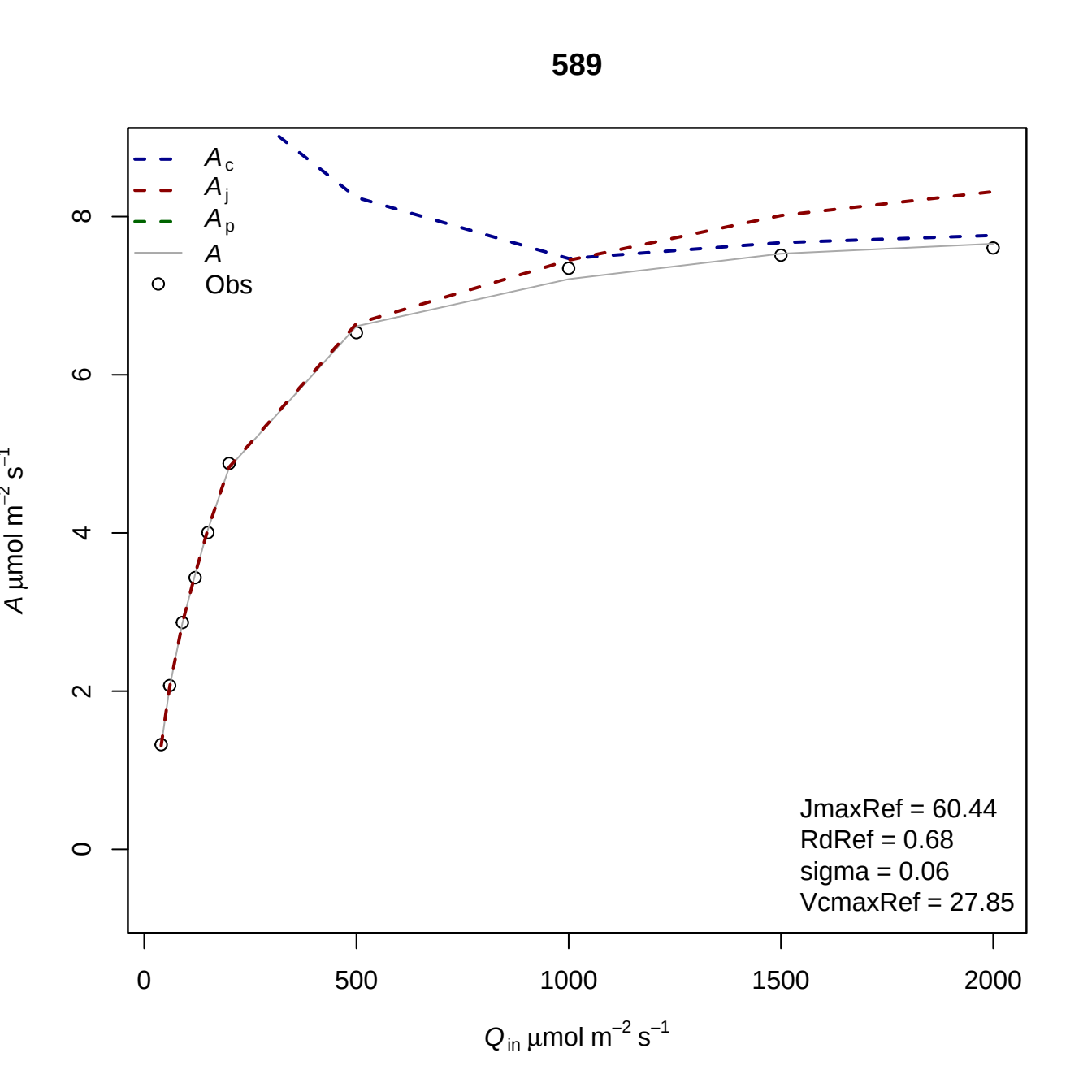

581
A_c
A_j
A_p
A
Obs
JmaxRef = 190.44
RdRef = 2.86
sigma = 0.48
VcmaxRef = 1437090.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
500
1000
1500
2000

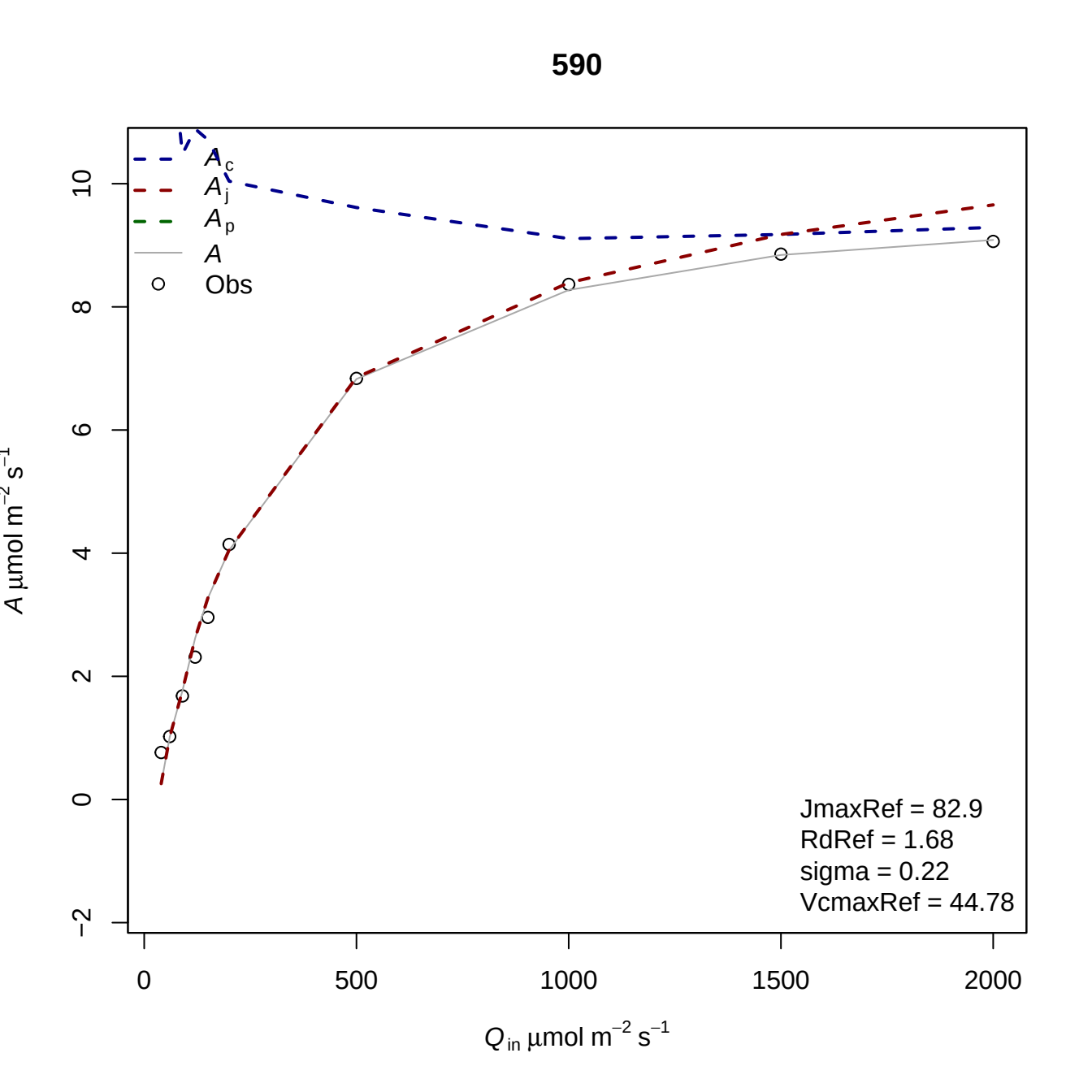

583
A_c
A_j
A_p
A
Obs
JmaxRef = 111.59
RdRef = 0.04
sigma = 0.15
VcmaxRef = 45.79
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

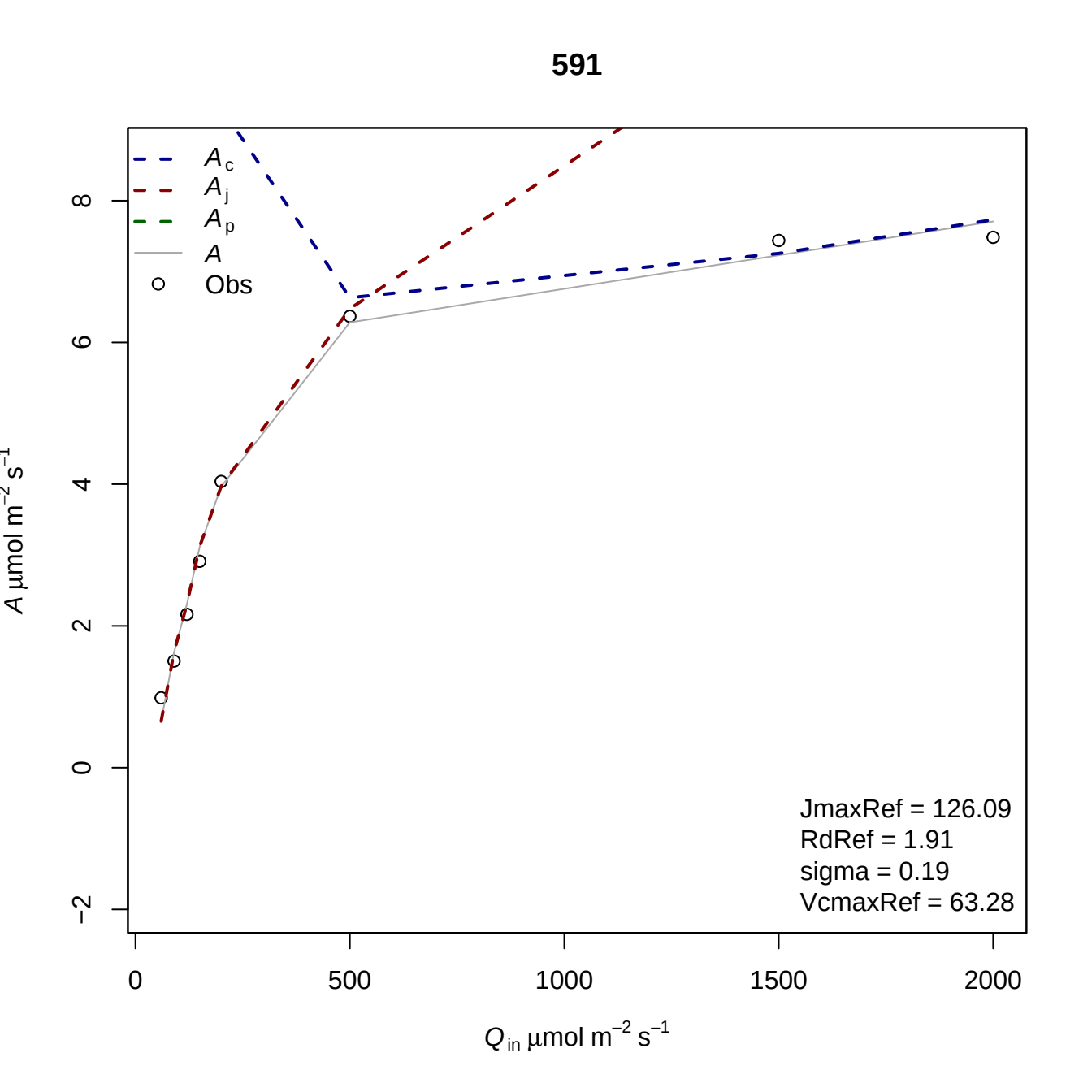

584
A_c
A_j
A_p
A
Obs
JmaxRef = 133.64
RdRef = 1.34
sigma = 0.14
VcmaxRef = 208681.01
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

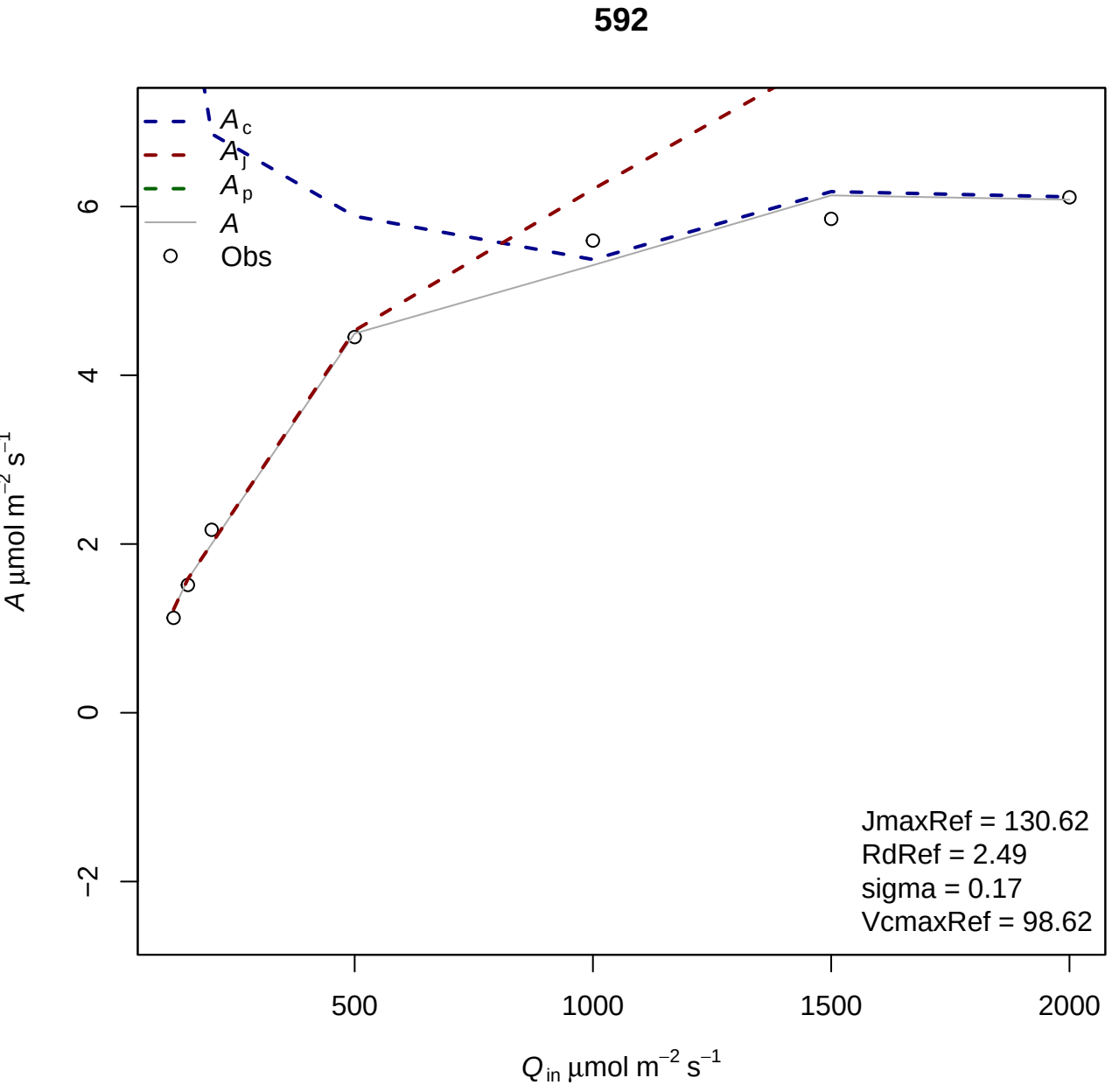

585
A_c
A_j
A_p
A
Obs
JmaxRef = 125.89
RdRef = 0.63
sigma = 0.13
VcmaxRef = 76.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500
2000

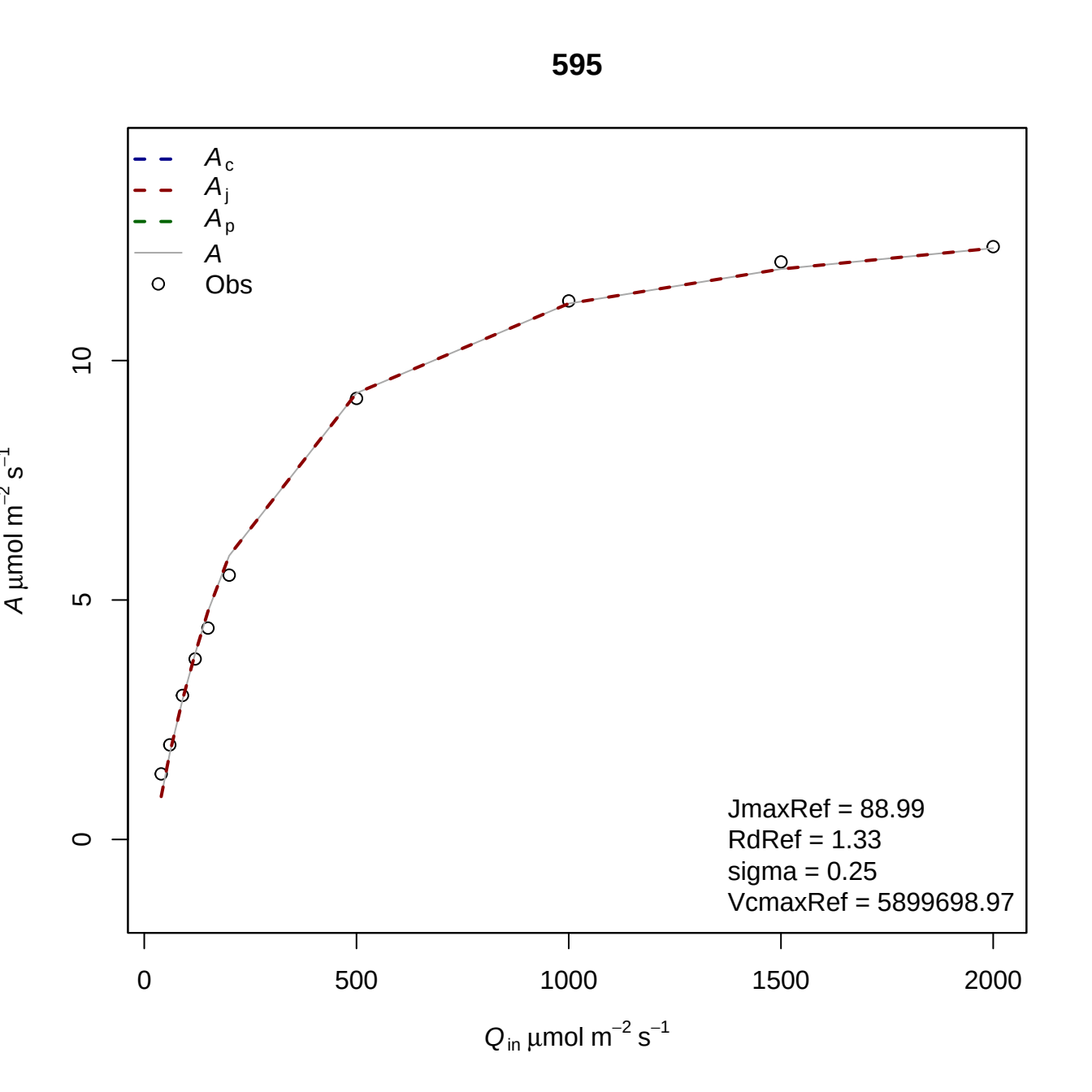

588
A_c
A_j
A_p
A
Obs
JmaxRef = 99.02
RdRef = 3.22
sigma = 0.41
VcmaxRef = 119105.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
500
1000
1500
2000

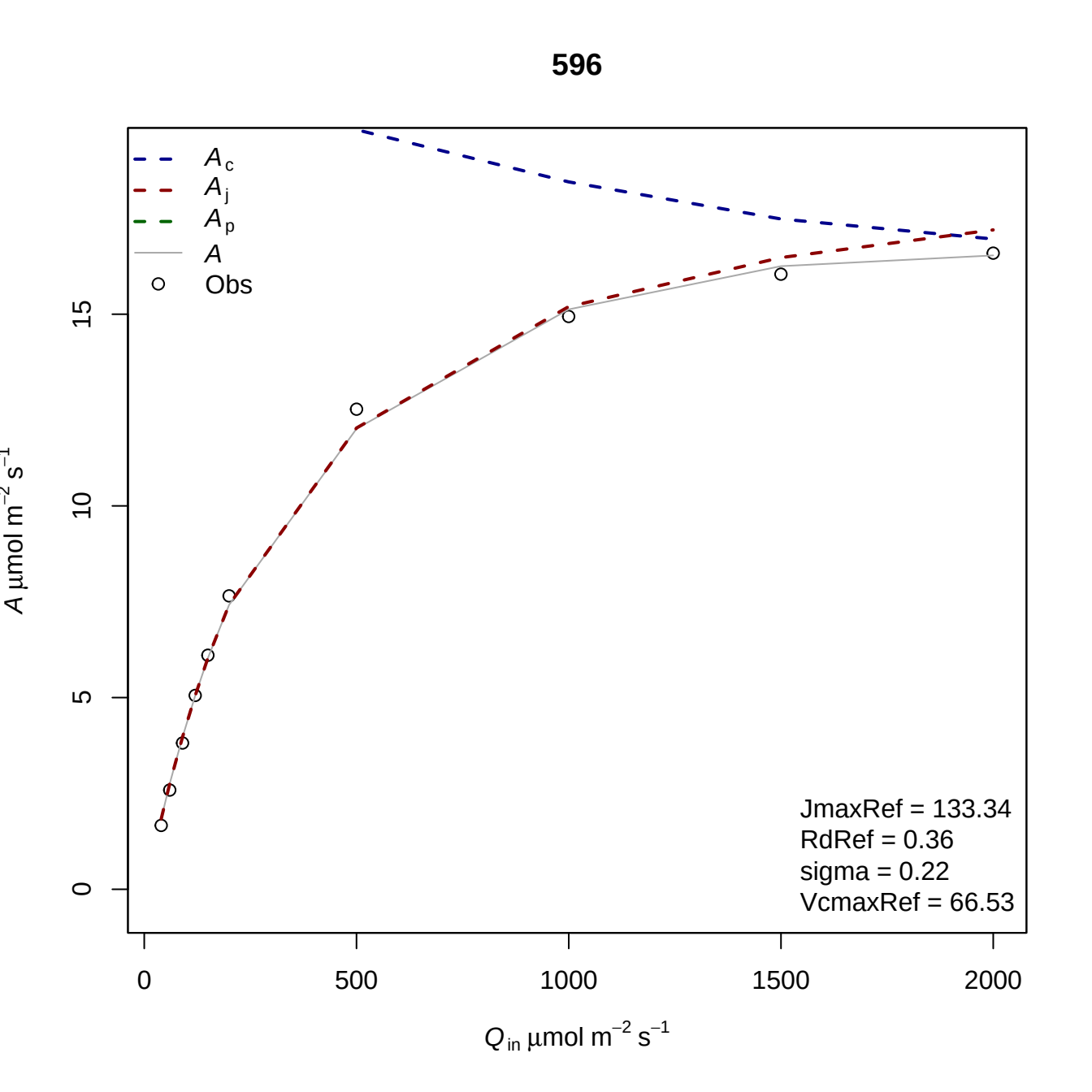

589
A_c
A_j
A_p
A
Obs
JmaxRef = 60.44
RdRef = 0.68
sigma = 0.06
VcmaxRef = 27.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

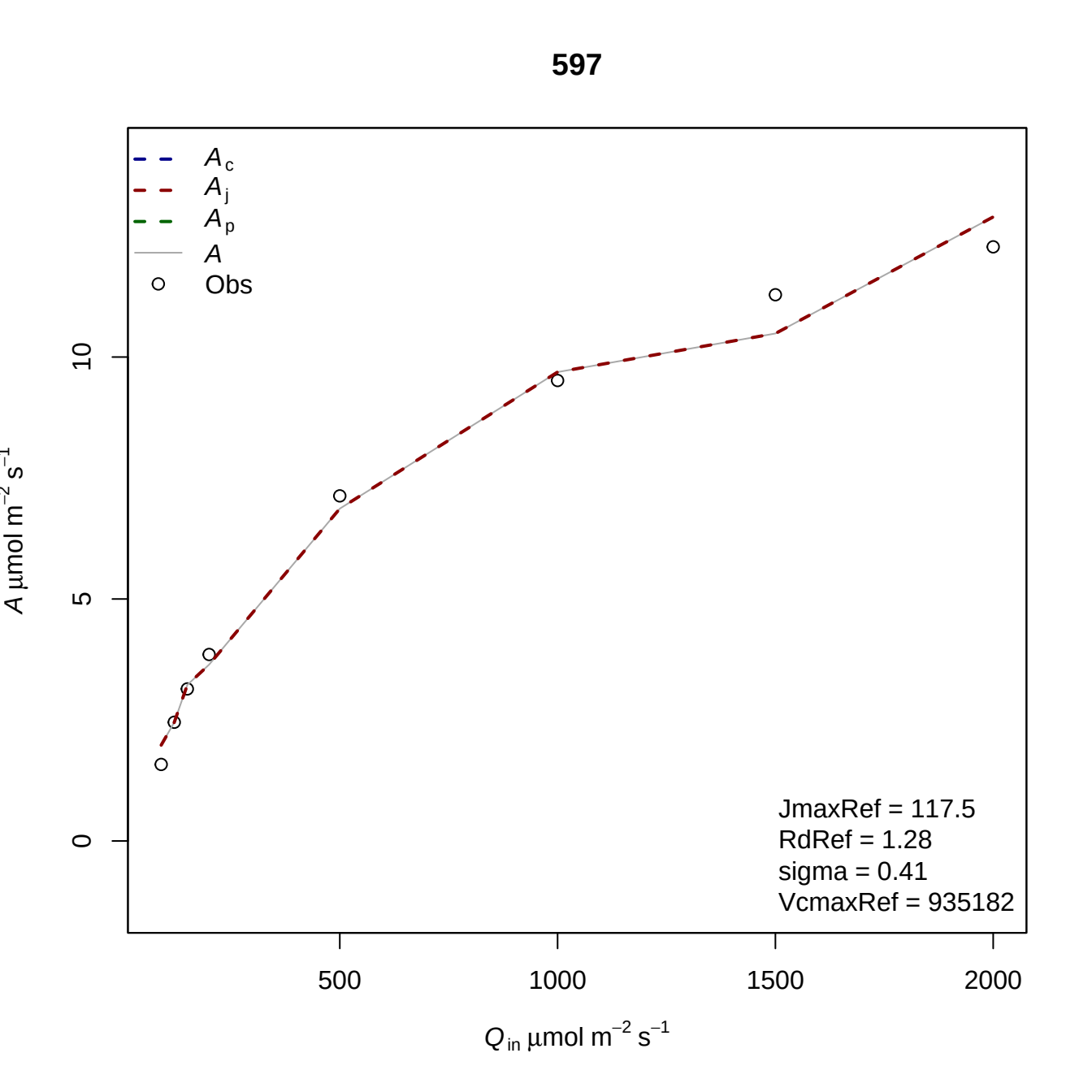

590
A_c
A_j
A_p
A
Obs
JmaxRef = 82.9
RdRef = 1.68
sigma = 0.22
VcmaxRef = 44.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

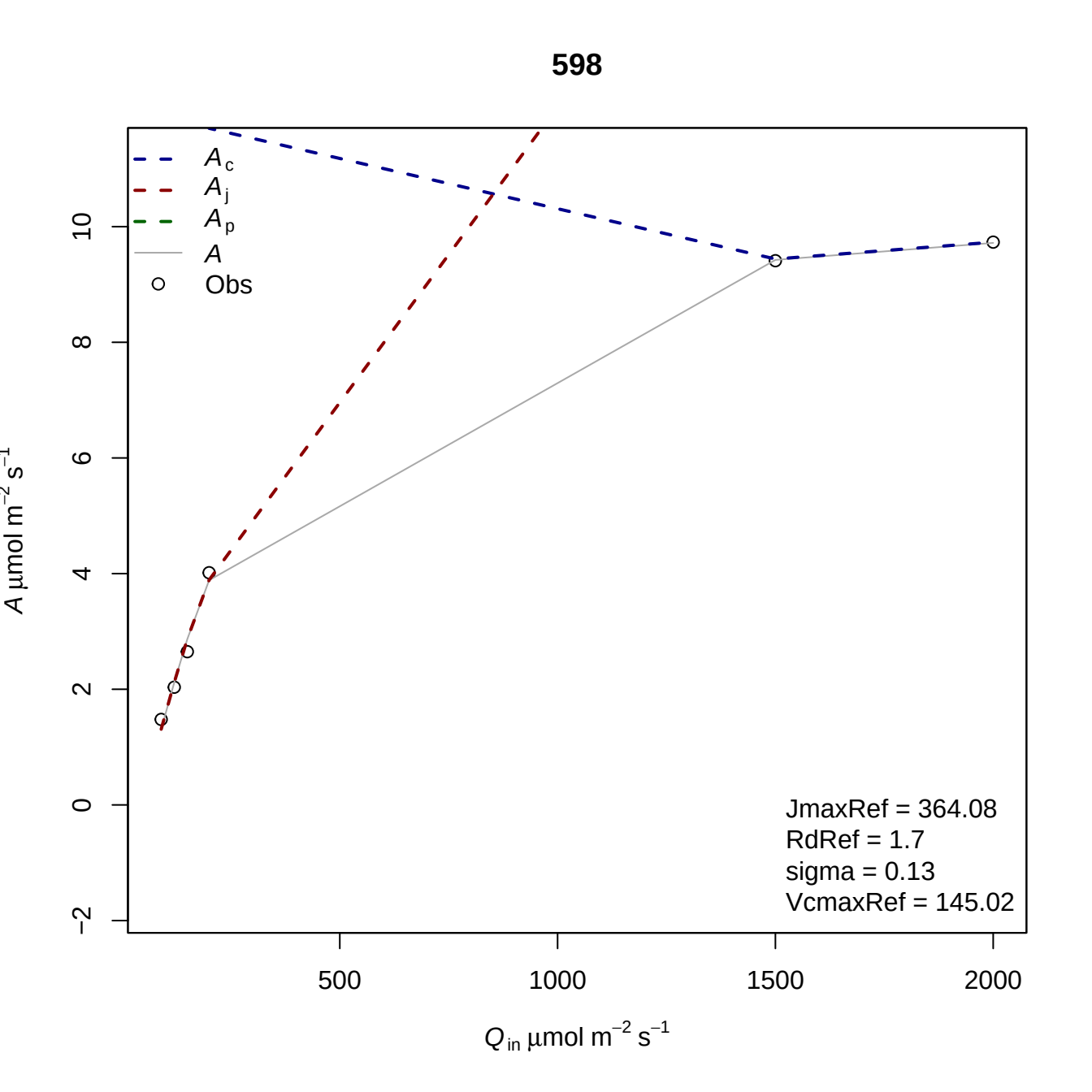

591
A_c
A_j
A_p
A
Obs
JmaxRef = 126.09
RdRef = 1.91
sigma = 0.19
VcmaxRef = 63.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

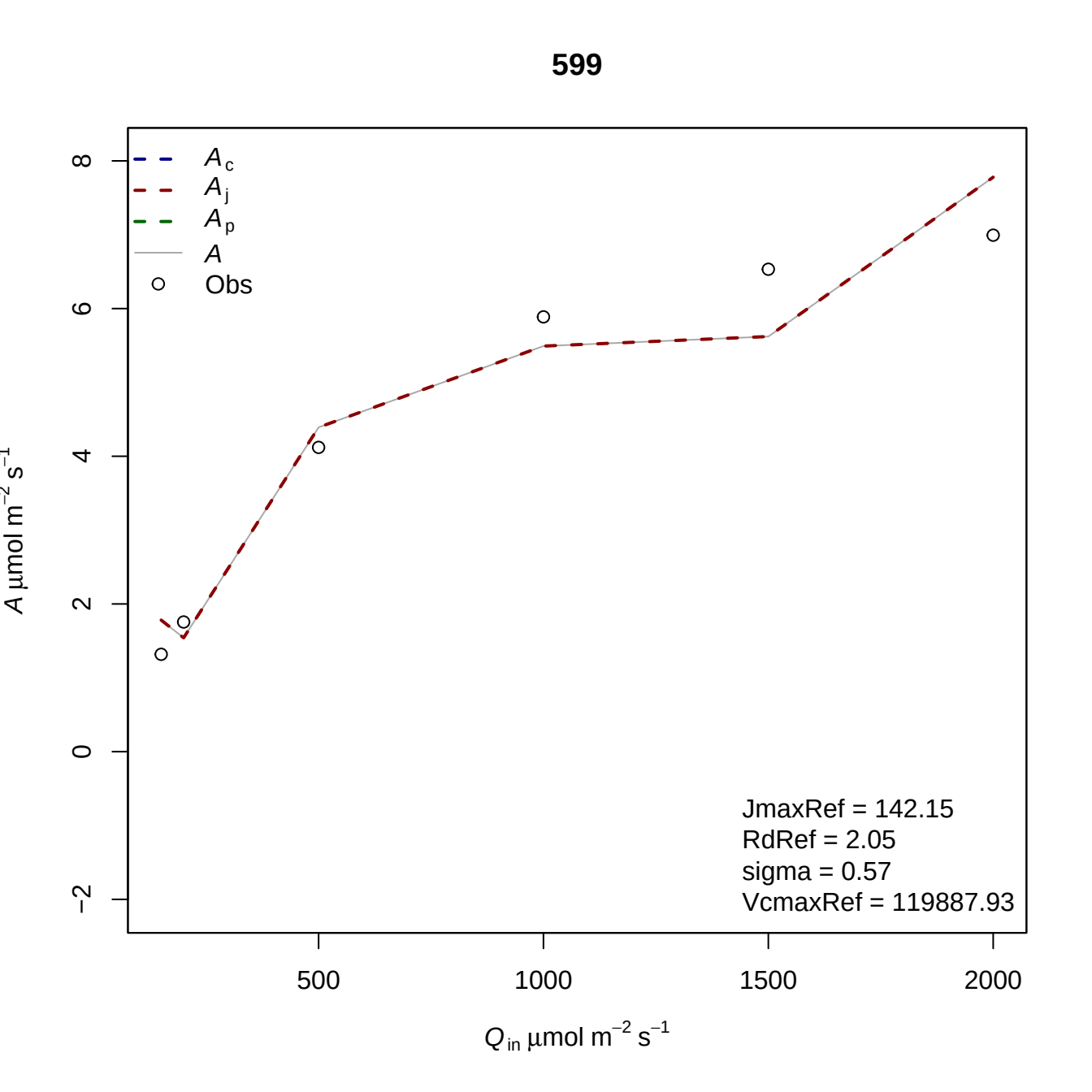

592
A_c
A_j
A_p
A
Obs
JmaxRef = 130.62
RdRef = 2.49
sigma = 0.17
VcmaxRef = 98.62
−2
0
2
4
6
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

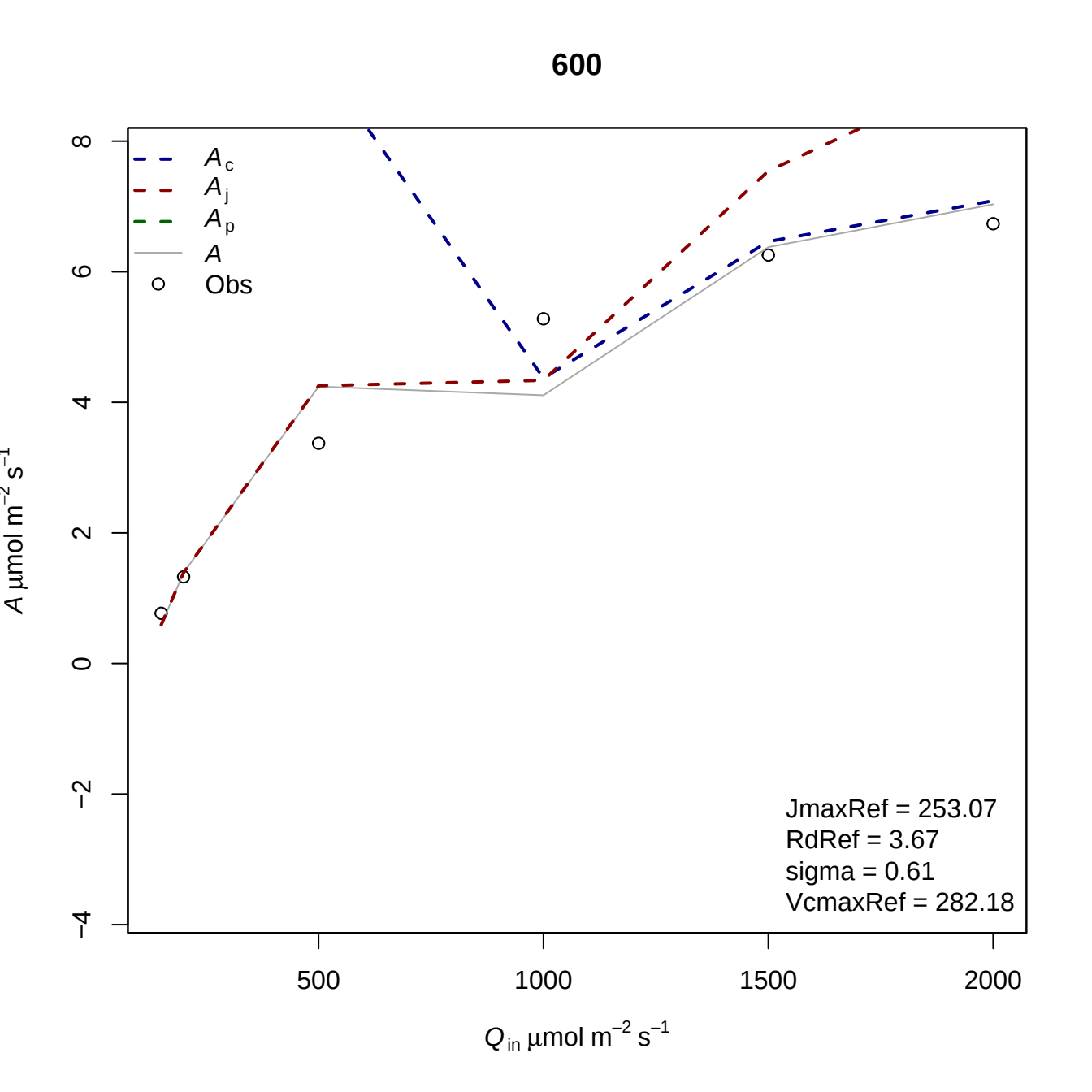

595
A_c
A_j
A_p
A
Obs
JmaxRef = 88.99
RdRef = 1.33
sigma = 0.25
VcmaxRef = 5899698.97
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

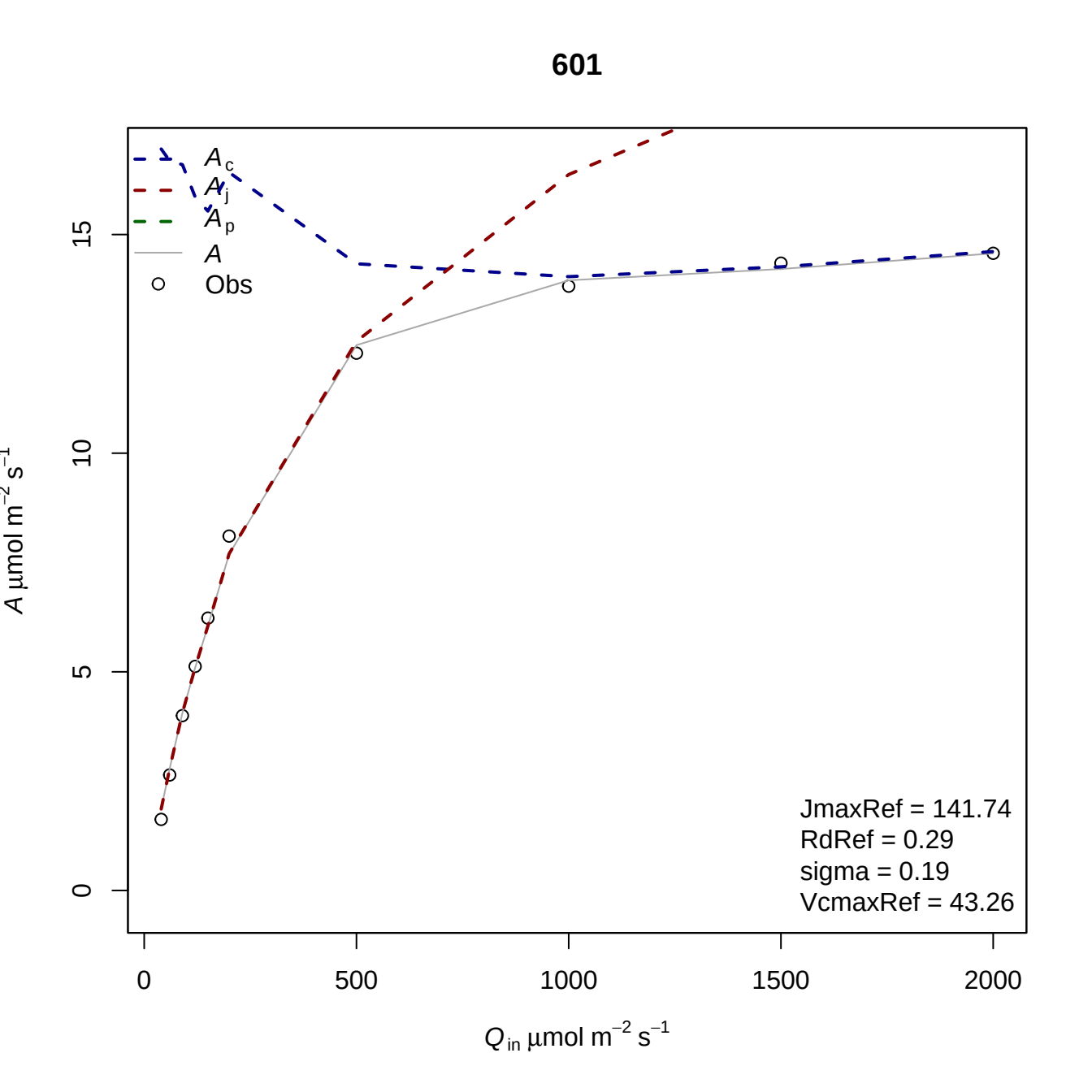

596
A_c
A_j
A_p
A
Obs
JmaxRef = 133.34
RdRef = 0.36
sigma = 0.22
VcmaxRef = 66.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

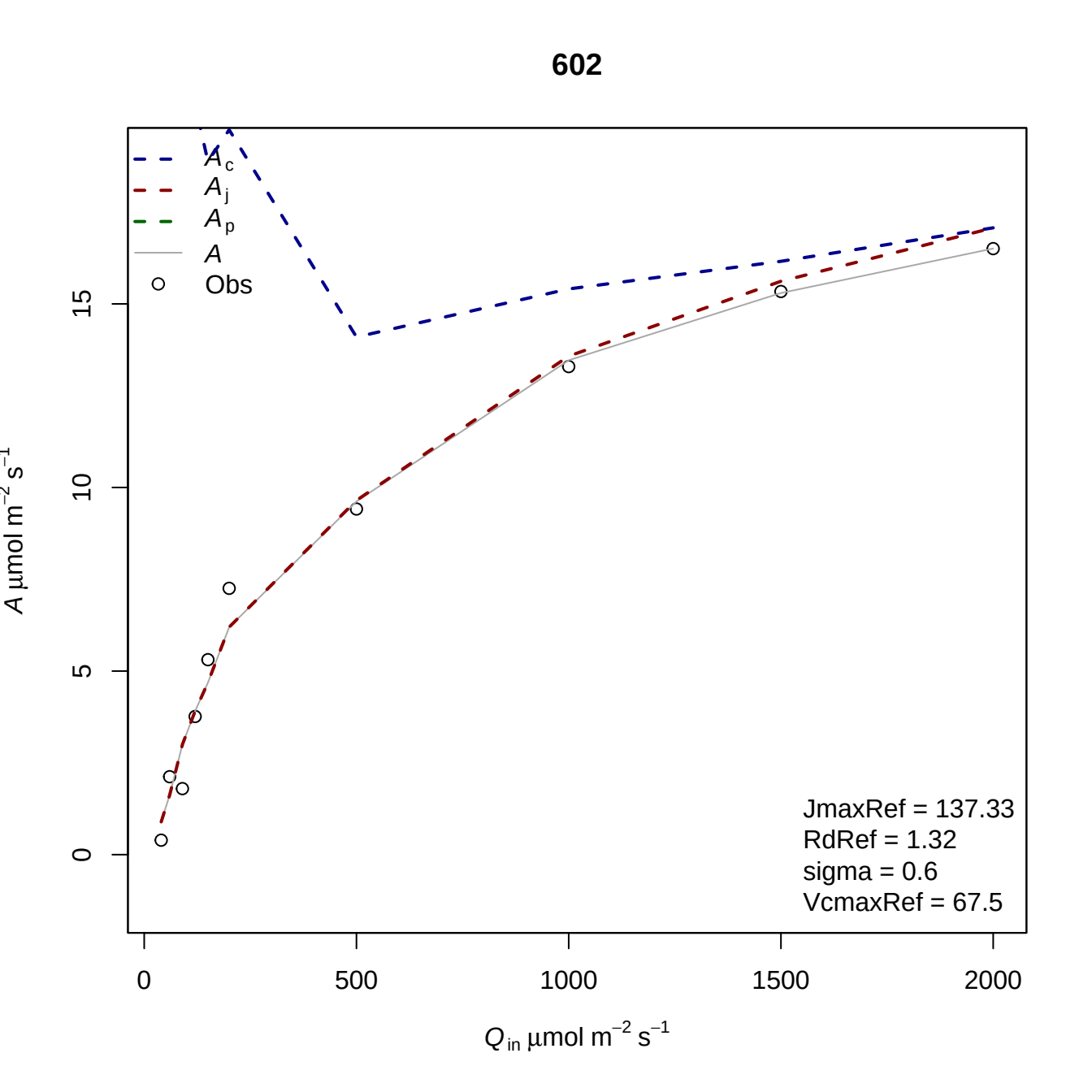

597
A_c
A_j
A_p
A
Obs
JmaxRef = 117.5
RdRef = 1.28
sigma = 0.41
VcmaxRef = 935182
0
5
10
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

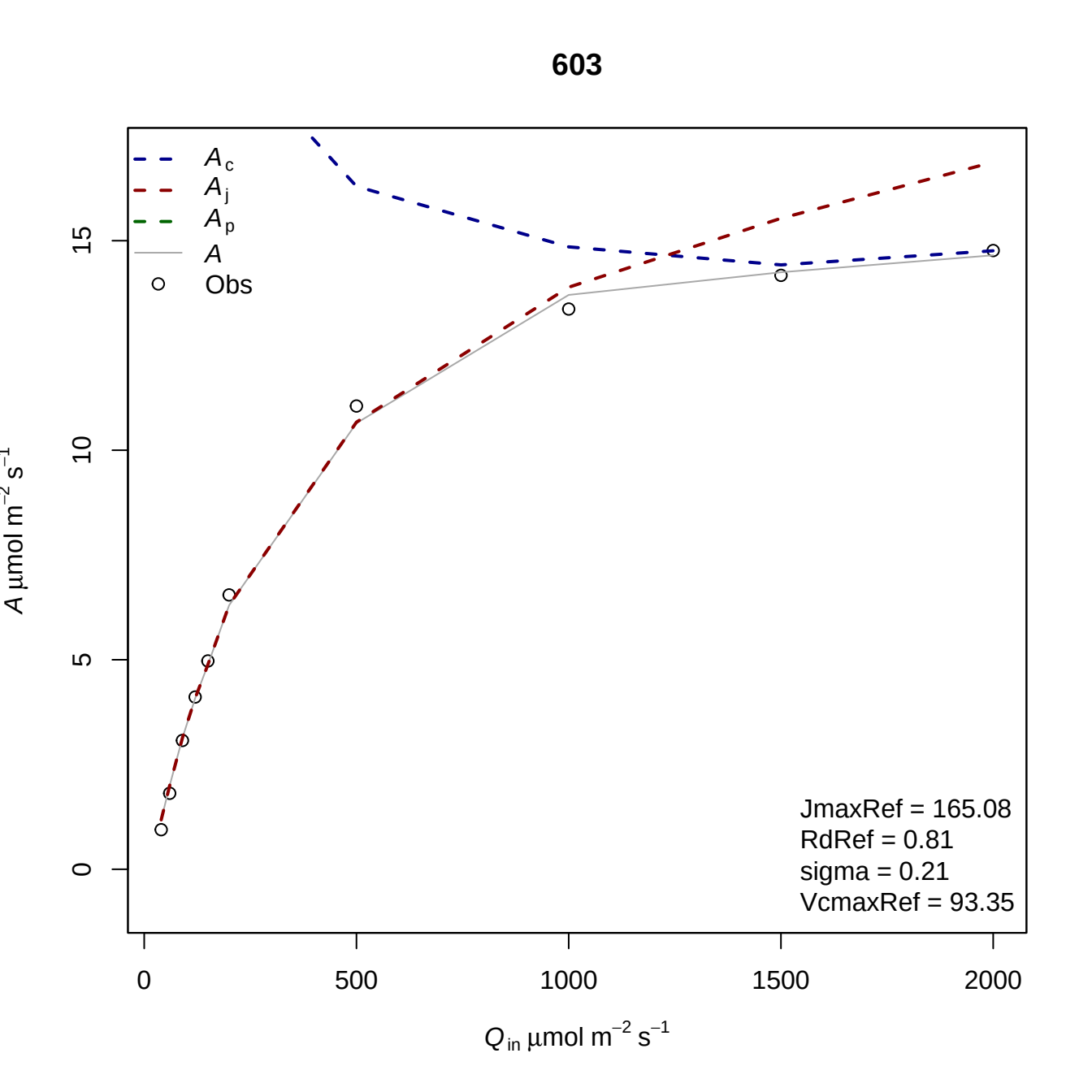

598
A_c
A_j
A_p
A
Obs
JmaxRef = 364.08
RdRef = 1.7
sigma = 0.13
VcmaxRef = 145.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
500
1000
1500
2000

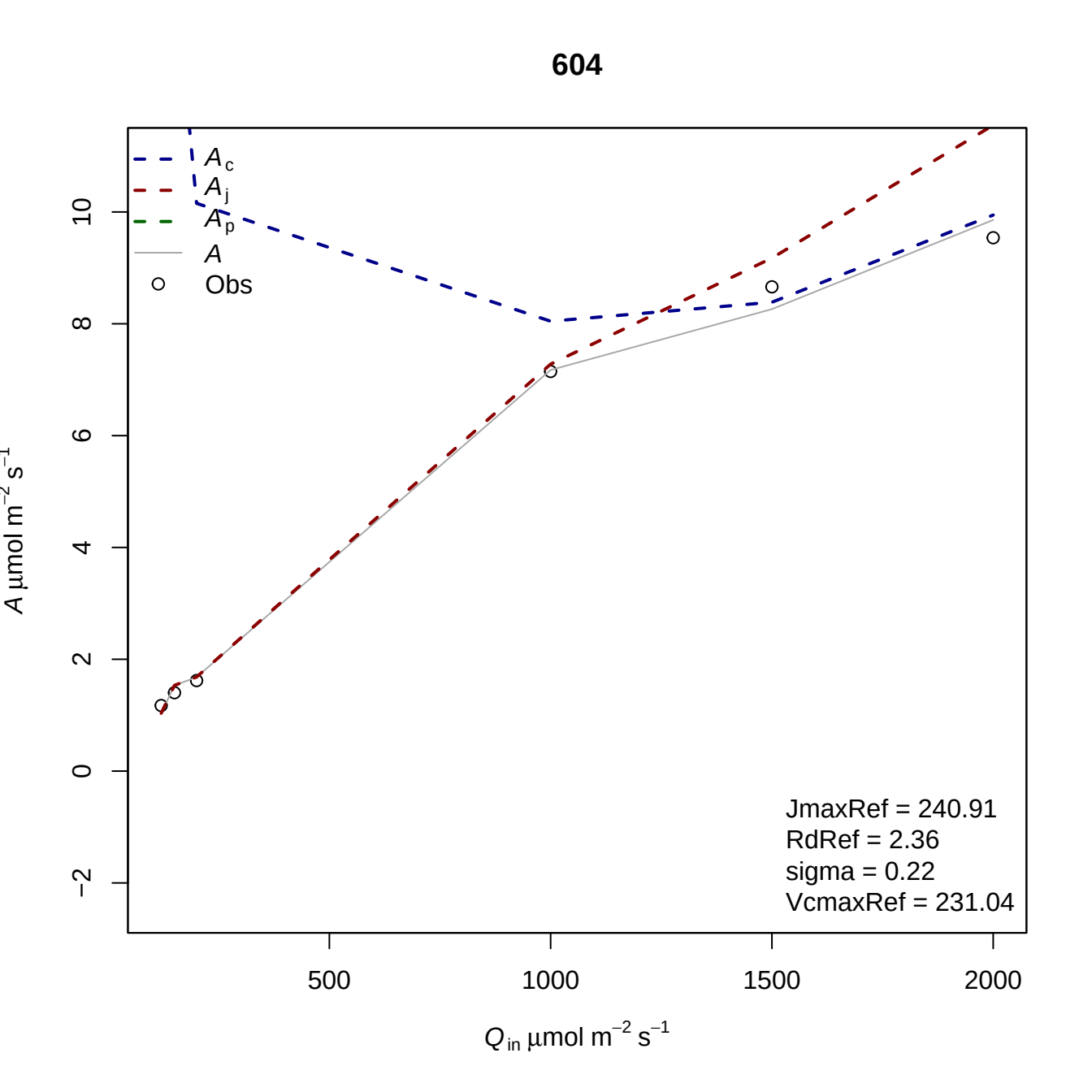

599
A_c
A_j
A_p
A
Obs
JmaxRef = 142.15
RdRef = 2.05
sigma = 0.57
VcmaxRef = 119887.93
8
6
4
2
0
−2
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

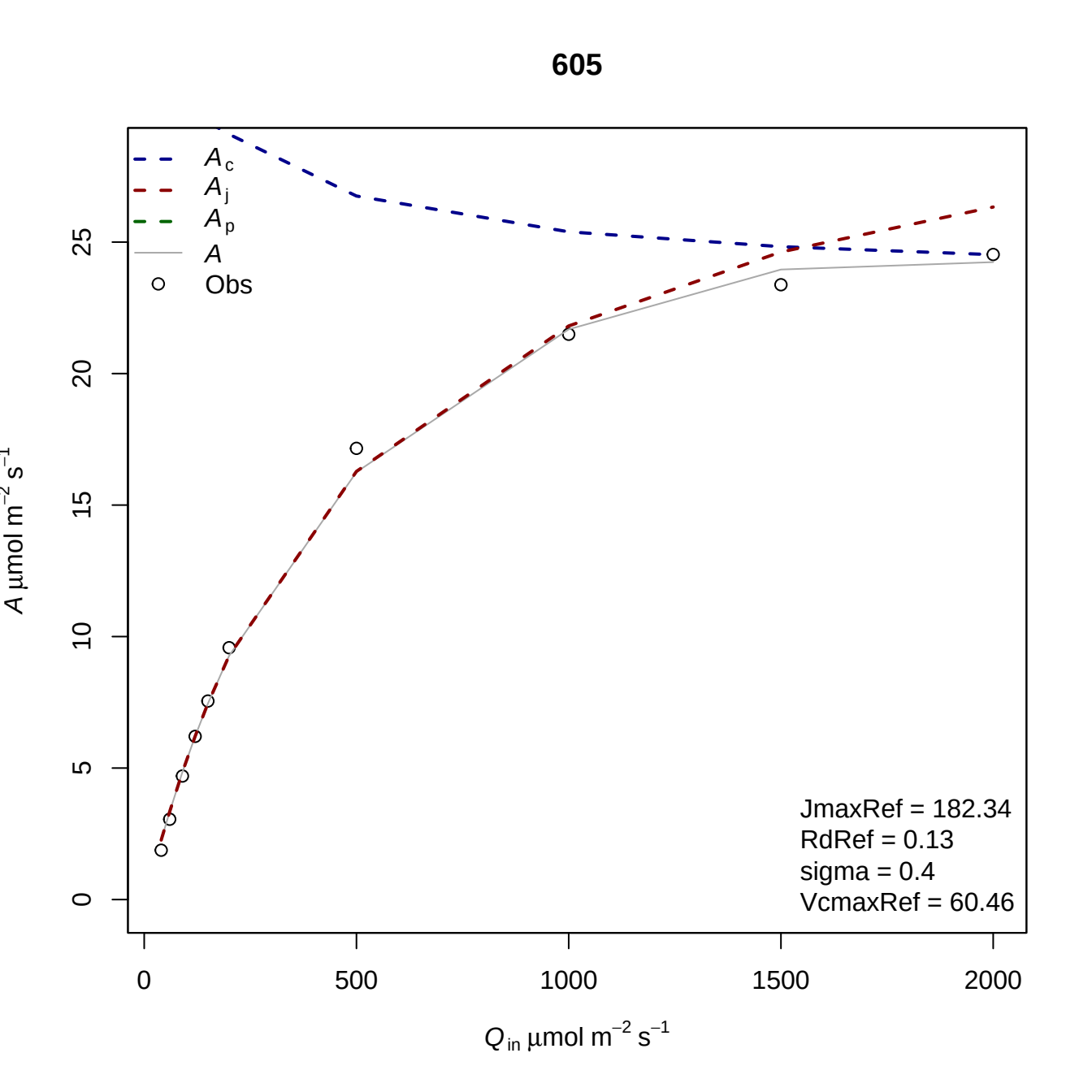

600
A_c
A_j
A_p
A
Obs
JmaxRef = 253.07
RdRef = 3.67
sigma = 0.61
VcmaxRef = 282.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
500
1000
1500
2000

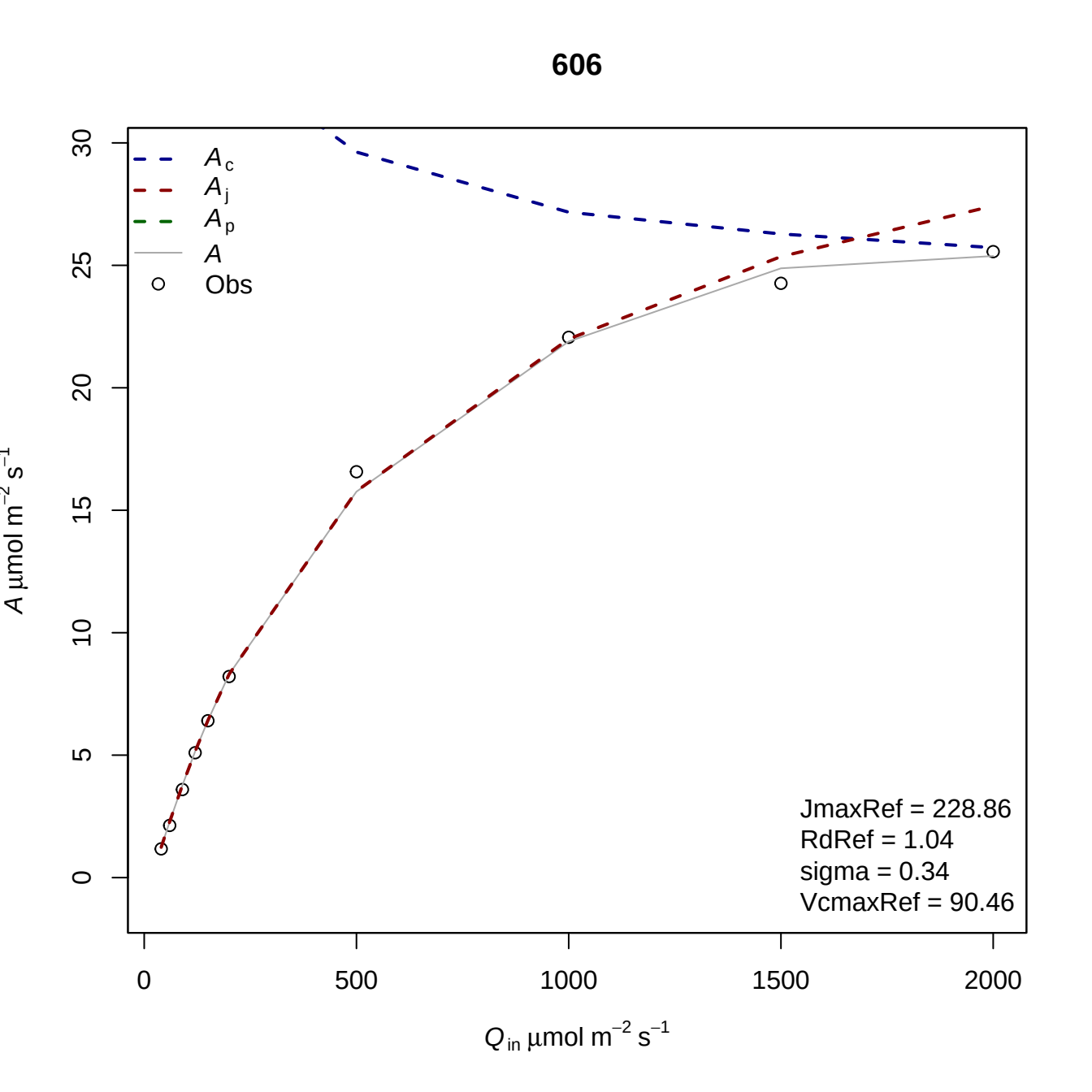

601
A_c
A_j
A_p
A
Obs
JmaxRef = 141.74
RdRef = 0.29
sigma = 0.19
VcmaxRef = 43.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

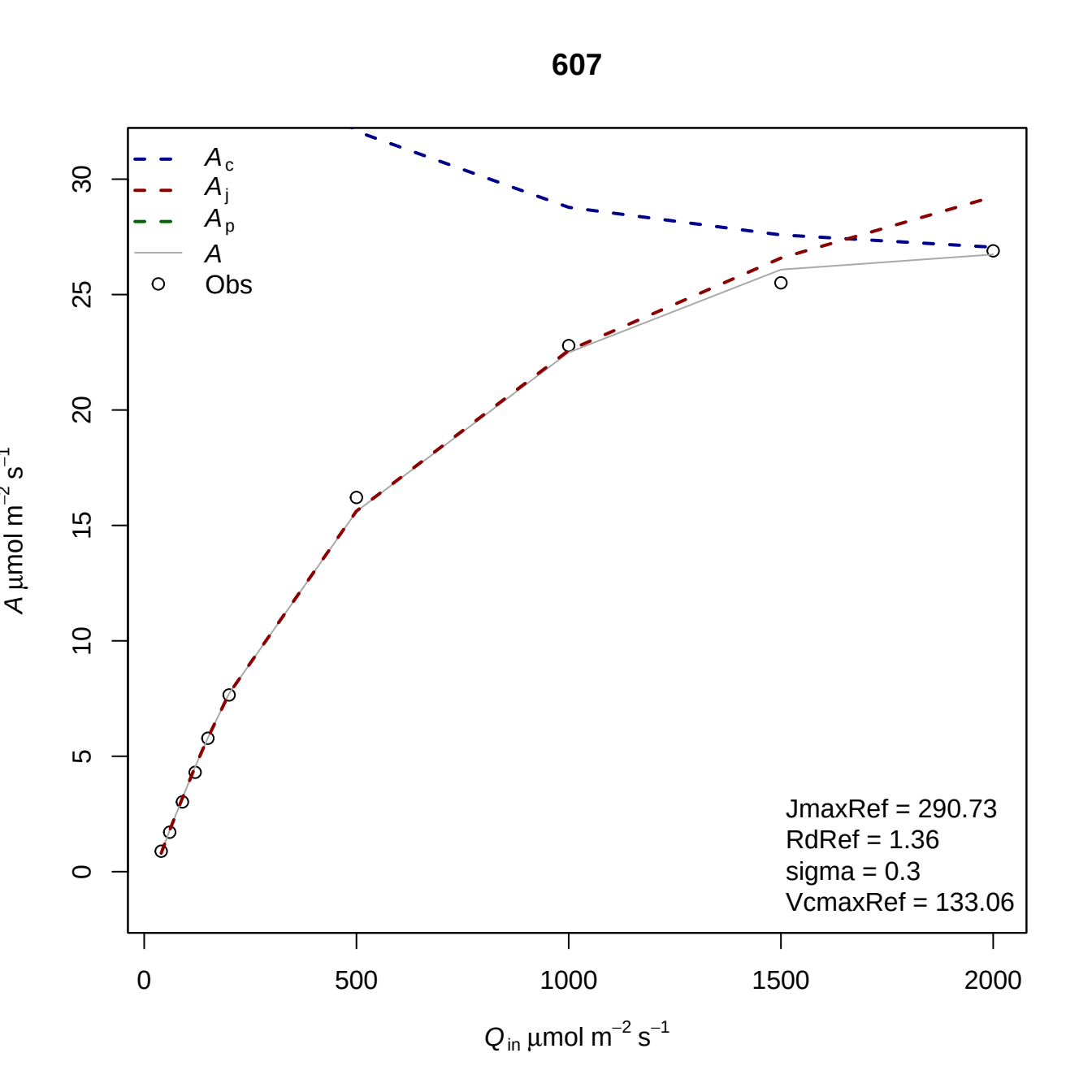

602
A_c
A_j
A_p
A
Obs
JmaxRef = 137.33
RdRef = 1.32
sigma = 0.6
VcmaxRef = 67.5
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

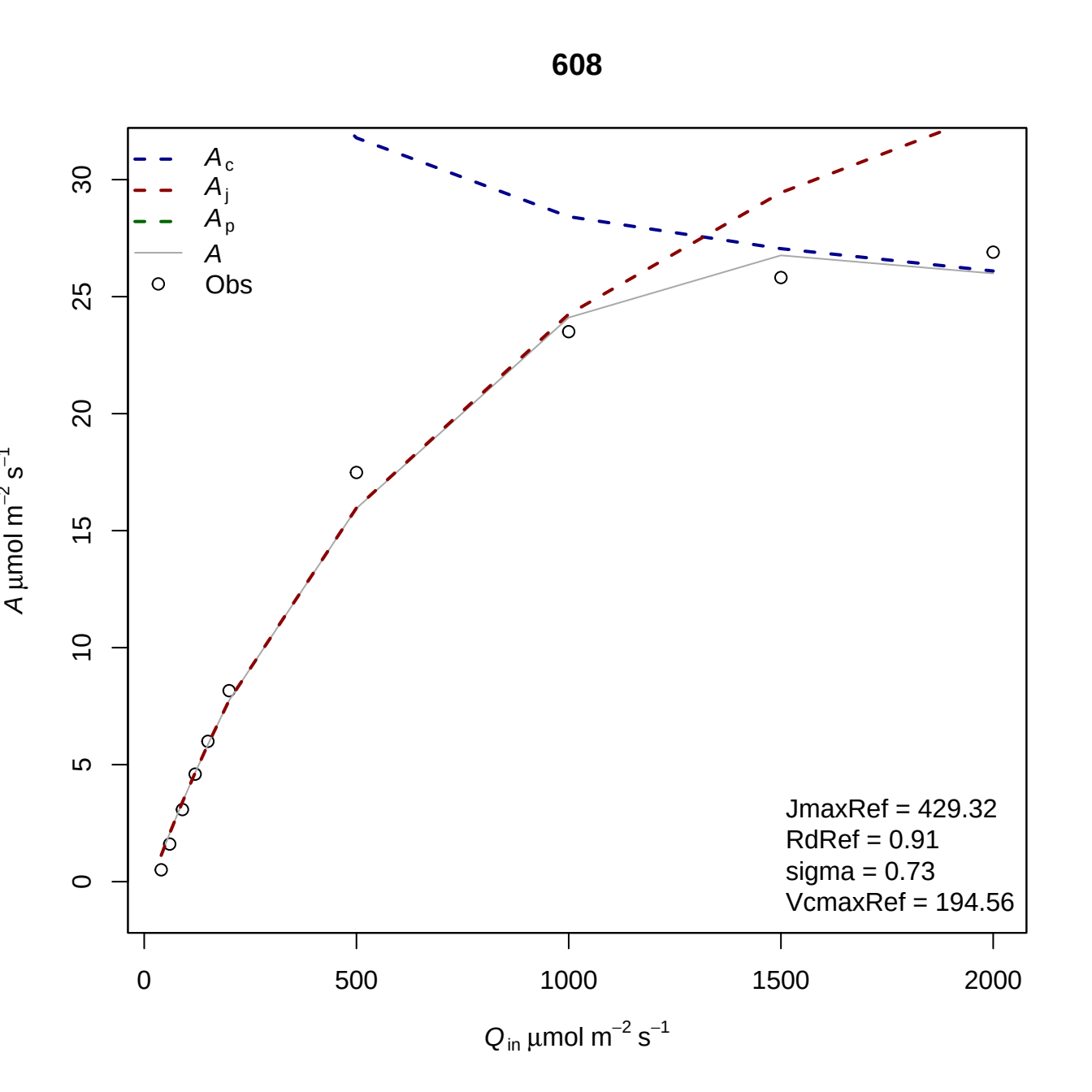

603
A_c
A_j
A_p
A
Obs
JmaxRef = 165.08
RdRef = 0.81
sigma = 0.21
VcmaxRef = 93.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

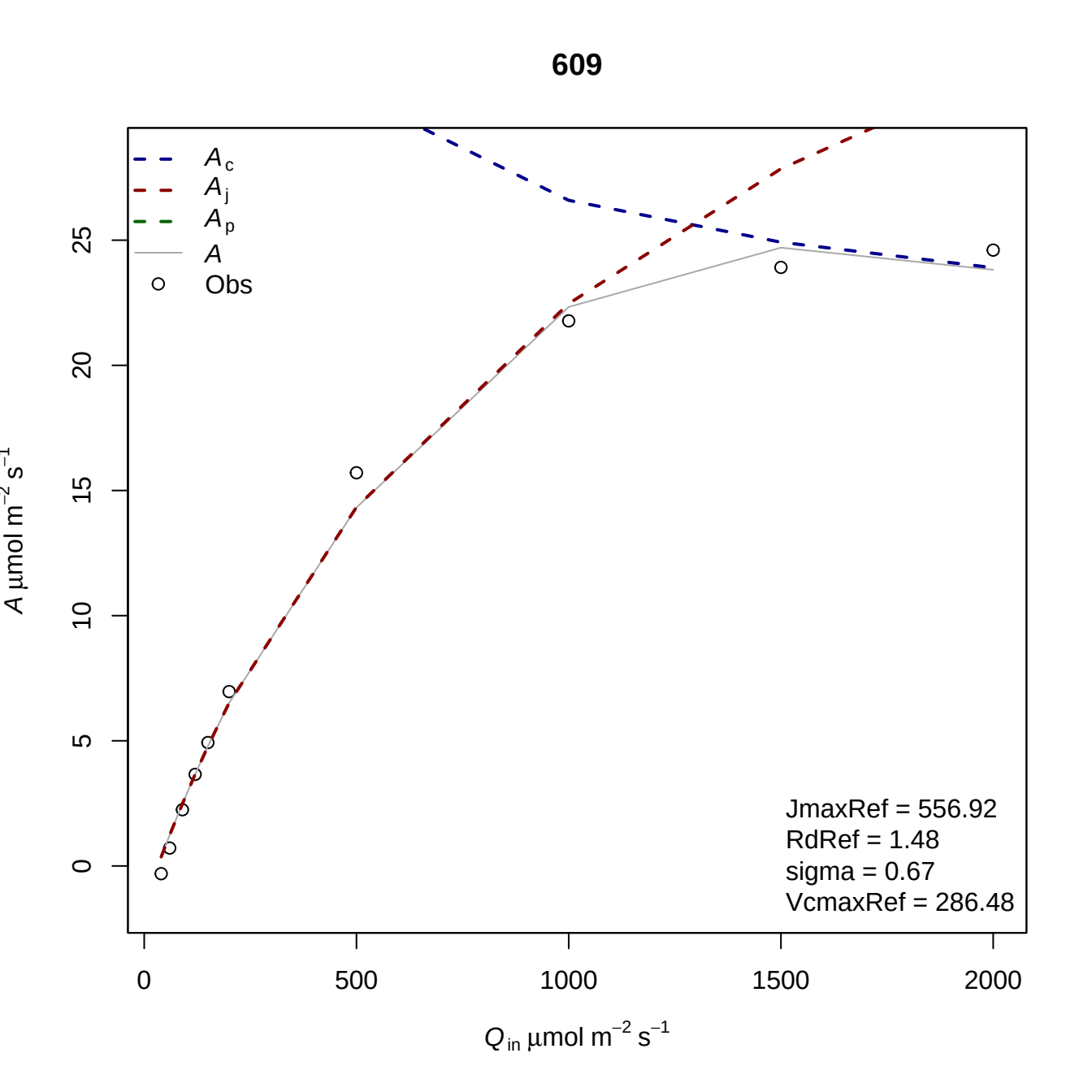

604
A_c
A_j
A_p
A
Obs
JmaxRef = 240.91
RdRef = 2.36
sigma = 0.22
VcmaxRef = 231.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
500
1000
1500
2000

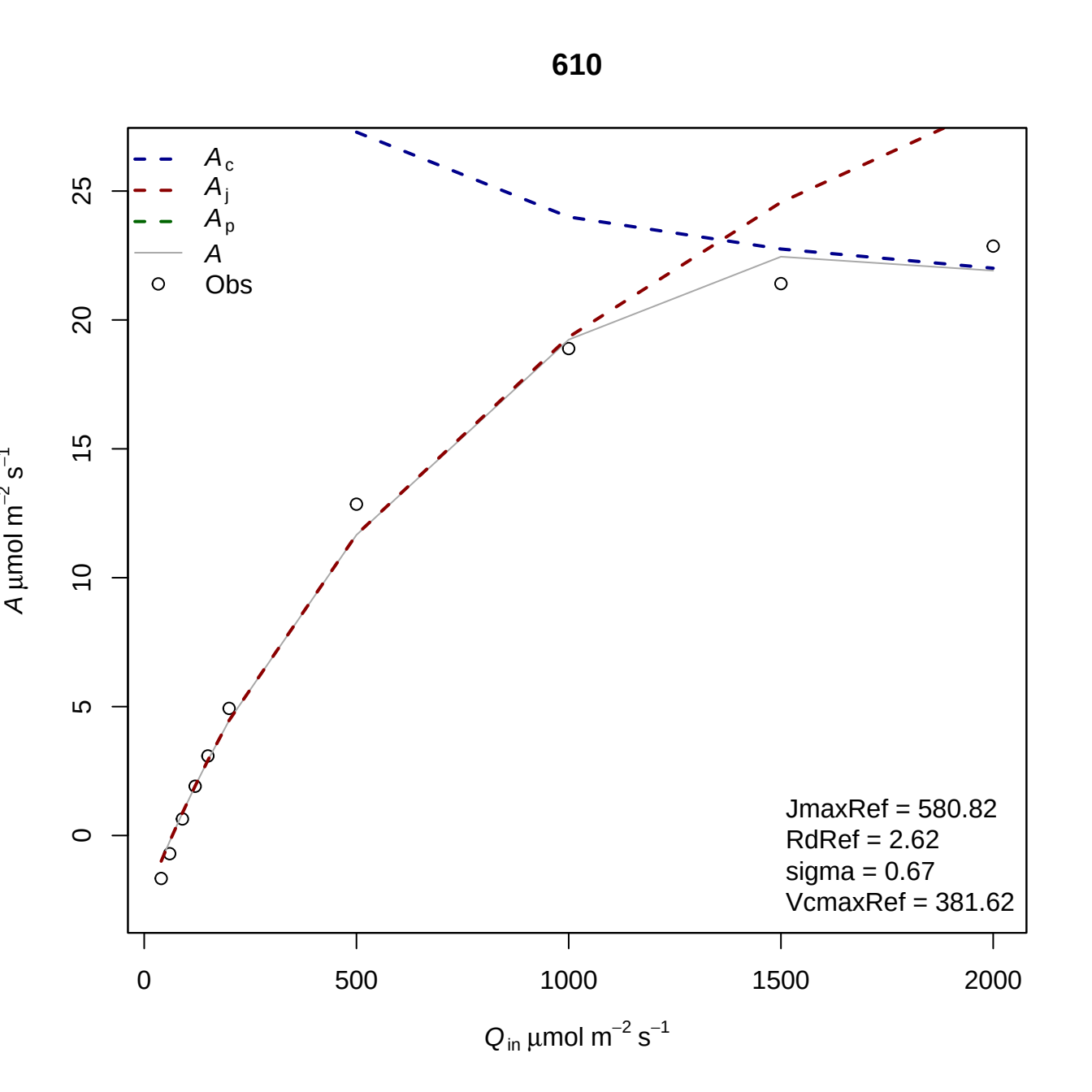

605
A_c
A_j
A_p
A
Obs
JmaxRef = 182.34
RdRef = 0.13
sigma = 0.4
VcmaxRef = 60.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

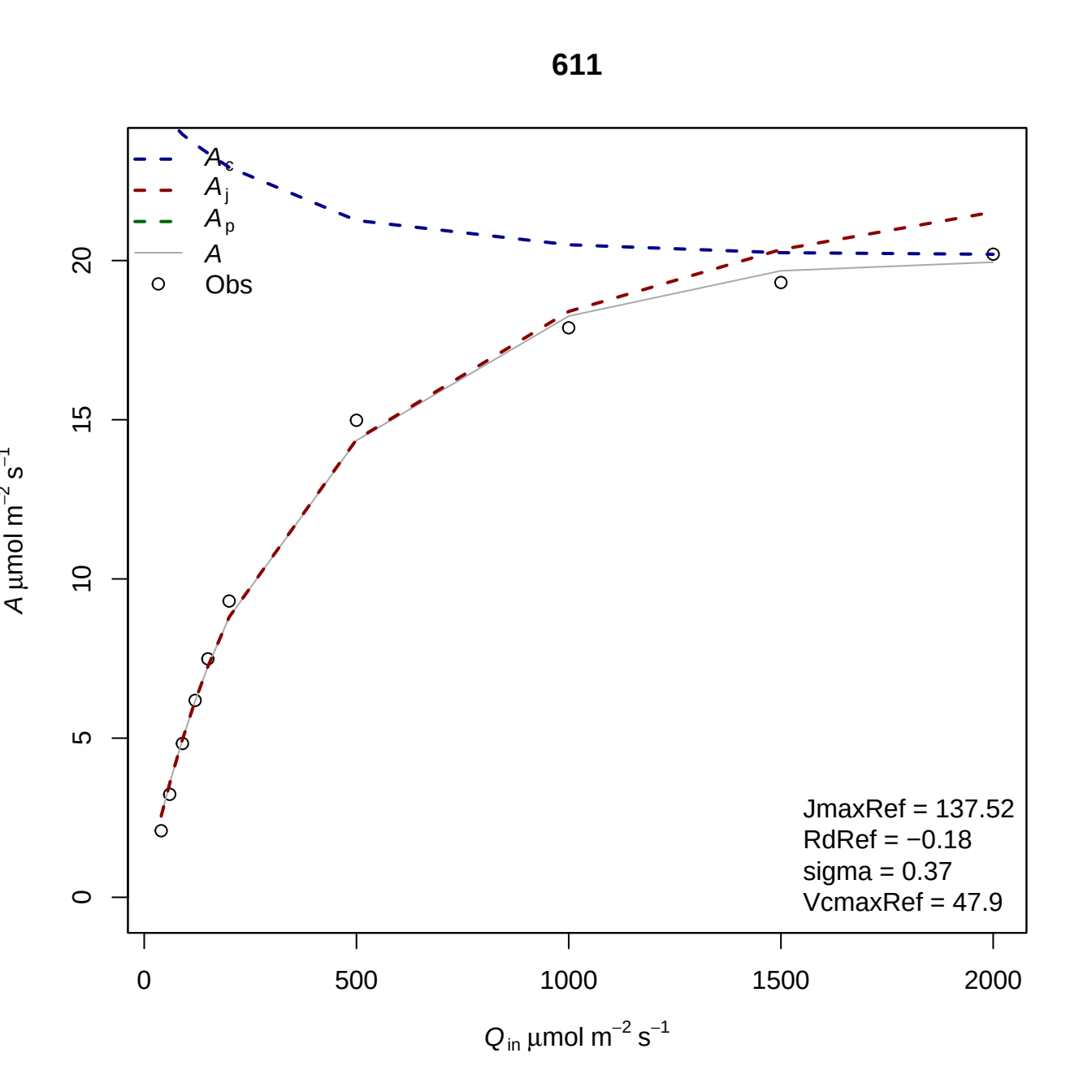

606
A_c
A_j
A_p
A
Obs
JmaxRef = 228.86
RdRef = 1.04
sigma = 0.34
VcmaxRef = 90.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

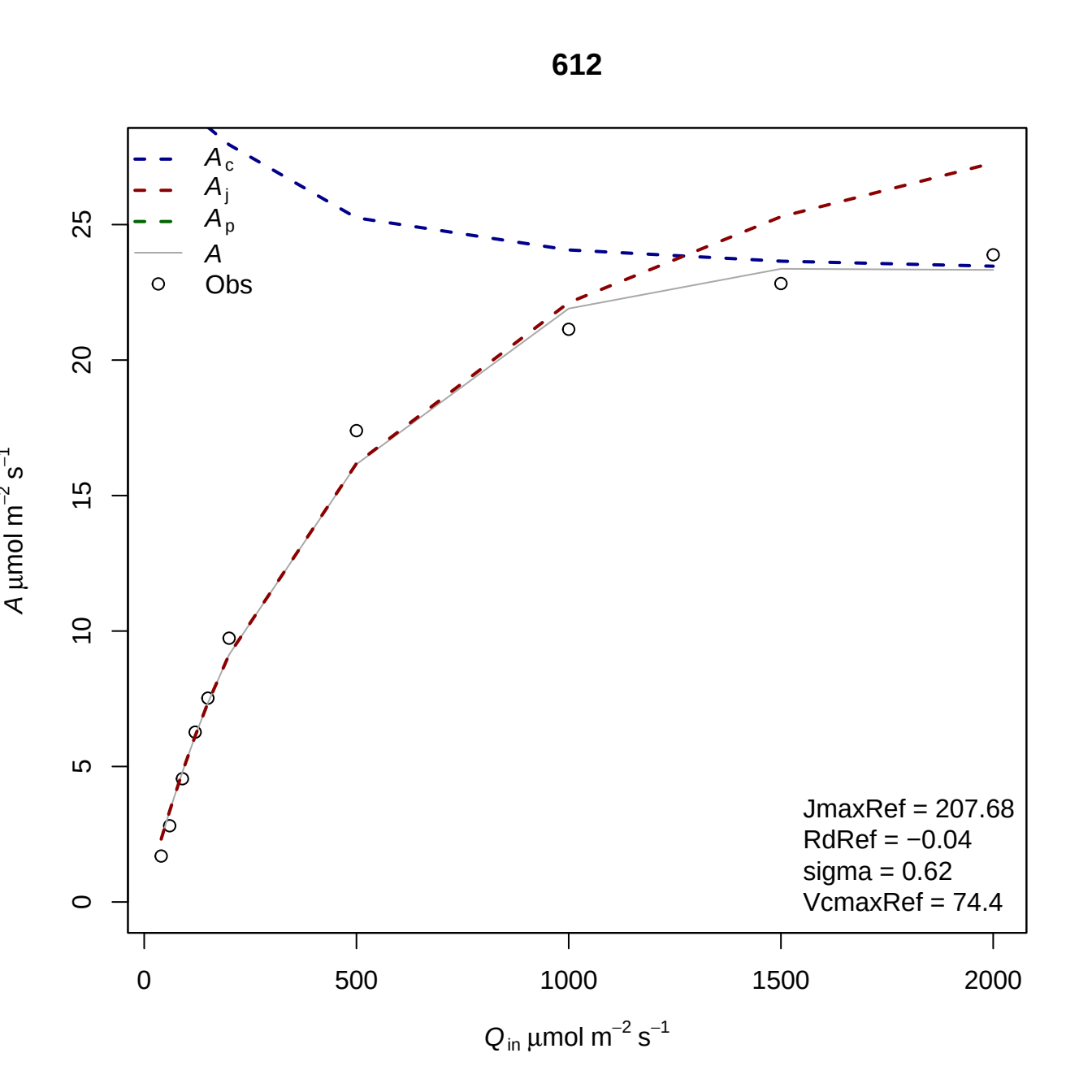

607
A_c
A_j
A_p
A
Obs
JmaxRef = 290.73
RdRef = 1.36
sigma = 0.3
VcmaxRef = 133.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

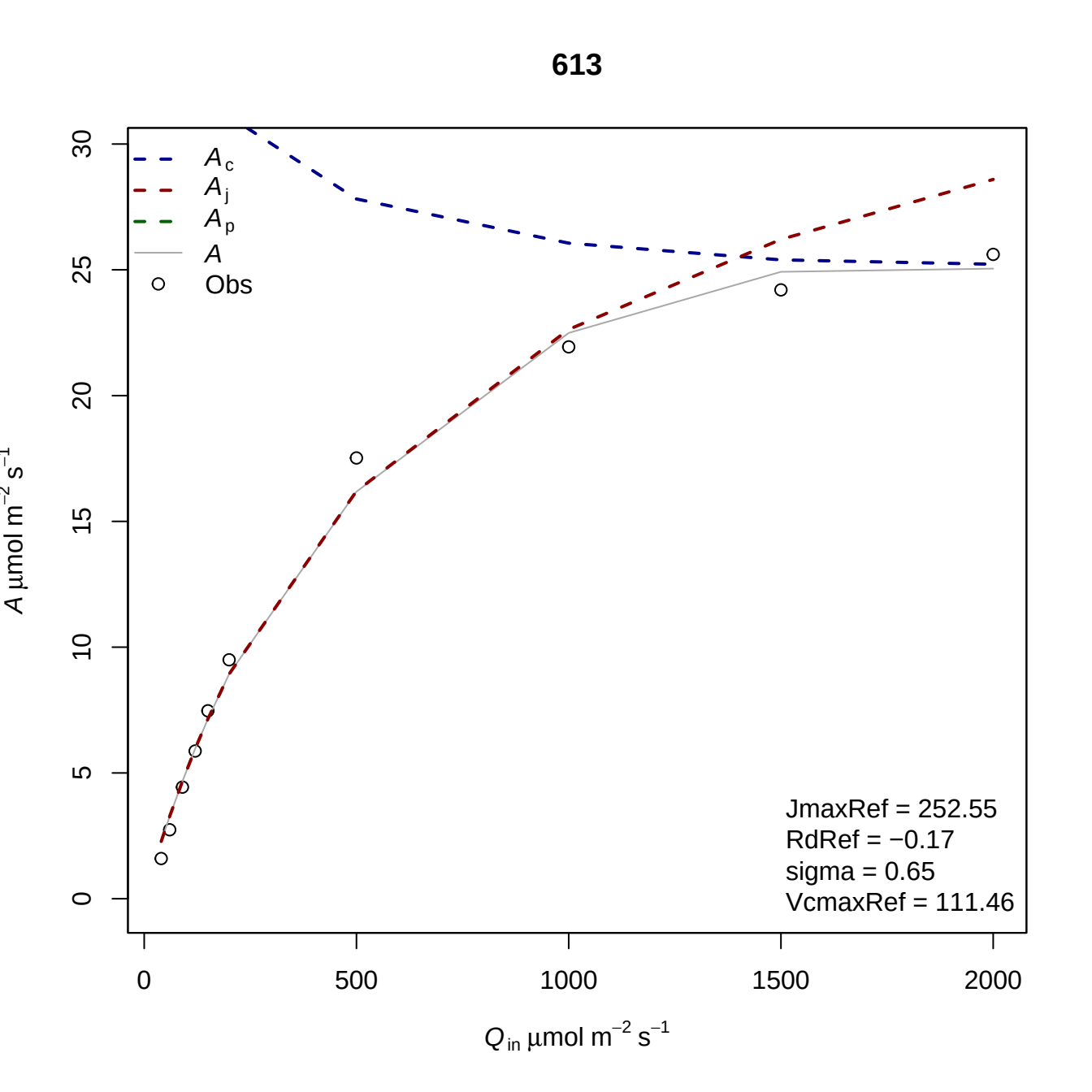

608
A_c
A_j
A_p
A
Obs
JmaxRef = 429.32
RdRef = 0.91
sigma = 0.73
VcmaxRef = 194.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

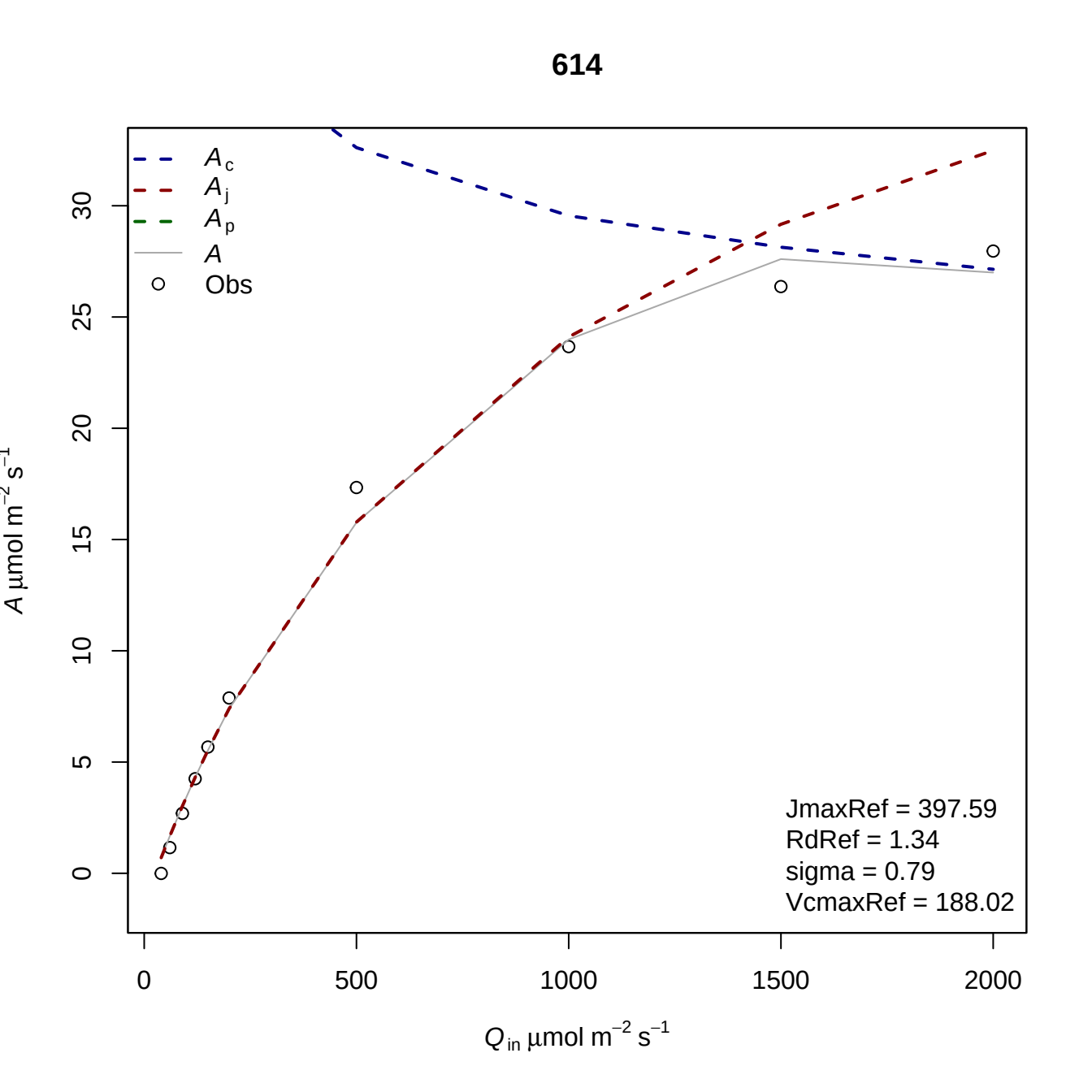

609
A_c
A_j
A_p
A
Obs
JmaxRef = 556.92
RdRef = 1.48
sigma = 0.67
VcmaxRef = 286.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

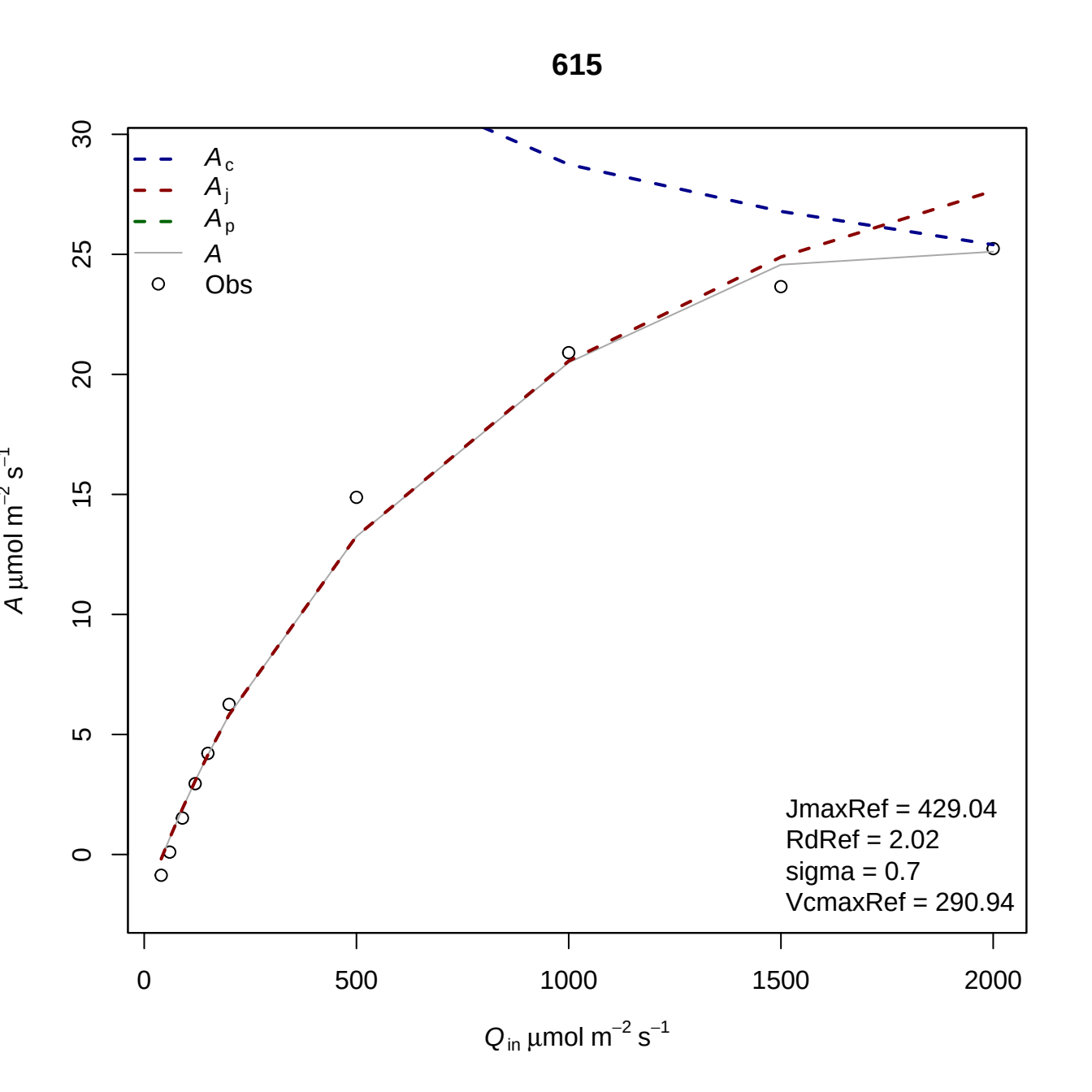

610
A_c
A_j
A_p
A
Obs
JmaxRef = 580.82
RdRef = 2.62
sigma = 0.67
VcmaxRef = 381.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

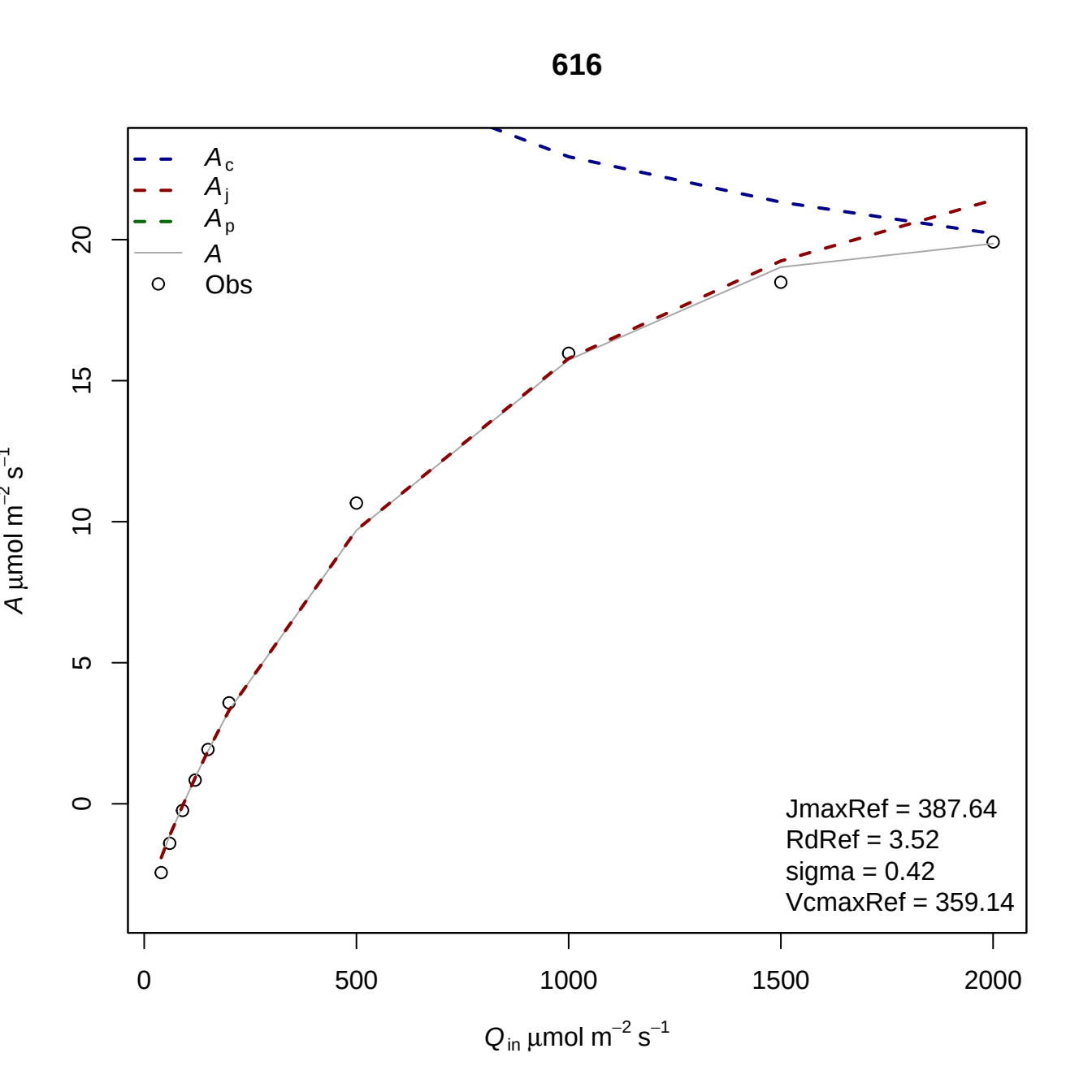

611
A_c
A_j
A_p
A
Obs
JmaxRef = 137.52
RdRef = −0.18
sigma = 0.37
VcmaxRef = 47.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

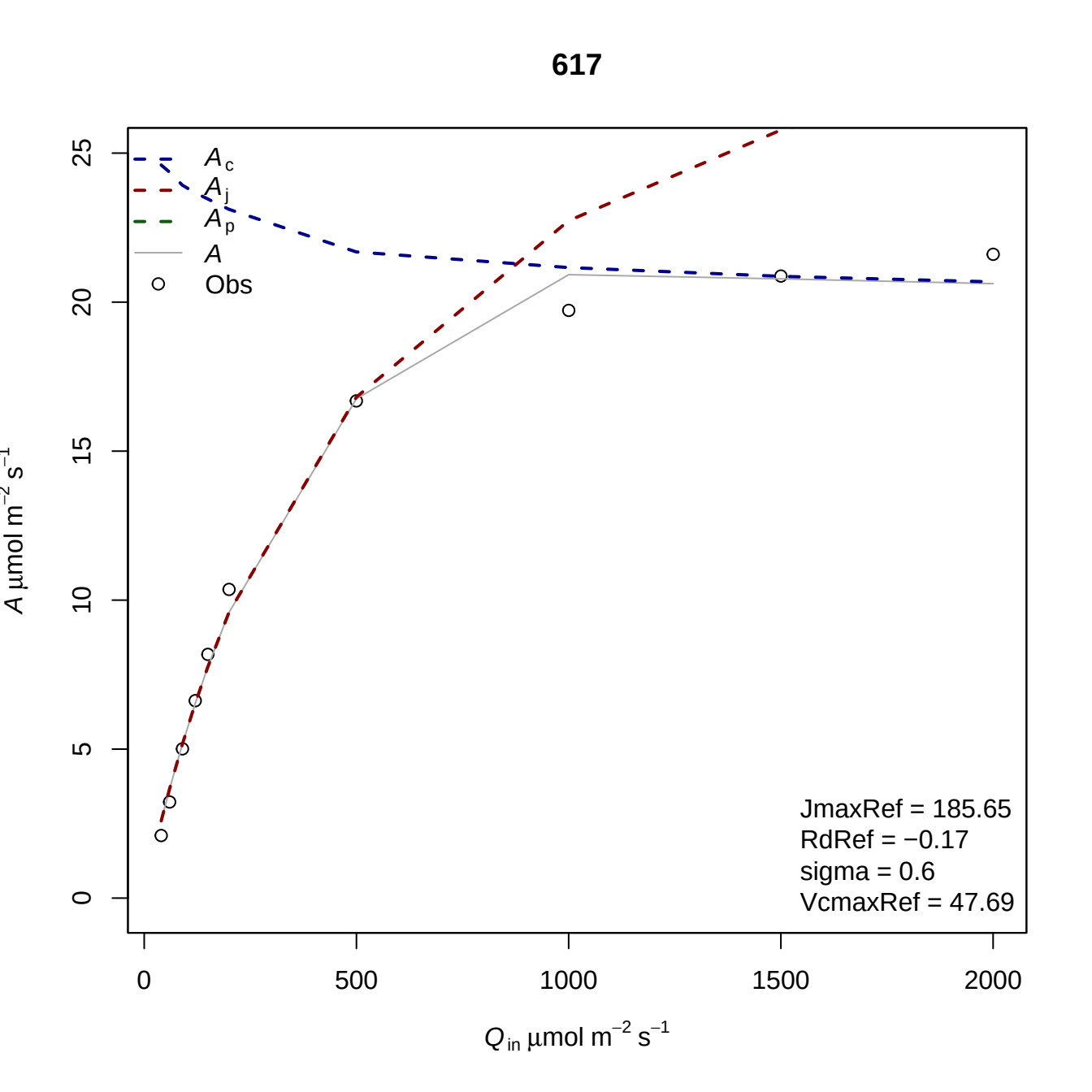

612
A_c
A_j
A_p
A
Obs
JmaxRef = 207.68
RdRef = −0.04
sigma = 0.62
VcmaxRef = 74.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

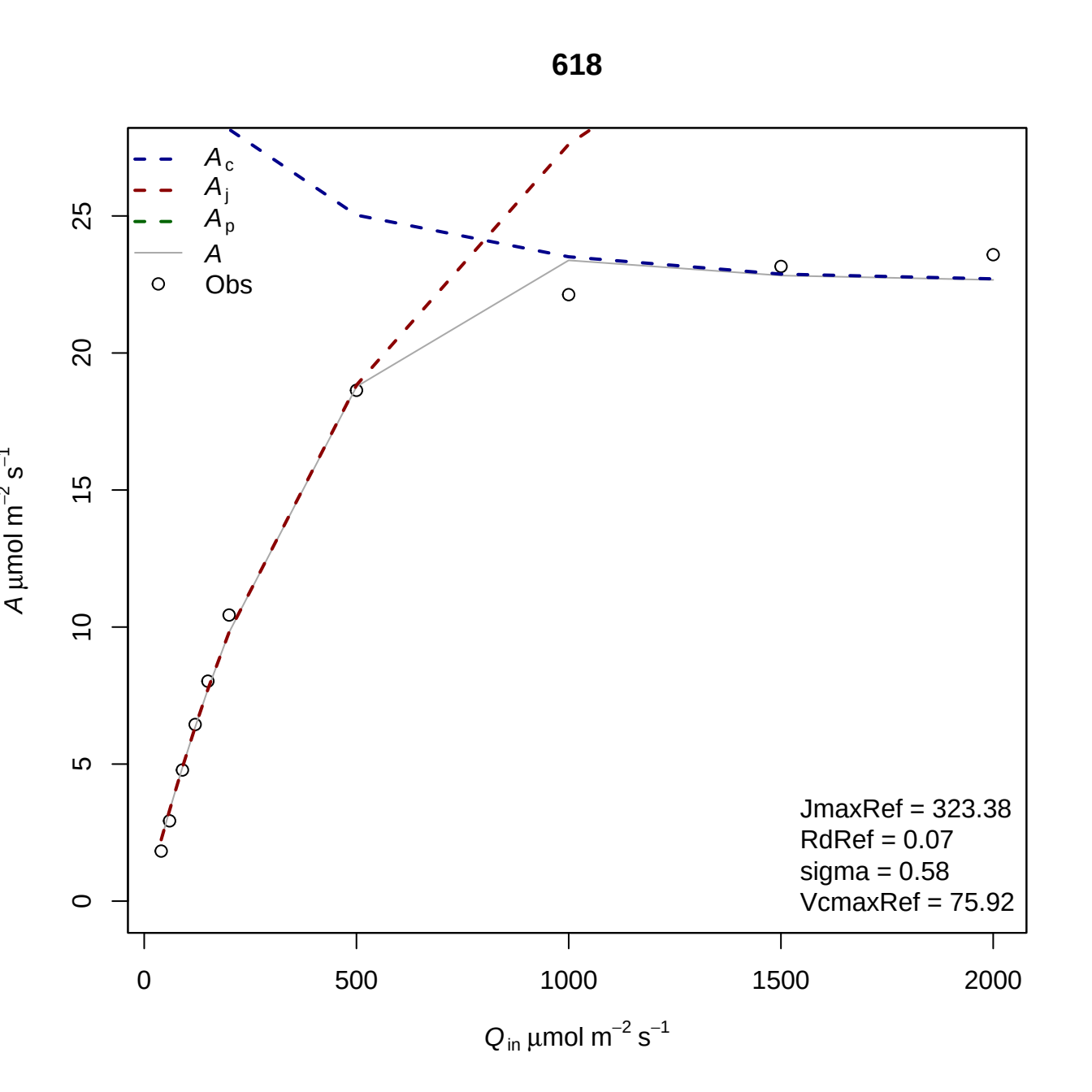

613
A_c
A_j
A_p
A
Obs
JmaxRef = 252.55
RdRef = −0.17
sigma = 0.65
VcmaxRef = 111.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

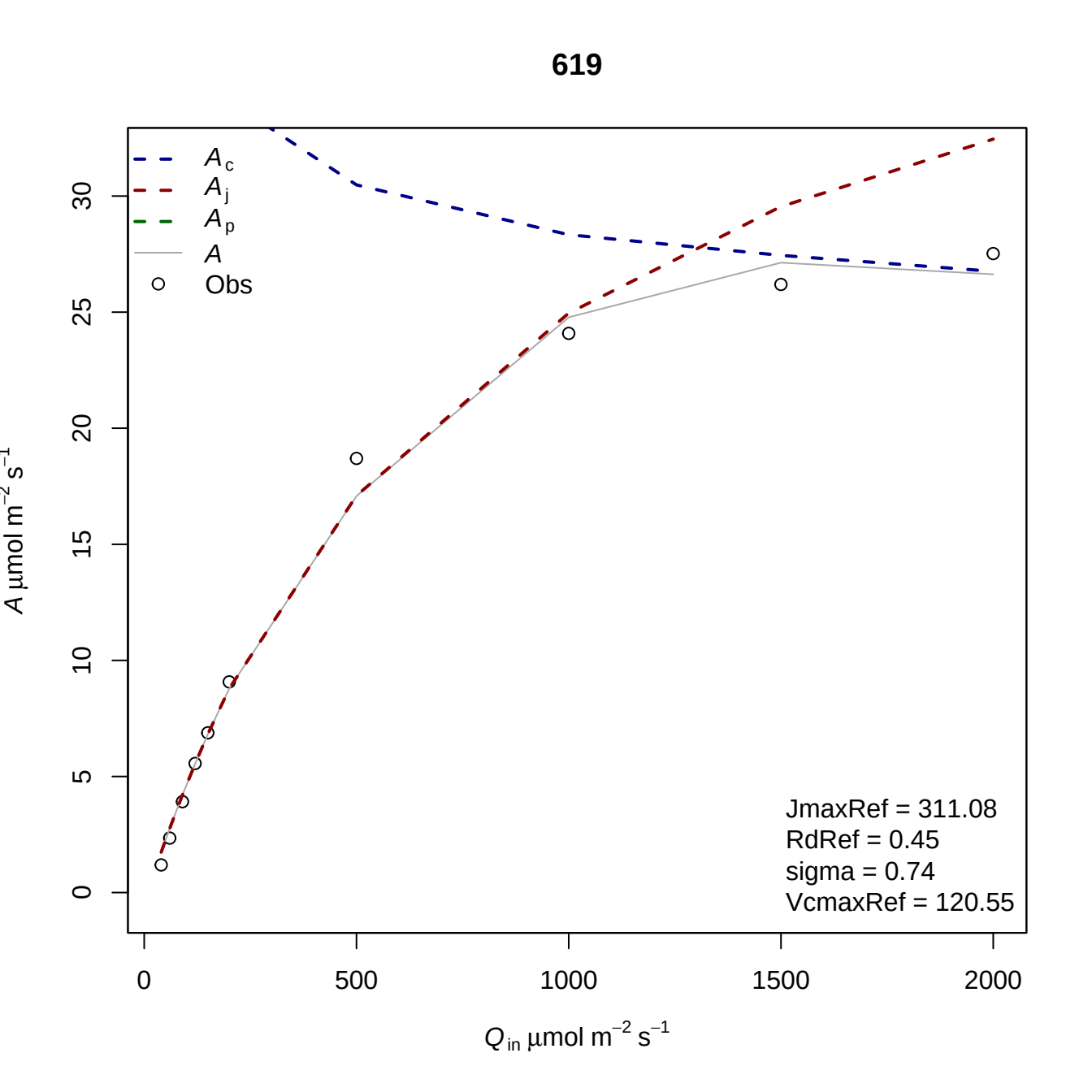

614
A_c
A_j
A_p
A
Obs
JmaxRef = 397.59
RdRef = 1.34
sigma = 0.79
VcmaxRef = 188.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

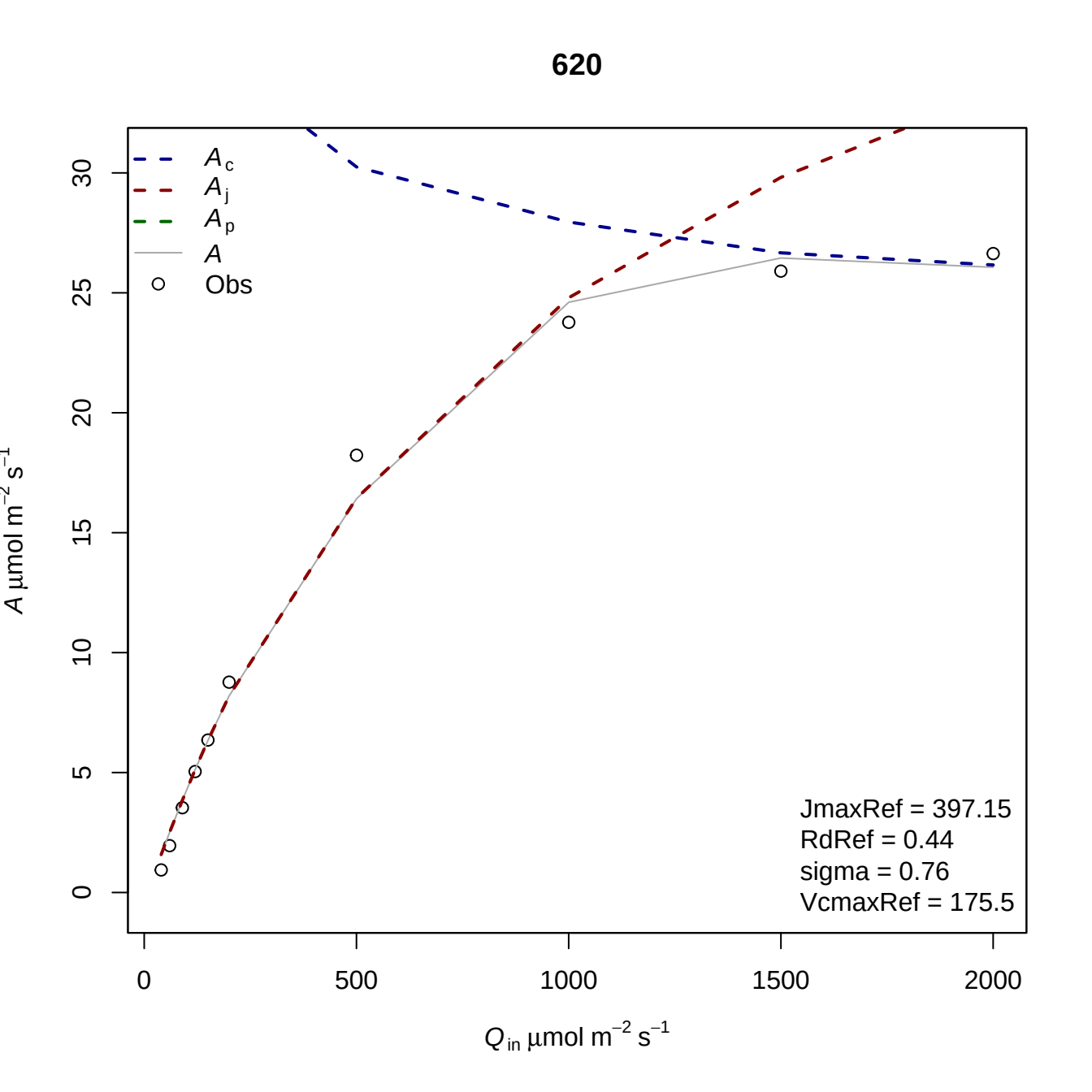

615
A_c
A_j
A_p
A
Obs
JmaxRef = 429.04
RdRef = 2.02
sigma = 0.7
VcmaxRef = 290.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

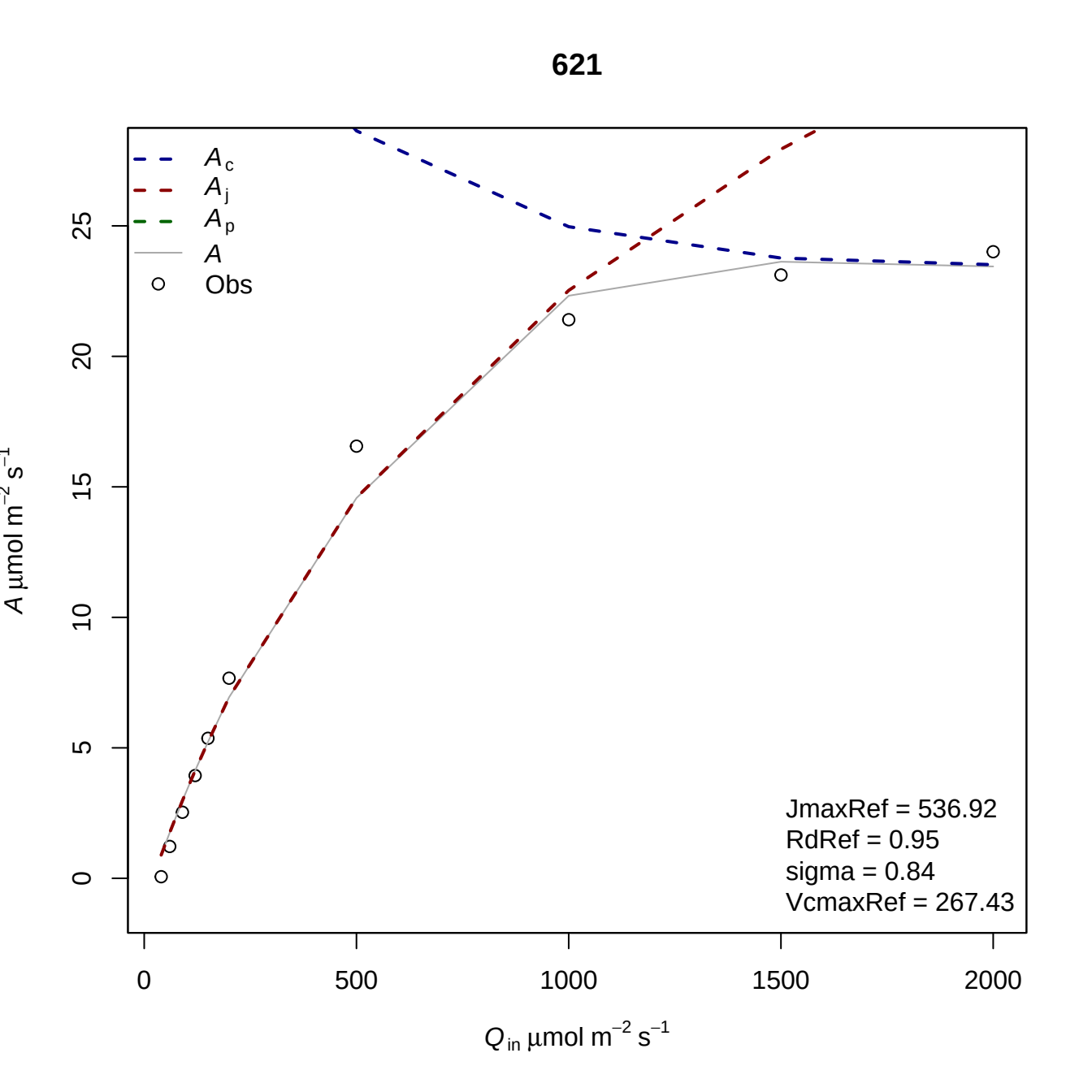

616
A_c
A_j
A_p
A
Obs
JmaxRef = 387.64
RdRef = 3.52
sigma = 0.42
VcmaxRef = 359.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

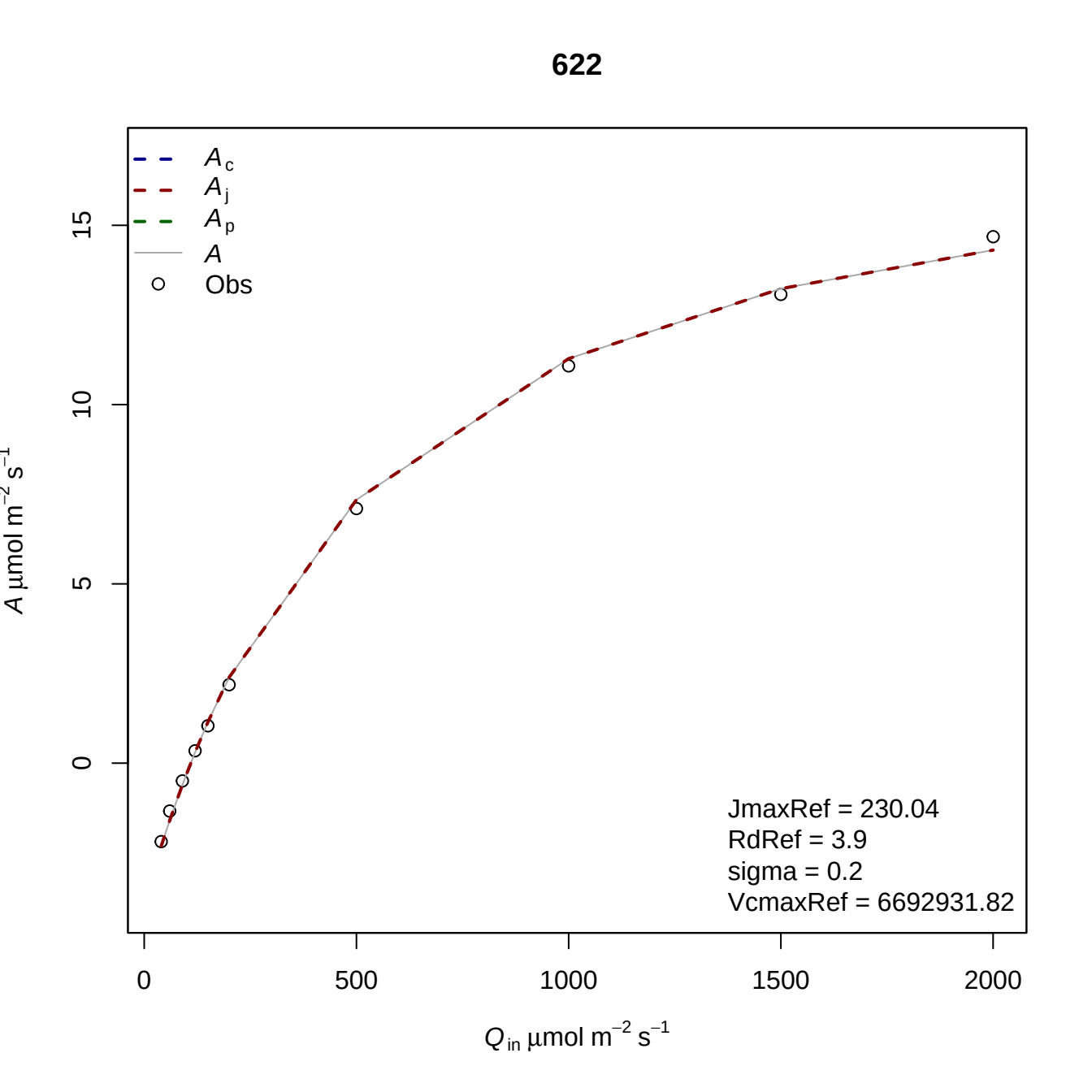

617
A_c
A_j
A_p
A
Obs
JmaxRef = 185.65
RdRef = −0.17
sigma = 0.6
VcmaxRef = 47.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

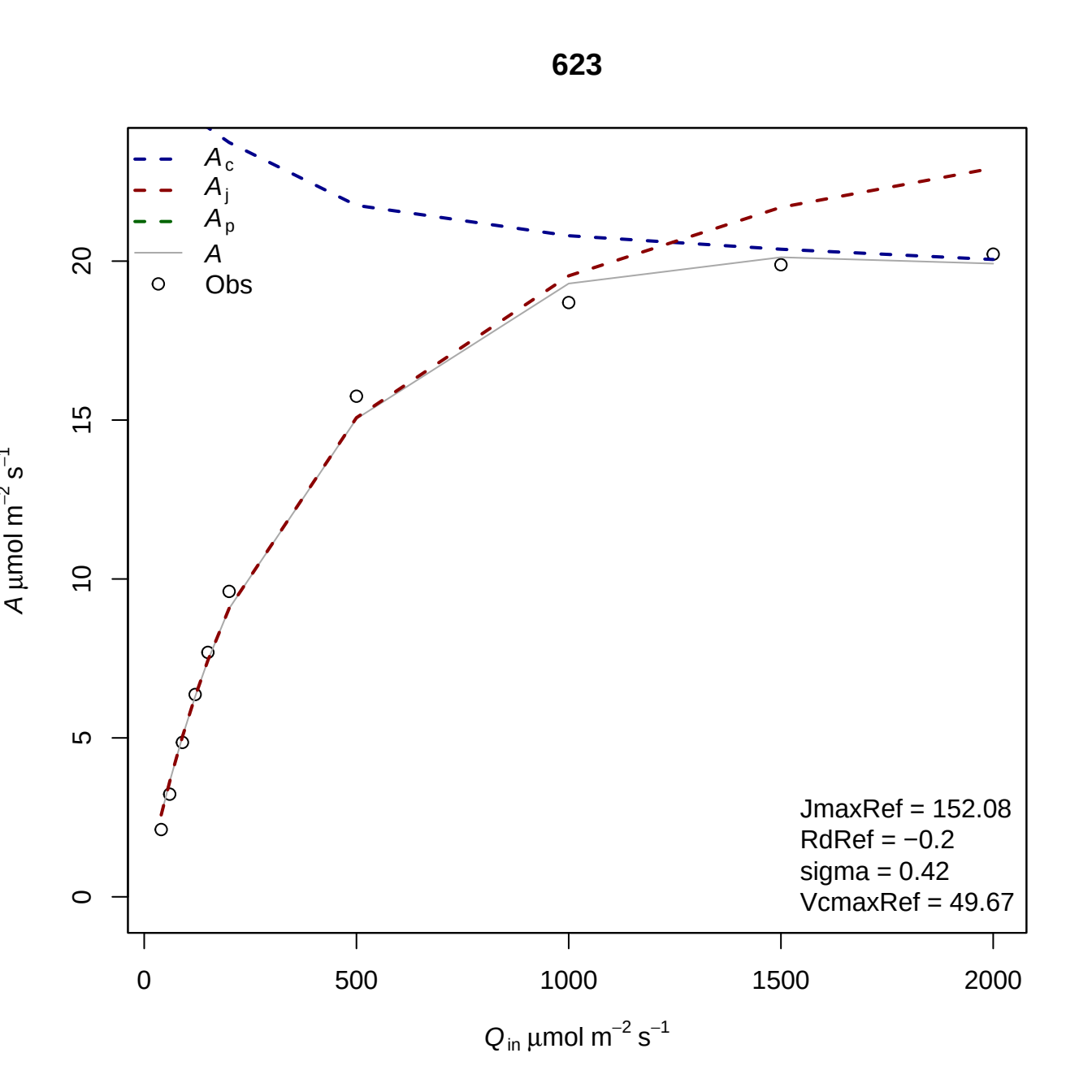

618
A_c
A_j
A_p
A
Obs
JmaxRef = 323.38
RdRef = 0.07
sigma = 0.58
VcmaxRef = 75.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

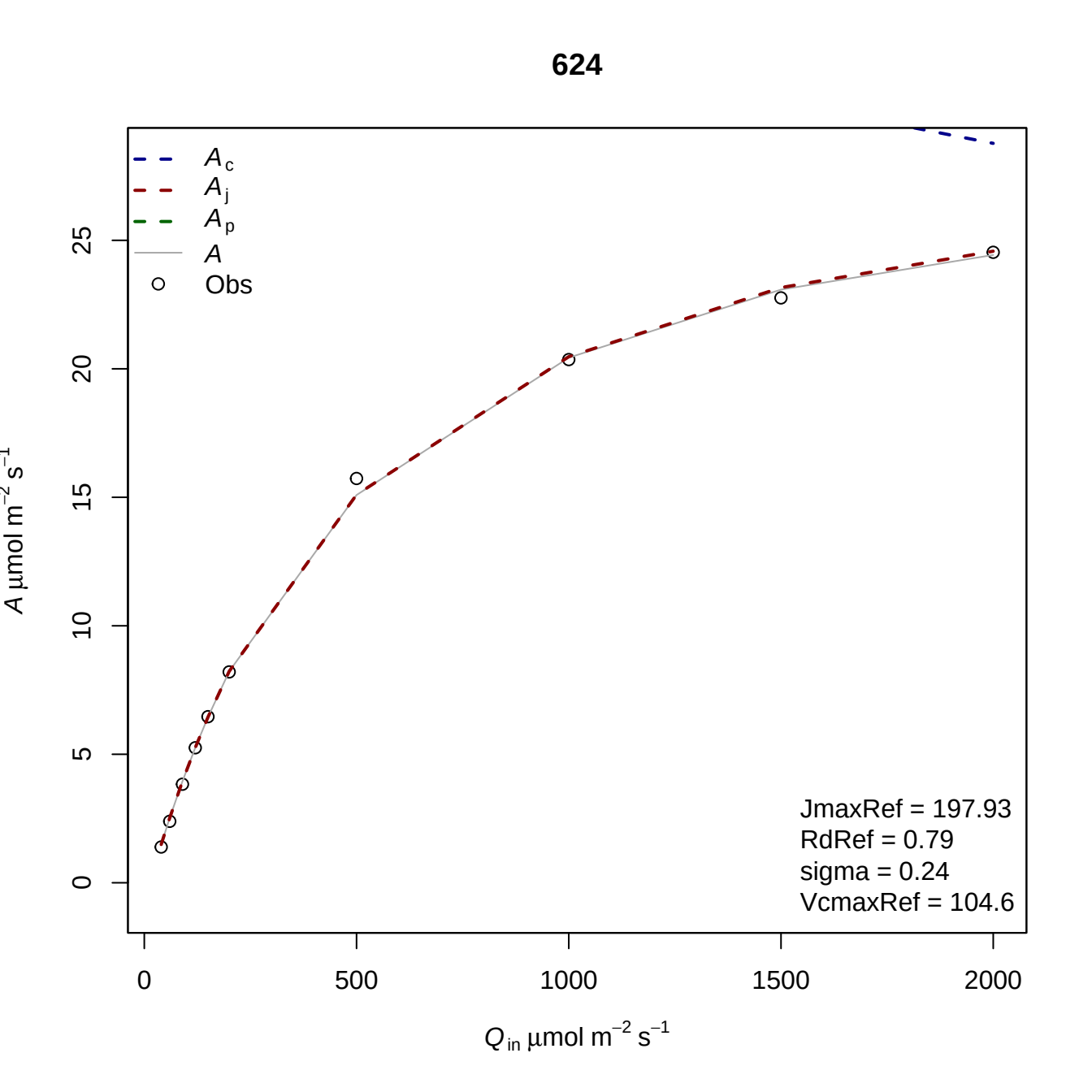

619
A_c
A_j
A_p
A
Obs
JmaxRef = 311.08
RdRef = 0.45
sigma = 0.74
VcmaxRef = 120.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

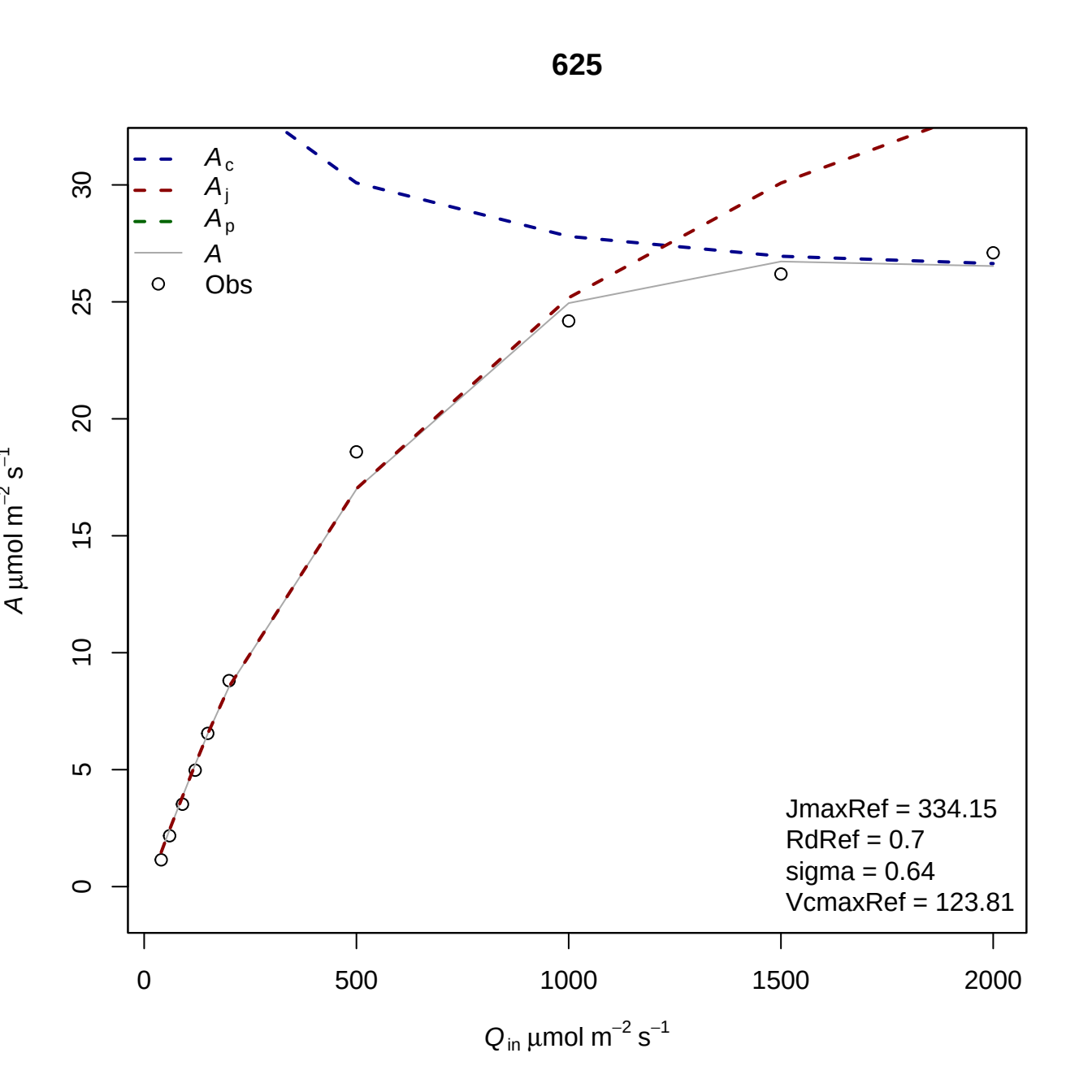

620
A_c
A_j
A_p
A
Obs
JmaxRef = 397.15
RdRef = 0.44
sigma = 0.76
VcmaxRef = 175.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

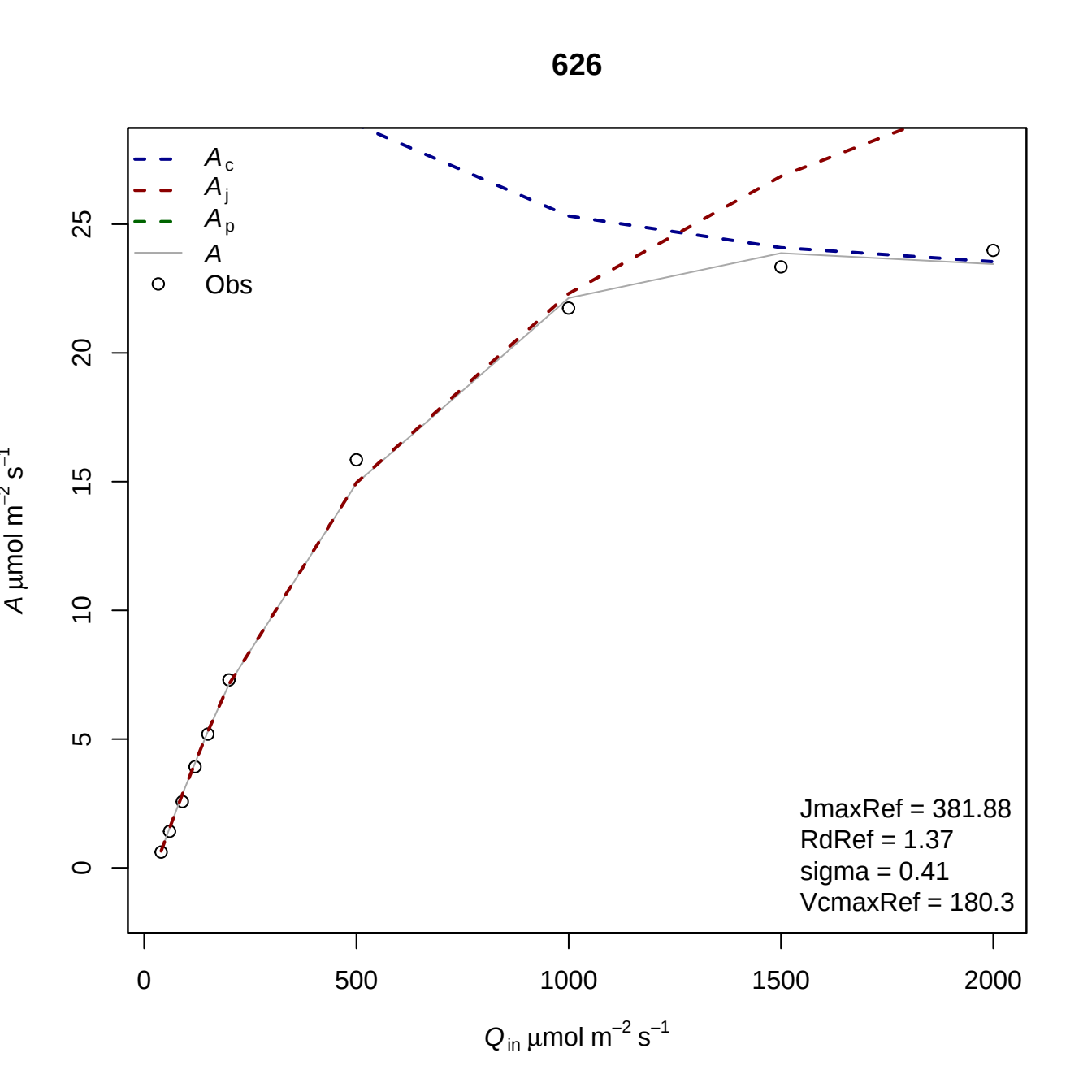

621
A_c
A_j
A_p
A
Obs
JmaxRef = 536.92
RdRef = 0.95
sigma = 0.84
VcmaxRef = 267.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

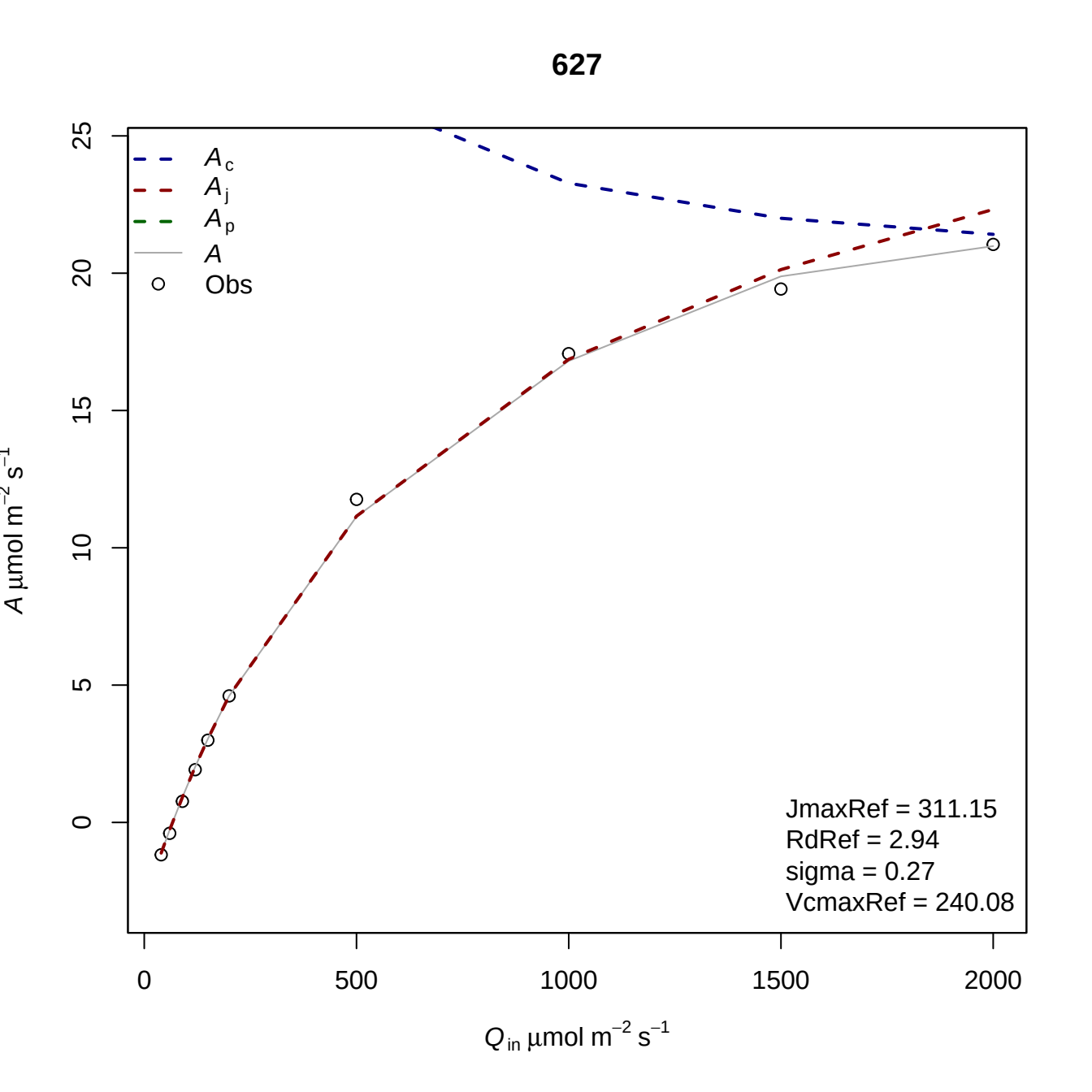

622
A_c
A_j
A_p
A
Obs
JmaxRef = 230.04
RdRef = 3.9
sigma = 0.2
VcmaxRef = 6692931.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

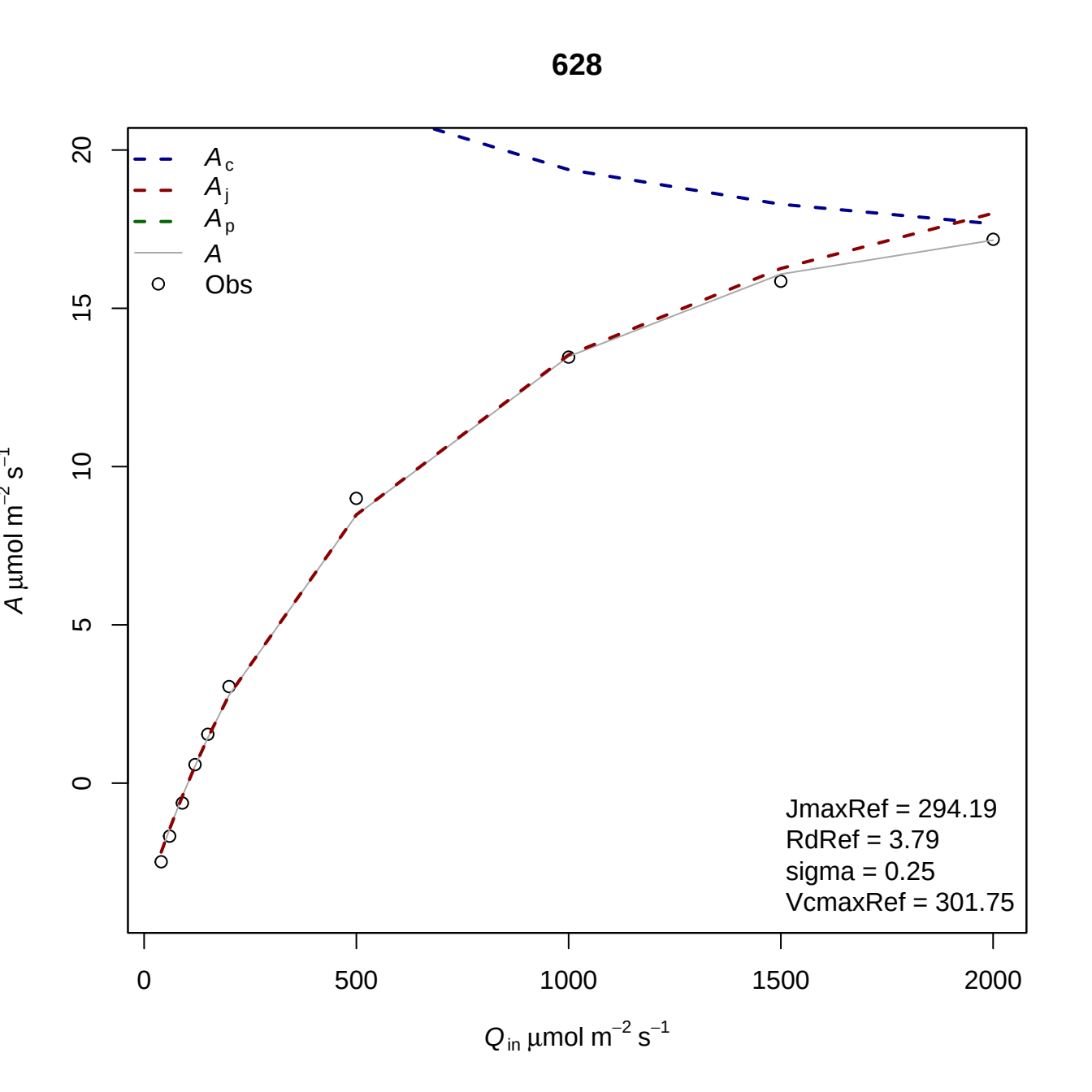

623
A_c
A_j
A_p
A
Obs
JmaxRef = 152.08
RdRef = −0.2
sigma = 0.42
VcmaxRef = 49.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

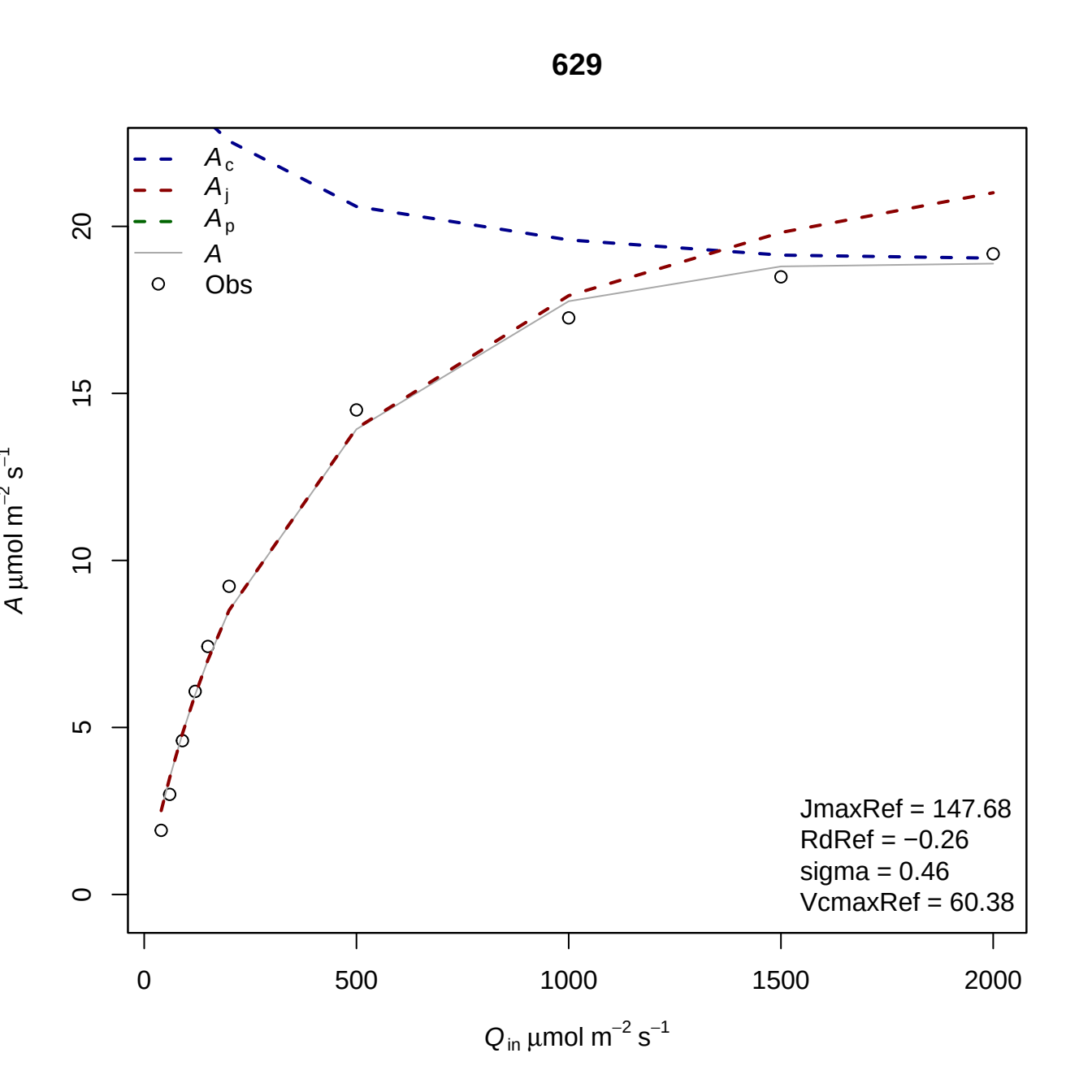

624
A_c
A_j
A_p
A
Obs
JmaxRef = 197.93
RdRef = 0.79
sigma = 0.24
VcmaxRef = 104.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

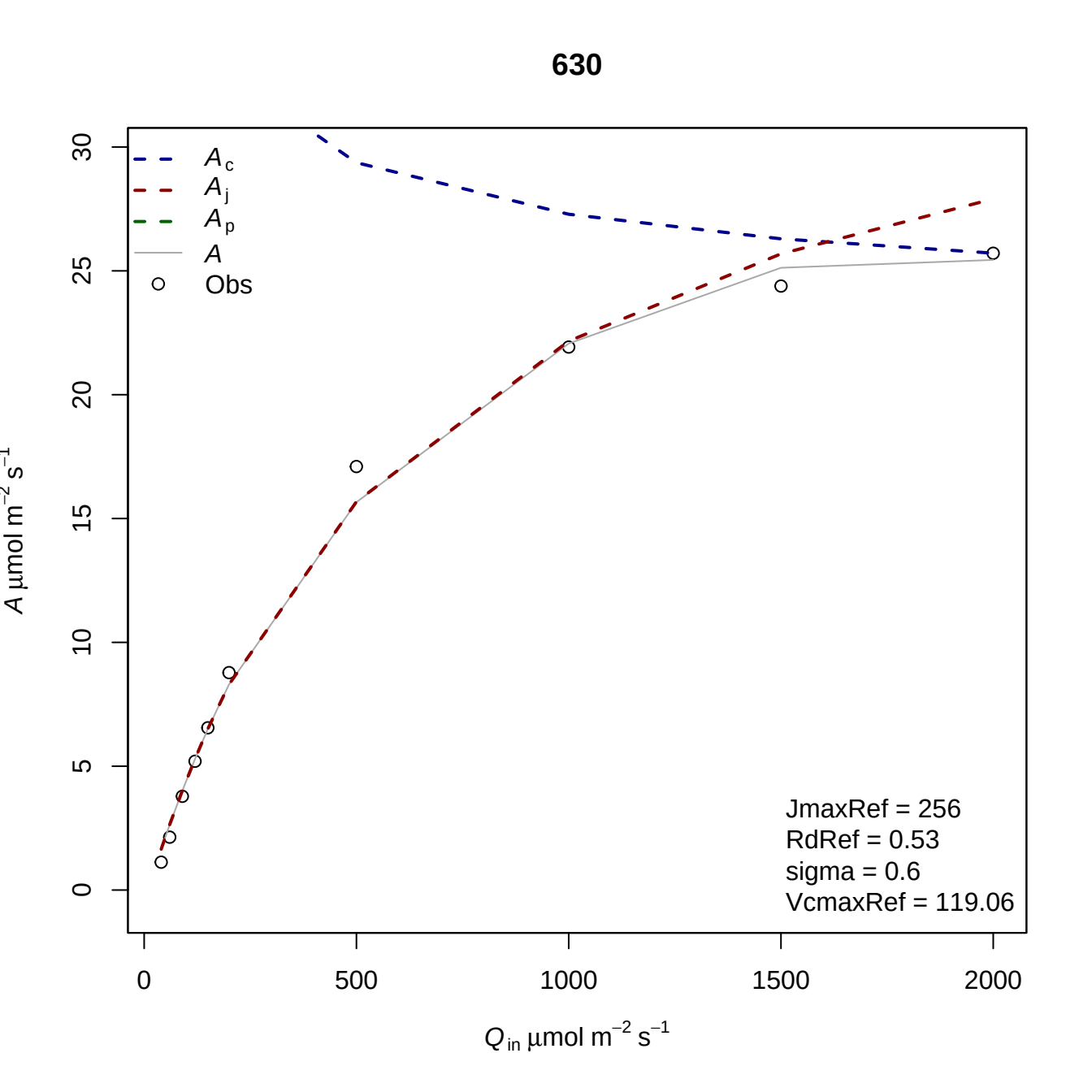

625
A_c
A_j
A_p
A
Obs
JmaxRef = 334.15
RdRef = 0.7
sigma = 0.64
VcmaxRef = 123.81
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

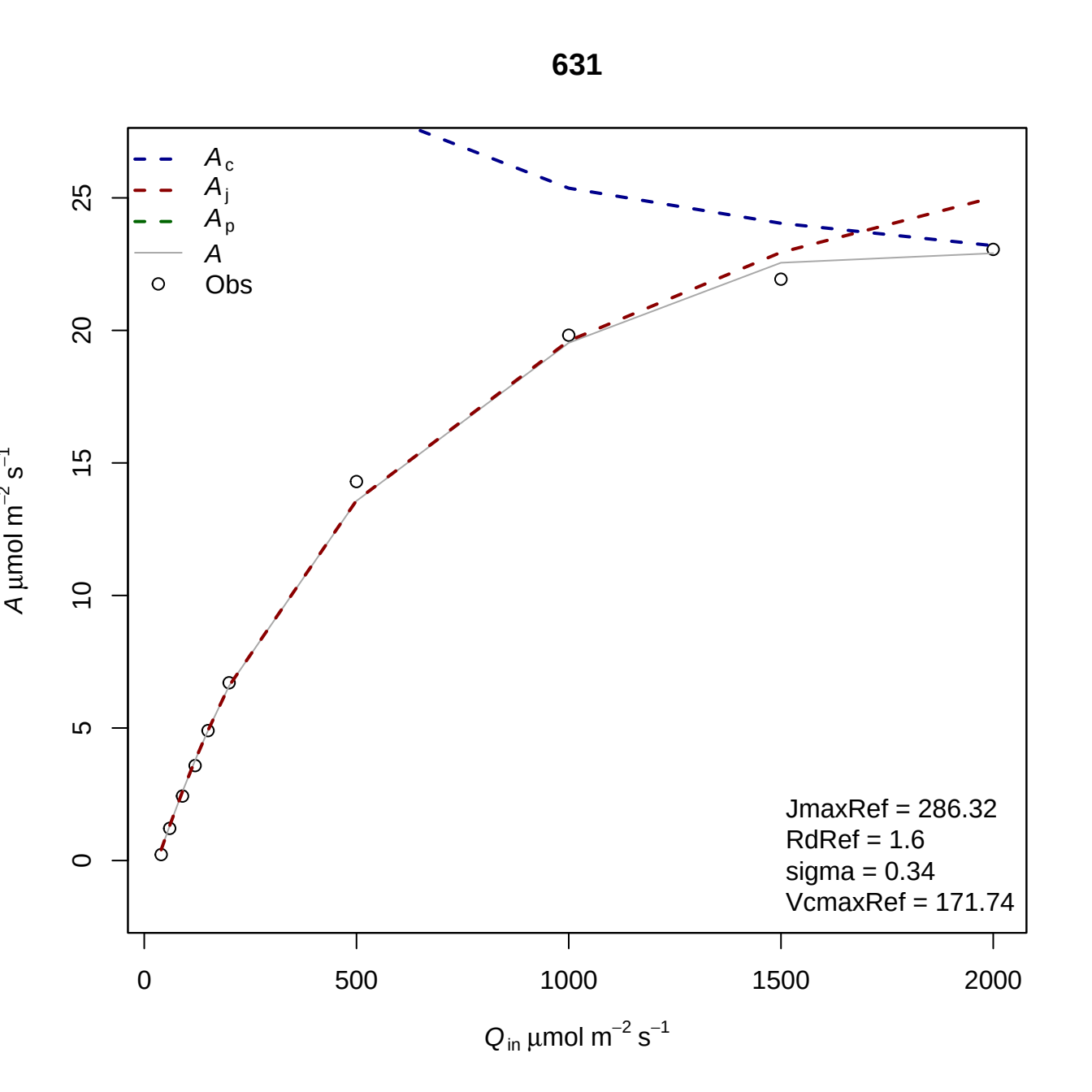

626
A_c
A_j
A_p
A
Obs
JmaxRef = 381.88
RdRef = 1.37
sigma = 0.41
VcmaxRef = 180.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

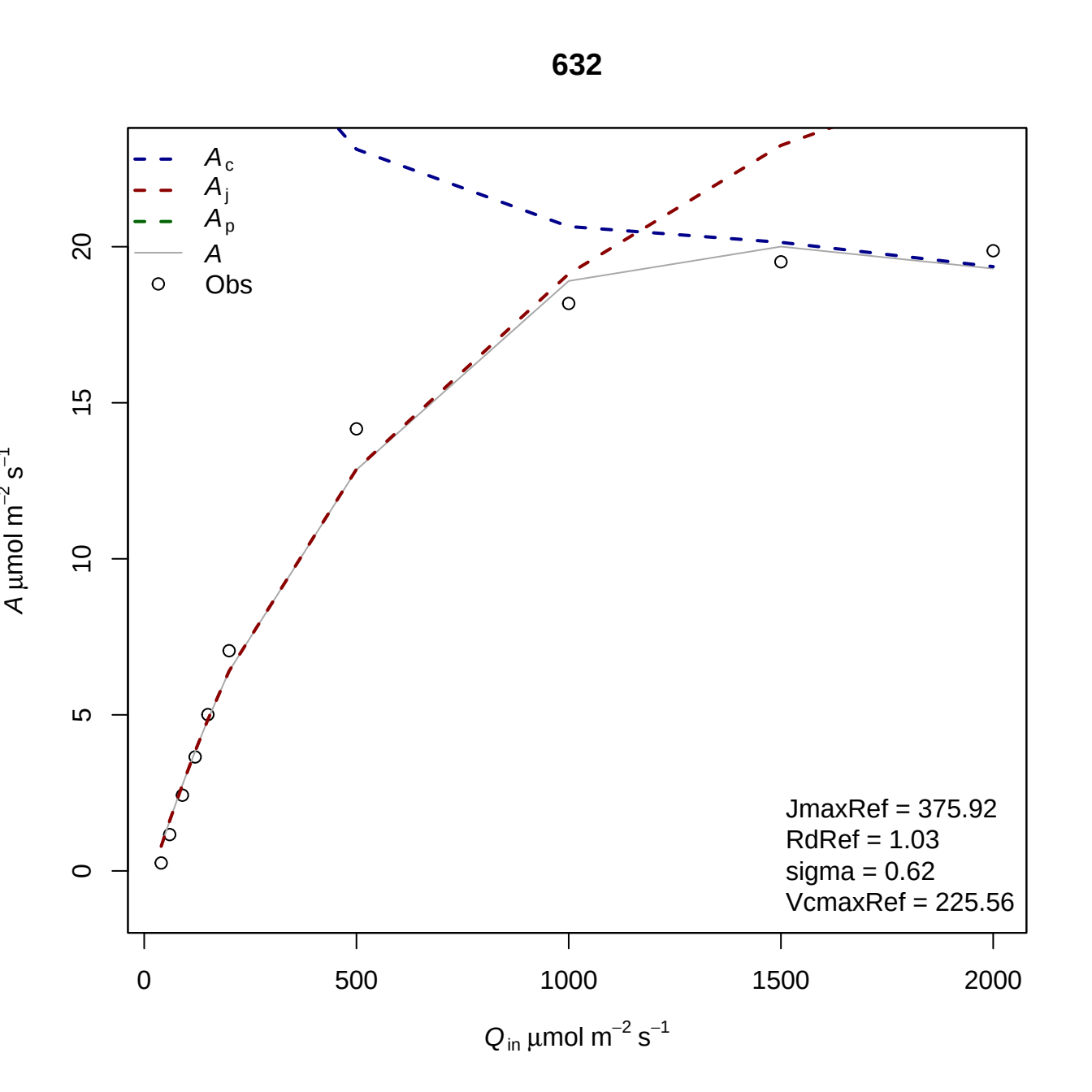

627
A_c
A_j
A_p
A
Obs
JmaxRef = 311.15
RdRef = 2.94
sigma = 0.27
VcmaxRef = 240.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

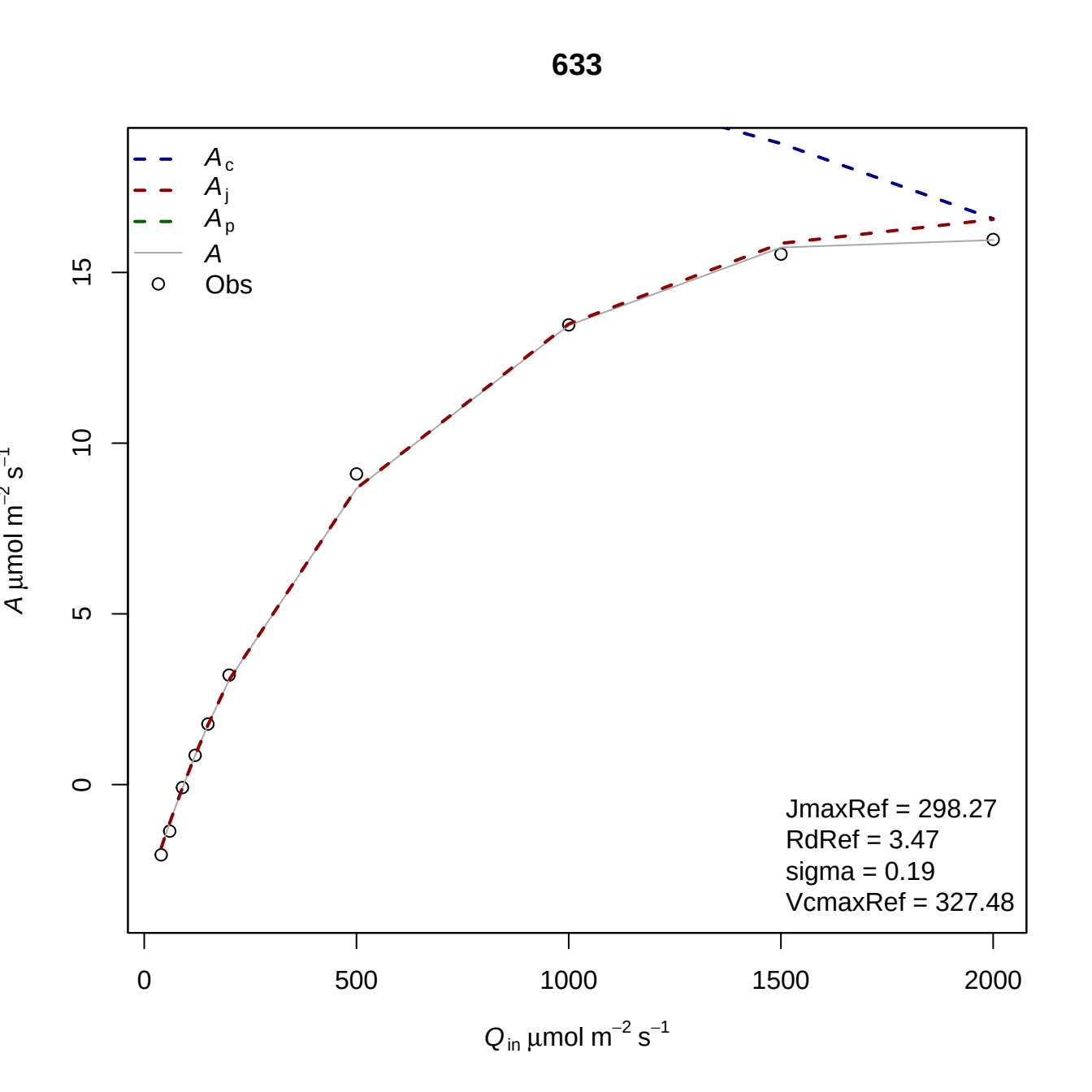

628
A_c
A_j
A_p
A
Obs
JmaxRef = 294.19
RdRef = 3.79
sigma = 0.25
VcmaxRef = 301.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

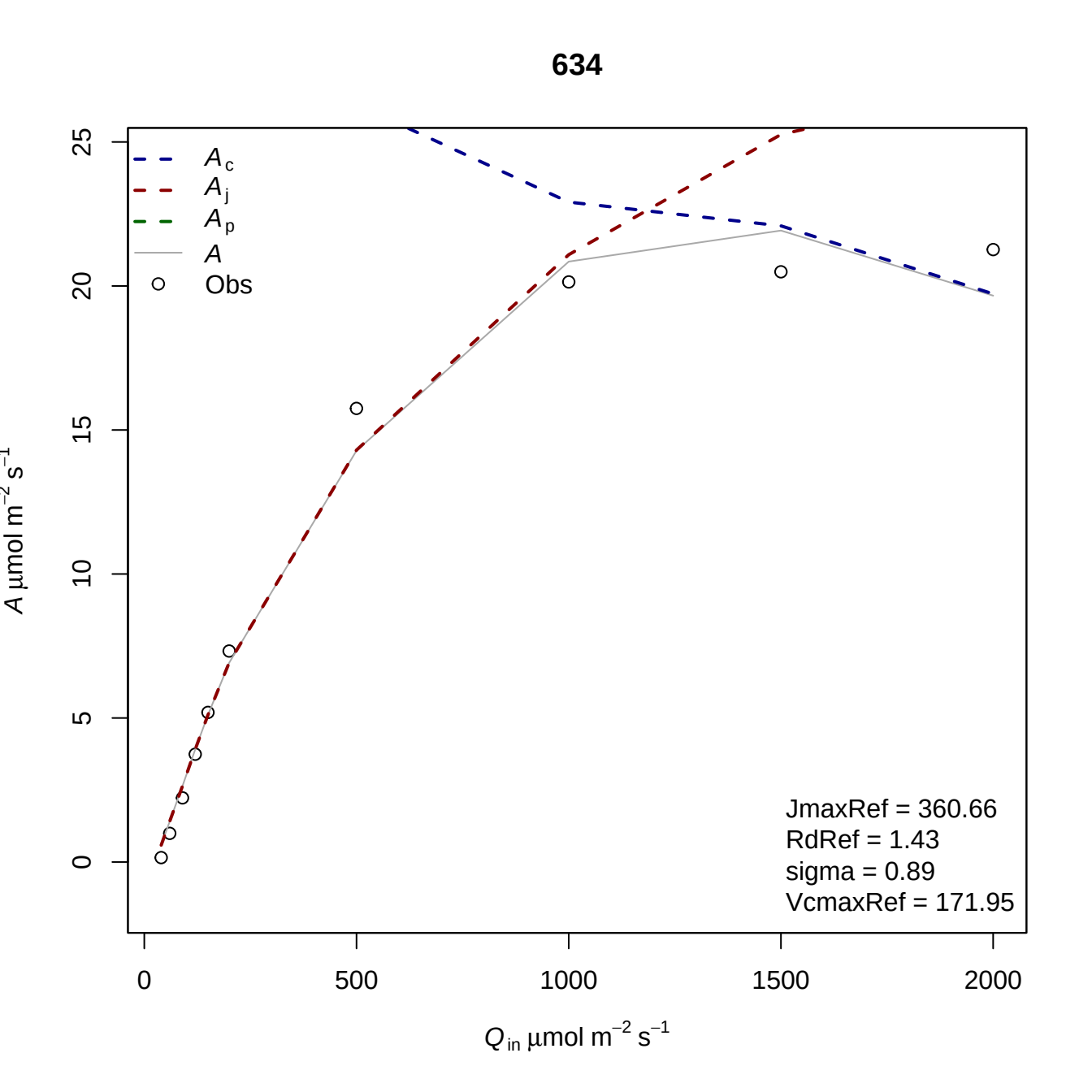

629
A_c
A_j
A_p
A
Obs
JmaxRef = 147.68
RdRef = −0.26
sigma = 0.46
VcmaxRef = 60.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

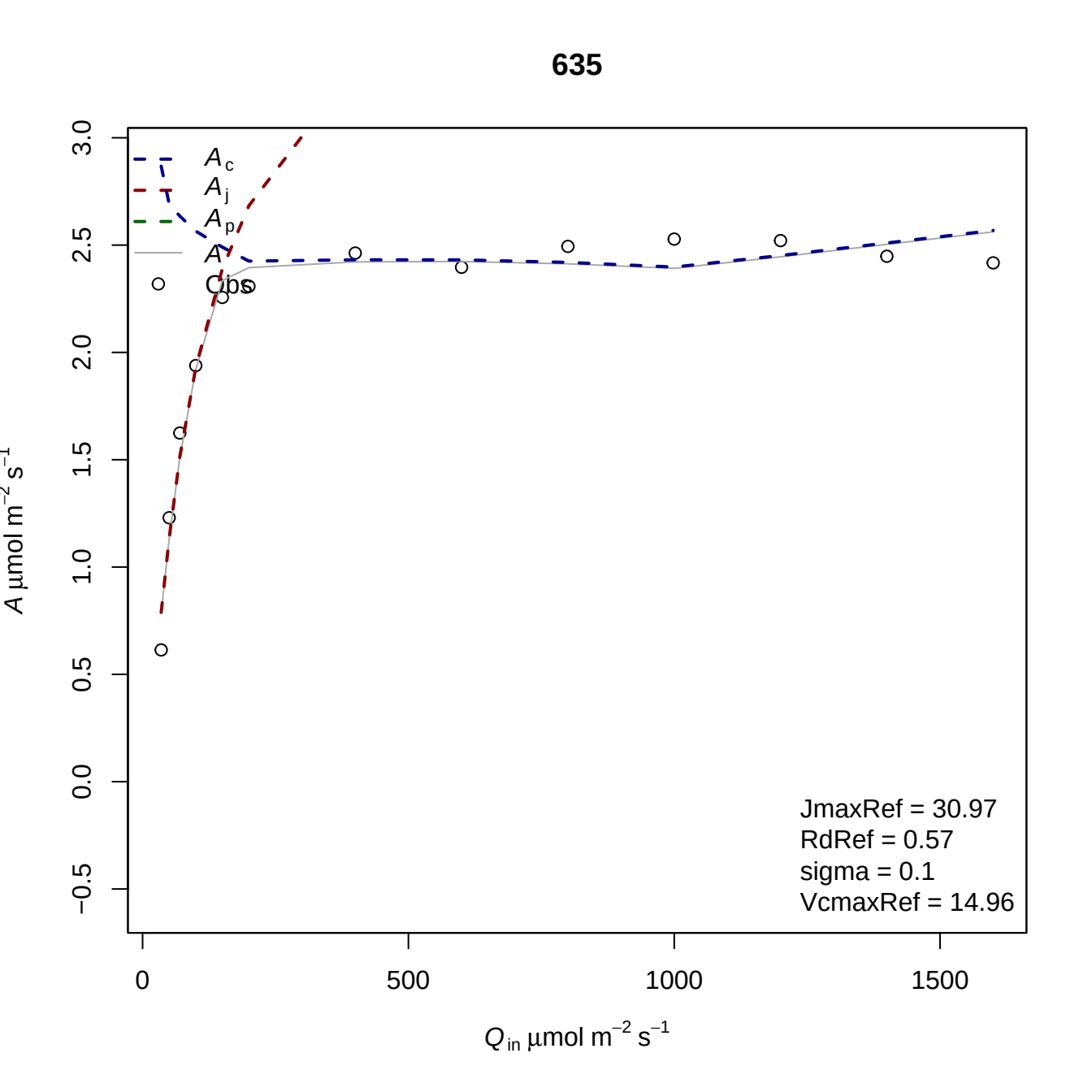

630
A_c
A_j
A_p
A
Obs
JmaxRef = 256
RdRef = 0.53
sigma = 0.6
VcmaxRef = 119.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

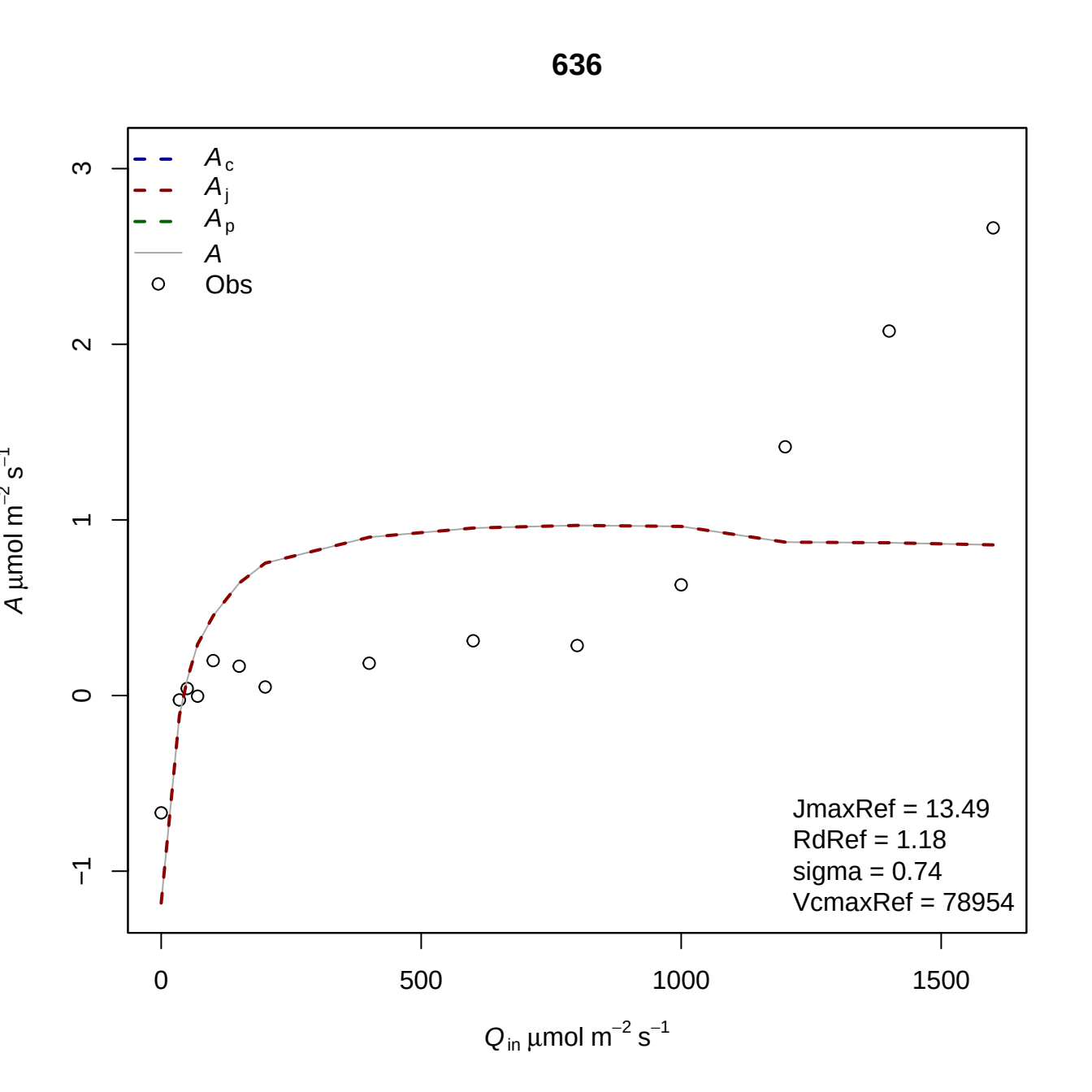

631
A_c
A_j
A_p
A
Obs
JmaxRef = 286.32
RdRef = 1.6
sigma = 0.34
VcmaxRef = 171.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

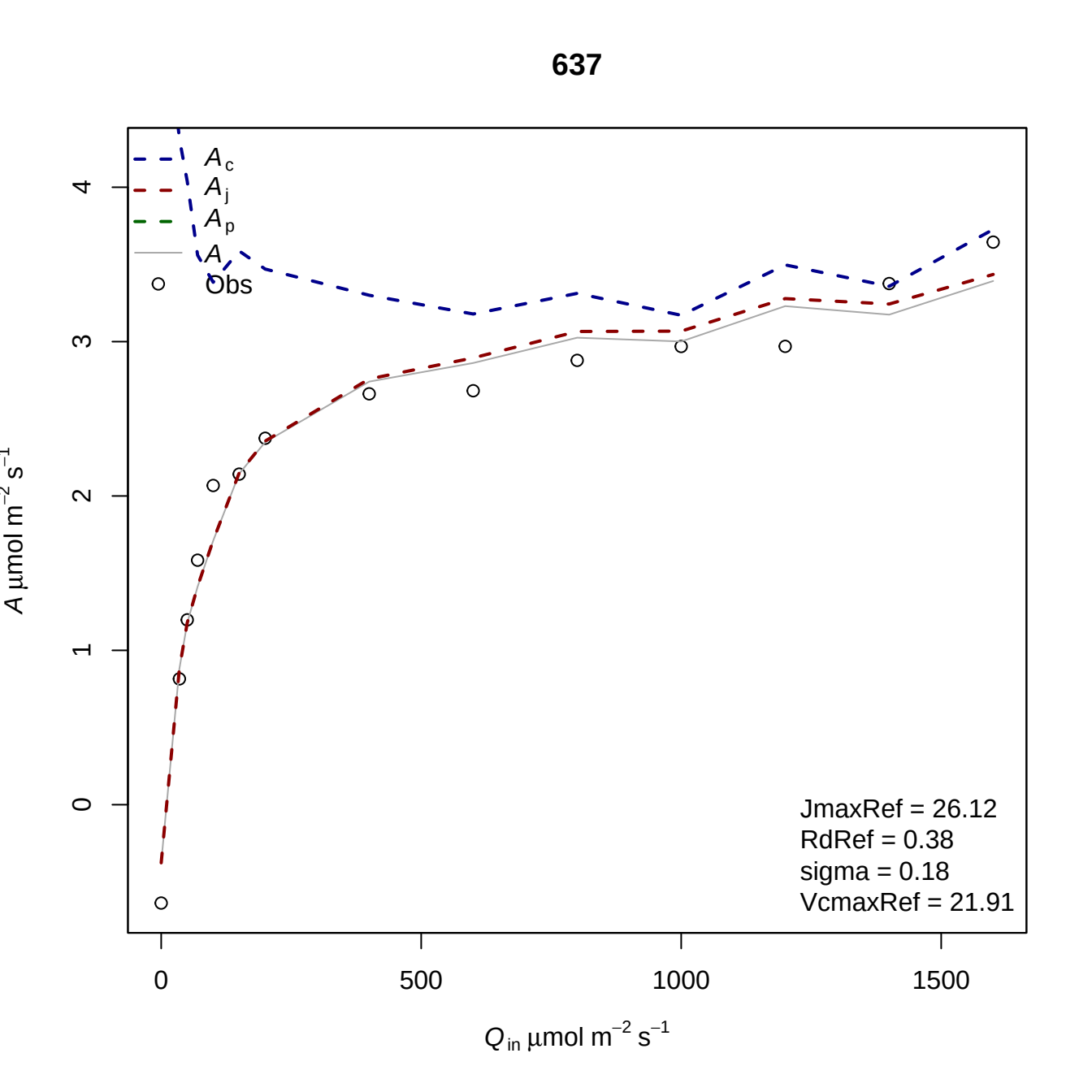

632
A_c
A_j
A_p
A
Obs
JmaxRef = 375.92
RdRef = 1.03
sigma = 0.62
VcmaxRef = 225.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

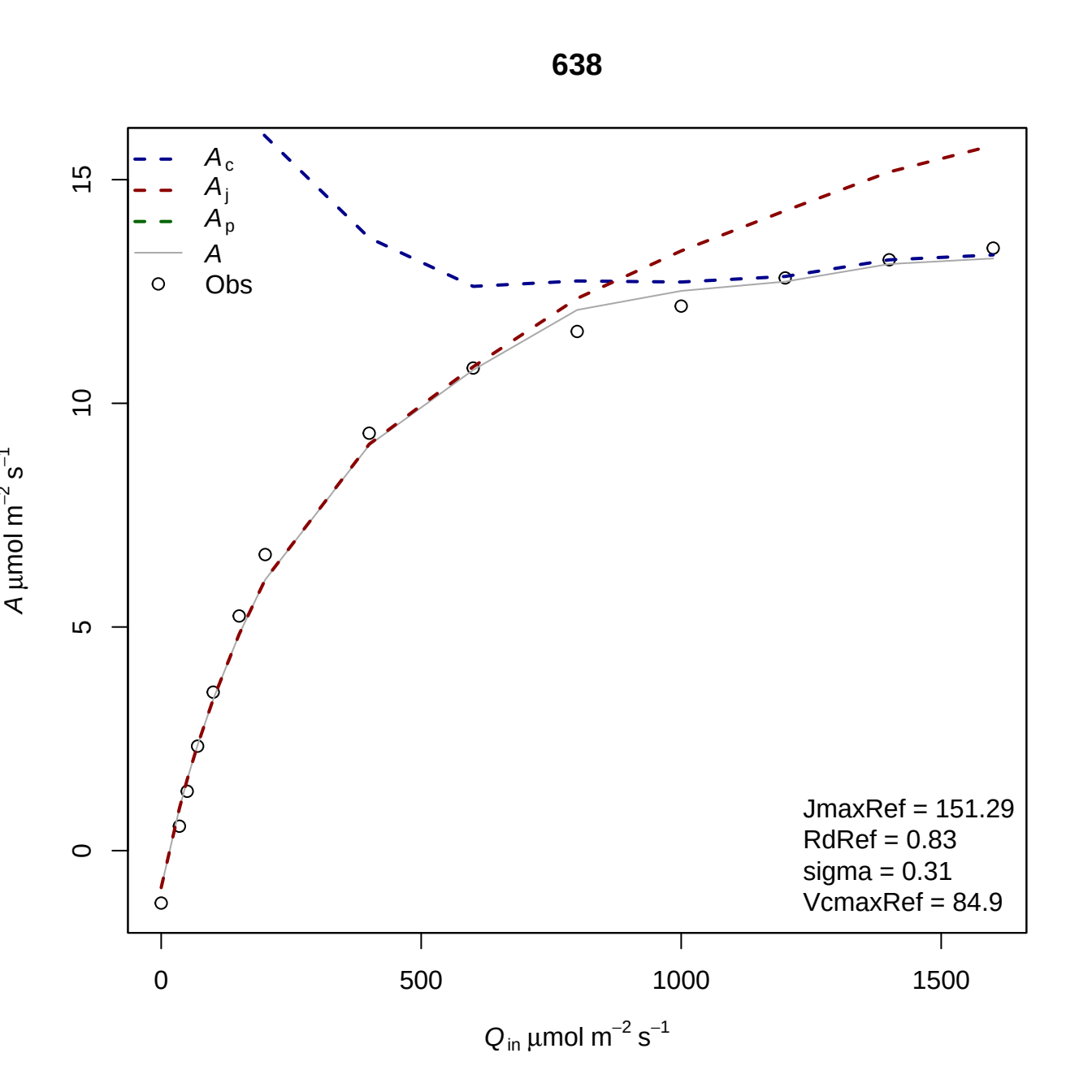

633
A_c
A_j
A_p
A
Obs
JmaxRef = 298.27
RdRef = 3.47
sigma = 0.19
VcmaxRef = 327.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

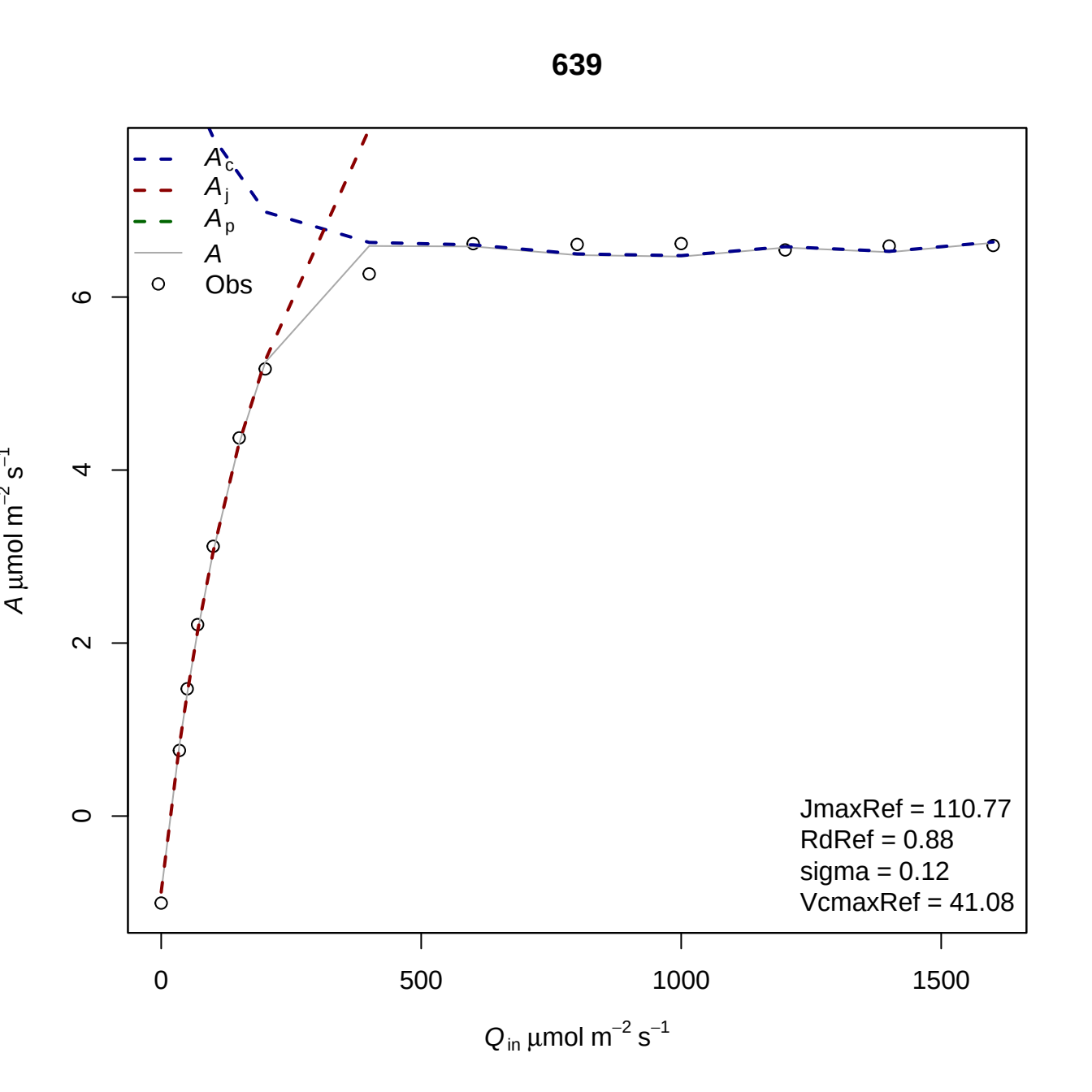

634
A_c
A_j
A_p
A
Obs
JmaxRef = 360.66
RdRef = 1.43
sigma = 0.89
VcmaxRef = 171.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

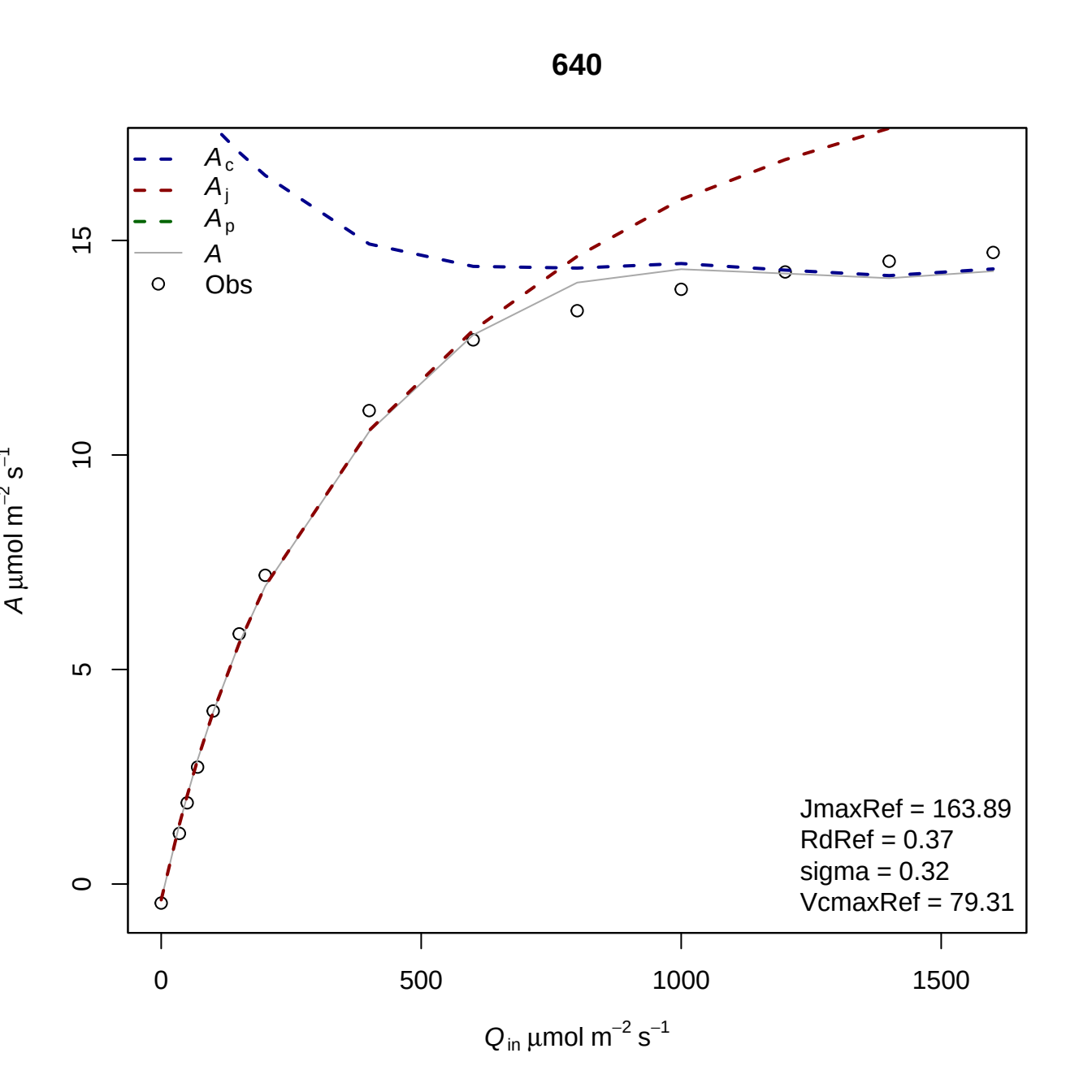

635
A_c
A_j
A_p
A
Obs
JmaxRef = 30.97
RdRef = 0.57
sigma = 0.1
VcmaxRef = 14.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500

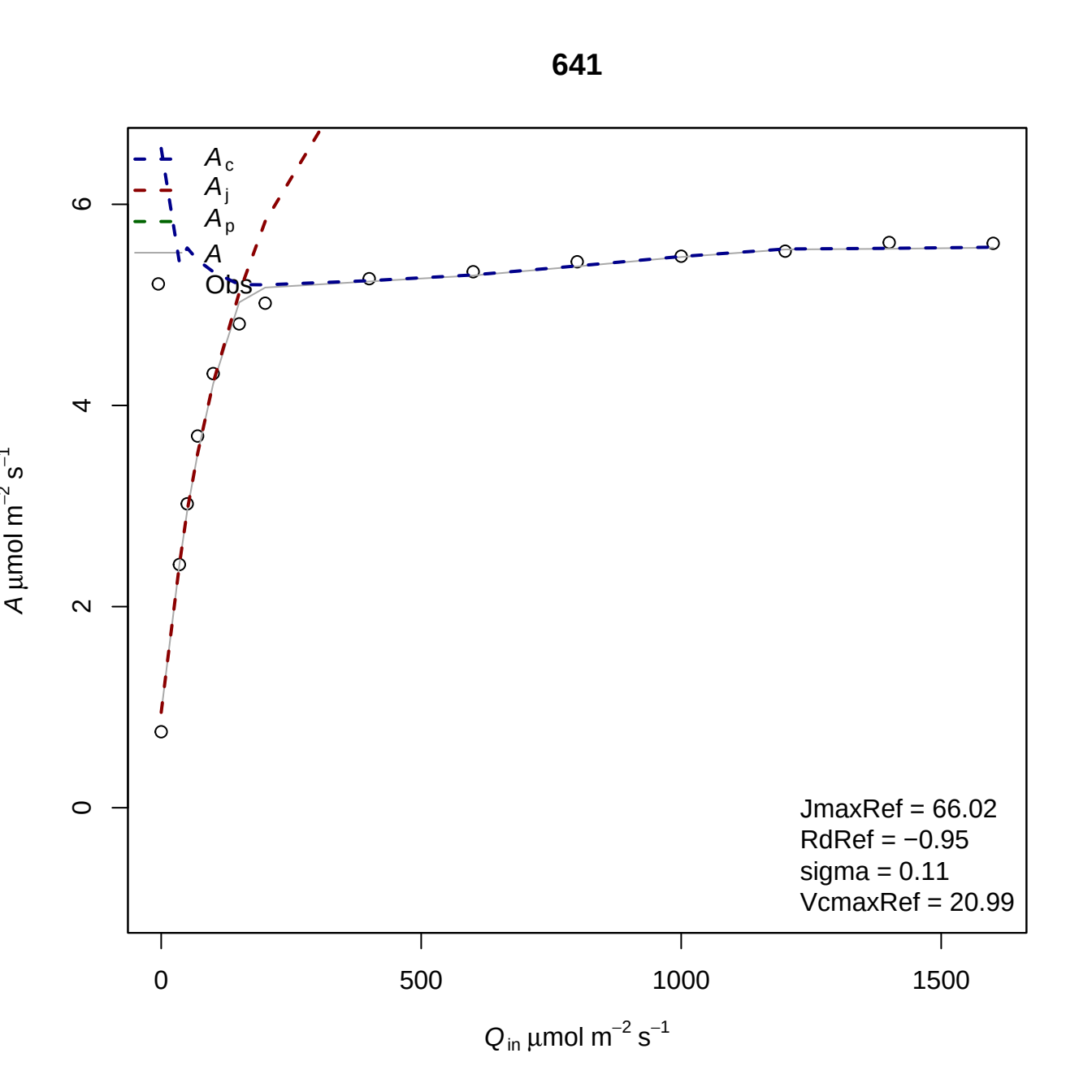

636
A_c
A_j
A_p
A
Obs
JmaxRef = 13.49
RdRef = 1.18
sigma = 0.74
VcmaxRef = 78954
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
500
1000
1500

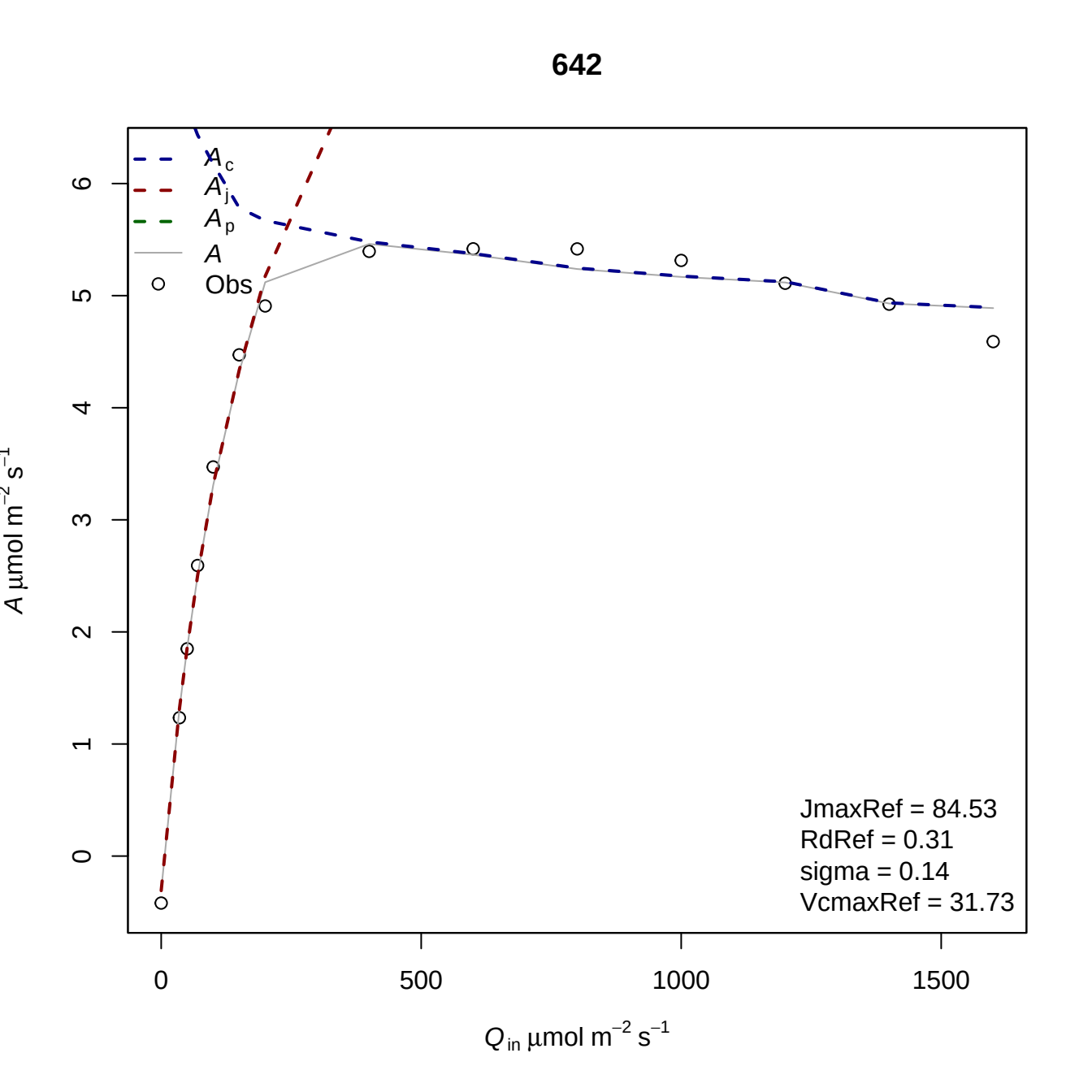

637
A_c
A_j
A_p
A
Obs
JmaxRef = 26.12
RdRef = 0.38
sigma = 0.18
VcmaxRef = 21.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

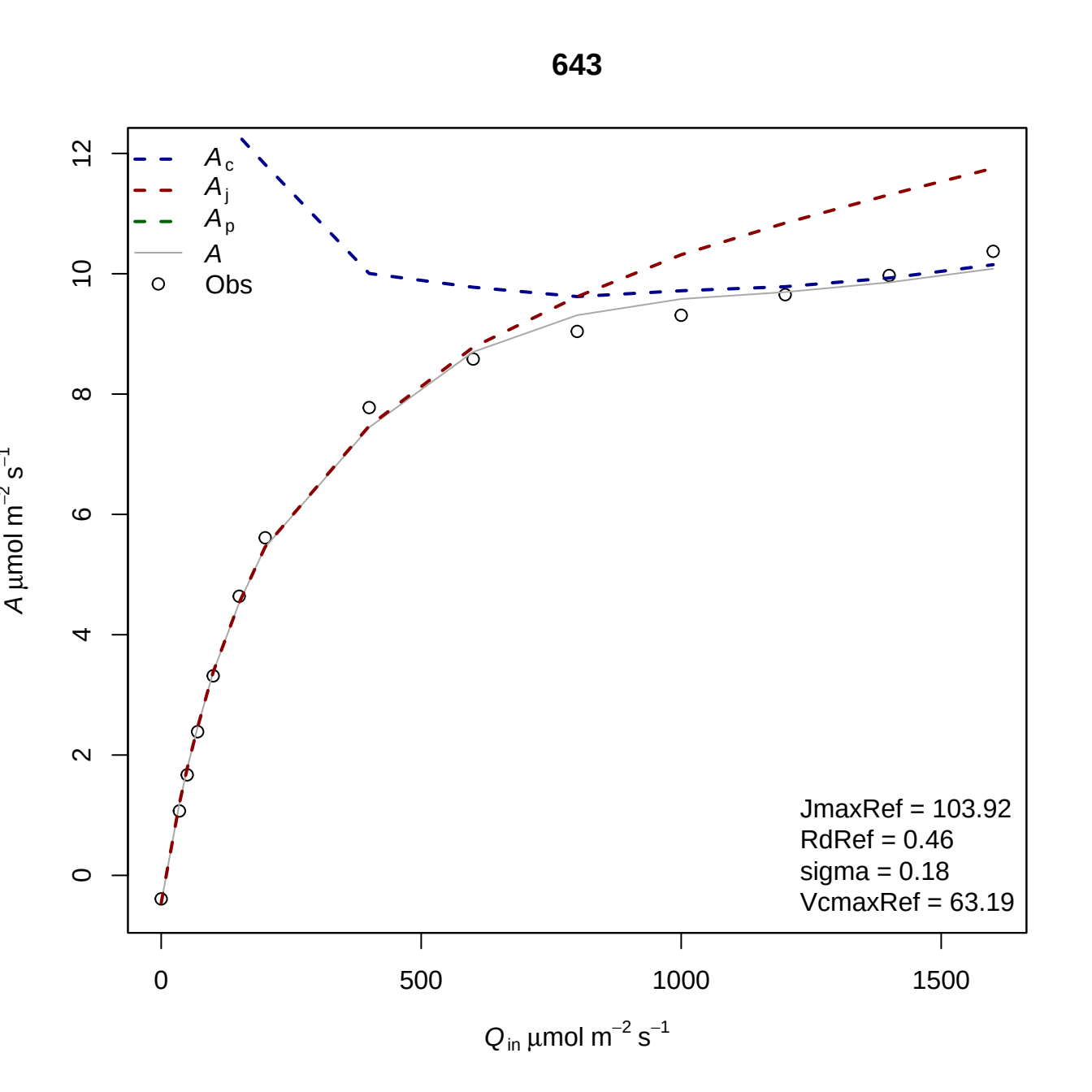

638
A_c
A_j
A_p
A
Obs
JmaxRef = 151.29
RdRef = 0.83
sigma = 0.31
VcmaxRef = 84.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

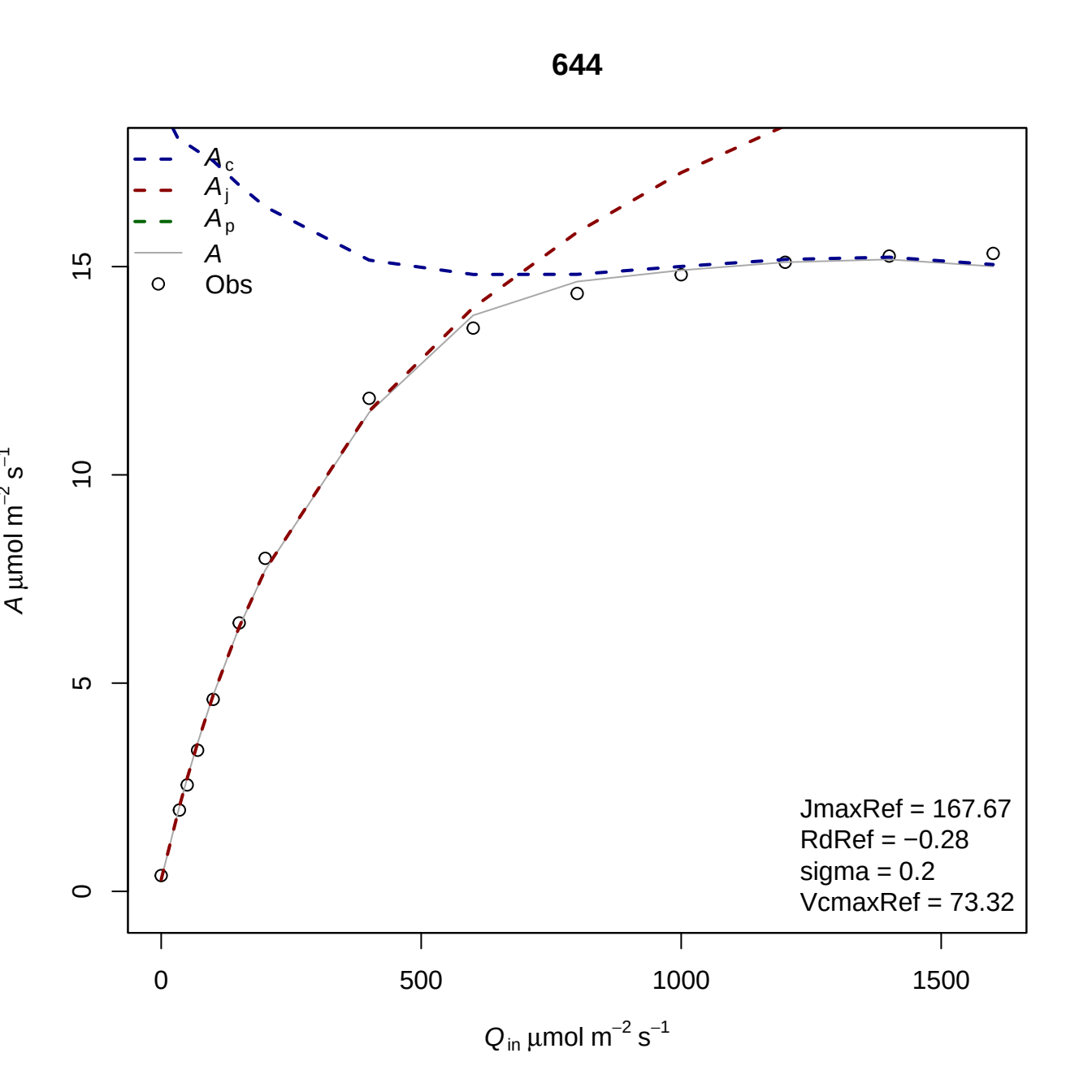

639
A_c
A_j
A_p
A
Obs
JmaxRef = 110.77
RdRef = 0.88
sigma = 0.12
VcmaxRef = 41.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

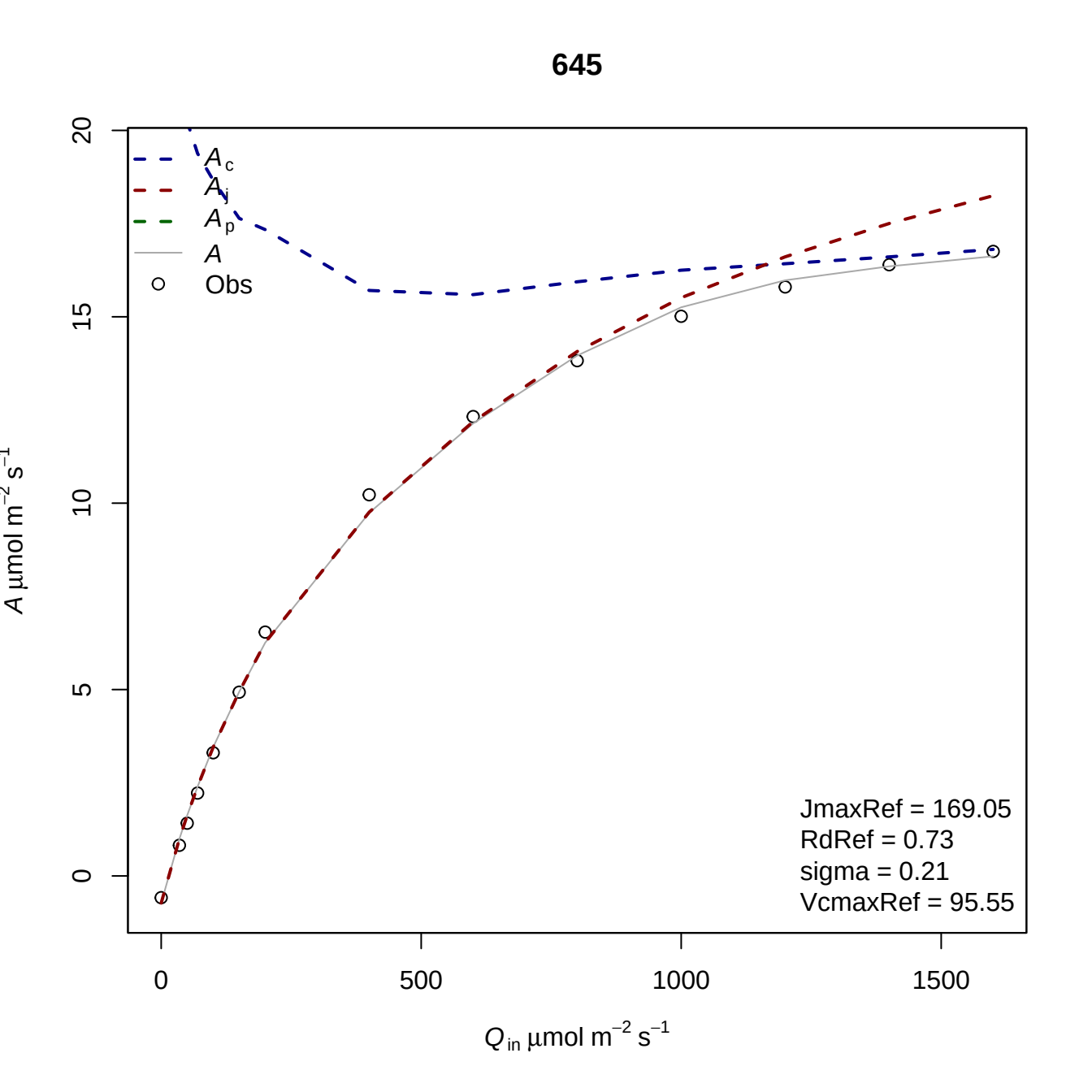

640
A_c
A_j
A_p
A
Obs
JmaxRef = 163.89
RdRef = 0.37
sigma = 0.32
VcmaxRef = 79.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

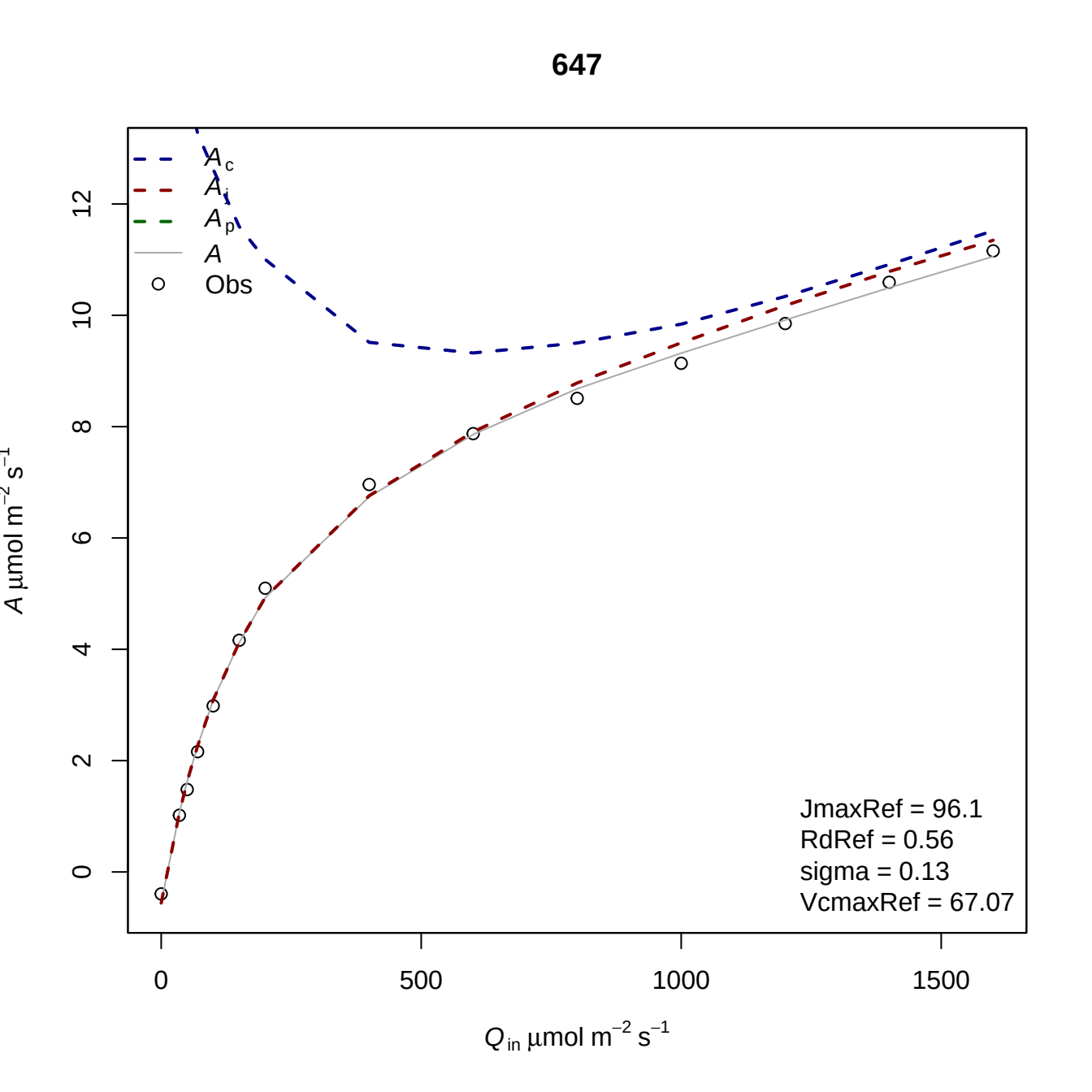

641
A_c
A_j
A_p
A
Obs
JmaxRef = 66.02
RdRef = −0.95
sigma = 0.11
VcmaxRef = 20.99
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

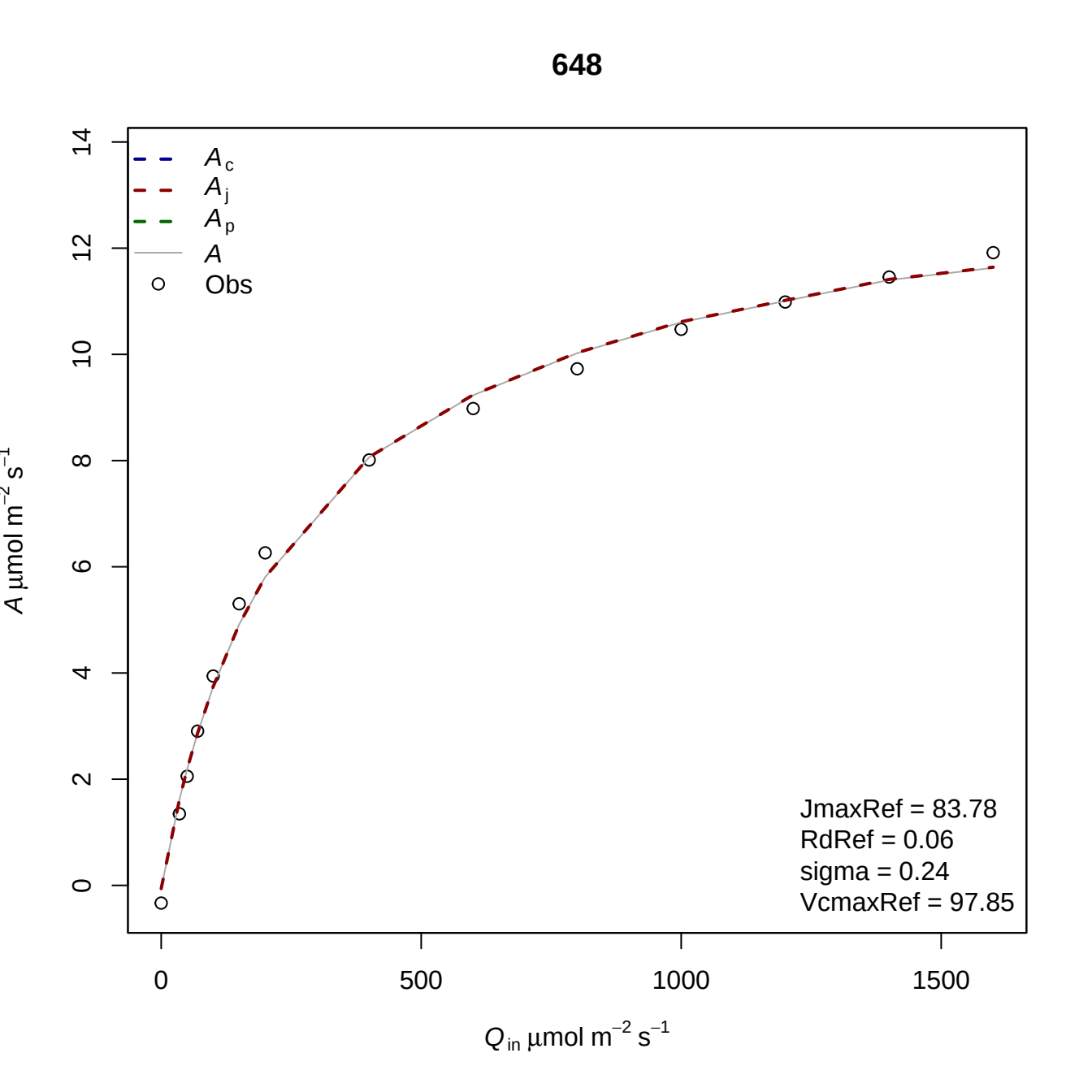

642
A_c
A_j
A_p
A
Obs
JmaxRef = 84.53
RdRef = 0.31
sigma = 0.14
VcmaxRef = 31.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

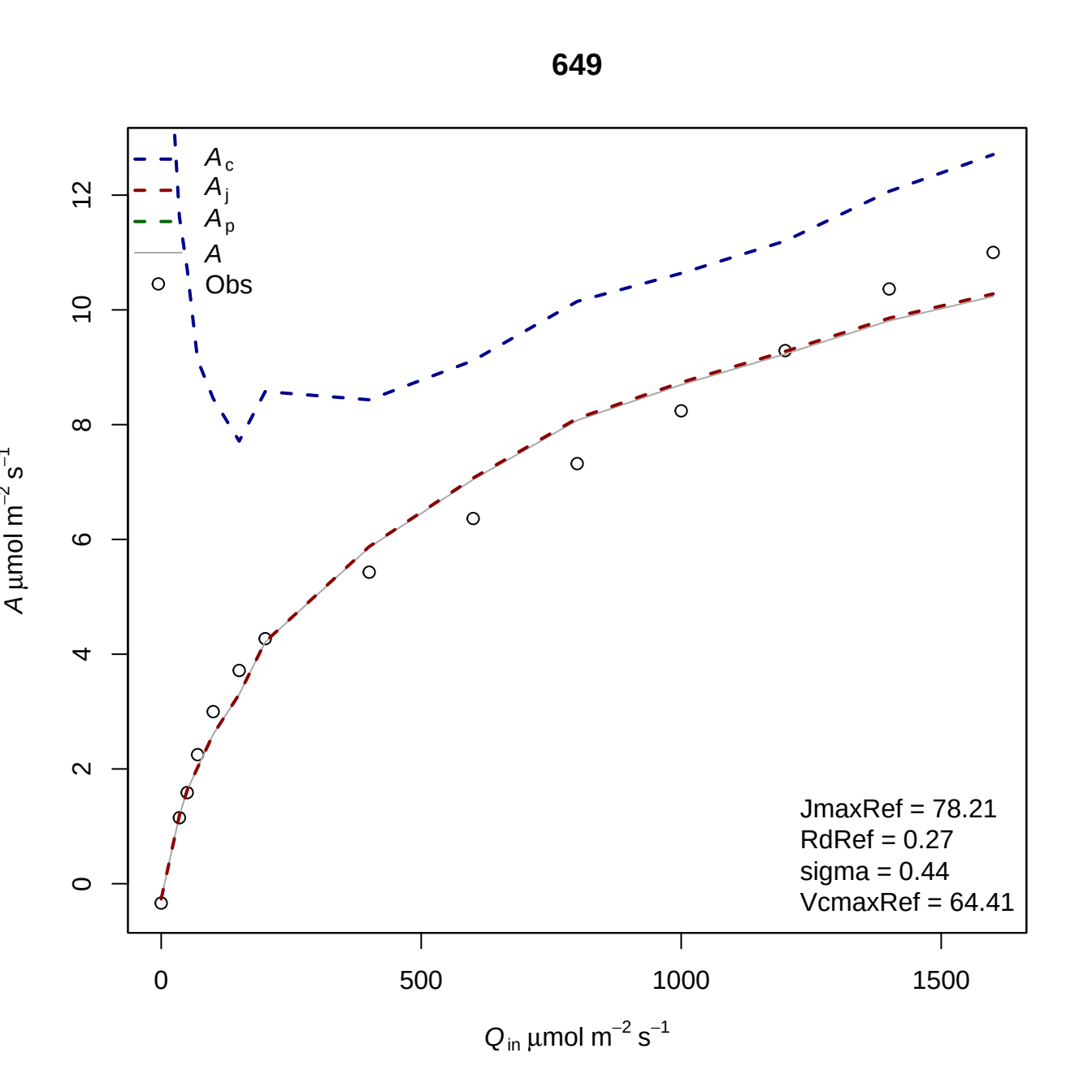

643
A_c
A_j
A_p
A
Obs
JmaxRef = 103.92
RdRef = 0.46
sigma = 0.18
VcmaxRef = 63.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

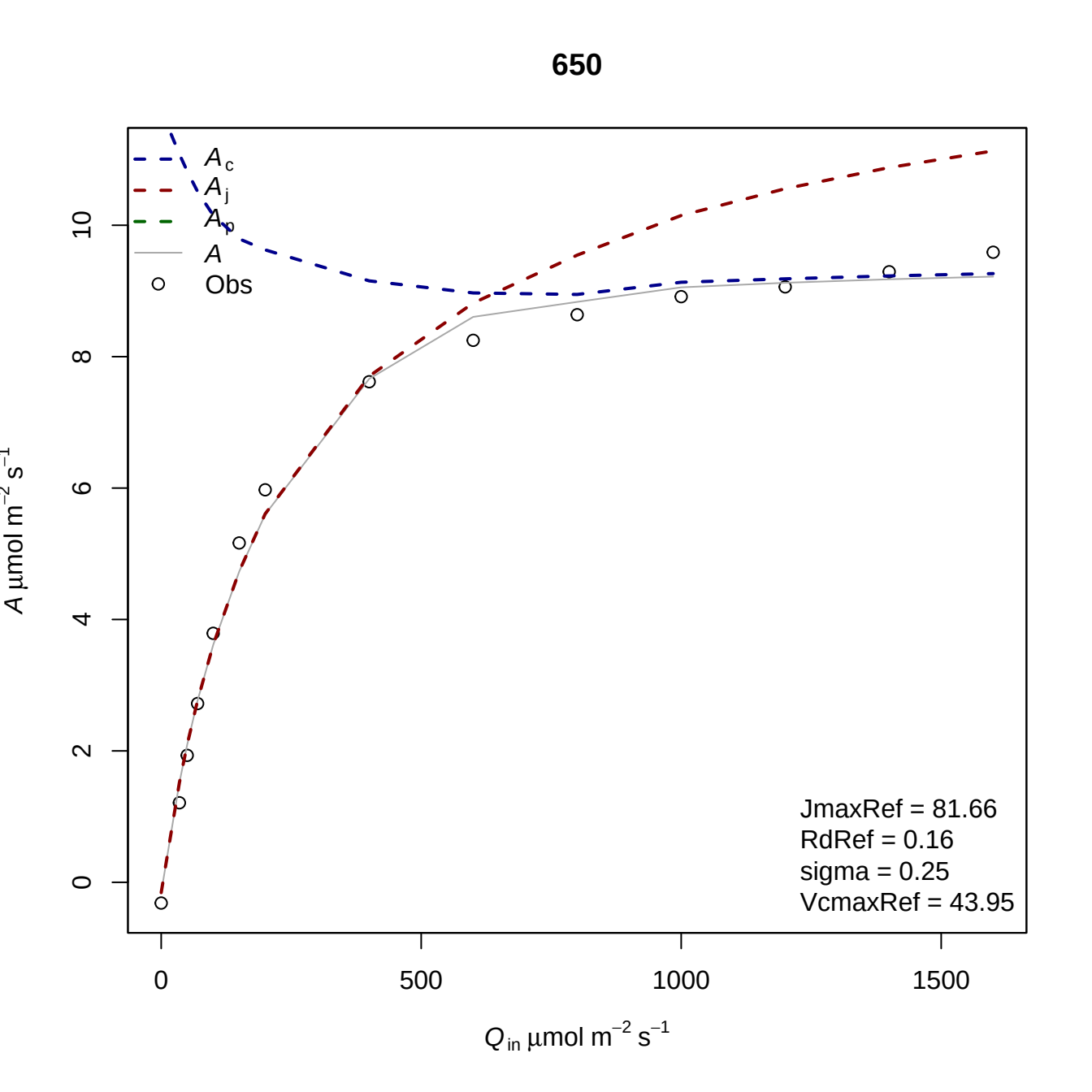

644
A_c
A_j
A_p
A
Obs
JmaxRef = 167.67
RdRef = −0.28
sigma = 0.2
VcmaxRef = 73.32
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

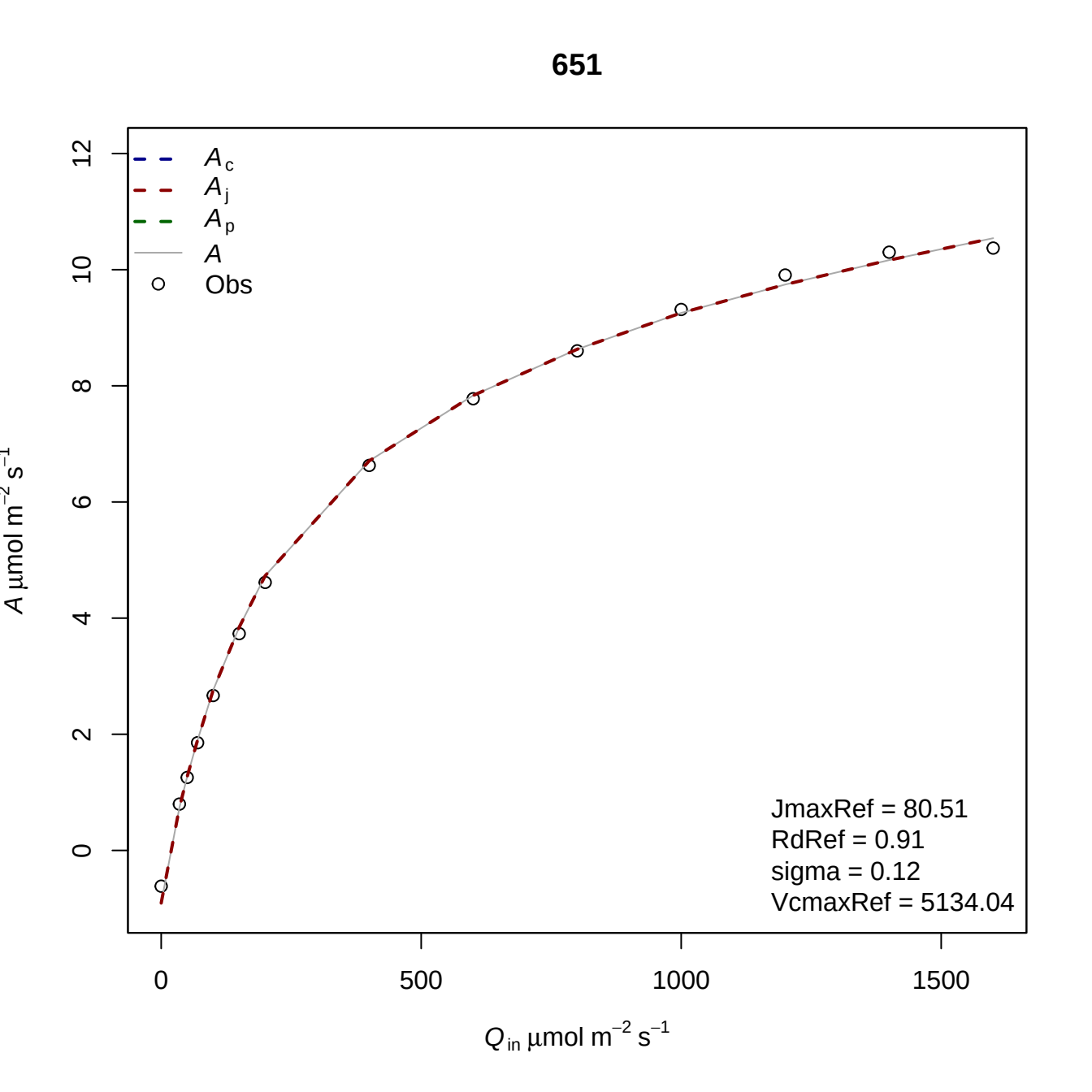

645
A_c
A_j
A_p
A
Obs
JmaxRef = 169.05
RdRef = 0.73
sigma = 0.21
VcmaxRef = 95.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

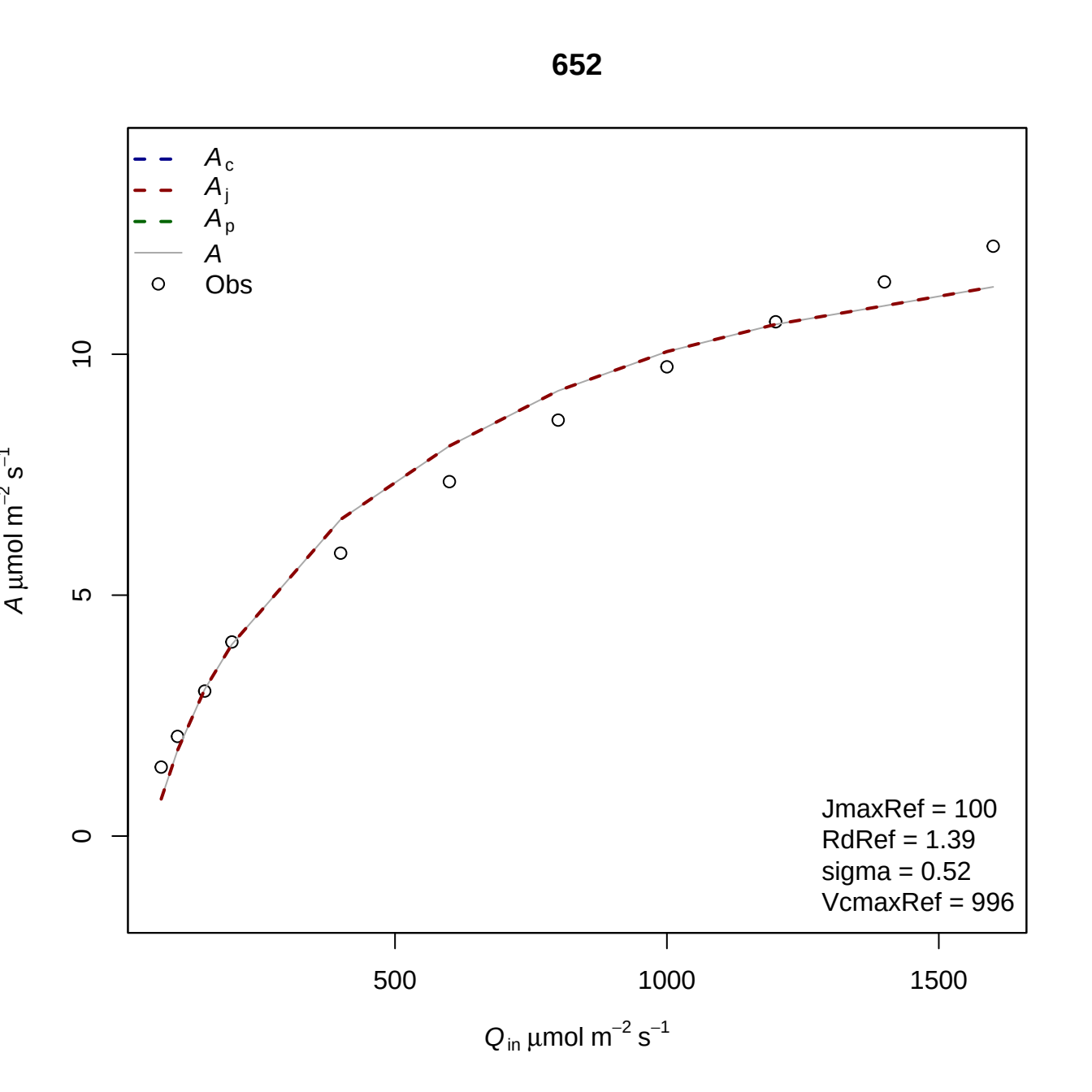

647
A_c
A_j
A_p
A
Obs
JmaxRef = 96.1
RdRef = 0.56
sigma = 0.13
VcmaxRef = 67.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

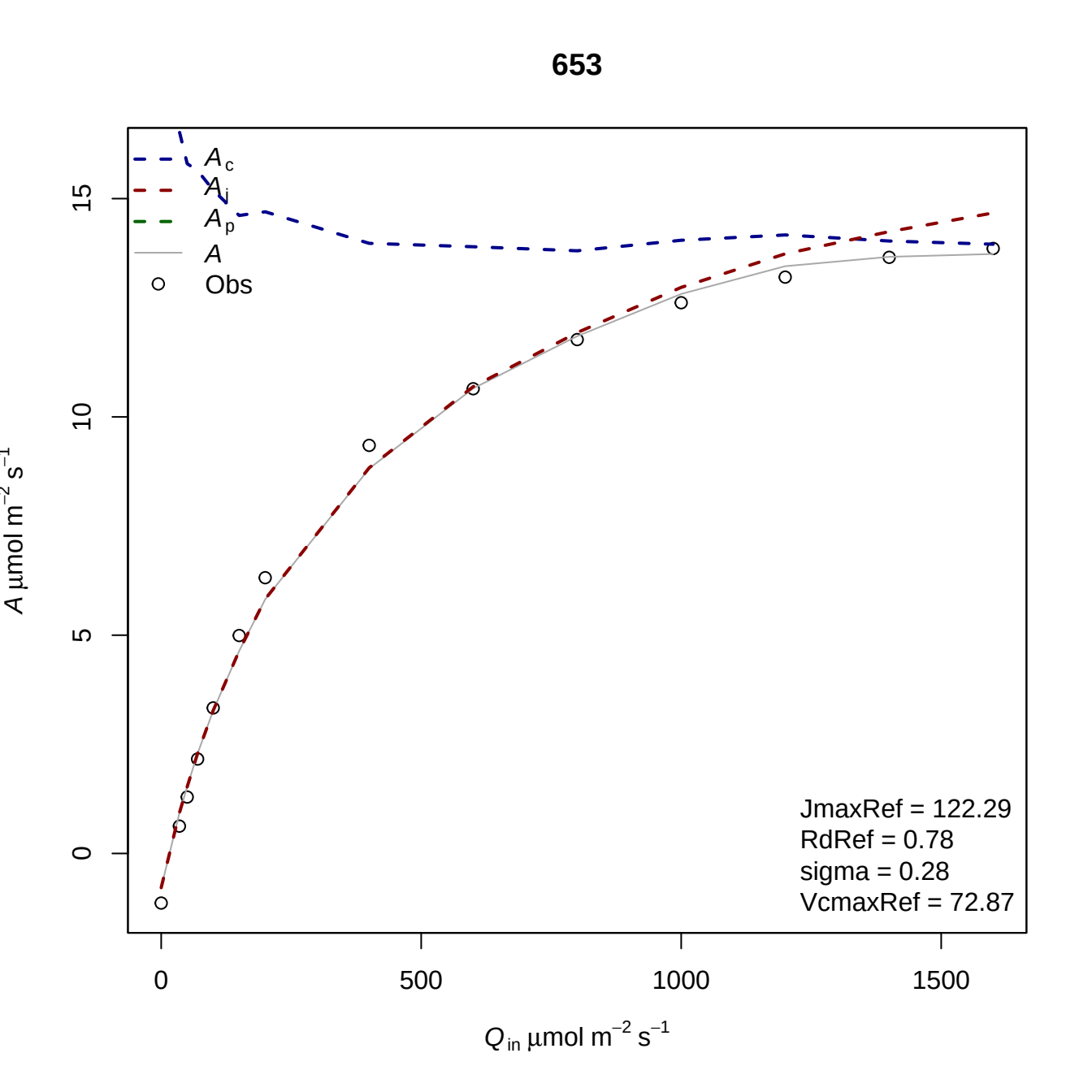

648
A_c
A_j
A_p
A
Obs
JmaxRef = 83.78
RdRef = 0.06
sigma = 0.24
VcmaxRef = 97.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

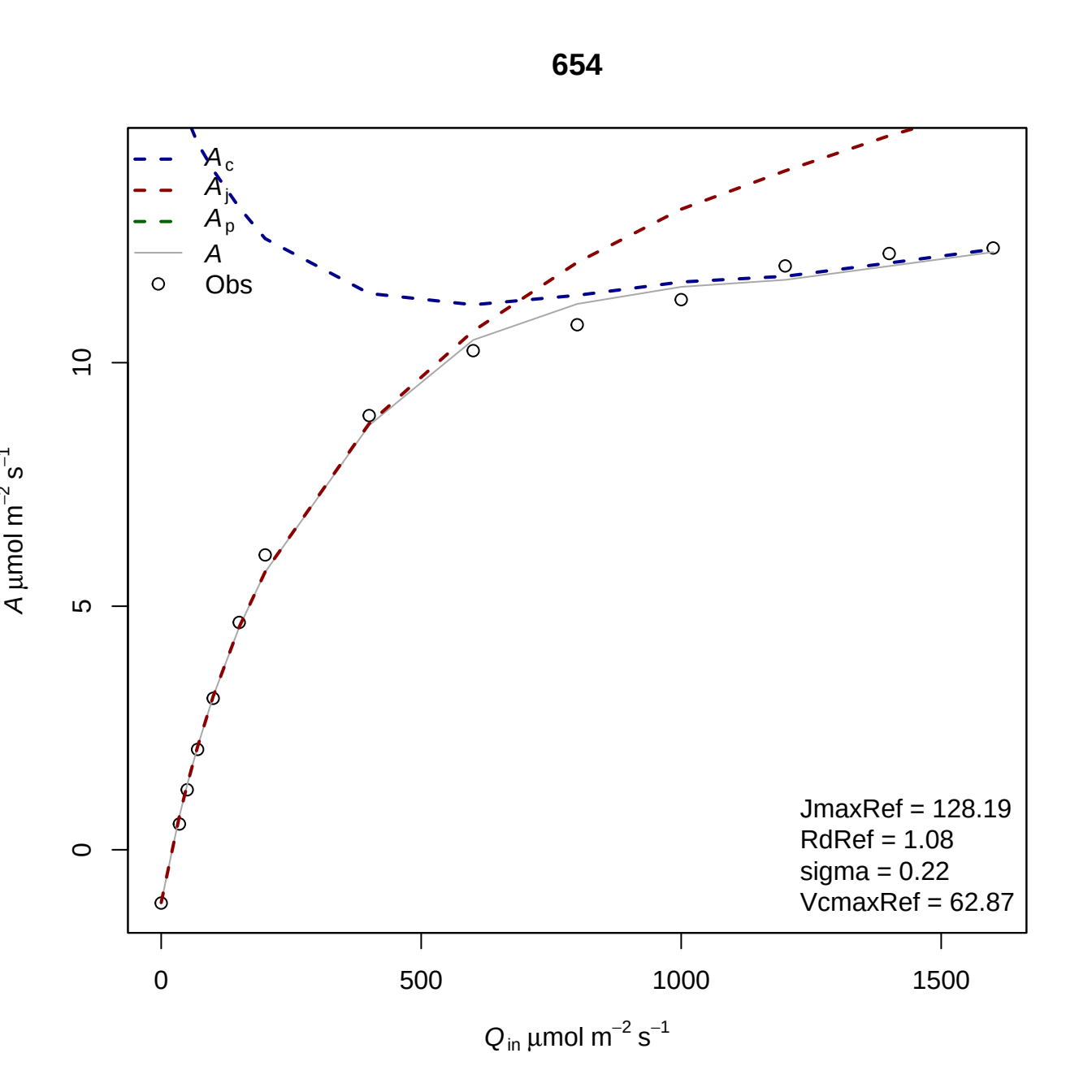

649
A_c
A_j
A_p
A
Obs
JmaxRef = 78.21
RdRef = 0.27
sigma = 0.44
VcmaxRef = 64.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

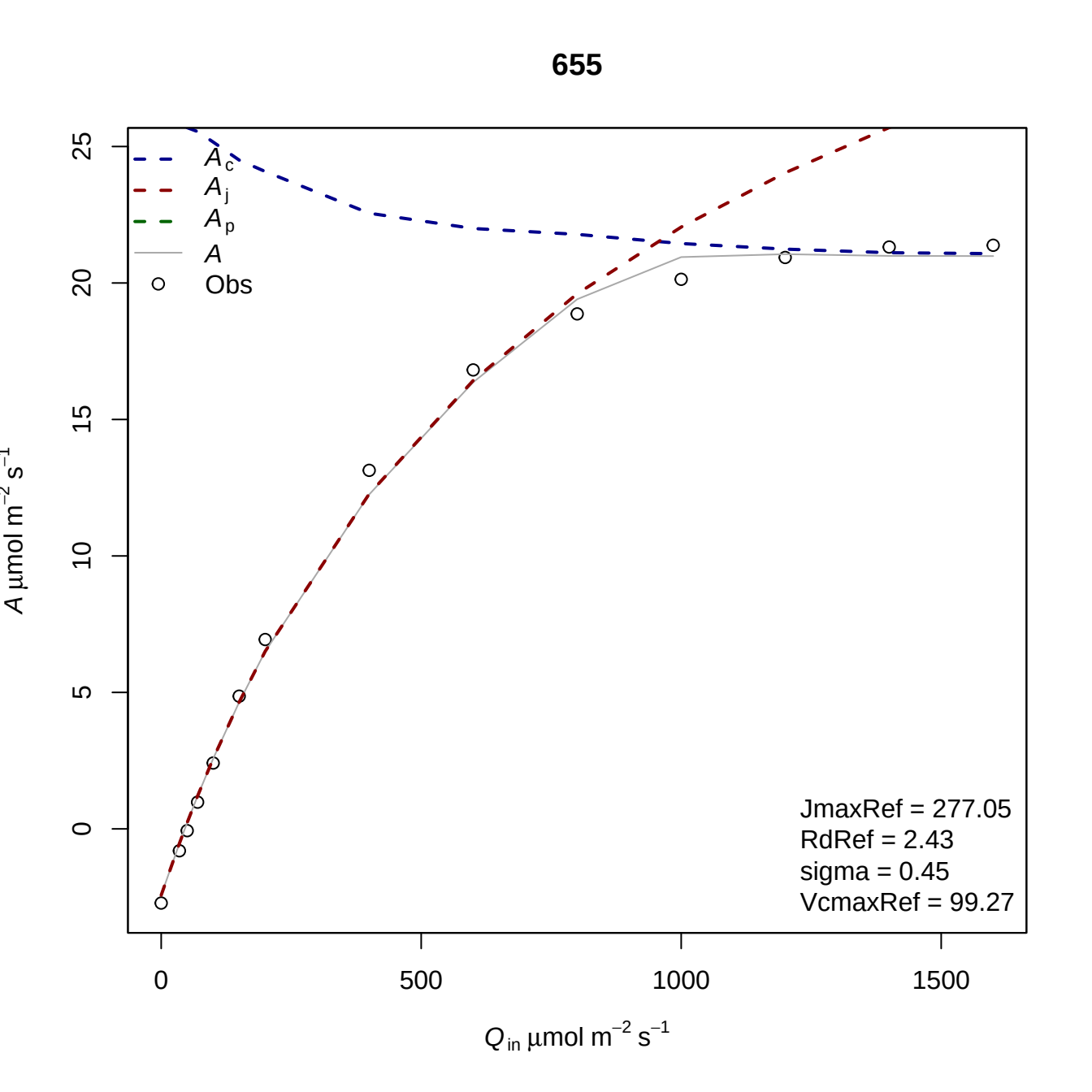

650
A_c
A_j
A_p
A
Obs
JmaxRef = 81.66
RdRef = 0.16
sigma = 0.25
VcmaxRef = 43.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

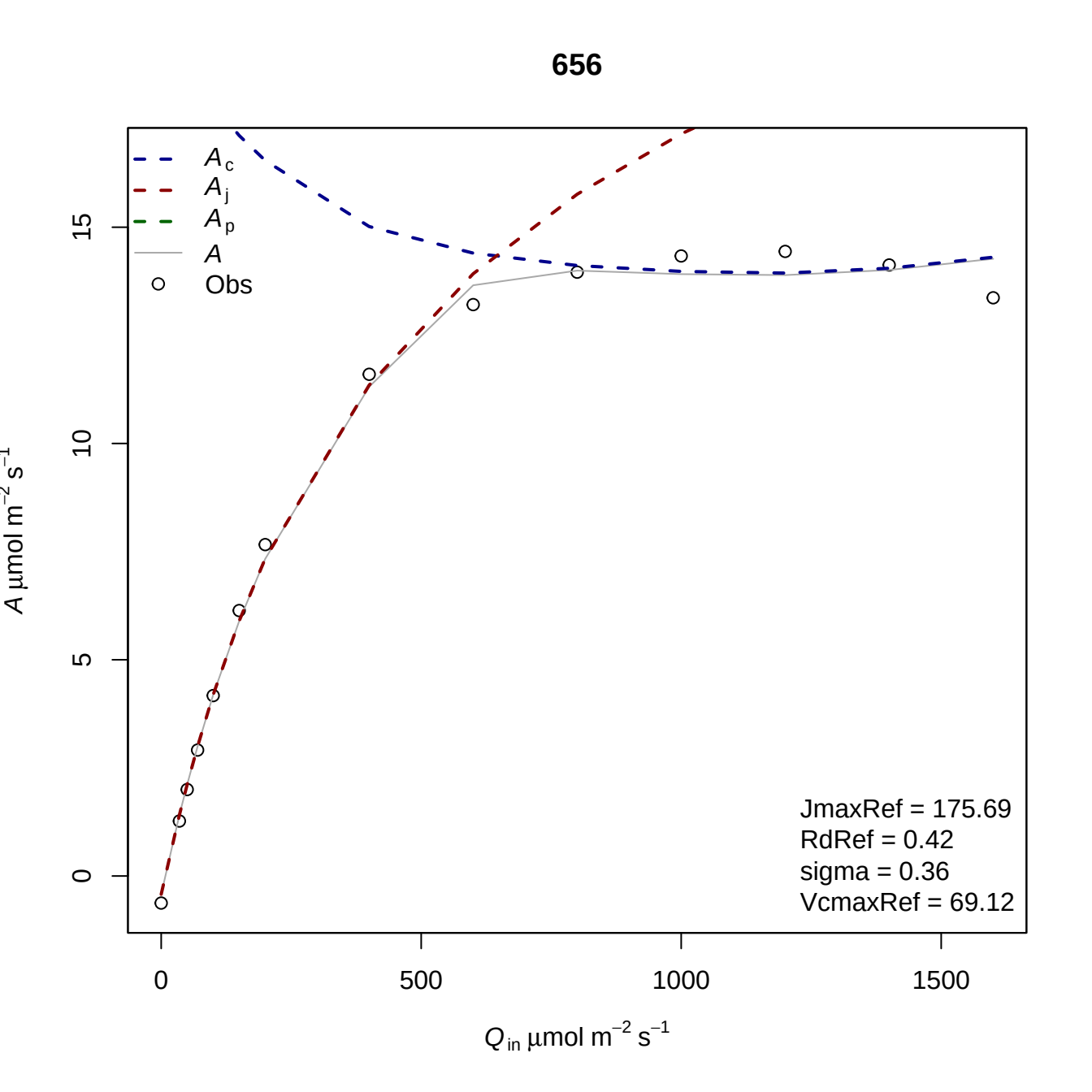

651
A_c
A_j
A_p
A
Obs
JmaxRef = 80.51
RdRef = 0.91
sigma = 0.12
VcmaxRef = 5134.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

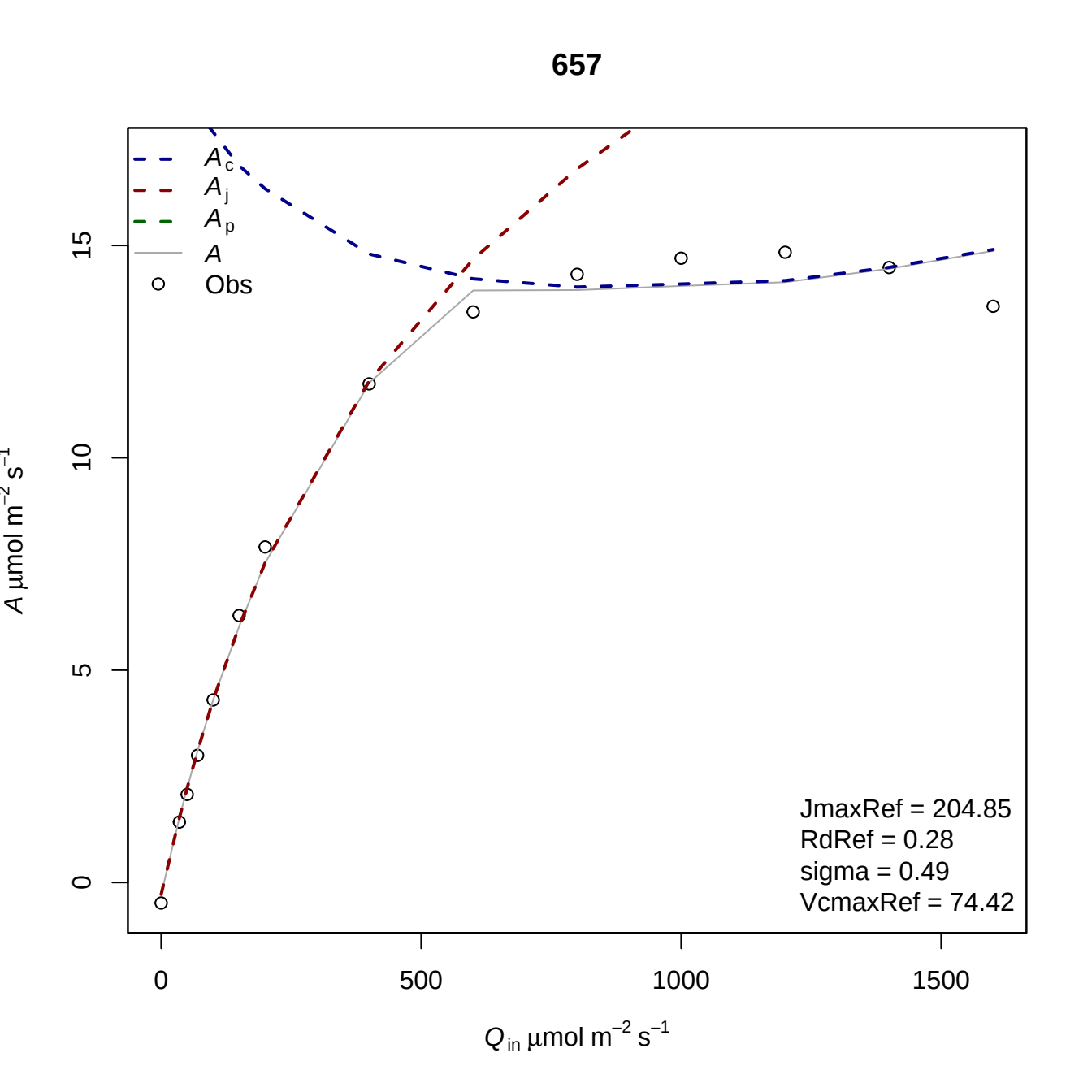

652
A_c
A_j
A_p
A
Obs
JmaxRef = 100
RdRef = 1.39
sigma = 0.52
VcmaxRef = 996
0
5
10
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

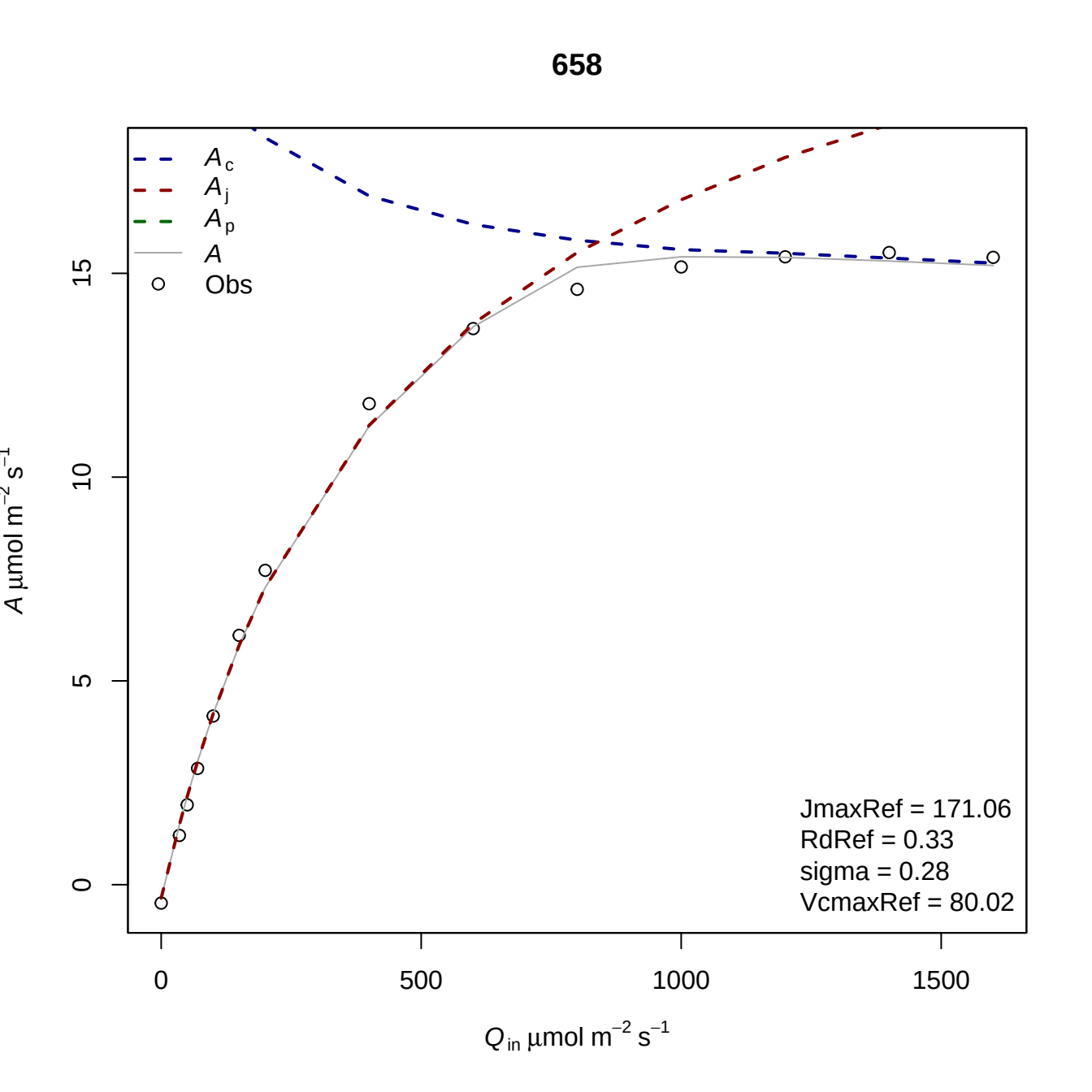

653
A_c
A_j
A_p
A
Obs
JmaxRef = 122.29
RdRef = 0.78
sigma = 0.28
VcmaxRef = 72.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

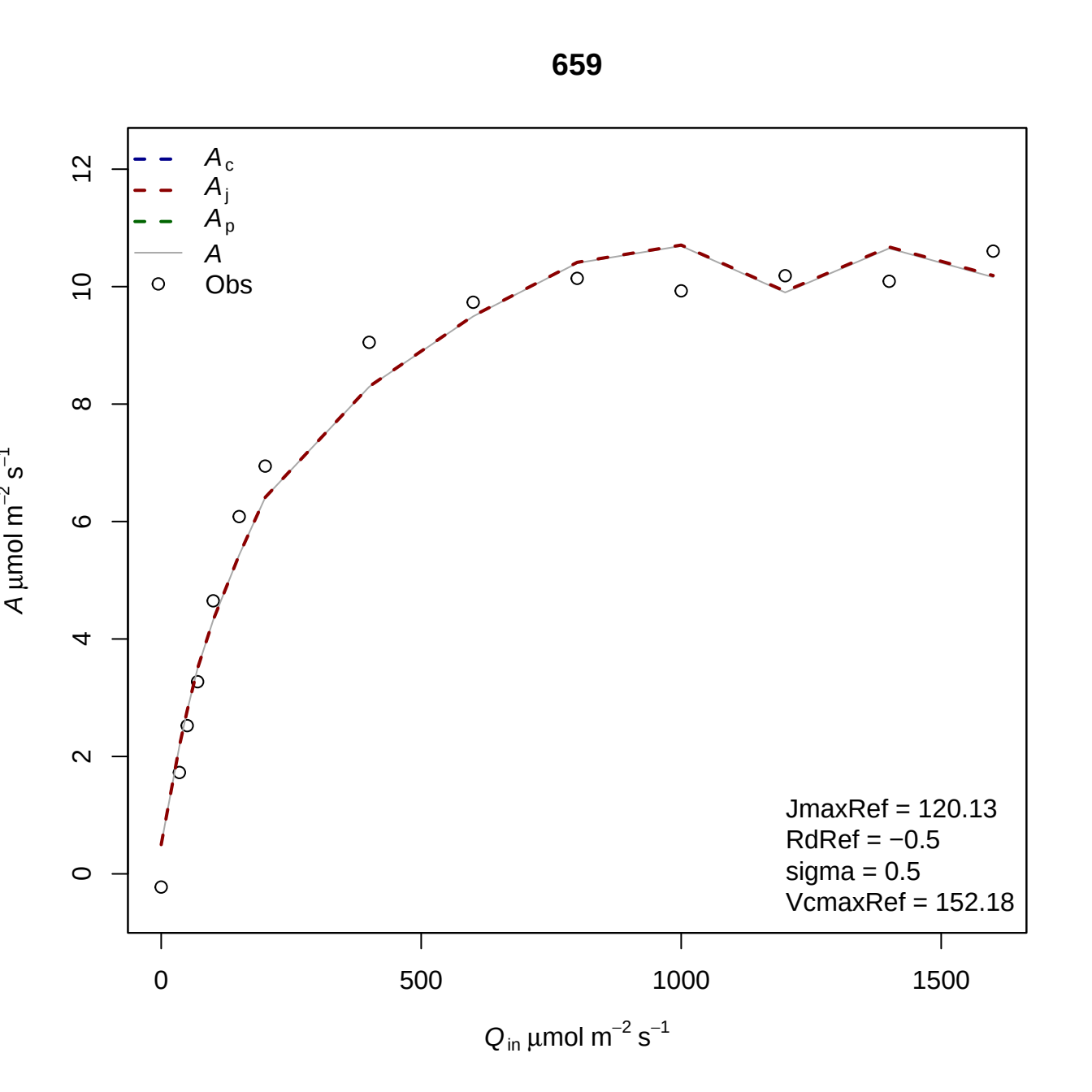

654
A_c
A_j
A_p
A
Obs
JmaxRef = 128.19
RdRef = 1.08
sigma = 0.22
VcmaxRef = 62.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

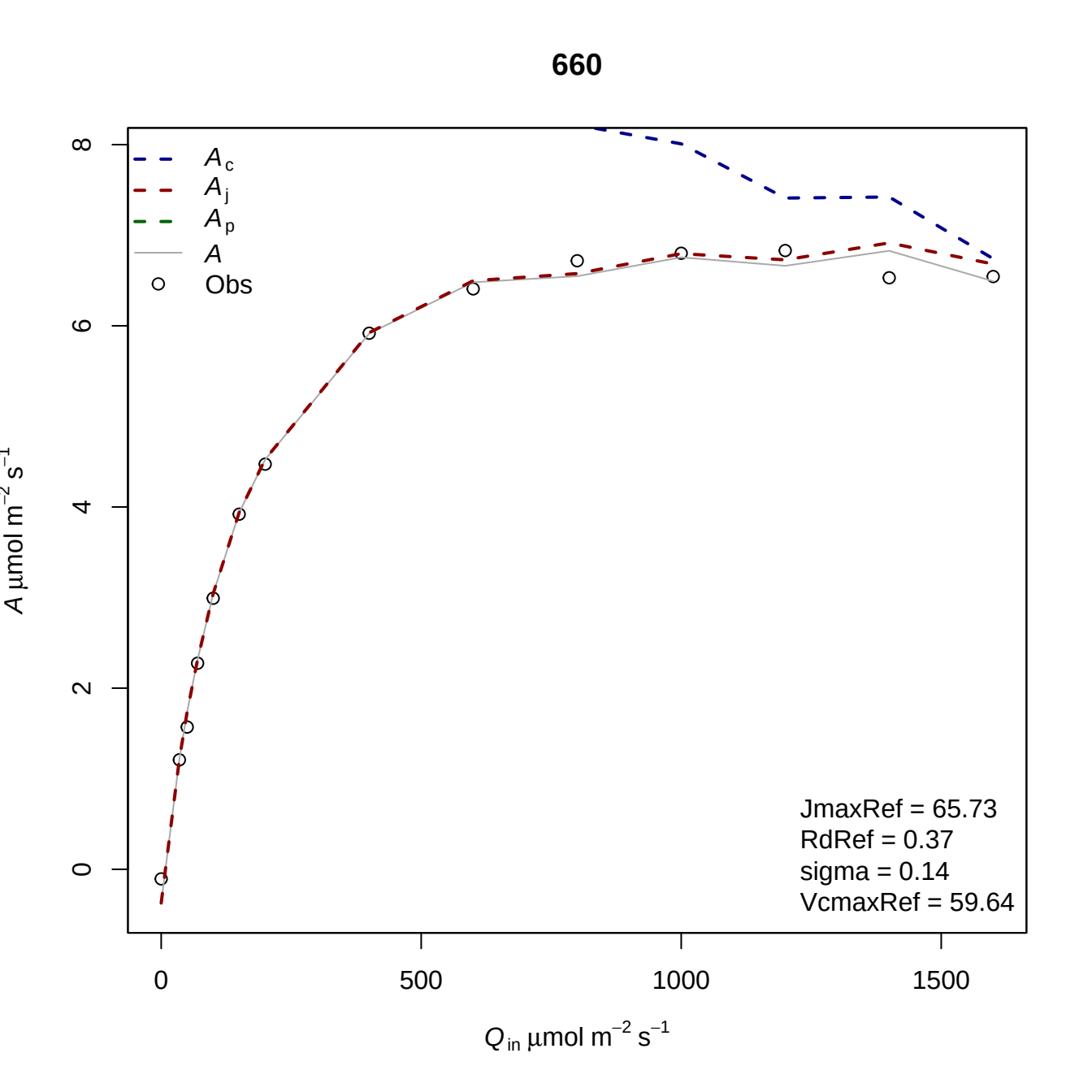

655
A_c
A_j
A_p
A
Obs
JmaxRef = 277.05
RdRef = 2.43
sigma = 0.45
VcmaxRef = 99.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

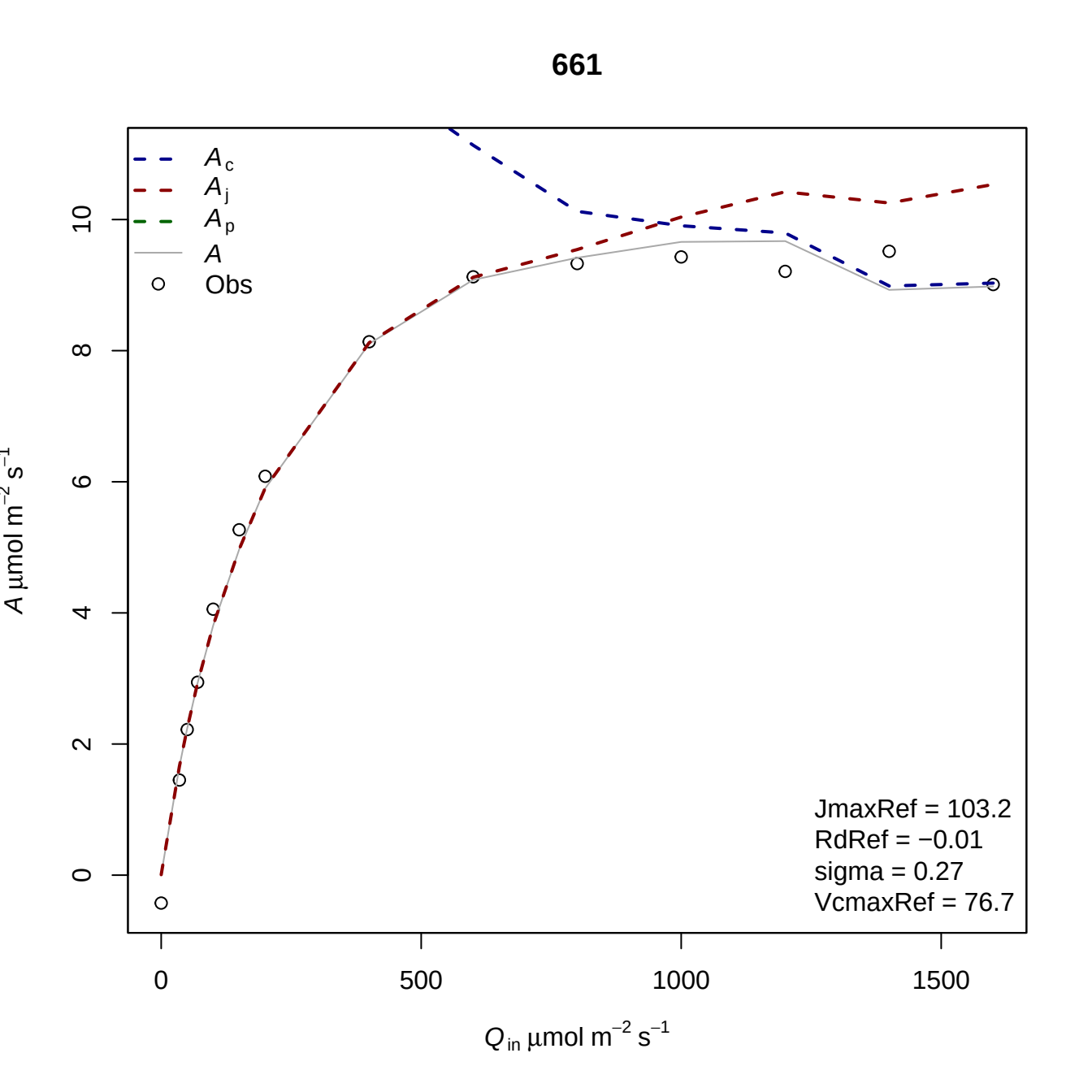

656
A_c
A_j
A_p
A
Obs
JmaxRef = 175.69
RdRef = 0.42
sigma = 0.36
VcmaxRef = 69.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

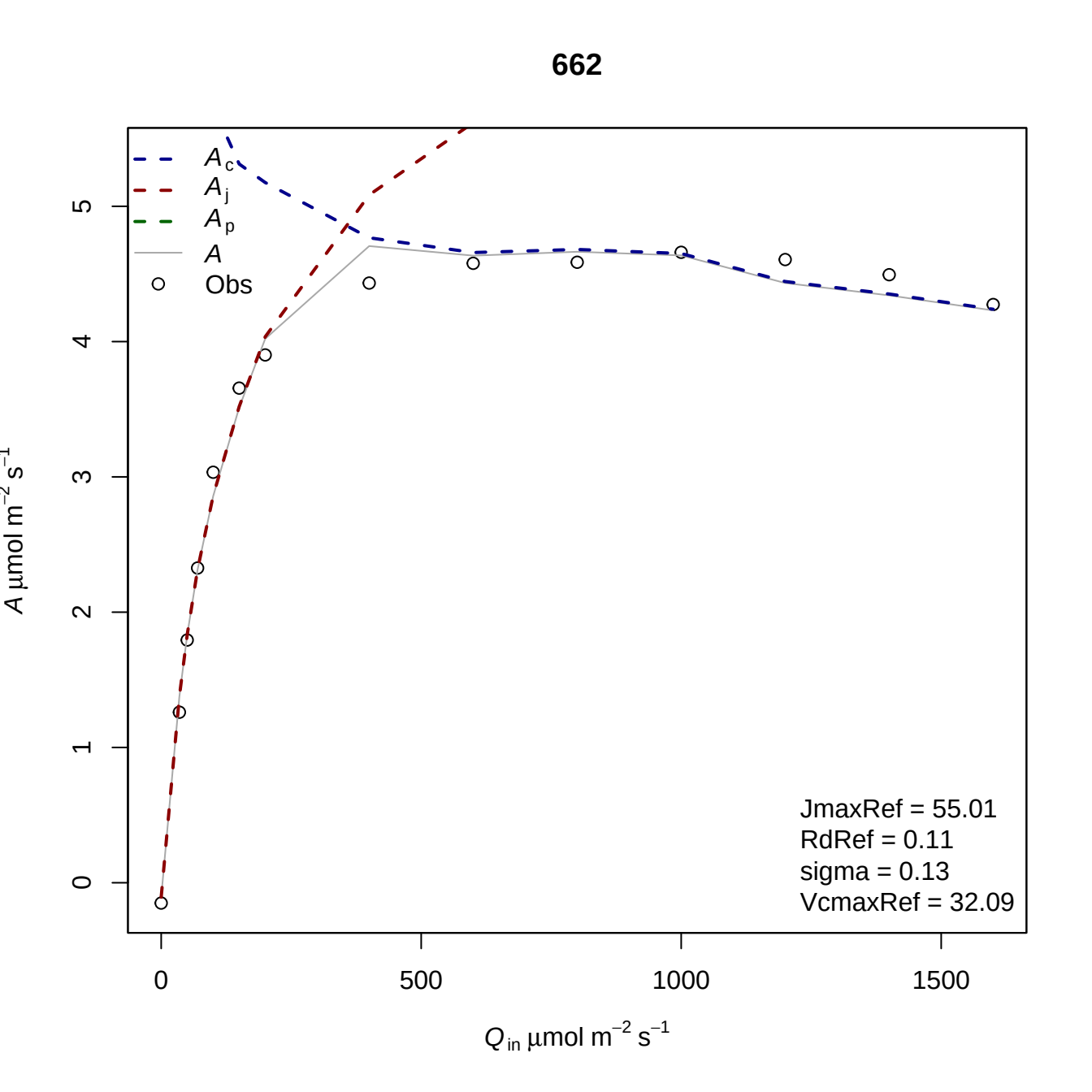

657
A_c
A_j
A_p
A
Obs
JmaxRef = 204.85
RdRef = 0.28
sigma = 0.49
VcmaxRef = 74.42
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

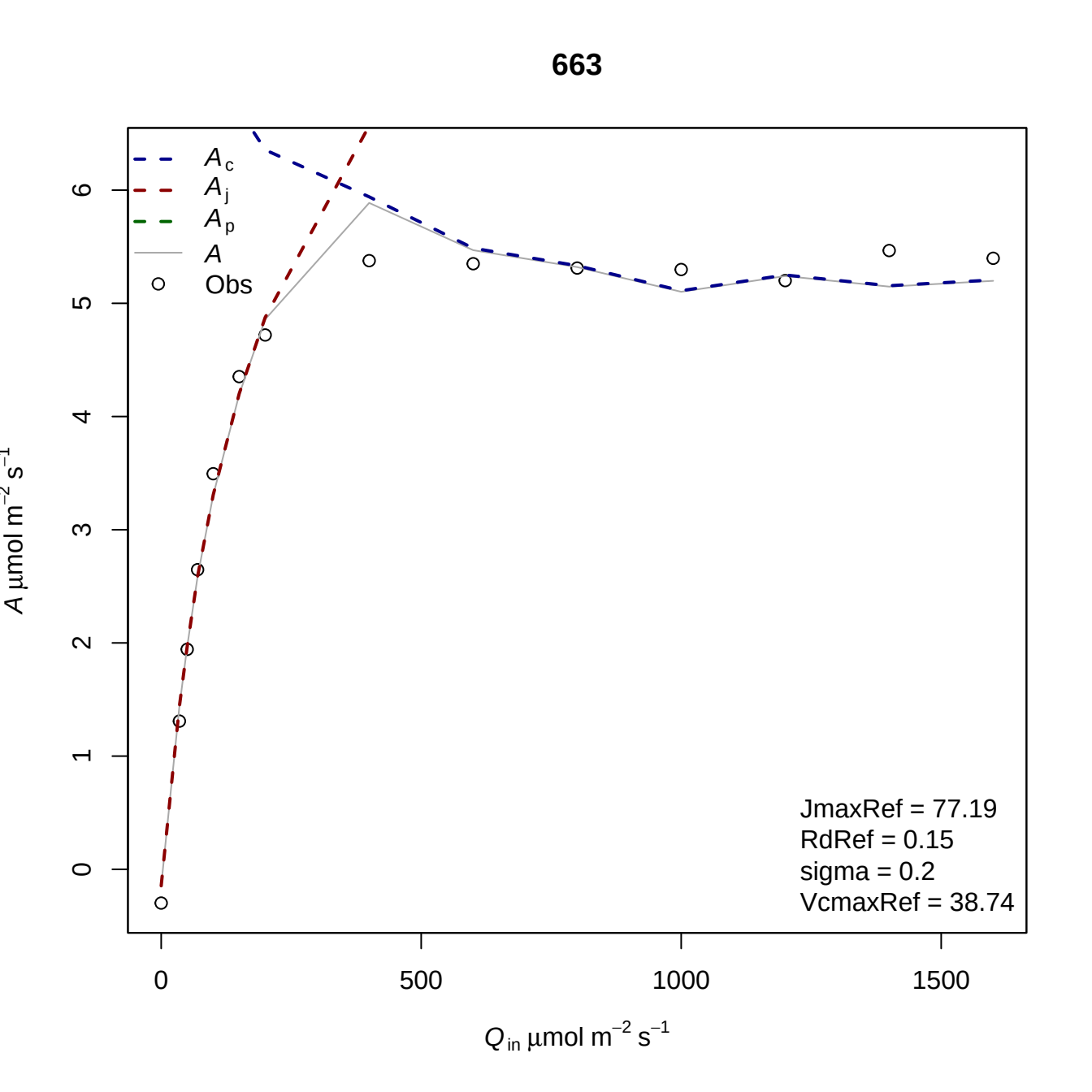

658
A_c
A_j
A_p
A
Obs
JmaxRef = 171.06
RdRef = 0.33
sigma = 0.28
VcmaxRef = 80.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

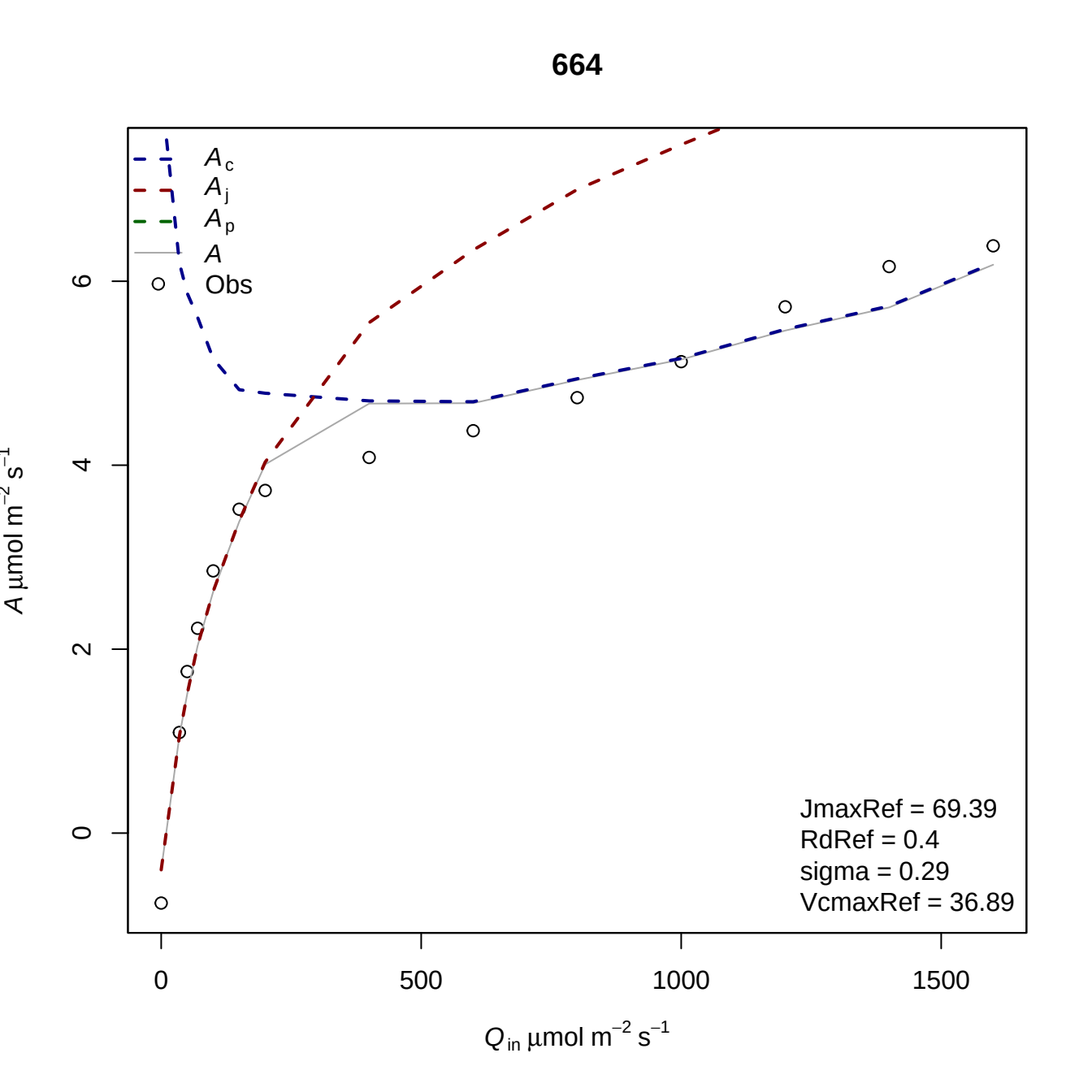

659
A_c
A_j
A_p
A
Obs
JmaxRef = 120.13
RdRef = −0.5
sigma = 0.5
VcmaxRef = 152.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

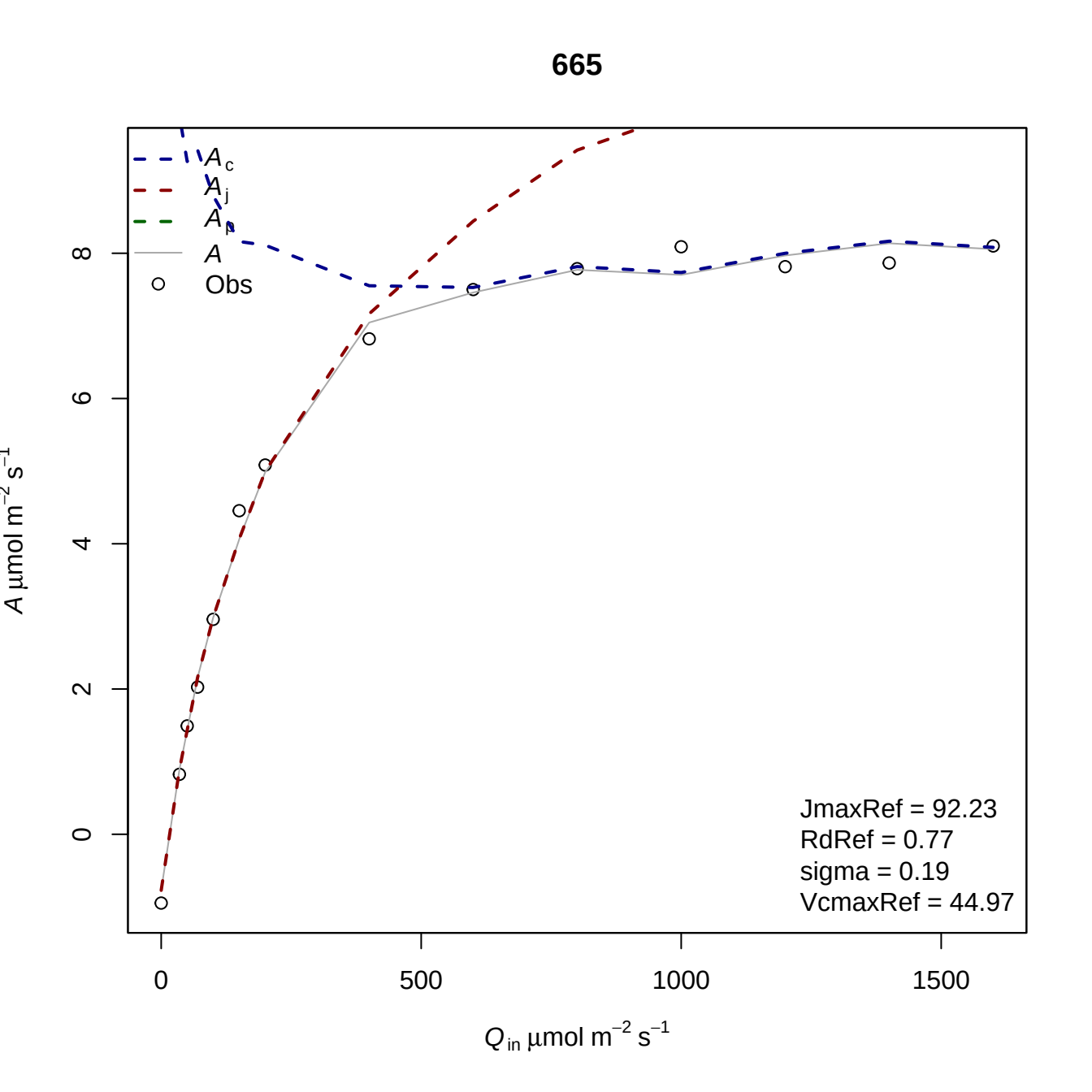

660
A_c
A_j
A_p
A
Obs
JmaxRef = 65.73
RdRef = 0.37
sigma = 0.14
VcmaxRef = 59.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

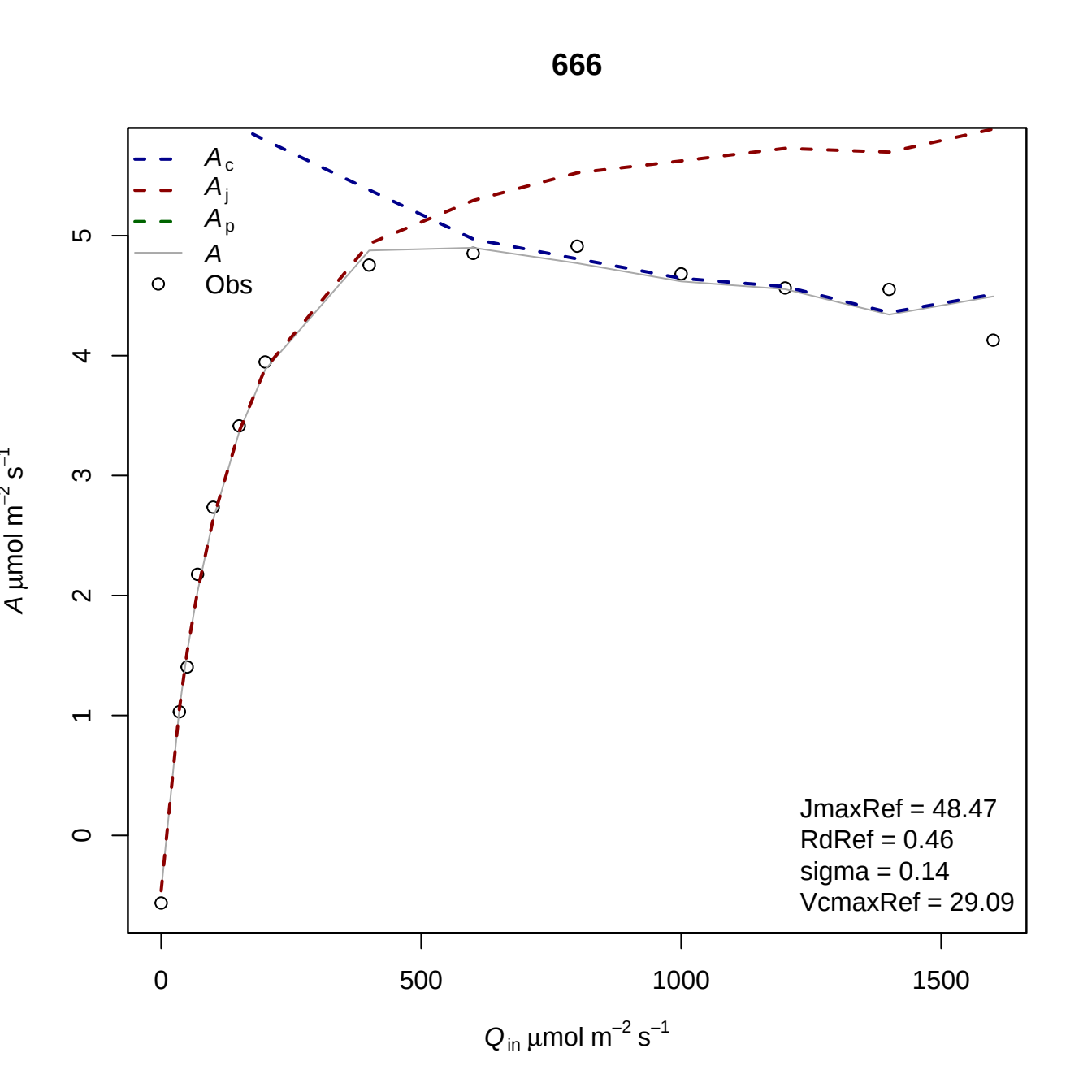

661
A_c
A_j
A_p
A
Obs
JmaxRef = 103.2
RdRef = −0.01
sigma = 0.27
VcmaxRef = 76.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

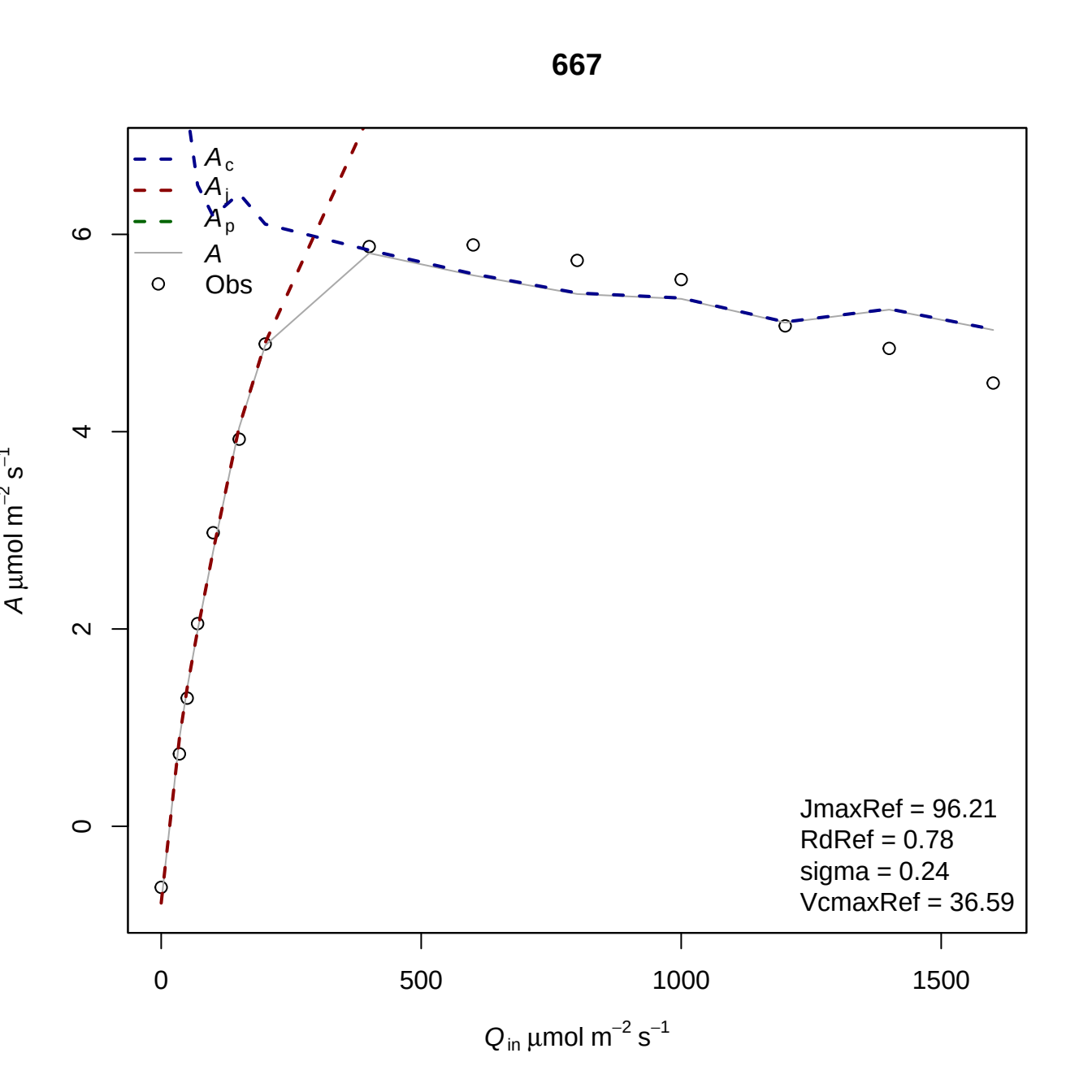

662
A_c
A_j
A_p
A
Obs
JmaxRef = 55.01
RdRef = 0.11
sigma = 0.13
VcmaxRef = 32.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

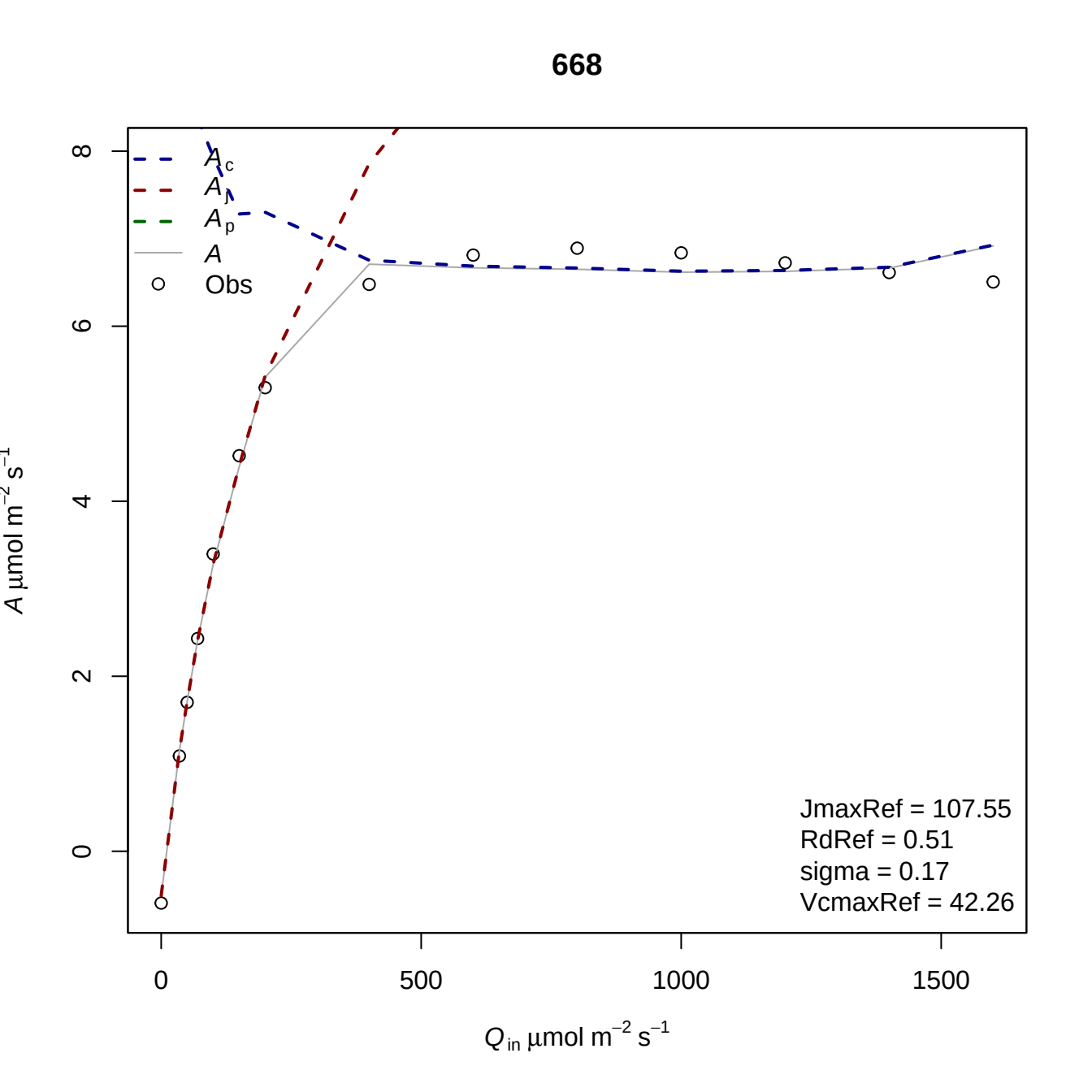

663
A_c
A_j
A_p
A
Obs
JmaxRef = 77.19
RdRef = 0.15
sigma = 0.2
VcmaxRef = 38.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

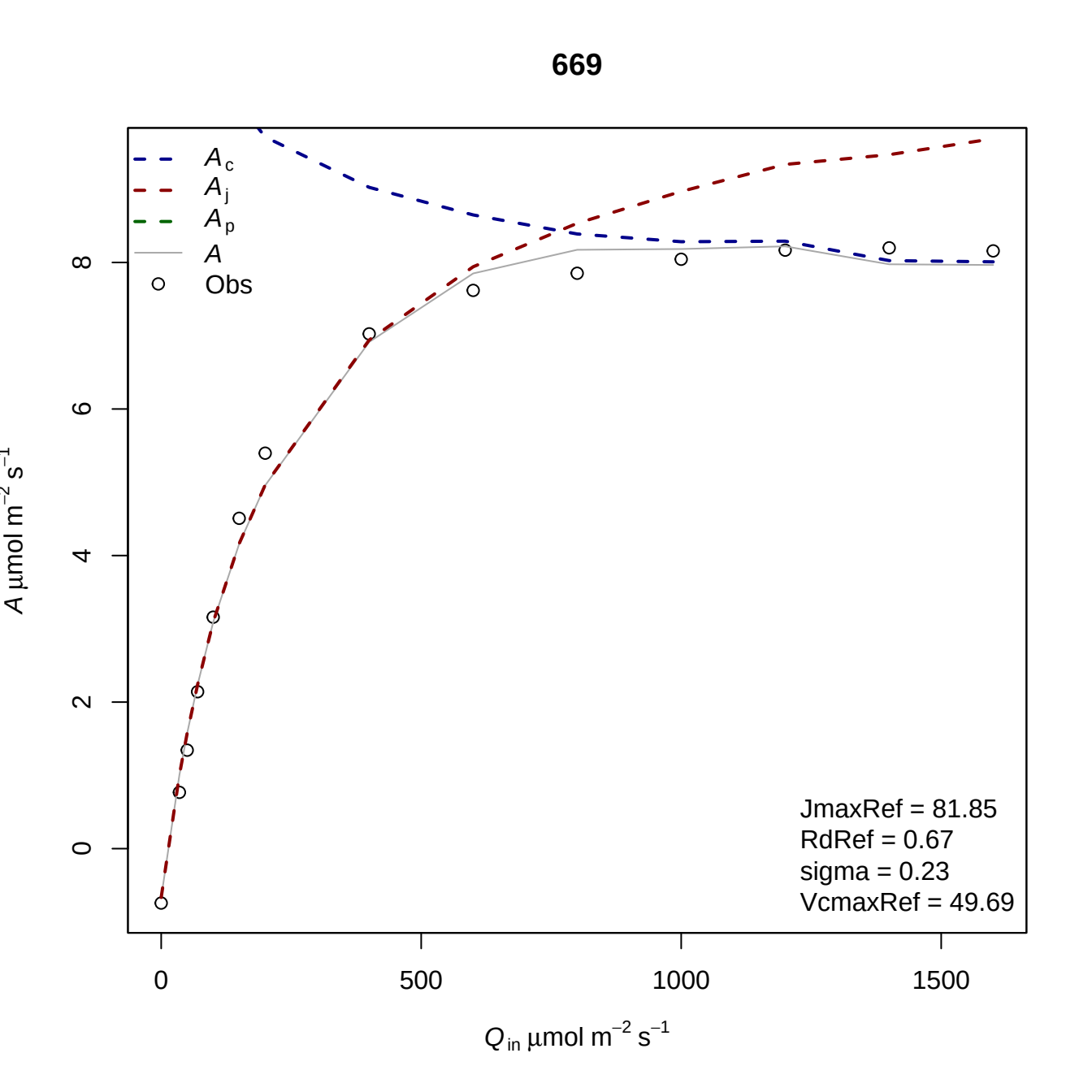

664
A_c
A_j
A_p
A
Obs
JmaxRef = 69.39
RdRef = 0.4
sigma = 0.29
VcmaxRef = 36.89
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

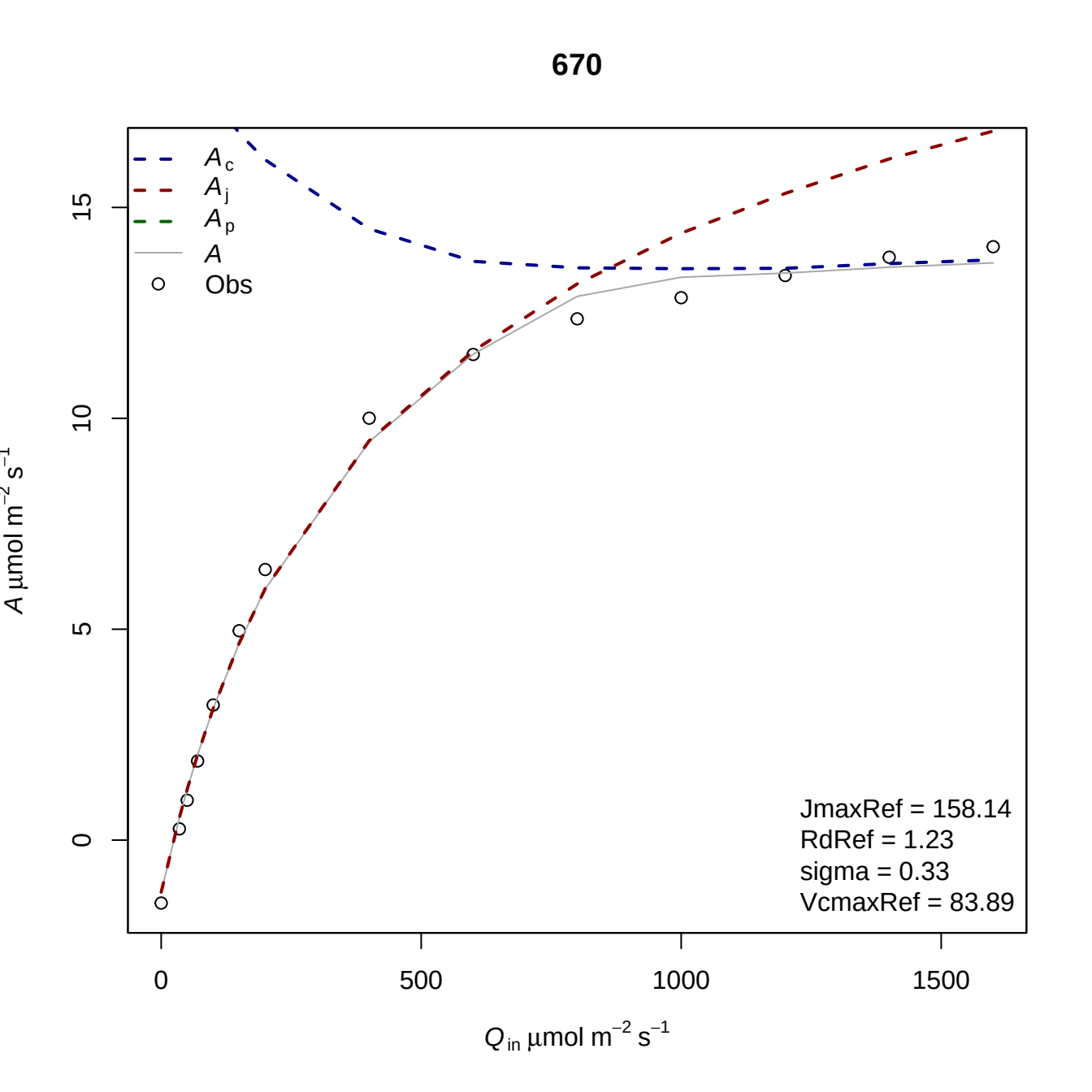

665
A_c
A_j
A_p
A
Obs
JmaxRef = 92.23
RdRef = 0.77
sigma = 0.19
VcmaxRef = 44.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

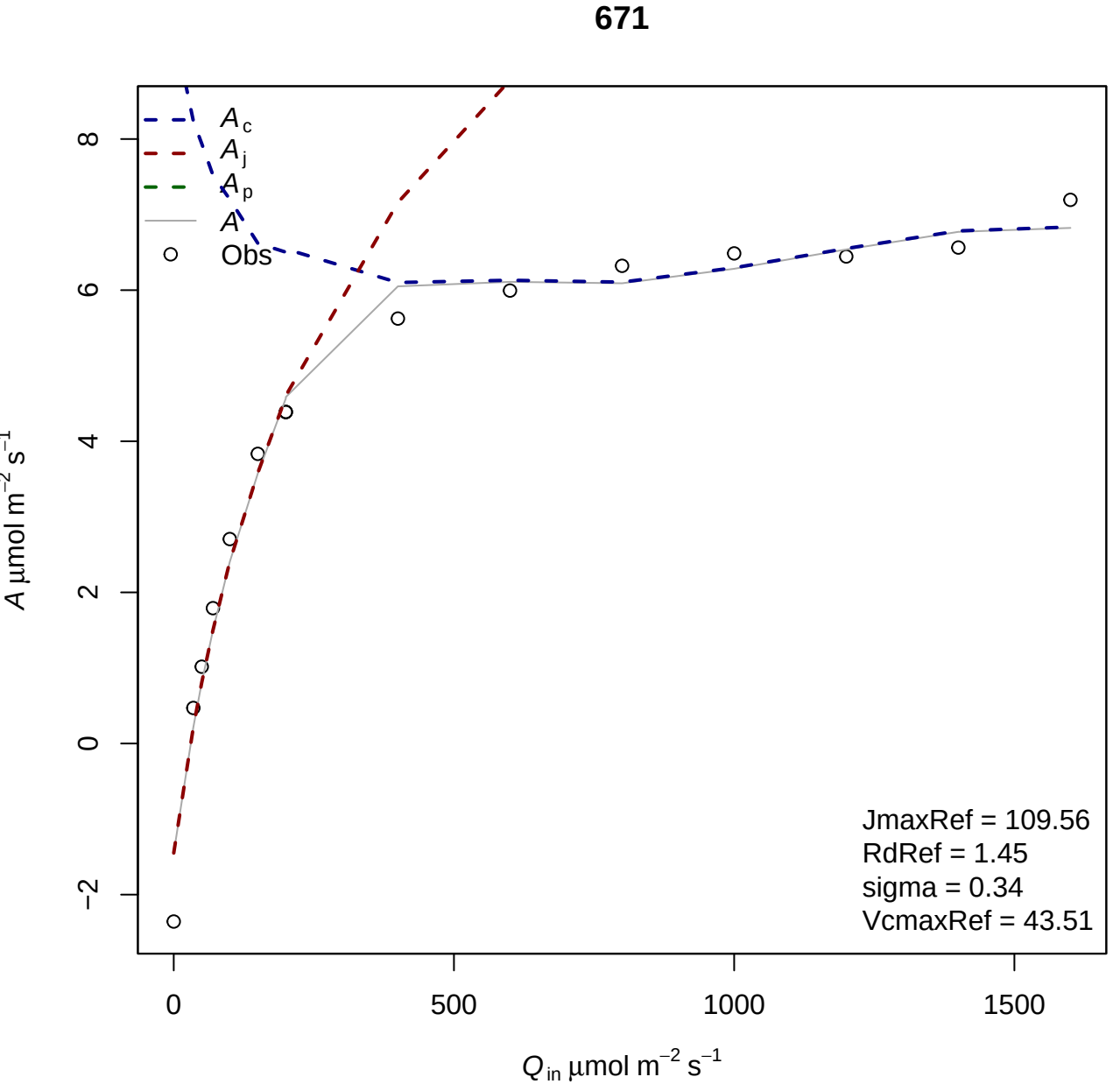

666
A_c
A_j
A_p
A
Obs
JmaxRef = 48.47
RdRef = 0.46
sigma = 0.14
VcmaxRef = 29.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

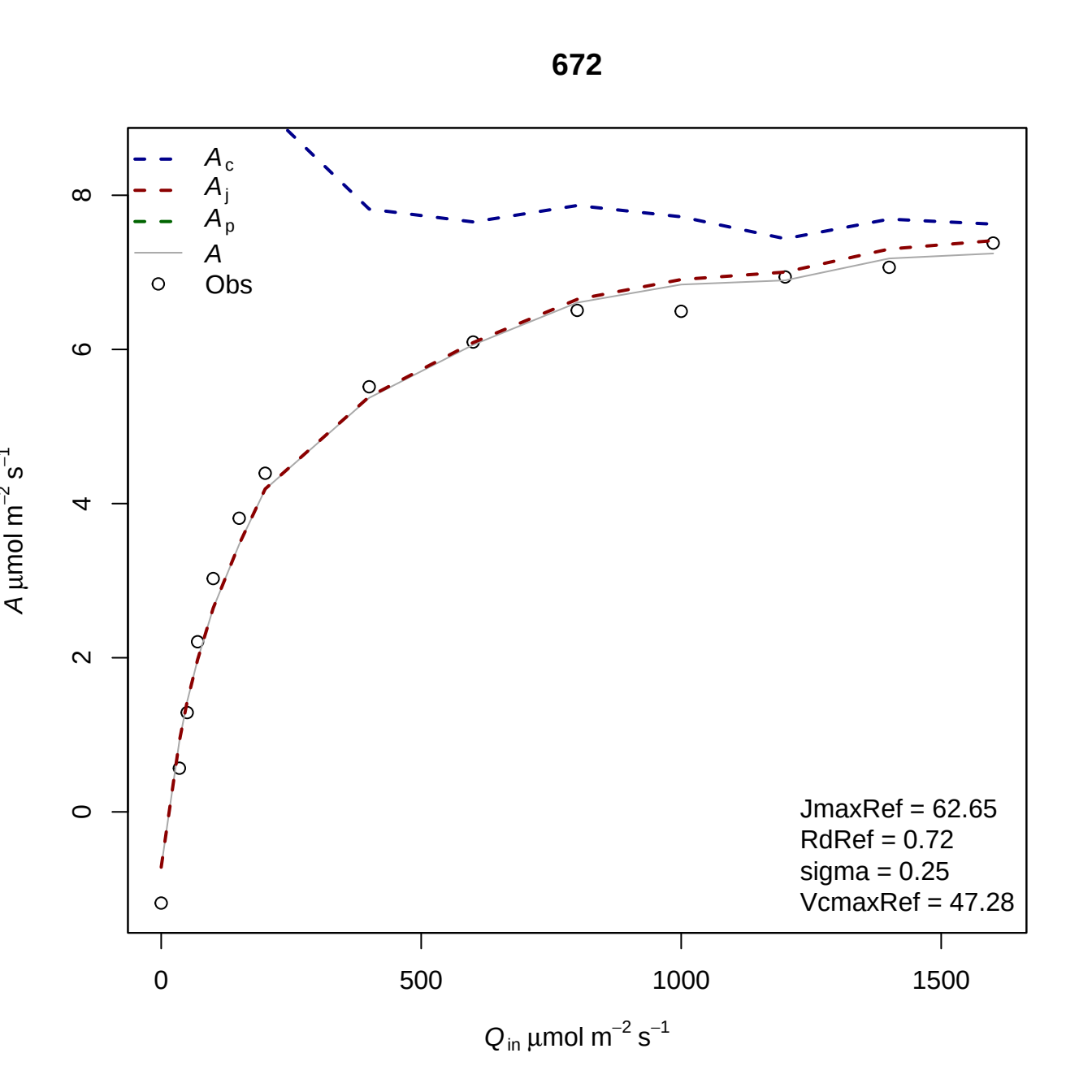

667
A_c
A_j
A_p
A
Obs
JmaxRef = 96.21
RdRef = 0.78
sigma = 0.24
VcmaxRef = 36.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

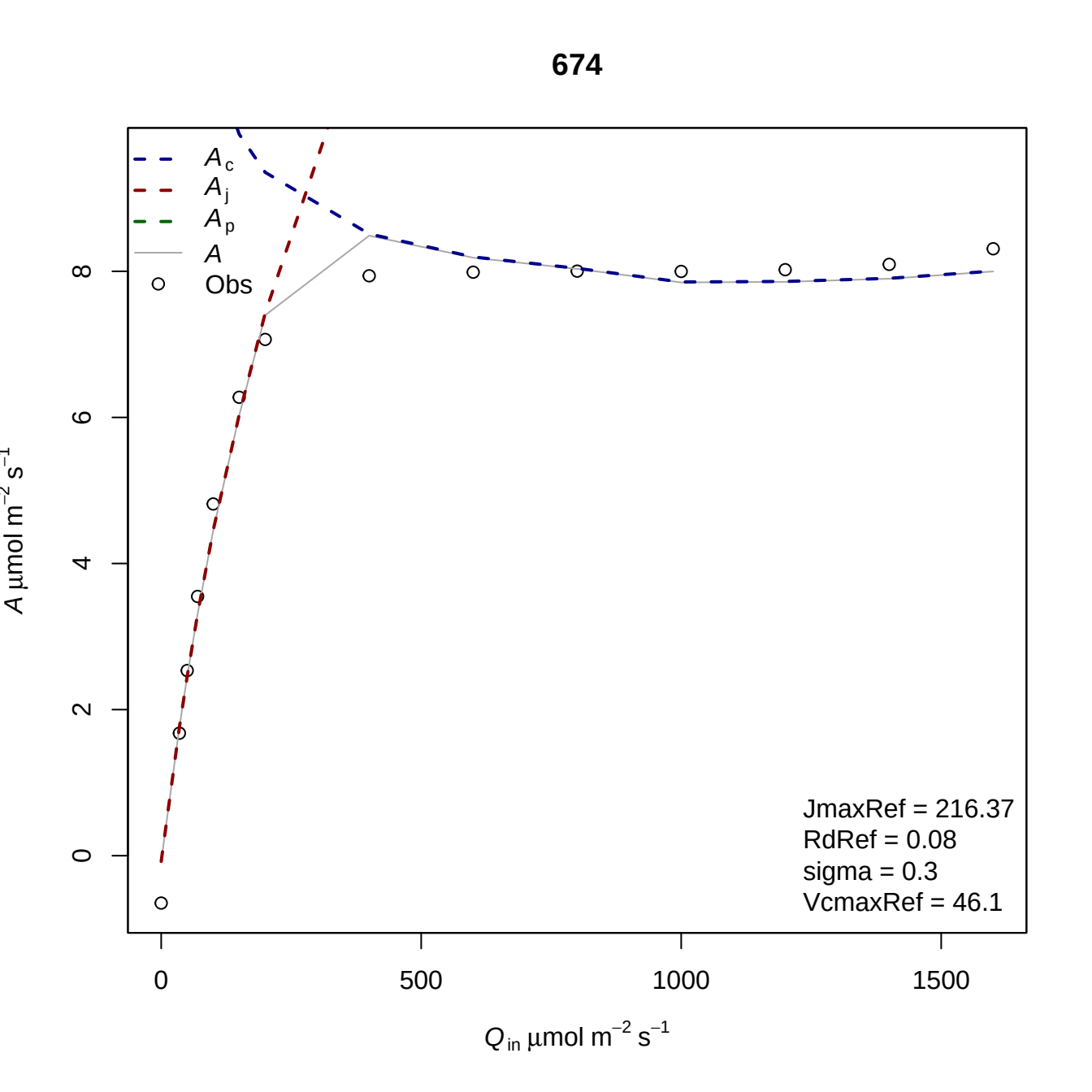

668
A_c
A_j
A_p
A
Obs
JmaxRef = 107.55
RdRef = 0.51
sigma = 0.17
VcmaxRef = 42.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

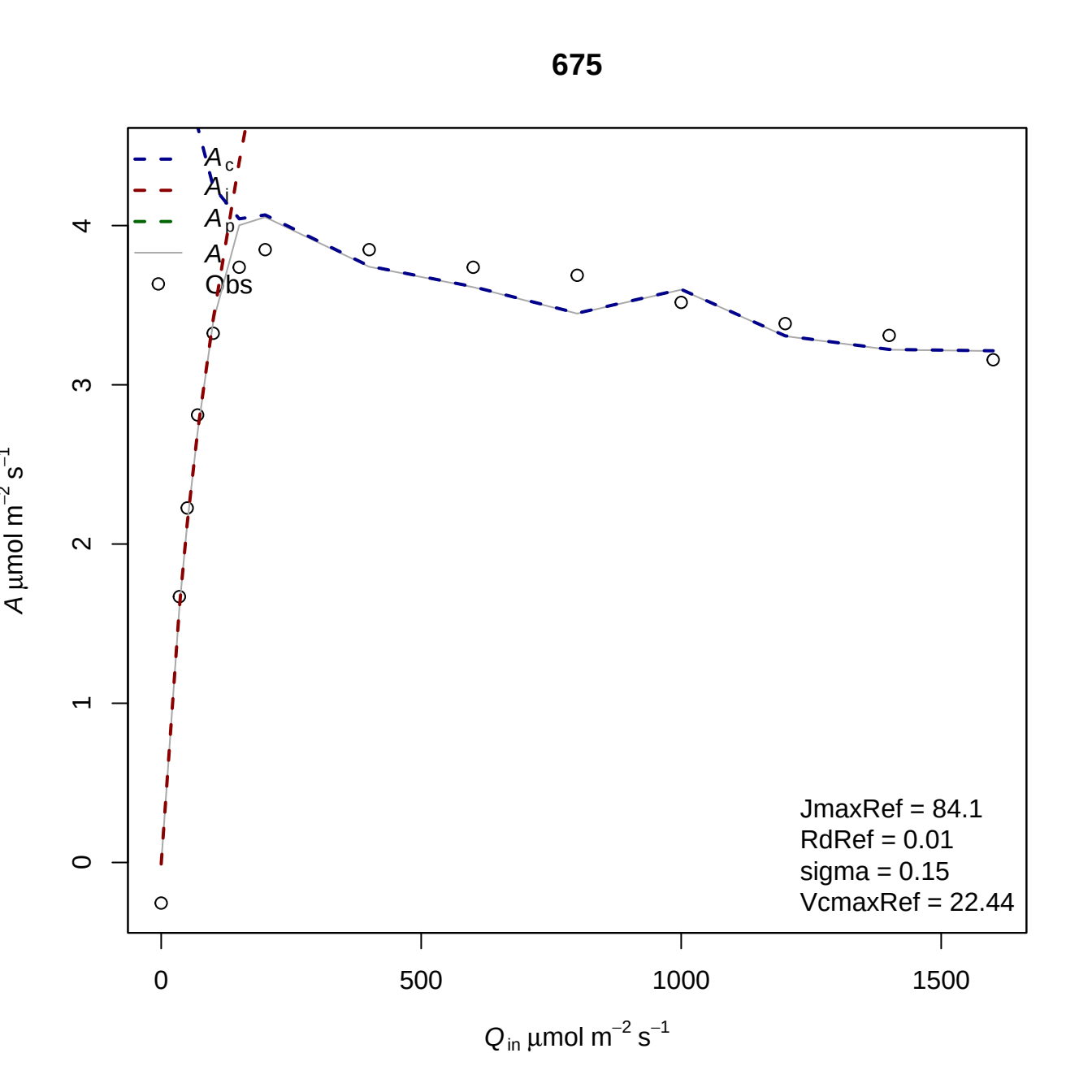

669
A_c
A_j
A_p
A
Obs
JmaxRef = 81.85
RdRef = 0.67
sigma = 0.23
VcmaxRef = 49.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

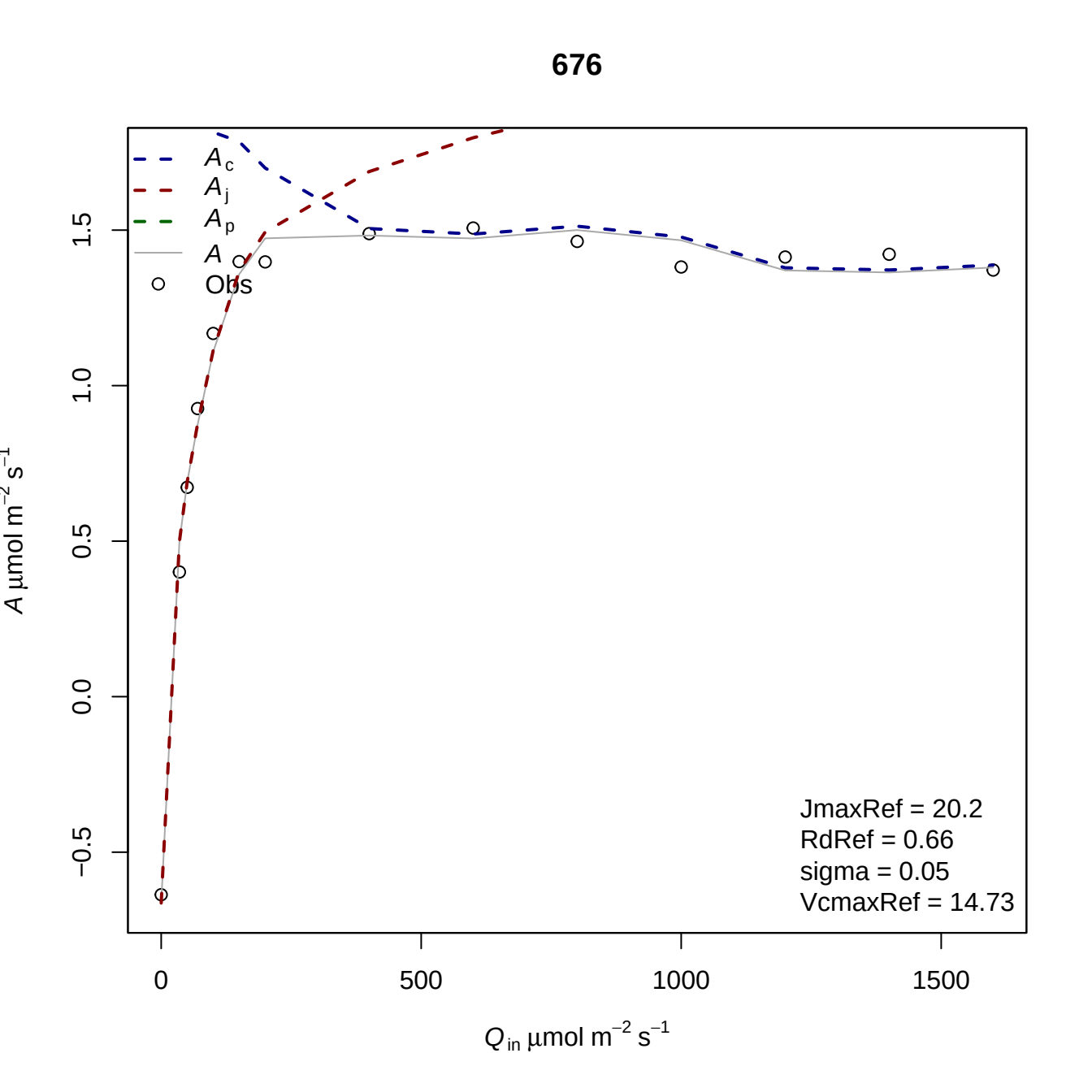

670
A_c
A_j
A_p
A
Obs
JmaxRef = 158.14
RdRef = 1.23
sigma = 0.33
VcmaxRef = 83.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

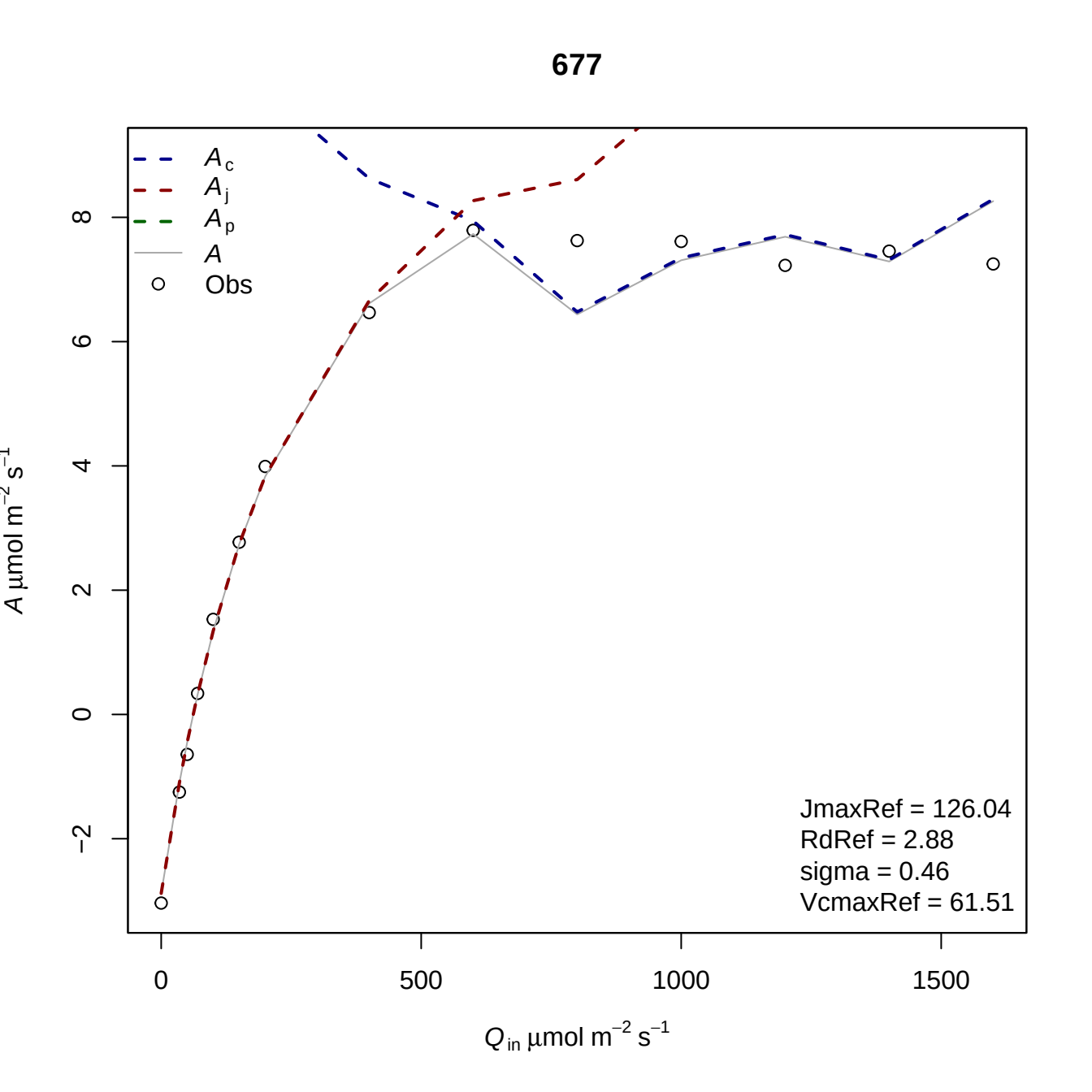

671
A_c
A_j
A_p
A
Obs
JmaxRef = 109.56
RdRef = 1.45
sigma = 0.34
VcmaxRef = 43.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

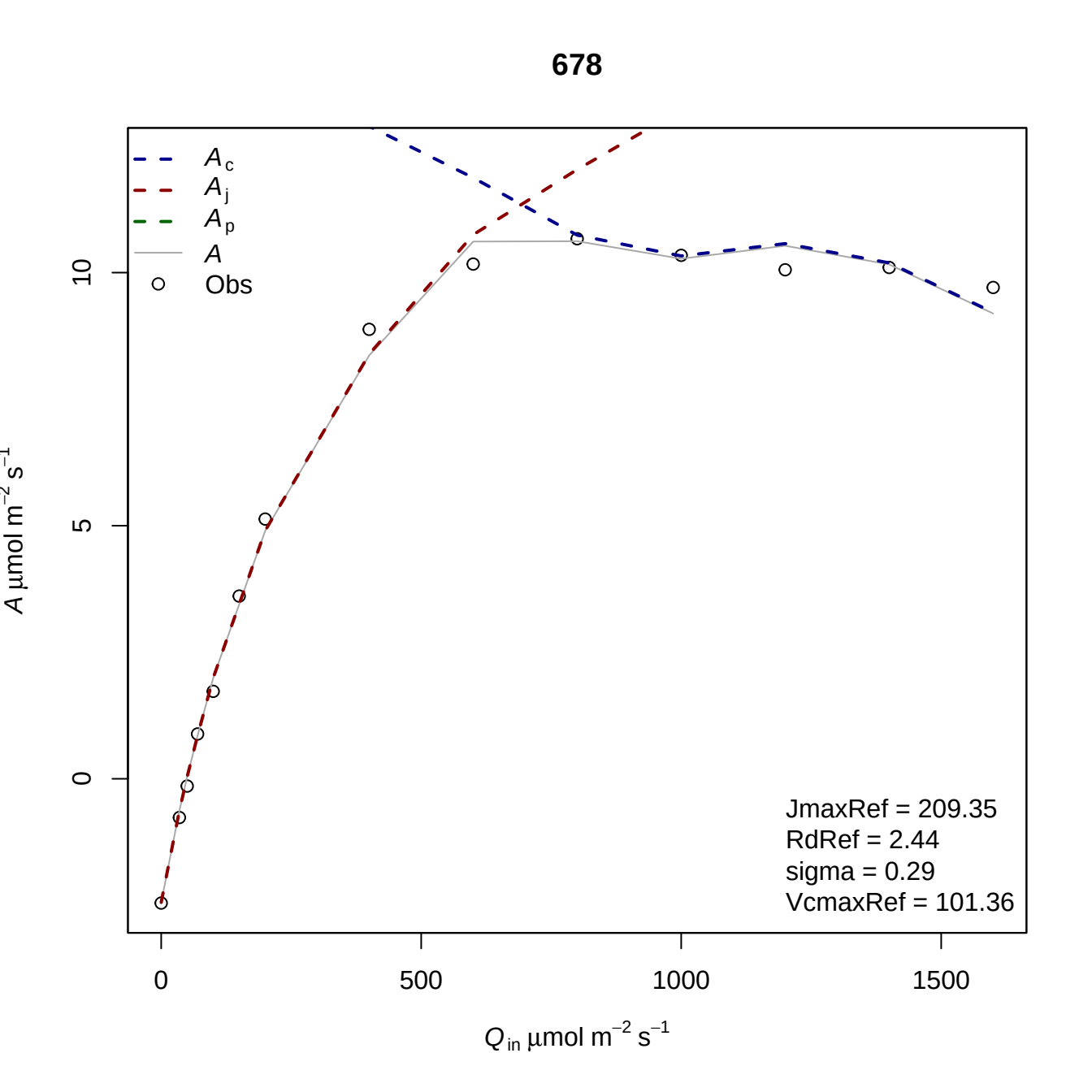

672
A_c
A_j
A_p
A
Obs
JmaxRef = 62.65
RdRef = 0.72
sigma = 0.25
VcmaxRef = 47.28
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

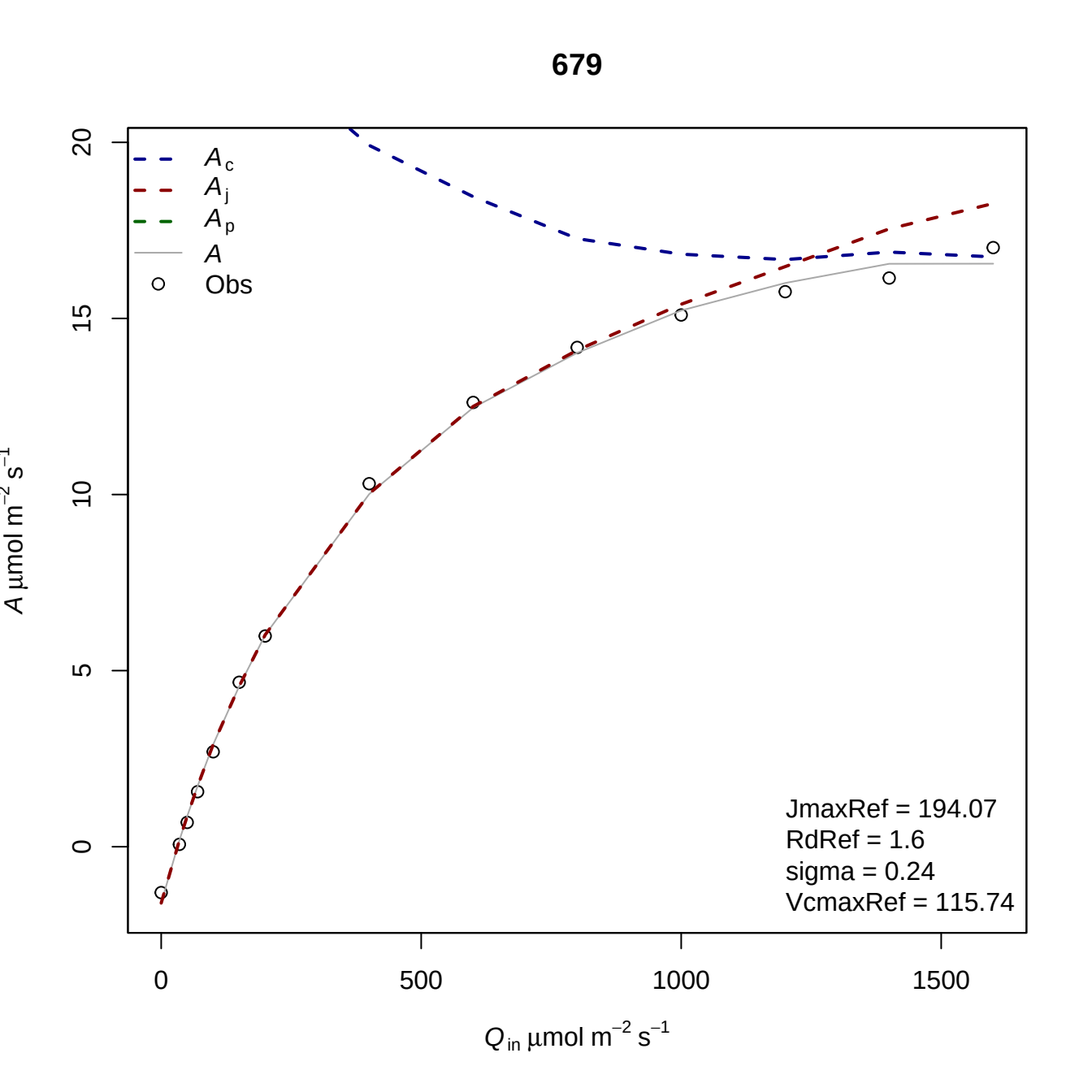

674
A_c
A_j
A_p
A
Obs
JmaxRef = 216.37
RdRef = 0.08
sigma = 0.3
VcmaxRef = 46.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

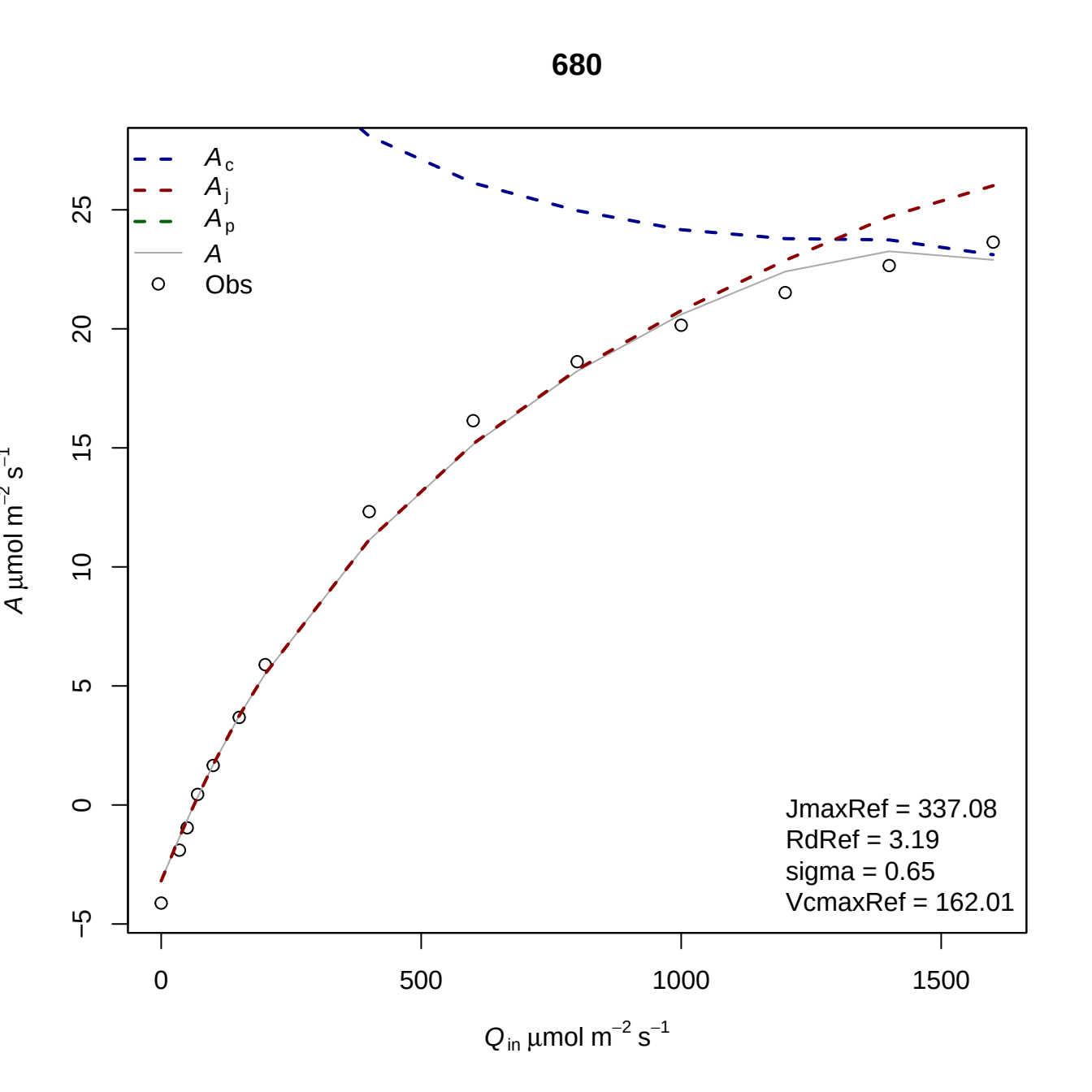

675
A_c
A_j
A_p
A_n
Obs
JmaxRef = 84.1
RdRef = 0.01
sigma = 0.15
VcmaxRef = 22.44
0
1
2
3
4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

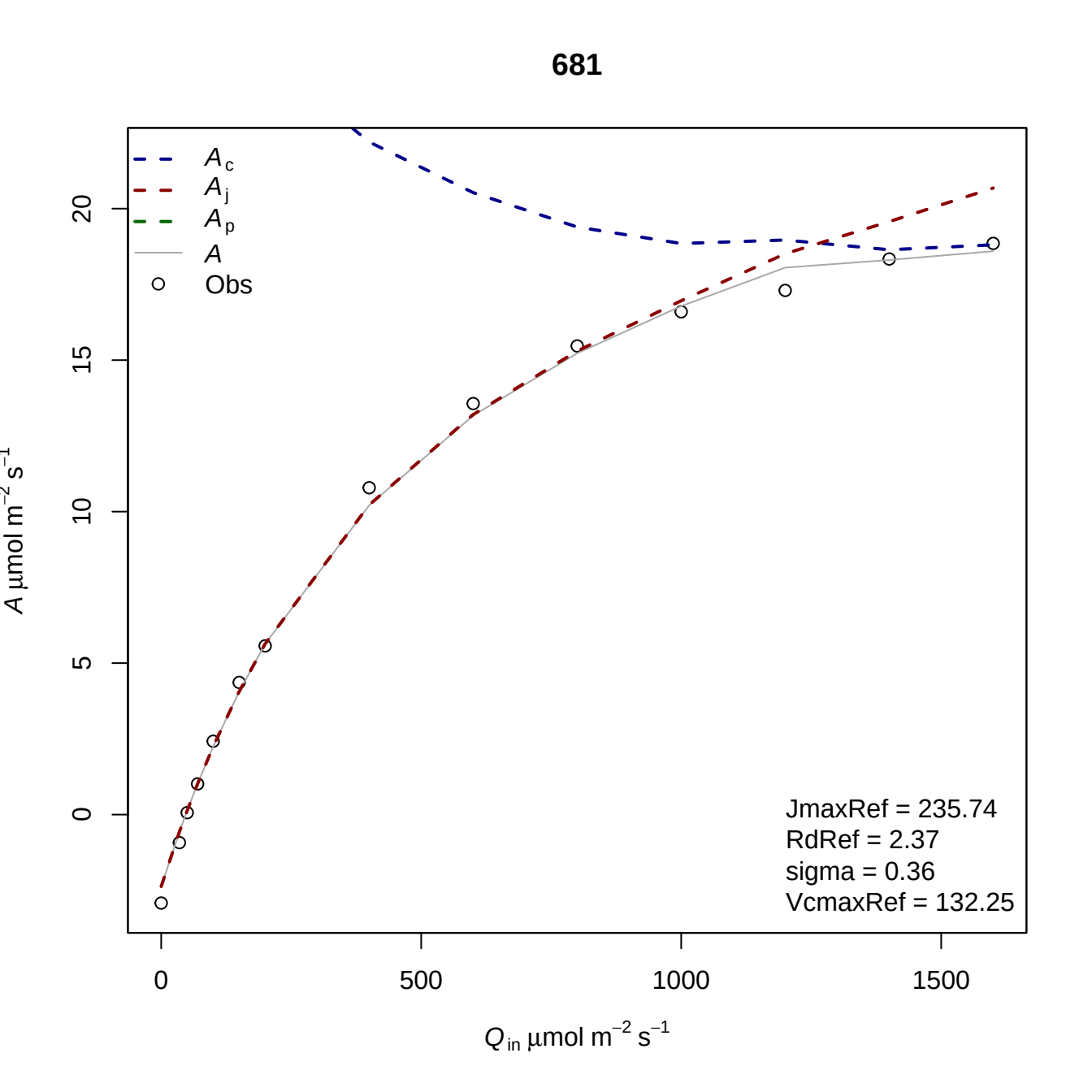

676
A_c
A_j
A_p
A
Obs
JmaxRef = 20.2
RdRef = 0.66
sigma = 0.05
VcmaxRef = 14.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
0
500
1000
1500

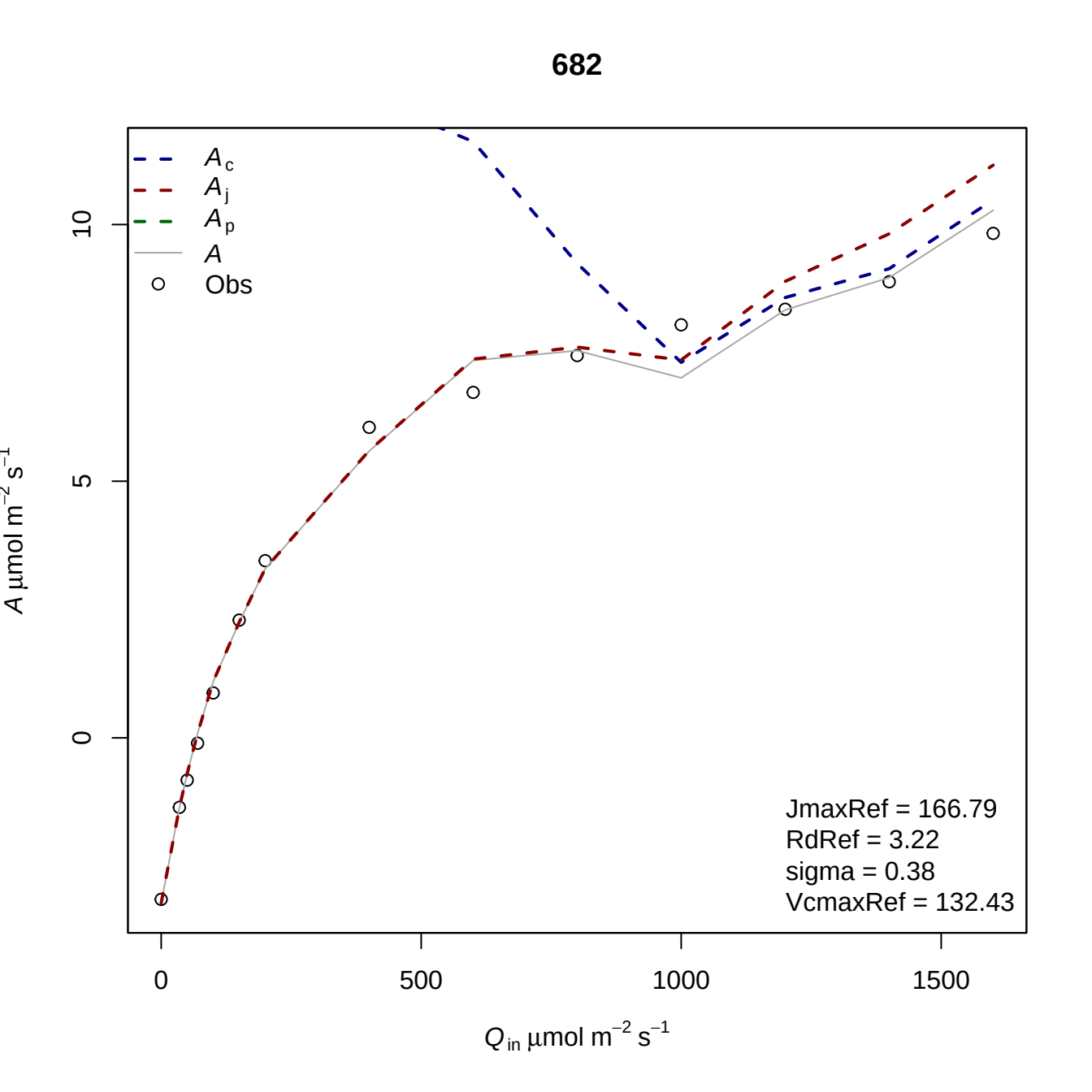

677
A_c
A_j
A_p
A
Obs
JmaxRef = 126.04
RdRef = 2.88
sigma = 0.46
VcmaxRef = 61.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

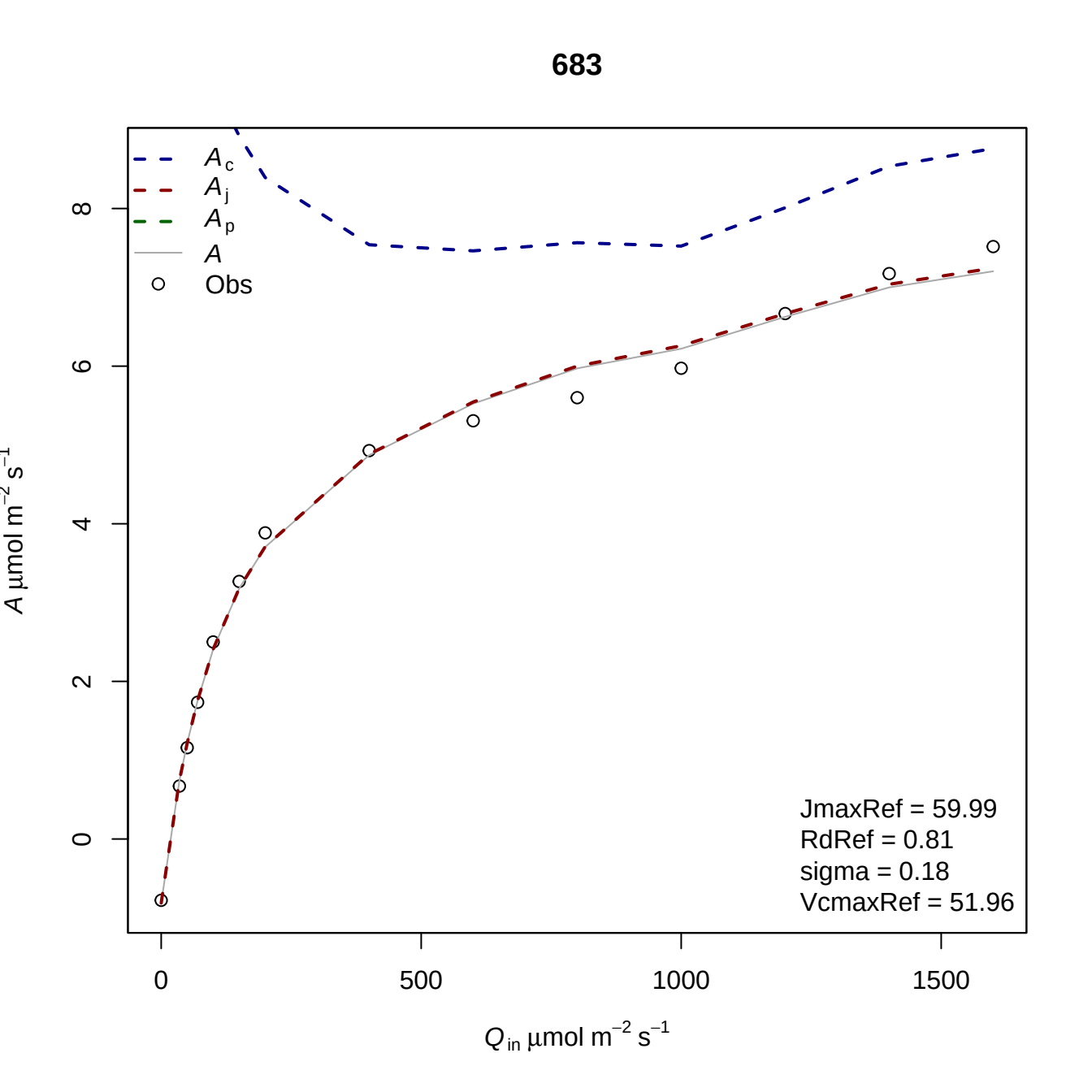

678
A_c
A_j
A_p
A
Obs
JmaxRef = 209.35
RdRef = 2.44
sigma = 0.29
VcmaxRef = 101.36
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

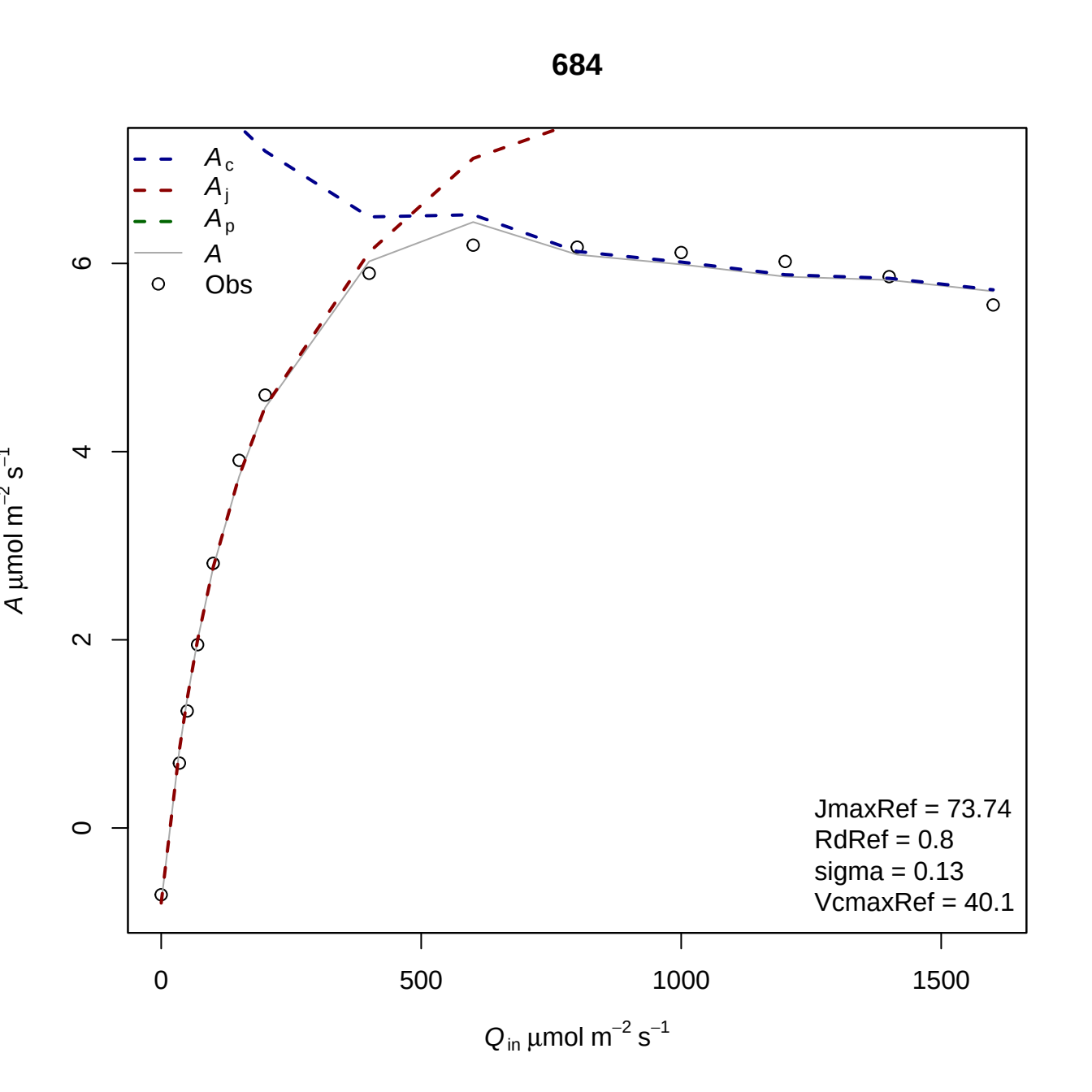

679
A_c
A_j
A_p
A
Obs
JmaxRef = 194.07
RdRef = 1.6
sigma = 0.24
VcmaxRef = 115.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

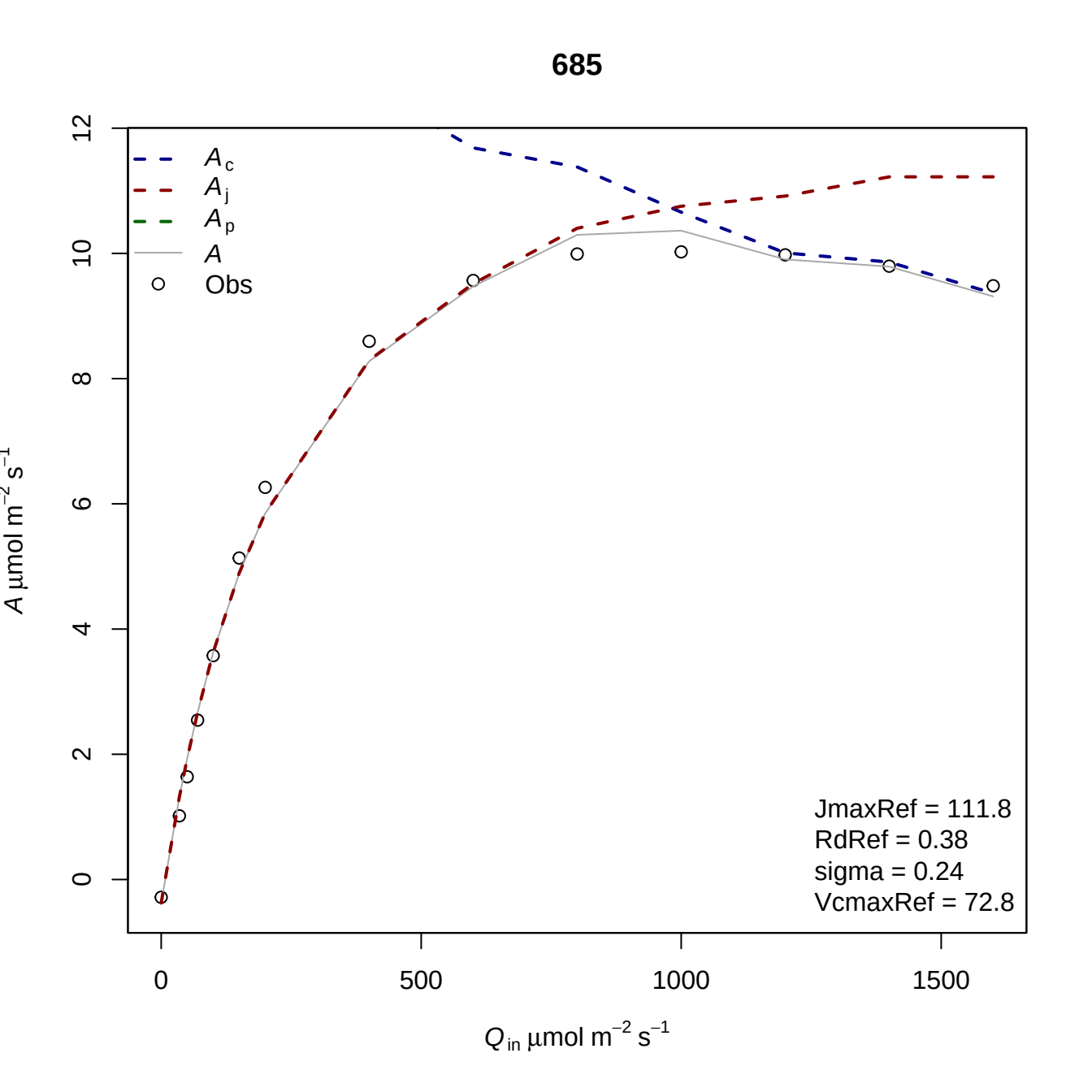

680
A_c
A_j
A_p
A
Obs
JmaxRef = 337.08
RdRef = 3.19
sigma = 0.65
VcmaxRef = 162.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
0
500
1000
1500

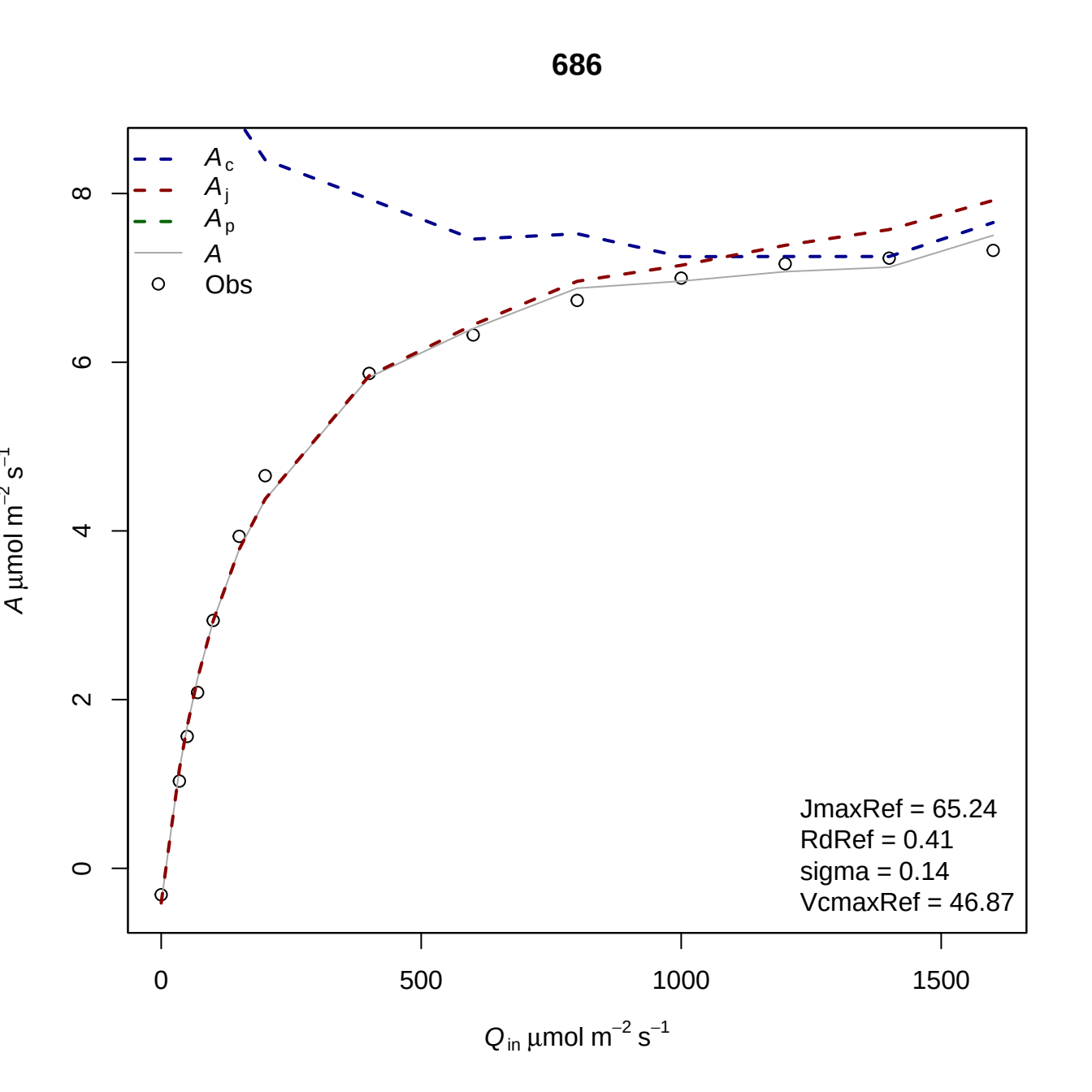

681
A_c
A_j
A_p
A
Obs
JmaxRef = 235.74
RdRef = 2.37
sigma = 0.36
VcmaxRef = 132.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

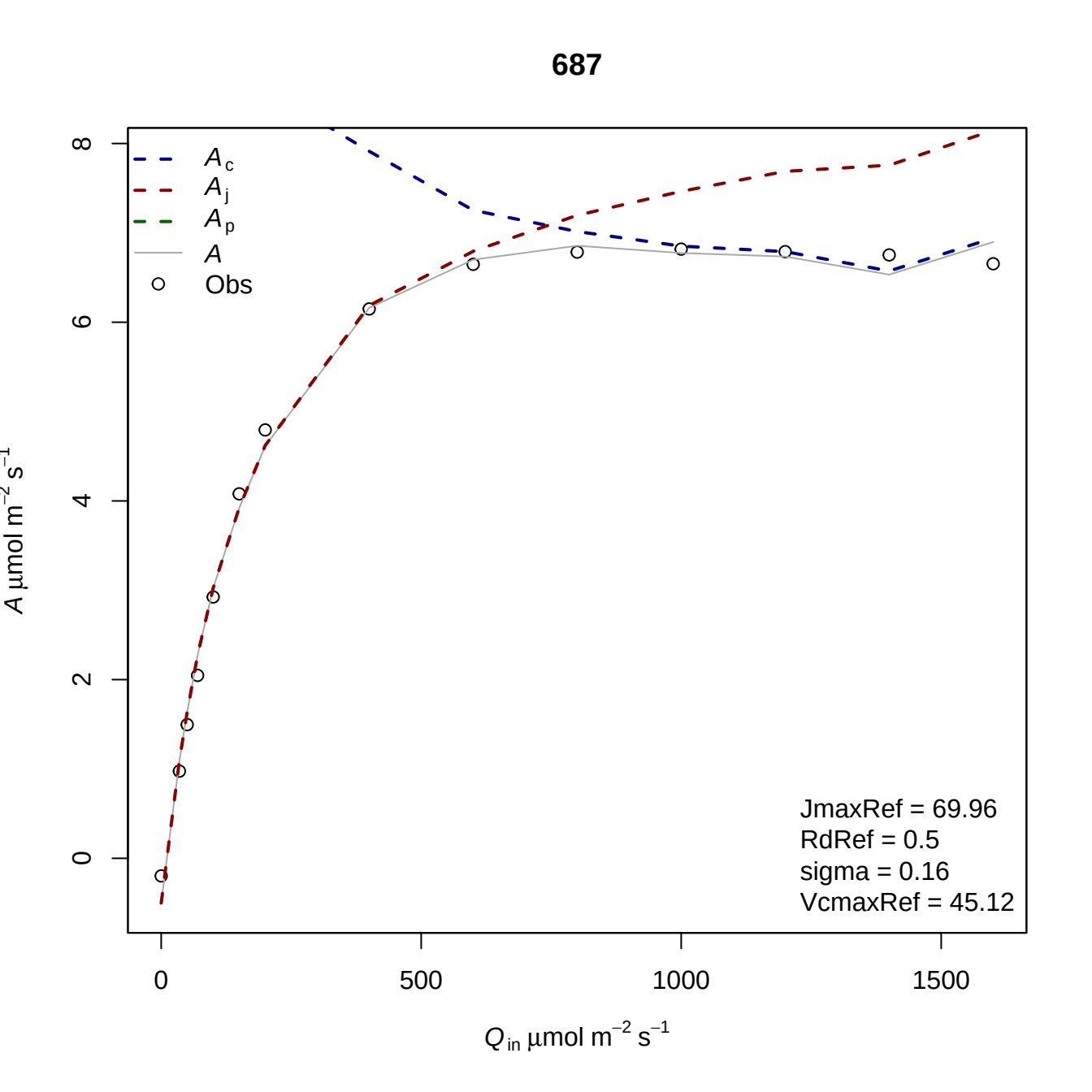

682
A_c
A_j
A_p
A
Obs
JmaxRef = 166.79
RdRef = 3.22
sigma = 0.38
VcmaxRef = 132.43
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

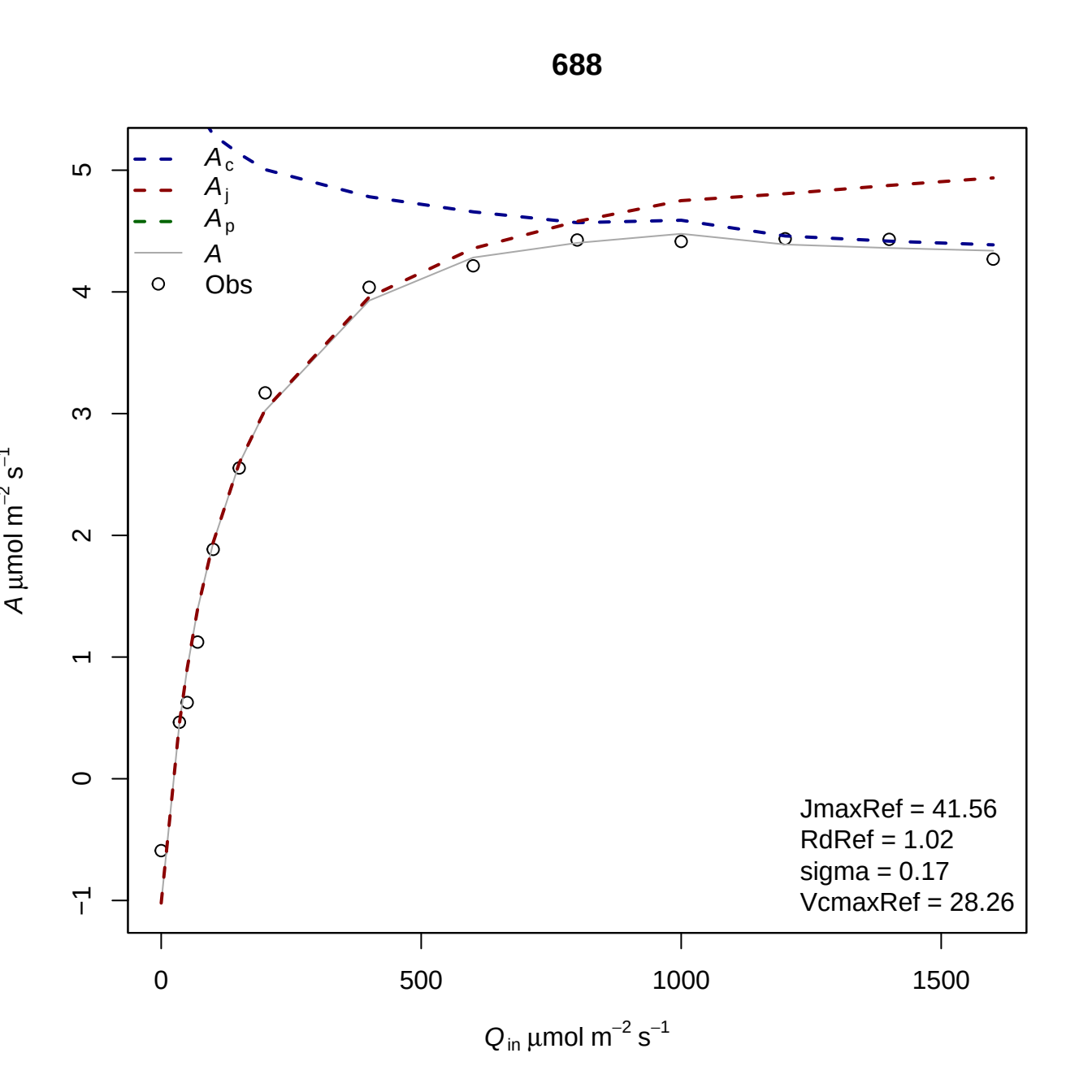

683
A_c
A_j
A_p
A
Obs
JmaxRef = 59.99
RdRef = 0.81
sigma = 0.18
VcmaxRef = 51.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

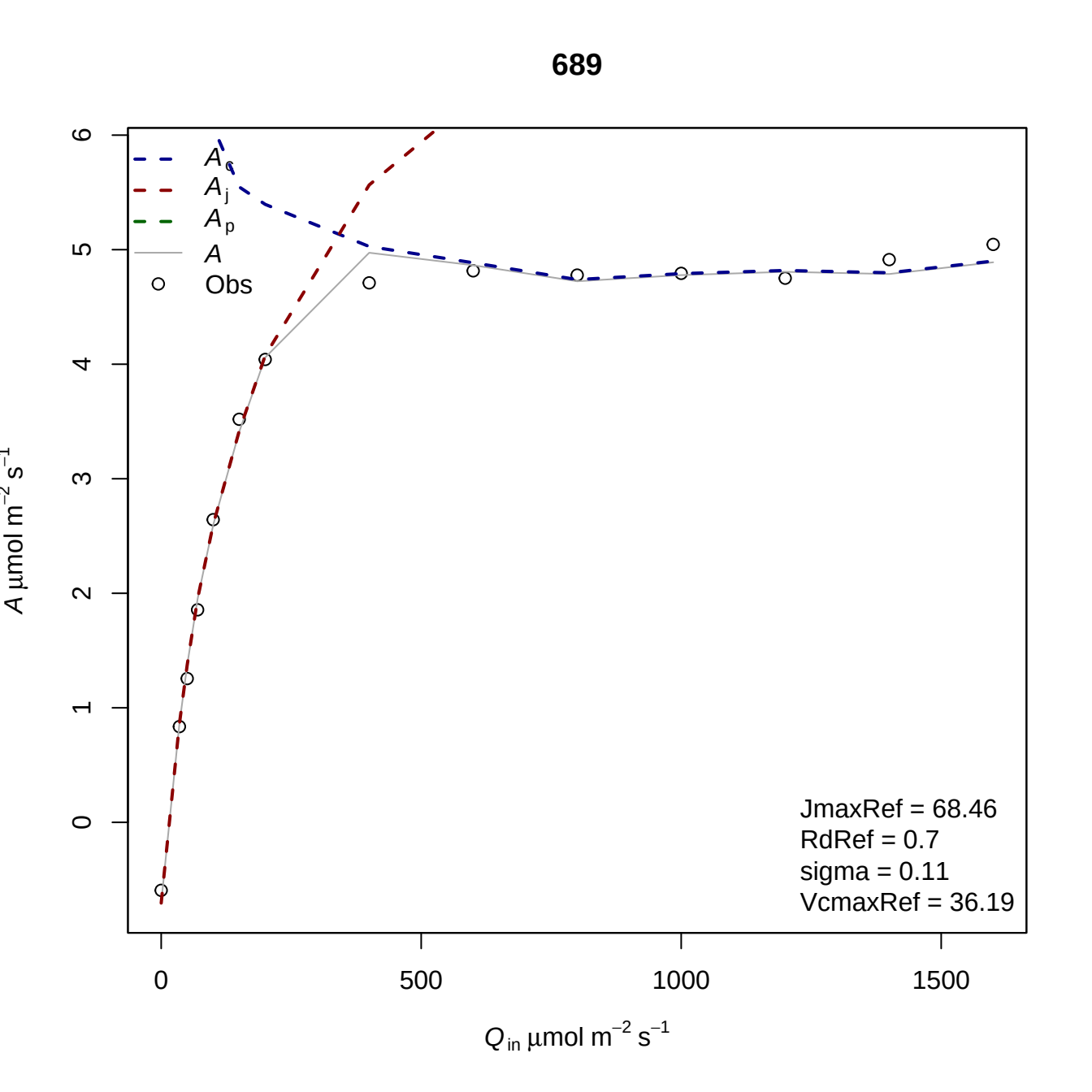

684
A_c
A_j
A_p
A
Obs
JmaxRef = 73.74
RdRef = 0.8
sigma = 0.13
VcmaxRef = 40.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

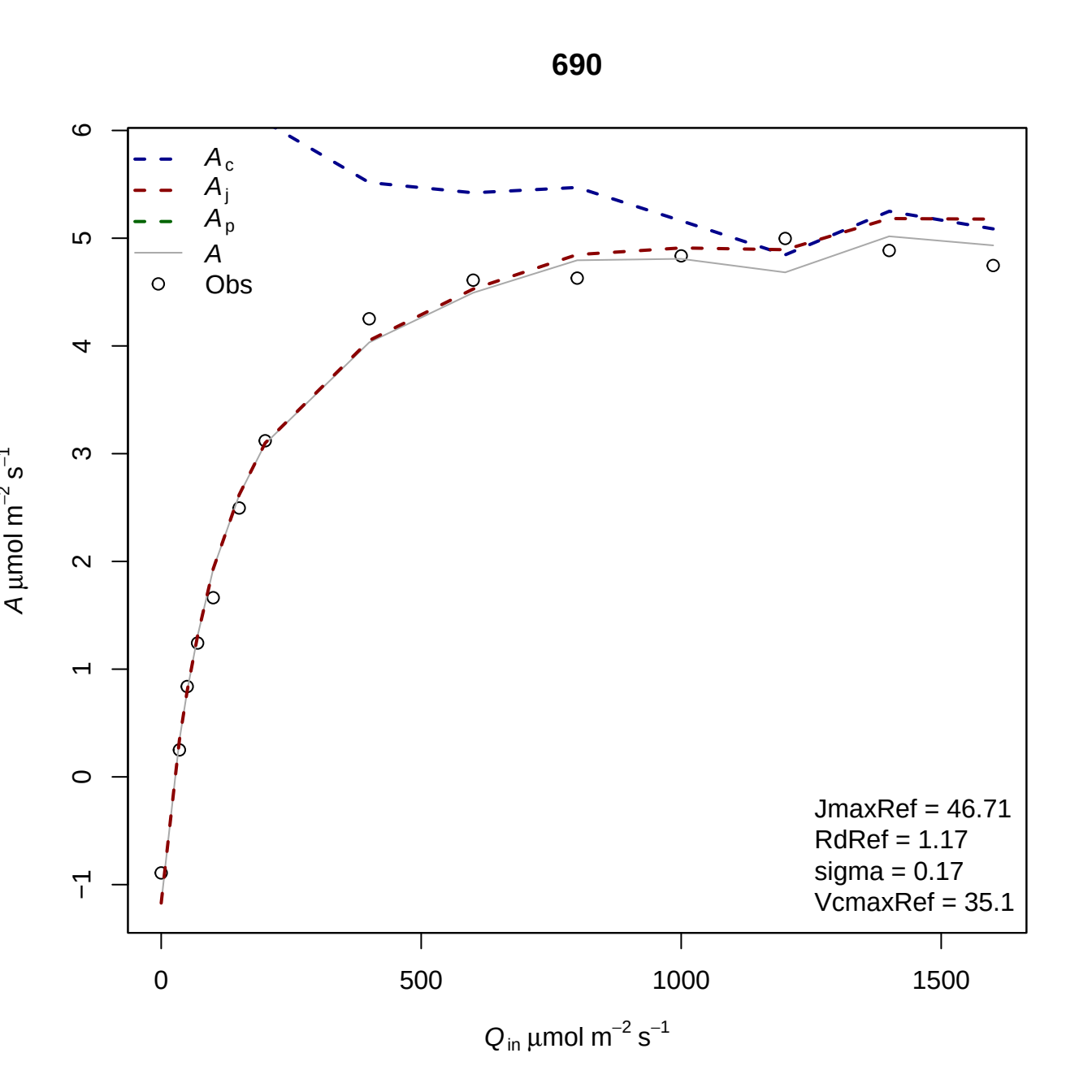

685
A_c
A_j
A_p
A
Obs
JmaxRef = 111.8
RdRef = 0.38
sigma = 0.24
VcmaxRef = 72.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

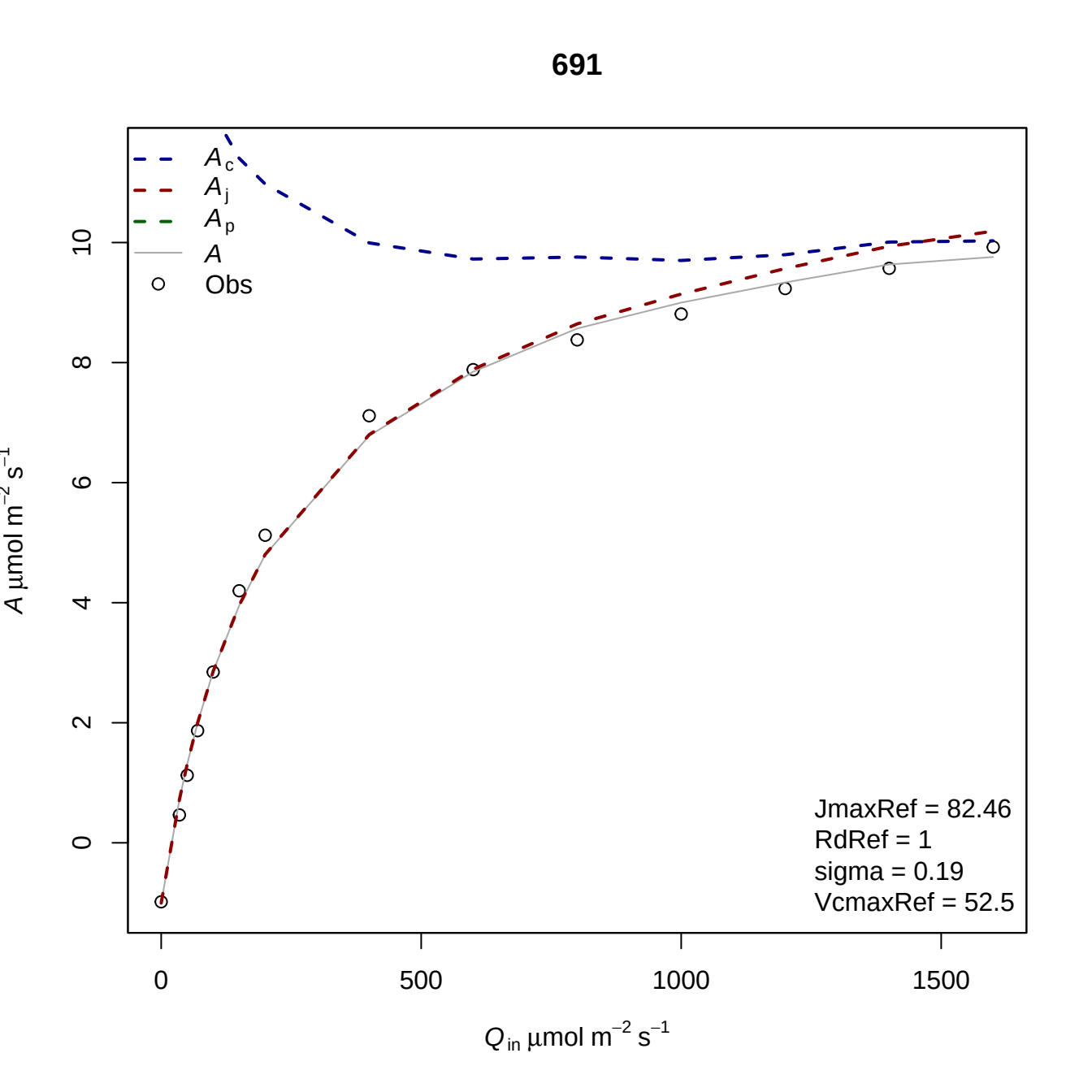

686
A_c
A_j
A_p
A
Obs
JmaxRef = 65.24
RdRef = 0.41
sigma = 0.14
VcmaxRef = 46.87
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

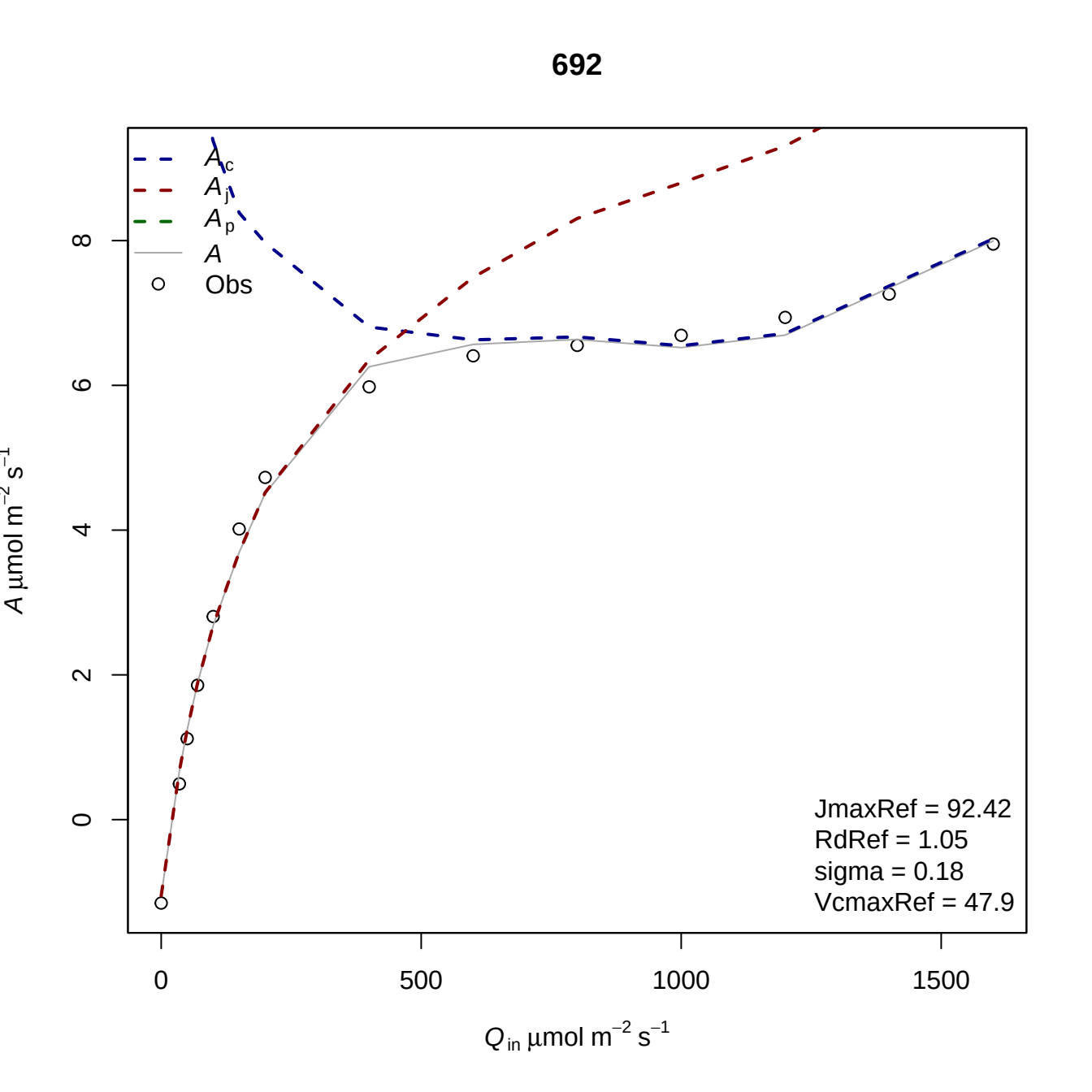

687
A_c
A_j
A_p
A
Obs
JmaxRef = 69.96
RdRef = 0.5
sigma = 0.16
VcmaxRef = 45.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

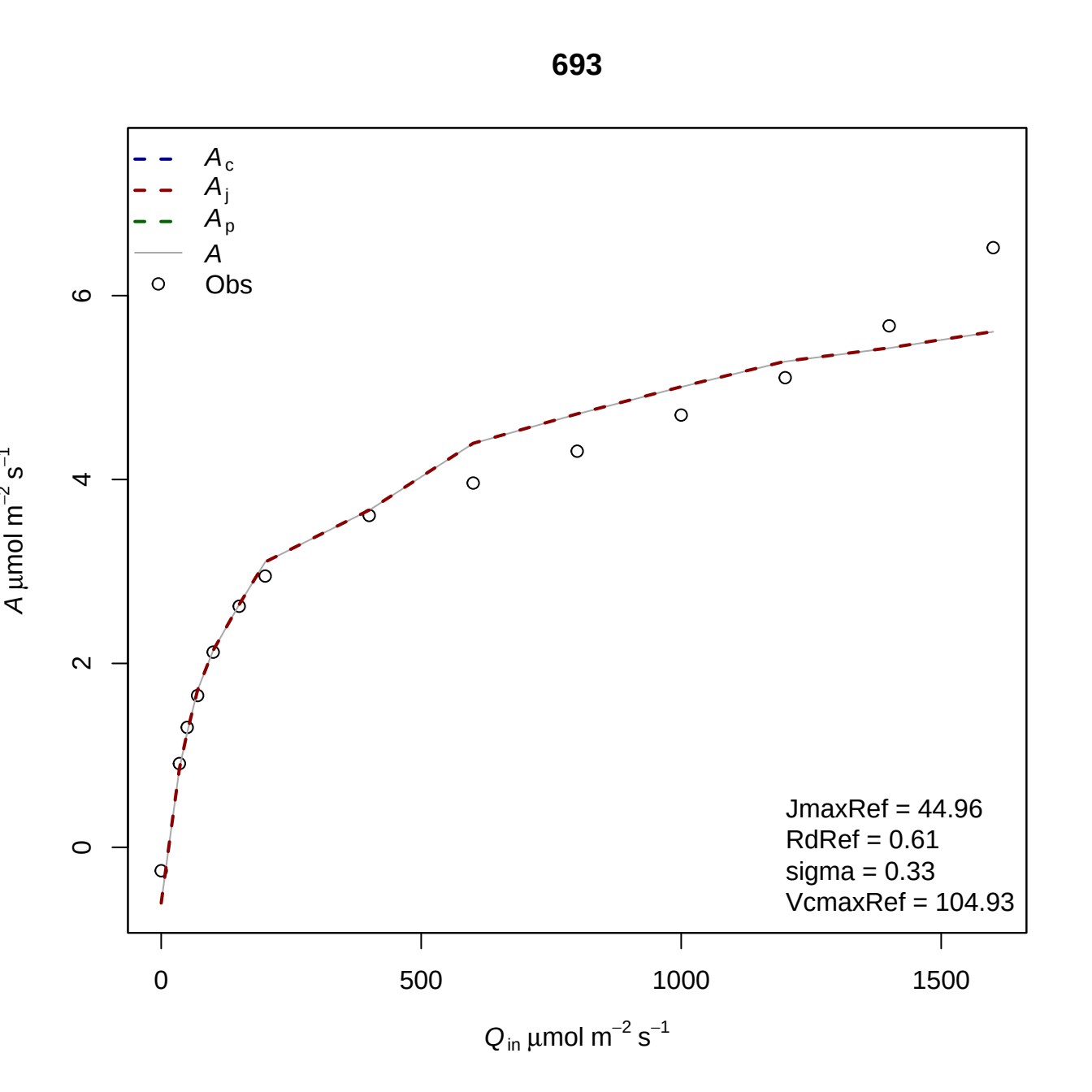

688
A_c
A_j
A_p
A
Obs
JmaxRef = 41.56
RdRef = 1.02
sigma = 0.17
VcmaxRef = 28.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

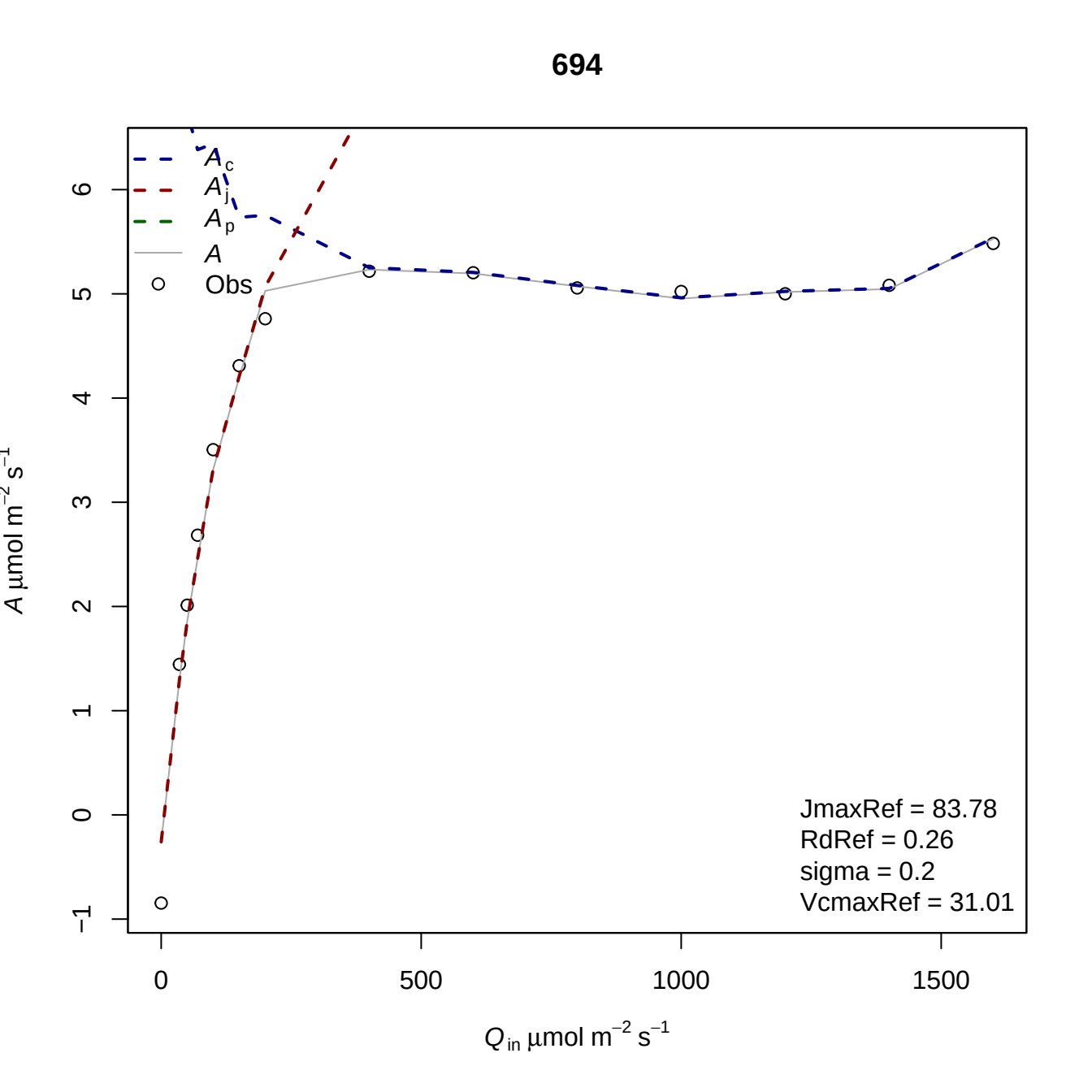

689
A_c
A_j
A_p
A
Obs
JmaxRef = 68.46
RdRef = 0.7
sigma = 0.11
VcmaxRef = 36.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

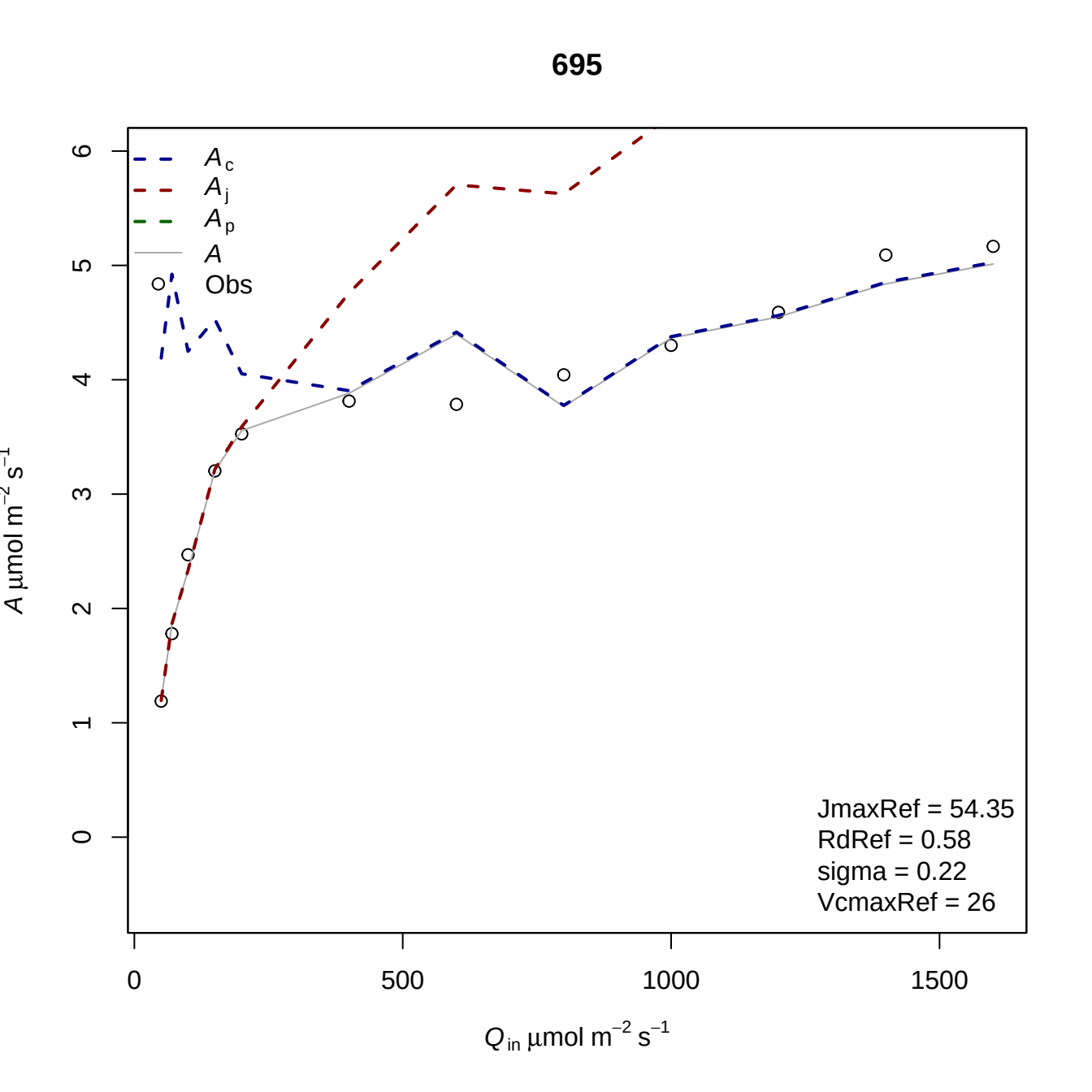

690
A_c
A_j
A_p
A
Obs
JmaxRef = 46.71
RdRef = 1.17
sigma = 0.17
VcmaxRef = 35.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

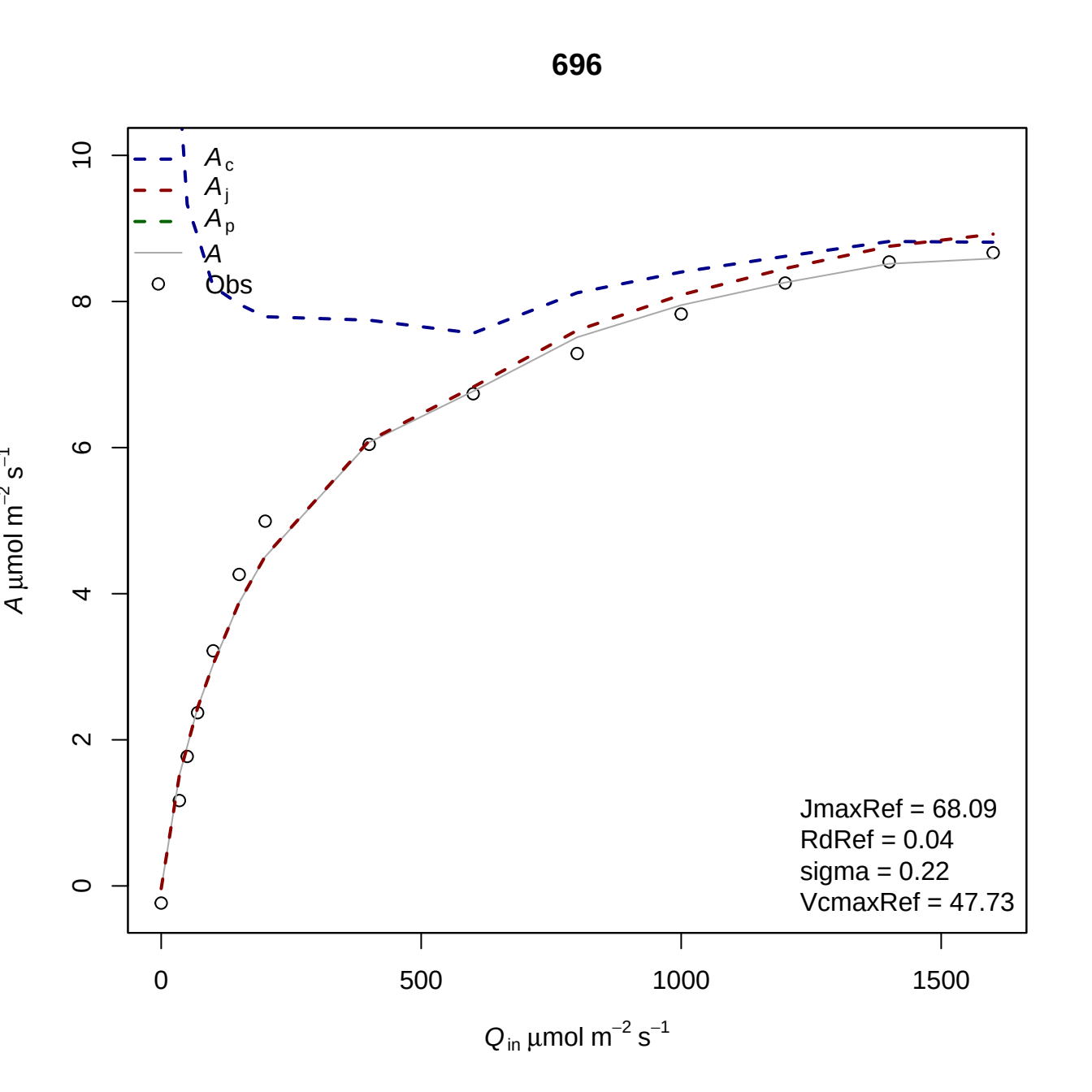

691
A_c
A_j
A_p
A
Obs
JmaxRef = 82.46
RdRef = 1
sigma = 0.19
VcmaxRef = 52.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

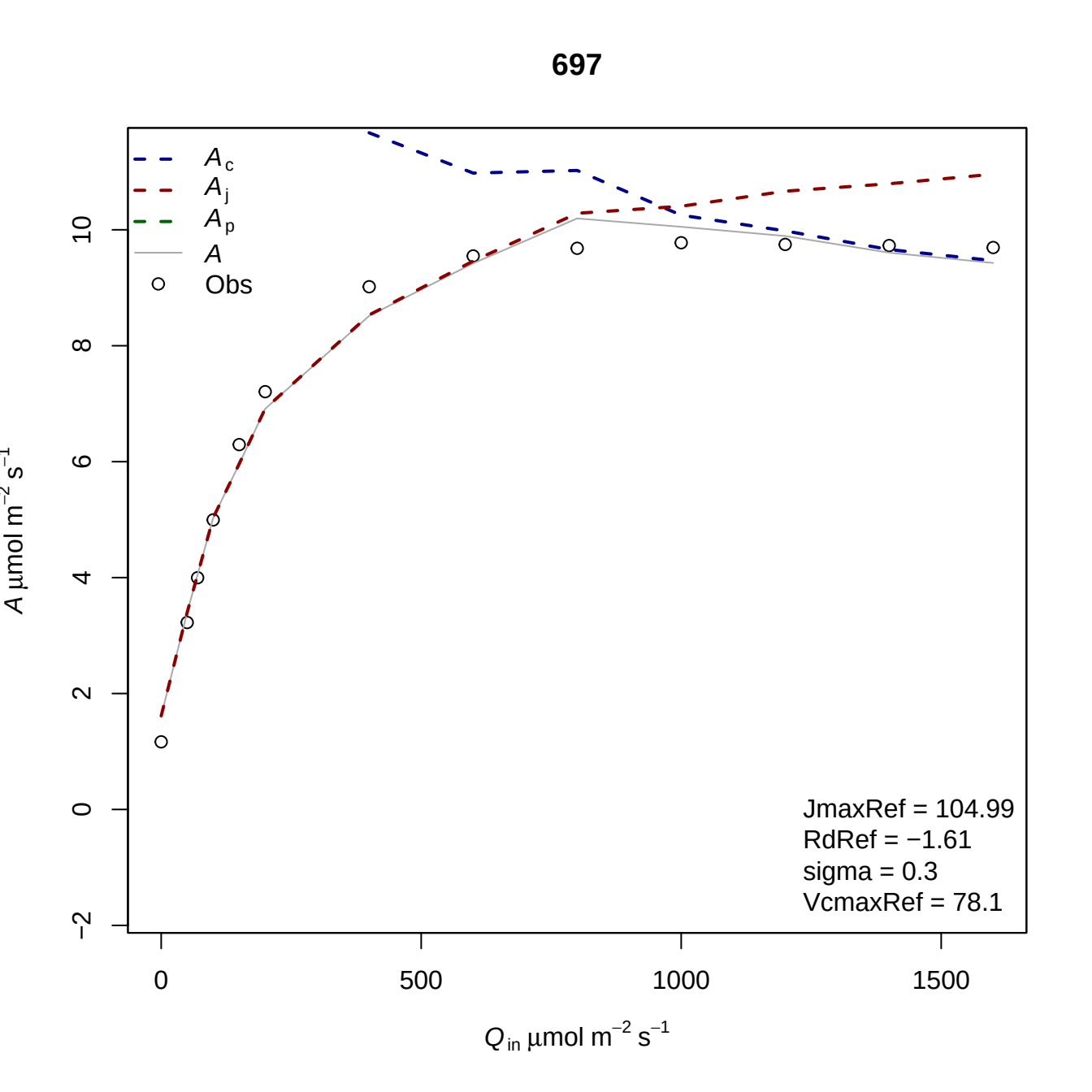

692
A_c
A_j
A_p
A
Obs
JmaxRef = 92.42
RdRef = 1.05
sigma = 0.18
VcmaxRef = 47.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

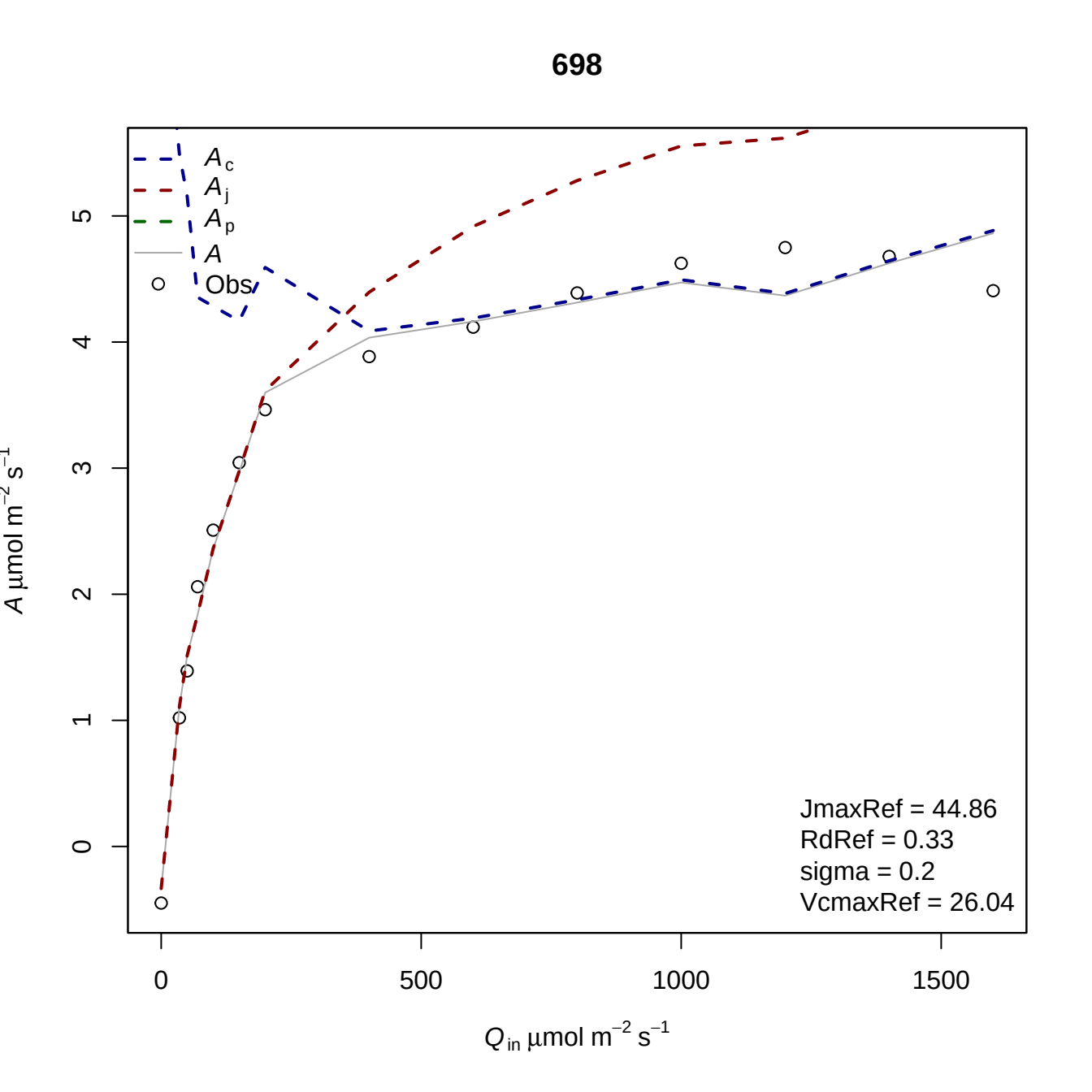

693
A_c
A_j
A_p
A
Obs
JmaxRef = 44.96
RdRef = 0.61
sigma = 0.33
VcmaxRef = 104.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

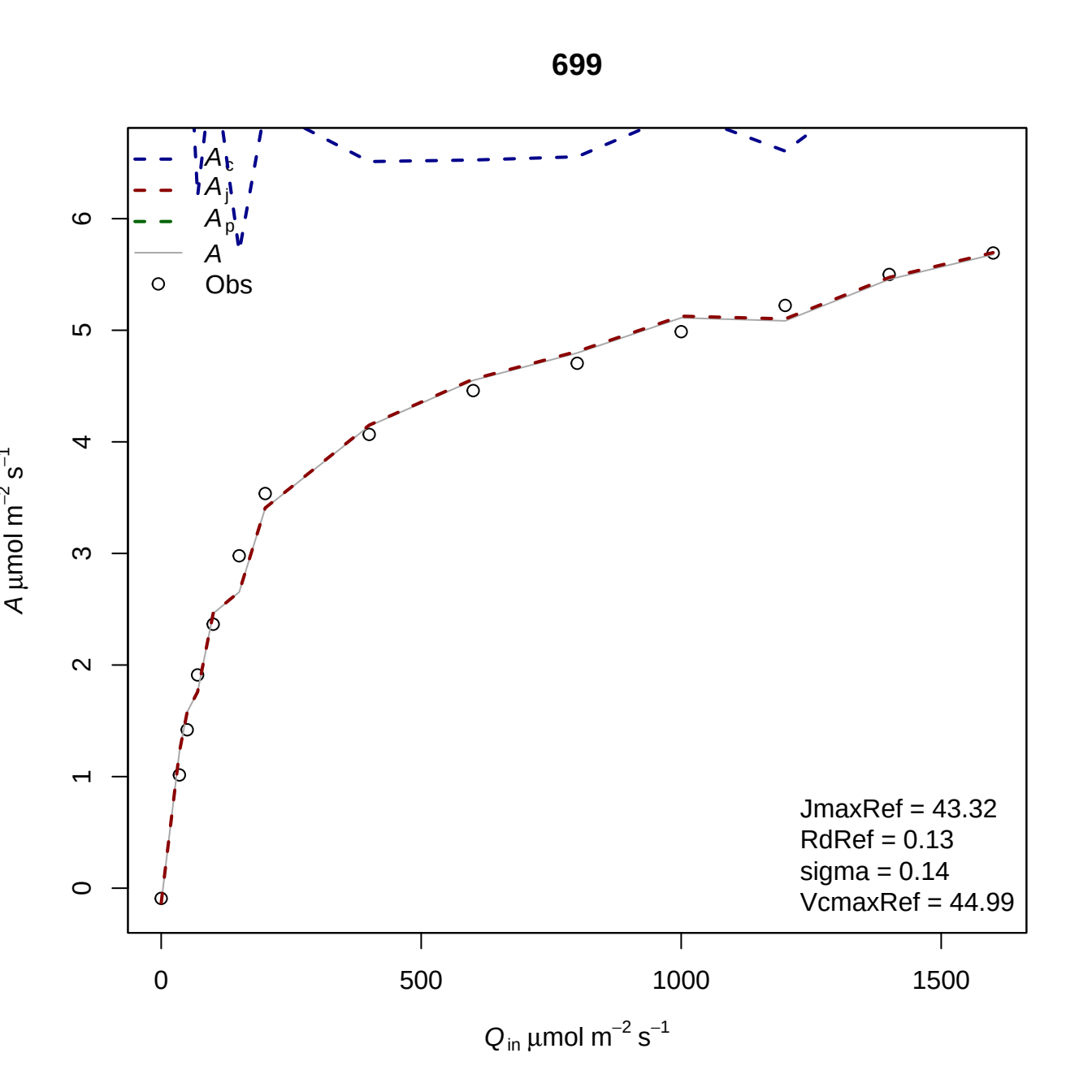

694
A_c
A_j
A_p
A
Obs
JmaxRef = 83.78
RdRef = 0.26
sigma = 0.2
VcmaxRef = 31.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

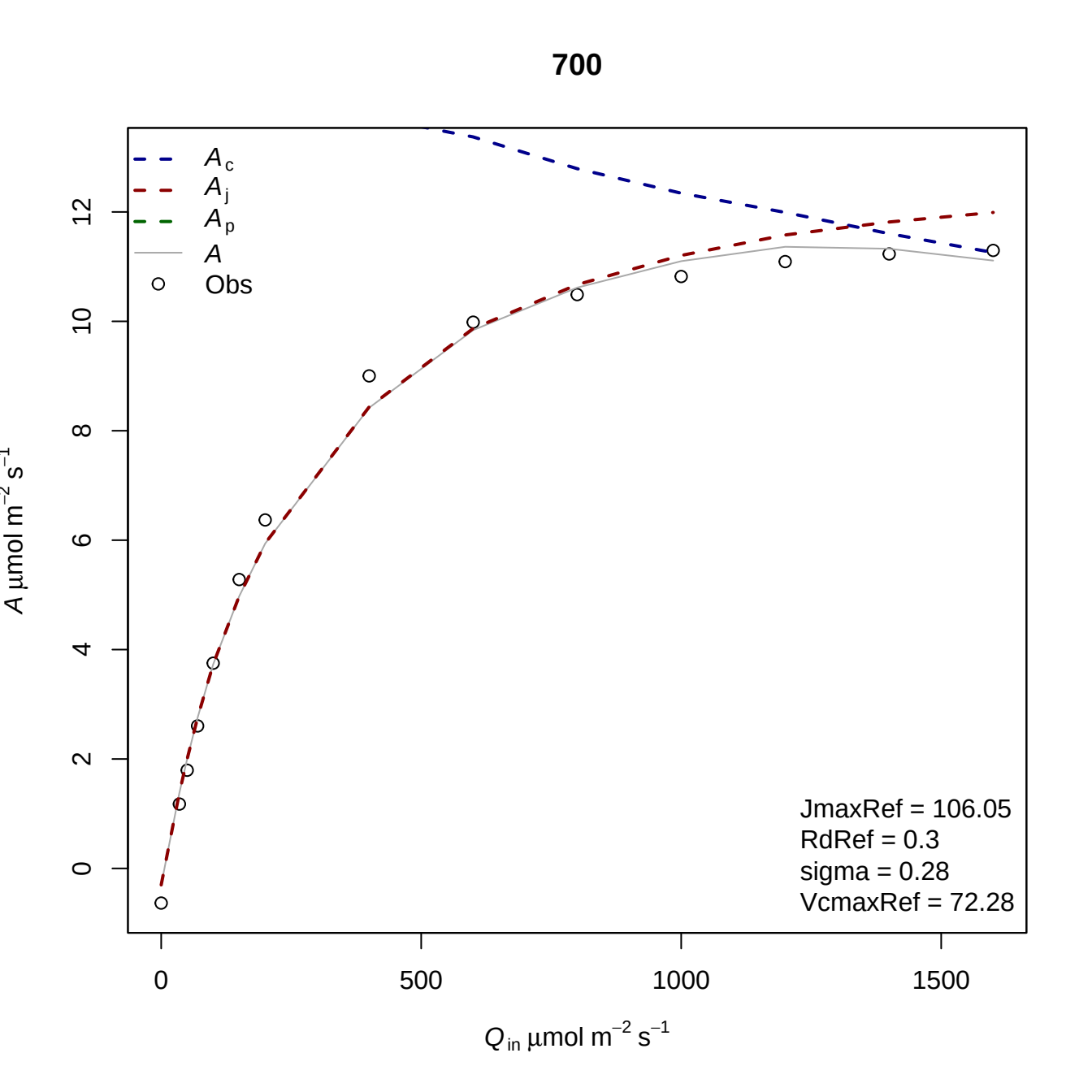

695
A_c
A_j
A_p
A
Obs
JmaxRef = 54.35
RdRef = 0.58
sigma = 0.22
VcmaxRef = 26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

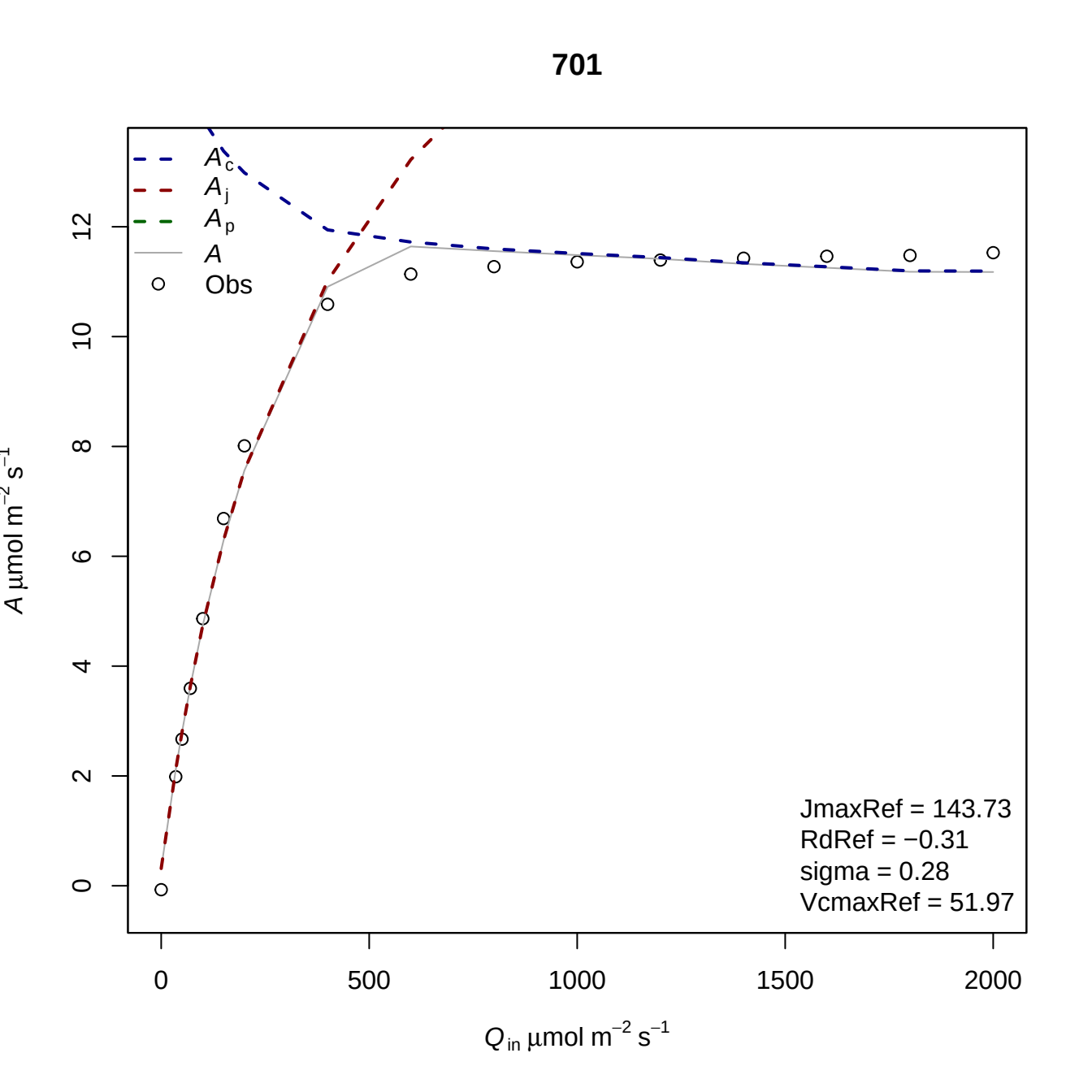

696
A_c
A_j
A_p
A
Obs
JmaxRef = 68.09
RdRef = 0.04
sigma = 0.22
VcmaxRef = 47.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

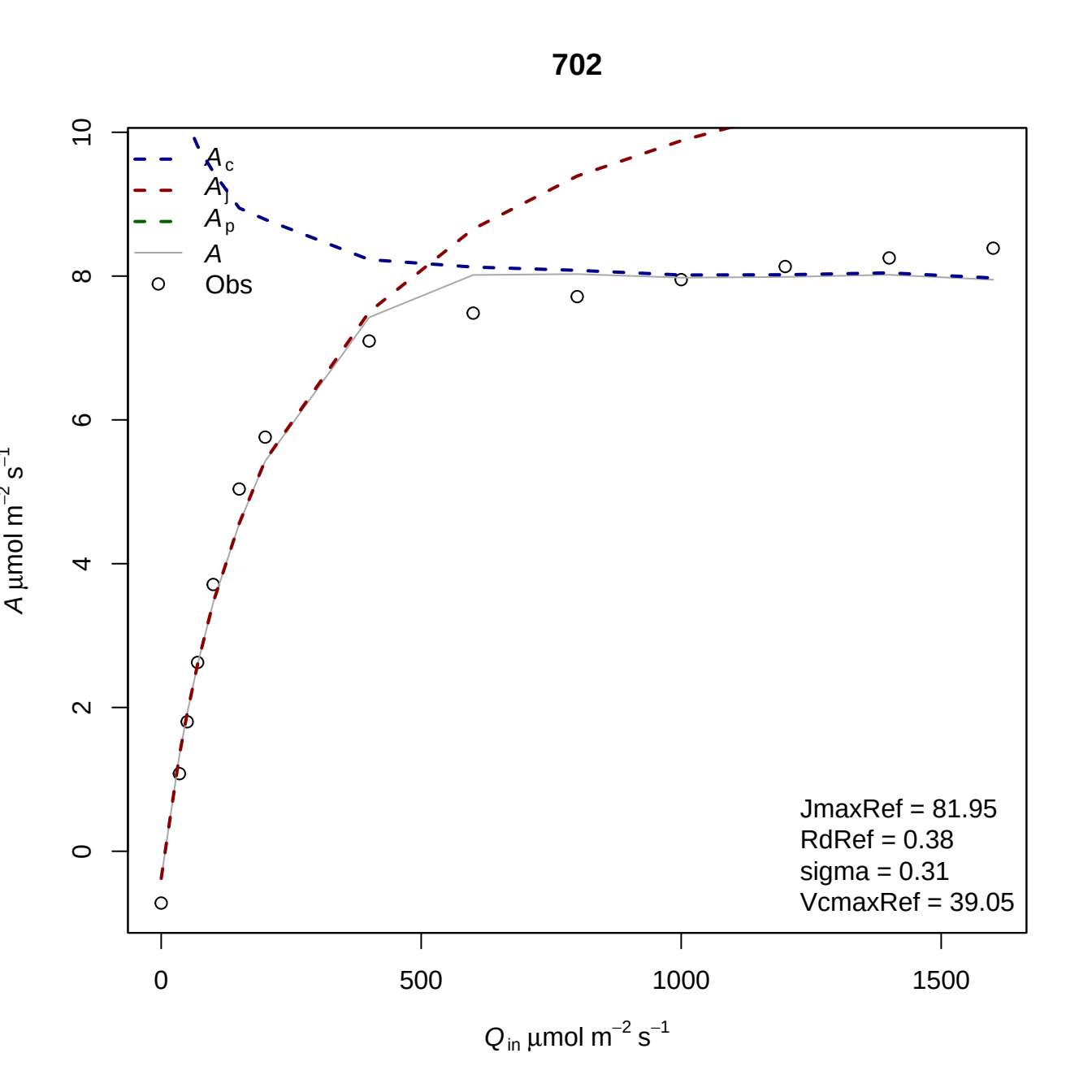

697
A_c
A_j
A_p
A
Obs
JmaxRef = 104.99
RdRef = −1.61
sigma = 0.3
VcmaxRef = 78.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

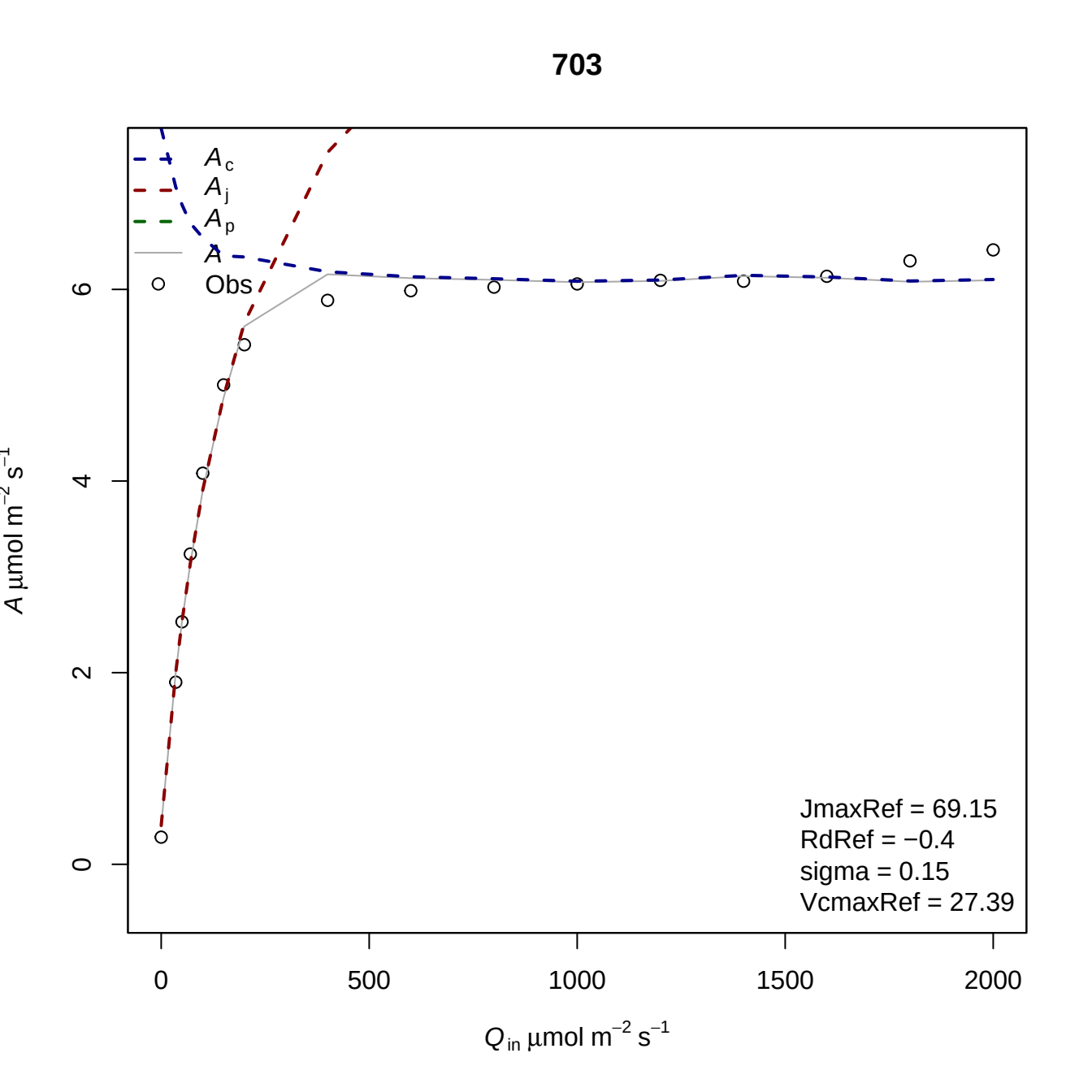

698
A_c
A_j
A_p
A
Obs
JmaxRef = 44.86
RdRef = 0.33
sigma = 0.2
VcmaxRef = 26.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

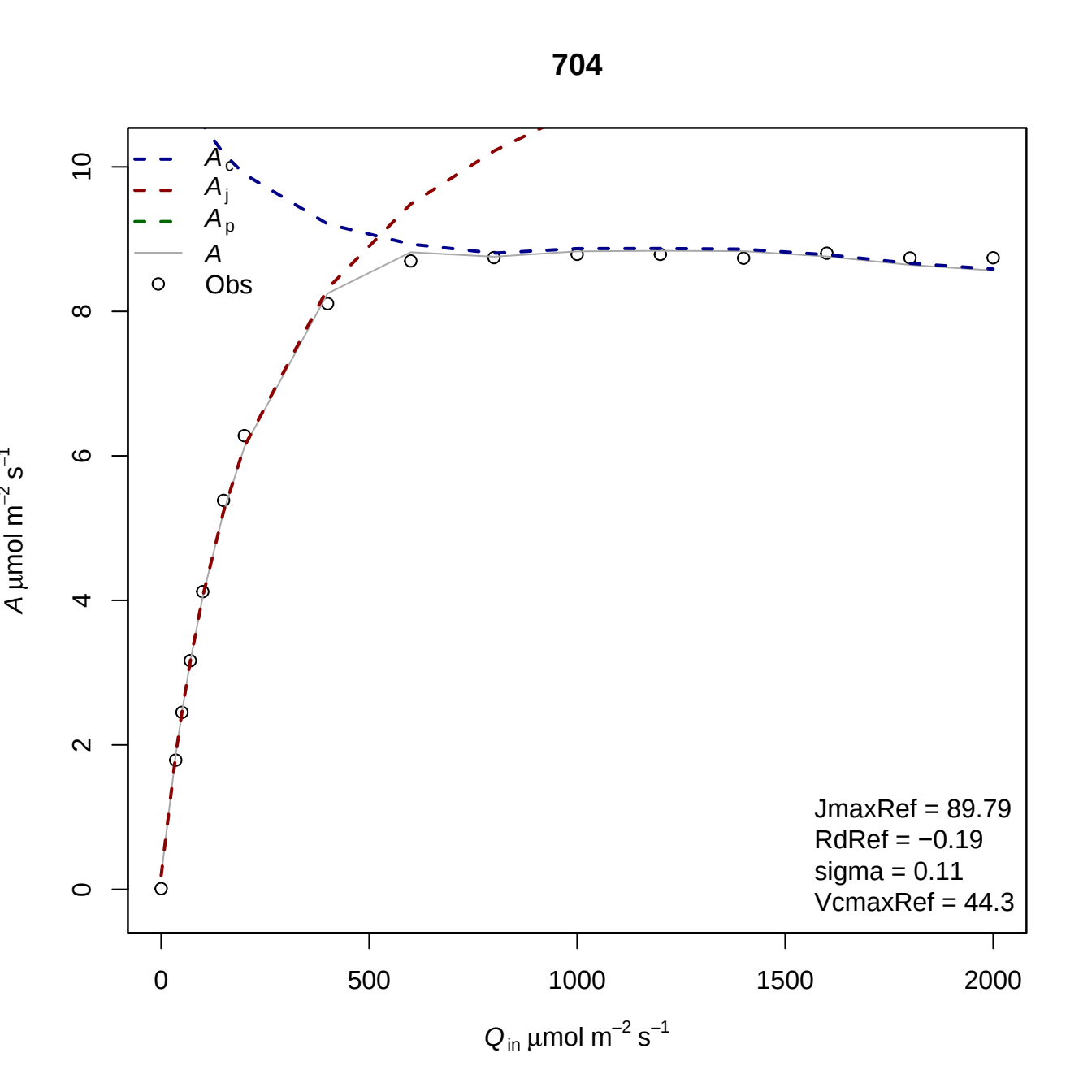

699
A_c
A_j
A_p
A
Obs
JmaxRef = 43.32
RdRef = 0.13
sigma = 0.14
VcmaxRef = 44.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

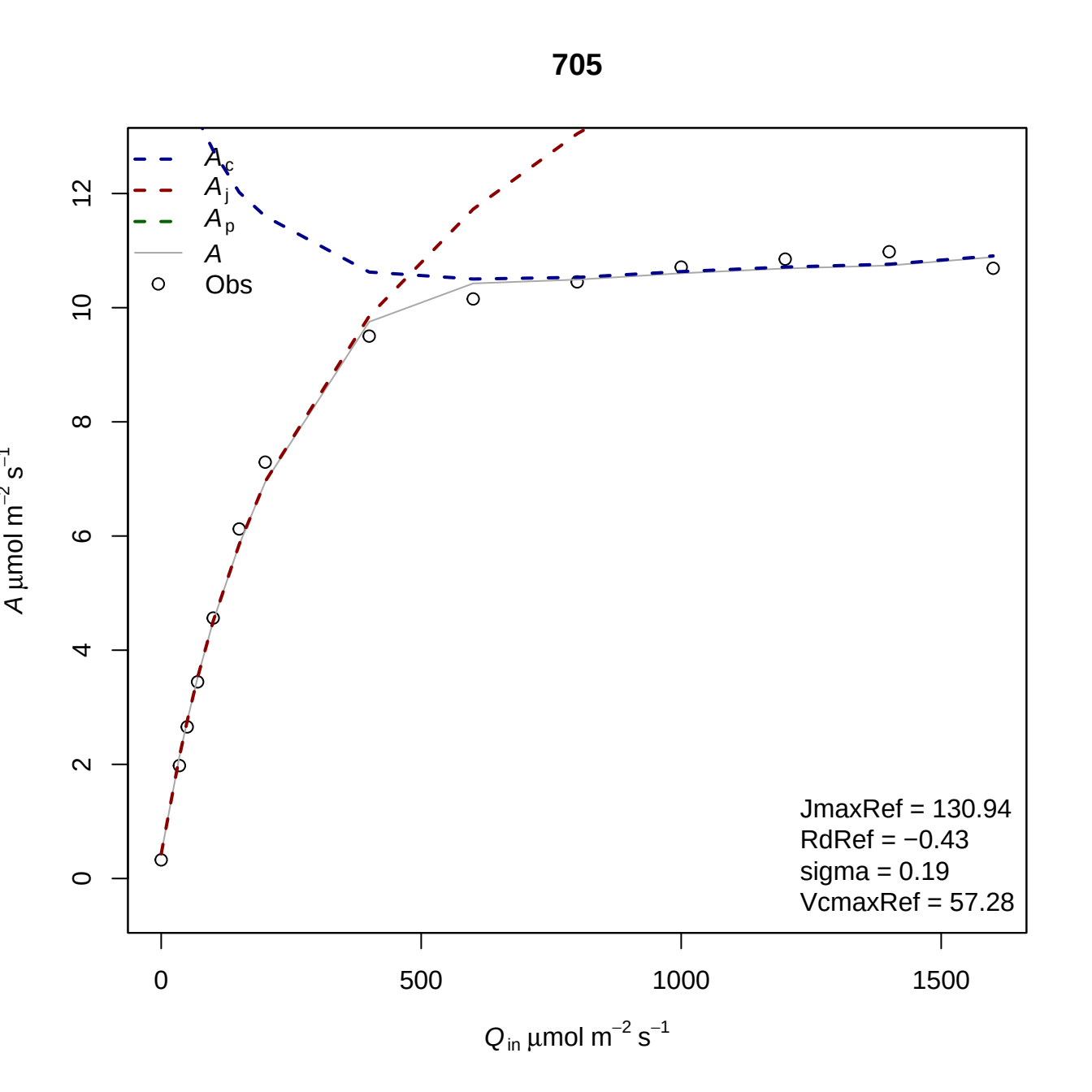

700
A_c
A_j
A_p
A
Obs
JmaxRef = 106.05
RdRef = 0.3
sigma = 0.28
VcmaxRef = 72.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

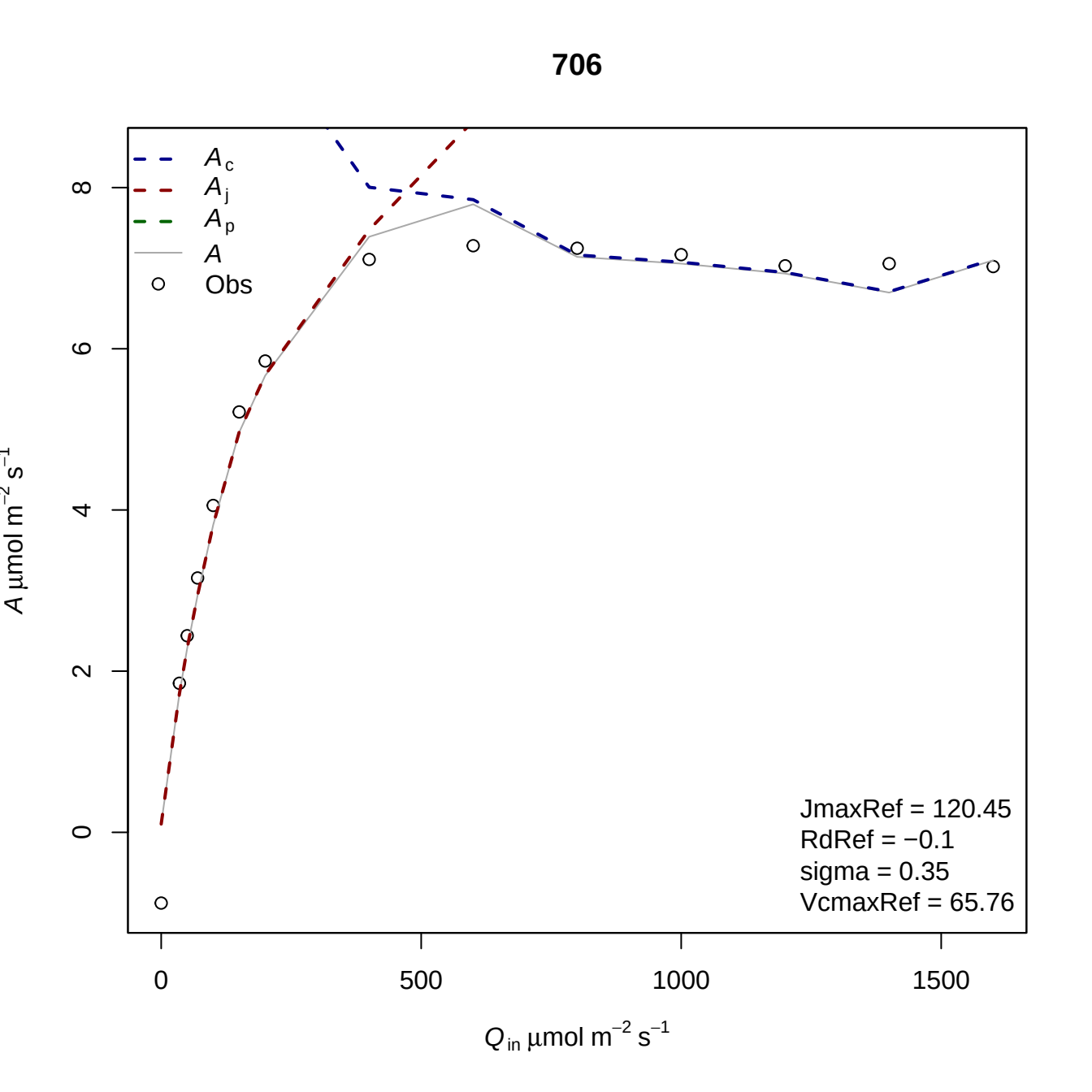

701
A_c
A_j
A_p
A
Obs
JmaxRef = 143.73
RdRef = −0.31
sigma = 0.28
VcmaxRef = 51.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

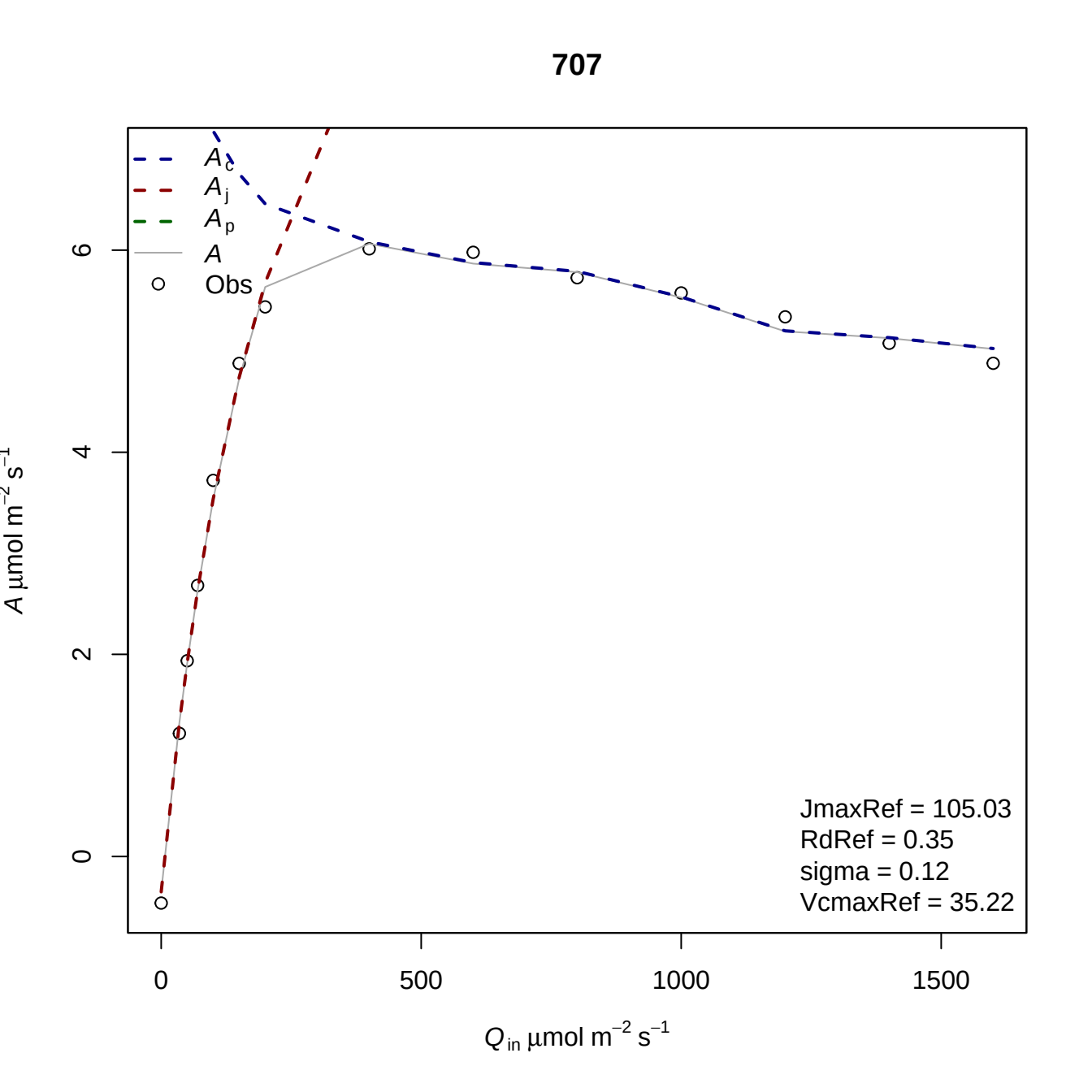

702
A_c
A_j
A_p
A
Obs
JmaxRef = 81.95
RdRef = 0.38
sigma = 0.31
VcmaxRef = 39.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

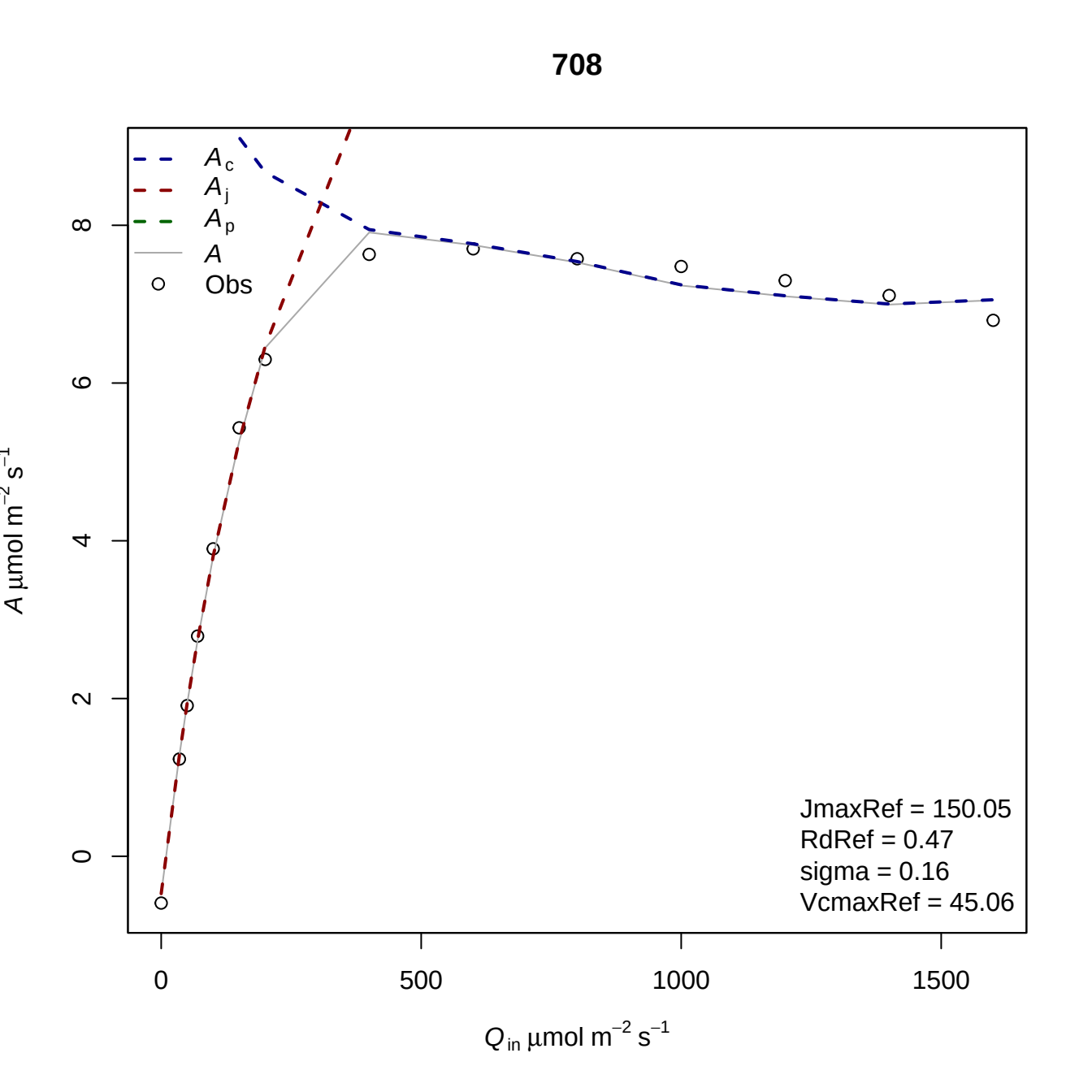

703
A_c
A_j
A_p
A
Obs
JmaxRef = 69.15
RdRef = −0.4
sigma = 0.15
VcmaxRef = 27.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

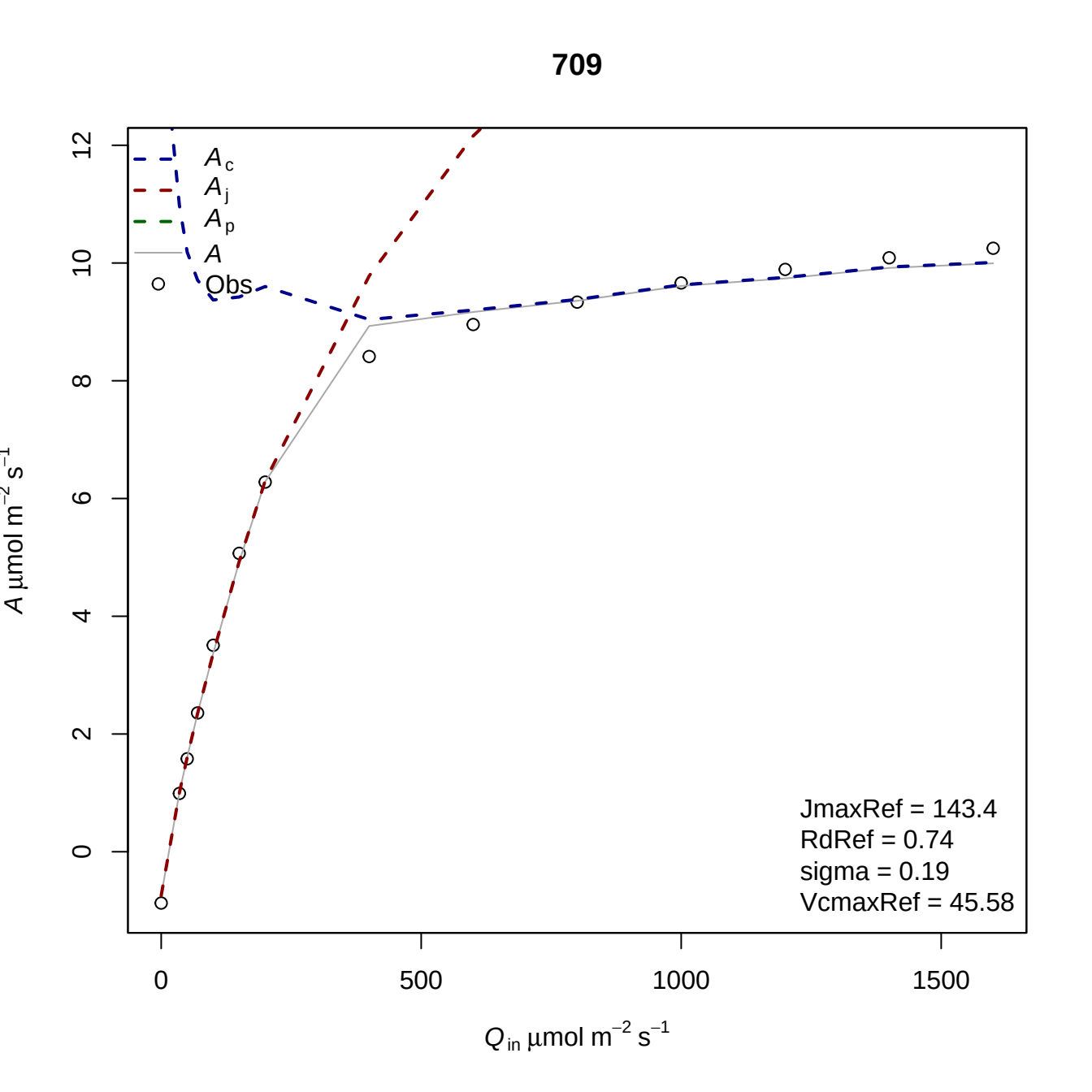

704
A_c
A_j
A_p
A
Obs
JmaxRef = 89.79
RdRef = −0.19
sigma = 0.11
VcmaxRef = 44.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

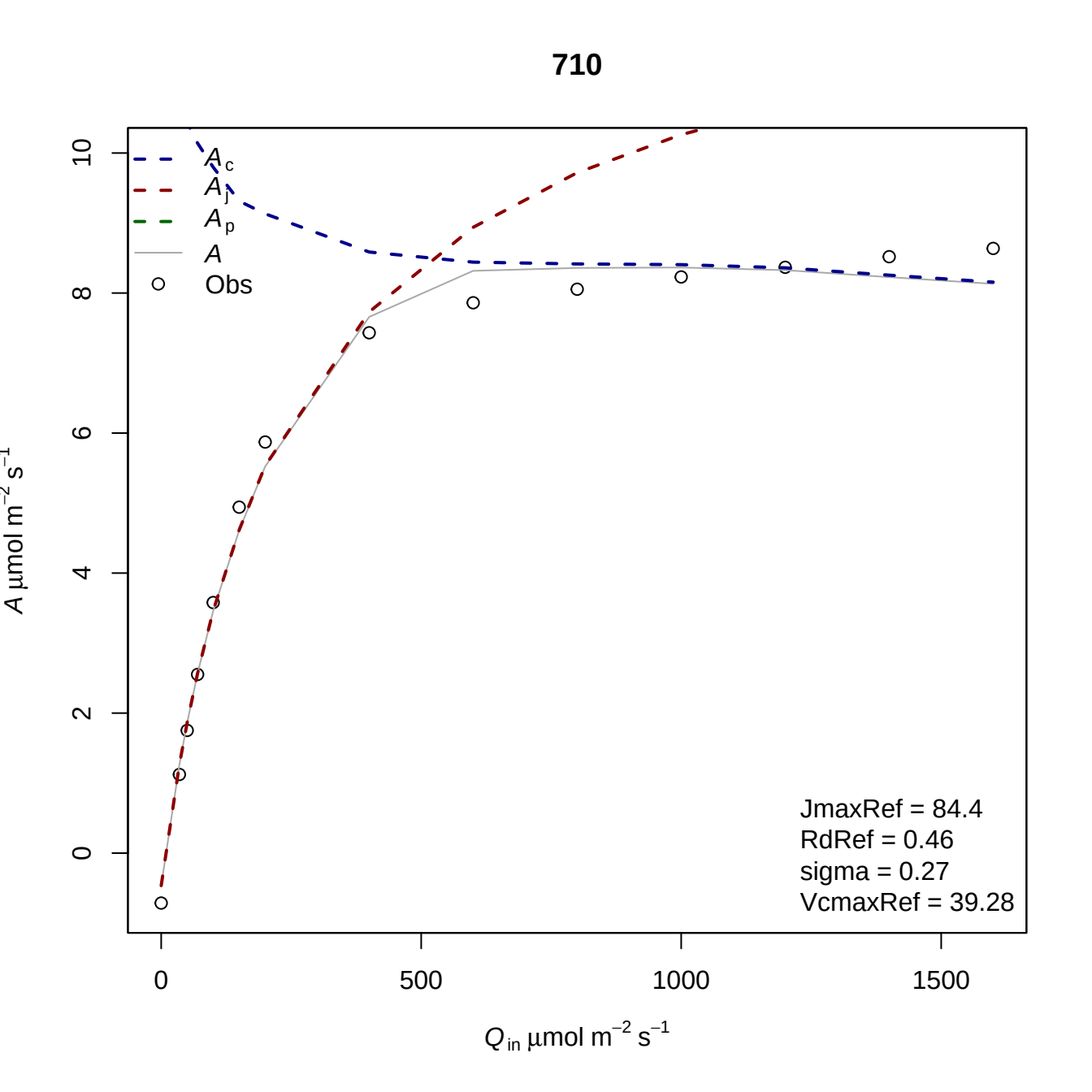

705
A_c
A_j
A_p
A
Obs
JmaxRef = 130.94
RdRef = −0.43
sigma = 0.19
VcmaxRef = 57.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

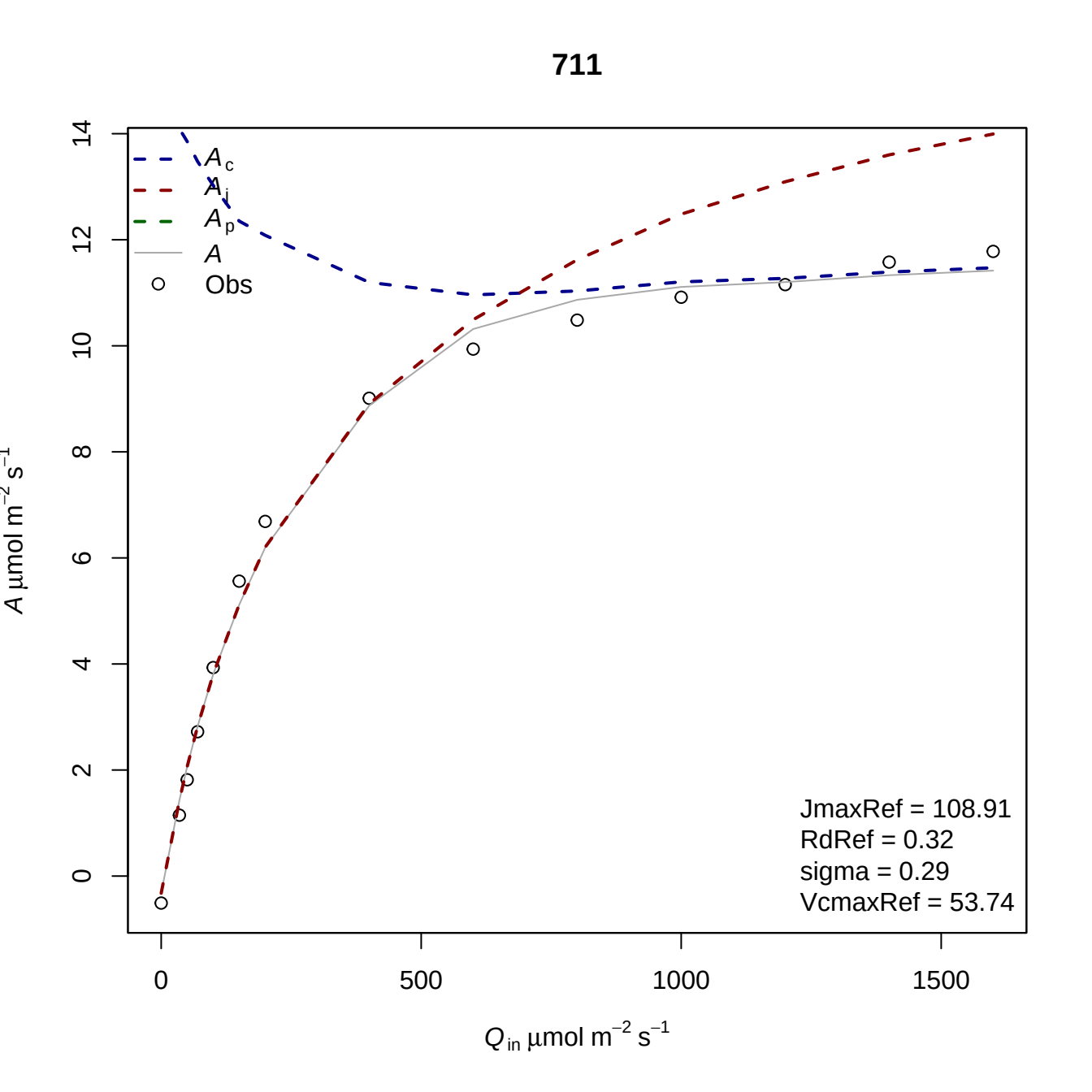

706
A_c
A_j
A_p
A
Obs
JmaxRef = 120.45
RdRef = −0.1
sigma = 0.35
VcmaxRef = 65.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

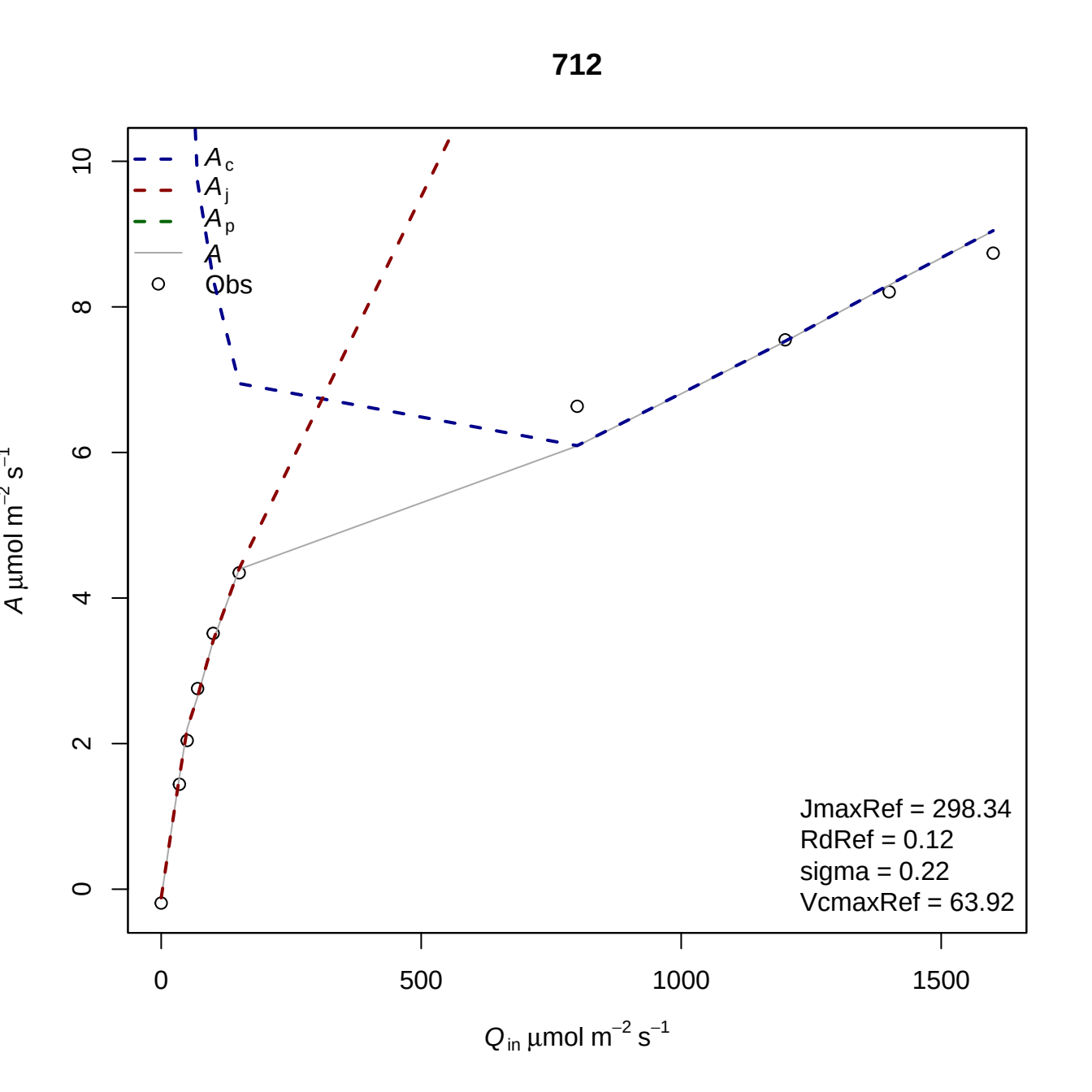

707
A_c
A_j
A_p
A
Obs
JmaxRef = 105.03
RdRef = 0.35
sigma = 0.12
VcmaxRef = 35.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

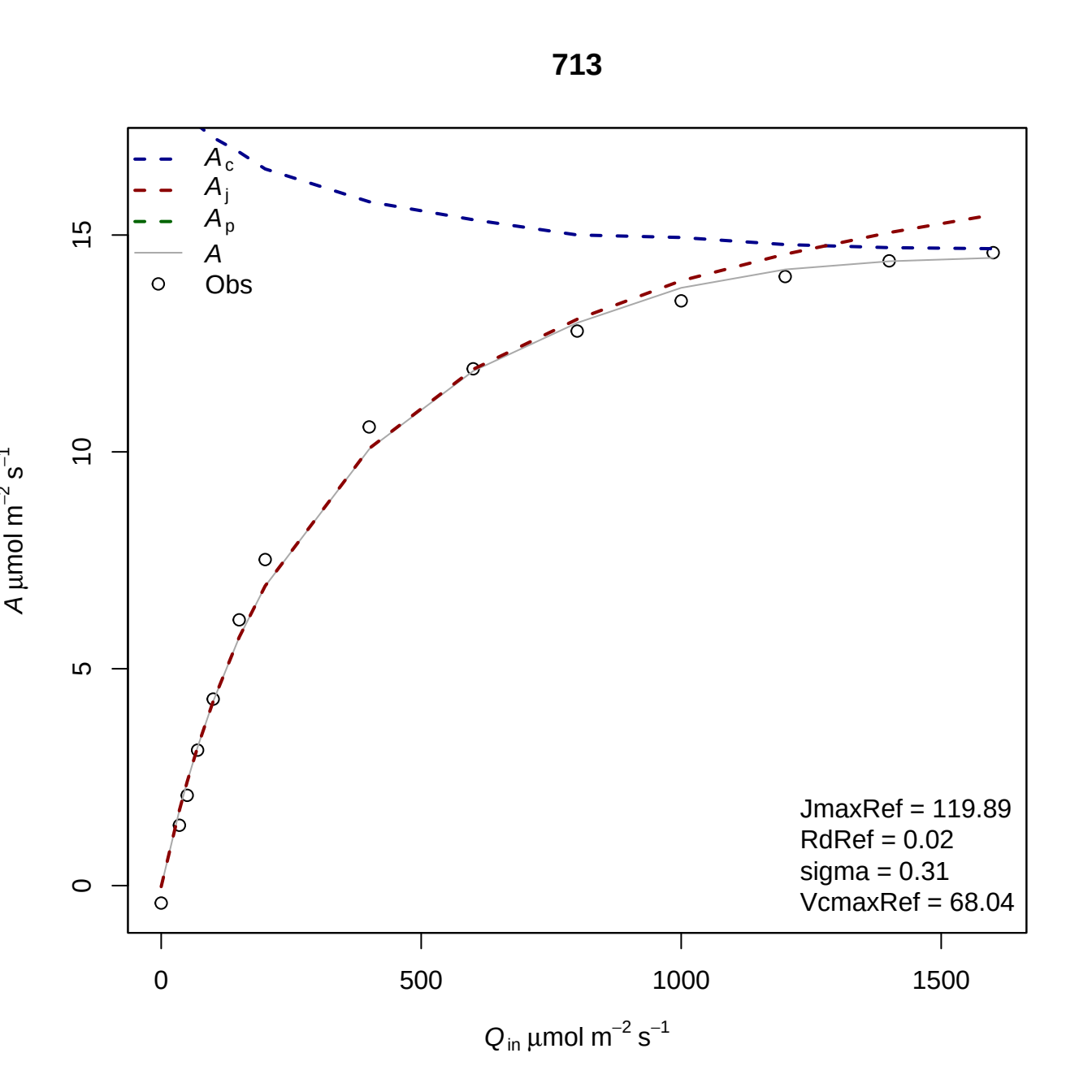

708
A_c
A_j
A_p
A
Obs
JmaxRef = 150.05
RdRef = 0.47
sigma = 0.16
VcmaxRef = 45.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

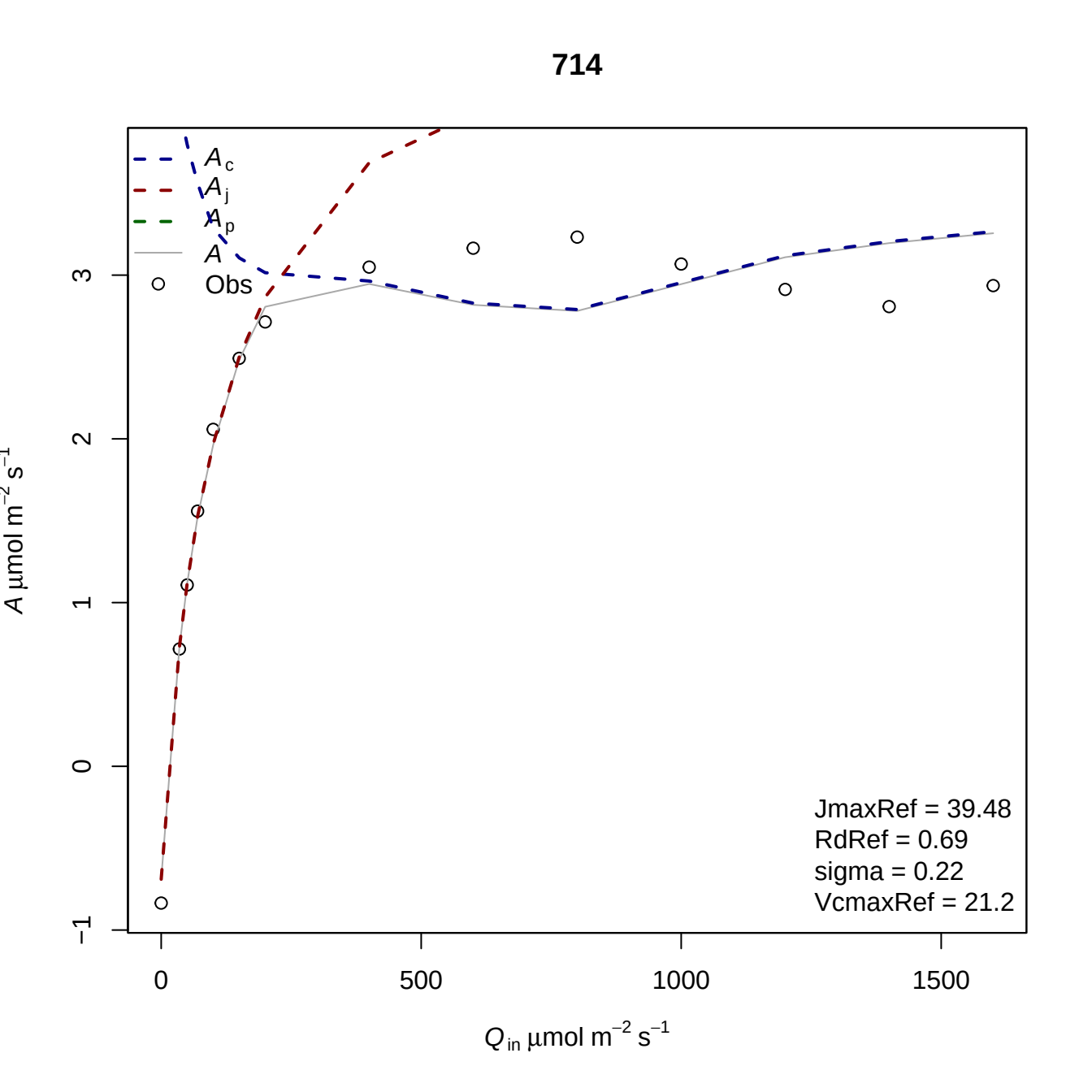

709
A_c
A_j
A_p
A
Obs
JmaxRef = 143.4
RdRef = 0.74
sigma = 0.19
VcmaxRef = 45.58
0
2
4
6
8
10
12
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

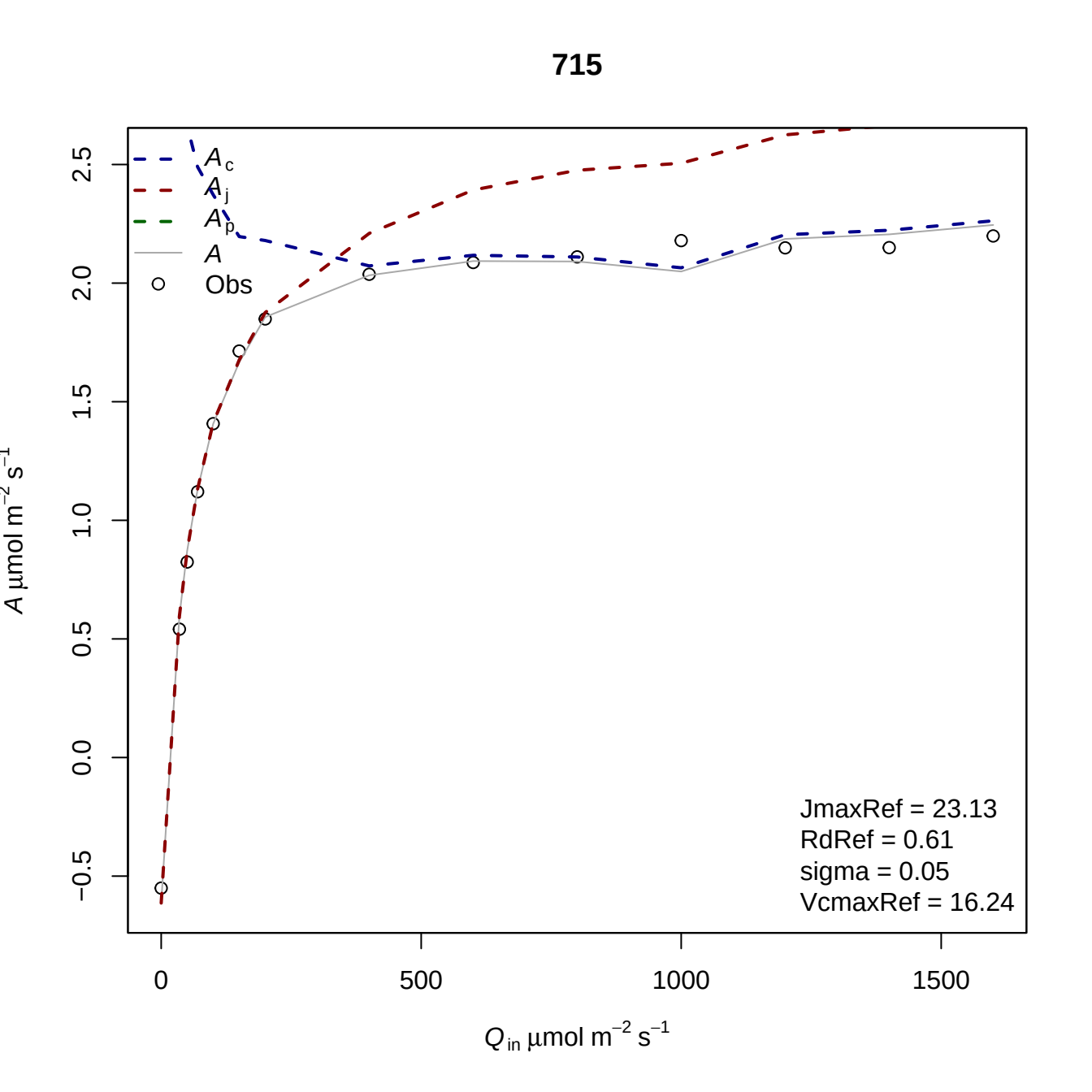

710
A_c
A_j
A_p
A
Obs
JmaxRef = 84.4
RdRef = 0.46
sigma = 0.27
VcmaxRef = 39.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

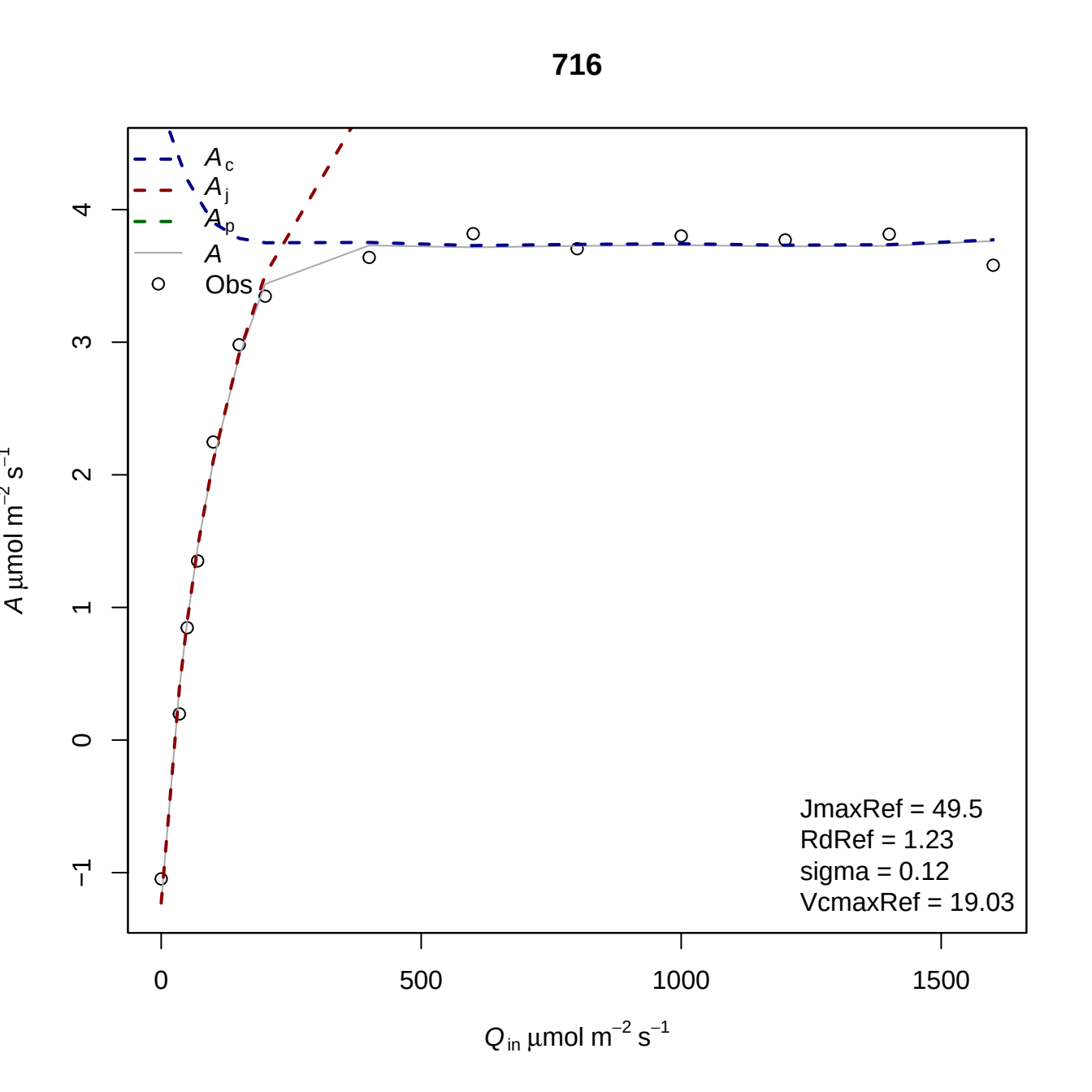

711
A_c
A_j
A_p
A
Obs
JmaxRef = 108.91
RdRef = 0.32
sigma = 0.29
VcmaxRef = 53.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

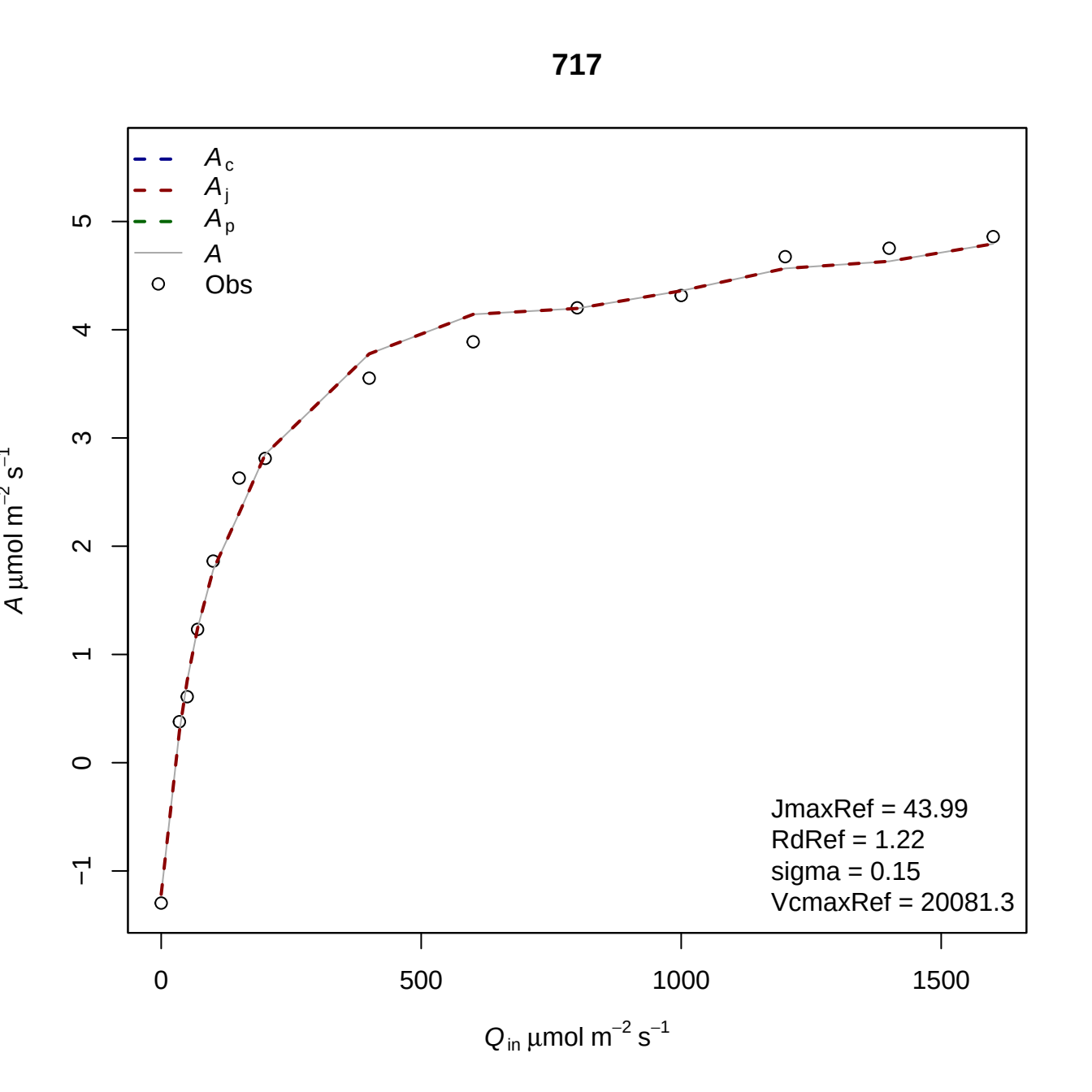

712
A_c
A_j
A_p
A
Obs
JmaxRef = 298.34
RdRef = 0.12
sigma = 0.22
VcmaxRef = 63.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

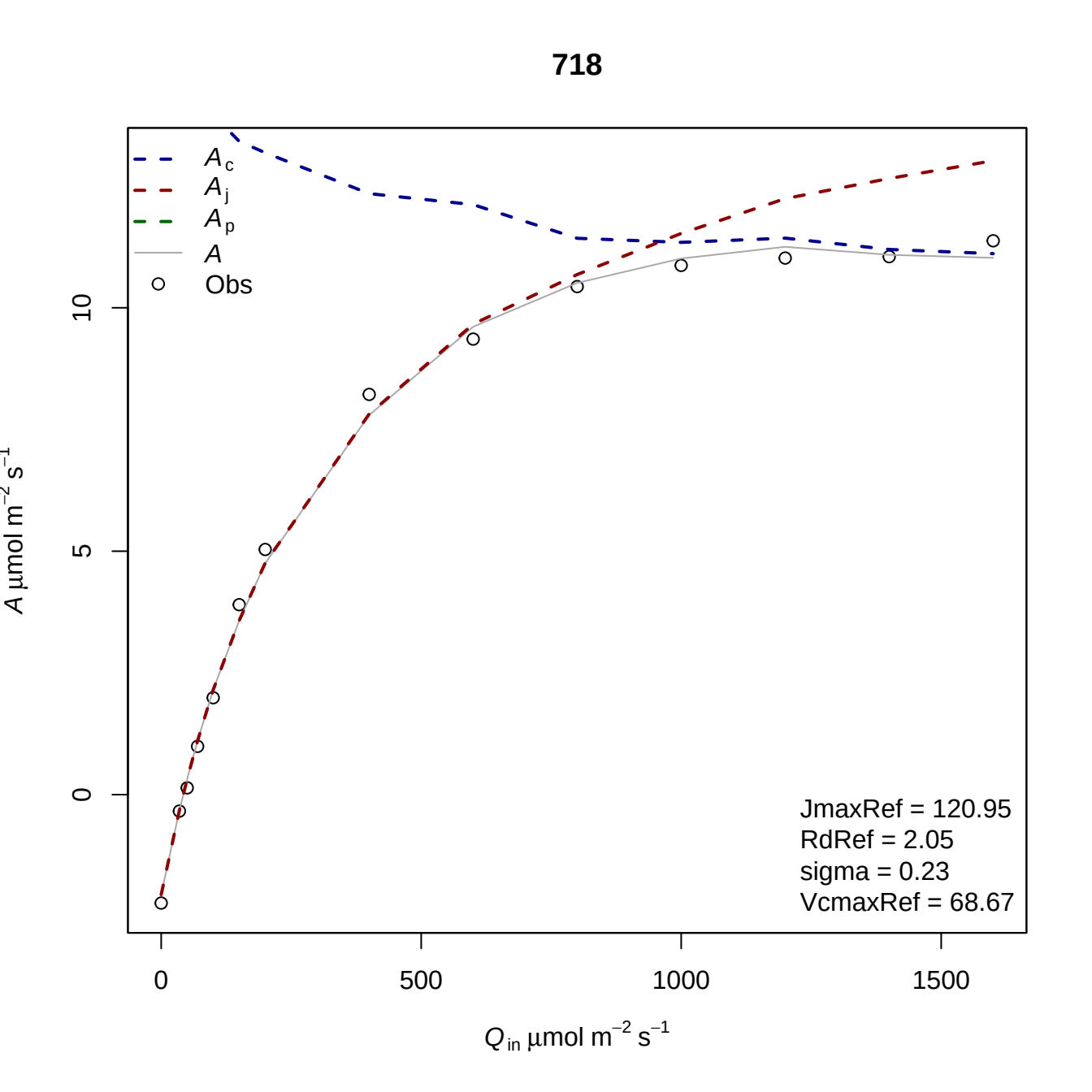

713
A_c
A_j
A_p
A
Obs
JmaxRef = 119.89
RdRef = 0.02
sigma = 0.31
VcmaxRef = 68.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

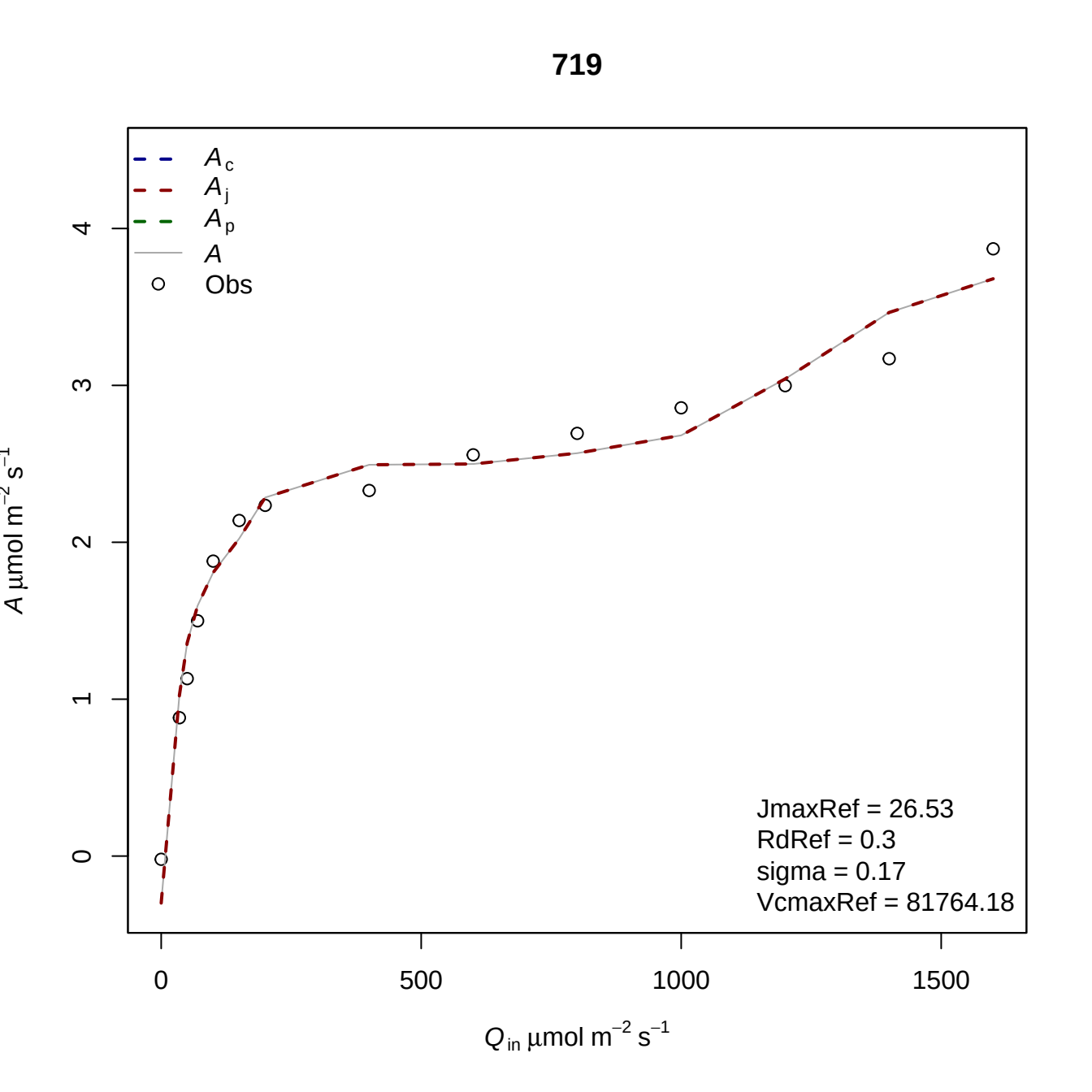

714
A_c
A_j
A_p
A
Obs
JmaxRef = 39.48
RdRef = 0.69
sigma = 0.22
VcmaxRef = 21.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

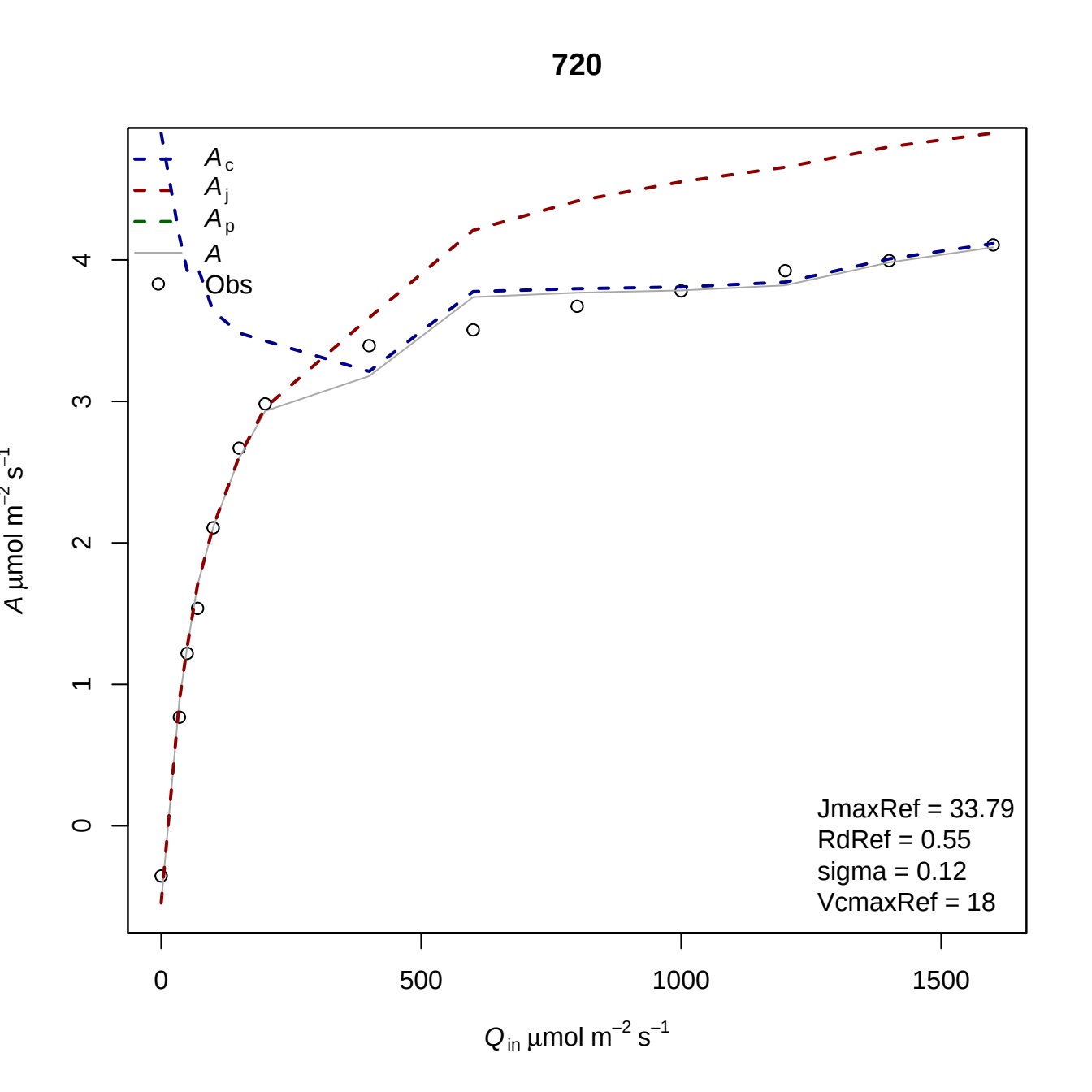

715
A_c
A_j
A_p
A
Obs
JmaxRef = 23.13
RdRef = 0.61
sigma = 0.05
VcmaxRef = 16.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500

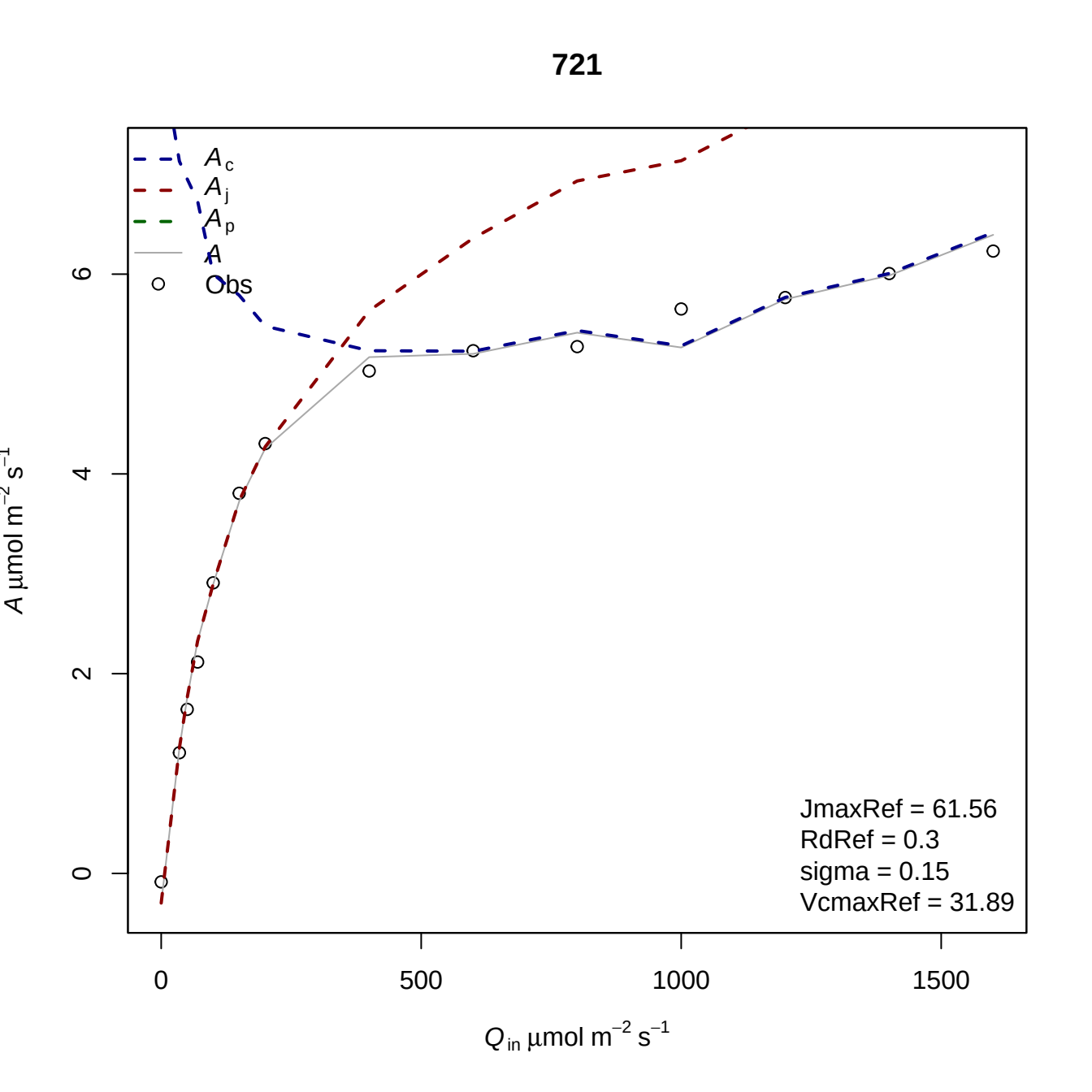

716
A_c
A_j
A_p
A
Obs
JmaxRef = 49.5
RdRef = 1.23
sigma = 0.12
VcmaxRef = 19.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

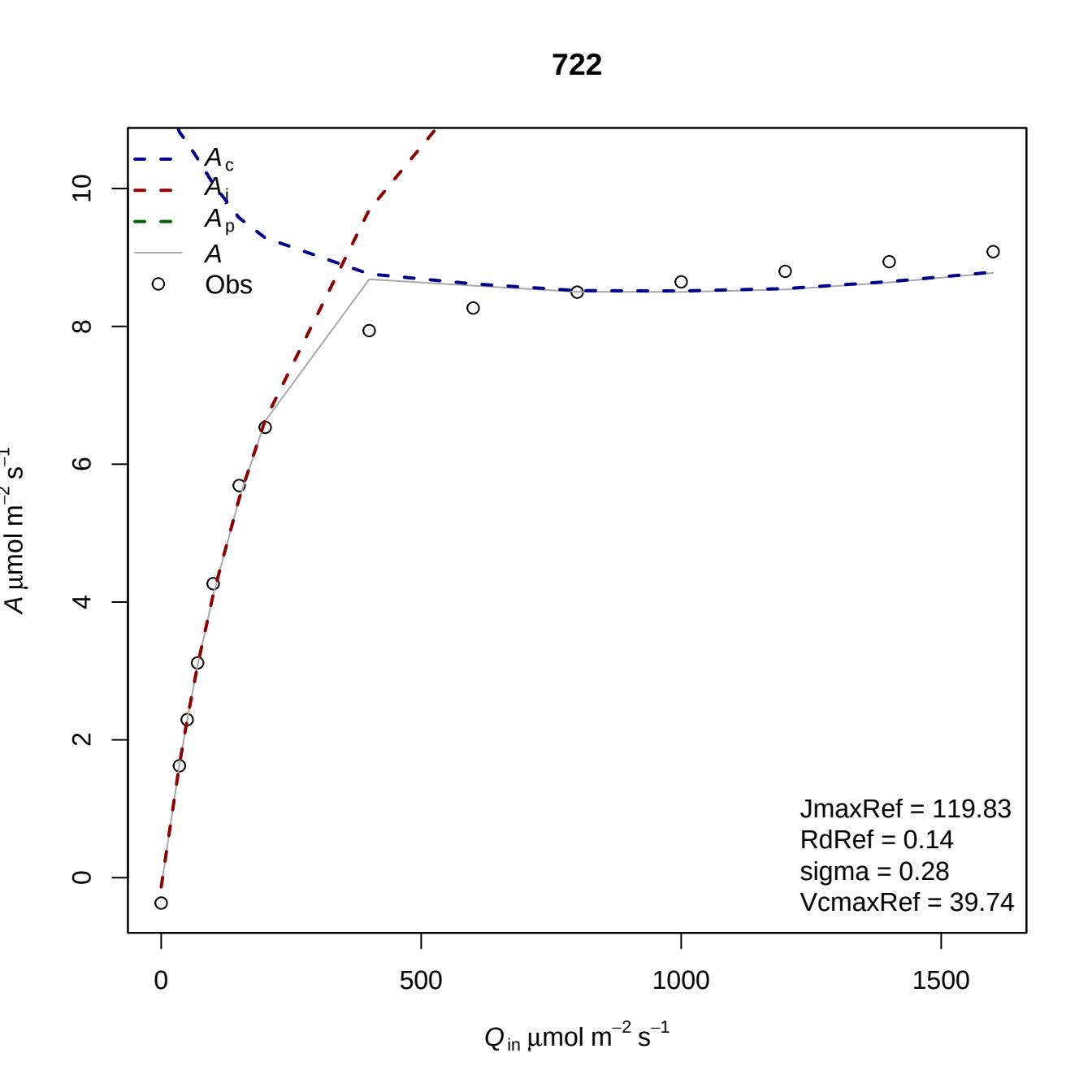

717
A_c
A_j
A_p
A
Obs
JmaxRef = 43.99
RdRef = 1.22
sigma = 0.15
VcmaxRef = 20081.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

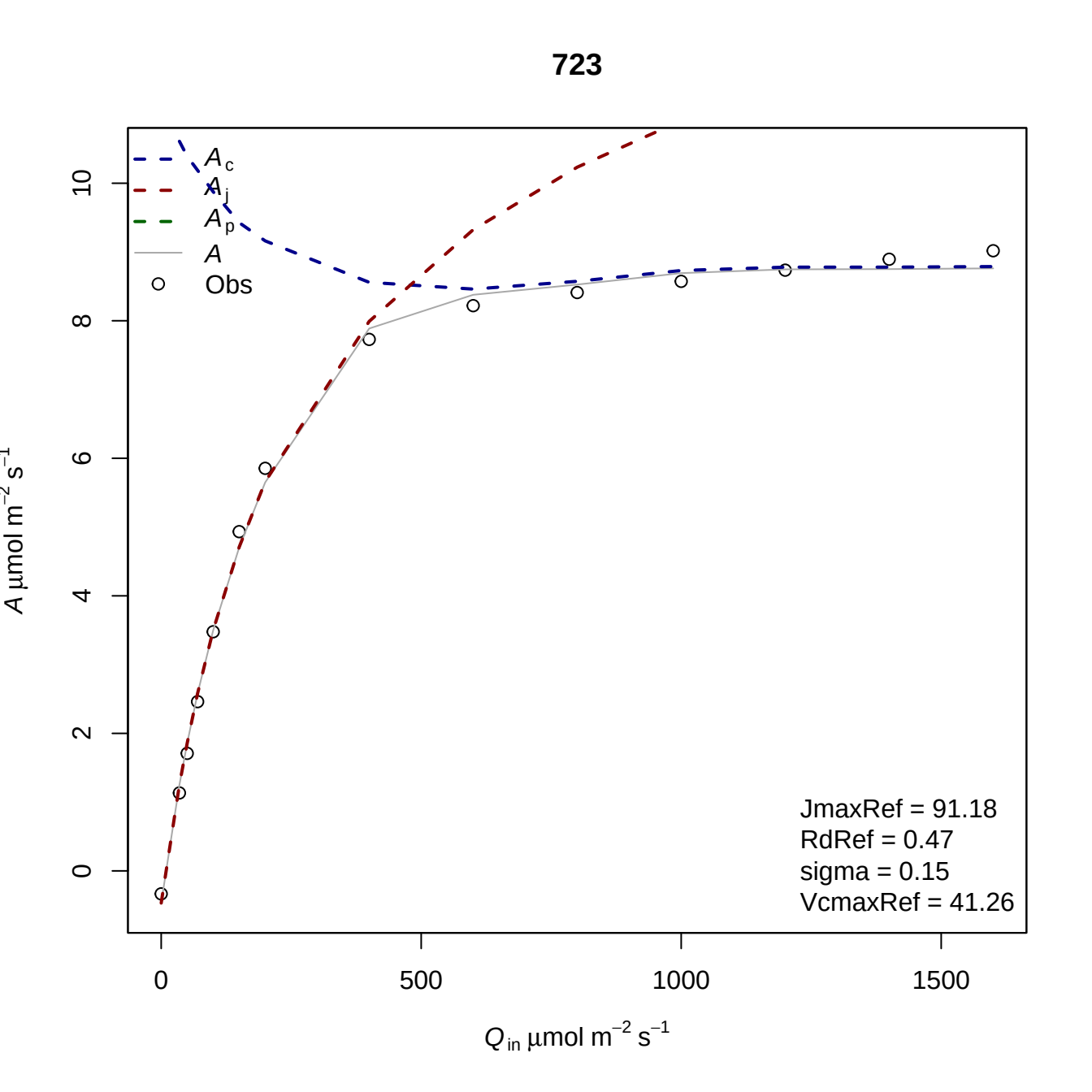

718
A_c
A_j
A_p
A
Obs
JmaxRef = 120.95
RdRef = 2.05
sigma = 0.23
VcmaxRef = 68.67
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

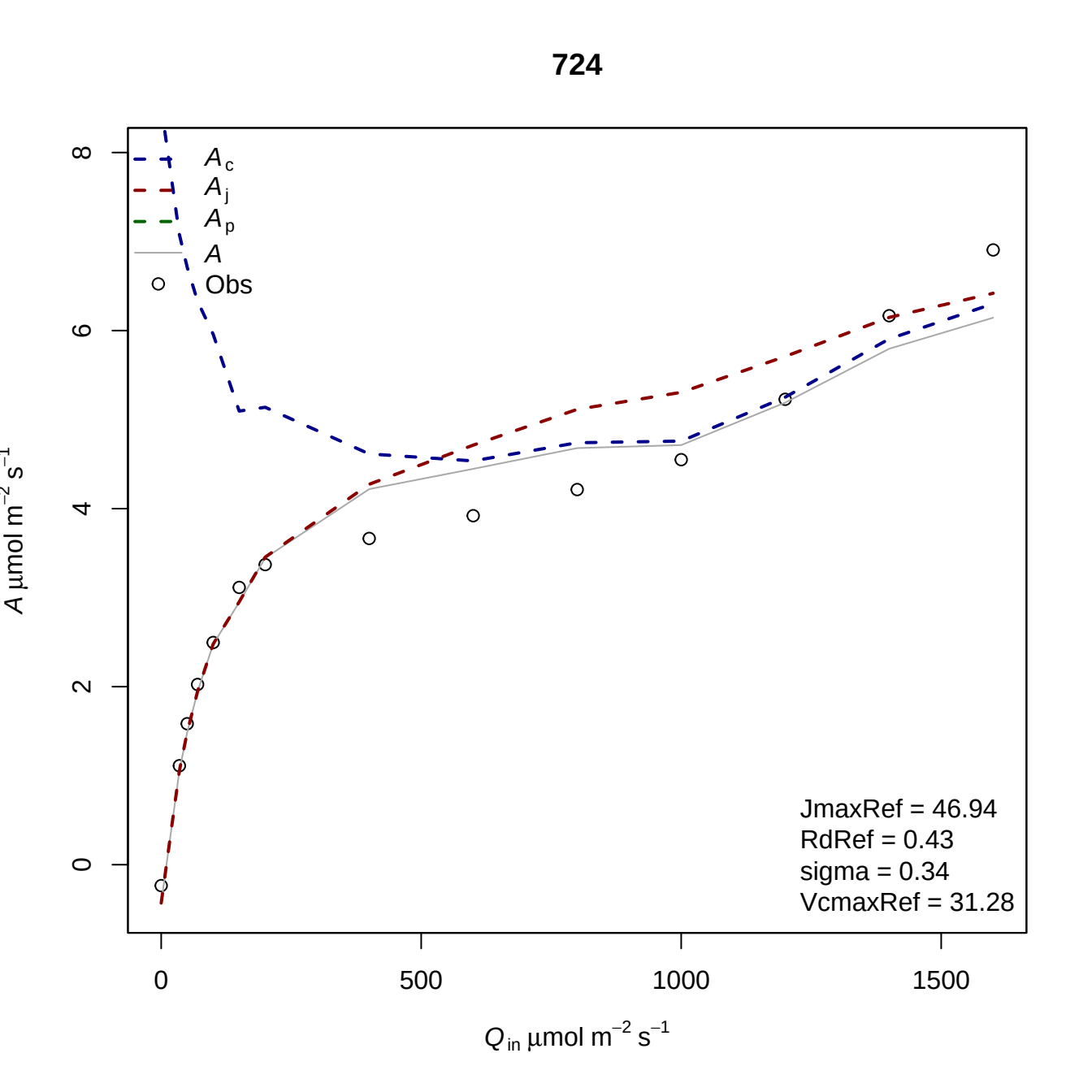

719
A_c
A_j
A_p
A
Obs
JmaxRef = 26.53
RdRef = 0.3
sigma = 0.17
VcmaxRef = 81764.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

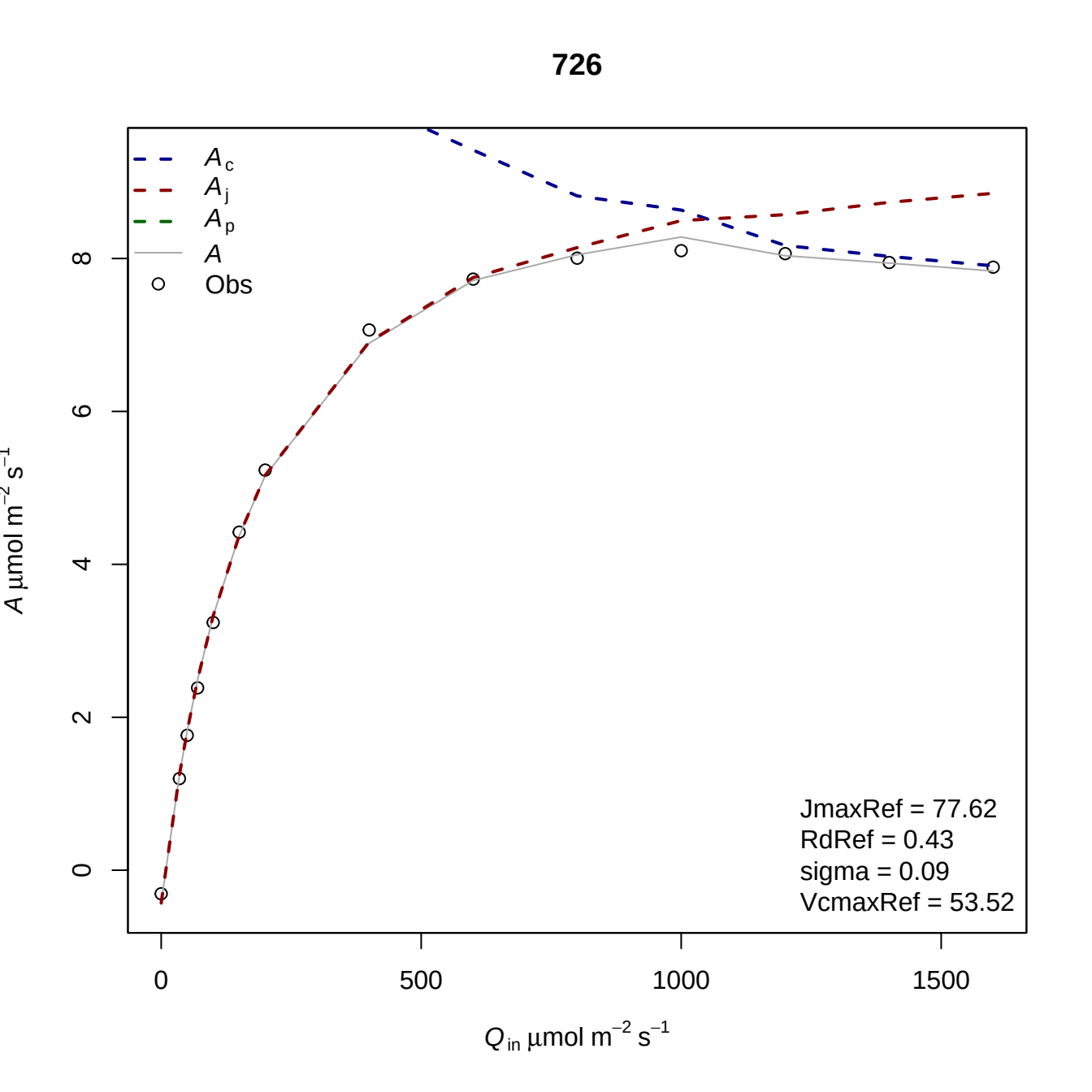

720
A_c
A_j
A_p
A
Obs
JmaxRef = 33.79
RdRef = 0.55
sigma = 0.12
VcmaxRef = 18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

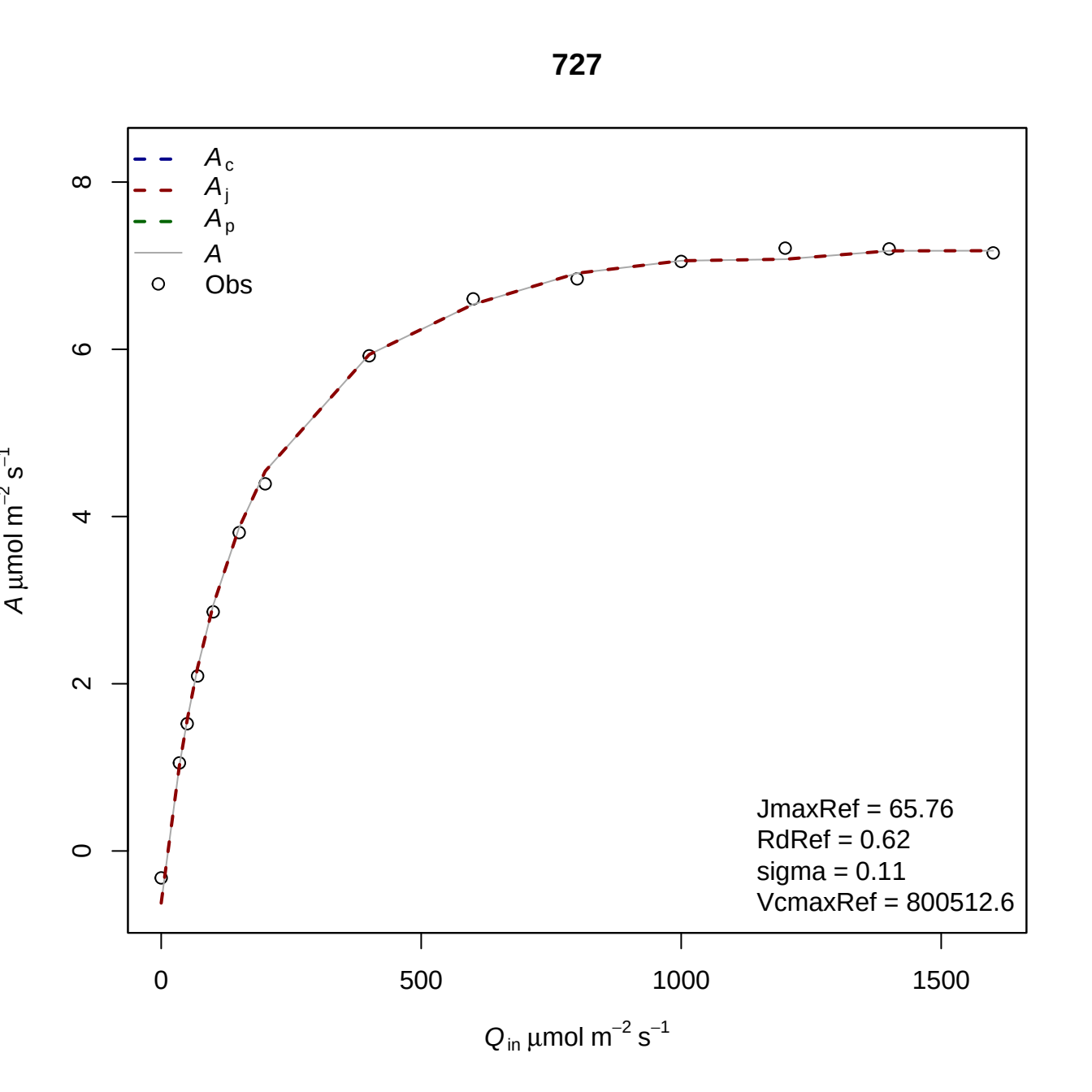

721
A_c
A_j
A_p
A
Obs
JmaxRef = 61.56
RdRef = 0.3
sigma = 0.15
VcmaxRef = 31.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

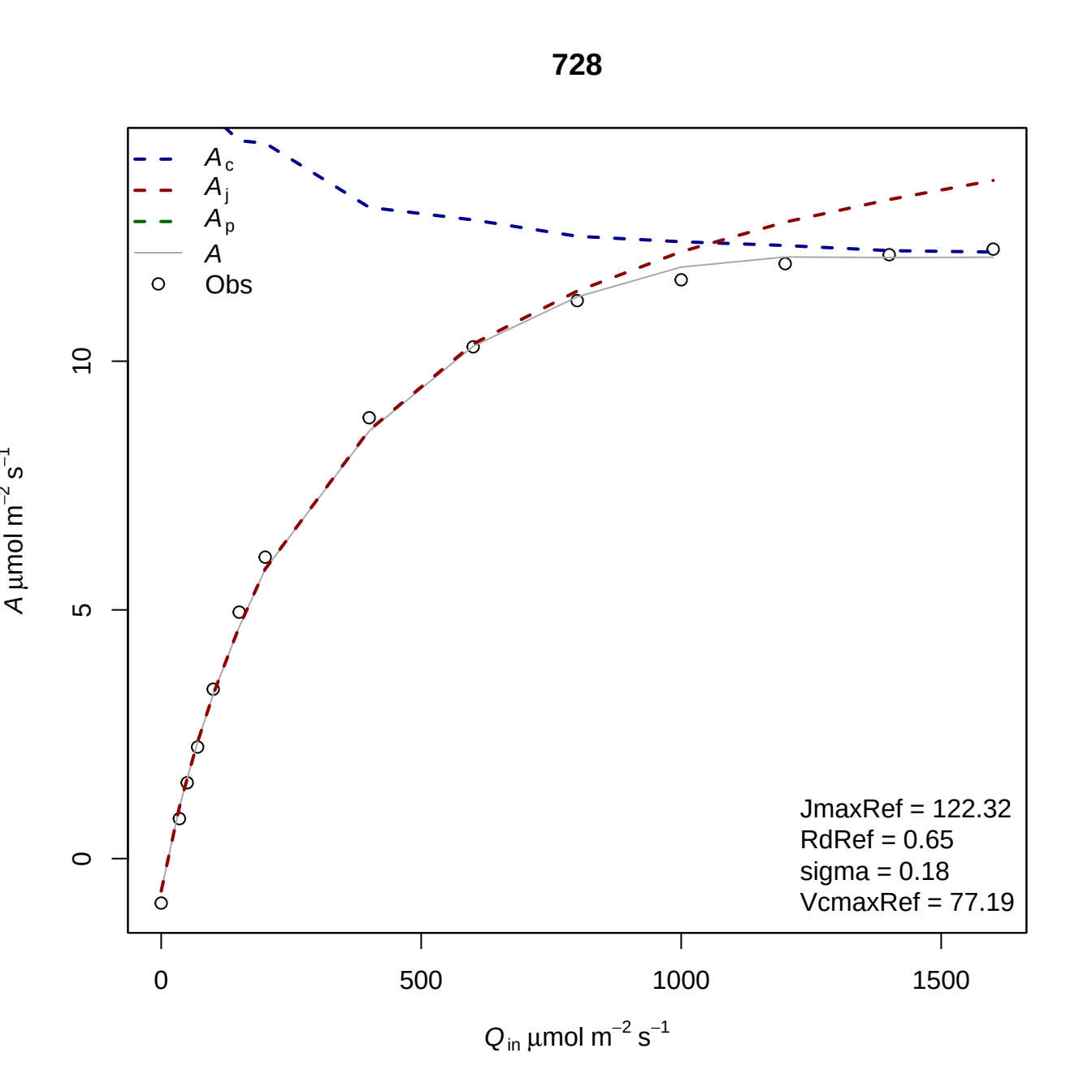

722
A_c
A_j
A_p
A
Obs
JmaxRef = 119.83
RdRef = 0.14
sigma = 0.28
VcmaxRef = 39.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

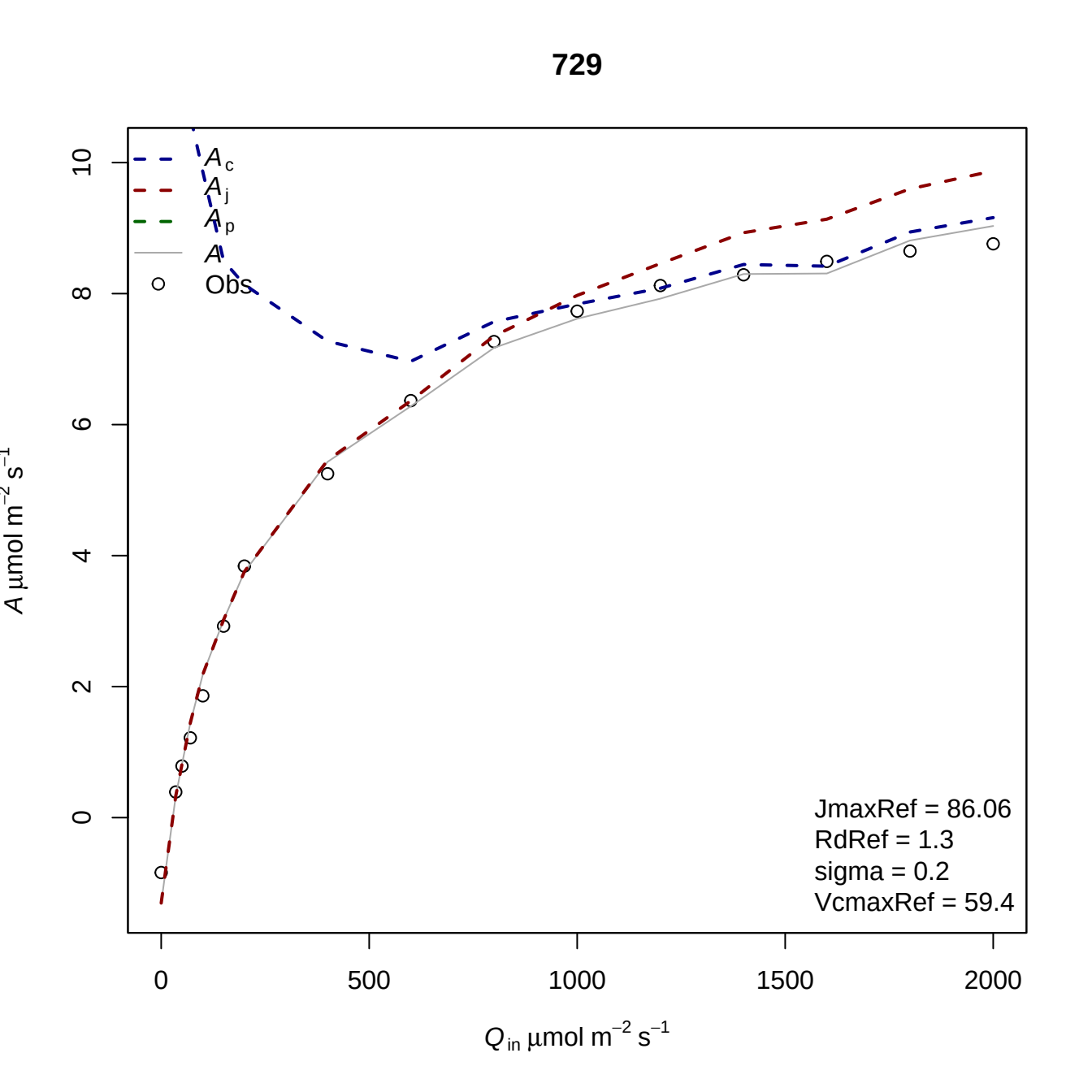

723
A_c
A_j
A_p
A
Obs
JmaxRef = 91.18
RdRef = 0.47
sigma = 0.15
VcmaxRef = 41.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

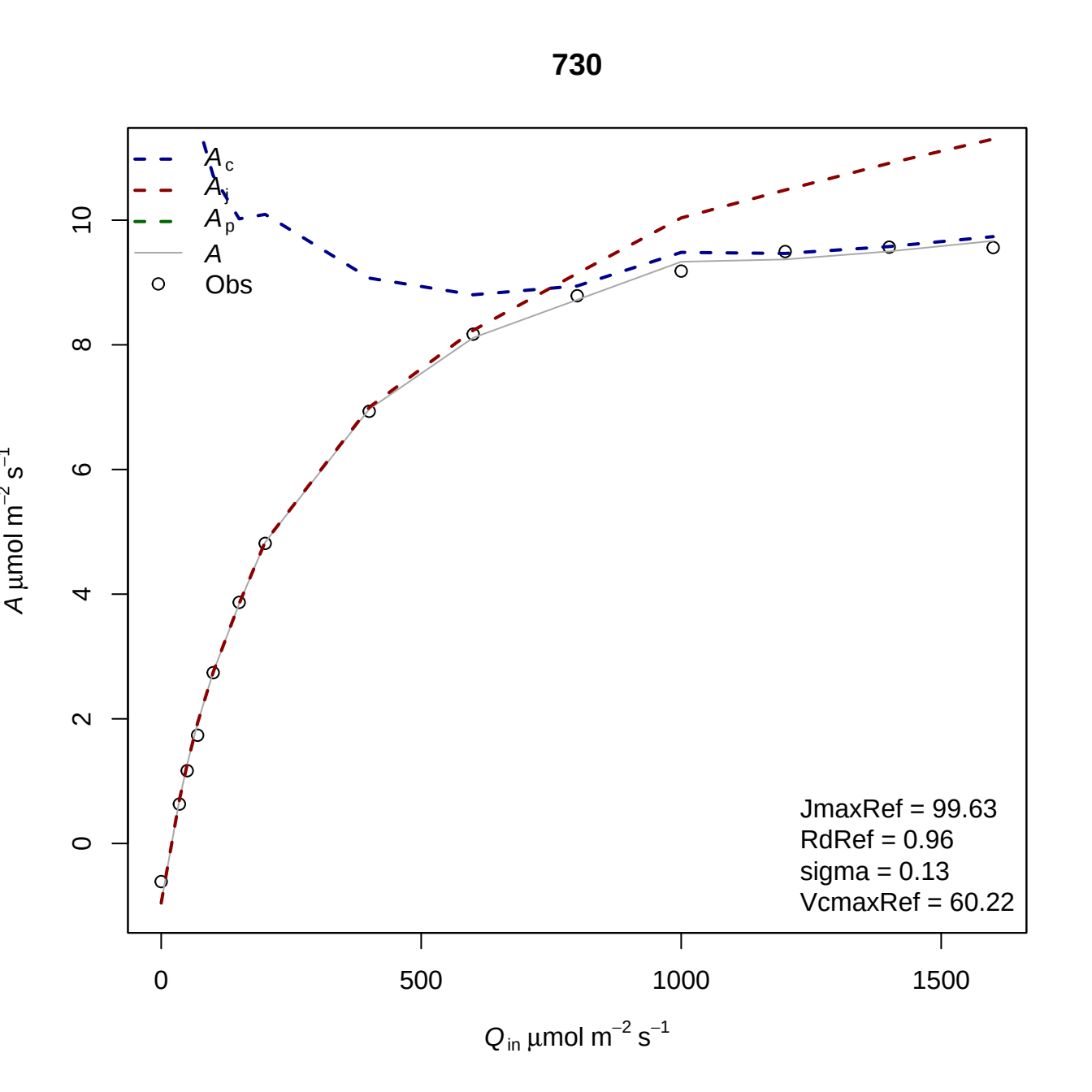

724
A_c
A_j
A_p
A
Obs
JmaxRef = 46.94
RdRef = 0.43
sigma = 0.34
VcmaxRef = 31.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

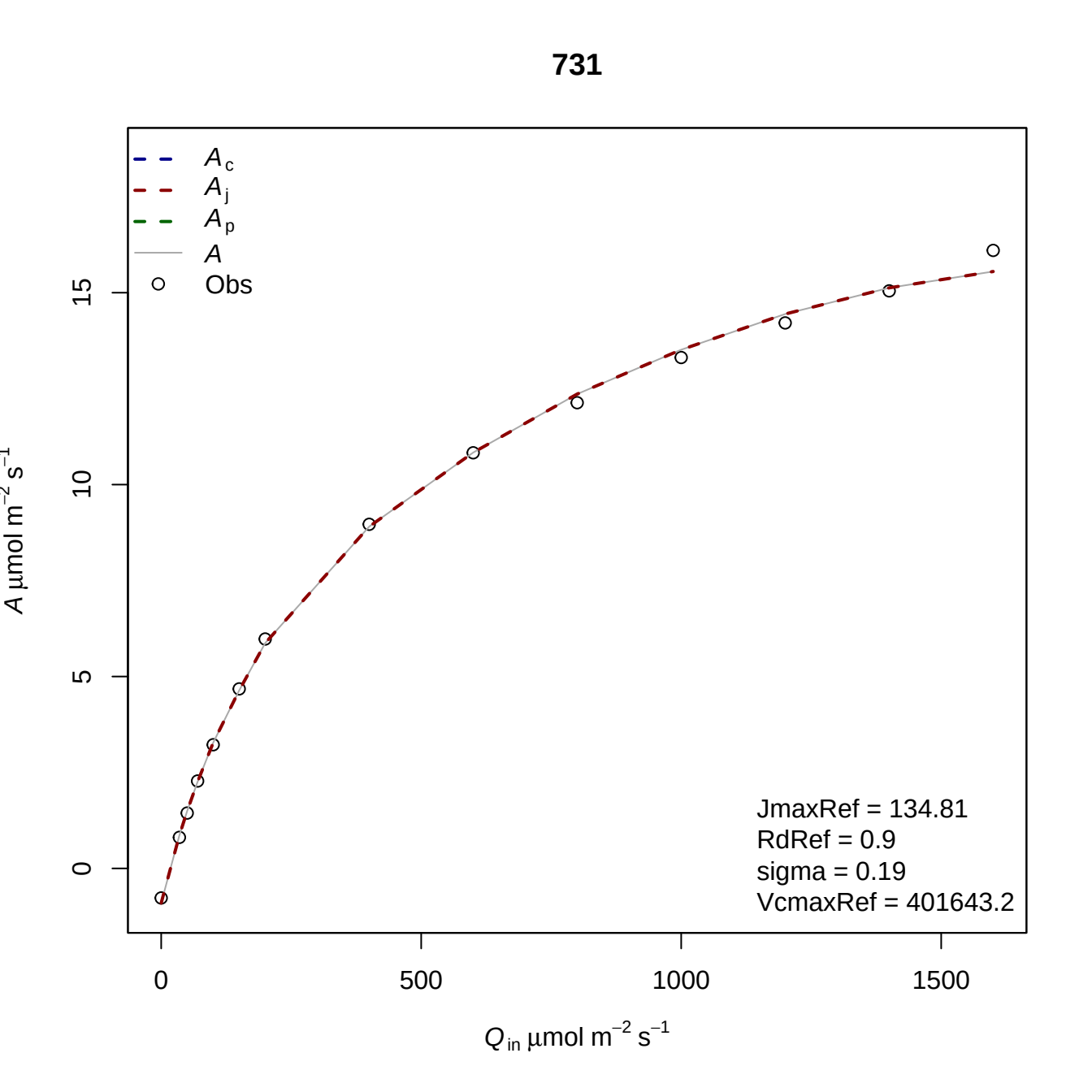

726
A_c
A_j
A_p
A
Obs
JmaxRef = 77.62
RdRef = 0.43
sigma = 0.09
VcmaxRef = 53.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

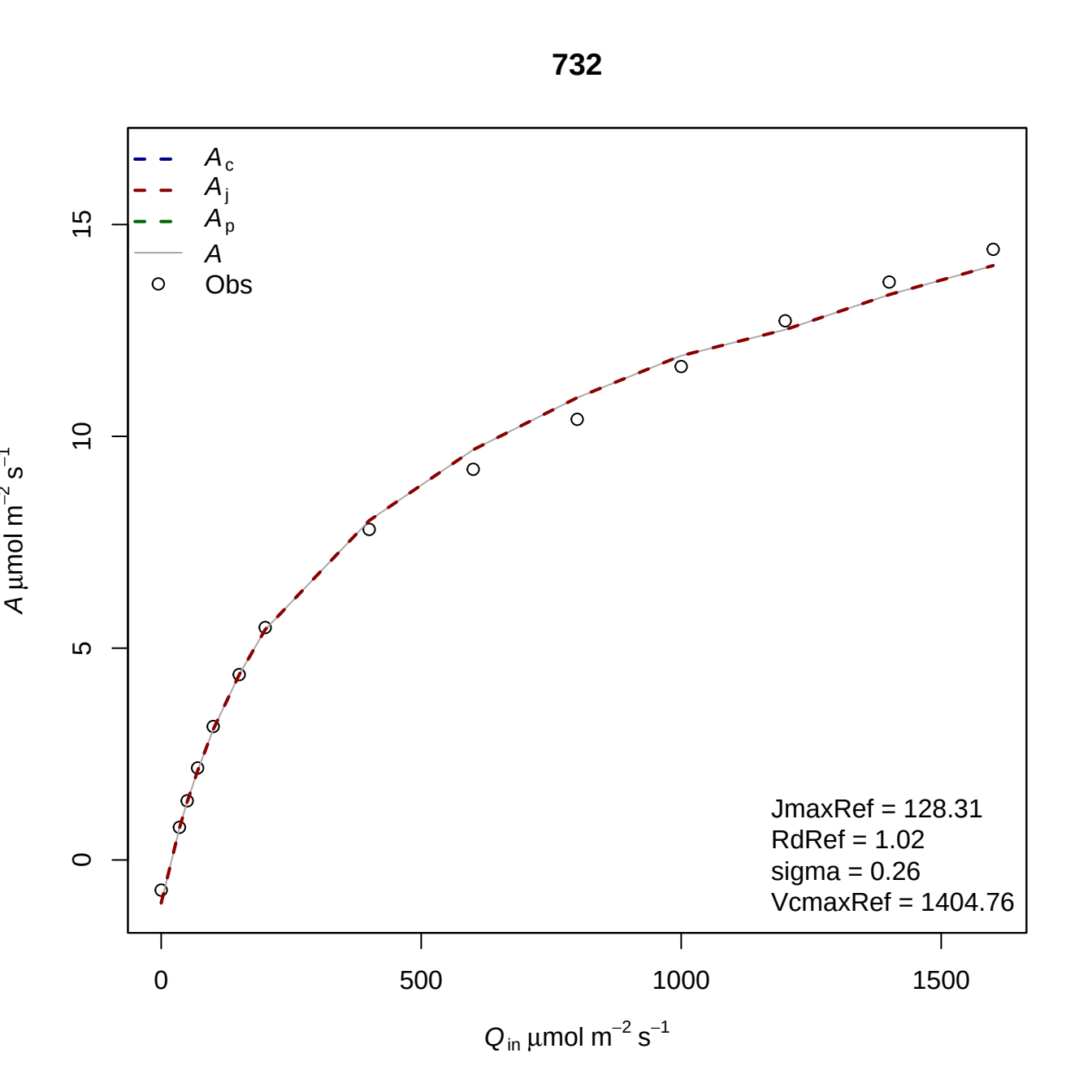

727
A_c
A_j
A_p
A
Obs
JmaxRef = 65.76
RdRef = 0.62
sigma = 0.11
VcmaxRef = 800512.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

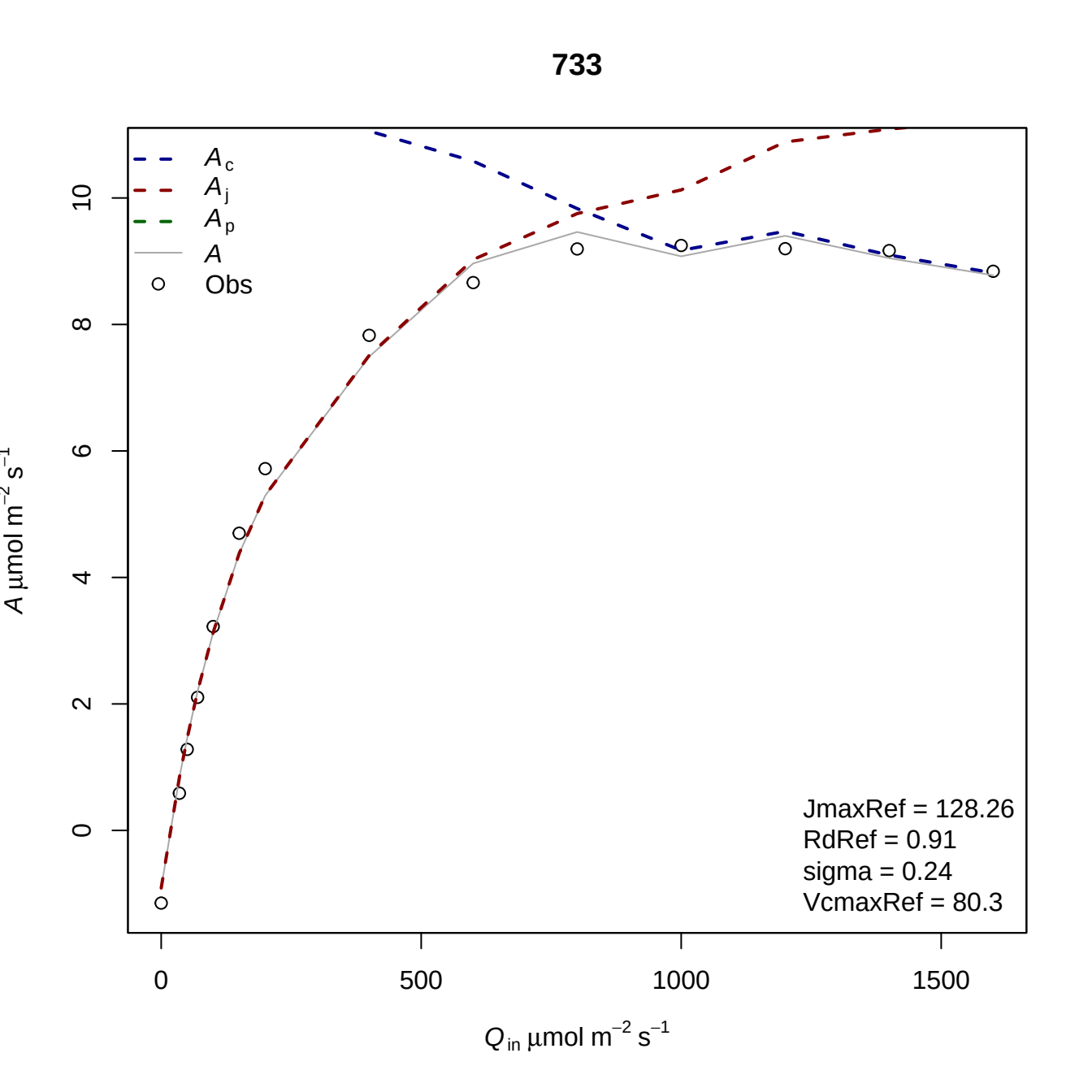

728
A_c
A_j
A_p
A
Obs
JmaxRef = 122.32
RdRef = 0.65
sigma = 0.18
VcmaxRef = 77.19
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

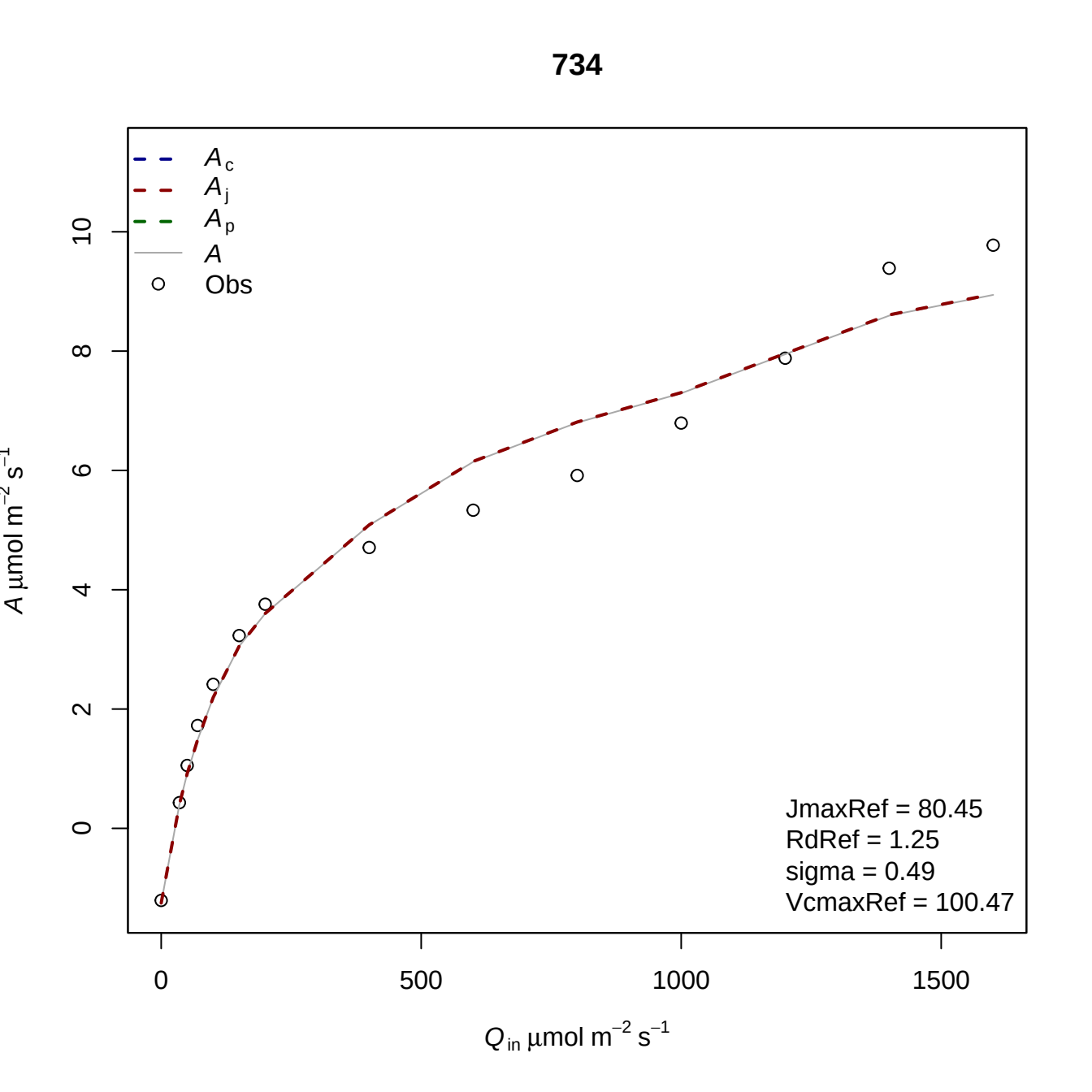

729
A_c
A_j
A_p
A
Obs
JmaxRef = 86.06
RdRef = 1.3
sigma = 0.2
VcmaxRef = 59.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

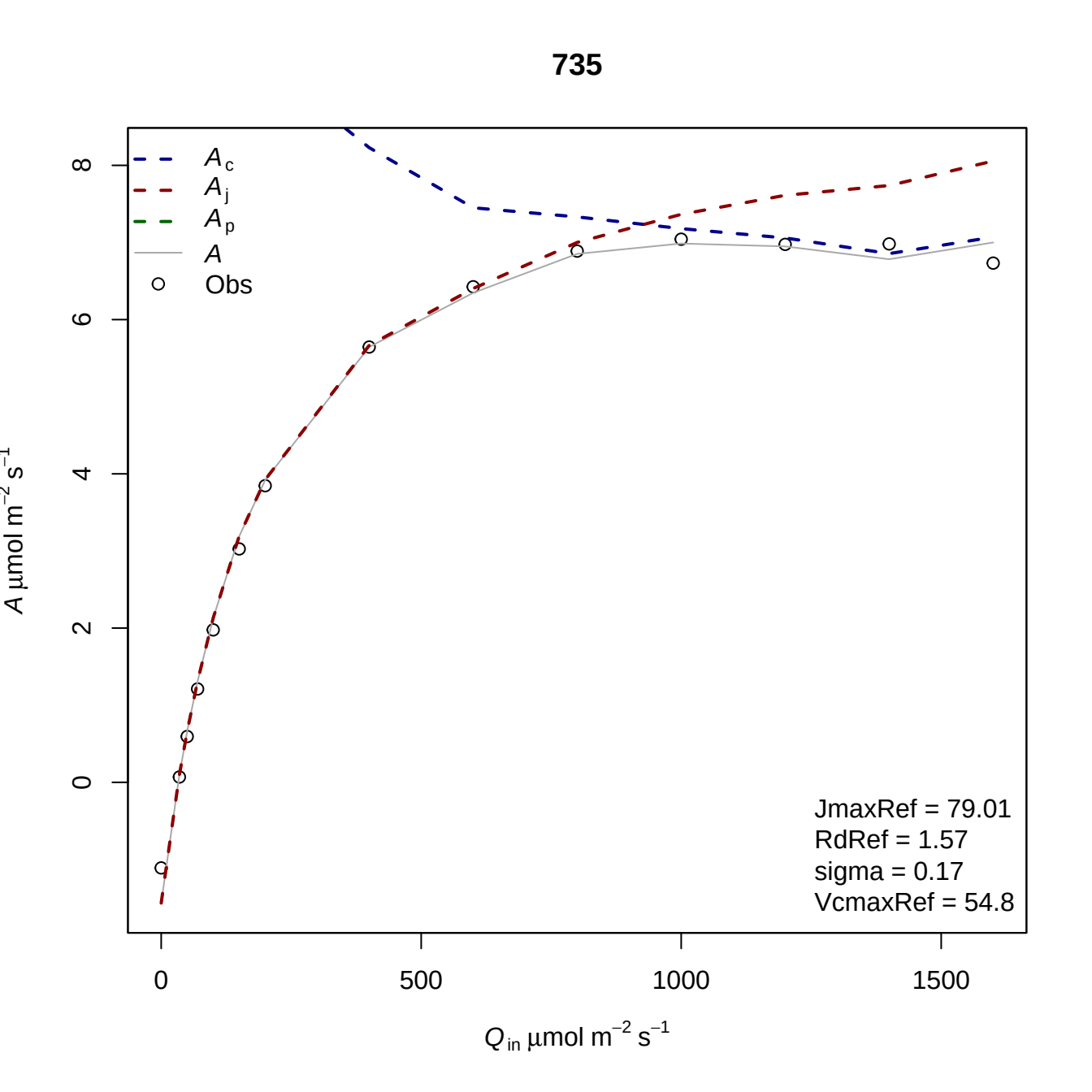

730
A_c
A_j
A_p
A
Obs
JmaxRef = 99.63
RdRef = 0.96
sigma = 0.13
VcmaxRef = 60.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

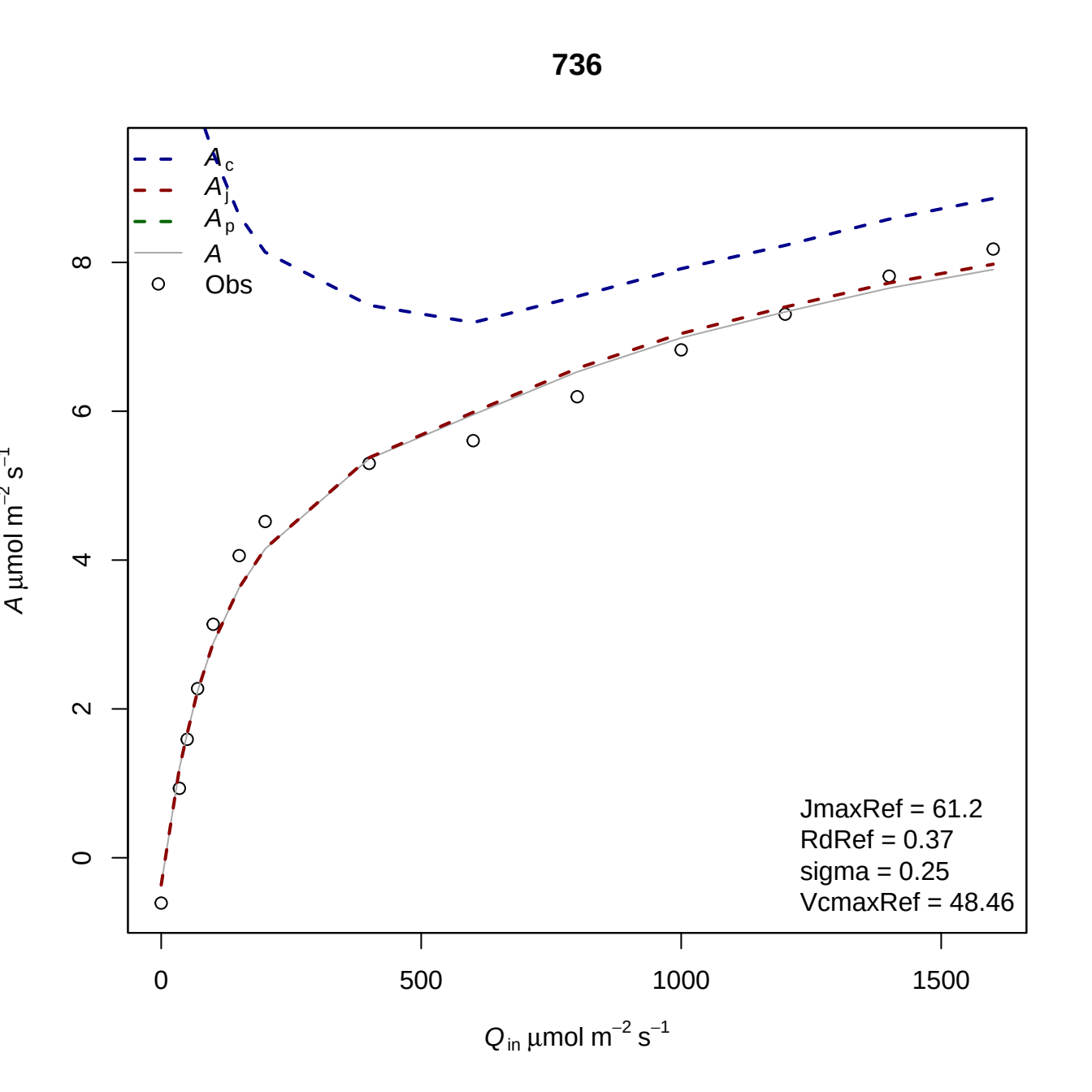

731
A_c
A_j
A_p
A
Obs
JmaxRef = 134.81
RdRef = 0.9
sigma = 0.19
VcmaxRef = 401643.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

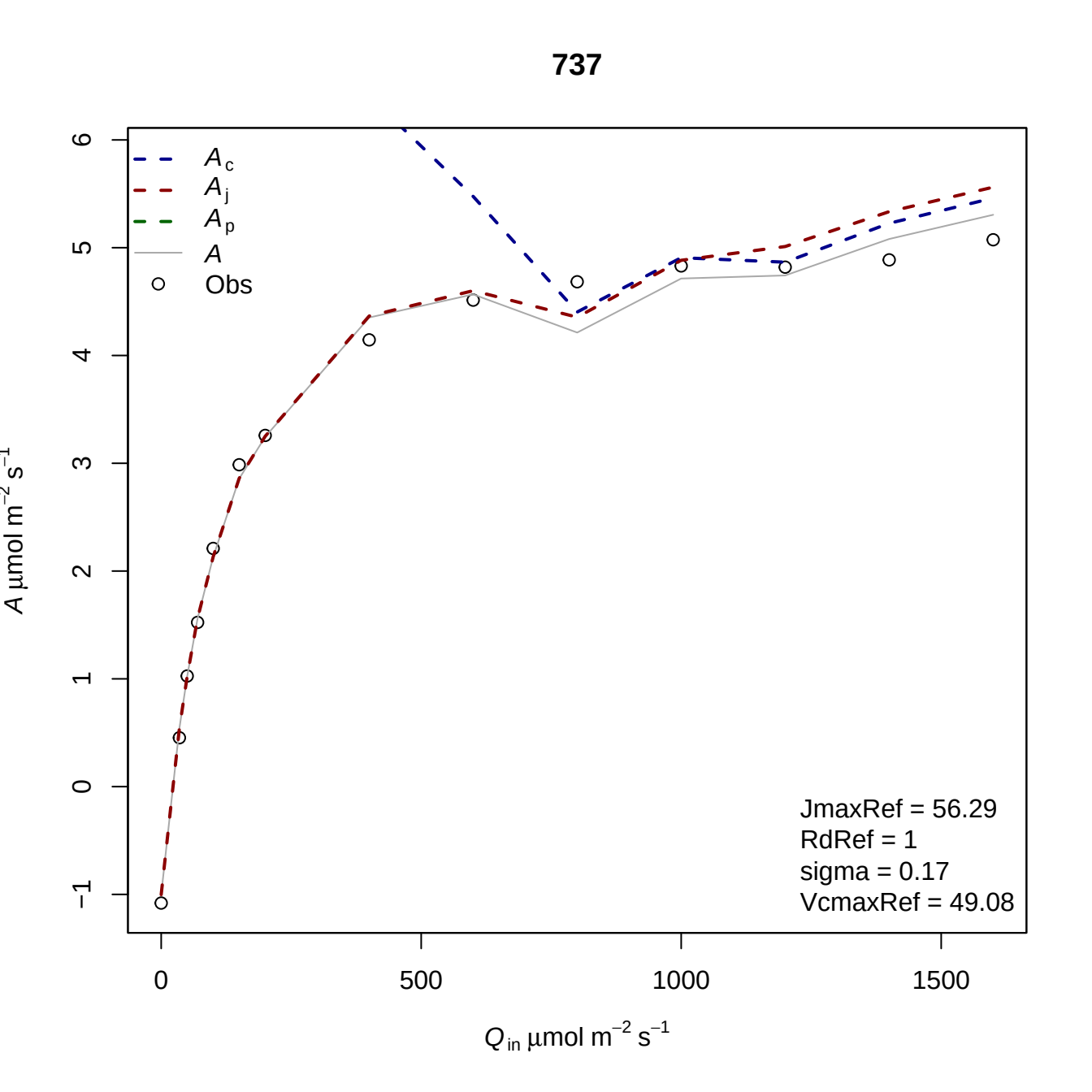

732
A_c
A_j
A_p
A
Obs
JmaxRef = 128.31
RdRef = 1.02
sigma = 0.26
VcmaxRef = 1404.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

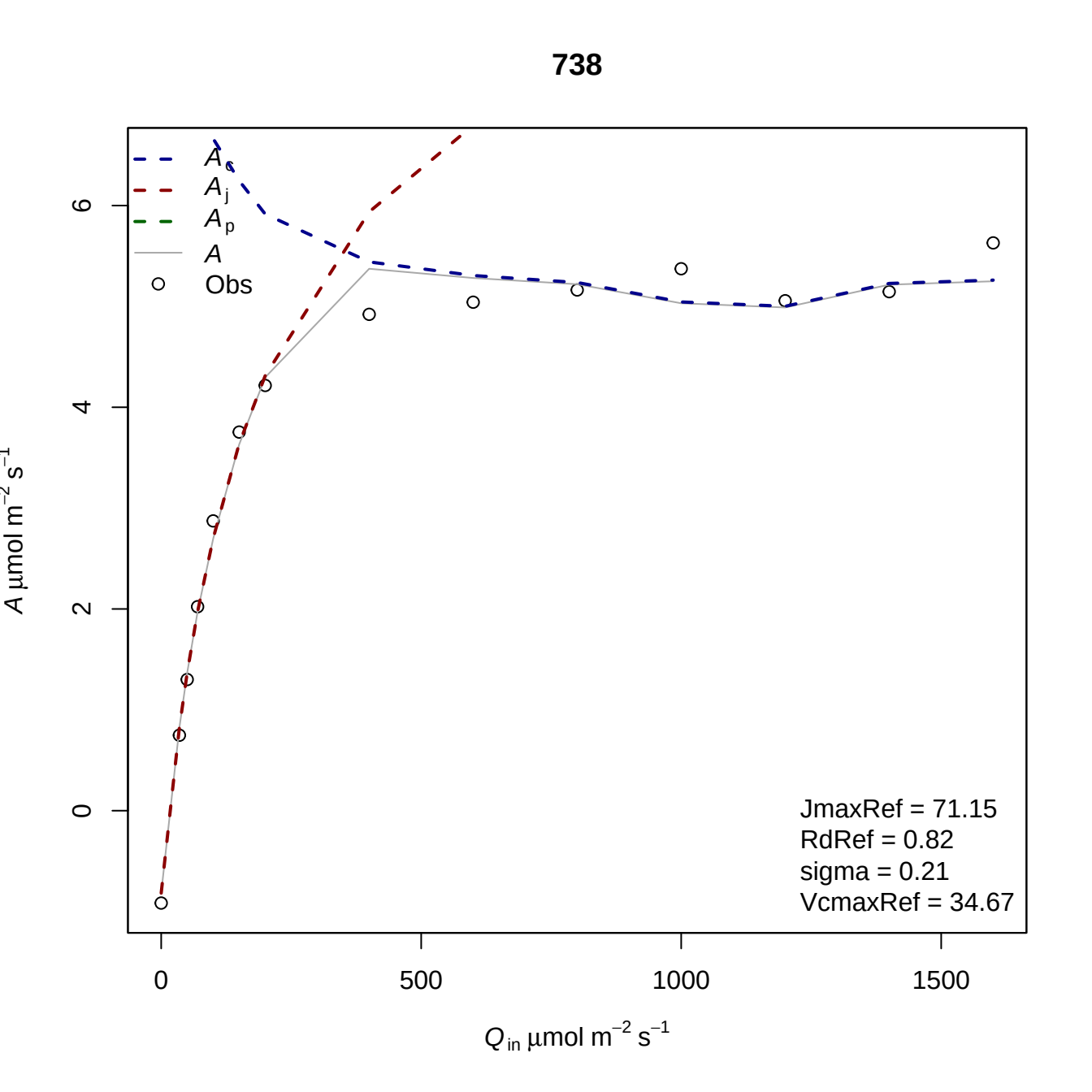

733
A_c
A_j
A_p
A
Obs
JmaxRef = 128.26
RdRef = 0.91
sigma = 0.24
VcmaxRef = 80.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

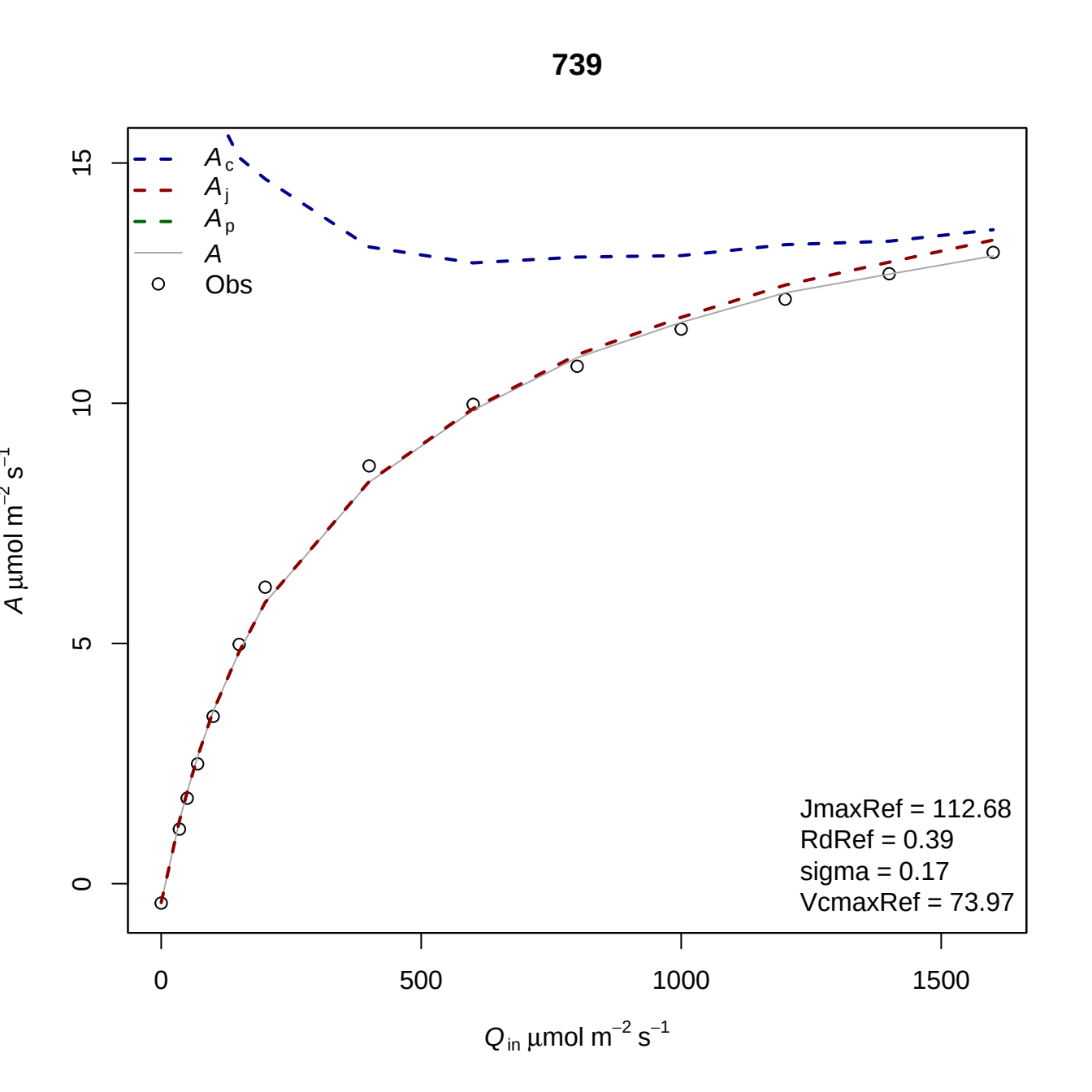

734
A_c
A_j
A_p
A
Obs
JmaxRef = 80.45
RdRef = 1.25
sigma = 0.49
VcmaxRef = 100.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

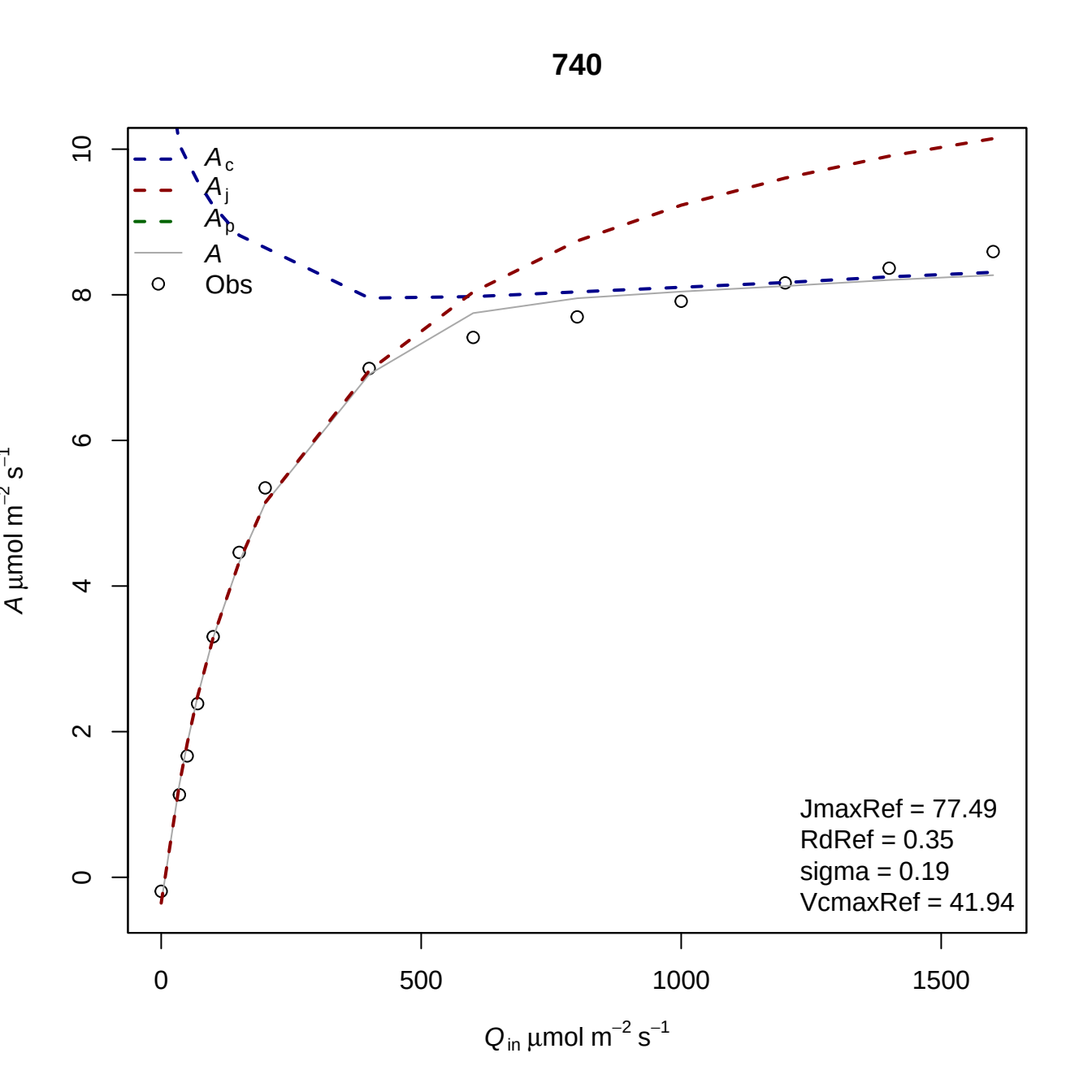

735
A_c
A_j
A_p
A
Obs
JmaxRef = 79.01
RdRef = 1.57
sigma = 0.17
VcmaxRef = 54.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

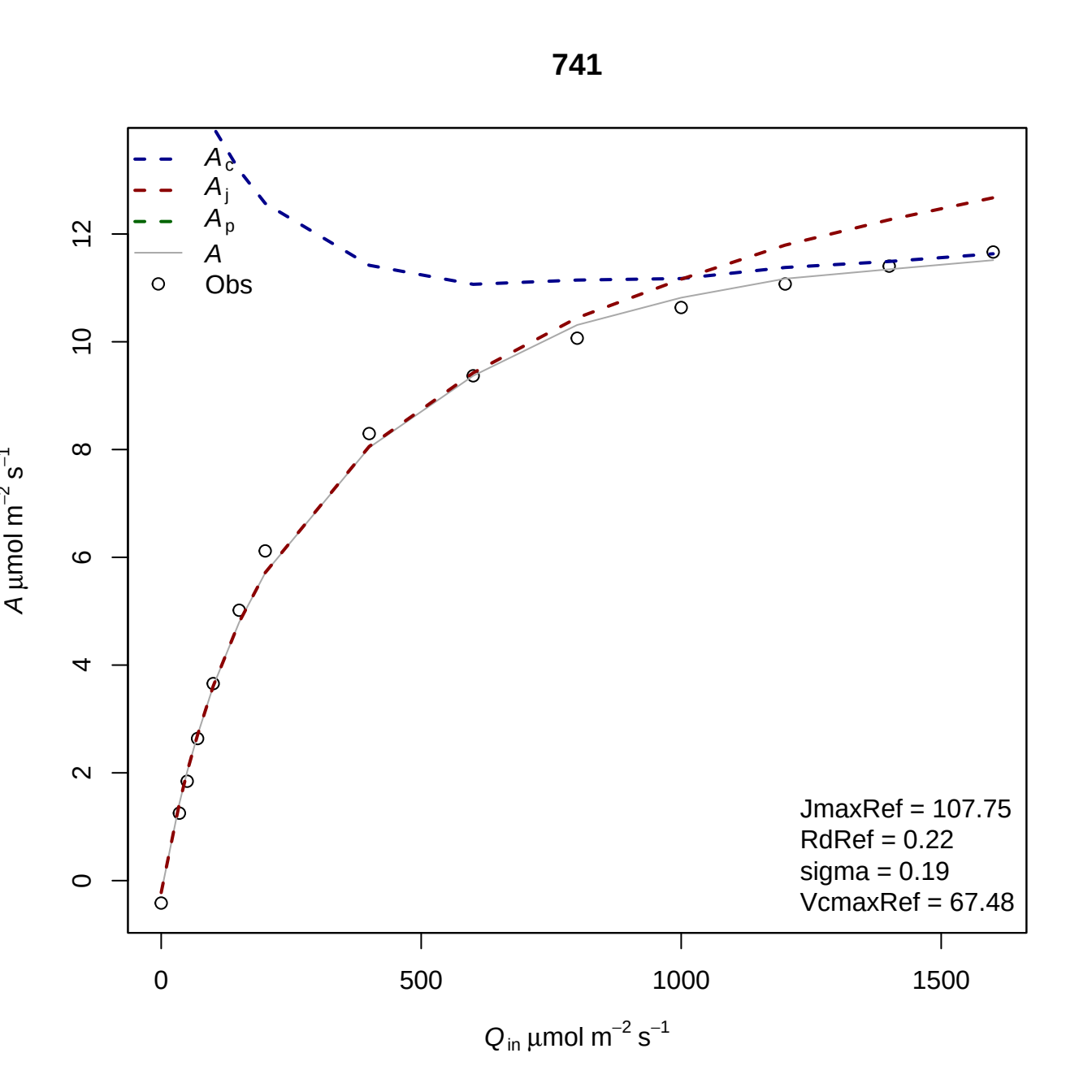

736
A_c
A_j
A_p
A
Obs
JmaxRef = 61.2
RdRef = 0.37
sigma = 0.25
VcmaxRef = 48.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

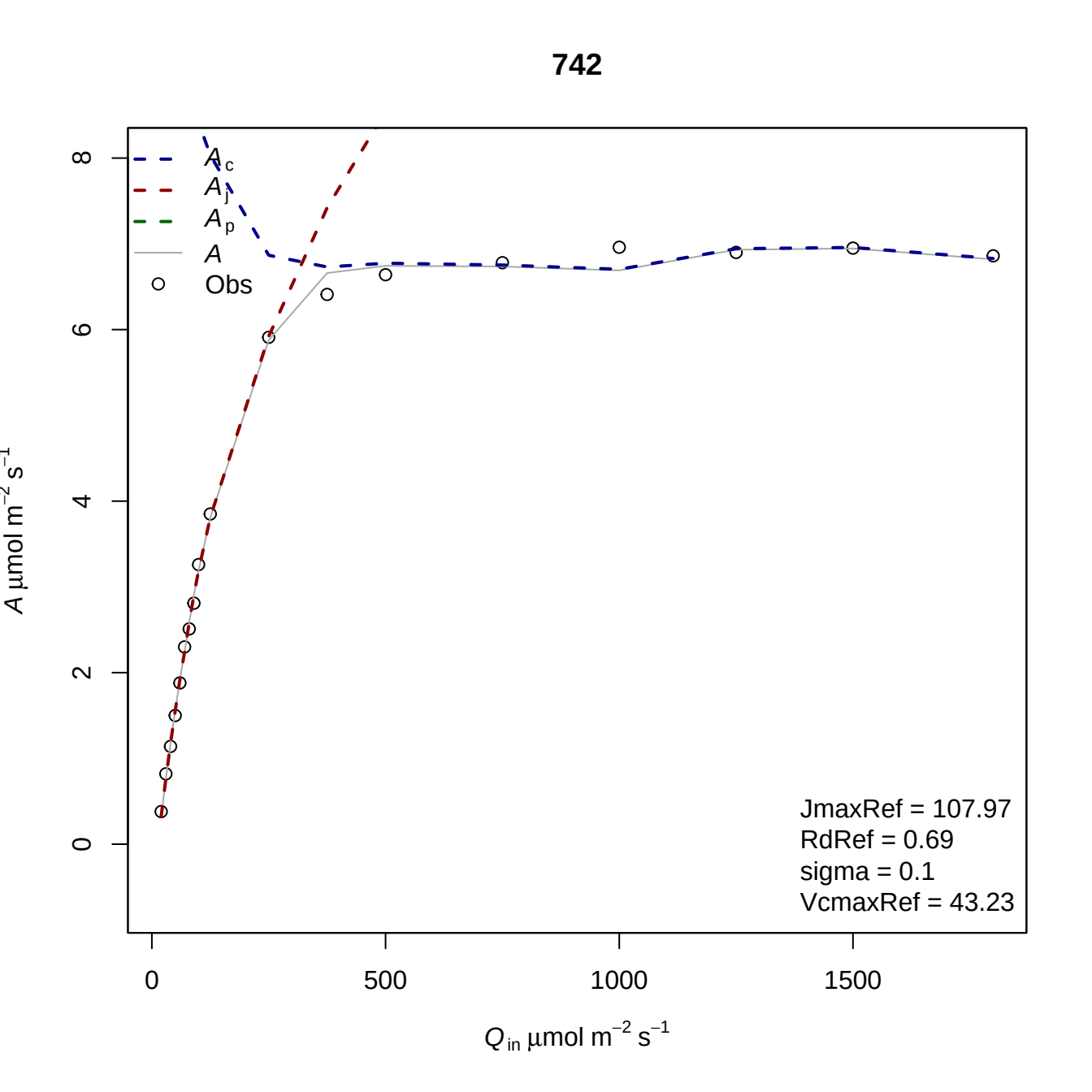

737
A_c
A_j
A_p
A
Obs
JmaxRef = 56.29
RdRef = 1
sigma = 0.17
VcmaxRef = 49.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

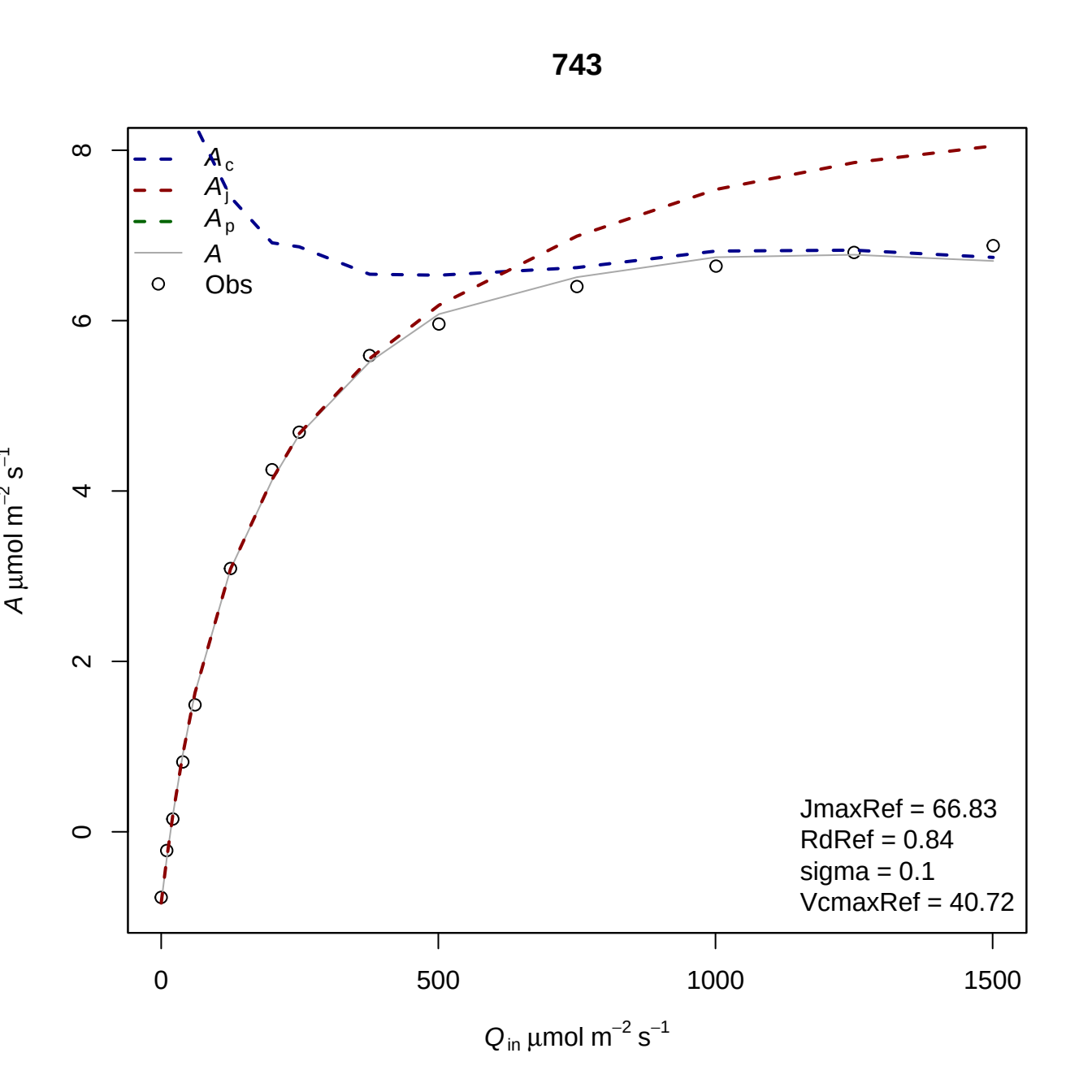

738
A_c
A_j
A_p
A
Obs
JmaxRef = 71.15
RdRef = 0.82
sigma = 0.21
VcmaxRef = 34.67
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

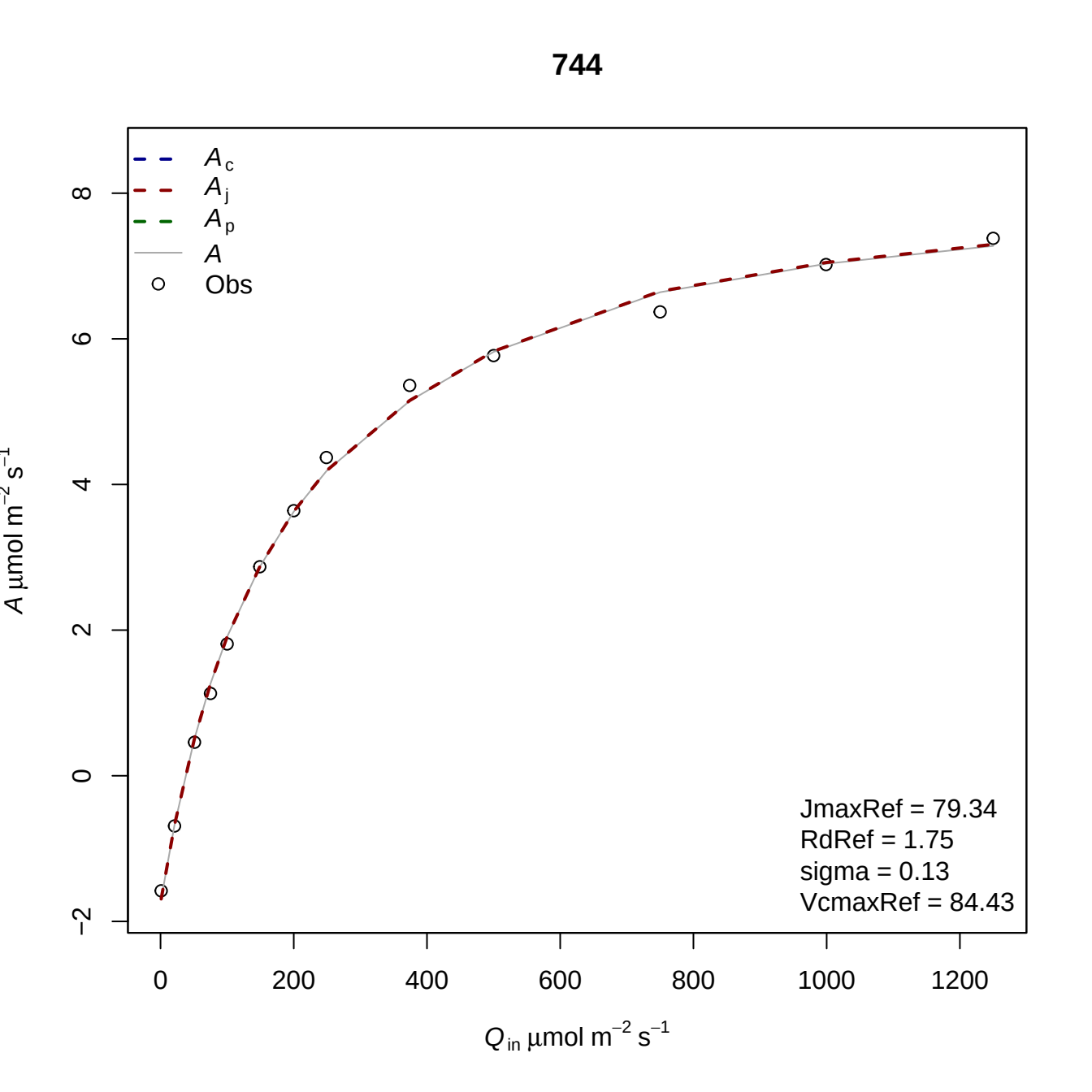

739
A_c
A_j
A_p
A
Obs
JmaxRef = 112.68
RdRef = 0.39
sigma = 0.17
VcmaxRef = 73.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

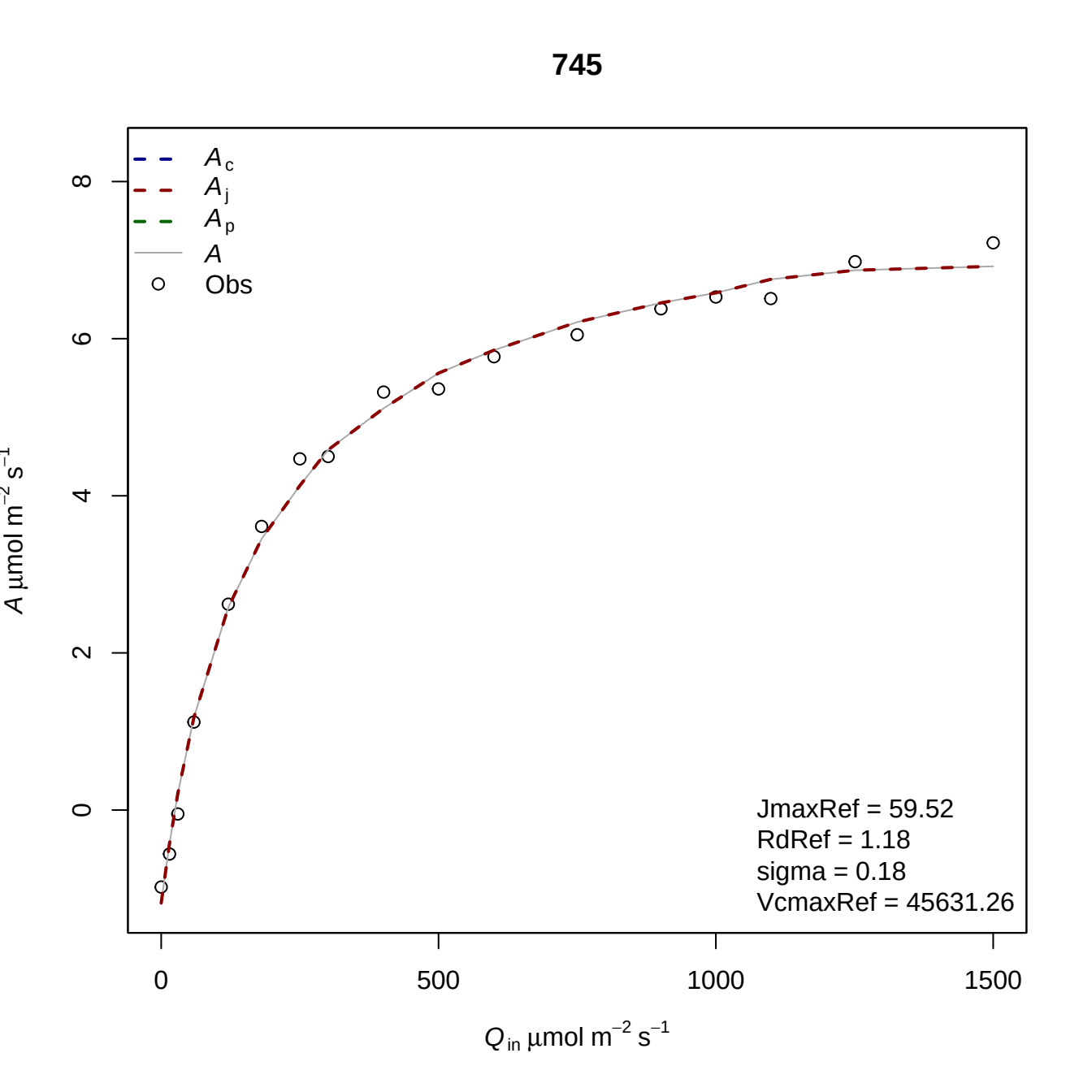

740
A_c
A_j
A_p
A
Obs
JmaxRef = 77.49
RdRef = 0.35
sigma = 0.19
VcmaxRef = 41.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

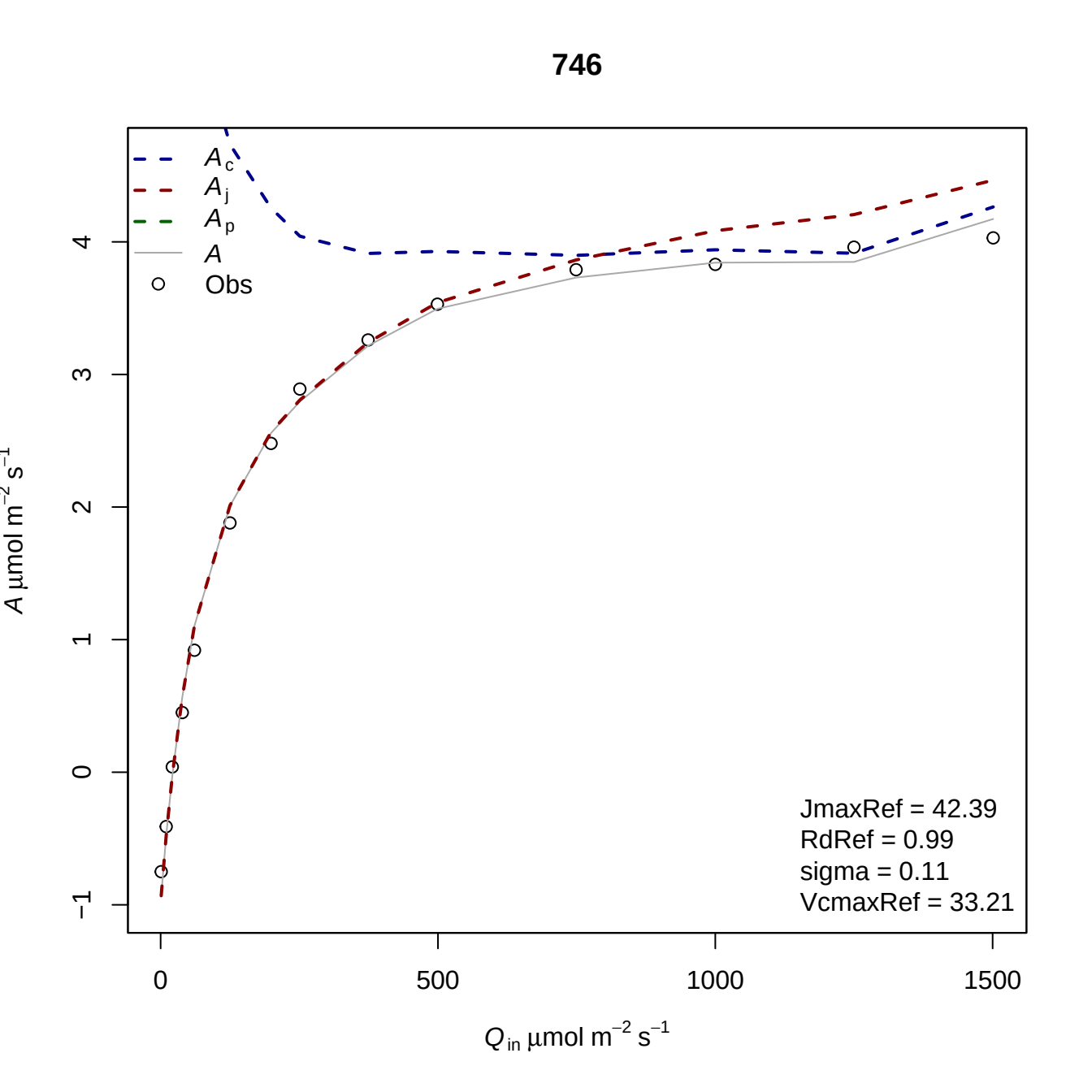

741
A_c
A_j
A_p
A
Obs
JmaxRef = 107.75
RdRef = 0.22
sigma = 0.19
VcmaxRef = 67.48
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

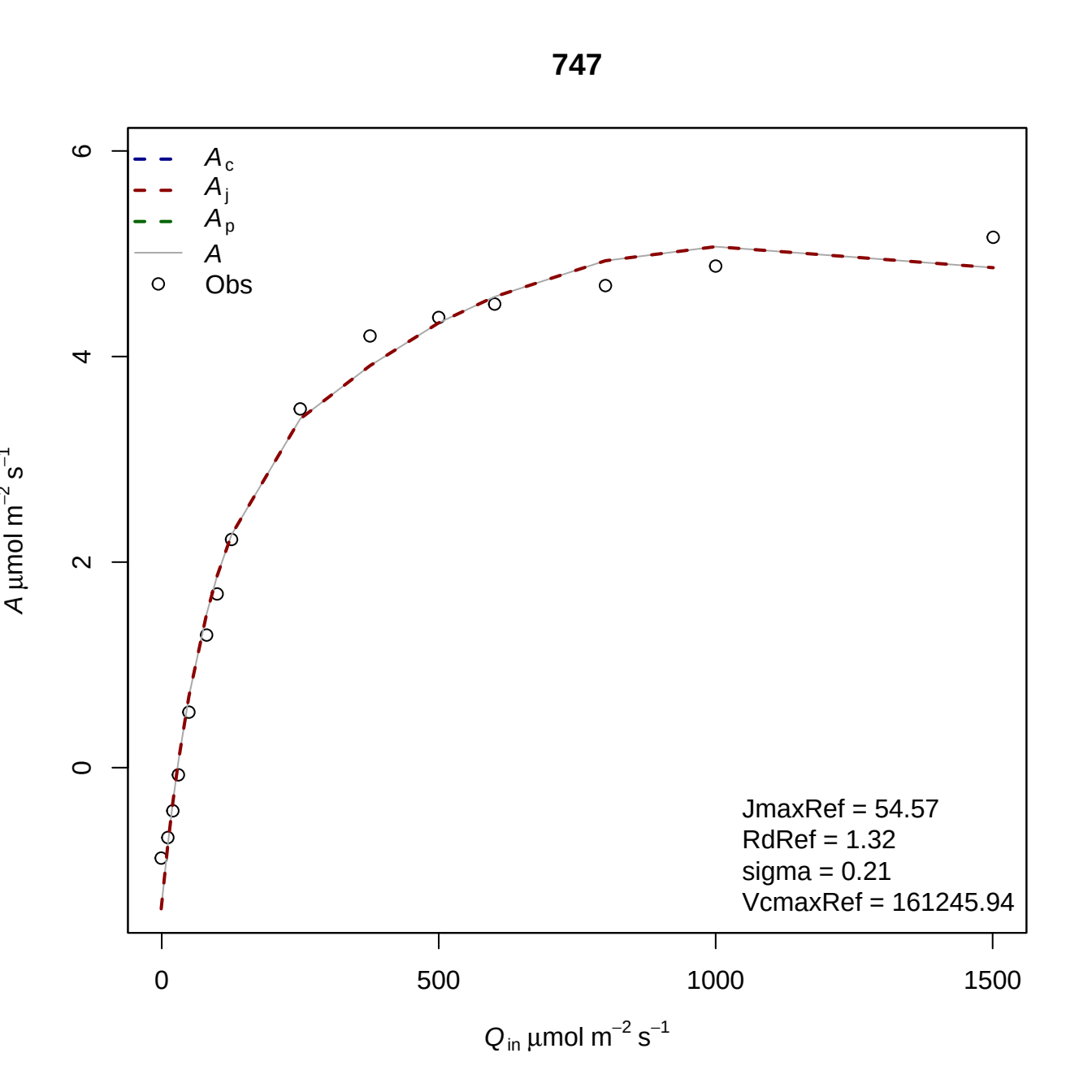

742
A_c
A_j
A_p
A
Obs
JmaxRef = 107.97
RdRef = 0.69
sigma = 0.1
VcmaxRef = 43.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

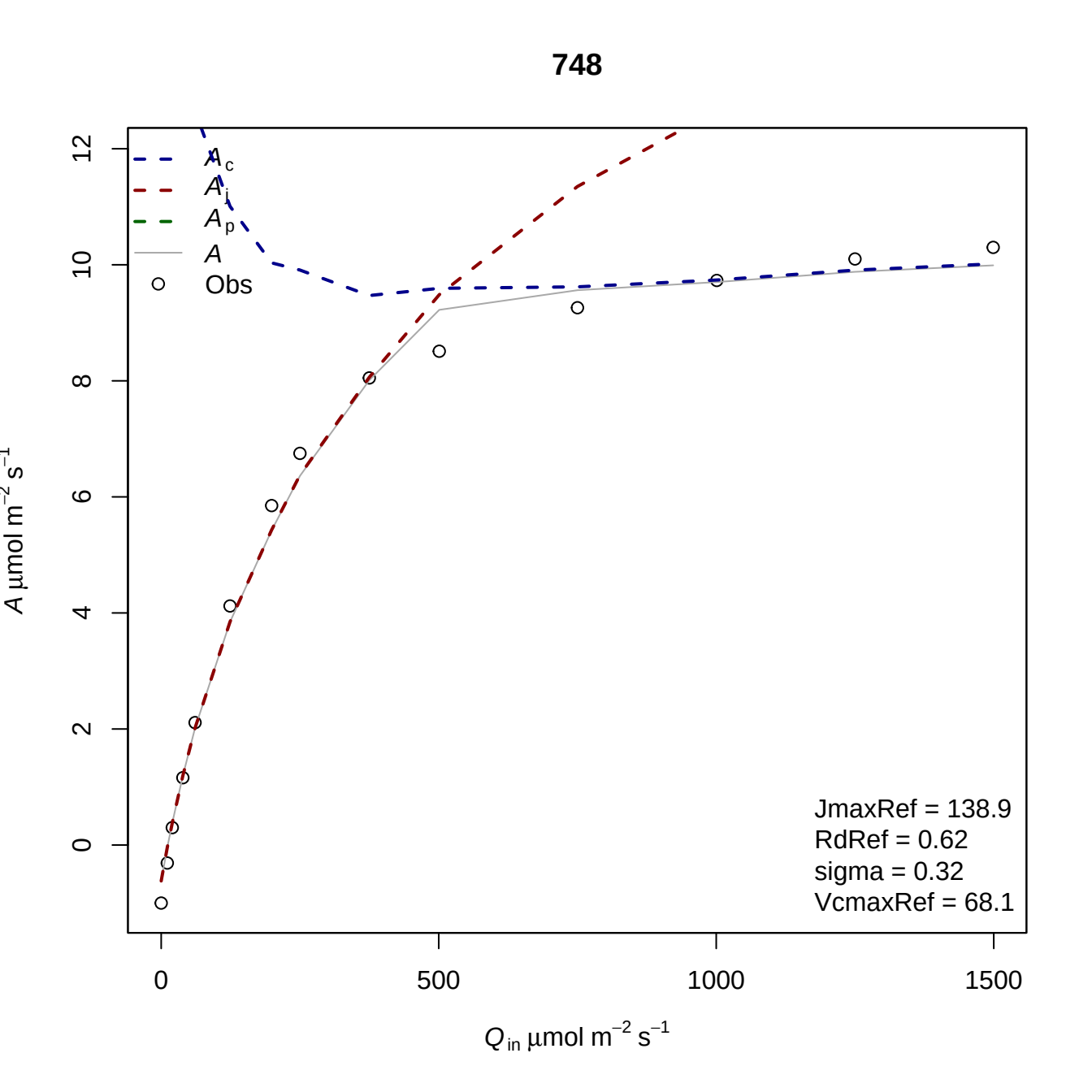

743
A_c
A_j
A_p
A
Obs
JmaxRef = 66.83
RdRef = 0.84
sigma = 0.1
VcmaxRef = 40.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

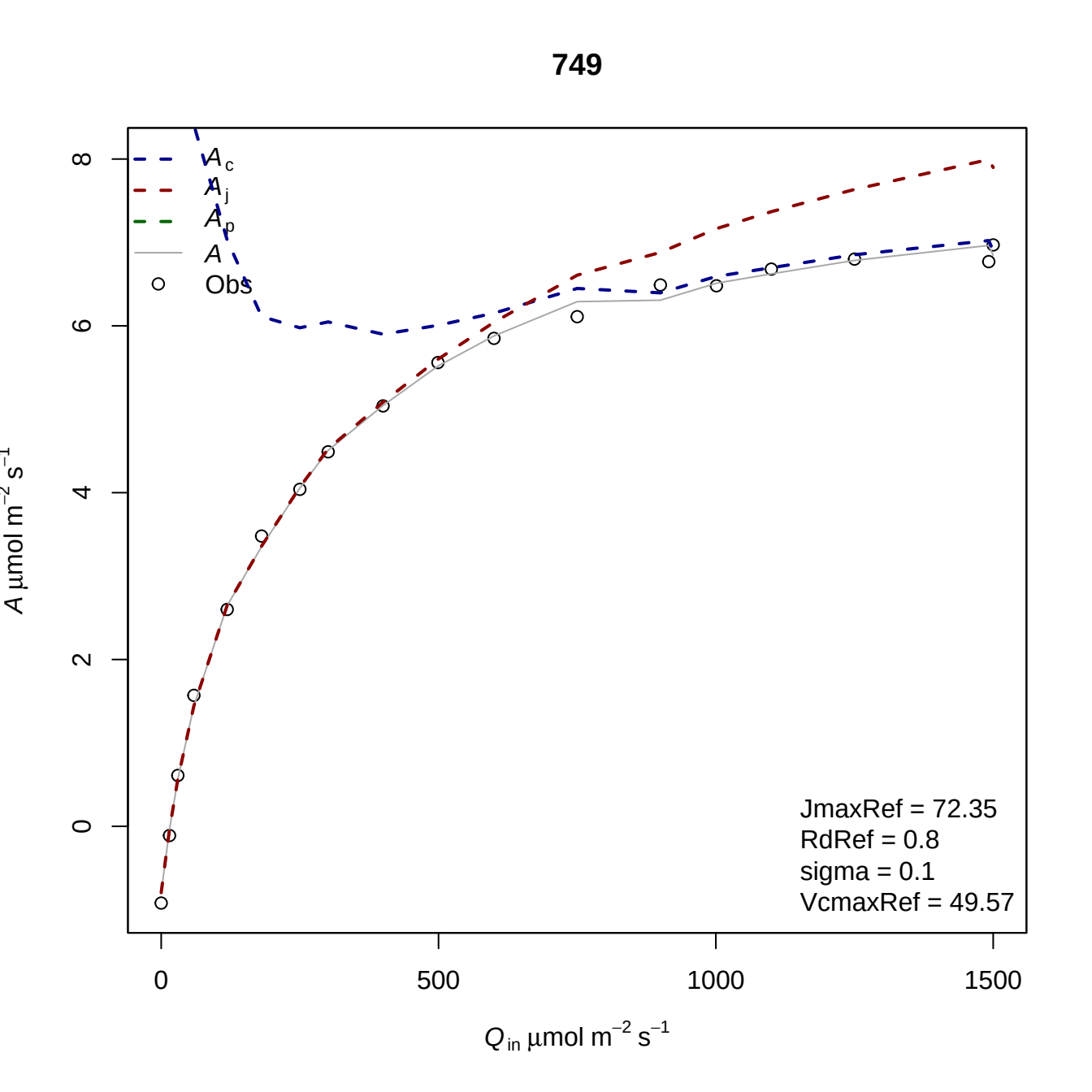

744
A_c
A_j
A_p
A
Obs
JmaxRef = 79.34
RdRef = 1.75
sigma = 0.13
VcmaxRef = 84.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000
1200

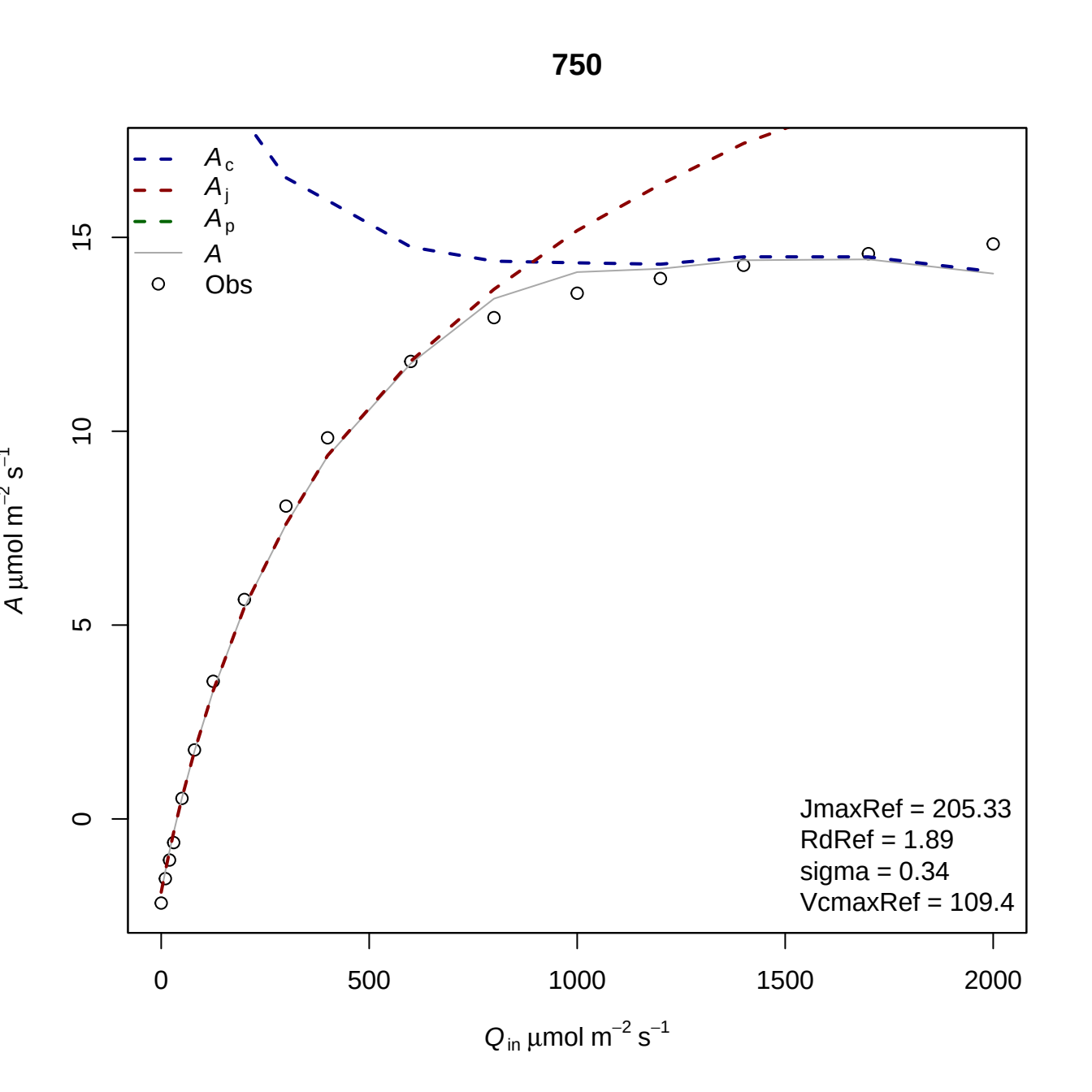

745
A_c
A_j
A_p
A
Obs
JmaxRef = 59.52
RdRef = 1.18
sigma = 0.18
VcmaxRef = 45631.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

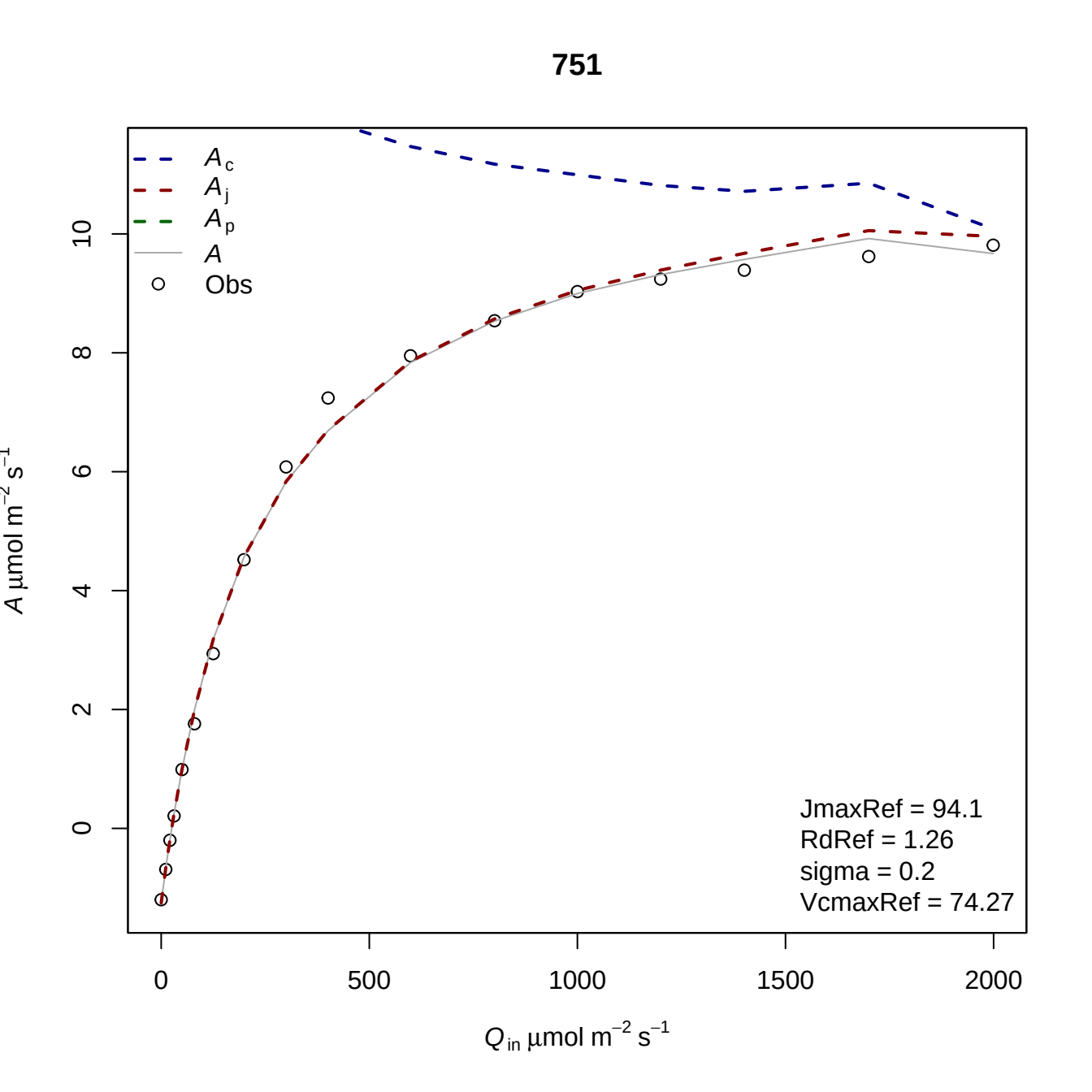

746
A_c
A_j
A_p
A
Obs
JmaxRef = 42.39
RdRef = 0.99
sigma = 0.11
VcmaxRef = 33.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

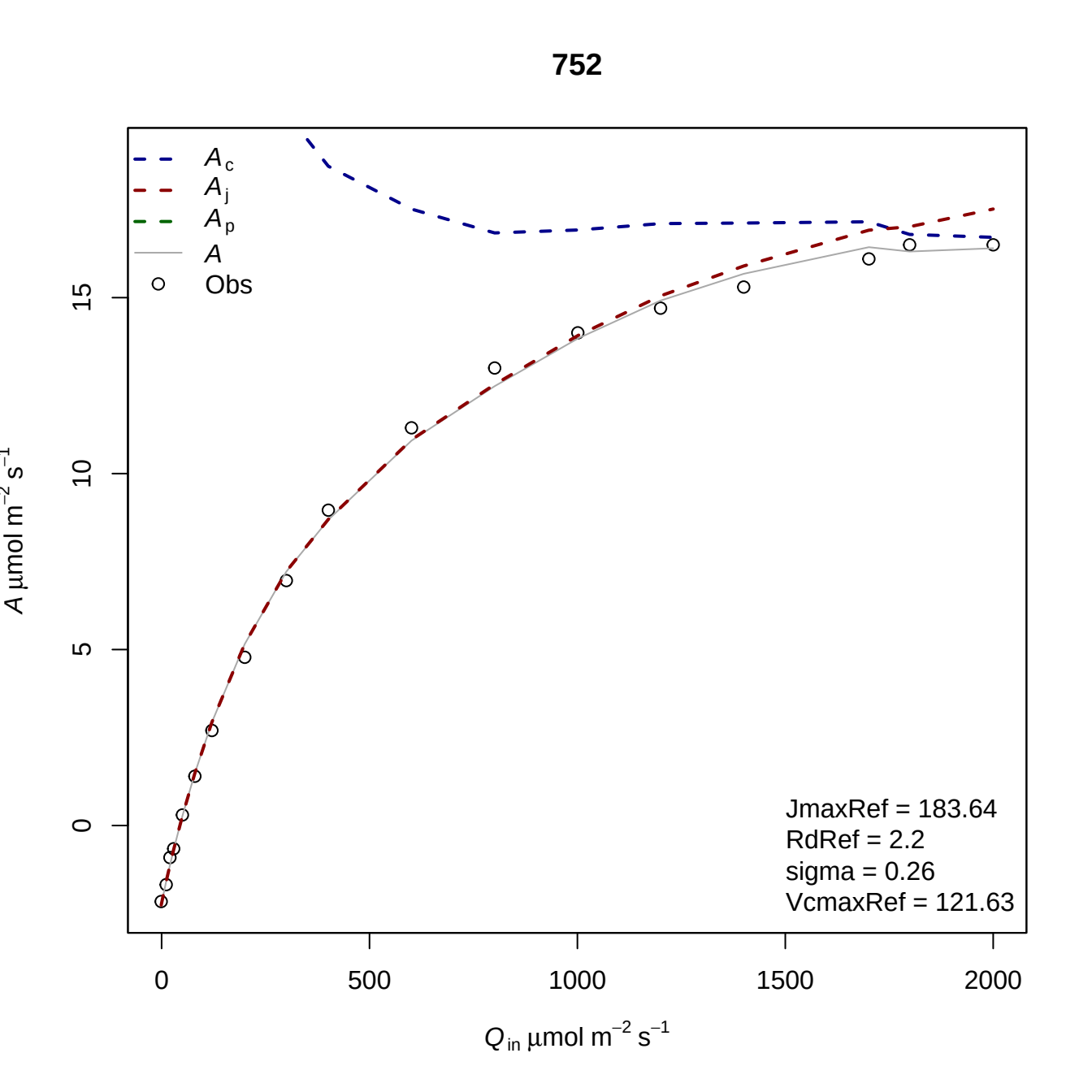

747
A_c
A_j
A_p
A
Obs
JmaxRef = 54.57
RdRef = 1.32
sigma = 0.21
VcmaxRef = 161245.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

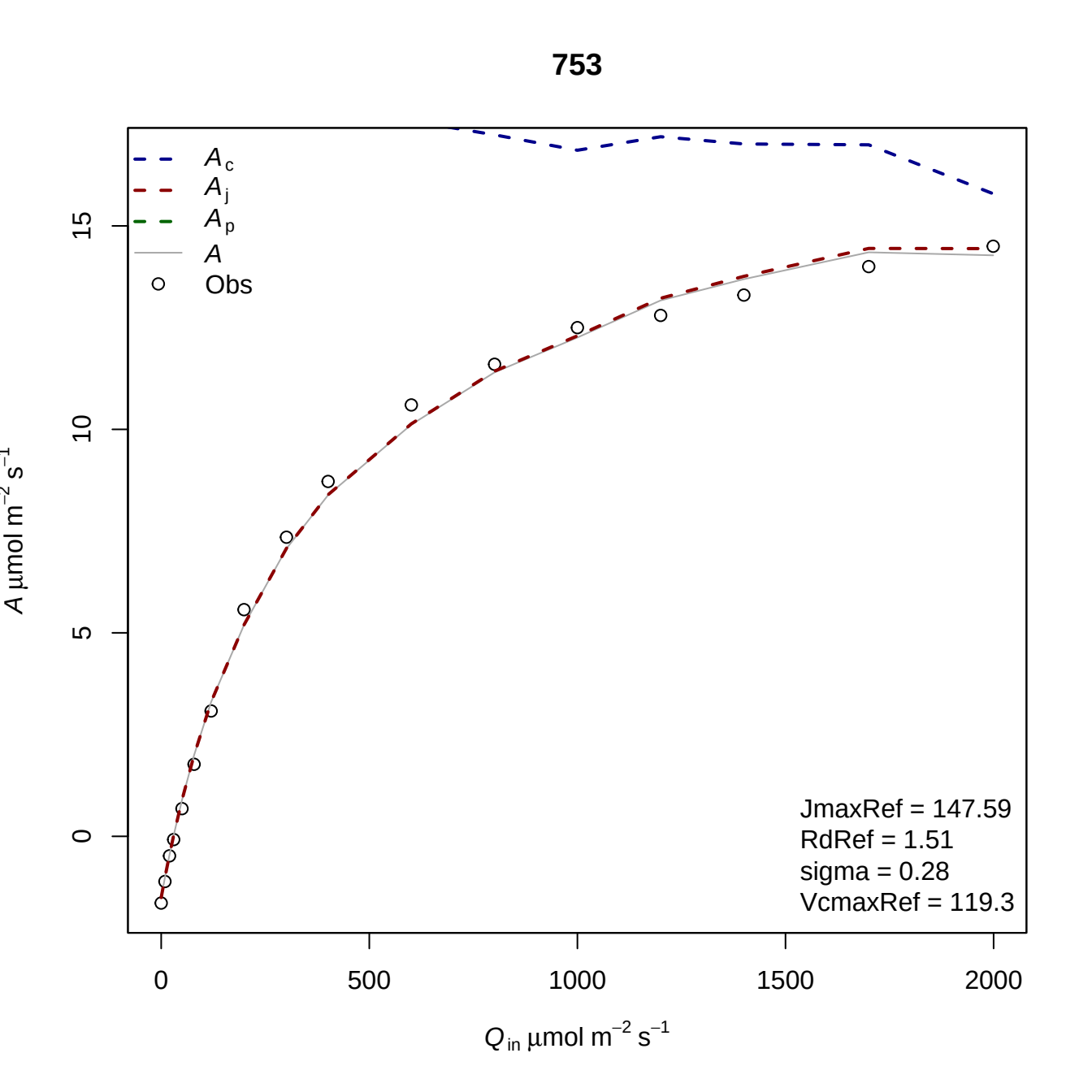

748
A_c
A_j
A_p
A
Obs
JmaxRef = 138.9
RdRef = 0.62
sigma = 0.32
VcmaxRef = 68.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

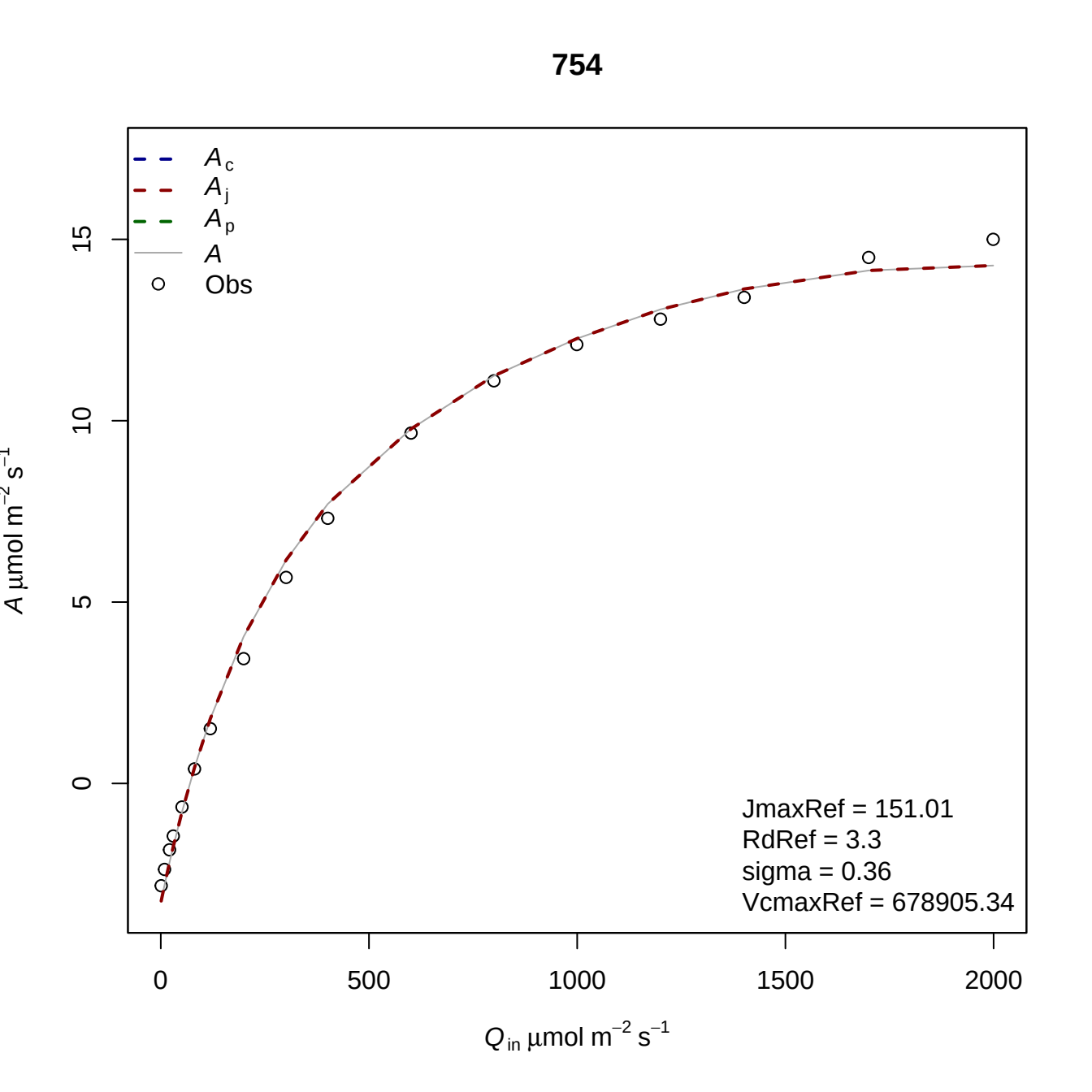

749
A_c
A_j
A_p
A
Obs
JmaxRef = 72.35
RdRef = 0.8
sigma = 0.1
VcmaxRef = 49.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

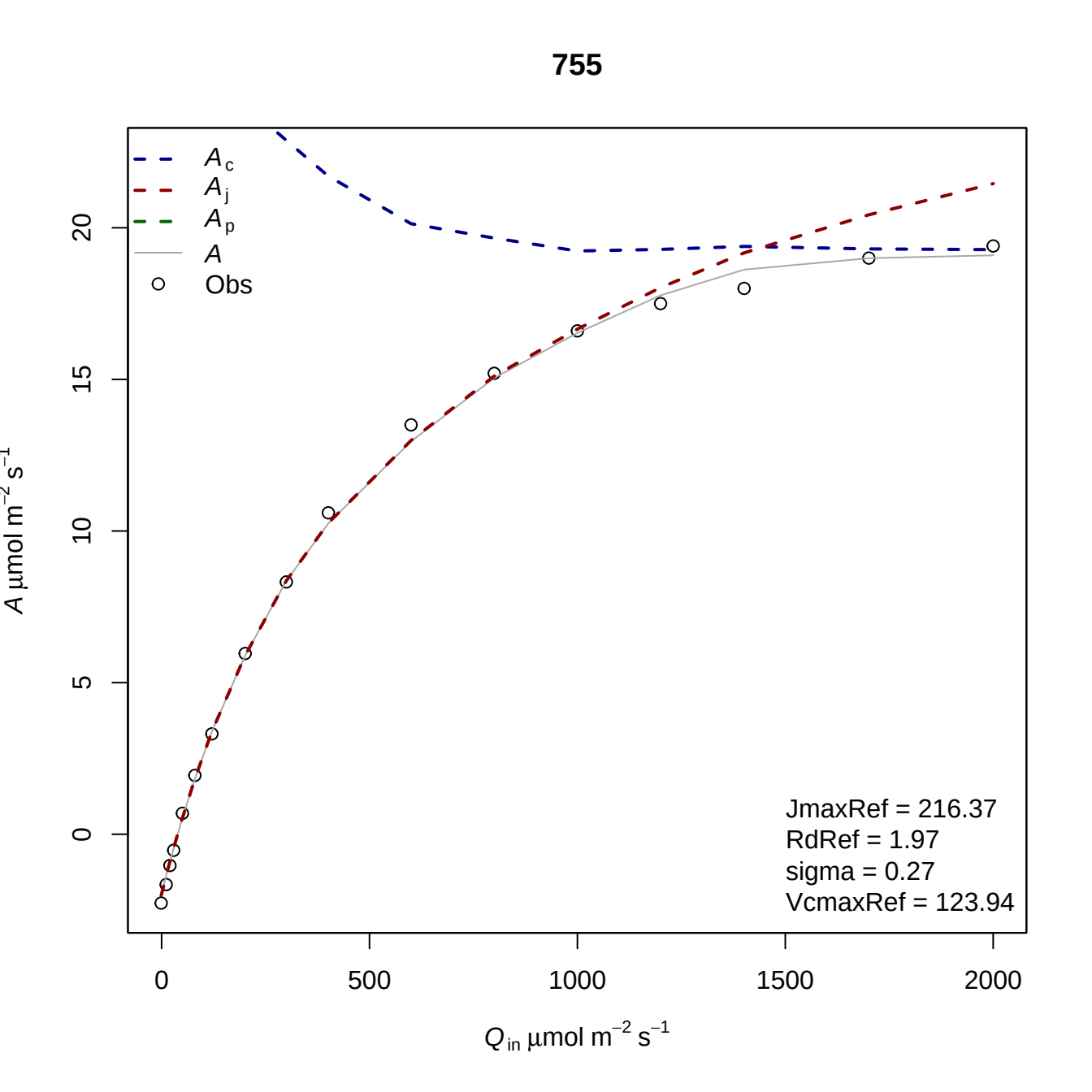

750
A_c
A_j
A_p
A
Obs
JmaxRef = 205.33
RdRef = 1.89
sigma = 0.34
VcmaxRef = 109.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

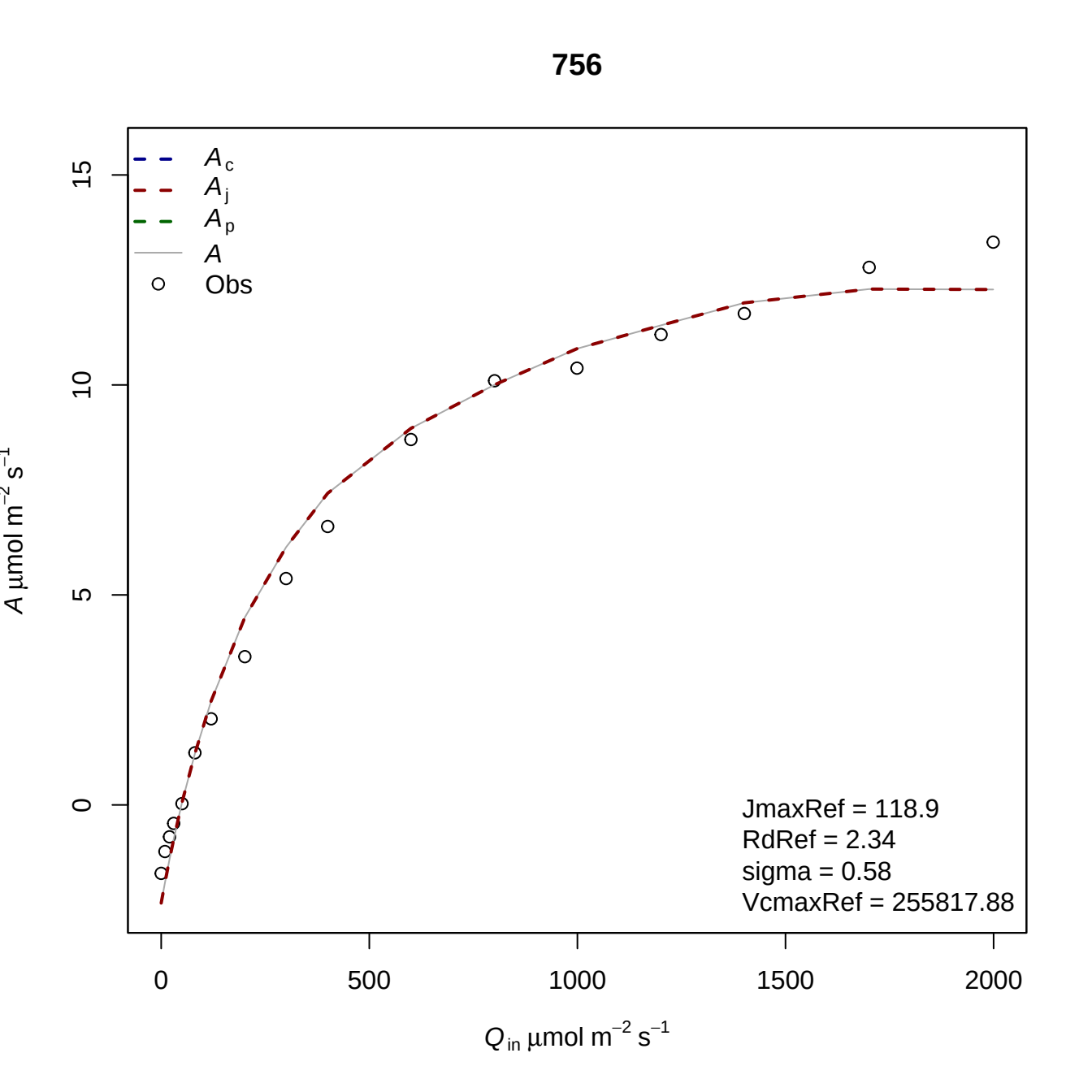

751
A_c
A_j
A_p
A
Obs
JmaxRef = 94.1
RdRef = 1.26
sigma = 0.2
VcmaxRef = 74.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

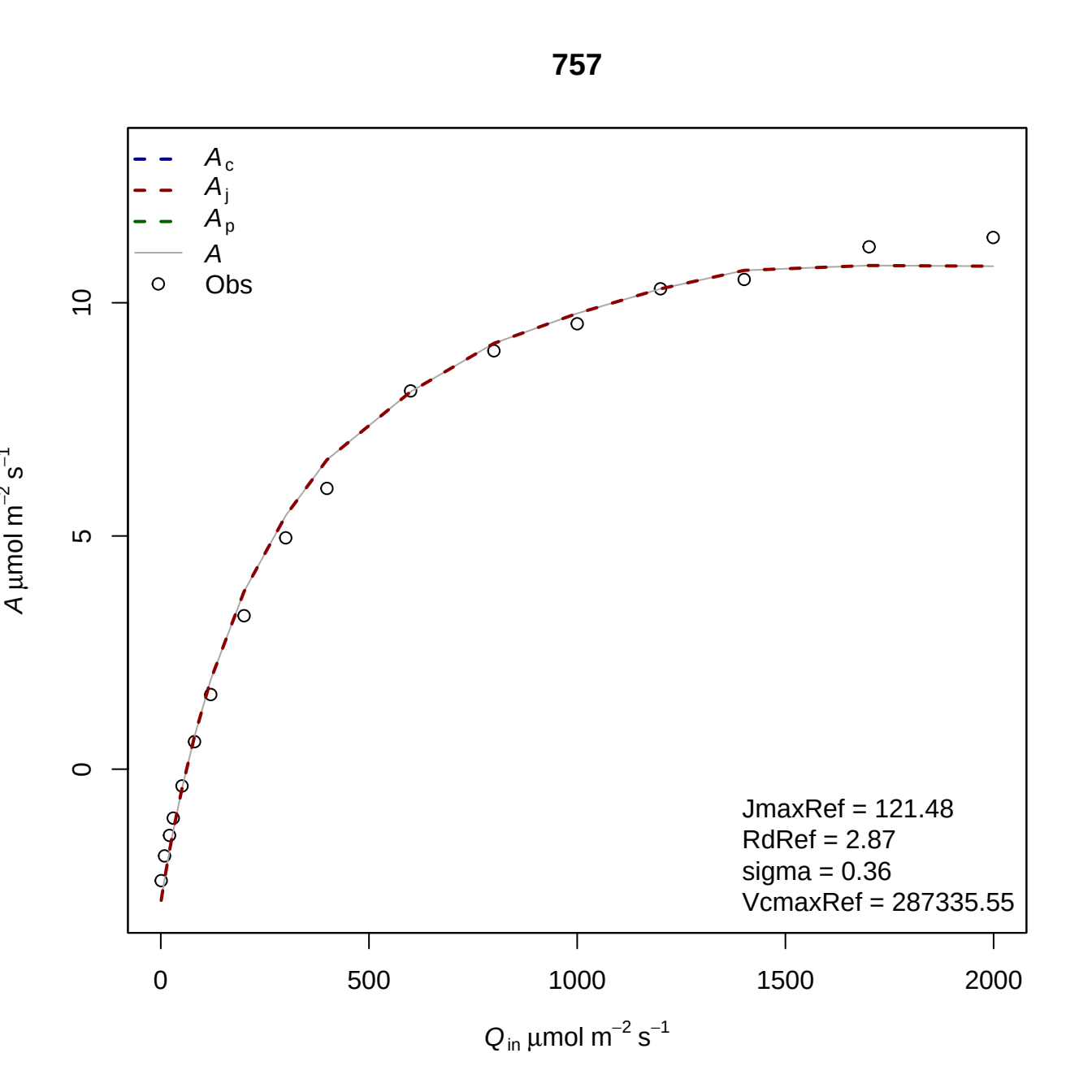

752
A_c
A_j
A_p
A
Obs
JmaxRef = 183.64
RdRef = 2.2
sigma = 0.26
VcmaxRef = 121.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

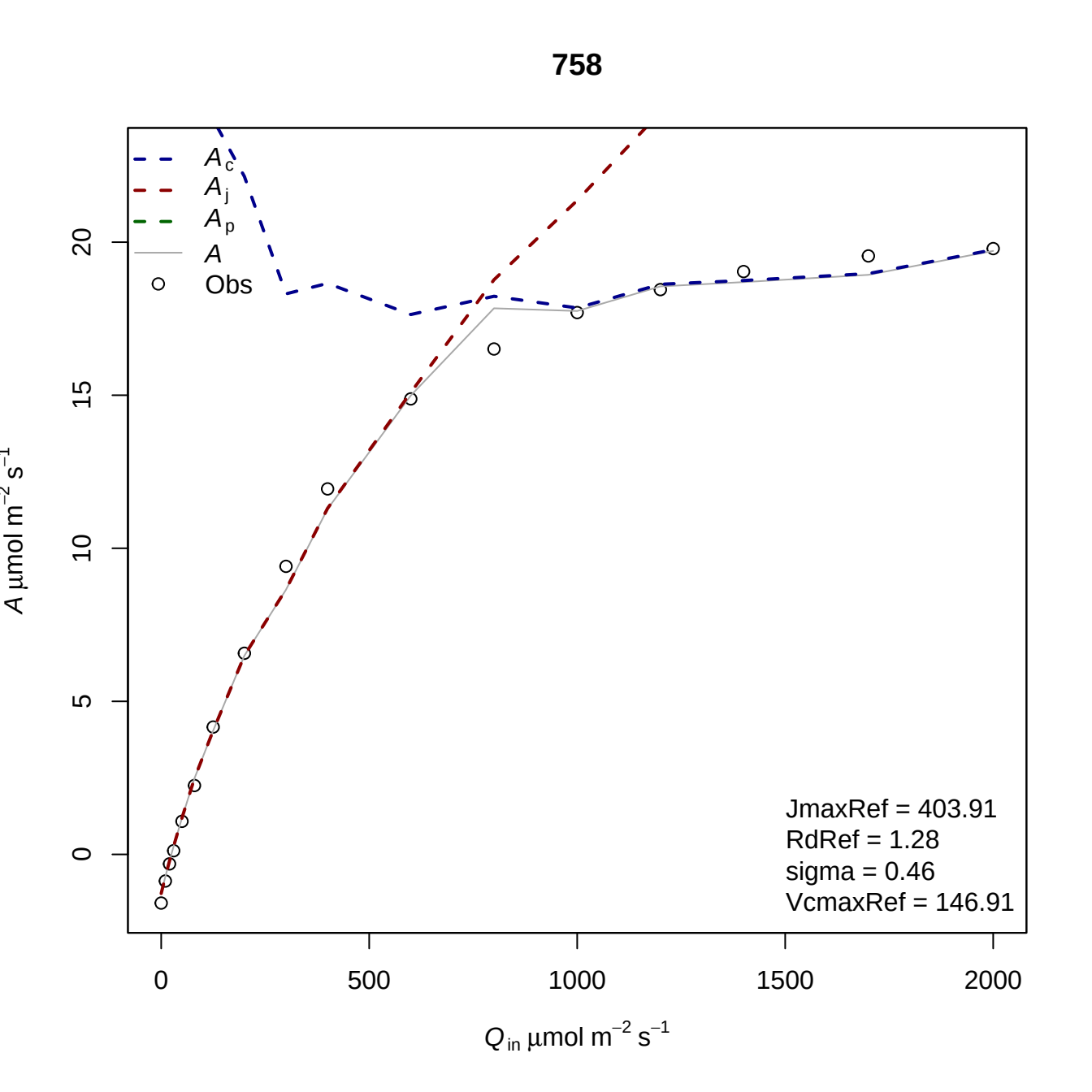

753
A_c
A_j
A_p
A
Obs
JmaxRef = 147.59
RdRef = 1.51
sigma = 0.28
VcmaxRef = 119.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

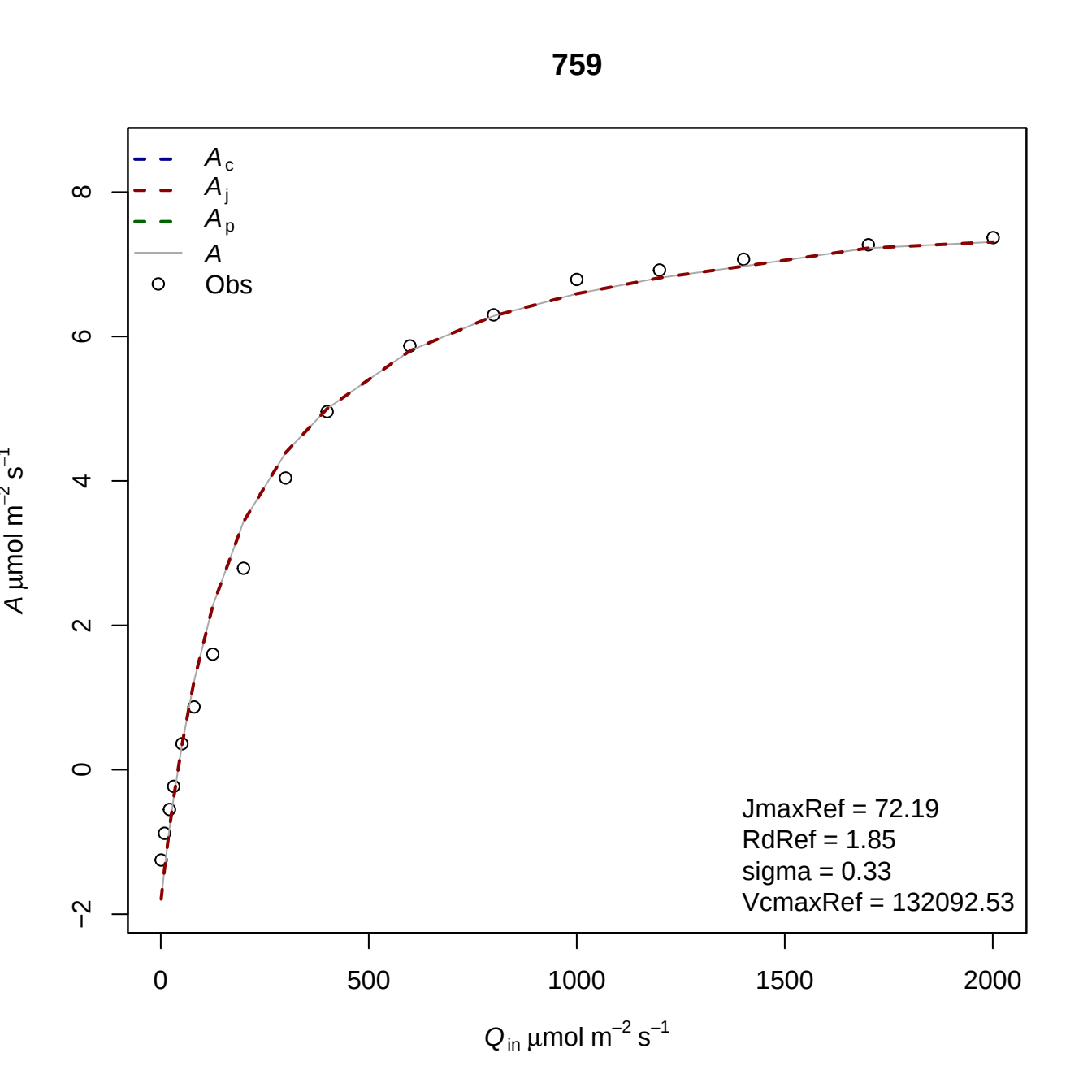

754
A_c
A_j
A_p
A
Obs
JmaxRef = 151.01
RdRef = 3.3
sigma = 0.36
VcmaxRef = 678905.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

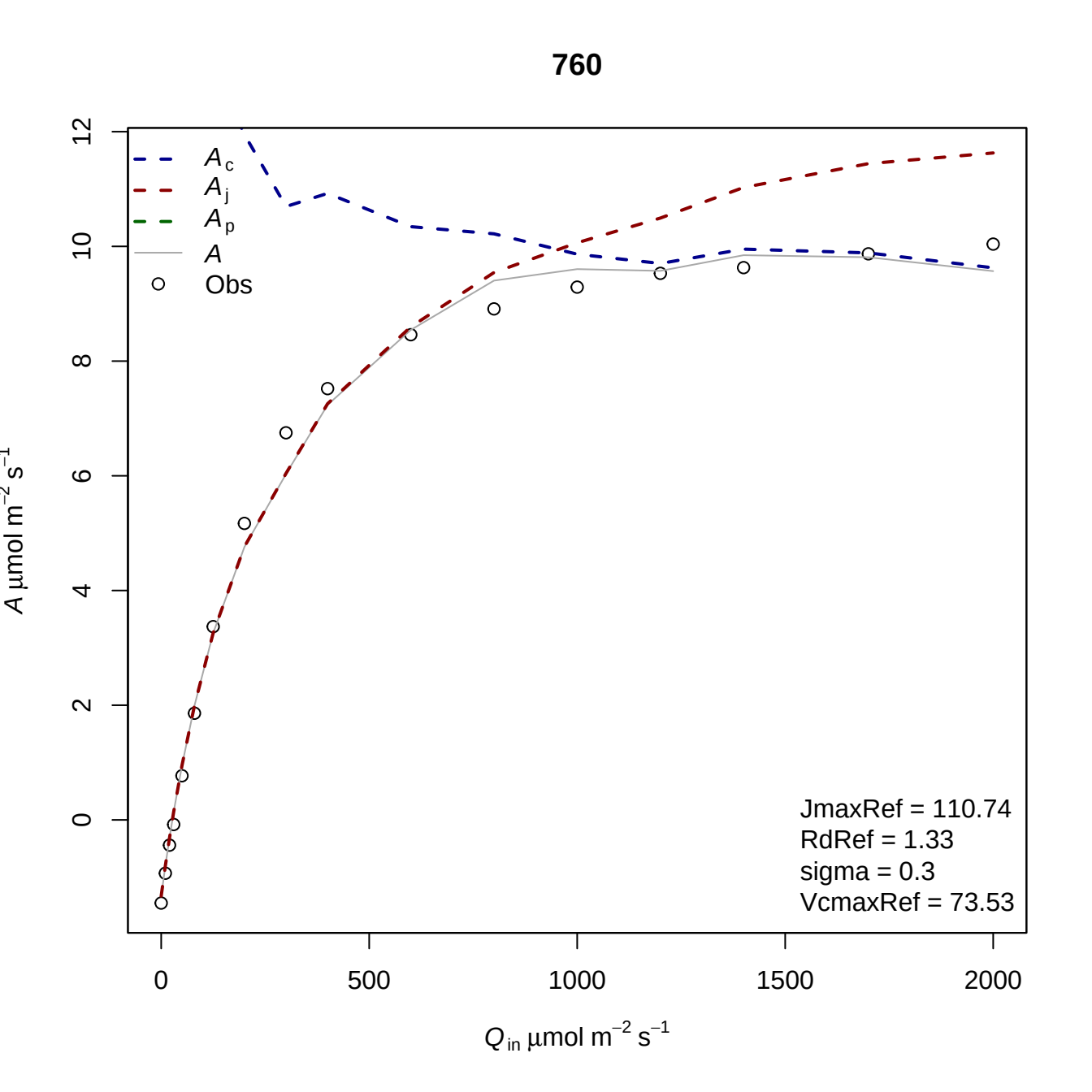

755
A_c
A_j
A_p
A
Obs
JmaxRef = 216.37
RdRef = 1.97
sigma = 0.27
VcmaxRef = 123.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

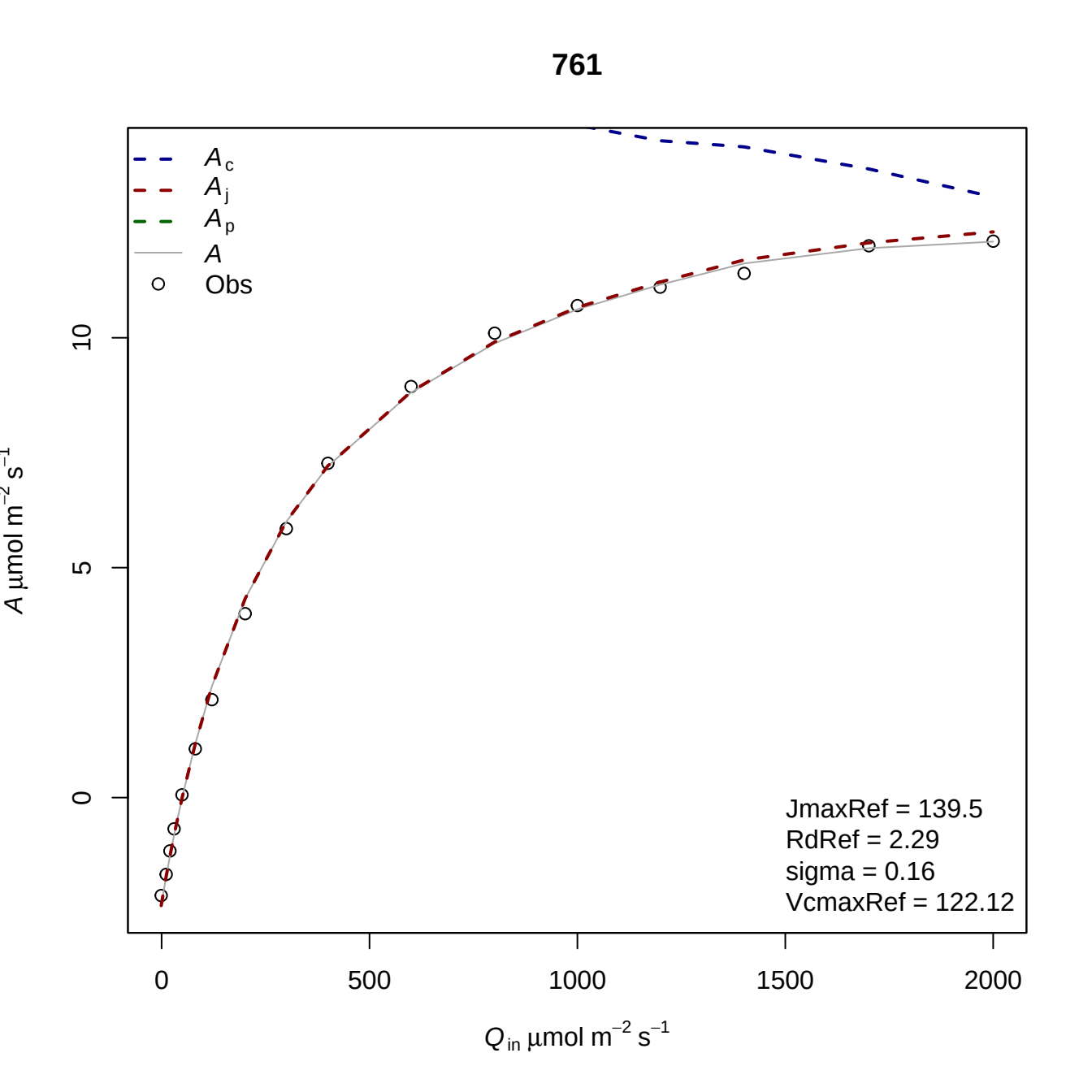

756
A_c
A_j
A_p
A
Obs
JmaxRef = 118.9
RdRef = 2.34
sigma = 0.58
VcmaxRef = 255817.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

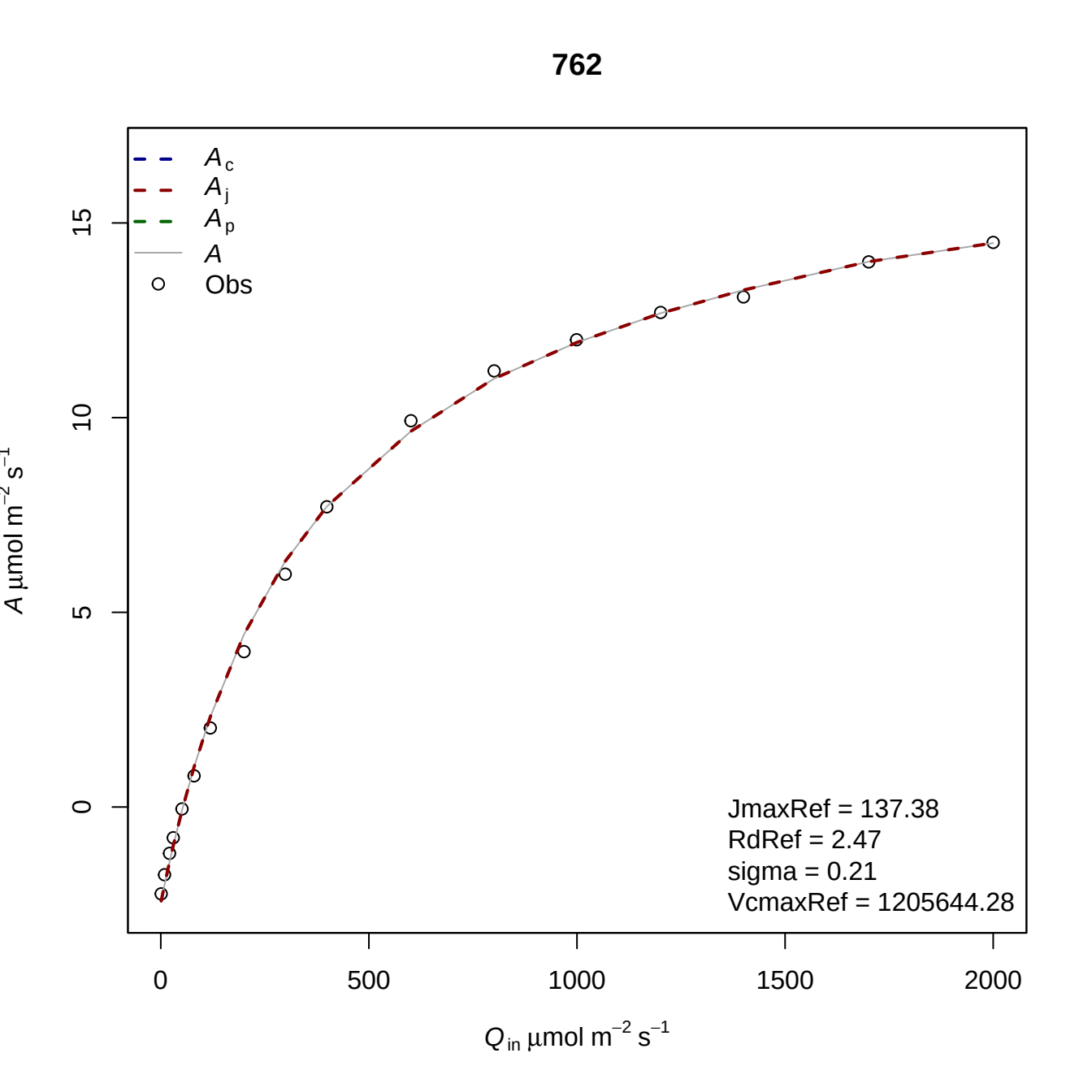

757
A_c
A_j
A_p
A
Obs
JmaxRef = 121.48
RdRef = 2.87
sigma = 0.36
VcmaxRef = 287335.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

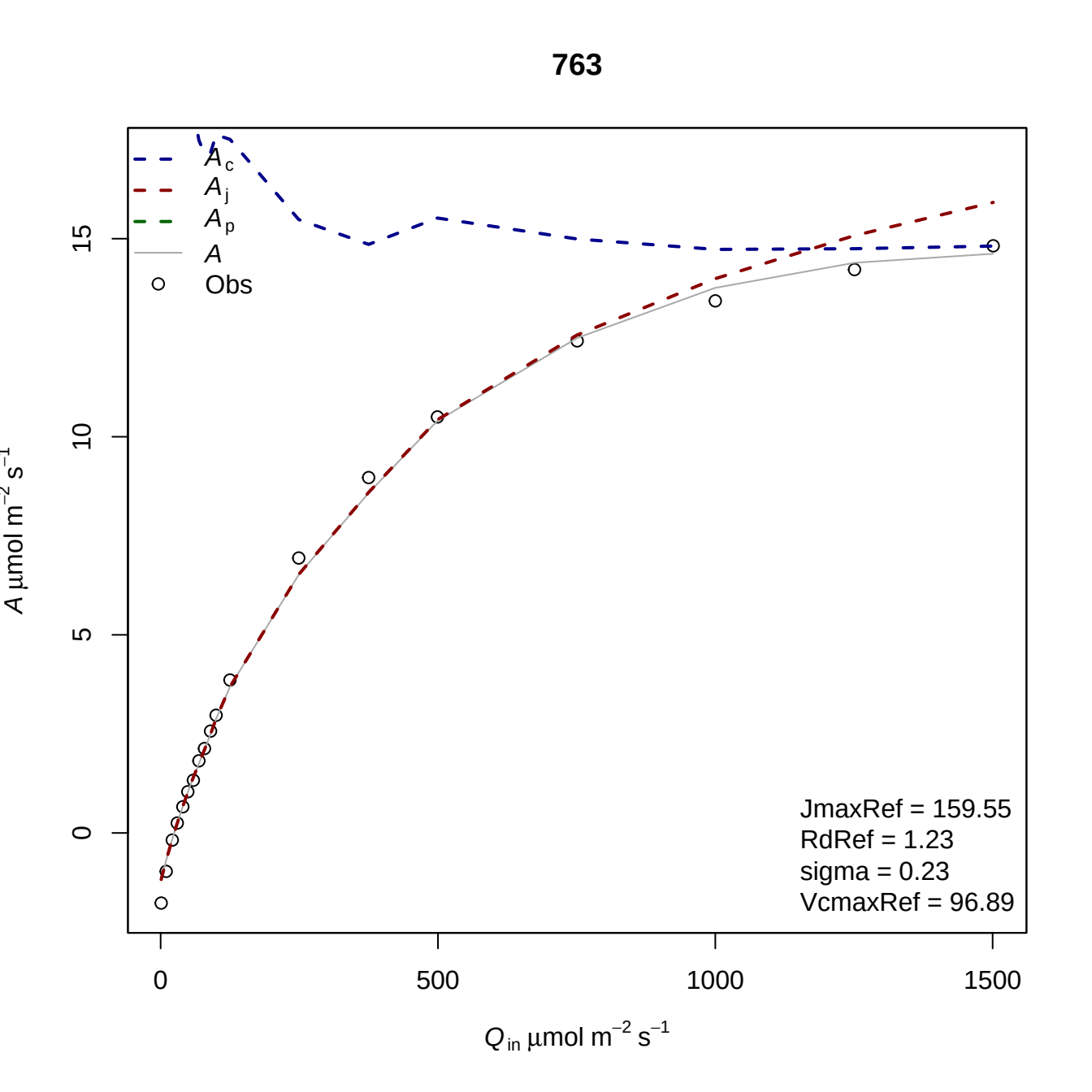

758
A_c
A_j
A_p
A
Obs
JmaxRef = 403.91
RdRef = 1.28
sigma = 0.46
VcmaxRef = 146.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

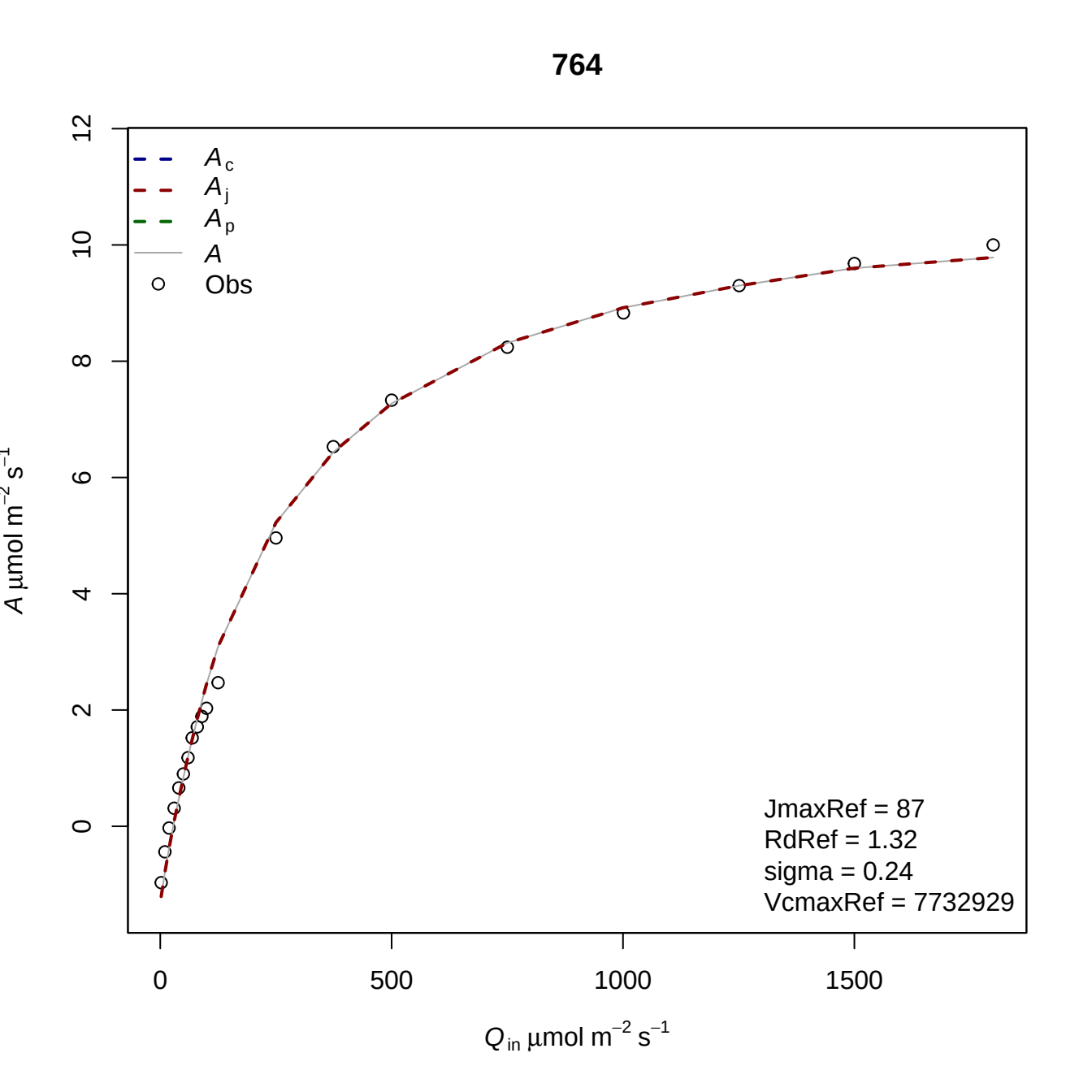

759
A_c
A_j
A_p
A
Obs
JmaxRef = 72.19
RdRef = 1.85
sigma = 0.33
VcmaxRef = 132092.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

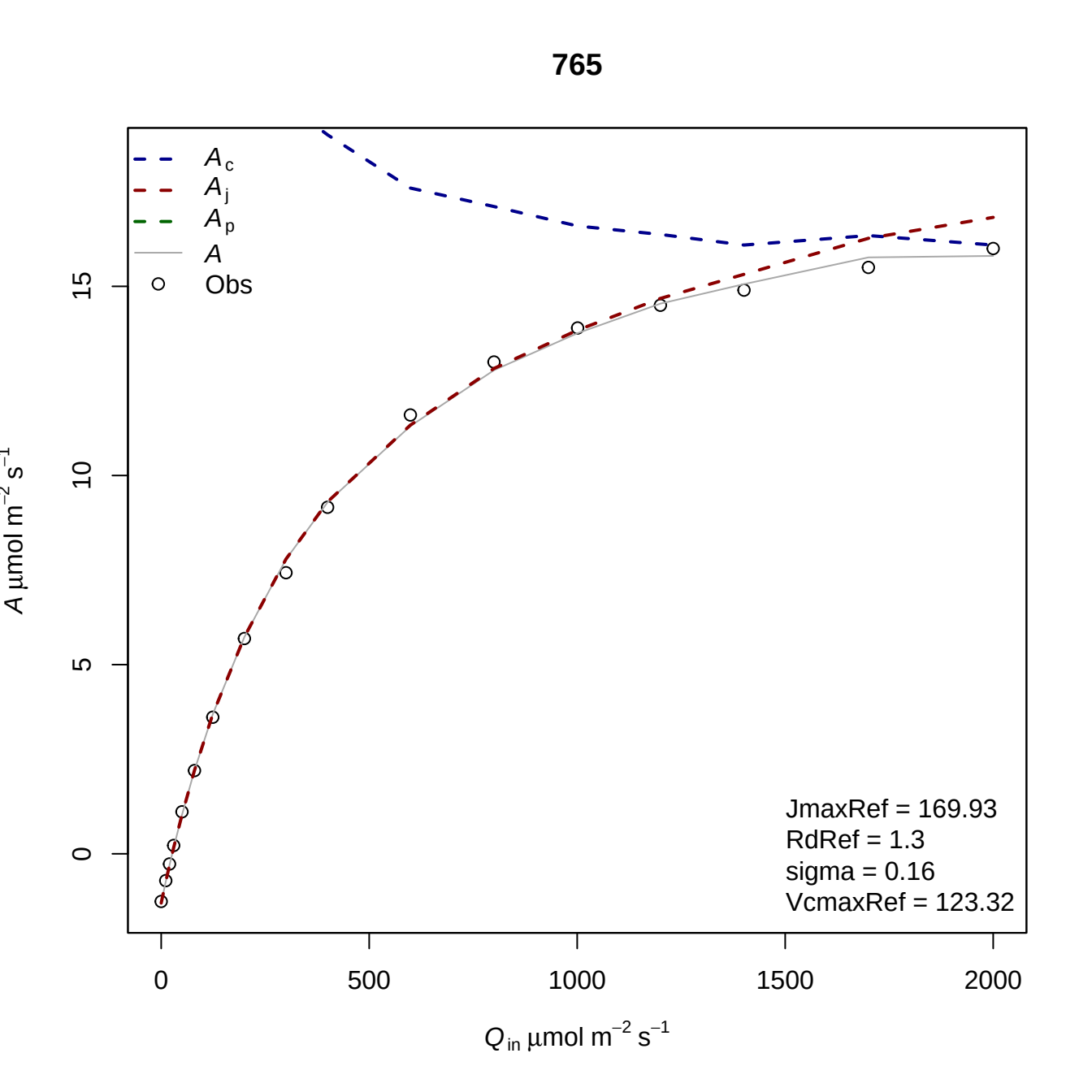

760
A_c
A_j
A_p
A
Obs
JmaxRef = 110.74
RdRef = 1.33
sigma = 0.3
VcmaxRef = 73.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

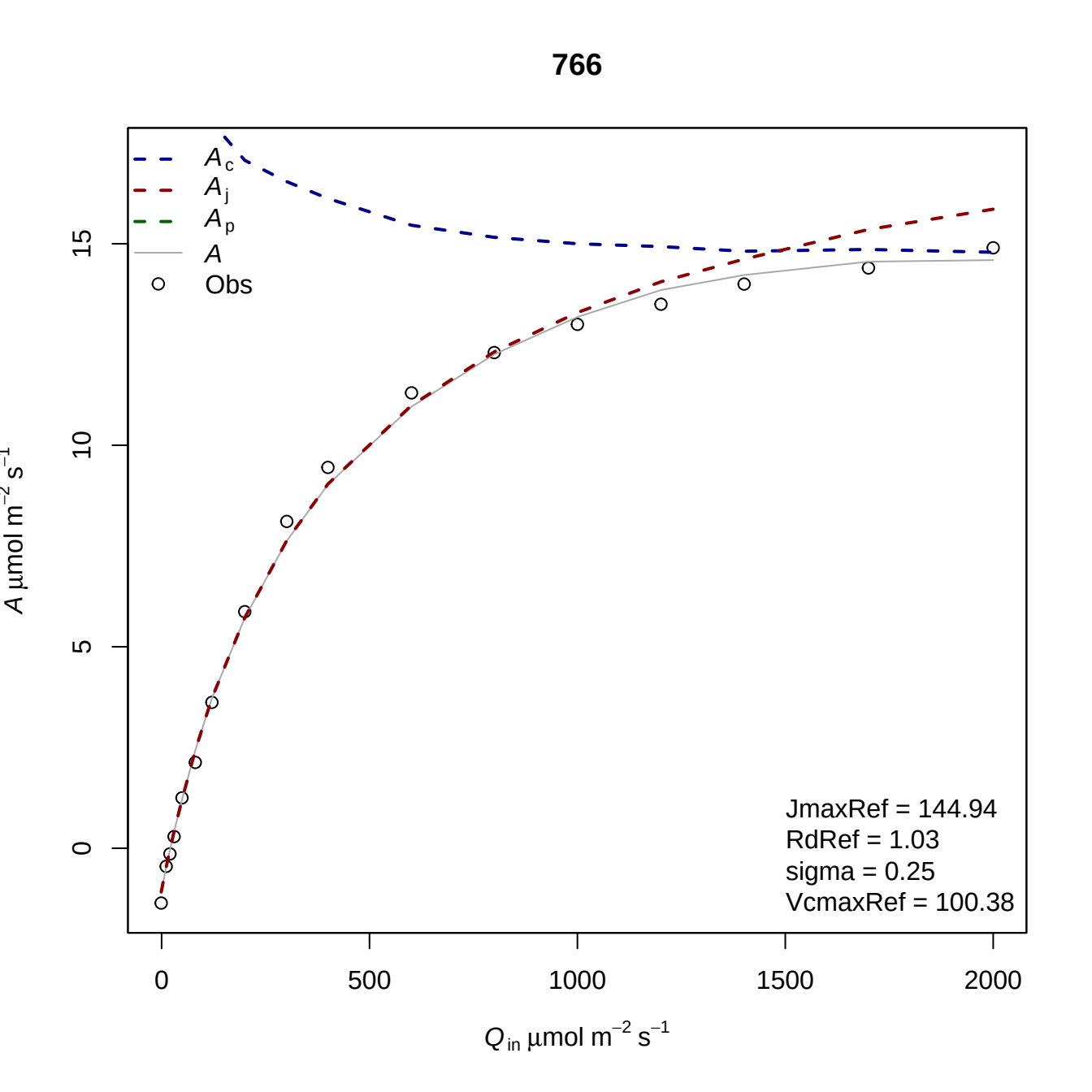

761
A_c
A_j
A_p
A
Obs
JmaxRef = 139.5
RdRef = 2.29
sigma = 0.16
VcmaxRef = 122.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

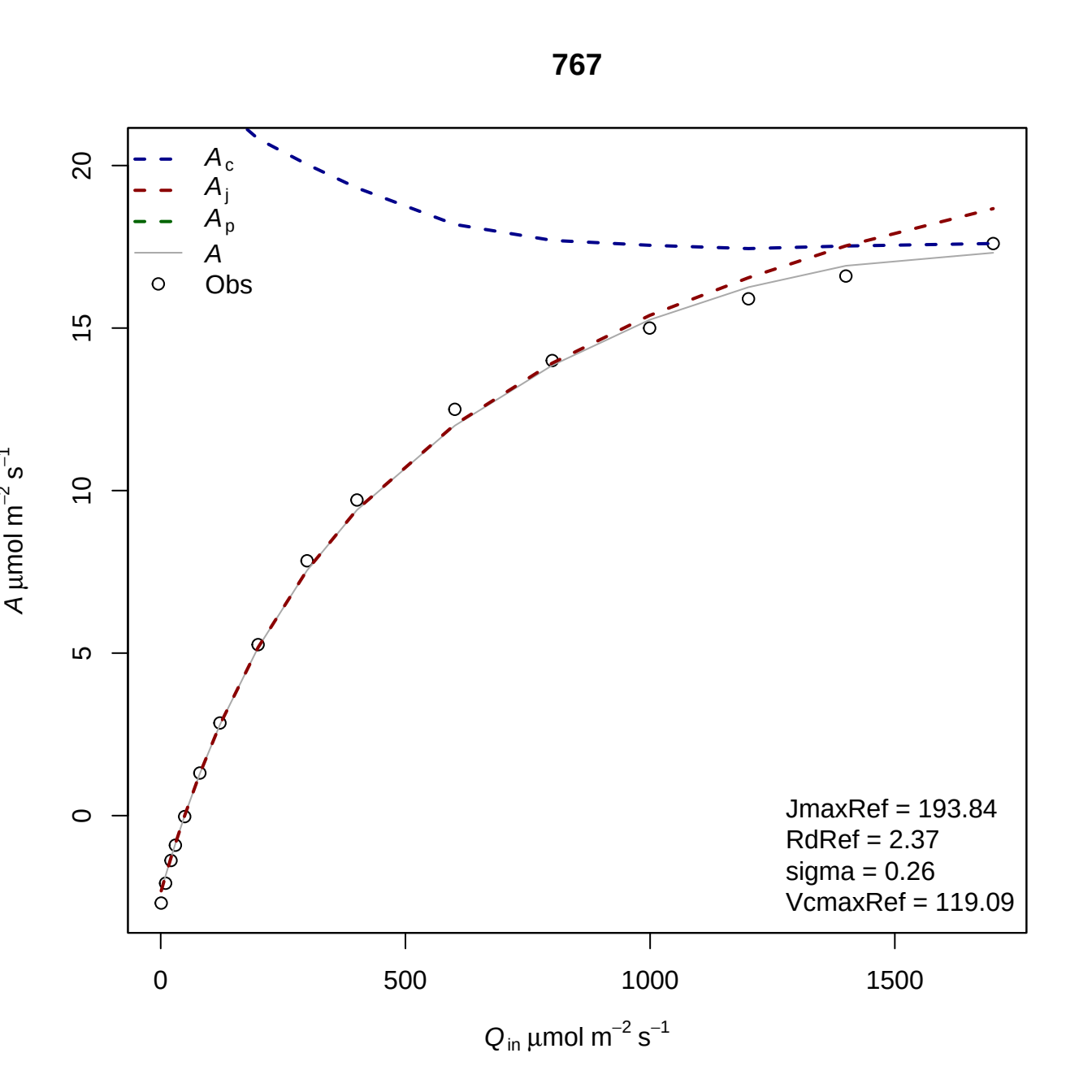

762
A_c
A_j
A_p
A
Obs
JmaxRef = 137.38
RdRef = 2.47
sigma = 0.21
VcmaxRef = 1205644.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

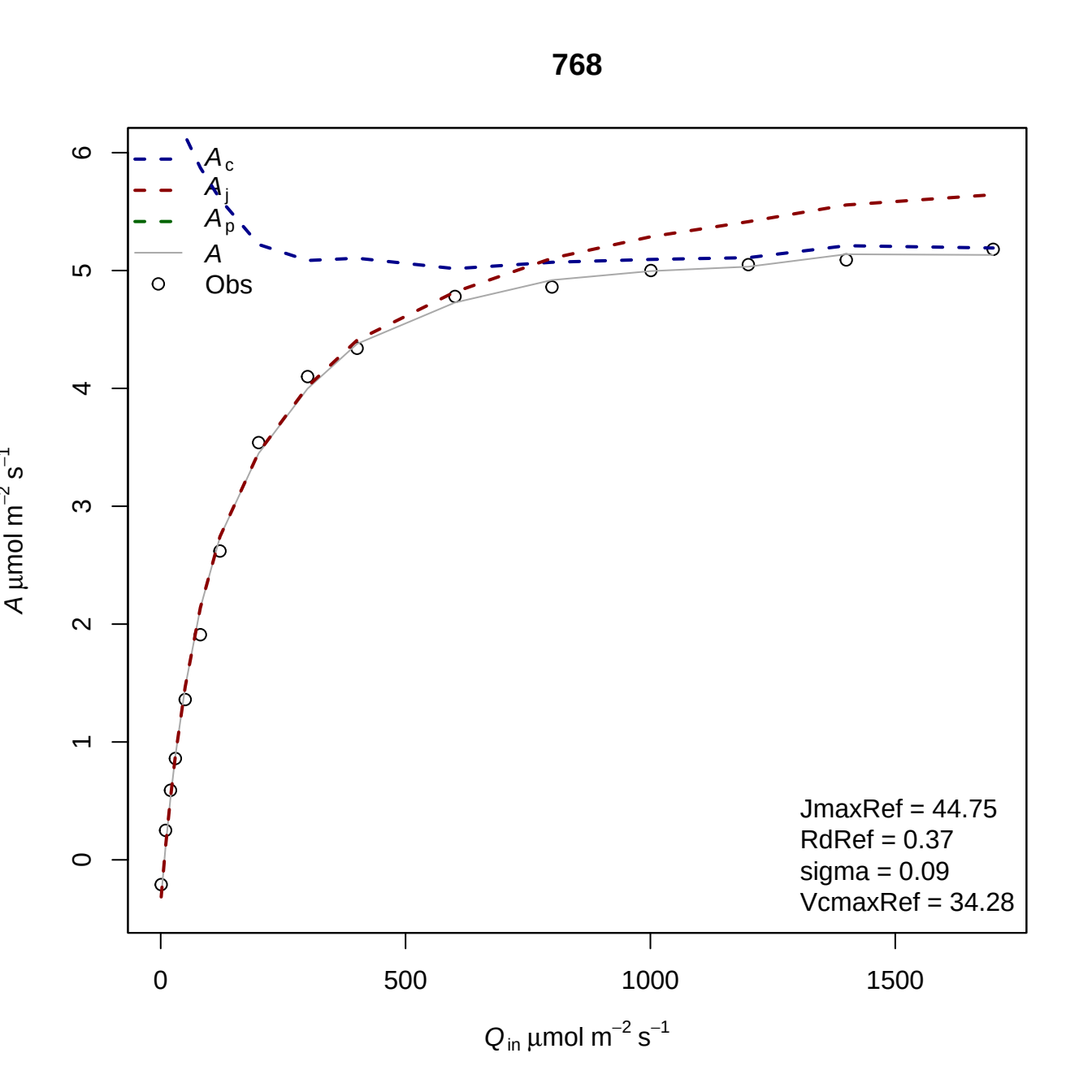

763
A_c
A_j
A_p
A
Obs
JmaxRef = 159.55
RdRef = 1.23
sigma = 0.23
VcmaxRef = 96.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

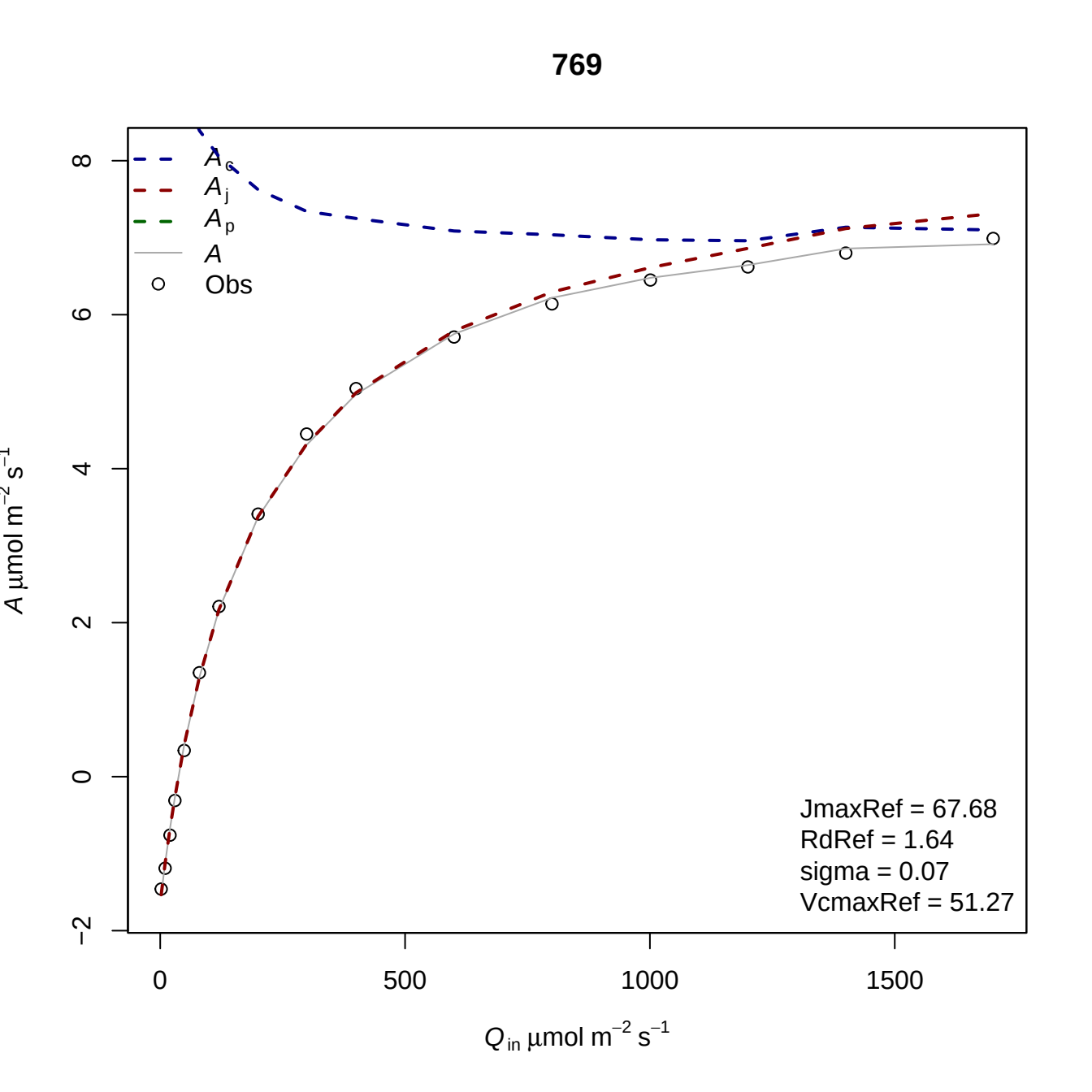

764
A_c
A_j
A_p
A
Obs
JmaxRef = 87
RdRef = 1.32
sigma = 0.24
VcmaxRef = 7732929
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

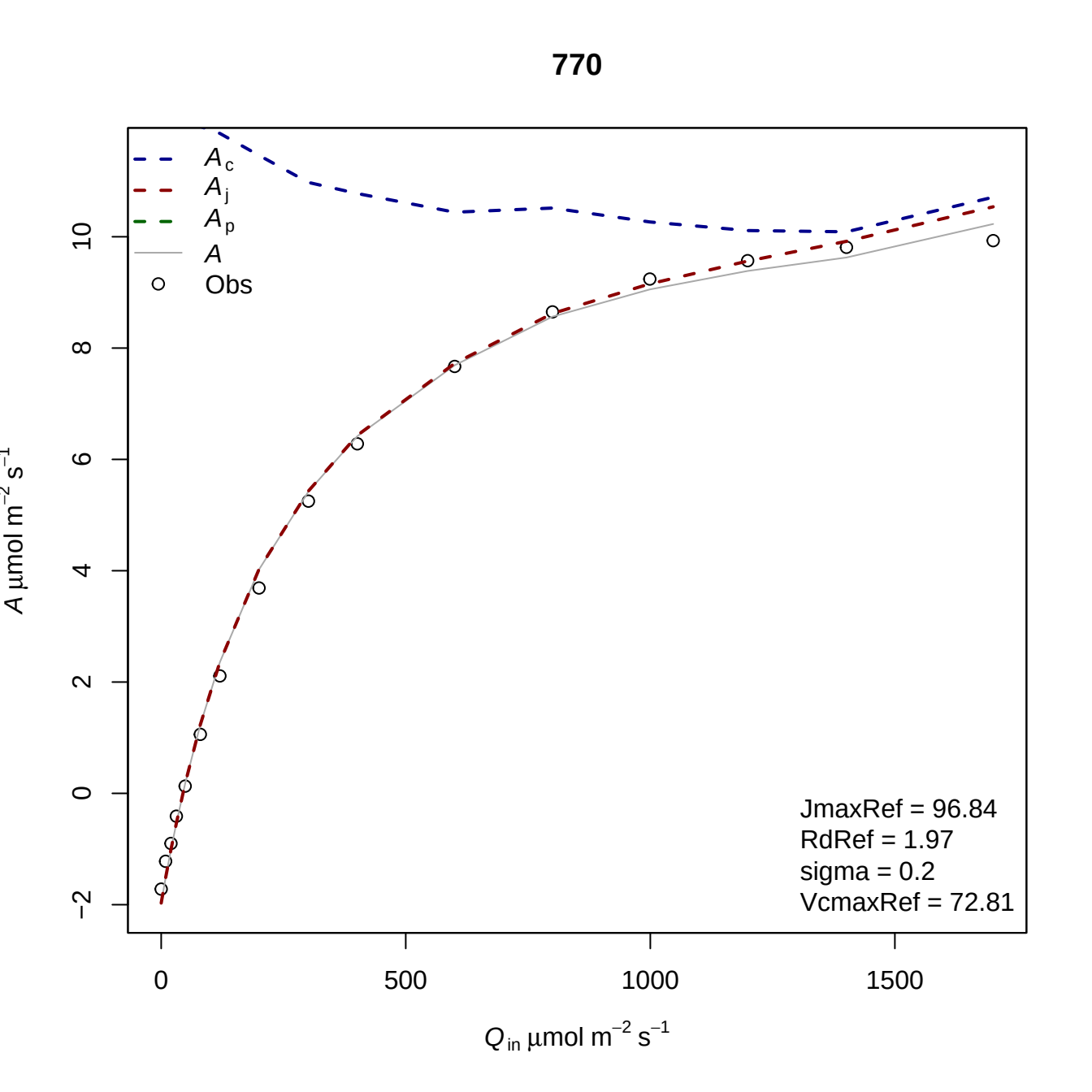

765
A_c
A_j
A_p
A
Obs
JmaxRef = 169.93
RdRef = 1.3
sigma = 0.16
VcmaxRef = 123.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

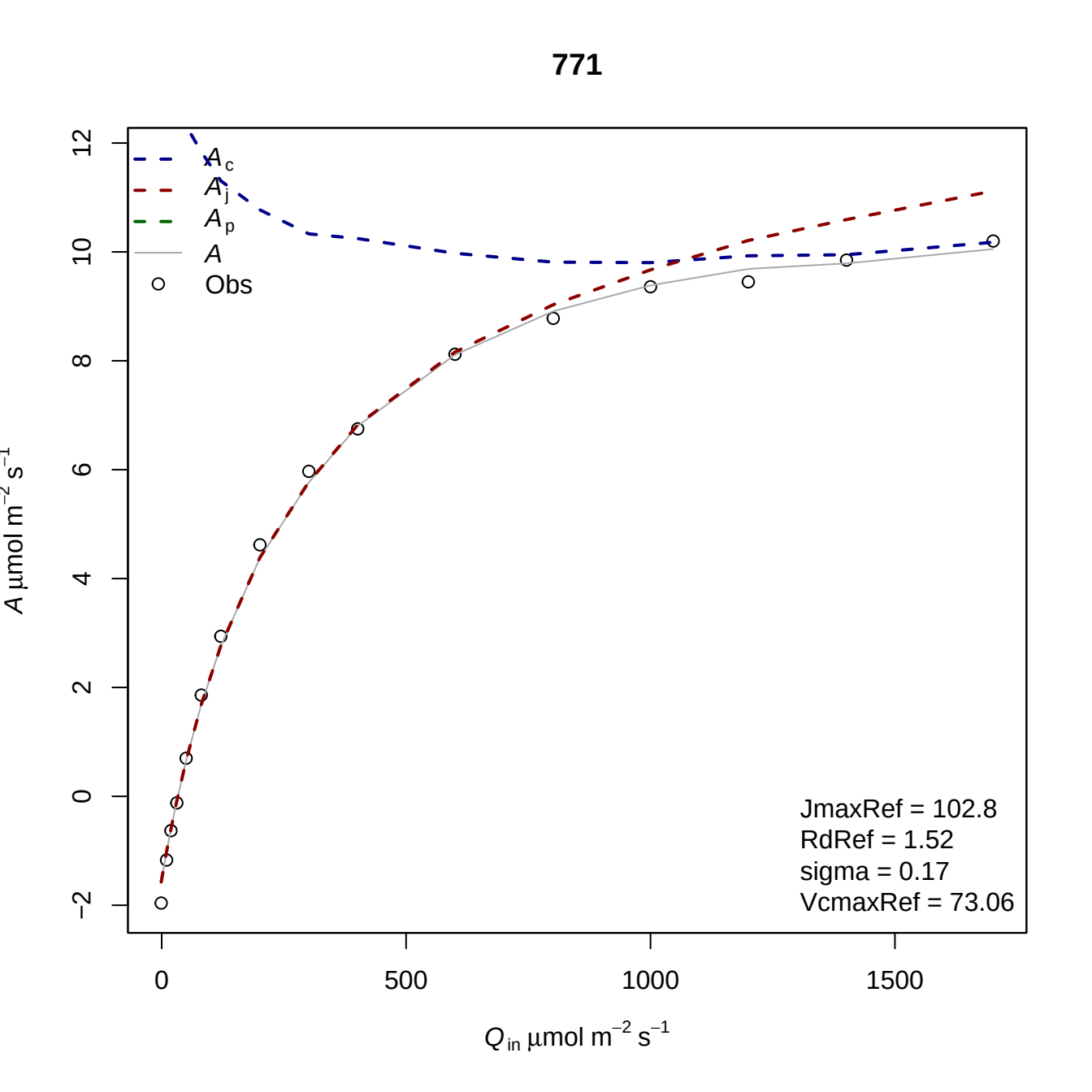

766
A_c
A_j
A_p
A
Obs
JmaxRef = 144.94
RdRef = 1.03
sigma = 0.25
VcmaxRef = 100.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

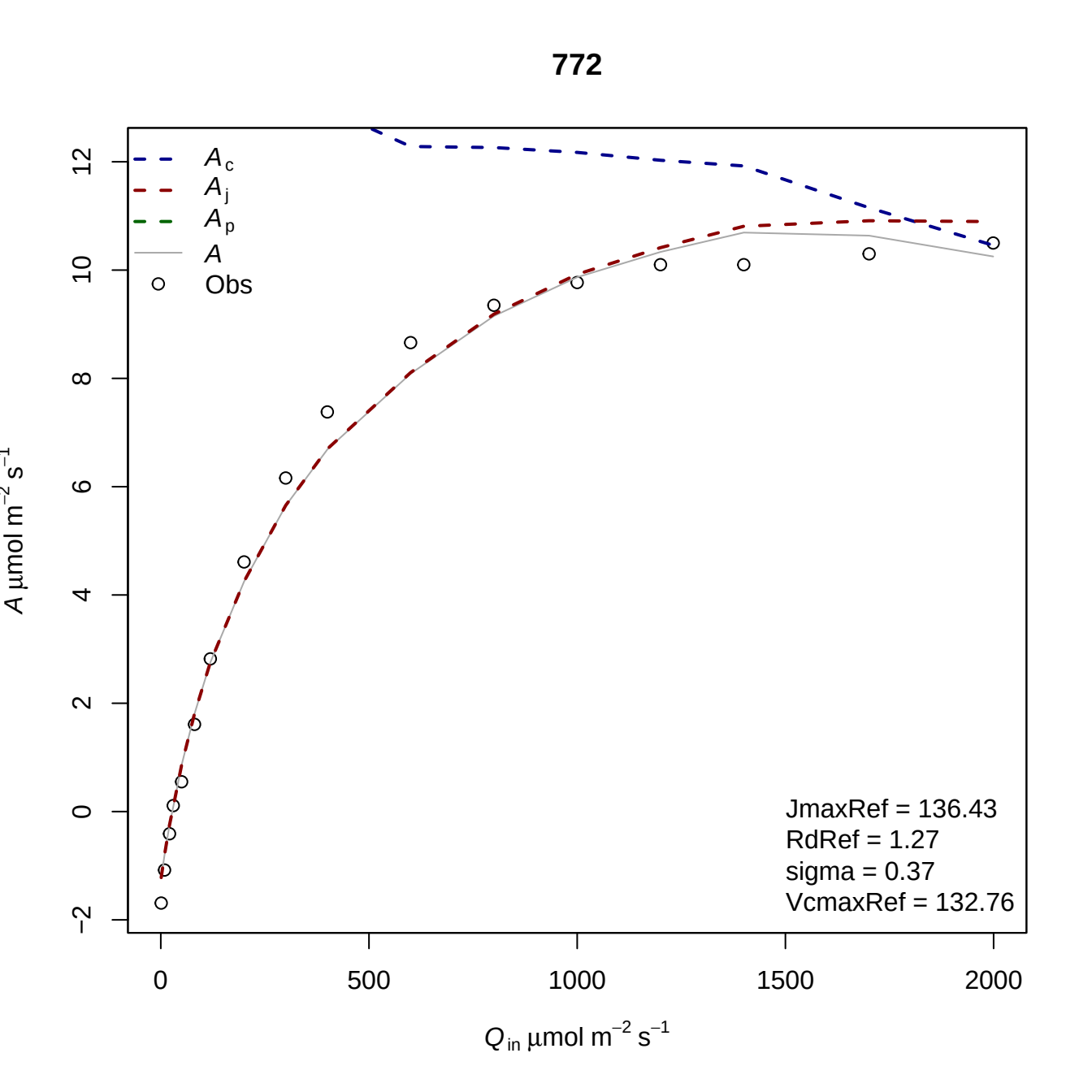

767
A_c
A_j
A_p
A
Obs
JmaxRef = 193.84
RdRef = 2.37
sigma = 0.26
VcmaxRef = 119.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

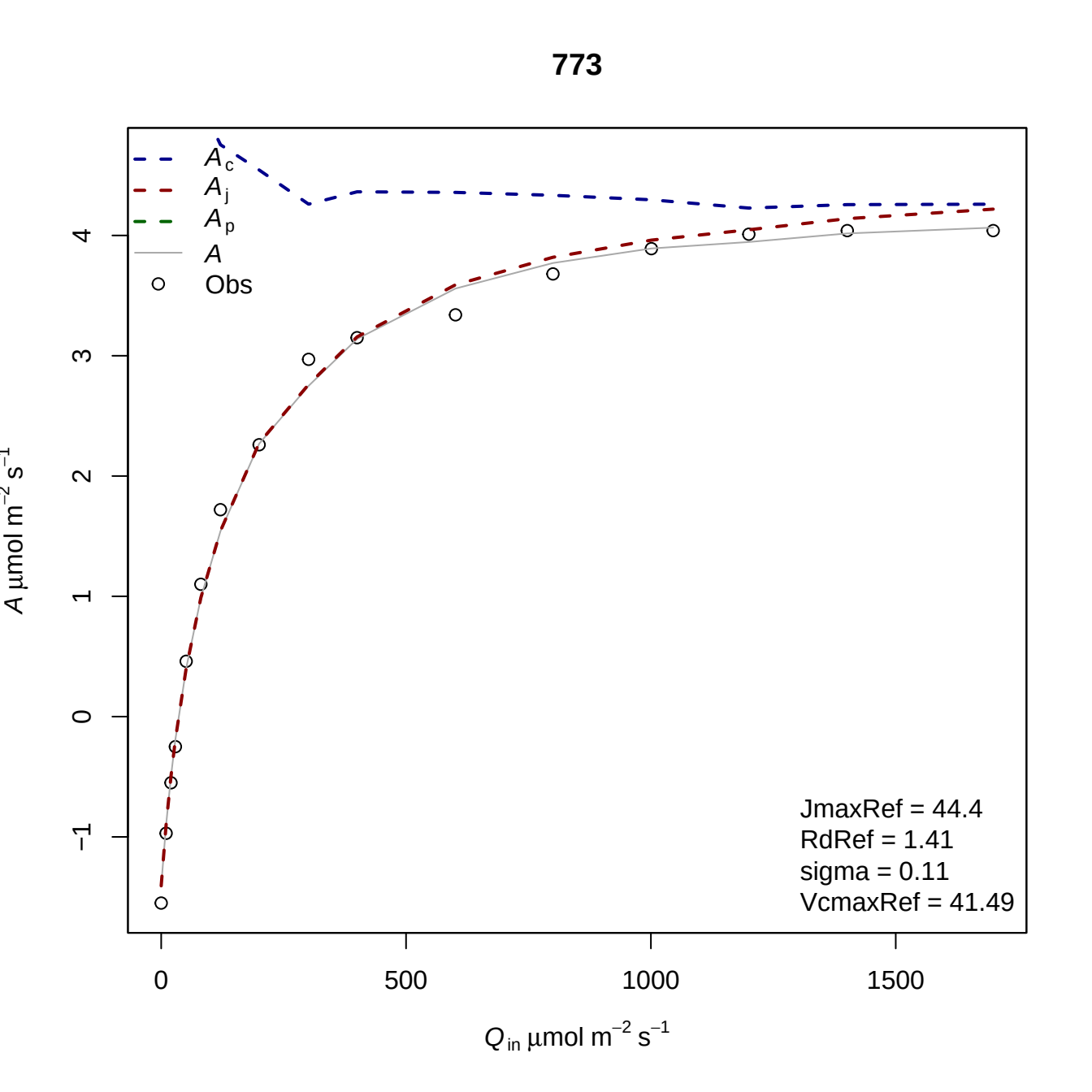

768
A_c
A_j
A_p
A
Obs
JmaxRef = 44.75
RdRef = 0.37
sigma = 0.09
VcmaxRef = 34.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

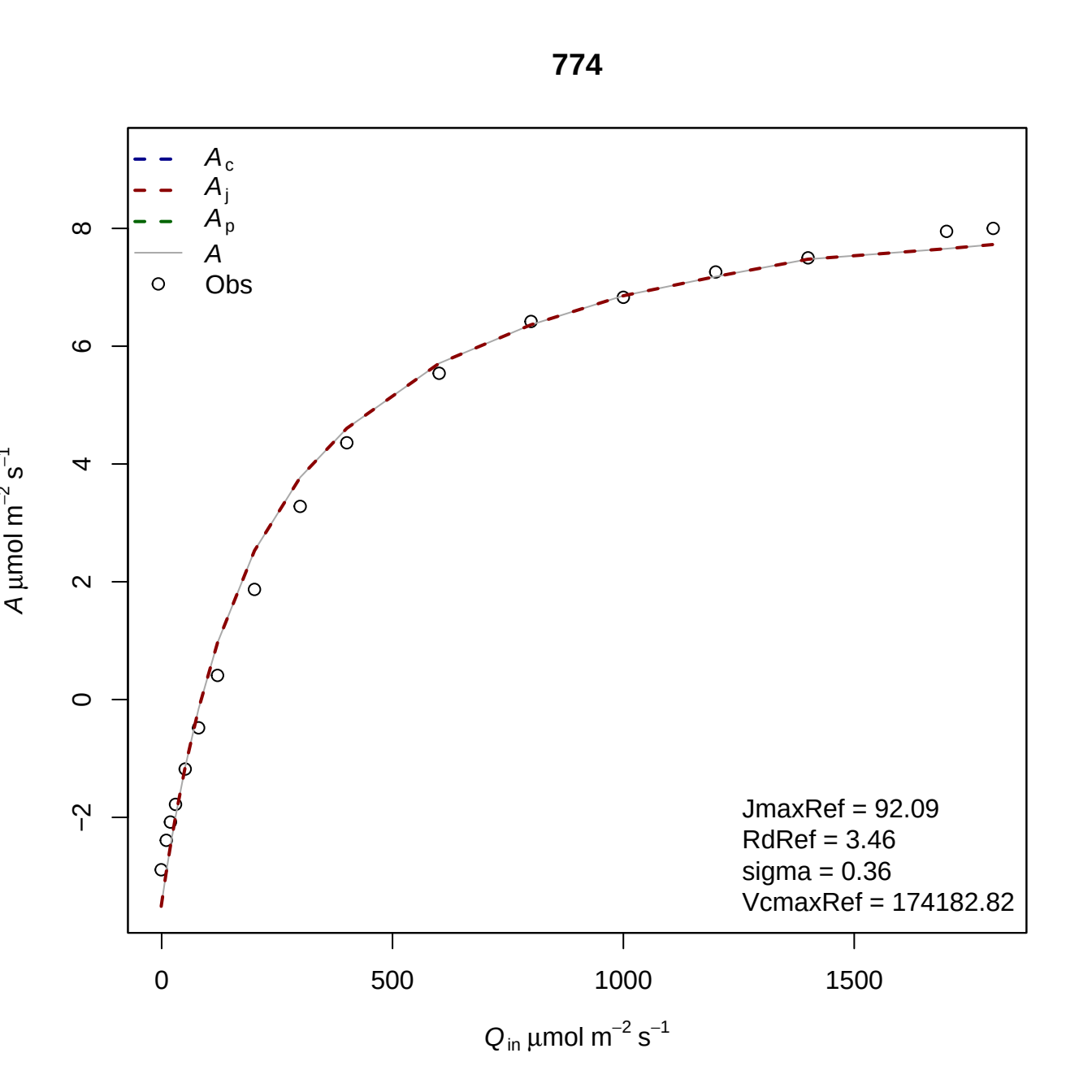

769
A_c
A_j
A_p
A
Obs
JmaxRef = 67.68
RdRef = 1.64
sigma = 0.07
VcmaxRef = 51.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

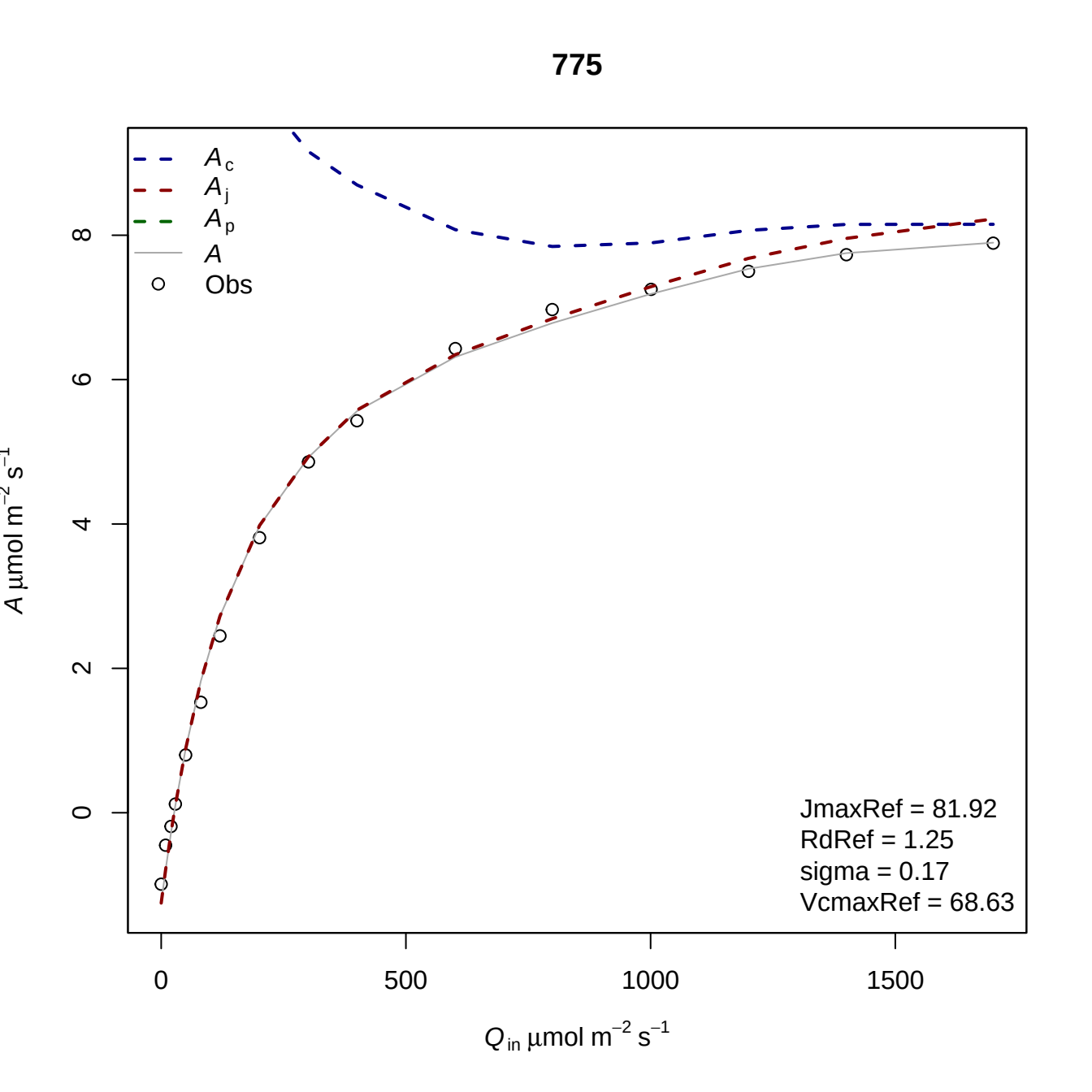

770
A_c
A_j
A_p
A
Obs
JmaxRef = 96.84
RdRef = 1.97
sigma = 0.2
VcmaxRef = 72.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

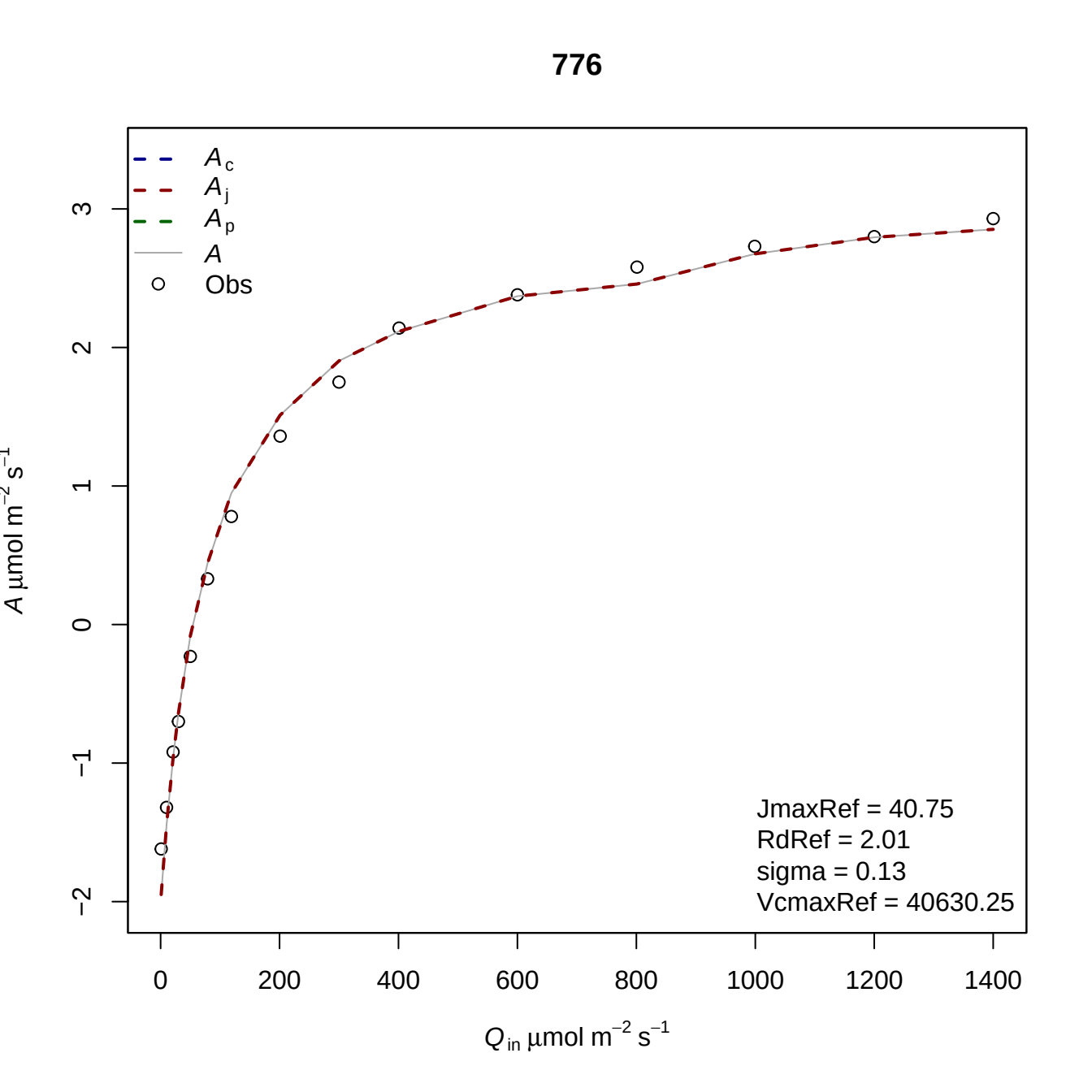

771
A_c
A_j
A_p
A
Obs
JmaxRef = 102.8
RdRef = 1.52
sigma = 0.17
VcmaxRef = 73.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

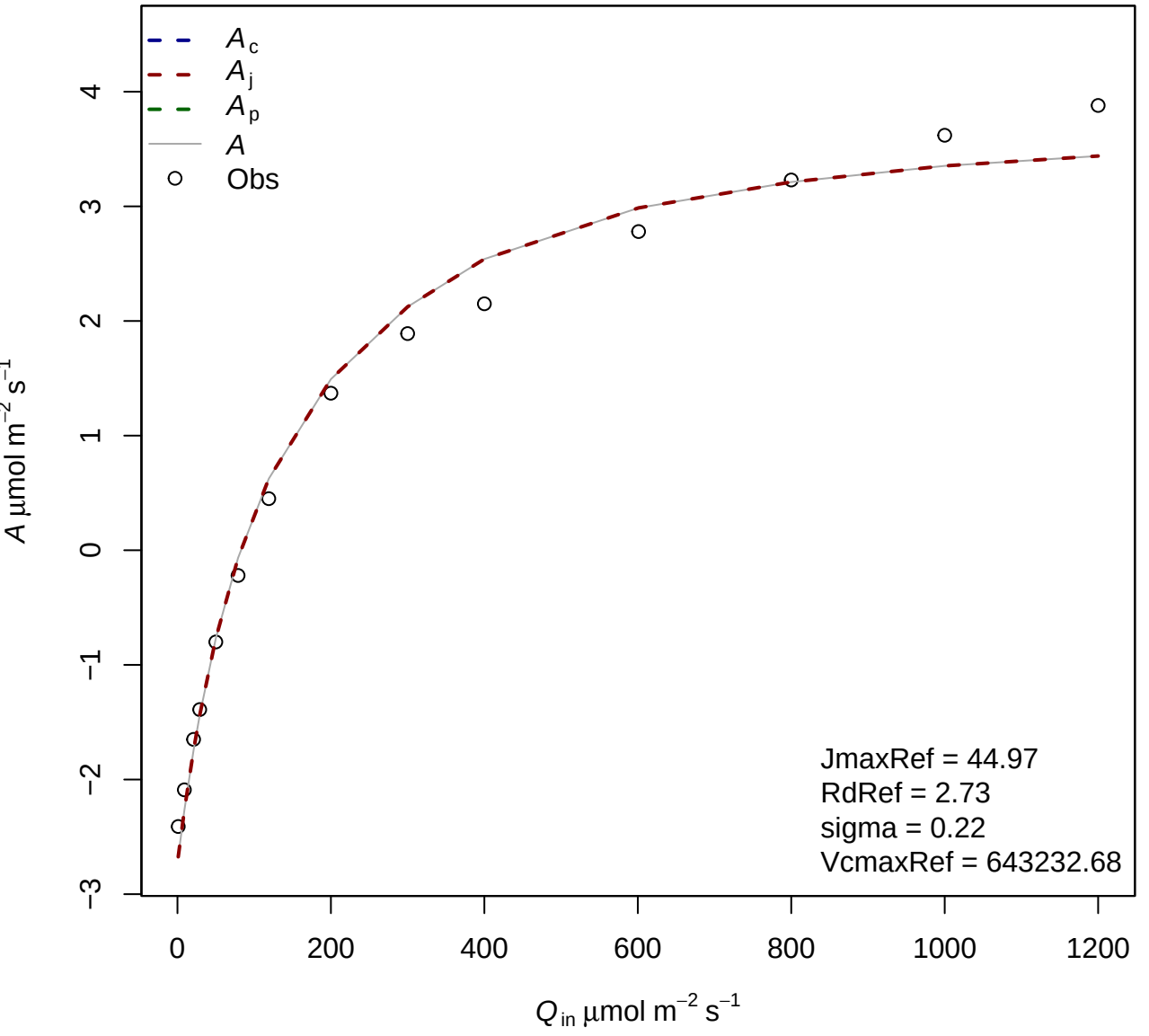

772
A_c
A_j
A_p
A
Obs
JmaxRef = 136.43
RdRef = 1.27
sigma = 0.37
VcmaxRef = 132.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

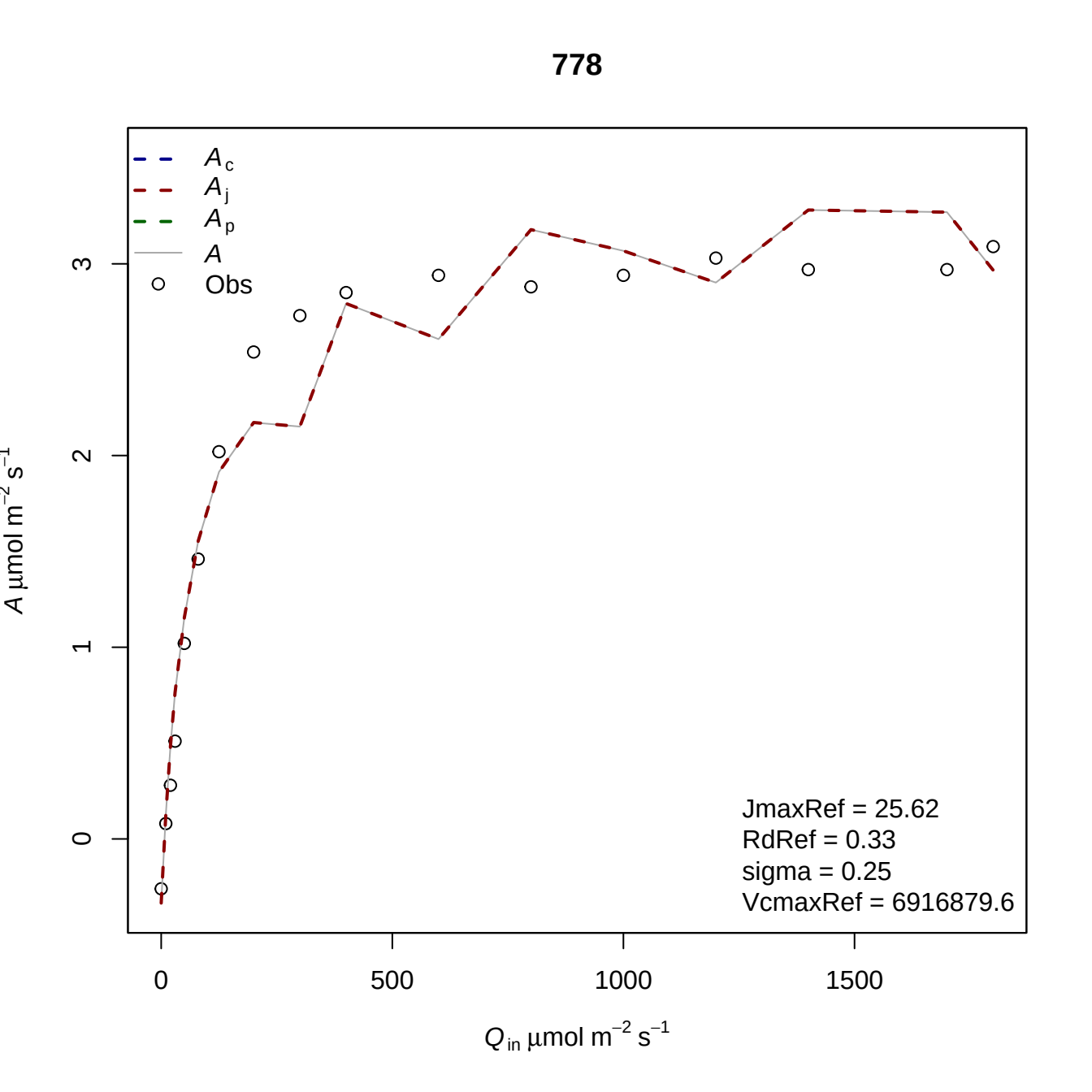

773
A_c
A_j
A_p
A
Obs
JmaxRef = 44.4
RdRef = 1.41
sigma = 0.11
VcmaxRef = 41.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

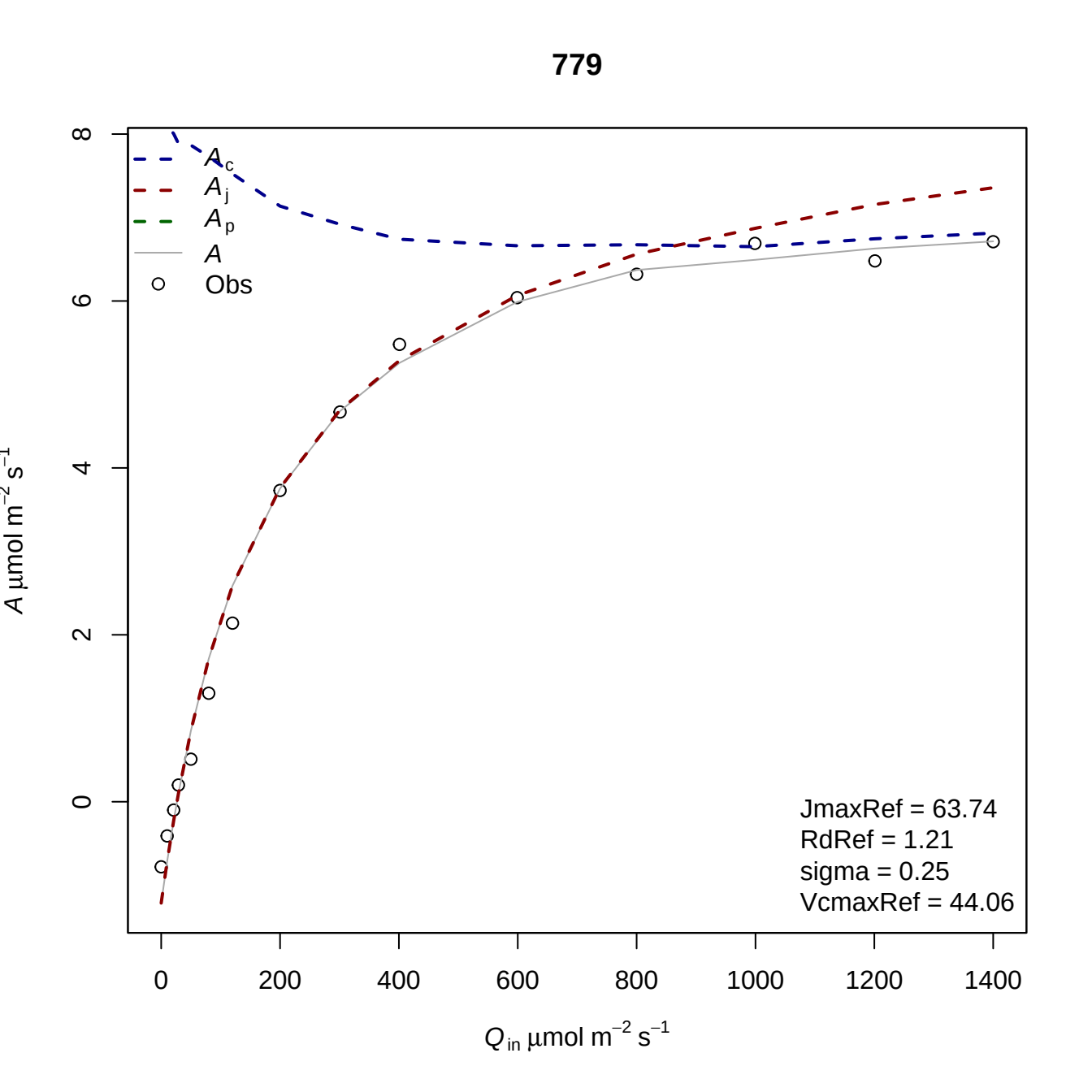

774
A_c
A_j
A_p
A
Obs
JmaxRef = 92.09
RdRef = 3.46
sigma = 0.36
VcmaxRef = 174182.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

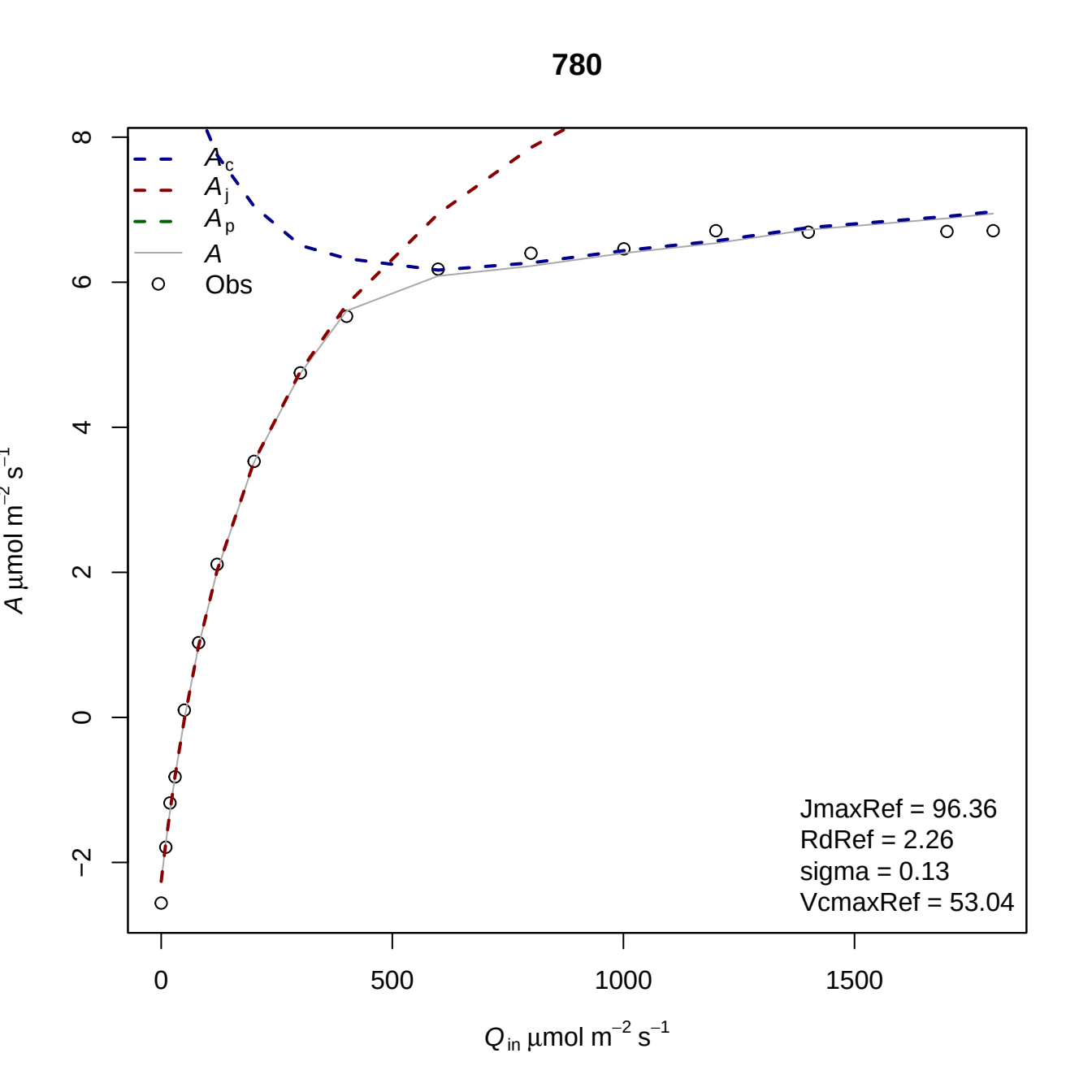

775
A_c
A_j
A_p
A
Obs
JmaxRef = 81.92
RdRef = 1.25
sigma = 0.17
VcmaxRef = 68.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

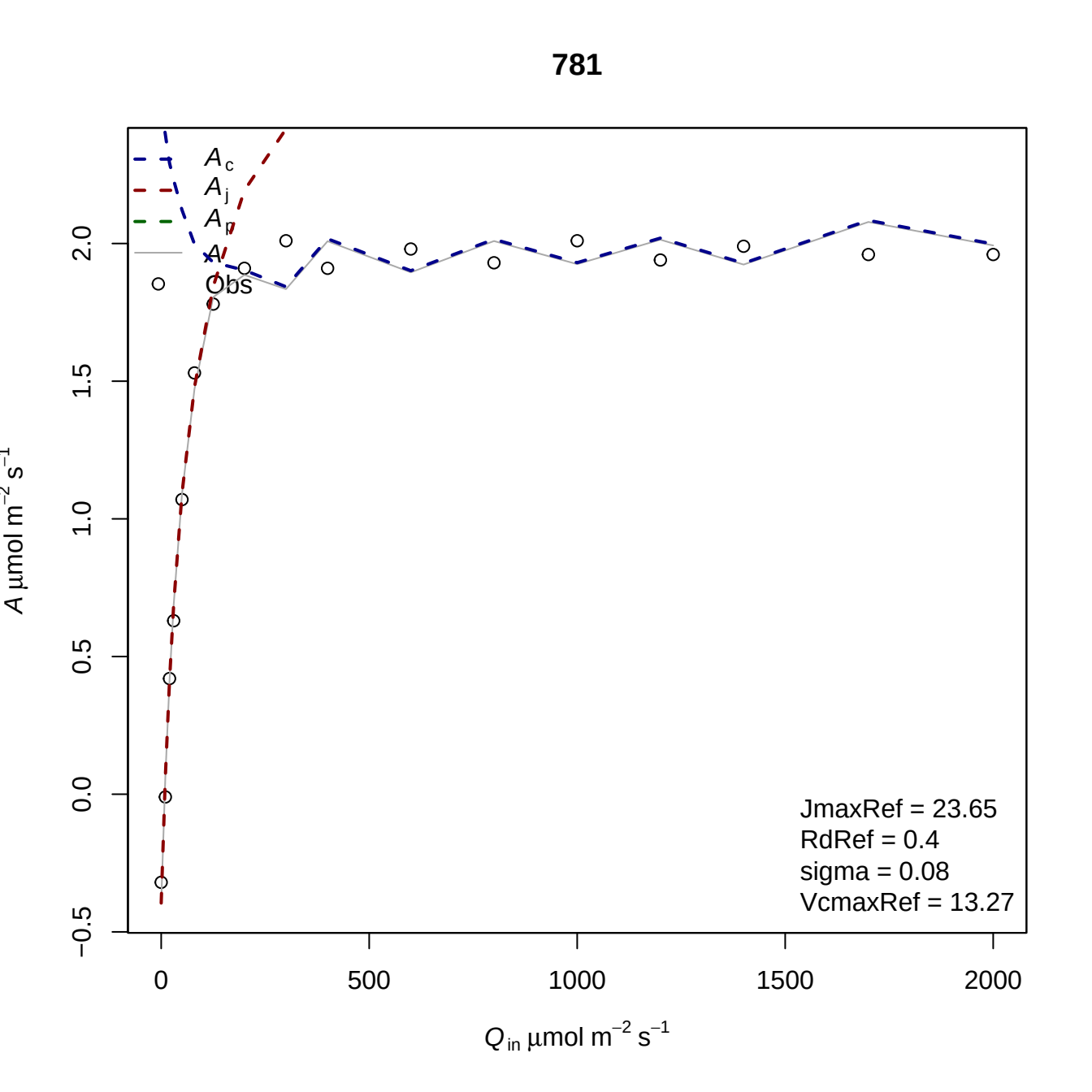

776
A_c
A_j
A_p
A
Obs
JmaxRef = 40.75
RdRef = 2.01
sigma = 0.13
VcmaxRef = 40630.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

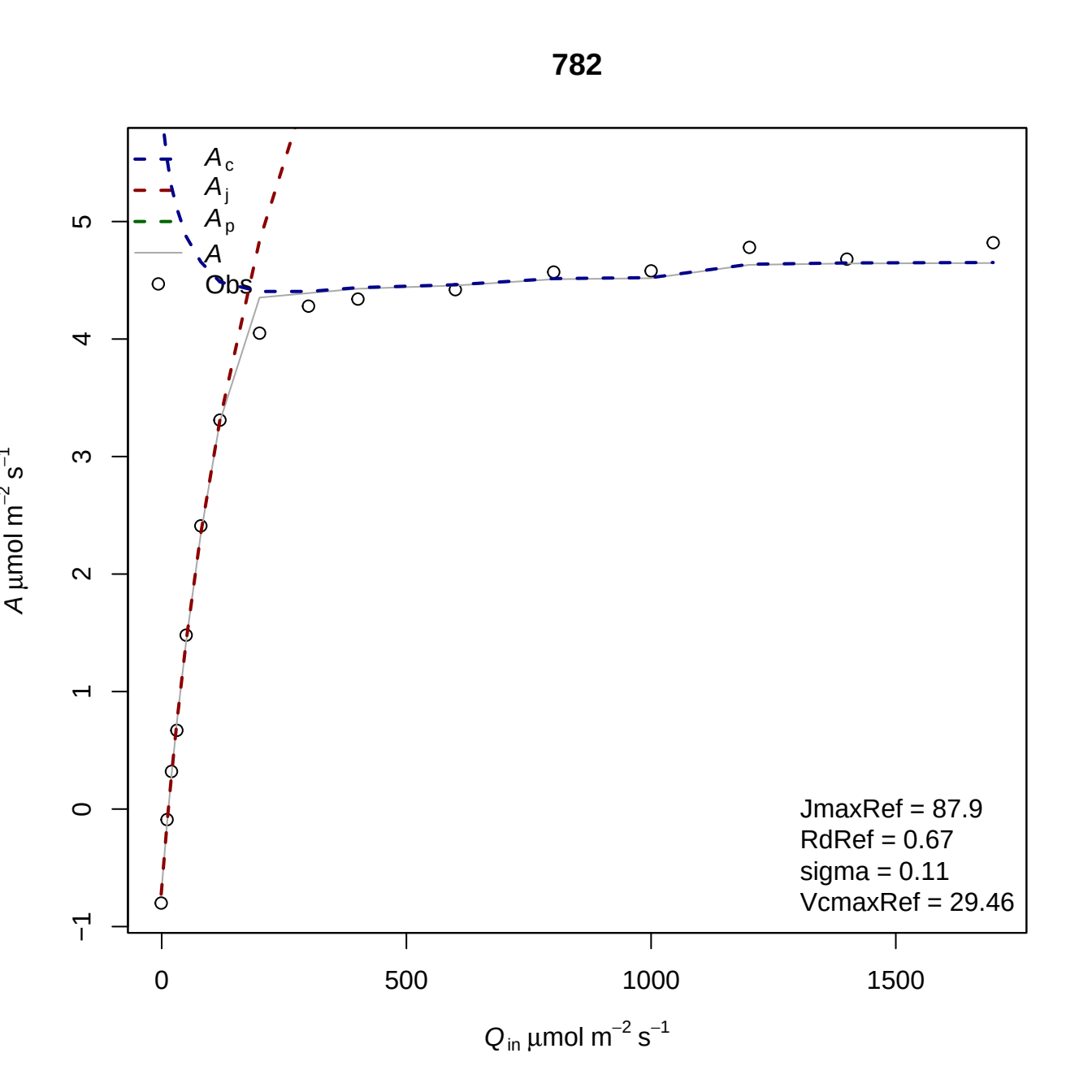

A_c
A_j
A_p
A
Obs
JmaxRef = 44.97
RdRef = 2.73
sigma = 0.22
VcmaxRef = 643232.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−3
−2
−1
0
1
2
3
4
0
200
400
600
800
1000
1200

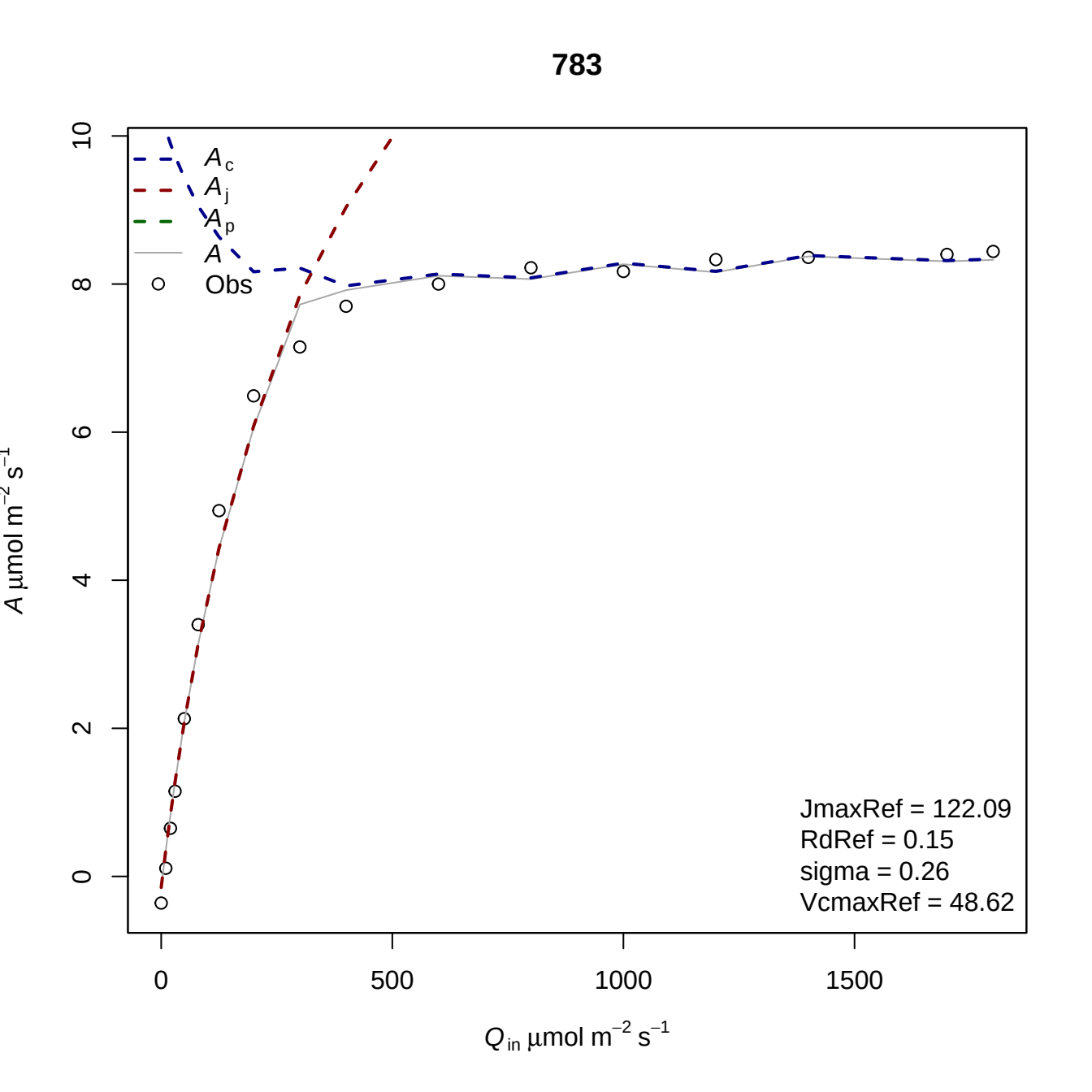

778
A_c
A_j
A_p
A
Obs
JmaxRef = 25.62
RdRef = 0.33
sigma = 0.25
VcmaxRef = 6916879.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500

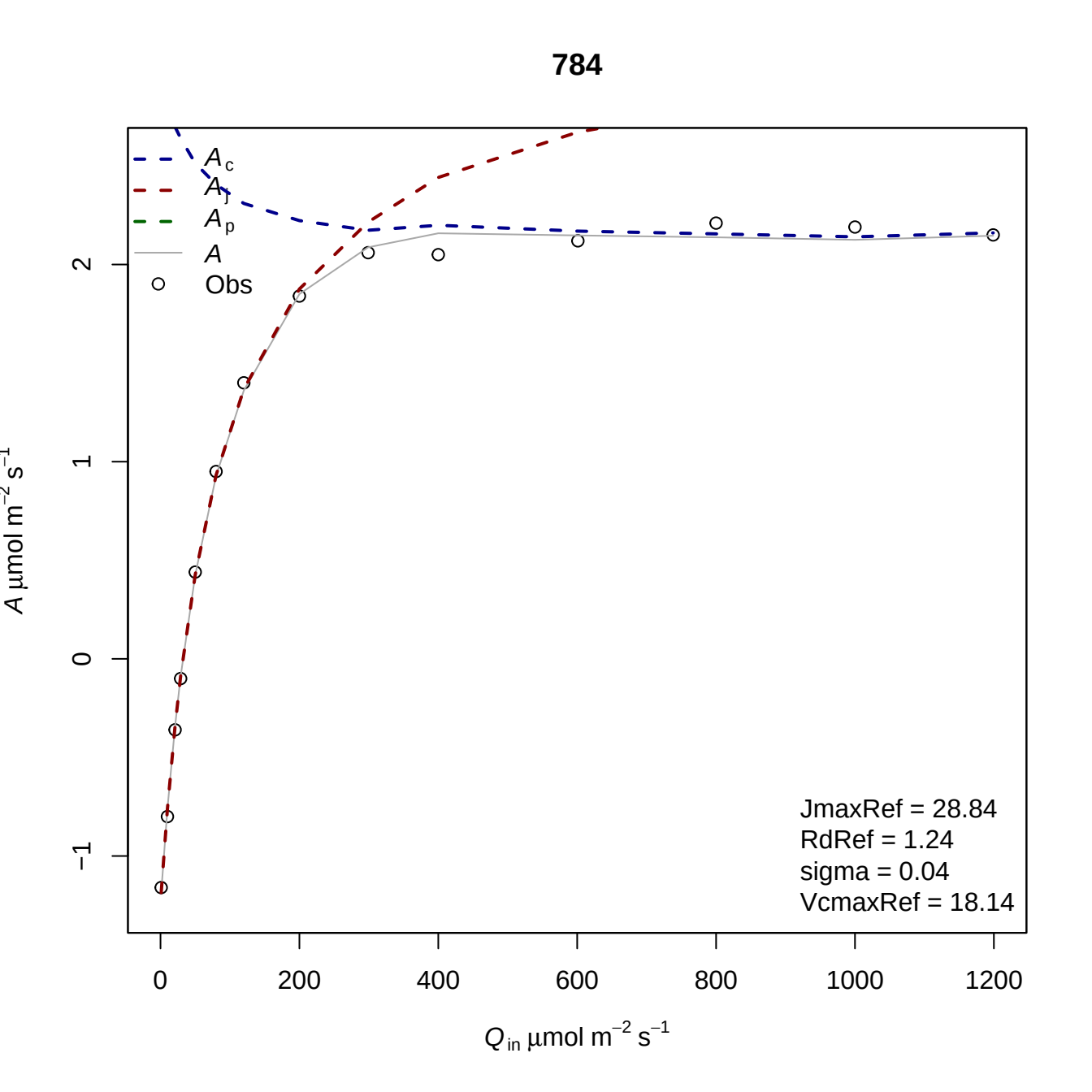

779
A_c
A_j
A_p
A
Obs
JmaxRef = 63.74
RdRef = 1.21
sigma = 0.25
VcmaxRef = 44.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
200
400
600
800
1000
1200
1400

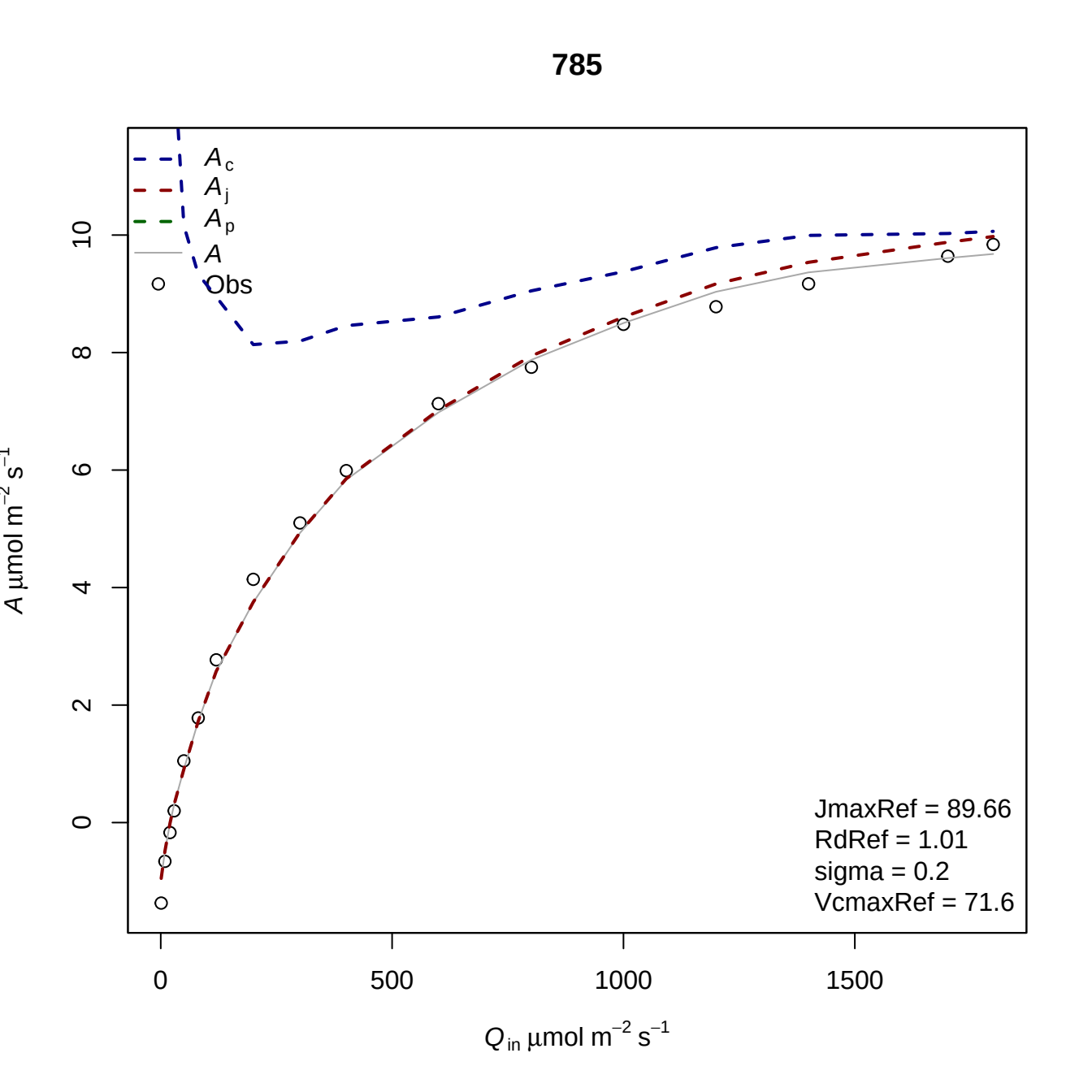

780
A_c
A_j
A_p
A
Obs
JmaxRef = 96.36
RdRef = 2.26
sigma = 0.13
VcmaxRef = 53.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

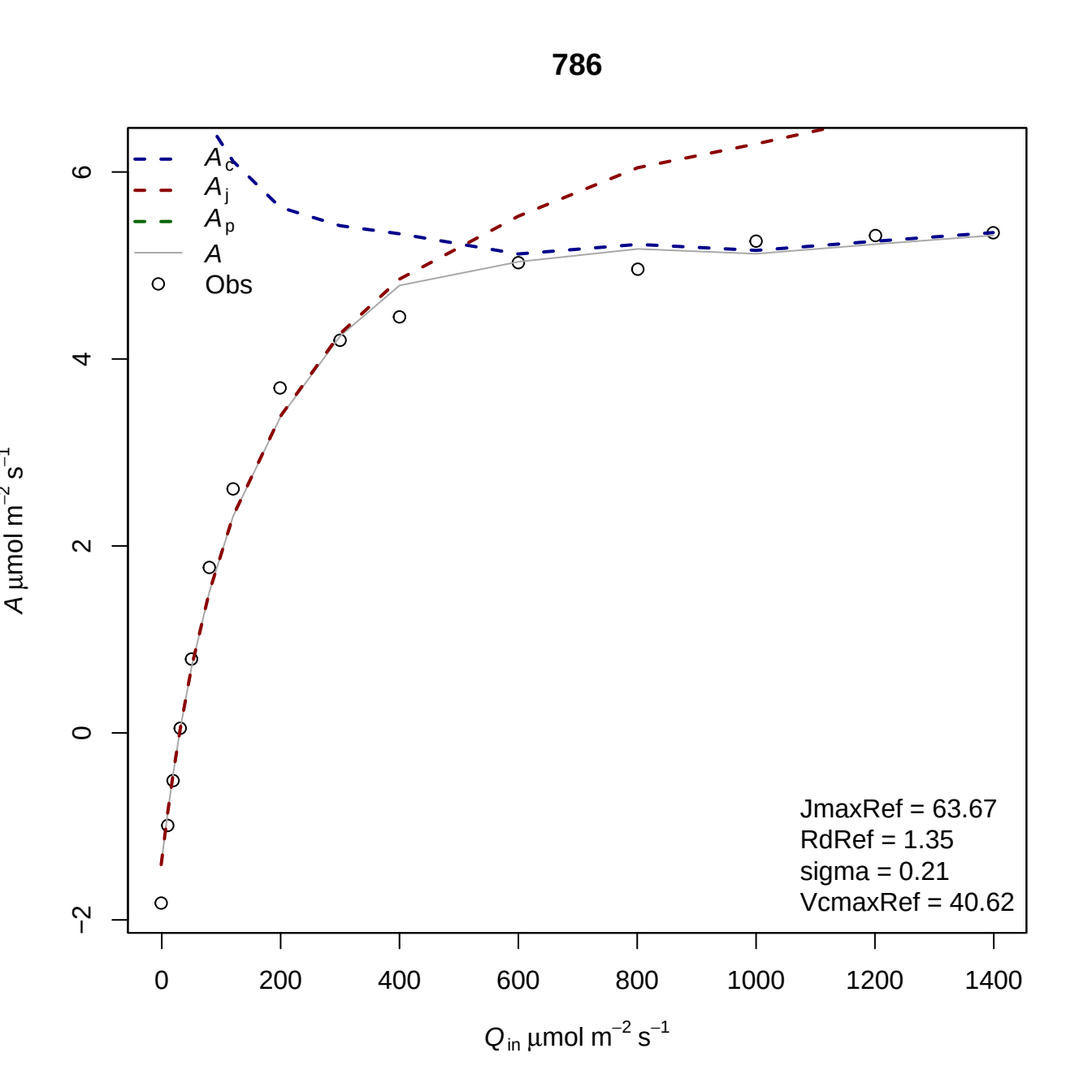

781
A_c
A_j
A_p
A
Obs
JmaxRef = 23.65
RdRef = 0.4
sigma = 0.08
VcmaxRef = 13.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500
2000

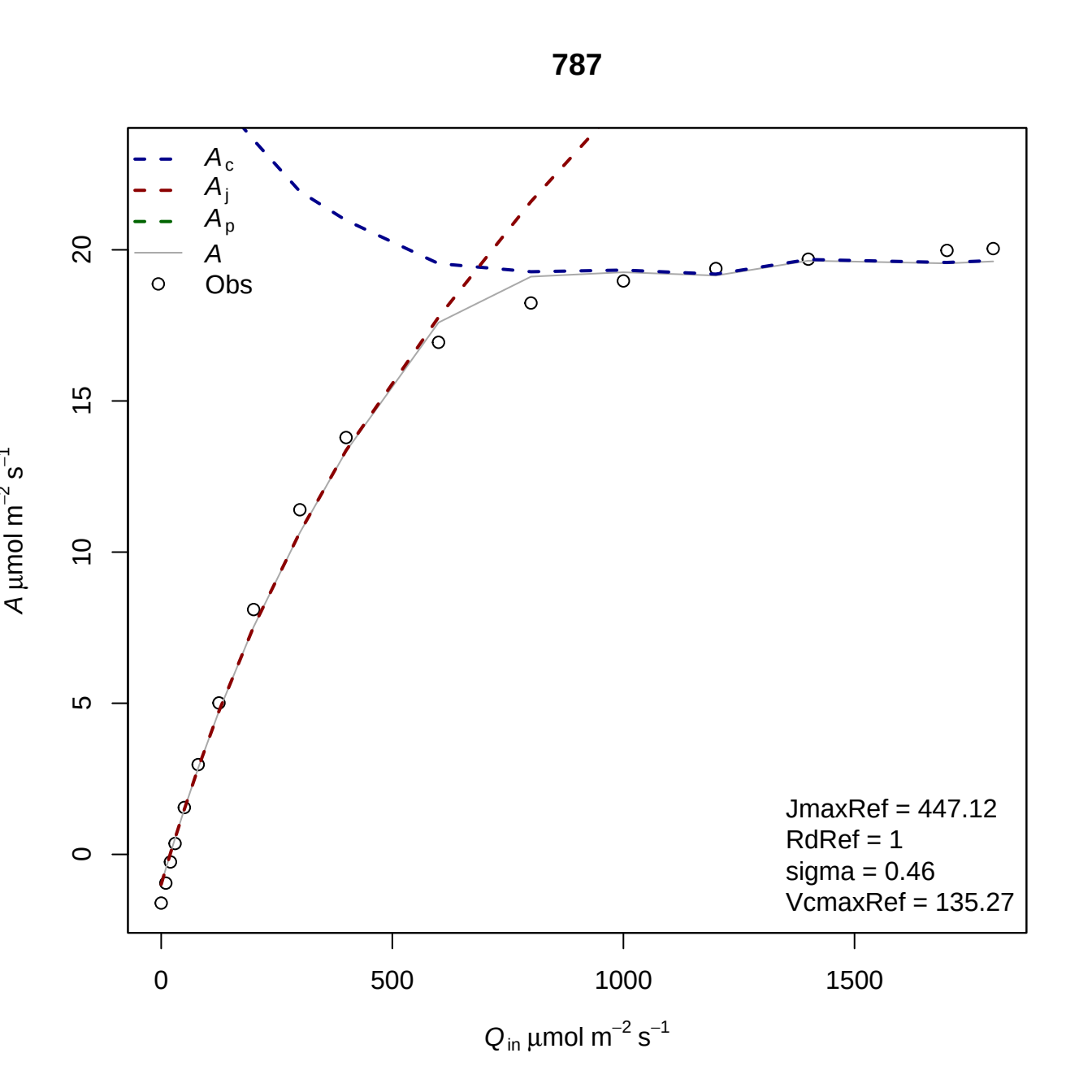

782
A_c
A_j
A_p
A
Obs
JmaxRef = 87.9
RdRef = 0.67
sigma = 0.11
VcmaxRef = 29.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

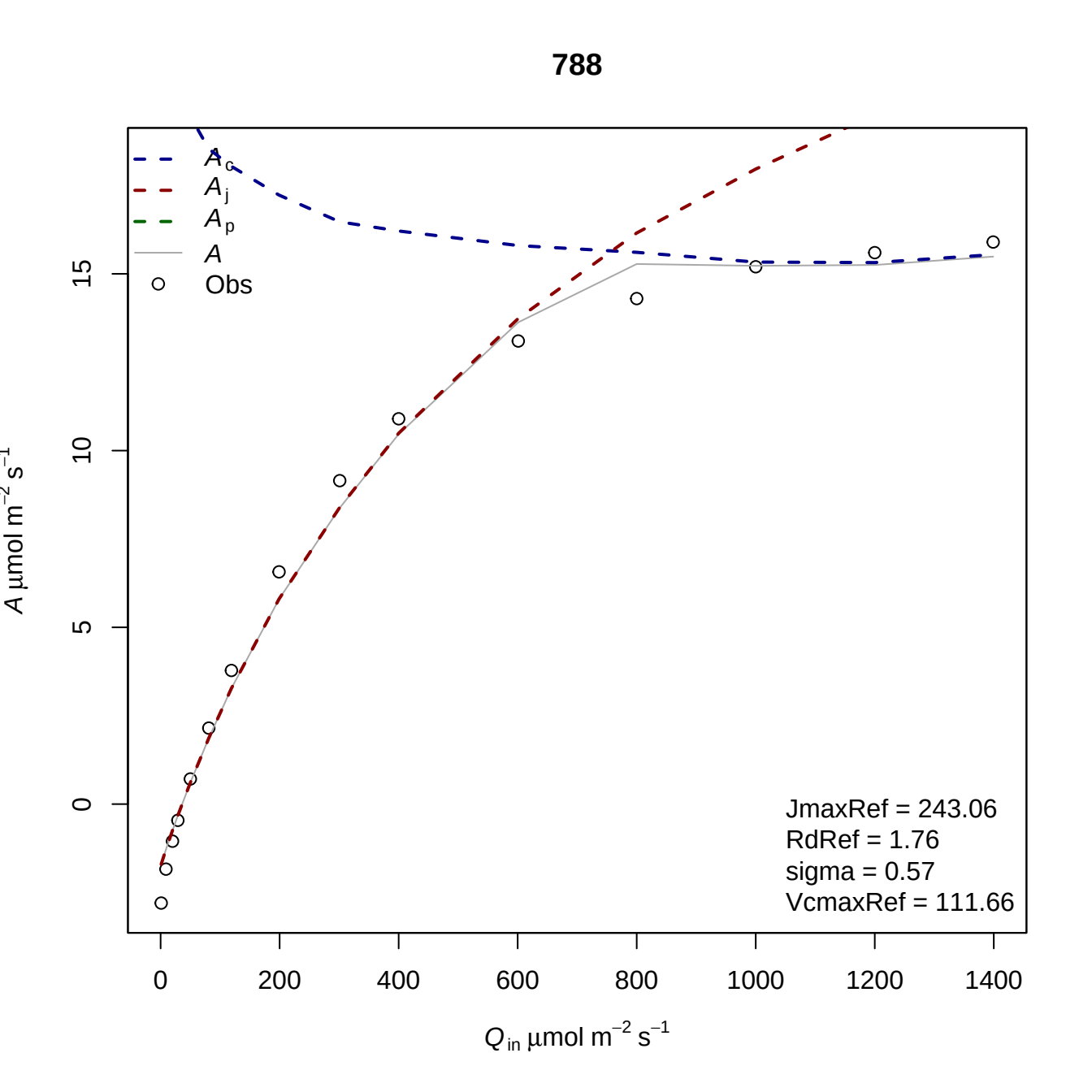

783
A_c
A_j
A_p
A
Obs
JmaxRef = 122.09
RdRef = 0.15
sigma = 0.26
VcmaxRef = 48.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

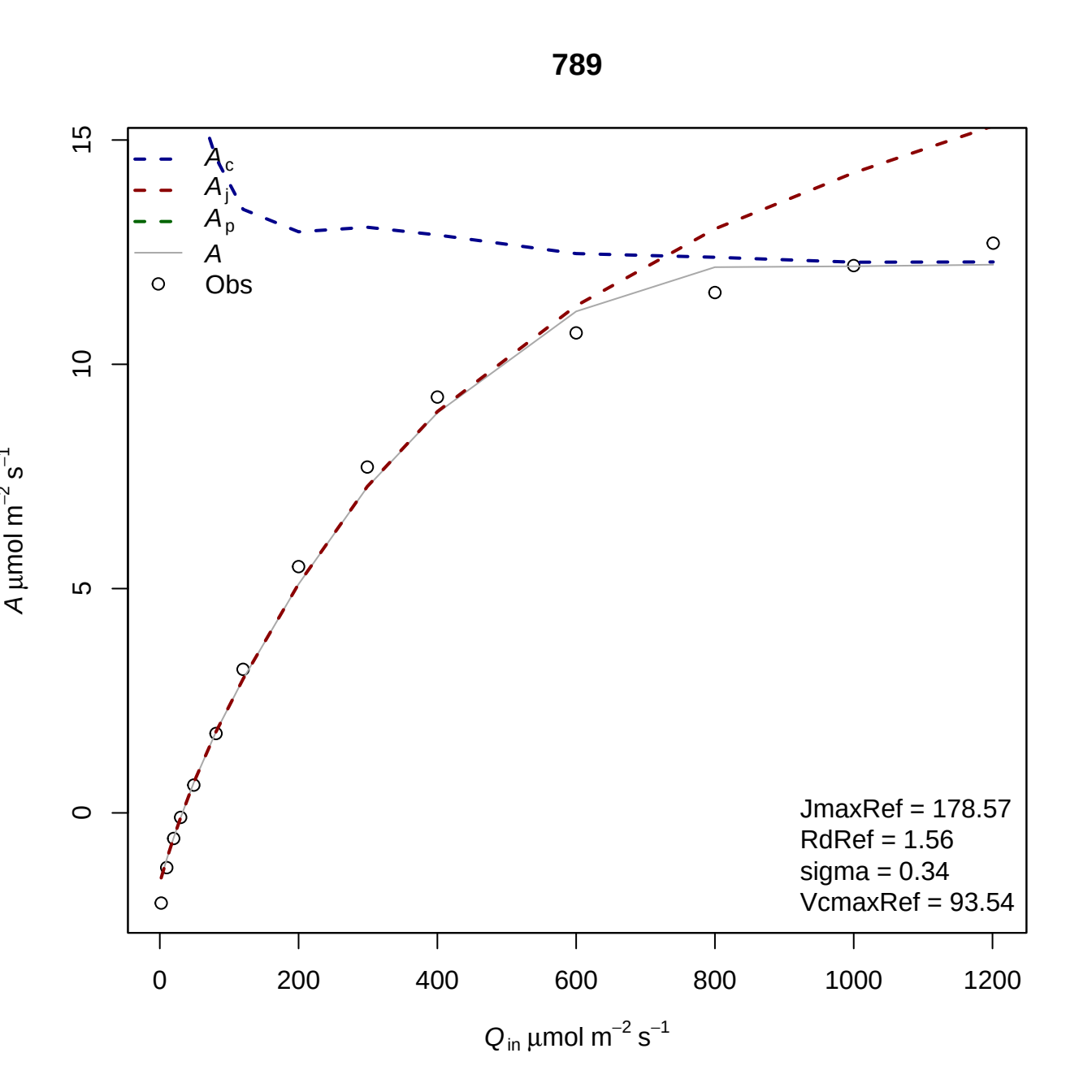

784
A_c
A_j
A_p
A
Obs
JmaxRef = 28.84
RdRef = 1.24
sigma = 0.04
VcmaxRef = 18.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
200
400
600
800
1000
1200

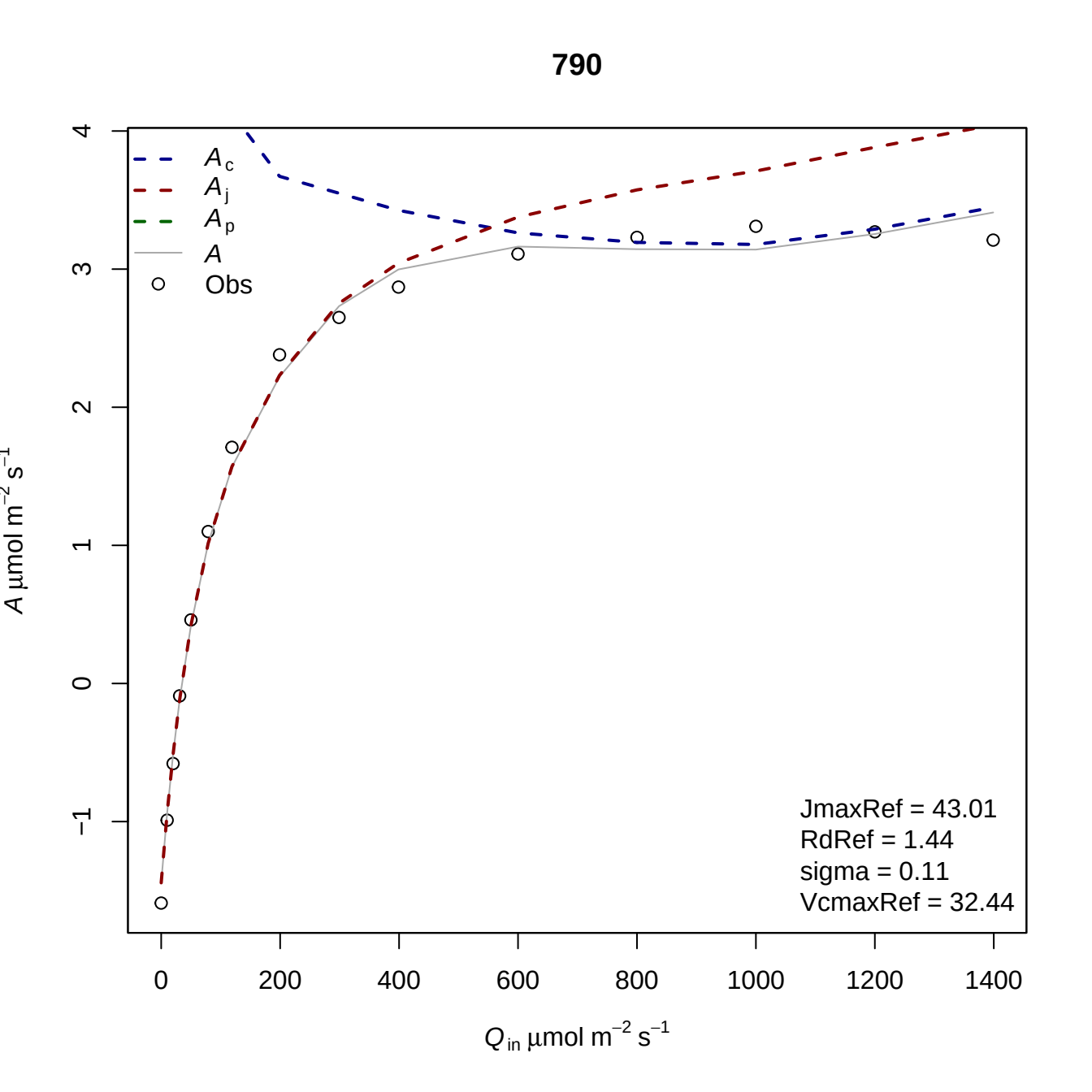

785
A_c
A_j
A_p
A
Obs
JmaxRef = 89.66
RdRef = 1.01
sigma = 0.2
VcmaxRef = 71.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

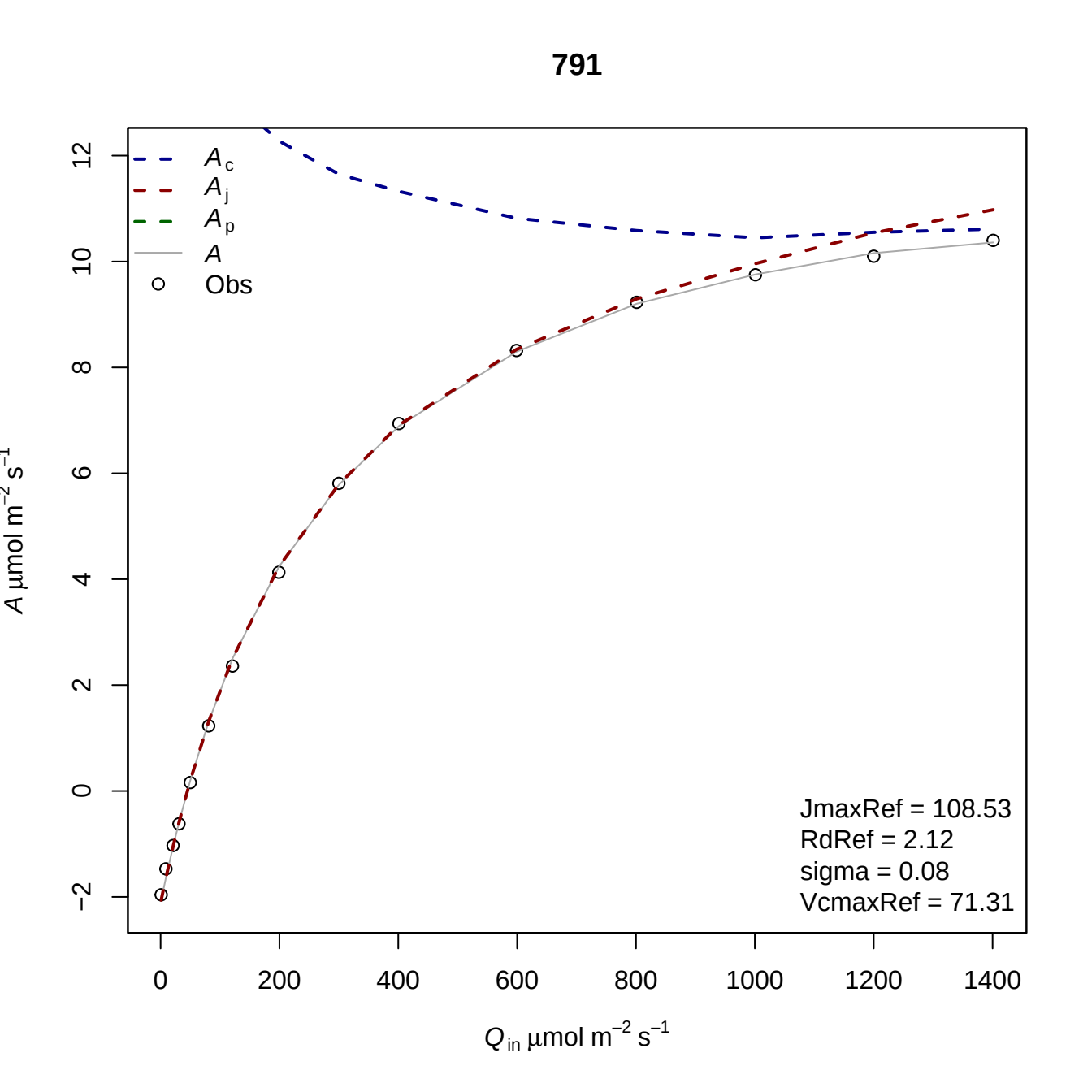

786
A_c
A_j
A_p
A
Obs
JmaxRef = 63.67
RdRef = 1.35
sigma = 0.21
VcmaxRef = 40.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200
1400

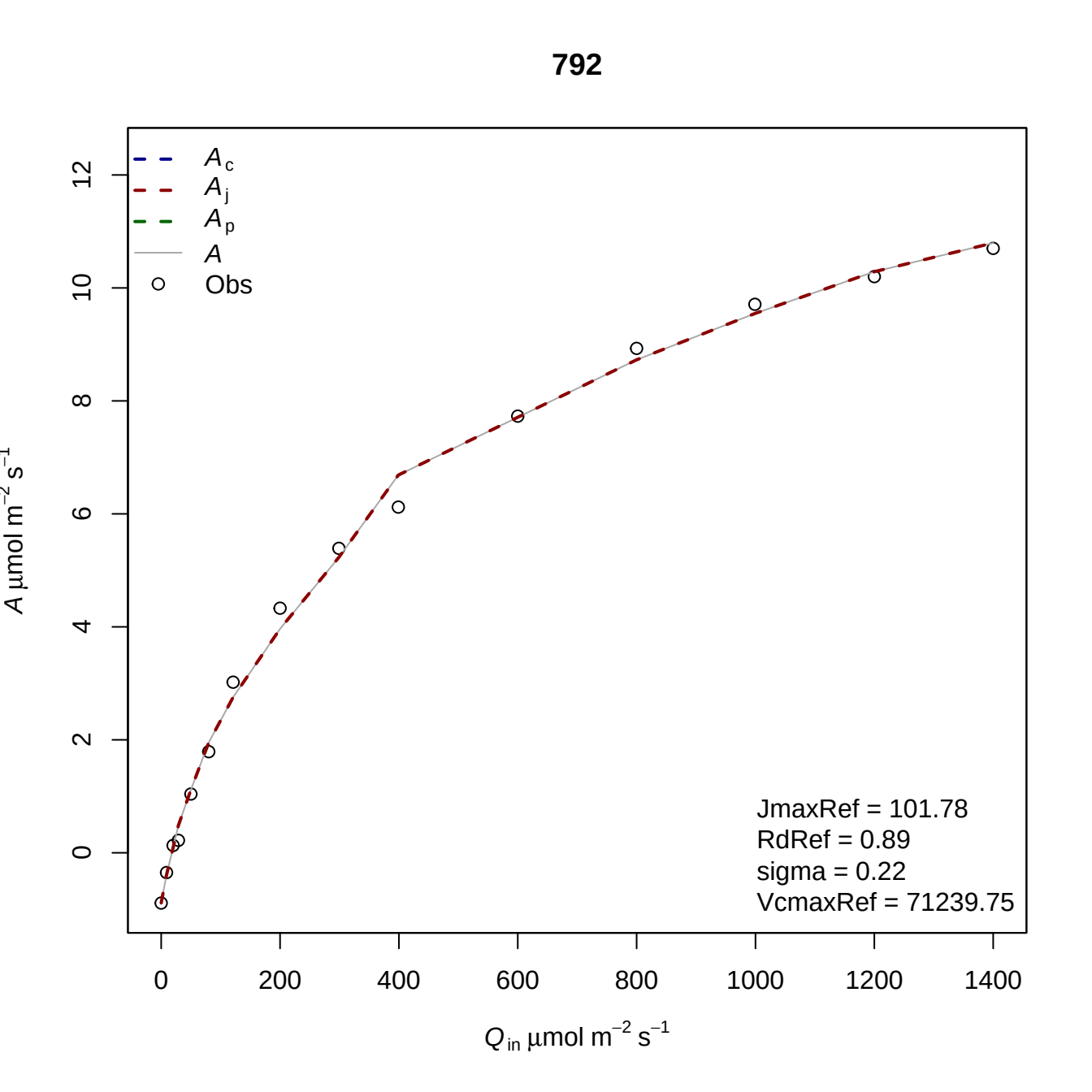

787
A_c
A_j
A_p
A
Obs
JmaxRef = 447.12
RdRef = 1
sigma = 0.46
VcmaxRef = 135.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

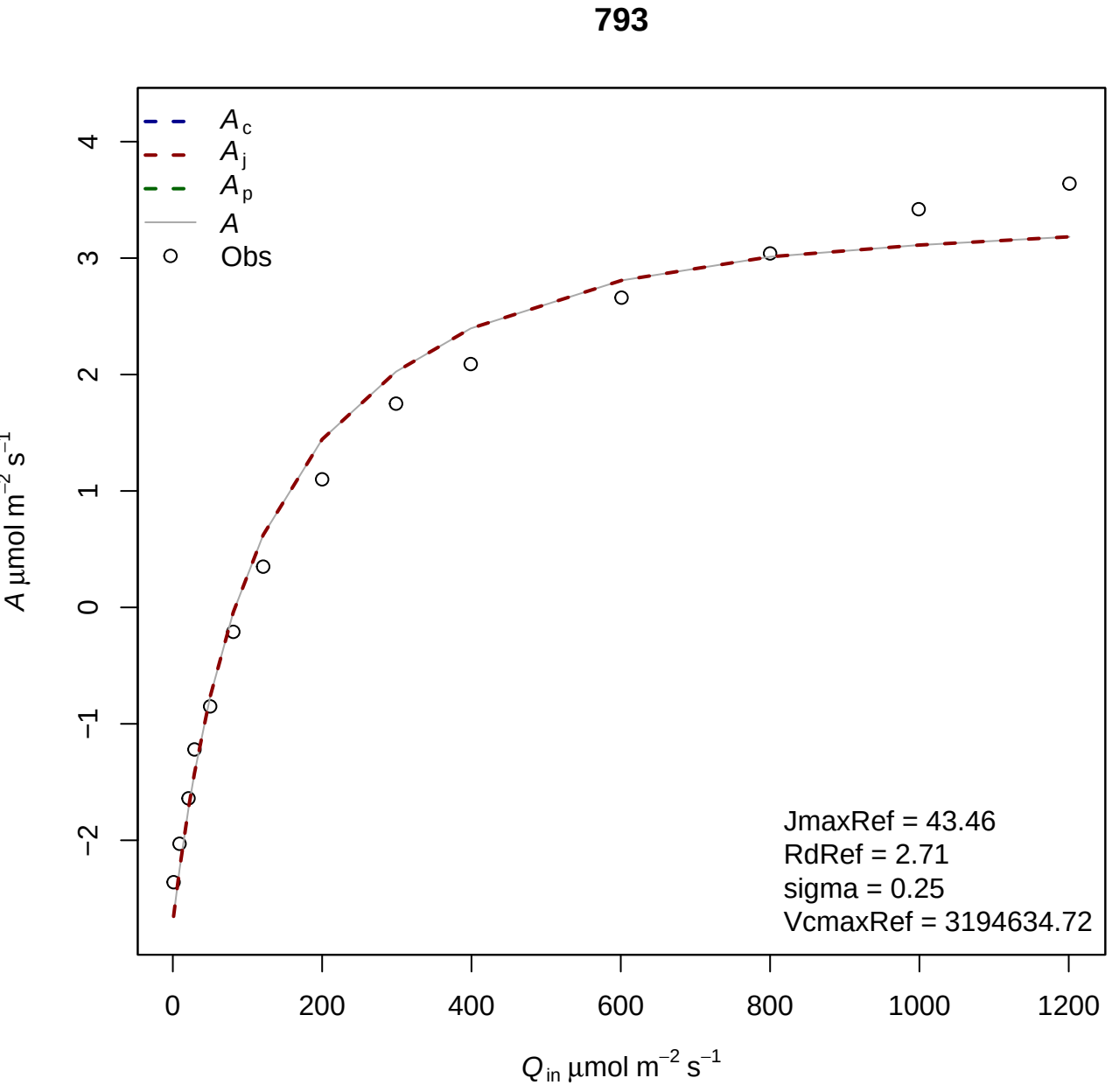

788
A_c
A_j
A_p
A
Obs
JmaxRef = 243.06
RdRef = 1.76
sigma = 0.57
VcmaxRef = 111.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200
1400

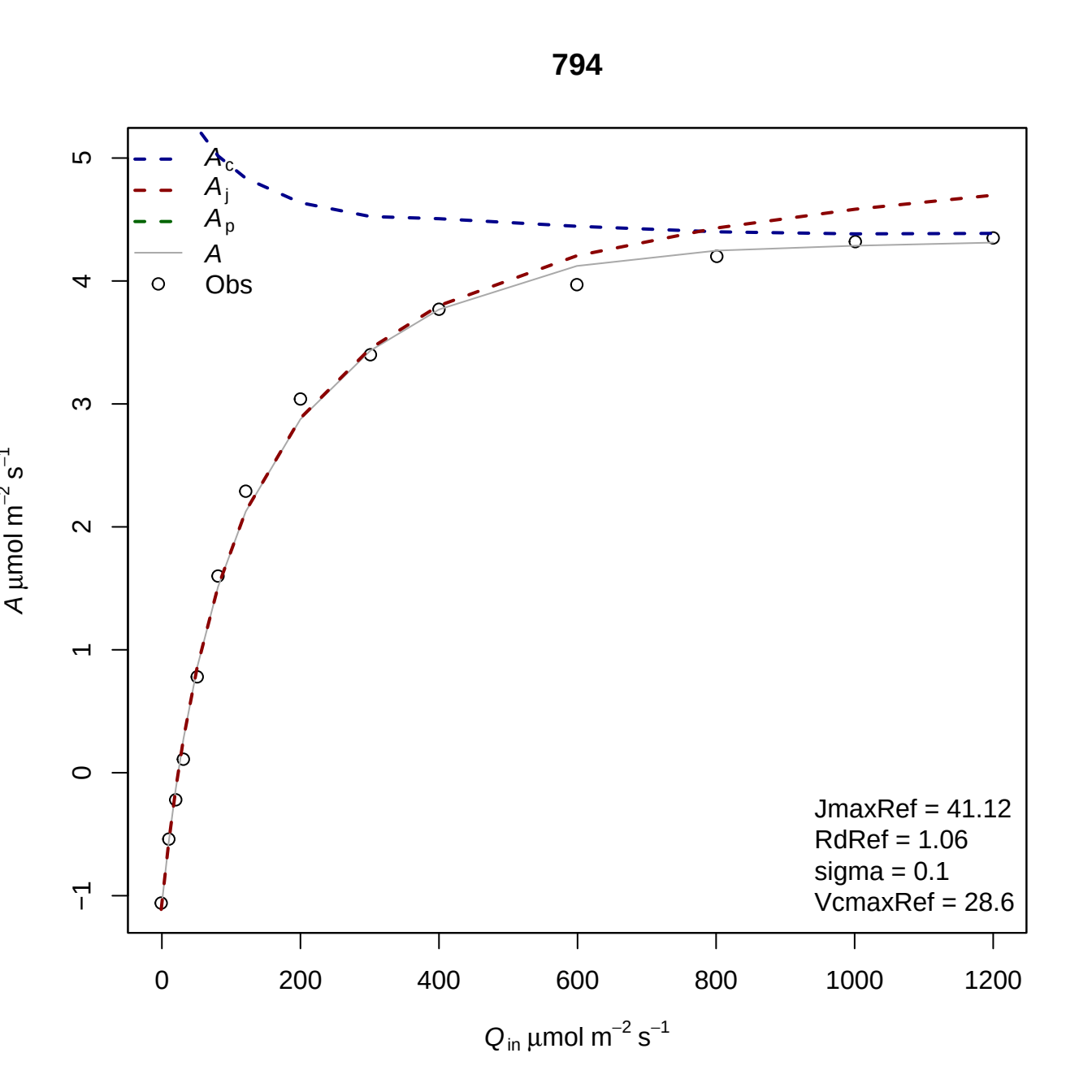

789
A_c
A_j
A_p
A
Obs
JmaxRef = 178.57
RdRef = 1.56
sigma = 0.34
VcmaxRef = 93.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200

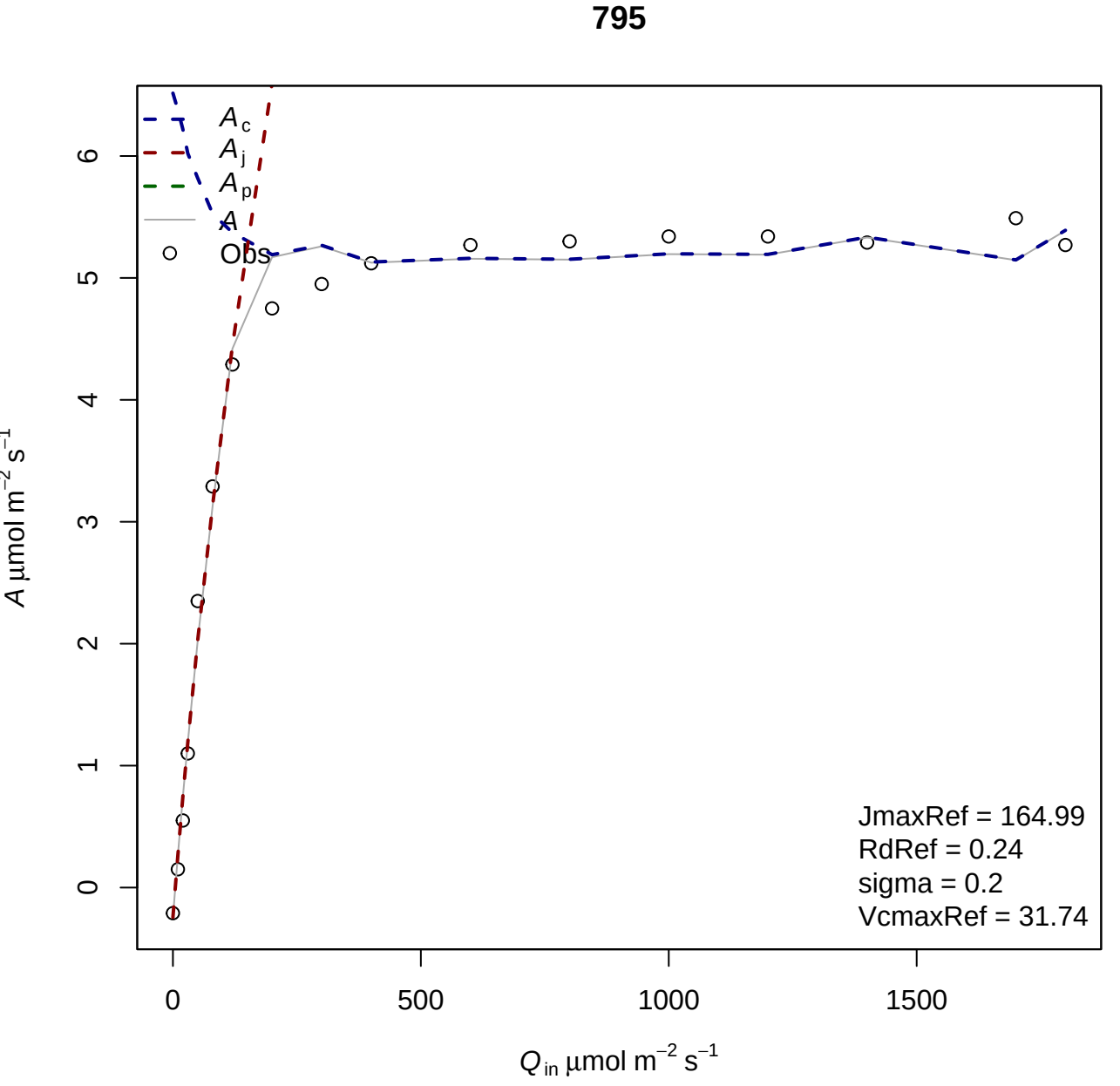

790
A_c
A_j
A_p
A
Obs
JmaxRef = 43.01
RdRef = 1.44
sigma = 0.11
VcmaxRef = 32.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000
1200
1400

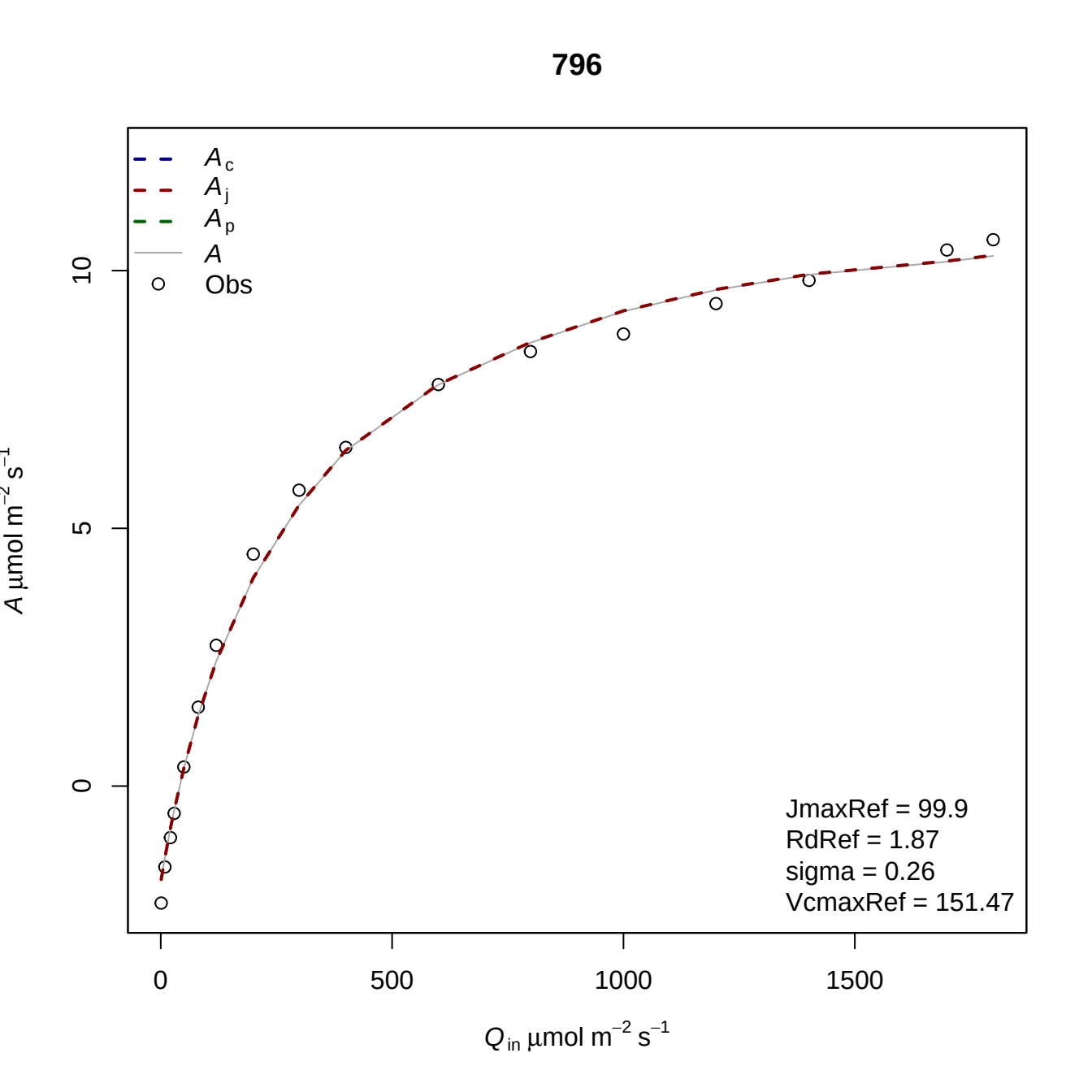

791
A_c
A_j
A_p
A
Obs
JmaxRef = 108.53
RdRef = 2.12
sigma = 0.08
VcmaxRef = 71.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

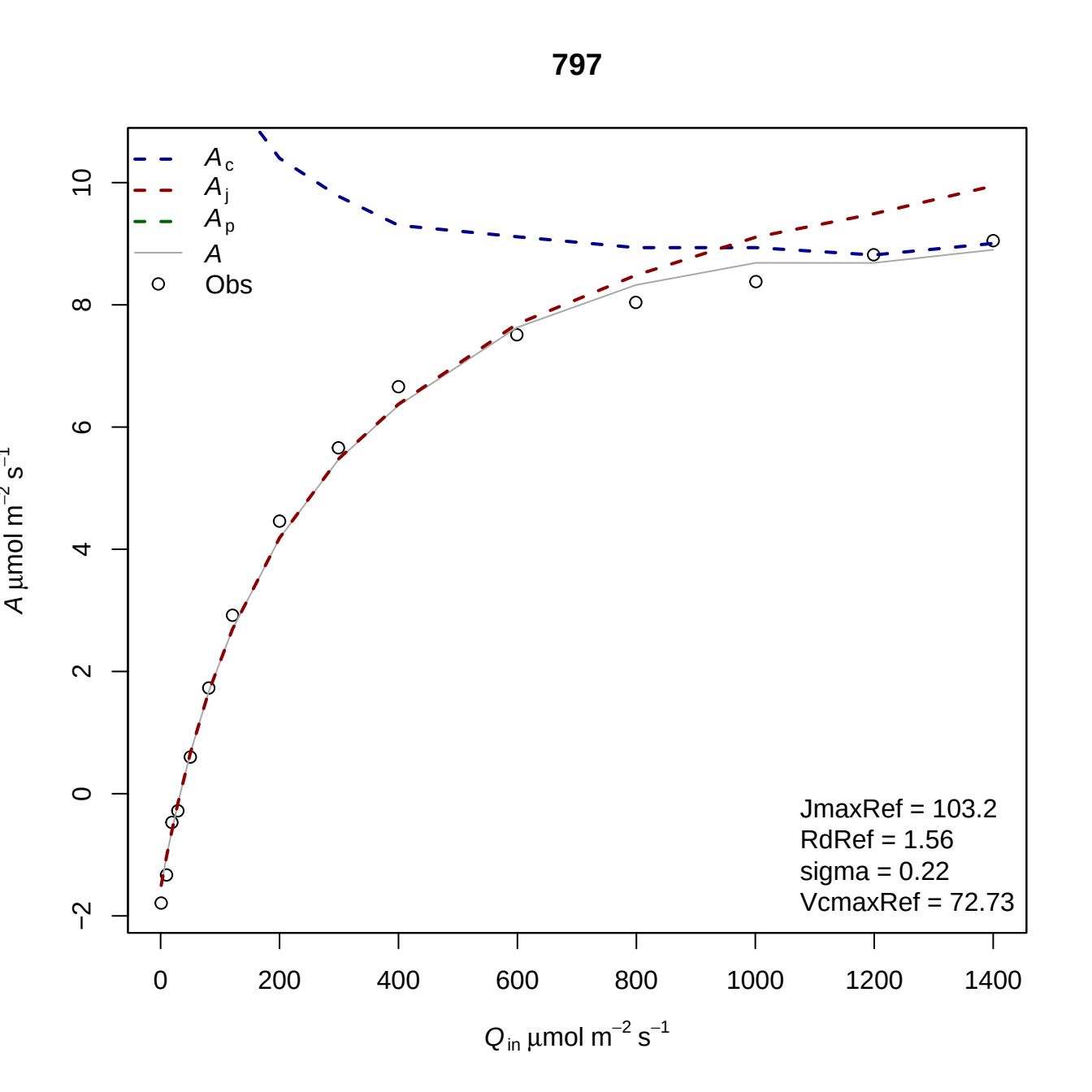

792
A_c
A_j
A_p
A
Obs
JmaxRef = 101.78
RdRef = 0.89
sigma = 0.22
VcmaxRef = 71239.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

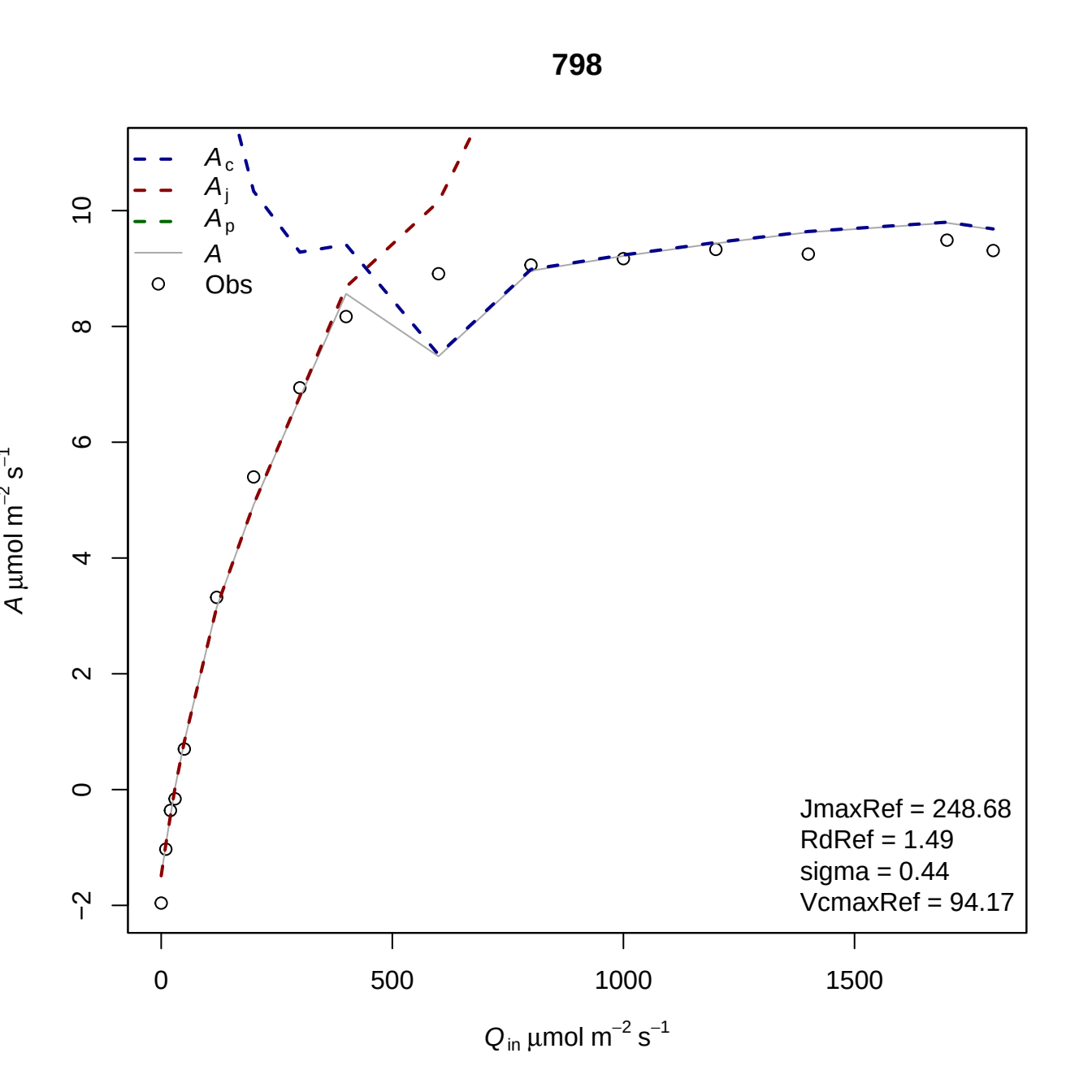

793
A_c
A_j
A_p
A
Obs
JmaxRef = 43.46
RdRef = 2.71
sigma = 0.25
VcmaxRef = 3194634.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
200
400
600
800
1000
1200

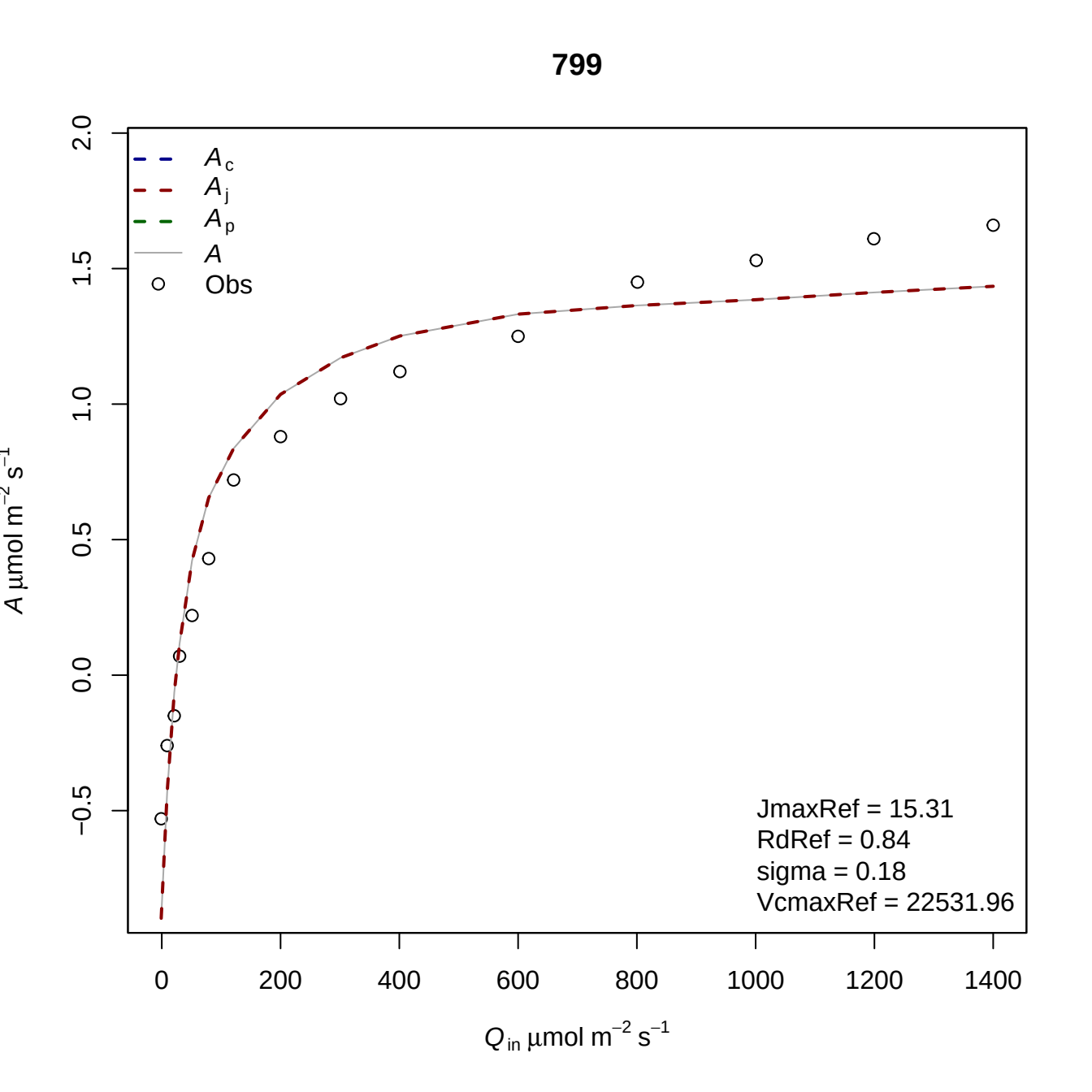

794
A_c
A_j
A_p
A
Obs
JmaxRef = 41.12
RdRef = 1.06
sigma = 0.1
VcmaxRef = 28.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
200
400
600
800
1000
1200

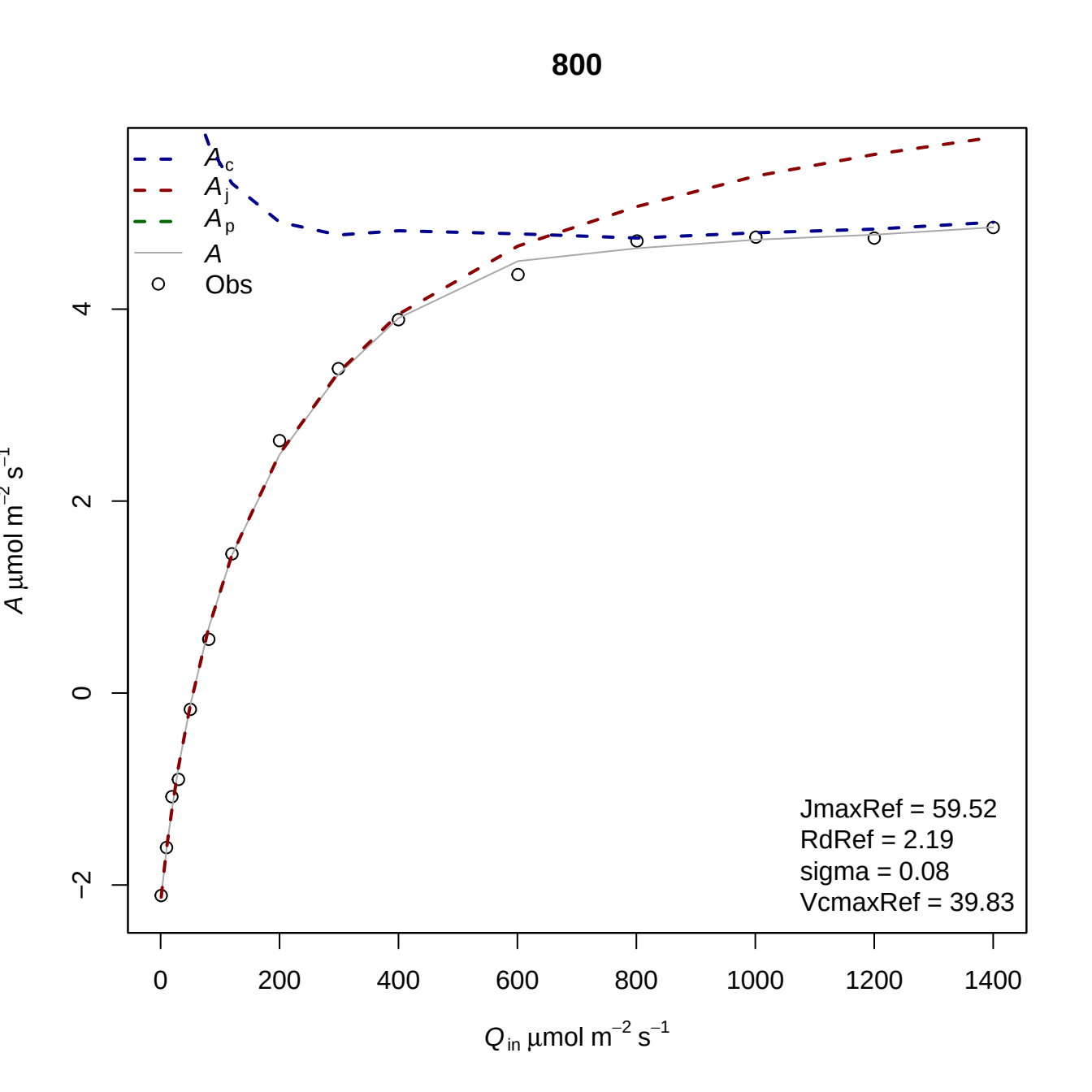

795
A_c
A_j
A_p
A
Obs
JmaxRef = 164.99
RdRef = 0.24
sigma = 0.2
VcmaxRef = 31.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

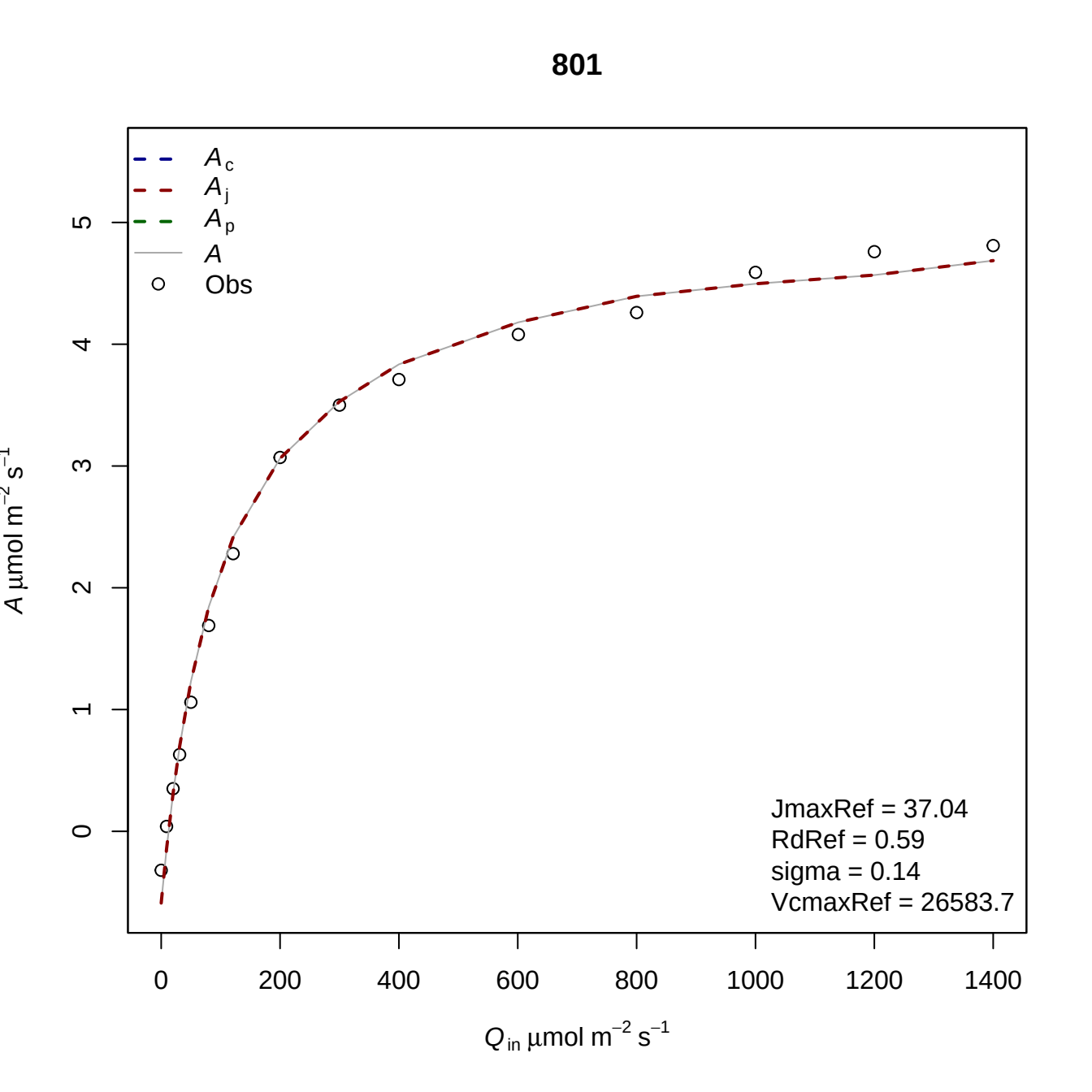

796
A_c
A_j
A_p
A
Obs
JmaxRef = 99.9
RdRef = 1.87
sigma = 0.26
VcmaxRef = 151.47
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

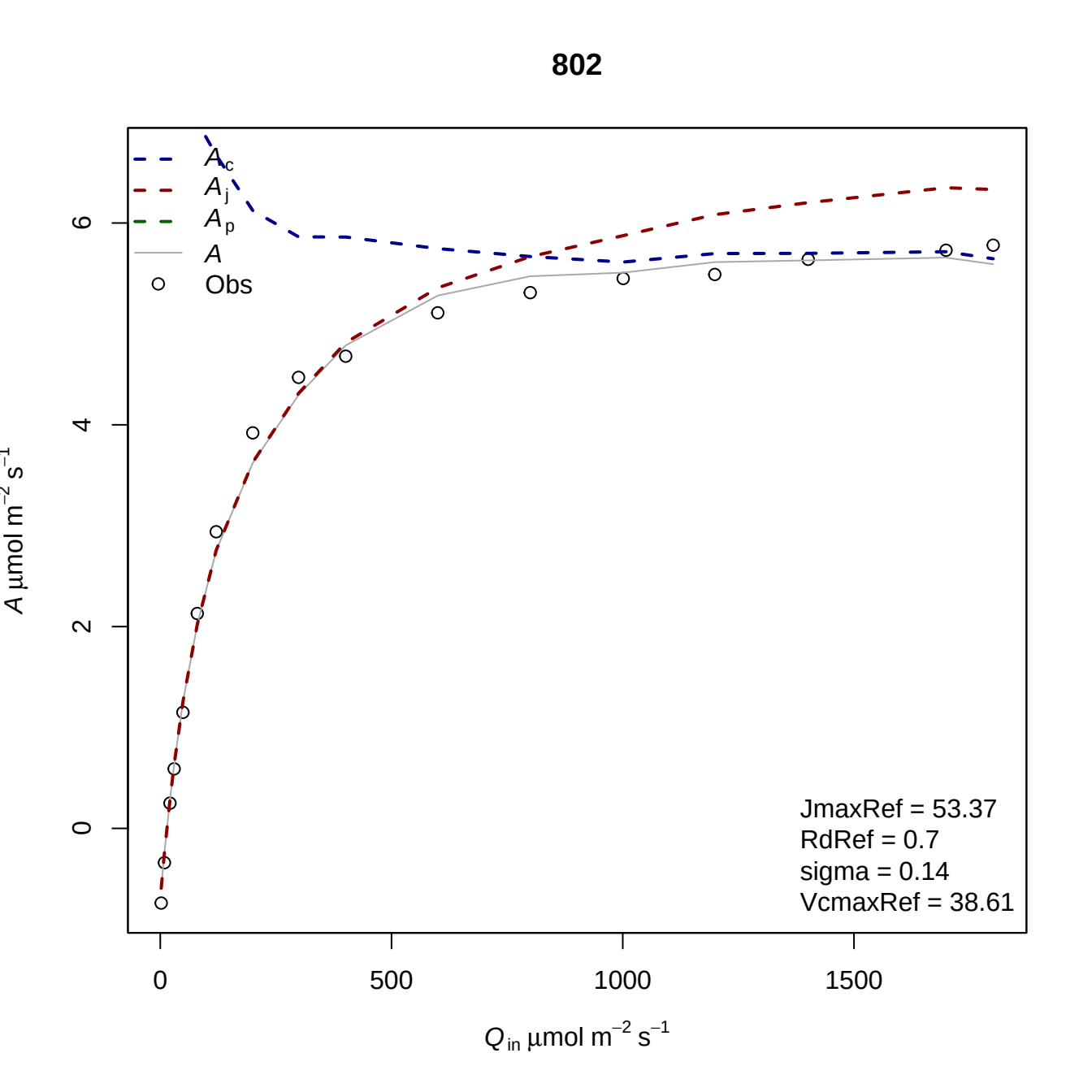

797
A_c
A_j
A_p
A
Obs
JmaxRef = 103.2
RdRef = 1.56
sigma = 0.22
VcmaxRef = 72.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

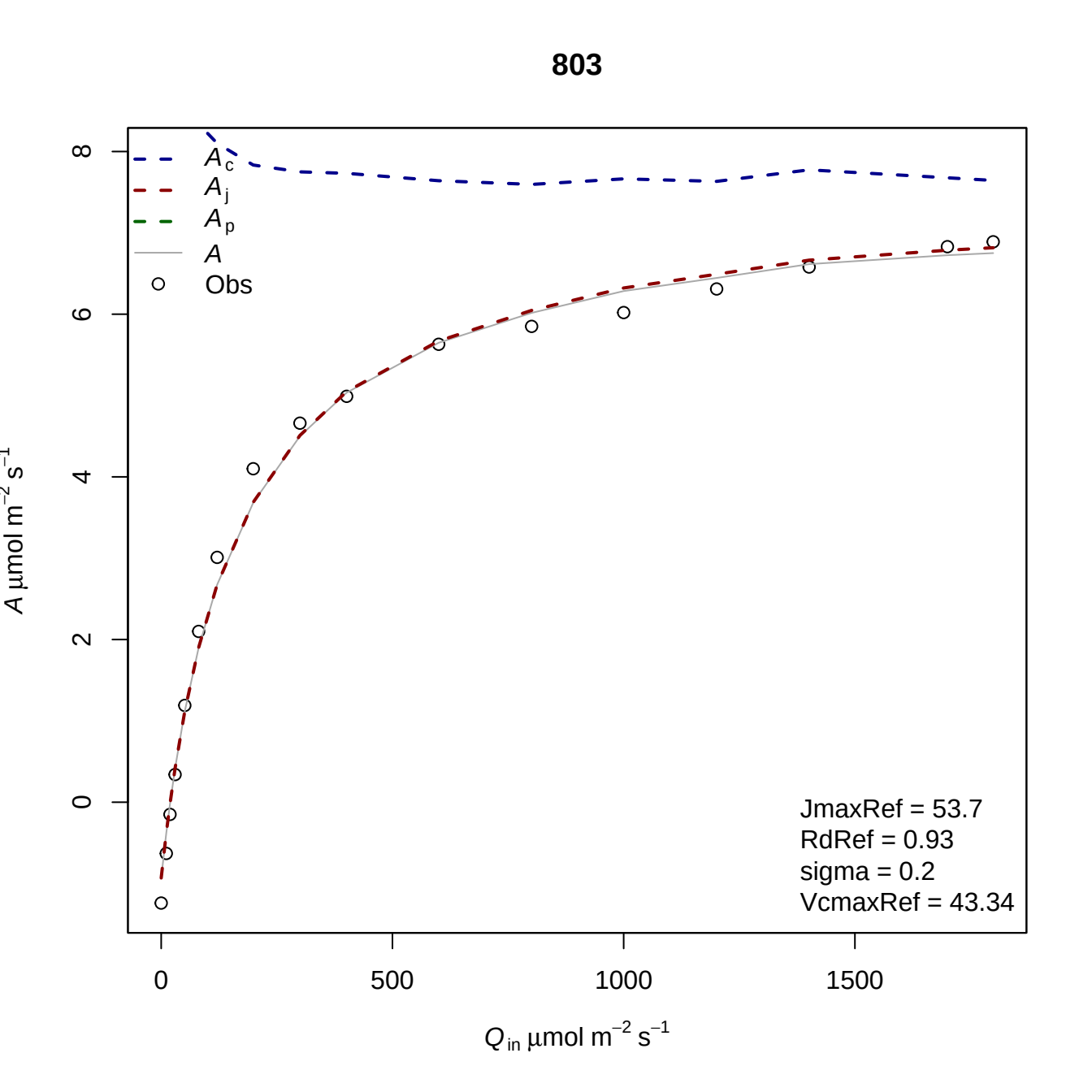

798
A_c
A_j
A_p
A
Obs
JmaxRef = 248.68
RdRef = 1.49
sigma = 0.44
VcmaxRef = 94.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

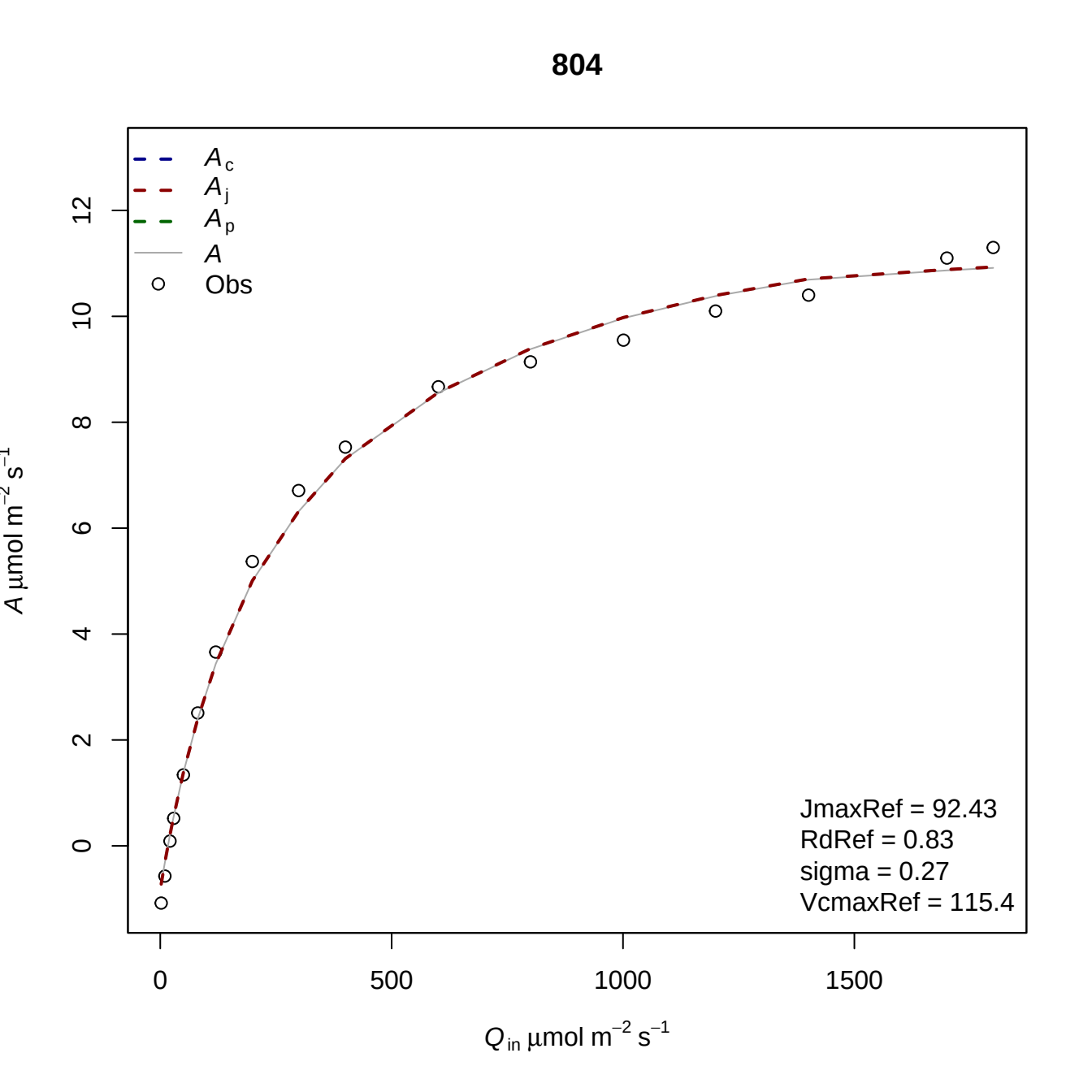

799
A_c
A_j
A_p
A
Obs
JmaxRef = 15.31
RdRef = 0.84
sigma = 0.18
VcmaxRef = 22531.96
2.0
1.5
1.0
0.5
0.0
−0.5
0
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

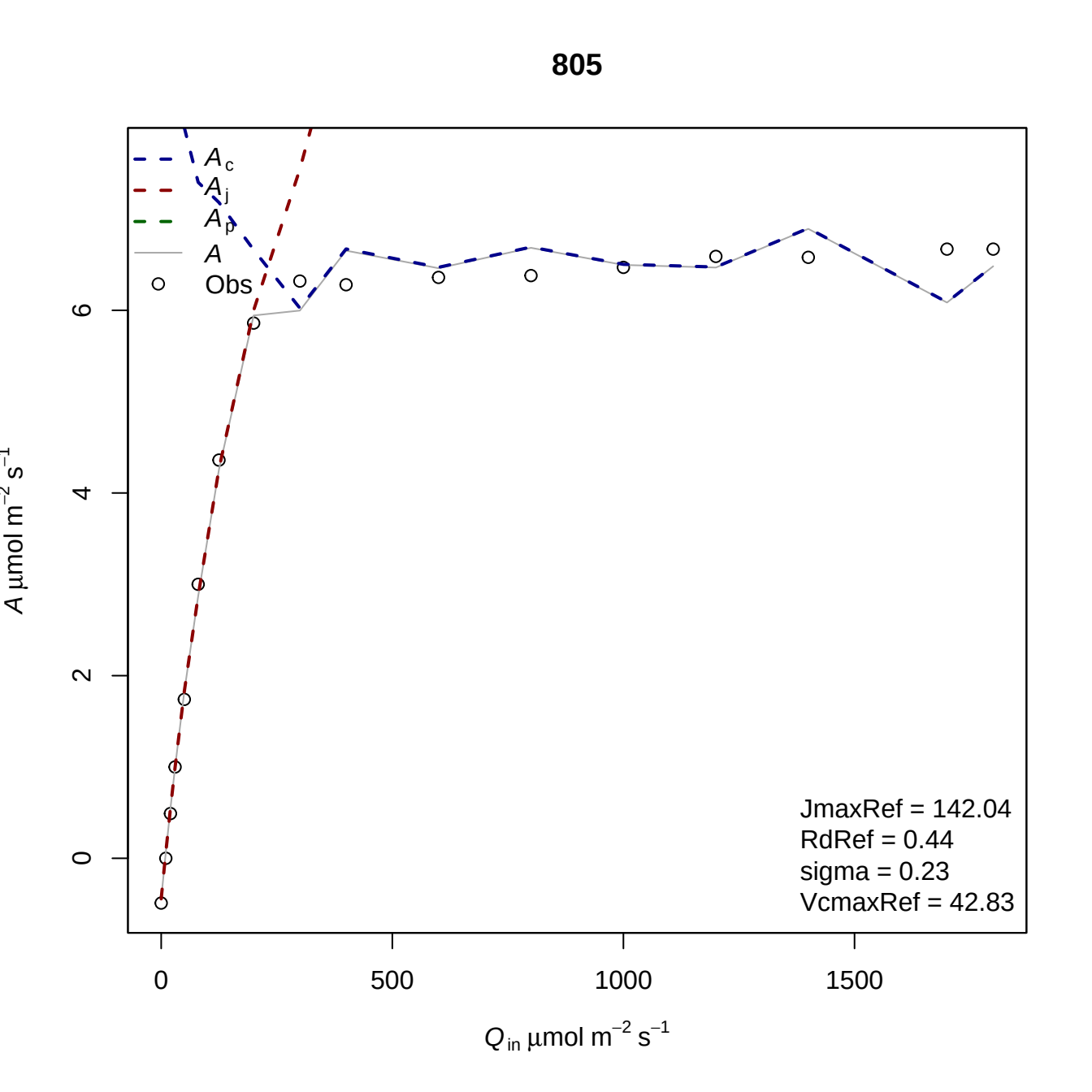

800
A_c
A_j
A_p
A
Obs
JmaxRef = 59.52
RdRef = 2.19
sigma = 0.08
VcmaxRef = 39.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000
1200
1400

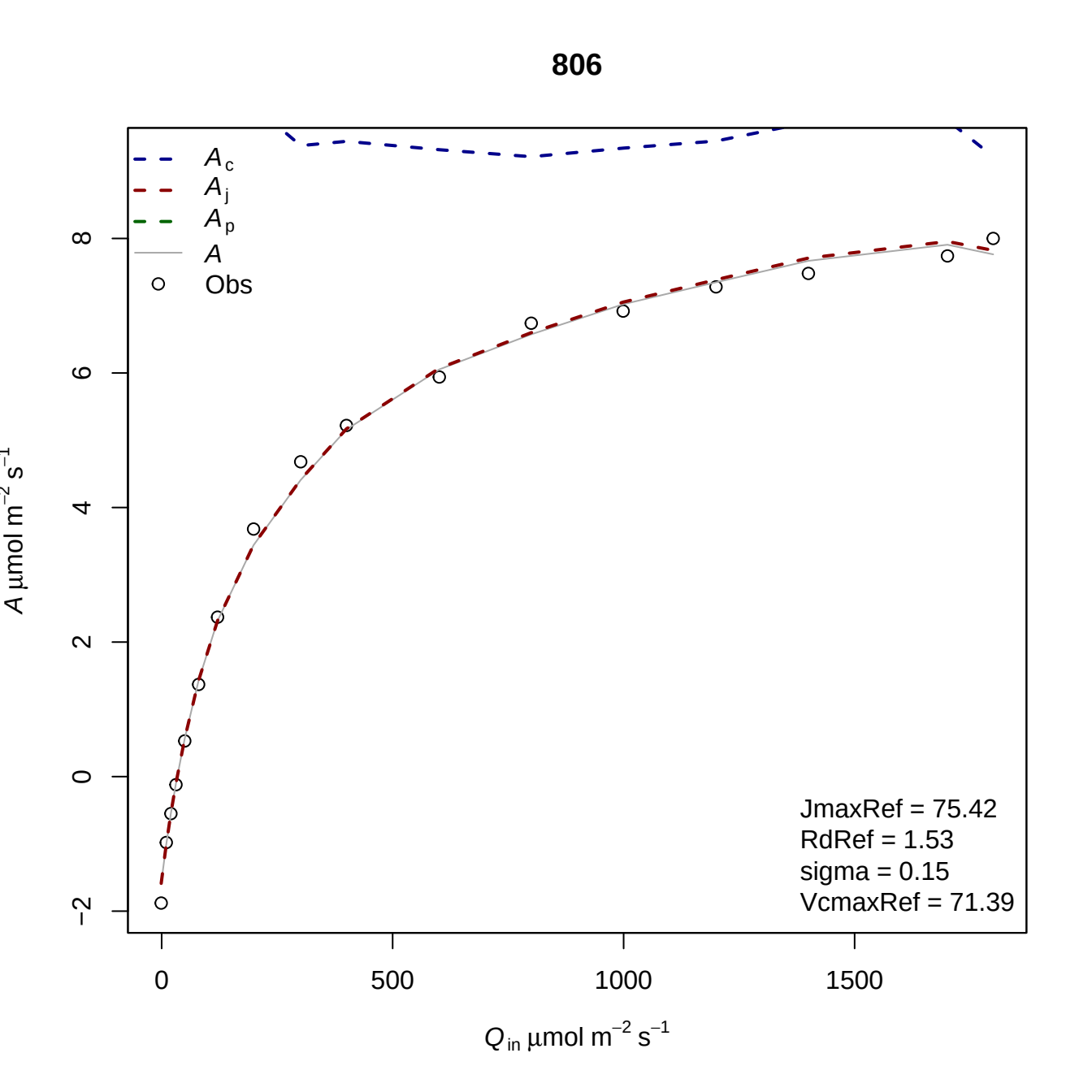

801
A_c
A_j
A_p
A
Obs
JmaxRef = 37.04
RdRef = 0.59
sigma = 0.14
VcmaxRef = 26583.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
200
400
600
800
1000
1200
1400

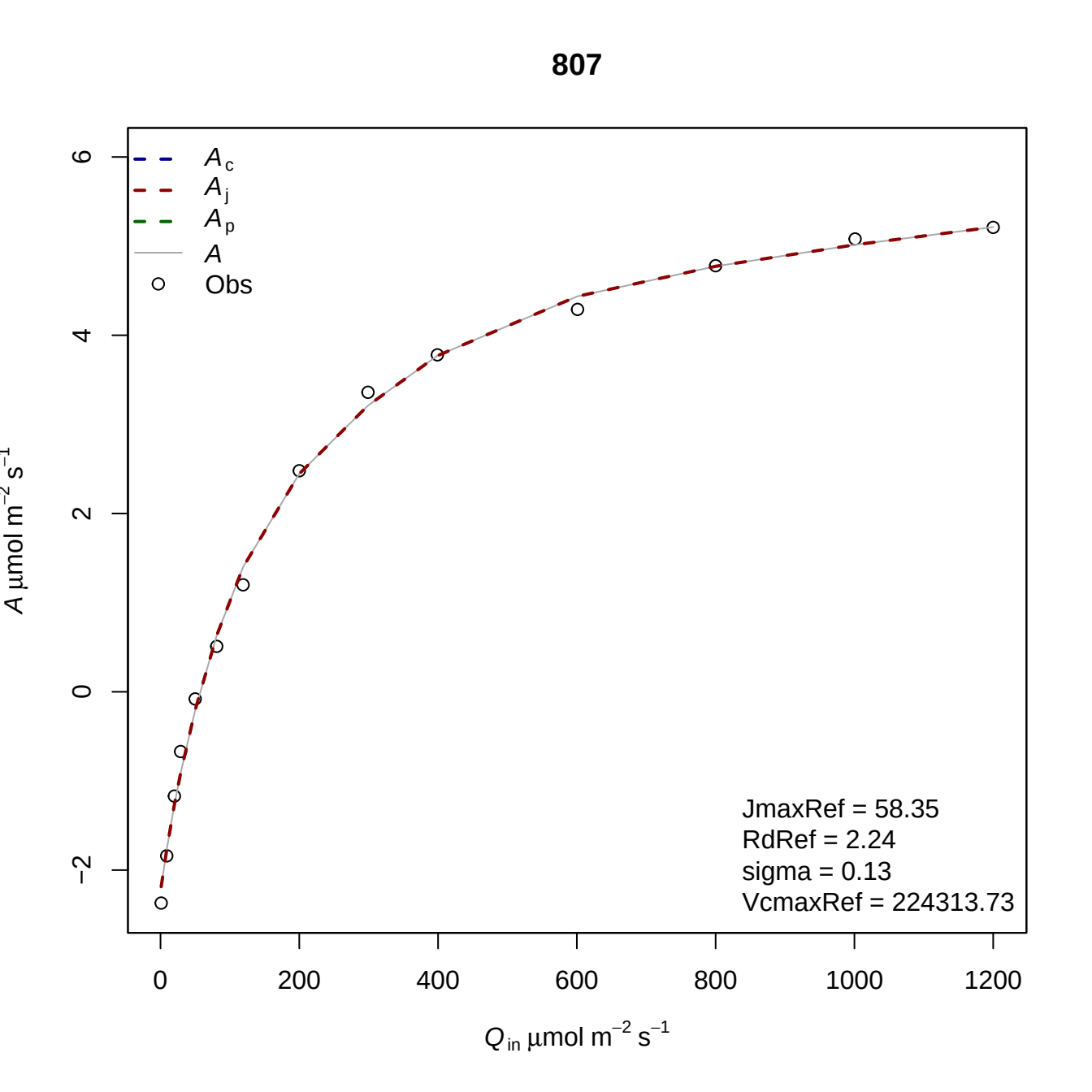

802
A_c
A_j
A_p
A
Obs
JmaxRef = 53.37
RdRef = 0.7
sigma = 0.14
VcmaxRef = 38.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

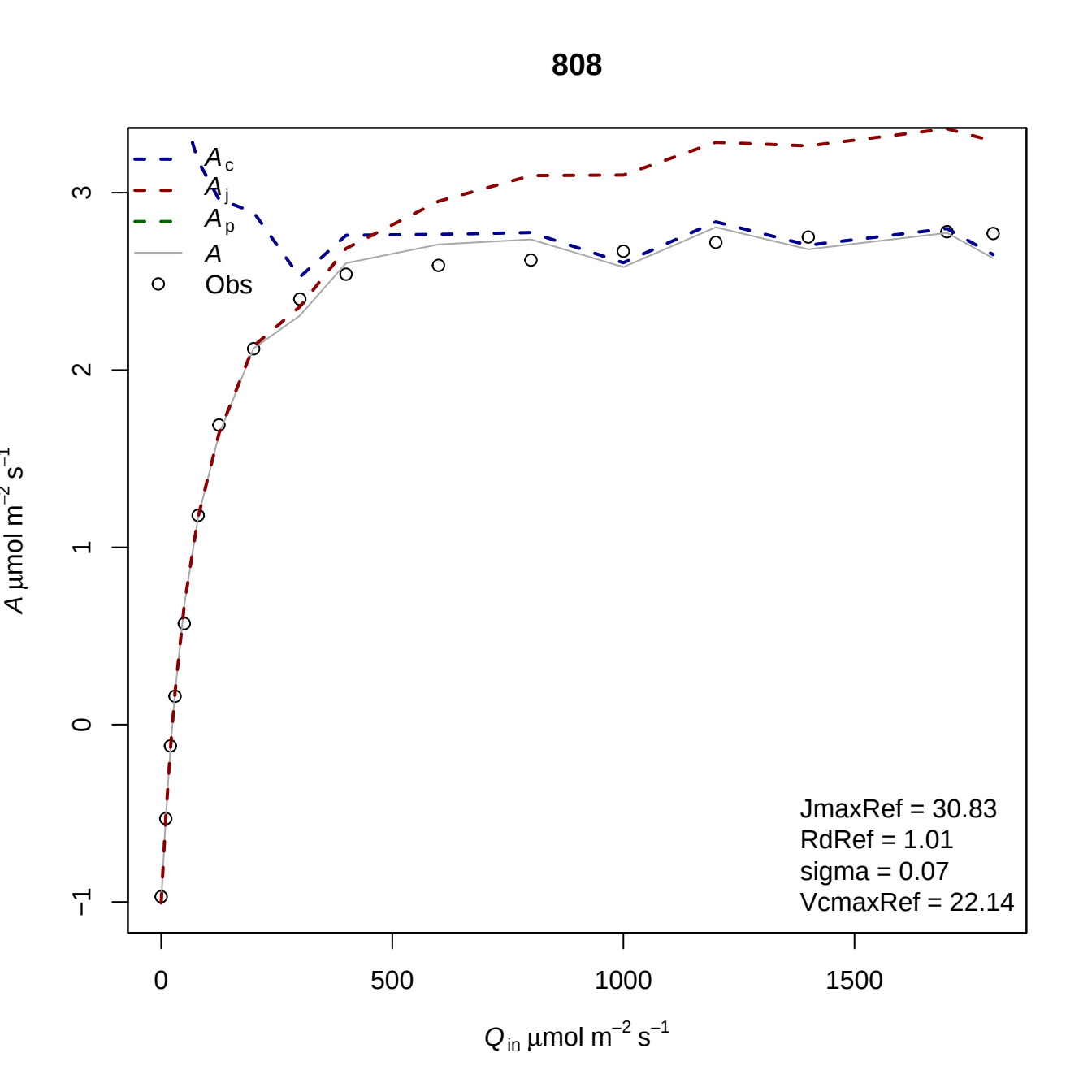

803
A_c
A_j
A_p
A
Obs
JmaxRef = 53.7
RdRef = 0.93
sigma = 0.2
VcmaxRef = 43.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

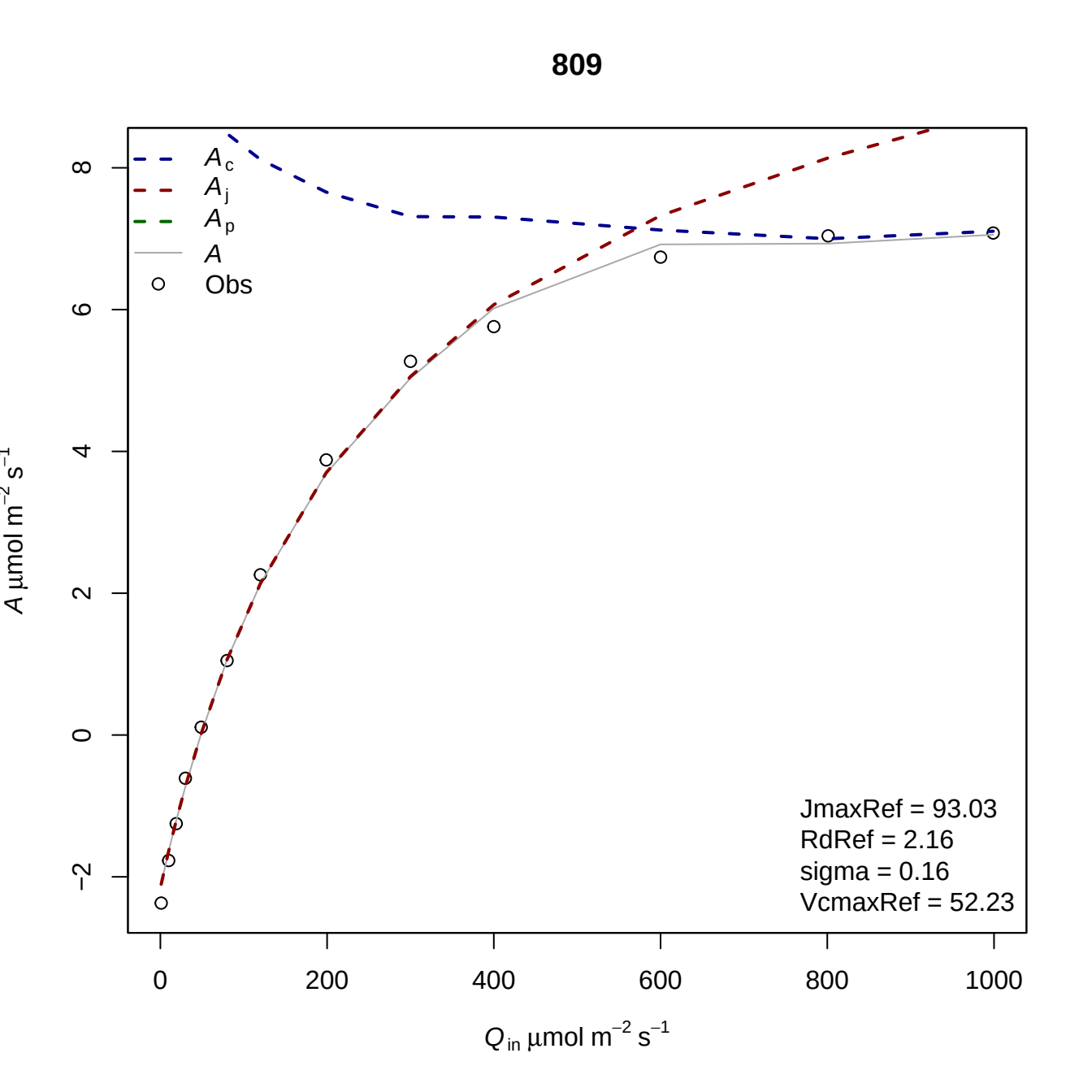

804
A_c
A_j
A_p
A
Obs
JmaxRef = 92.43
RdRef = 0.83
sigma = 0.27
VcmaxRef = 115.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

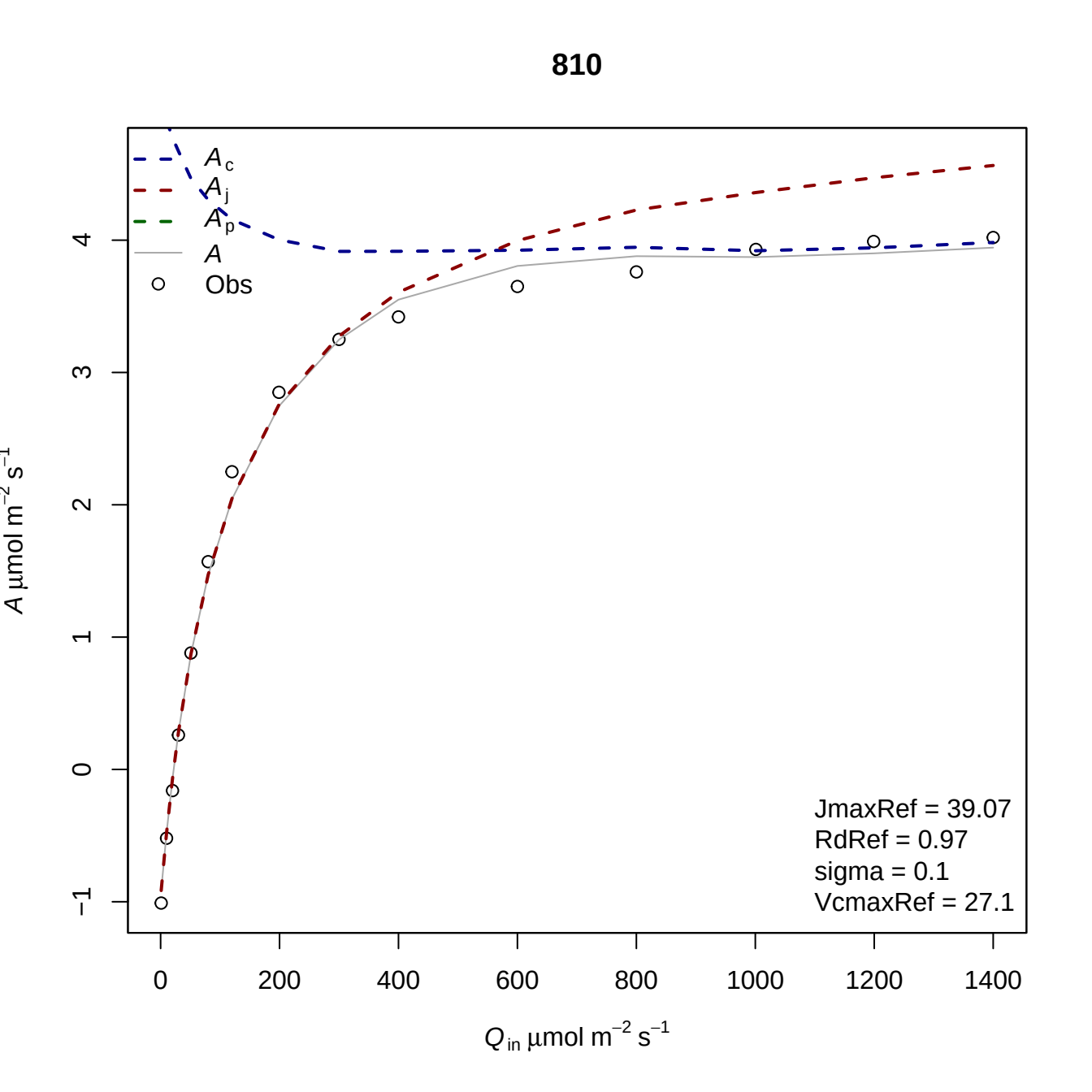

805
A_c
A_j
A_p
A
Obs
JmaxRef = 142.04
RdRef = 0.44
sigma = 0.23
VcmaxRef = 42.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

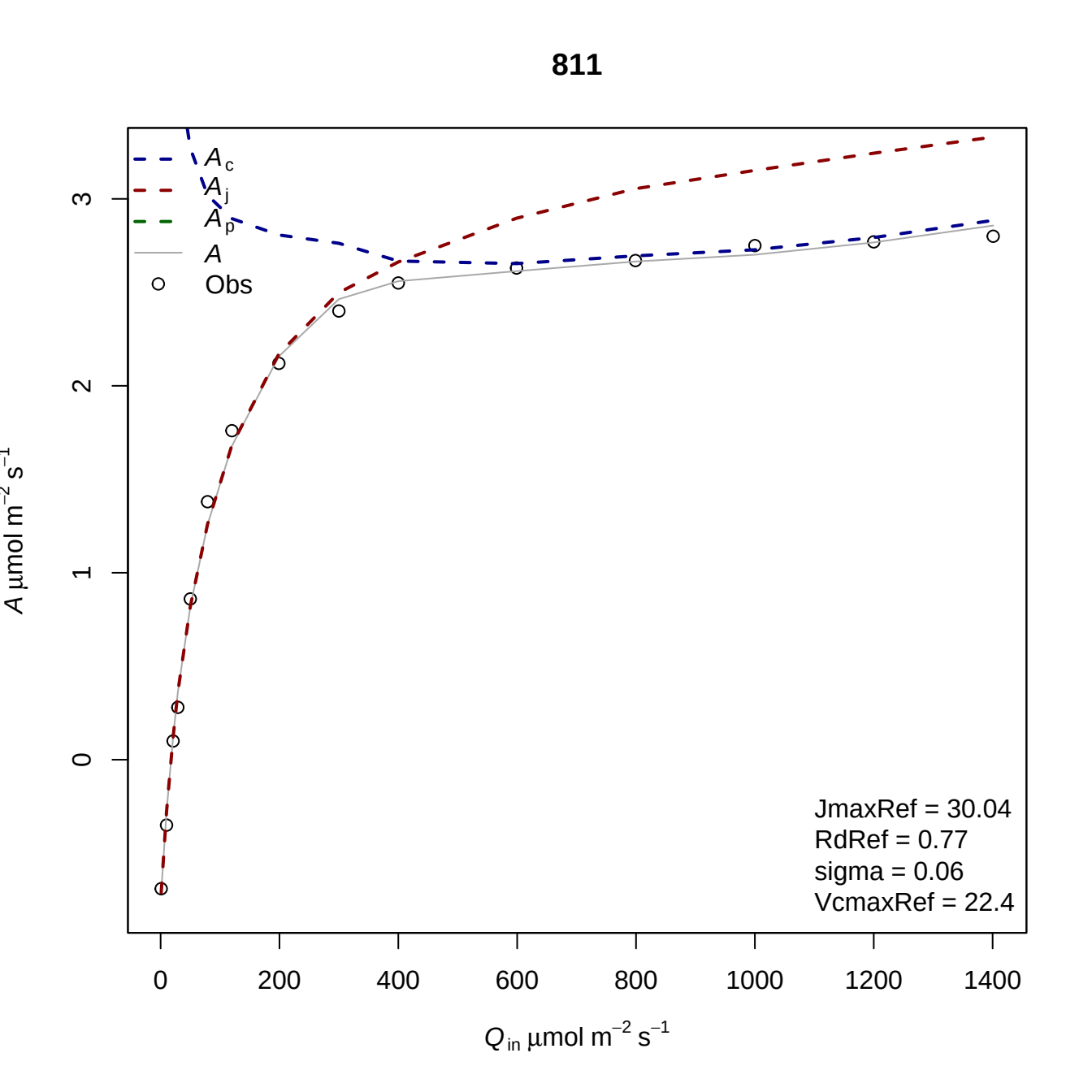

806
A_c
A_j
A_p
A
Obs
JmaxRef = 75.42
RdRef = 1.53
sigma = 0.15
VcmaxRef = 71.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

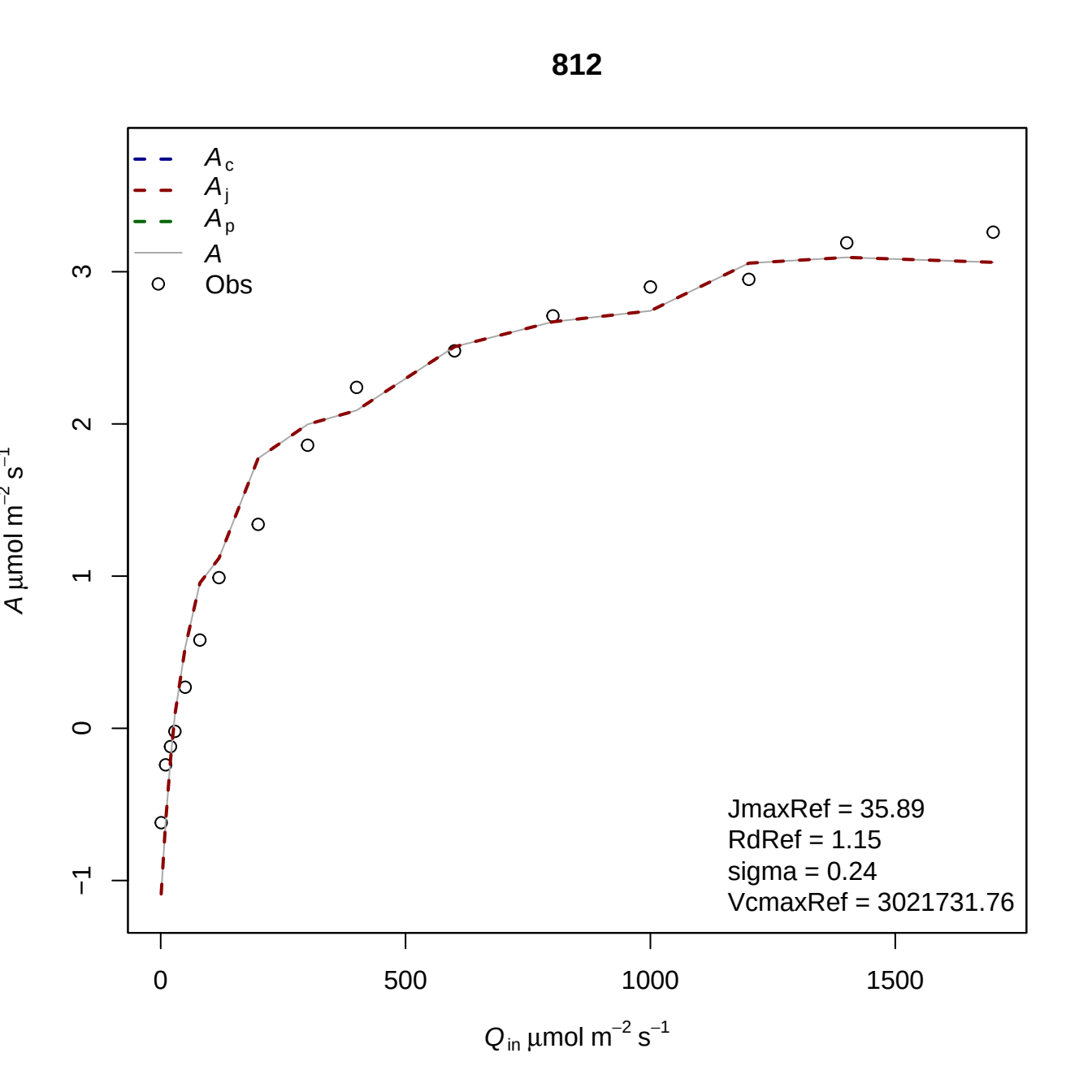

807
A_c
A_j
A_p
A
Obs
JmaxRef = 58.35
RdRef = 2.24
sigma = 0.13
VcmaxRef = 224313.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000
1200

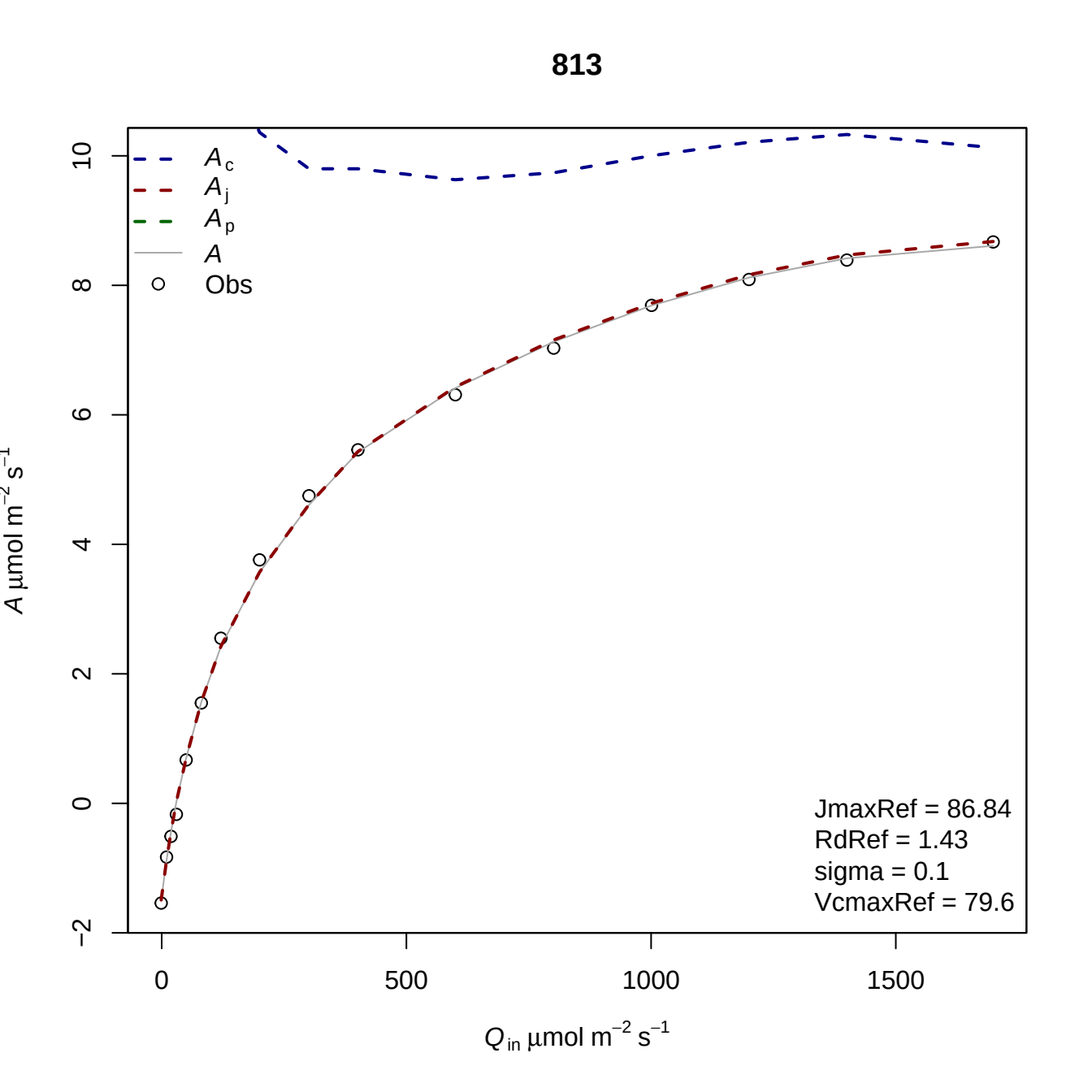

808
A_c
A_j
A_p
A
Obs
JmaxRef = 30.83
RdRef = 1.01
sigma = 0.07
VcmaxRef = 22.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

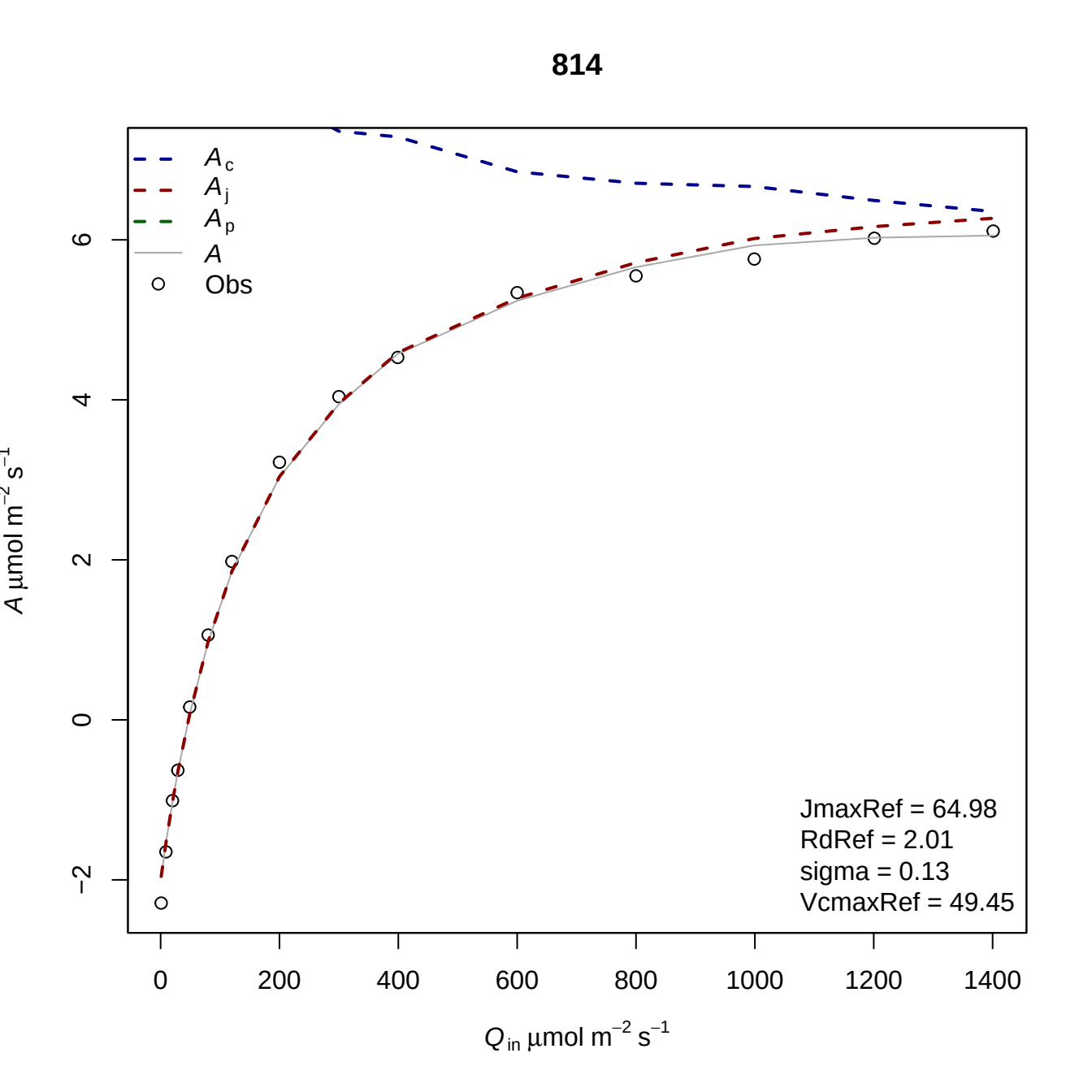

809
A_c
A_j
A_p
A
Obs
JmaxRef = 93.03
RdRef = 2.16
sigma = 0.16
VcmaxRef = 52.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000

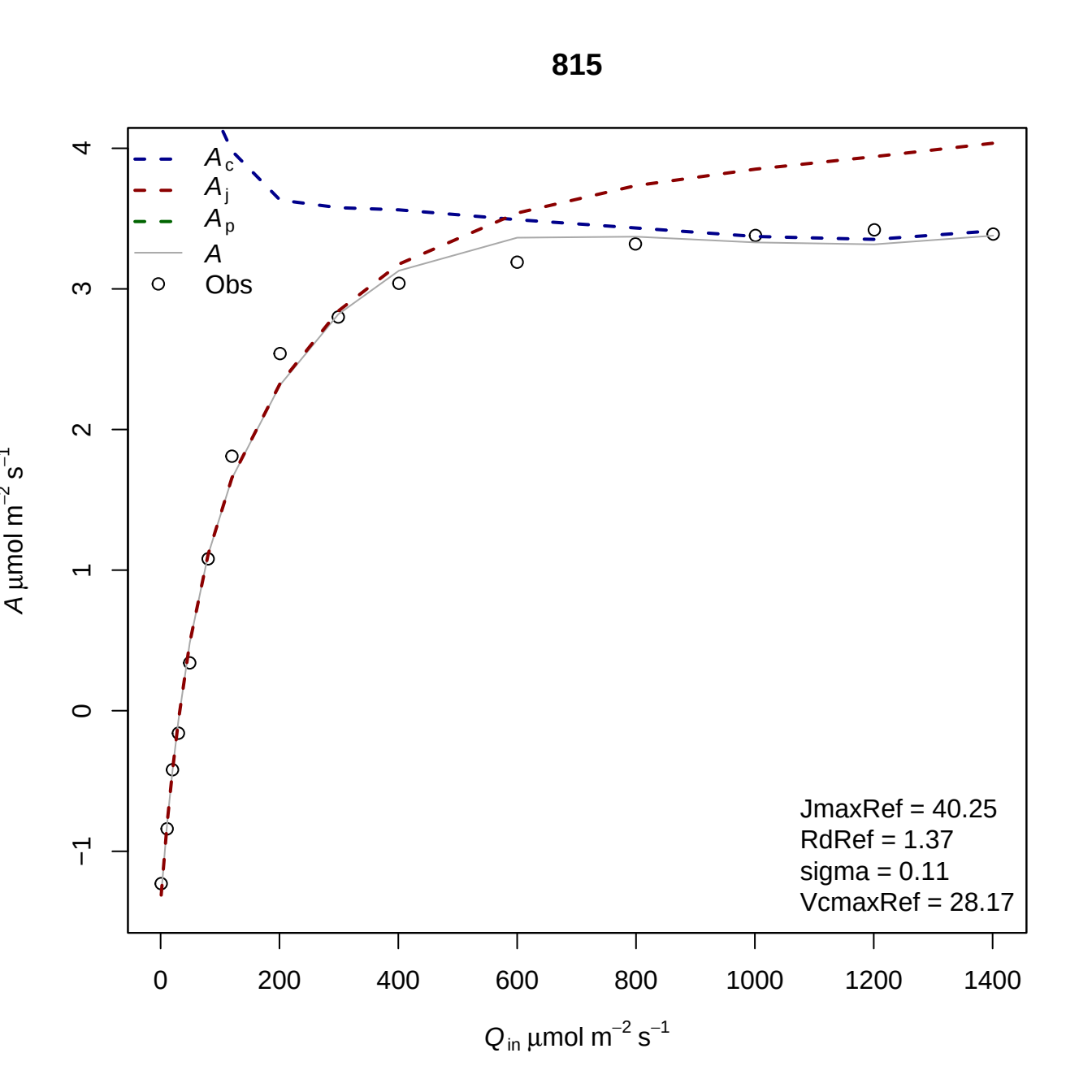

810
A_c
A_j
A_p
A
Obs
JmaxRef = 39.07
RdRef = 0.97
sigma = 0.1
VcmaxRef = 27.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000
1200
1400

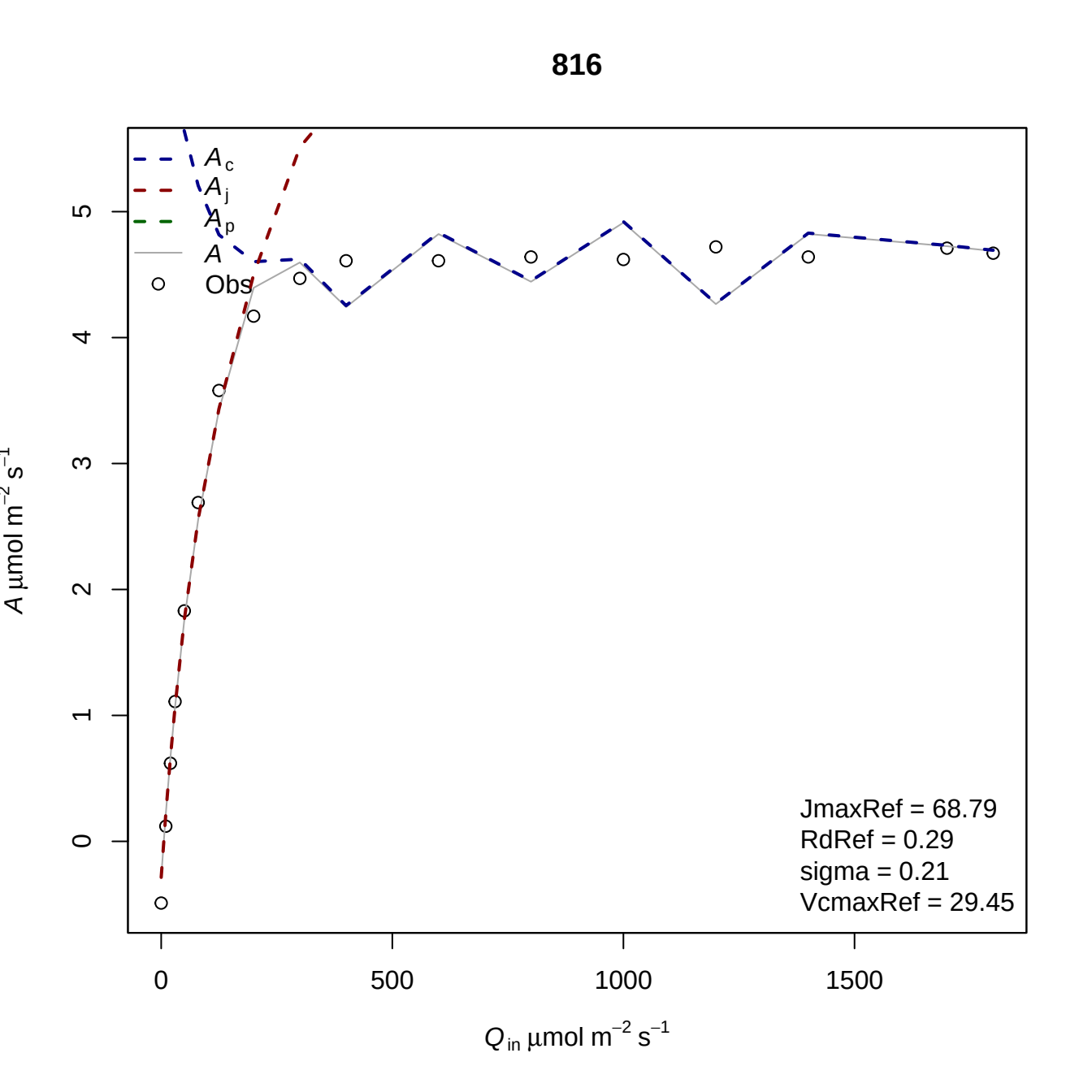

811
A_c
A_j
A_p
A
Obs
JmaxRef = 30.04
RdRef = 0.77
sigma = 0.06
VcmaxRef = 22.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
200
400
600
800
1000
1200
1400

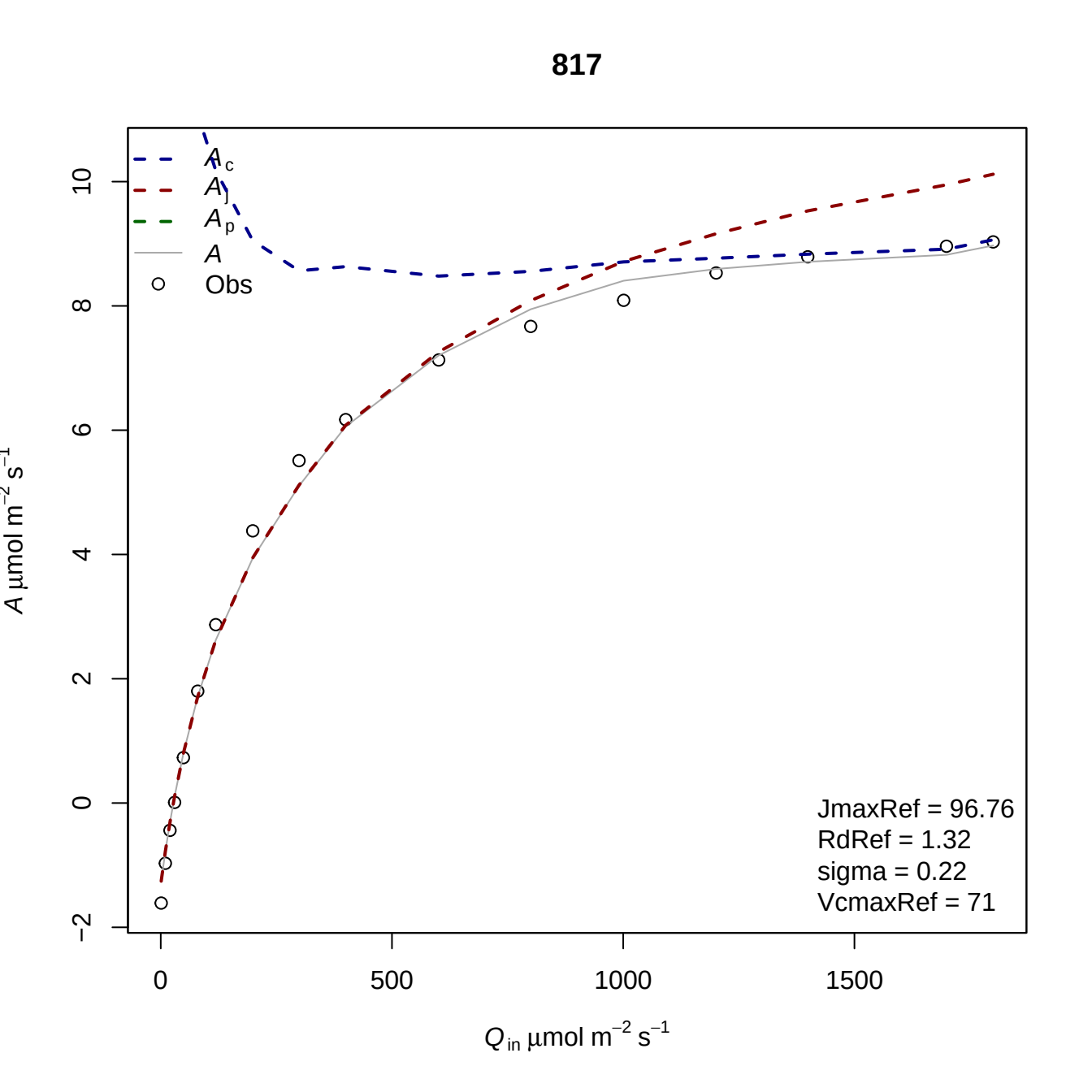

812
A_c
A_j
A_p
A
Obs
JmaxRef = 35.89
RdRef = 1.15
sigma = 0.24
VcmaxRef = 3021731.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

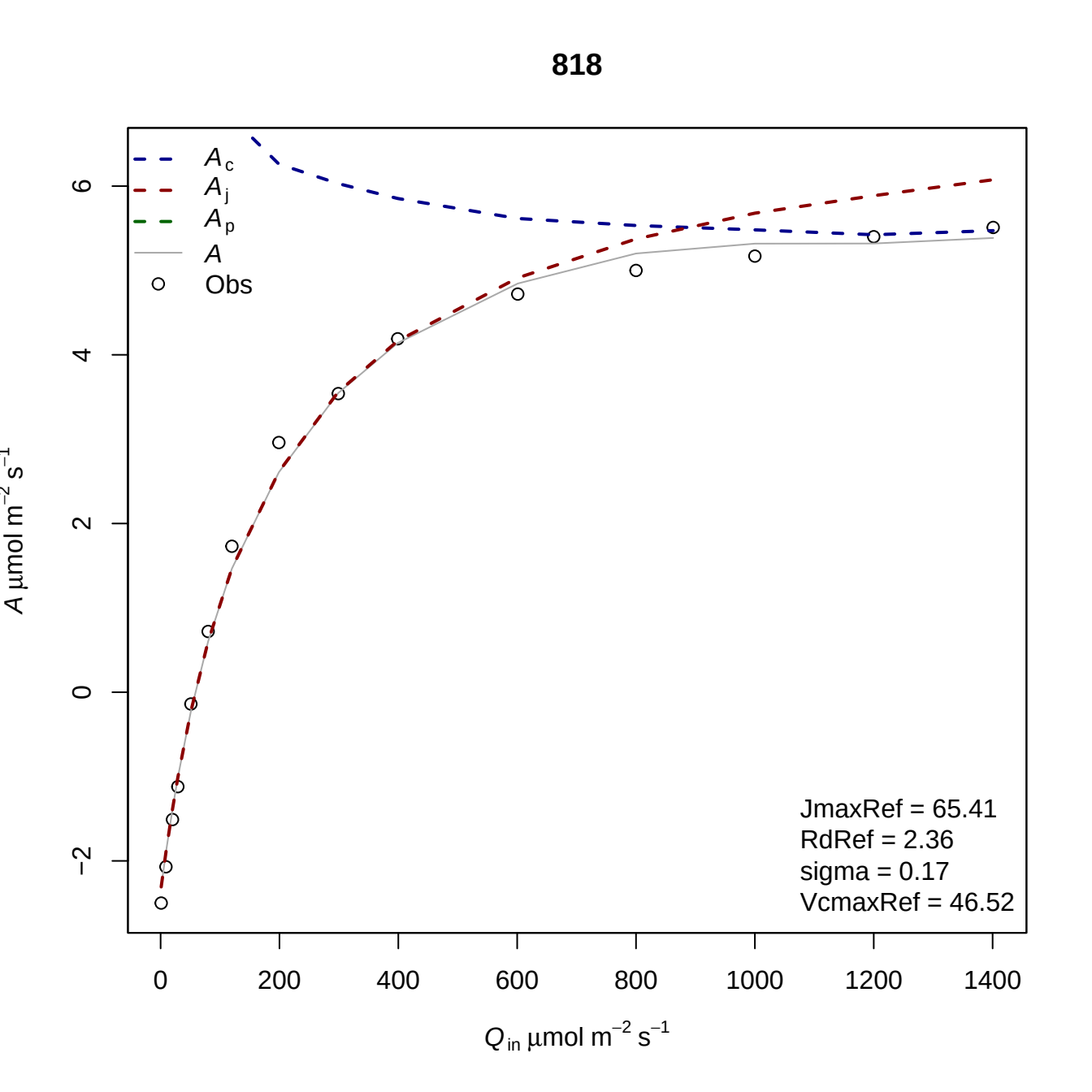

813
A_c
A_j
A_p
A
Obs
JmaxRef = 86.84
RdRef = 1.43
sigma = 0.1
VcmaxRef = 79.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

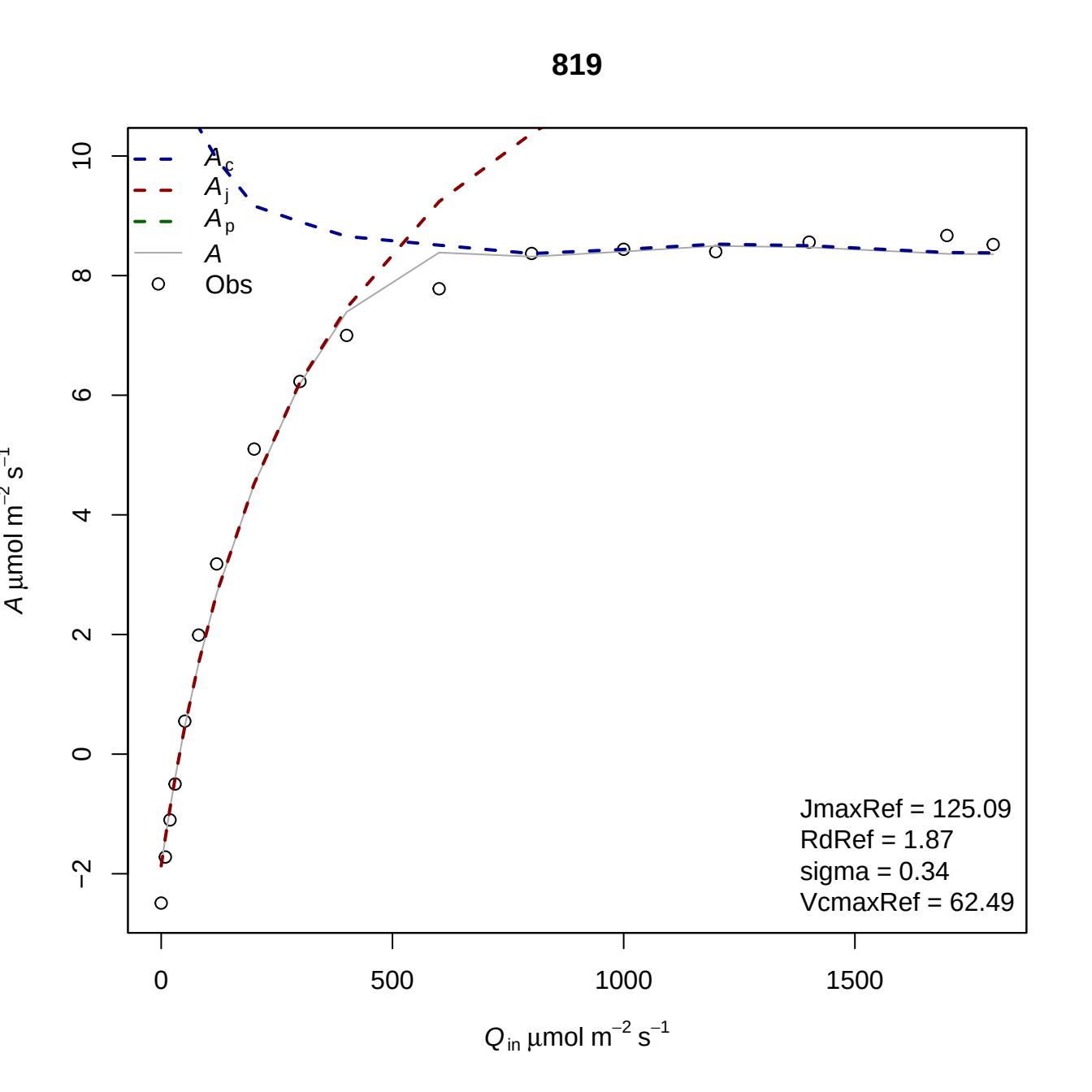

814
A_c
A_j
A_p
A
Obs
JmaxRef = 64.98
RdRef = 2.01
sigma = 0.13
VcmaxRef = 49.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200
1400

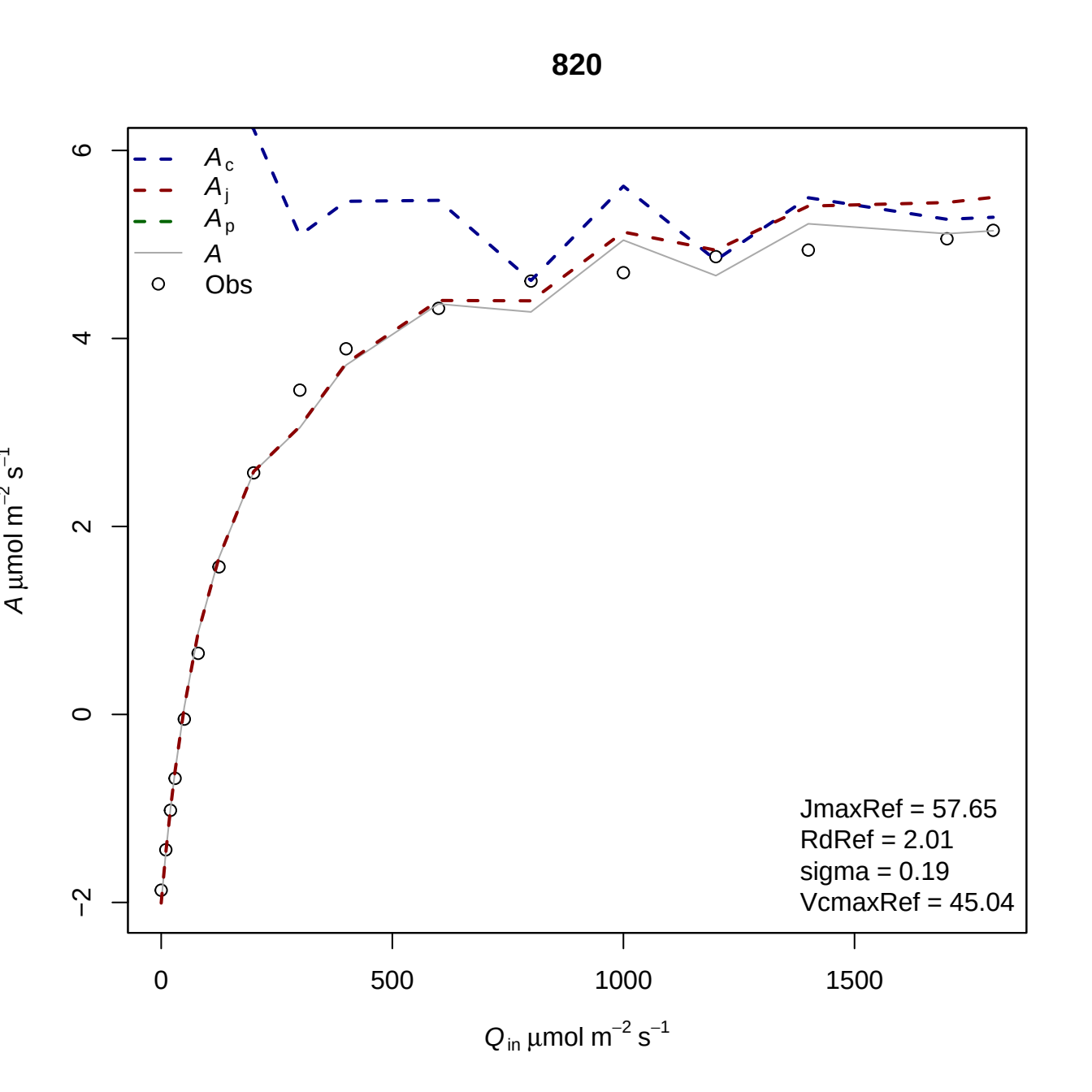

815
A_c
A_j
A_p
A
Obs
JmaxRef = 40.25
RdRef = 1.37
sigma = 0.11
VcmaxRef = 28.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000
1200
1400

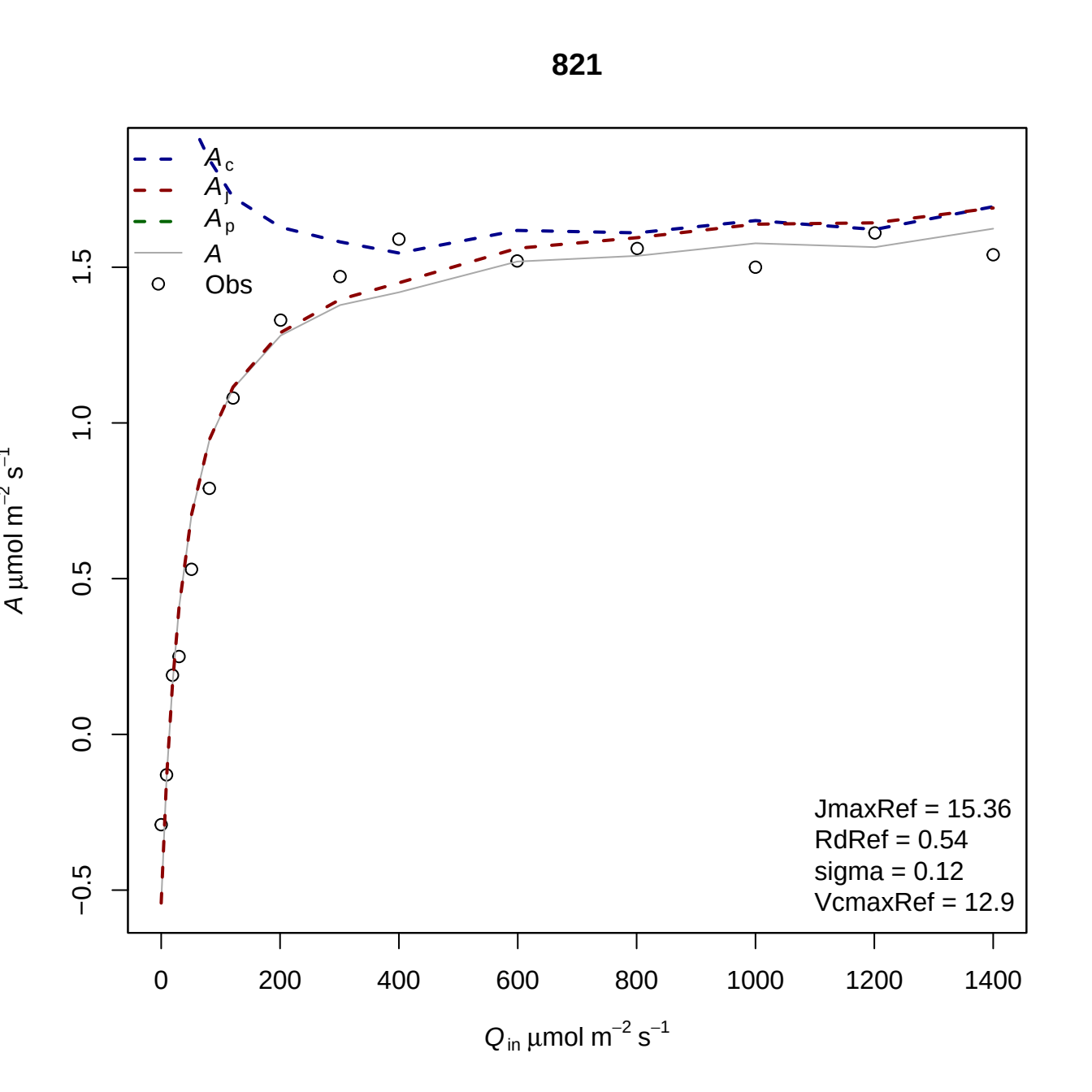

816
A_c
A_j
A_p
A
Obs
JmaxRef = 68.79
RdRef = 0.29
sigma = 0.21
VcmaxRef = 29.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

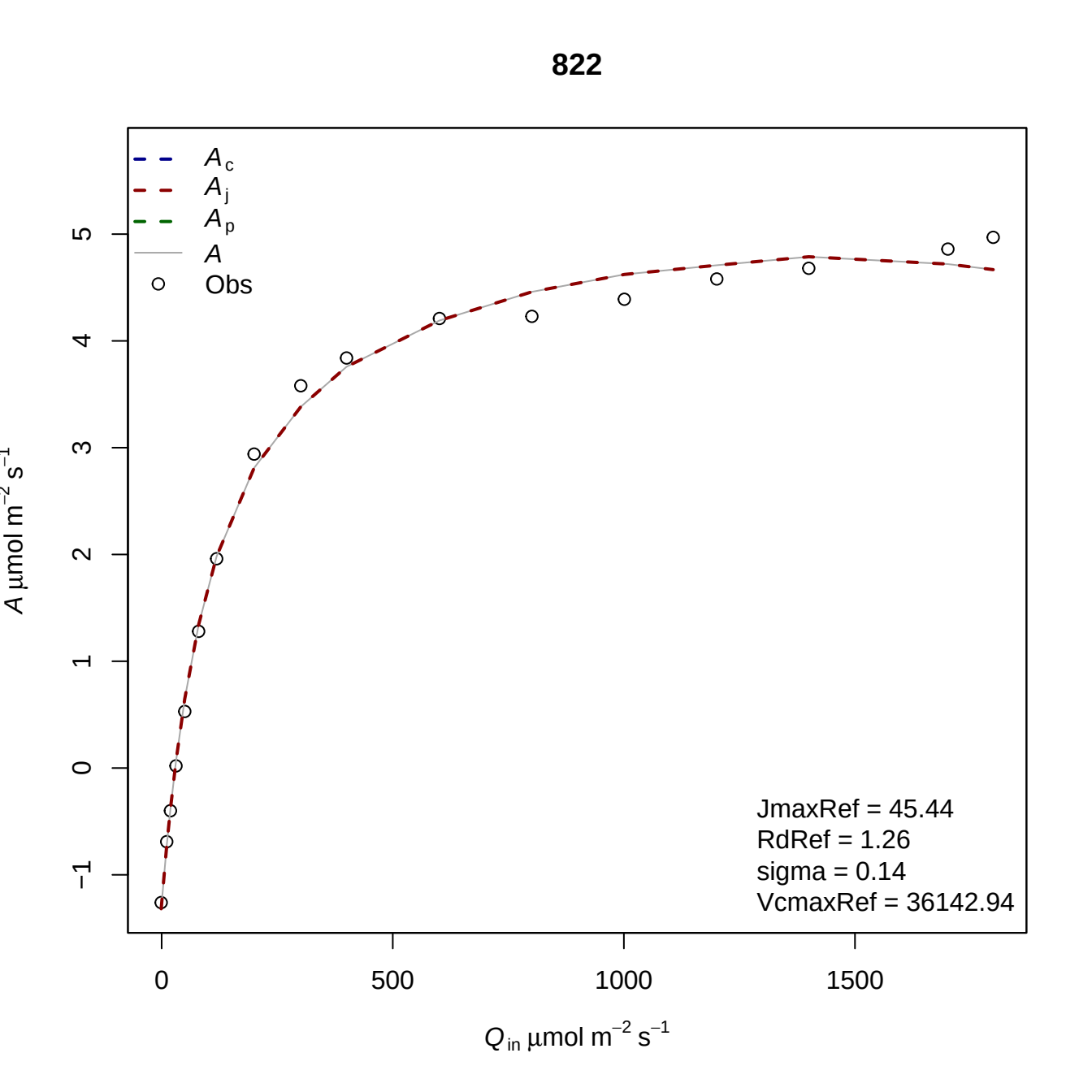

817
A_c
A_j
A_p
A
Obs
JmaxRef = 96.76
RdRef = 1.32
sigma = 0.22
VcmaxRef = 71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

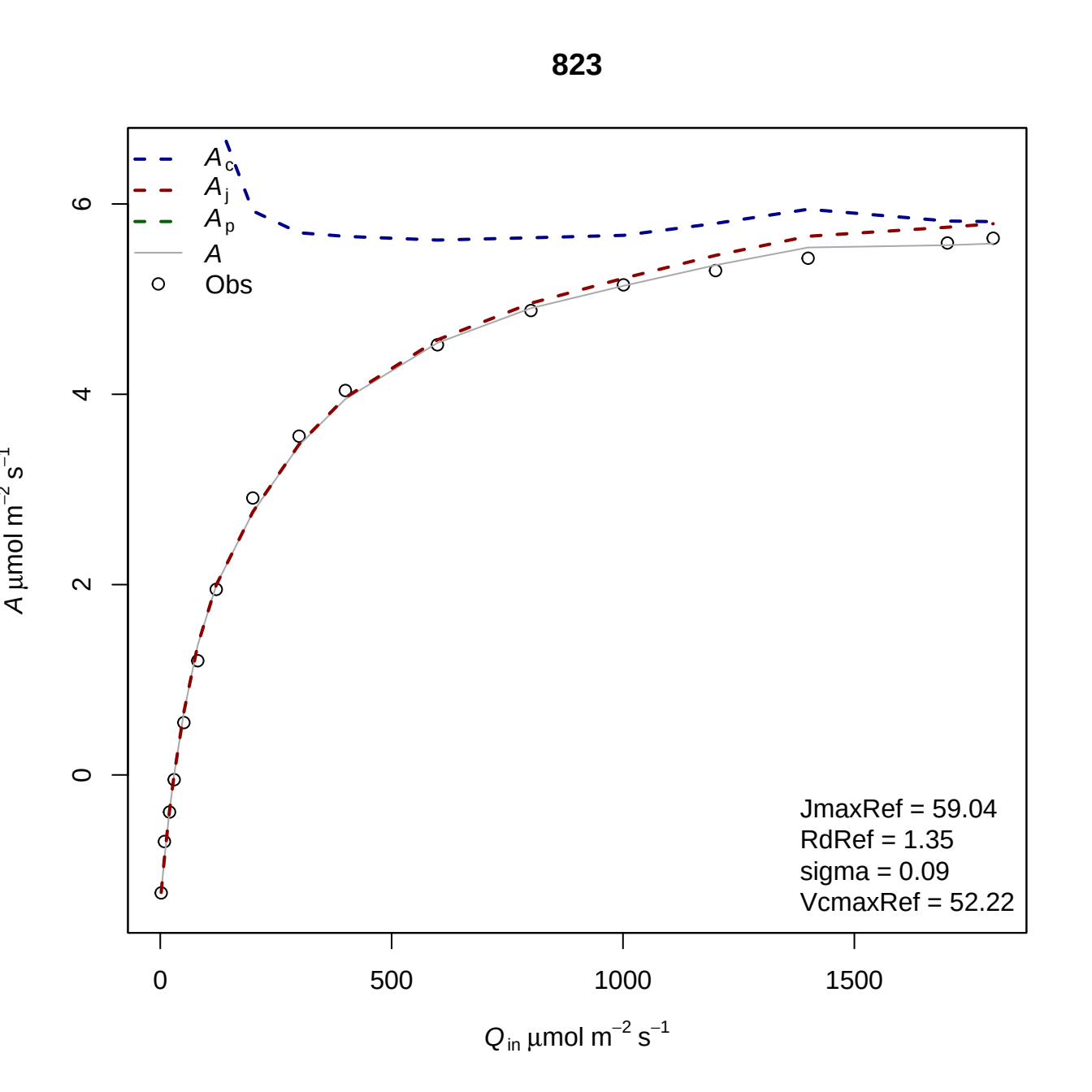

818
A_c
A_j
A_p
A
Obs
JmaxRef = 65.41
RdRef = 2.36
sigma = 0.17
VcmaxRef = 46.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200
1400

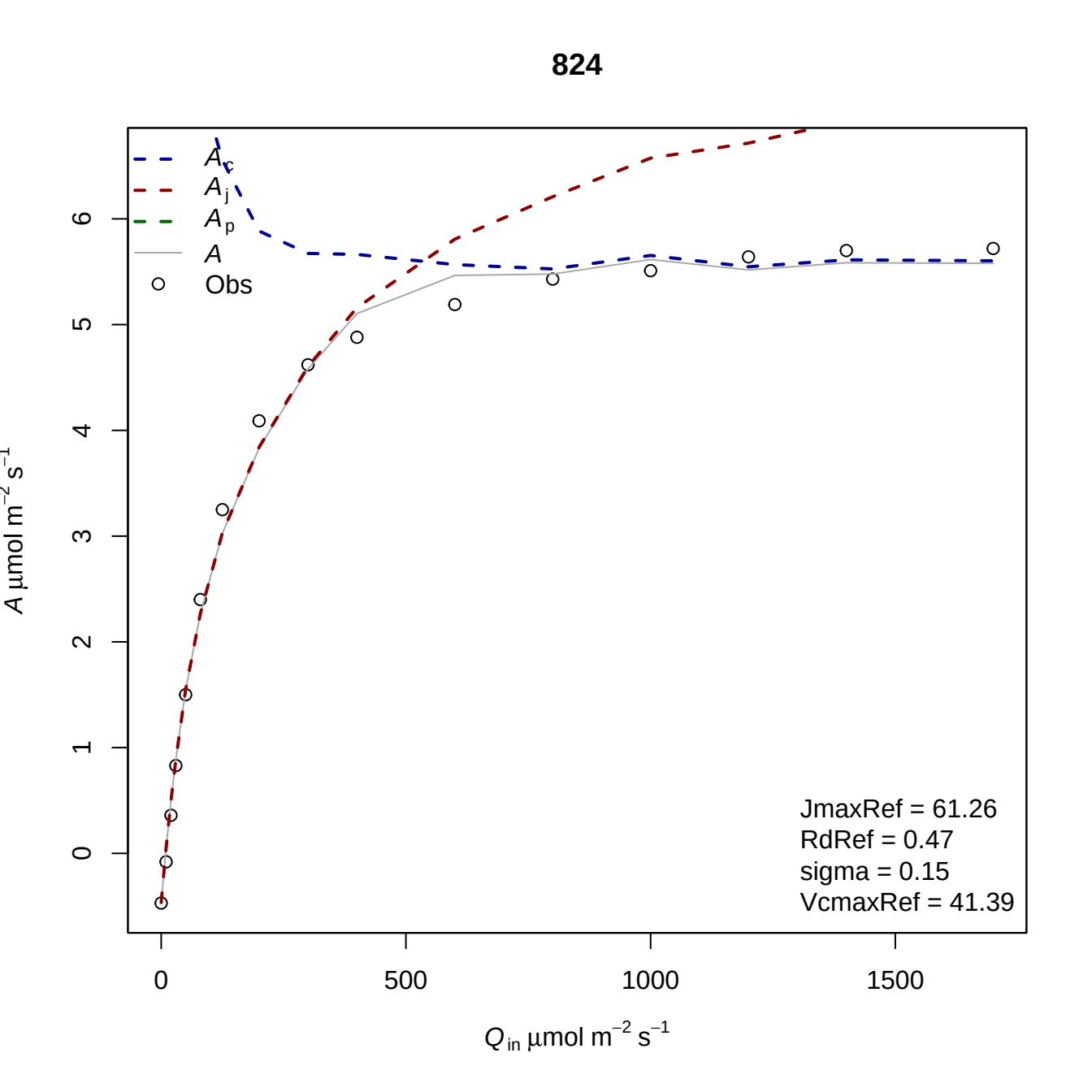

819
A_c
A_j
A_p
A
Obs
JmaxRef = 125.09
RdRef = 1.87
sigma = 0.34
VcmaxRef = 62.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

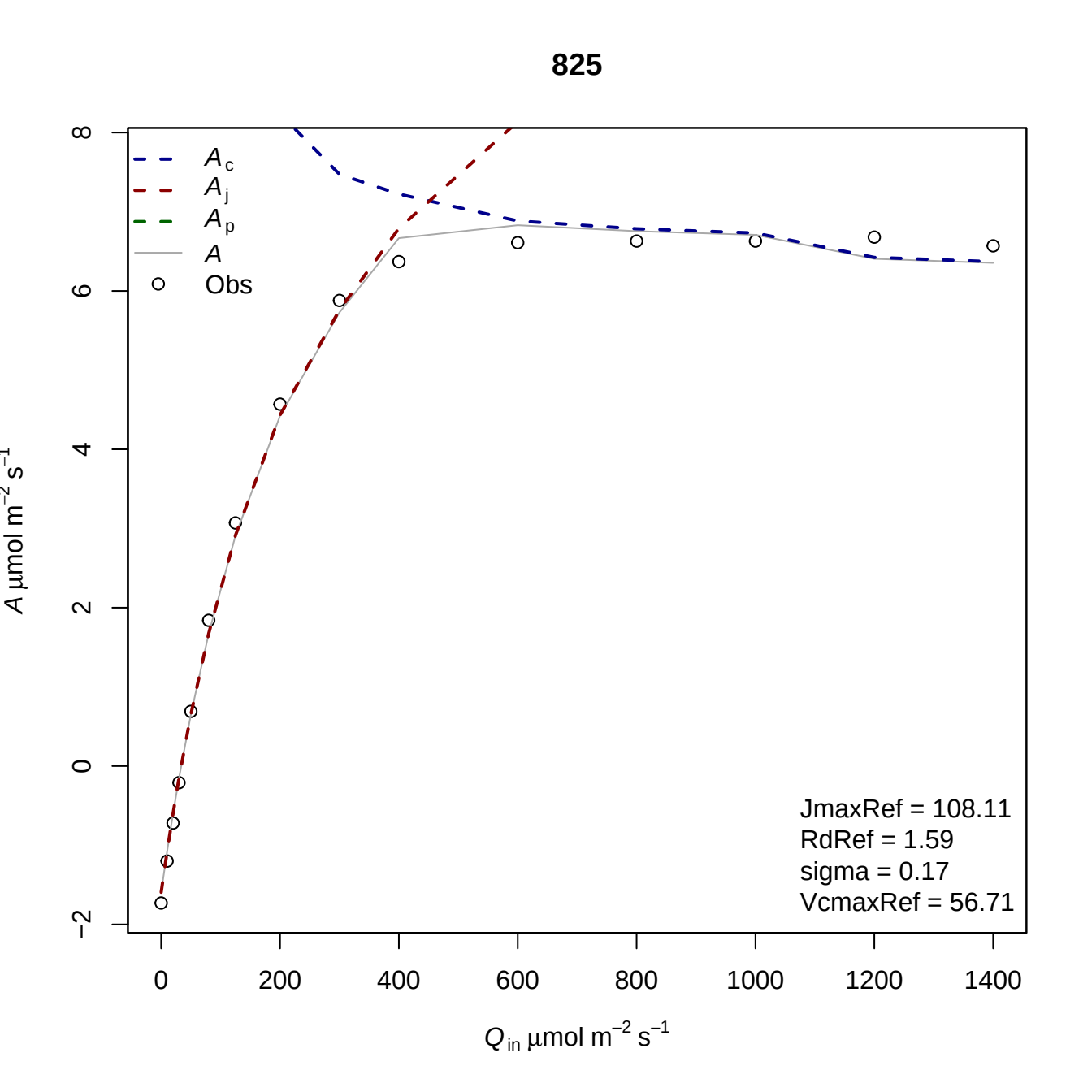

820
A_c
A_j
A_p
A
Obs
JmaxRef = 57.65
RdRef = 2.01
sigma = 0.19
VcmaxRef = 45.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

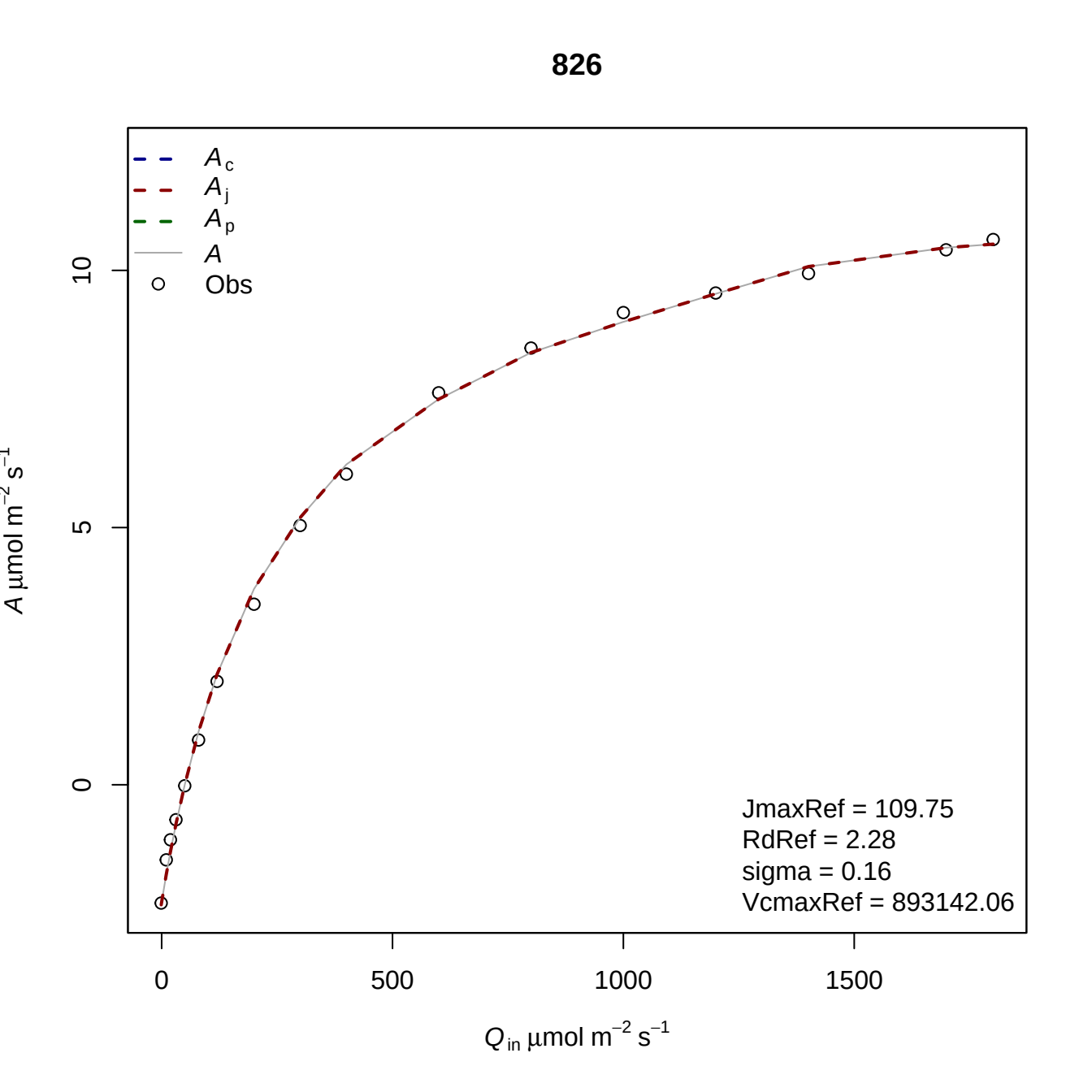

821
A_c
A_j
A_p
A
Obs
JmaxRef = 15.36
RdRef = 0.54
sigma = 0.12
VcmaxRef = 12.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
0
200
400
600
800
1000
1200
1400

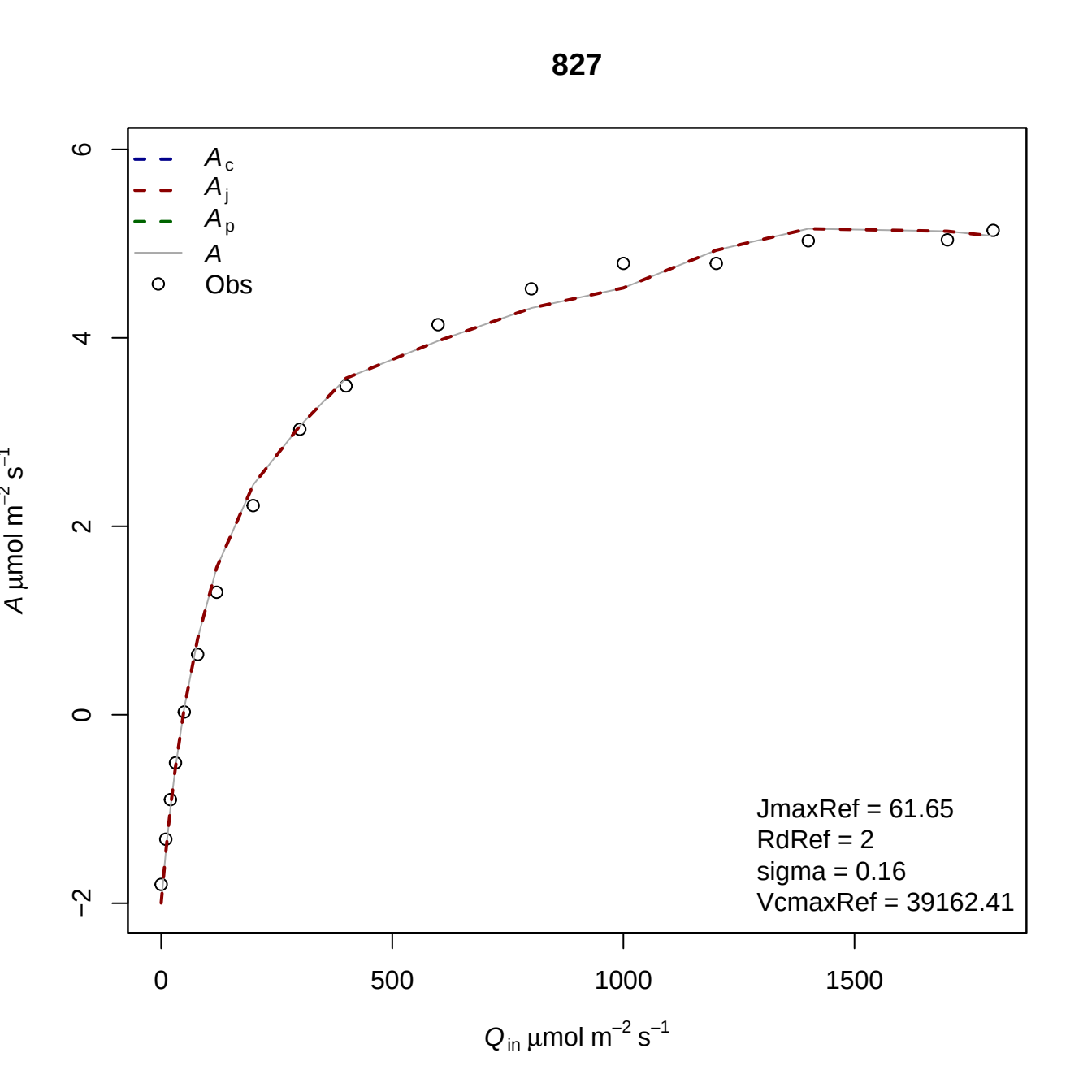

822
A_c
A_j
A_p
A
Obs
JmaxRef = 45.44
RdRef = 1.26
sigma = 0.14
VcmaxRef = 36142.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

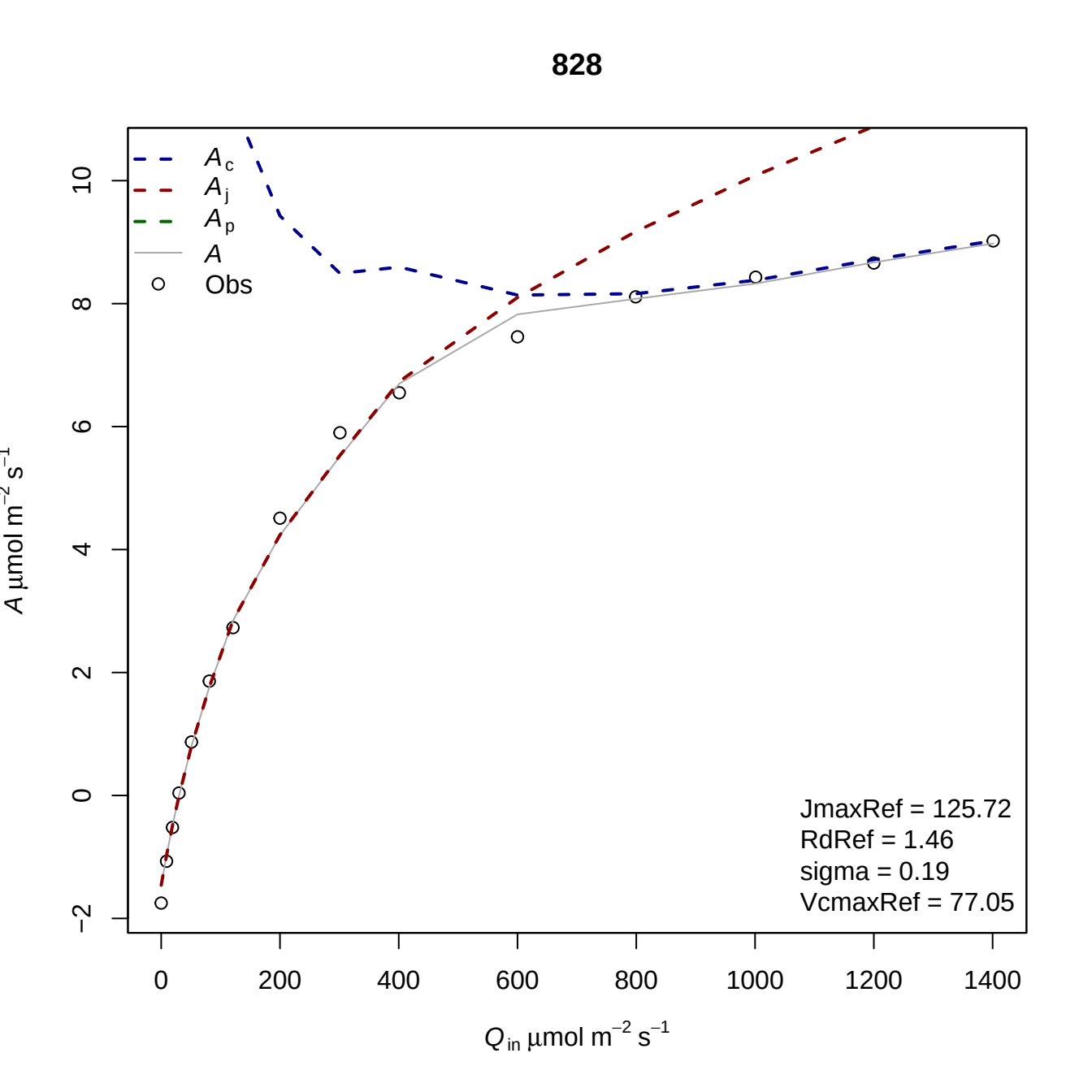

823
A_c
A_j
A_p
A
Obs
JmaxRef = 59.04
RdRef = 1.35
sigma = 0.09
VcmaxRef = 52.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

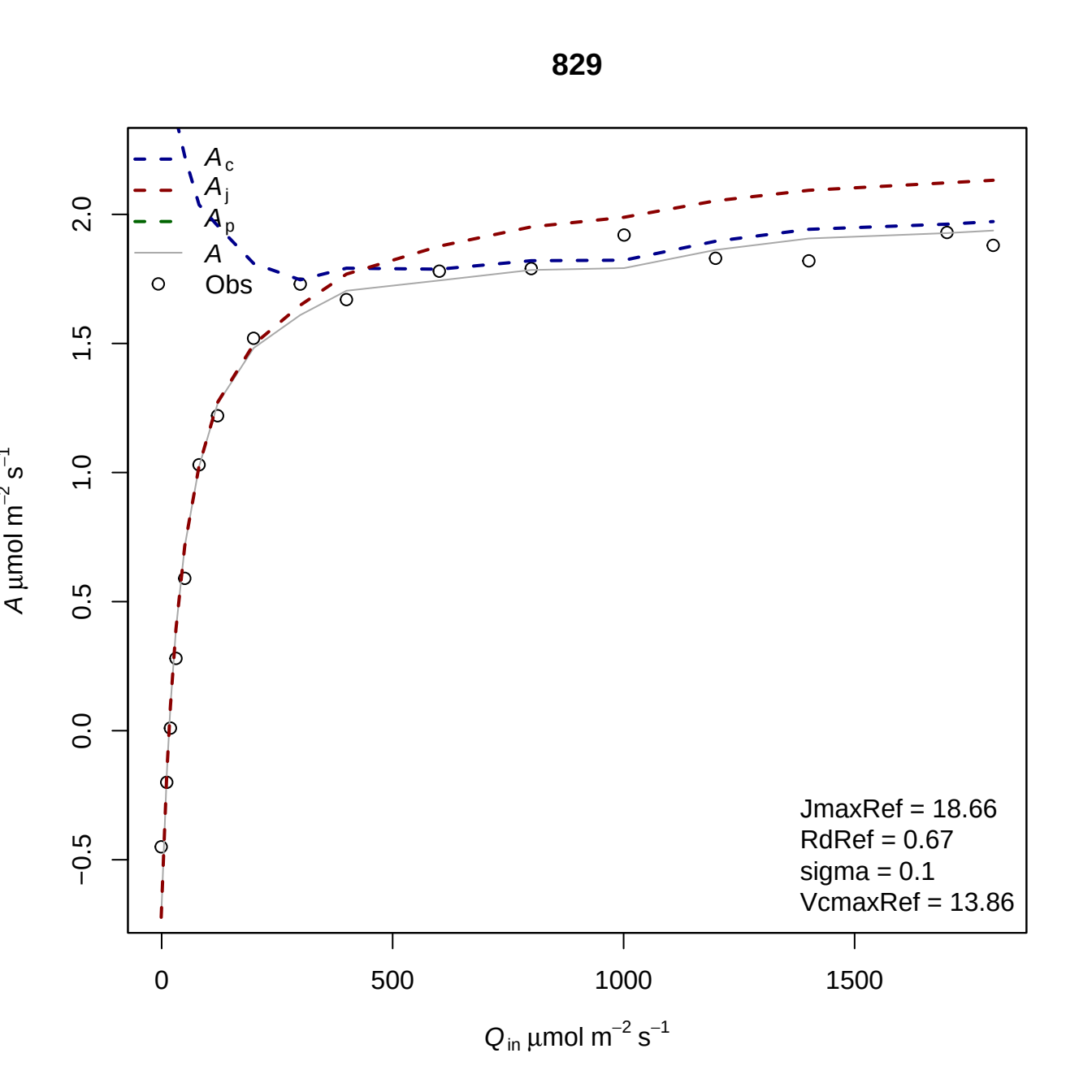

824
A_c
A_j
A_p
A
Obs
JmaxRef = 61.26
RdRef = 0.47
sigma = 0.15
VcmaxRef = 41.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

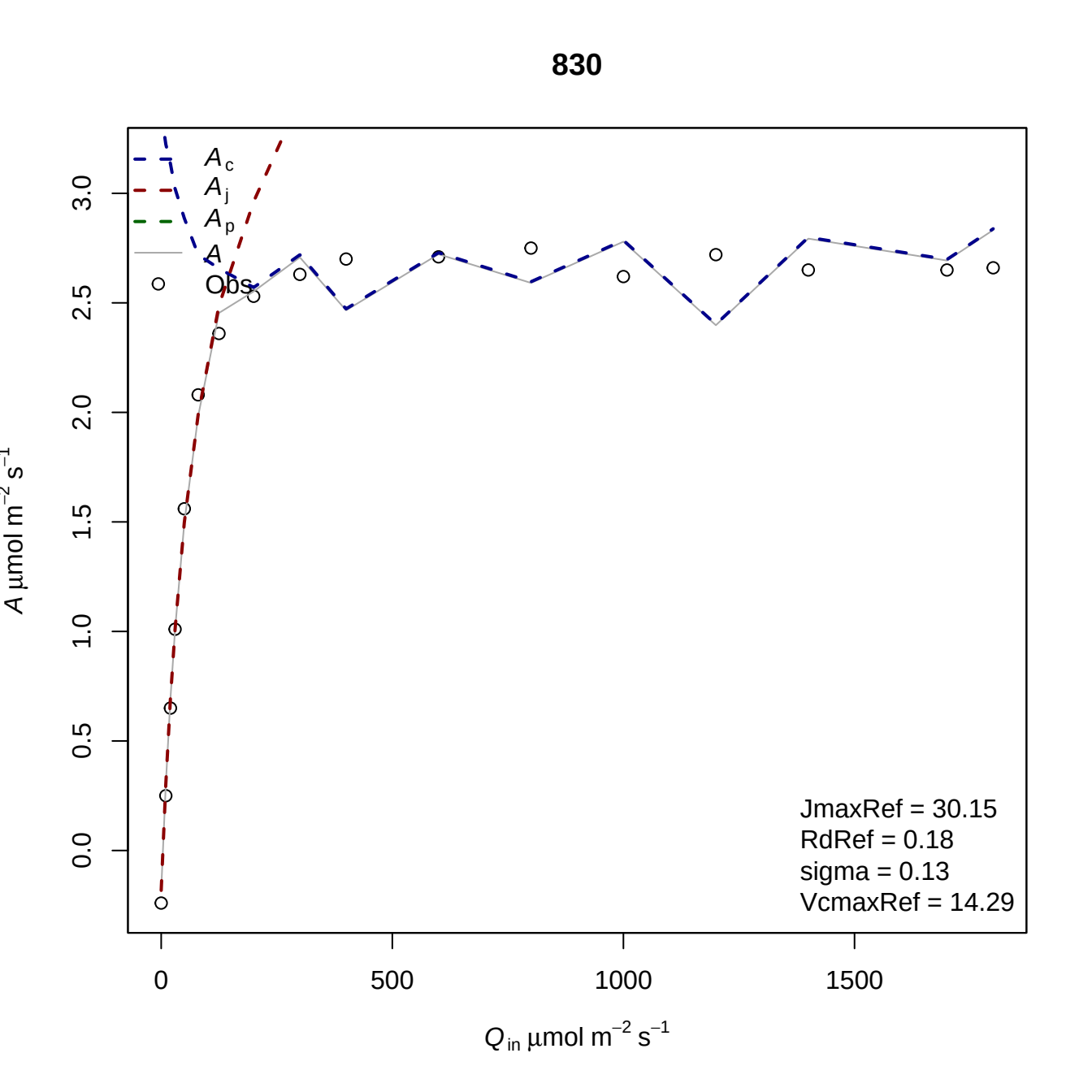

825
A_c
A_j
A_p
A
Obs
JmaxRef = 108.11
RdRef = 1.59
sigma = 0.17
VcmaxRef = 56.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000
1200
1400

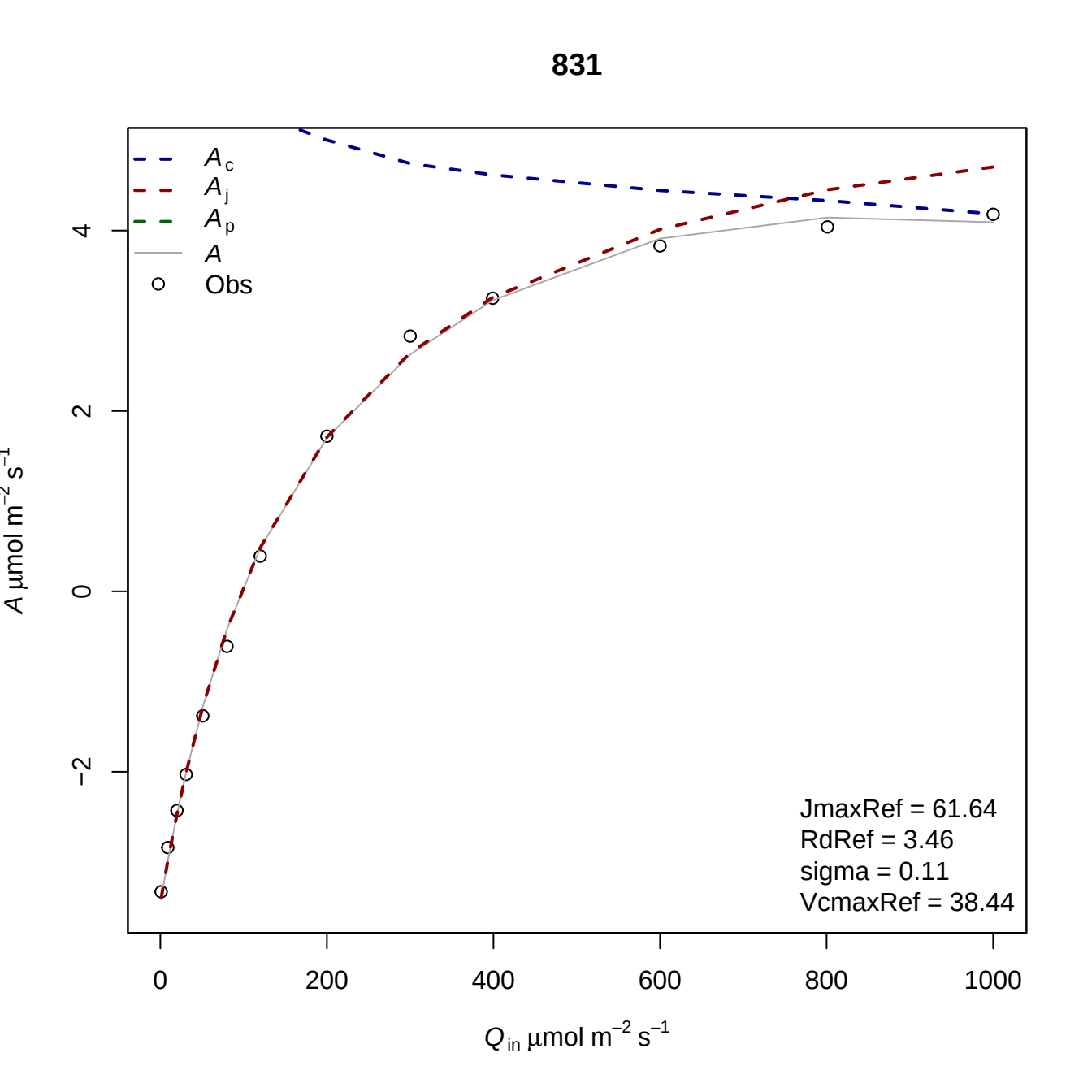

826
A_c
A_j
A_p
A
Obs
JmaxRef = 109.75
RdRef = 2.28
sigma = 0.16
VcmaxRef = 893142.06
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

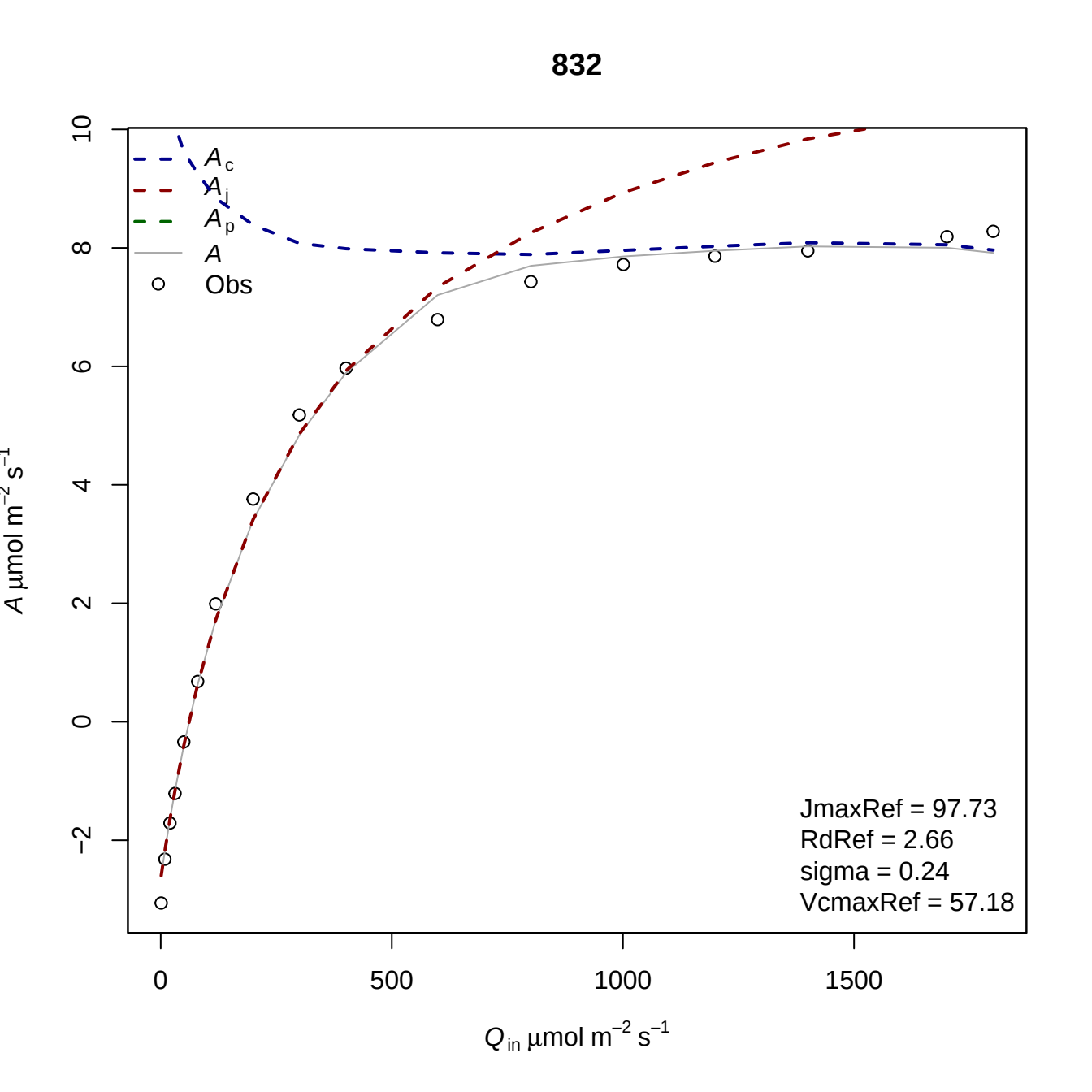

827
A_c
A_j
A_p
A
Obs
JmaxRef = 61.65
RdRef = 2
sigma = 0.16
VcmaxRef = 39162.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

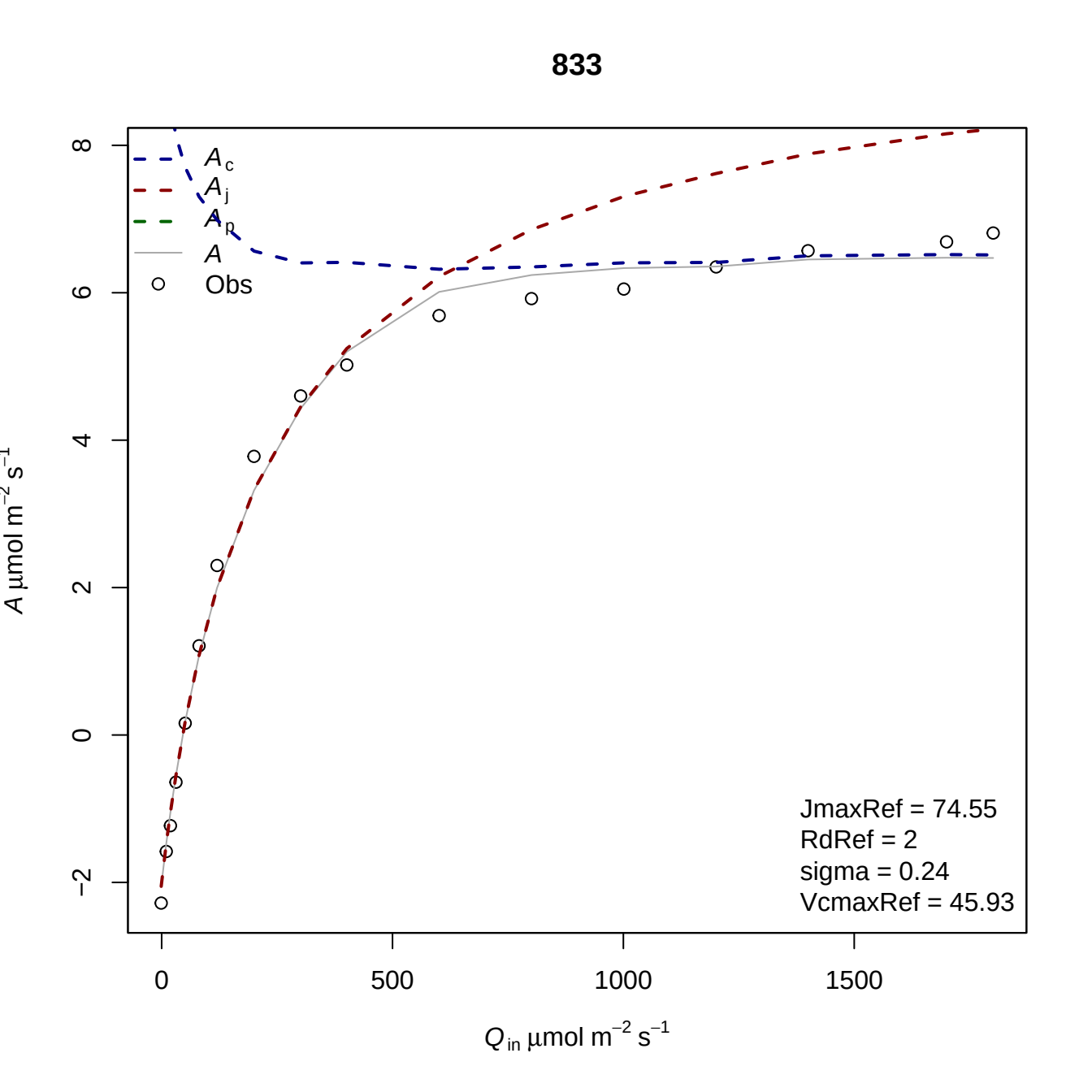

828
A_c
A_j
A_p
A
Obs
JmaxRef = 125.72
RdRef = 1.46
sigma = 0.19
VcmaxRef = 77.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

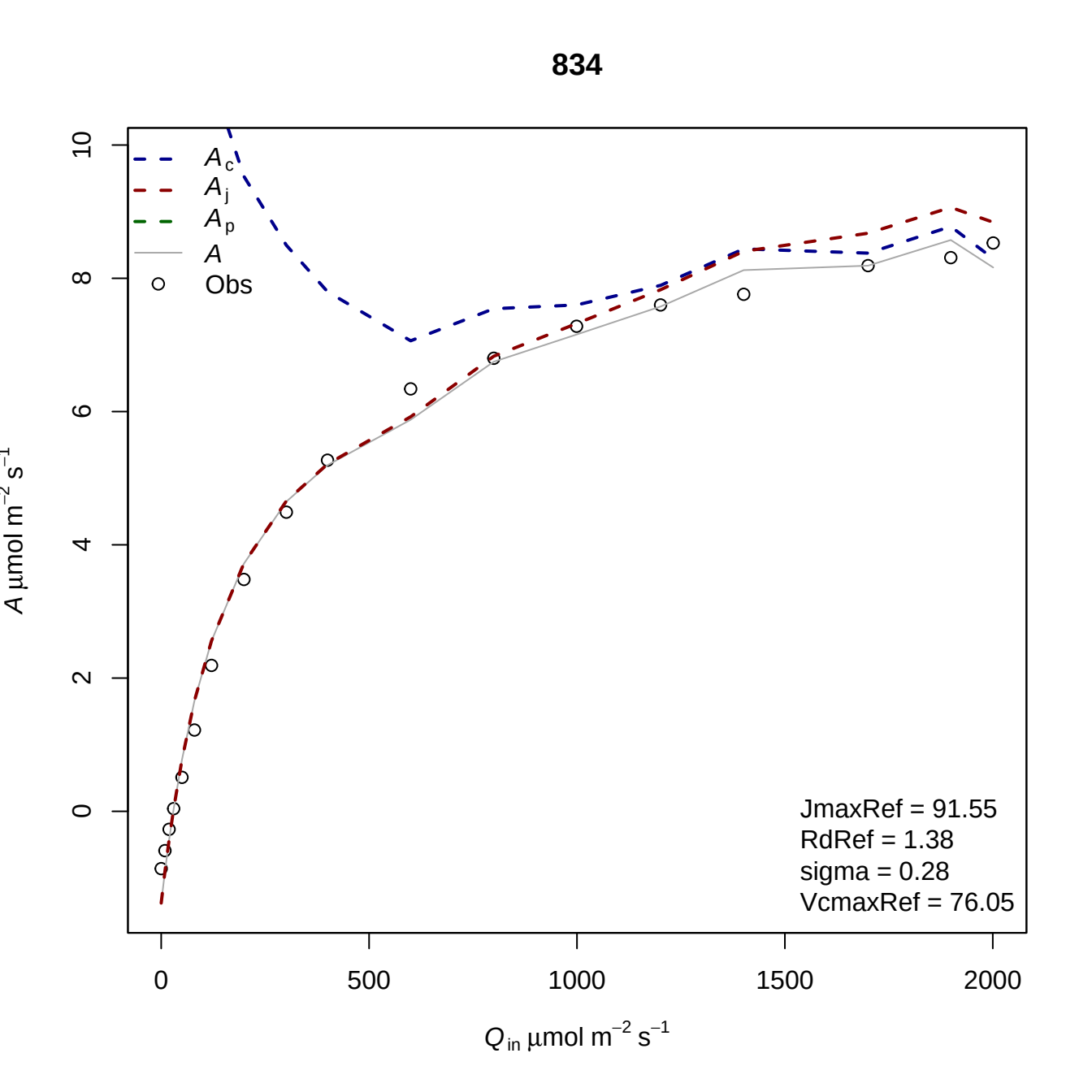

829
A_c
A_j
A_p
A
Obs
JmaxRef = 18.66
RdRef = 0.67
sigma = 0.1
VcmaxRef = 13.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500

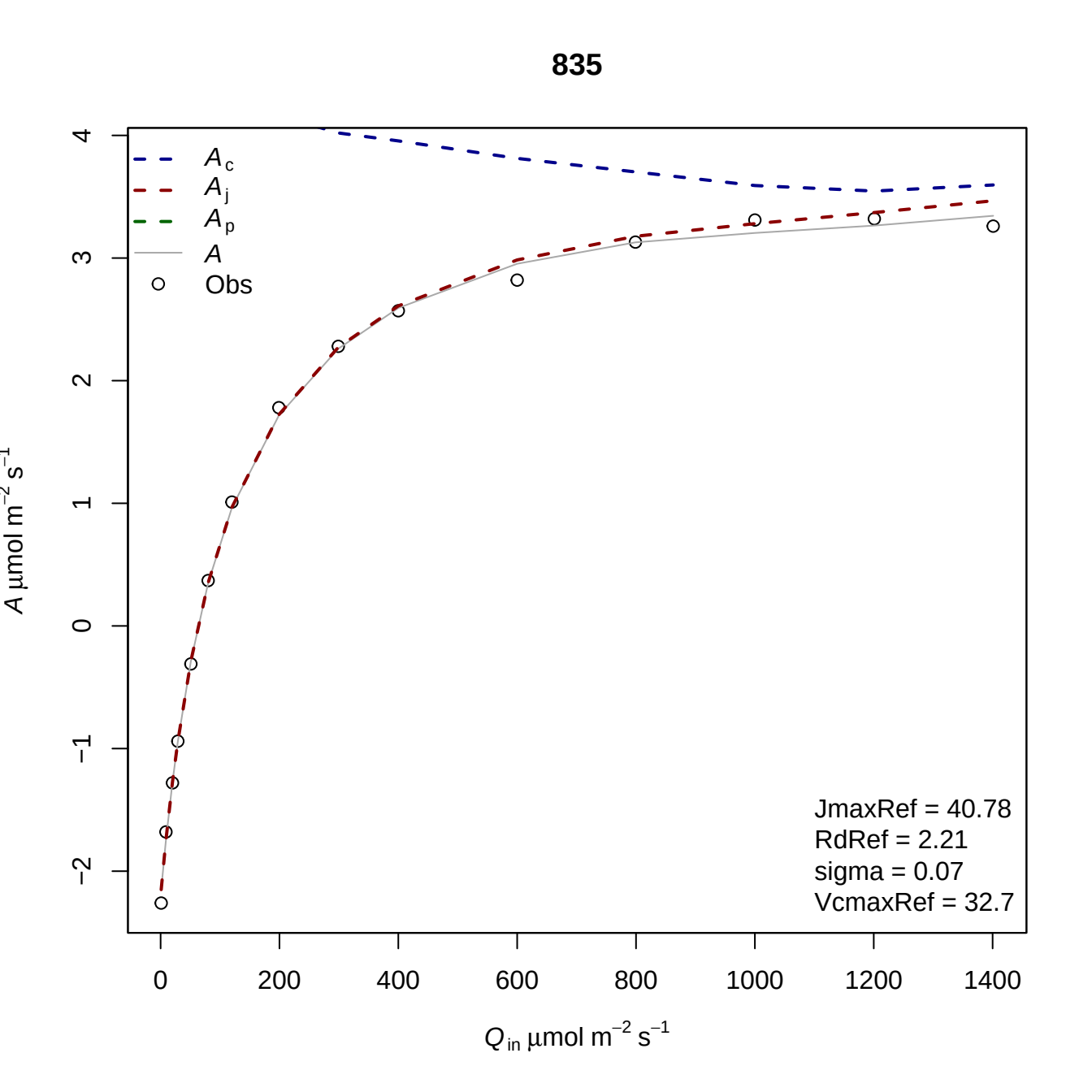

830
A_c
A_j
A_p
A
Obs
JmaxRef = 30.15
RdRef = 0.18
sigma = 0.13
VcmaxRef = 14.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
500
1000
1500

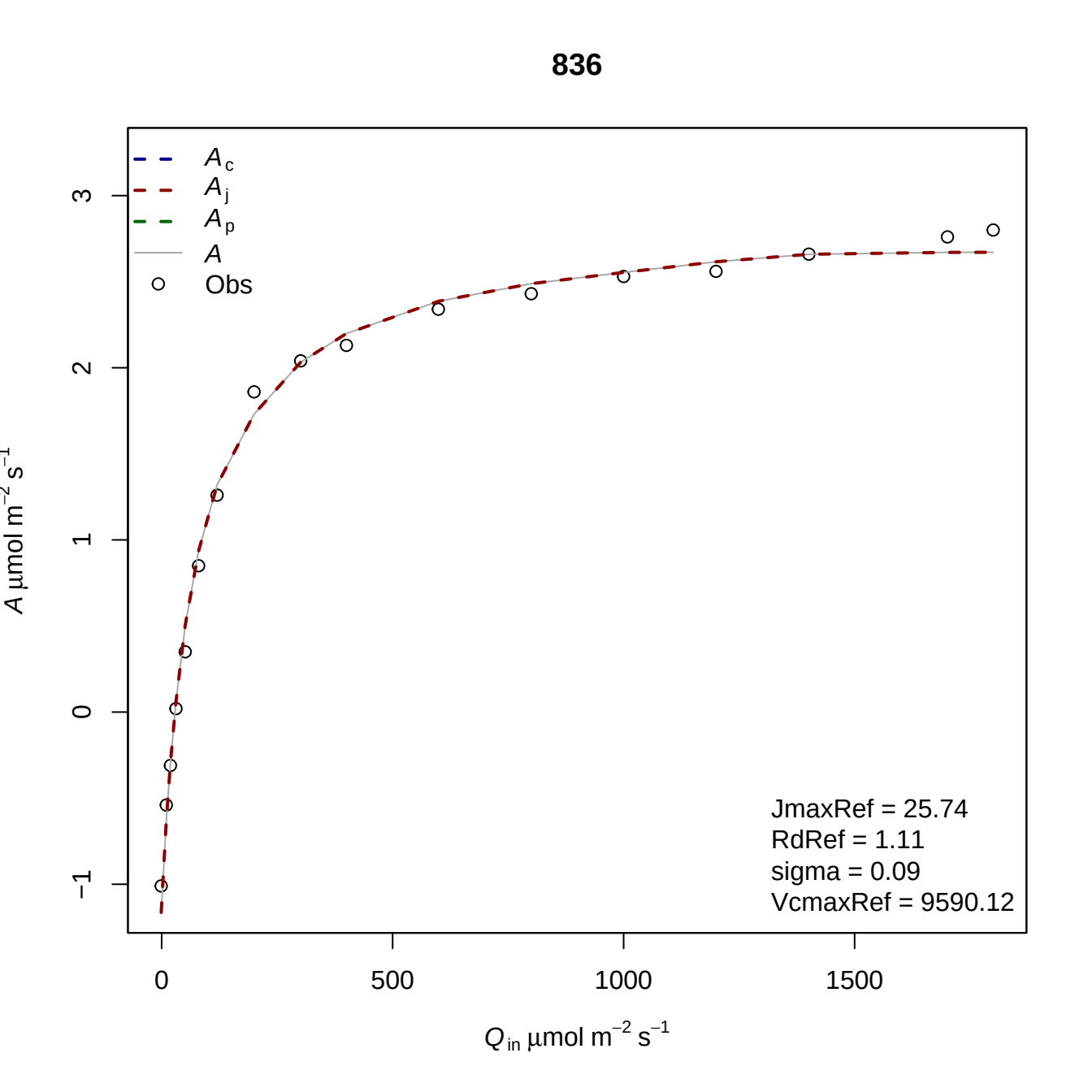

831
A_c
A_j
A_p
A
Obs
JmaxRef = 61.64
RdRef = 3.46
sigma = 0.11
VcmaxRef = 38.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000

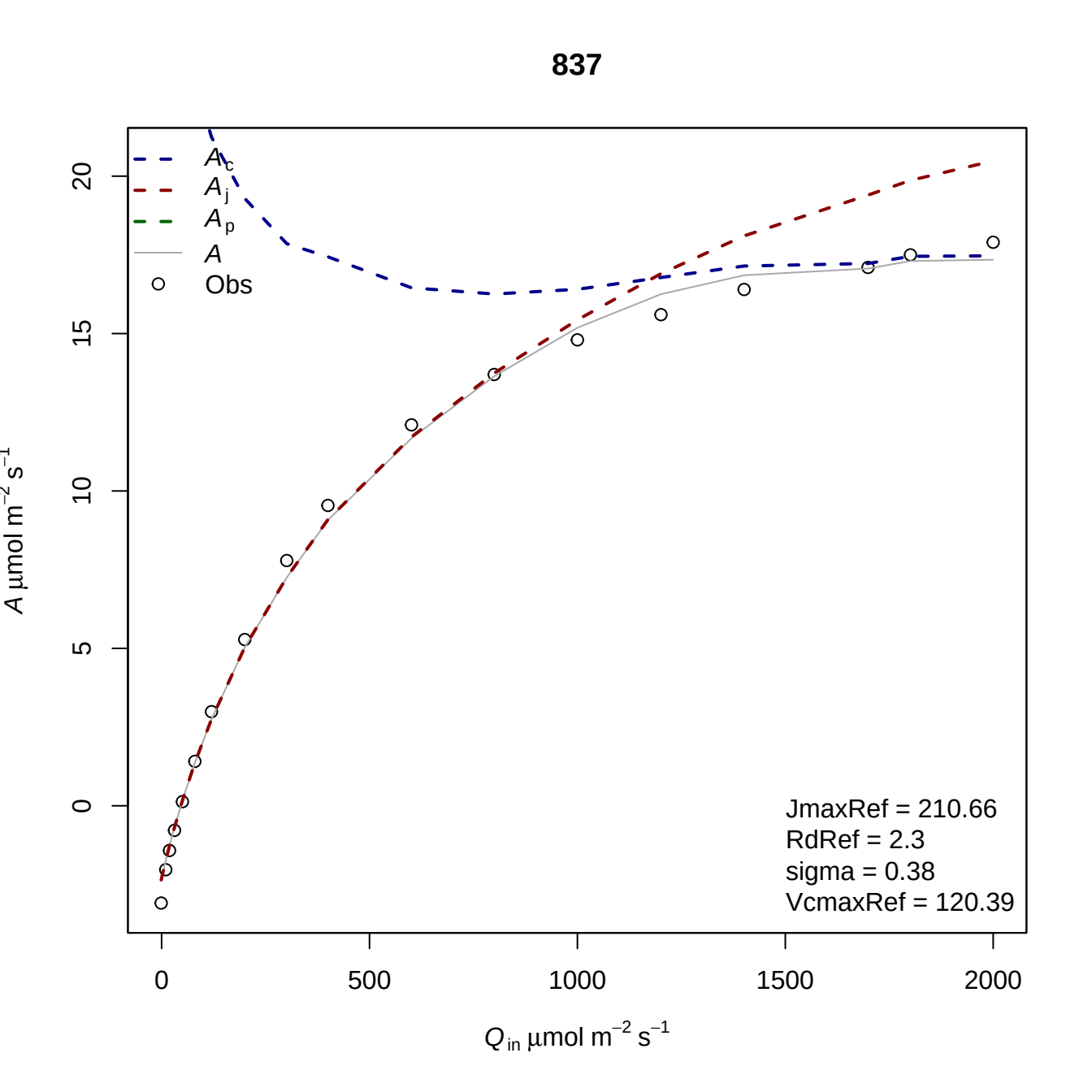

832
A_c
A_j
A_p
A
Obs
JmaxRef = 97.73
RdRef = 2.66
sigma = 0.24
VcmaxRef = 57.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

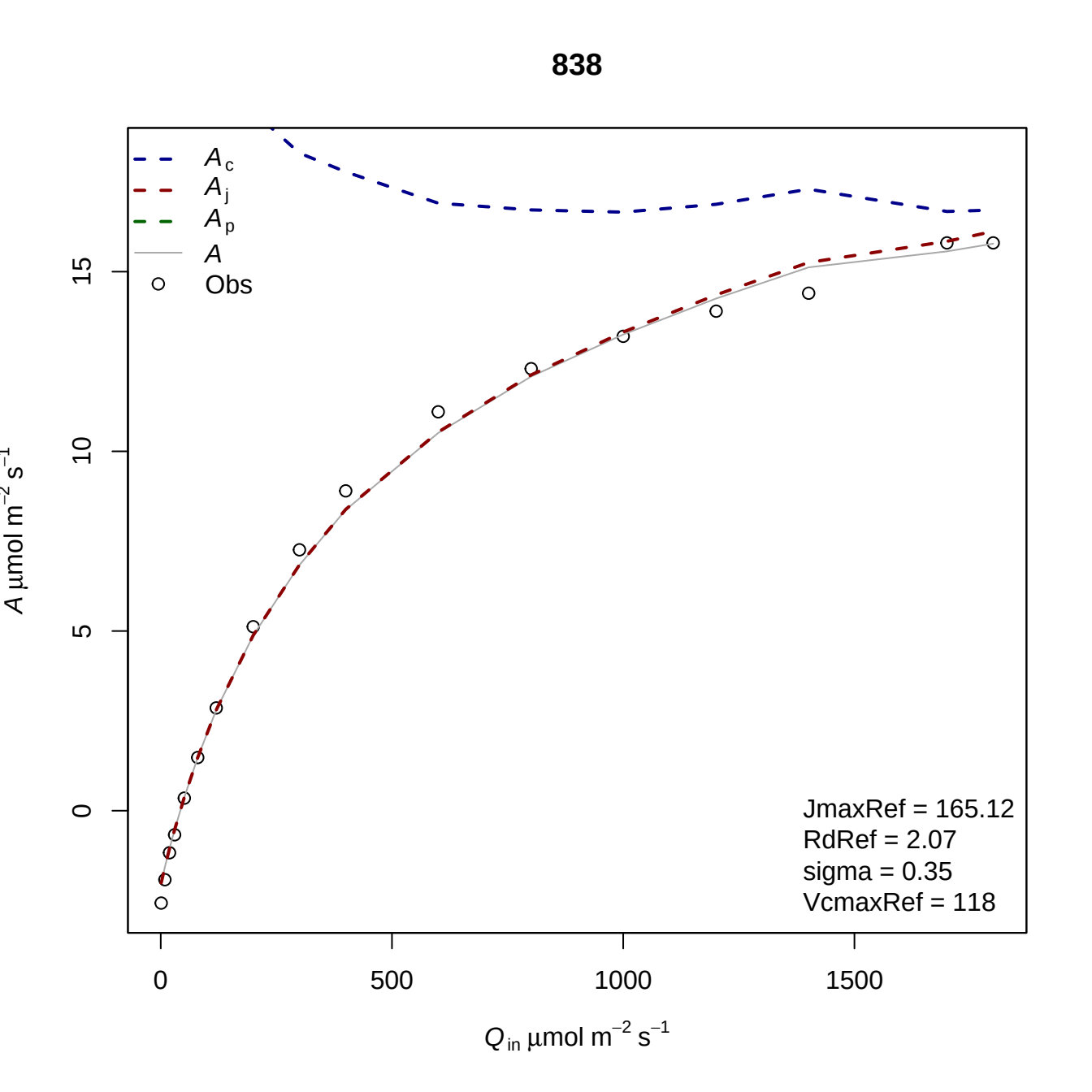

833
A_c
A_j
A_p
A
Obs
JmaxRef = 74.55
RdRef = 2
sigma = 0.24
VcmaxRef = 45.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

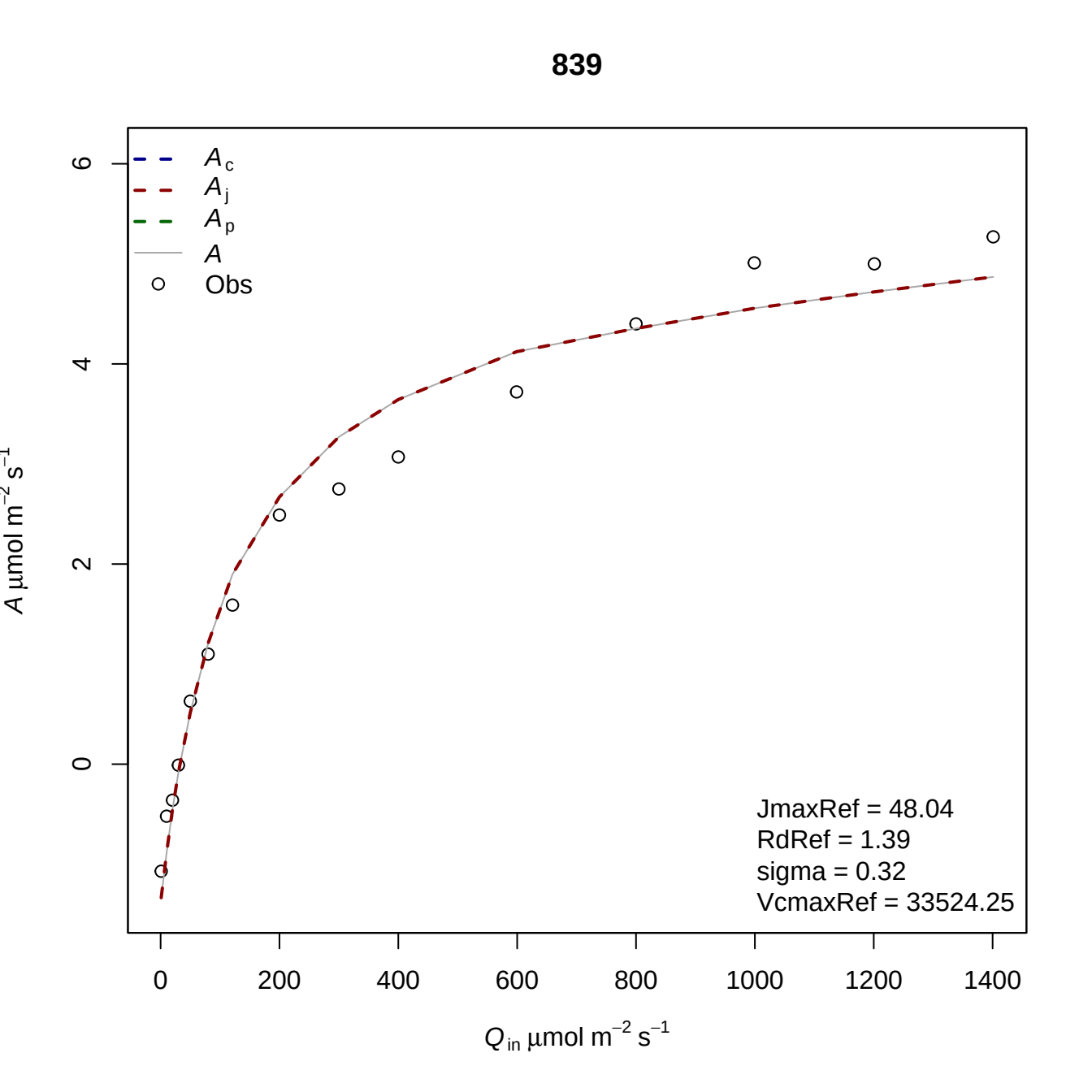

834
A_c
A_j
A_p
A
Obs
JmaxRef = 91.55
RdRef = 1.38
sigma = 0.28
VcmaxRef = 76.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

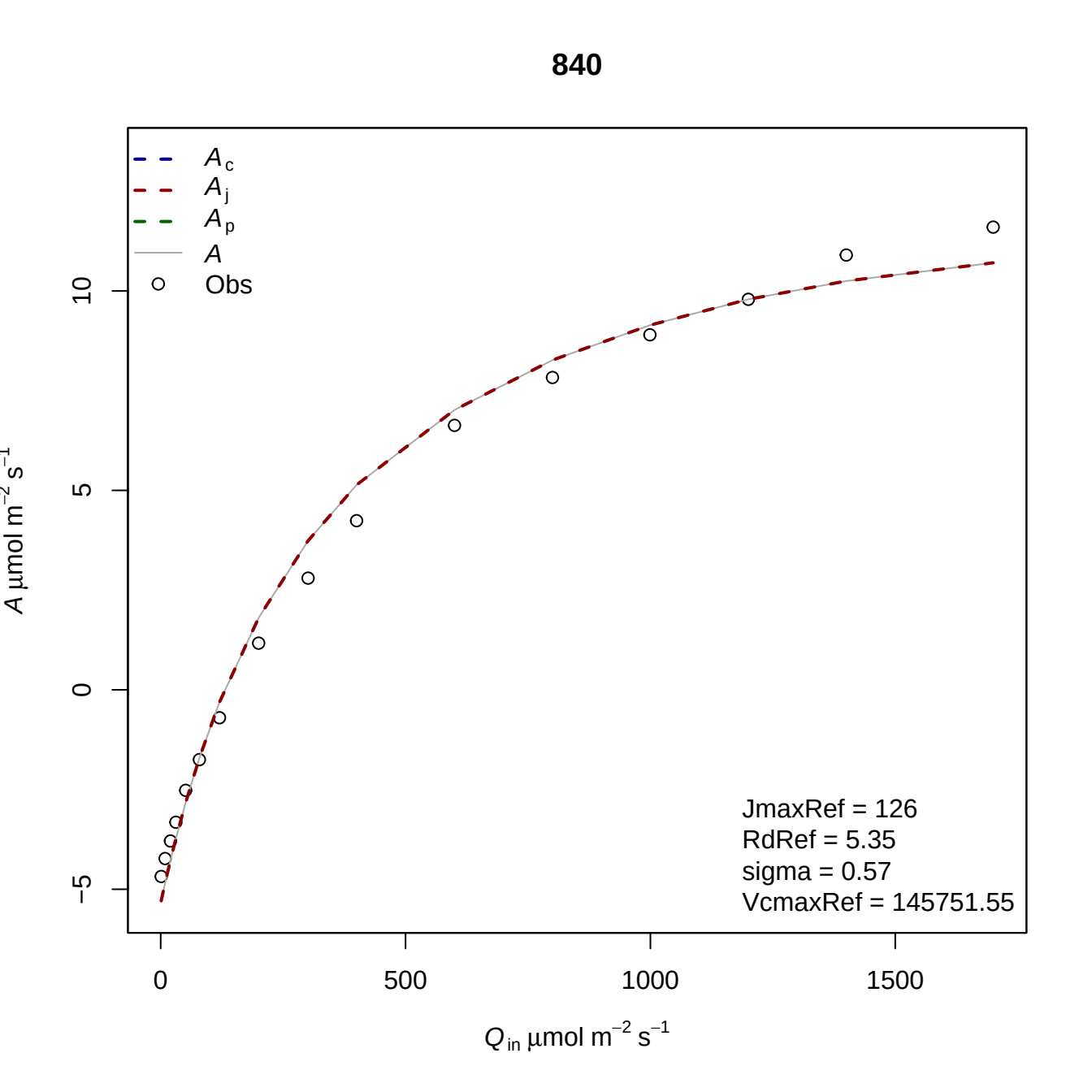

835
A_c
A_j
A_p
A
Obs
JmaxRef = 40.78
RdRef = 2.21
sigma = 0.07
VcmaxRef = 32.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
200
400
600
800
1000
1200
1400

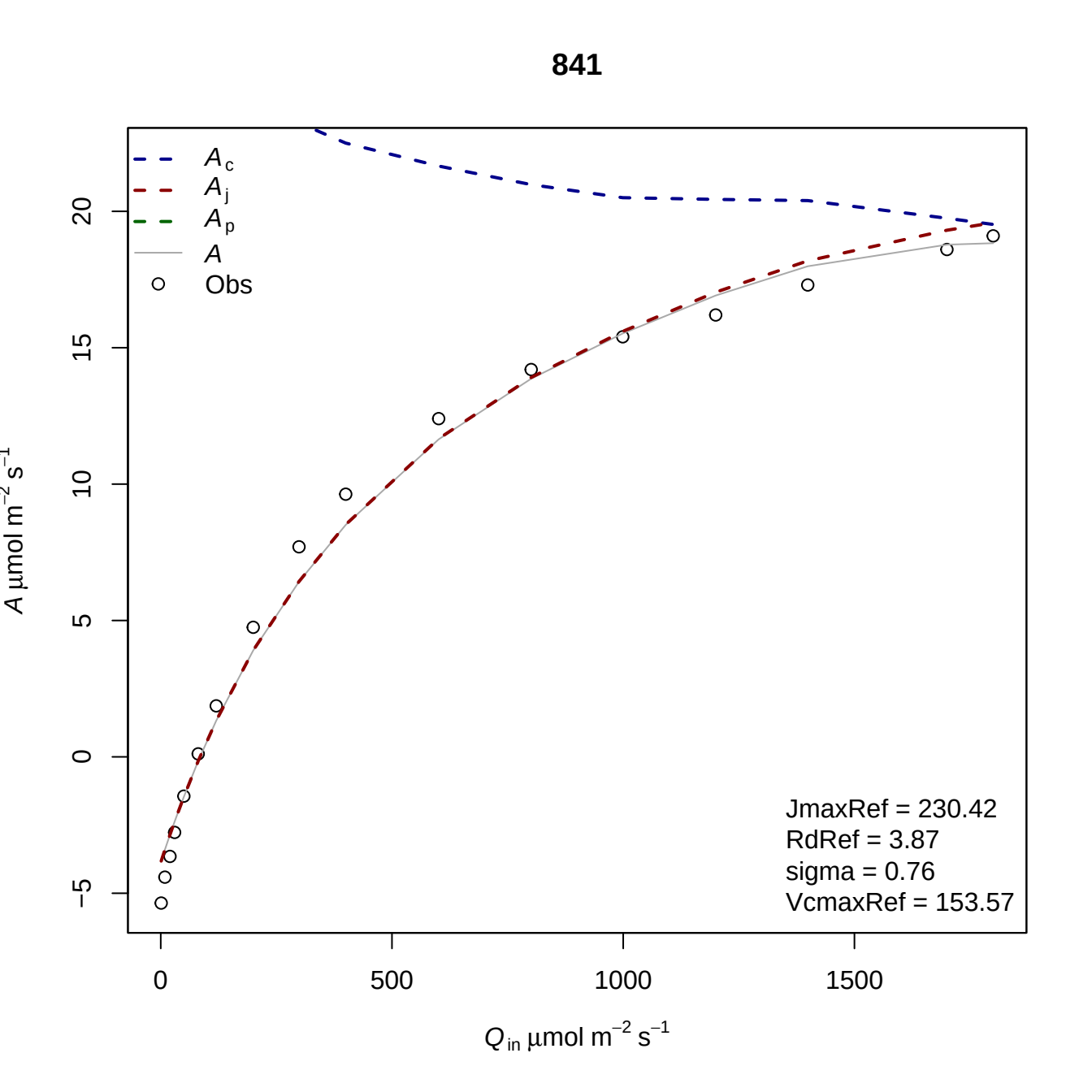

836
A_c
A_j
A_p
A
Obs
JmaxRef = 25.74
RdRef = 1.11
sigma = 0.09
VcmaxRef = 9590.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

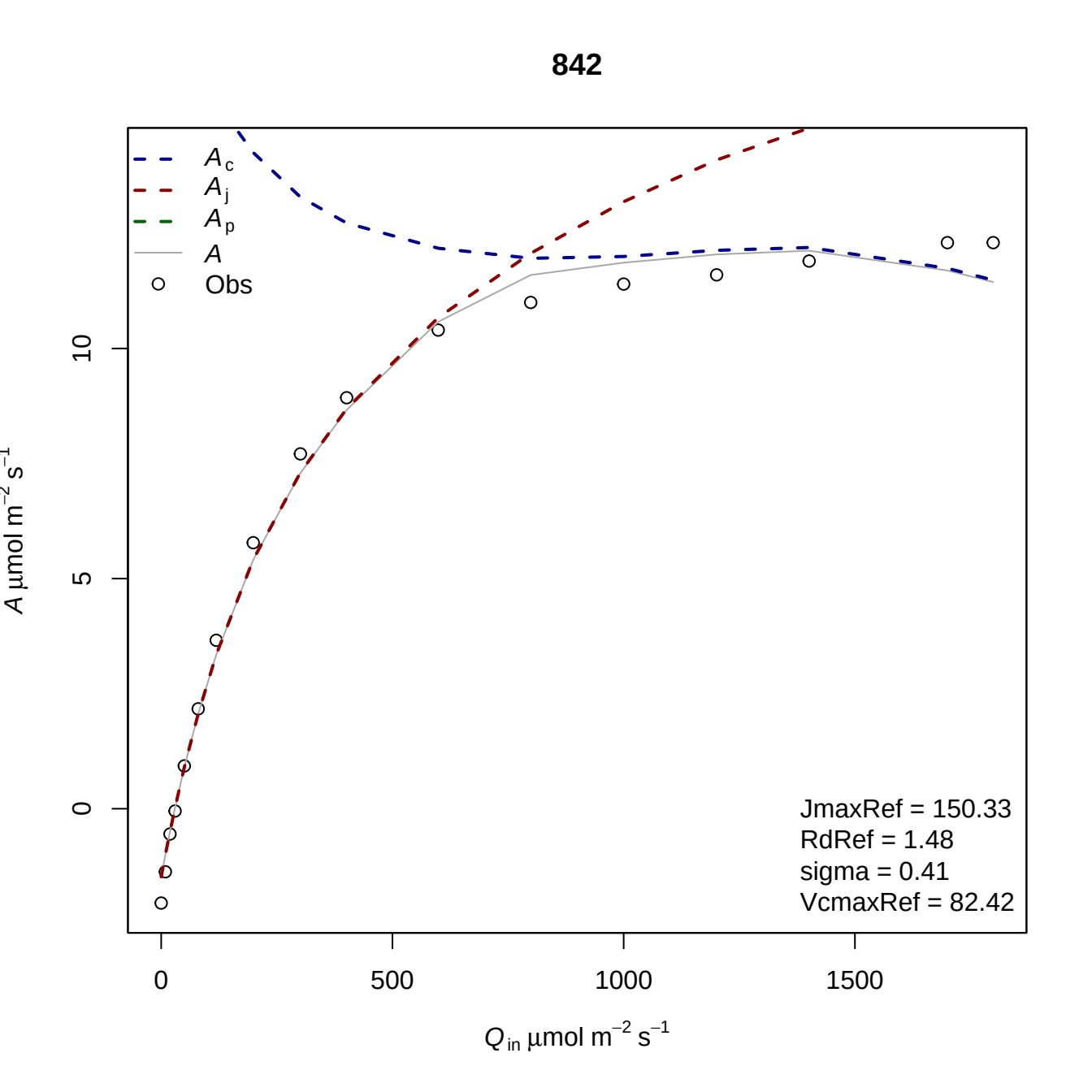

837
A_c
A_j
A_p
A
Obs
JmaxRef = 210.66
RdRef = 2.3
sigma = 0.38
VcmaxRef = 120.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

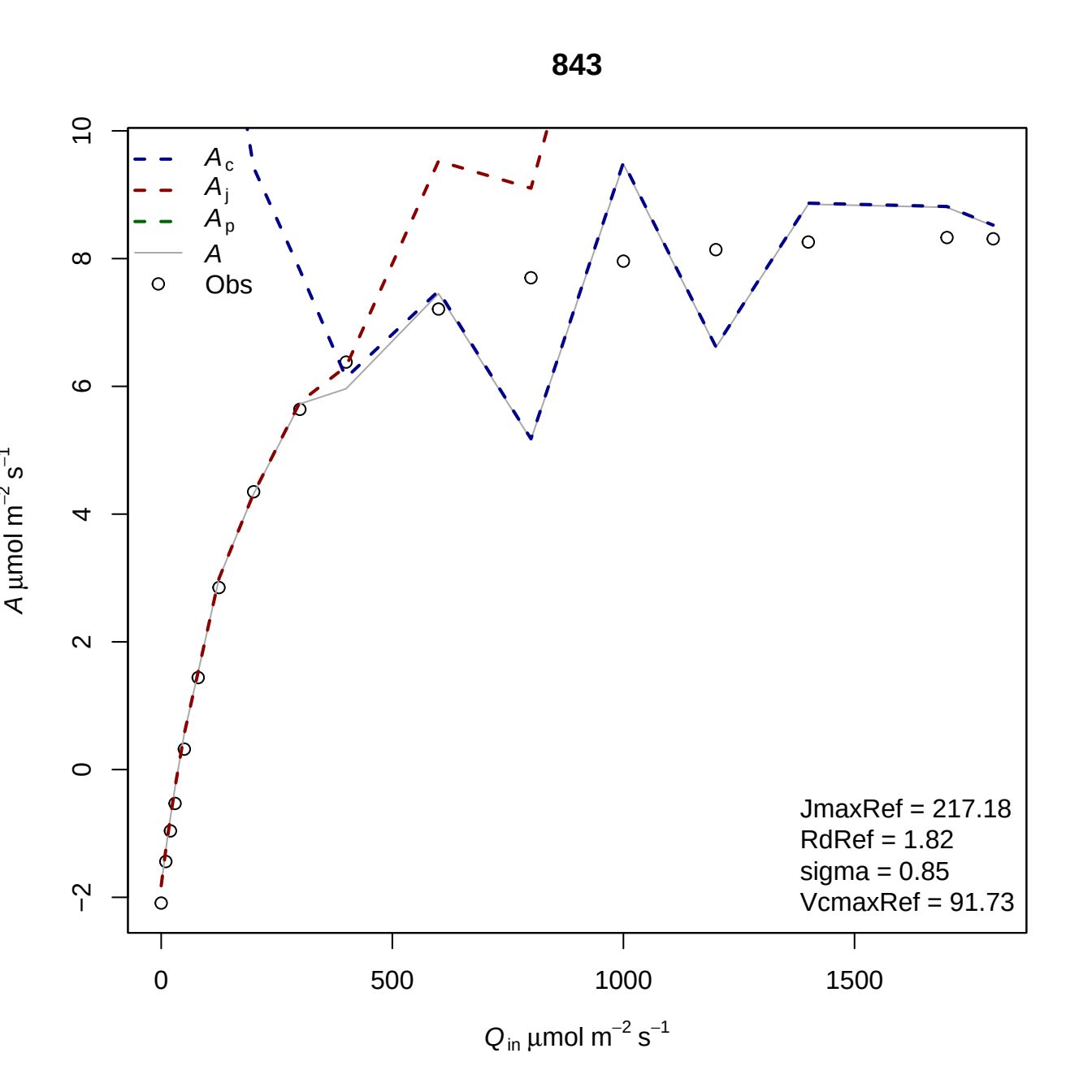

838
A_c
A_j
A_p
A
Obs
JmaxRef = 165.12
RdRef = 2.07
sigma = 0.35
VcmaxRef = 118
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

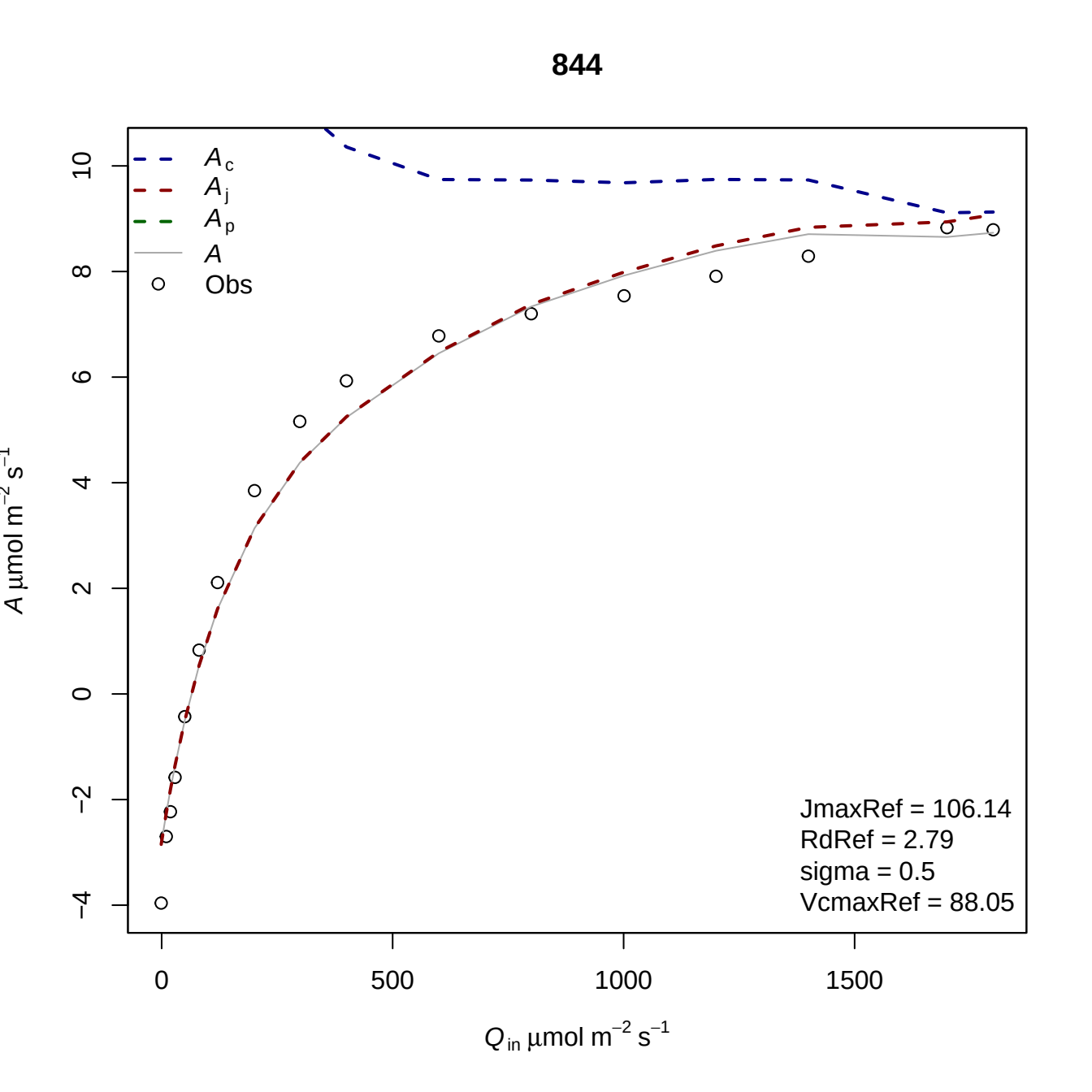

839
A_c
A_j
A_p
A
Obs
JmaxRef = 48.04
RdRef = 1.39
sigma = 0.32
VcmaxRef = 33524.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800
1000
1200
1400

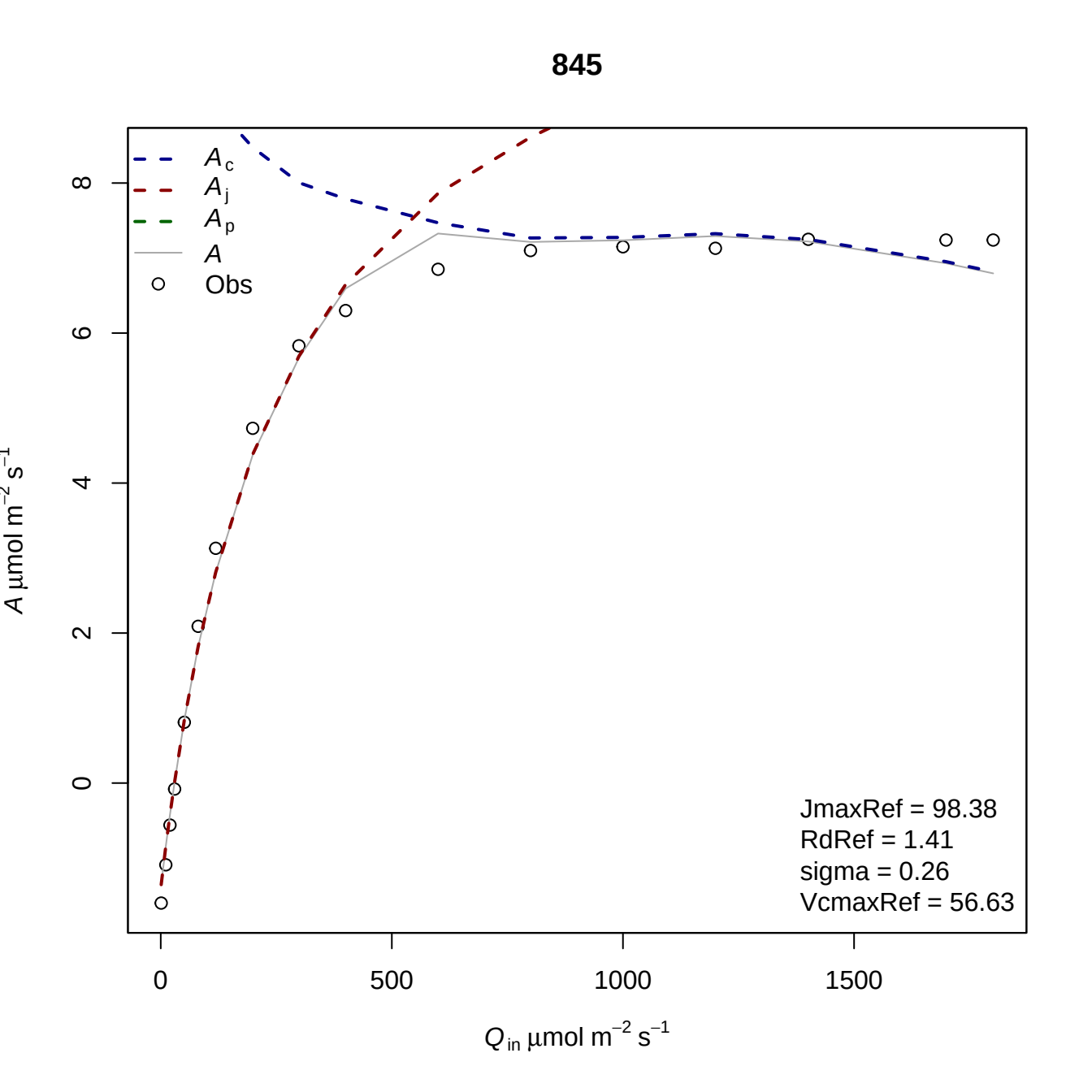

840
A_c
A_j
A_p
A
Obs
JmaxRef = 126
RdRef = 5.35
sigma = 0.57
VcmaxRef = 145751.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500

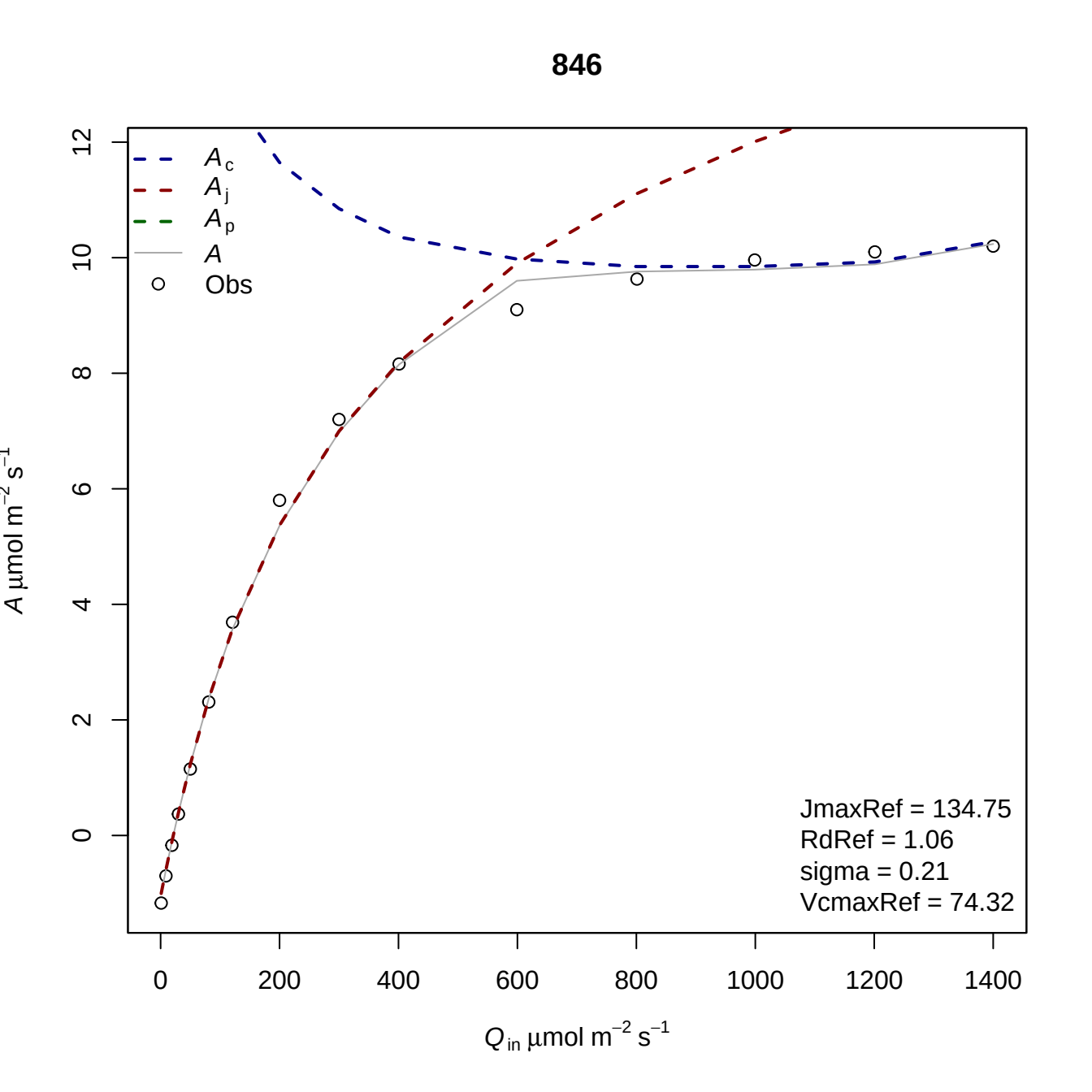

841
A_c
A_j
A_p
A
Obs
JmaxRef = 230.42
RdRef = 3.87
sigma = 0.76
VcmaxRef = 153.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500

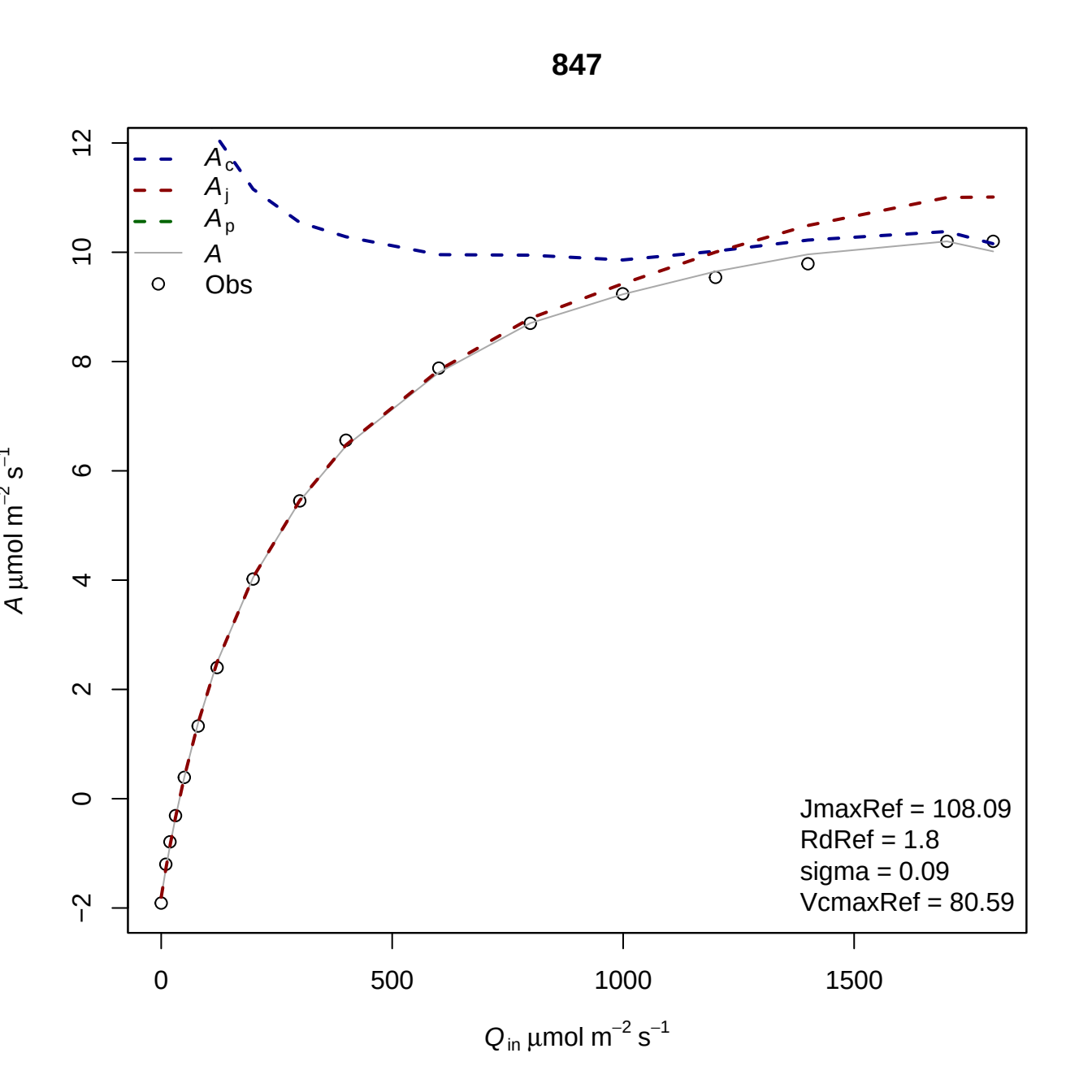

842
A_c
A_j
A_p
A
Obs
JmaxRef = 150.33
RdRef = 1.48
sigma = 0.41
VcmaxRef = 82.42
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

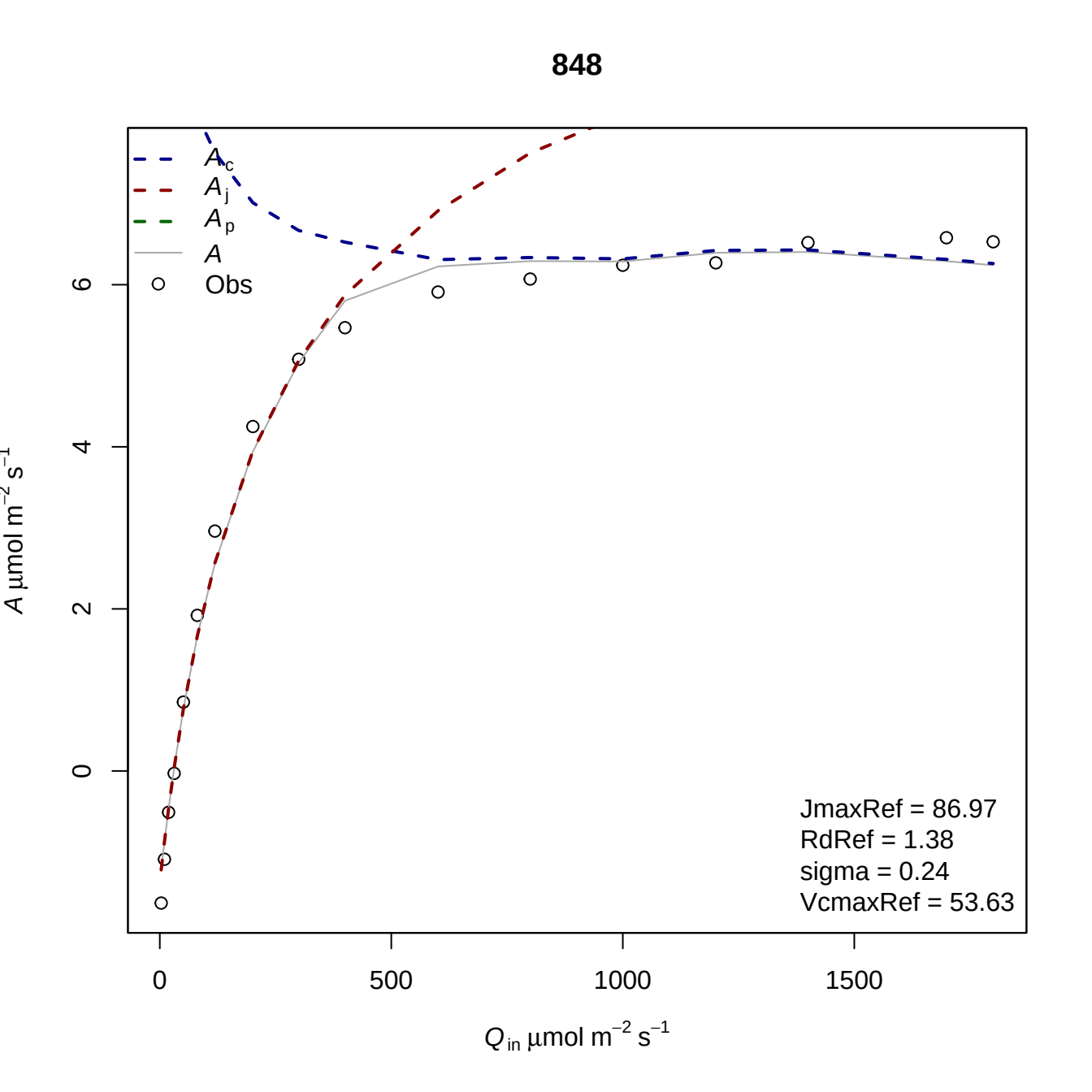

843
A_c
A_j
A_p
A
Obs
JmaxRef = 217.18
RdRef = 1.82
sigma = 0.85
VcmaxRef = 91.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

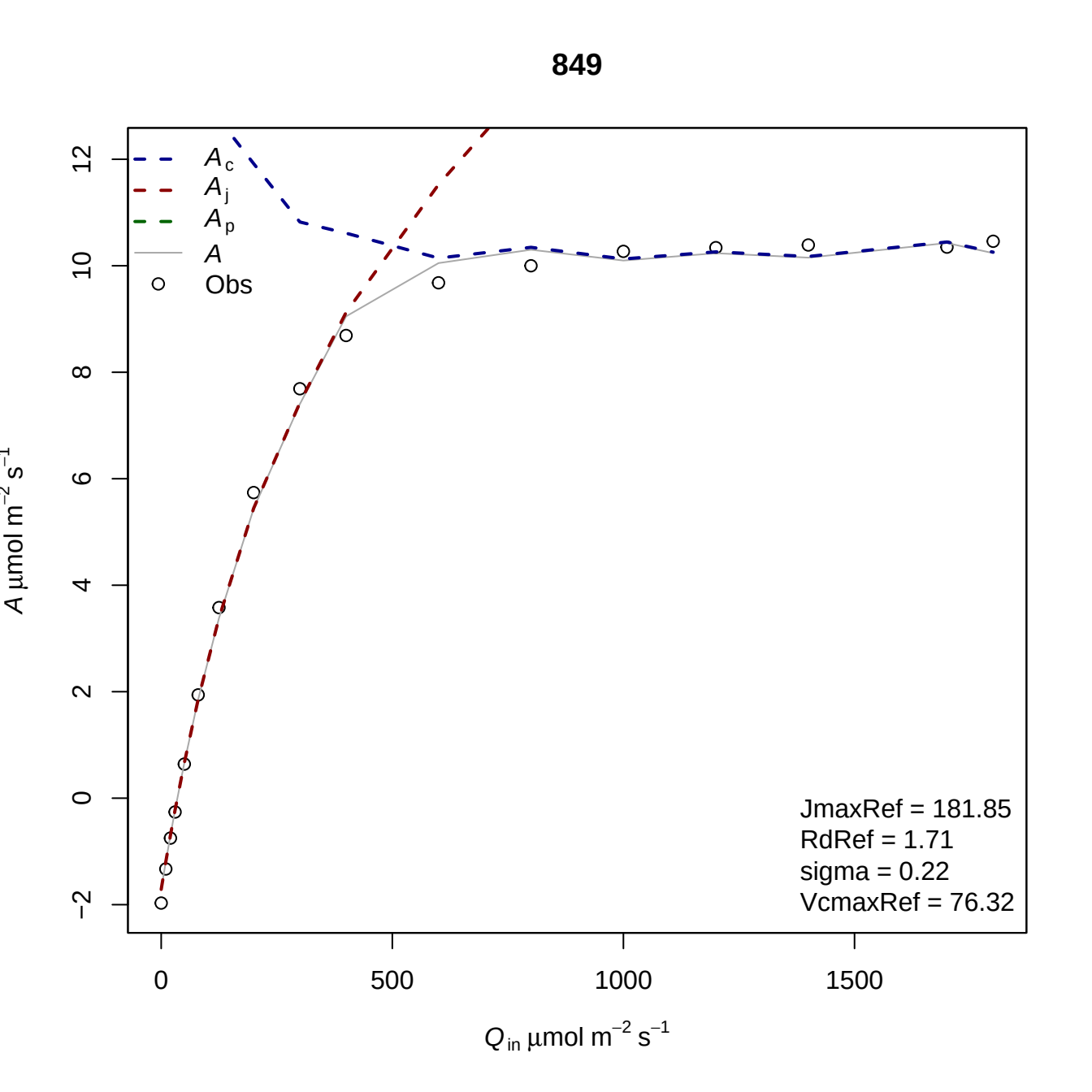

844
A_c
A_j
A_p
A
Obs
JmaxRef = 106.14
RdRef = 2.79
sigma = 0.5
VcmaxRef = 88.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500

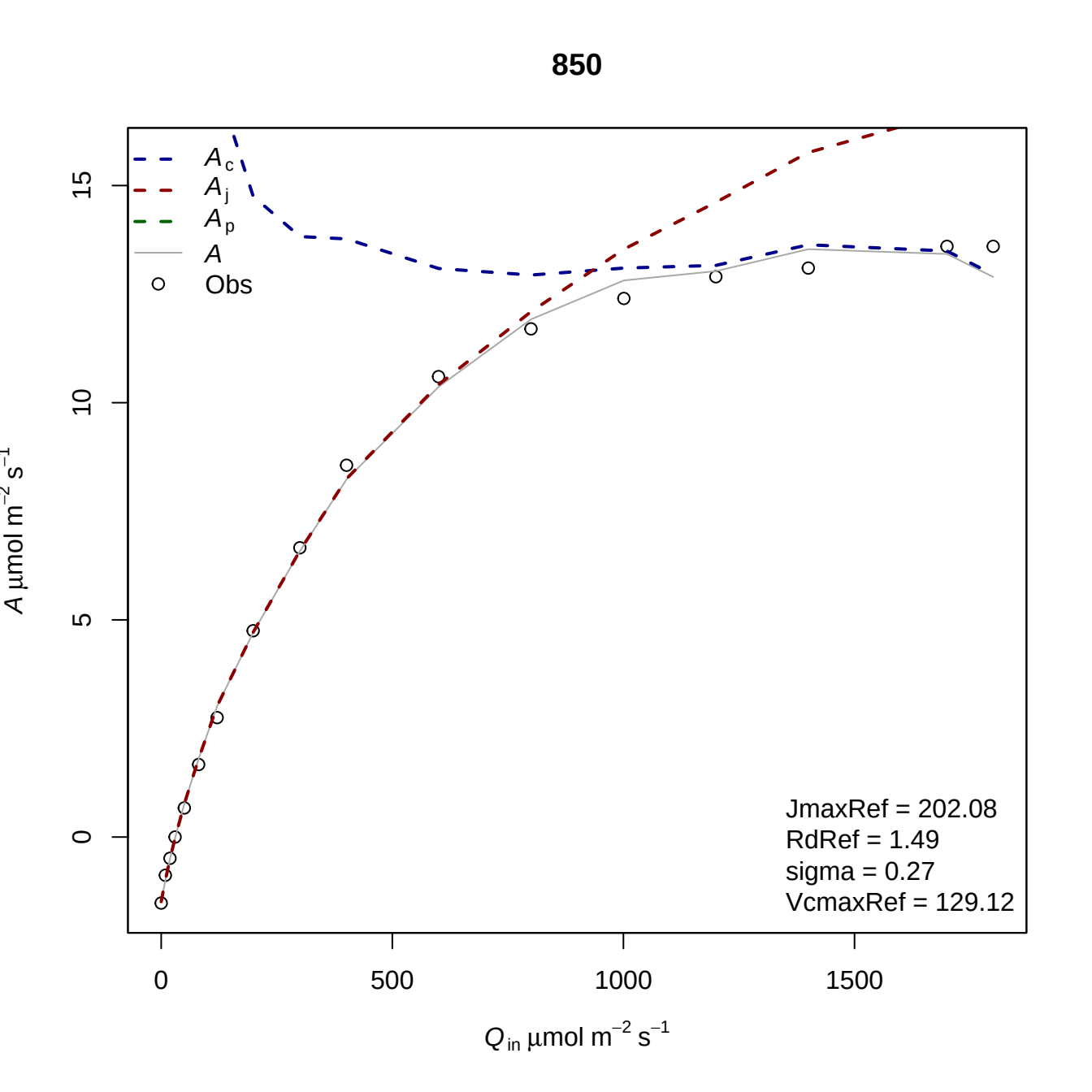

845
A_c
A_j
A_p
A
Obs
JmaxRef = 98.38
RdRef = 1.41
sigma = 0.26
VcmaxRef = 56.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

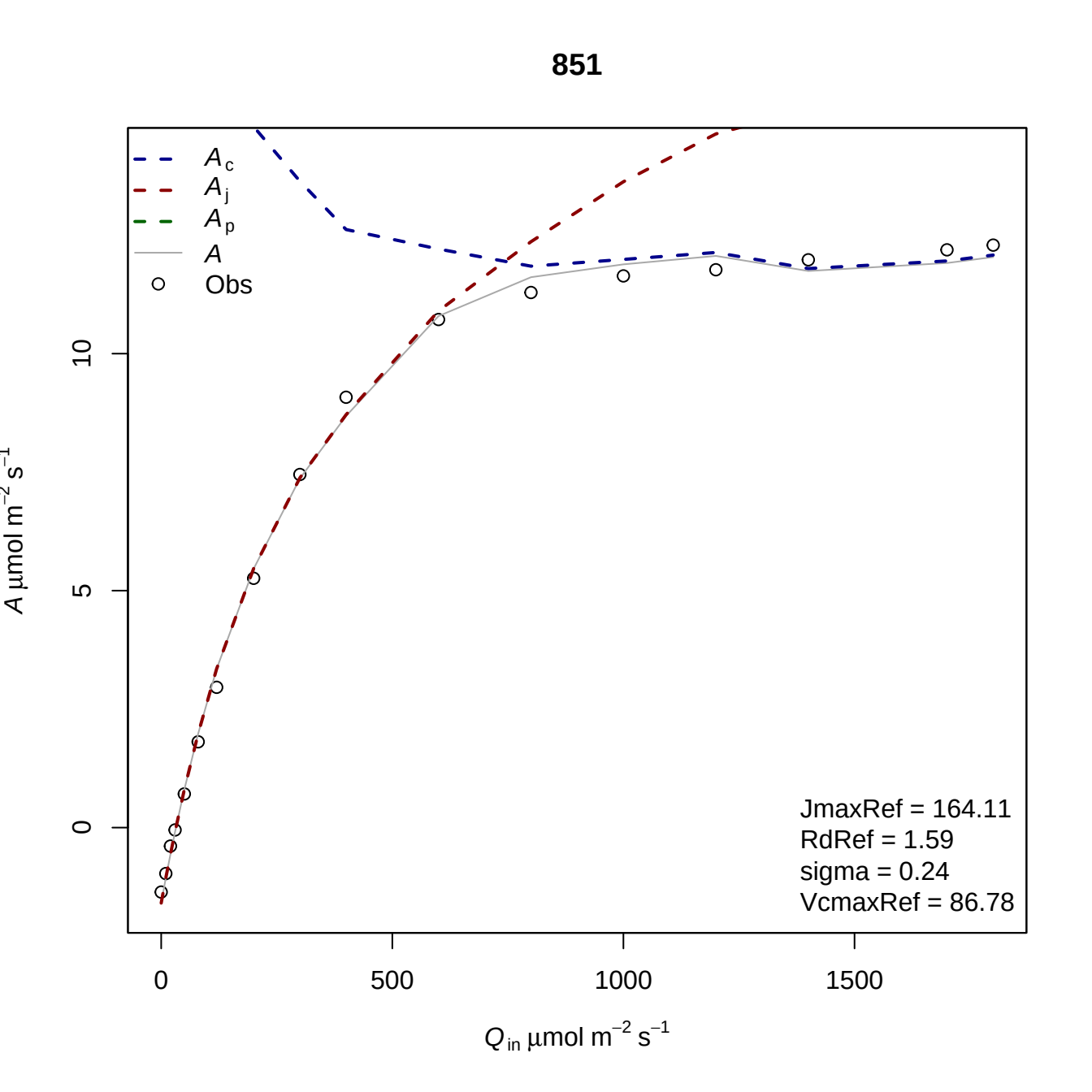

846
A_c
A_j
A_p
A
Obs
JmaxRef = 134.75
RdRef = 1.06
sigma = 0.21
VcmaxRef = 74.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

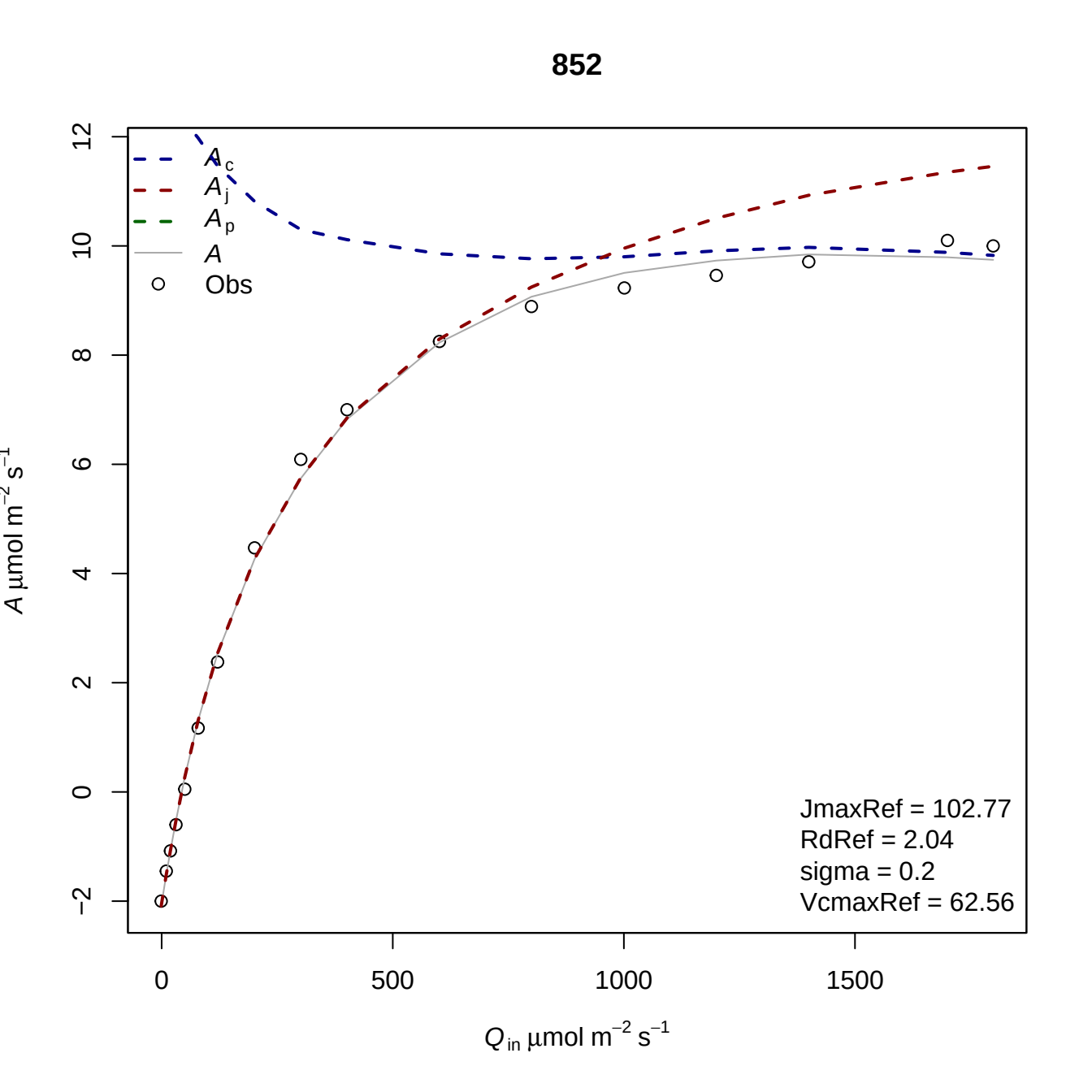

847
A_c
A_j
A_p
A
Obs
JmaxRef = 108.09
RdRef = 1.8
sigma = 0.09
VcmaxRef = 80.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

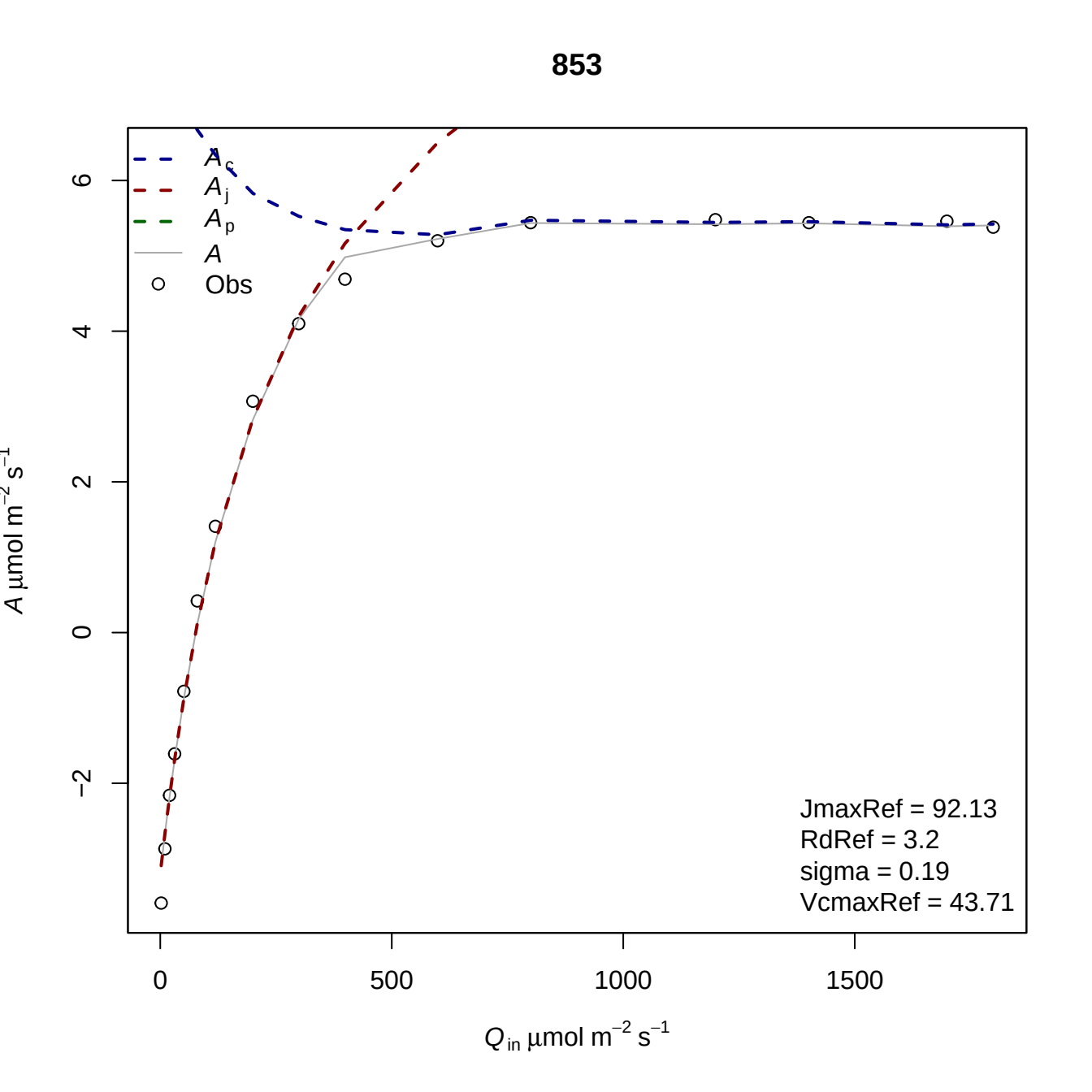

848
A_c
A_j
A_p
A
Obs
JmaxRef = 86.97
RdRef = 1.38
sigma = 0.24
VcmaxRef = 53.63
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

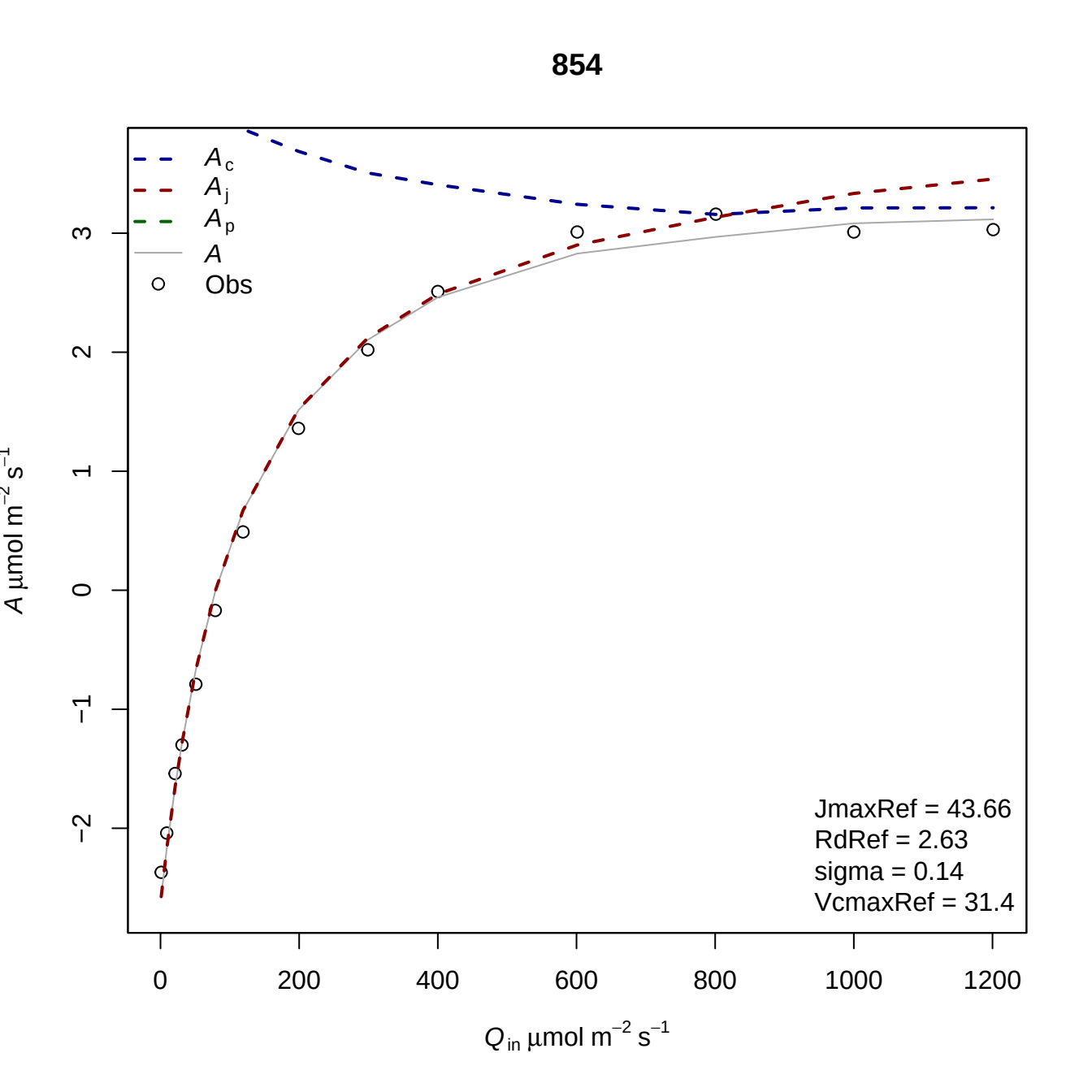

849
A_c
A_j
A_p
A
Obs
JmaxRef = 181.85
RdRef = 1.71
sigma = 0.22
VcmaxRef = 76.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

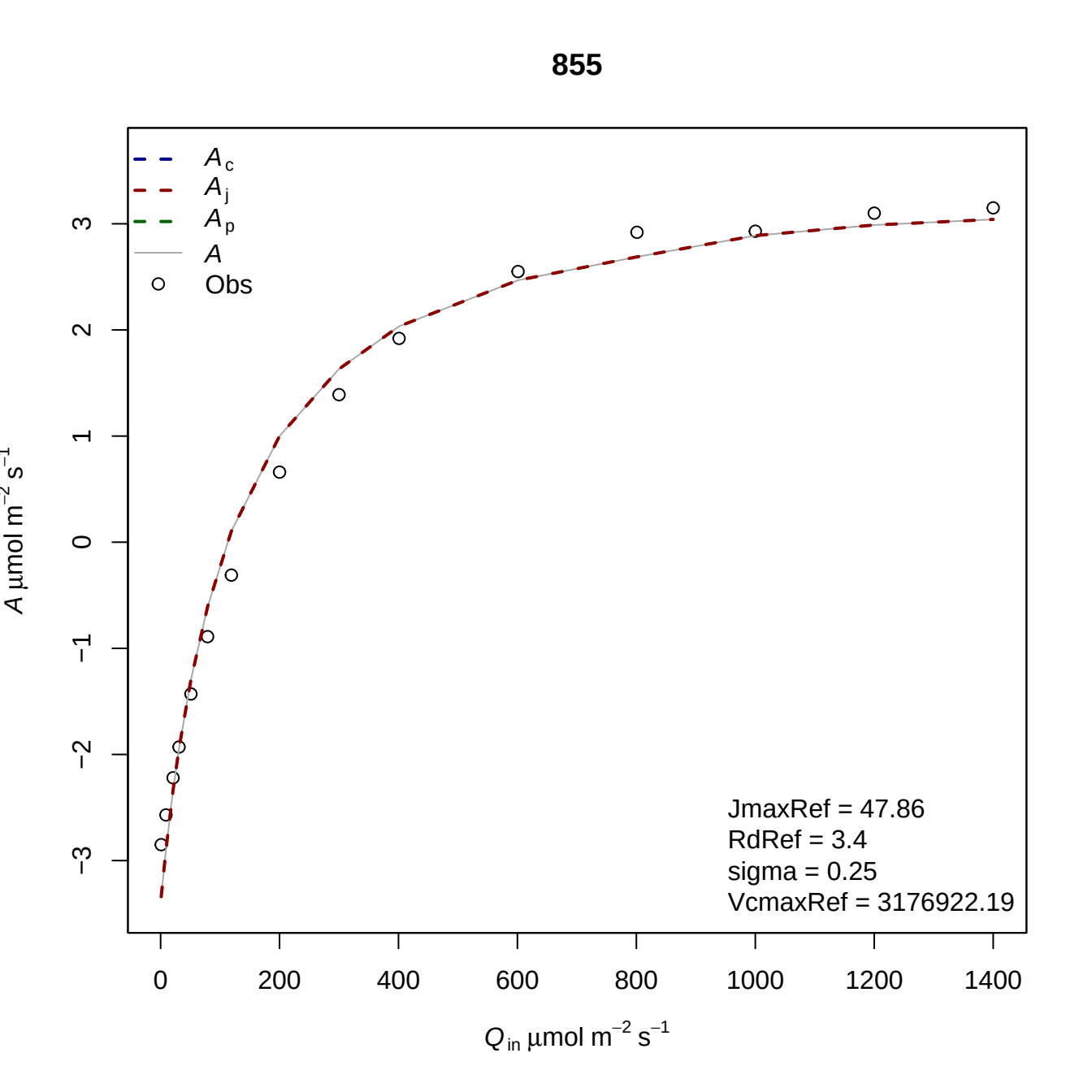

850
A_c
A_j
A_p
A
Obs
JmaxRef = 202.08
RdRef = 1.49
sigma = 0.27
VcmaxRef = 129.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

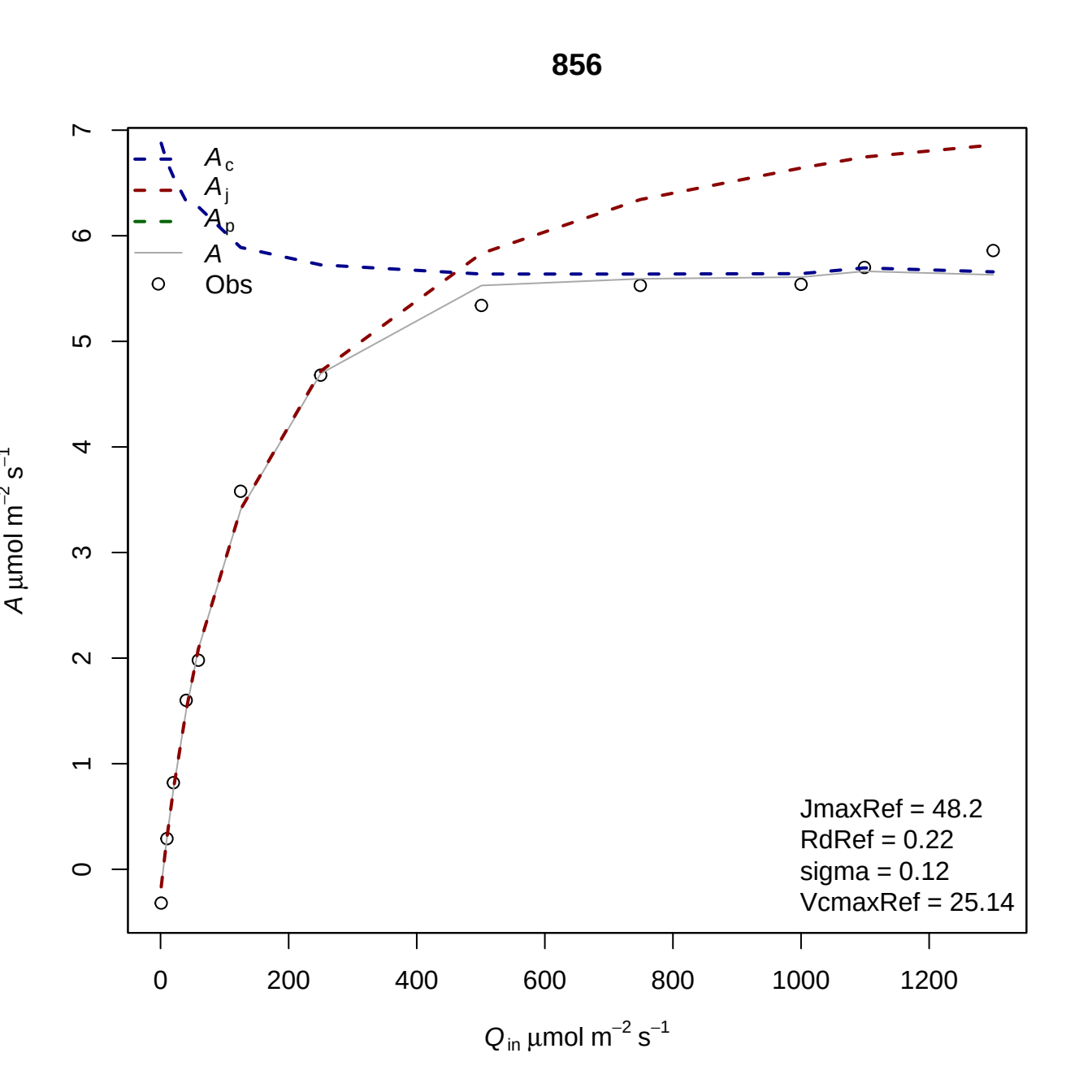

851
A_c
A_j
A_p
A
Obs
JmaxRef = 164.11
RdRef = 1.59
sigma = 0.24
VcmaxRef = 86.78
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

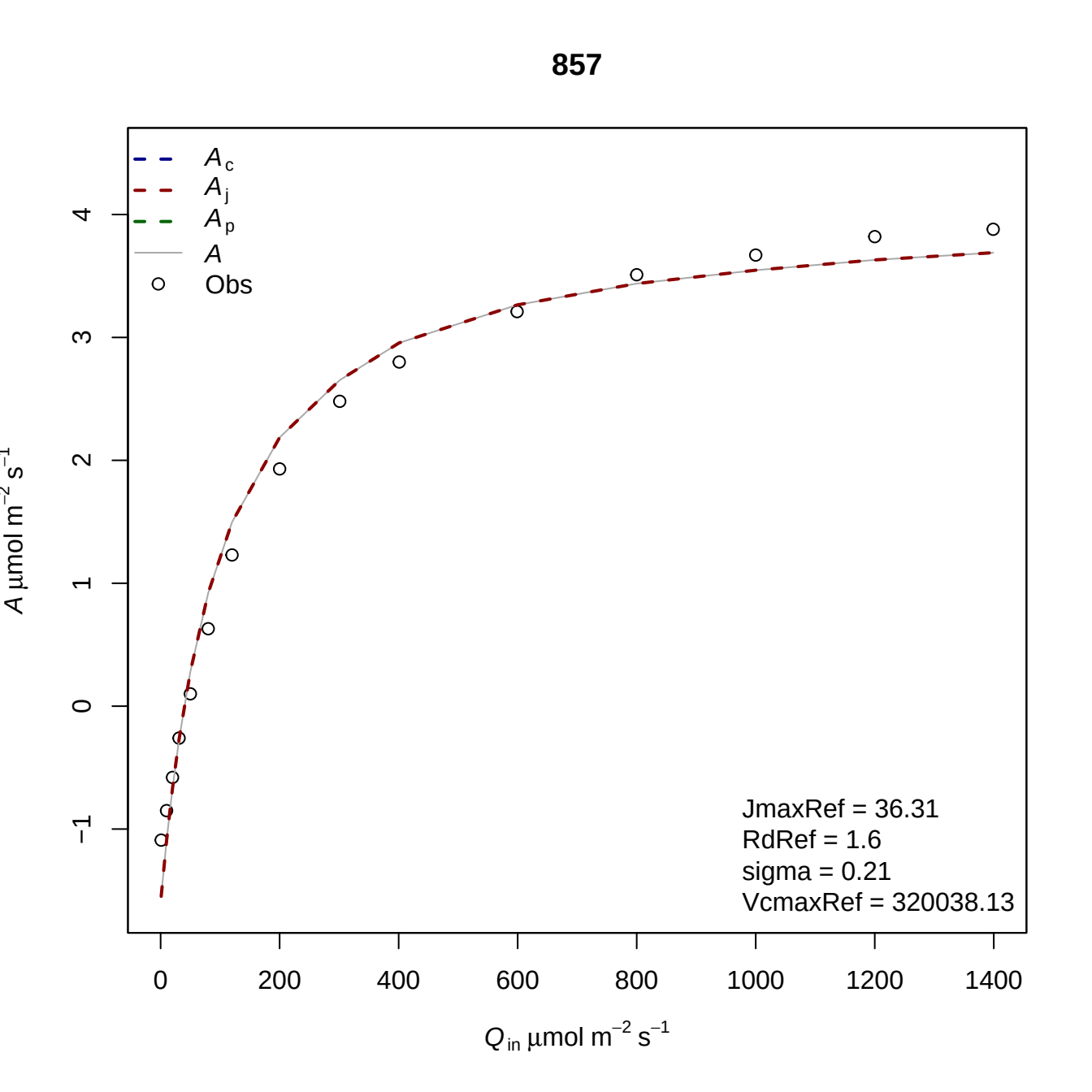

852
A_c
A_j
A_p
A
Obs
JmaxRef = 102.77
RdRef = 2.04
sigma = 0.2
VcmaxRef = 62.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

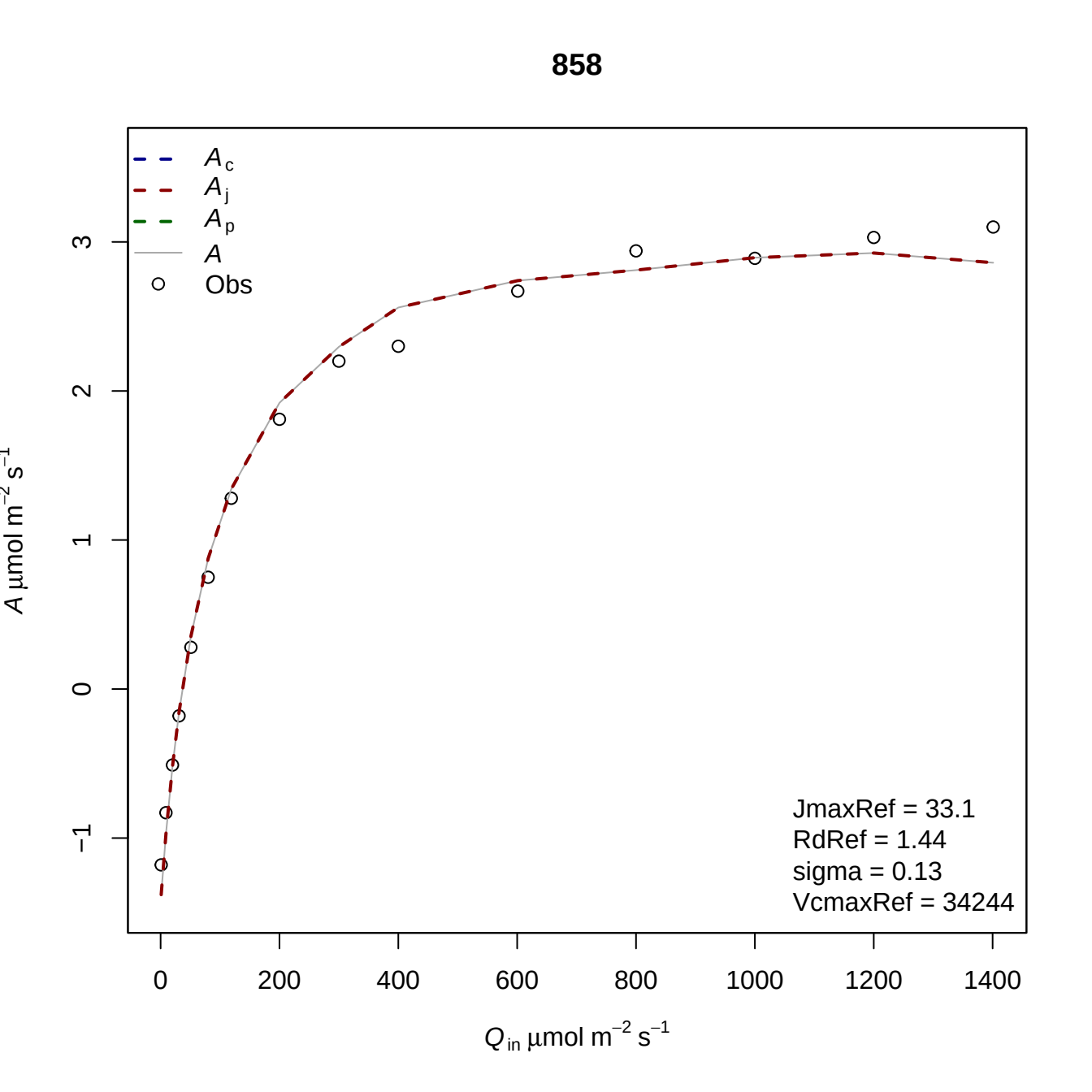

853
A_c
A_j
A_p
A
Obs
JmaxRef = 92.13
RdRef = 3.2
sigma = 0.19
VcmaxRef = 43.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

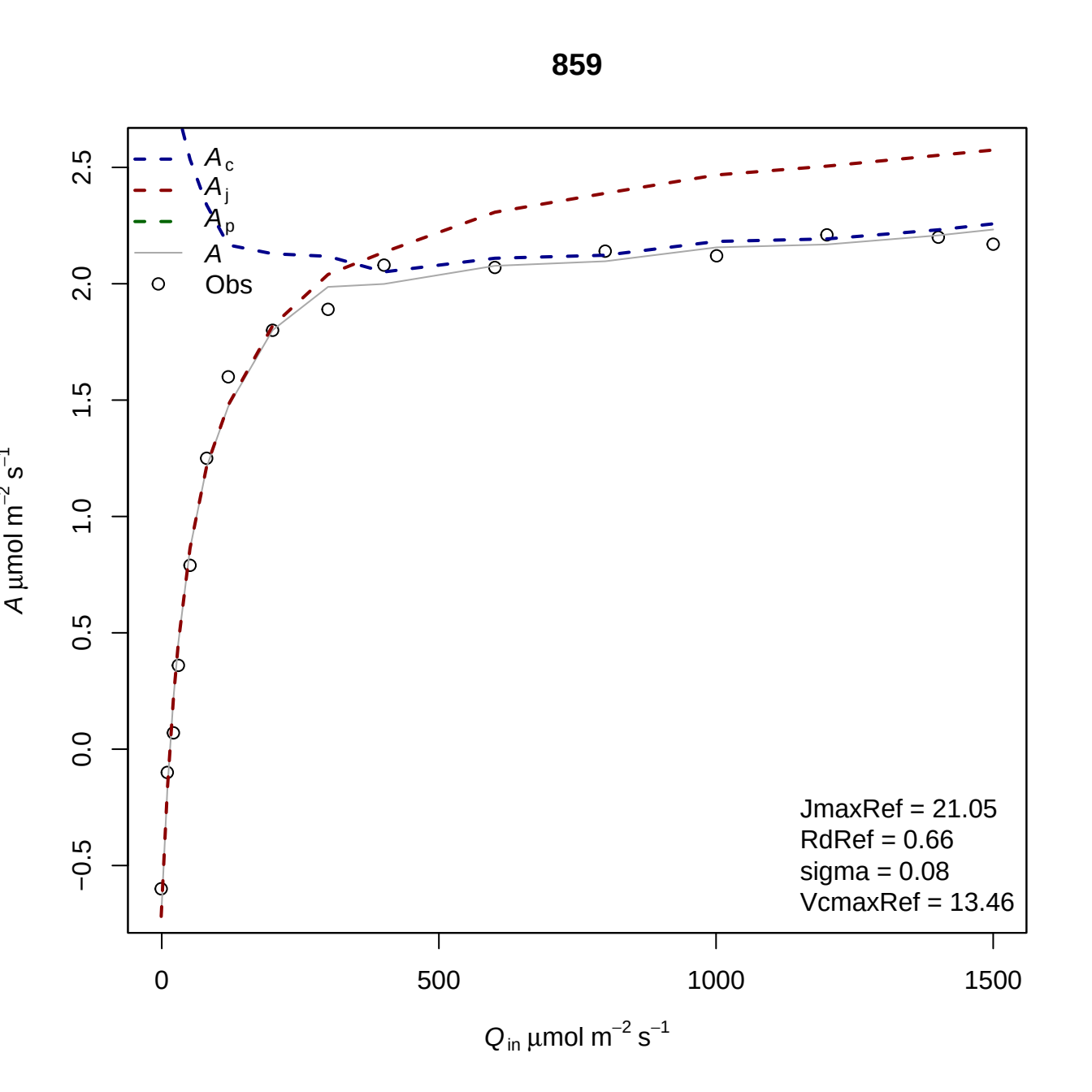

854
A_c
A_j
A_p
A
Obs
JmaxRef = 43.66
RdRef = 2.63
sigma = 0.14
VcmaxRef = 31.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
0
200
400
600
800
1000
1200

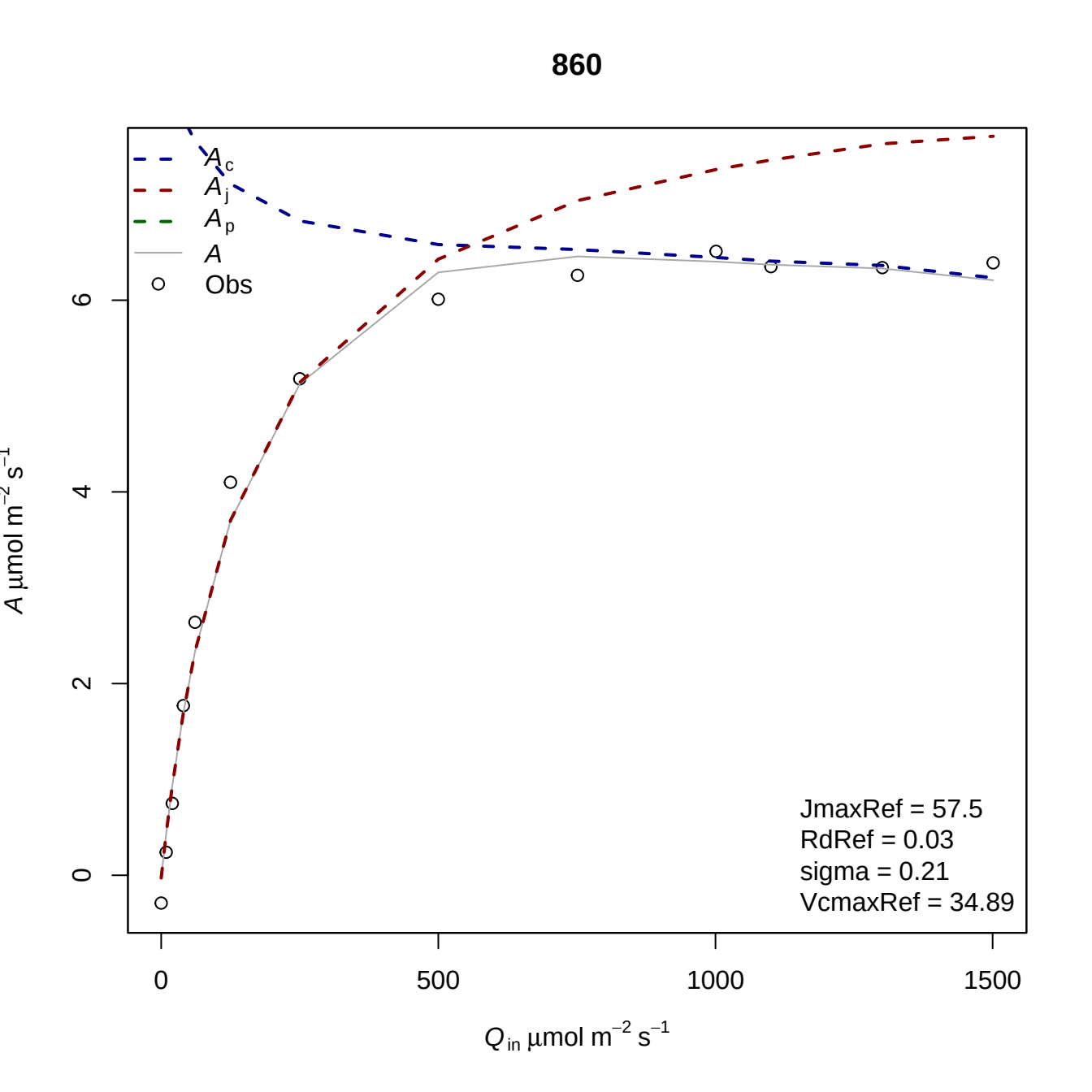

855
A_c
A_j
A_p
A
Obs
JmaxRef = 47.86
RdRef = 3.4
sigma = 0.25
VcmaxRef = 3176922.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−3
−2
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

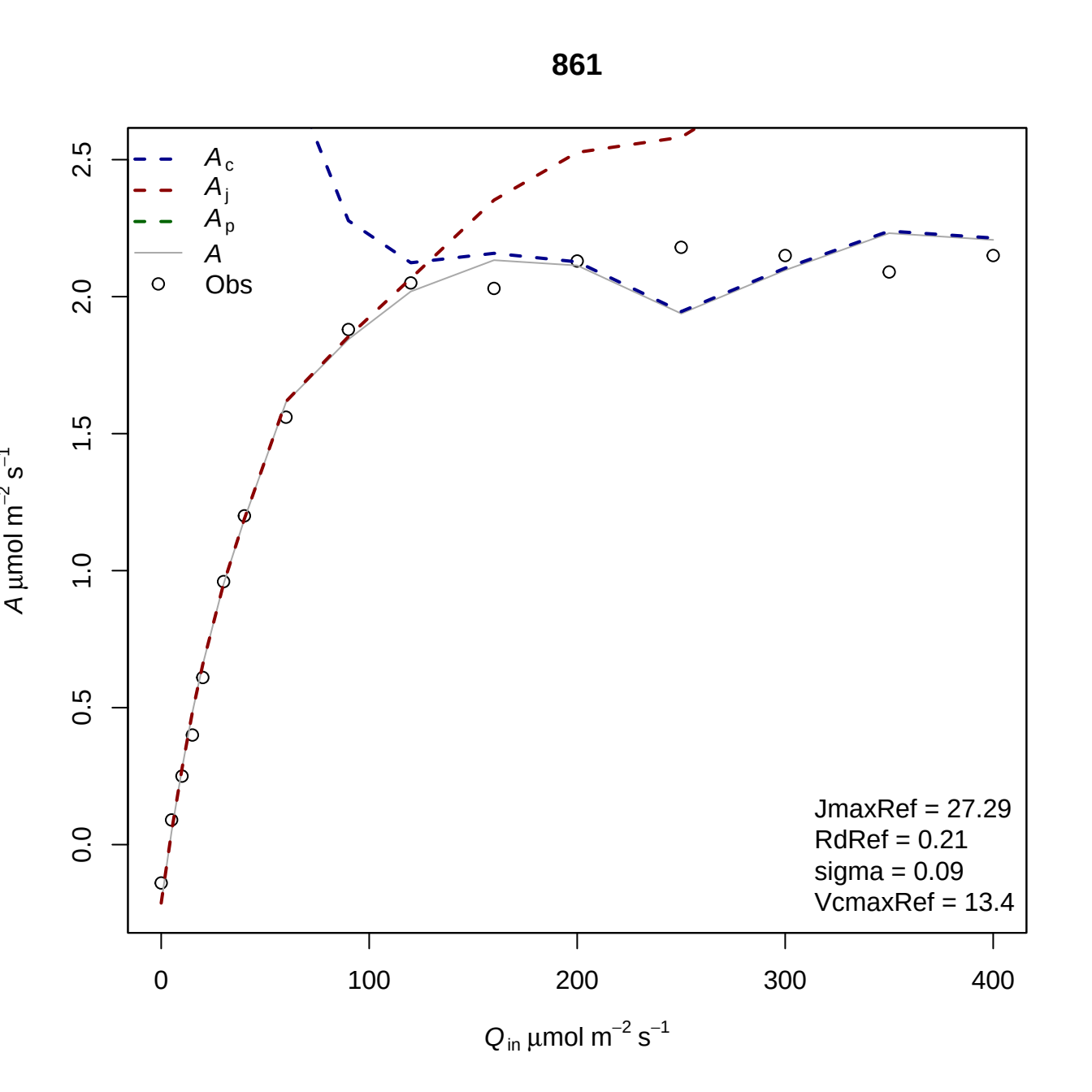

856
A_c
A_j
A_p
A
Obs
JmaxRef = 48.2
RdRef = 0.22
sigma = 0.12
VcmaxRef = 25.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
7
0
200
400
600
800
1000
1200

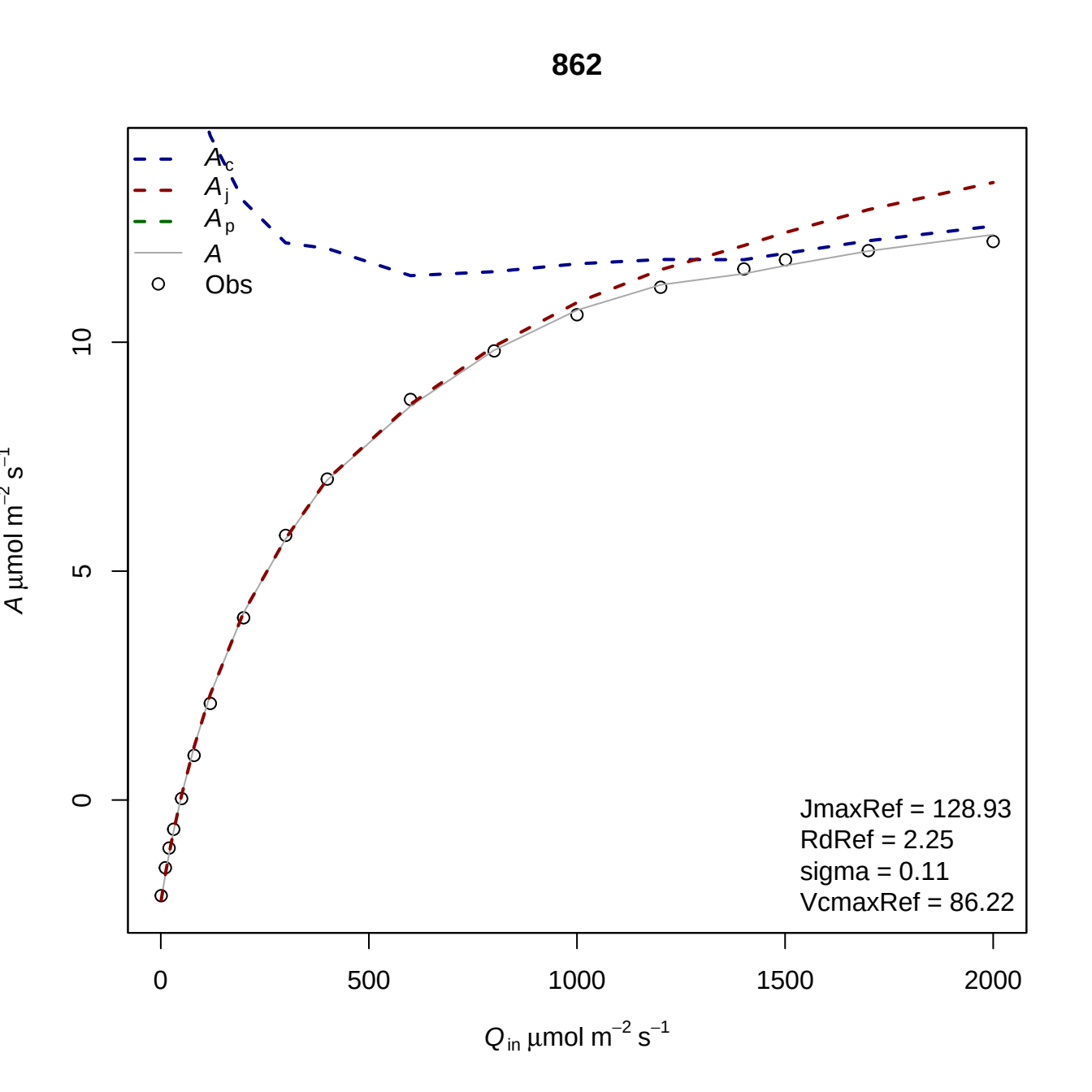

857
A_c
A_j
A_p
A
Obs
JmaxRef = 36.31
RdRef = 1.6
sigma = 0.21
VcmaxRef = 320038.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000
1200
1400

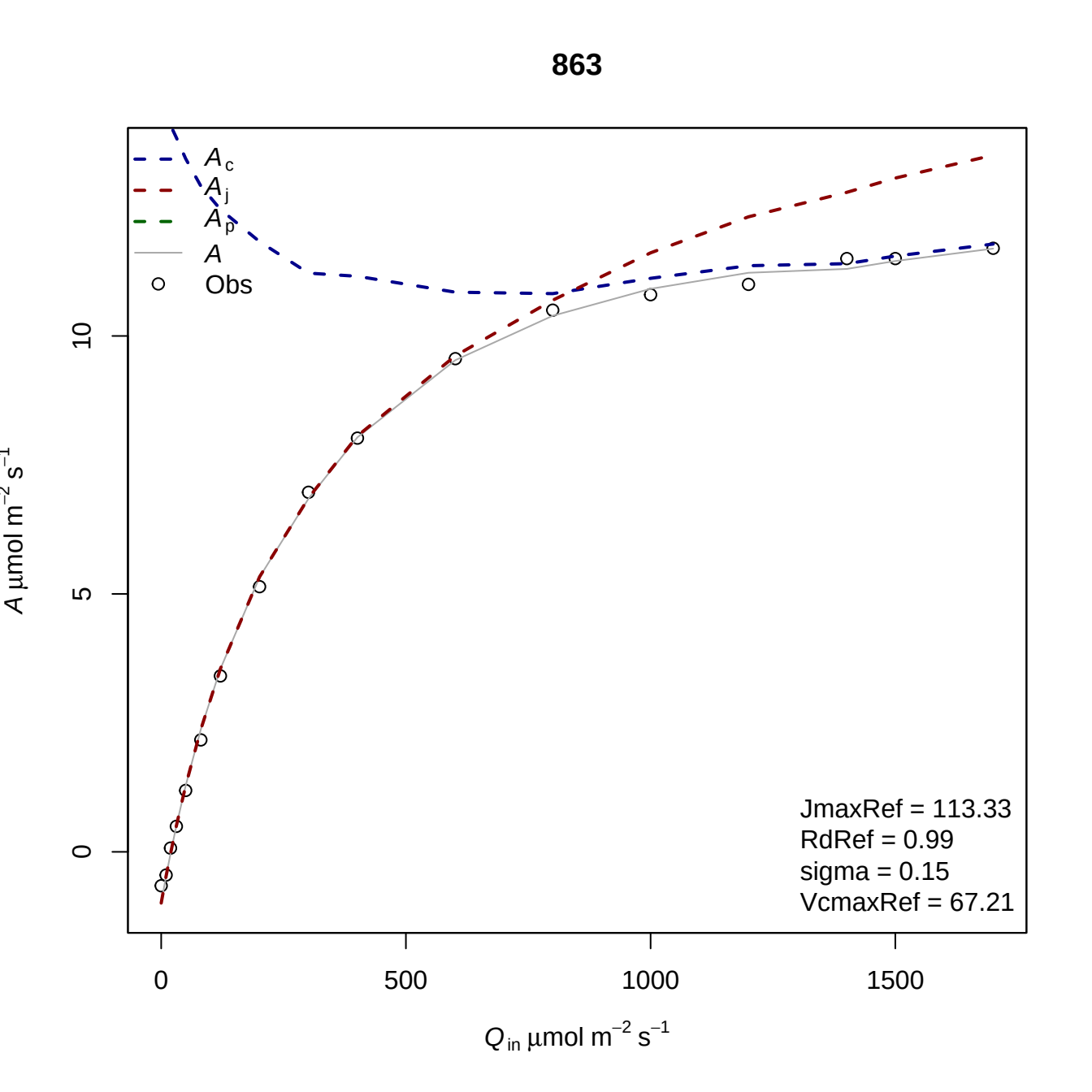

858
A_c
A_j
A_p
A
Obs
JmaxRef = 33.1
RdRef = 1.44
sigma = 0.13
VcmaxRef = 34244
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

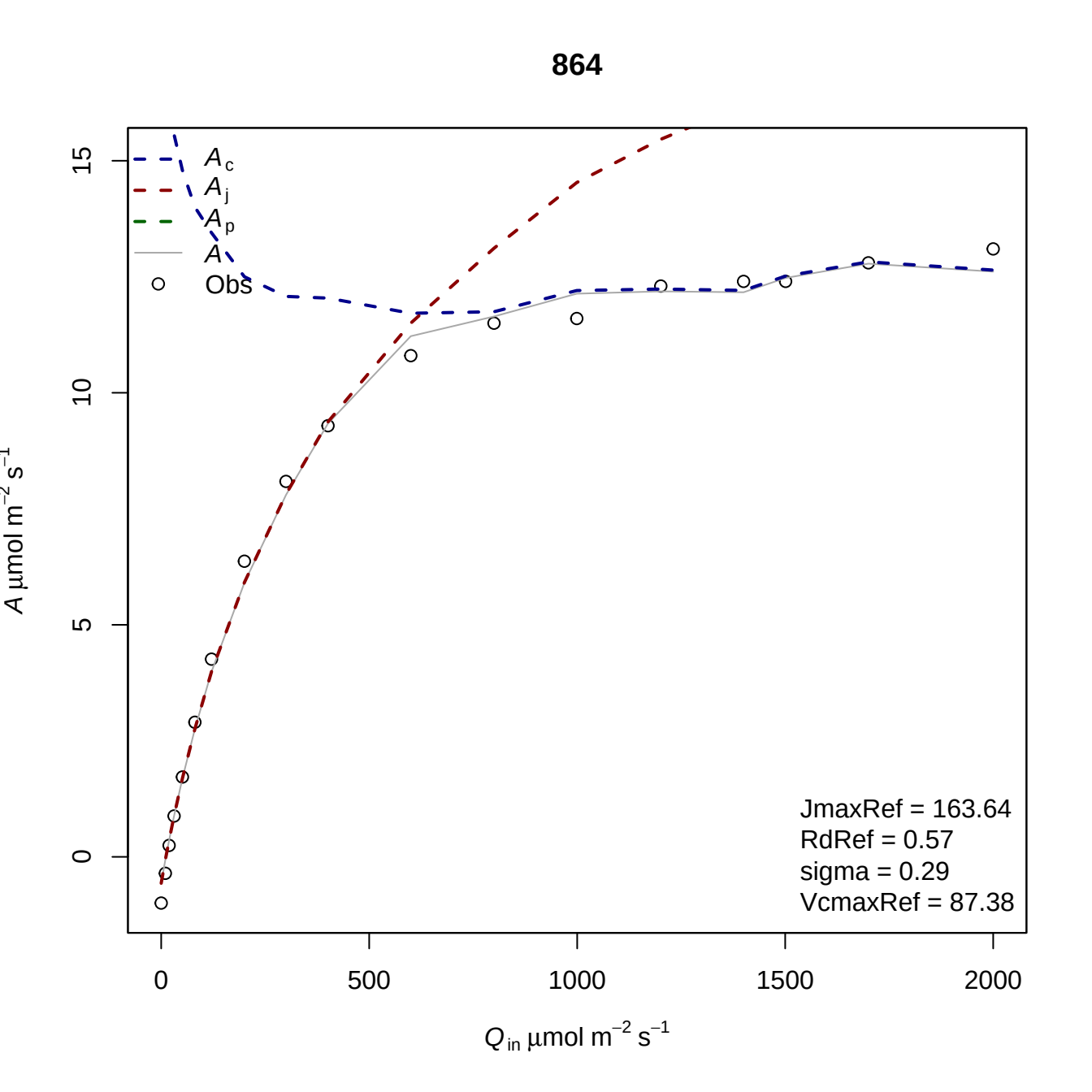

859
A_c
A_j
A_p
A
Obs
JmaxRef = 21.05
RdRef = 0.66
sigma = 0.08
VcmaxRef = 13.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500

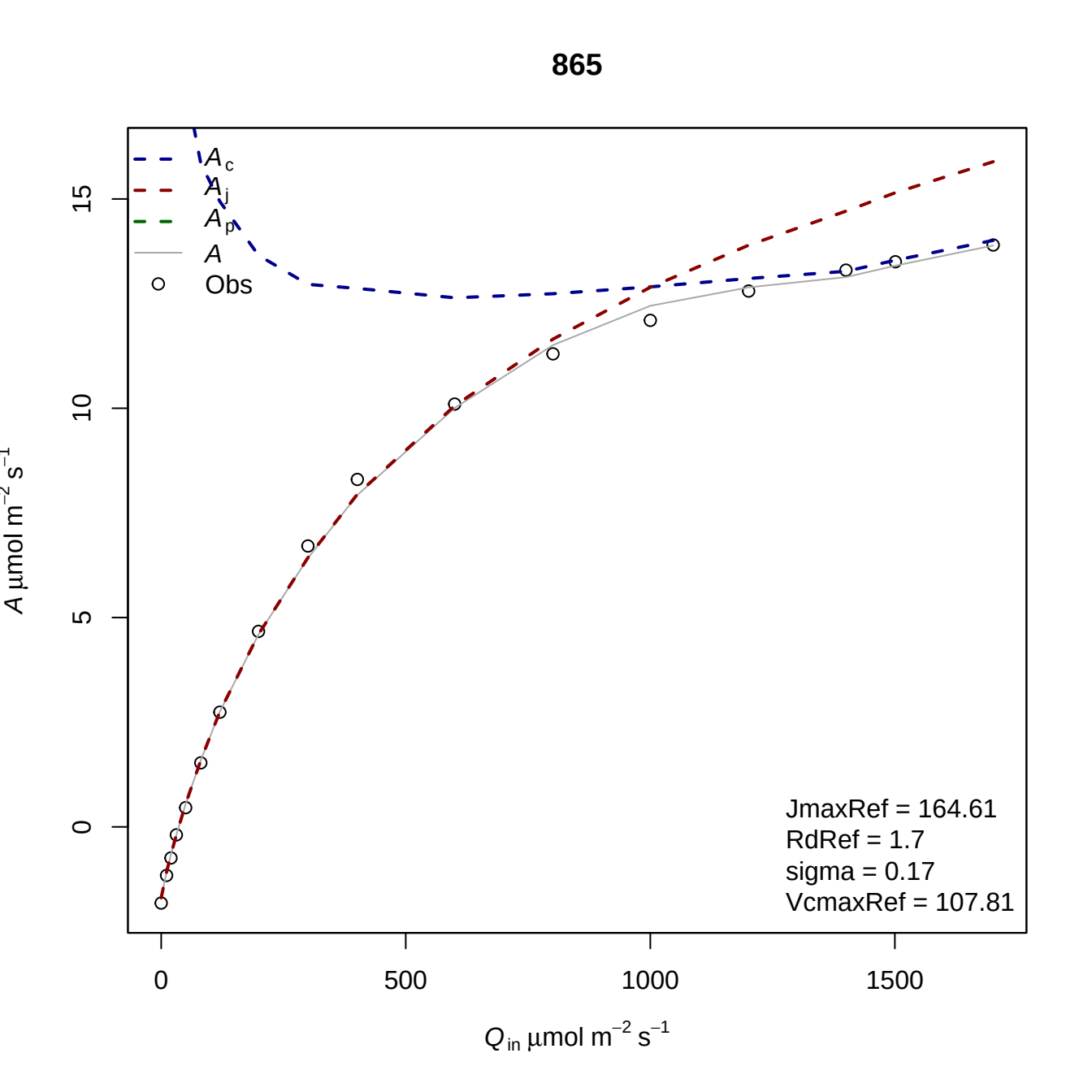

860
A_c
A_j
A_p
A
Obs
JmaxRef = 57.5
RdRef = 0.03
sigma = 0.21
VcmaxRef = 34.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

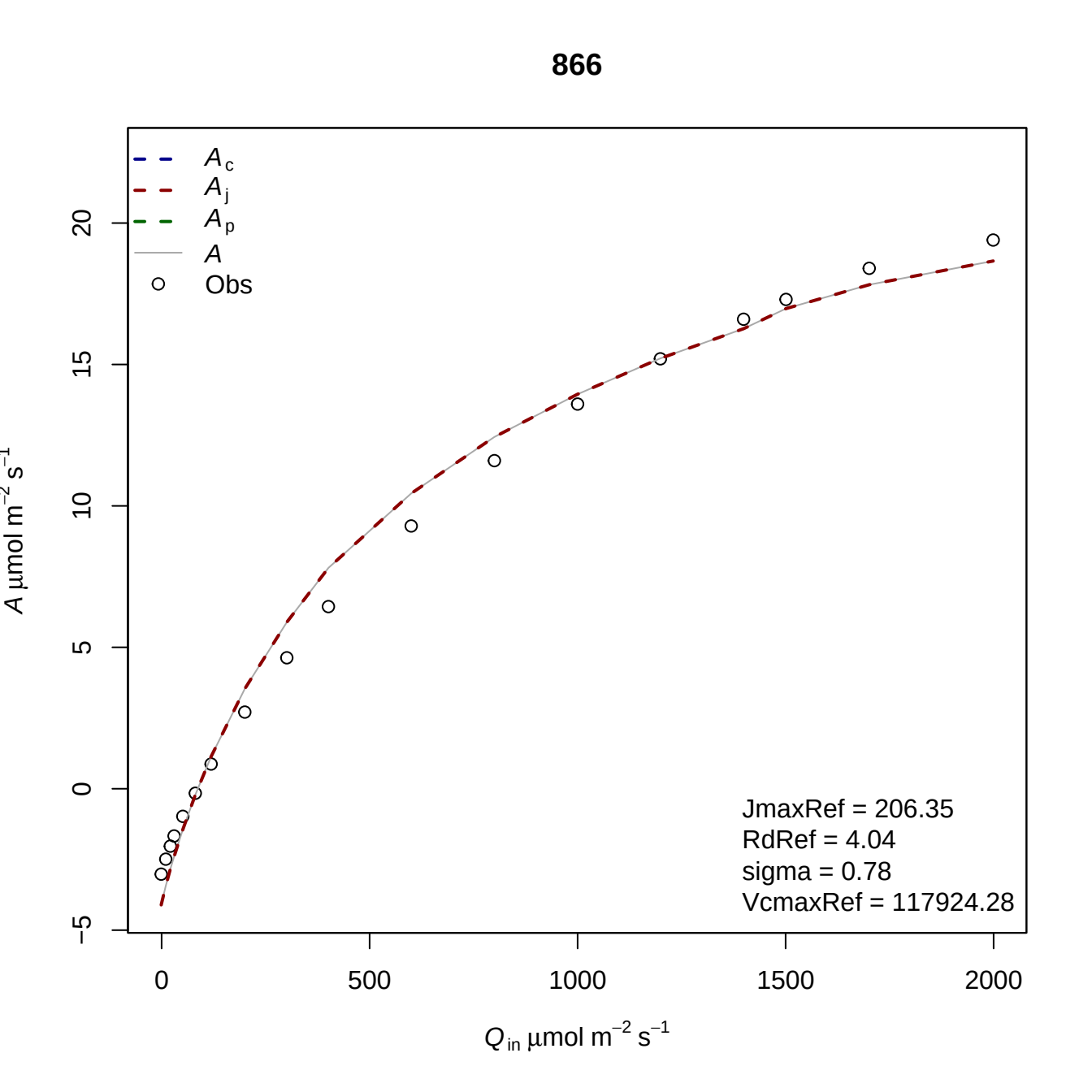

861
A_c
A_j
A_p
A
Obs
JmaxRef = 27.29
RdRef = 0.21
sigma = 0.09
VcmaxRef = 13.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0.0
0.5
1.0
1.5
2.0
2.5
0
100
200
300
400

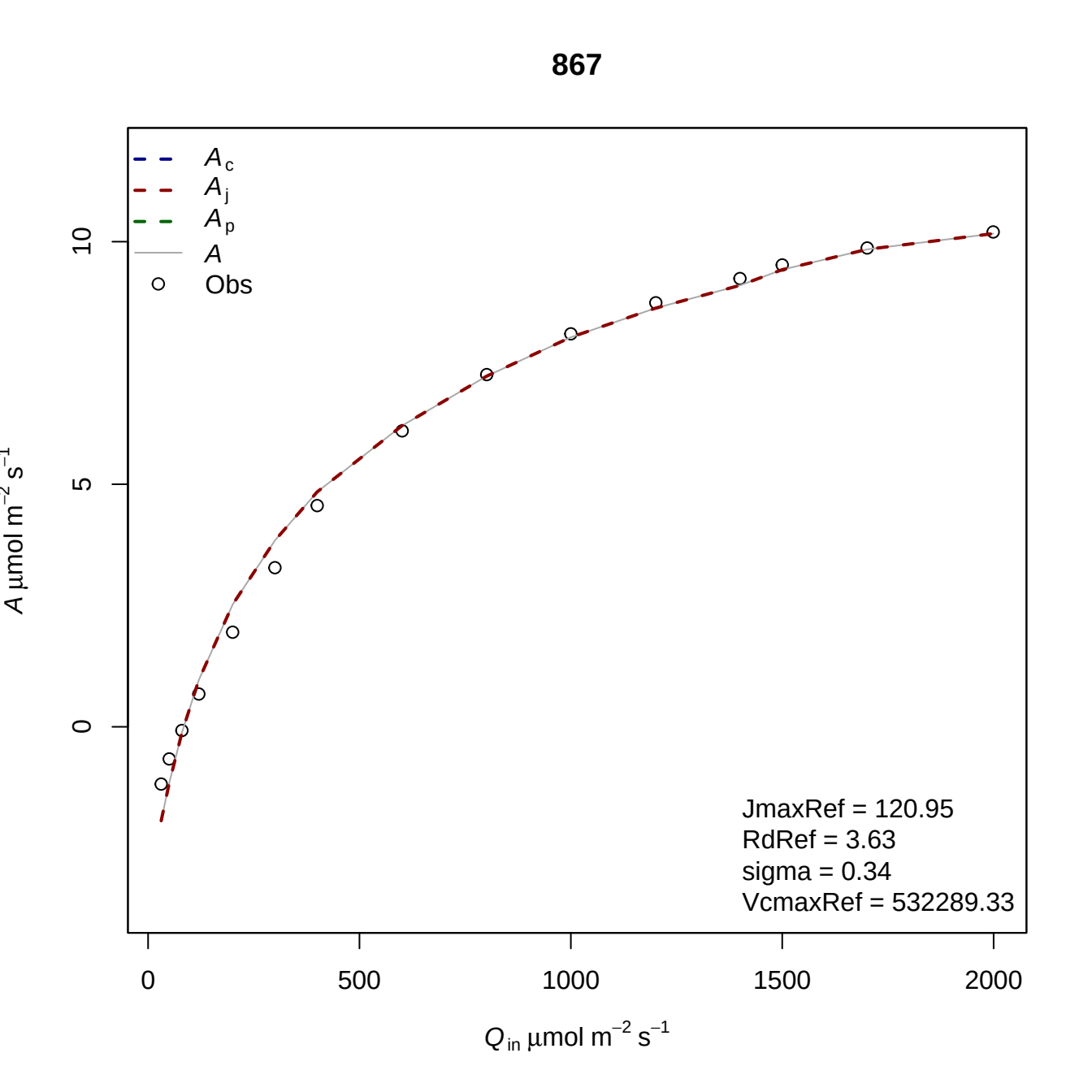

862
A_c
A_j
A_p
A
Obs
JmaxRef = 128.93
RdRef = 2.25
sigma = 0.11
VcmaxRef = 86.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

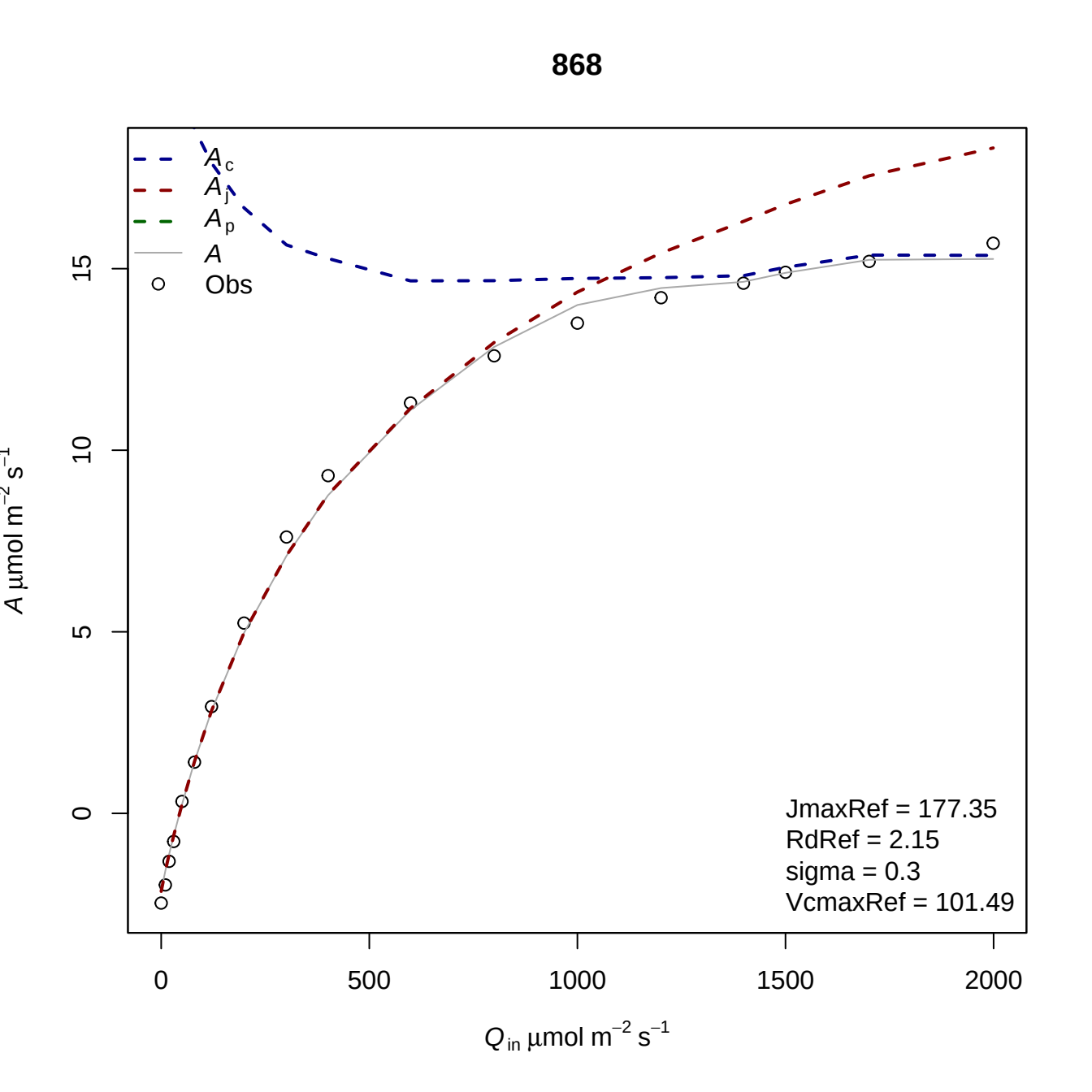

863
A_c
A_j
A_p
A
Obs
JmaxRef = 113.33
RdRef = 0.99
sigma = 0.15
VcmaxRef = 67.21
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

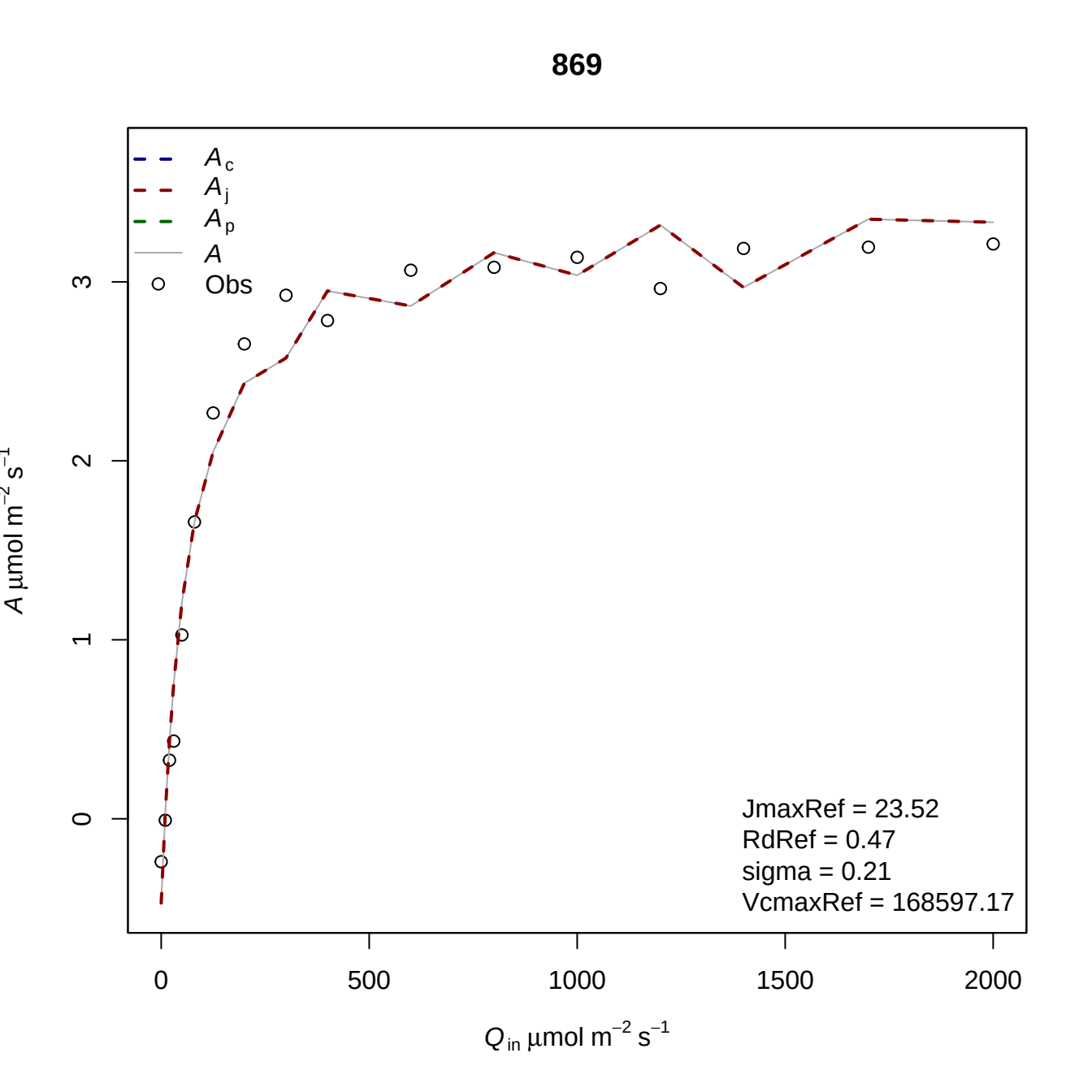

864
A_c
A_j
A_p
A
Obs
JmaxRef = 163.64
RdRef = 0.57
sigma = 0.29
VcmaxRef = 87.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

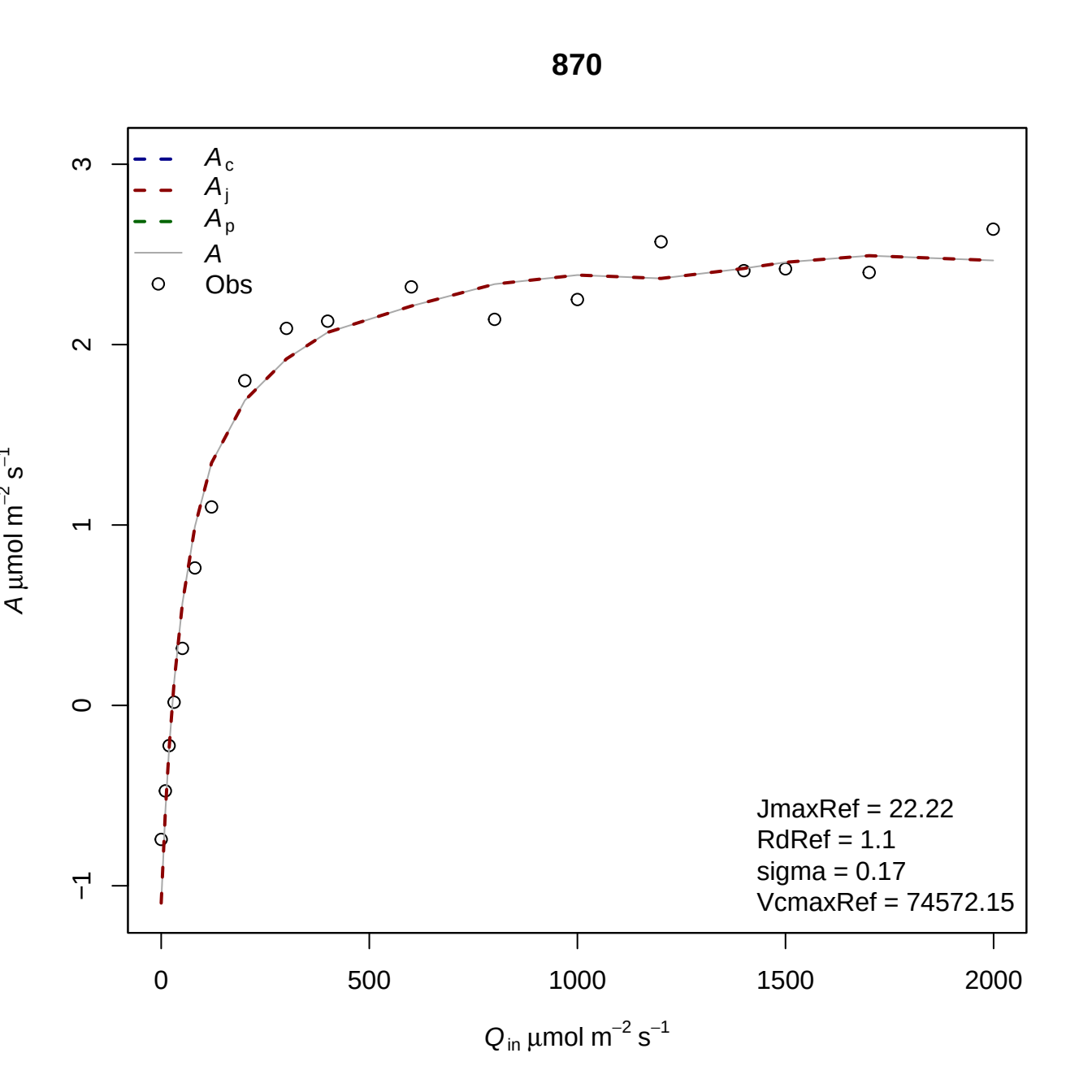

865
A_c
A_j
A_p
A
Obs
JmaxRef = 164.61
RdRef = 1.7
sigma = 0.17
VcmaxRef = 107.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

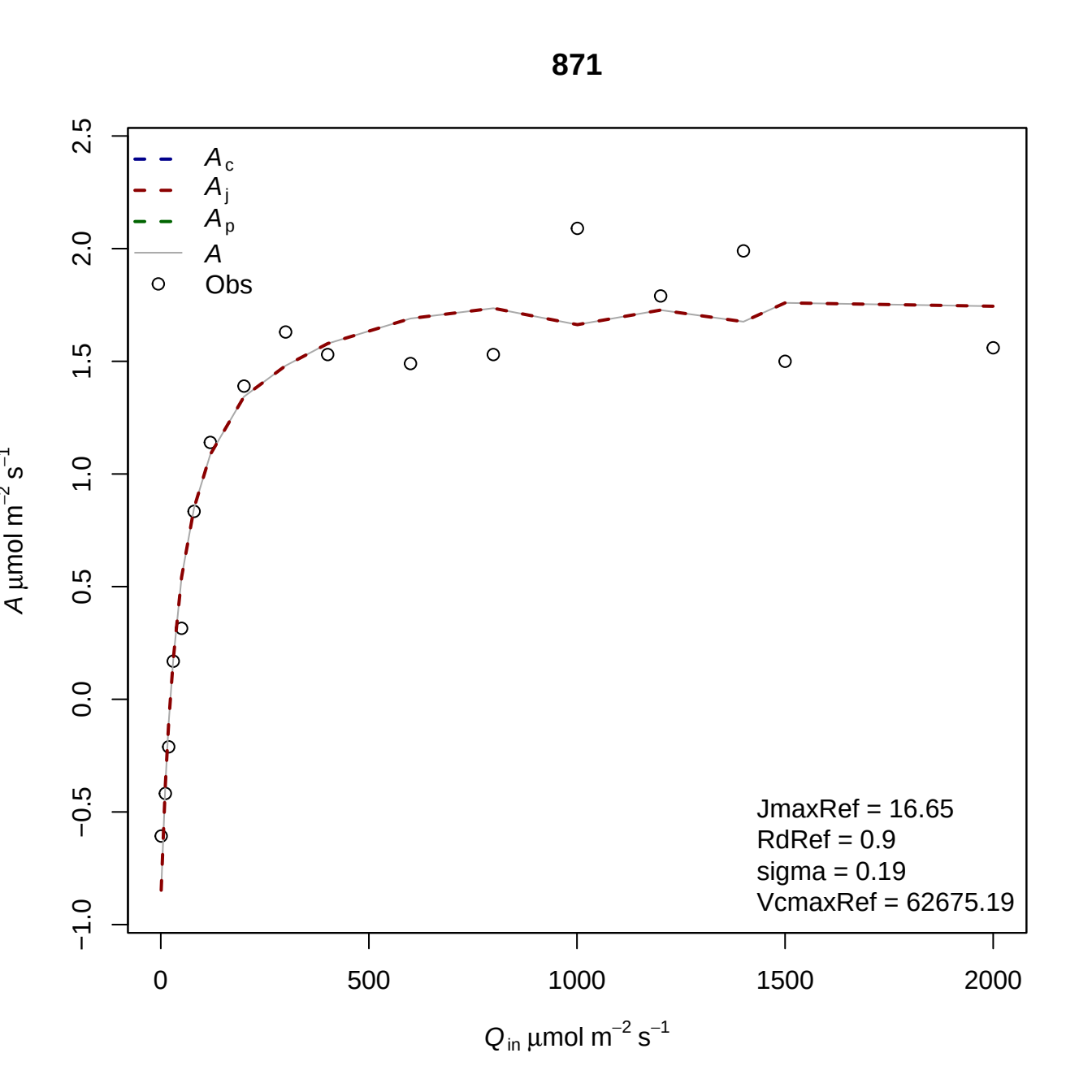

866
A_c
A_j
A_p
A
Obs
JmaxRef = 206.35
RdRef = 4.04
sigma = 0.78
VcmaxRef = 117924.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

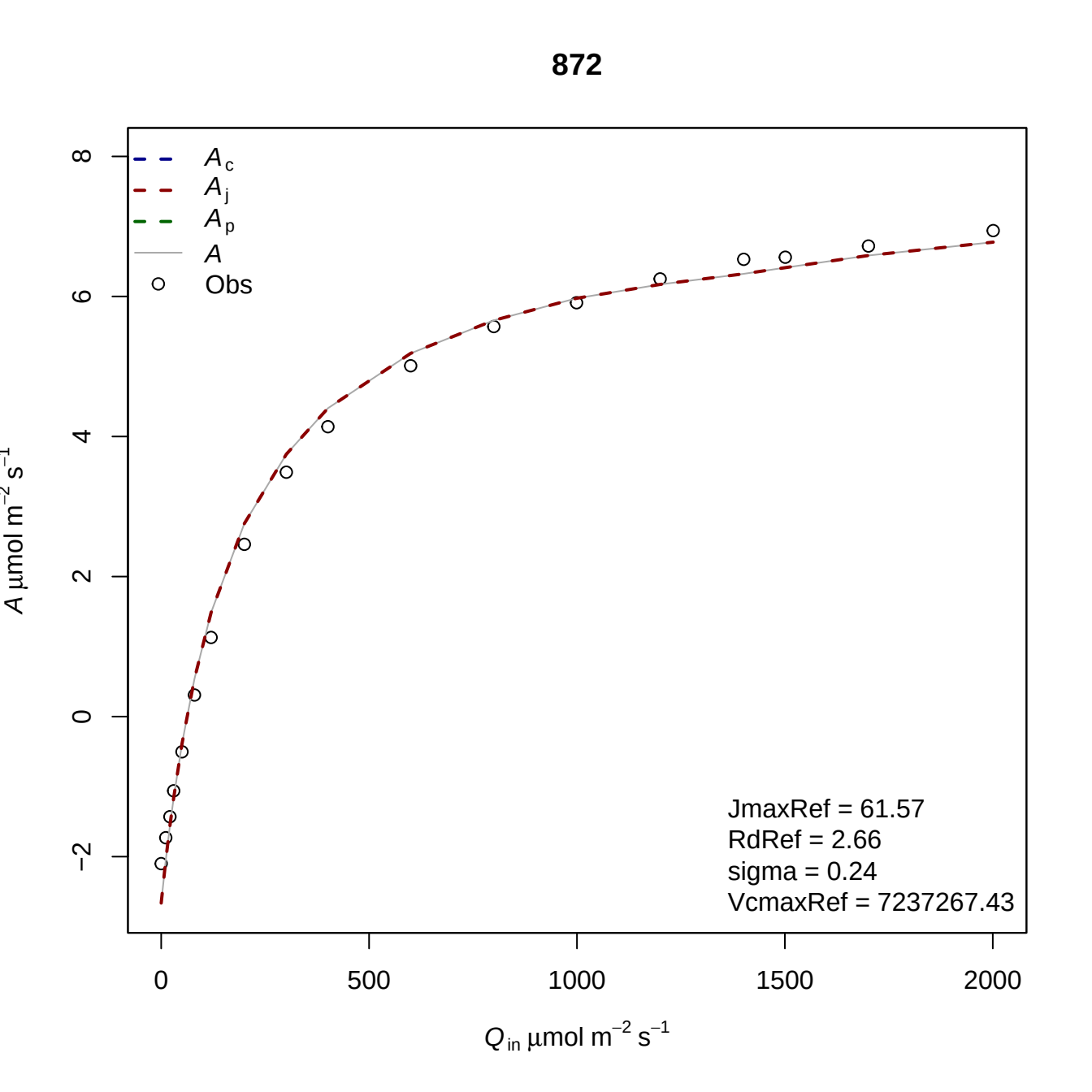

867
A_c
A_j
A_p
A
Obs
JmaxRef = 120.95
RdRef = 3.63
sigma = 0.34
VcmaxRef = 532289.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

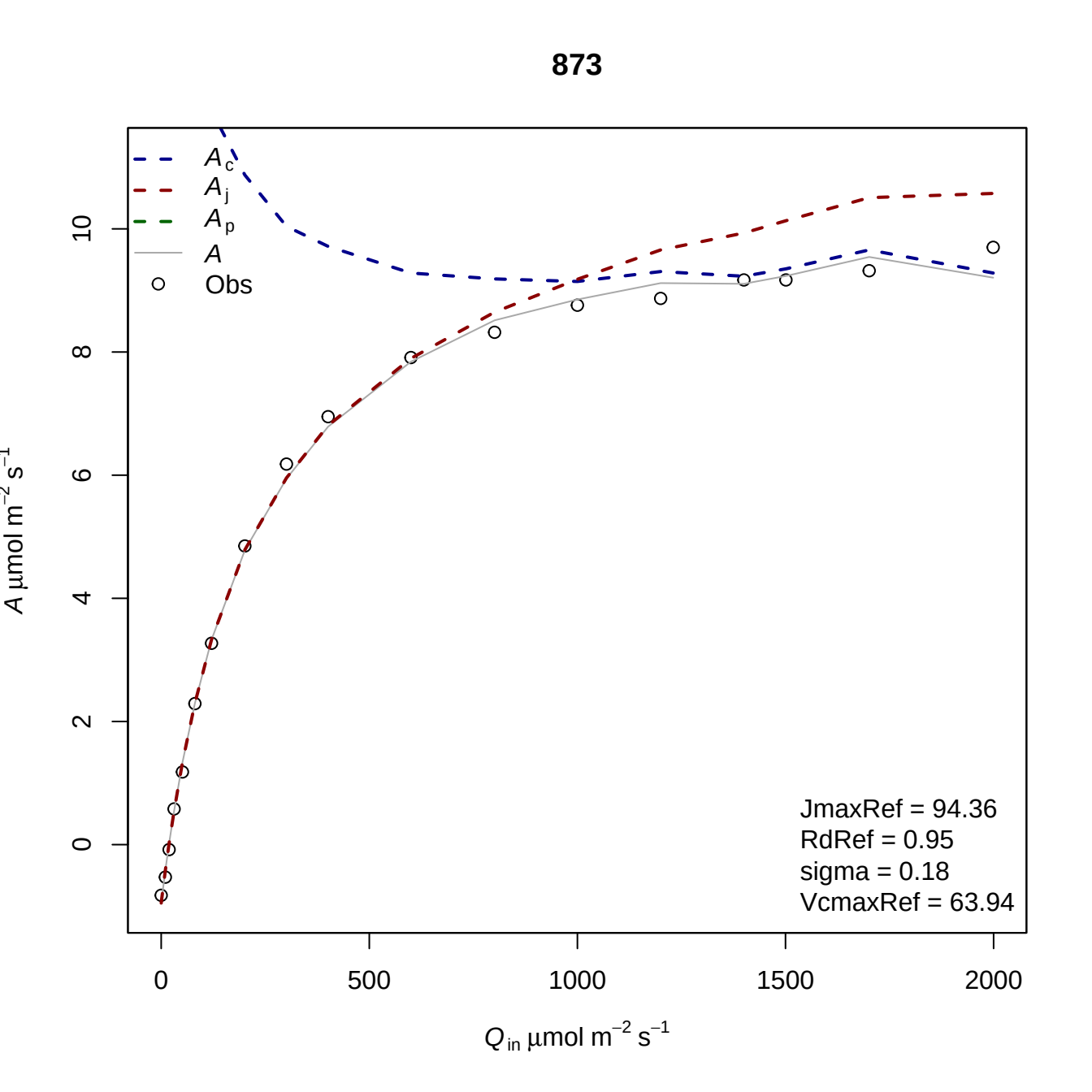

868
A_c
A_j
A_p
A
Obs
JmaxRef = 177.35
RdRef = 2.15
sigma = 0.3
VcmaxRef = 101.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

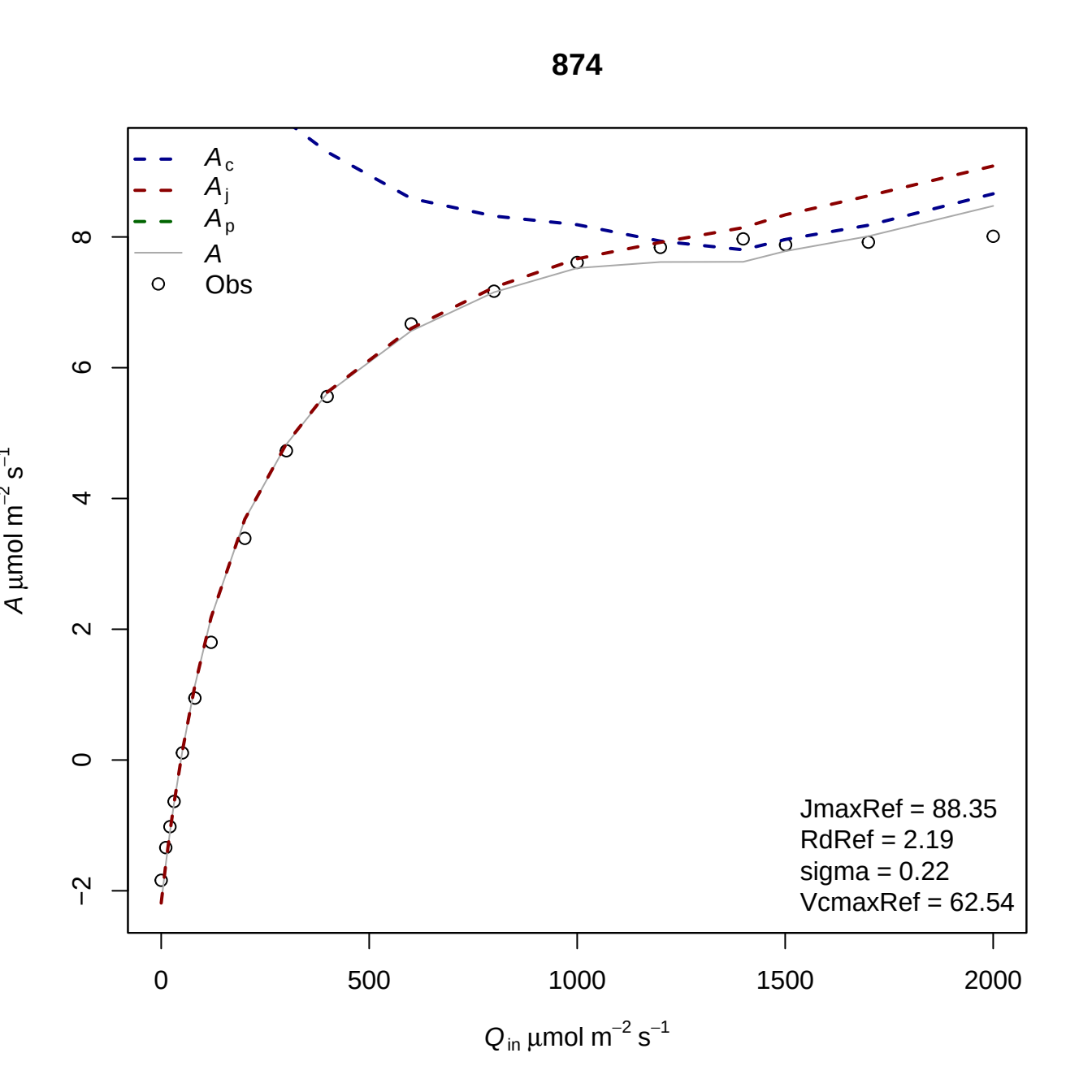

869
A_c
A_j
A_p
A
Obs
JmaxRef = 23.52
RdRef = 0.47
sigma = 0.21
VcmaxRef = 168597.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500
2000

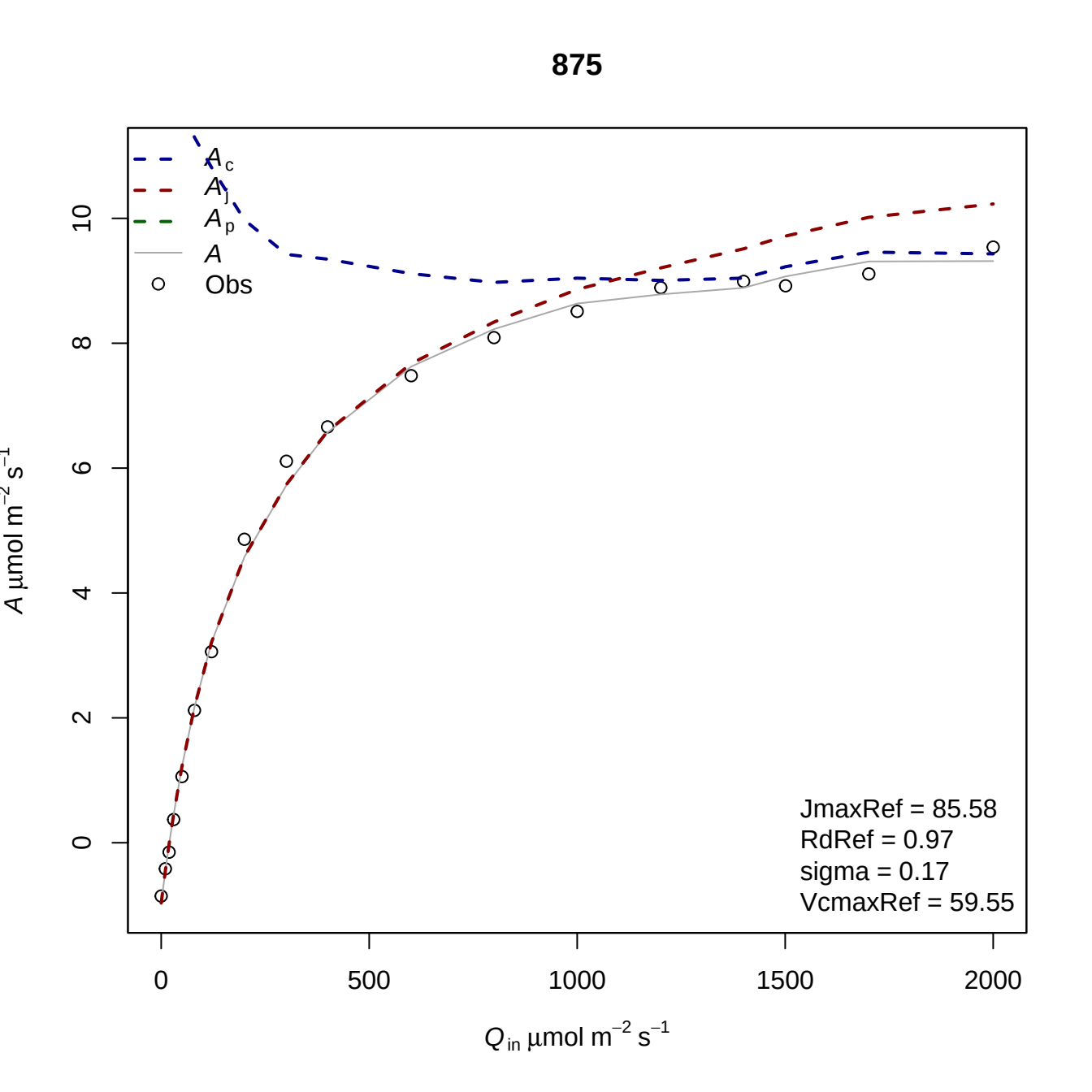

870
A_c
A_j
A_p
A
Obs
JmaxRef = 22.22
RdRef = 1.1
sigma = 0.17
VcmaxRef = 74572.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
500
1000
1500
2000

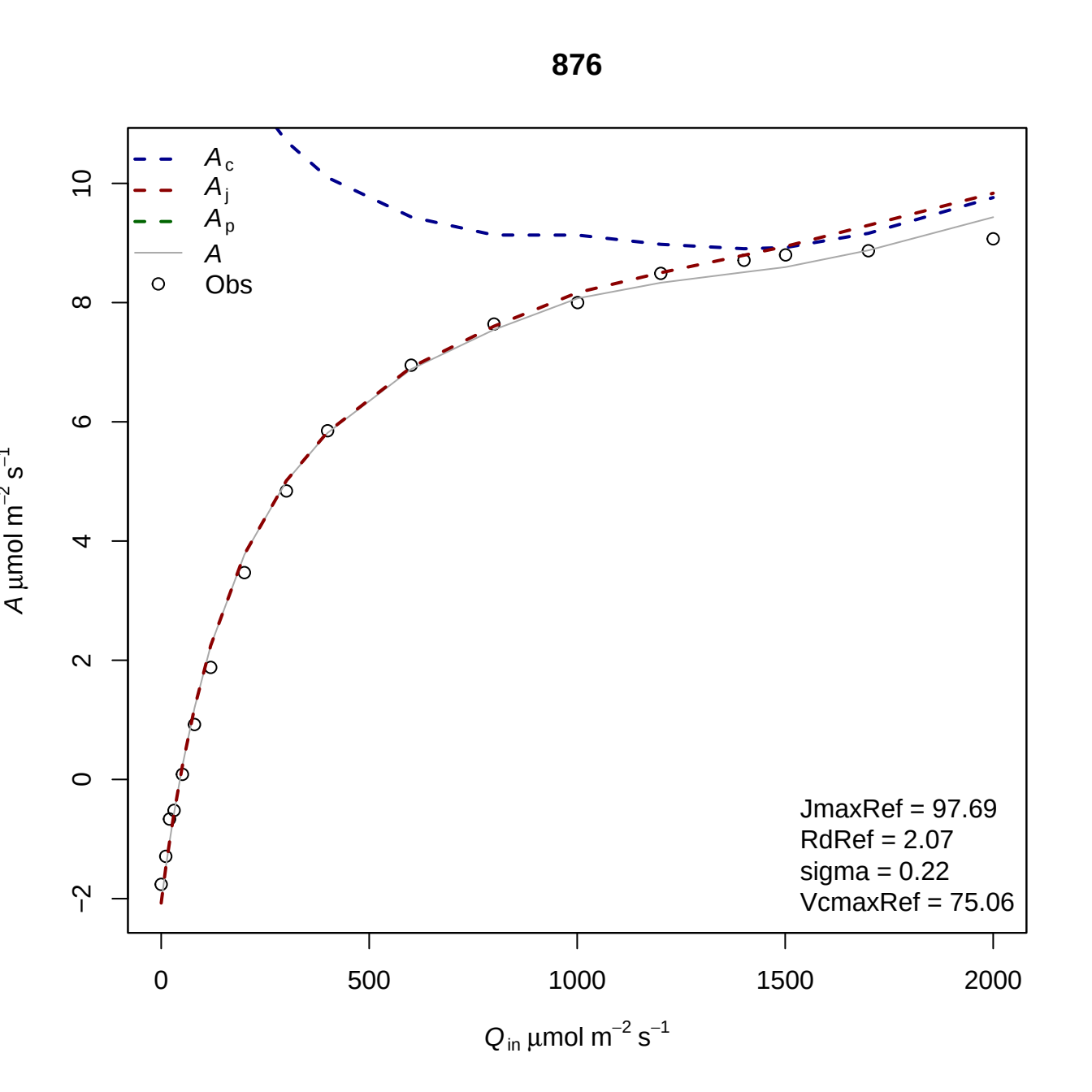

871
A_c
A_j
A_p
A
Obs
JmaxRef = 16.65
RdRef = 0.9
sigma = 0.19
VcmaxRef = 62675.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500
2000

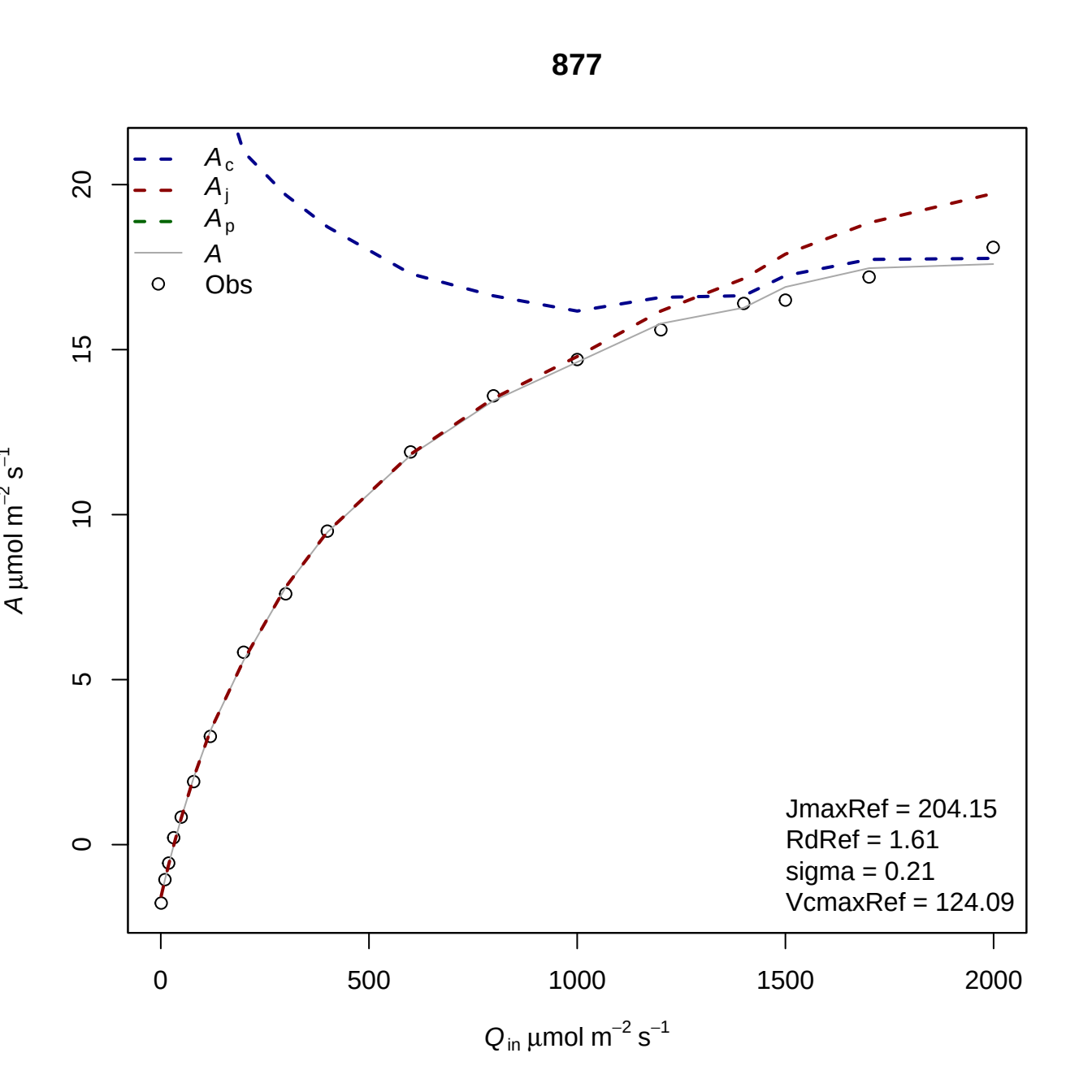

872
A_c
A_j
A_p
A
Obs
JmaxRef = 61.57
RdRef = 2.66
sigma = 0.24
VcmaxRef = 7237267.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

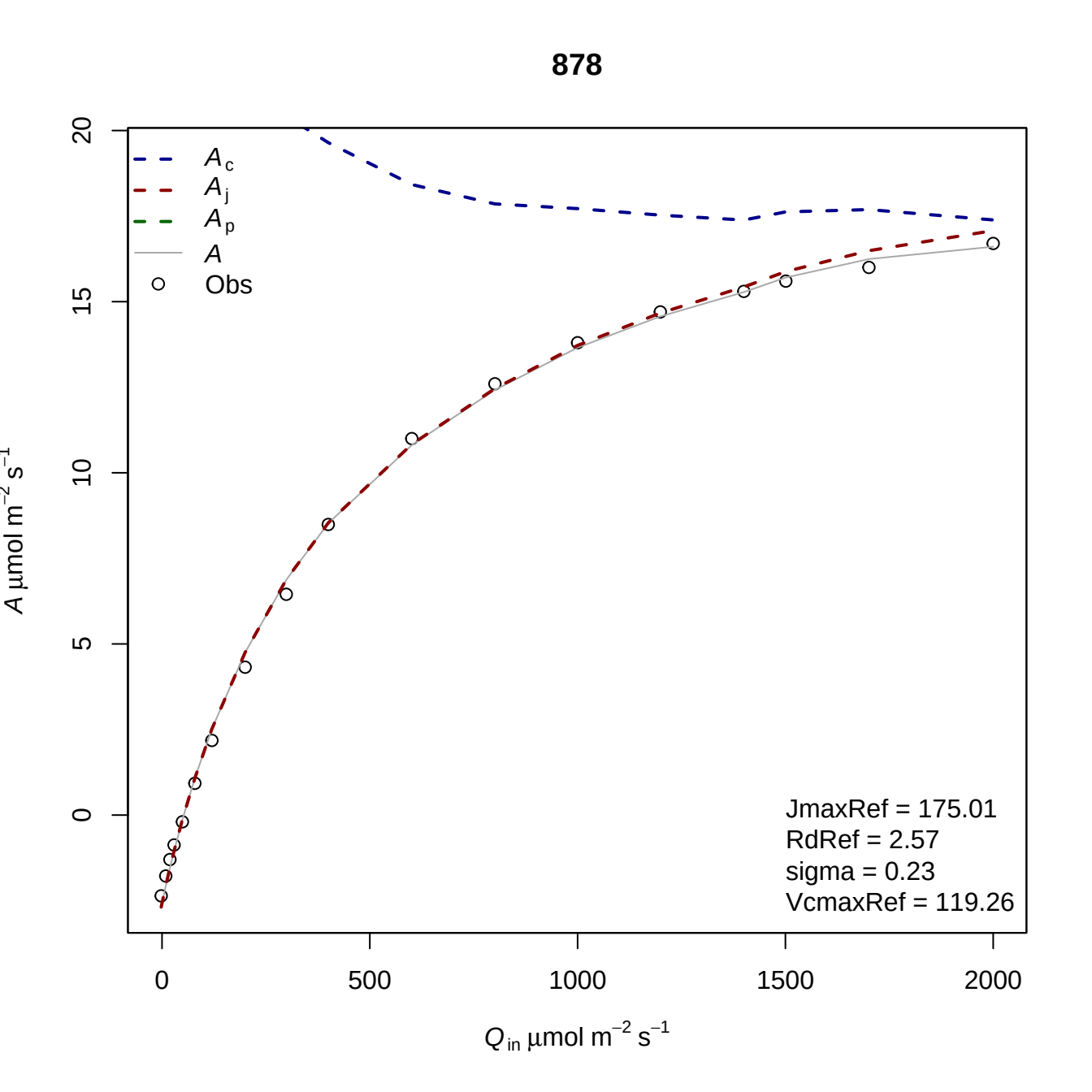

873
A_c
A_j
A_p
A
Obs
JmaxRef = 94.36
RdRef = 0.95
sigma = 0.18
VcmaxRef = 63.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

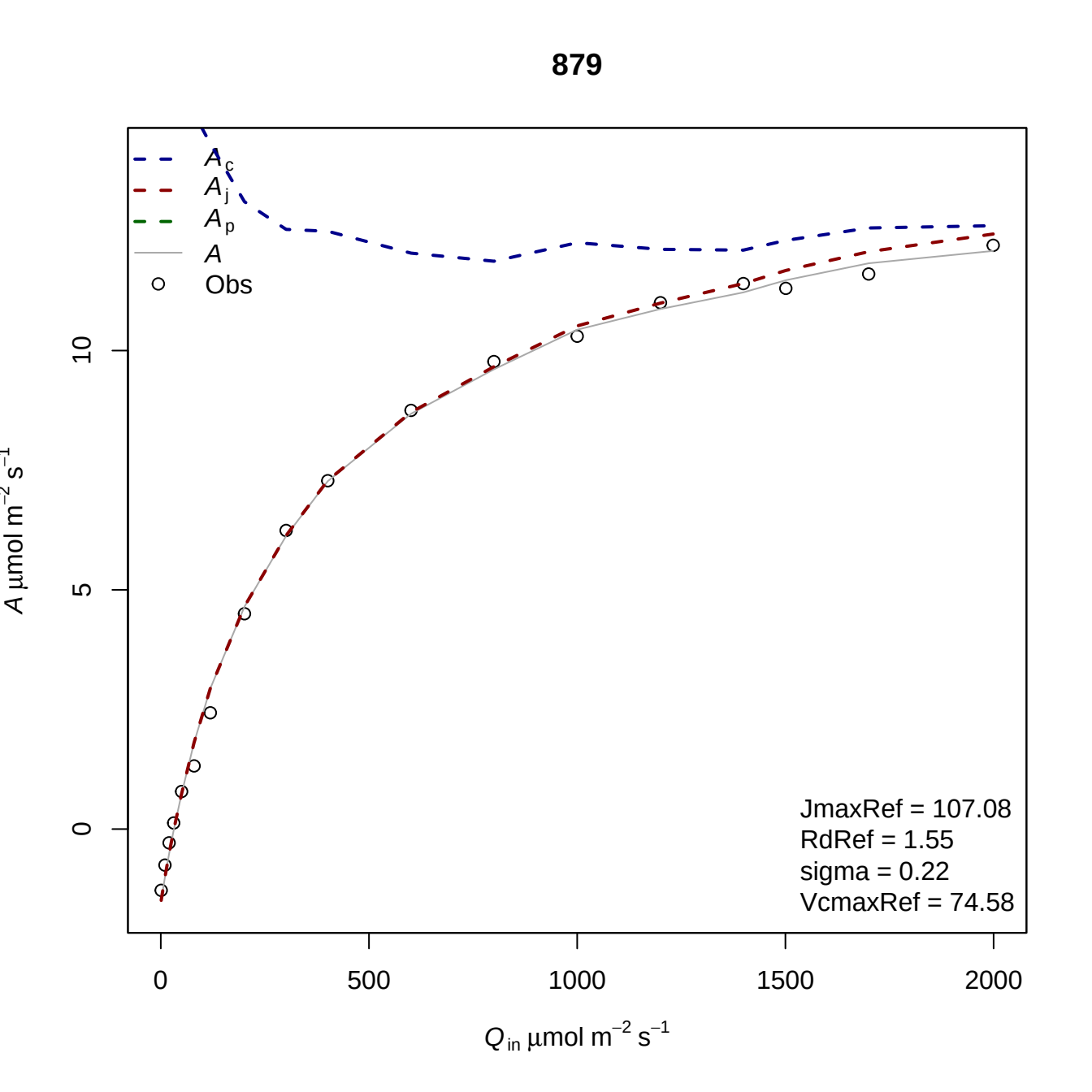

874
A_c
A_j
A_p
A
Obs
JmaxRef = 88.35
RdRef = 2.19
sigma = 0.22
VcmaxRef = 62.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

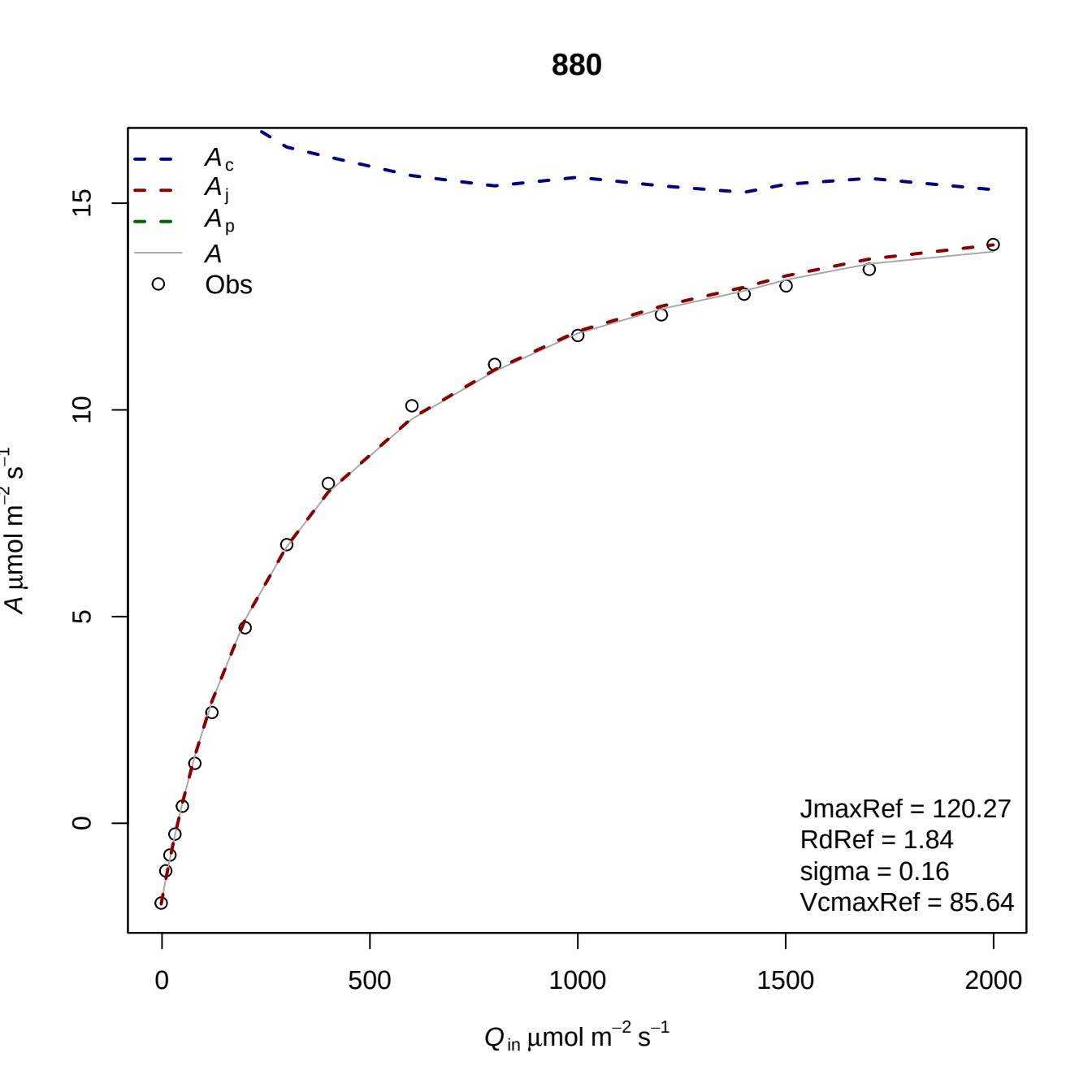

875
A_c
A_j
A_p
A
Obs
JmaxRef = 85.58
RdRef = 0.97
sigma = 0.17
VcmaxRef = 59.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

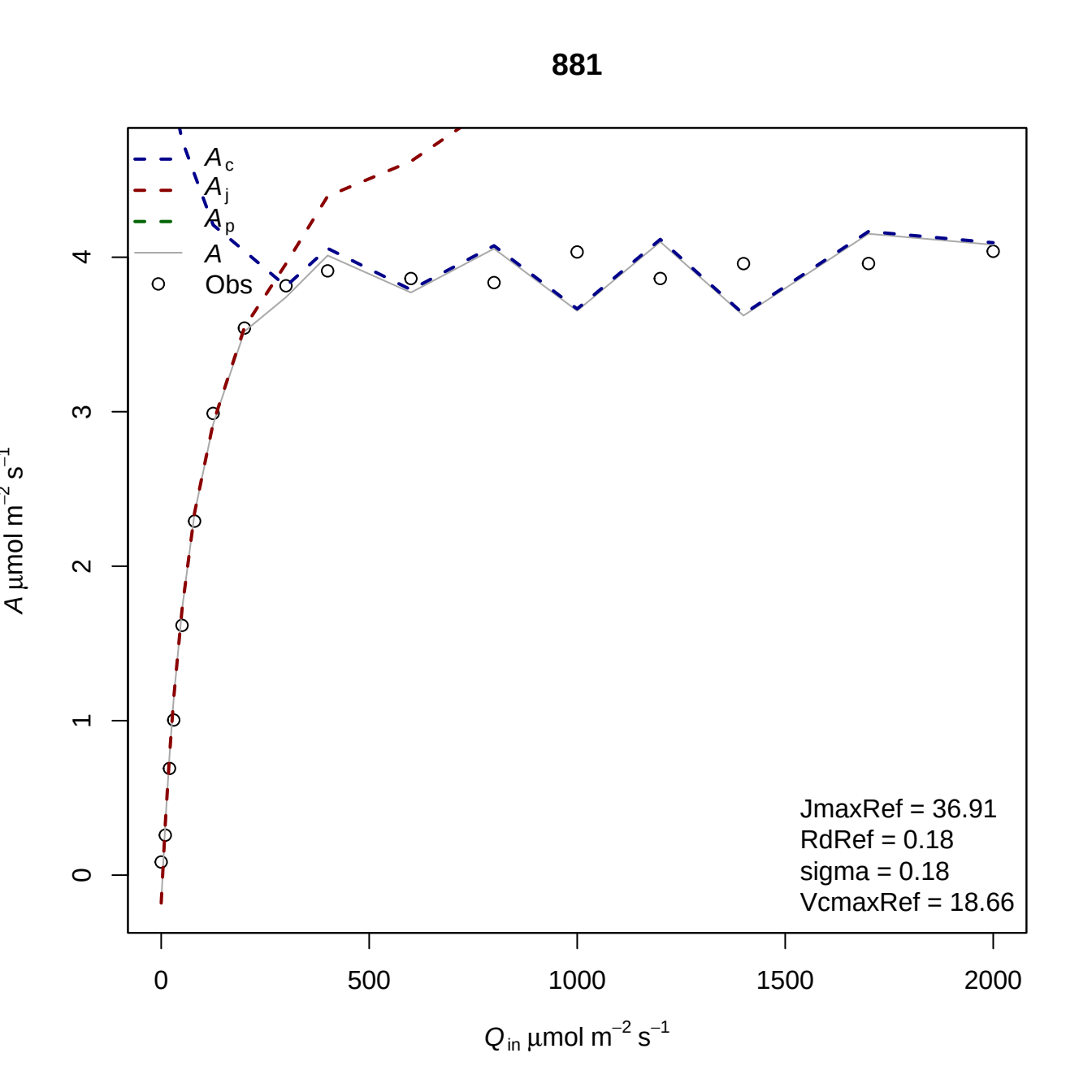

876
A_c
A_j
A_p
A
Obs
JmaxRef = 97.69
RdRef = 2.07
sigma = 0.22
VcmaxRef = 75.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

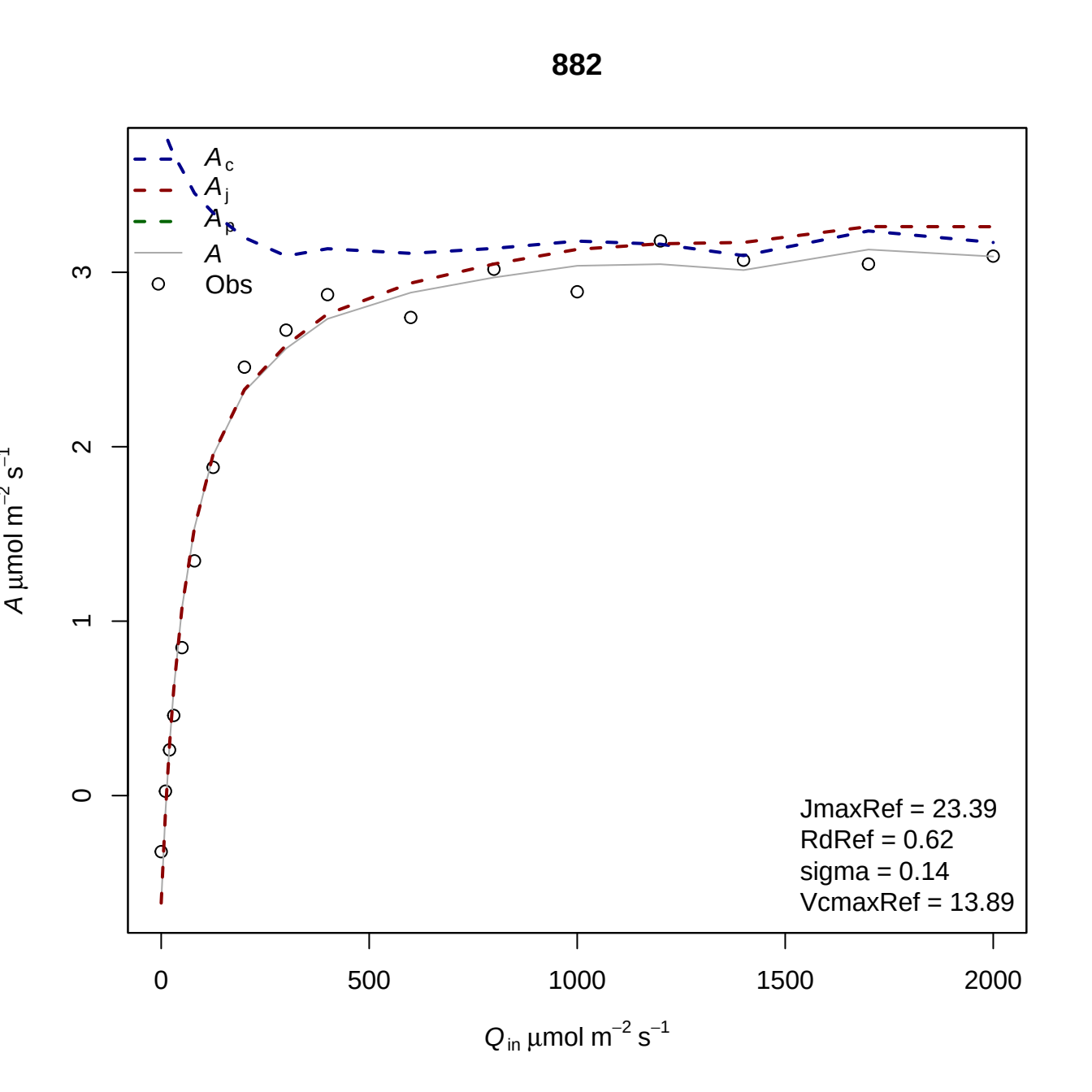

877
A_c
A_j
A_p
A
Obs
JmaxRef = 204.15
RdRef = 1.61
sigma = 0.21
VcmaxRef = 124.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

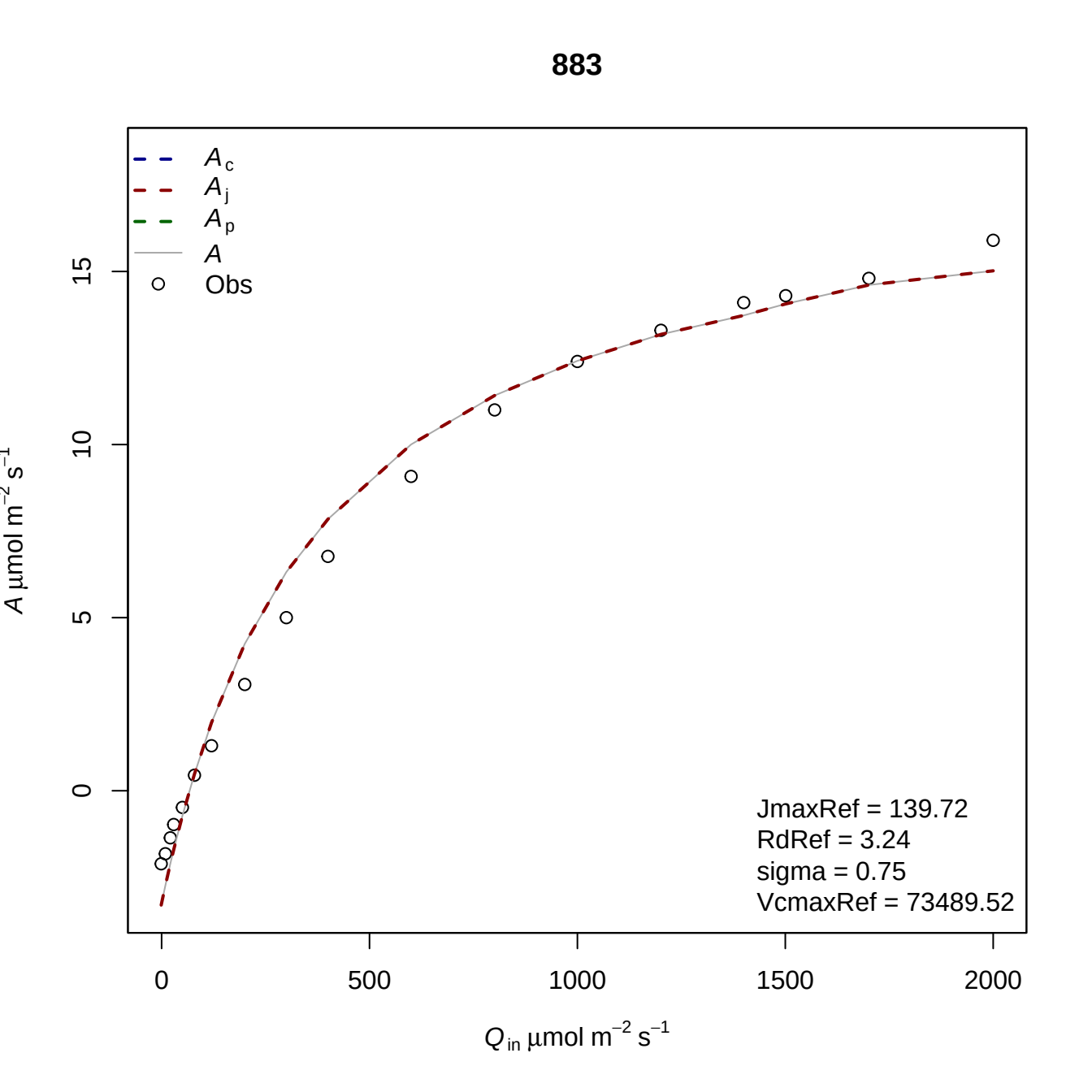

878
A_c
A_j
A_p
A
Obs
JmaxRef = 175.01
RdRef = 2.57
sigma = 0.23
VcmaxRef = 119.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

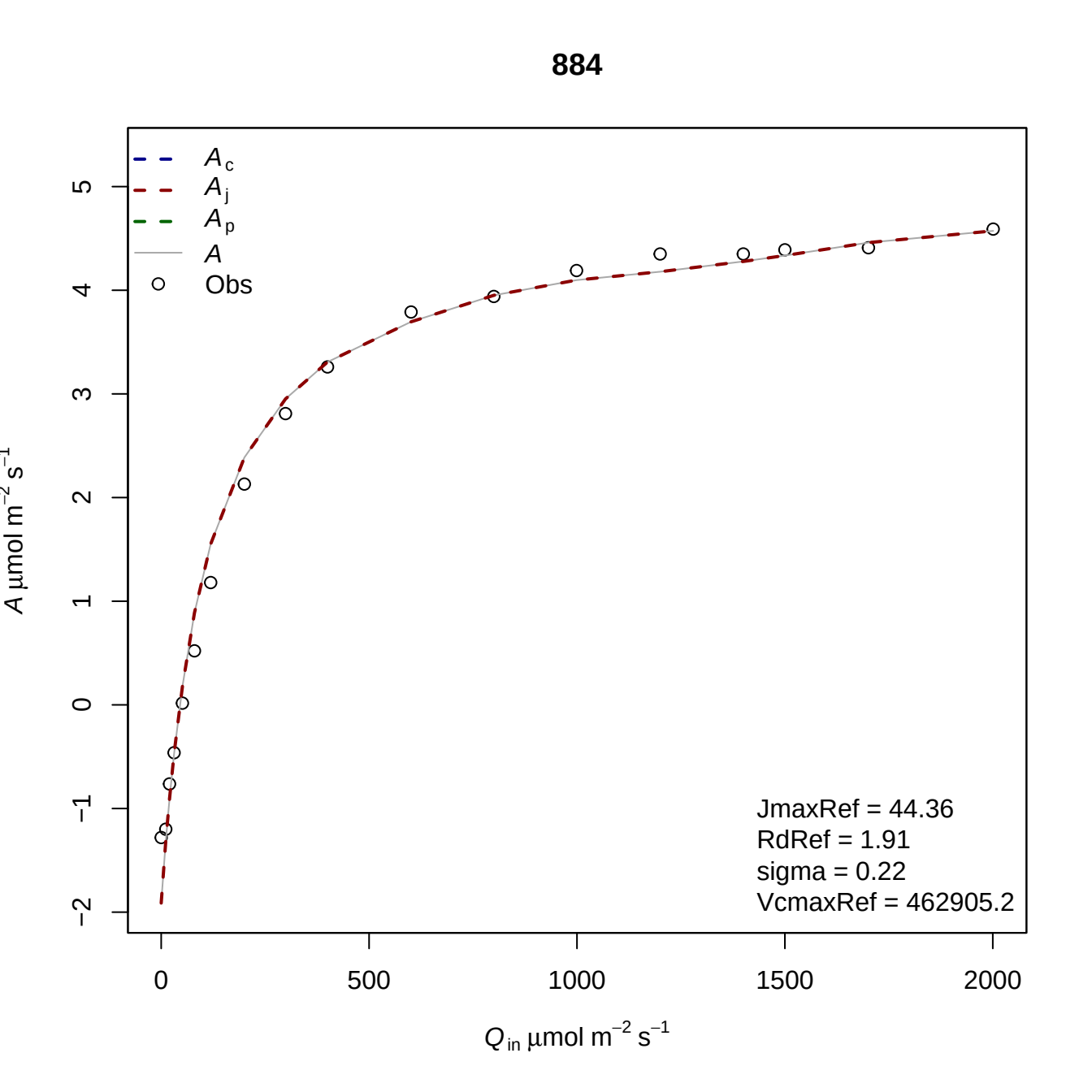

879
A_c
A_j
A_p
A
Obs
JmaxRef = 107.08
RdRef = 1.55
sigma = 0.22
VcmaxRef = 74.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

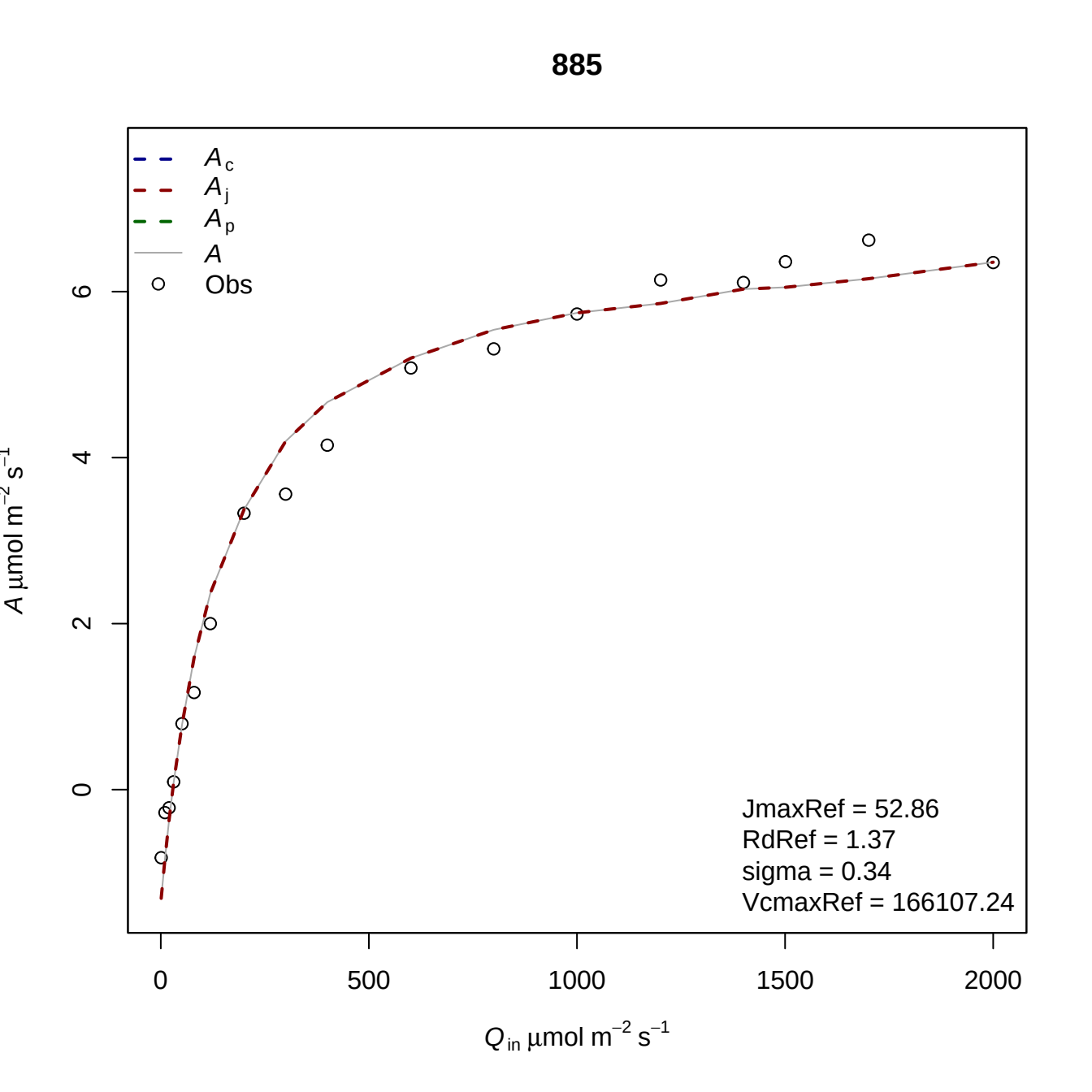

880
A_c
A_j
A_p
A
Obs
JmaxRef = 120.27
RdRef = 1.84
sigma = 0.16
VcmaxRef = 85.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

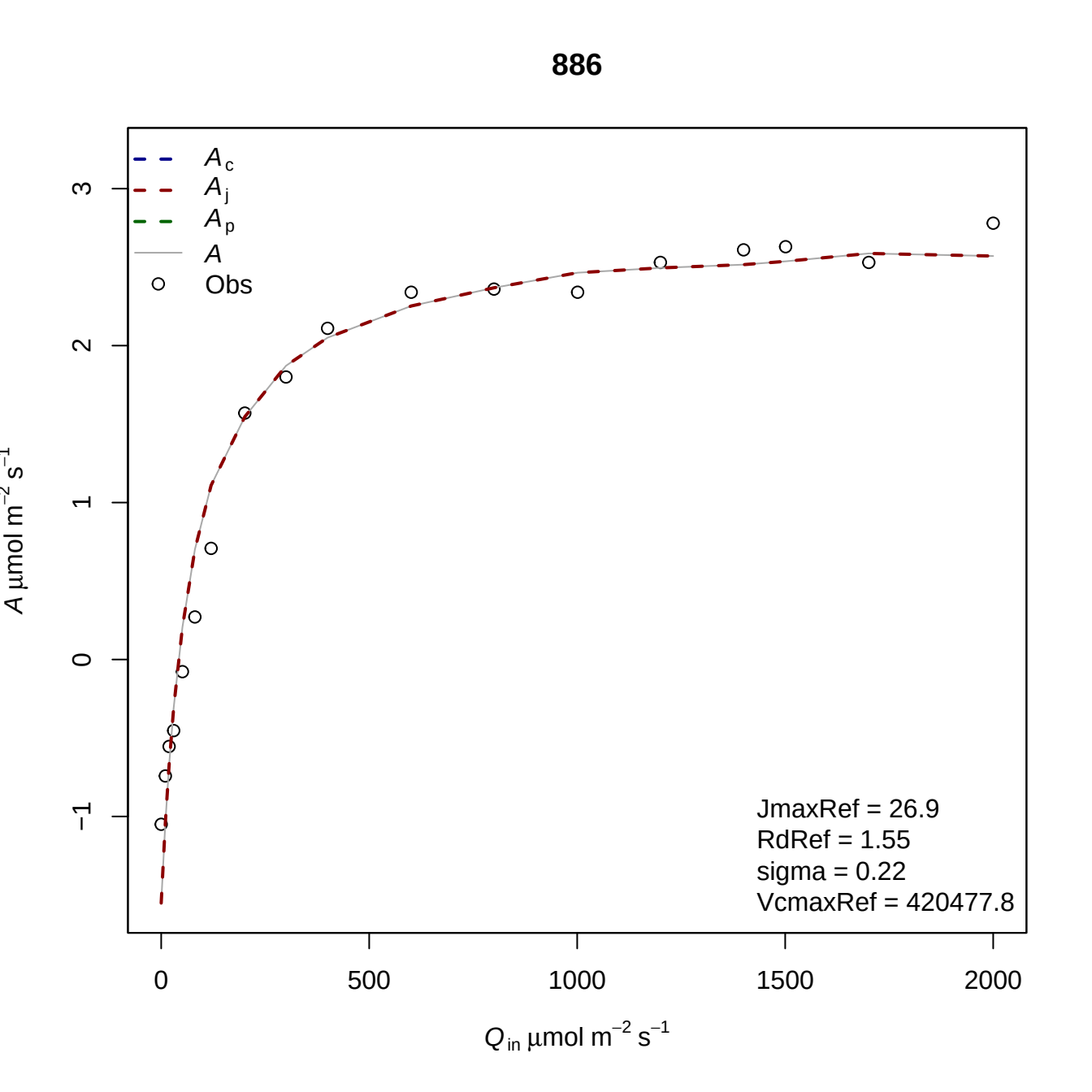

881
A_c
A_j
A_p
A
Obs
JmaxRef = 36.91
RdRef = 0.18
sigma = 0.18
VcmaxRef = 18.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500
2000

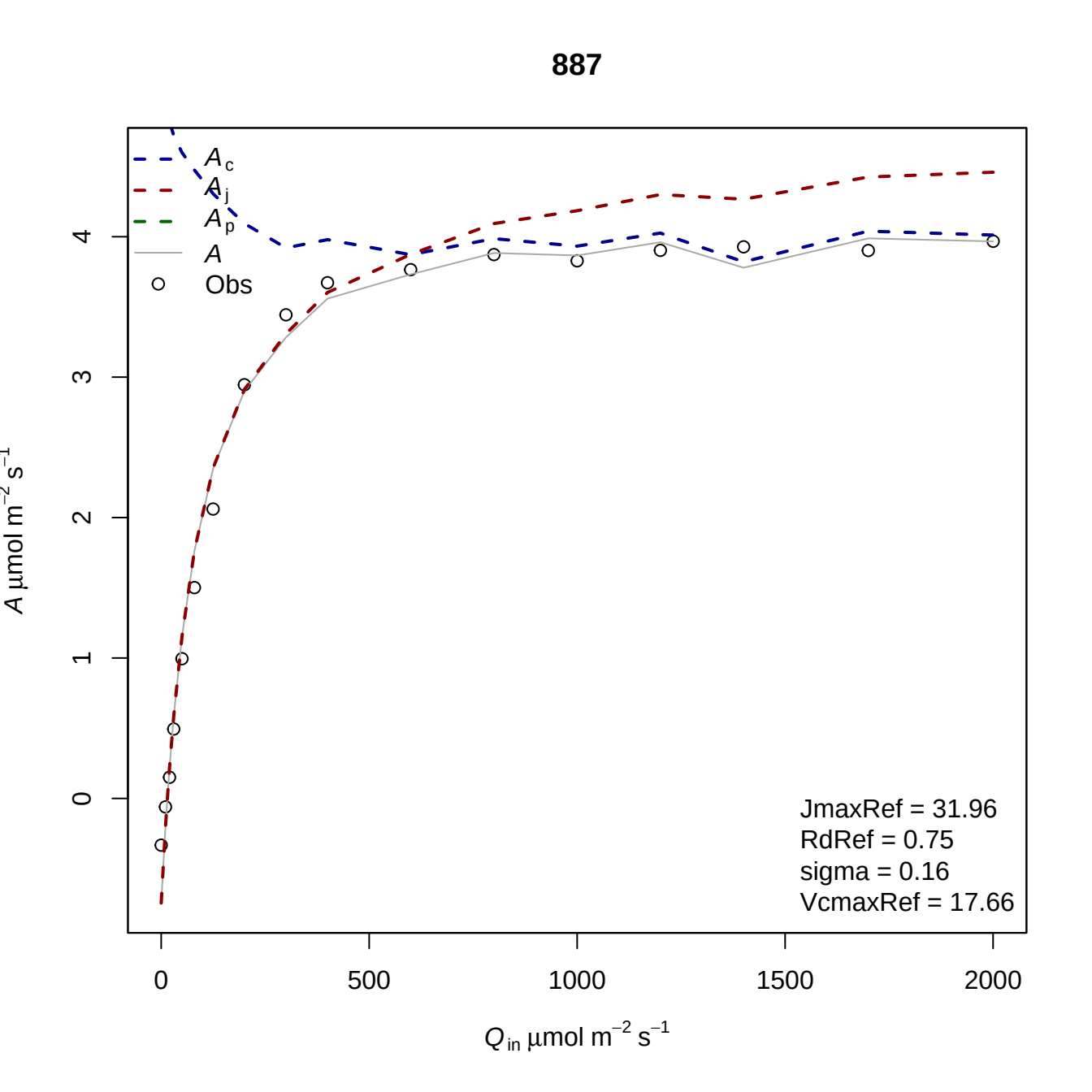

882
A_c
A_j
A_p
A
Obs
JmaxRef = 23.39
RdRef = 0.62
sigma = 0.14
VcmaxRef = 13.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500
2000

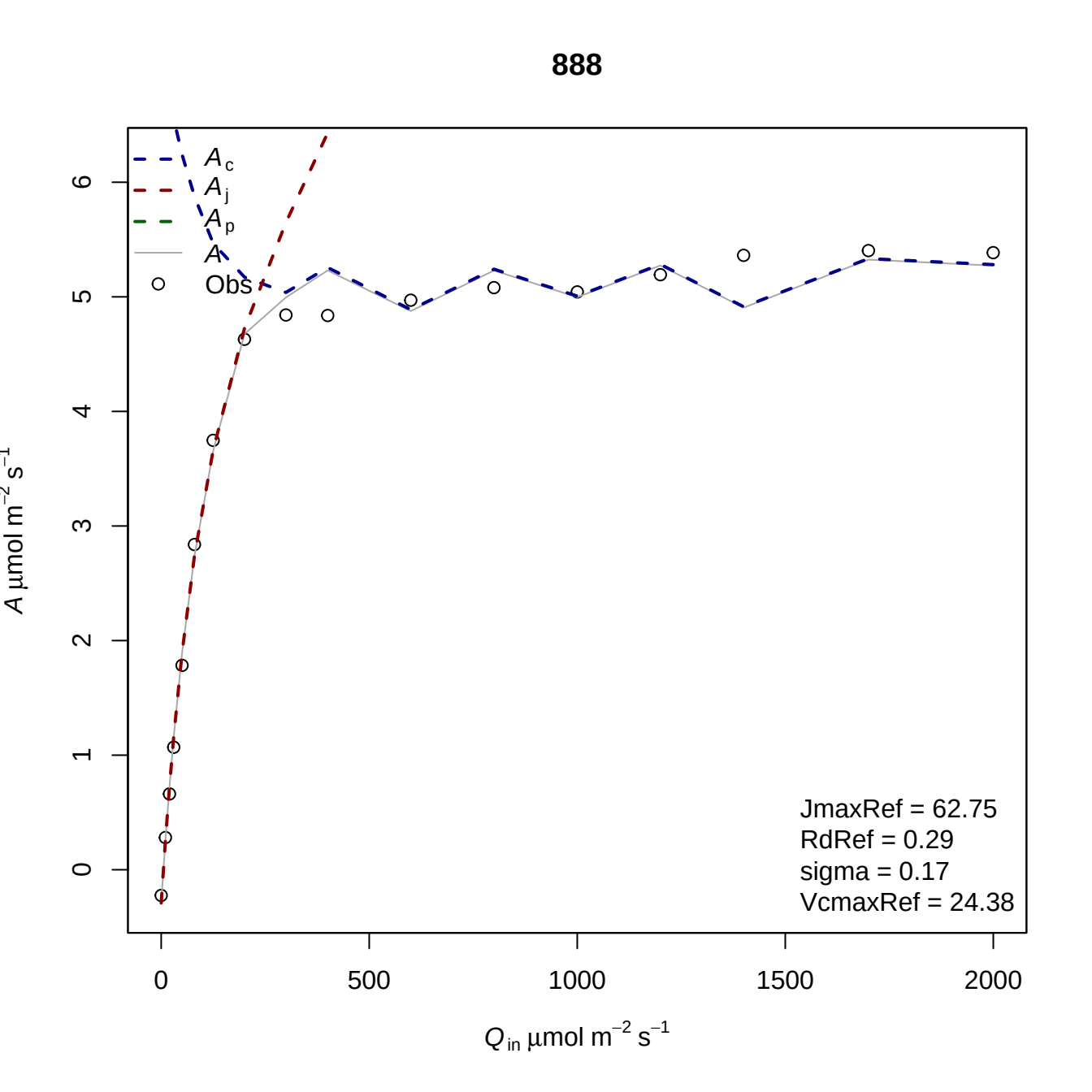

883
A_c
A_j
A_p
A
Obs
JmaxRef = 139.72
RdRef = 3.24
sigma = 0.75
VcmaxRef = 73489.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

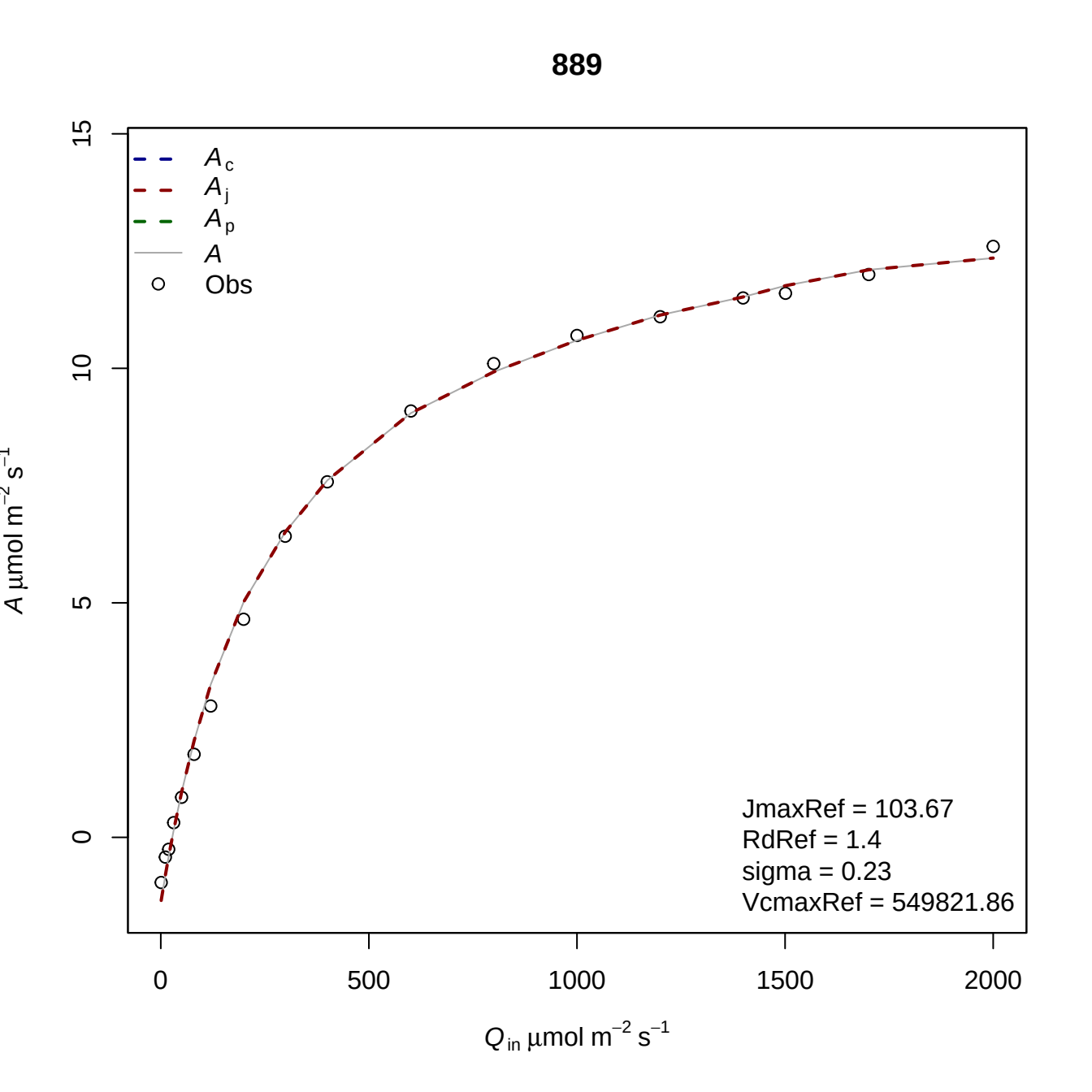

884
A_c
A_j
A_p
A
Obs
JmaxRef = 44.36
RdRef = 1.91
sigma = 0.22
VcmaxRef = 462905.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
5
0
500
1000
1500
2000

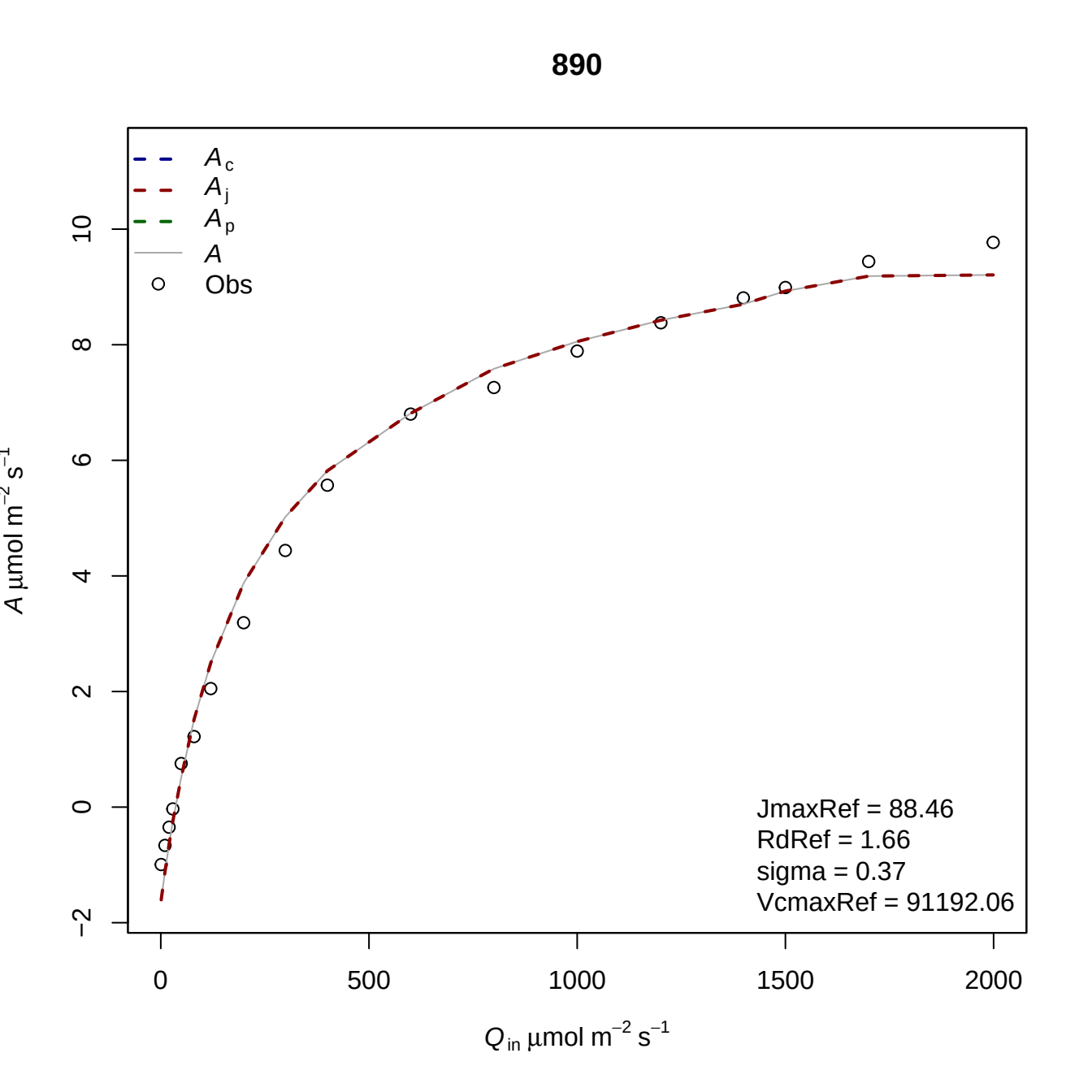

885
A_c
A_j
A_p
A
Obs
JmaxRef = 52.86
RdRef = 1.37
sigma = 0.34
VcmaxRef = 166107.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

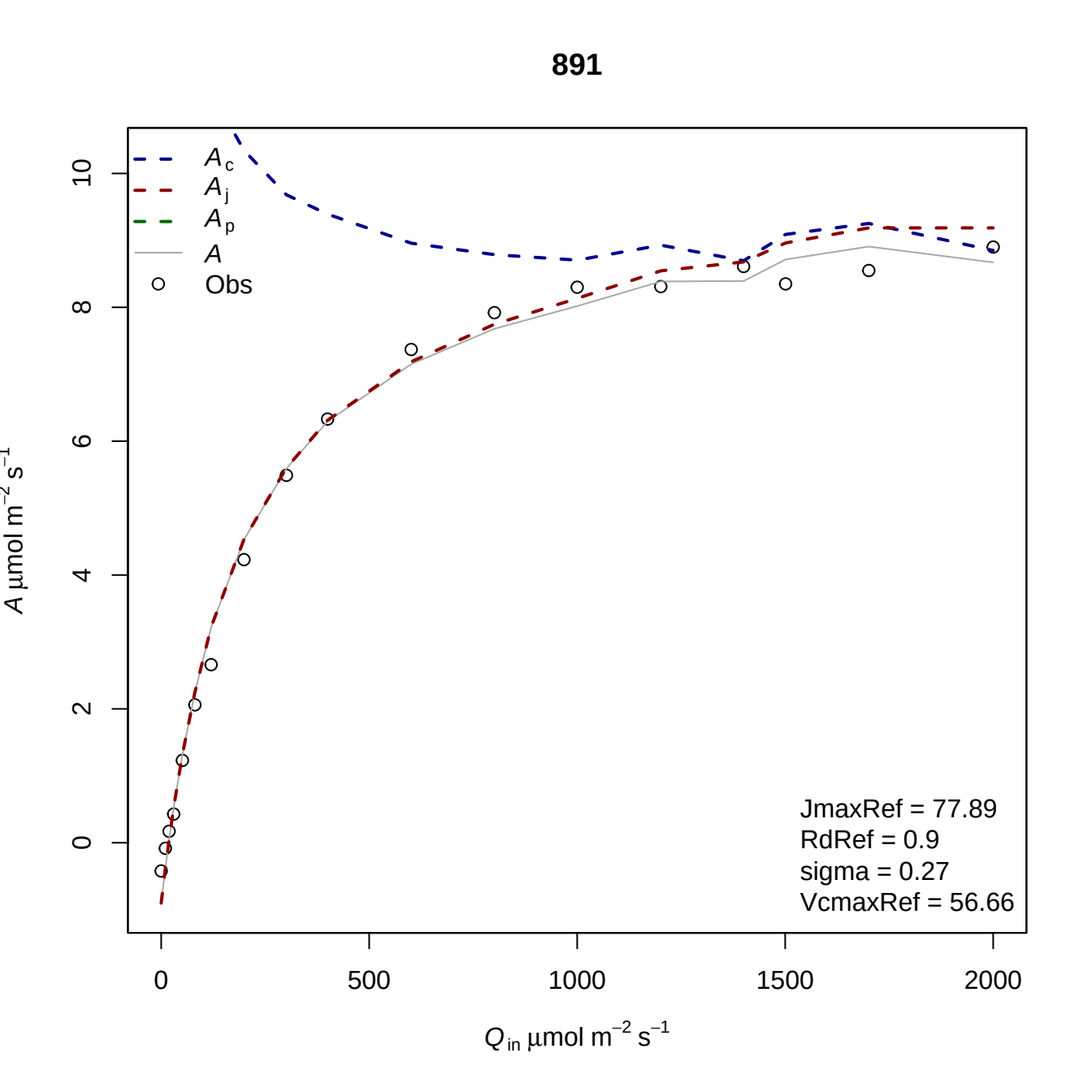

886
A_c
A_j
A_p
A
Obs
JmaxRef = 26.9
RdRef = 1.55
sigma = 0.22
VcmaxRef = 420477.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
500
1000
1500
2000

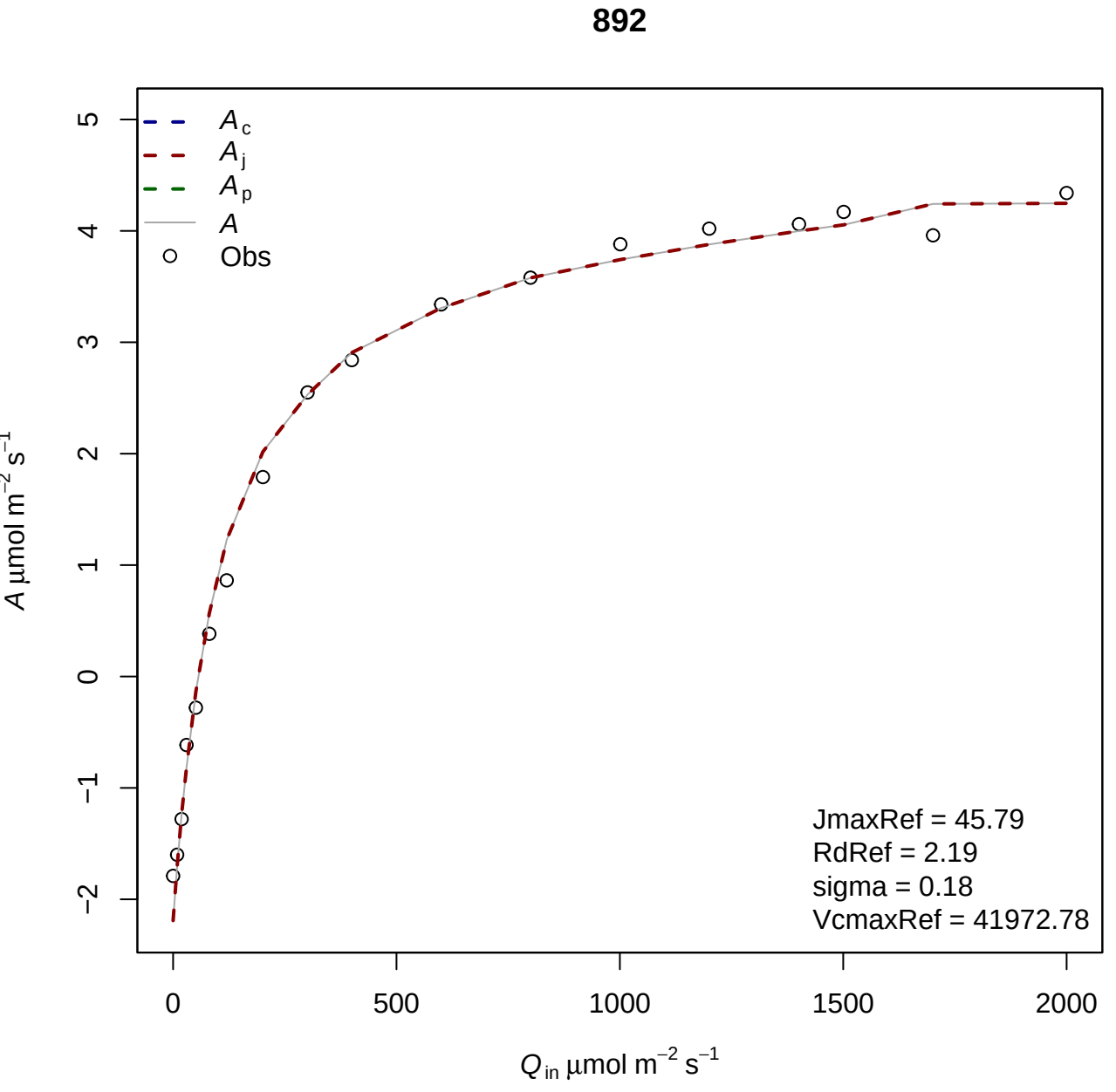

887
A_c
A_j
A_p
A
Obs
JmaxRef = 31.96
RdRef = 0.75
sigma = 0.16
VcmaxRef = 17.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500
2000

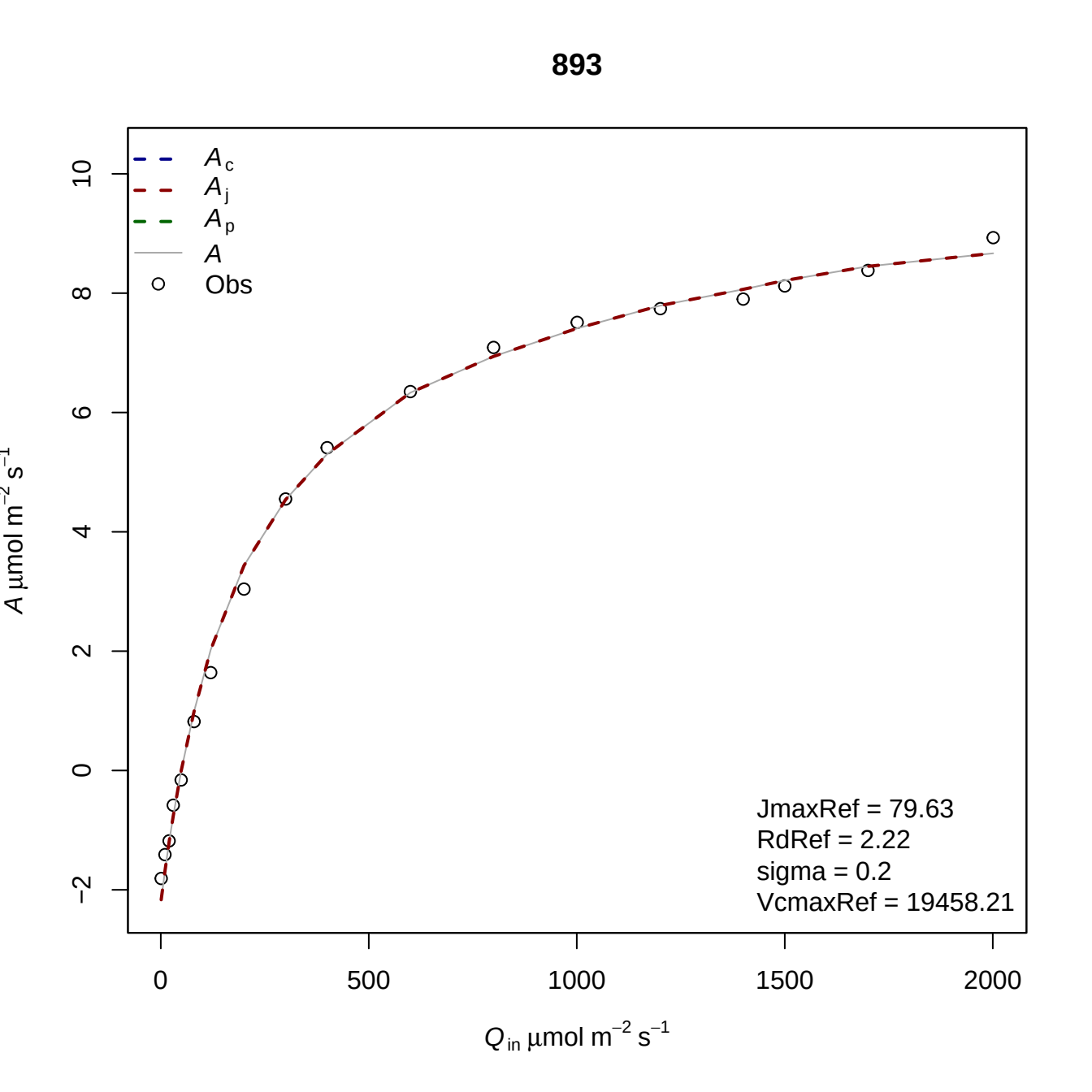

888
A_c
A_j
A_p
A
Obs
JmaxRef = 62.75
RdRef = 0.29
sigma = 0.17
VcmaxRef = 24.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500
2000

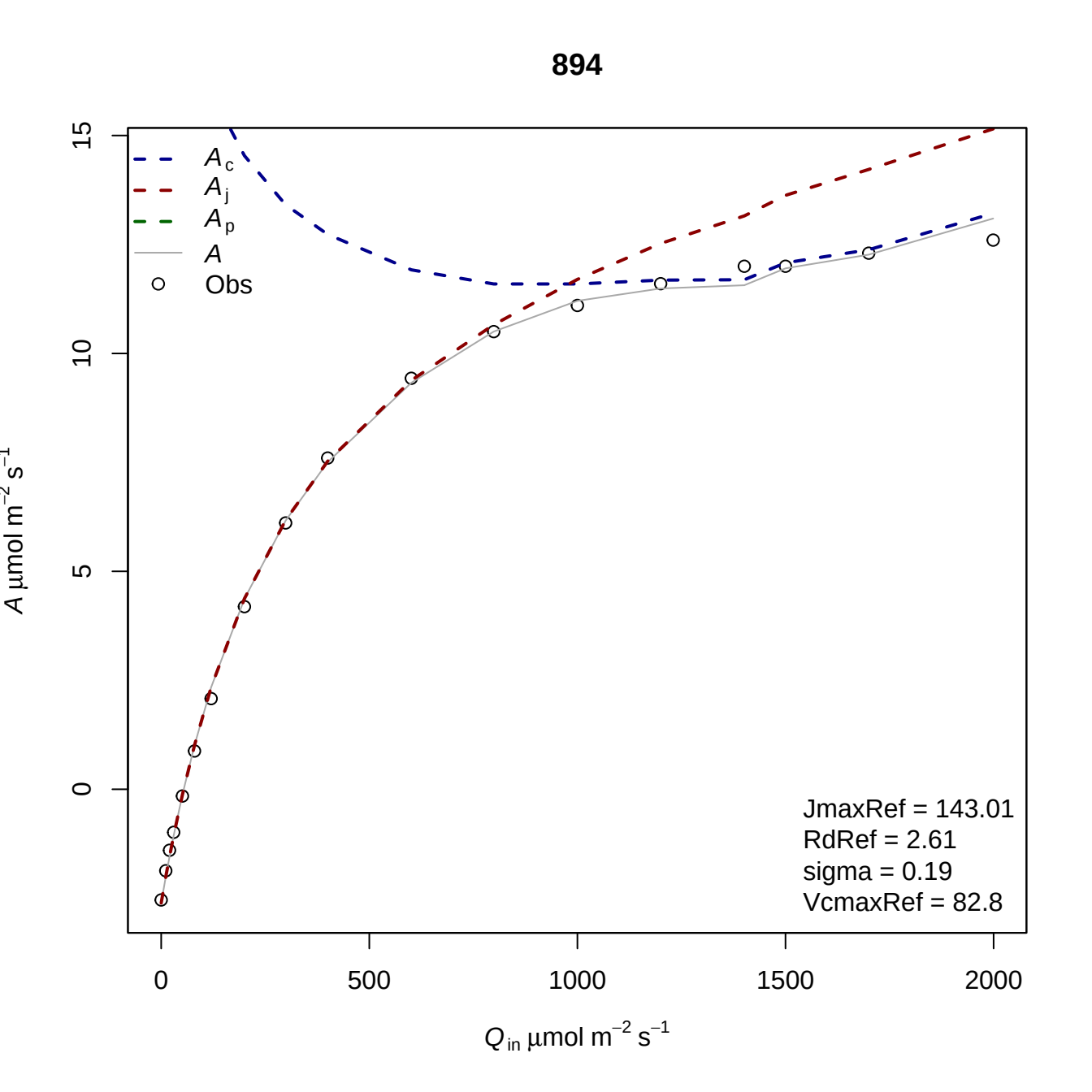

889
A_c
A_j
A_p
A
Obs
JmaxRef = 103.67
RdRef = 1.4
sigma = 0.23
VcmaxRef = 549821.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

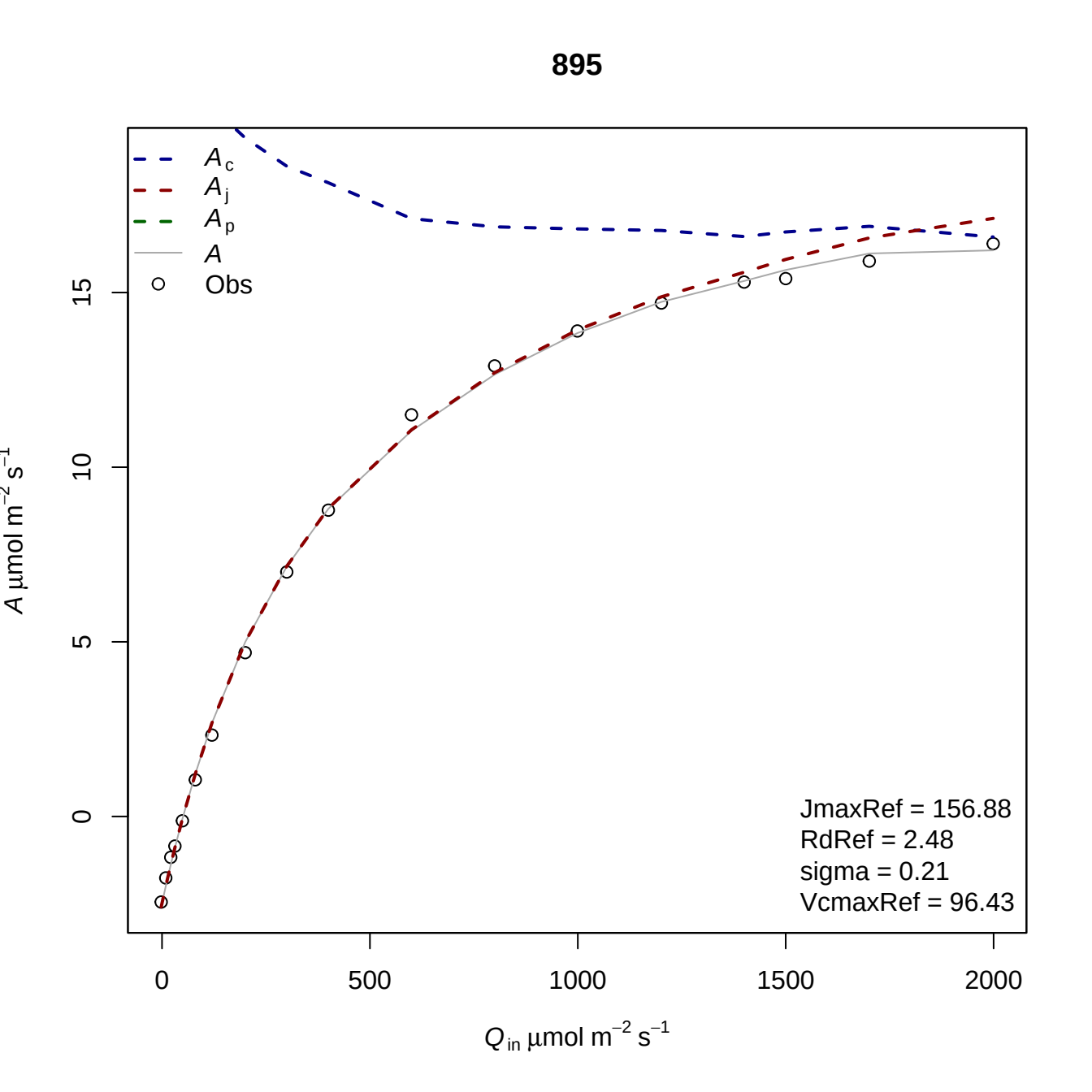

890
A_c
A_j
A_p
A
Obs
JmaxRef = 88.46
RdRef = 1.66
sigma = 0.37
VcmaxRef = 91192.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

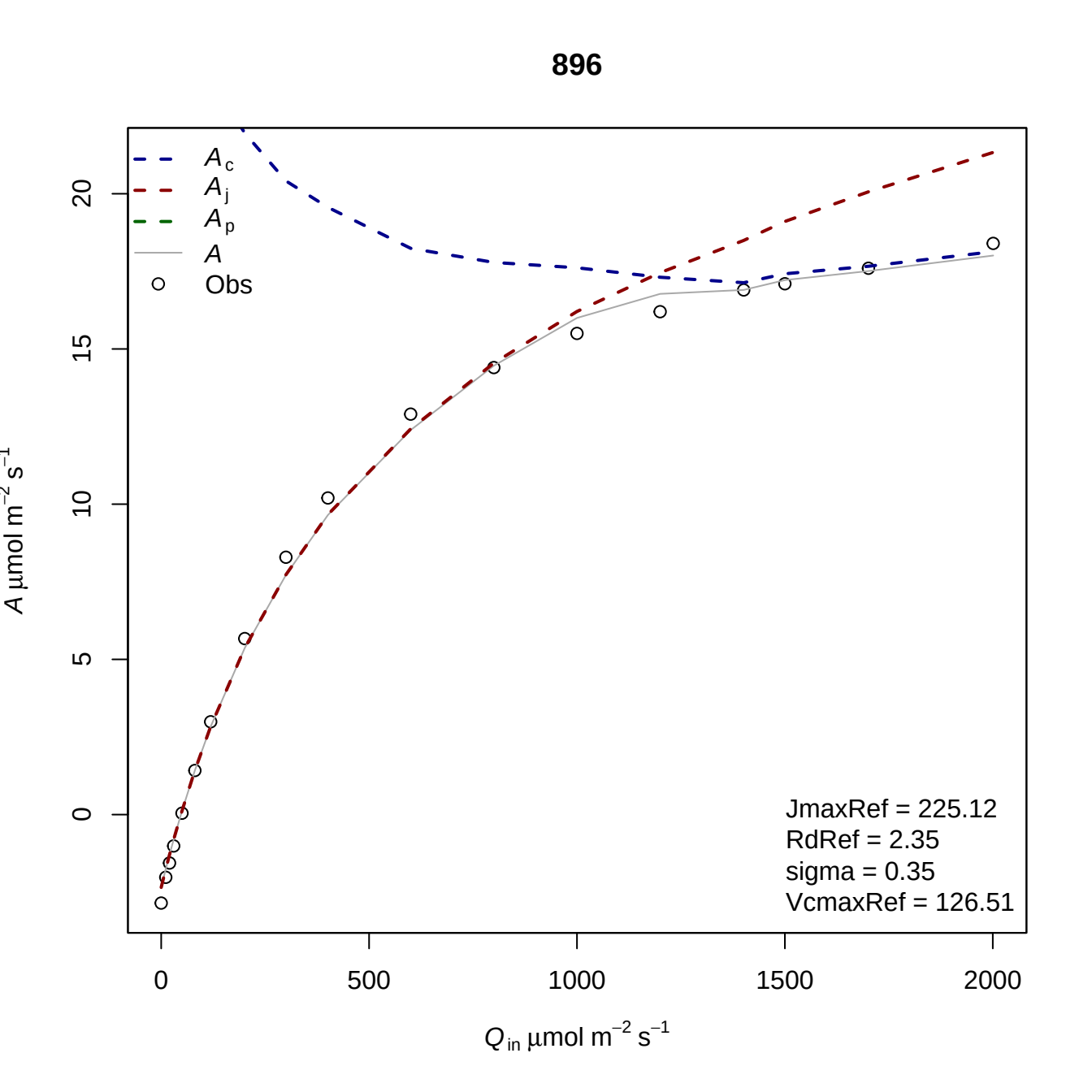

891
A_c
A_j
A_p
A
Obs
JmaxRef = 77.89
RdRef = 0.9
sigma = 0.27
VcmaxRef = 56.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

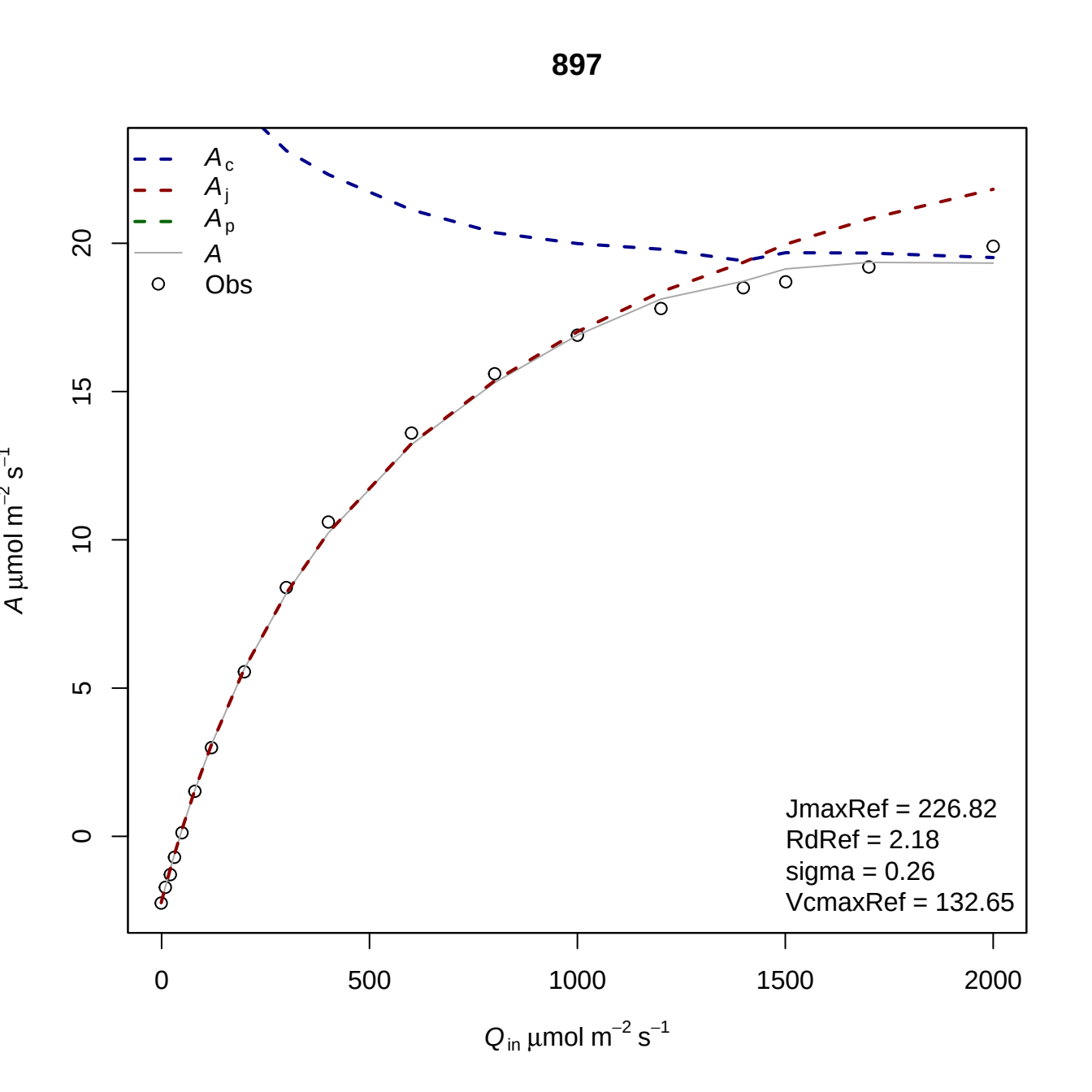

892
A_c
A_j
A_p
A
Obs
JmaxRef = 45.79
RdRef = 2.19
sigma = 0.18
VcmaxRef = 41972.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
5
4
3
2
1
0
−1
−2
0
500
1000
1500
2000

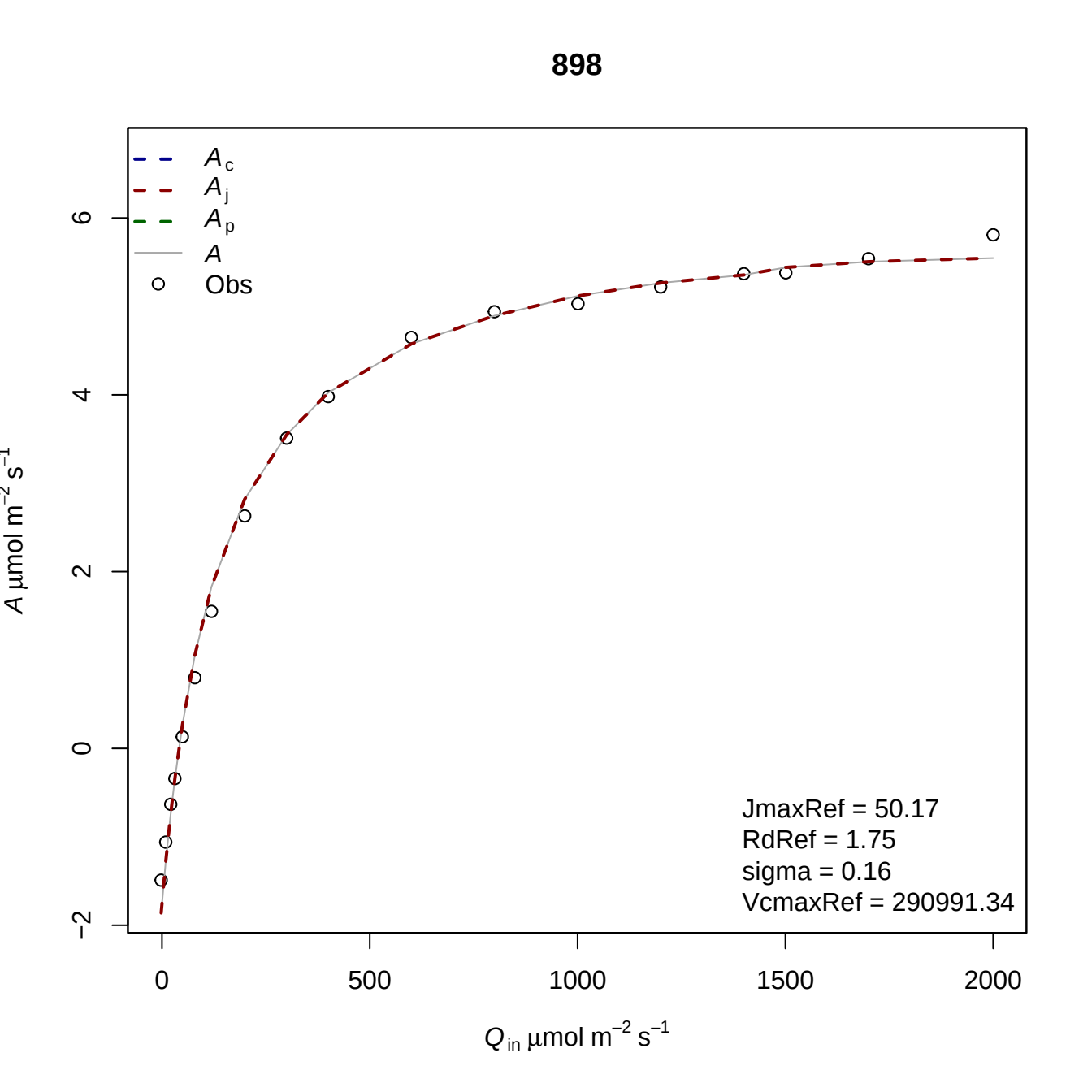

893
A_c
A_j
A_p
A
Obs
JmaxRef = 79.63
RdRef = 2.22
sigma = 0.2
VcmaxRef = 19458.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

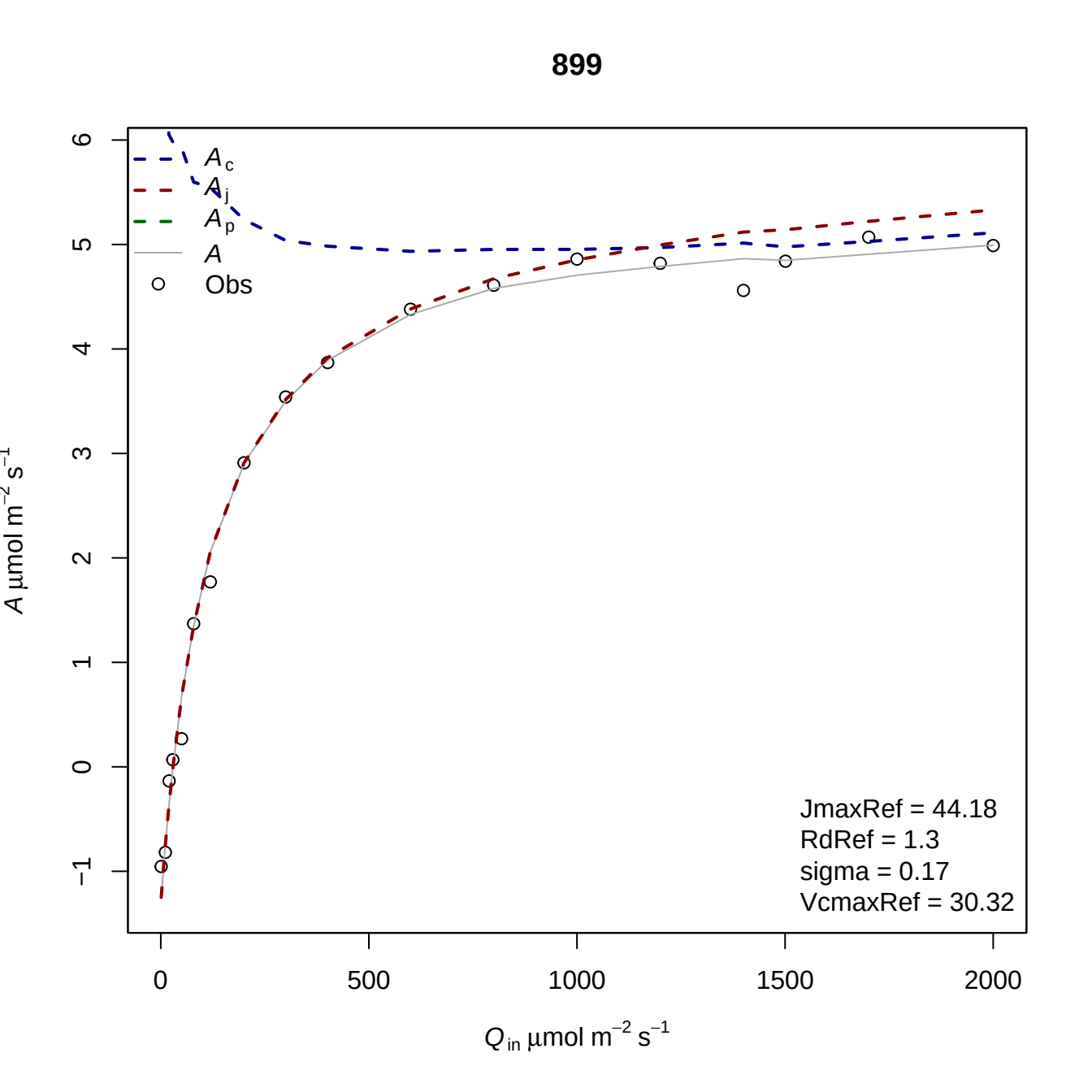

894
A_c
A_j
A_p
A
Obs
JmaxRef = 143.01
RdRef = 2.61
sigma = 0.19
VcmaxRef = 82.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

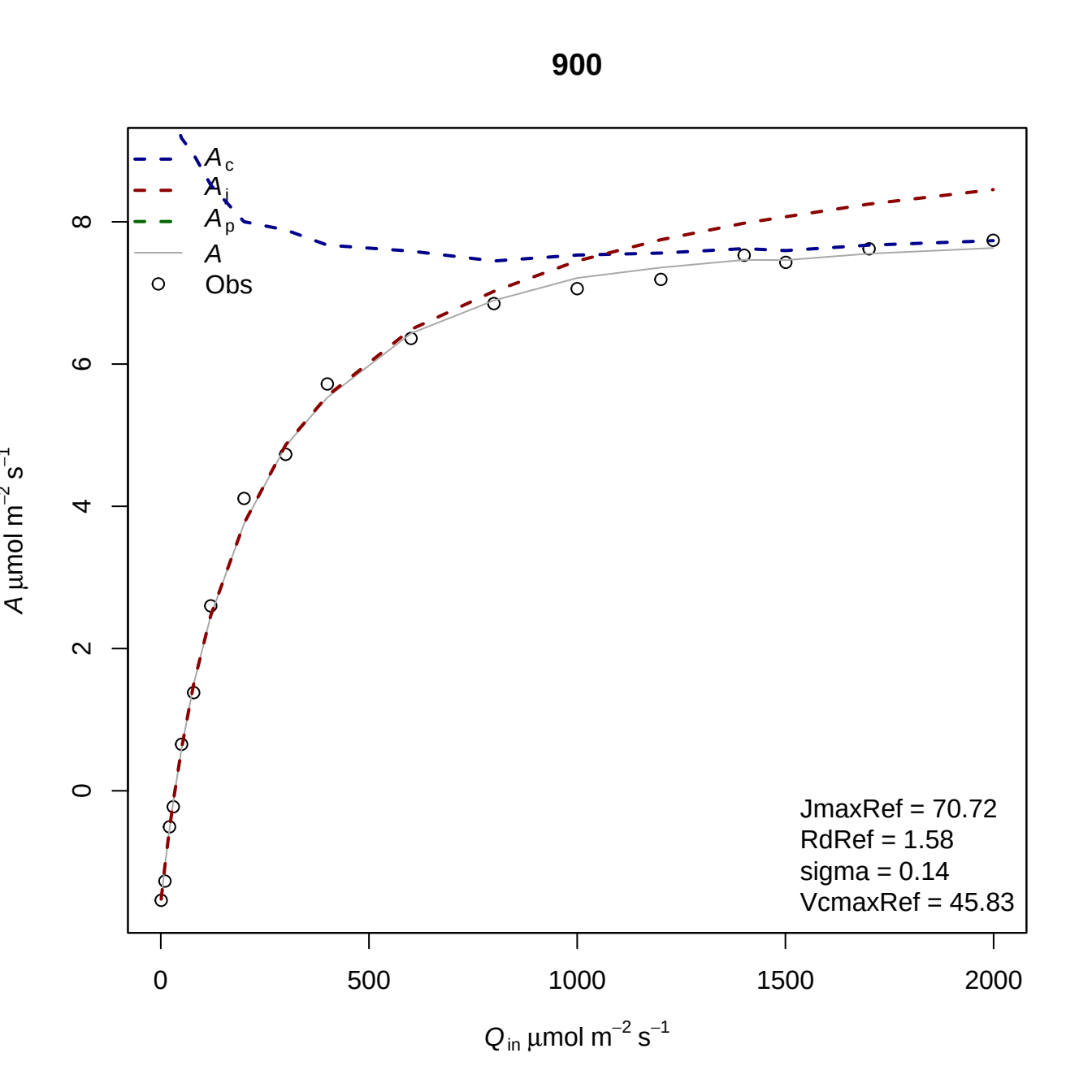

895
A_c
A_j
A_p
A
Obs
JmaxRef = 156.88
RdRef = 2.48
sigma = 0.21
VcmaxRef = 96.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

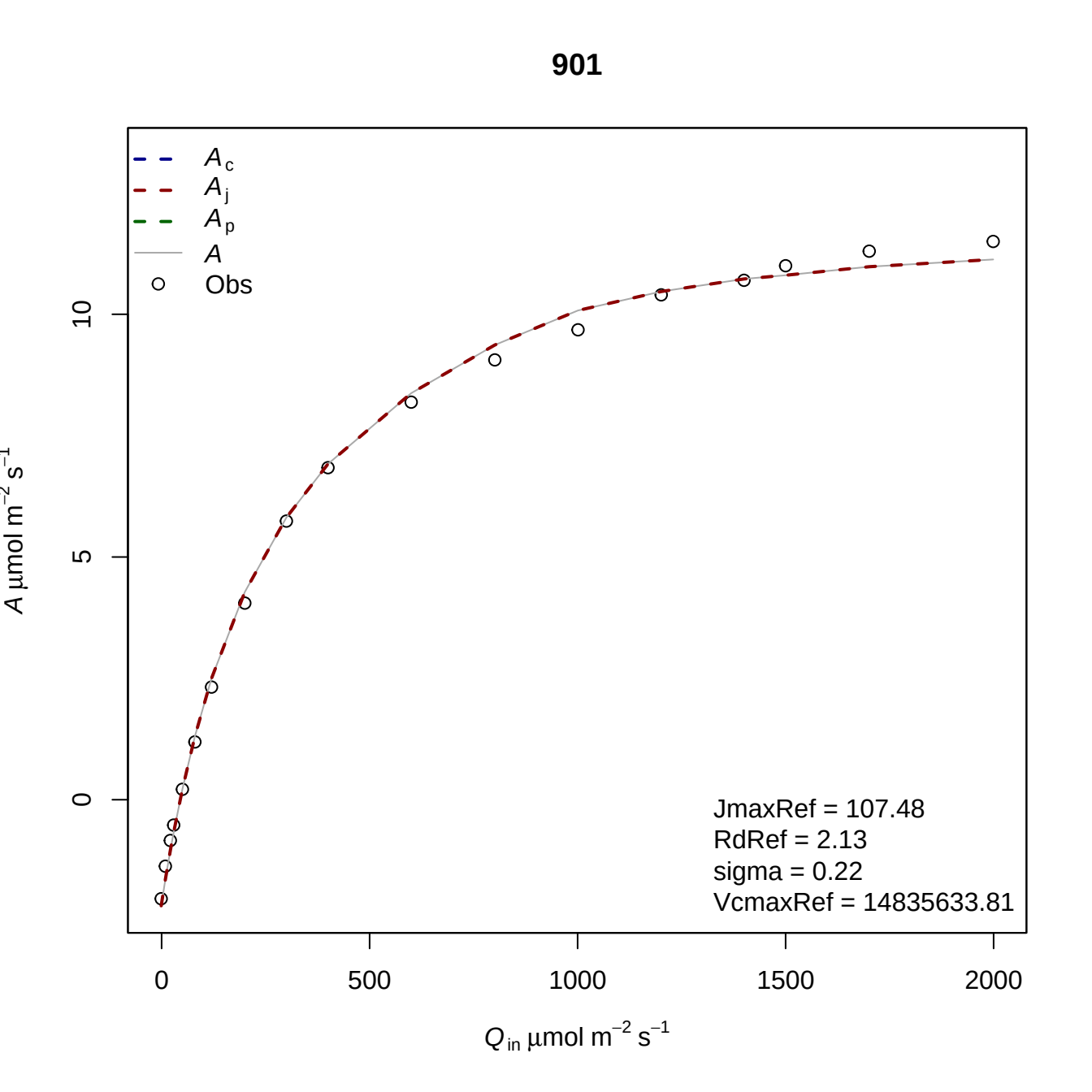

896
A_c
A_j
A_p
A
Obs
JmaxRef = 225.12
RdRef = 2.35
sigma = 0.35
VcmaxRef = 126.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

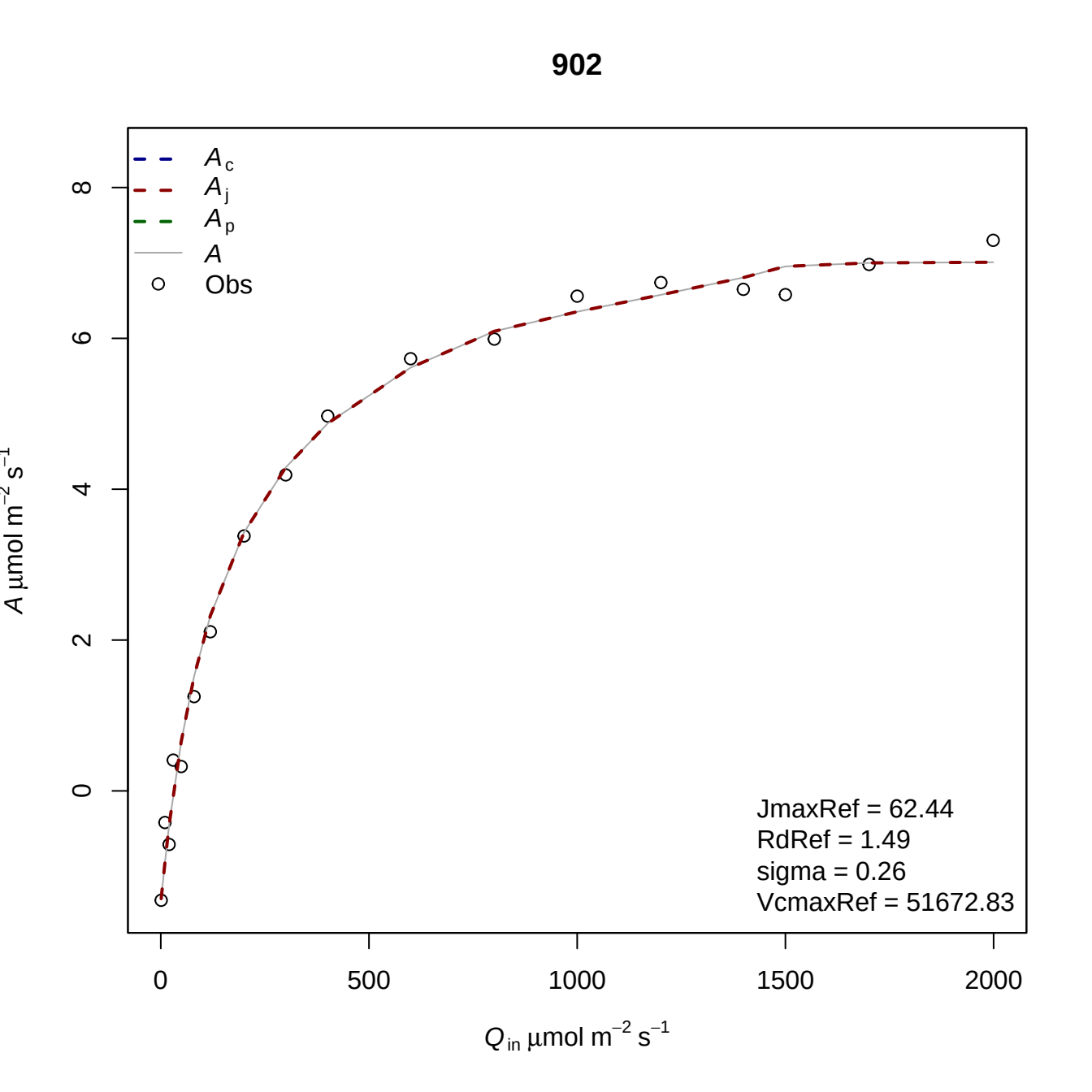

897
A_c
A_j
A_p
A
Obs
JmaxRef = 226.82
RdRef = 2.18
sigma = 0.26
VcmaxRef = 132.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

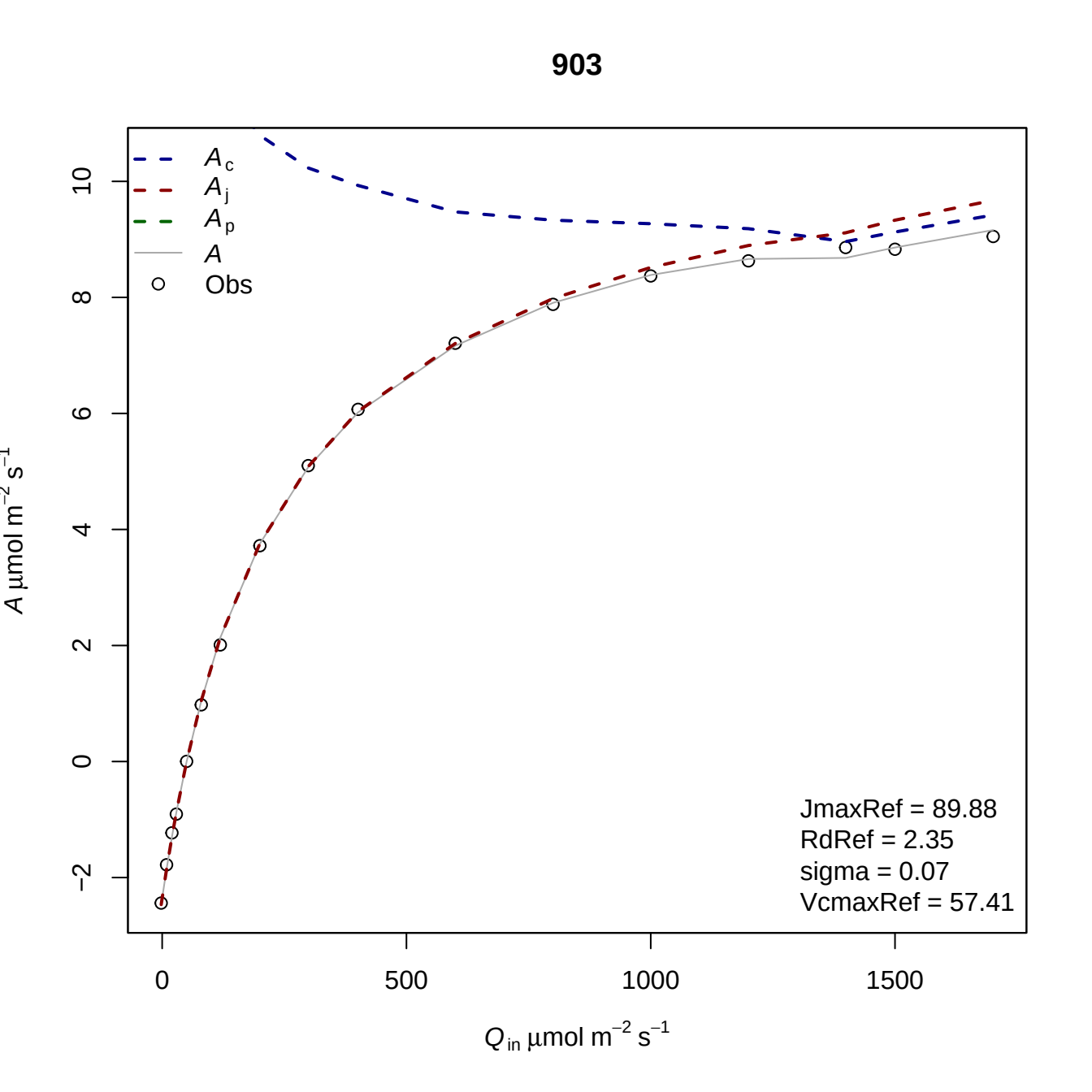

898
A_c
A_j
A_p
A
Obs
JmaxRef = 50.17
RdRef = 1.75
sigma = 0.16
VcmaxRef = 290991.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

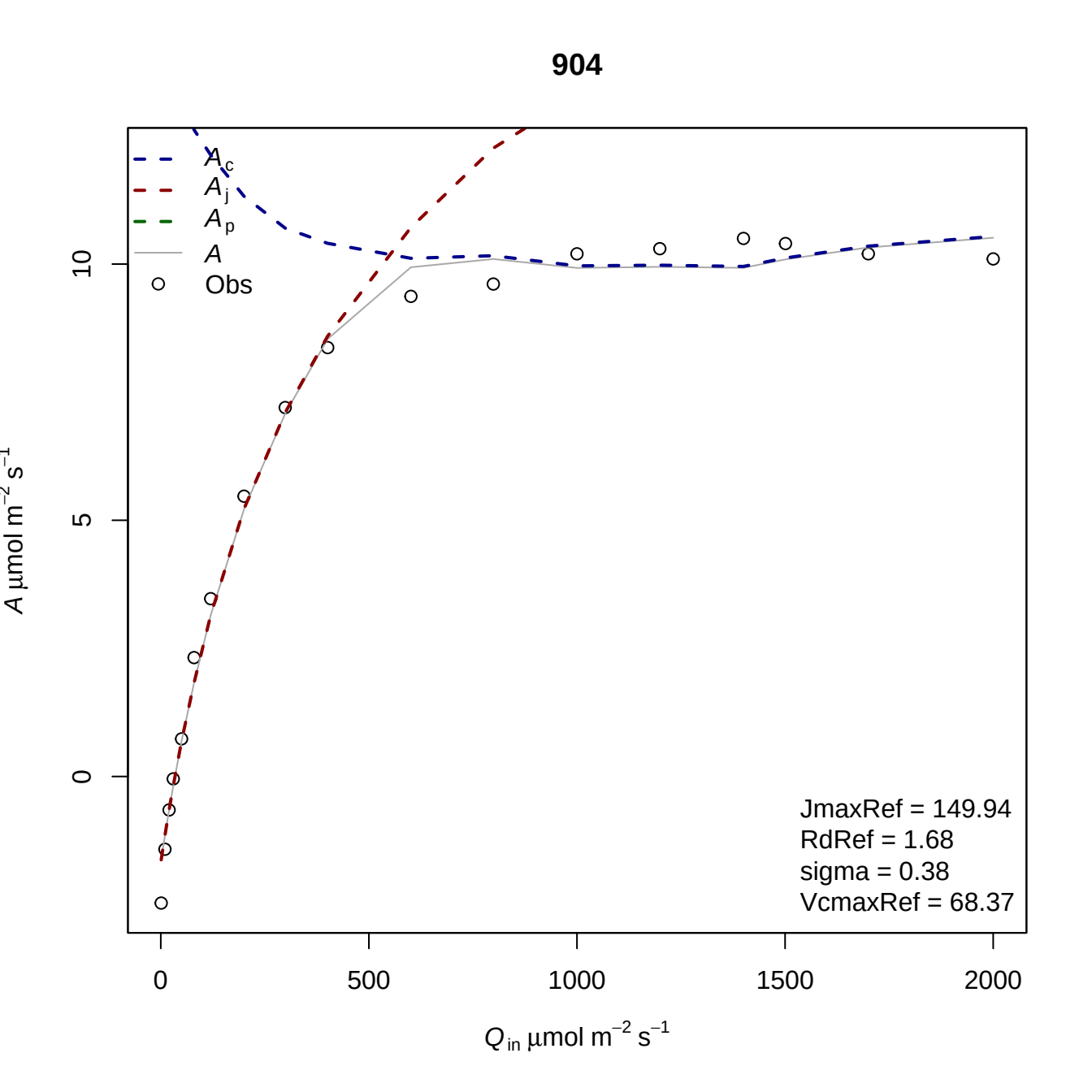

899
A_c
A_j
A_p
A
Obs
JmaxRef = 44.18
RdRef = 1.3
sigma = 0.17
VcmaxRef = 30.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

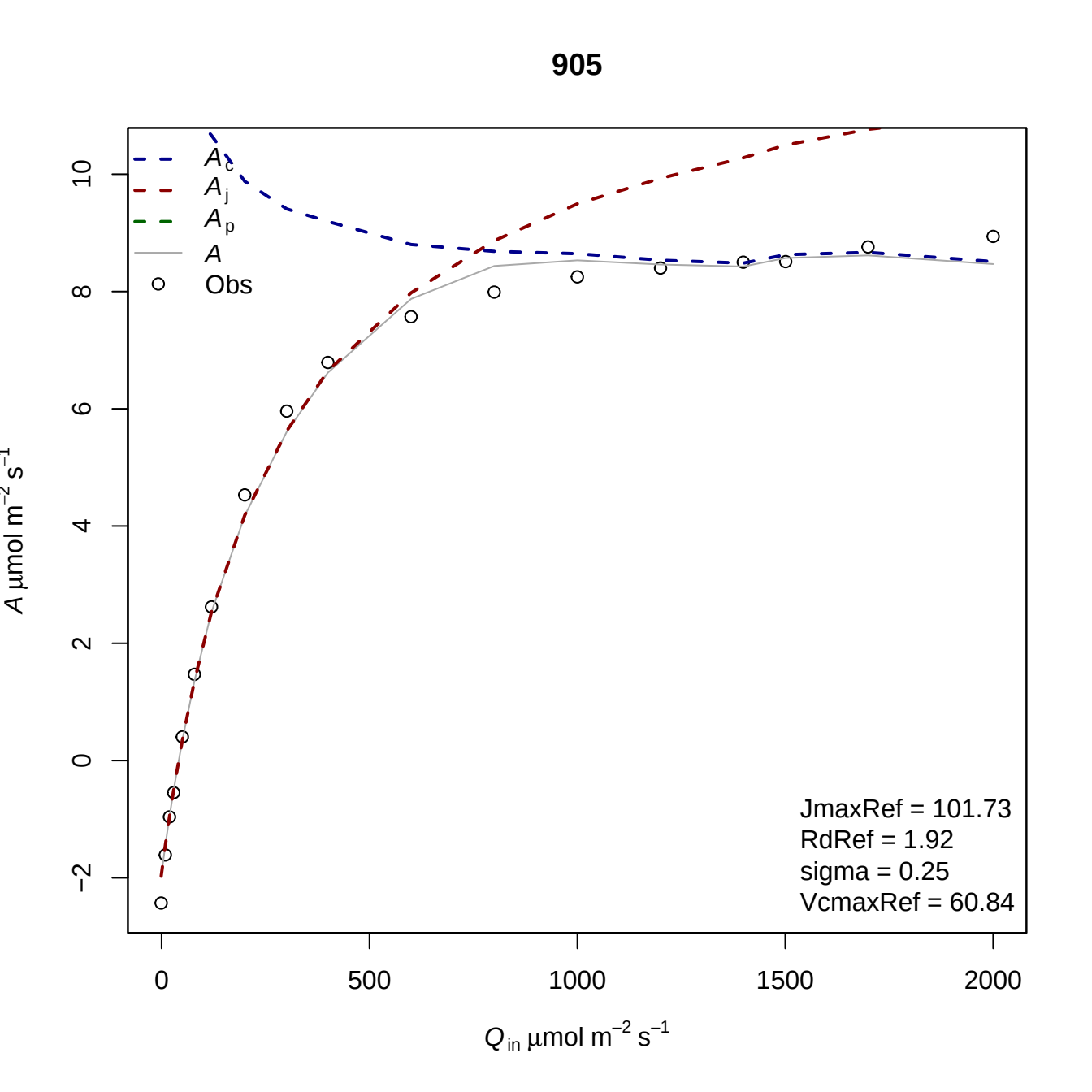

900
A_c
A_j
A_p
A
Obs
JmaxRef = 70.72
RdRef = 1.58
sigma = 0.14
VcmaxRef = 45.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

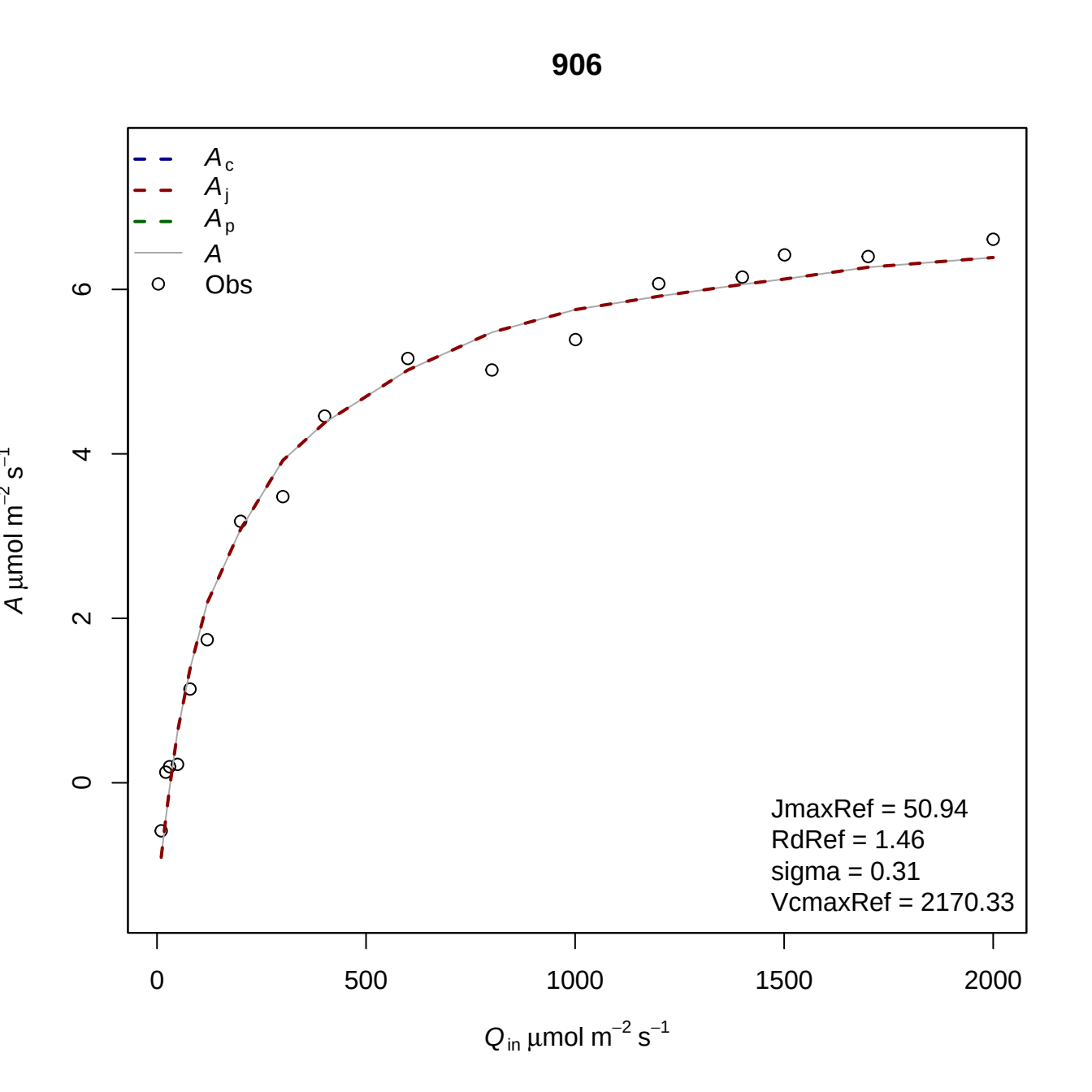

901
A_c
A_j
A_p
A
Obs
JmaxRef = 107.48
RdRef = 2.13
sigma = 0.22
VcmaxRef = 14835633.81
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

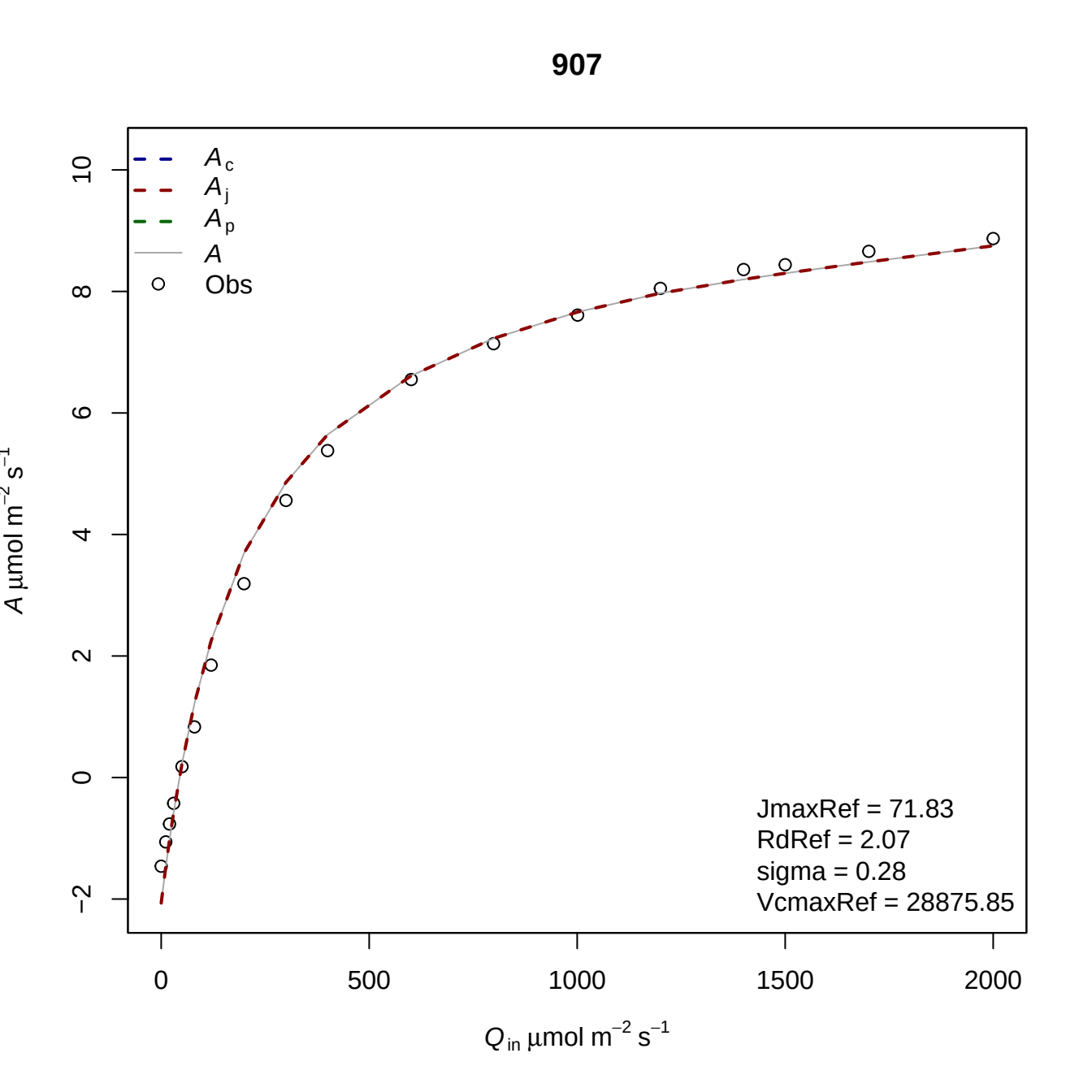

902
A_c
A_j
A_p
A
Obs
JmaxRef = 62.44
RdRef = 1.49
sigma = 0.26
VcmaxRef = 51672.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

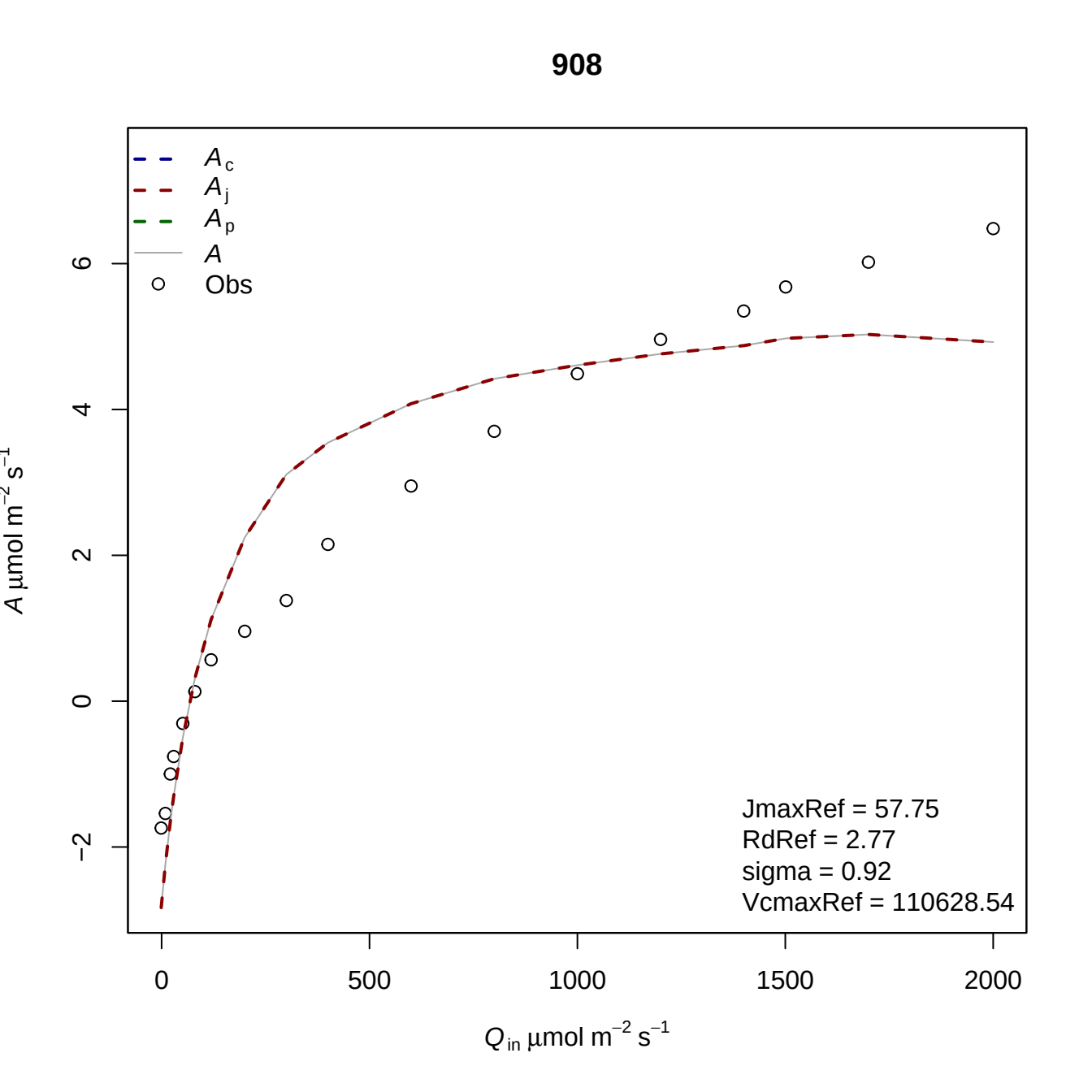

903
A_c
A_j
A_p
A
Obs
JmaxRef = 89.88
RdRef = 2.35
sigma = 0.07
VcmaxRef = 57.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

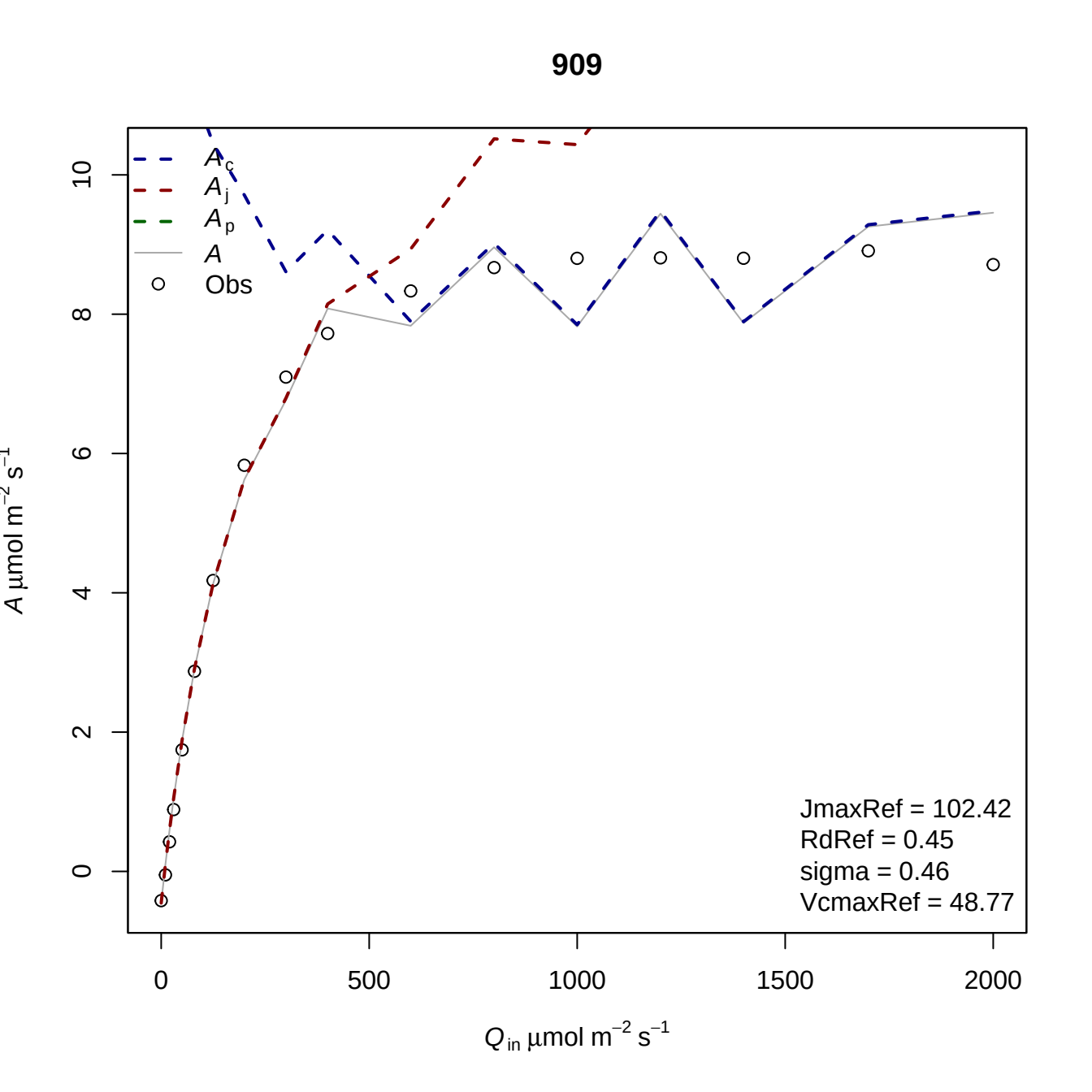

904
A_c
A_j
A_p
A
Obs
JmaxRef = 149.94
RdRef = 1.68
sigma = 0.38
VcmaxRef = 68.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

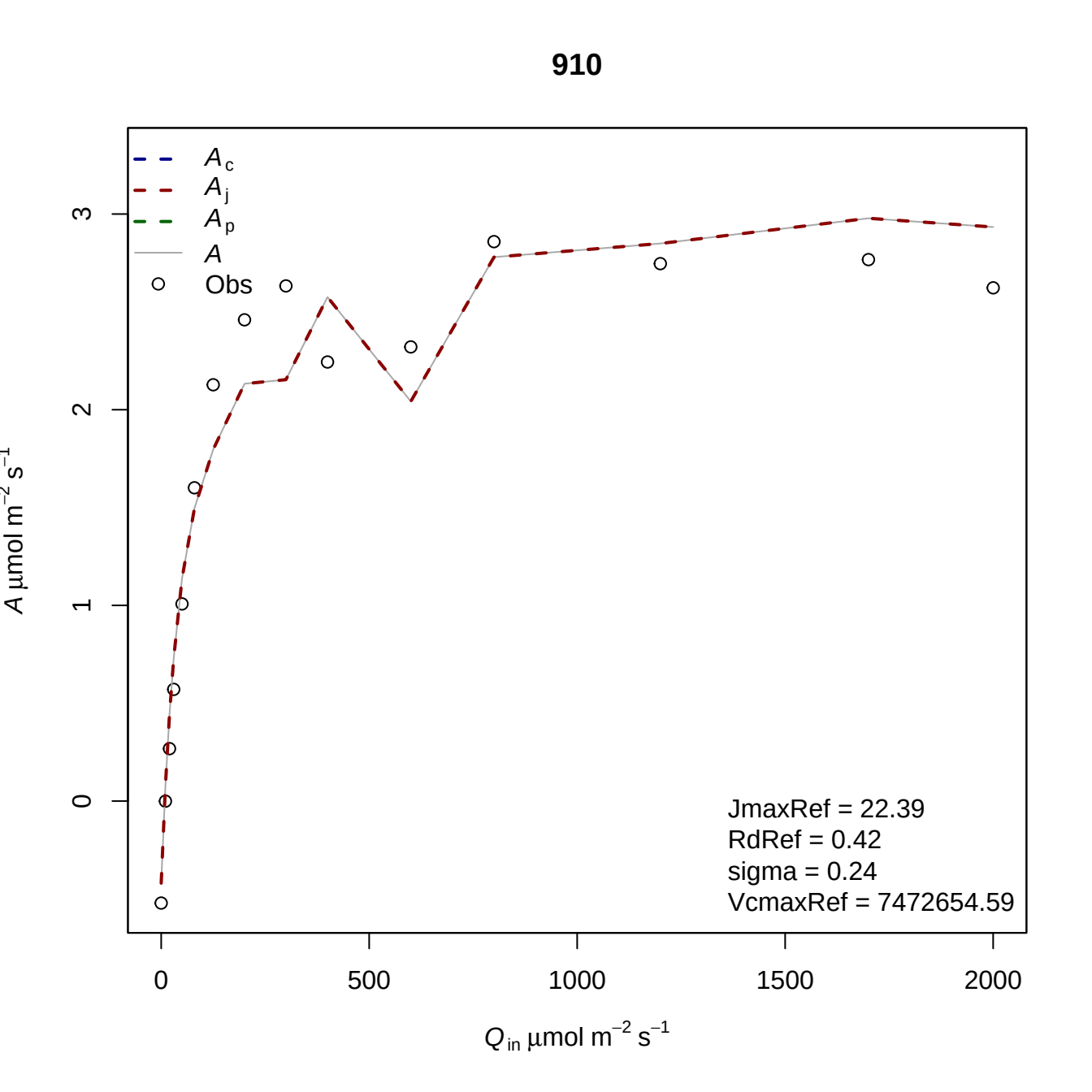

905
A_c
A_j
A_p
A
Obs
JmaxRef = 101.73
RdRef = 1.92
sigma = 0.25
VcmaxRef = 60.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

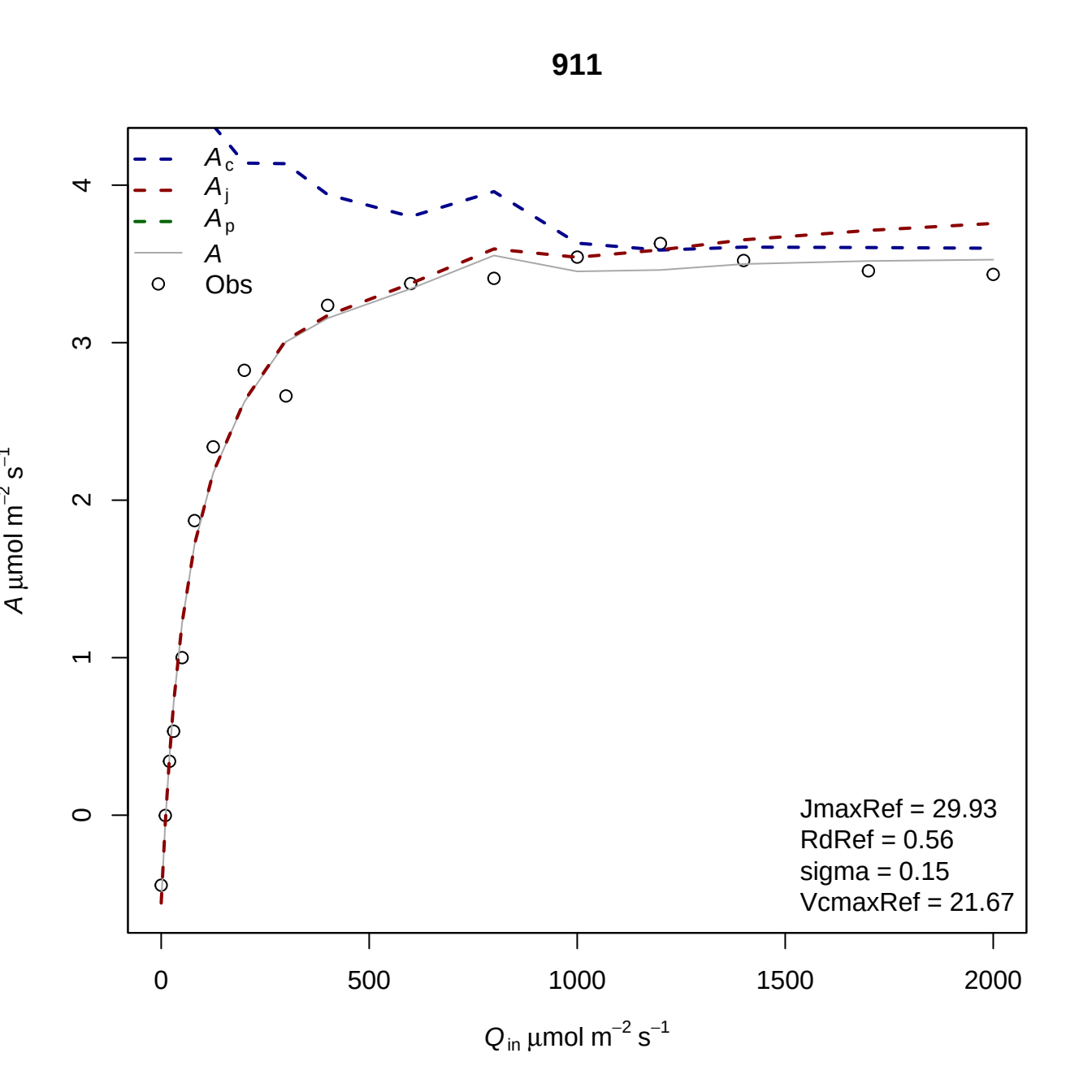

906
A_c
A_j
A_p
A
Obs
JmaxRef = 50.94
RdRef = 1.46
sigma = 0.31
VcmaxRef = 2170.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

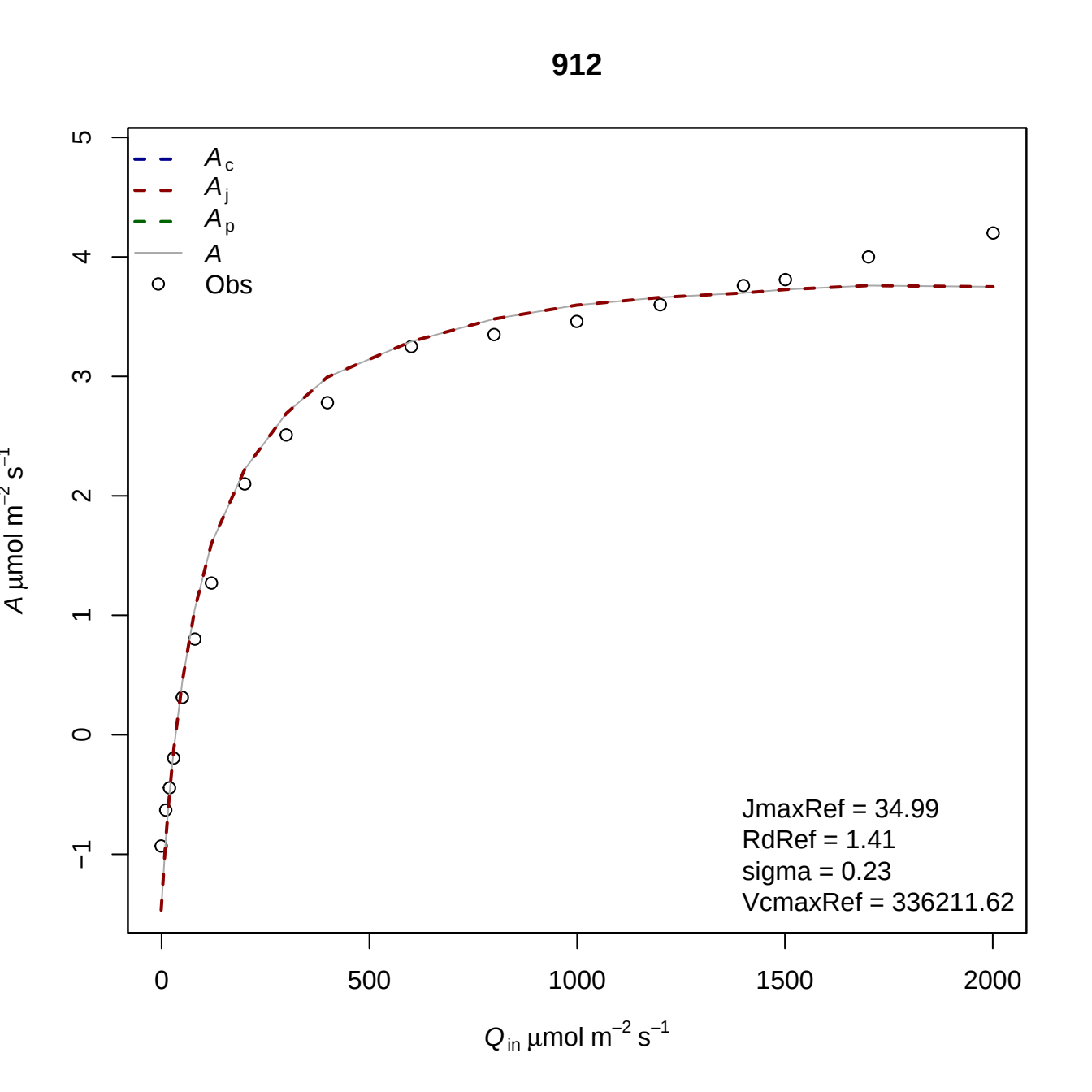

907
A_c
A_j
A_p
A
Obs
JmaxRef = 71.83
RdRef = 2.07
sigma = 0.28
VcmaxRef = 28875.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

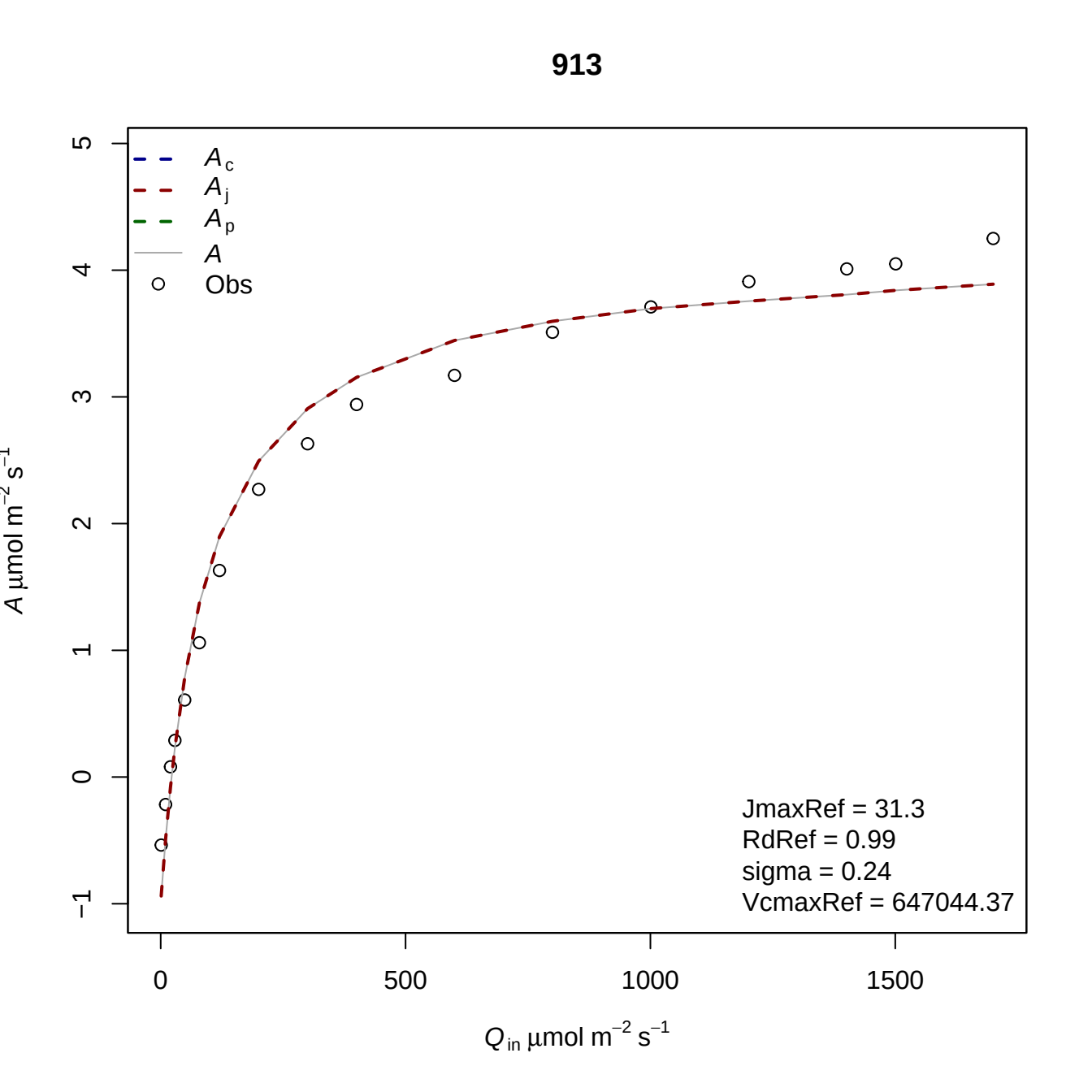

908
A_c
A_j
A_p
A
Obs
JmaxRef = 57.75
RdRef = 2.77
sigma = 0.92
VcmaxRef = 110628.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

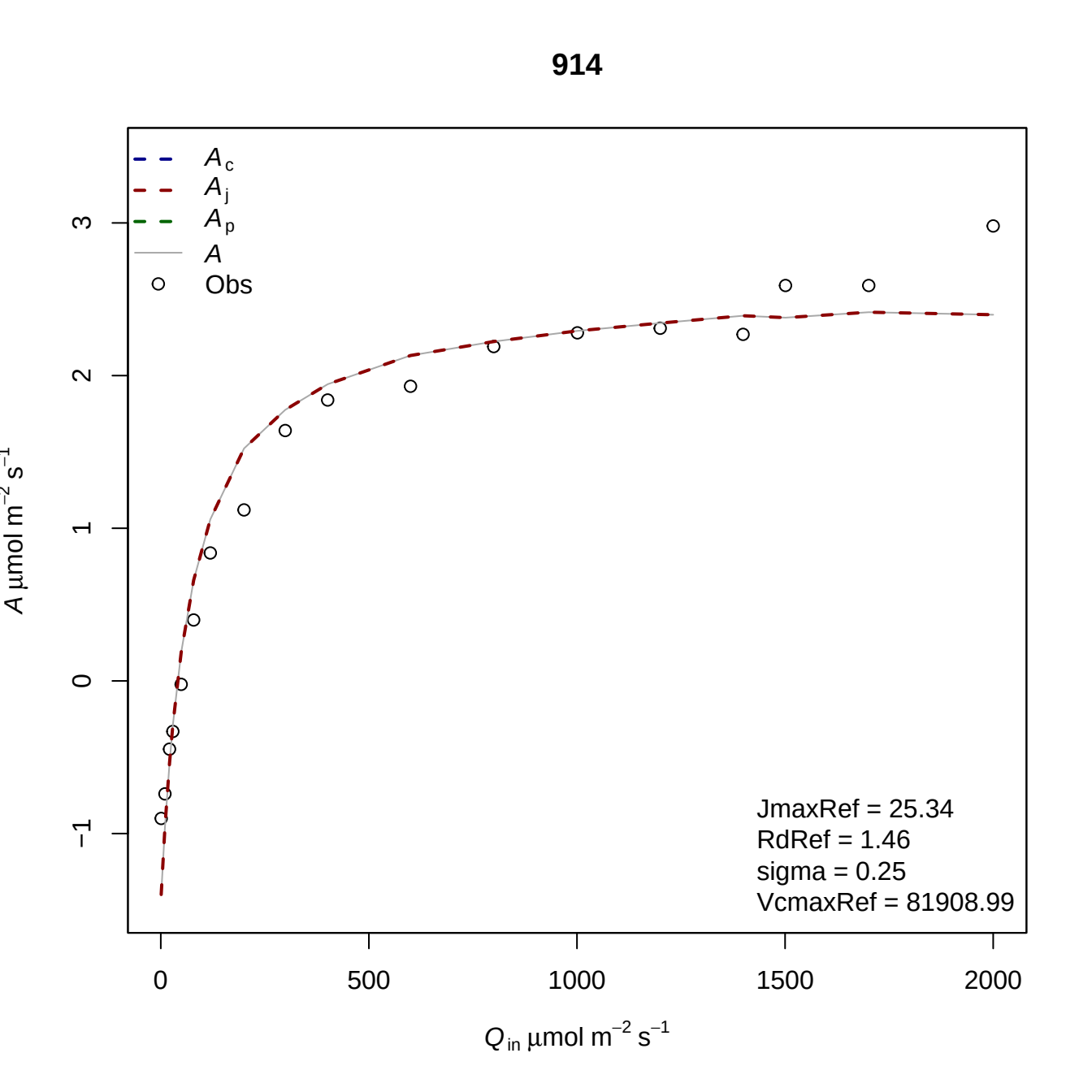

909
A_c
A_j
A_p
A
Obs
JmaxRef = 102.42
RdRef = 0.45
sigma = 0.46
VcmaxRef = 48.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

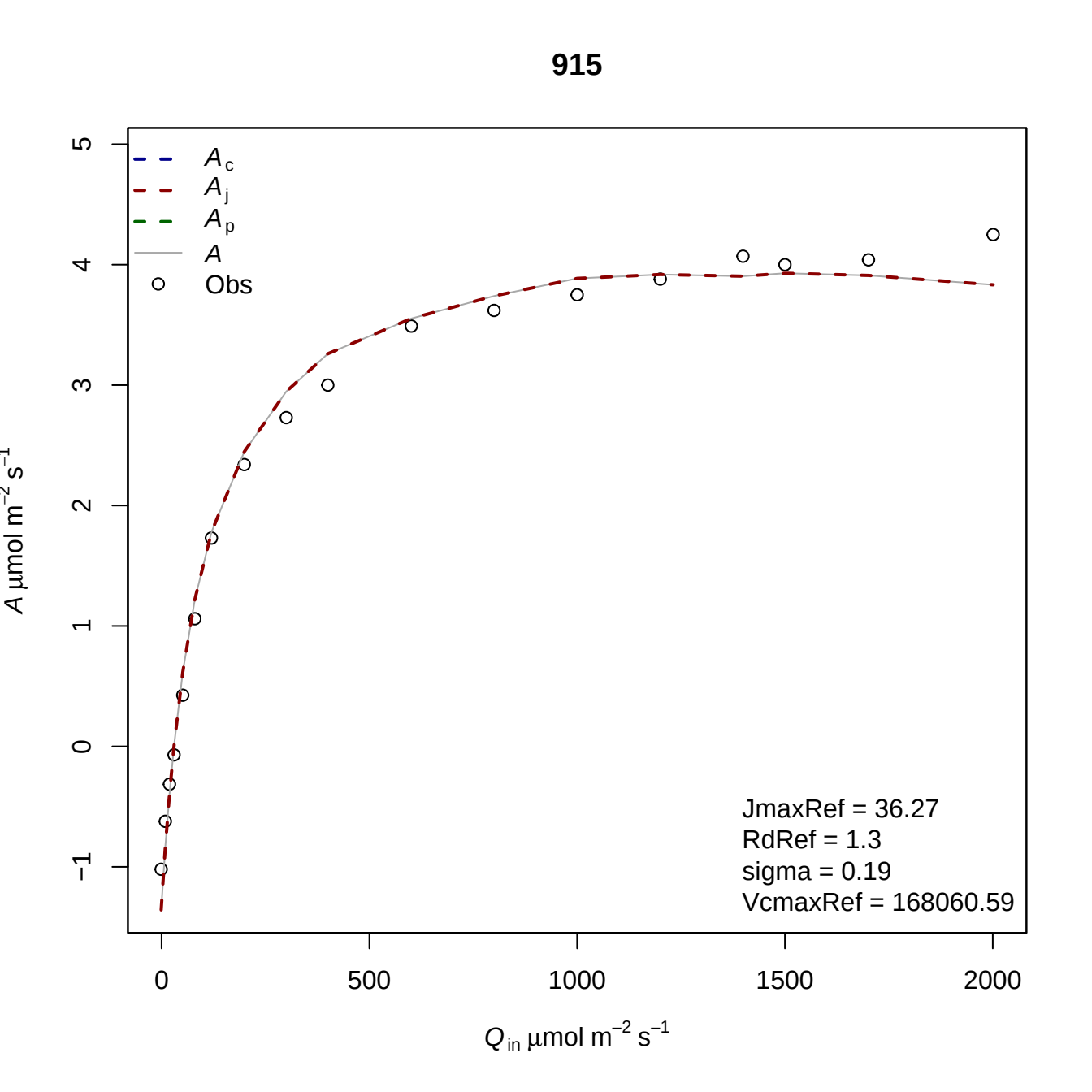

910
A_c
A_j
A_p
A
Obs
JmaxRef = 22.39
RdRef = 0.42
sigma = 0.24
VcmaxRef = 7472654.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500
2000

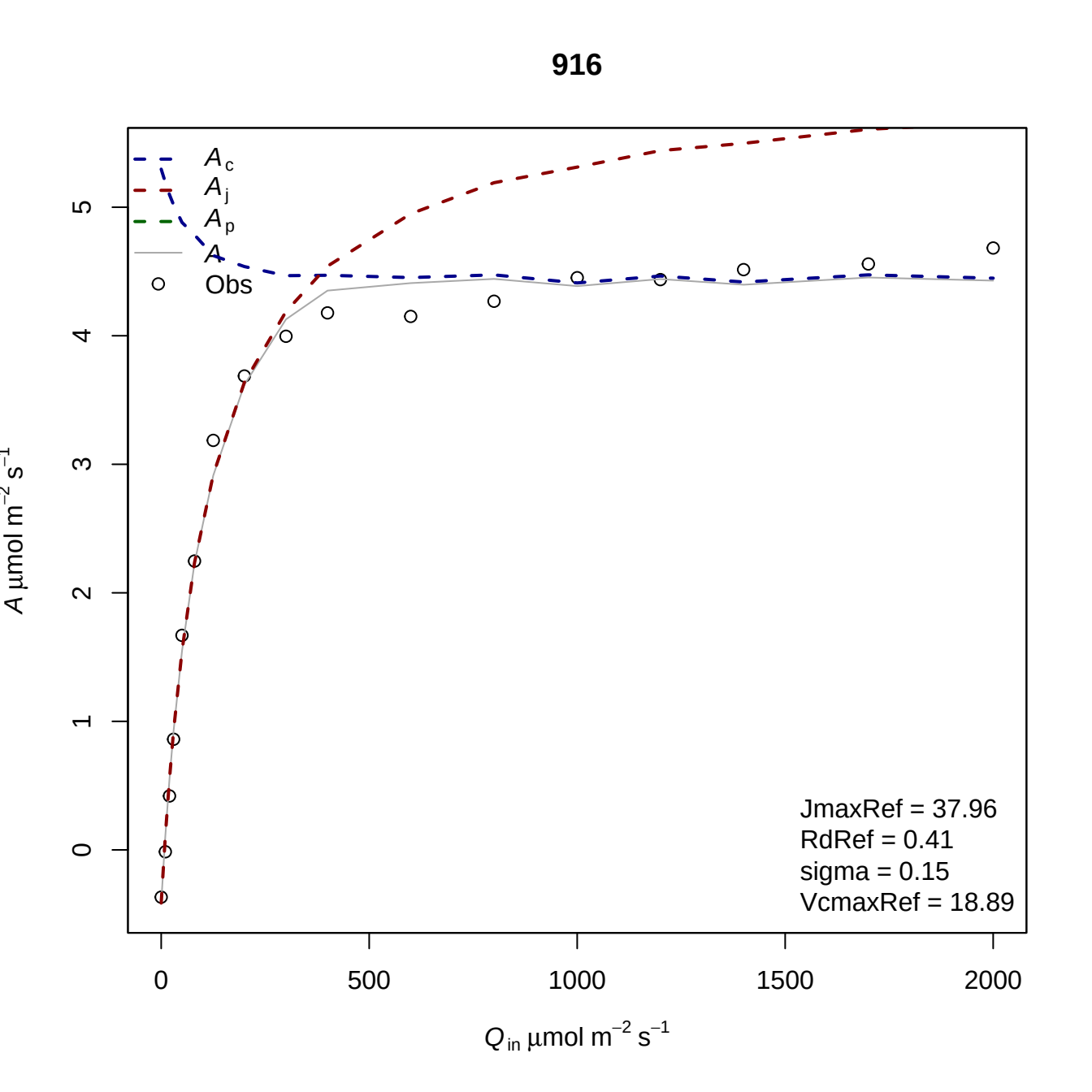

911
A_c
A_j
A_p
A
Obs
JmaxRef = 29.93
RdRef = 0.56
sigma = 0.15
VcmaxRef = 21.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500
2000

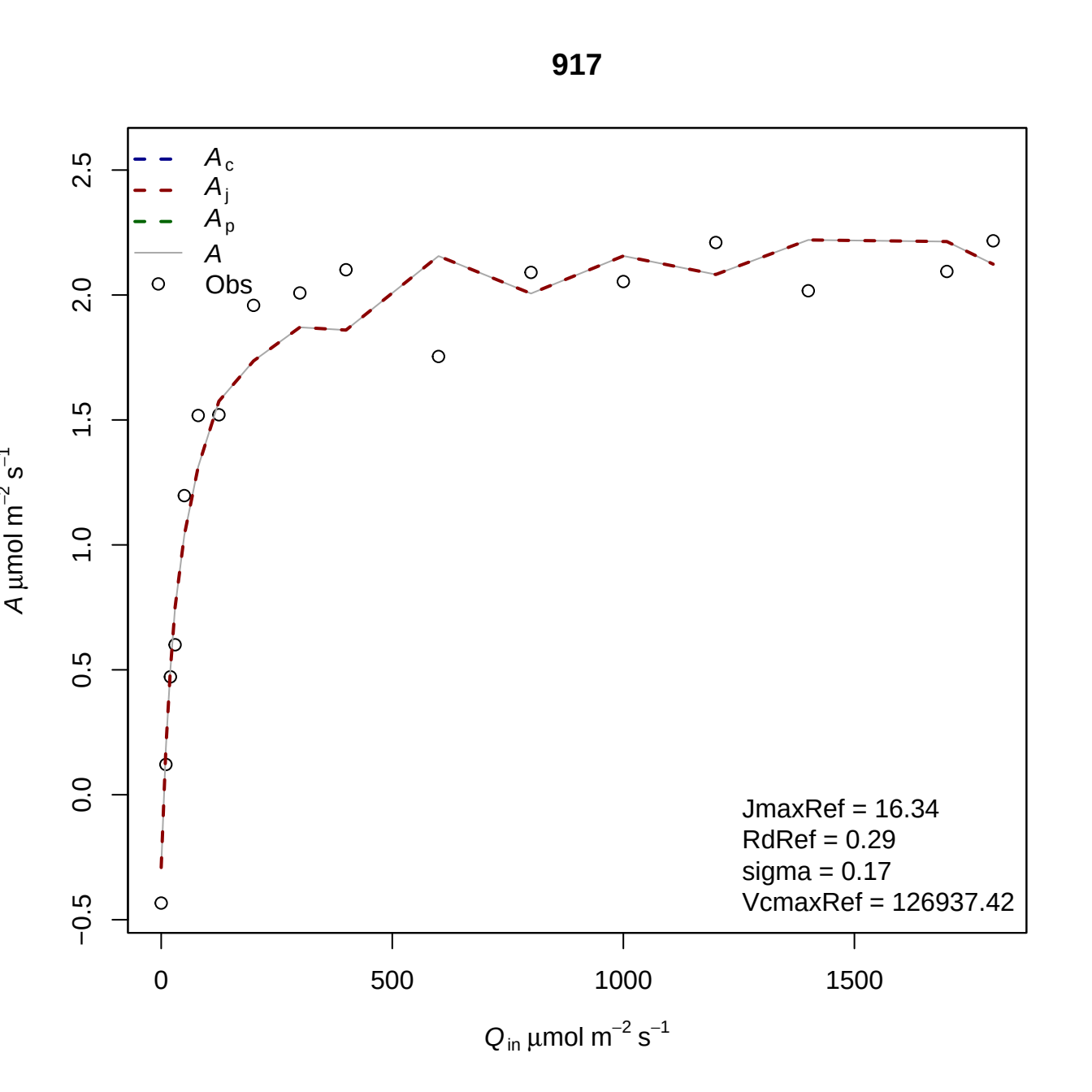

912
A_c
A_j
A_p
A
Obs
JmaxRef = 34.99
RdRef = 1.41
sigma = 0.23
VcmaxRef = 336211.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

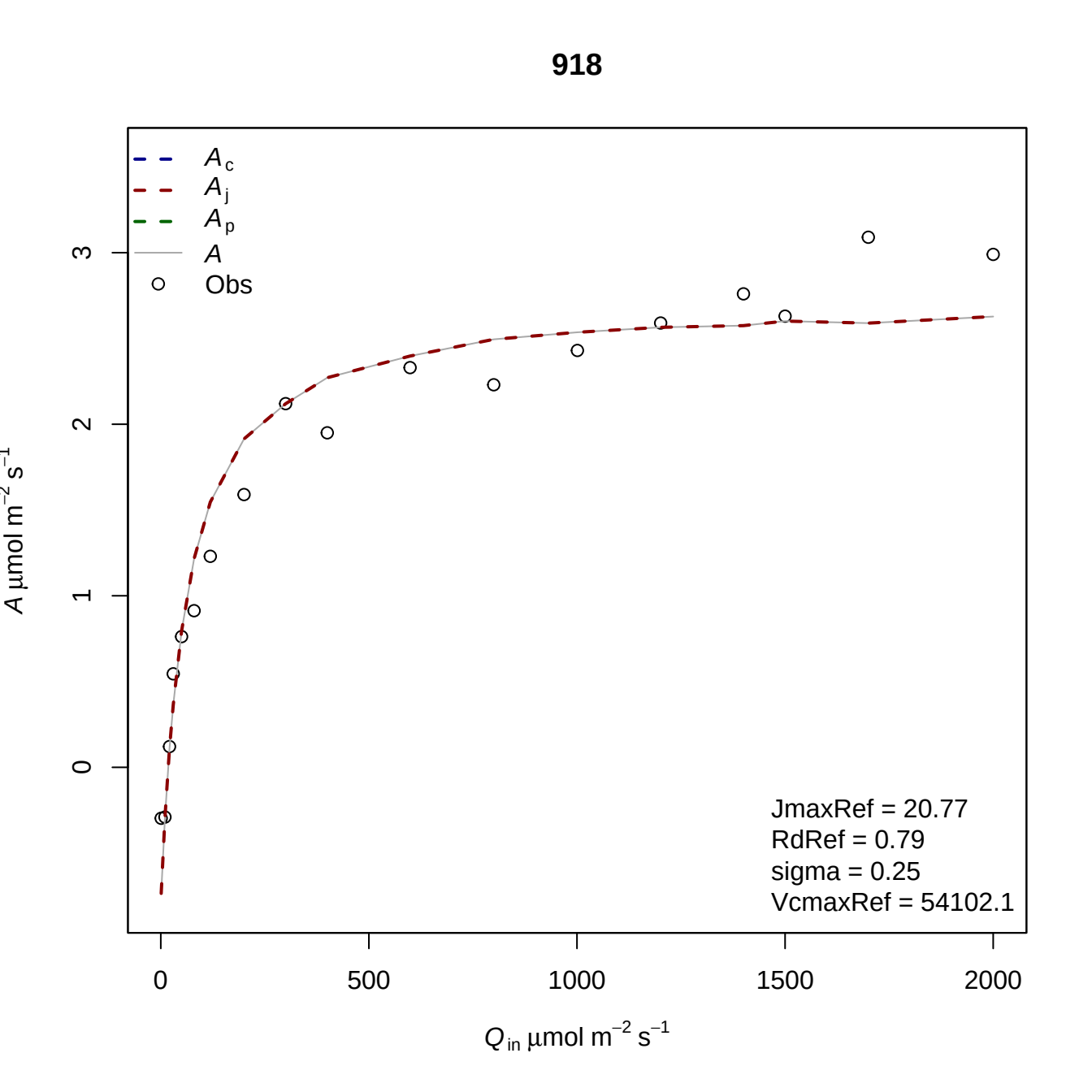

913
A_c
A_j
A_p
A
Obs
JmaxRef = 31.3
RdRef = 0.99
sigma = 0.24
VcmaxRef = 647044.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

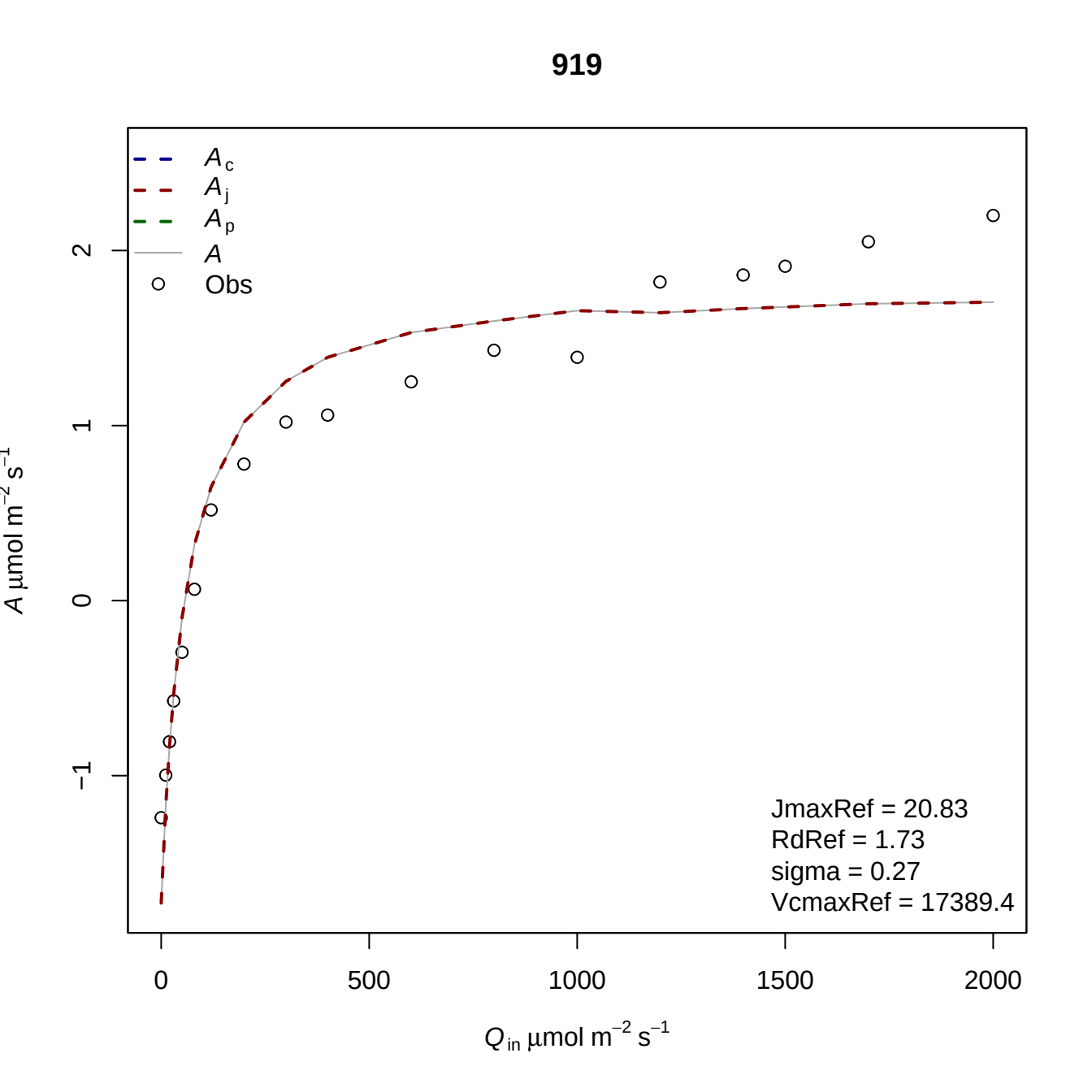

914
A_c
A_j
A_p
A
Obs
JmaxRef = 25.34
RdRef = 1.46
sigma = 0.25
VcmaxRef = 81908.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

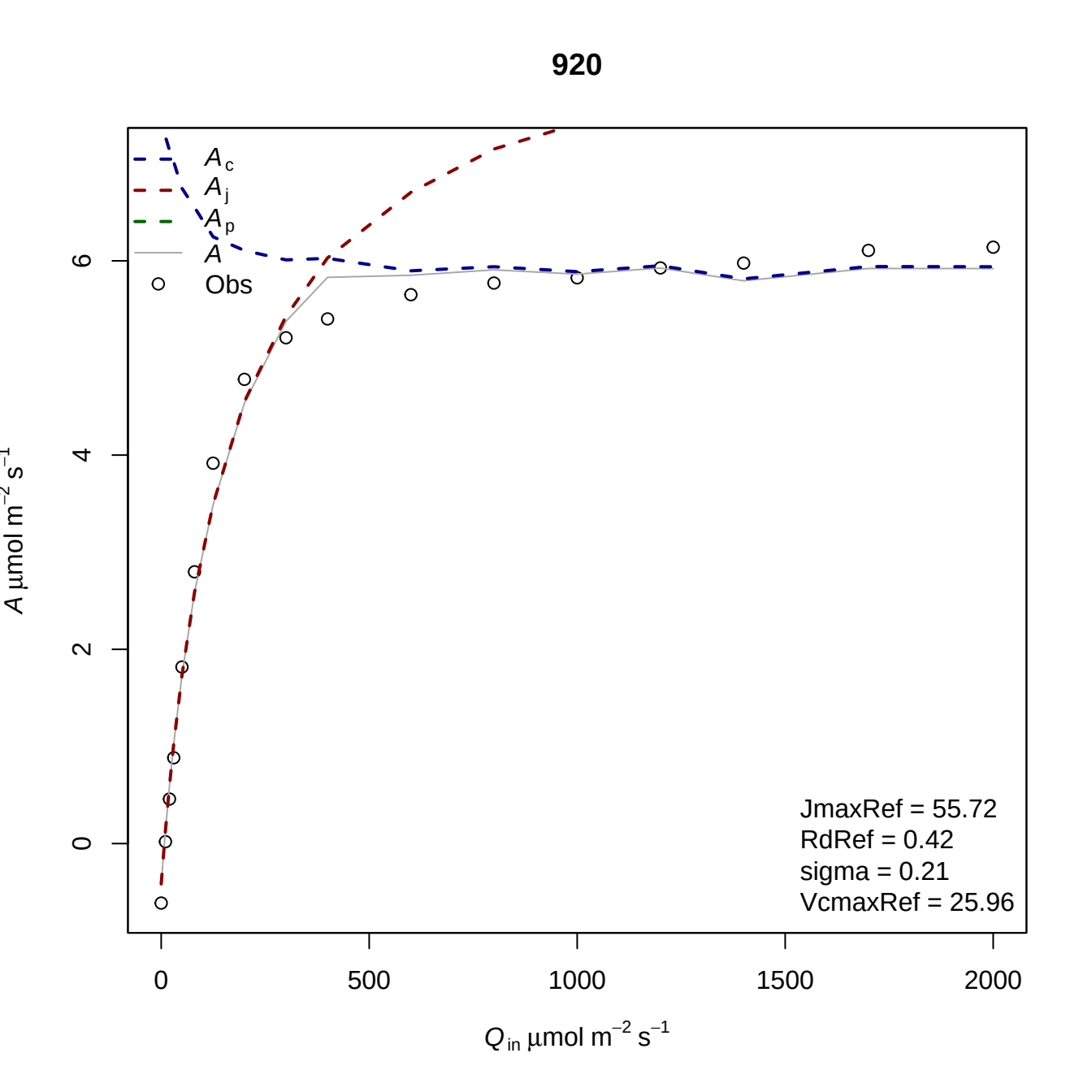

915
A_c
A_j
A_p
A
Obs
JmaxRef = 36.27
RdRef = 1.3
sigma = 0.19
VcmaxRef = 168060.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

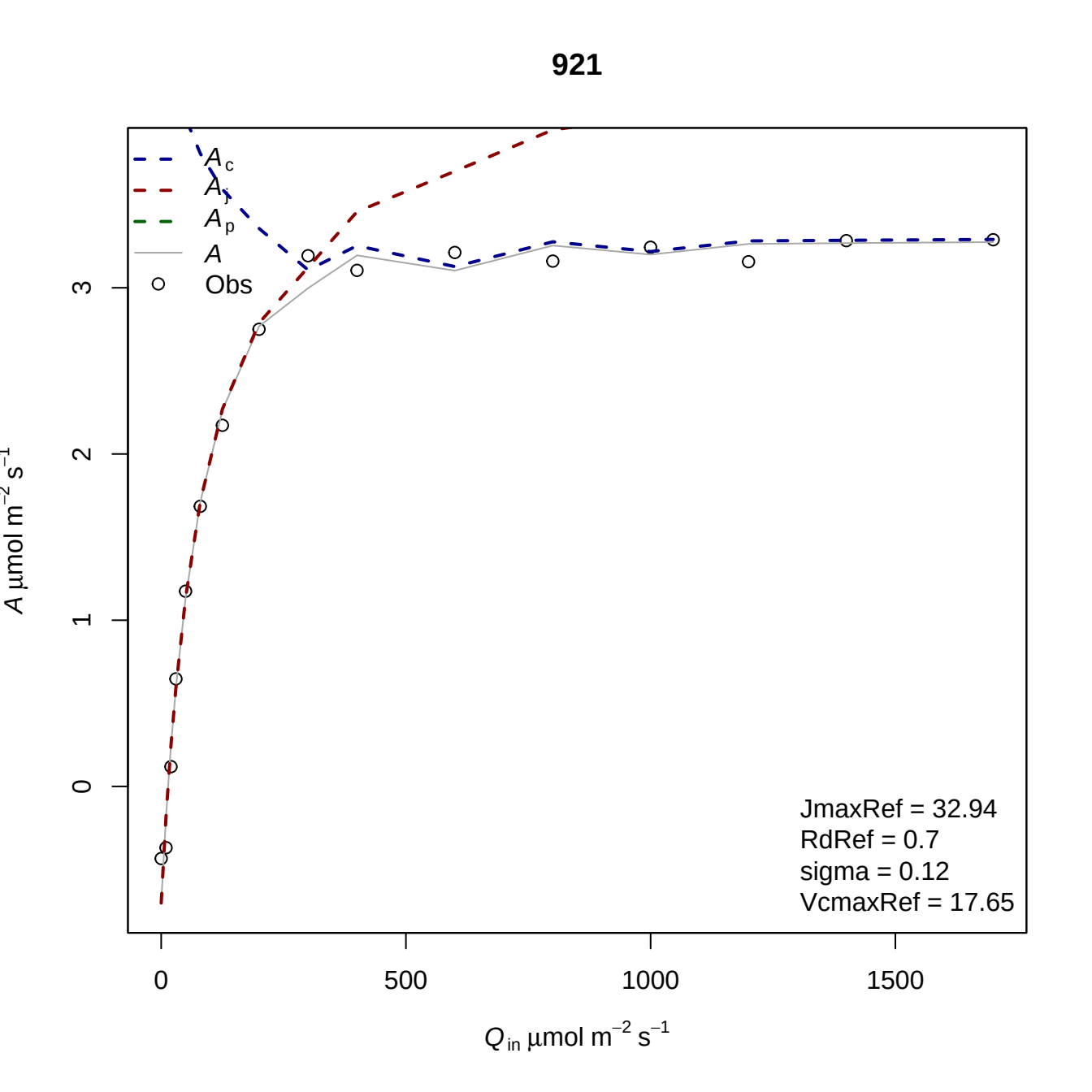

916
A_c
A_j
A_p
A
Obs
JmaxRef = 37.96
RdRef = 0.41
sigma = 0.15
VcmaxRef = 18.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500
2000

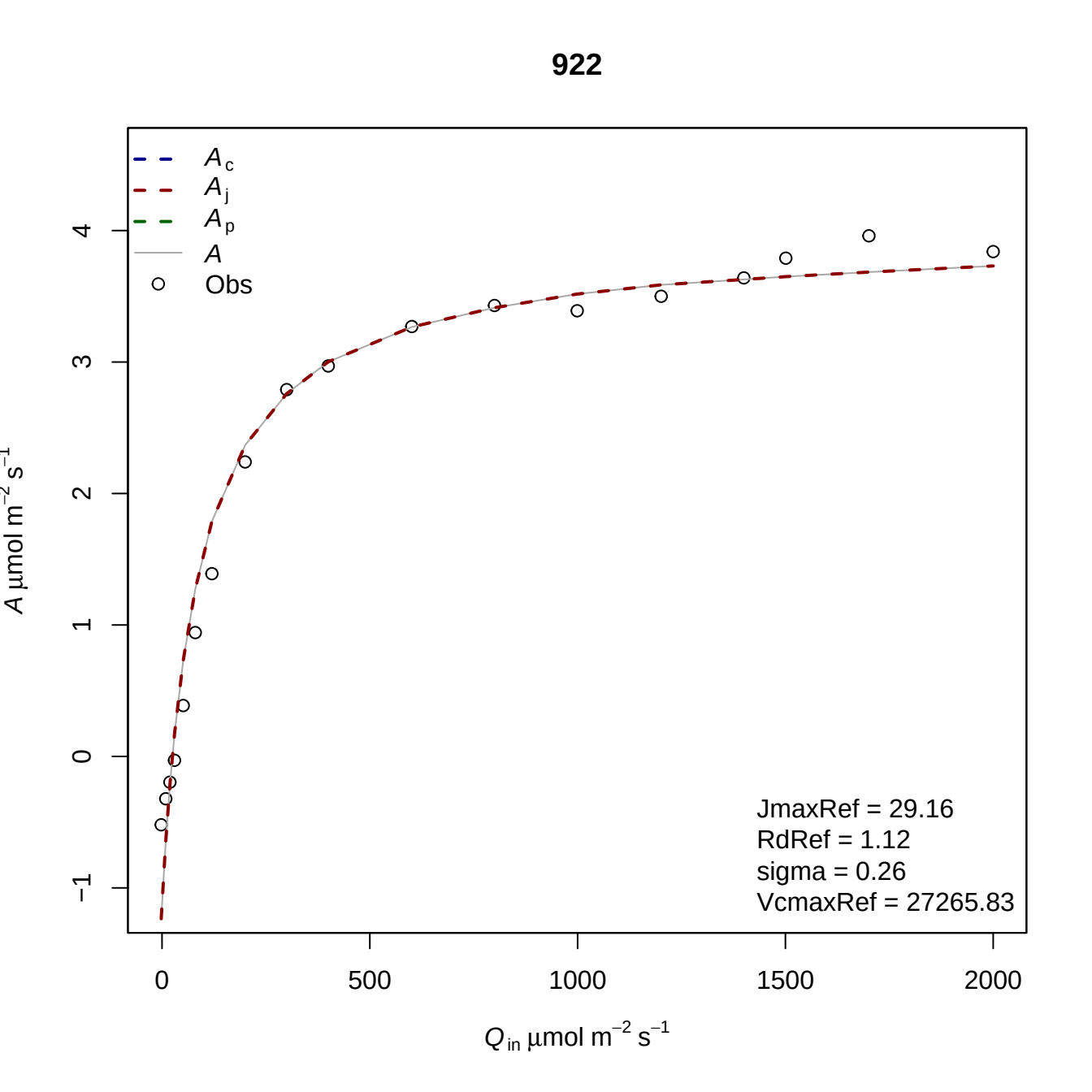

917
A_c
A_j
A_p
A
Obs
JmaxRef = 16.34
RdRef = 0.29
sigma = 0.17
VcmaxRef = 126937.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
0
500
1000
1500

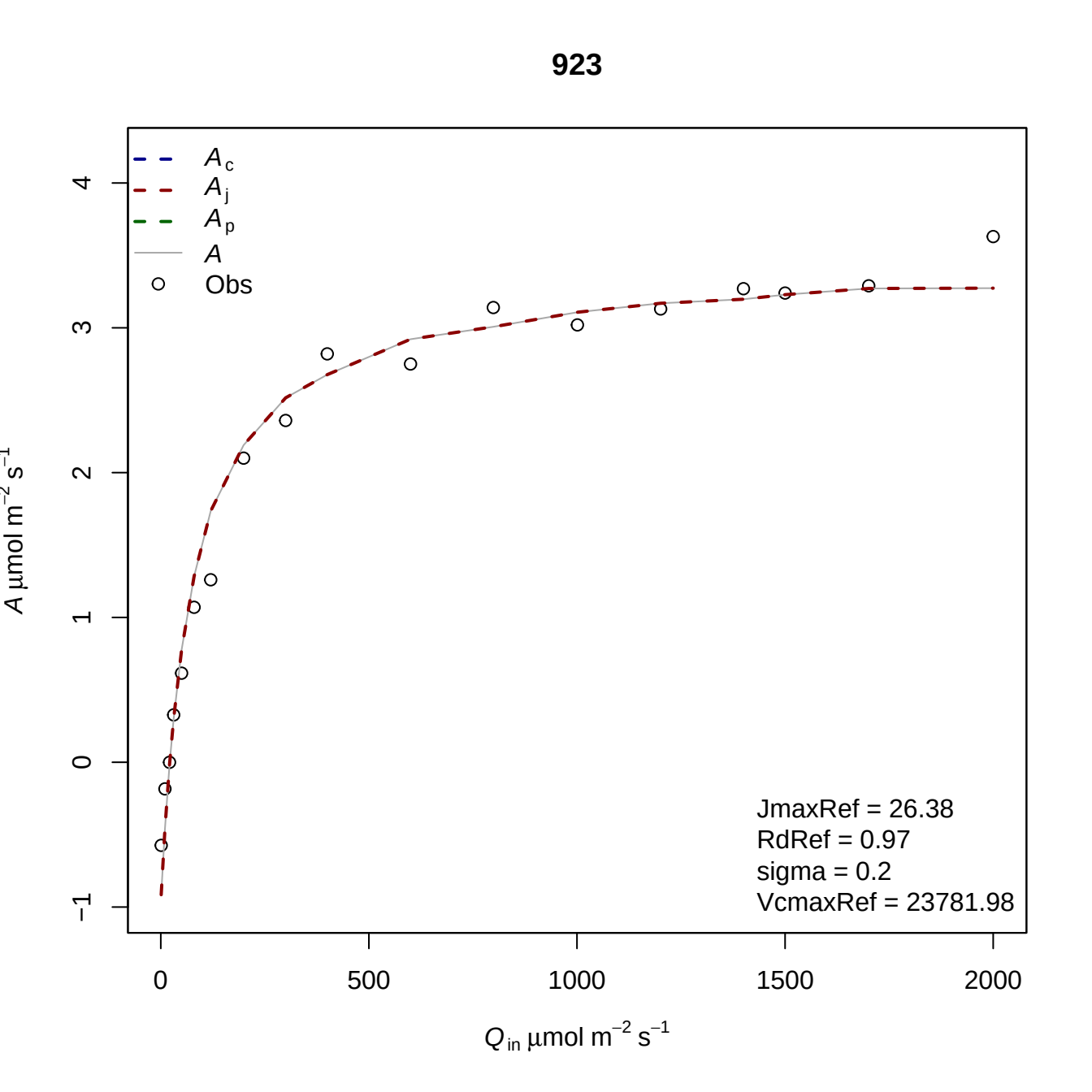

918
A_c
A_j
A_p
A
Obs
JmaxRef = 20.77
RdRef = 0.79
sigma = 0.25
VcmaxRef = 54102.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500
2000

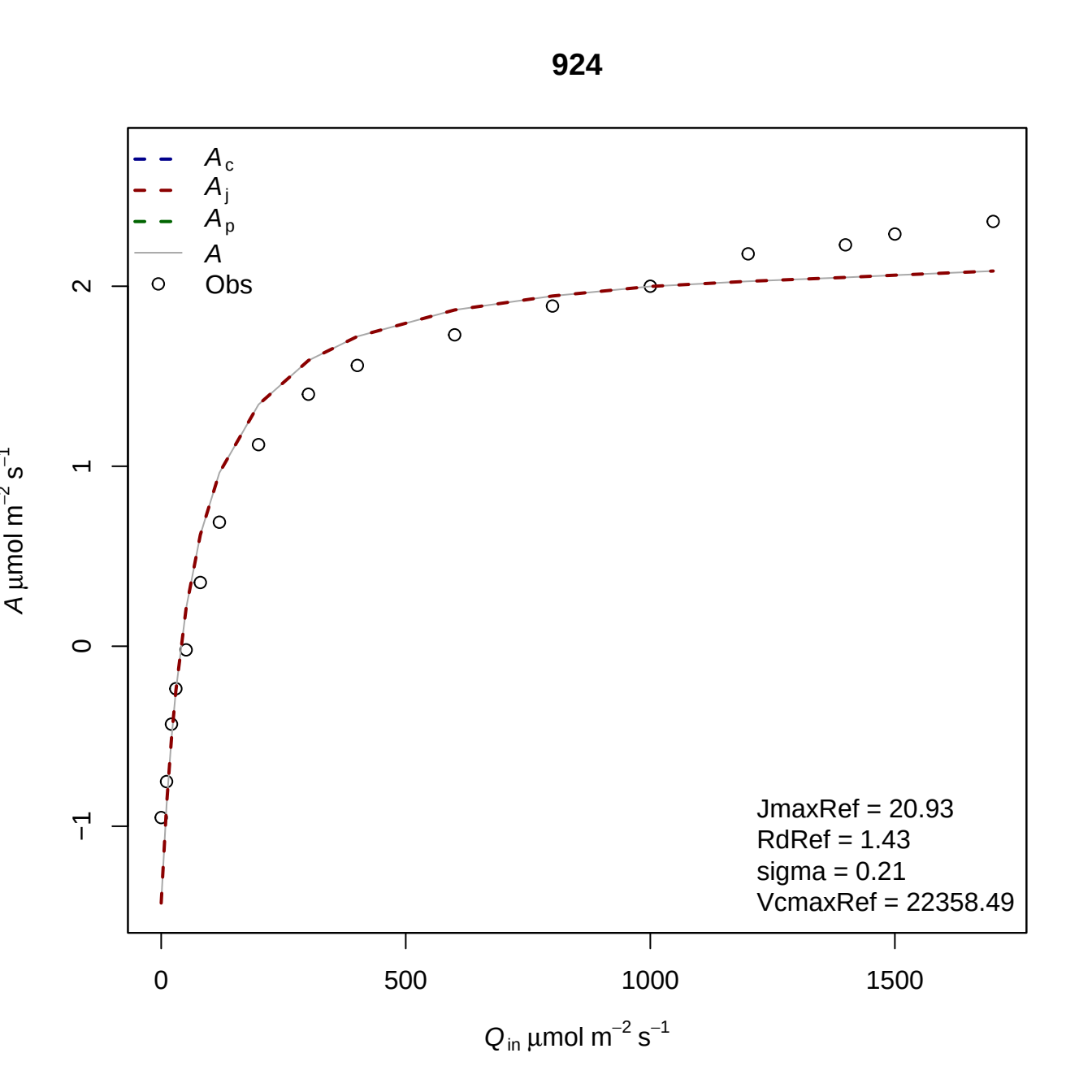

919
A_c
A_j
A_p
A
Obs
JmaxRef = 20.83
RdRef = 1.73
sigma = 0.27
VcmaxRef = 17389.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500
2000

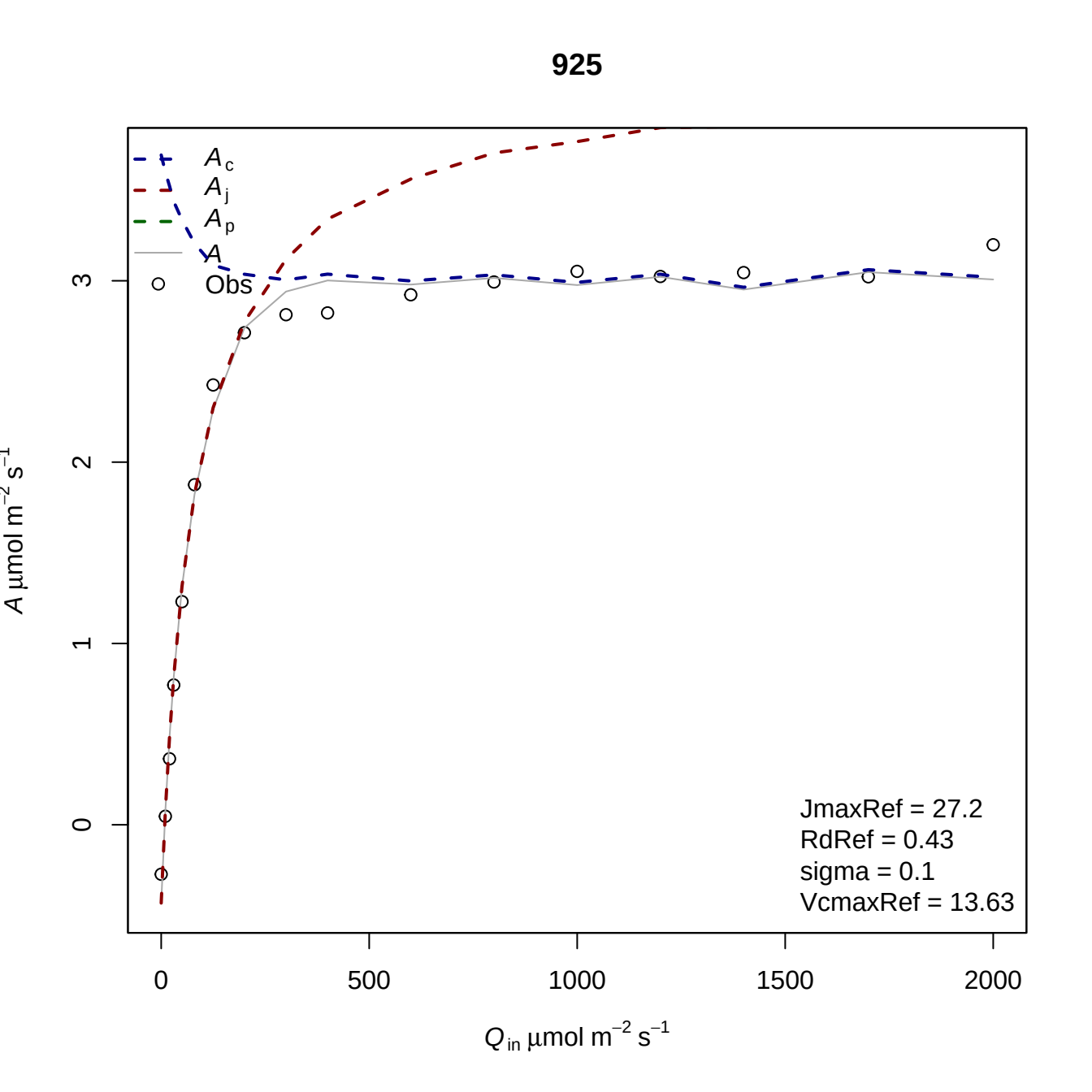

920
A_c
A_j
A_p
A
Obs
JmaxRef = 55.72
RdRef = 0.42
sigma = 0.21
VcmaxRef = 25.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

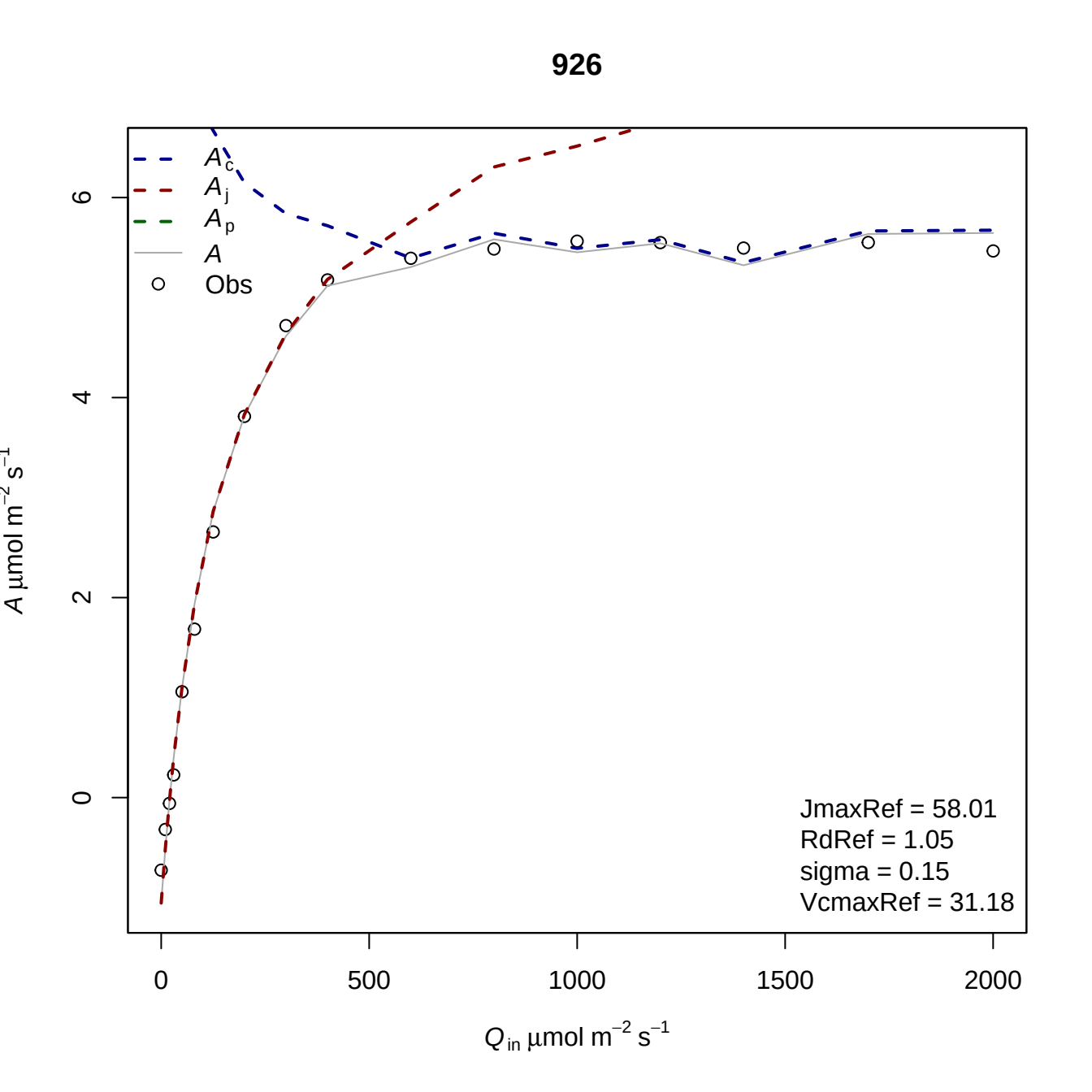

921
A_c
A_j
A_p
A
Obs
JmaxRef = 32.94
RdRef = 0.7
sigma = 0.12
VcmaxRef = 17.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500

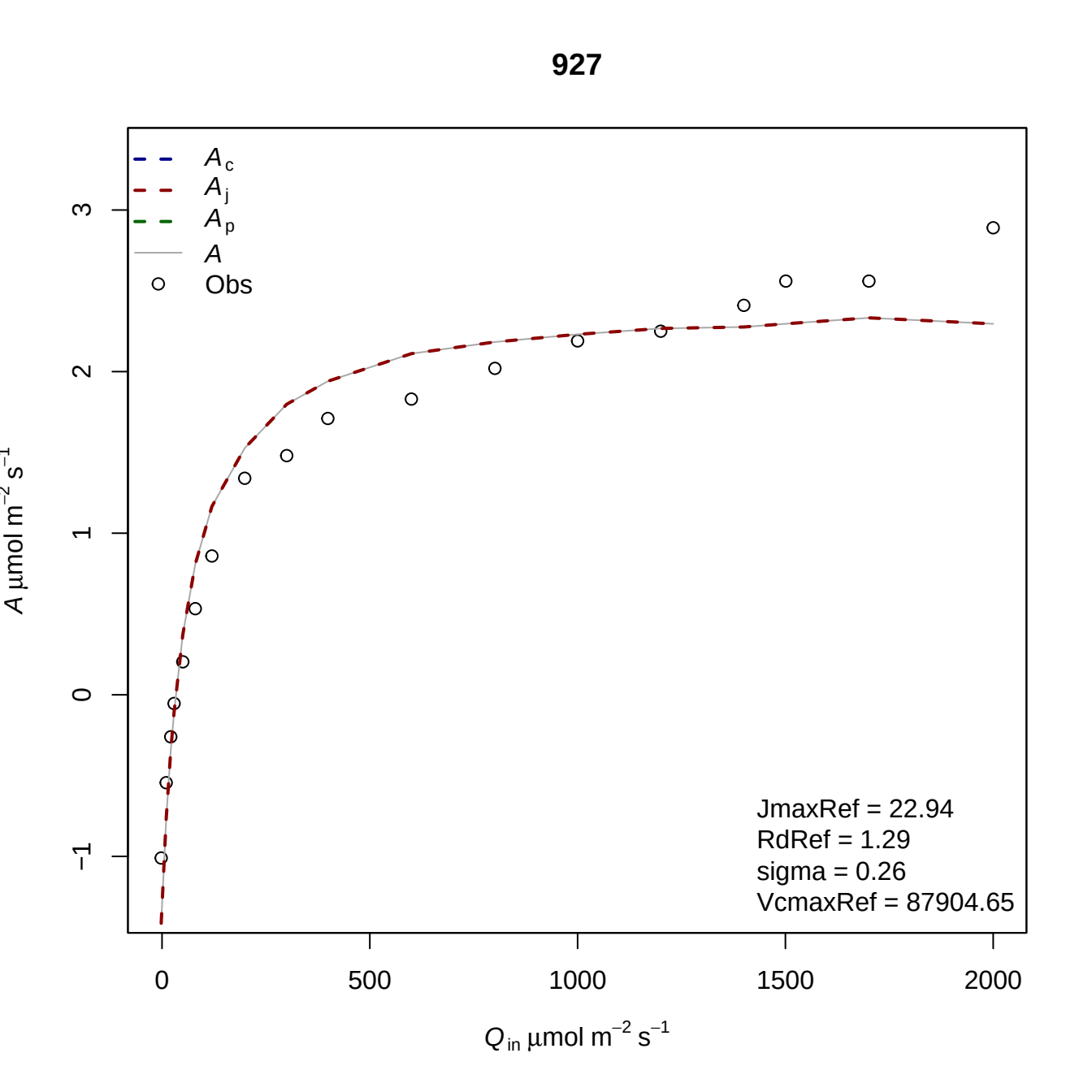

922
A_c
A_j
A_p
A
Obs
JmaxRef = 29.16
RdRef = 1.12
sigma = 0.26
VcmaxRef = 27265.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

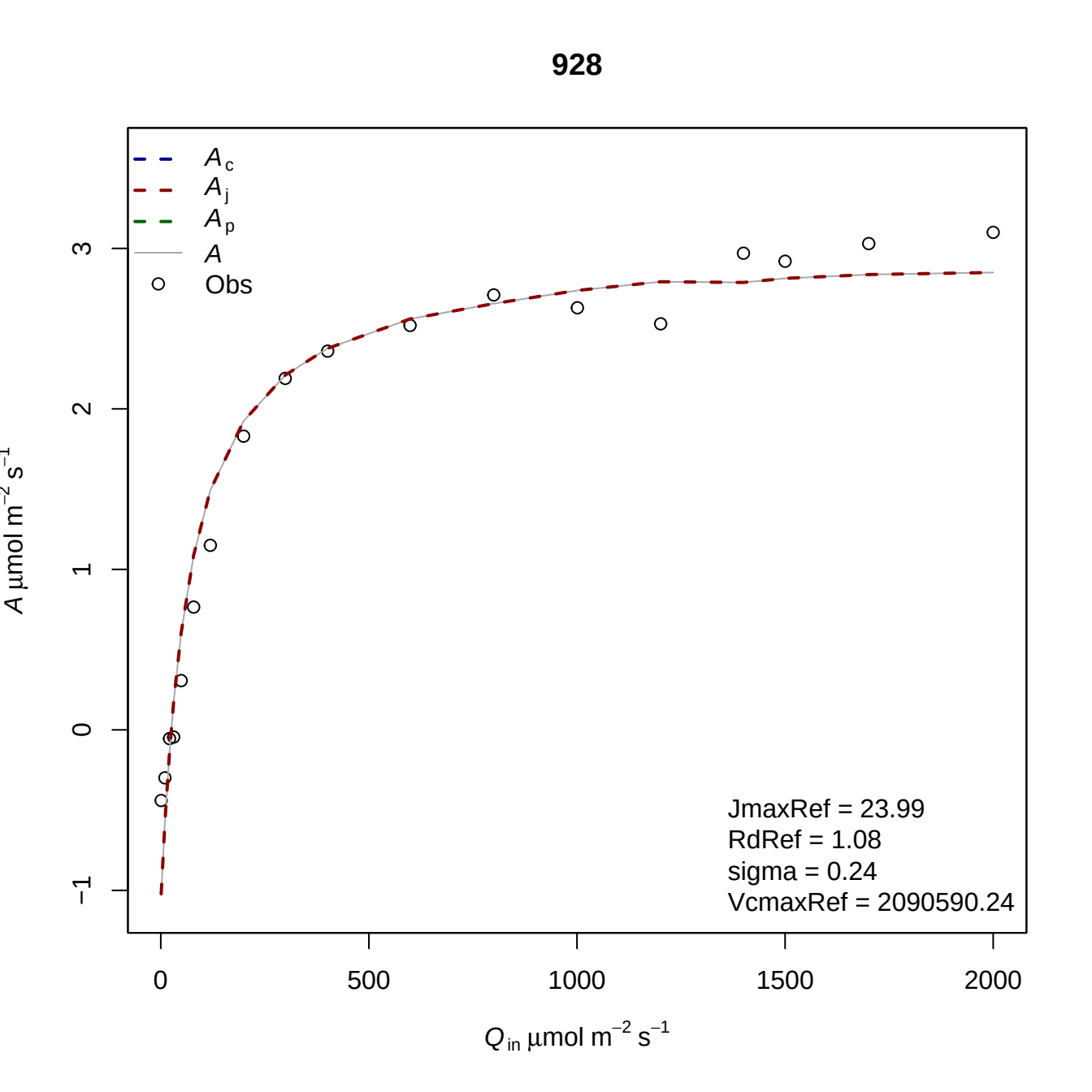

923
A_c
A_j
A_p
A
Obs
JmaxRef = 26.38
RdRef = 0.97
sigma = 0.2
VcmaxRef = 23781.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

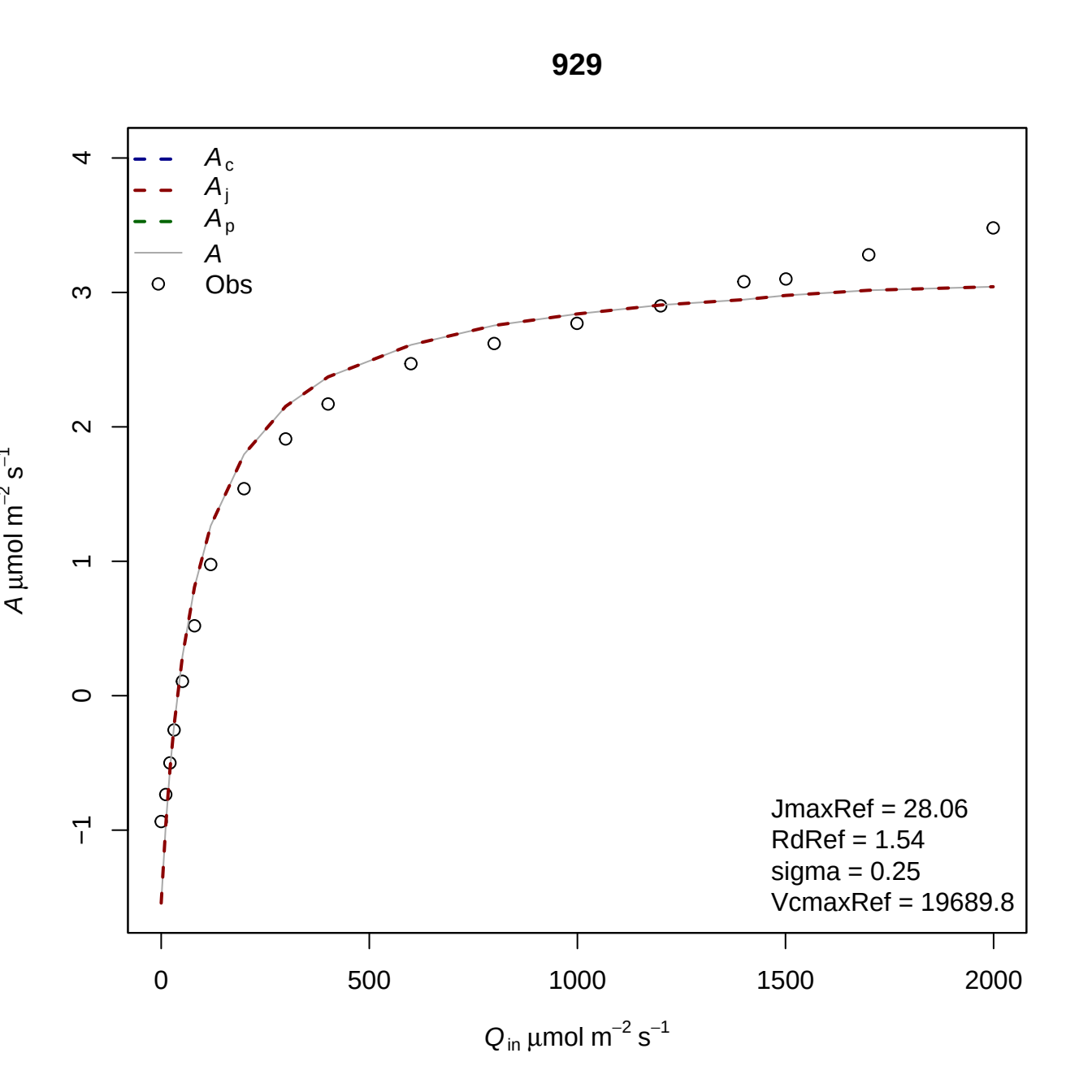

924
A_c
A_j
A_p
A
Obs
JmaxRef = 20.93
RdRef = 1.43
sigma = 0.21
VcmaxRef = 22358.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500

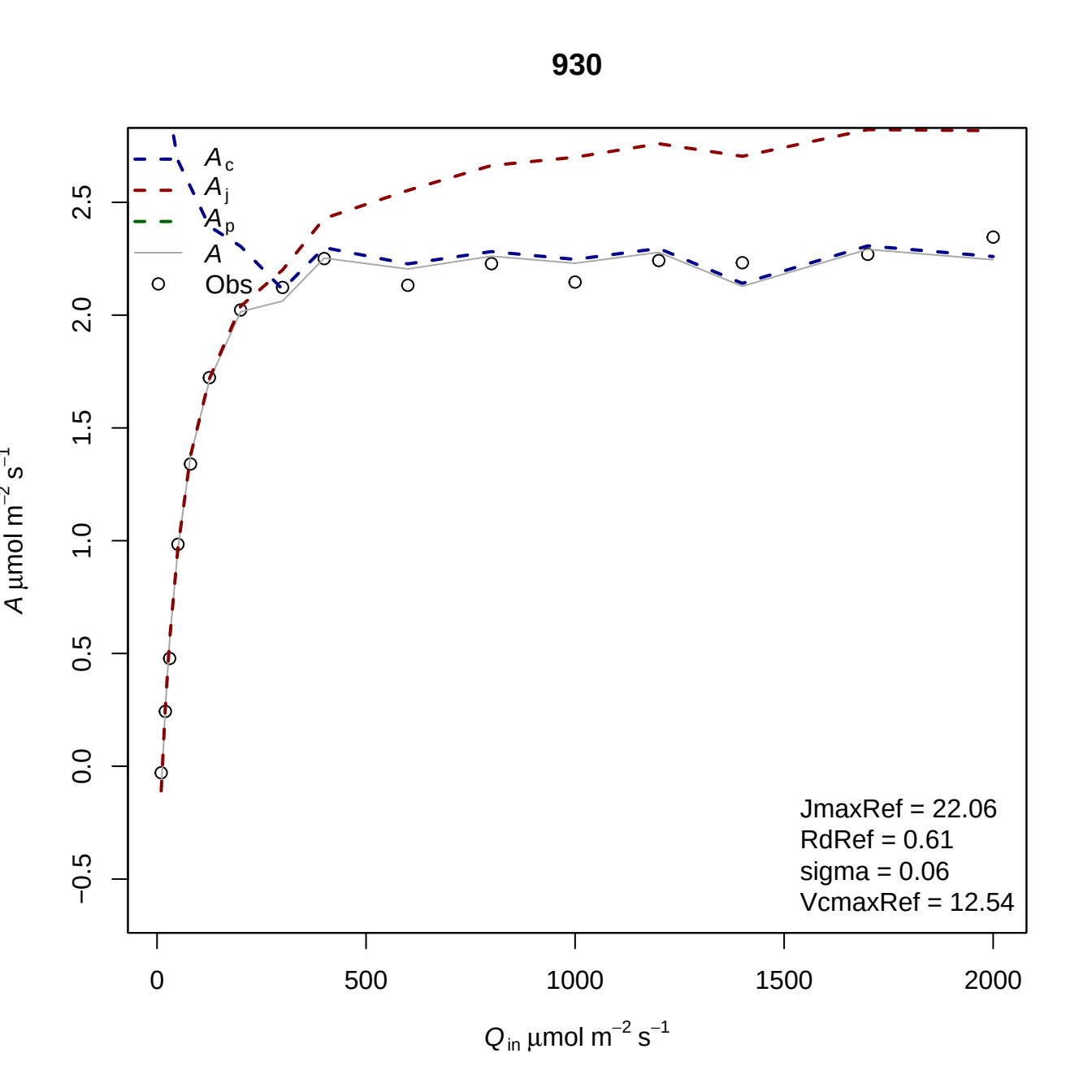

925
A_c
A_j
A_p
A
Obs
JmaxRef = 27.2
RdRef = 0.43
sigma = 0.1
VcmaxRef = 13.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500
2000

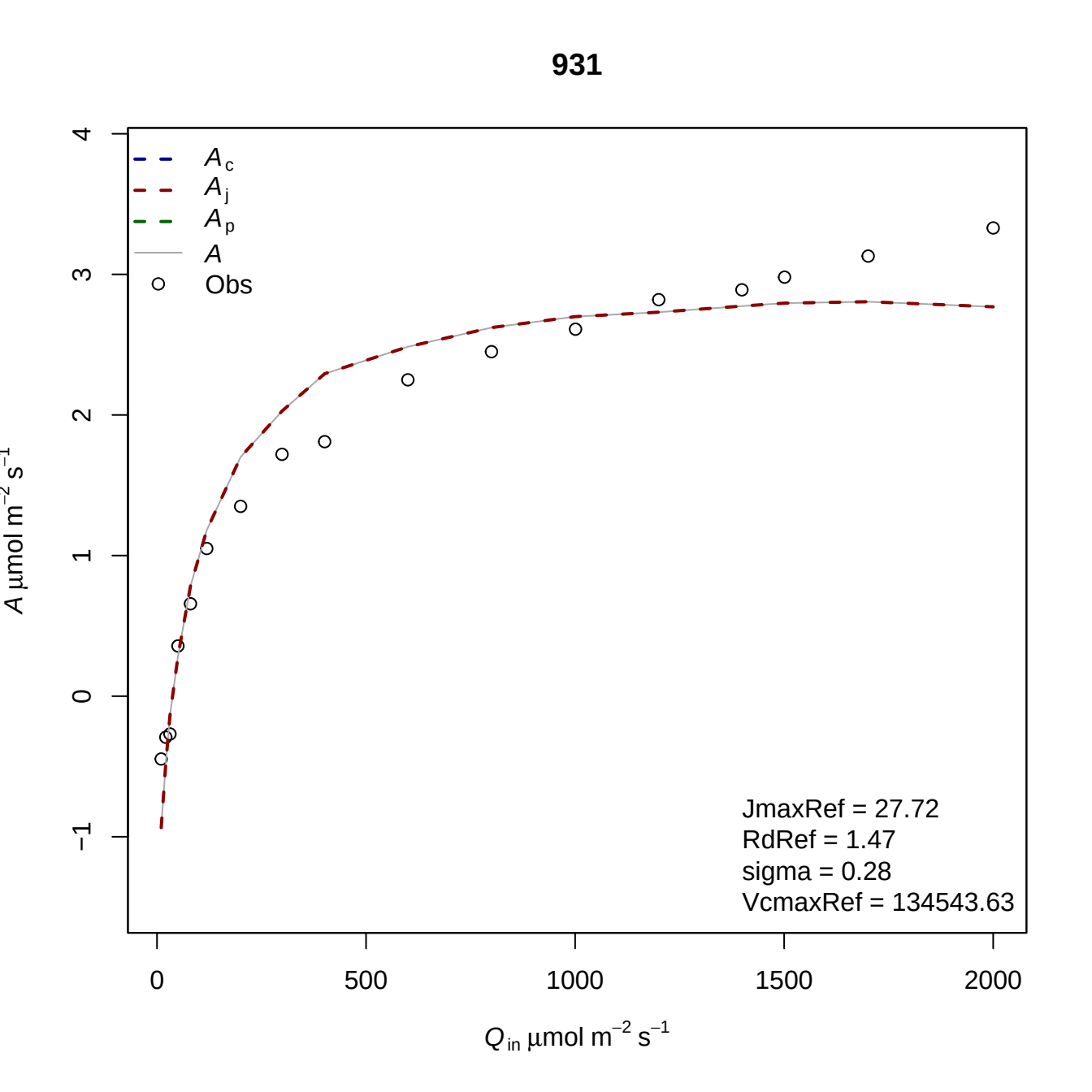

926
A_c
A_j
A_p
A
Obs
JmaxRef = 58.01
RdRef = 1.05
sigma = 0.15
VcmaxRef = 31.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

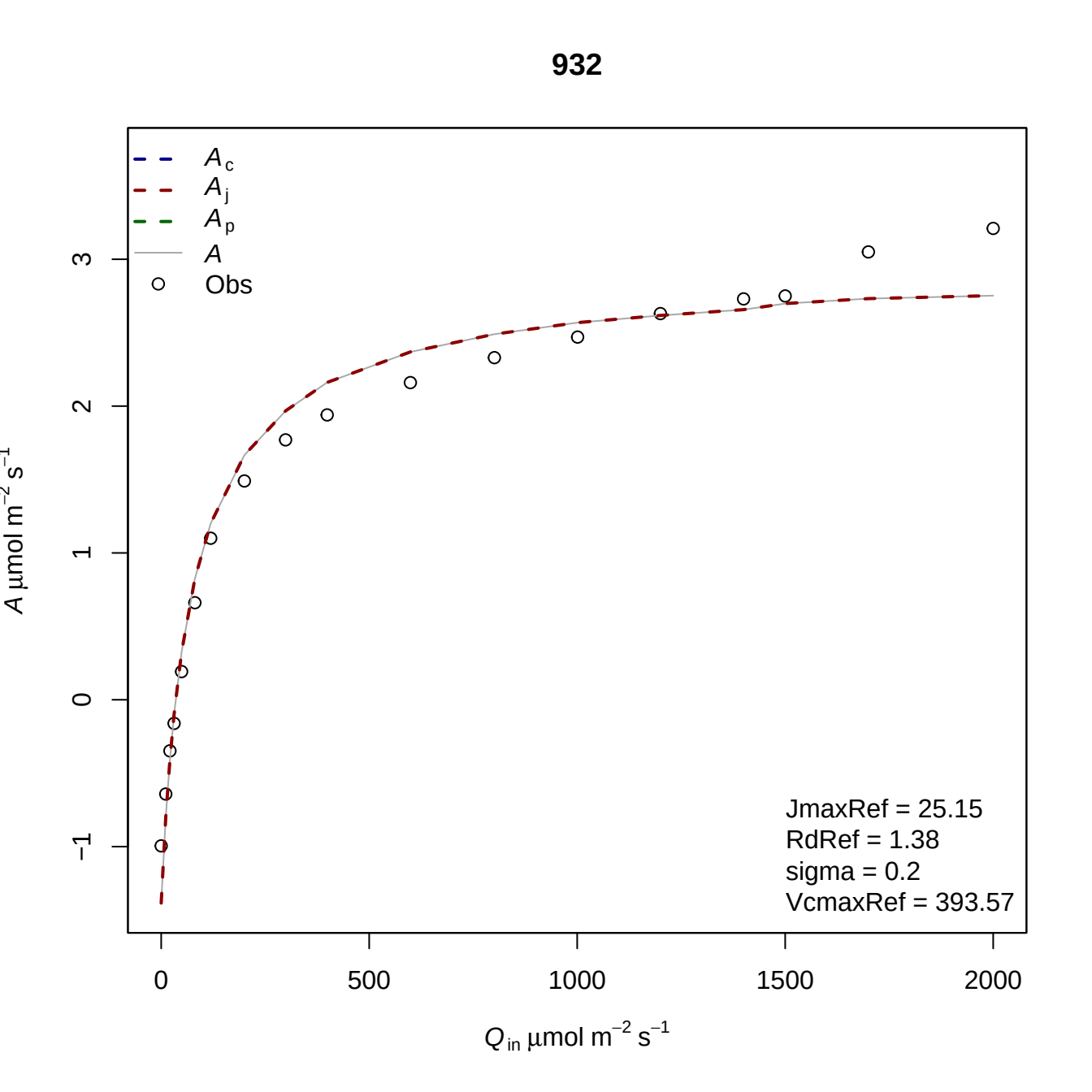

927
A_c
A_j
A_p
A
Obs
JmaxRef = 22.94
RdRef = 1.29
sigma = 0.26
VcmaxRef = 87904.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

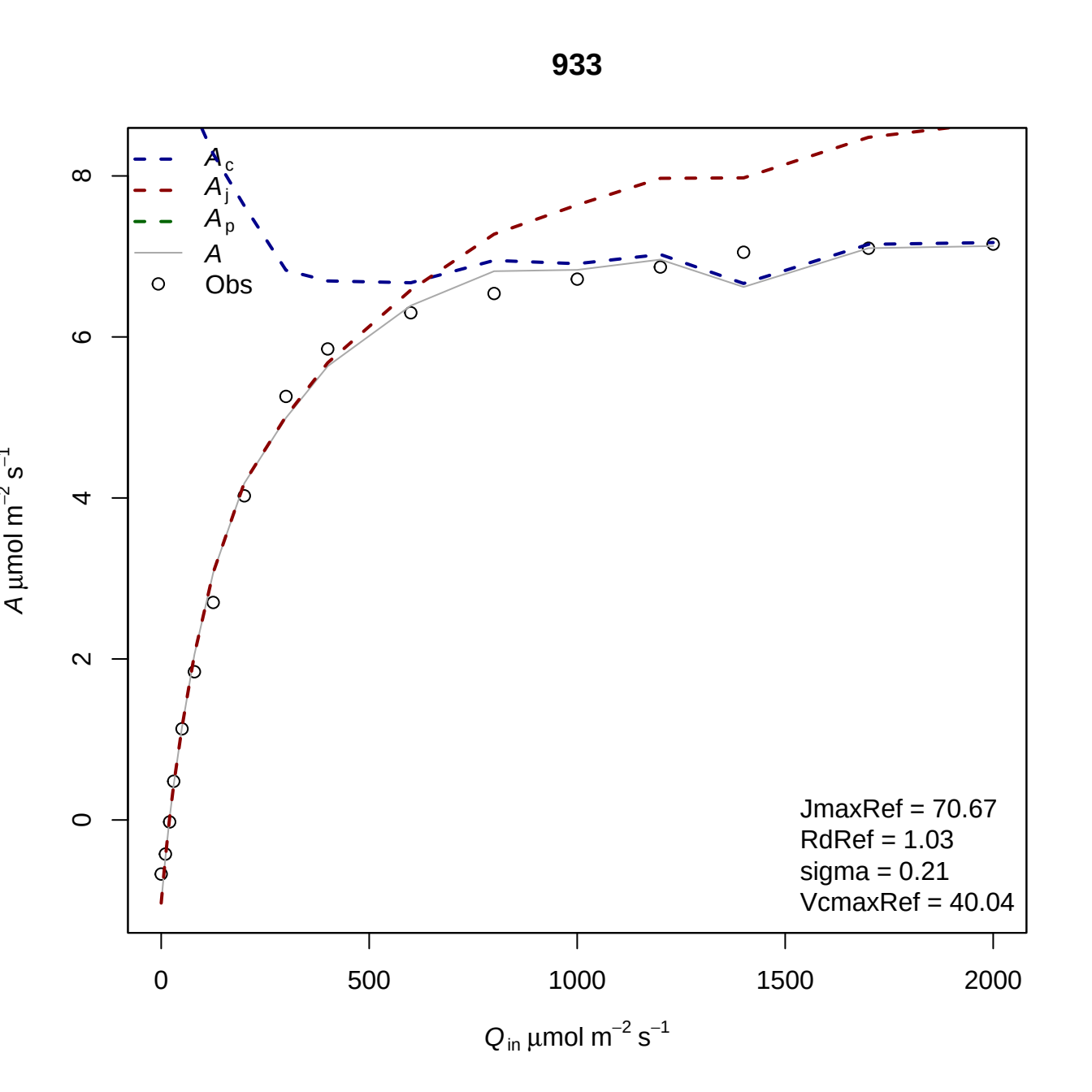

928
A_c
A_j
A_p
A
Obs
JmaxRef = 23.99
RdRef = 1.08
sigma = 0.24
VcmaxRef = 2090590.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

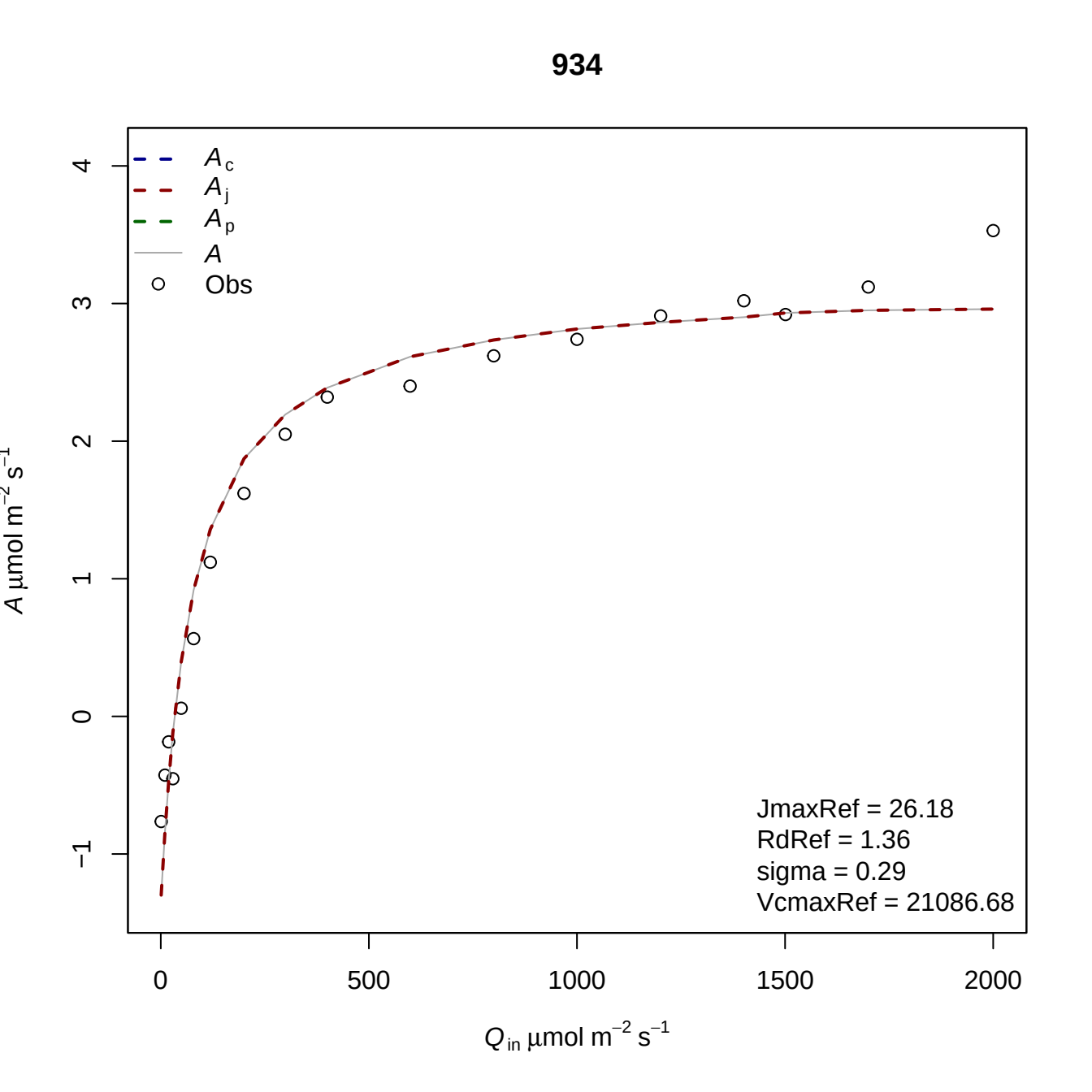

929
A_c
A_j
A_p
A
Obs
JmaxRef = 28.06
RdRef = 1.54
sigma = 0.25
VcmaxRef = 19689.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
0
500
1000
1500
2000

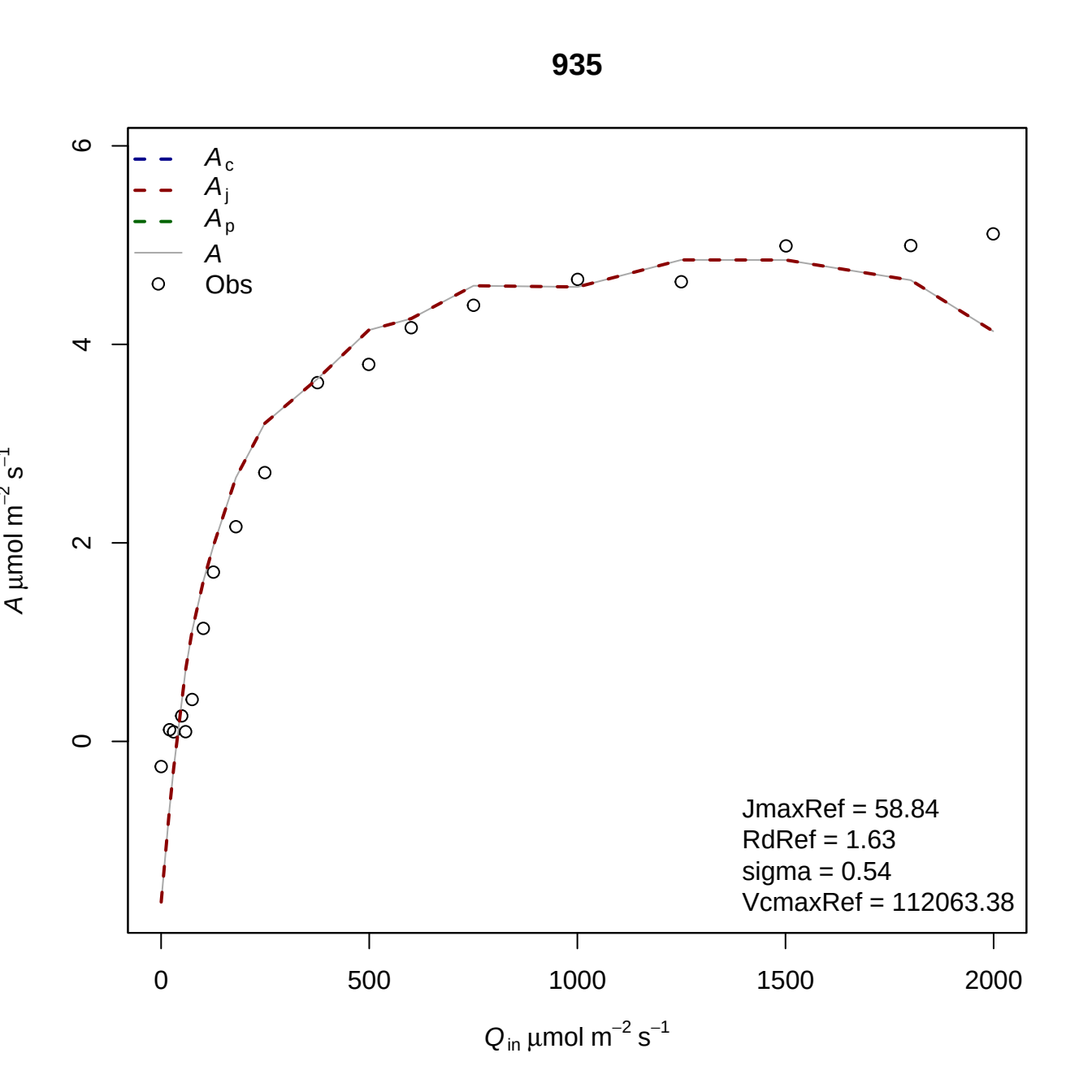

930
A_c
A_j
A_p
A
Obs
JmaxRef = 22.06
RdRef = 0.61
sigma = 0.06
VcmaxRef = 12.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
0
500
1000
1500
2000

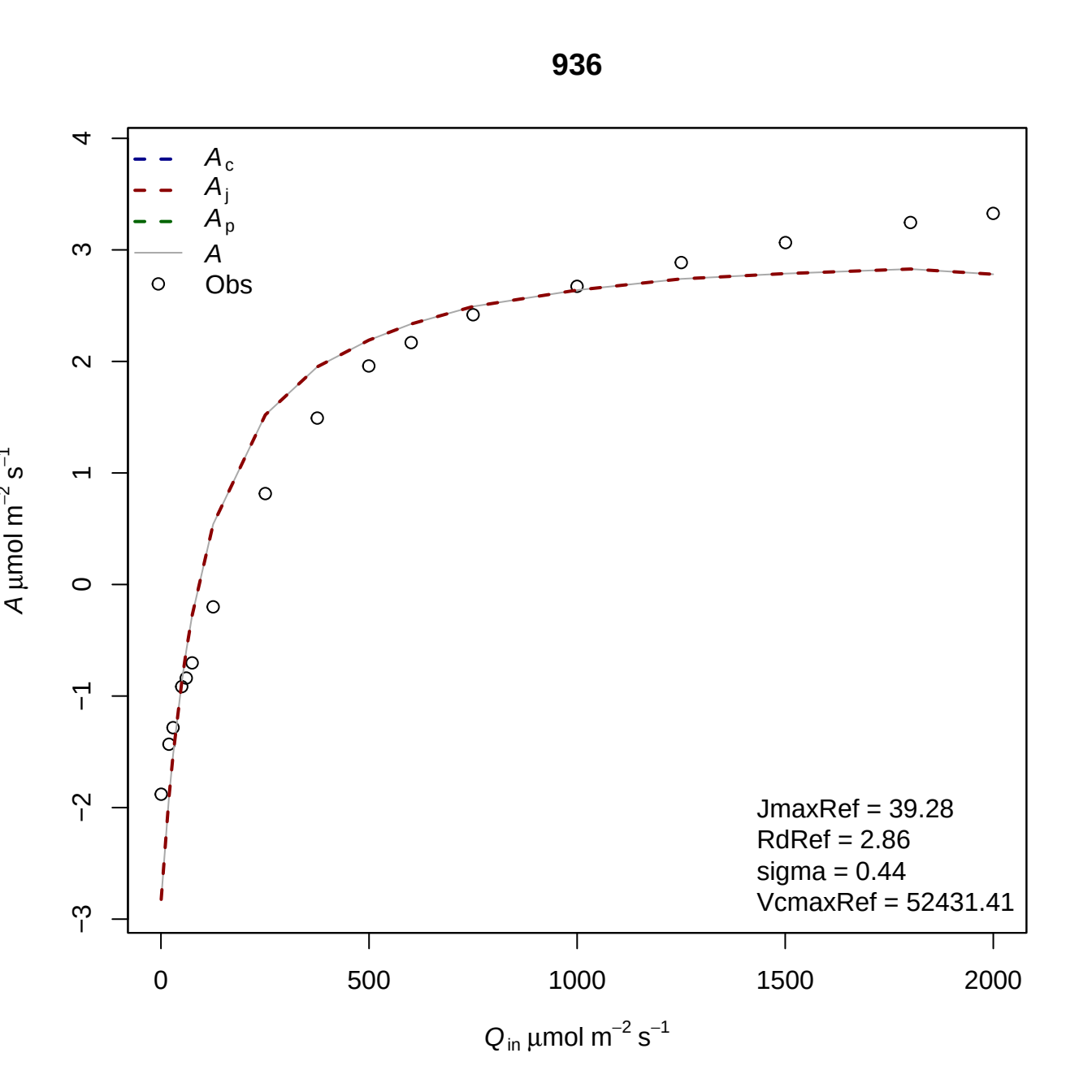

931
A_c
A_j
A_p
A
Obs
JmaxRef = 27.72
RdRef = 1.47
sigma = 0.28
VcmaxRef = 134543.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
0
500
1000
1500
2000

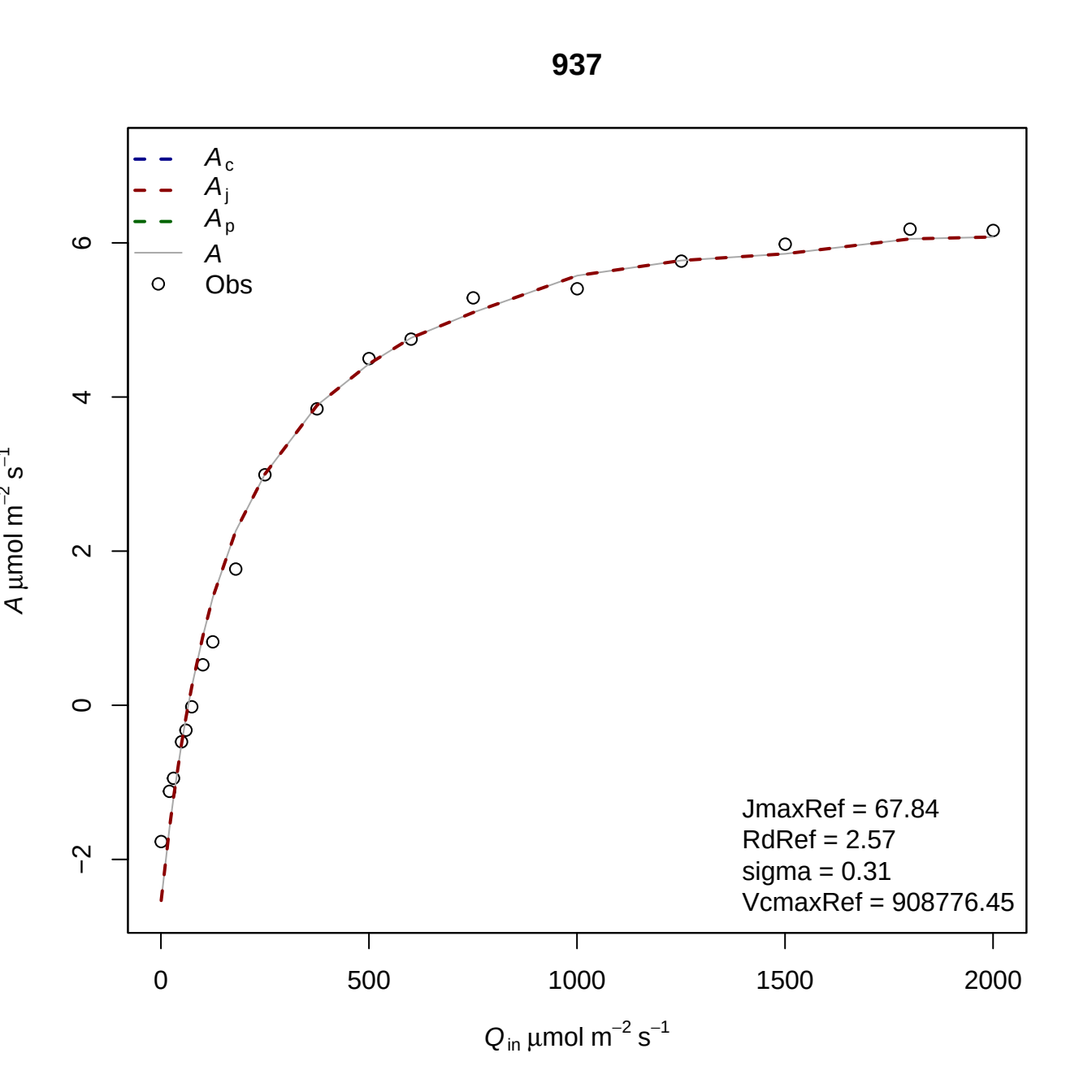

932
A_c
A_j
A_p
A
Obs
JmaxRef = 25.15
RdRef = 1.38
sigma = 0.2
VcmaxRef = 393.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

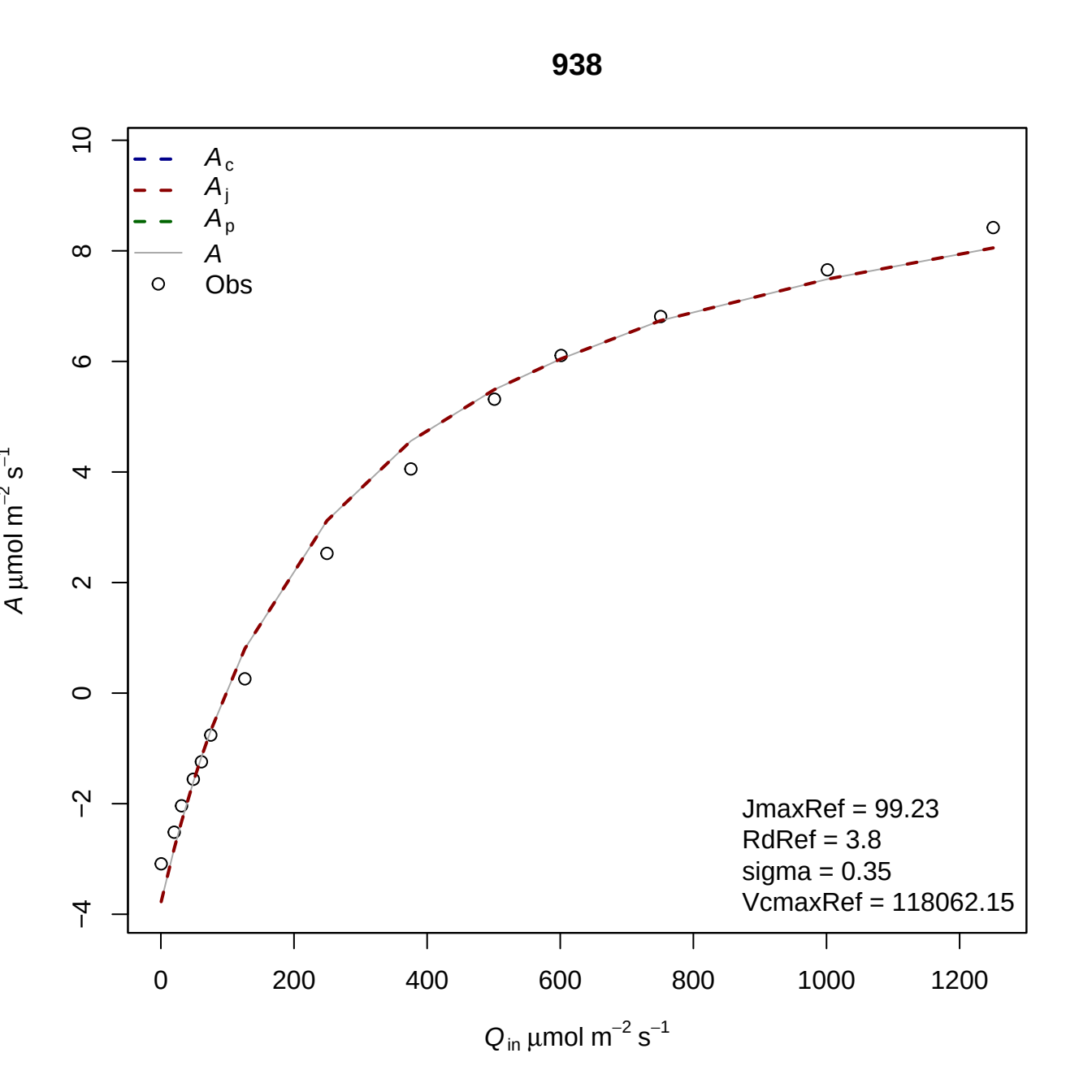

933
A_c
A_j
A_p
A
Obs
JmaxRef = 70.67
RdRef = 1.03
sigma = 0.21
VcmaxRef = 40.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

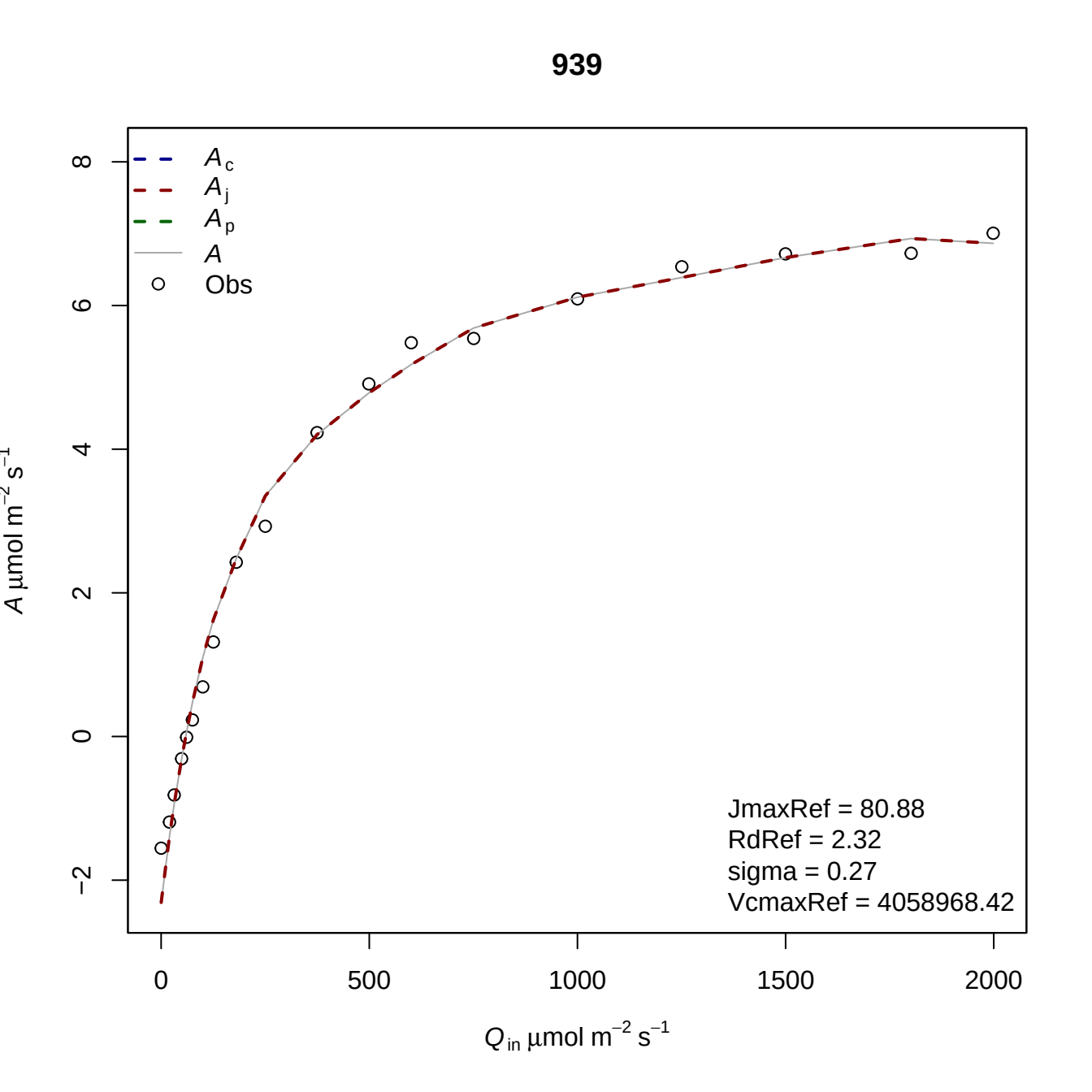

934
A_c
A_j
A_p
A
Obs
JmaxRef = 26.18
RdRef = 1.36
sigma = 0.29
VcmaxRef = 21086.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

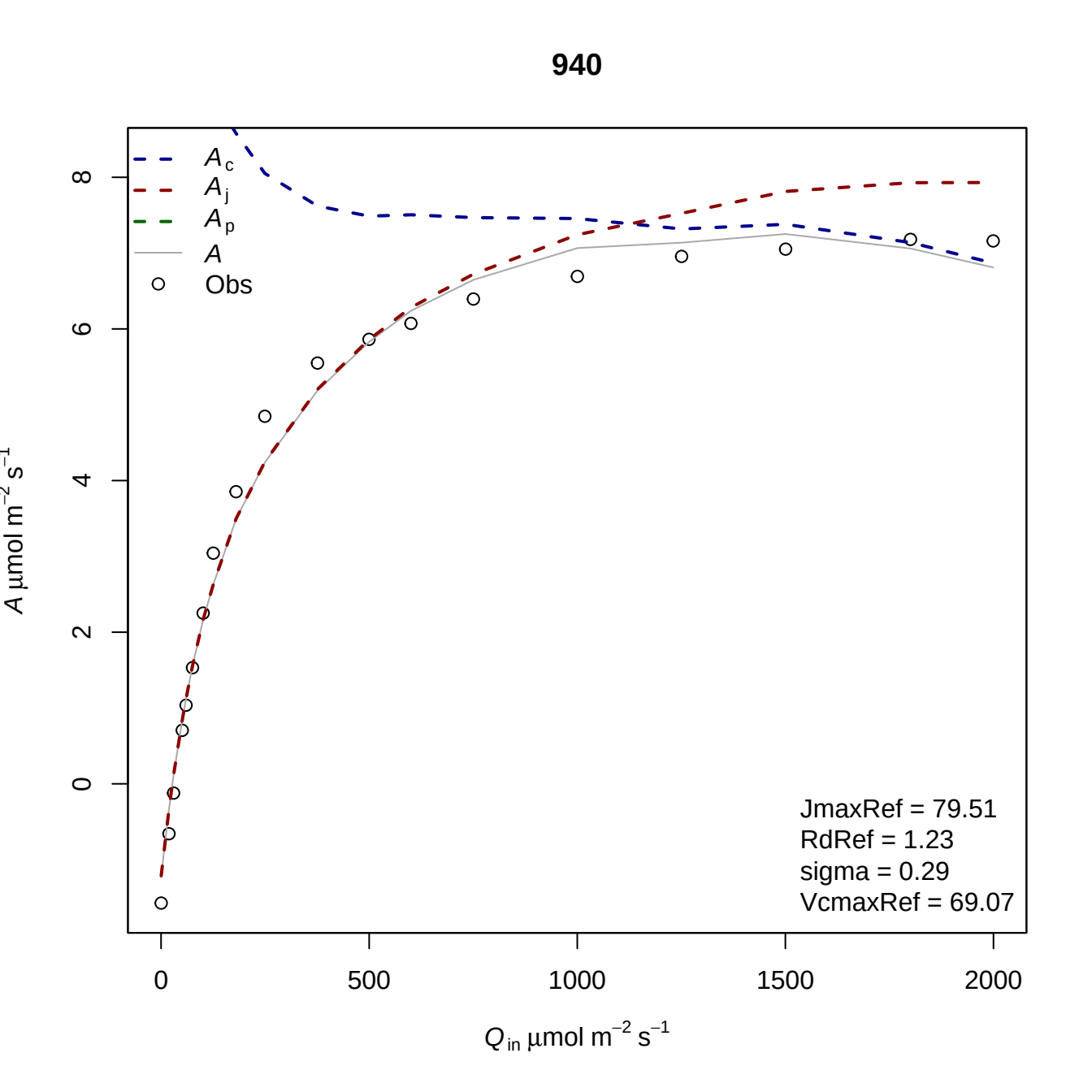

935
A_c
A_j
A_p
A
Obs
JmaxRef = 58.84
RdRef = 1.63
sigma = 0.54
VcmaxRef = 112063.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

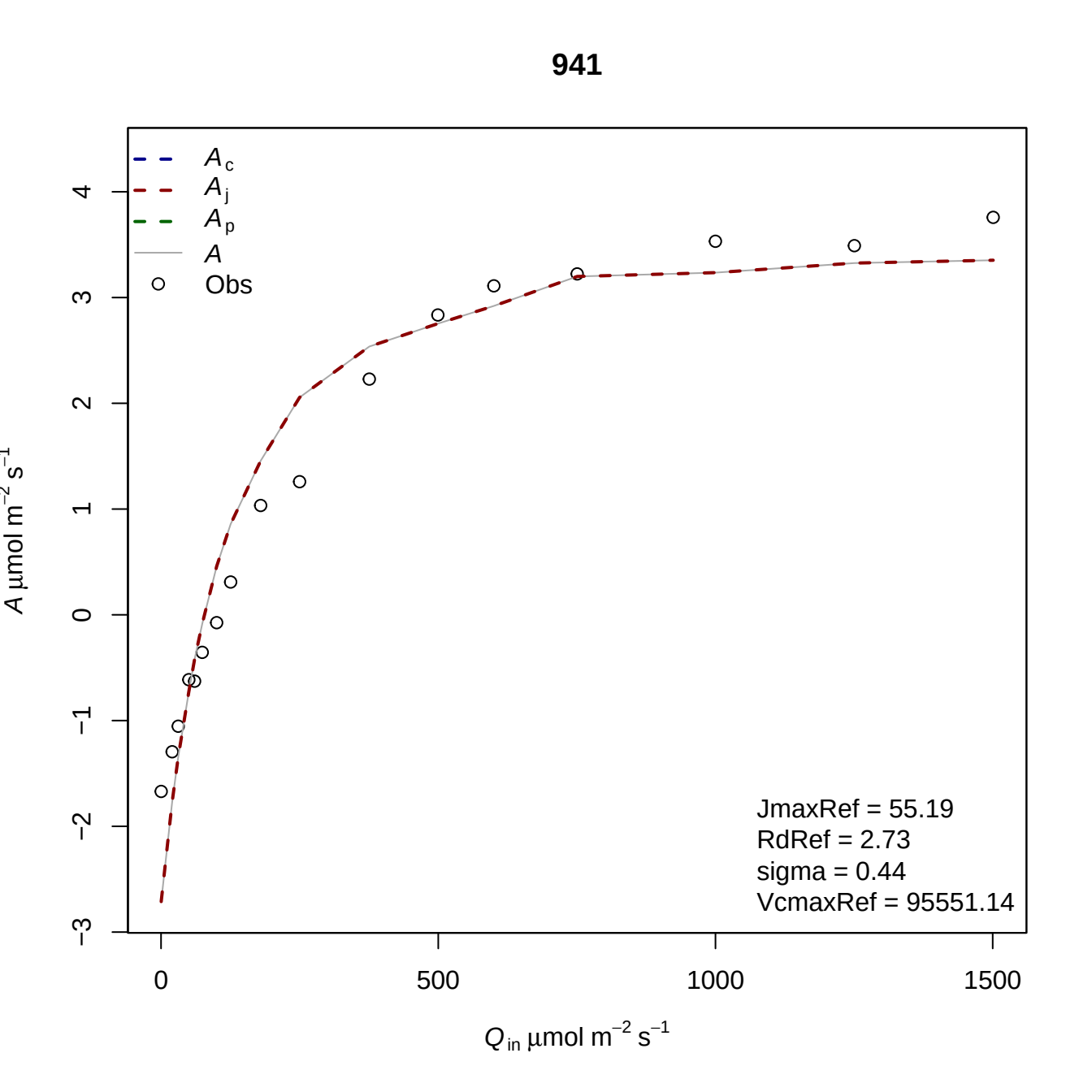

936
A_c
A_j
A_p
A
Obs
JmaxRef = 39.28
RdRef = 2.86
sigma = 0.44
VcmaxRef = 52431.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
−3
0
500
1000
1500
2000

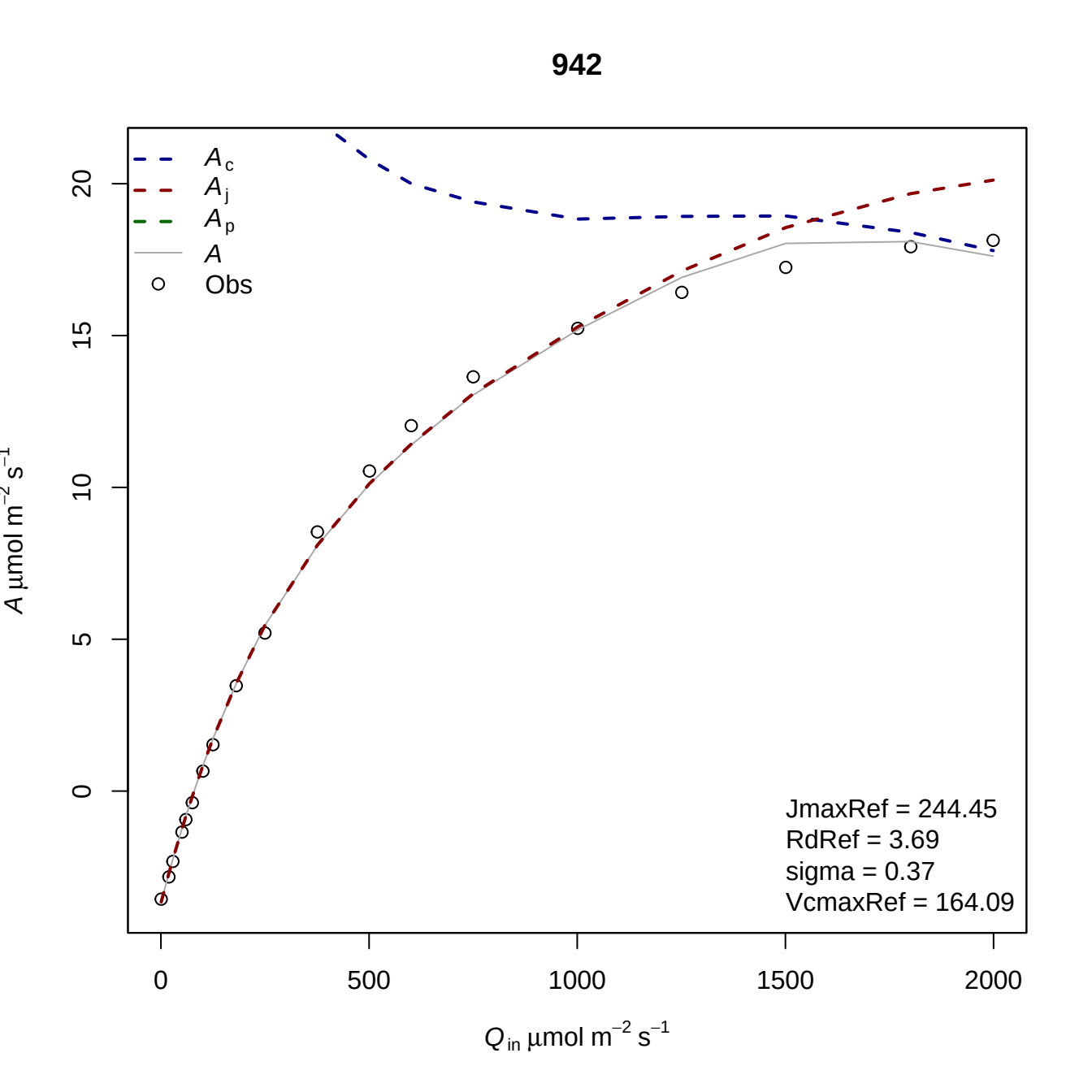

937
A_c
A_j
A_p
A
Obs
JmaxRef = 67.84
RdRef = 2.57
sigma = 0.31
VcmaxRef = 908776.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

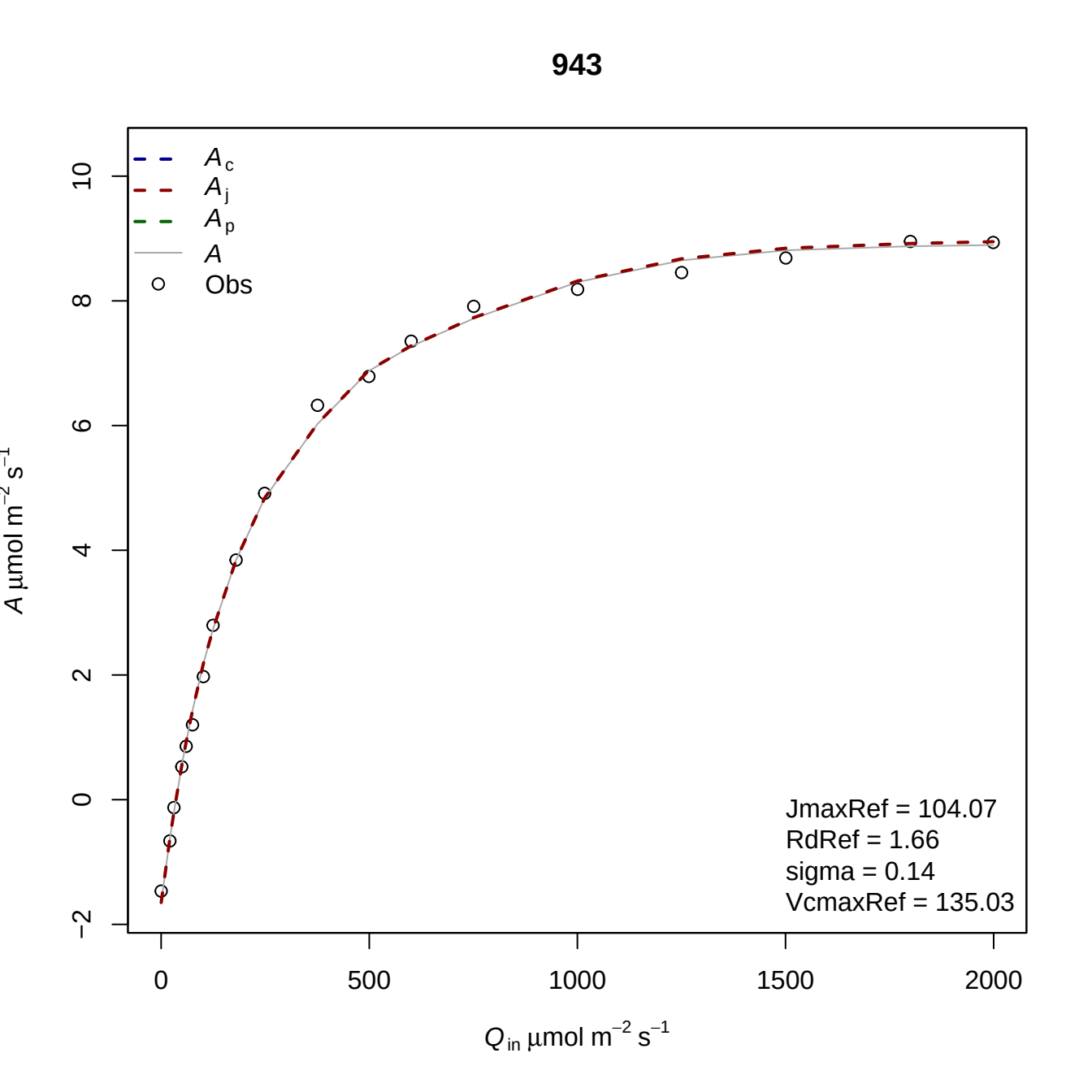

938
A_c
A_j
A_p
A
Obs
JmaxRef = 99.23
RdRef = 3.8
sigma = 0.35
VcmaxRef = 118062.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

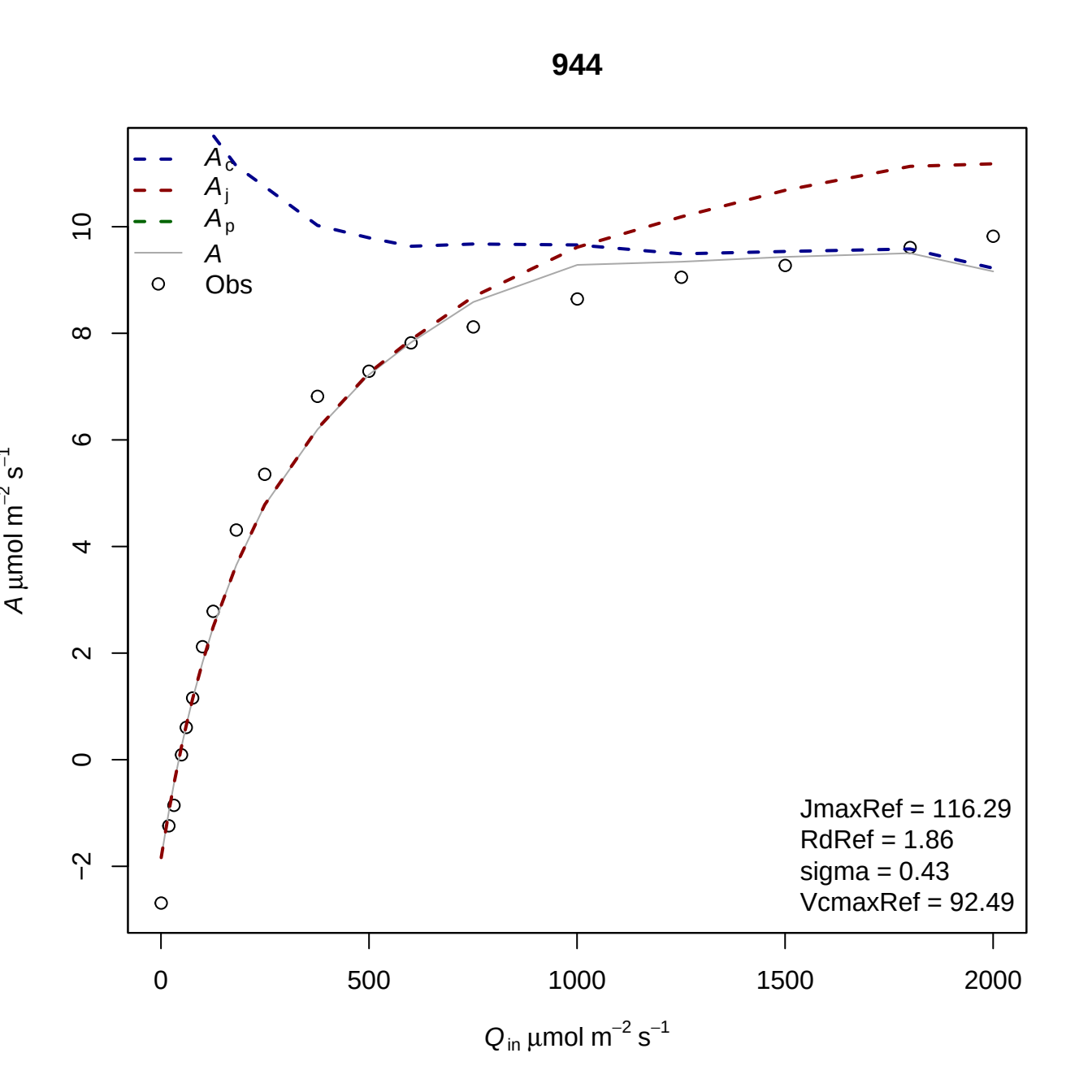

939
A_c
A_j
A_p
A
Obs
JmaxRef = 80.88
RdRef = 2.32
sigma = 0.27
VcmaxRef = 4058968.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

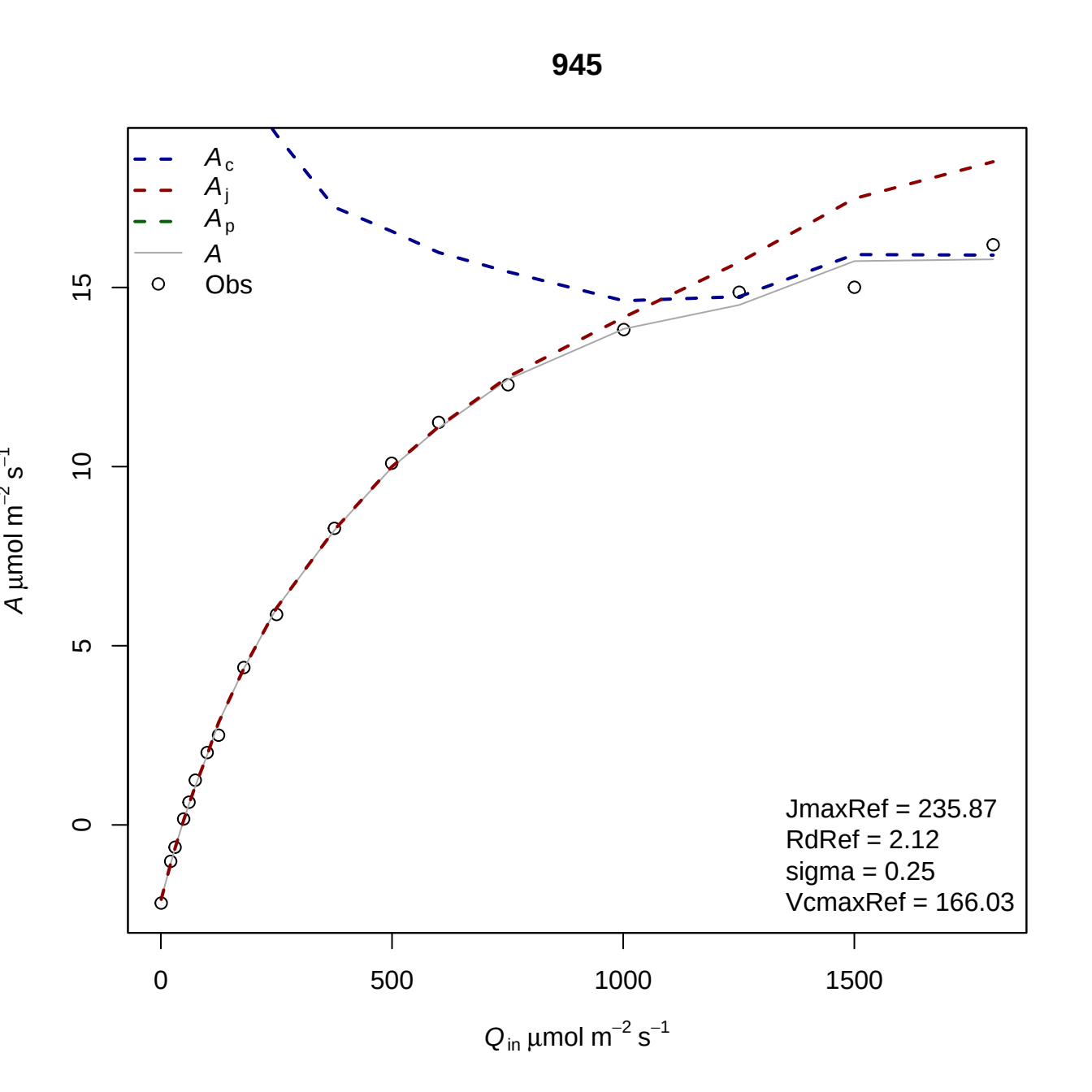

940
A_c
A_j
A_p
A
Obs
JmaxRef = 79.51
RdRef = 1.23
sigma = 0.29
VcmaxRef = 69.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

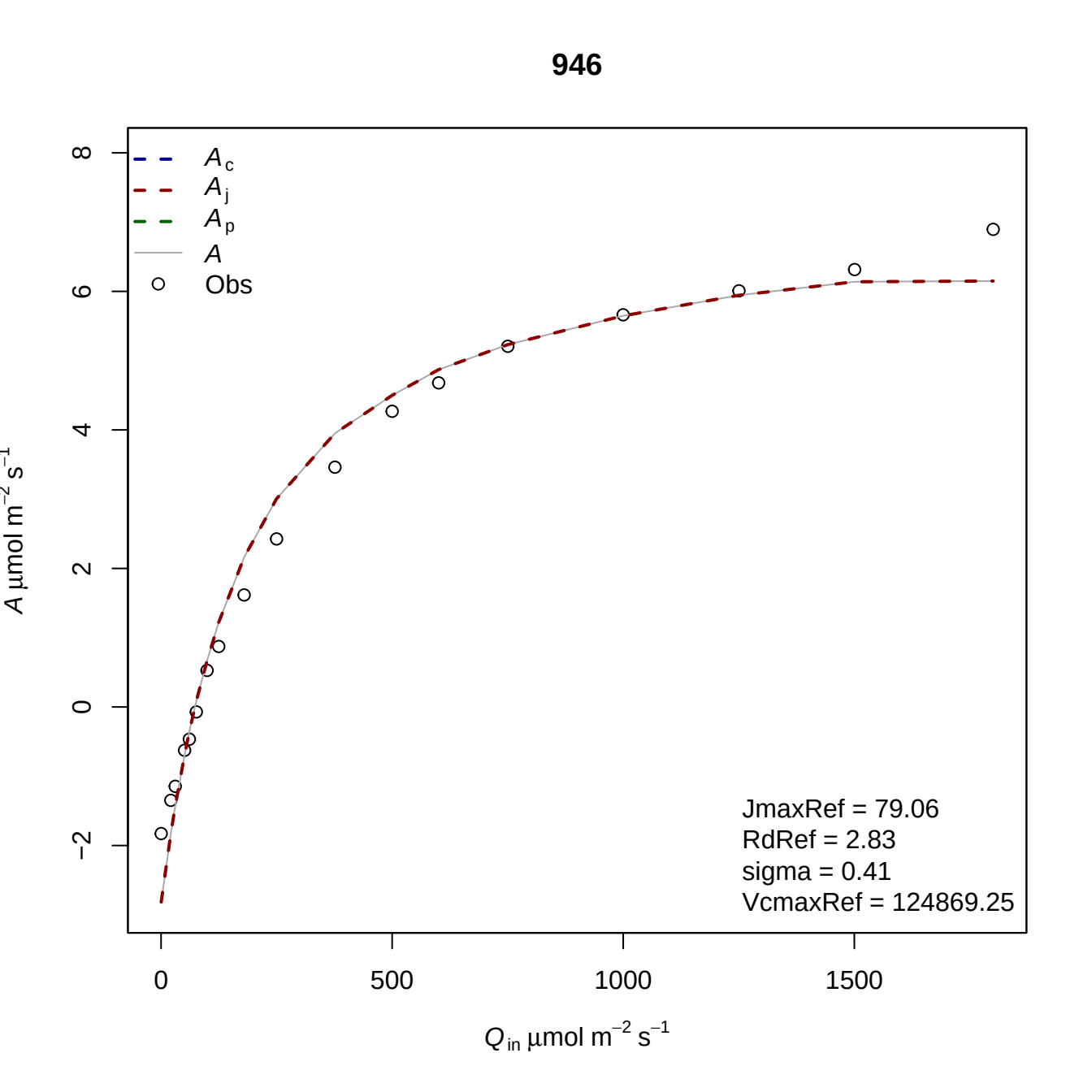

941
A_c
A_j
A_p
A
Obs
JmaxRef = 55.19
RdRef = 2.73
sigma = 0.44
VcmaxRef = 95551.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
−3
0
500
1000
1500

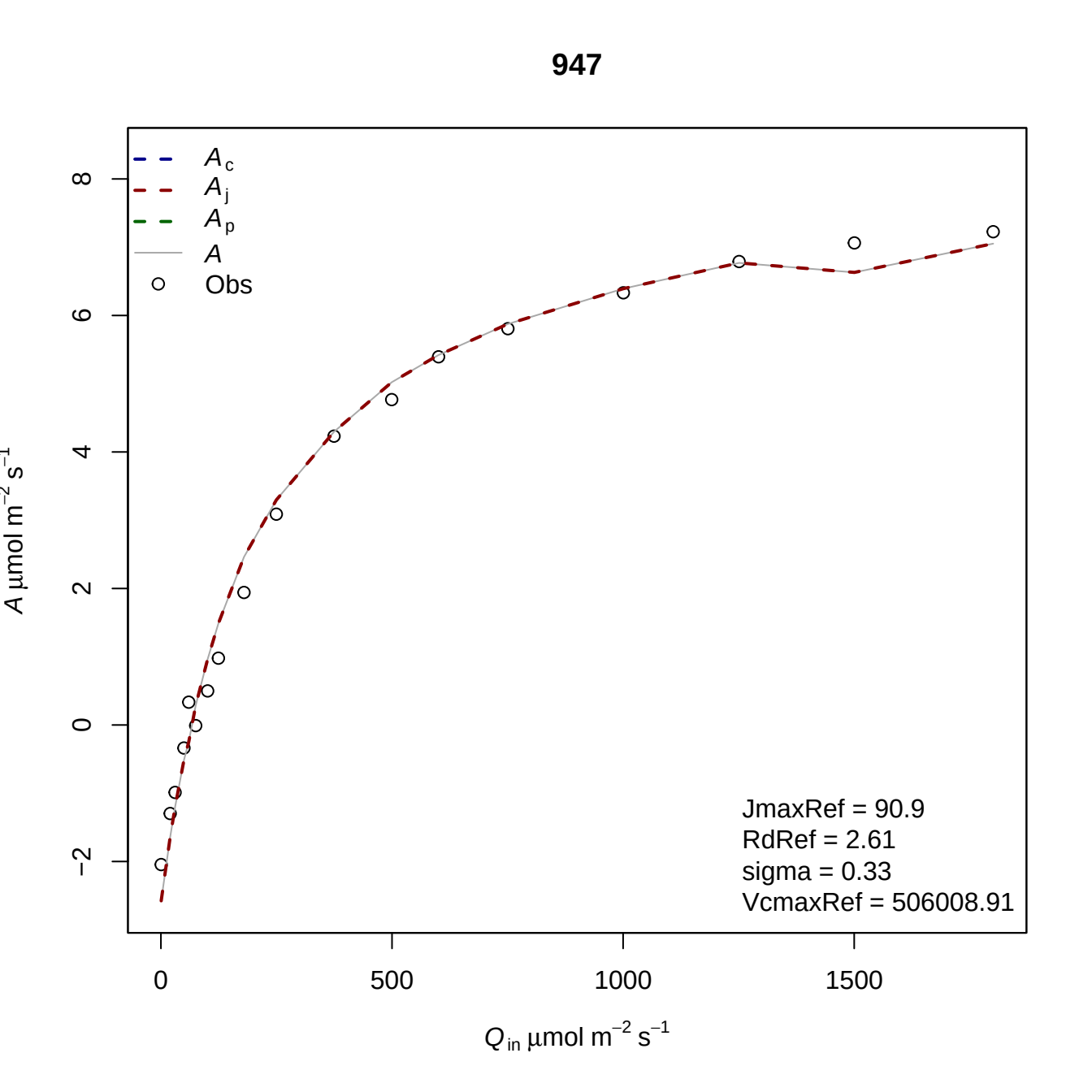

942
A_c
A_j
A_p
A
Obs
JmaxRef = 244.45
RdRef = 3.69
sigma = 0.37
VcmaxRef = 164.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

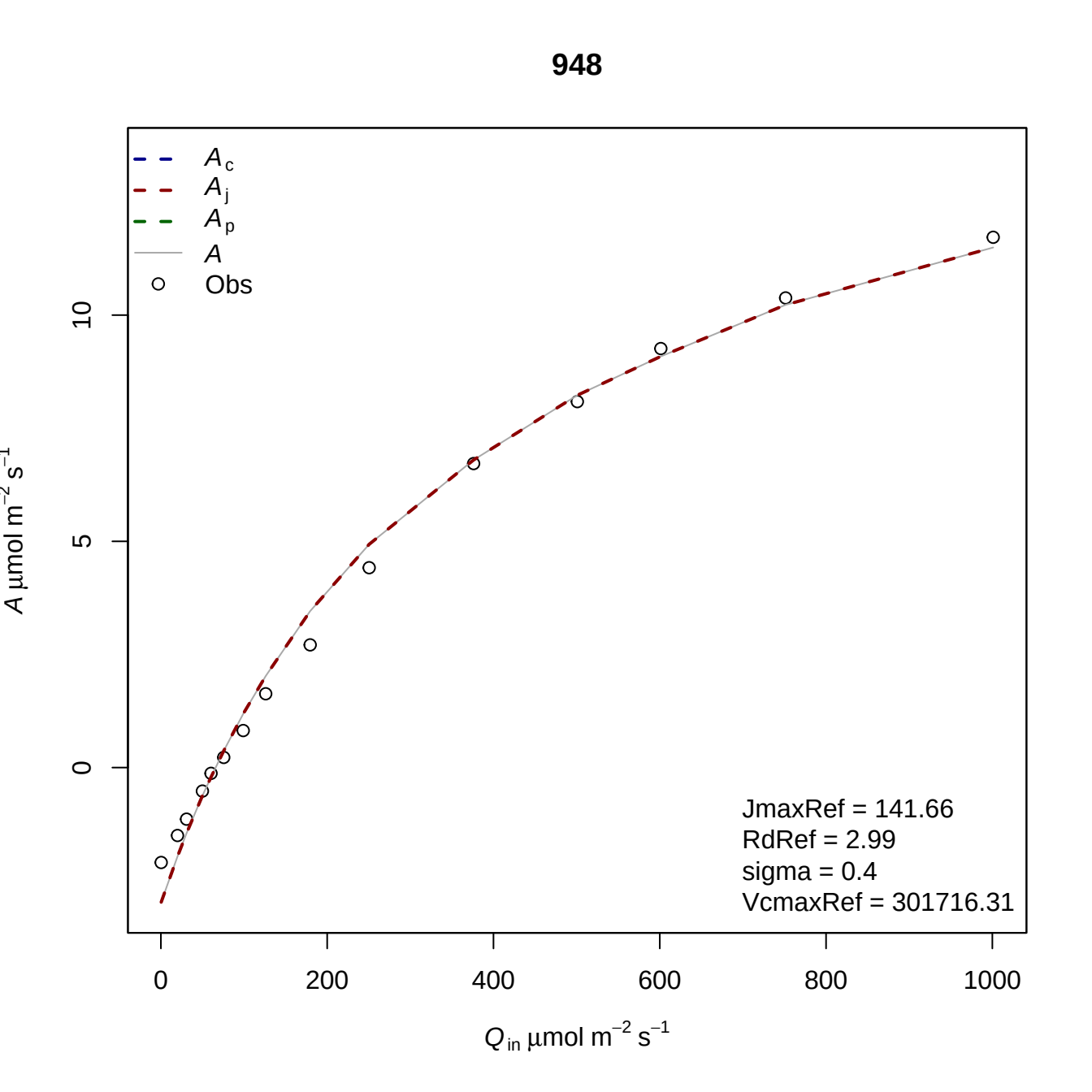

943
A_c
A_j
A_p
A
Obs
JmaxRef = 104.07
RdRef = 1.66
sigma = 0.14
VcmaxRef = 135.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

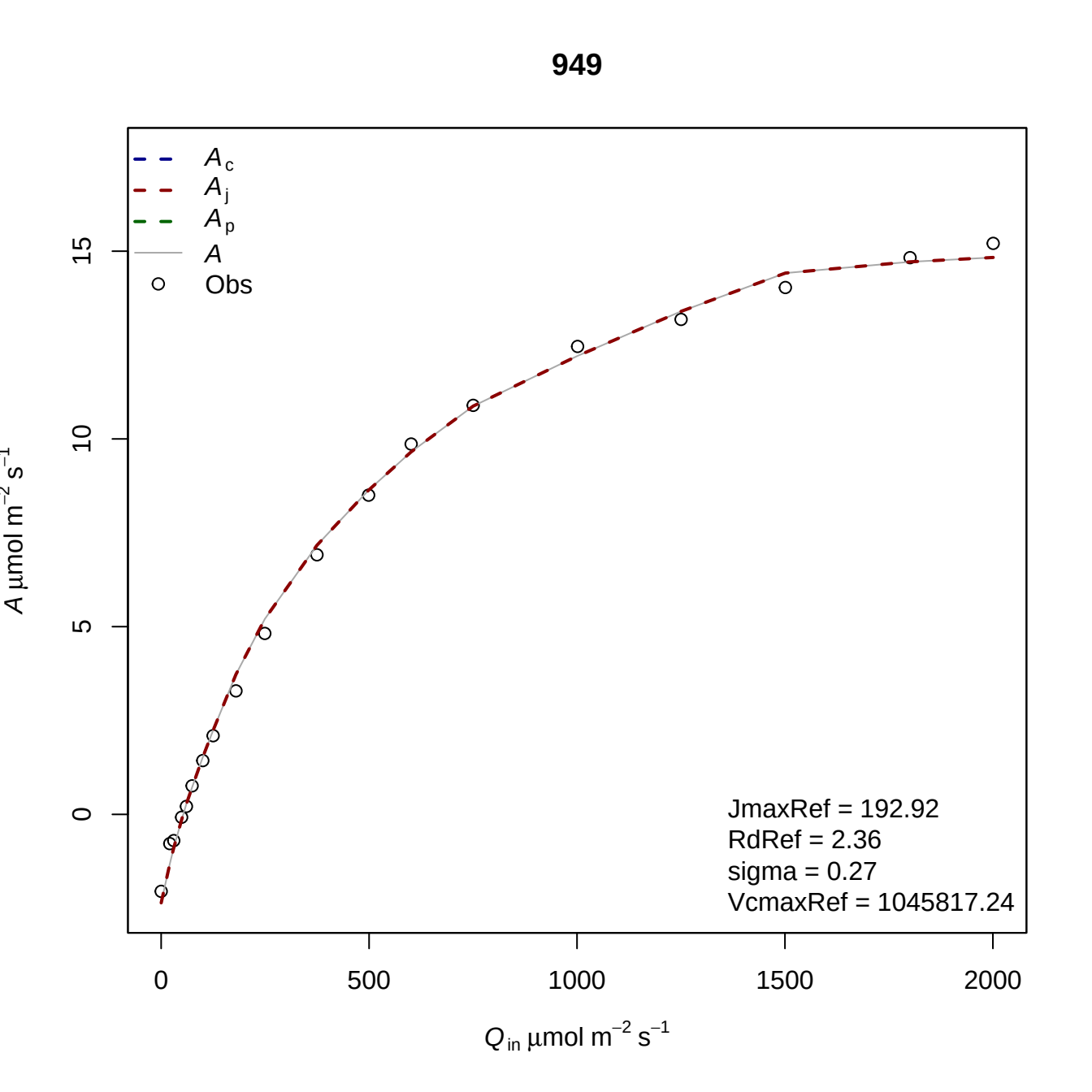

944
A_c
A_j
A_p
A
Obs
JmaxRef = 116.29
RdRef = 1.86
sigma = 0.43
VcmaxRef = 92.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

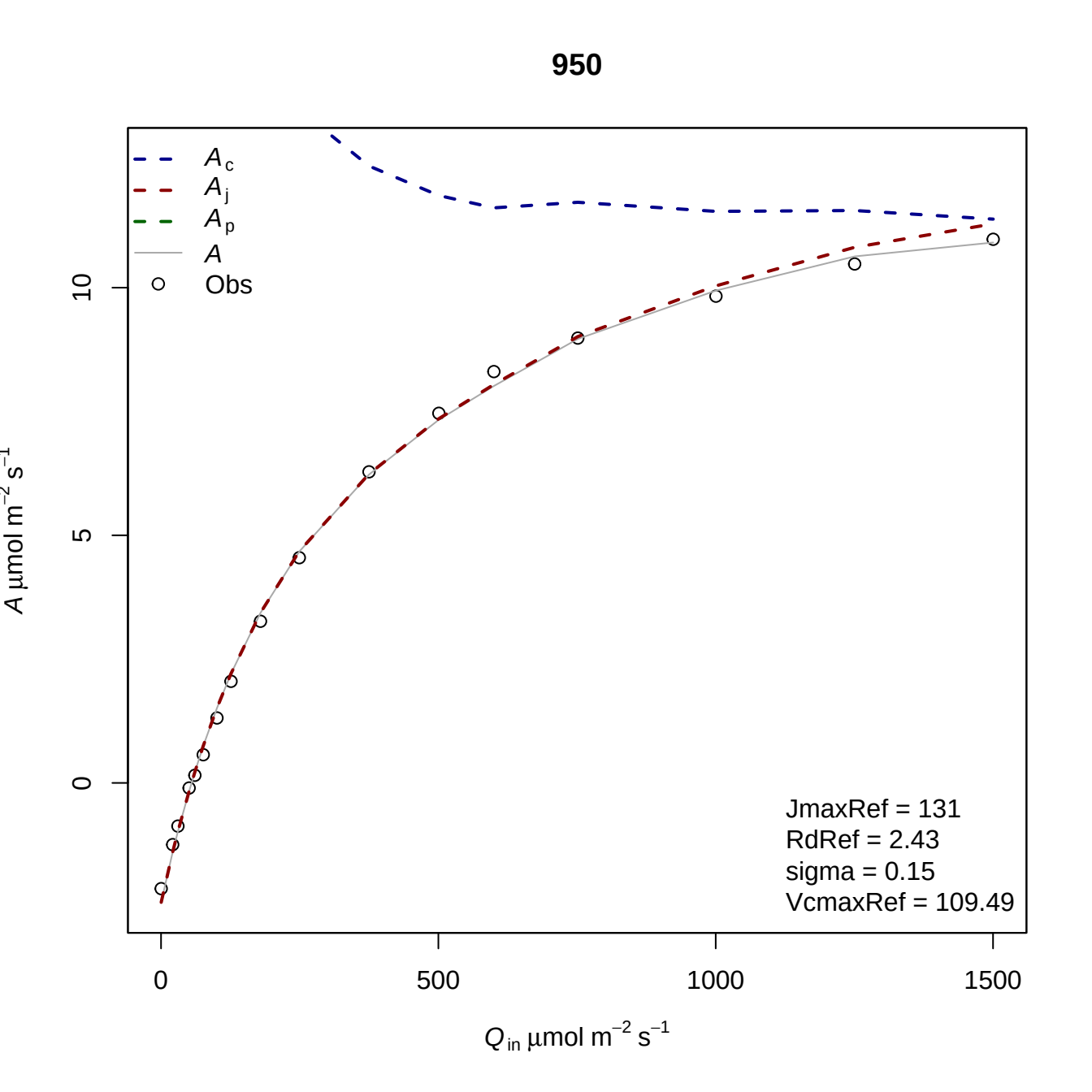

945
A_c
A_j
A_p
A
Obs
JmaxRef = 235.87
RdRef = 2.12
sigma = 0.25
VcmaxRef = 166.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

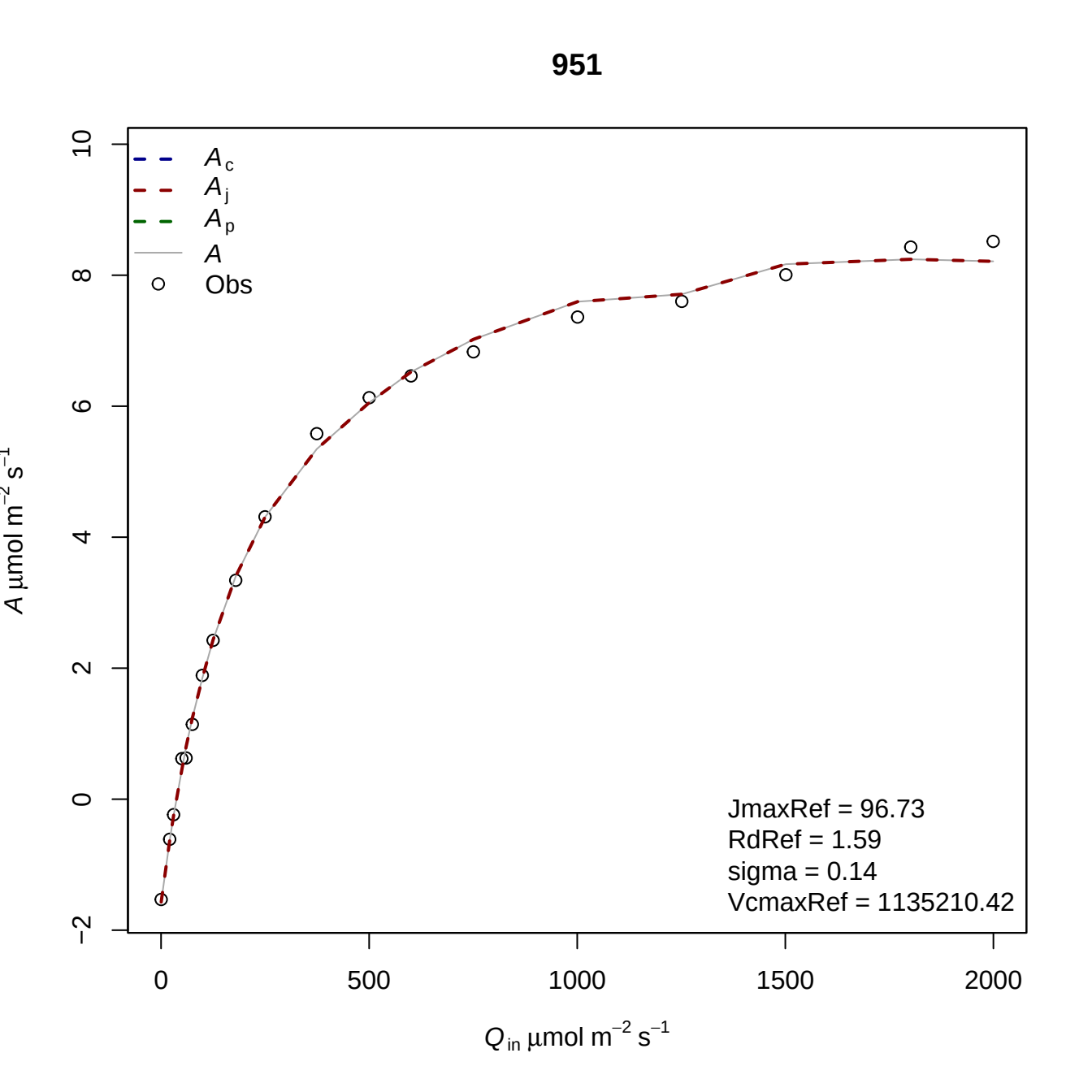

946
A_c
A_j
A_p
A
Obs
JmaxRef = 79.06
RdRef = 2.83
sigma = 0.41
VcmaxRef = 124869.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

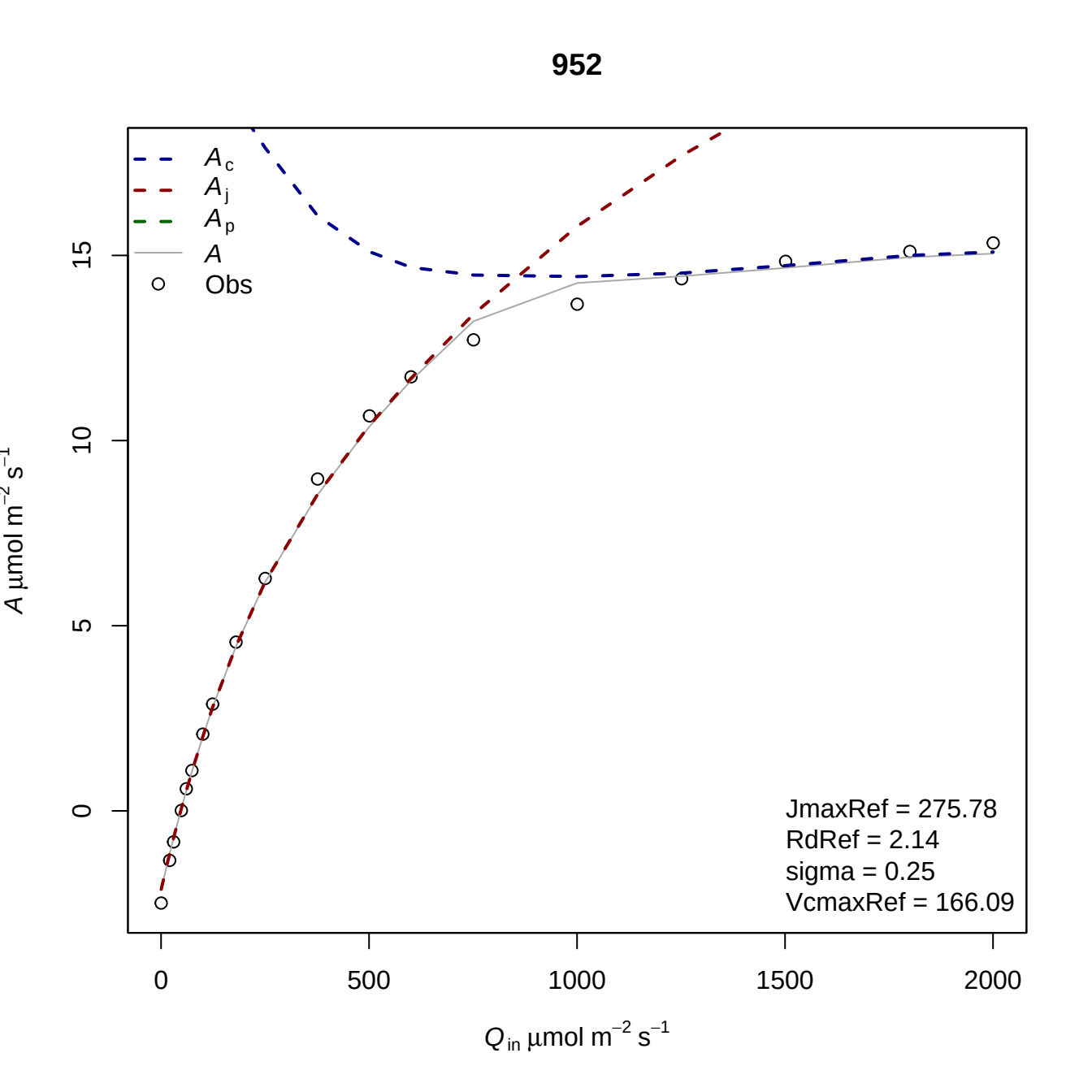

947
A_c
A_j
A_p
A
Obs
JmaxRef = 90.9
RdRef = 2.61
sigma = 0.33
VcmaxRef = 506008.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

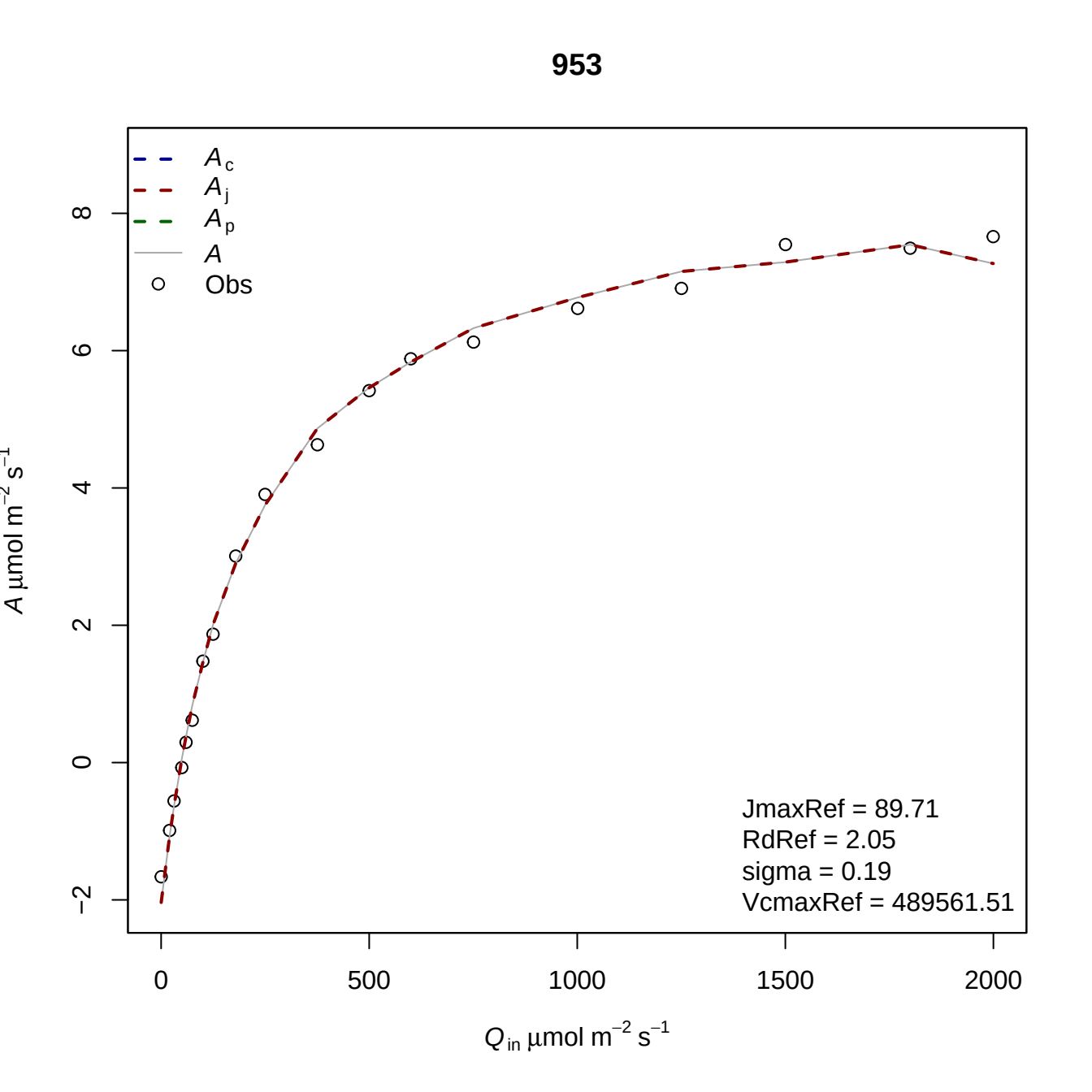

948
A_c
A_j
A_p
A
Obs
JmaxRef = 141.66
RdRef = 2.99
sigma = 0.4
VcmaxRef = 301716.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000

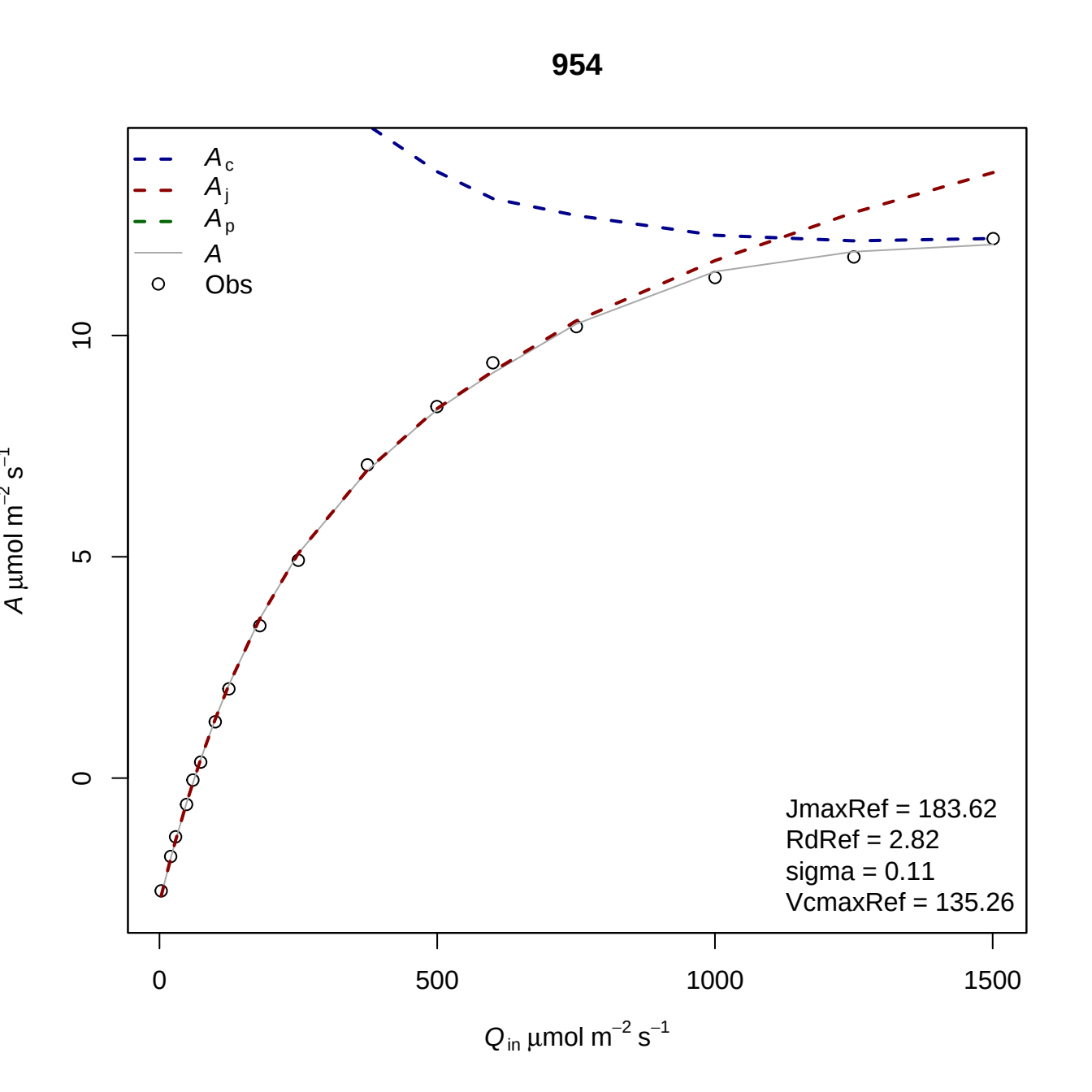

949
A_c
A_j
A_p
A
Obs
JmaxRef = 192.92
RdRef = 2.36
sigma = 0.27
VcmaxRef = 1045817.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

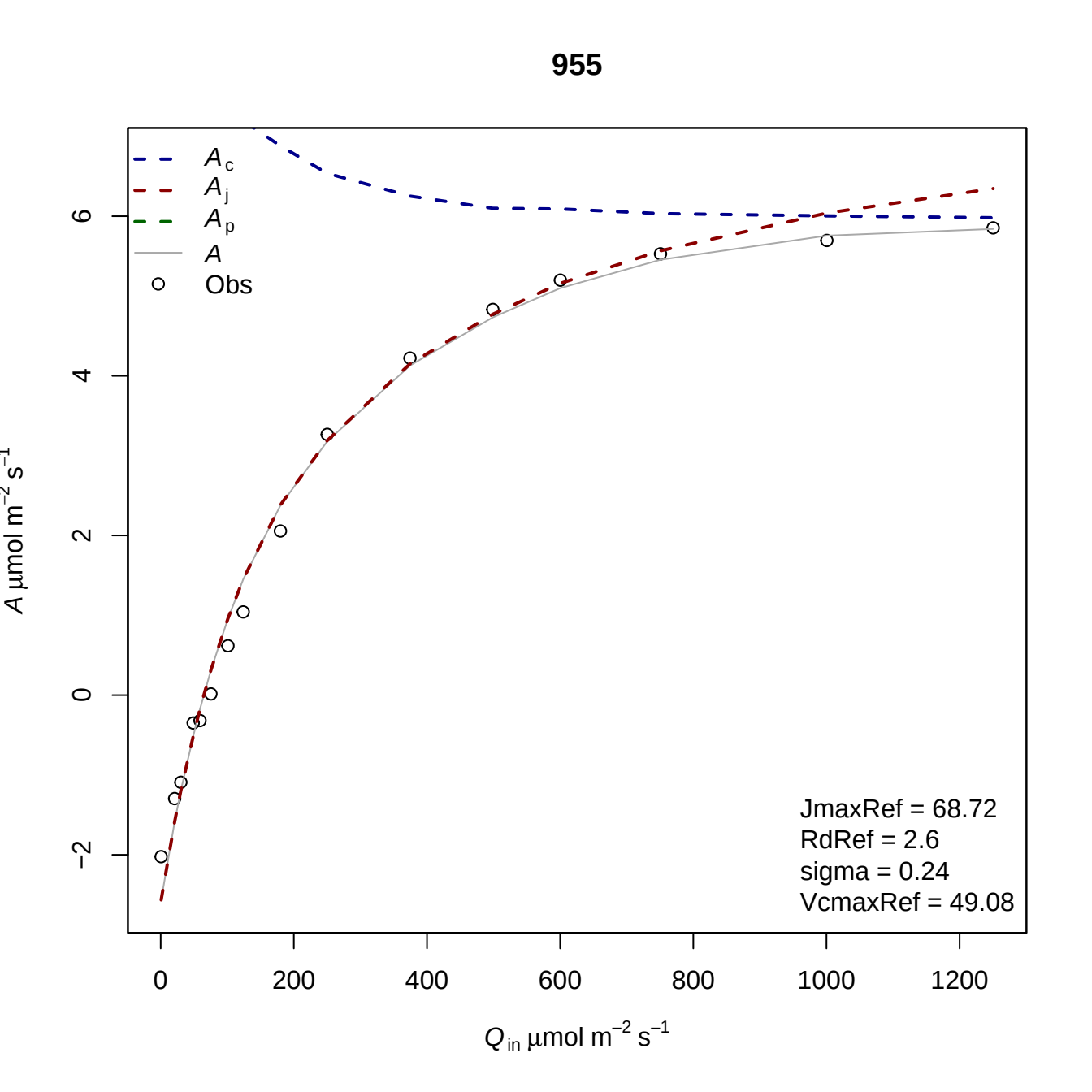

950
A_c
A_j
A_p
A
Obs
JmaxRef = 131
RdRef = 2.43
sigma = 0.15
VcmaxRef = 109.49
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

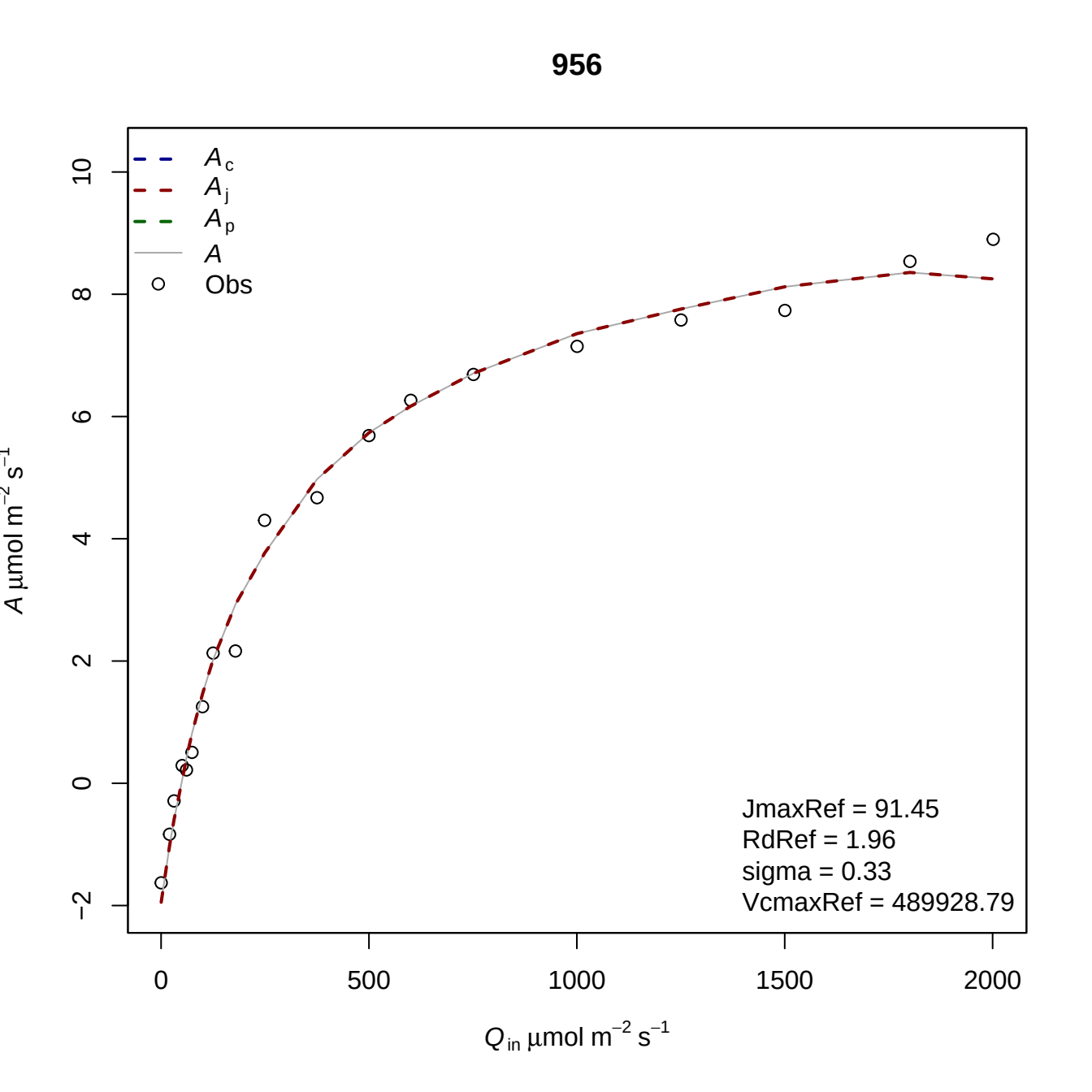

951
A_c
A_j
A_p
A
Obs
JmaxRef = 96.73
RdRef = 1.59
sigma = 0.14
VcmaxRef = 1135210.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

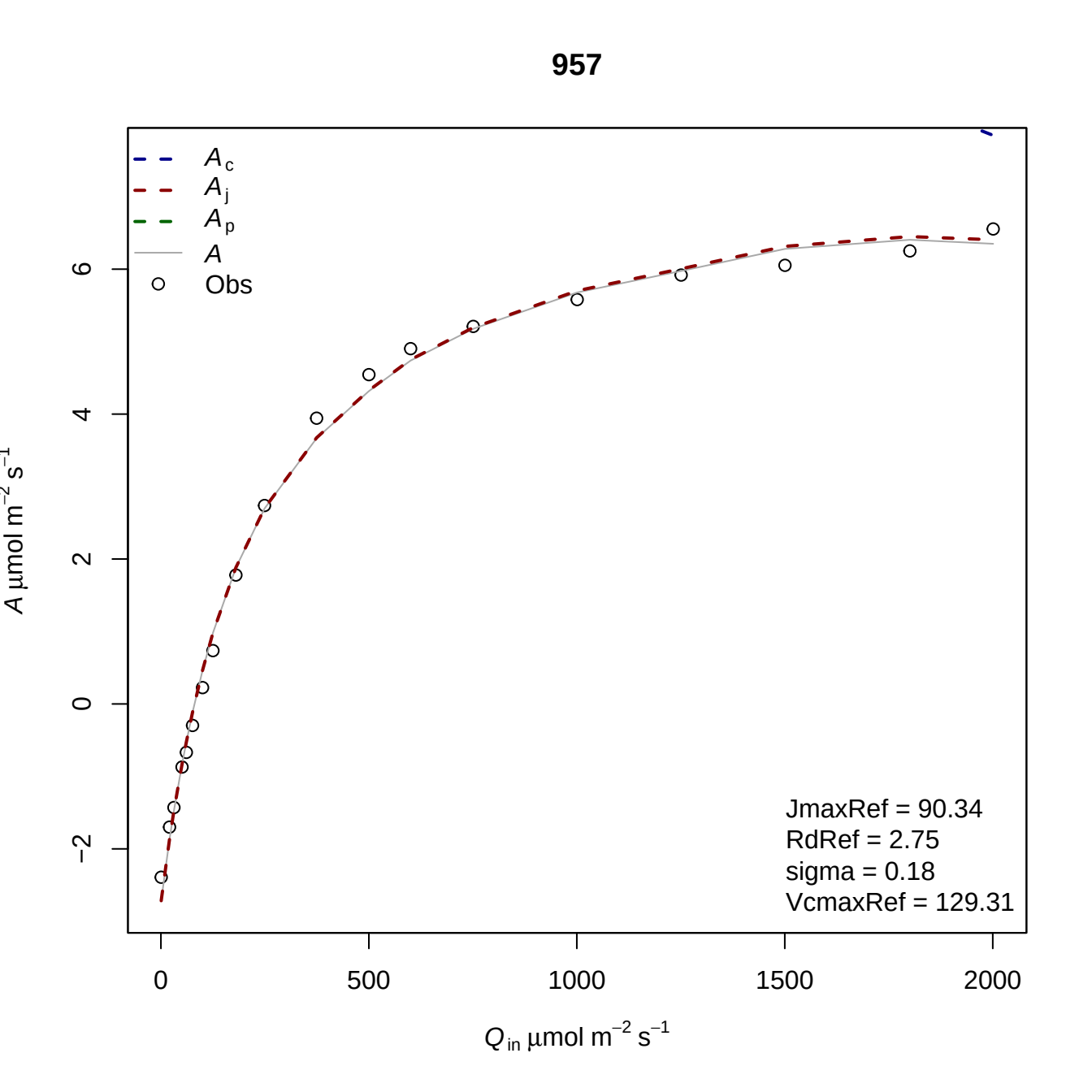

952
A_c
A_j
A_p
A
Obs
JmaxRef = 275.78
RdRef = 2.14
sigma = 0.25
VcmaxRef = 166.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

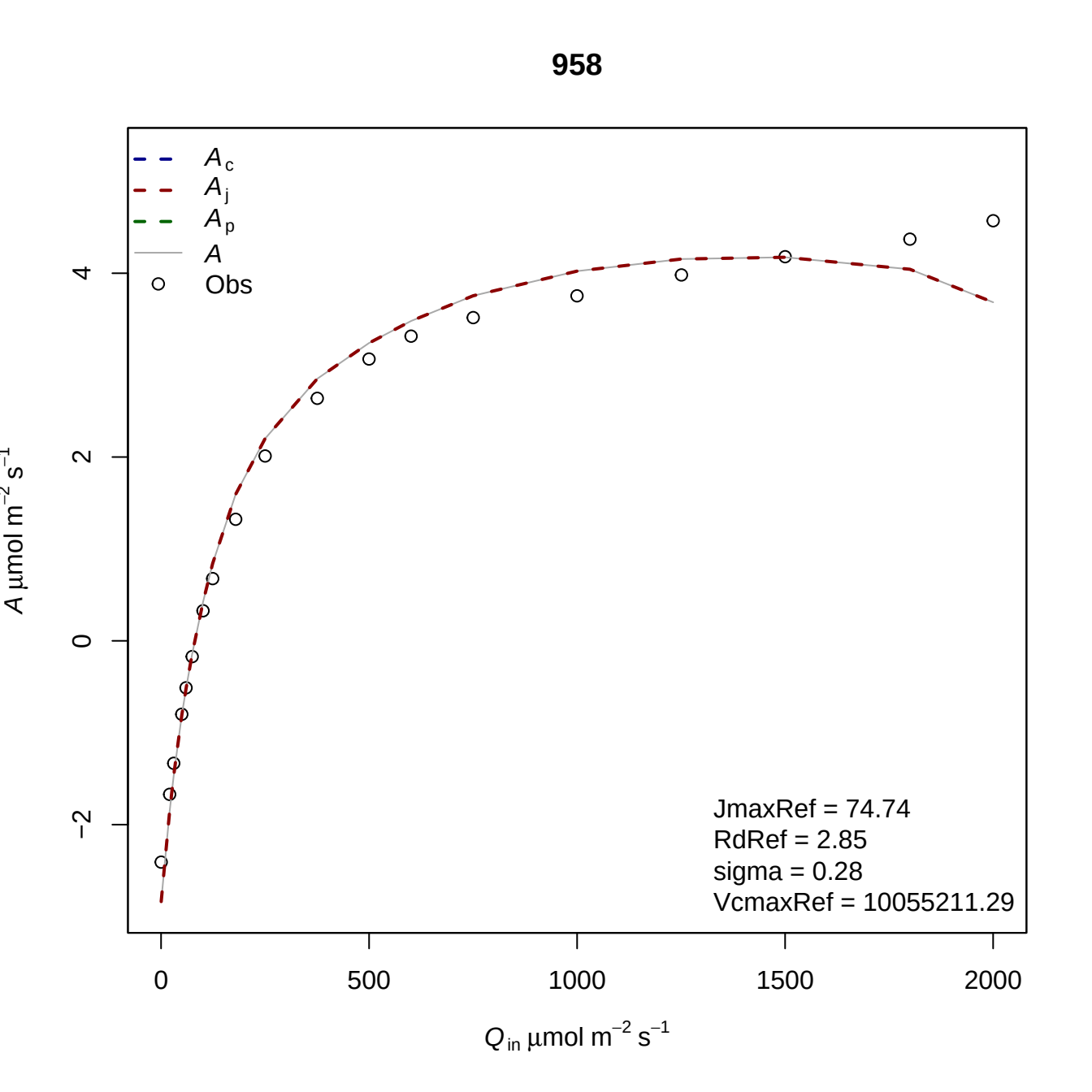

953
A_c
A_j
A_p
A
Obs
JmaxRef = 89.71
RdRef = 2.05
sigma = 0.19
VcmaxRef = 489561.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

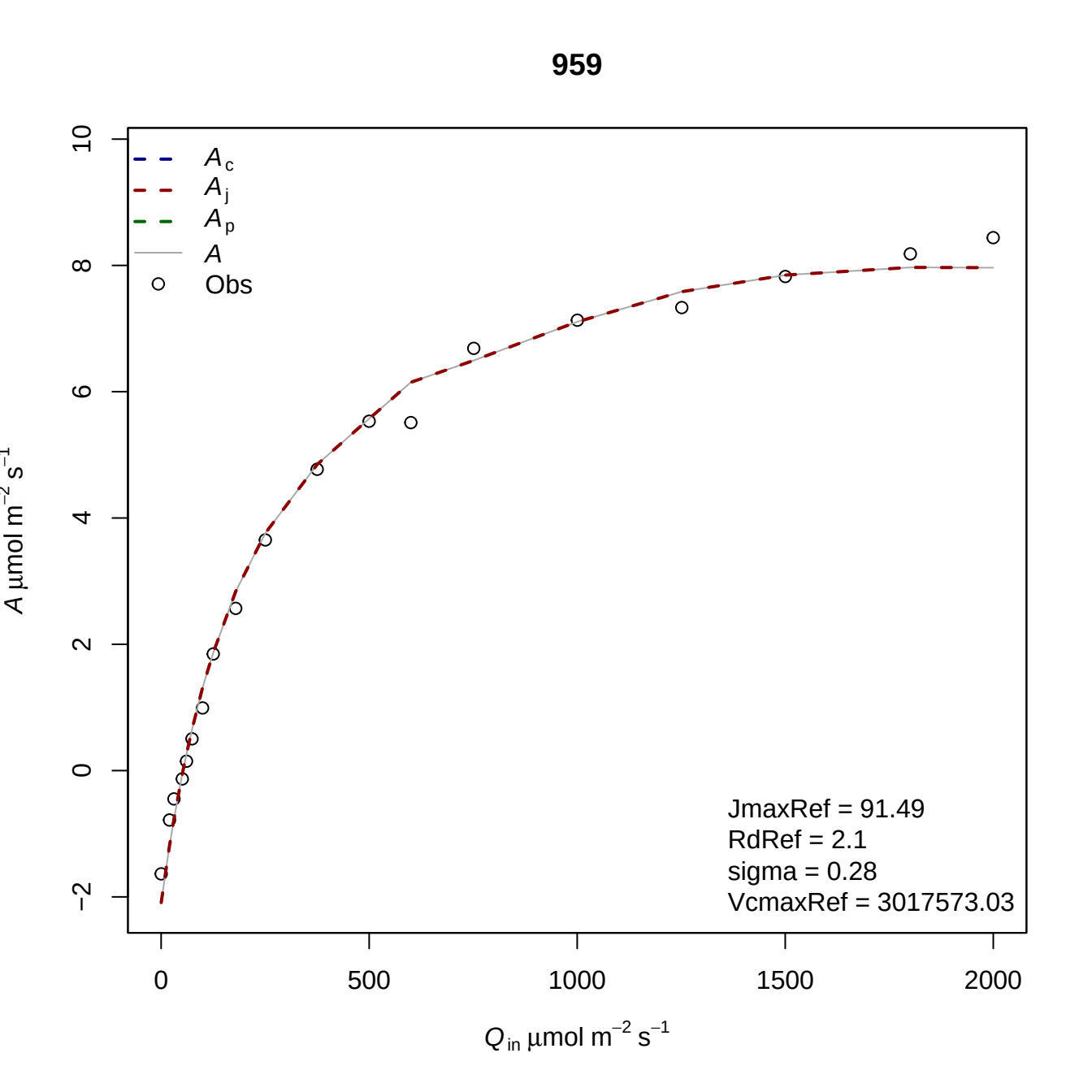

954
A_c
A_j
A_p
A
Obs
JmaxRef = 183.62
RdRef = 2.82
sigma = 0.11
VcmaxRef = 135.26
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

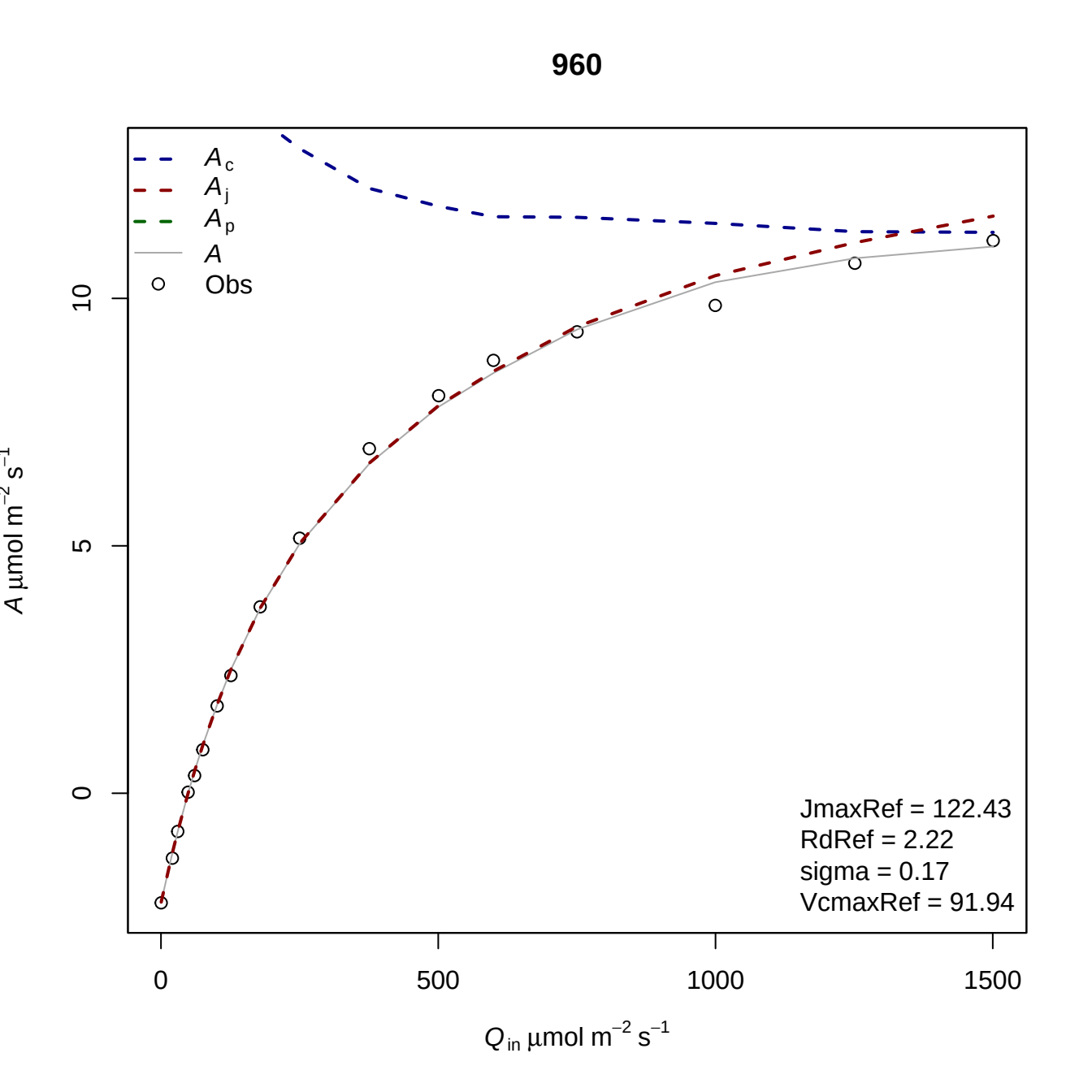

955
A_c
A_j
A_p
A
Obs
JmaxRef = 68.72
RdRef = 2.6
sigma = 0.24
VcmaxRef = 49.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200

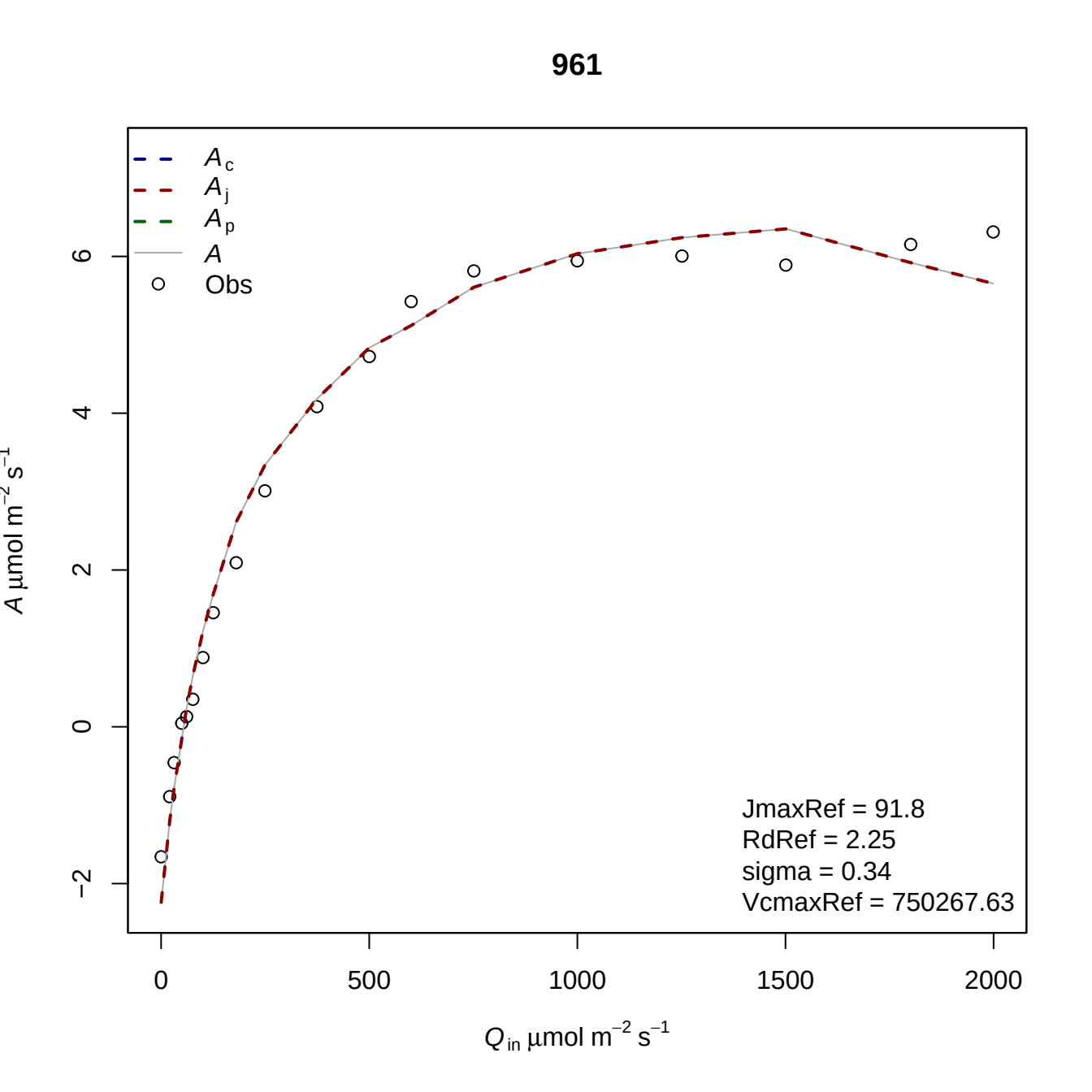

956
A_c
A_j
A_p
A
Obs
JmaxRef = 91.45
RdRef = 1.96
sigma = 0.33
VcmaxRef = 489928.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

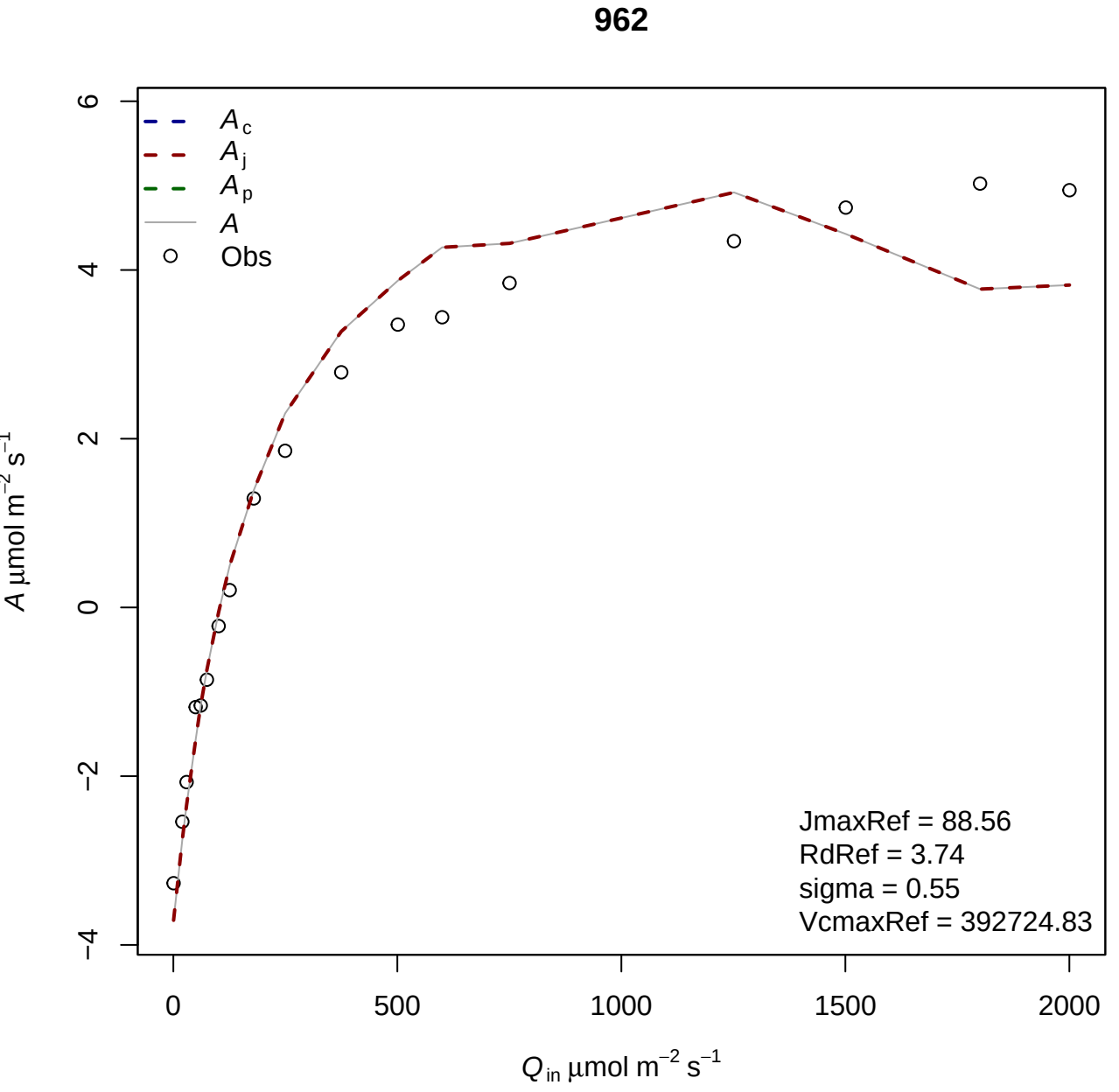

957
A_c
A_j
A_p
A
Obs
JmaxRef = 90.34
RdRef = 2.75
sigma = 0.18
VcmaxRef = 129.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

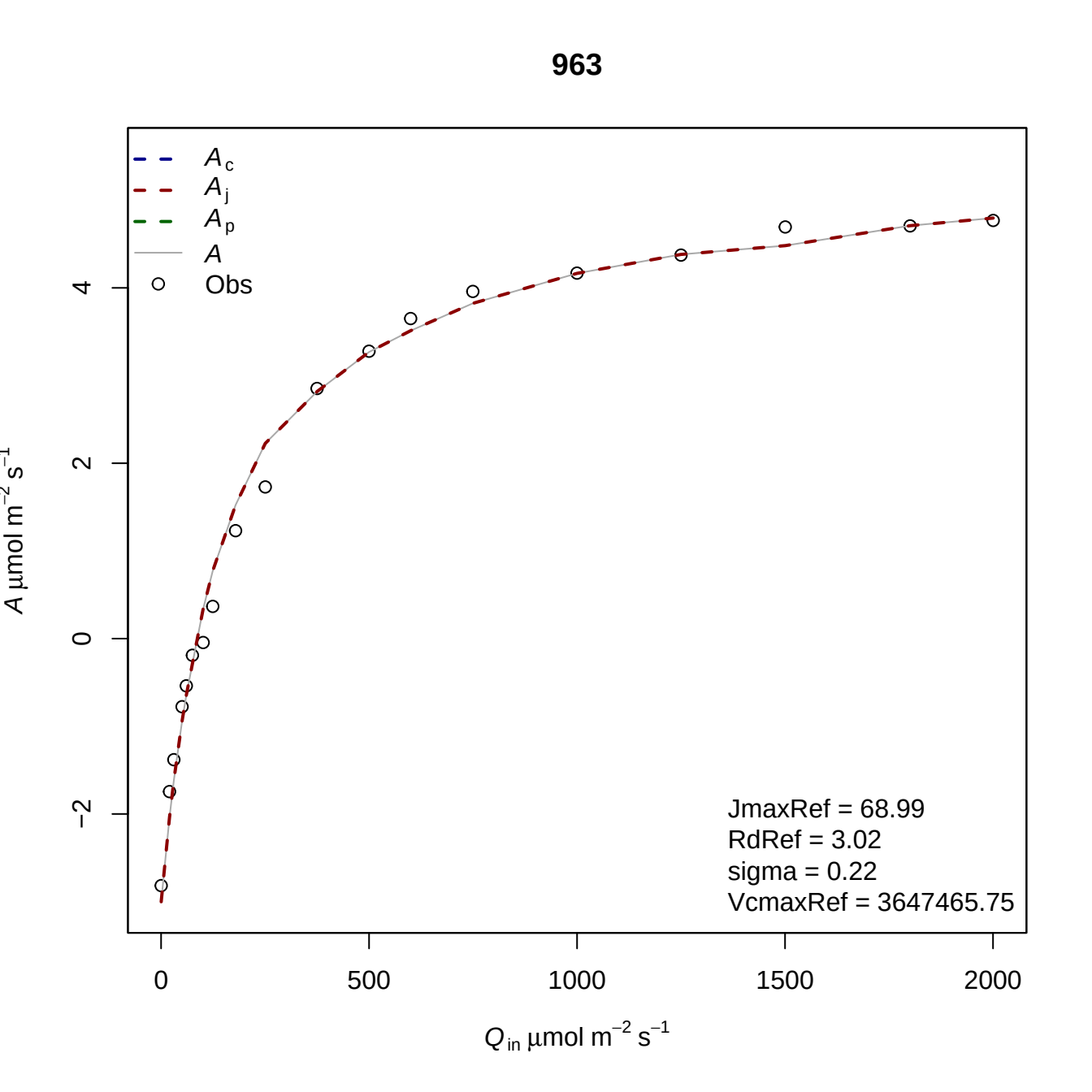

958
A_c
A_j
A_p
A
Obs
JmaxRef = 74.74
RdRef = 2.85
sigma = 0.28
VcmaxRef = 10055211.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

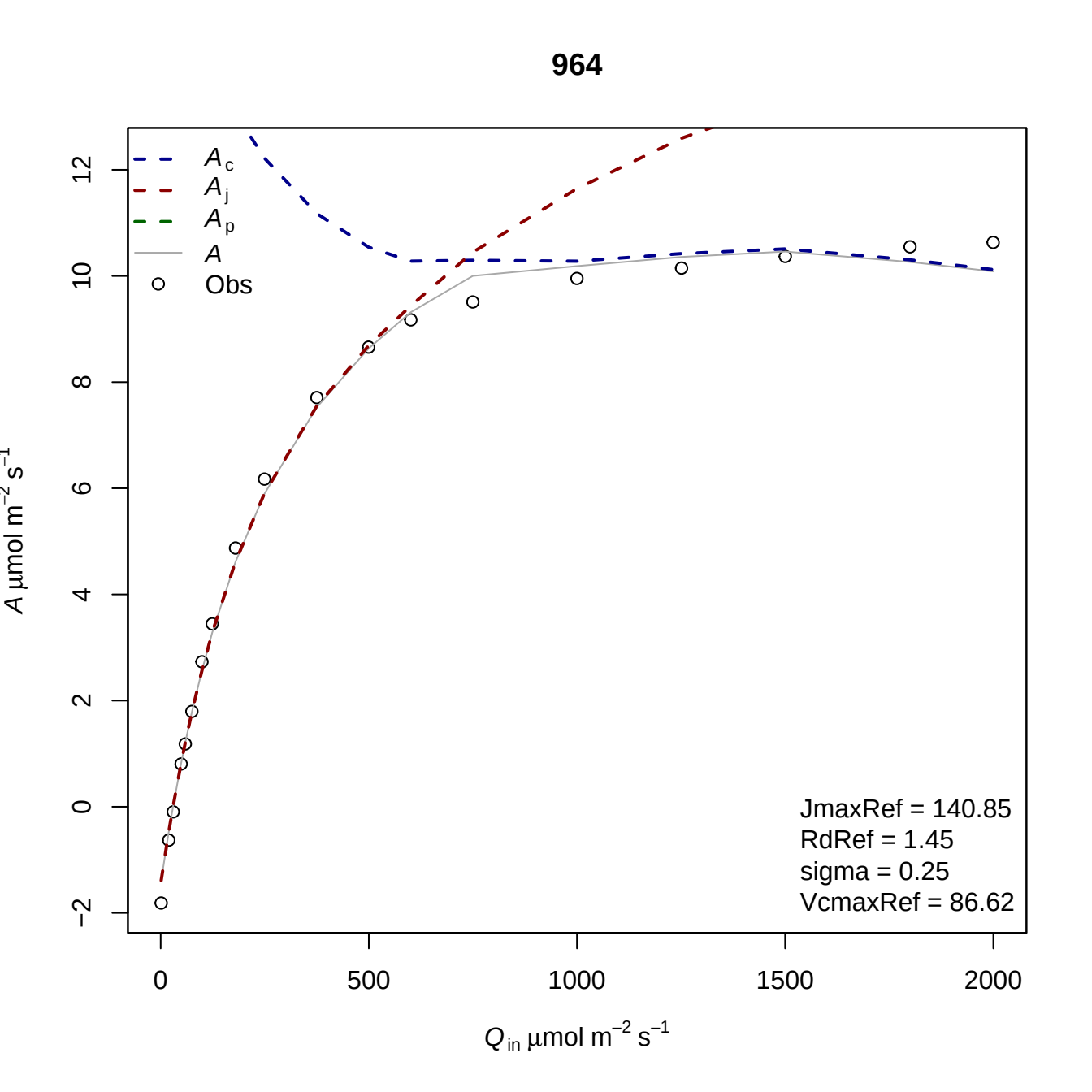

959
A_c
A_j
A_p
A
Obs
JmaxRef = 91.49
RdRef = 2.1
sigma = 0.28
VcmaxRef = 3017573.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

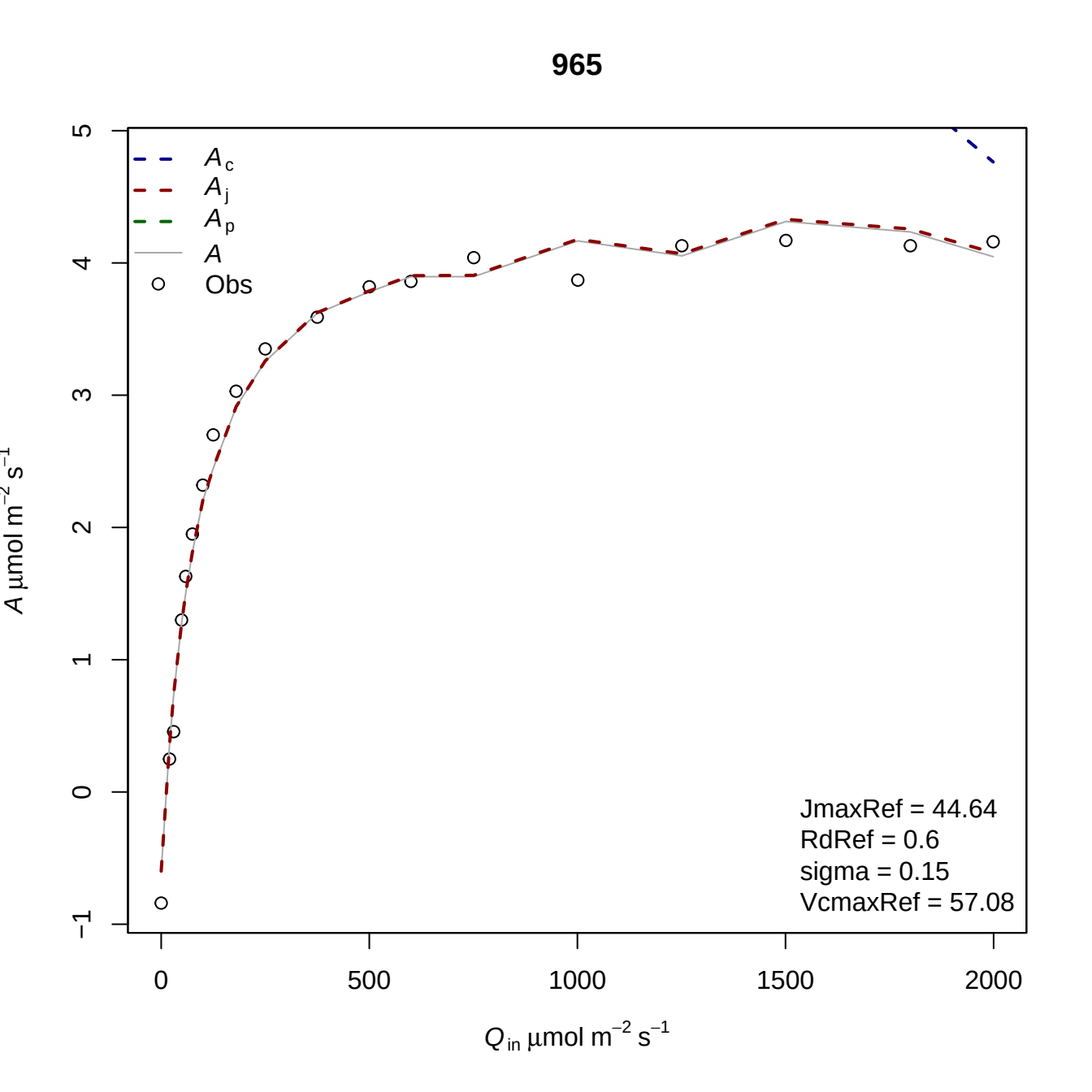

960
A_c
A_j
A_p
A
Obs
JmaxRef = 122.43
RdRef = 2.22
sigma = 0.17
VcmaxRef = 91.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

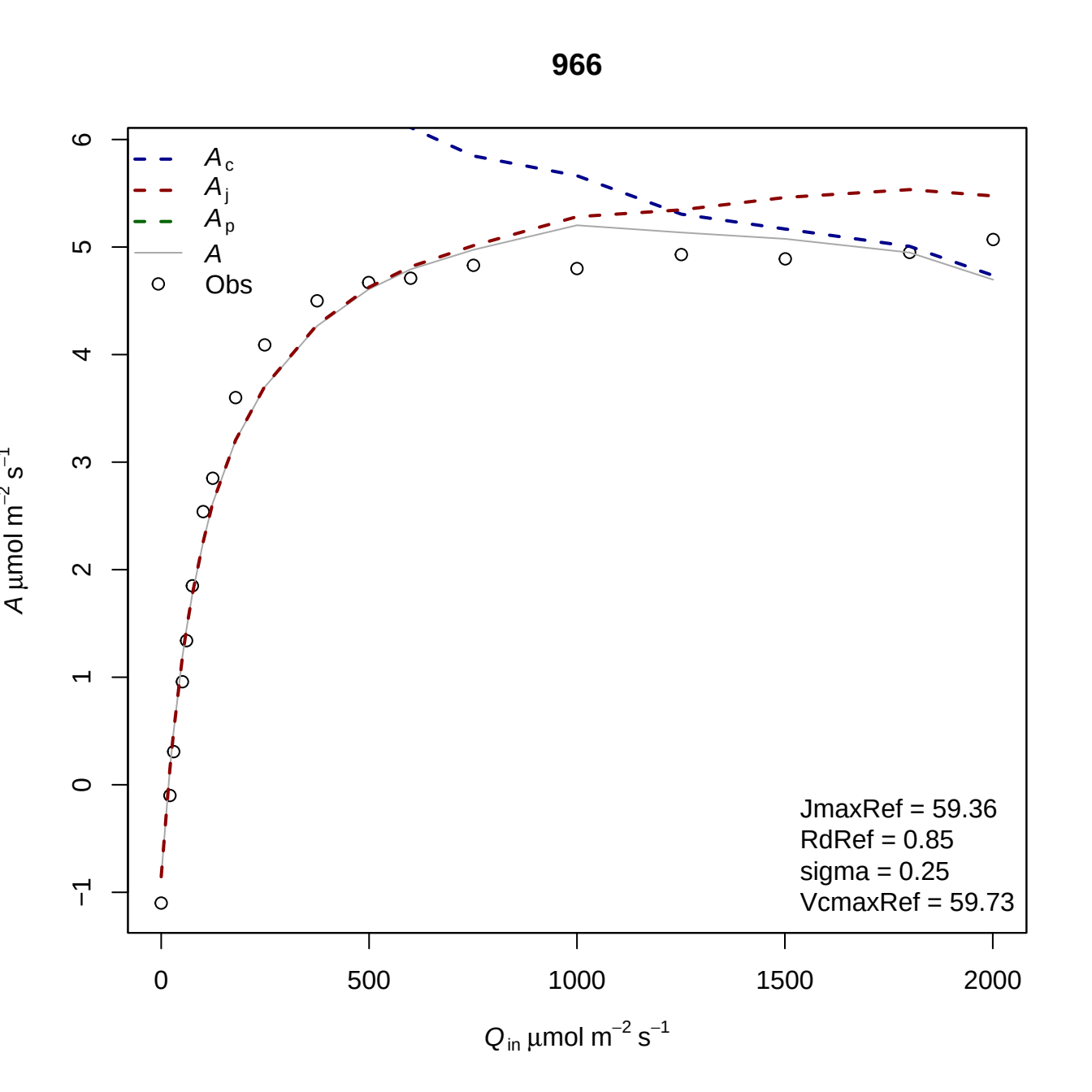

961
A_c
A_j
A_p
A
Obs
JmaxRef = 91.8
RdRef = 2.25
sigma = 0.34
VcmaxRef = 750267.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

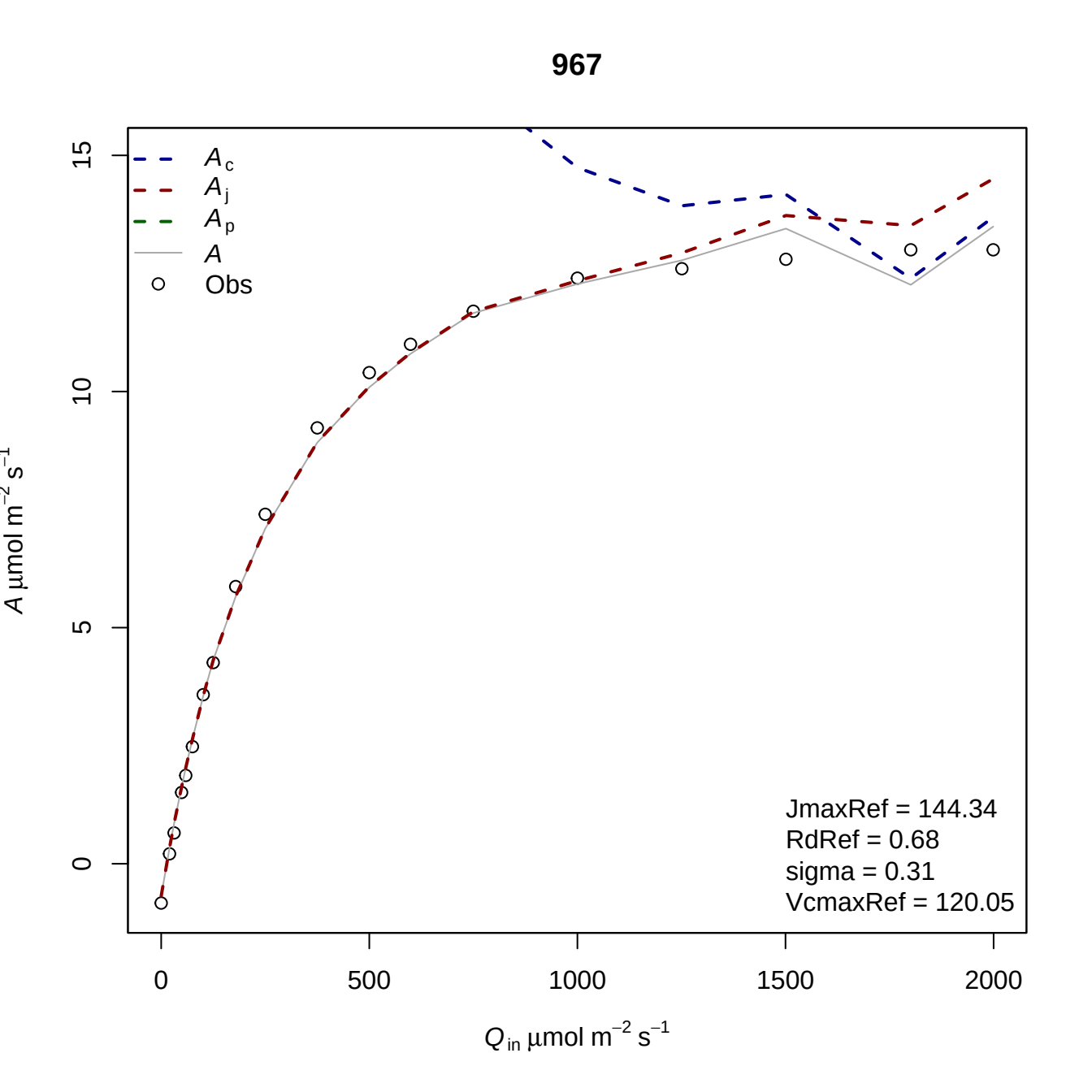

962
A_c
A_j
A_p
A
Obs
JmaxRef = 88.56
RdRef = 3.74
sigma = 0.55
VcmaxRef = 392724.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

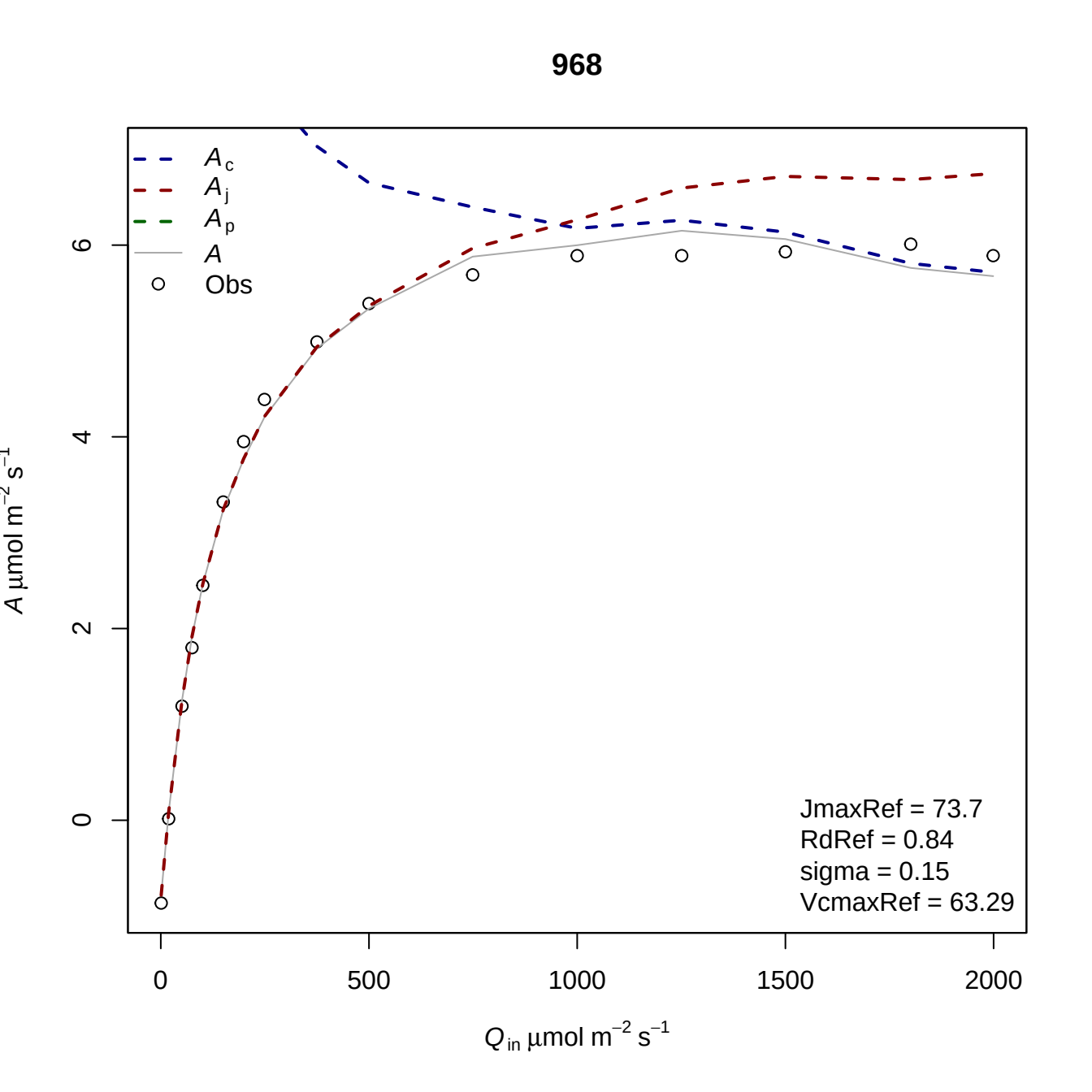

963
A_c
A_j
A_p
A
Obs
JmaxRef = 68.99
RdRef = 3.02
sigma = 0.22
VcmaxRef = 3647465.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

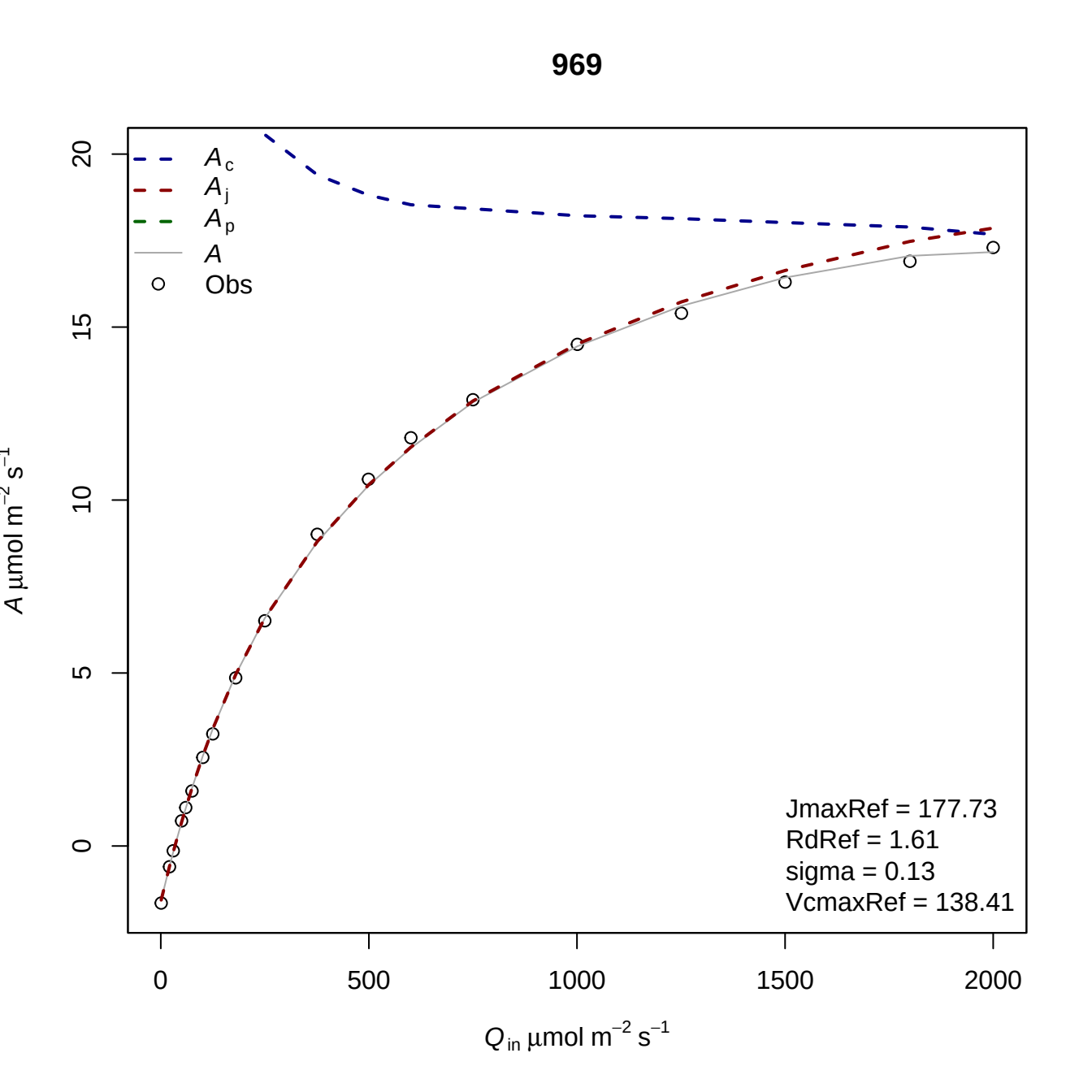

964
A_c
A_j
A_p
A
Obs
JmaxRef = 140.85
RdRef = 1.45
sigma = 0.25
VcmaxRef = 86.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

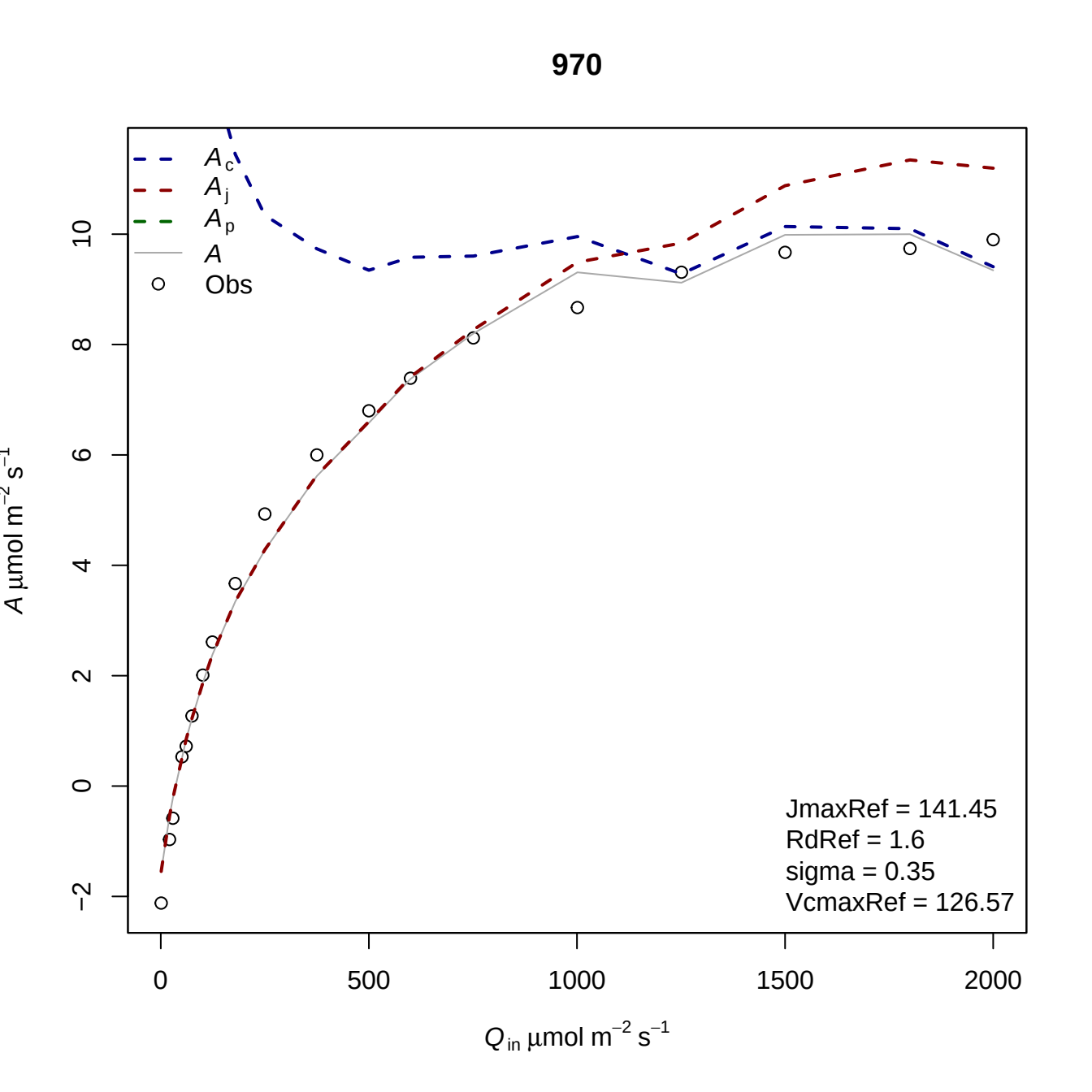

965
A_c
A_j
A_p
A
Obs
JmaxRef = 44.64
RdRef = 0.6
sigma = 0.15
VcmaxRef = 57.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

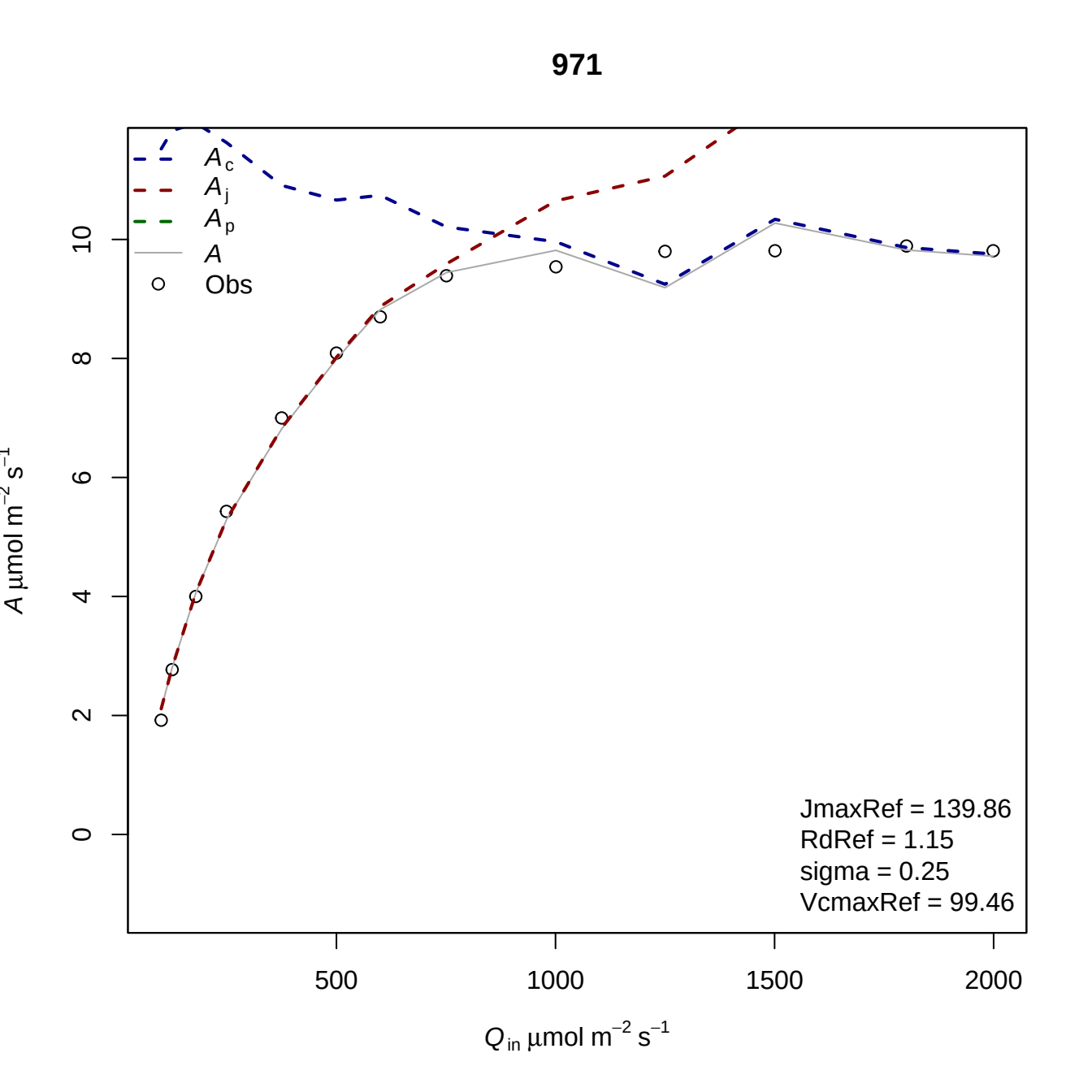

966
A_c
A_j
A_p
A
Obs
JmaxRef = 59.36
RdRef = 0.85
sigma = 0.25
VcmaxRef = 59.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

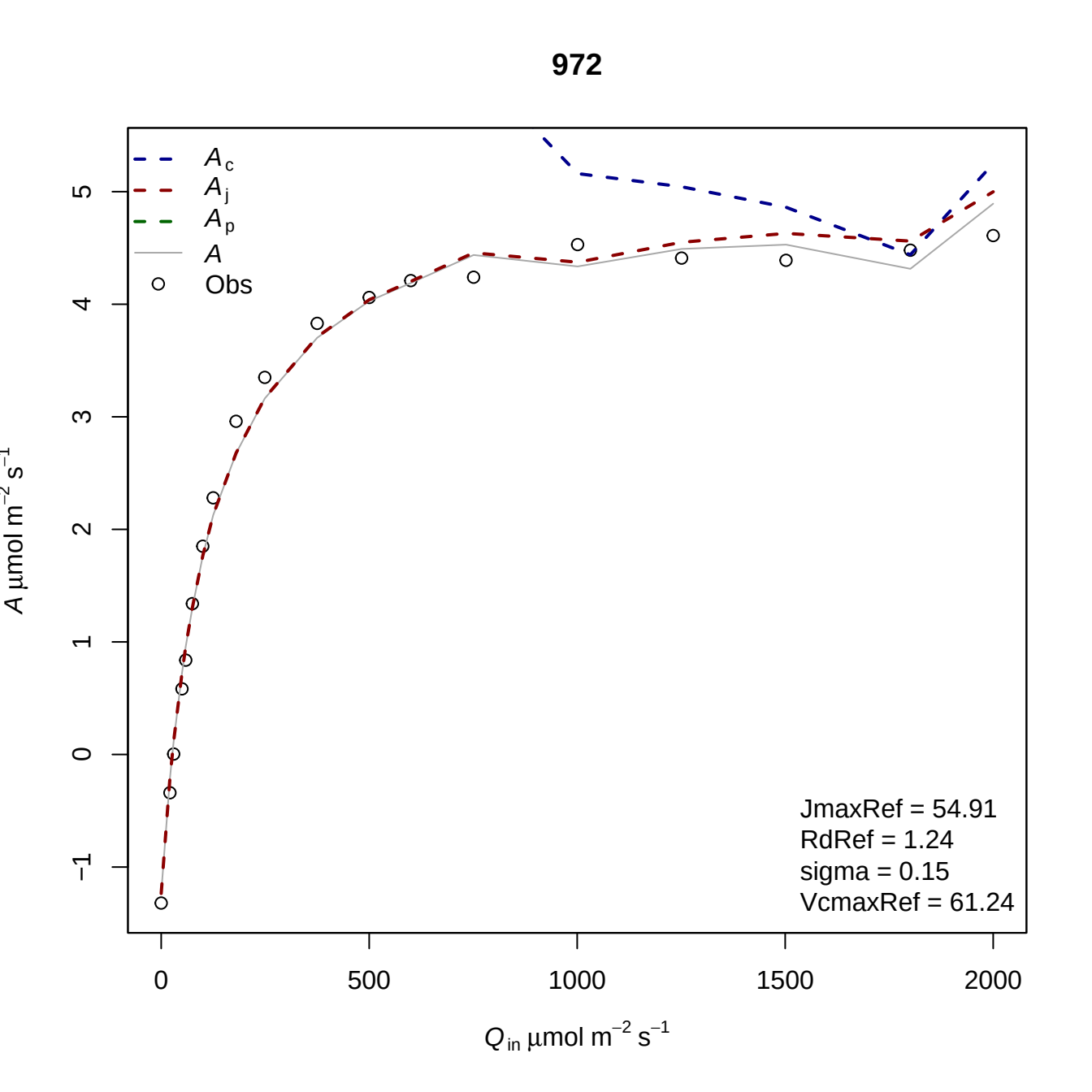

967
A_c
A_j
A_p
A
Obs
JmaxRef = 144.34
RdRef = 0.68
sigma = 0.31
VcmaxRef = 120.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

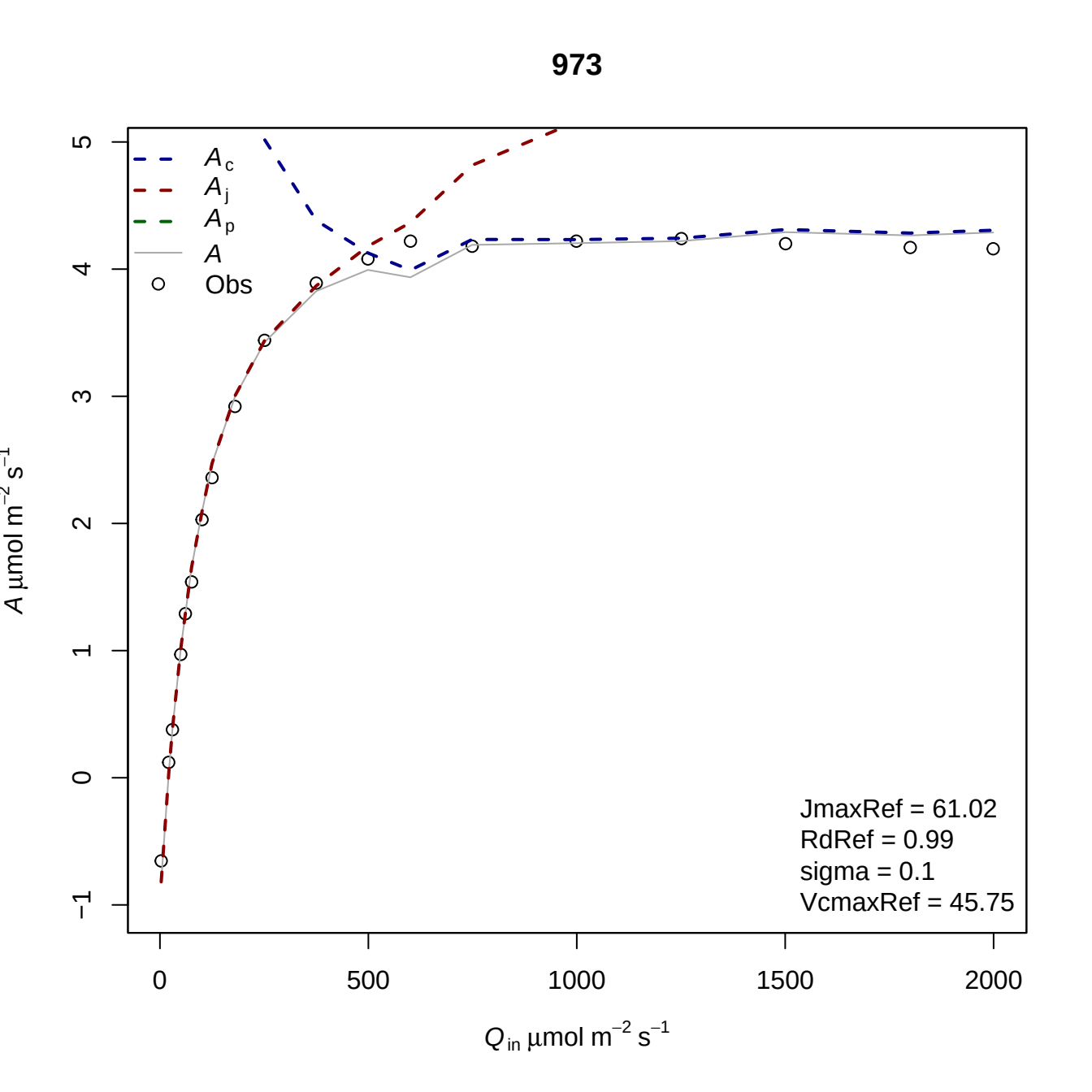

968
A_c
A_j
A_p
A
Obs
JmaxRef = 73.7
RdRef = 0.84
sigma = 0.15
VcmaxRef = 63.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

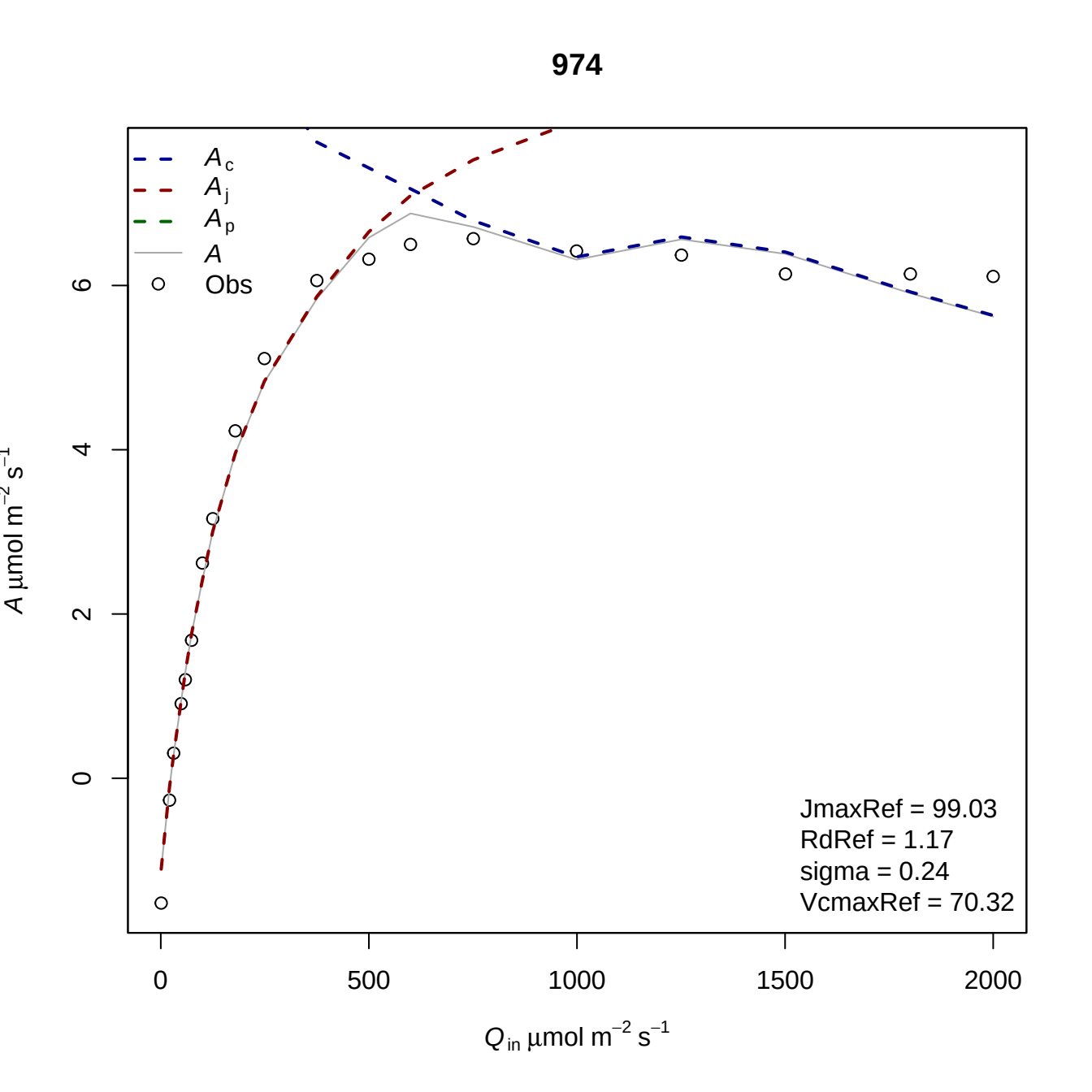

969
A_c
A_j
A_p
A
Obs
JmaxRef = 177.73
RdRef = 1.61
sigma = 0.13
VcmaxRef = 138.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

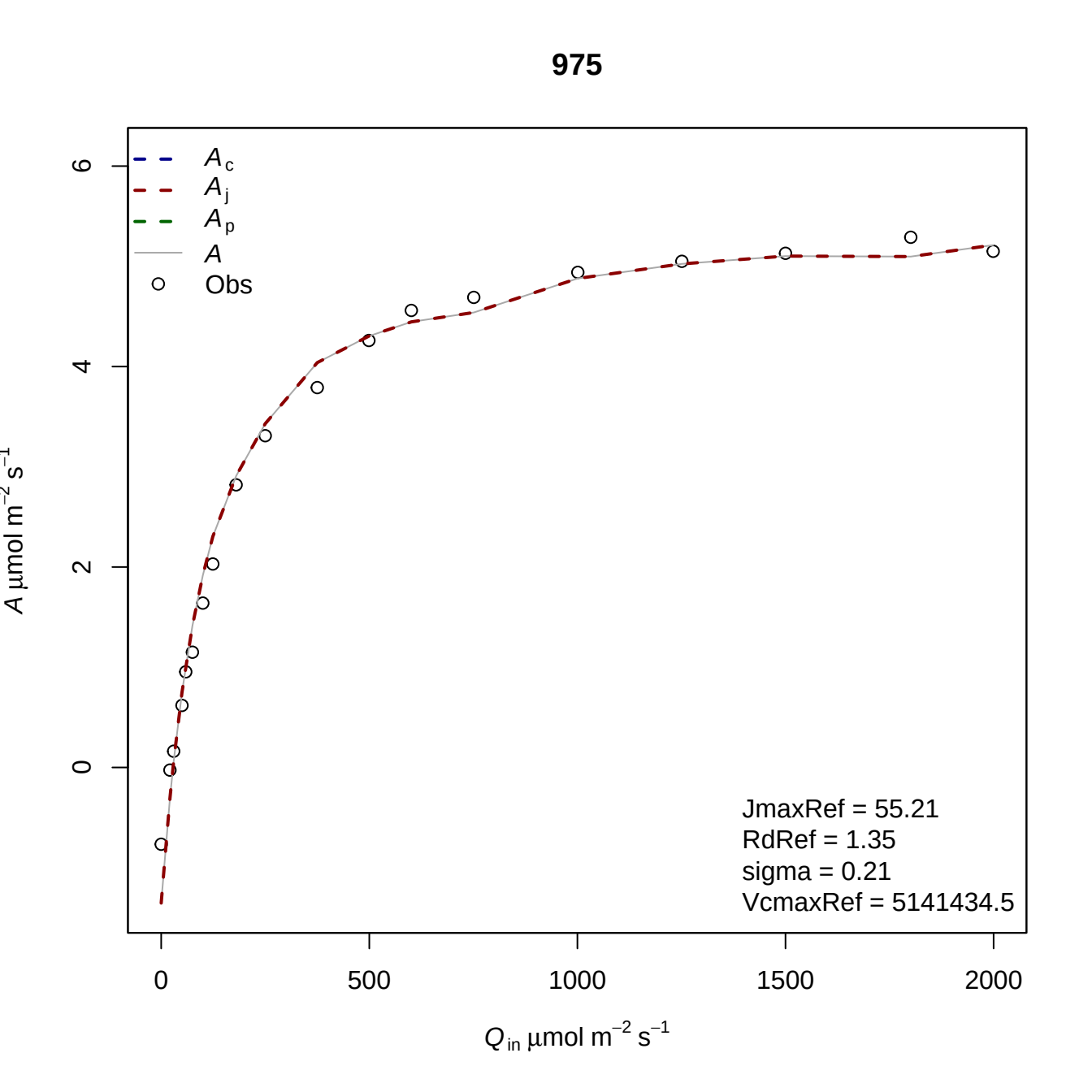

970
A_c
A_j
A_p
A
Obs
JmaxRef = 141.45
RdRef = 1.6
sigma = 0.35
VcmaxRef = 126.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

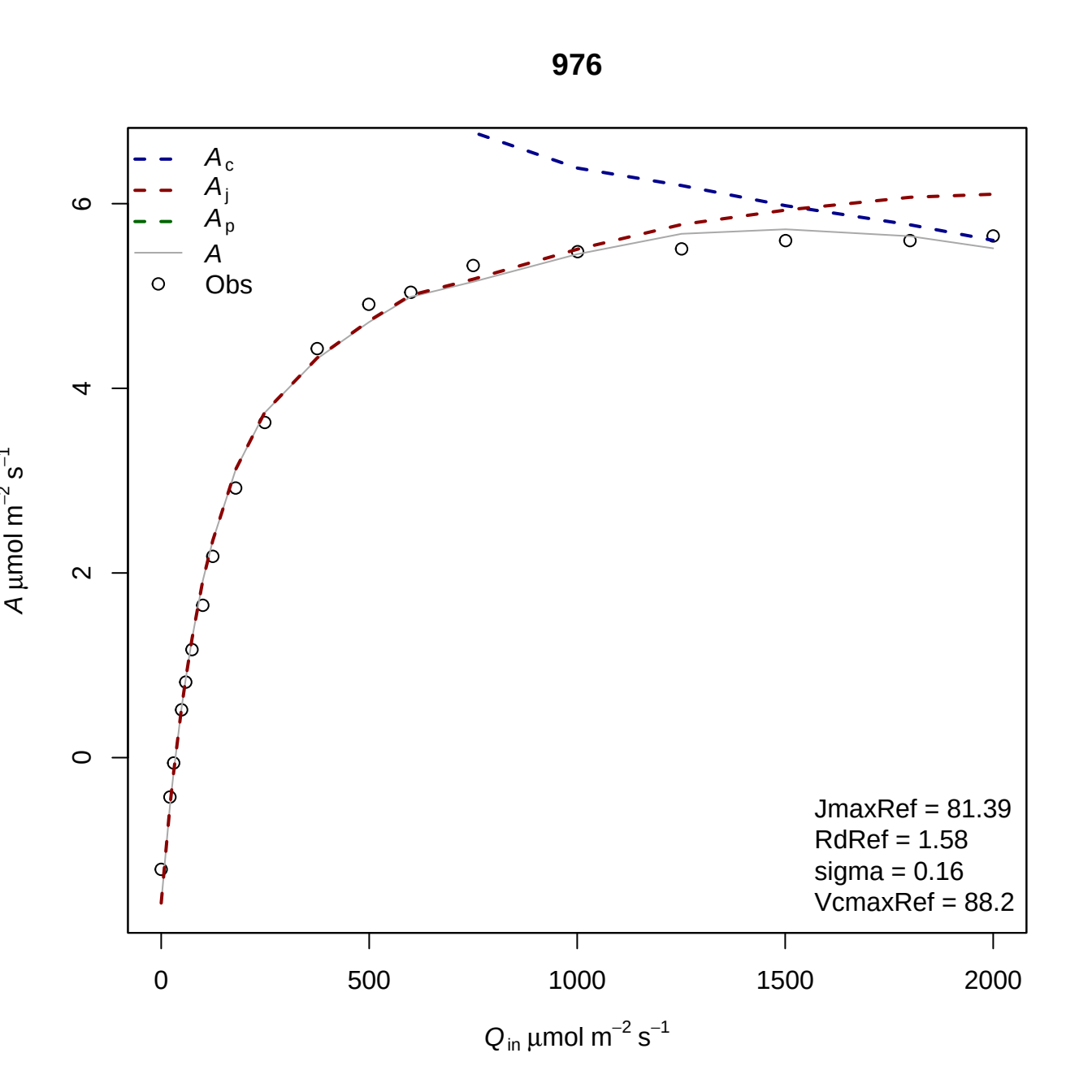

971
A_c
A_j
A_p
A
Obs
JmaxRef = 139.86
RdRef = 1.15
sigma = 0.25
VcmaxRef = 99.46
0
2
4
6
8
10
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

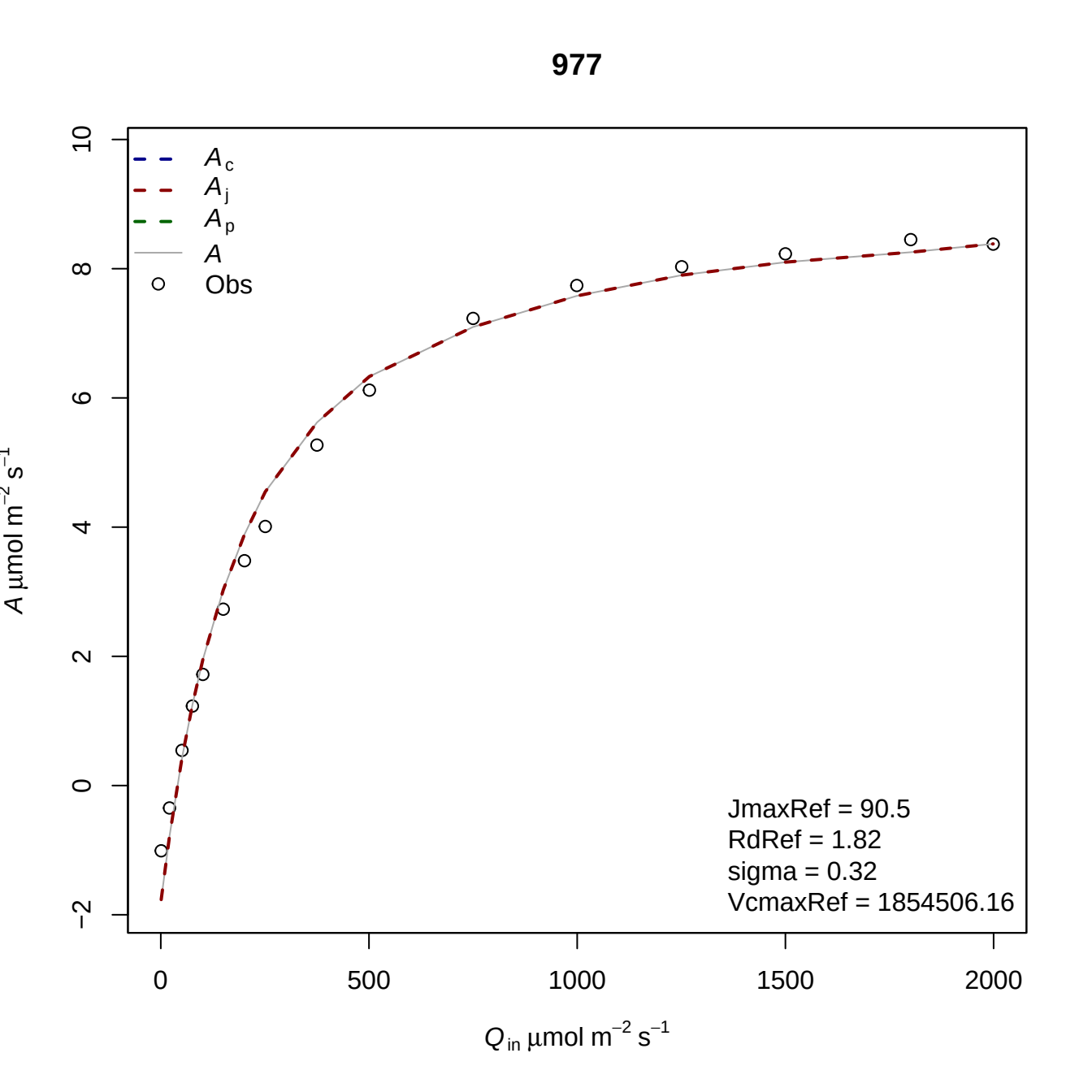

972
A_c
A_j
A_p
A
Obs
JmaxRef = 54.91
RdRef = 1.24
sigma = 0.15
VcmaxRef = 61.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

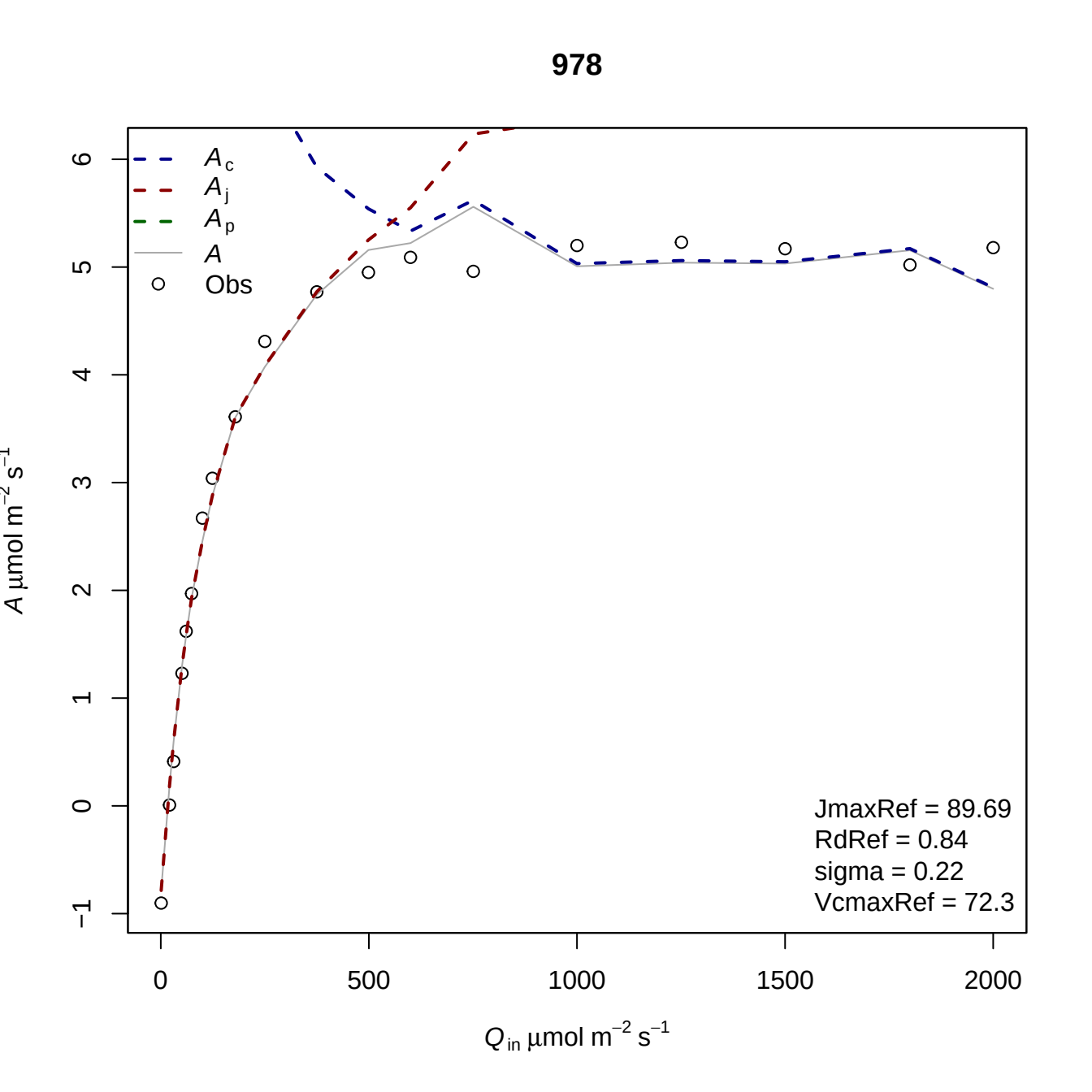

973
A_c
A_j
A_p
A
Obs
JmaxRef = 61.02
RdRef = 0.99
sigma = 0.1
VcmaxRef = 45.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

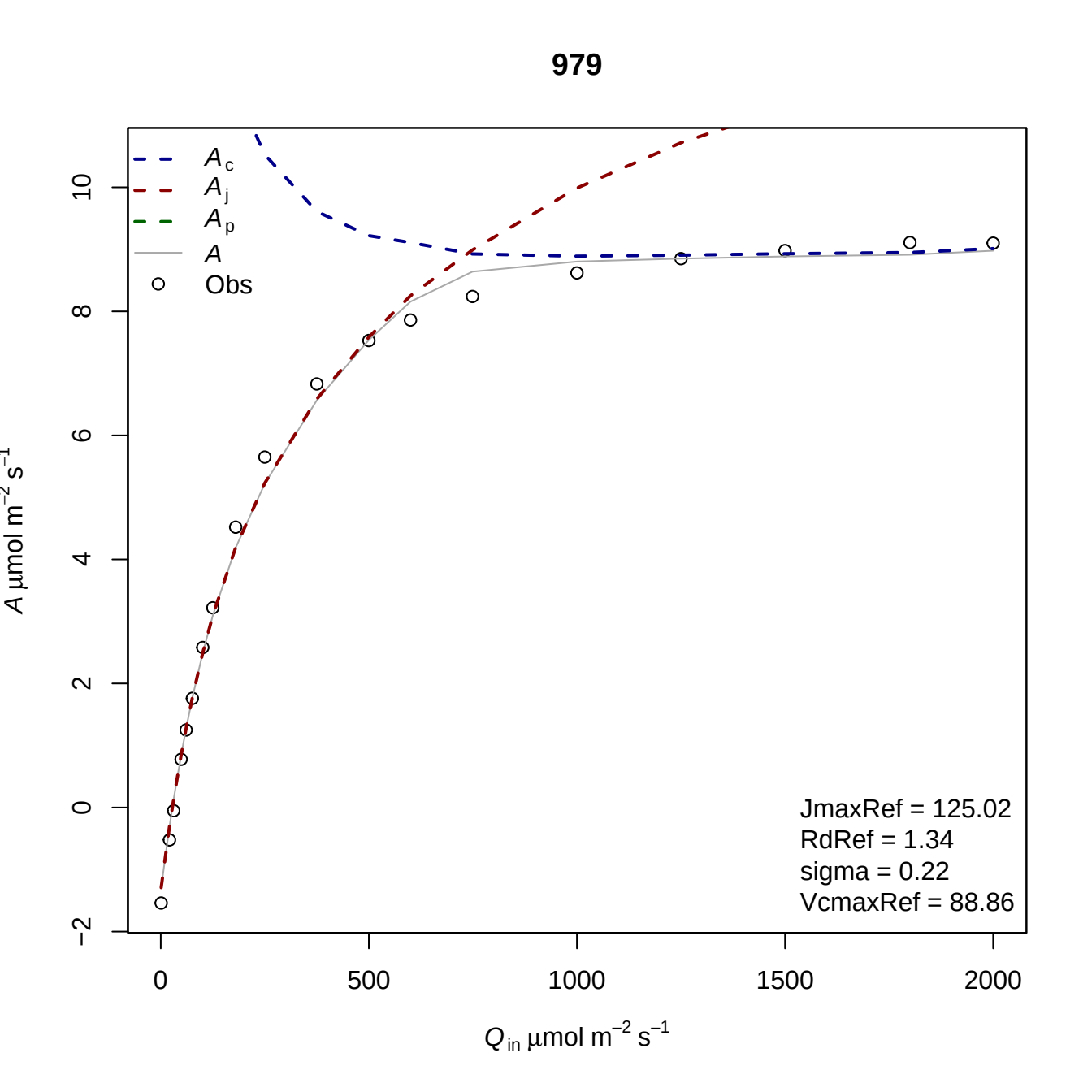

974
A_c
A_j
A_p
A
Obs
JmaxRef = 99.03
RdRef = 1.17
sigma = 0.24
VcmaxRef = 70.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

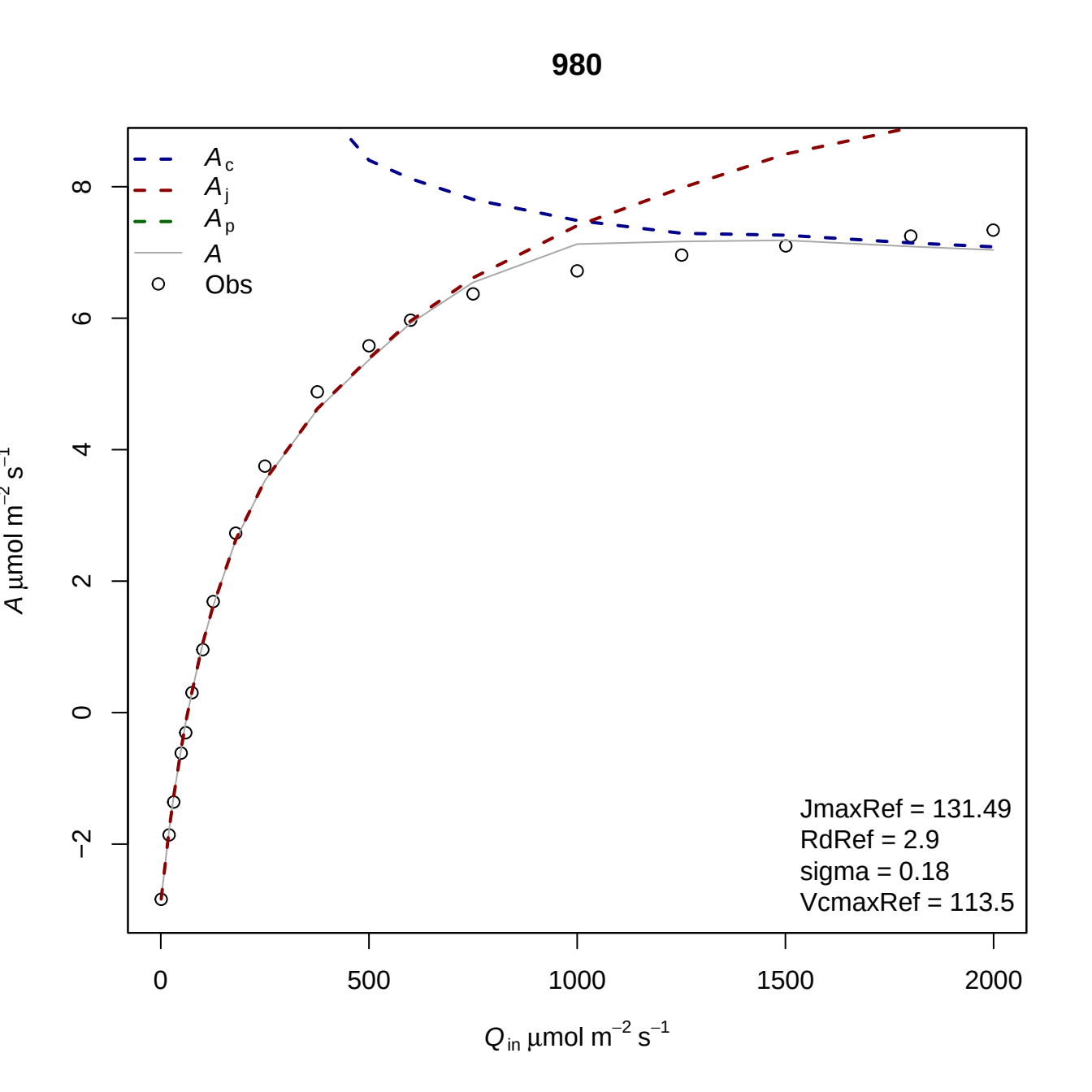

975
A_c
A_j
A_p
A
Obs
JmaxRef = 55.21
RdRef = 1.35
sigma = 0.21
VcmaxRef = 5141434.5
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

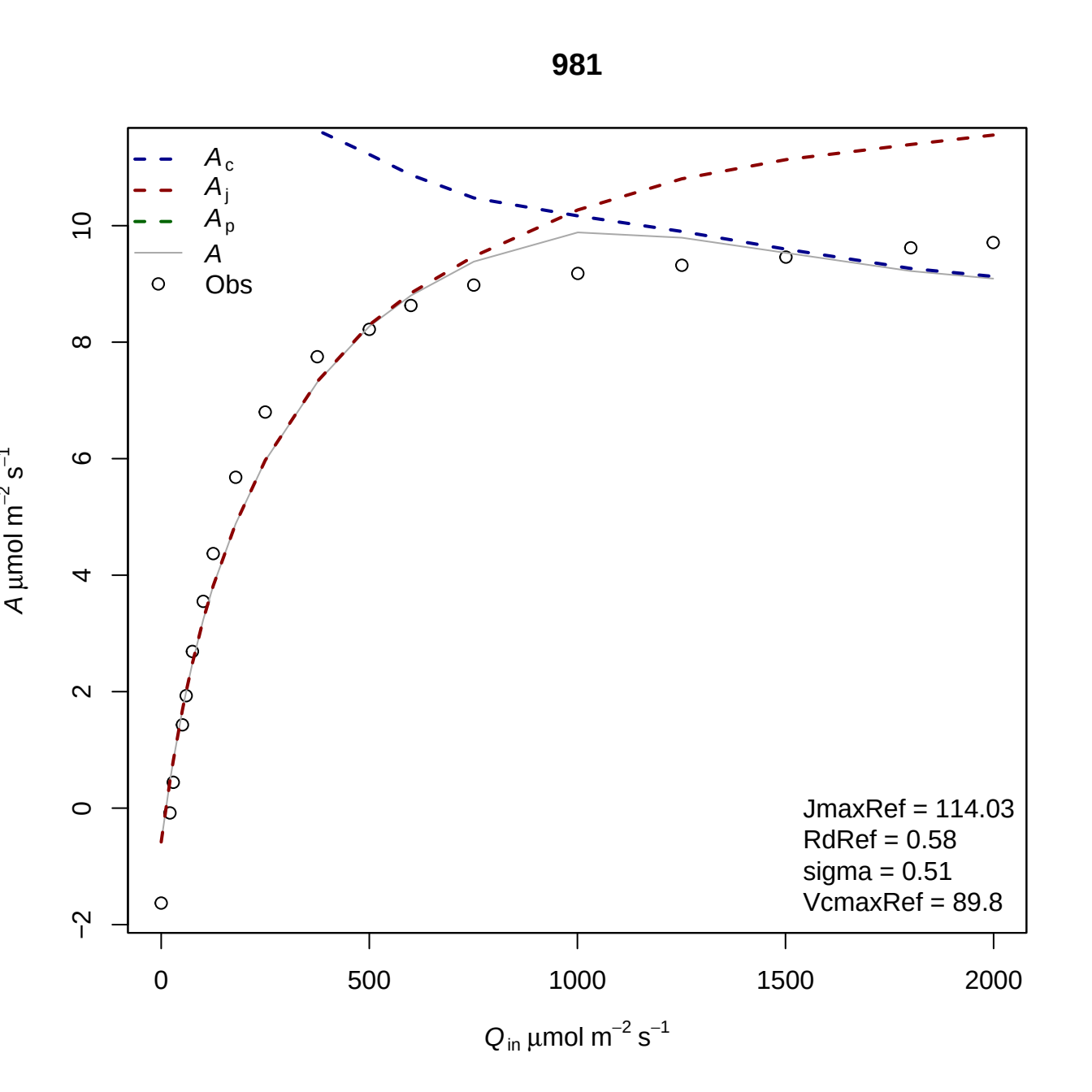

976
A_c
A_j
A_p
A
Obs
JmaxRef = 81.39
RdRef = 1.58
sigma = 0.16
VcmaxRef = 88.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

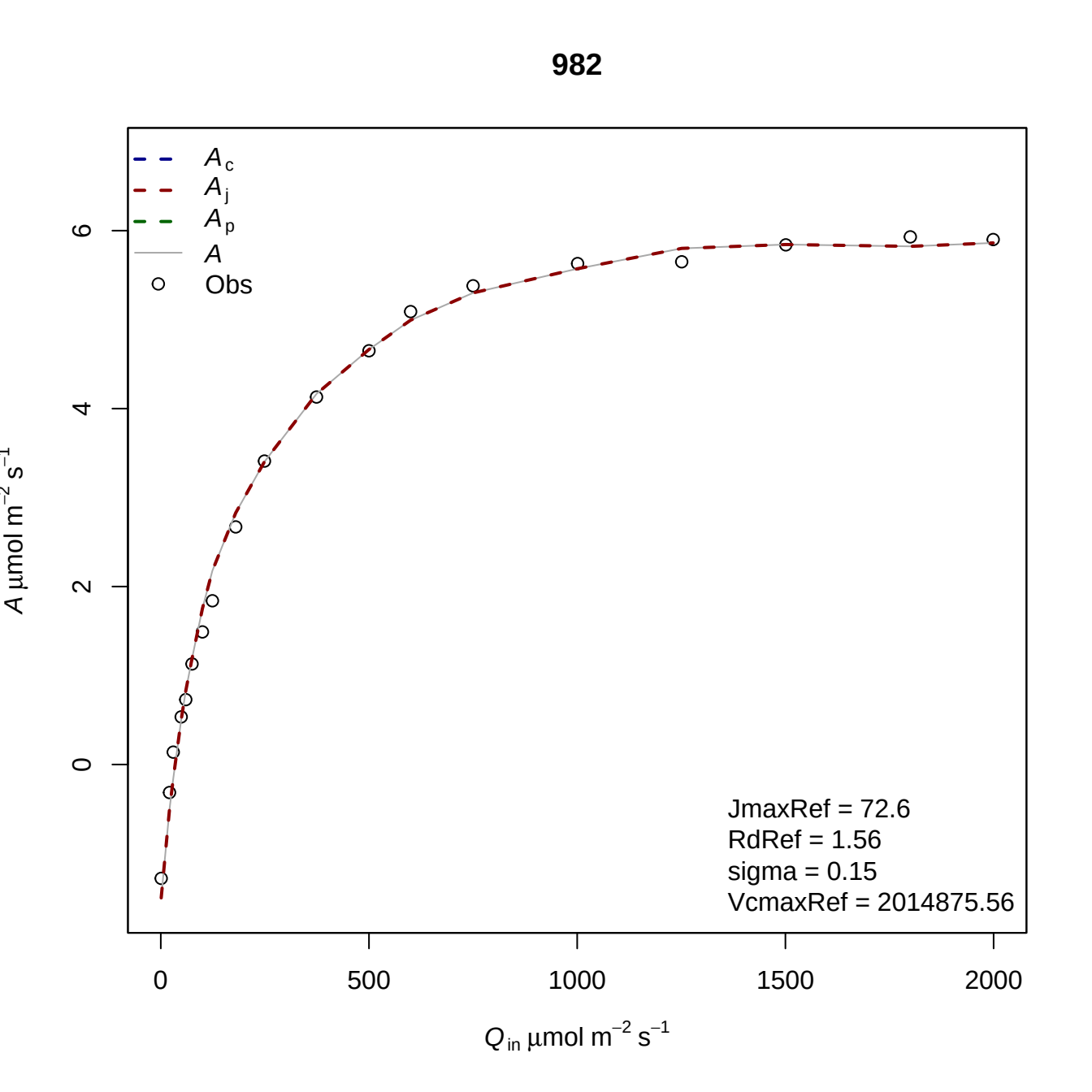

977
A_c
A_j
A_p
A
Obs
JmaxRef = 90.5
RdRef = 1.82
sigma = 0.32
VcmaxRef = 1854506.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

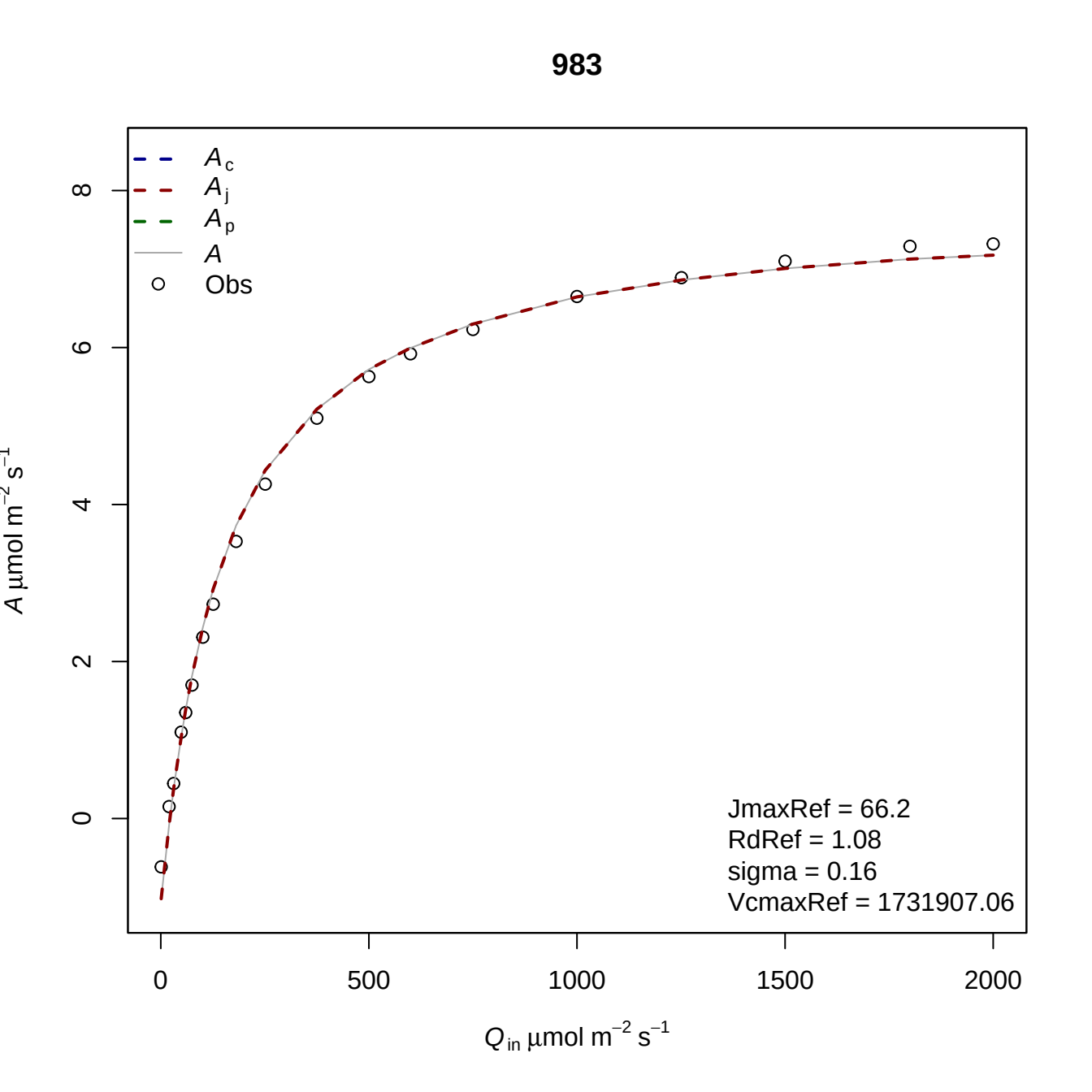

978
A_c
A_j
A_p
A
Obs
JmaxRef = 89.69
RdRef = 0.84
sigma = 0.22
VcmaxRef = 72.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

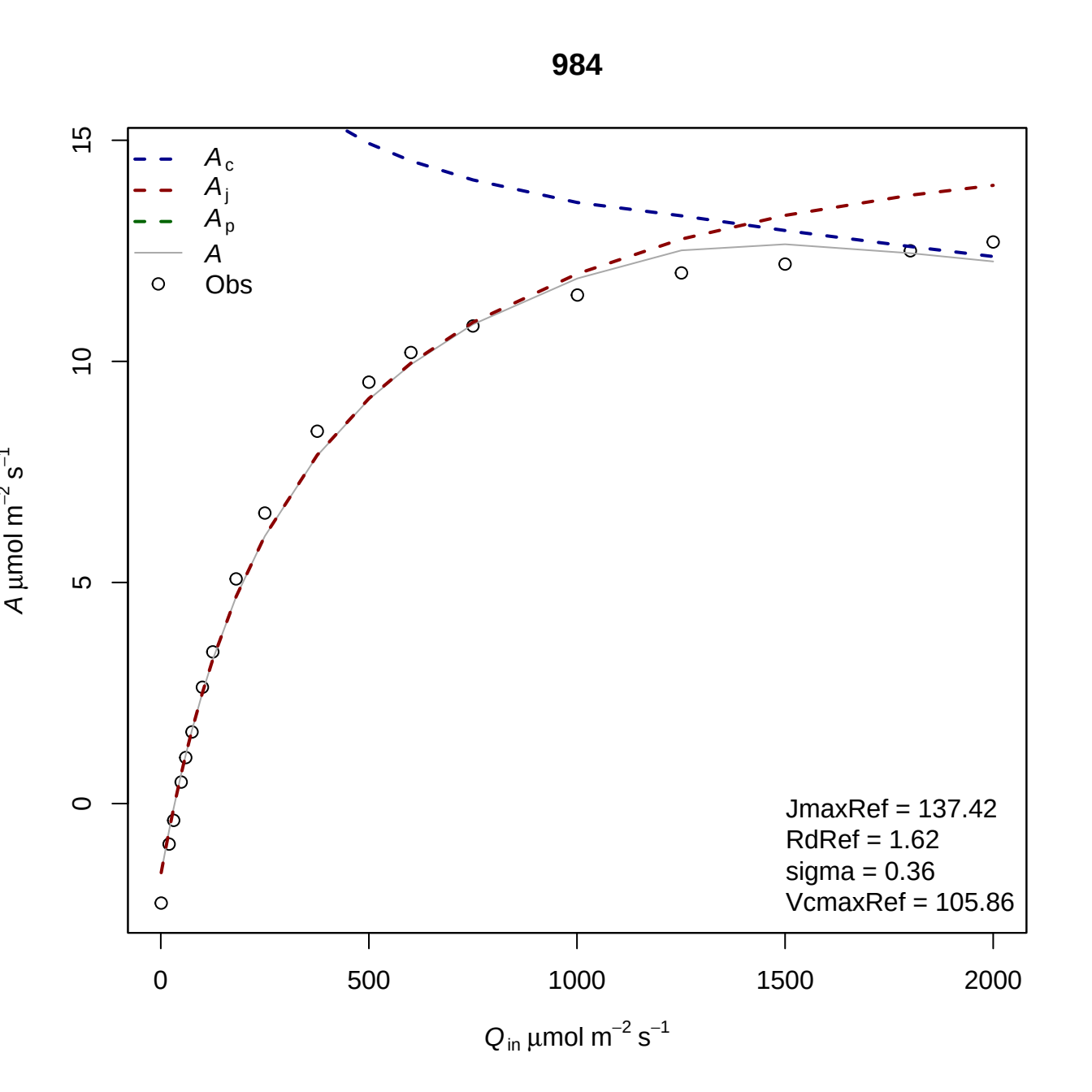

979
A_c
A_j
A_p
A
Obs
JmaxRef = 125.02
RdRef = 1.34
sigma = 0.22
VcmaxRef = 88.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

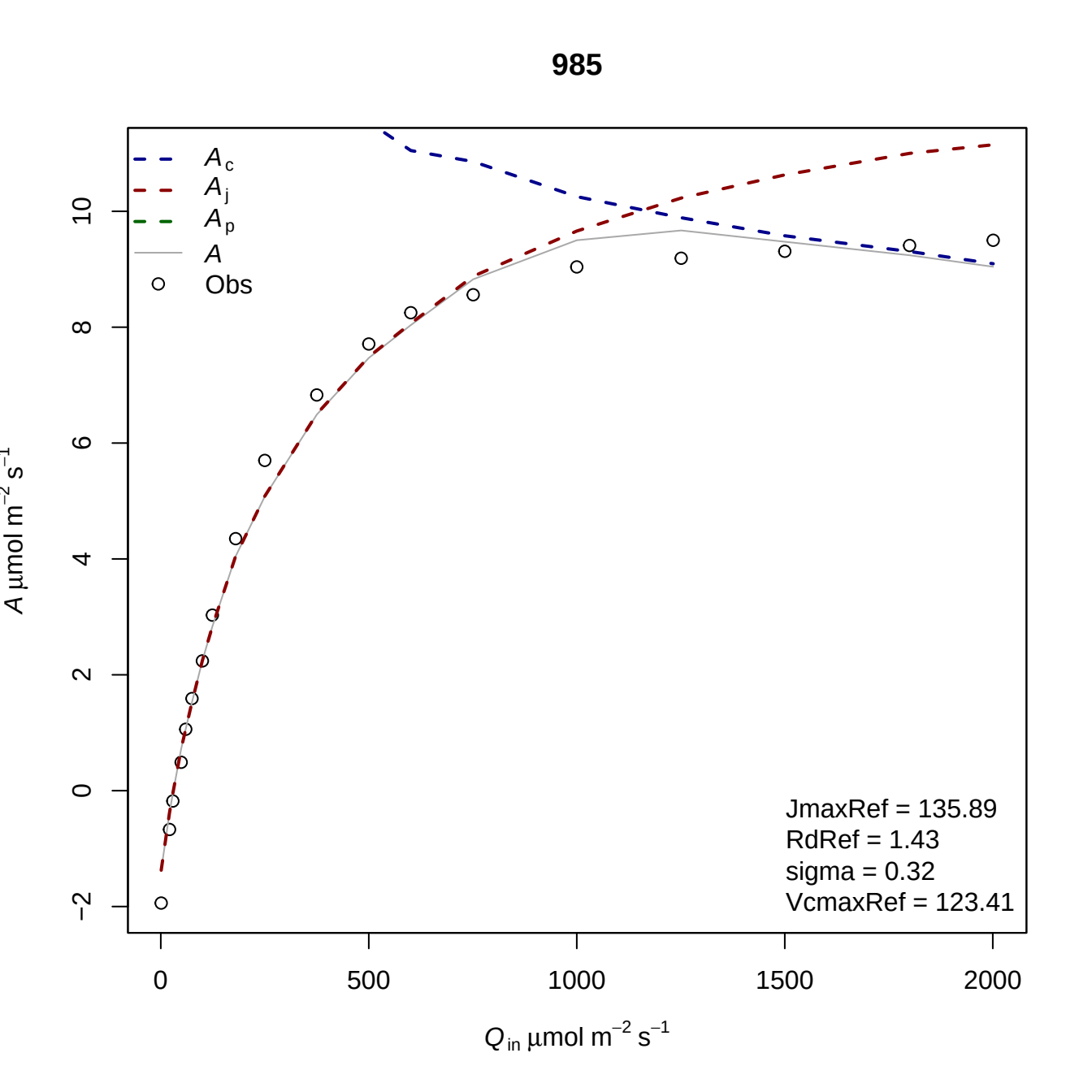

980
A_c
A_j
A_p
A
Obs
JmaxRef = 131.49
RdRef = 2.9
sigma = 0.18
VcmaxRef = 113.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

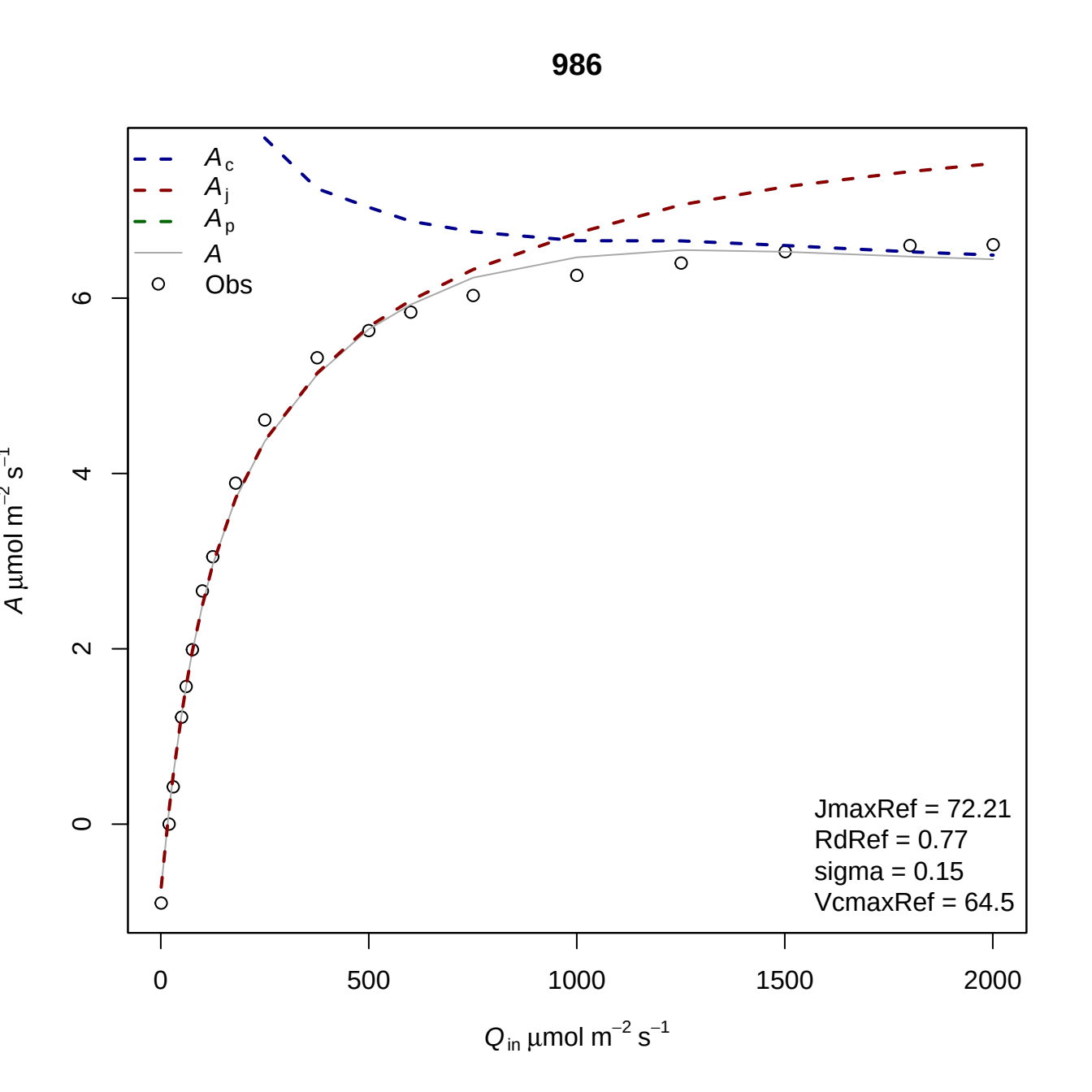

981
A_c
A_j
A_p
A
Obs
JmaxRef = 114.03
RdRef = 0.58
sigma = 0.51
VcmaxRef = 89.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

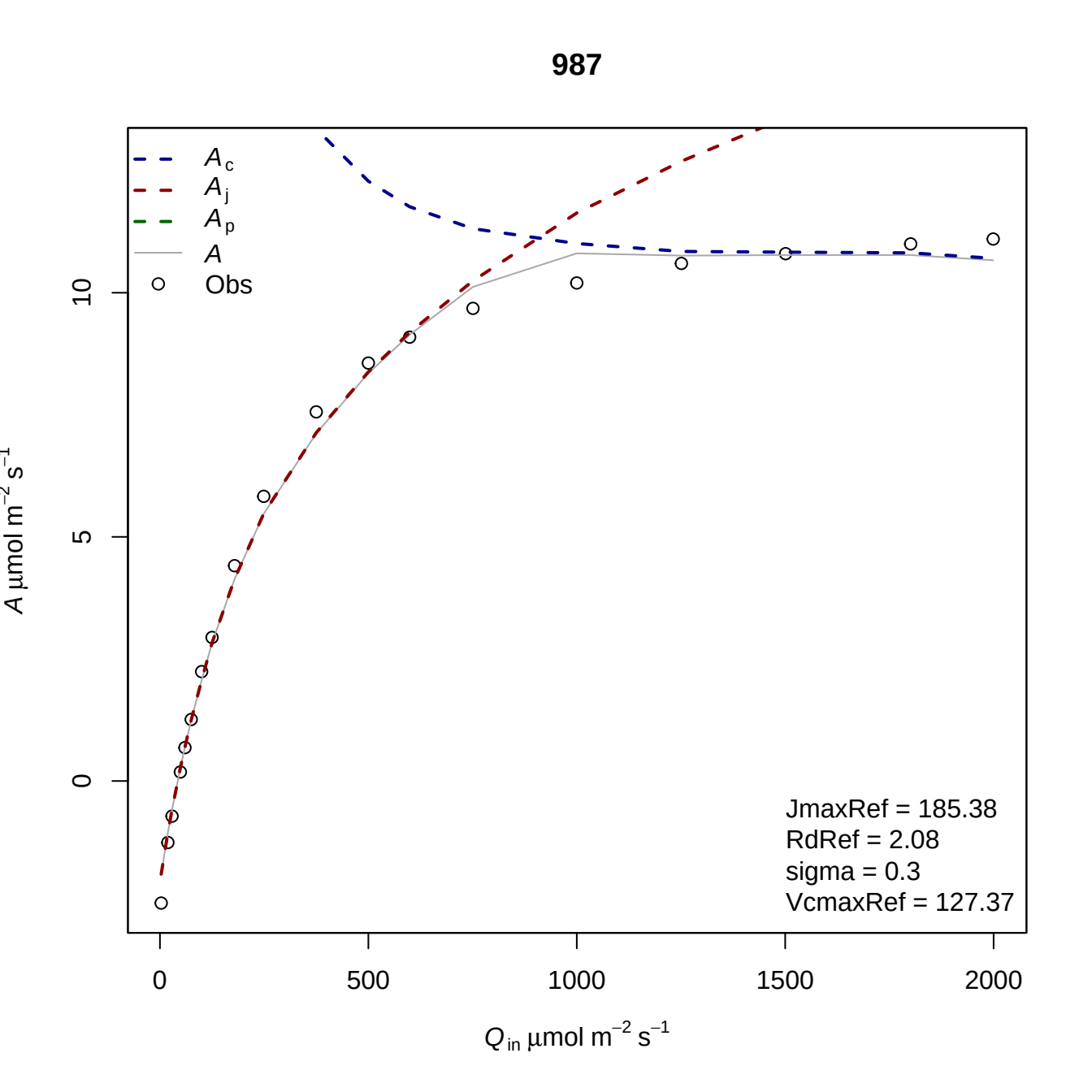

982
A_c
A_j
A_p
A
Obs
JmaxRef = 72.6
RdRef = 1.56
sigma = 0.15
VcmaxRef = 2014875.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

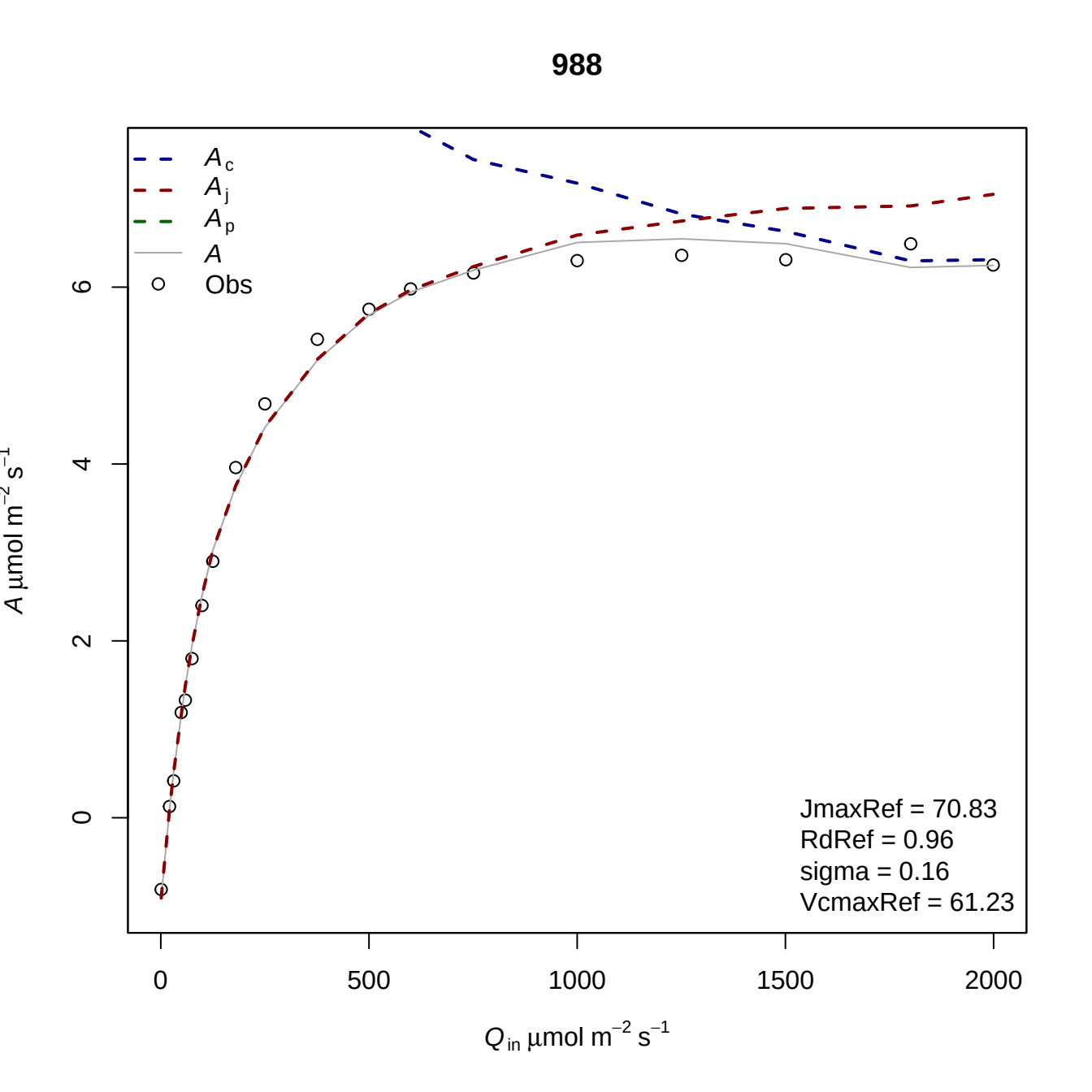

983
A_c
A_j
A_p
A
Obs
JmaxRef = 66.2
RdRef = 1.08
sigma = 0.16
VcmaxRef = 1731907.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

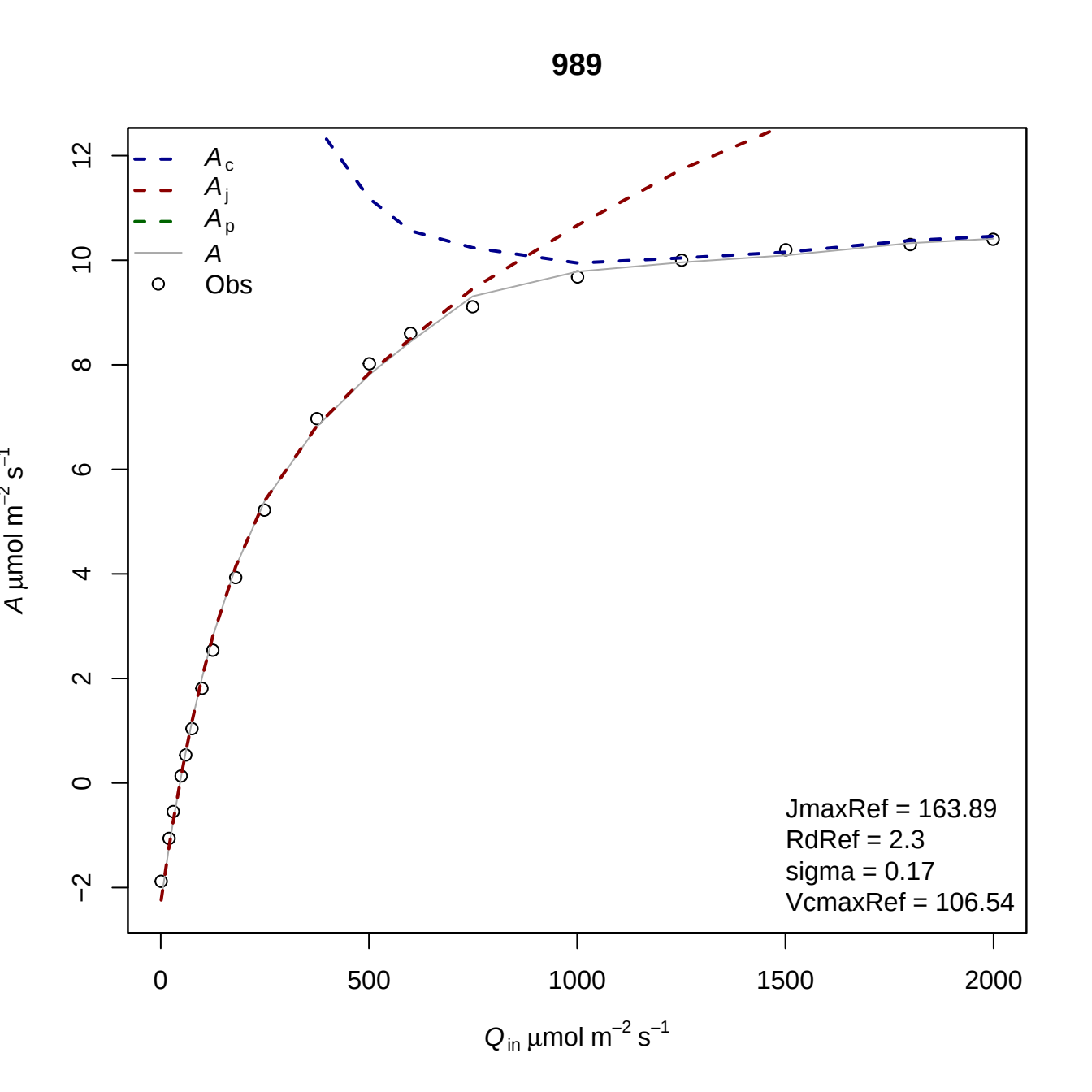

984
A_c
A_j
A_p
A
Obs
JmaxRef = 137.42
RdRef = 1.62
sigma = 0.36
VcmaxRef = 105.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

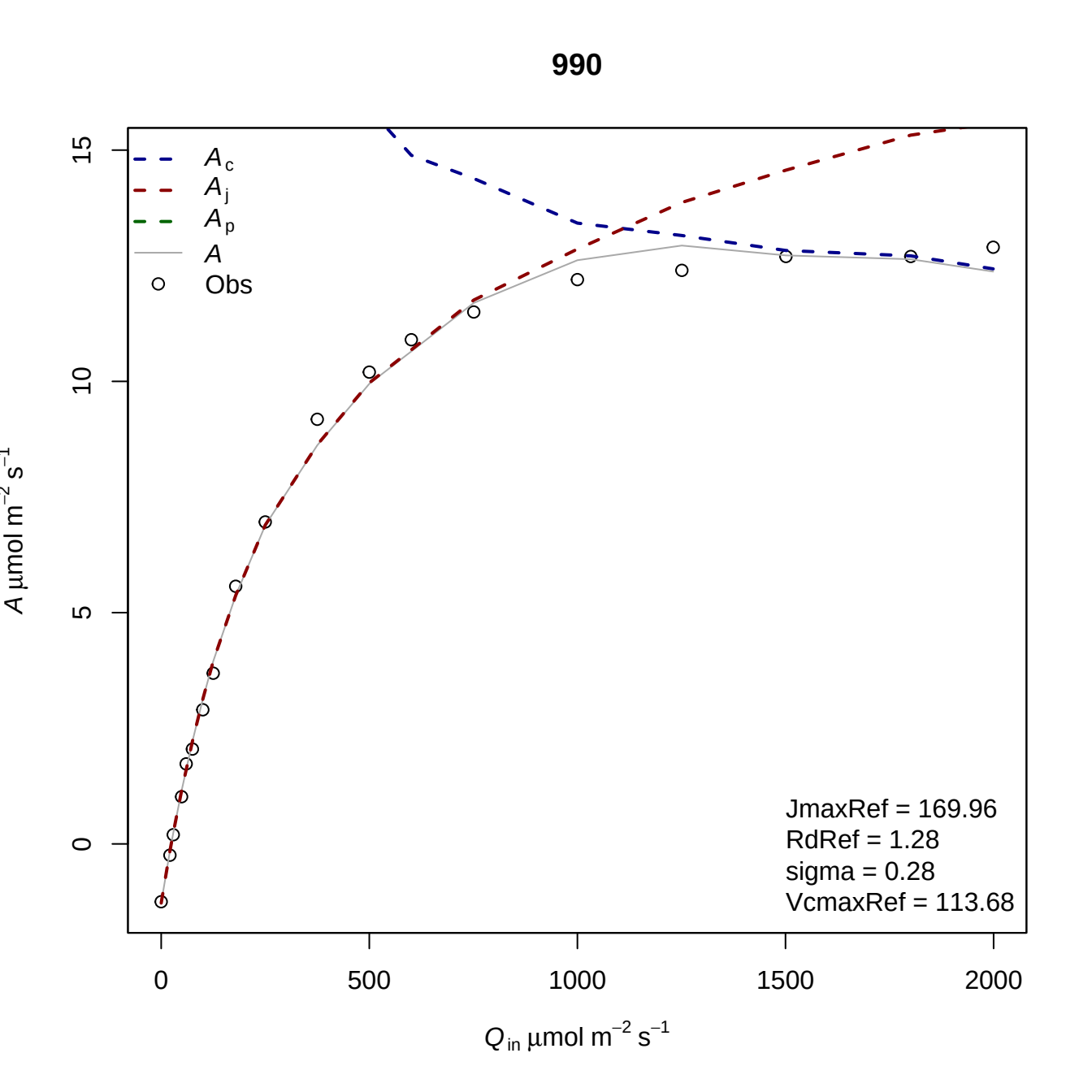

985
A_c
A_j
A_p
A
Obs
JmaxRef = 135.89
RdRef = 1.43
sigma = 0.32
VcmaxRef = 123.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

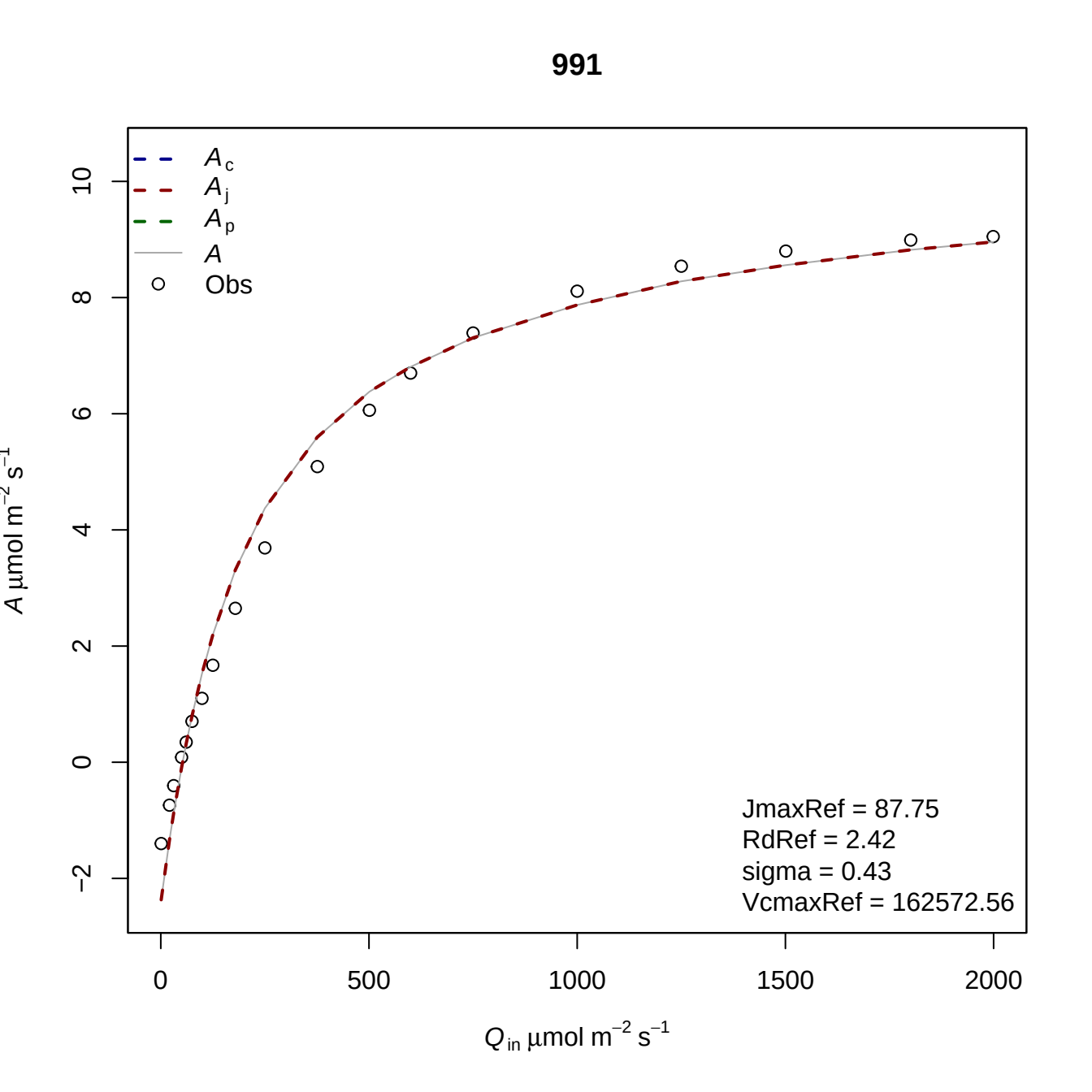

986
A_c
A_j
A_p
A
Obs
JmaxRef = 72.21
RdRef = 0.77
sigma = 0.15
VcmaxRef = 64.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

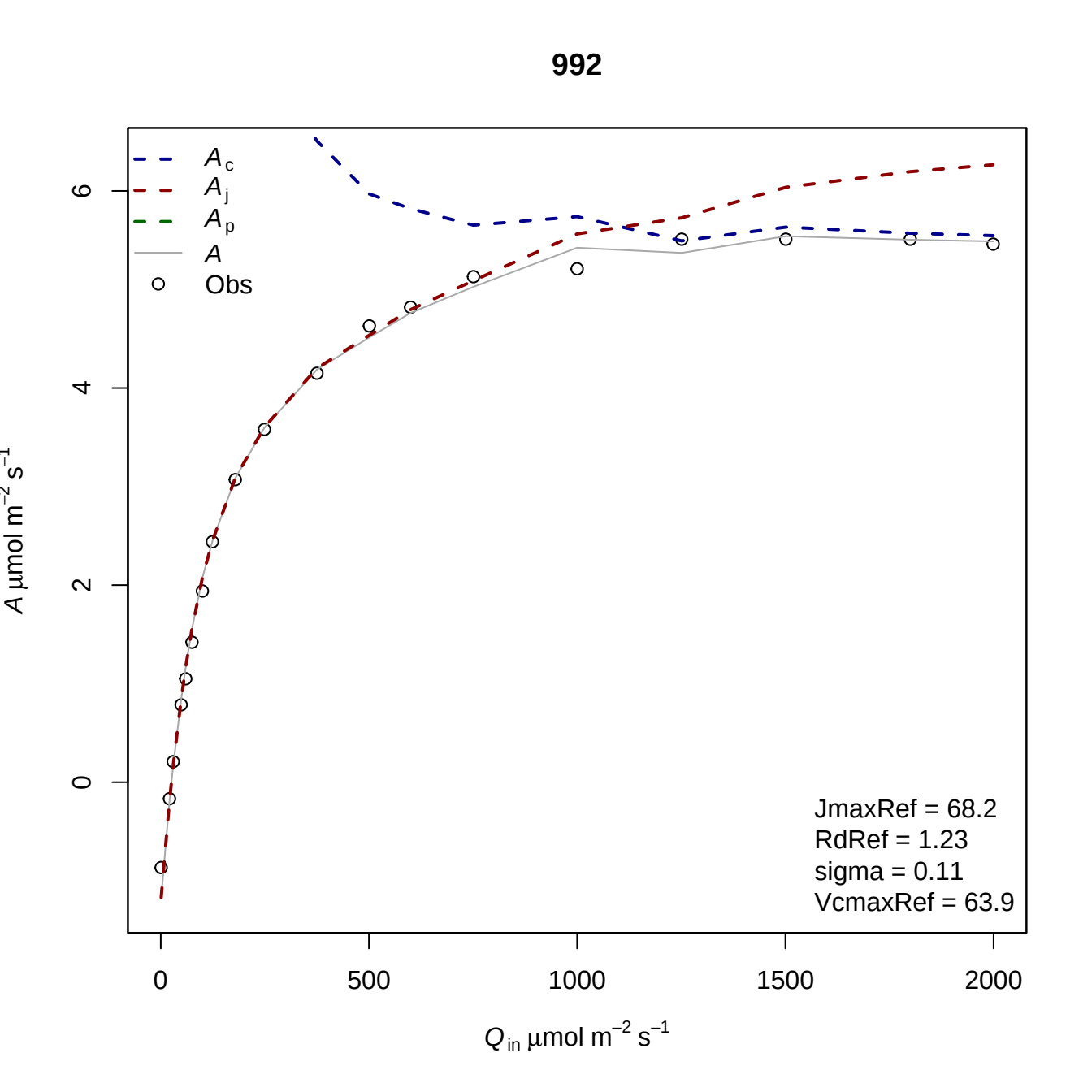

987
A_c
A_j
A_p
A
Obs
JmaxRef = 185.38
RdRef = 2.08
sigma = 0.3
VcmaxRef = 127.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

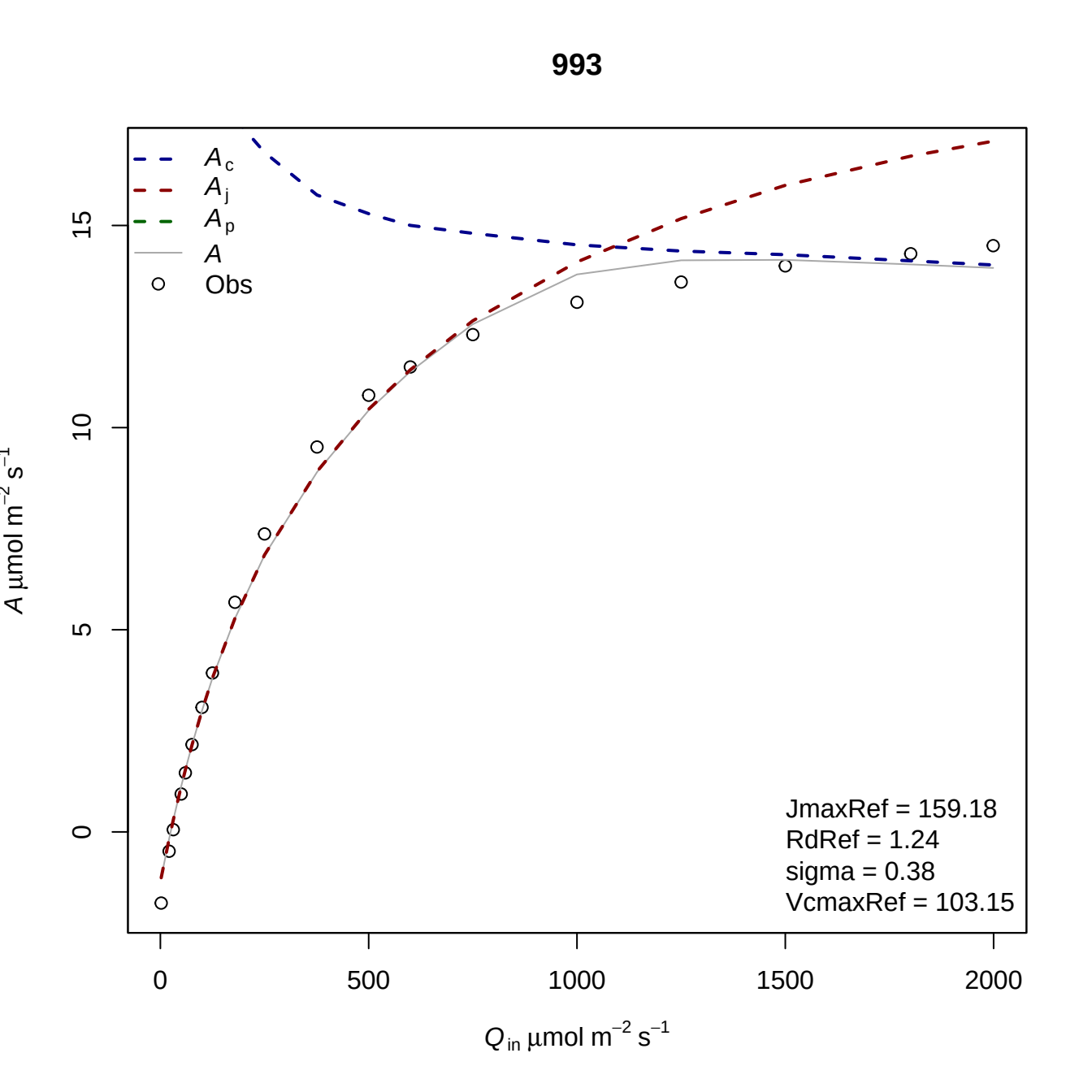

988
A_c
A_j
A_p
A
Obs
JmaxRef = 70.83
RdRef = 0.96
sigma = 0.16
VcmaxRef = 61.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

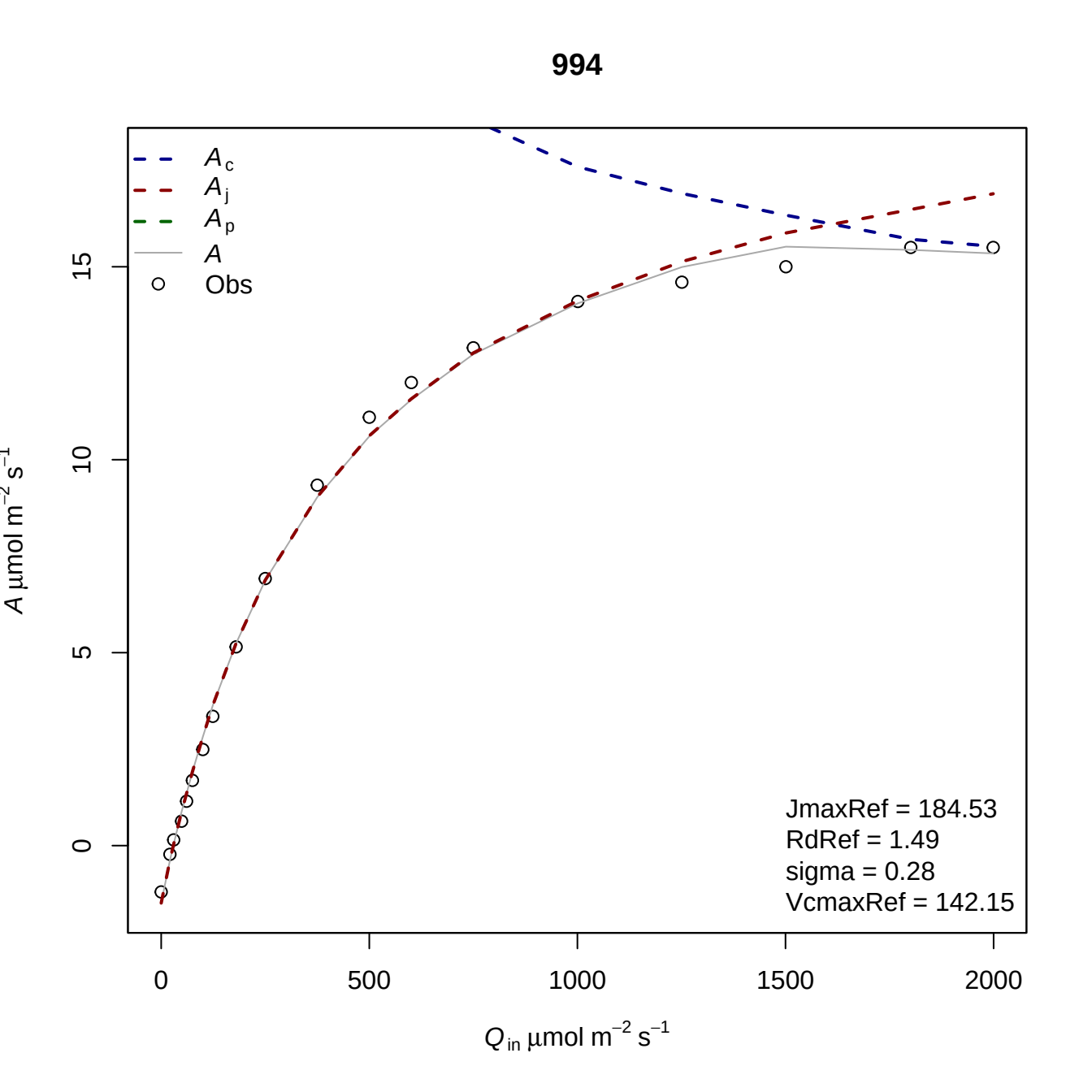

989
A_c
A_j
A_p
A
Obs
JmaxRef = 163.89
RdRef = 2.3
sigma = 0.17
VcmaxRef = 106.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

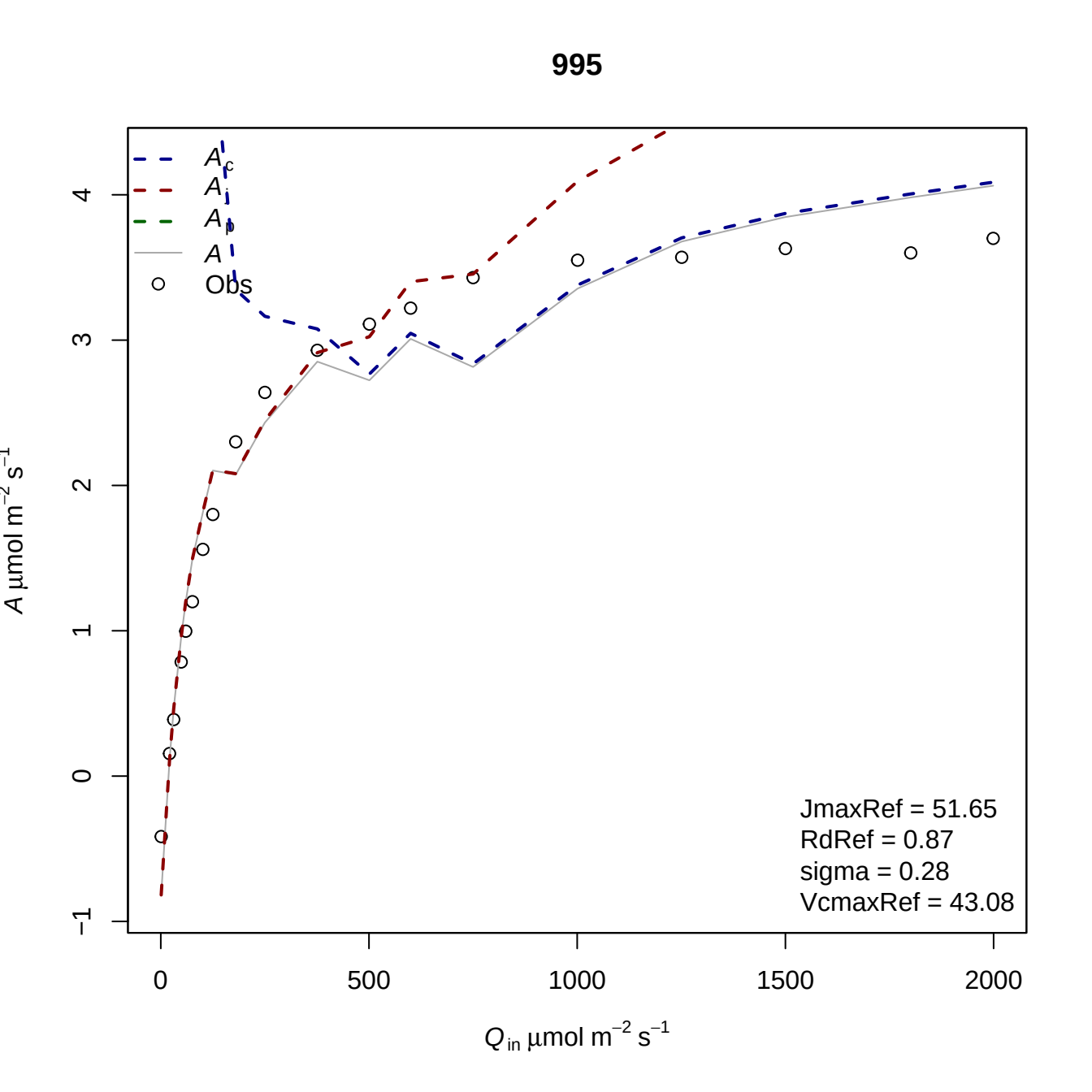

990
A_c
A_j
A_p
A
Obs
JmaxRef = 169.96
RdRef = 1.28
sigma = 0.28
VcmaxRef = 113.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

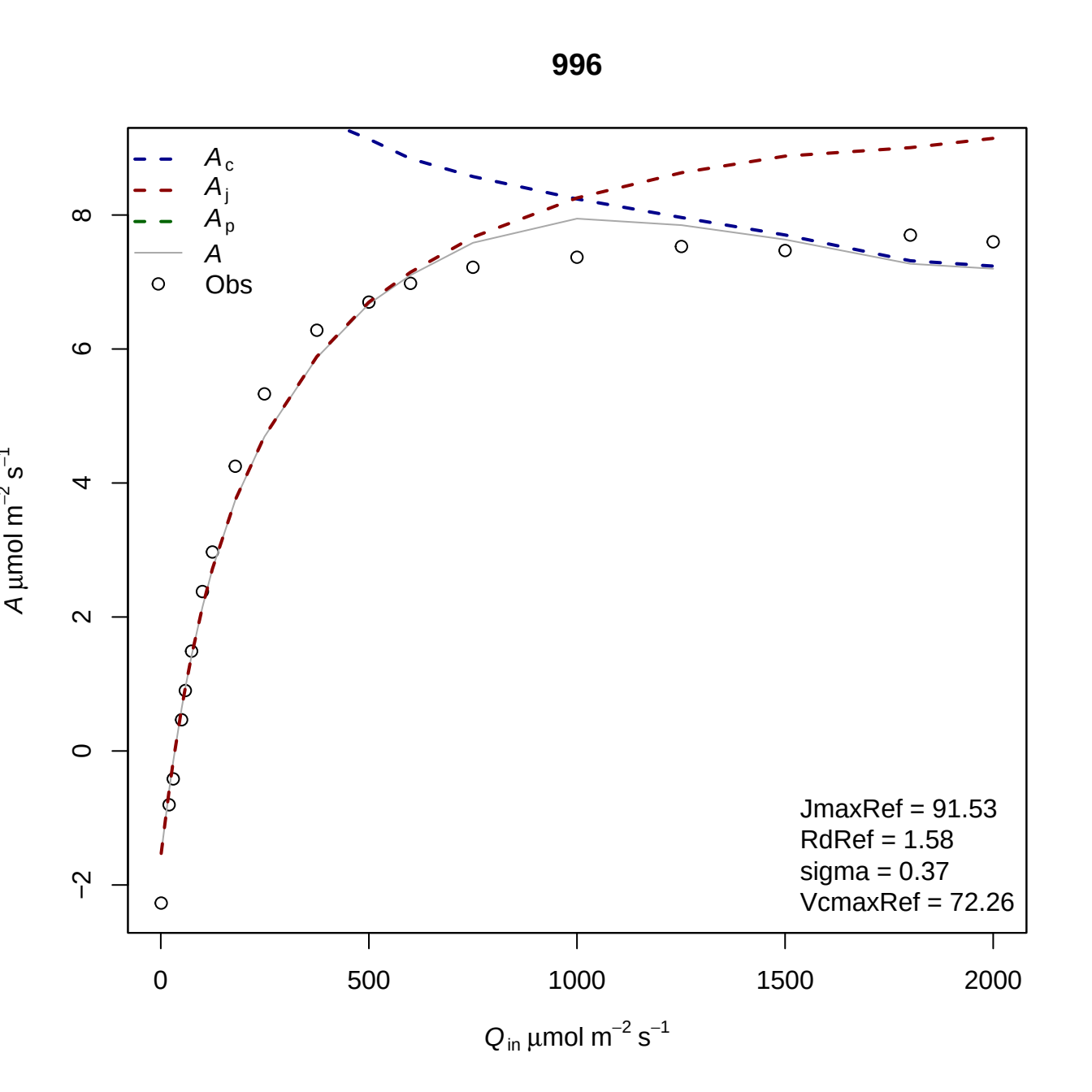

991
A_c
A_j
A_p
A
Obs
JmaxRef = 87.75
RdRef = 2.42
sigma = 0.43
VcmaxRef = 162572.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

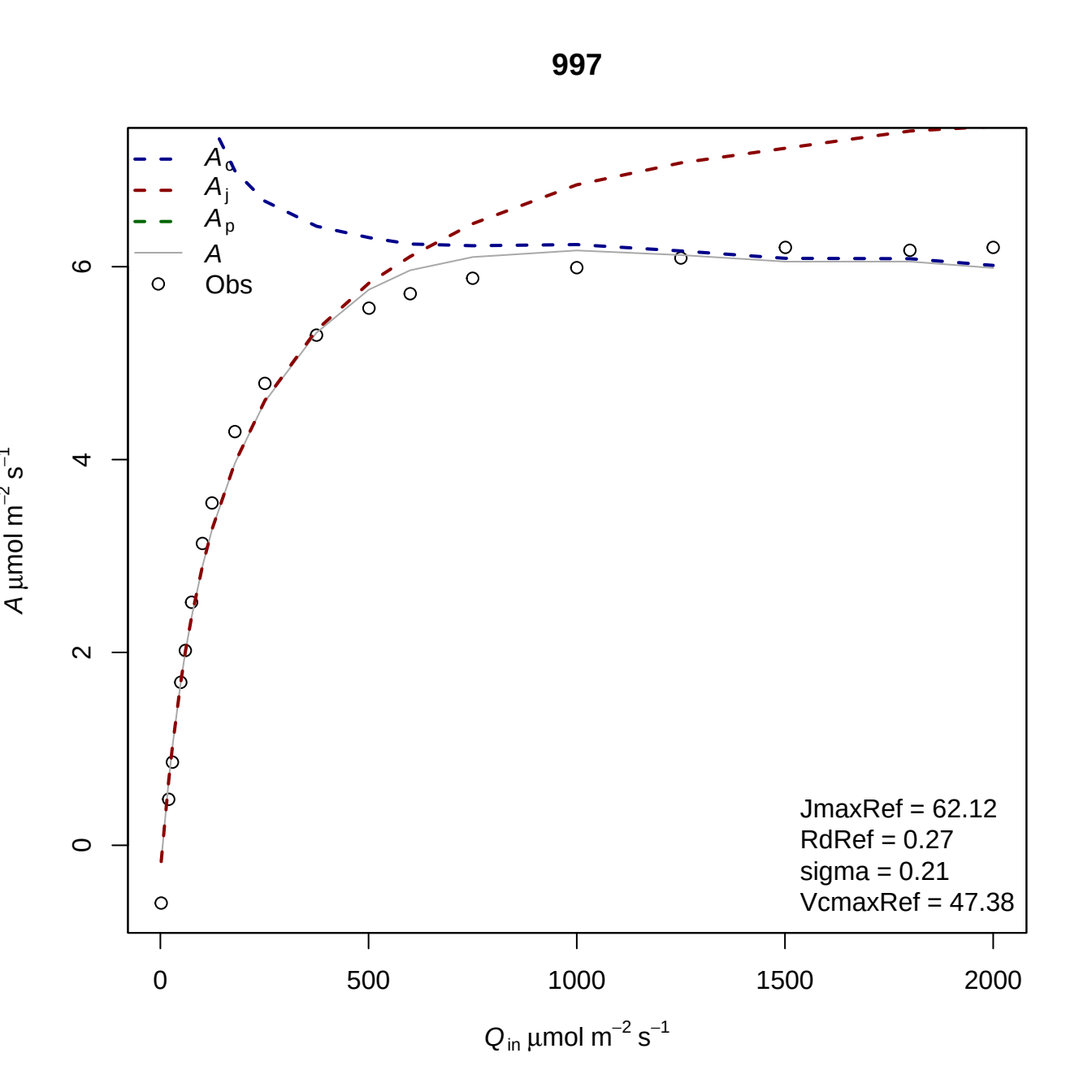

992
A_c
A_j
A_p
A
Obs
JmaxRef = 68.2
RdRef = 1.23
sigma = 0.11
VcmaxRef = 63.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

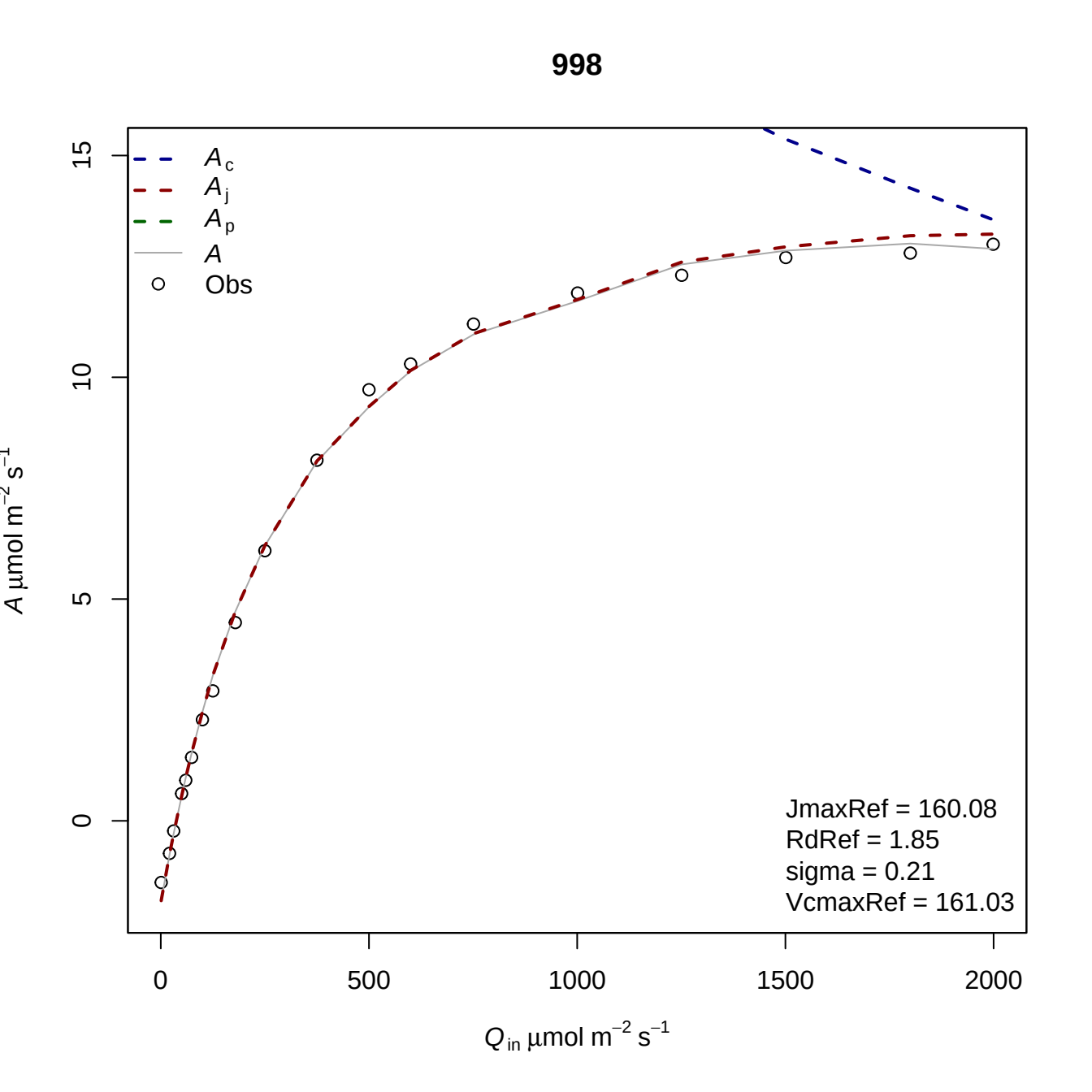

993
A_c
A_j
A_p
A
Obs
JmaxRef = 159.18
RdRef = 1.24
sigma = 0.38
VcmaxRef = 103.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

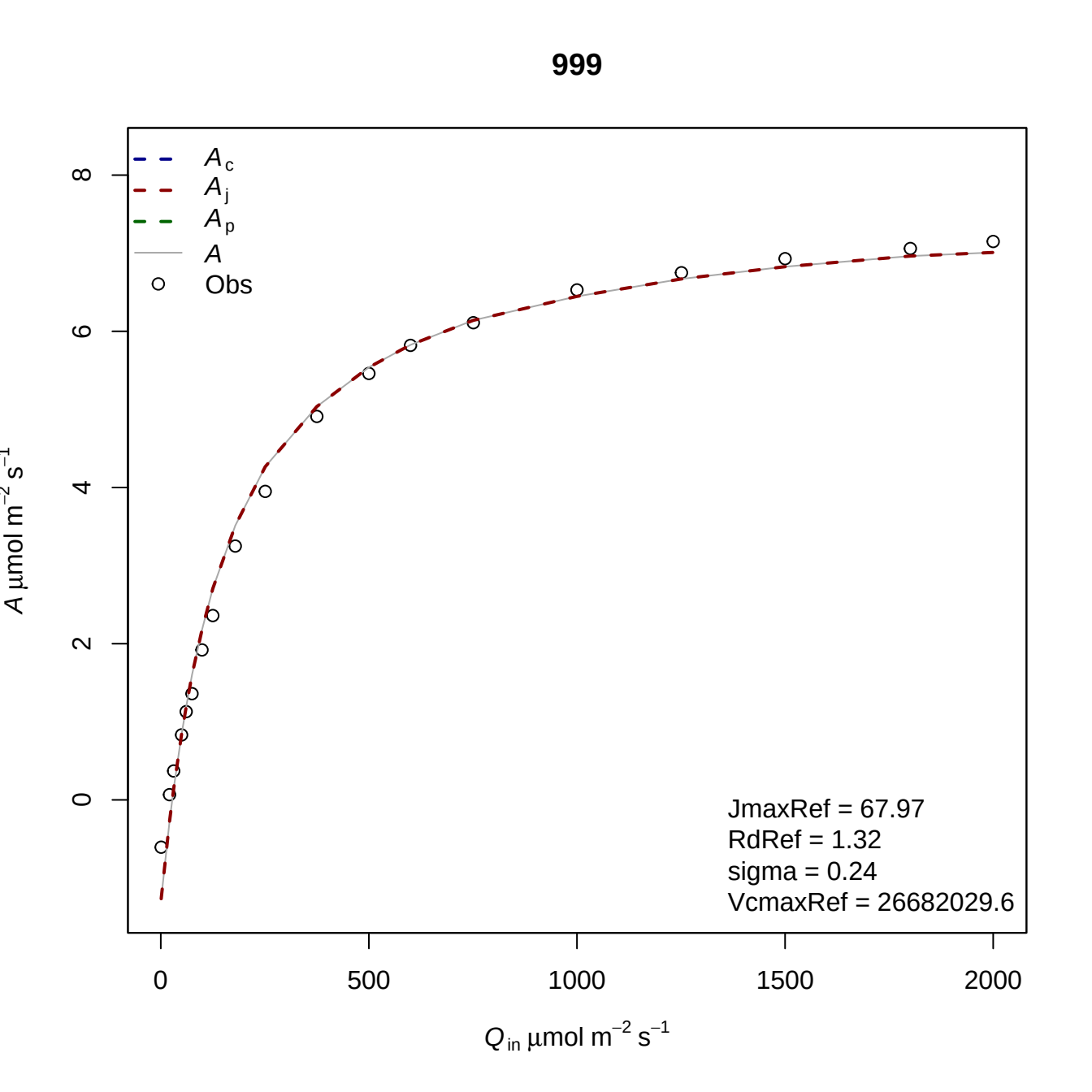

994
A_c
A_j
A_p
A
Obs
JmaxRef = 184.53
RdRef = 1.49
sigma = 0.28
VcmaxRef = 142.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

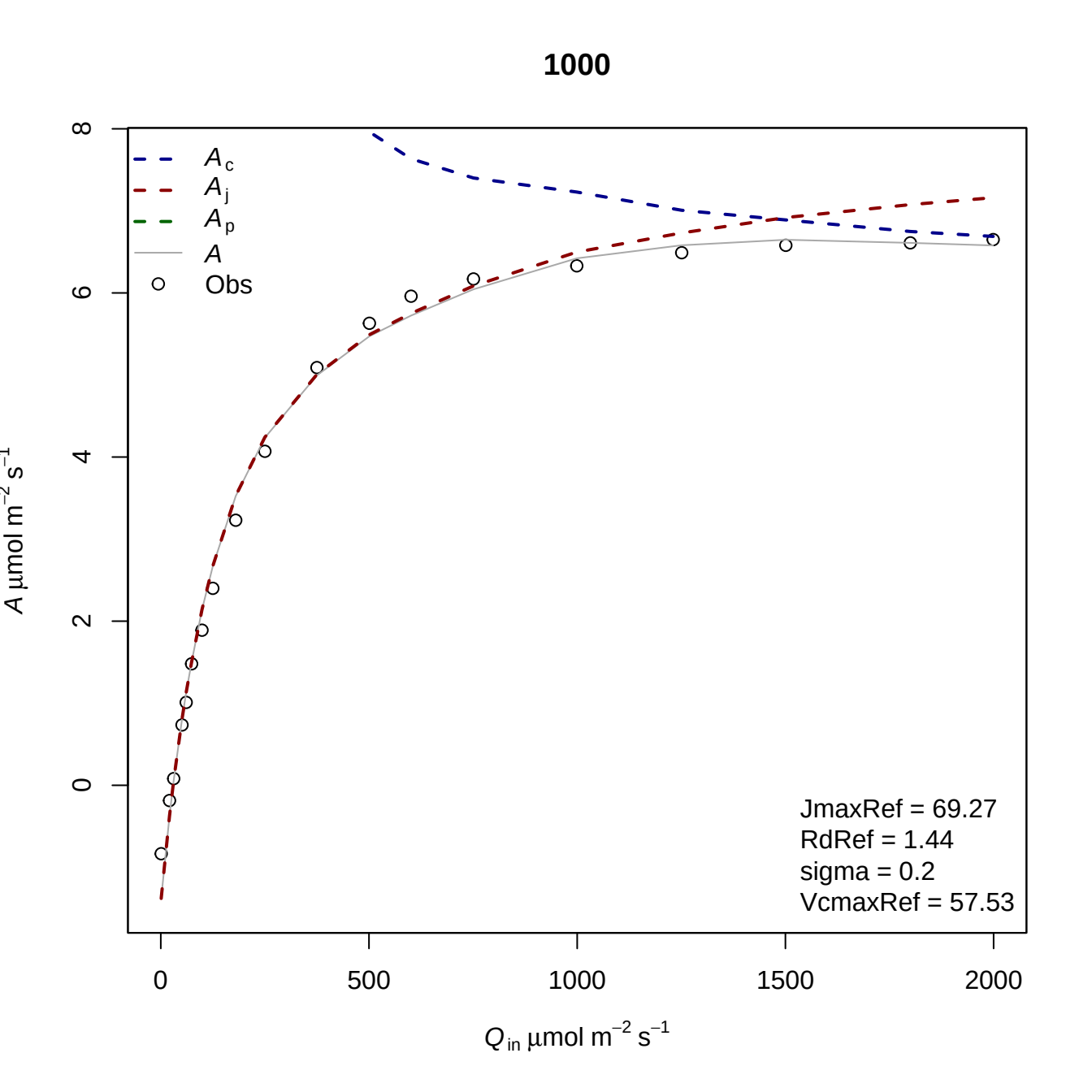

995
A_c
A_j
A_p
A
Obs
JmaxRef = 51.65
RdRef = 0.87
sigma = 0.28
VcmaxRef = 43.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

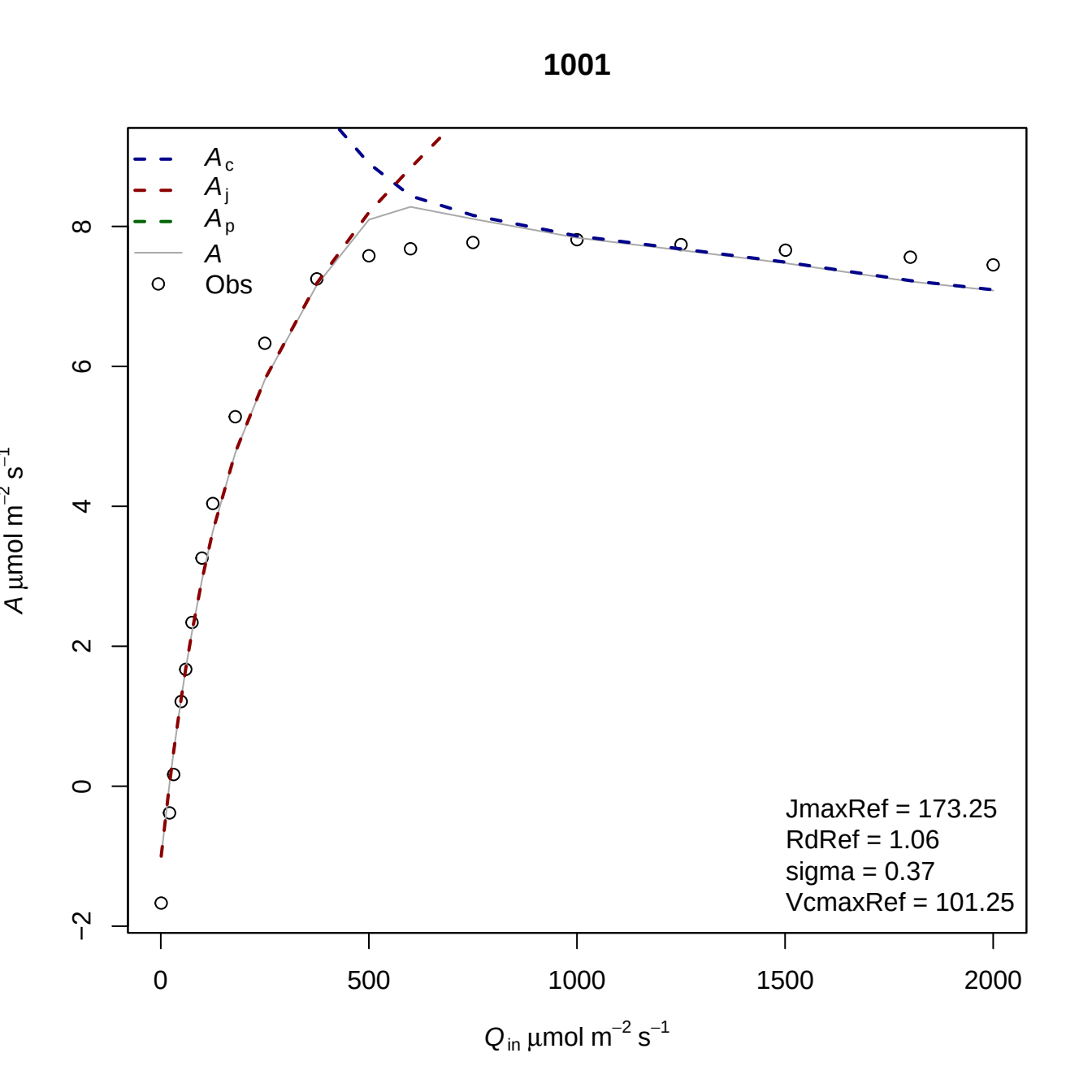

996
A_c
A_j
A_p
A
Obs
JmaxRef = 91.53
RdRef = 1.58
sigma = 0.37
VcmaxRef = 72.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

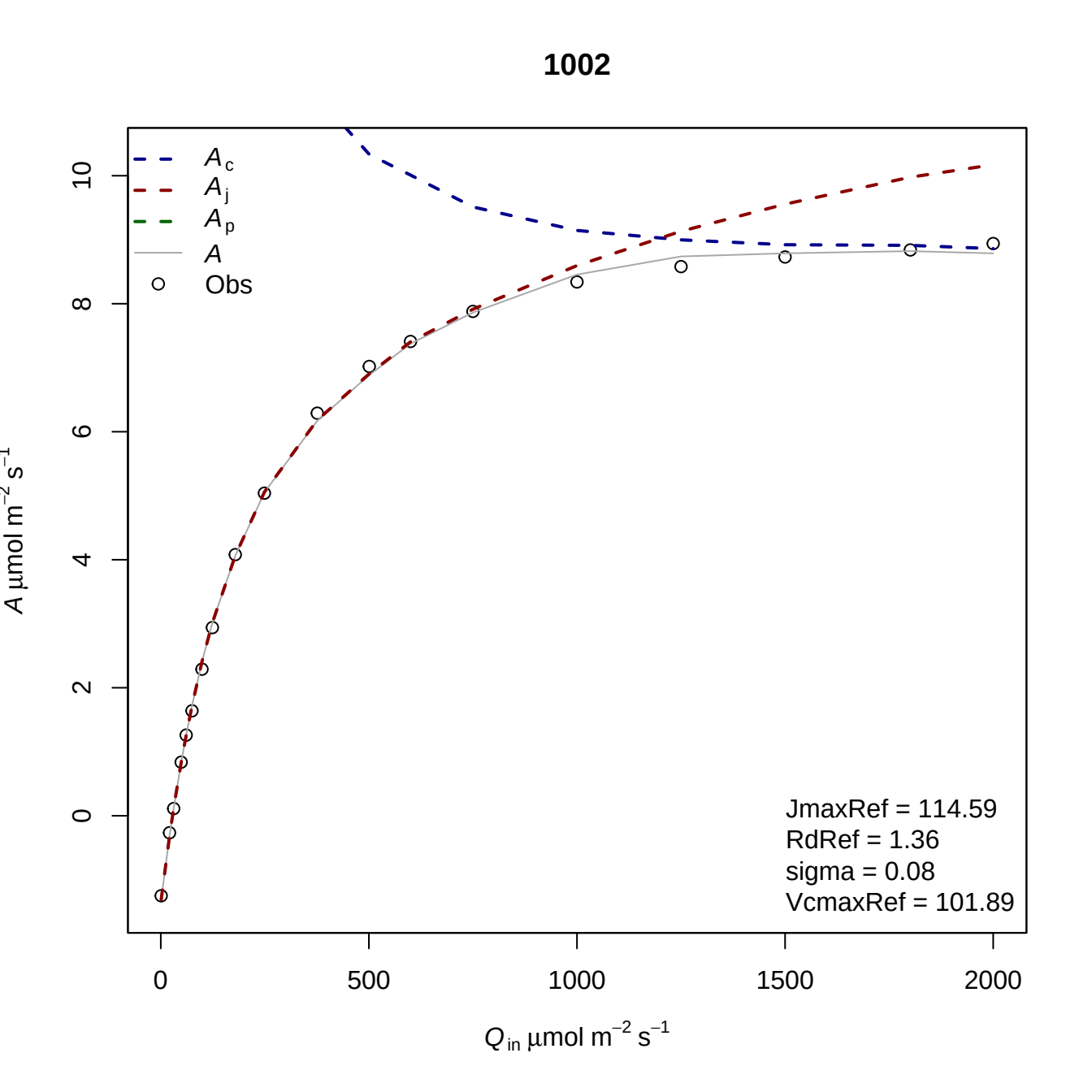

997
A_c
A_j
A_p
A
Obs
JmaxRef = 62.12
RdRef = 0.27
sigma = 0.21
VcmaxRef = 47.38
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

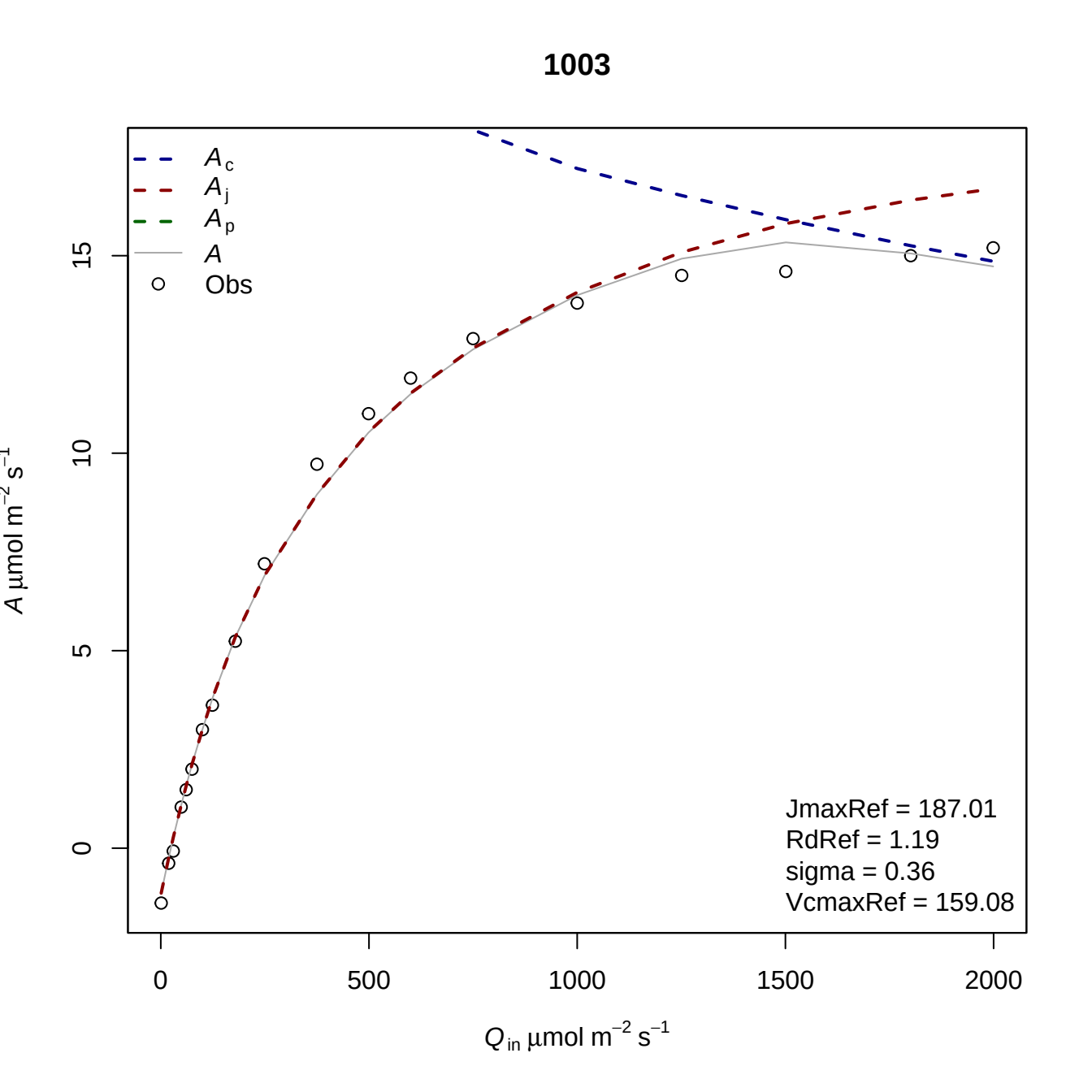

998
A_c
A_j
A_p
A
Obs
JmaxRef = 160.08
RdRef = 1.85
sigma = 0.21
VcmaxRef = 161.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

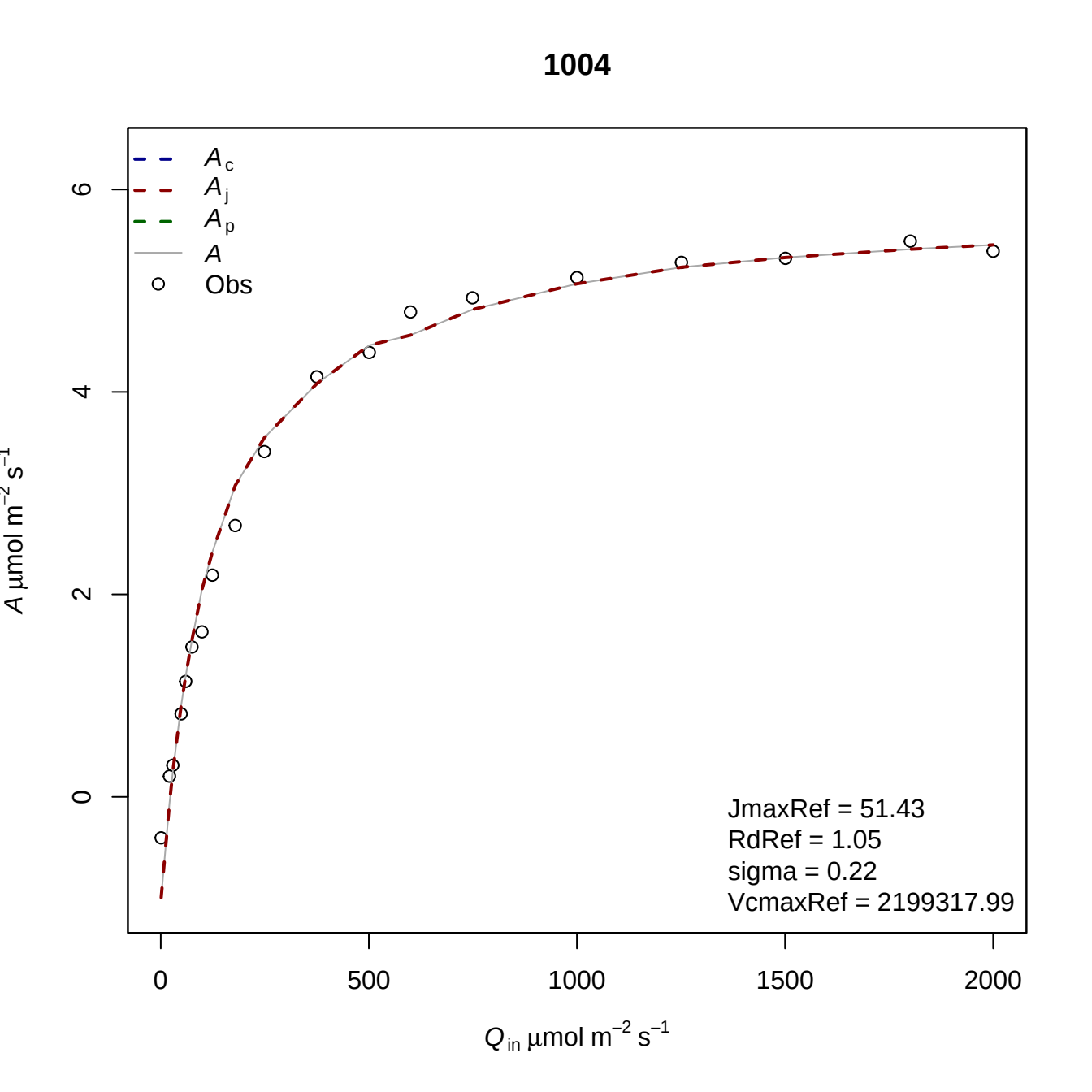

999
A_c
A_j
A_p
A
Obs
JmaxRef = 67.97
RdRef = 1.32
sigma = 0.24
VcmaxRef = 26682029.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

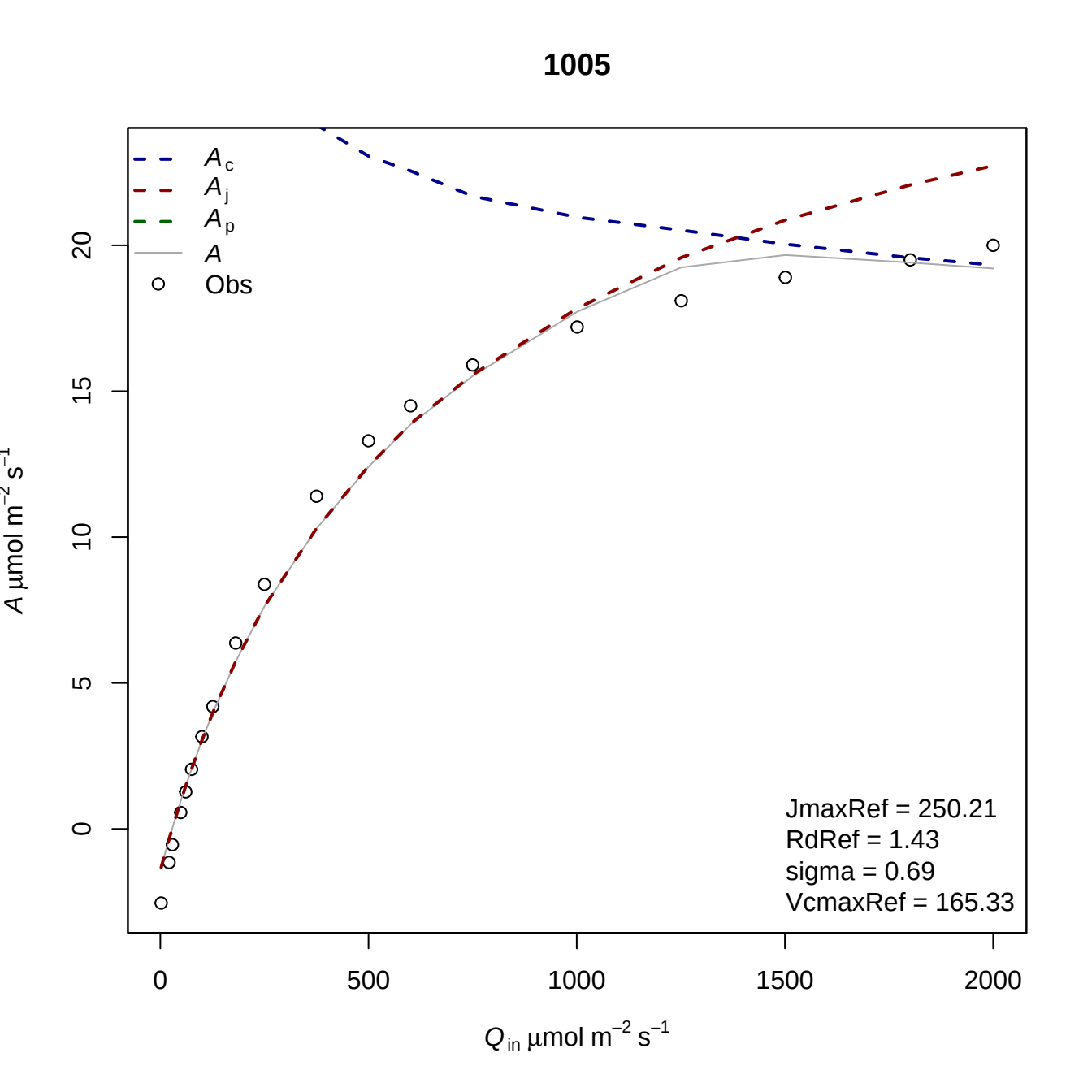

1000
A_c
A_j
A_p
A
Obs
JmaxRef = 69.27
RdRef = 1.44
sigma = 0.2
VcmaxRef = 57.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

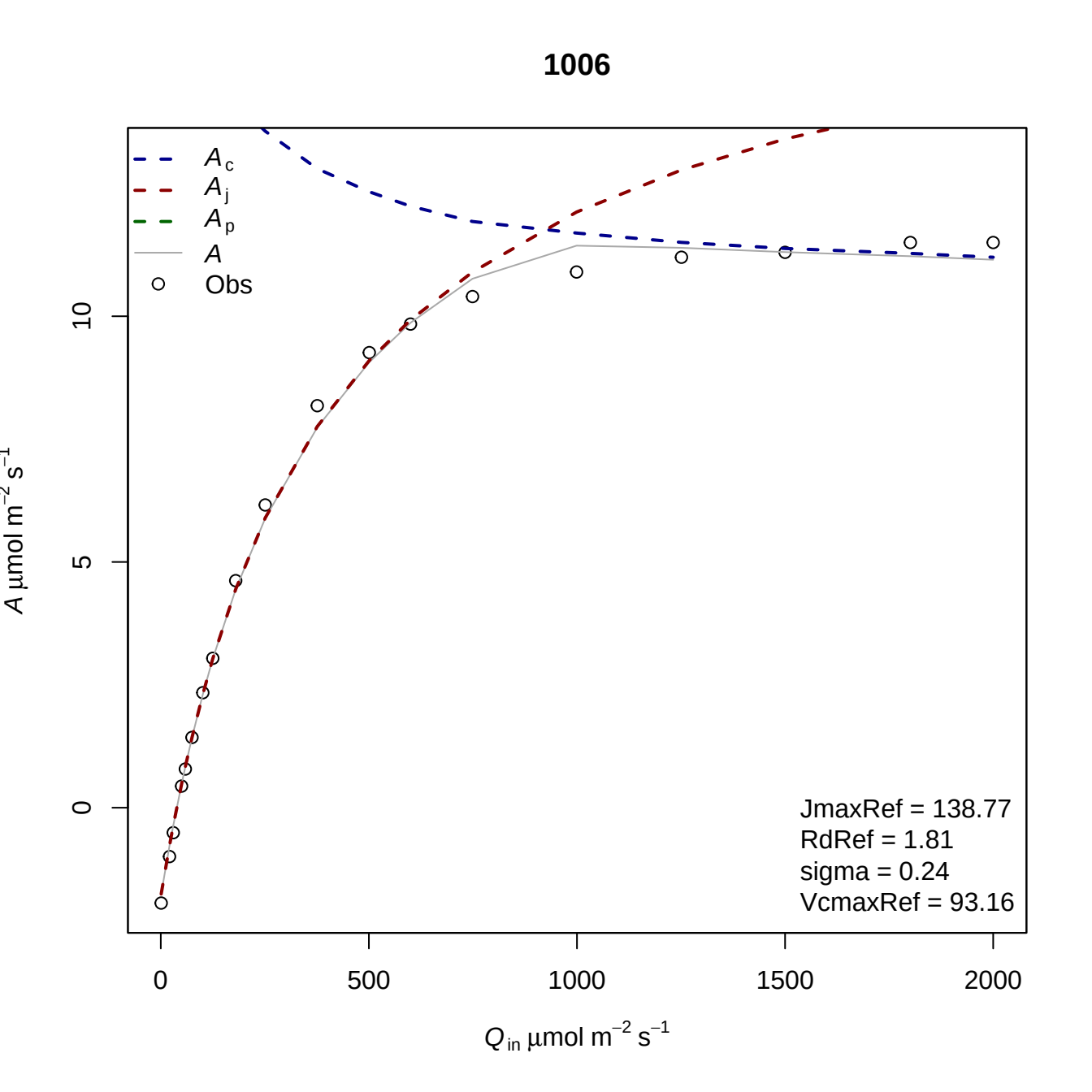

1001
A_c
A_j
A_p
A
Obs
JmaxRef = 173.25
RdRef = 1.06
sigma = 0.37
VcmaxRef = 101.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

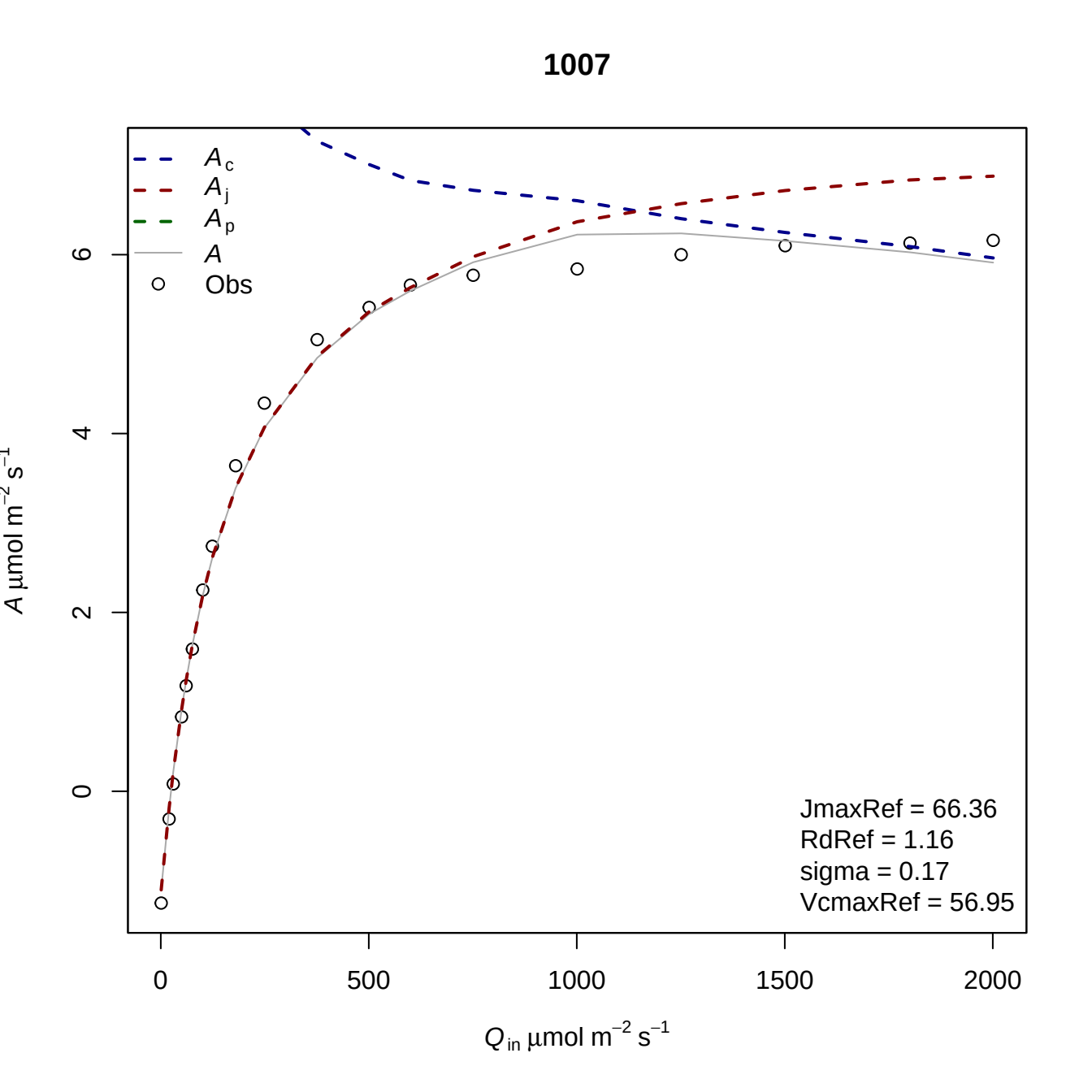

1002
A_c
A_j
A_p
A
Obs
JmaxRef = 114.59
RdRef = 1.36
sigma = 0.08
VcmaxRef = 101.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

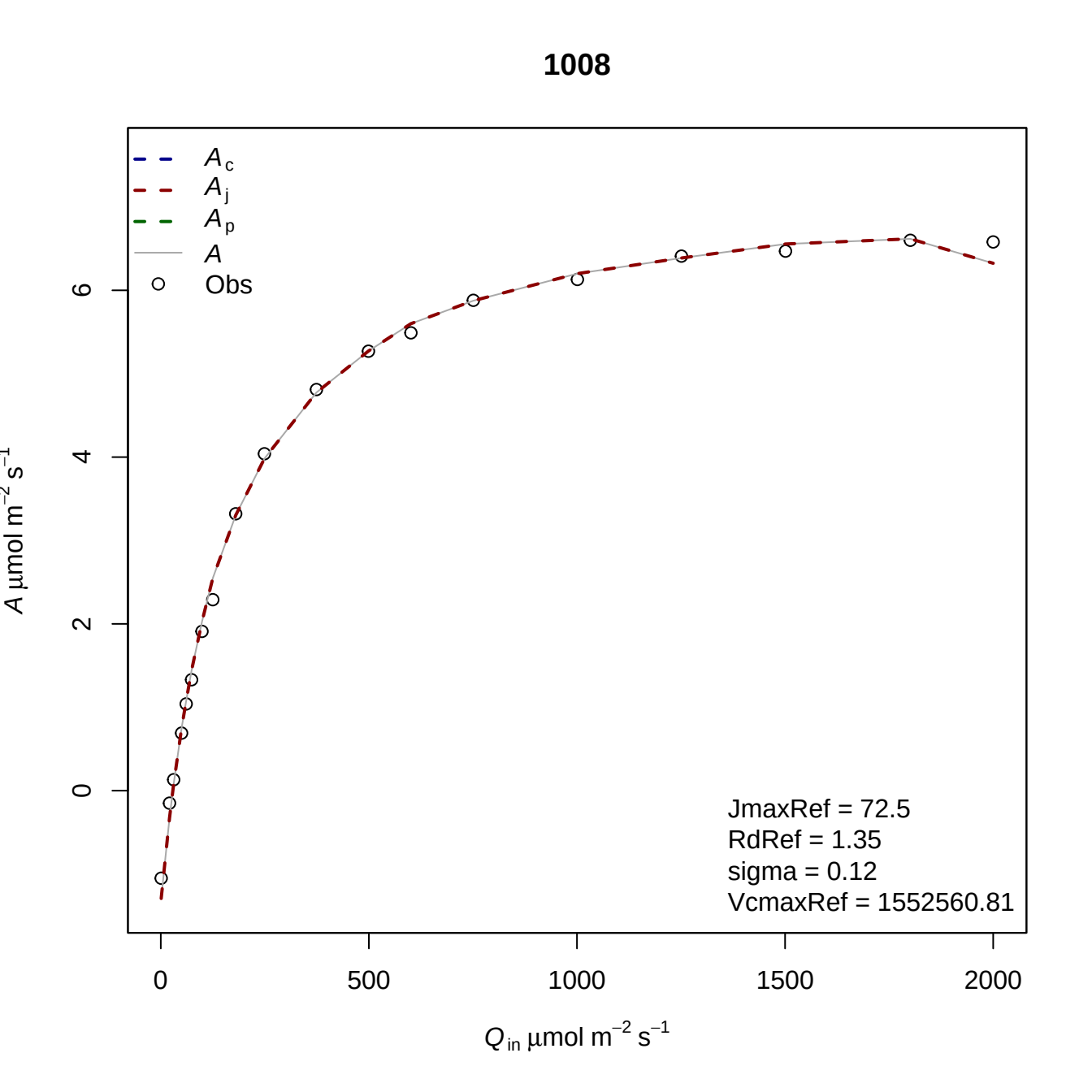

1003
A_c
A_j
A_p
A
Obs
JmaxRef = 187.01
RdRef = 1.19
sigma = 0.36
VcmaxRef = 159.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

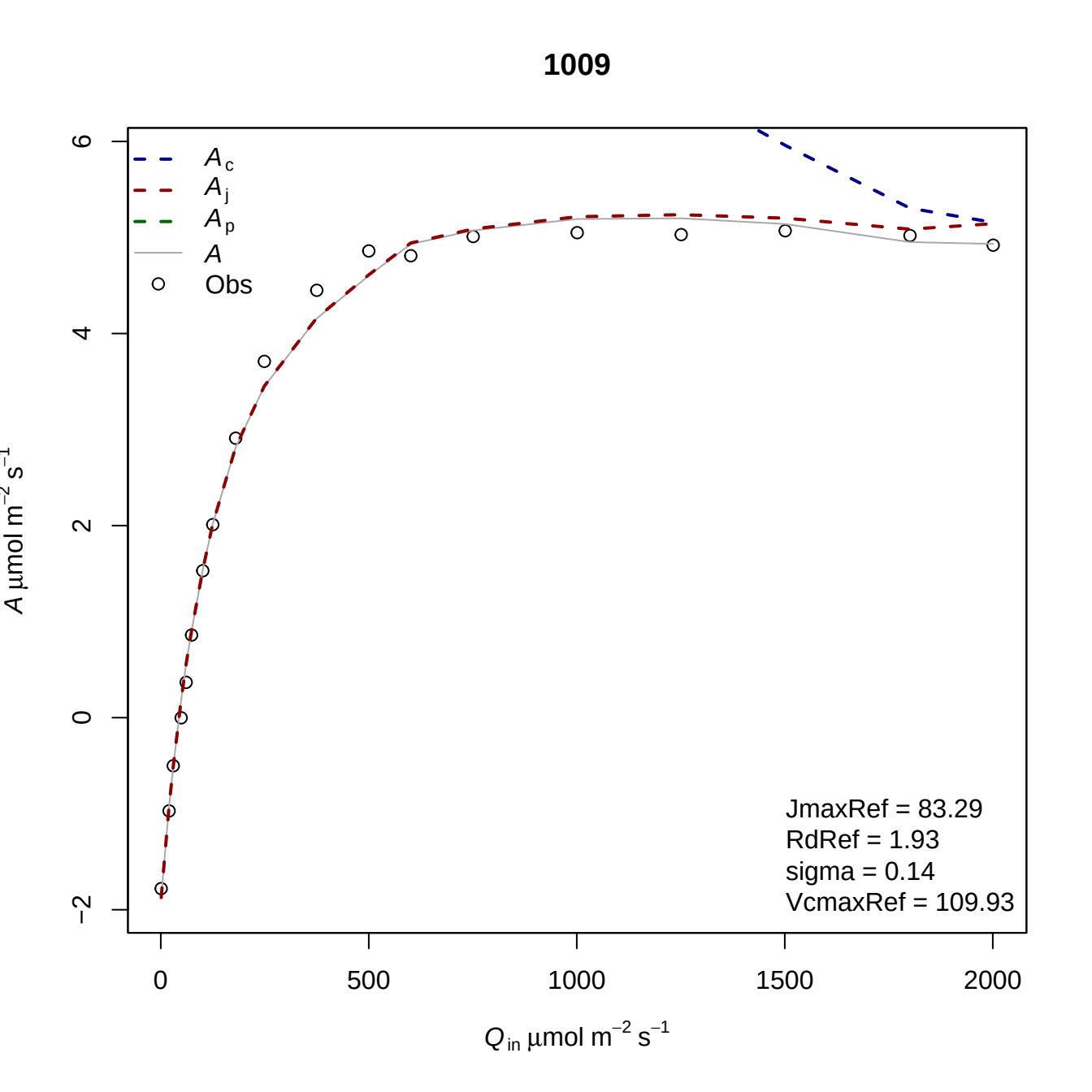

1004
A_c
A_j
A_p
A
Obs
JmaxRef = 51.43
RdRef = 1.05
sigma = 0.22
VcmaxRef = 2199317.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

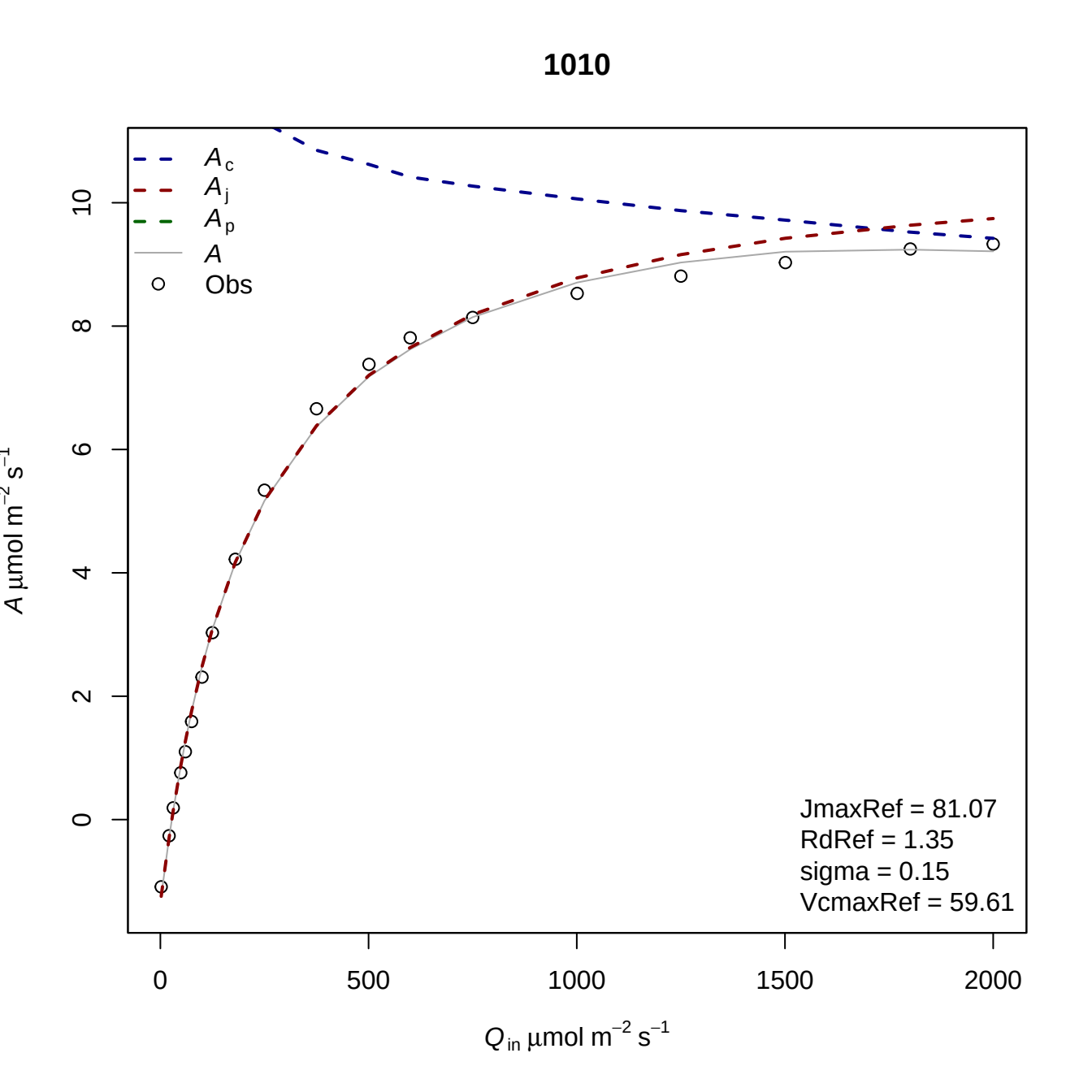

1005
A_c
A_j
A_p
A
Obs
JmaxRef = 250.21
RdRef = 1.43
sigma = 0.69
VcmaxRef = 165.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

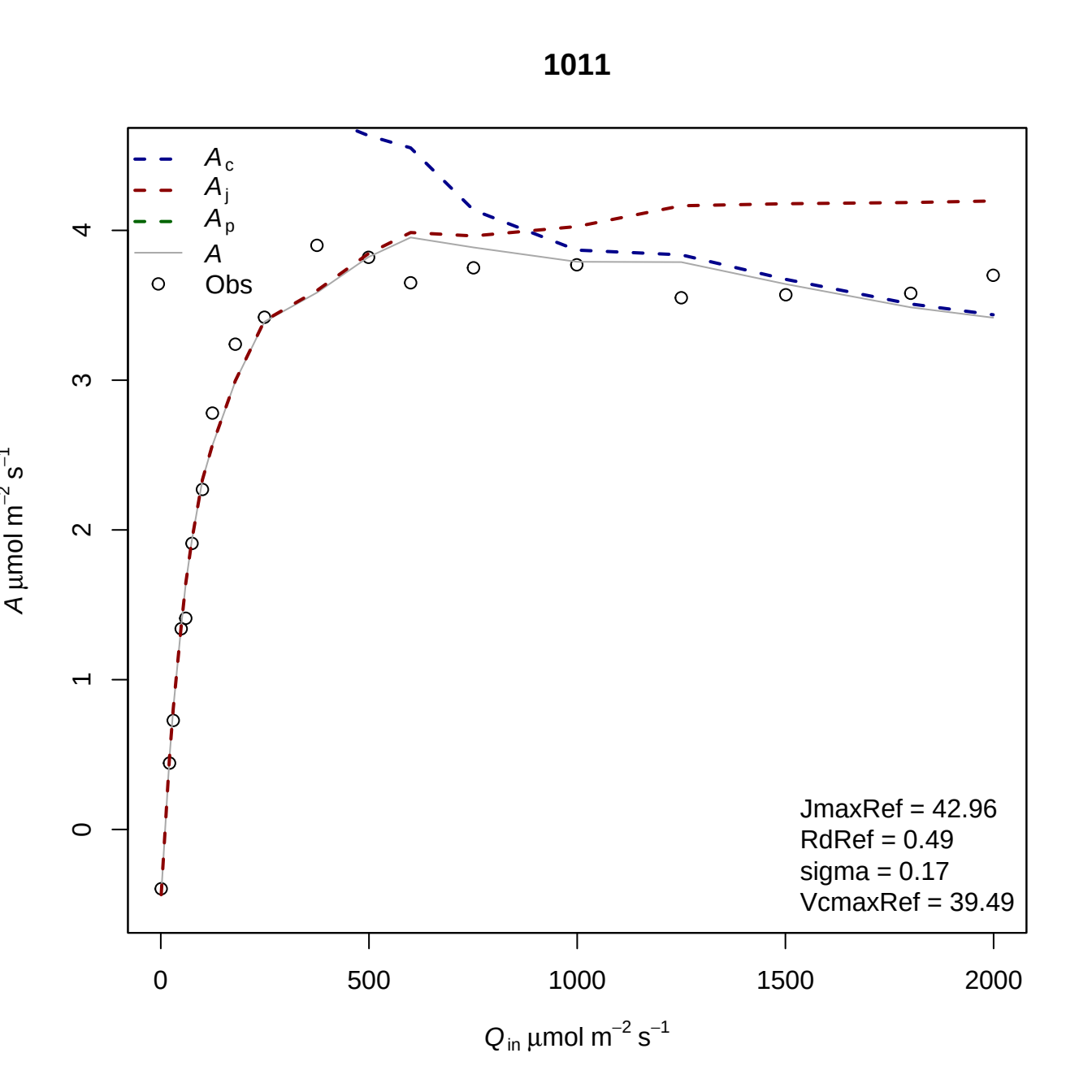

1006
A_c
A_j
A_p
A
Obs
JmaxRef = 138.77
RdRef = 1.81
sigma = 0.24
VcmaxRef = 93.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

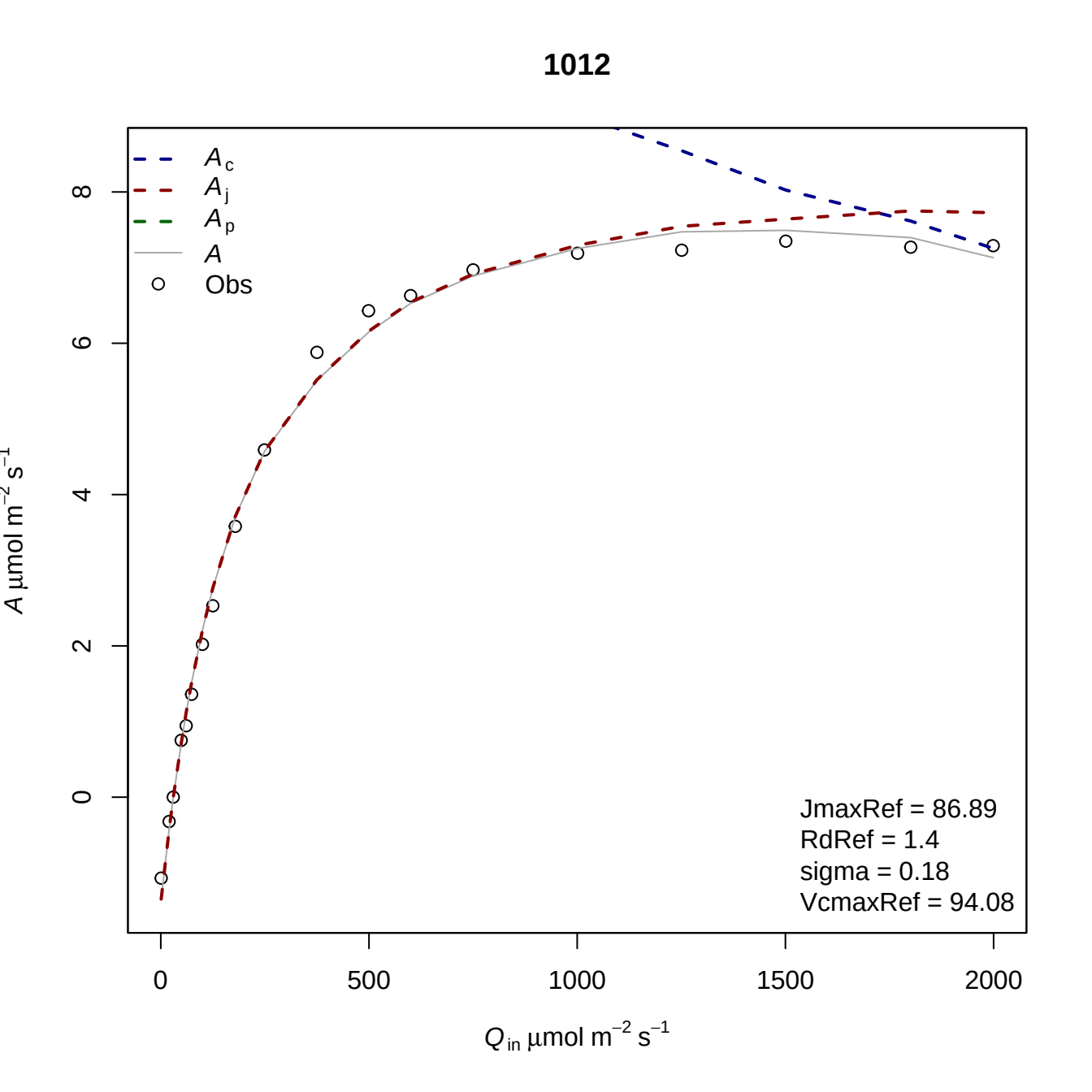

1007
A_c
A_j
A_p
A
Obs
JmaxRef = 66.36
RdRef = 1.16
sigma = 0.17
VcmaxRef = 56.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

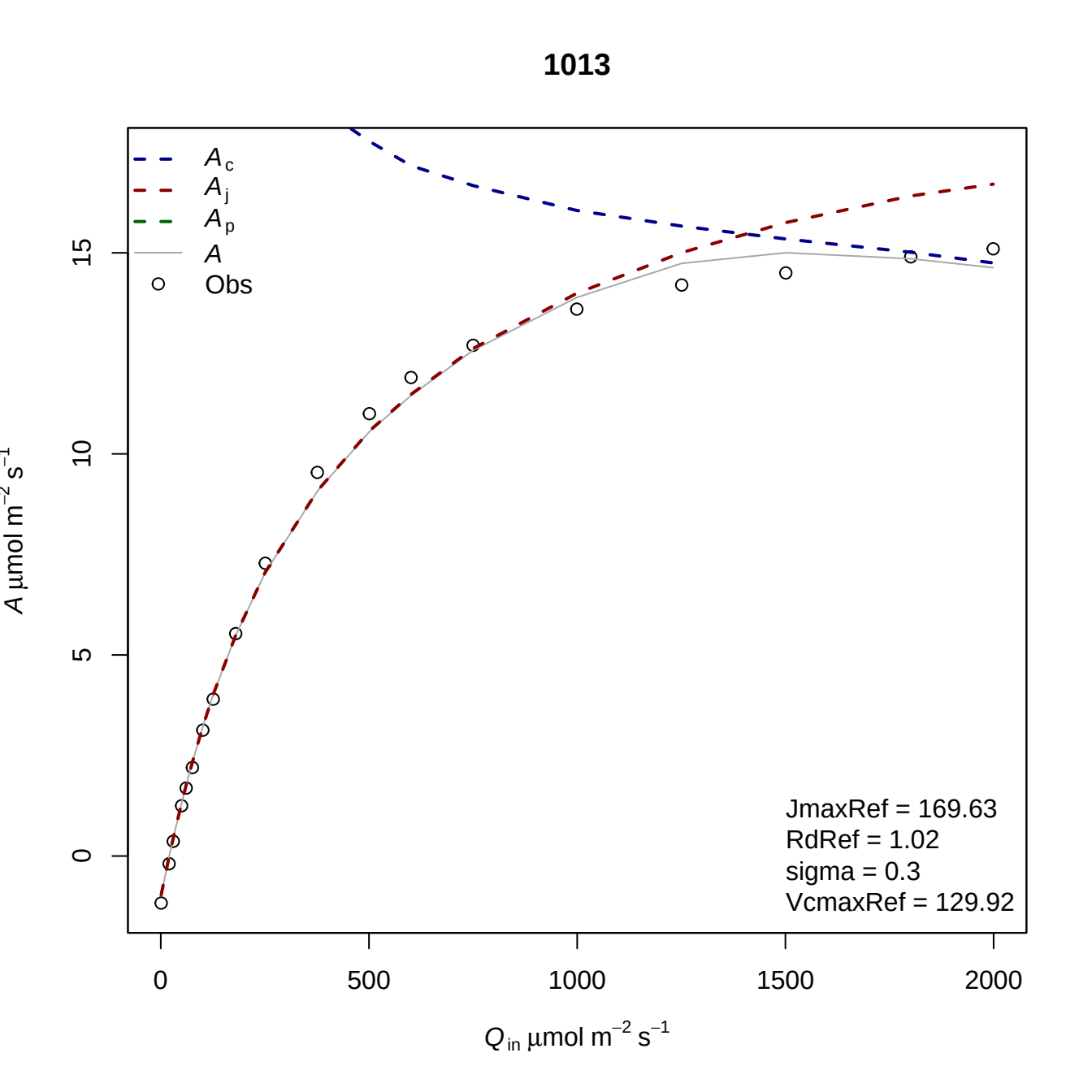

1008
A_c
A_j
A_p
A
Obs
JmaxRef = 72.5
RdRef = 1.35
sigma = 0.12
VcmaxRef = 1552560.81
A µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

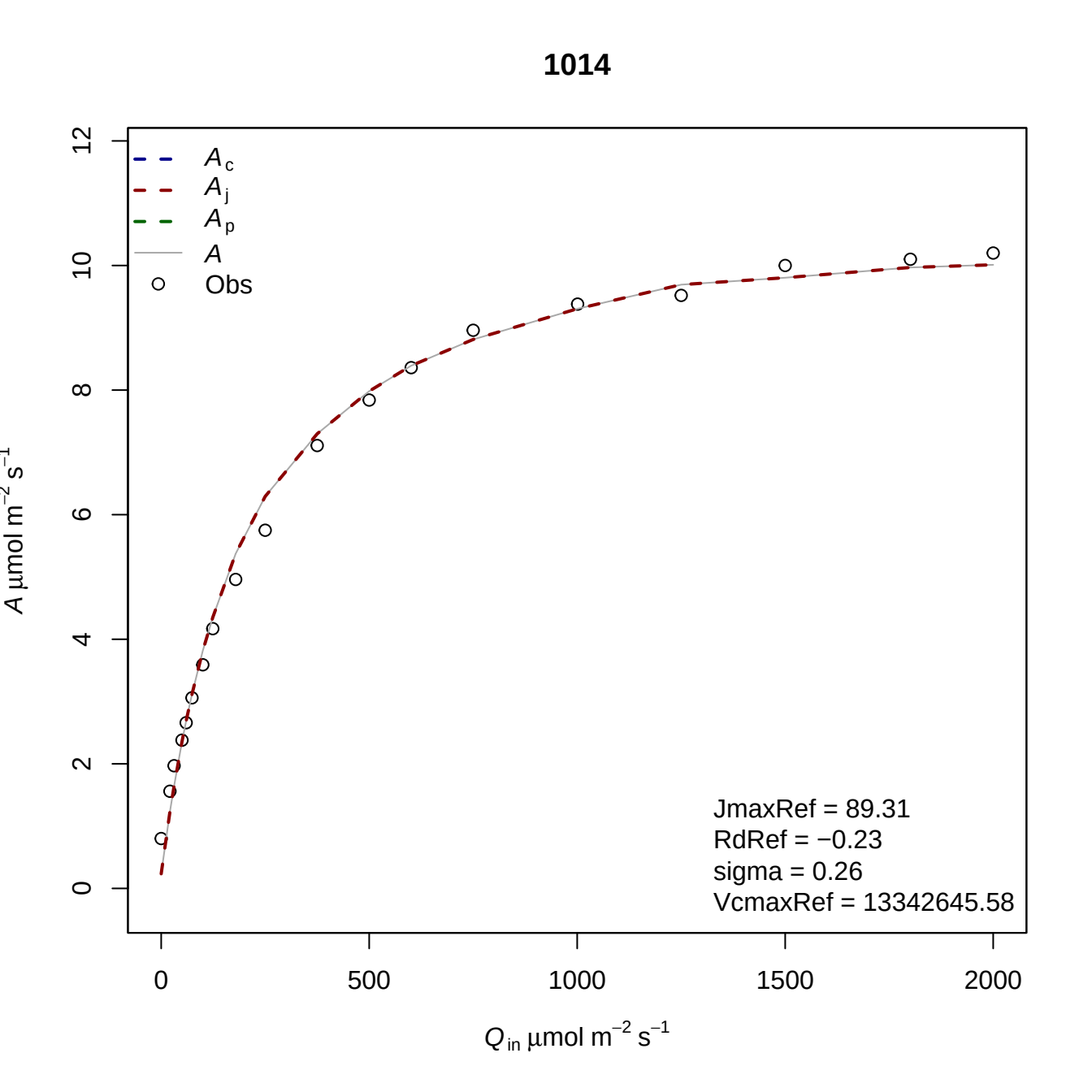

1009
A_c
A_j
A_p
A
Obs
JmaxRef = 83.29
RdRef = 1.93
sigma = 0.14
VcmaxRef = 109.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

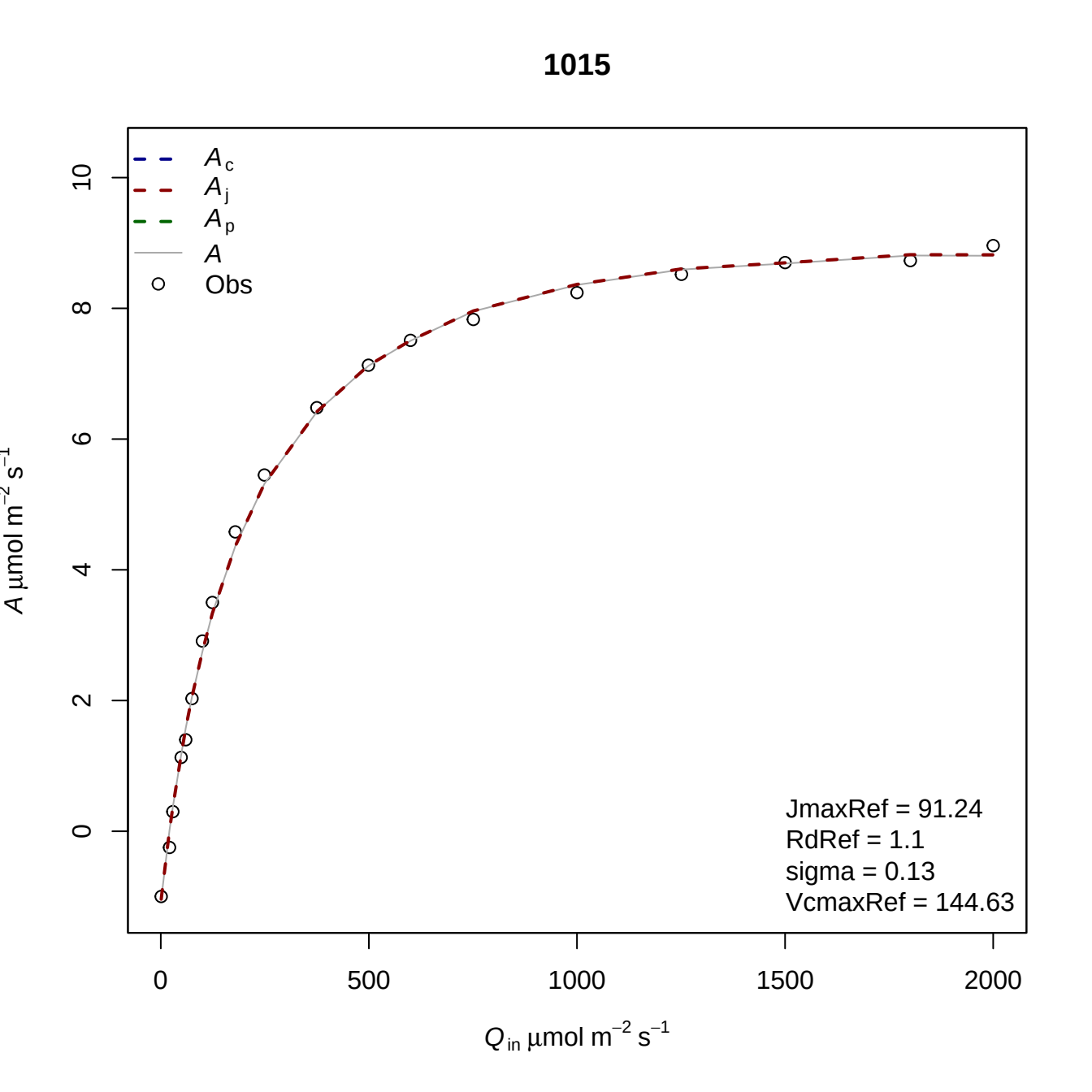

1010
A_c
A_j
A_p
A
Obs
JmaxRef = 81.07
RdRef = 1.35
sigma = 0.15
VcmaxRef = 59.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

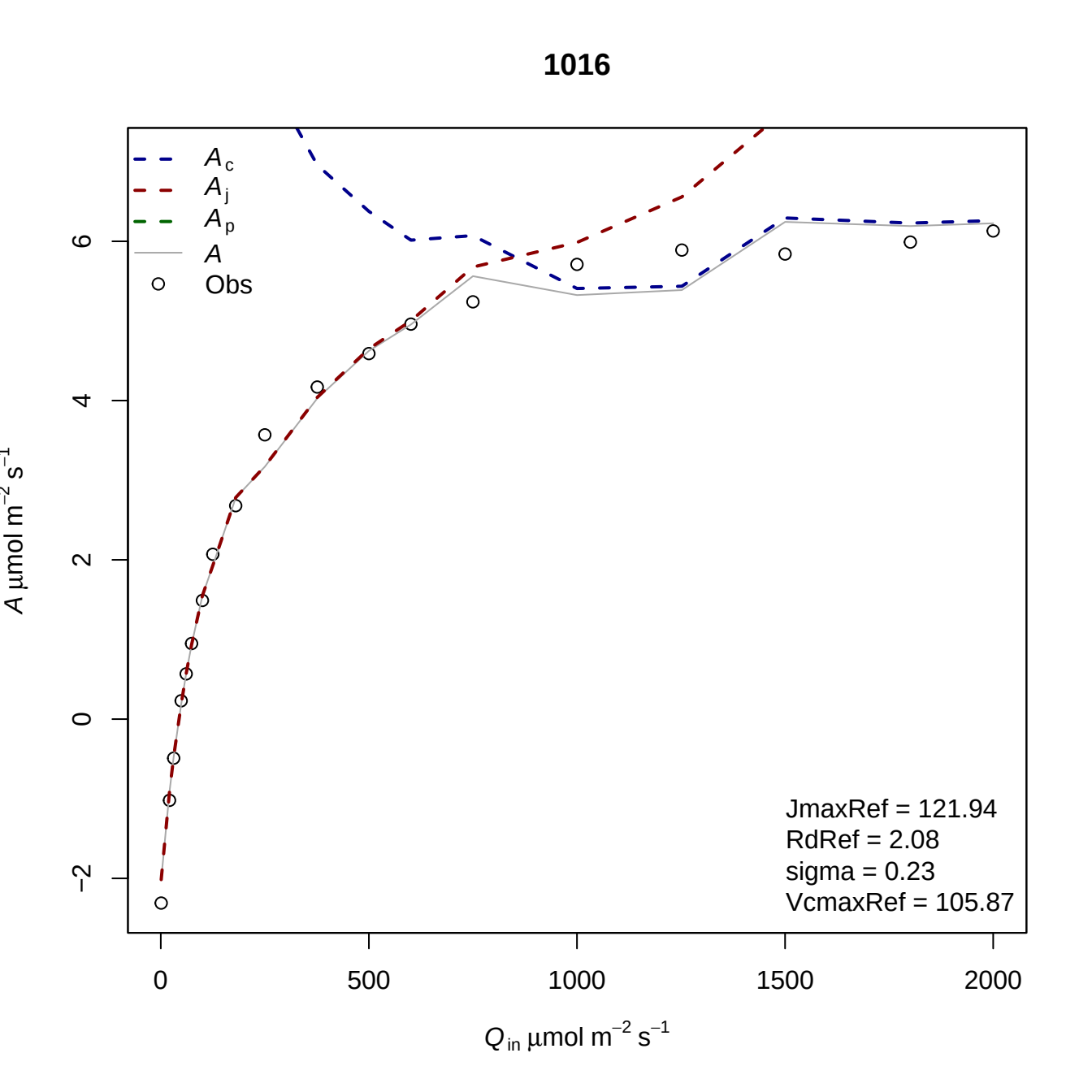

1011
A_c
A_j
A_p
A
Obs
JmaxRef = 42.96
RdRef = 0.49
sigma = 0.17
VcmaxRef = 39.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500
2000

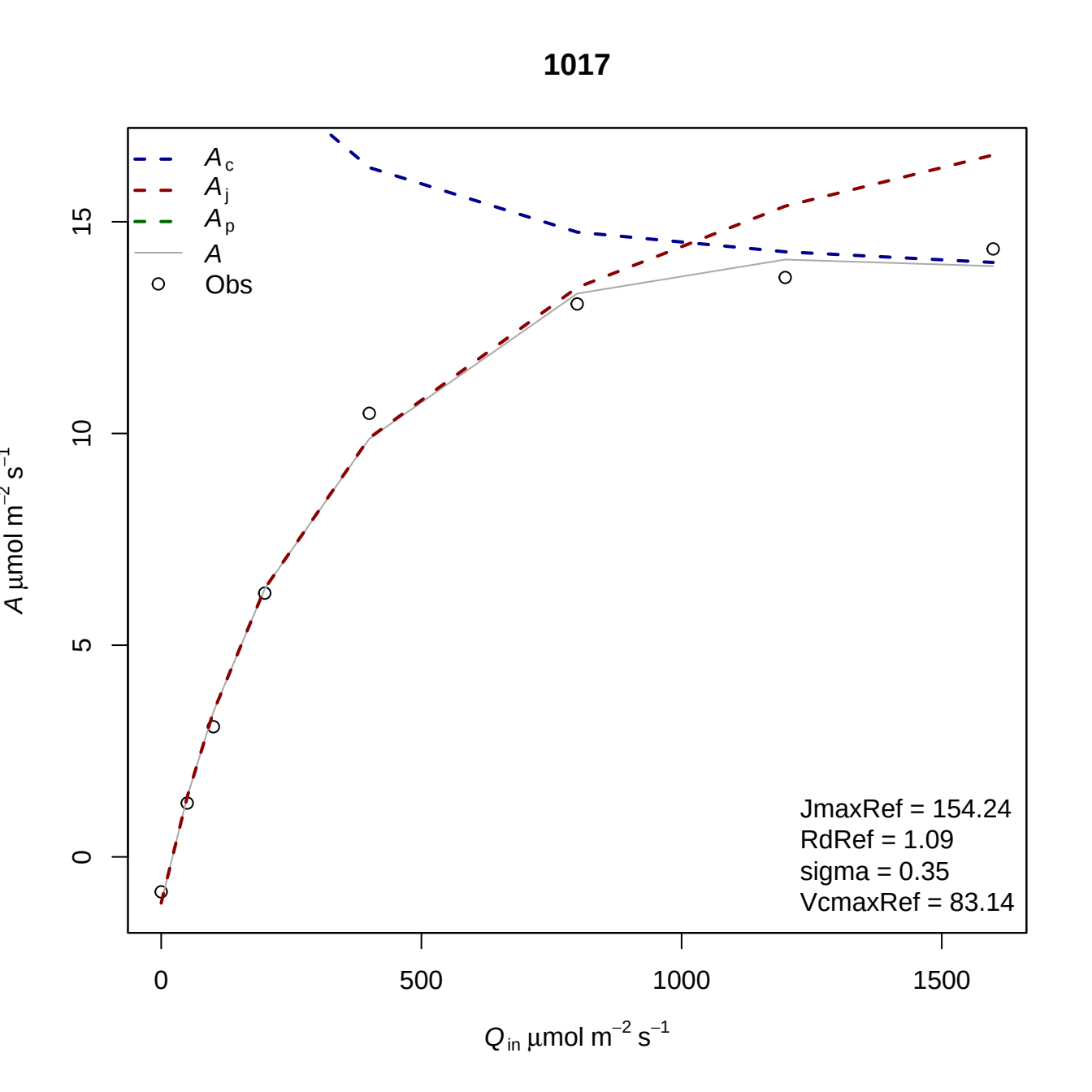

1012
A_c
A_j
A_p
A
Obs
JmaxRef = 86.89
RdRef = 1.4
sigma = 0.18
VcmaxRef = 94.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

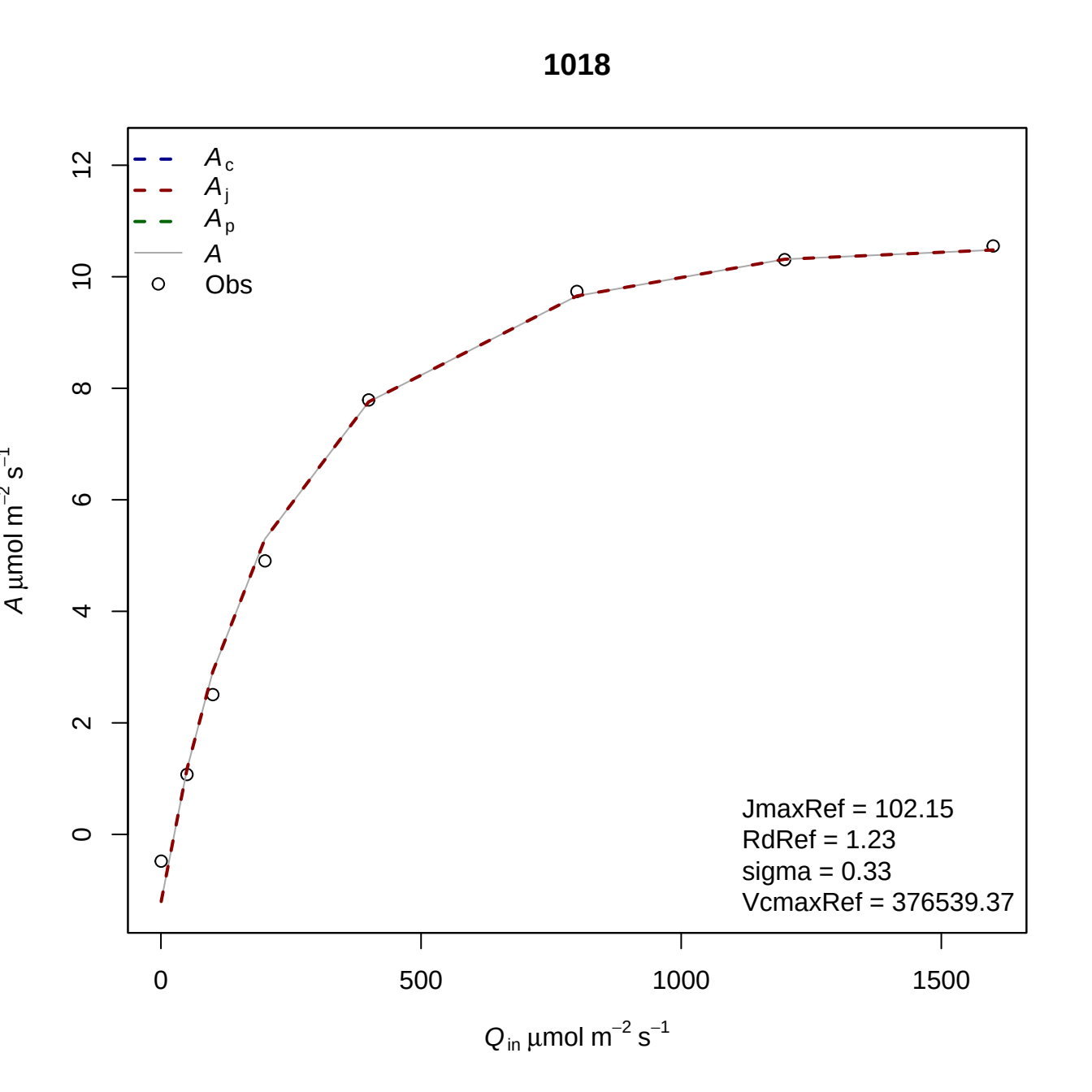

1013
A_c
A_j
A_p
A
Obs
JmaxRef = 169.63
RdRef = 1.02
sigma = 0.3
VcmaxRef = 129.92
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

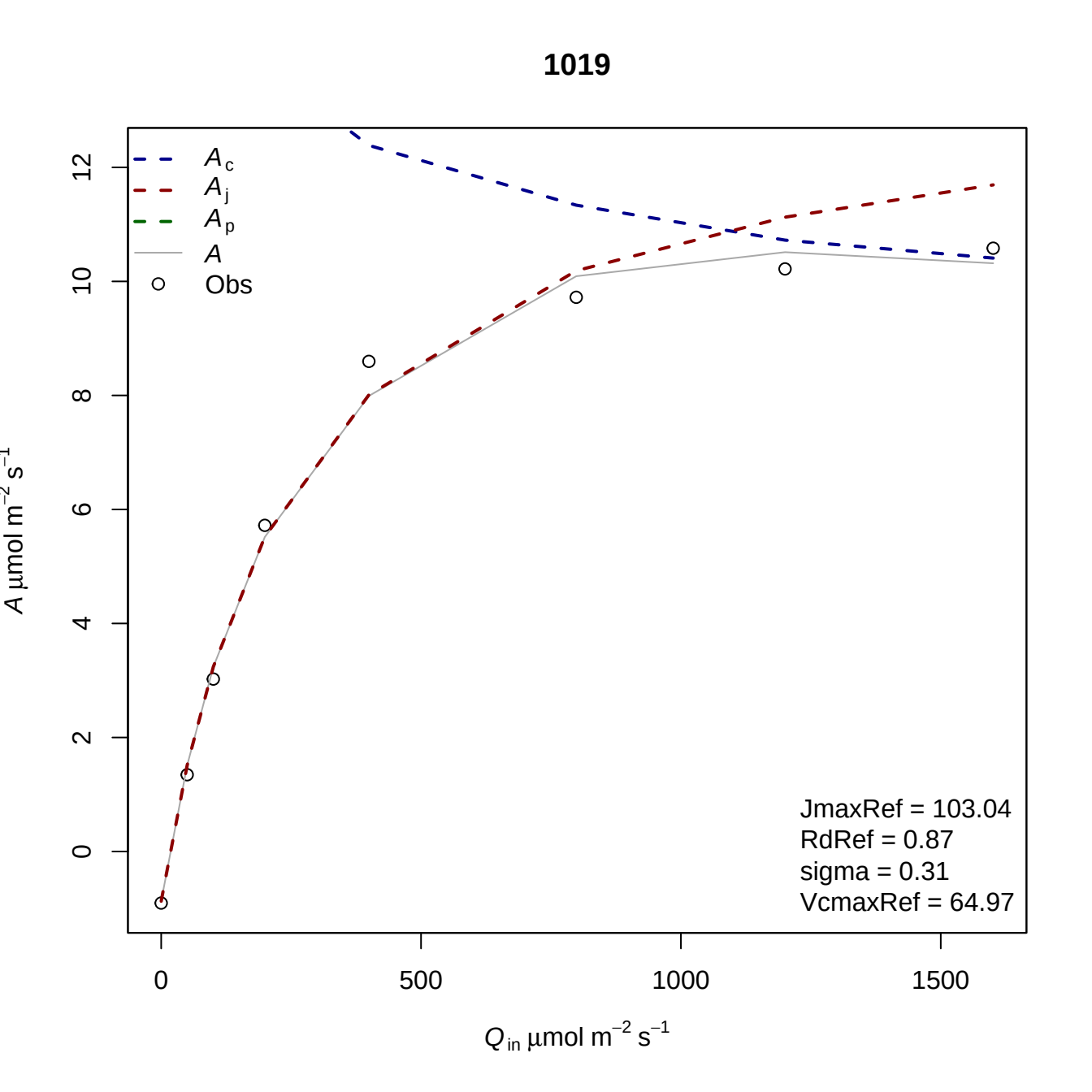

1014
A_c
A_j
A_p
A
Obs
JmaxRef = 89.31
RdRef = −0.23
sigma = 0.26
VcmaxRef = 13342645.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

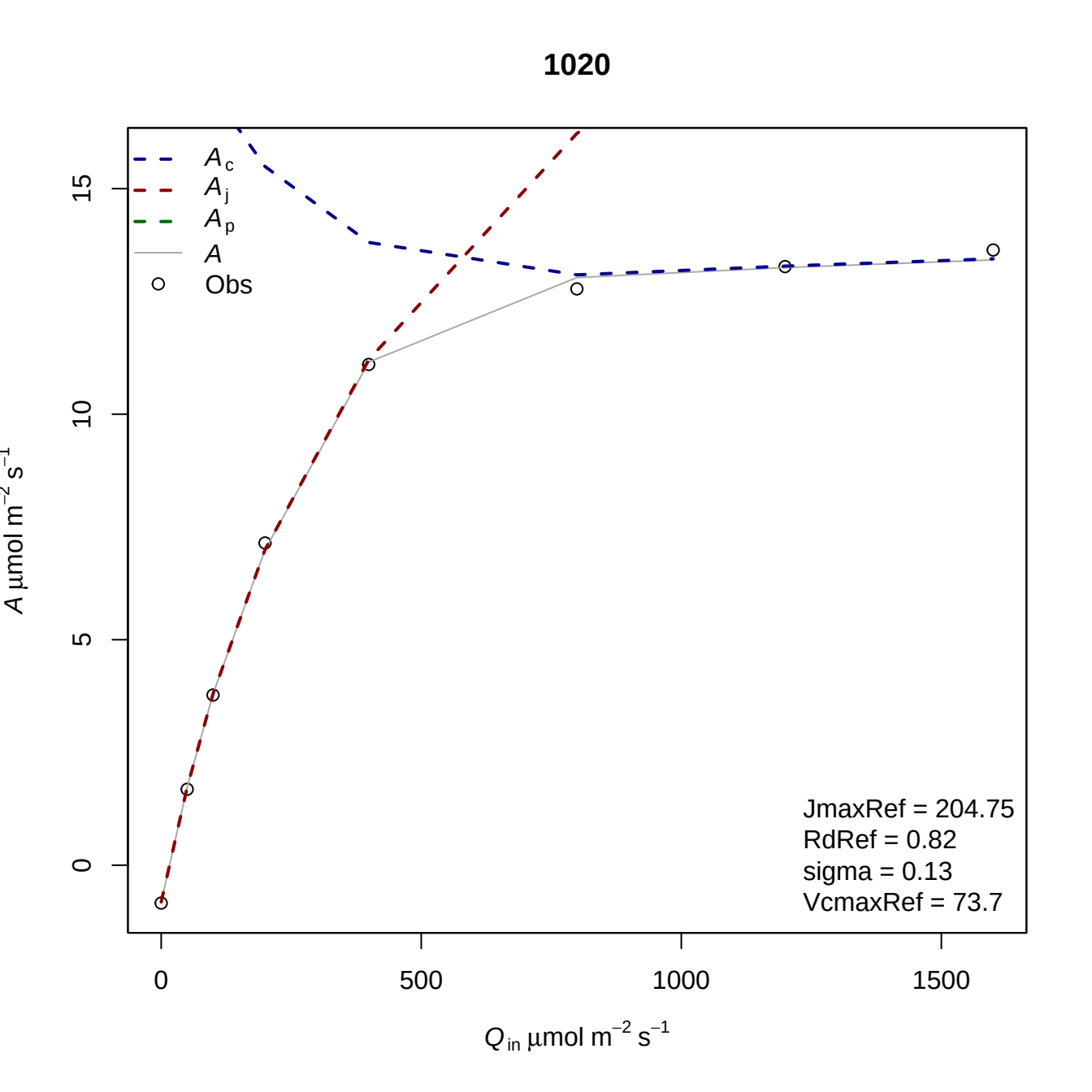

1015
A_c
A_j
A_p
A
Obs
JmaxRef = 91.24
RdRef = 1.1
sigma = 0.13
VcmaxRef = 144.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

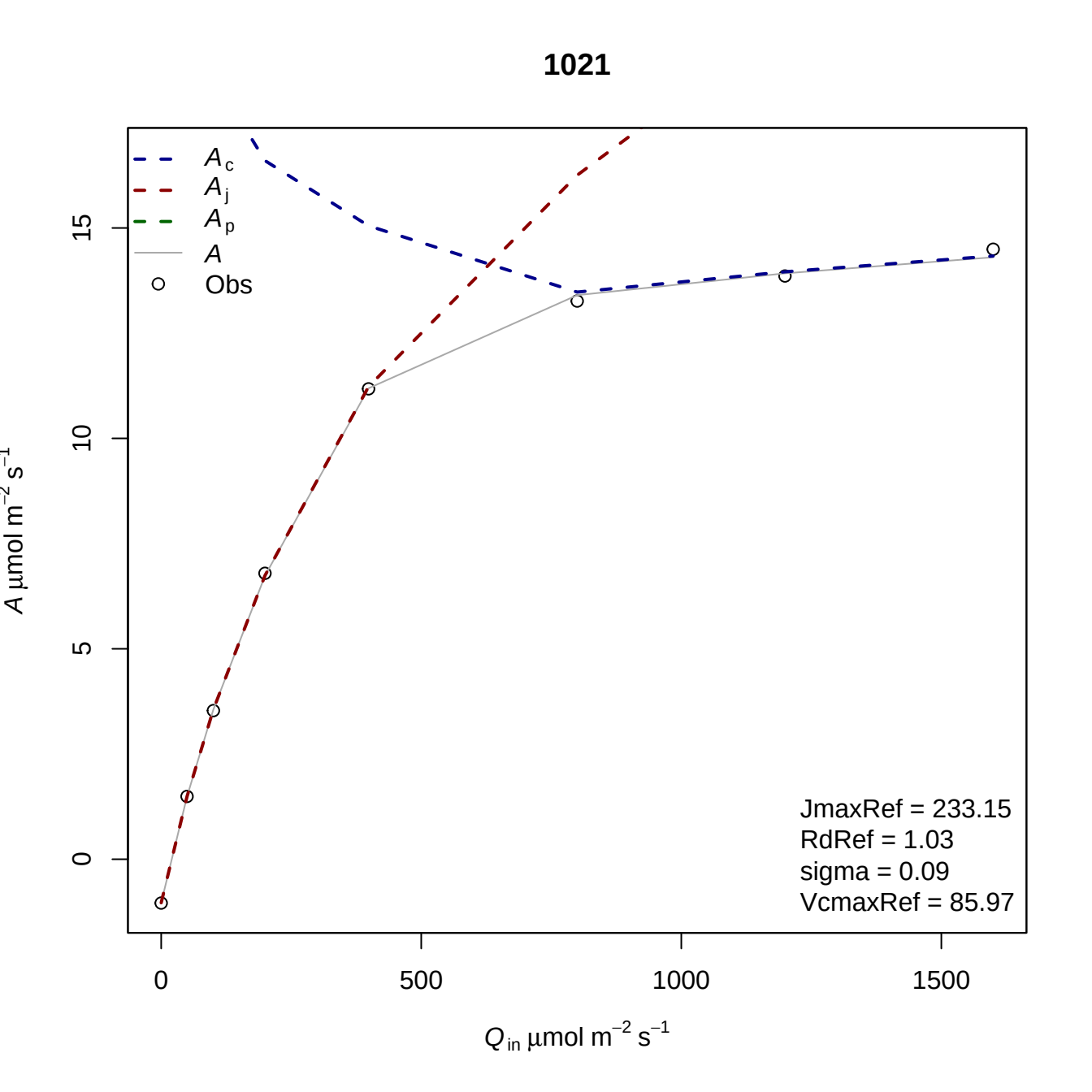

1016
A_c
A_j
A_p
A
Obs
JmaxRef = 121.94
RdRef = 2.08
sigma = 0.23
VcmaxRef = 105.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

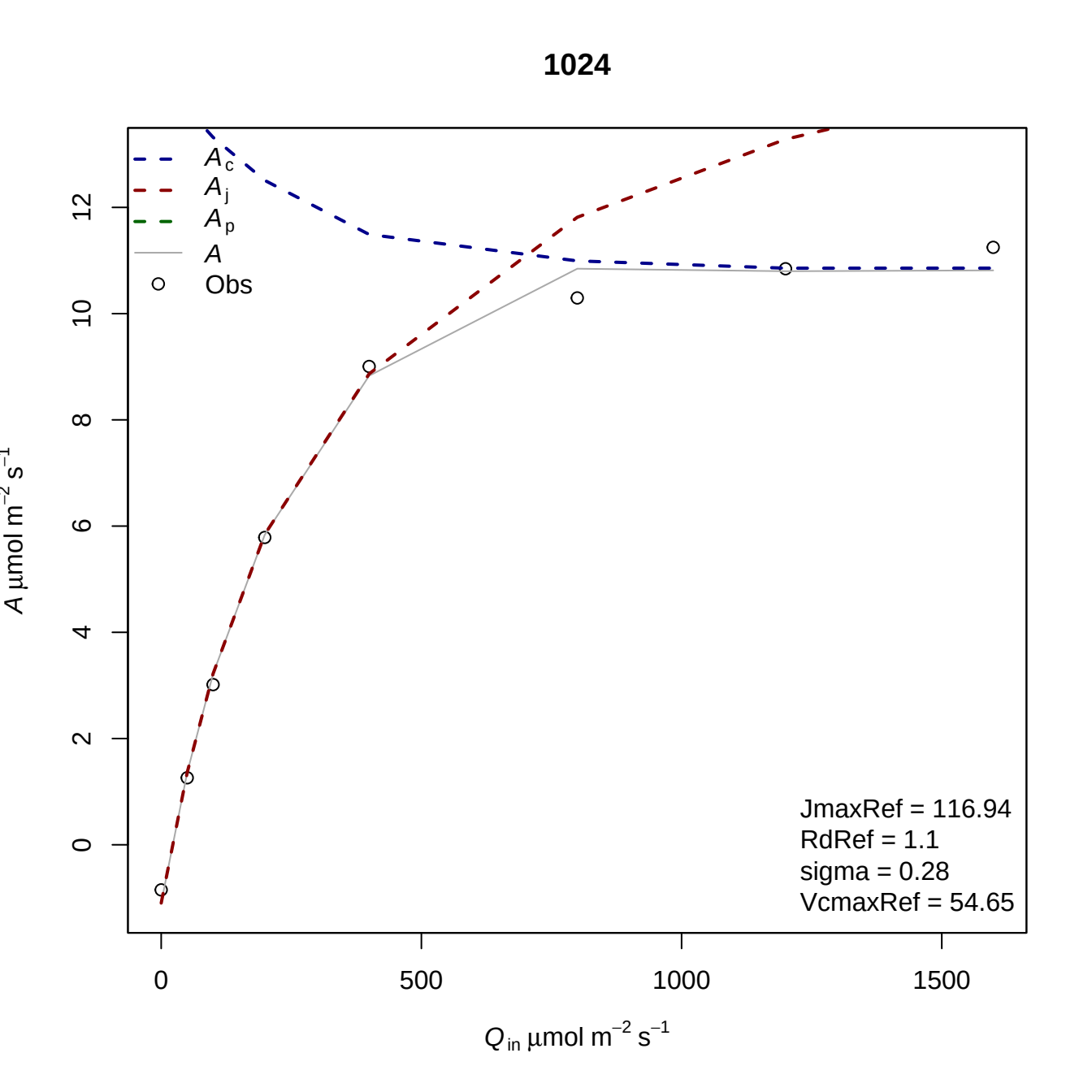

1017
A_c
A_j
A_p
A
Obs
JmaxRef = 154.24
RdRef = 1.09
sigma = 0.35
VcmaxRef = 83.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

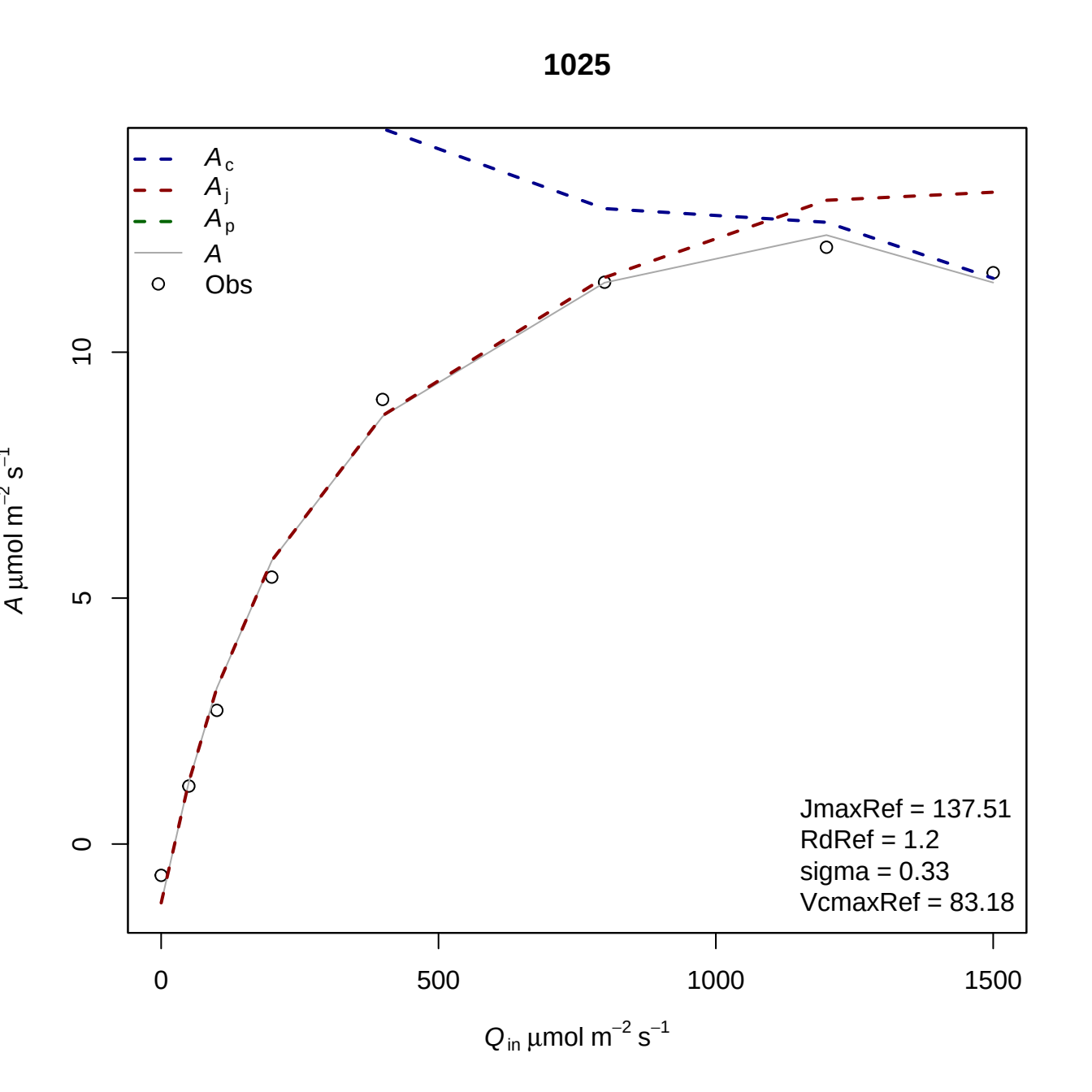

1018
A_c
A_j
A_p
A
Obs
JmaxRef = 102.15
RdRef = 1.23
sigma = 0.33
VcmaxRef = 376539.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

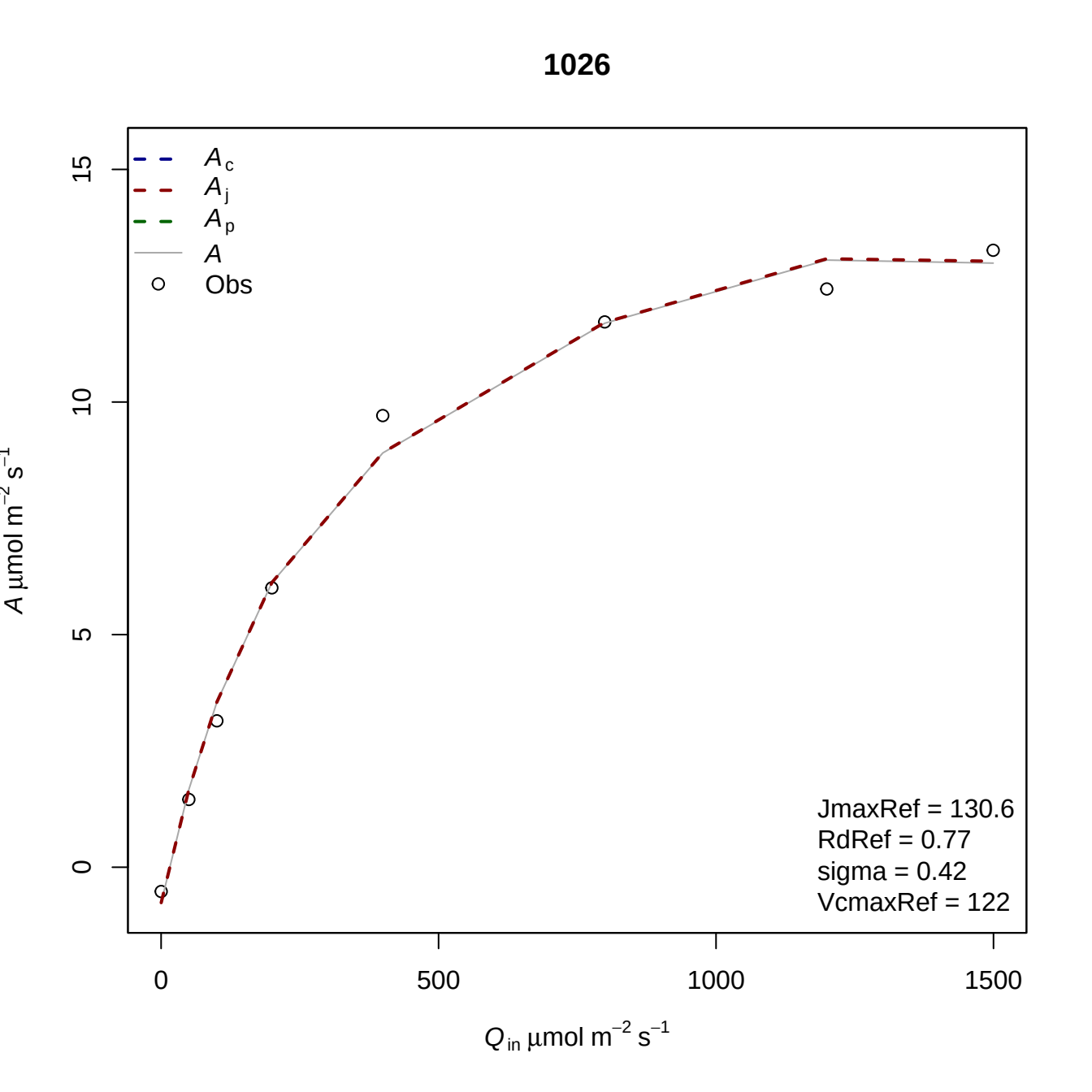

1019
A_c
A_j
A_p
A
Obs
JmaxRef = 103.04
RdRef = 0.87
sigma = 0.31
VcmaxRef = 64.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

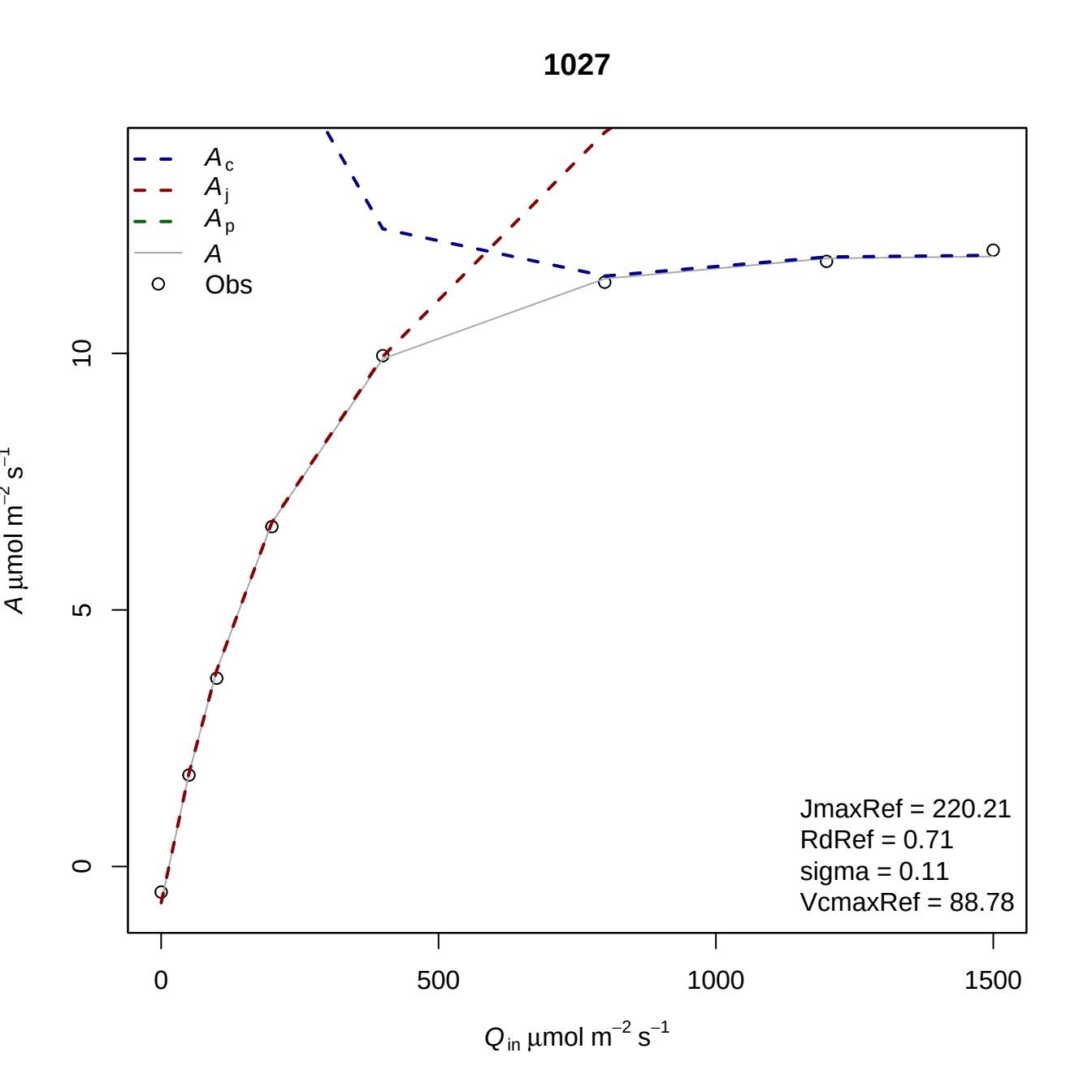

1020
A_c
A_j
A_p
A
Obs
JmaxRef = 204.75
RdRef = 0.82
sigma = 0.13
VcmaxRef = 73.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

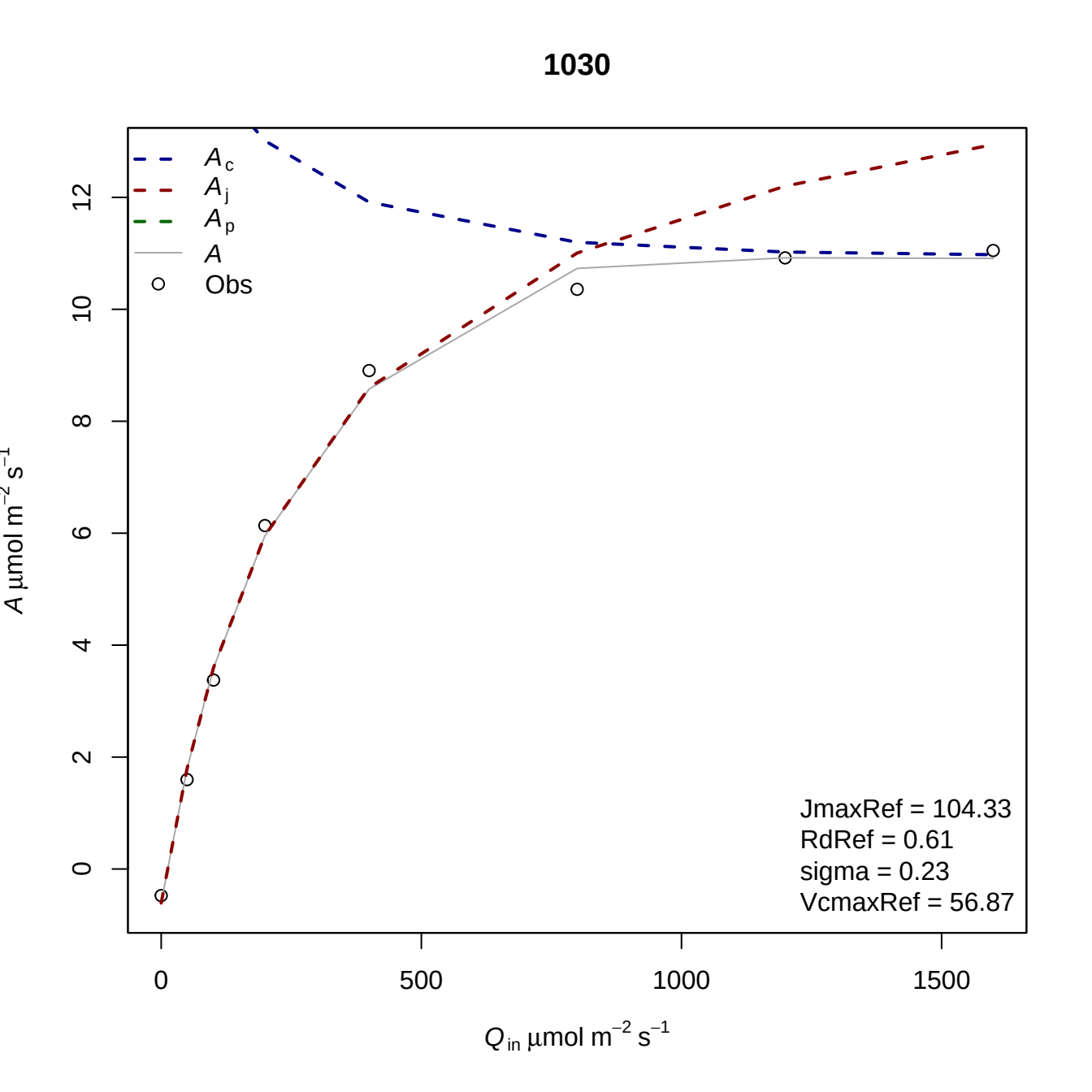

1021
A_c
A_j
A_p
A
Obs
JmaxRef = 233.15
RdRef = 1.03
sigma = 0.09
VcmaxRef = 85.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

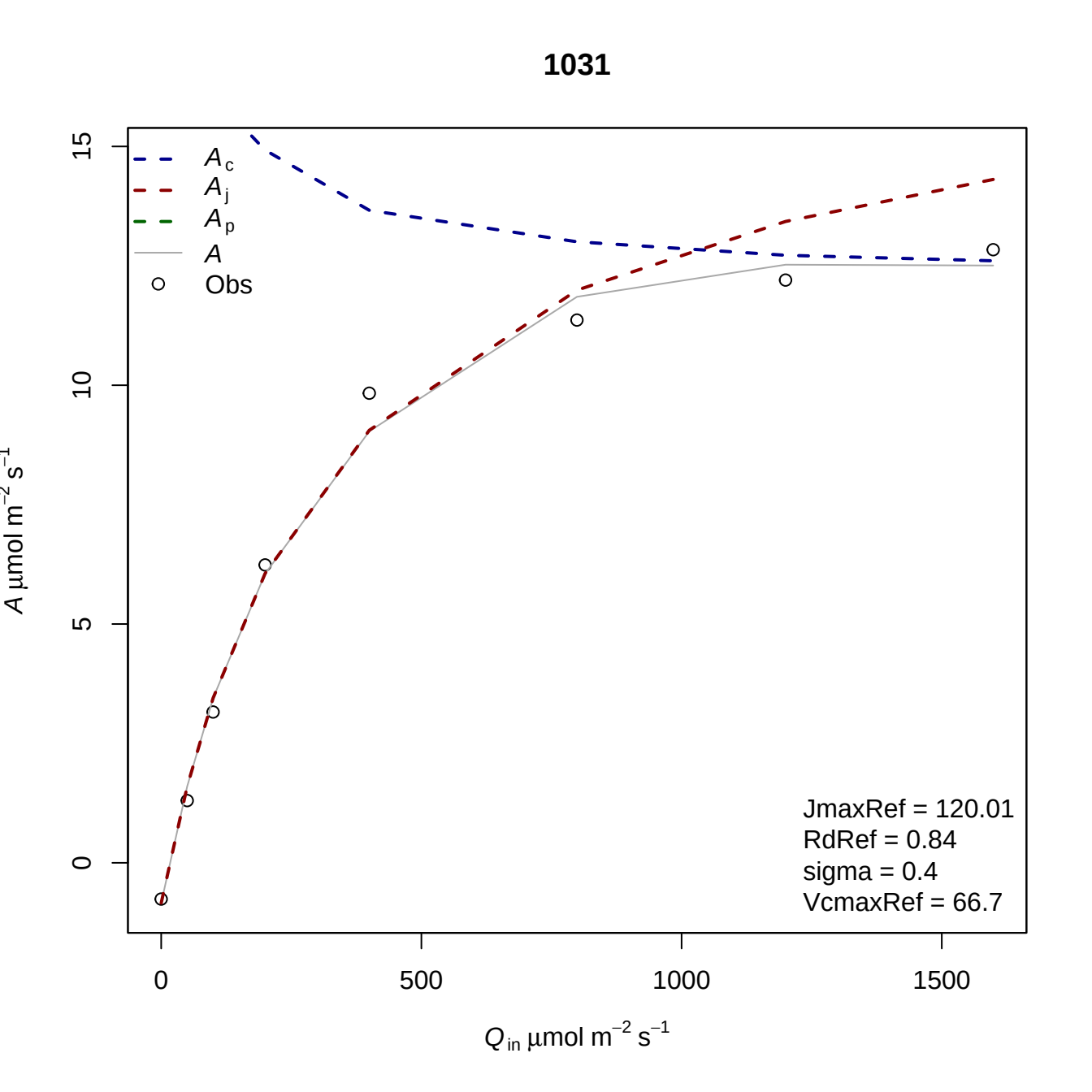

1024
A_c
A_j
A_p
A
Obs
JmaxRef = 116.94
RdRef = 1.1
sigma = 0.28
VcmaxRef = 54.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

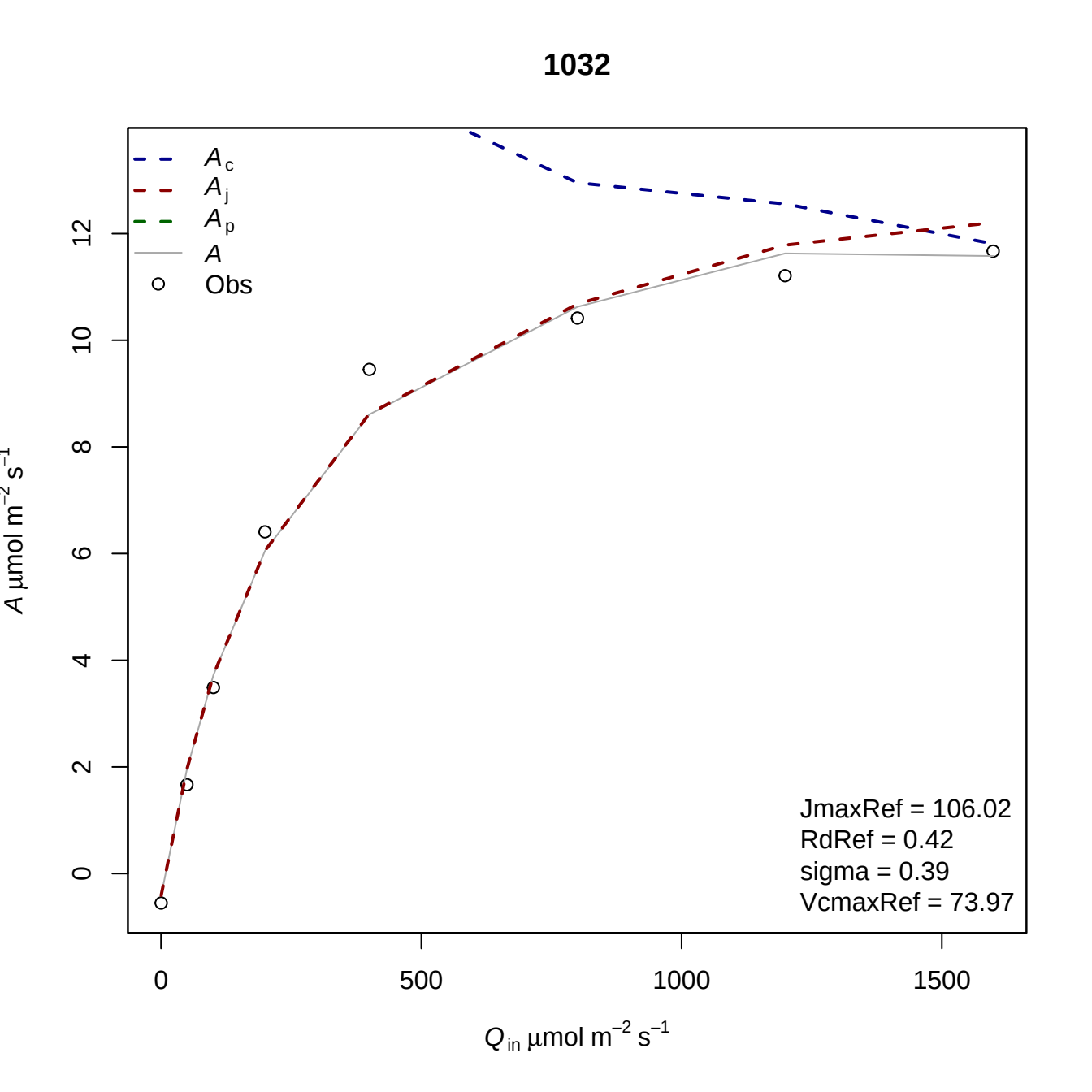

1025
A_c
A_j
A_p
A
Obs
JmaxRef = 137.51
RdRef = 1.2
sigma = 0.33
VcmaxRef = 83.18
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

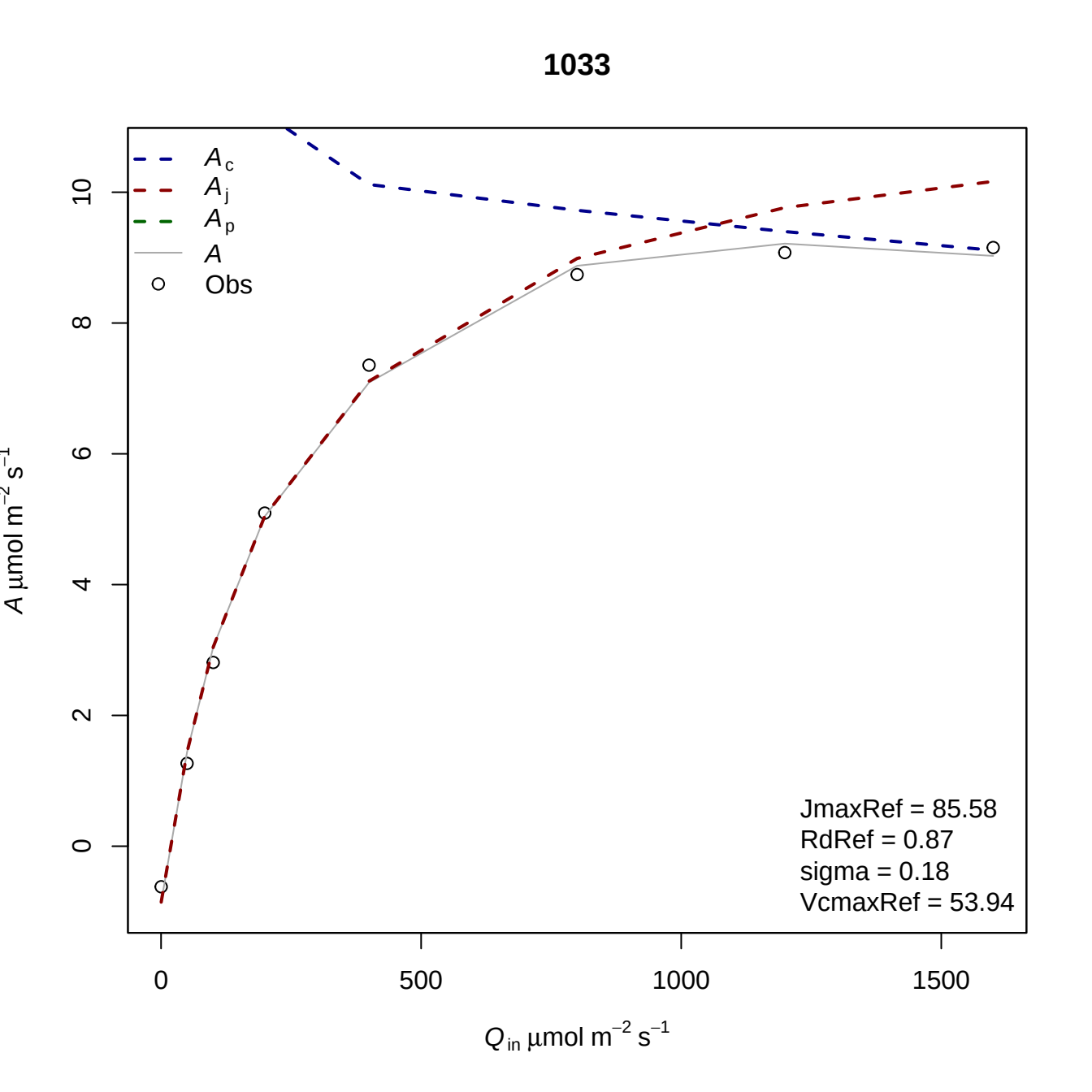

1026
A_c
A_j
A_p
A
Obs
JmaxRef = 130.6
RdRef = 0.77
sigma = 0.42
VcmaxRef = 122
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

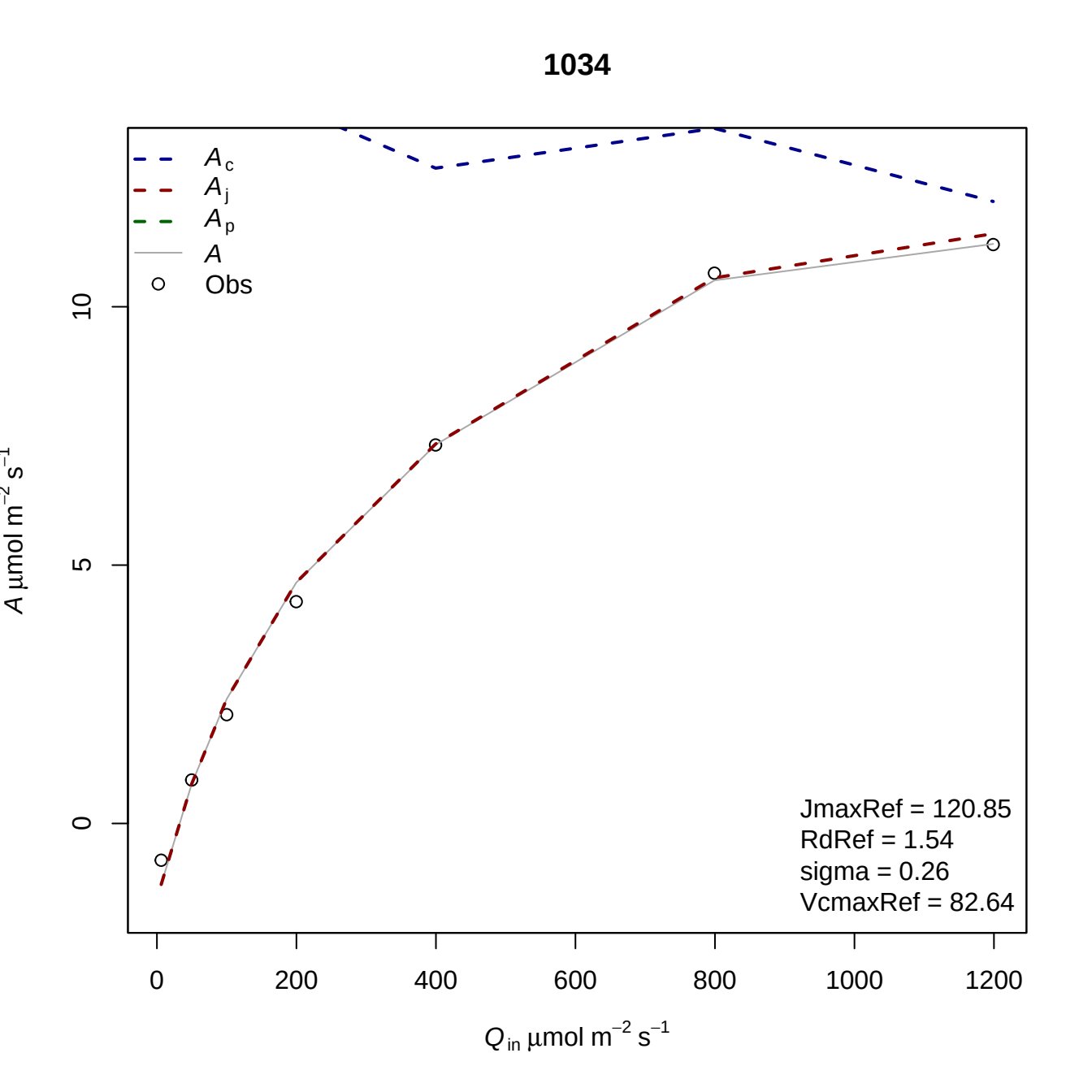

1027
A_c
A_j
A_p
A
Obs
JmaxRef = 220.21
RdRef = 0.71
sigma = 0.11
VcmaxRef = 88.78
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

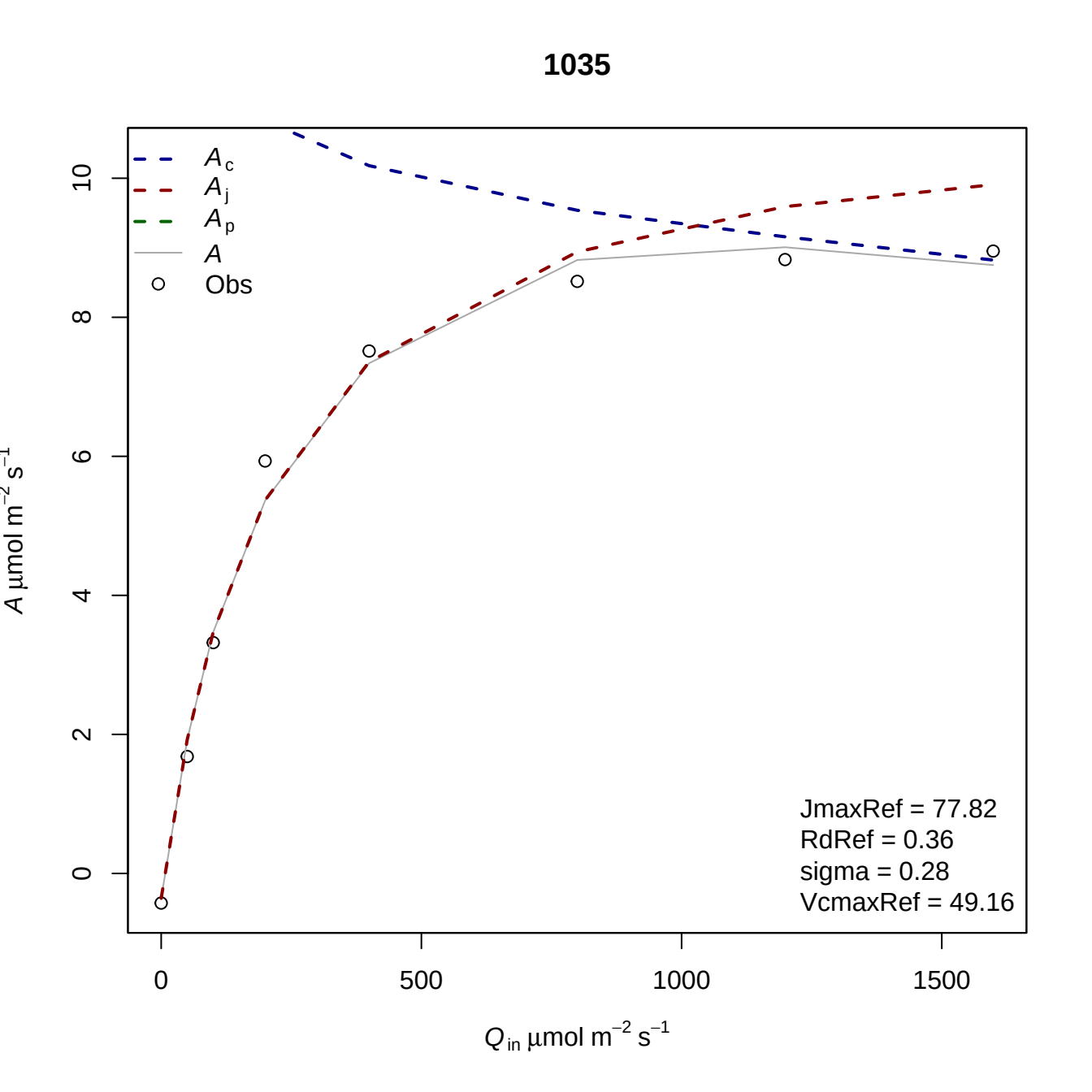

1030
A_c
A_j
A_p
A
Obs
JmaxRef = 104.33
RdRef = 0.61
sigma = 0.23
VcmaxRef = 56.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

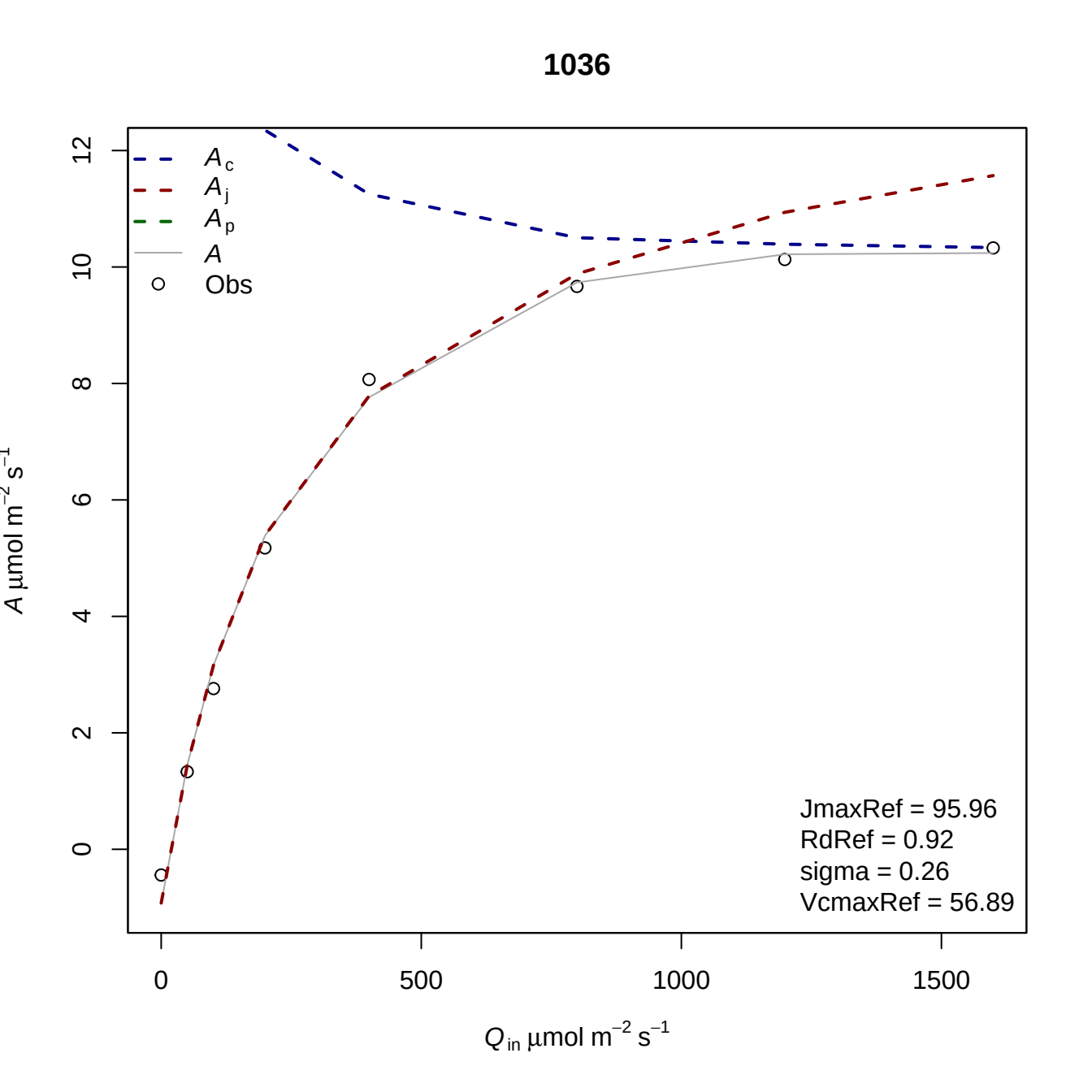

1031
A_c
A_j
A_p
A
Obs
JmaxRef = 120.01
RdRef = 0.84
sigma = 0.4
VcmaxRef = 66.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

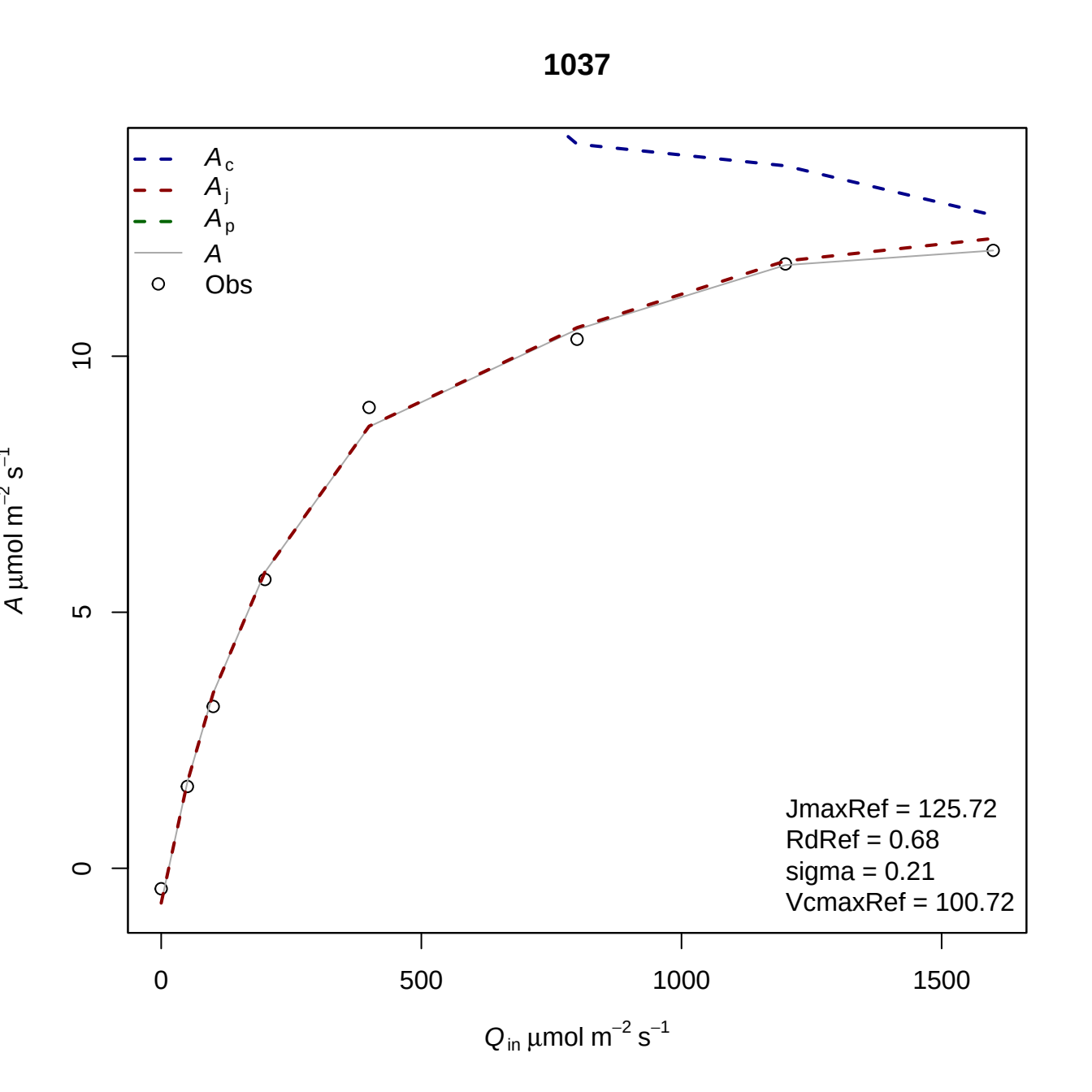

1032
A_c
A_j
A_p
A
Obs
JmaxRef = 106.02
RdRef = 0.42
sigma = 0.39
VcmaxRef = 73.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

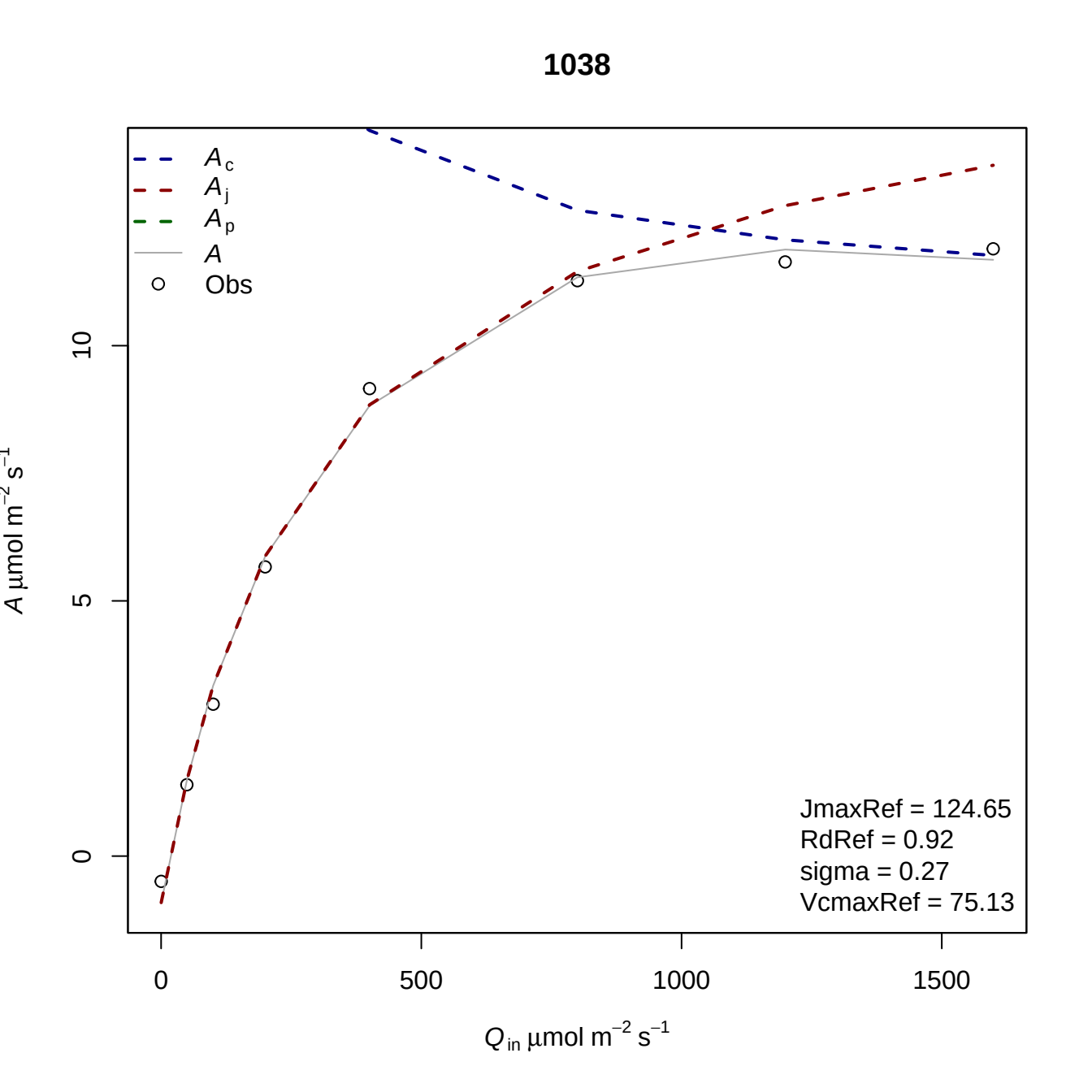

1033
A_c
A_j
A_p
A
Obs
JmaxRef = 85.58
RdRef = 0.87
sigma = 0.18
VcmaxRef = 53.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

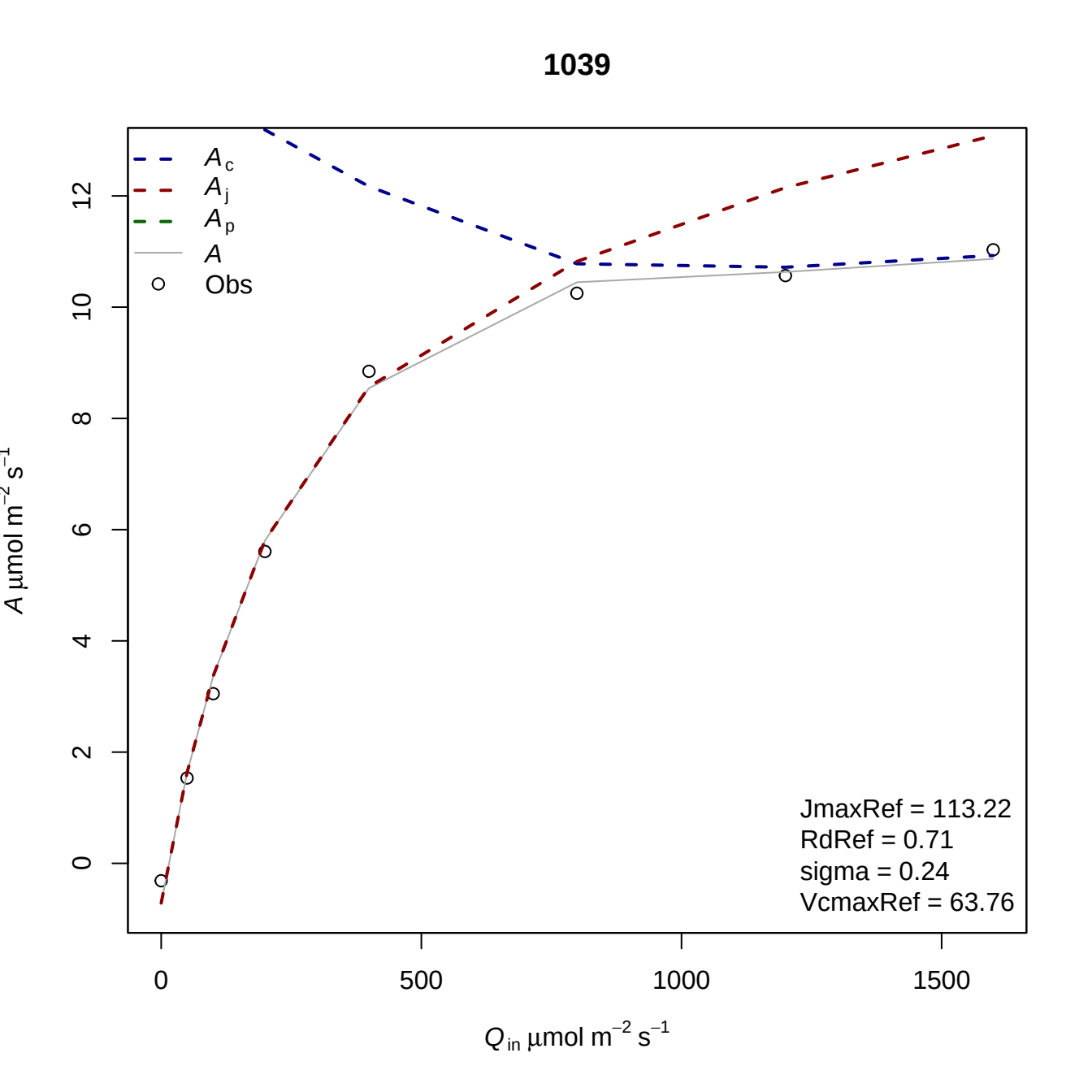

1034
A_c
A_j
A_p
A
Obs
JmaxRef = 120.85
RdRef = 1.54
sigma = 0.26
VcmaxRef = 82.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000
1200

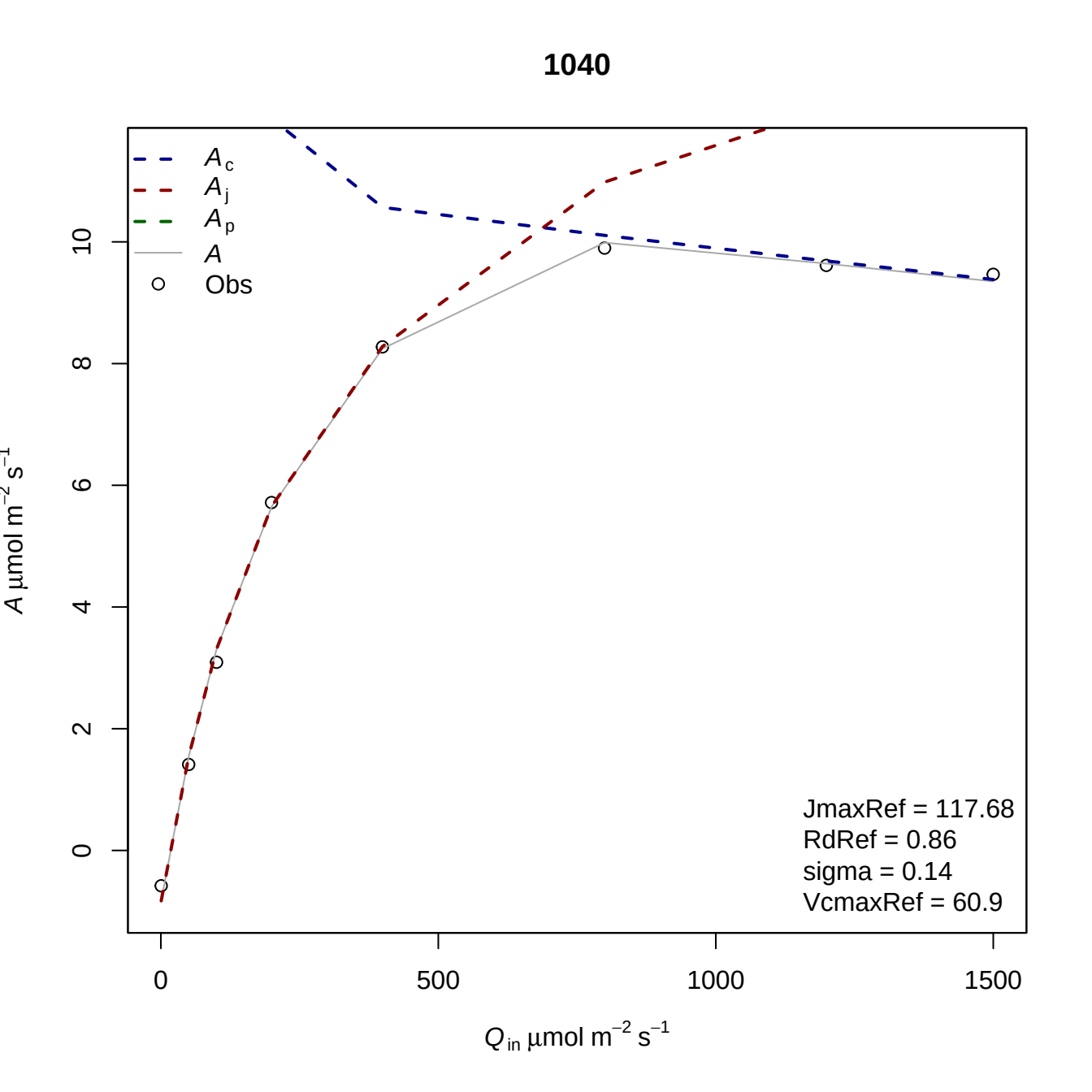

1035
A_c
A_j
A_p
A
Obs
JmaxRef = 77.82
RdRef = 0.36
sigma = 0.28
VcmaxRef = 49.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

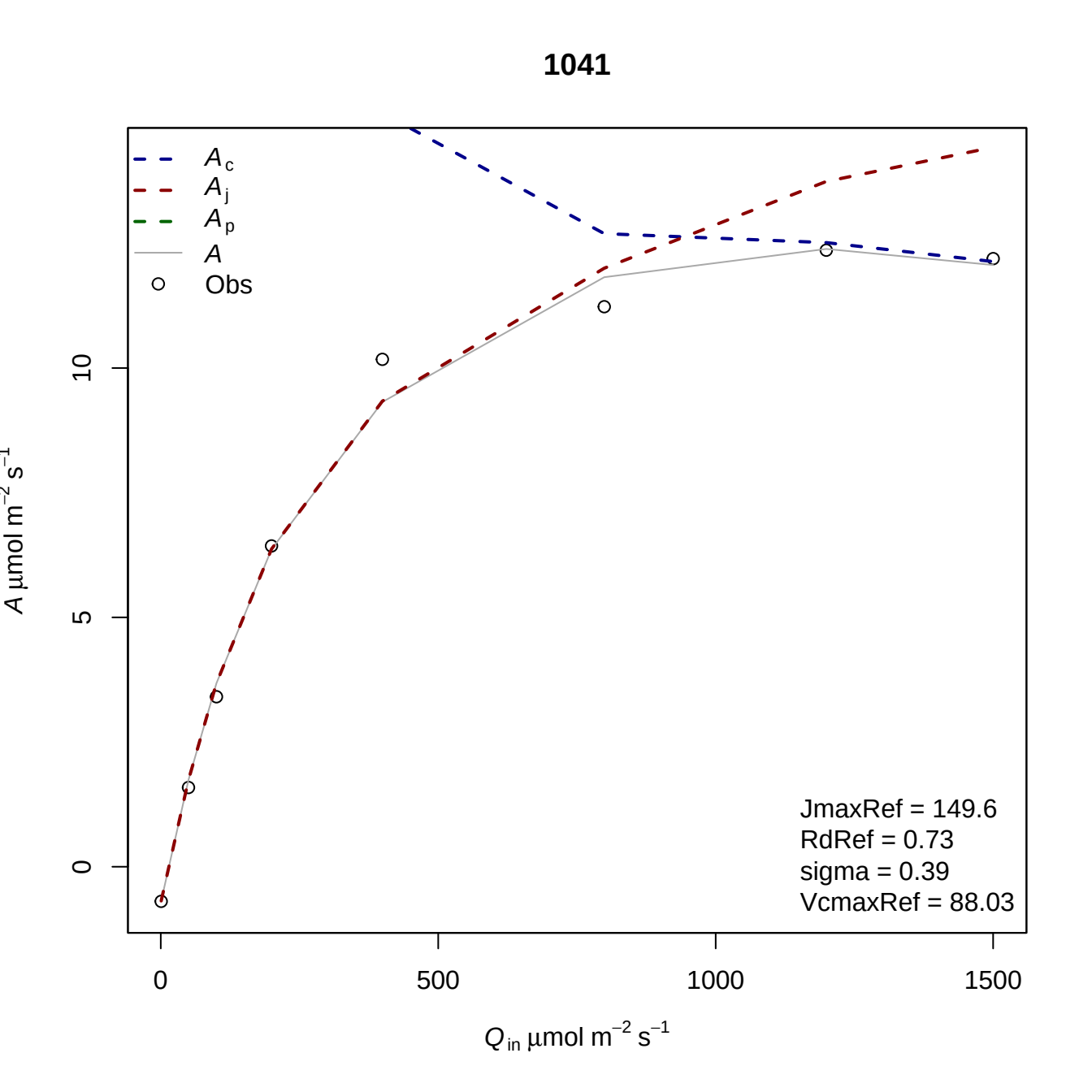

1036
A_c
A_j
A_p
A
Obs
JmaxRef = 95.96
RdRef = 0.92
sigma = 0.26
VcmaxRef = 56.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

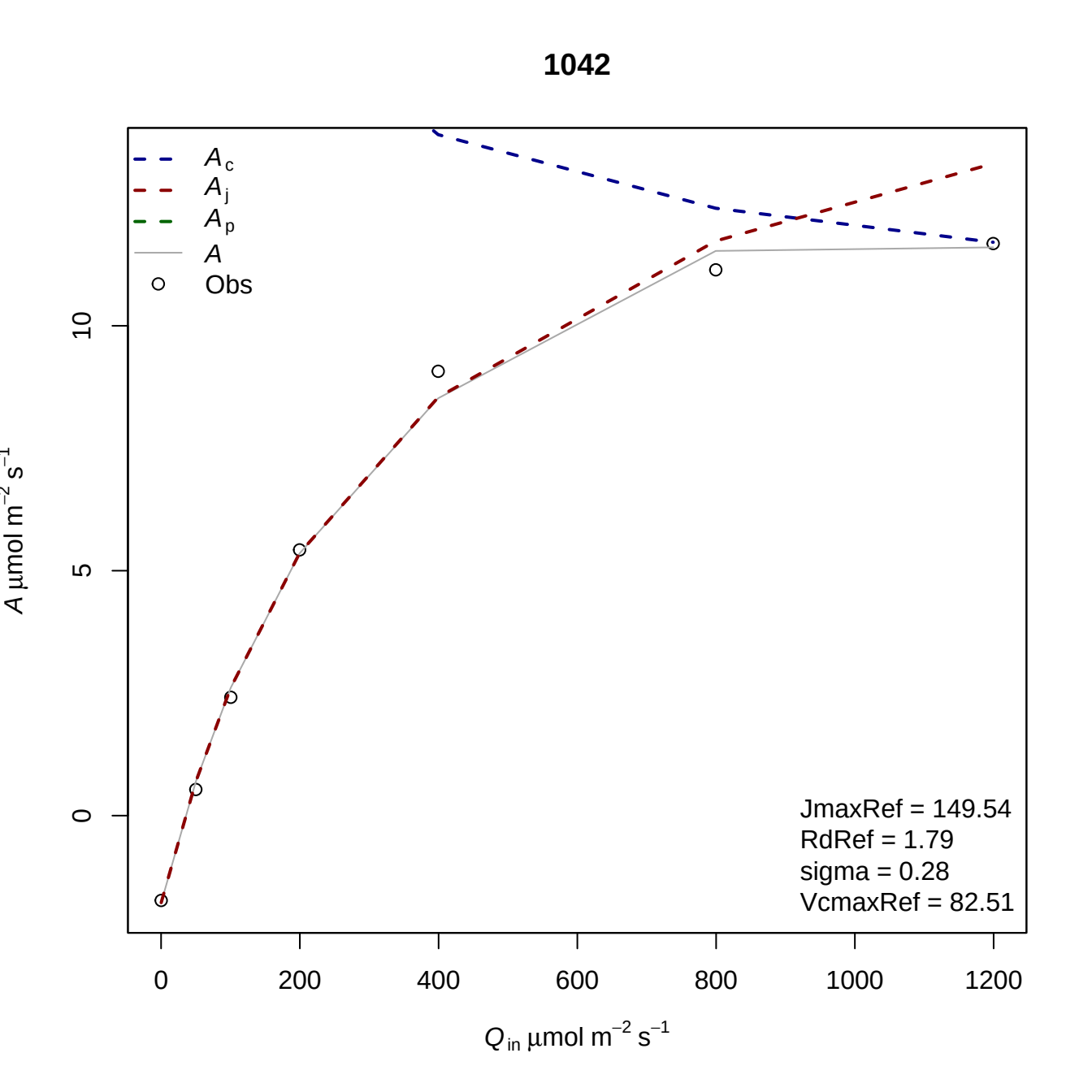

1037
A_c
A_j
A_p
A
Obs
JmaxRef = 125.72
RdRef = 0.68
sigma = 0.21
VcmaxRef = 100.72
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

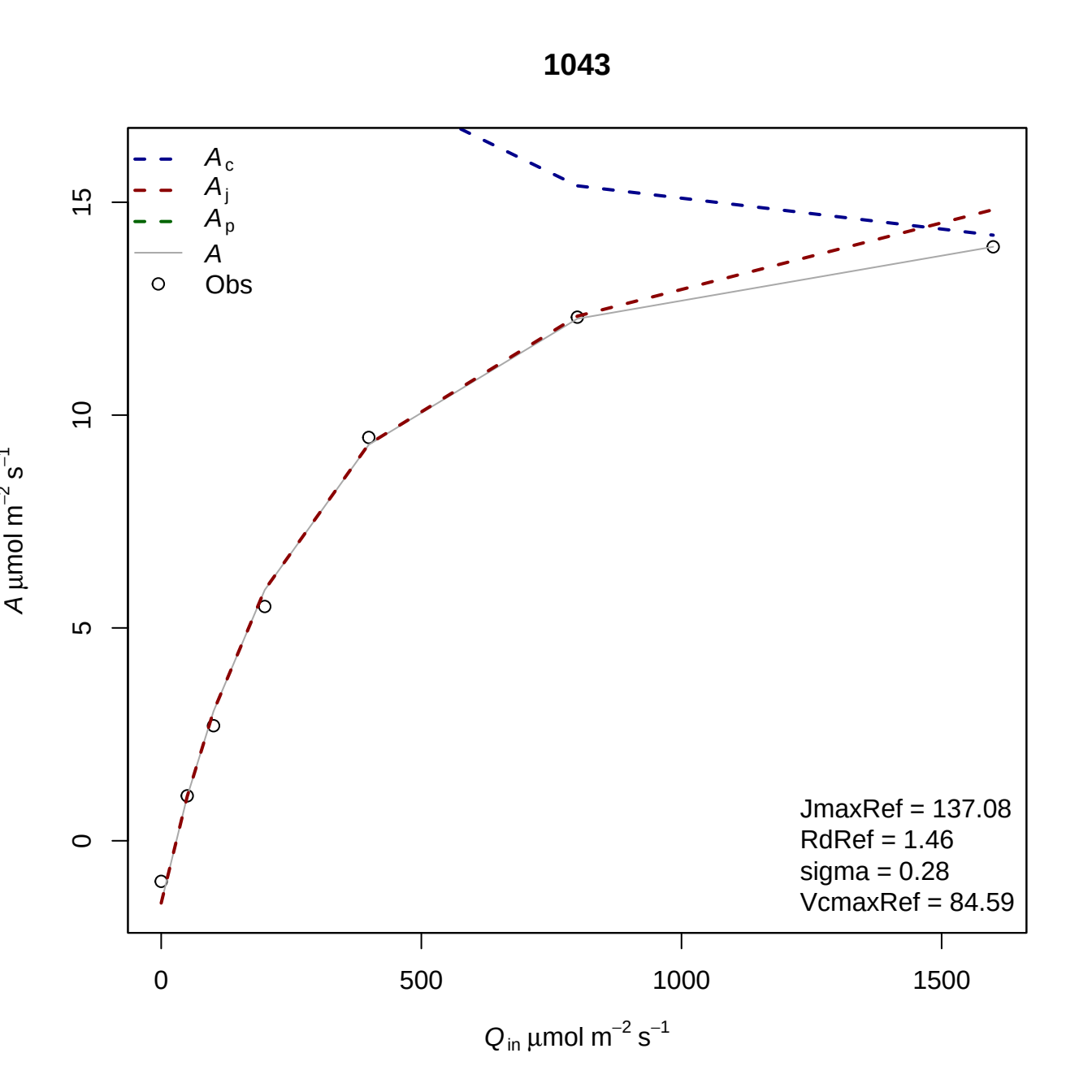

1038
A_c
A_j
A_p
A
Obs
JmaxRef = 124.65
RdRef = 0.92
sigma = 0.27
VcmaxRef = 75.13
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

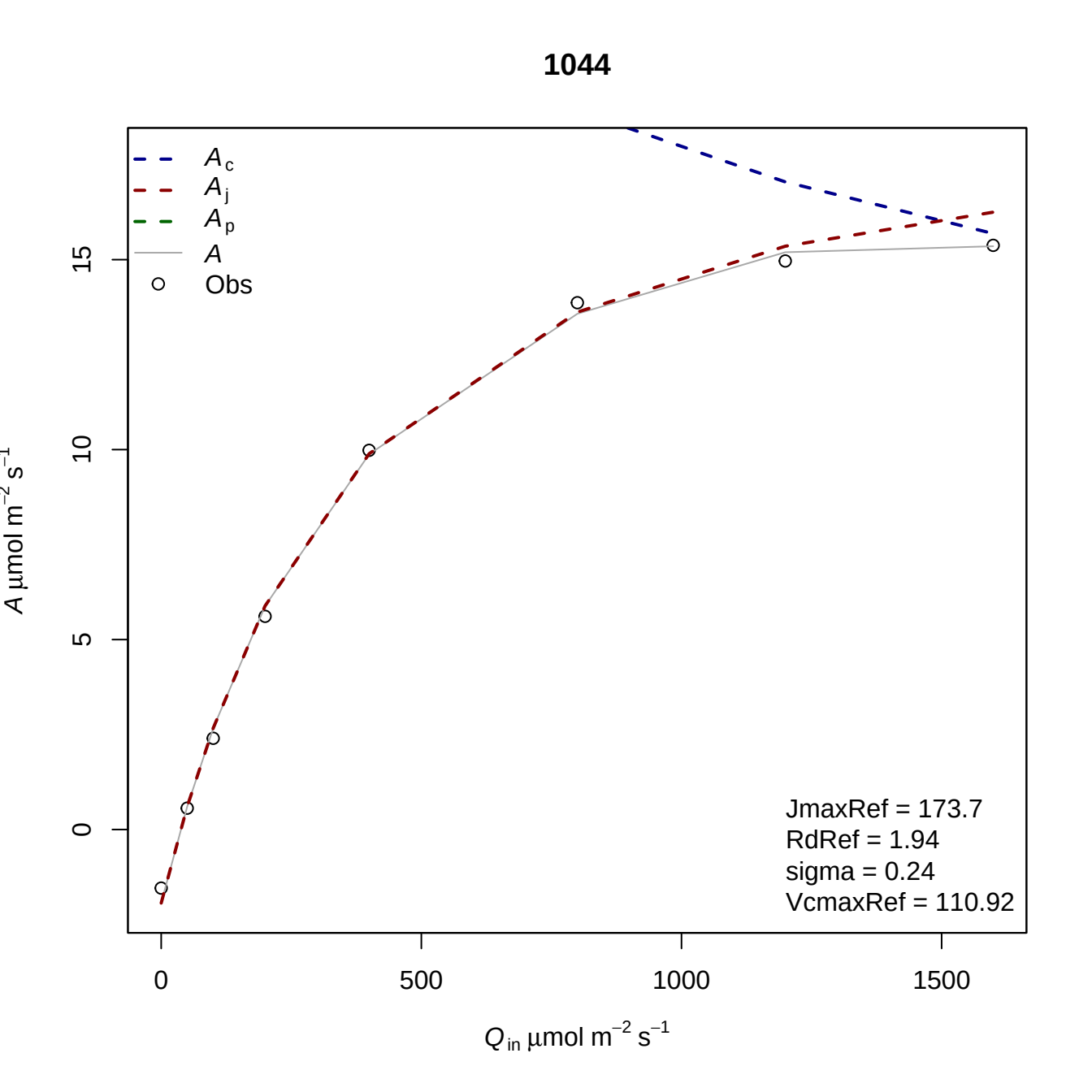

1039
A_c
A_j
A_p
A
Obs
JmaxRef = 113.22
RdRef = 0.71
sigma = 0.24
VcmaxRef = 63.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

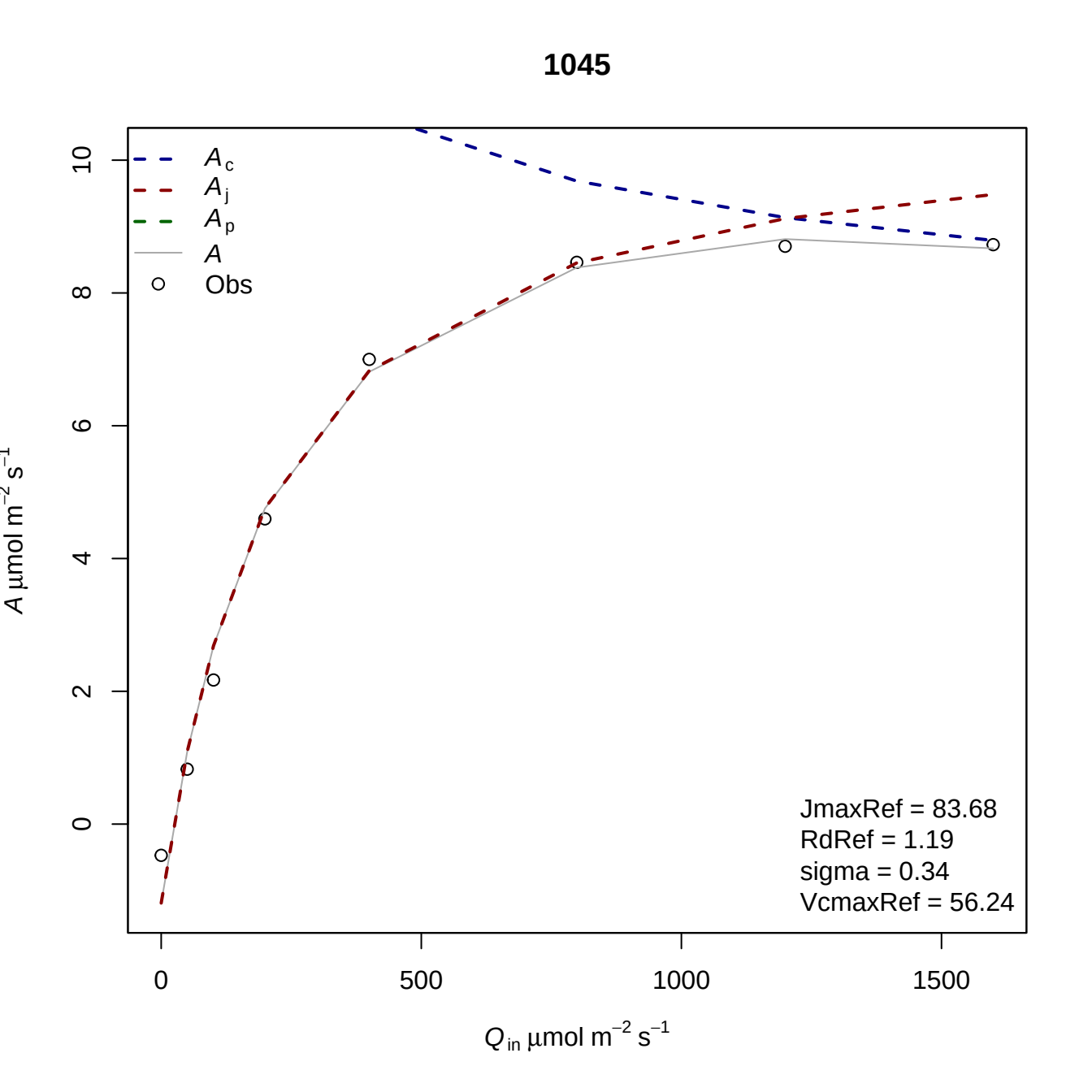

1040
A_c
A_j
A_p
A
Obs
JmaxRef = 117.68
RdRef = 0.86
sigma = 0.14
VcmaxRef = 60.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

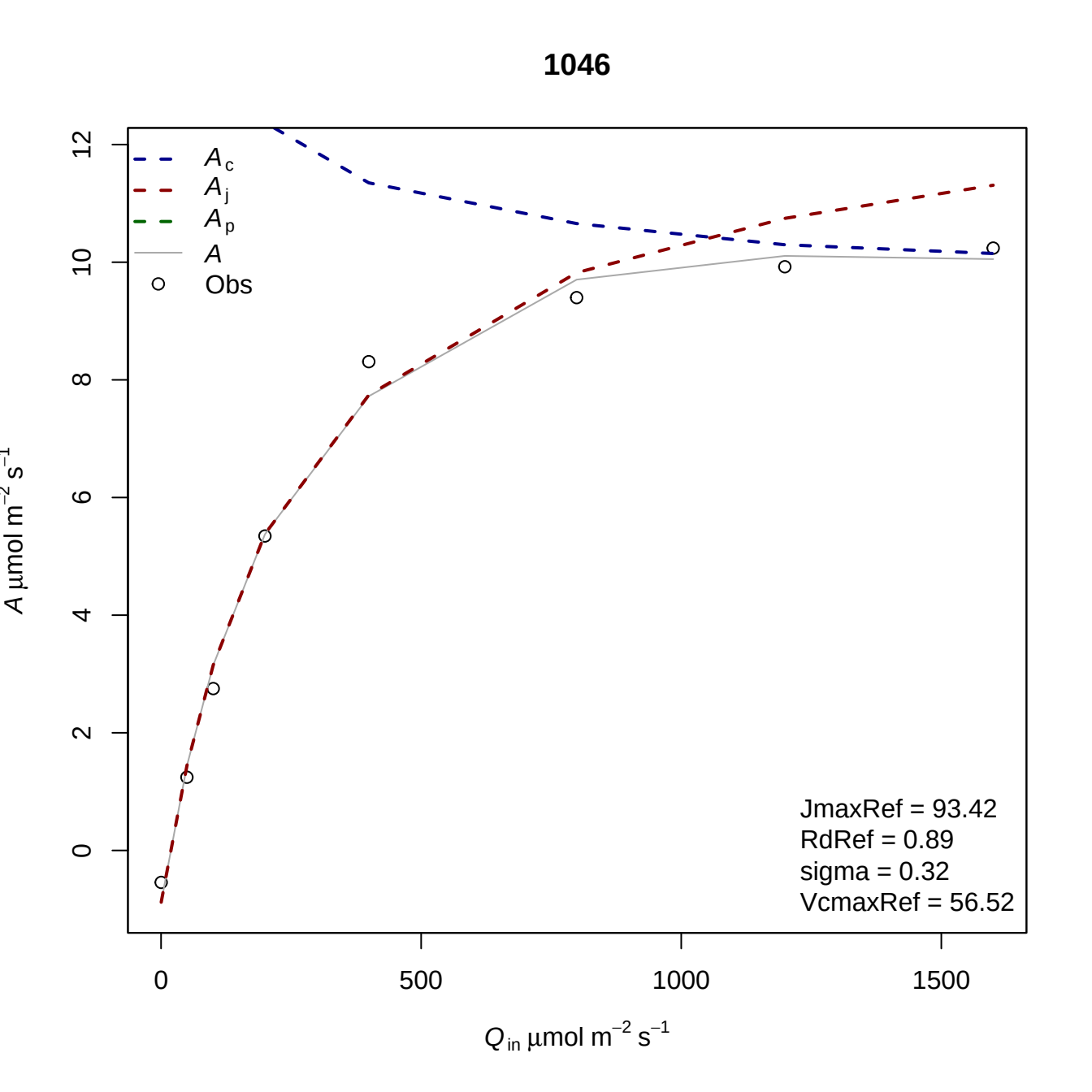

1041
A_c
A_j
A_p
A
Obs
JmaxRef = 149.6
RdRef = 0.73
sigma = 0.39
VcmaxRef = 88.03
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

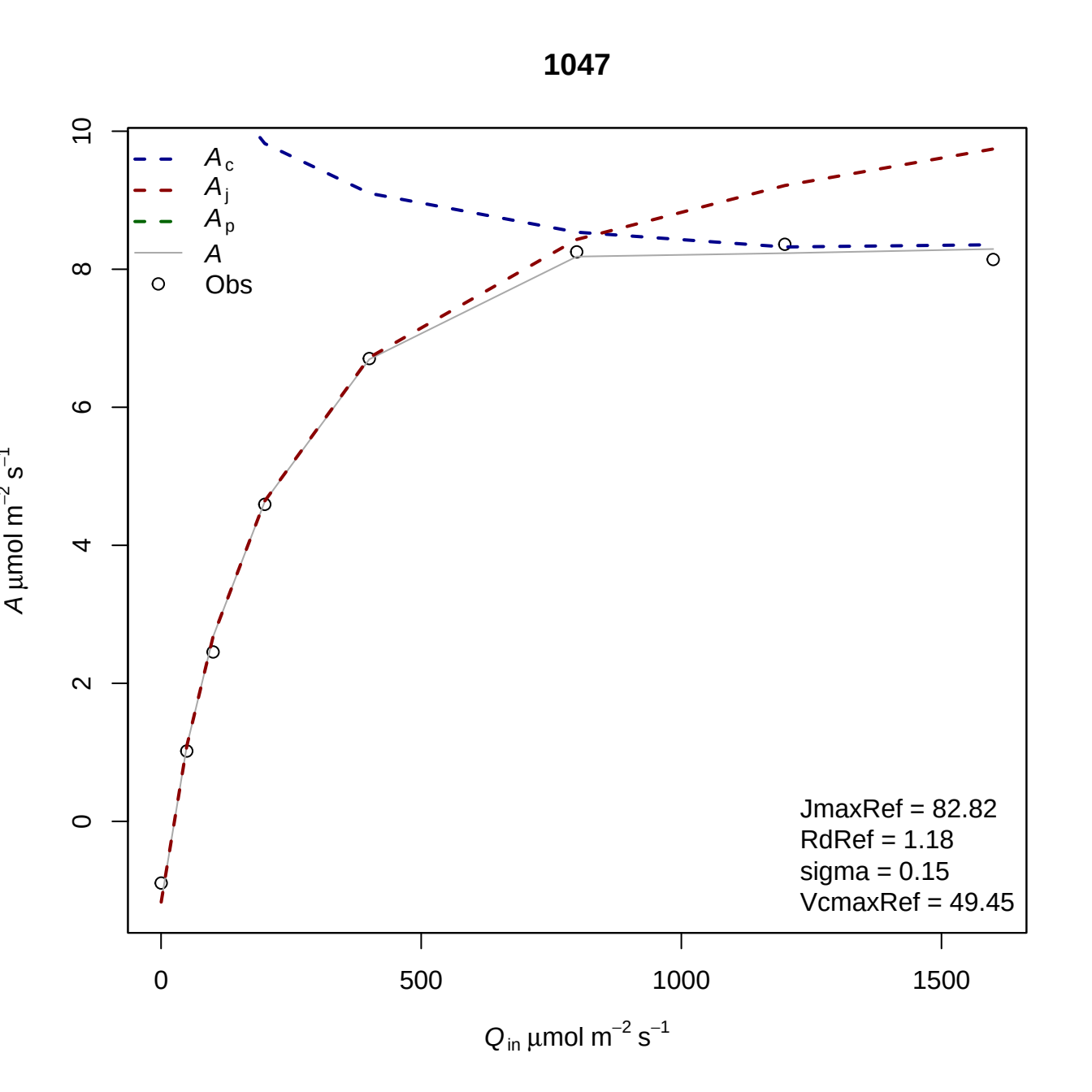

1042
A_c
A_j
A_p
A
Obs
JmaxRef = 149.54
RdRef = 1.79
sigma = 0.28
VcmaxRef = 82.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000
1200

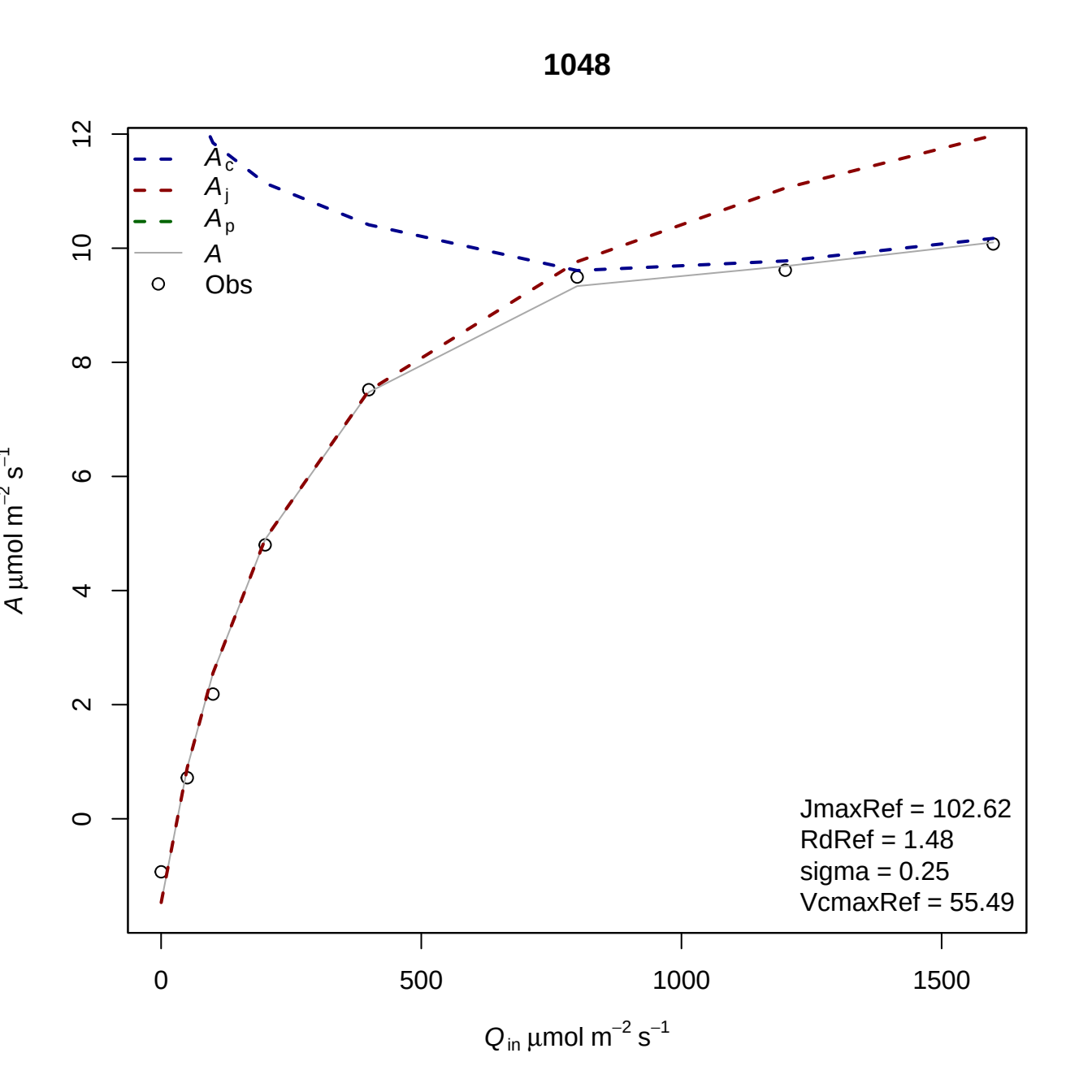

1043
A_c
A_j
A_p
A
Obs
JmaxRef = 137.08
RdRef = 1.46
sigma = 0.28
VcmaxRef = 84.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

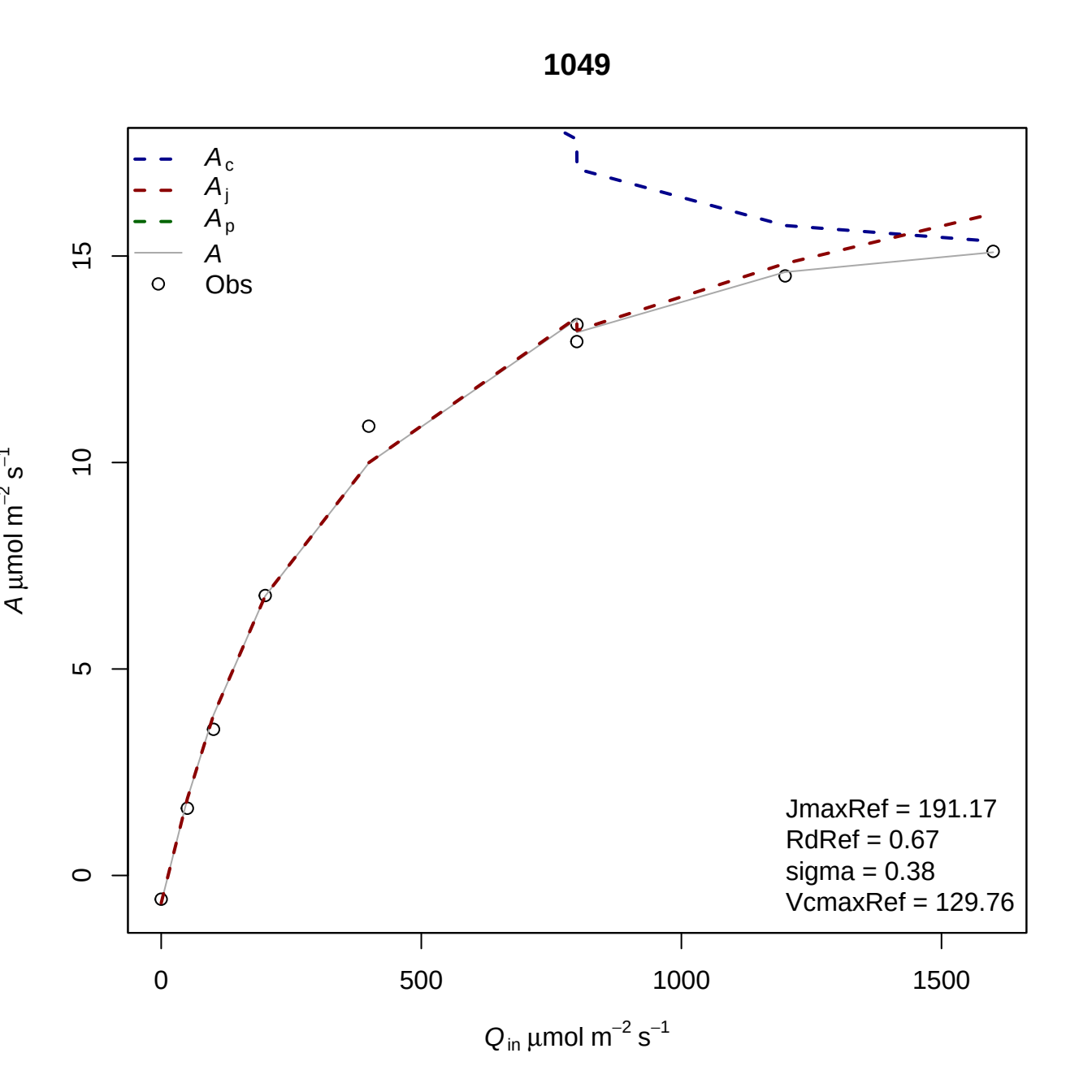

1044
A_c
A_j
A_p
A
Obs
JmaxRef = 173.7
RdRef = 1.94
sigma = 0.24
VcmaxRef = 110.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

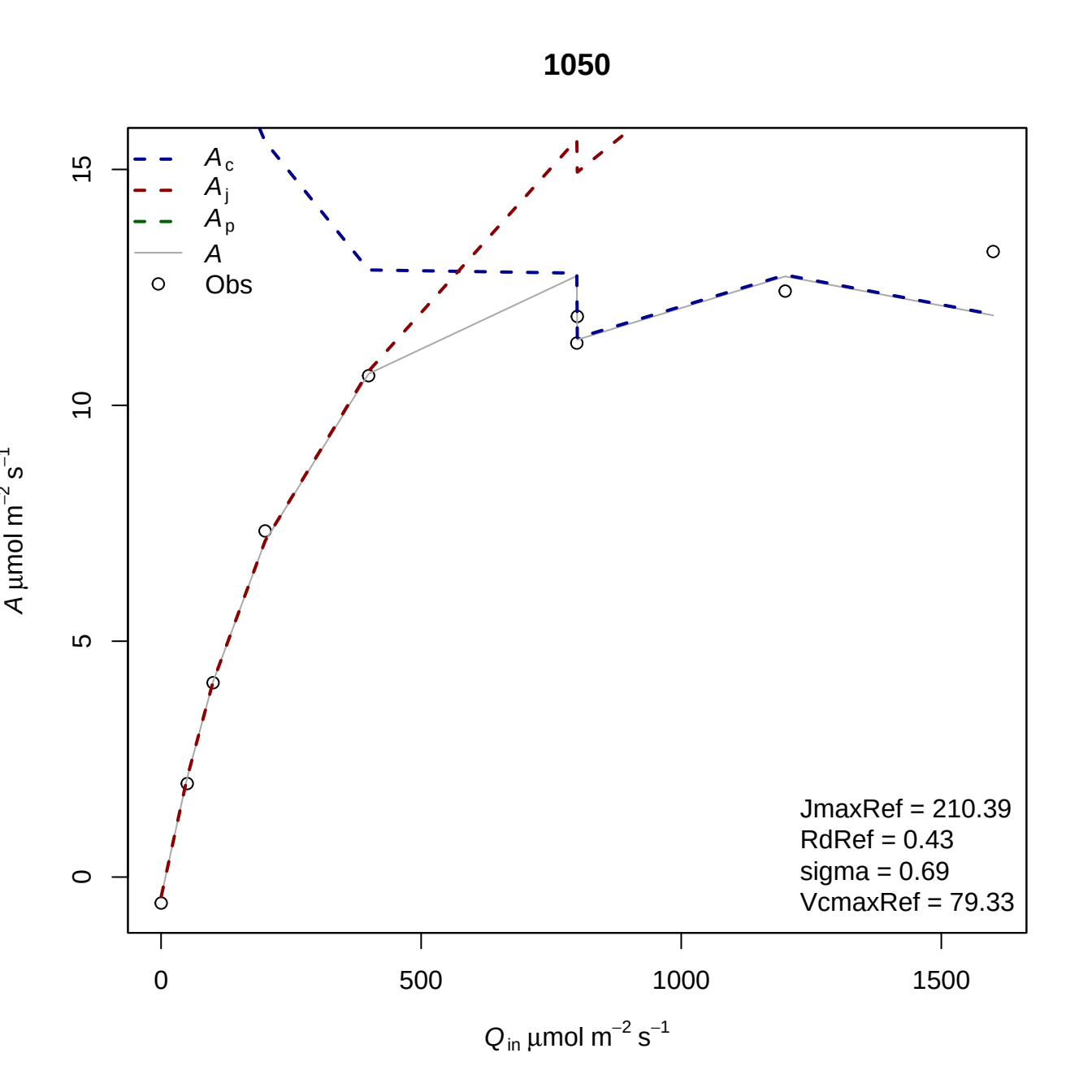

1045
A_c
A_j
A_p
A
Obs
JmaxRef = 83.68
RdRef = 1.19
sigma = 0.34
VcmaxRef = 56.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

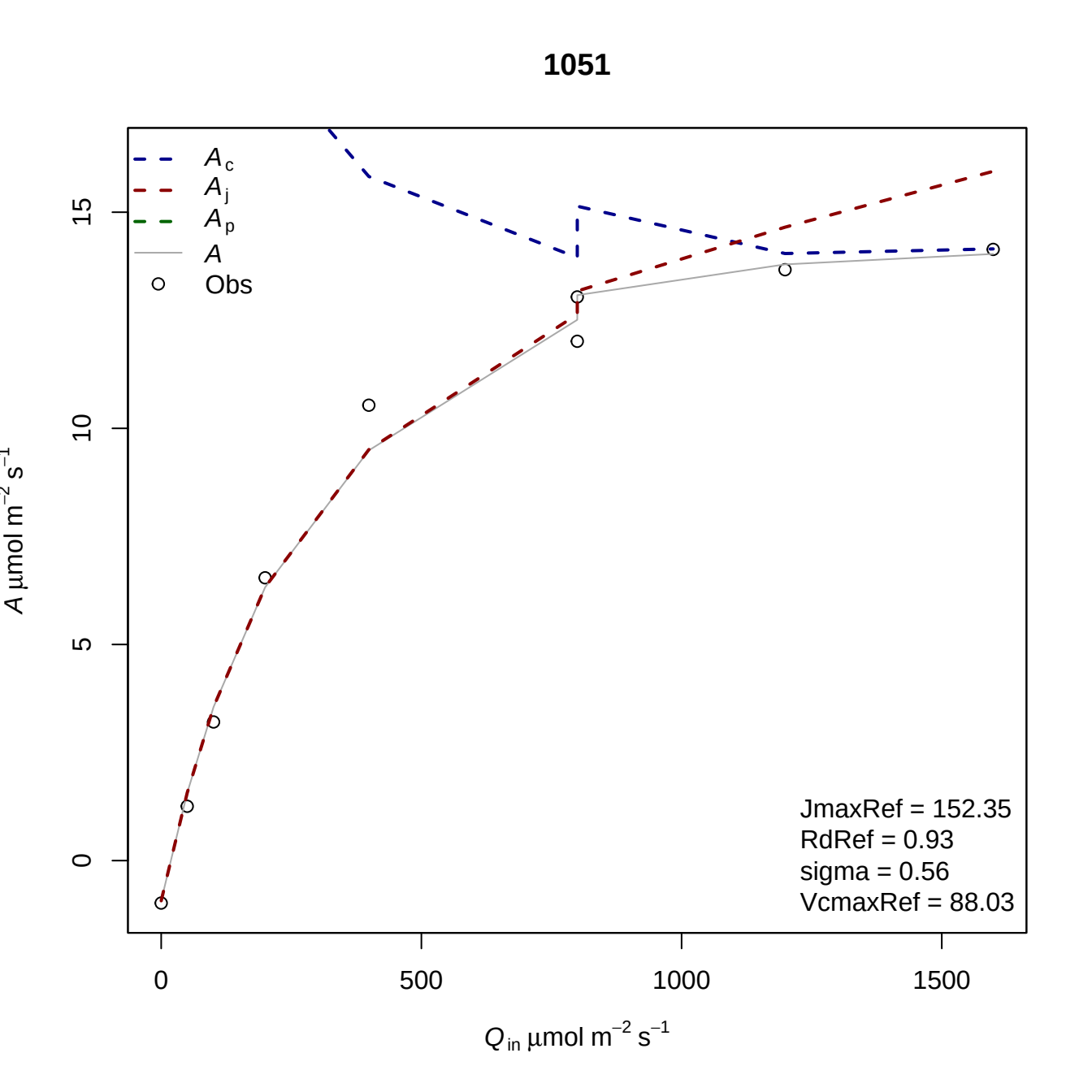

1046
A_c
A_j
A_p
A
Obs
JmaxRef = 93.42
RdRef = 0.89
sigma = 0.32
VcmaxRef = 56.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

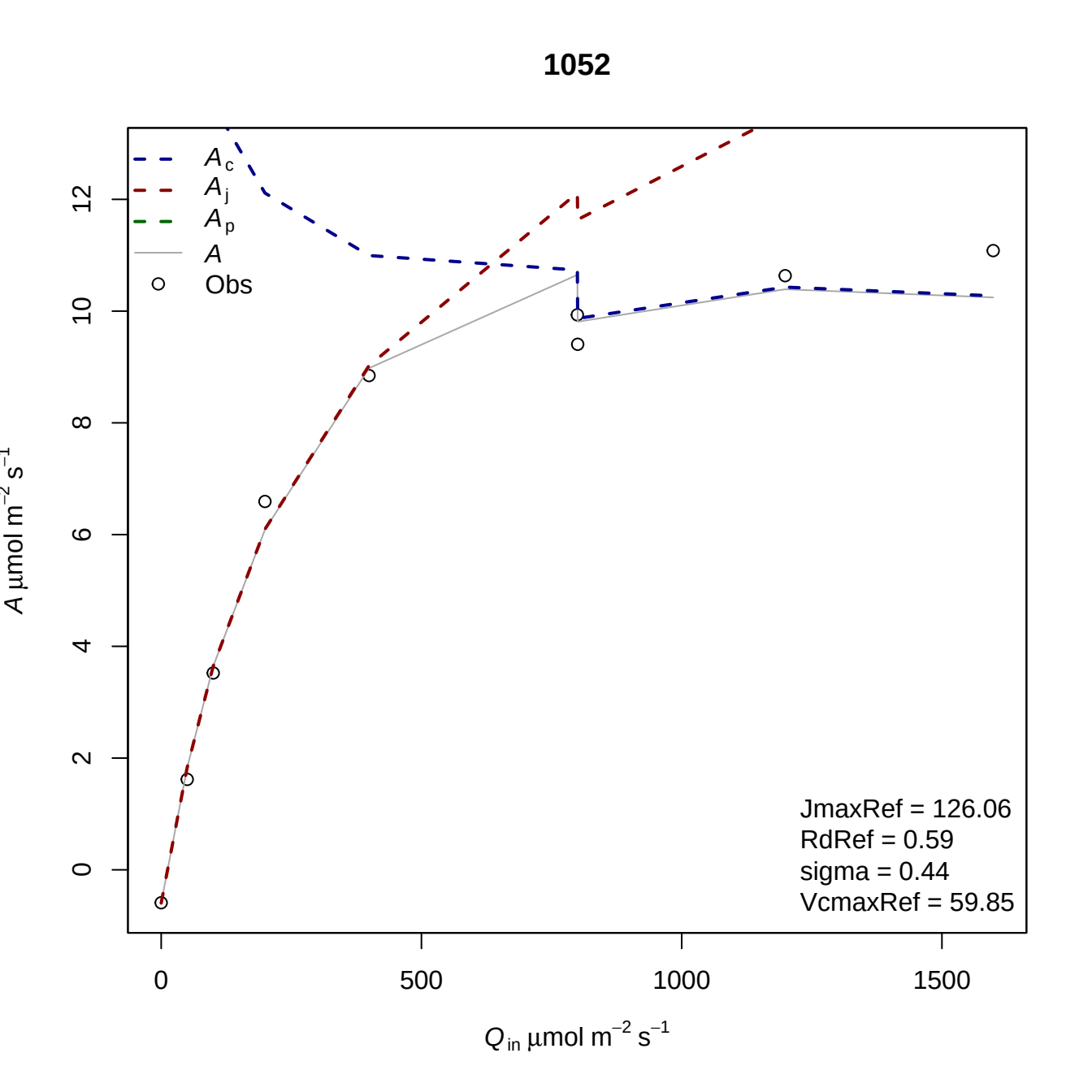

1047
A_c
A_j
A_p
A
Obs
JmaxRef = 82.82
RdRef = 1.18
sigma = 0.15
VcmaxRef = 49.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

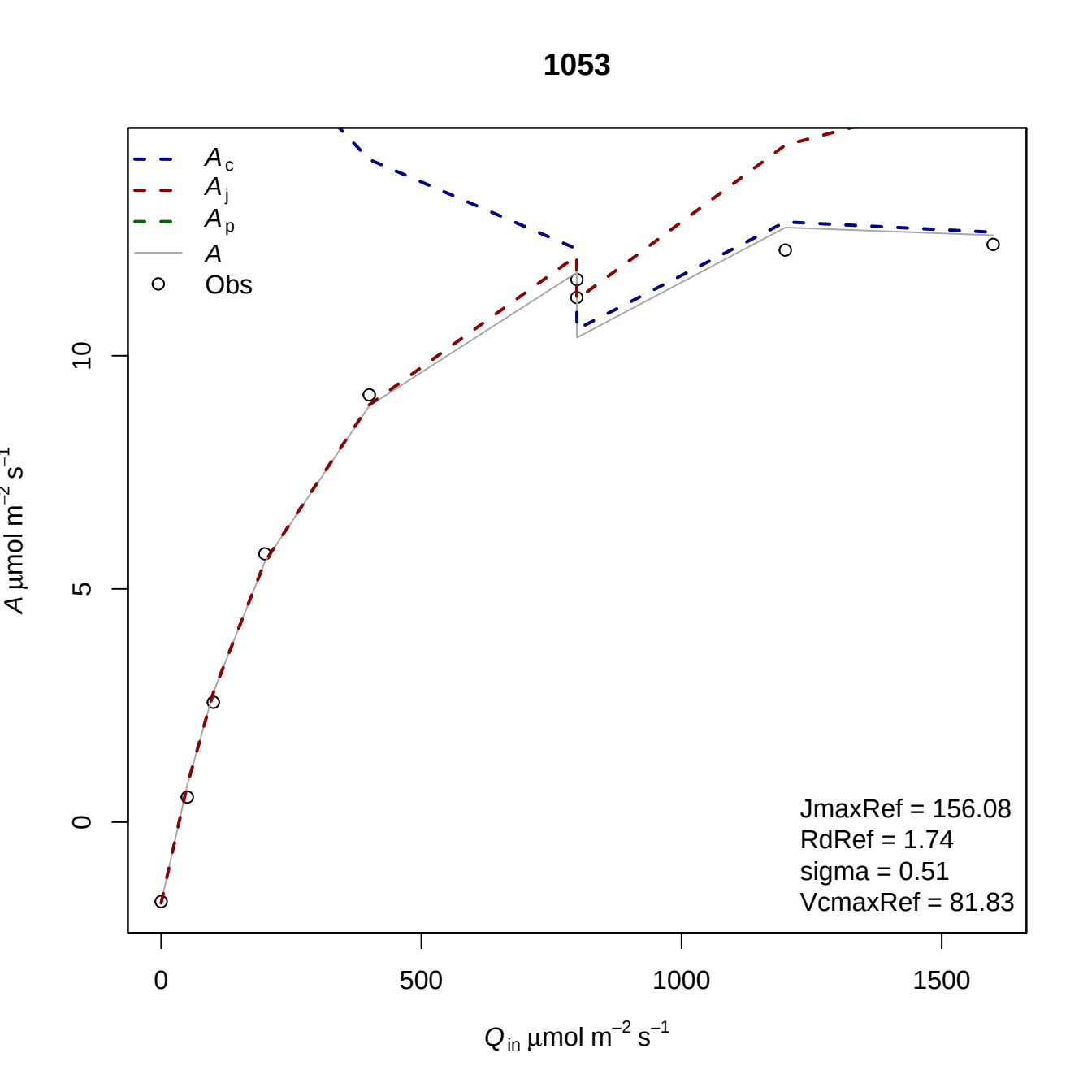

1048
A_c
A_j
A_p
A
Obs
JmaxRef = 102.62
RdRef = 1.48
sigma = 0.25
VcmaxRef = 55.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

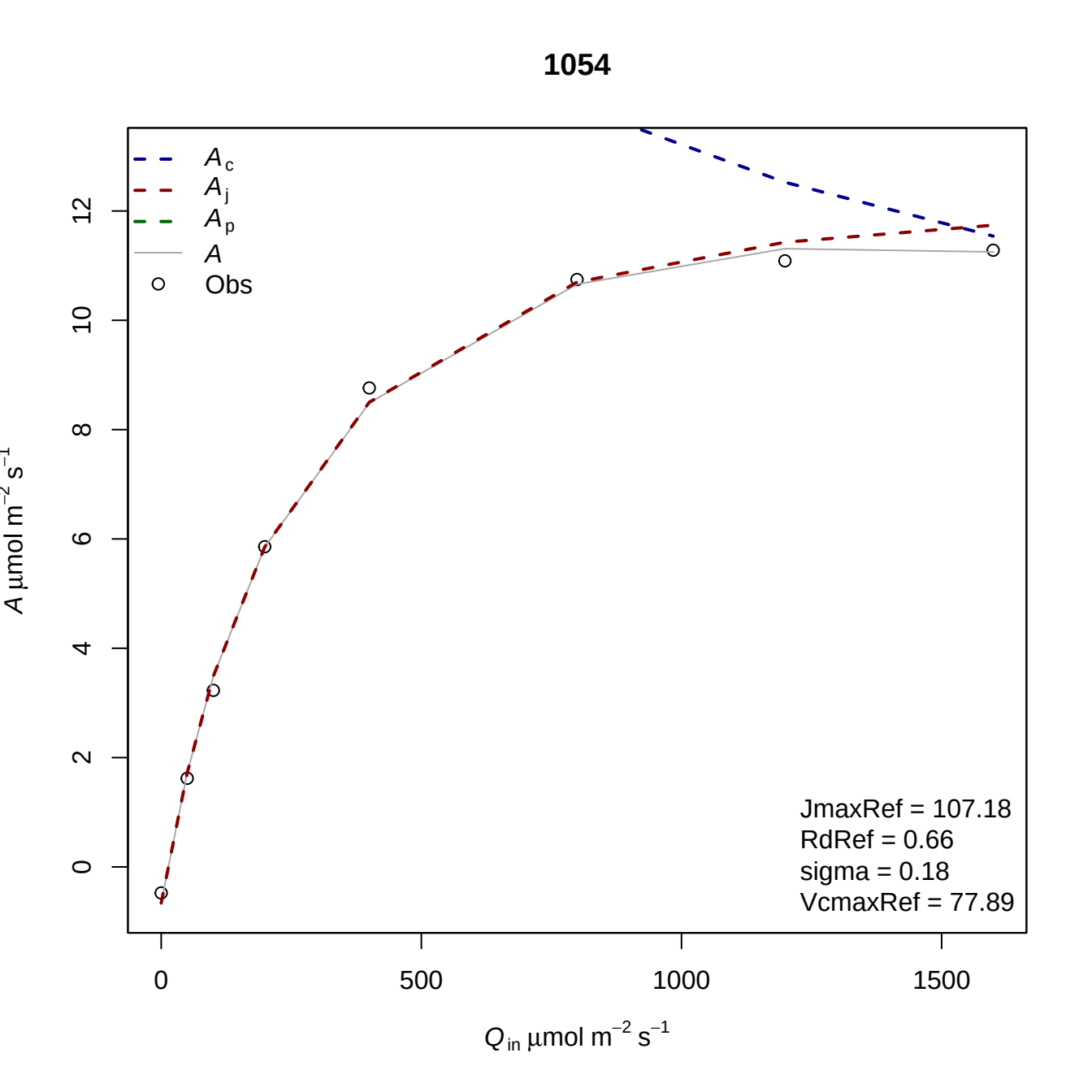

1049
A_c
A_j
A_p
A
Obs
JmaxRef = 191.17
RdRef = 0.67
sigma = 0.38
VcmaxRef = 129.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

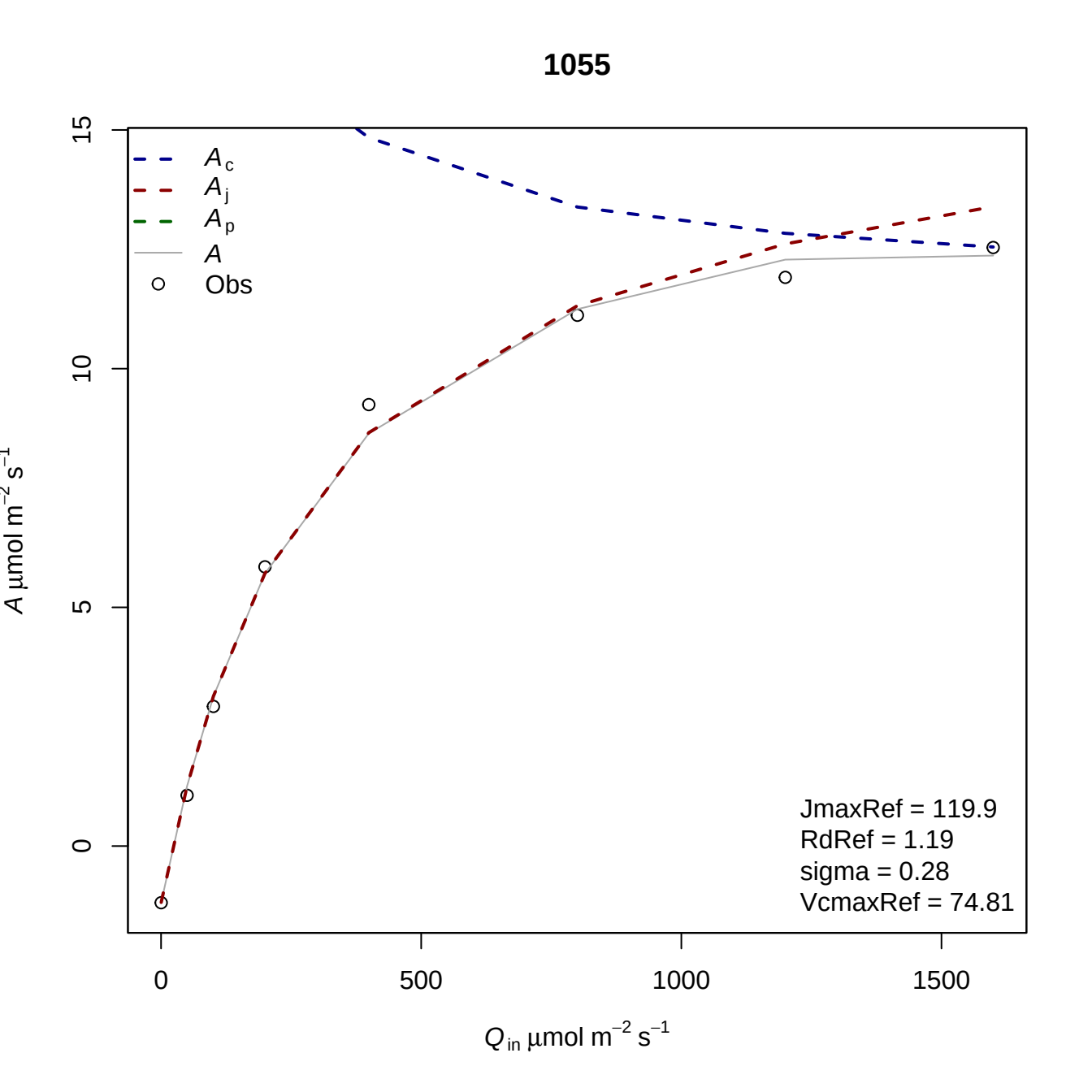

1050
A_c
A_j
A_p
A
Obs
JmaxRef = 210.39
RdRef = 0.43
sigma = 0.69
VcmaxRef = 79.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

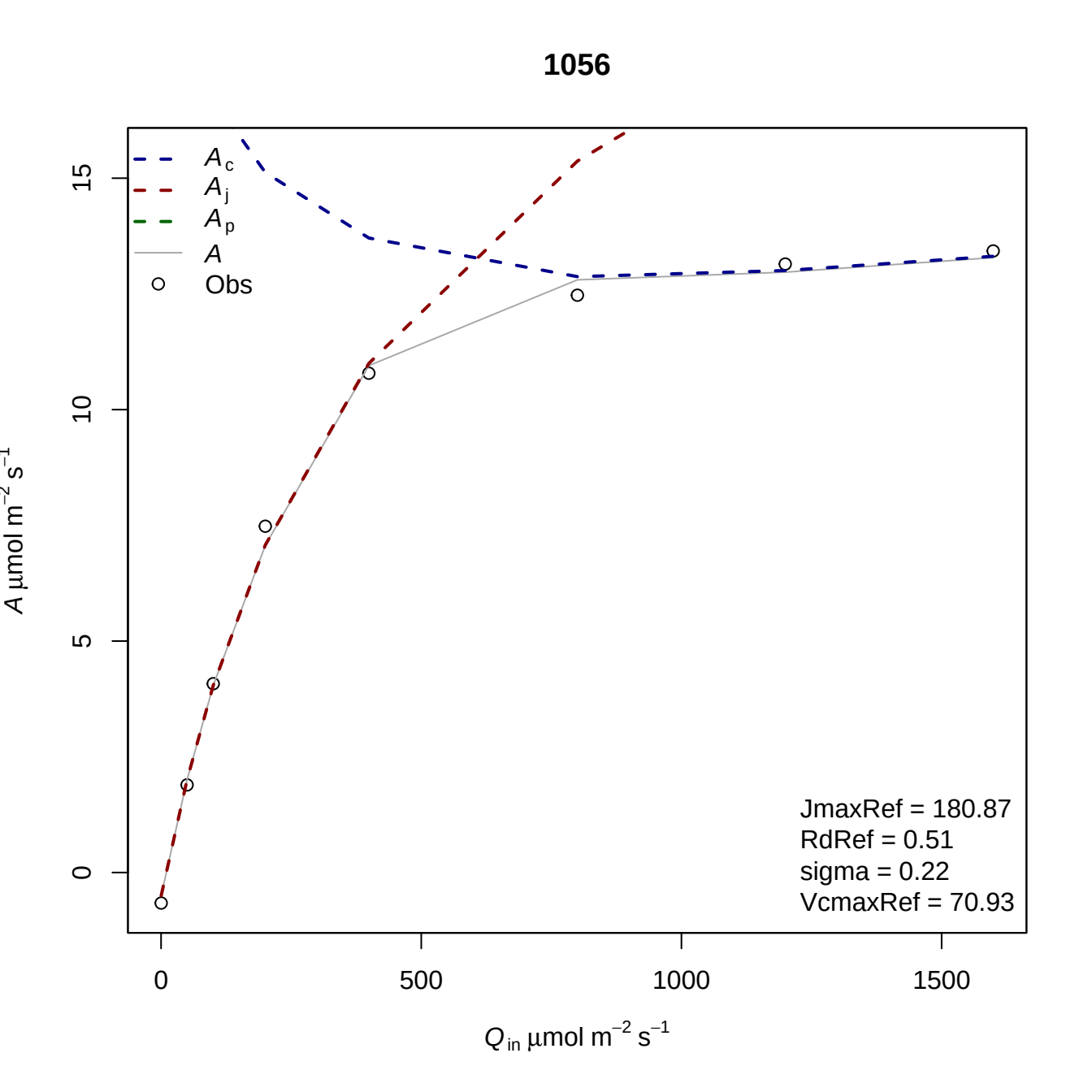

1051
A_c
A_j
A_p
A
Obs
JmaxRef = 152.35
RdRef = 0.93
sigma = 0.56
VcmaxRef = 88.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

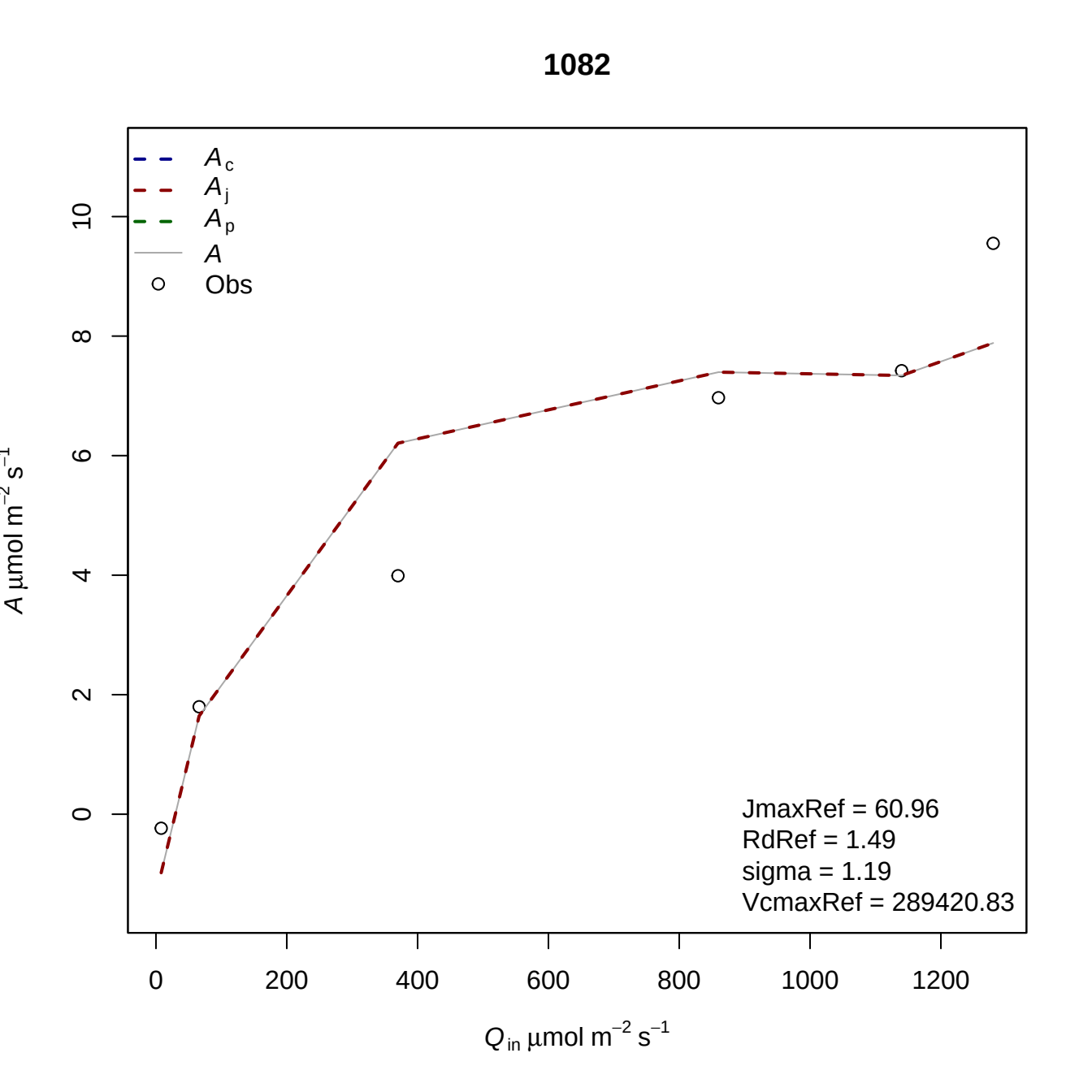

1052
A_c
A_j
A_p
A
Obs
JmaxRef = 126.06
RdRef = 0.59
sigma = 0.44
VcmaxRef = 59.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

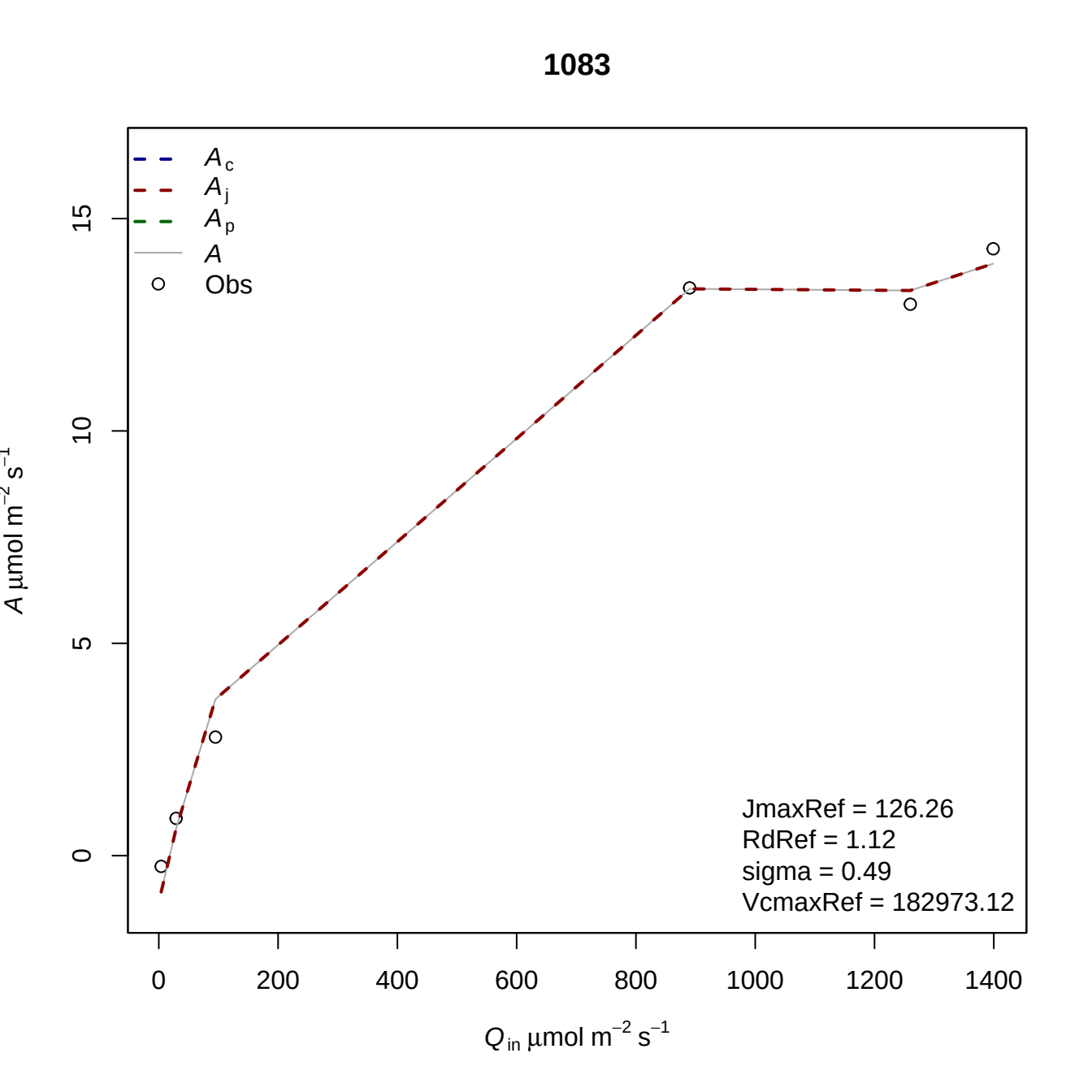

1053
A_c
A_j
A_p
A
Obs
JmaxRef = 156.08
RdRef = 1.74
sigma = 0.51
VcmaxRef = 81.83
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

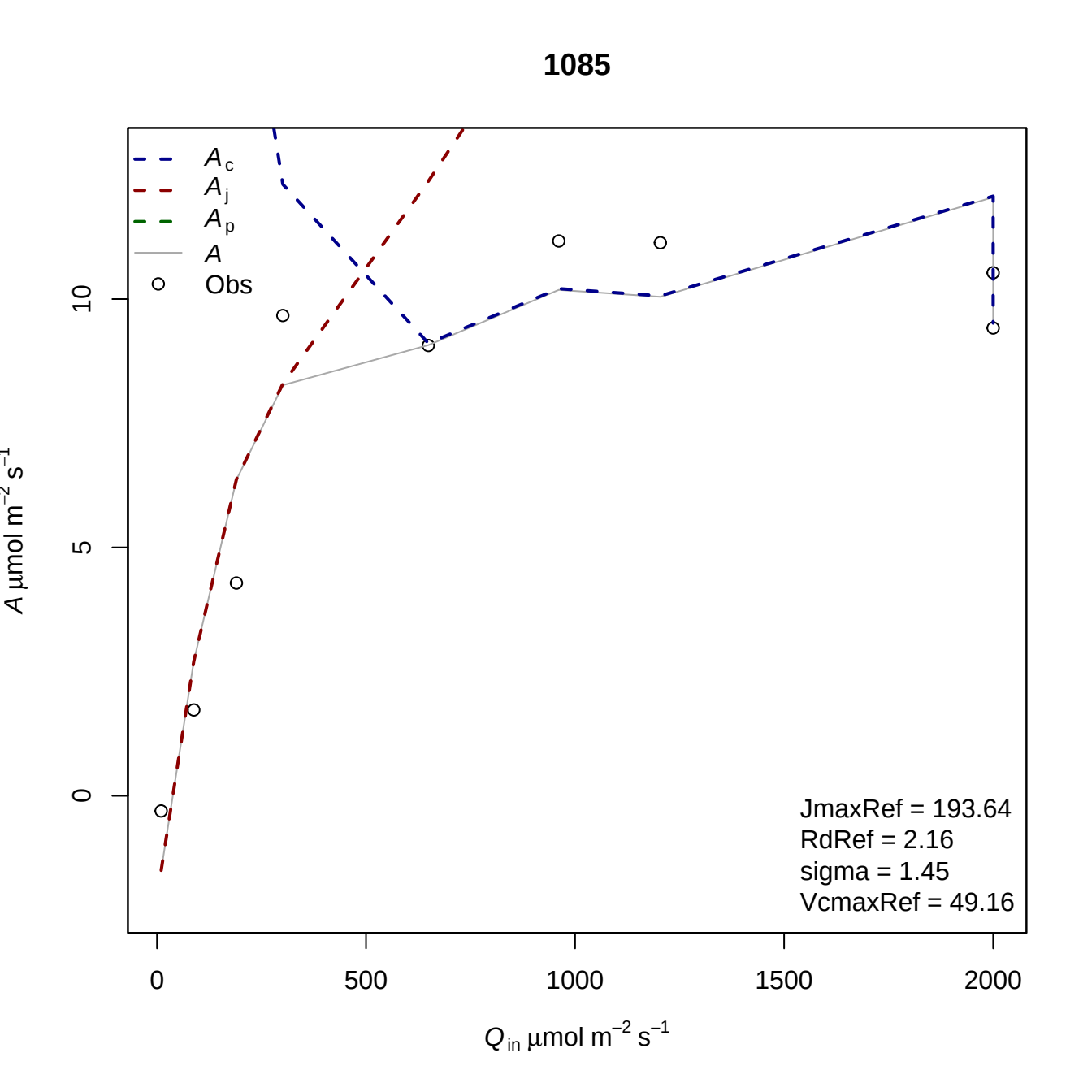

1054
A_c
A_j
A_p
A
Obs
JmaxRef = 107.18
RdRef = 0.66
sigma = 0.18
VcmaxRef = 77.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

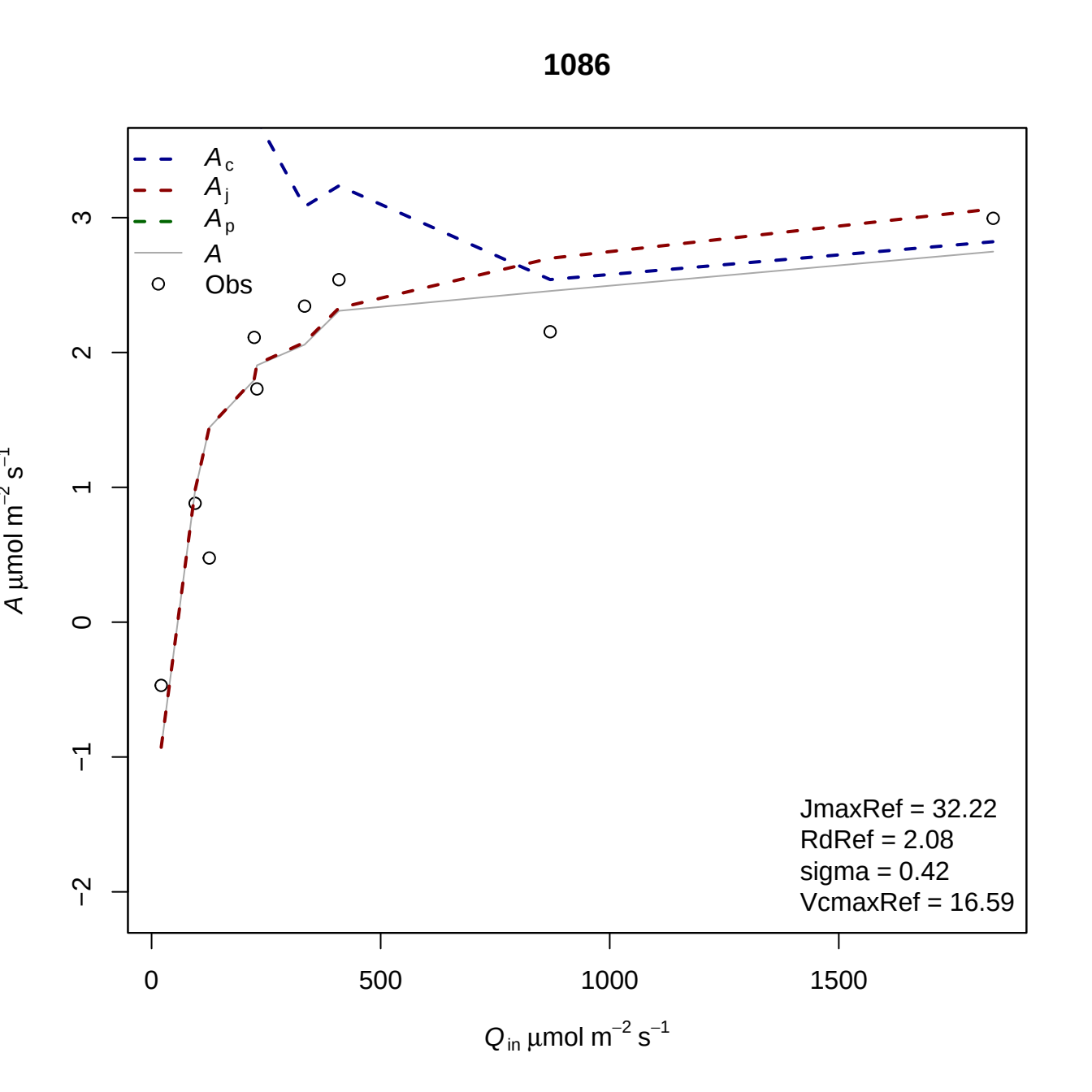

1055
A_c
A_j
A_p
A
Obs
JmaxRef = 119.9
RdRef = 1.19
sigma = 0.28
VcmaxRef = 74.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

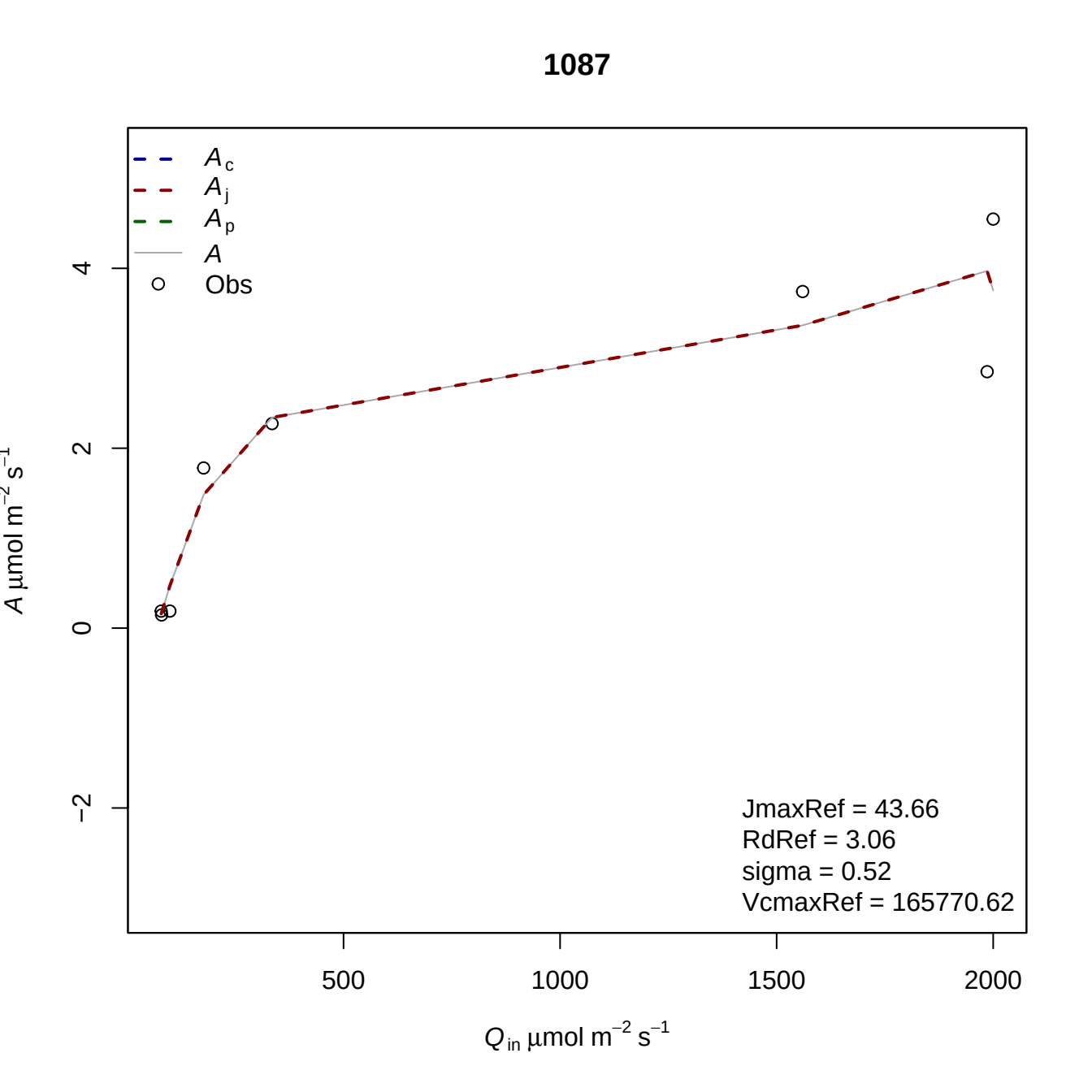

1056
A_c
A_j
A_p
A
Obs
JmaxRef = 180.87
RdRef = 0.51
sigma = 0.22
VcmaxRef = 70.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

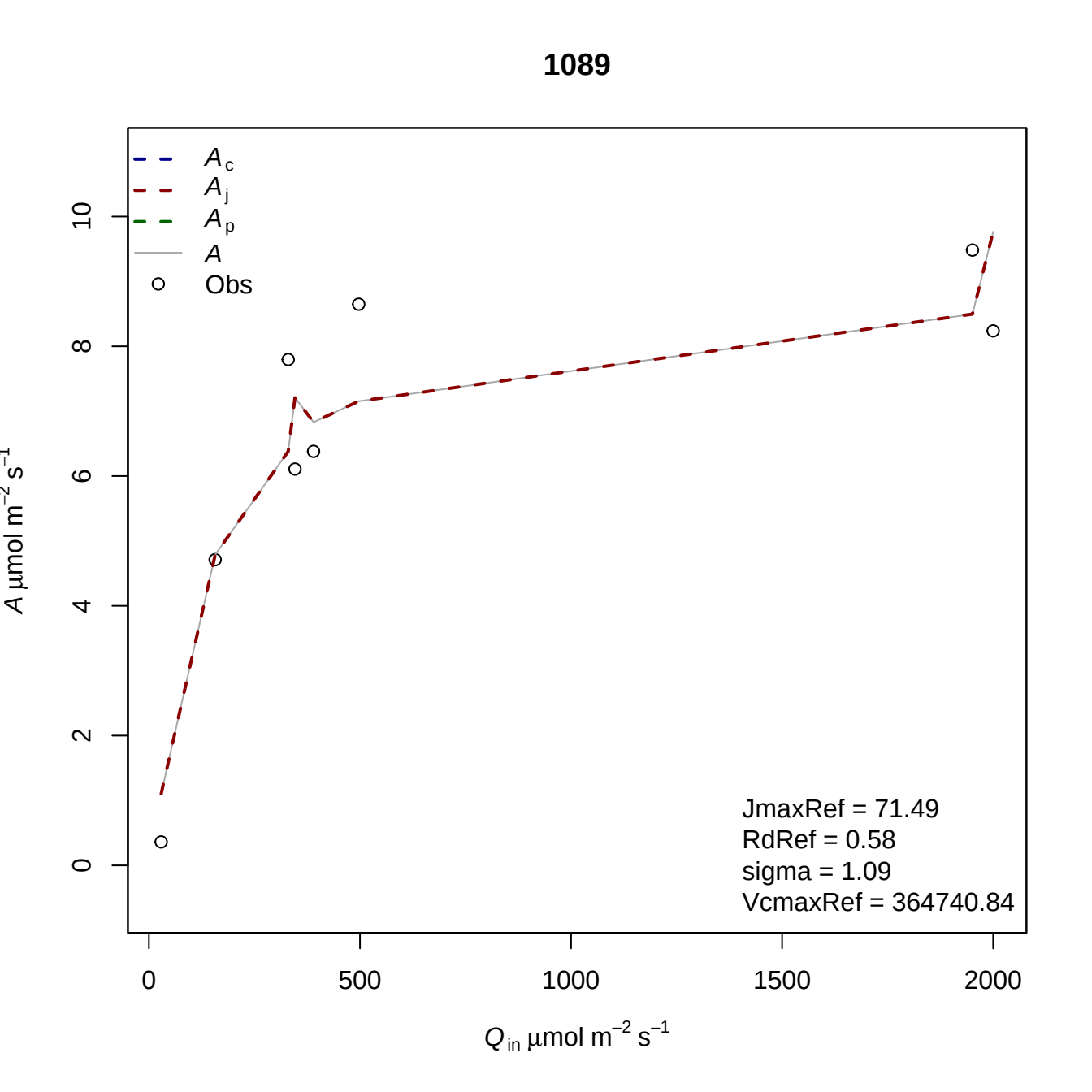

1082
A_c
A_j
A_p
A
Obs
JmaxRef = 60.96
RdRef = 1.49
sigma = 1.19
VcmaxRef = 289420.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
200
400
600
800
1000
1200

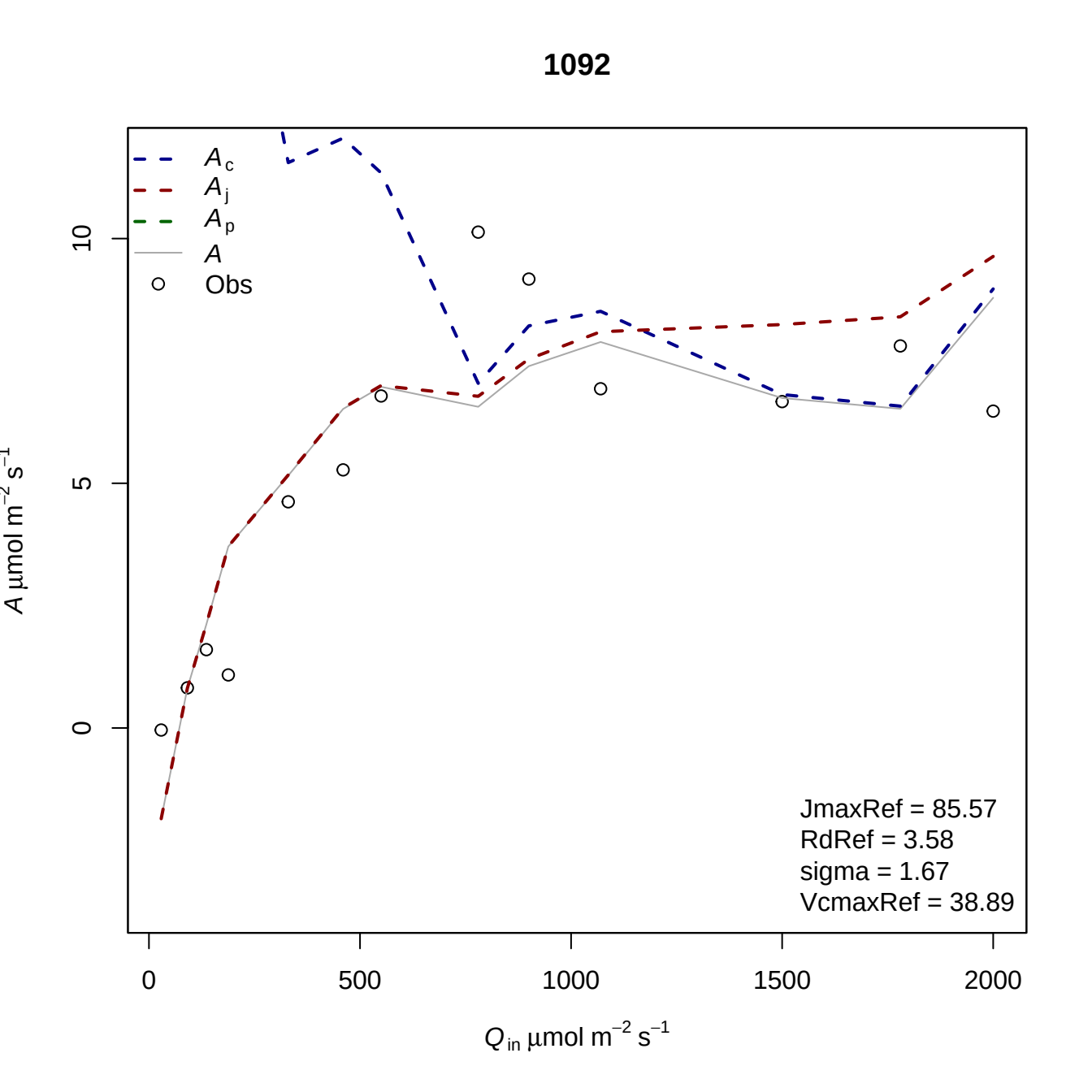

1083
A_c
A_j
A_p
A
Obs
JmaxRef = 126.26
RdRef = 1.12
sigma = 0.49
VcmaxRef = 182973.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200
1400

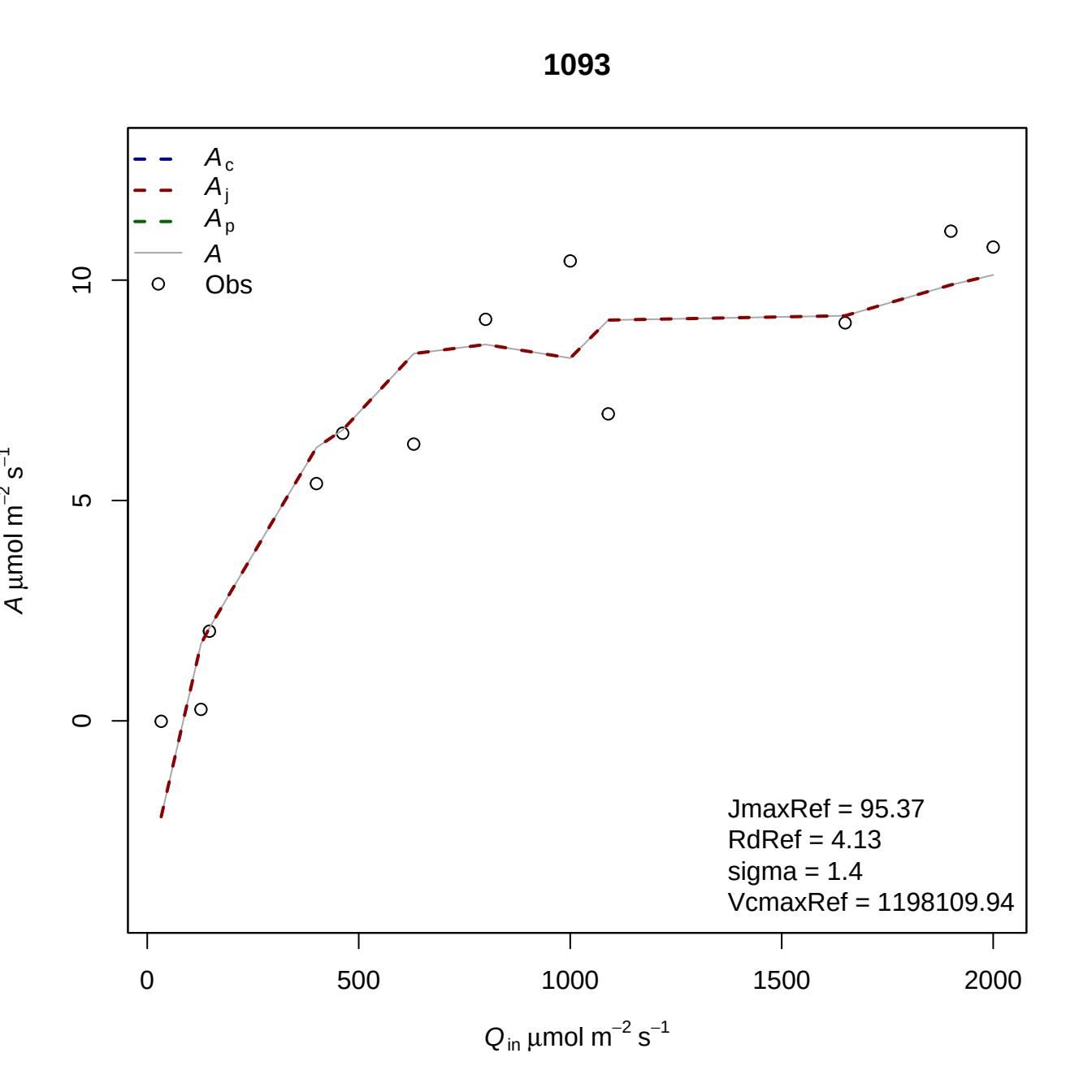

1085
A_c
A_j
A_p
A
Obs
JmaxRef = 193.64
RdRef = 2.16
sigma = 1.45
VcmaxRef = 49.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

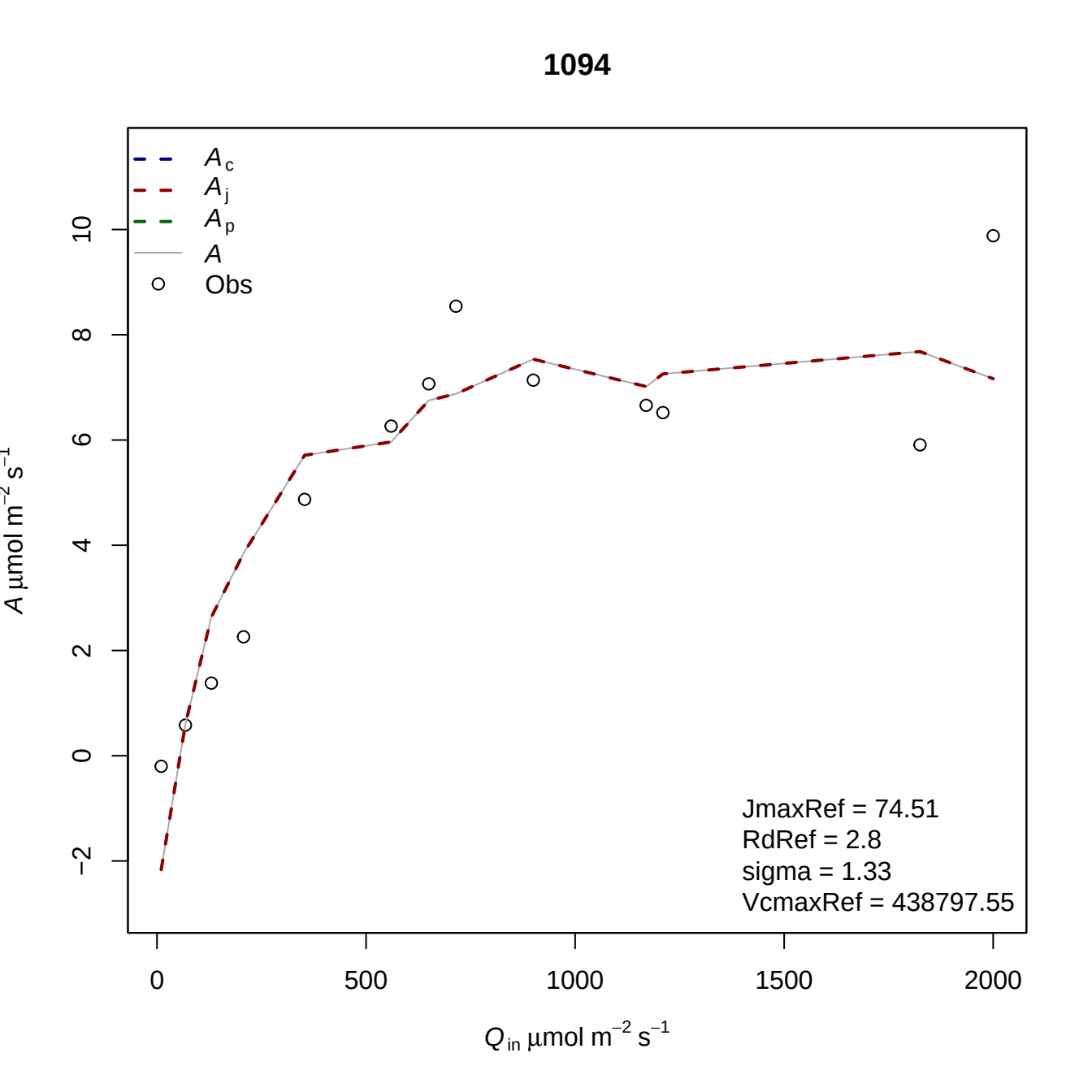

1086
A_c
A_j
A_p
A
Obs
JmaxRef = 32.22
RdRef = 2.08
sigma = 0.42
VcmaxRef = 16.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
−2
0
500
1000
1500

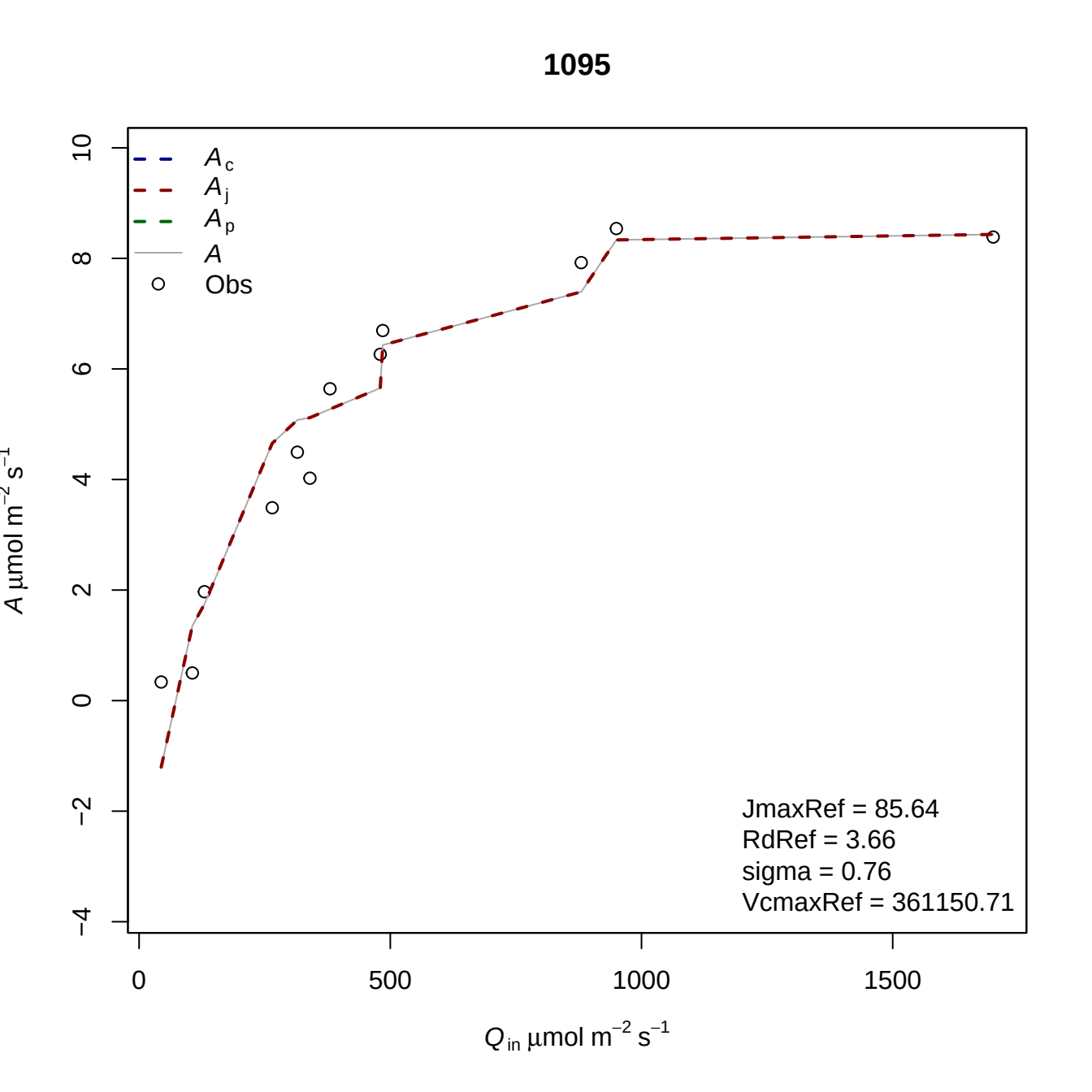

1087
A_c
A_j
A_p
A
Obs
JmaxRef = 43.66
RdRef = 3.06
sigma = 0.52
VcmaxRef = 165770.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
500
1000
1500
2000

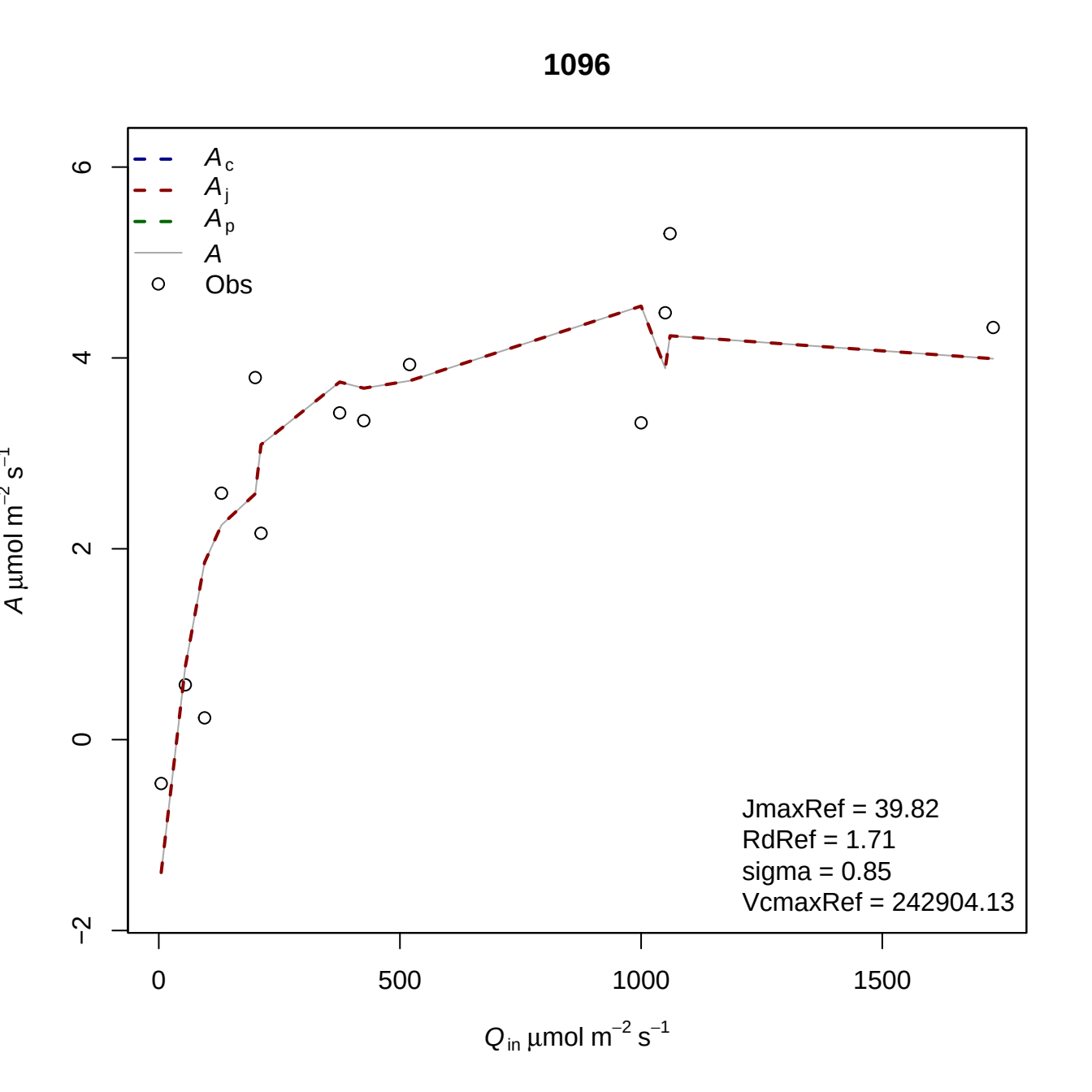

1089
A_c
A_j
A_p
A
Obs
JmaxRef = 71.49
RdRef = 0.58
sigma = 1.09
VcmaxRef = 364740.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

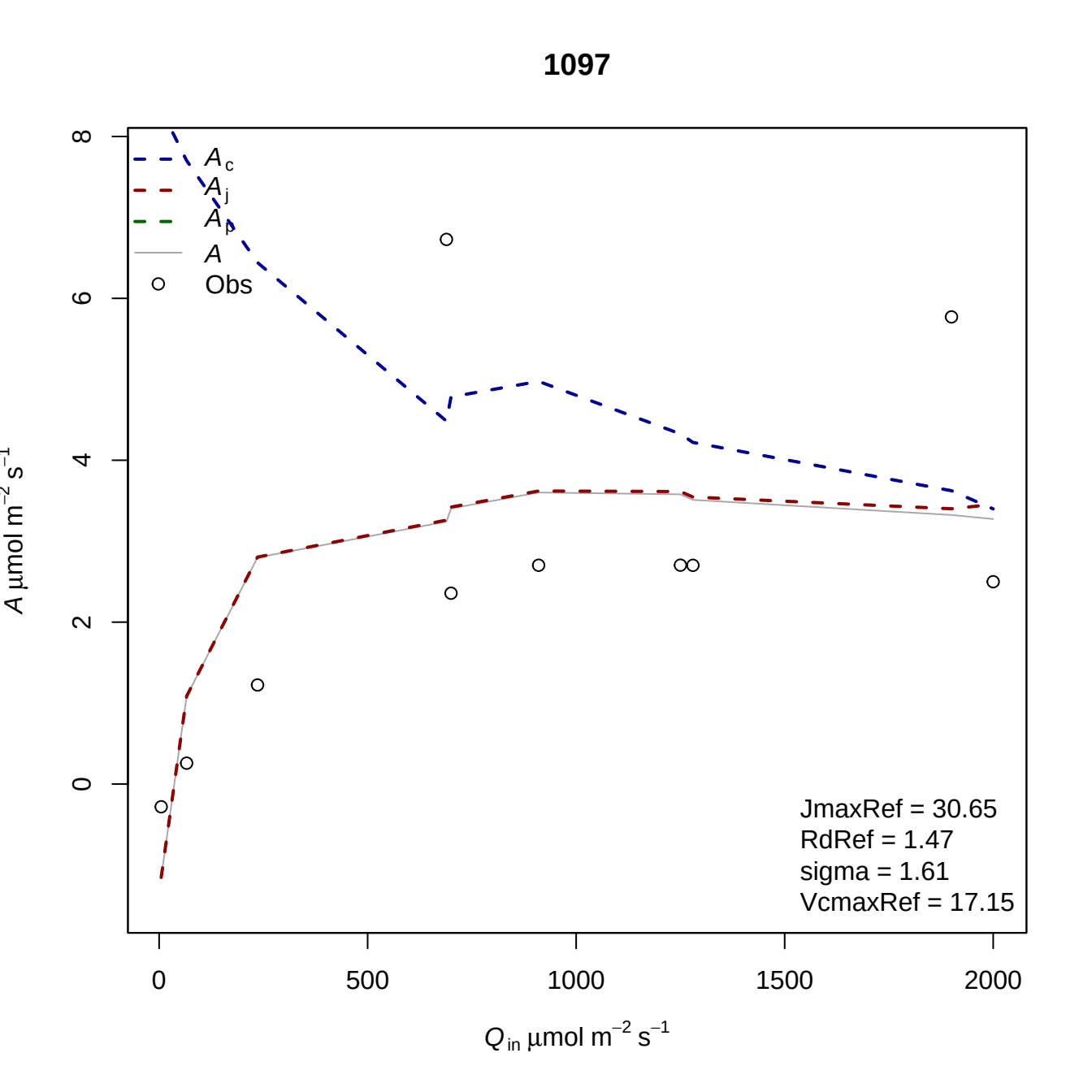

1092
A_c
A_j
A_p
A
Obs
JmaxRef = 85.57
RdRef = 3.58
sigma = 1.67
VcmaxRef = 38.89
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

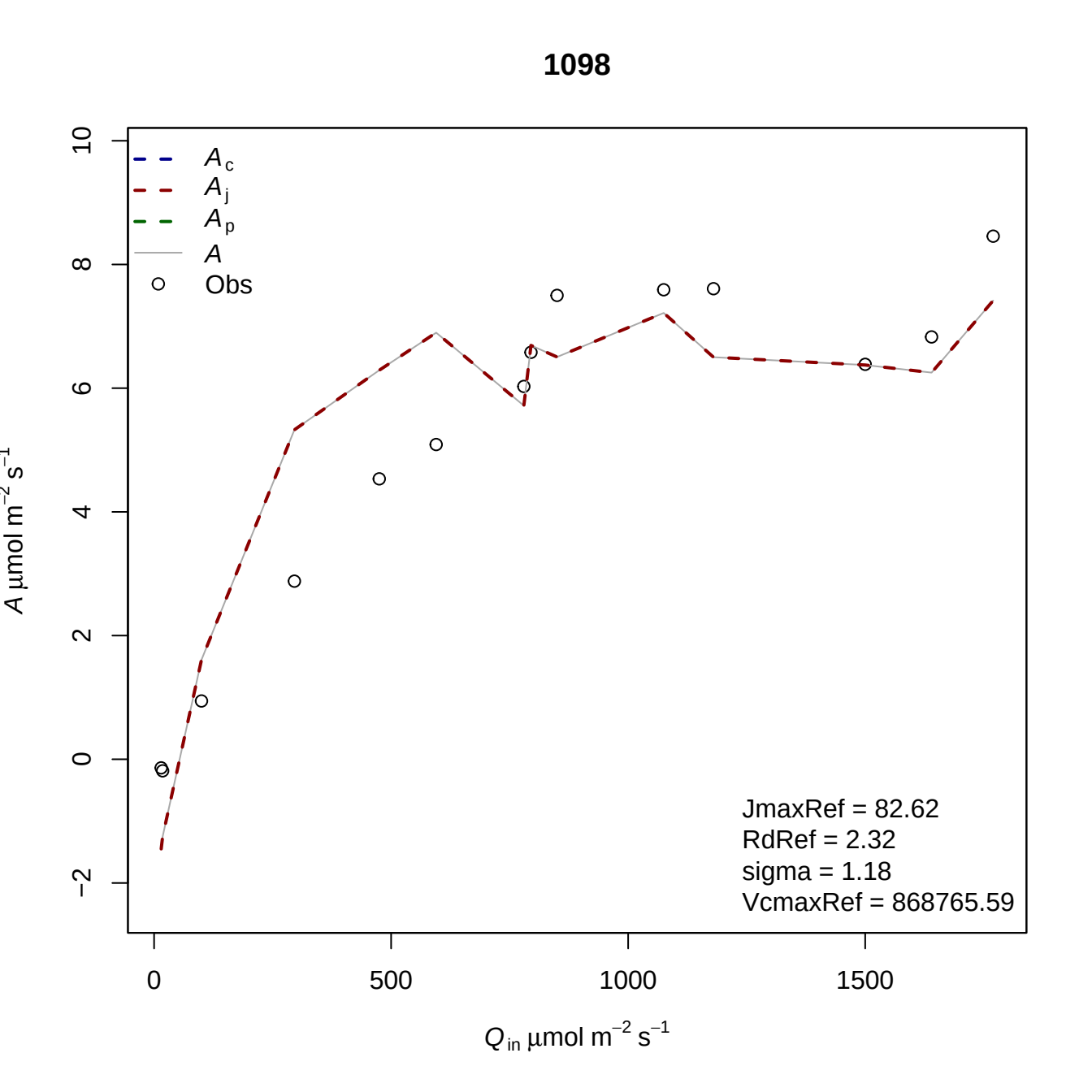

1093
A_c
A_j
A_p
A
Obs
JmaxRef = 95.37
RdRef = 4.13
sigma = 1.4
VcmaxRef = 1198109.94
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

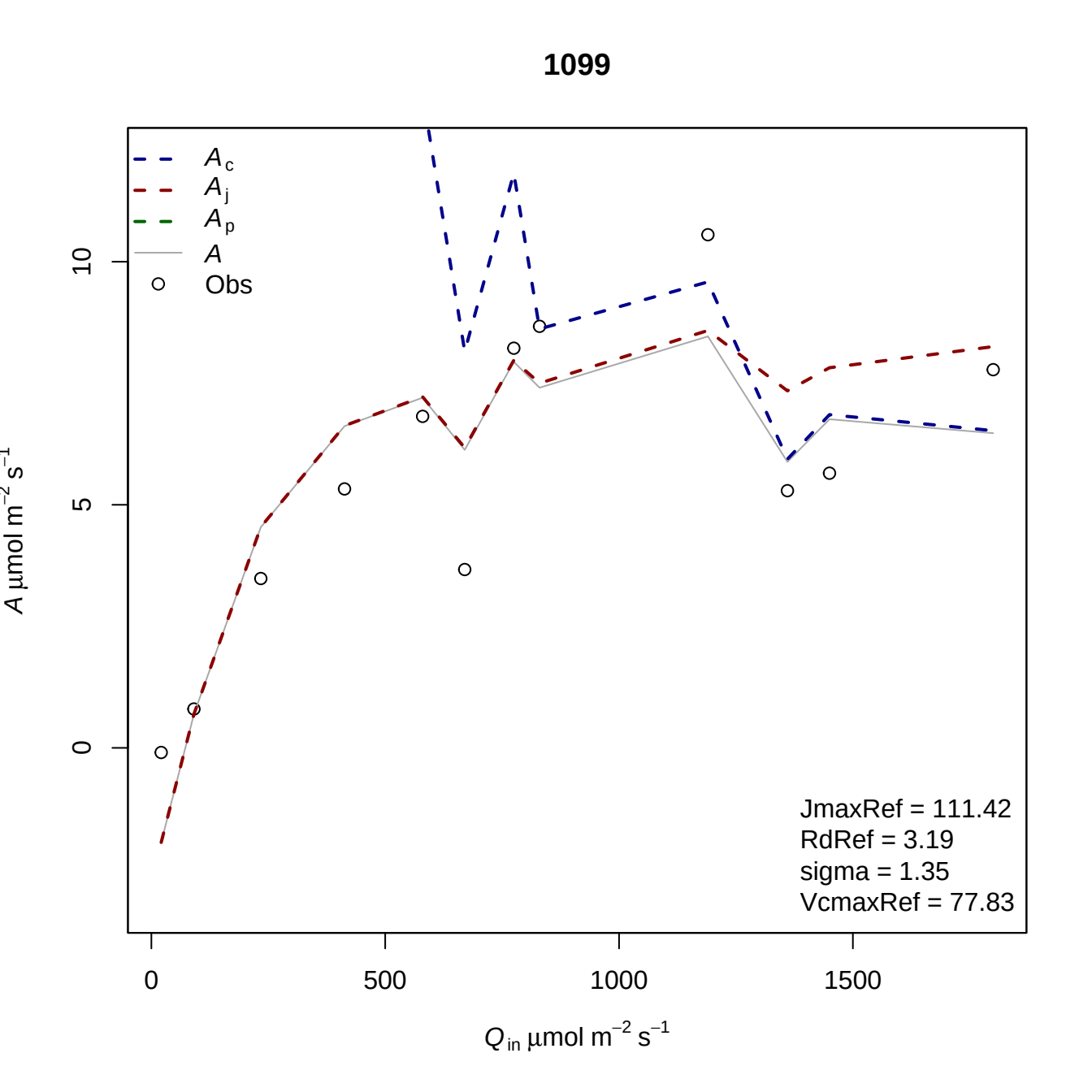

1094
A_c
A_j
A_p
A
Obs
JmaxRef = 74.51
RdRef = 2.8
sigma = 1.33
VcmaxRef = 438797.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

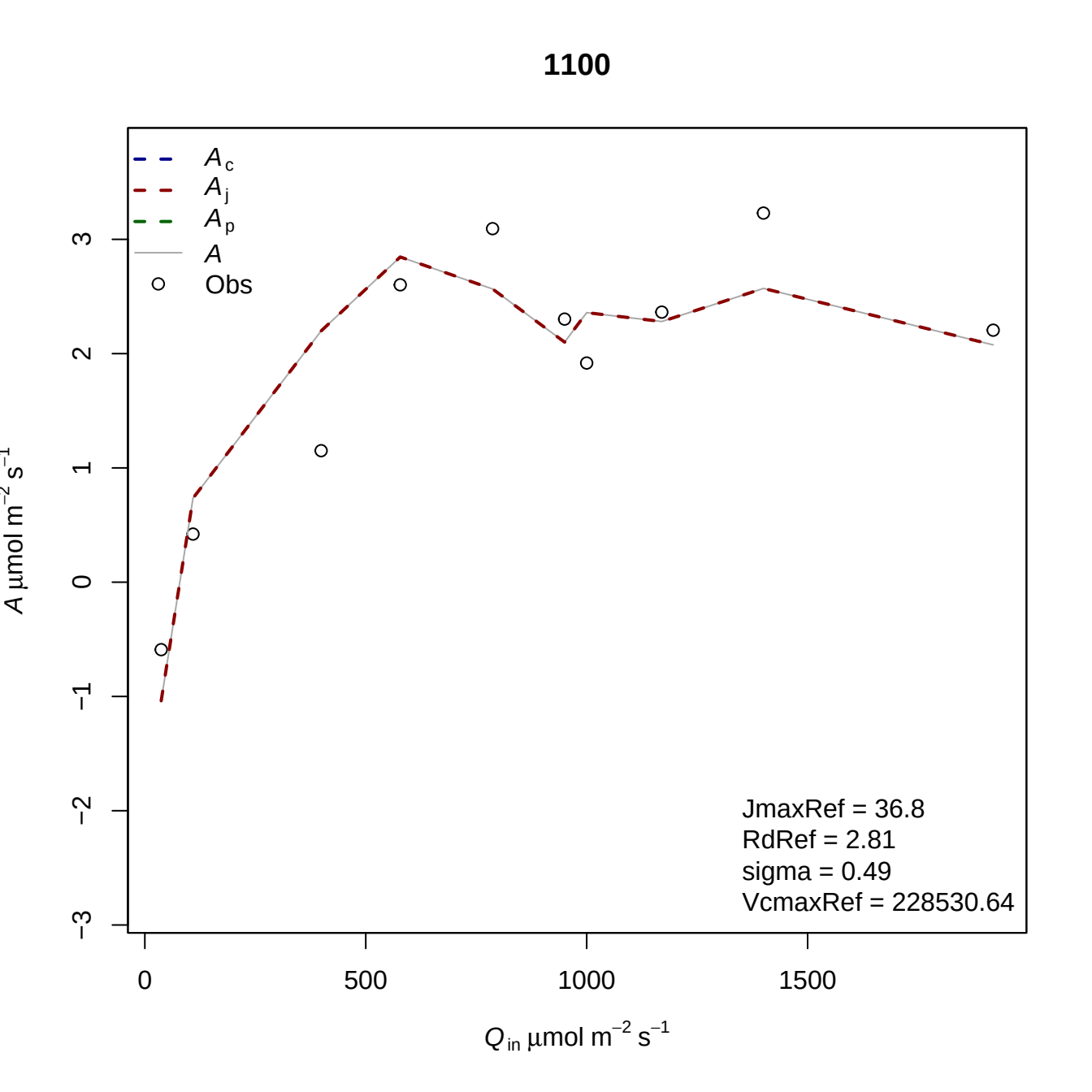

1095
A_c
A_j
A_p
A
Obs
JmaxRef = 85.64
RdRef = 3.66
sigma = 0.76
VcmaxRef = 361150.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500

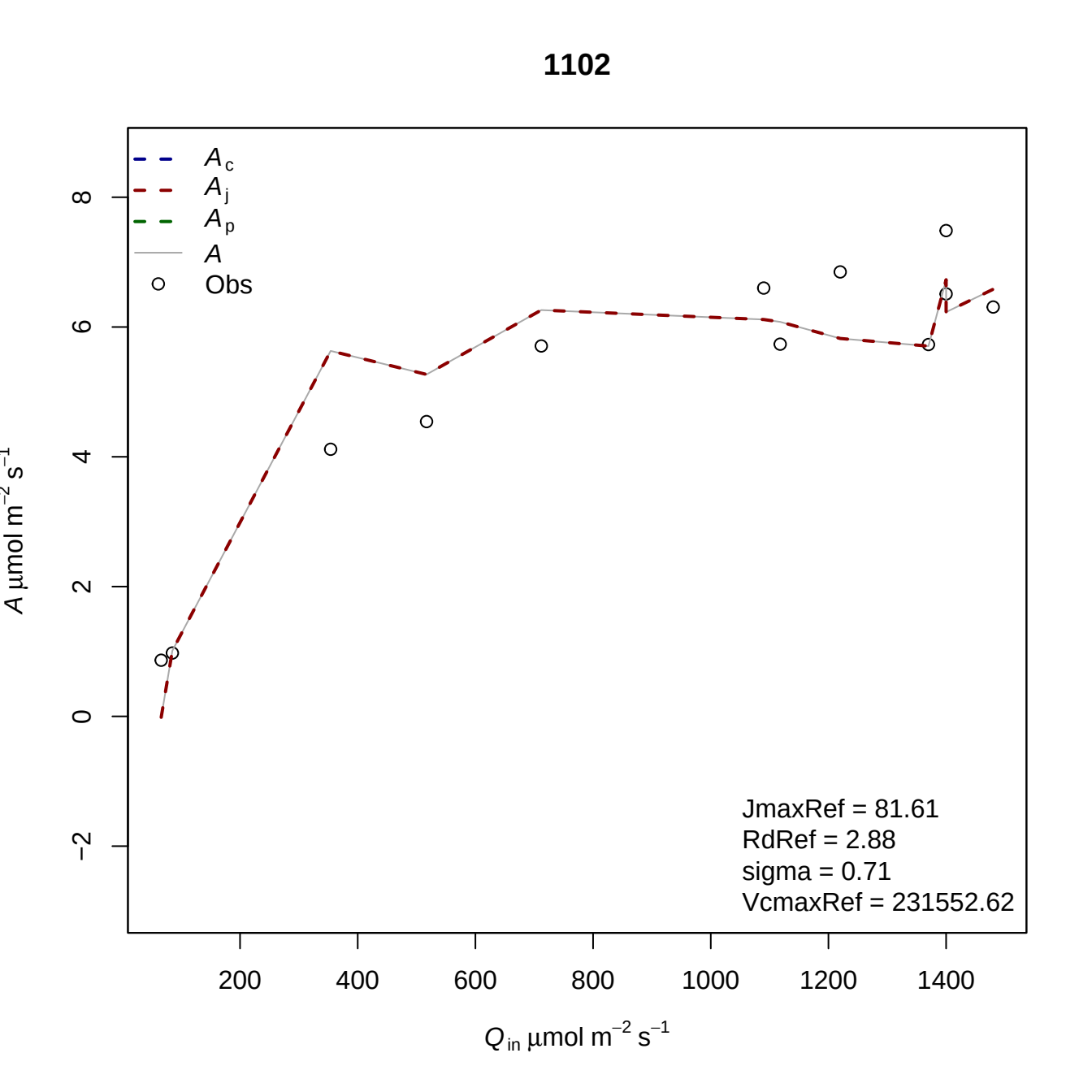

1096
A_c
A_j
A_p
A
Obs
JmaxRef = 39.82
RdRef = 1.71
sigma = 0.85
VcmaxRef = 242904.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

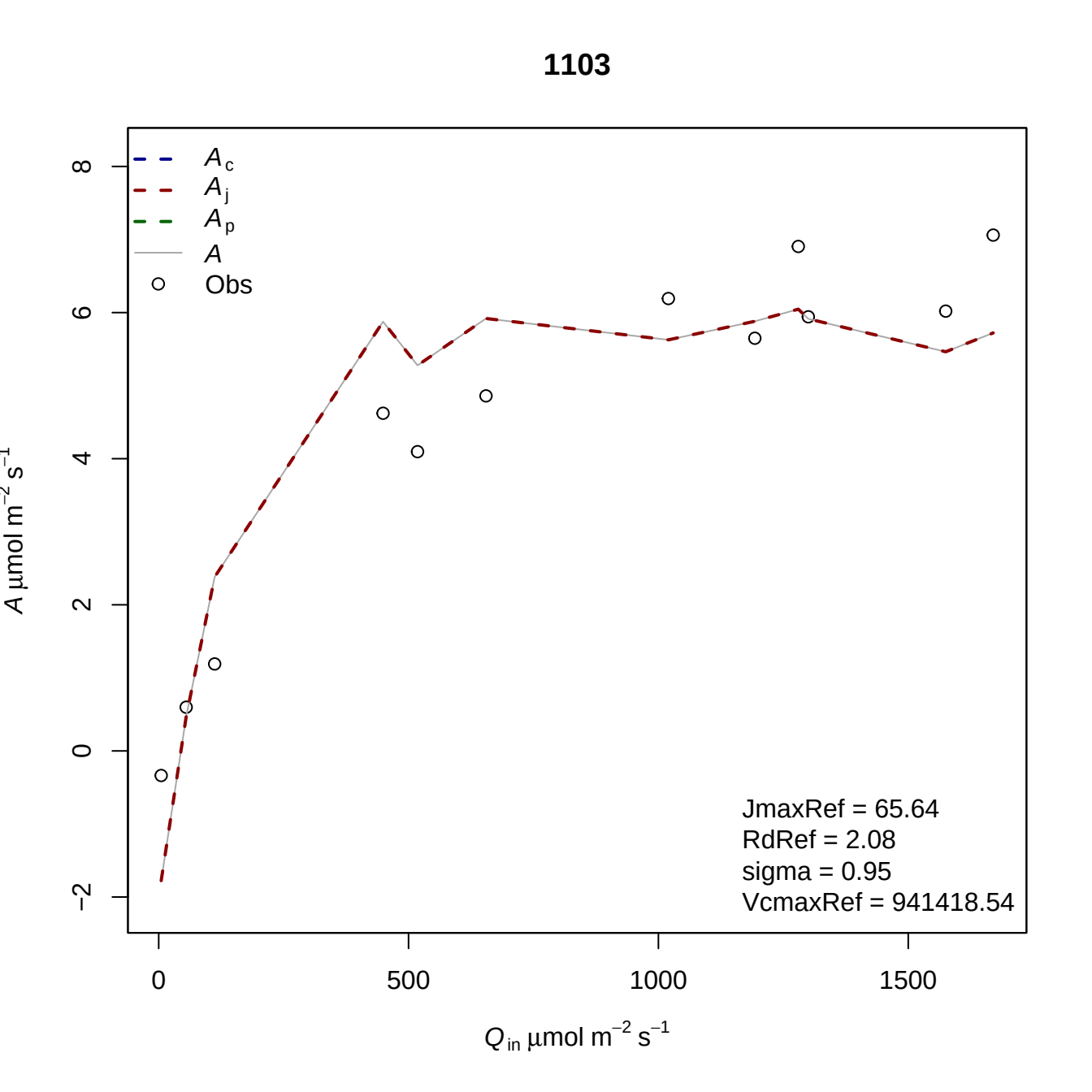

1097
A_c
A_j
A_p
A
Obs
JmaxRef = 30.65
RdRef = 1.47
sigma = 1.61
VcmaxRef = 17.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

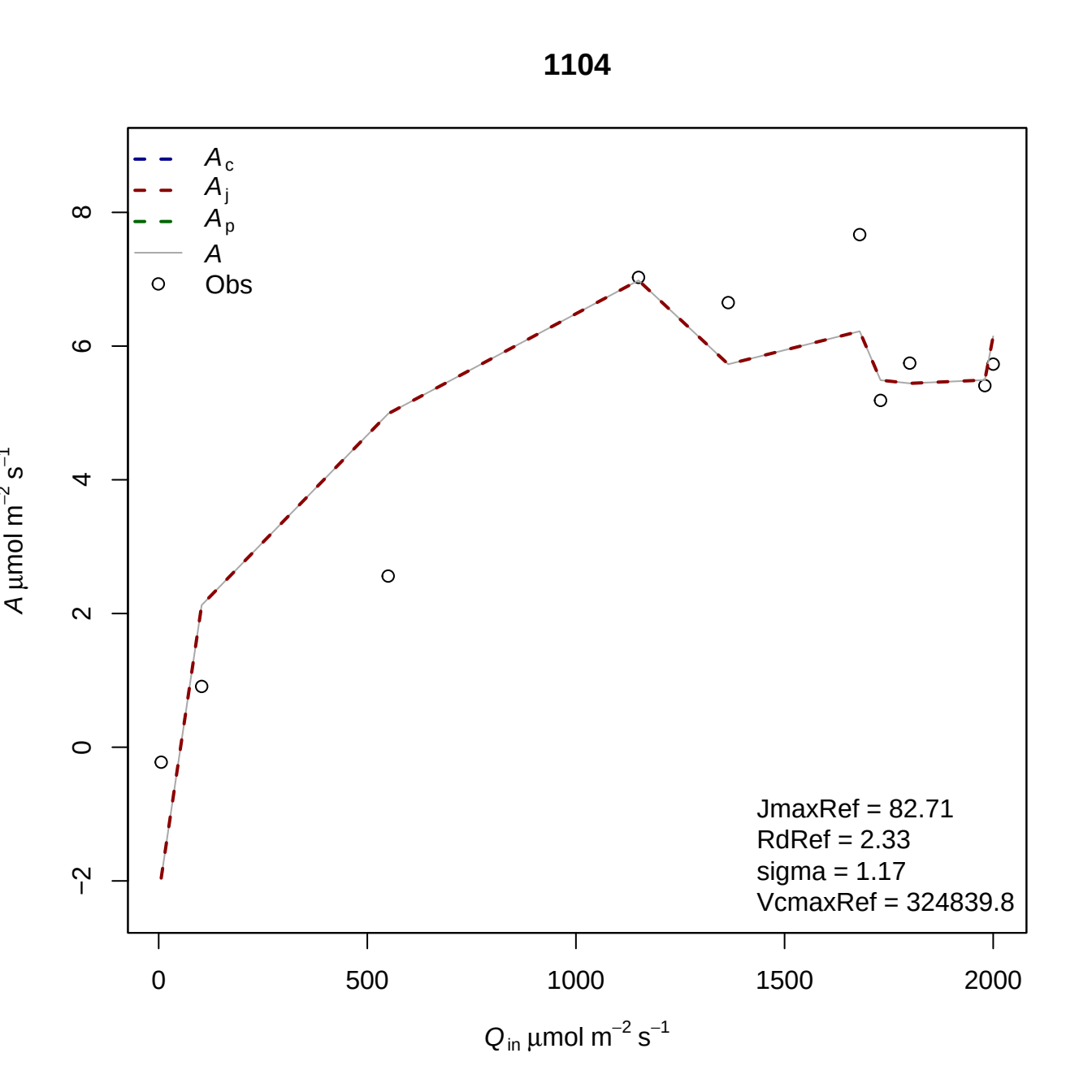

1098
A_c
A_j
A_p
A
Obs
JmaxRef = 82.62
RdRef = 2.32
sigma = 1.18
VcmaxRef = 868765.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

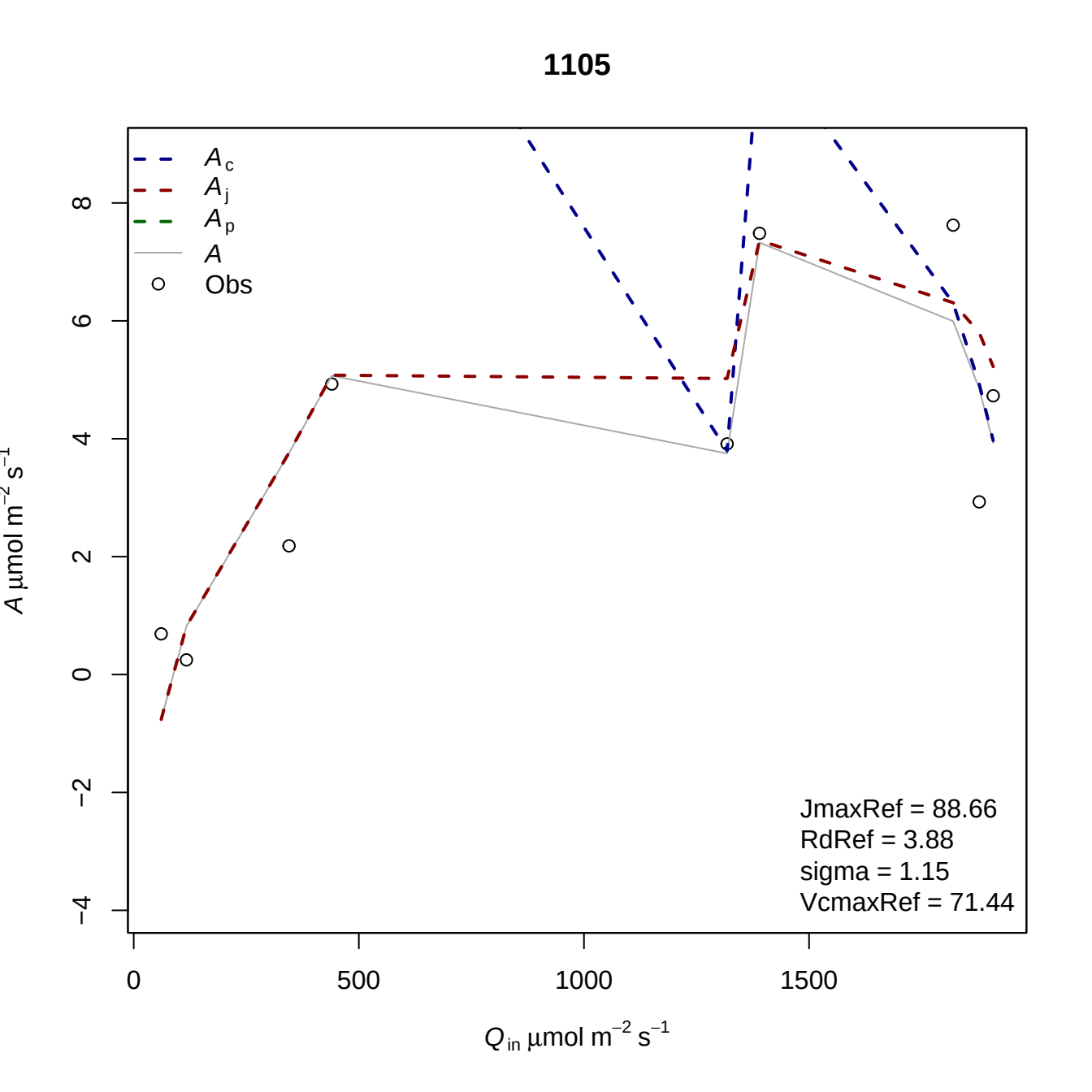

1099
A_c
A_j
A_p
A
Obs
JmaxRef = 111.42
RdRef = 3.19
sigma = 1.35
VcmaxRef = 77.83
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

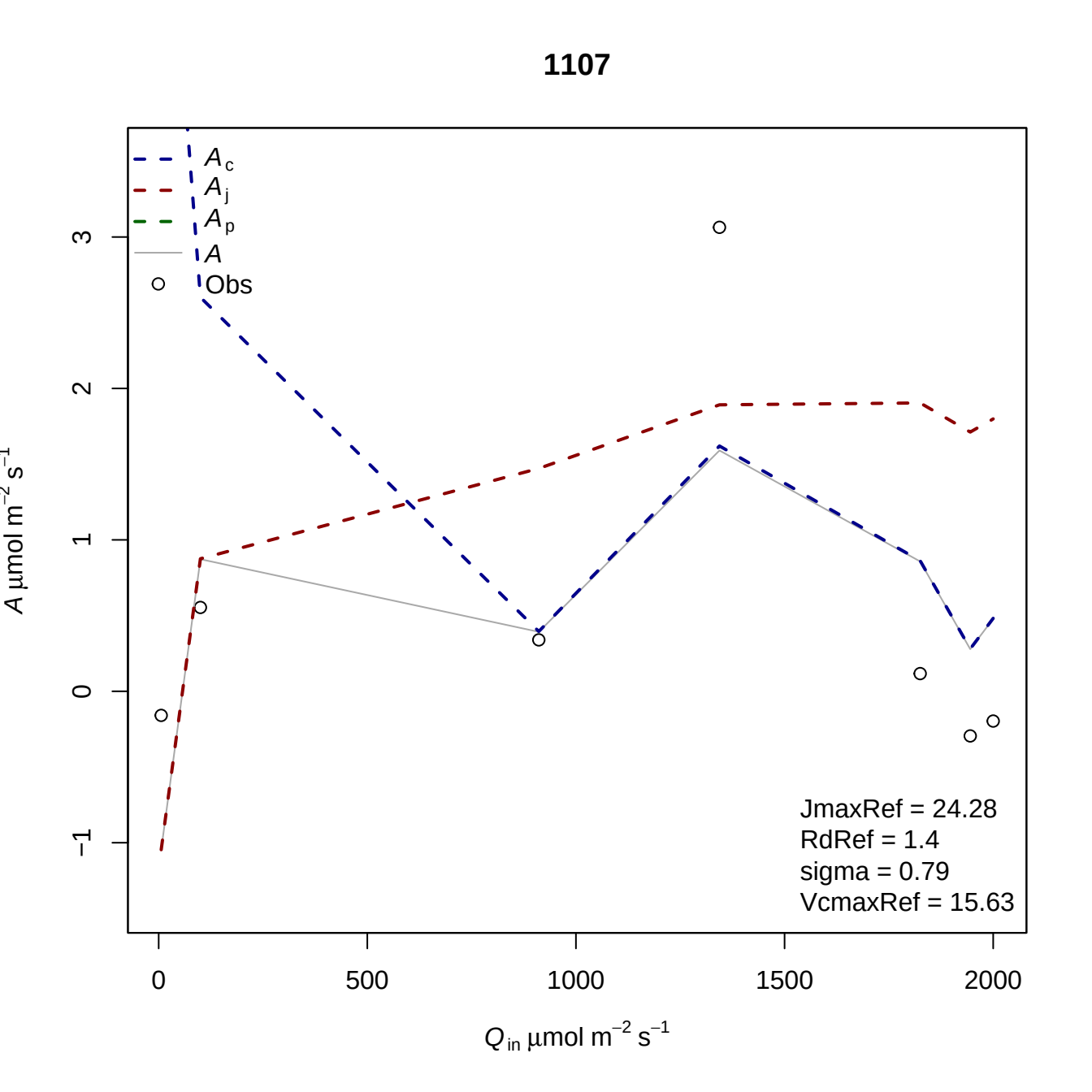

1100
A_c
A_j
A_p
A
Obs
JmaxRef = 36.8
RdRef = 2.81
sigma = 0.49
VcmaxRef = 228530.64
A µmol m^{-2} s^{-1}
3
2
1
0
−1
−2
−3
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

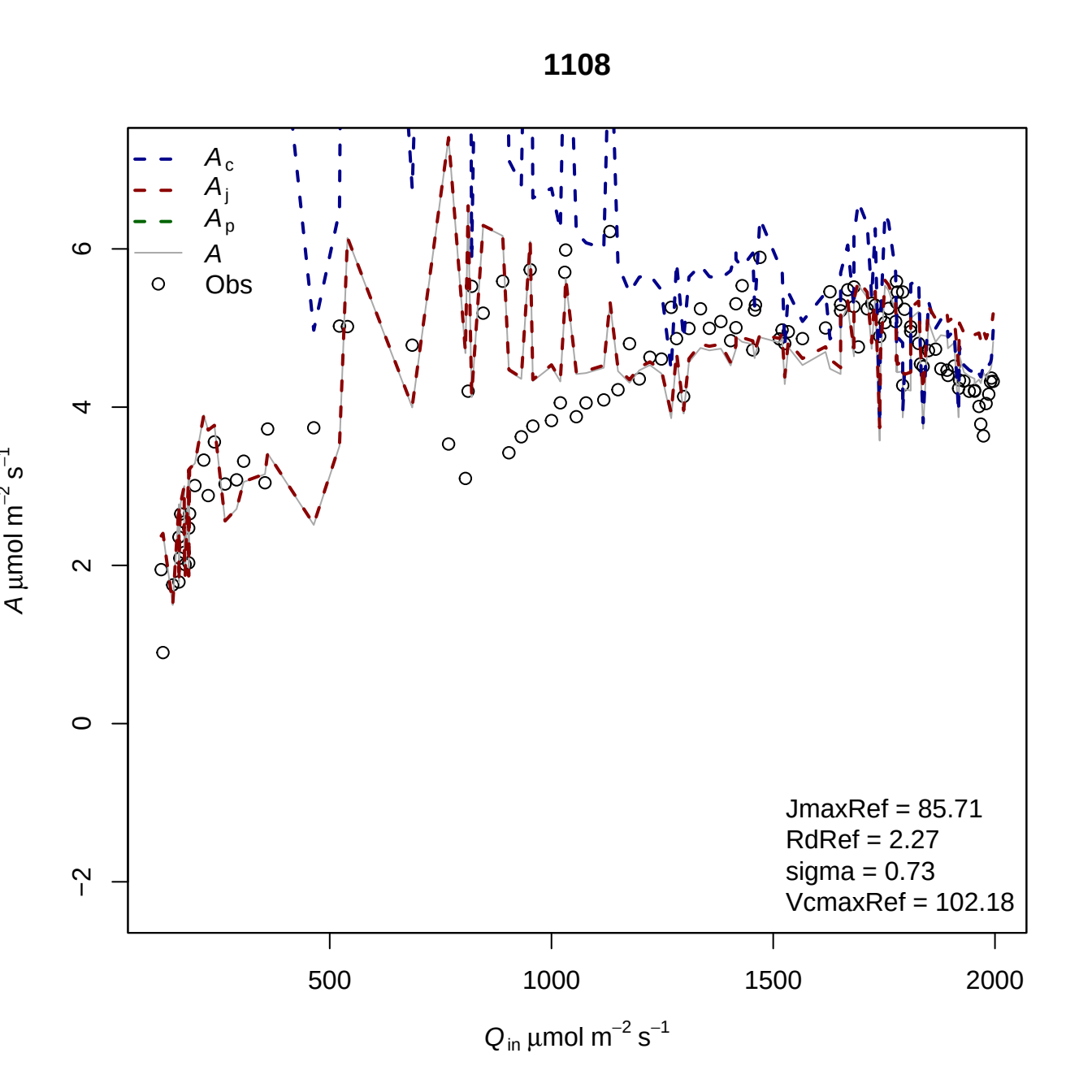

1102
A_c
A_j
A_p
A
Obs
JmaxRef = 81.61
RdRef = 2.88
sigma = 0.71
VcmaxRef = 231552.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
200
400
600
800
1000
1200
1400

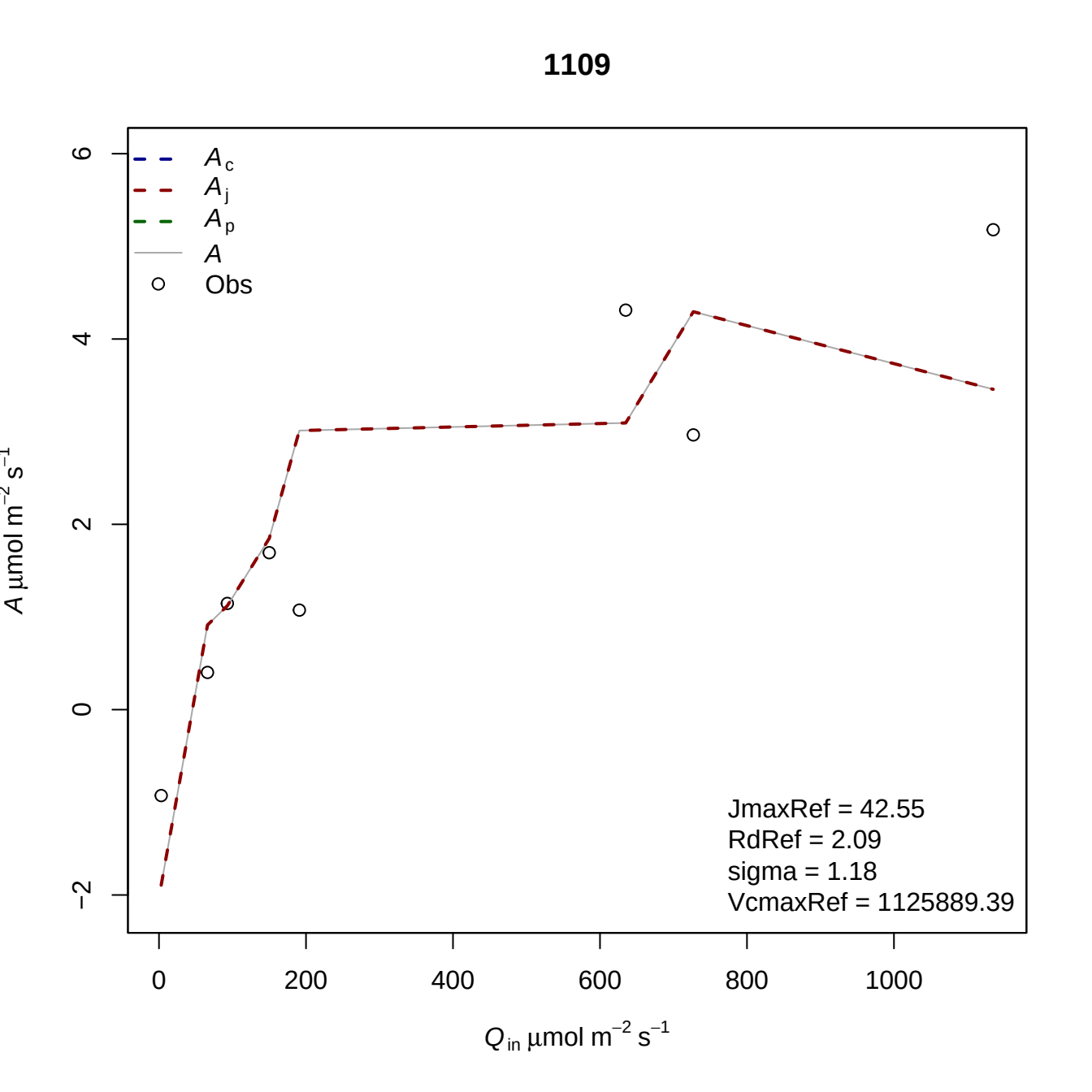

1103
A_c
A_j
A_p
A
Obs
JmaxRef = 65.64
RdRef = 2.08
sigma = 0.95
VcmaxRef = 941418.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

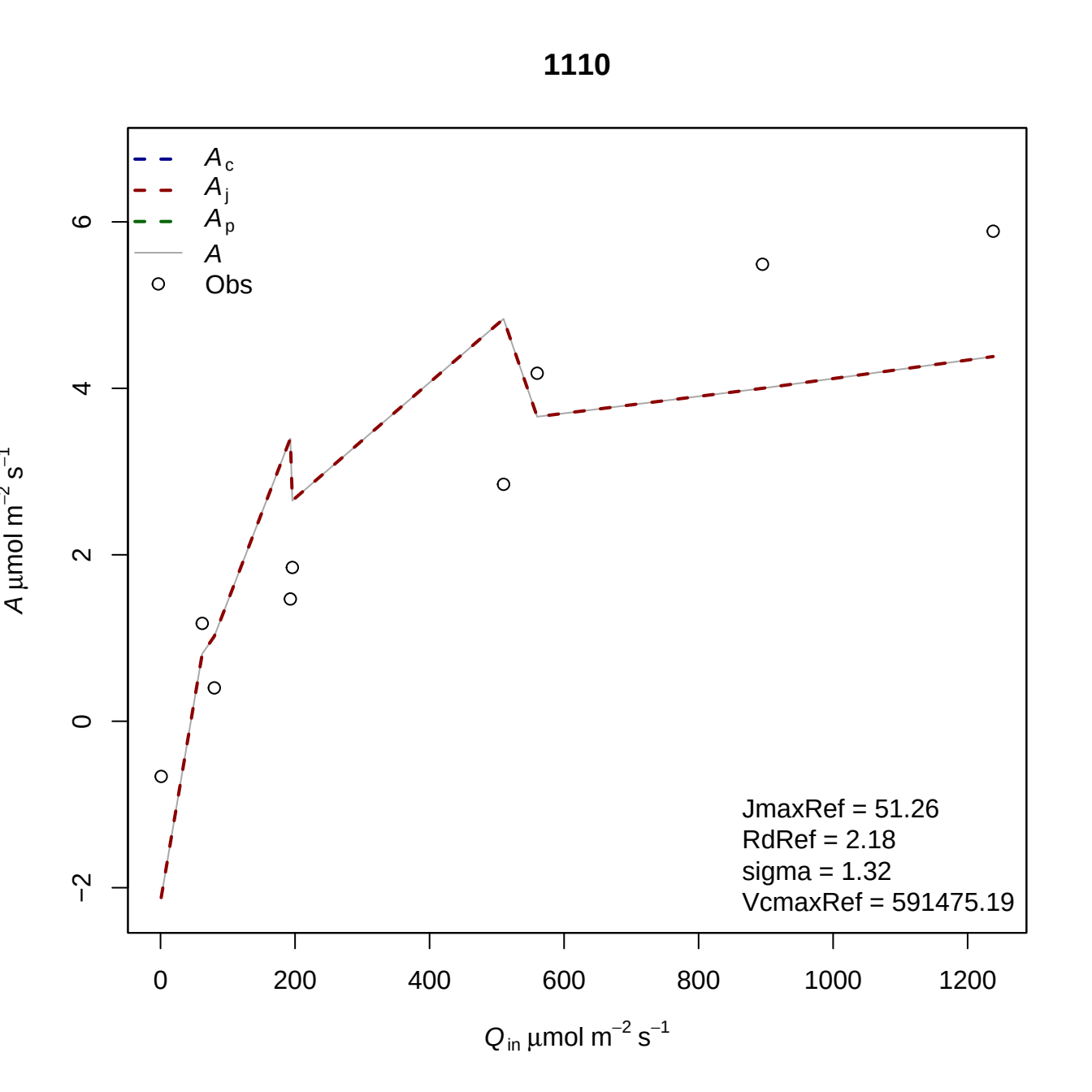

1104
A_c
A_j
A_p
A
Obs
JmaxRef = 82.71
RdRef = 2.33
sigma = 1.17
VcmaxRef = 324839.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

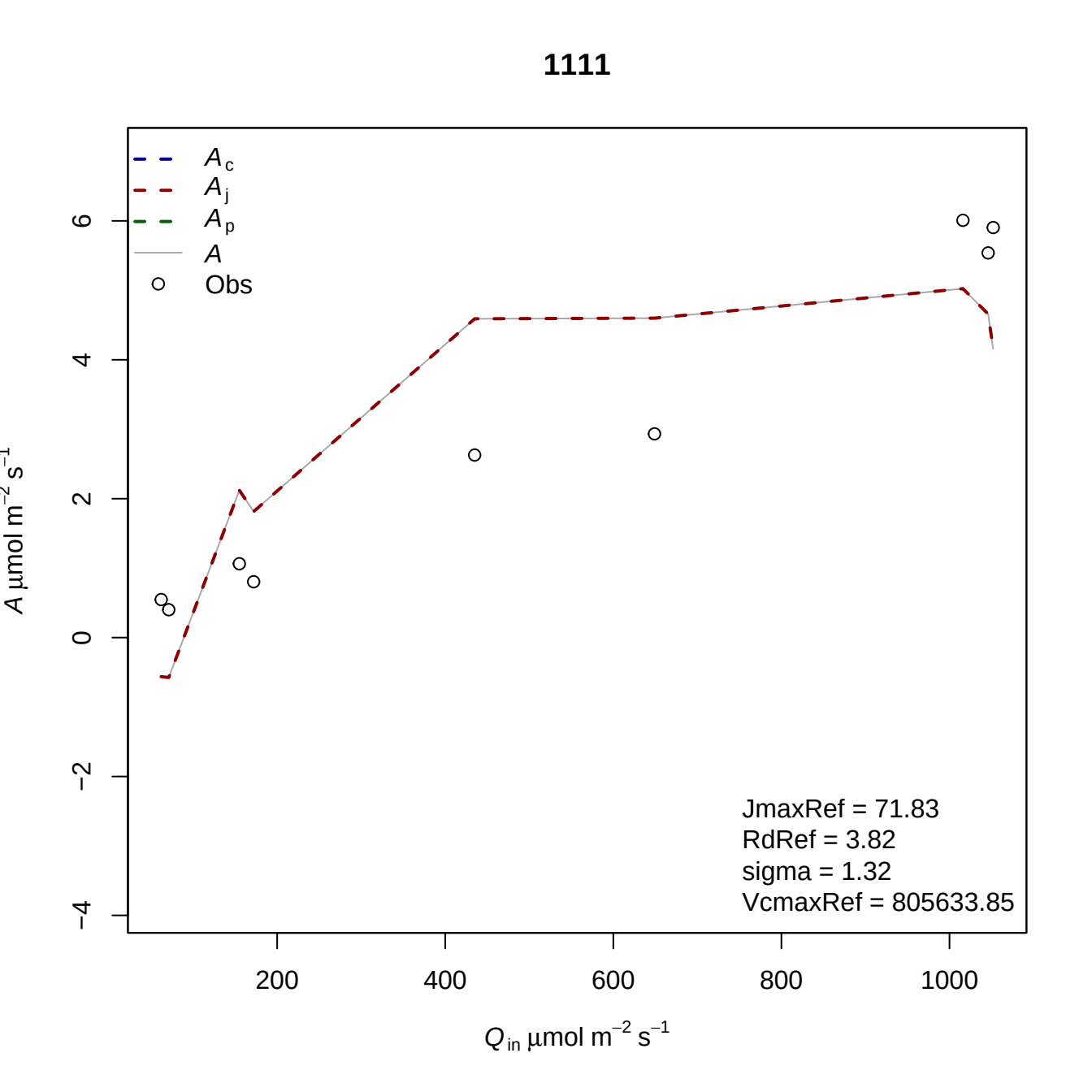

1105
A_c
A_j
A_p
A
Obs
JmaxRef = 88.66
RdRef = 3.88
sigma = 1.15
VcmaxRef = 71.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

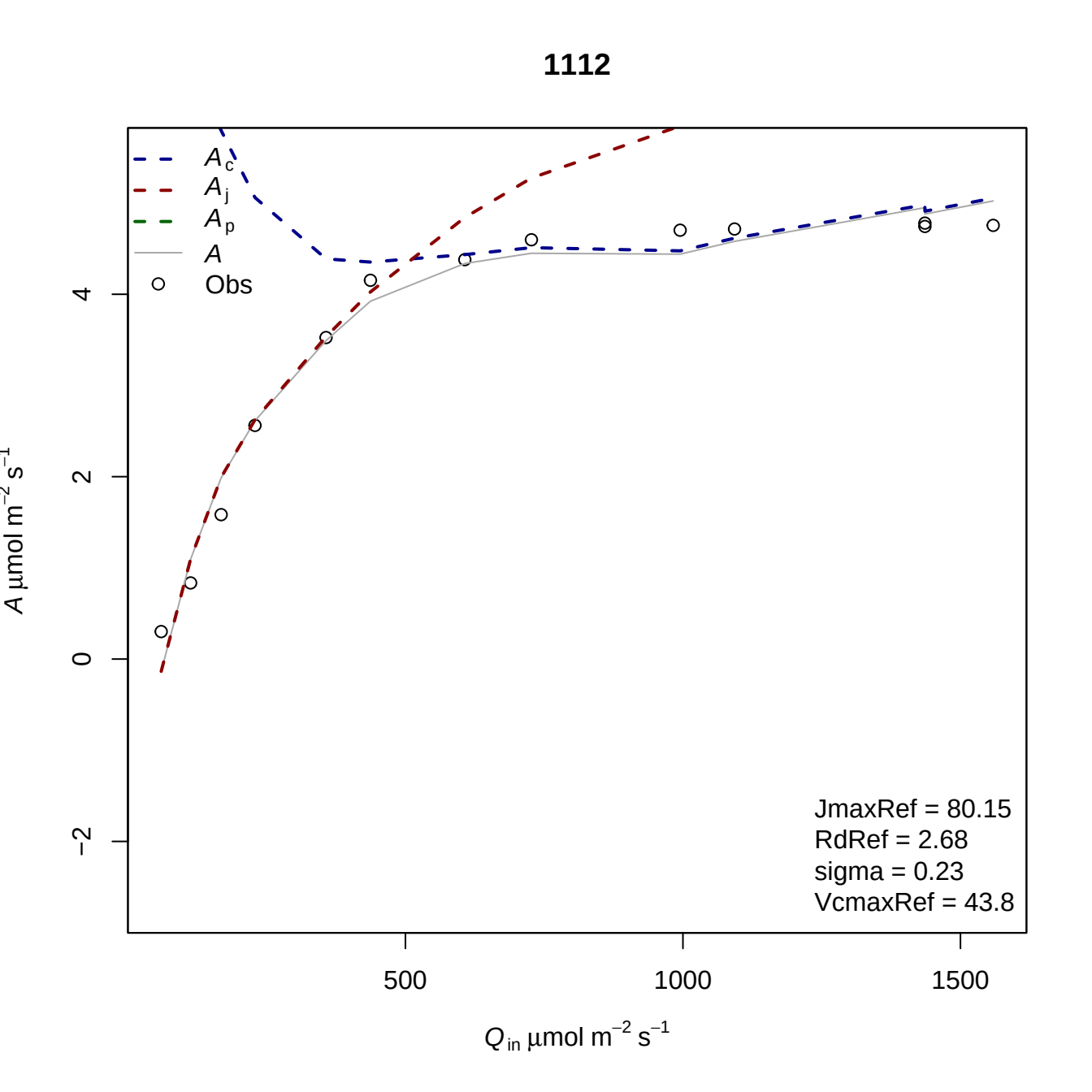

1107
A_c
A_j
A_p
A
Obs
JmaxRef = 24.28
RdRef = 1.4
sigma = 0.79
VcmaxRef = 15.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

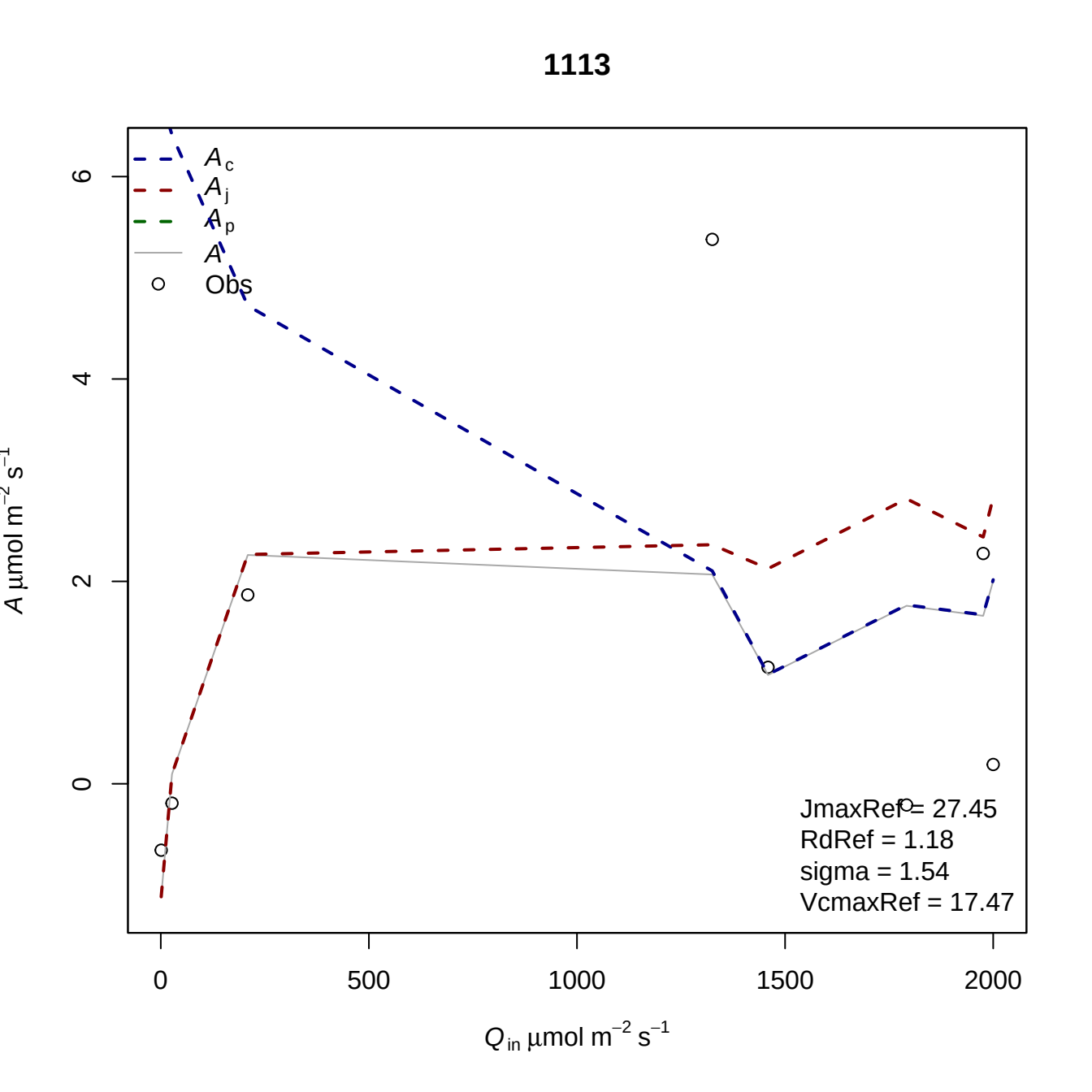

1108
A_c
A_j
A_p
A
Obs
JmaxRef = 85.71
RdRef = 2.27
sigma = 0.73
VcmaxRef = 102.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
500
1000
1500
2000

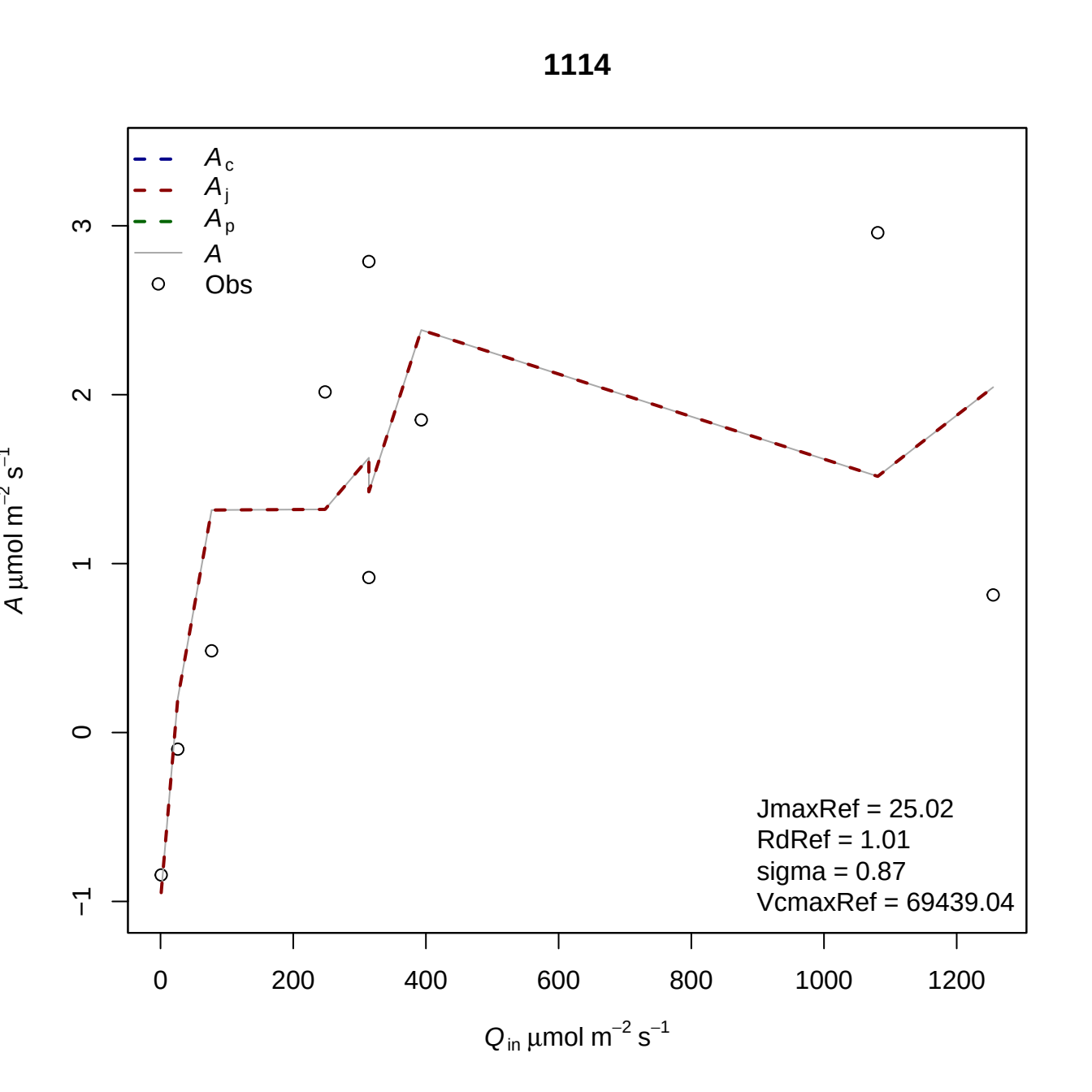

1109
A_c
A_j
A_p
A
Obs
JmaxRef = 42.55
RdRef = 2.09
sigma = 1.18
VcmaxRef = 1125889.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000

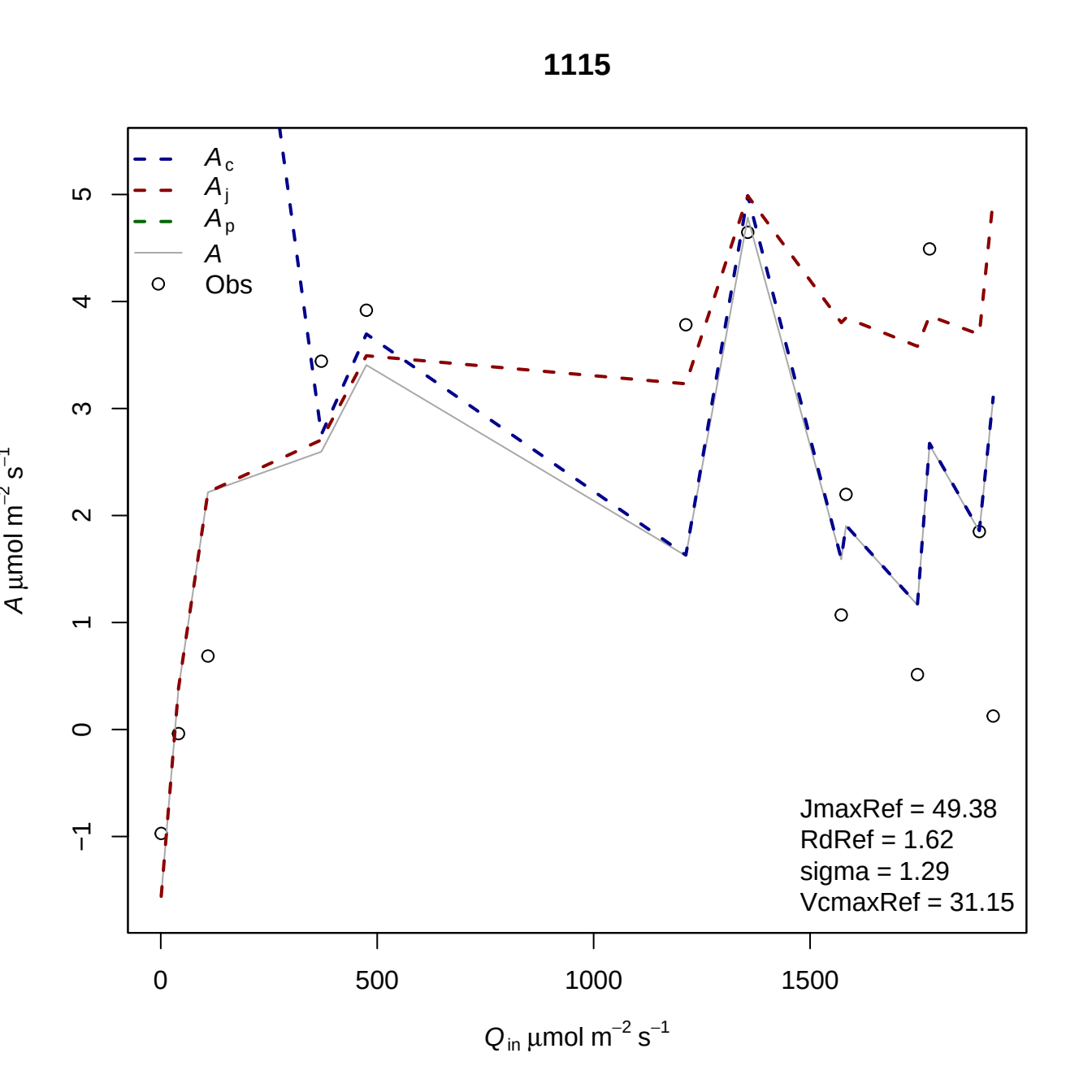

1110
A_c
A_j
A_p
A
Obs
JmaxRef = 51.26
RdRef = 2.18
sigma = 1.32
VcmaxRef = 591475.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200

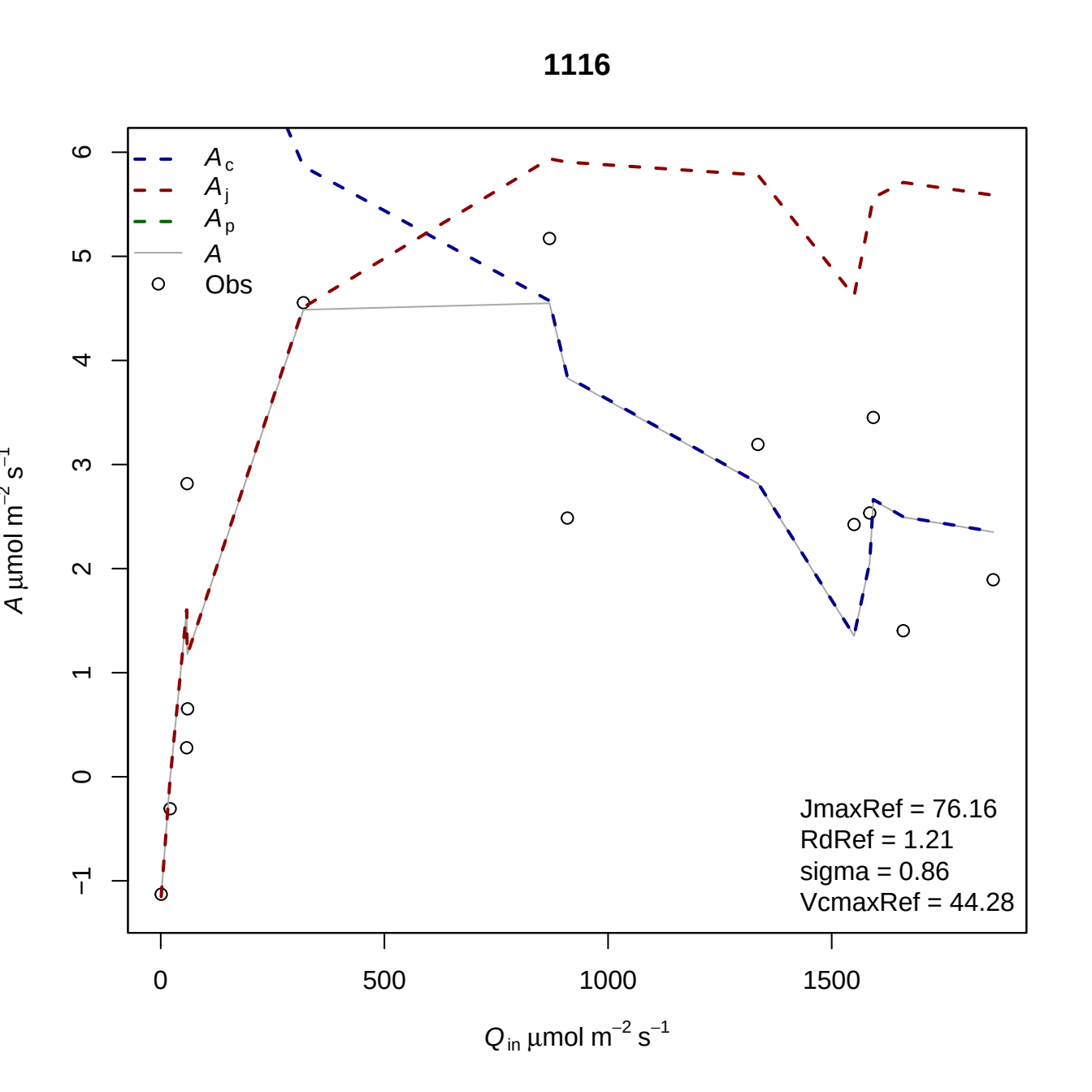

1111
A_c
A_j
A_p
A
Obs
JmaxRef = 71.83
RdRef = 3.82
sigma = 1.32
VcmaxRef = 805633.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
200
400
600
800
1000

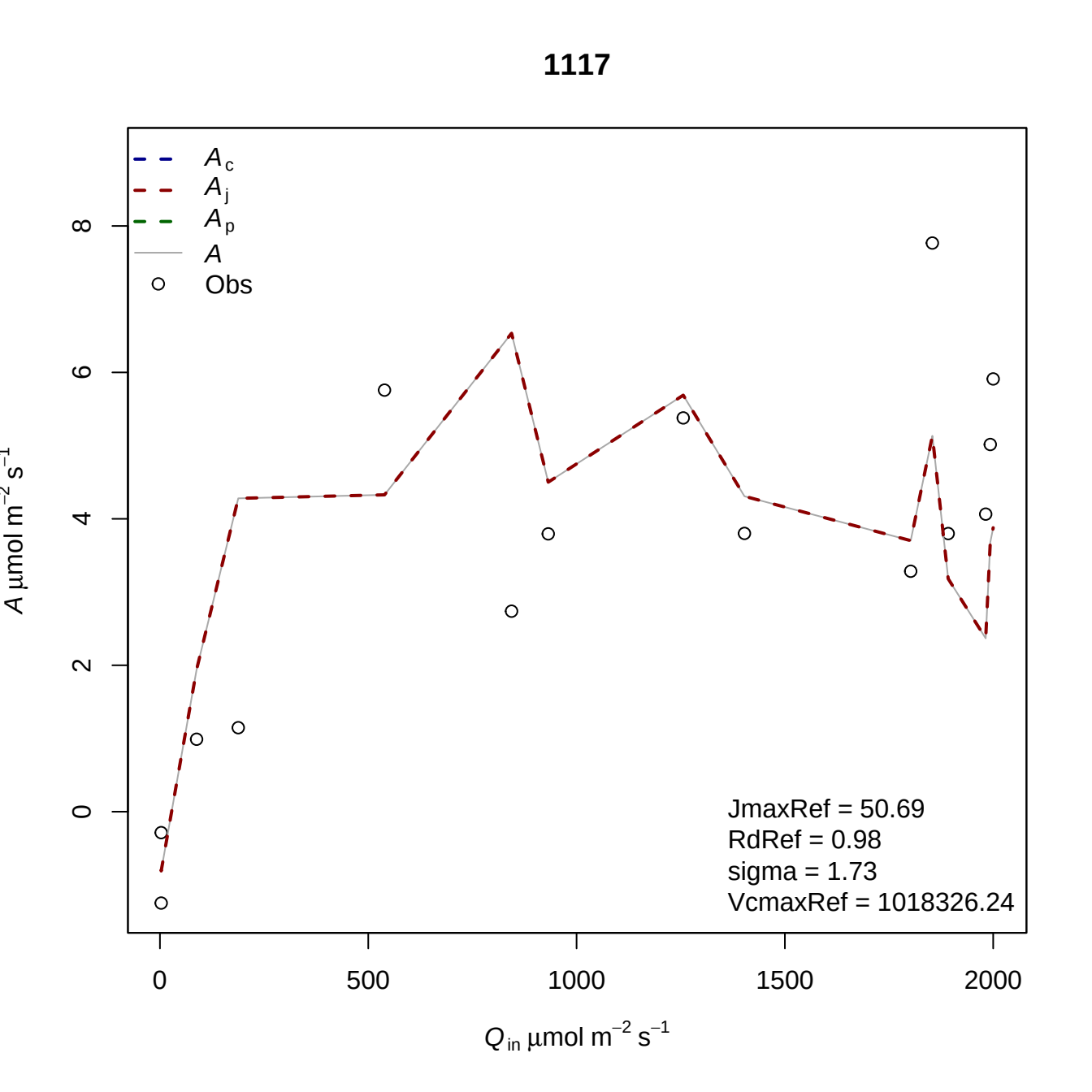

1112
A_c
A_j
A_p
A
Obs
JmaxRef = 80.15
RdRef = 2.68
sigma = 0.23
VcmaxRef = 43.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
500
1000
1500

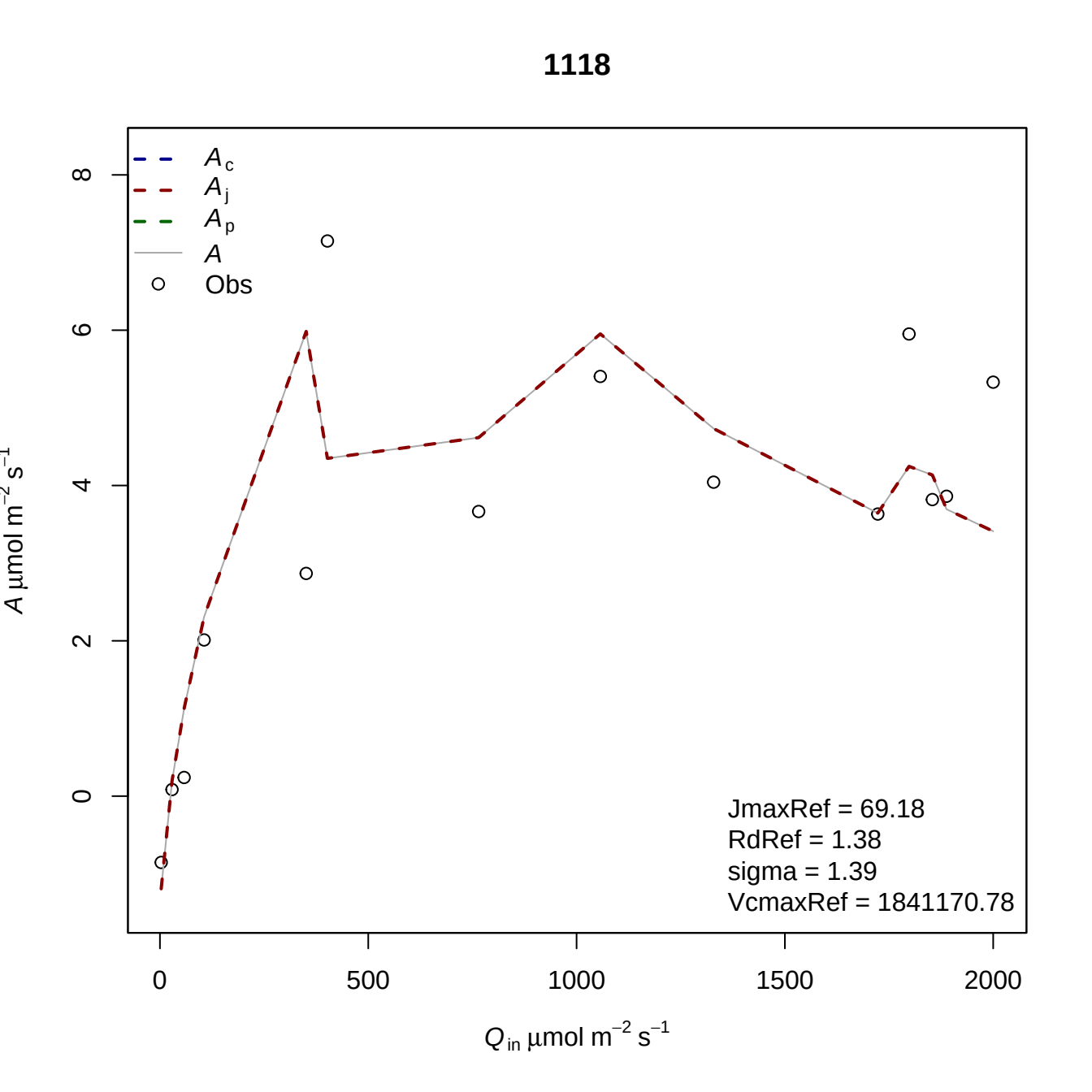

1113
A_c
A_j
A_p
A
Obs
JmaxRef = 27.45
RdRef = 1.18
sigma = 1.54
VcmaxRef = 17.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

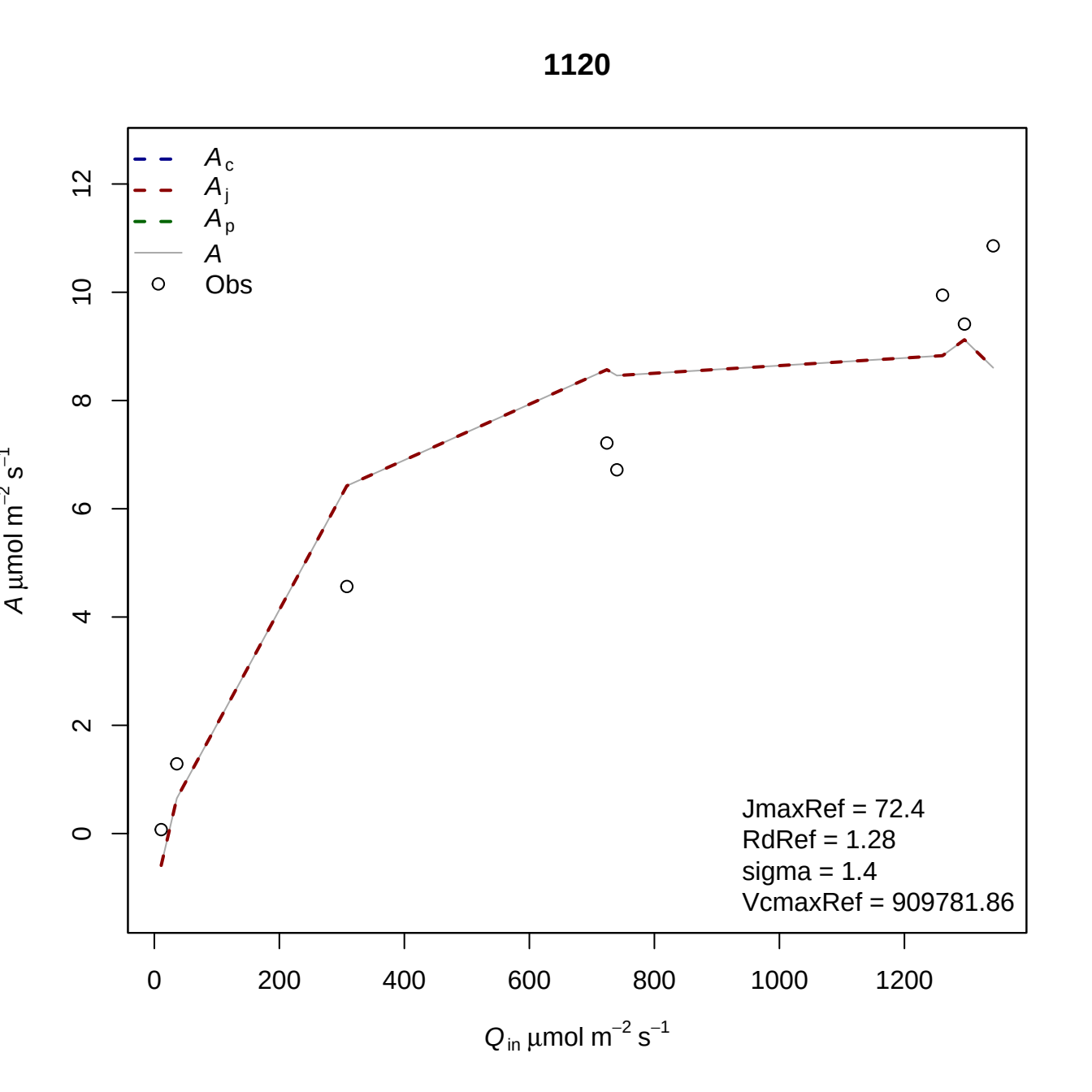

1114
A_c
A_j
A_p
A
Obs
JmaxRef = 25.02
RdRef = 1.01
sigma = 0.87
VcmaxRef = 69439.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000
1200

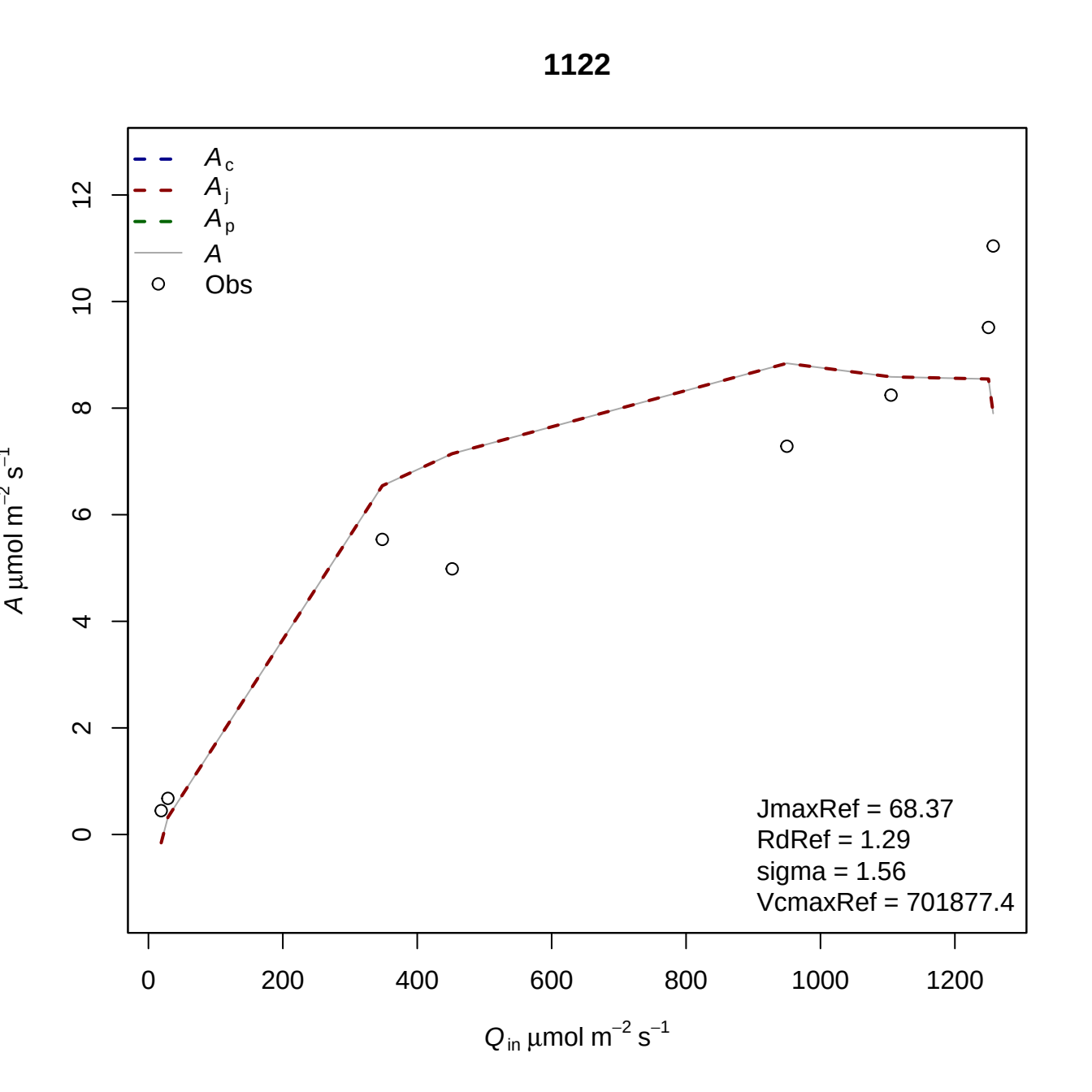

1115
A_c
A_j
A_p
A
Obs
JmaxRef = 49.38
RdRef = 1.62
sigma = 1.29
VcmaxRef = 31.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

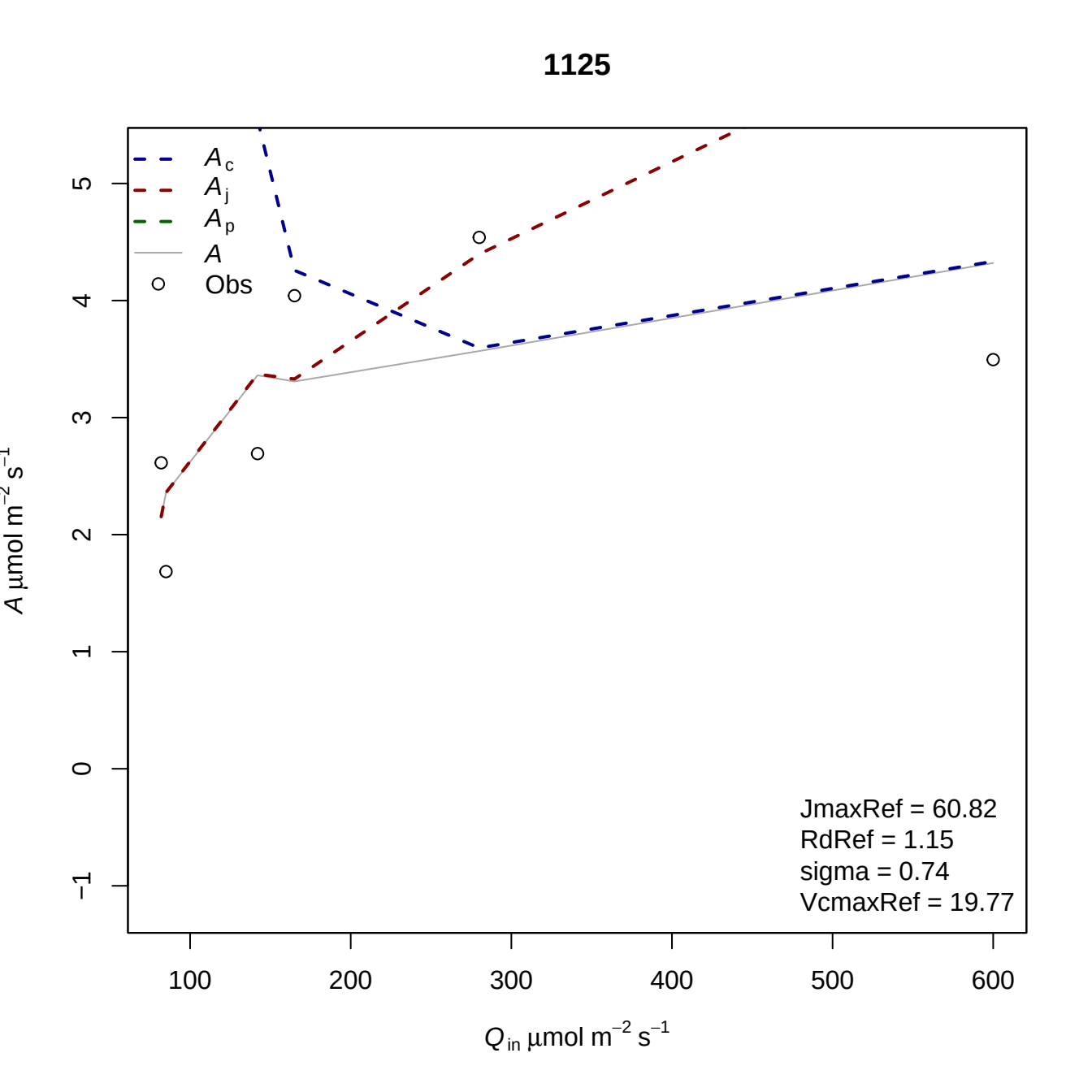

1116
A_c
A_j
A_p
A
Obs
JmaxRef = 76.16
RdRef = 1.21
sigma = 0.86
VcmaxRef = 44.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

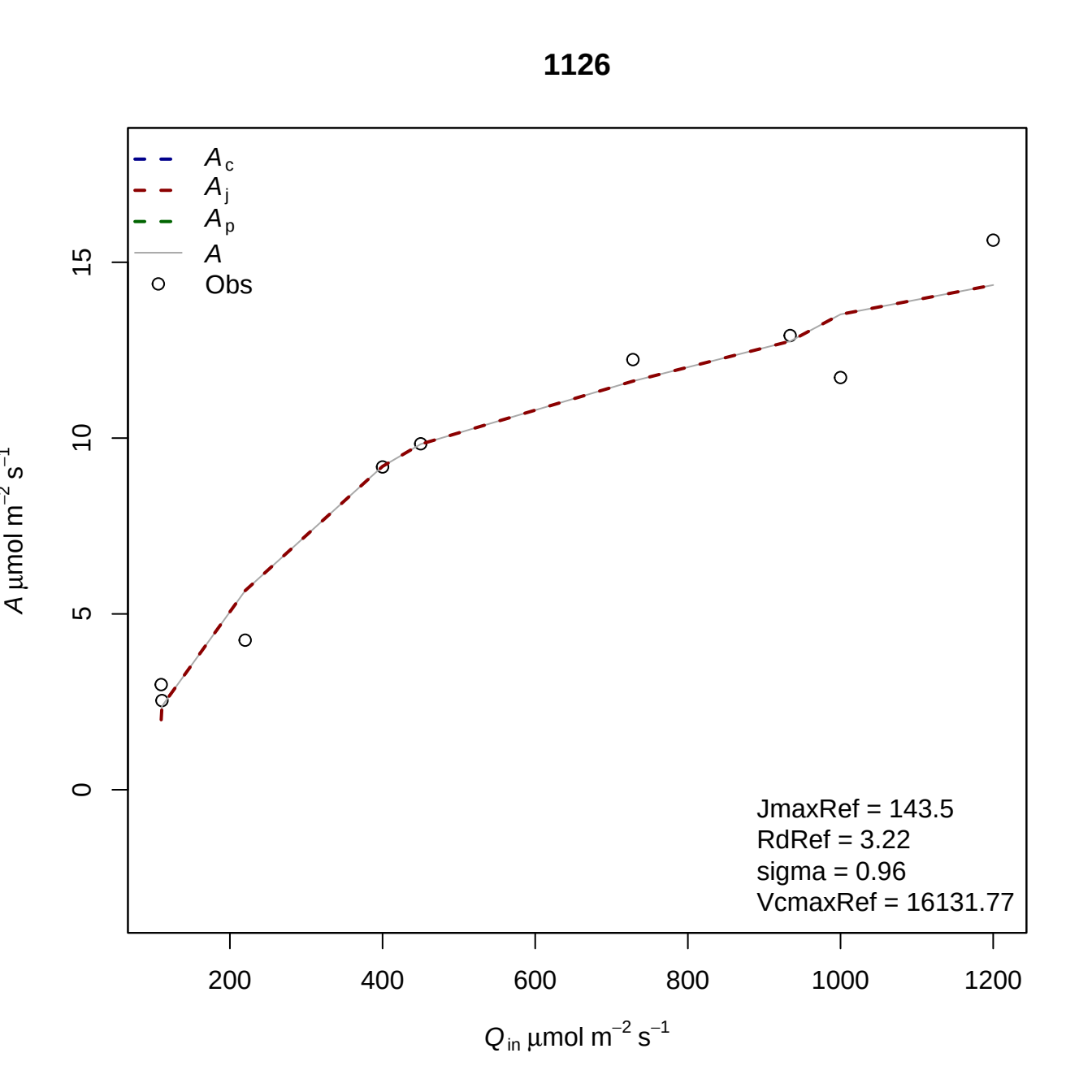

1117
A_c
A_j
A_p
A
Obs
JmaxRef = 50.69
RdRef = 0.98
sigma = 1.73
VcmaxRef = 1018326.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

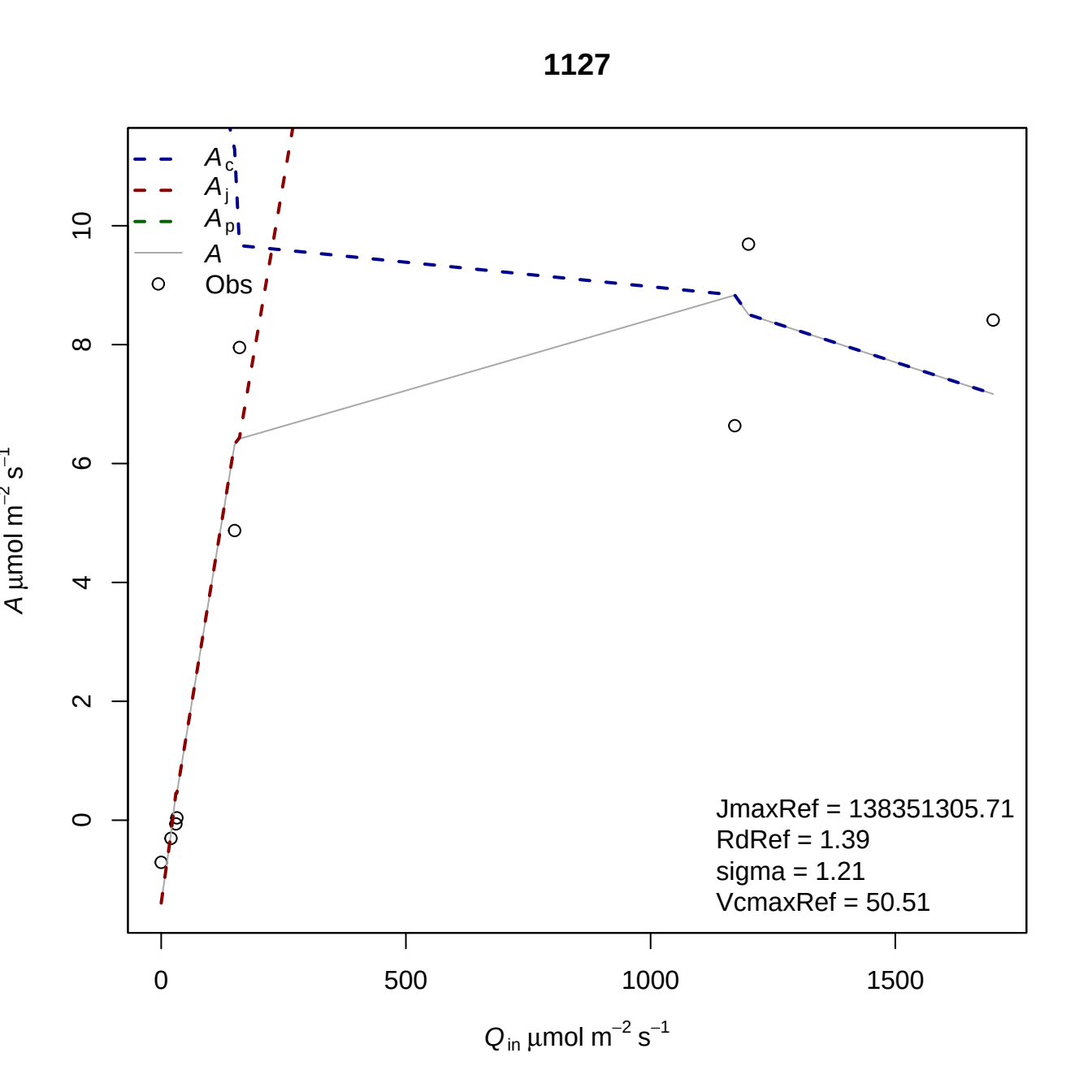

1118
A_c
A_j
A_p
A
Obs
JmaxRef = 69.18
RdRef = 1.38
sigma = 1.39
VcmaxRef = 1841170.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

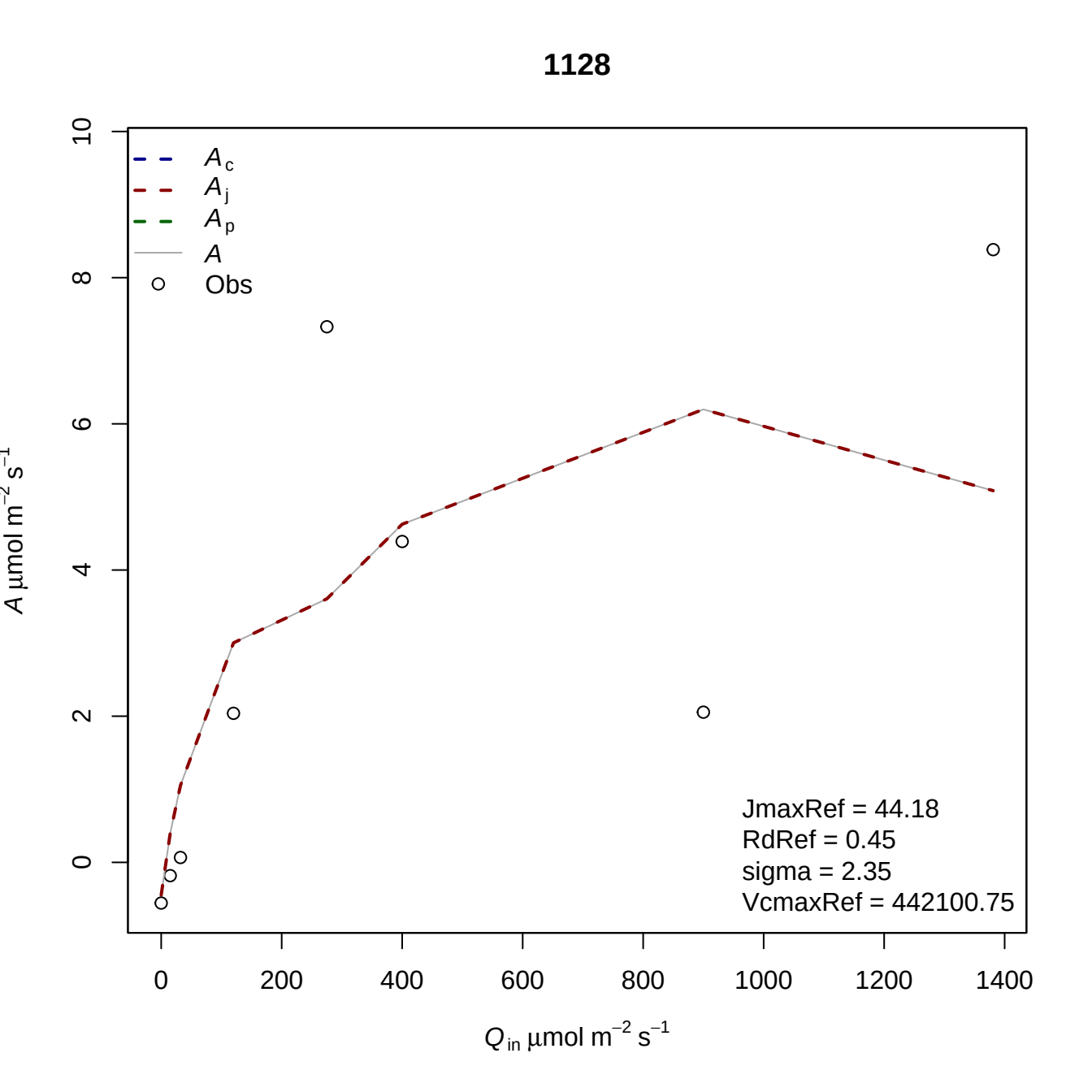

1120
A_c
A_j
A_p
A
Obs
JmaxRef = 72.4
RdRef = 1.28
sigma = 1.4
VcmaxRef = 909781.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

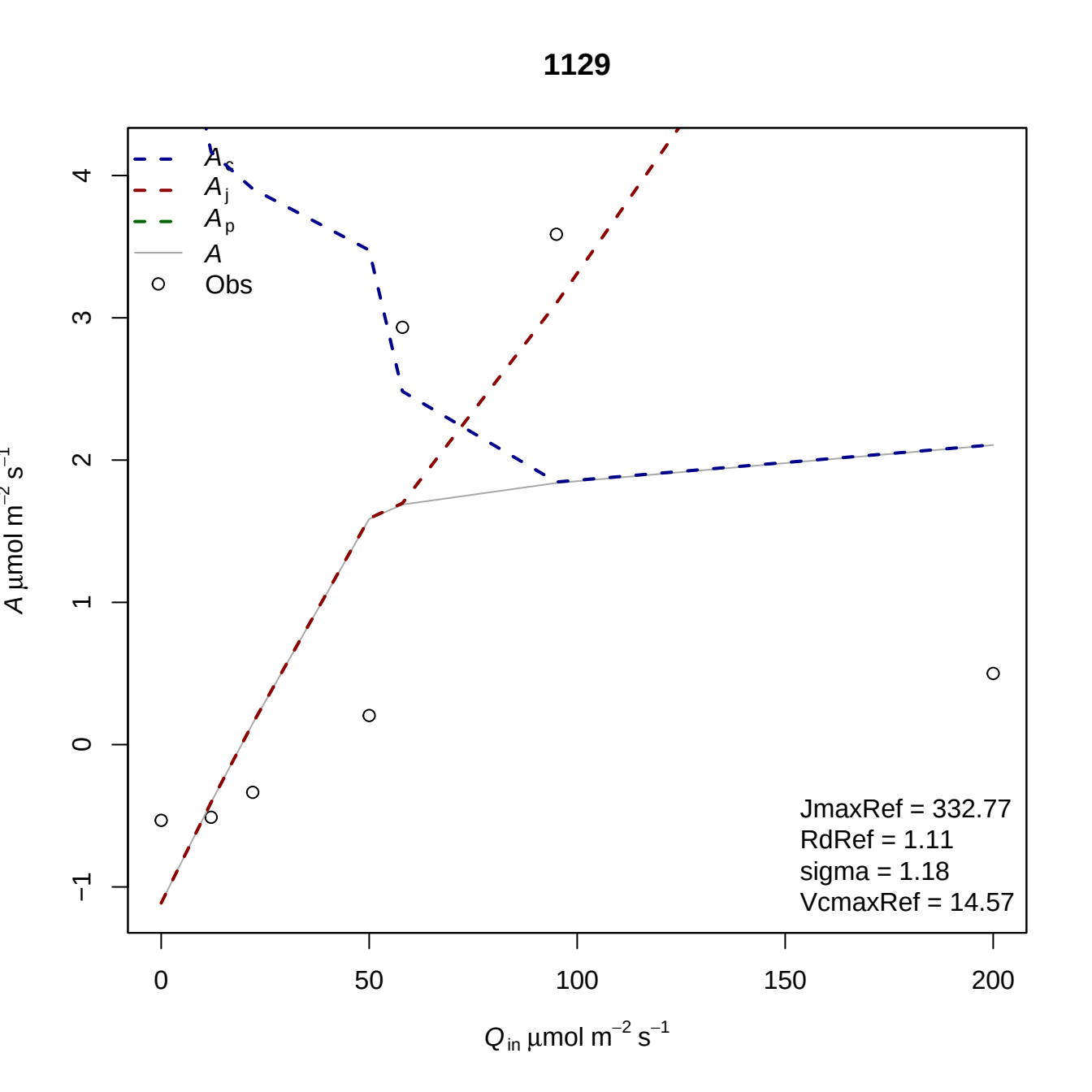

1122
A_c
A_j
A_p
A
Obs
JmaxRef = 68.37
RdRef = 1.29
sigma = 1.56
VcmaxRef = 701877.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

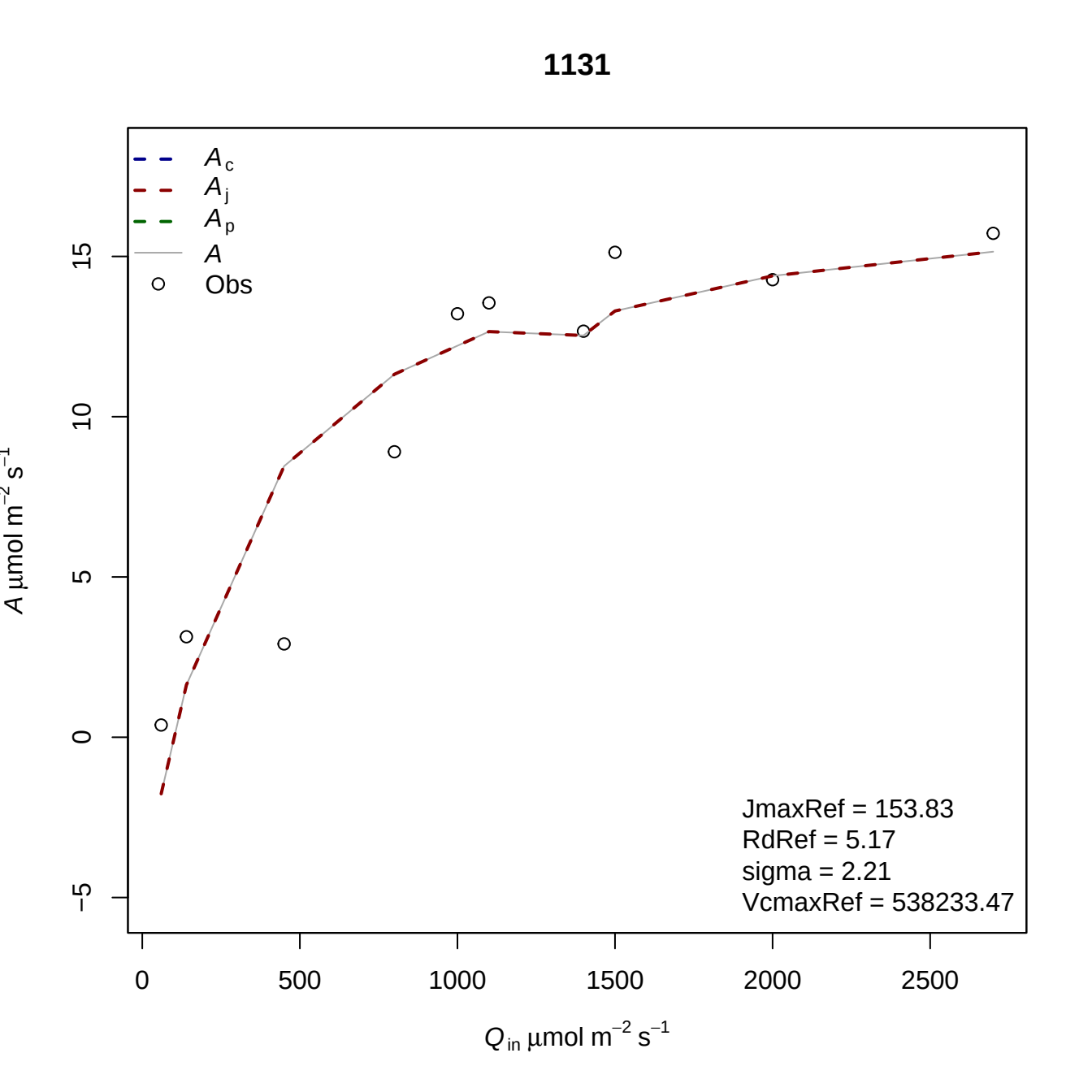

1125
A_c
A_j
A_p
A
Obs
JmaxRef = 60.82
RdRef = 1.15
sigma = 0.74
VcmaxRef = 19.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
100
200
300
400
500
600

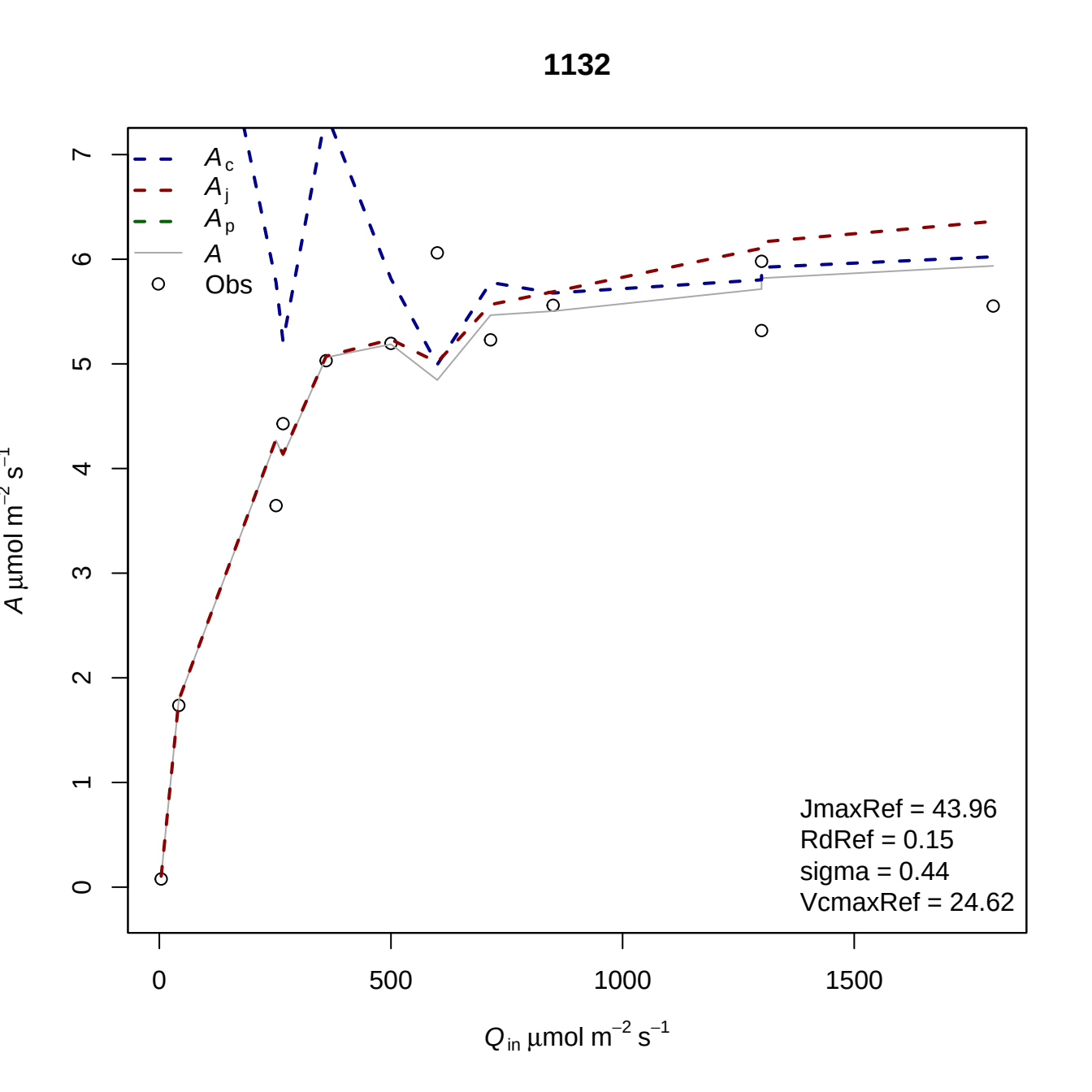

1126
A_c
A_j
A_p
A
Obs
JmaxRef = 143.5
RdRef = 3.22
sigma = 0.96
VcmaxRef = 16131.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
200
400
600
800
1000
1200

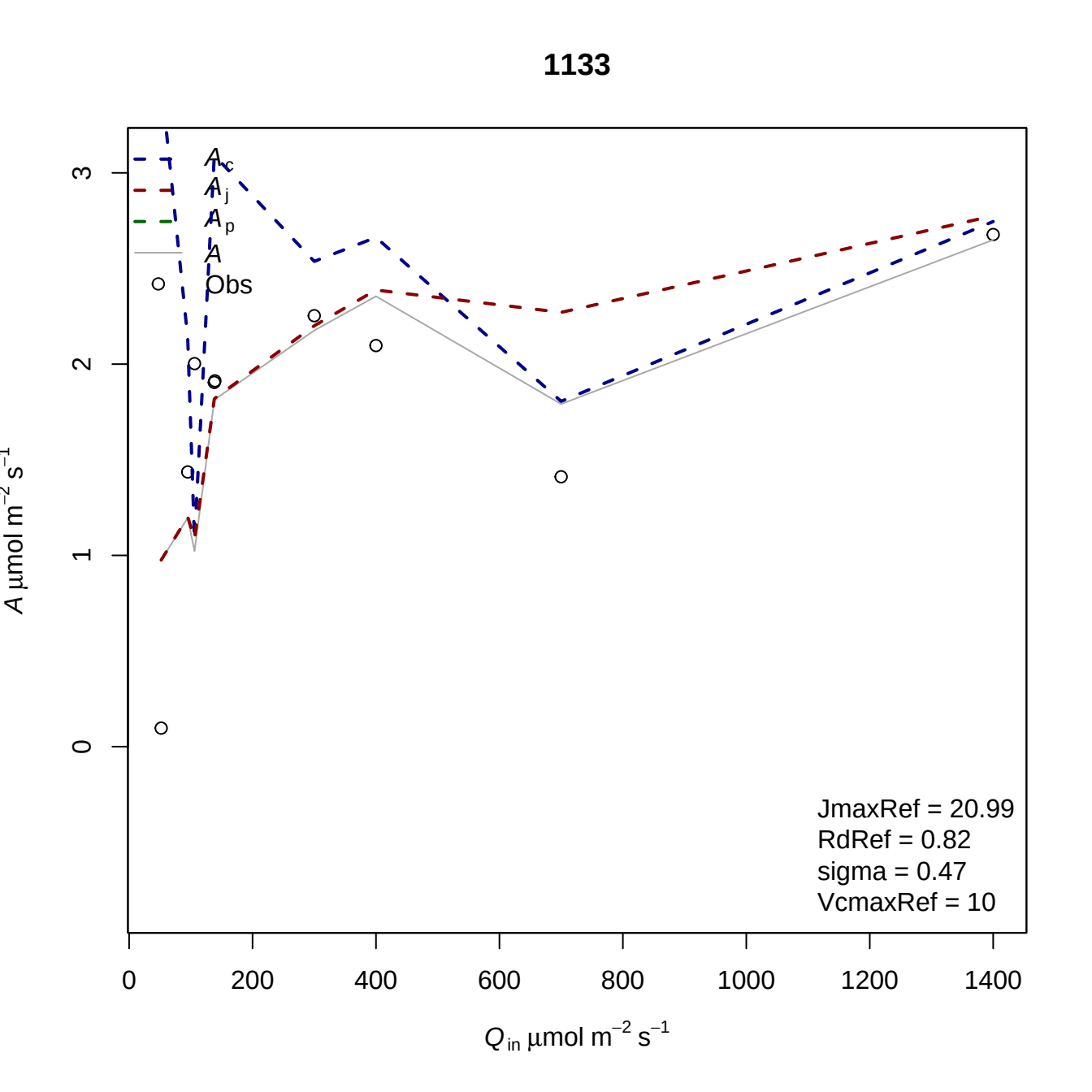

1127
A_c
A_j
A_p
A
Obs
JmaxRef = 138351305.71
RdRef = 1.39
sigma = 1.21
VcmaxRef = 50.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

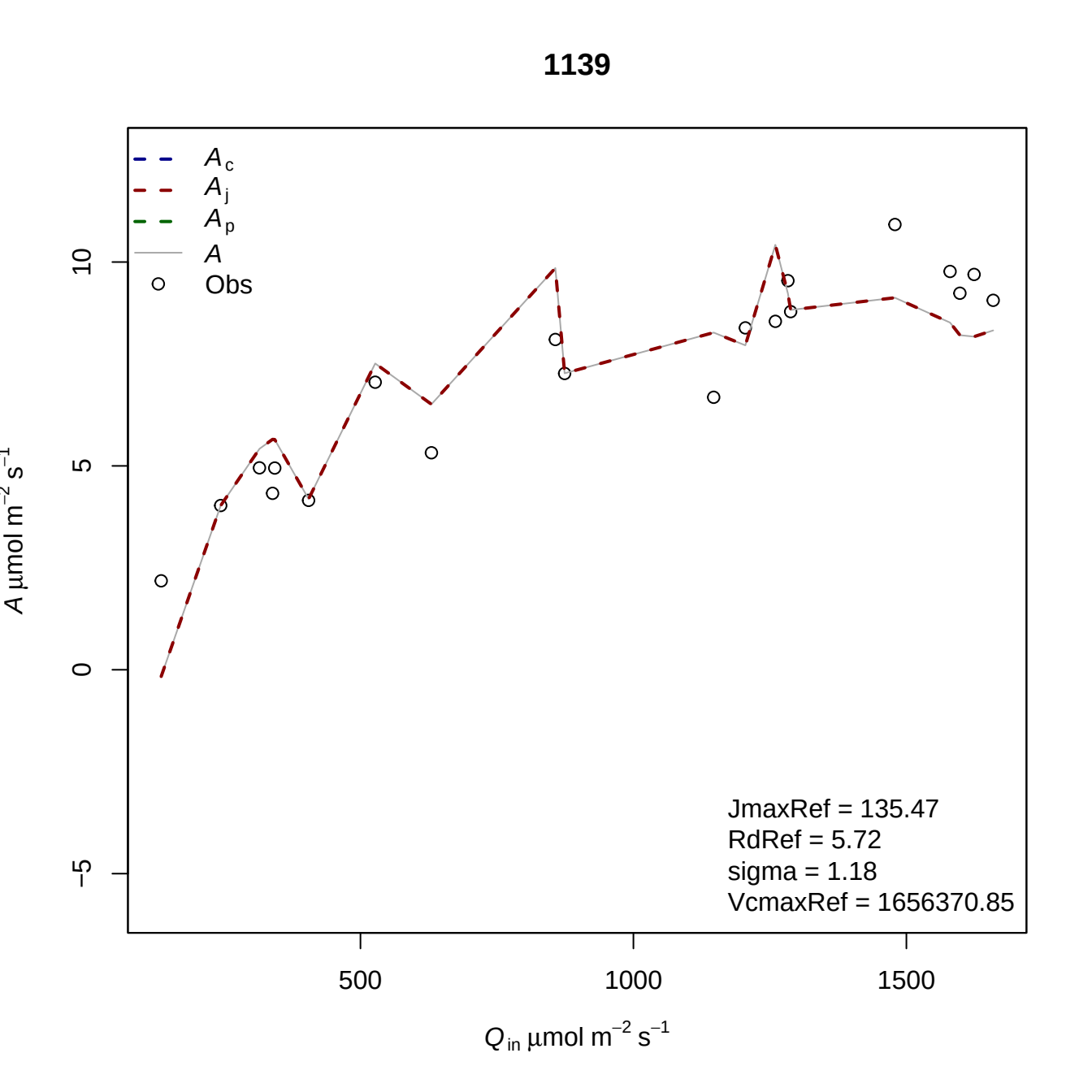

1128
A_c
A_j
A_p
A
Obs
JmaxRef = 44.18
RdRef = 0.45
sigma = 2.35
VcmaxRef = 442100.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

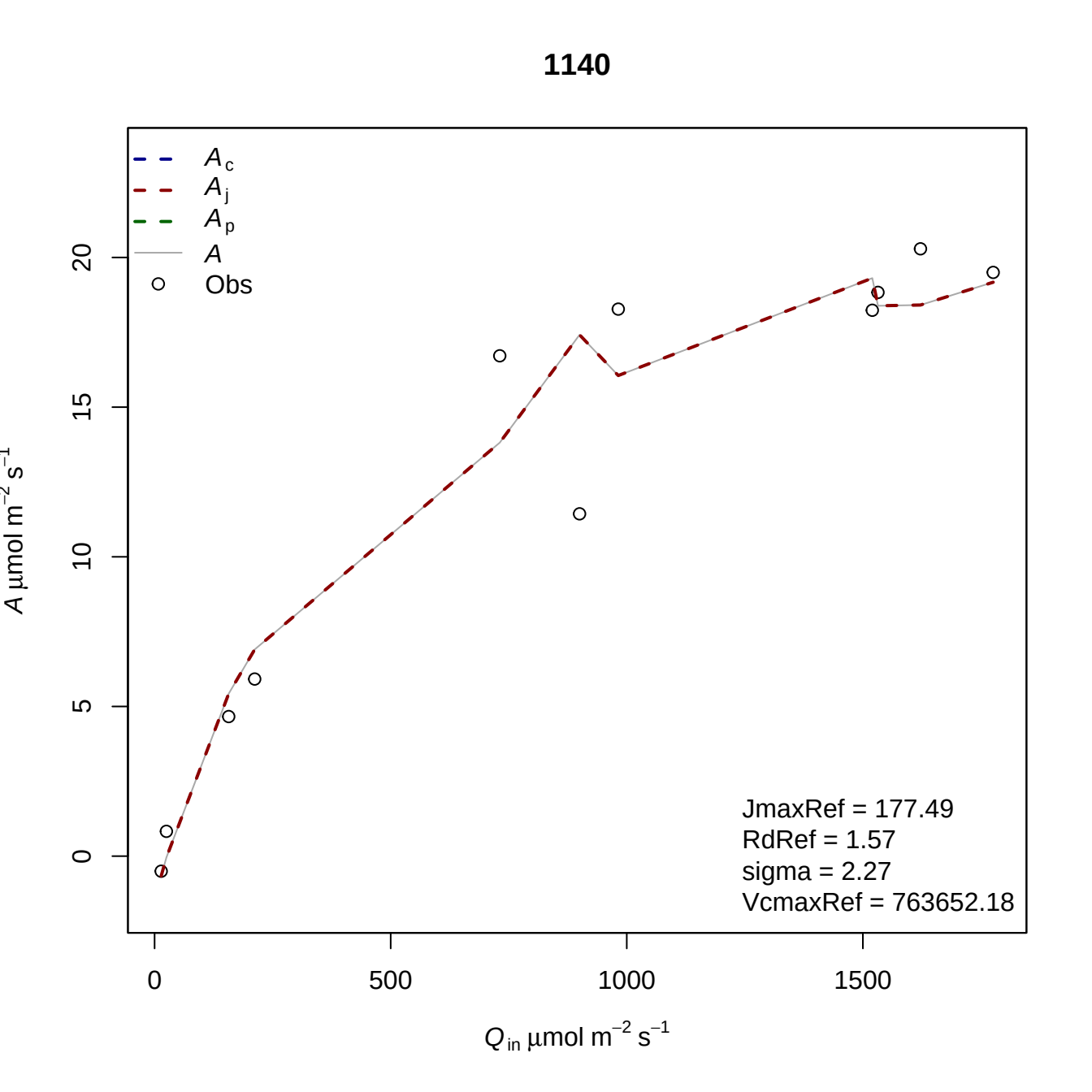

1129
A_c
A_j
A_p
A
Obs
JmaxRef = 332.77
RdRef = 1.11
sigma = 1.18
VcmaxRef = 14.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
50
100
150
200

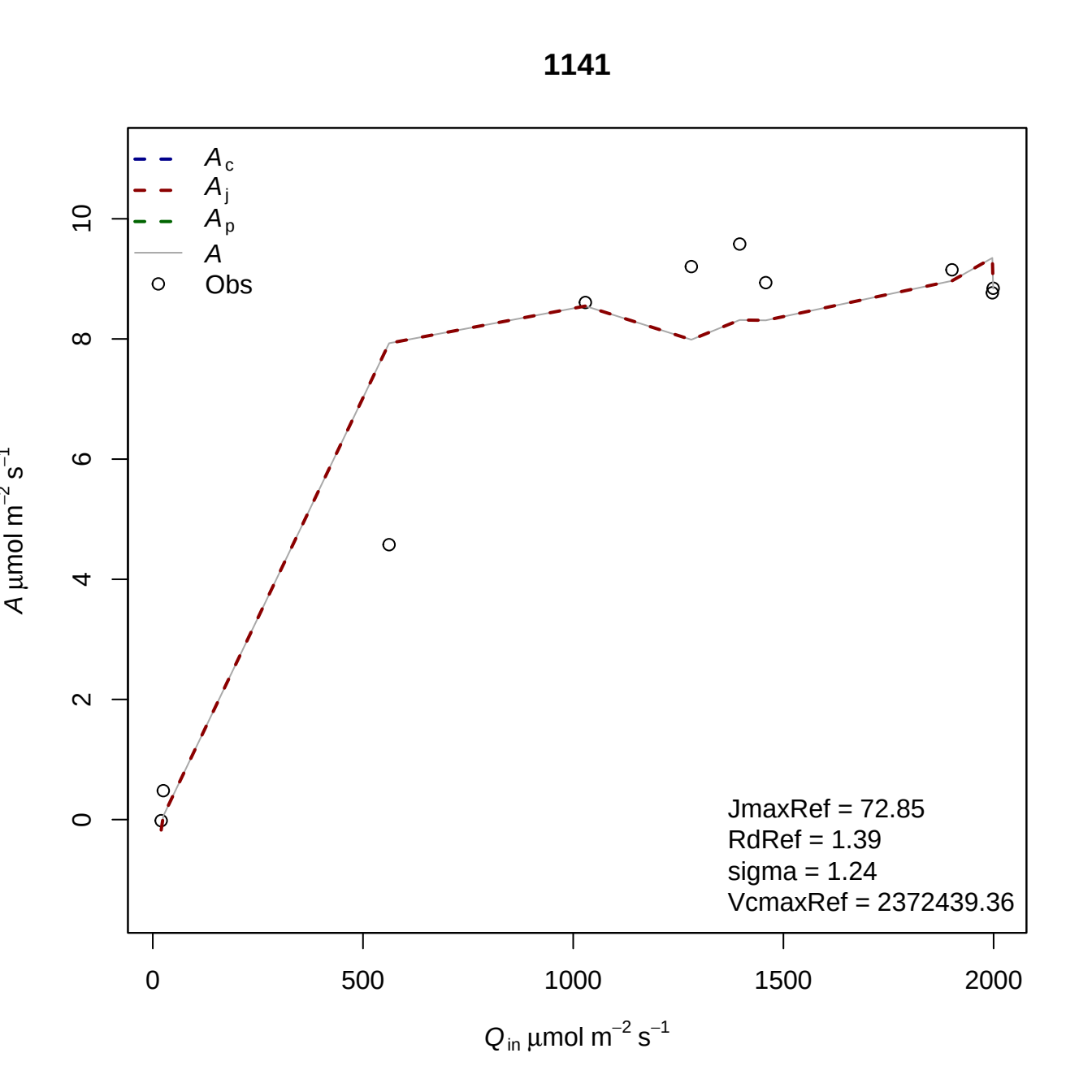

1131
A_c
A_j
A_p
A
Obs
JmaxRef = 153.83
RdRef = 5.17
sigma = 2.21
VcmaxRef = 538233.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000
2500

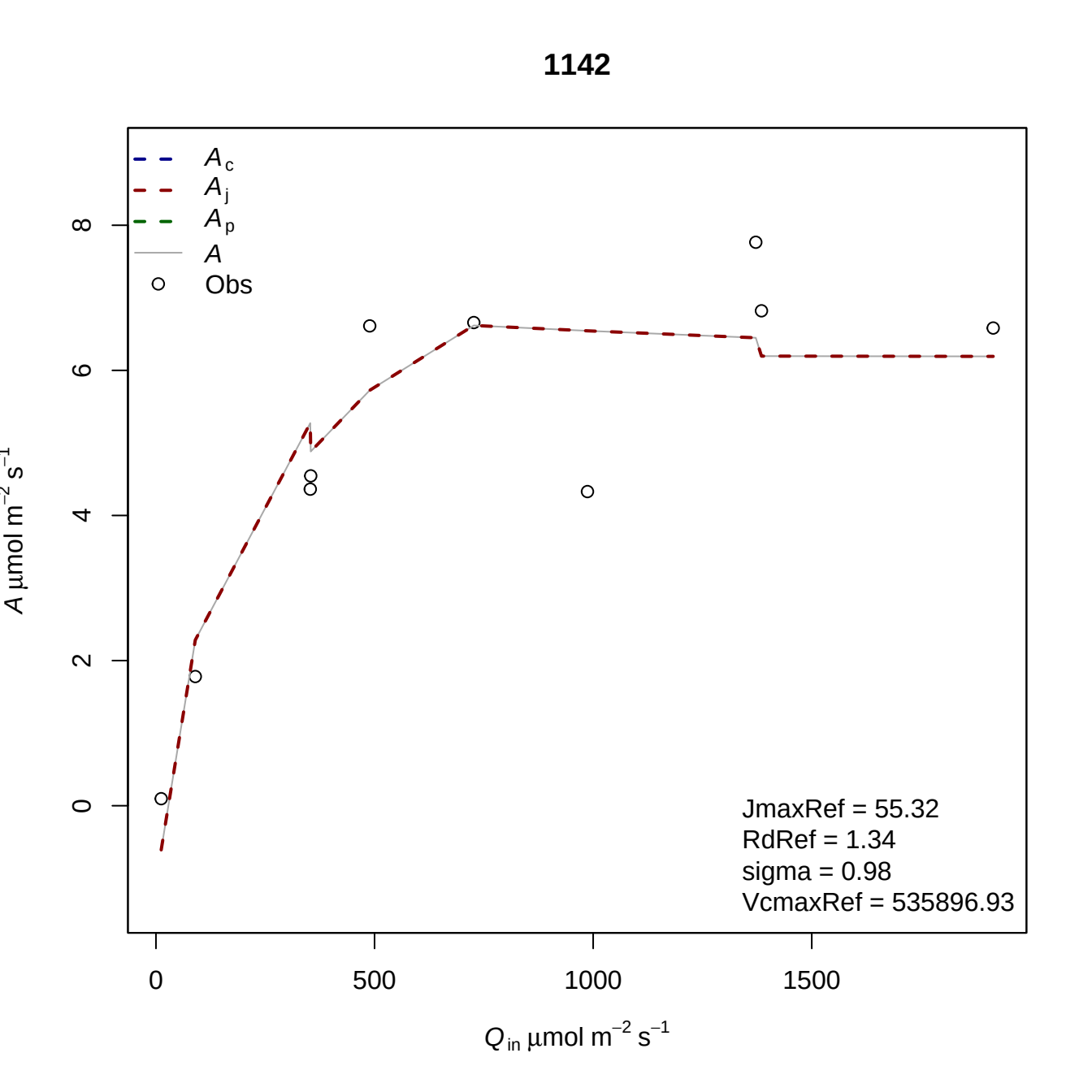

1132
A_c
A_j
A_p
A
Obs
JmaxRef = 43.96
RdRef = 0.15
sigma = 0.44
VcmaxRef = 24.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
7
0
500
1000
1500

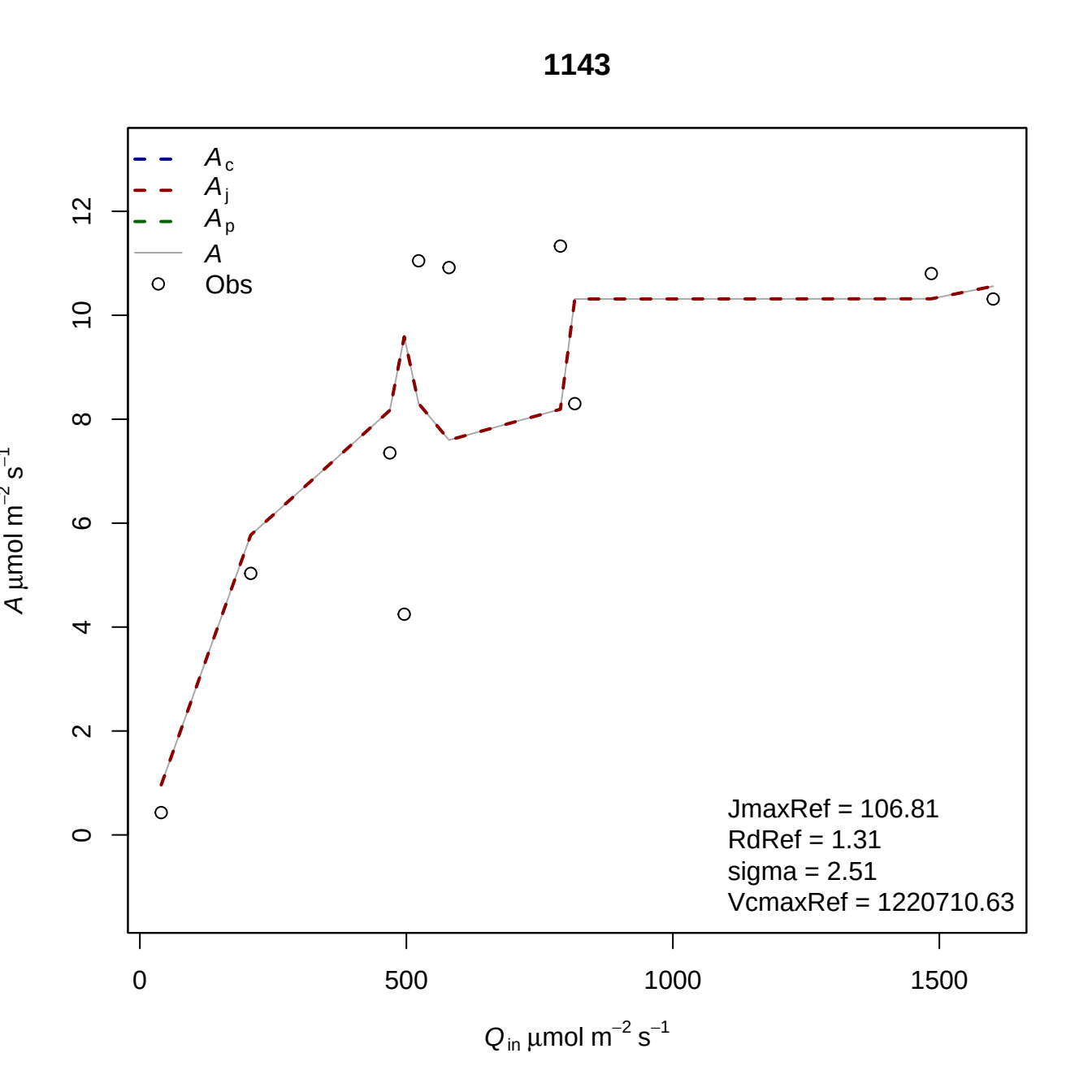

1133
A_c
A_j
A_p
A
Obs
JmaxRef = 20.99
RdRef = 0.82
sigma = 0.47
VcmaxRef = 10
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
200
400
600
800
1000
1200
1400

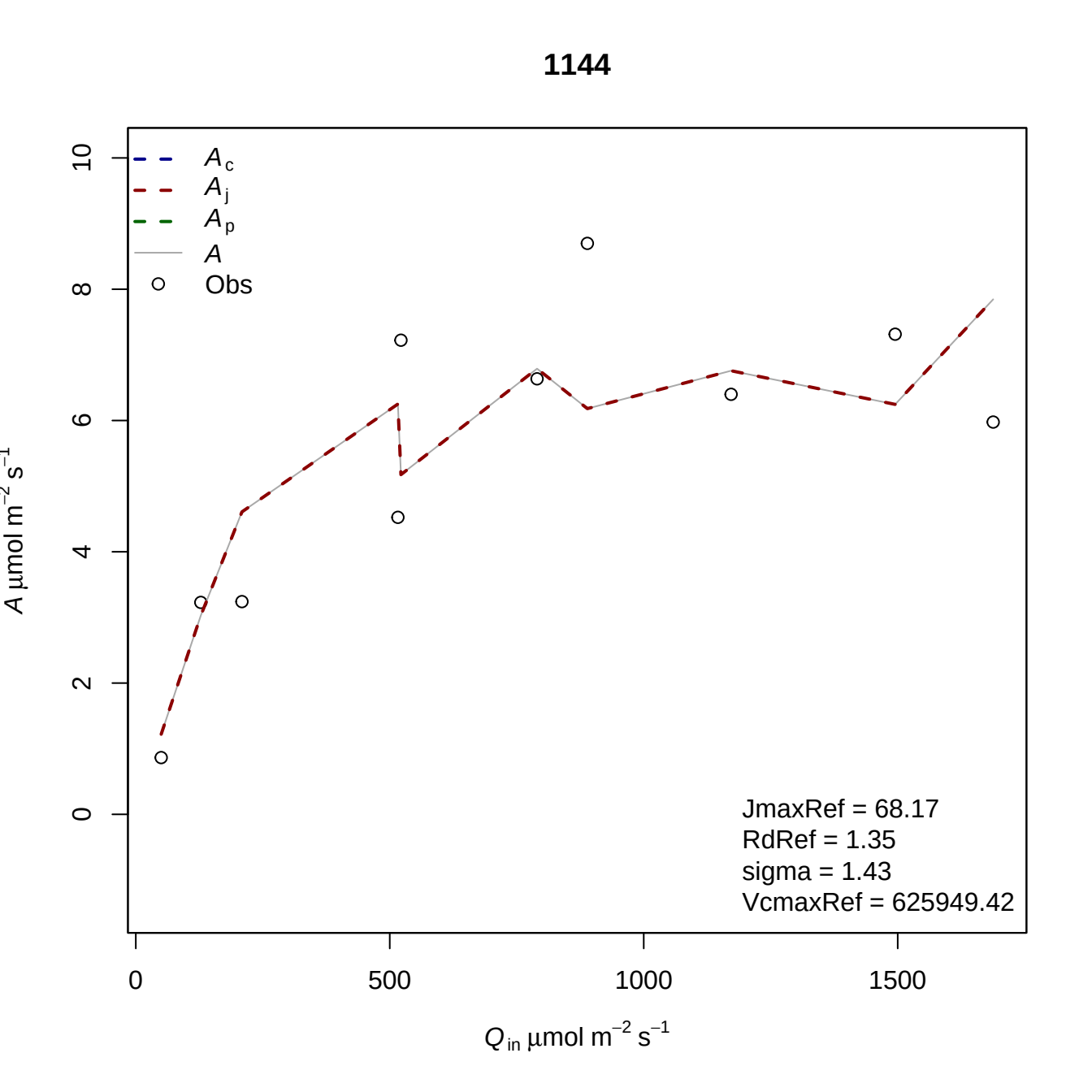

1139
A_c
A_j
A_p
A
Obs
JmaxRef = 135.47
RdRef = 5.72
sigma = 1.18
VcmaxRef = 1656370.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
500
1000
1500

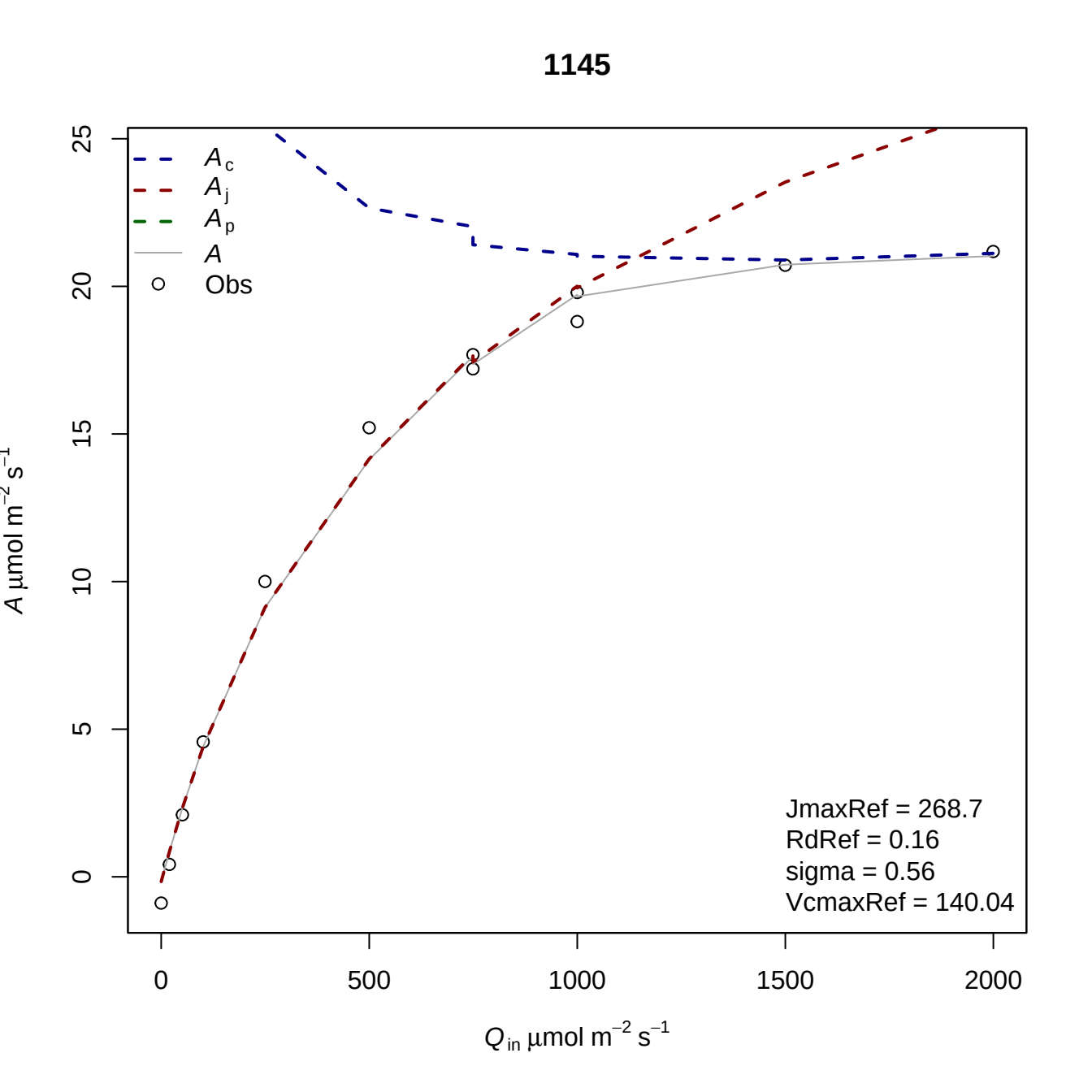

1140
A_c
A_j
A_p
A
Obs
JmaxRef = 177.49
RdRef = 1.57
sigma = 2.27
VcmaxRef = 763652.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

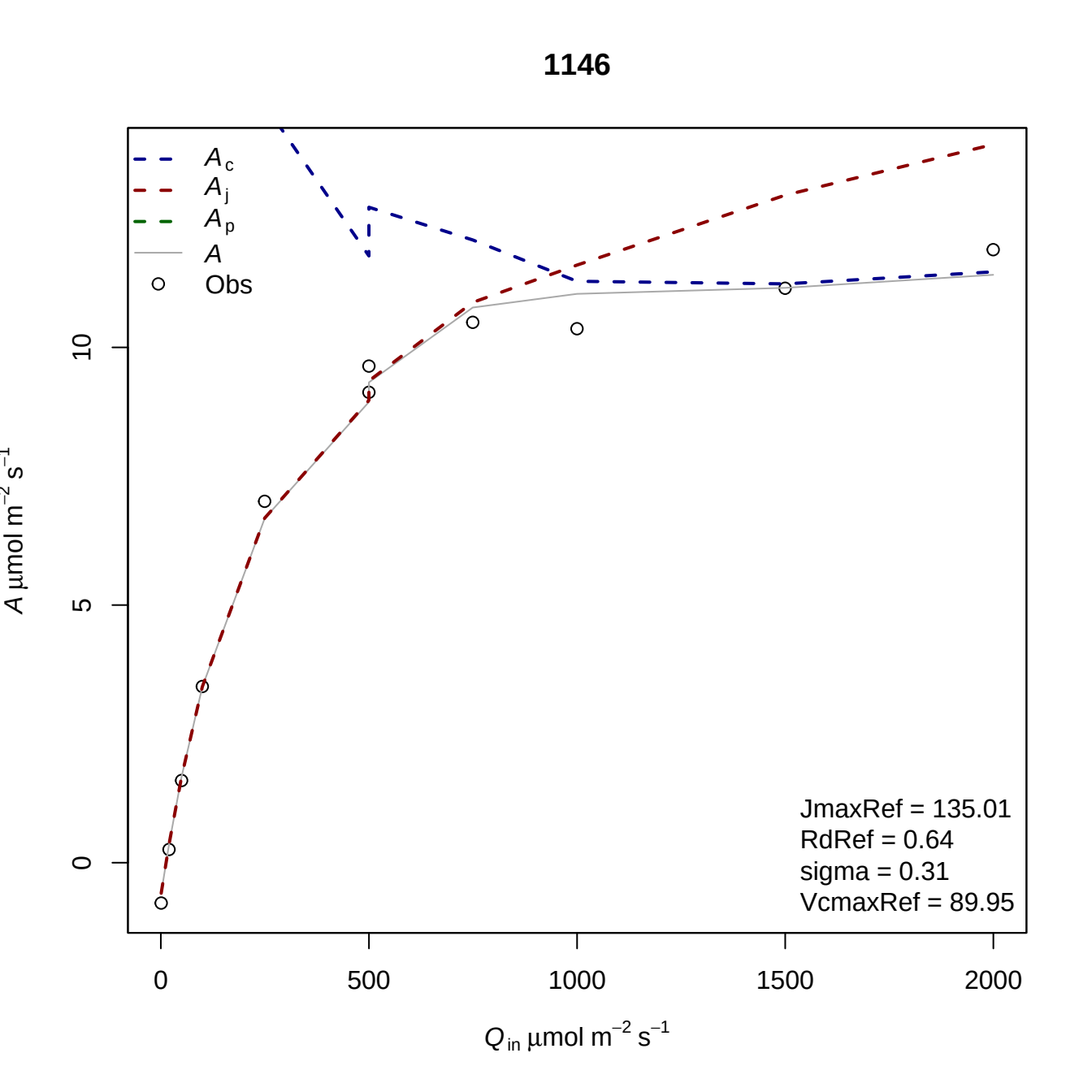

1141
A_c
A_j
A_p
A
Obs
JmaxRef = 72.85
RdRef = 1.39
sigma = 1.24
VcmaxRef = 2372439.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

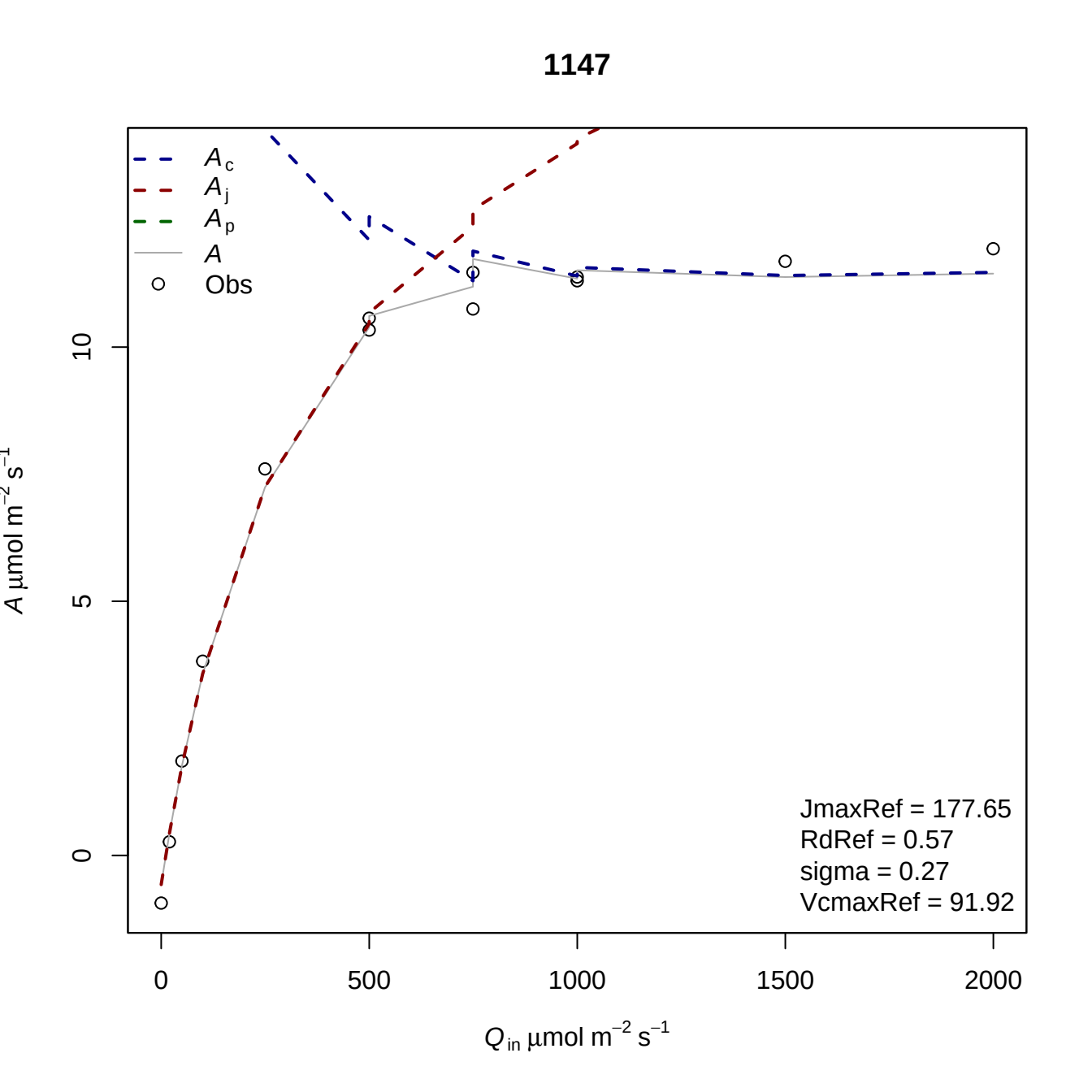

1142
A_c
A_j
A_p
A
Obs
JmaxRef = 55.32
RdRef = 1.34
sigma = 0.98
VcmaxRef = 535896.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

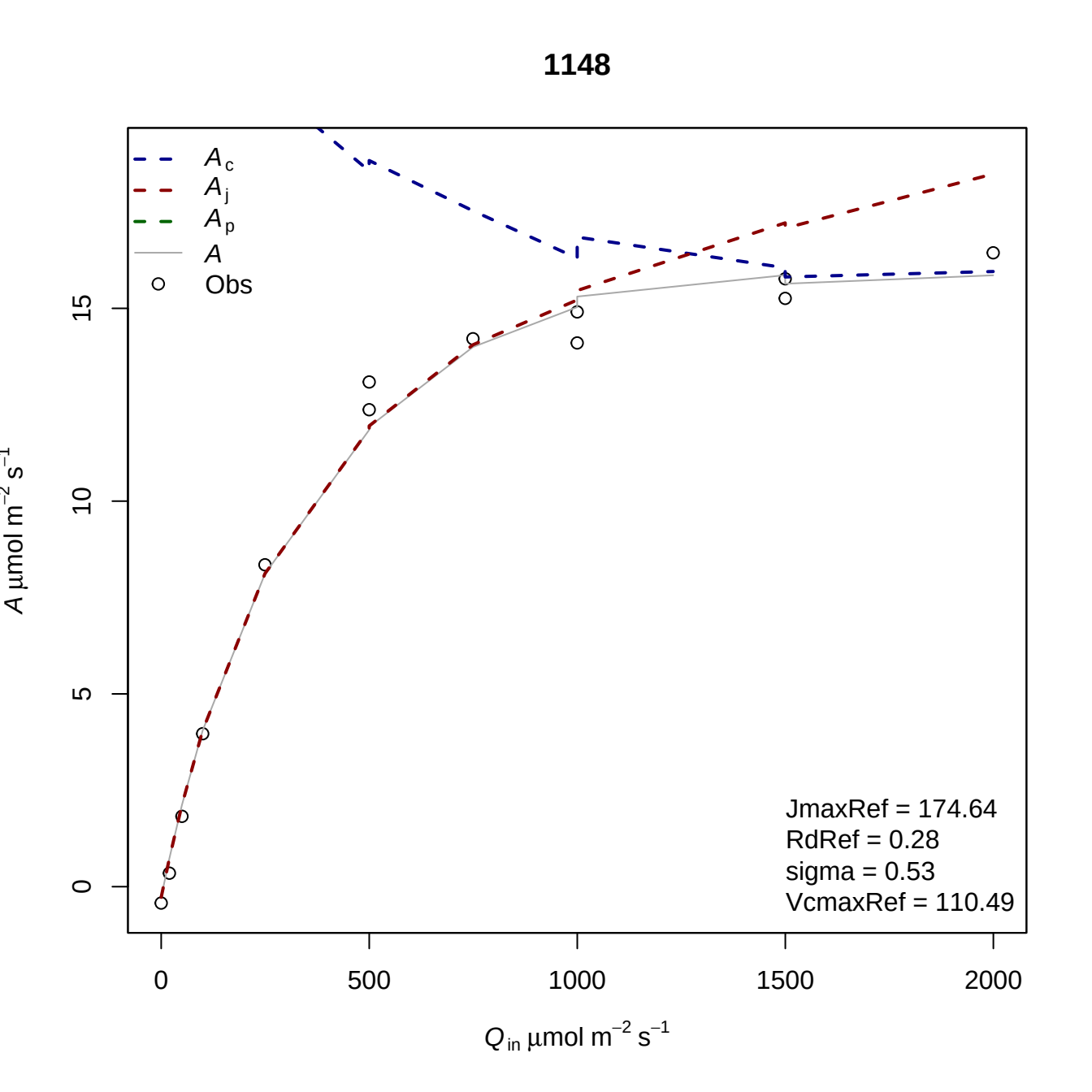

1143
A_c
A_j
A_p
A
Obs
JmaxRef = 106.81
RdRef = 1.31
sigma = 2.51
VcmaxRef = 1220710.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

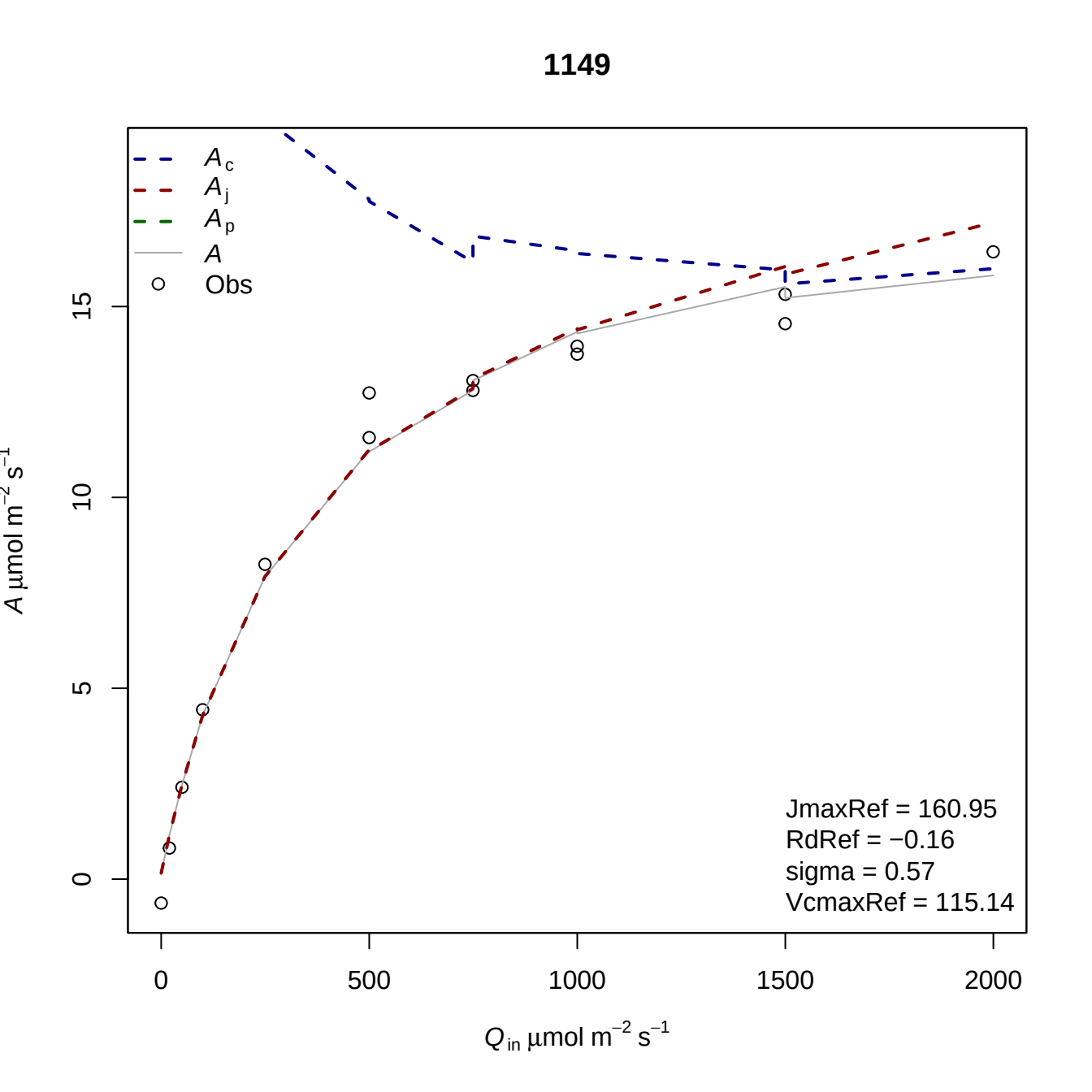

1144
A_c
A_j
A_p
A
Obs
JmaxRef = 68.17
RdRef = 1.35
sigma = 1.43
VcmaxRef = 625949.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

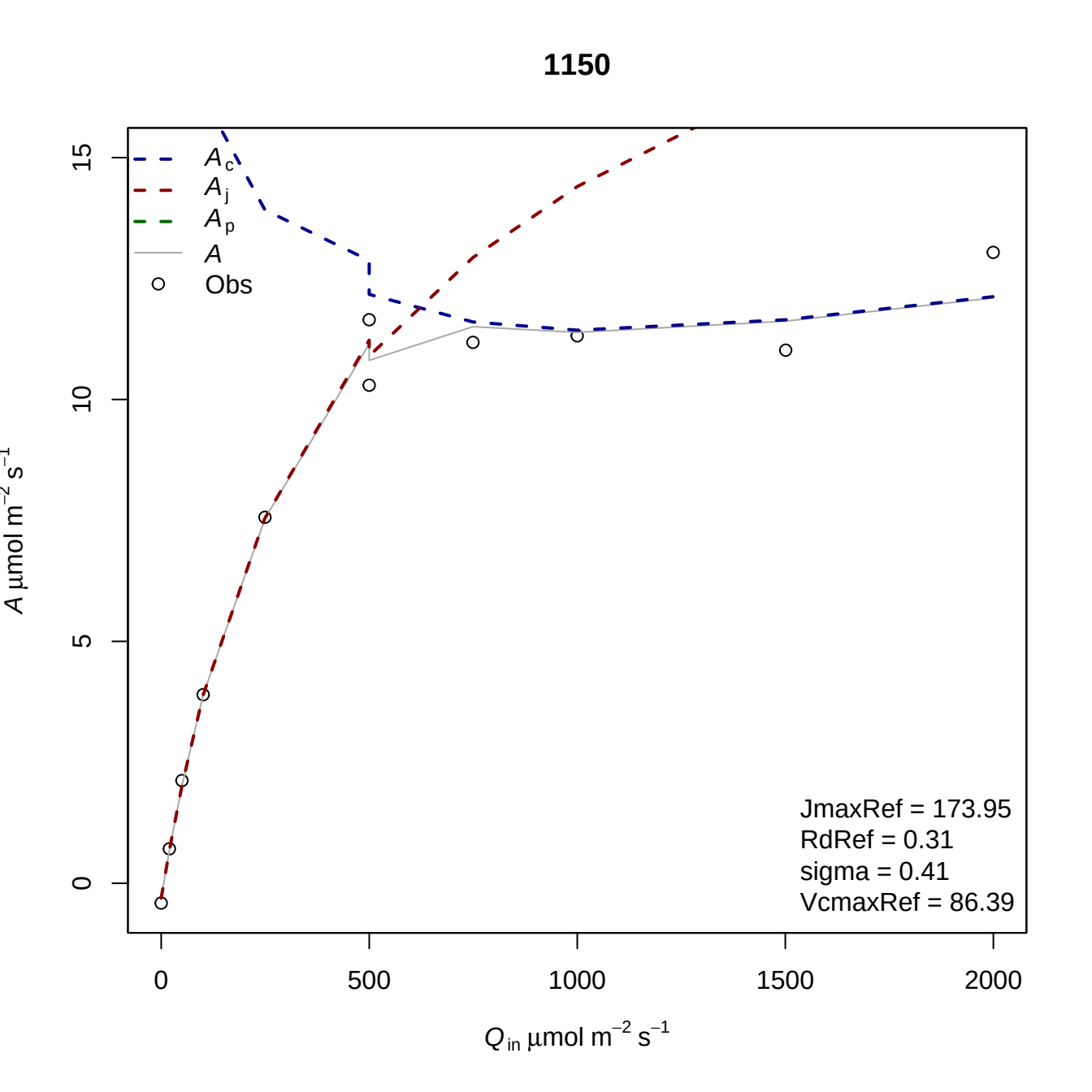

1145
A_c
A_j
A_p
A
Obs
JmaxRef = 268.7
RdRef = 0.16
sigma = 0.56
VcmaxRef = 140.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

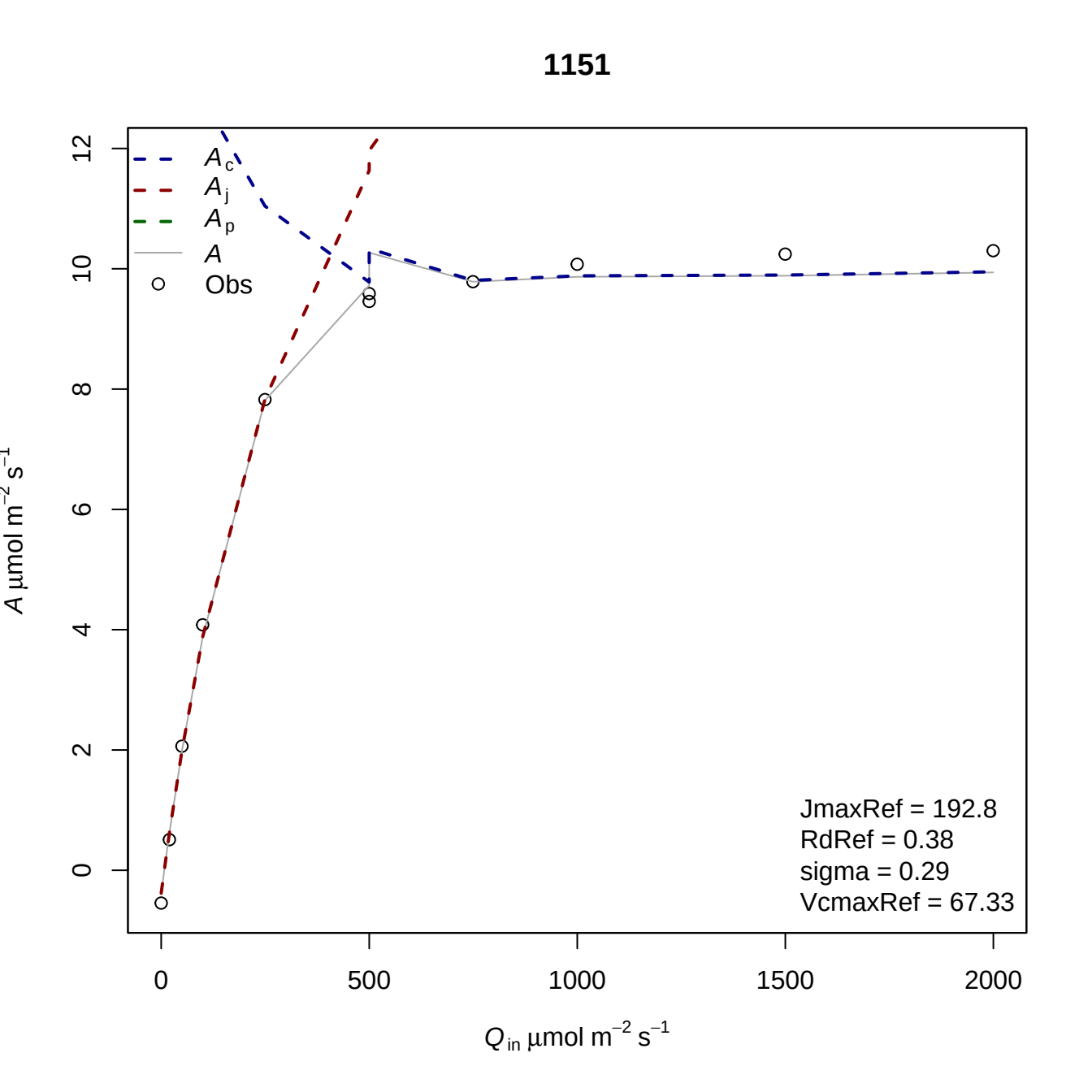

1146
A_c
A_j
A_p
A
Obs
JmaxRef = 135.01
RdRef = 0.64
sigma = 0.31
VcmaxRef = 89.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

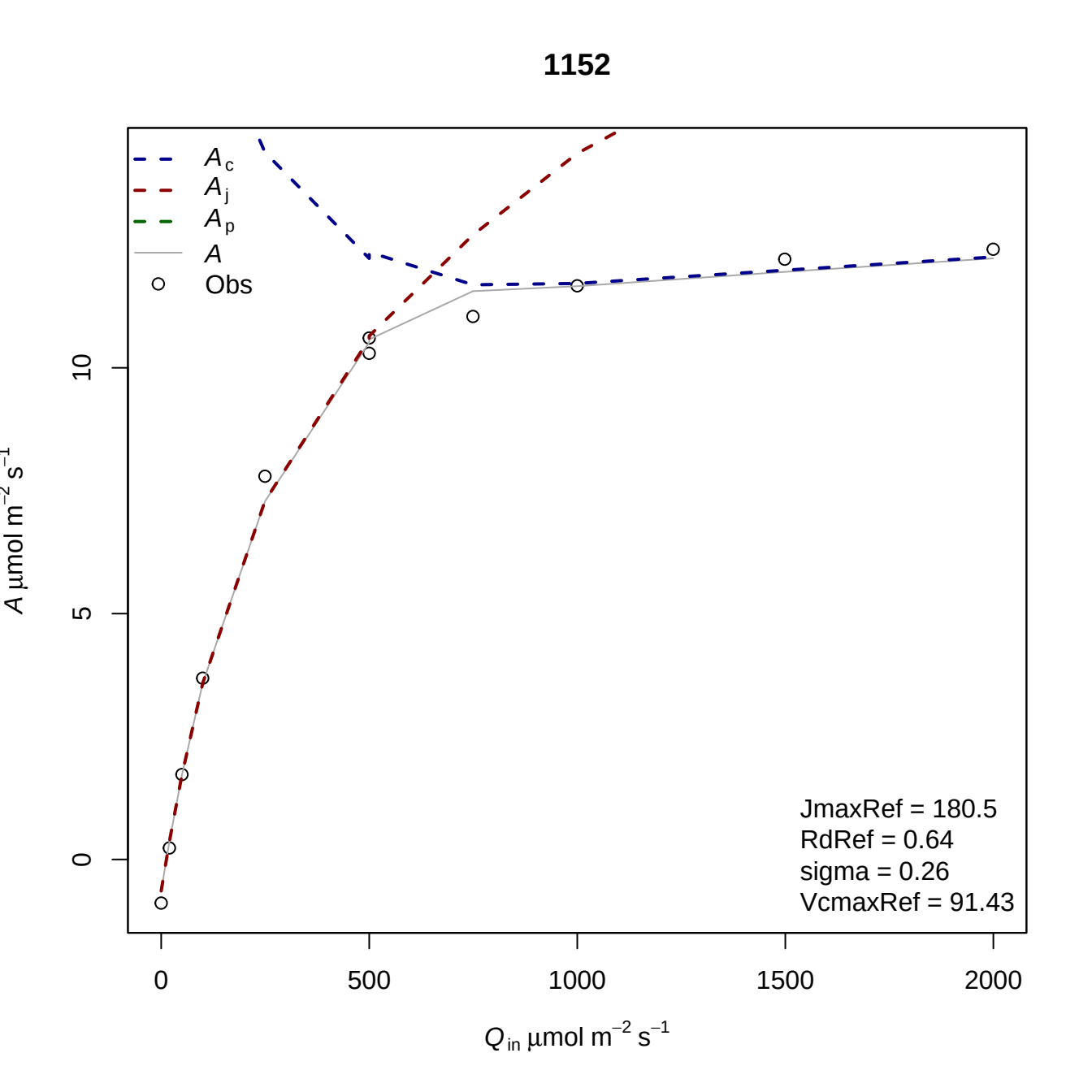

1147
A_c
A_j
A_p
A
Obs
JmaxRef = 177.65
RdRef = 0.57
sigma = 0.27
VcmaxRef = 91.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

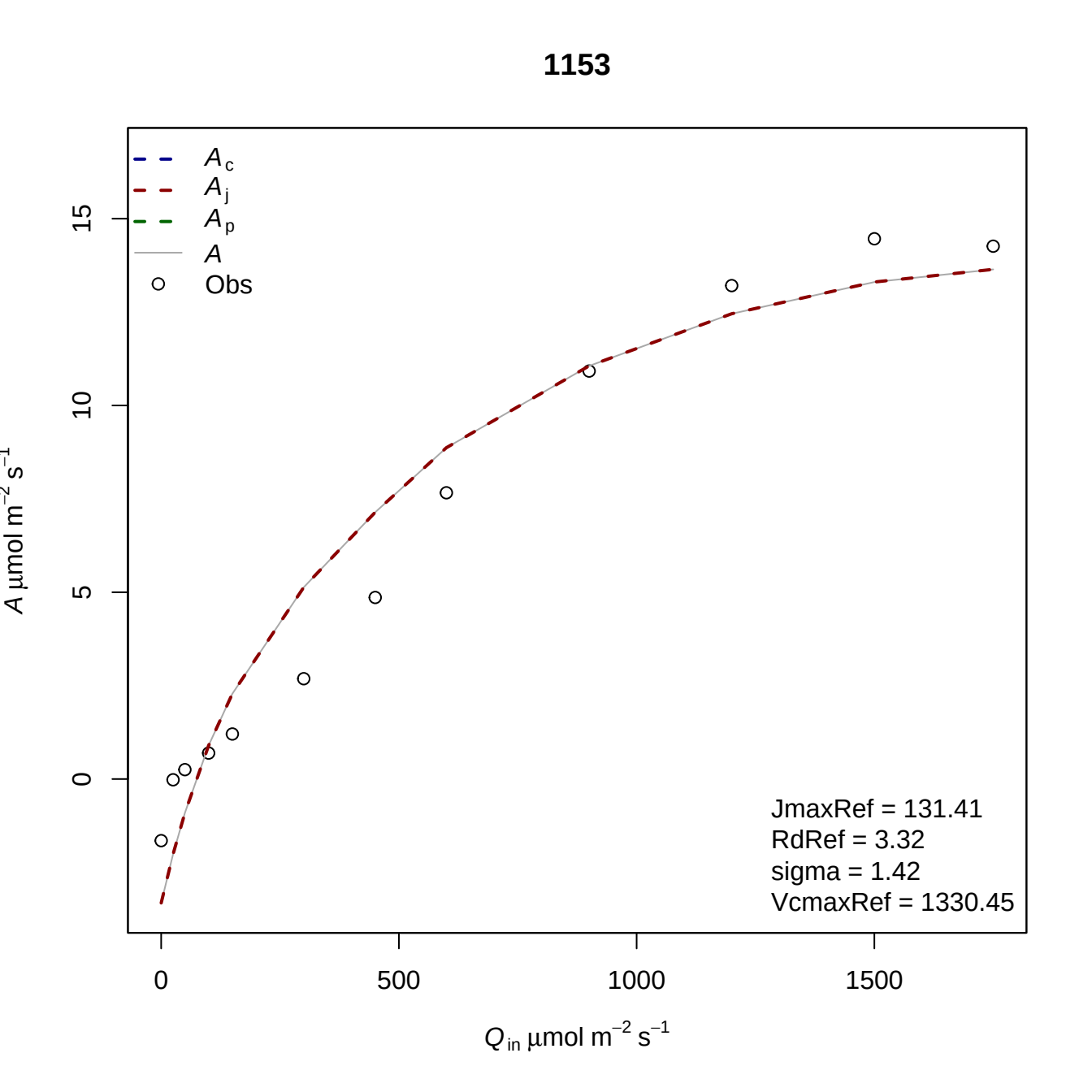

1148
A_c
A_j
A_p
A
Obs
JmaxRef = 174.64
RdRef = 0.28
sigma = 0.53
VcmaxRef = 110.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

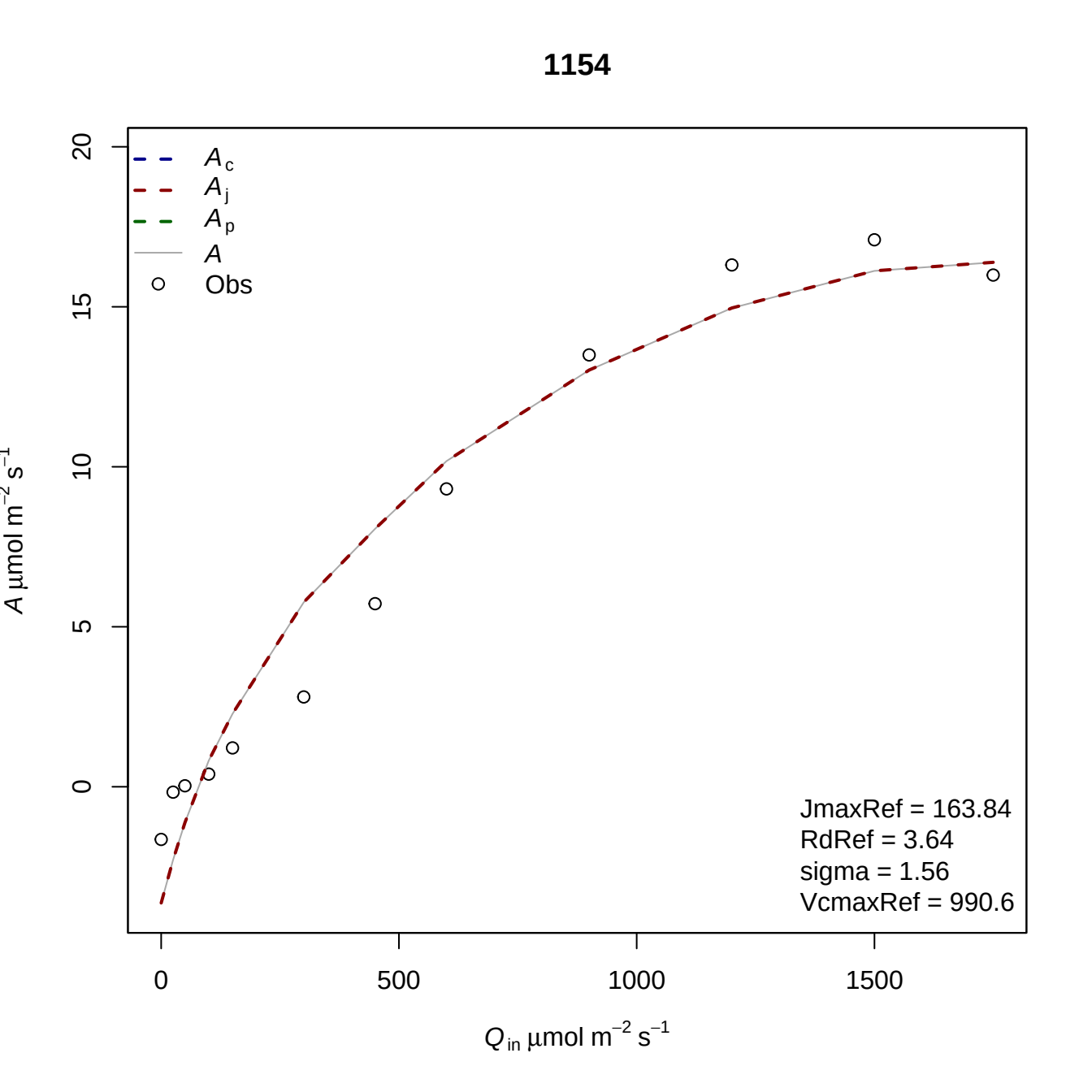

1149
A_c
A_j
A_p
A
Obs
JmaxRef = 160.95
RdRef = −0.16
sigma = 0.57
VcmaxRef = 115.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

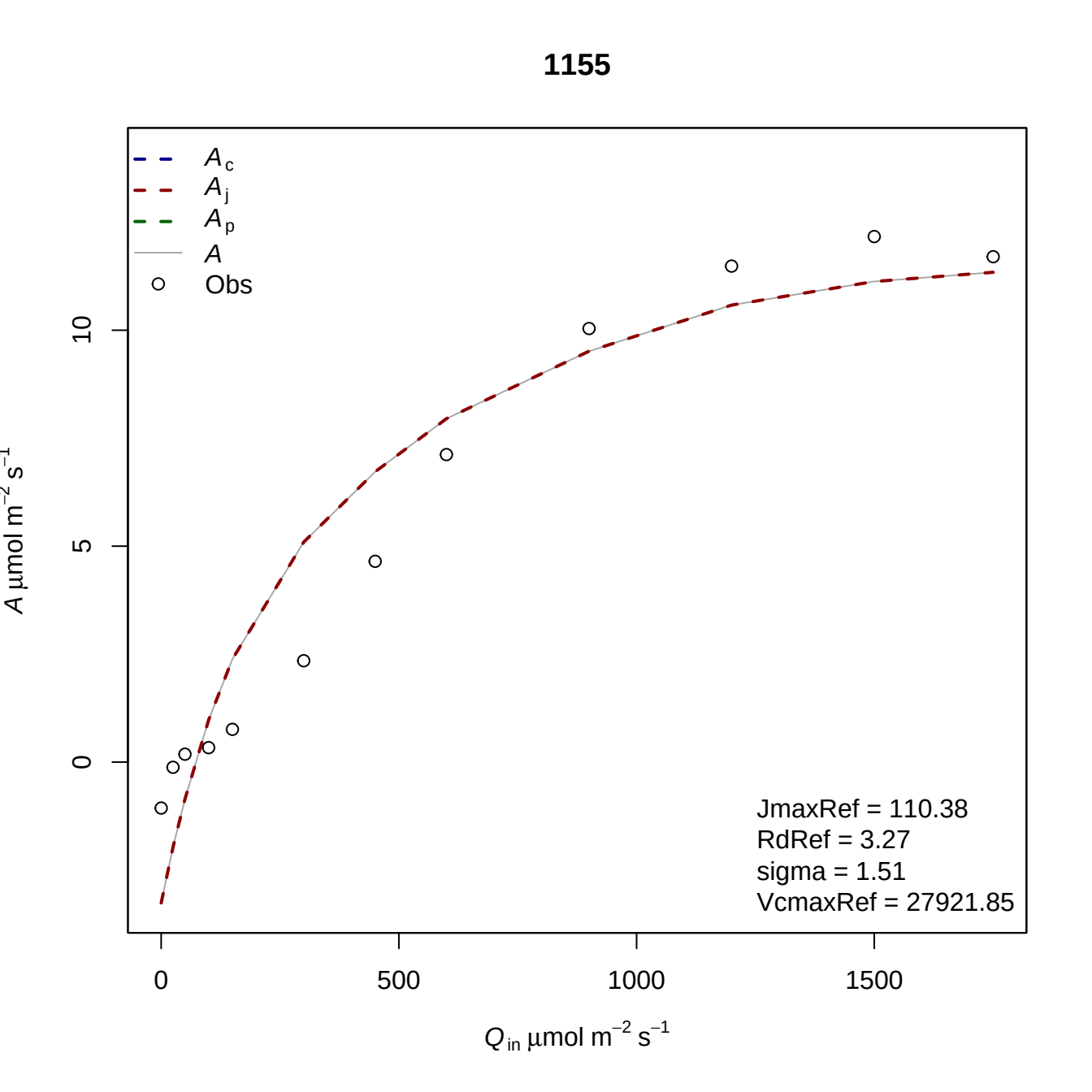

1150
A_c
A_j
A_p
A
Obs
JmaxRef = 173.95
RdRef = 0.31
sigma = 0.41
VcmaxRef = 86.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

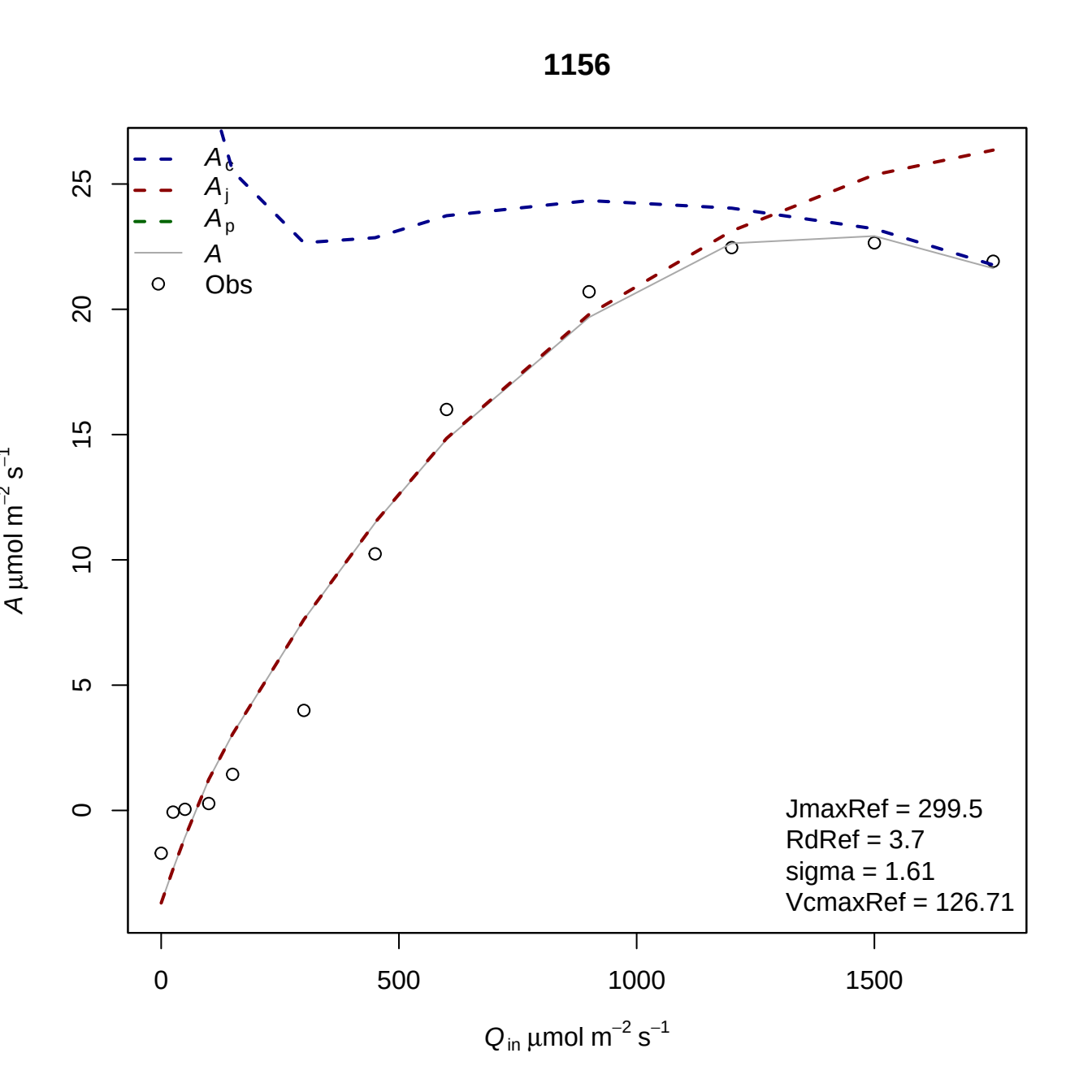

1151
A_c
A_j
A_p
A
Obs
JmaxRef = 192.8
RdRef = 0.38
sigma = 0.29
VcmaxRef = 67.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

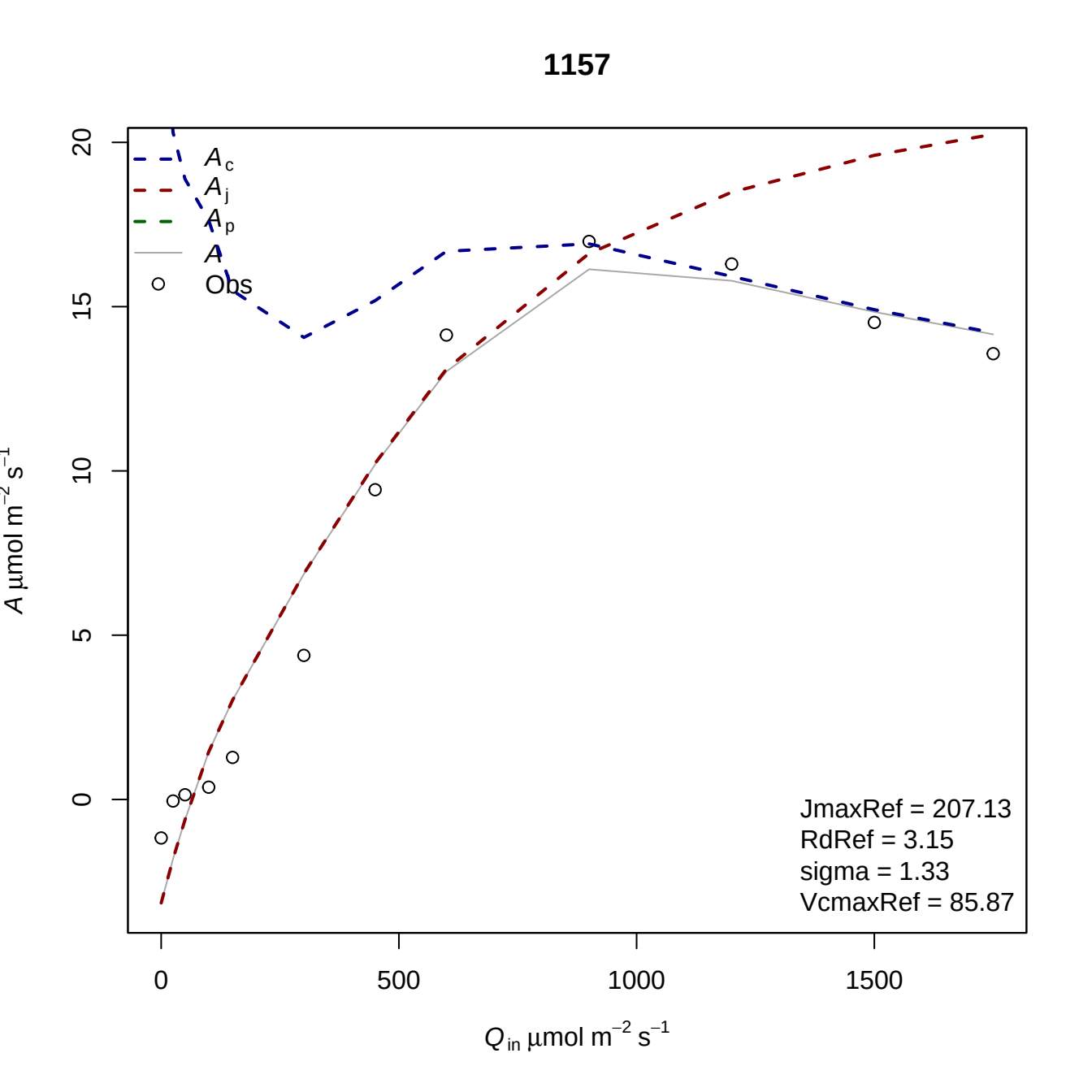

1152
A_c
A_j
A_p
A
Obs
JmaxRef = 180.5
RdRef = 0.64
sigma = 0.26
VcmaxRef = 91.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

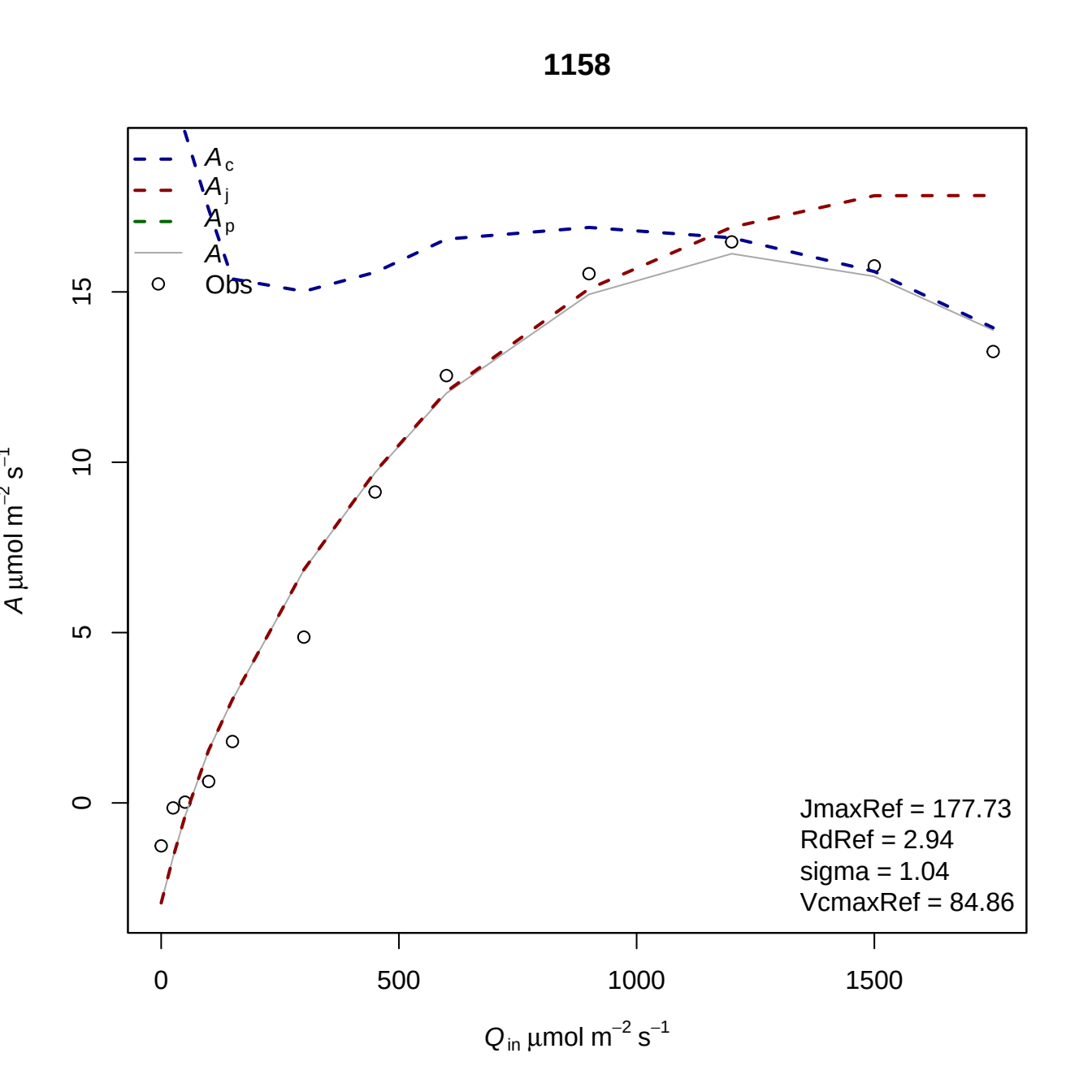

1153
A_c
A_j
A_p
A
Obs
JmaxRef = 131.41
RdRef = 3.32
sigma = 1.42
VcmaxRef = 1330.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

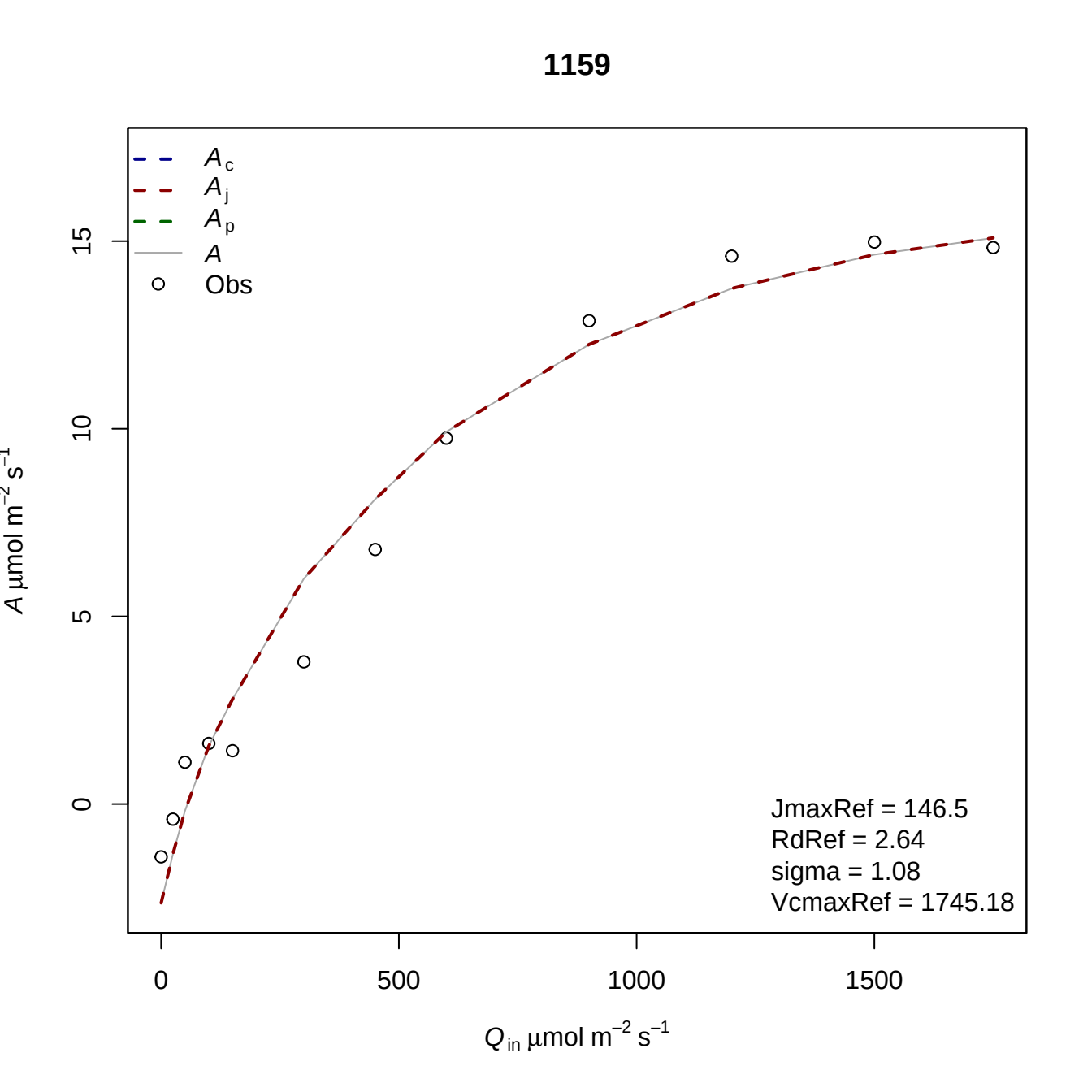

1154
A_c
A_j
A_p
A
Obs
JmaxRef = 163.84
RdRef = 3.64
sigma = 1.56
VcmaxRef = 990.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

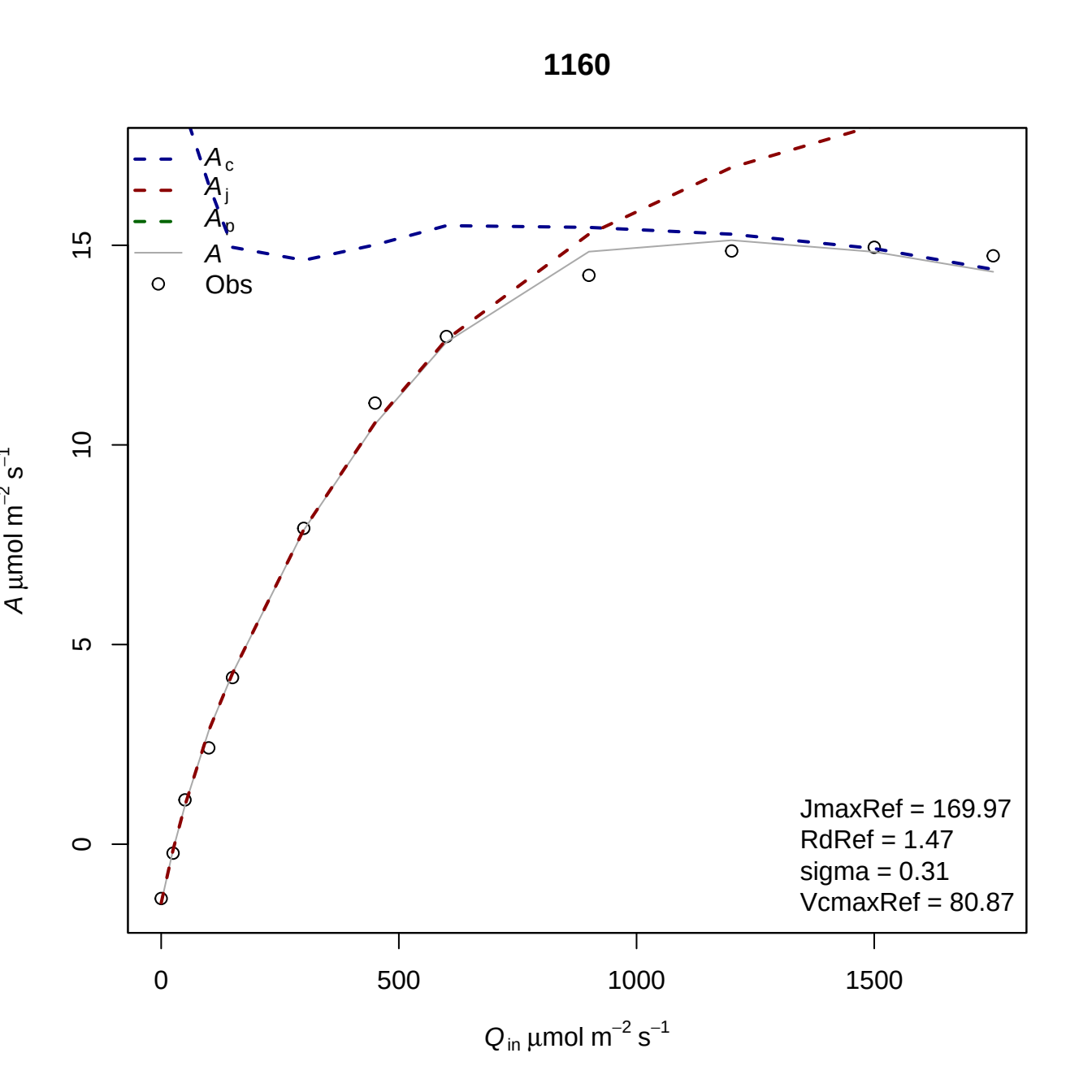

1155
A_c
A_j
A_p
A
Obs
JmaxRef = 110.38
RdRef = 3.27
sigma = 1.51
VcmaxRef = 27921.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

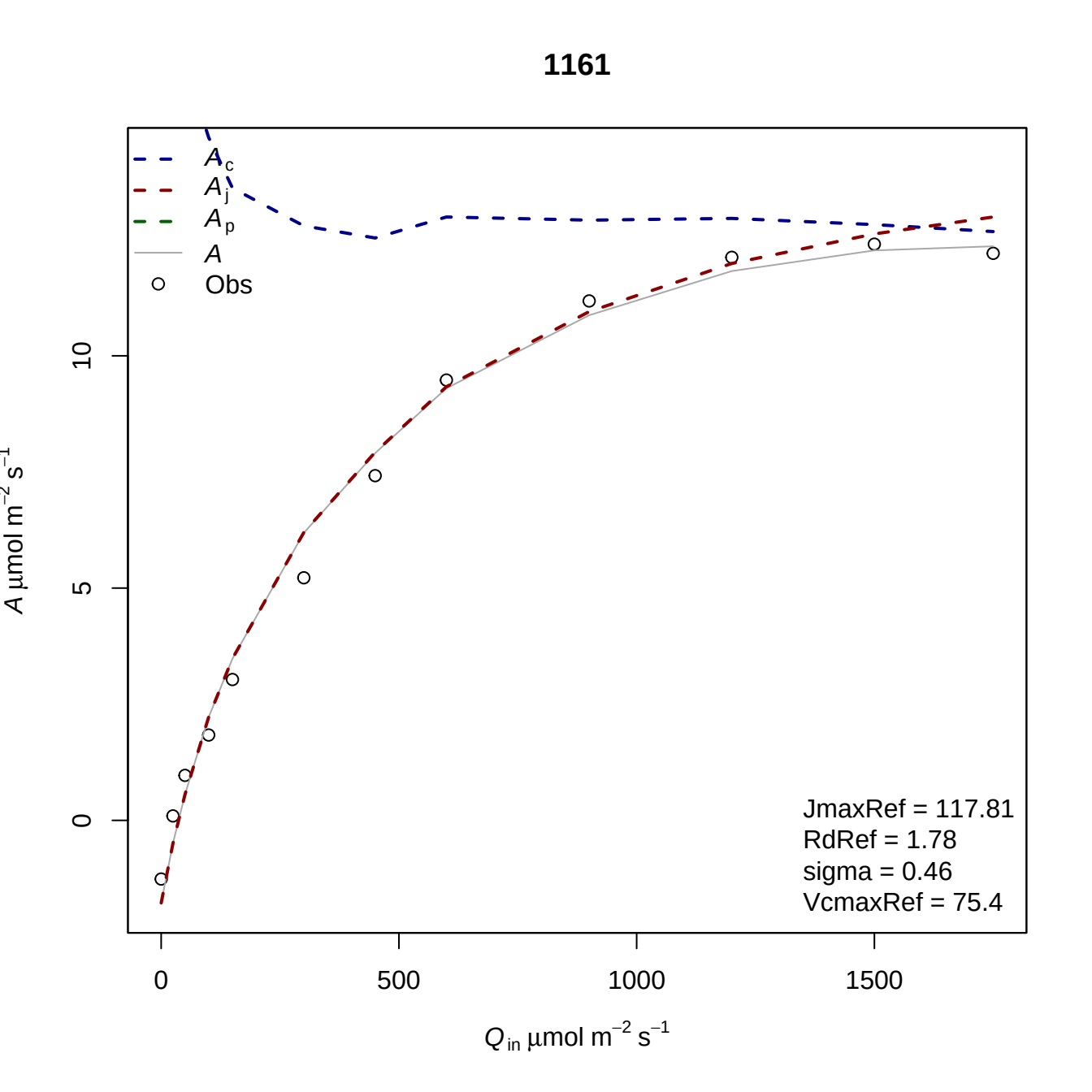

1156
A_c
A_j
A_p
A
Obs
JmaxRef = 299.5
RdRef = 3.7
sigma = 1.61
VcmaxRef = 126.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

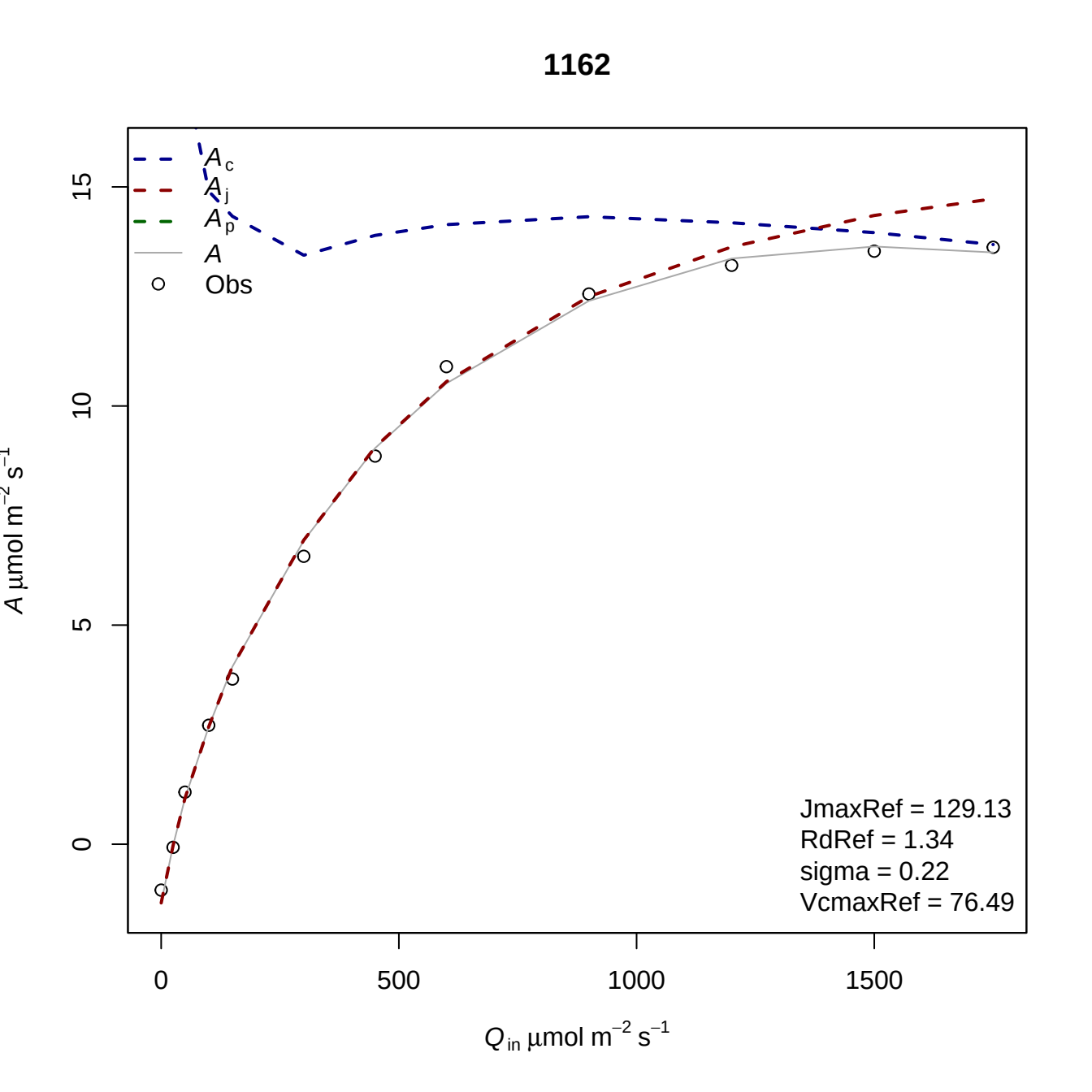

1157
A_c
A_j
A_p
A
Obs
JmaxRef = 207.13
RdRef = 3.15
sigma = 1.33
VcmaxRef = 85.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

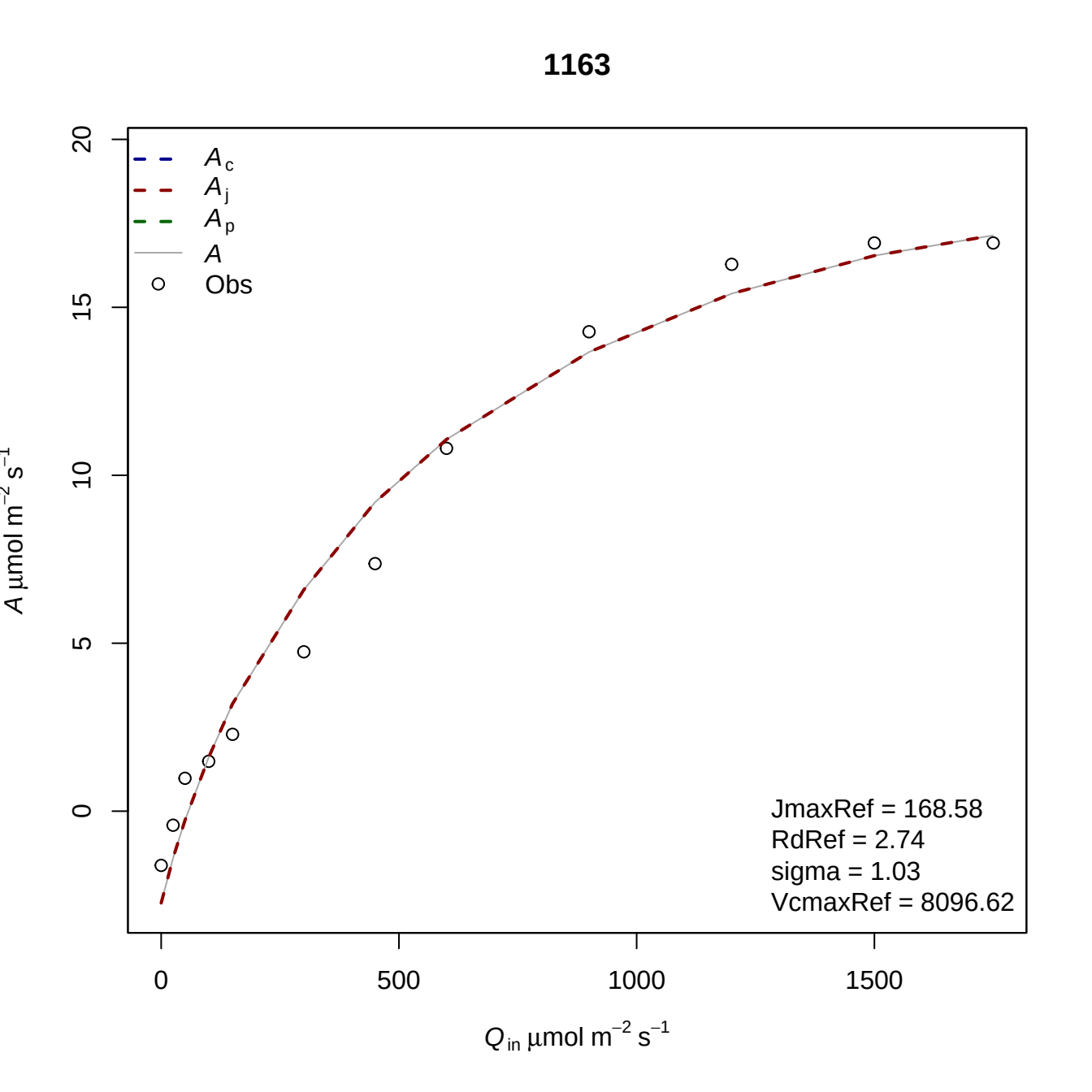

1158
A_c
A_j
A_p
A
Obs
JmaxRef = 177.73
RdRef = 2.94
sigma = 1.04
VcmaxRef = 84.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

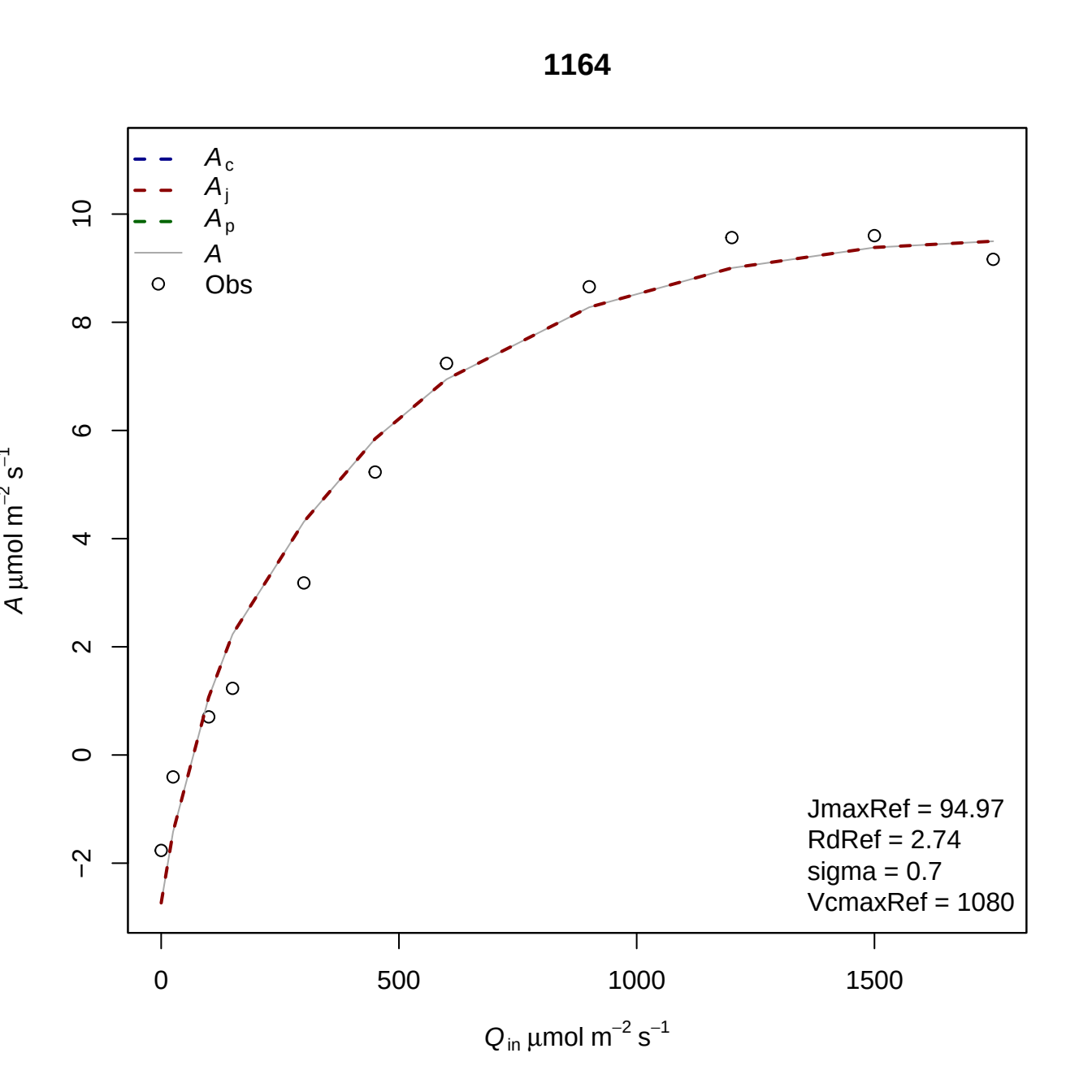

1159
A_c
A_j
A_p
A
Obs
JmaxRef = 146.5
RdRef = 2.64
sigma = 1.08
VcmaxRef = 1745.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

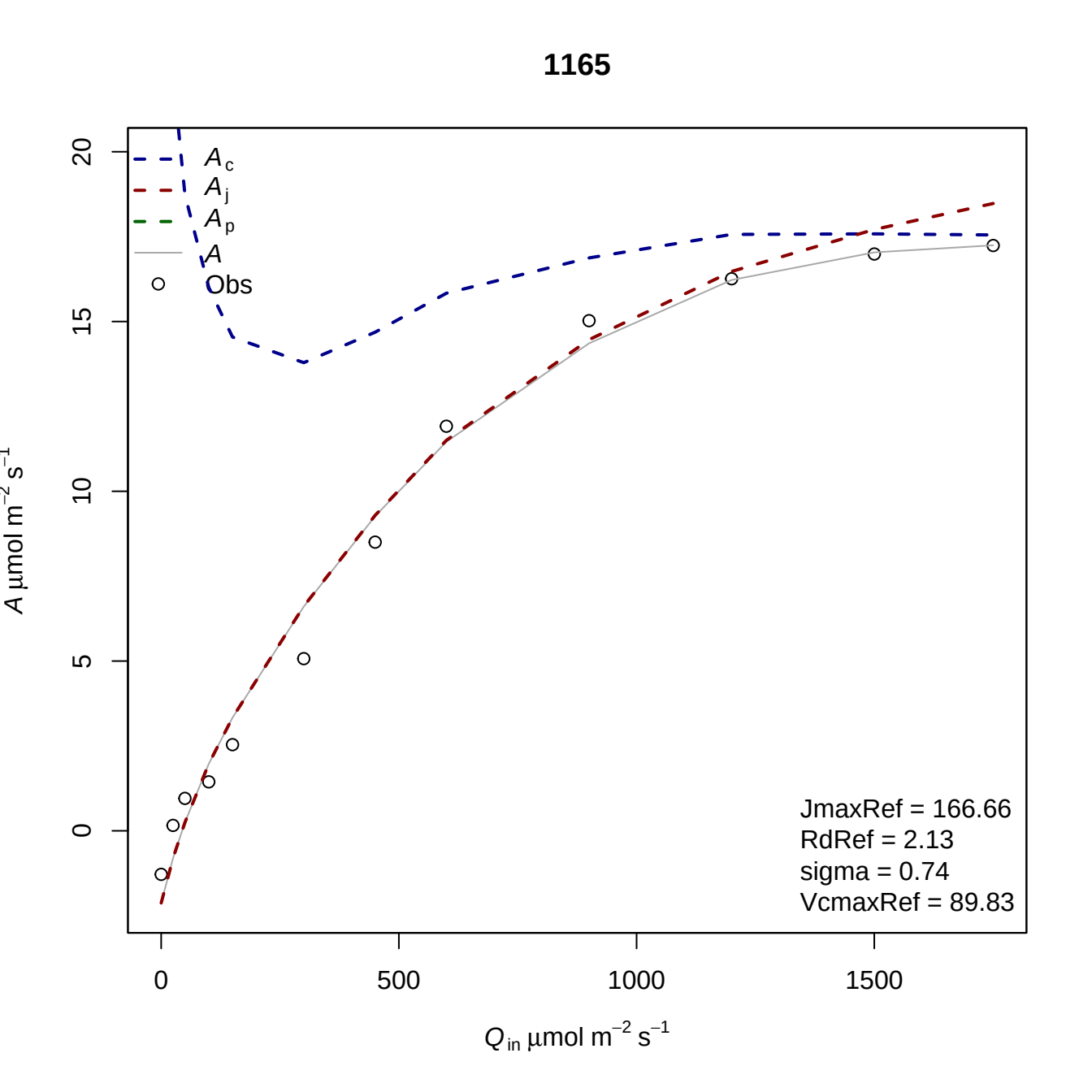

1160
A_c
A_j
A_p
A
Obs
JmaxRef = 169.97
RdRef = 1.47
sigma = 0.31
VcmaxRef = 80.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

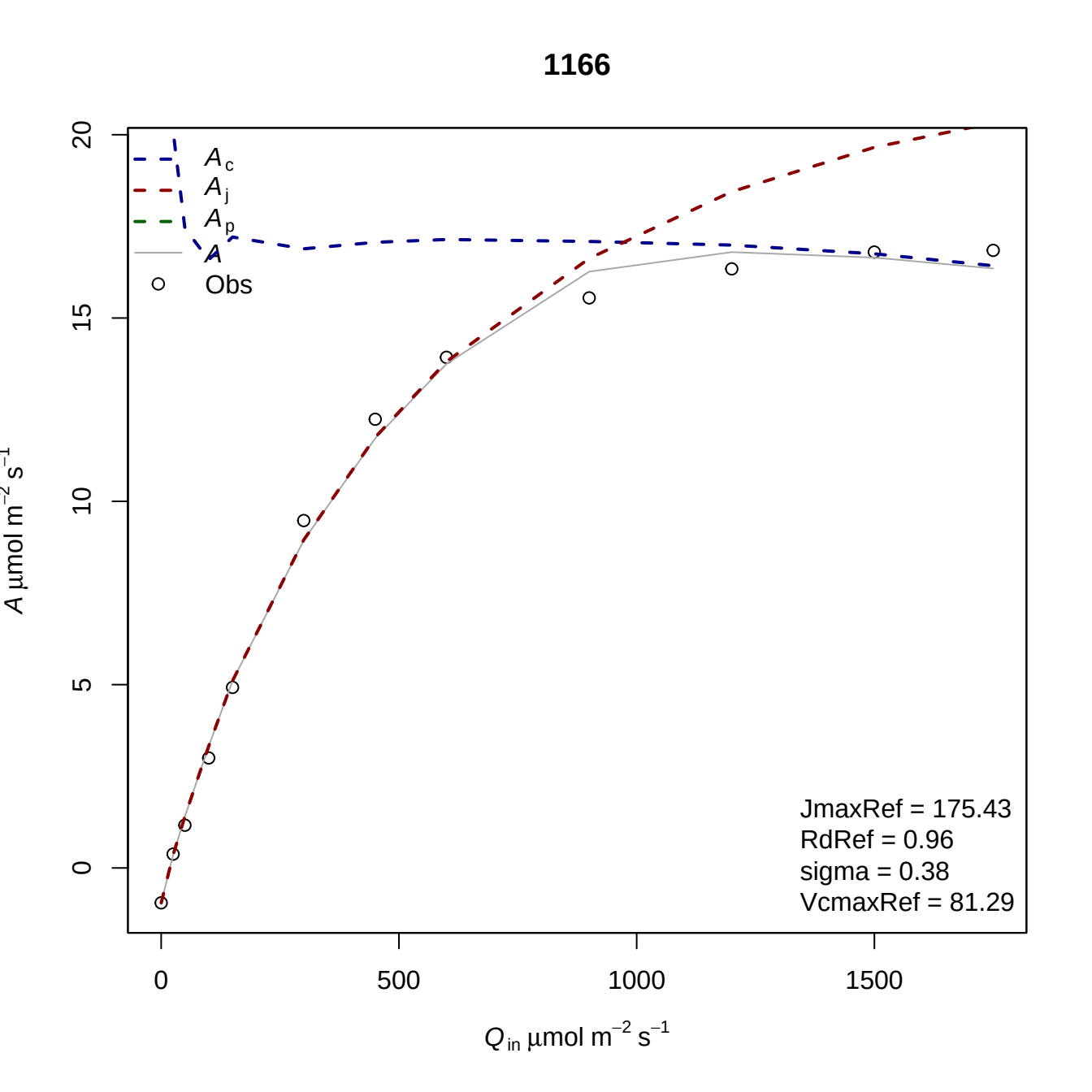

1161
A_c
A_j
A_p
A
Obs
JmaxRef = 117.81
RdRef = 1.78
sigma = 0.46
VcmaxRef = 75.4
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

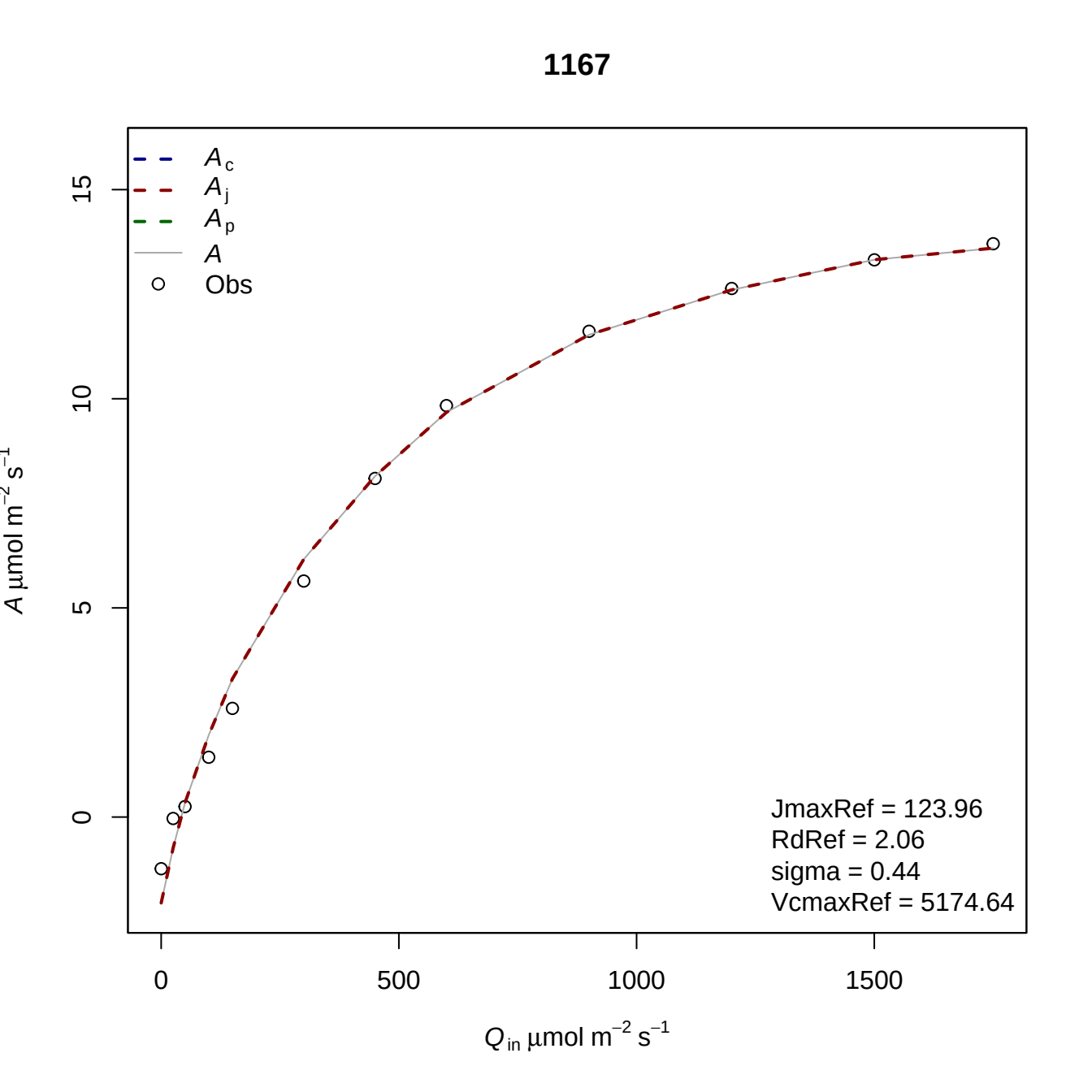

1162
A_c
A_j
A_p
A
Obs
JmaxRef = 129.13
RdRef = 1.34
sigma = 0.22
VcmaxRef = 76.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

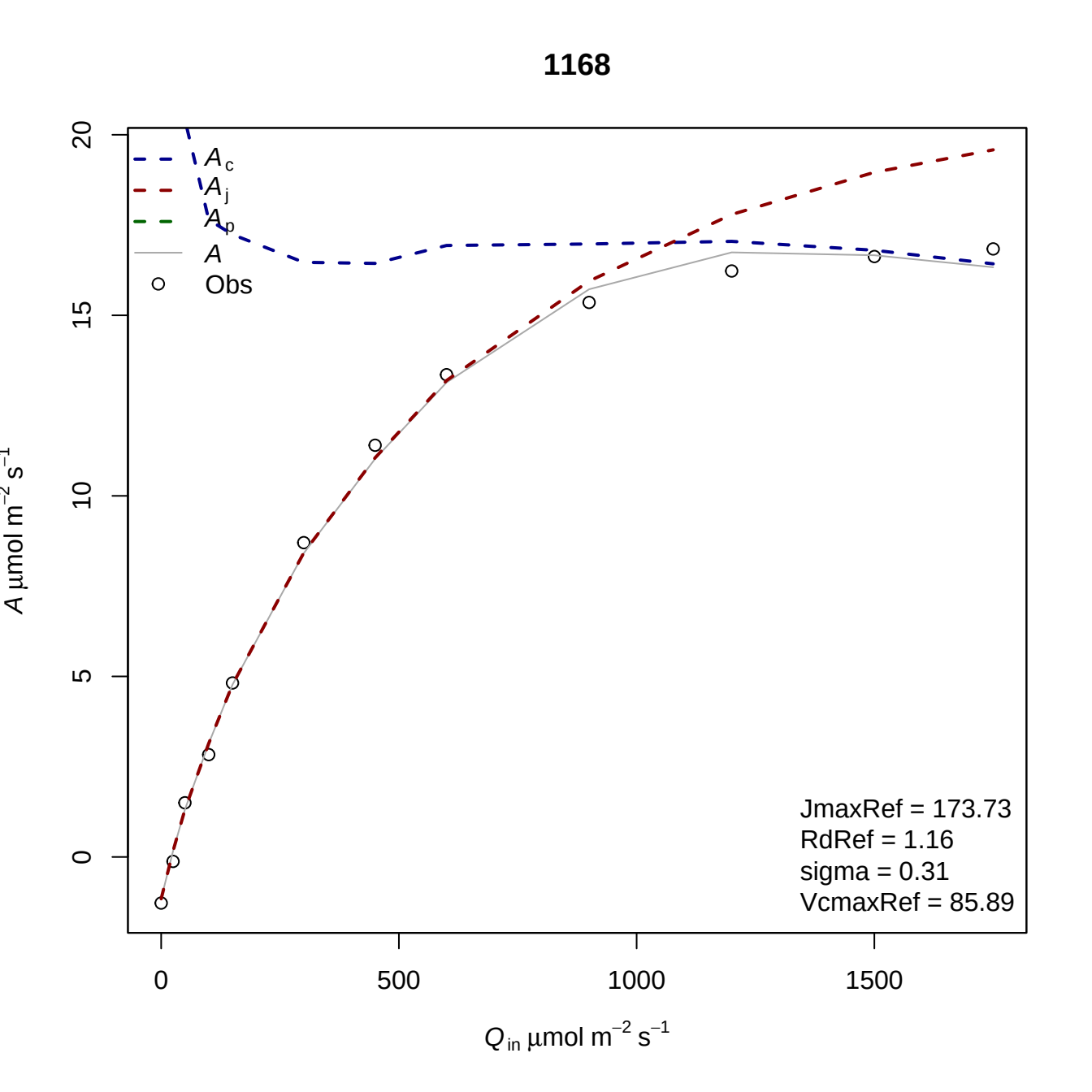

1163
A_c
A_j
A_p
A
Obs
JmaxRef = 168.58
RdRef = 2.74
sigma = 1.03
VcmaxRef = 8096.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

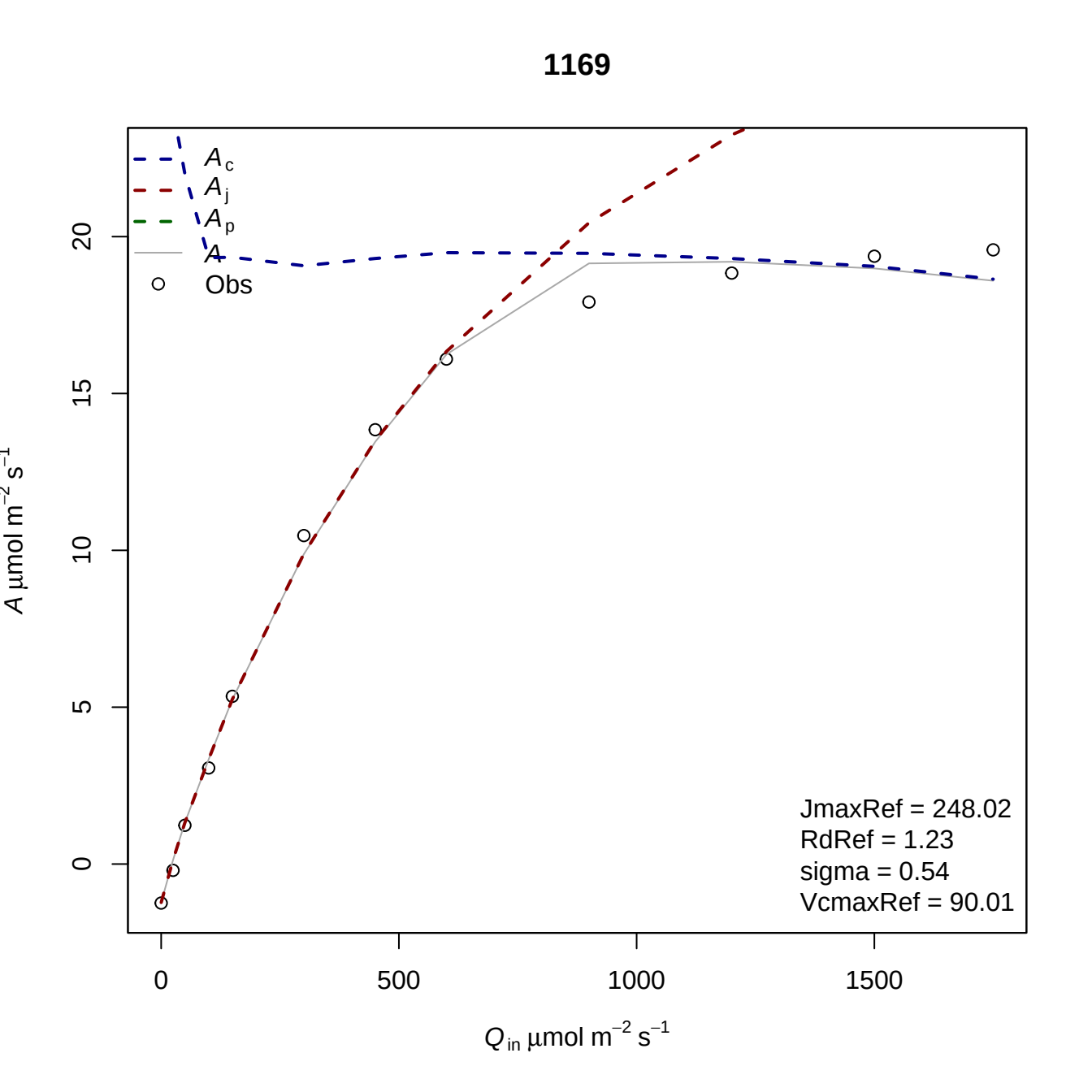

1164
A_c
A_j
A_p
A
Obs
JmaxRef = 94.97
RdRef = 2.74
sigma = 0.7
VcmaxRef = 1080
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

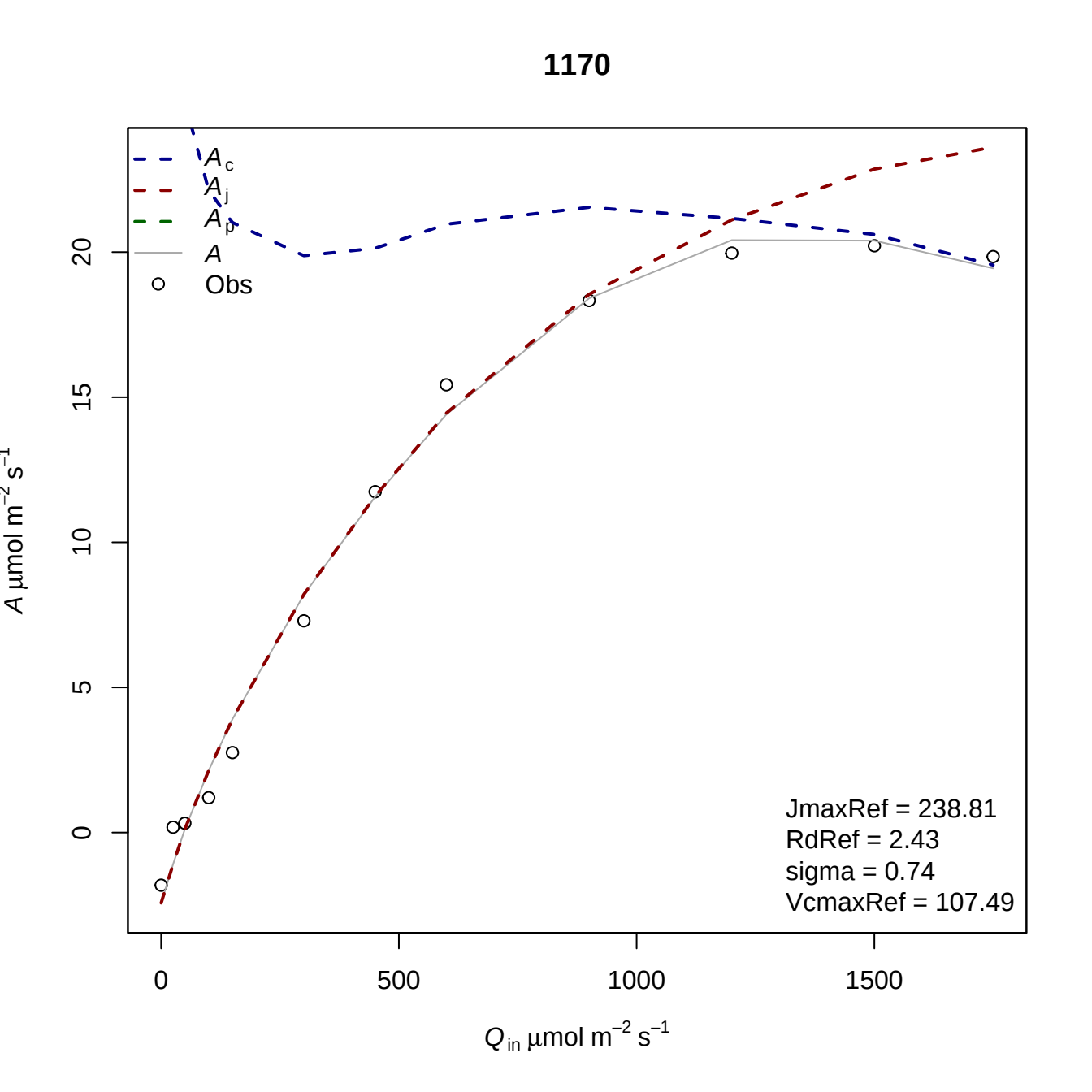

1165
A_c
A_j
A_p
A
Obs
JmaxRef = 166.66
RdRef = 2.13
sigma = 0.74
VcmaxRef = 89.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

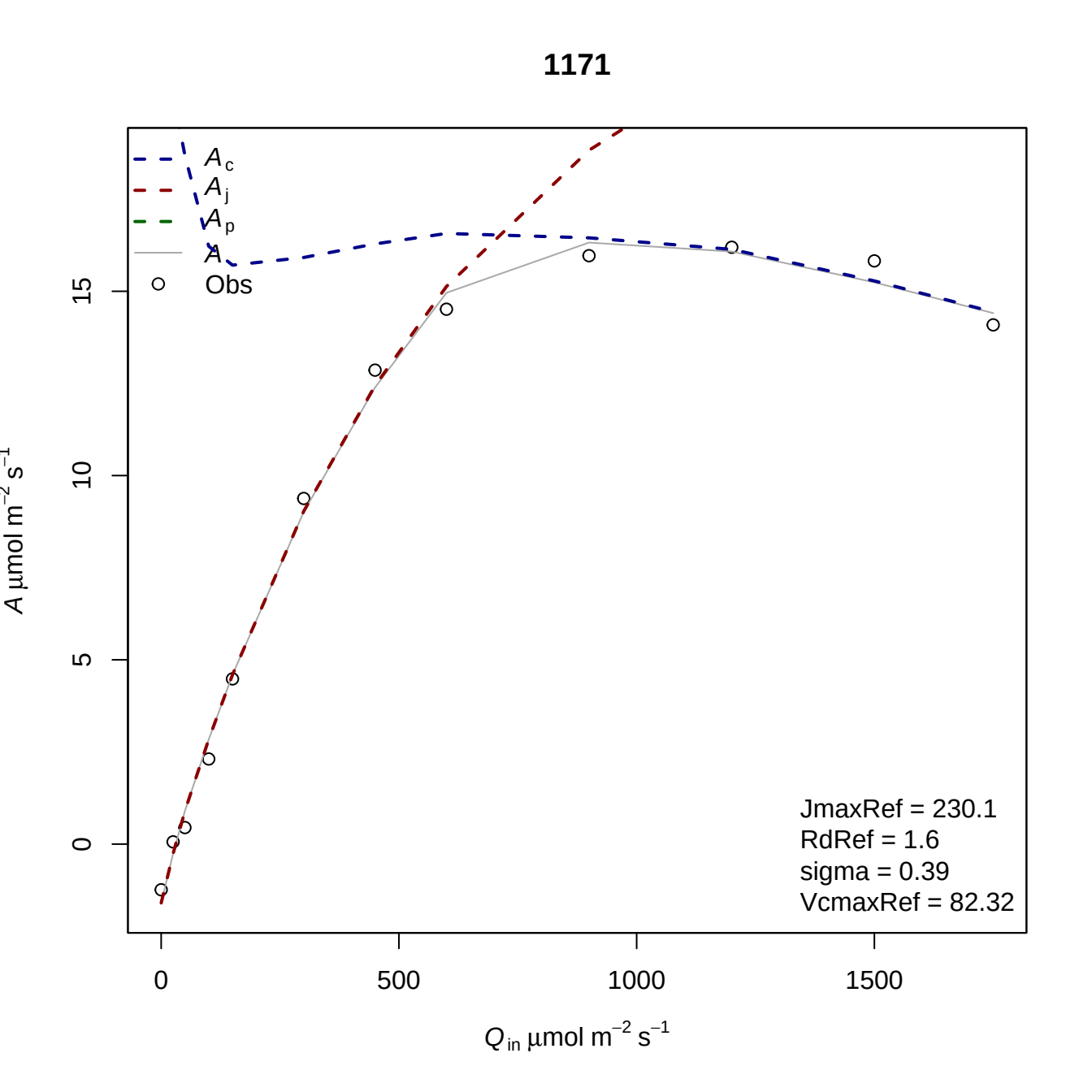

1166
A_c
A_j
A_p
A
Obs
JmaxRef = 175.43
RdRef = 0.96
sigma = 0.38
VcmaxRef = 81.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

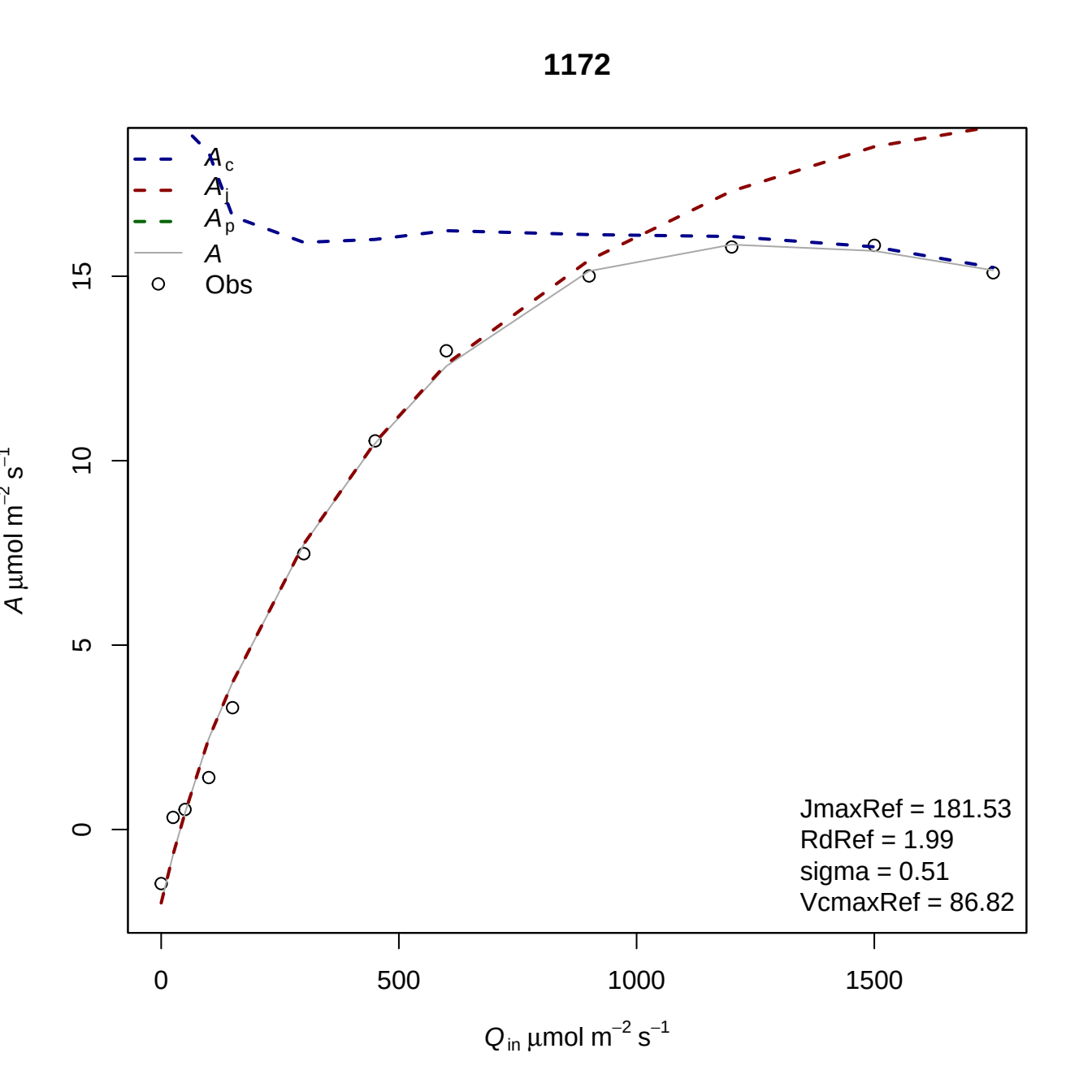

1167
A_c
A_j
A_p
A
Obs
JmaxRef = 123.96
RdRef = 2.06
sigma = 0.44
VcmaxRef = 5174.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

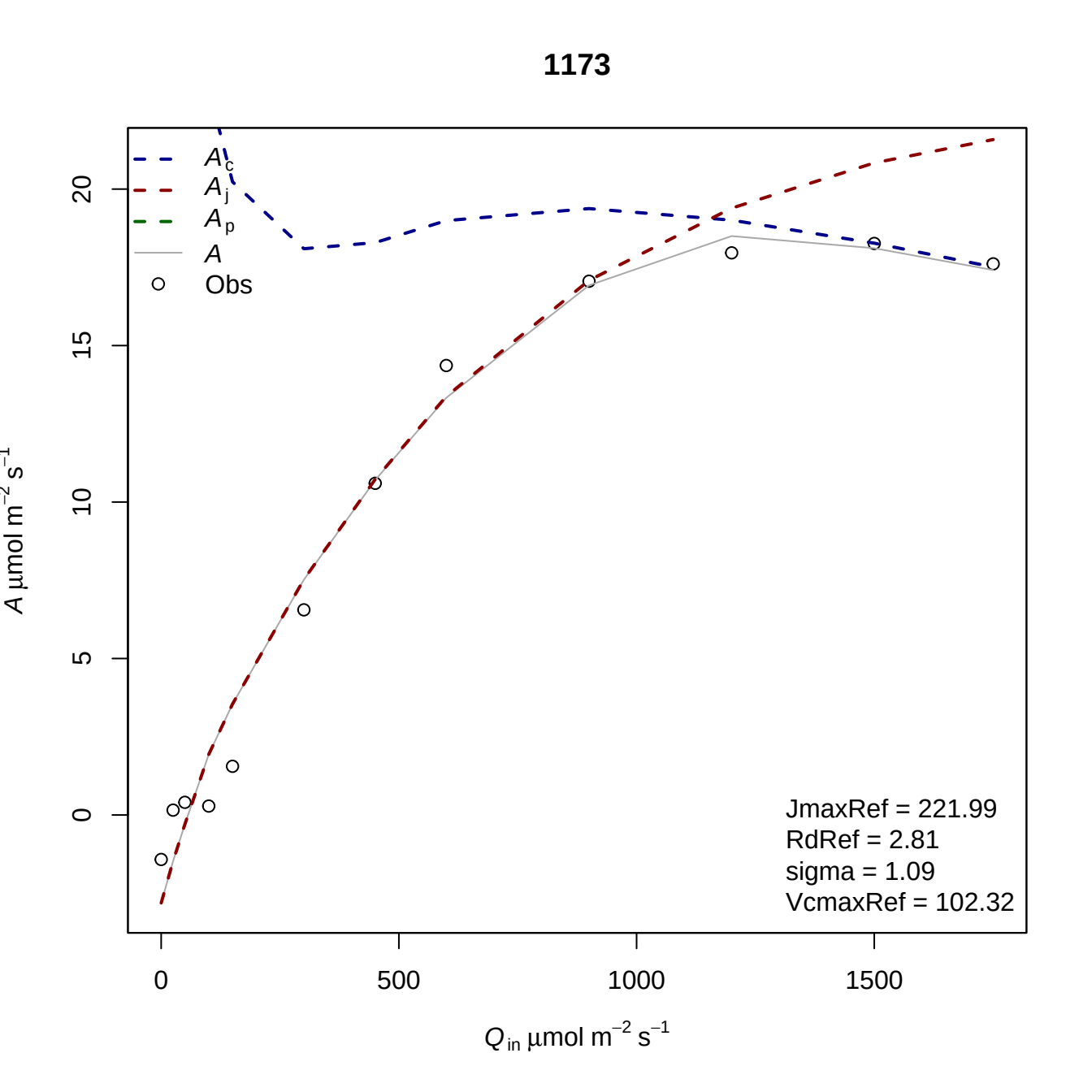

1168
A_c
A_j
A_p
A
Obs
JmaxRef = 173.73
RdRef = 1.16
sigma = 0.31
VcmaxRef = 85.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

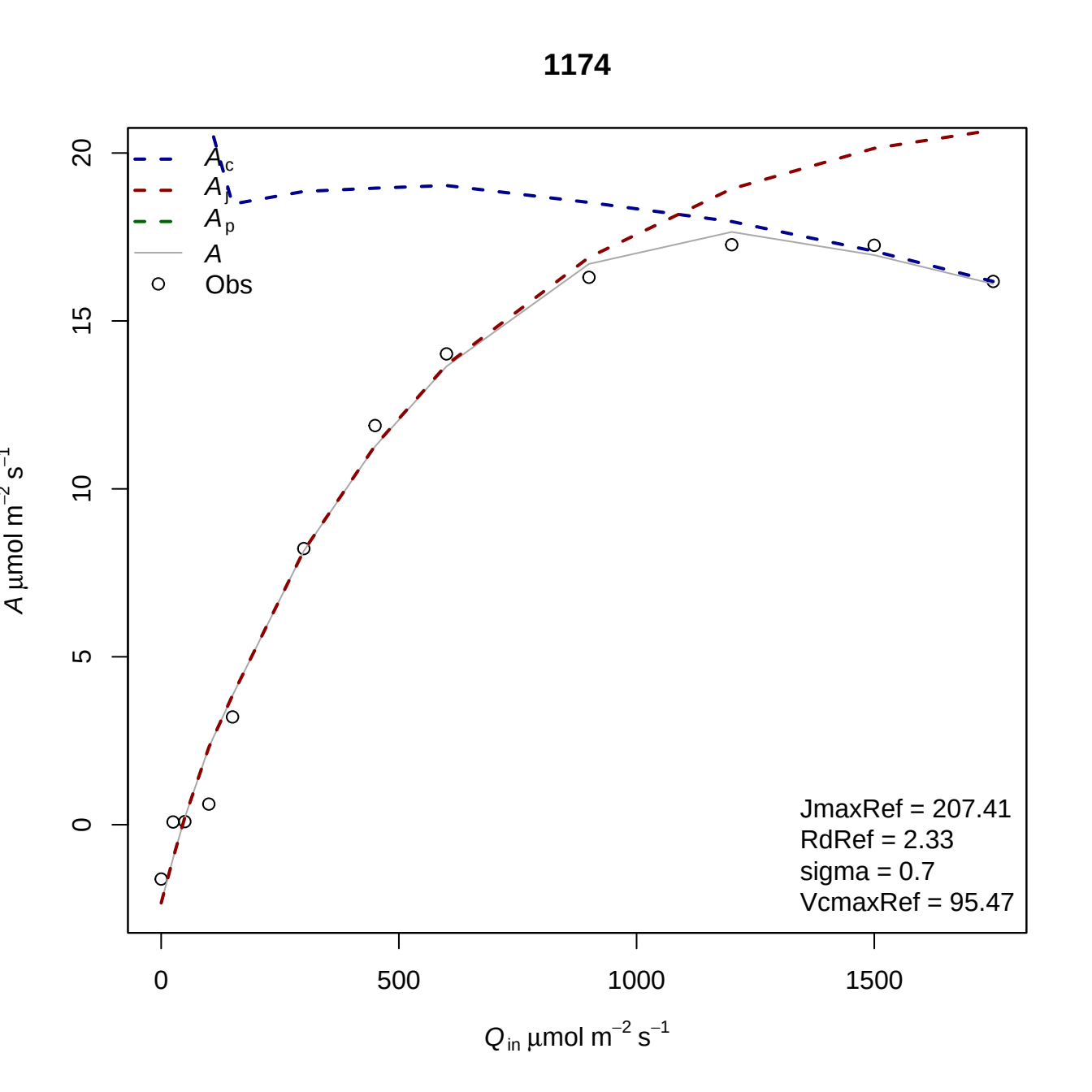

1169
A_c
A_j
A_p
A
Obs
JmaxRef = 248.02
RdRef = 1.23
sigma = 0.54
VcmaxRef = 90.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

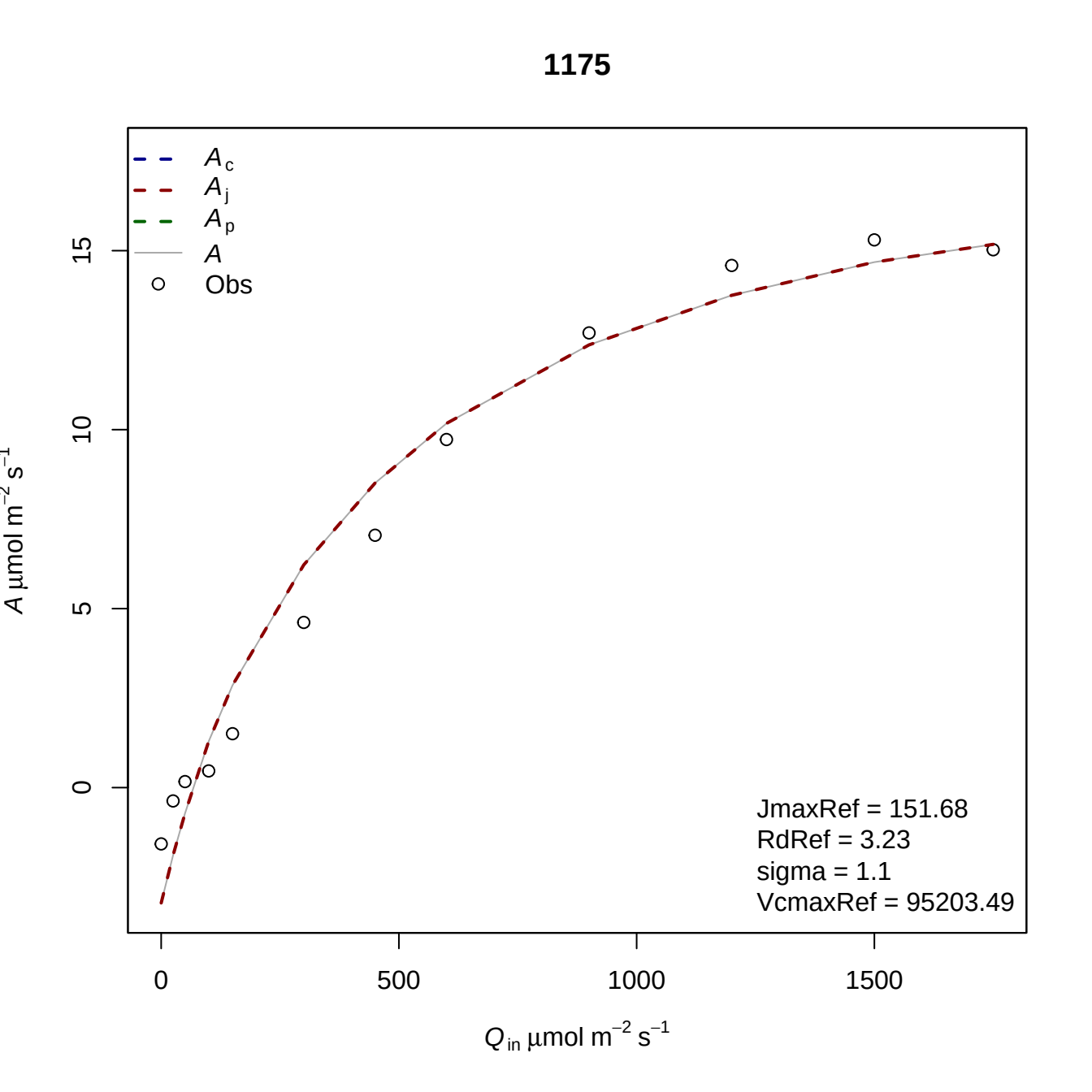

1170
A_c
A_j
A_p
A
Obs
JmaxRef = 238.81
RdRef = 2.43
sigma = 0.74
VcmaxRef = 107.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

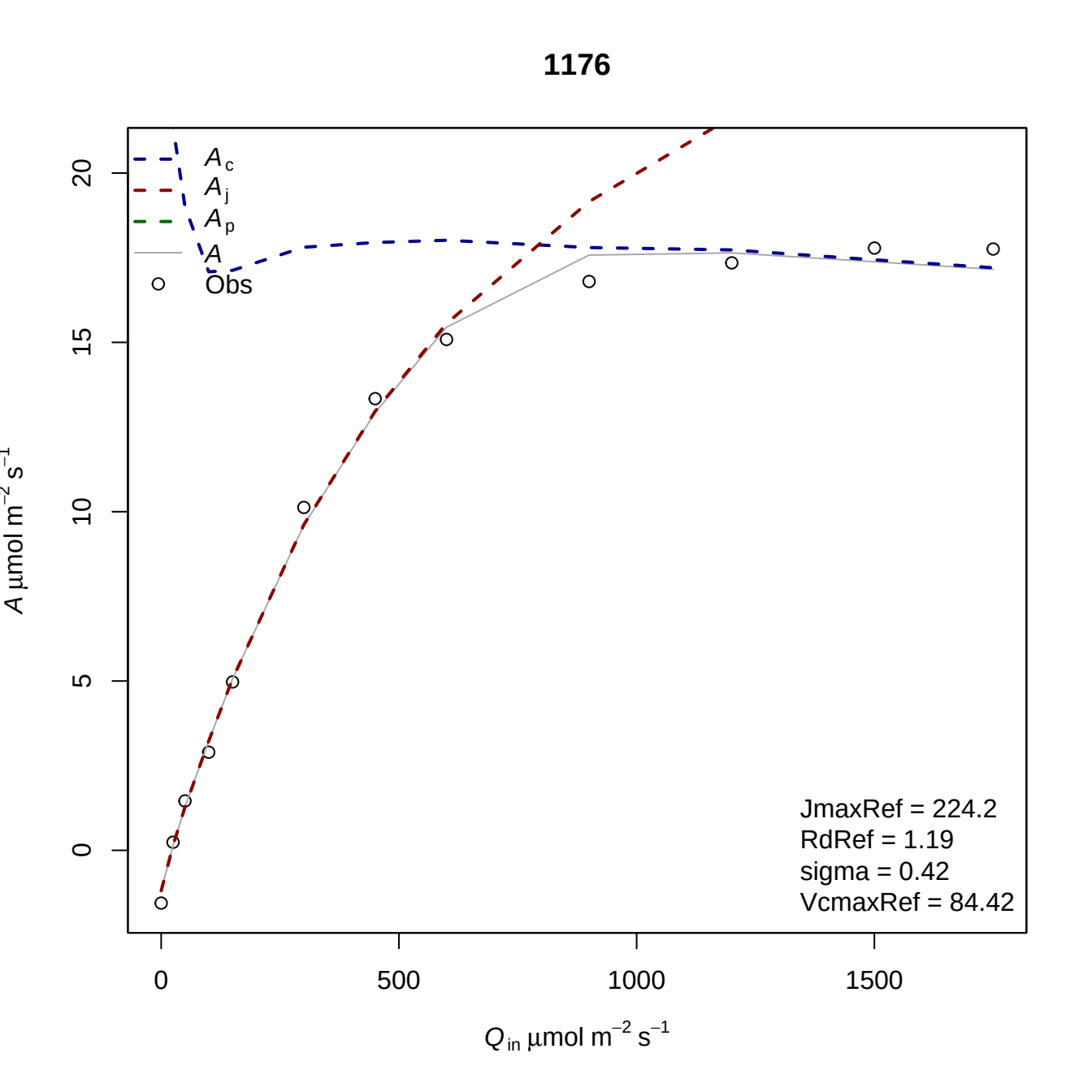

1171
A_c
A_j
A_p
A
Obs
JmaxRef = 230.1
RdRef = 1.6
sigma = 0.39
VcmaxRef = 82.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

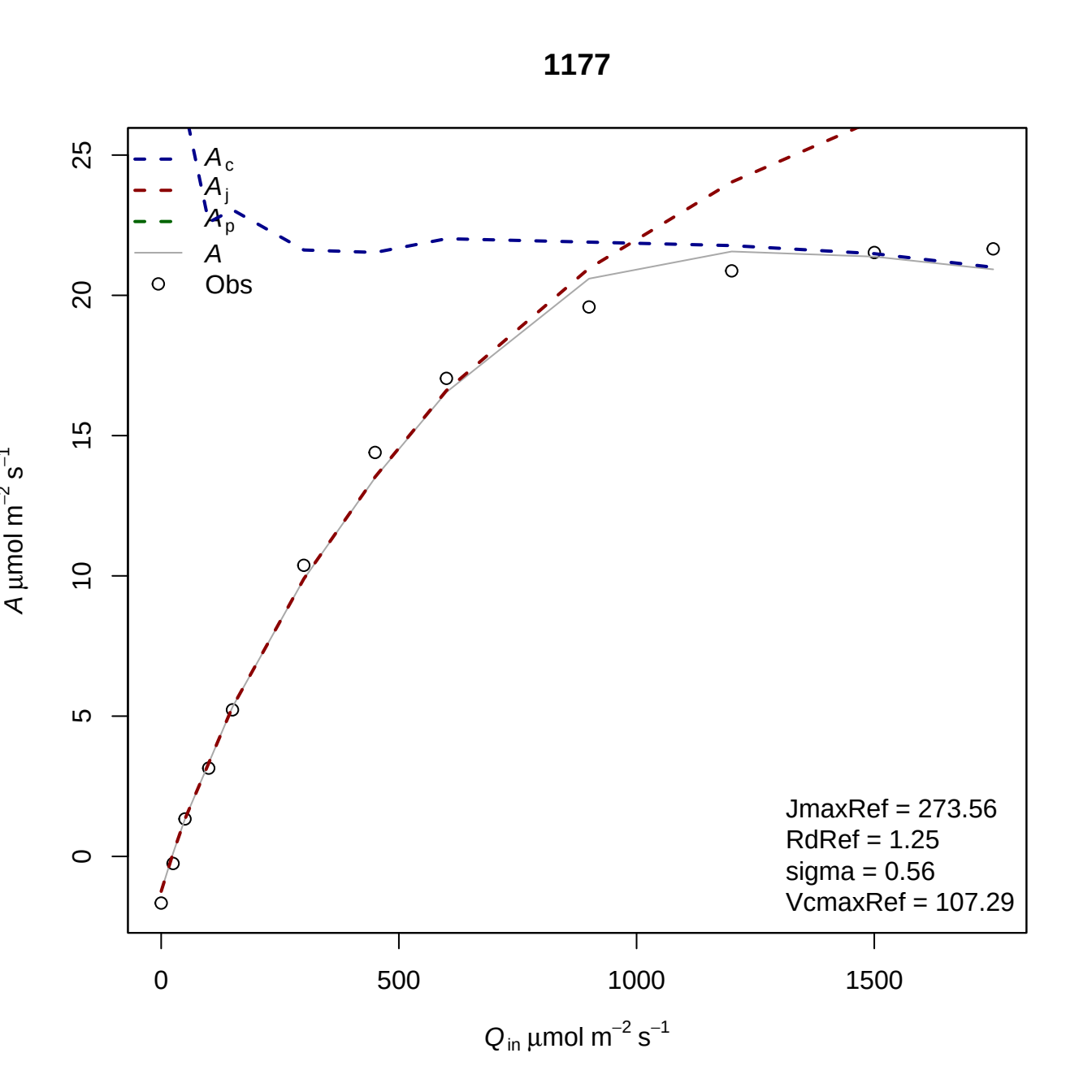

1172
A_c
A_j
A_p
A
Obs
JmaxRef = 181.53
RdRef = 1.99
sigma = 0.51
VcmaxRef = 86.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

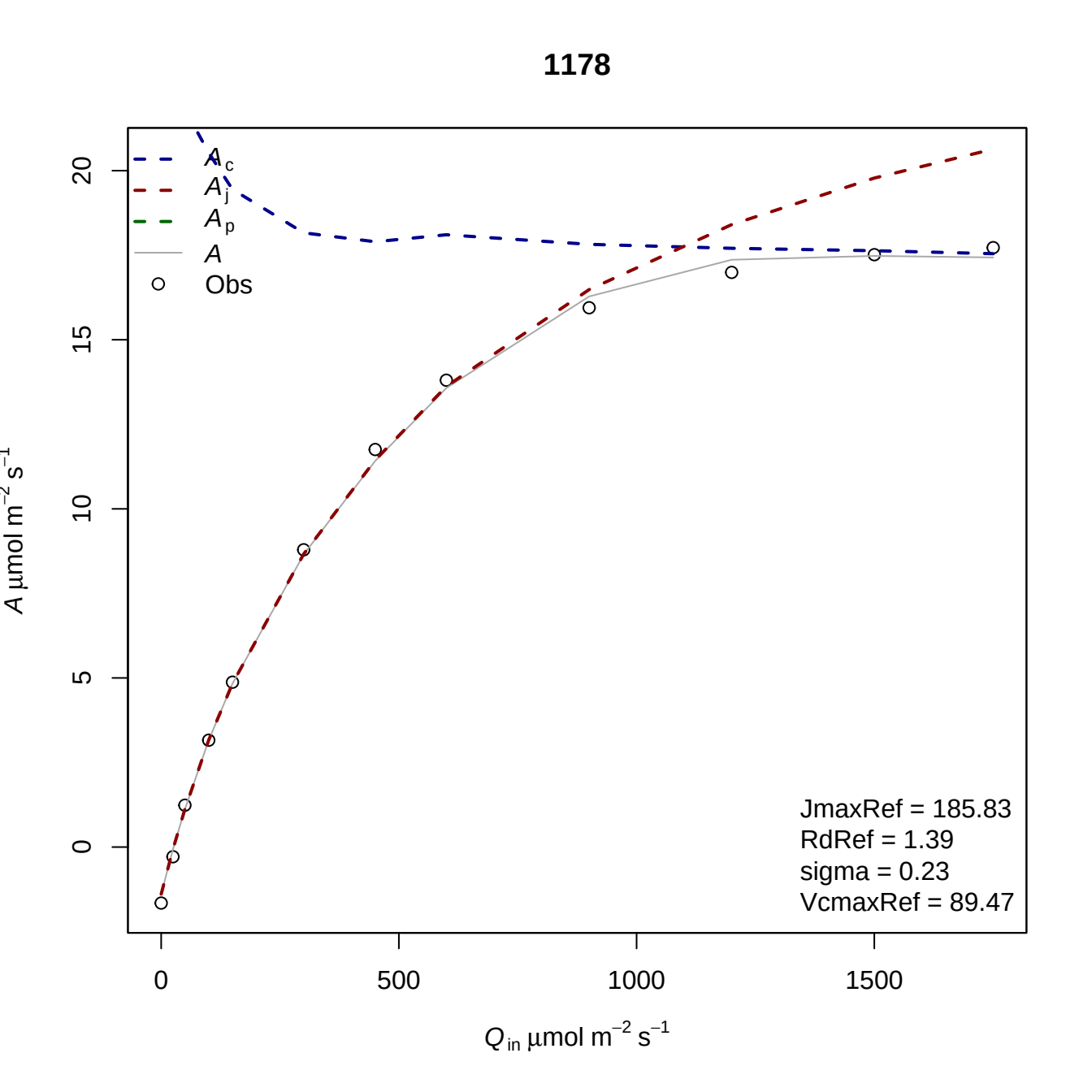

1173
A_c
A_j
A_p
A
Obs
JmaxRef = 221.99
RdRef = 2.81
sigma = 1.09
VcmaxRef = 102.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

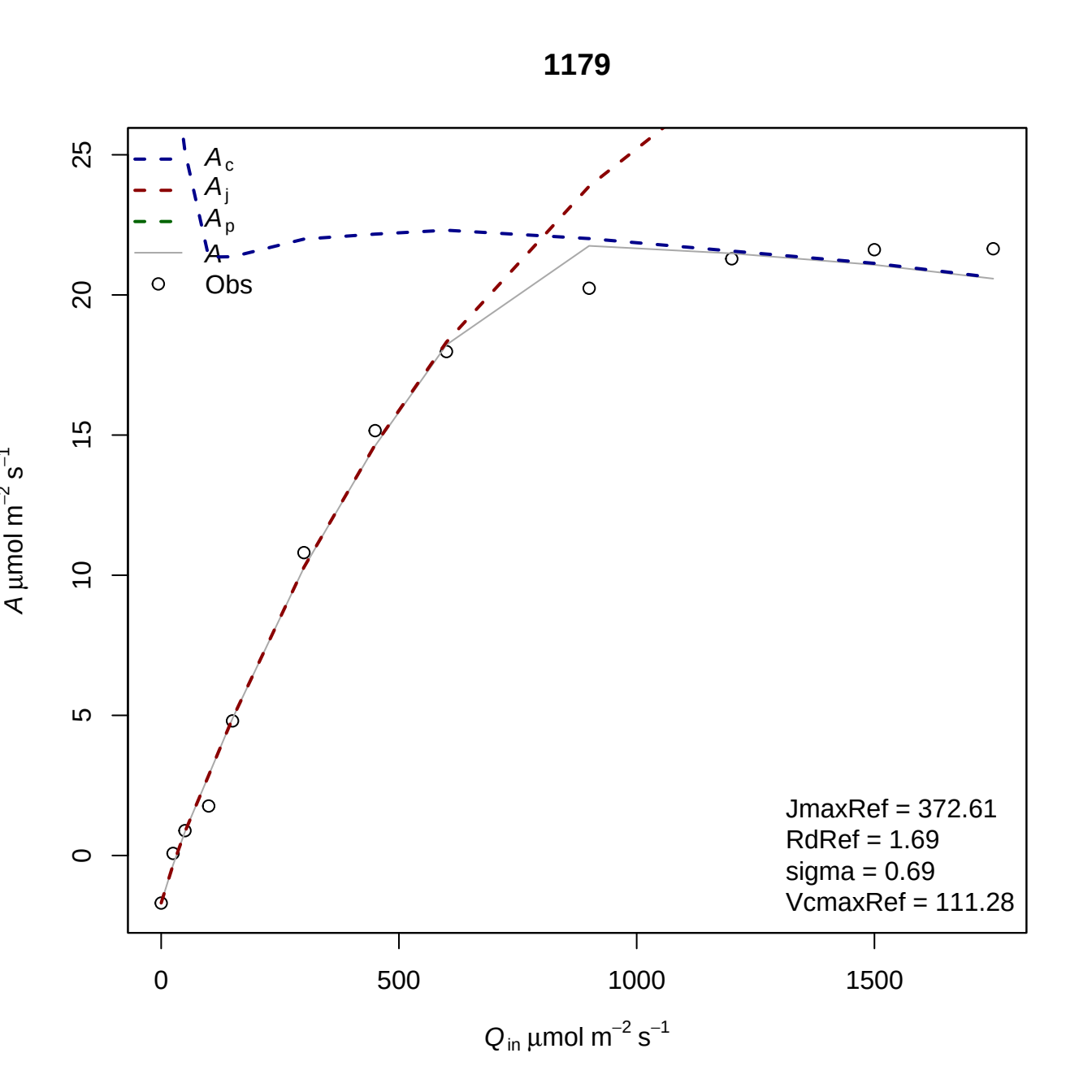

1174
A_c
A_j
A_p
A
Obs
JmaxRef = 207.41
RdRef = 2.33
sigma = 0.7
VcmaxRef = 95.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

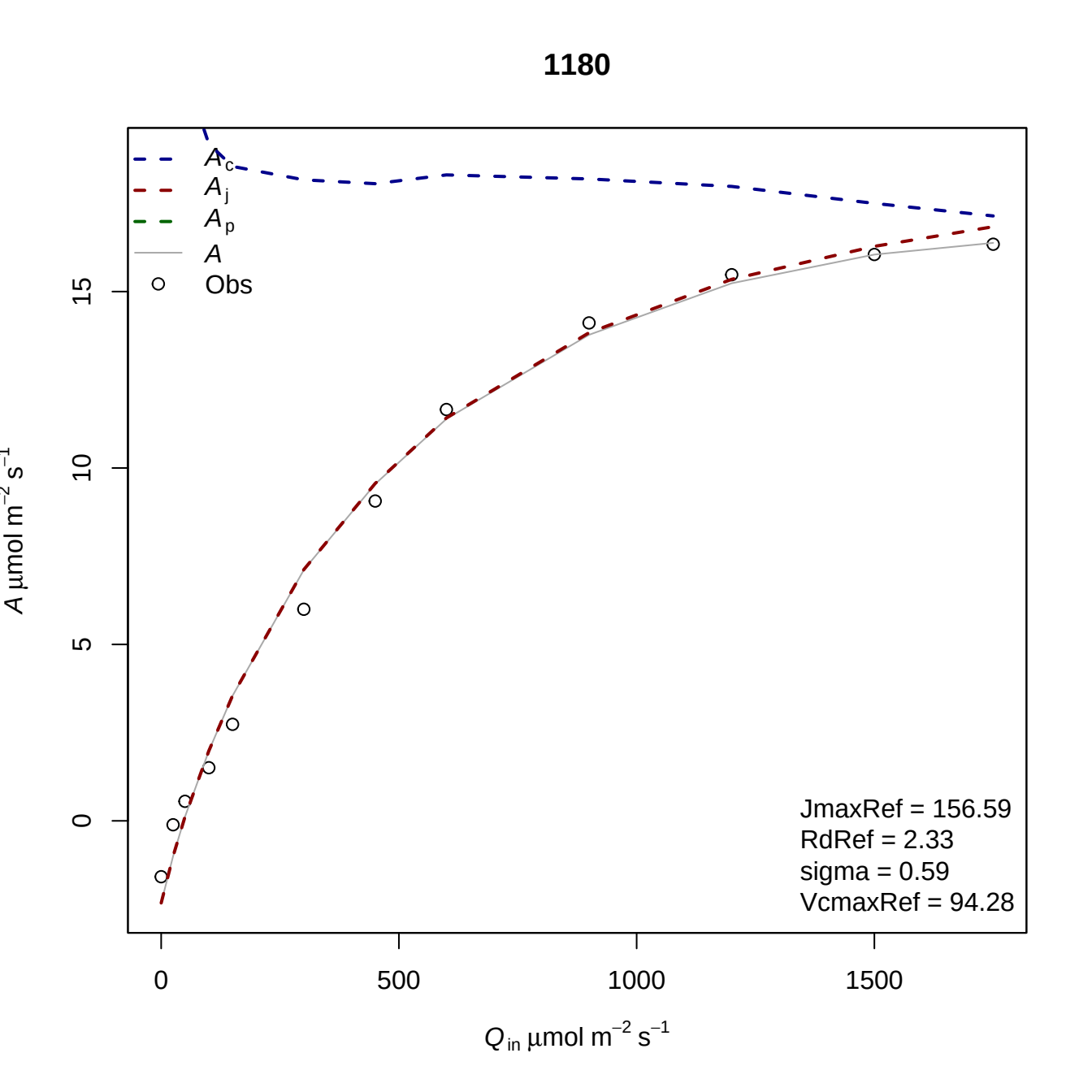

1175
A_c
A_j
A_p
A
Obs
JmaxRef = 151.68
RdRef = 3.23
sigma = 1.1
VcmaxRef = 95203.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

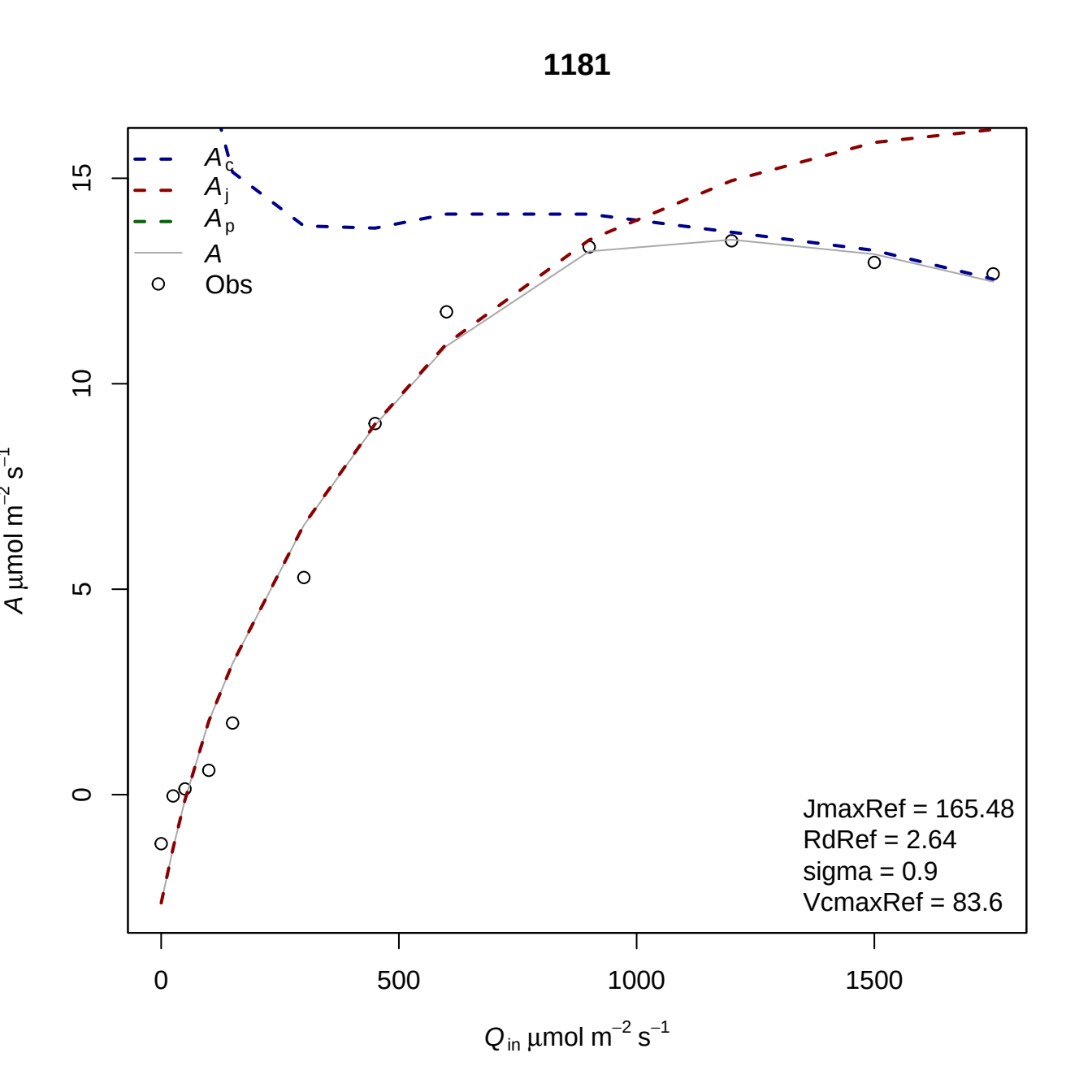

1176
A_c
A_j
A_p
A
Obs
JmaxRef = 224.2
RdRef = 1.19
sigma = 0.42
VcmaxRef = 84.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

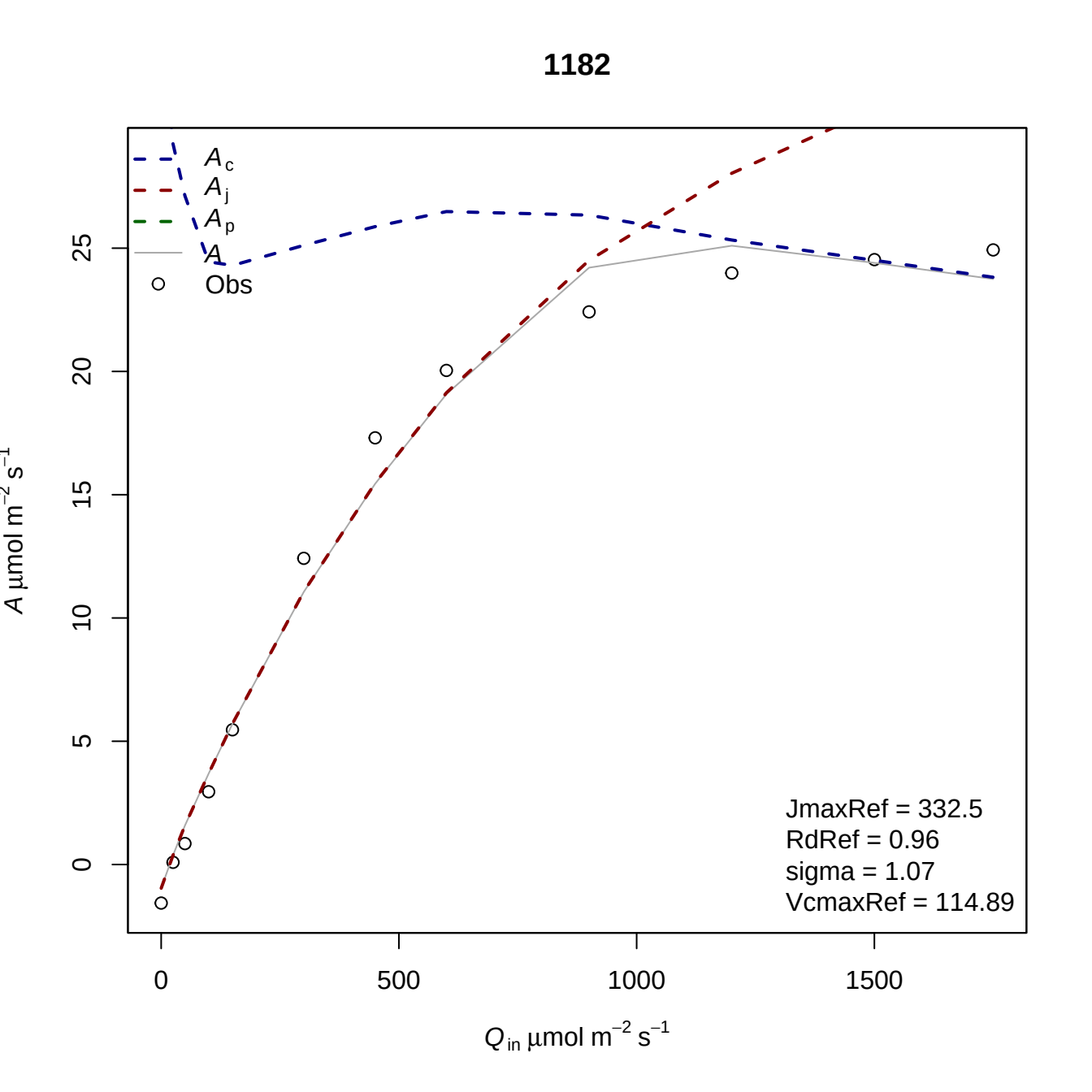

1177
A_c
A_j
A_p
A
Obs
JmaxRef = 273.56
RdRef = 1.25
sigma = 0.56
VcmaxRef = 107.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

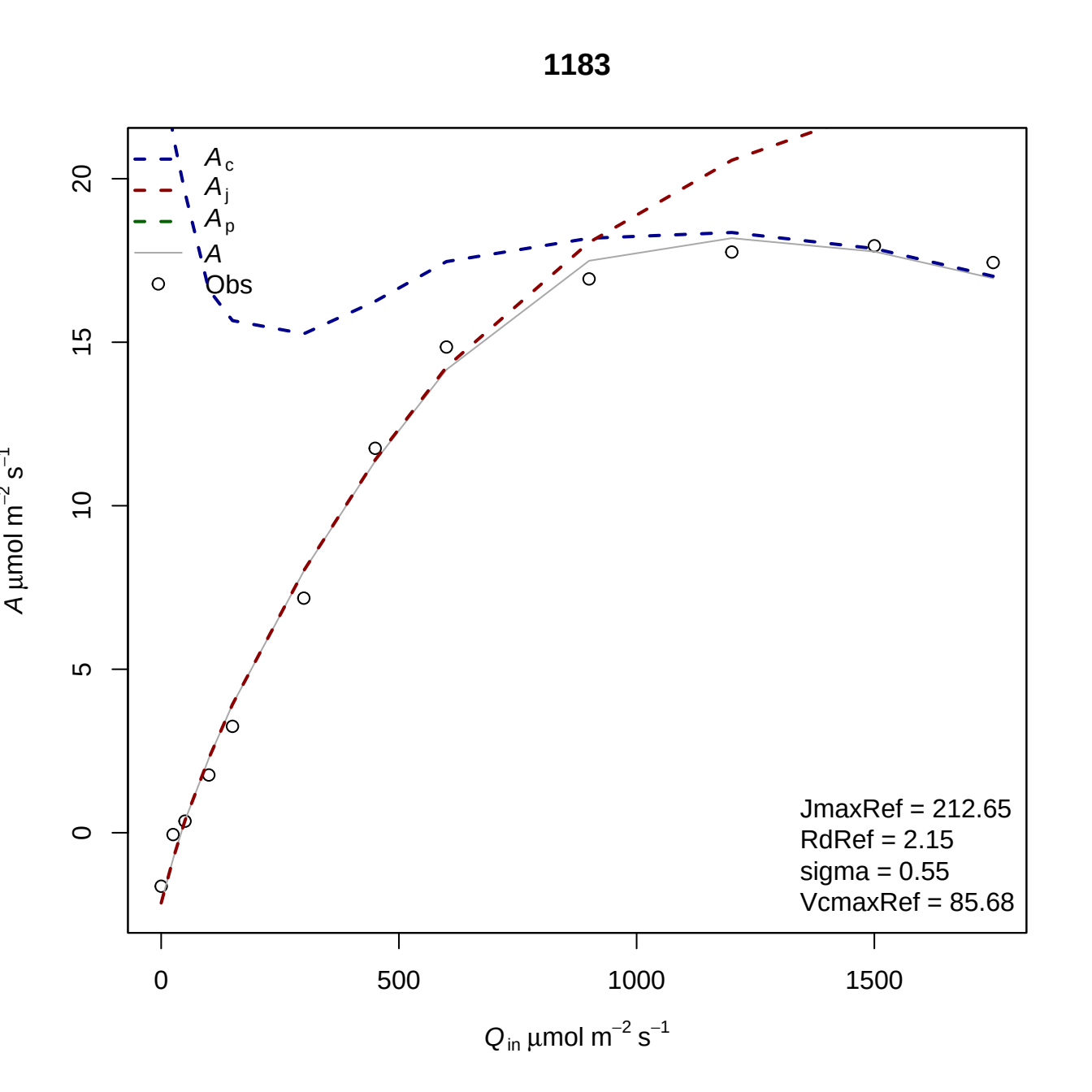

1178
A_c
A_j
A_p
A
Obs
JmaxRef = 185.83
RdRef = 1.39
sigma = 0.23
VcmaxRef = 89.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

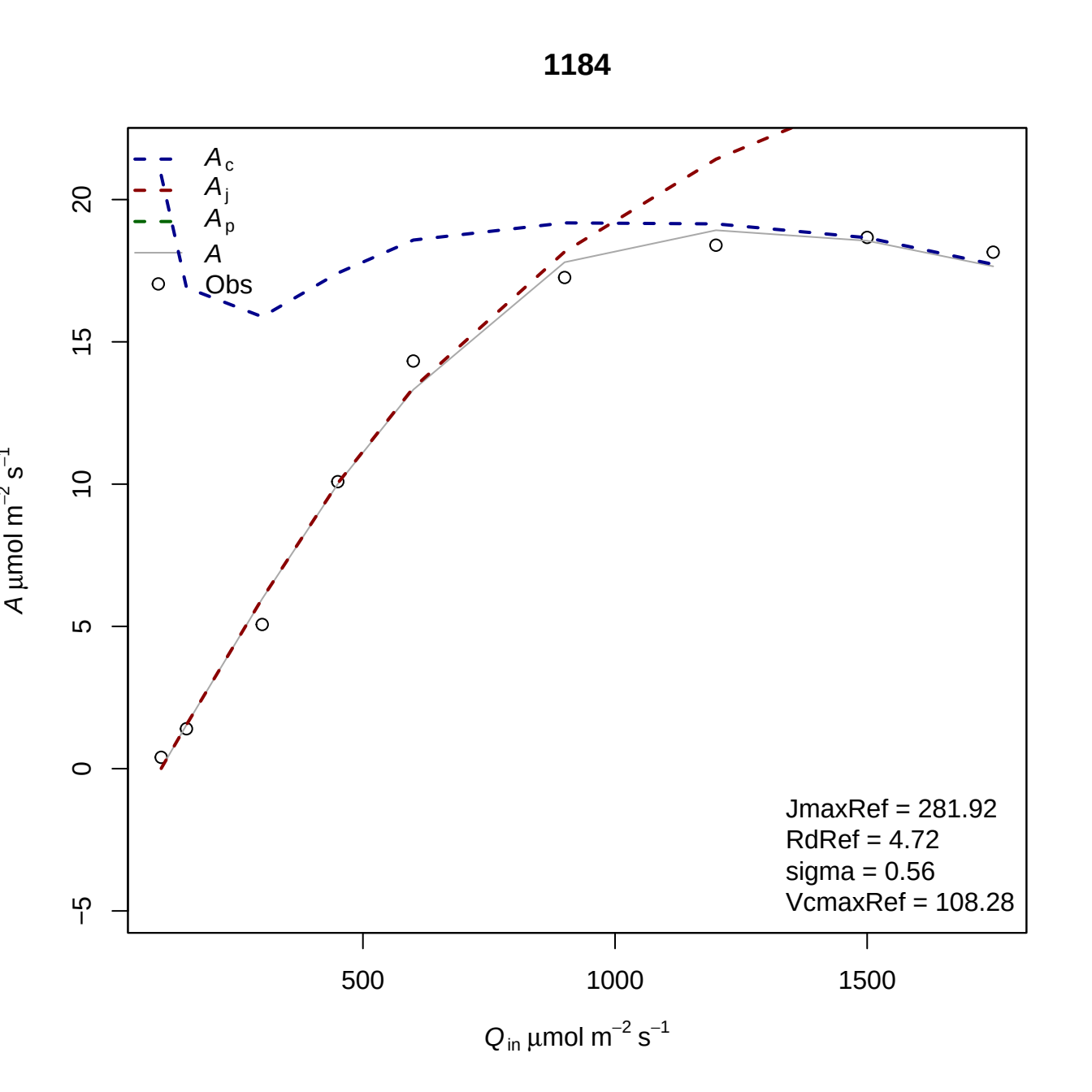

1179
A_c
A_j
A_p
A
Obs
JmaxRef = 372.61
RdRef = 1.69
sigma = 0.69
VcmaxRef = 111.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

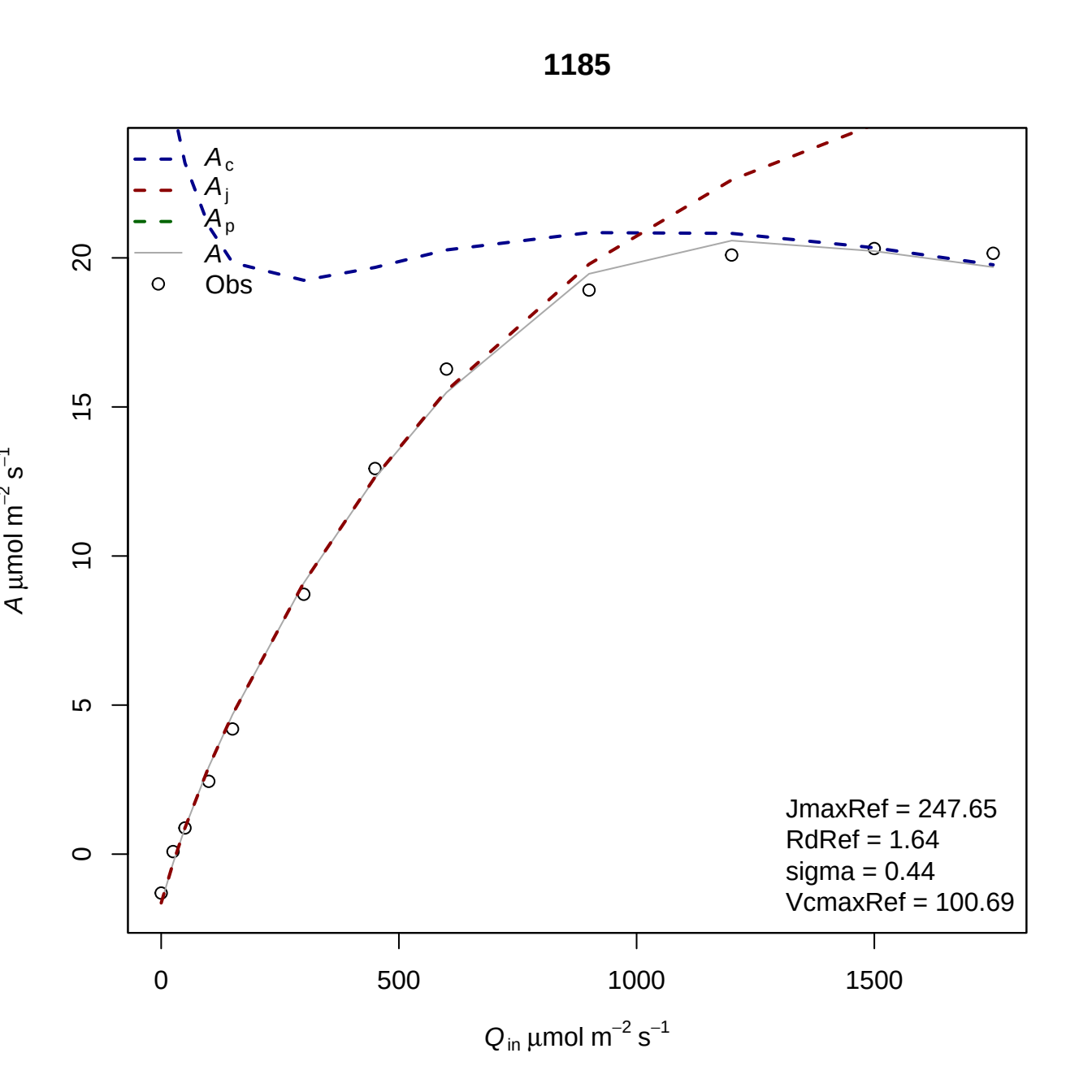

1180
A_c
A_j
A_p
A
Obs
JmaxRef = 156.59
RdRef = 2.33
sigma = 0.59
VcmaxRef = 94.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

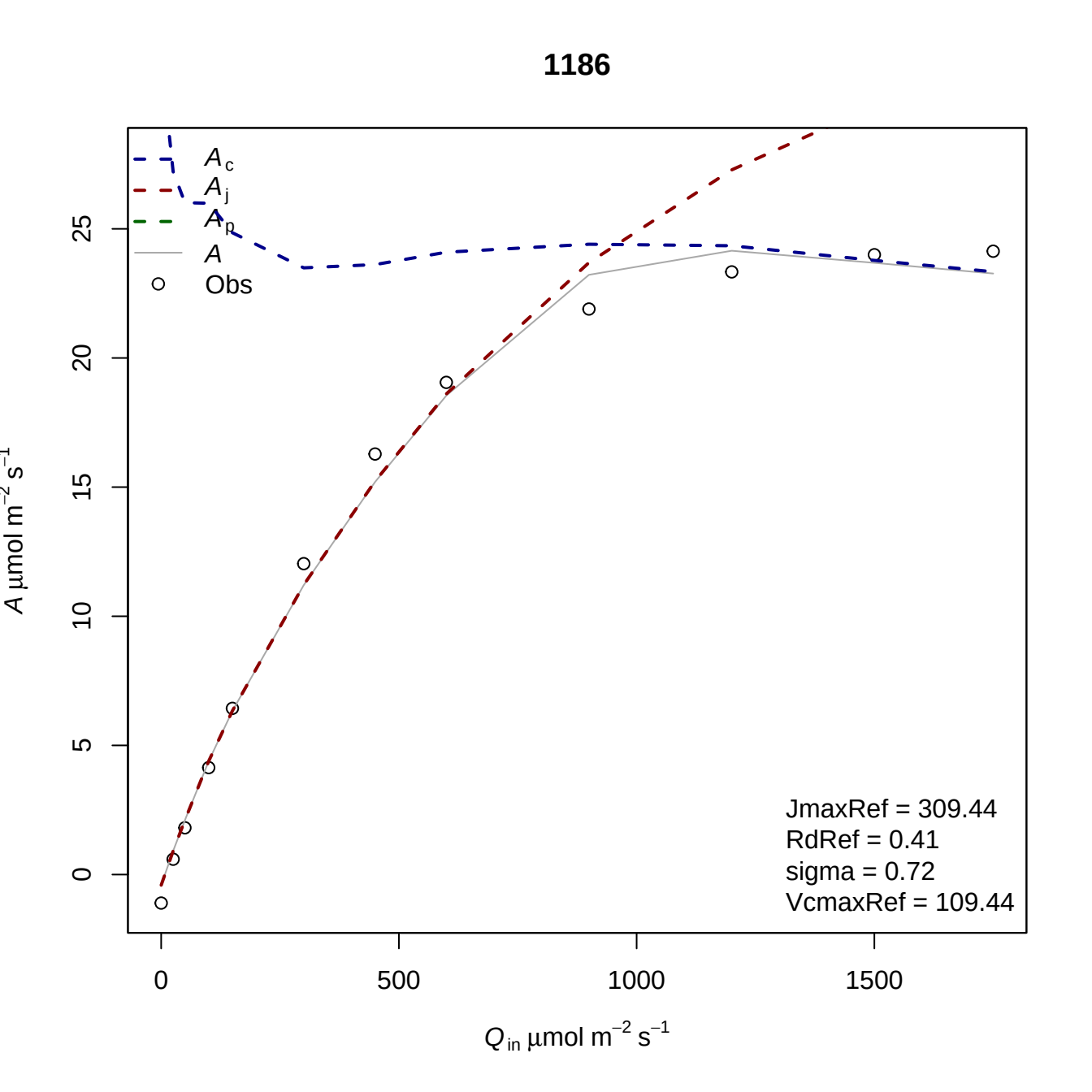

1181
A_c
A_j
A_p
A
Obs
JmaxRef = 165.48
RdRef = 2.64
sigma = 0.9
VcmaxRef = 83.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

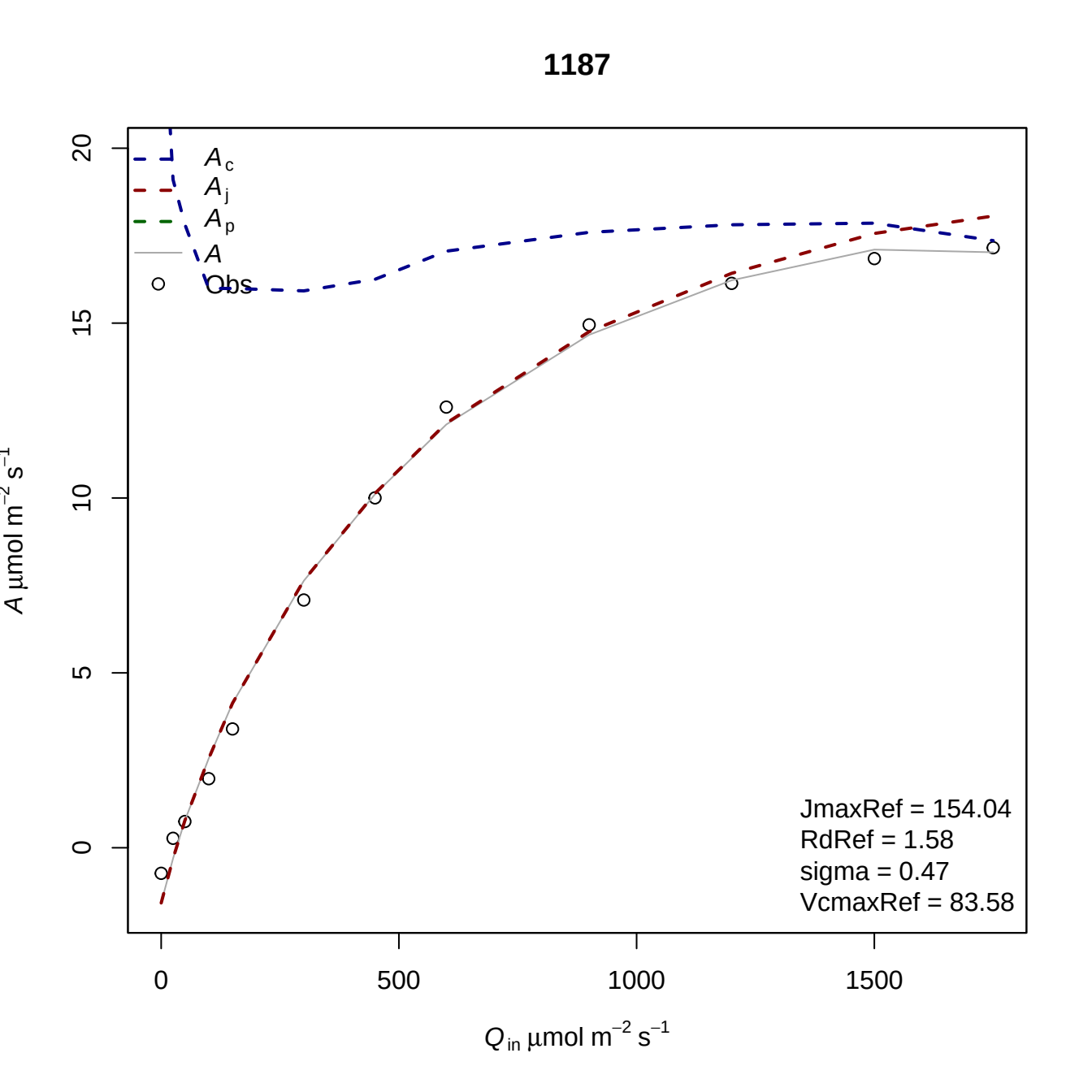

1182
A_c
A_j
A_p
A
Obs
JmaxRef = 332.5
RdRef = 0.96
sigma = 1.07
VcmaxRef = 114.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

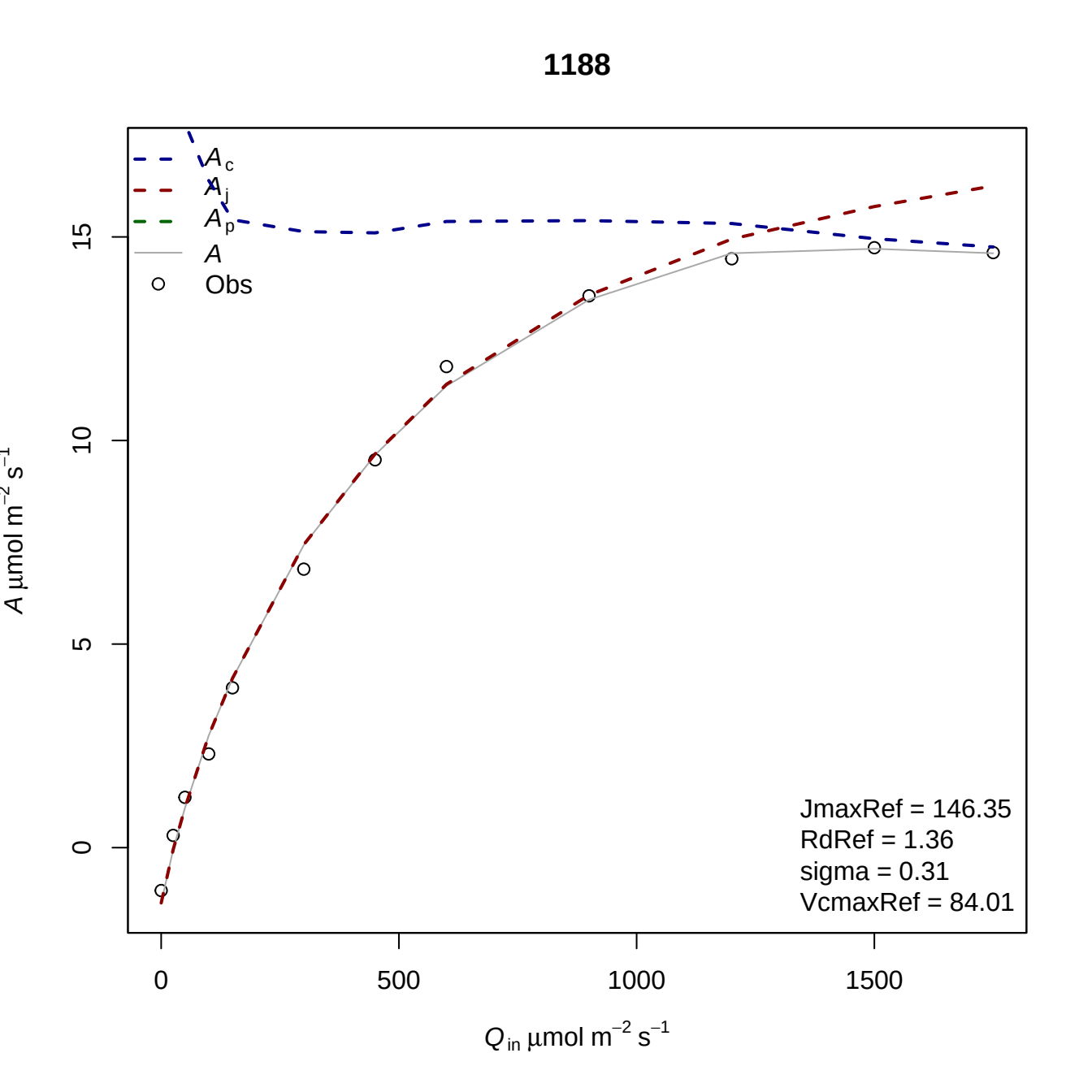

1183
A_c
A_j
A_p
A
Obs
JmaxRef = 212.65
RdRef = 2.15
sigma = 0.55
VcmaxRef = 85.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

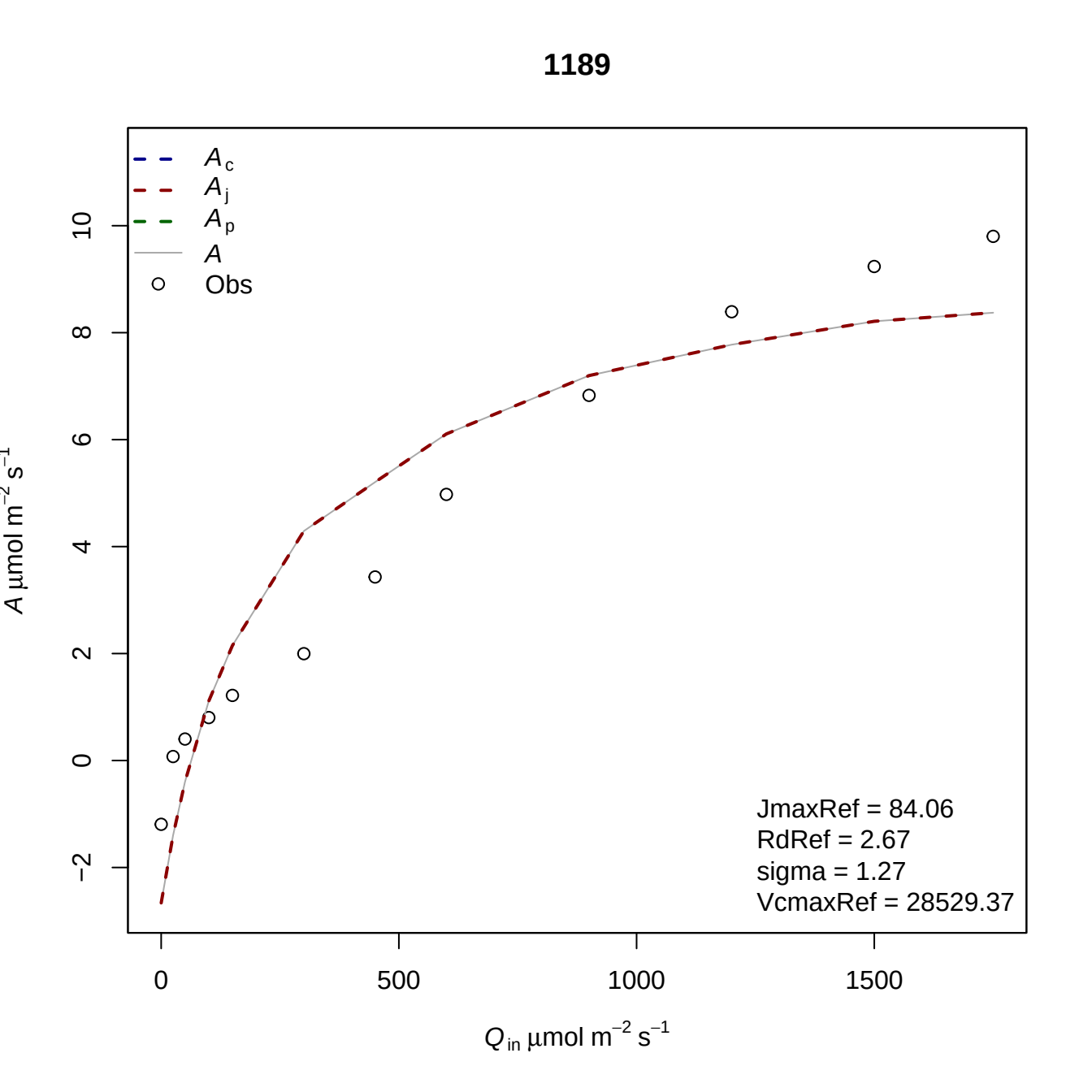

1184
A_c
A_j
A_p
A
Obs
JmaxRef = 281.92
RdRef = 4.72
sigma = 0.56
VcmaxRef = 108.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
500
1000
1500

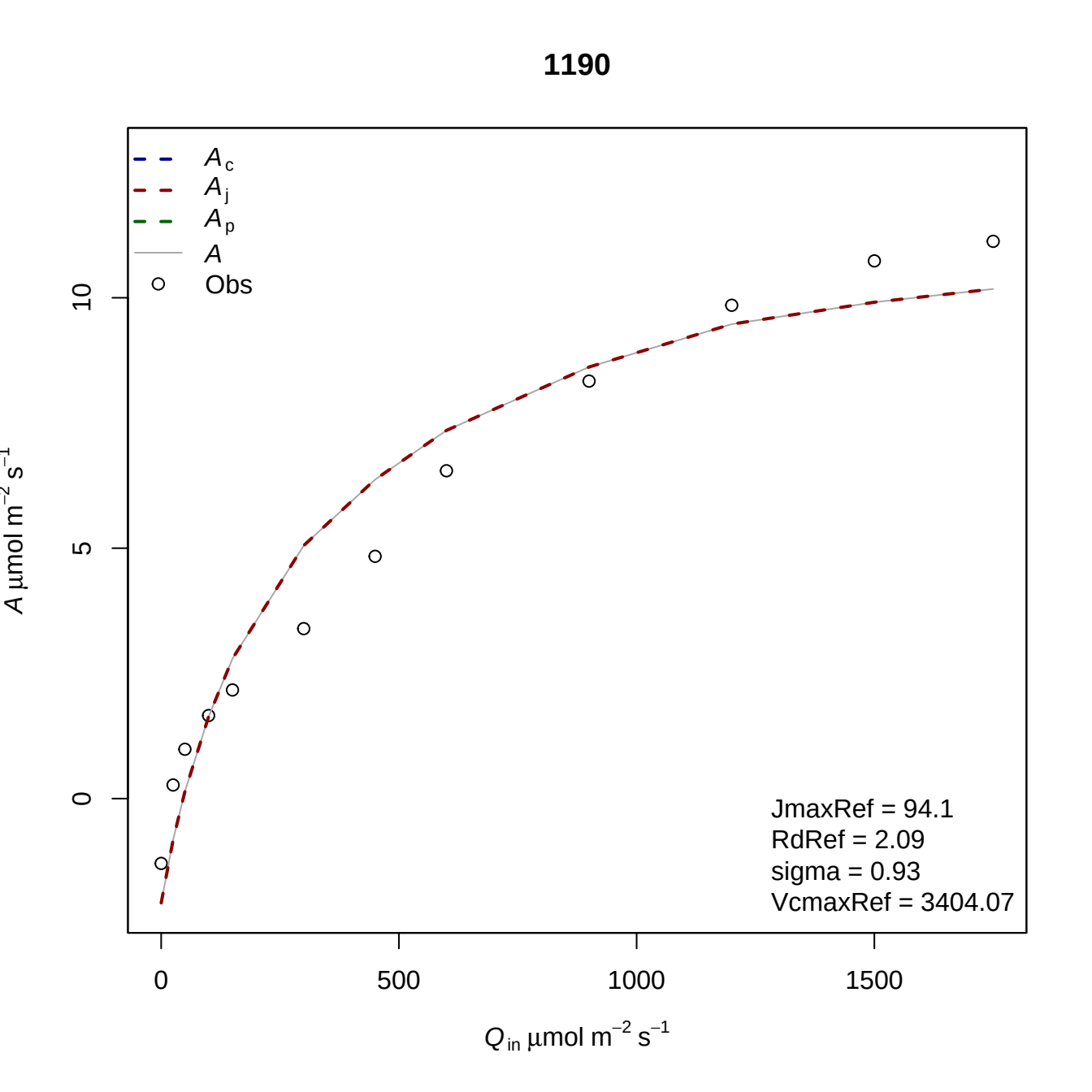

1185
A_c
A_j
A_p
A
Obs
JmaxRef = 247.65
RdRef = 1.64
sigma = 0.44
VcmaxRef = 100.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

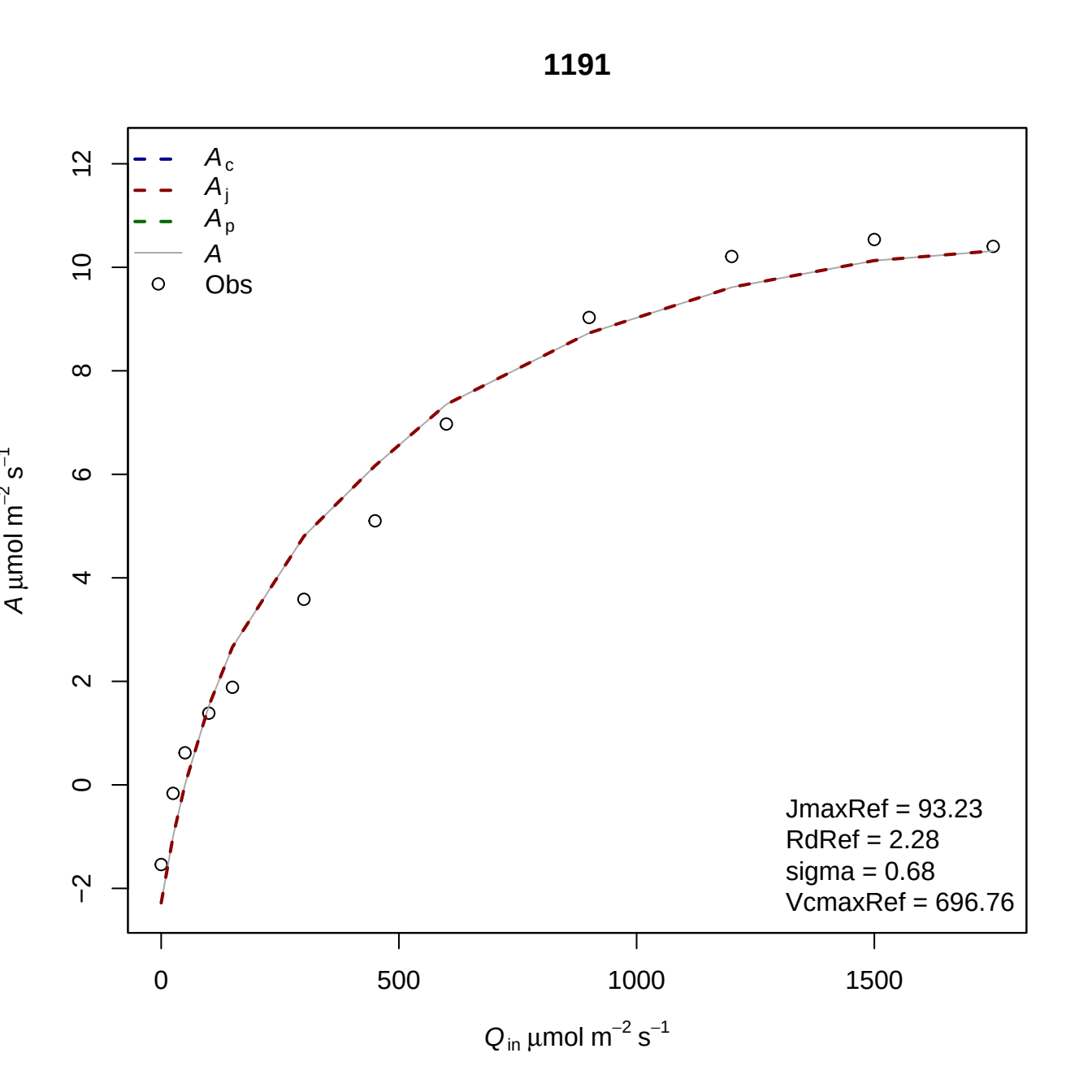

1186
A_c
A_j
A_p
A
Obs
JmaxRef = 309.44
RdRef = 0.41
sigma = 0.72
VcmaxRef = 109.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

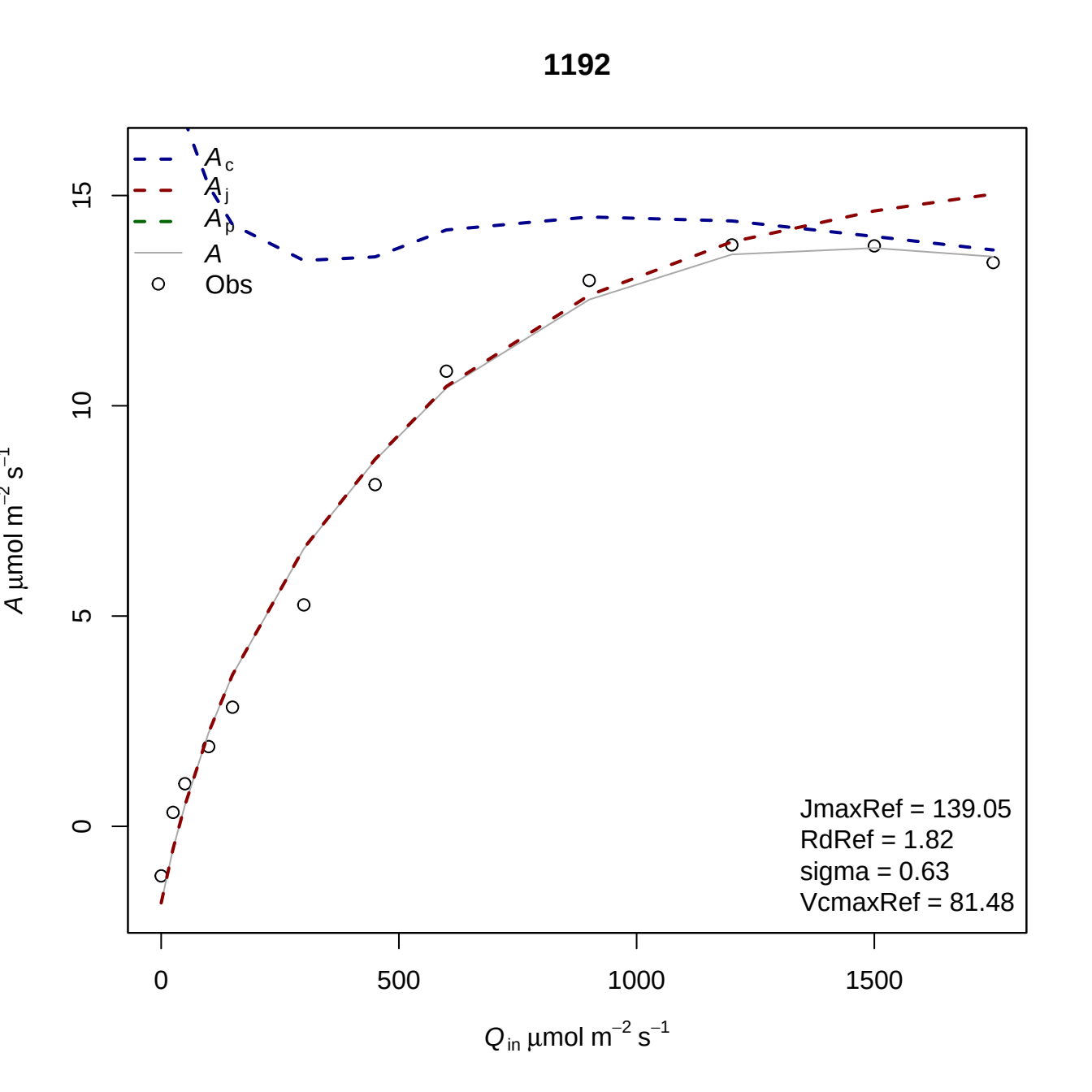

1187
A_c
A_j
A_p
A
Obs
JmaxRef = 154.04
RdRef = 1.58
sigma = 0.47
VcmaxRef = 83.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

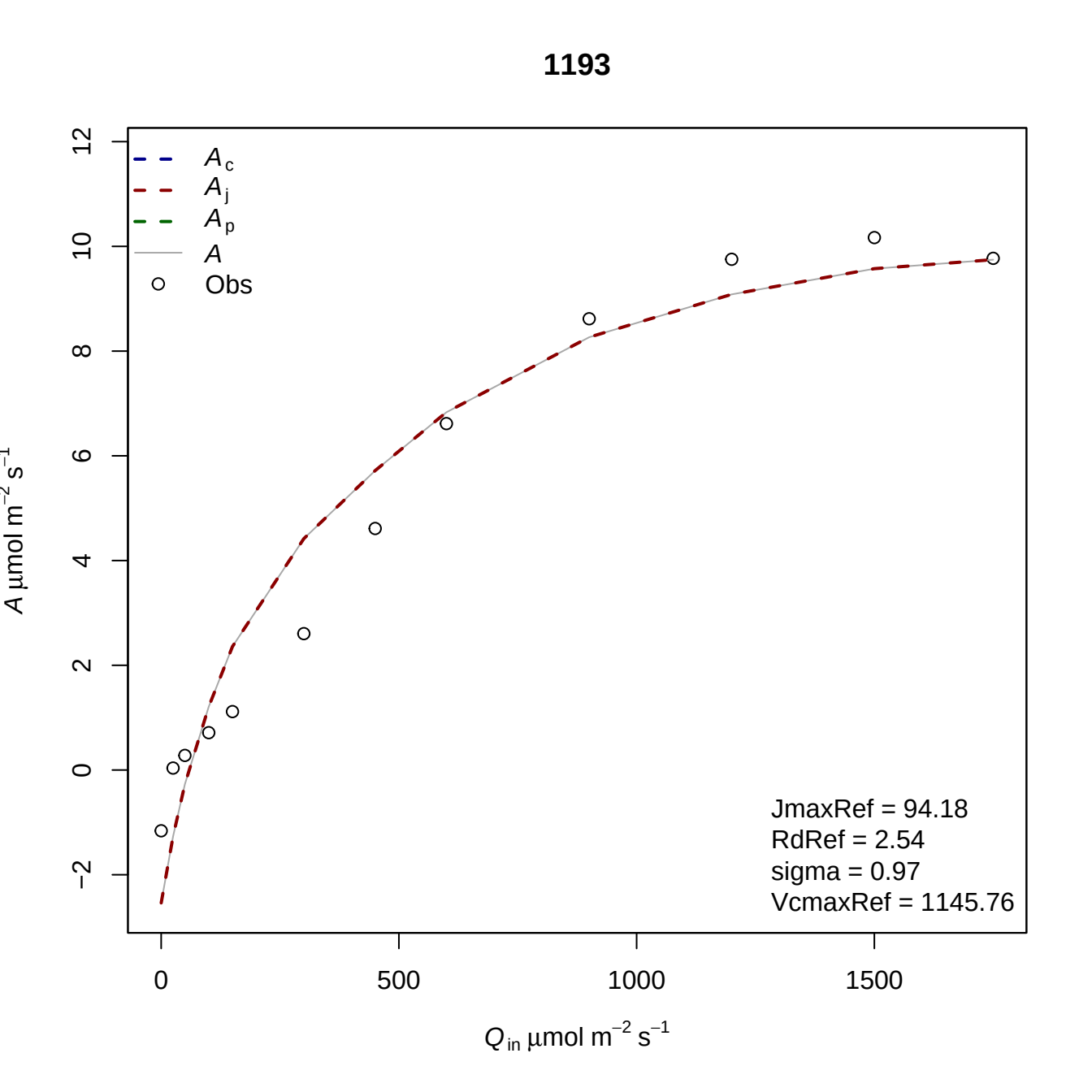

1188
A_c
A_j
A_p
A
Obs
JmaxRef = 146.35
RdRef = 1.36
sigma = 0.31
VcmaxRef = 84.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

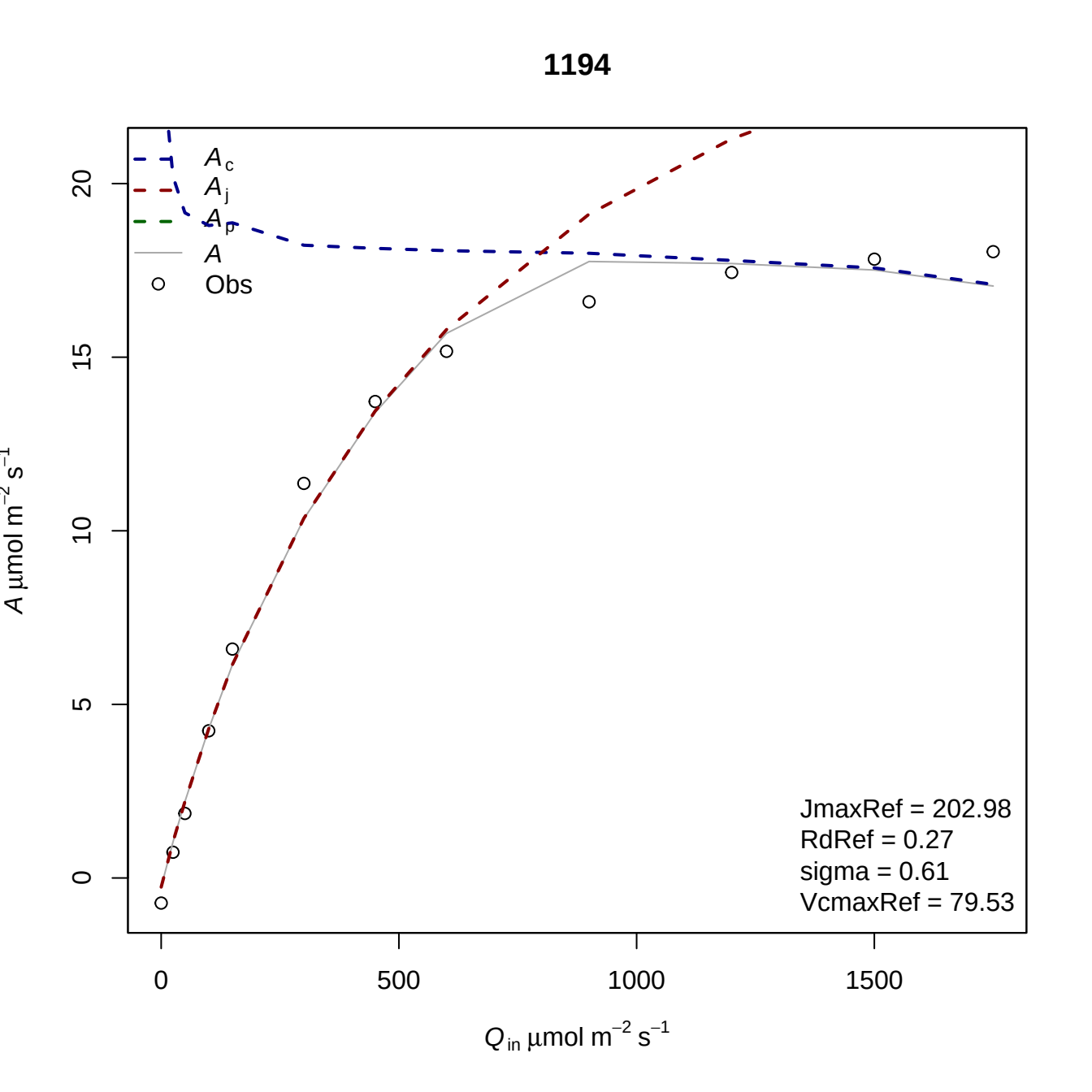

1189
A_c
A_j
A_p
A
Obs
JmaxRef = 84.06
RdRef = 2.67
sigma = 1.27
VcmaxRef = 28529.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

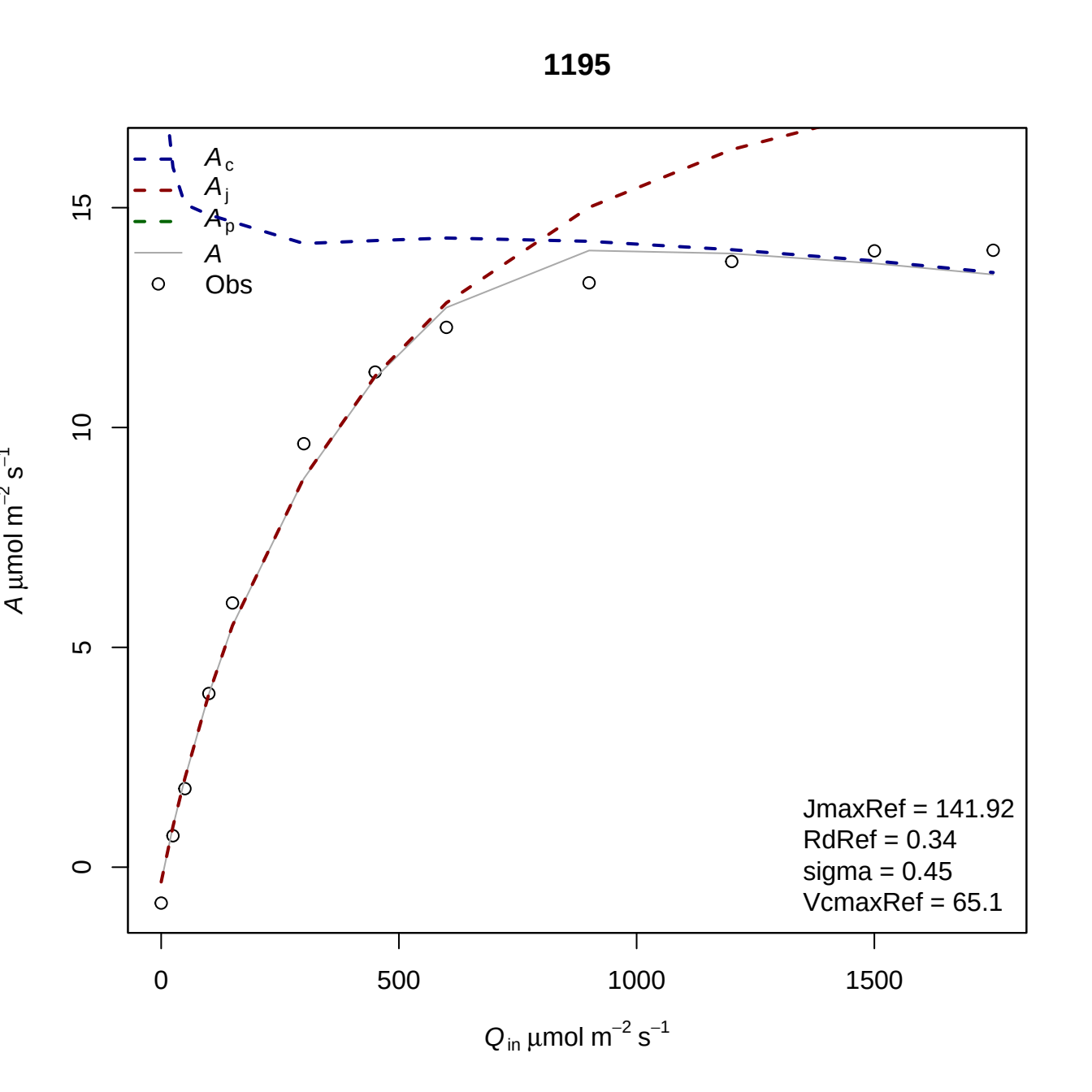

1190
A_c
A_j
A_p
A
Obs
JmaxRef = 94.1
RdRef = 2.09
sigma = 0.93
VcmaxRef = 3404.07
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

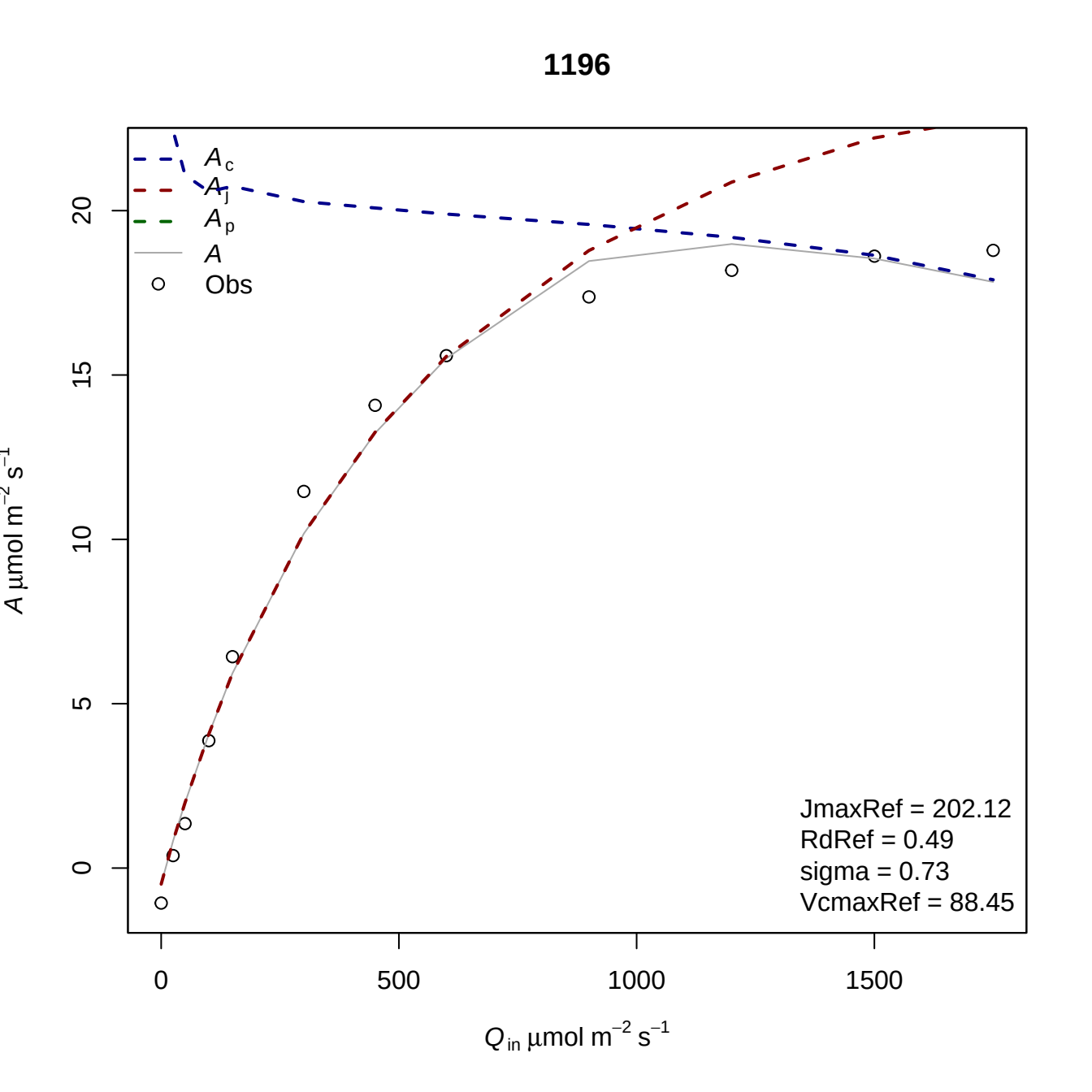

1191
A_c
A_j
A_p
A
Obs
JmaxRef = 93.23
RdRef = 2.28
sigma = 0.68
VcmaxRef = 696.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

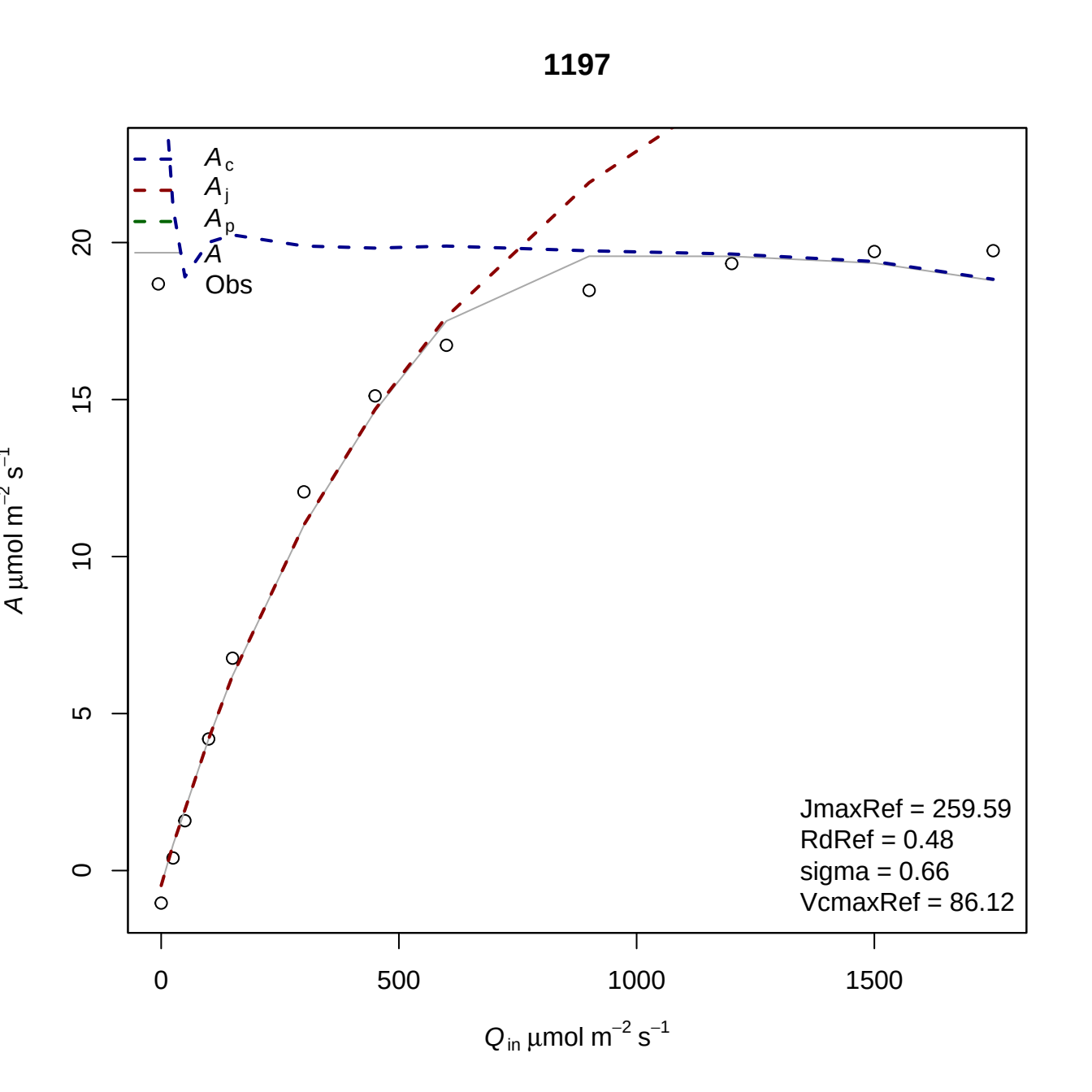

1192
A_c
A_j
A_p
A
Obs
JmaxRef = 139.05
RdRef = 1.82
sigma = 0.63
VcmaxRef = 81.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

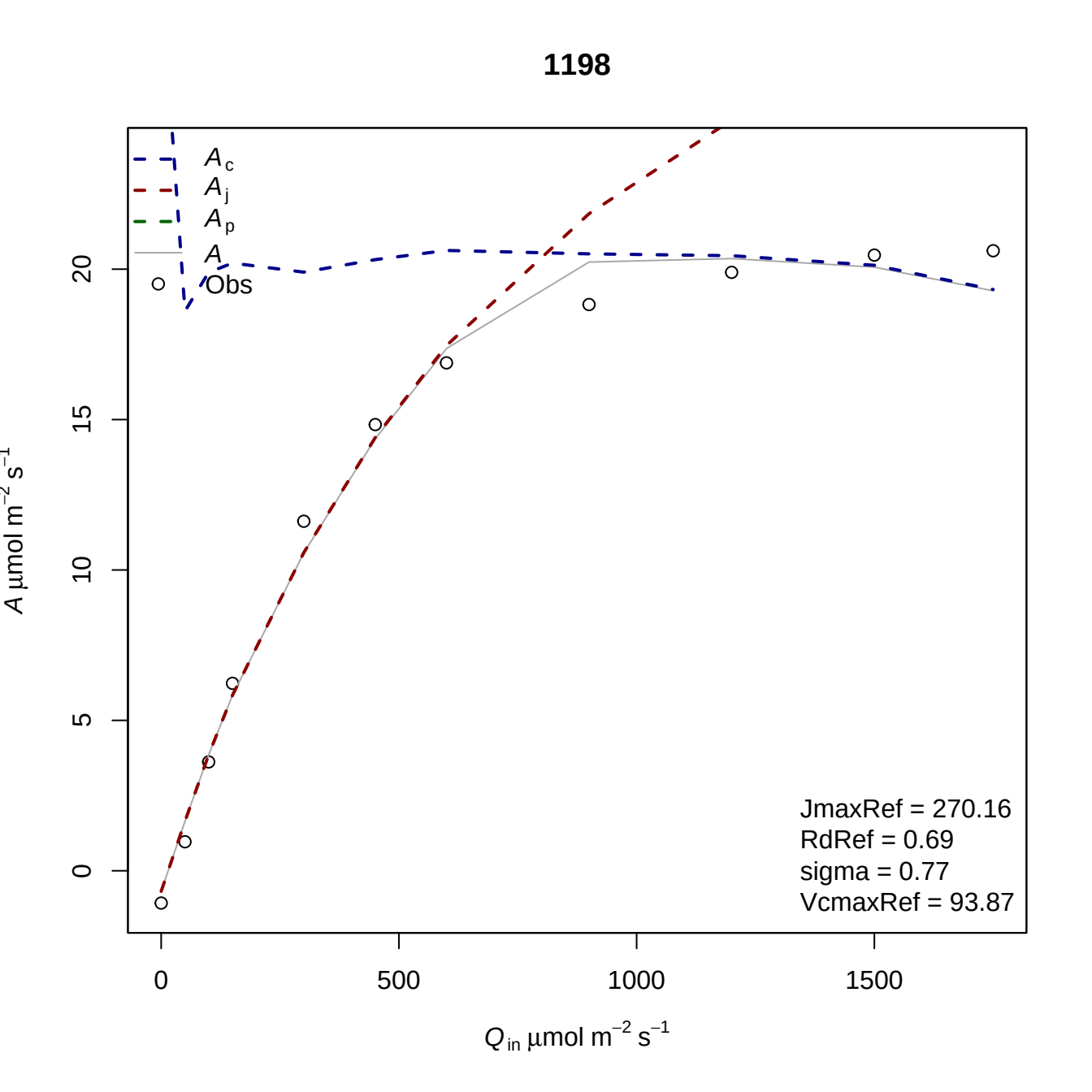

1193
A_c
A_j
A_p
A
Obs
JmaxRef = 94.18
RdRef = 2.54
sigma = 0.97
VcmaxRef = 1145.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

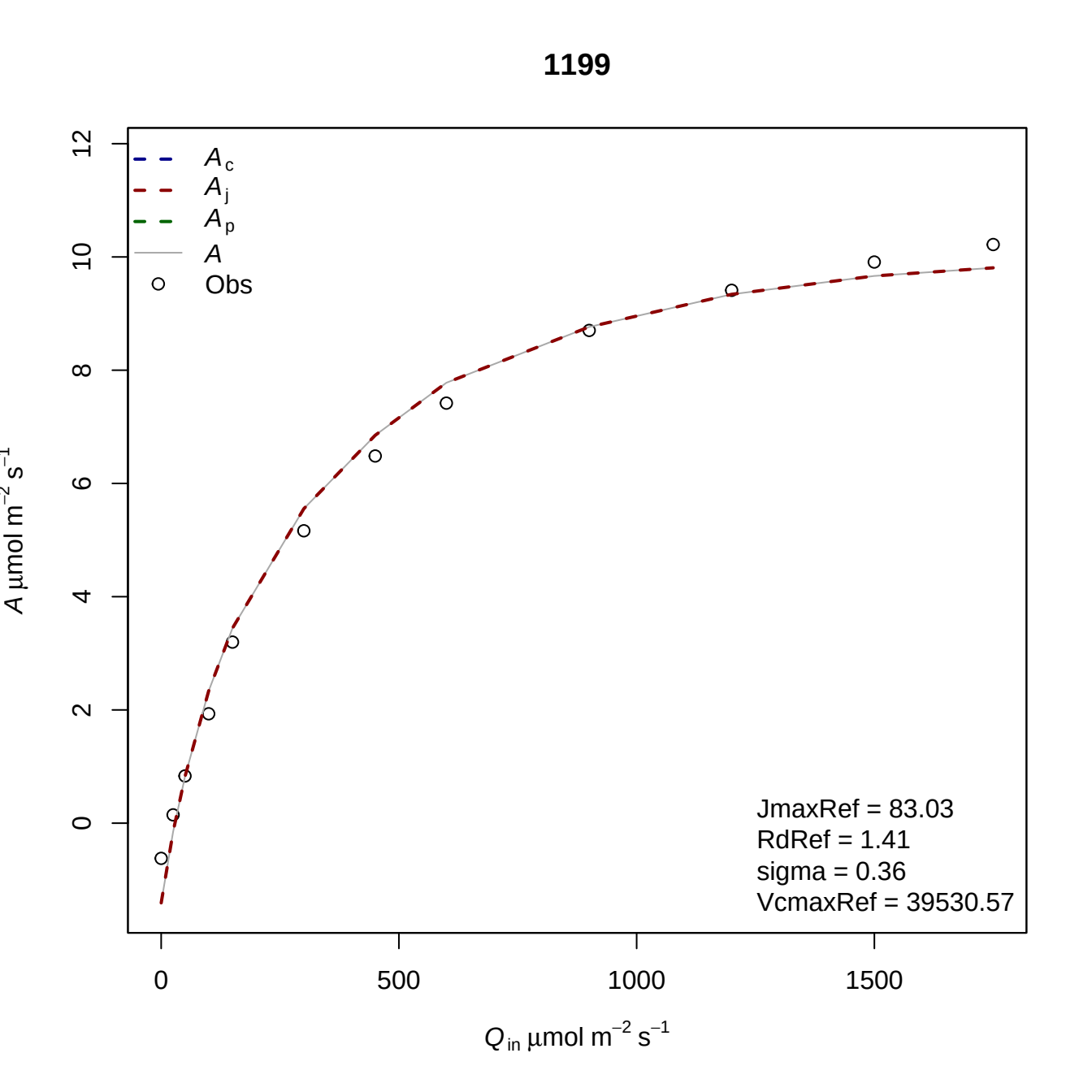

1194
A_c
A_j
A_p
A
Obs
JmaxRef = 202.98
RdRef = 0.27
sigma = 0.61
VcmaxRef = 79.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

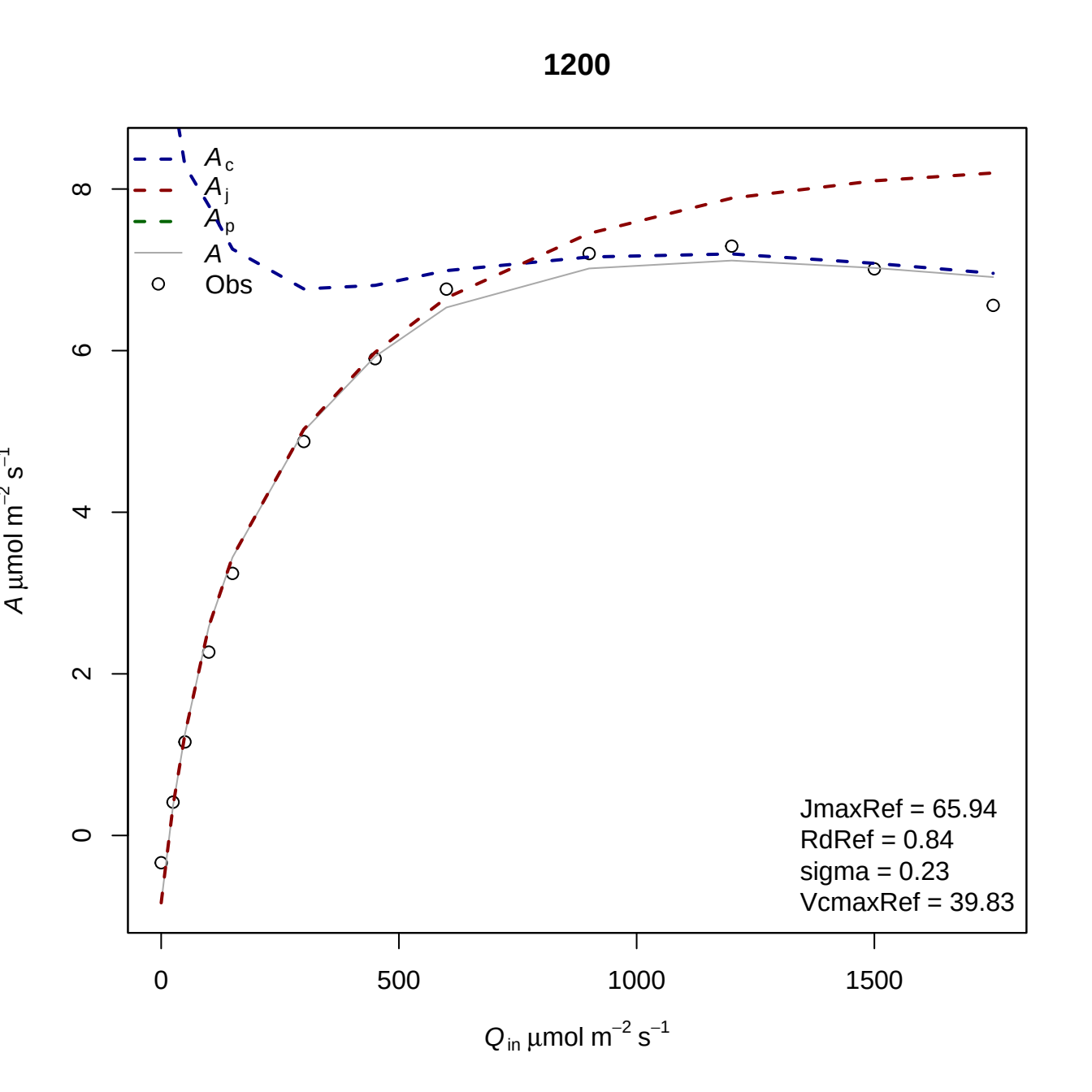

1195
A_c
A_j
A_p
A
Obs
JmaxRef = 141.92
RdRef = 0.34
sigma = 0.45
VcmaxRef = 65.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

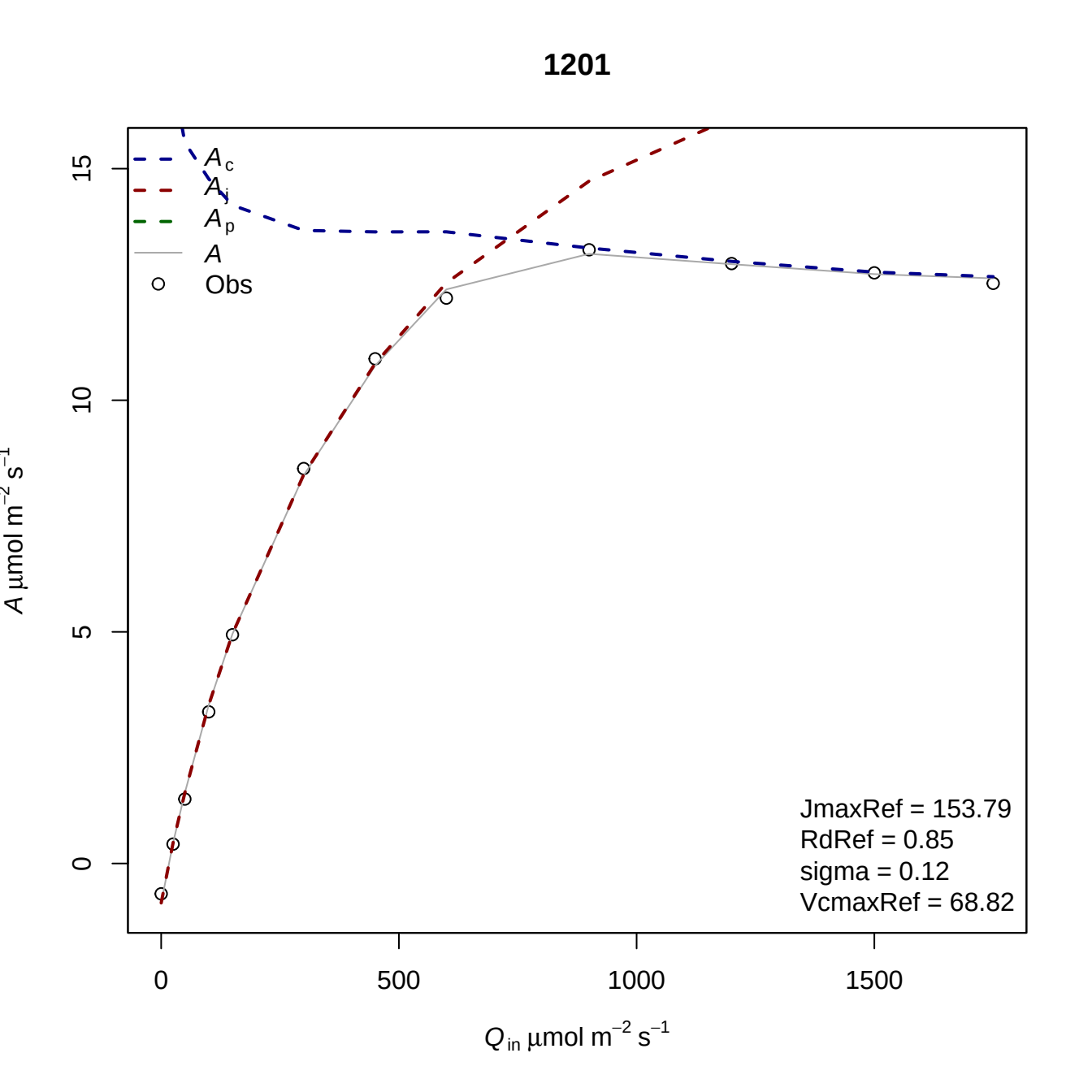

1196
A_c
A_j
A_p
A
Obs
JmaxRef = 202.12
RdRef = 0.49
sigma = 0.73
VcmaxRef = 88.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

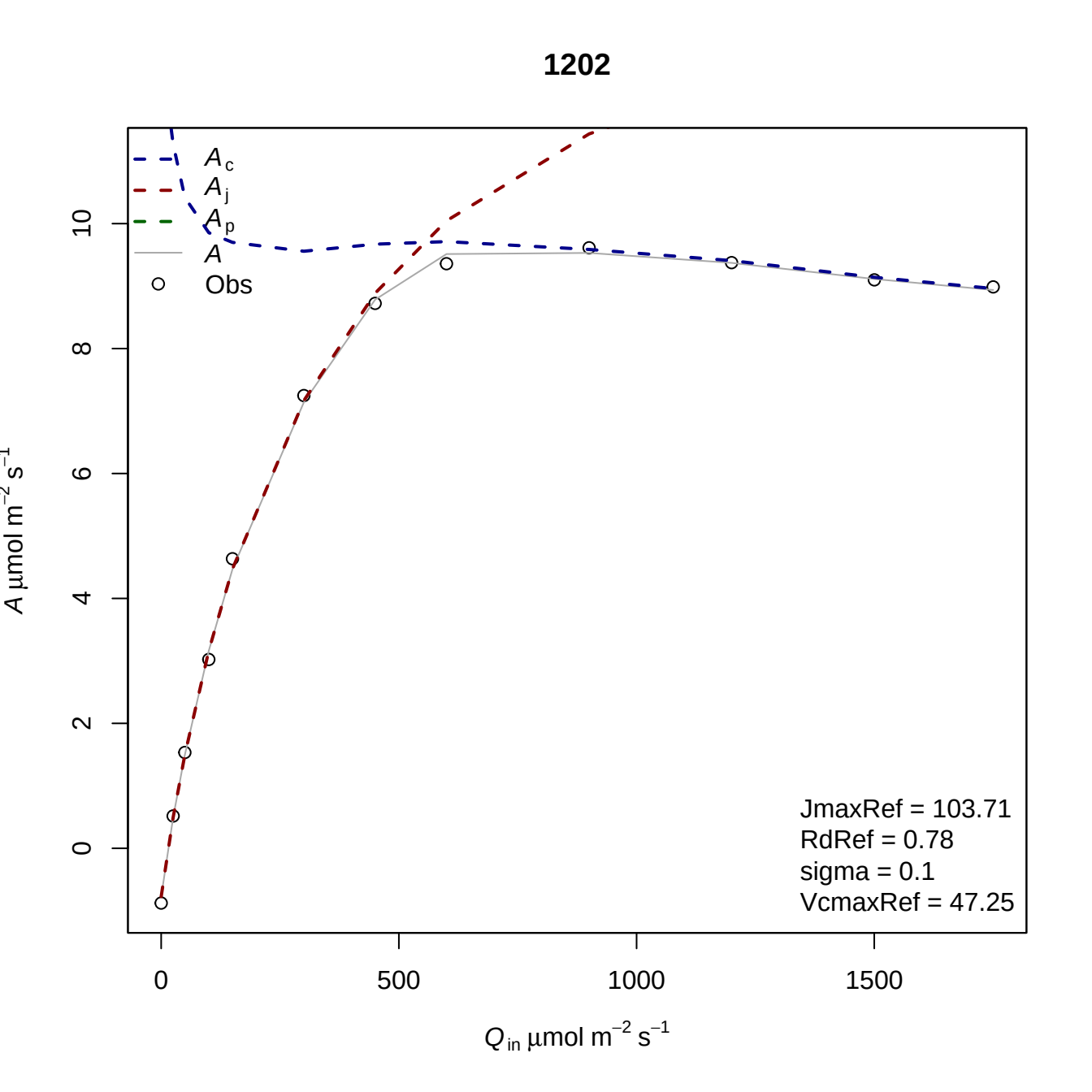

1197
A_c
A_j
A_p
A
Obs
JmaxRef = 259.59
RdRef = 0.48
sigma = 0.66
VcmaxRef = 86.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

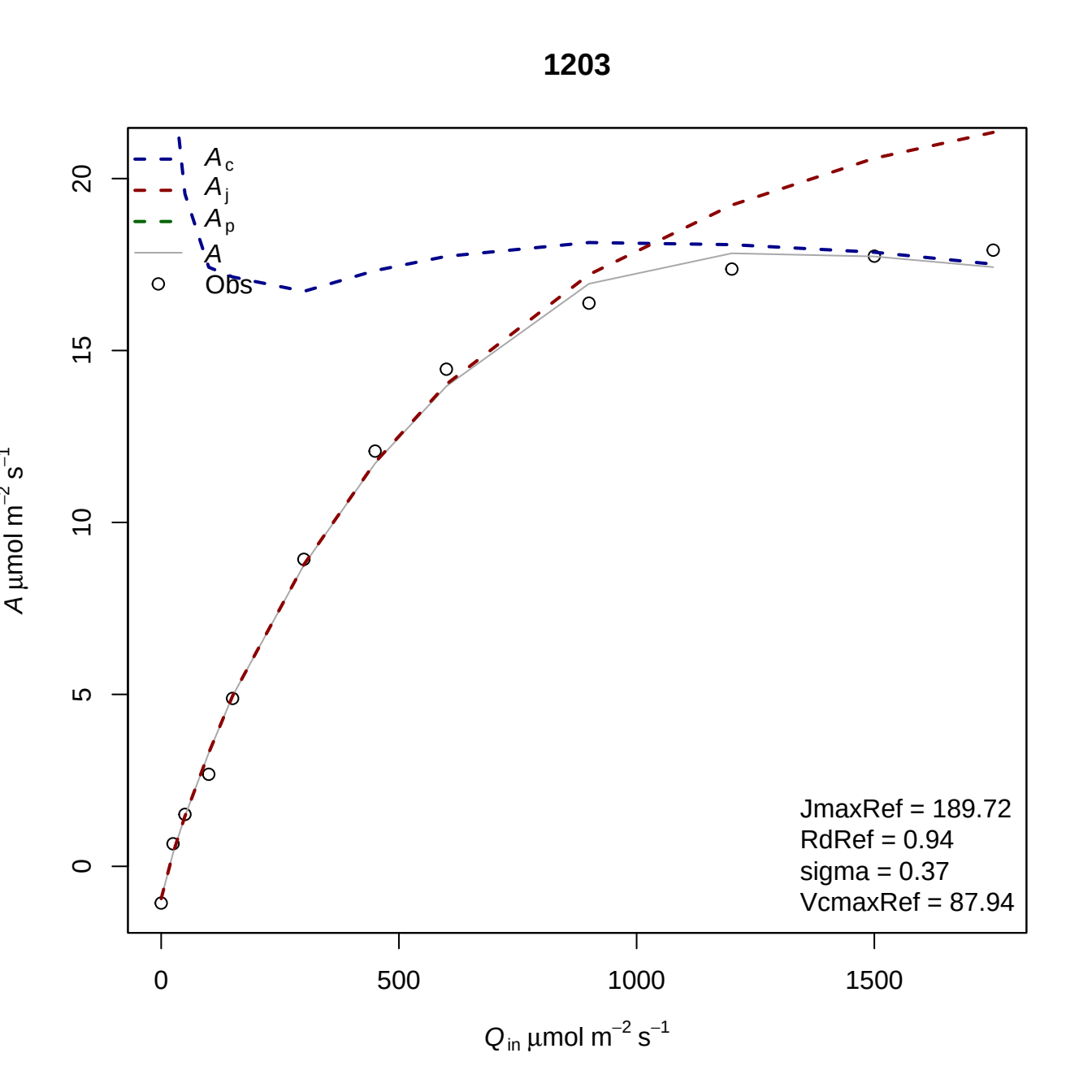

1198
A_c
A_j
A_p
A
Obs
JmaxRef = 270.16
RdRef = 0.69
sigma = 0.77
VcmaxRef = 93.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

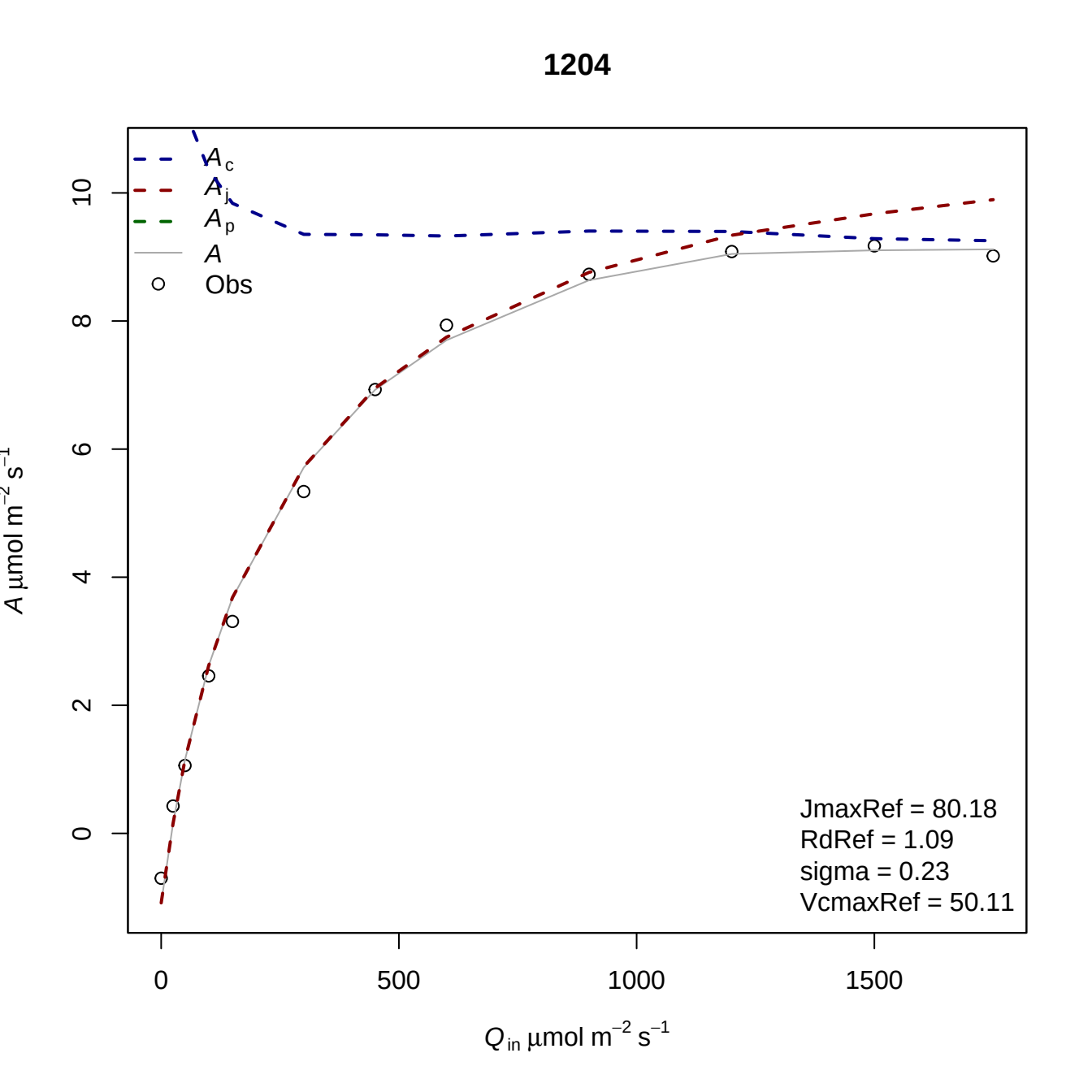

1199
A_c
A_j
A_p
A
Obs
JmaxRef = 83.03
RdRef = 1.41
sigma = 0.36
VcmaxRef = 39530.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

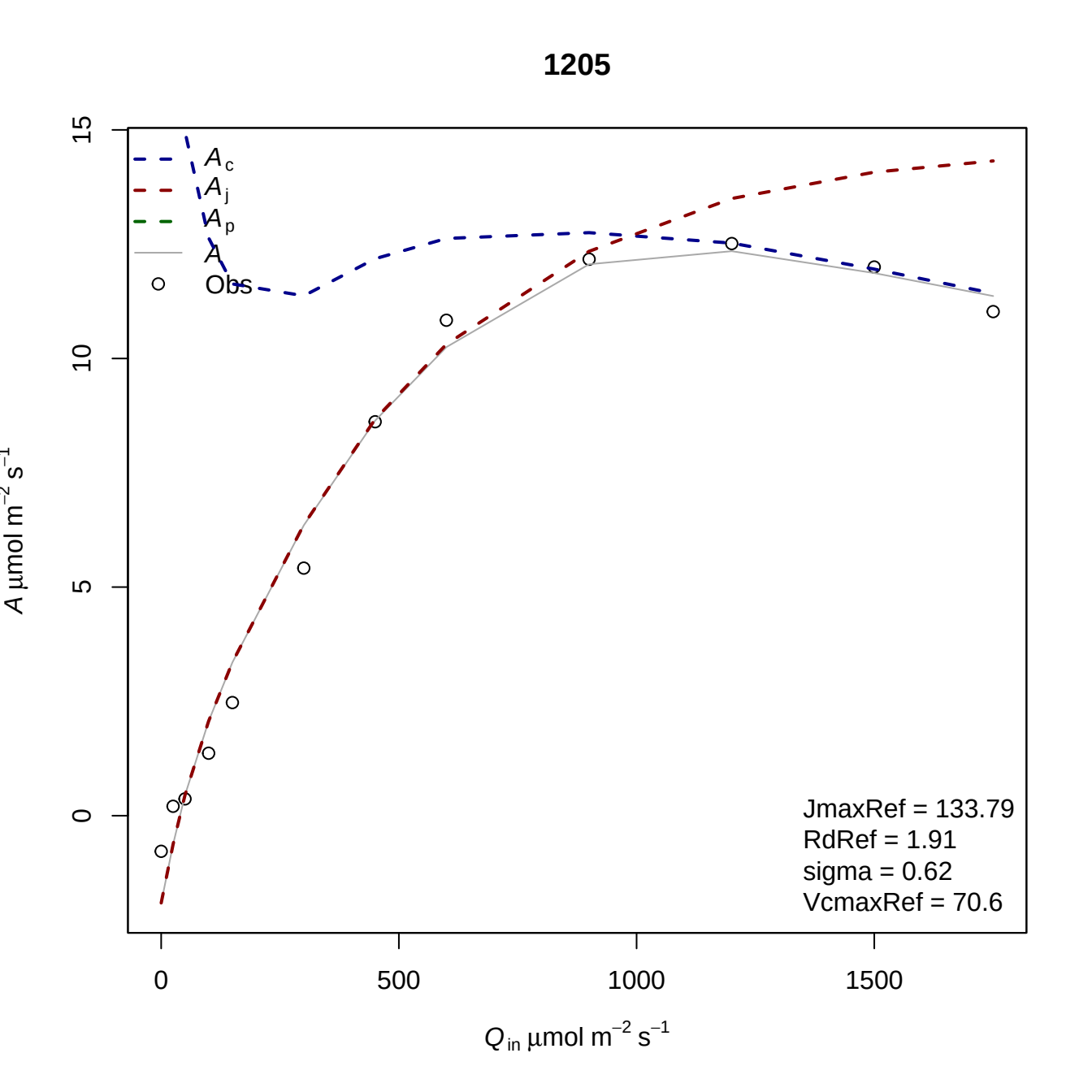

1200
A_c
A_j
A_p
A
Obs
JmaxRef = 65.94
RdRef = 0.84
sigma = 0.23
VcmaxRef = 39.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

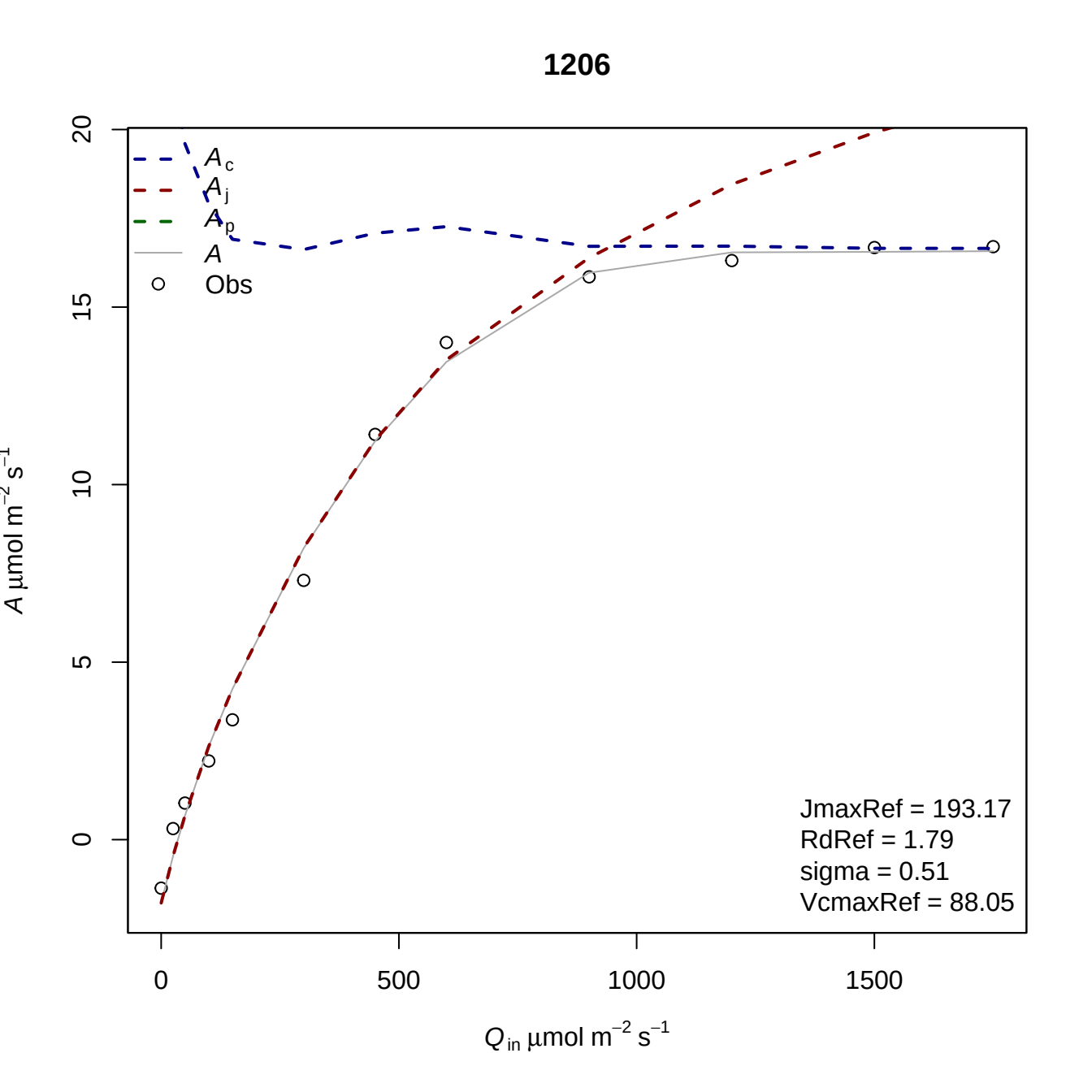

1201
A_c
A_j
A_p
A
Obs
JmaxRef = 153.79
RdRef = 0.85
sigma = 0.12
VcmaxRef = 68.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

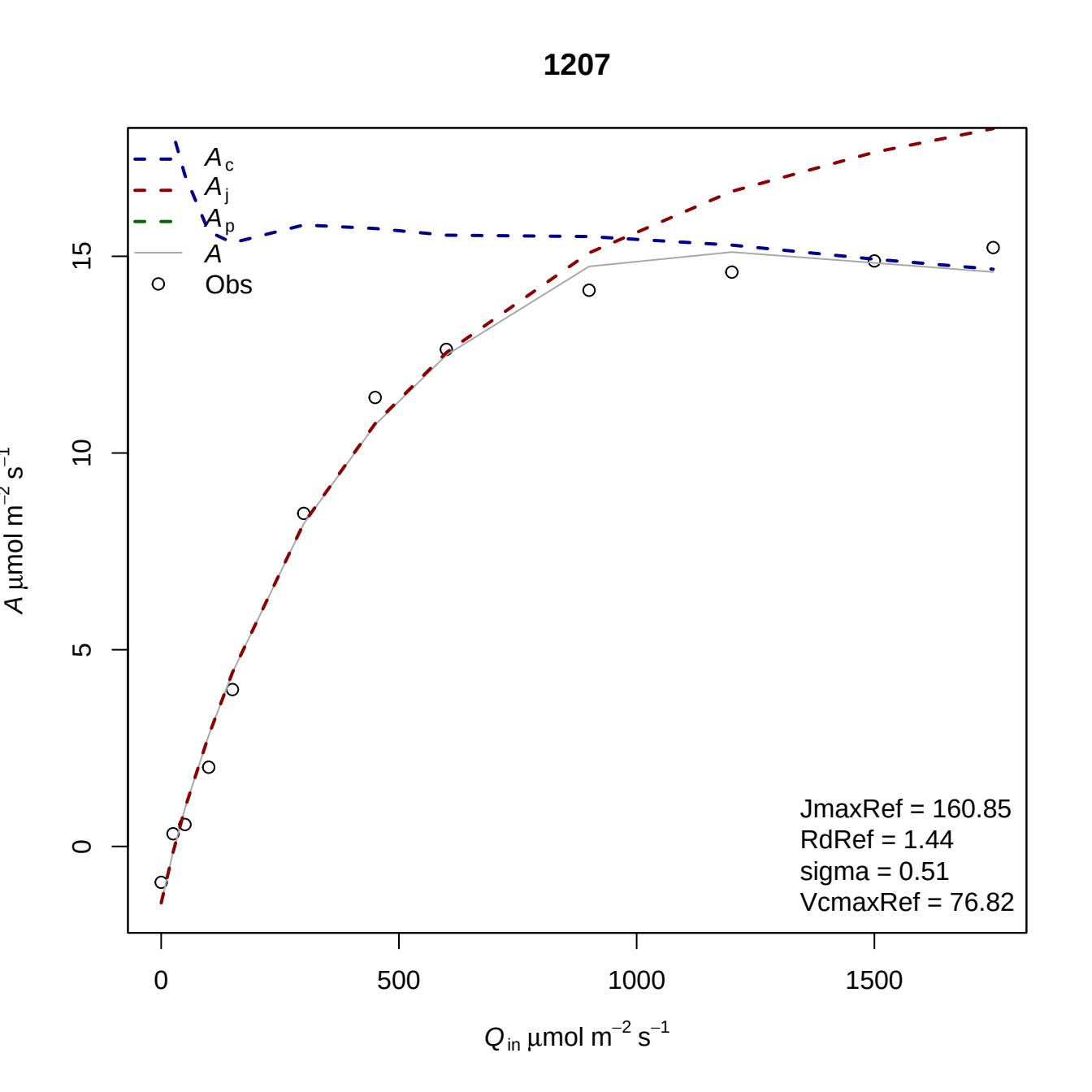

1202
A_c
A_j
A_p
A
Obs
JmaxRef = 103.71
RdRef = 0.78
sigma = 0.1
VcmaxRef = 47.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

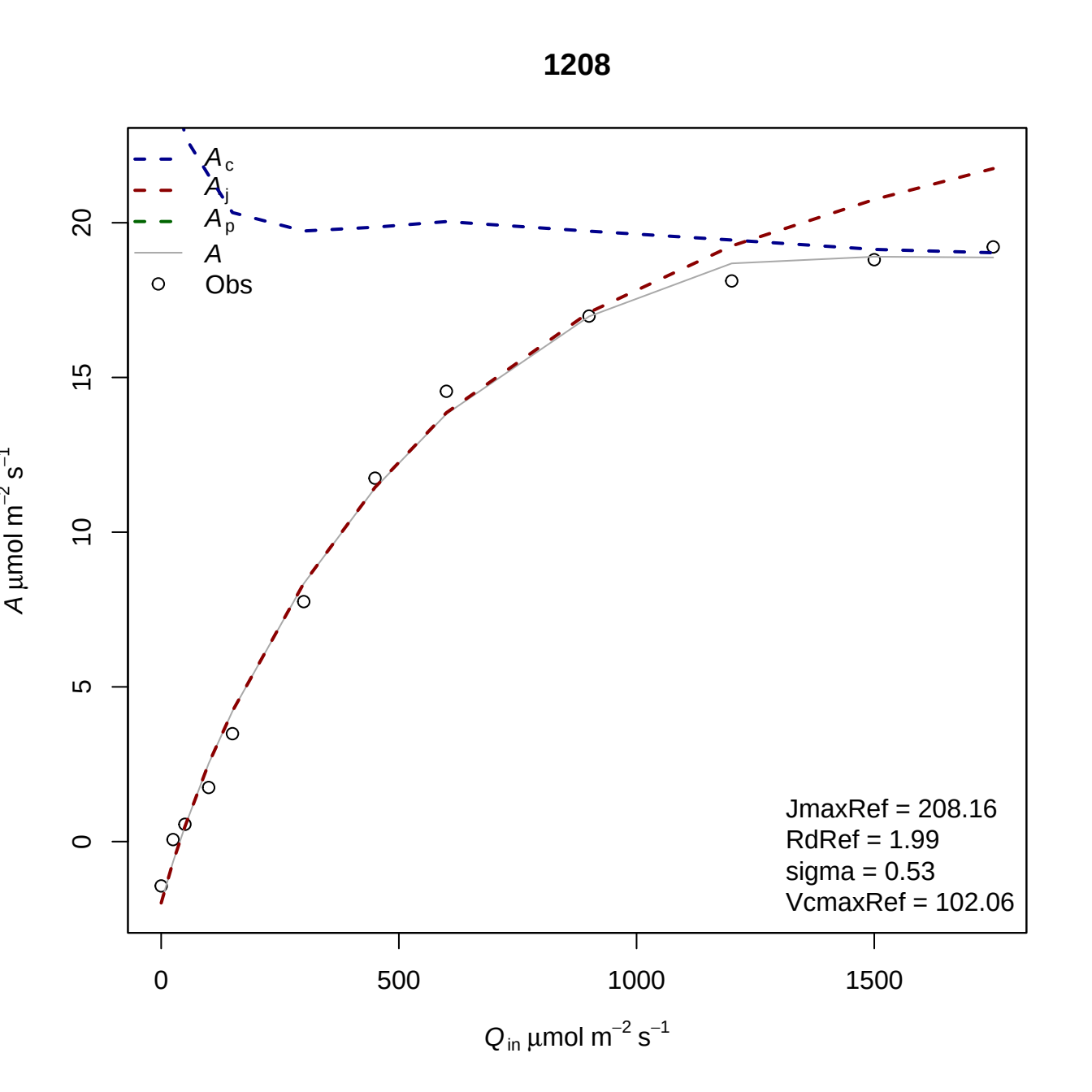

1203
A_c
A_j
A_p
A
Obs
JmaxRef = 189.72
RdRef = 0.94
sigma = 0.37
VcmaxRef = 87.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

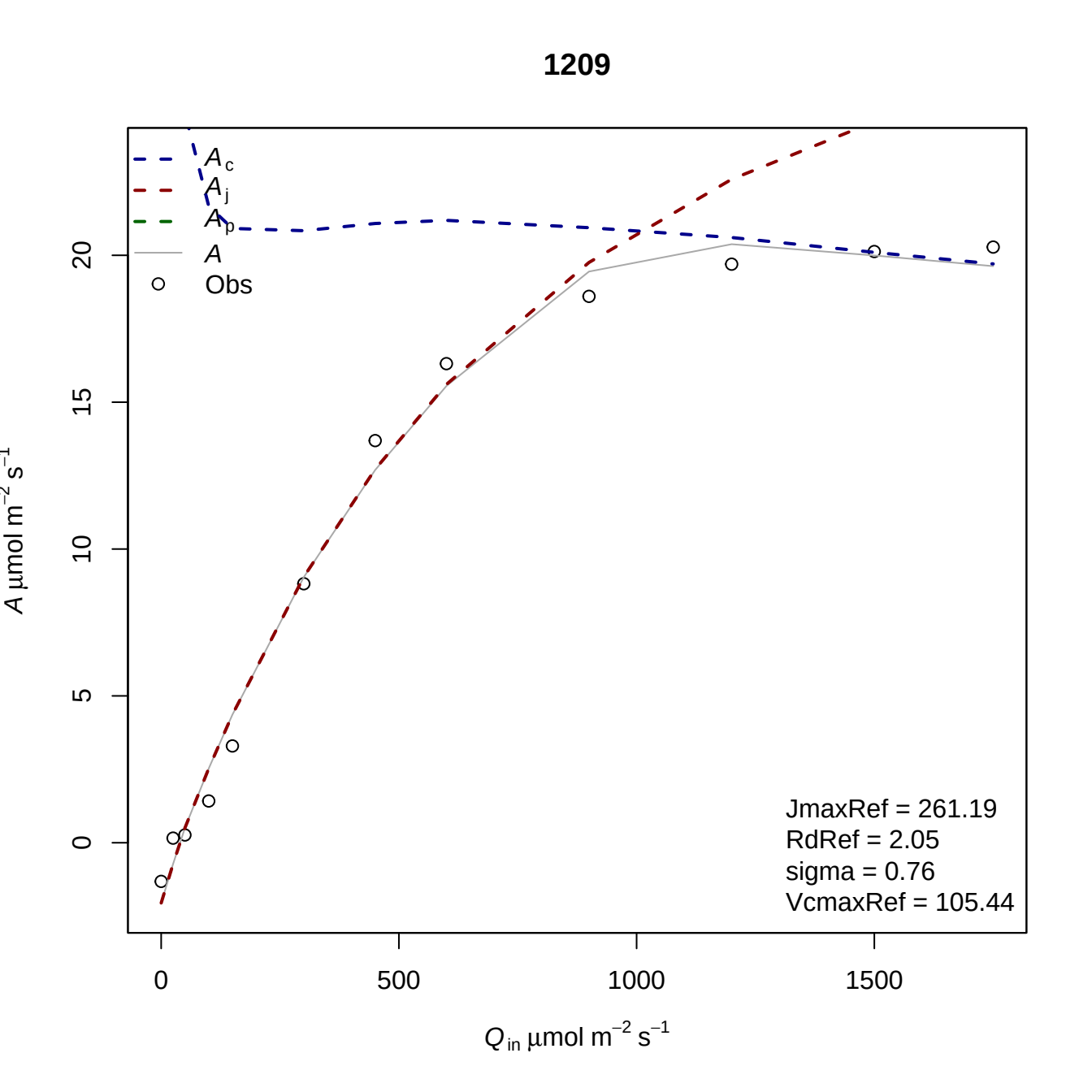

1204
A_c
A_j
A_p
A
Obs
JmaxRef = 80.18
RdRef = 1.09
sigma = 0.23
VcmaxRef = 50.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

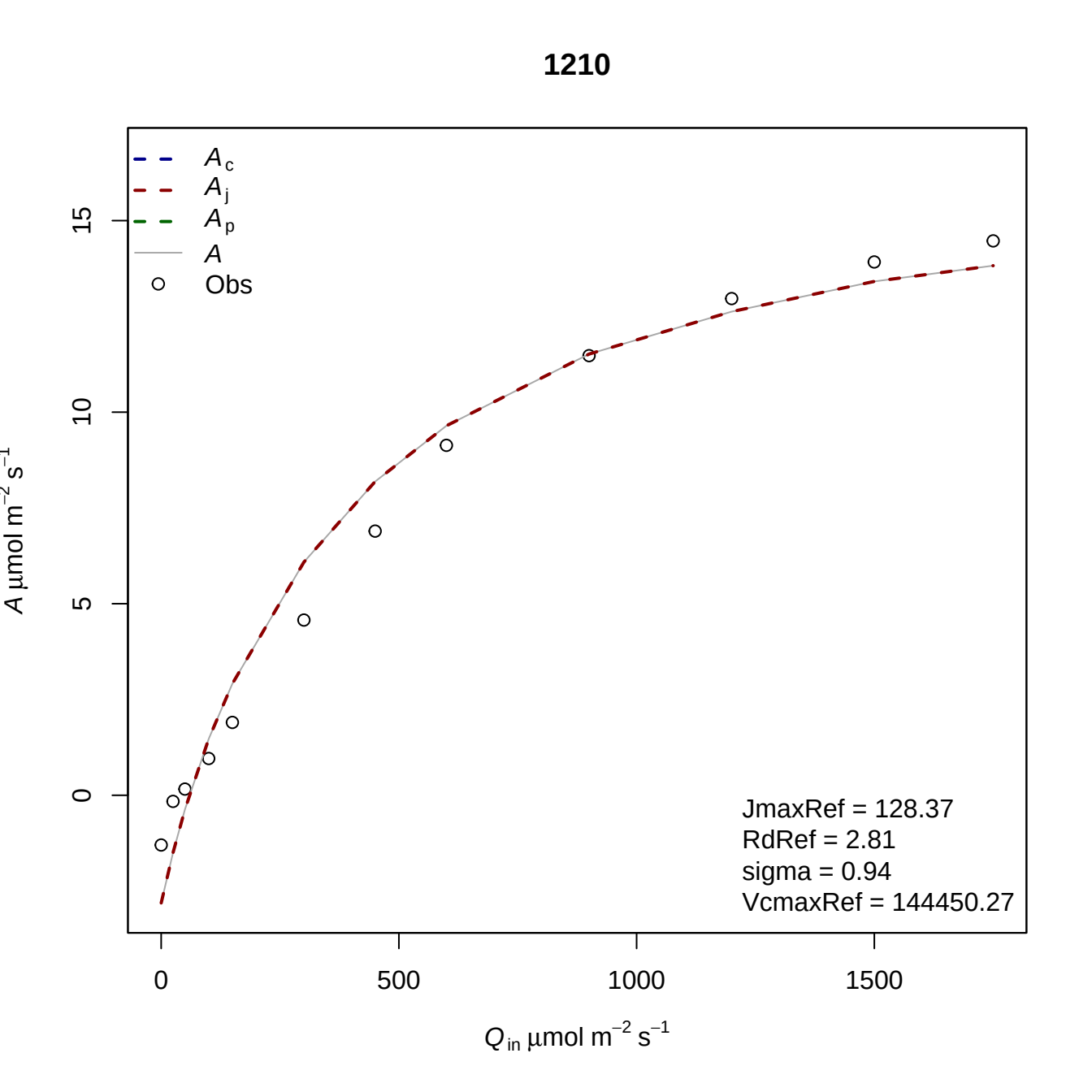

1205
A_c
A_j
A_p
A
Obs
JmaxRef = 133.79
RdRef = 1.91
sigma = 0.62
VcmaxRef = 70.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

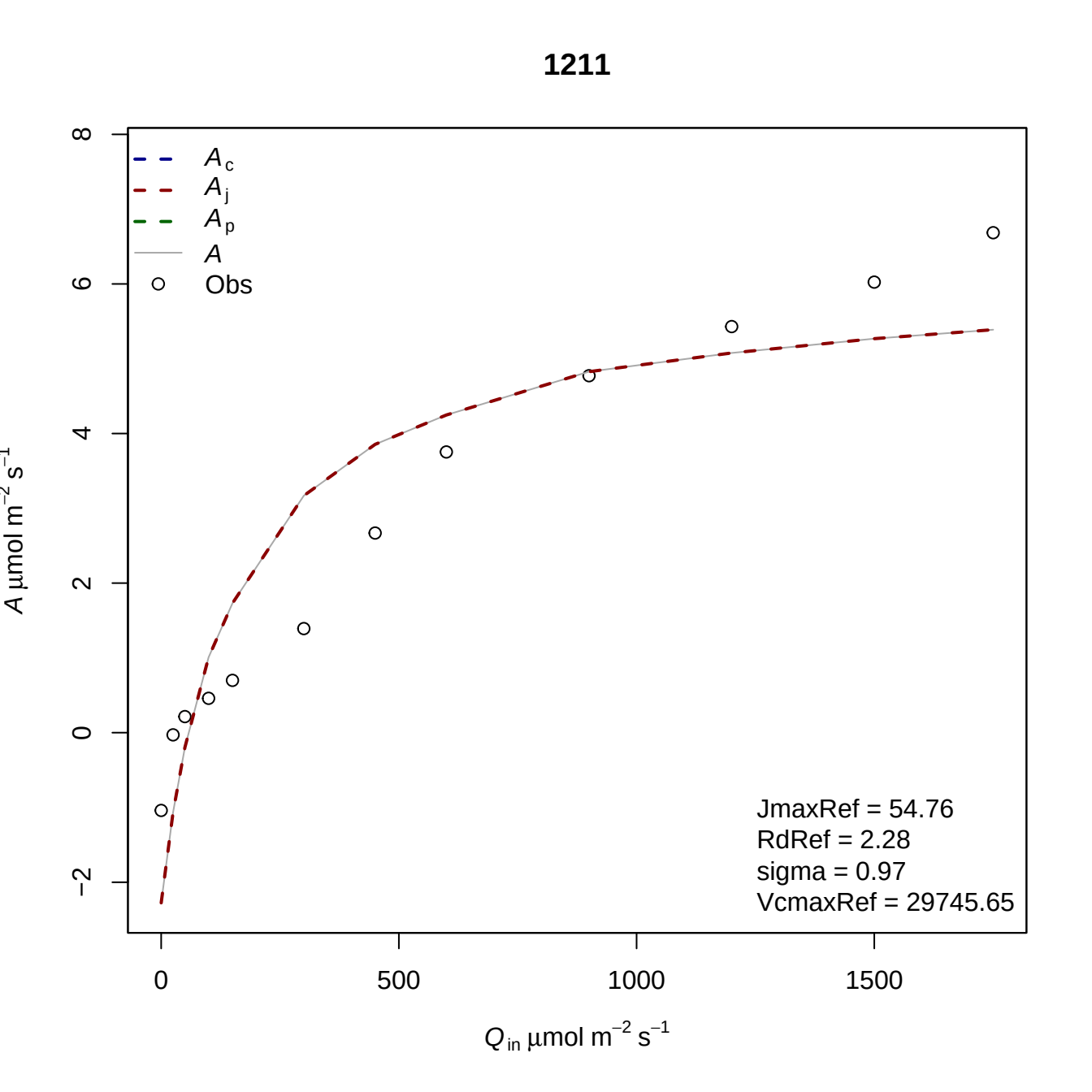

1206
A_c
A_j
A_p
A
Obs
JmaxRef = 193.17
RdRef = 1.79
sigma = 0.51
VcmaxRef = 88.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

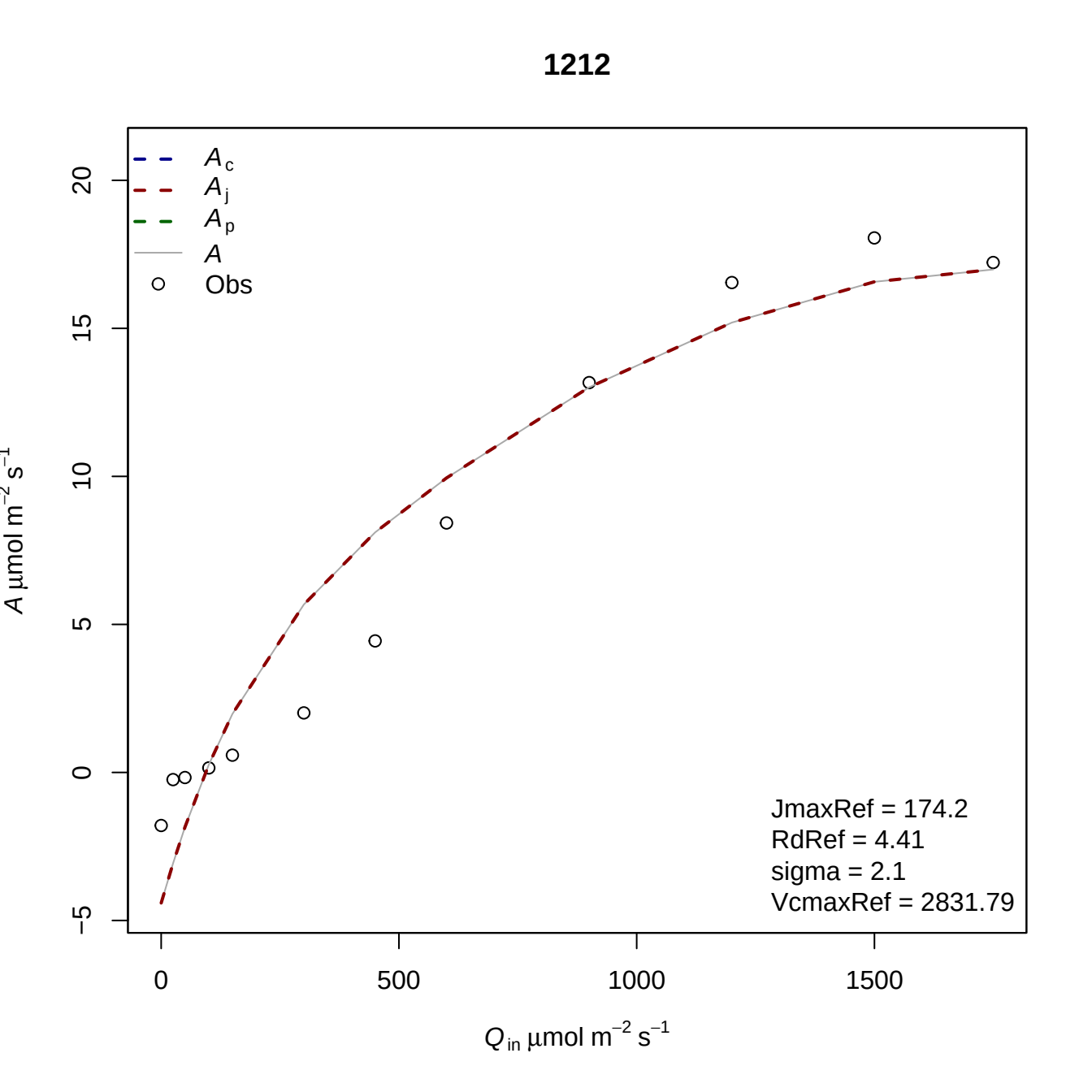

1207
A_c
A_j
A_p
A
Obs
JmaxRef = 160.85
RdRef = 1.44
sigma = 0.51
VcmaxRef = 76.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

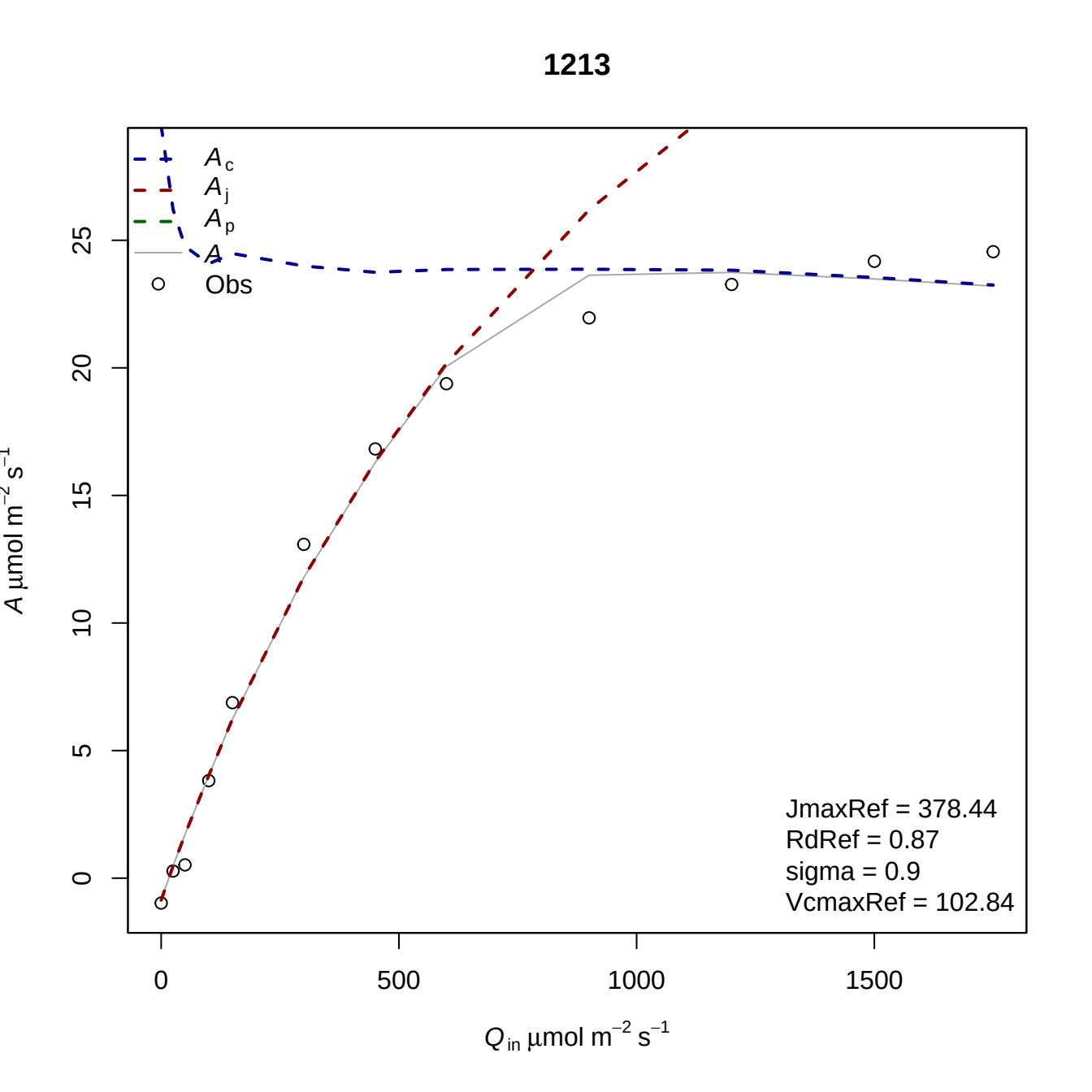

1208
A_c
A_j
A_p
A
Obs
JmaxRef = 208.16
RdRef = 1.99
sigma = 0.53
VcmaxRef = 102.06
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

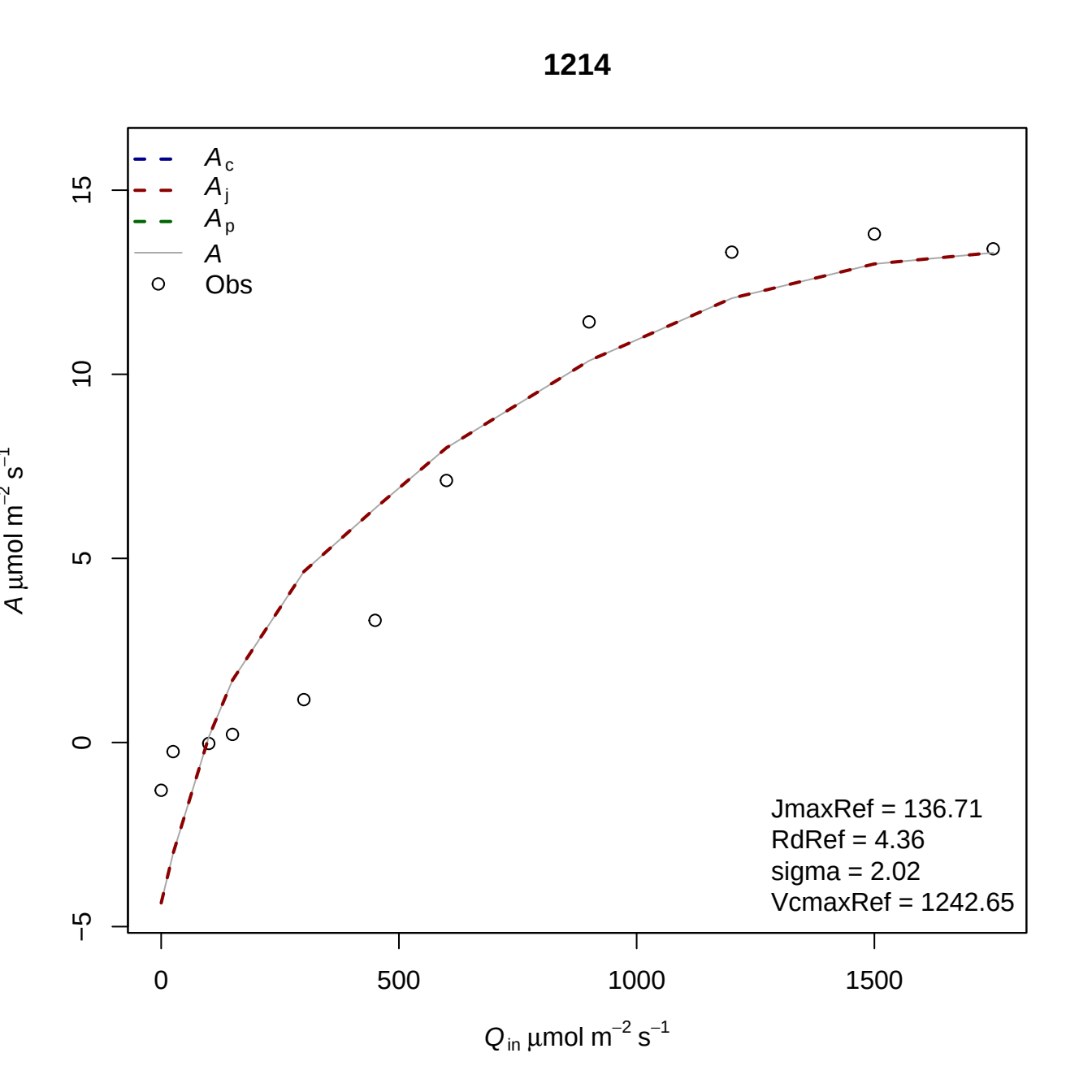

1209
A_c
A_j
A_p
A
Obs
JmaxRef = 261.19
RdRef = 2.05
sigma = 0.76
VcmaxRef = 105.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

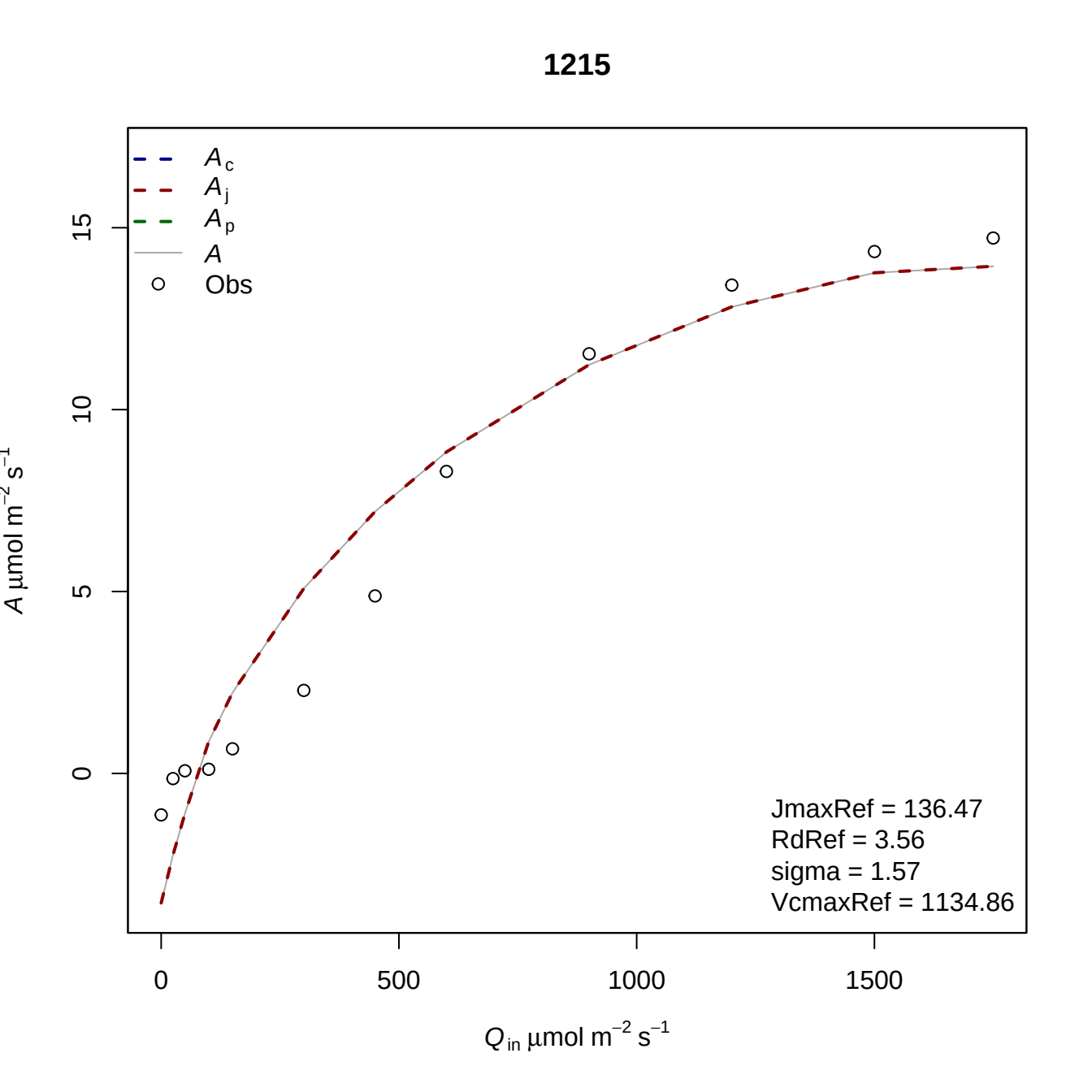

1210
A_c
A_j
A_p
A
Obs
JmaxRef = 128.37
RdRef = 2.81
sigma = 0.94
VcmaxRef = 144450.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

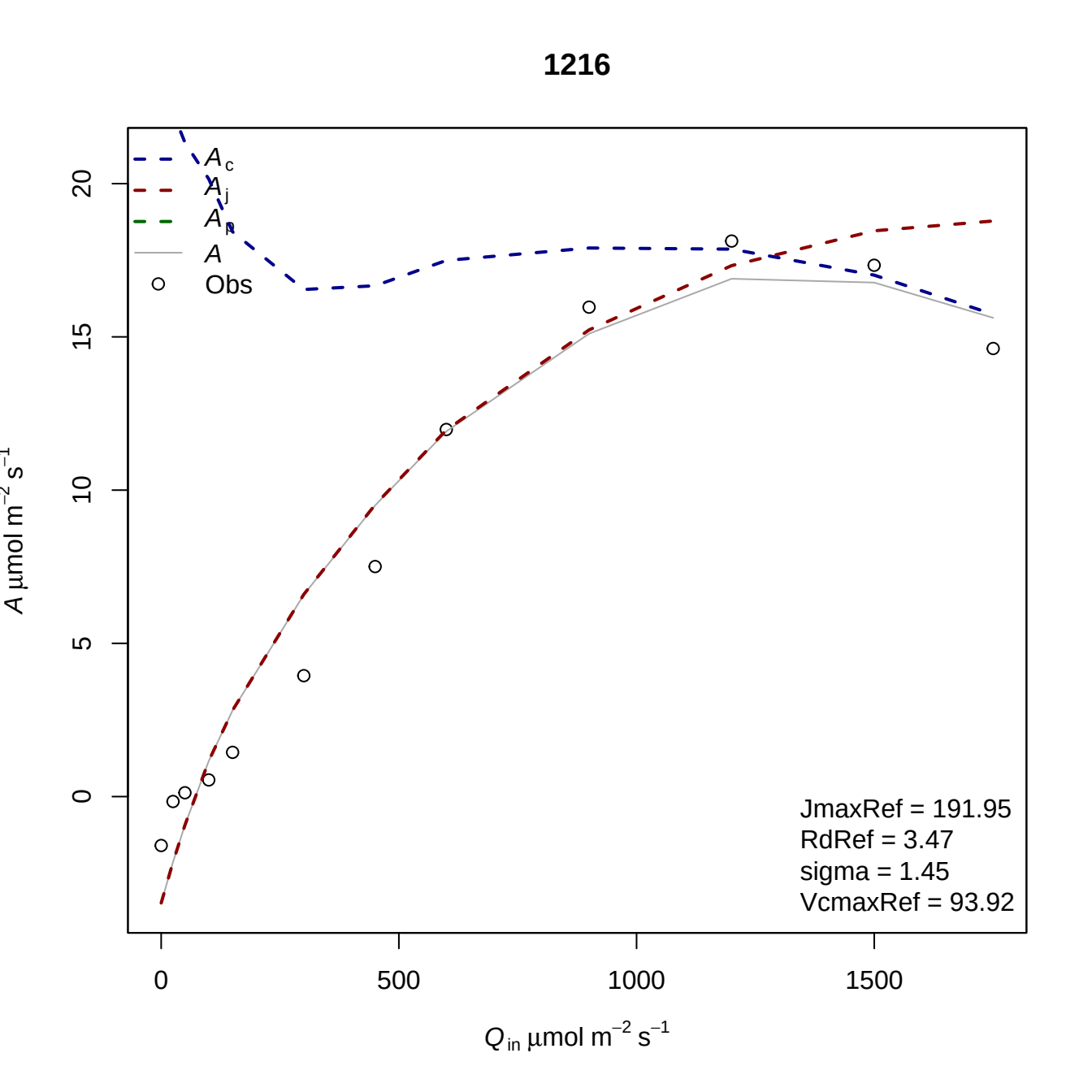

1211
A_c
A_j
A_p
A
Obs
JmaxRef = 54.76
RdRef = 2.28
sigma = 0.97
VcmaxRef = 29745.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

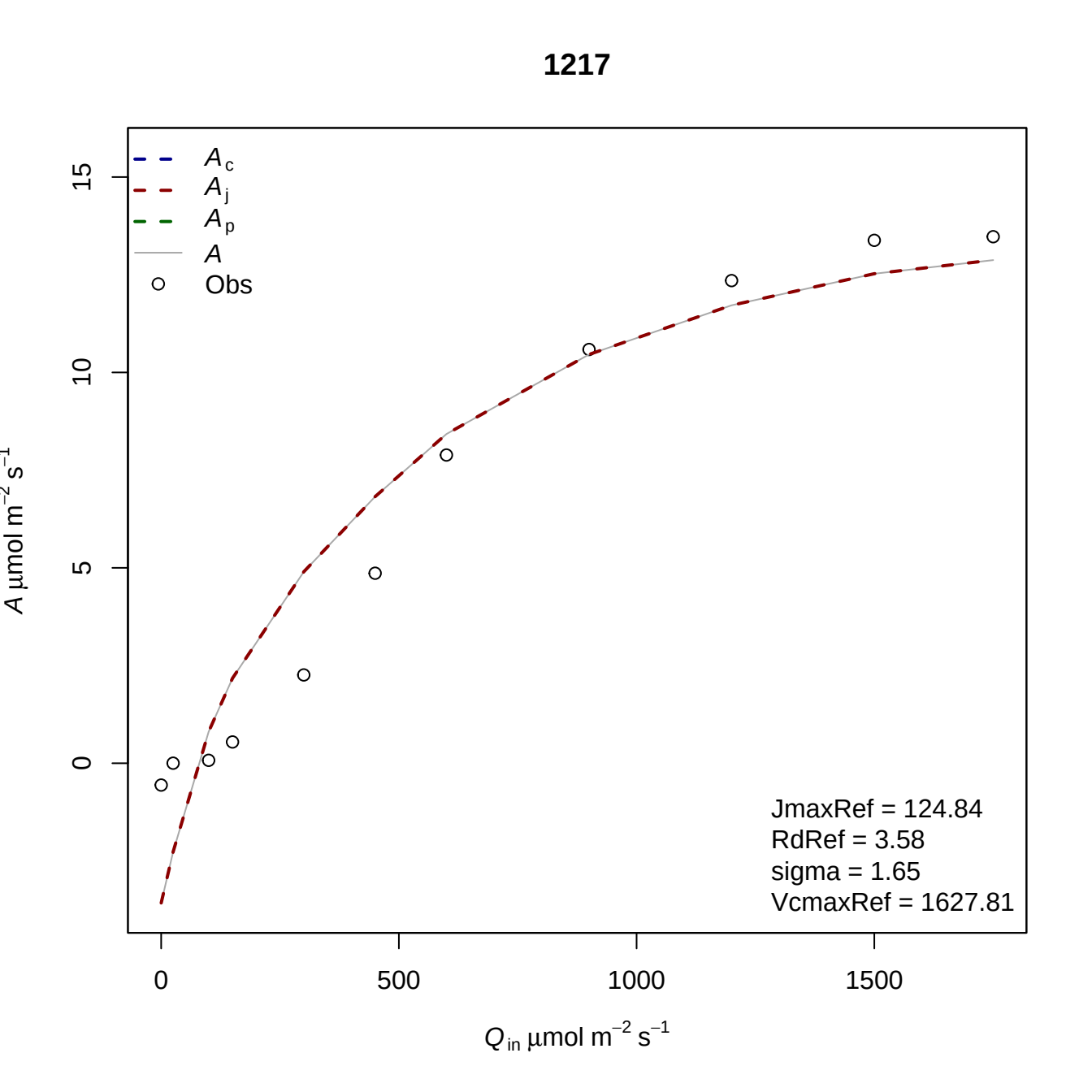

1212
A_c
A_j
A_p
A
Obs
JmaxRef = 174.2
RdRef = 4.41
sigma = 2.1
VcmaxRef = 2831.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500

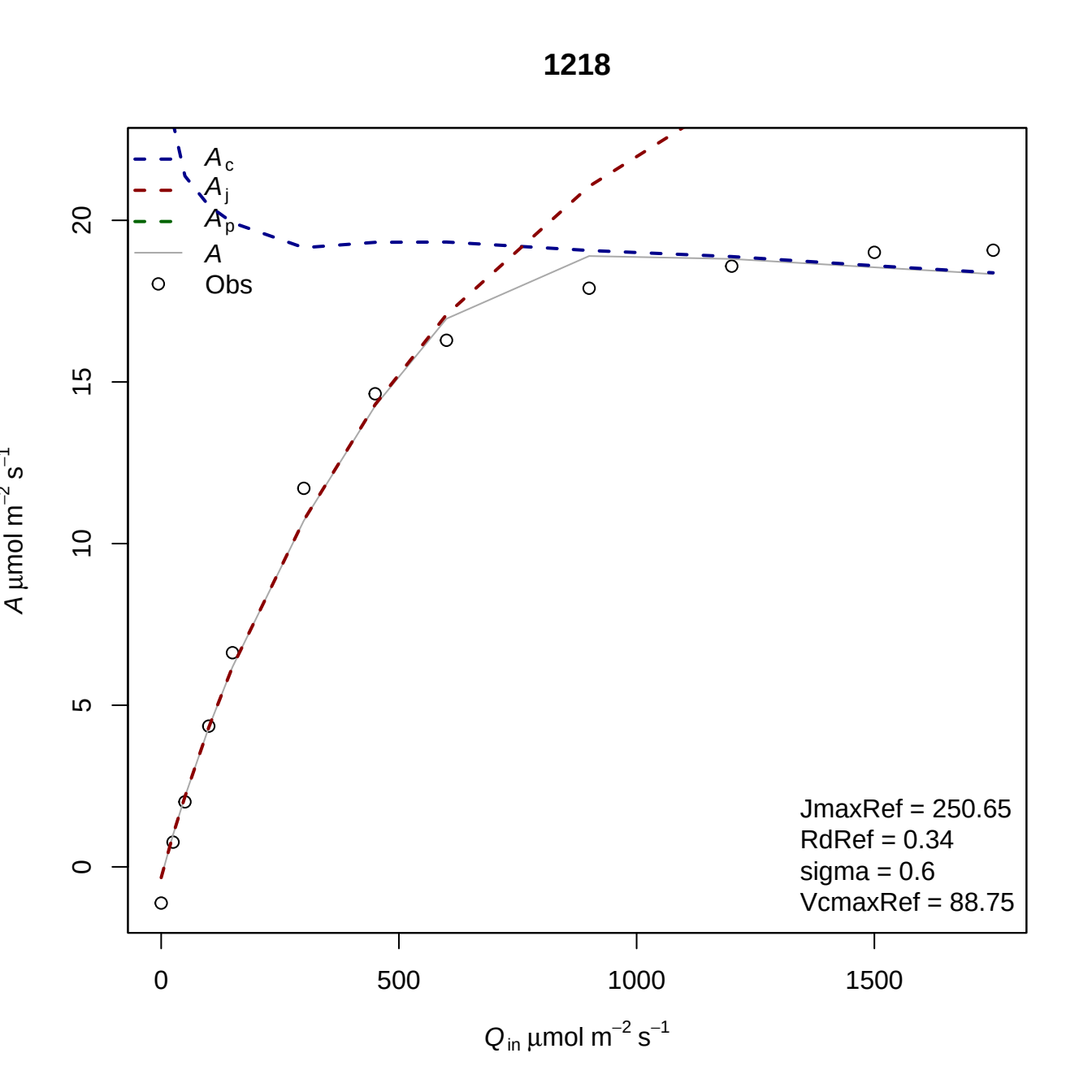

1213
A_c
A_j
A_p
A
Obs
JmaxRef = 378.44
RdRef = 0.87
sigma = 0.9
VcmaxRef = 102.84
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

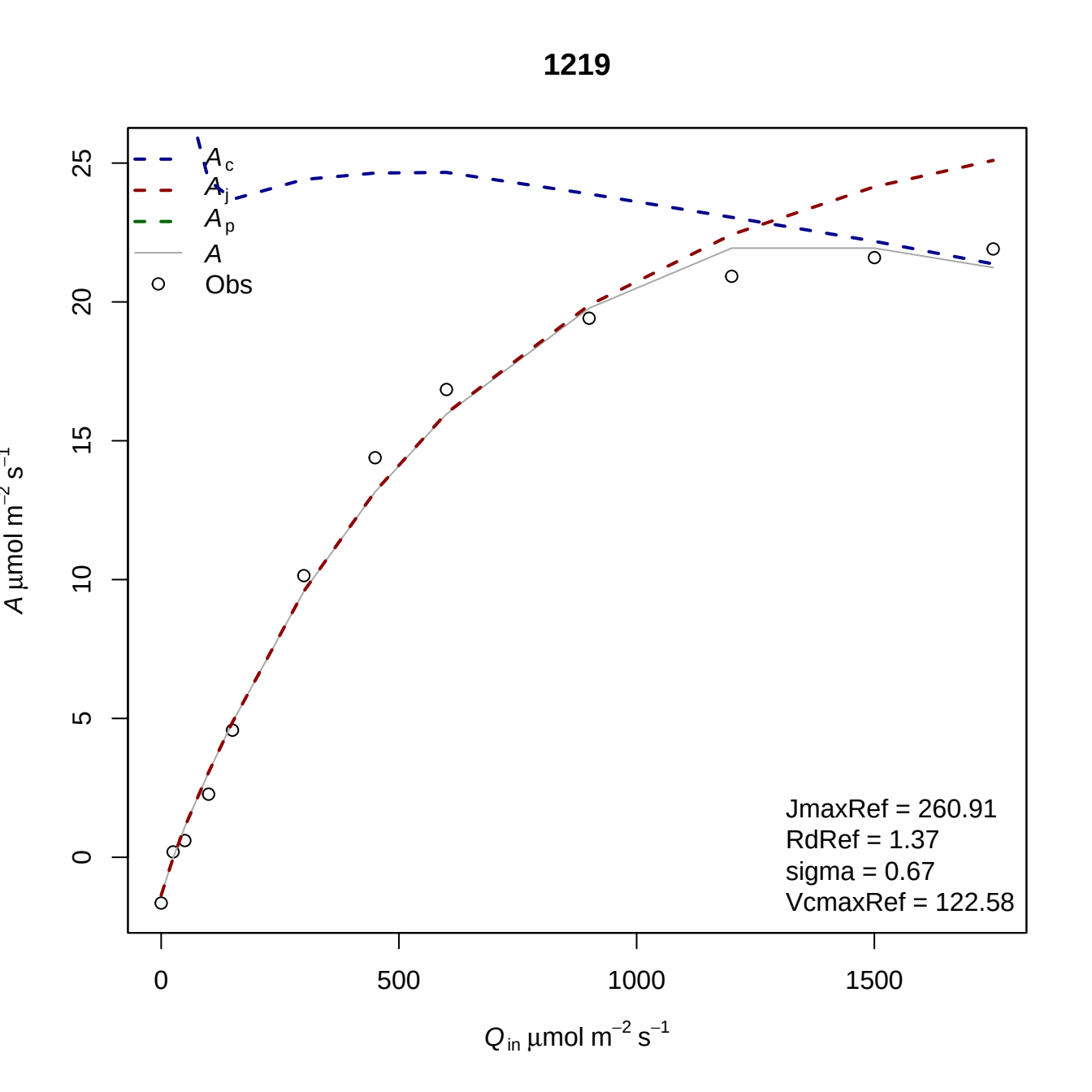

1214
A_c
A_j
A_p
A
Obs
JmaxRef = 136.71
RdRef = 4.36
sigma = 2.02
VcmaxRef = 1242.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500

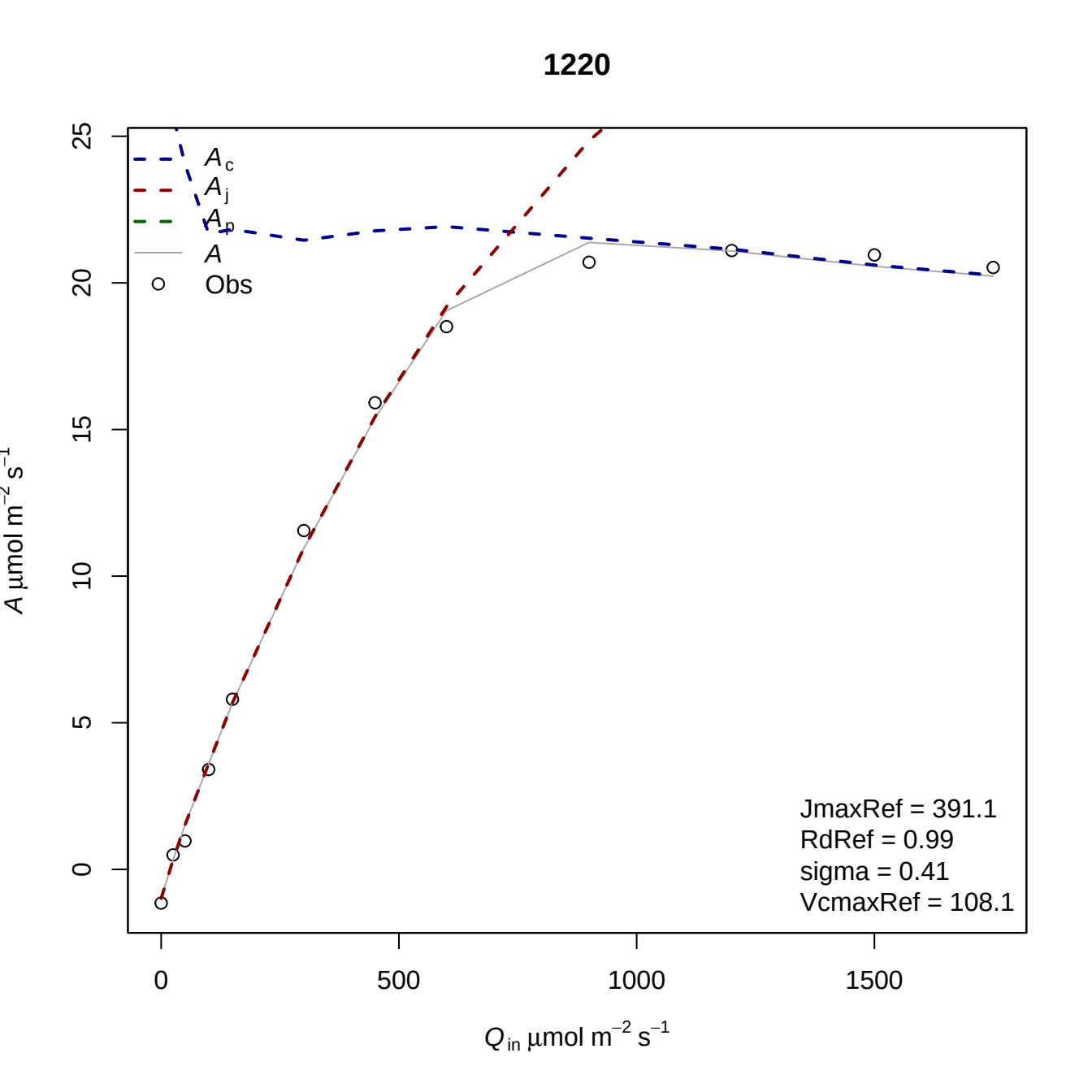

1215
A_c
A_j
A_p
A
Obs
JmaxRef = 136.47
RdRef = 3.56
sigma = 1.57
VcmaxRef = 1134.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

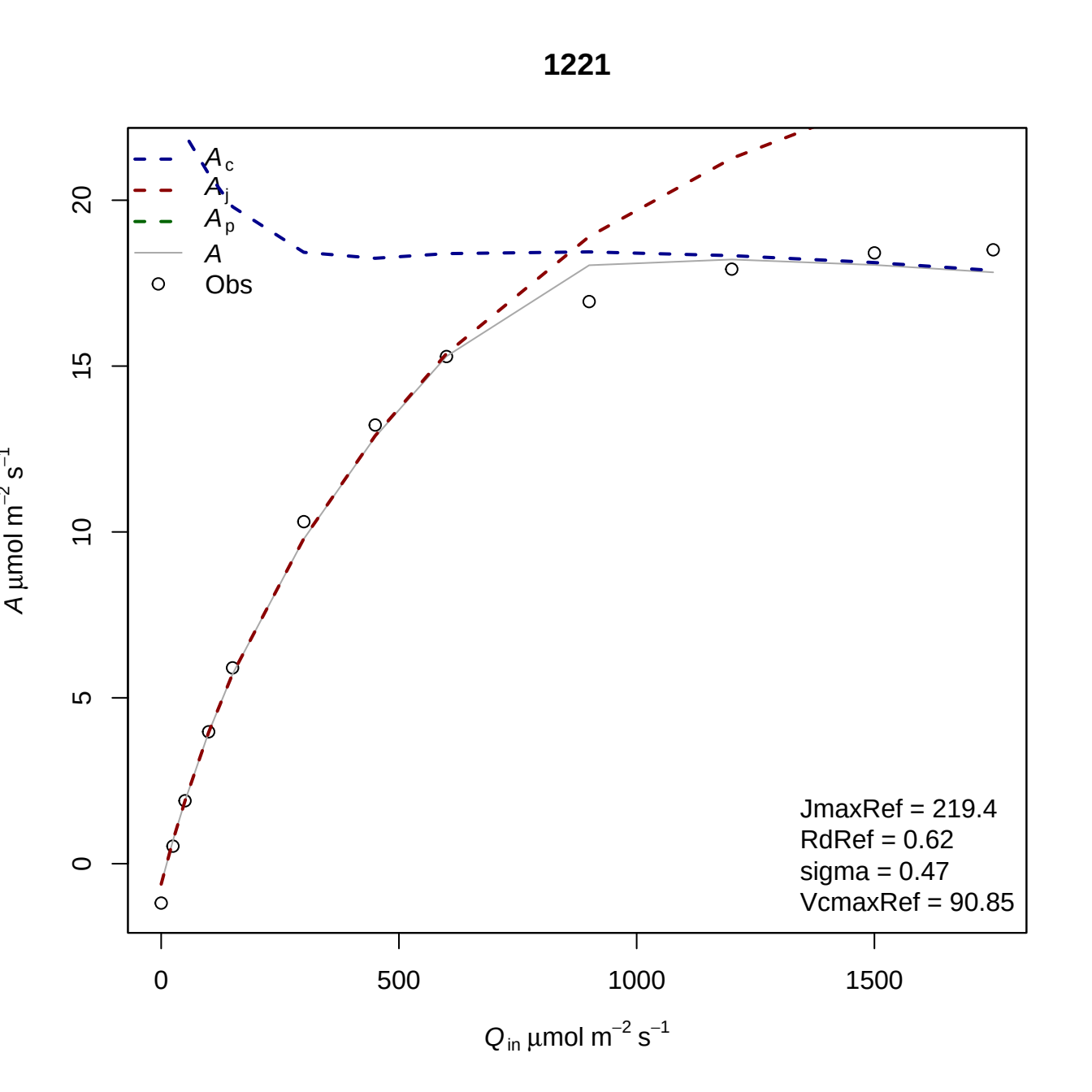

1216
A_c
A_j
A_p
A
Obs
JmaxRef = 191.95
RdRef = 3.47
sigma = 1.45
VcmaxRef = 93.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

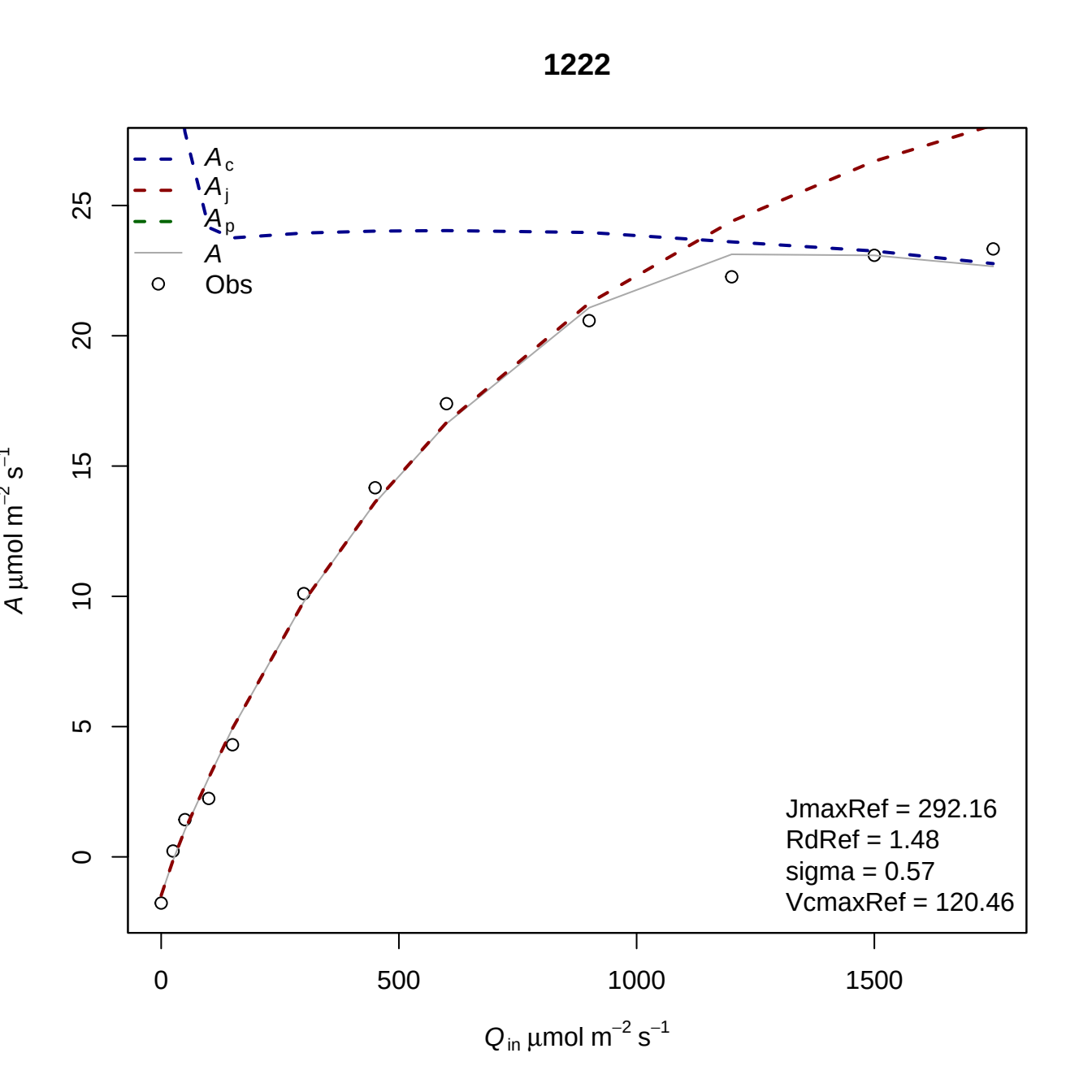

1217
A_c
A_j
A_p
A
Obs
JmaxRef = 124.84
RdRef = 3.58
sigma = 1.65
VcmaxRef = 1627.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

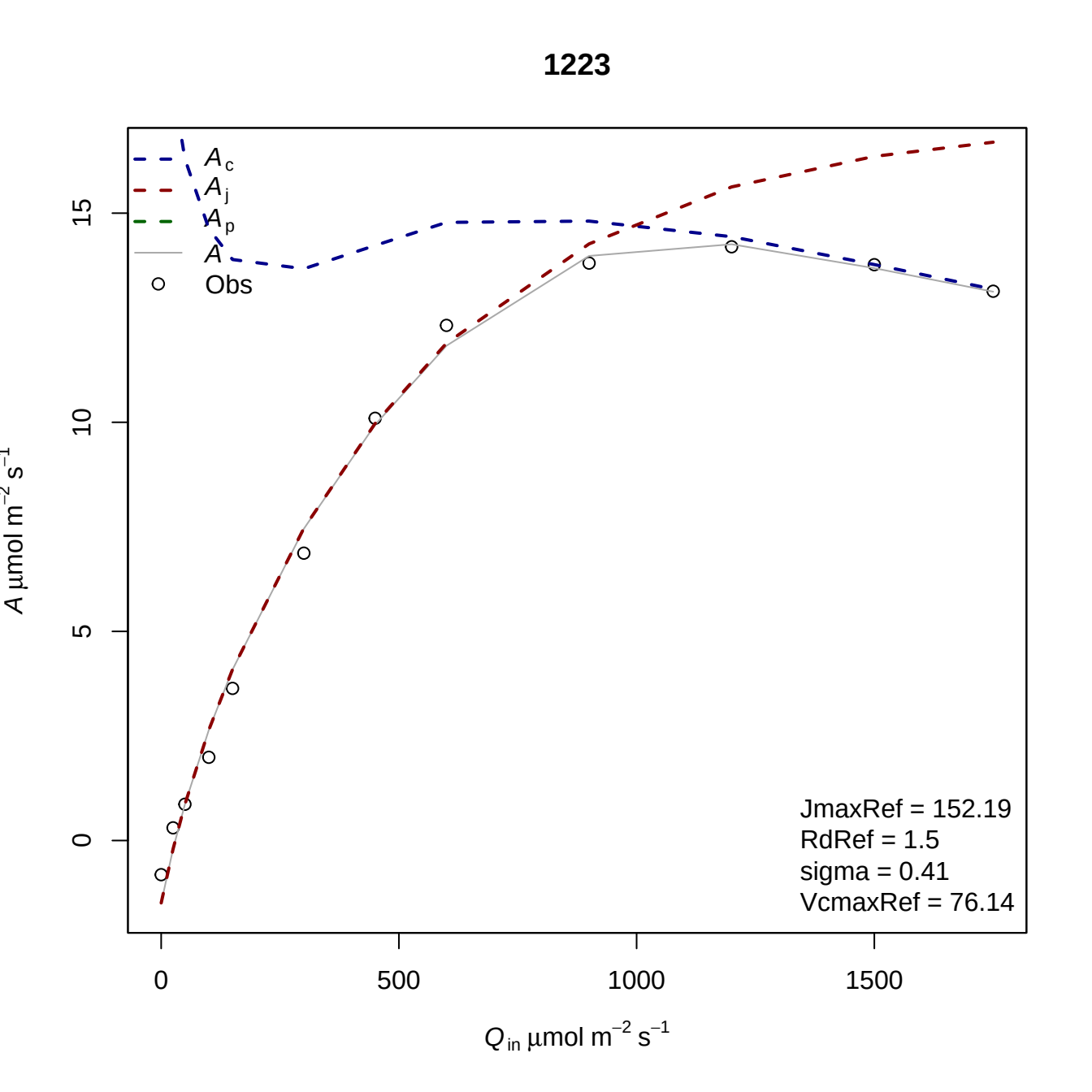

1218
A_c
A_j
A_p
A
Obs
JmaxRef = 250.65
RdRef = 0.34
sigma = 0.6
VcmaxRef = 88.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

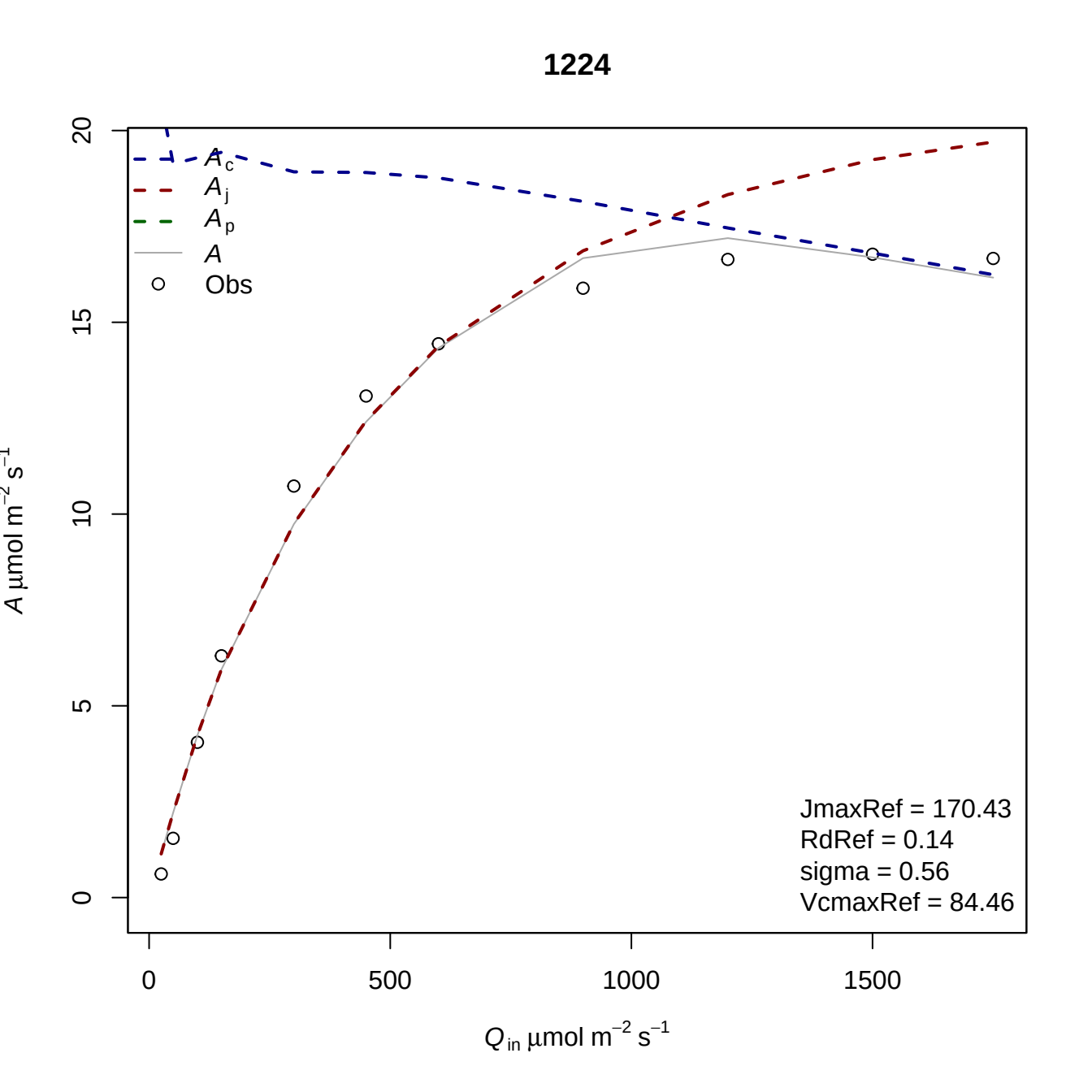

1219
A_c
A_j
A_p
A
Obs
JmaxRef = 260.91
RdRef = 1.37
sigma = 0.67
VcmaxRef = 122.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

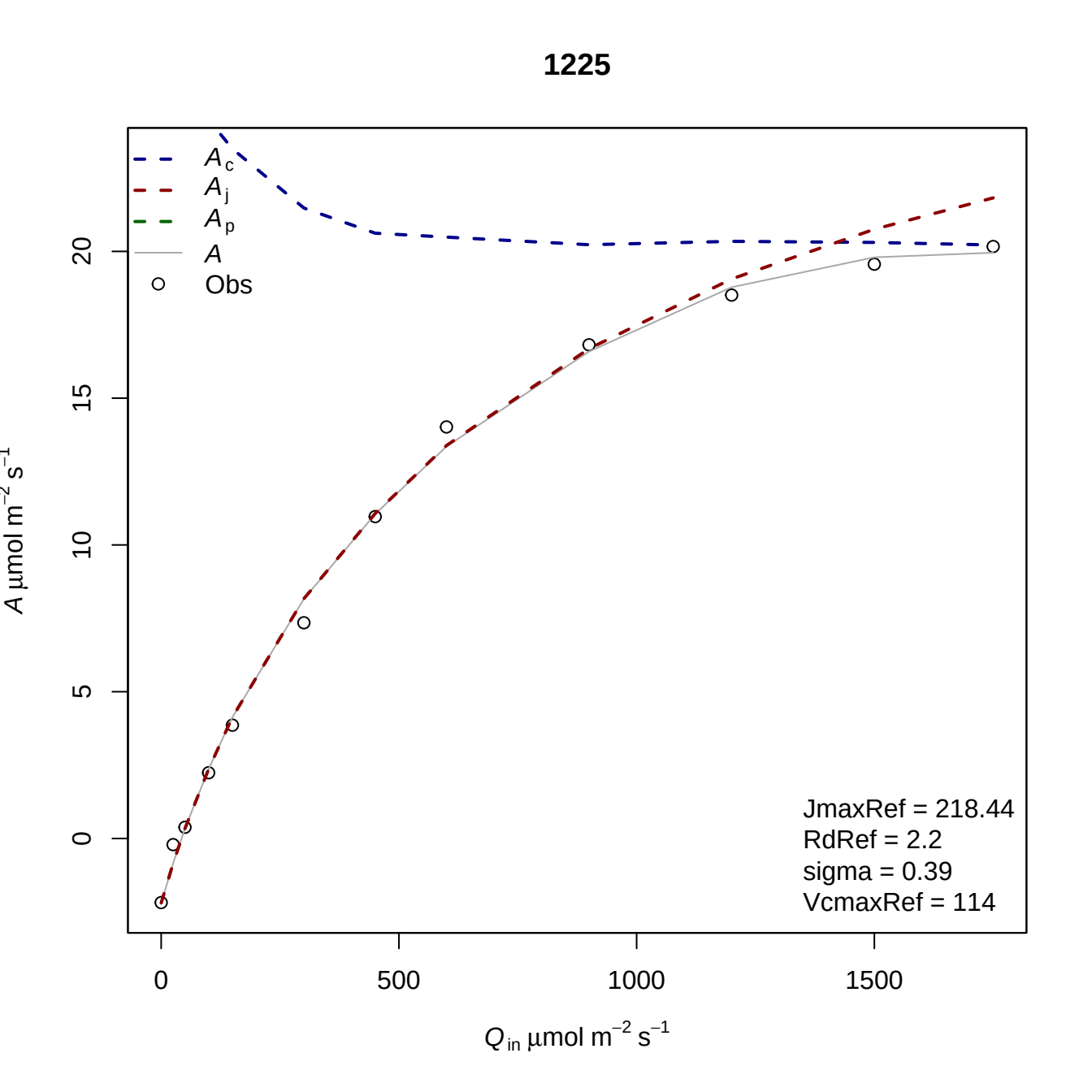

1220
A_c
A_j
A_p
A
Obs
JmaxRef = 391.1
RdRef = 0.99
sigma = 0.41
VcmaxRef = 108.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

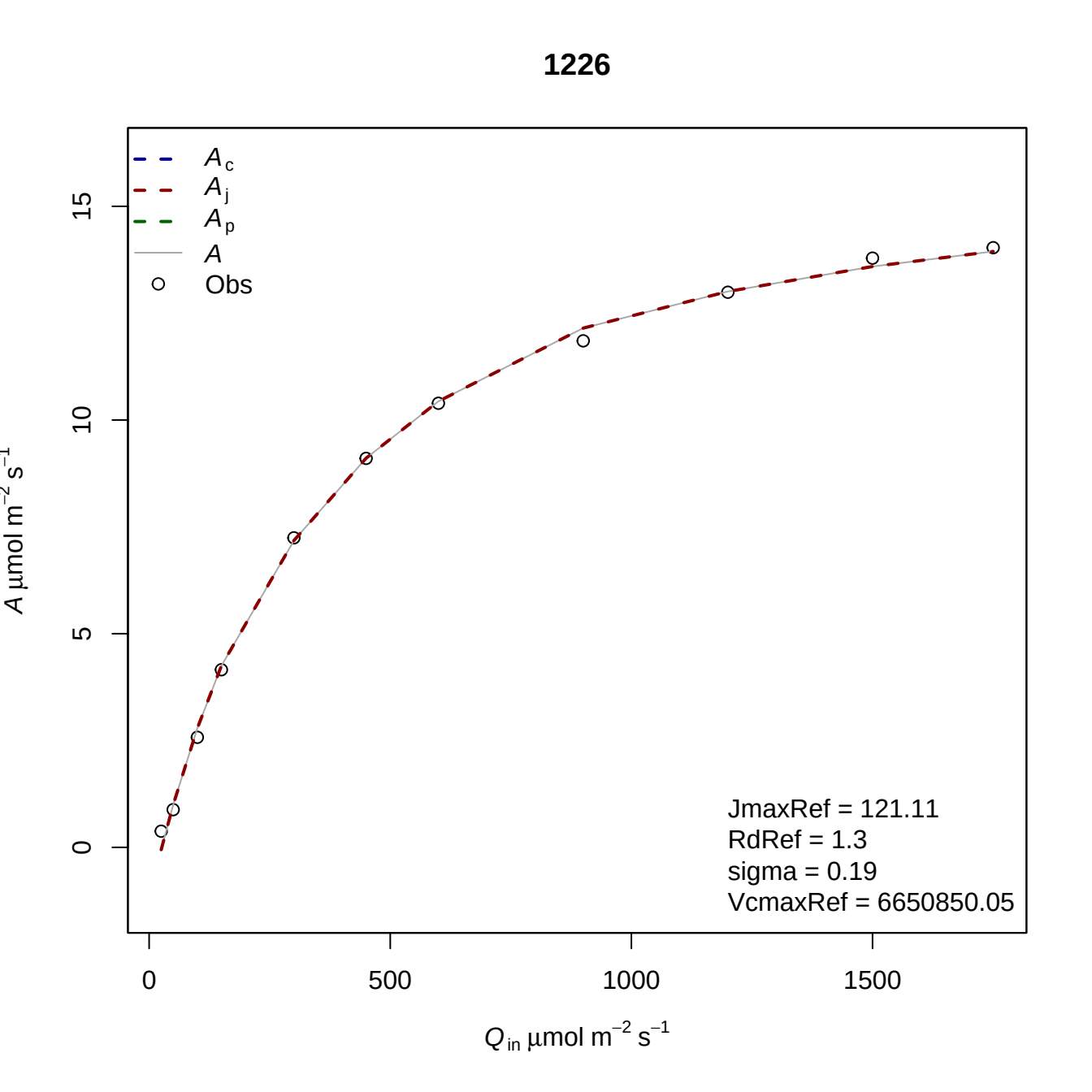

1221
A_c
A_j
A_p
A
Obs
JmaxRef = 219.4
RdRef = 0.62
sigma = 0.47
VcmaxRef = 90.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

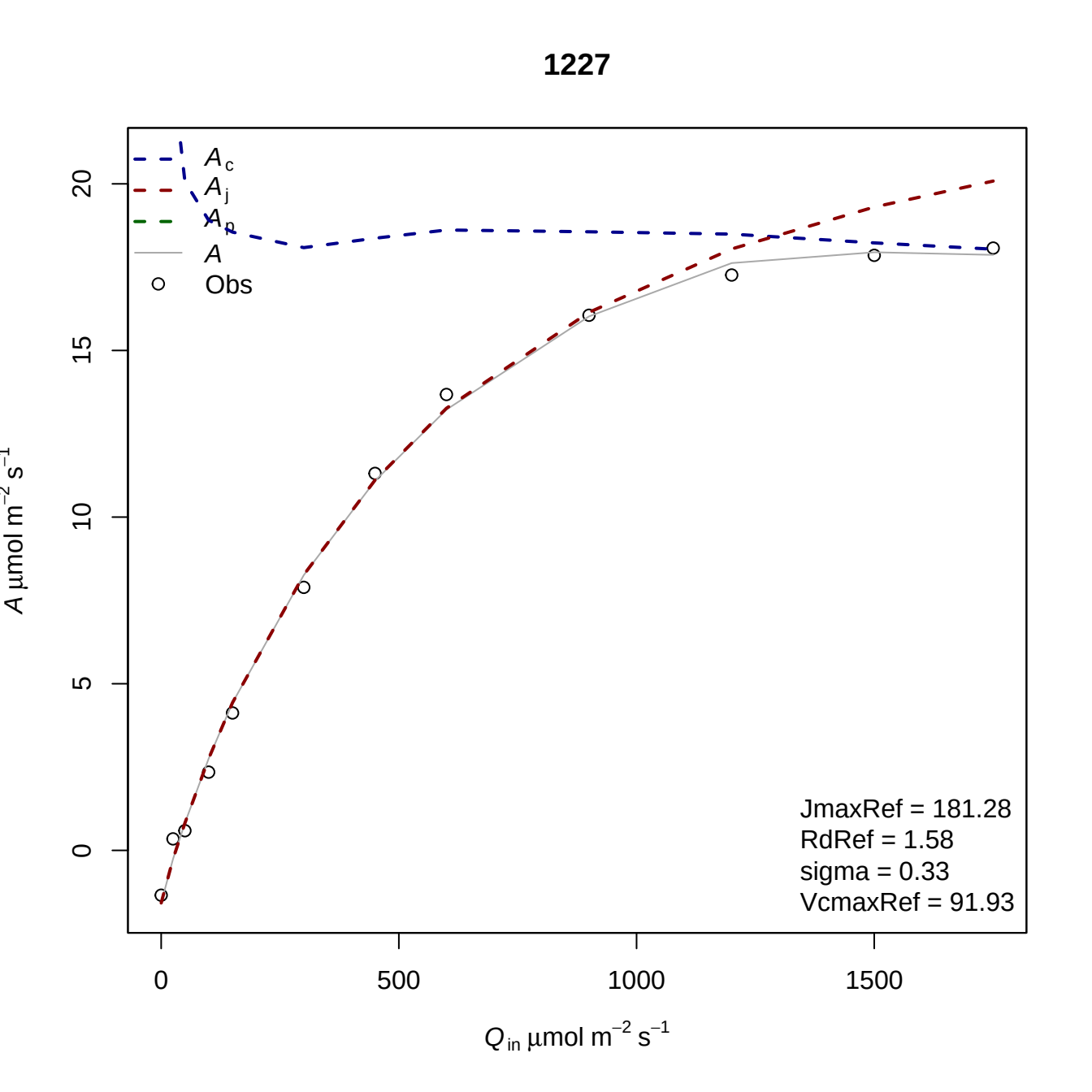

1222
A_c
A_j
A_p
A
Obs
JmaxRef = 292.16
RdRef = 1.48
sigma = 0.57
VcmaxRef = 120.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

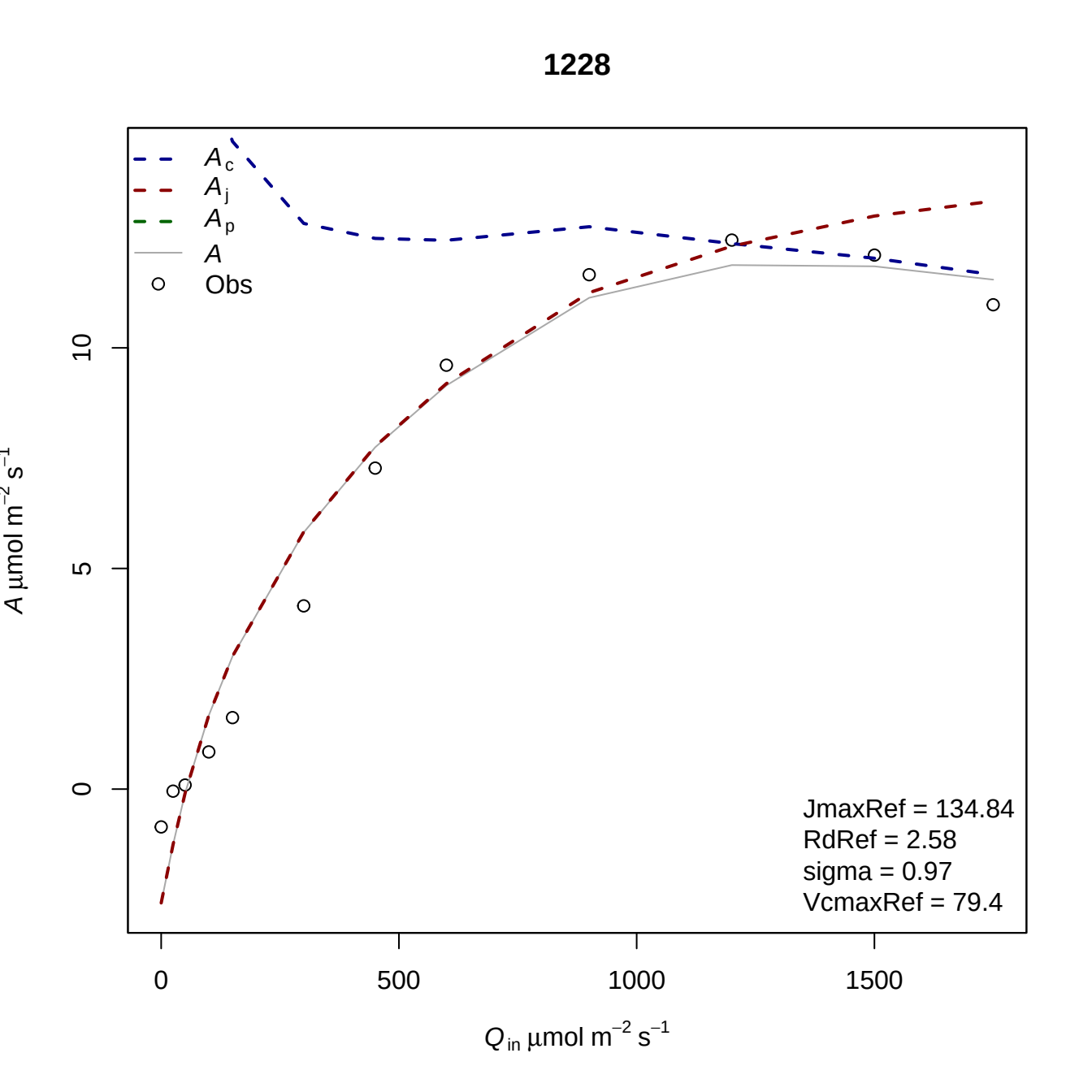

1223
A_c
A_j
A_p
A
Obs
JmaxRef = 152.19
RdRef = 1.5
sigma = 0.41
VcmaxRef = 76.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

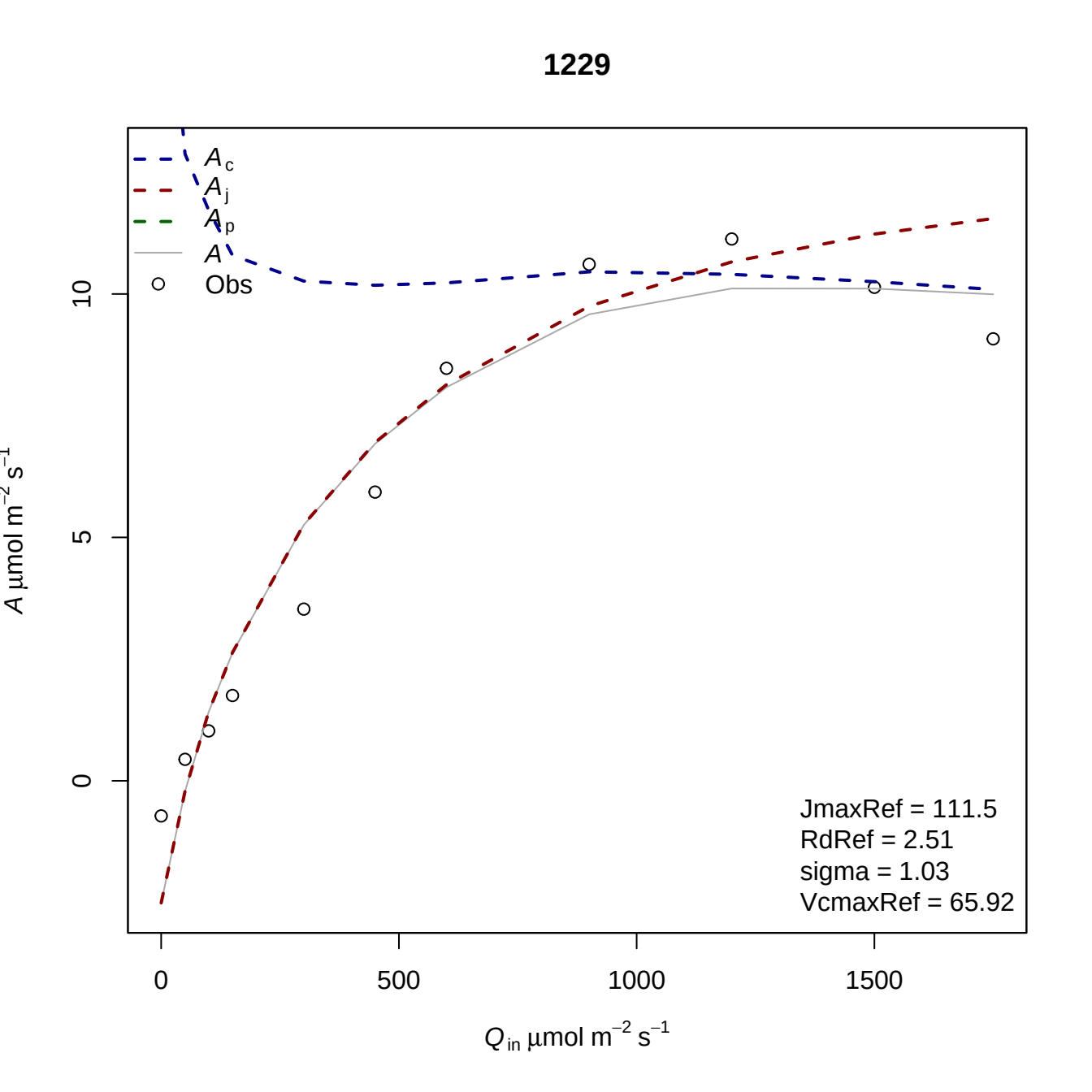

1224
A_c
A_j
A_p
A
Obs
JmaxRef = 170.43
RdRef = 0.14
sigma = 0.56
VcmaxRef = 84.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

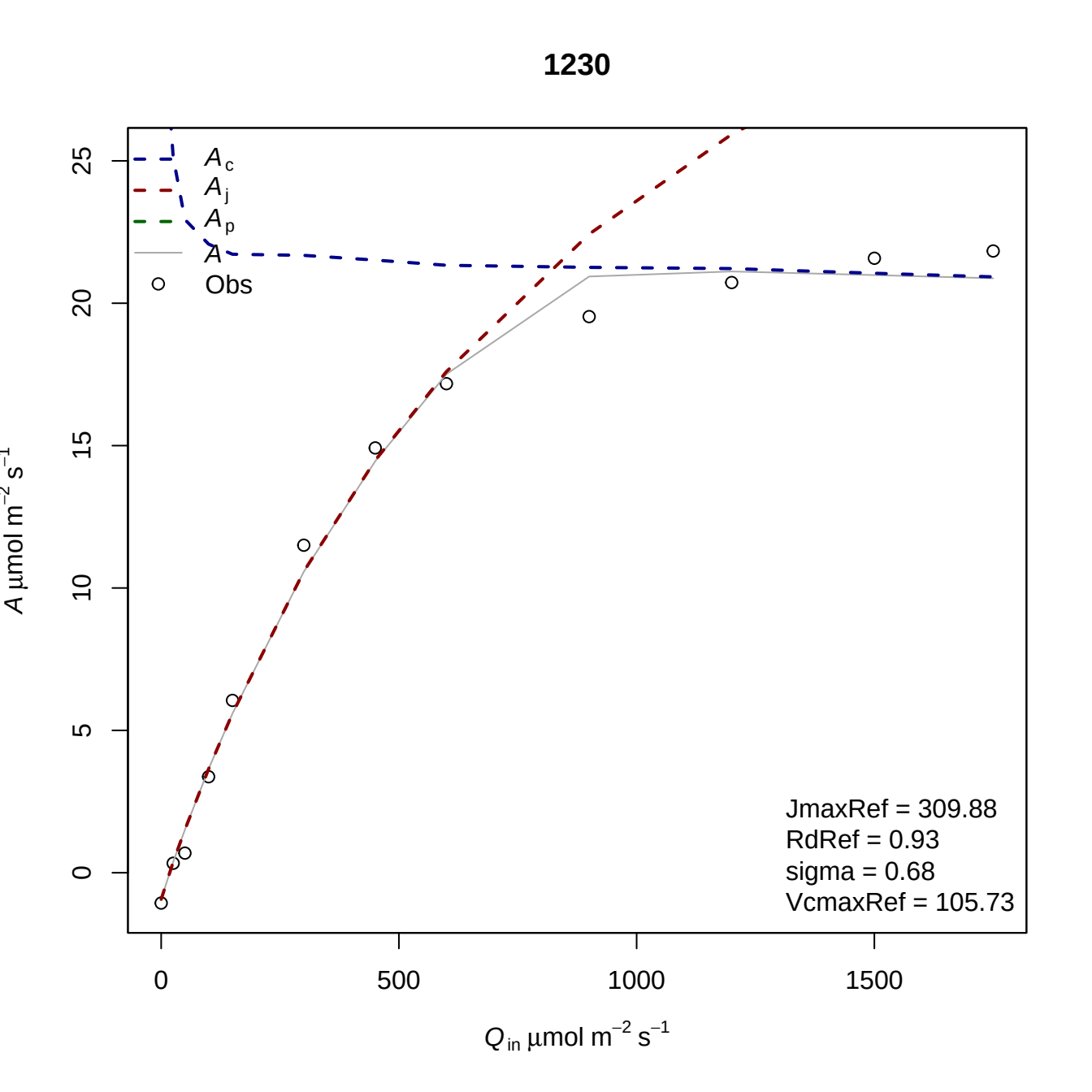

1225
A_c
A_j
A_p
A
Obs
JmaxRef = 218.44
RdRef = 2.2
sigma = 0.39
VcmaxRef = 114
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

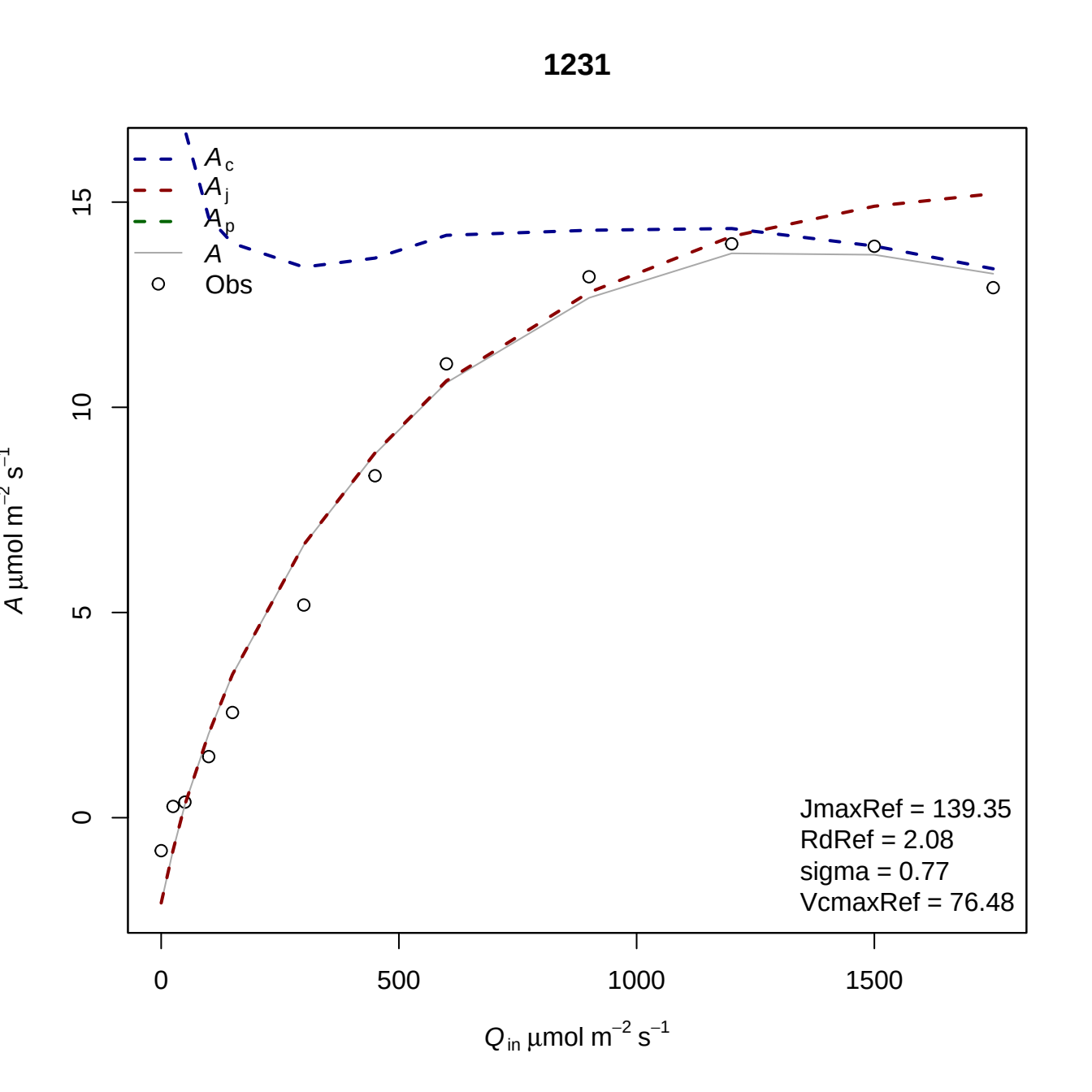

1226
A_c
A_j
A_p
A
Obs
JmaxRef = 121.11
RdRef = 1.3
sigma = 0.19
VcmaxRef = 6650850.05
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

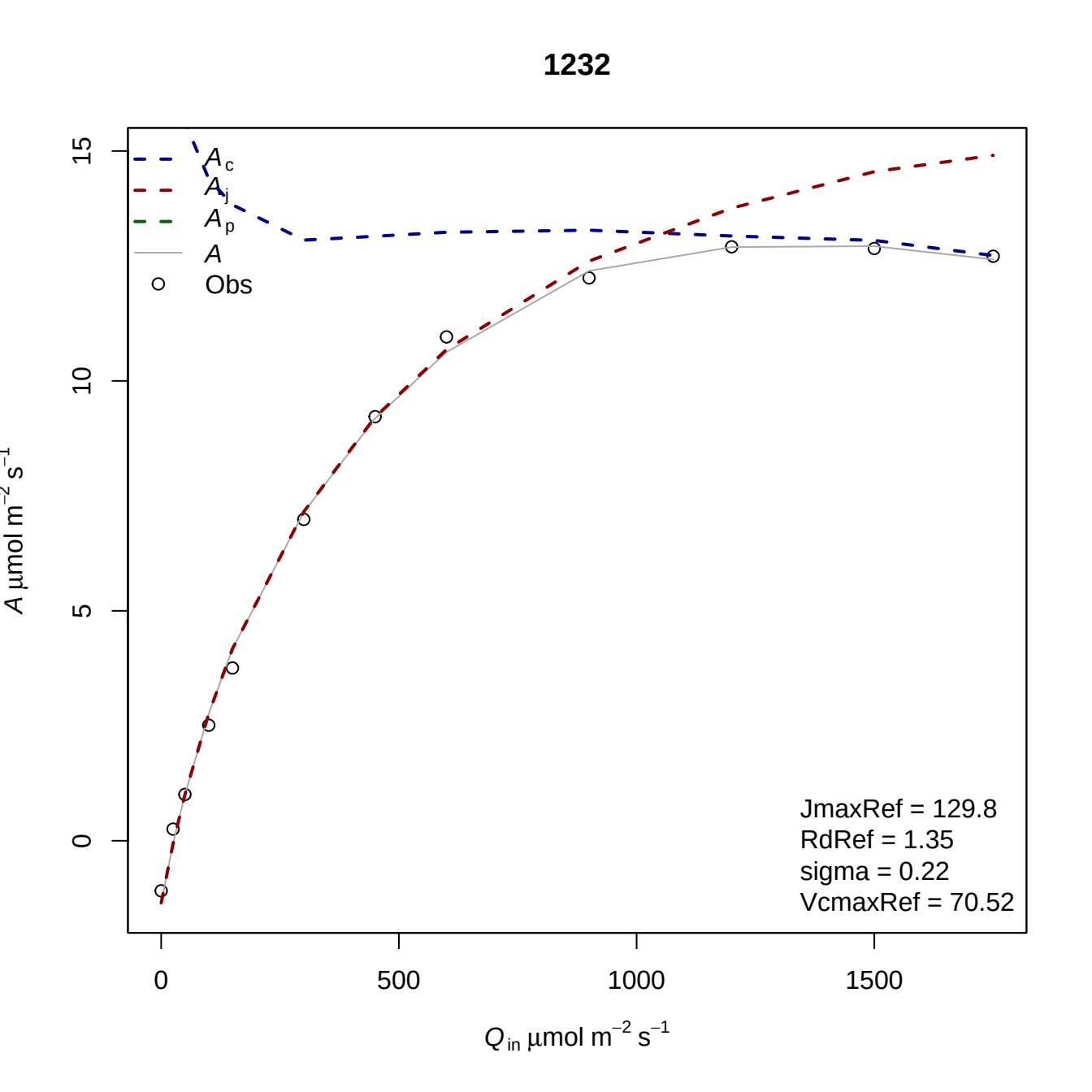

1227
A_c
A_j
A_p
A
Obs
JmaxRef = 181.28
RdRef = 1.58
sigma = 0.33
VcmaxRef = 91.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

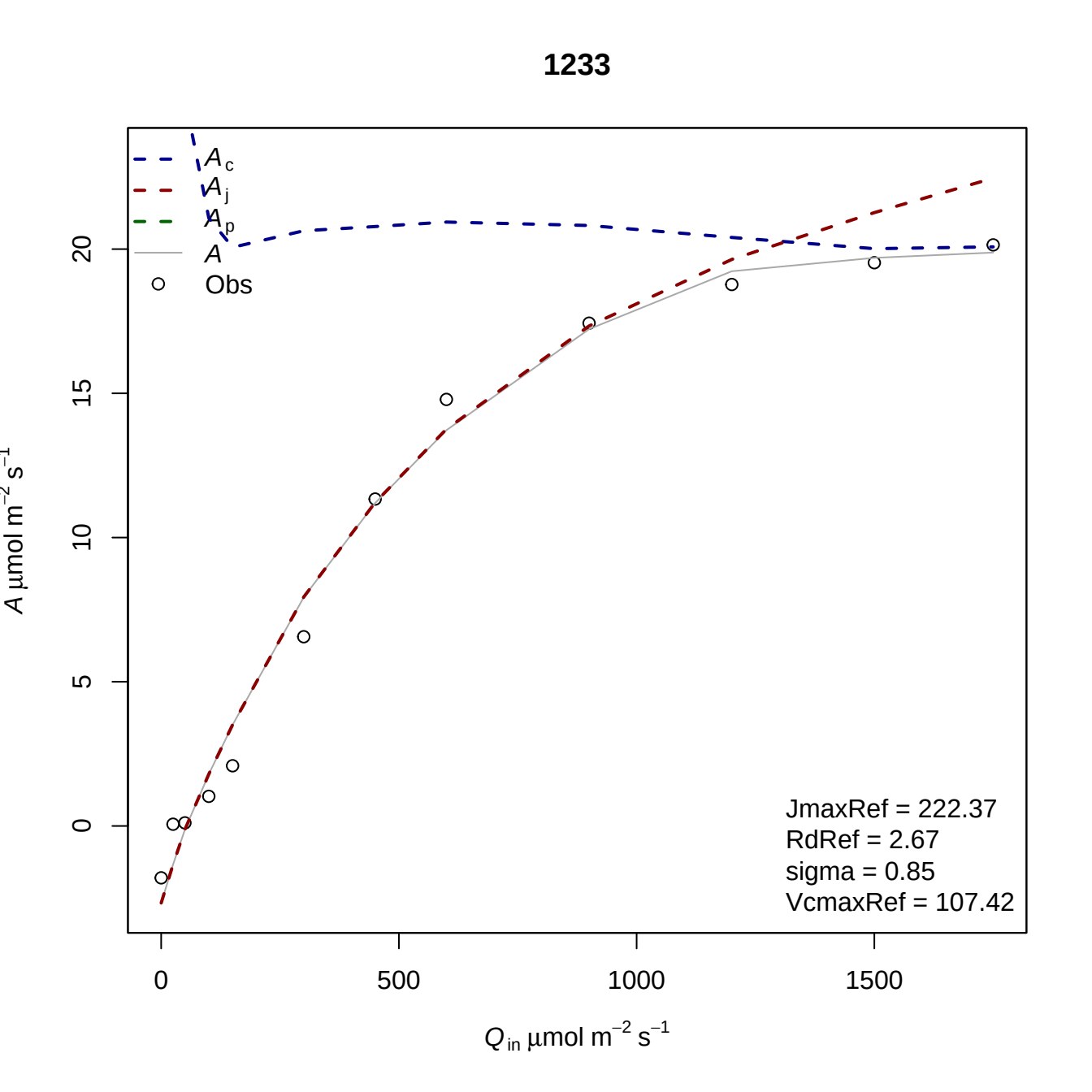

1228
A_c
A_j
A_p
A
Obs
JmaxRef = 134.84
RdRef = 2.58
sigma = 0.97
VcmaxRef = 79.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

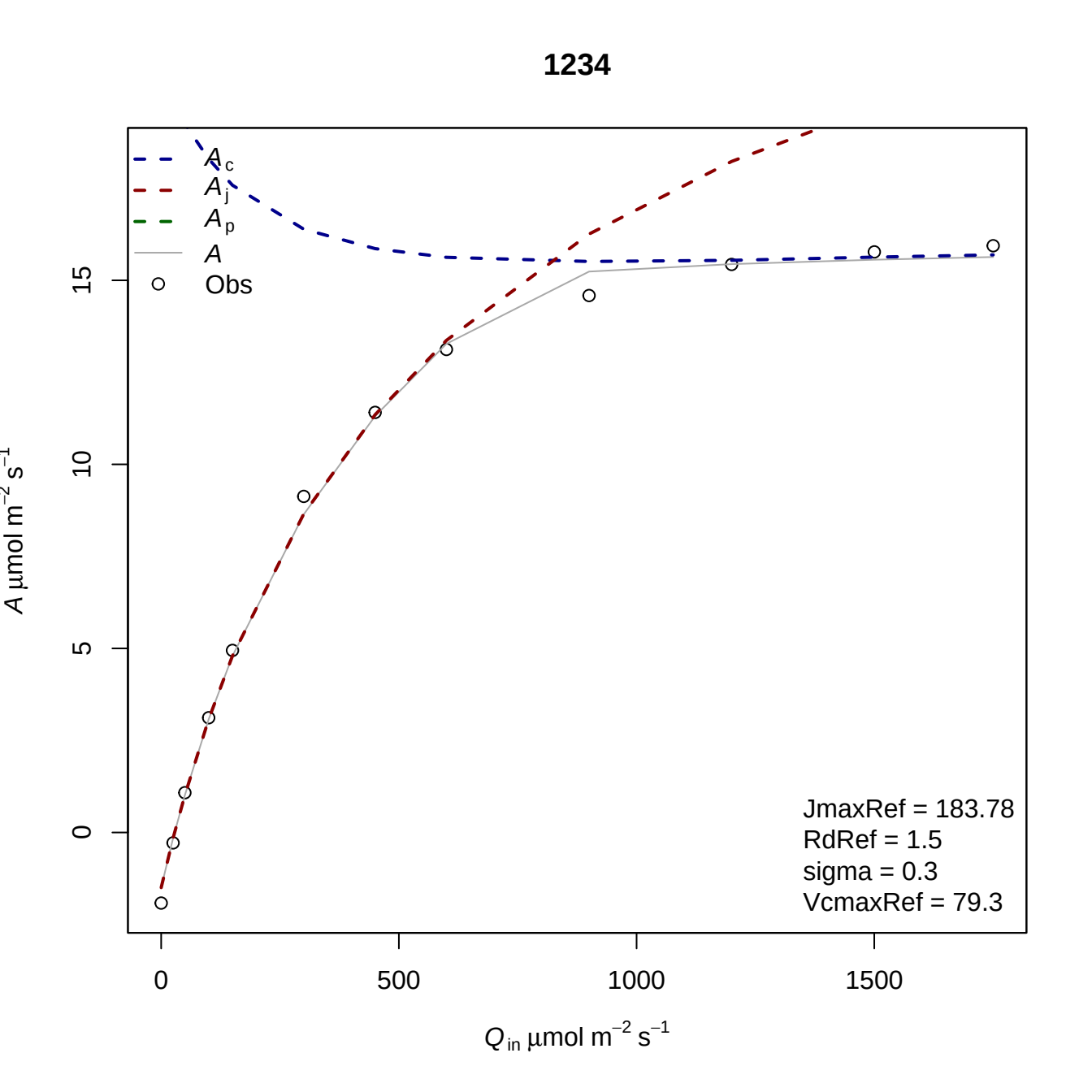

1229
A_c
A_j
A_p
A
Obs
JmaxRef = 111.5
RdRef = 2.51
sigma = 1.03
VcmaxRef = 65.92
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

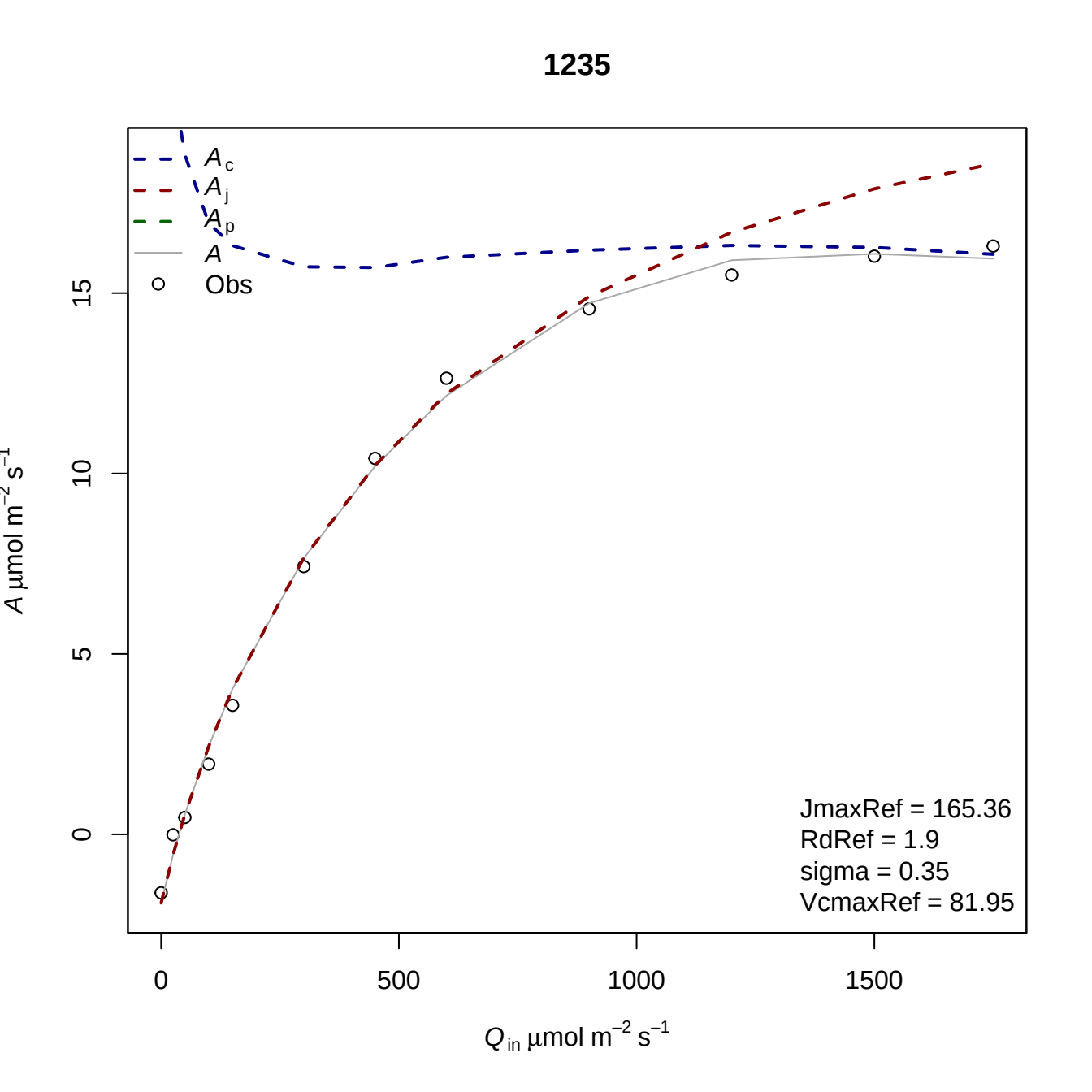

1230
A_c
A_j
A_p
A
Obs
JmaxRef = 309.88
RdRef = 0.93
sigma = 0.68
VcmaxRef = 105.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

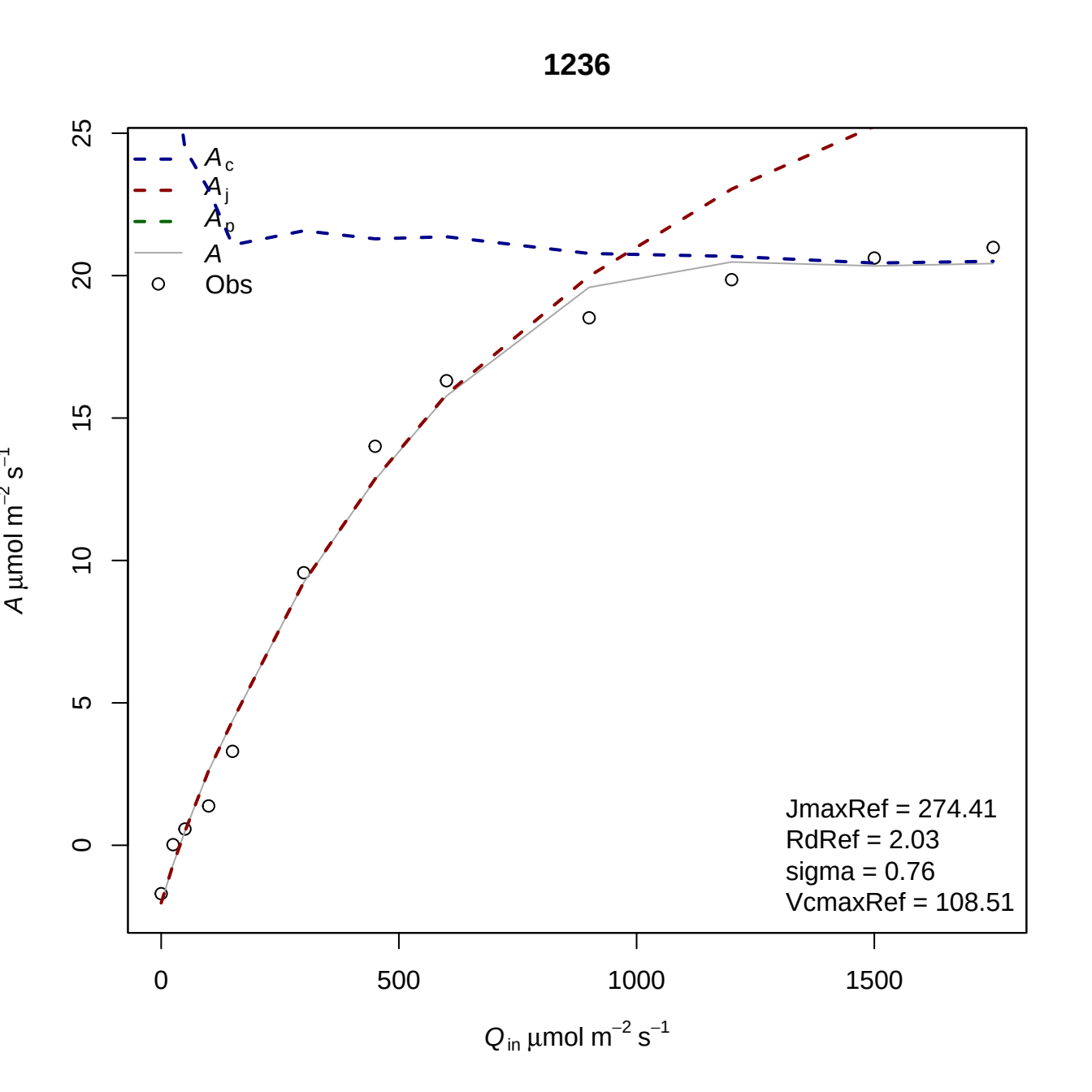

1231
A_c
A_j
A_p
A
Obs
JmaxRef = 139.35
RdRef = 2.08
sigma = 0.77
VcmaxRef = 76.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

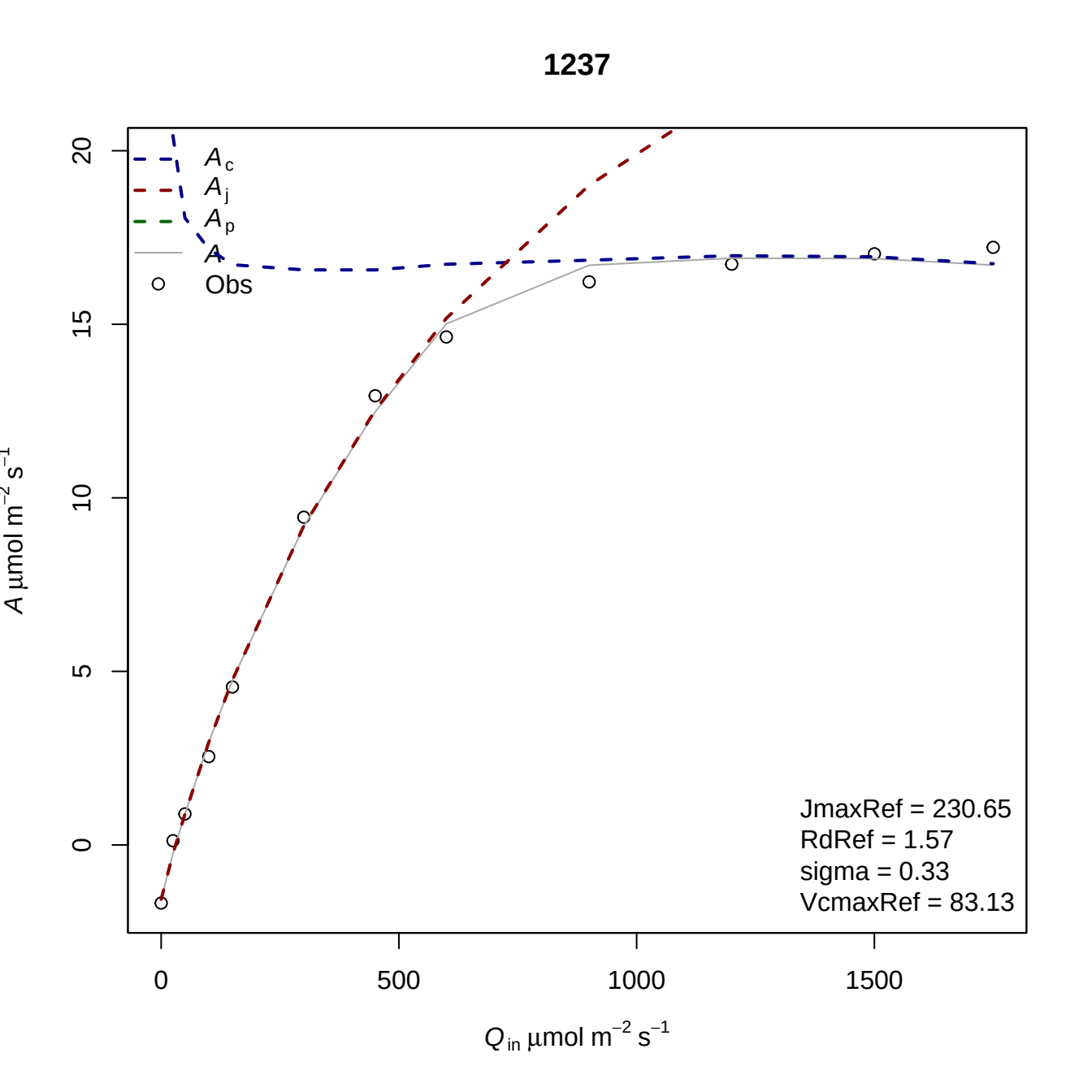

1232
A_c
A_j
A_p
A
Obs
JmaxRef = 129.8
RdRef = 1.35
sigma = 0.22
VcmaxRef = 70.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

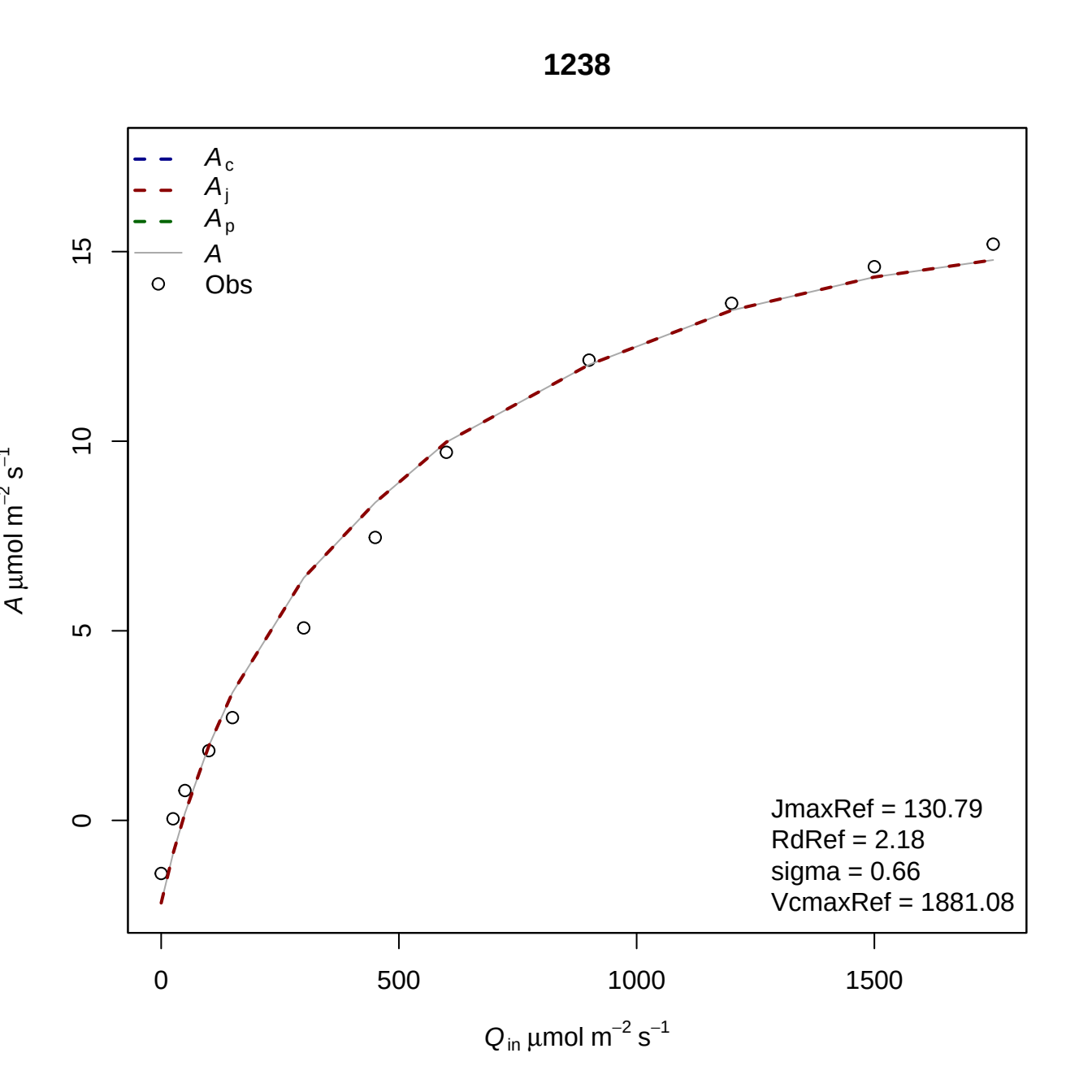

1233
A_c
A_j
A_p
A
Obs
JmaxRef = 222.37
RdRef = 2.67
sigma = 0.85
VcmaxRef = 107.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

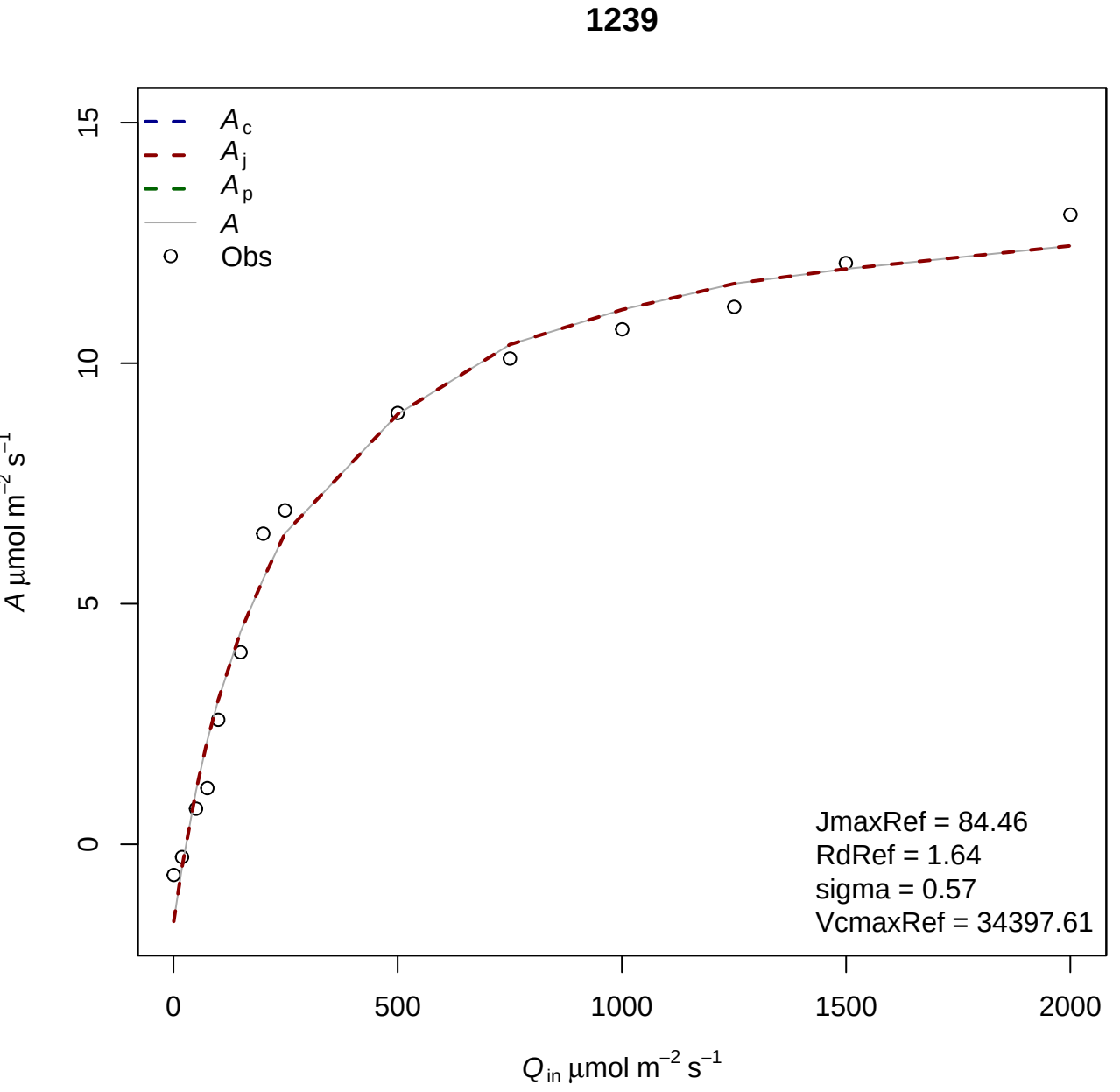

1234
A_c
A_j
A_p
A
Obs
JmaxRef = 183.78
RdRef = 1.5
sigma = 0.3
VcmaxRef = 79.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

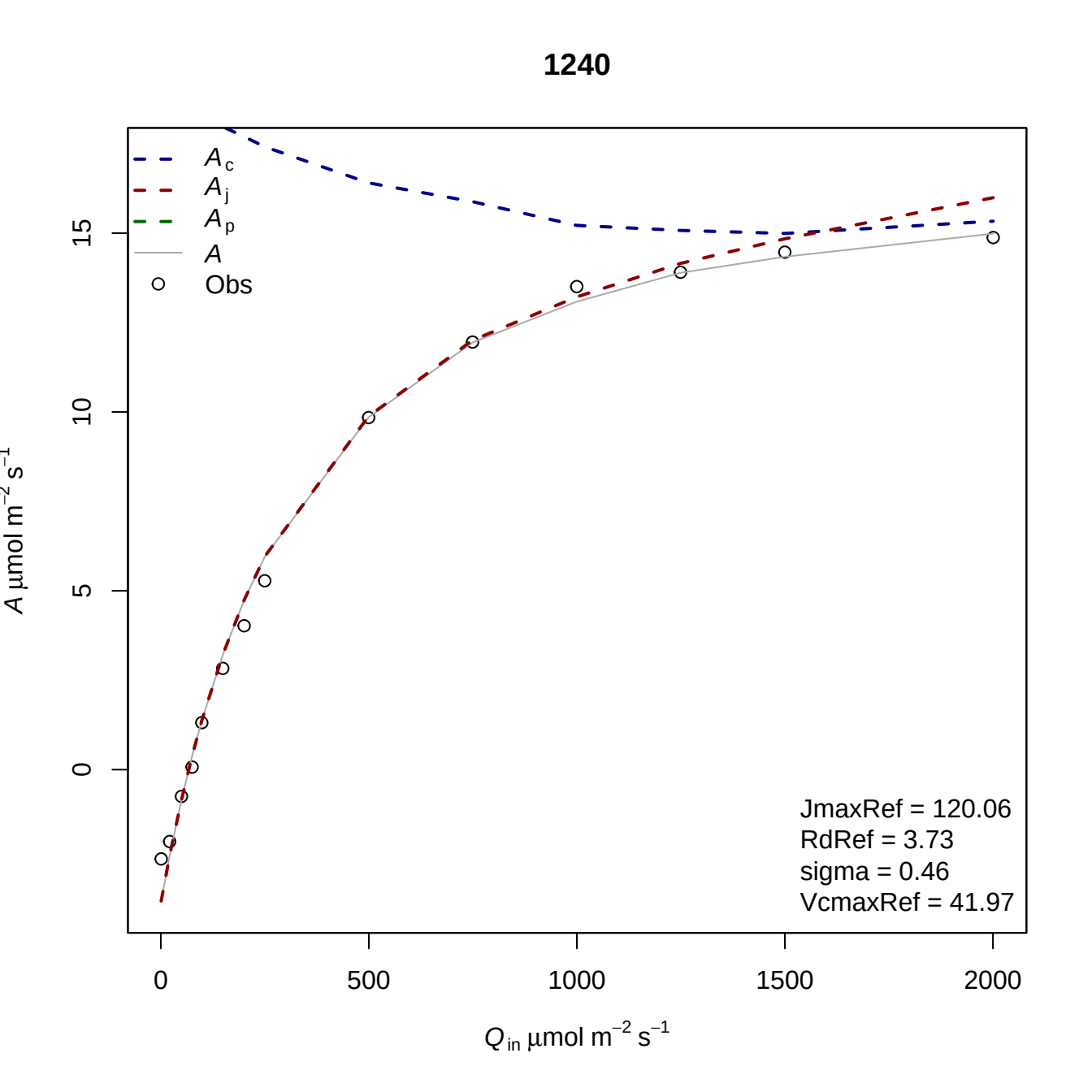

1235
A_c
A_j
A_p
A
Obs
JmaxRef = 165.36
RdRef = 1.9
sigma = 0.35
VcmaxRef = 81.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

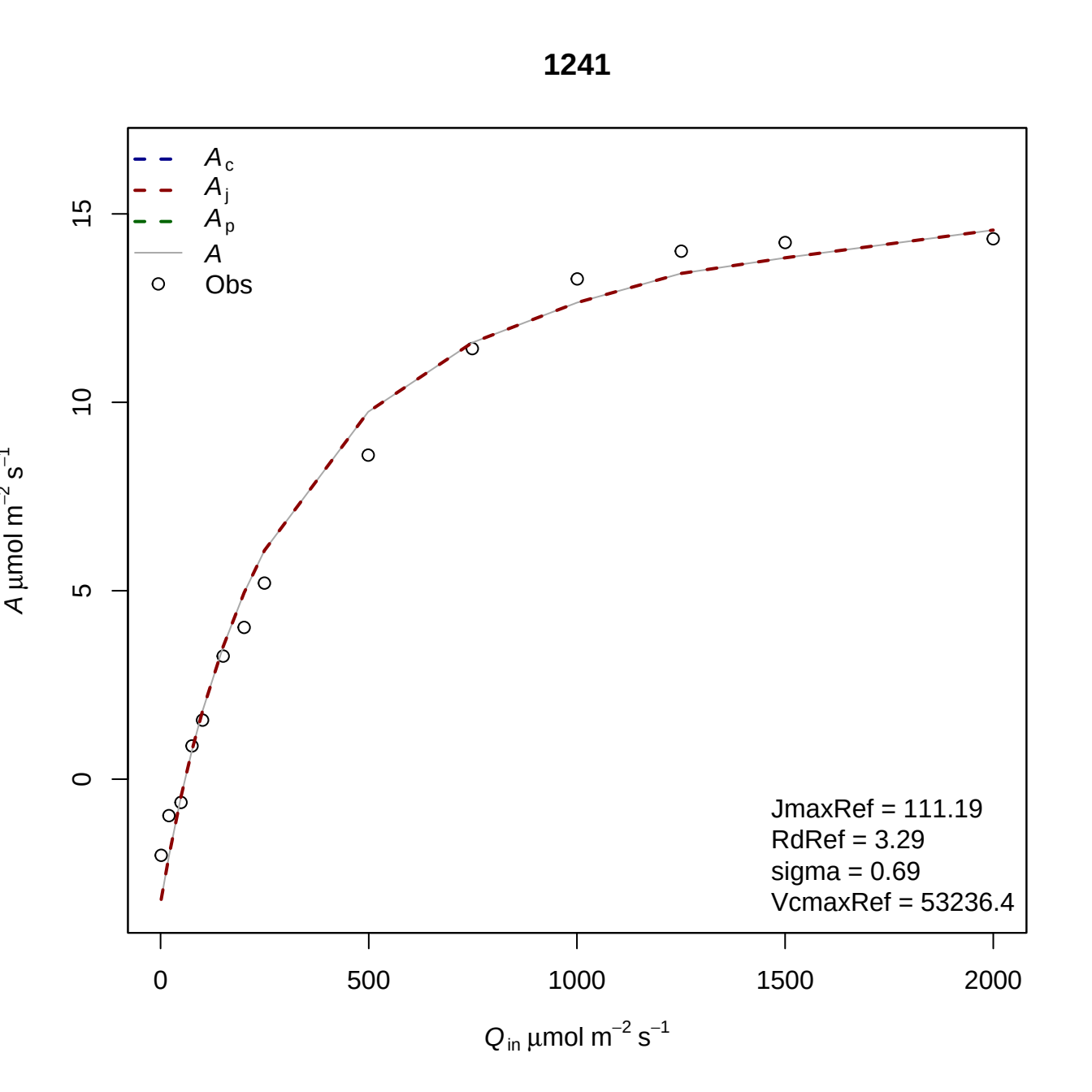

1236
A_c
A_j
A_p
A
Obs
JmaxRef = 274.41
RdRef = 2.03
sigma = 0.76
VcmaxRef = 108.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

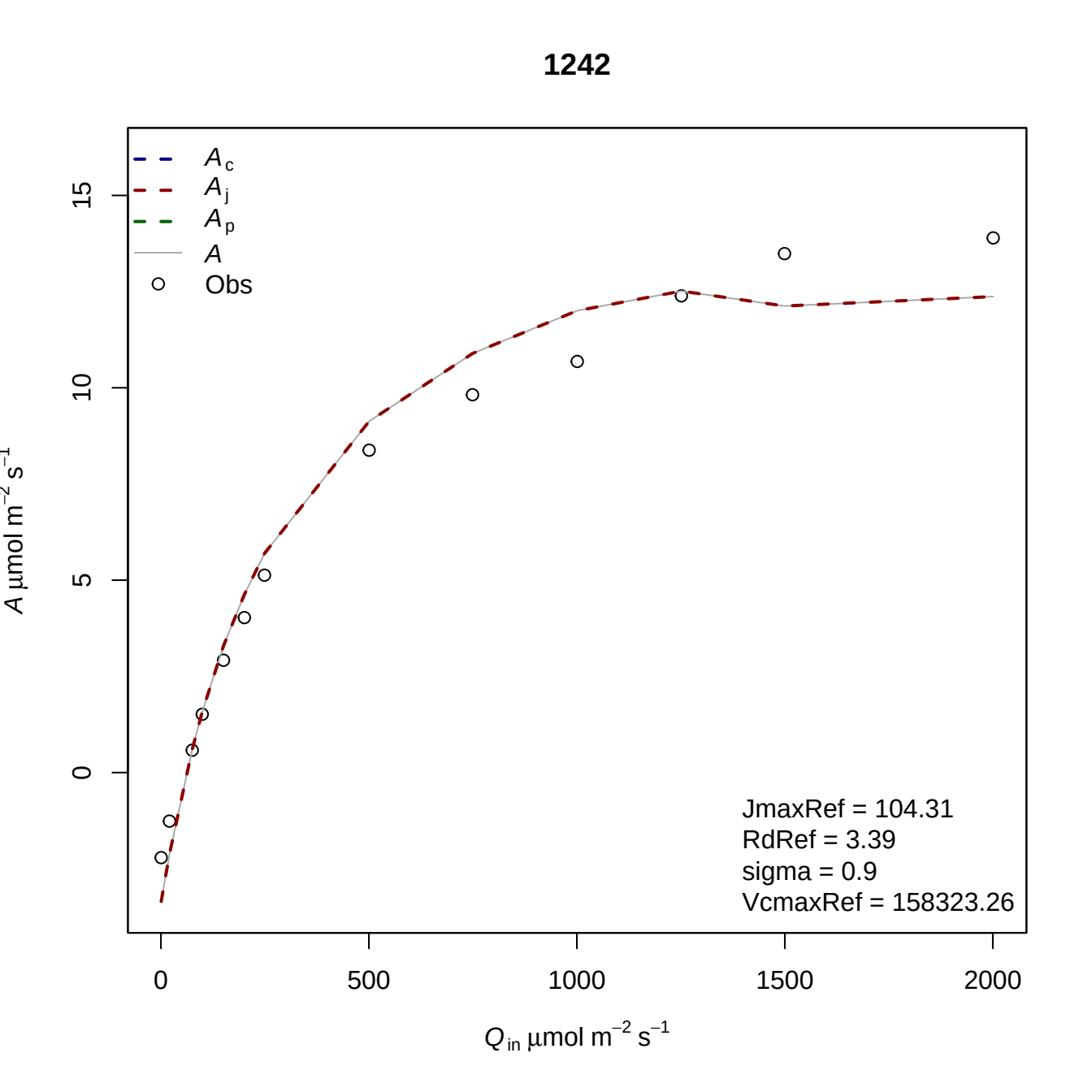

1237
A_c
A_j
A_p
A
Obs
JmaxRef = 230.65
RdRef = 1.57
sigma = 0.33
VcmaxRef = 83.13
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

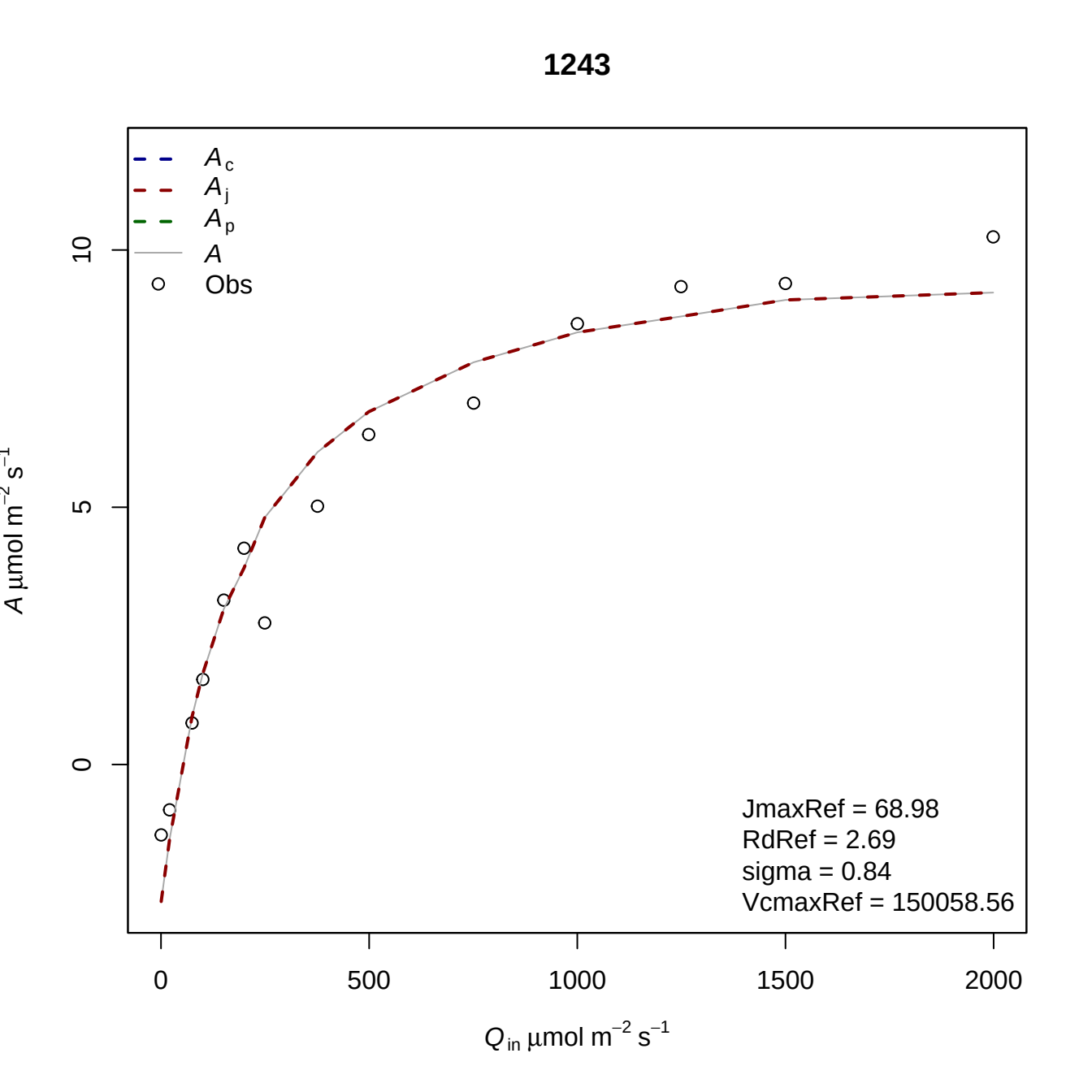

1238
A_c
A_j
A_p
A
Obs
JmaxRef = 130.79
RdRef = 2.18
sigma = 0.66
VcmaxRef = 1881.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

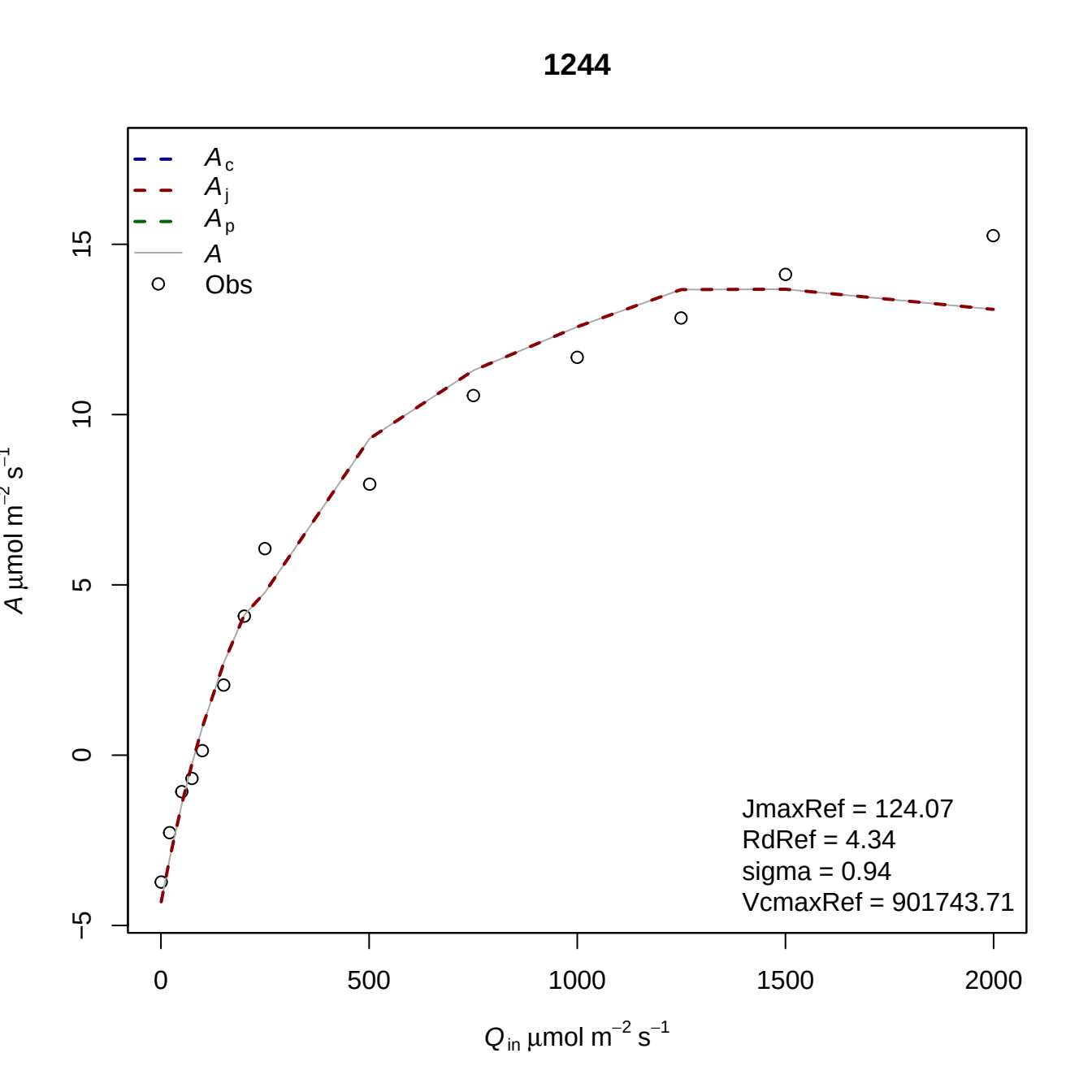

1239
A_c
A_j
A_p
A
Obs
JmaxRef = 84.46
RdRef = 1.64
sigma = 0.57
VcmaxRef = 34397.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

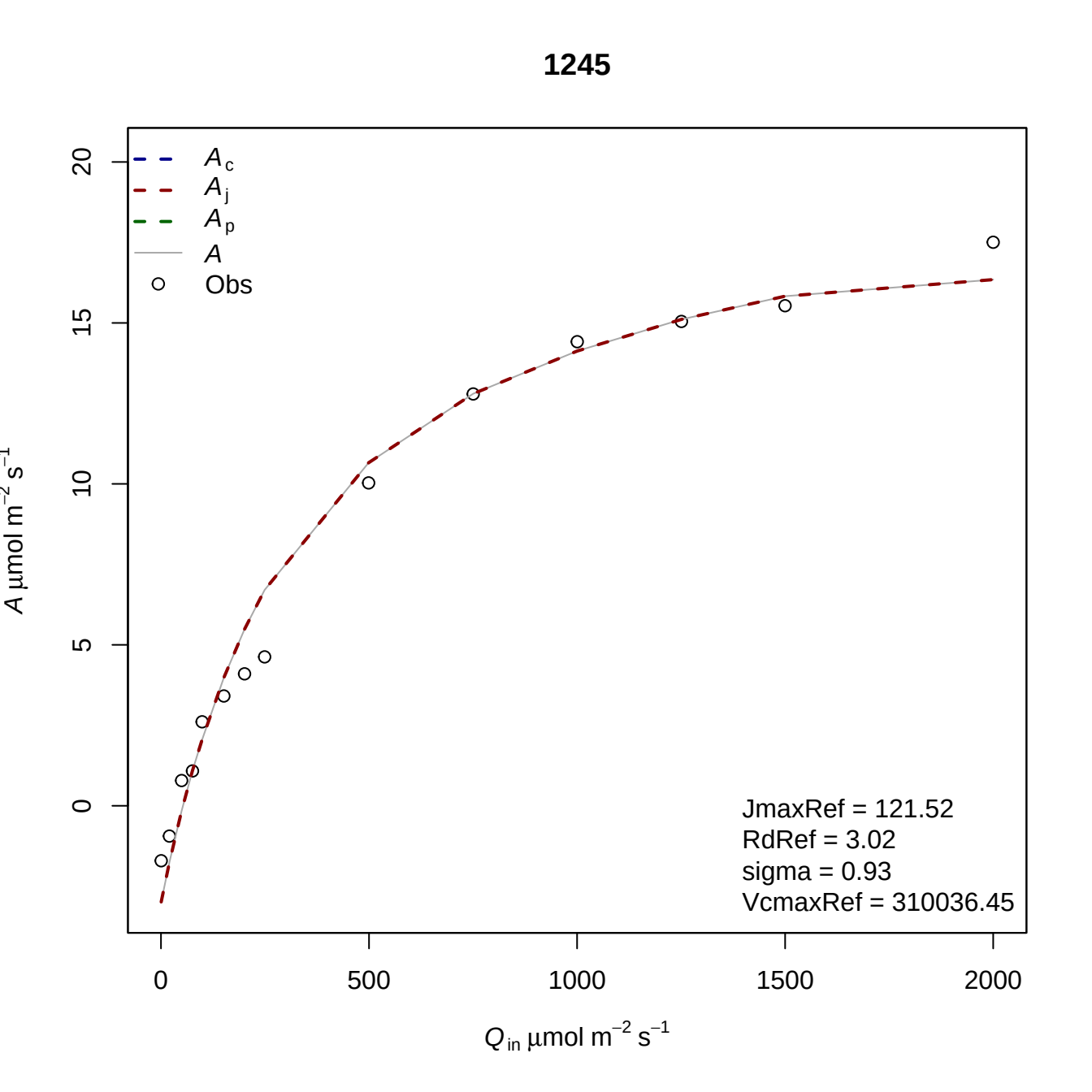

1240
A_c
A_j
A_p
A
Obs
JmaxRef = 120.06
RdRef = 3.73
sigma = 0.46
VcmaxRef = 41.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

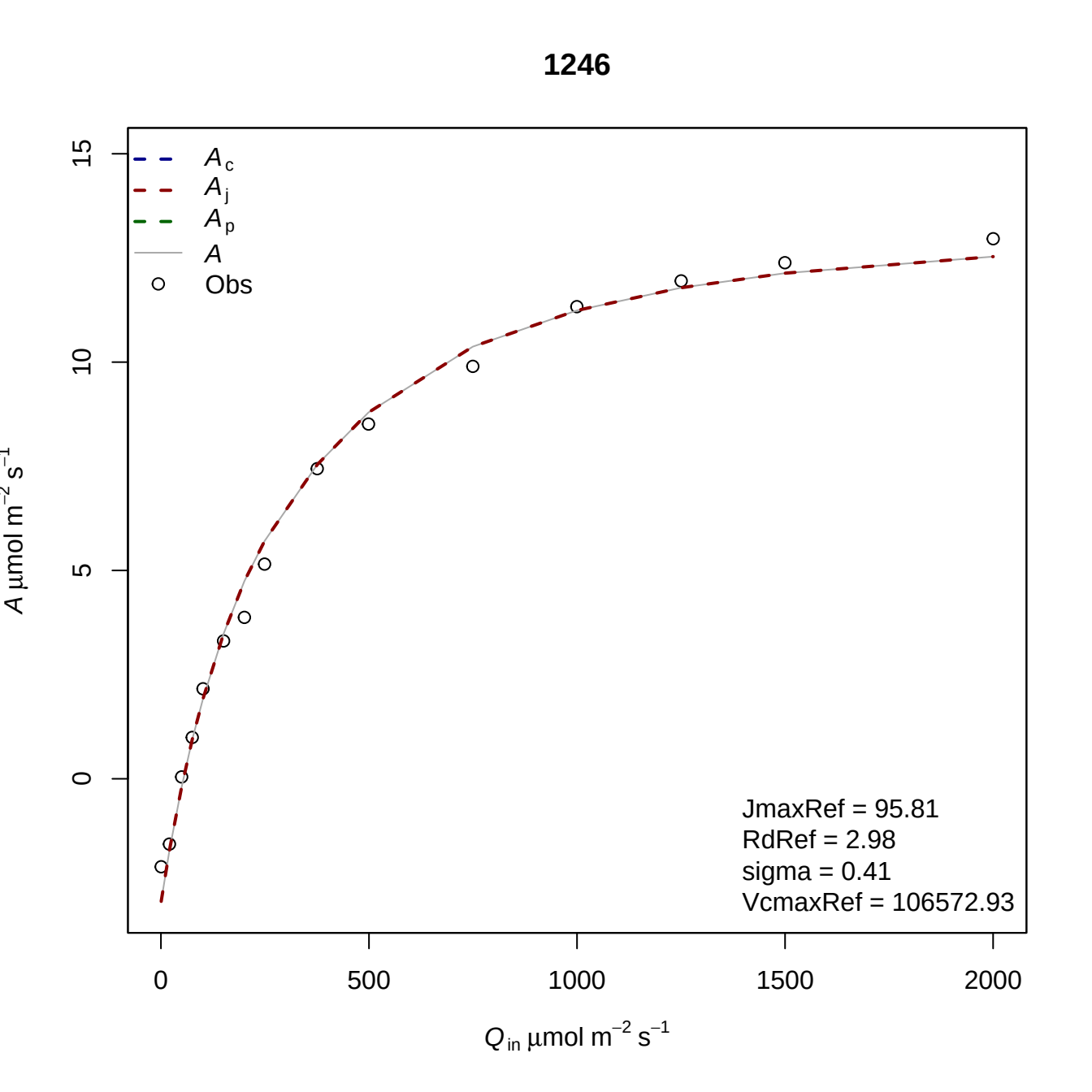

1241
A_c
A_j
A_p
A
Obs
JmaxRef = 111.19
RdRef = 3.29
sigma = 0.69
VcmaxRef = 53236.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

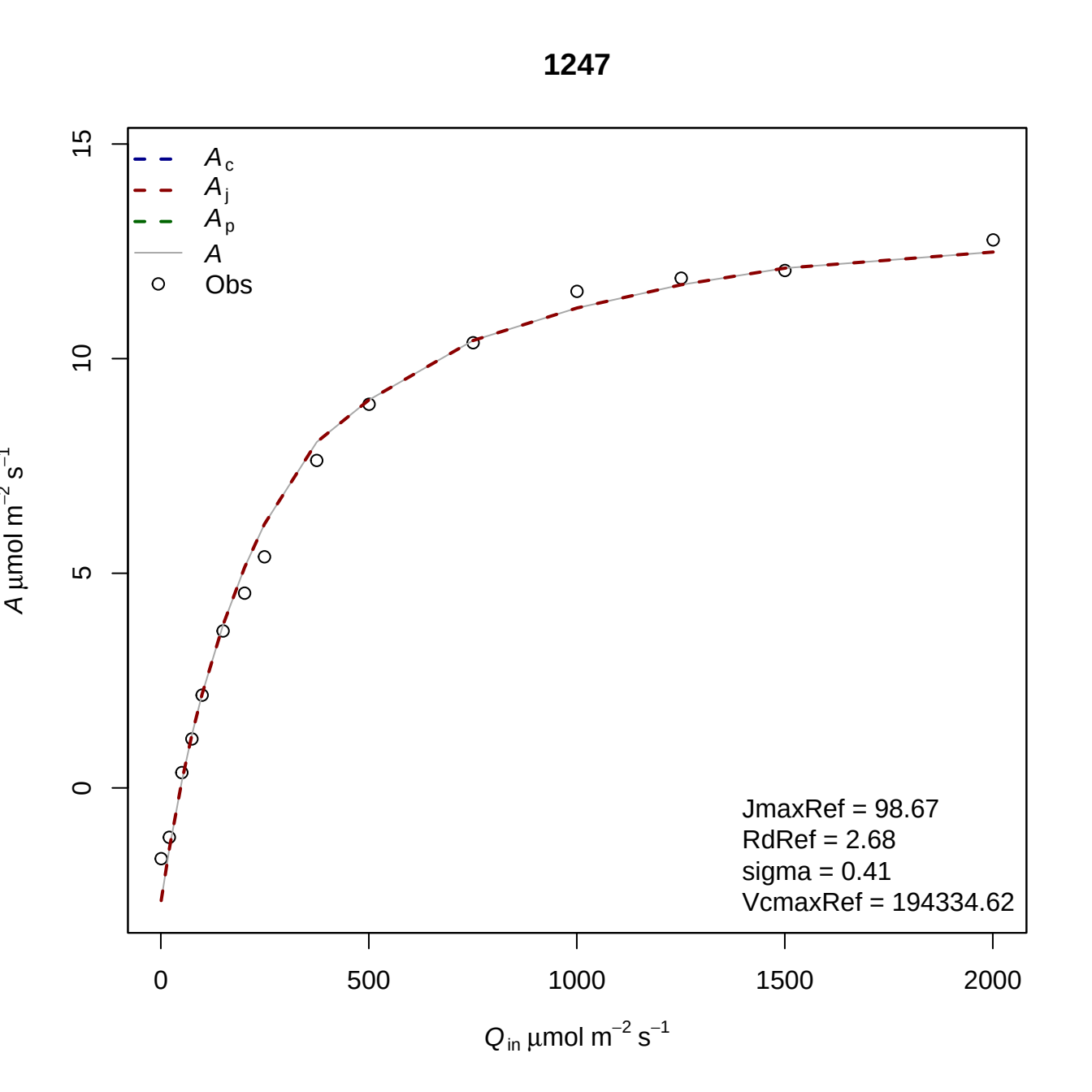

1242
A_c
A_j
A_p
A
Obs
JmaxRef = 104.31
RdRef = 3.39
sigma = 0.9
VcmaxRef = 158323.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

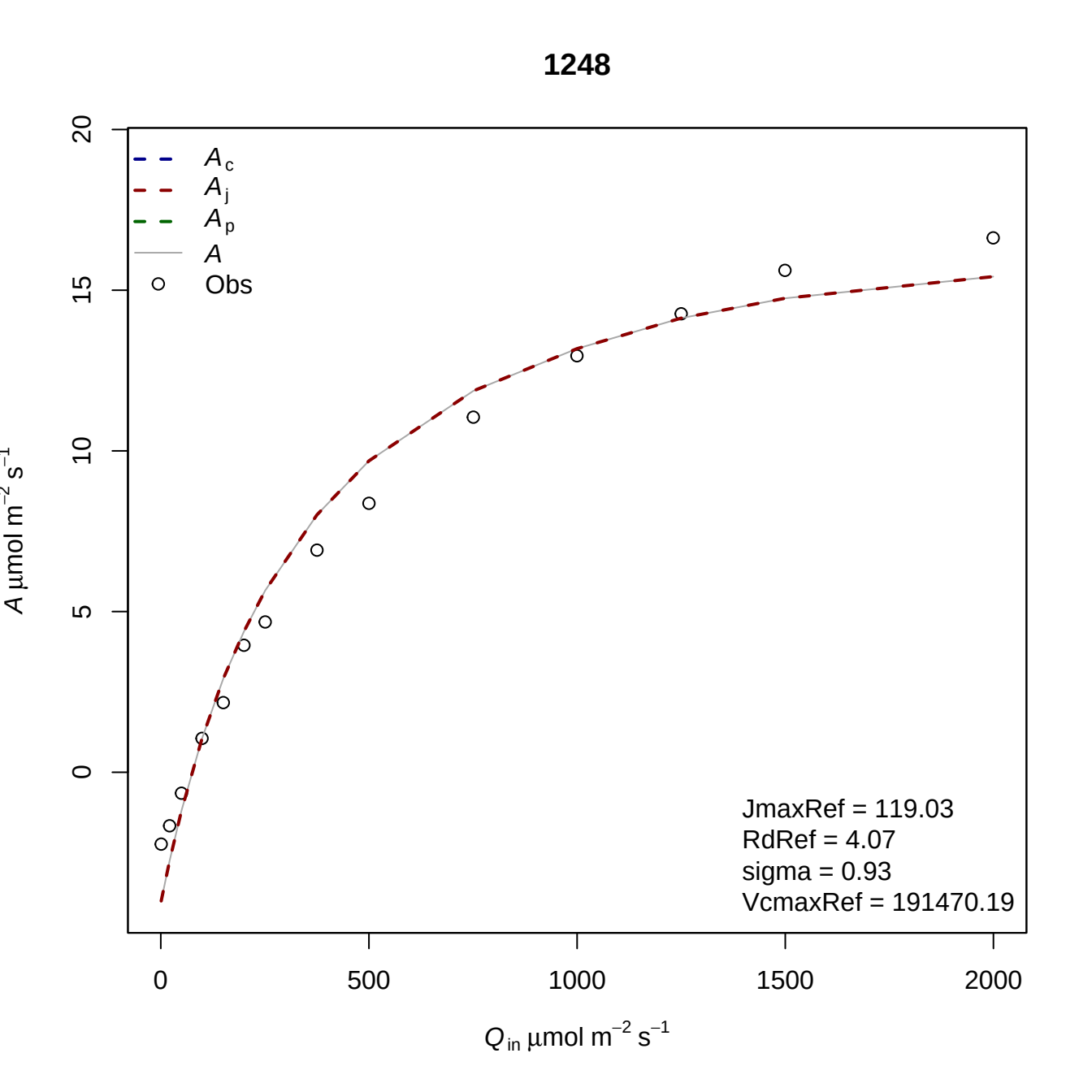

1243
A_c
A_j
A_p
A
Obs
JmaxRef = 68.98
RdRef = 2.69
sigma = 0.84
VcmaxRef = 150058.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

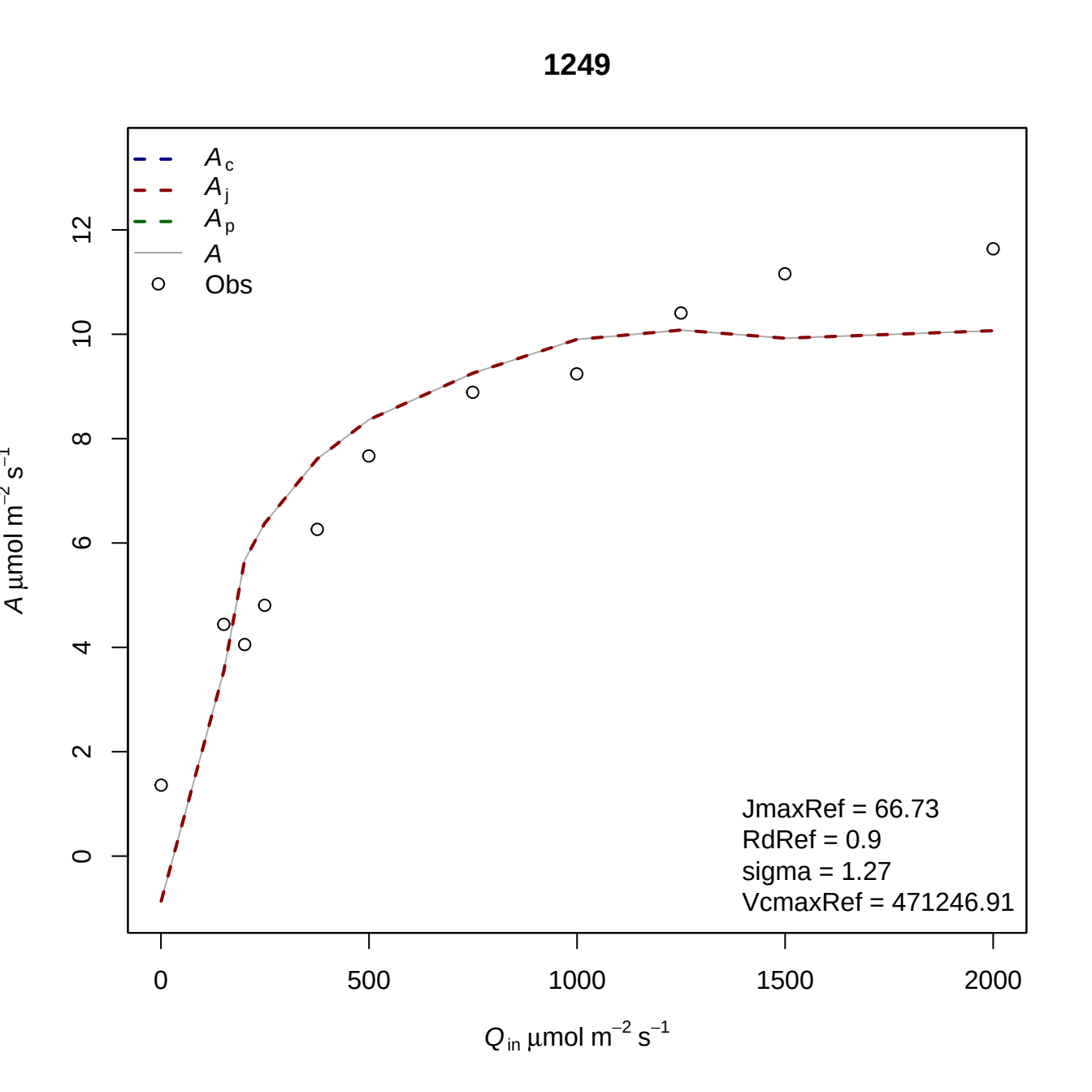

1244
A_c
A_j
A_p
A
Obs
JmaxRef = 124.07
RdRef = 4.34
sigma = 0.94
VcmaxRef = 901743.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

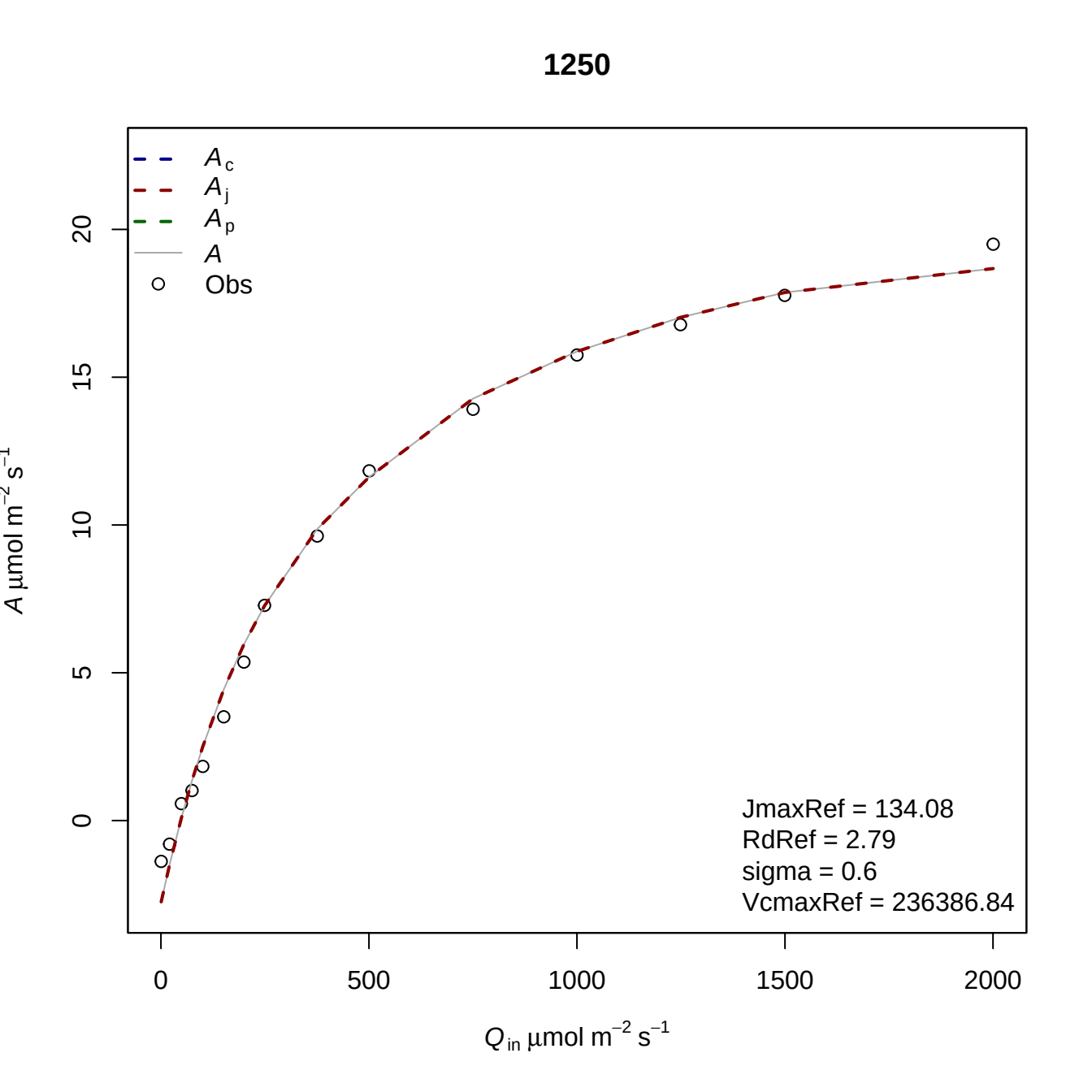

1245
A_c
A_j
A_p
A
Obs
JmaxRef = 121.52
RdRef = 3.02
sigma = 0.93
VcmaxRef = 310036.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

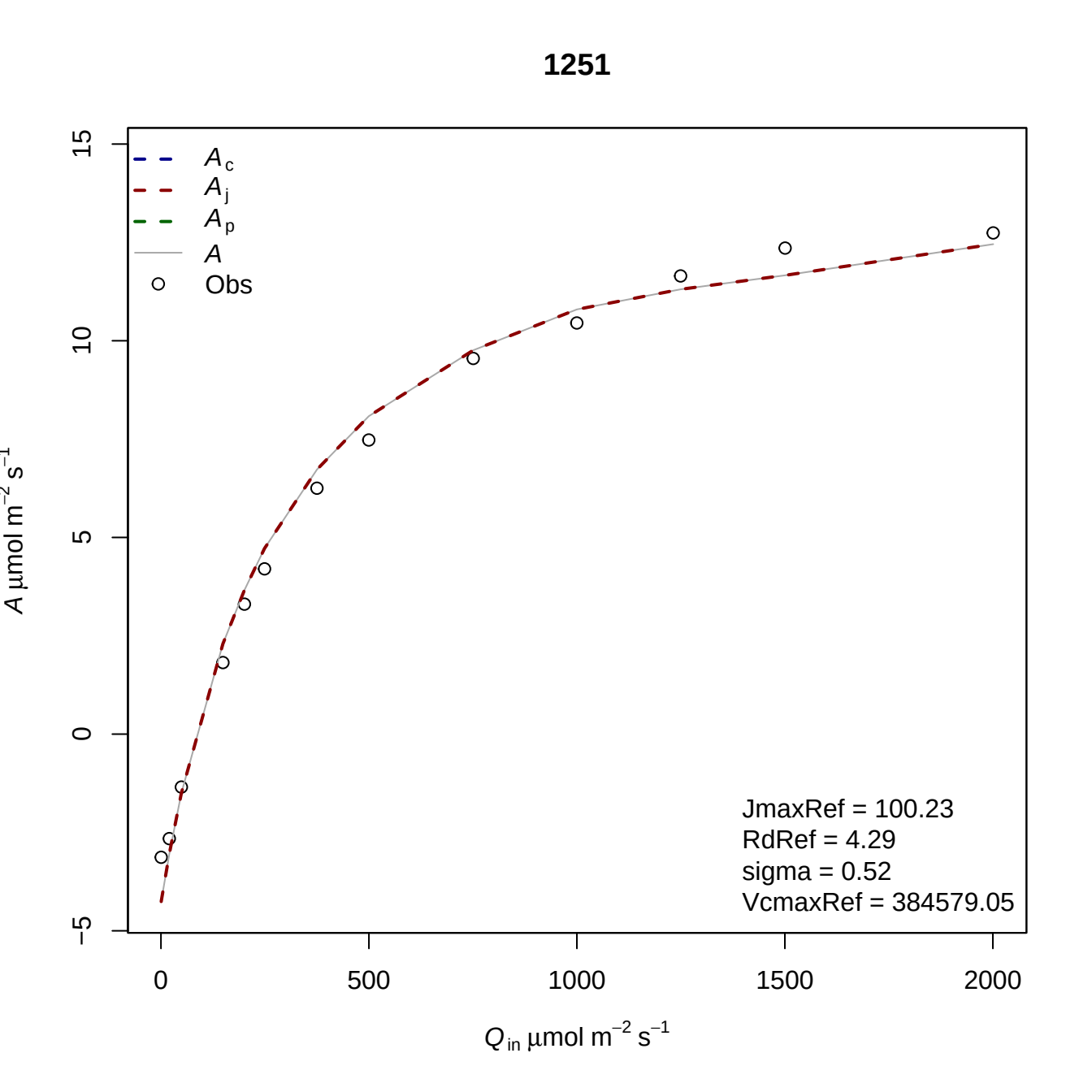

1246
A_c
A_j
A_p
A
Obs
JmaxRef = 95.81
RdRef = 2.98
sigma = 0.41
VcmaxRef = 106572.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

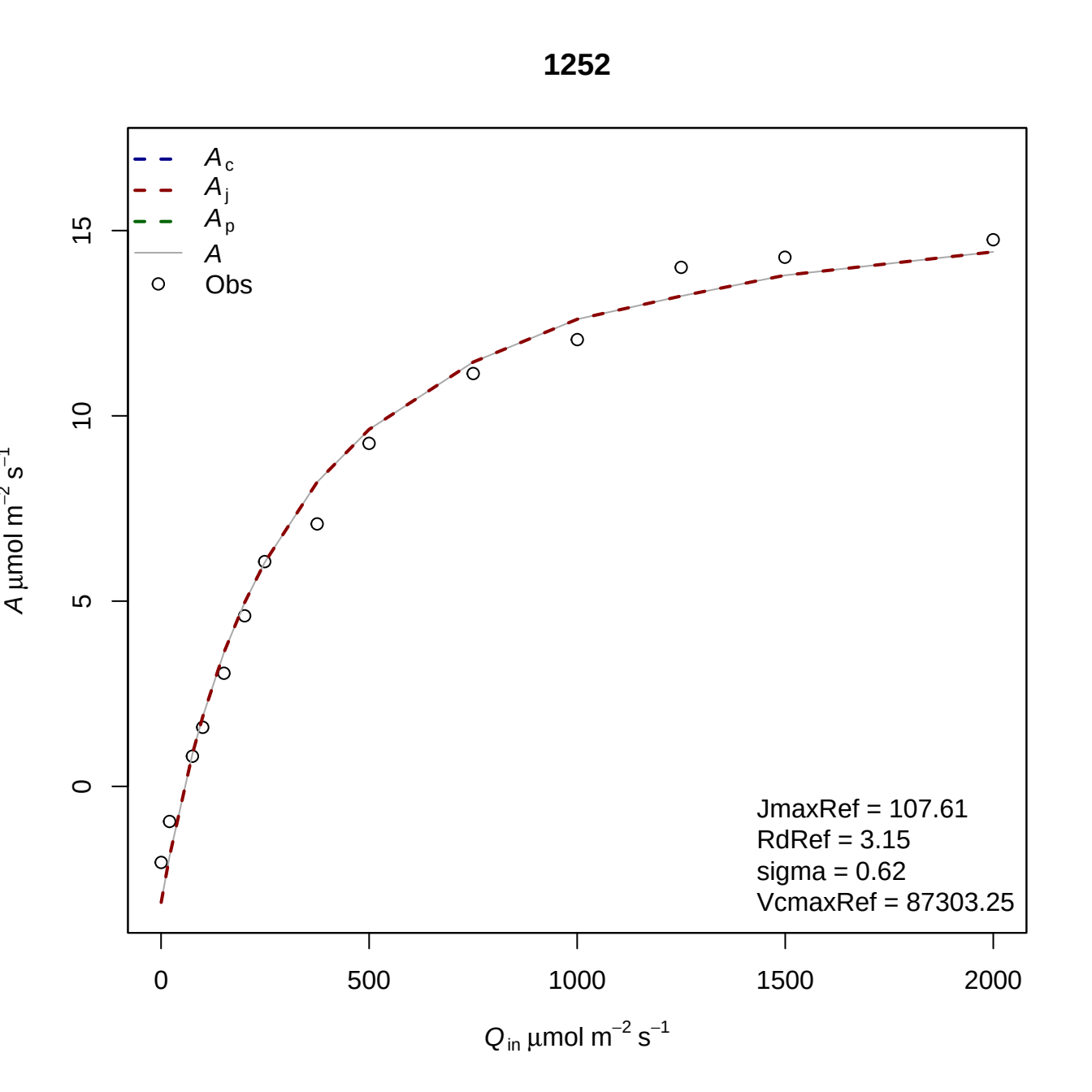

1247
A_c
A_j
A_p
A
Obs
JmaxRef = 98.67
RdRef = 2.68
sigma = 0.41
VcmaxRef = 194334.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

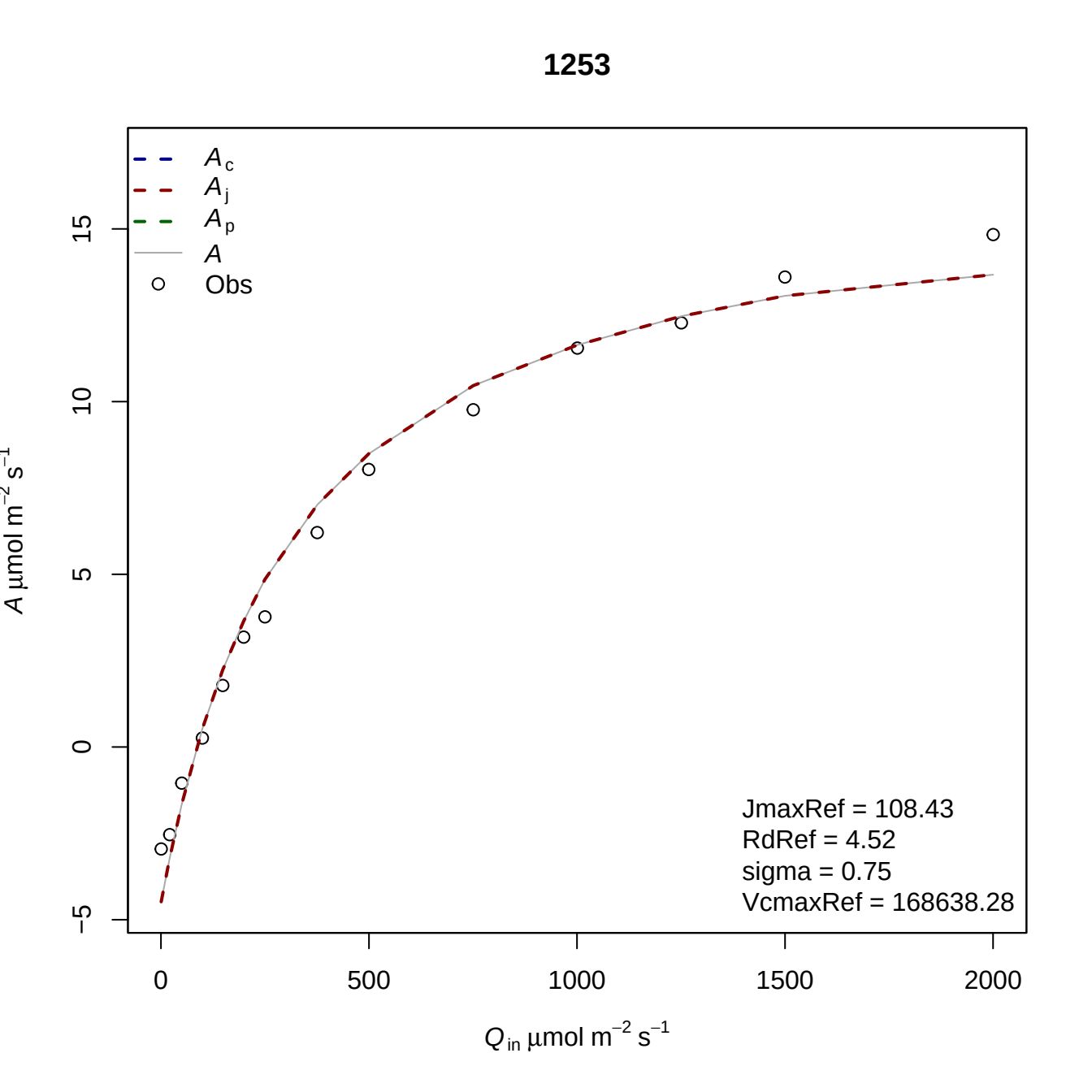

1248
A_c
A_j
A_p
A
Obs
JmaxRef = 119.03
RdRef = 4.07
sigma = 0.93
VcmaxRef = 191470.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

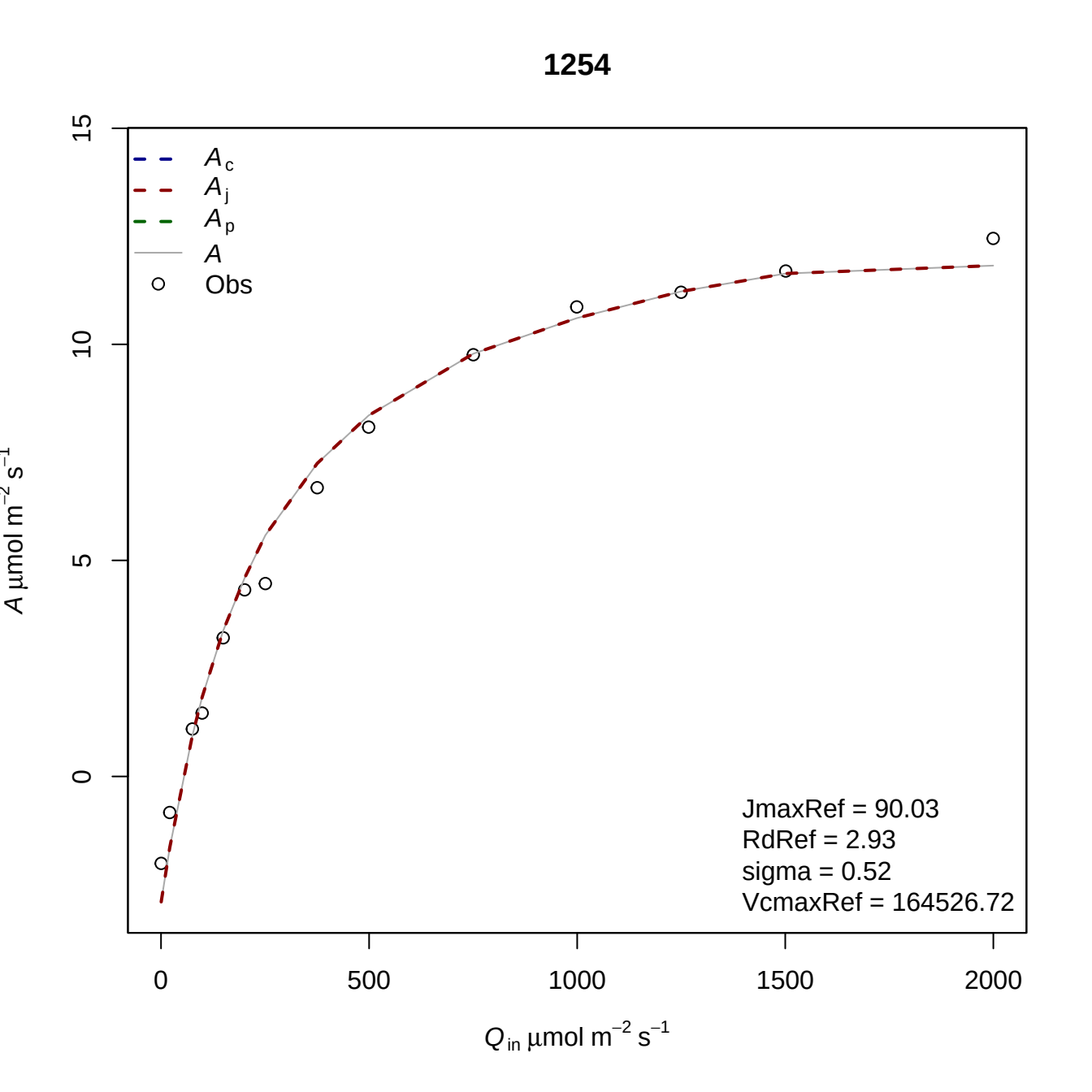

1249
A_c
A_j
A_p
A
Obs
JmaxRef = 66.73
RdRef = 0.9
sigma = 1.27
VcmaxRef = 471246.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

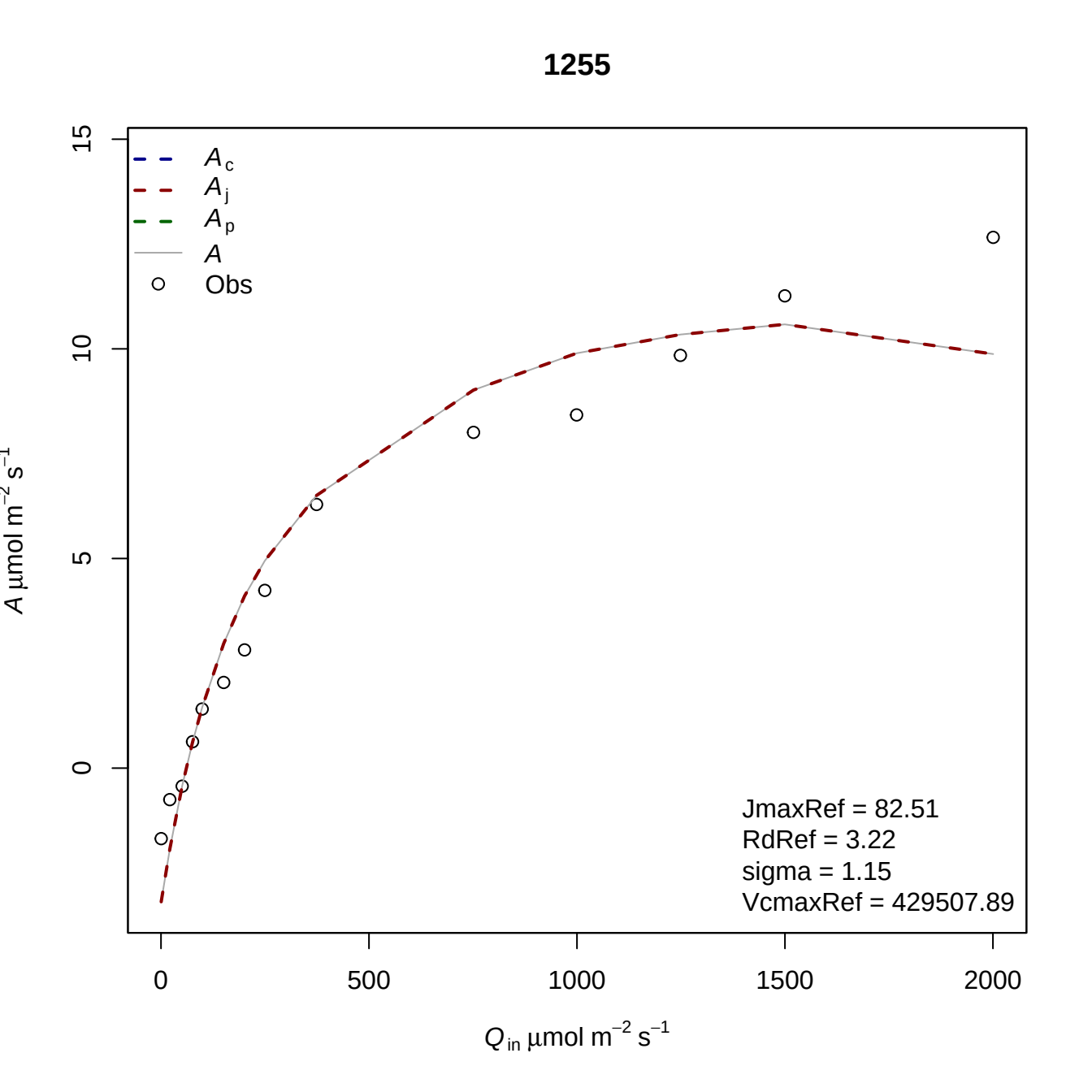

1250
A_c
A_j
A_p
A
Obs
JmaxRef = 134.08
RdRef = 2.79
sigma = 0.6
VcmaxRef = 236386.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

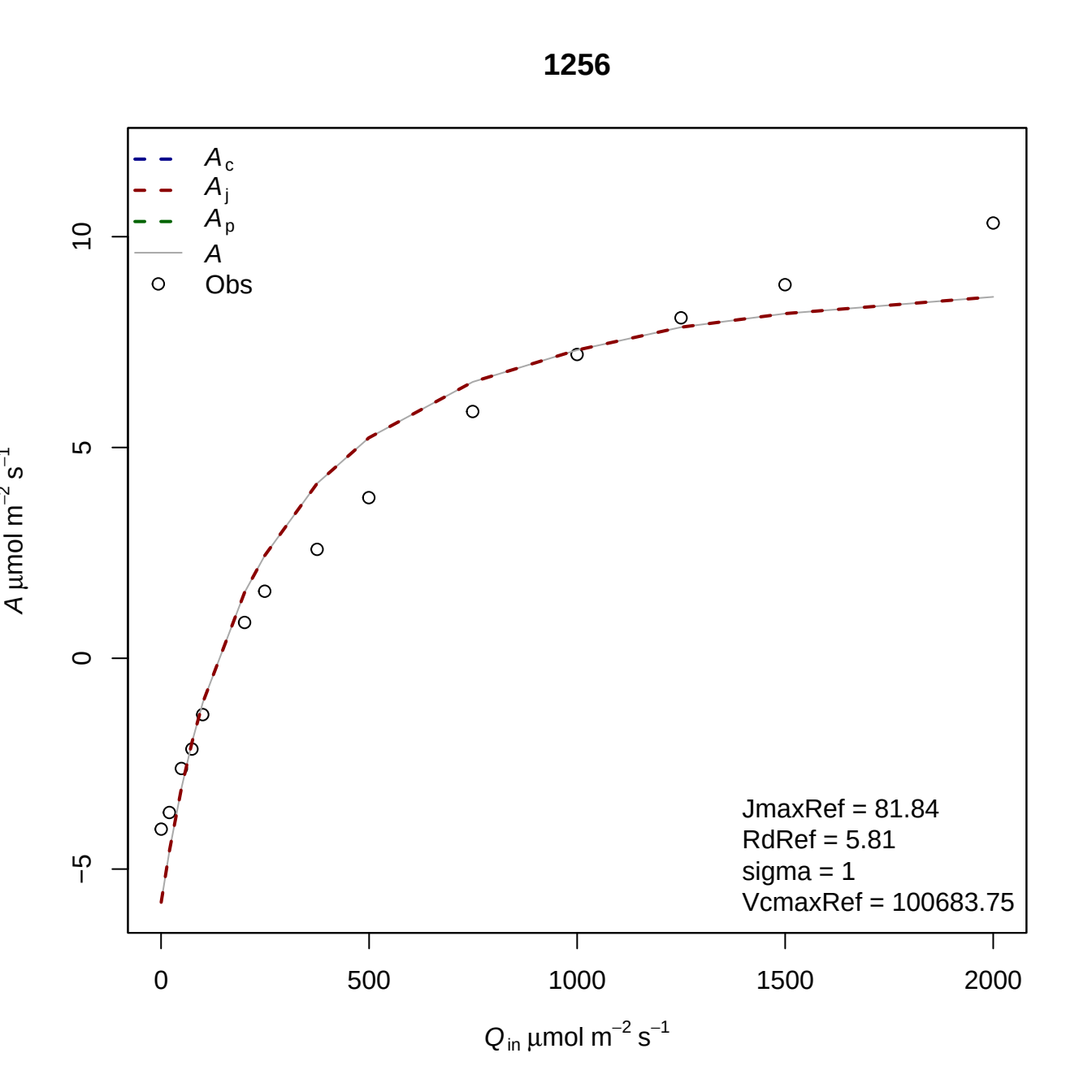

1251
A_c
A_j
A_p
A
Obs
JmaxRef = 100.23
RdRef = 4.29
sigma = 0.52
VcmaxRef = 384579.05
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

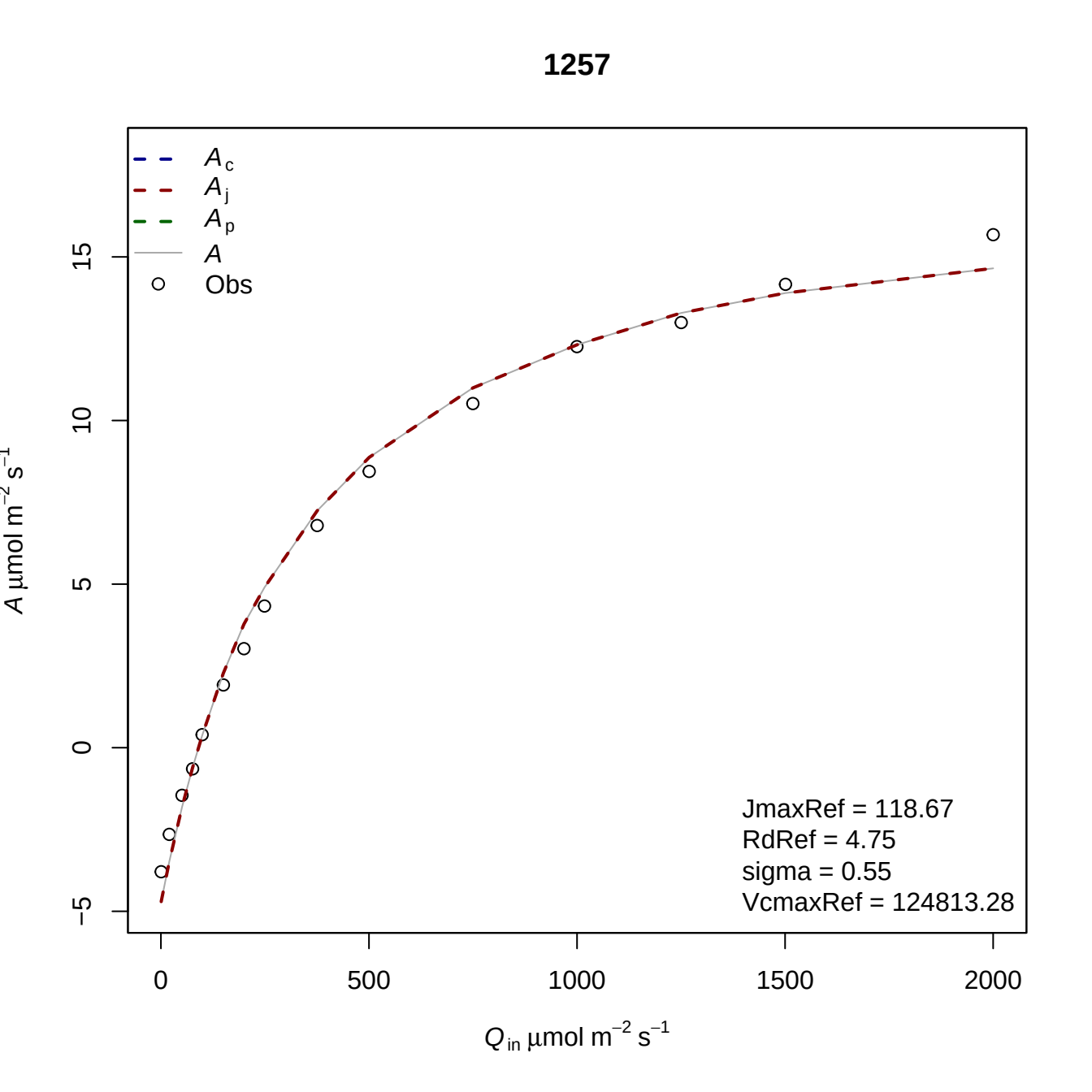

1252
A_c
A_j
A_p
A
Obs
JmaxRef = 107.61
RdRef = 3.15
sigma = 0.62
VcmaxRef = 87303.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

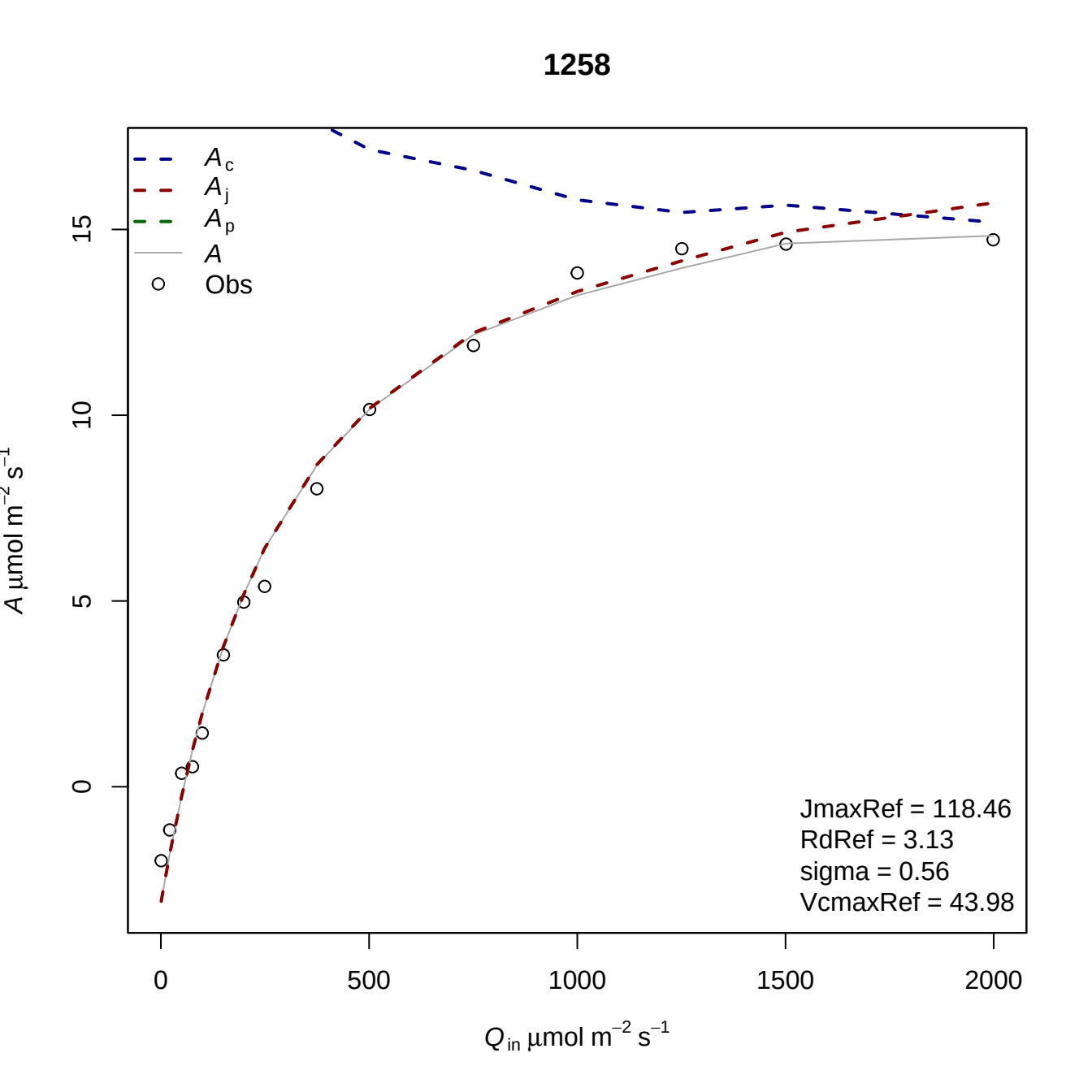

1253
A_c
A_j
A_p
A
Obs
JmaxRef = 108.43
RdRef = 4.52
sigma = 0.75
VcmaxRef = 168638.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

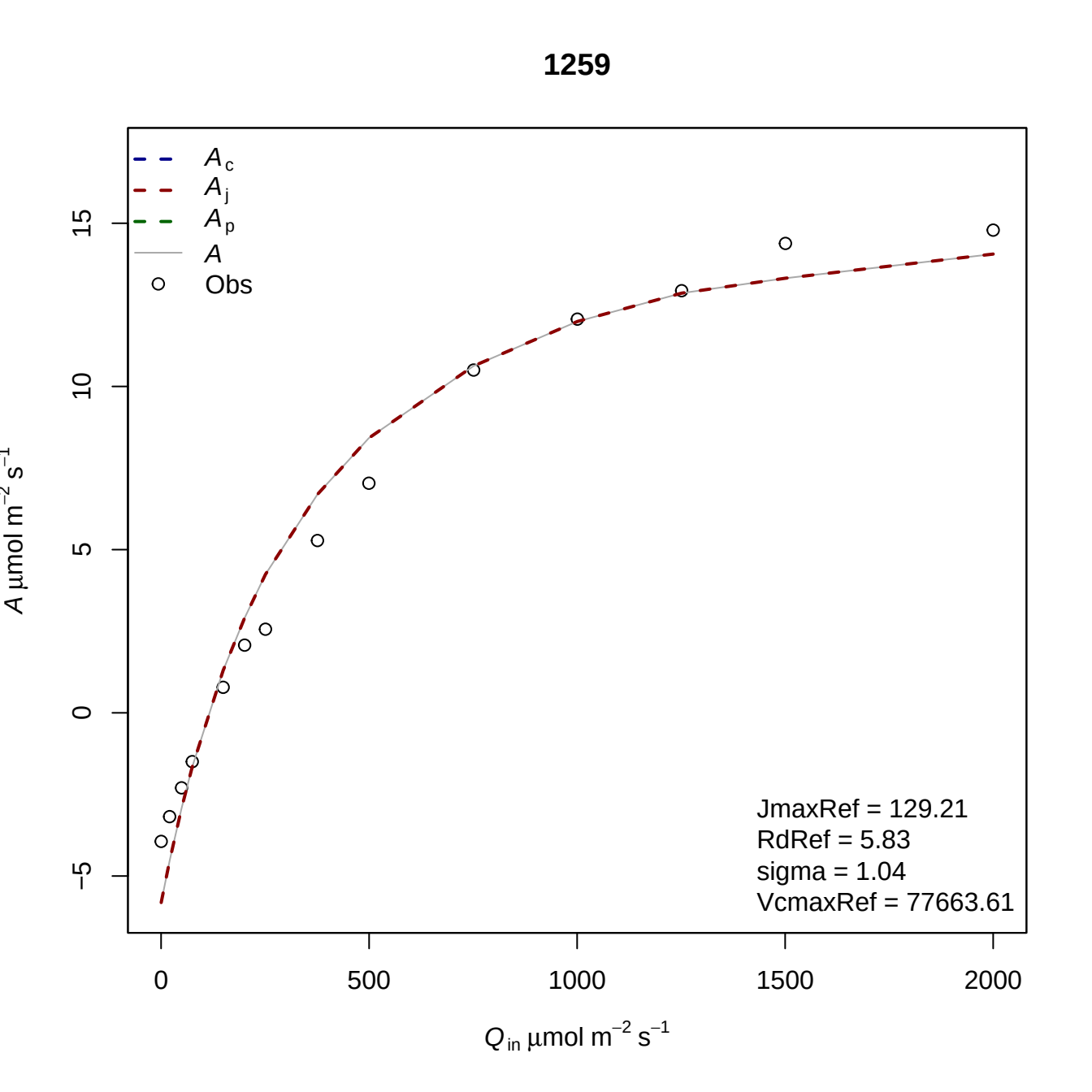

1254
A_c
A_j
A_p
A
Obs
JmaxRef = 90.03
RdRef = 2.93
sigma = 0.52
VcmaxRef = 164526.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

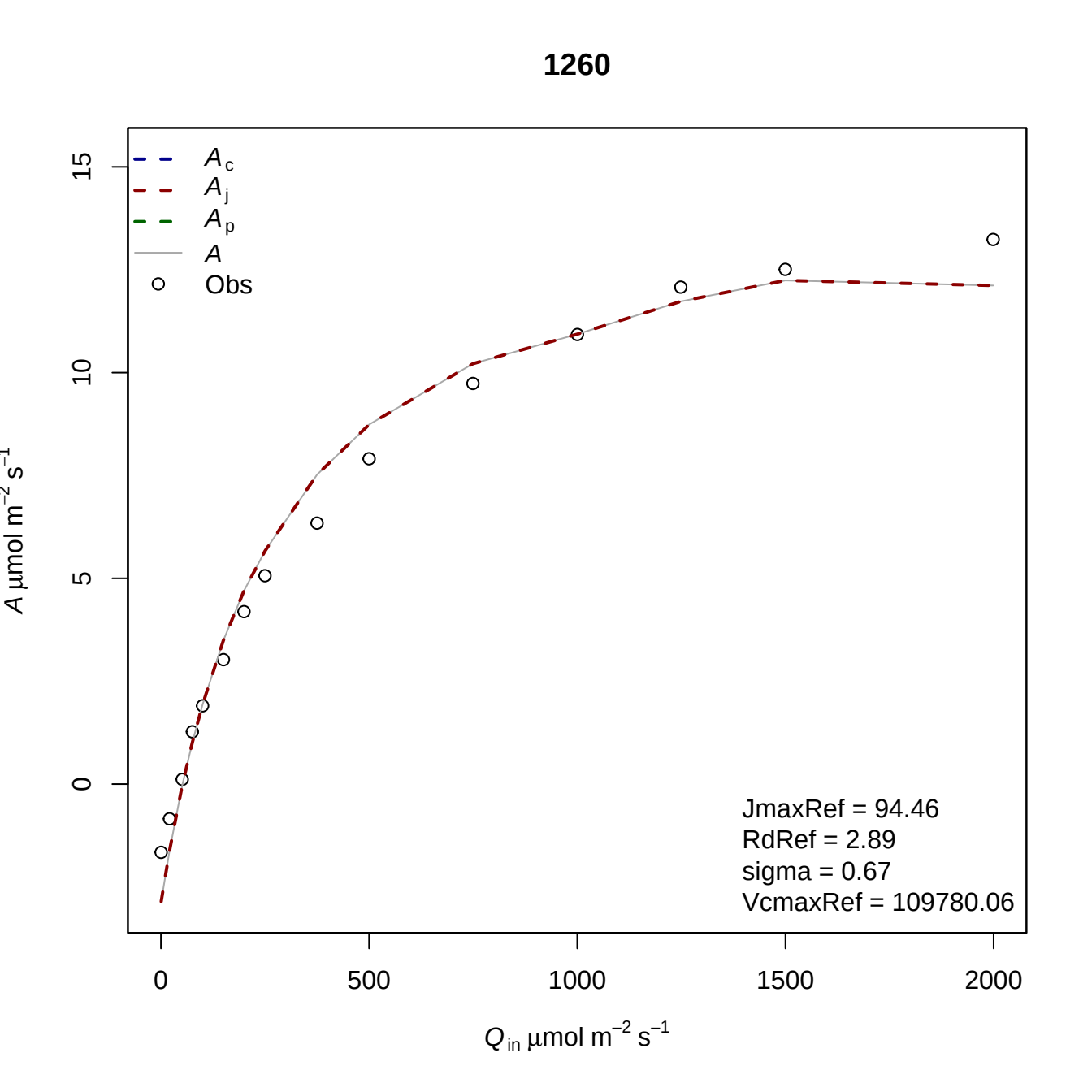

1255
A_c
A_j
A_p
A
Obs
JmaxRef = 82.51
RdRef = 3.22
sigma = 1.15
VcmaxRef = 429507.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

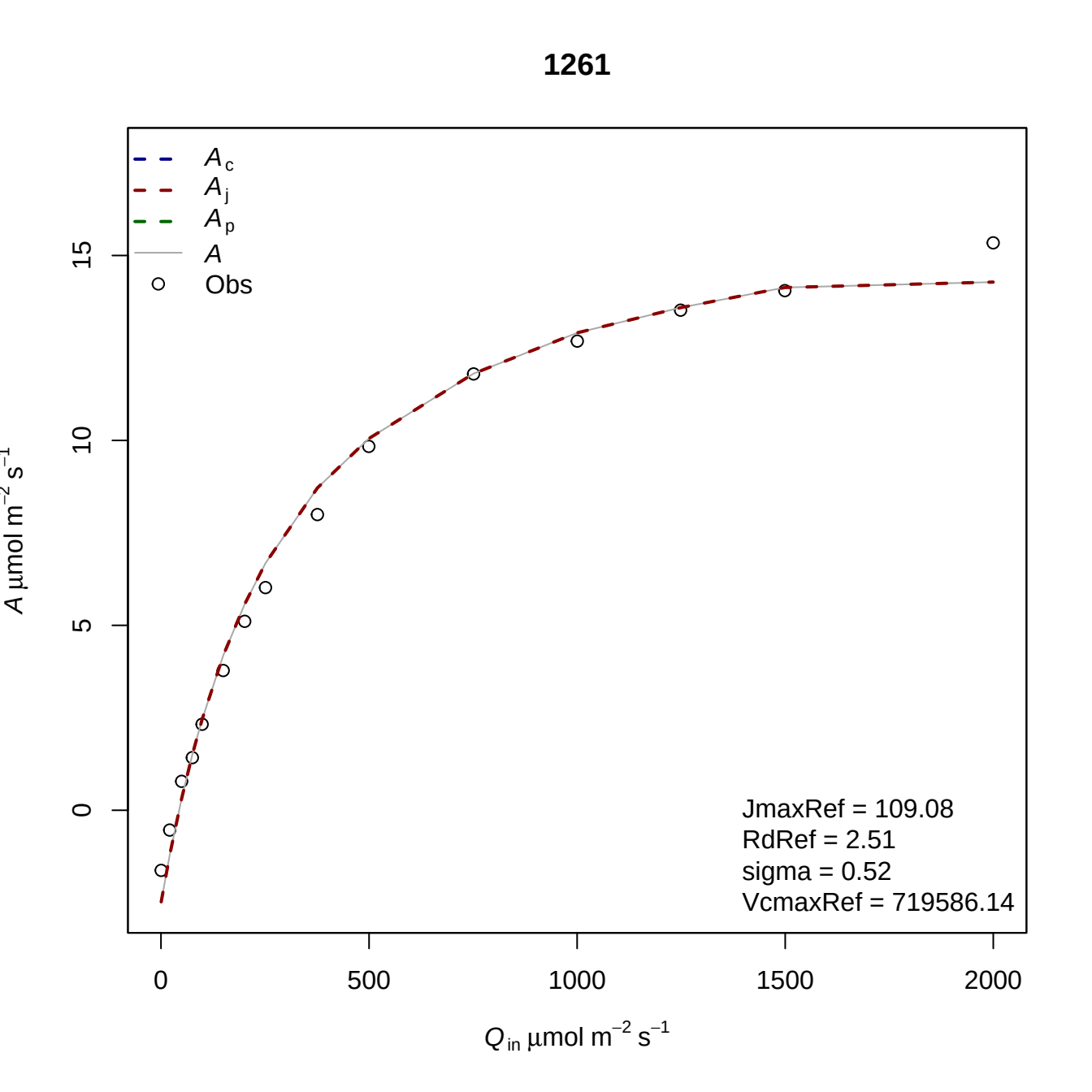

1256
A_c
A_j
A_p
A
Obs
JmaxRef = 81.84
RdRef = 5.81
sigma = 1
VcmaxRef = 100683.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
5
0
−5
0
500
1000
1500
2000

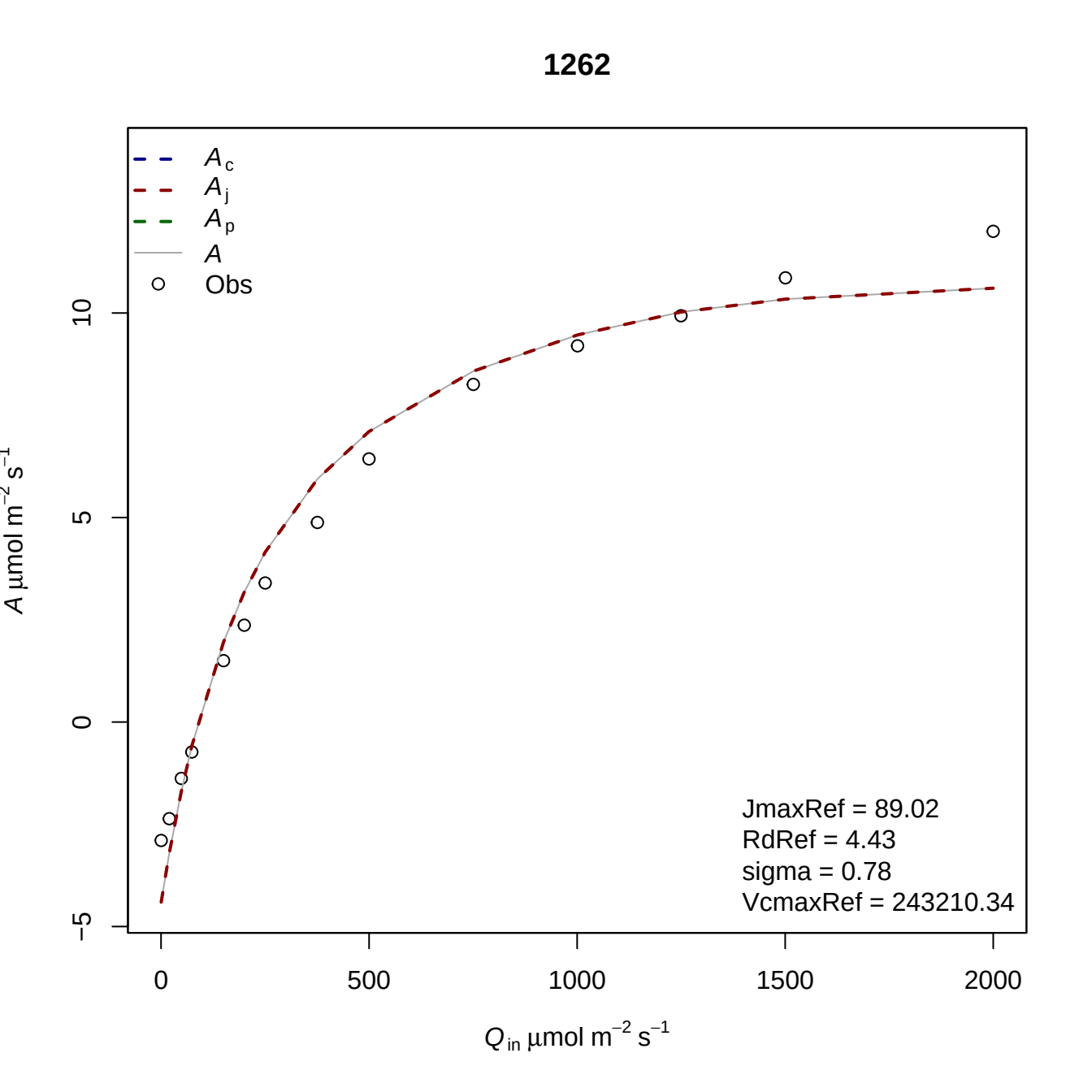

1257
A_c
A_j
A_p
A
Obs
JmaxRef = 118.67
RdRef = 4.75
sigma = 0.55
VcmaxRef = 124813.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

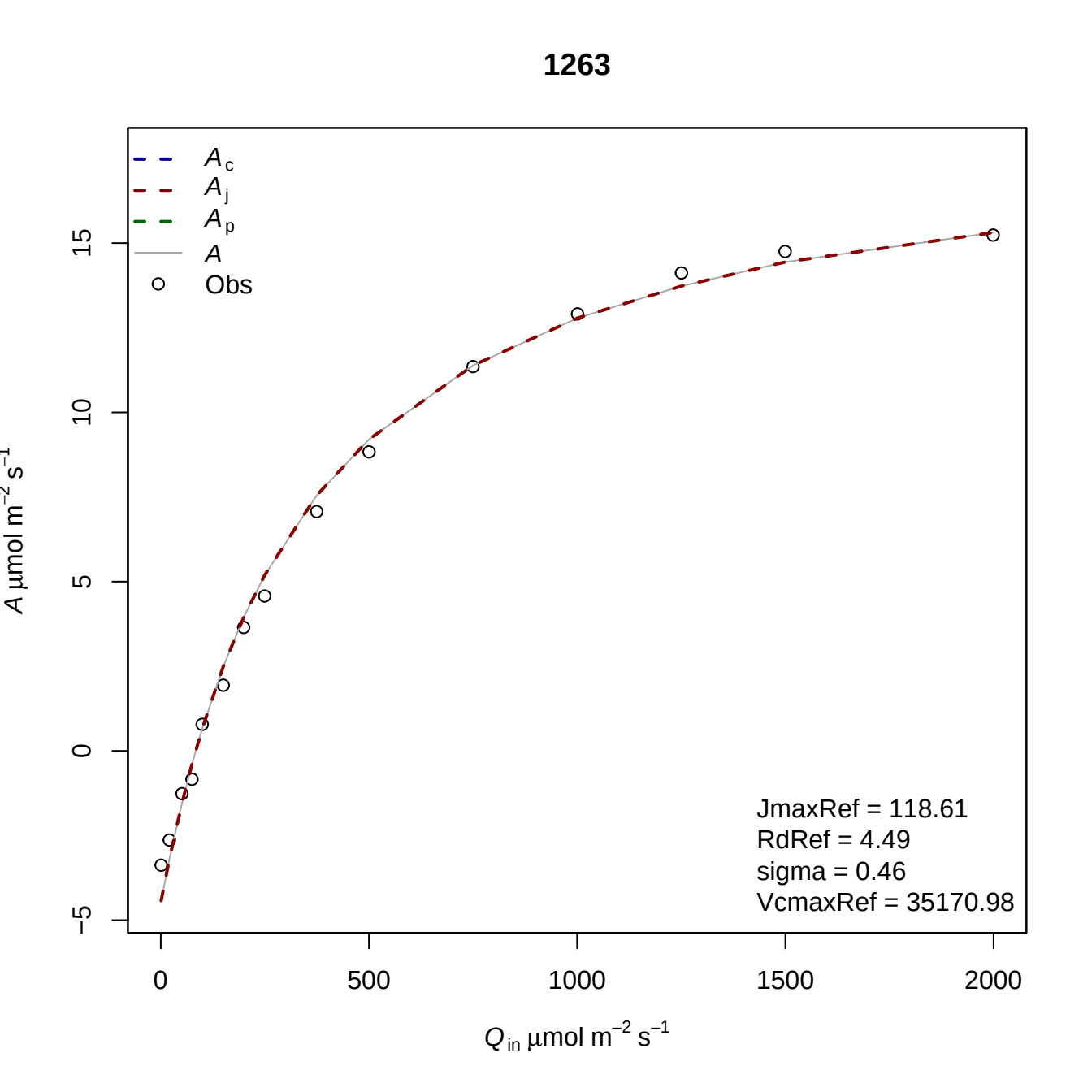

1258
A_c
A_j
A_p
A
Obs
JmaxRef = 118.46
RdRef = 3.13
sigma = 0.56
VcmaxRef = 43.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

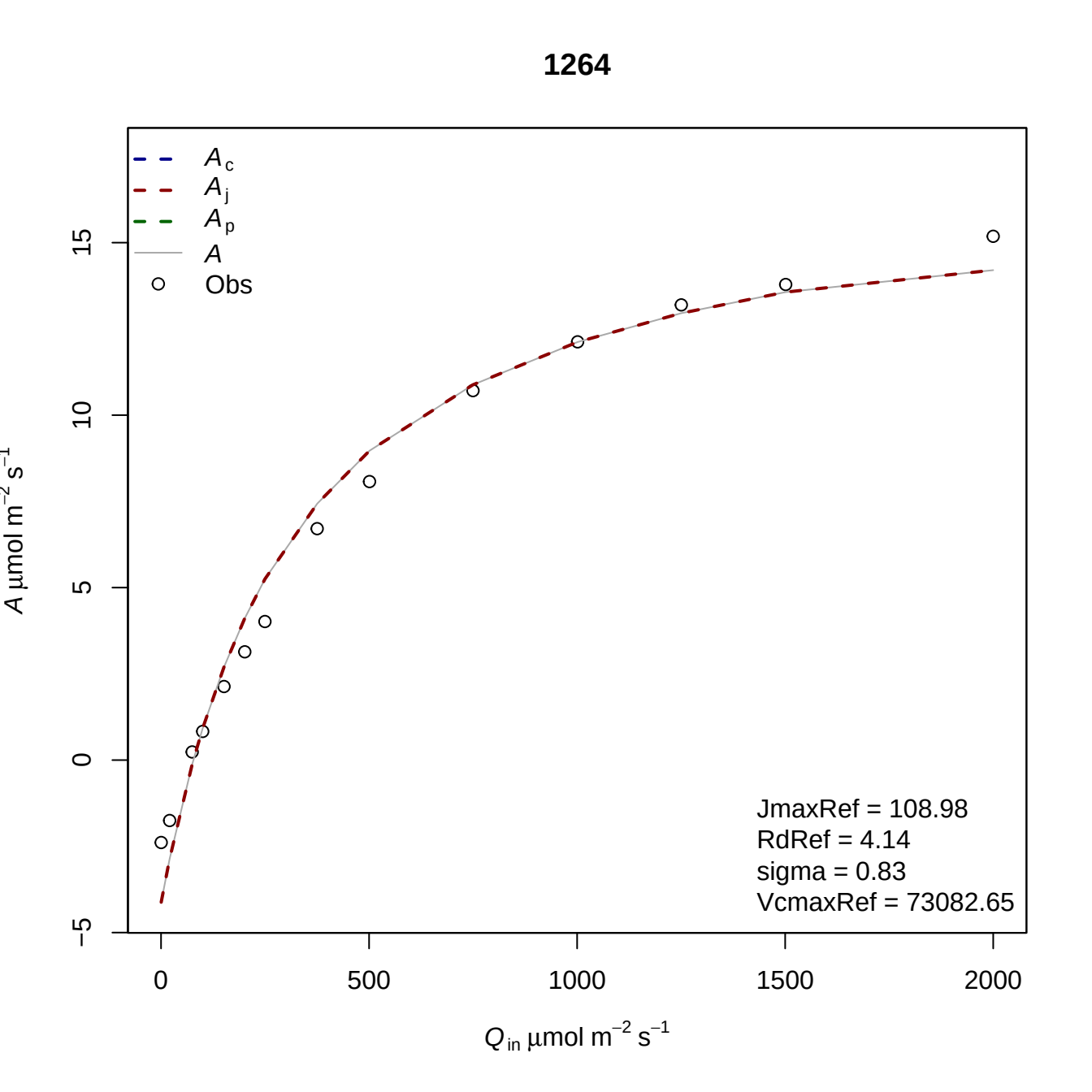

1259
A_c
A_j
A_p
A
Obs
JmaxRef = 129.21
RdRef = 5.83
sigma = 1.04
VcmaxRef = 77663.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

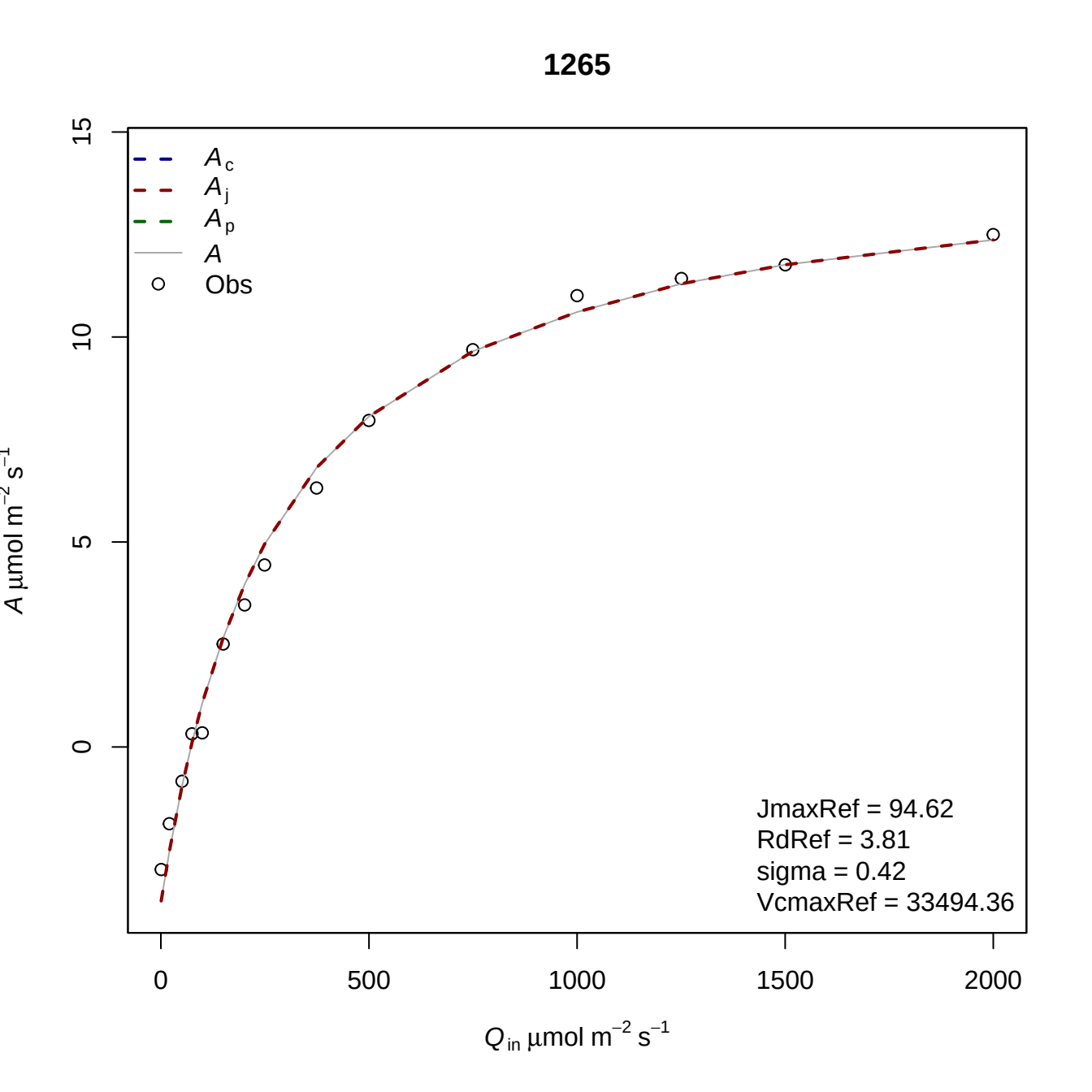

1260
A_c
A_j
A_p
A
Obs
JmaxRef = 94.46
RdRef = 2.89
sigma = 0.67
VcmaxRef = 109780.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

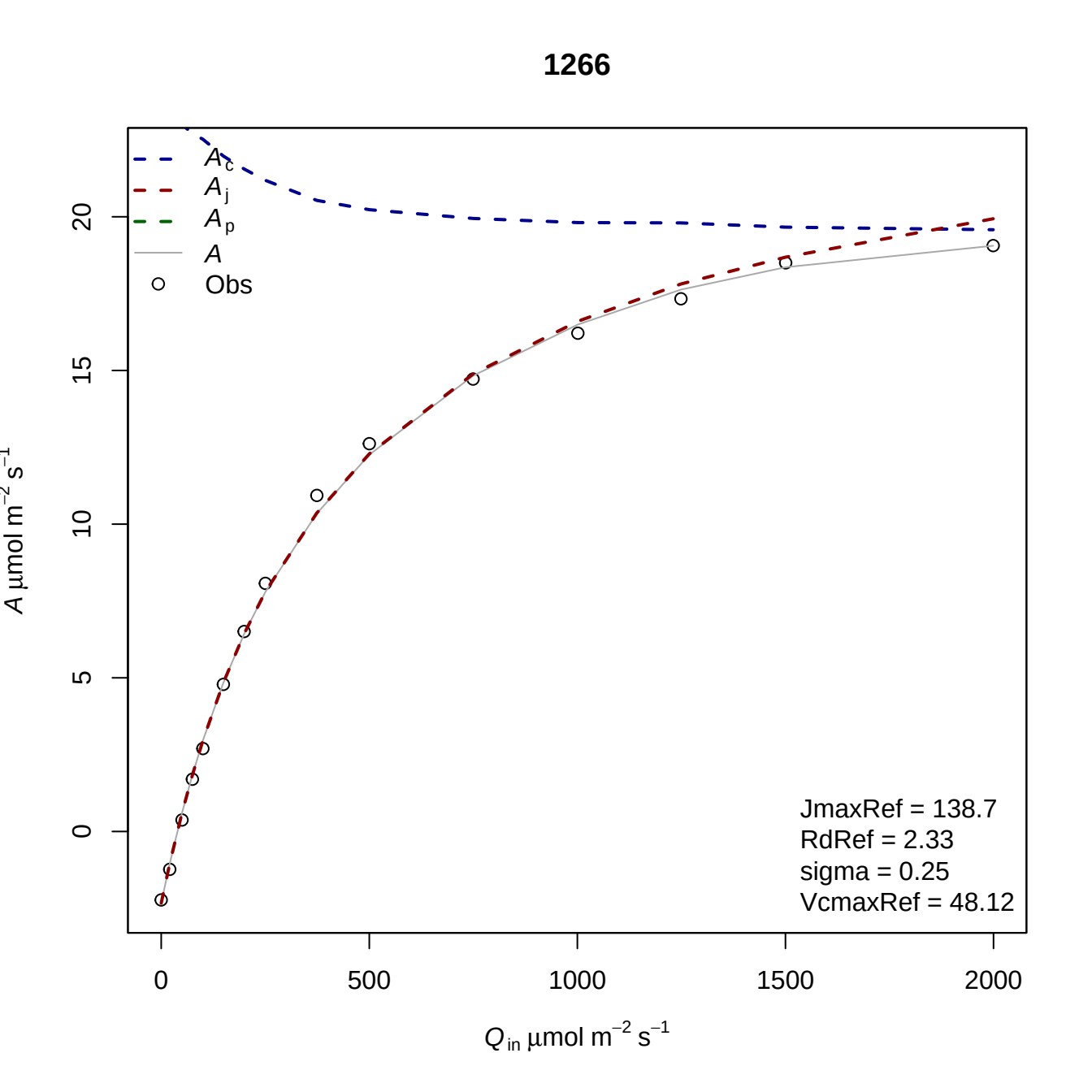

1261
A_c
A_j
A_p
A
Obs
JmaxRef = 109.08
RdRef = 2.51
sigma = 0.52
VcmaxRef = 719586.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

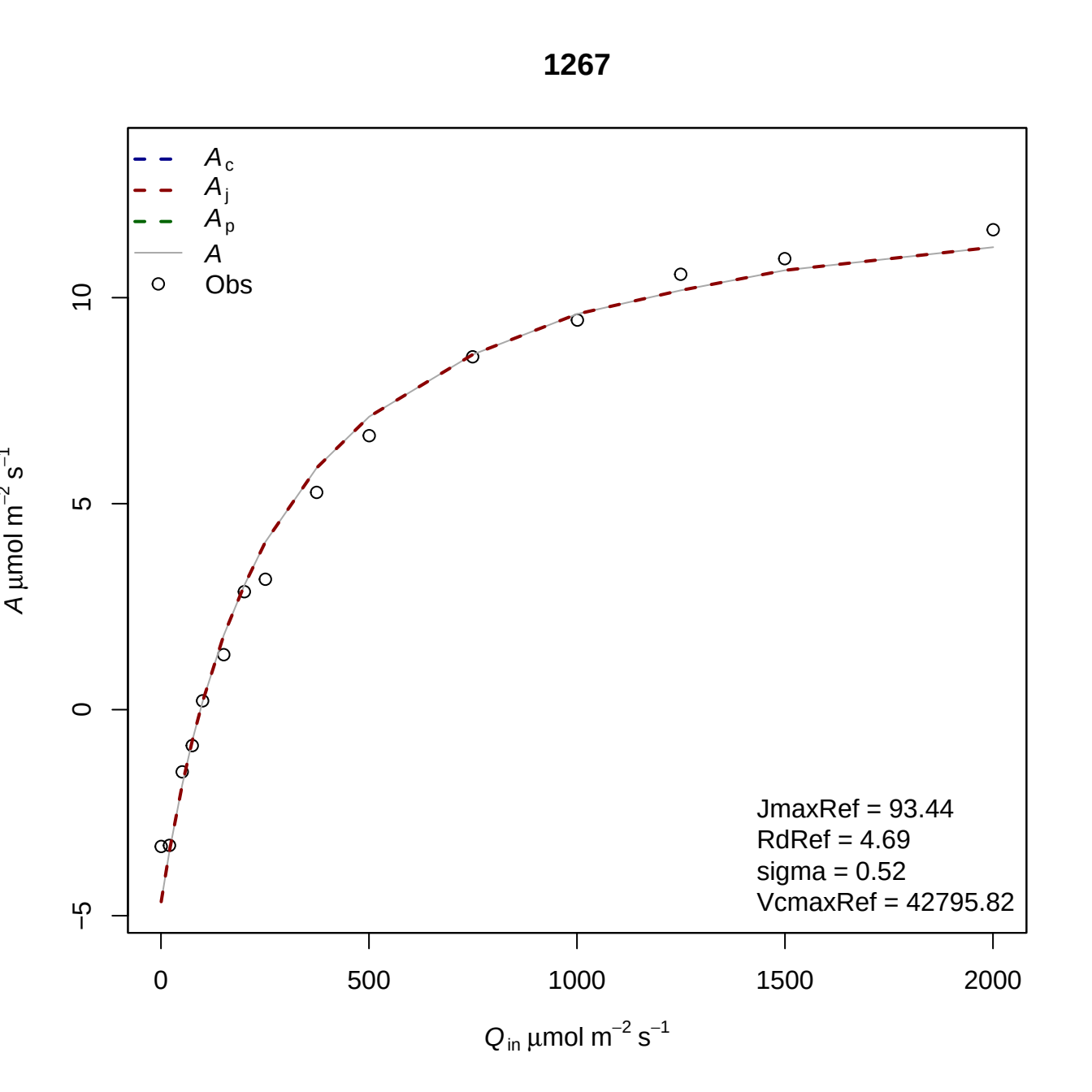

1262
A_c
A_j
A_p
A
Obs
JmaxRef = 89.02
RdRef = 4.43
sigma = 0.78
VcmaxRef = 243210.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

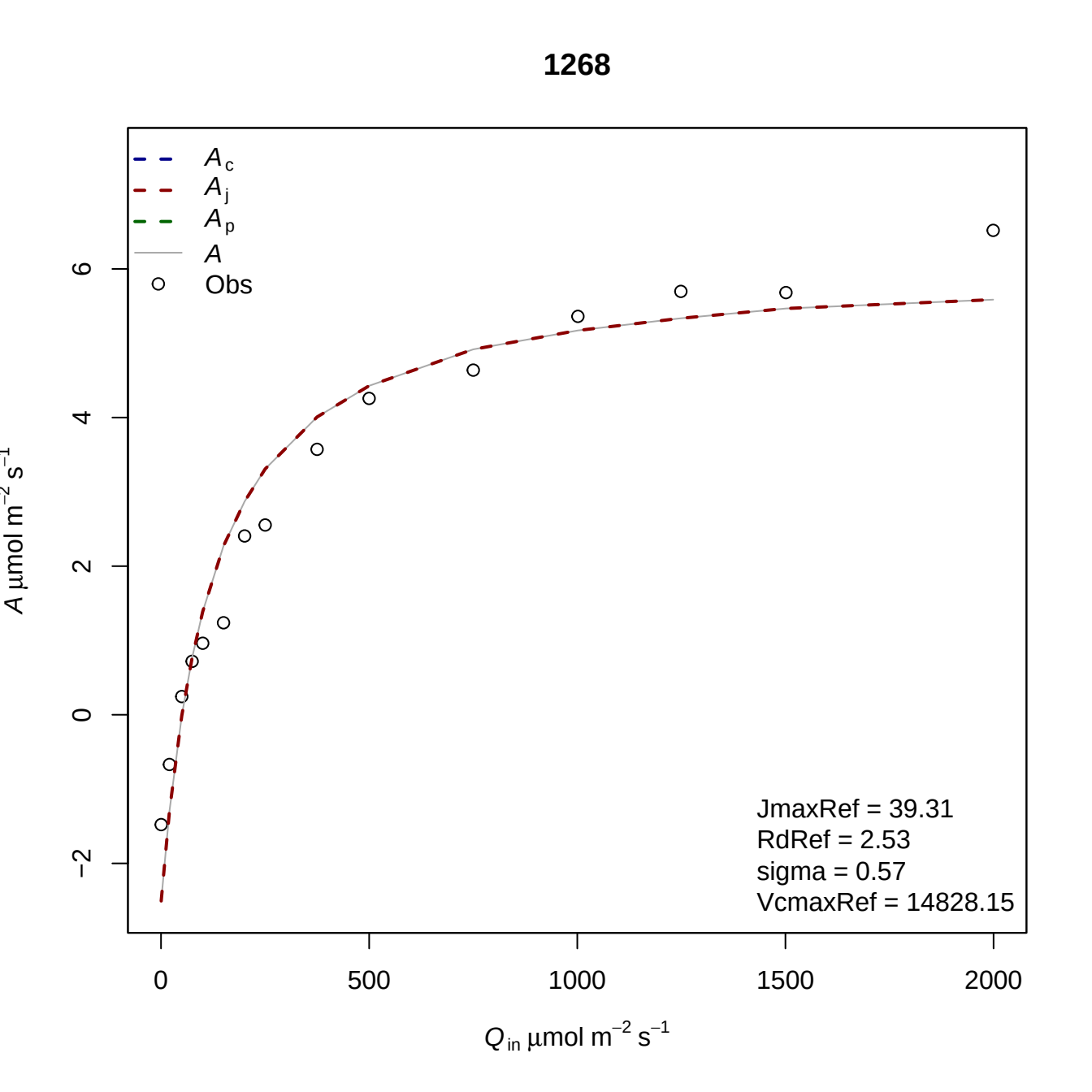

1263
A_c
A_j
A_p
A
Obs
JmaxRef = 118.61
RdRef = 4.49
sigma = 0.46
VcmaxRef = 35170.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

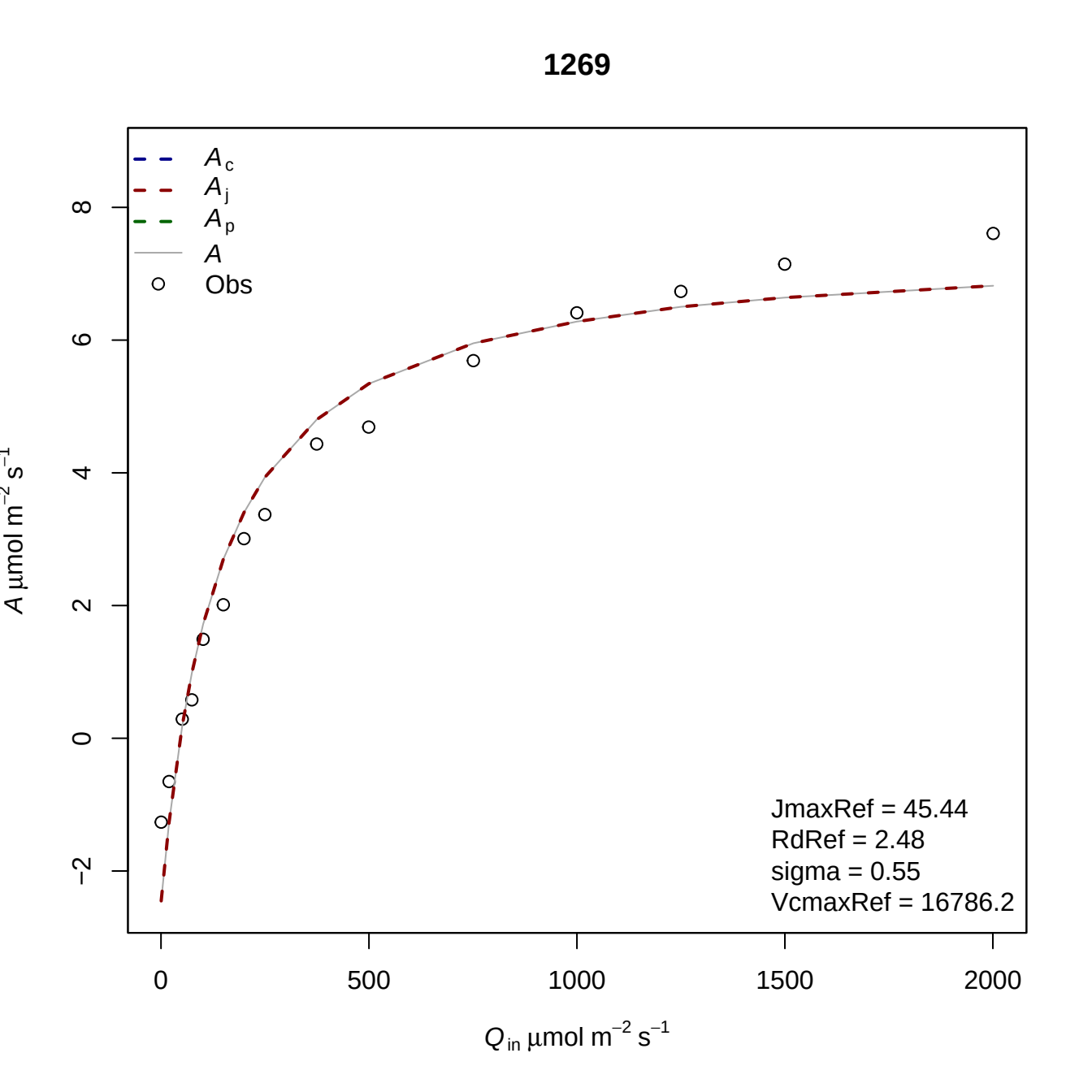

1264
A_c
A_j
A_p
A
Obs
JmaxRef = 108.98
RdRef = 4.14
sigma = 0.83
VcmaxRef = 73082.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

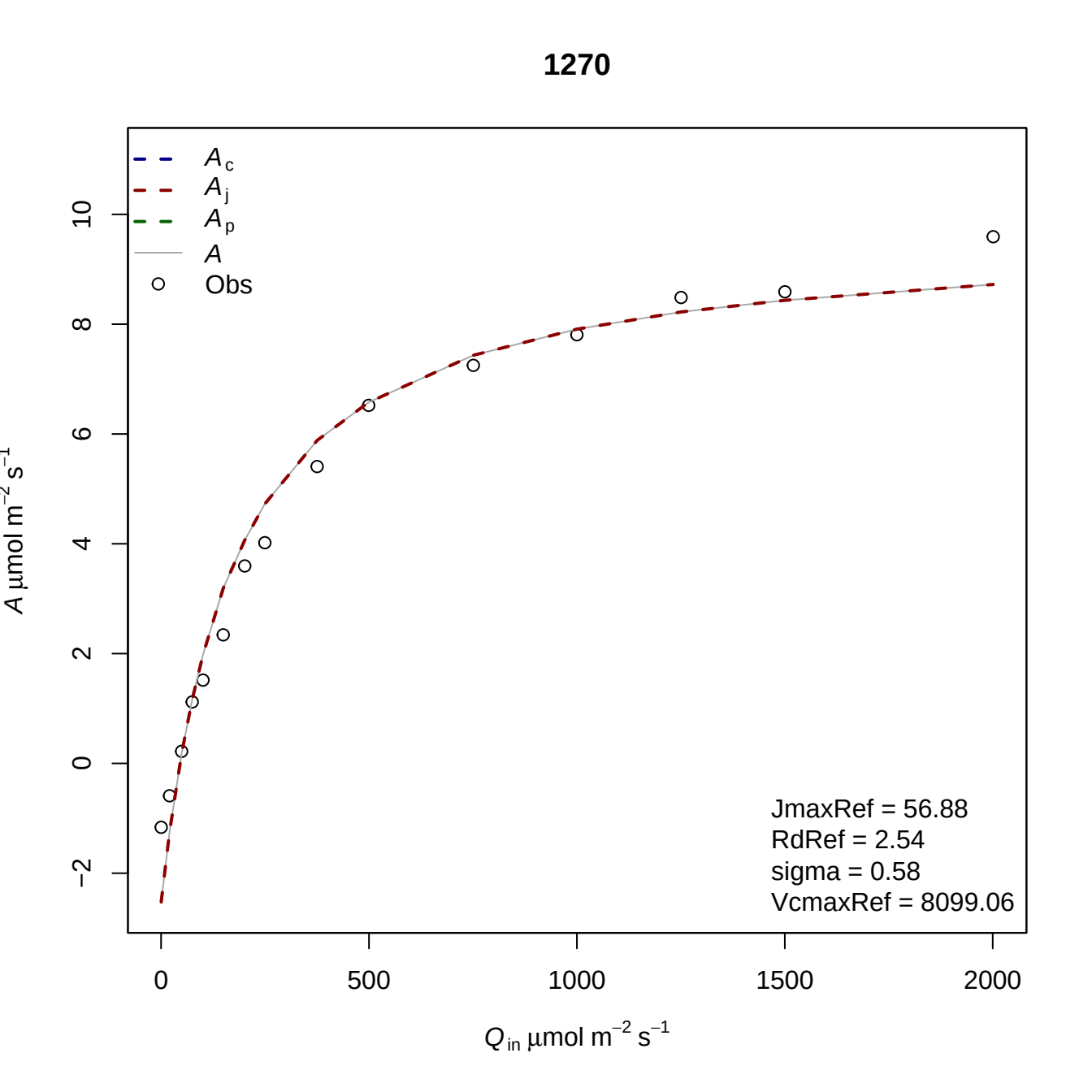

1265
A_c
A_j
A_p
A
Obs
JmaxRef = 94.62
RdRef = 3.81
sigma = 0.42
VcmaxRef = 33494.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

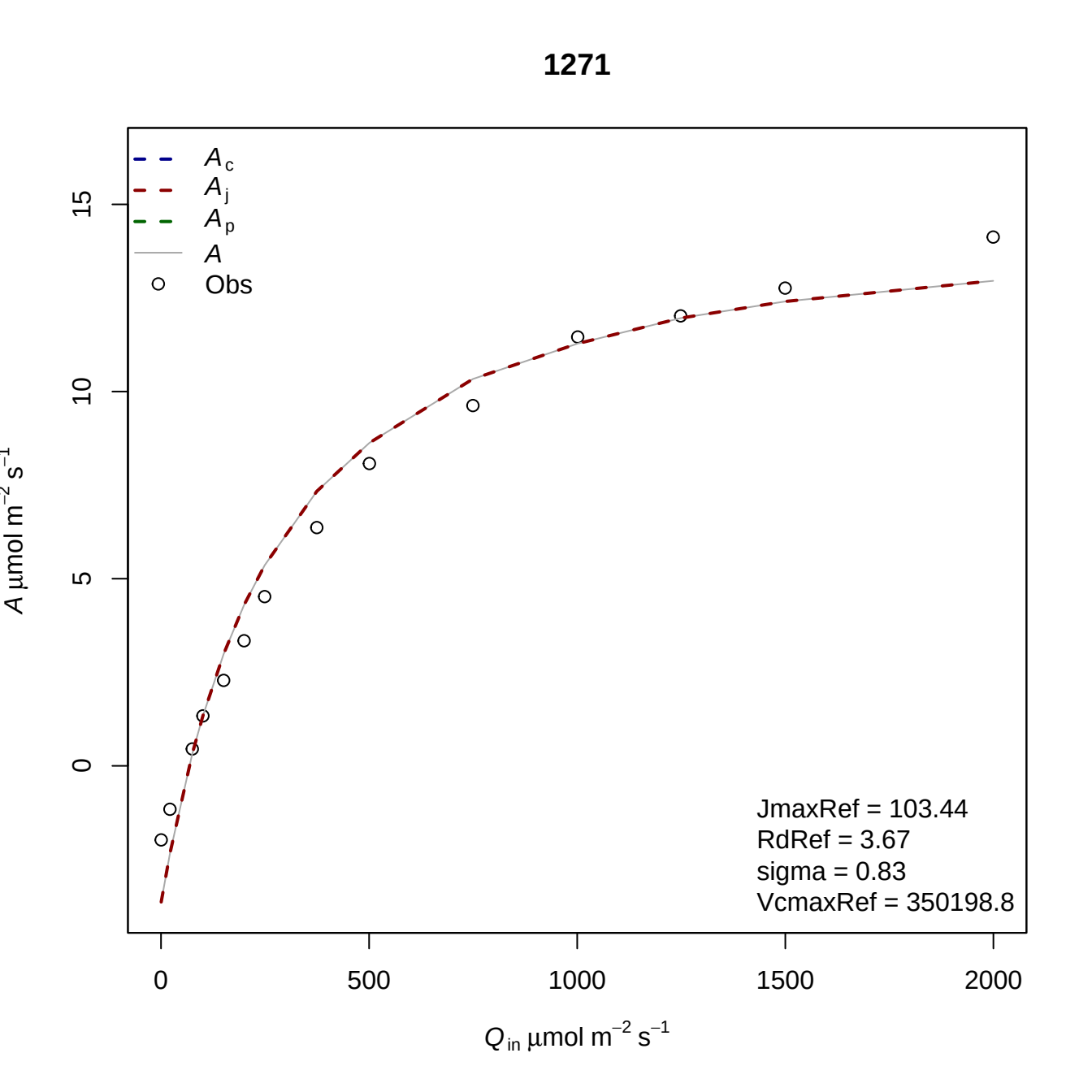

1266
A_c
A_j
A_p
A
Obs
JmaxRef = 138.7
RdRef = 2.33
sigma = 0.25
VcmaxRef = 48.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

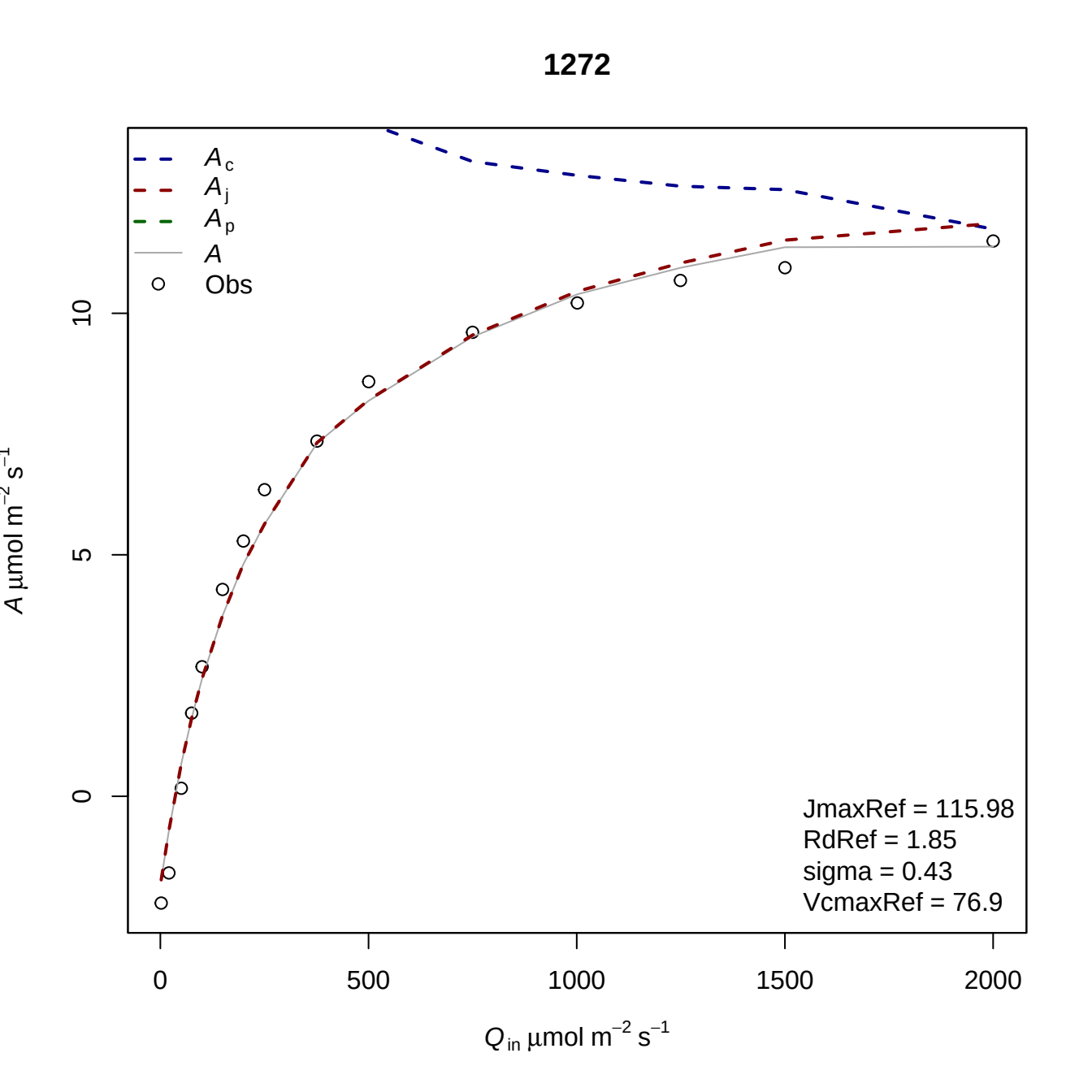

1267
A_c
A_j
A_p
A
Obs
JmaxRef = 93.44
RdRef = 4.69
sigma = 0.52
VcmaxRef = 42795.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

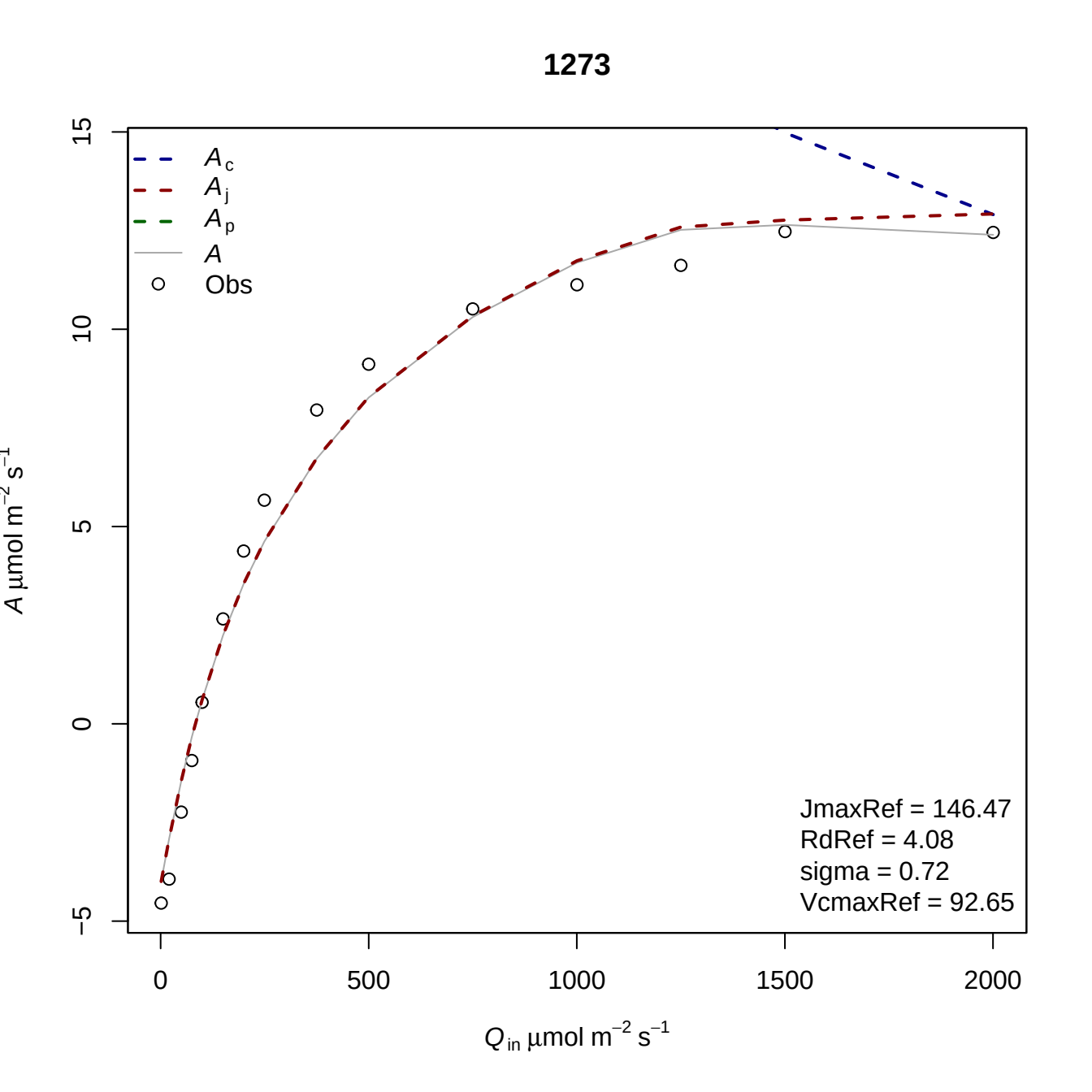

1268
A_c
A_j
A_p
A
Obs
JmaxRef = 39.31
RdRef = 2.53
sigma = 0.57
VcmaxRef = 14828.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

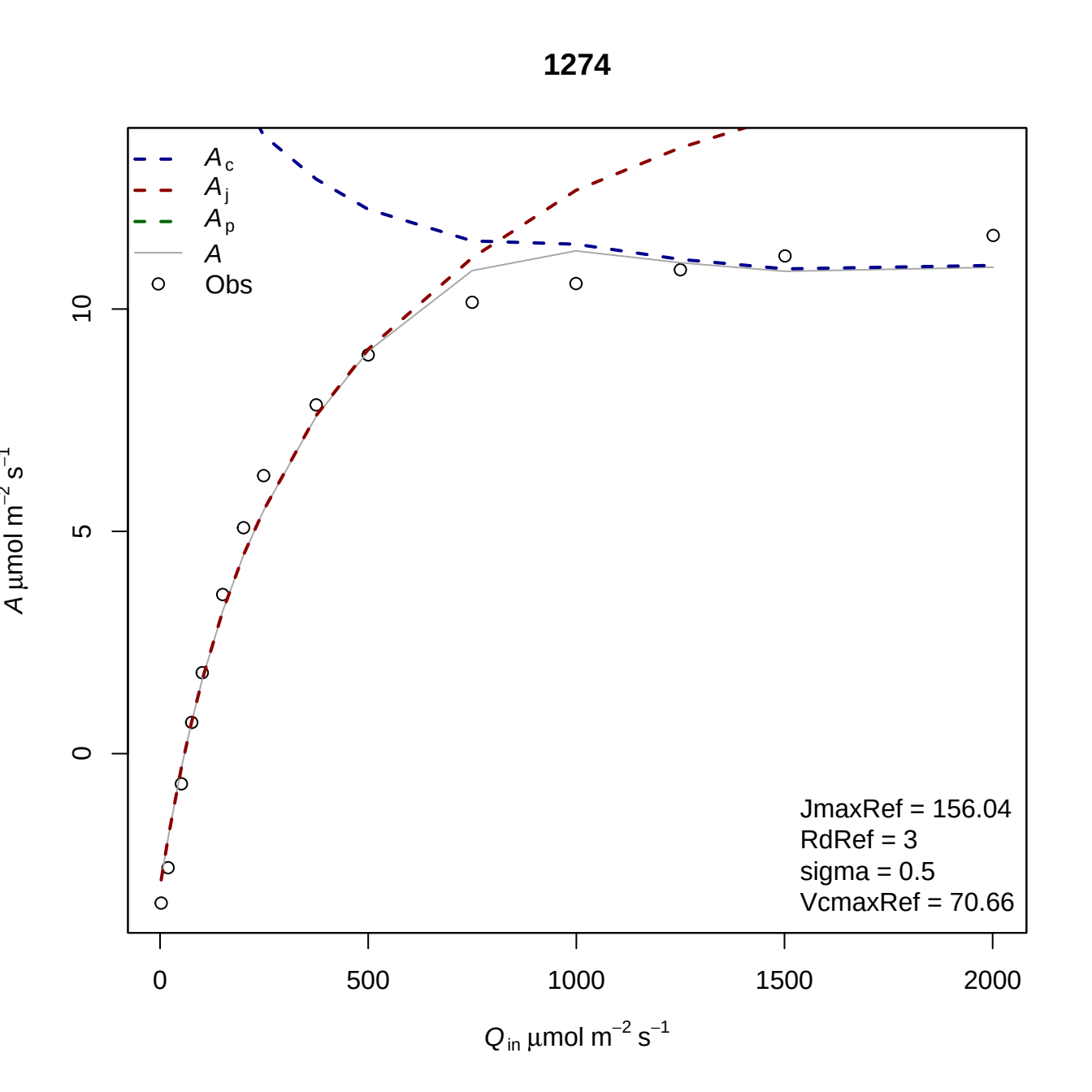

1269
A_c
A_j
A_p
A
Obs
JmaxRef = 45.44
RdRef = 2.48
sigma = 0.55
VcmaxRef = 16786.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

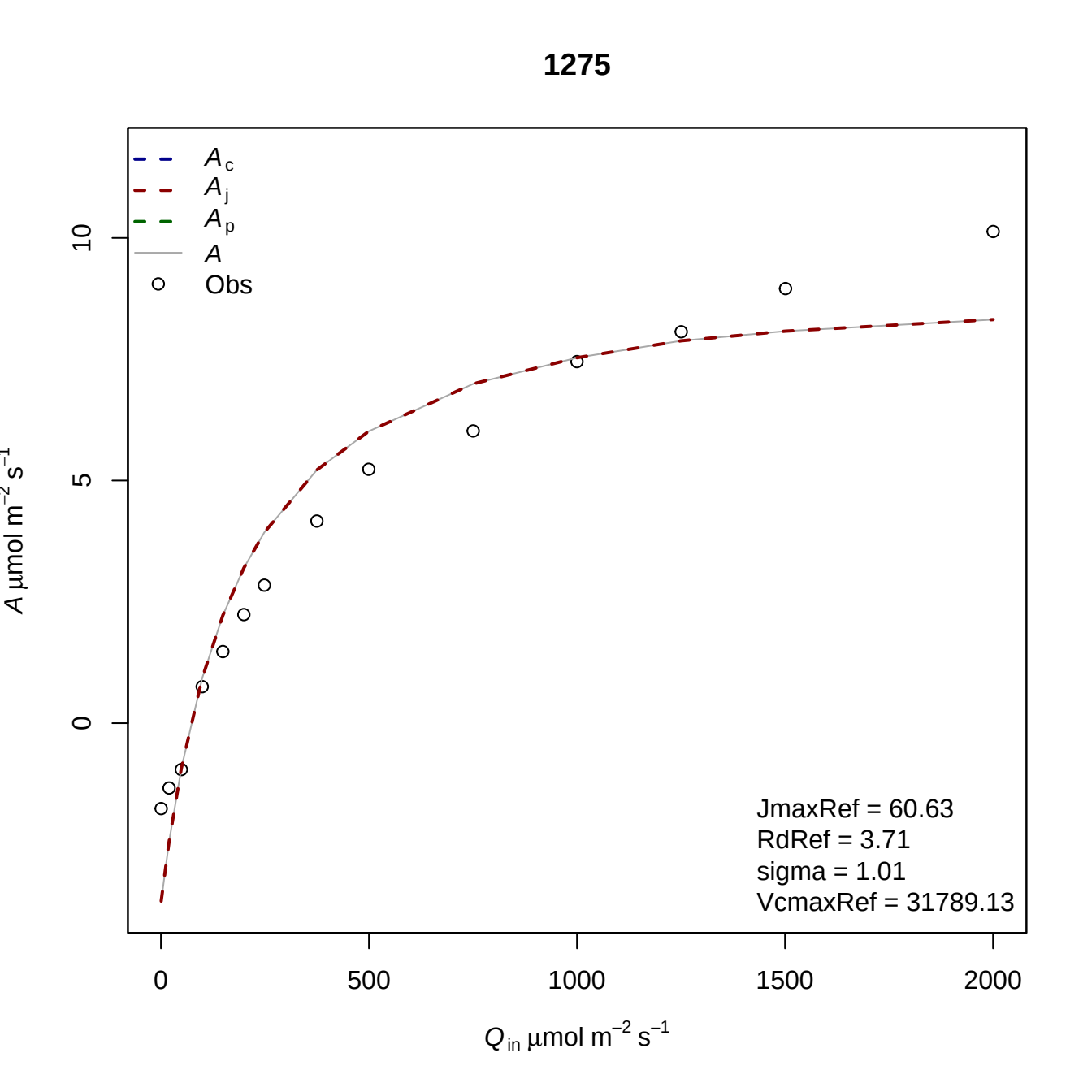

1270
A_c
A_j
A_p
A
Obs
JmaxRef = 56.88
RdRef = 2.54
sigma = 0.58
VcmaxRef = 8099.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

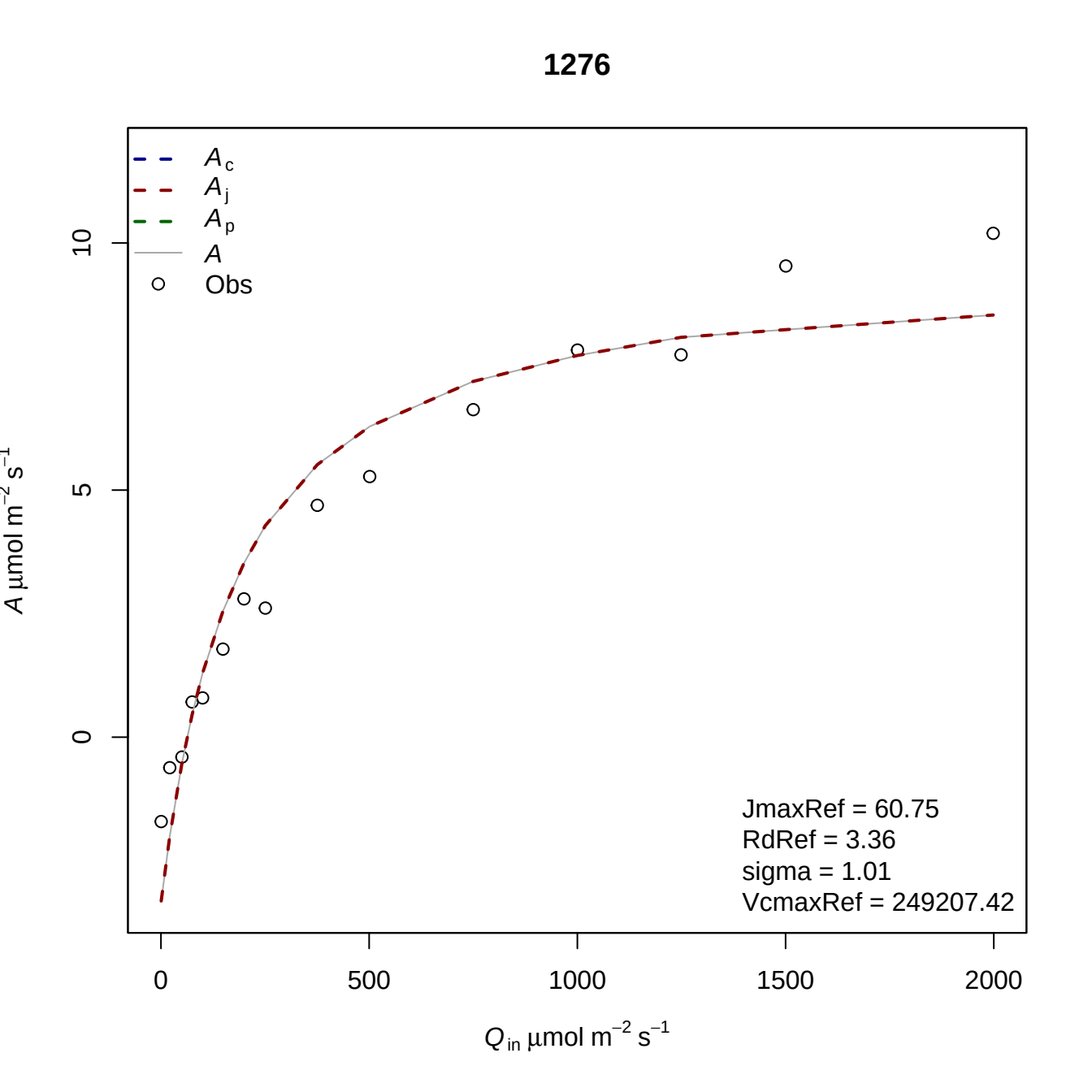

1271
A_c
A_j
A_p
A
Obs
JmaxRef = 103.44
RdRef = 3.67
sigma = 0.83
VcmaxRef = 350198.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

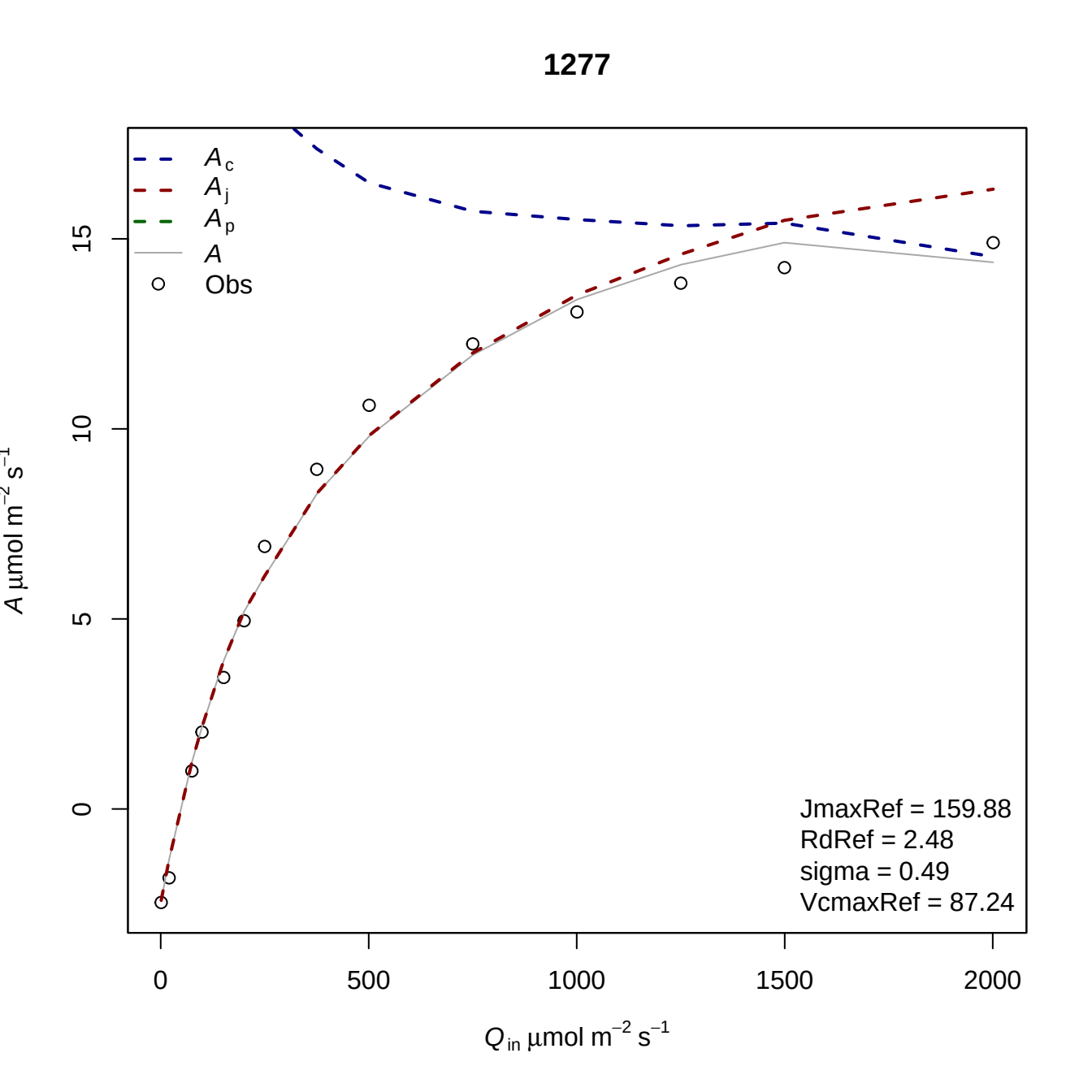

1272
A_c
A_j
A_p
A
Obs
JmaxRef = 115.98
RdRef = 1.85
sigma = 0.43
VcmaxRef = 76.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

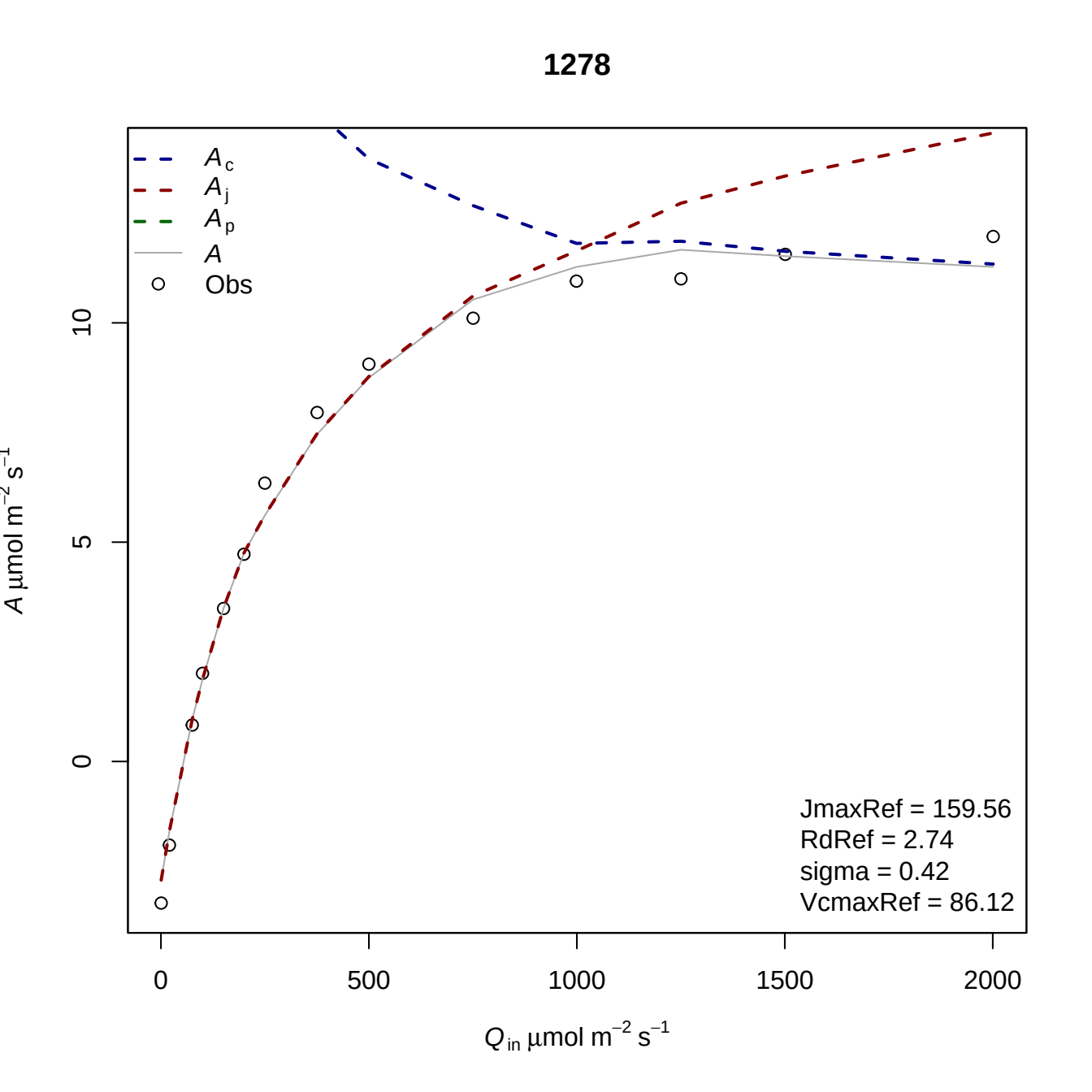

1273
A_c
A_j
A_p
A
Obs
JmaxRef = 146.47
RdRef = 4.08
sigma = 0.72
VcmaxRef = 92.65
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

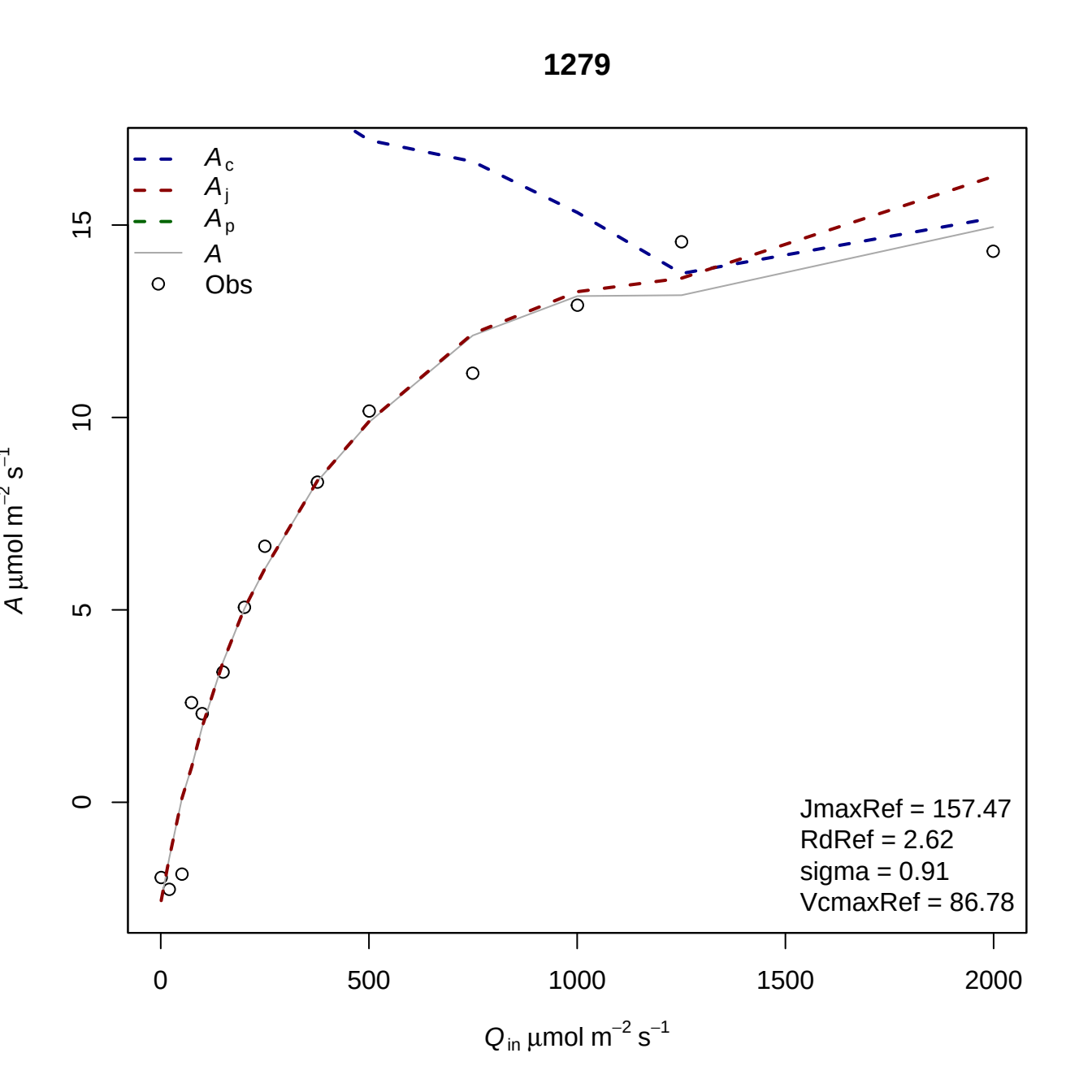

1274
A_c
A_j
A_p
A
Obs
JmaxRef = 156.04
RdRef = 3
sigma = 0.5
VcmaxRef = 70.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

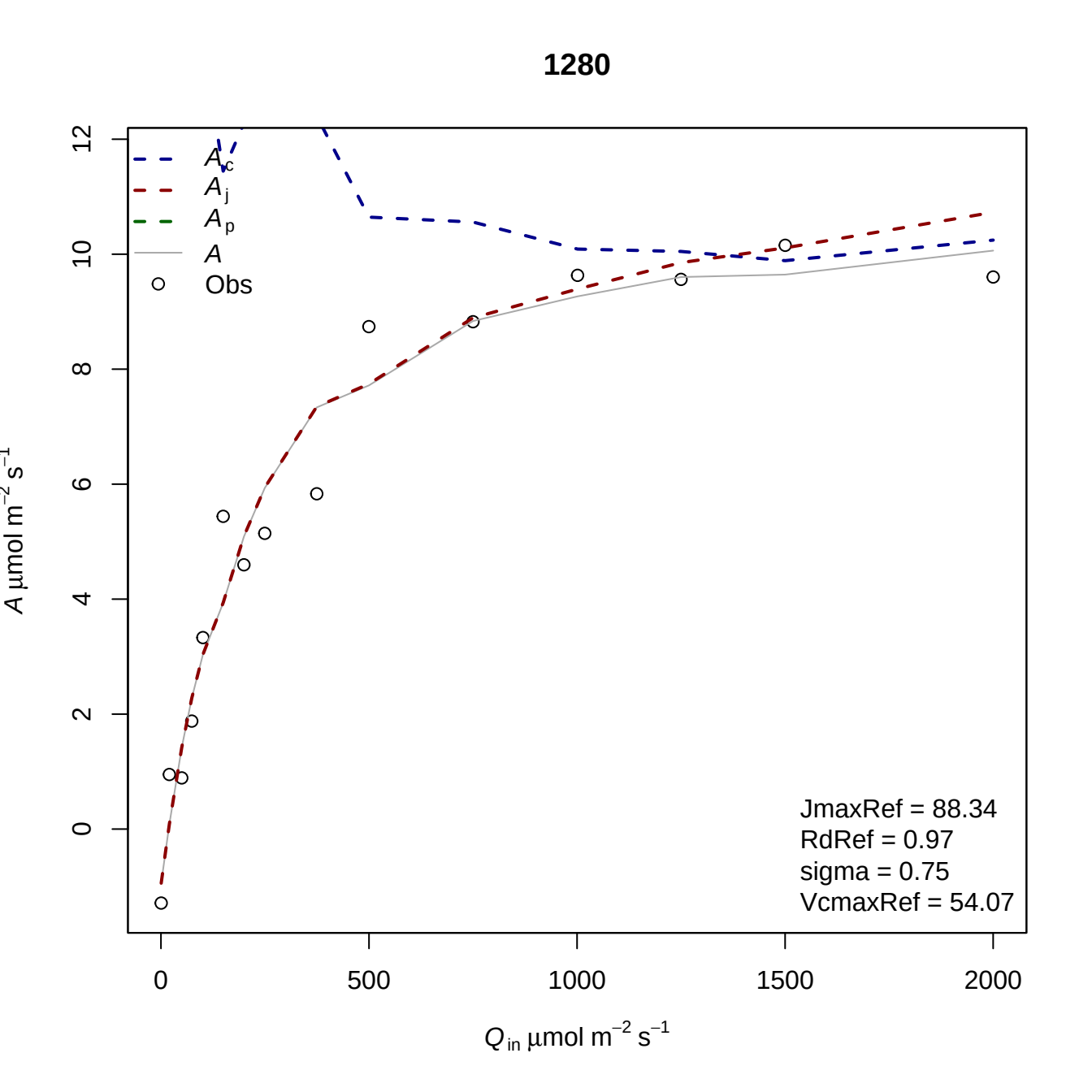

1275
A_c
A_j
A_p
A
Obs
JmaxRef = 60.63
RdRef = 3.71
sigma = 1.01
VcmaxRef = 31789.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

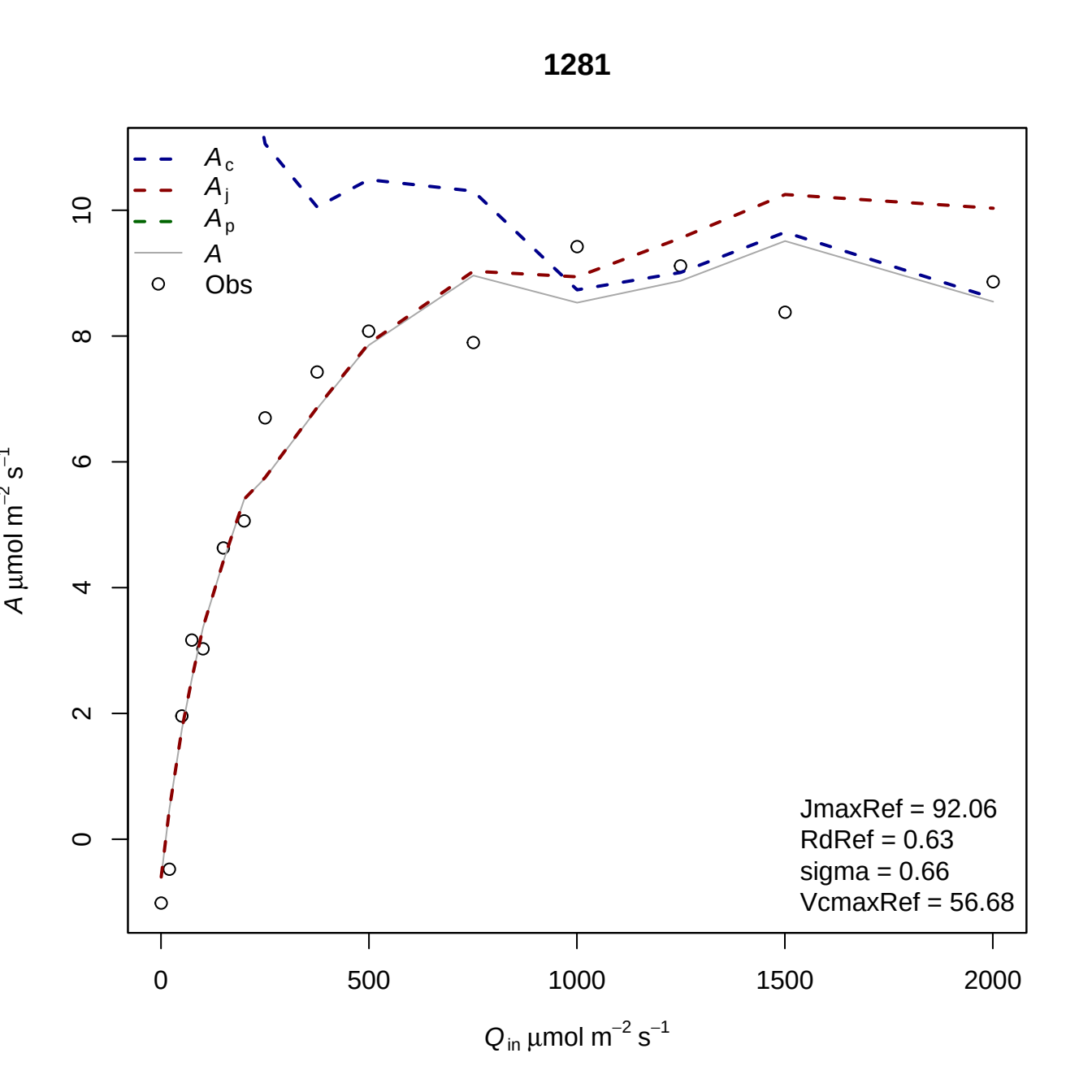

1276
A_c
A_j
A_p
A
Obs
JmaxRef = 60.75
RdRef = 3.36
sigma = 1.01
VcmaxRef = 249207.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

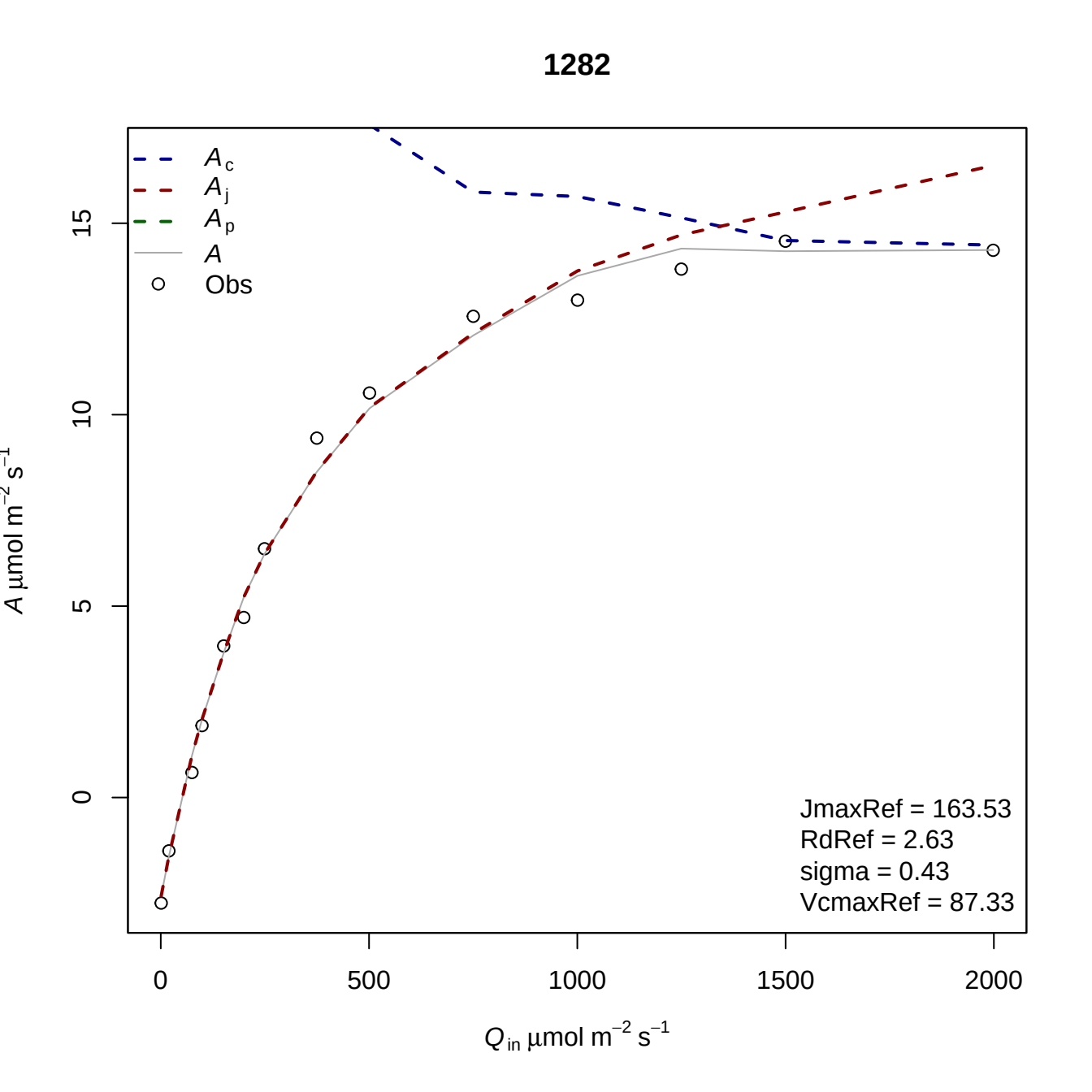

1277
A_c
A_j
A_p
A
Obs
JmaxRef = 159.88
RdRef = 2.48
sigma = 0.49
VcmaxRef = 87.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

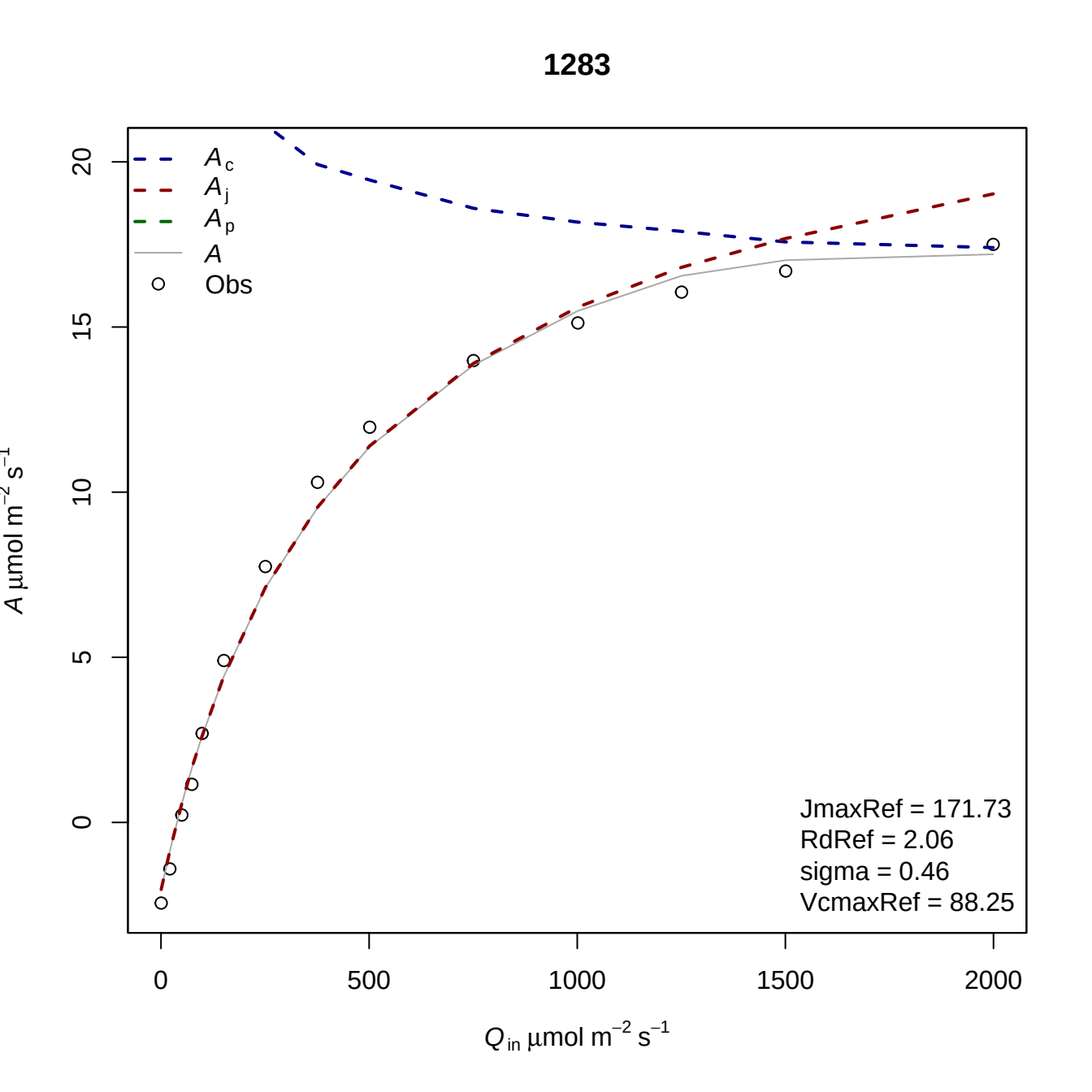

1278
A_c
A_j
A_p
A
Obs
JmaxRef = 159.56
RdRef = 2.74
sigma = 0.42
VcmaxRef = 86.12
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

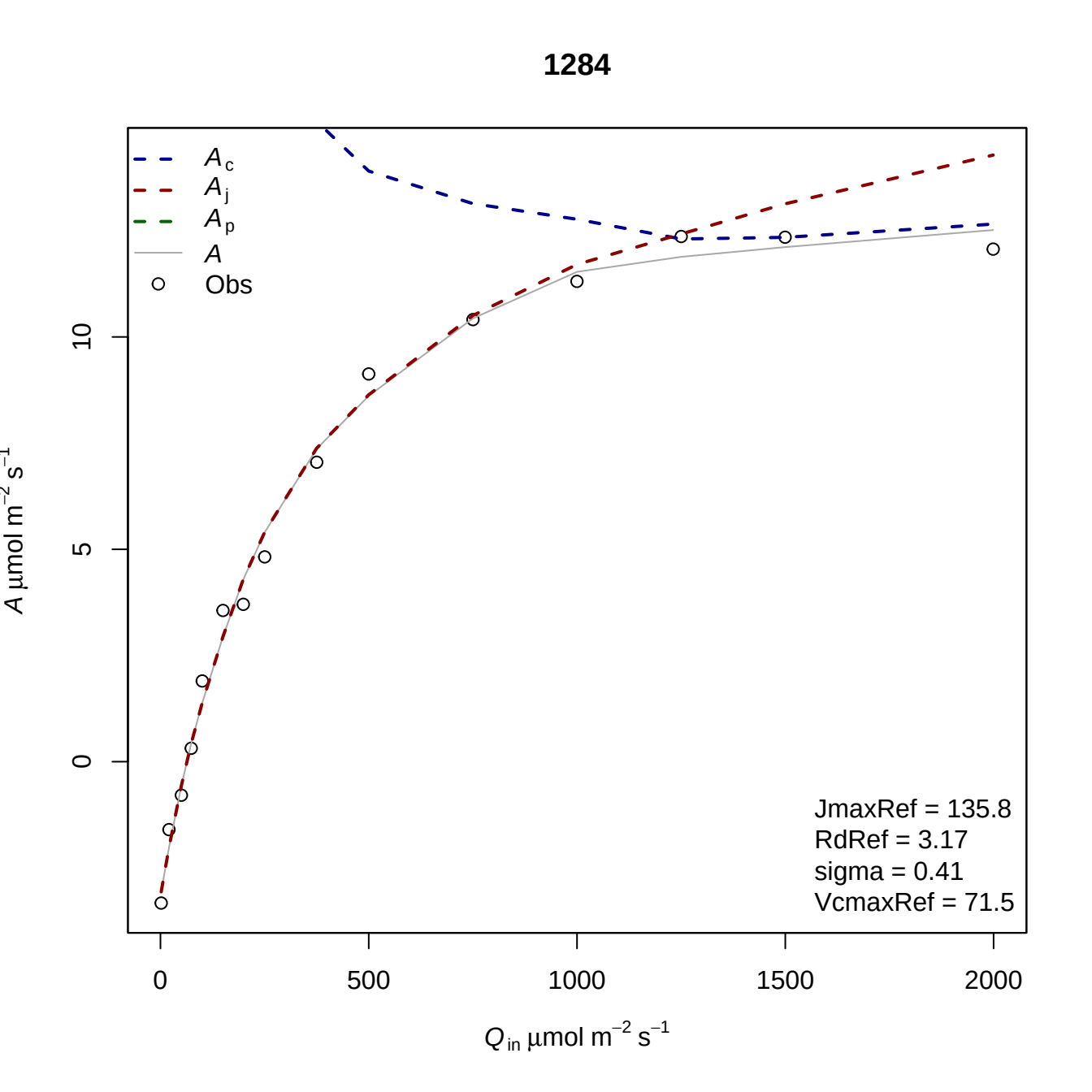

1279
A_c
A_j
A_p
A
Obs
JmaxRef = 157.47
RdRef = 2.62
sigma = 0.91
VcmaxRef = 86.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

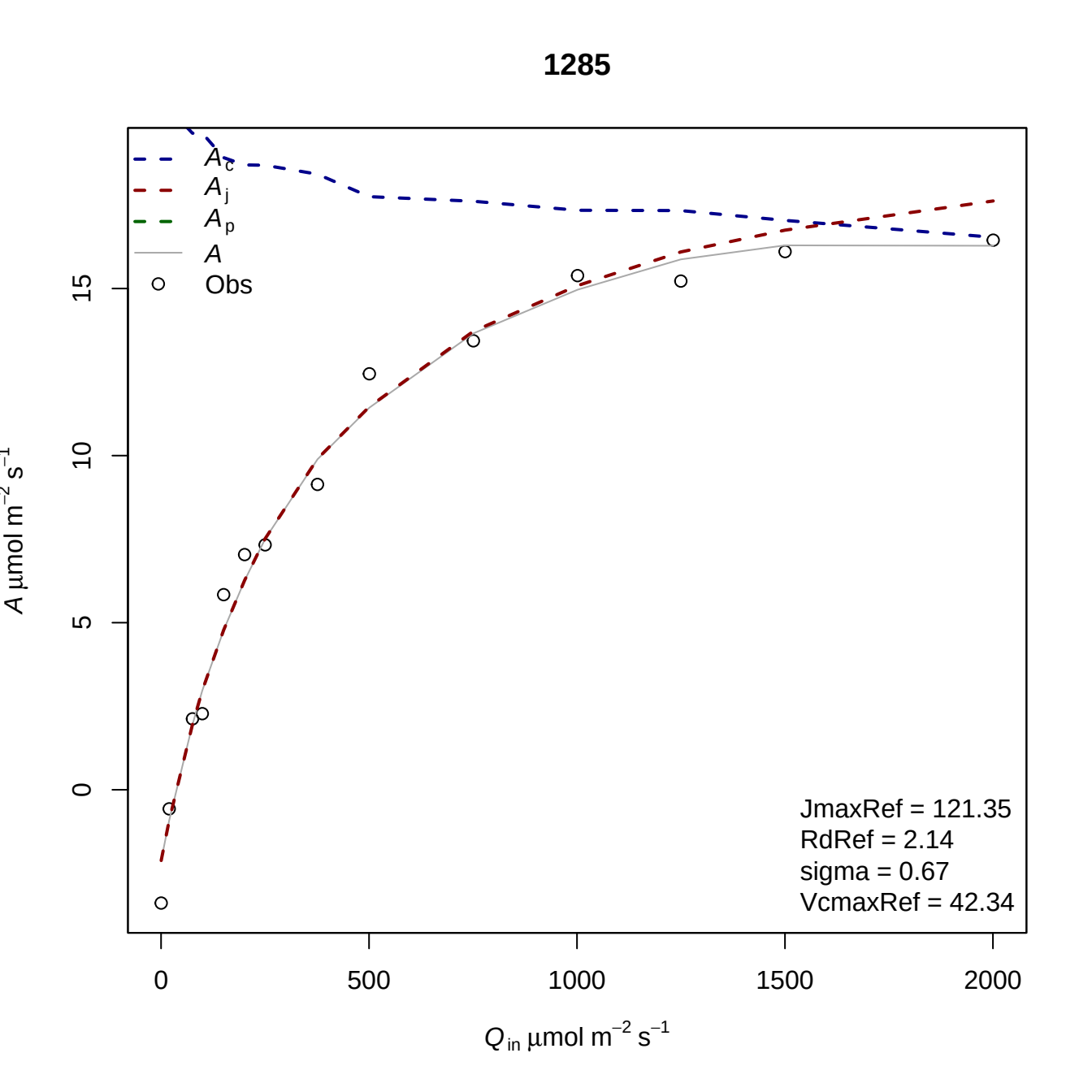

1280
A_c
A_j
A_p
A
Obs
JmaxRef = 88.34
RdRef = 0.97
sigma = 0.75
VcmaxRef = 54.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

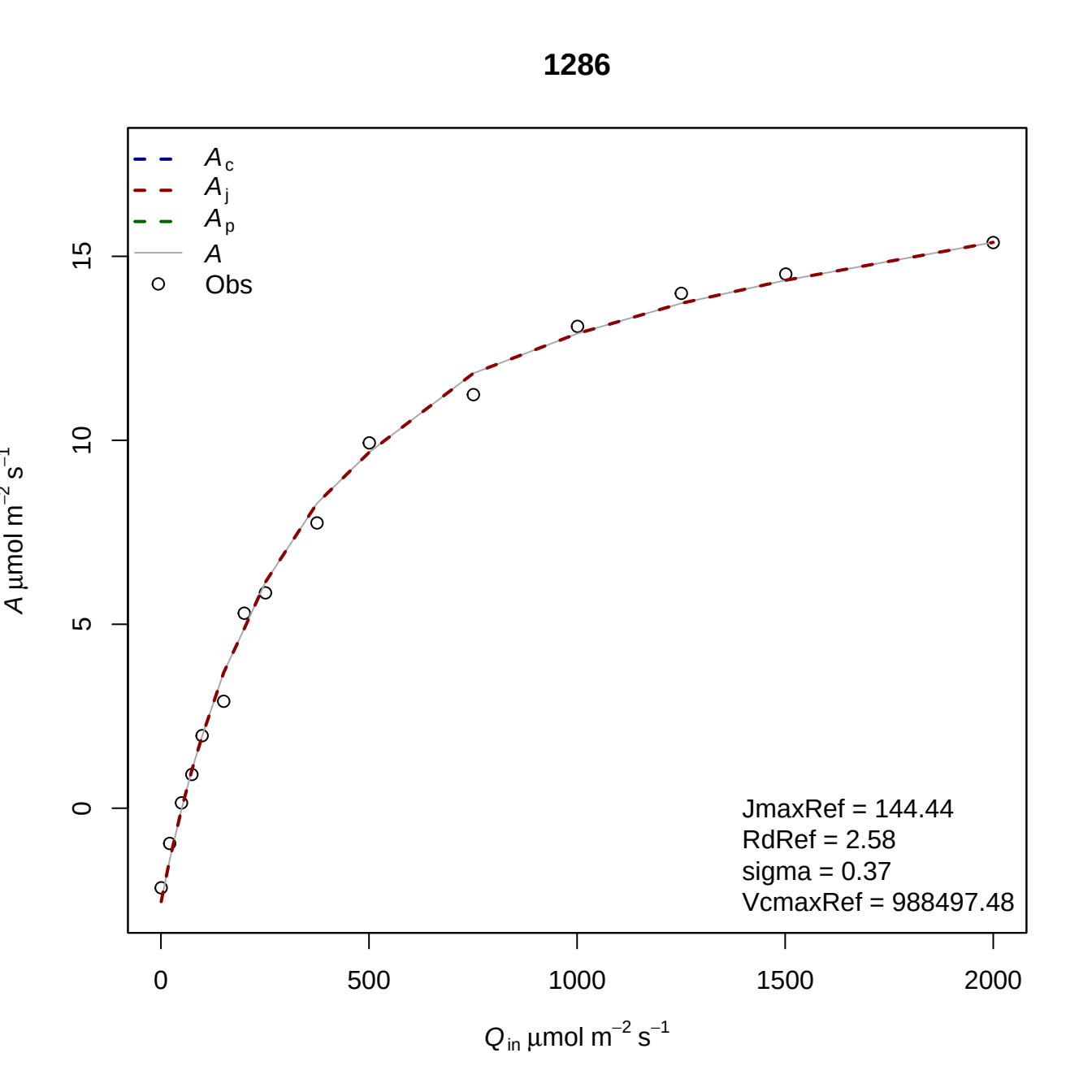

1281
A_c
A_j
A_p
A
Obs
JmaxRef = 92.06
RdRef = 0.63
sigma = 0.66
VcmaxRef = 56.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

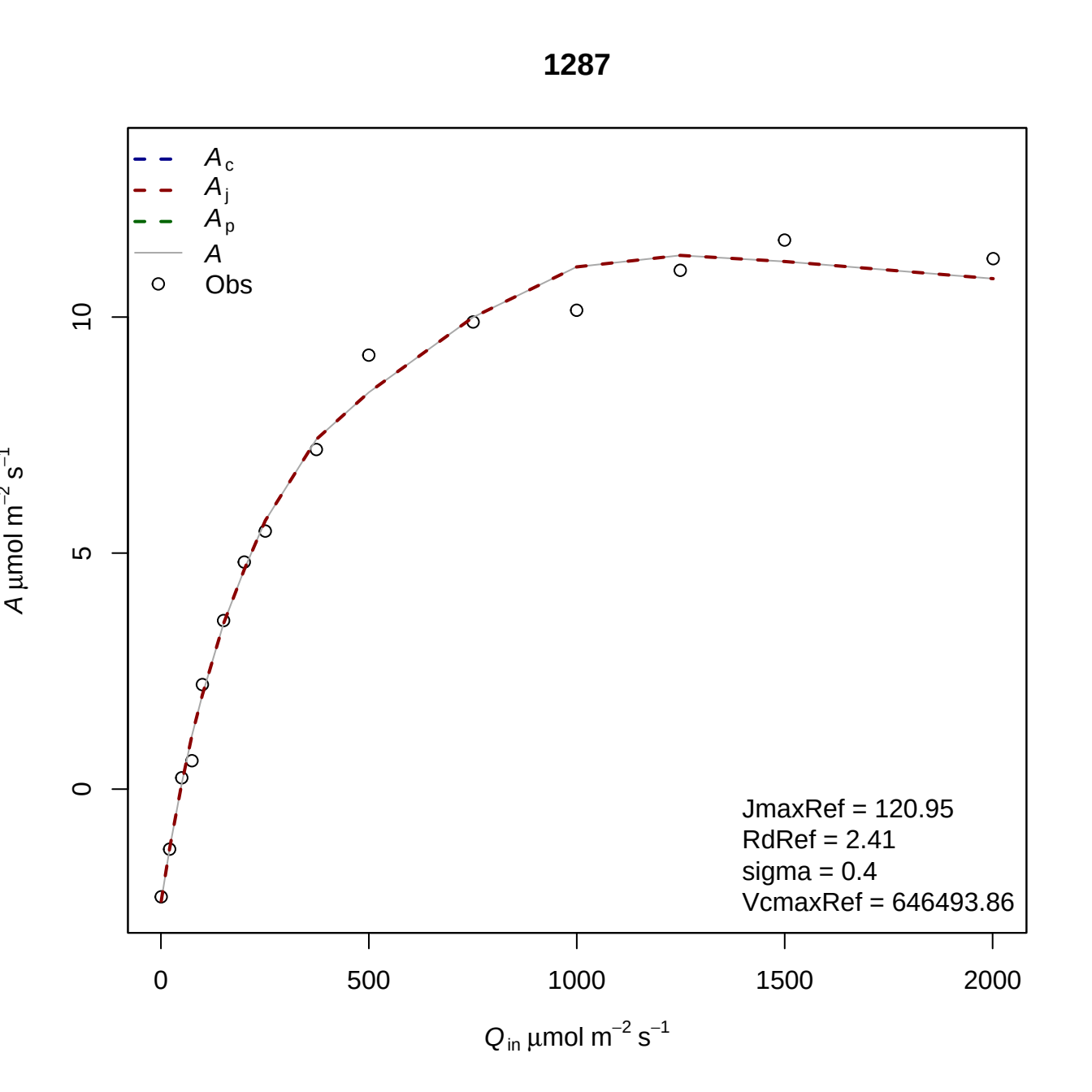

1282
A_c
A_j
A_p
A
Obs
JmaxRef = 163.53
RdRef = 2.63
sigma = 0.43
VcmaxRef = 87.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

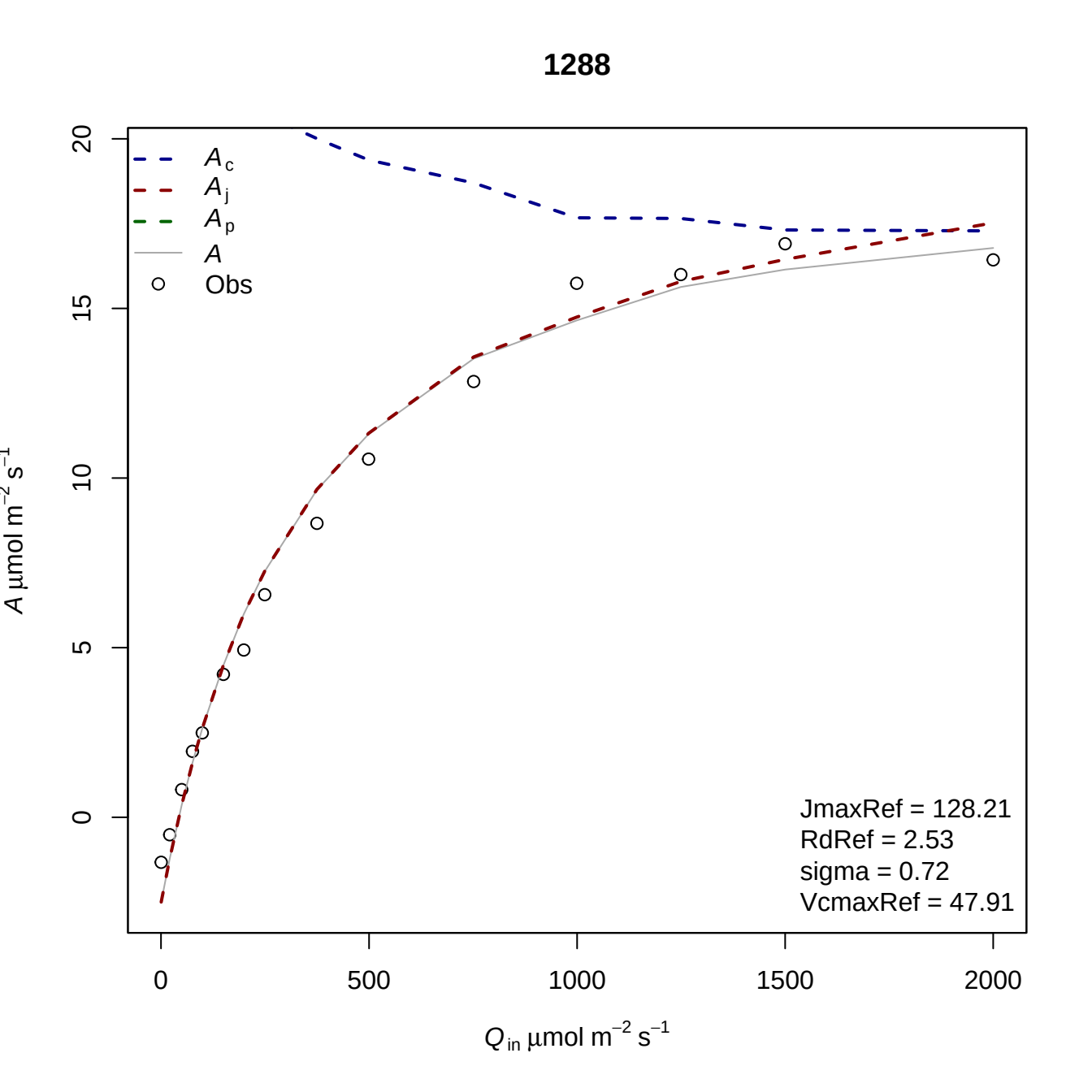

1283
A_c
A_j
A_p
A
Obs
JmaxRef = 171.73
RdRef = 2.06
sigma = 0.46
VcmaxRef = 88.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

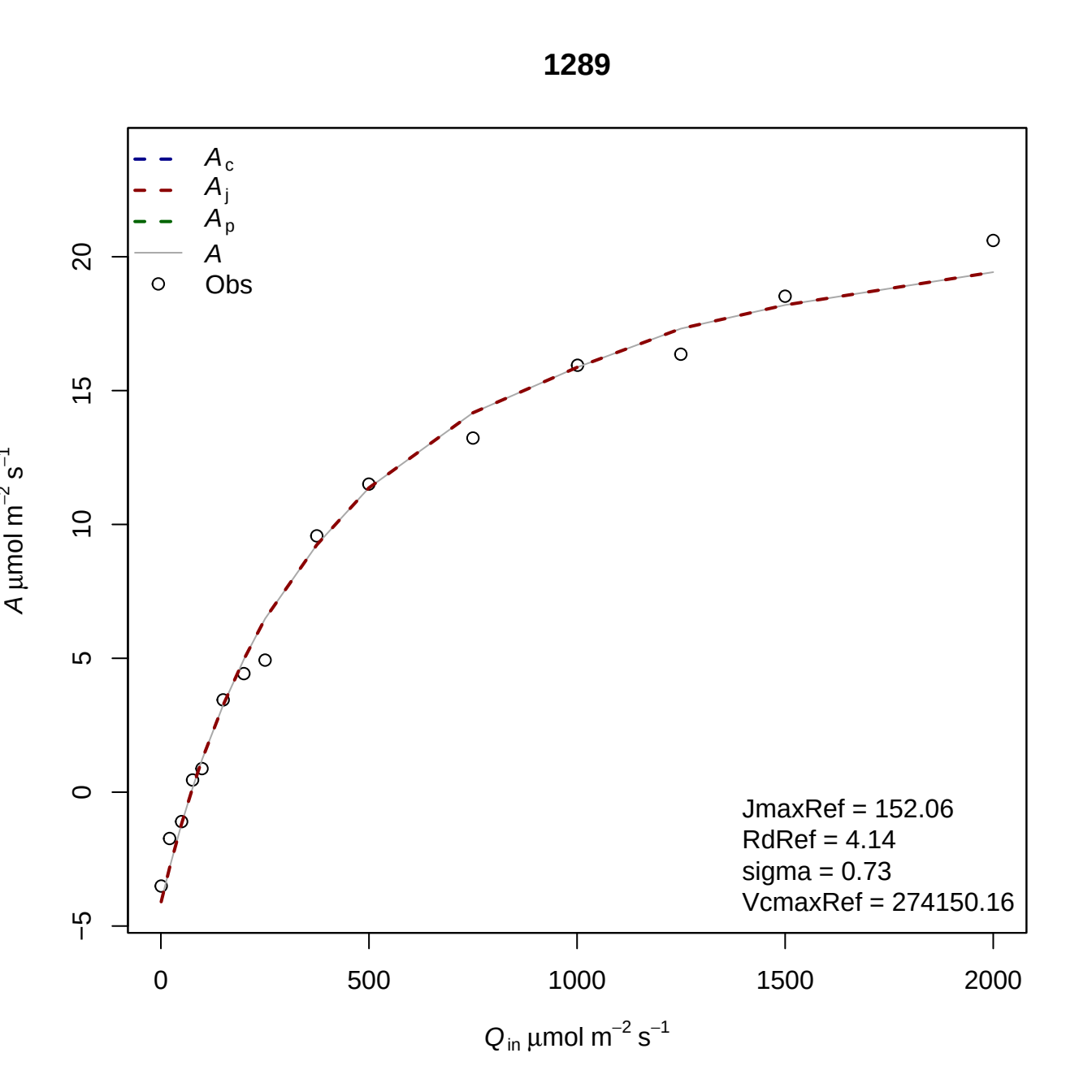

1284
A_c
A_j
A_p
A
Obs
JmaxRef = 135.8
RdRef = 3.17
sigma = 0.41
VcmaxRef = 71.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

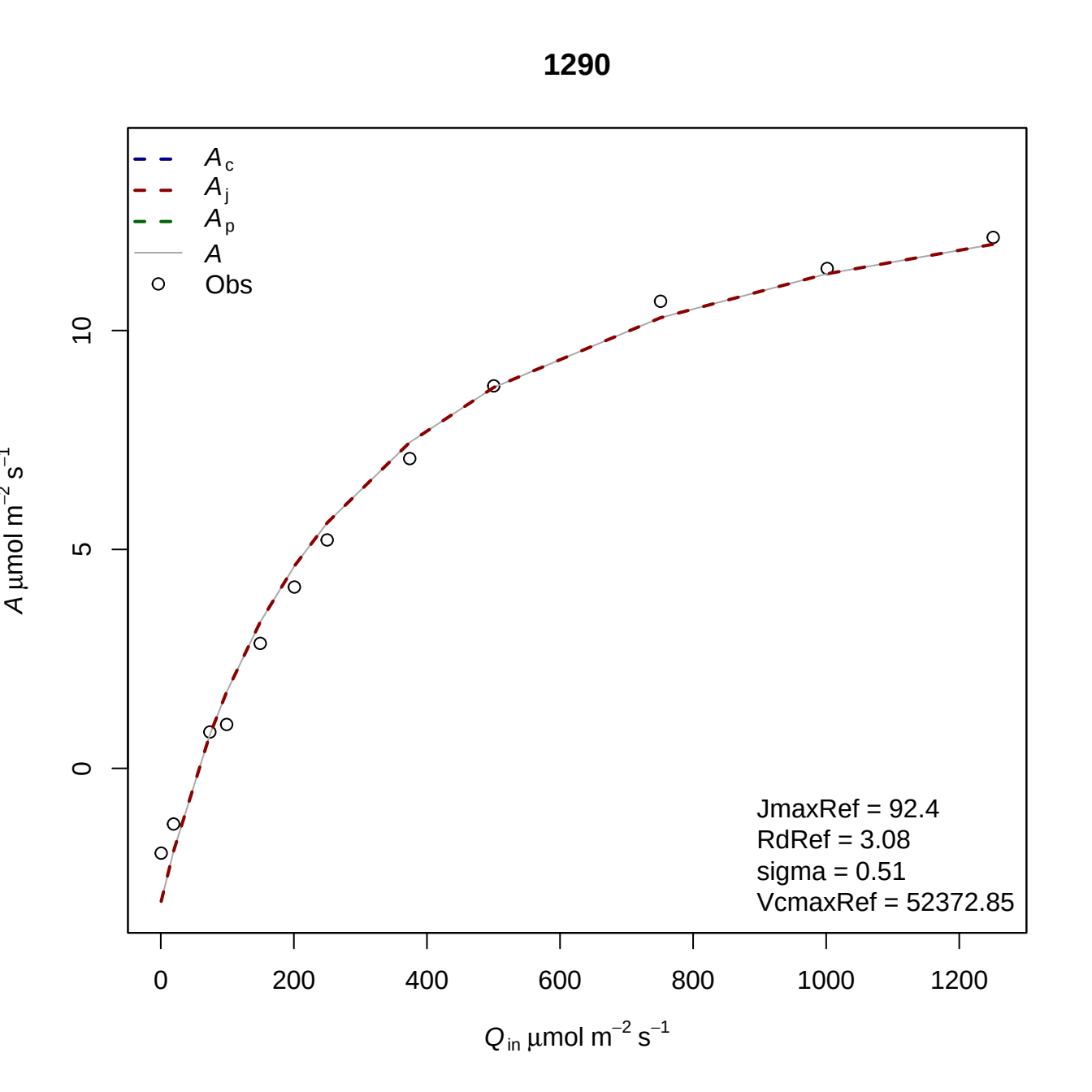

1285
A_c
A_j
A_p
A
Obs
JmaxRef = 121.35
RdRef = 2.14
sigma = 0.67
VcmaxRef = 42.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

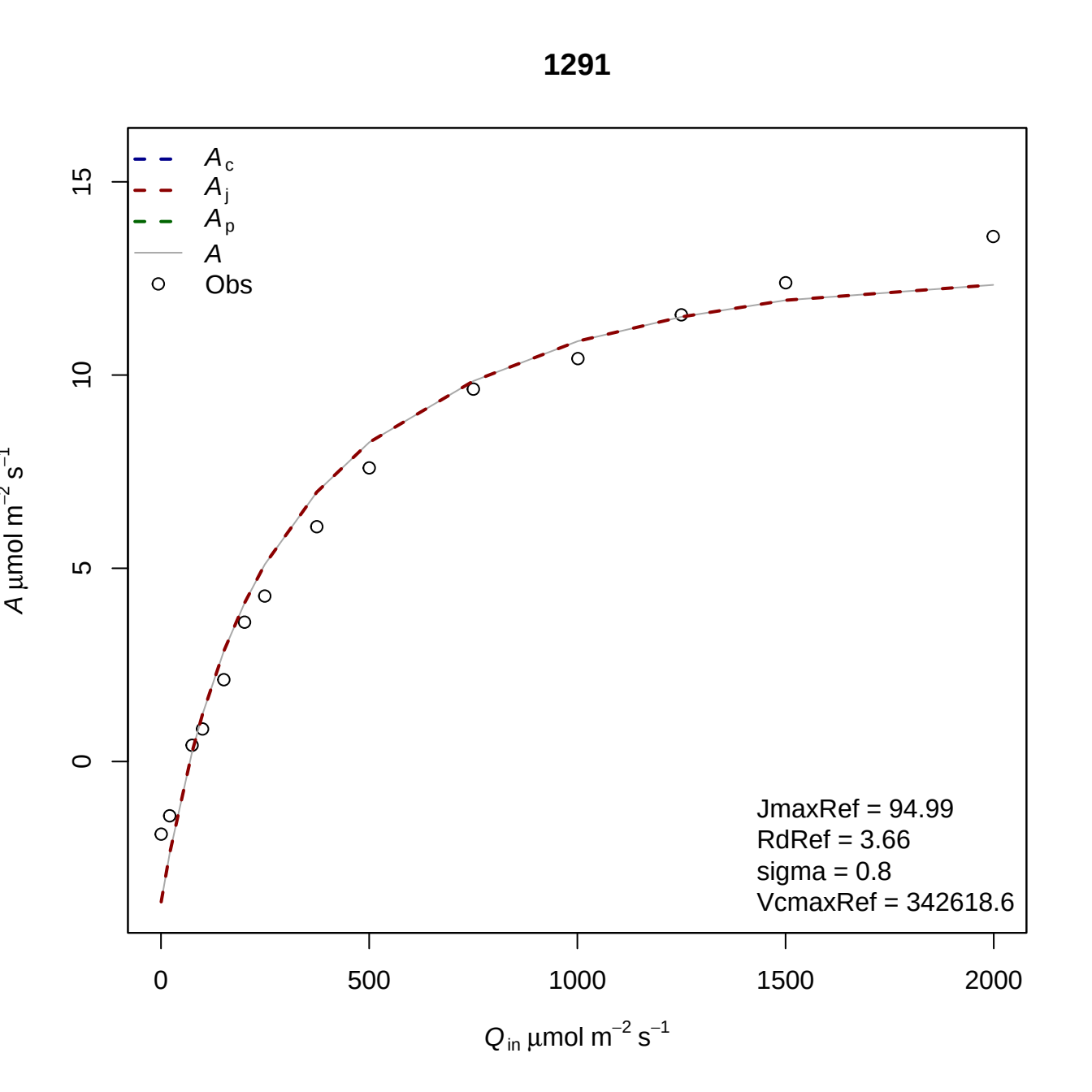

1286
A_c
A_j
A_p
A
Obs
JmaxRef = 144.44
RdRef = 2.58
sigma = 0.37
VcmaxRef = 988497.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

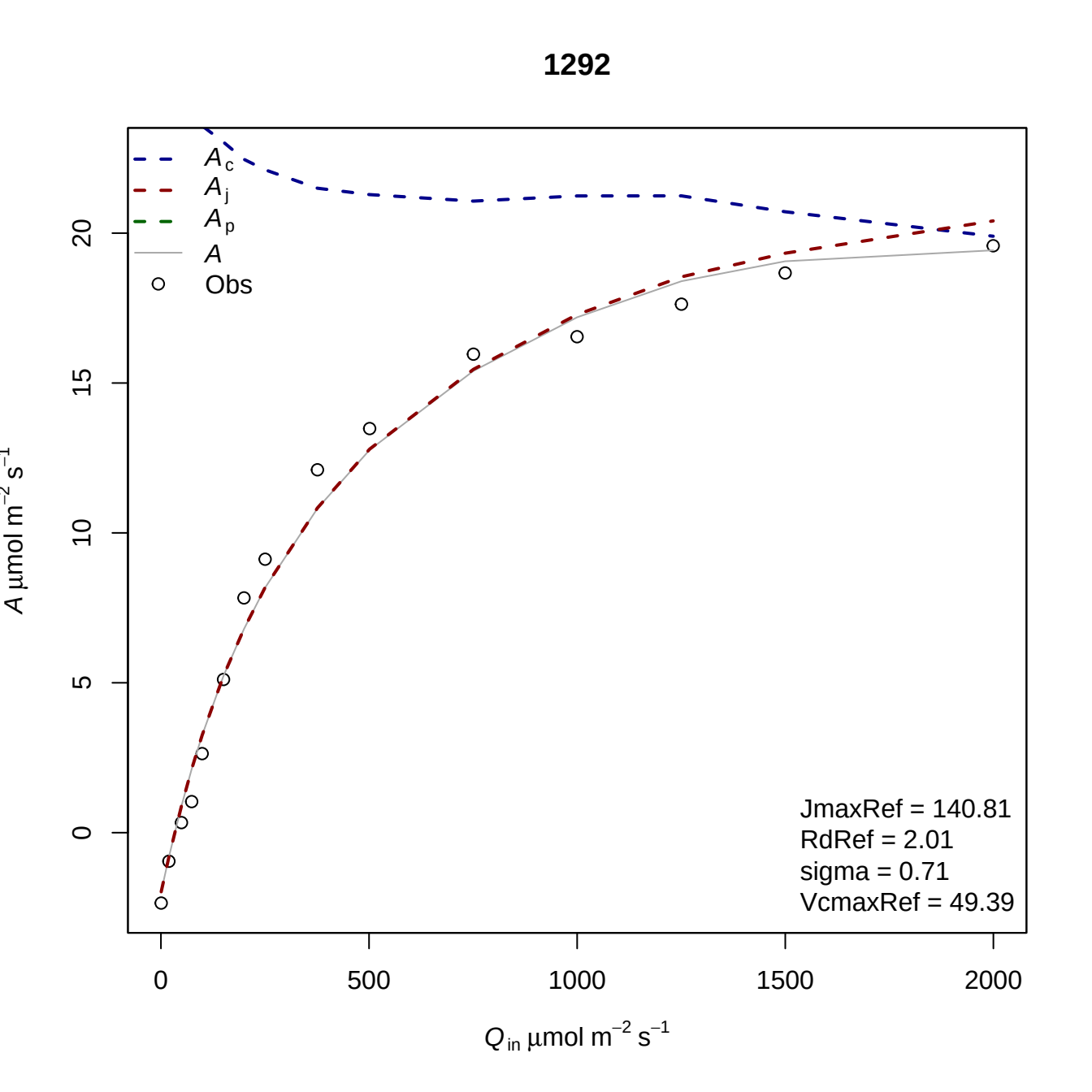

1287
A_c
A_j
A_p
A
Obs
JmaxRef = 120.95
RdRef = 2.41
sigma = 0.4
VcmaxRef = 646493.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

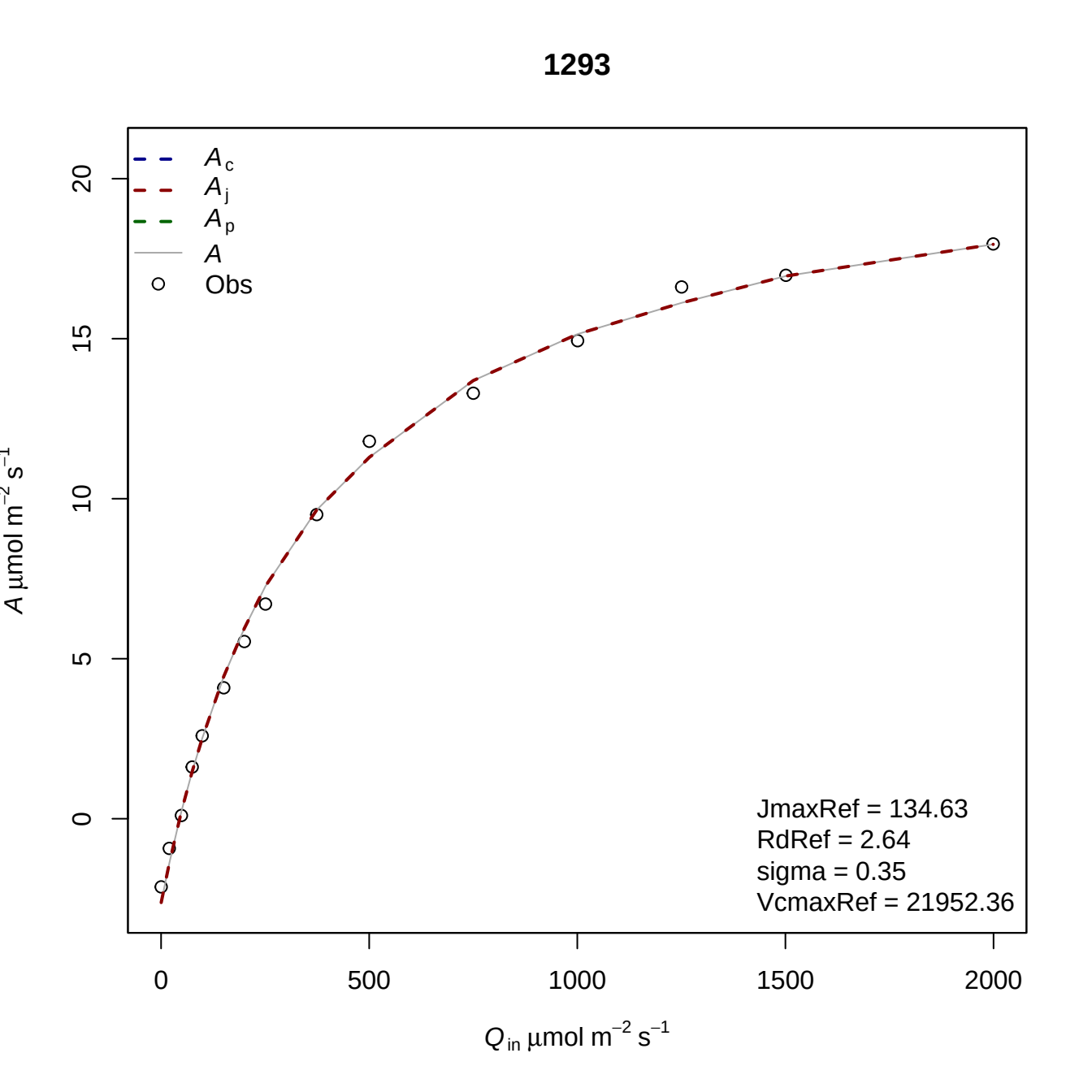

1288
A_c
A_j
A_p
A
Obs
JmaxRef = 128.21
RdRef = 2.53
sigma = 0.72
VcmaxRef = 47.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

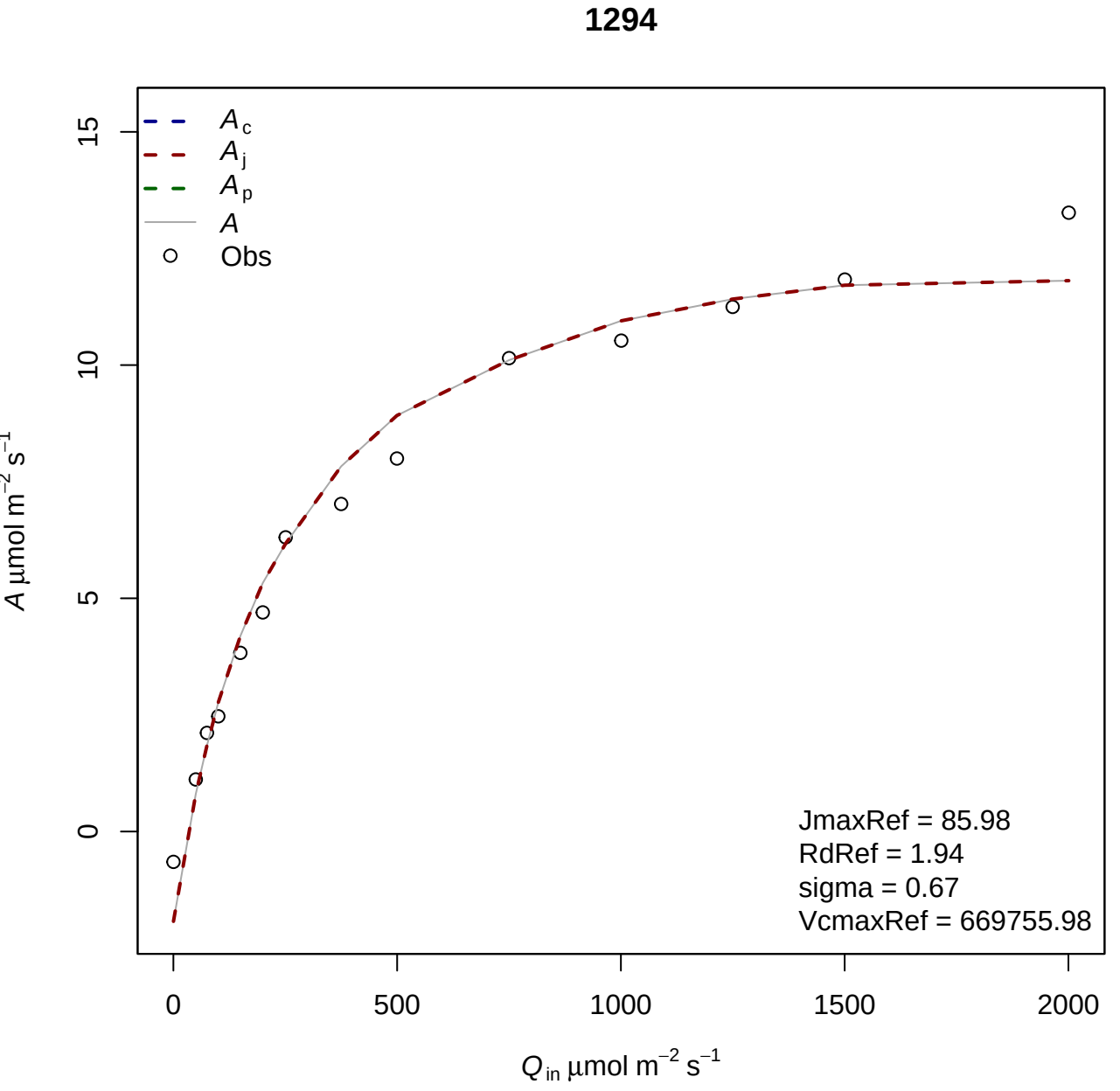

1289
A_c
A_j
A_p
A
Obs
JmaxRef = 152.06
RdRef = 4.14
sigma = 0.73
VcmaxRef = 274150.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

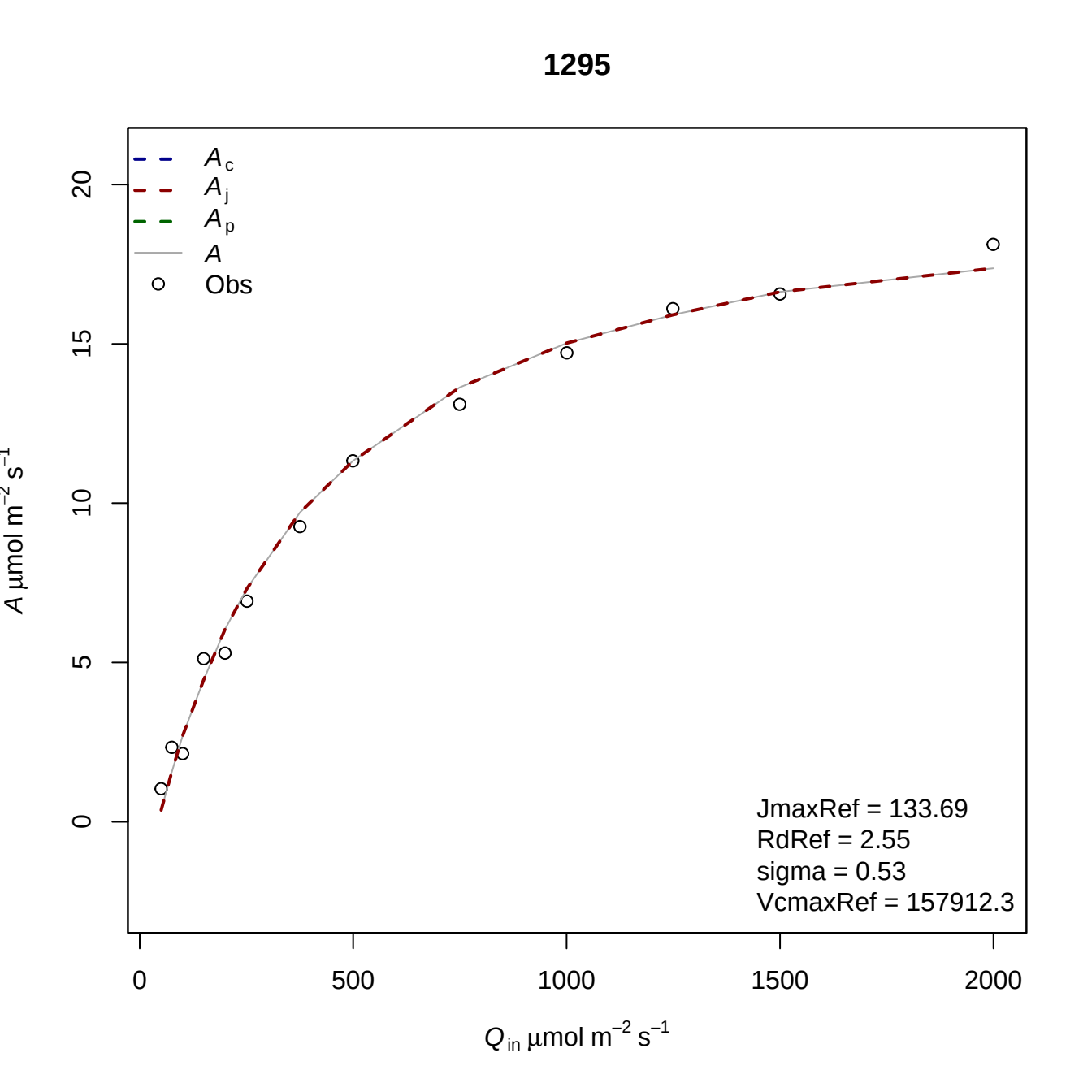

1290
A_c
A_j
A_p
A
Obs
JmaxRef = 92.4
RdRef = 3.08
sigma = 0.51
VcmaxRef = 52372.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000
1200

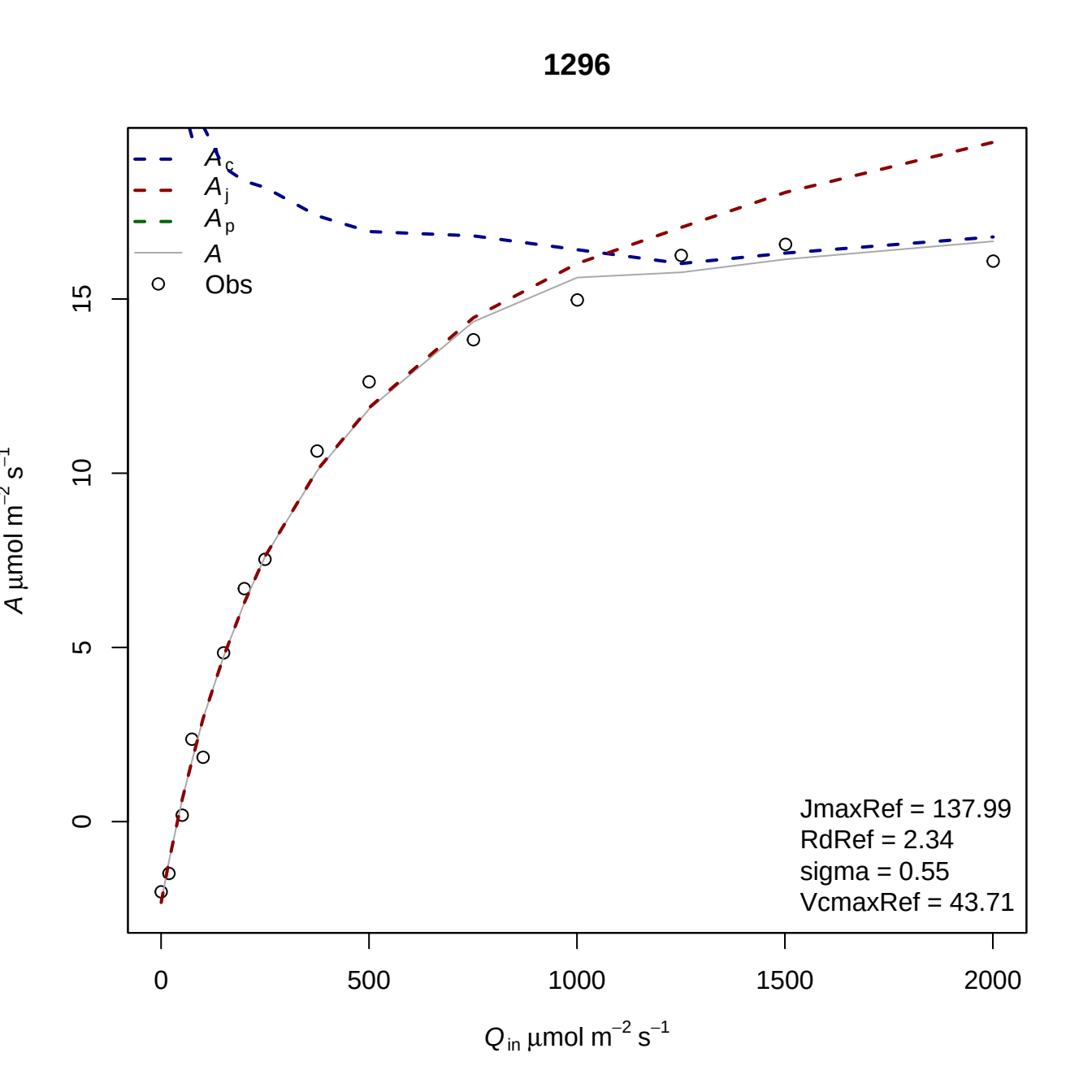

1291
A_c
A_j
A_p
A
Obs
JmaxRef = 94.99
RdRef = 3.66
sigma = 0.8
VcmaxRef = 342618.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

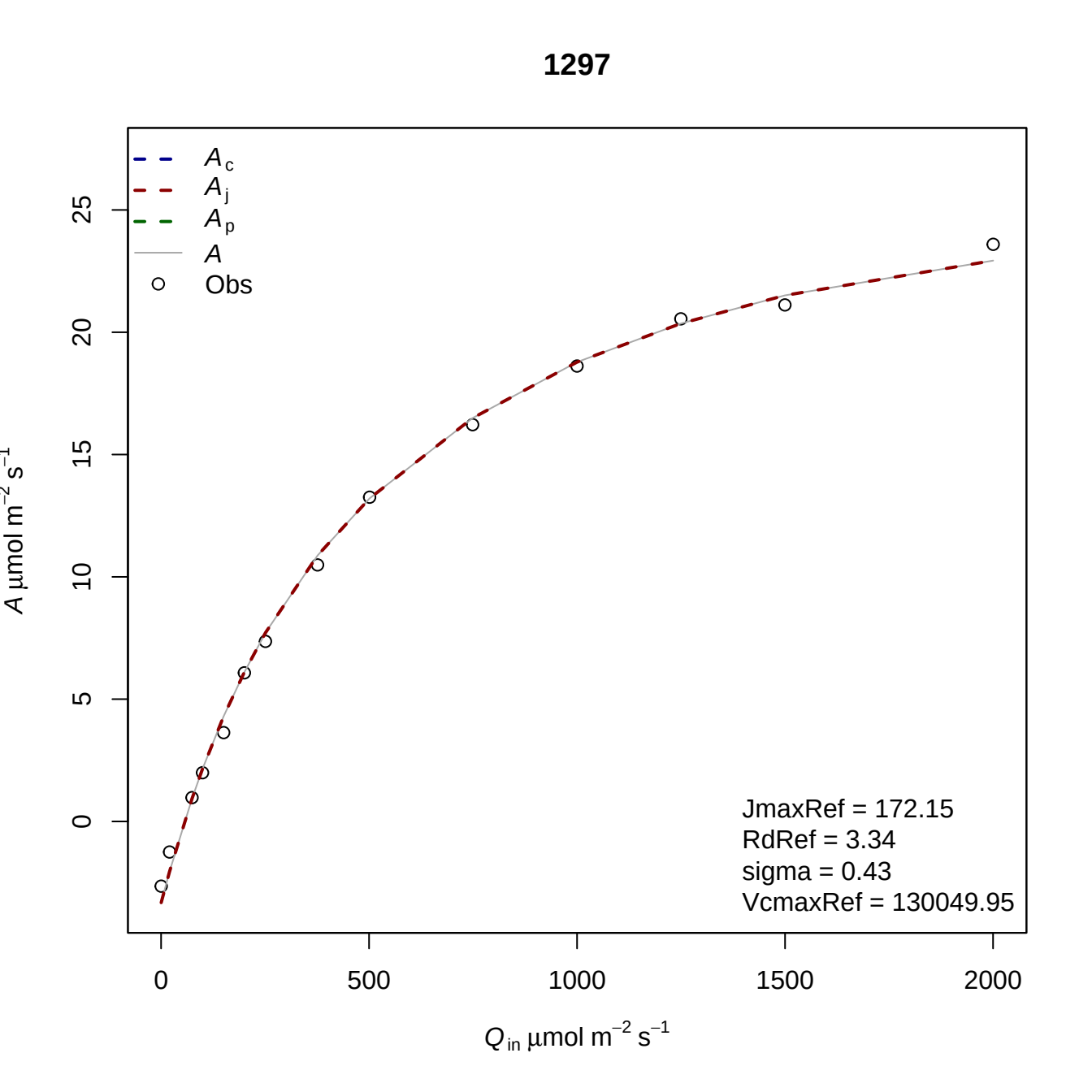

1292
A_c
A_j
A_p
A
Obs
JmaxRef = 140.81
RdRef = 2.01
sigma = 0.71
VcmaxRef = 49.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

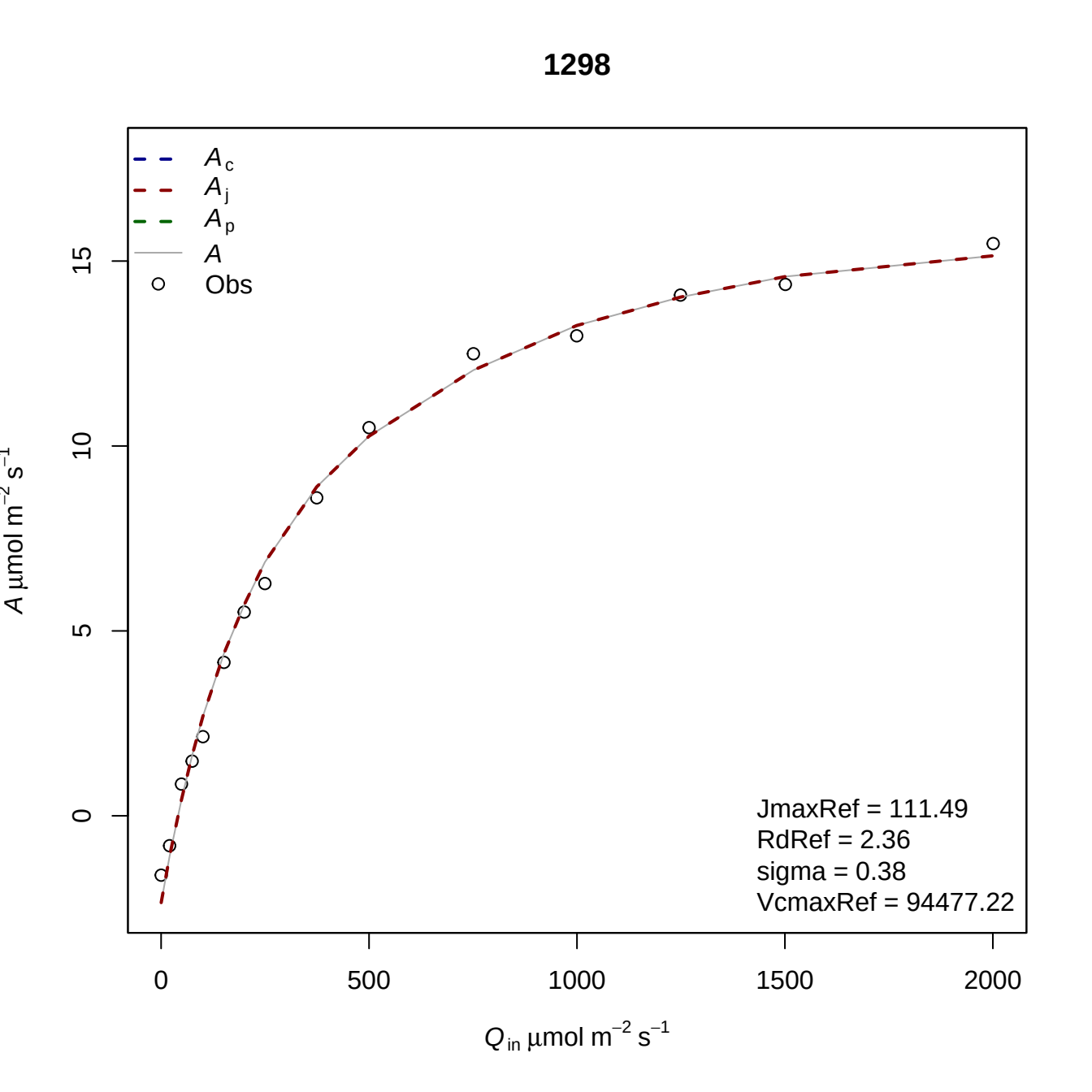

1293
A_c
A_j
A_p
A
Obs
JmaxRef = 134.63
RdRef = 2.64
sigma = 0.35
VcmaxRef = 21952.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

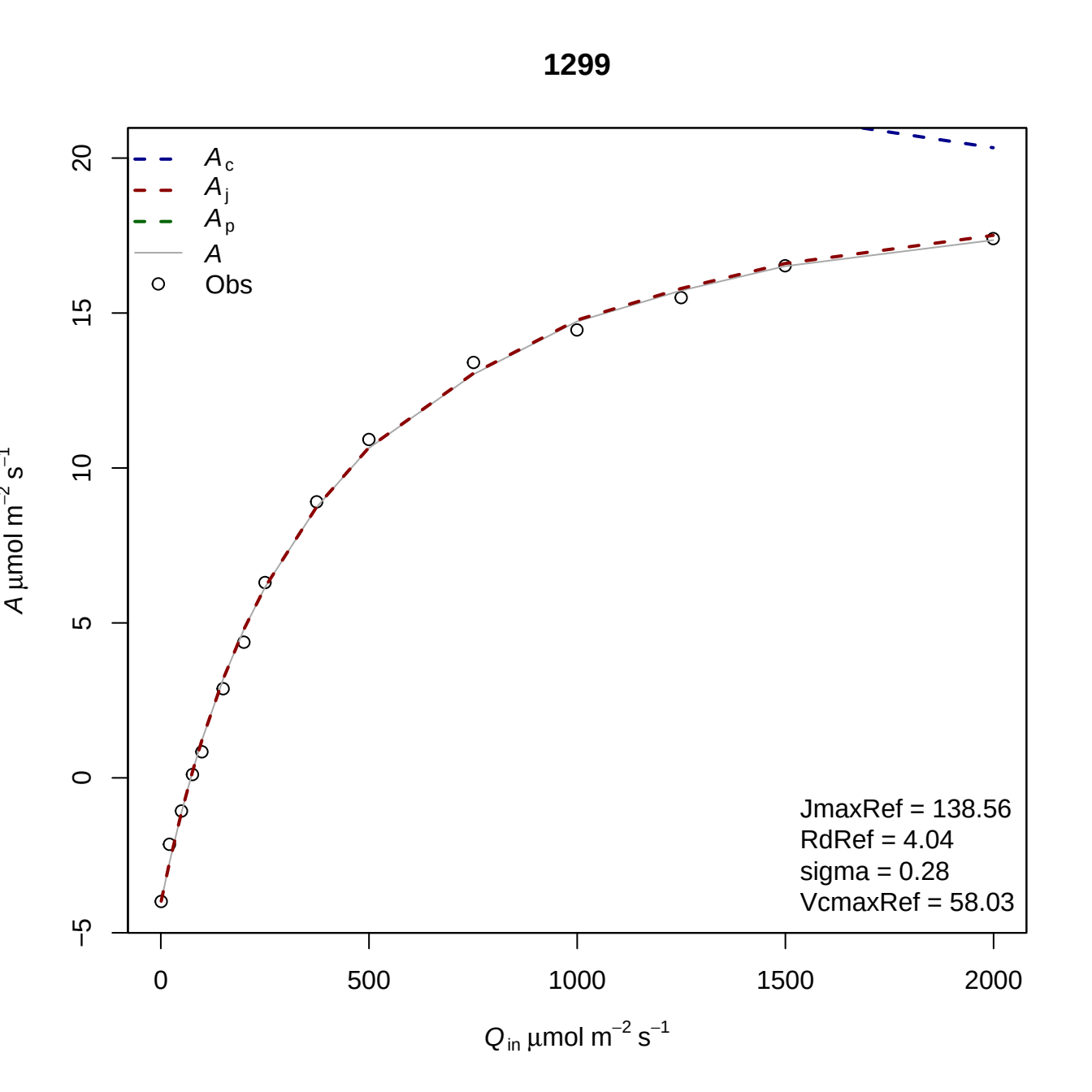

1294
A_c
A_j
A_p
A
Obs
JmaxRef = 85.98
RdRef = 1.94
sigma = 0.67
VcmaxRef = 669755.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

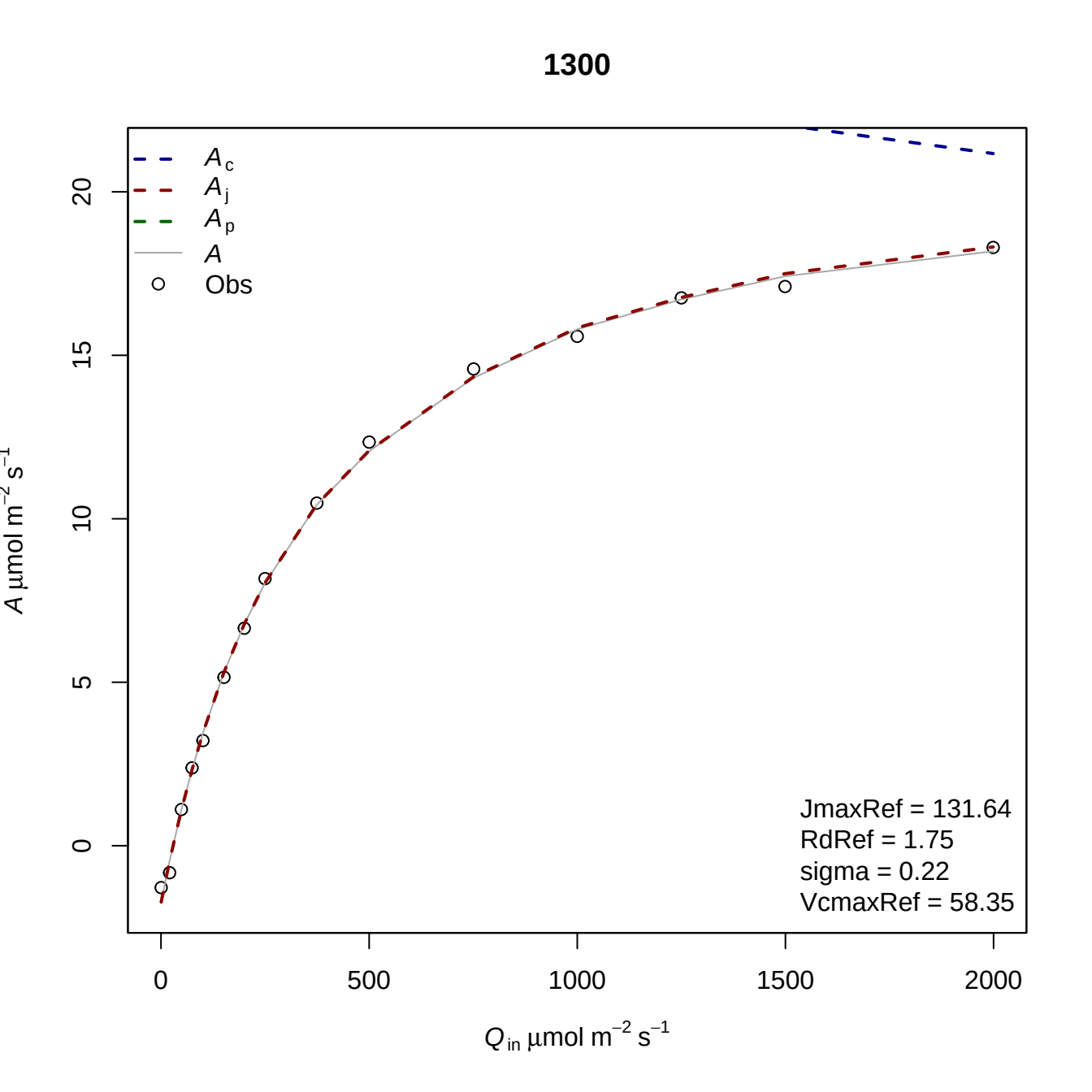

1295
A_c
A_j
A_p
A
Obs
JmaxRef = 133.69
RdRef = 2.55
sigma = 0.53
VcmaxRef = 157912.3
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

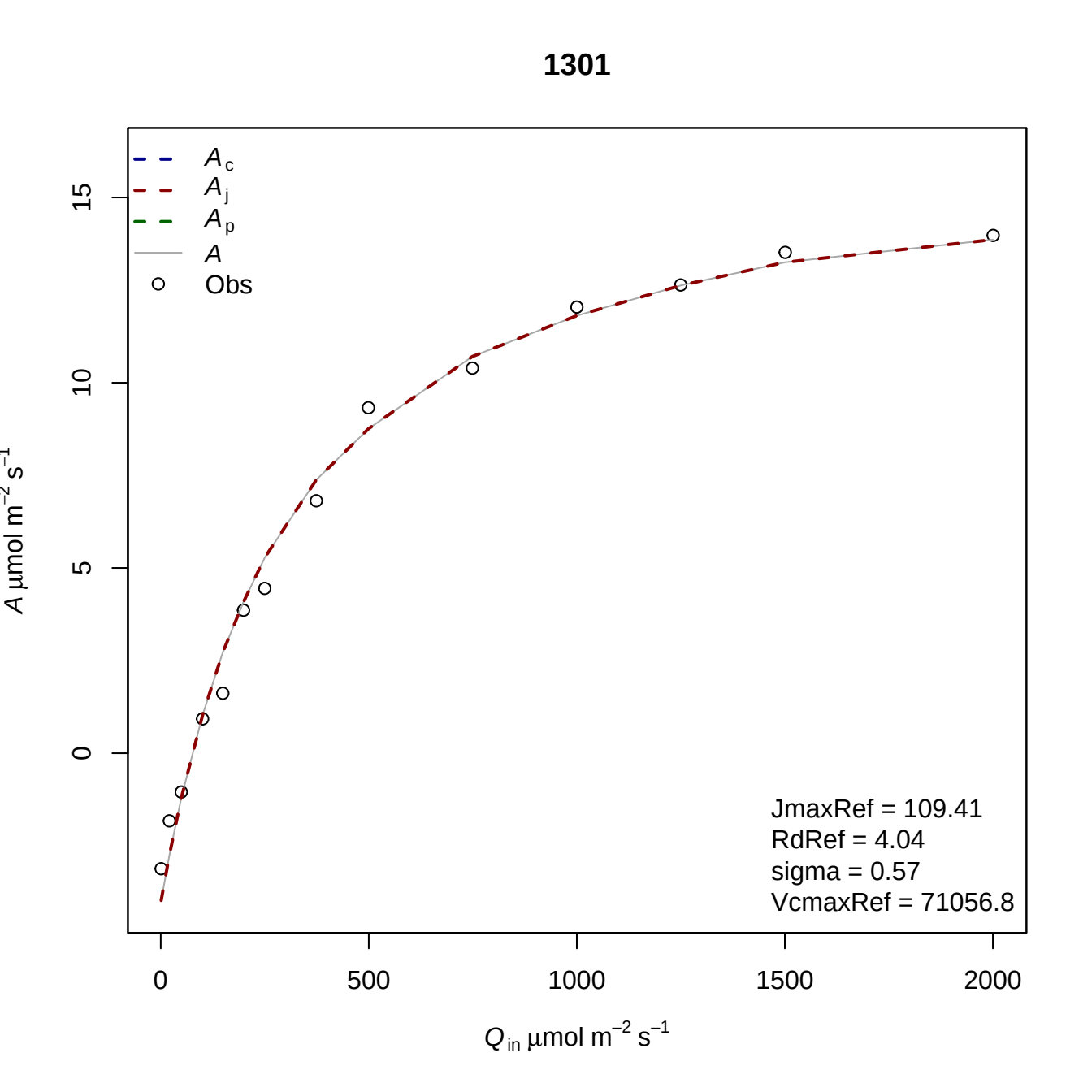

1296
A_c
A_j
A_p
A
Obs
JmaxRef = 137.99
RdRef = 2.34
sigma = 0.55
VcmaxRef = 43.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

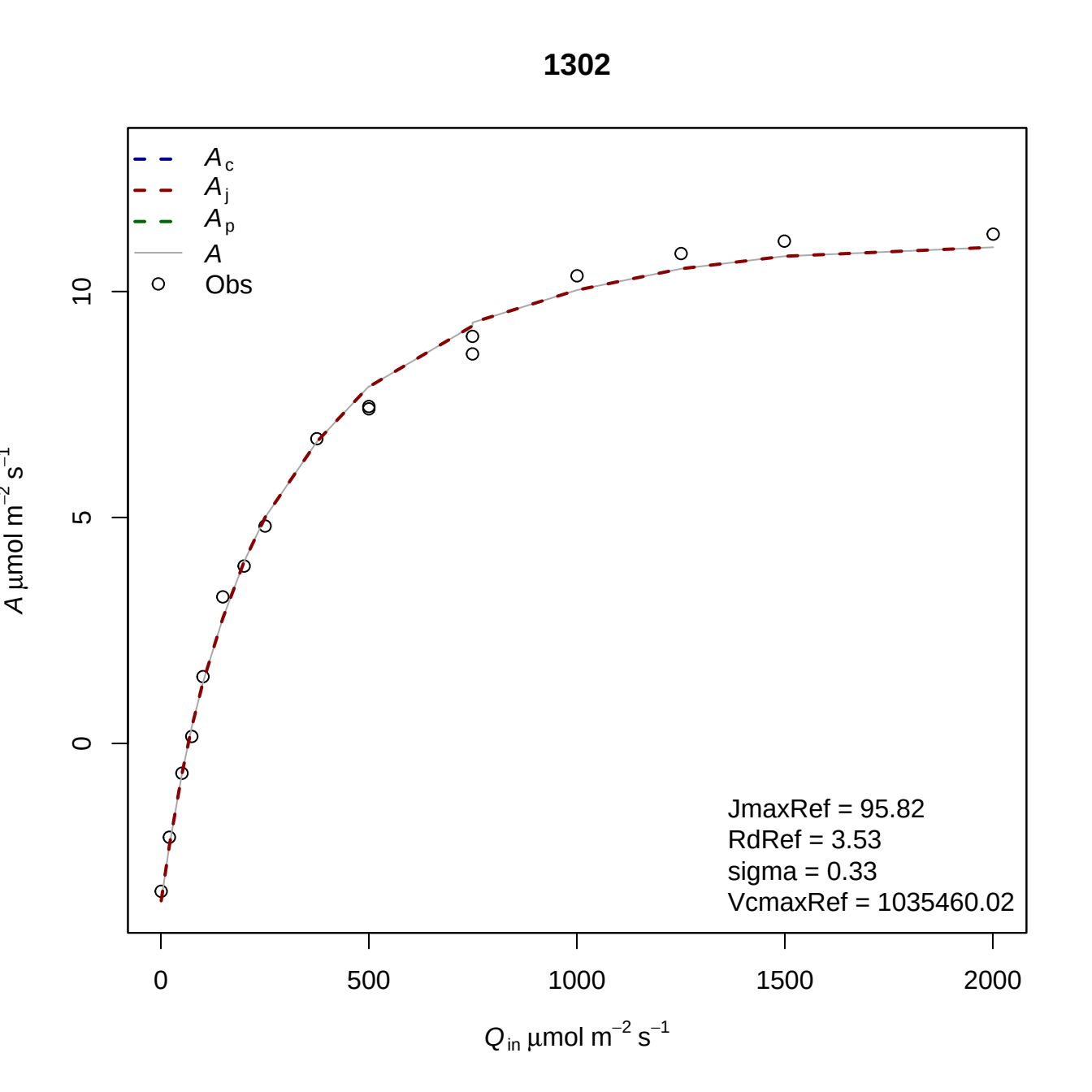

1297
A_c
A_j
A_p
A
Obs
JmaxRef = 172.15
RdRef = 3.34
sigma = 0.43
VcmaxRef = 130049.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

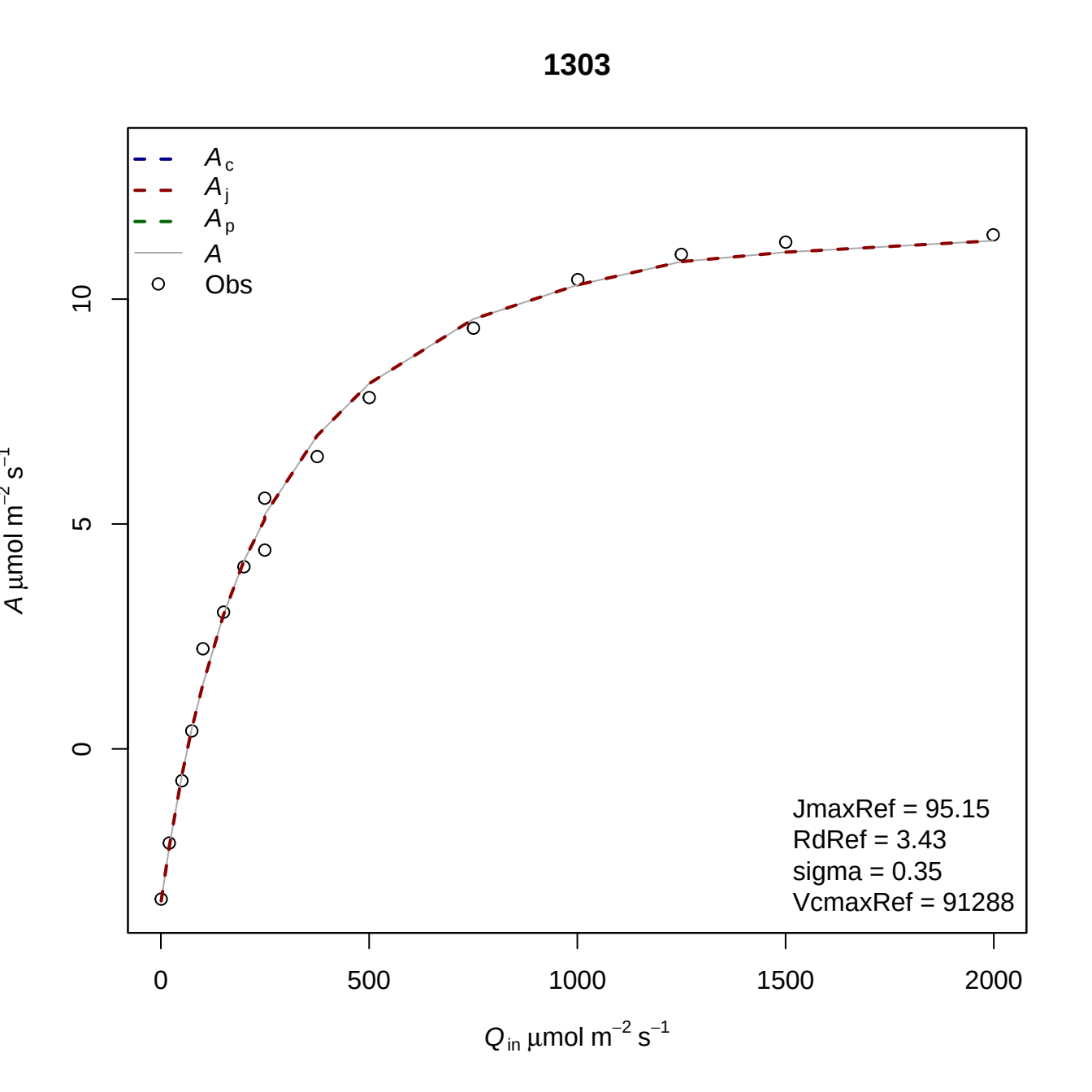

1298
A_c
A_j
A_p
A
Obs
JmaxRef = 111.49
RdRef = 2.36
sigma = 0.38
VcmaxRef = 94477.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

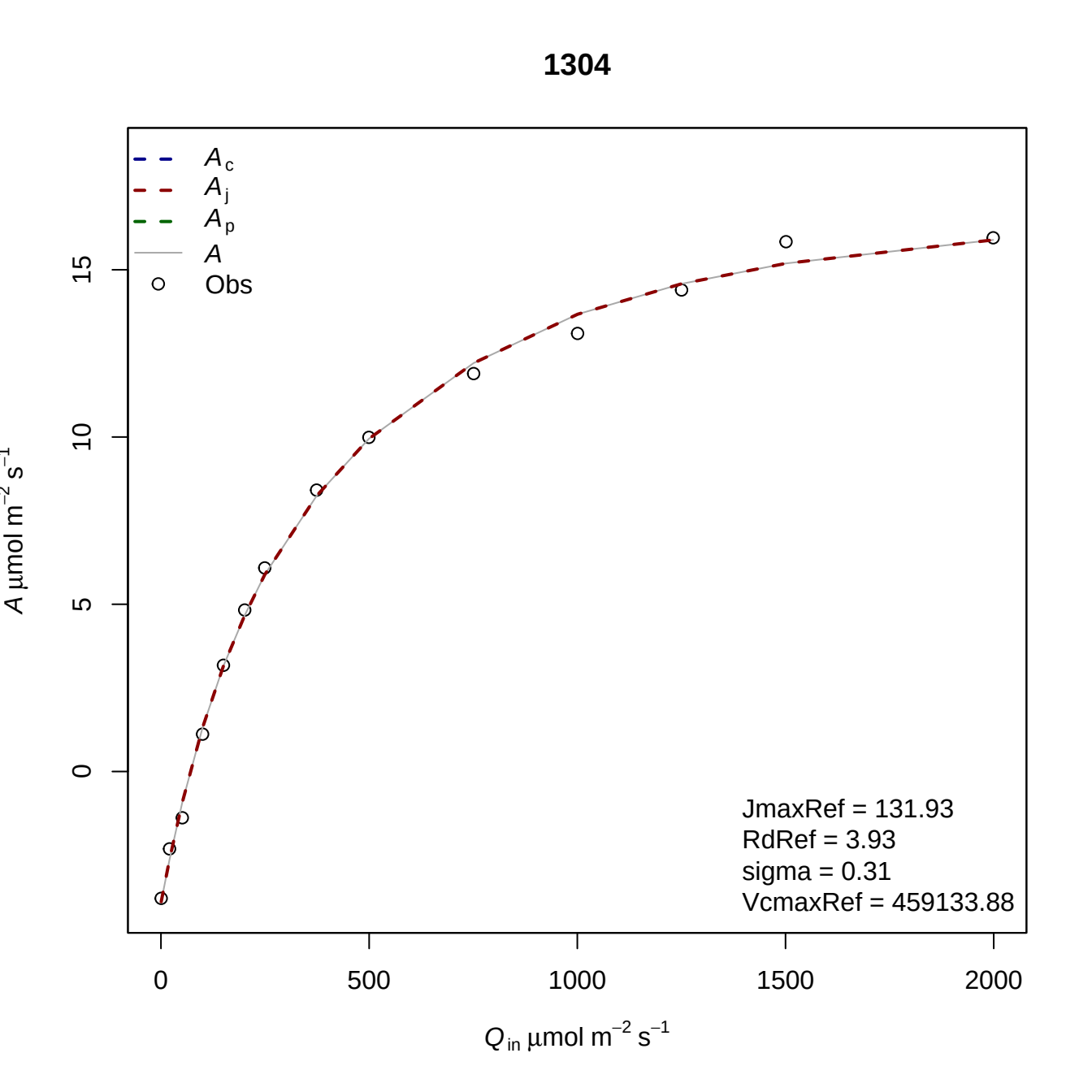

1299
A_c
A_j
A_p
A
Obs
JmaxRef = 138.56
RdRef = 4.04
sigma = 0.28
VcmaxRef = 58.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

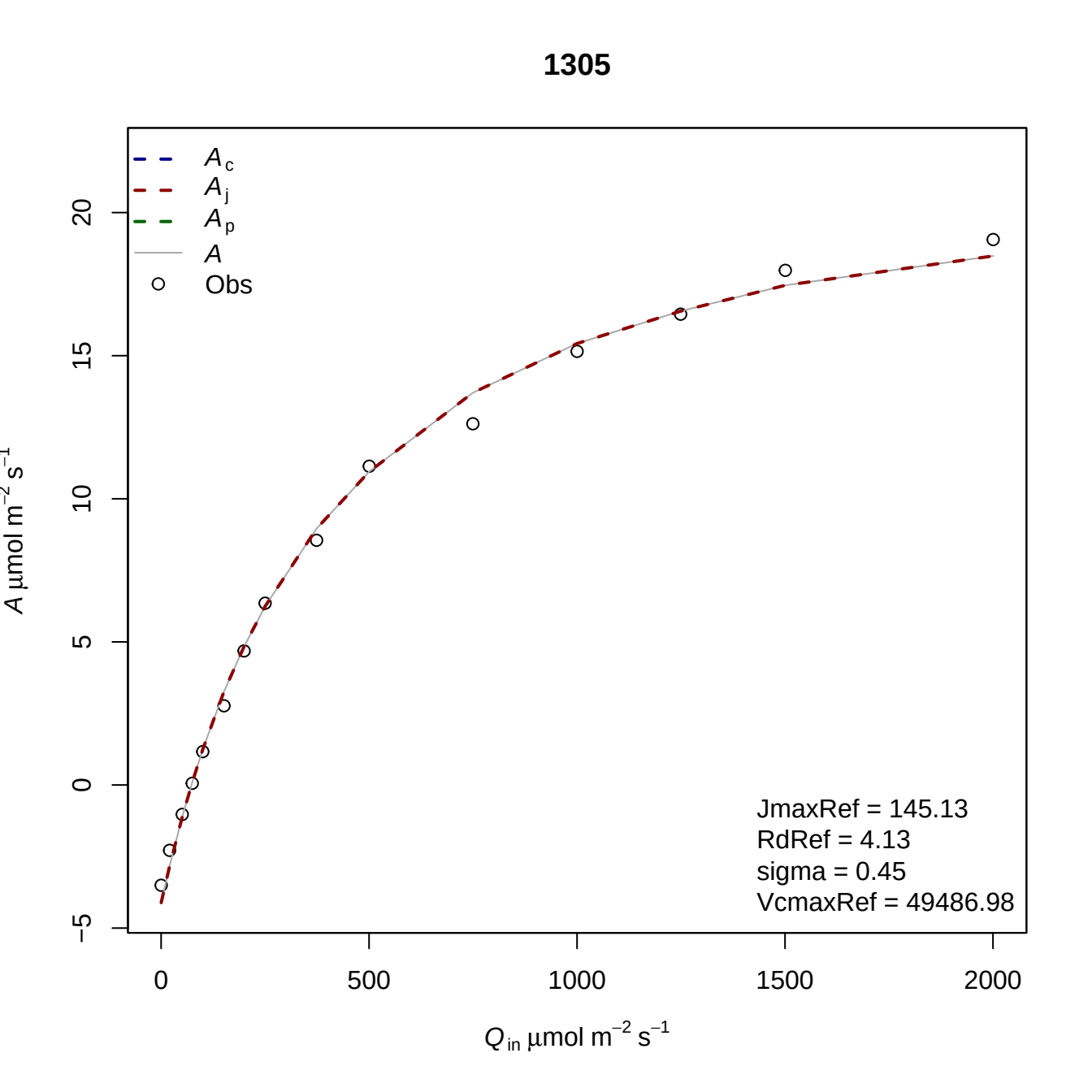

1300
A_c
A_j
A_p
A
Obs
JmaxRef = 131.64
RdRef = 1.75
sigma = 0.22
VcmaxRef = 58.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

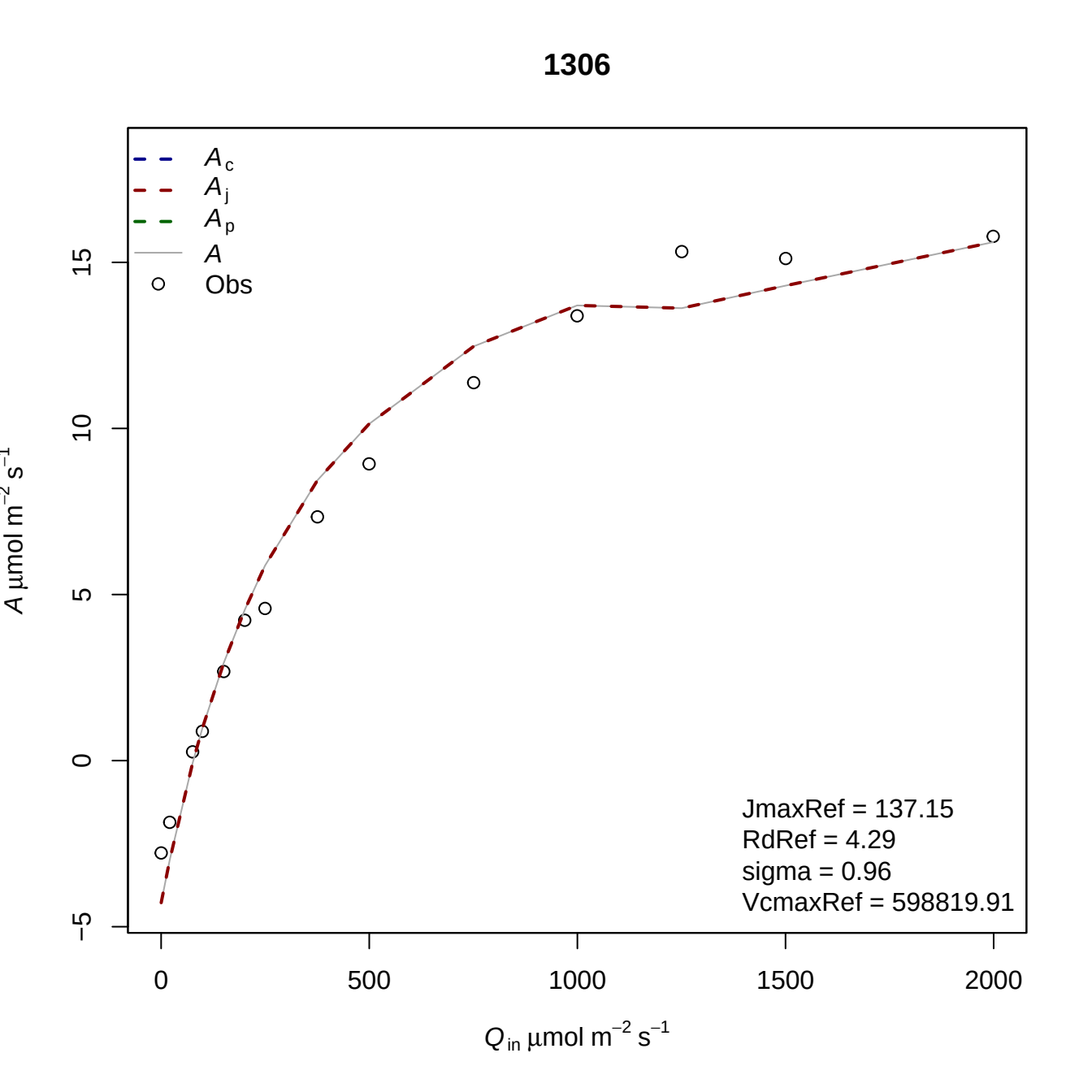

1301
A_c
A_j
A_p
A
Obs
JmaxRef = 109.41
RdRef = 4.04
sigma = 0.57
VcmaxRef = 71056.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

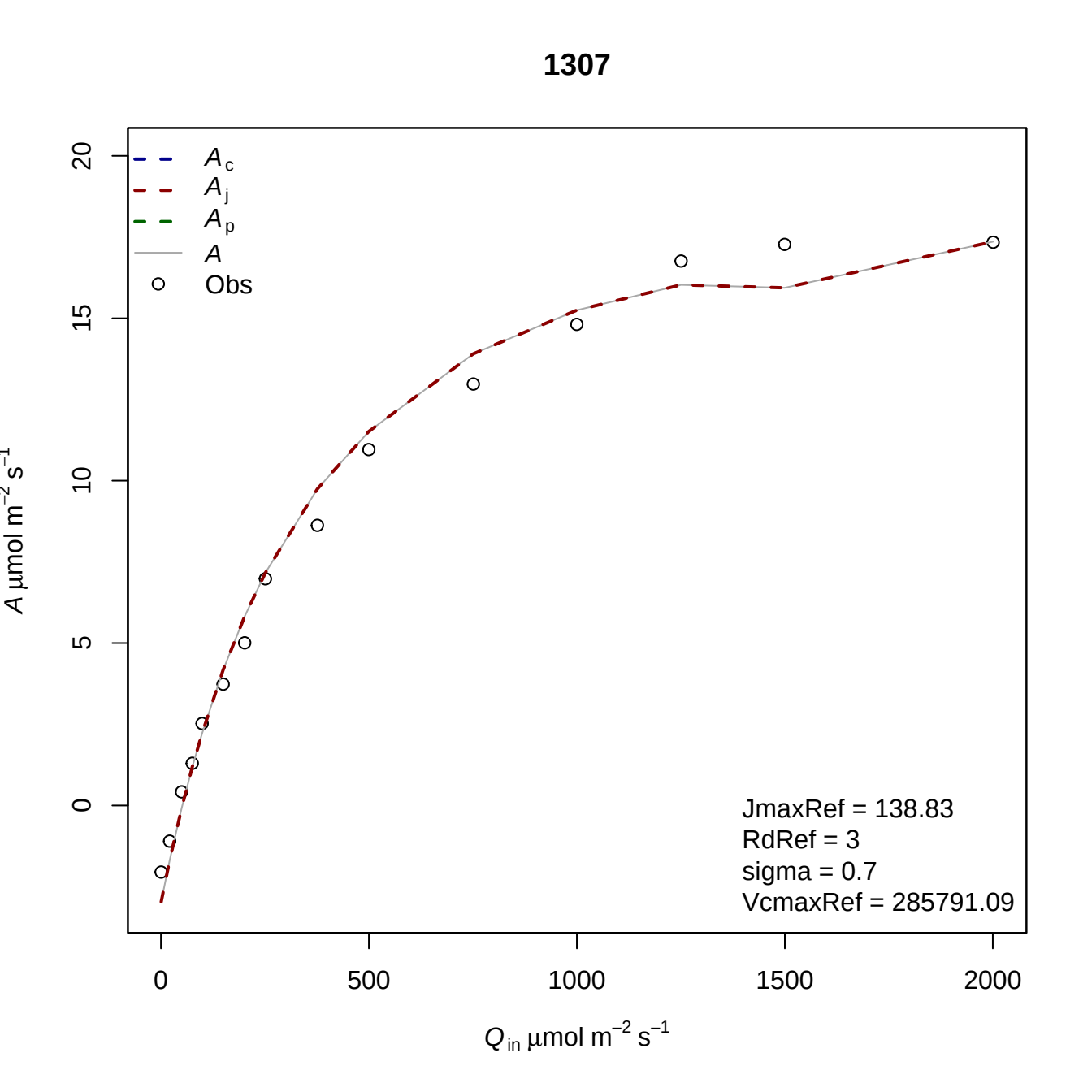

1302
A_c
A_j
A_p
A
Obs
JmaxRef = 95.82
RdRef = 3.53
sigma = 0.33
VcmaxRef = 1035460.02
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

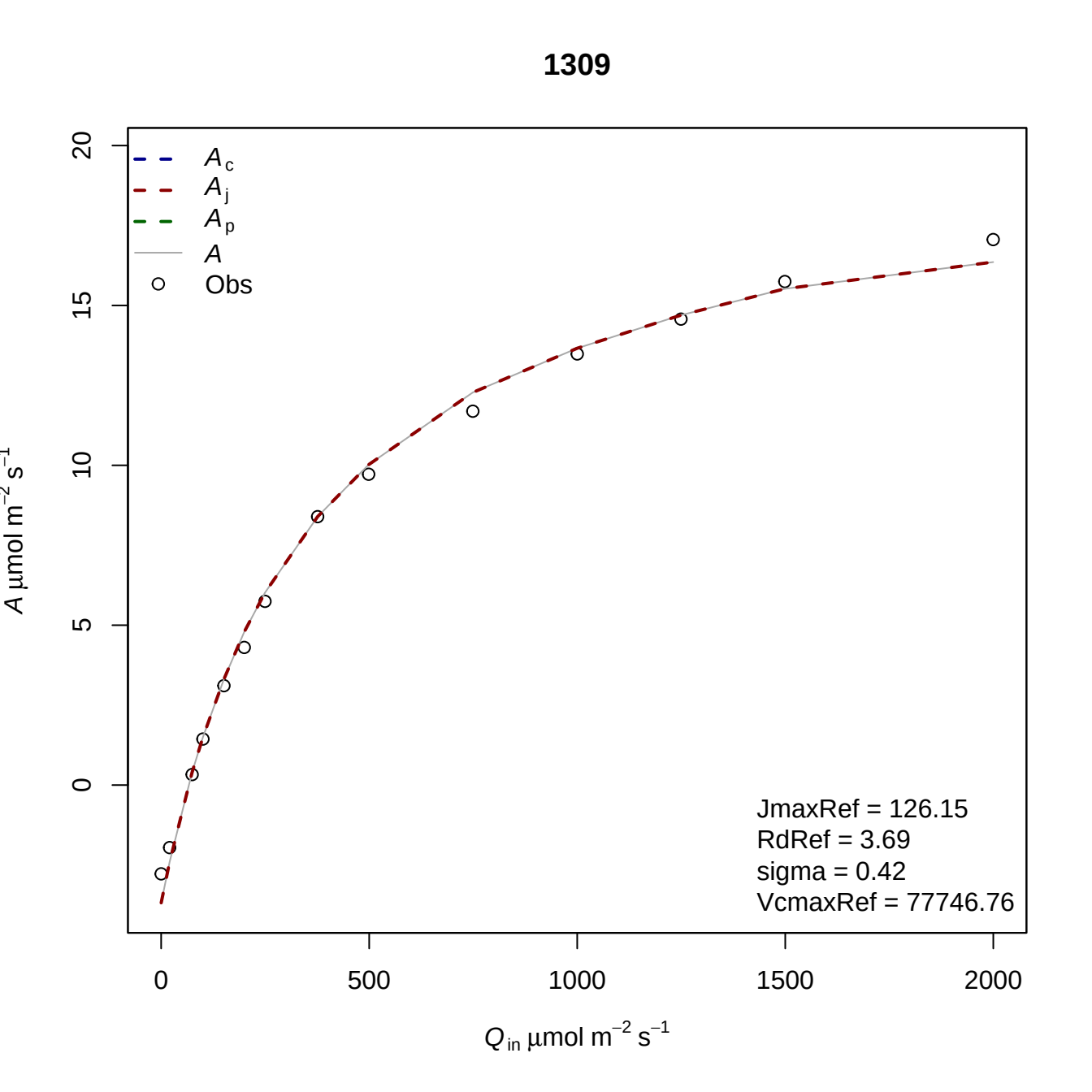

1303
A_c
A_j
A_p
A
Obs
JmaxRef = 95.15
RdRef = 3.43
sigma = 0.35
VcmaxRef = 91288
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

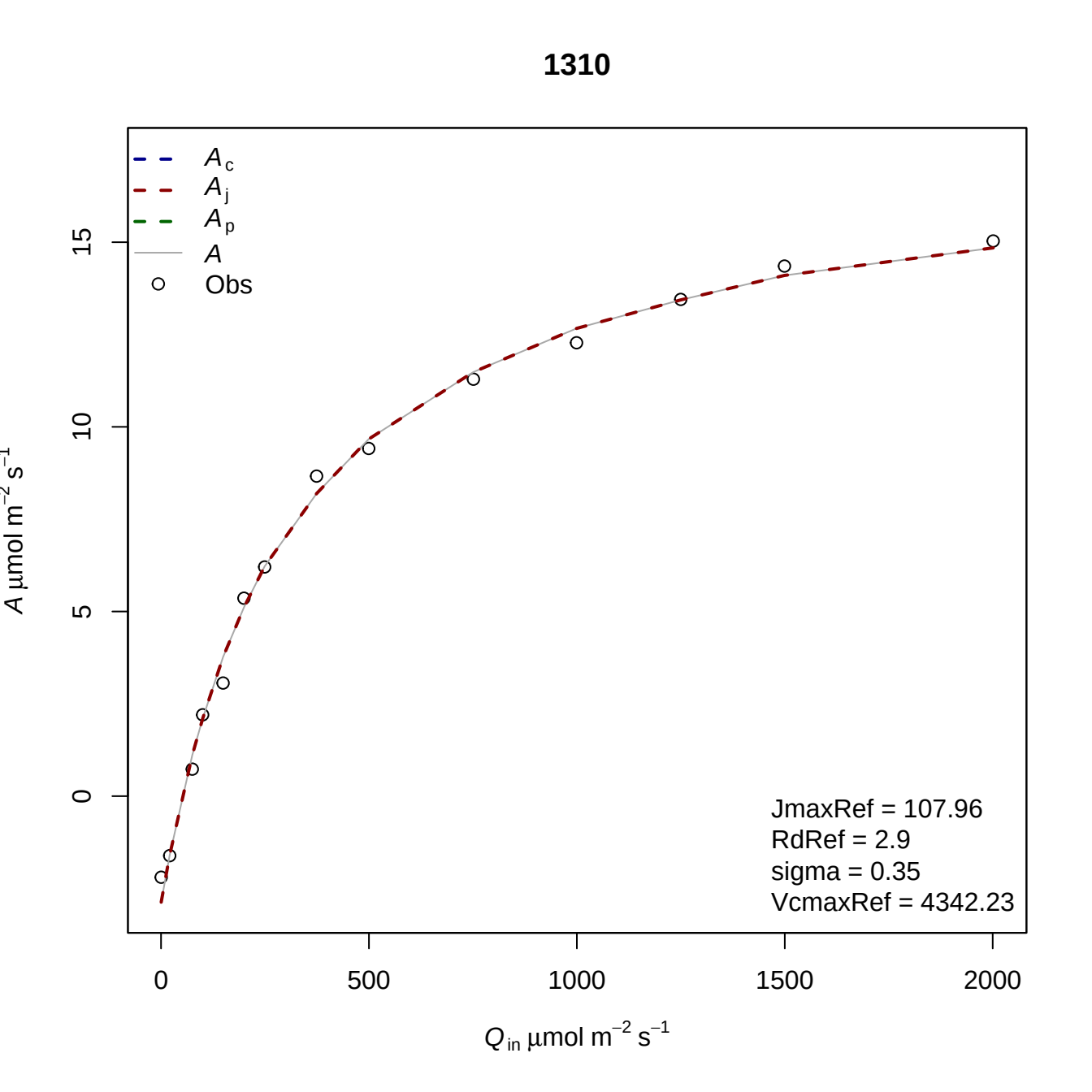

1304
A_c
A_j
A_p
A
Obs
JmaxRef = 131.93
RdRef = 3.93
sigma = 0.31
VcmaxRef = 459133.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

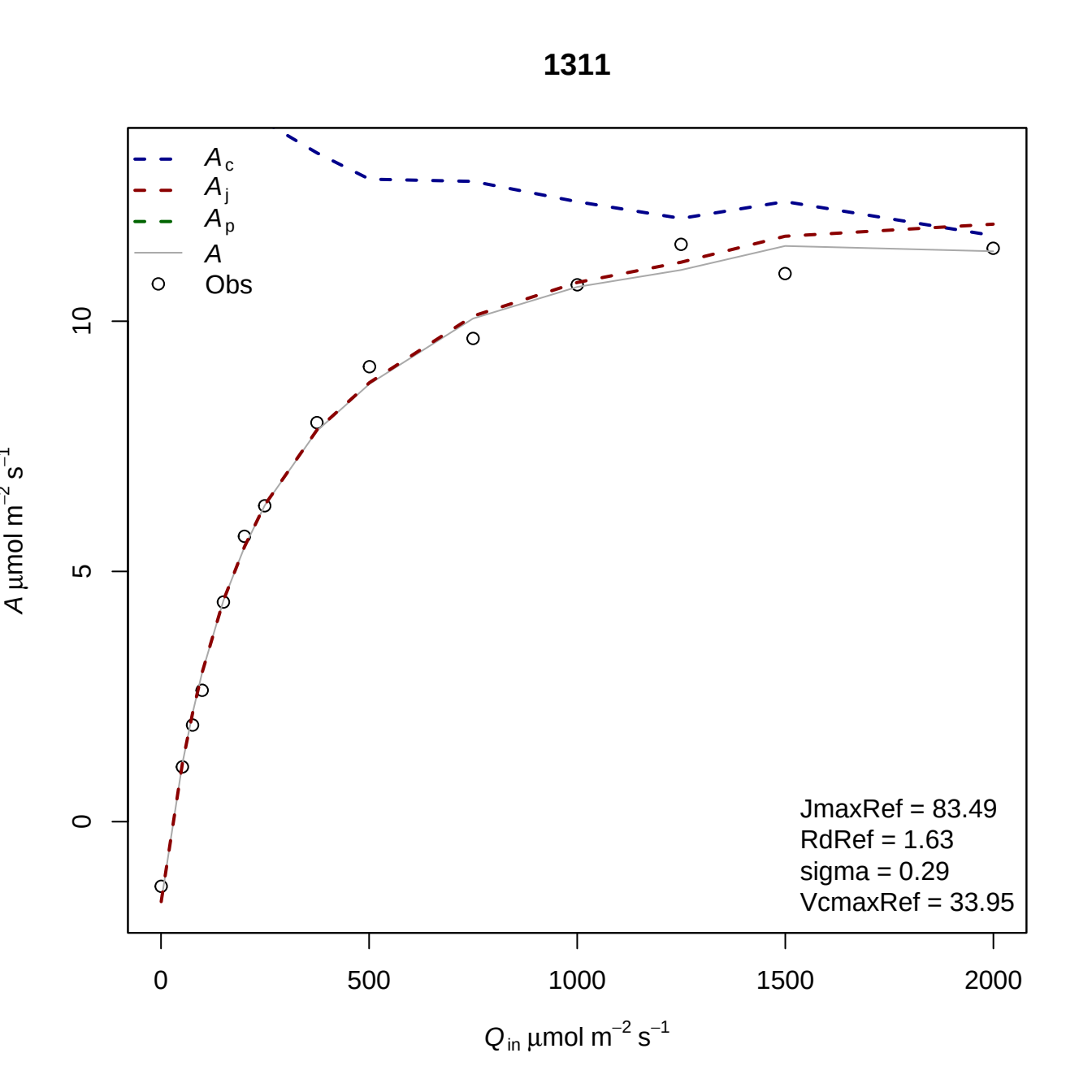

1305
A_c
A_j
A_p
A
Obs
JmaxRef = 145.13
RdRef = 4.13
sigma = 0.45
VcmaxRef = 49486.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

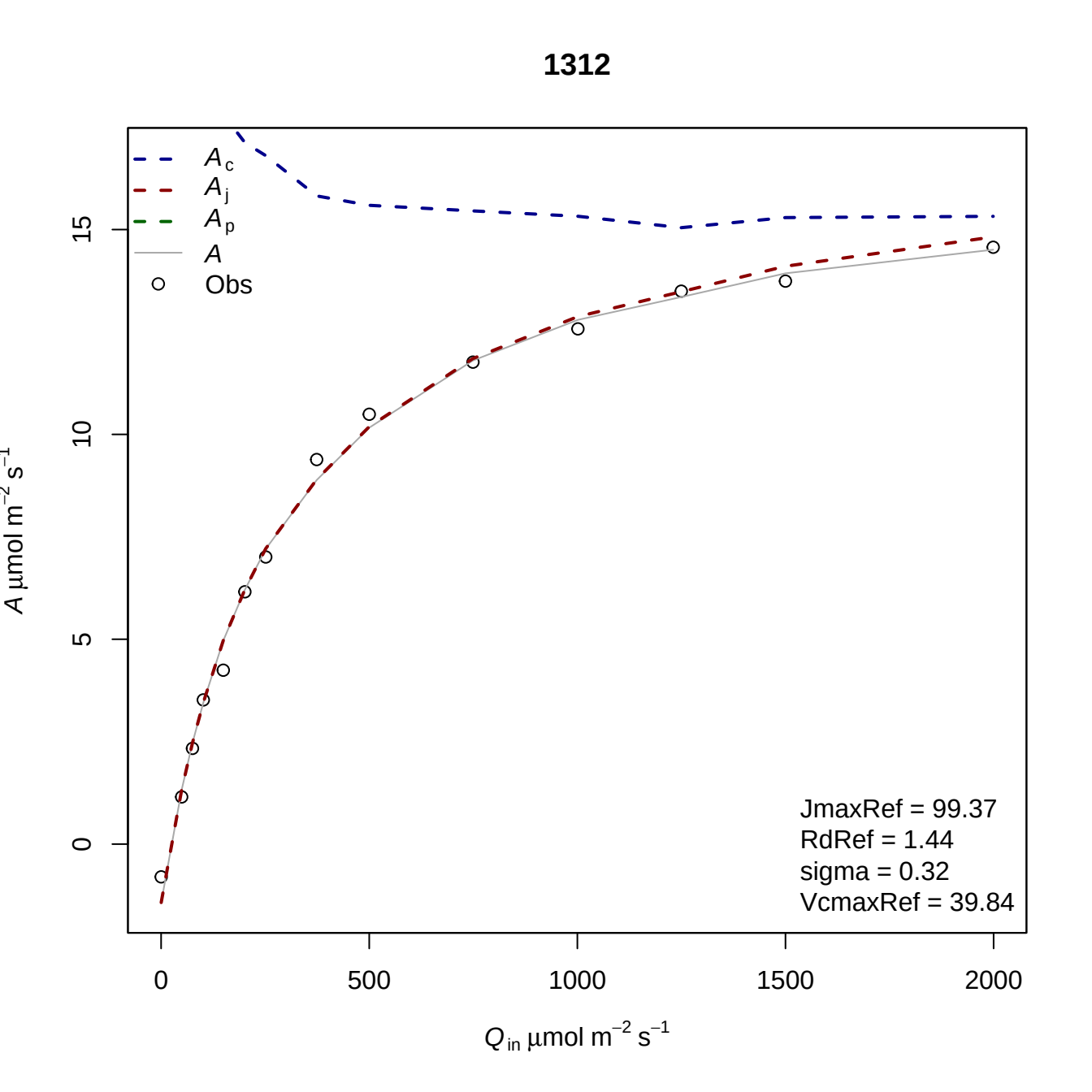

1306
A_c
A_j
A_p
A
Obs
JmaxRef = 137.15
RdRef = 4.29
sigma = 0.96
VcmaxRef = 598819.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

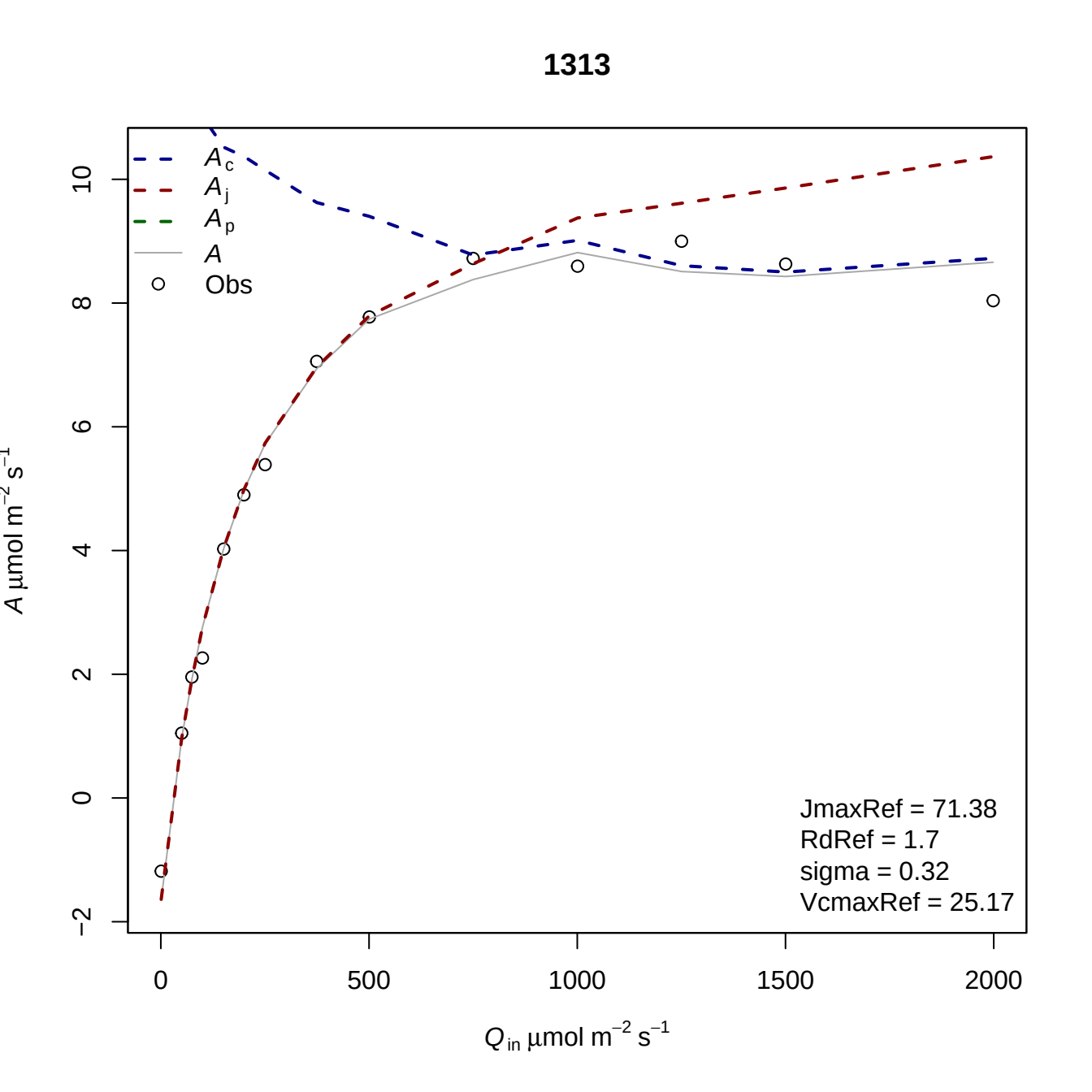

1307
A_c
A_j
A_p
A
Obs
JmaxRef = 138.83
RdRef = 3
sigma = 0.7
VcmaxRef = 285791.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

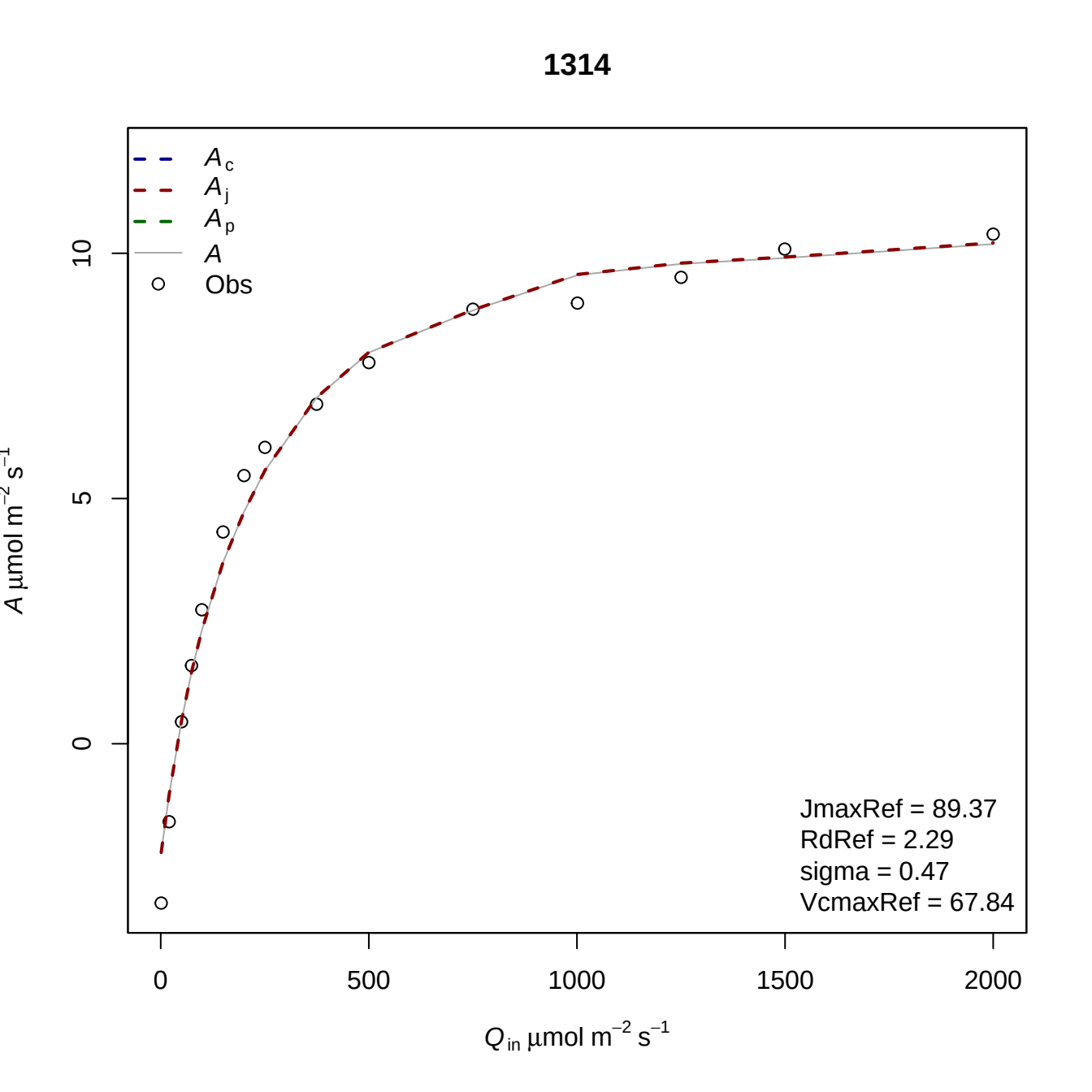

1309
A_c
A_j
A_p
A
Obs
JmaxRef = 126.15
RdRef = 3.69
sigma = 0.42
VcmaxRef = 77746.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

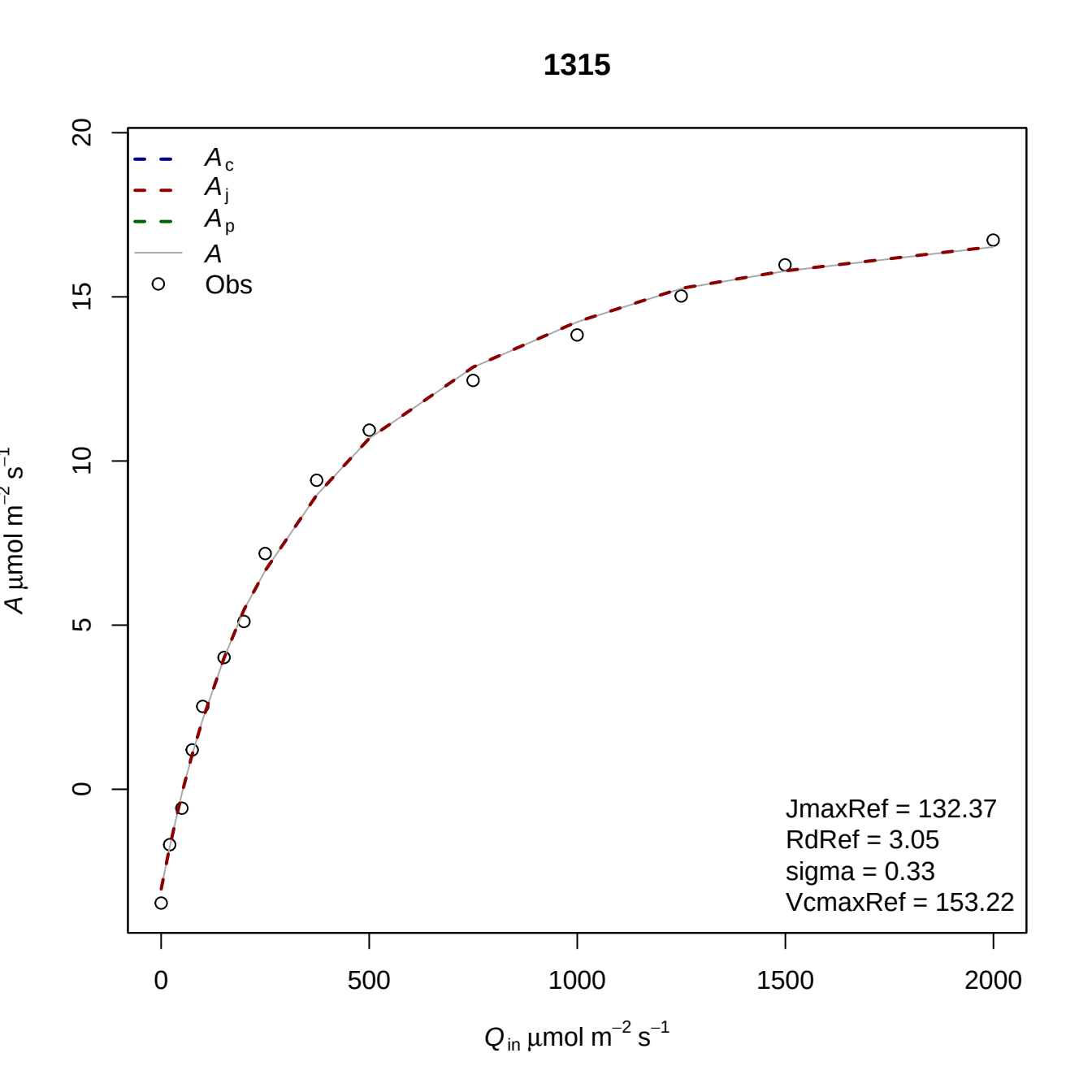

1310
A_c
A_j
A_p
A
Obs
JmaxRef = 107.96
RdRef = 2.9
sigma = 0.35
VcmaxRef = 4342.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

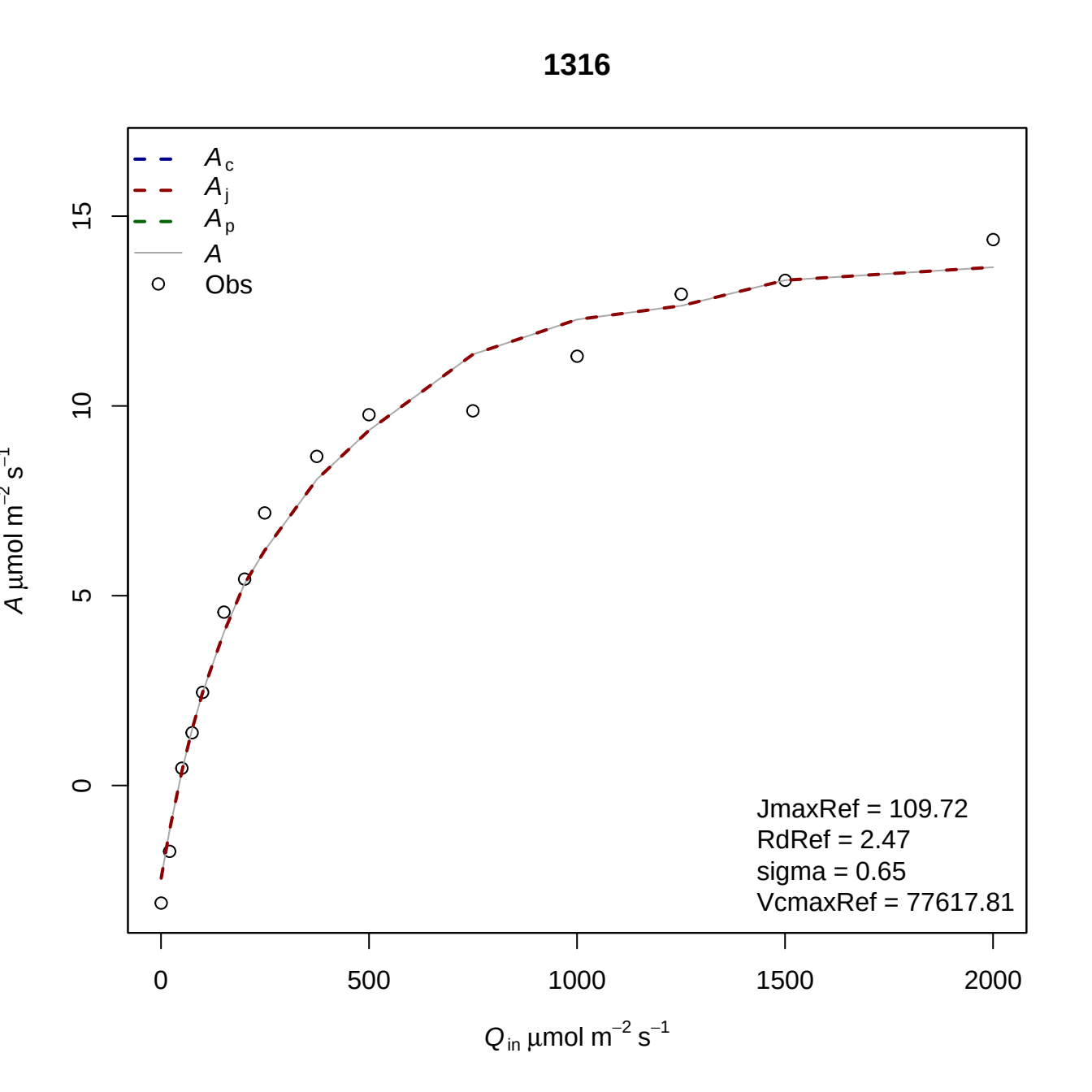

1311
A_c
A_j
A_p
A
Obs
JmaxRef = 83.49
RdRef = 1.63
sigma = 0.29
VcmaxRef = 33.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

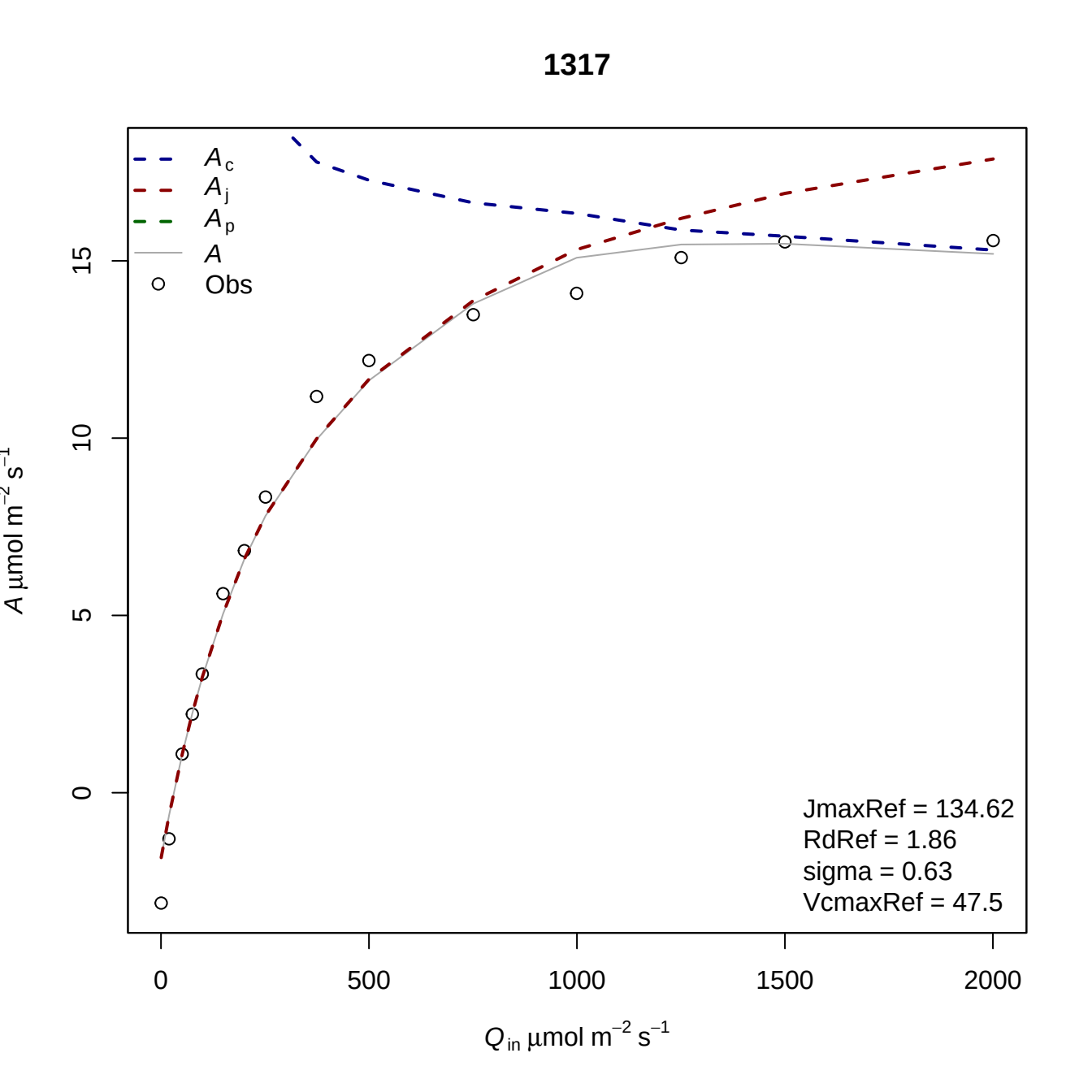

1312
A_c
A_j
A_p
A
Obs
JmaxRef = 99.37
RdRef = 1.44
sigma = 0.32
VcmaxRef = 39.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

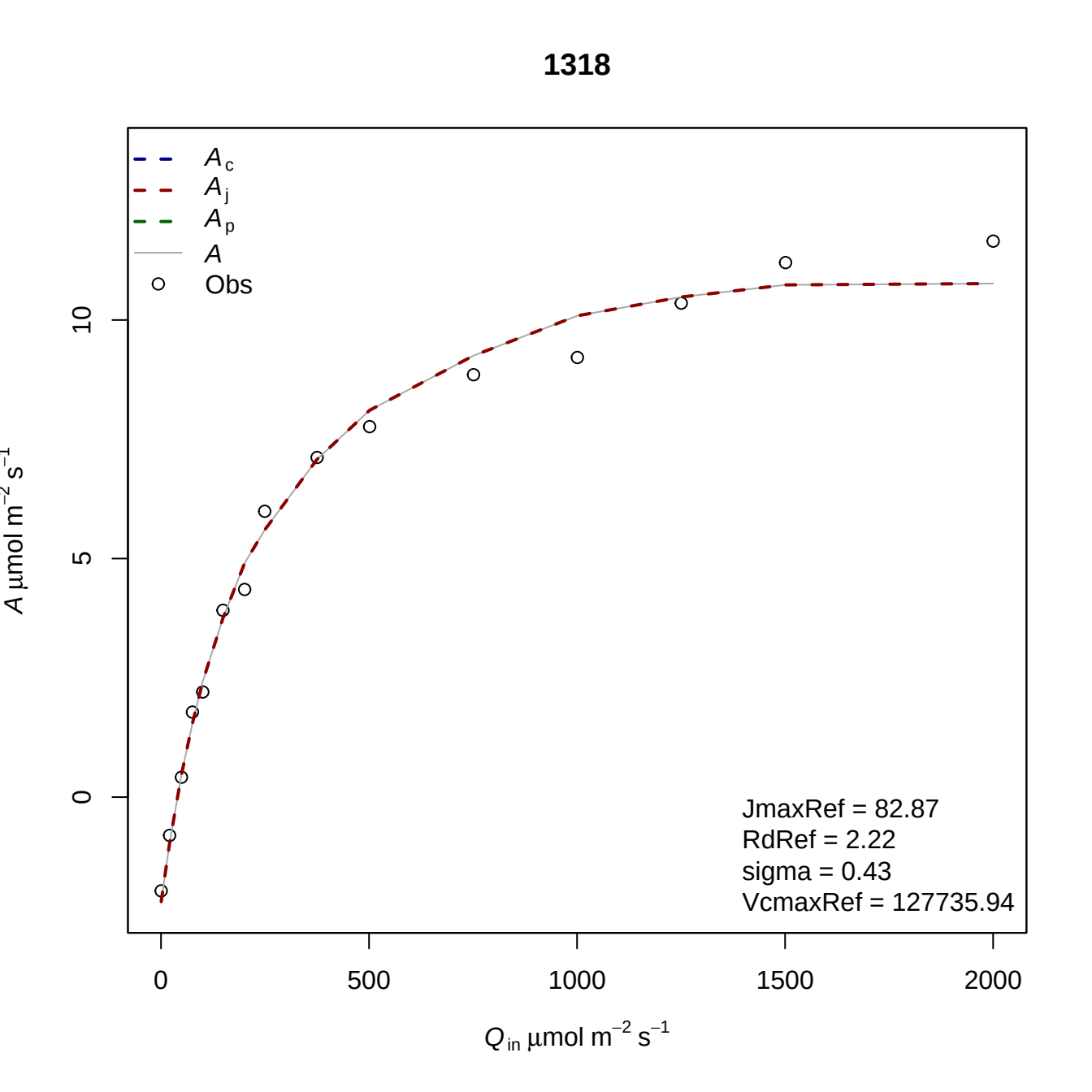

1313
A_c
A_j
A_p
A
Obs
JmaxRef = 71.38
RdRef = 1.7
sigma = 0.32
VcmaxRef = 25.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

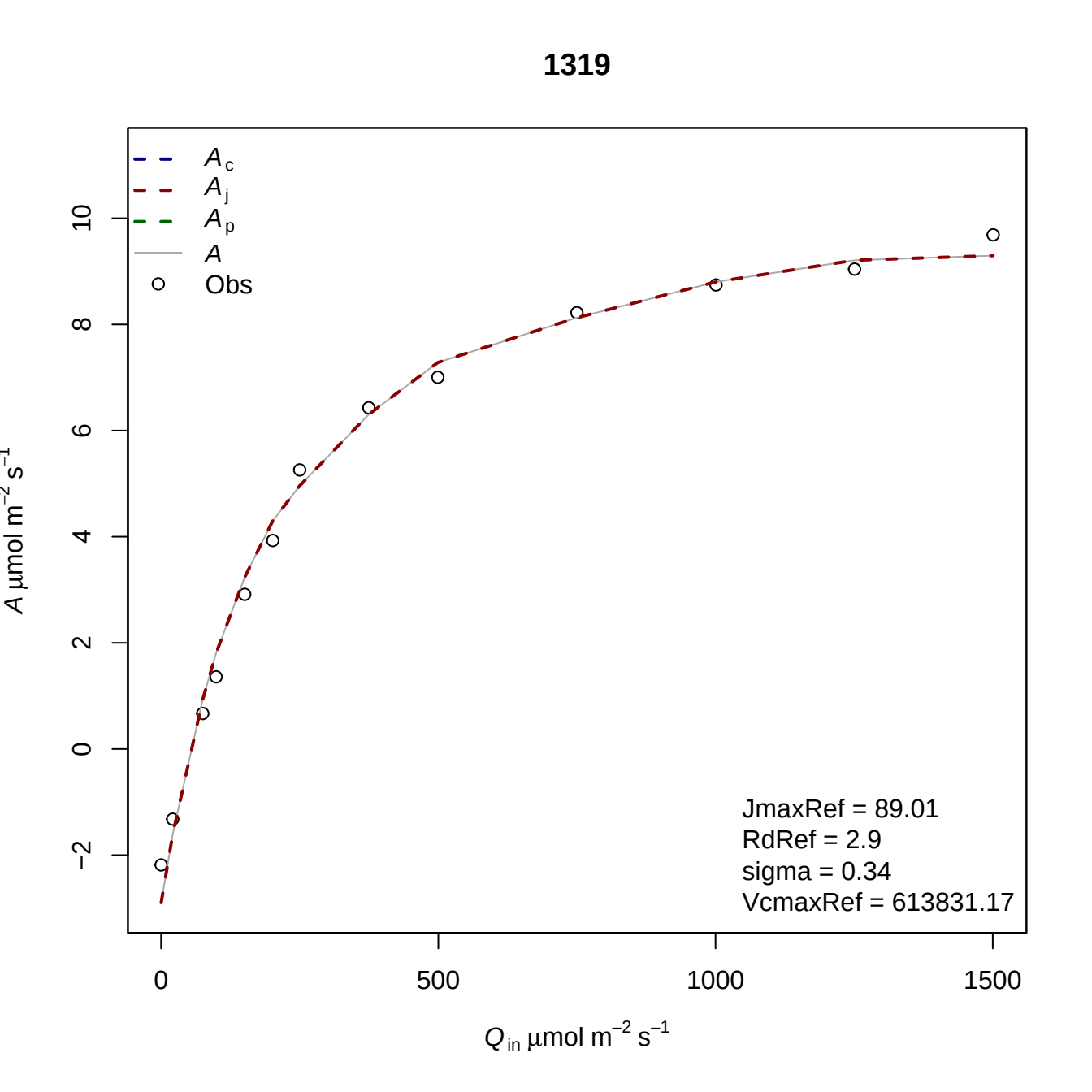

1314
A_c
A_j
A_p
A
Obs
JmaxRef = 89.37
RdRef = 2.29
sigma = 0.47
VcmaxRef = 67.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

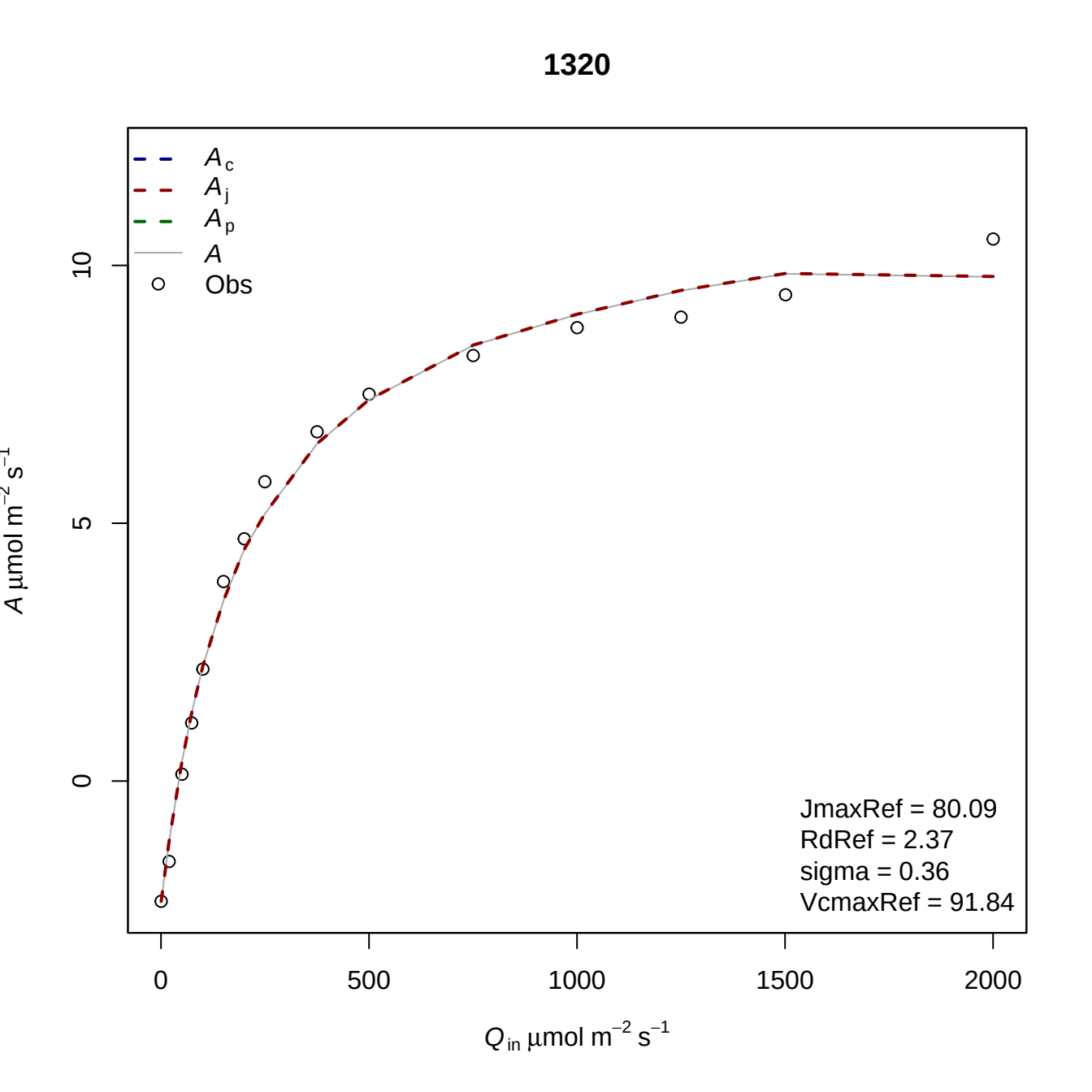

1315
A_c
A_j
A_p
A
Obs
JmaxRef = 132.37
RdRef = 3.05
sigma = 0.33
VcmaxRef = 153.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

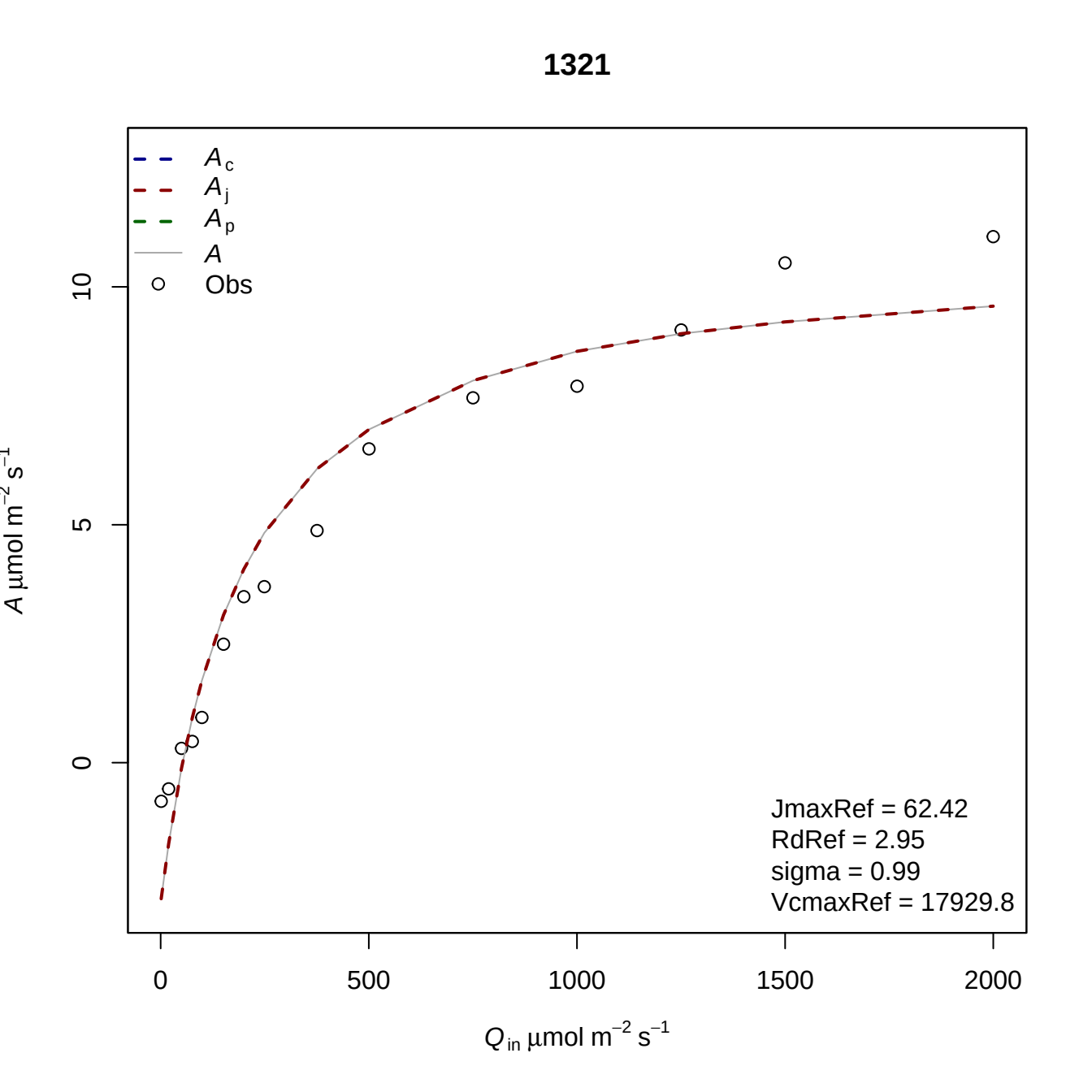

1316
A_c
A_j
A_p
A
Obs
JmaxRef = 109.72
RdRef = 2.47
sigma = 0.65
VcmaxRef = 77617.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

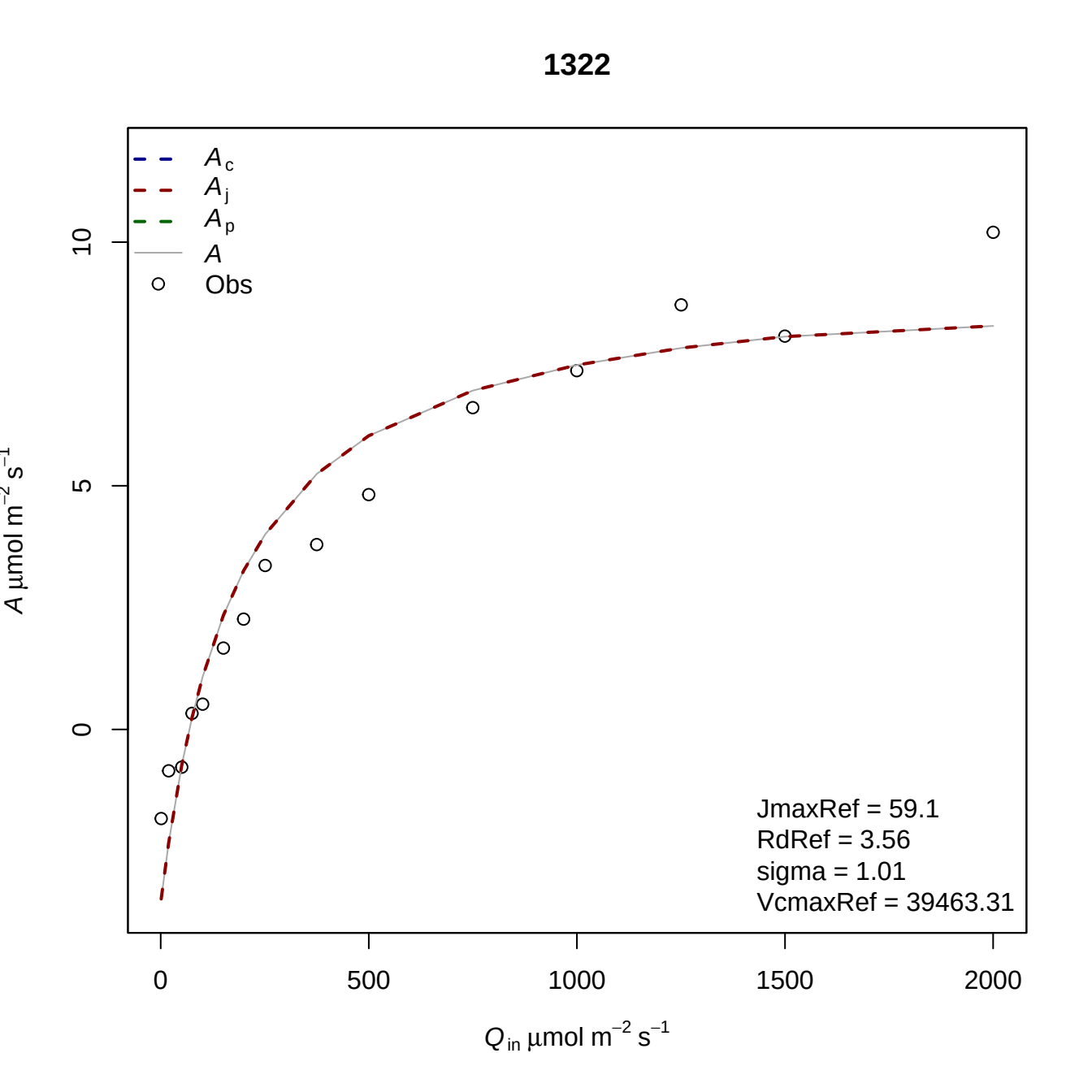

1317
A_c
A_j
A_p
A
Obs
JmaxRef = 134.62
RdRef = 1.86
sigma = 0.63
VcmaxRef = 47.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

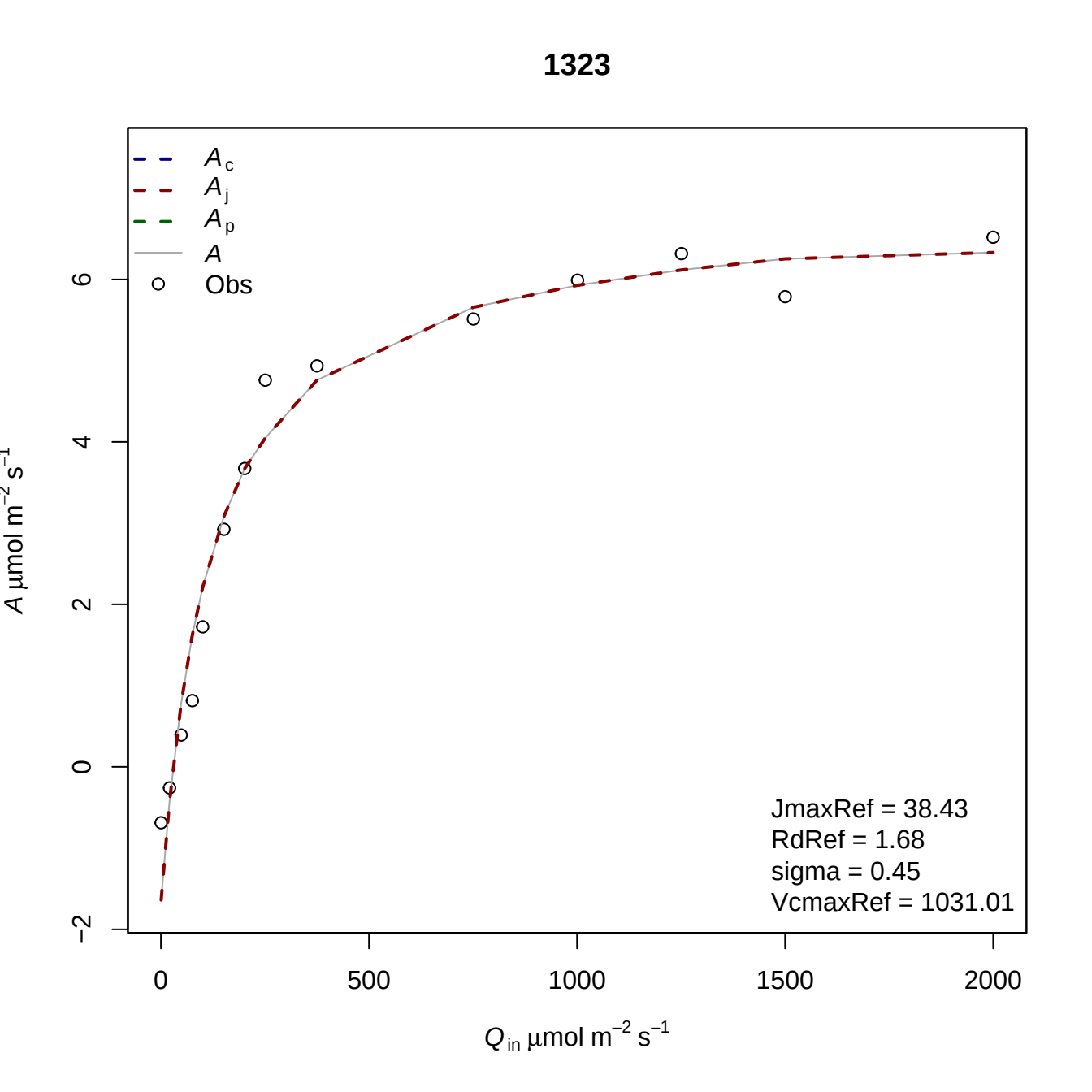

1318
A_c
A_j
A_p
A
Obs
JmaxRef = 82.87
RdRef = 2.22
sigma = 0.43
VcmaxRef = 127735.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

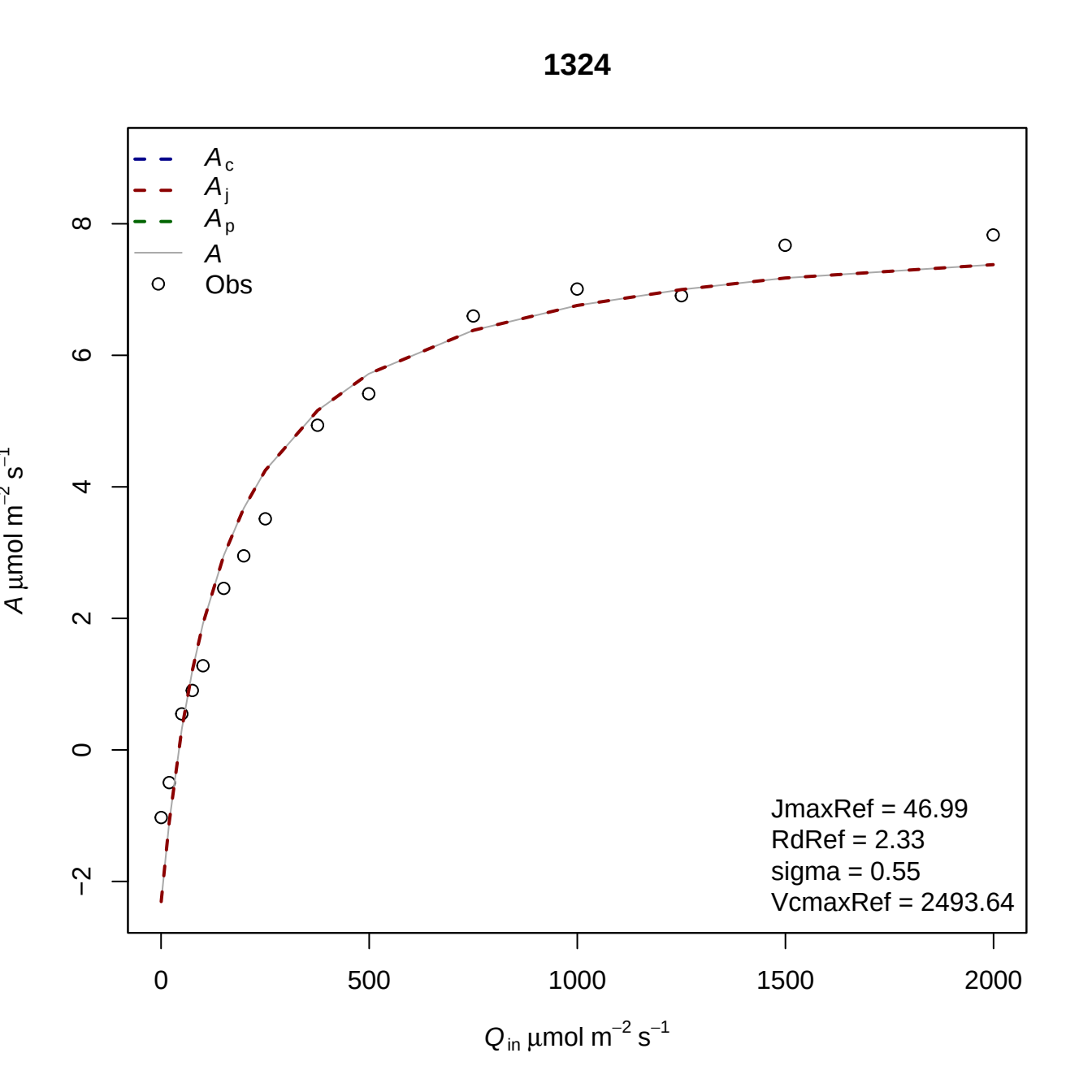

1319
A_c
A_j
A_p
A
Obs
JmaxRef = 89.01
RdRef = 2.9
sigma = 0.34
VcmaxRef = 613831.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

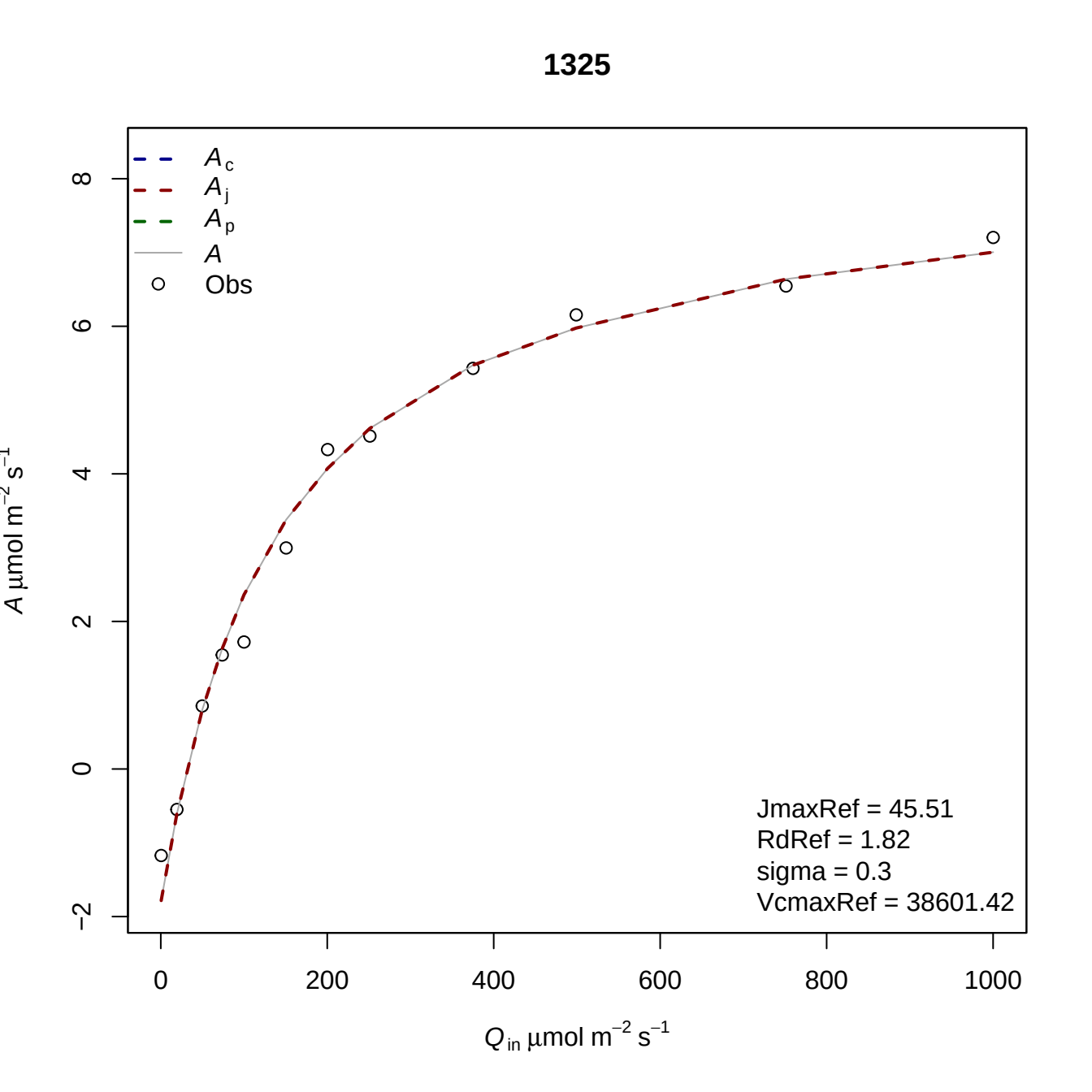

1320
A_c
A_j
A_p
A
Obs
JmaxRef = 80.09
RdRef = 2.37
sigma = 0.36
VcmaxRef = 91.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

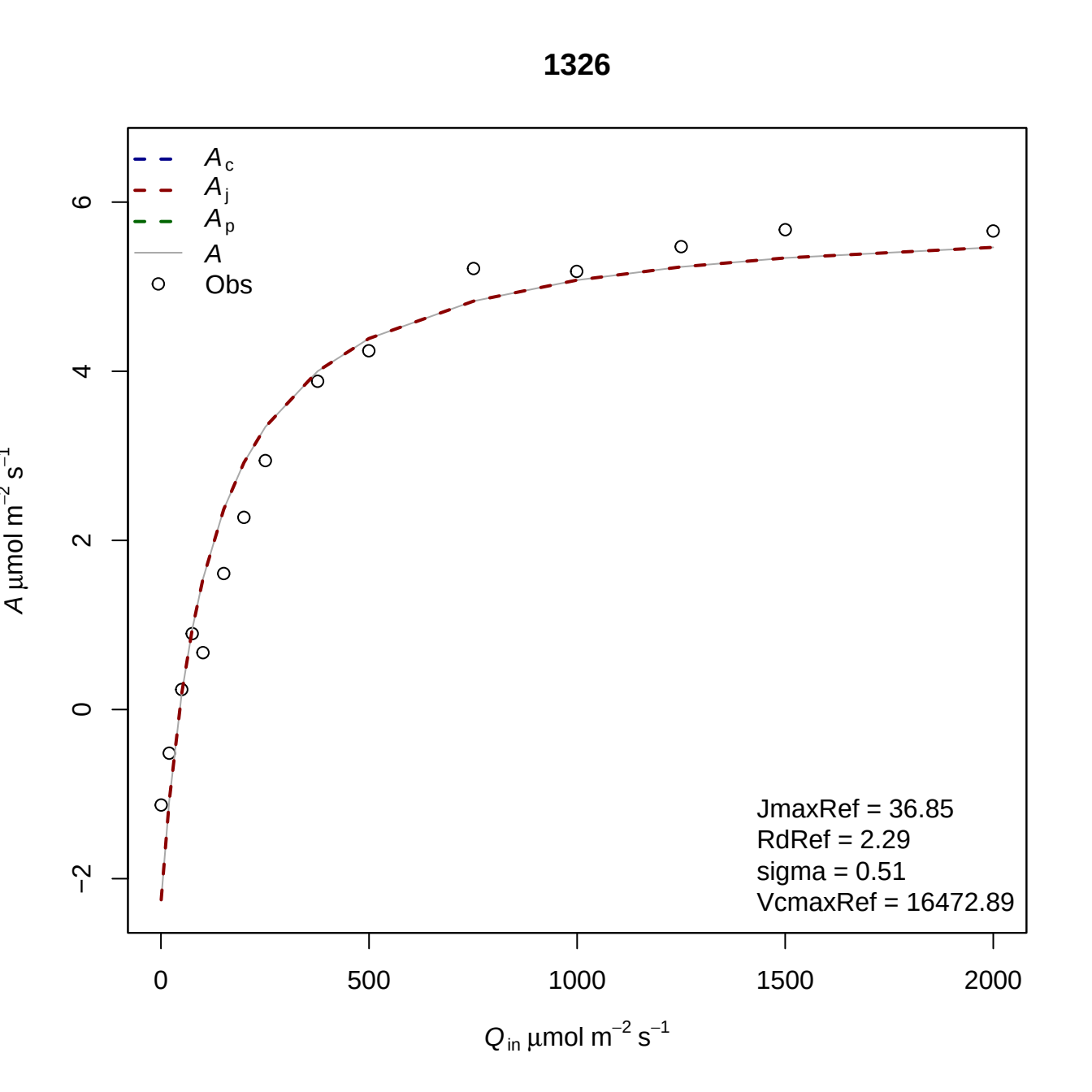

1321
A_c
A_j
A_p
A
Obs
JmaxRef = 62.42
RdRef = 2.95
sigma = 0.99
VcmaxRef = 17929.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

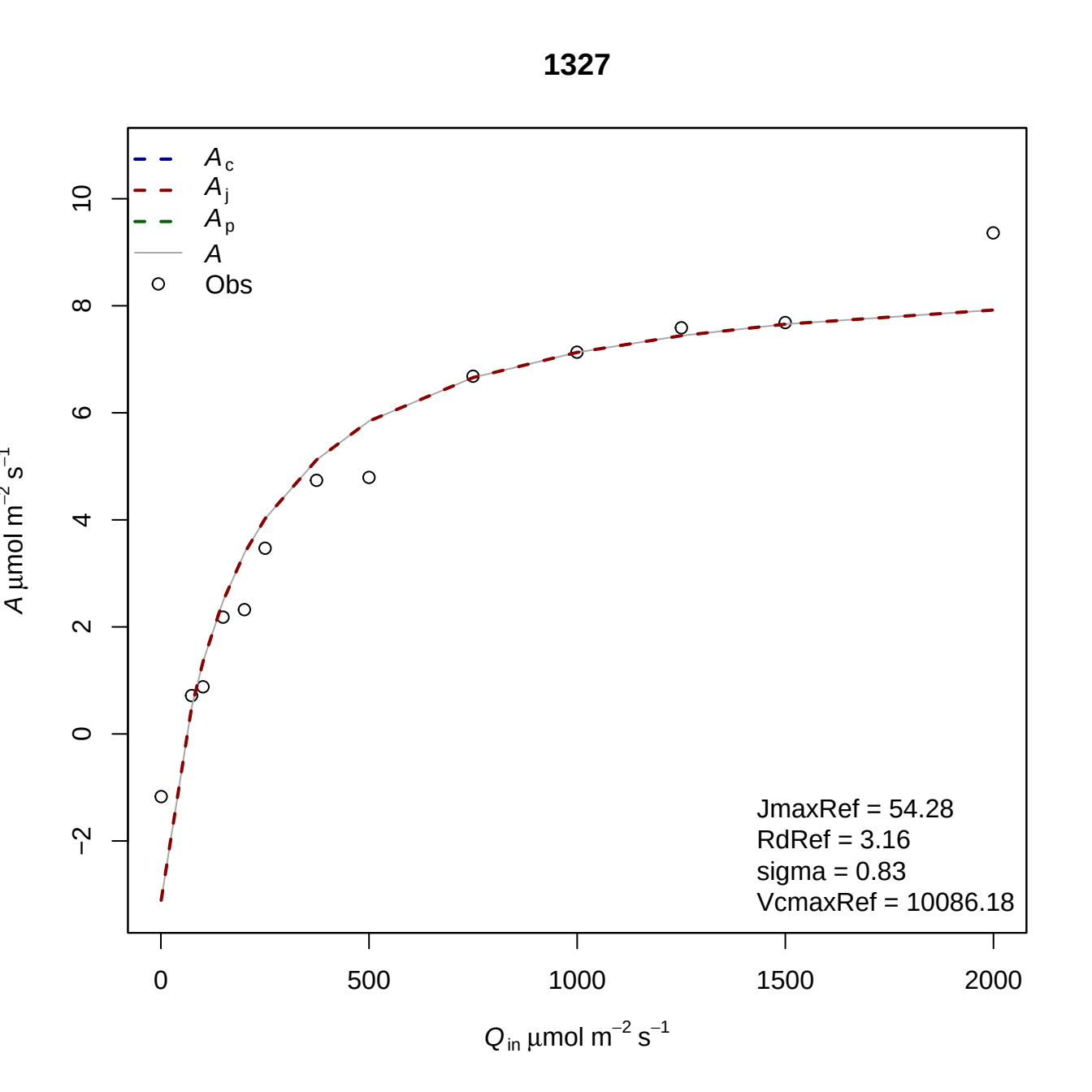

1322
A_c
A_j
A_p
A
Obs
JmaxRef = 59.1
RdRef = 3.56
sigma = 1.01
VcmaxRef = 39463.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

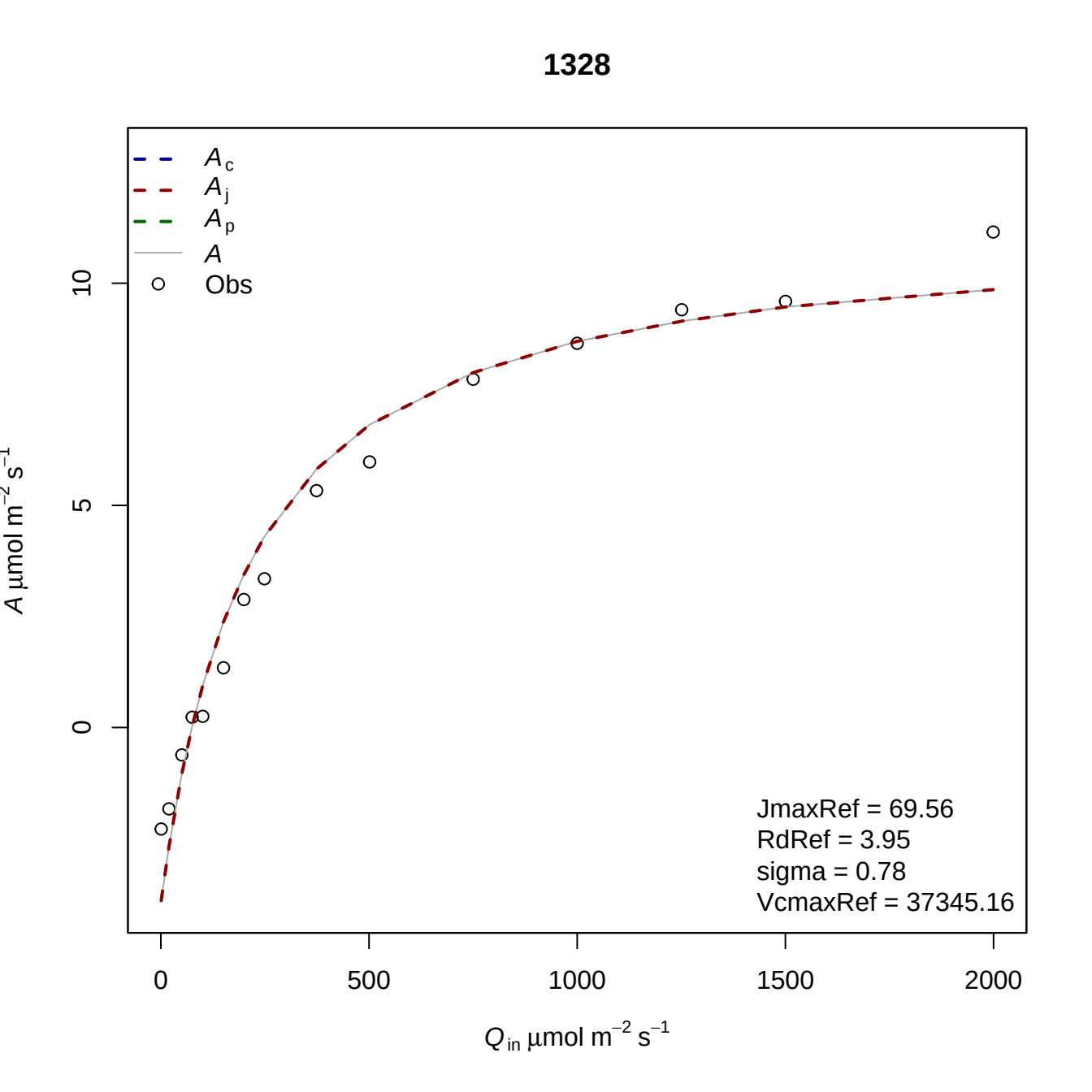

1323
A_c
A_j
A_p
A
Obs
JmaxRef = 38.43
RdRef = 1.68
sigma = 0.45
VcmaxRef = 1031.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

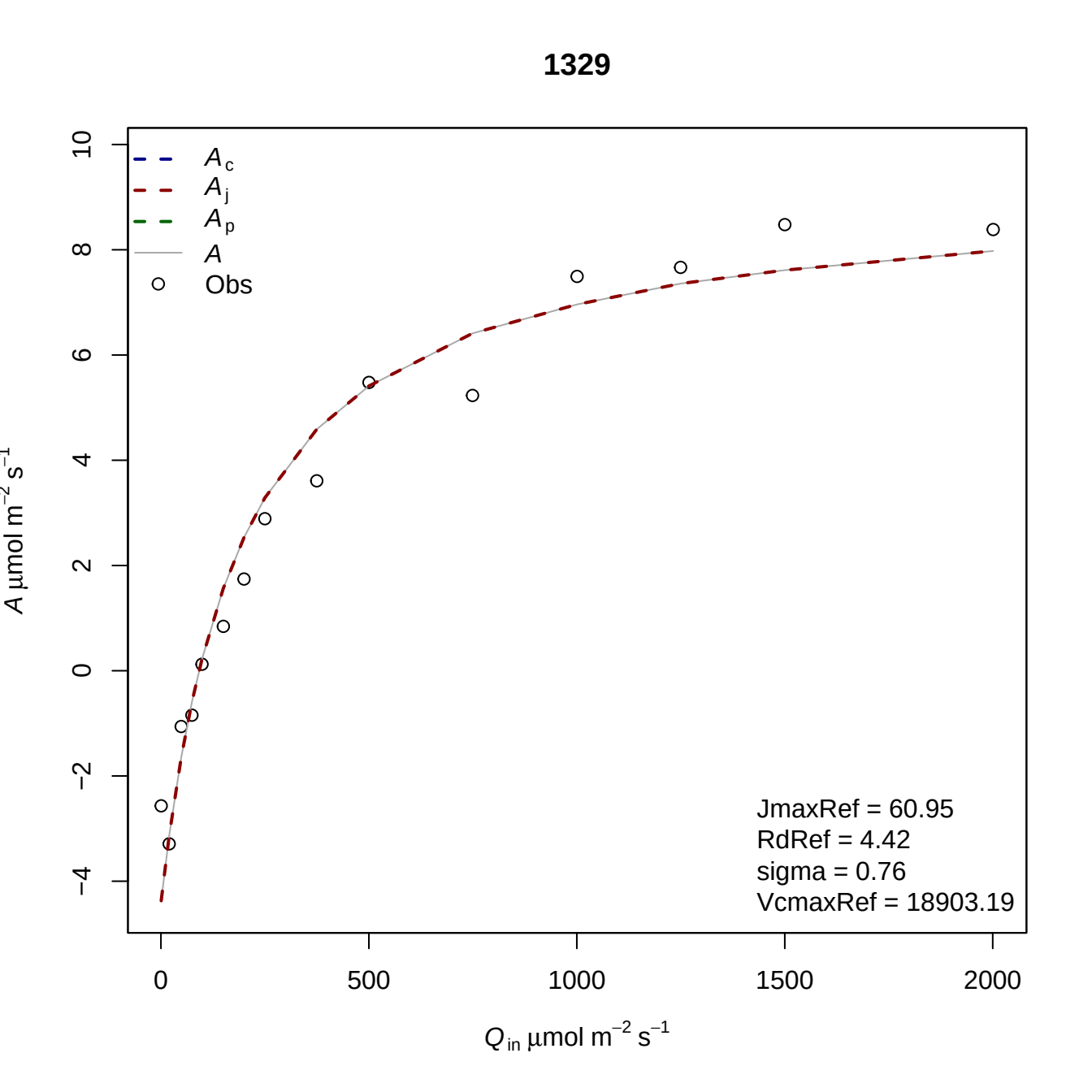

1324
A_c
A_j
A_p
A
Obs
JmaxRef = 46.99
RdRef = 2.33
sigma = 0.55
VcmaxRef = 2493.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

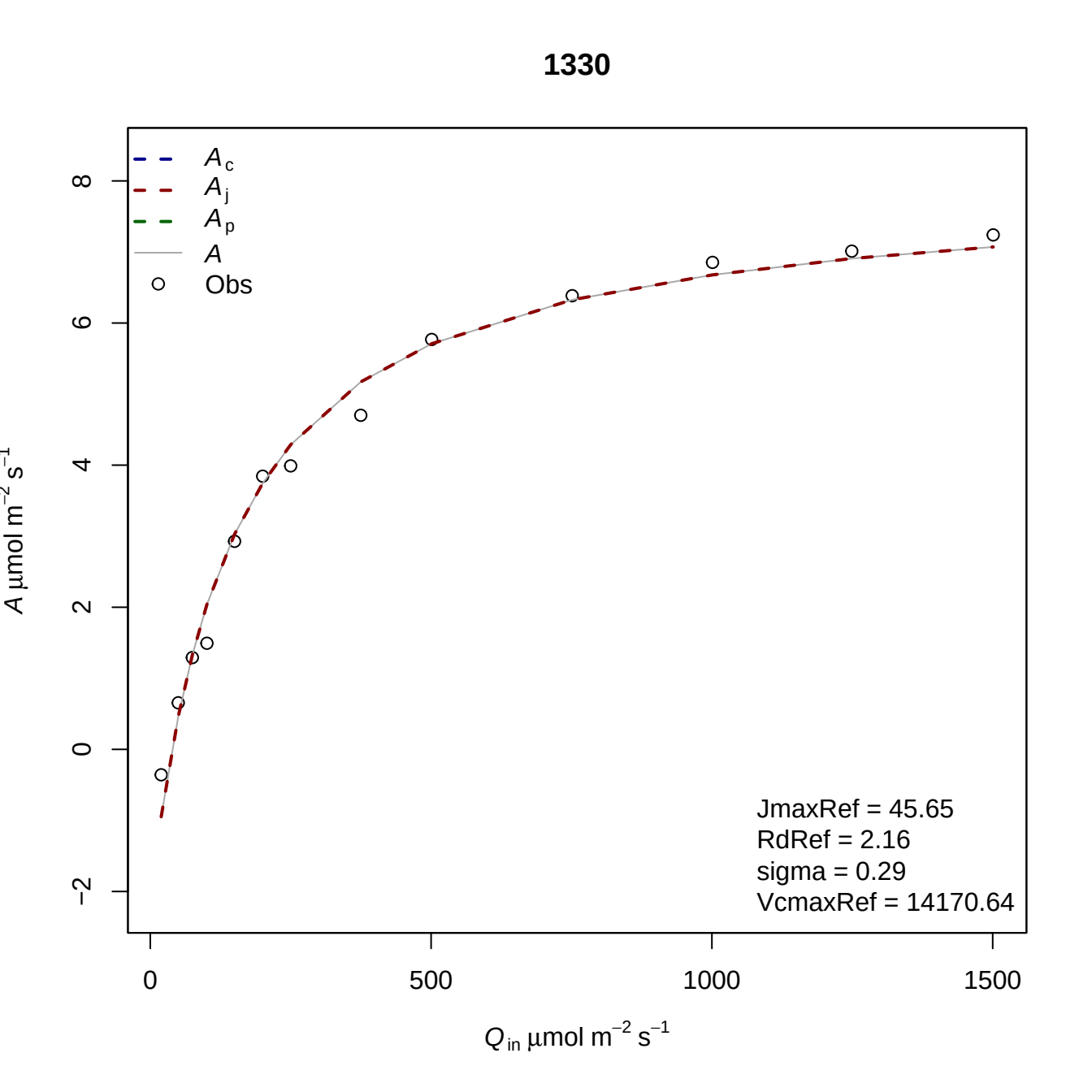

1325
A_c
A_j
A_p
A
Obs
JmaxRef = 45.51
RdRef = 1.82
sigma = 0.3
VcmaxRef = 38601.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000

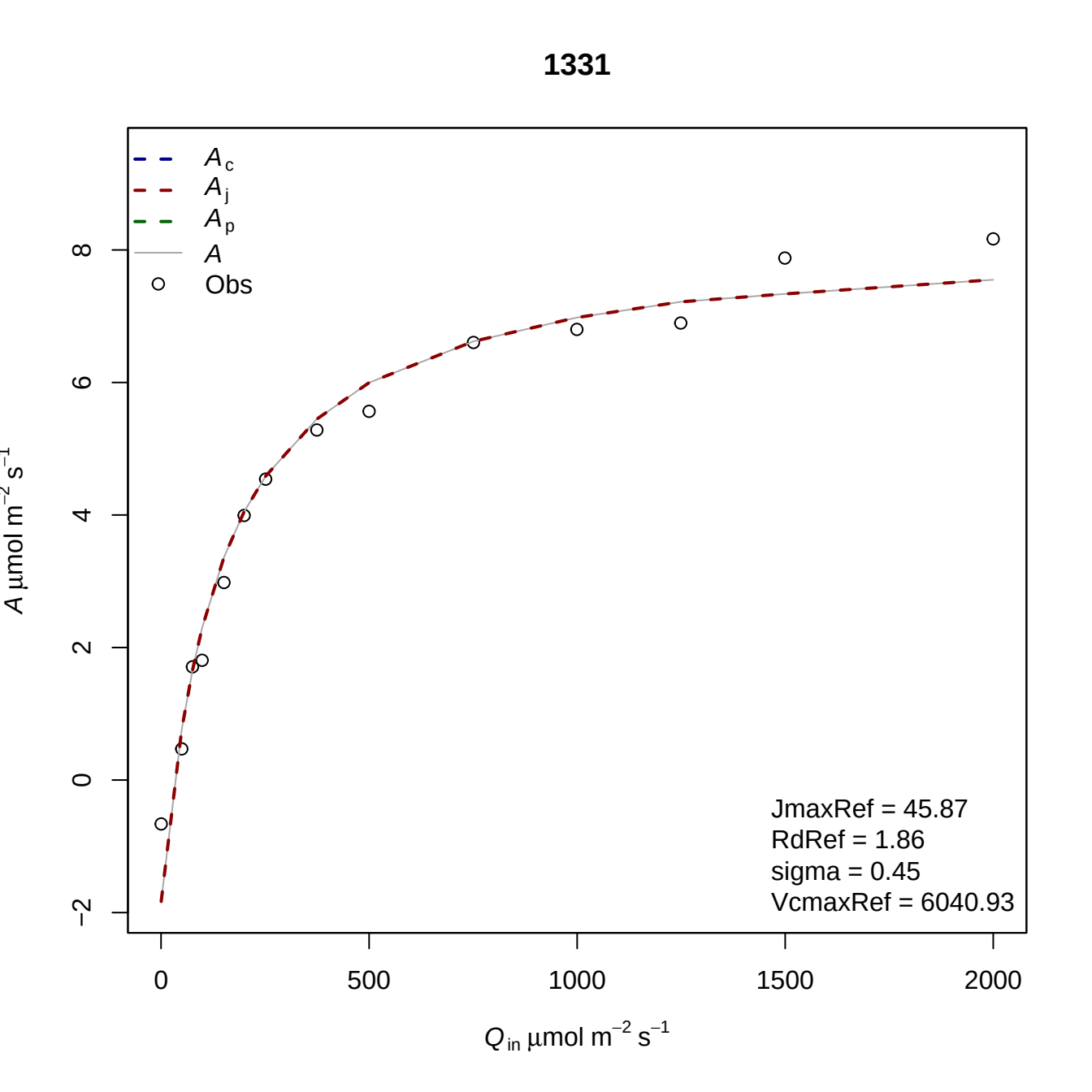

1326
A_c
A_j
A_p
A
Obs
JmaxRef = 36.85
RdRef = 2.29
sigma = 0.51
VcmaxRef = 16472.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

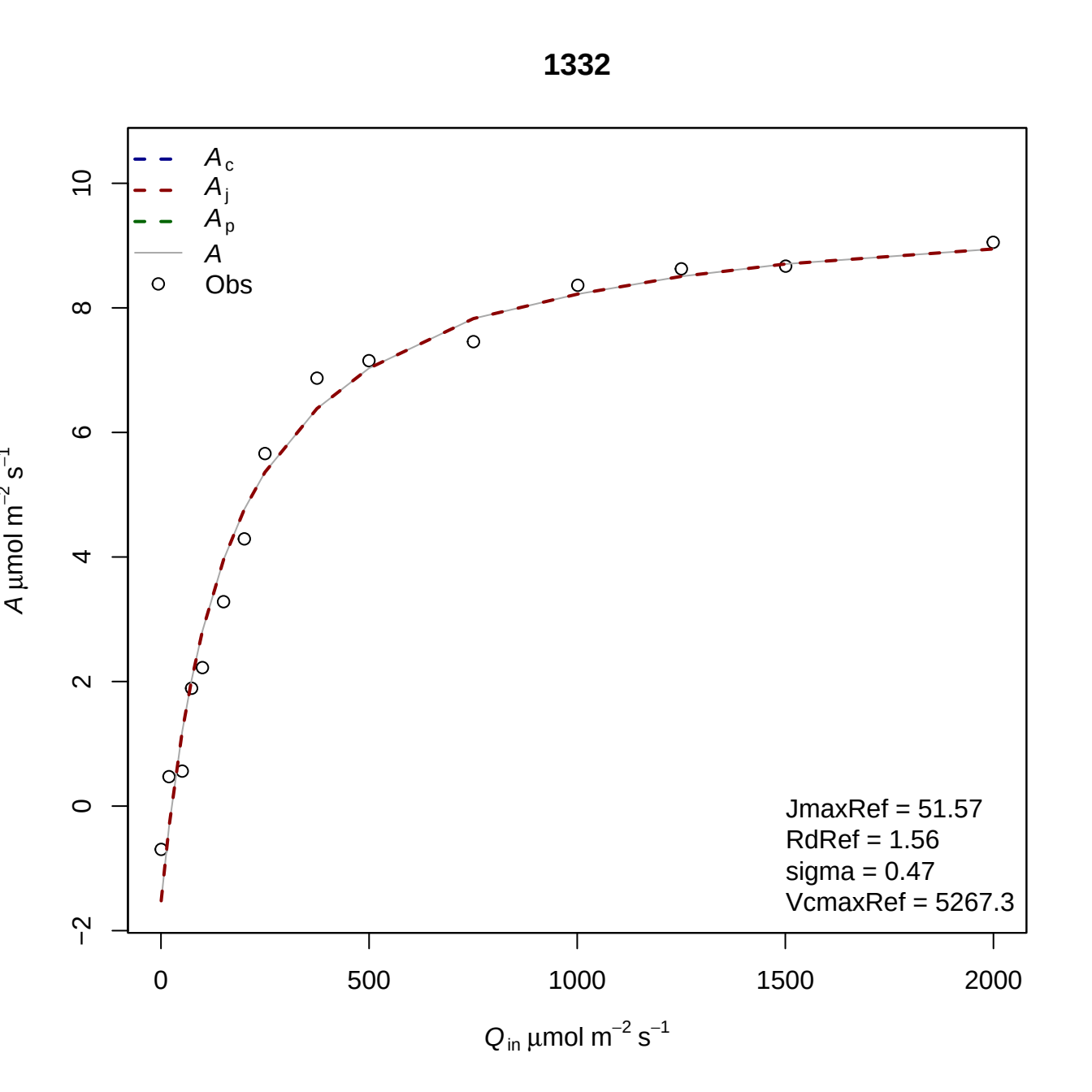

1327
A_c
A_j
A_p
A
Obs
JmaxRef = 54.28
RdRef = 3.16
sigma = 0.83
VcmaxRef = 10086.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

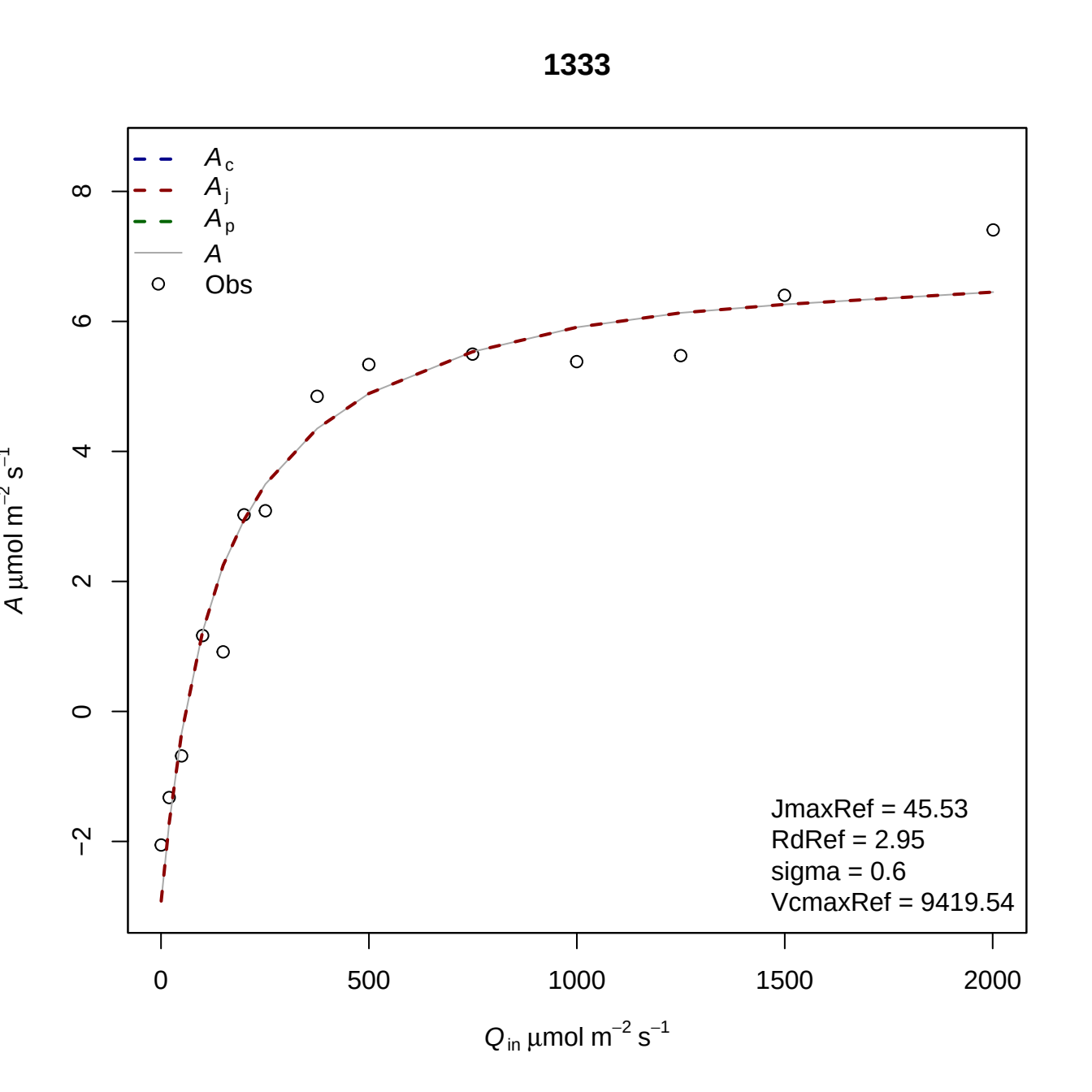

1328
A_c
A_j
A_p
A
Obs
JmaxRef = 69.56
RdRef = 3.95
sigma = 0.78
VcmaxRef = 37345.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

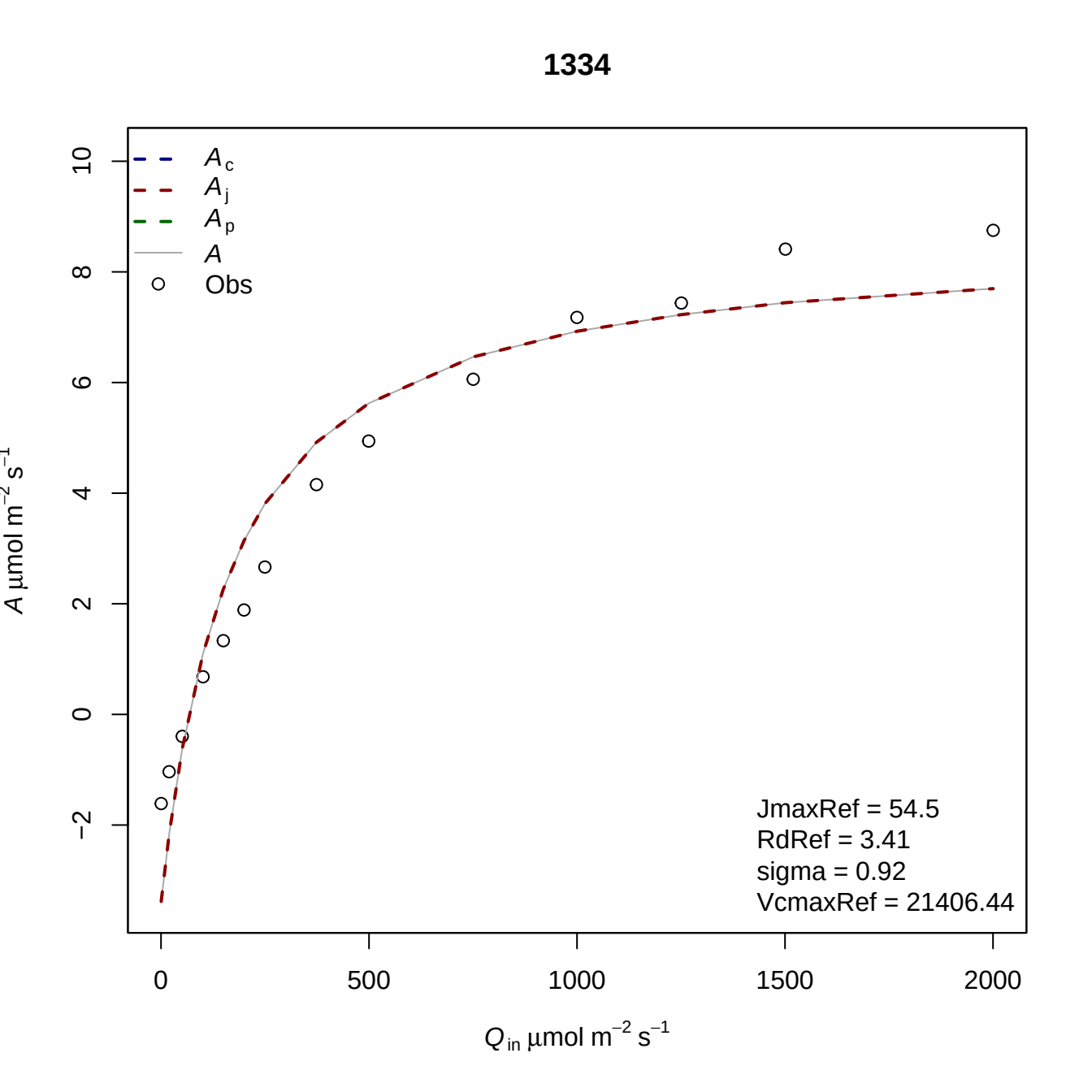

1329
A_c
A_j
A_p
A
Obs
JmaxRef = 60.95
RdRef = 4.42
sigma = 0.76
VcmaxRef = 18903.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

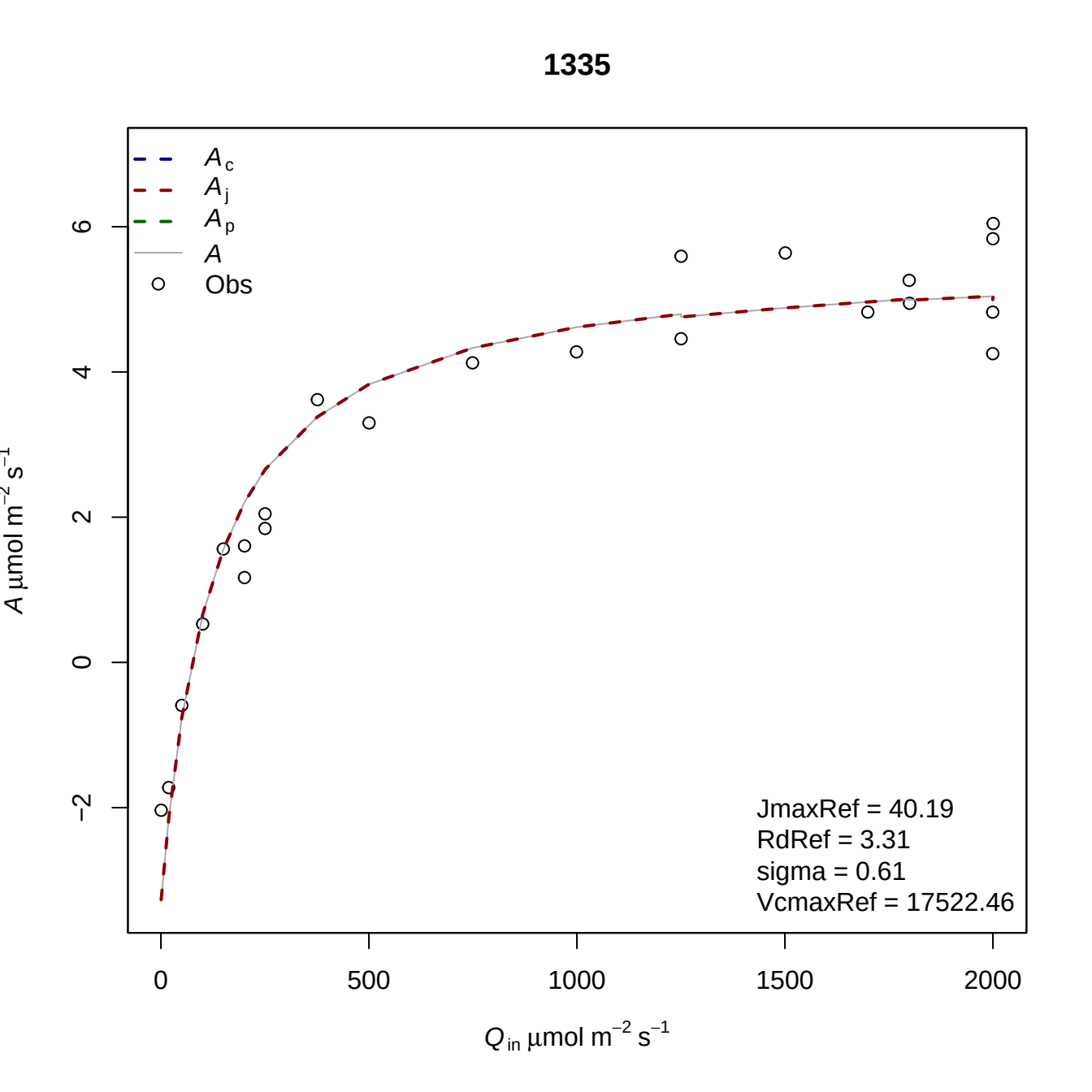

1330
A_c
A_j
A_p
A
Obs
JmaxRef = 45.65
RdRef = 2.16
sigma = 0.29
VcmaxRef = 14170.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

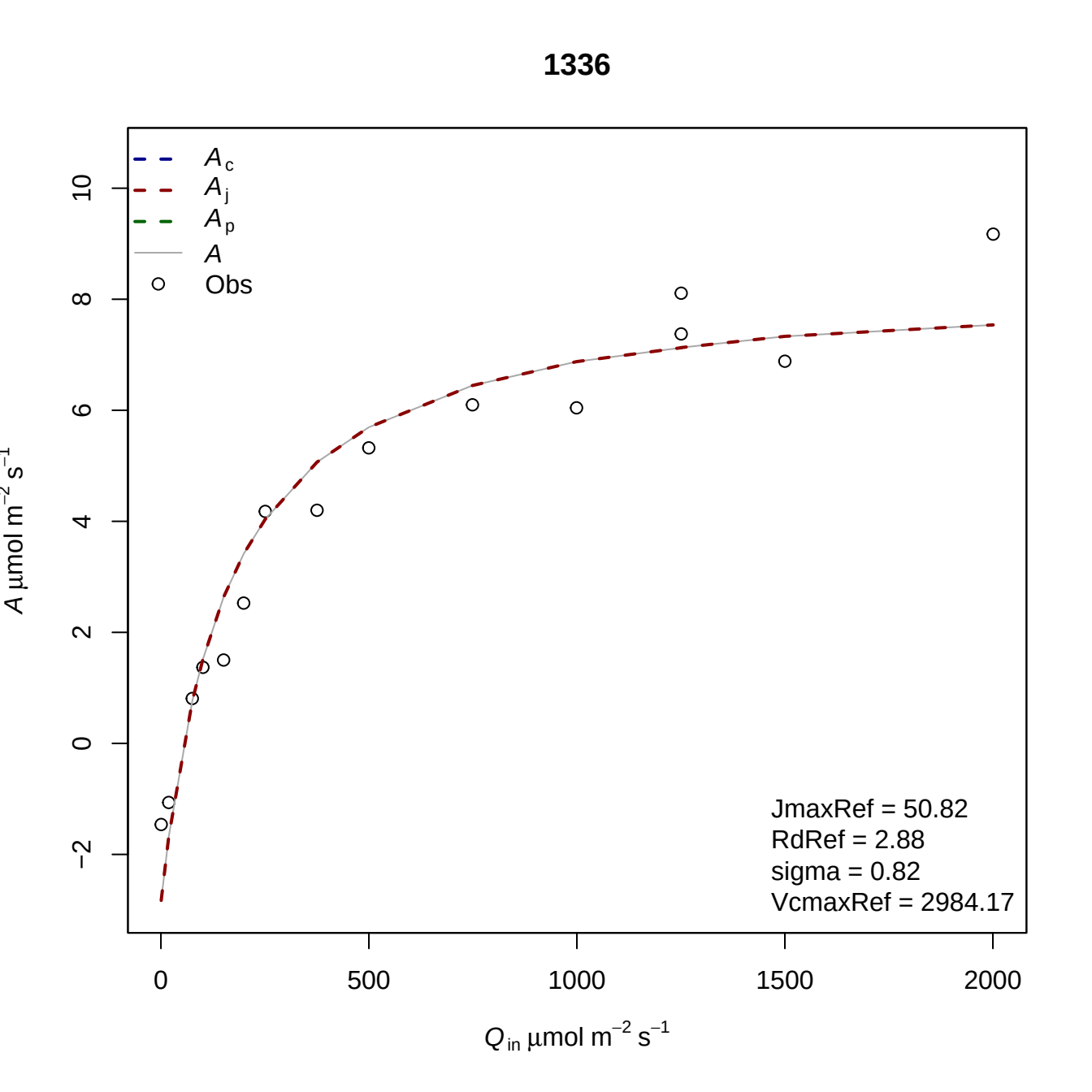

1331
A_c
A_j
A_p
A
Obs
JmaxRef = 45.87
RdRef = 1.86
sigma = 0.45
VcmaxRef = 6040.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

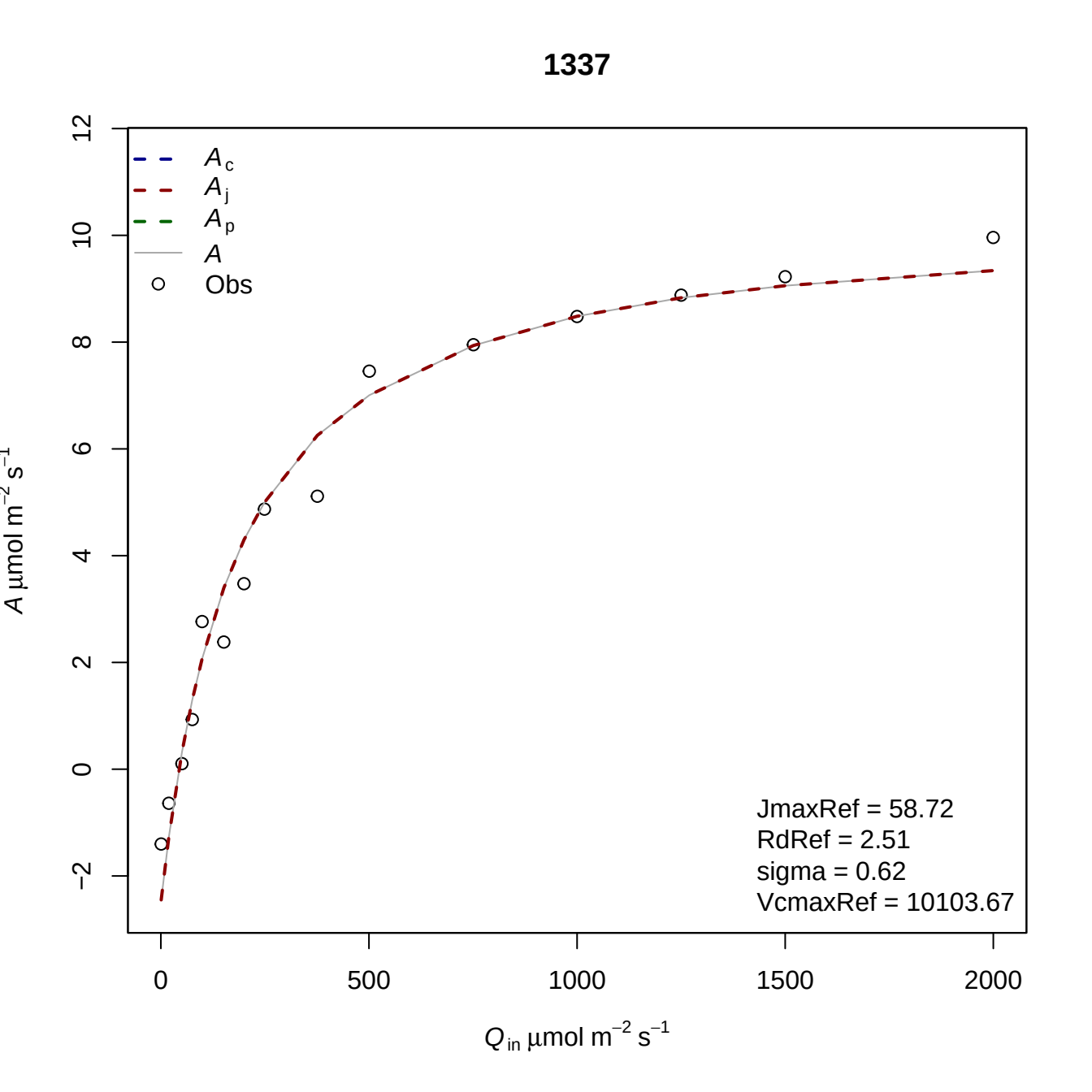

1332
A_c
A_j
A_p
A
Obs
JmaxRef = 51.57
RdRef = 1.56
sigma = 0.47
VcmaxRef = 5267.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

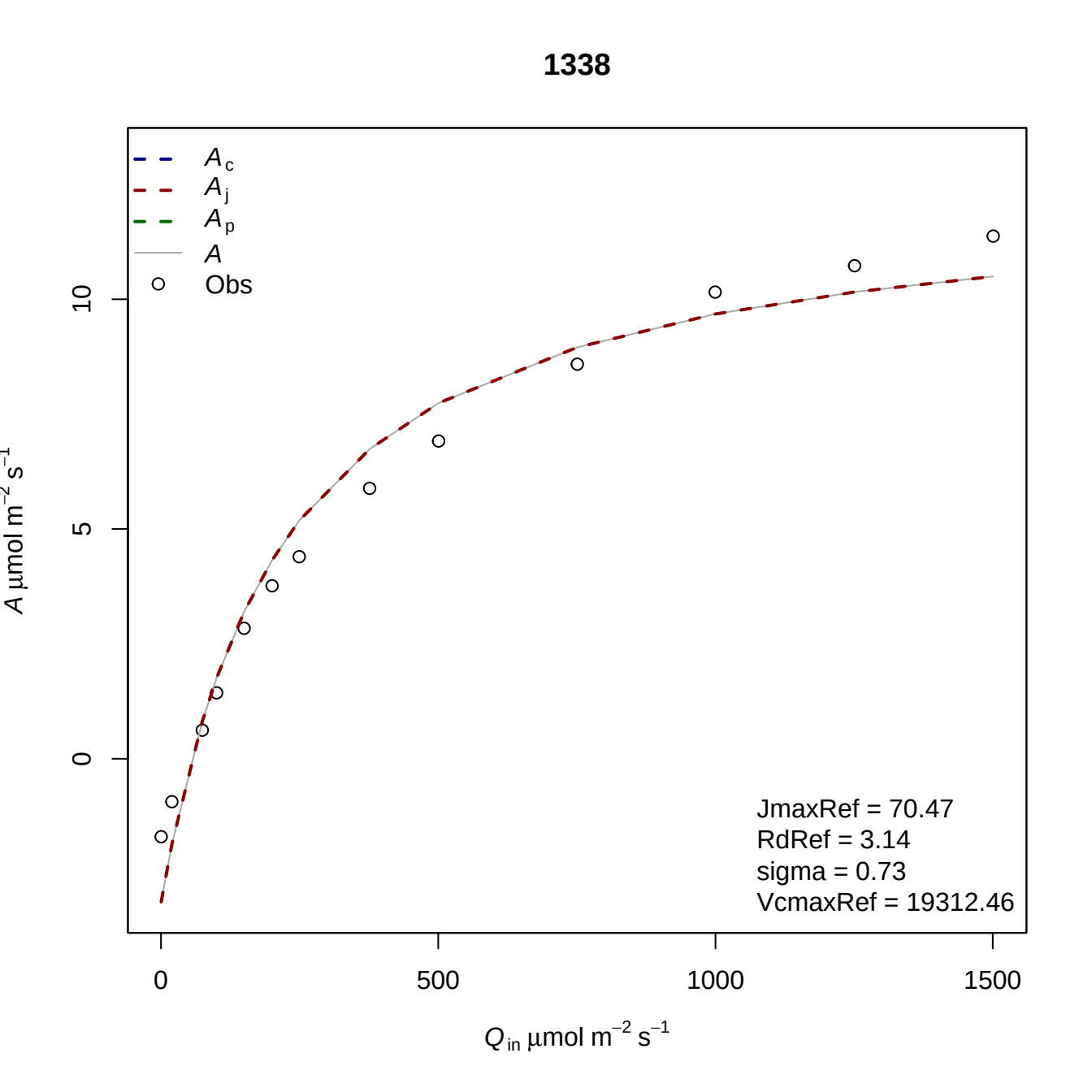

1333
A_c
A_j
A_p
A
Obs
JmaxRef = 45.53
RdRef = 2.95
sigma = 0.6
VcmaxRef = 9419.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

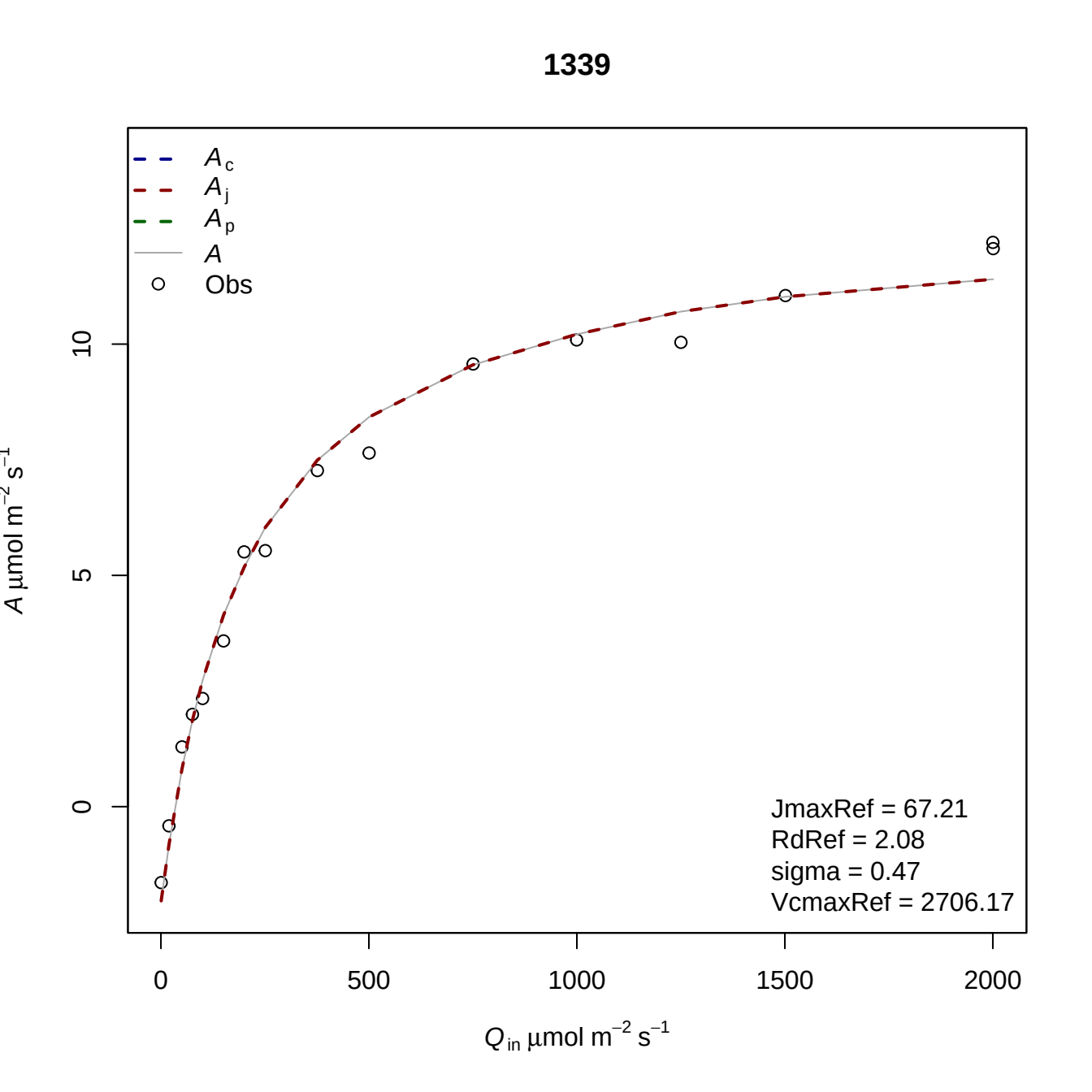

1334
A_c
A_j
A_p
A
Obs
JmaxRef = 54.5
RdRef = 3.41
sigma = 0.92
VcmaxRef = 21406.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

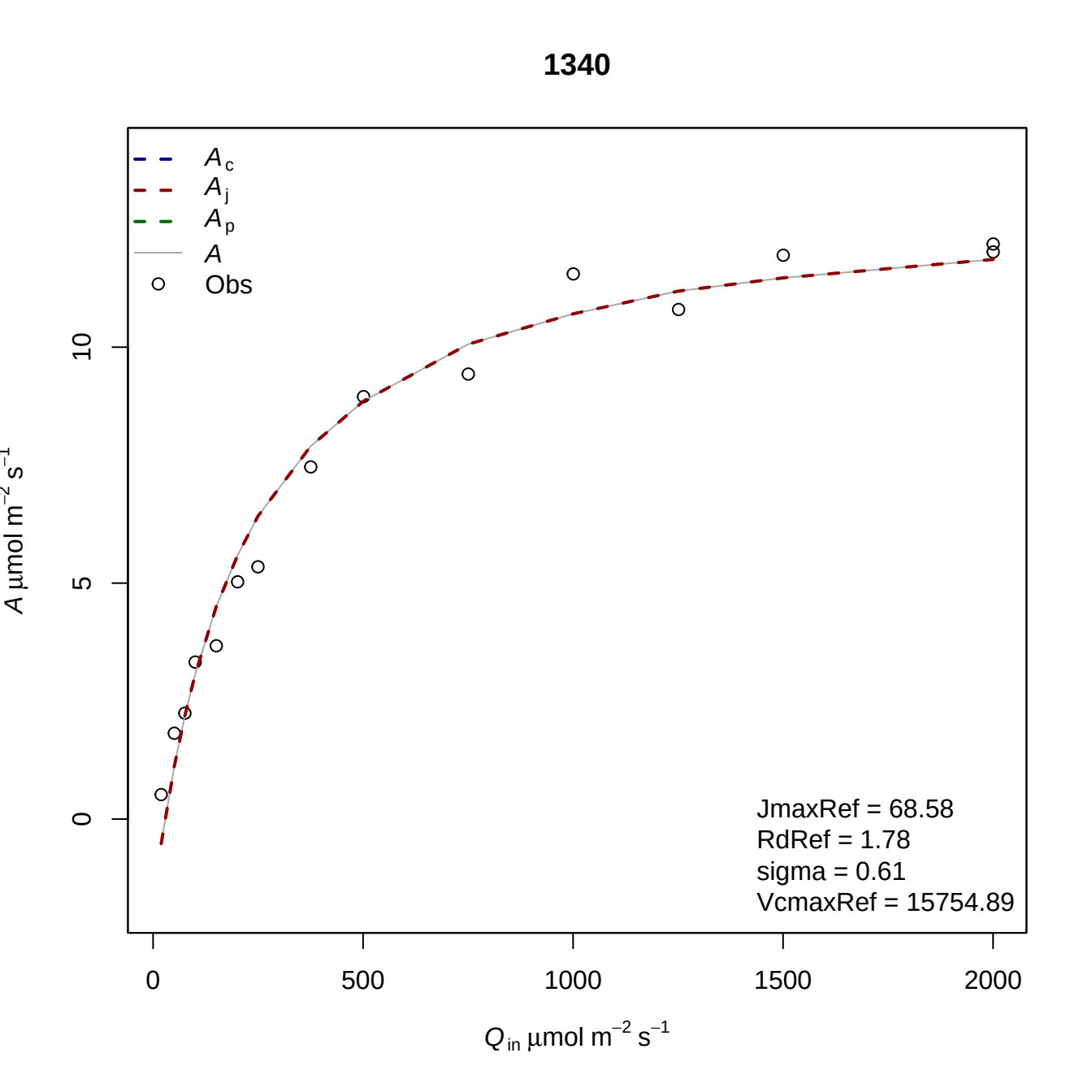

1335
A_c
A_j
A_p
A
Obs
JmaxRef = 40.19
RdRef = 3.31
sigma = 0.61
VcmaxRef = 17522.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

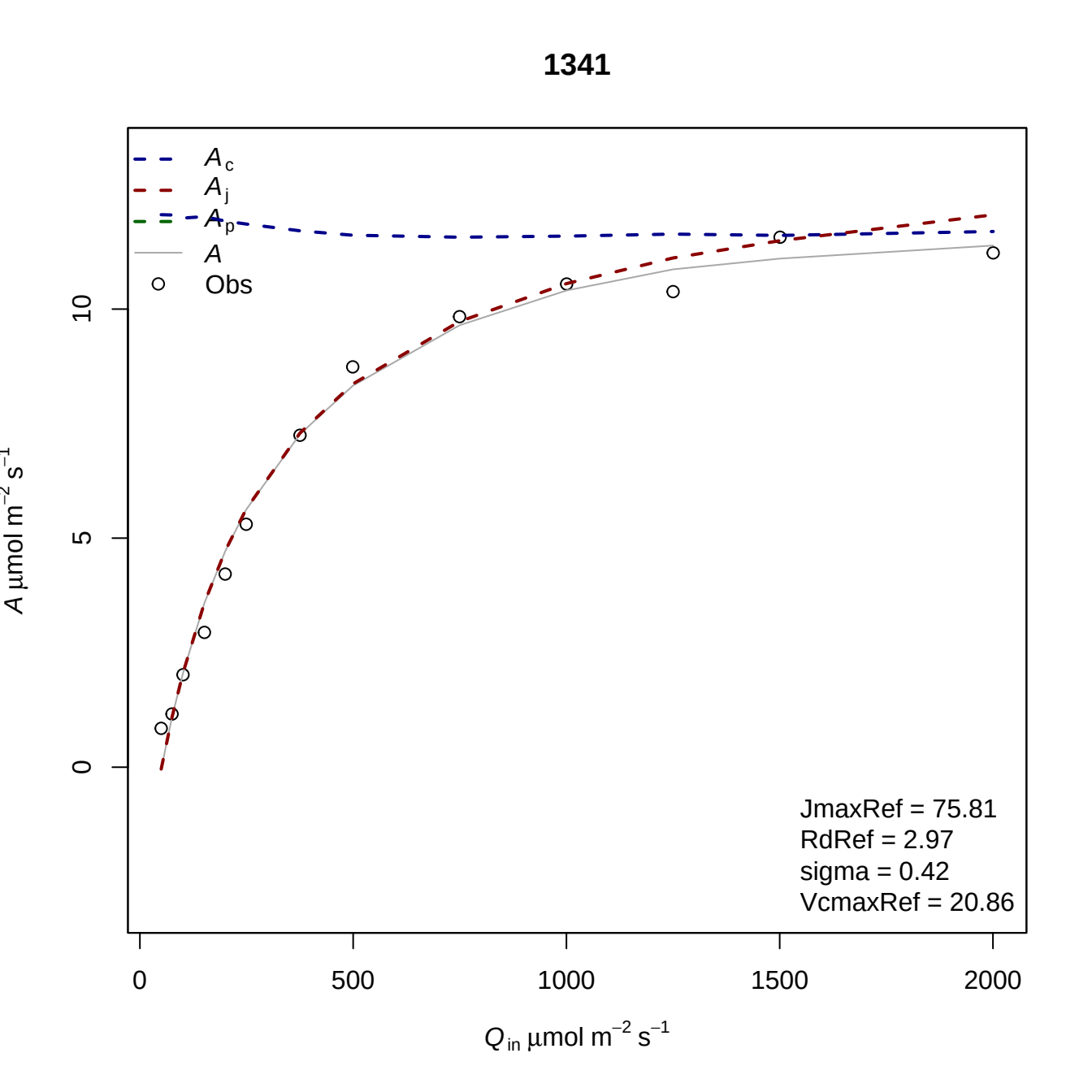

1336
A_c
A_j
A_p
A
Obs
JmaxRef = 50.82
RdRef = 2.88
sigma = 0.82
VcmaxRef = 2984.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

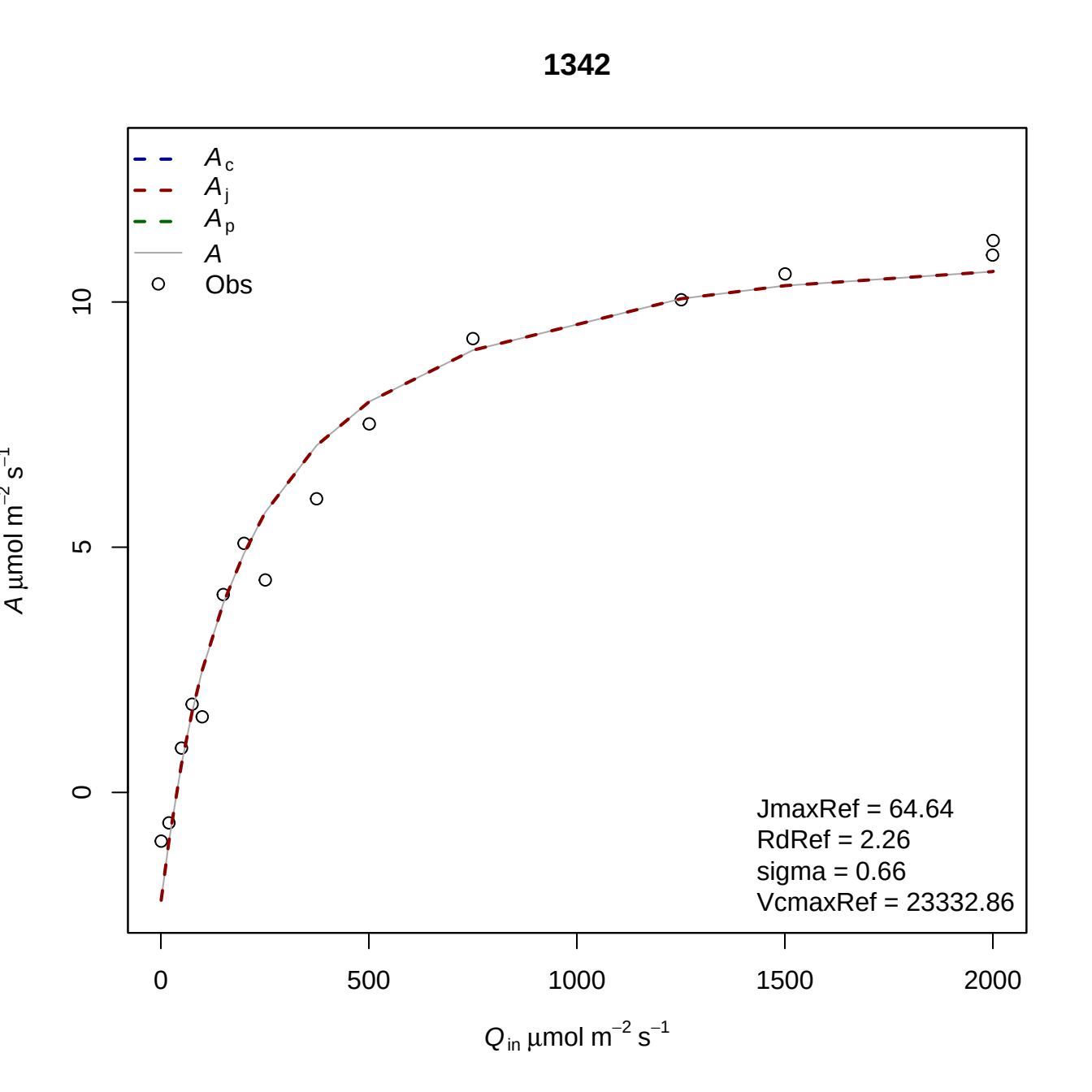

1337
A_c
A_j
A_p
A
Obs
JmaxRef = 58.72
RdRef = 2.51
sigma = 0.62
VcmaxRef = 10103.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

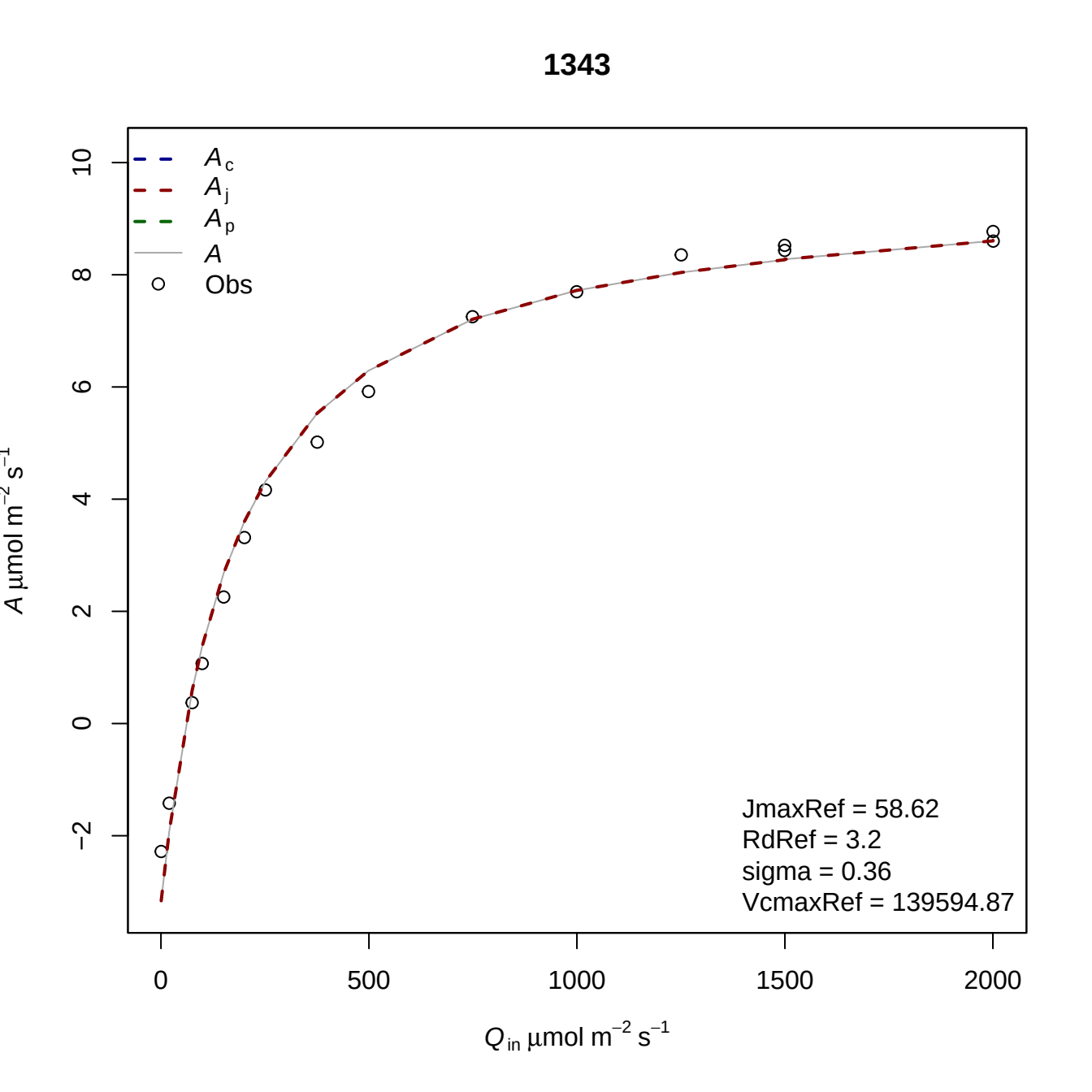

1338
A_c
A_j
A_p
A
Obs
JmaxRef = 70.47
RdRef = 3.14
sigma = 0.73
VcmaxRef = 19312.46
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

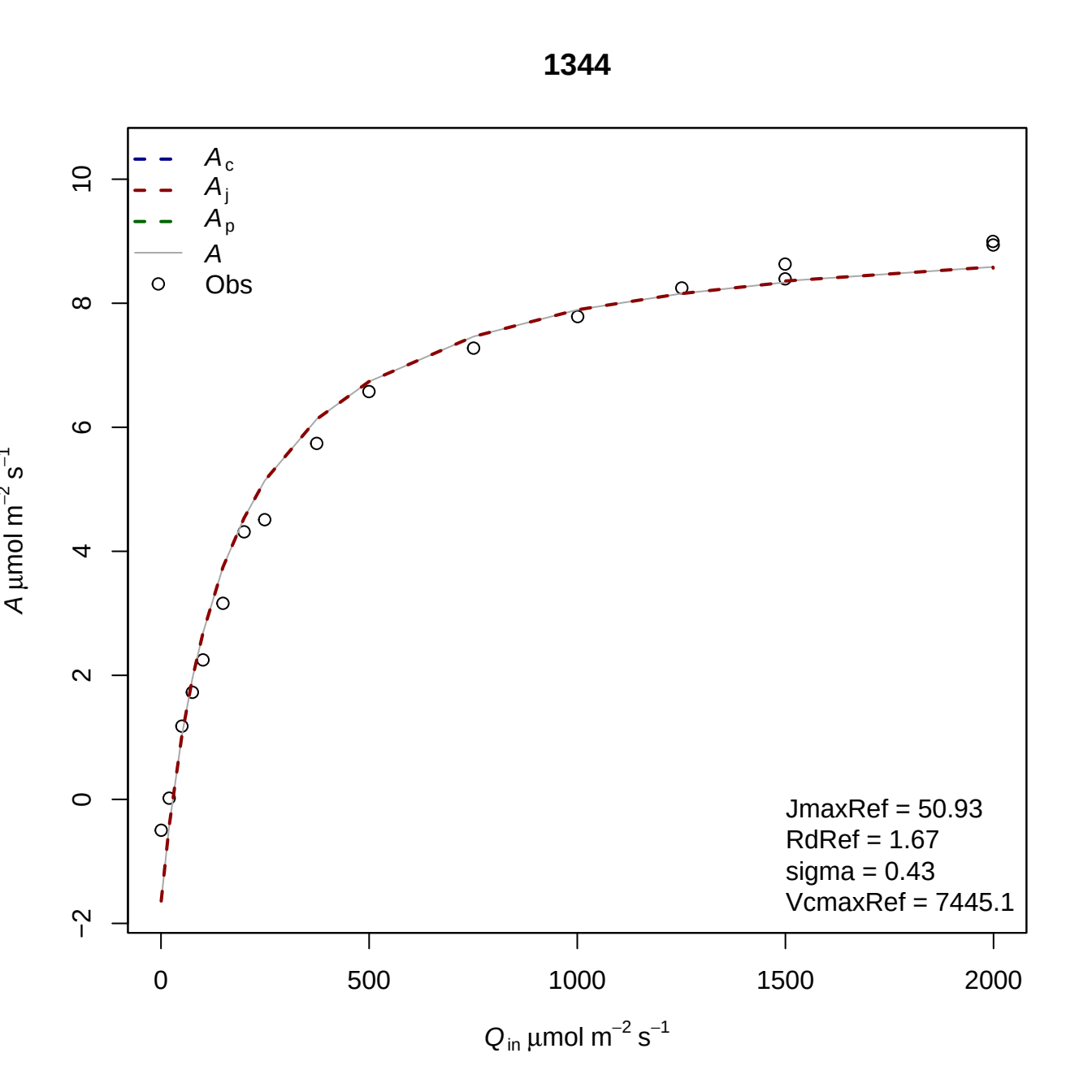

1339
A_c
A_j
A_p
A
Obs
JmaxRef = 67.21
RdRef = 2.08
sigma = 0.47
VcmaxRef = 2706.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

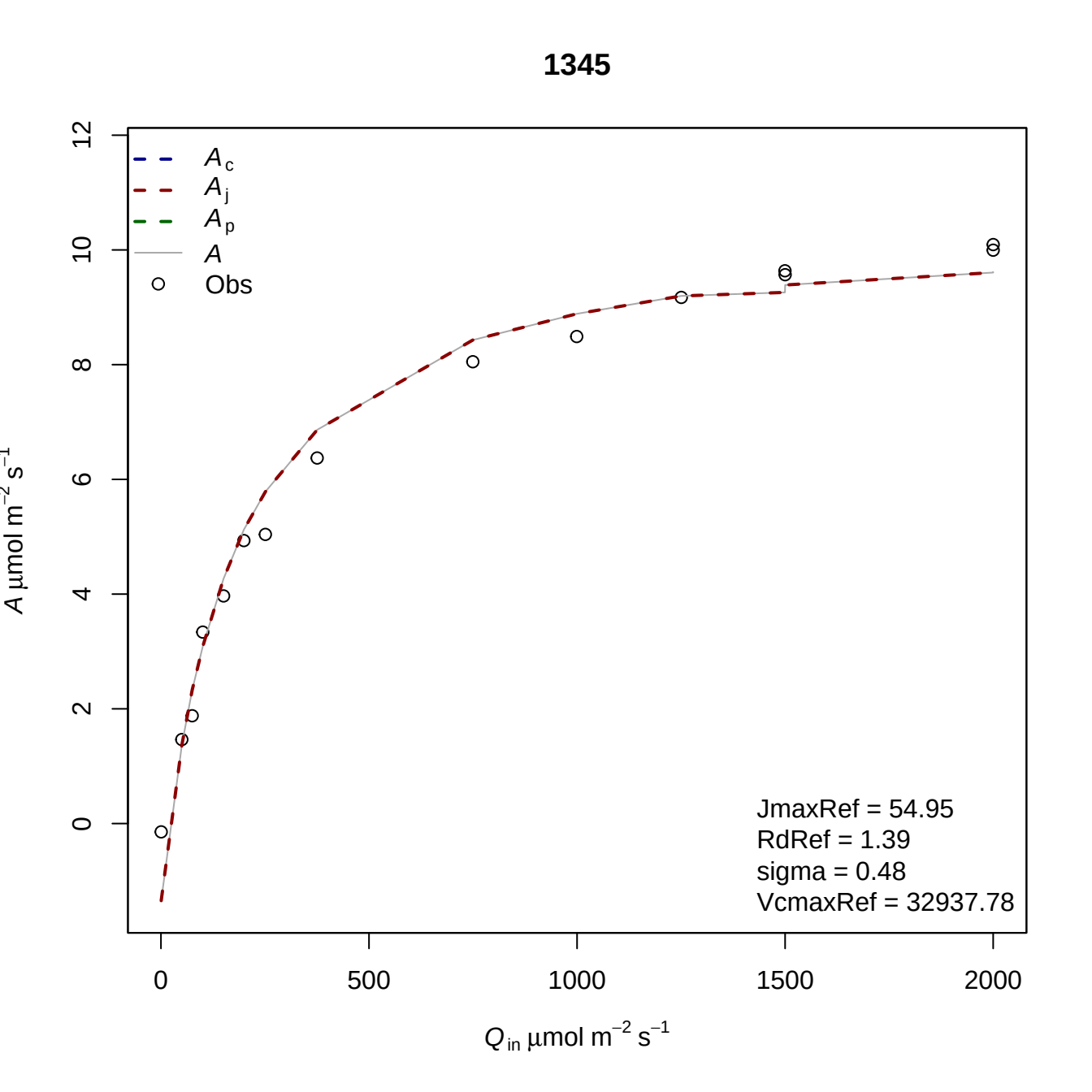

1340
A_c
A_j
A_p
A
Obs
JmaxRef = 68.58
RdRef = 1.78
sigma = 0.61
VcmaxRef = 15754.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

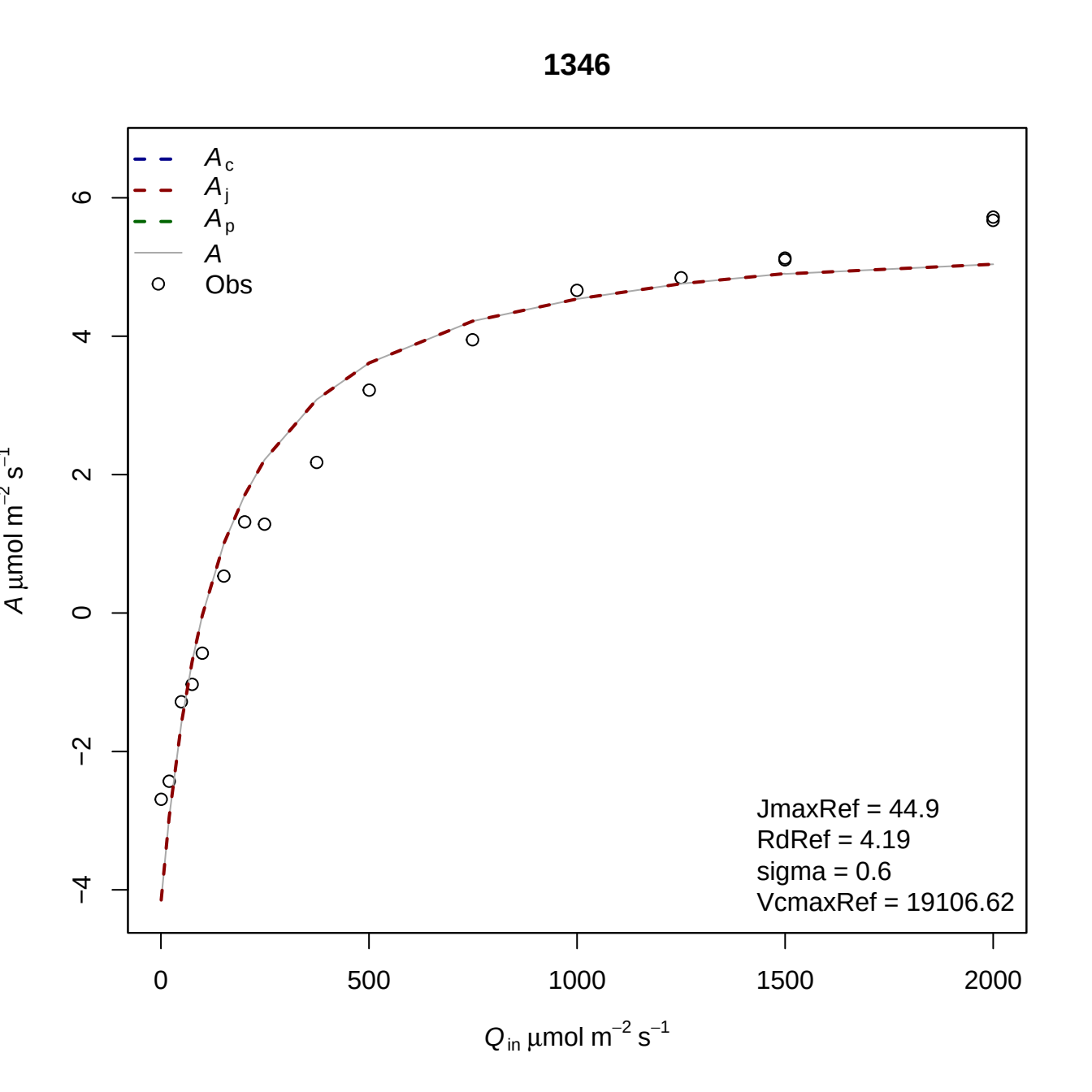

1341
A_c
A_j
A_p
A
Obs
JmaxRef = 75.81
RdRef = 2.97
sigma = 0.42
VcmaxRef = 20.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

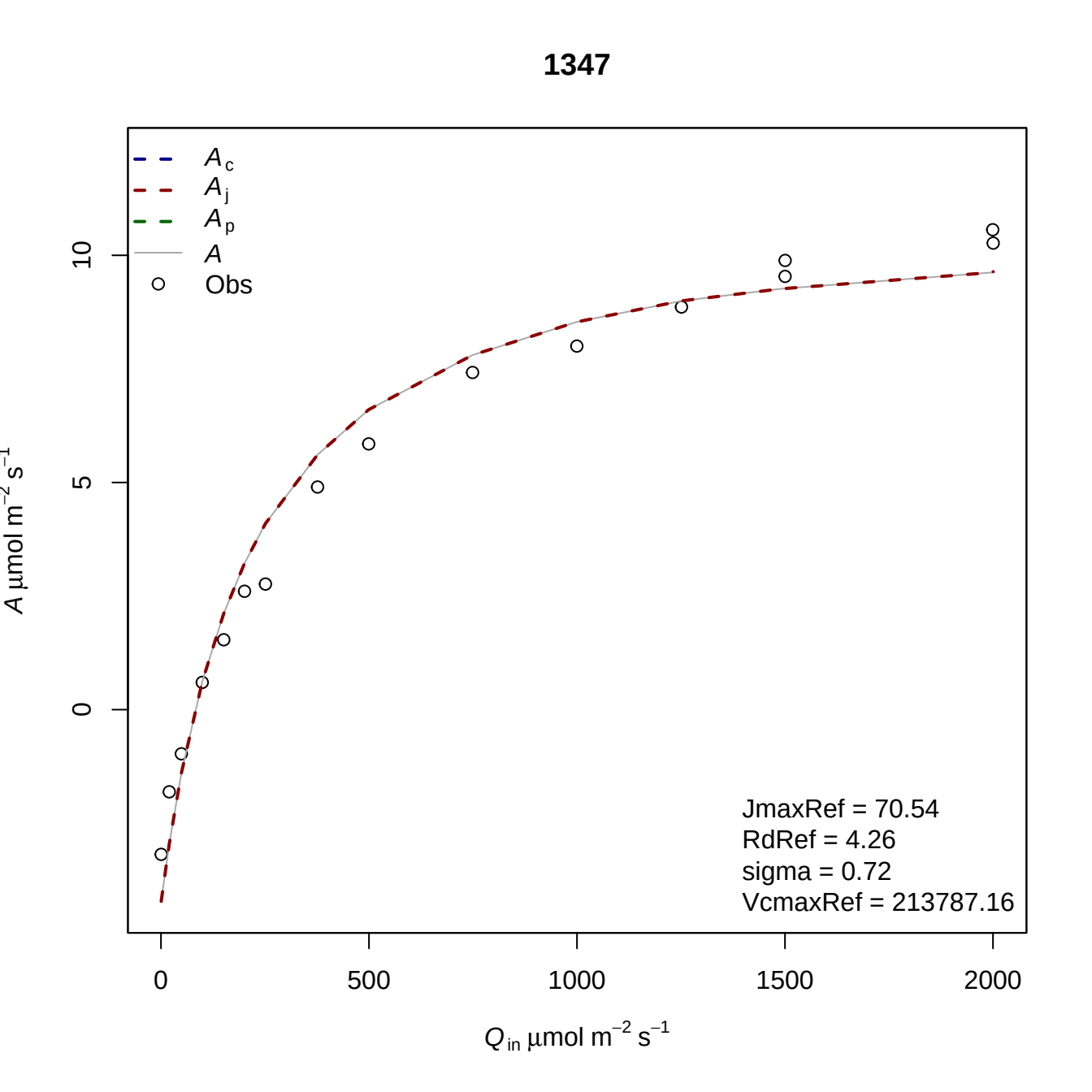

1342
A_c
A_j
A_p
A
Obs
JmaxRef = 64.64
RdRef = 2.26
sigma = 0.66
VcmaxRef = 23332.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

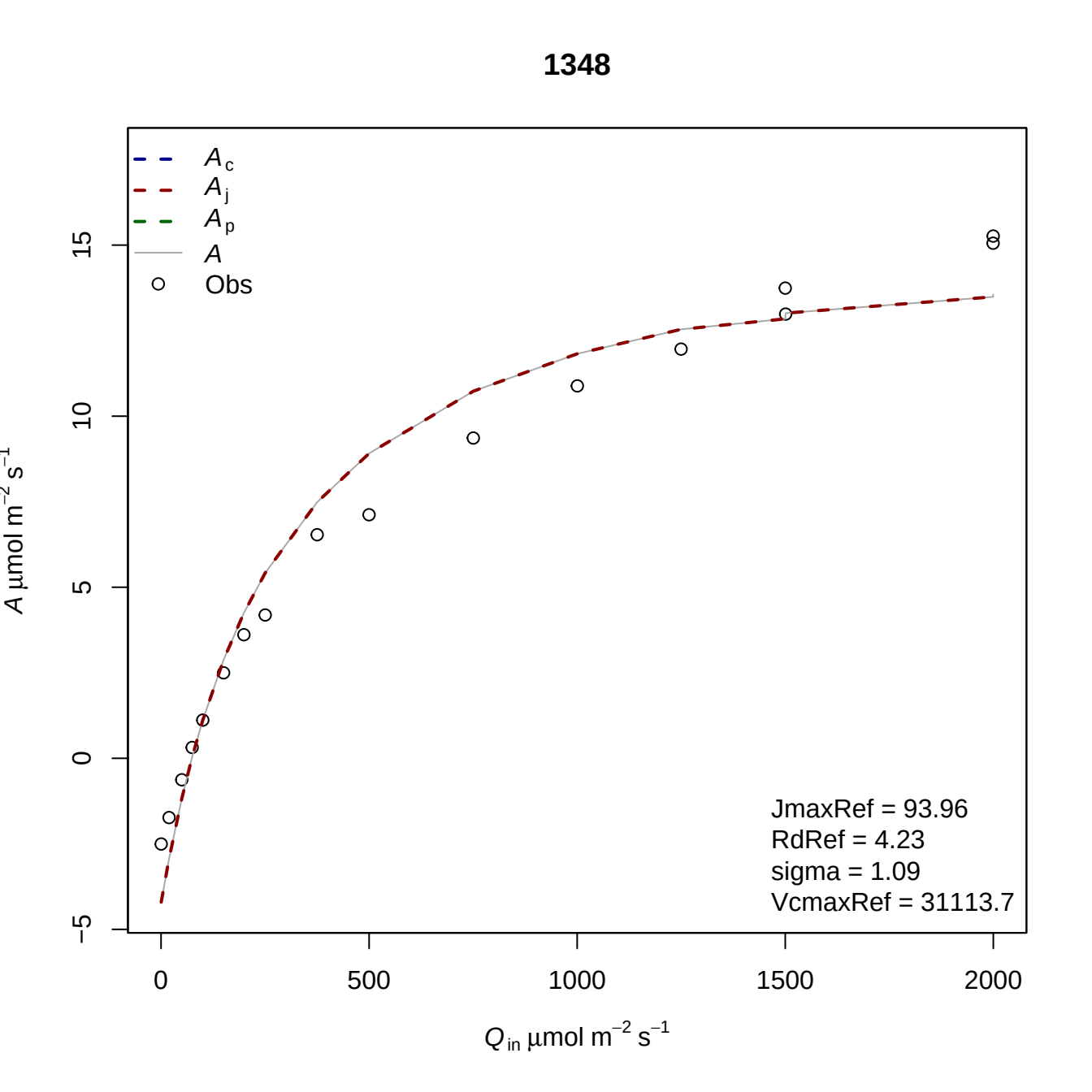

1343
A_c
A_j
A_p
A
Obs
JmaxRef = 58.62
RdRef = 3.2
sigma = 0.36
VcmaxRef = 139594.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

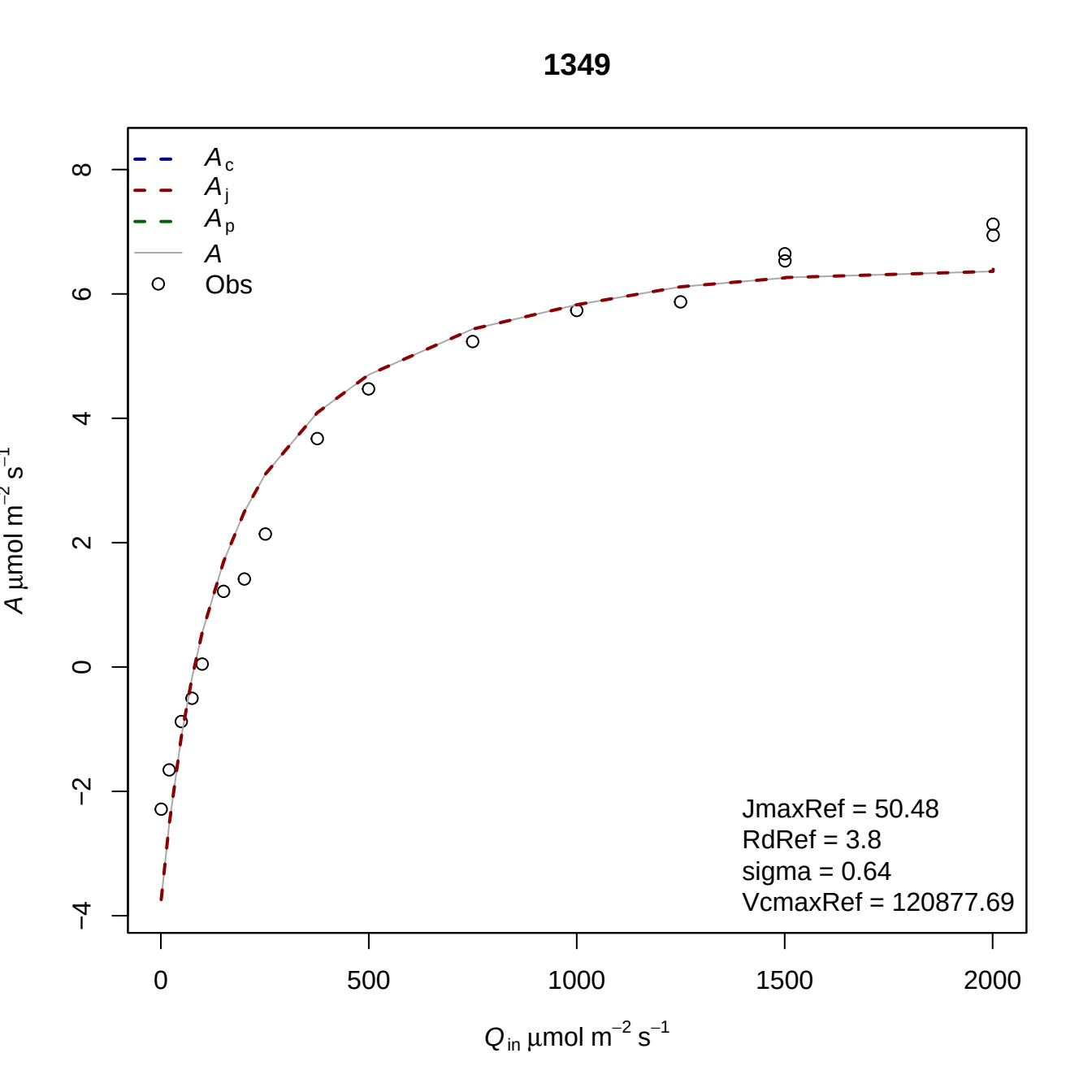

1344
A_c
A_j
A_p
A
Obs
JmaxRef = 50.93
RdRef = 1.67
sigma = 0.43
VcmaxRef = 7445.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

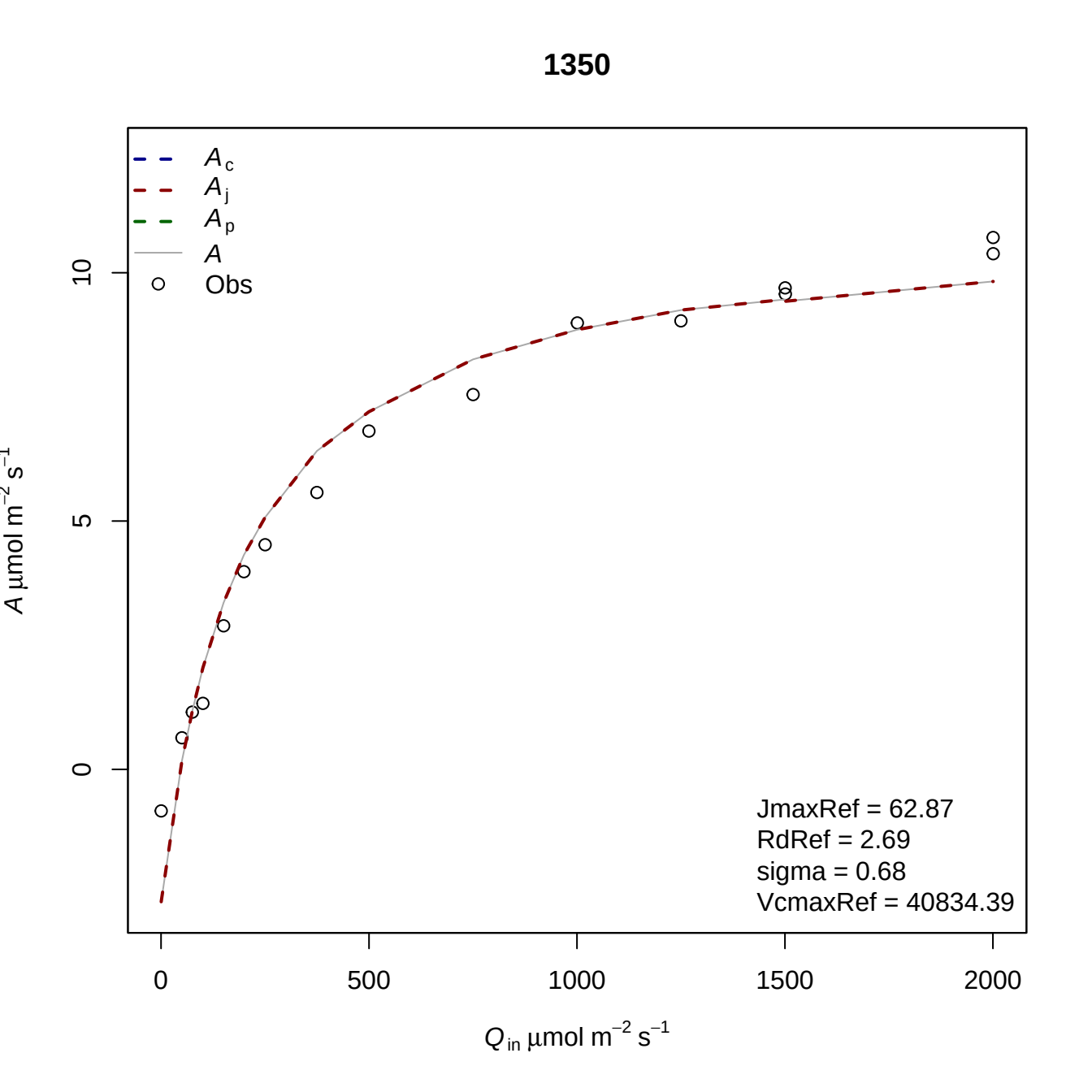

1345
A_c
A_j
A_p
A
Obs
JmaxRef = 54.95
RdRef = 1.39
sigma = 0.48
VcmaxRef = 32937.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

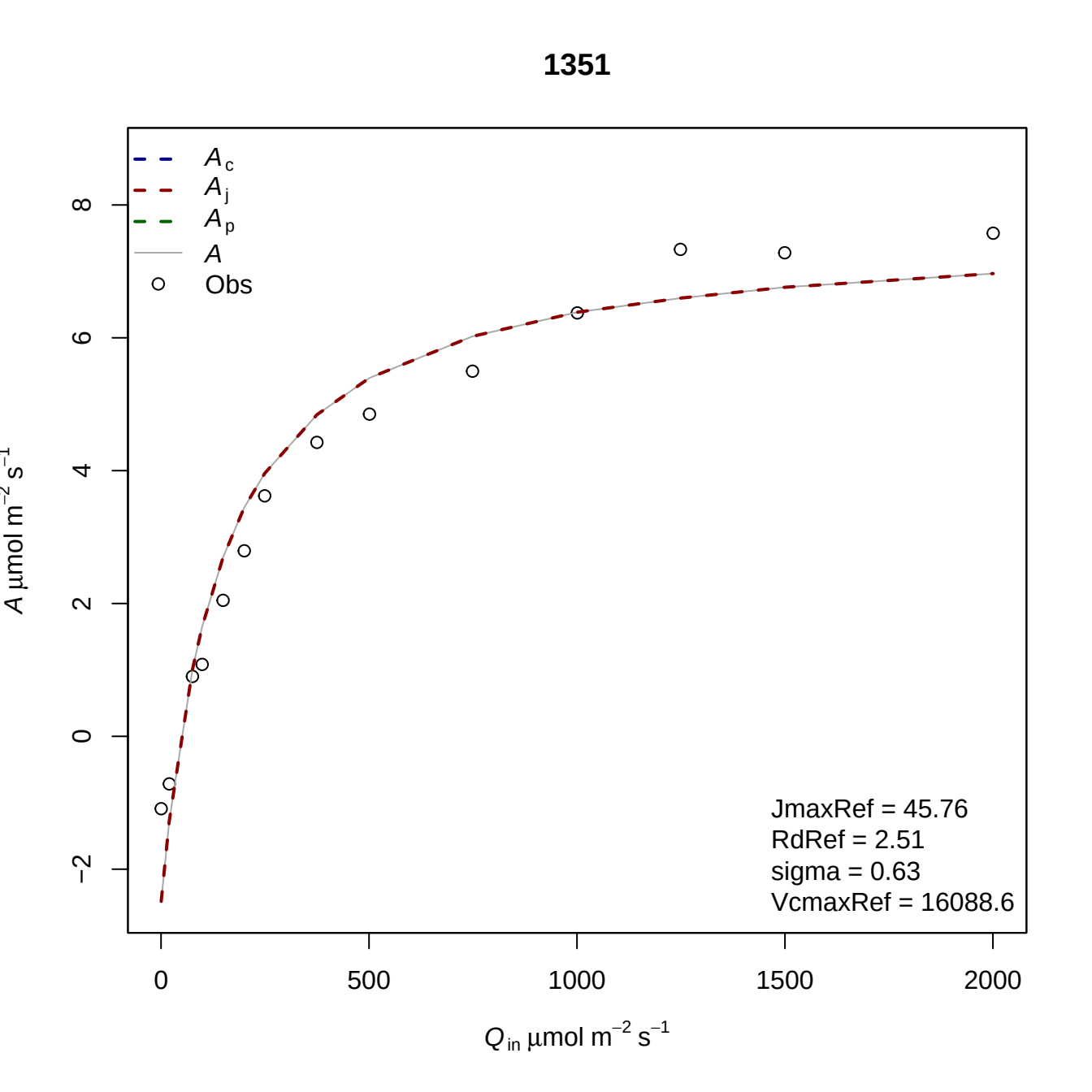

1346
A_c
A_j
A_p
A
Obs
JmaxRef = 44.9
RdRef = 4.19
sigma = 0.6
VcmaxRef = 19106.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

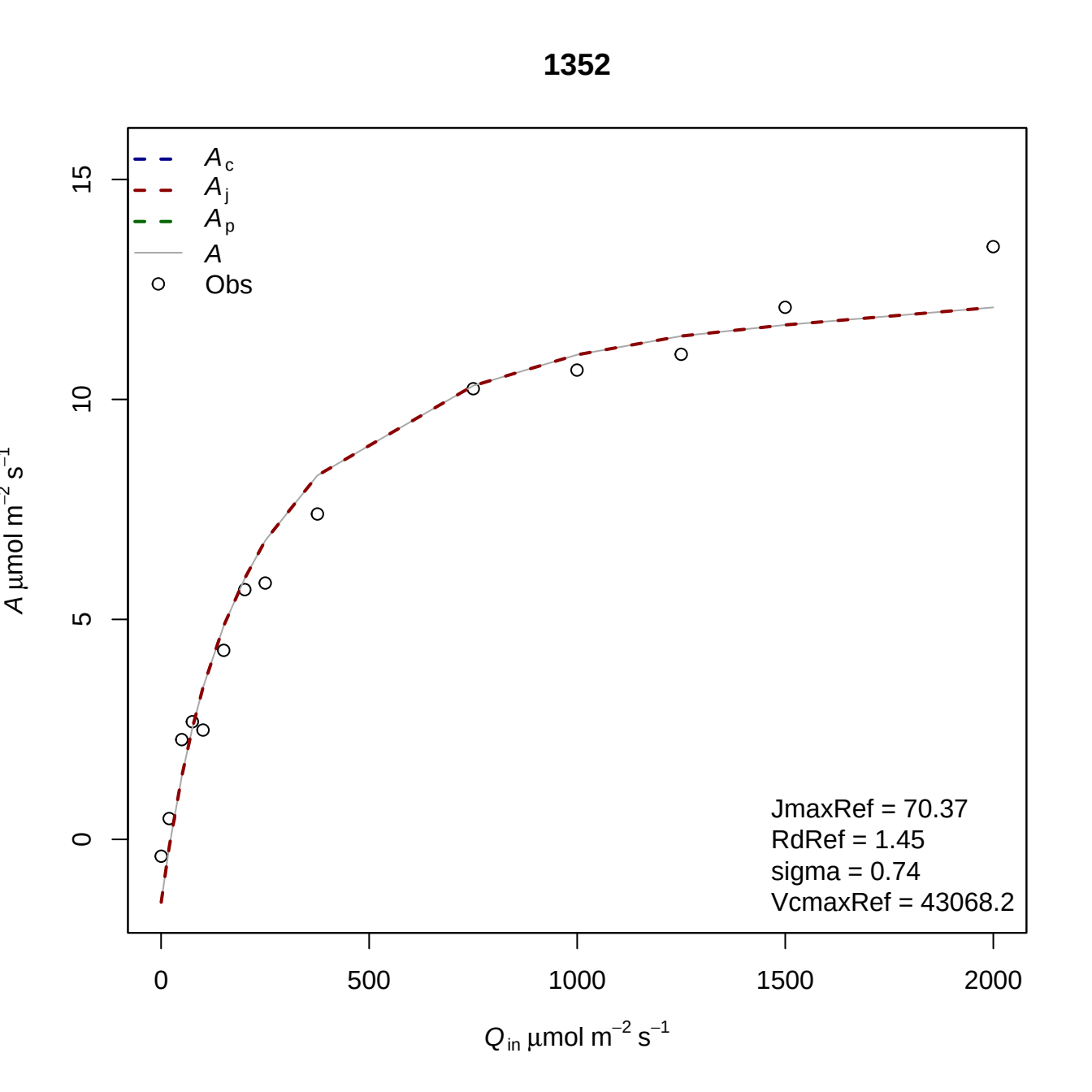

1347
A_c
A_j
A_p
A
Obs
JmaxRef = 70.54
RdRef = 4.26
sigma = 0.72
VcmaxRef = 213787.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

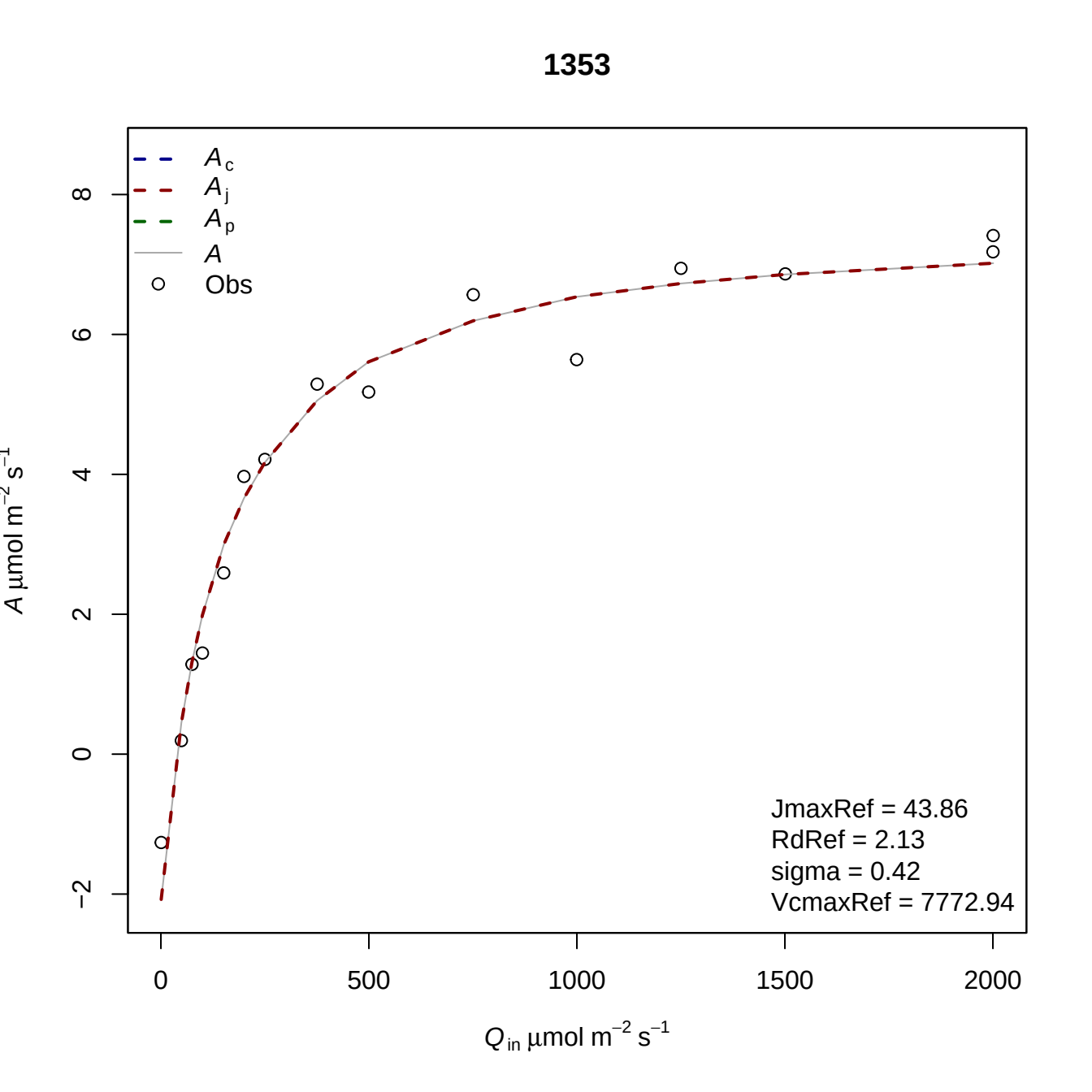

1348
A_c
A_j
A_p
A
Obs
JmaxRef = 93.96
RdRef = 4.23
sigma = 1.09
VcmaxRef = 31113.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

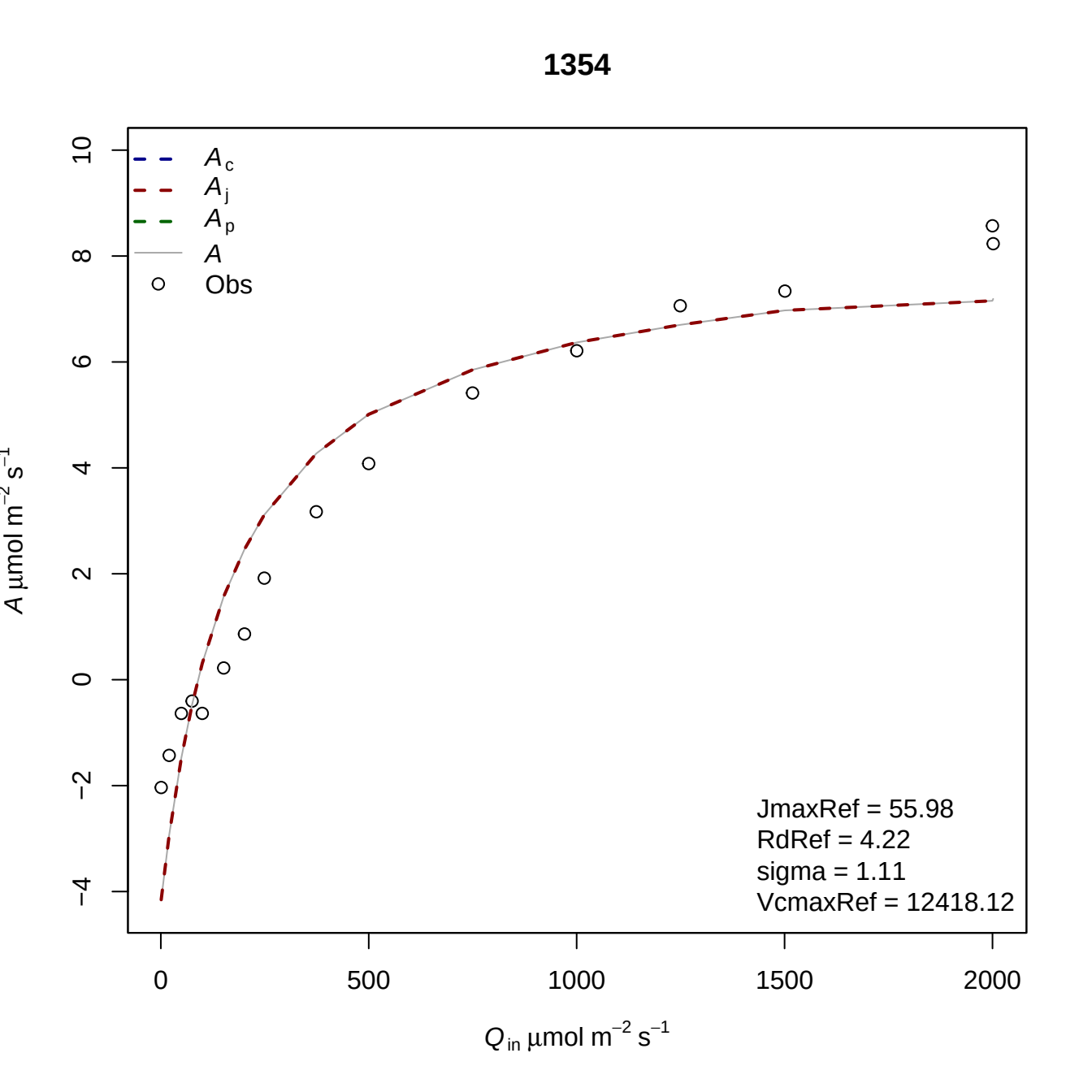

1349
A_c
A_j
A_p
A
Obs
JmaxRef = 50.48
RdRef = 3.8
sigma = 0.64
VcmaxRef = 120877.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

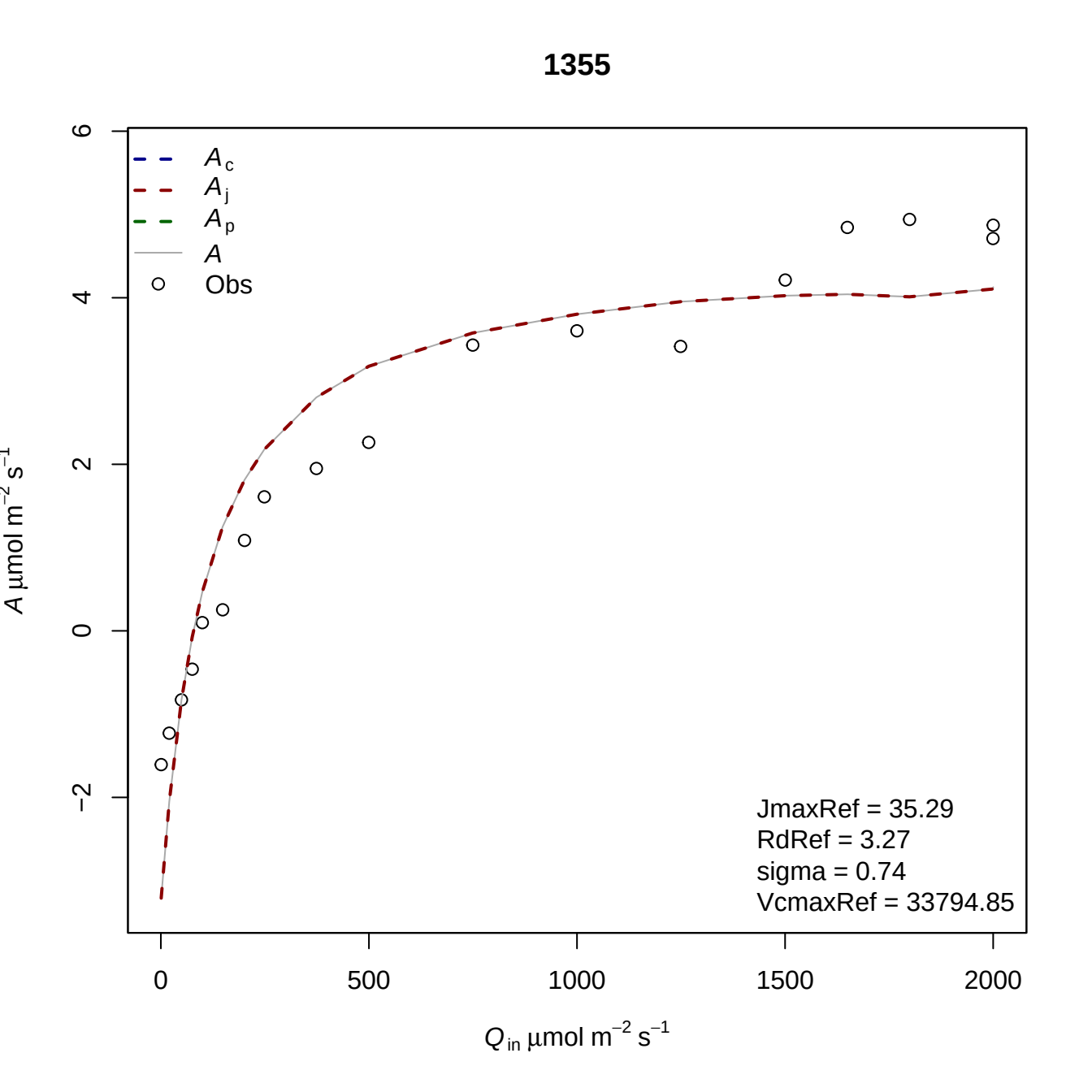

1350
A_c
A_j
A_p
A
Obs
JmaxRef = 62.87
RdRef = 2.69
sigma = 0.68
VcmaxRef = 40834.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

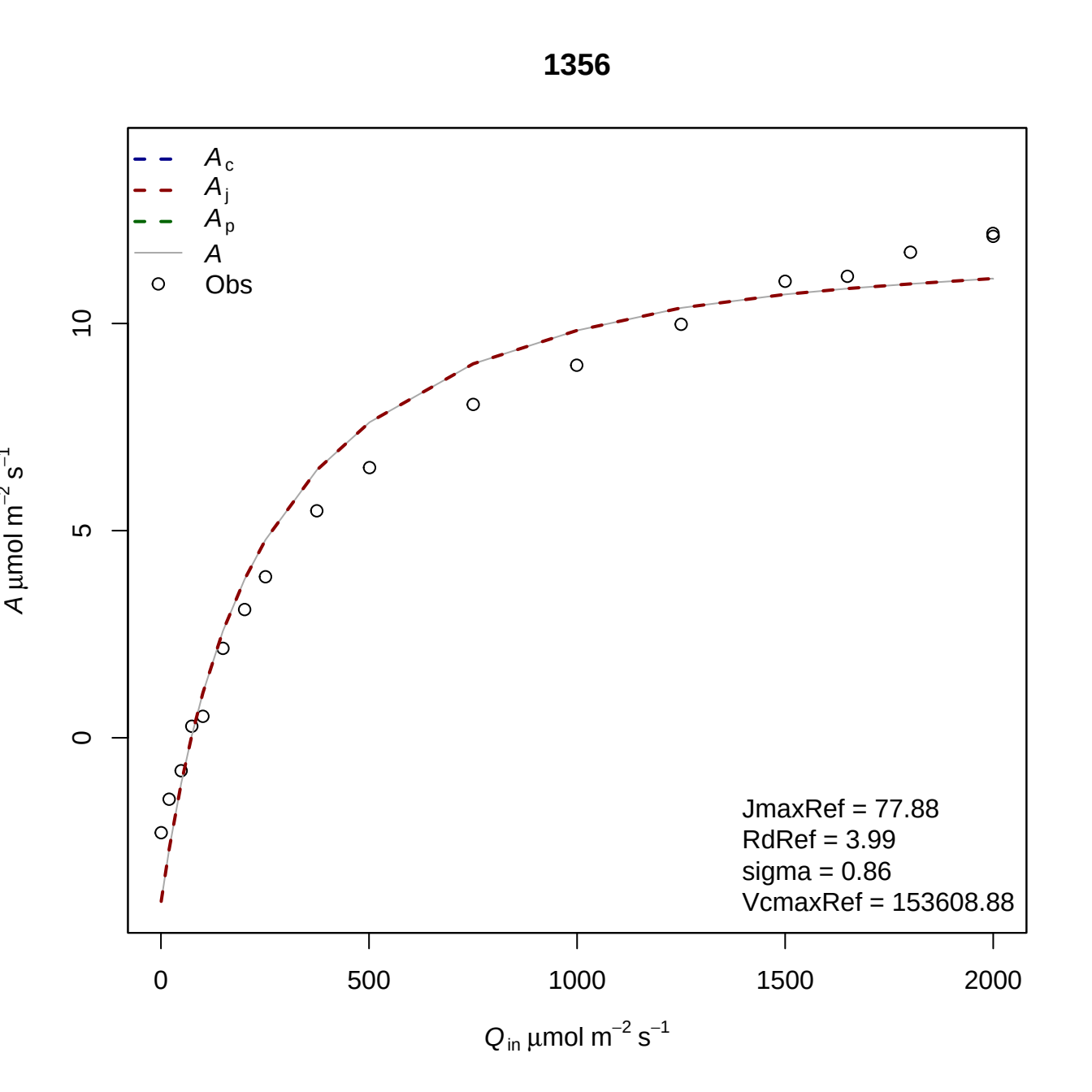

1351
A_c
A_j
A_p
A
Obs
JmaxRef = 45.76
RdRef = 2.51
sigma = 0.63
VcmaxRef = 16088.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

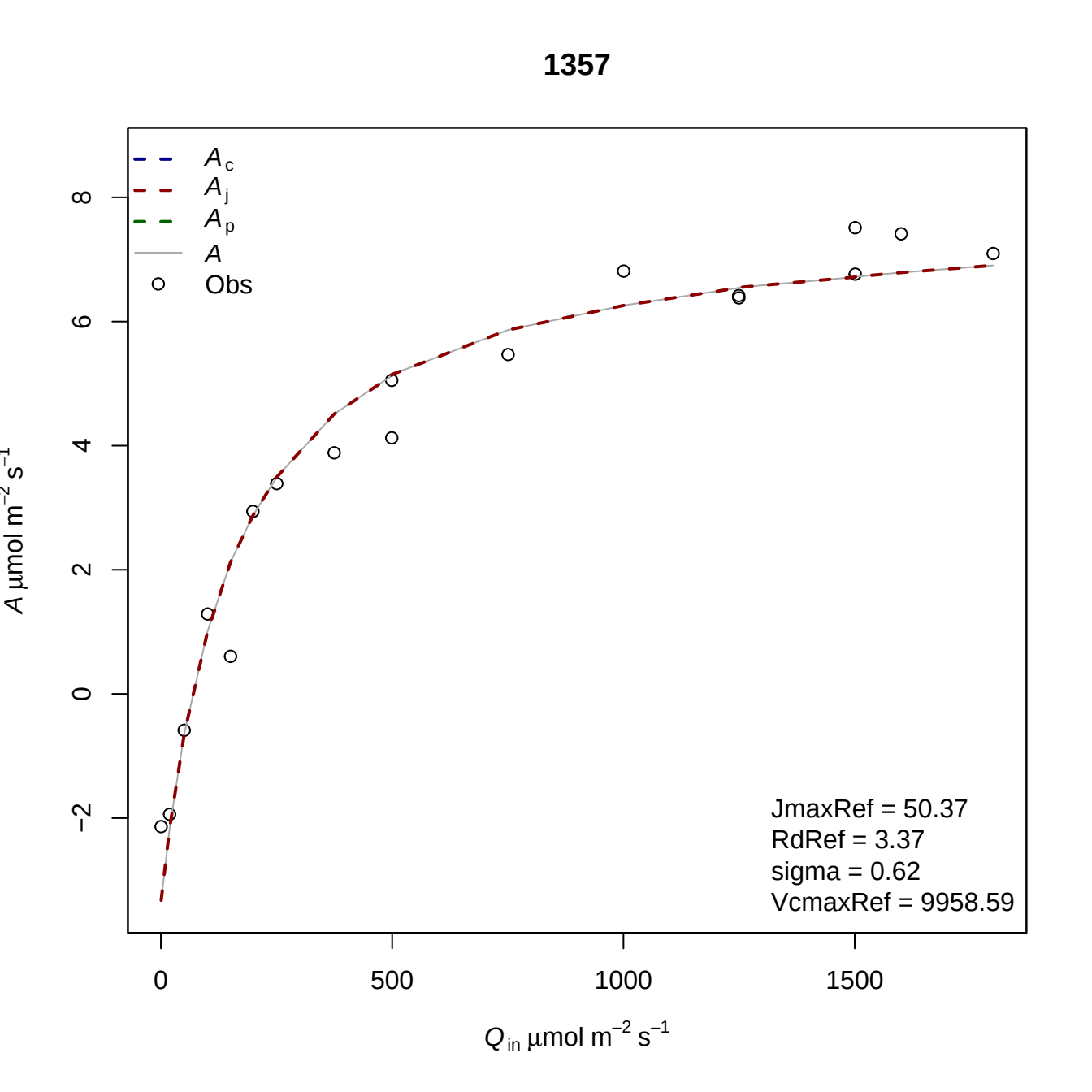

1352
A_c
A_j
A_p
A
Obs
JmaxRef = 70.37
RdRef = 1.45
sigma = 0.74
VcmaxRef = 43068.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

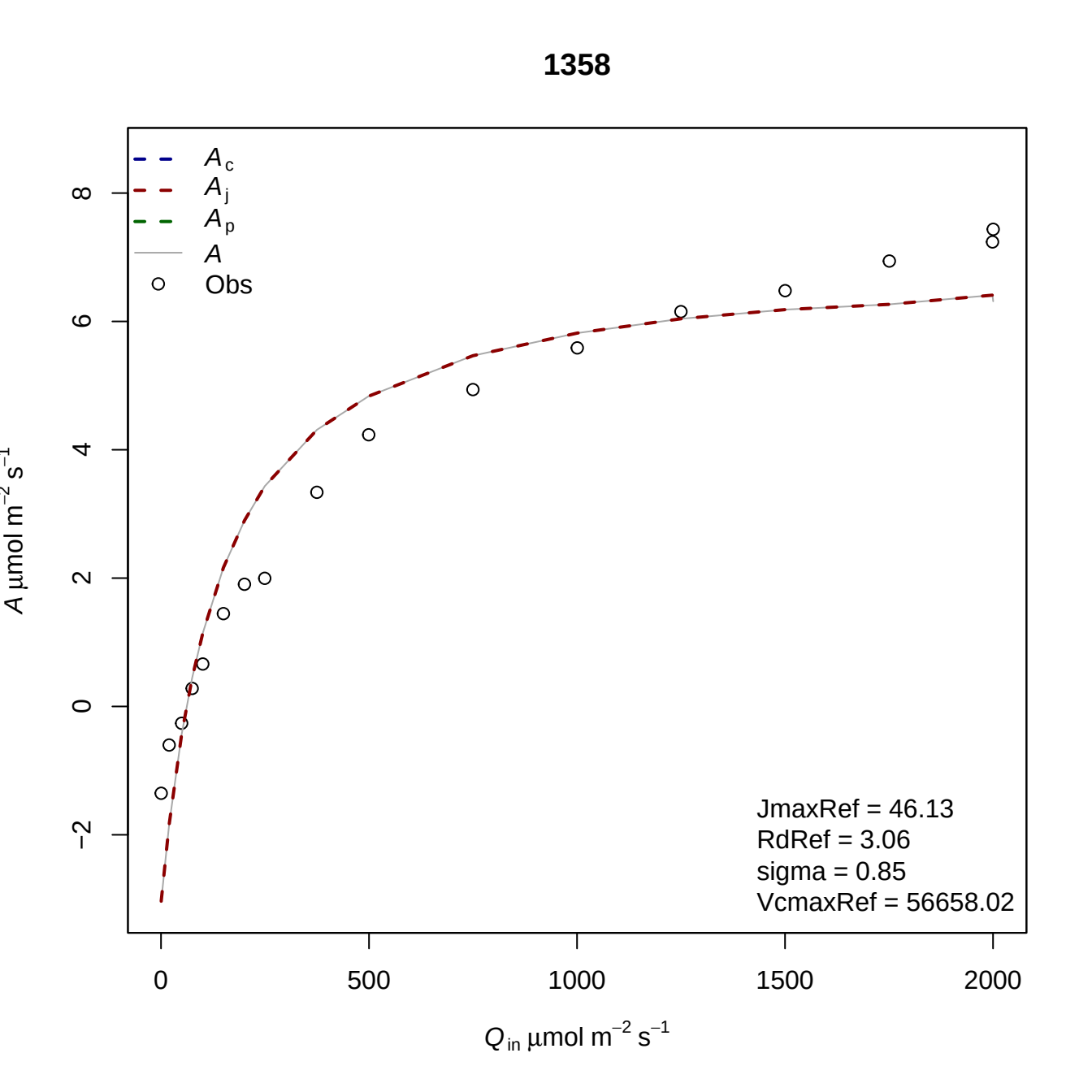

1353
A_c
A_j
A_p
A
Obs
JmaxRef = 43.86
RdRef = 2.13
sigma = 0.42
VcmaxRef = 7772.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

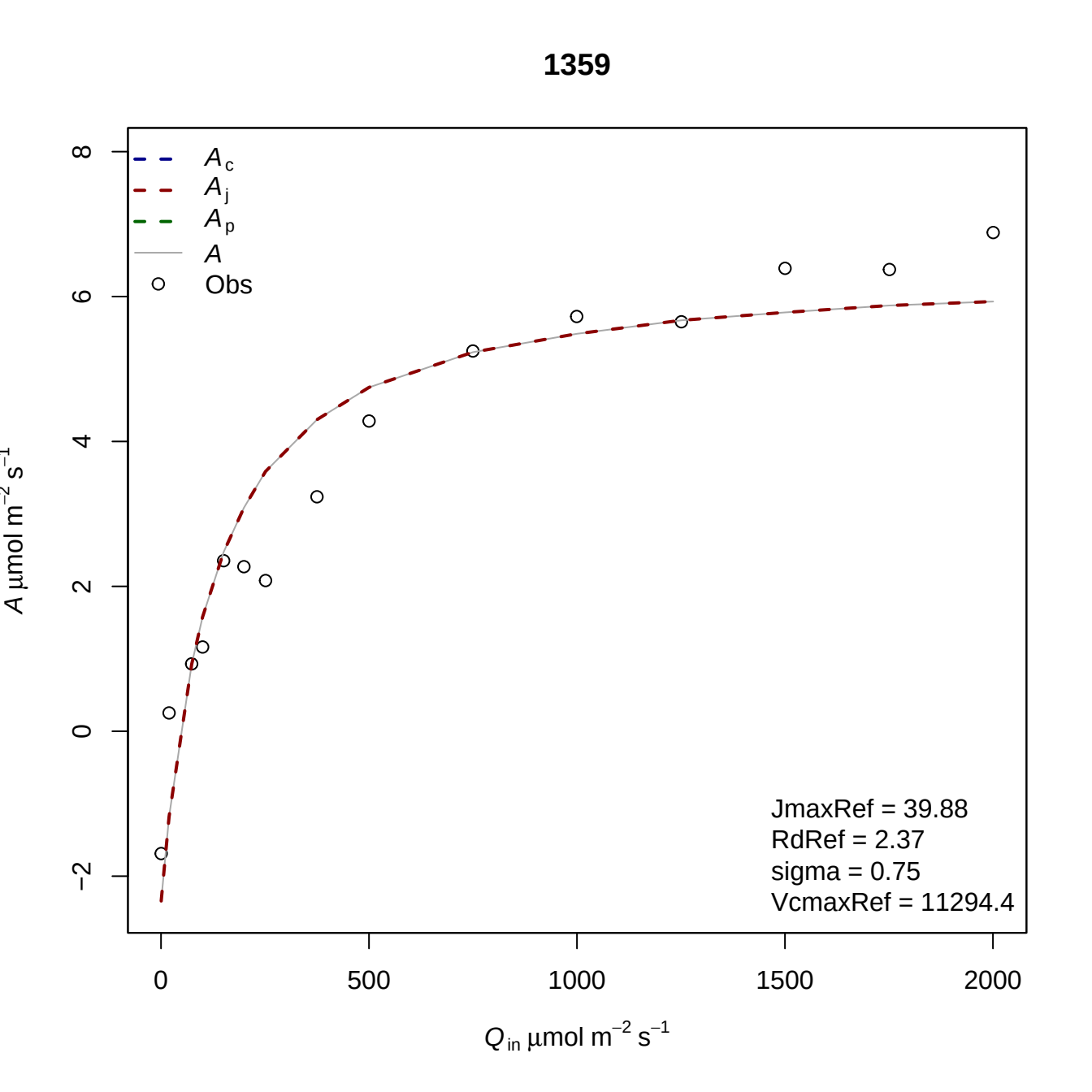

1354
A_c
A_j
A_p
A
Obs
JmaxRef = 55.98
RdRef = 4.22
sigma = 1.11
VcmaxRef = 12418.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

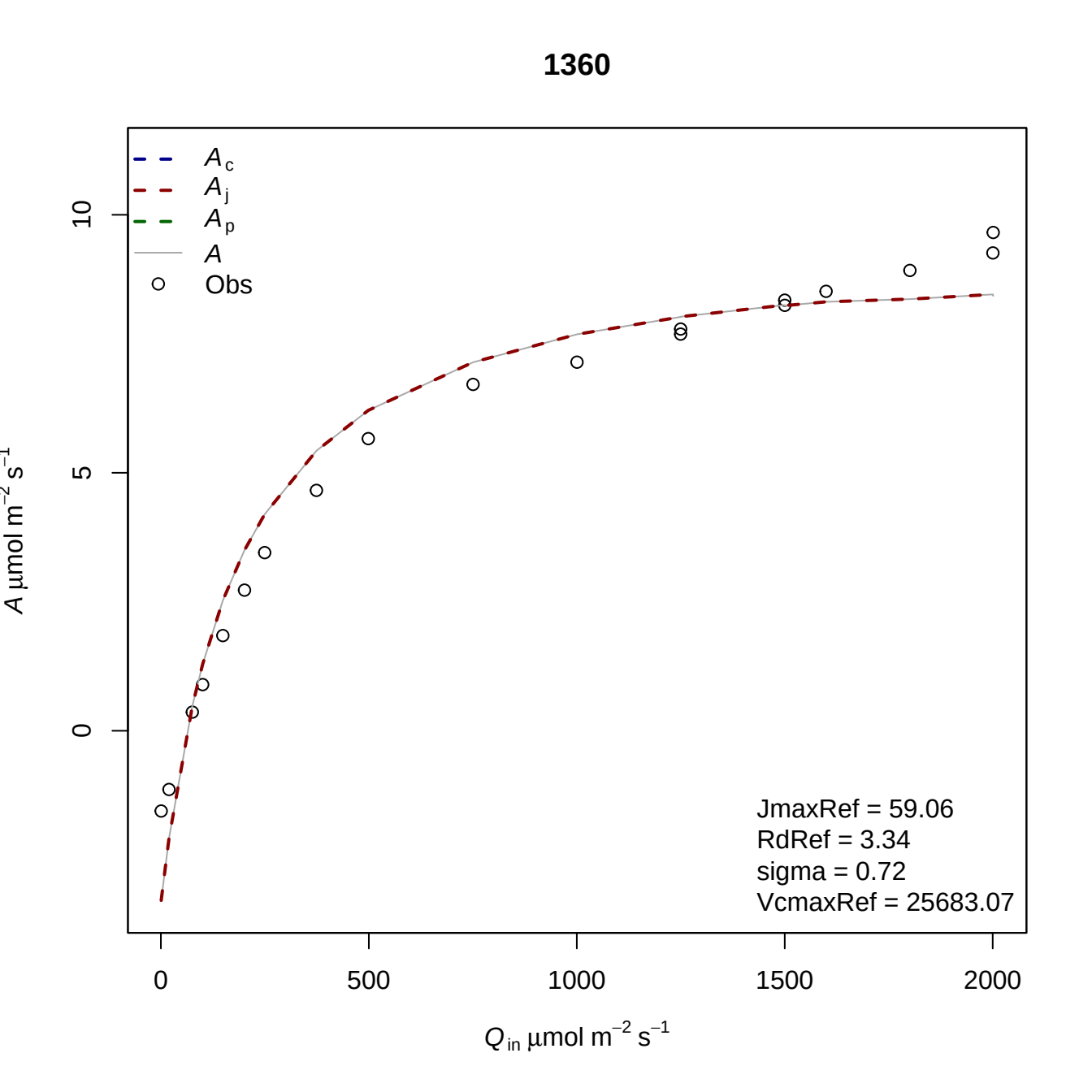

1355
A_c
A_j
A_p
A
Obs
JmaxRef = 35.29
RdRef = 3.27
sigma = 0.74
VcmaxRef = 33794.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

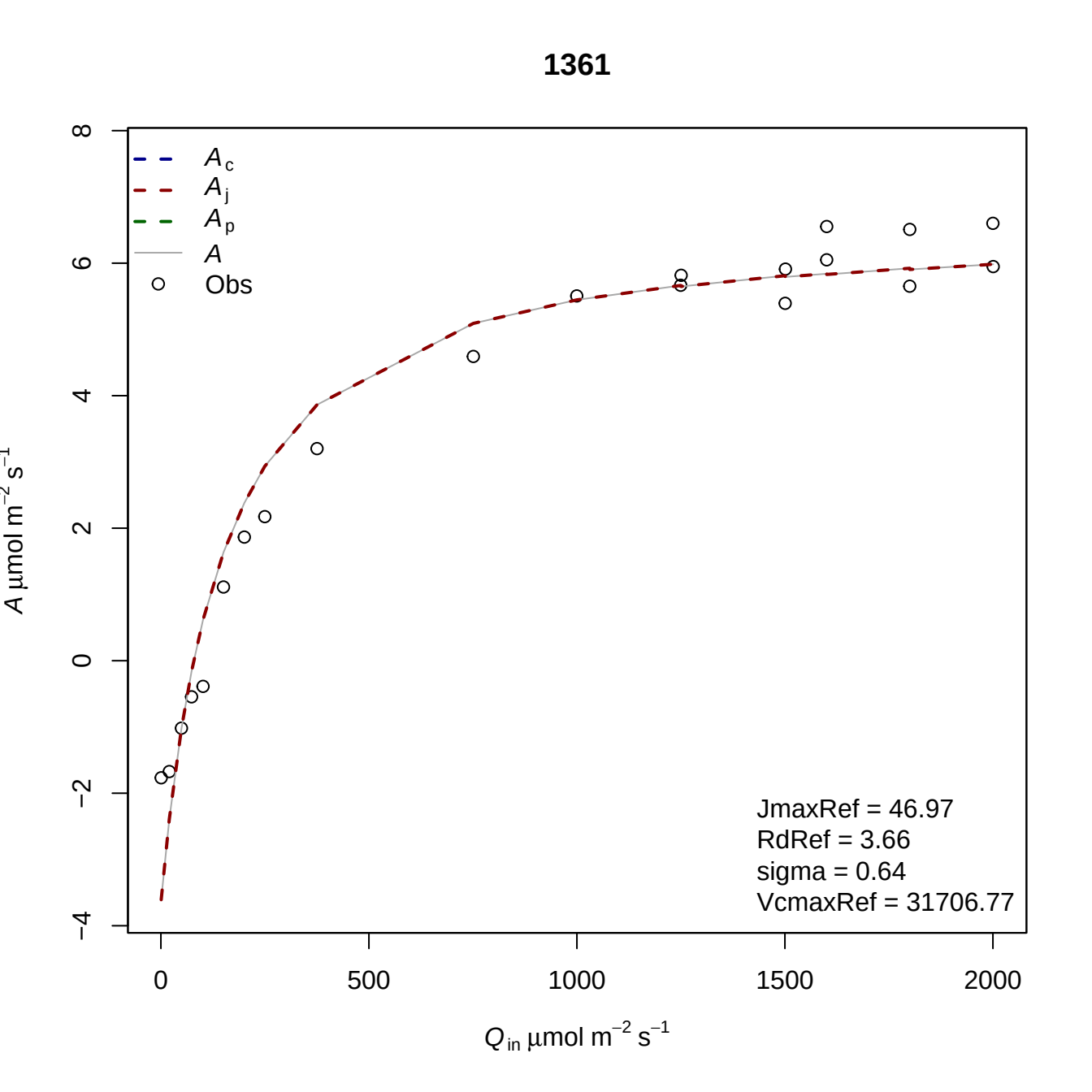

1356
A_c
A_j
A_p
A
Obs
JmaxRef = 77.88
RdRef = 3.99
sigma = 0.86
VcmaxRef = 153608.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

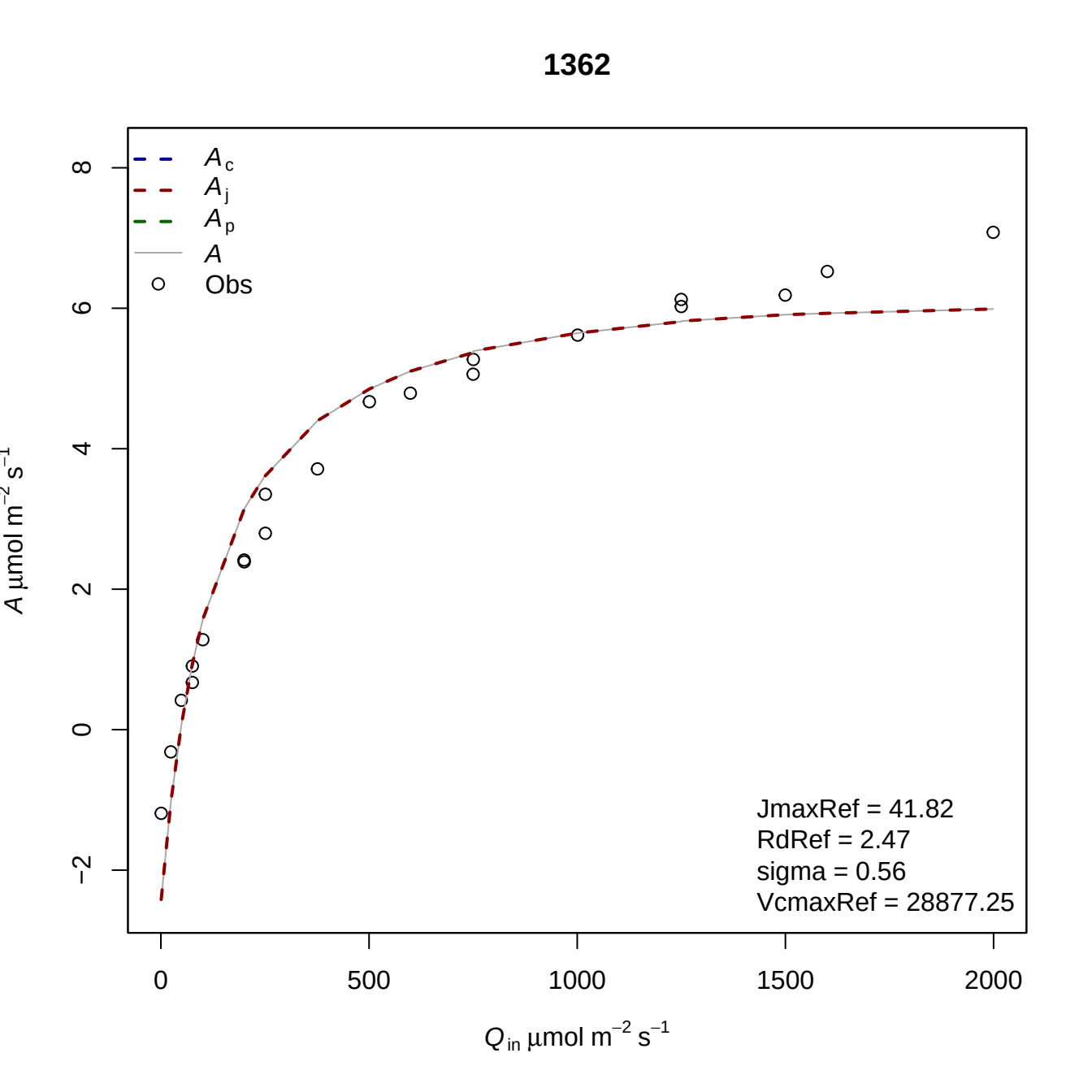

1357
A_c
A_j
A_p
A
Obs
JmaxRef = 50.37
RdRef = 3.37
sigma = 0.62
VcmaxRef = 9958.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

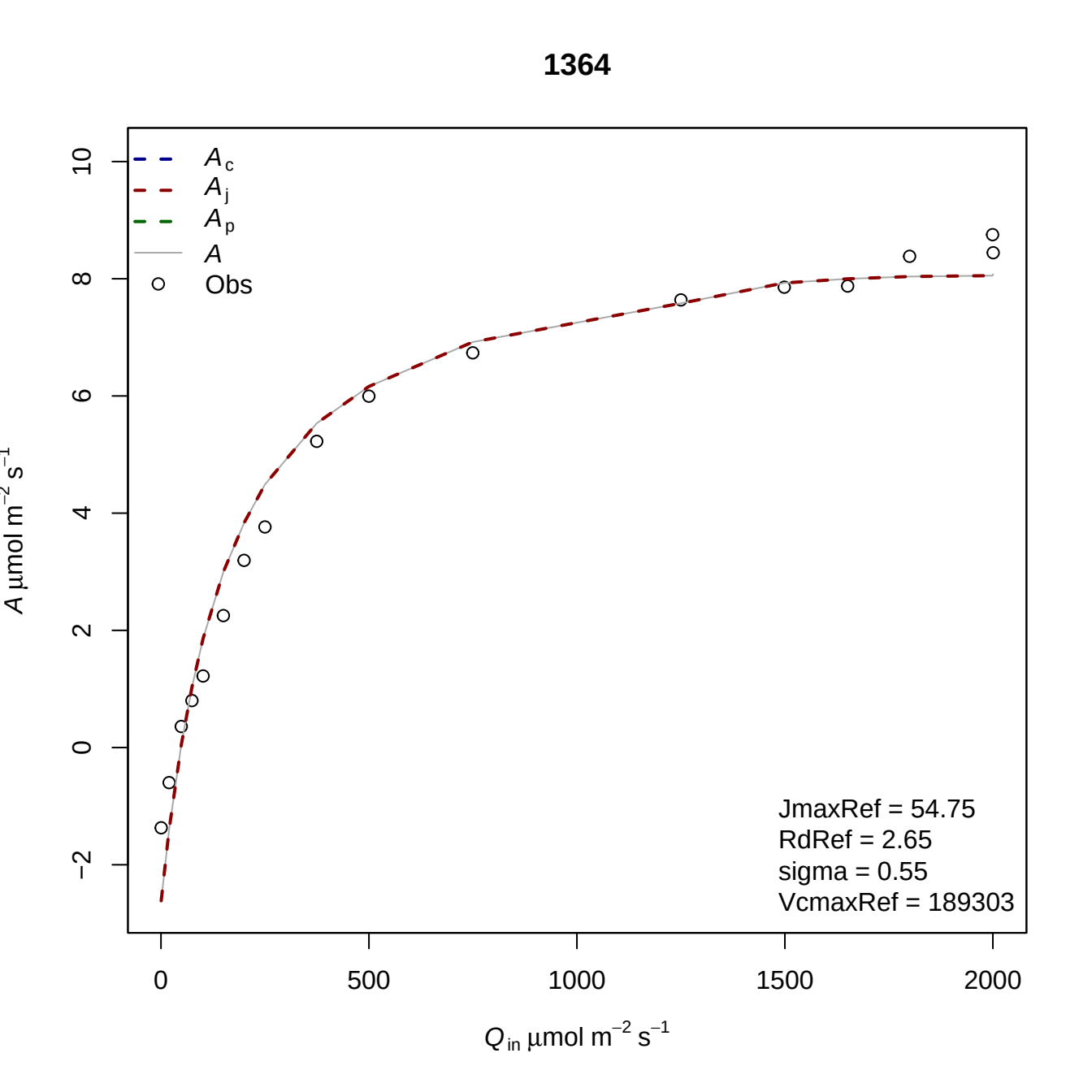

1358
A_c
A_j
A_p
A
Obs
JmaxRef = 46.13
RdRef = 3.06
sigma = 0.85
VcmaxRef = 56658.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

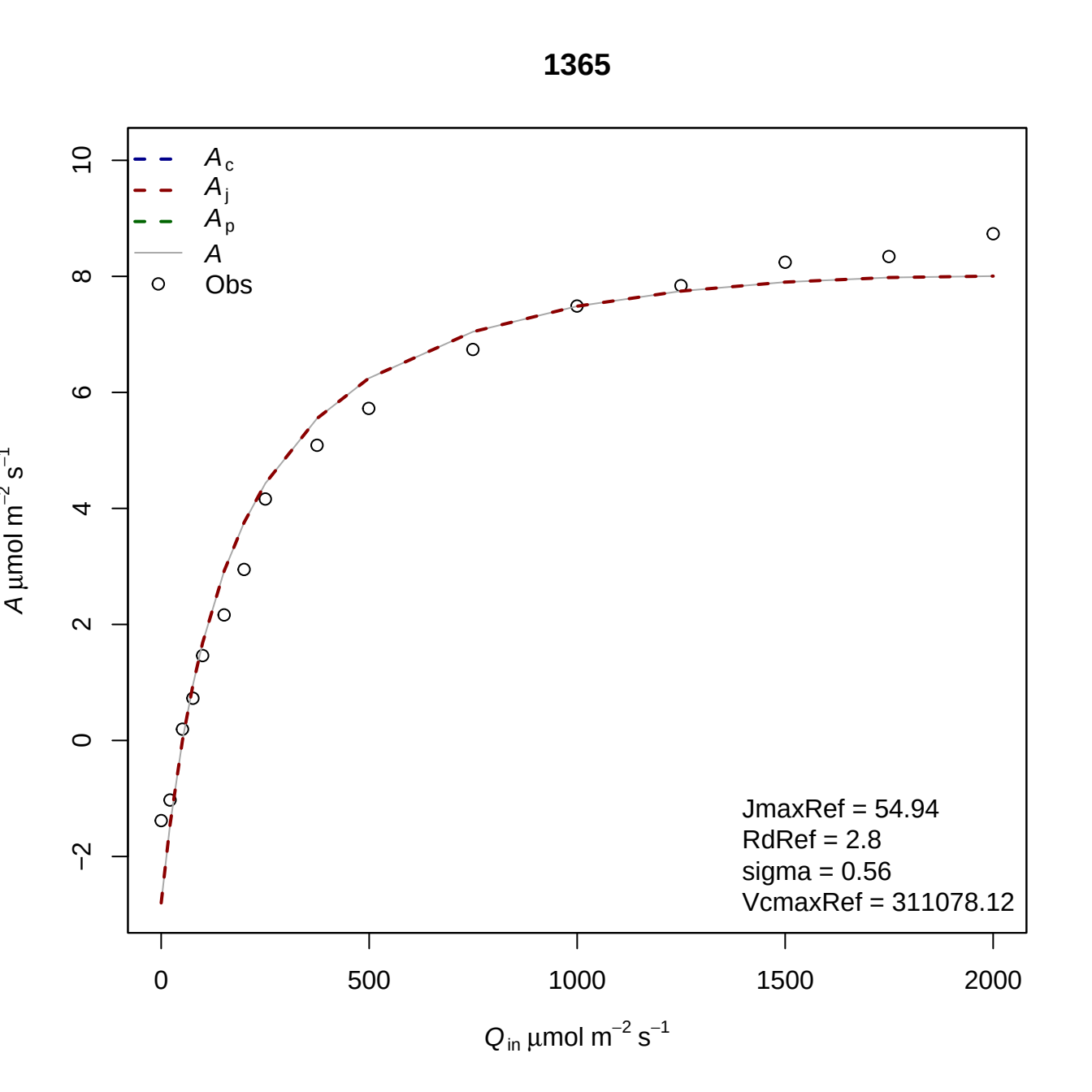

1359
A_c
A_j
A_p
A
Obs
JmaxRef = 39.88
RdRef = 2.37
sigma = 0.75
VcmaxRef = 11294.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

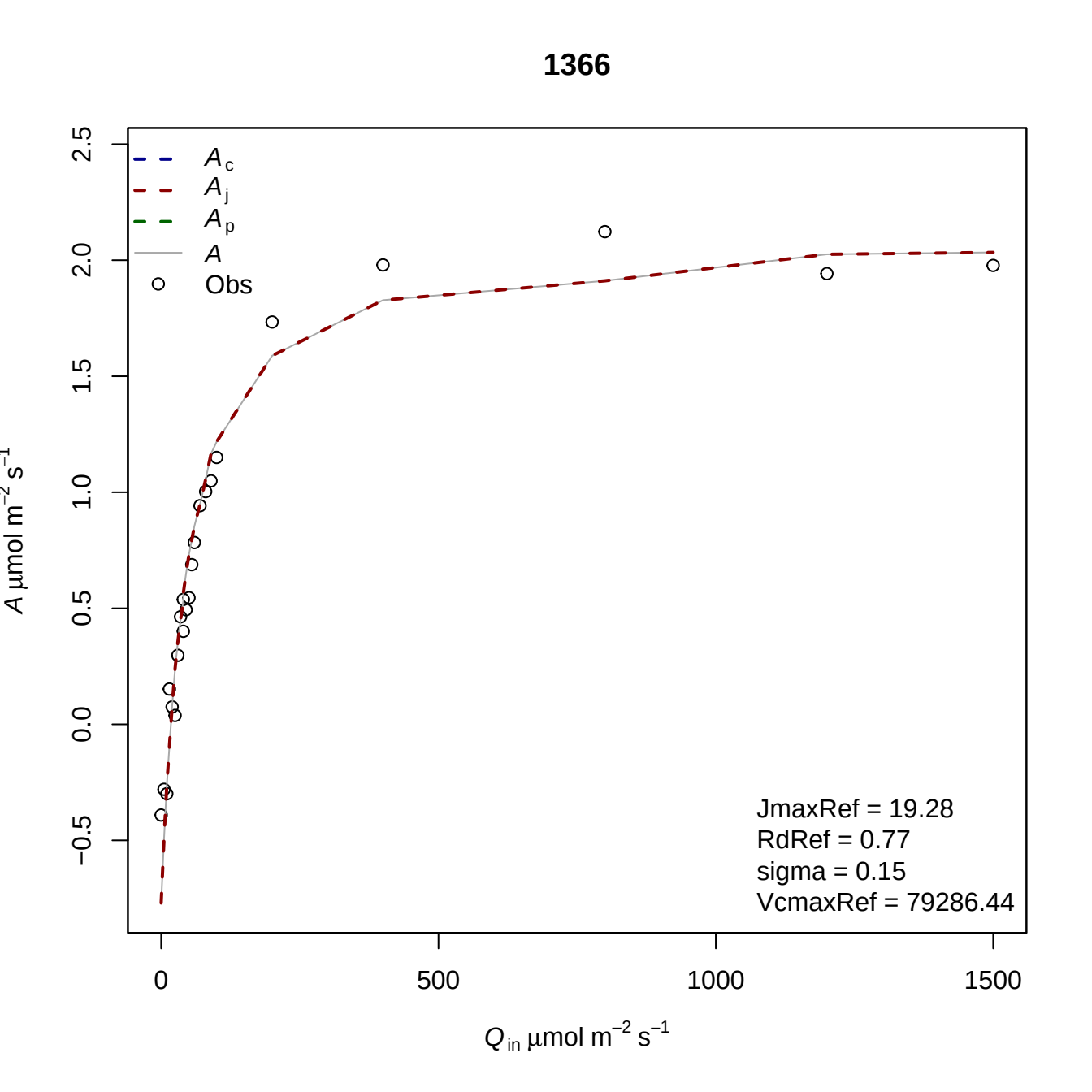

1360
A_c
A_j
A_p
A
Obs
JmaxRef = 59.06
RdRef = 3.34
sigma = 0.72
VcmaxRef = 25683.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

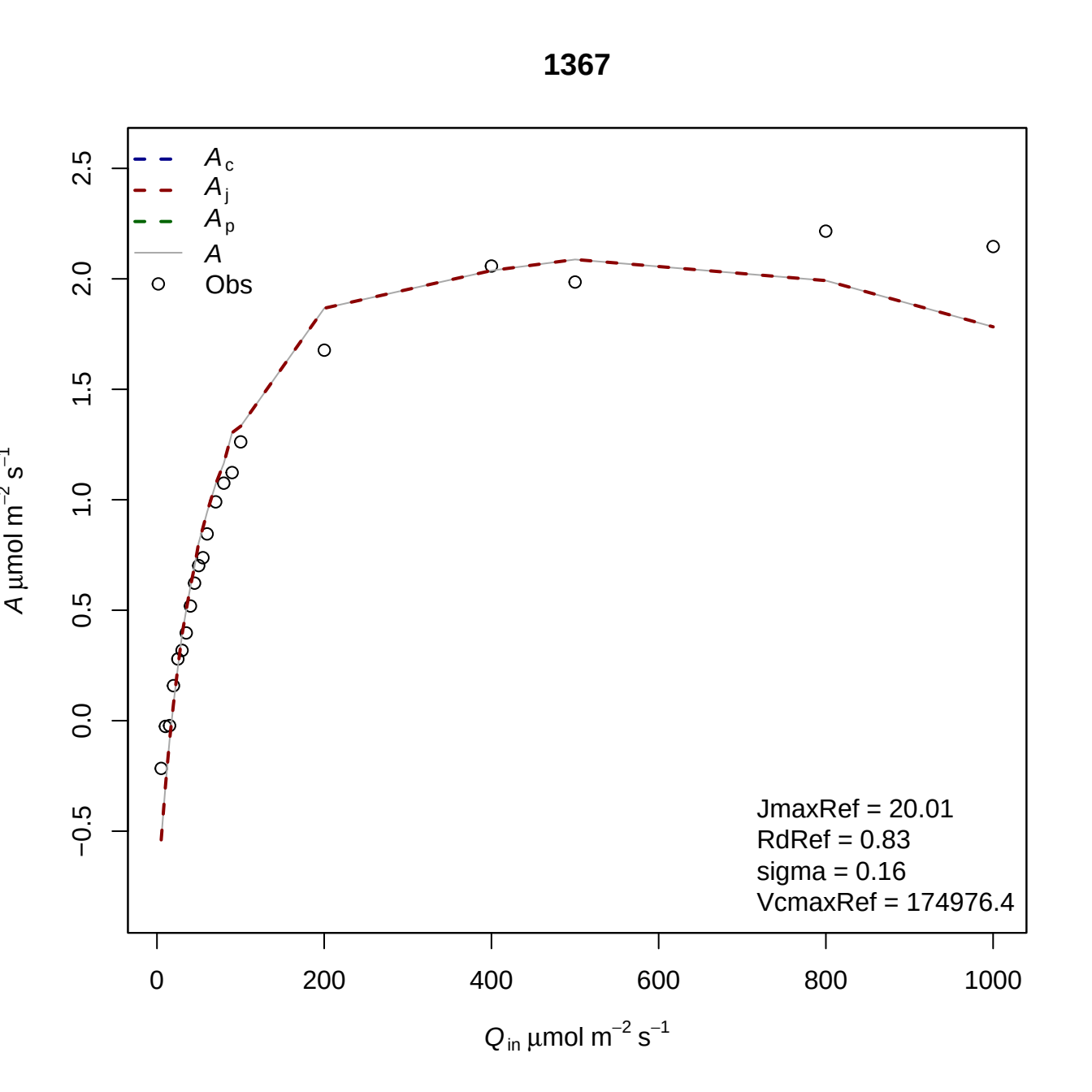

1361
A_c
A_j
A_p
A
Obs
JmaxRef = 46.97
RdRef = 3.66
sigma = 0.64
VcmaxRef = 31706.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

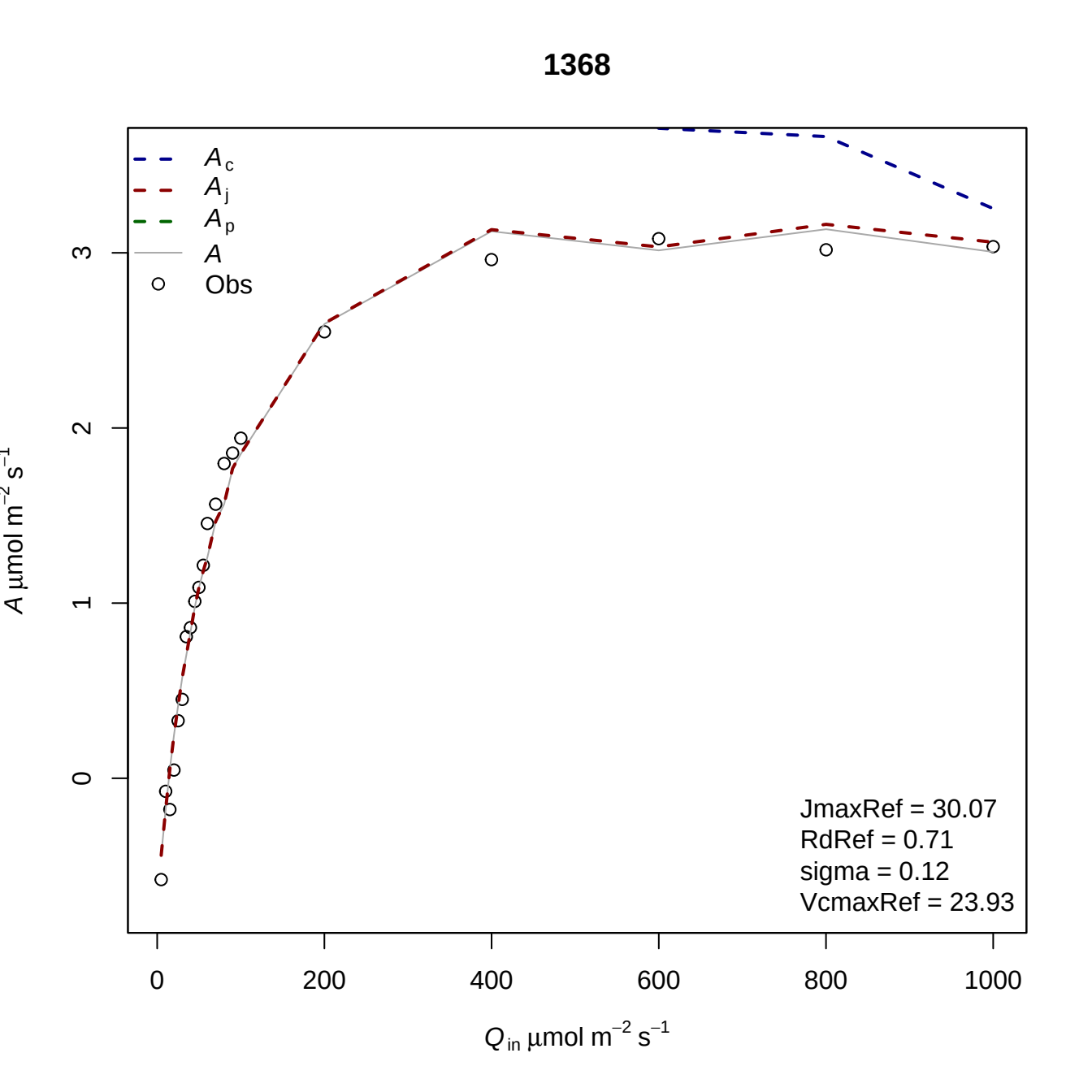

1362
A_c
A_j
A_p
A
Obs
JmaxRef = 41.82
RdRef = 2.47
sigma = 0.56
VcmaxRef = 28877.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

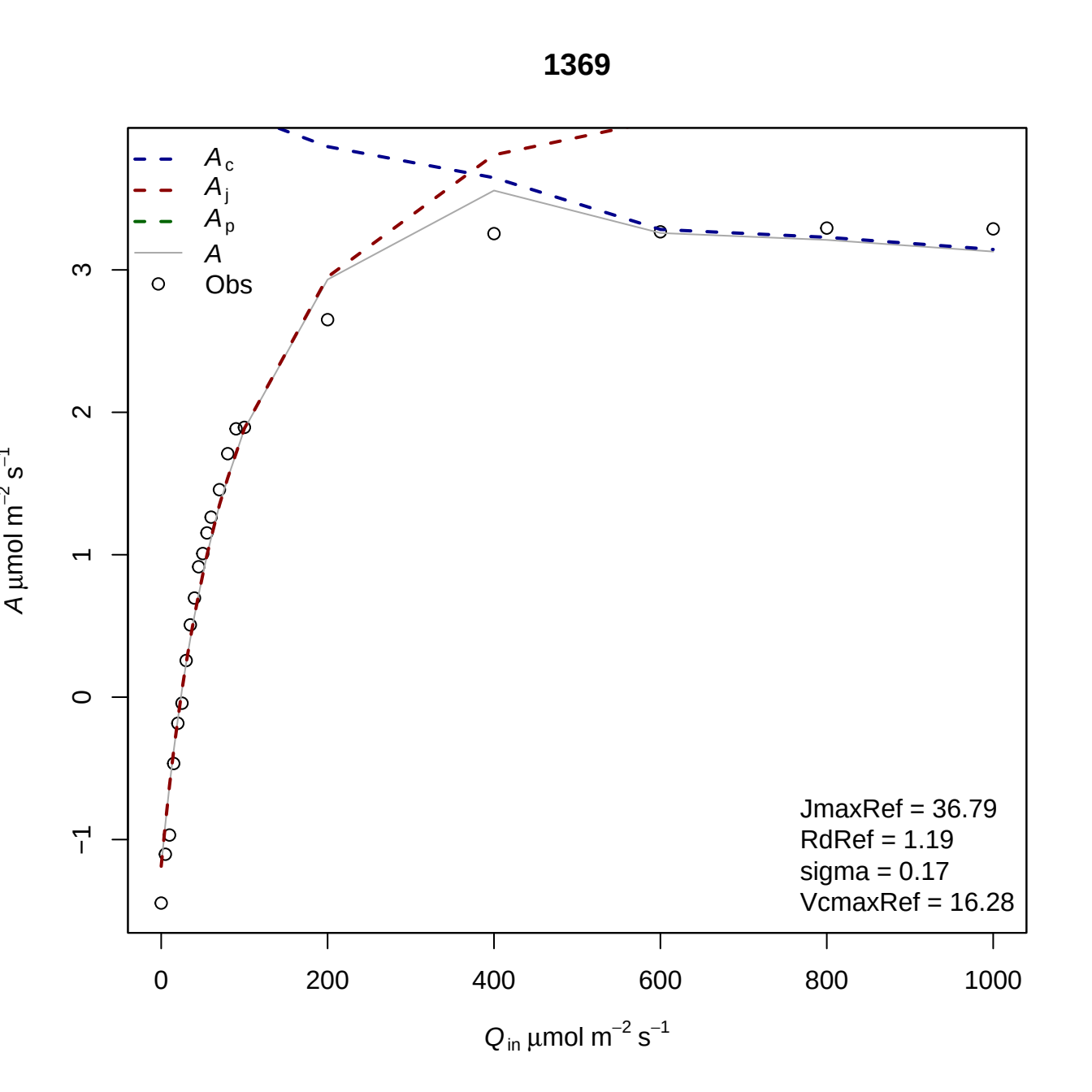

1364
A_c
A_j
A_p
A
Obs
JmaxRef = 54.75
RdRef = 2.65
sigma = 0.55
VcmaxRef = 189303
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

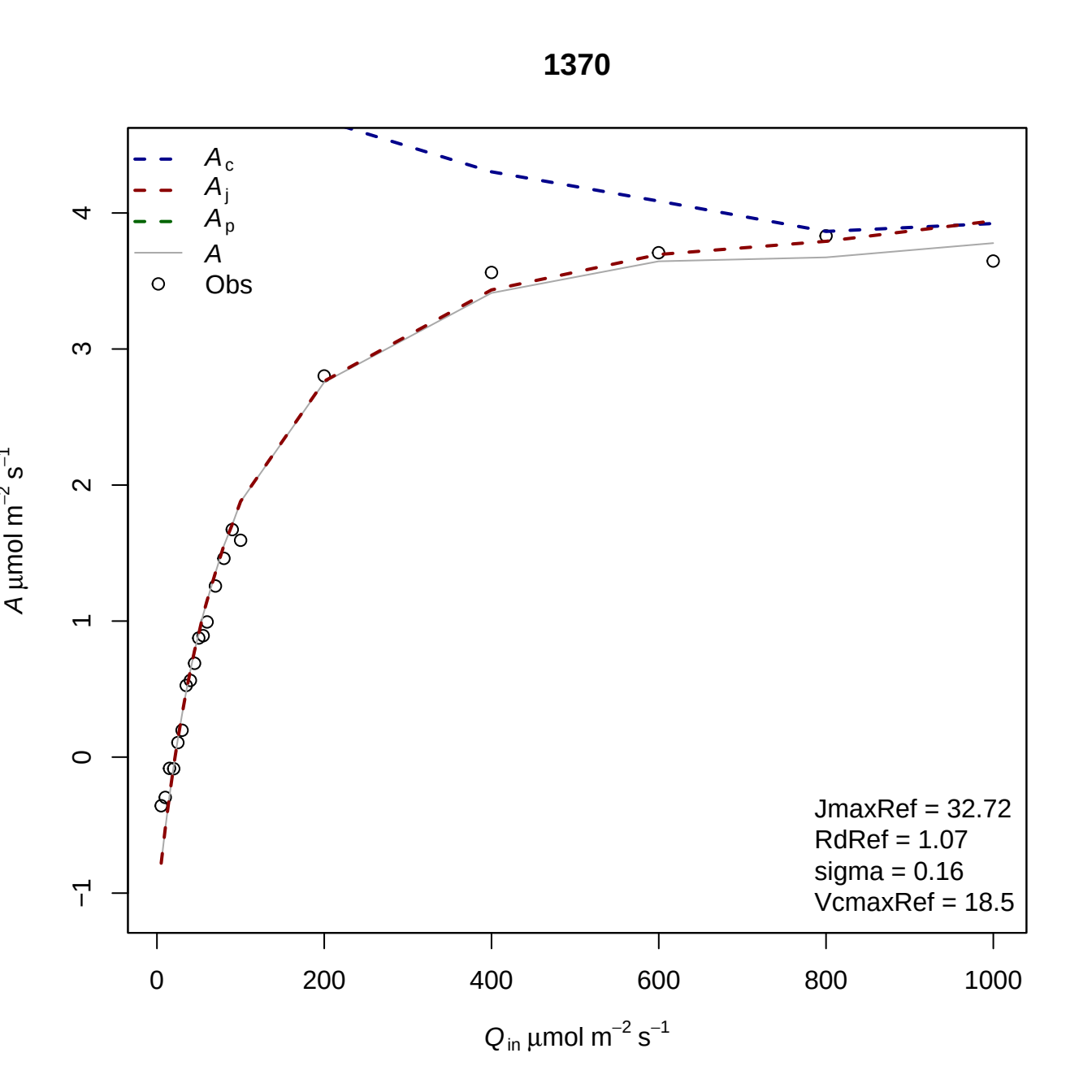

1365
A_c
A_j
A_p
A
Obs
JmaxRef = 54.94
RdRef = 2.8
sigma = 0.56
VcmaxRef = 311078.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

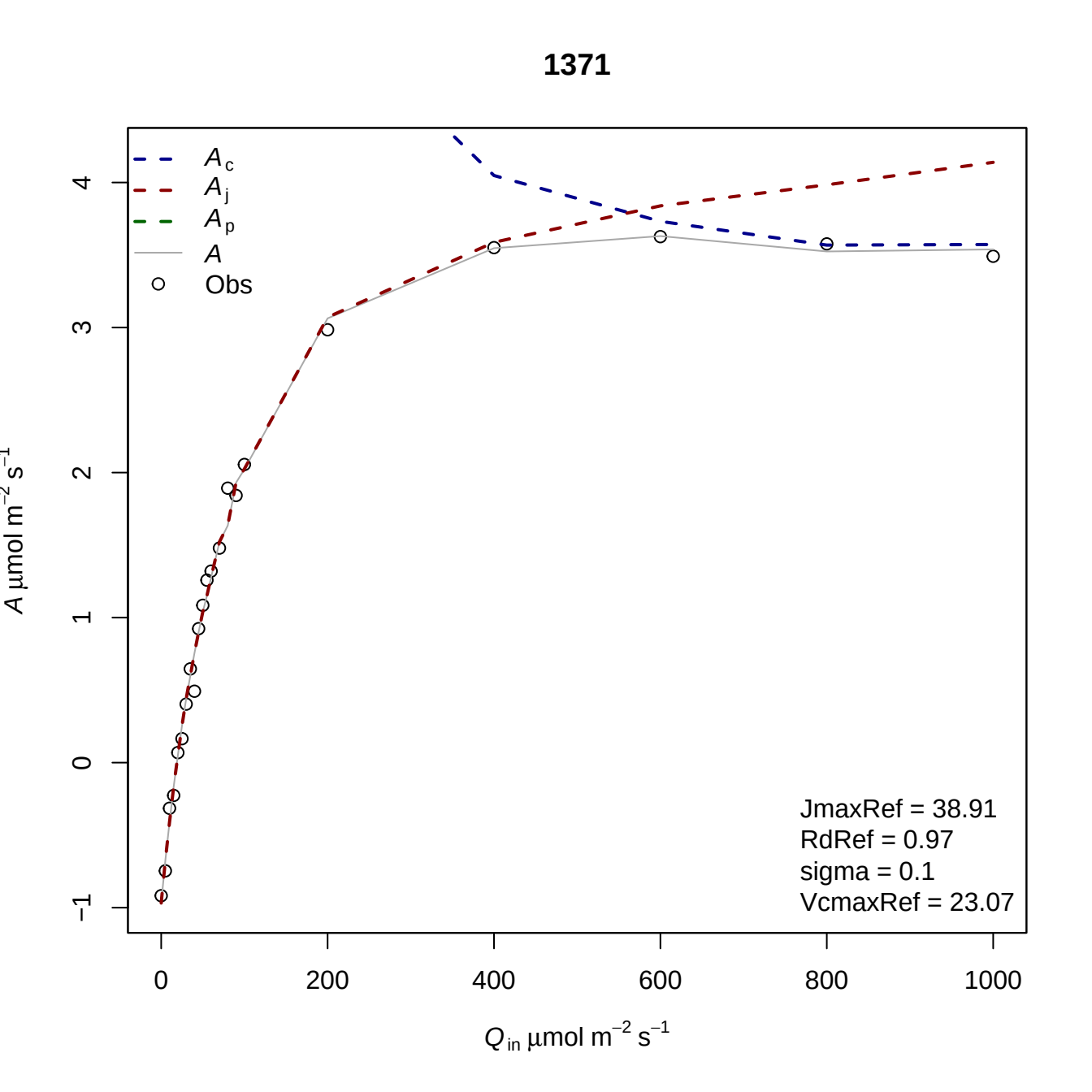

1366
A_c
A_j
A_p
A
Obs
JmaxRef = 19.28
RdRef = 0.77
sigma = 0.15
VcmaxRef = 79286.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500

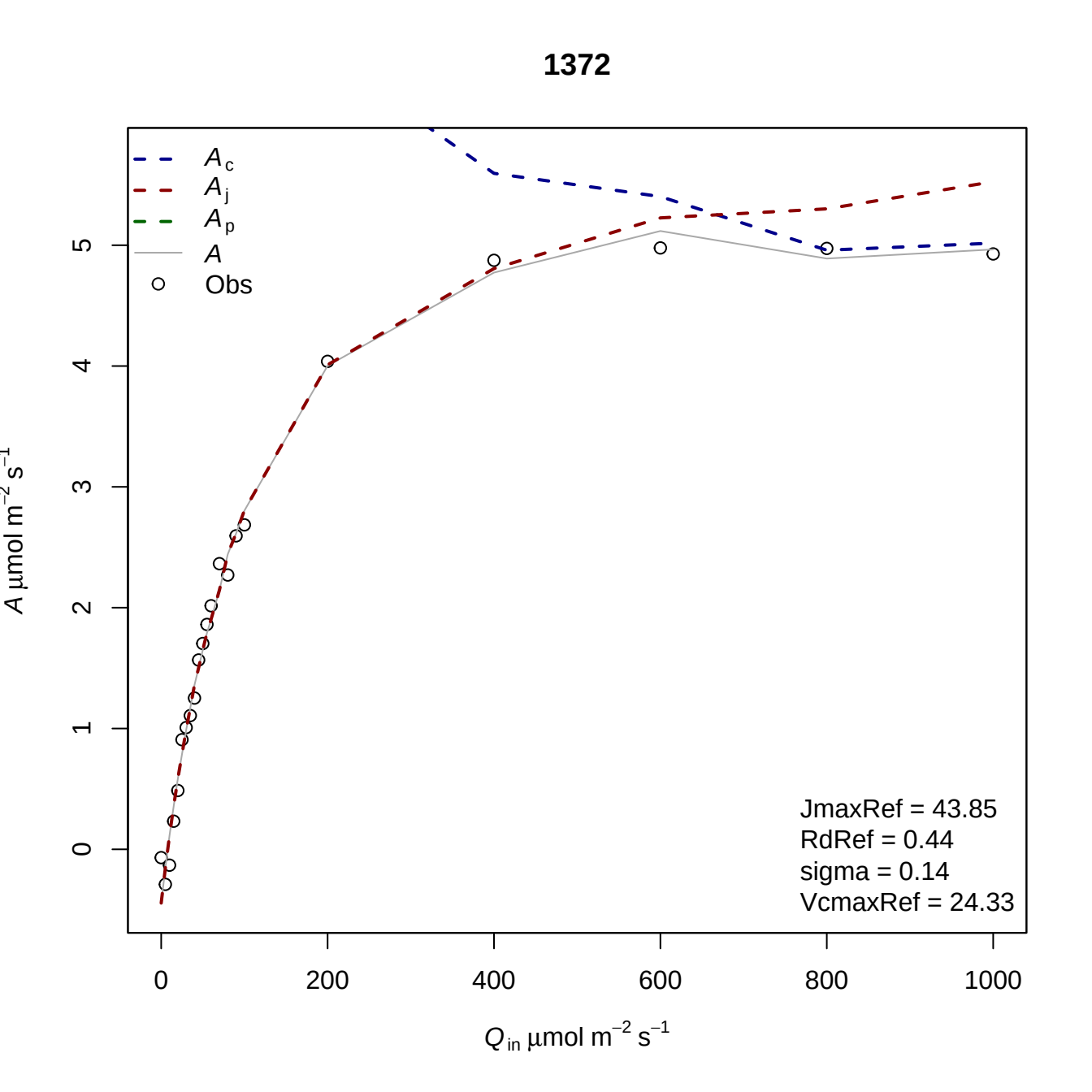

1367
A_c
A_j
A_p
A
Obs
JmaxRef = 20.01
RdRef = 0.83
sigma = 0.16
VcmaxRef = 174976.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
200
400
600
800
1000

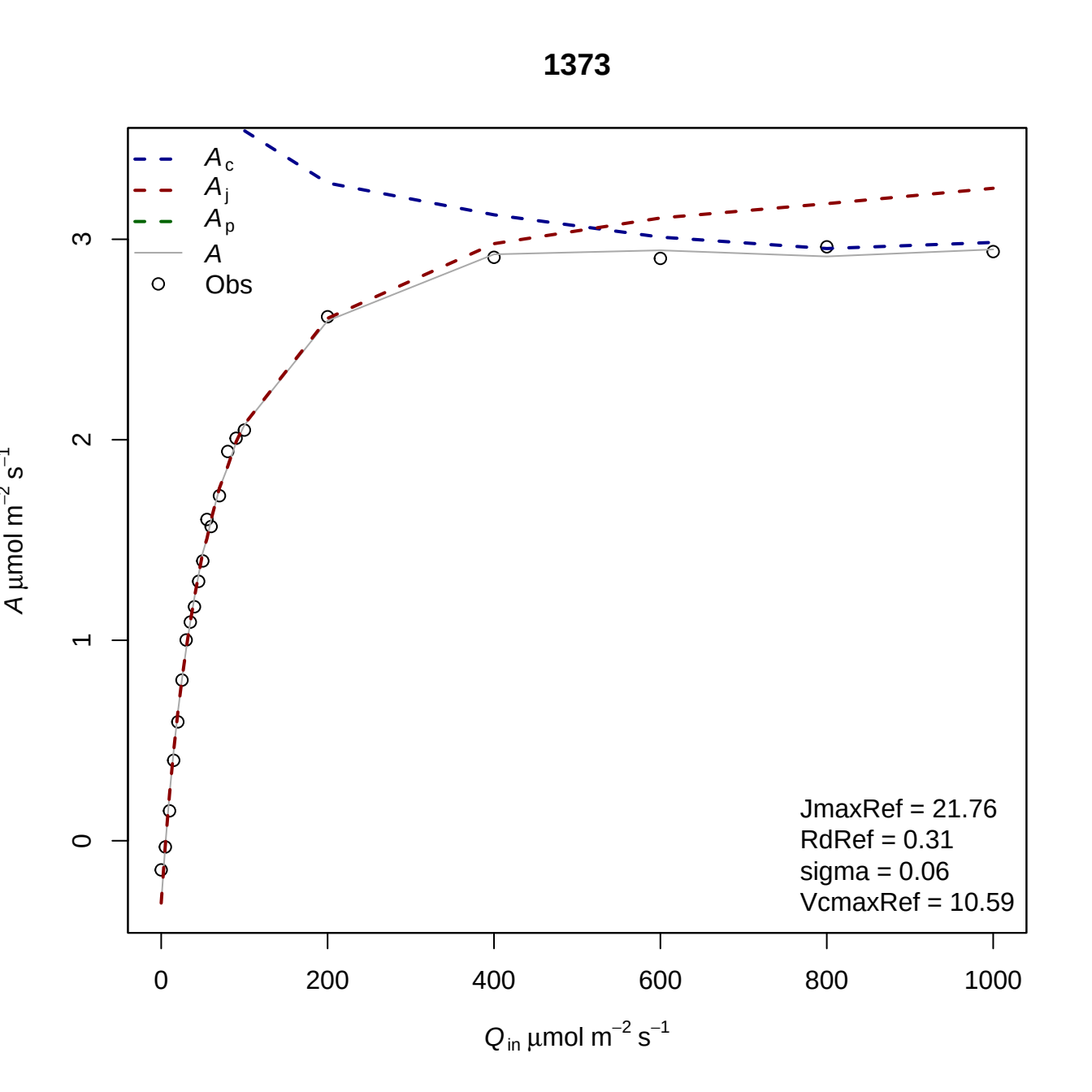

1368
A_c
A_j
A_p
A
Obs
JmaxRef = 30.07
RdRef = 0.71
sigma = 0.12
VcmaxRef = 23.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
200
400
600
800
1000

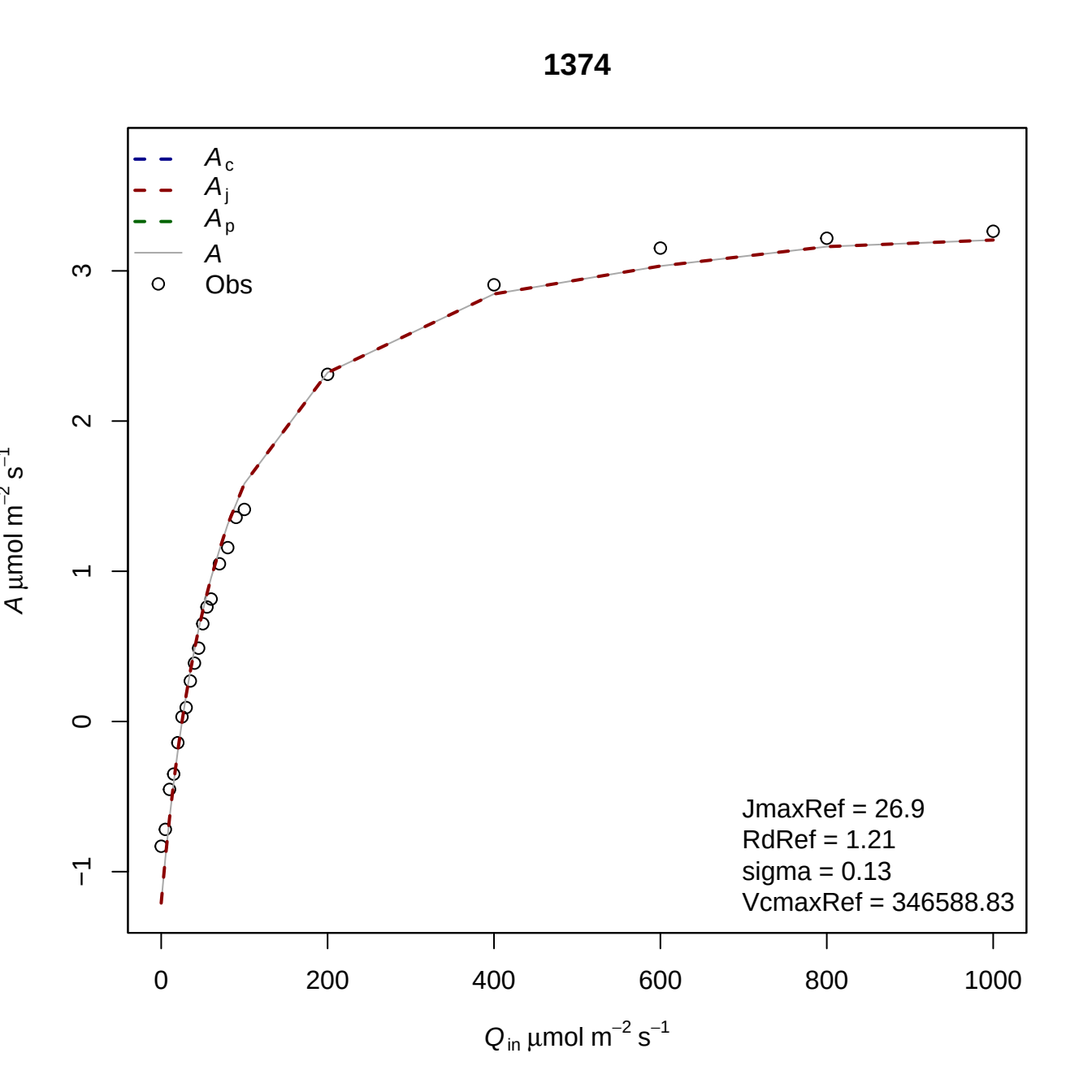

1369
A_c
A_j
A_p
A
Obs
JmaxRef = 36.79
RdRef = 1.19
sigma = 0.17
VcmaxRef = 16.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000

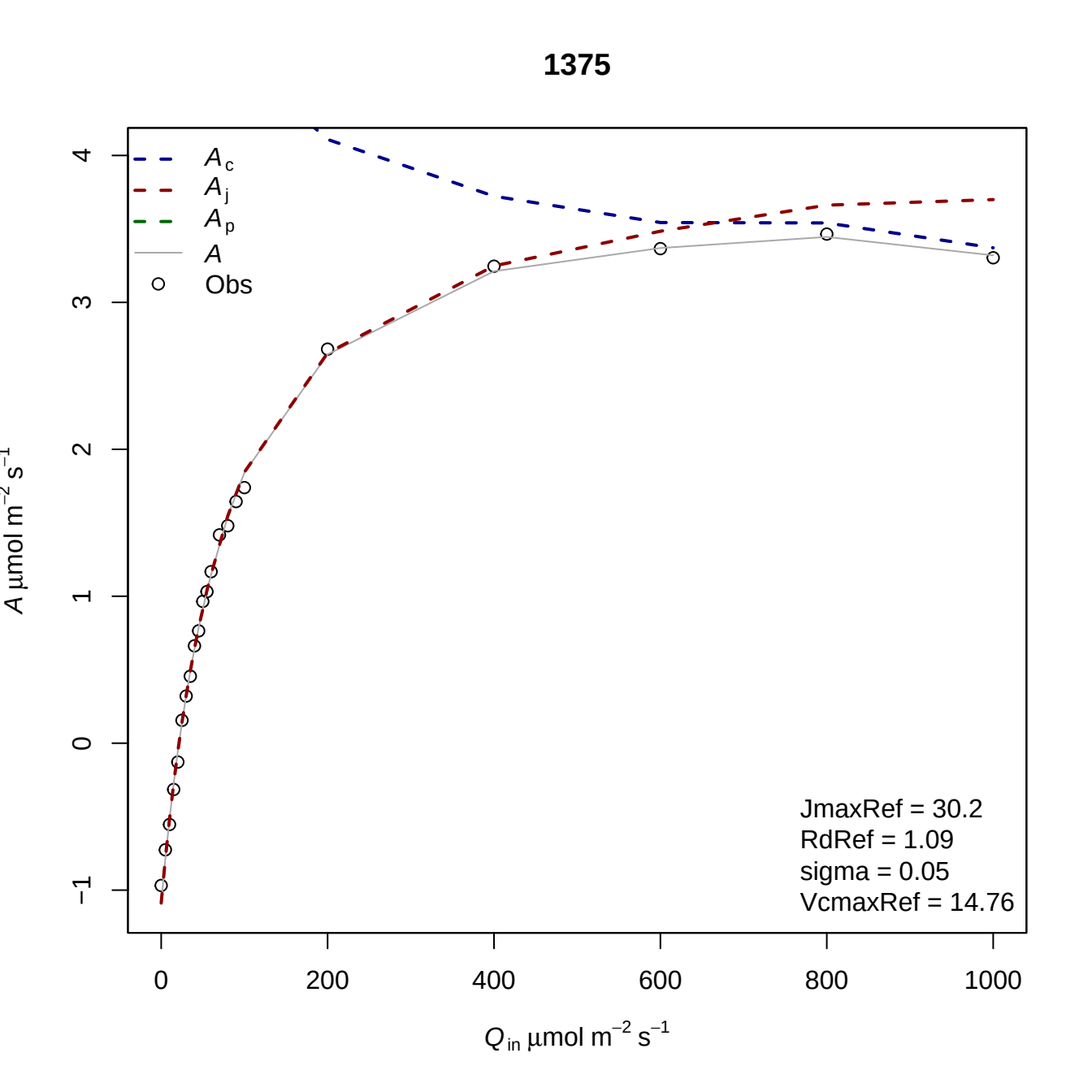

1370
A_c
A_j
A_p
A
Obs
JmaxRef = 32.72
RdRef = 1.07
sigma = 0.16
VcmaxRef = 18.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

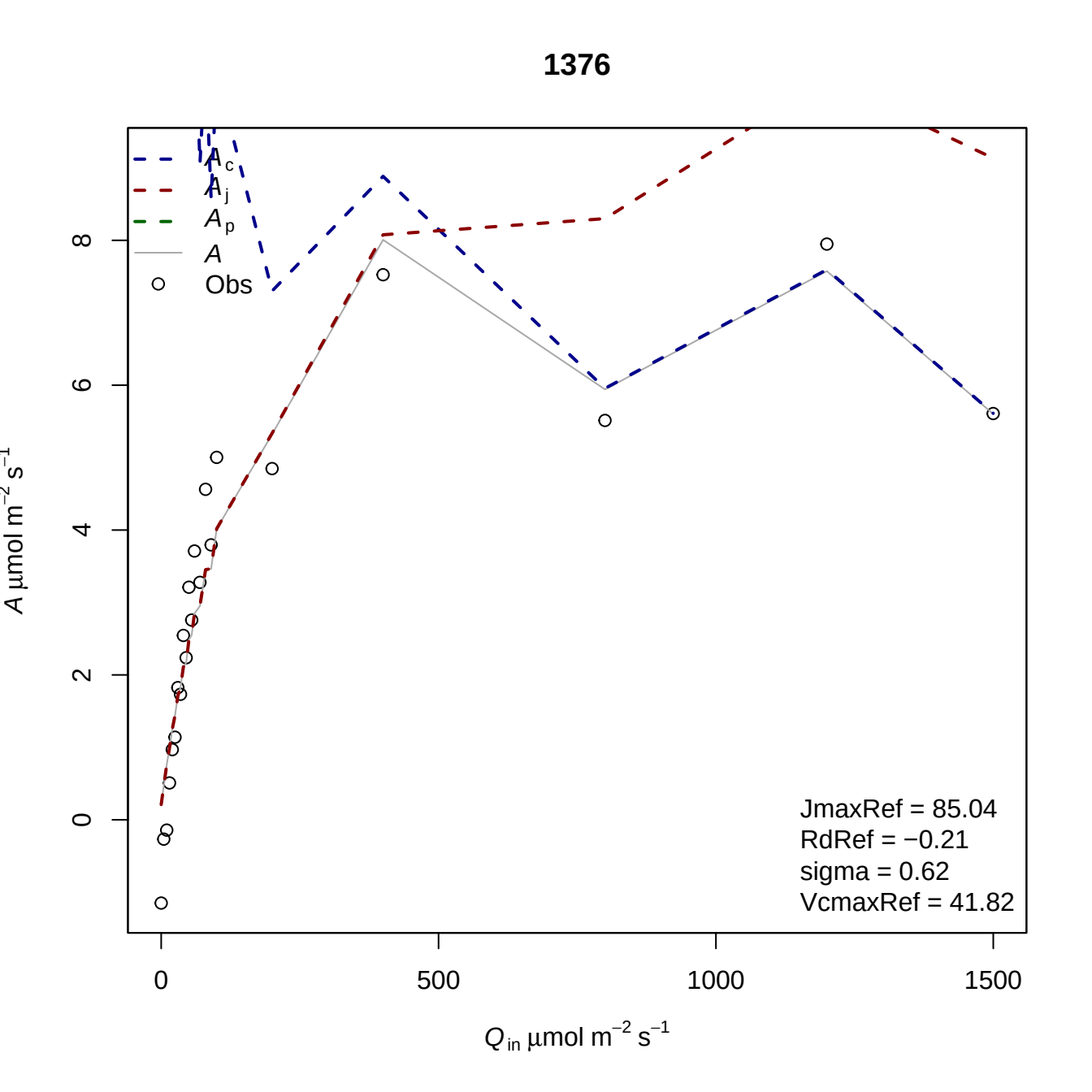

1371
A_c
A_j
A_p
A
Obs
JmaxRef = 38.91
RdRef = 0.97
sigma = 0.1
VcmaxRef = 23.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

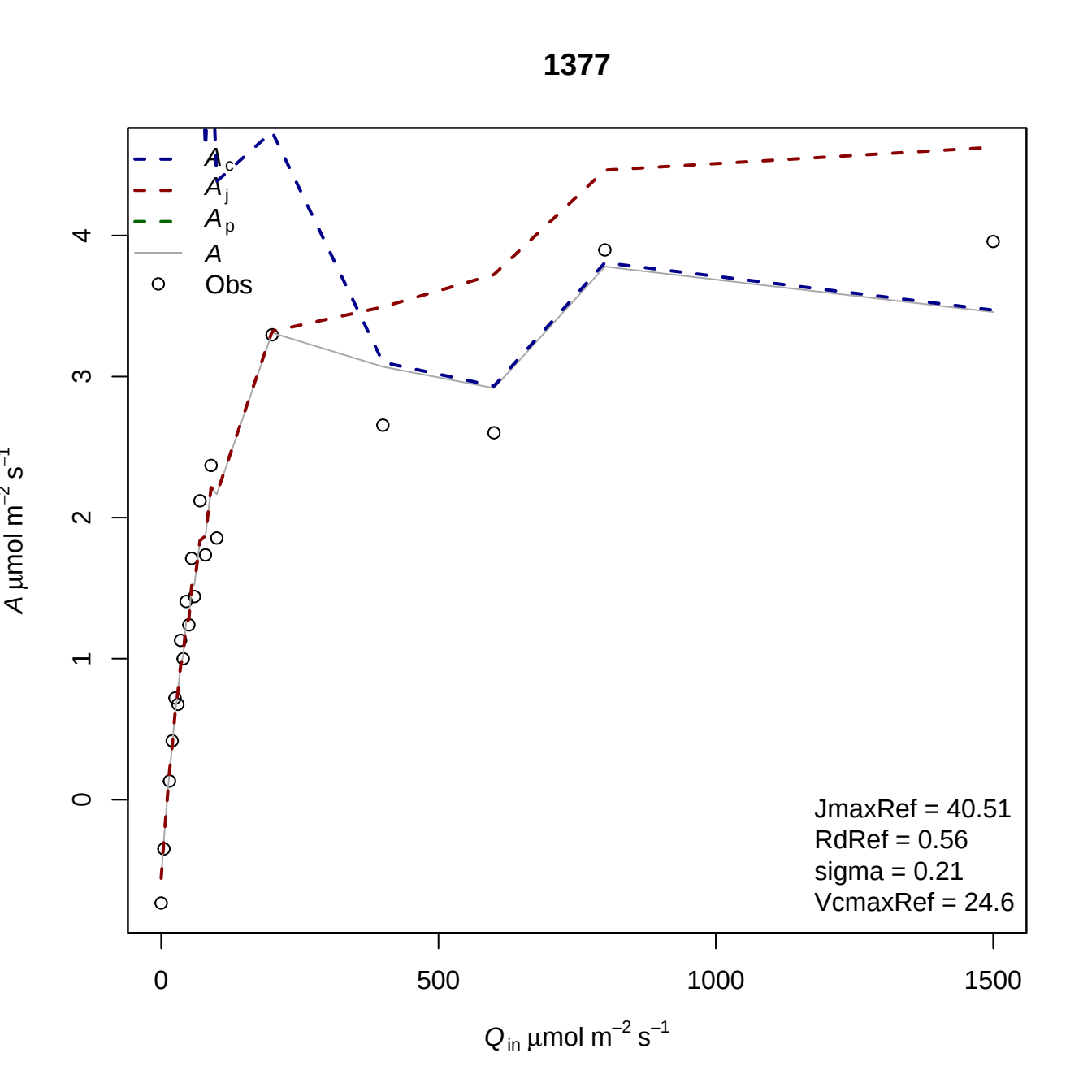

1372
A_c
A_j
A_p
A
Obs
JmaxRef = 43.85
RdRef = 0.44
sigma = 0.14
VcmaxRef = 24.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
200
400
600
800
1000

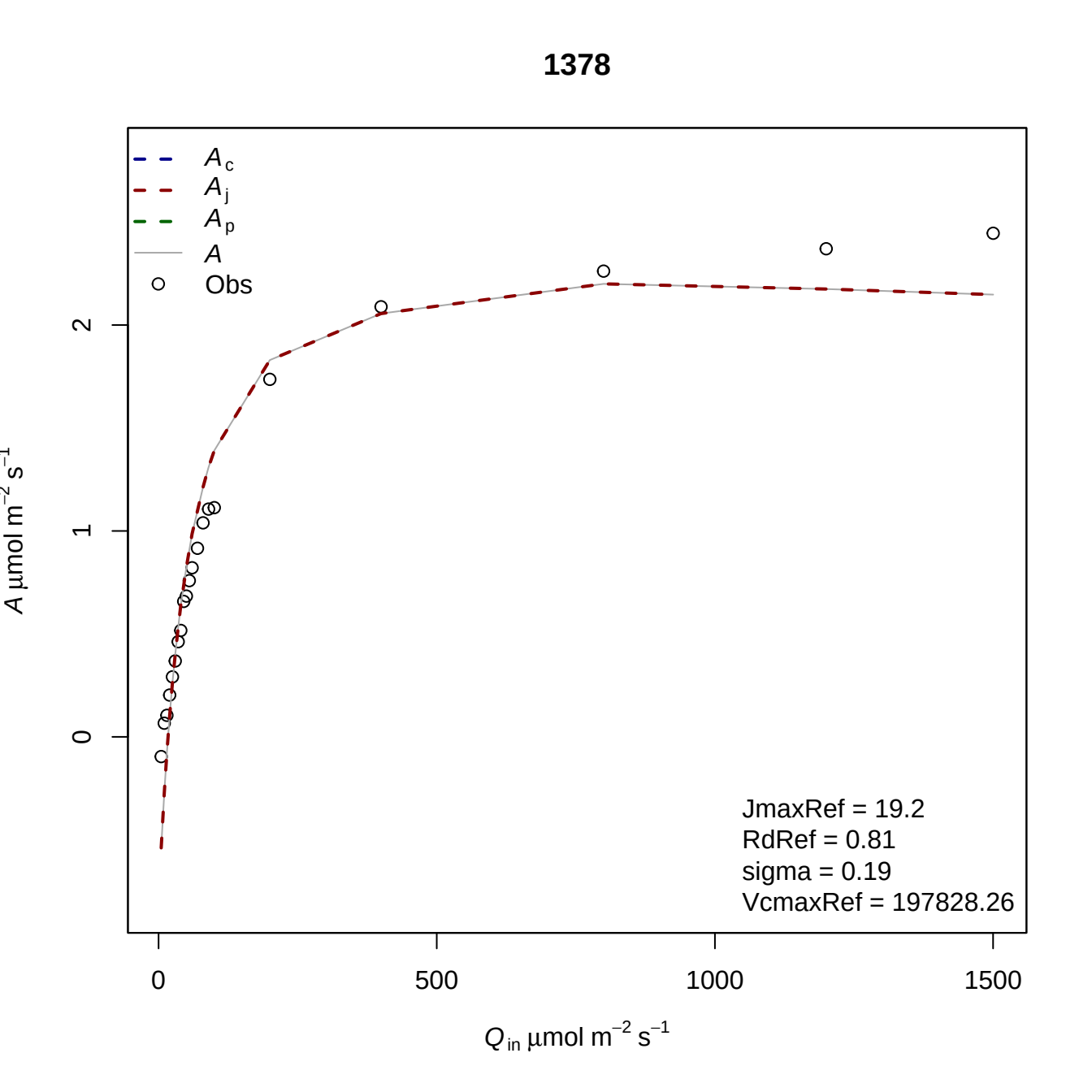

1373
A_c
A_j
A_p
A
Obs
JmaxRef = 21.76
RdRef = 0.31
sigma = 0.06
VcmaxRef = 10.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
200
400
600
800
1000

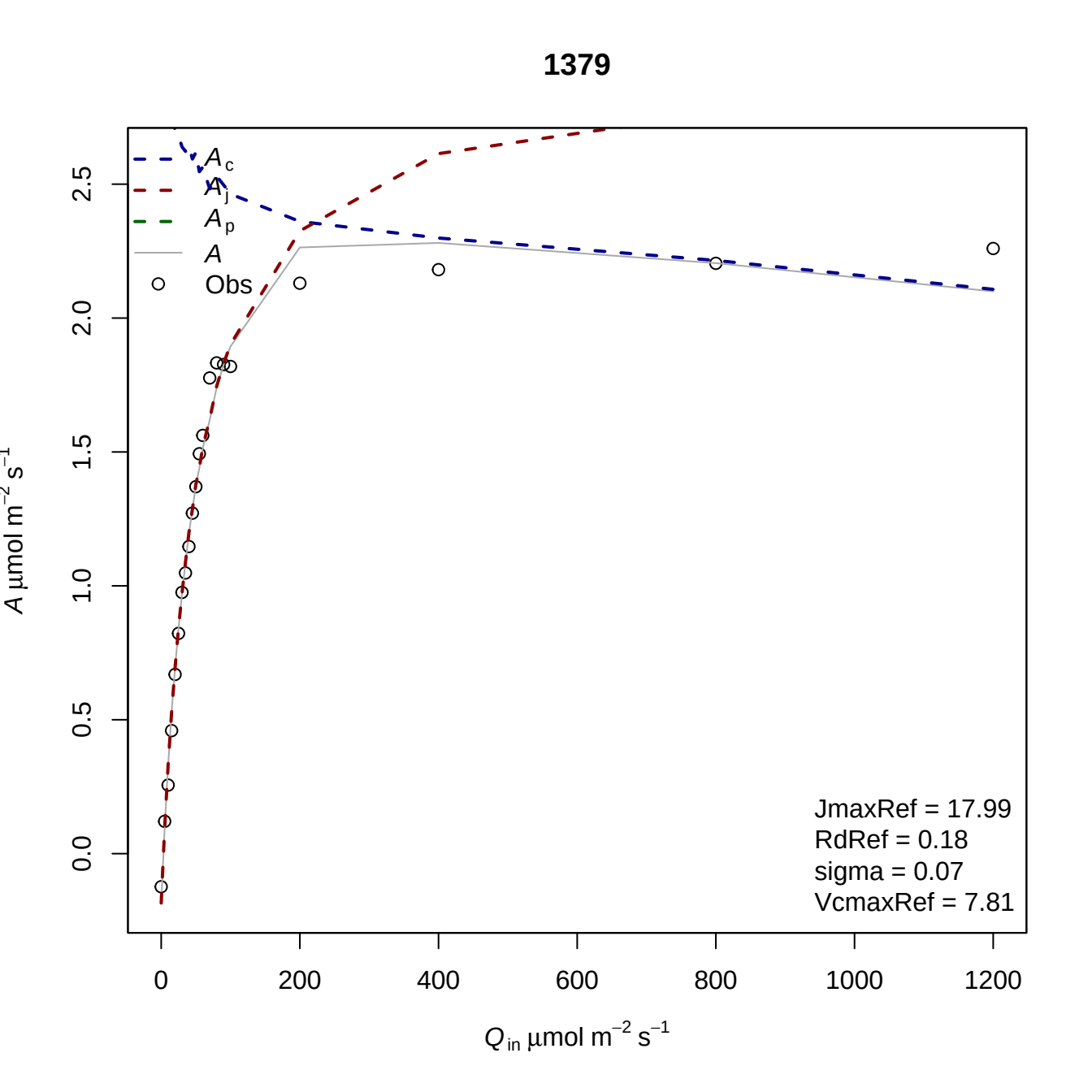

1374
A_c
A_j
A_p
A
Obs
JmaxRef = 26.9
RdRef = 1.21
sigma = 0.13
VcmaxRef = 346588.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000

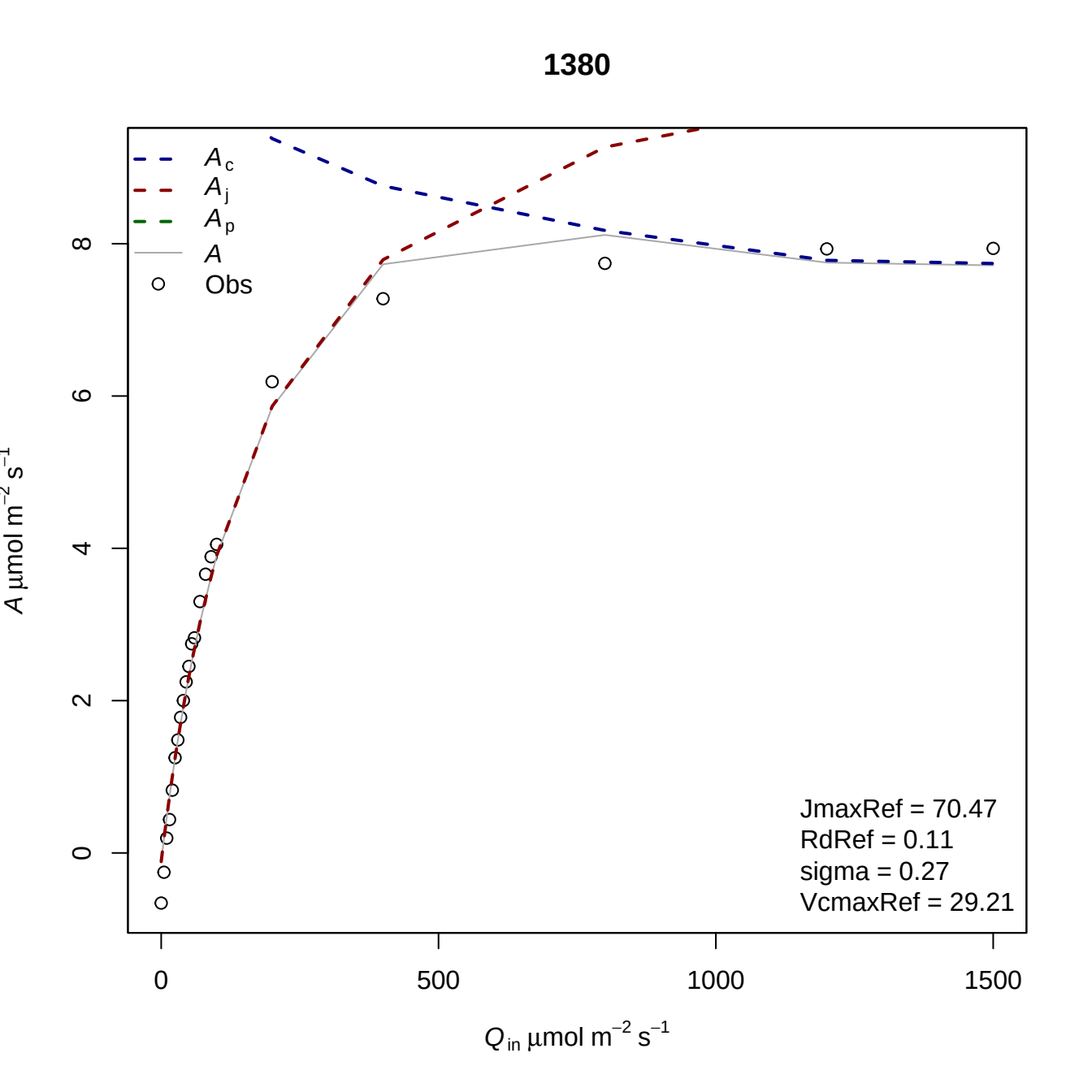

1375
A_c
A_j
A_p
A
Obs
JmaxRef = 30.2
RdRef = 1.09
sigma = 0.05
VcmaxRef = 14.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

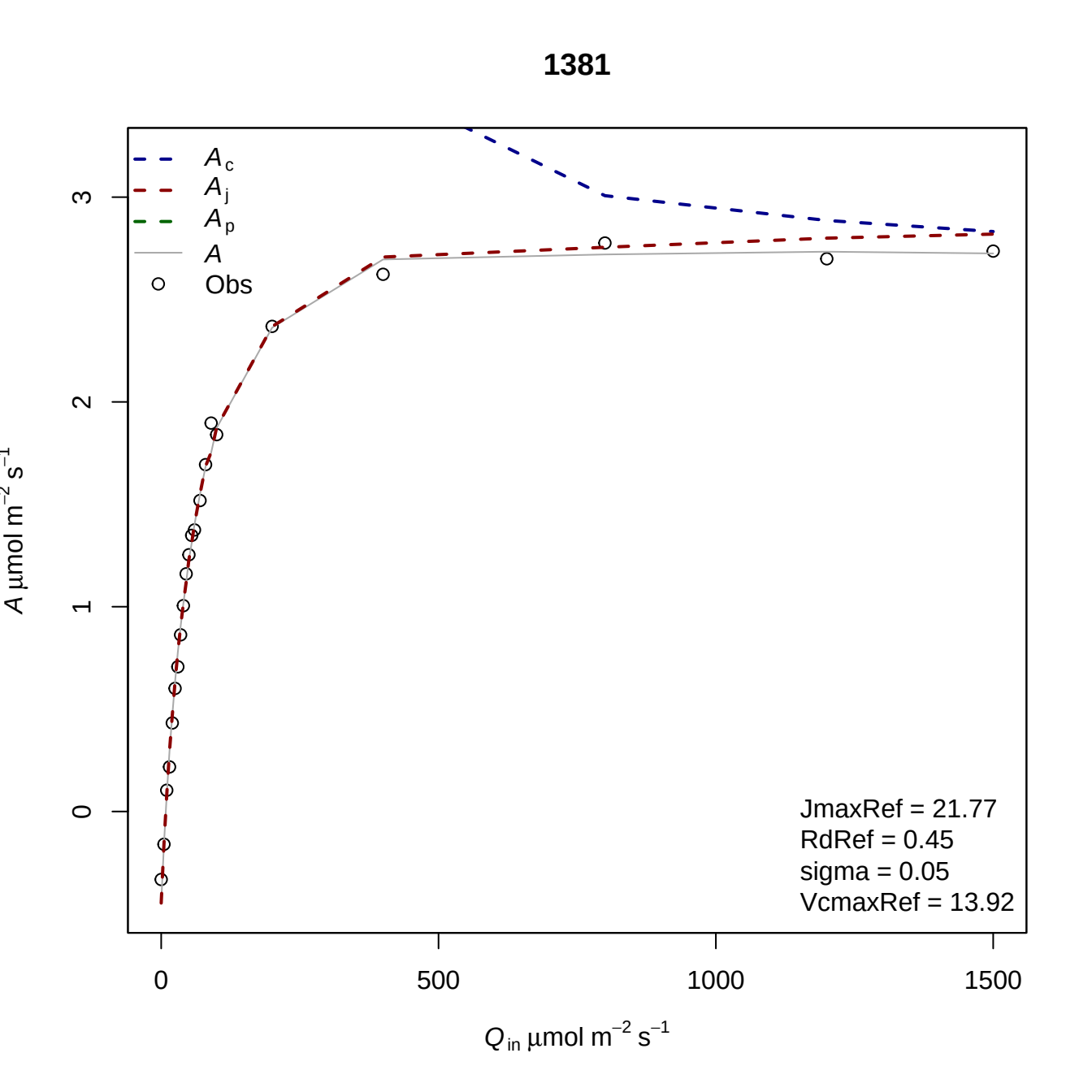

1376
A_c
A_j
A_p
A
Obs
JmaxRef = 85.04
RdRef = −0.21
sigma = 0.62
VcmaxRef = 41.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

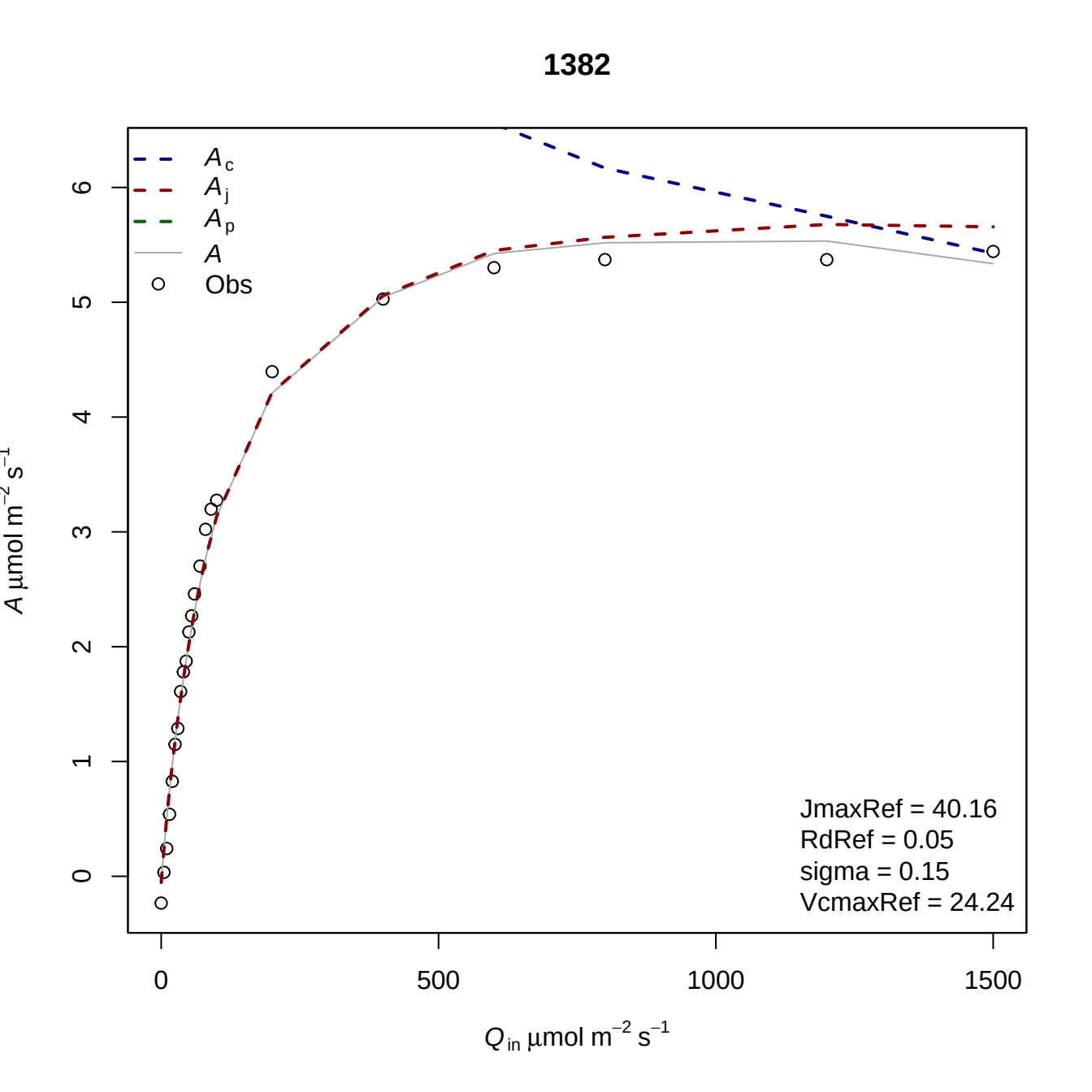

1377
A_c
A_j
A_p
A
Obs
JmaxRef = 40.51
RdRef = 0.56
sigma = 0.21
VcmaxRef = 24.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

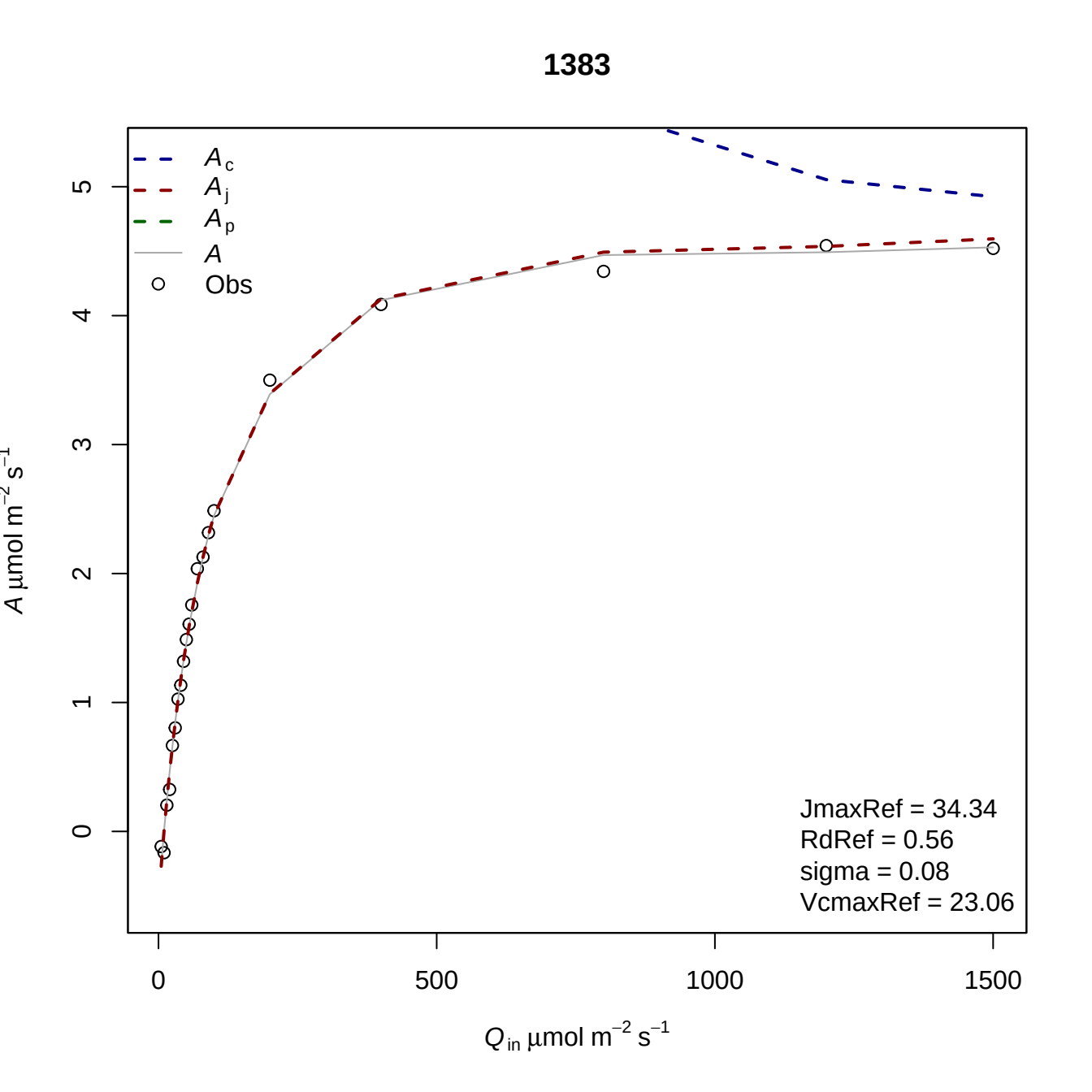

1378
A_c
A_j
A_p
A
Obs
JmaxRef = 19.2
RdRef = 0.81
sigma = 0.19
VcmaxRef = 197828.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
0
500
1000
1500

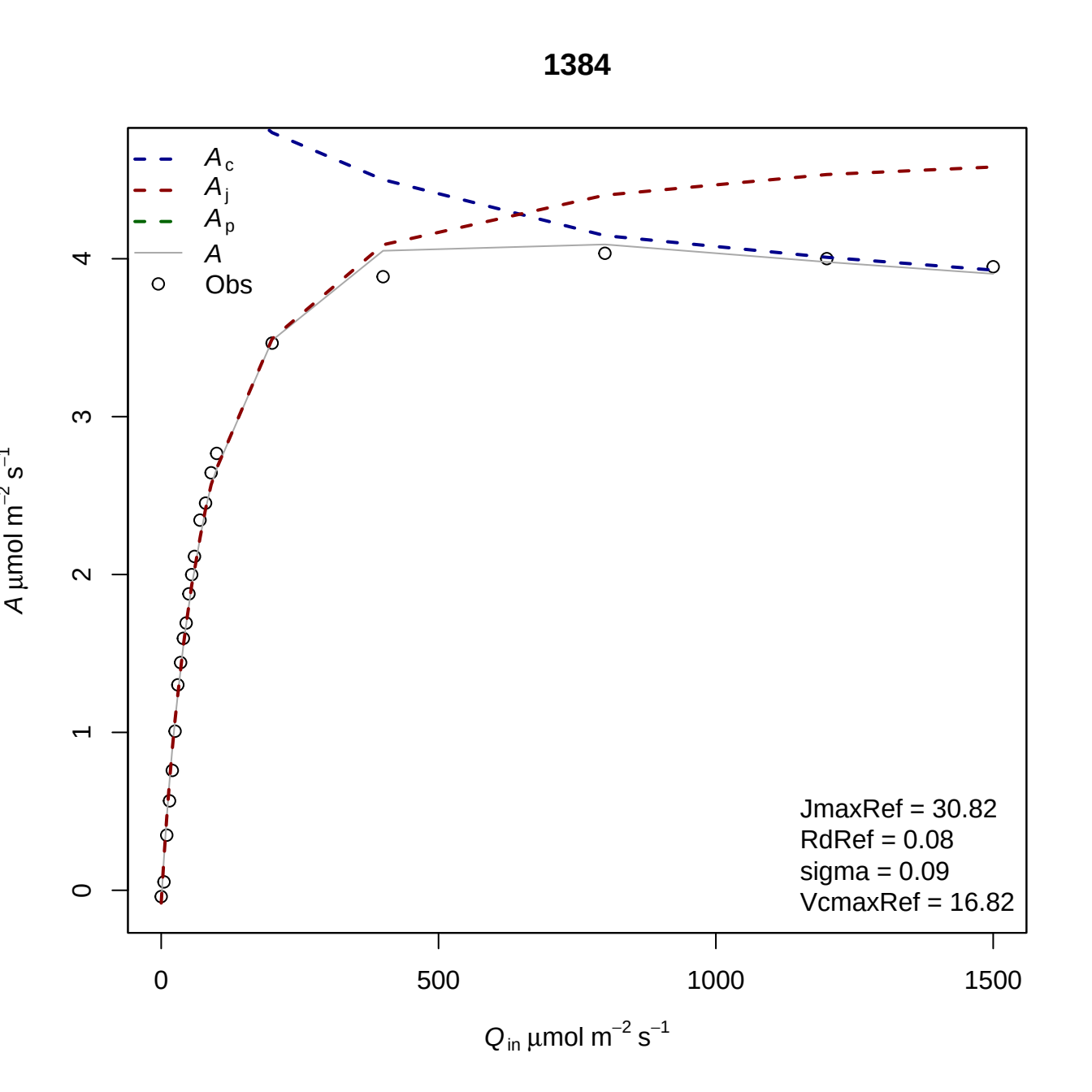

1379
A_c
A_j
A_p
A
Obs
JmaxRef = 17.99
RdRef = 0.18
sigma = 0.07
VcmaxRef = 7.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0.0
0.5
1.0
1.5
2.0
2.5
0
200
400
600
800
1000
1200

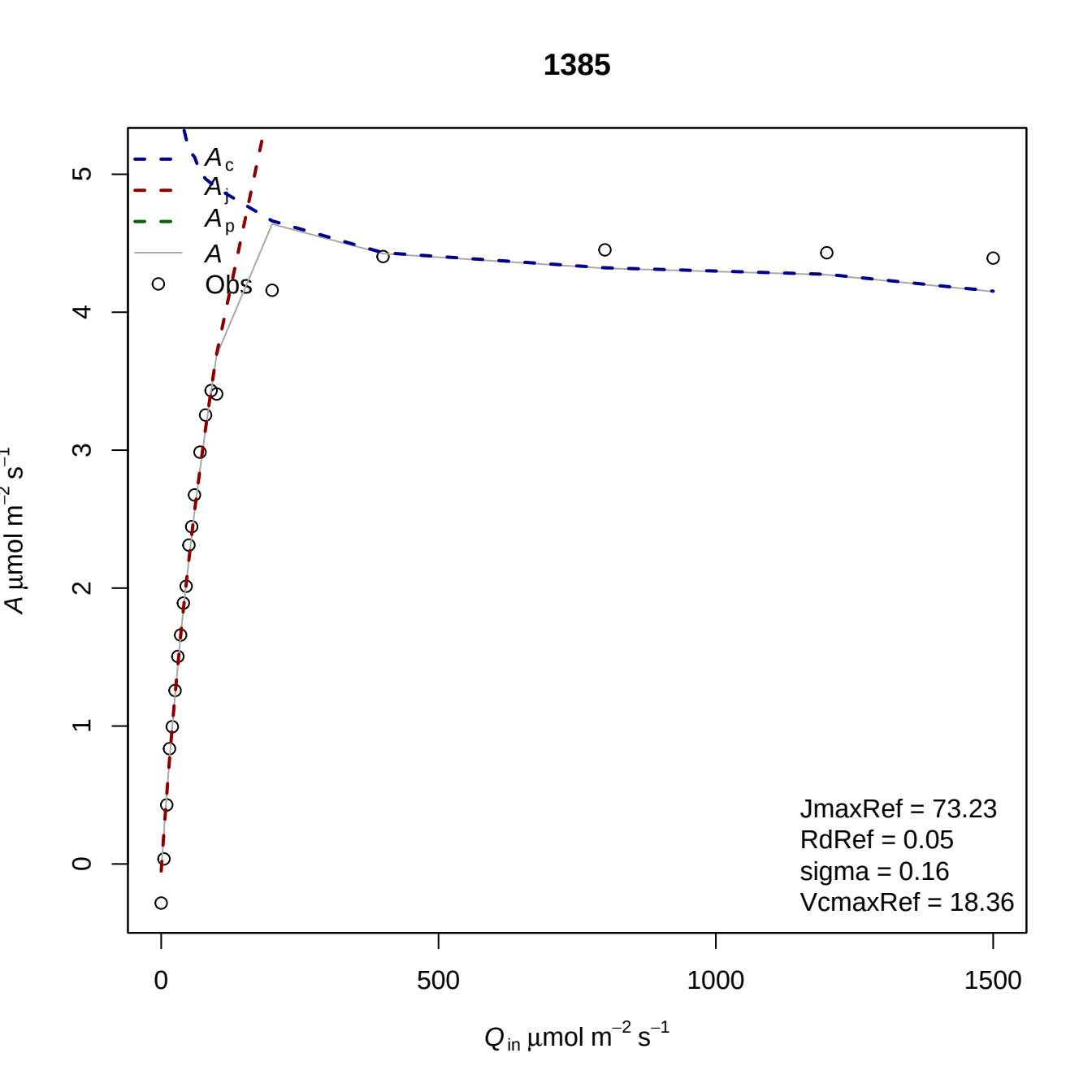

1380
A_c
A_j
A_p
A
Obs
JmaxRef = 70.47
RdRef = 0.11
sigma = 0.27
VcmaxRef = 29.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

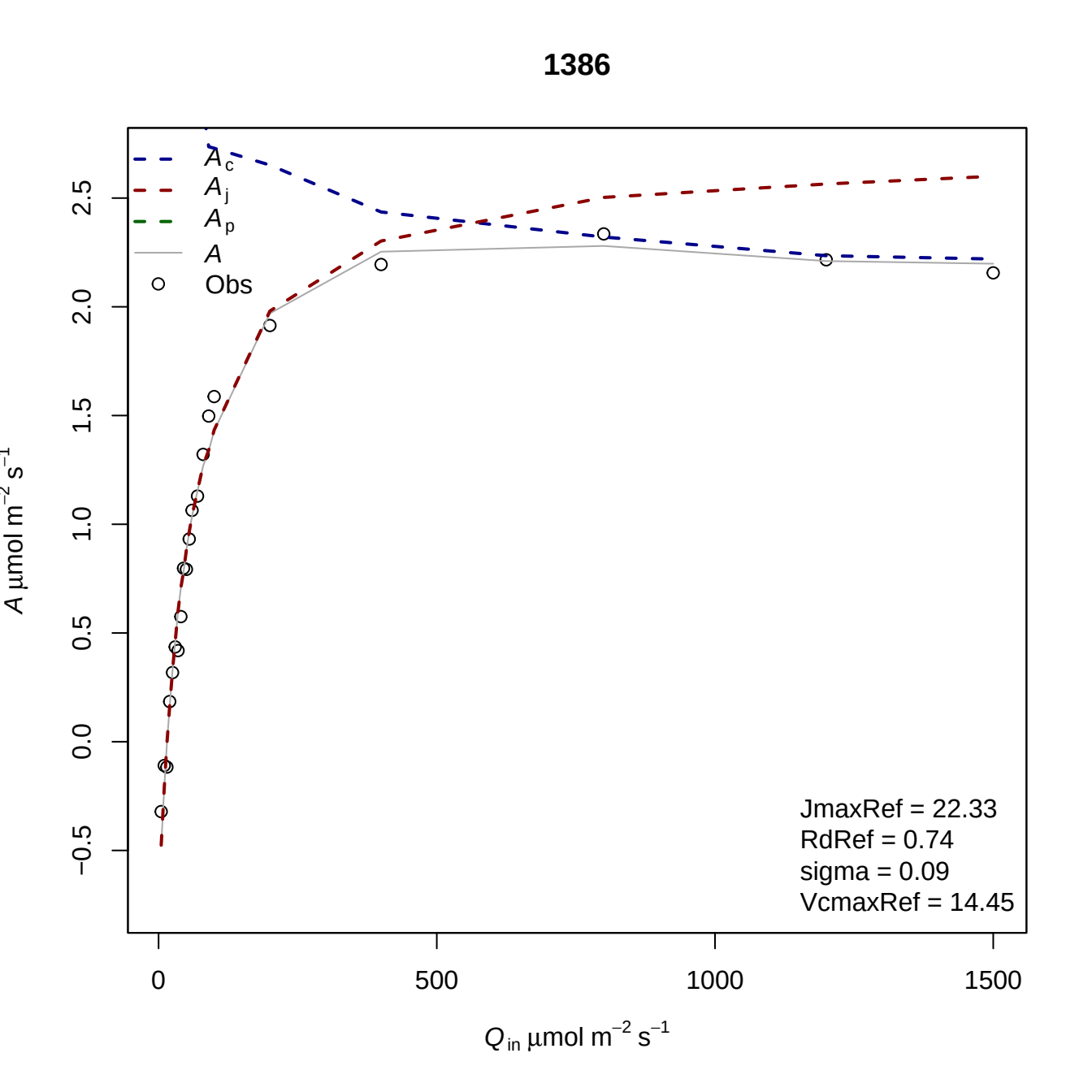

1381
A_c
A_j
A_p
A
Obs
JmaxRef = 21.77
RdRef = 0.45
sigma = 0.05
VcmaxRef = 13.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500

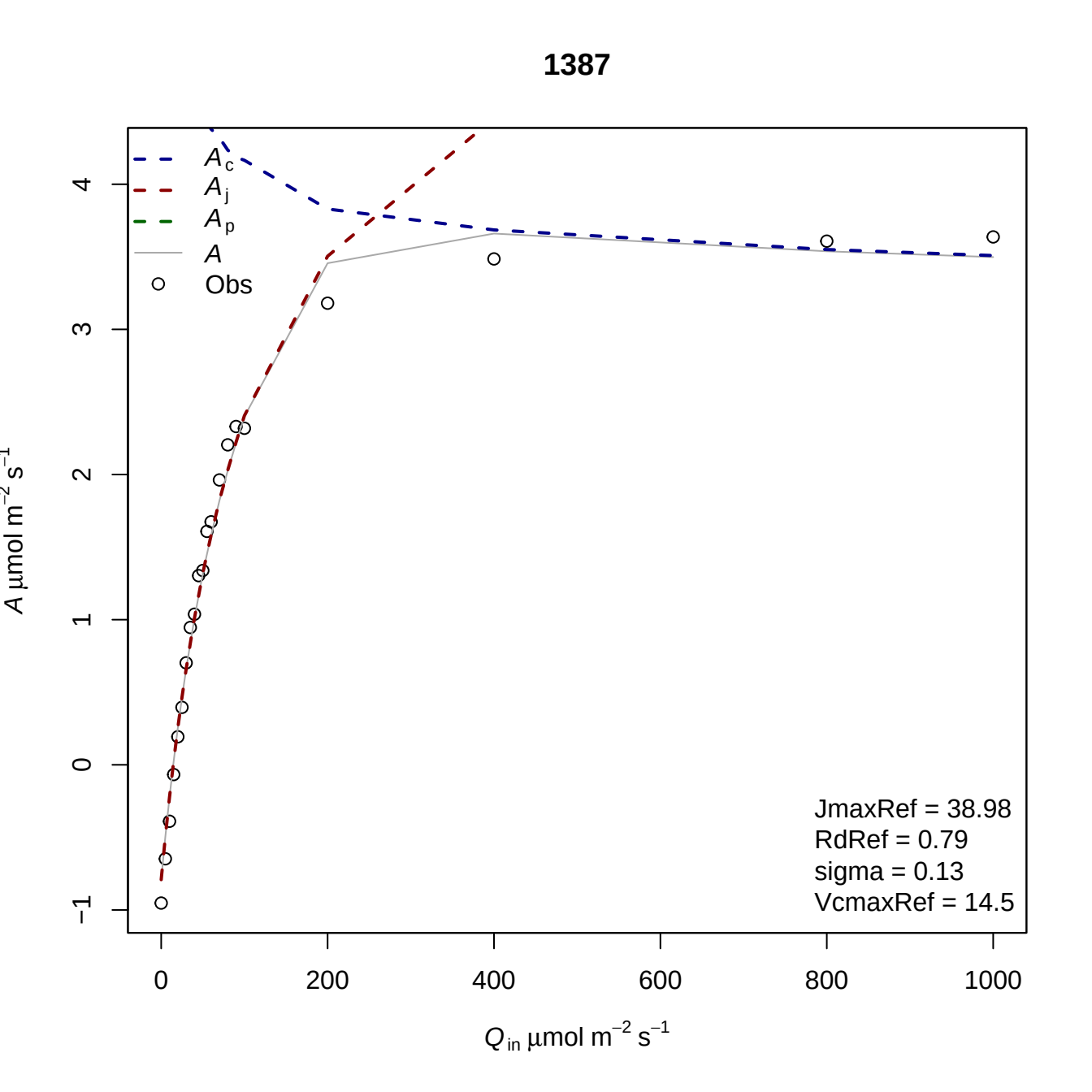

1382
A_c
A_j
A_p
A
Obs
JmaxRef = 40.16
RdRef = 0.05
sigma = 0.15
VcmaxRef = 24.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

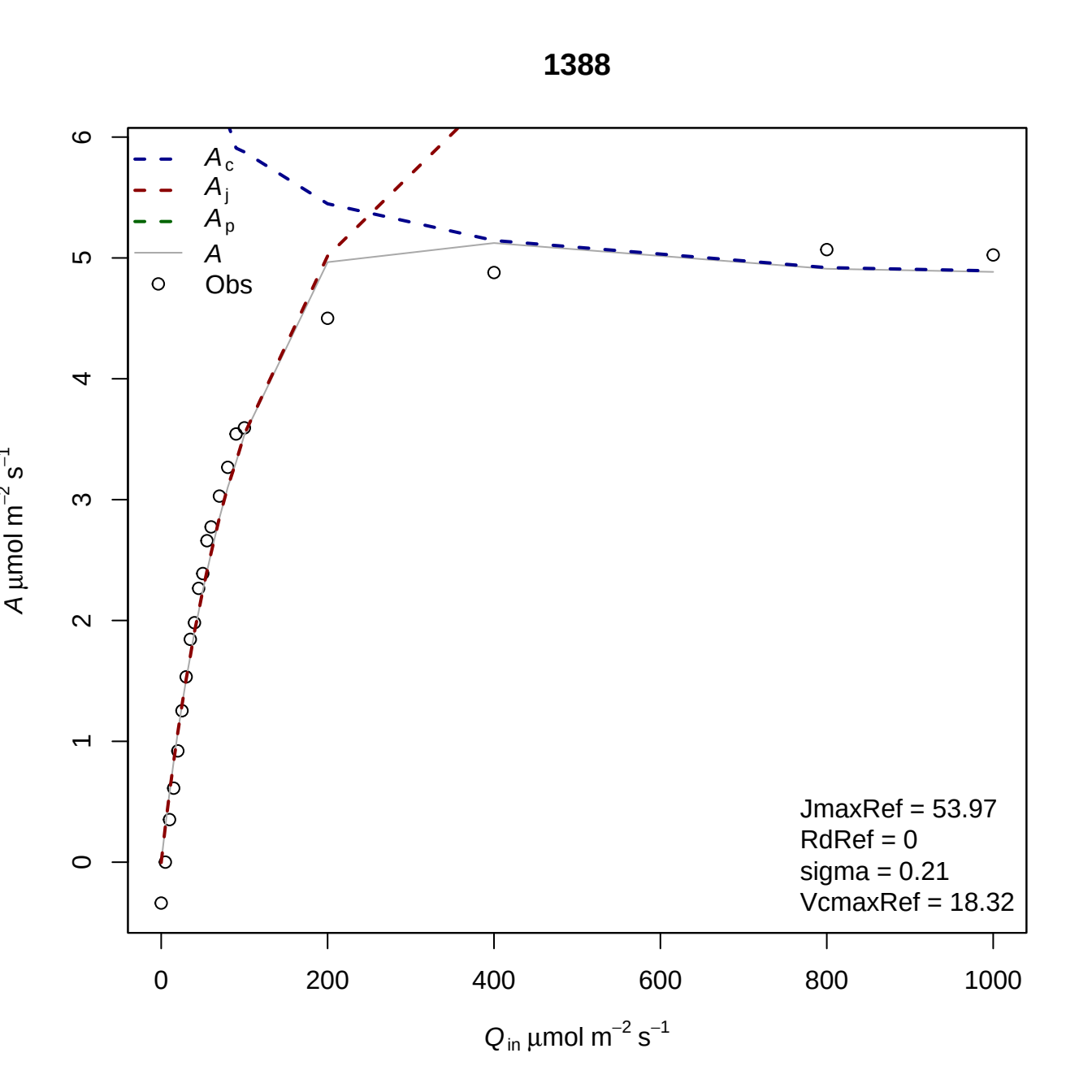

1383
A_c
A_j
A_p
A
Obs
JmaxRef = 34.34
RdRef = 0.56
sigma = 0.08
VcmaxRef = 23.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

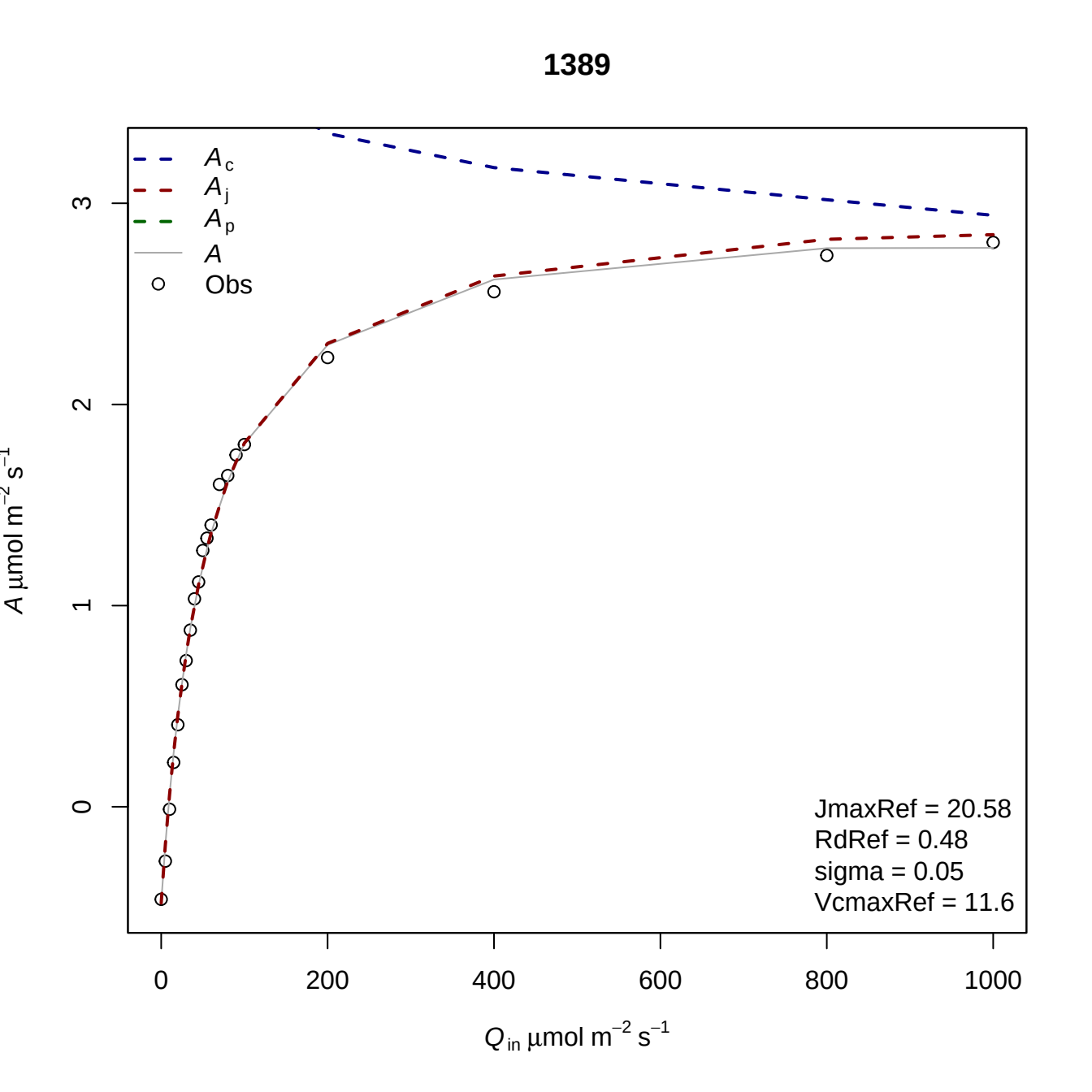

1384
A_c
A_j
A_p
A
Obs
JmaxRef = 30.82
RdRef = 0.08
sigma = 0.09
VcmaxRef = 16.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

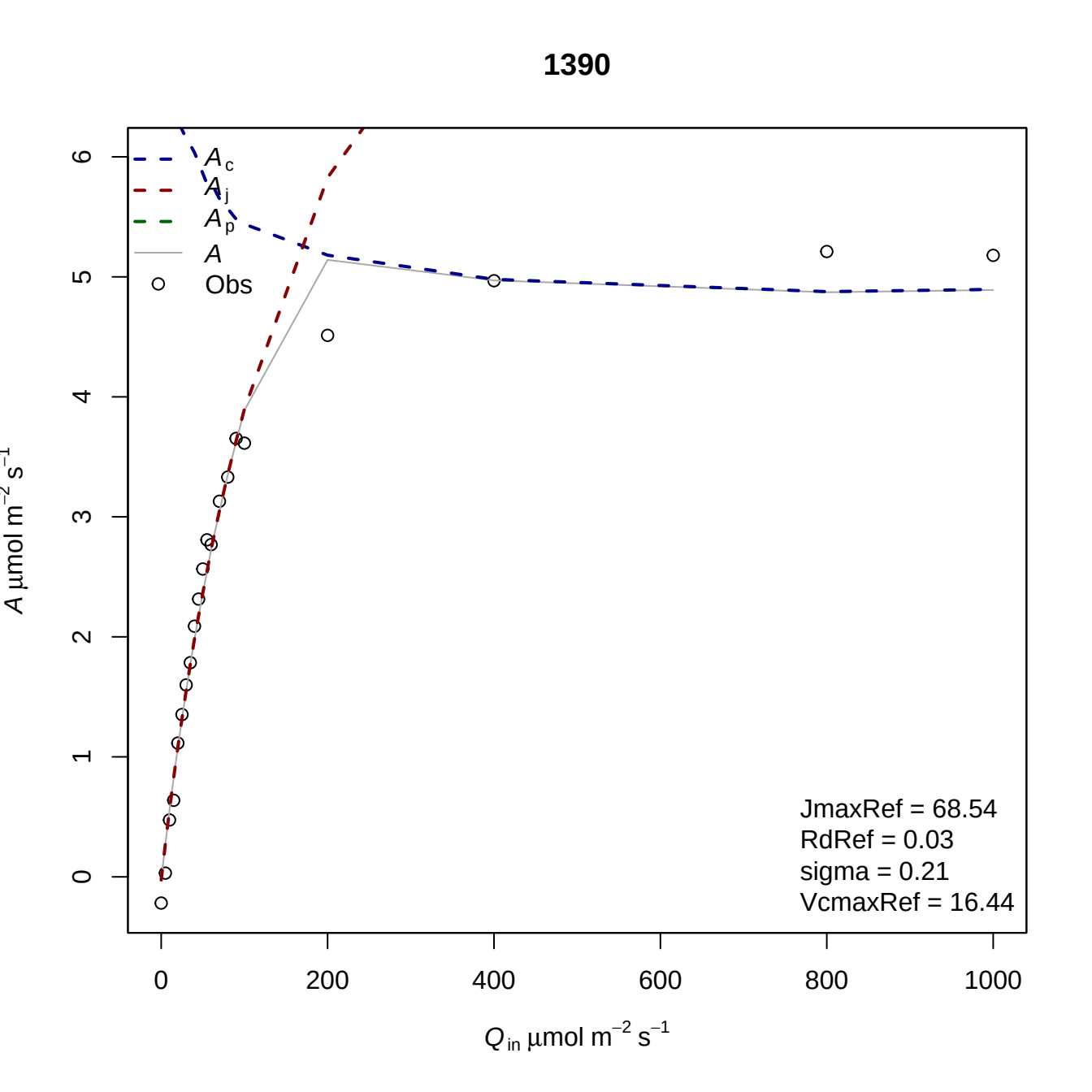

1385
A_c
A_j
A_p
A
Obs
JmaxRef = 73.23
RdRef = 0.05
sigma = 0.16
VcmaxRef = 18.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

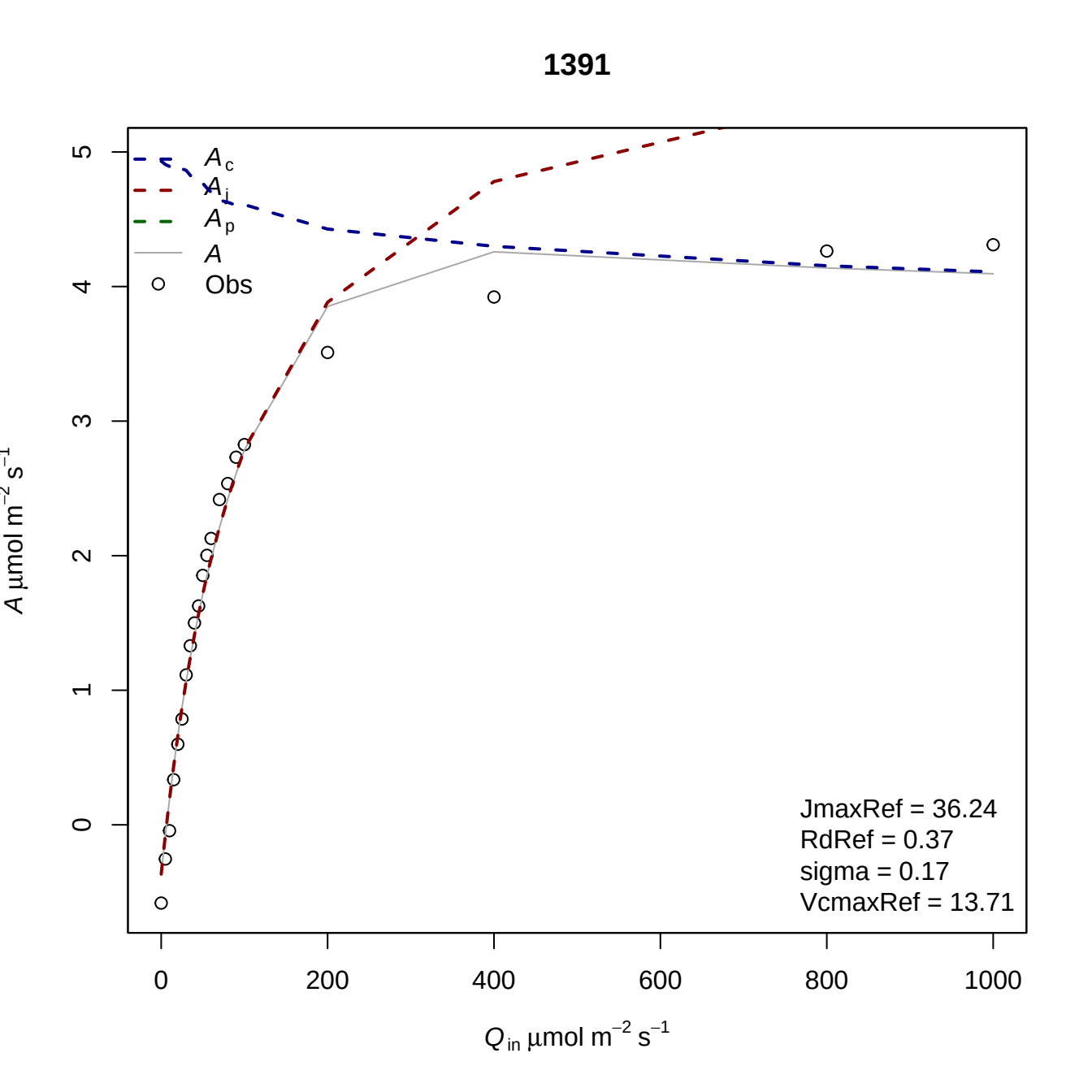

1386
A_c
A_j
A_p
A
Obs
JmaxRef = 22.33
RdRef = 0.74
sigma = 0.09
VcmaxRef = 14.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
0
500
1000
1500

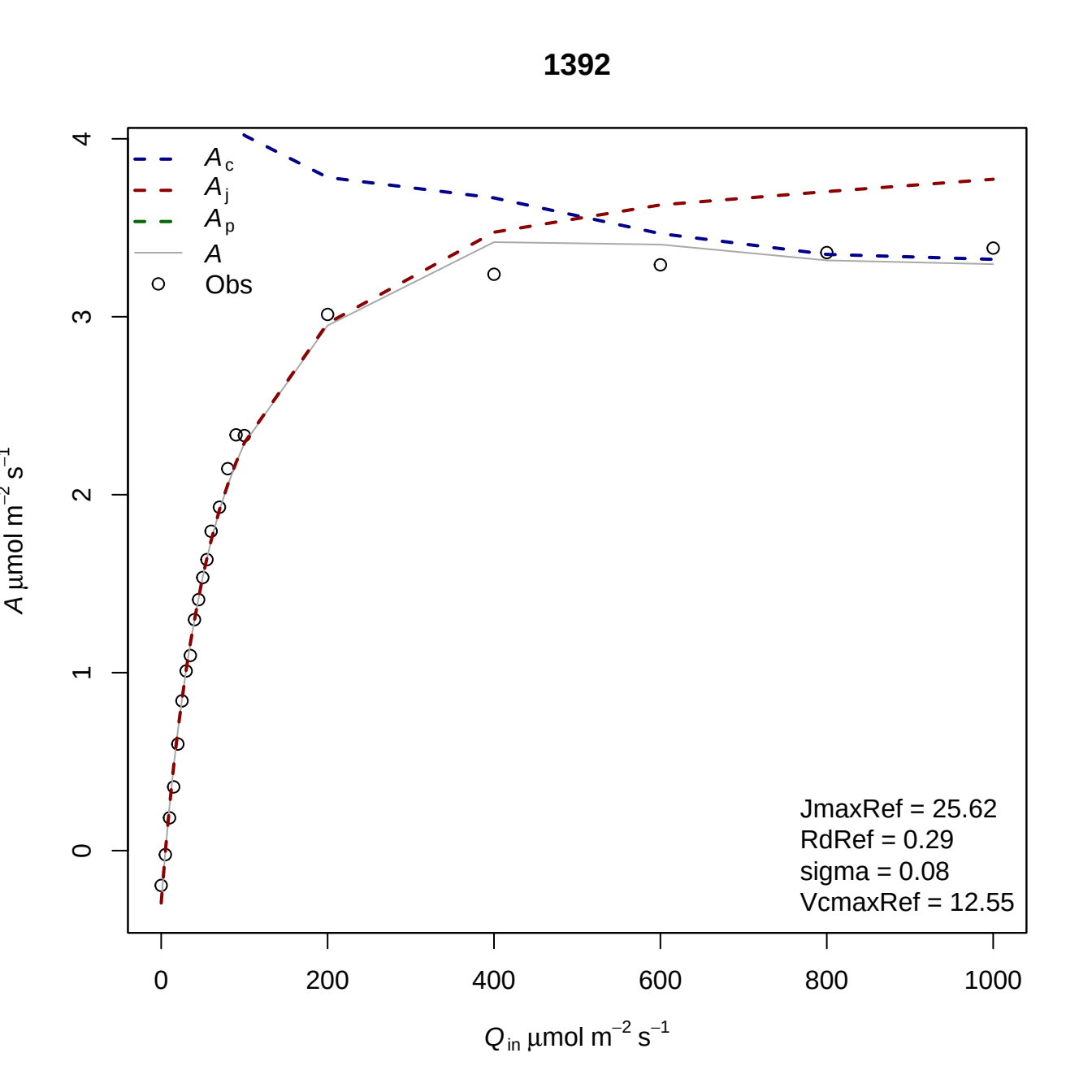

1387
A_c
A_j
A_p
A
Obs
JmaxRef = 38.98
RdRef = 0.79
sigma = 0.13
VcmaxRef = 14.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

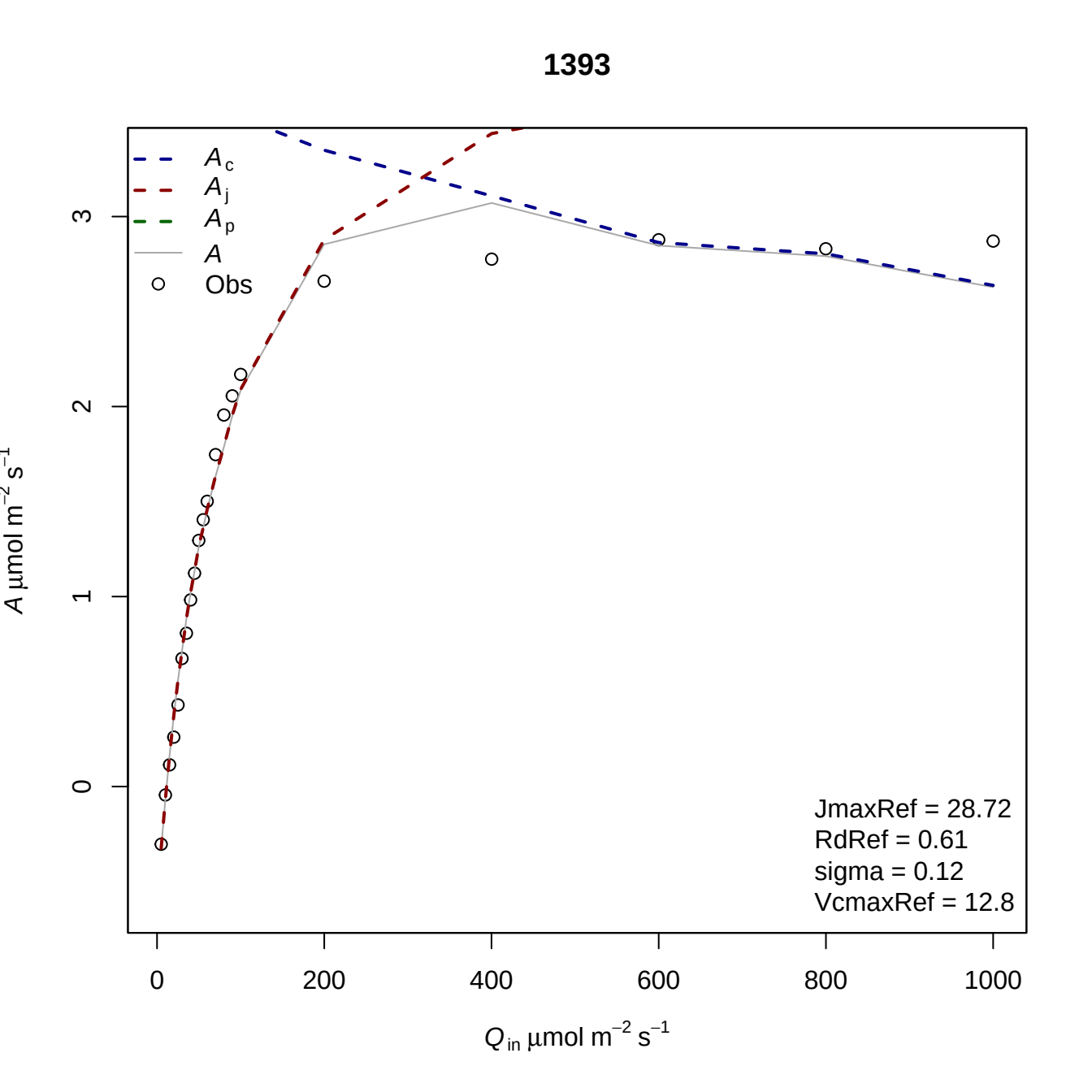

1388
A_c
A_j
A_p
A
Obs
JmaxRef = 53.97
RdRef = 0
sigma = 0.21
VcmaxRef = 18.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

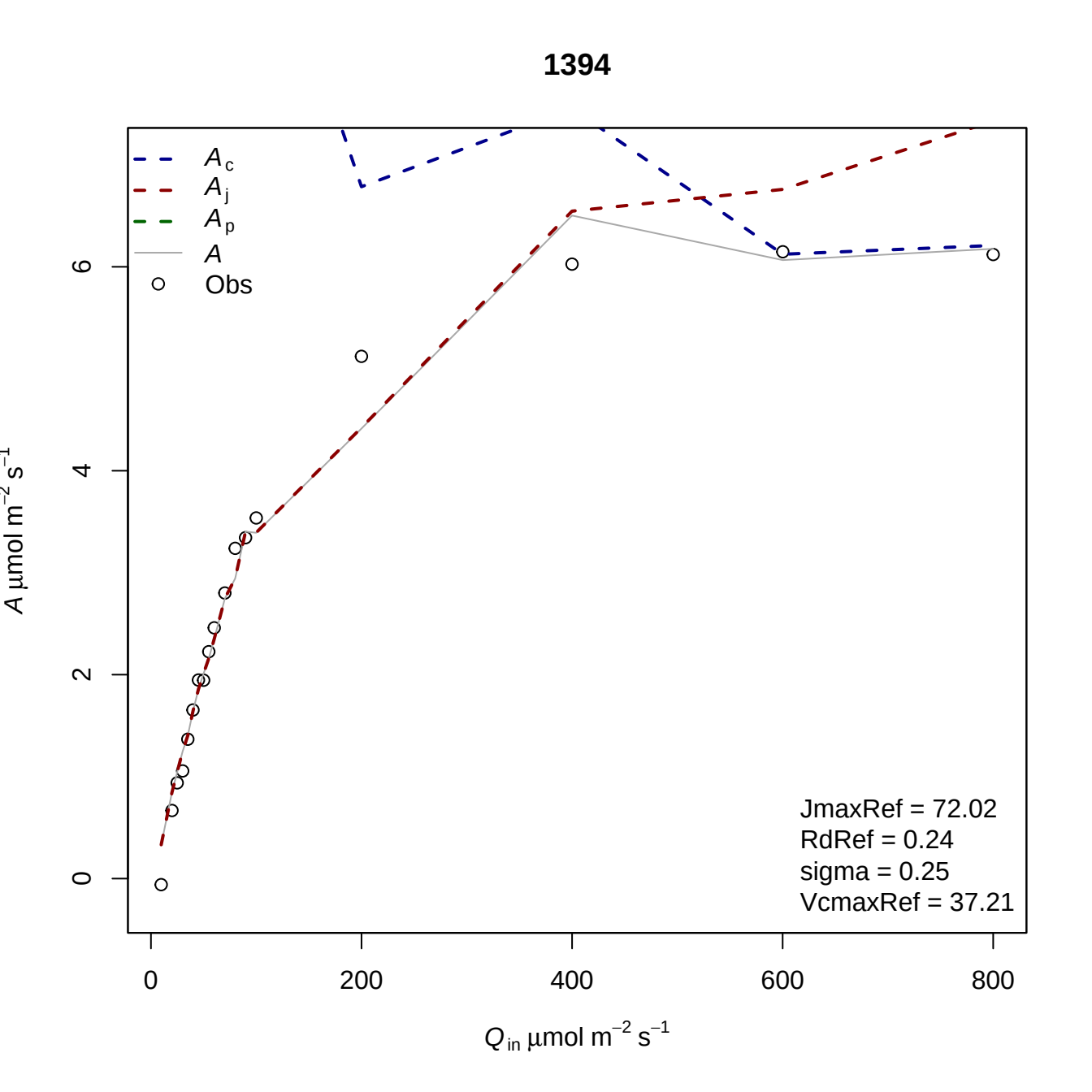

1389
A_c
A_j
A_p
A
Obs
JmaxRef = 20.58
RdRef = 0.48
sigma = 0.05
VcmaxRef = 11.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
200
400
600
800
1000

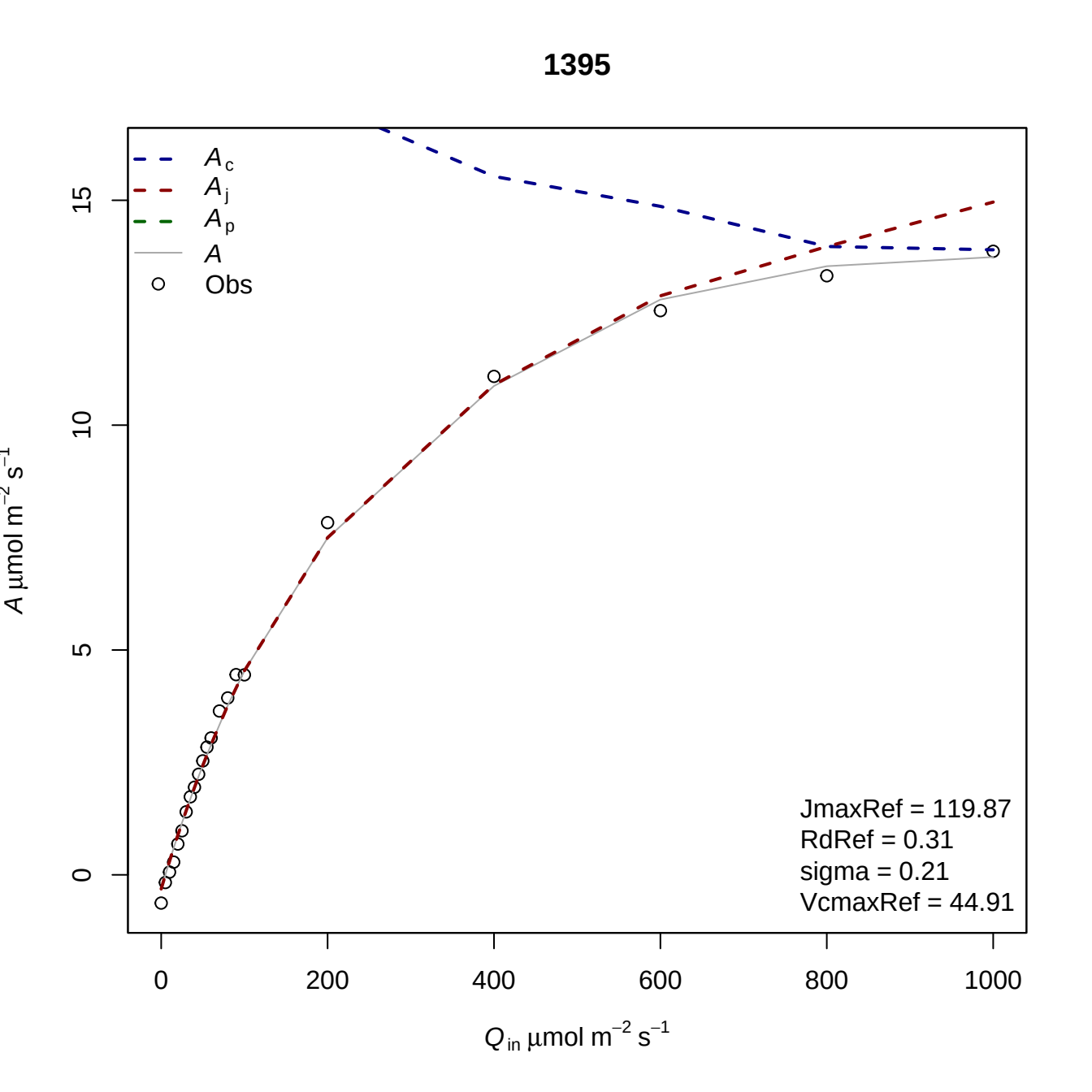

1390
A_c
A_j
A_p
A
Obs
JmaxRef = 68.54
RdRef = 0.03
sigma = 0.21
VcmaxRef = 16.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

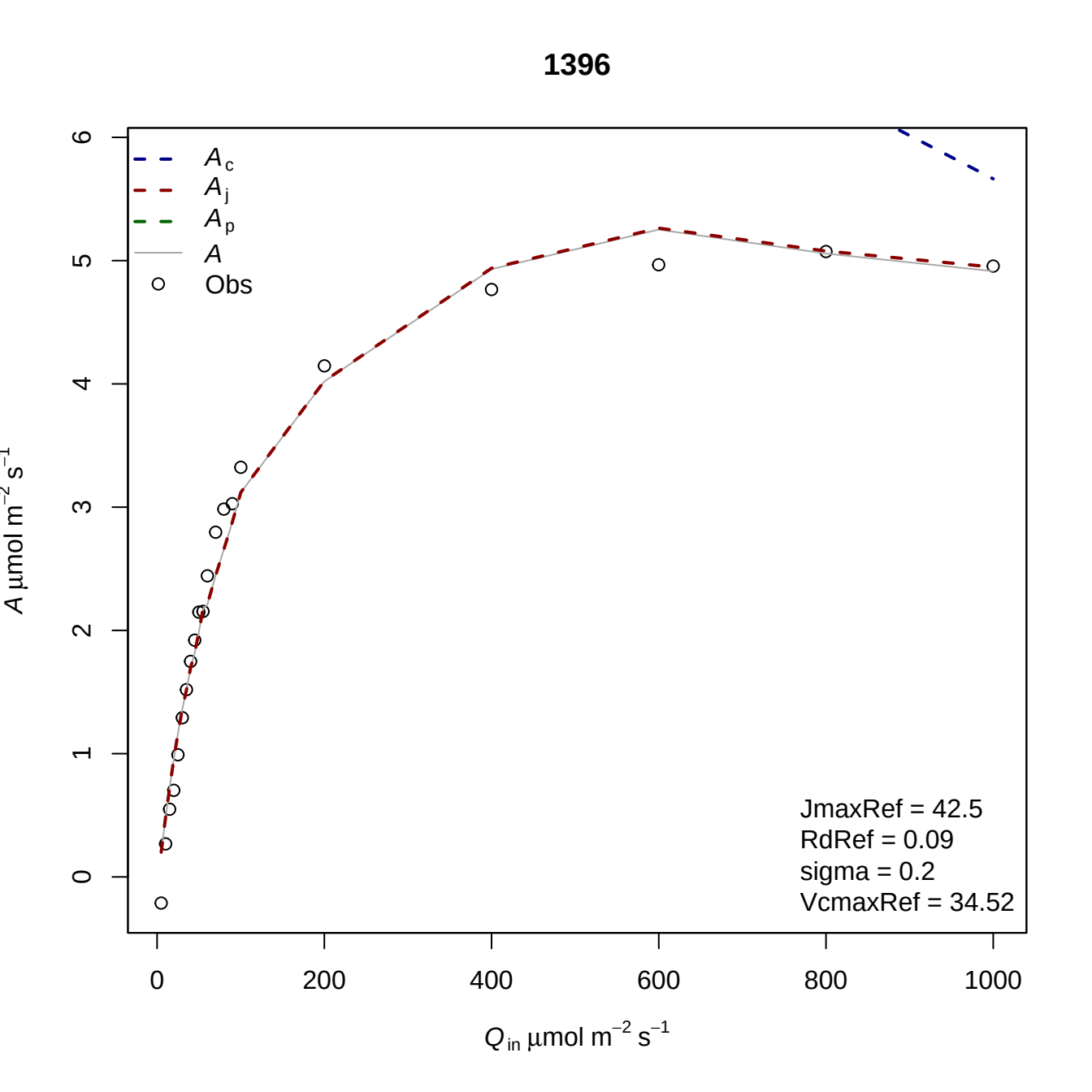

1391
A_c
A_j
A_p
A
Obs
JmaxRef = 36.24
RdRef = 0.37
sigma = 0.17
VcmaxRef = 13.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
200
400
600
800
1000

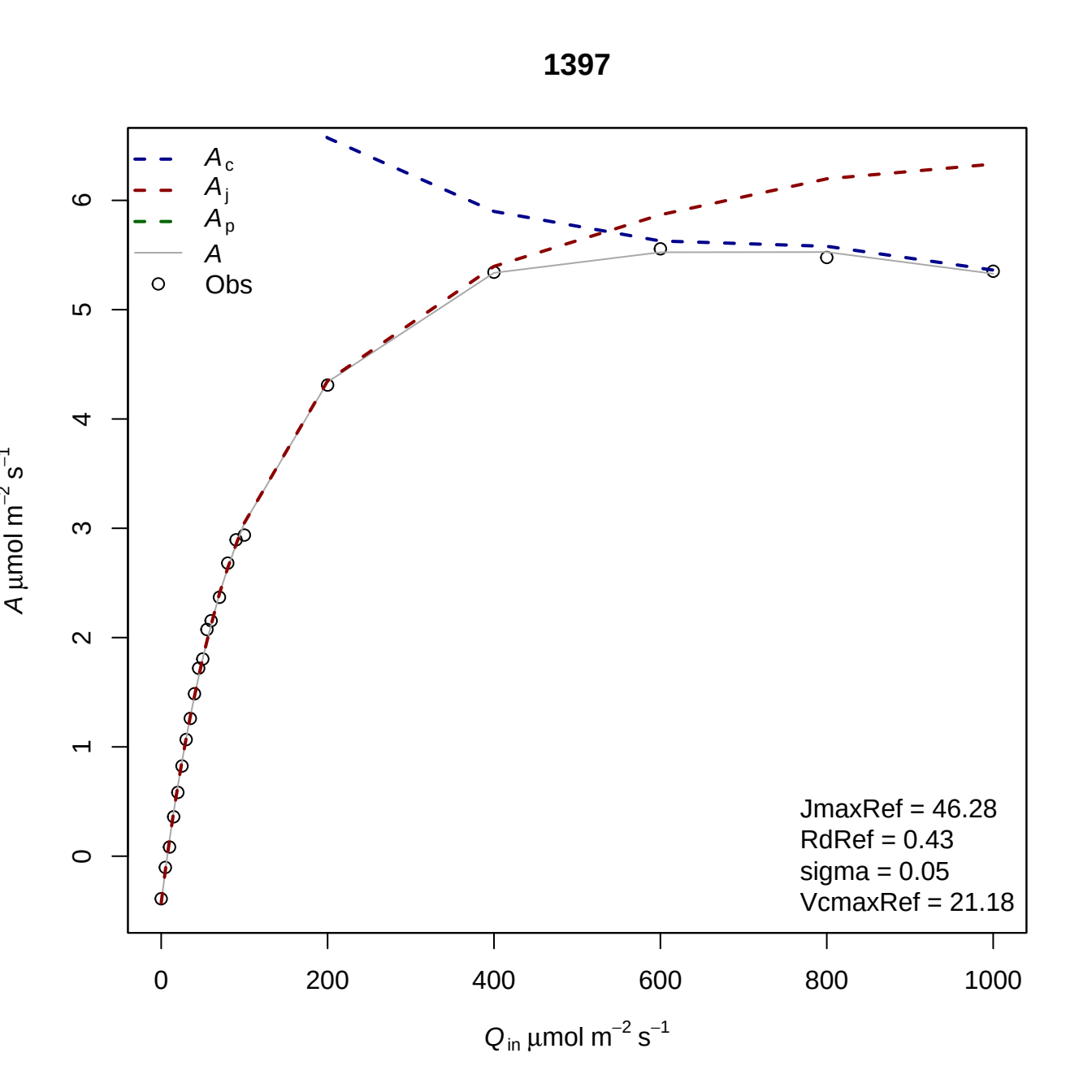

1392
A_c
A_j
A_p
A
Obs
JmaxRef = 25.62
RdRef = 0.29
sigma = 0.08
VcmaxRef = 12.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
200
400
600
800
1000

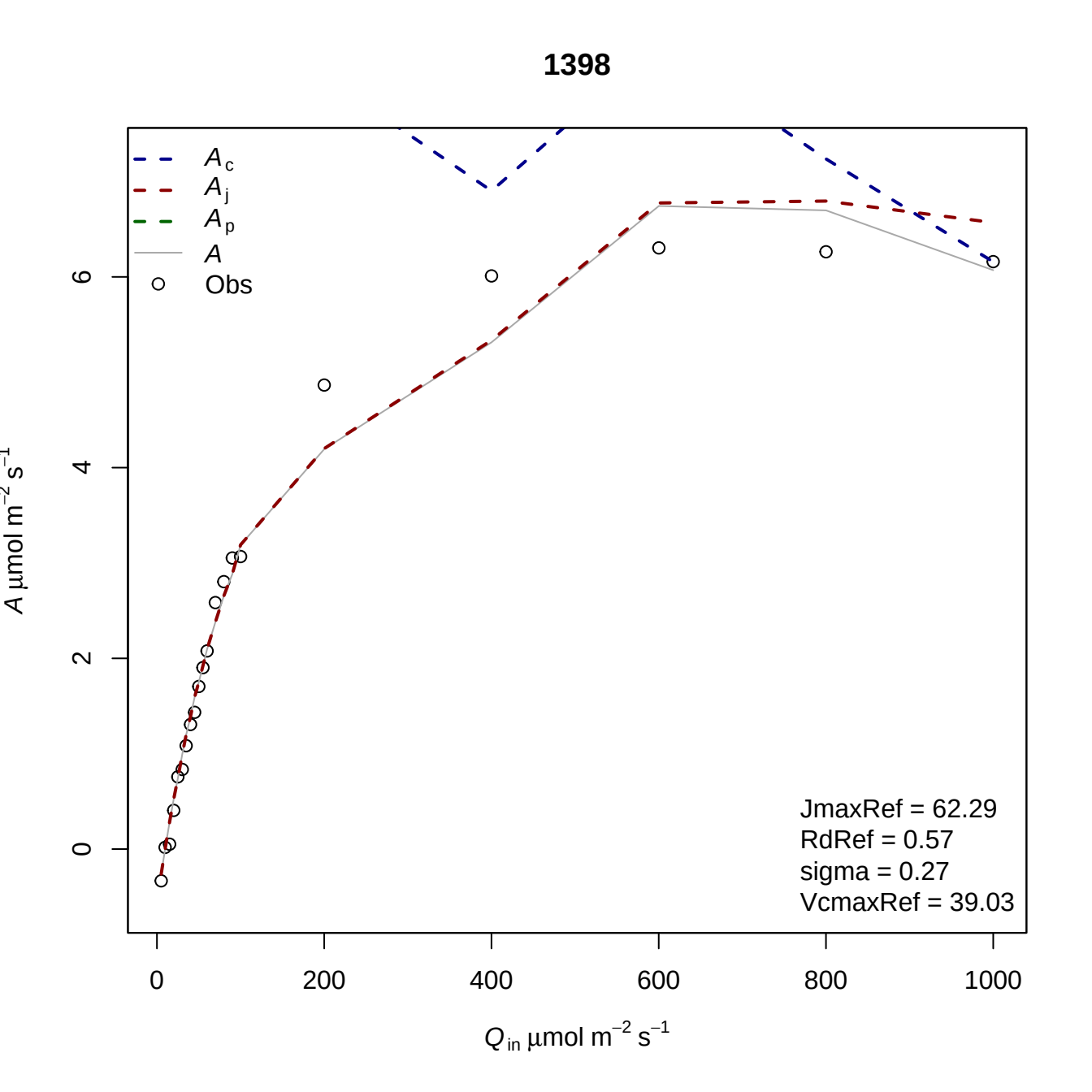

1393
A_c
A_j
A_p
A
Obs
JmaxRef = 28.72
RdRef = 0.61
sigma = 0.12
VcmaxRef = 12.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
200
400
600
800
1000

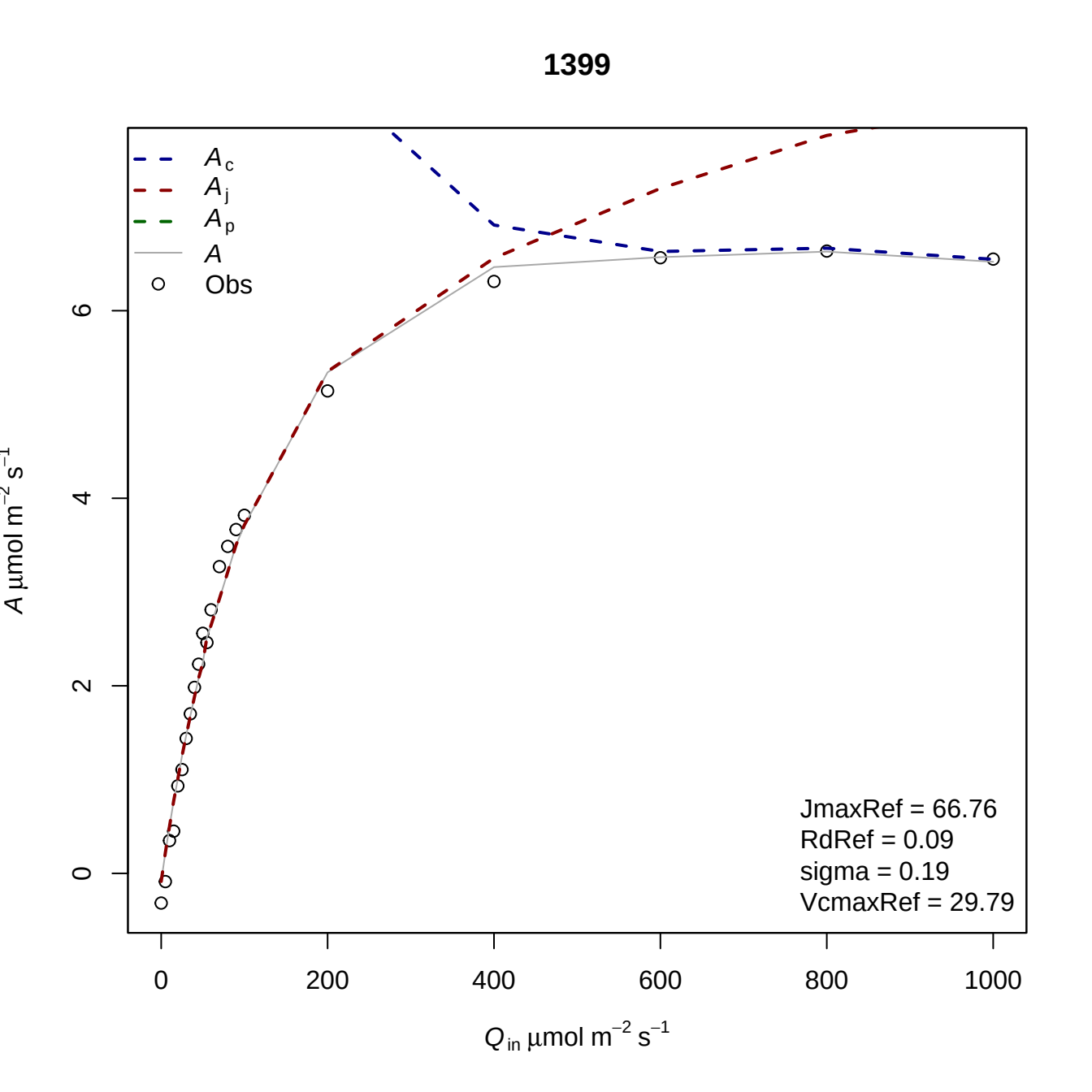

1394
A_c
A_j
A_p
A
Obs
JmaxRef = 72.02
RdRef = 0.24
sigma = 0.25
VcmaxRef = 37.21
0
2
4
6
0
200
400
600
800
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

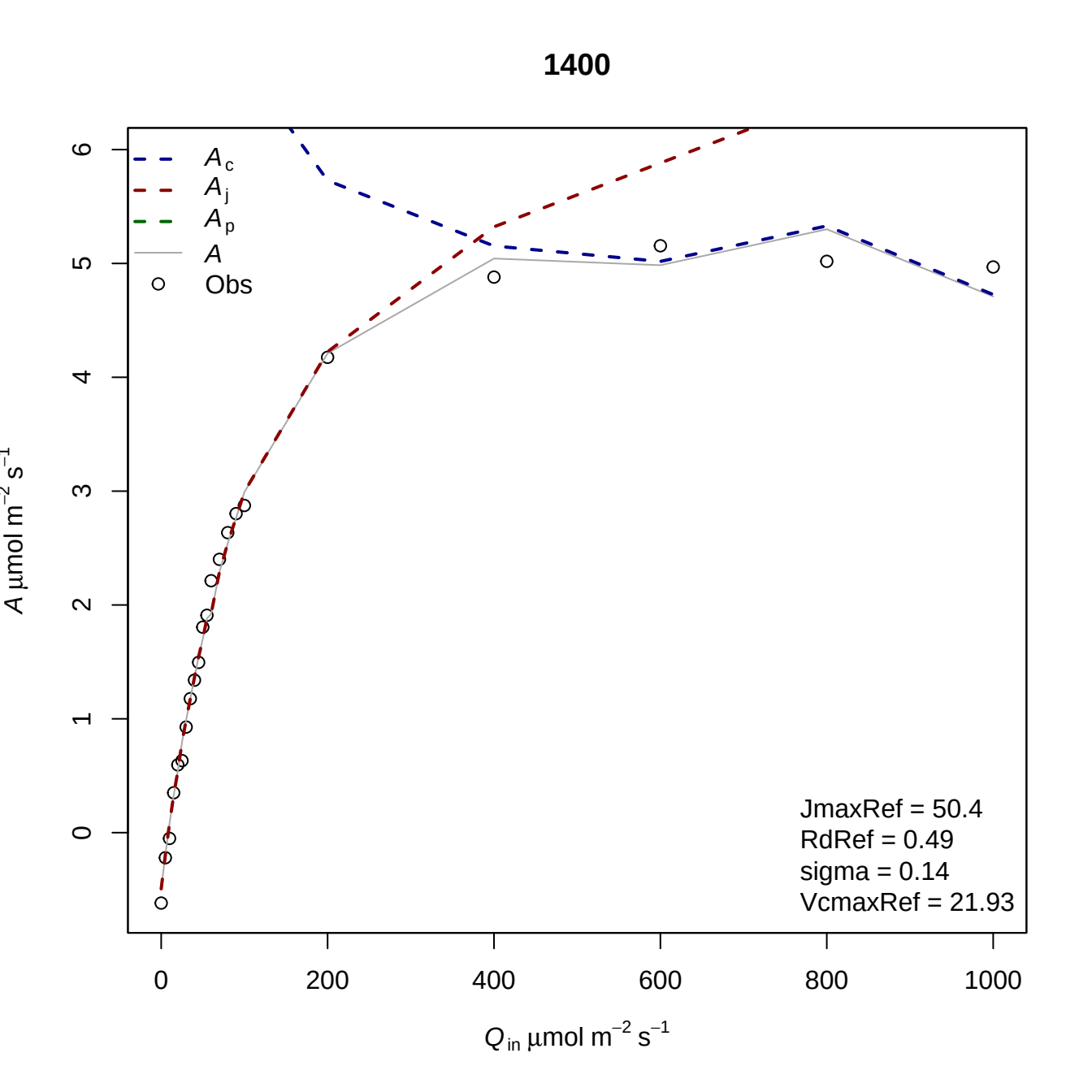

1395
A_c
A_j
A_p
A
Obs
JmaxRef = 119.87
RdRef = 0.31
sigma = 0.21
VcmaxRef = 44.91
0
5
10
15
0
200
400
600
800
1000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

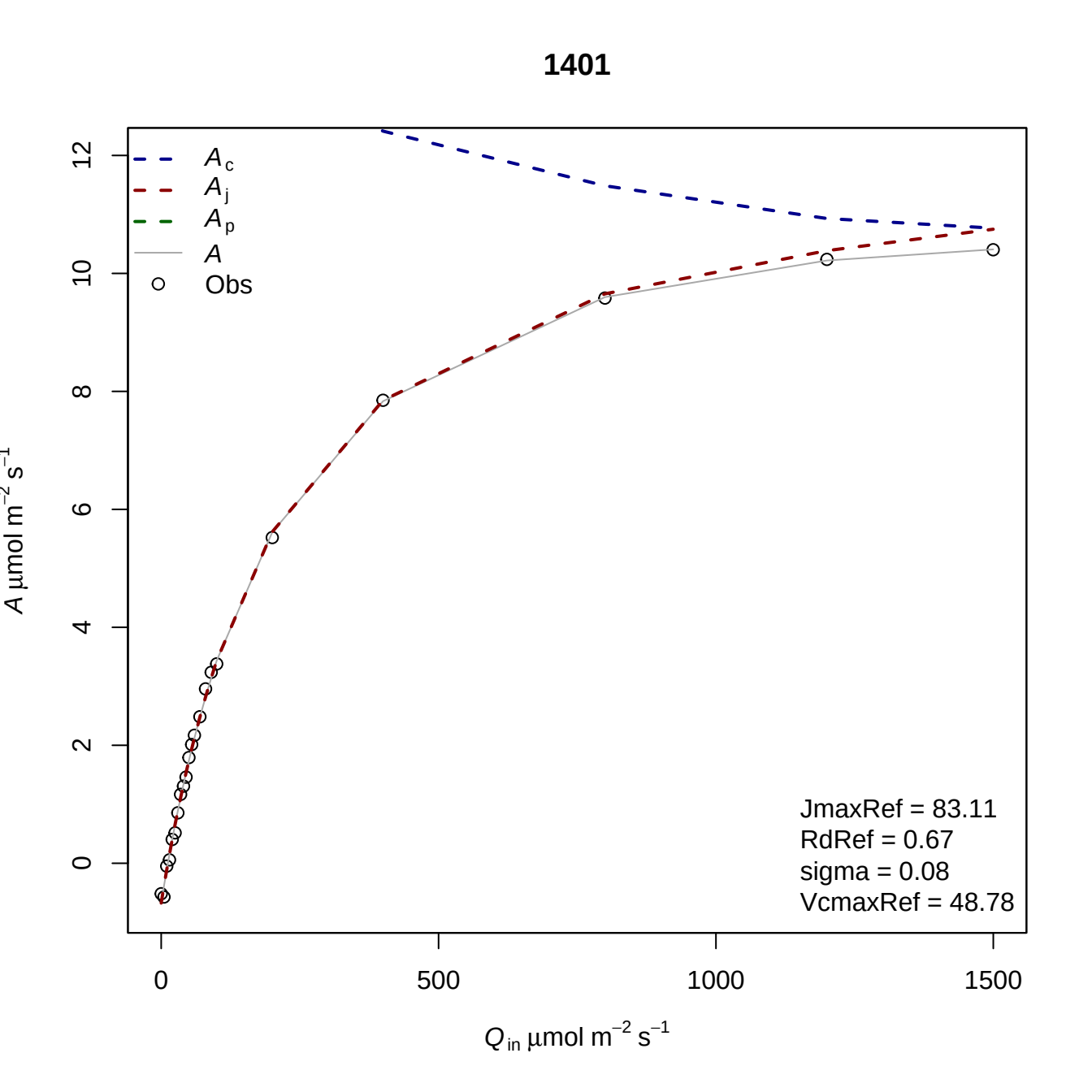

1396
A_c
A_j
A_p
A
Obs
JmaxRef = 42.5
RdRef = 0.09
sigma = 0.2
VcmaxRef = 34.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

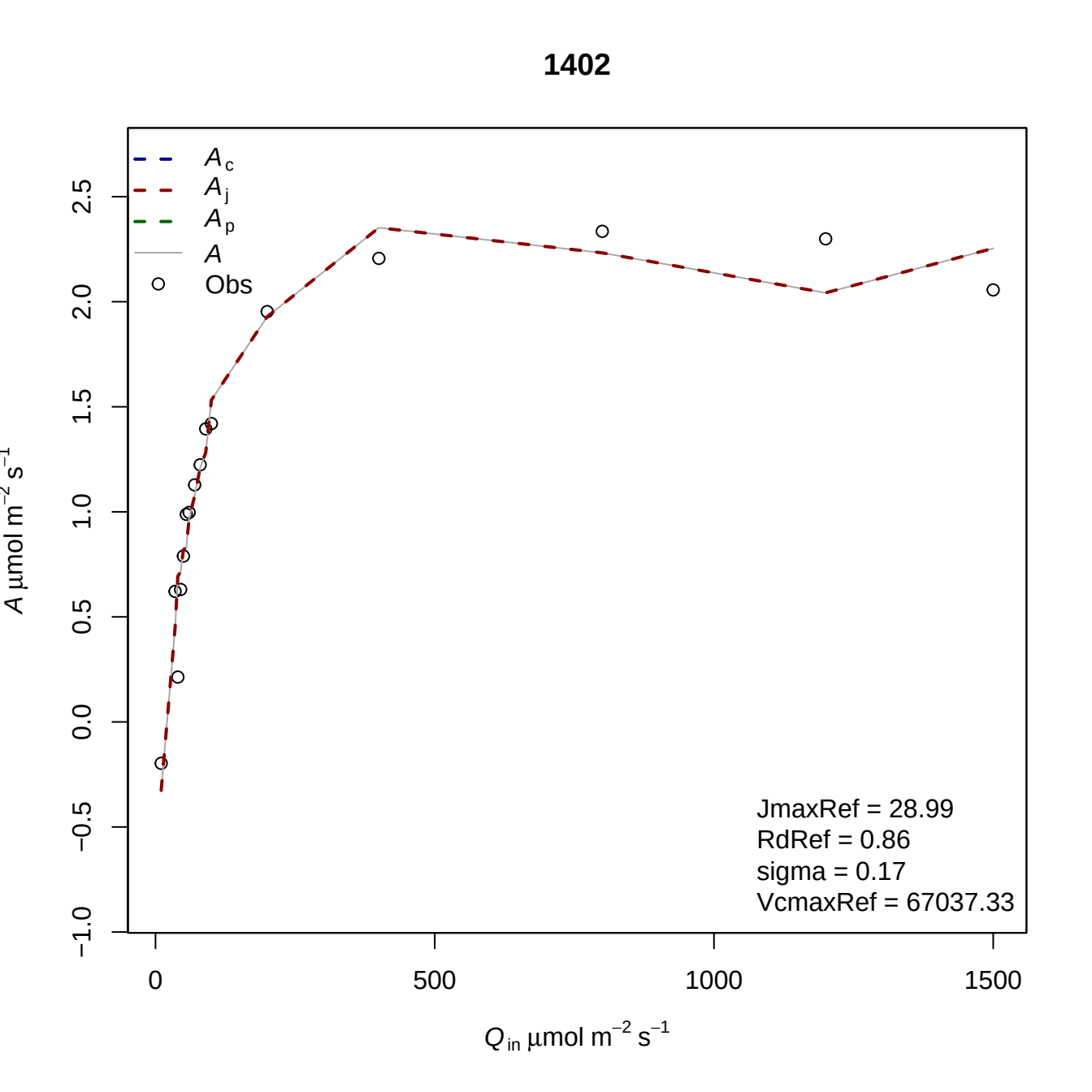

1397
A_c
A_j
A_p
A
Obs
JmaxRef = 46.28
RdRef = 0.43
sigma = 0.05
VcmaxRef = 21.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

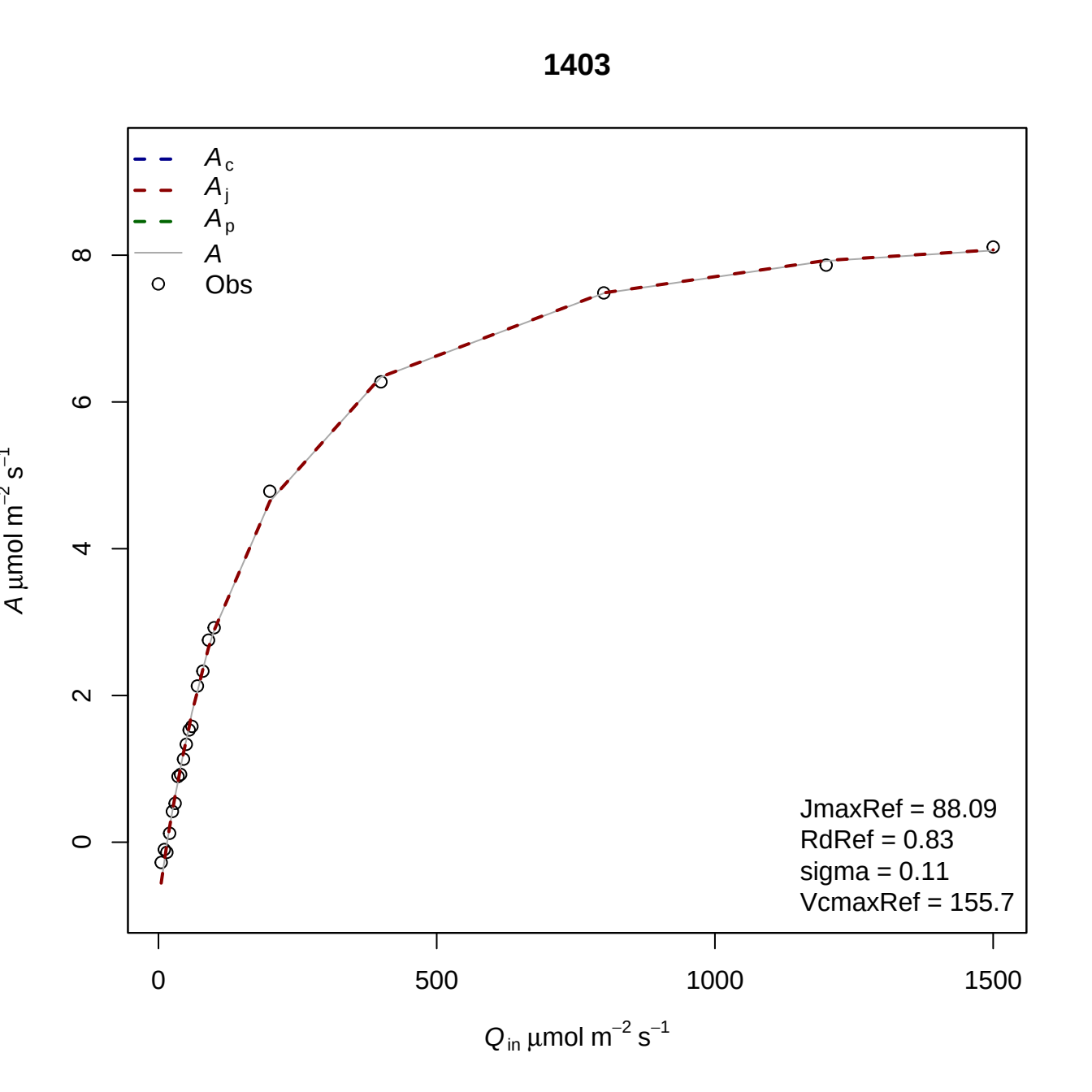

1398
A_c
A_j
A_p
A
Obs
JmaxRef = 62.29
RdRef = 0.57
sigma = 0.27
VcmaxRef = 39.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800
1000

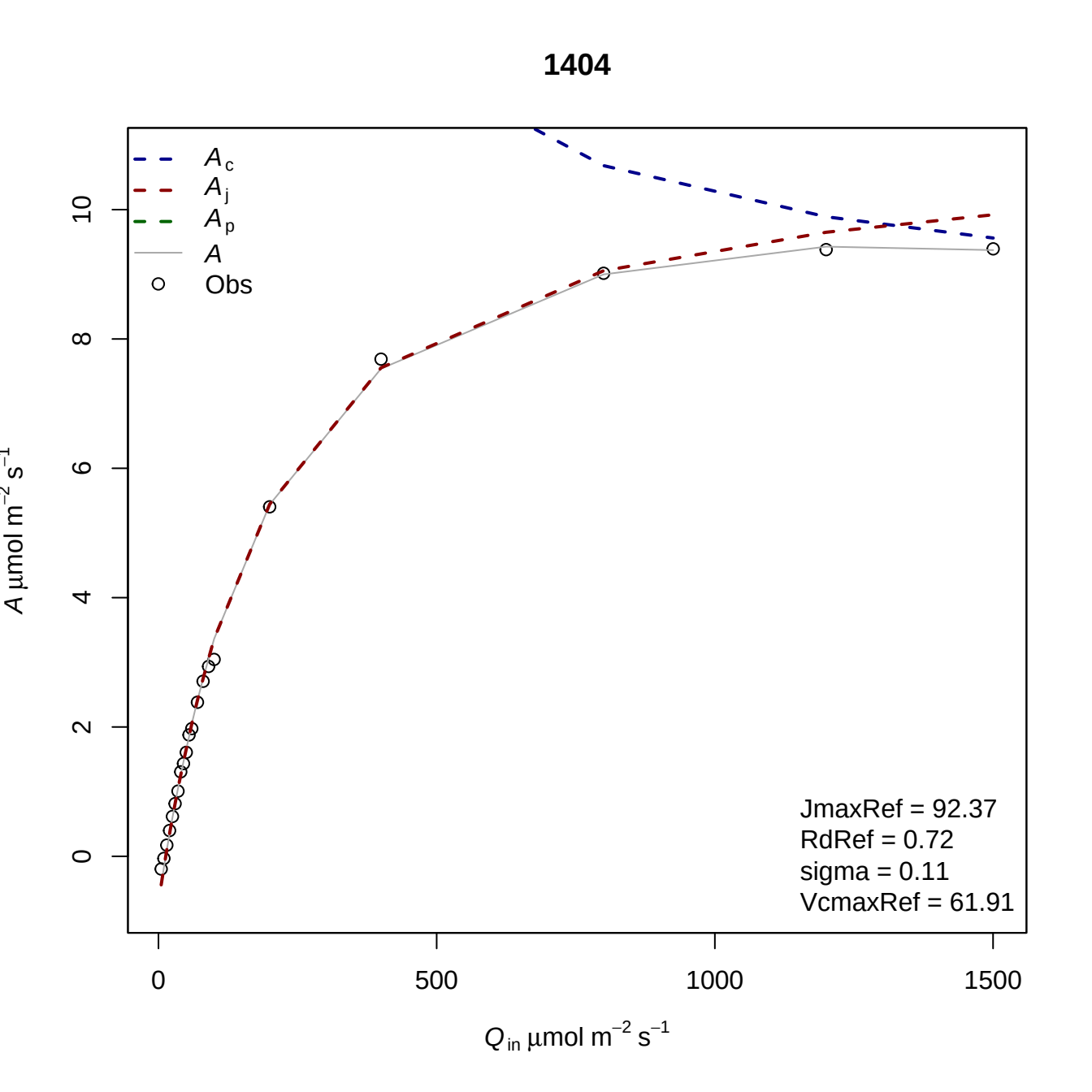

1399
A_c
A_j
A_p
A
Obs
JmaxRef = 66.76
RdRef = 0.09
sigma = 0.19
VcmaxRef = 29.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800
1000

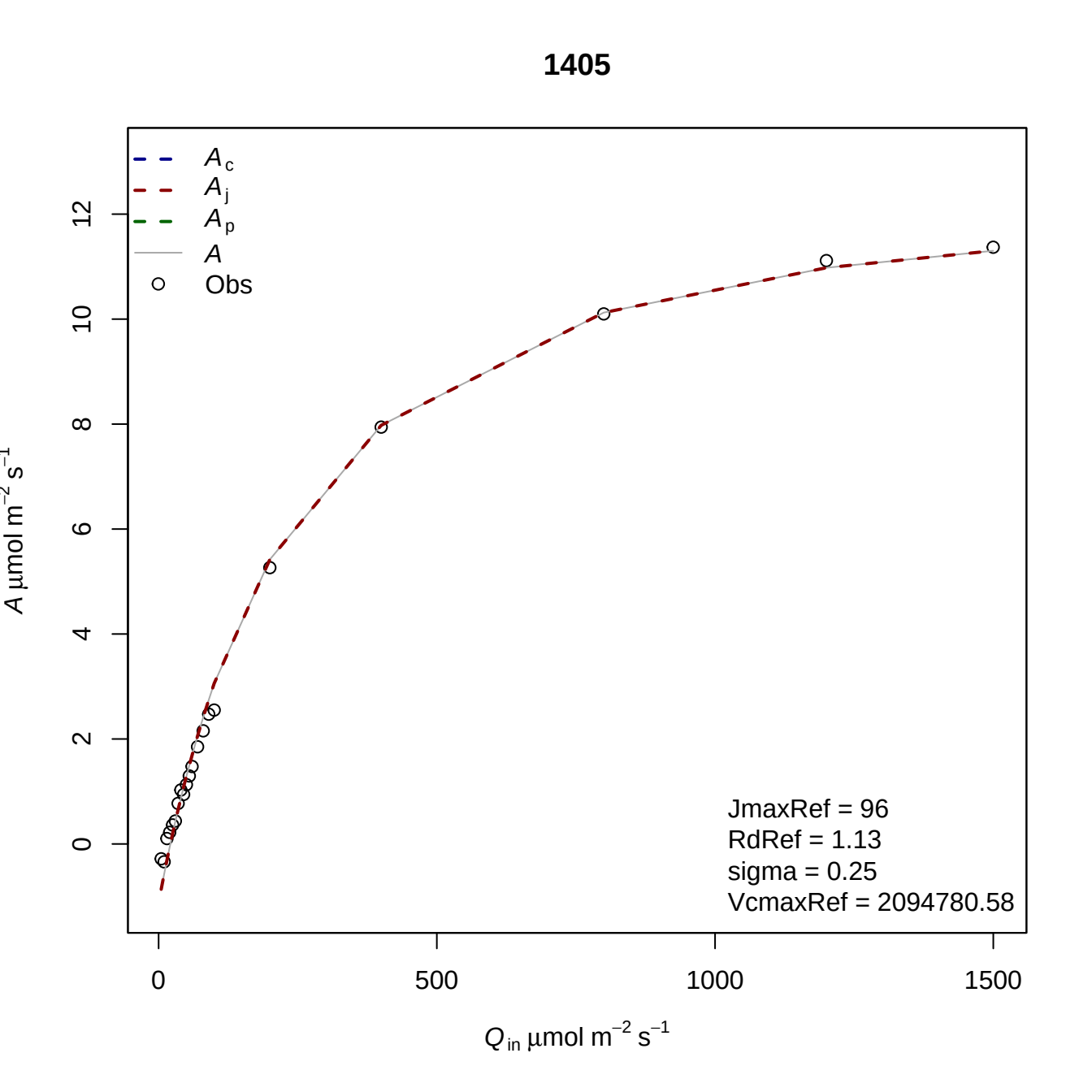

1400
A_c
A_j
A_p
A
Obs
JmaxRef = 50.4
RdRef = 0.49
sigma = 0.14
VcmaxRef = 21.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

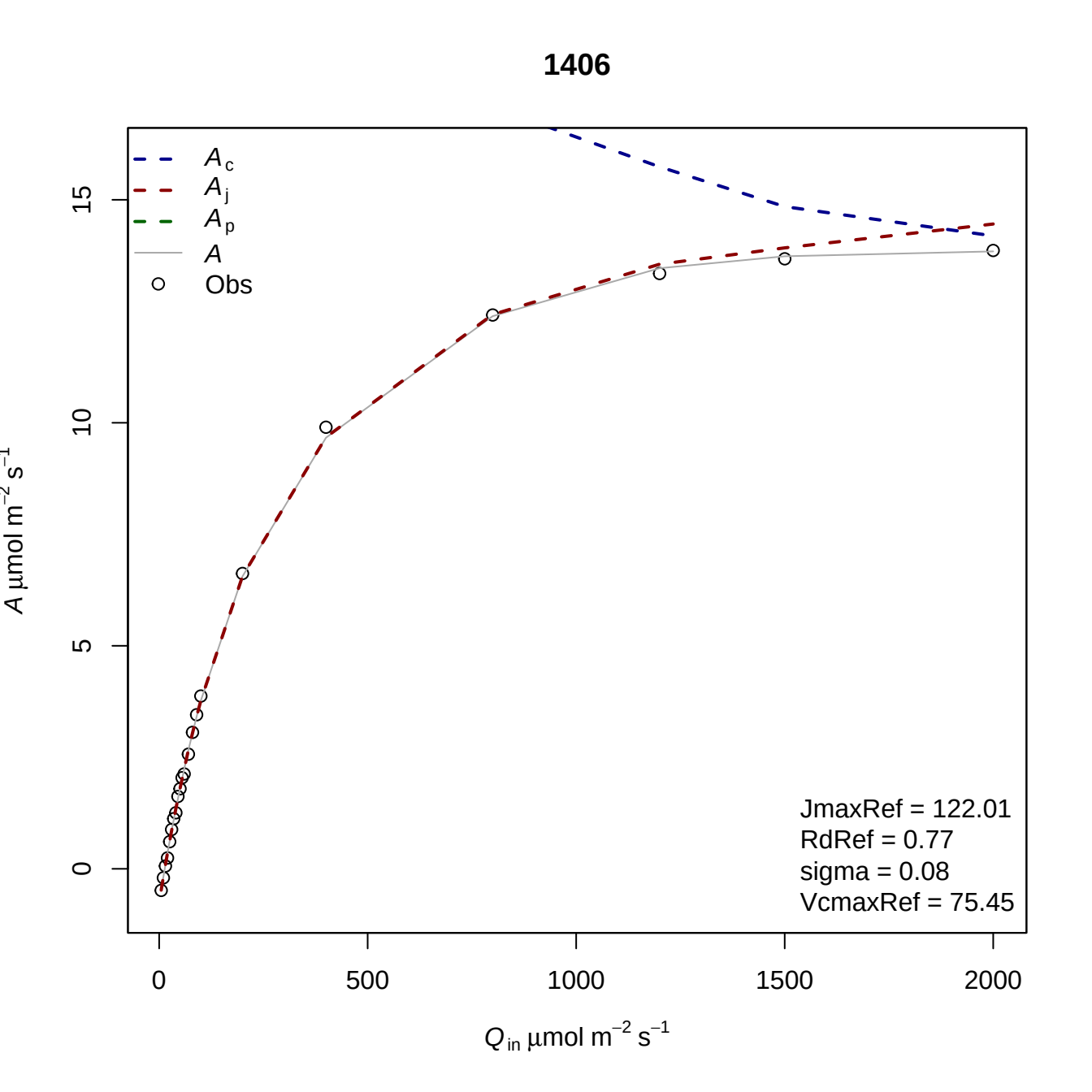

1401
A_c
A_j
A_p
A
Obs
JmaxRef = 83.11
RdRef = 0.67
sigma = 0.08
VcmaxRef = 48.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

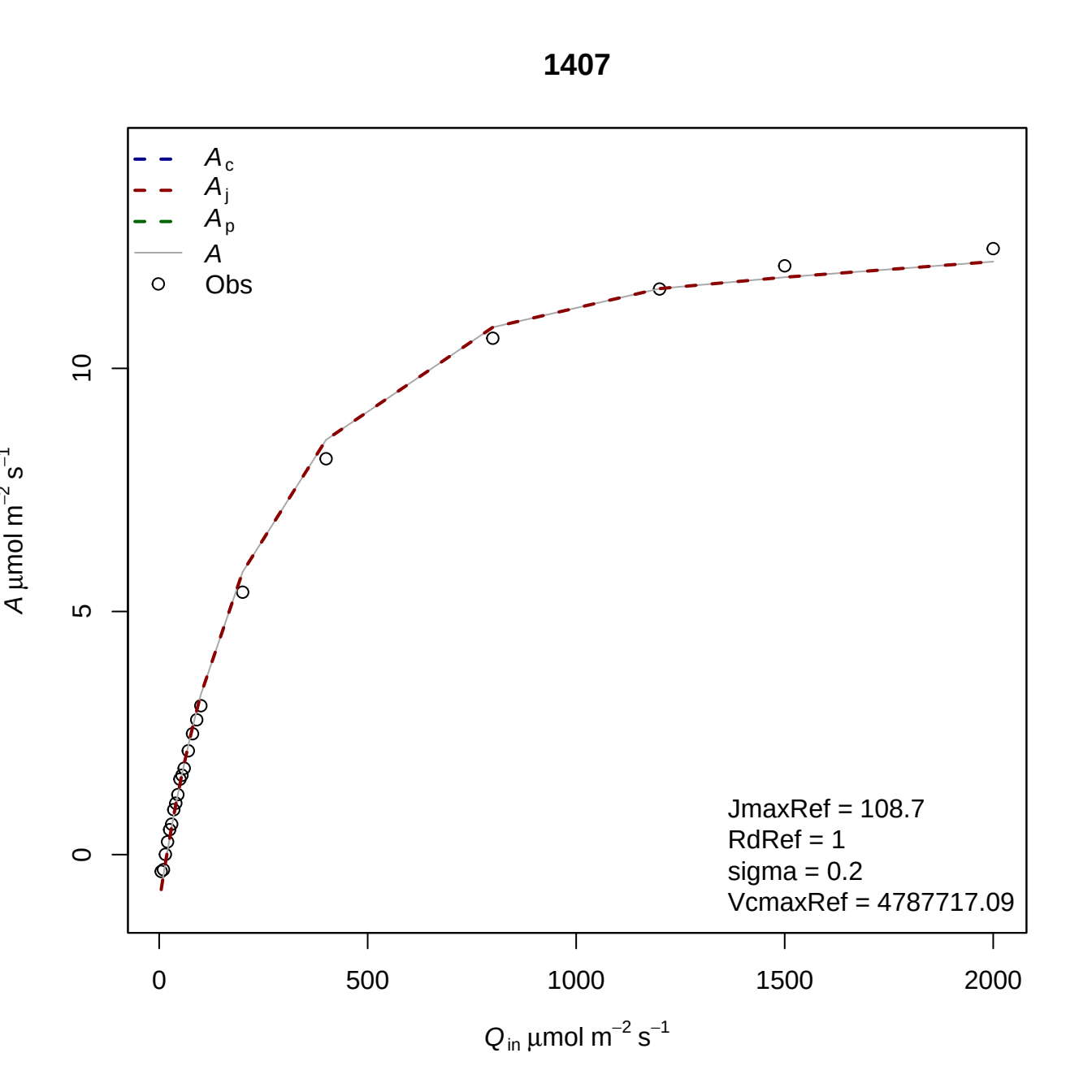

1402
A_c
A_j
A_p
A
Obs
JmaxRef = 28.99
RdRef = 0.86
sigma = 0.17
VcmaxRef = 67037.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
0
500
1000
1500

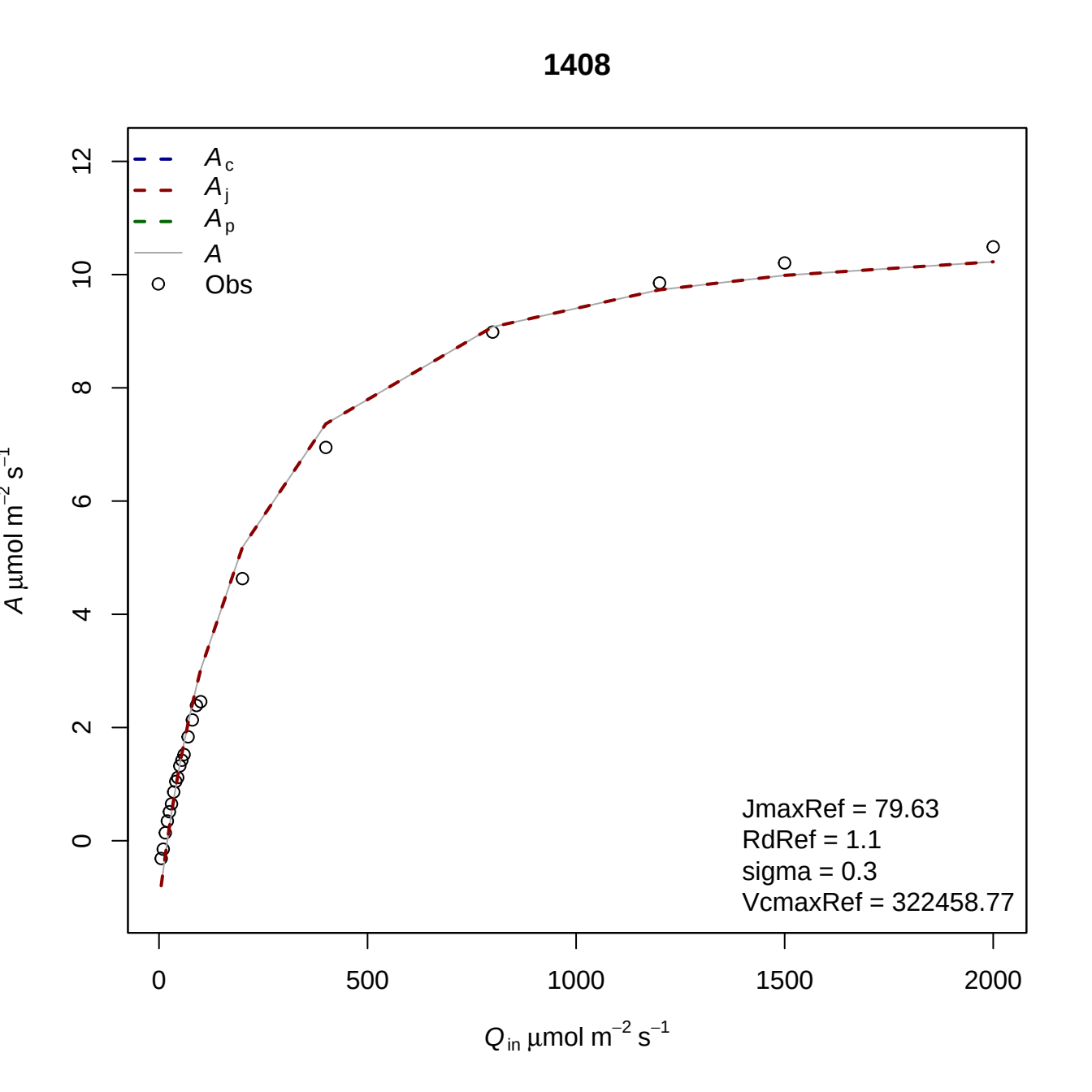

1403
A_c
A_j
A_p
A
Obs
JmaxRef = 88.09
RdRef = 0.83
sigma = 0.11
VcmaxRef = 155.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

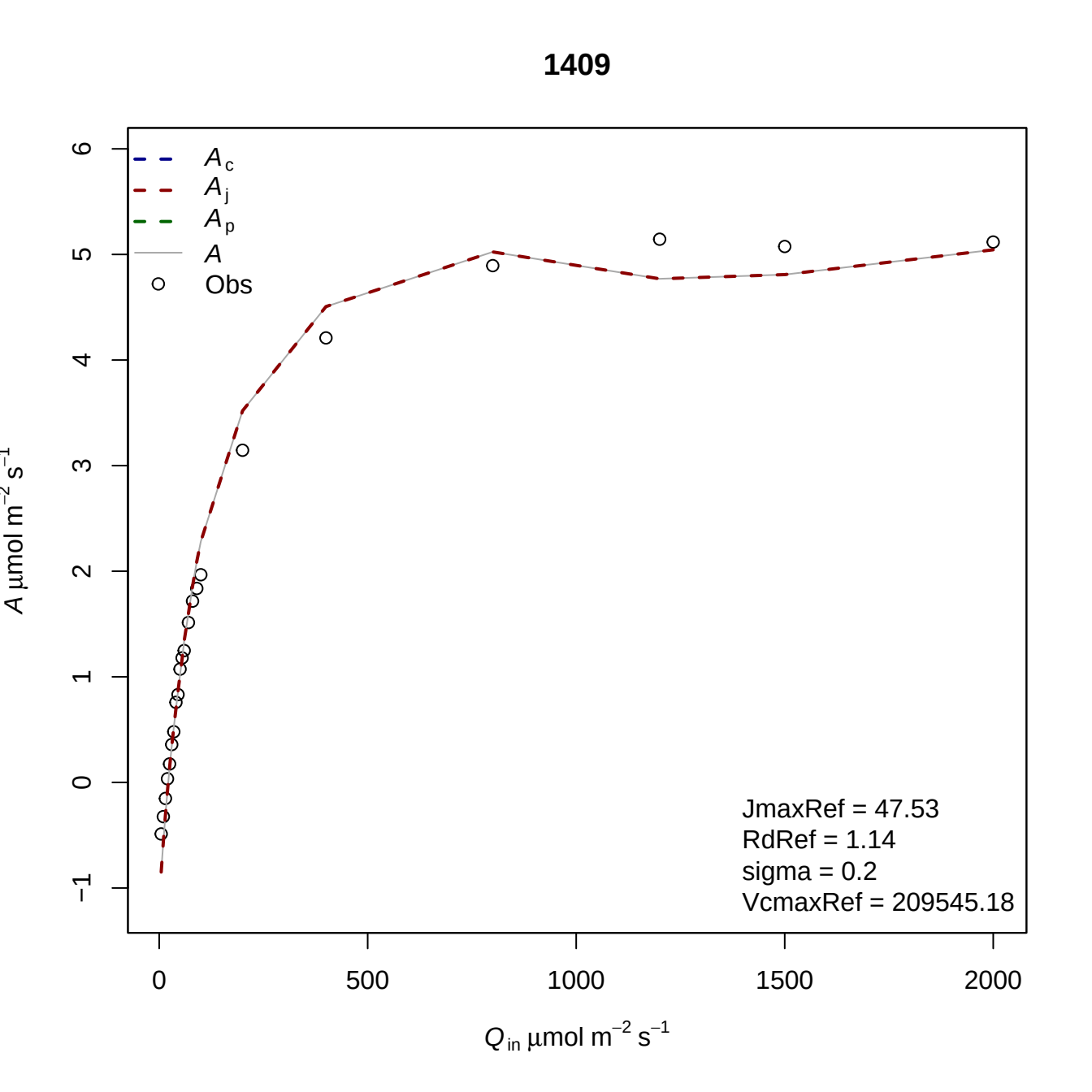

1404
A_c
A_j
A_p
A
Obs
JmaxRef = 92.37
RdRef = 0.72
sigma = 0.11
VcmaxRef = 61.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

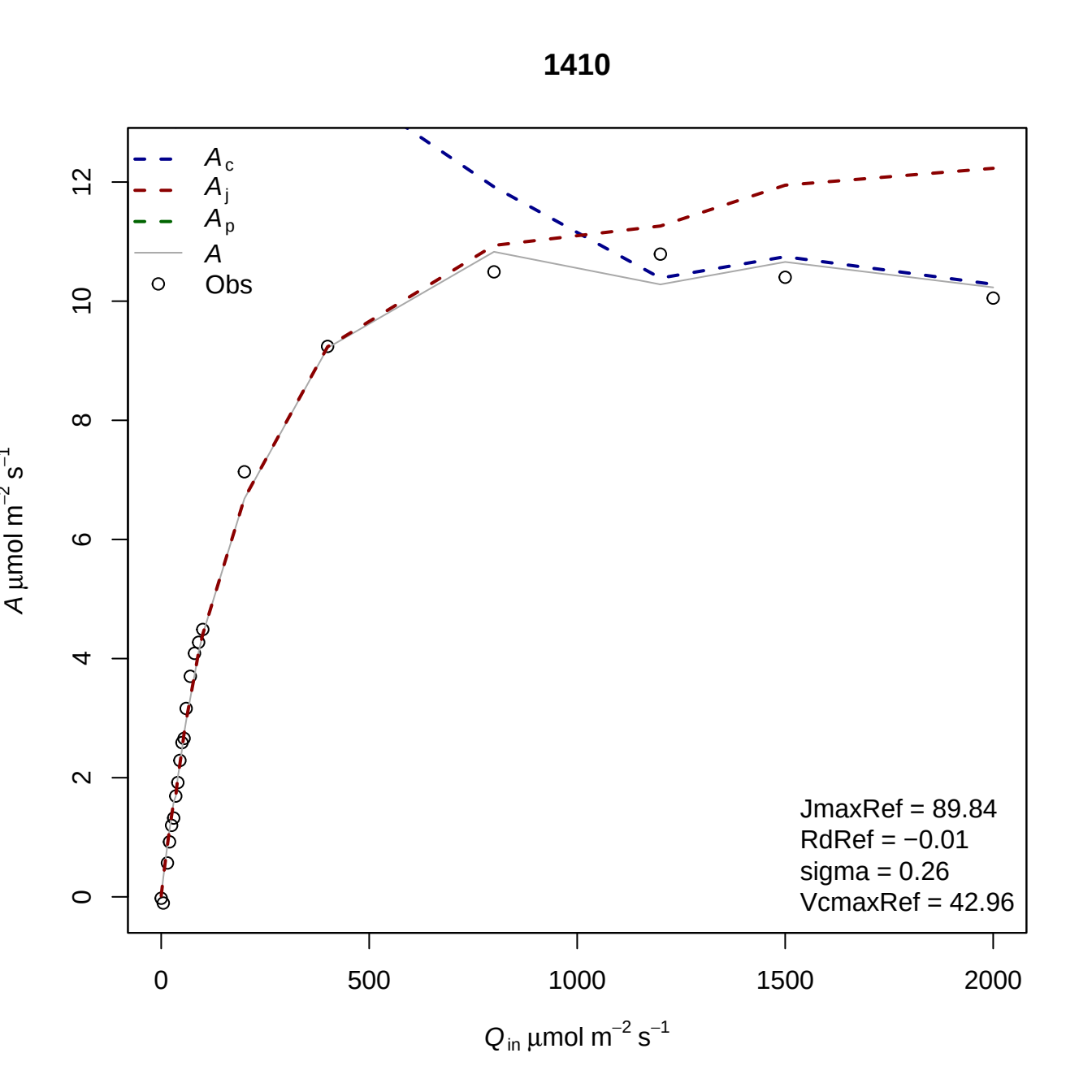

1405
A_c
A_j
A_p
A
Obs
JmaxRef = 96
RdRef = 1.13
sigma = 0.25
VcmaxRef = 2094780.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

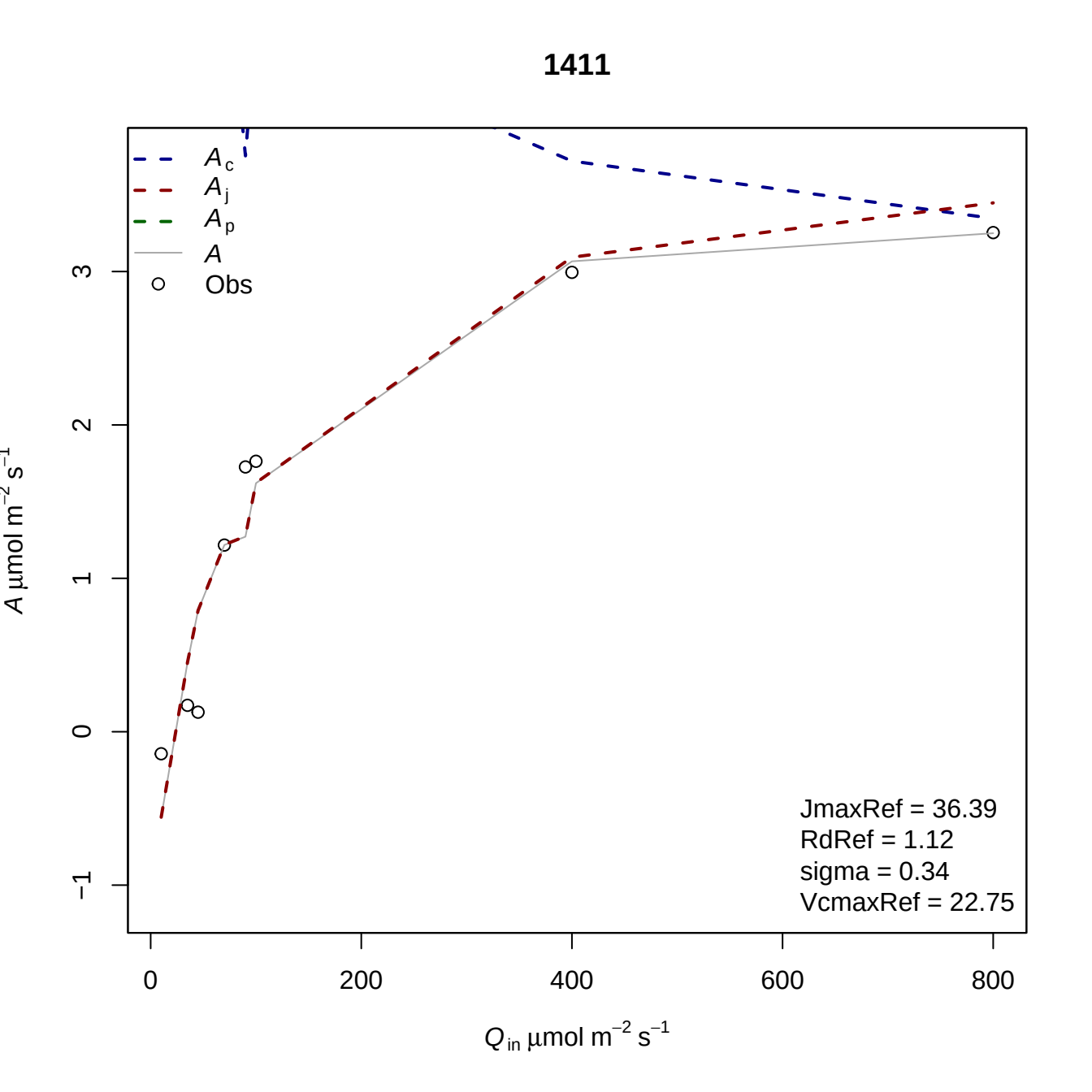

1406
A_c
A_j
A_p
A
Obs
JmaxRef = 122.01
RdRef = 0.77
sigma = 0.08
VcmaxRef = 75.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

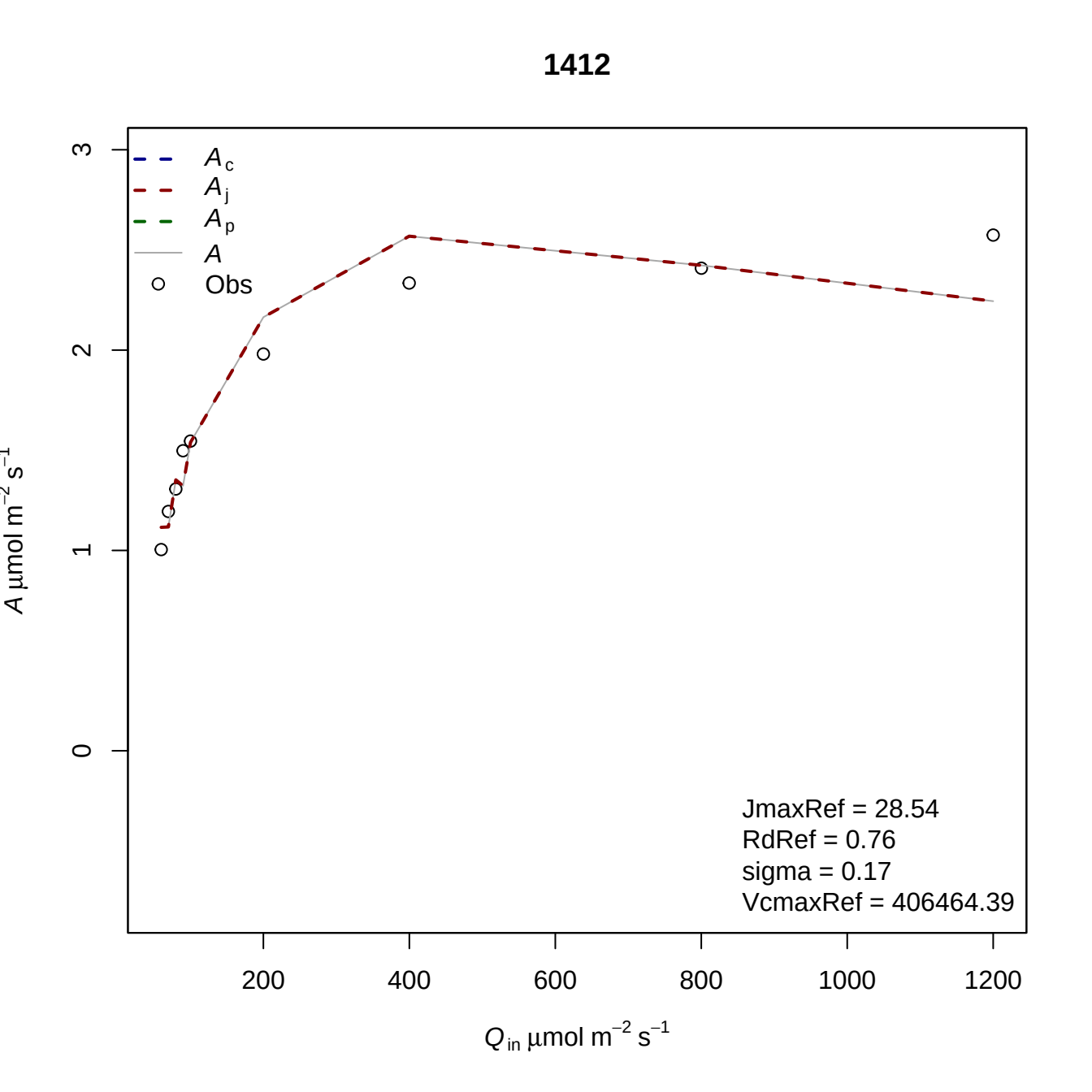

1407
A_c
A_j
A_p
A
Obs
JmaxRef = 108.7
RdRef = 1
sigma = 0.2
VcmaxRef = 4787717.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

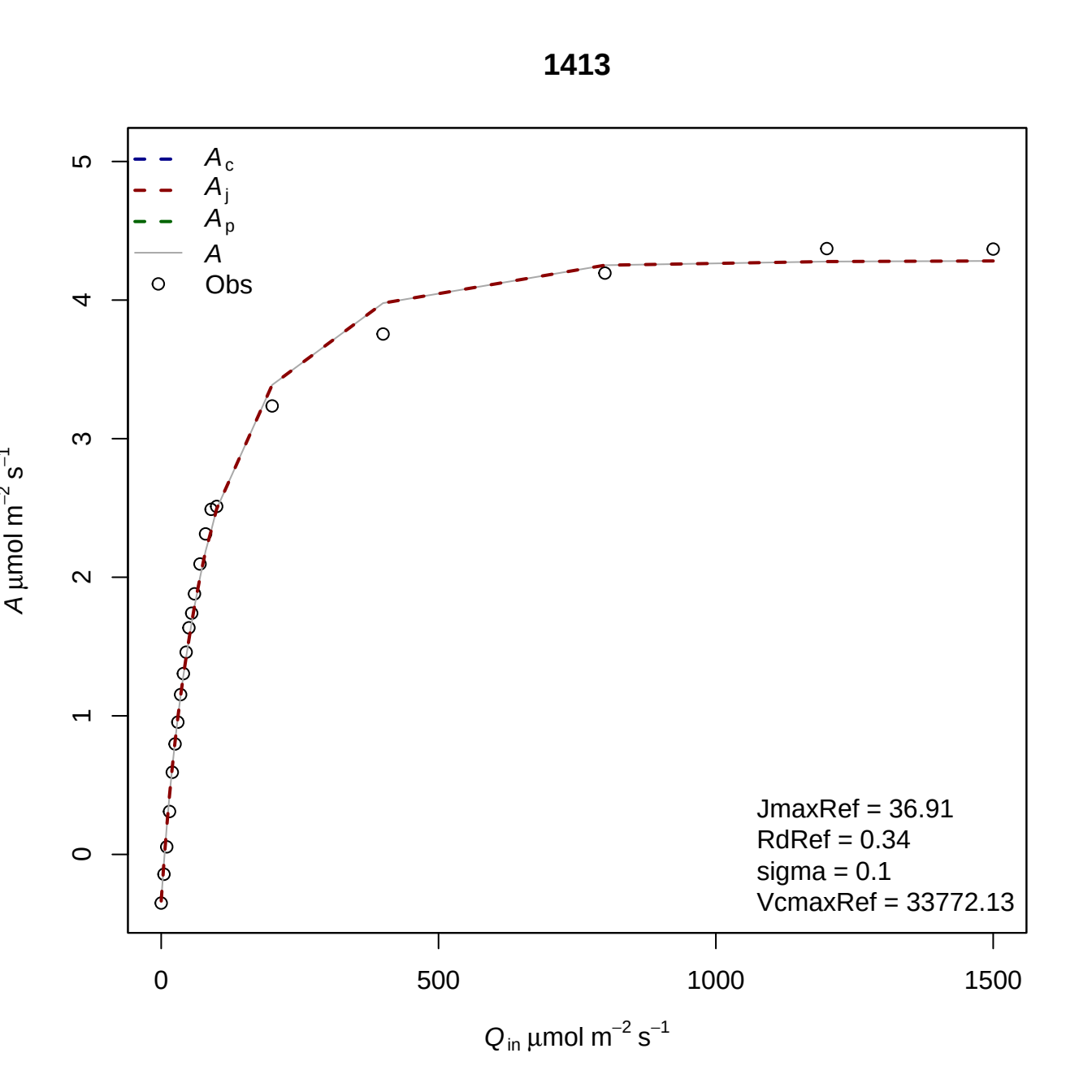

1408
A_c
A_j
A_p
A
Obs
JmaxRef = 79.63
RdRef = 1.1
sigma = 0.3
VcmaxRef = 322458.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

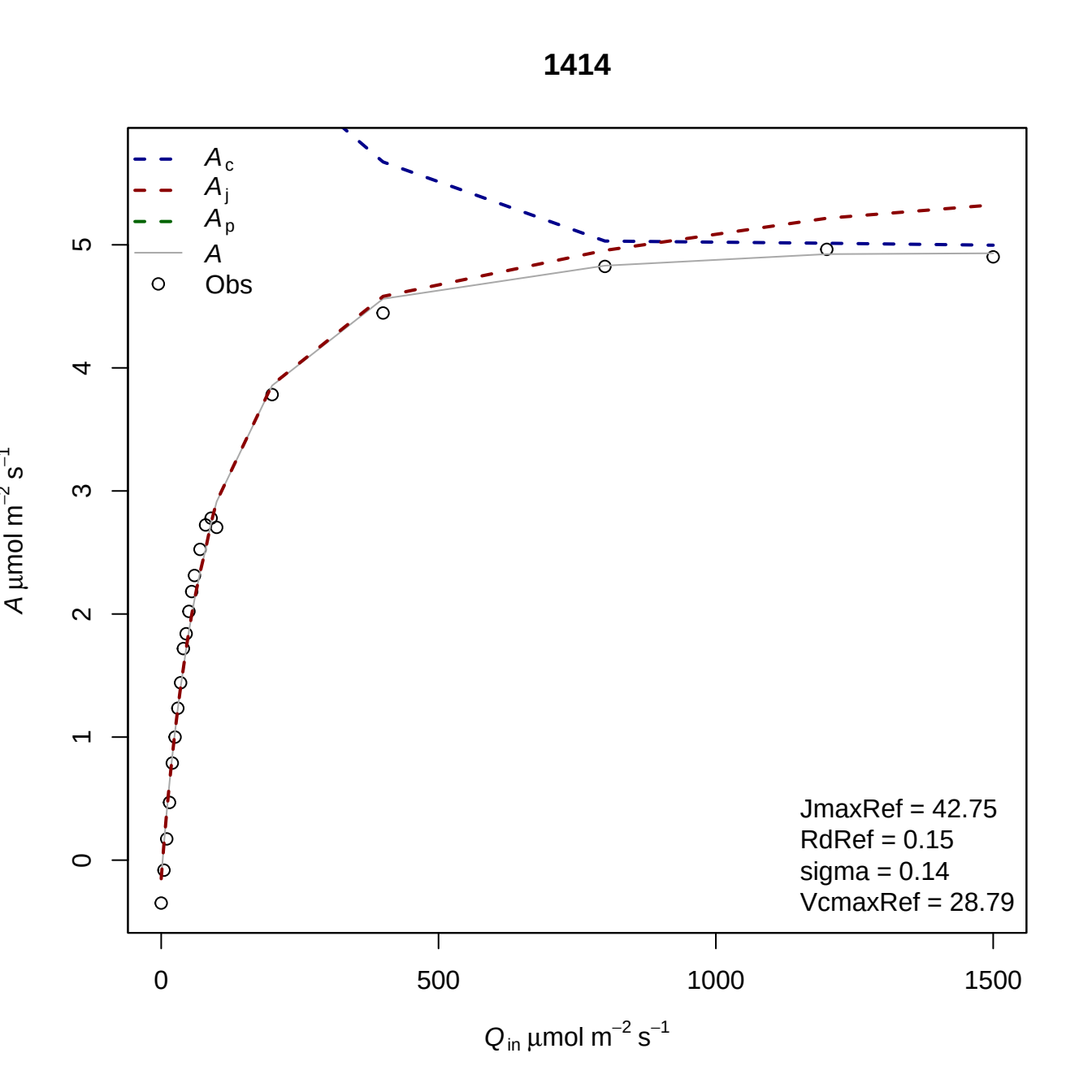

1409
A_c
A_j
A_p
A
Obs
JmaxRef = 47.53
RdRef = 1.14
sigma = 0.2
VcmaxRef = 209545.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

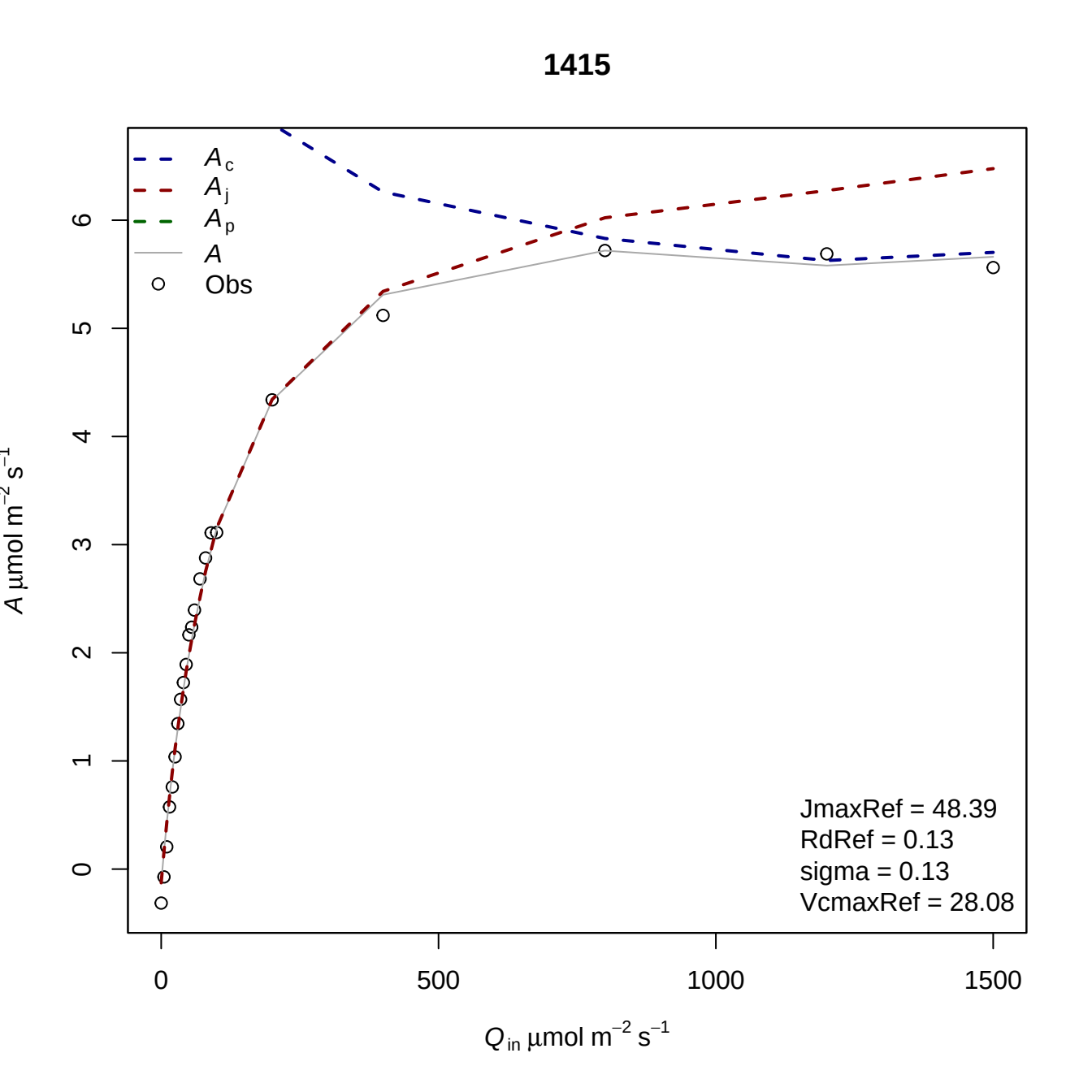

1410
A_c
A_j
A_p
A
Obs
JmaxRef = 89.84
RdRef = −0.01
sigma = 0.26
VcmaxRef = 42.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

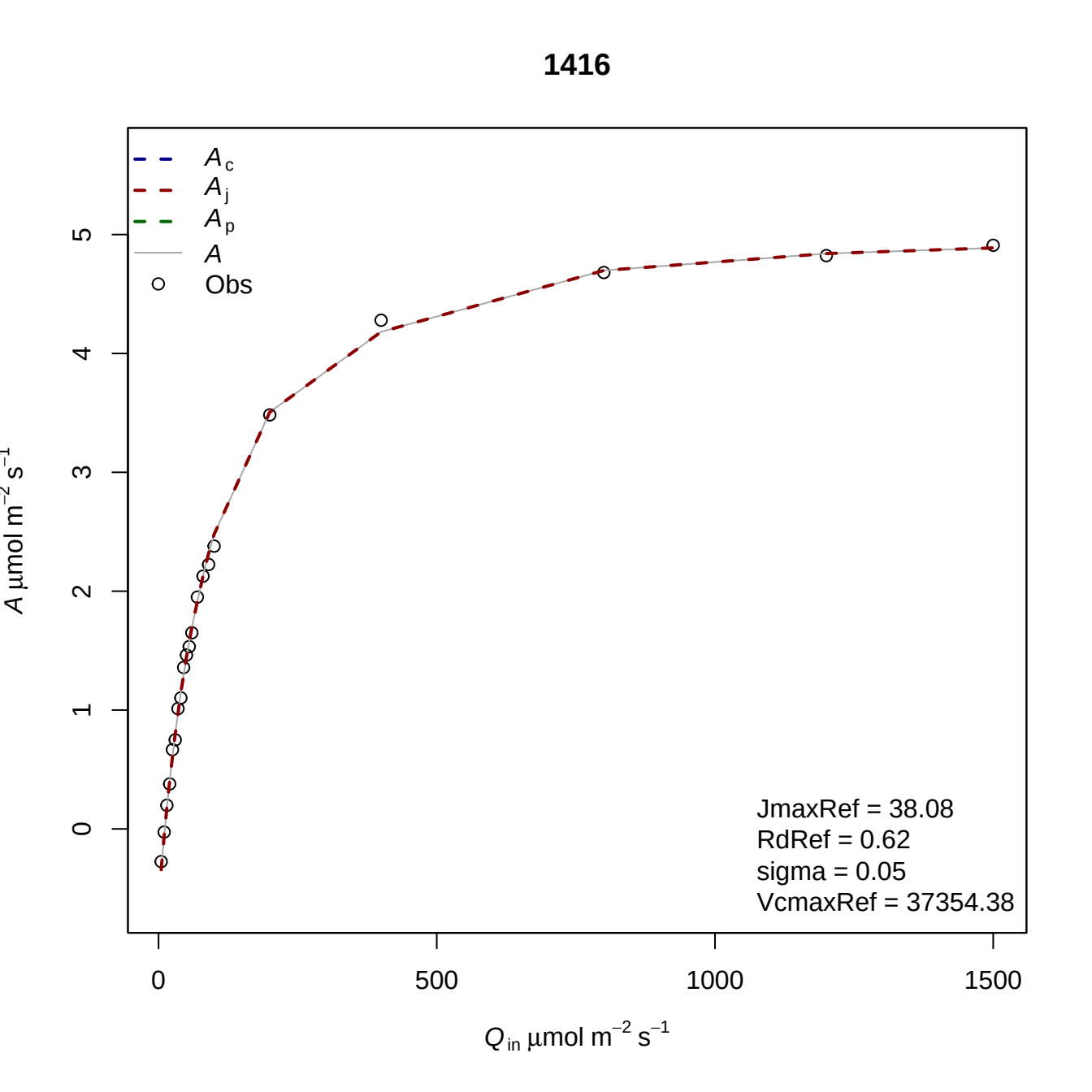

1411
A_c
A_j
A_p
A
Obs
JmaxRef = 36.39
RdRef = 1.12
sigma = 0.34
VcmaxRef = 22.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800

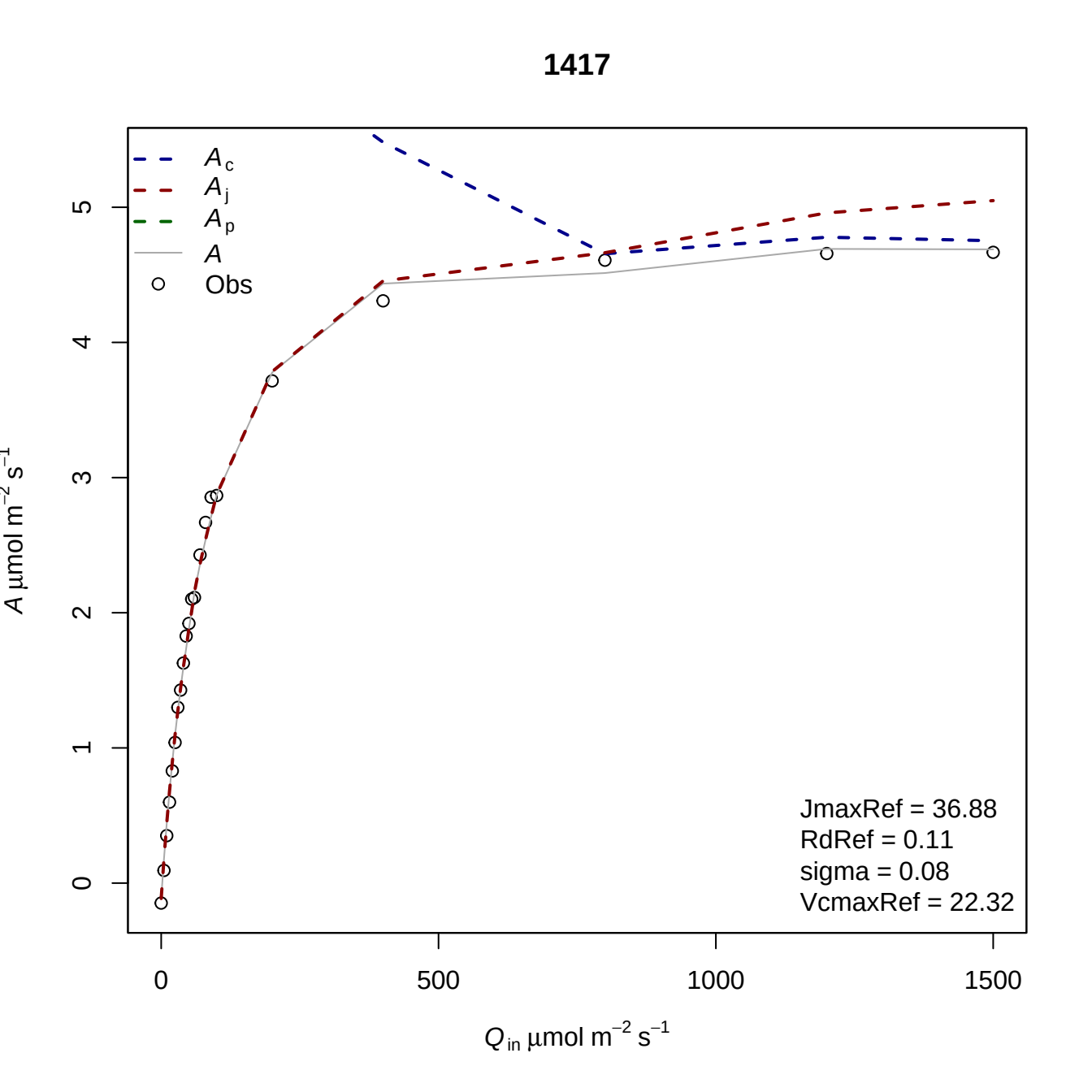

1412
A_c
A_j
A_p
A
Obs
JmaxRef = 28.54
RdRef = 0.76
sigma = 0.17
VcmaxRef = 406464.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
200
400
600
800
1000
1200

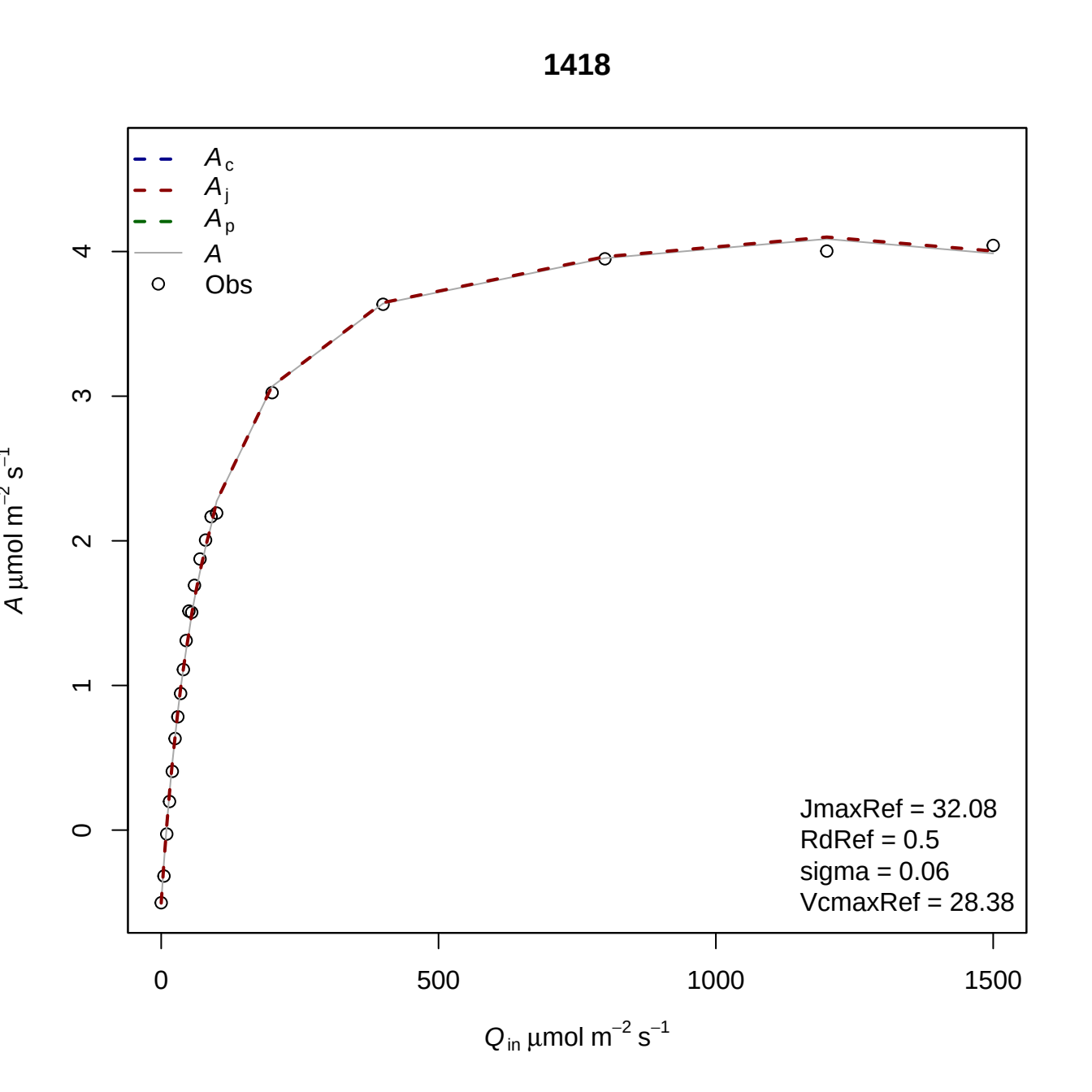

1413
A_c
A_j
A_p
A
Obs
JmaxRef = 36.91
RdRef = 0.34
sigma = 0.1
VcmaxRef = 33772.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

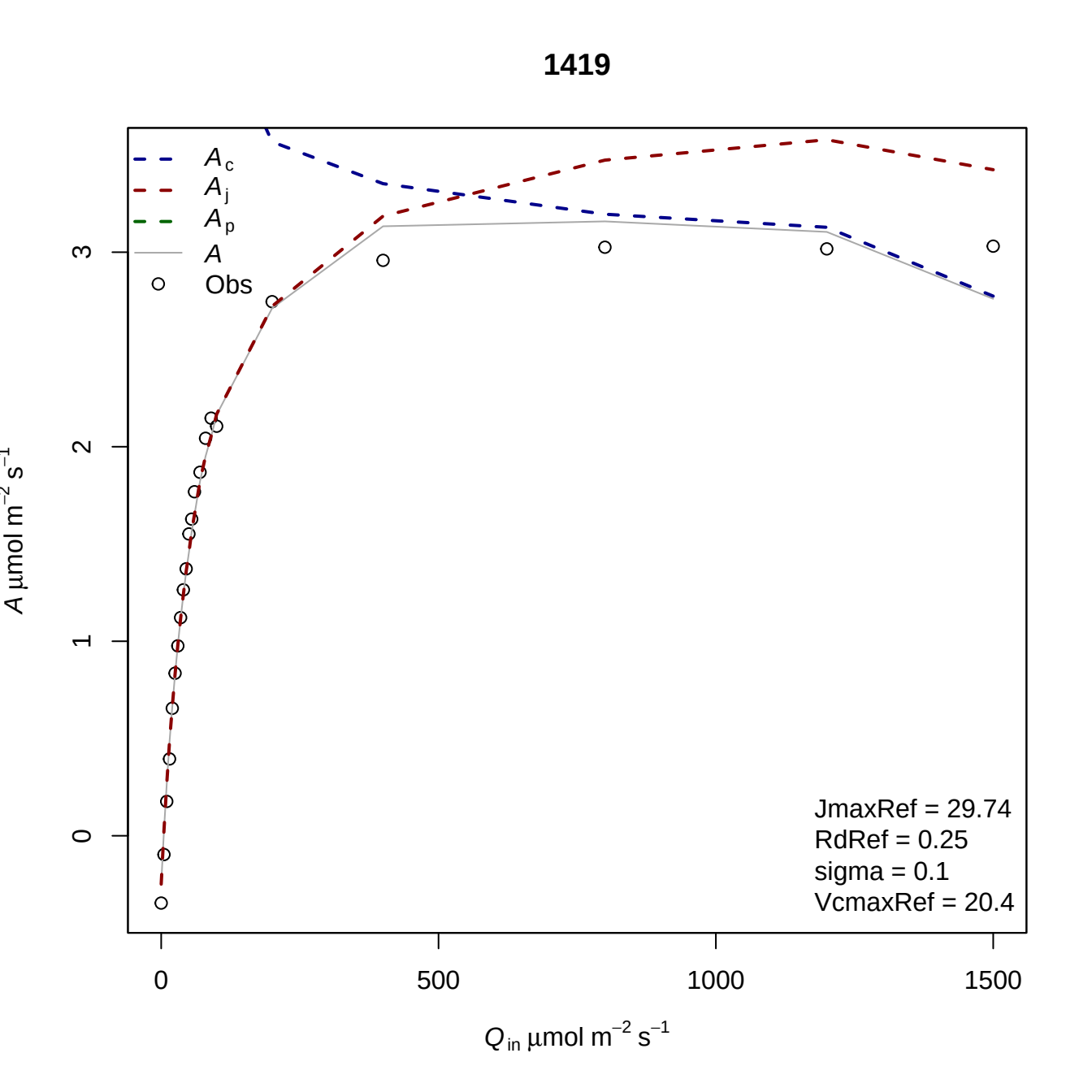

1414
A_c
A_j
A_p
A
Obs
JmaxRef = 42.75
RdRef = 0.15
sigma = 0.14
VcmaxRef = 28.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

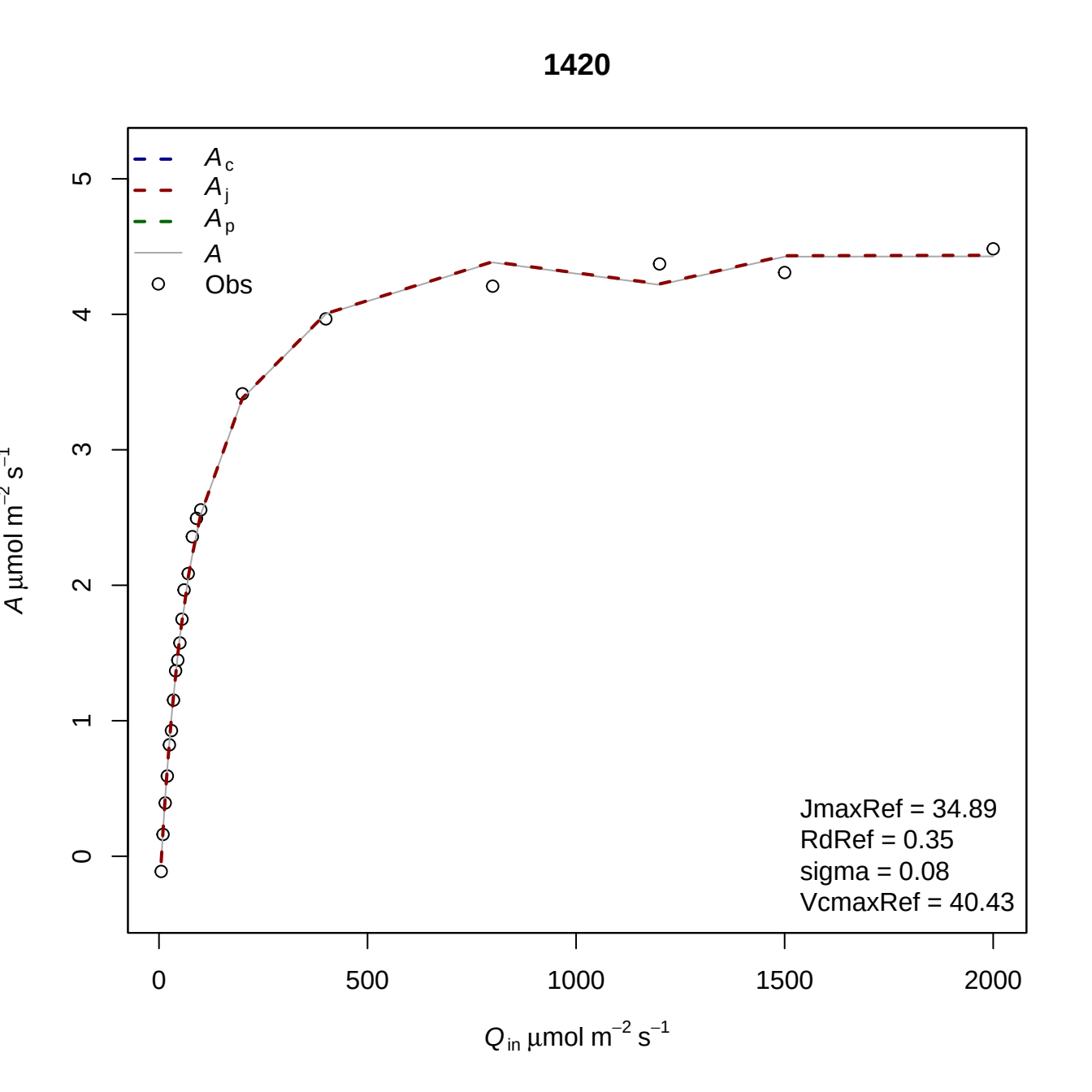

1415
A_c
A_j
A_p
A
Obs
JmaxRef = 48.39
RdRef = 0.13
sigma = 0.13
VcmaxRef = 28.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

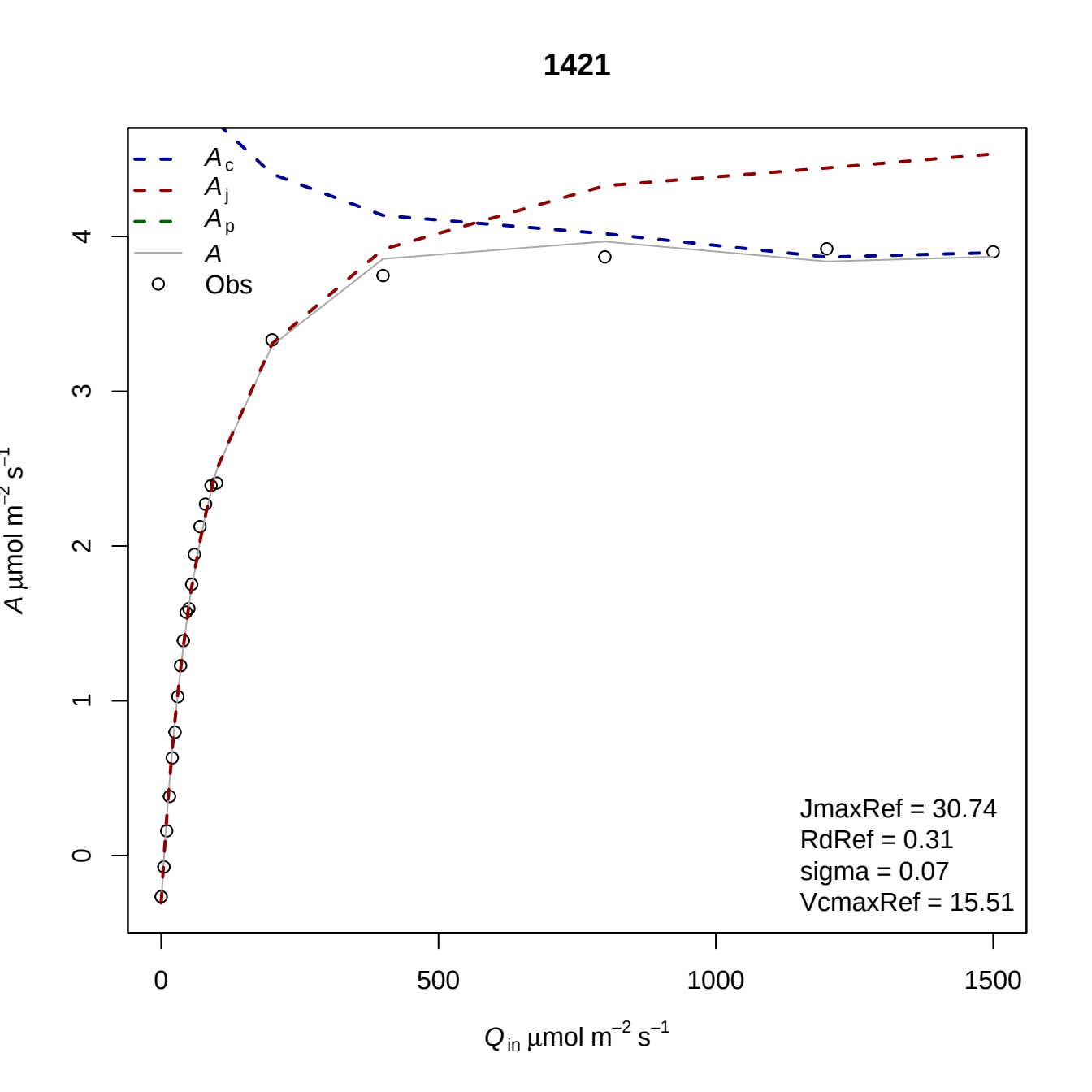

1416
A_c
A_j
A_p
A
Obs
JmaxRef = 38.08
RdRef = 0.62
sigma = 0.05
VcmaxRef = 37354.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

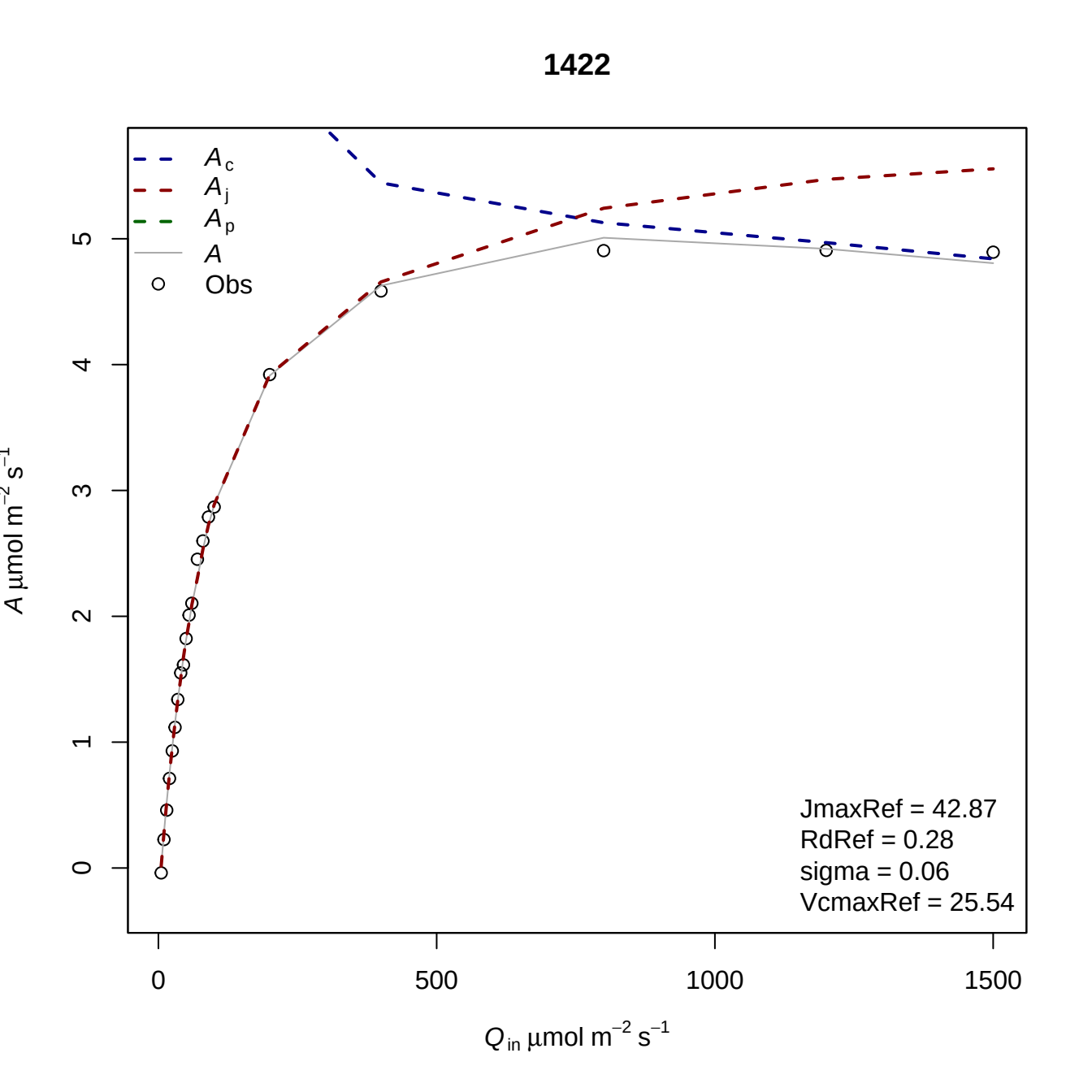

1417
A_c
A_j
A_p
A
Obs
JmaxRef = 36.88
RdRef = 0.11
sigma = 0.08
VcmaxRef = 22.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

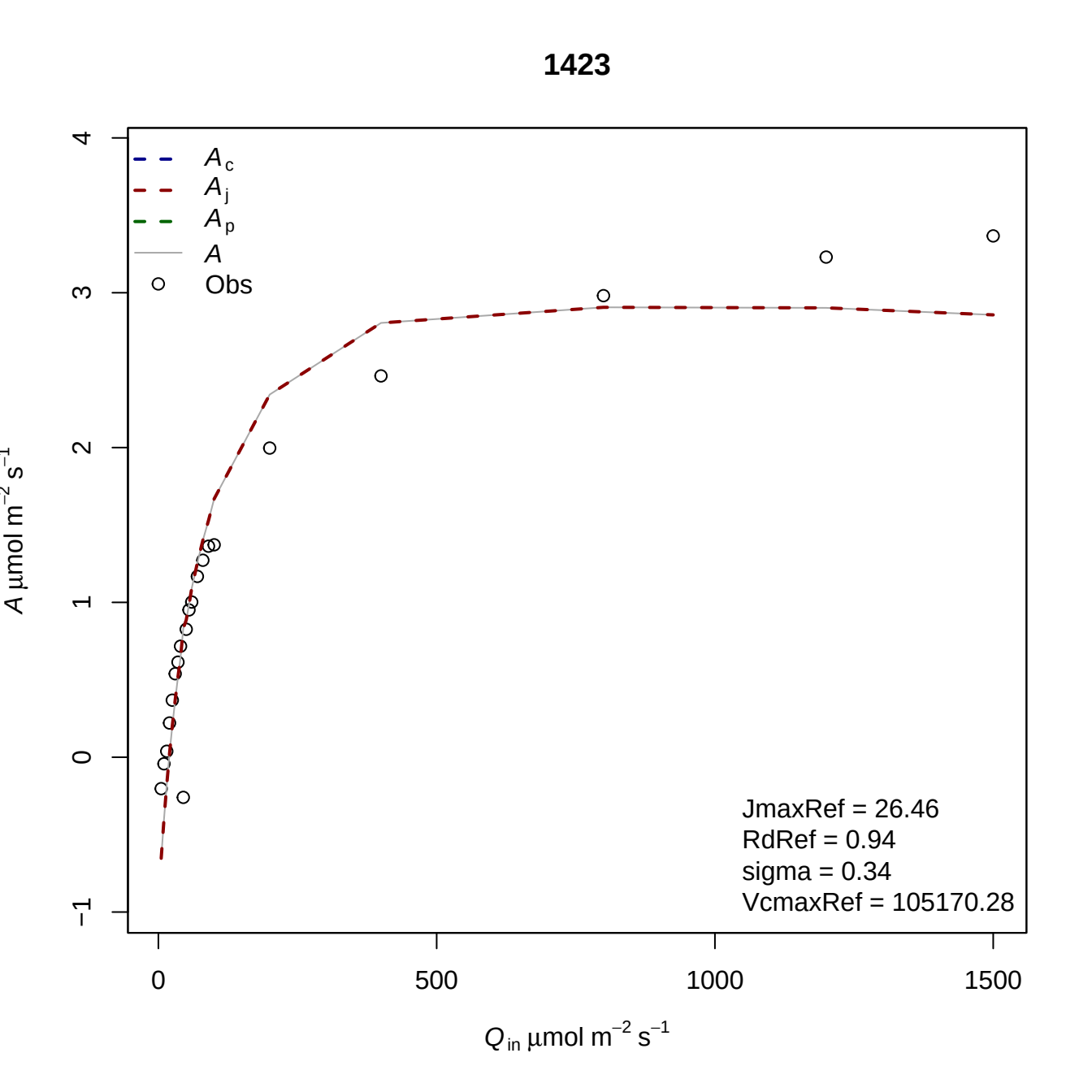

1418
A_c
A_j
A_p
A
Obs
JmaxRef = 32.08
RdRef = 0.5
sigma = 0.06
VcmaxRef = 28.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

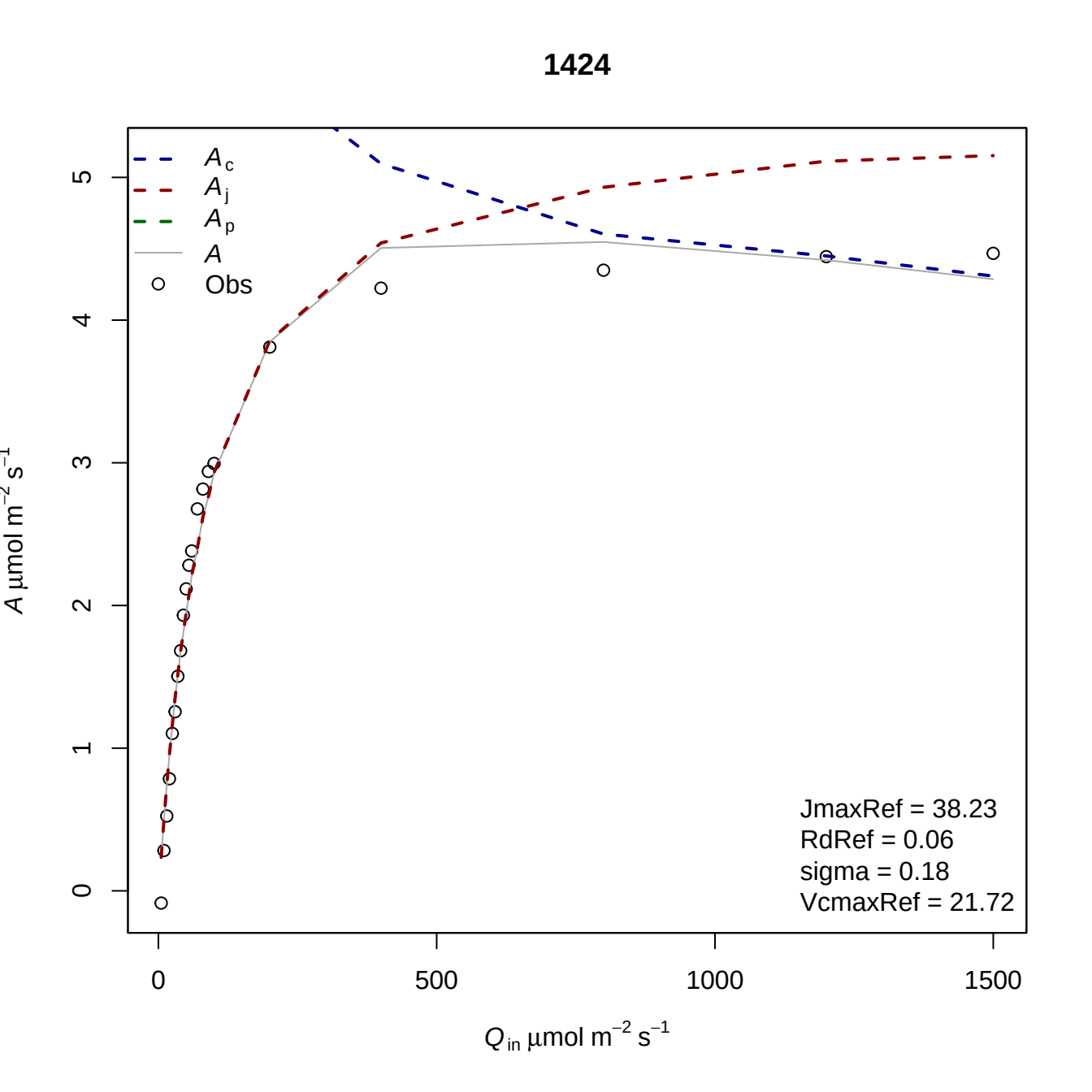

1419
A_c
A_j
A_p
A
Obs
JmaxRef = 29.74
RdRef = 0.25
sigma = 0.1
VcmaxRef = 20.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500

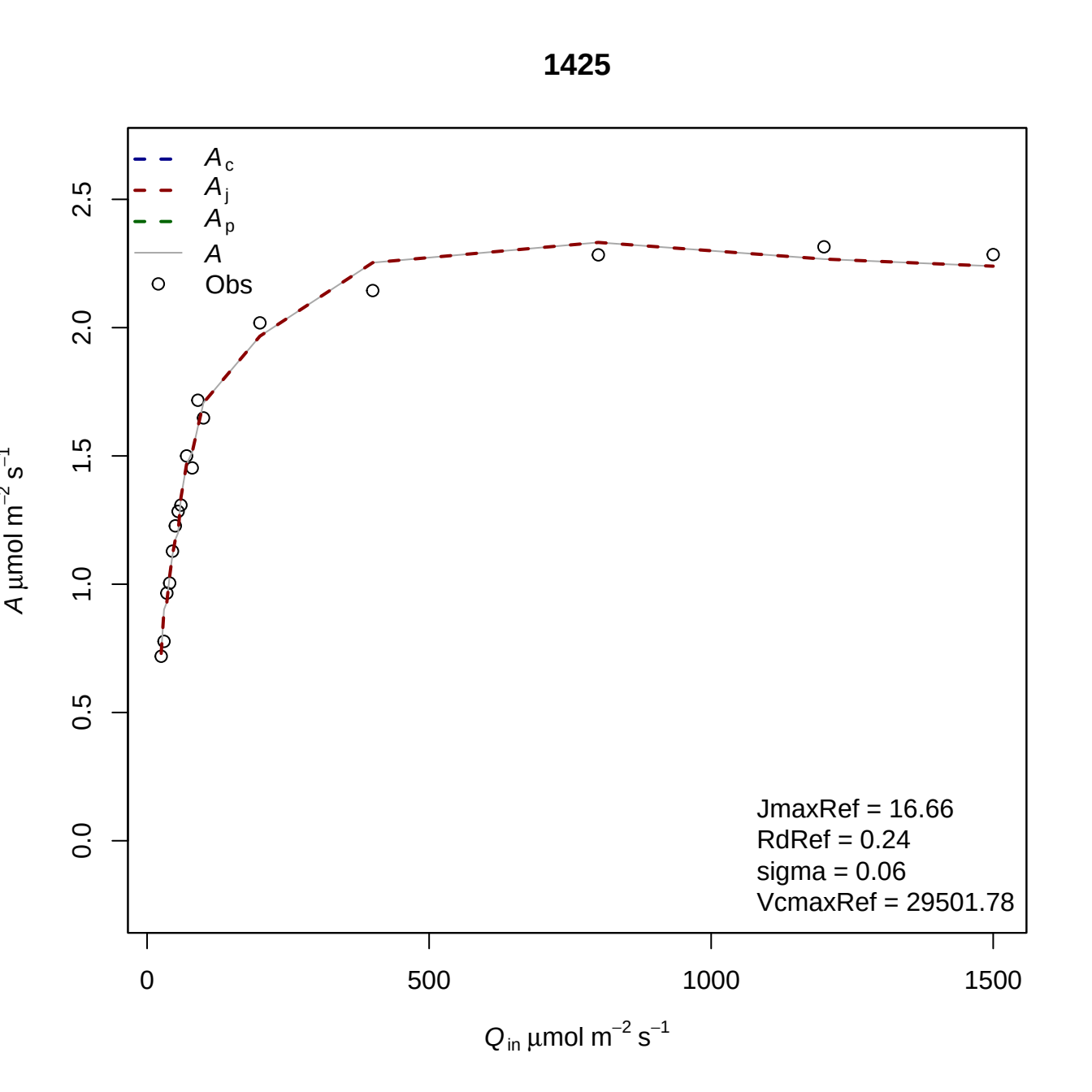

1420
A_c
A_j
A_p
A
Obs
JmaxRef = 34.89
RdRef = 0.35
sigma = 0.08
VcmaxRef = 40.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500
2000

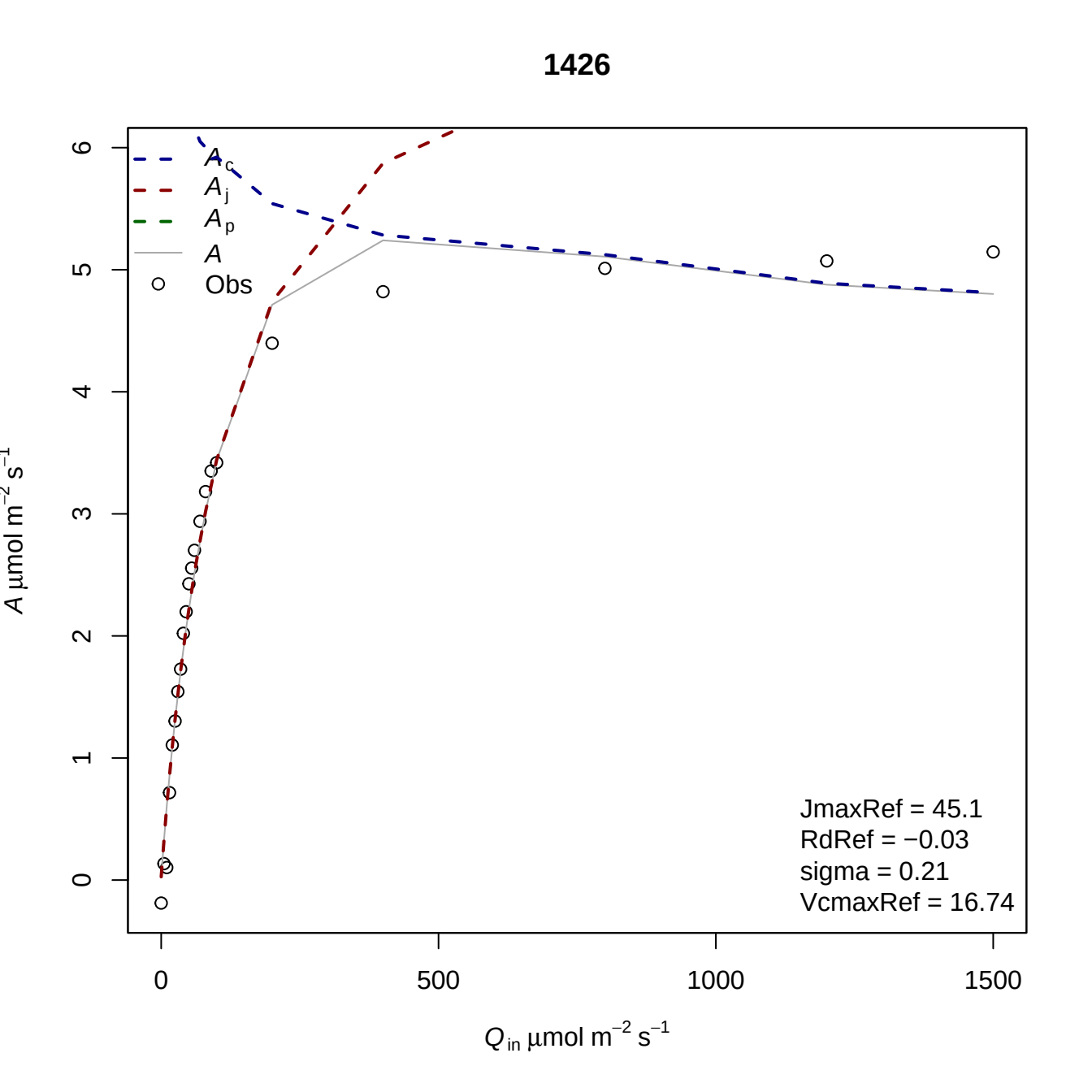

1421
A_c
A_j
A_p
A
Obs
JmaxRef = 30.74
RdRef = 0.31
sigma = 0.07
VcmaxRef = 15.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

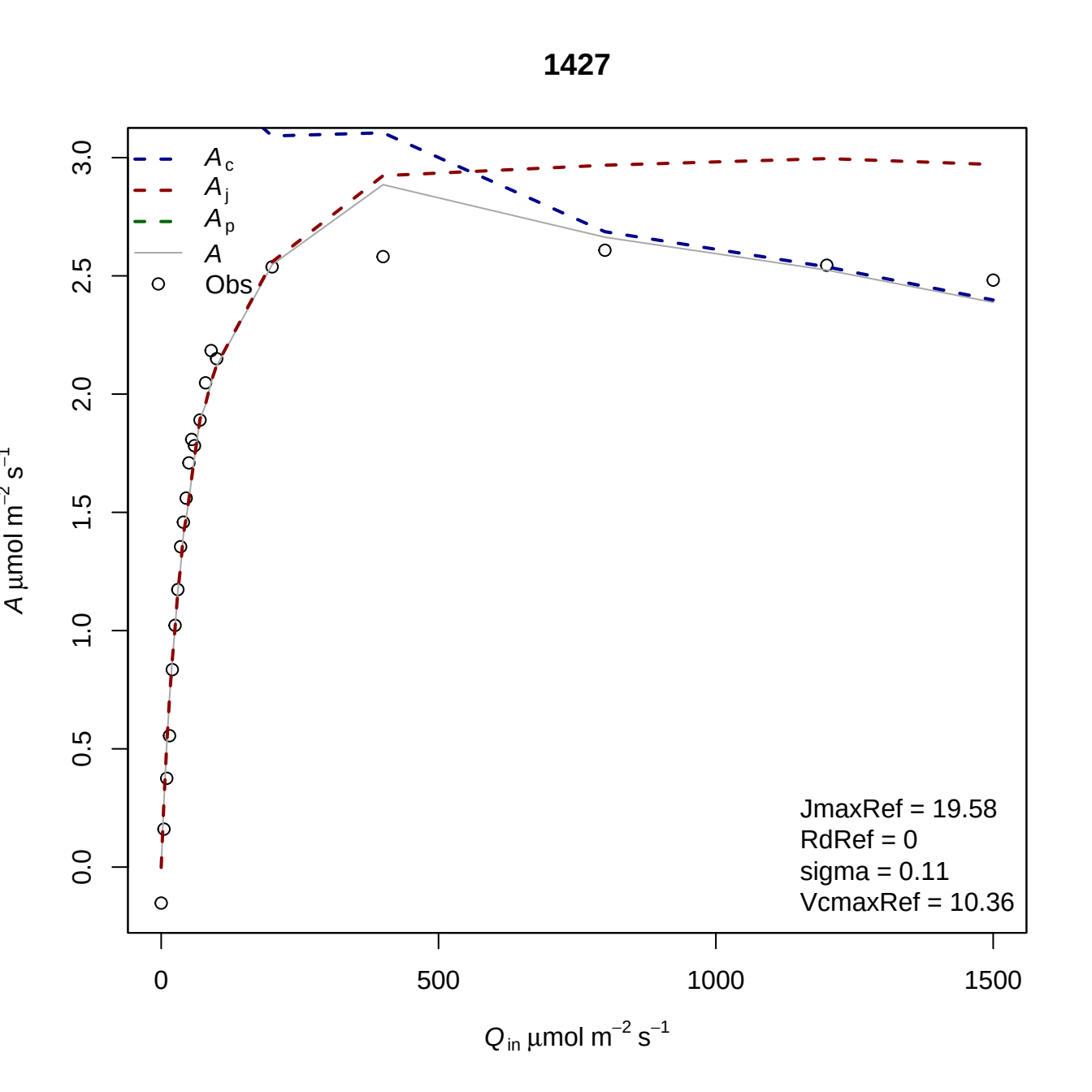

1422
A_c
A_j
A_p
A
Obs
JmaxRef = 42.87
RdRef = 0.28
sigma = 0.06
VcmaxRef = 25.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

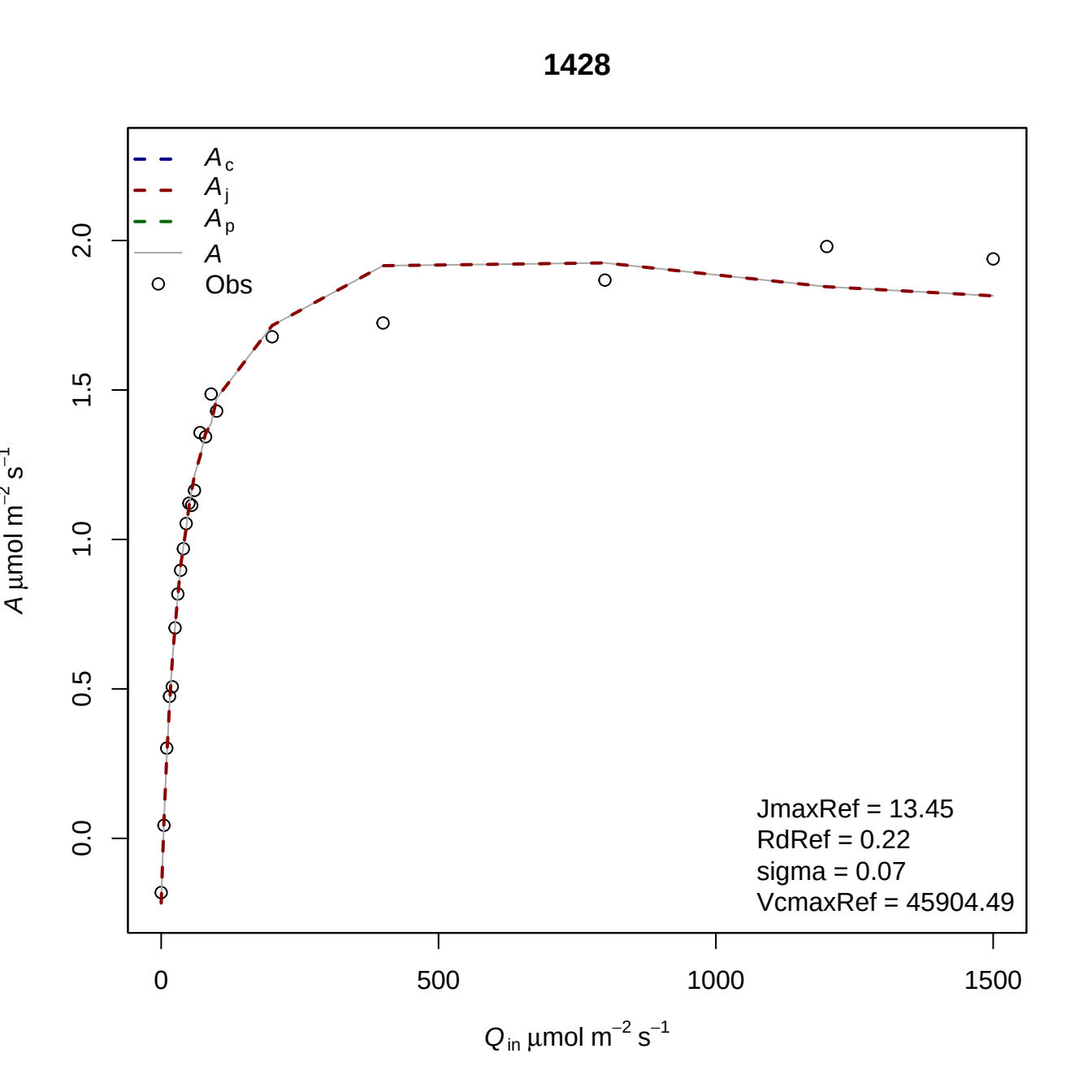

1423
A_c
A_j
A_p
A
Obs
JmaxRef = 26.46
RdRef = 0.94
sigma = 0.34
VcmaxRef = 105170.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

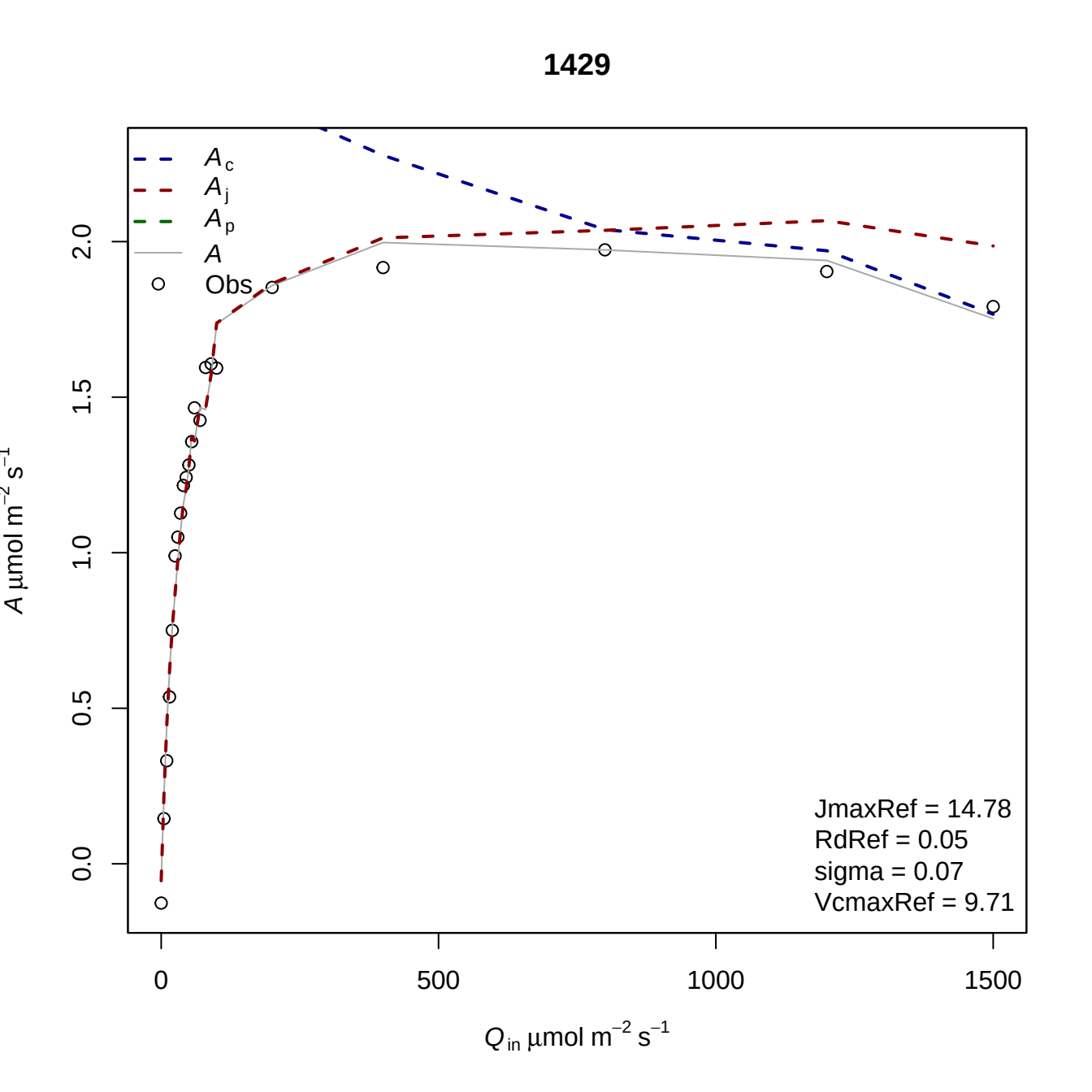

1424
A_c
A_j
A_p
A
Obs
JmaxRef = 38.23
RdRef = 0.06
sigma = 0.18
VcmaxRef = 21.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

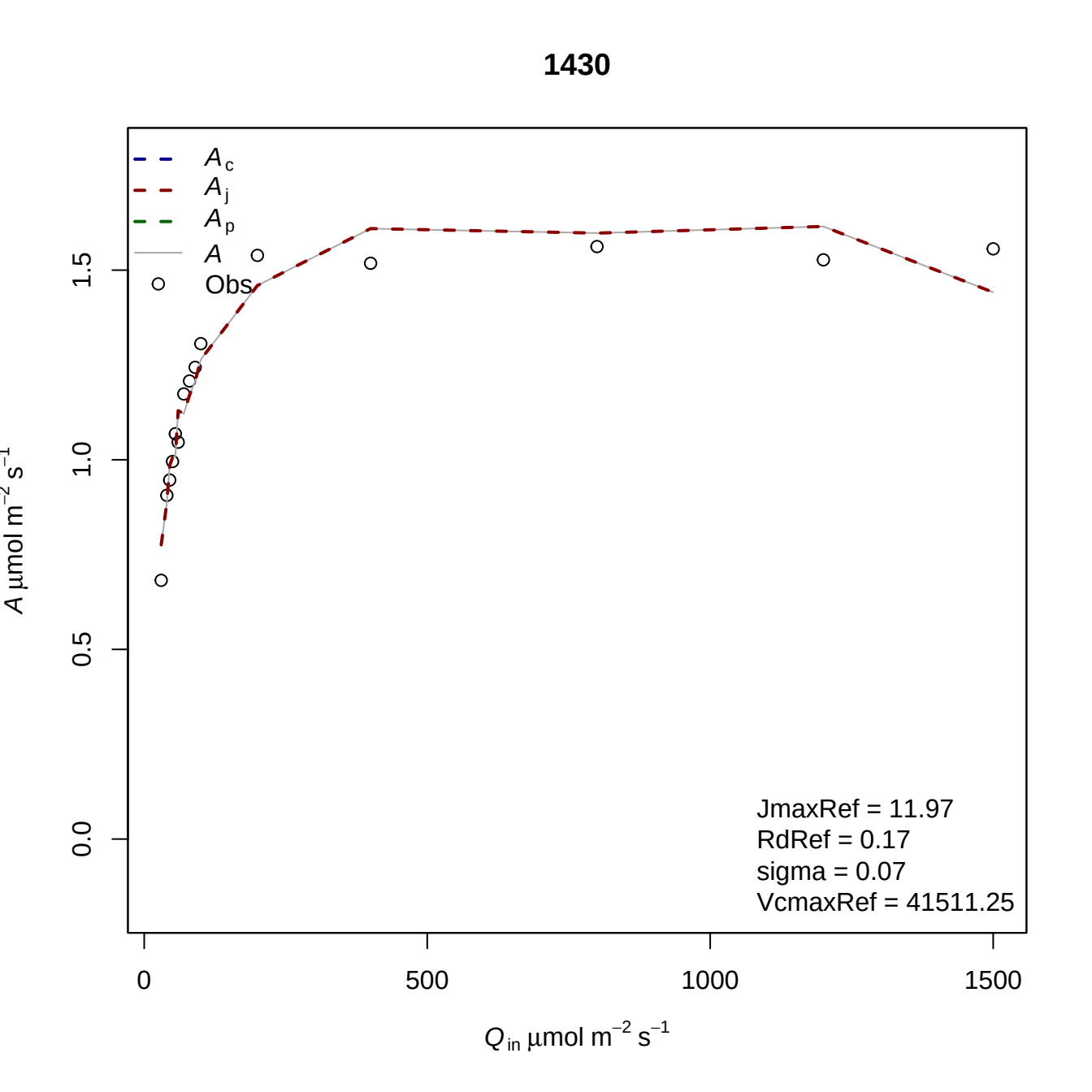

1425
A_c
A_j
A_p
A
Obs
JmaxRef = 16.66
RdRef = 0.24
sigma = 0.06
VcmaxRef = 29501.78
0.0
0.5
1.0
1.5
2.0
2.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

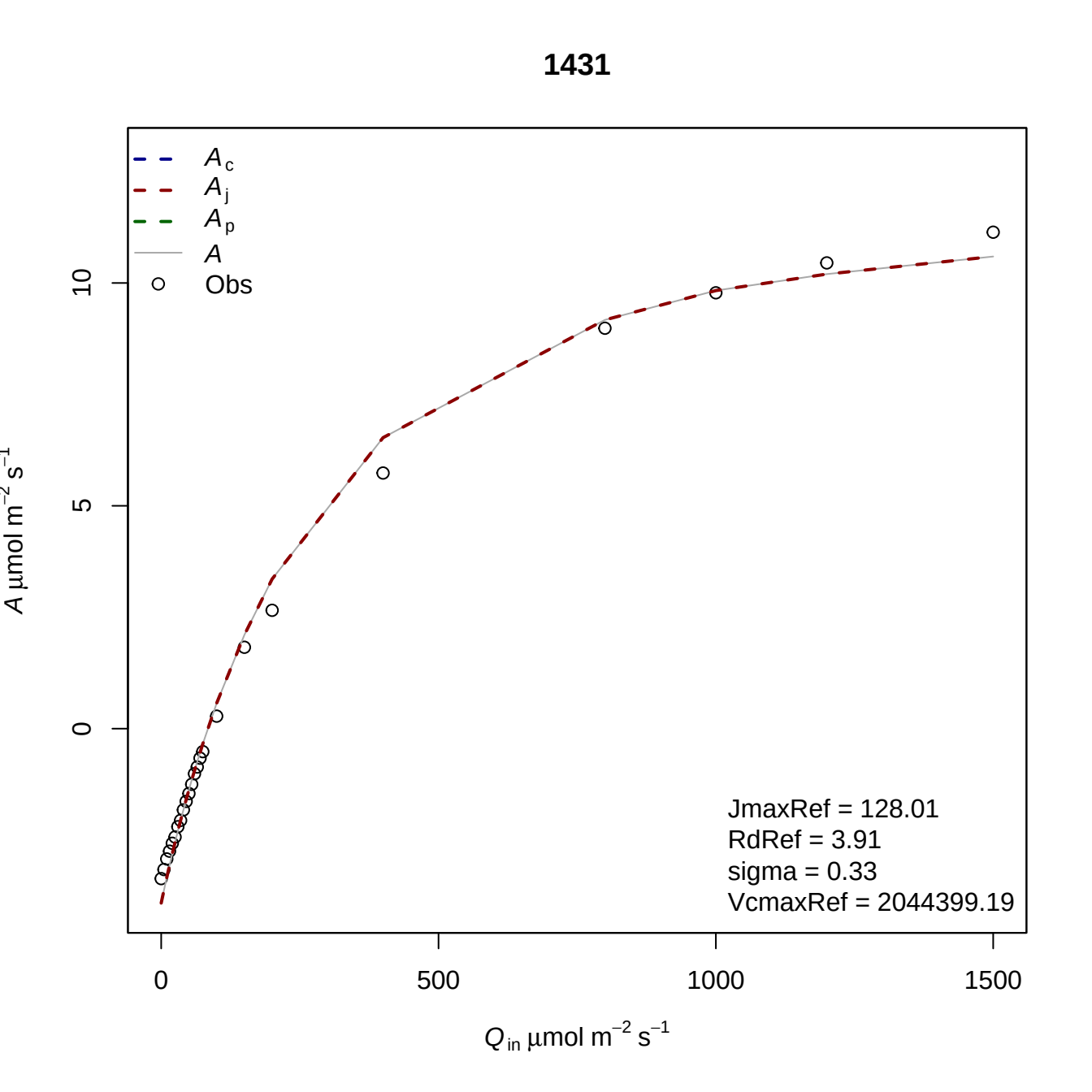

1426
A_c
A_j
A_p
A
Obs
JmaxRef = 45.1
RdRef = −0.03
sigma = 0.21
VcmaxRef = 16.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

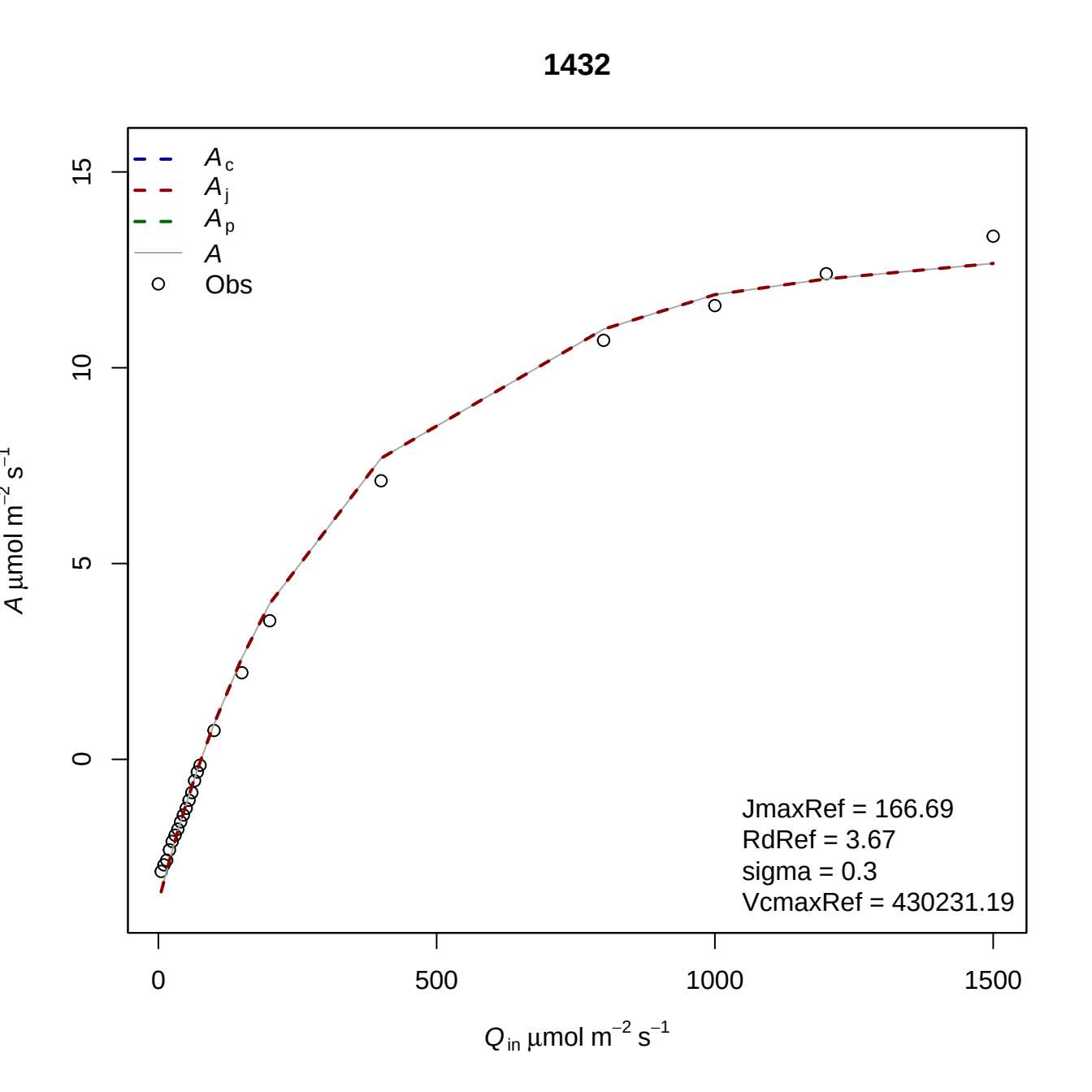

1427
A_c
A_j
A_p
A
Obs
JmaxRef = 19.58
RdRef = 0
sigma = 0.11
VcmaxRef = 10.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
500
1000
1500

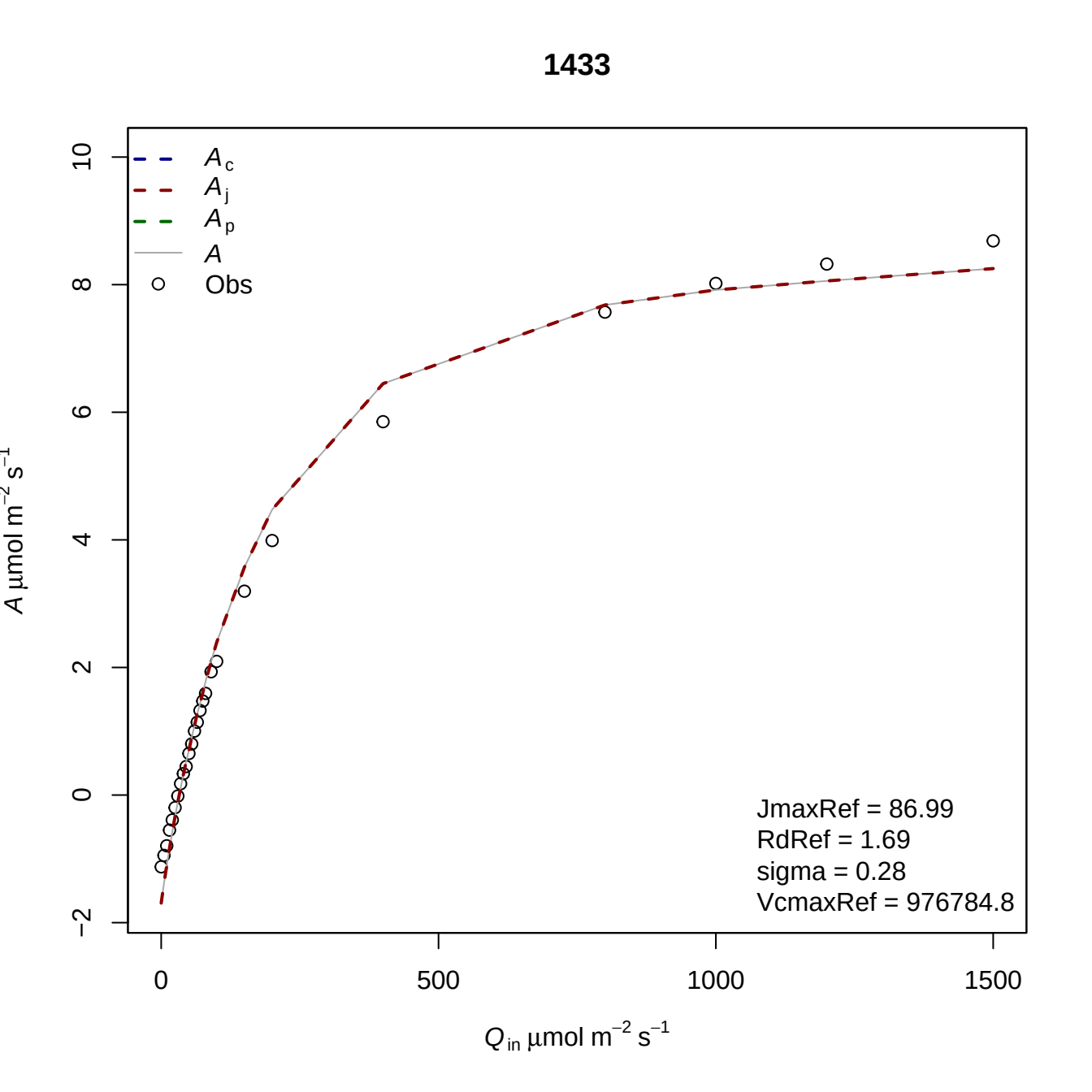

1428
A_c
A_j
A_p
A
Obs
JmaxRef = 13.45
RdRef = 0.22
sigma = 0.07
VcmaxRef = 45904.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500

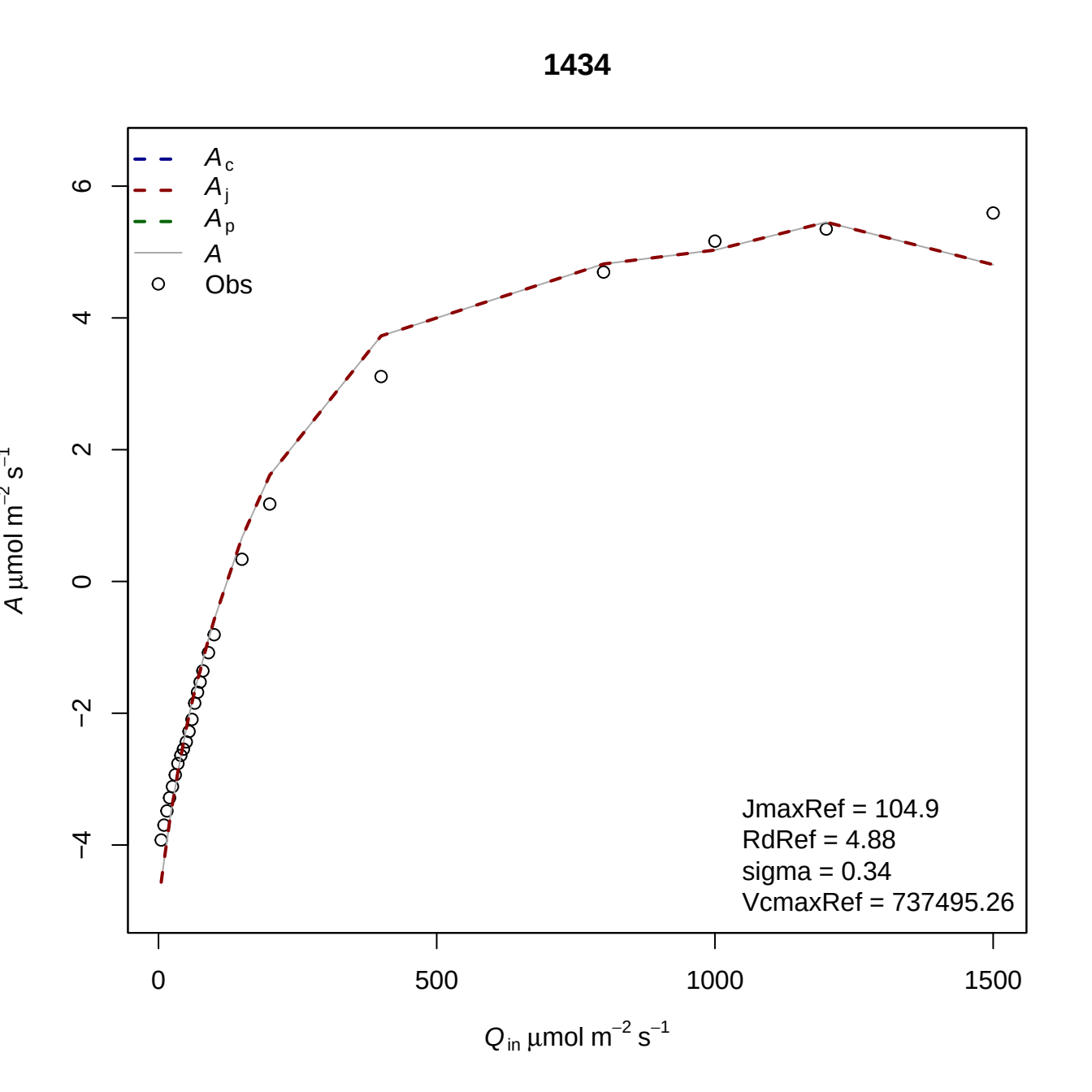

1429
A_c
A_j
A_p
A
Obs
JmaxRef = 14.78
RdRef = 0.05
sigma = 0.07
VcmaxRef = 9.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500

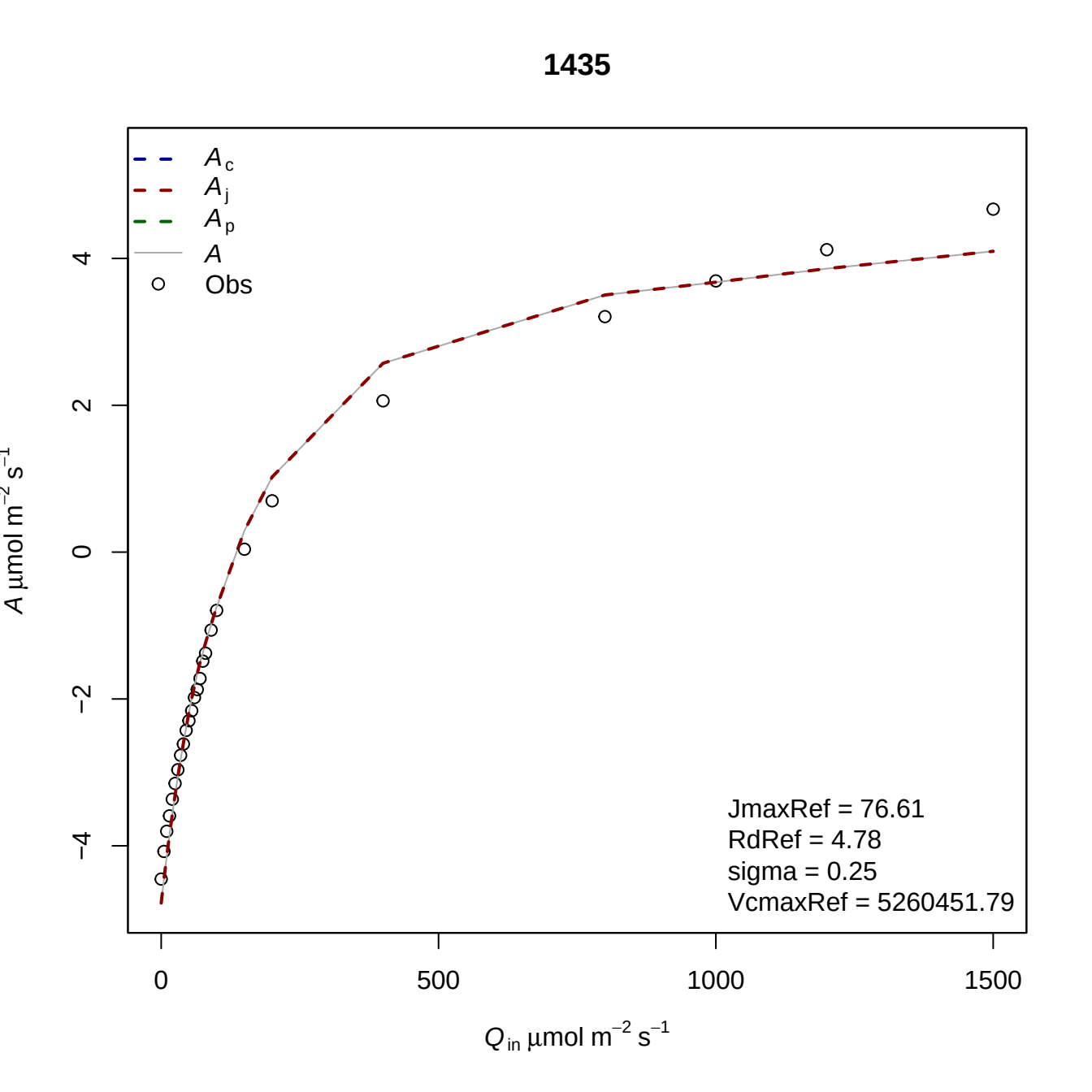

1430
A_c
A_j
A_p
A
Obs
JmaxRef = 11.97
RdRef = 0.17
sigma = 0.07
VcmaxRef = 41511.25
0.0
0.5
1.0
1.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

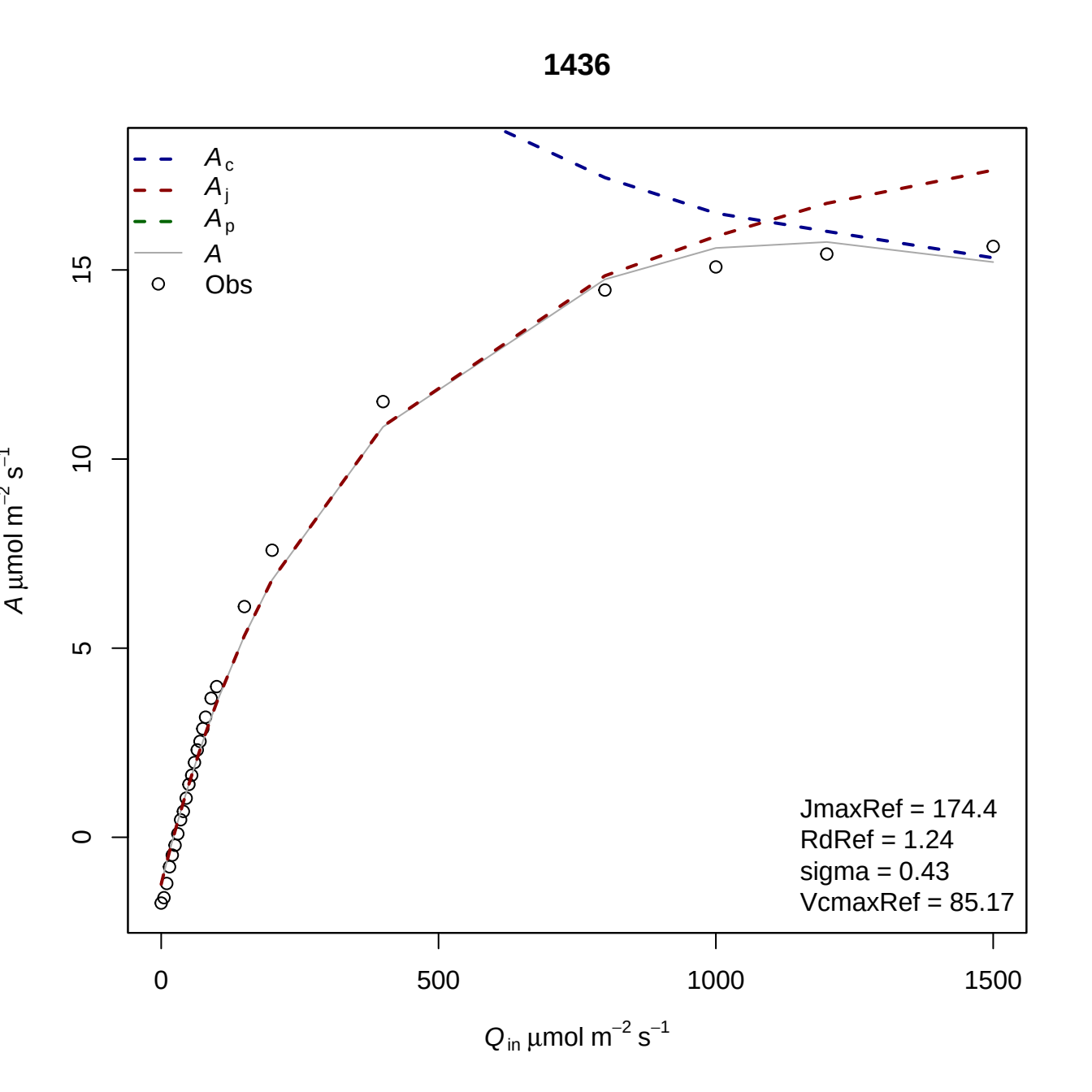

1431
A_c
A_j
A_p
A
Obs
JmaxRef = 128.01
RdRef = 3.91
sigma = 0.33
VcmaxRef = 2044399.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

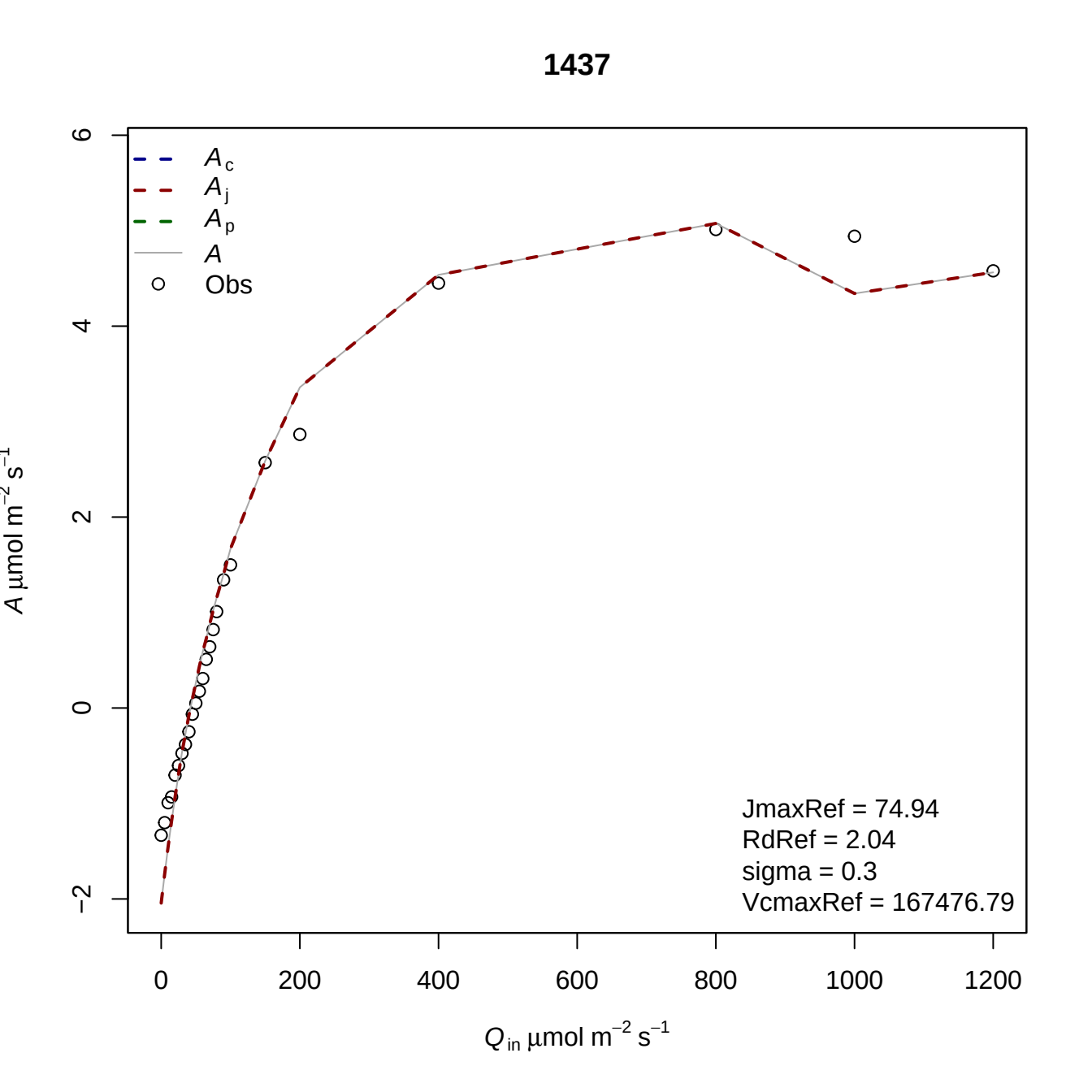

1432
A_c
A_j
A_p
A
Obs
JmaxRef = 166.69
RdRef = 3.67
sigma = 0.3
VcmaxRef = 430231.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

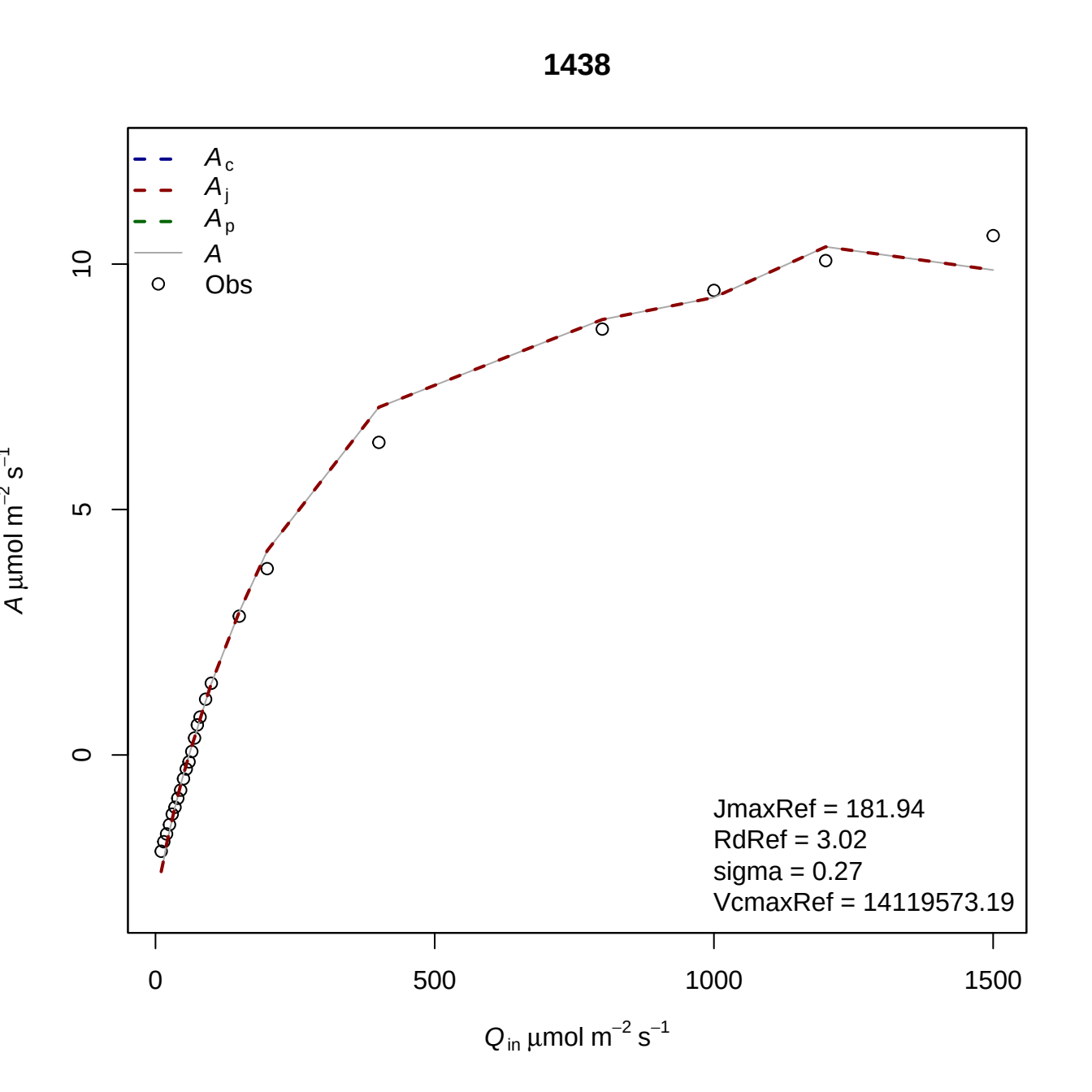

1433
A_c
A_j
A_p
A
Obs
JmaxRef = 86.99
RdRef = 1.69
sigma = 0.28
VcmaxRef = 976784.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

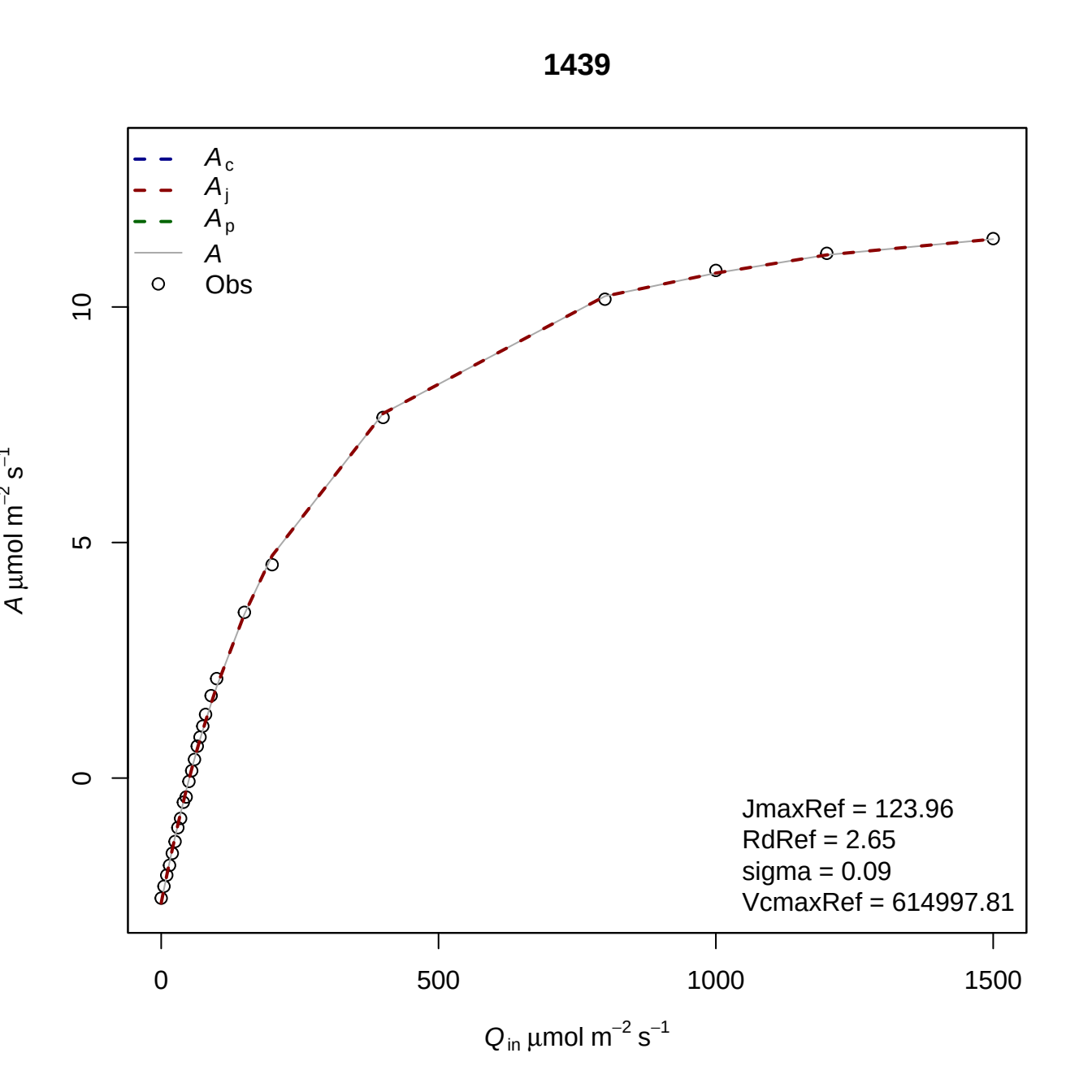

1434
A_c
A_j
A_p
A
Obs
JmaxRef = 104.9
RdRef = 4.88
sigma = 0.34
VcmaxRef = 737495.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

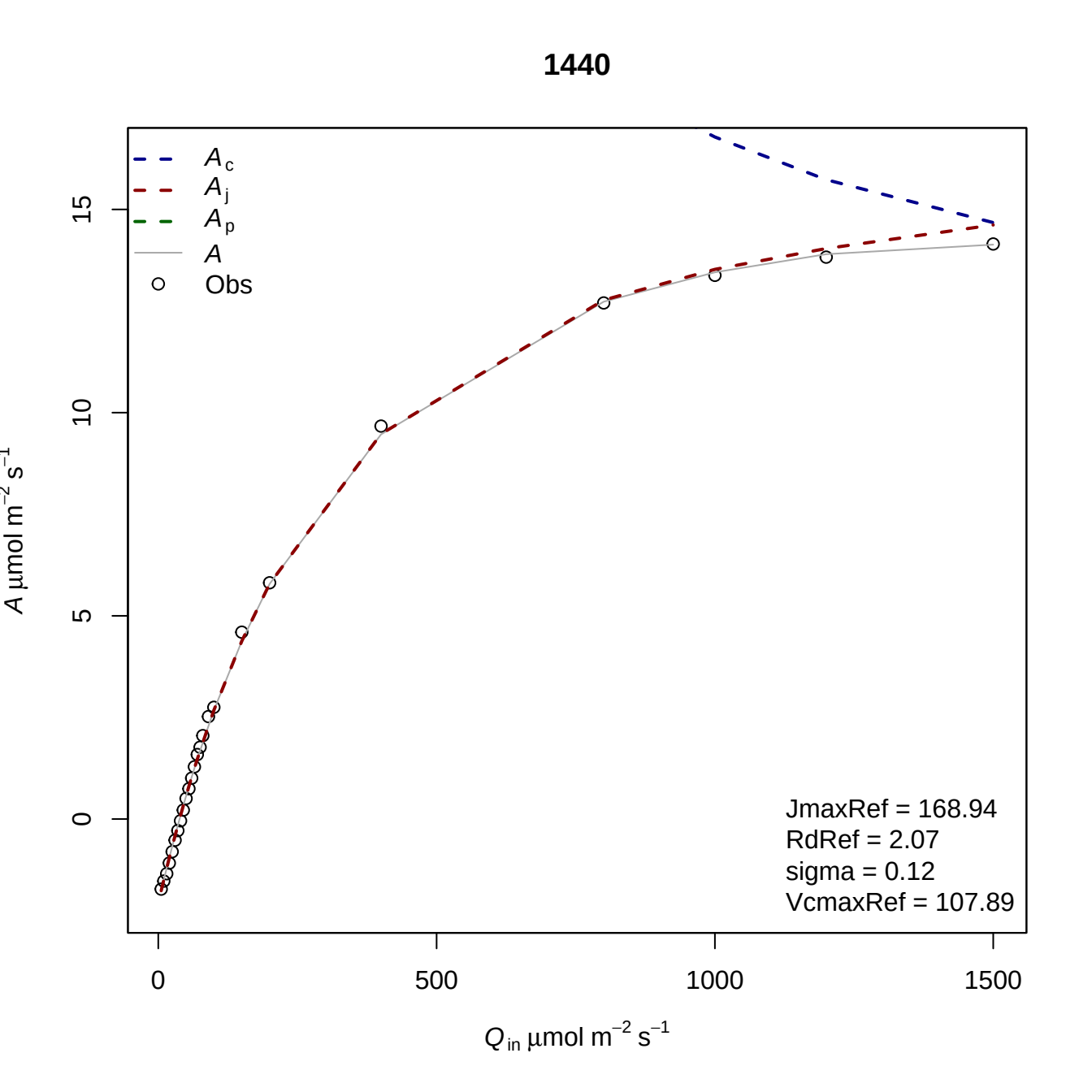

1435
A_c
A_j
A_p
A
Obs
JmaxRef = 76.61
RdRef = 4.78
sigma = 0.25
VcmaxRef = 5260451.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
500
1000
1500

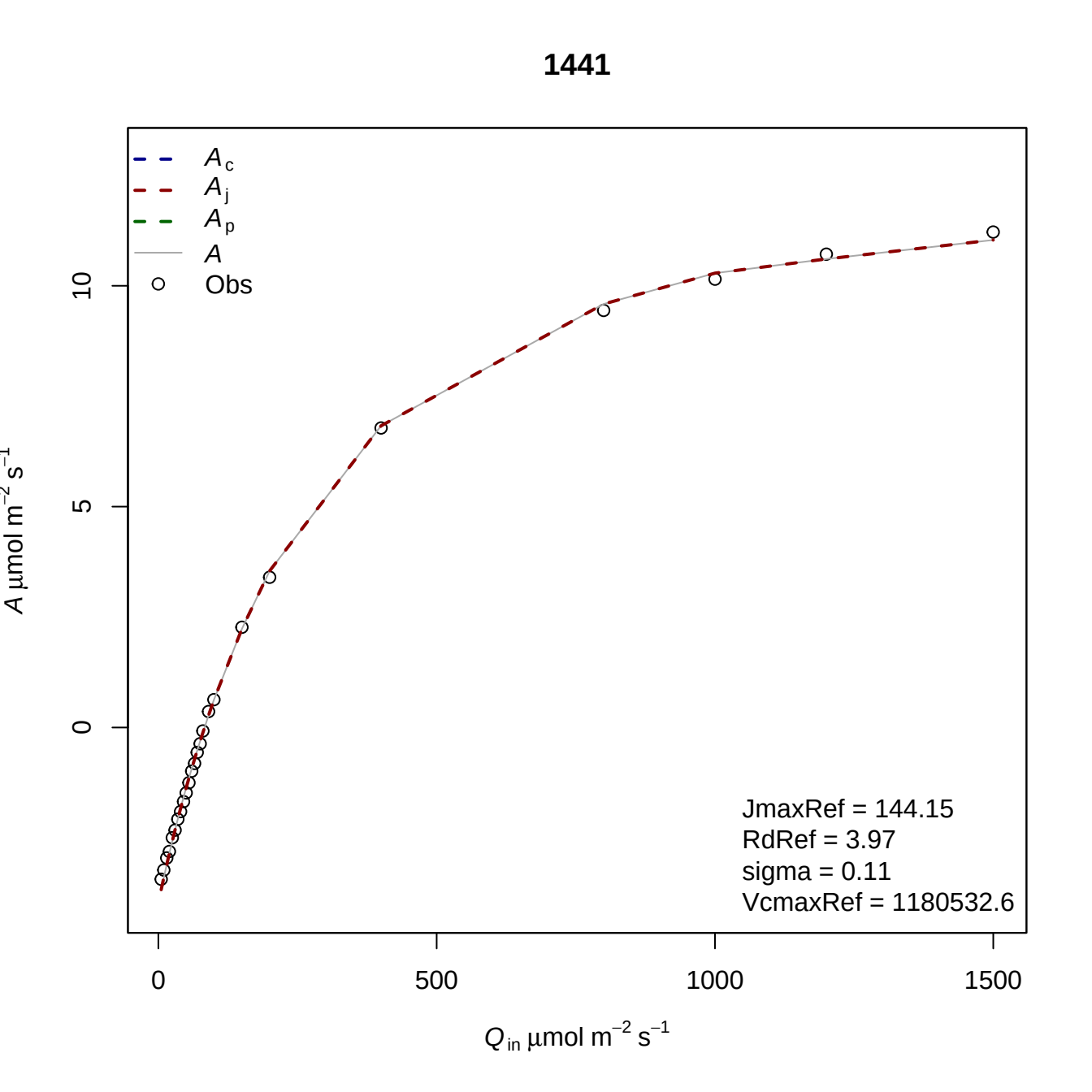

1436
A_c
A_j
A_p
A
Obs
JmaxRef = 174.4
RdRef = 1.24
sigma = 0.43
VcmaxRef = 85.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

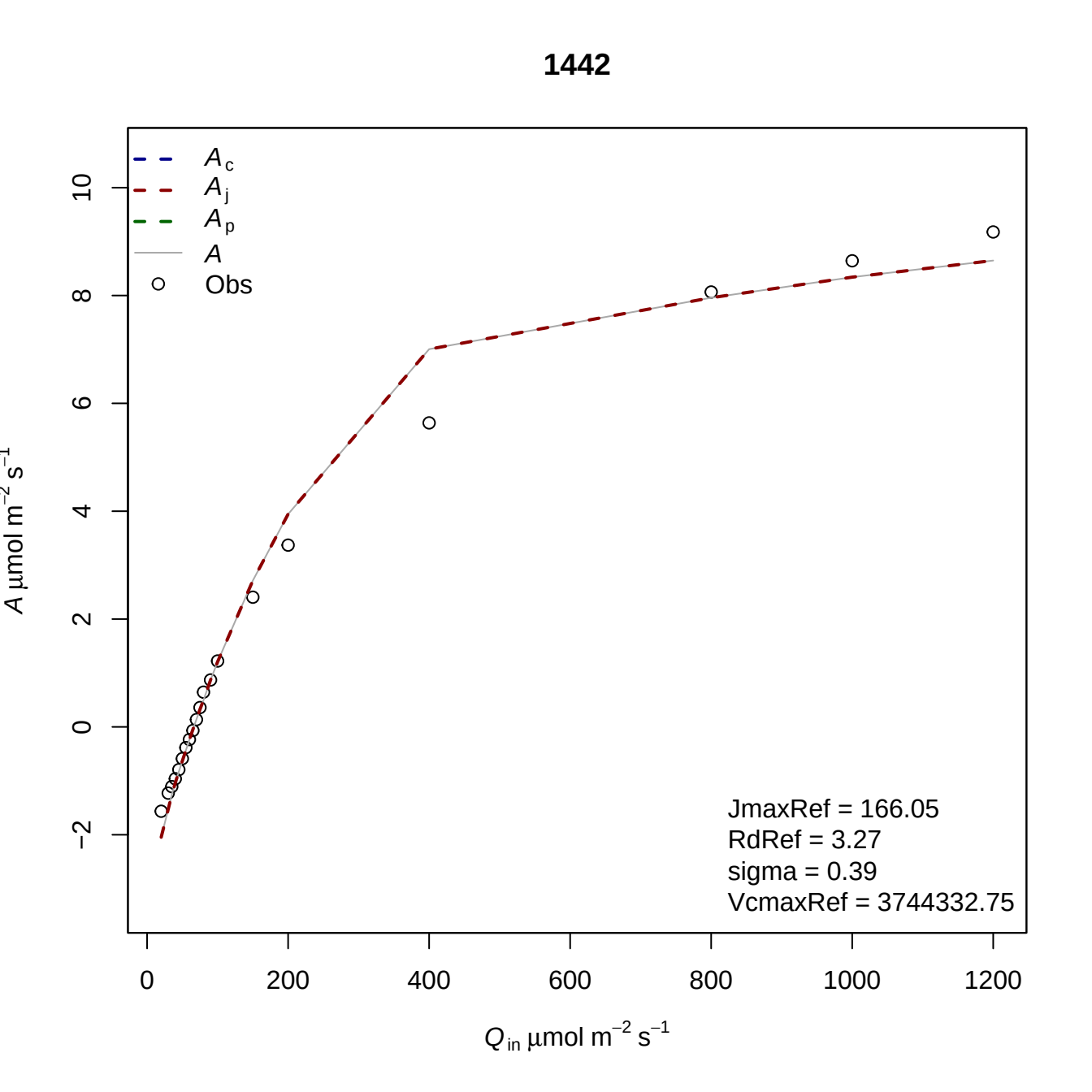

1437
A_c
A_j
A_p
A
Obs
JmaxRef = 74.94
RdRef = 2.04
sigma = 0.3
VcmaxRef = 167476.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000
1200

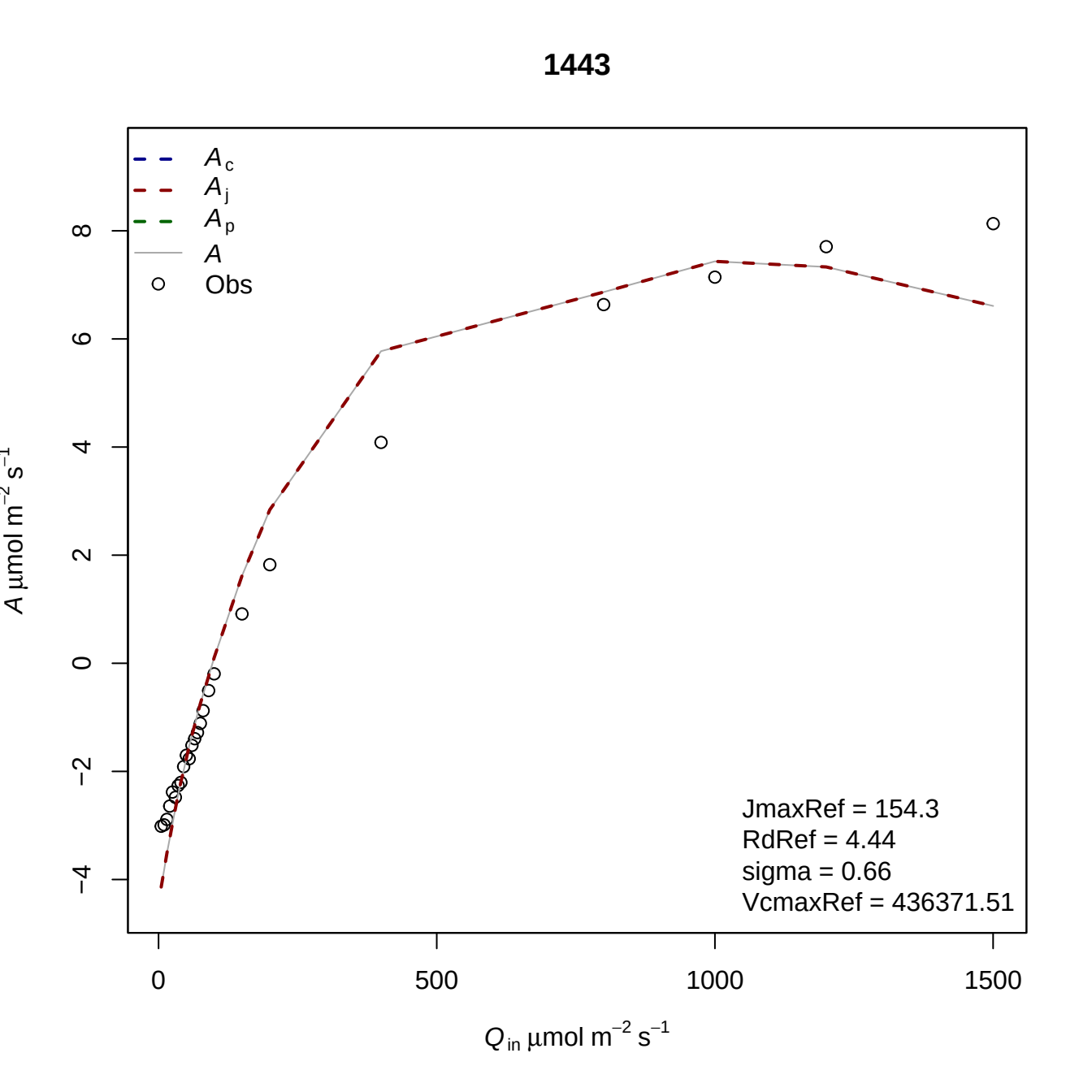

1438
A_c
A_j
A_p
A
Obs
JmaxRef = 181.94
RdRef = 3.02
sigma = 0.27
VcmaxRef = 14119573.19
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

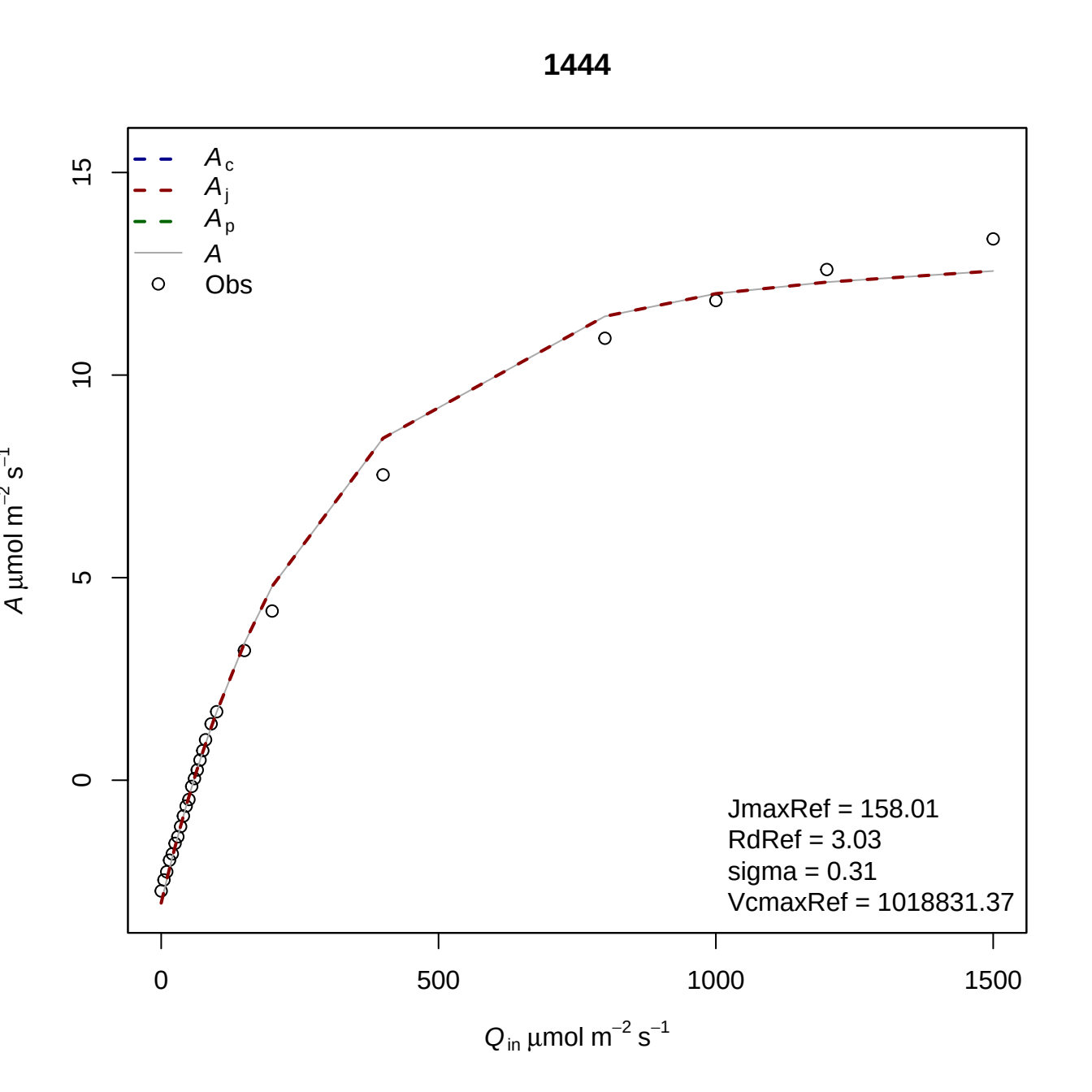

1439
A_c
A_j
A_p
A
Obs
JmaxRef = 123.96
RdRef = 2.65
sigma = 0.09
VcmaxRef = 614997.81
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

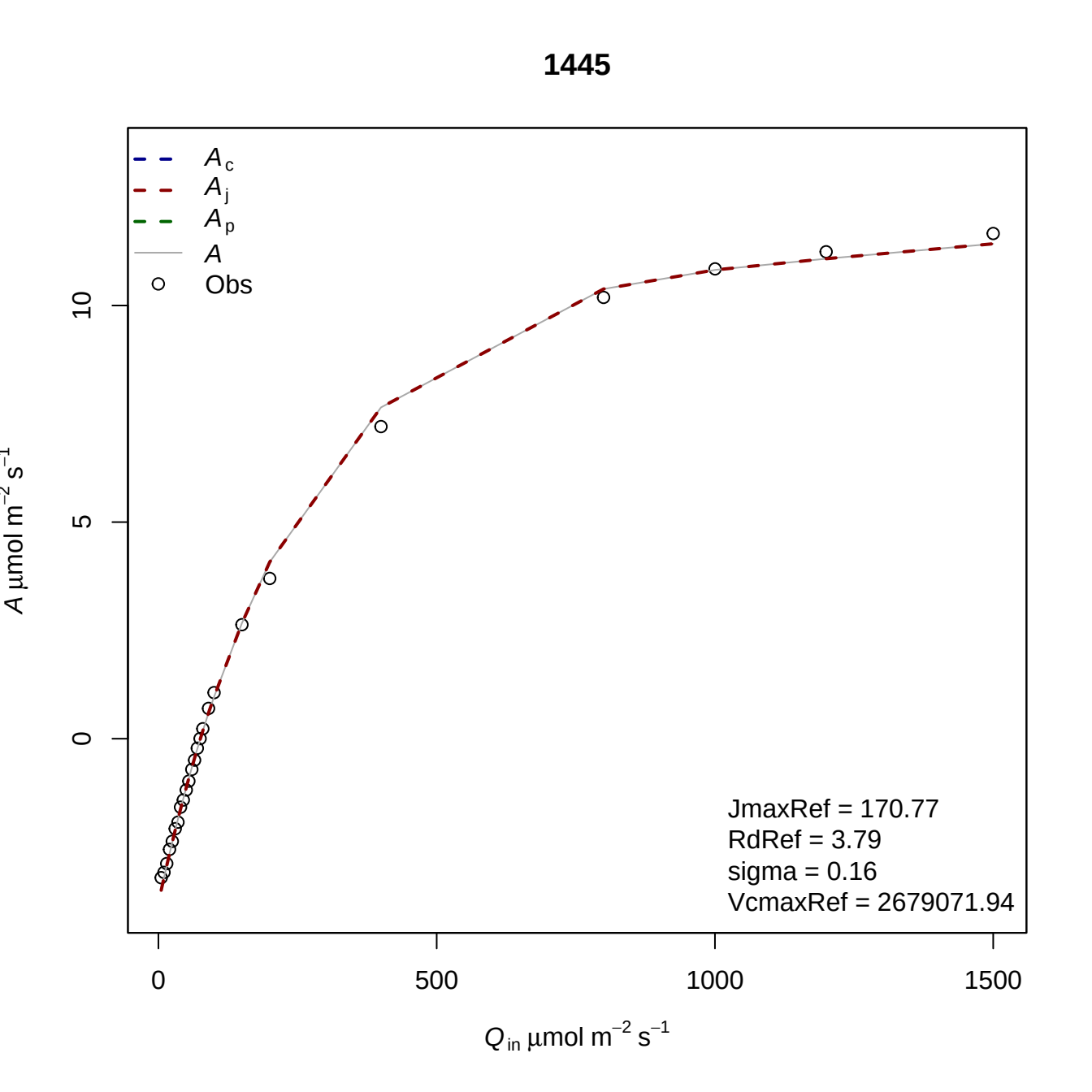

1440
A_c
A_j
A_p
A
Obs
JmaxRef = 168.94
RdRef = 2.07
sigma = 0.12
VcmaxRef = 107.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

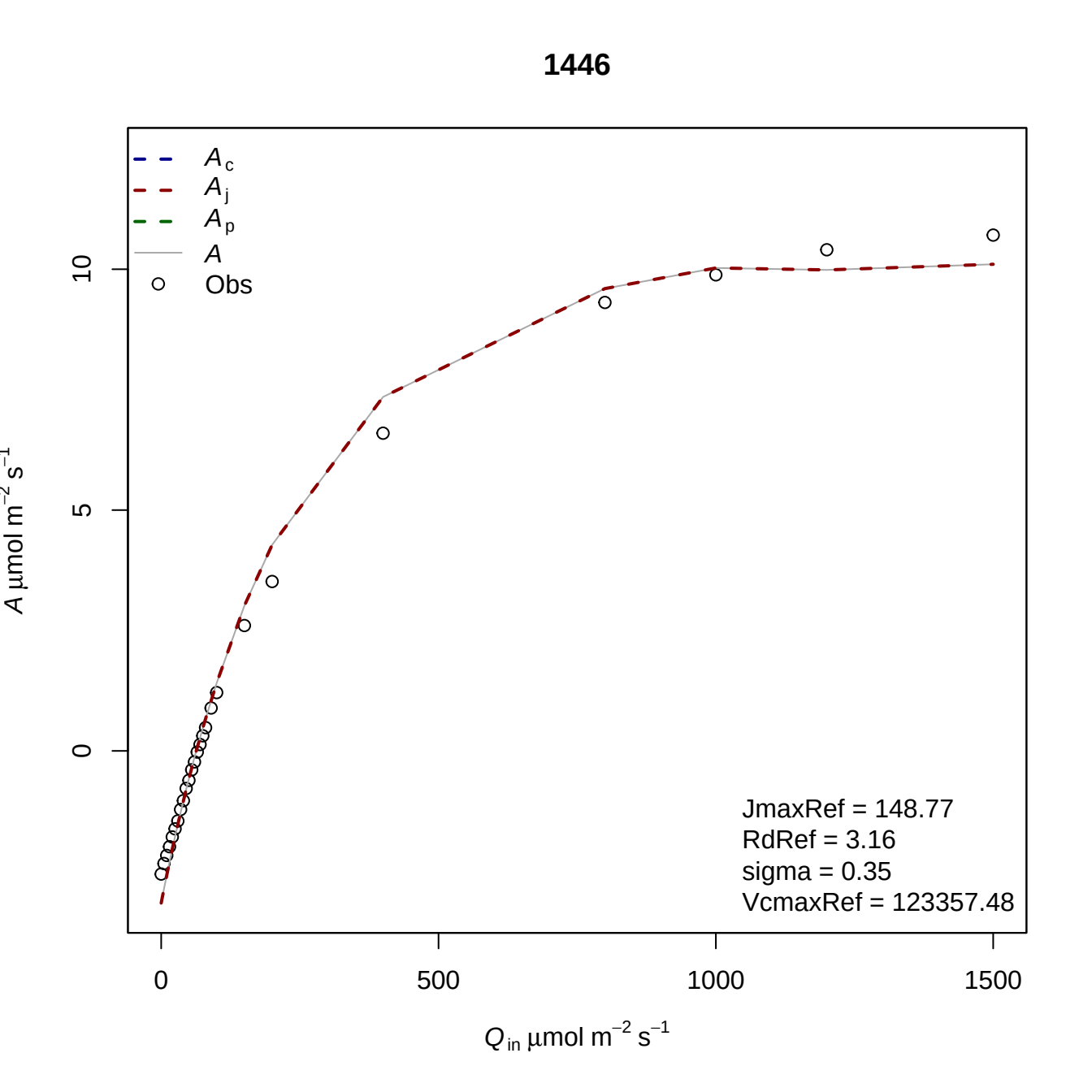

1441
A_c
A_j
A_p
A
Obs
JmaxRef = 144.15
RdRef = 3.97
sigma = 0.11
VcmaxRef = 1180532.6
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

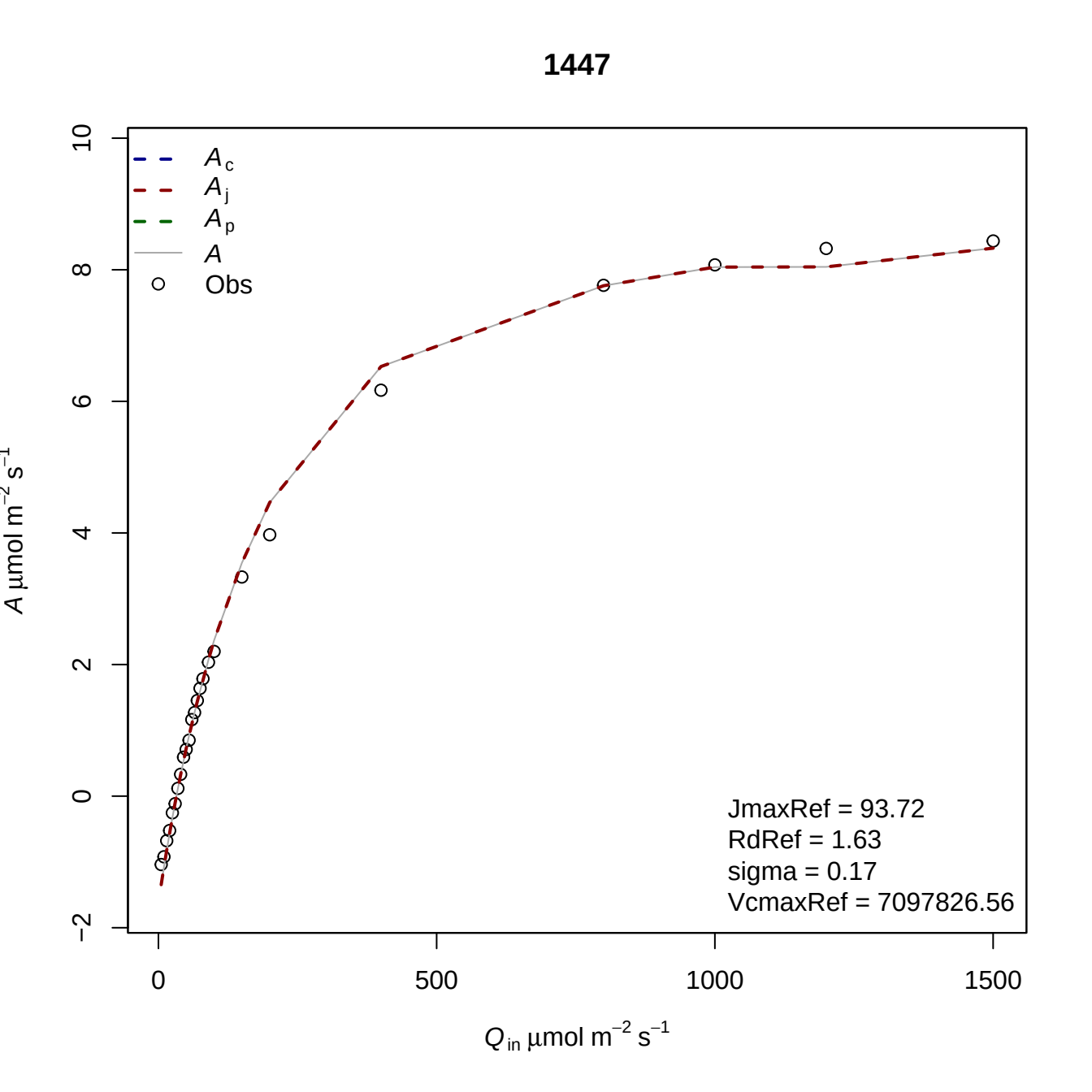

1442
A_c
A_j
A_p
A
Obs
JmaxRef = 166.05
RdRef = 3.27
sigma = 0.39
VcmaxRef = 3744332.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200

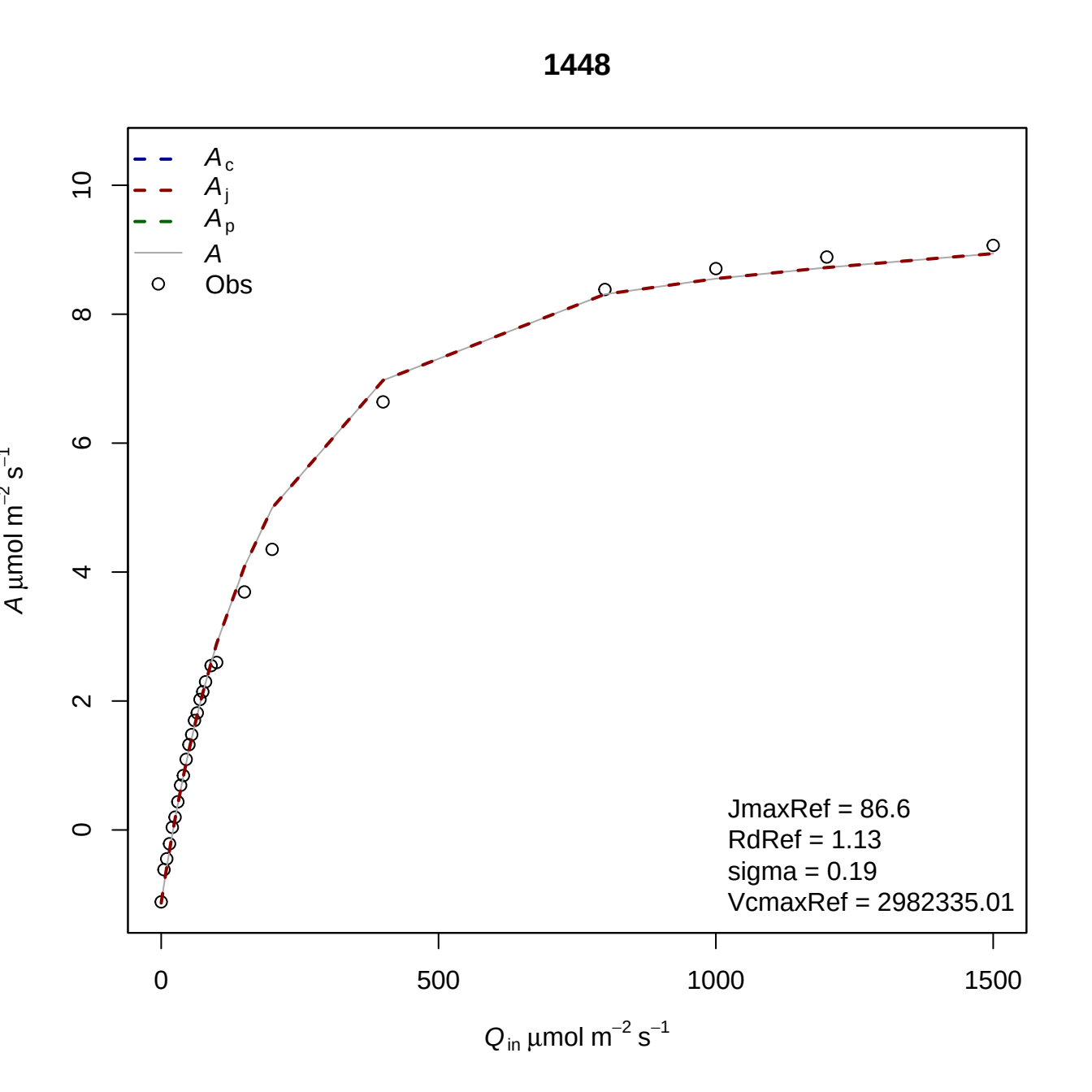

1443
A_c
A_j
A_p
A
Obs
JmaxRef = 154.3
RdRef = 4.44
sigma = 0.66
VcmaxRef = 436371.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

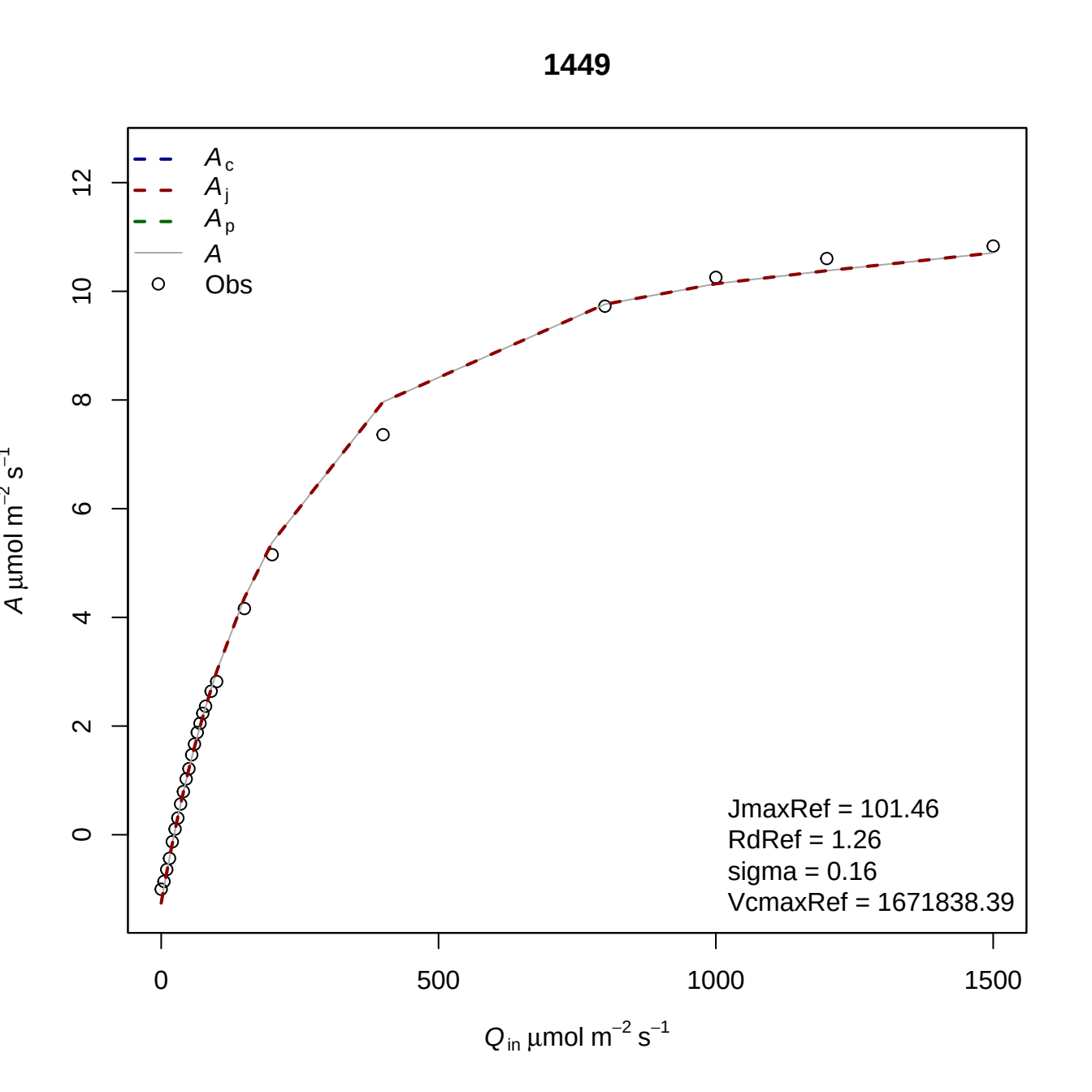

1444
A_c
A_j
A_p
A
Obs
JmaxRef = 158.01
RdRef = 3.03
sigma = 0.31
VcmaxRef = 1018831.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

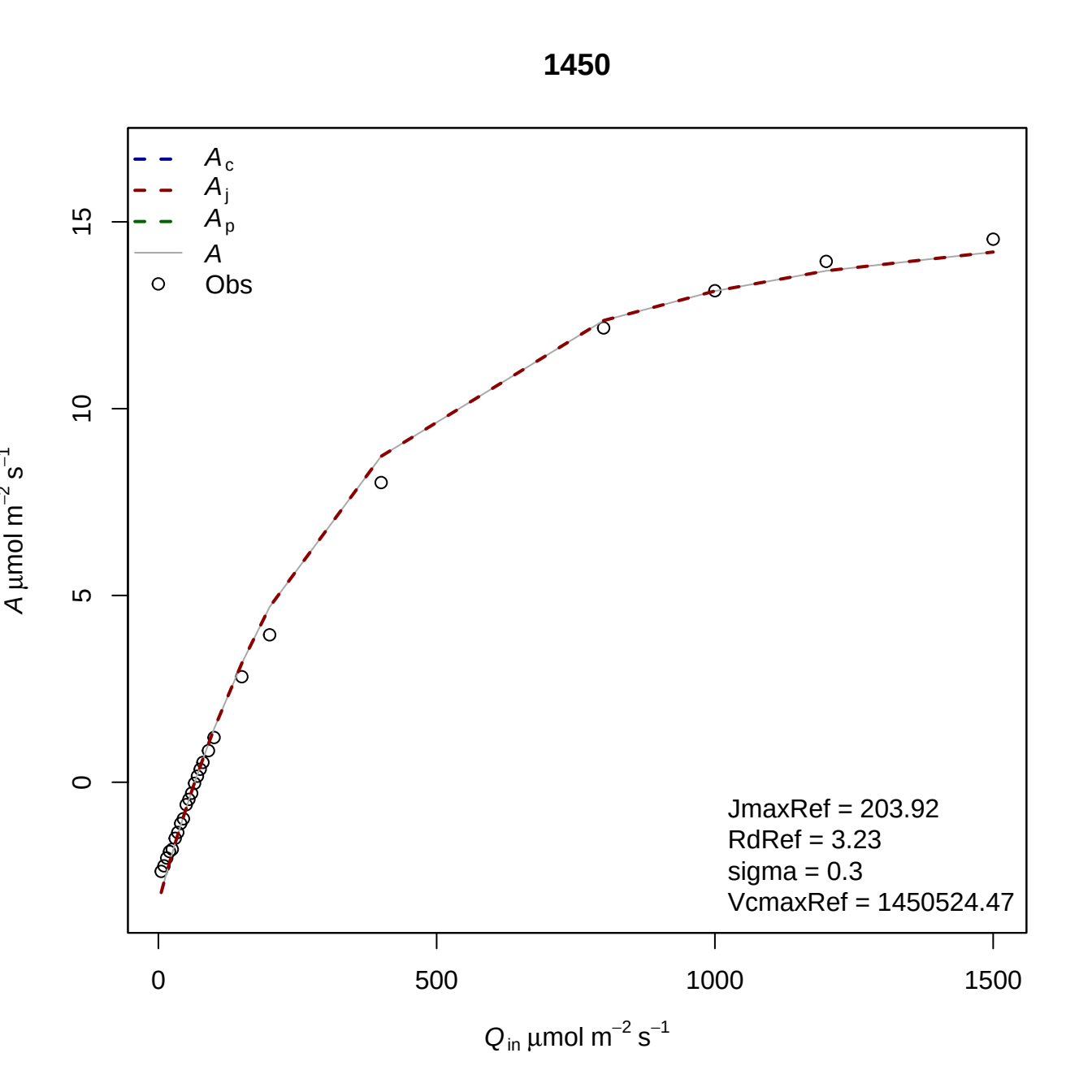

1445
A_c
A_j
A_p
A
Obs
JmaxRef = 170.77
RdRef = 3.79
sigma = 0.16
VcmaxRef = 2679071.94
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

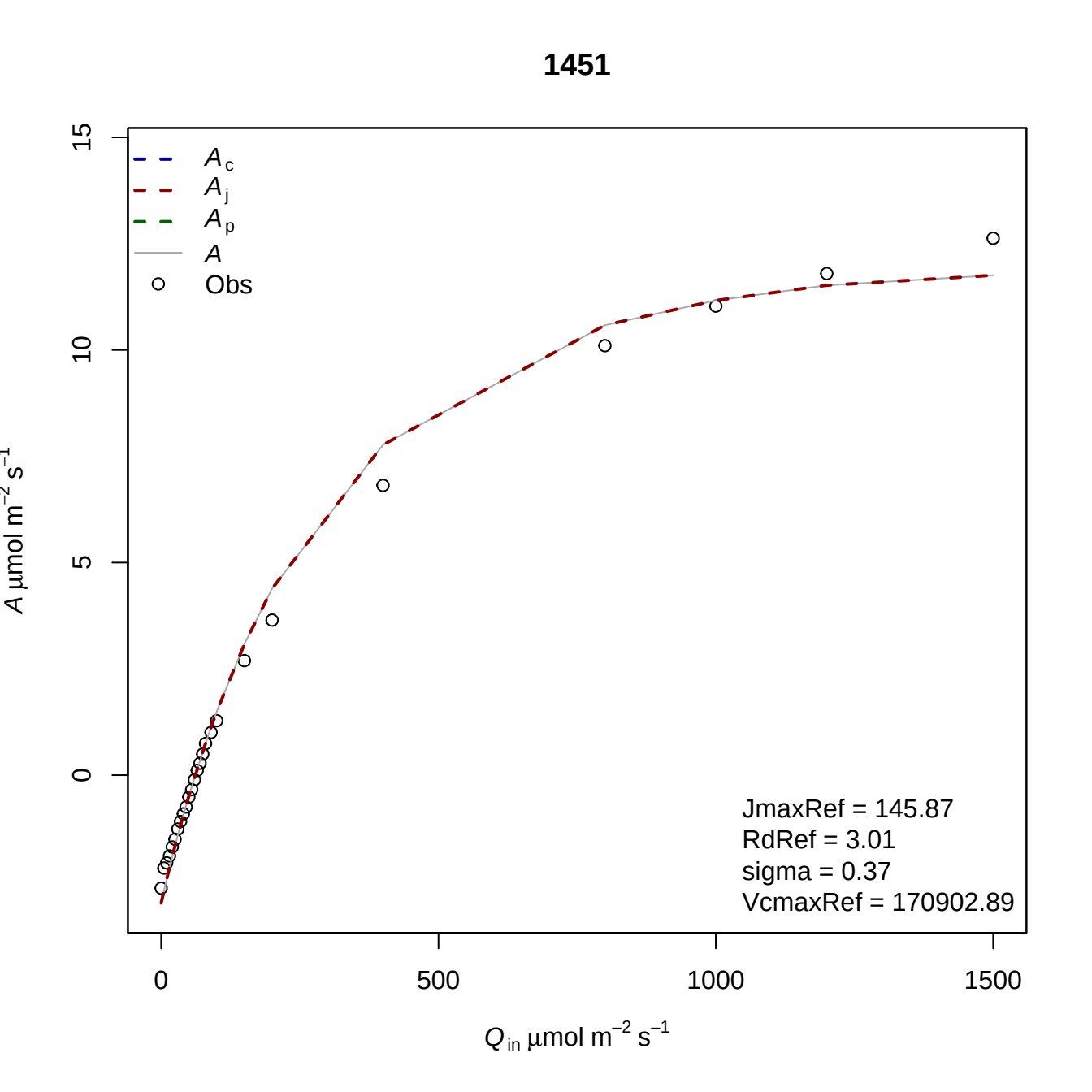

1446
A_c
A_j
A_p
A
Obs
JmaxRef = 148.77
RdRef = 3.16
sigma = 0.35
VcmaxRef = 123357.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

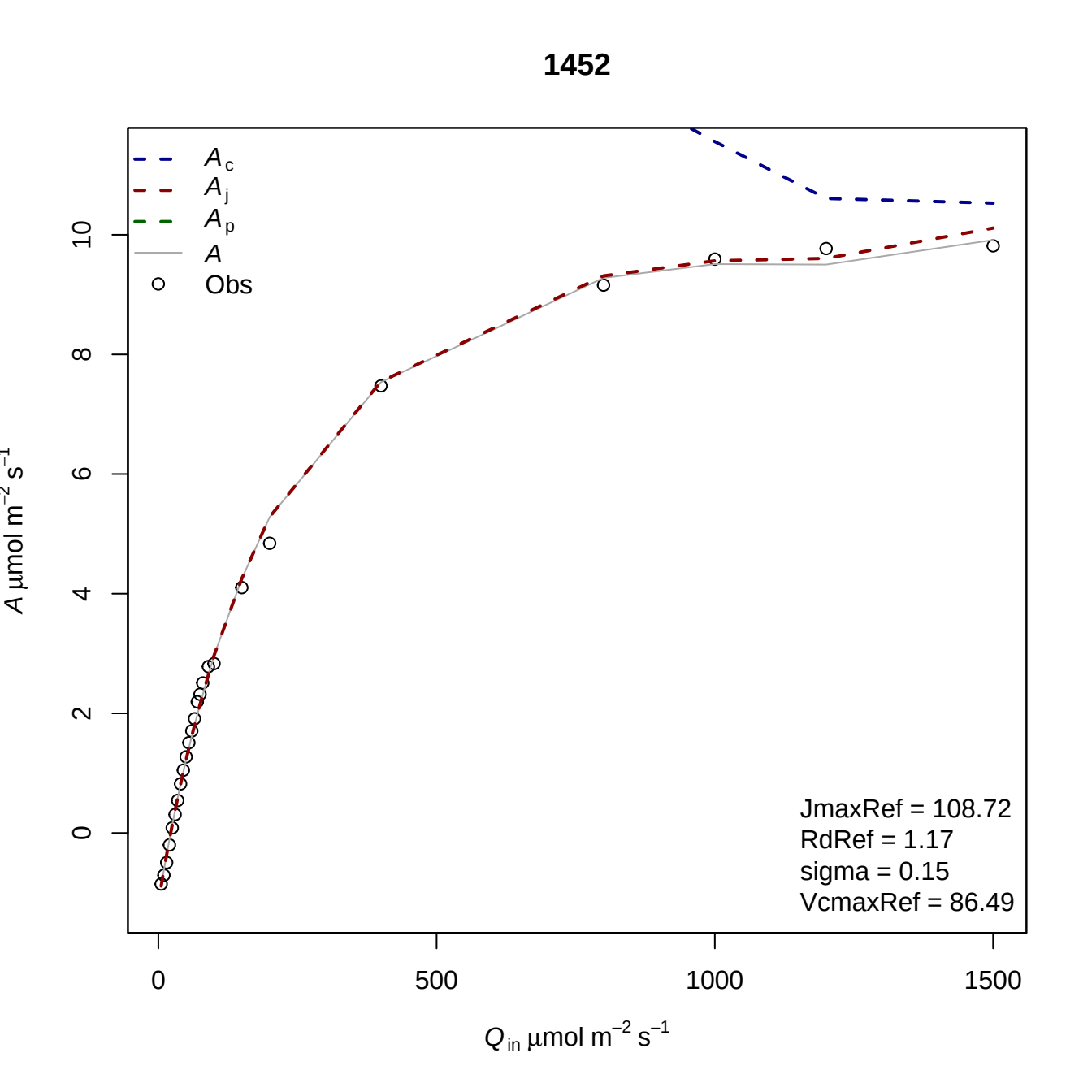

1447
A_c
A_j
A_p
A
Obs
JmaxRef = 93.72
RdRef = 1.63
sigma = 0.17
VcmaxRef = 7097826.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

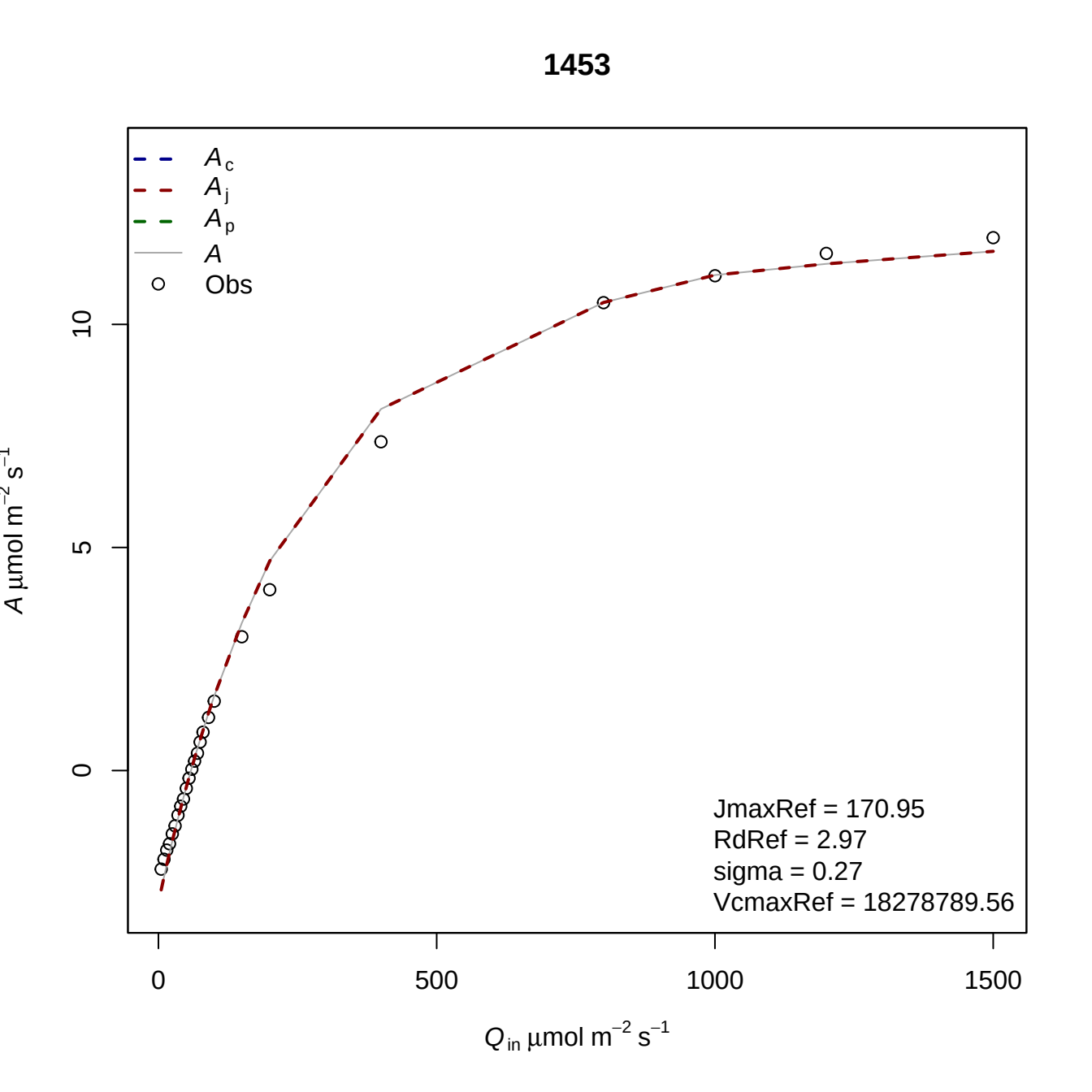

1448
A_c
A_j
A_p
A
Obs
JmaxRef = 86.6
RdRef = 1.13
sigma = 0.19
VcmaxRef = 2982335.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

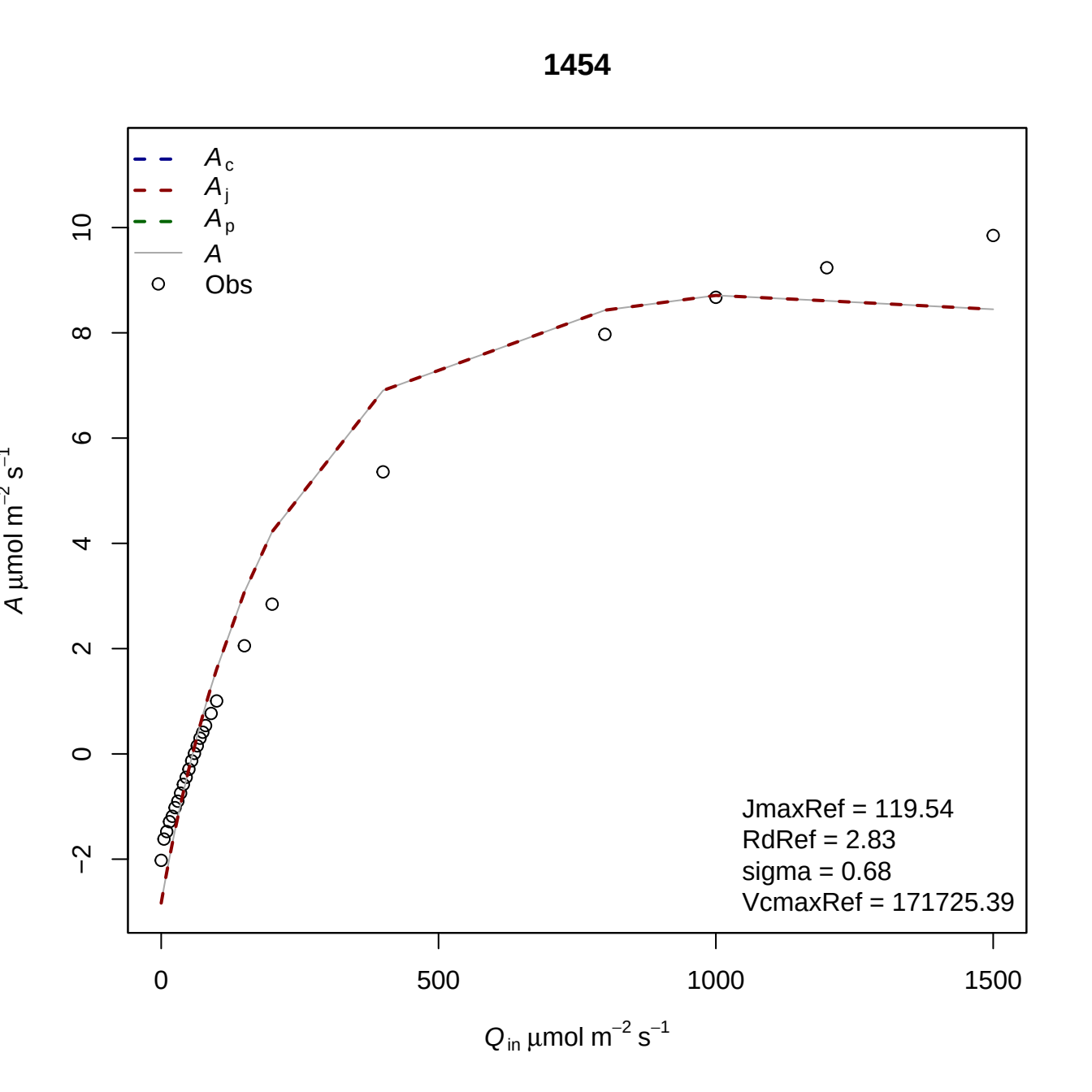

1449
A_c
A_j
A_p
A
Obs
JmaxRef = 101.46
RdRef = 1.26
sigma = 0.16
VcmaxRef = 1671838.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

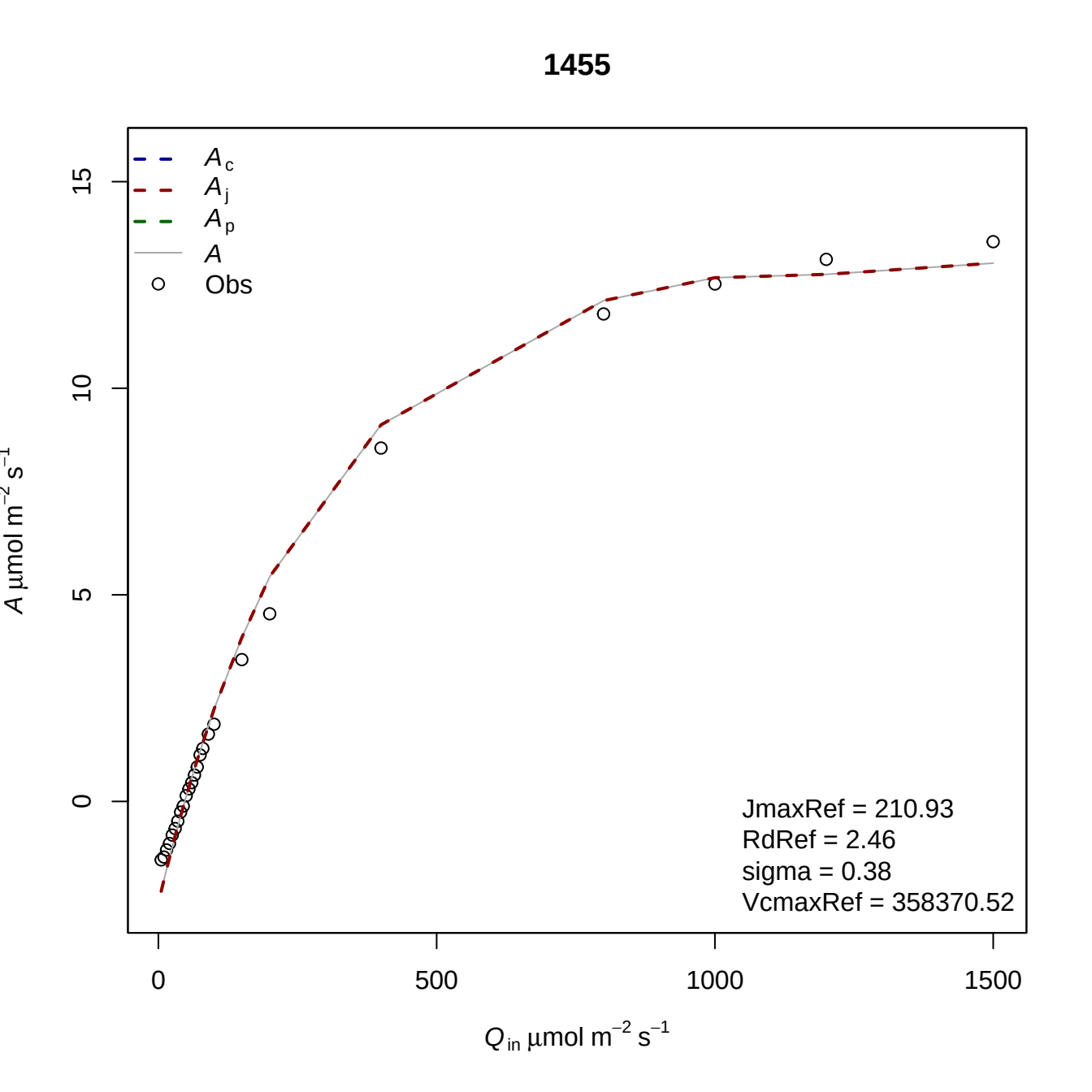

1450
A_c
A_j
A_p
A
Obs
JmaxRef = 203.92
RdRef = 3.23
sigma = 0.3
VcmaxRef = 1450524.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

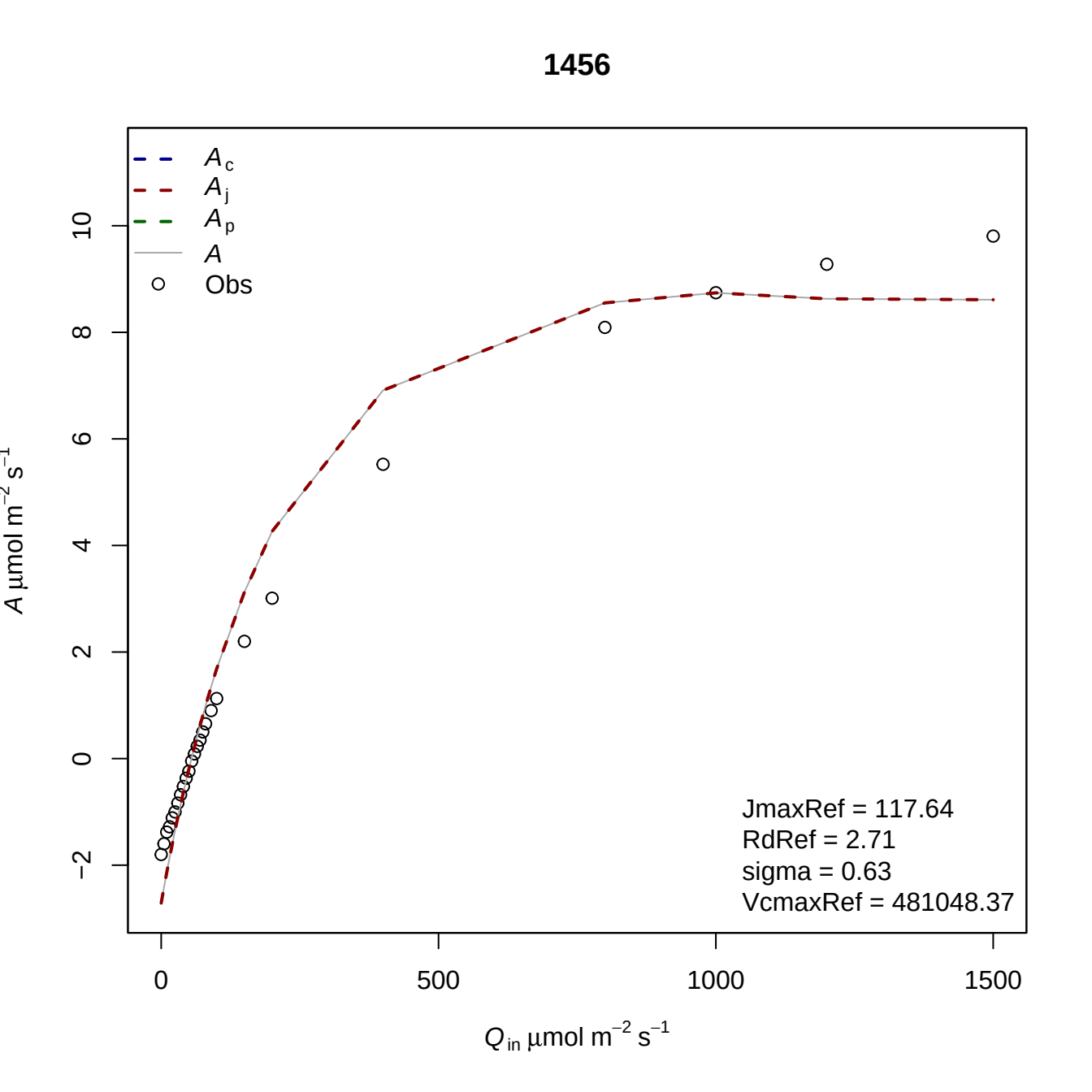

1451
A_c
A_j
A_p
A
Obs
JmaxRef = 145.87
RdRef = 3.01
sigma = 0.37
VcmaxRef = 170902.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

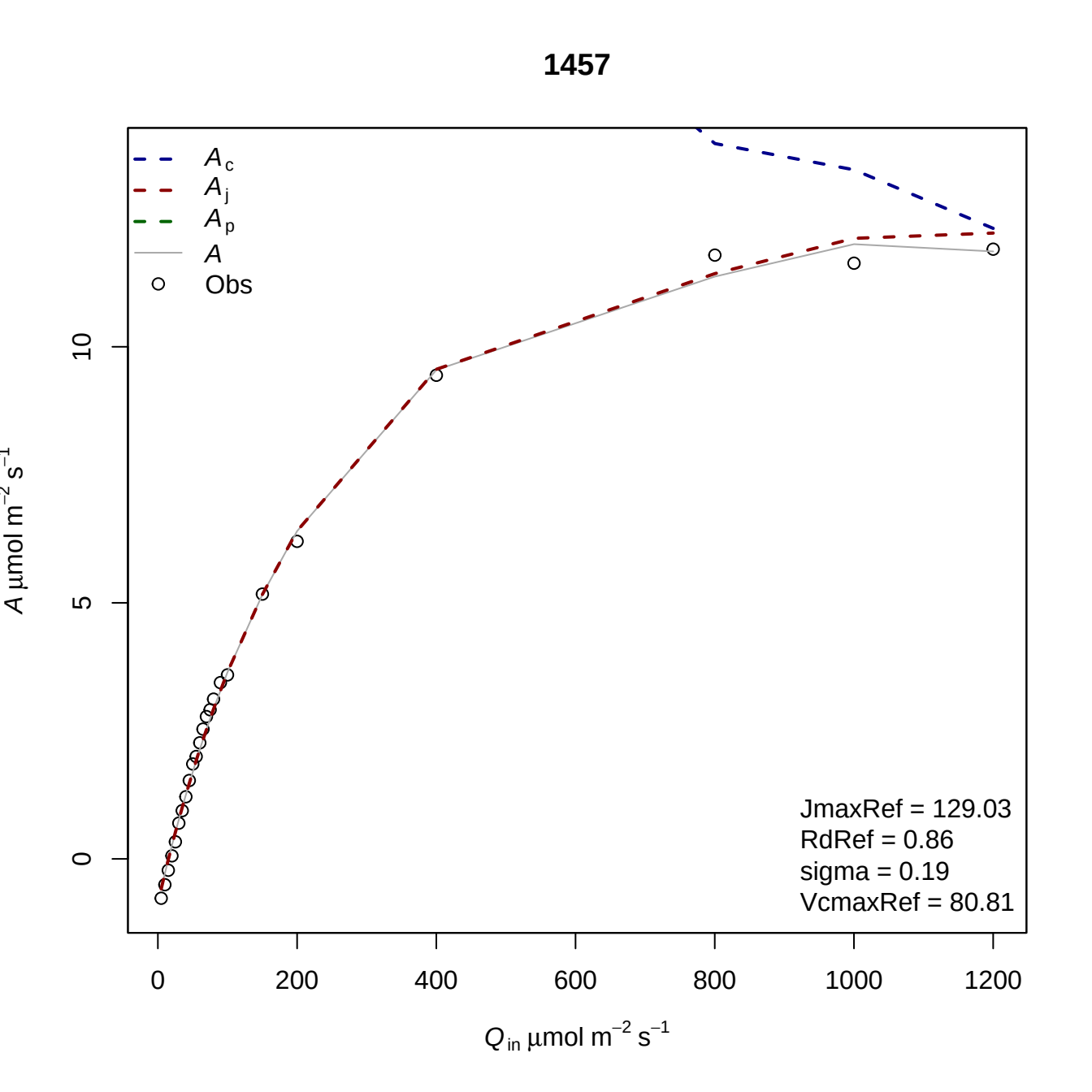

1452
A_c
A_j
A_p
A
Obs
JmaxRef = 108.72
RdRef = 1.17
sigma = 0.15
VcmaxRef = 86.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

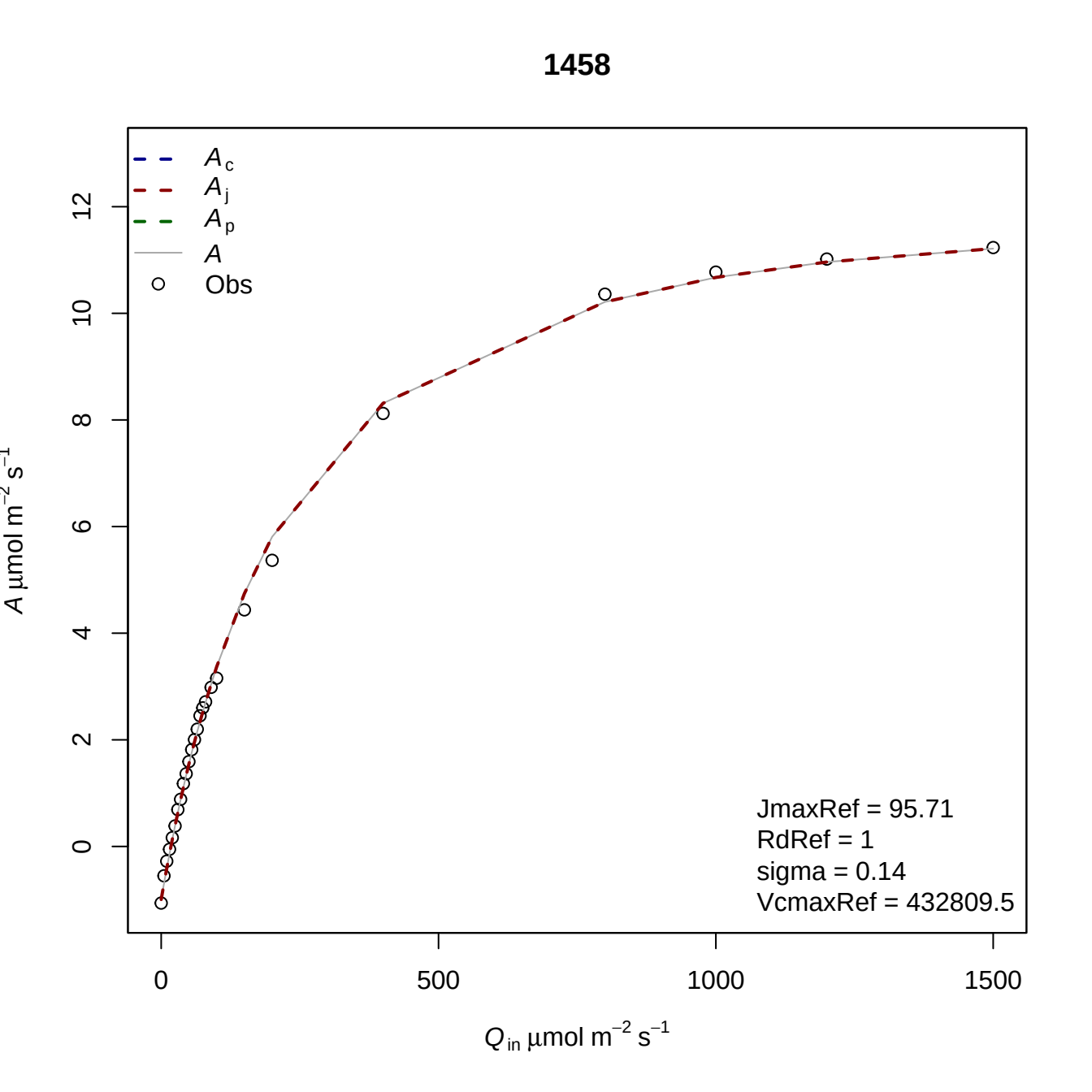

1453
A_c
A_j
A_p
A
Obs
JmaxRef = 170.95
RdRef = 2.97
sigma = 0.27
VcmaxRef = 18278789.56
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

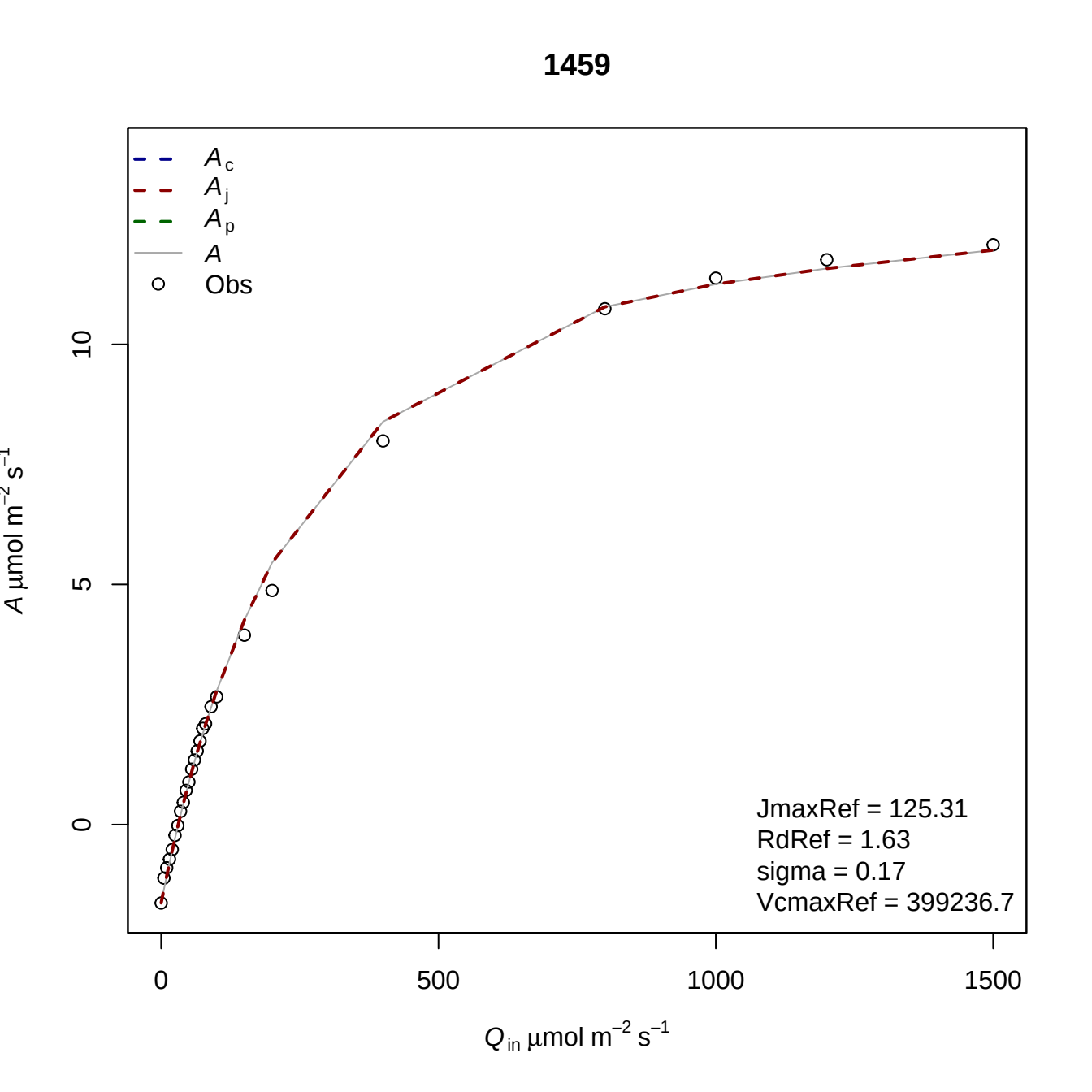

1454
A_c
A_j
A_p
A
Obs
JmaxRef = 119.54
RdRef = 2.83
sigma = 0.68
VcmaxRef = 171725.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

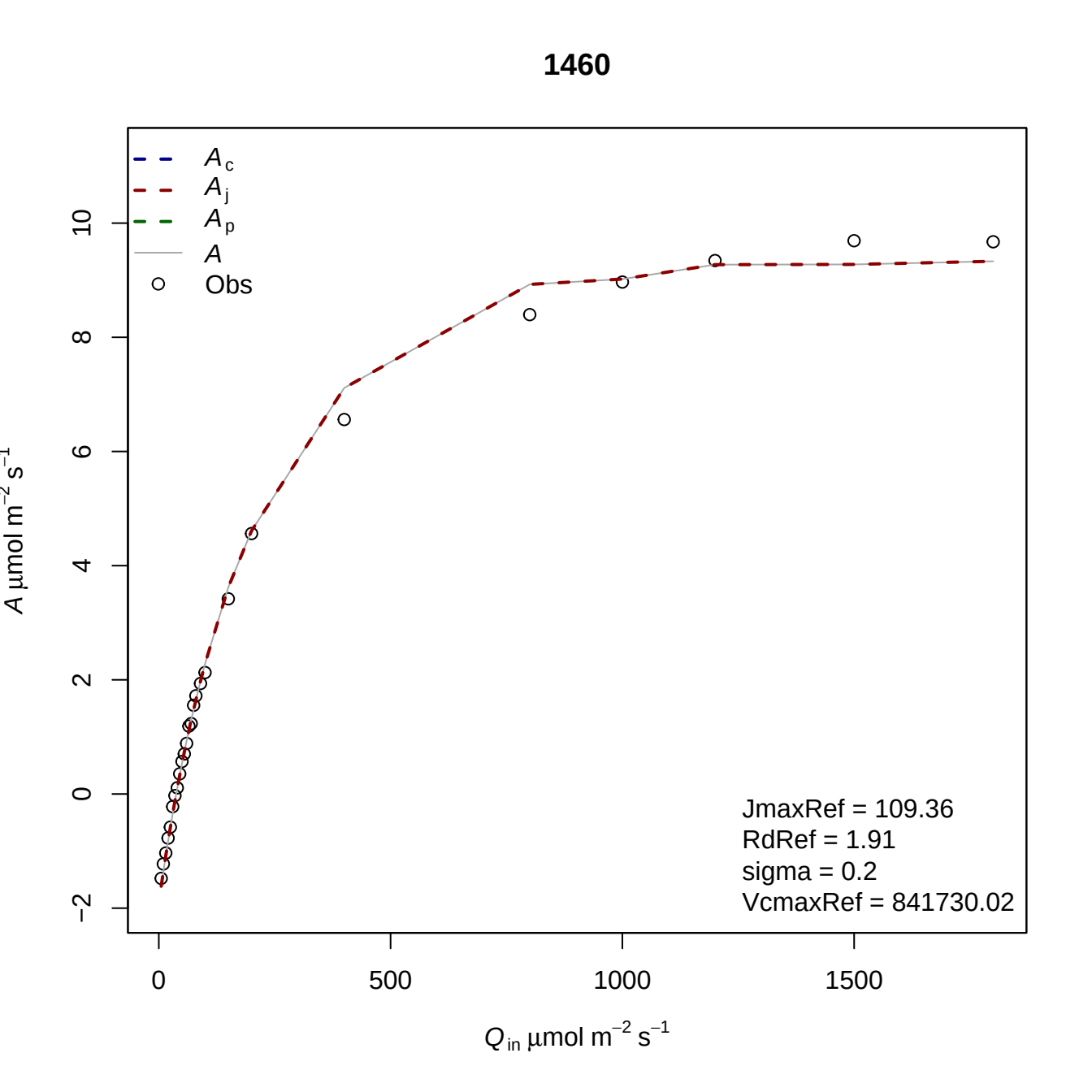

1455
A_c
A_j
A_p
A
Obs
JmaxRef = 210.93
RdRef = 2.46
sigma = 0.38
VcmaxRef = 358370.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

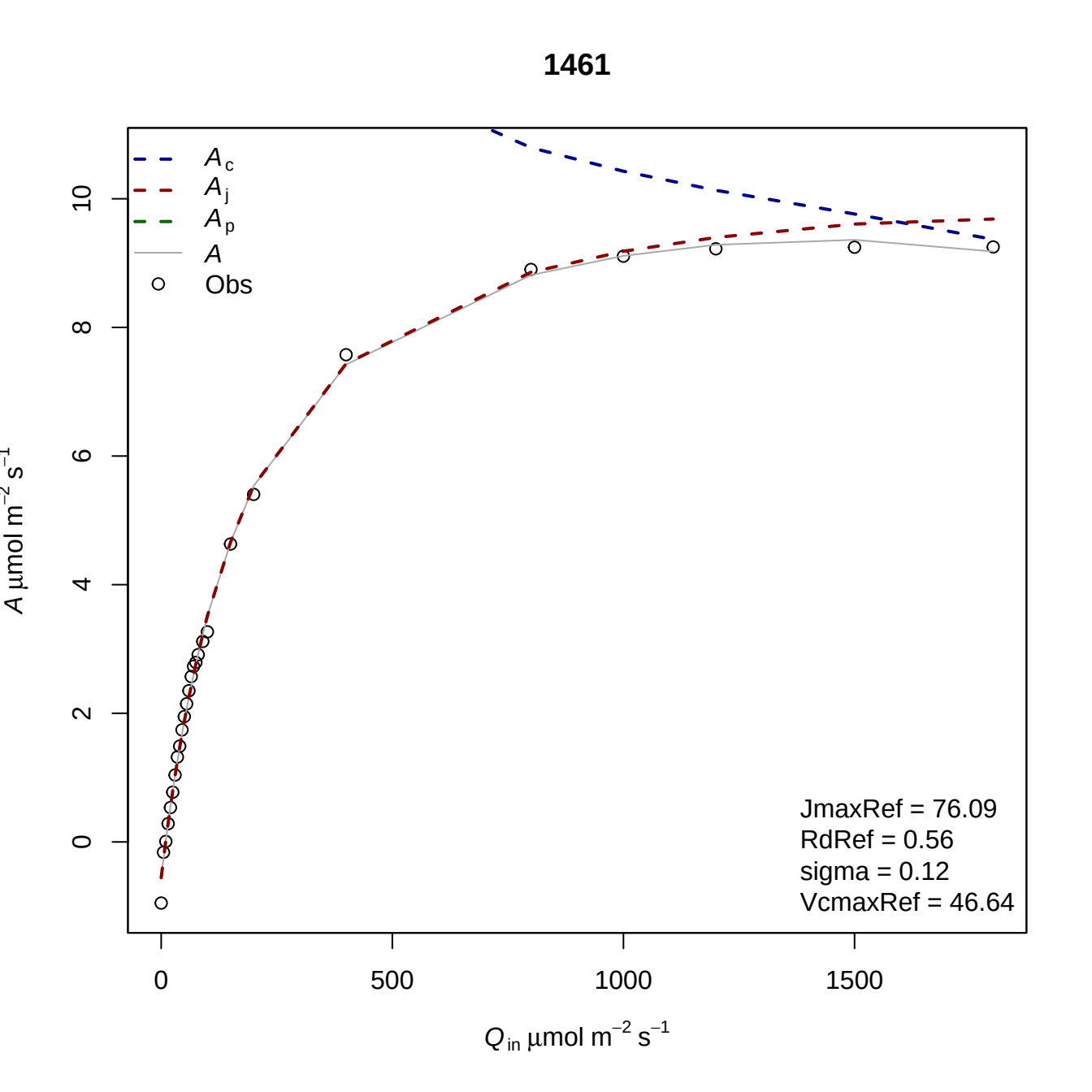

1456
A_c
A_j
A_p
A
Obs
JmaxRef = 117.64
RdRef = 2.71
sigma = 0.63
VcmaxRef = 481048.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

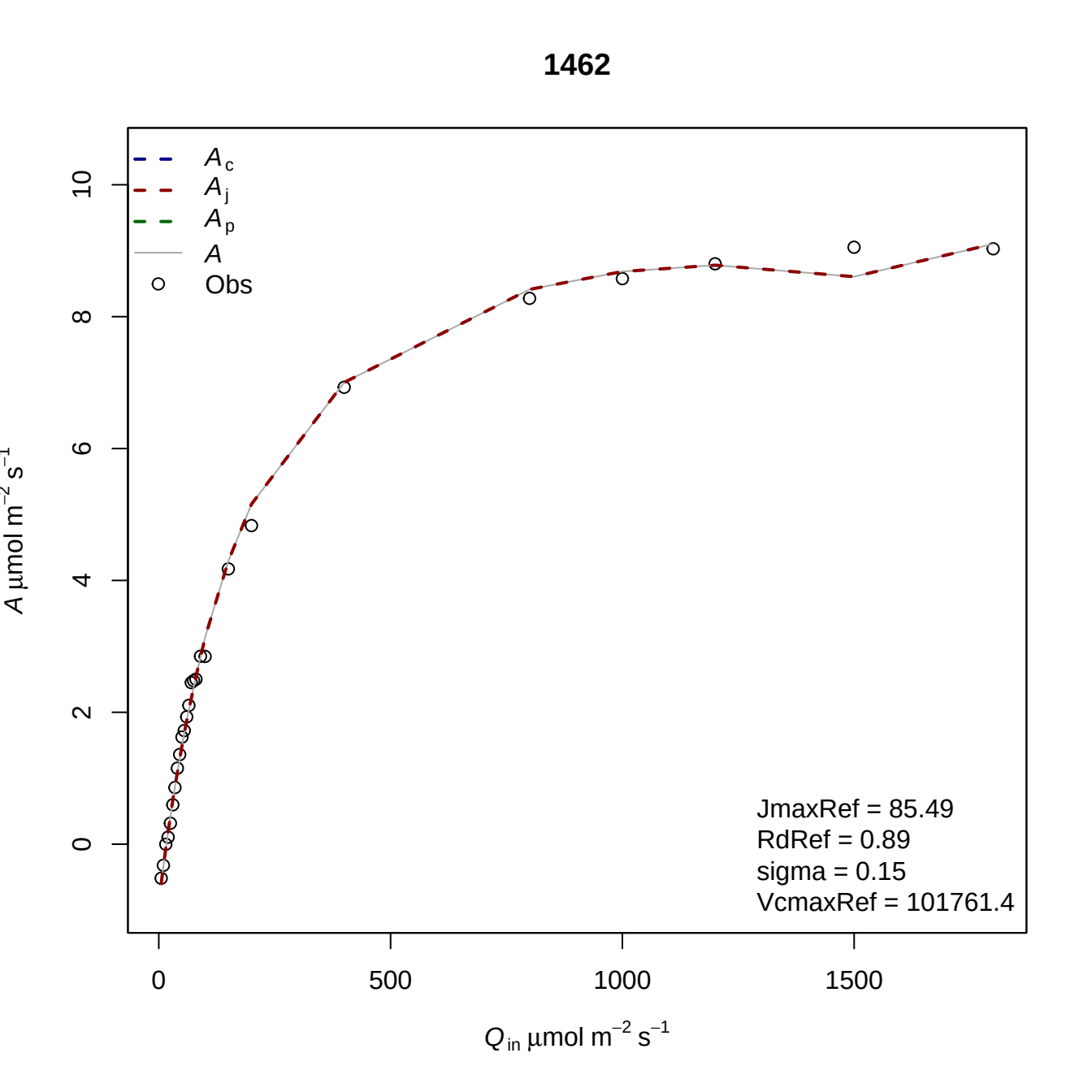

1457
A_c
A_j
A_p
A
Obs
JmaxRef = 129.03
RdRef = 0.86
sigma = 0.19
VcmaxRef = 80.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000
1200

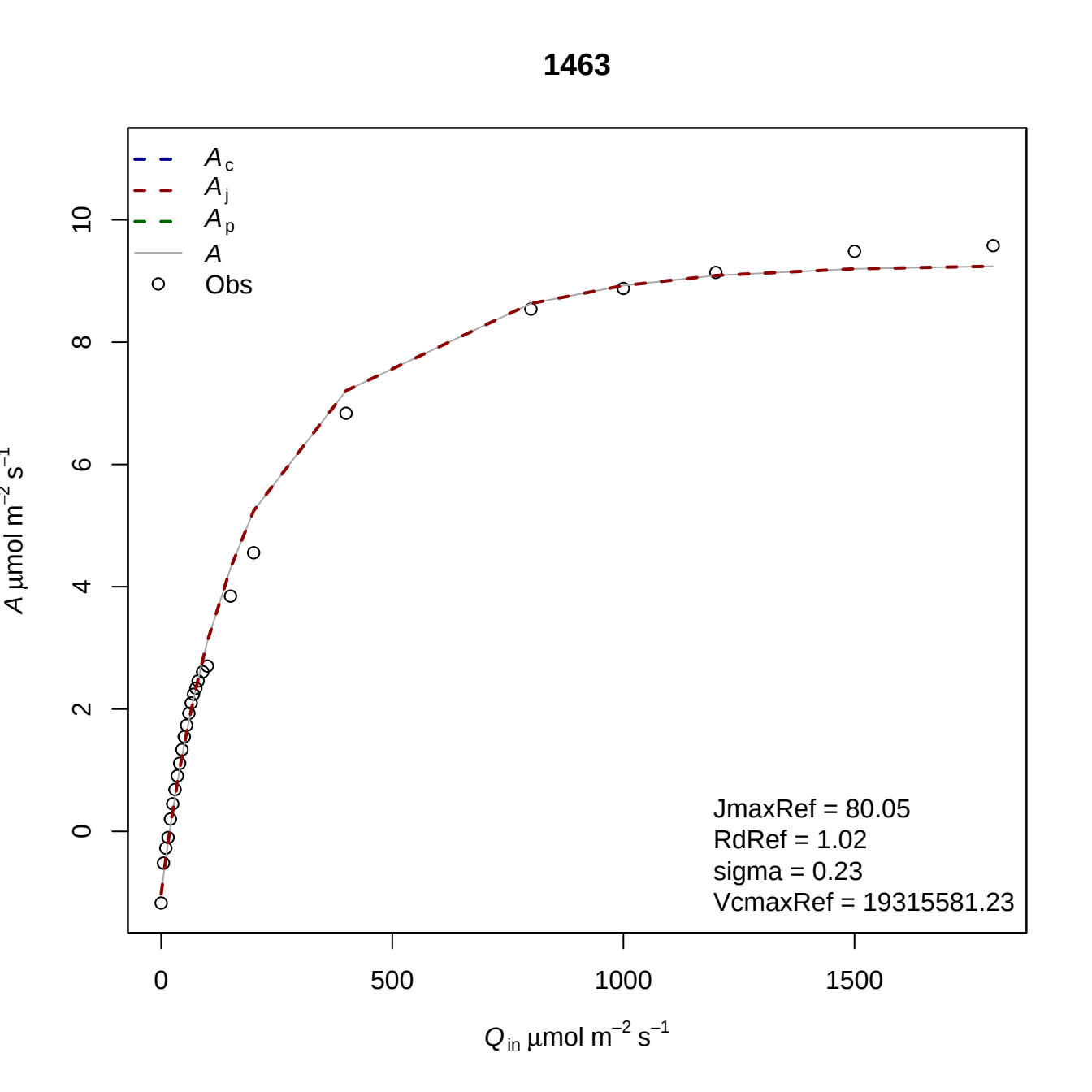

1458
A_c
A_j
A_p
A
Obs
JmaxRef = 95.71
RdRef = 1
sigma = 0.14
VcmaxRef = 432809.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

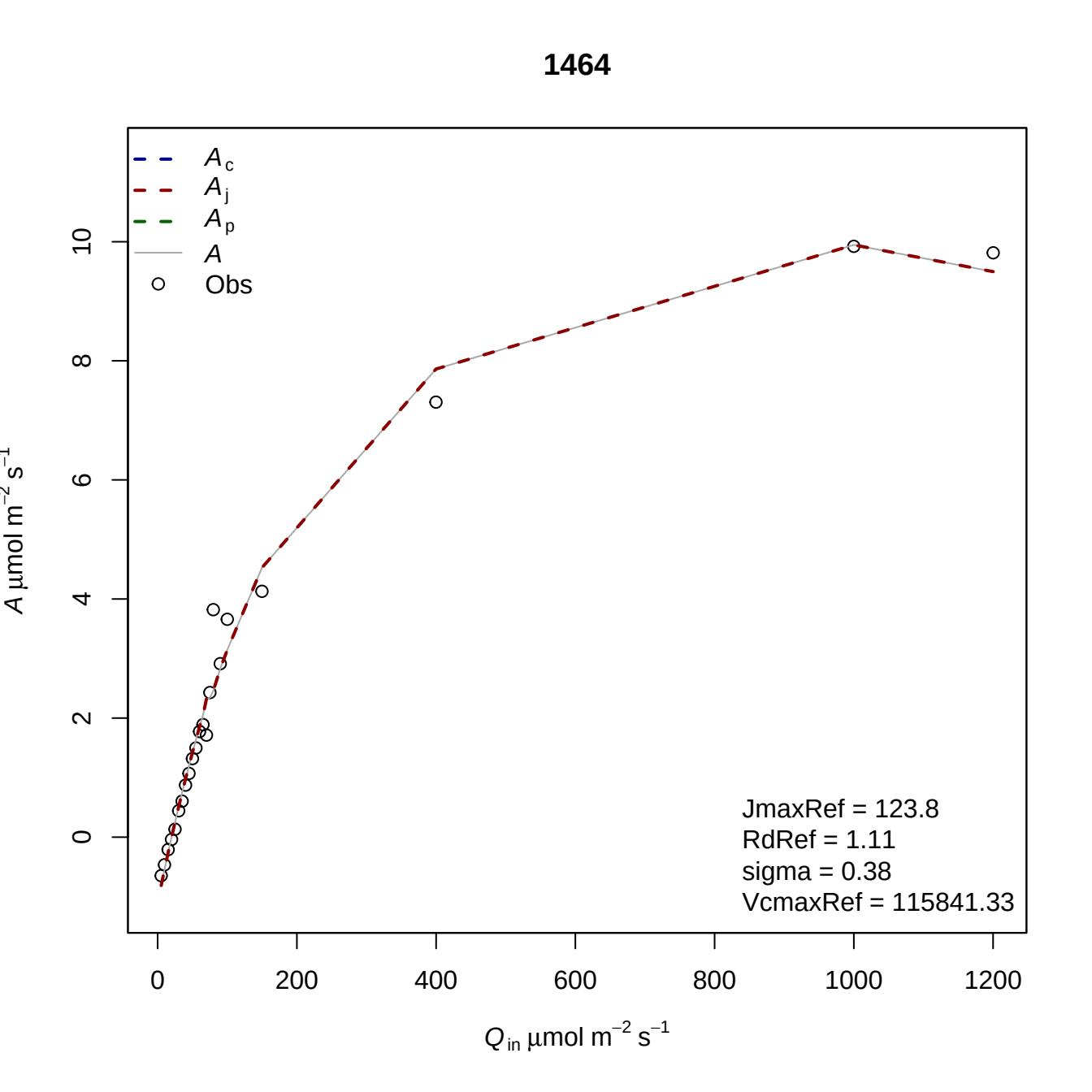

1459
A_c
A_j
A_p
A
Obs
JmaxRef = 125.31
RdRef = 1.63
sigma = 0.17
VcmaxRef = 399236.7
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

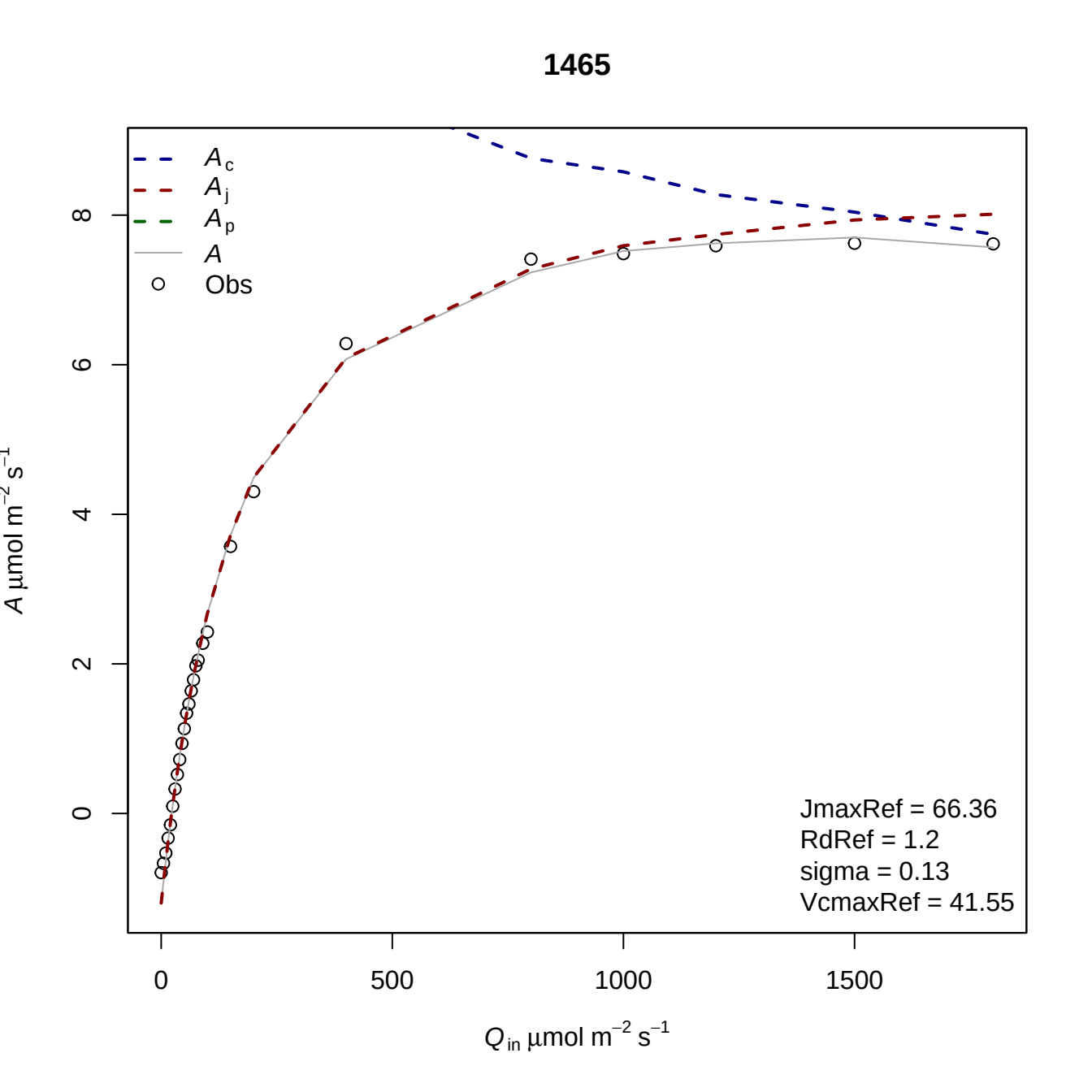

1460
A_c
A_j
A_p
A
Obs
JmaxRef = 109.36
RdRef = 1.91
sigma = 0.2
VcmaxRef = 841730.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

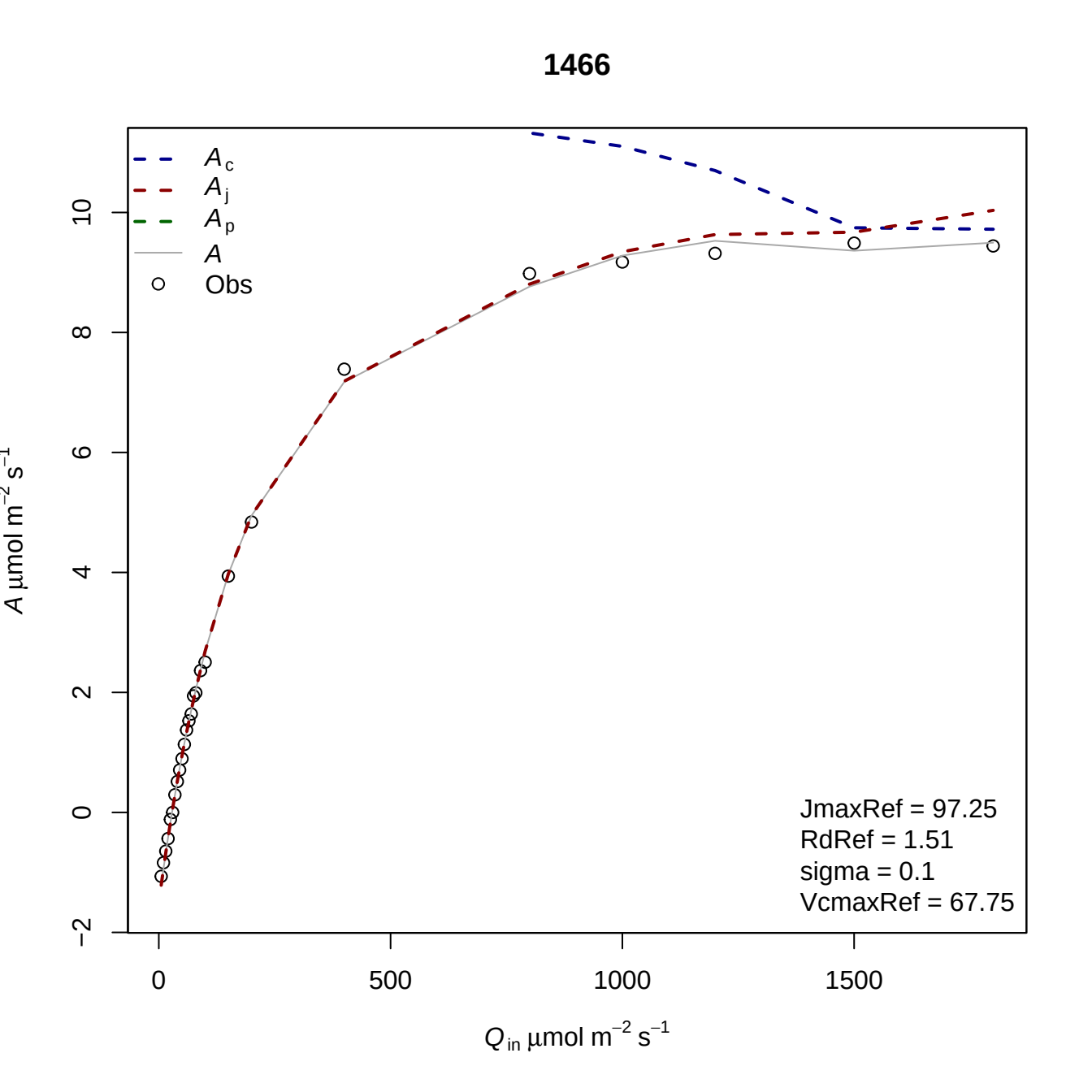

1461
A_c
A_j
A_p
A
Obs
JmaxRef = 76.09
RdRef = 0.56
sigma = 0.12
VcmaxRef = 46.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

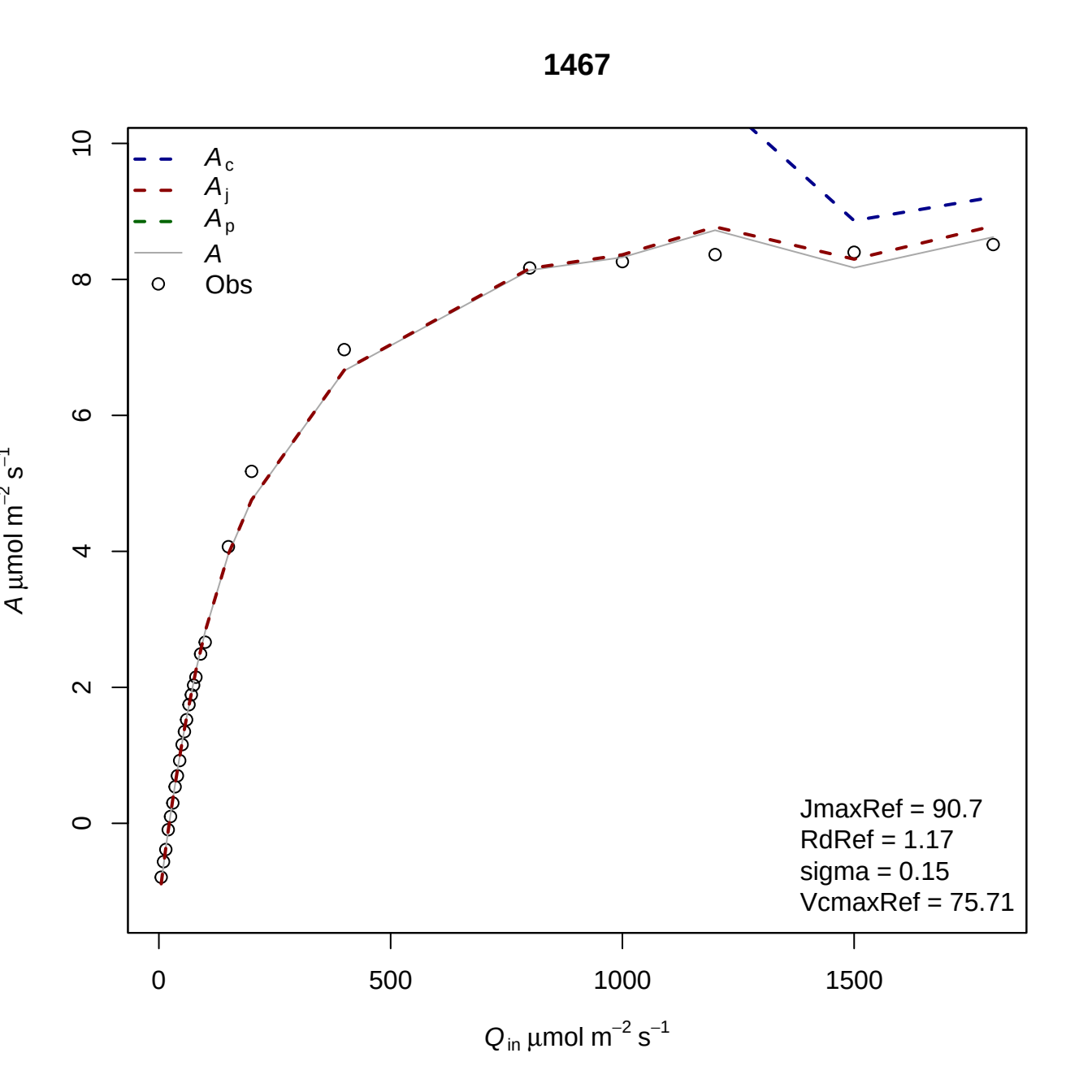

1462
A_c
A_j
A_p
A
Obs
JmaxRef = 85.49
RdRef = 0.89
sigma = 0.15
VcmaxRef = 101761.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

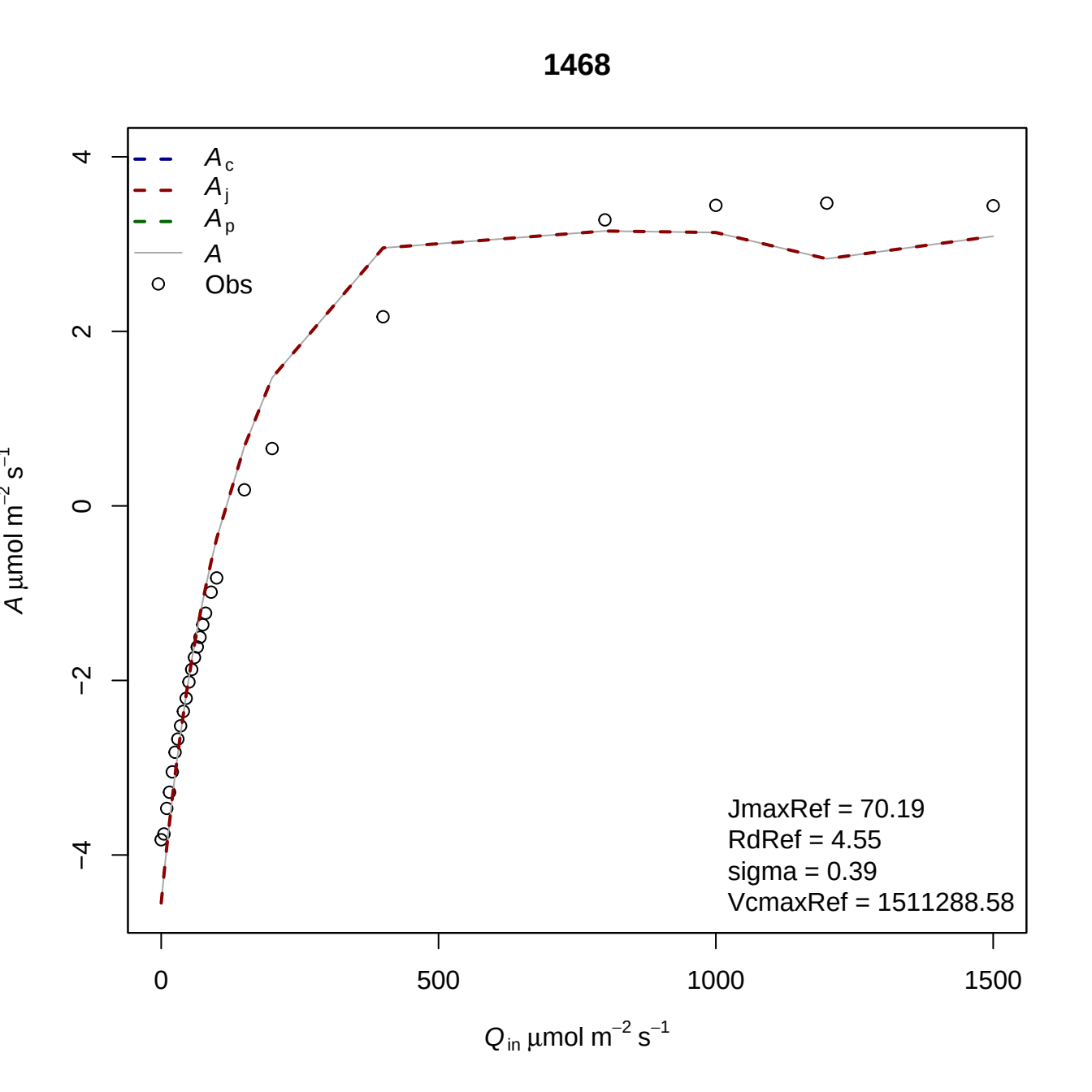

1463
A_c
A_j
A_p
A
Obs
JmaxRef = 80.05
RdRef = 1.02
sigma = 0.23
VcmaxRef = 19315581.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

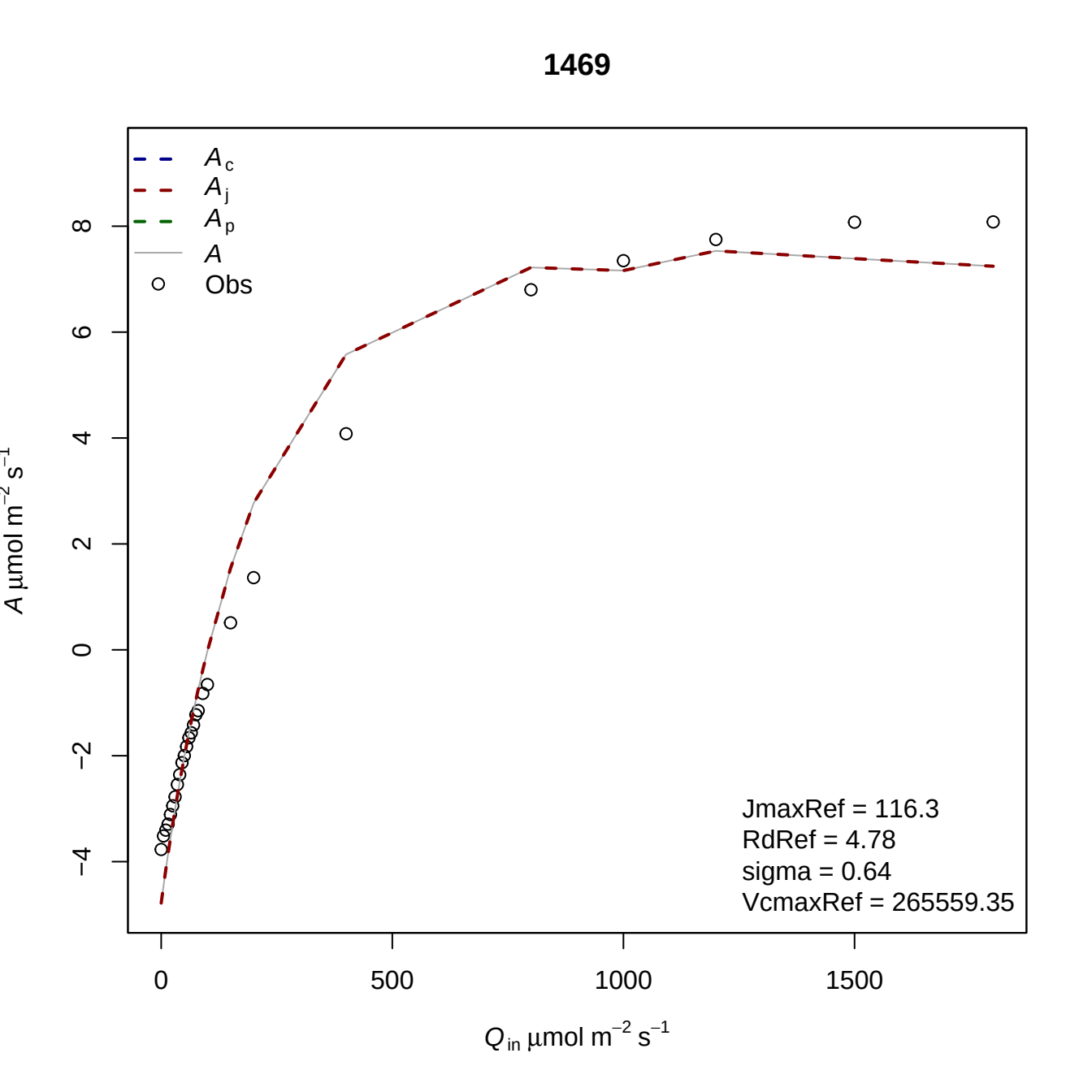

1464
A_c
A_j
A_p
A
Obs
JmaxRef = 123.8
RdRef = 1.11
sigma = 0.38
VcmaxRef = 115841.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
200
400
600
800
1000
1200

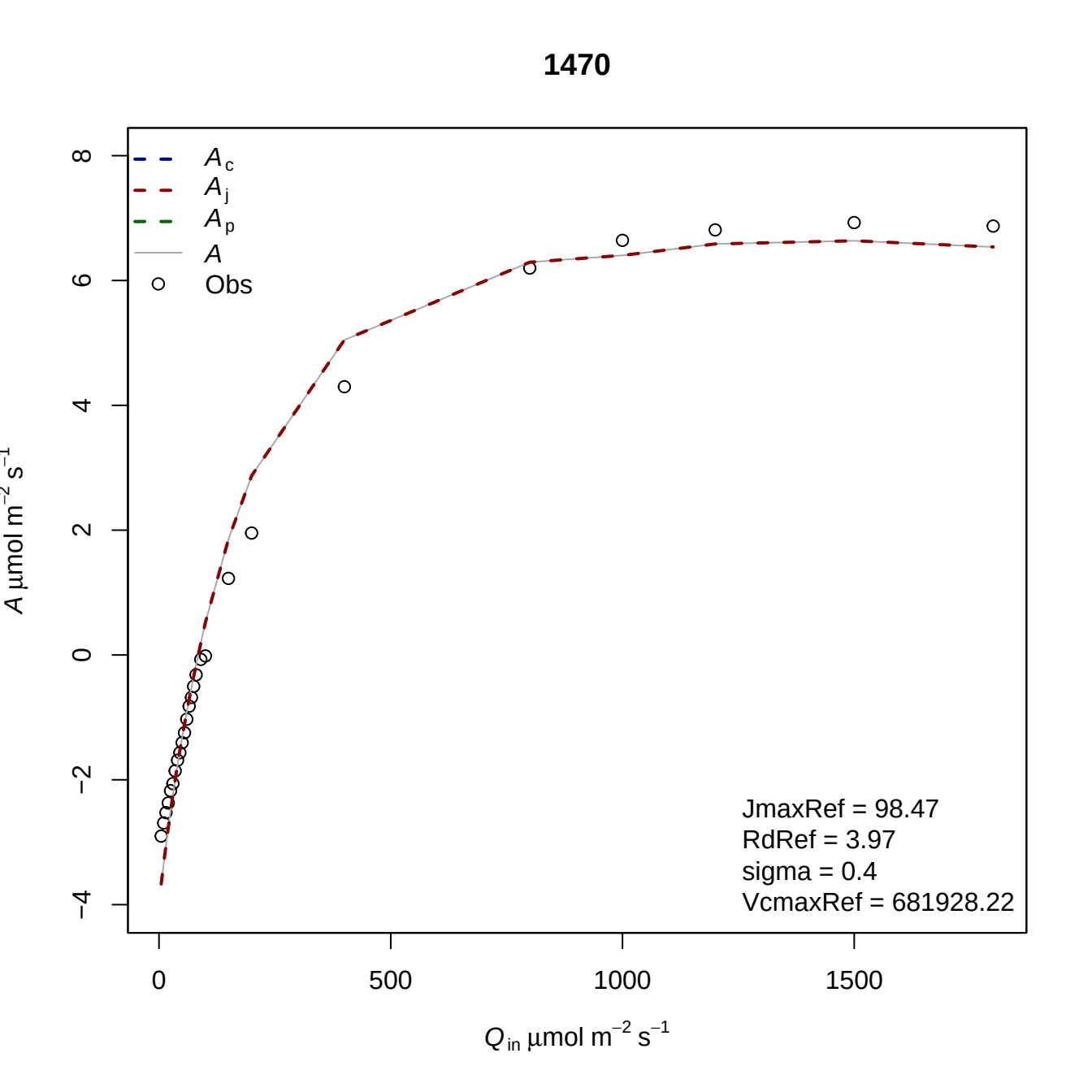

1465
A_c
A_j
A_p
A
Obs
JmaxRef = 66.36
RdRef = 1.2
sigma = 0.13
VcmaxRef = 41.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

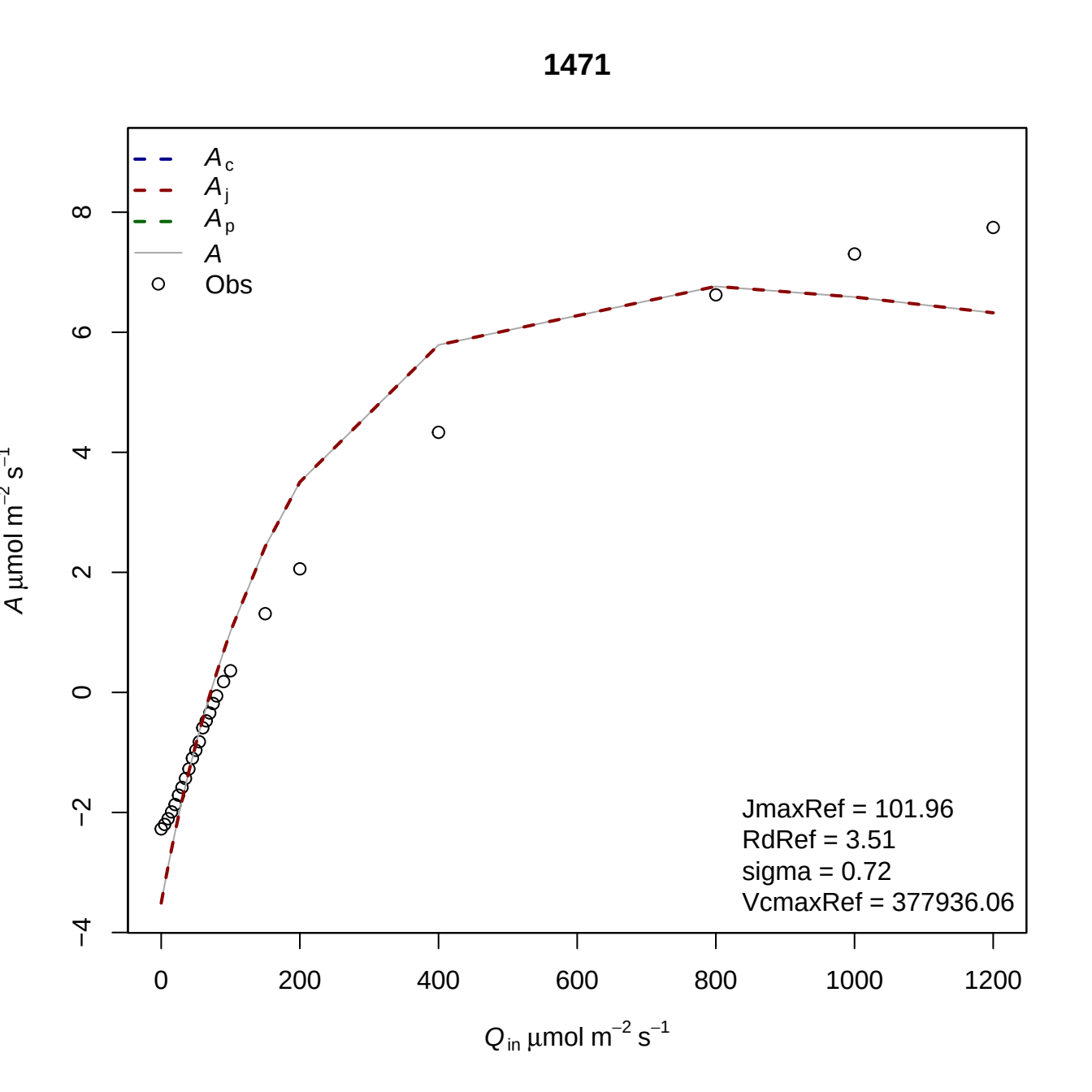

1466
A_c
A_j
A_p
A
Obs
JmaxRef = 97.25
RdRef = 1.51
sigma = 0.1
VcmaxRef = 67.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

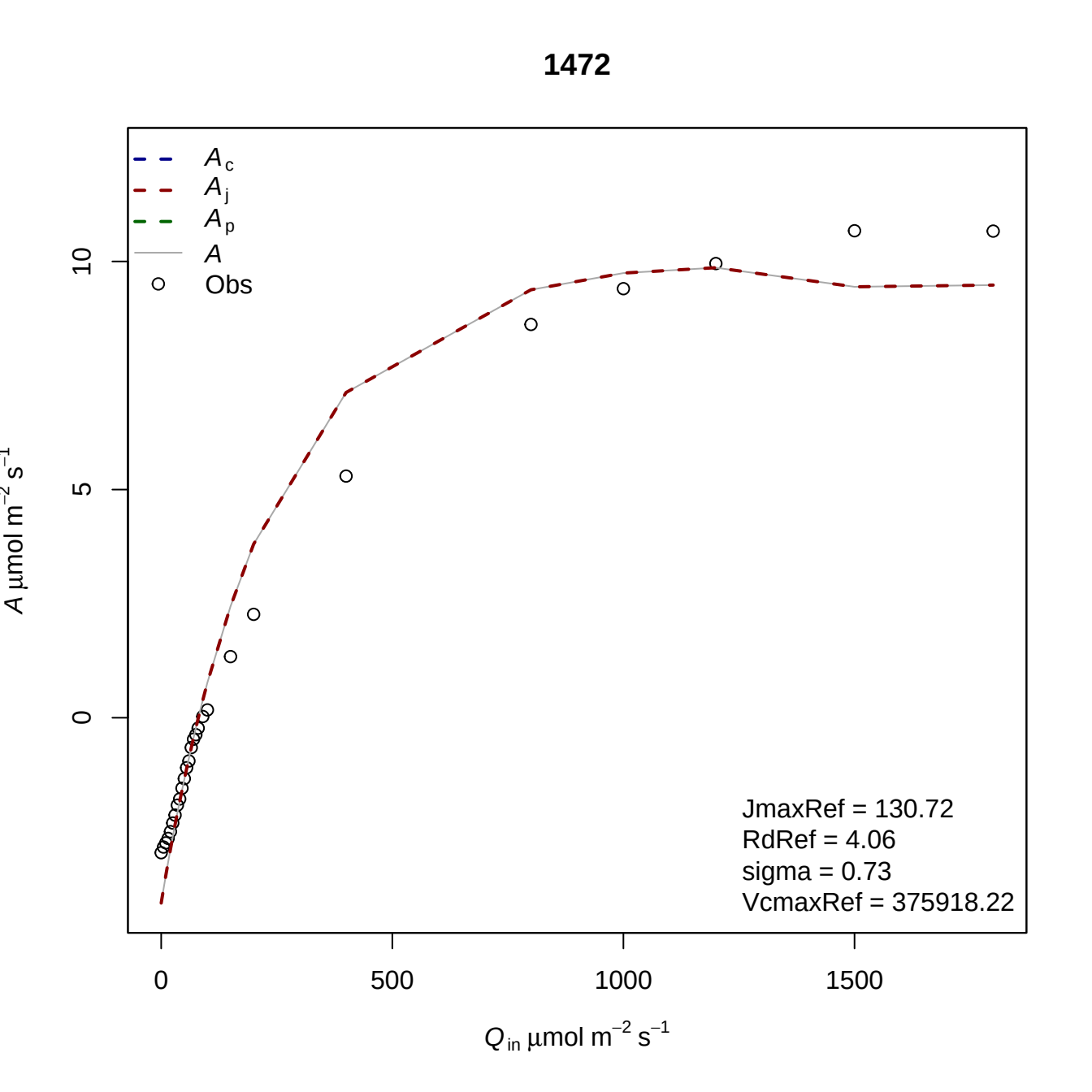

1467
A_c
A_j
A_p
A
Obs
JmaxRef = 90.7
RdRef = 1.17
sigma = 0.15
VcmaxRef = 75.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

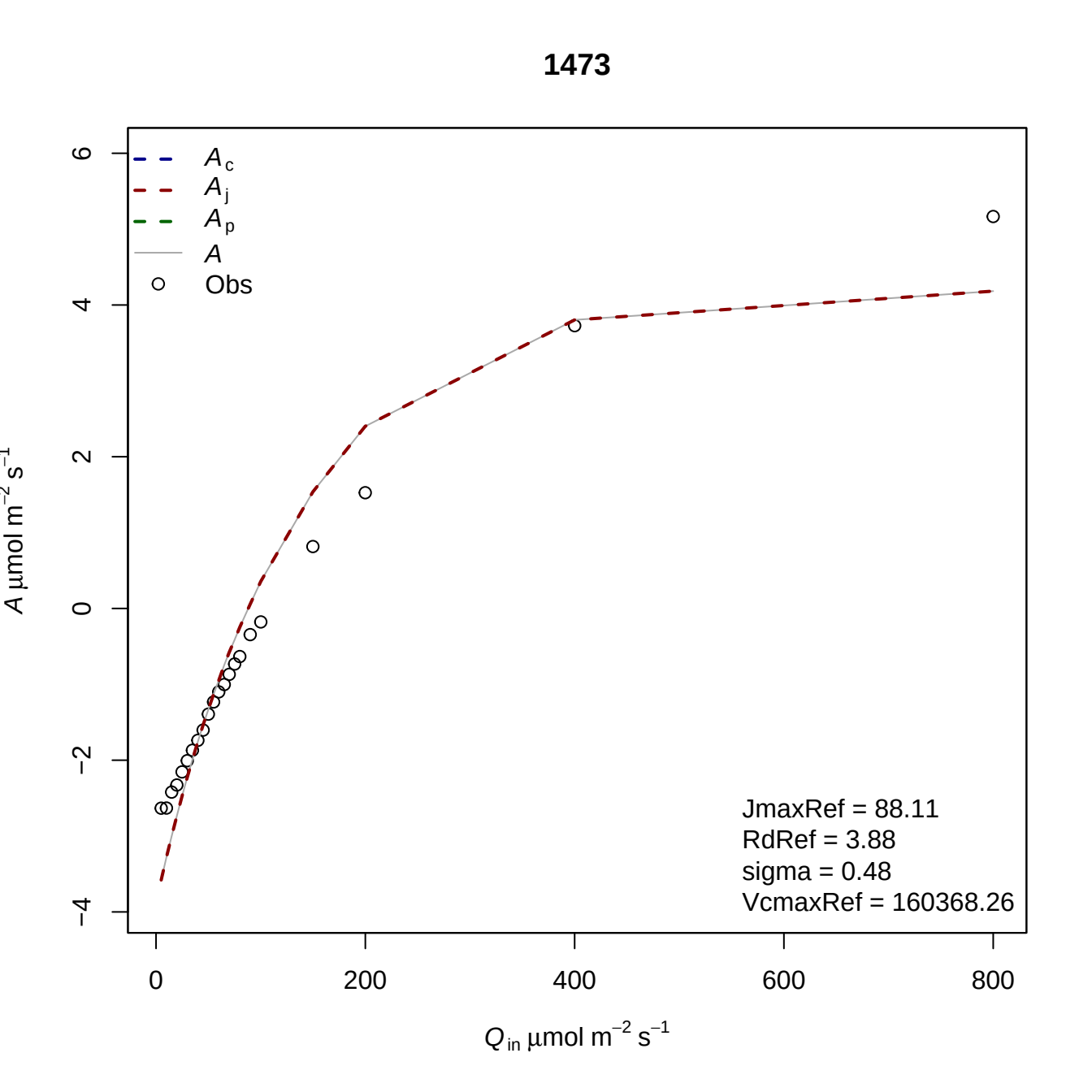

1468
A_c
A_j
A_p
A
Obs
JmaxRef = 70.19
RdRef = 4.55
sigma = 0.39
VcmaxRef = 1511288.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
500
1000
1500

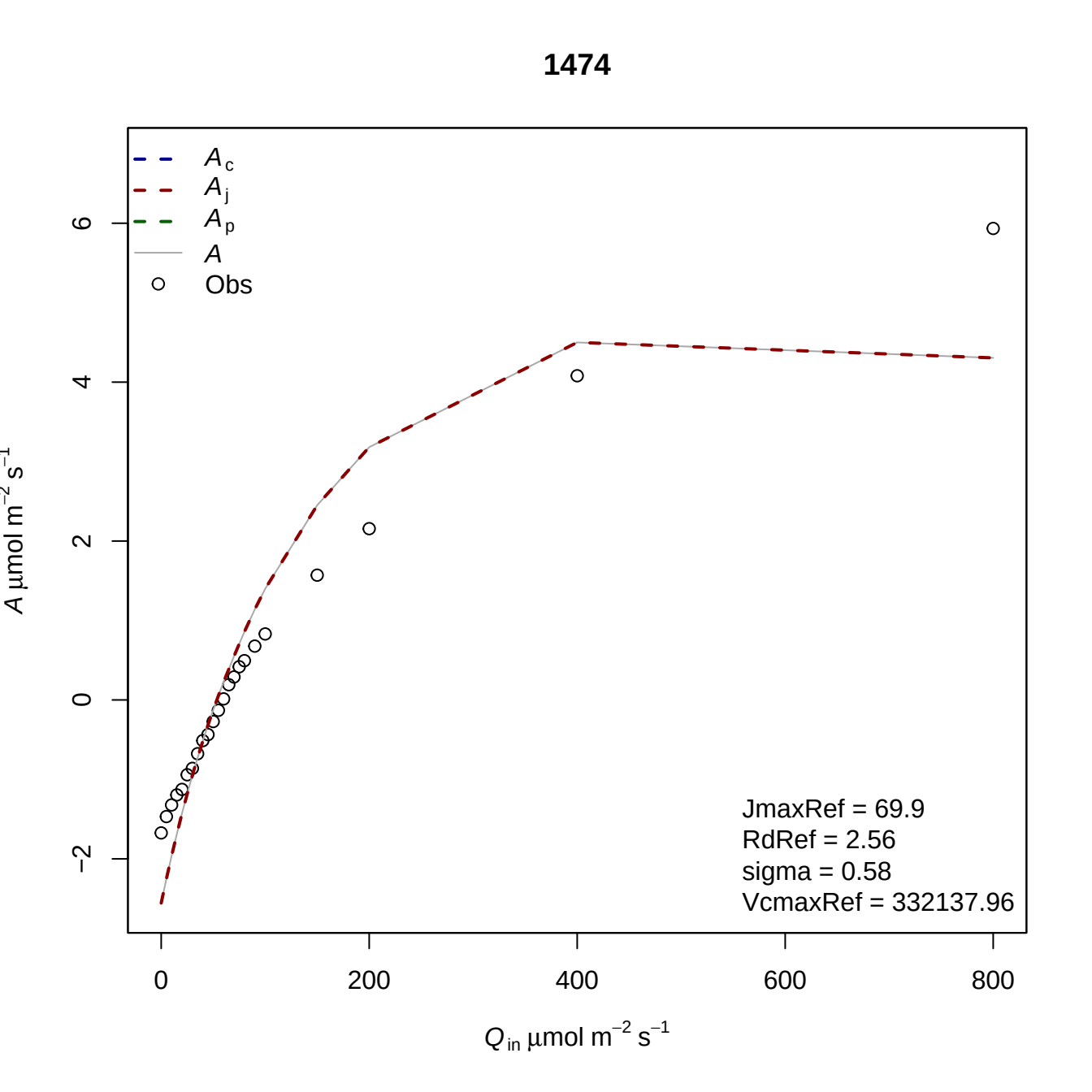

1469
A_c
A_j
A_p
A
Obs
JmaxRef = 116.3
RdRef = 4.78
sigma = 0.64
VcmaxRef = 265559.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

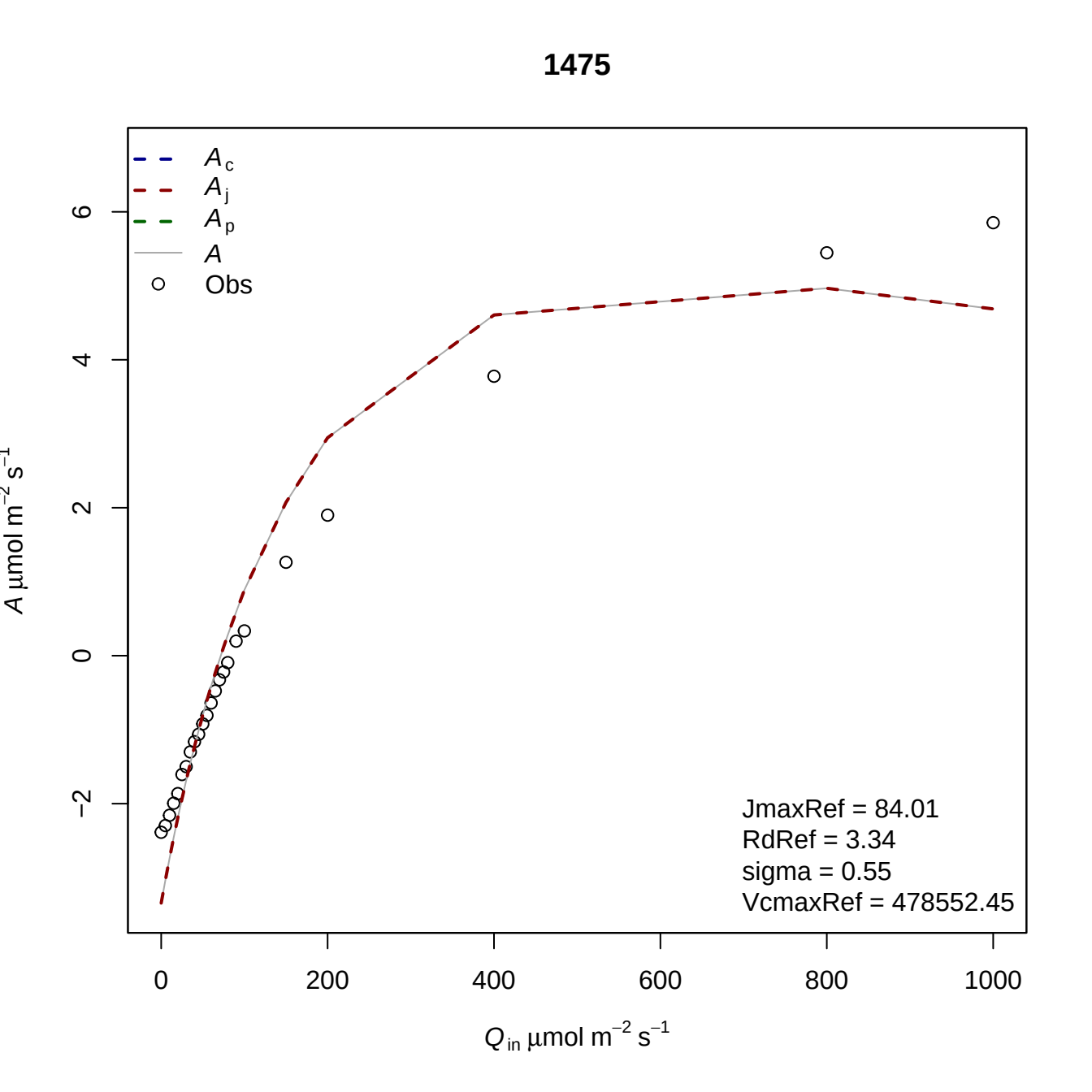

1470
A_c
A_j
A_p
A
Obs
JmaxRef = 98.47
RdRef = 3.97
sigma = 0.4
VcmaxRef = 681928.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

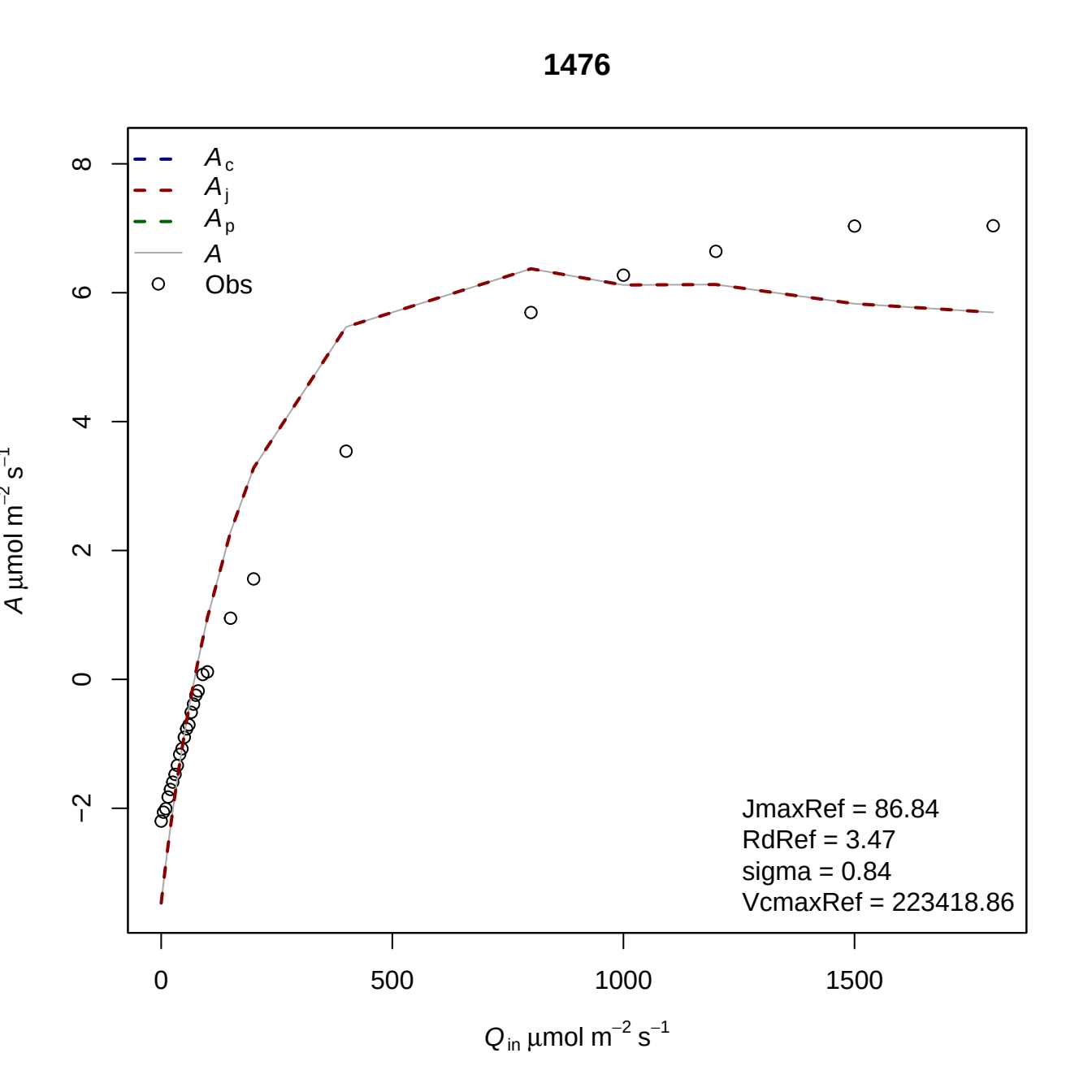

1471
A_c
A_j
A_p
A
Obs
JmaxRef = 101.96
RdRef = 3.51
sigma = 0.72
VcmaxRef = 377936.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

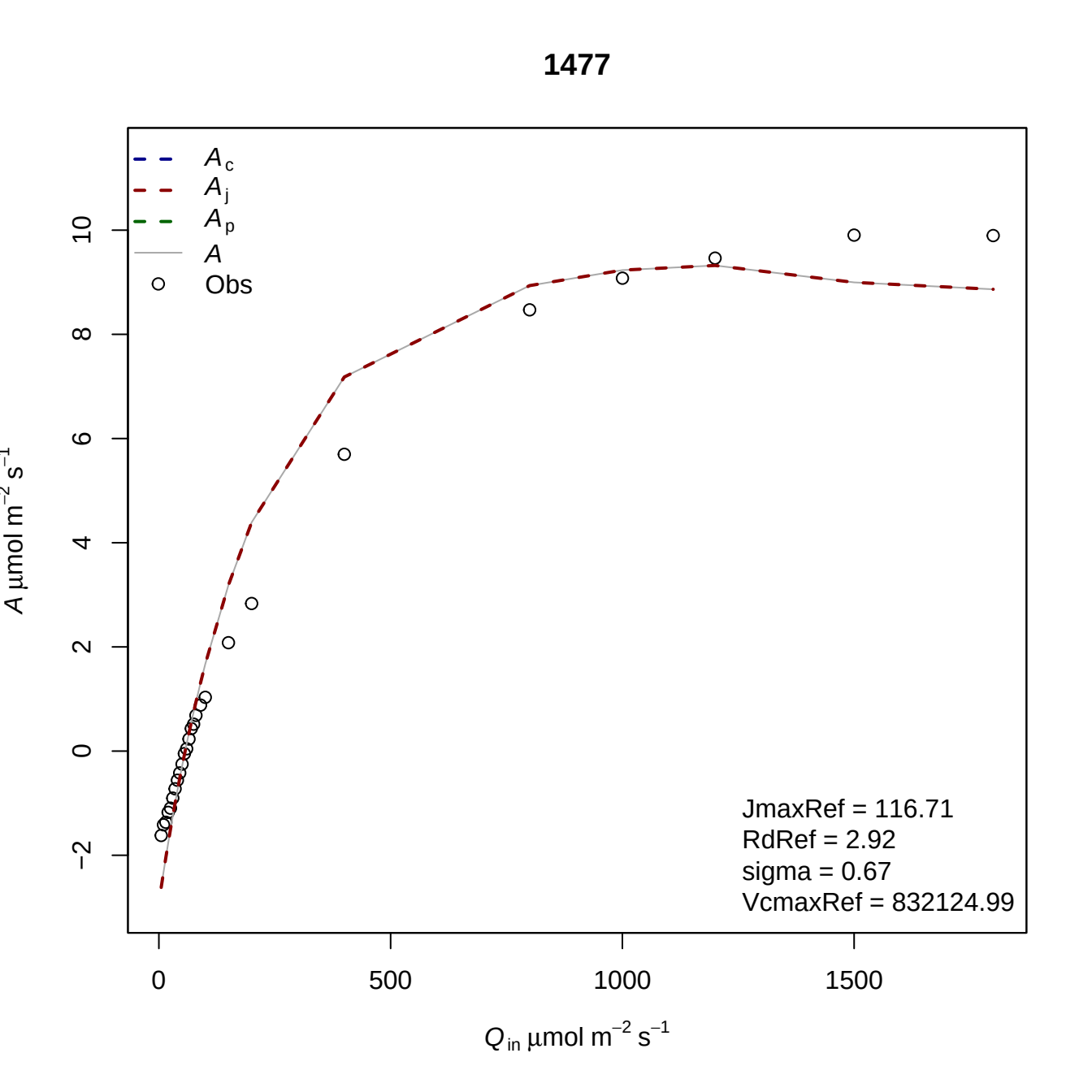

1472
A_c
A_j
A_p
A
Obs
JmaxRef = 130.72
RdRef = 4.06
sigma = 0.73
VcmaxRef = 375918.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

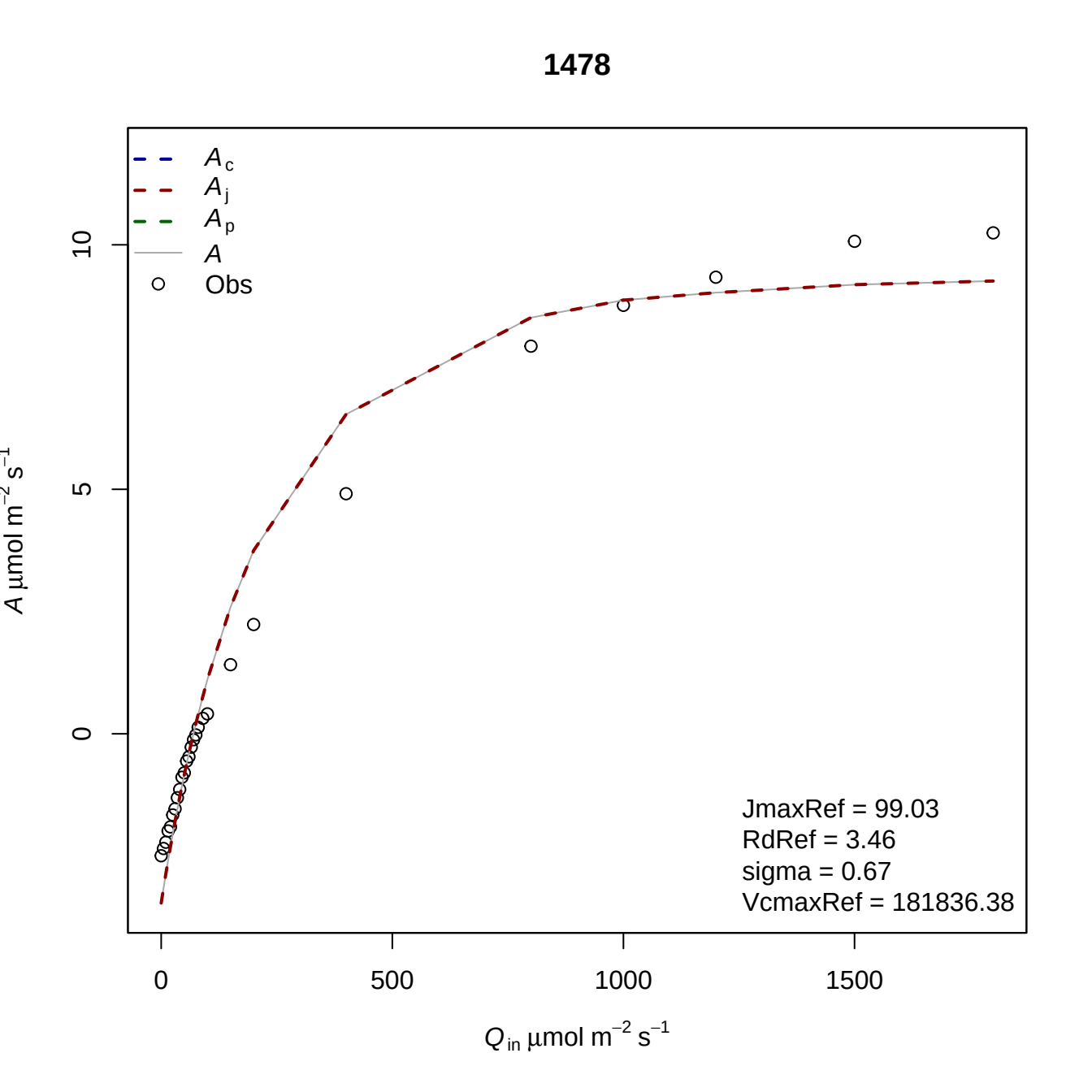

1473
A_c
A_j
A_p
A
Obs
JmaxRef = 88.11
RdRef = 3.88
sigma = 0.48
VcmaxRef = 160368.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
200
400
600
800

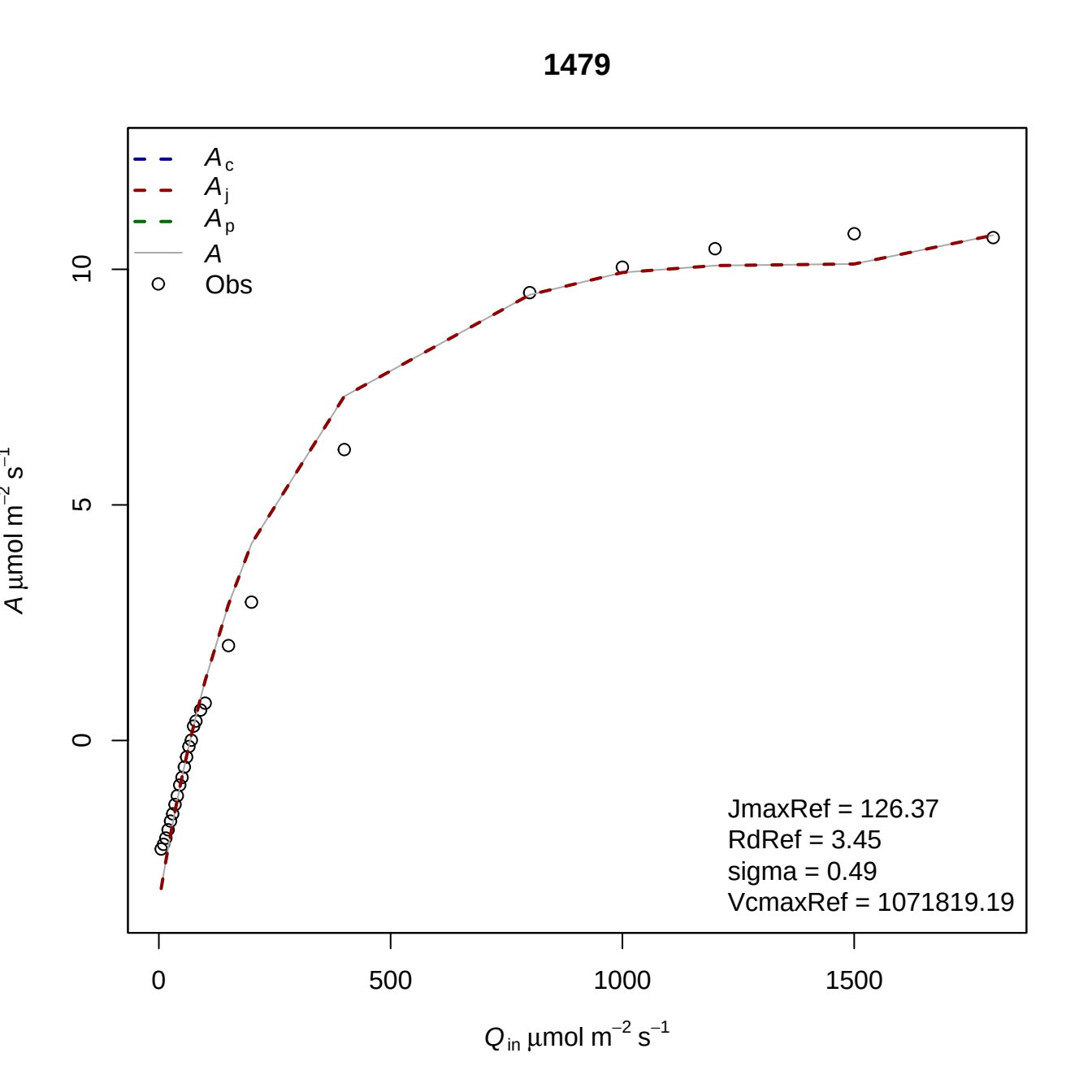

1474
A_c
A_j
A_p
A
Obs
JmaxRef = 69.9
RdRef = 2.56
sigma = 0.58
VcmaxRef = 332137.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800

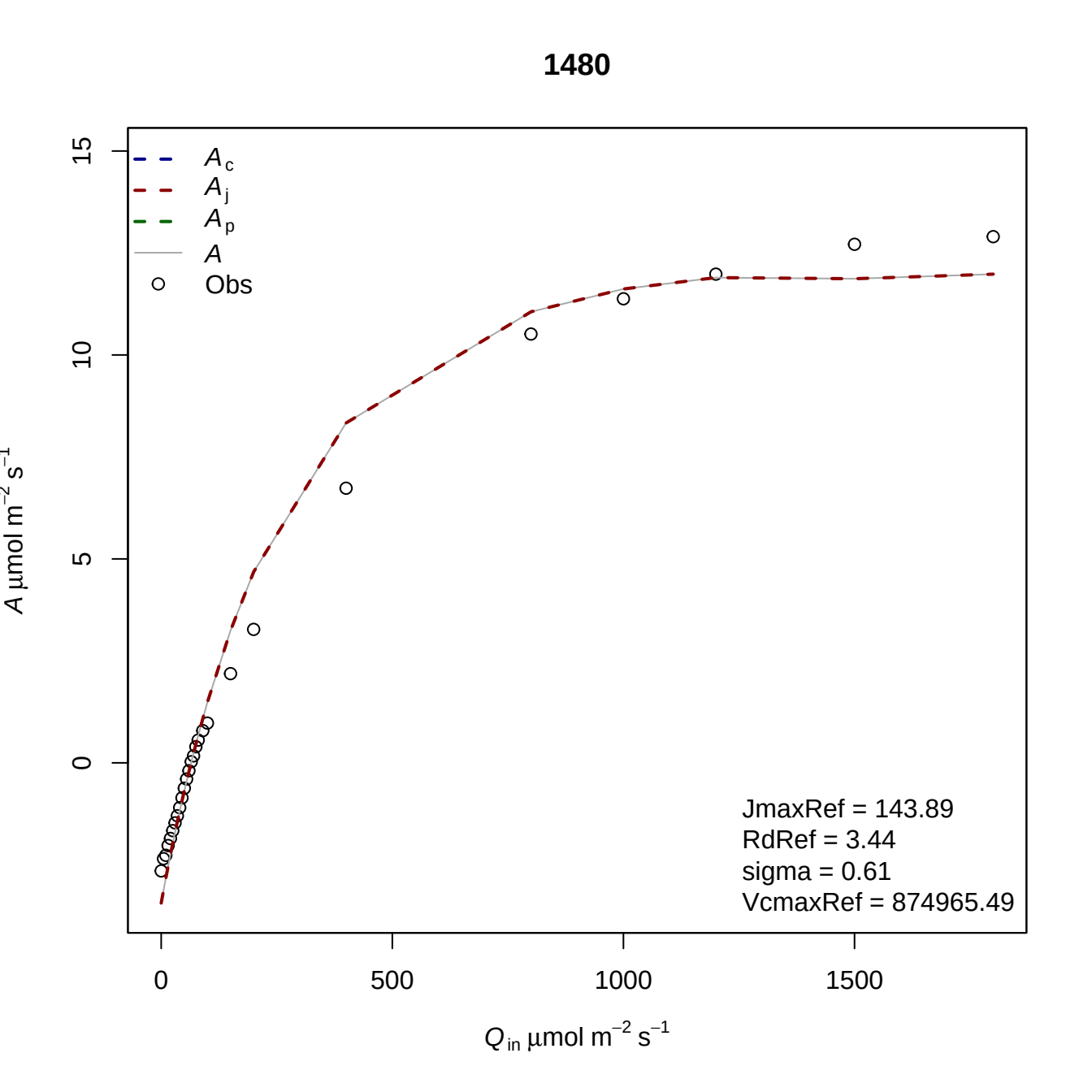

1475
A_c
A_j
A_p
A
Obs
JmaxRef = 84.01
RdRef = 3.34
sigma = 0.55
VcmaxRef = 478552.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000

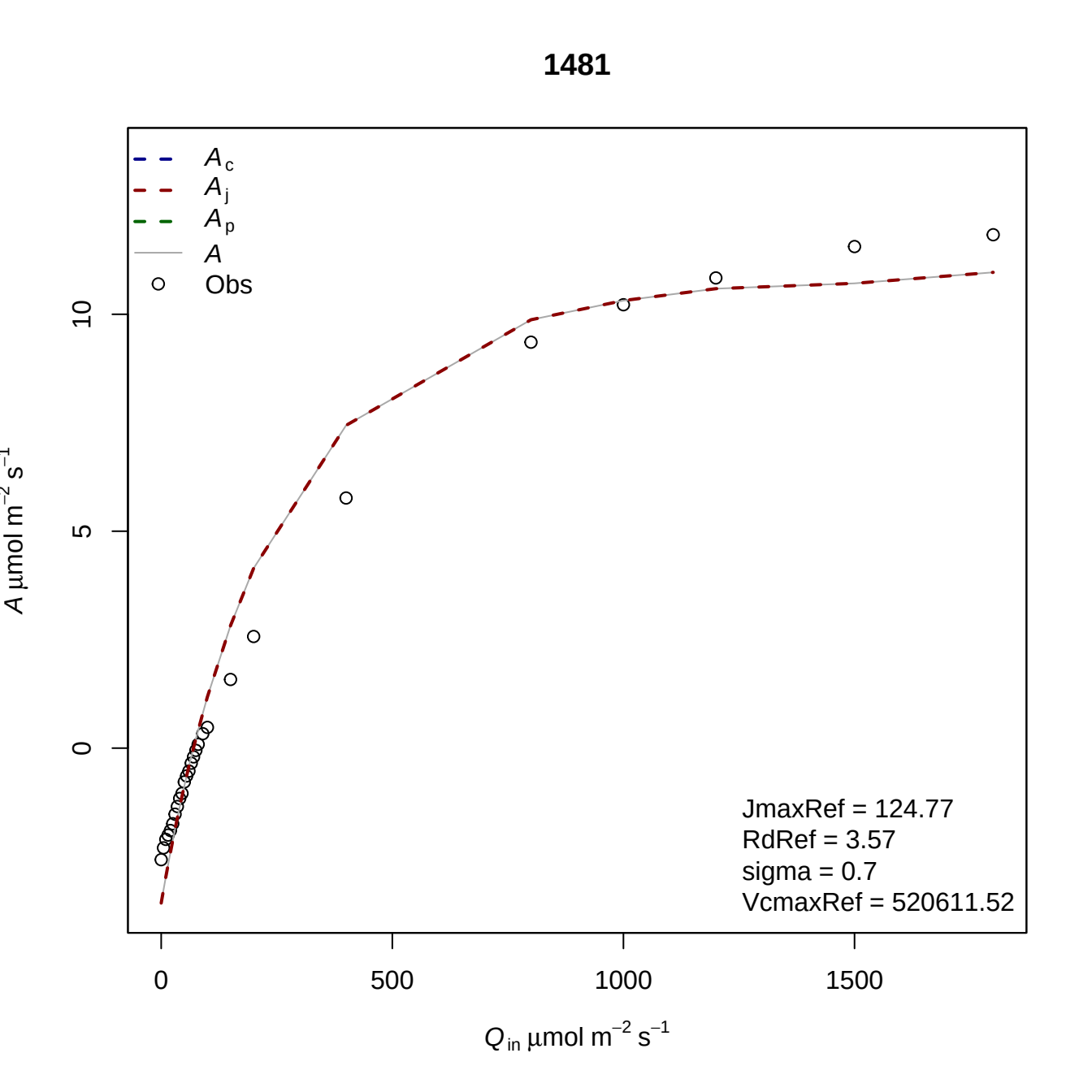

1476
A_c
A_j
A_p
A
Obs
JmaxRef = 86.84
RdRef = 3.47
sigma = 0.84
VcmaxRef = 223418.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

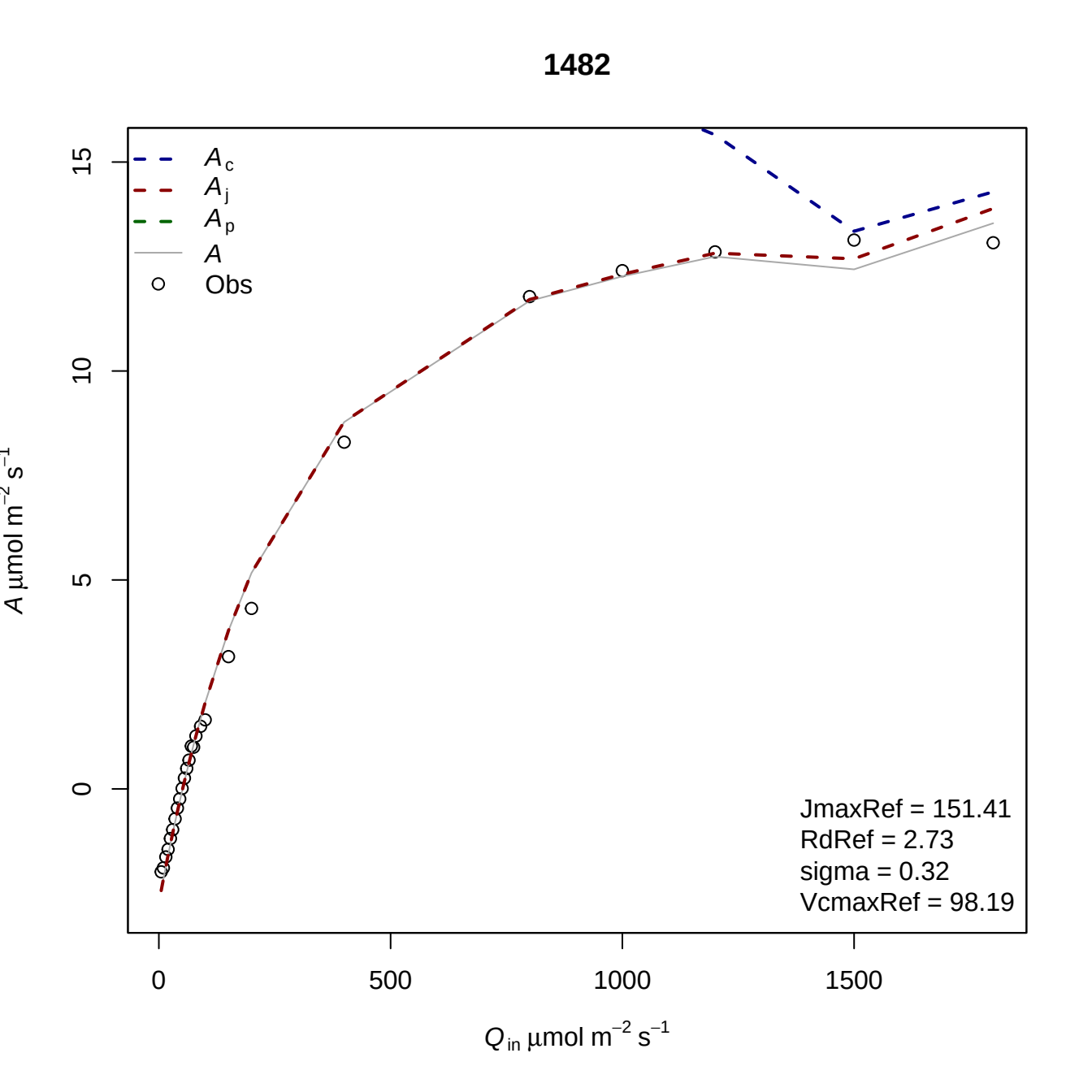

1477
A_c
A_j
A_p
A
Obs
JmaxRef = 116.71
RdRef = 2.92
sigma = 0.67
VcmaxRef = 832124.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

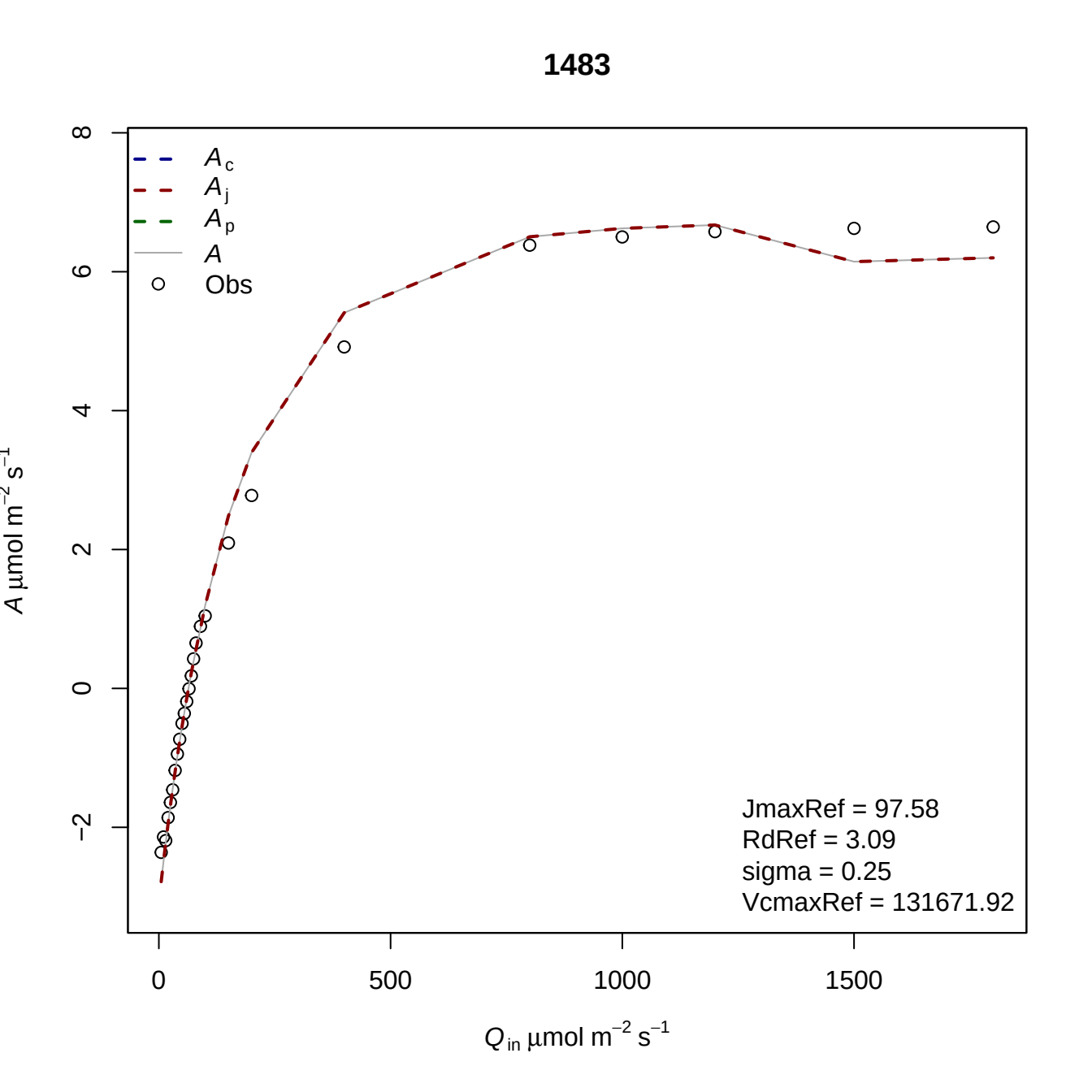

1478
A_c
A_j
A_p
A
Obs
JmaxRef = 99.03
RdRef = 3.46
sigma = 0.67
VcmaxRef = 181836.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

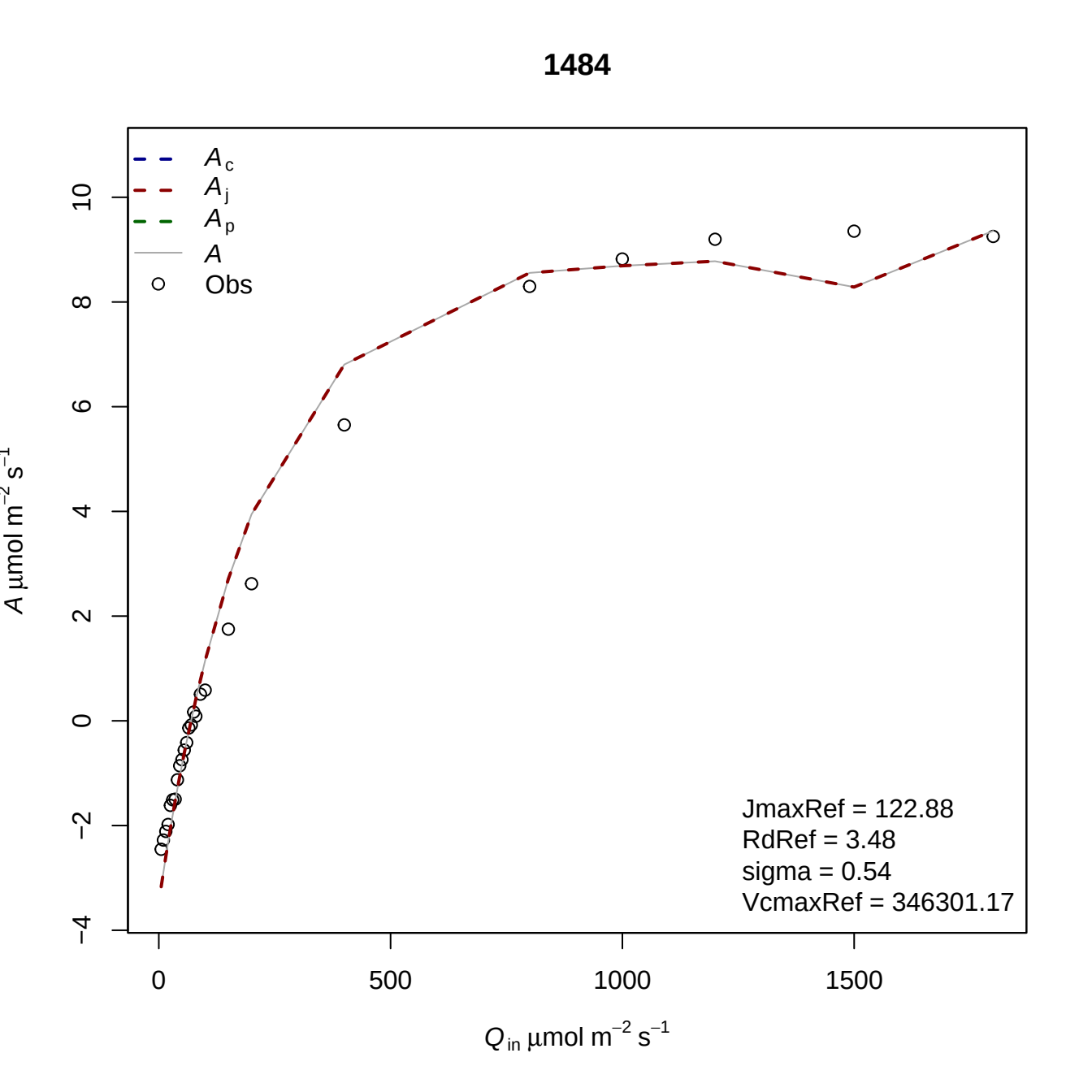

1479
A_c
A_j
A_p
A
Obs
JmaxRef = 126.37
RdRef = 3.45
sigma = 0.49
VcmaxRef = 1071819.19
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

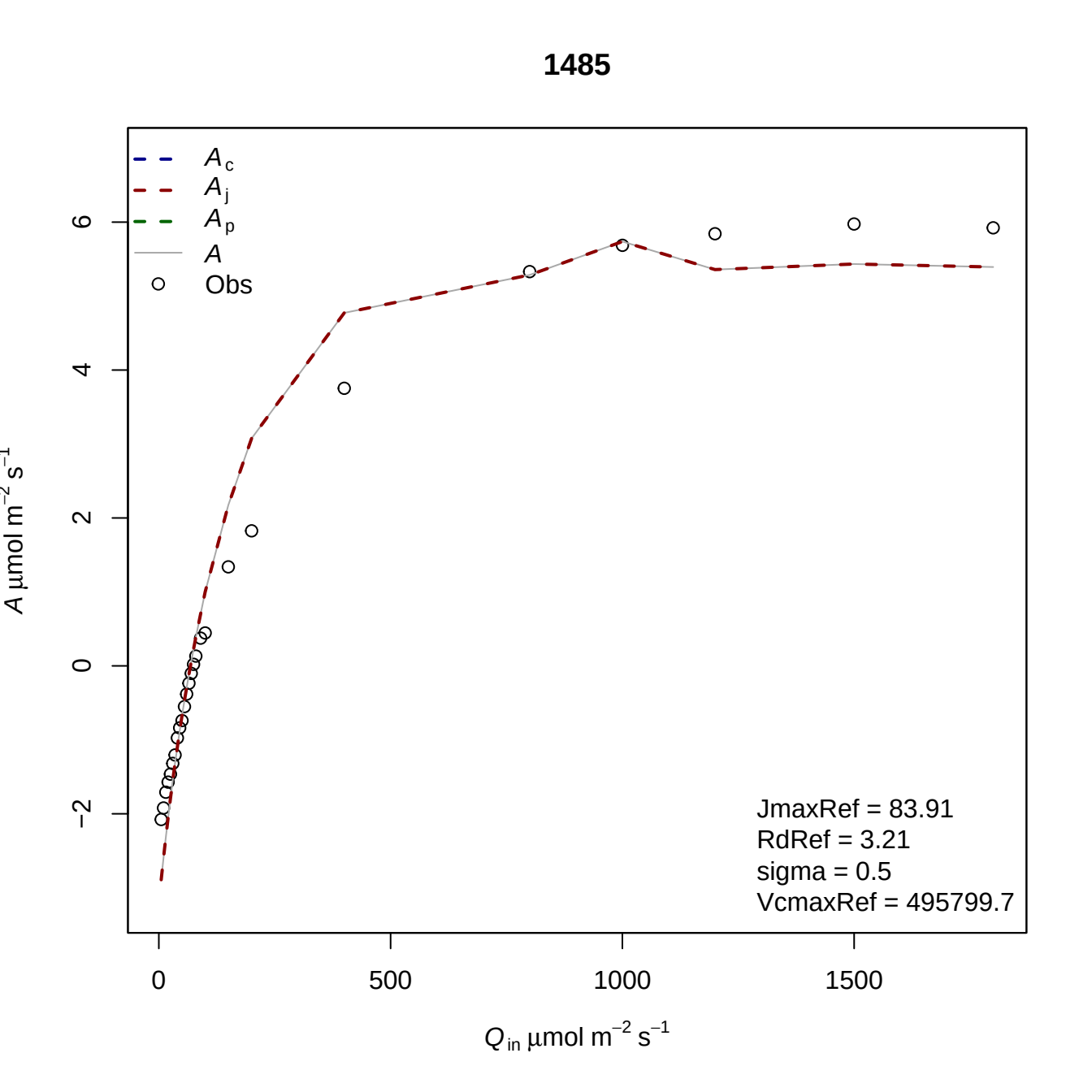

1480
A_c
A_j
A_p
A
Obs
JmaxRef = 143.89
RdRef = 3.44
sigma = 0.61
VcmaxRef = 874965.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

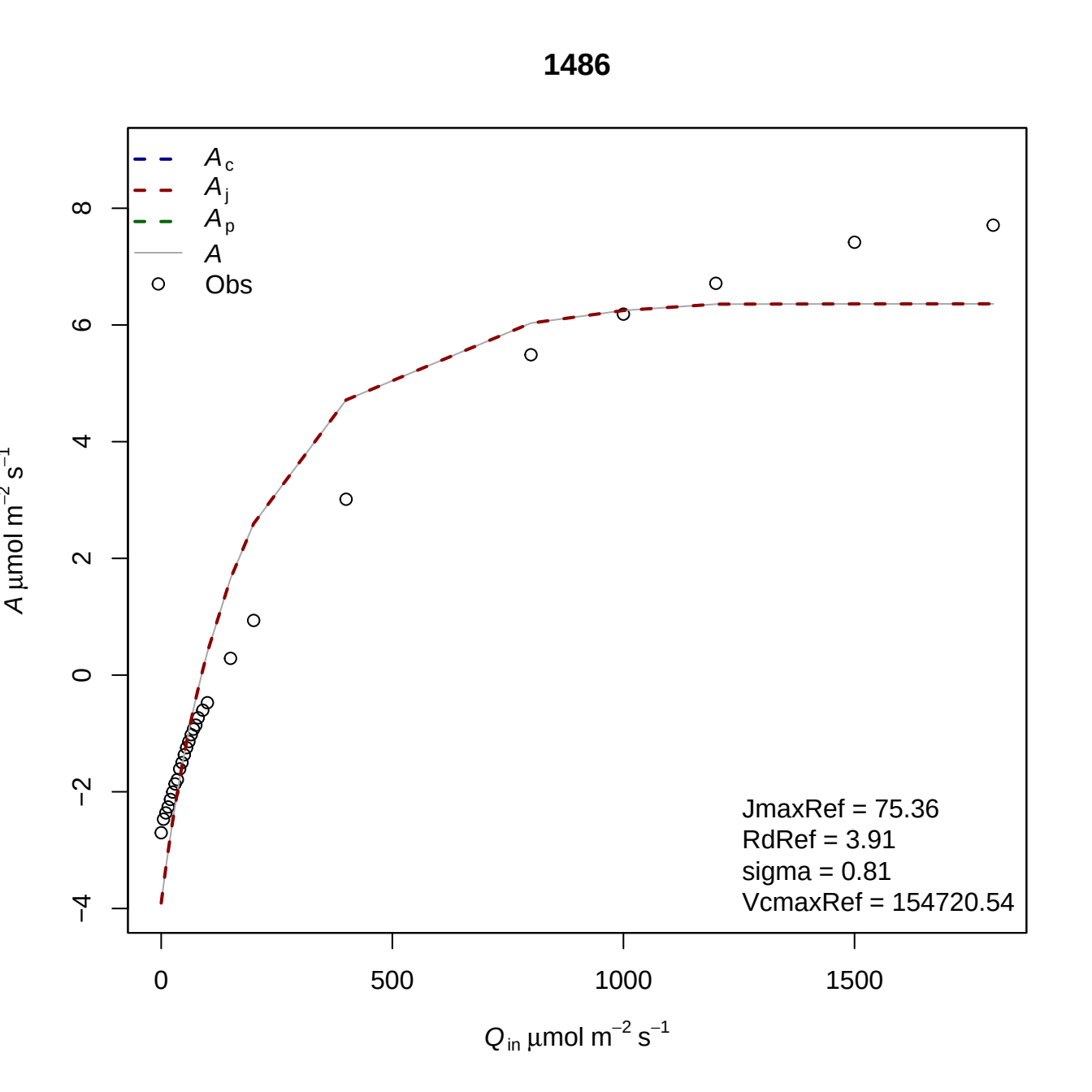

1481
A_c
A_j
A_p
A
Obs
JmaxRef = 124.77
RdRef = 3.57
sigma = 0.7
VcmaxRef = 520611.52
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

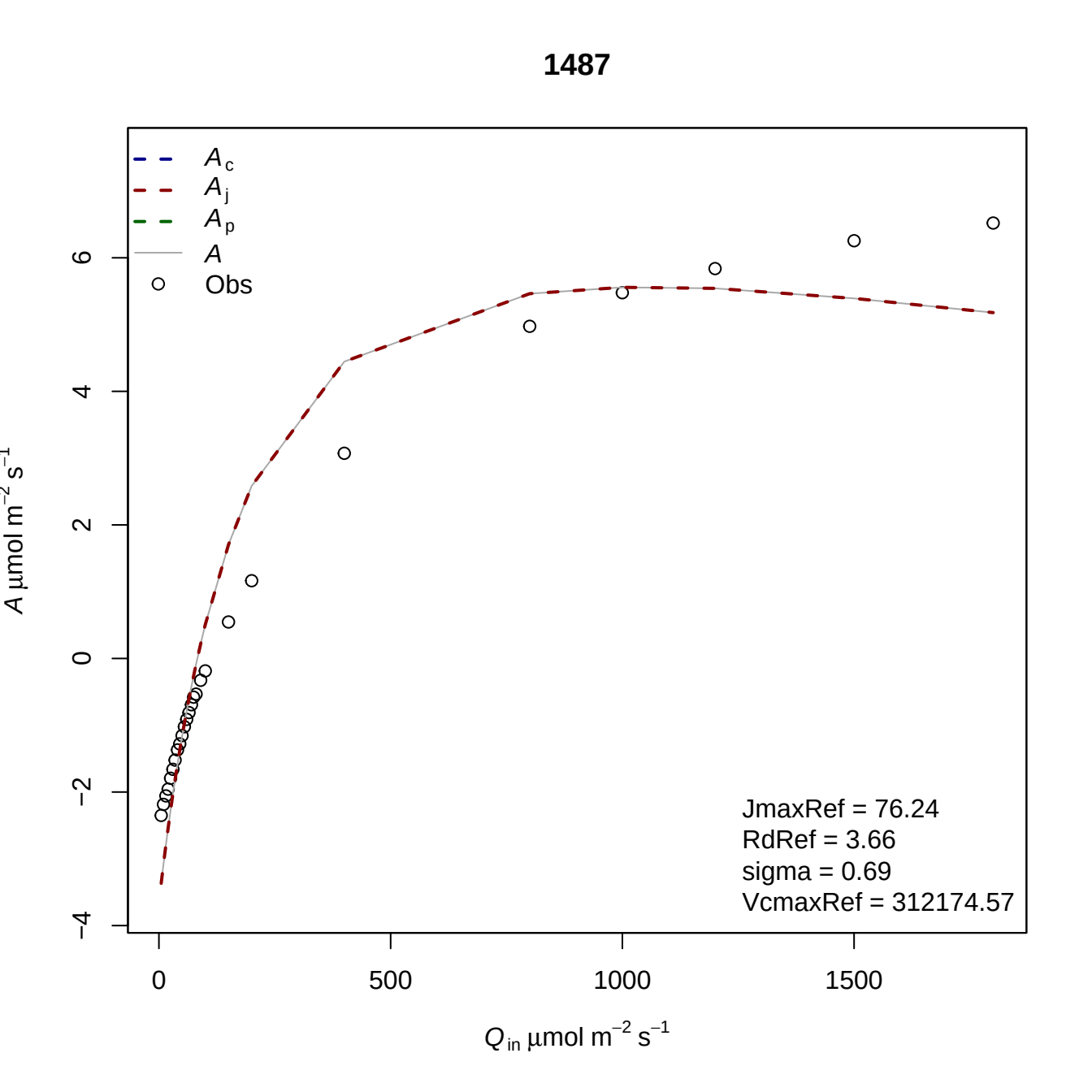

1482
A_c
A_j
A_p
A
Obs
JmaxRef = 151.41
RdRef = 2.73
sigma = 0.32
VcmaxRef = 98.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

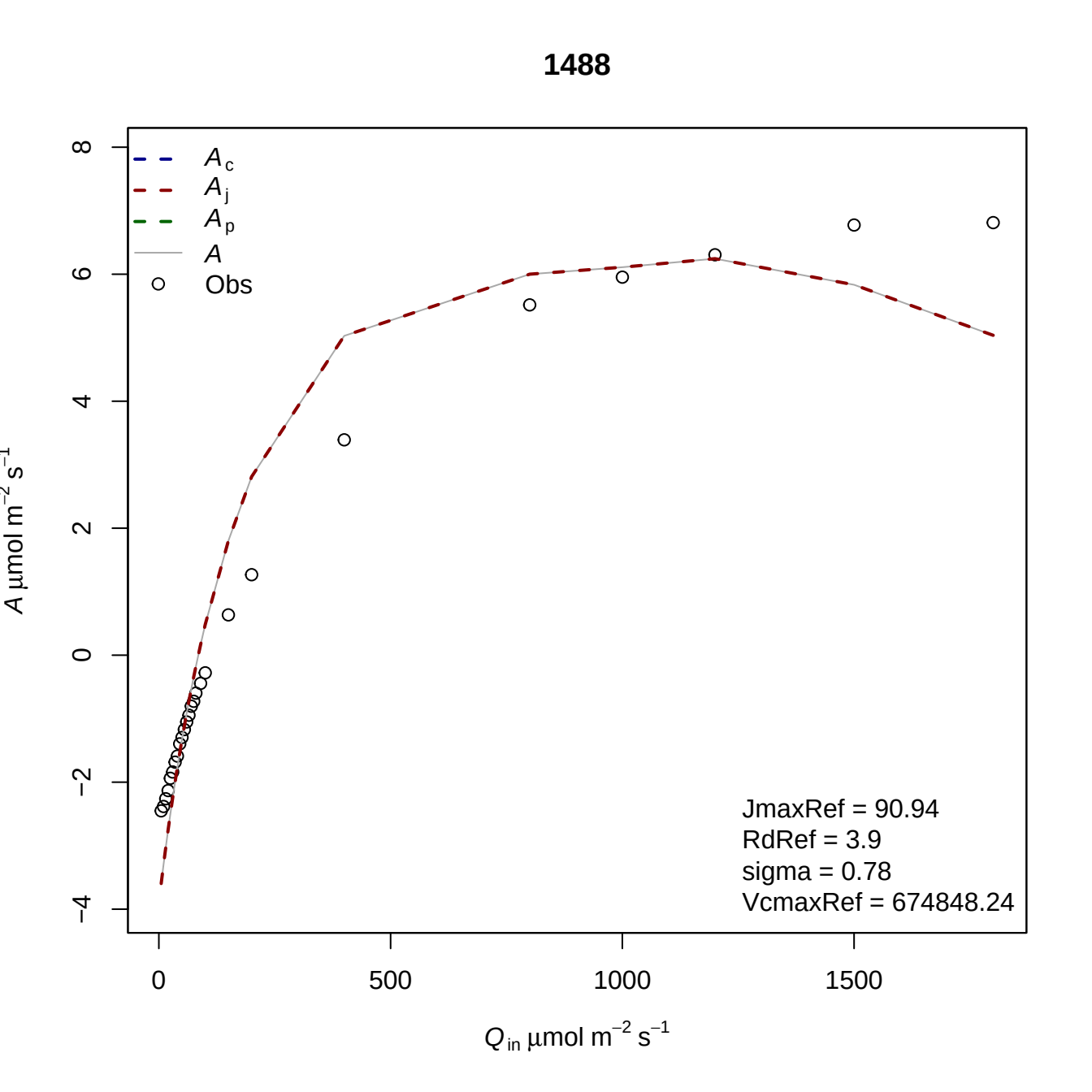

1483
A_c
A_j
A_p
A
Obs
JmaxRef = 97.58
RdRef = 3.09
sigma = 0.25
VcmaxRef = 131671.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

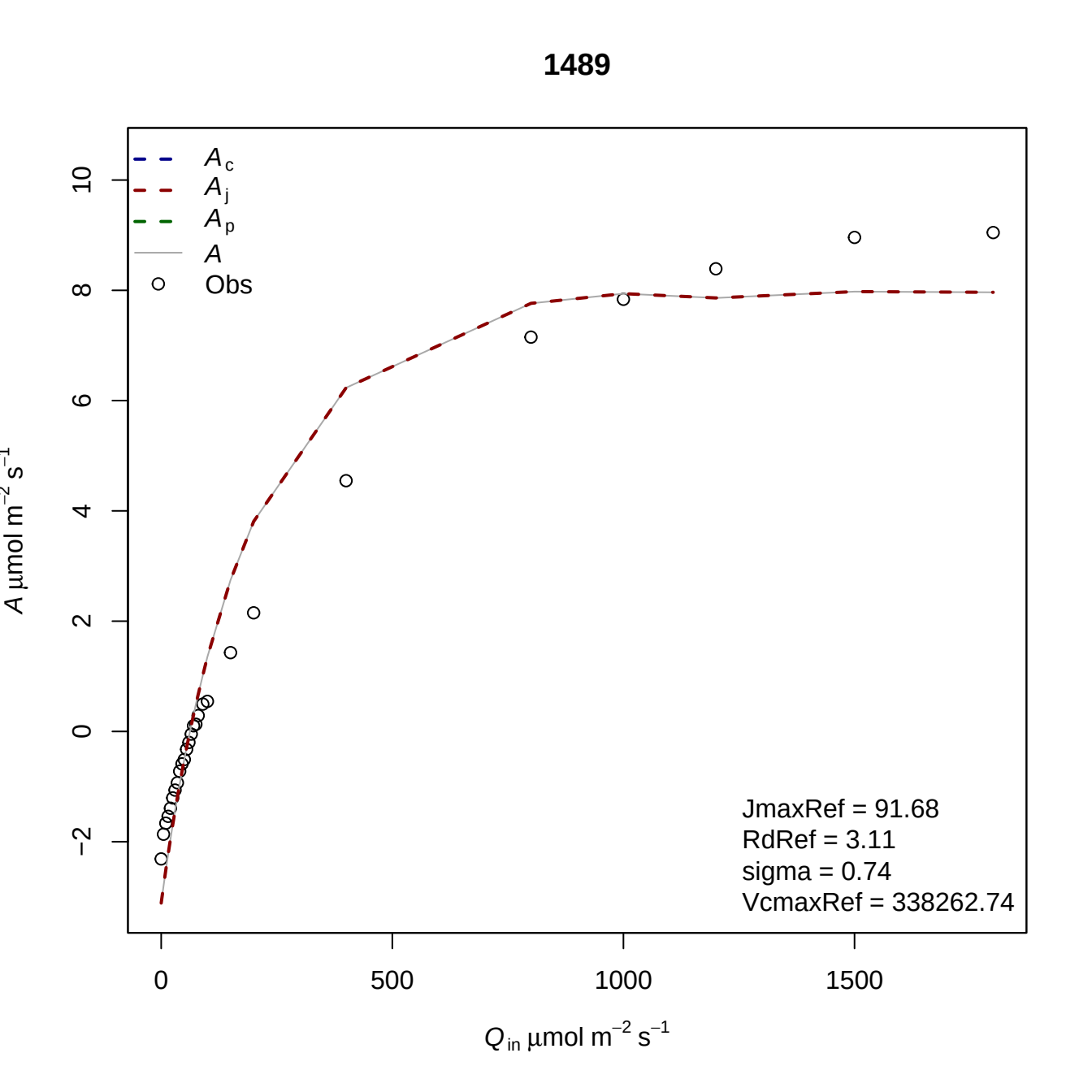

1484
A_c
A_j
A_p
A
Obs
JmaxRef = 122.88
RdRef = 3.48
sigma = 0.54
VcmaxRef = 346301.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500

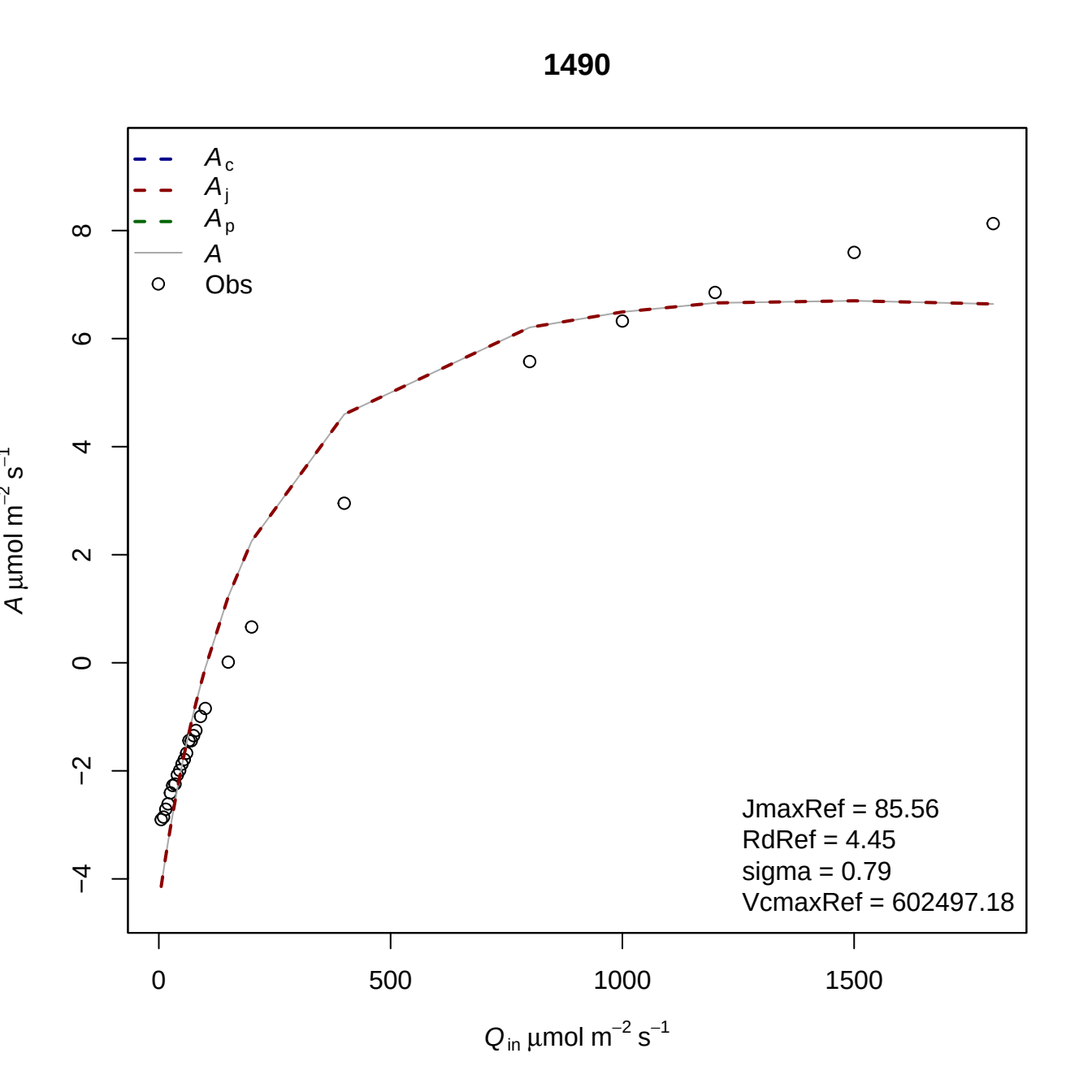

1485
A_c
A_j
A_p
A
Obs
JmaxRef = 83.91
RdRef = 3.21
sigma = 0.5
VcmaxRef = 495799.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

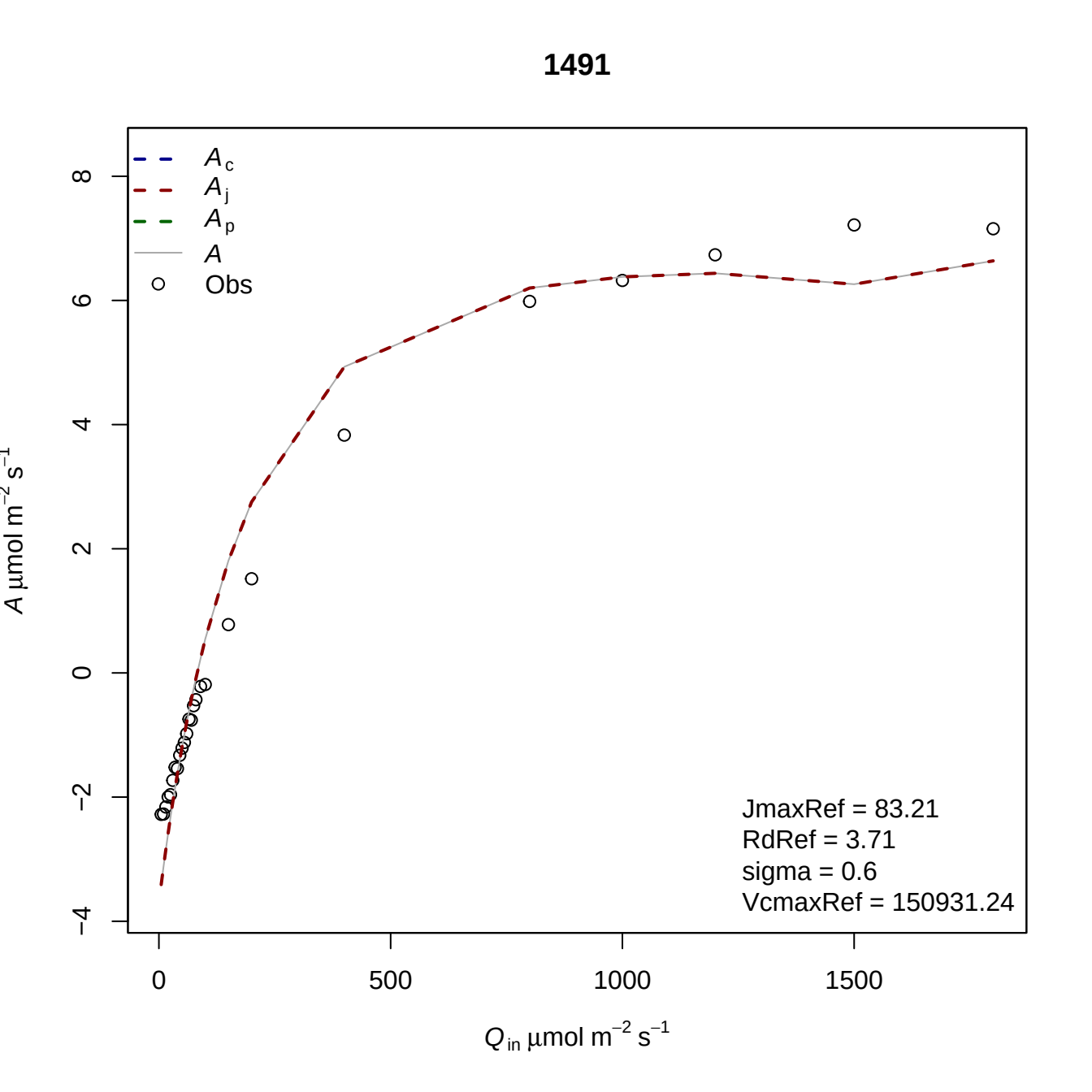

1486
A_c
A_j
A_p
A
Obs
JmaxRef = 75.36
RdRef = 3.91
sigma = 0.81
VcmaxRef = 154720.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

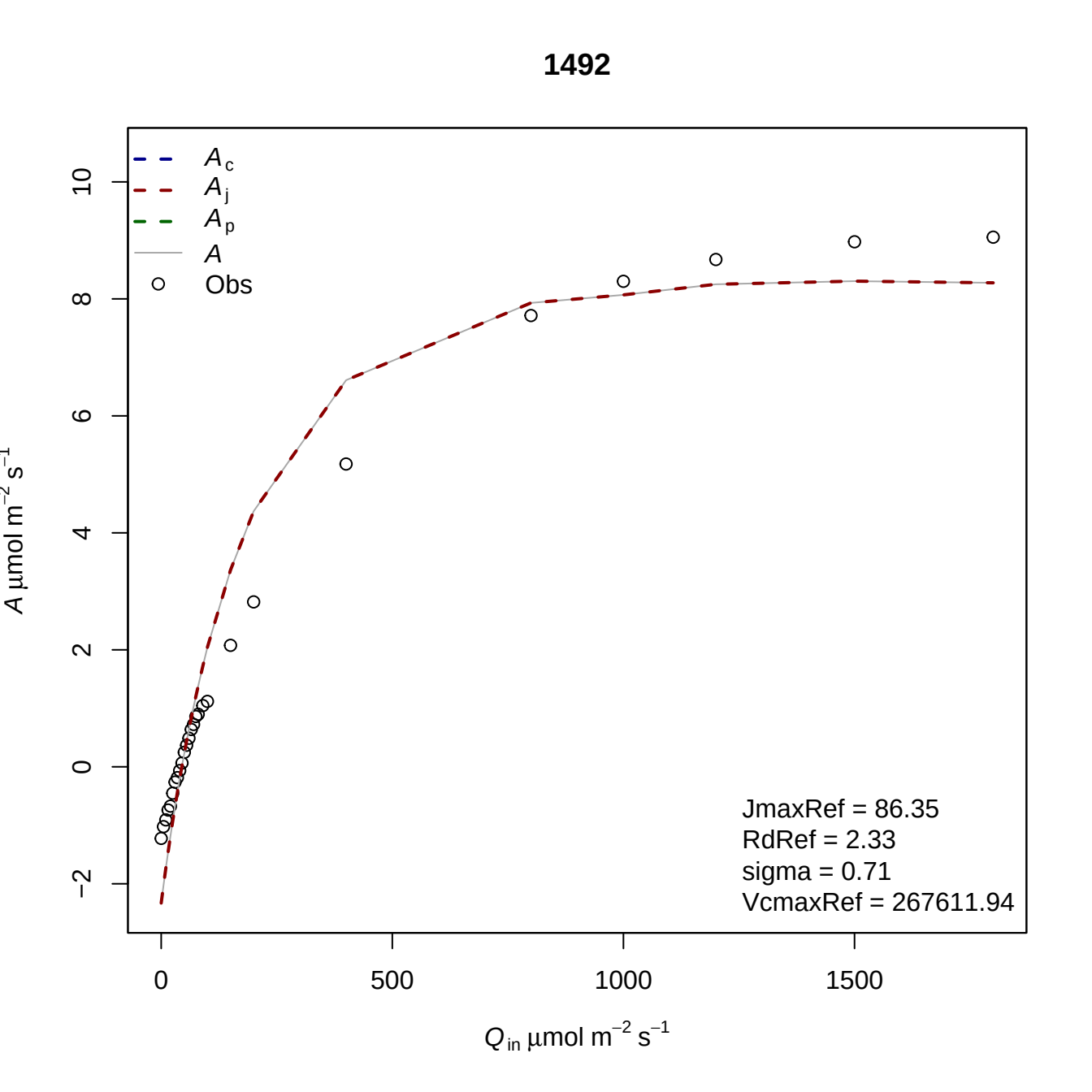

1487
A_c
A_j
A_p
A
Obs
JmaxRef = 76.24
RdRef = 3.66
sigma = 0.69
VcmaxRef = 312174.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

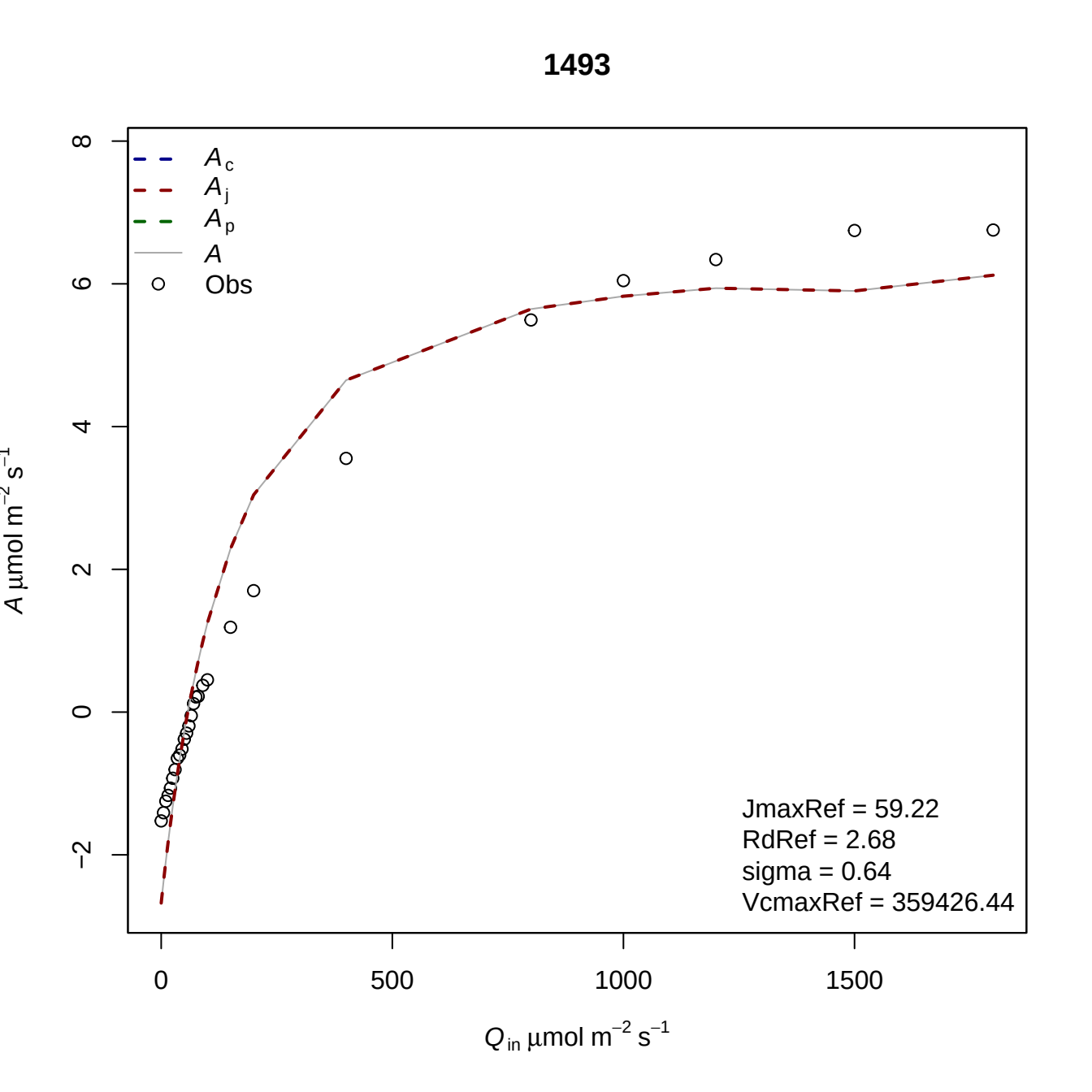

1488
A_c
A_j
A_p
A
Obs
JmaxRef = 90.94
RdRef = 3.9
sigma = 0.78
VcmaxRef = 674848.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

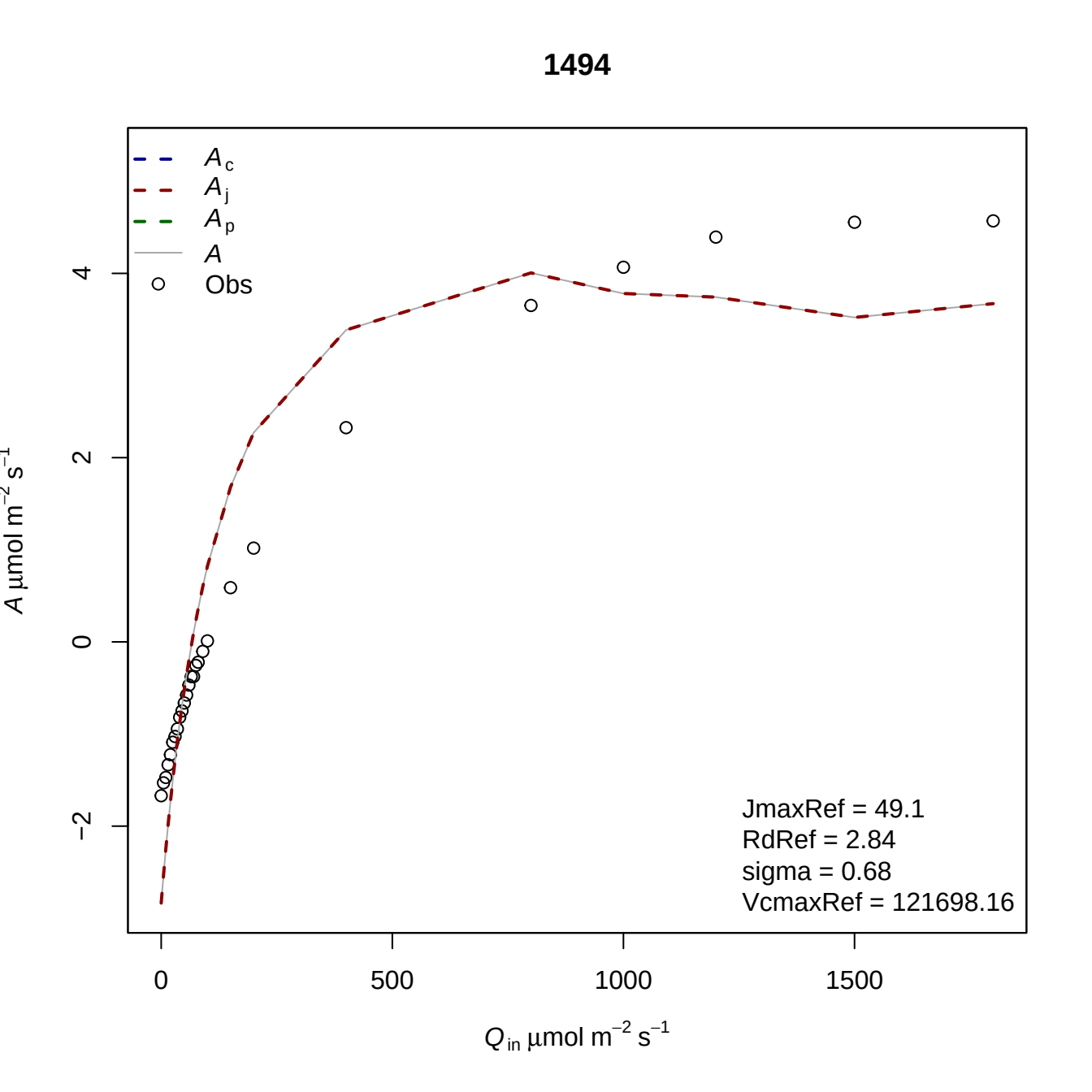

1489
A_c
A_j
A_p
A
Obs
JmaxRef = 91.68
RdRef = 3.11
sigma = 0.74
VcmaxRef = 338262.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

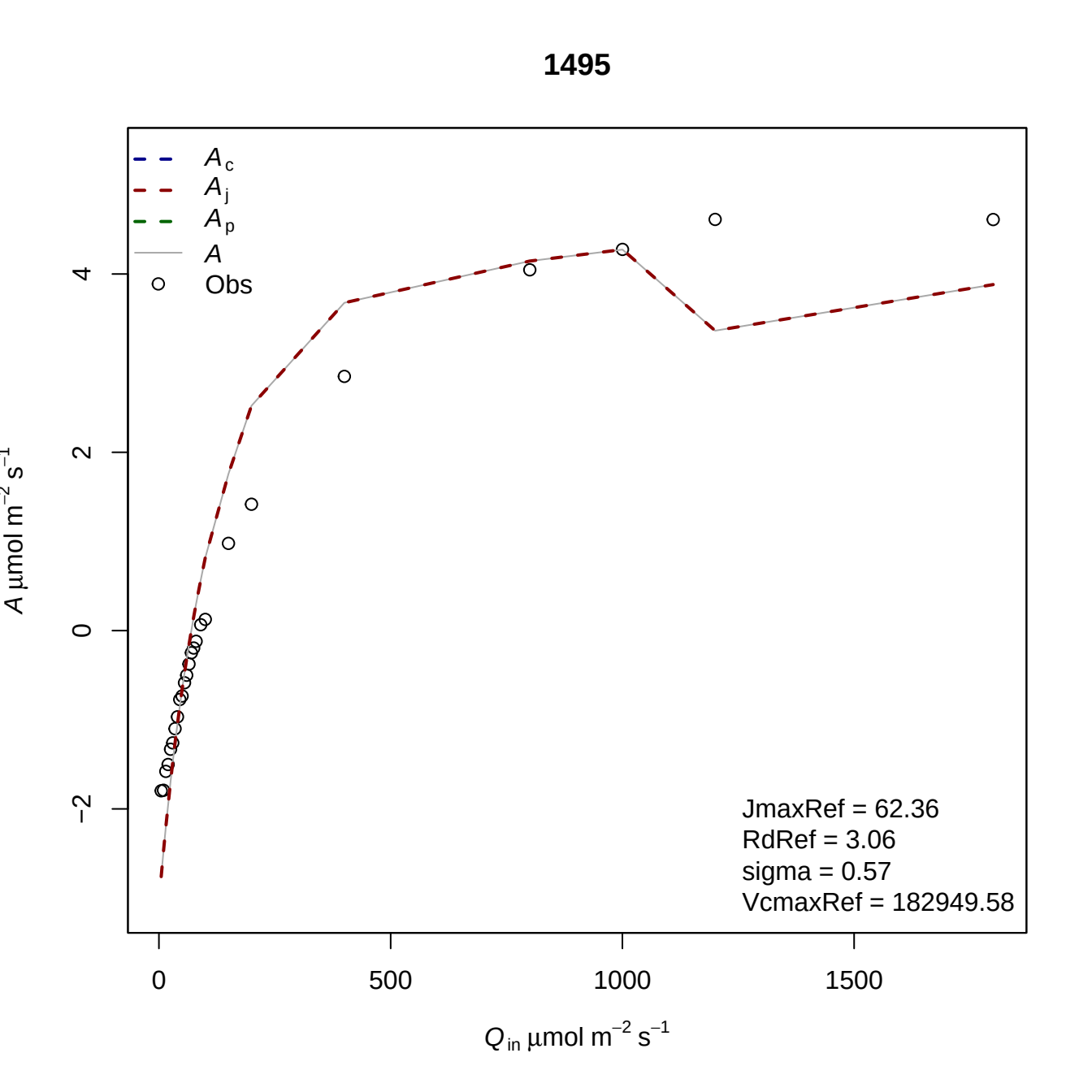

1490
A_c
A_j
A_p
A
Obs
JmaxRef = 85.56
RdRef = 4.45
sigma = 0.79
VcmaxRef = 602497.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

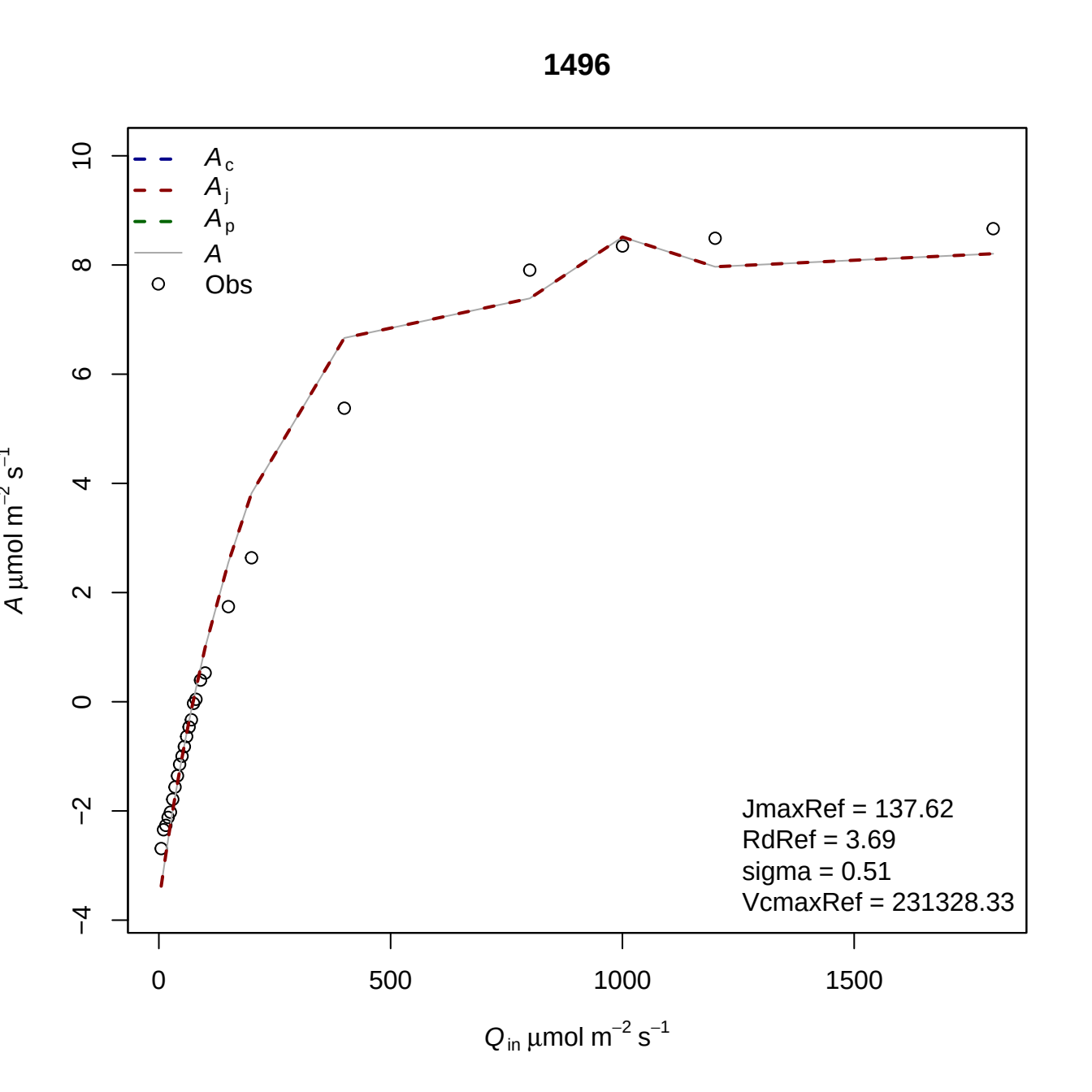

1491
A_c
A_j
A_p
A
Obs
JmaxRef = 83.21
RdRef = 3.71
sigma = 0.6
VcmaxRef = 150931.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

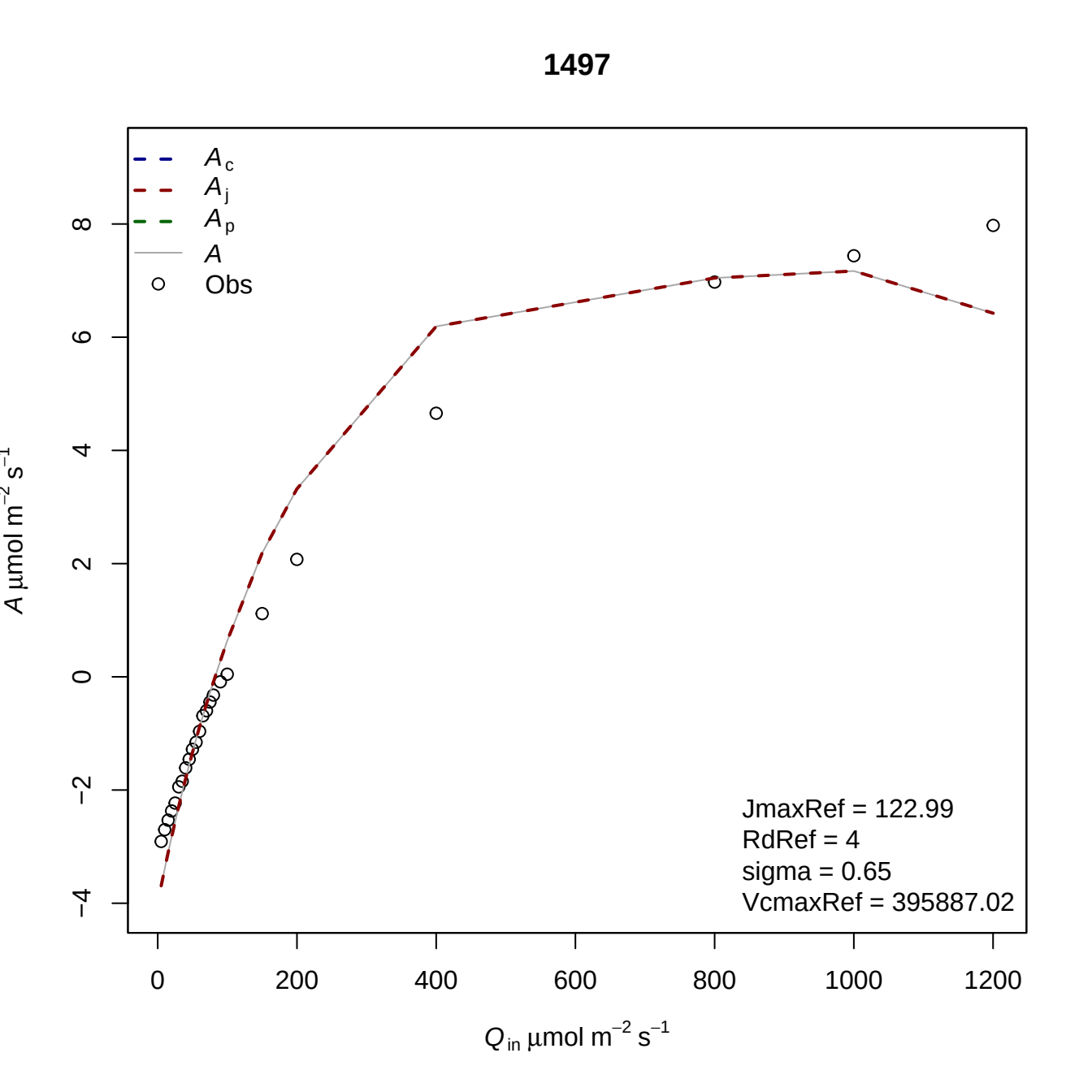

1492
A_c
A_j
A_p
A
Obs
JmaxRef = 86.35
RdRef = 2.33
sigma = 0.71
VcmaxRef = 267611.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

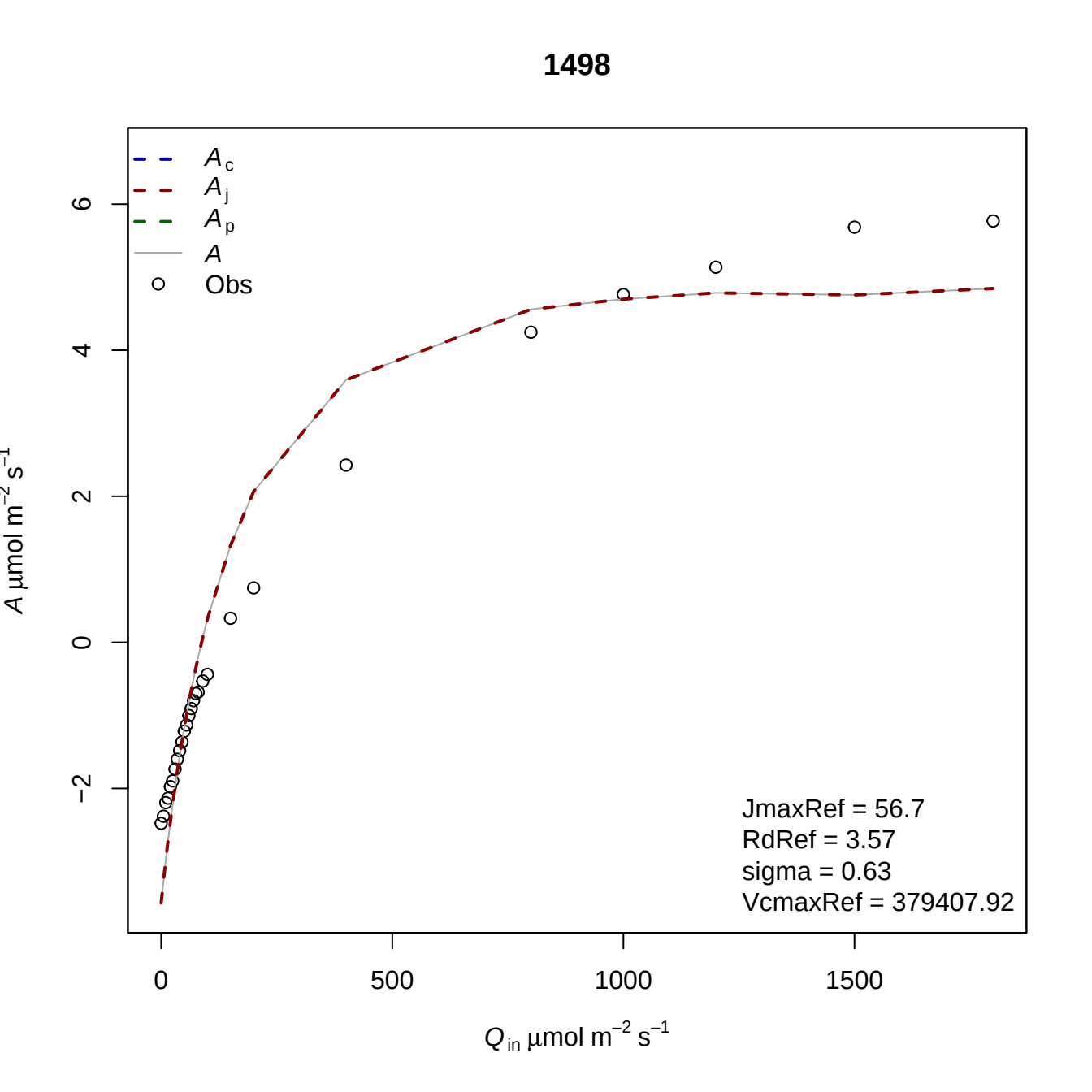

1493
A_c
A_j
A_p
A
Obs
JmaxRef = 59.22
RdRef = 2.68
sigma = 0.64
VcmaxRef = 359426.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

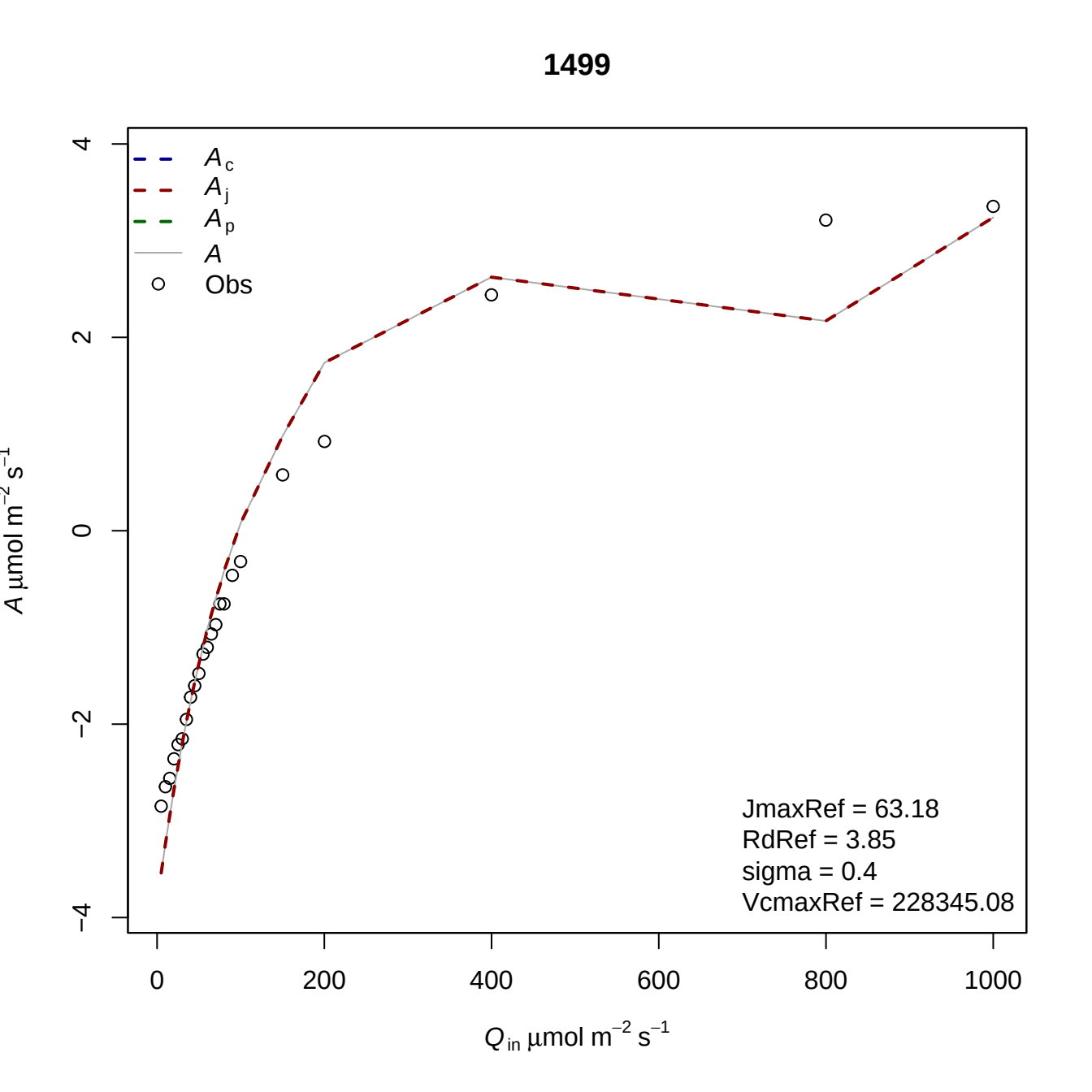

1494
A_c
A_j
A_p
A
Obs
JmaxRef = 49.1
RdRef = 2.84
sigma = 0.68
VcmaxRef = 121698.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

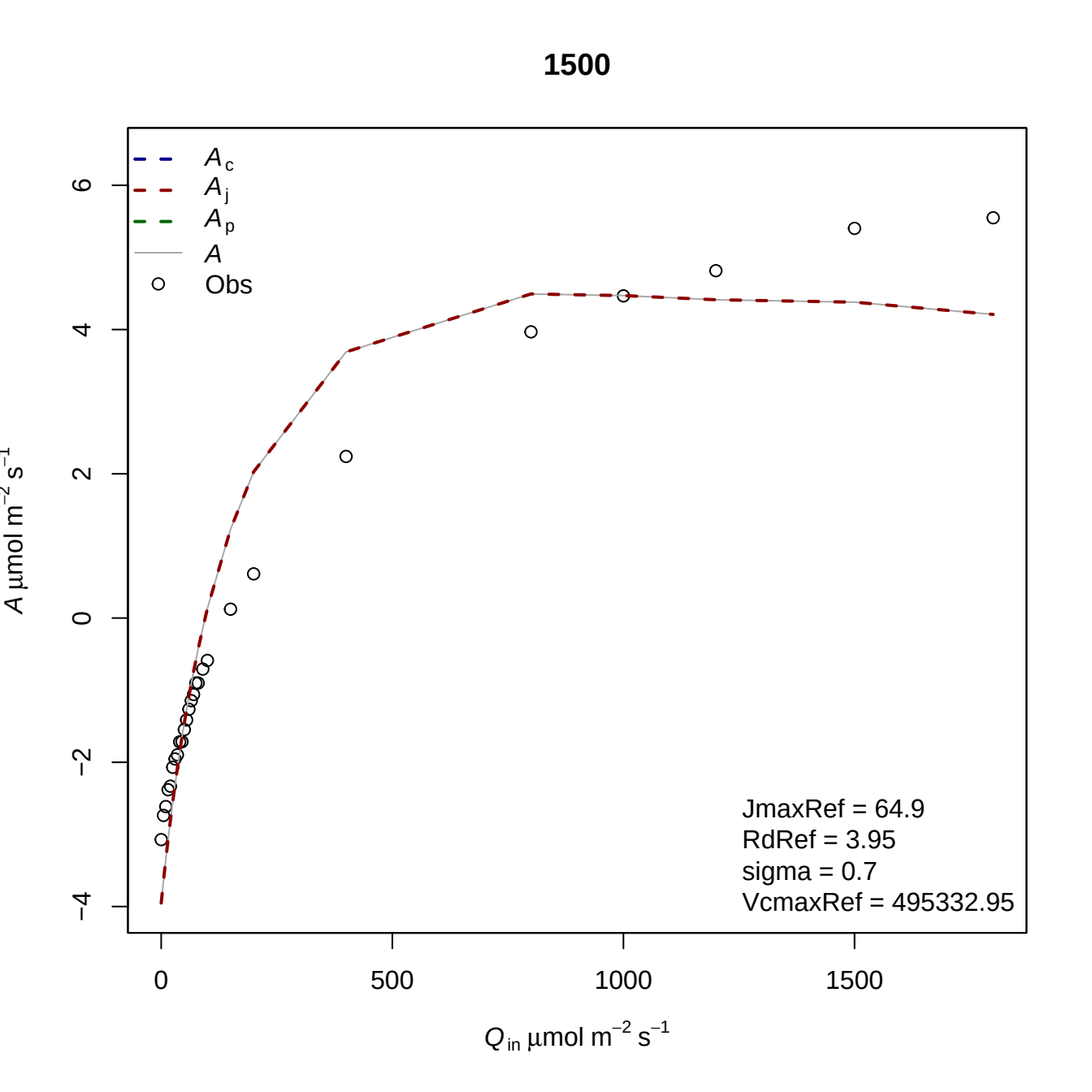

1495
A_c
A_j
A_p
A
Obs
JmaxRef = 62.36
RdRef = 3.06
sigma = 0.57
VcmaxRef = 182949.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

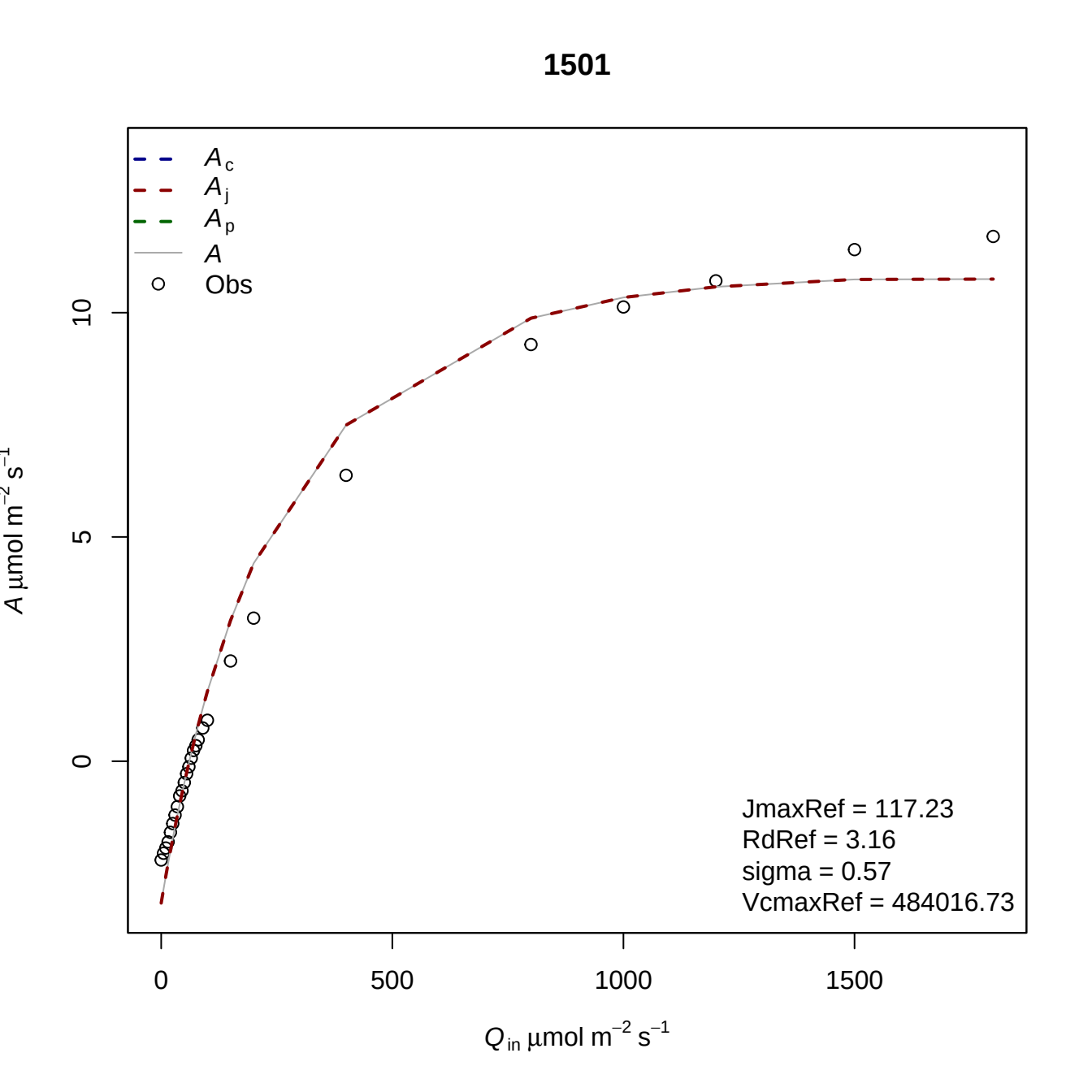

1496
A_c
A_j
A_p
A
Obs
JmaxRef = 137.62
RdRef = 3.69
sigma = 0.51
VcmaxRef = 231328.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500

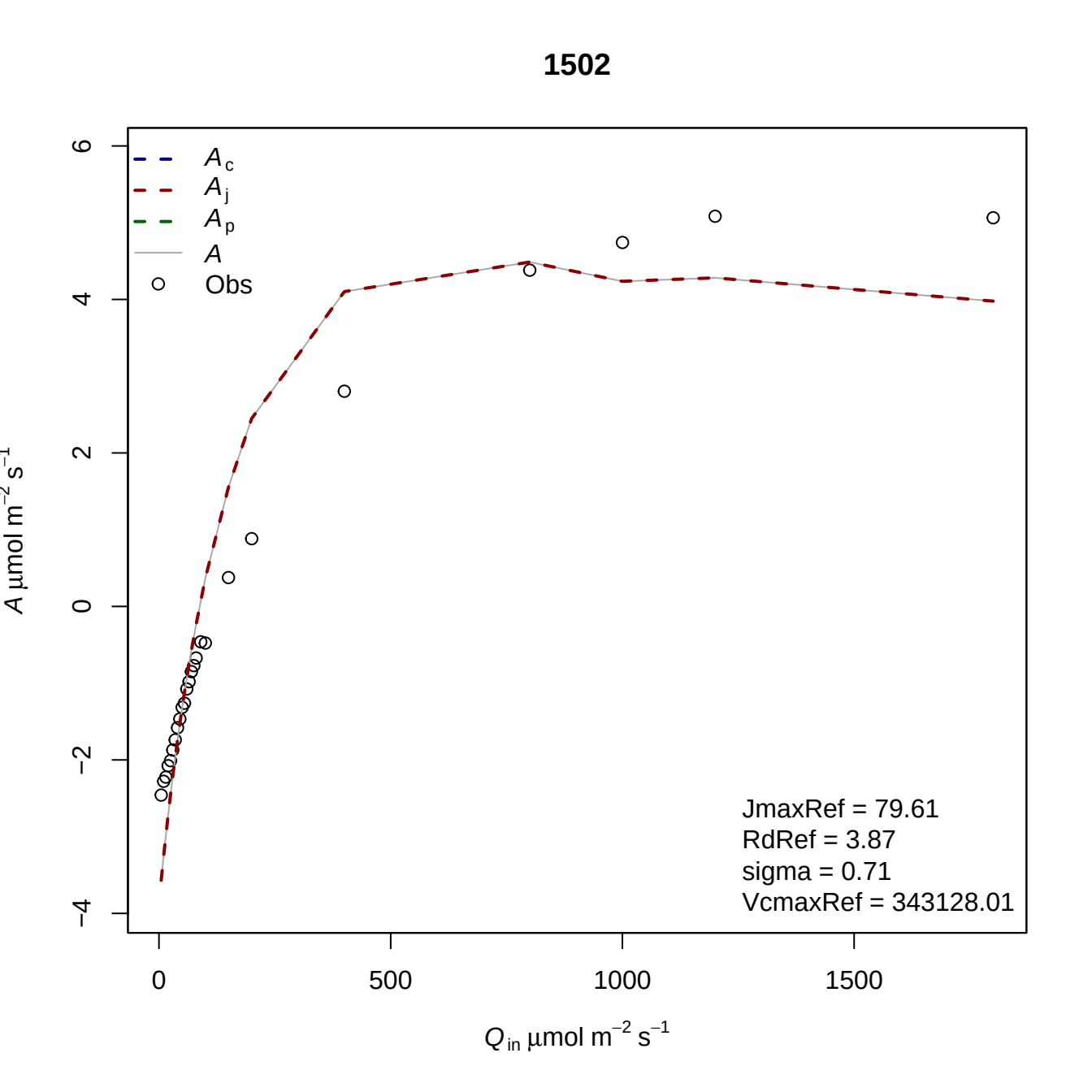

1497
A_c
A_j
A_p
A
Obs
JmaxRef = 122.99
RdRef = 4
sigma = 0.65
VcmaxRef = 395887.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

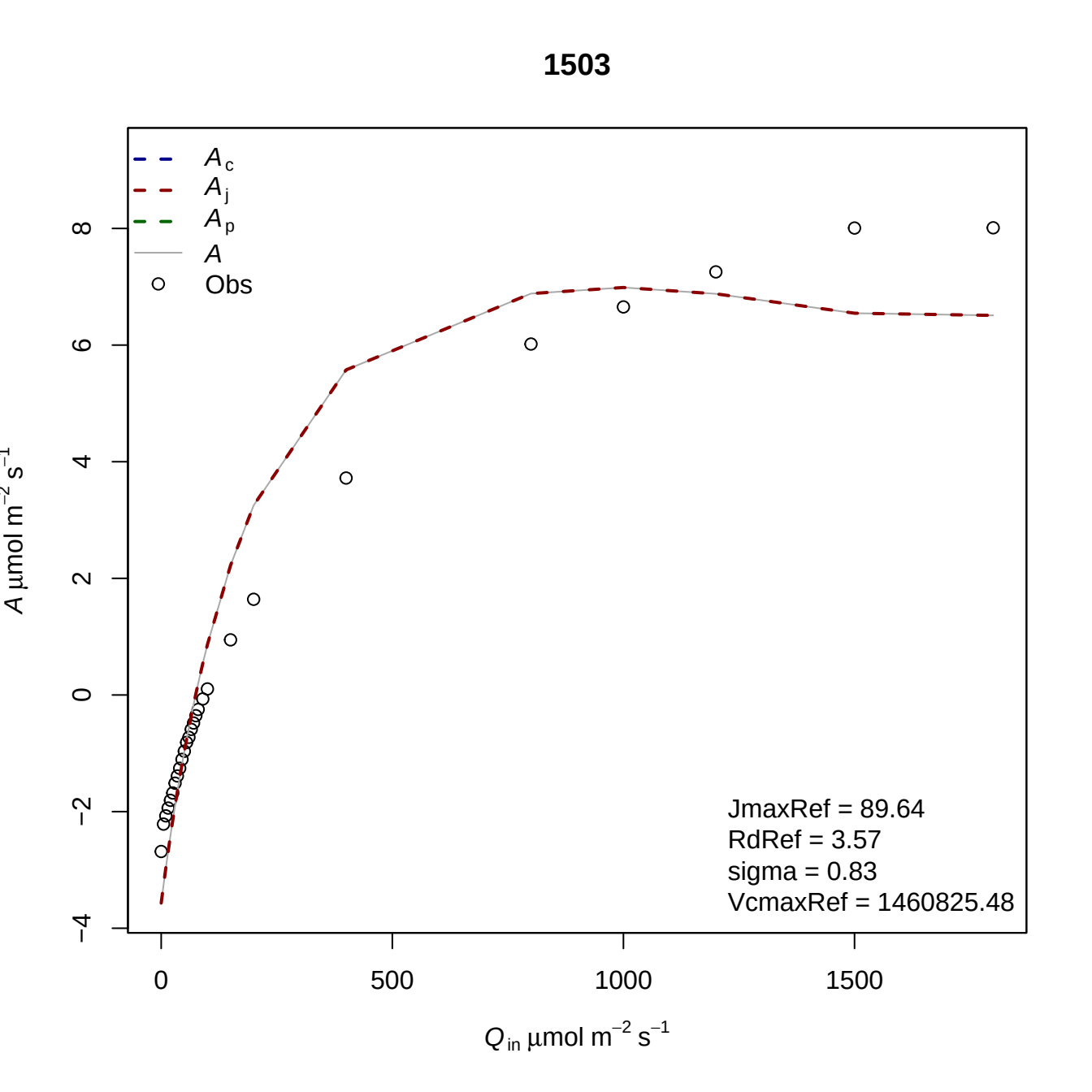

1498
A_c
A_j
A_p
A
Obs
JmaxRef = 56.7
RdRef = 3.57
sigma = 0.63
VcmaxRef = 379407.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

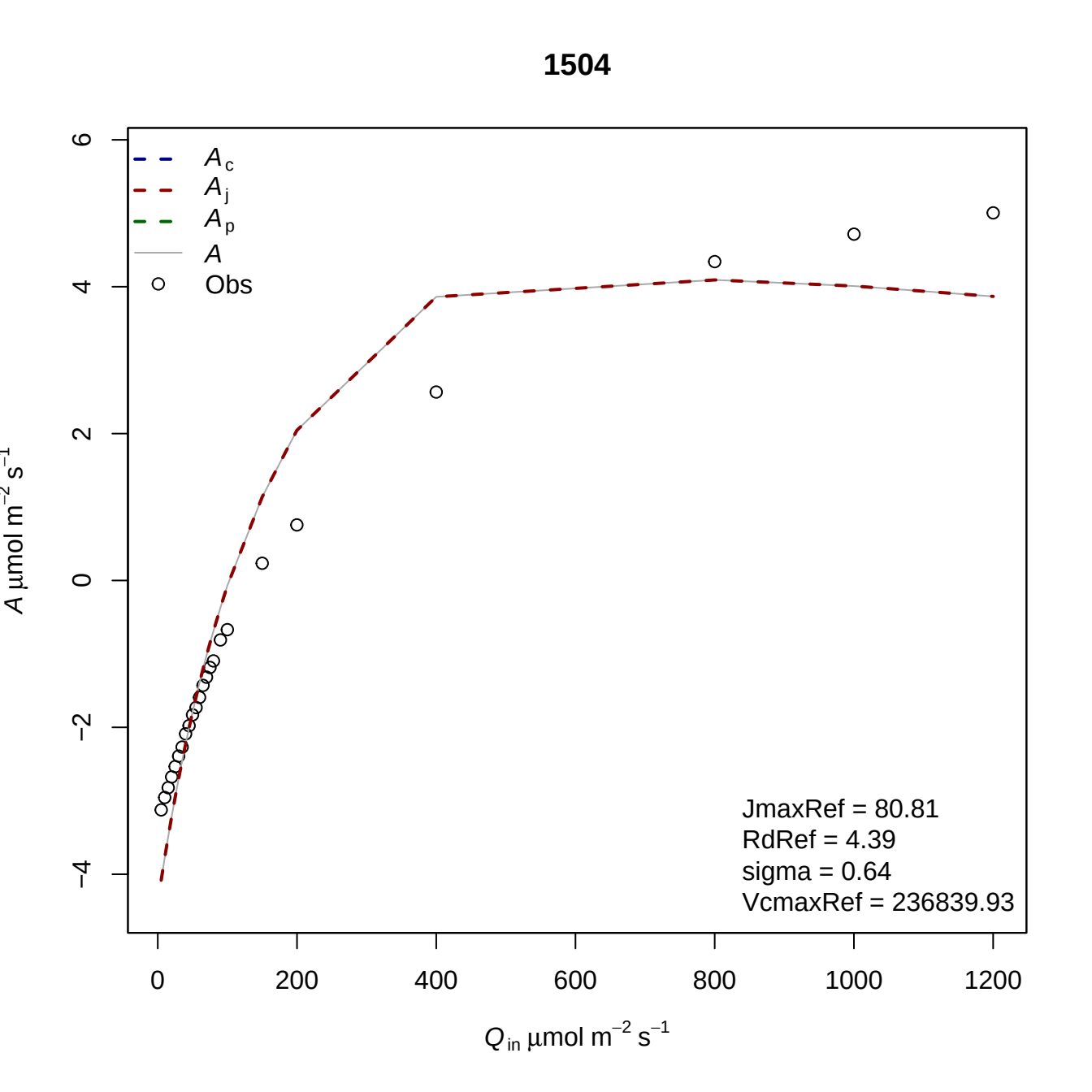

1499
A_c
A_j
A_p
A
Obs
JmaxRef = 63.18
RdRef = 3.85
sigma = 0.4
VcmaxRef = 228345.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
200
400
600
800
1000

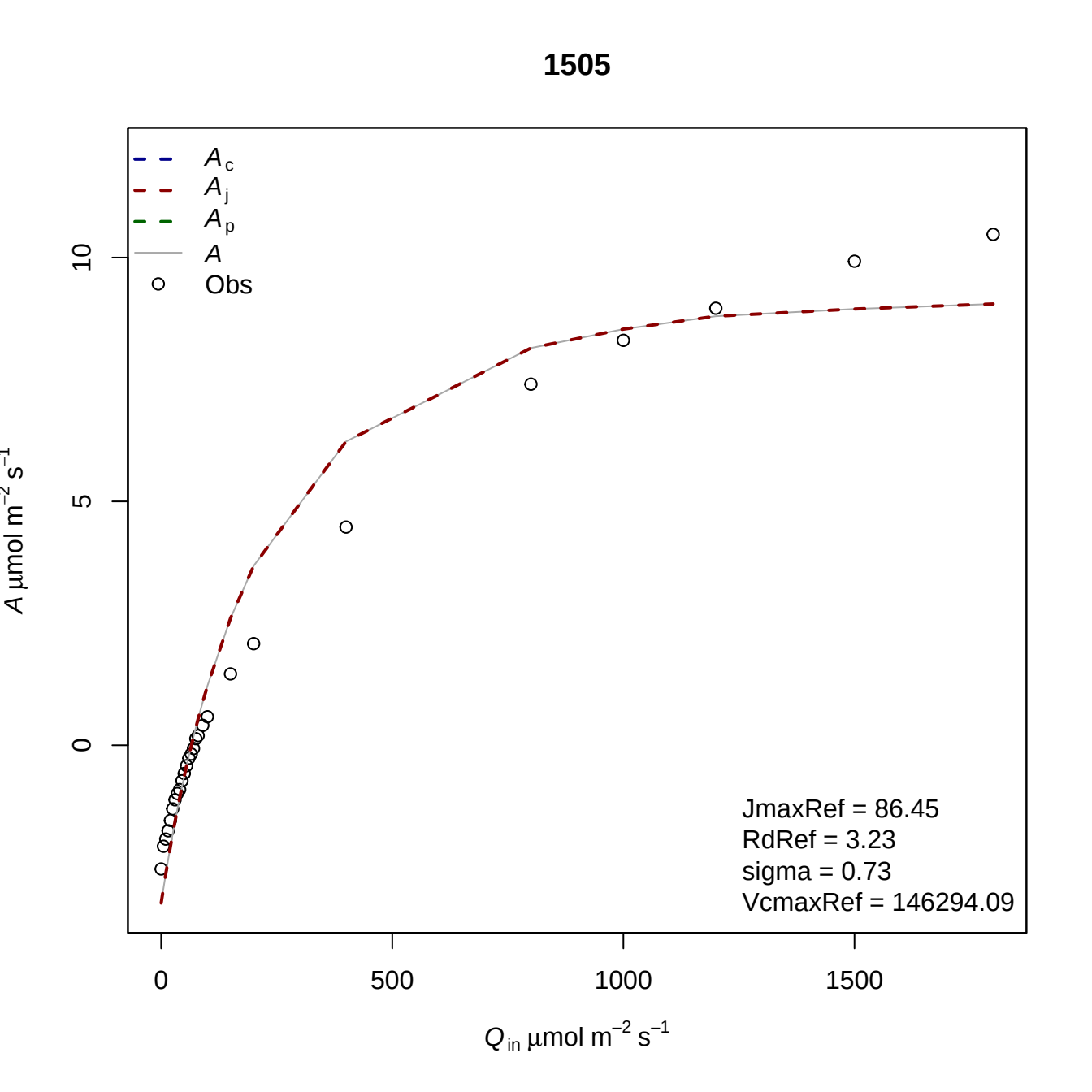

1500
A_c
A_j
A_p
A
Obs
JmaxRef = 64.9
RdRef = 3.95
sigma = 0.7
VcmaxRef = 495332.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

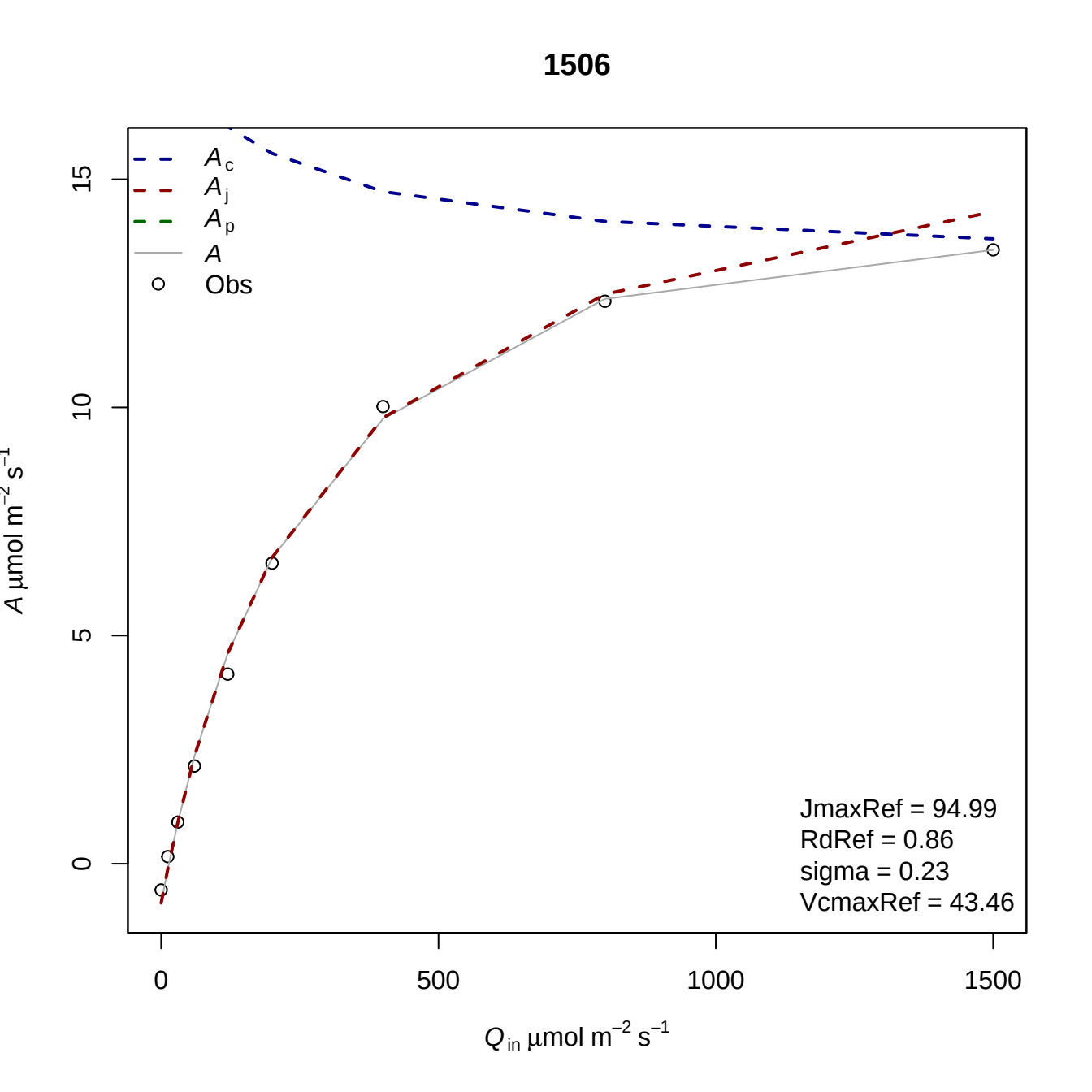

1501
A_c
A_j
A_p
A
Obs
JmaxRef = 117.23
RdRef = 3.16
sigma = 0.57
VcmaxRef = 484016.73
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

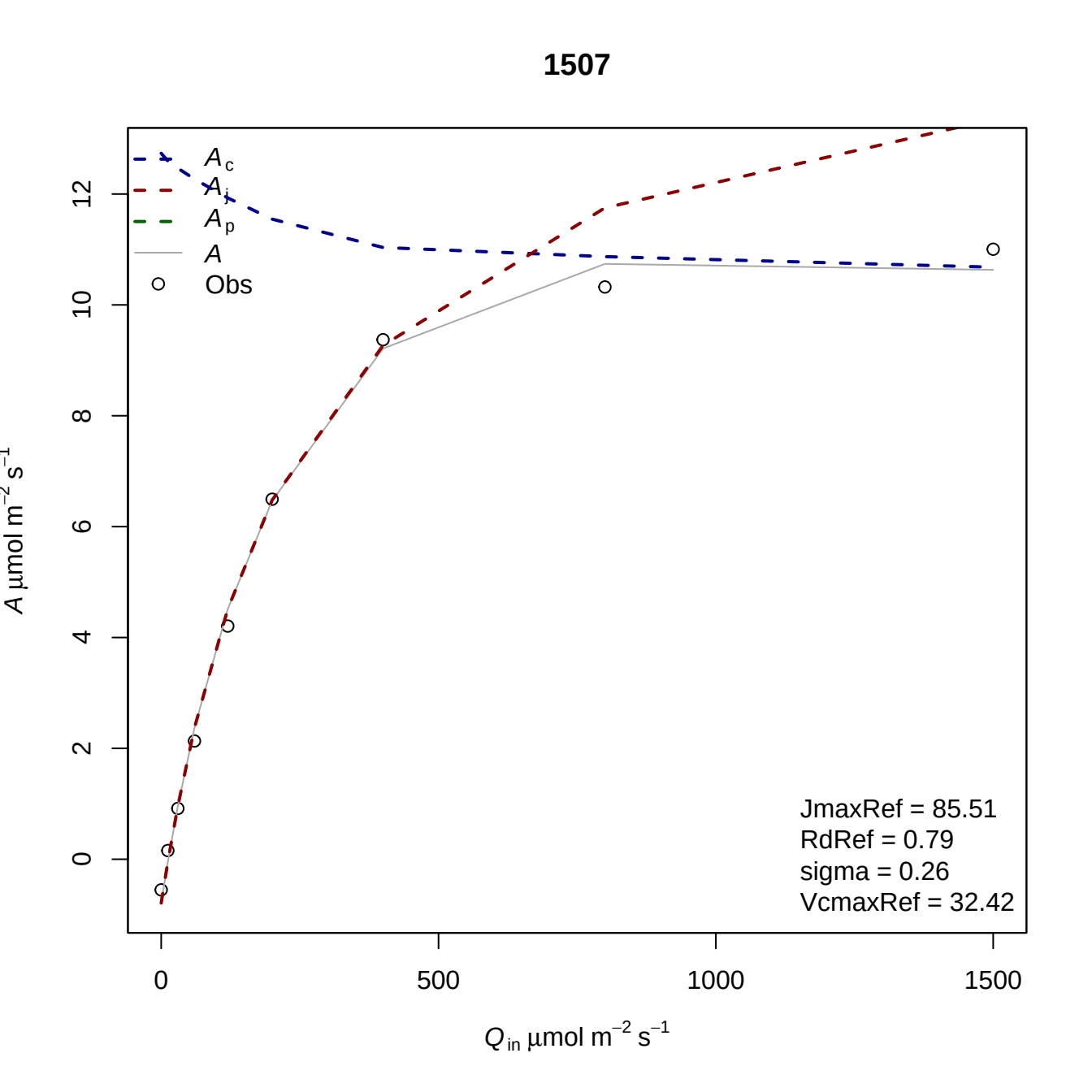

1502
A_c
A_j
A_p
A
Obs
JmaxRef = 79.61
RdRef = 3.87
sigma = 0.71
VcmaxRef = 343128.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

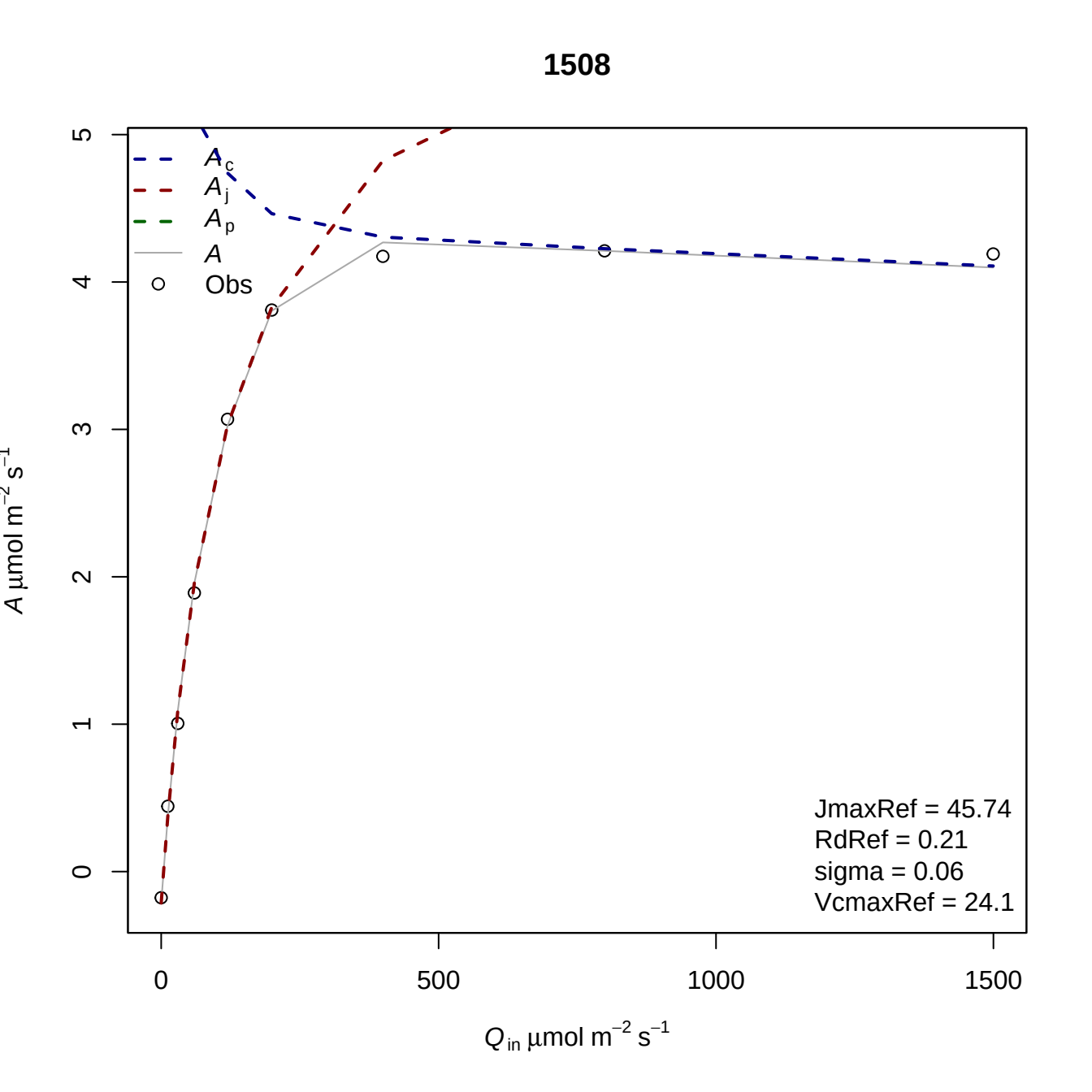

1503
A_c
A_j
A_p
A
Obs
JmaxRef = 89.64
RdRef = 3.57
sigma = 0.83
VcmaxRef = 1460825.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−4
−2
0
2
4
6
8
0
500
1000
1500

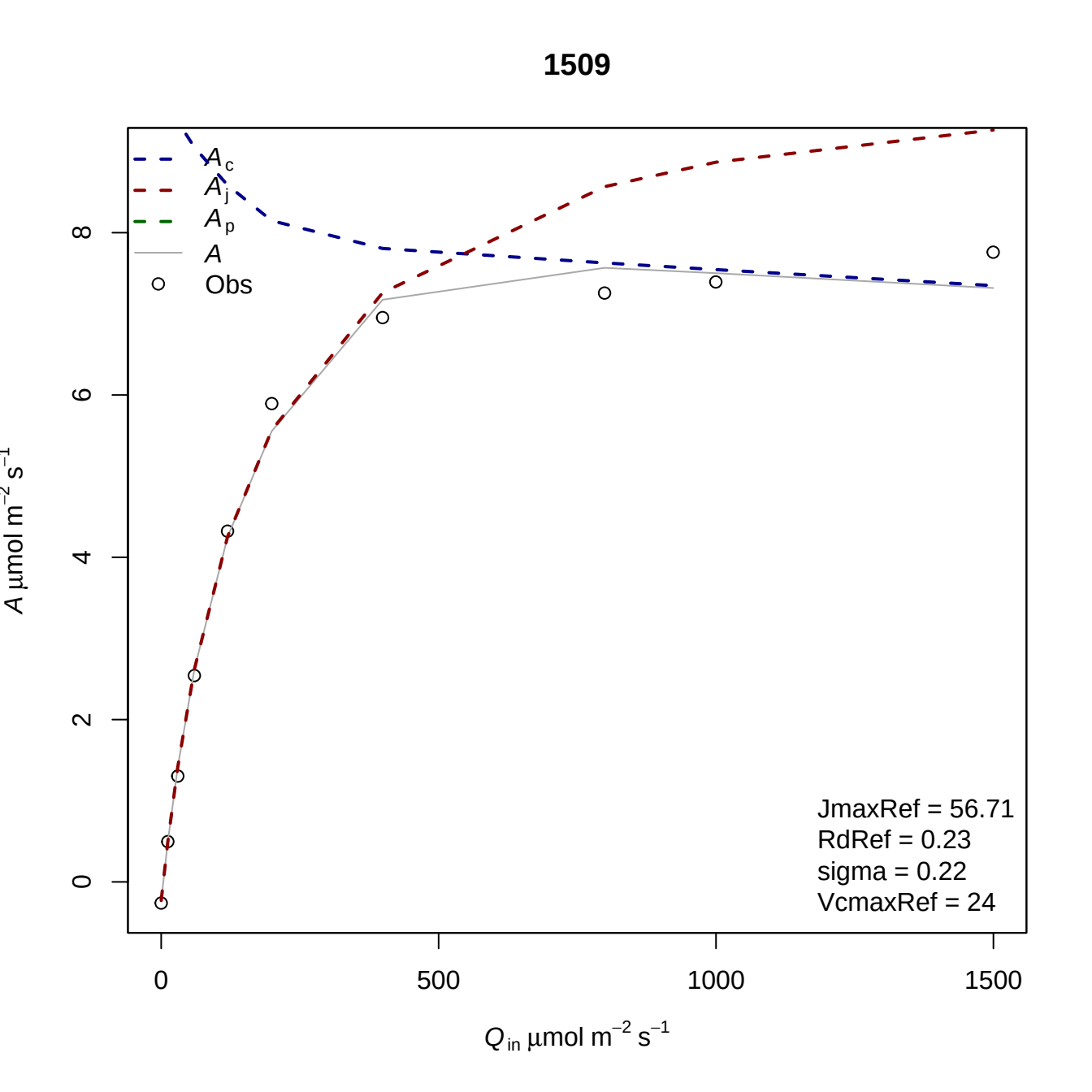

1504
A_c
A_j
A_p
A
Obs
JmaxRef = 80.81
RdRef = 4.39
sigma = 0.64
VcmaxRef = 236839.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

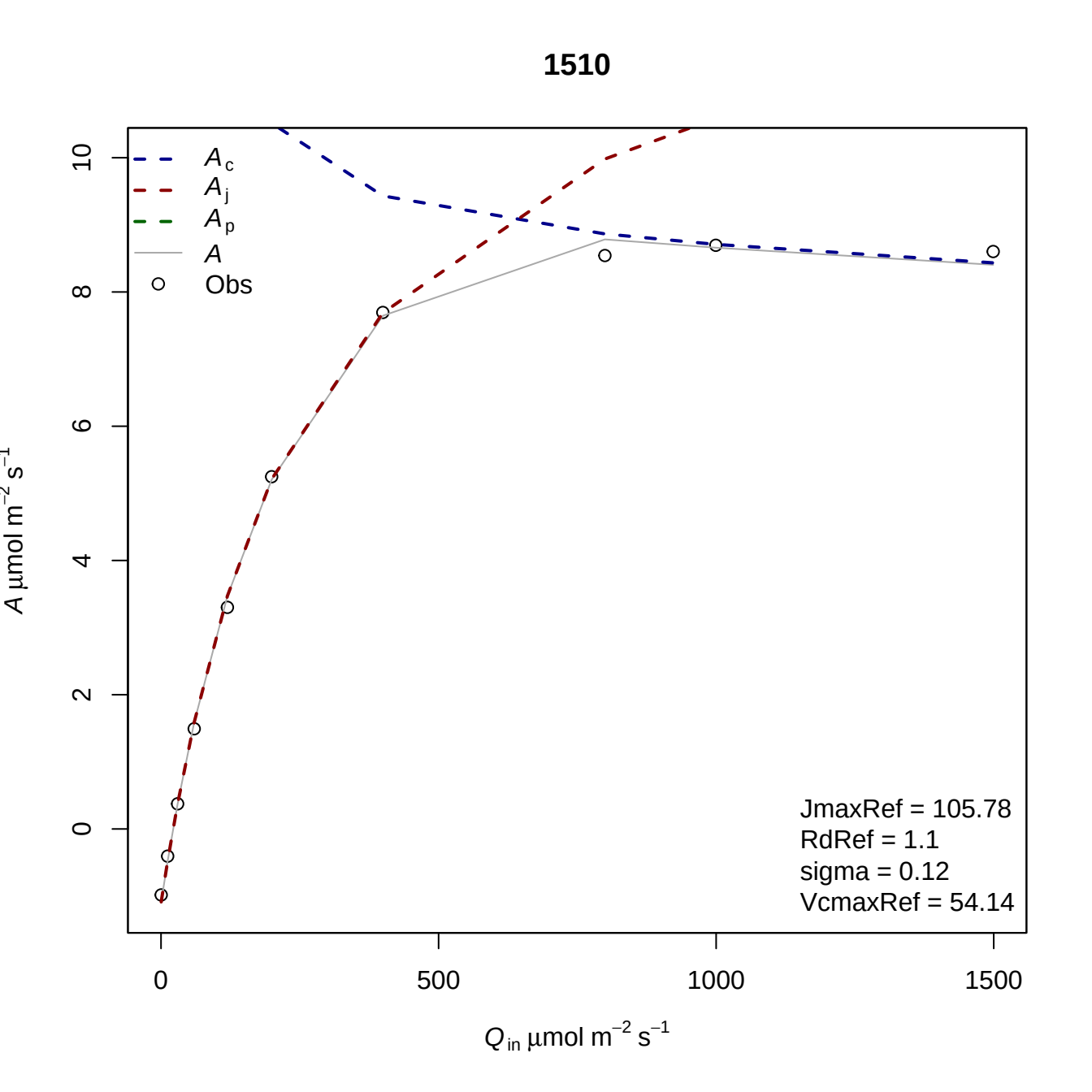

1505
A_c
A_j
A_p
A
Obs
JmaxRef = 86.45
RdRef = 3.23
sigma = 0.73
VcmaxRef = 146294.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

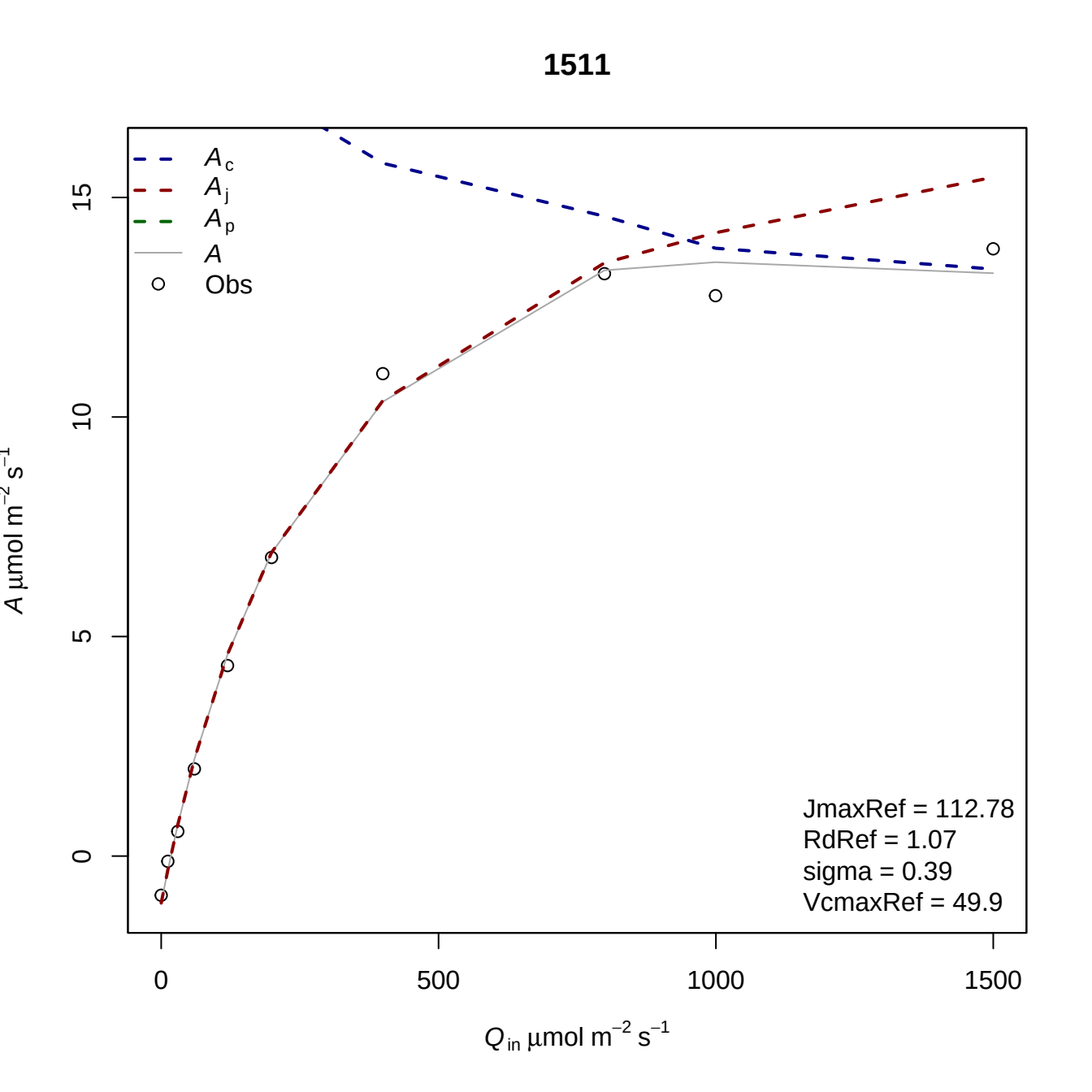

1506
A_c
A_j
A_p
A
Obs
JmaxRef = 94.99
RdRef = 0.86
sigma = 0.23
VcmaxRef = 43.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

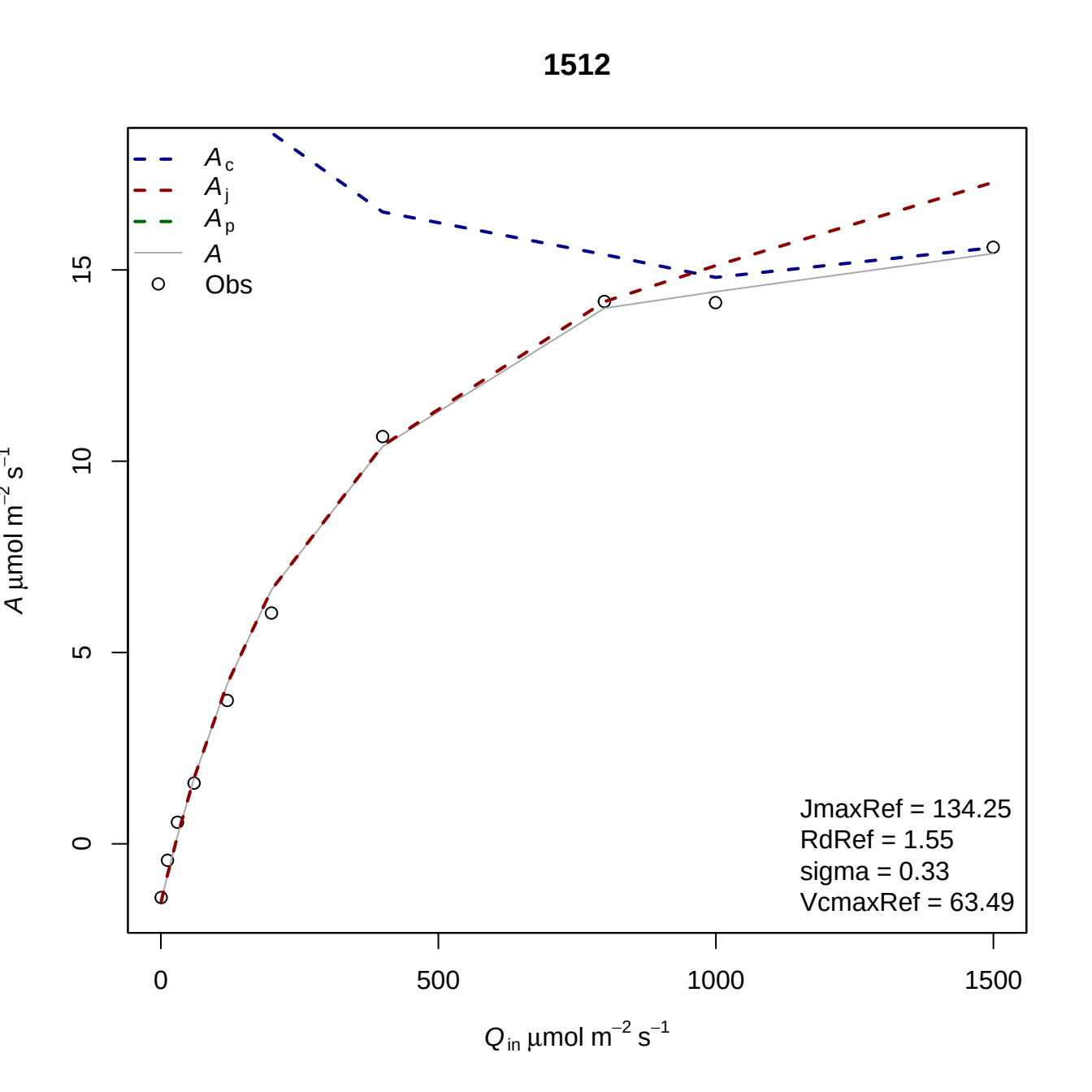

1507
A_c
A_j
A_p
A
Obs
JmaxRef = 85.51
RdRef = 0.79
sigma = 0.26
VcmaxRef = 32.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

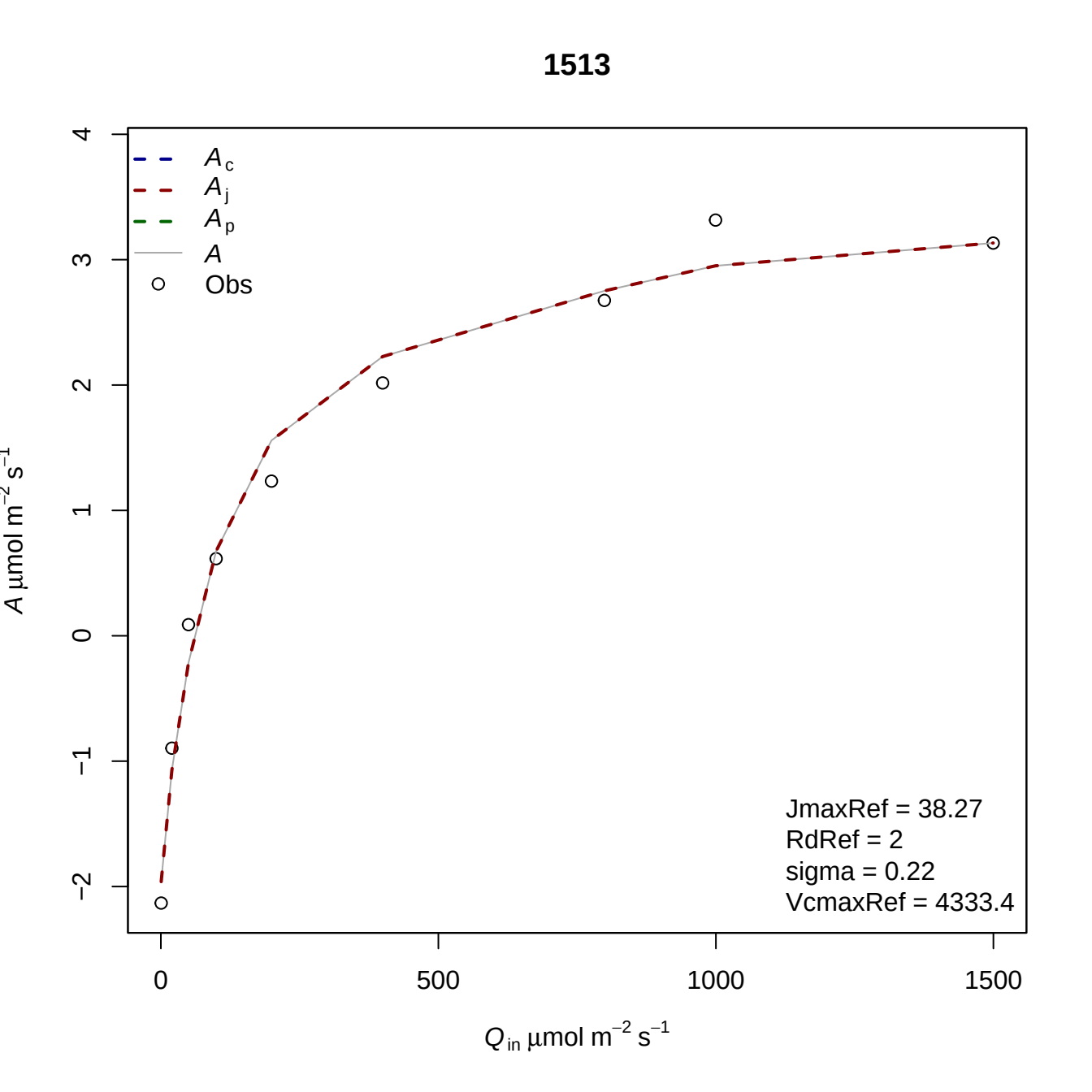

1508
A_c
A_j
A_p
A
Obs
JmaxRef = 45.74
RdRef = 0.21
sigma = 0.06
VcmaxRef = 24.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

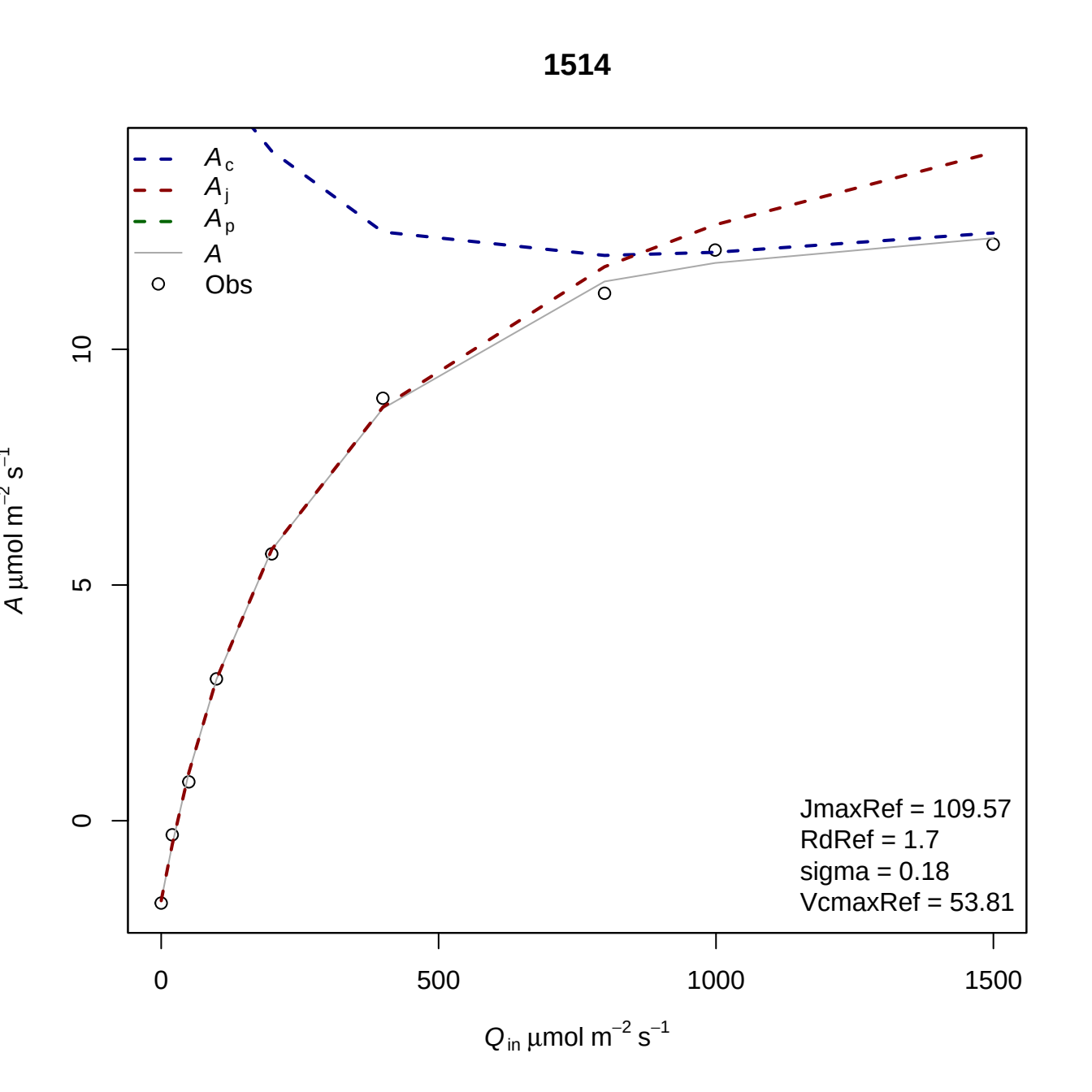

1509
A_c
A_j
A_p
A
Obs
JmaxRef = 56.71
RdRef = 0.23
sigma = 0.22
VcmaxRef = 24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

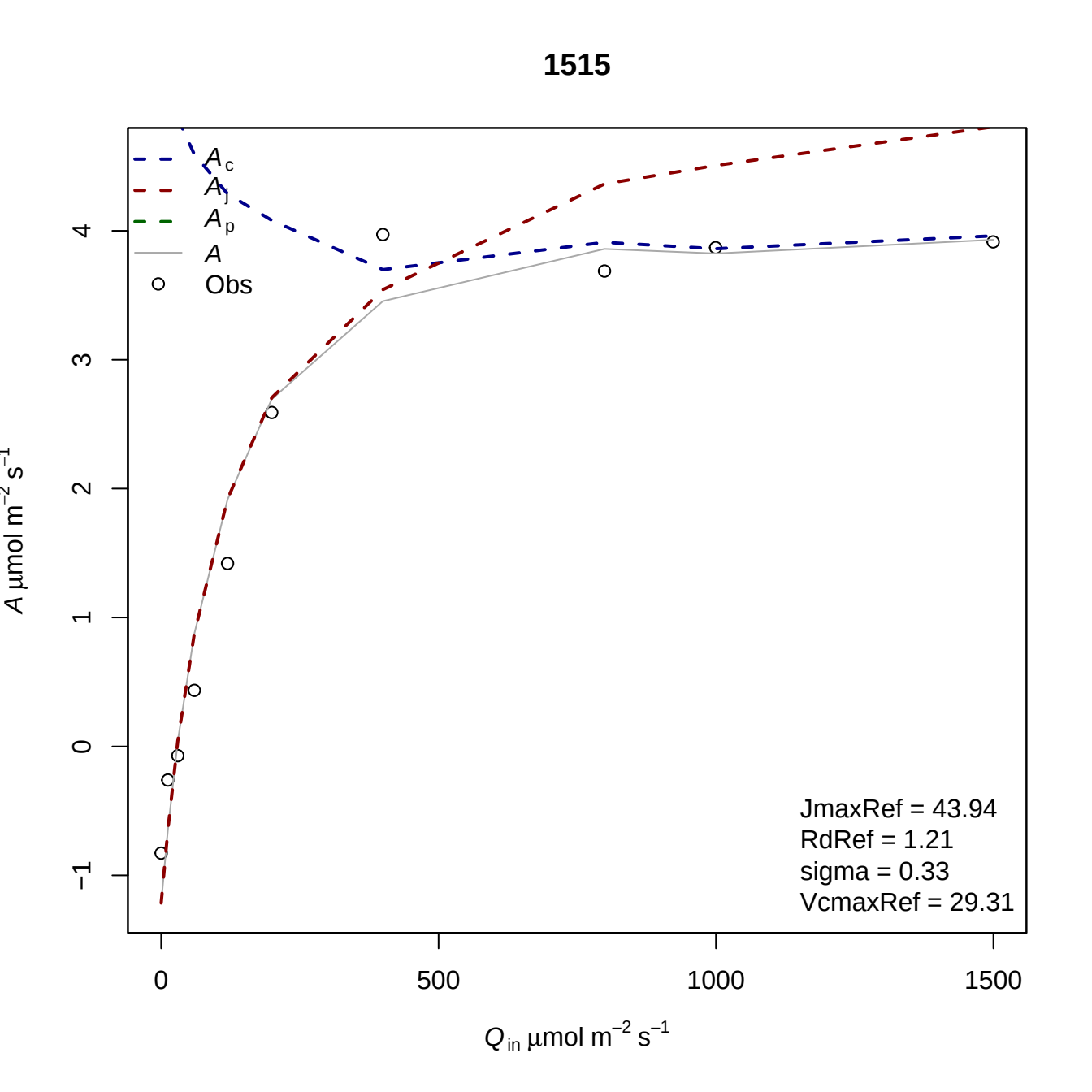

1510
A_c
A_j
A_p
A
Obs
JmaxRef = 105.78
RdRef = 1.1
sigma = 0.12
VcmaxRef = 54.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

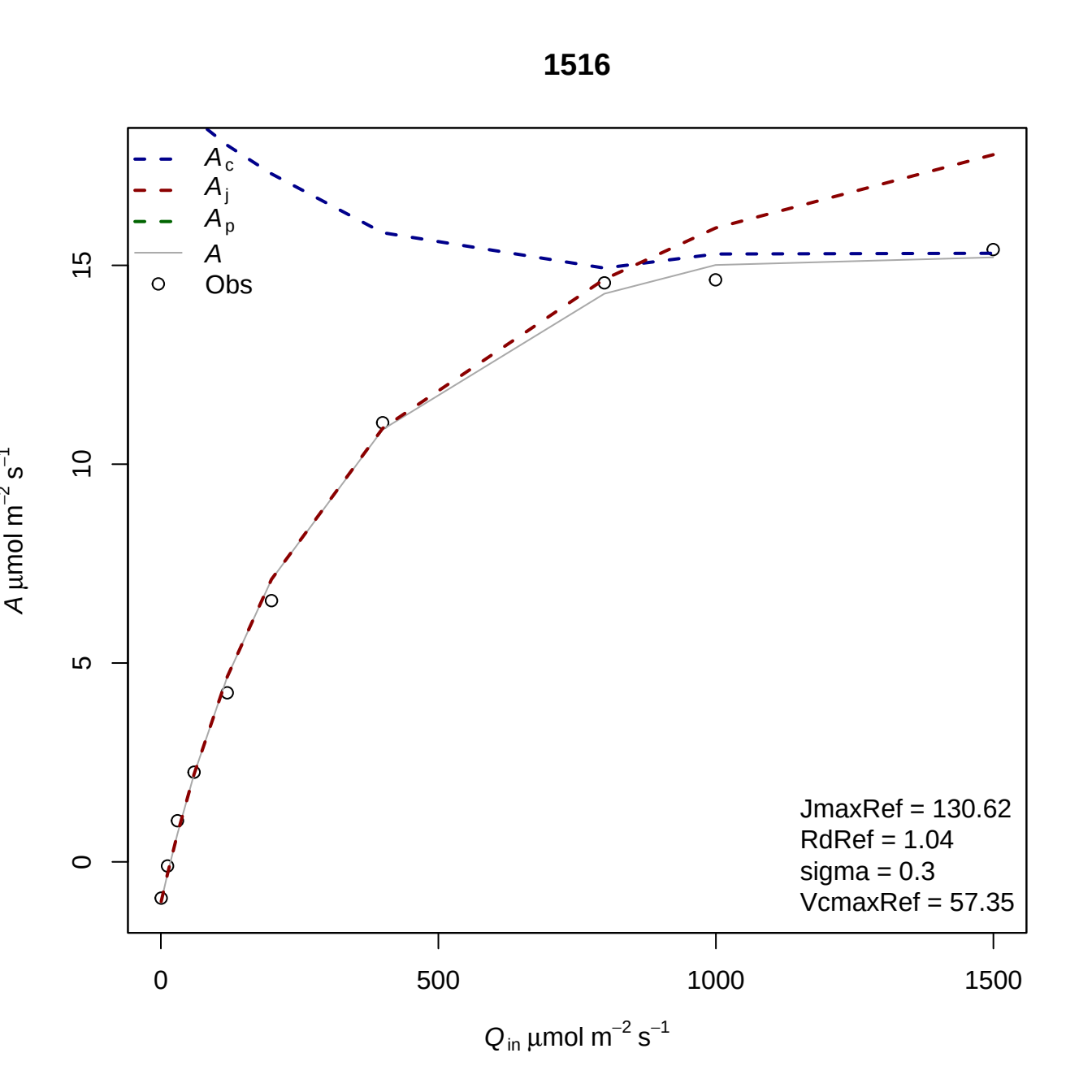

1511
A_c
A_j
A_p
A
Obs
JmaxRef = 112.78
RdRef = 1.07
sigma = 0.39
VcmaxRef = 49.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

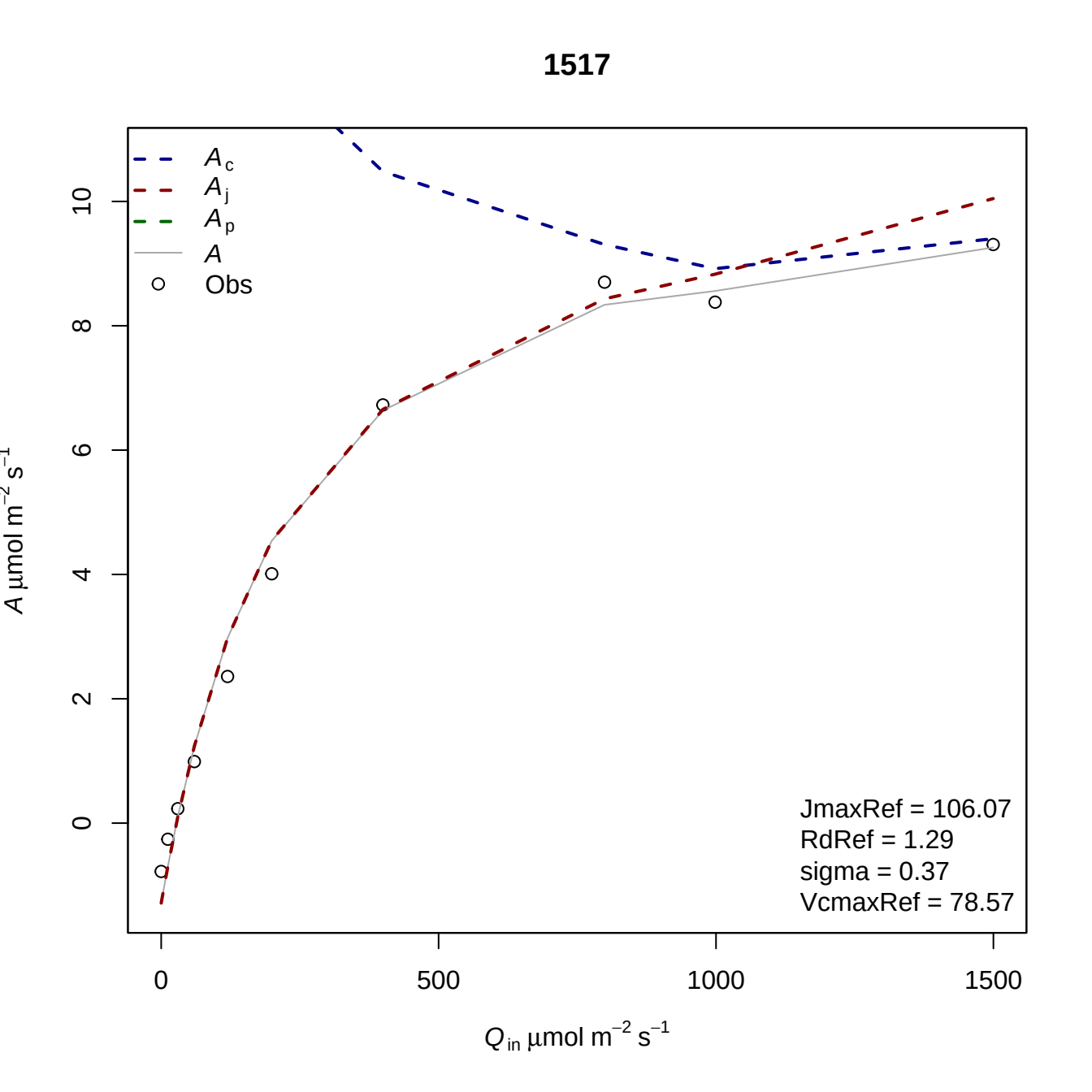

1512
A_c
A_j
A_p
A
Obs
JmaxRef = 134.25
RdRef = 1.55
sigma = 0.33
VcmaxRef = 63.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

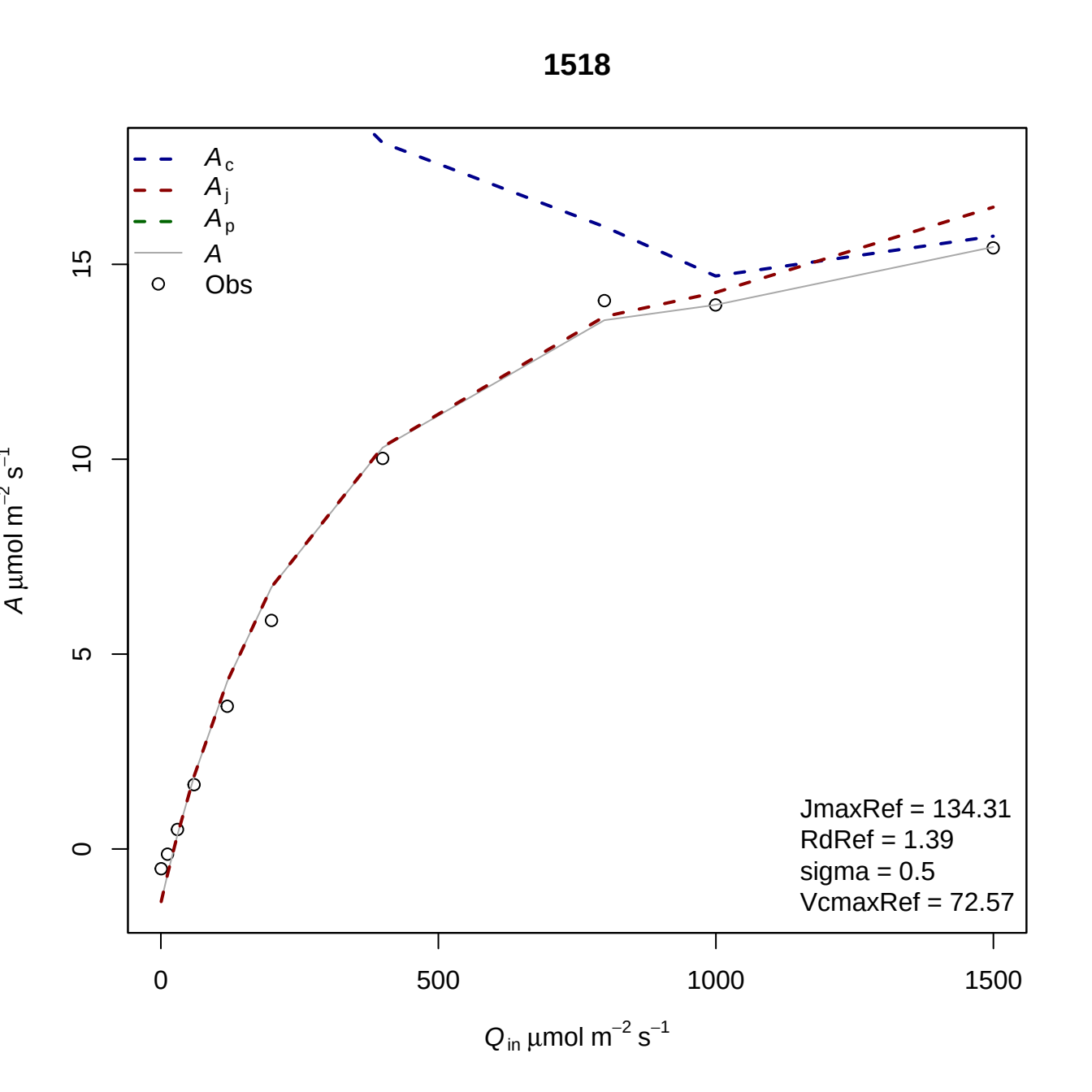

1513
A_c
A_j
A_p
A
Obs
JmaxRef = 38.27
RdRef = 2
sigma = 0.22
VcmaxRef = 4333.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500

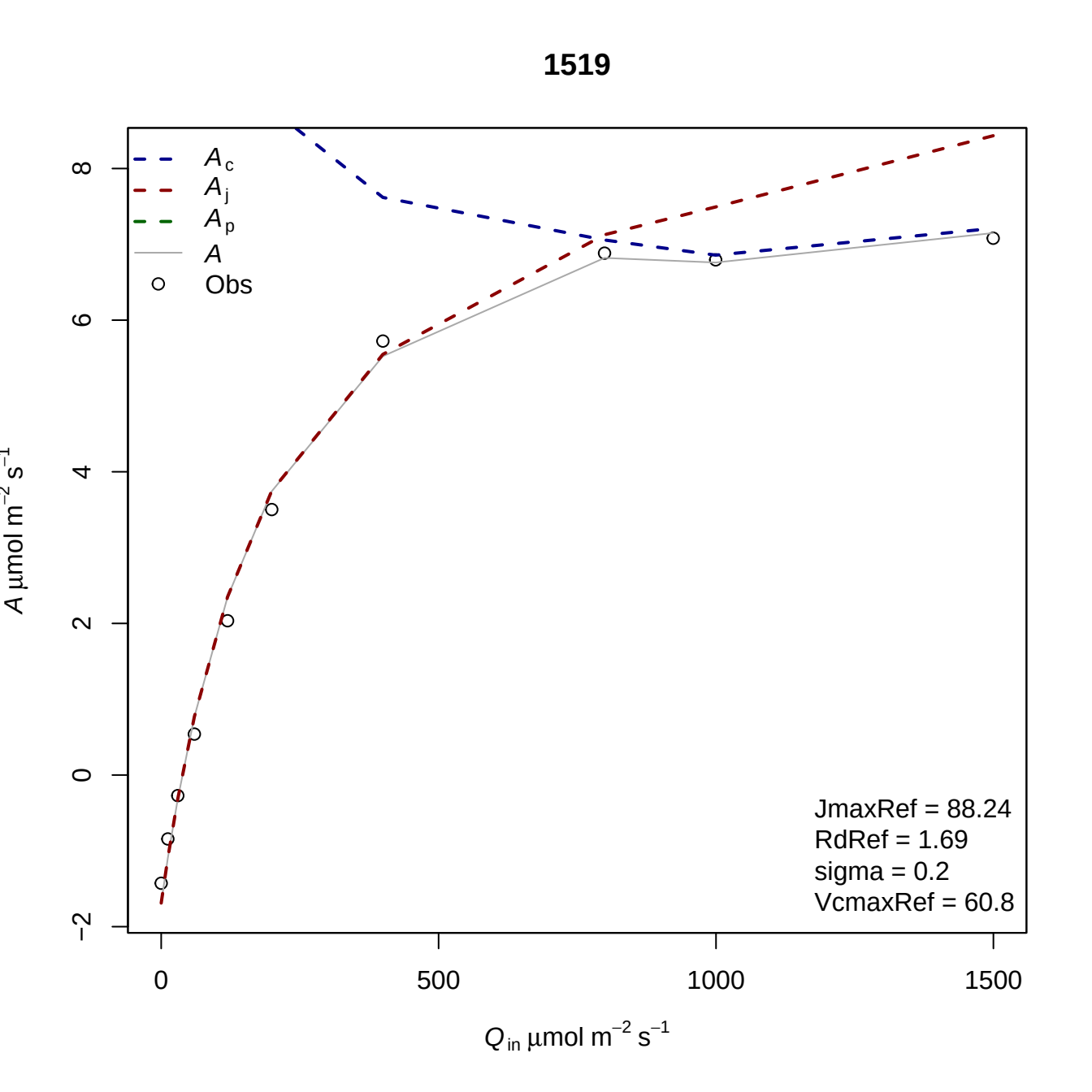

1514
A_c
A_j
A_p
A
Obs
JmaxRef = 109.57
RdRef = 1.7
sigma = 0.18
VcmaxRef = 53.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

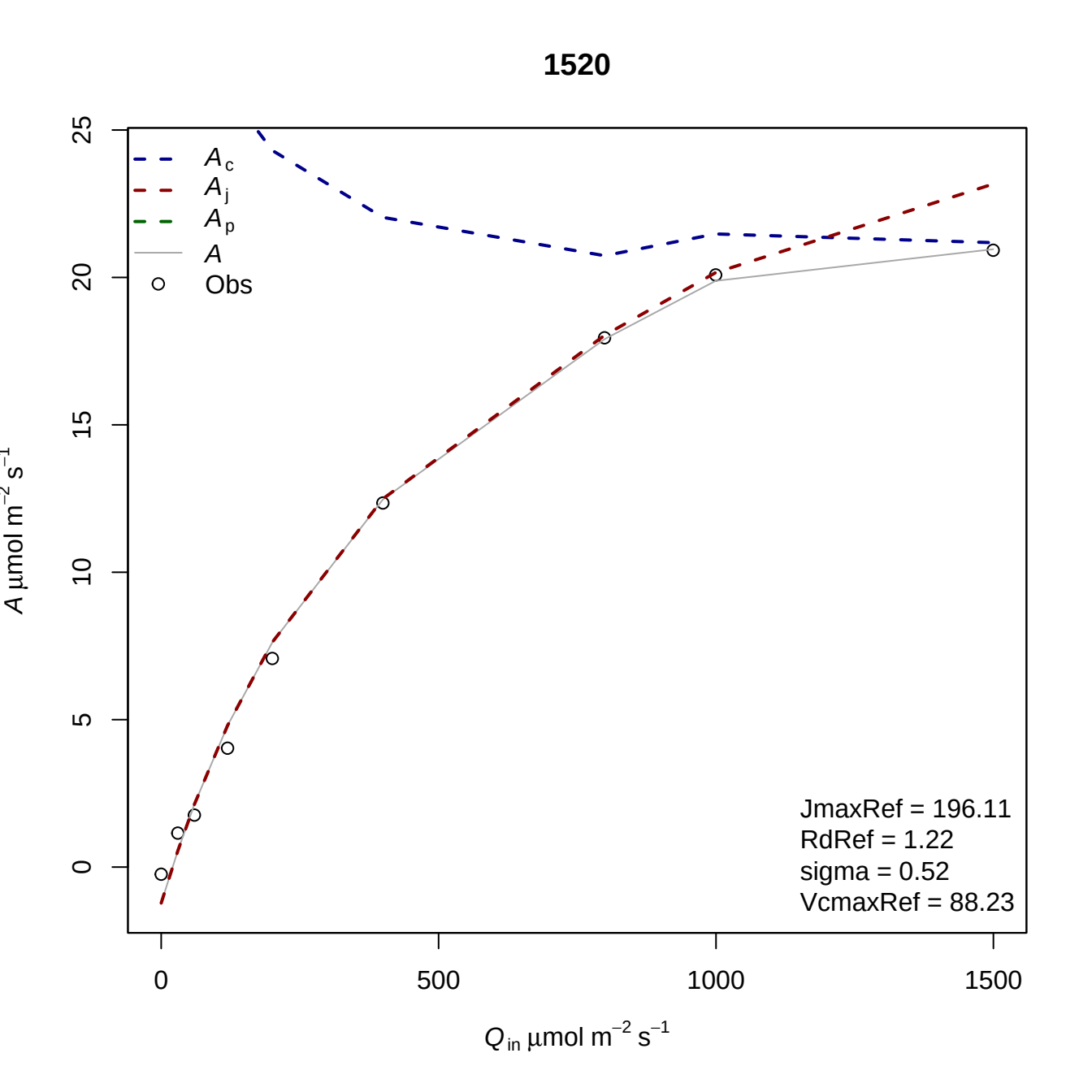

1515
A_c
A_j
A_p
A
Obs
JmaxRef = 43.94
RdRef = 1.21
sigma = 0.33
VcmaxRef = 29.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

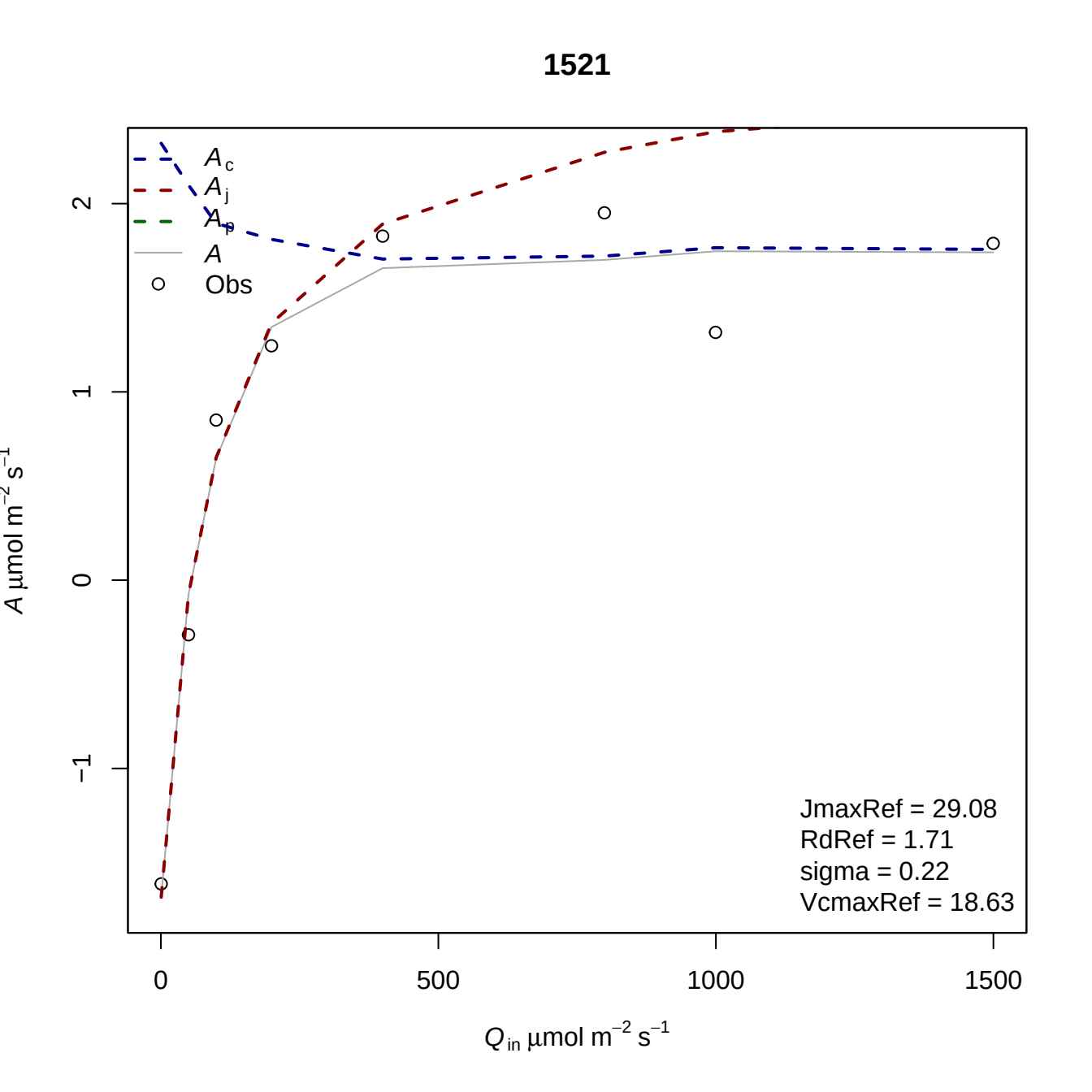

1516
A_c
A_j
A_p
A
Obs
JmaxRef = 130.62
RdRef = 1.04
sigma = 0.3
VcmaxRef = 57.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

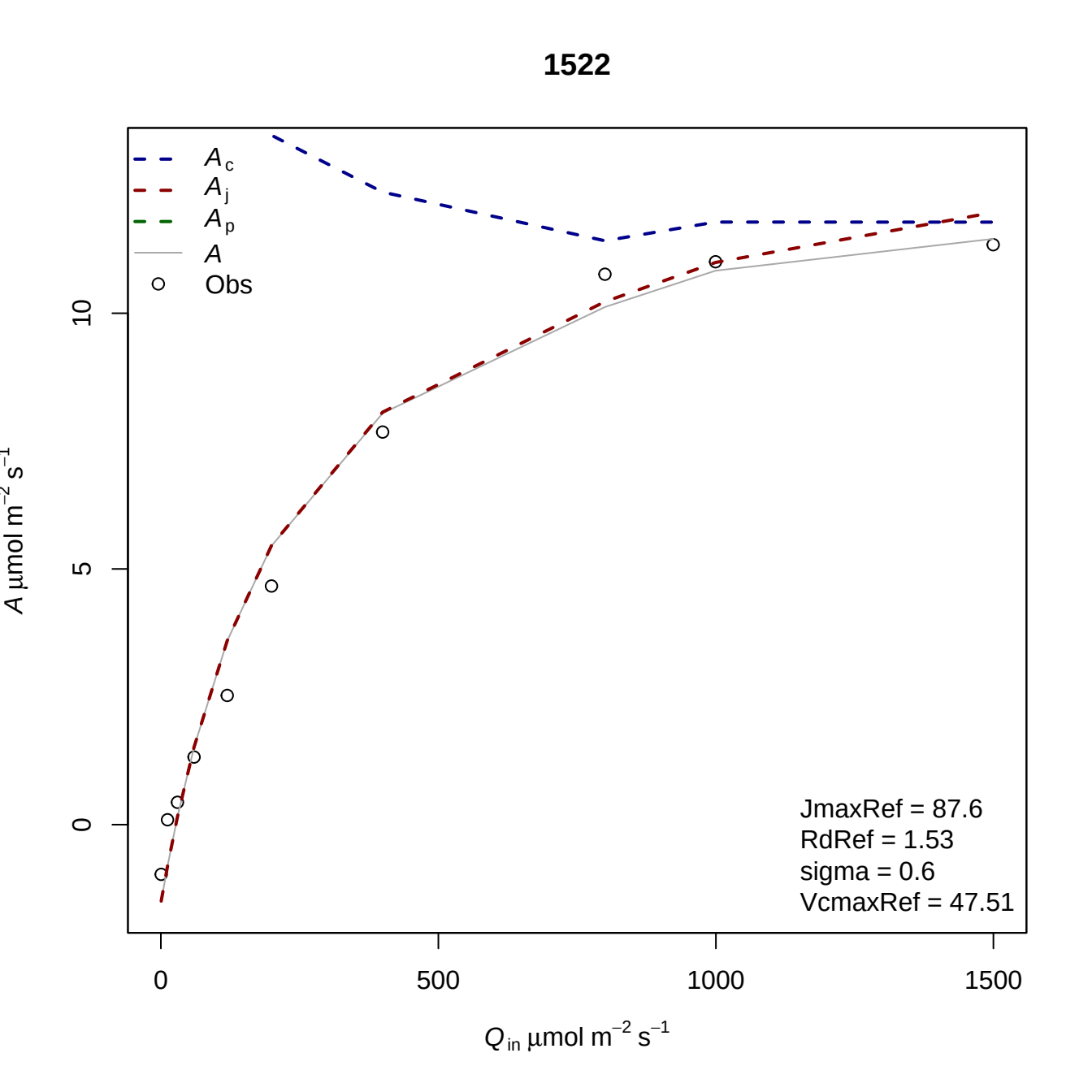

1517
A_c
A_j
A_p
A
Obs
JmaxRef = 106.07
RdRef = 1.29
sigma = 0.37
VcmaxRef = 78.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

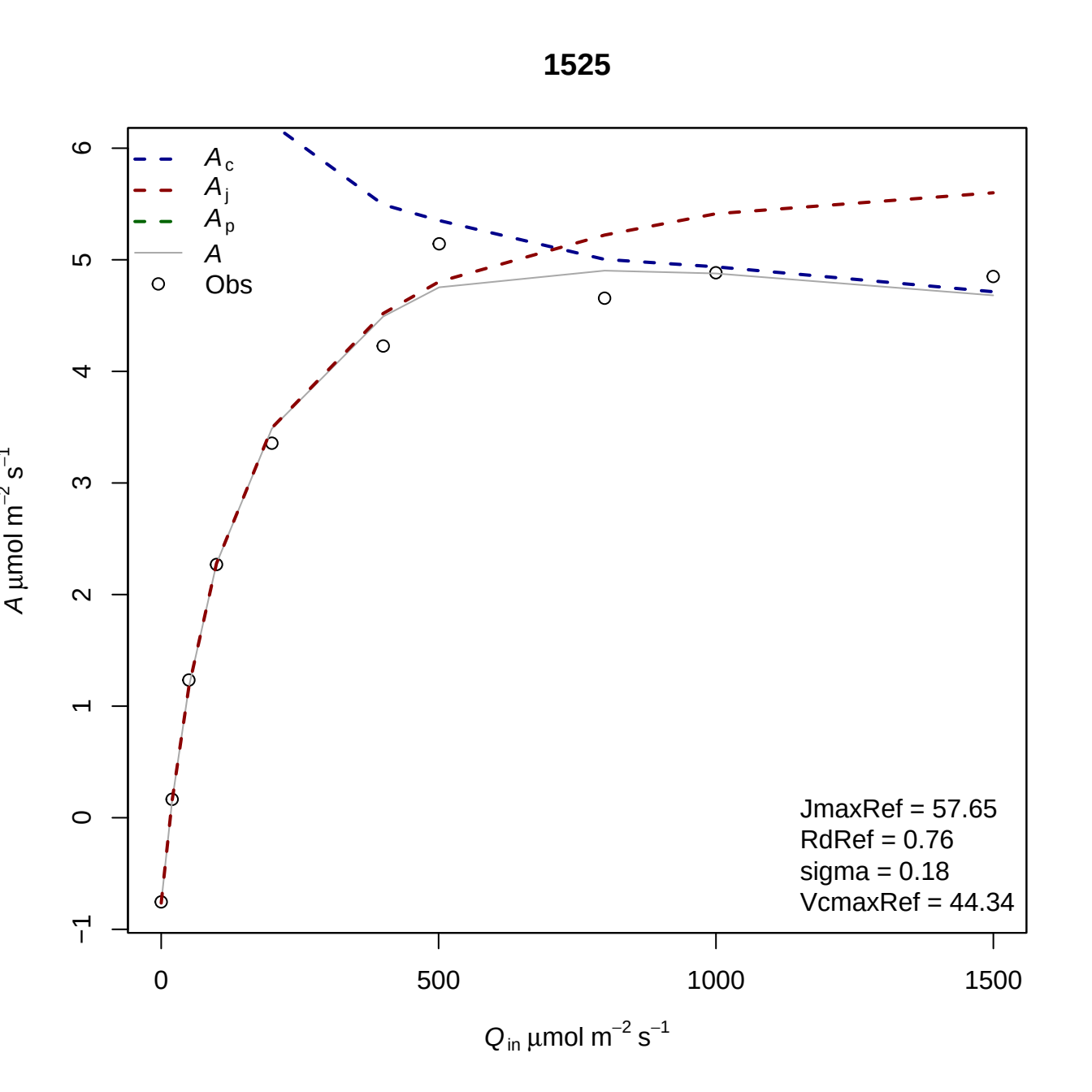

1518
A_c
A_j
A_p
A
Obs
JmaxRef = 134.31
RdRef = 1.39
sigma = 0.5
VcmaxRef = 72.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

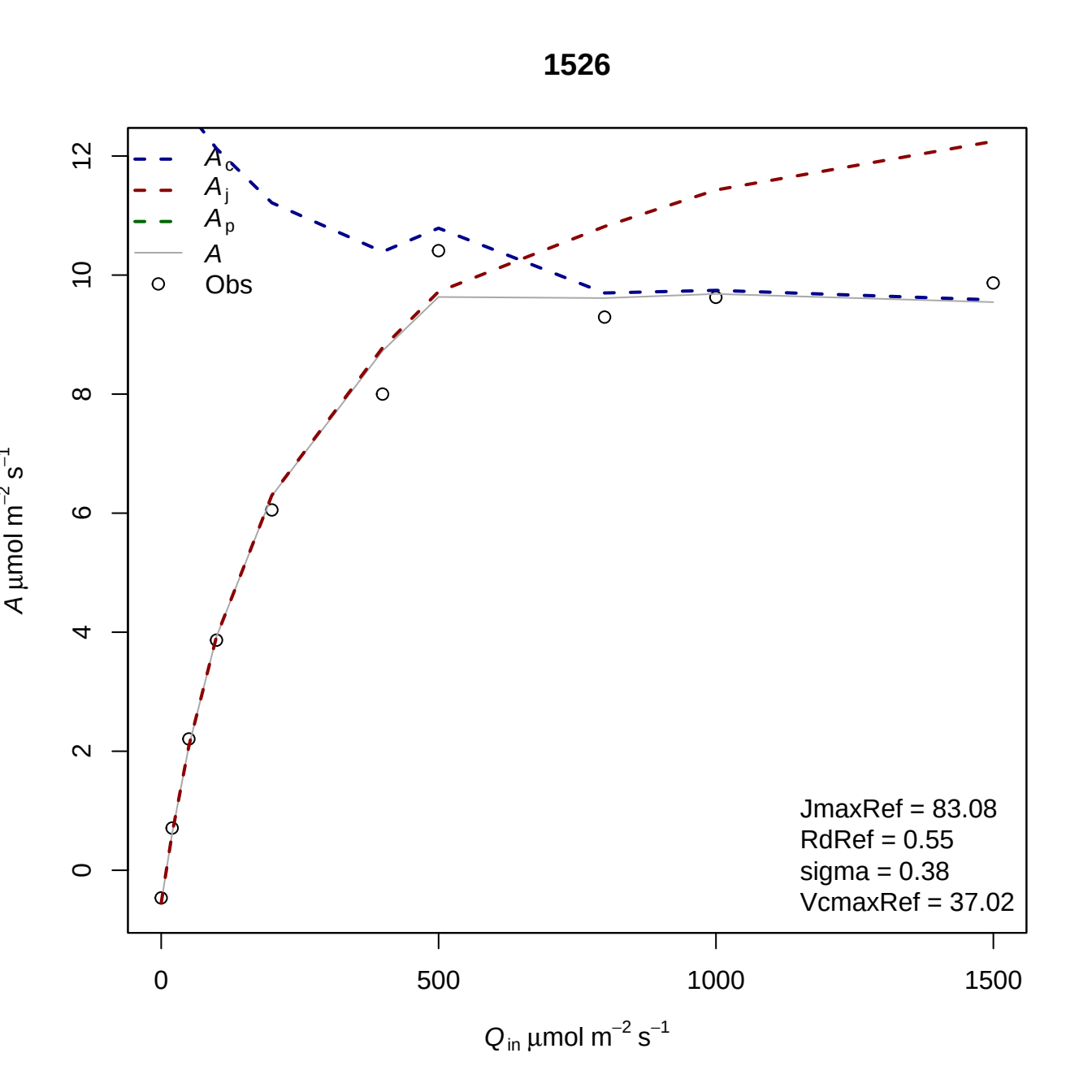

1519
A_c
A_j
A_p
A
Obs
JmaxRef = 88.24
RdRef = 1.69
sigma = 0.2
VcmaxRef = 60.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

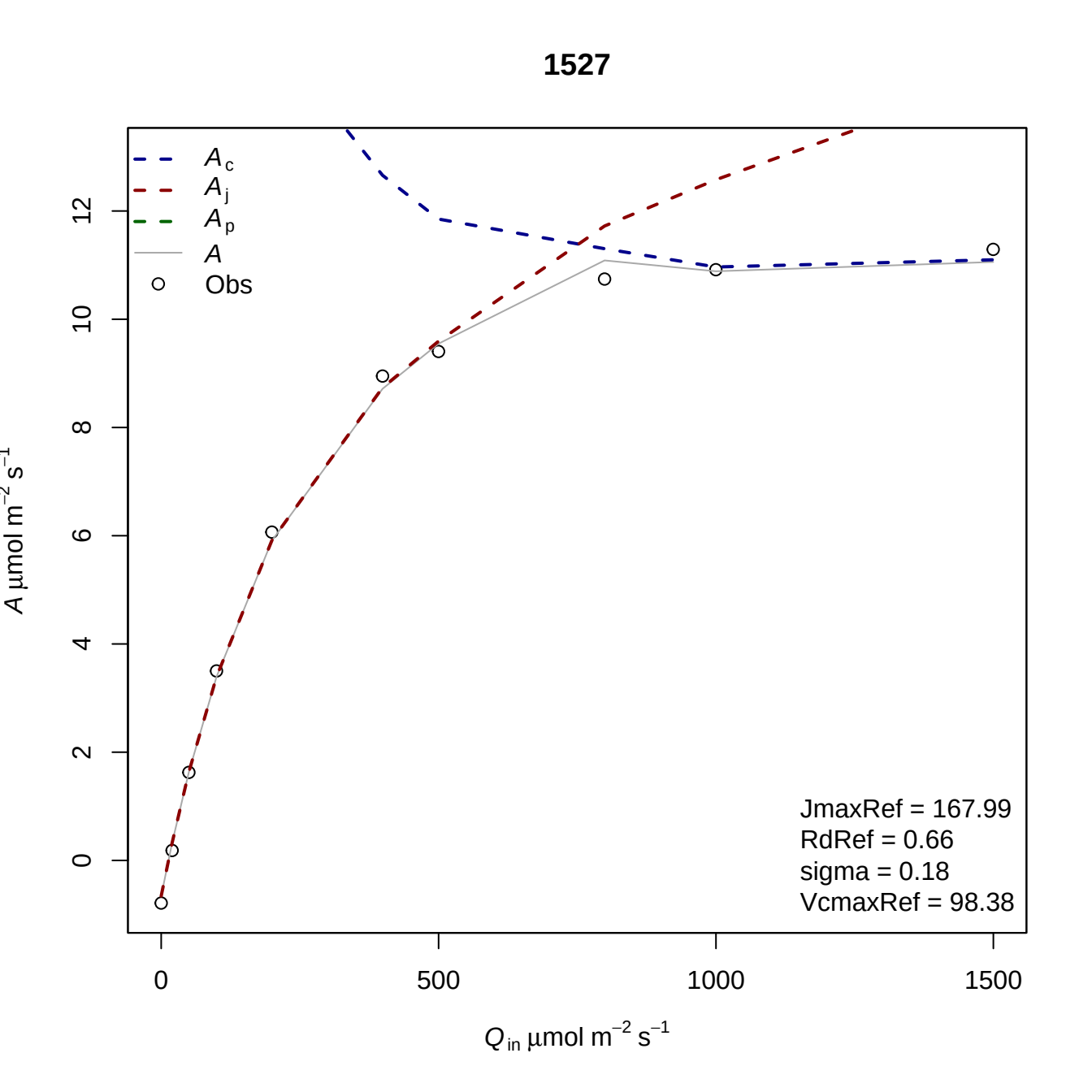

1520
A_c
A_j
A_p
A
Obs
JmaxRef = 196.11
RdRef = 1.22
sigma = 0.52
VcmaxRef = 88.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

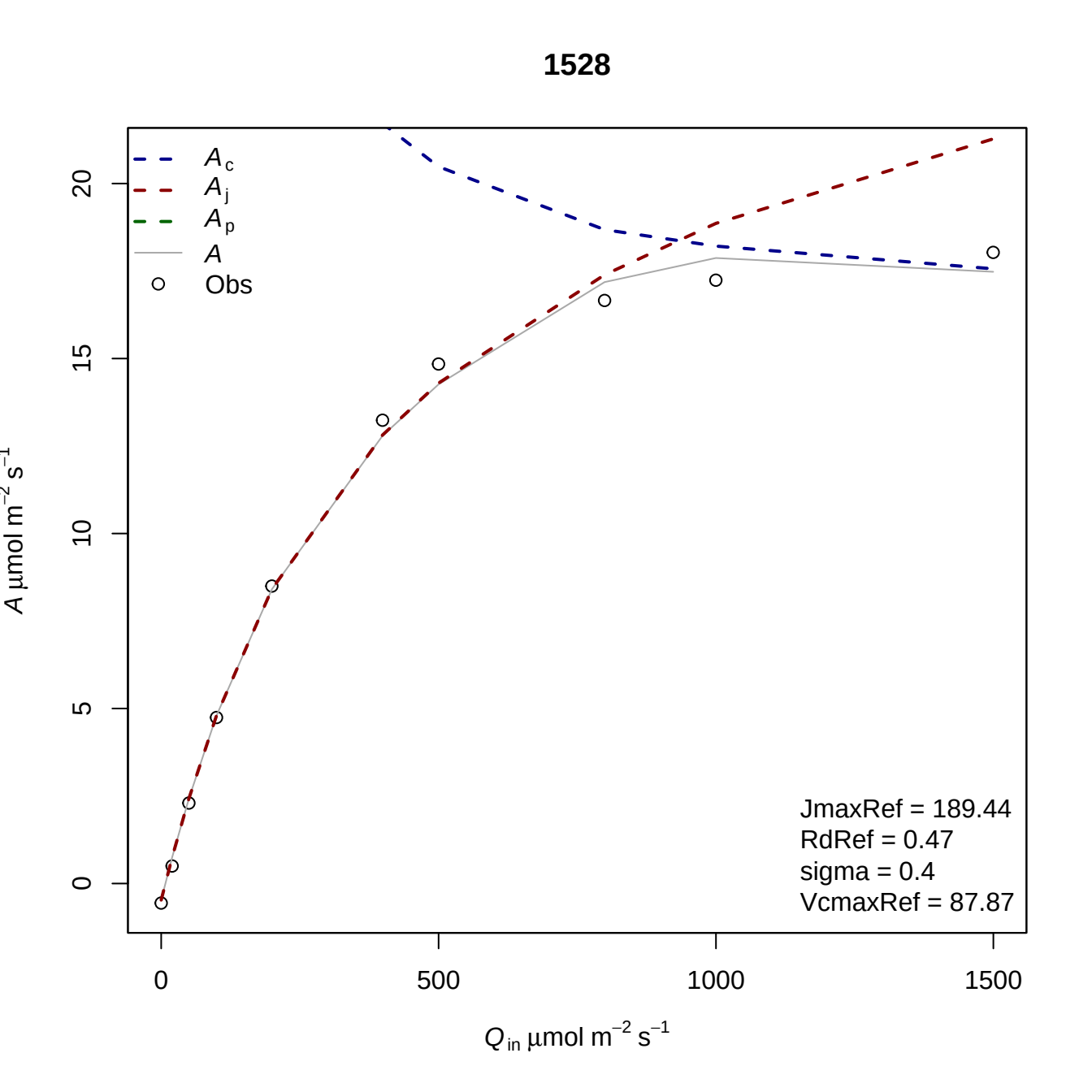

1521
A_c
A_j
A_p
A
Obs
JmaxRef = 29.08
RdRef = 1.71
sigma = 0.22
VcmaxRef = 18.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
0
500
1000
1500

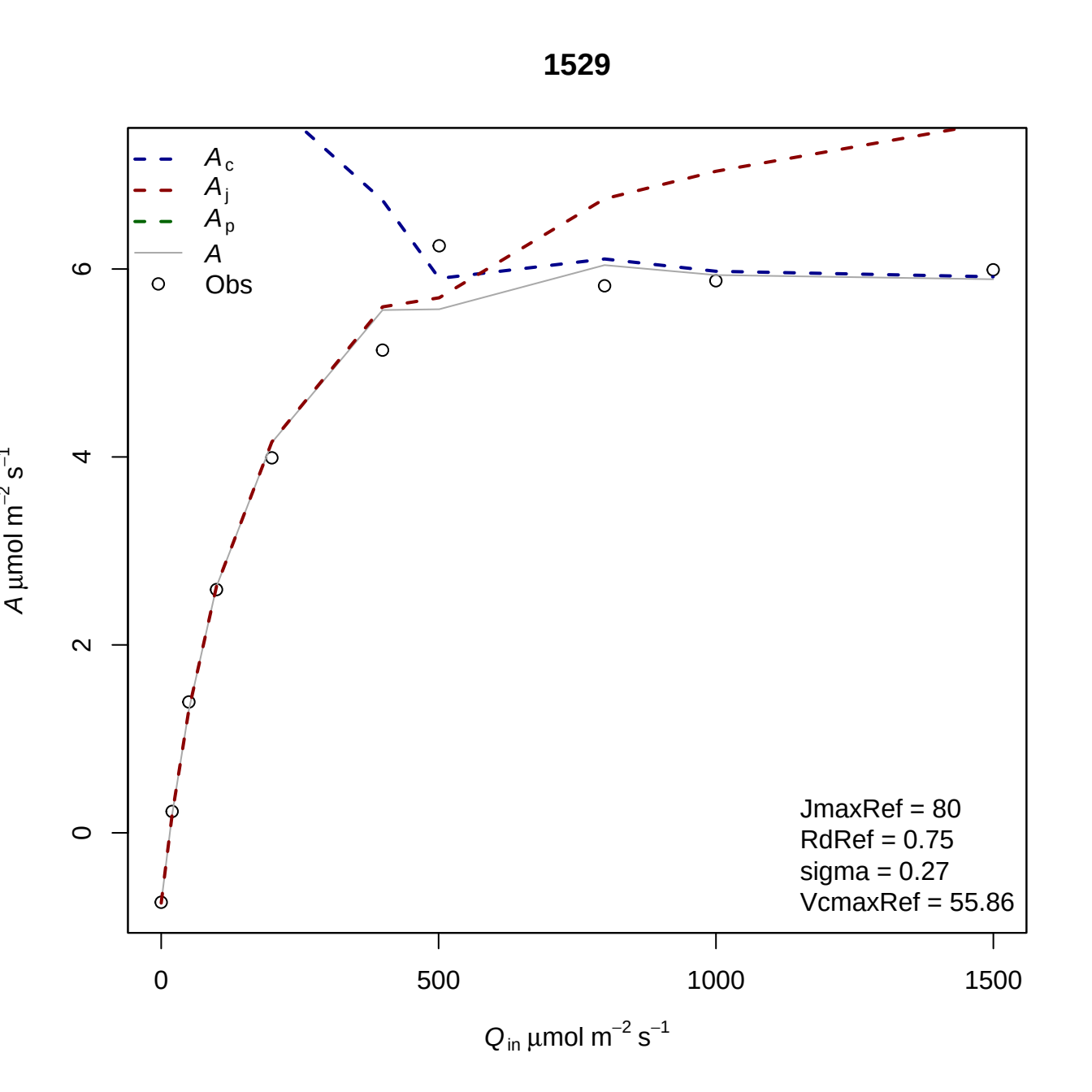

1522
A_c
A_j
A_p
A
Obs
JmaxRef = 87.6
RdRef = 1.53
sigma = 0.6
VcmaxRef = 47.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

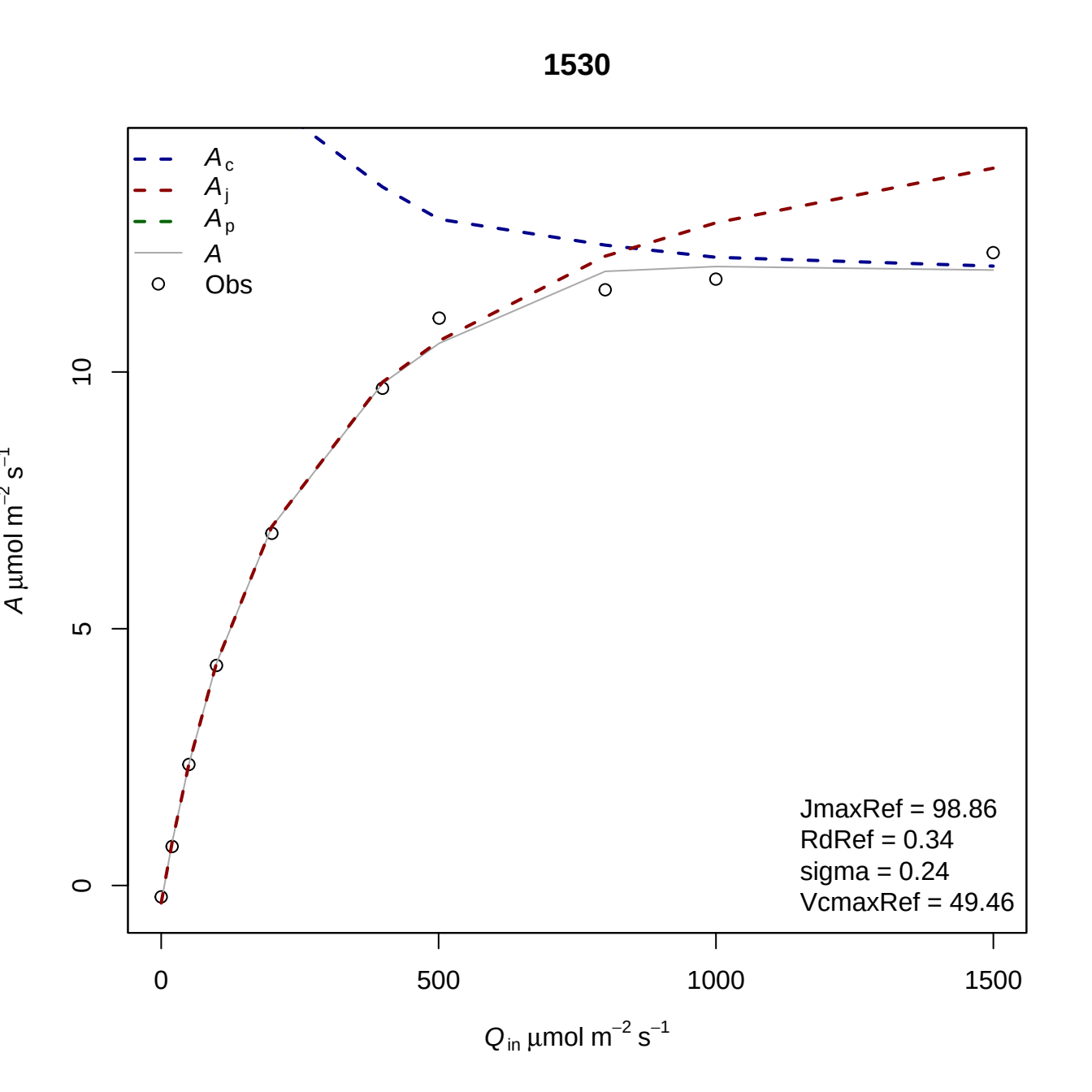

1525
A_c
A_j
A_p
A
Obs
JmaxRef = 57.65
RdRef = 0.76
sigma = 0.18
VcmaxRef = 44.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

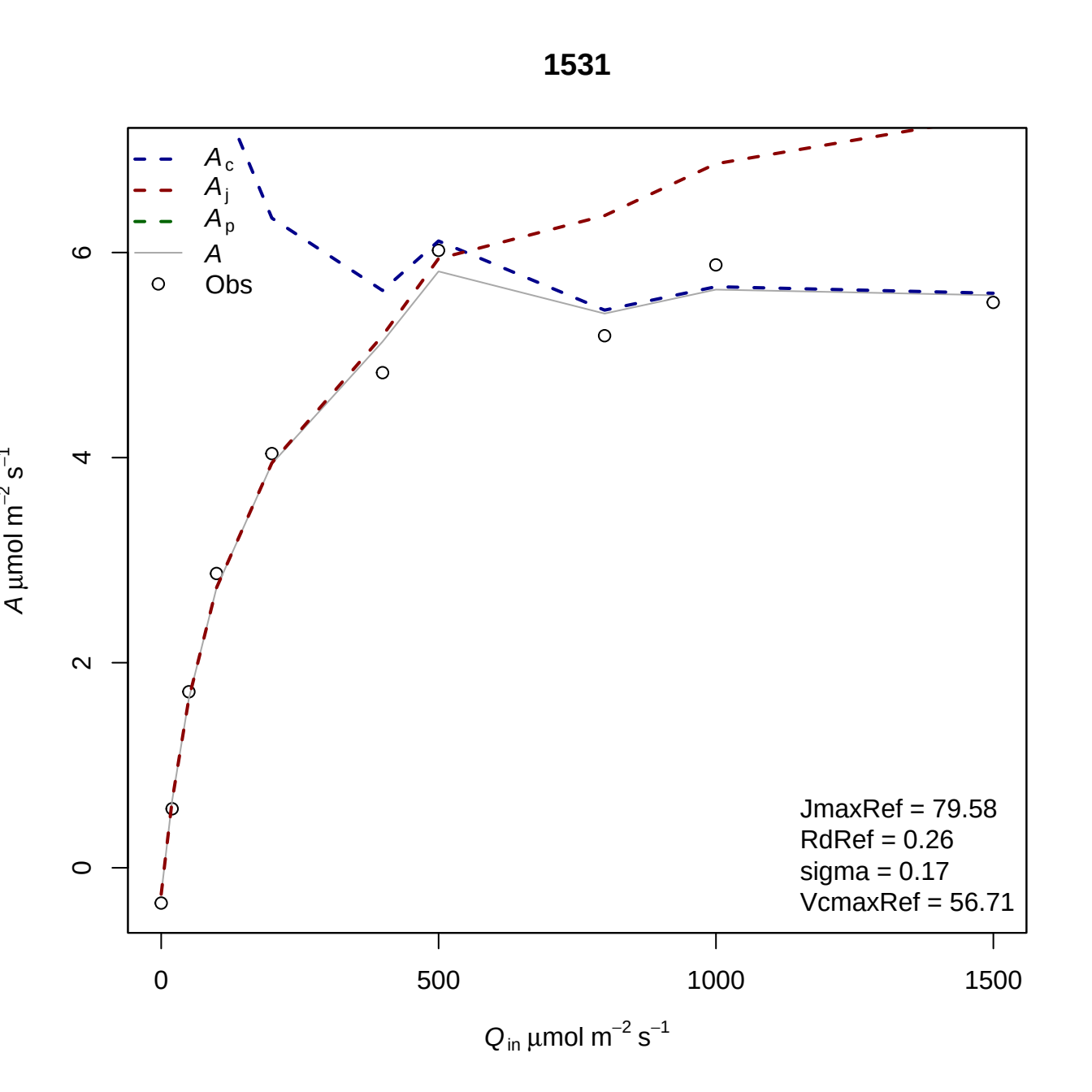

1526
A_c
A_j
A_p
A
Obs
JmaxRef = 83.08
RdRef = 0.55
sigma = 0.38
VcmaxRef = 37.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

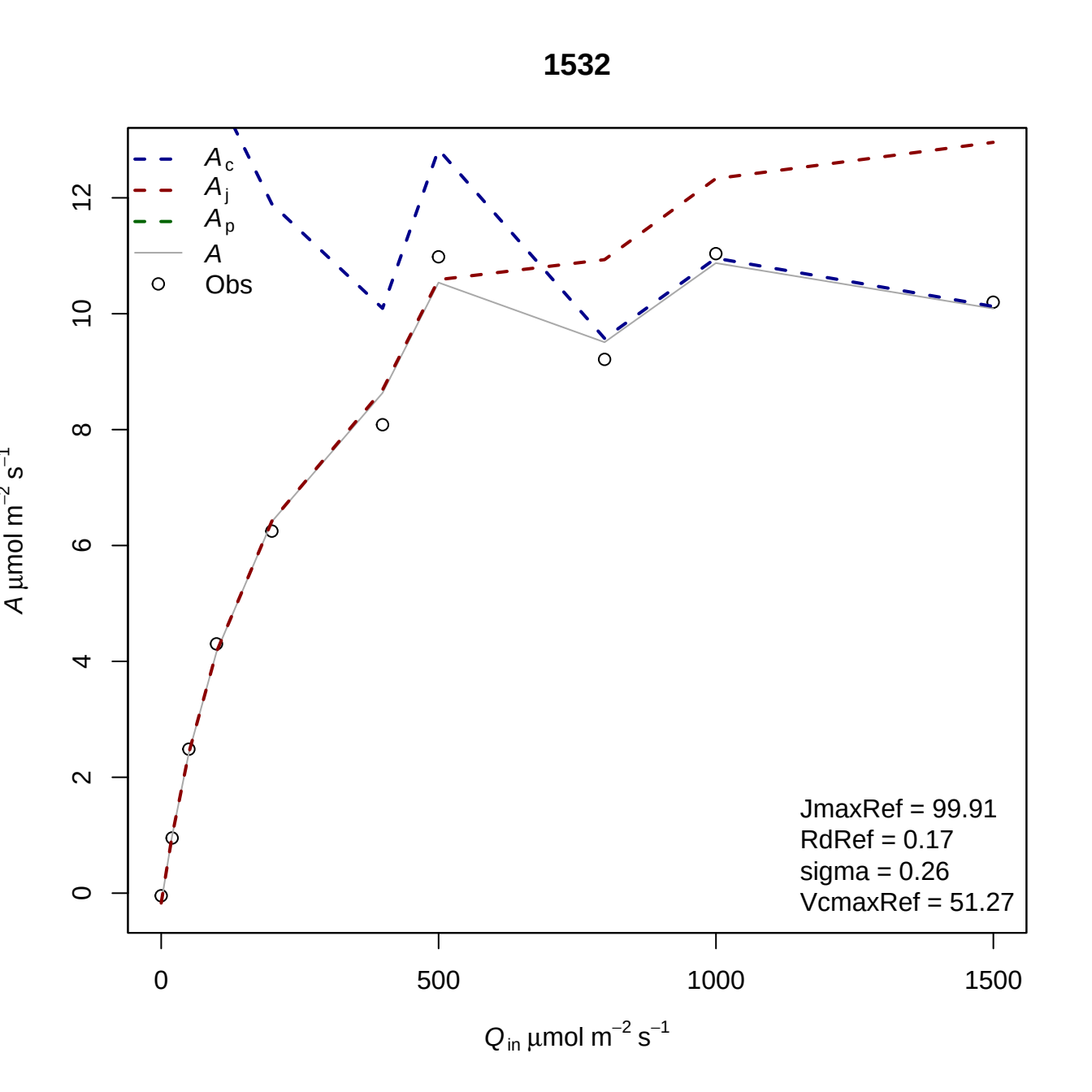

1527
A_c
A_j
A_p
A
Obs
JmaxRef = 167.99
RdRef = 0.66
sigma = 0.18
VcmaxRef = 98.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

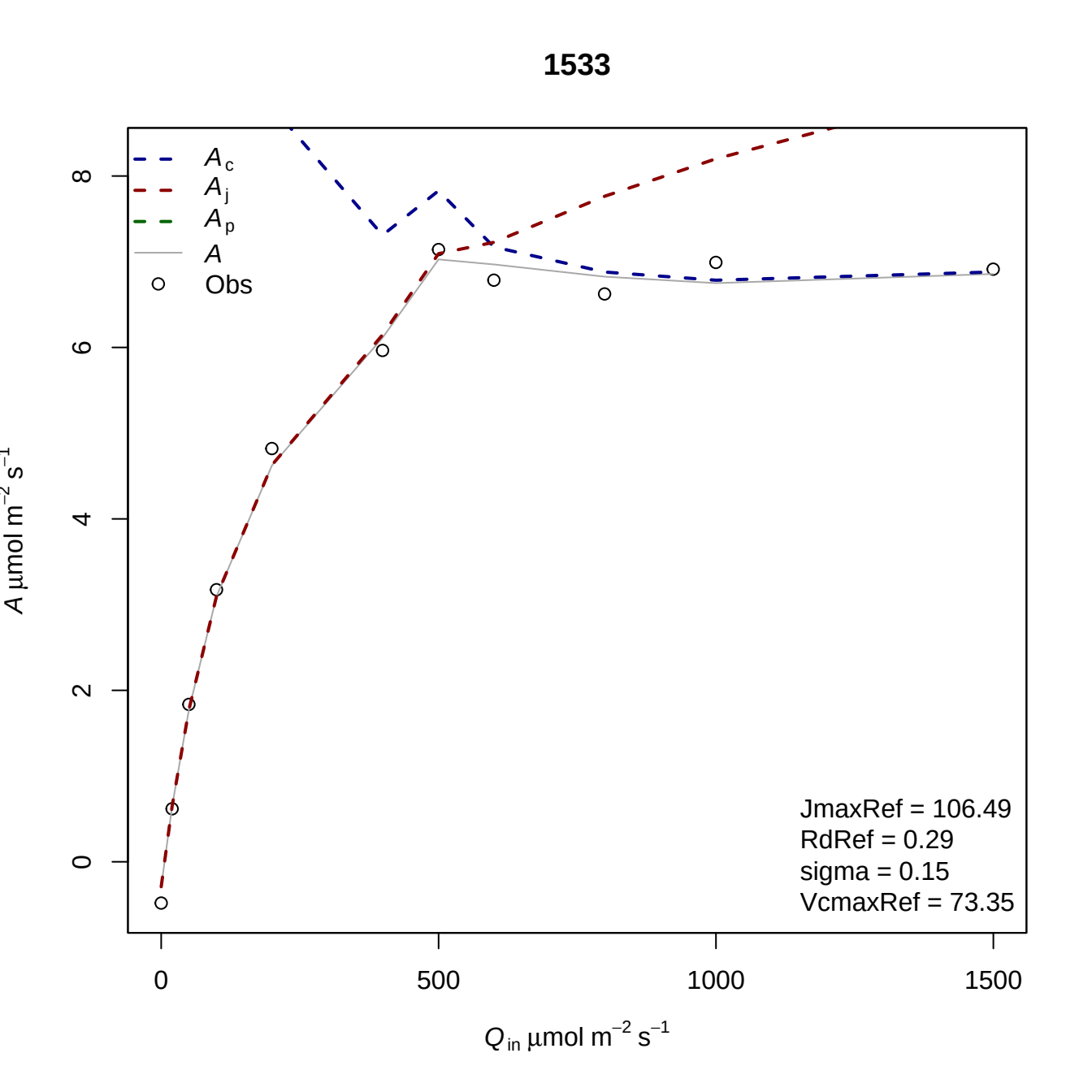

1528
A_c
A_j
A_p
A
Obs
JmaxRef = 189.44
RdRef = 0.47
sigma = 0.4
VcmaxRef = 87.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

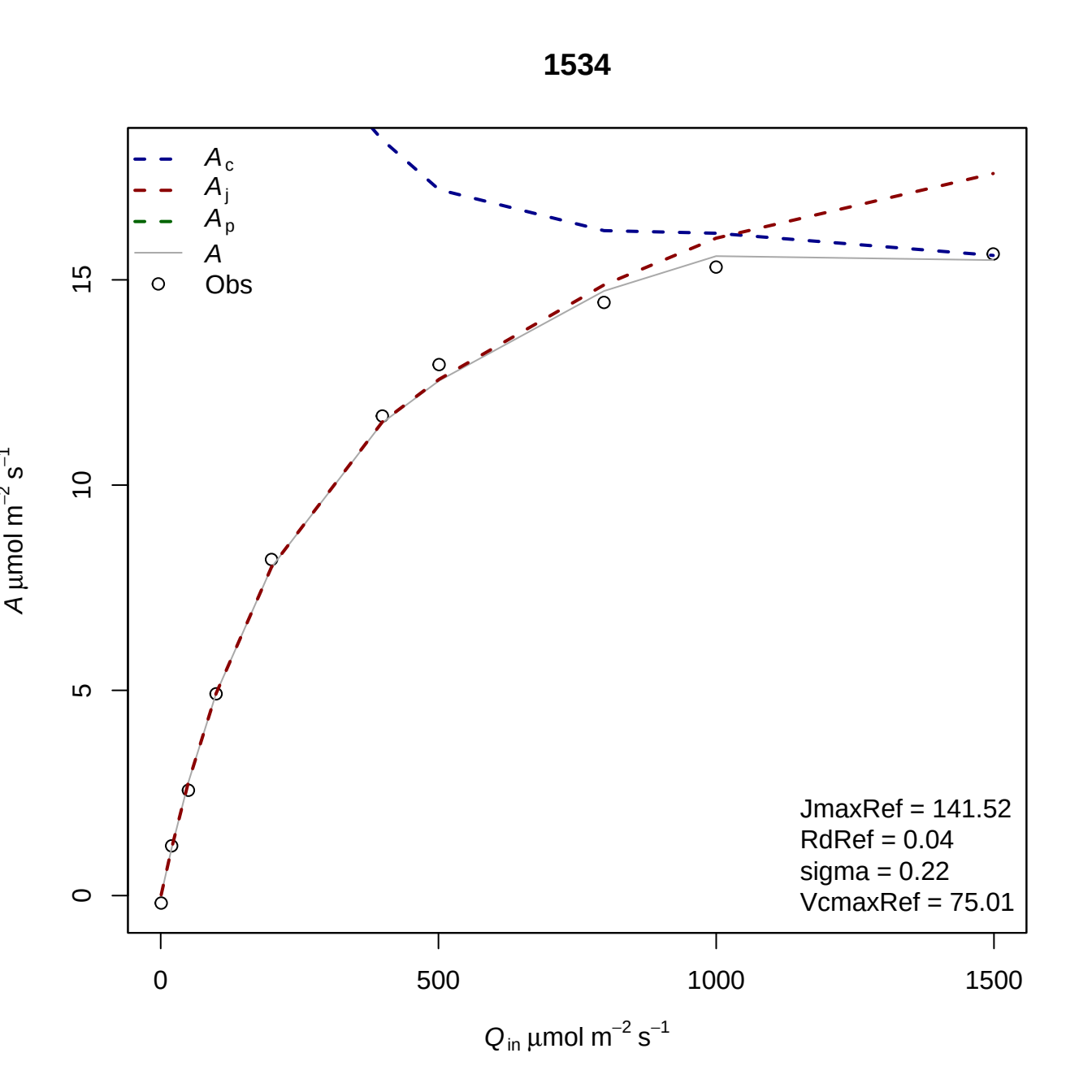

1529
A_c
A_j
A_p
A
Obs
JmaxRef = 80
RdRef = 0.75
sigma = 0.27
VcmaxRef = 55.86
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

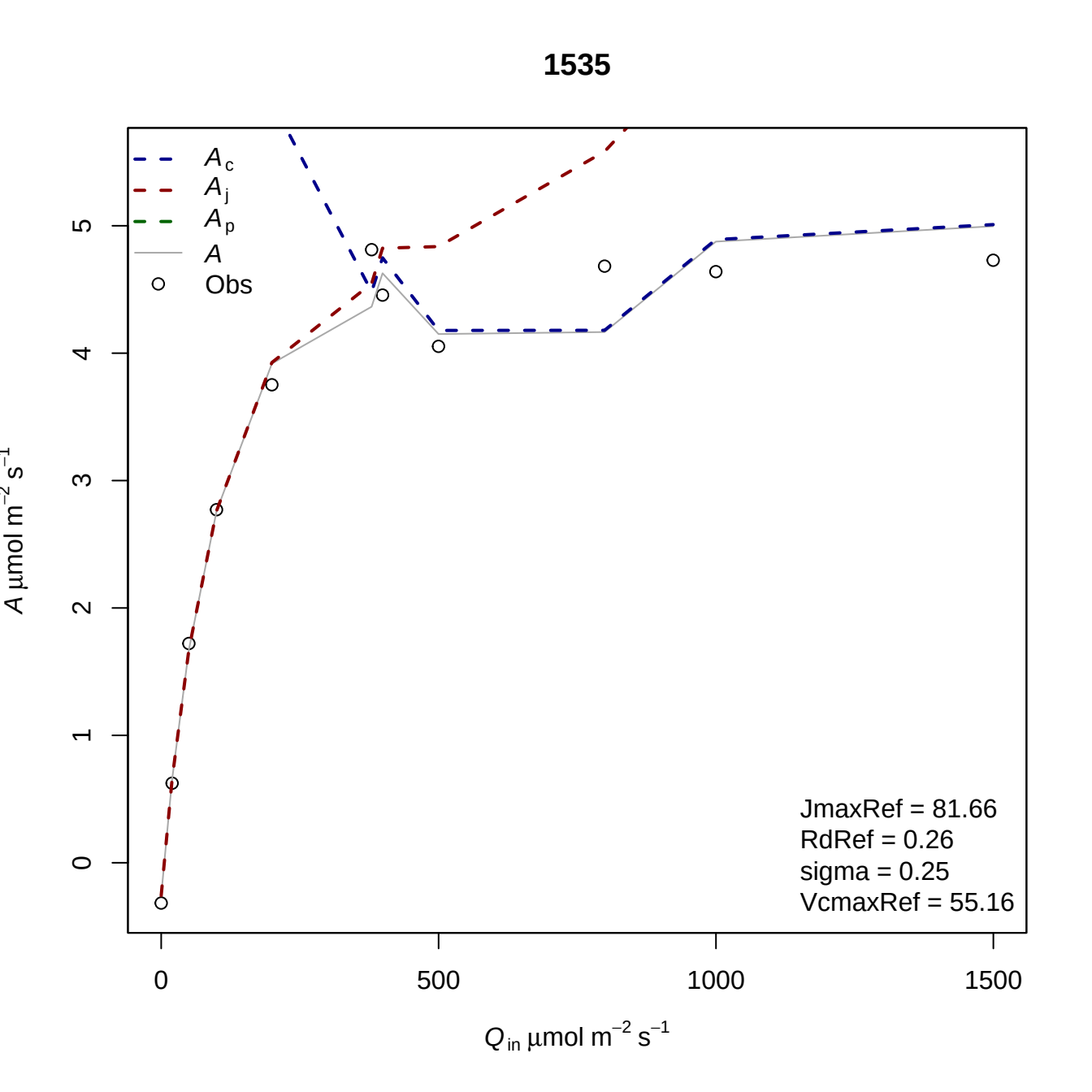

1530
A_c
A_j
A_p
A
Obs
JmaxRef = 98.86
RdRef = 0.34
sigma = 0.24
VcmaxRef = 49.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

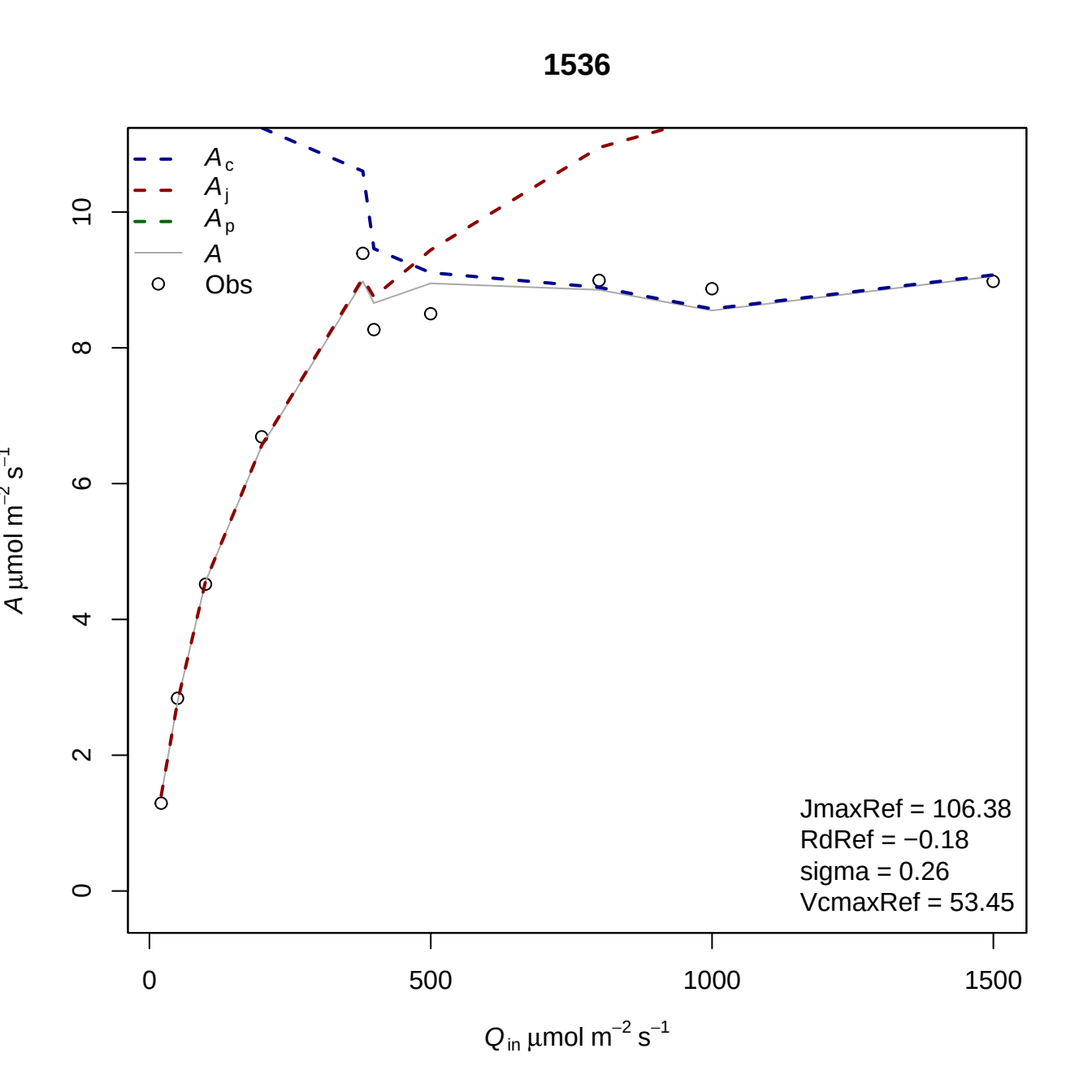

1531
A_c
A_j
A_p
A
Obs
JmaxRef = 79.58
RdRef = 0.26
sigma = 0.17
VcmaxRef = 56.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

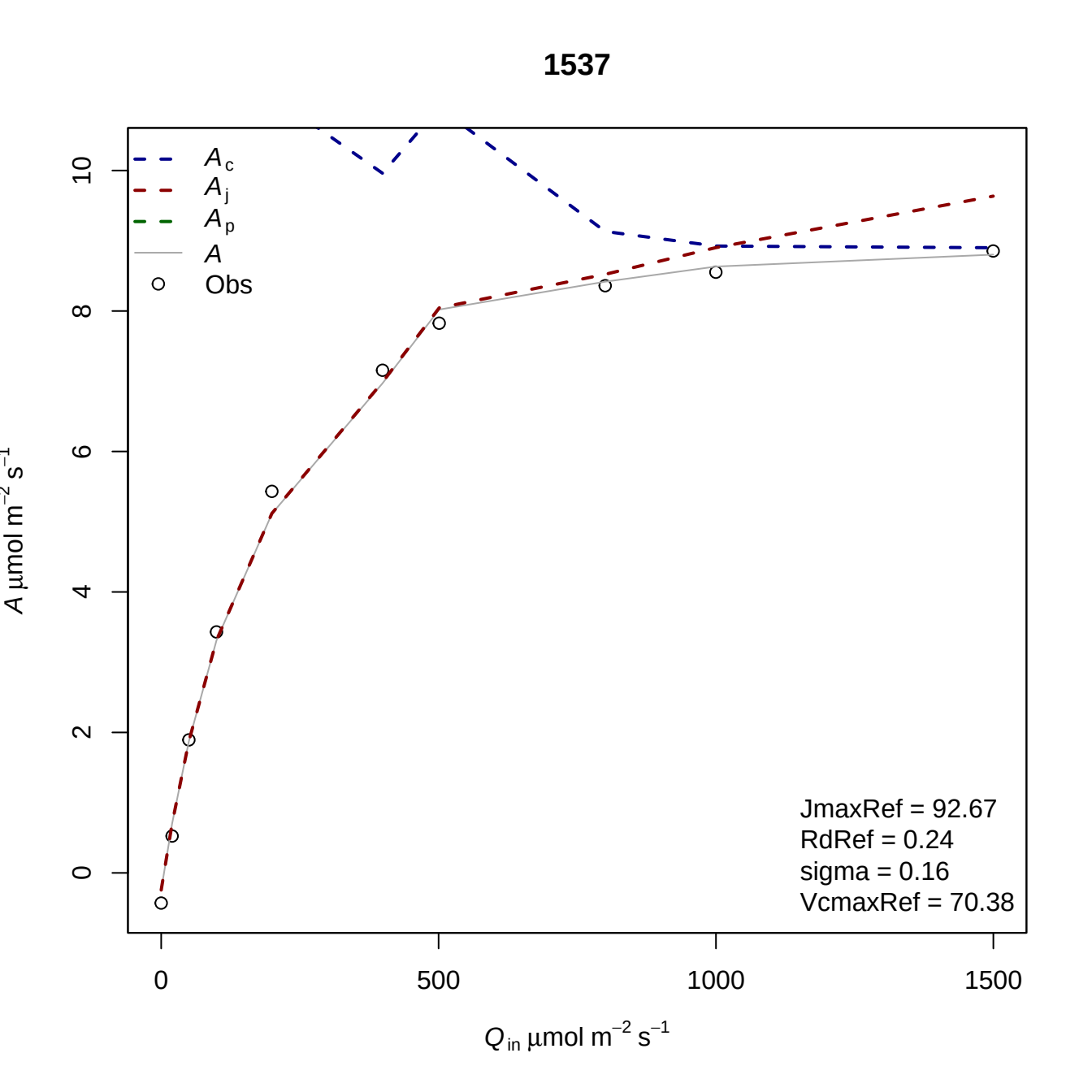

1532
A_c
A_j
A_p
A
Obs
JmaxRef = 99.91
RdRef = 0.17
sigma = 0.26
VcmaxRef = 51.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

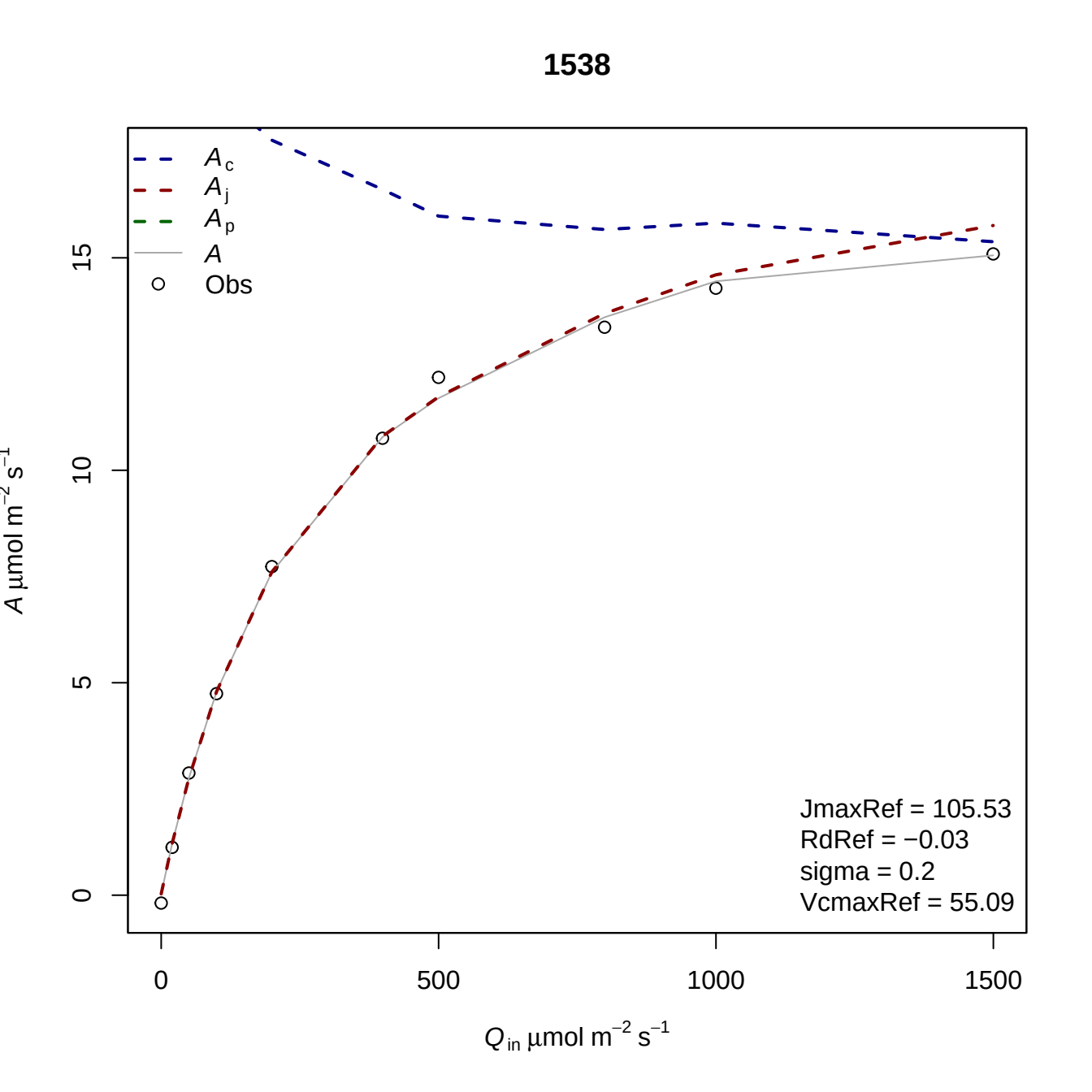

1533
A_c
A_j
A_p
A
Obs
JmaxRef = 106.49
RdRef = 0.29
sigma = 0.15
VcmaxRef = 73.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

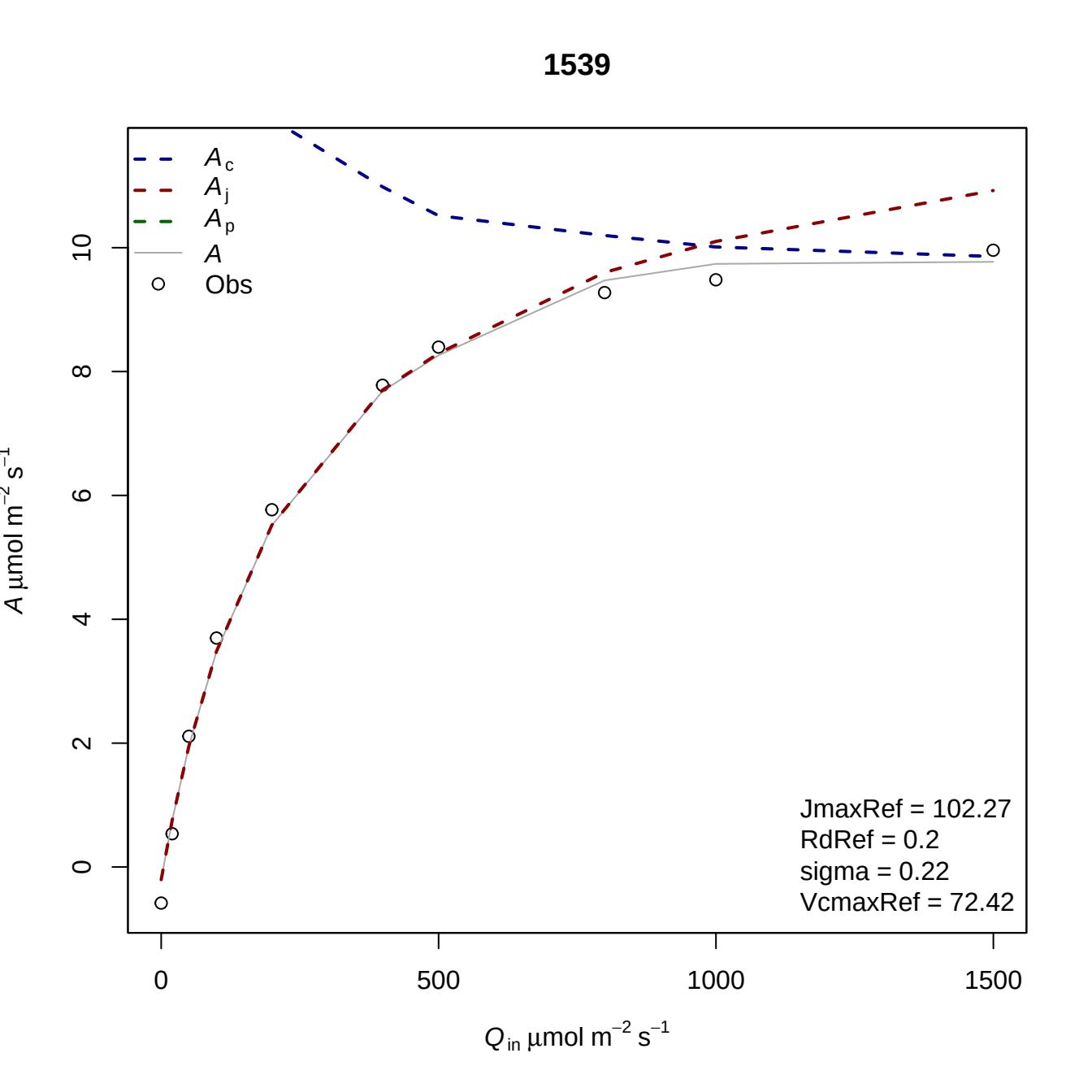

1534
A_c
A_j
A_p
A
Obs
JmaxRef = 141.52
RdRef = 0.04
sigma = 0.22
VcmaxRef = 75.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

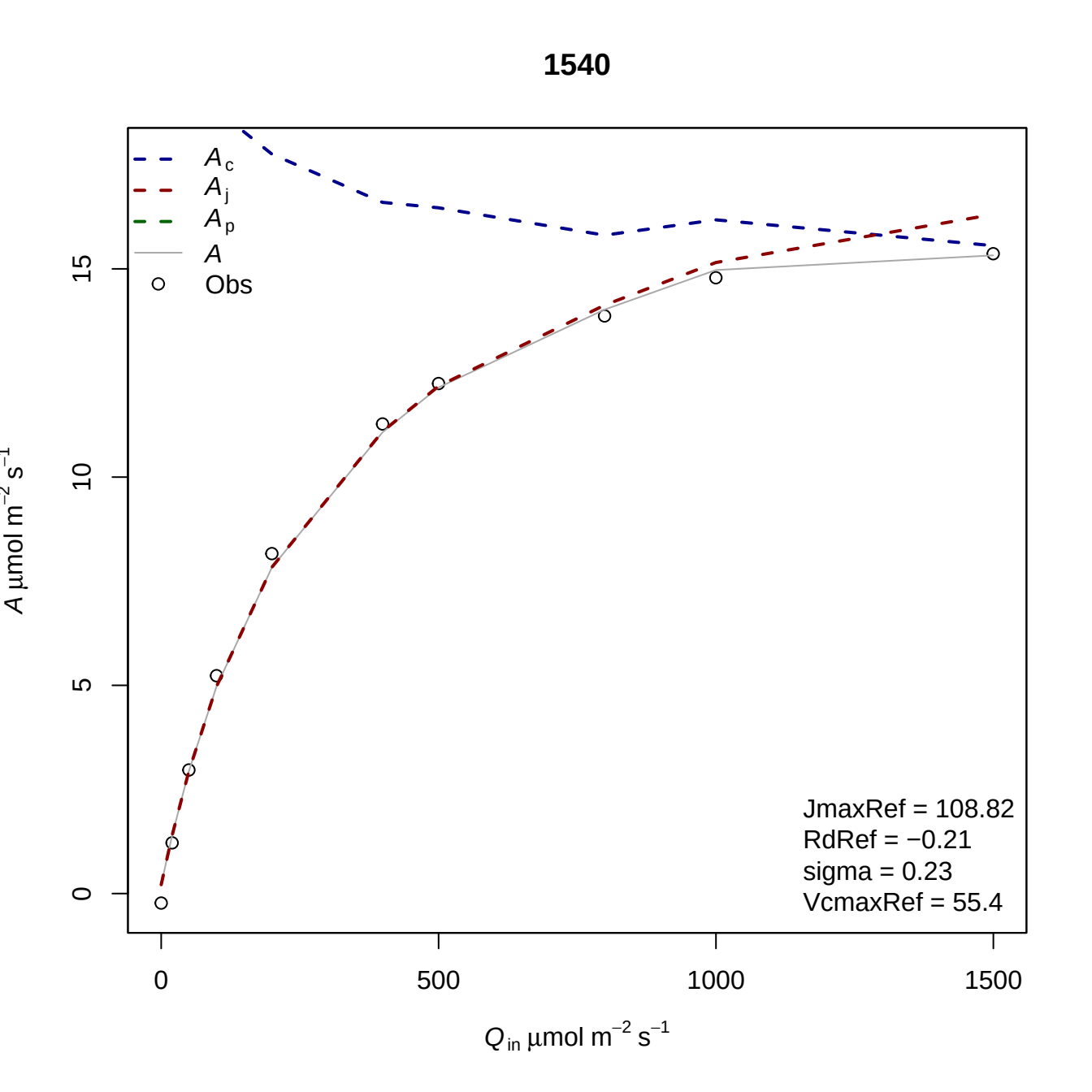

1535
A_c
A_j
A_p
A
Obs
JmaxRef = 81.66
RdRef = 0.26
sigma = 0.25
VcmaxRef = 55.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

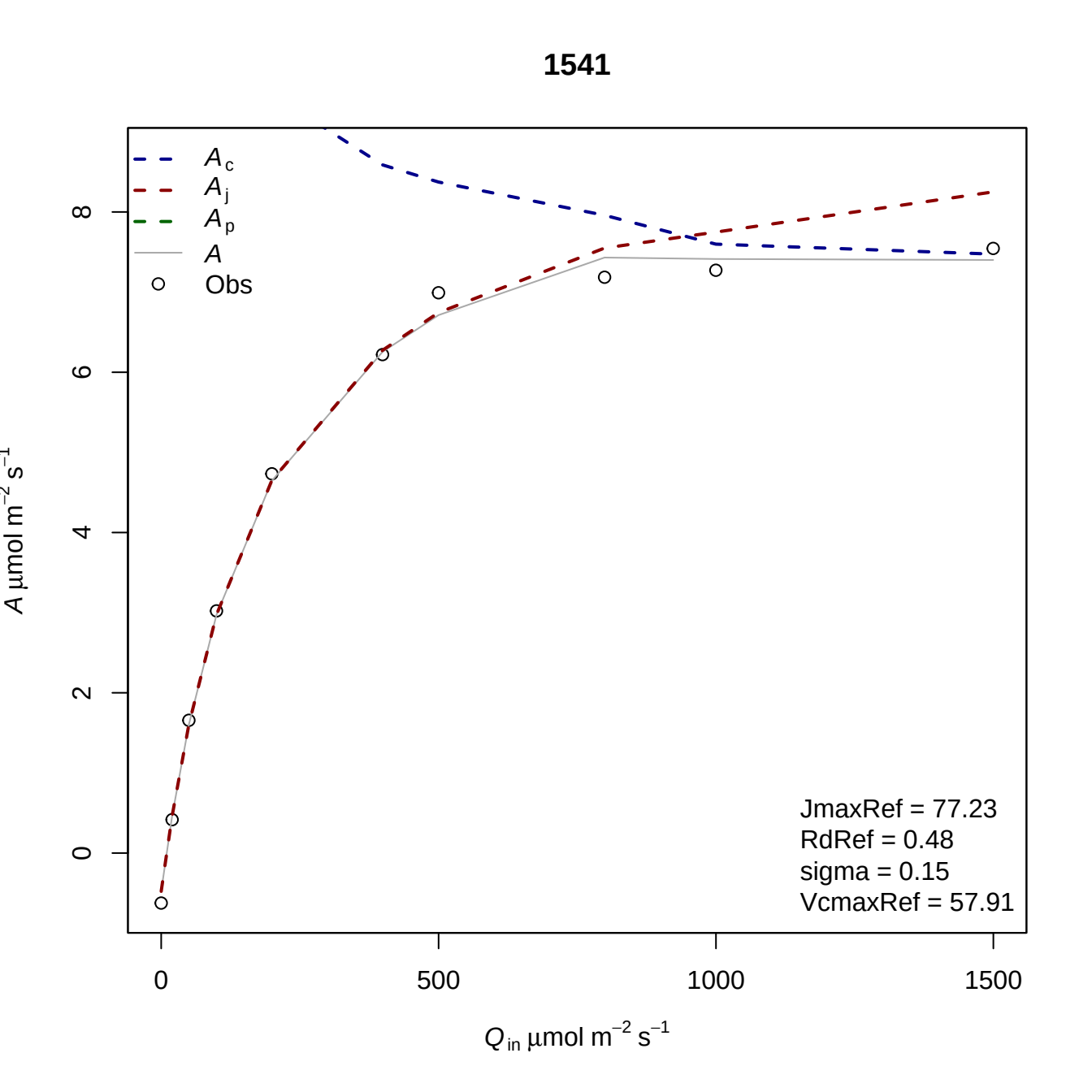

1536
A_c
A_j
A_p
A
Obs
JmaxRef = 106.38
RdRef = −0.18
sigma = 0.26
VcmaxRef = 53.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

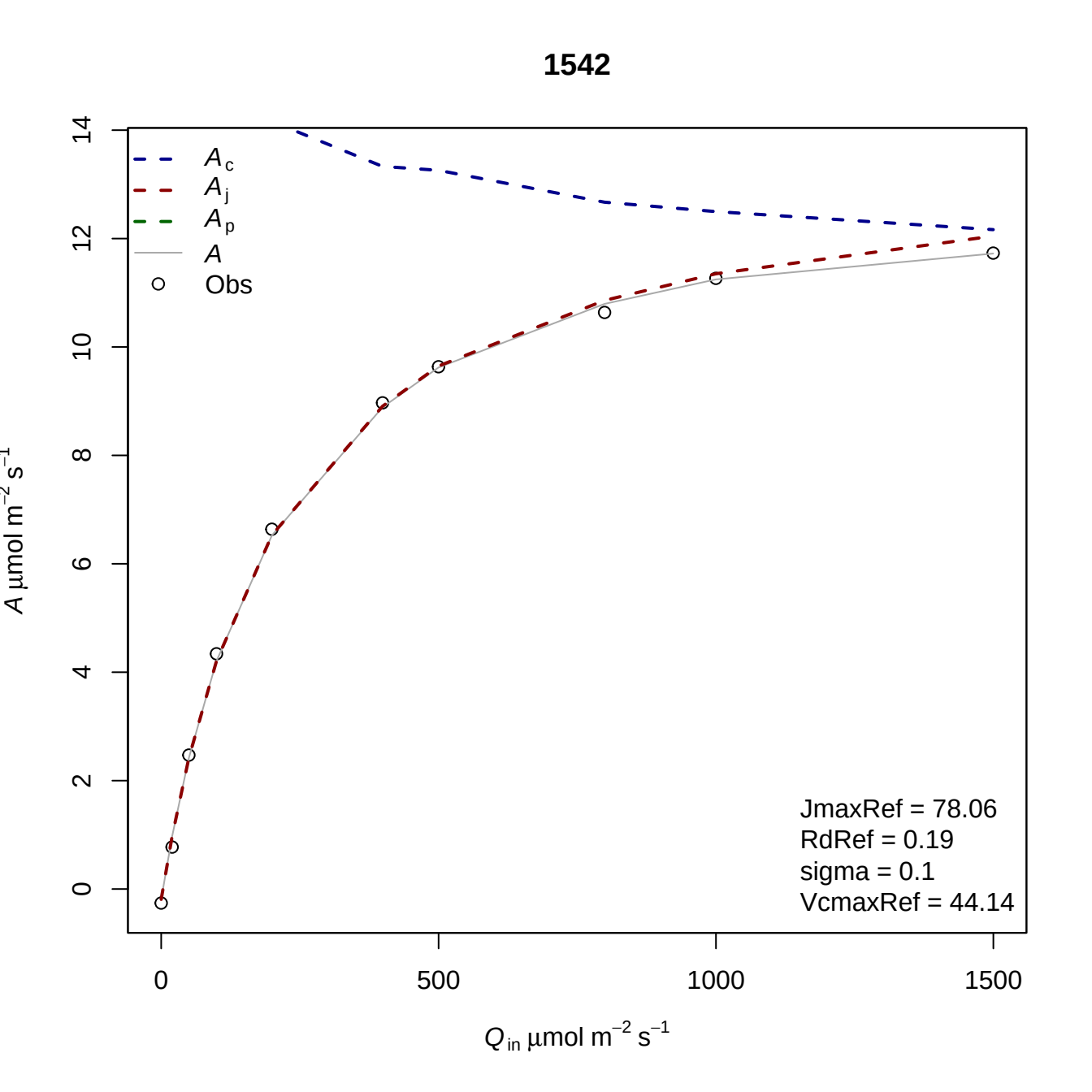

1537
A_c
A_j
A_p
A
Obs
JmaxRef = 92.67
RdRef = 0.24
sigma = 0.16
VcmaxRef = 70.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

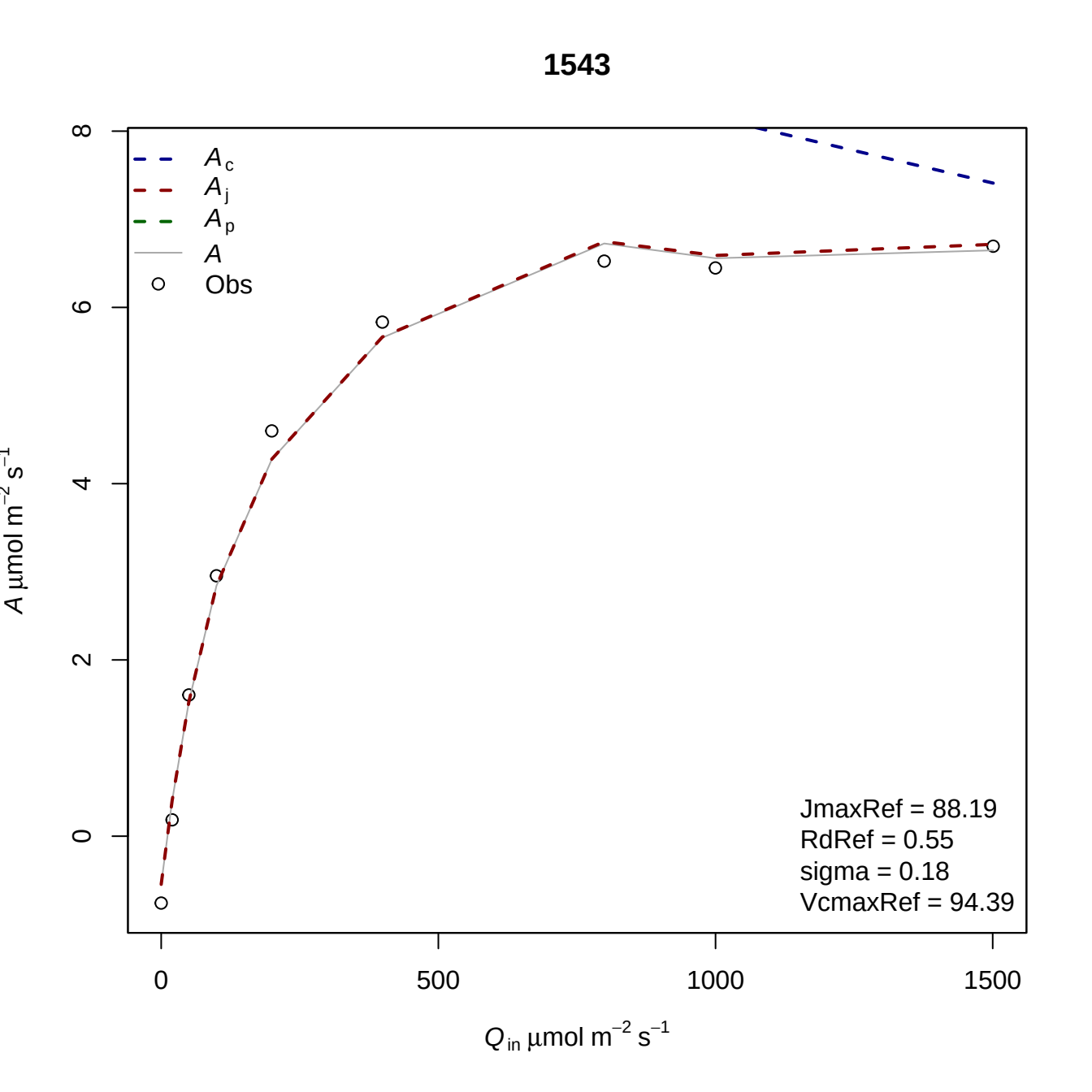

1538
A_c
A_j
A_p
A
Obs
JmaxRef = 105.53
RdRef = −0.03
sigma = 0.2
VcmaxRef = 55.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

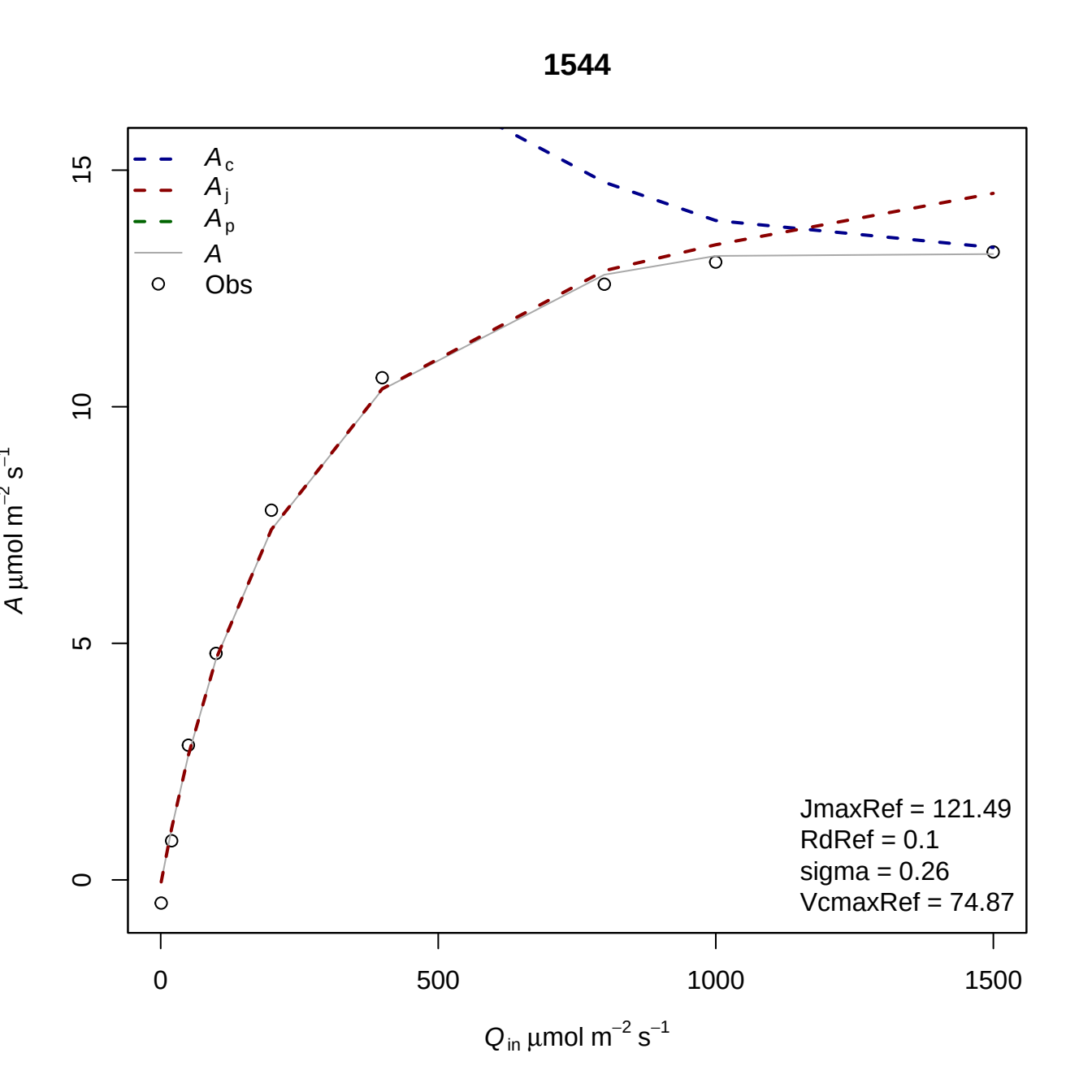

1539
A_c
A_j
A_p
A
Obs
JmaxRef = 102.27
RdRef = 0.2
sigma = 0.22
VcmaxRef = 72.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

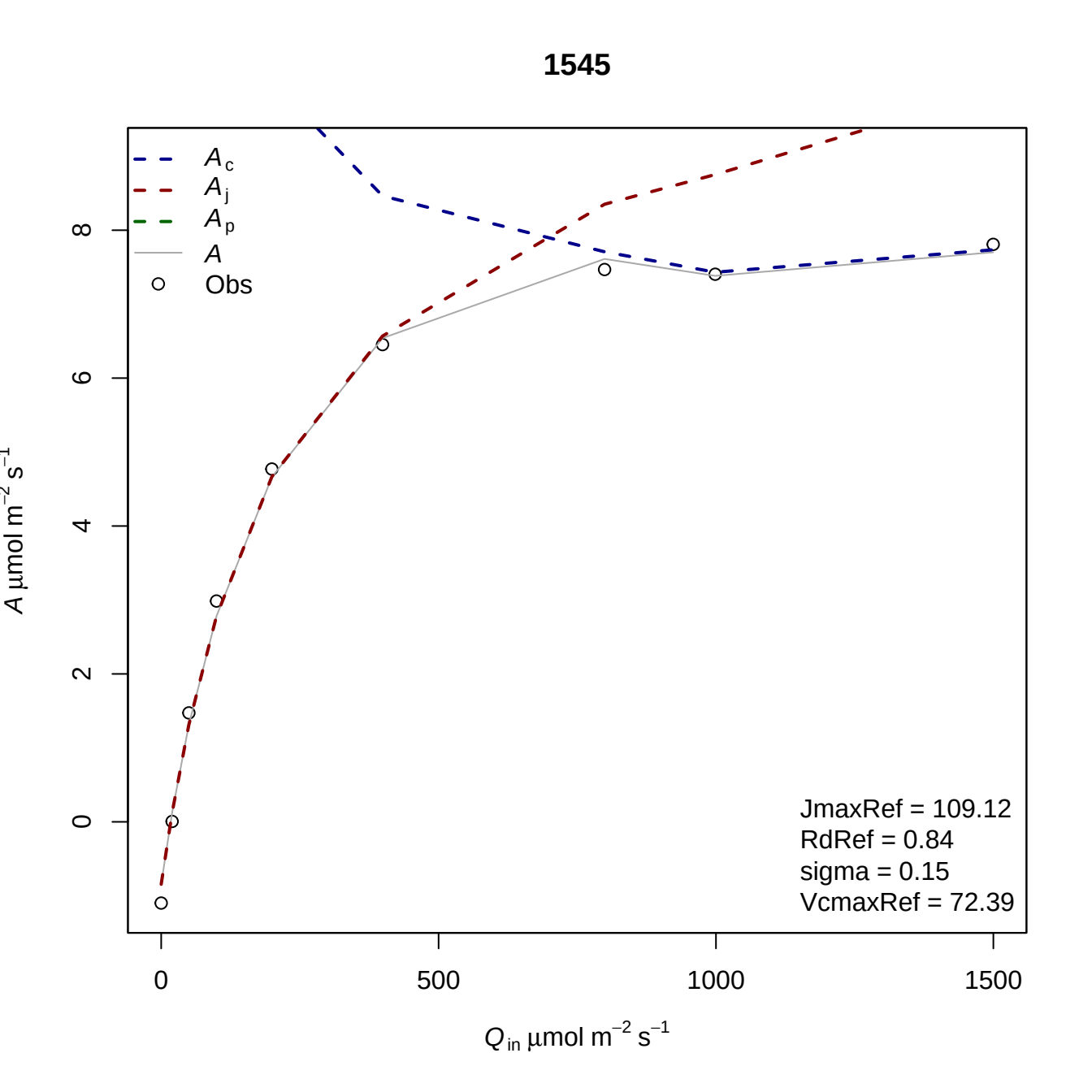

1540
A_c
A_j
A_p
A
Obs
JmaxRef = 108.82
RdRef = −0.21
sigma = 0.23
VcmaxRef = 55.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

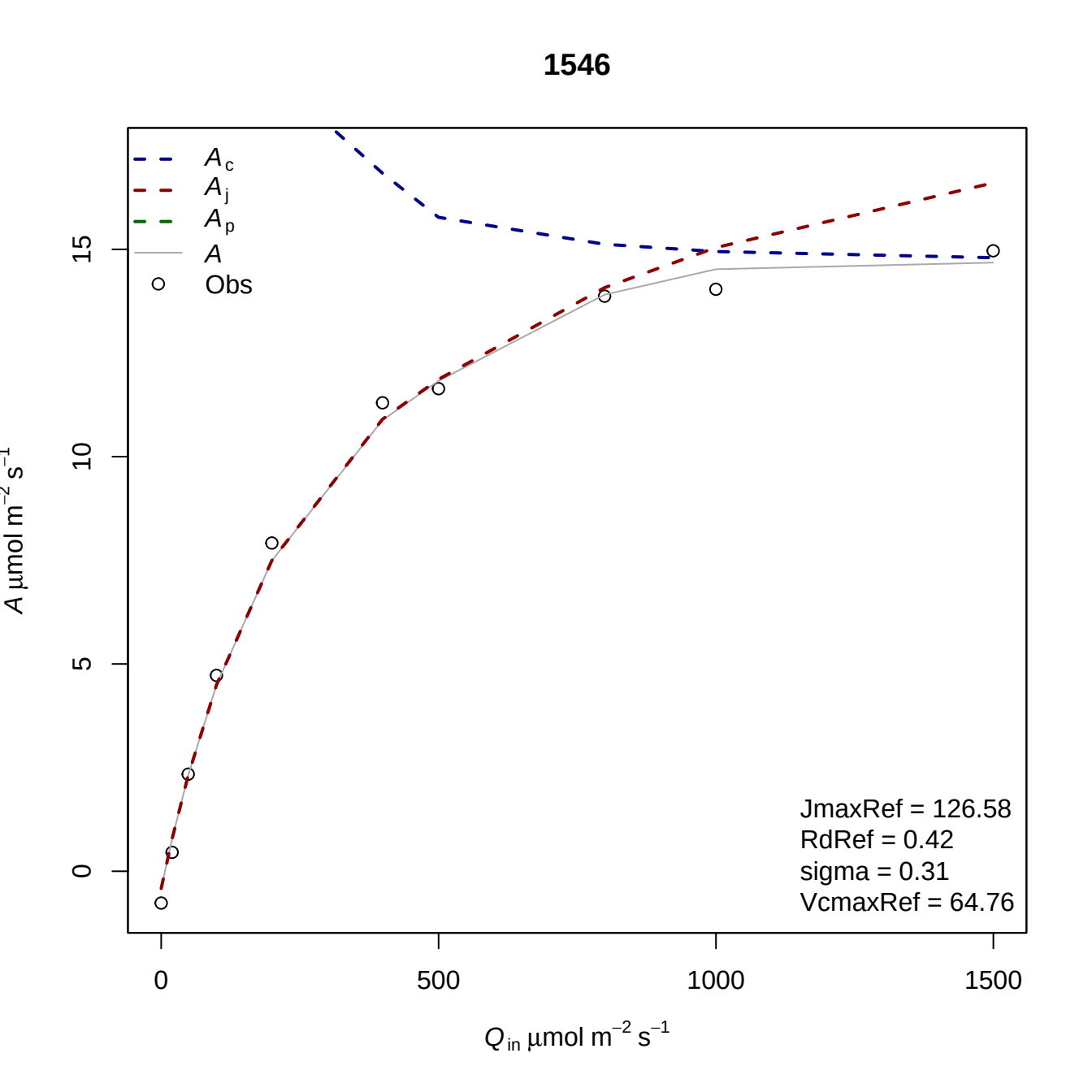

1541
A_c
A_j
A_p
A
Obs
JmaxRef = 77.23
RdRef = 0.48
sigma = 0.15
VcmaxRef = 57.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

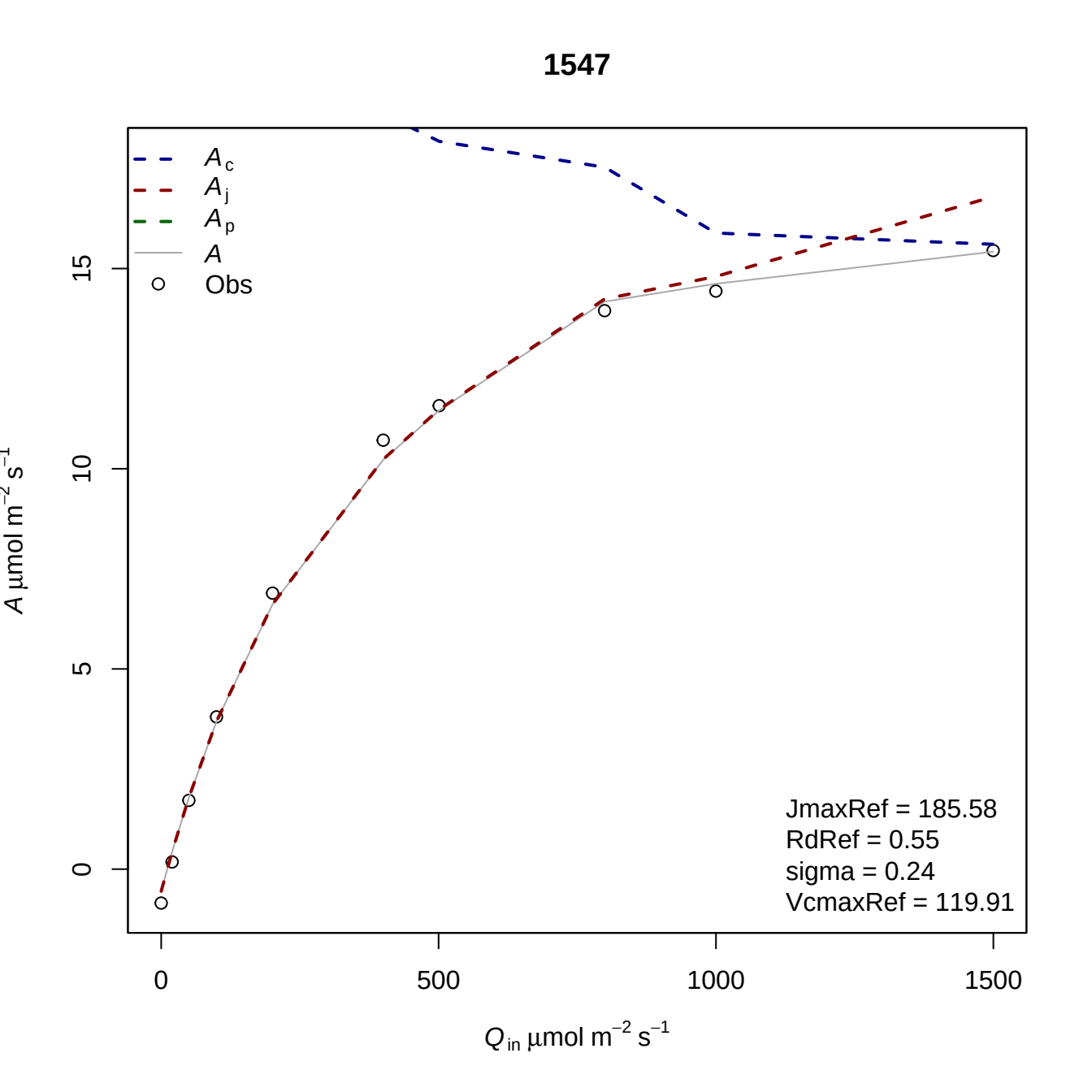

1542
A_c
A_j
A_p
A
Obs
JmaxRef = 78.06
RdRef = 0.19
sigma = 0.1
VcmaxRef = 44.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

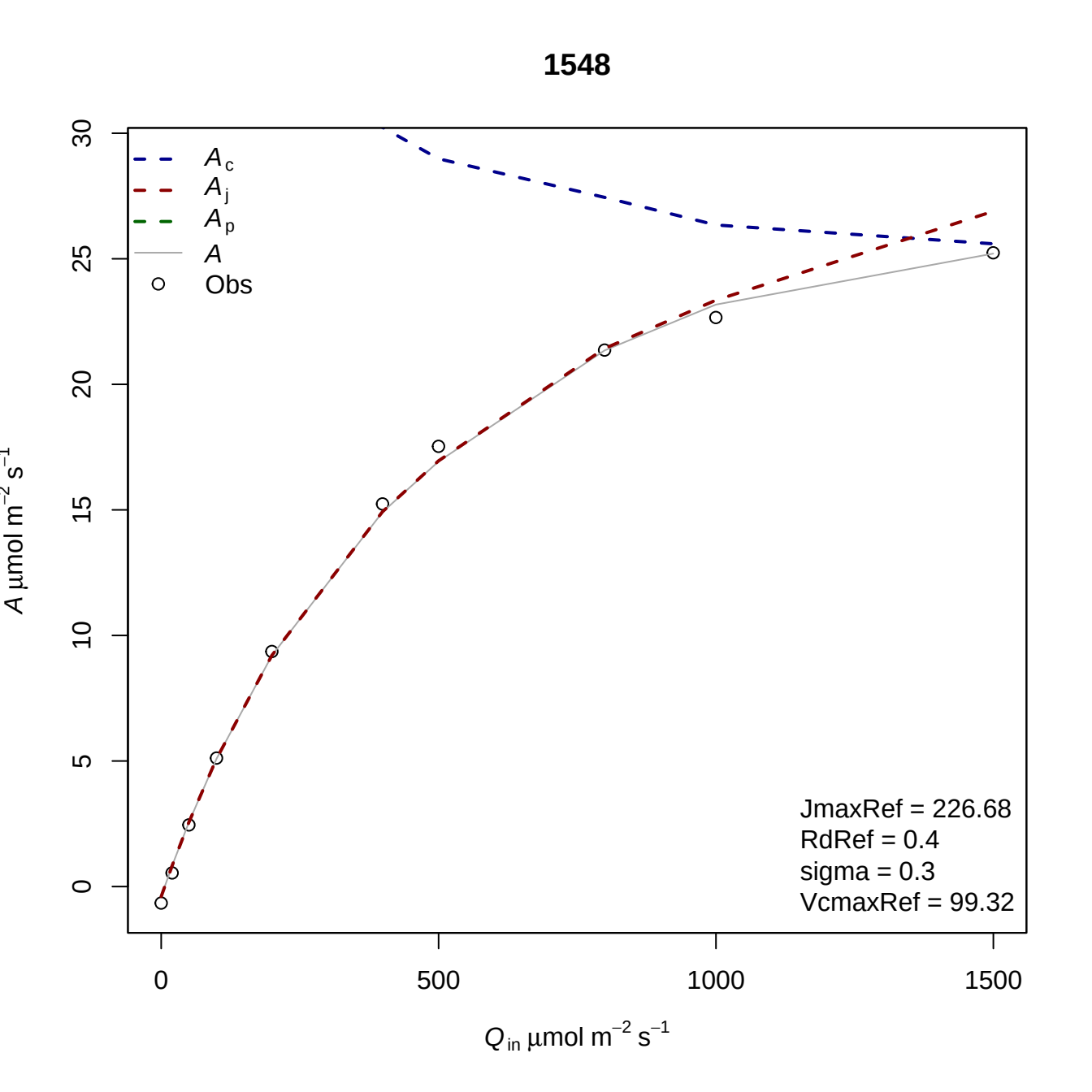

1543
A_c
A_j
A_p
A
Obs
JmaxRef = 88.19
RdRef = 0.55
sigma = 0.18
VcmaxRef = 94.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

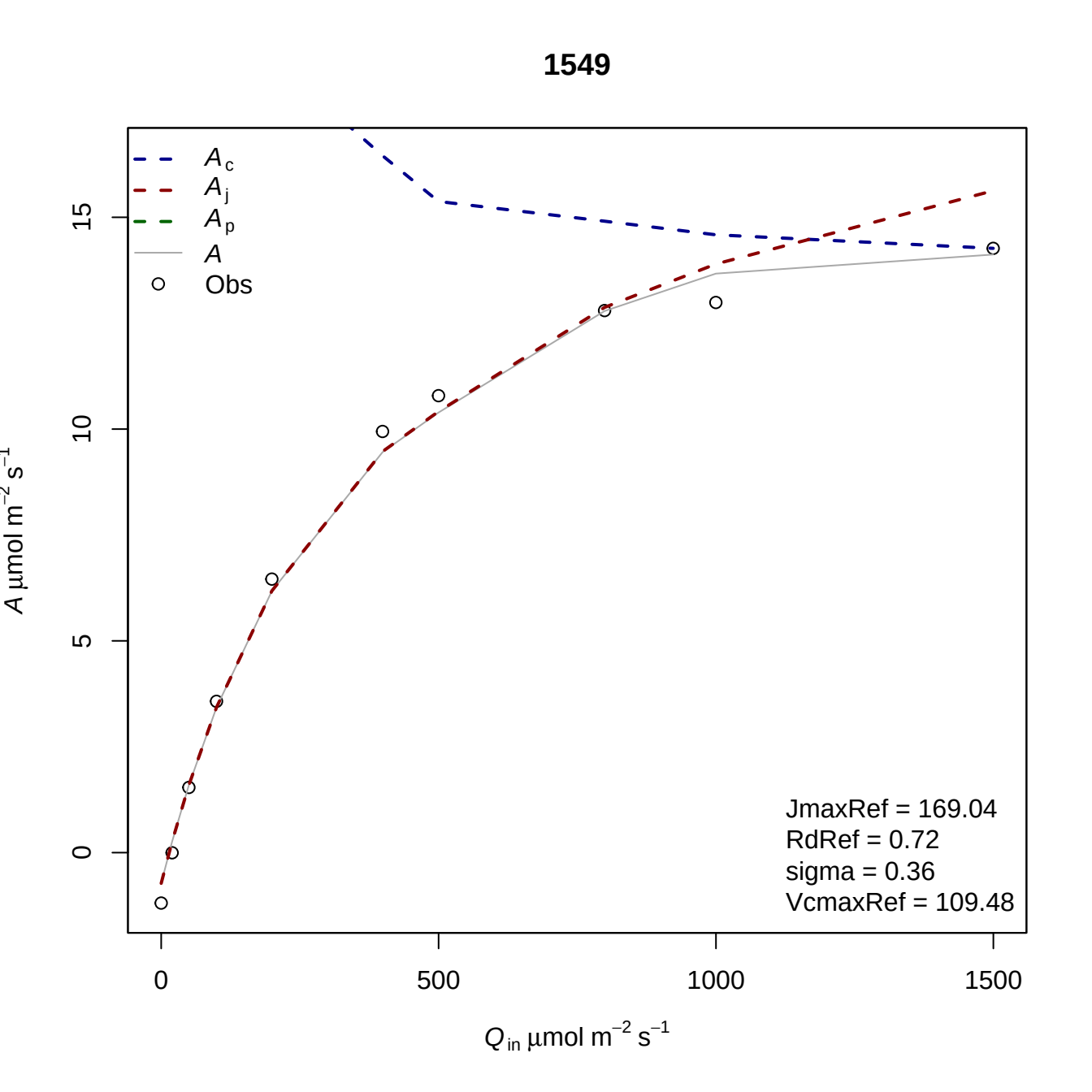

1544
A_c
A_j
A_p
A
Obs
JmaxRef = 121.49
RdRef = 0.1
sigma = 0.26
VcmaxRef = 74.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

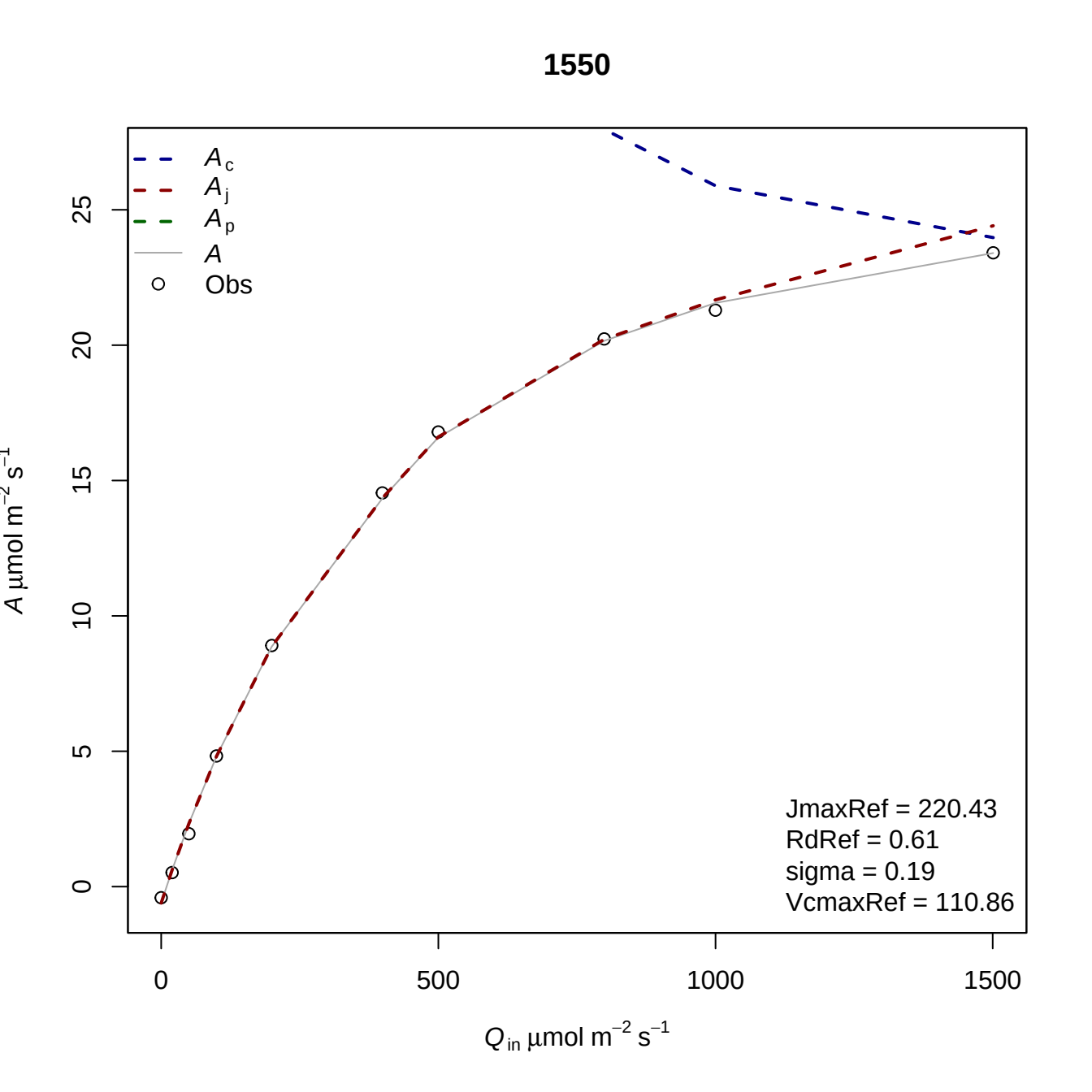

1545
A_c
A_j
A_p
A
Obs
JmaxRef = 109.12
RdRef = 0.84
sigma = 0.15
VcmaxRef = 72.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

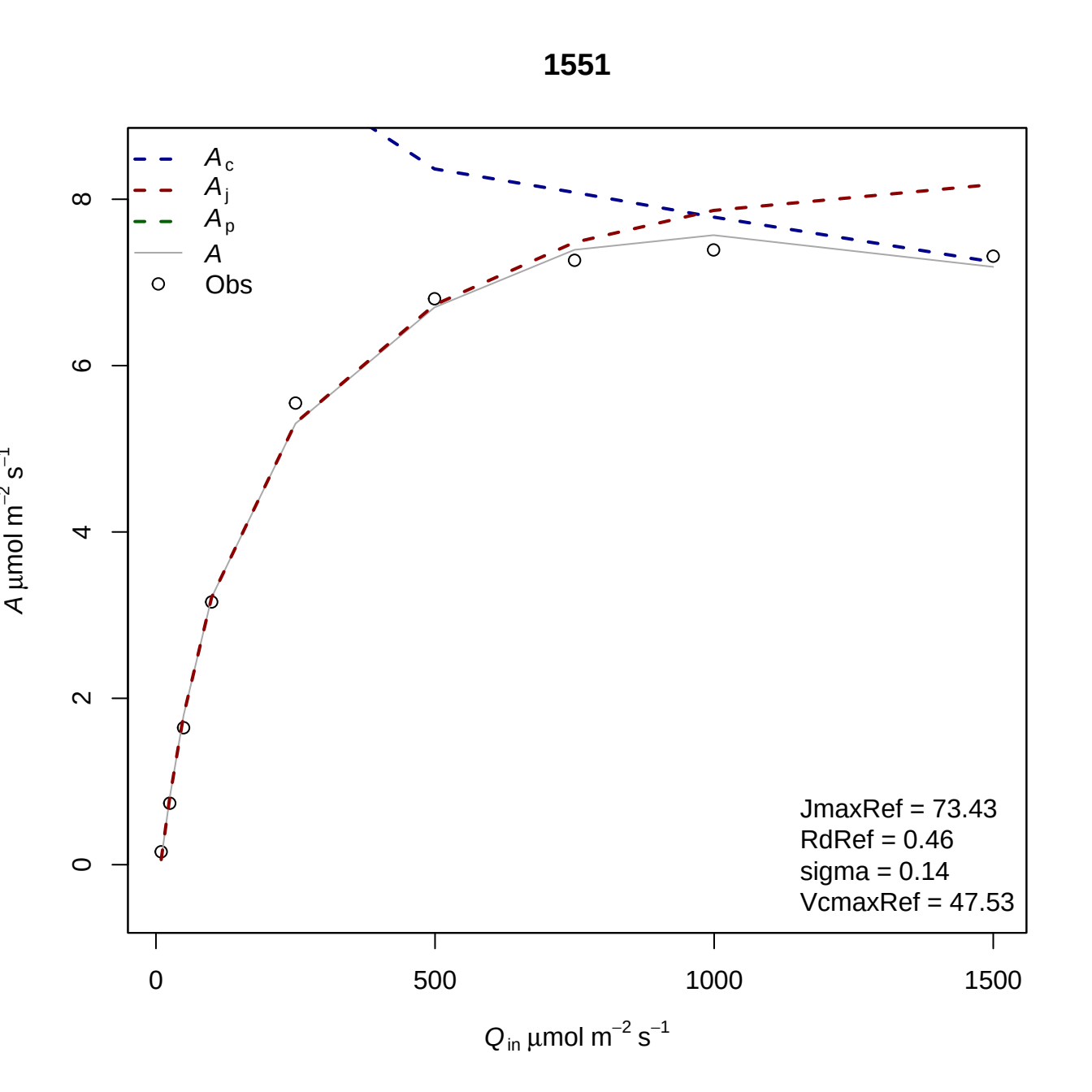

1546
A_c
A_j
A_p
A
Obs
JmaxRef = 126.58
RdRef = 0.42
sigma = 0.31
VcmaxRef = 64.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

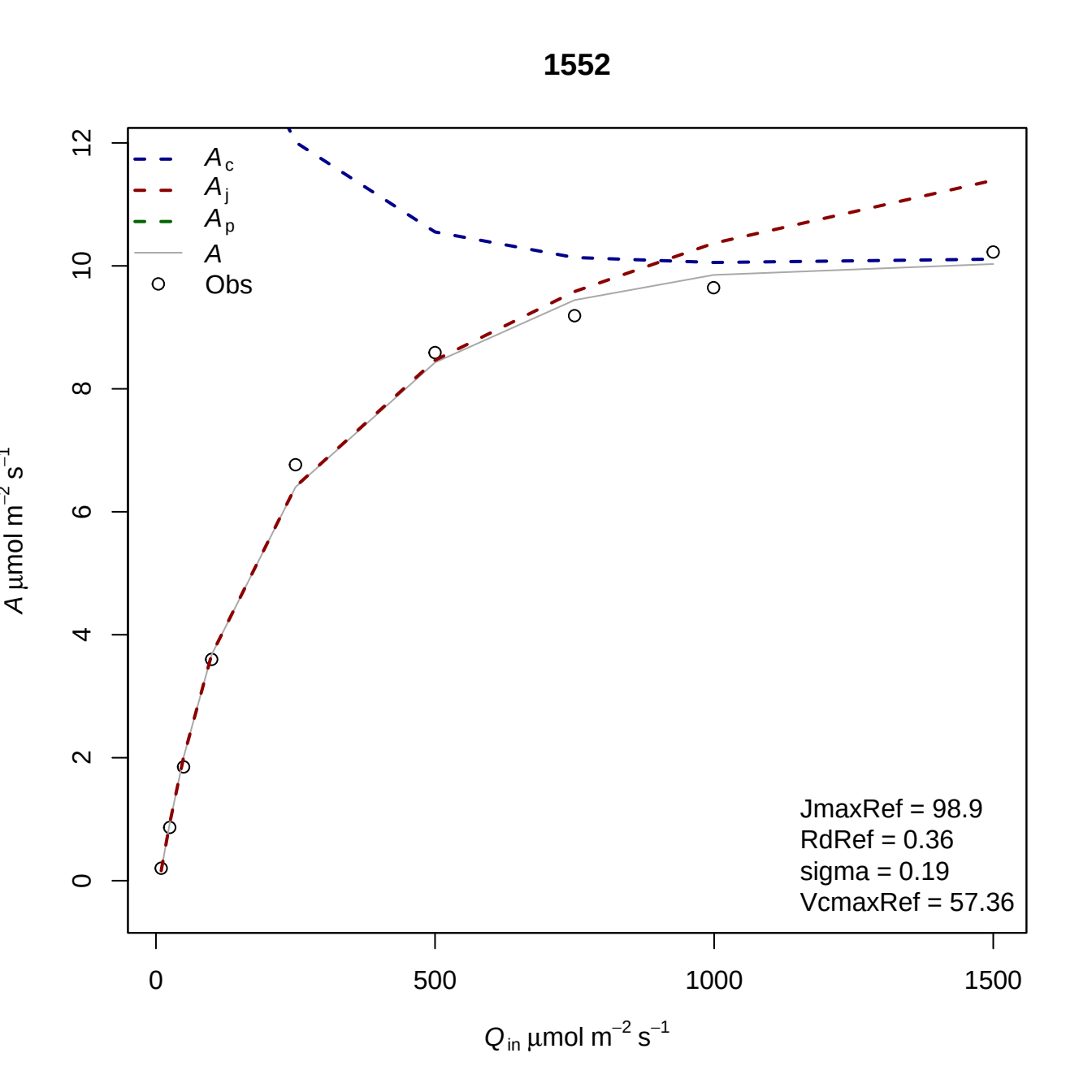

1547
A_c
A_j
A_p
A
Obs
JmaxRef = 185.58
RdRef = 0.55
sigma = 0.24
VcmaxRef = 119.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

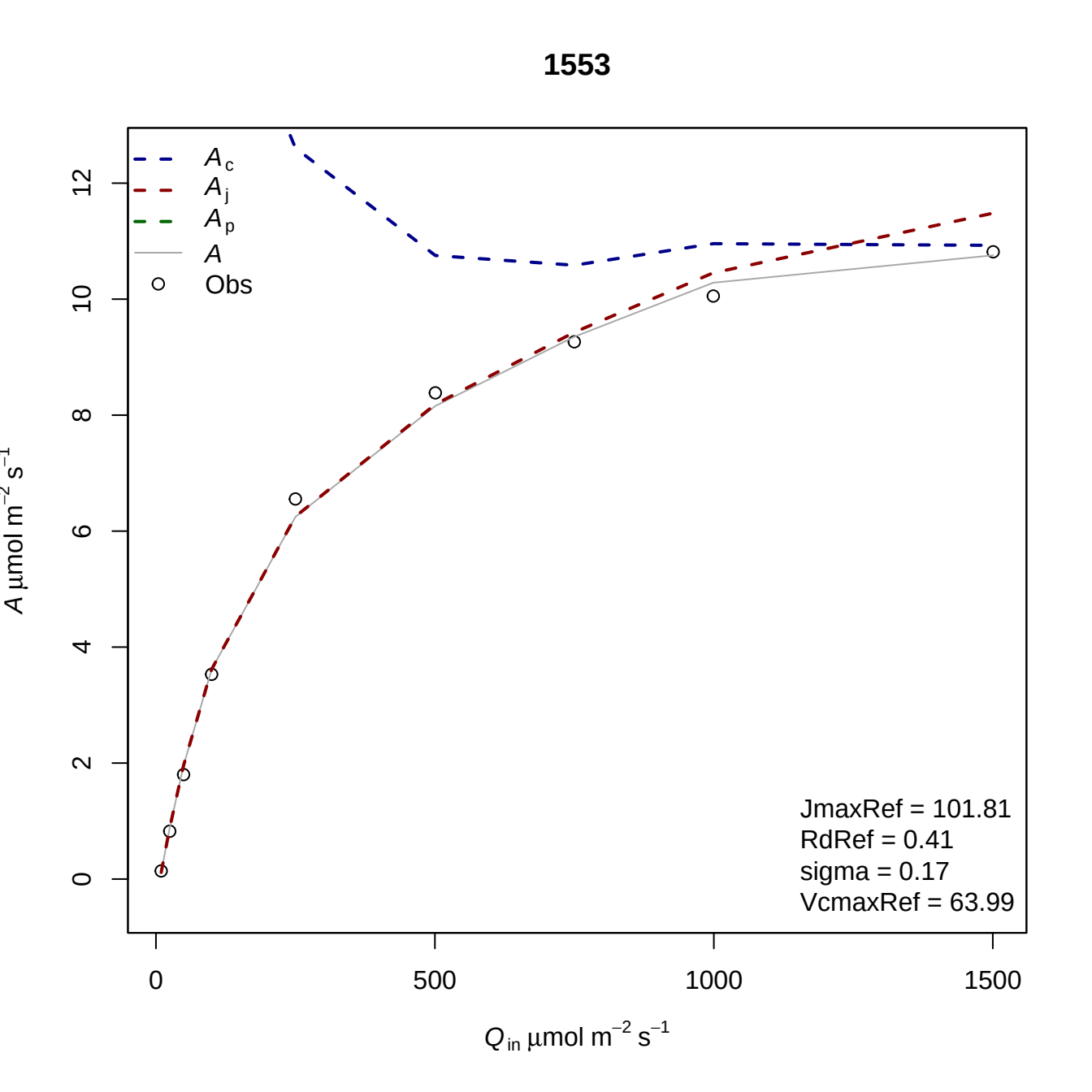

1548
A_c
A_j
A_p
A
Obs
JmaxRef = 226.68
RdRef = 0.4
sigma = 0.3
VcmaxRef = 99.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500

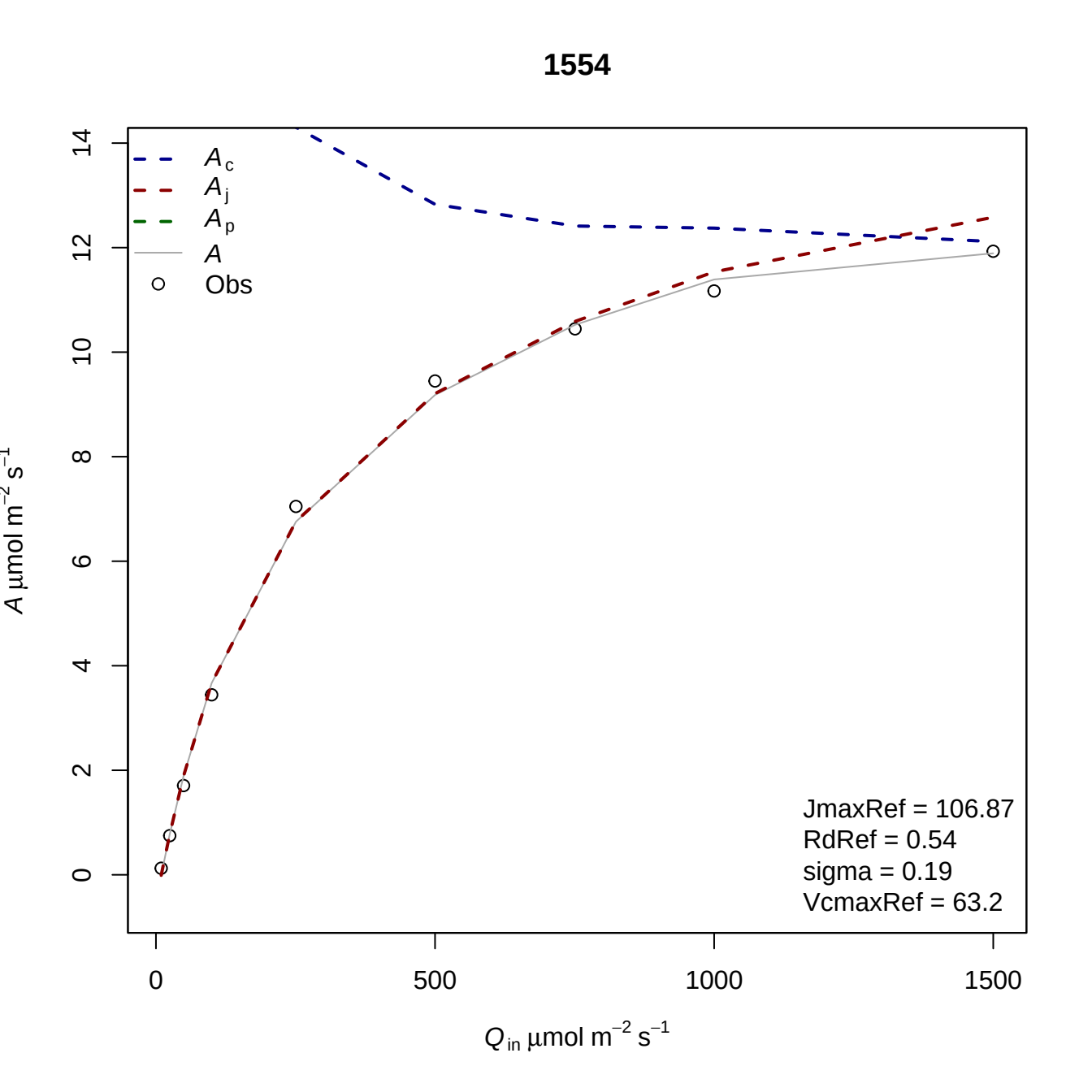

1549
A_c
A_j
A_p
A
Obs
JmaxRef = 169.04
RdRef = 0.72
sigma = 0.36
VcmaxRef = 109.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

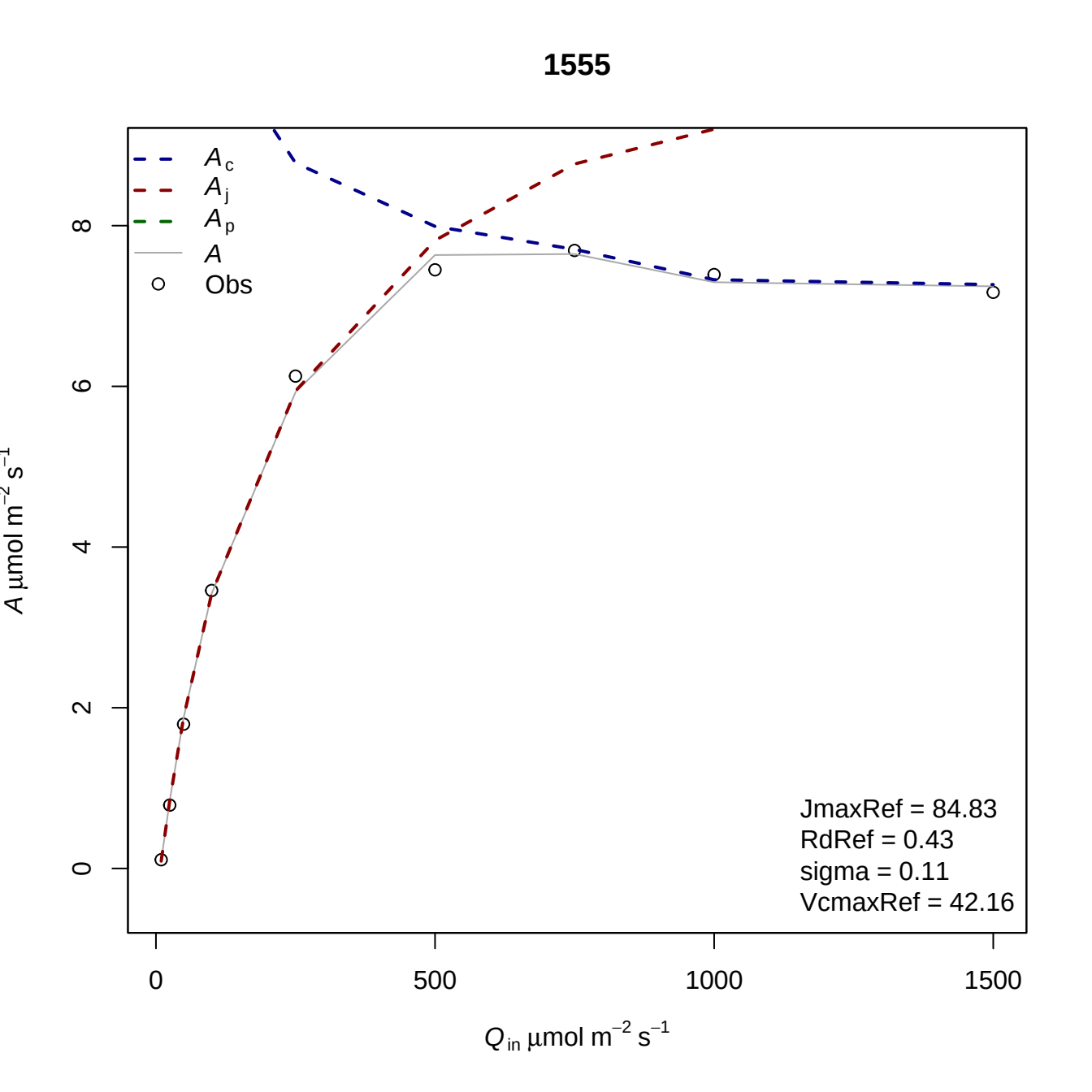

1550
A_c
A_j
A_p
A
Obs
JmaxRef = 220.43
RdRef = 0.61
sigma = 0.19
VcmaxRef = 110.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

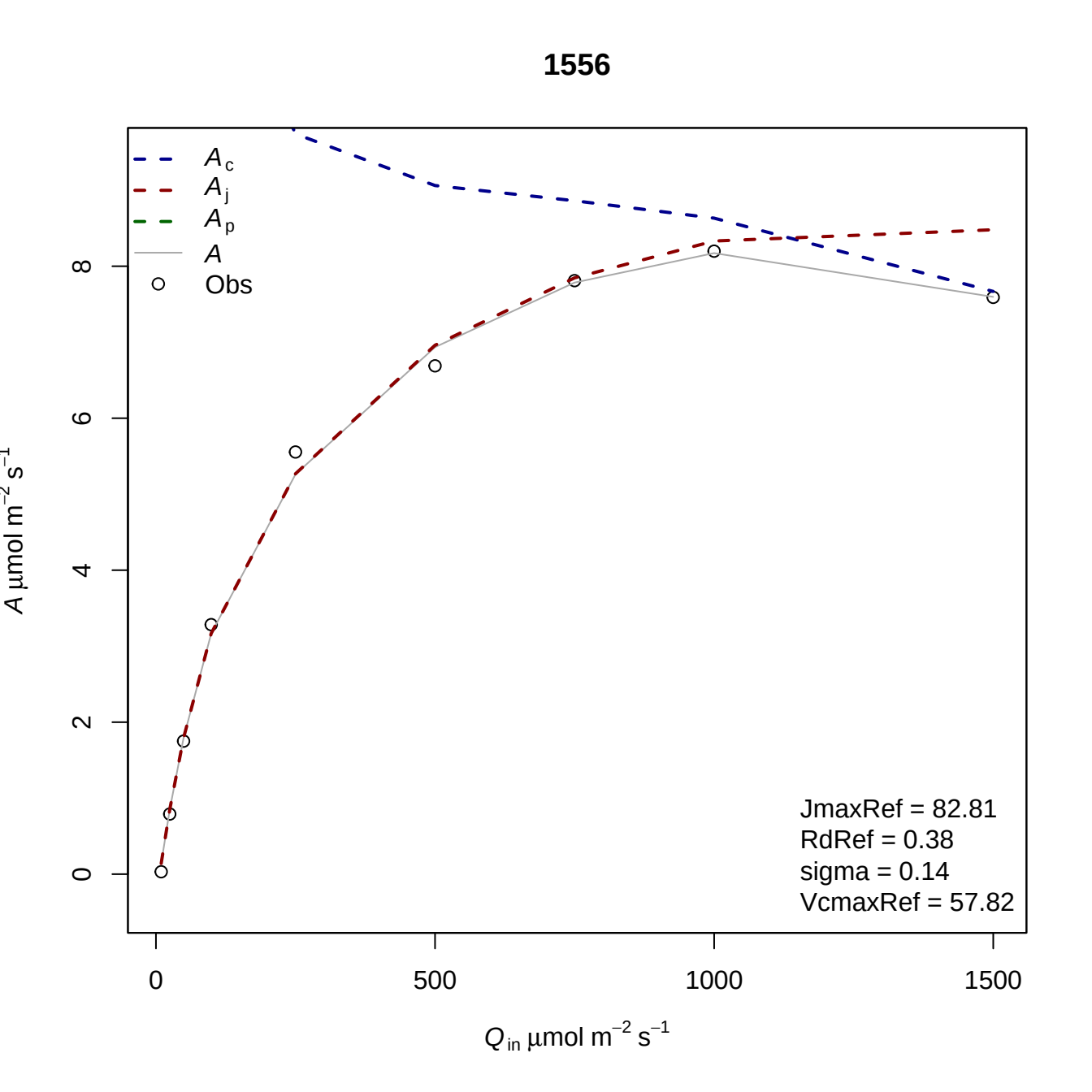

1551
A_c
A_j
A_p
A
Obs
JmaxRef = 73.43
RdRef = 0.46
sigma = 0.14
VcmaxRef = 47.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

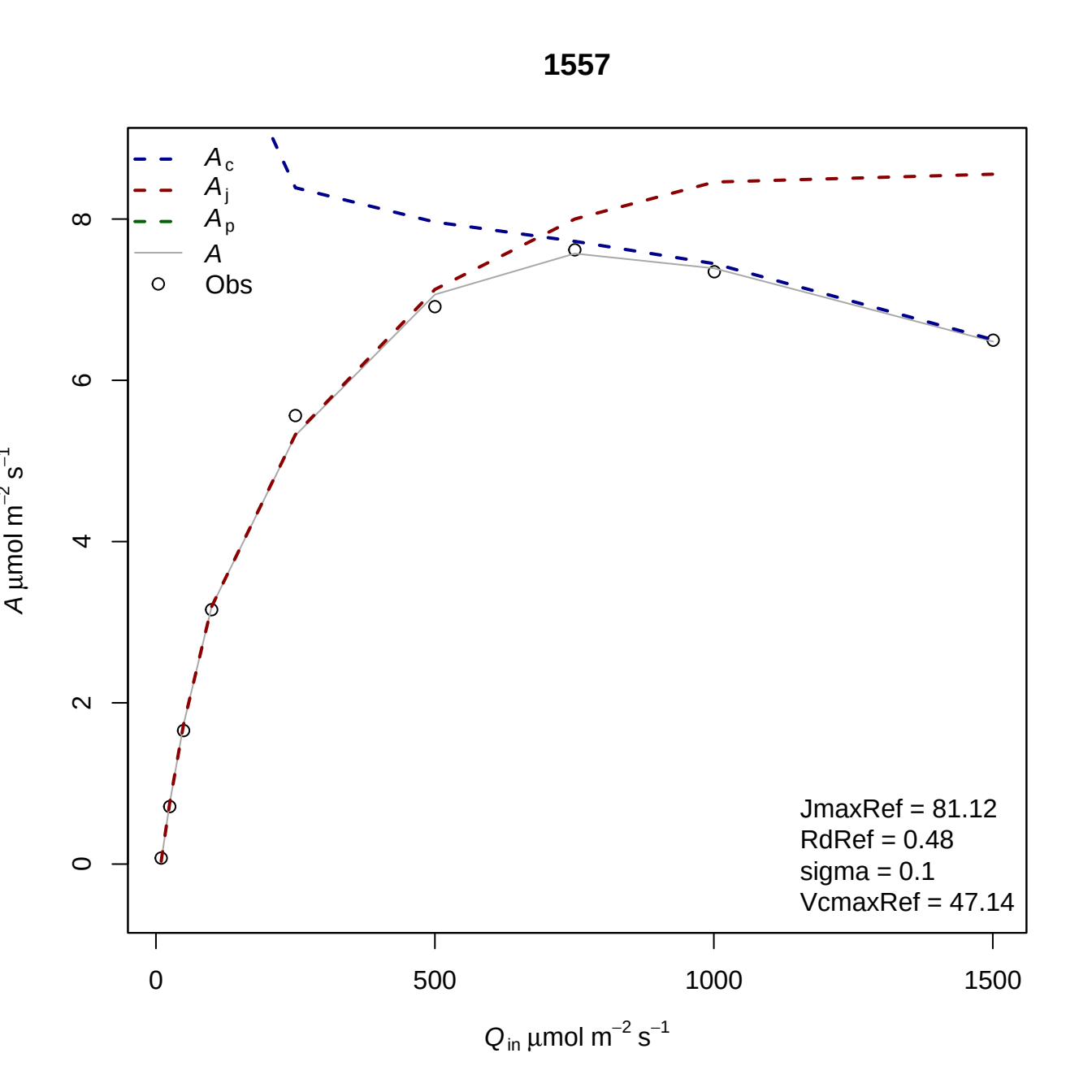

1552
A_c
A_j
A_p
A
Obs
JmaxRef = 98.9
RdRef = 0.36
sigma = 0.19
VcmaxRef = 57.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

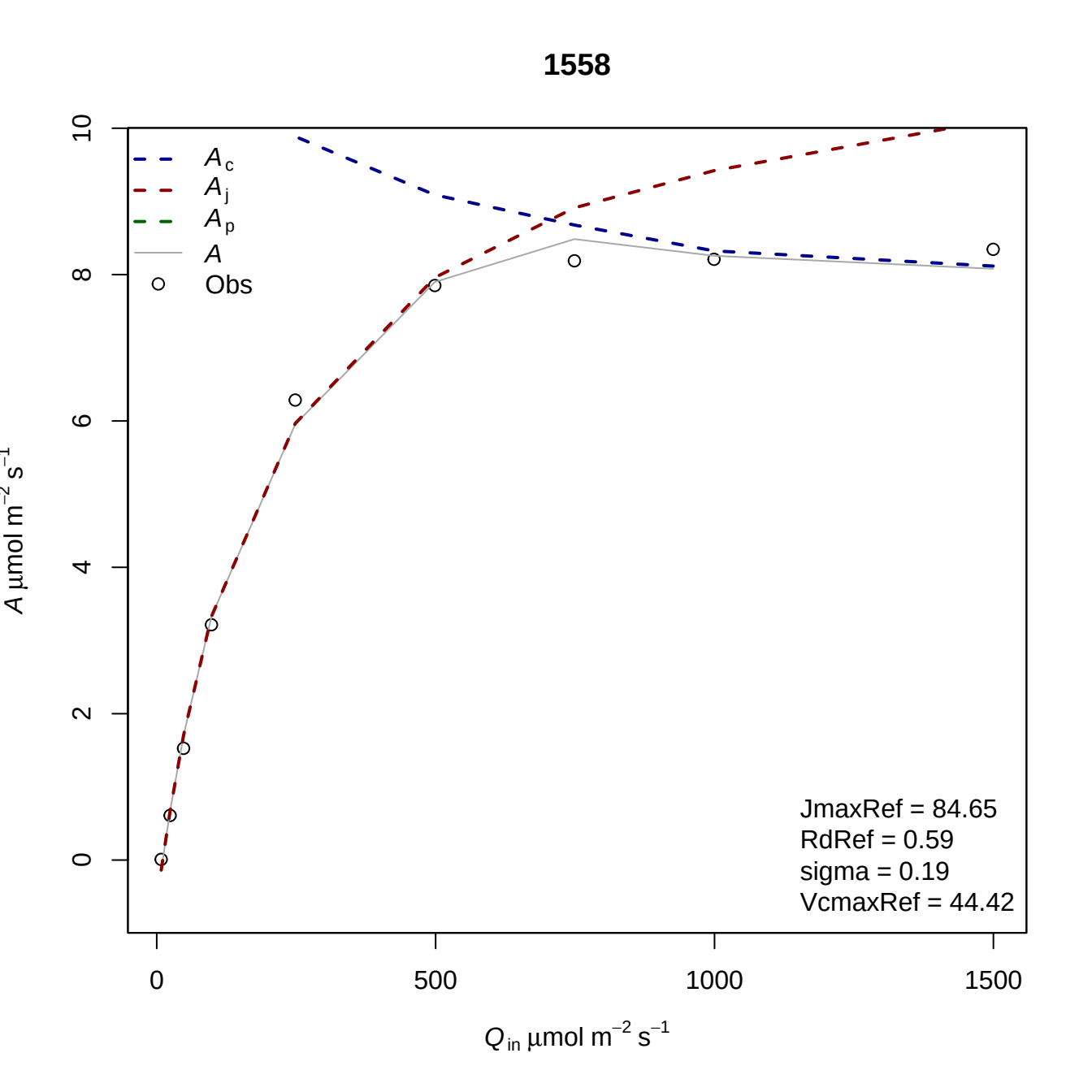

1553
A_c
A_j
A_p
A
Obs
JmaxRef = 101.81
RdRef = 0.41
sigma = 0.17
VcmaxRef = 63.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

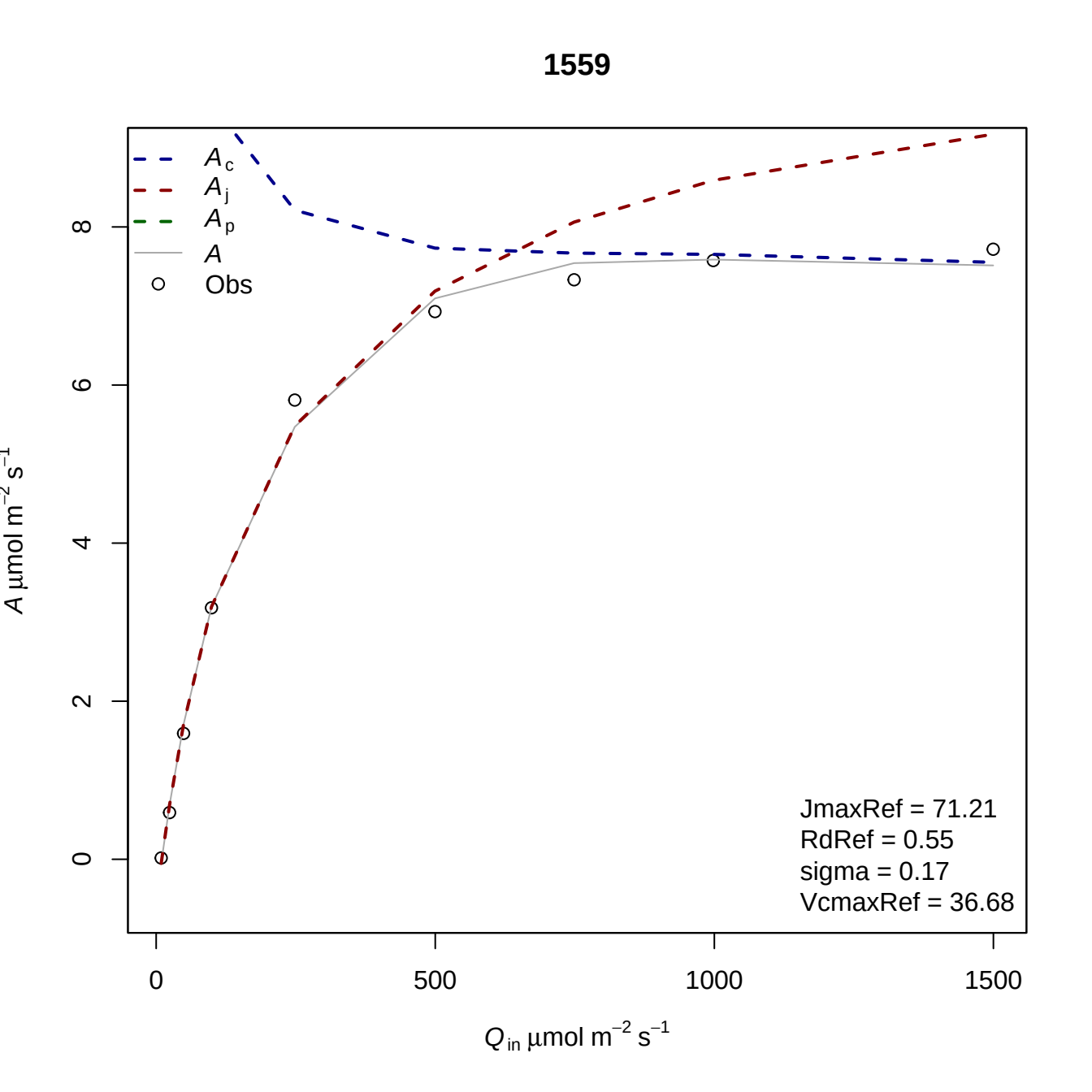

1554
A_c
A_j
A_p
A
Obs
JmaxRef = 106.87
RdRef = 0.54
sigma = 0.19
VcmaxRef = 63.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

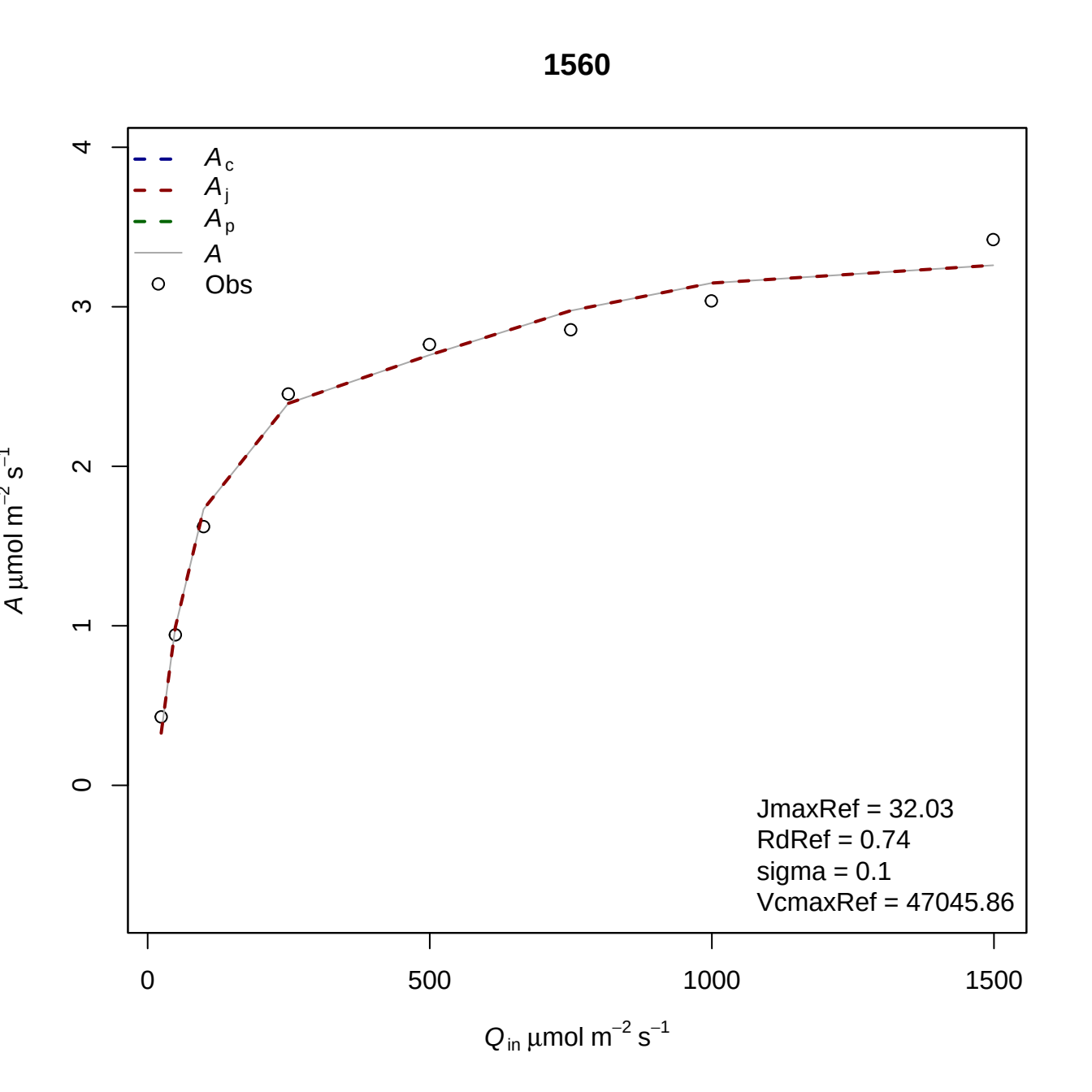

1555
A_c
A_j
A_p
A
Obs
JmaxRef = 84.83
RdRef = 0.43
sigma = 0.11
VcmaxRef = 42.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

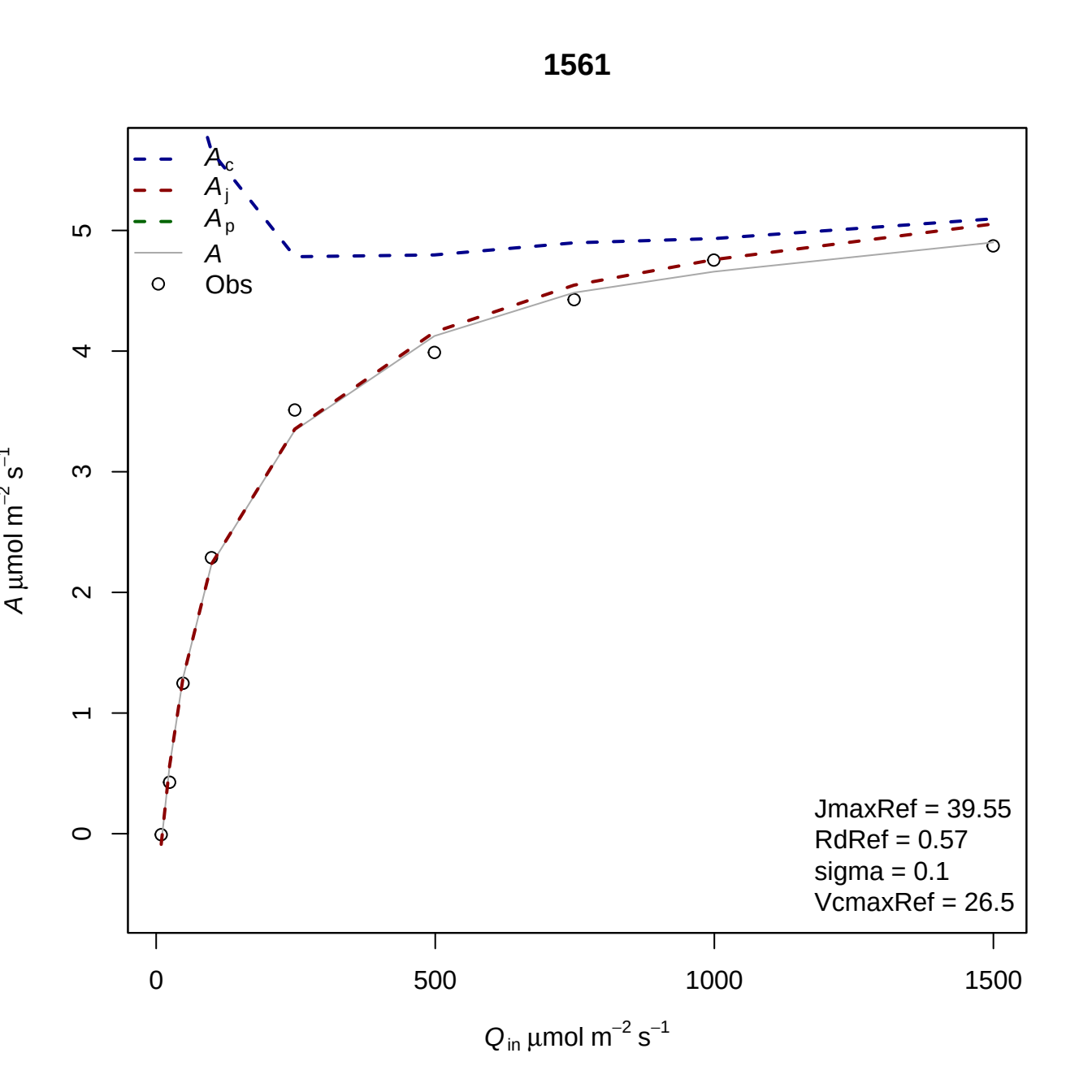

1556
A_c
A_j
A_p
A
Obs
JmaxRef = 82.81
RdRef = 0.38
sigma = 0.14
VcmaxRef = 57.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

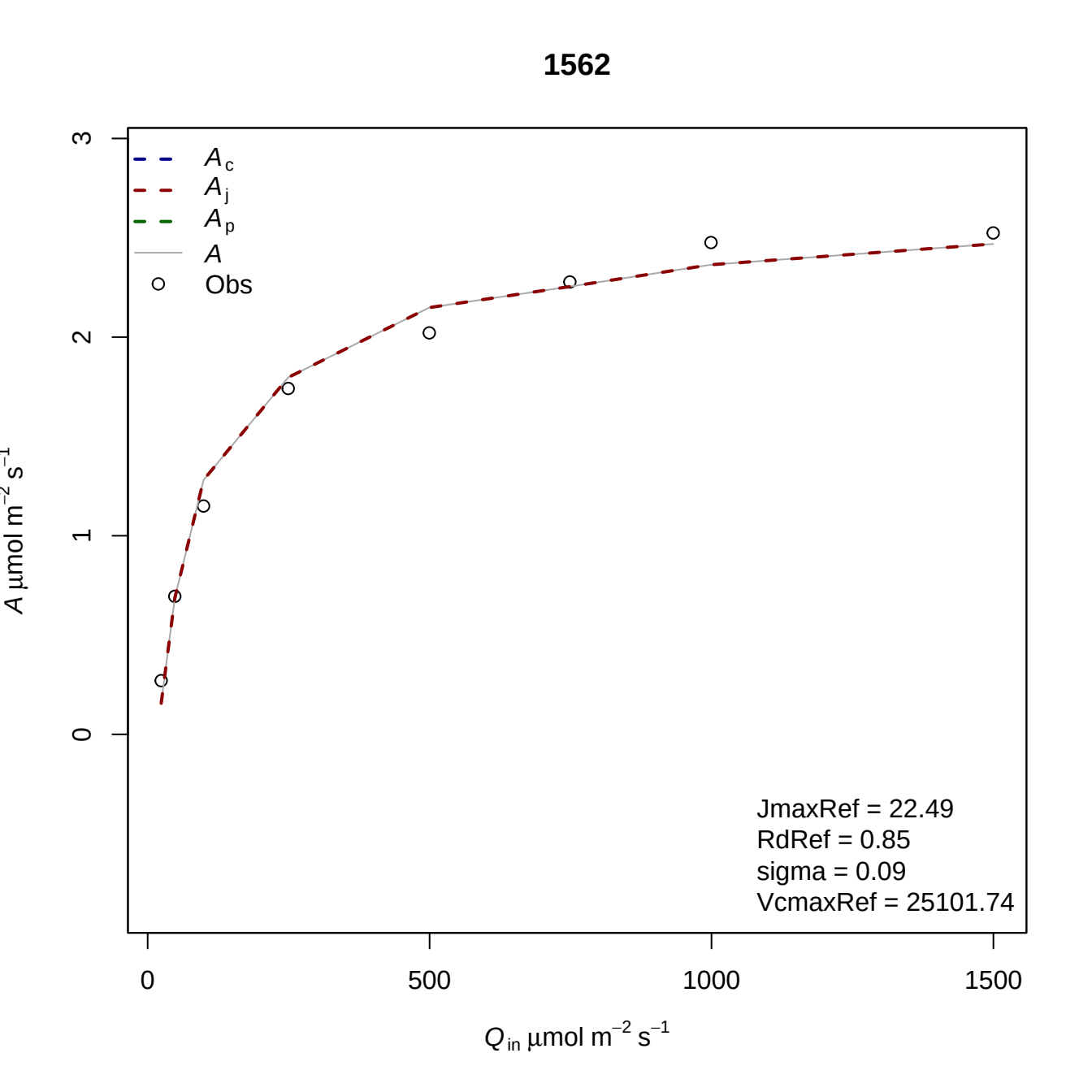

1557
A_c
A_j
A_p
A
Obs
JmaxRef = 81.12
RdRef = 0.48
sigma = 0.1
VcmaxRef = 47.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

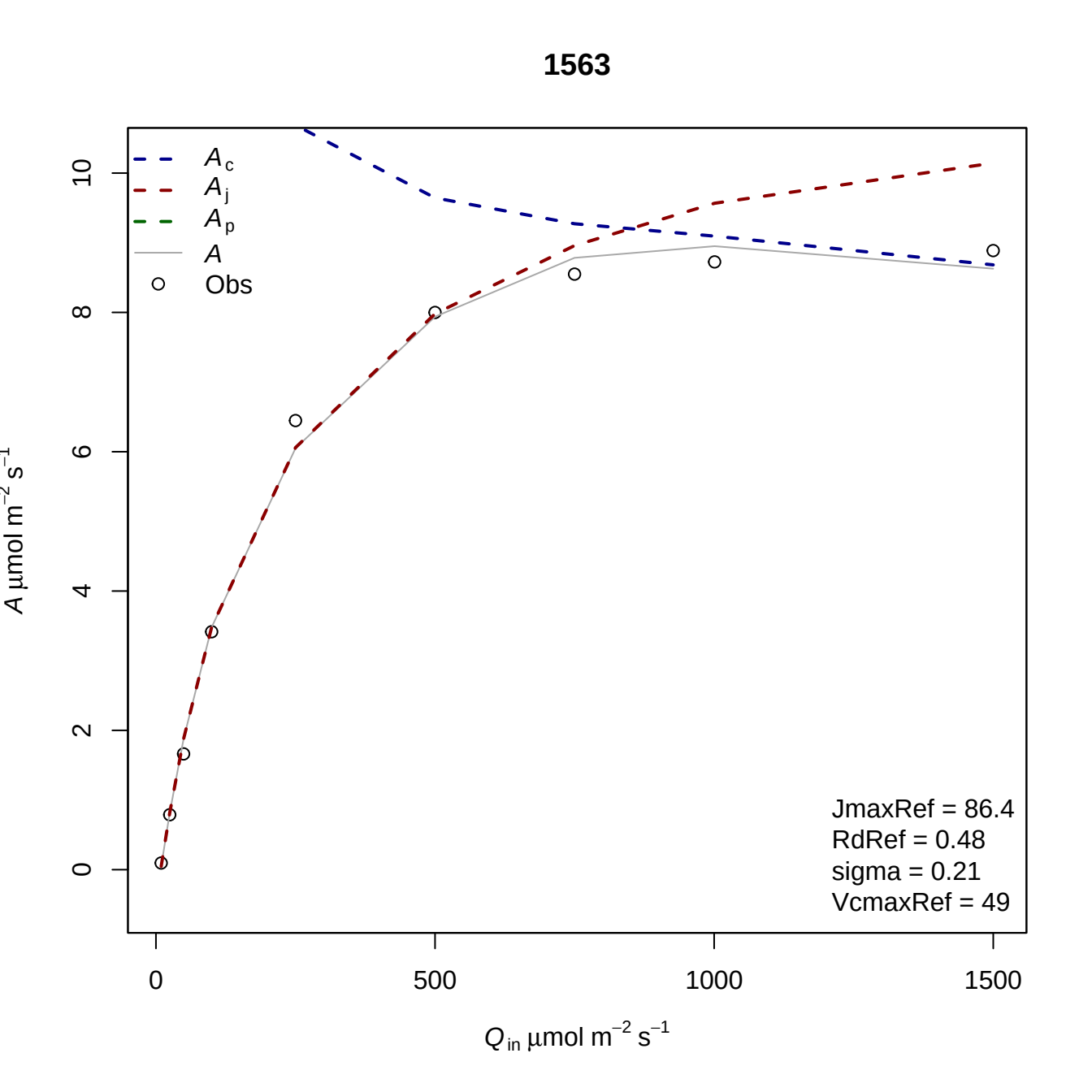

1558
A_c
A_j
A_p
A
Obs
JmaxRef = 84.65
RdRef = 0.59
sigma = 0.19
VcmaxRef = 44.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

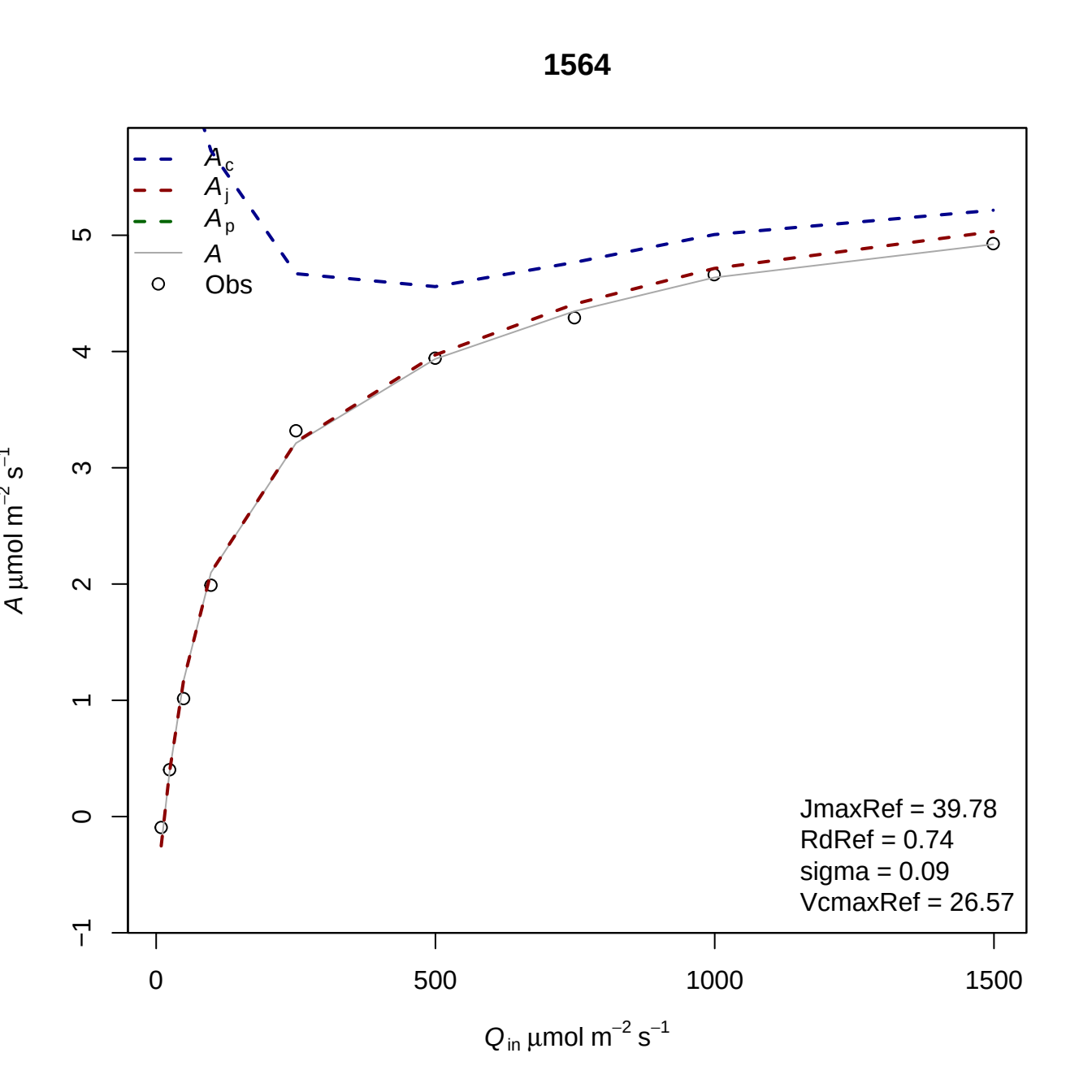

1559
A_c
A_j
A_p
A
Obs
JmaxRef = 71.21
RdRef = 0.55
sigma = 0.17
VcmaxRef = 36.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

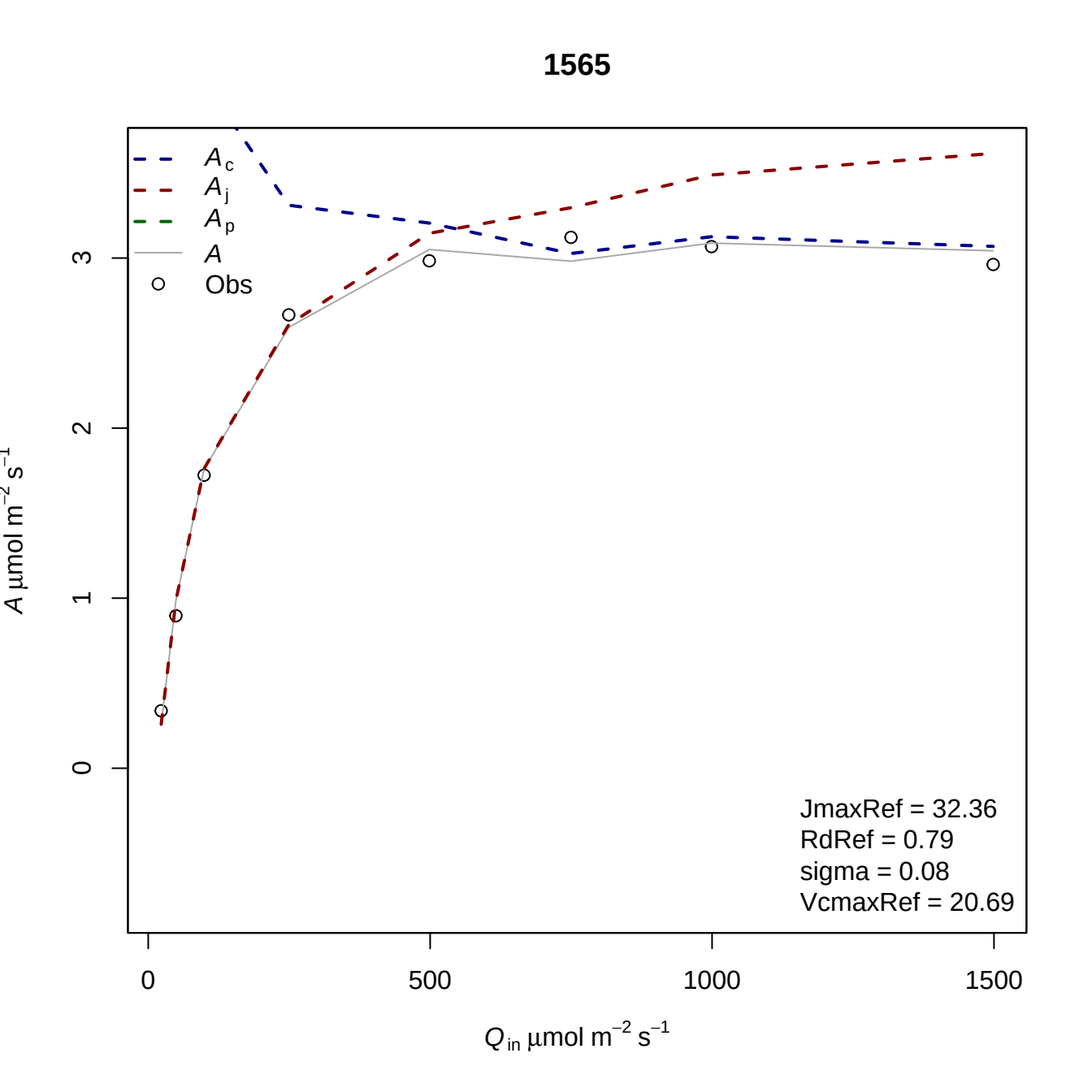

1560
A_c
A_j
A_p
A
Obs
JmaxRef = 32.03
RdRef = 0.74
sigma = 0.1
VcmaxRef = 47045.86
0
1
2
3
4
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

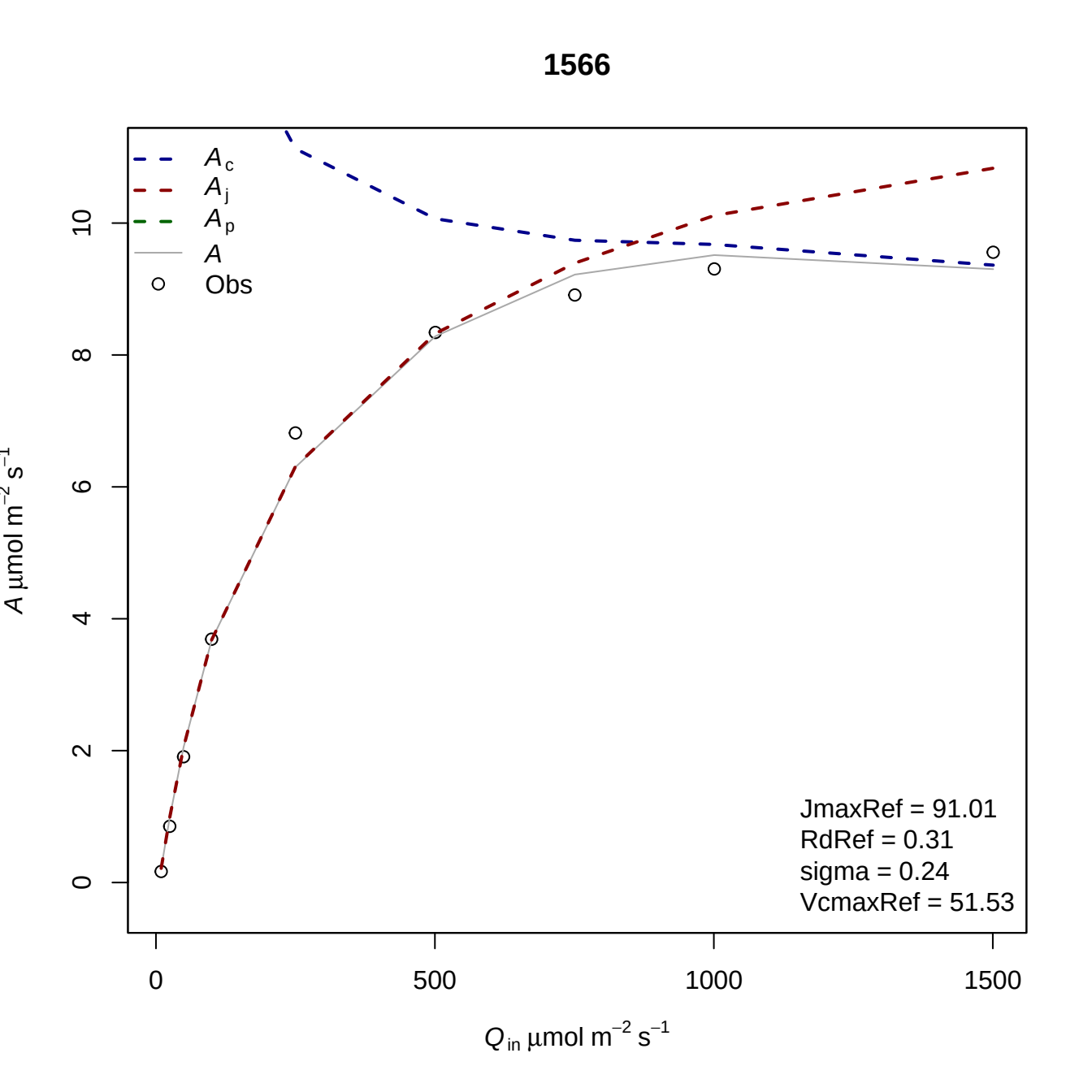

1561
A_c
A_j
A_p
A
Obs
JmaxRef = 39.55
RdRef = 0.57
sigma = 0.1
VcmaxRef = 26.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

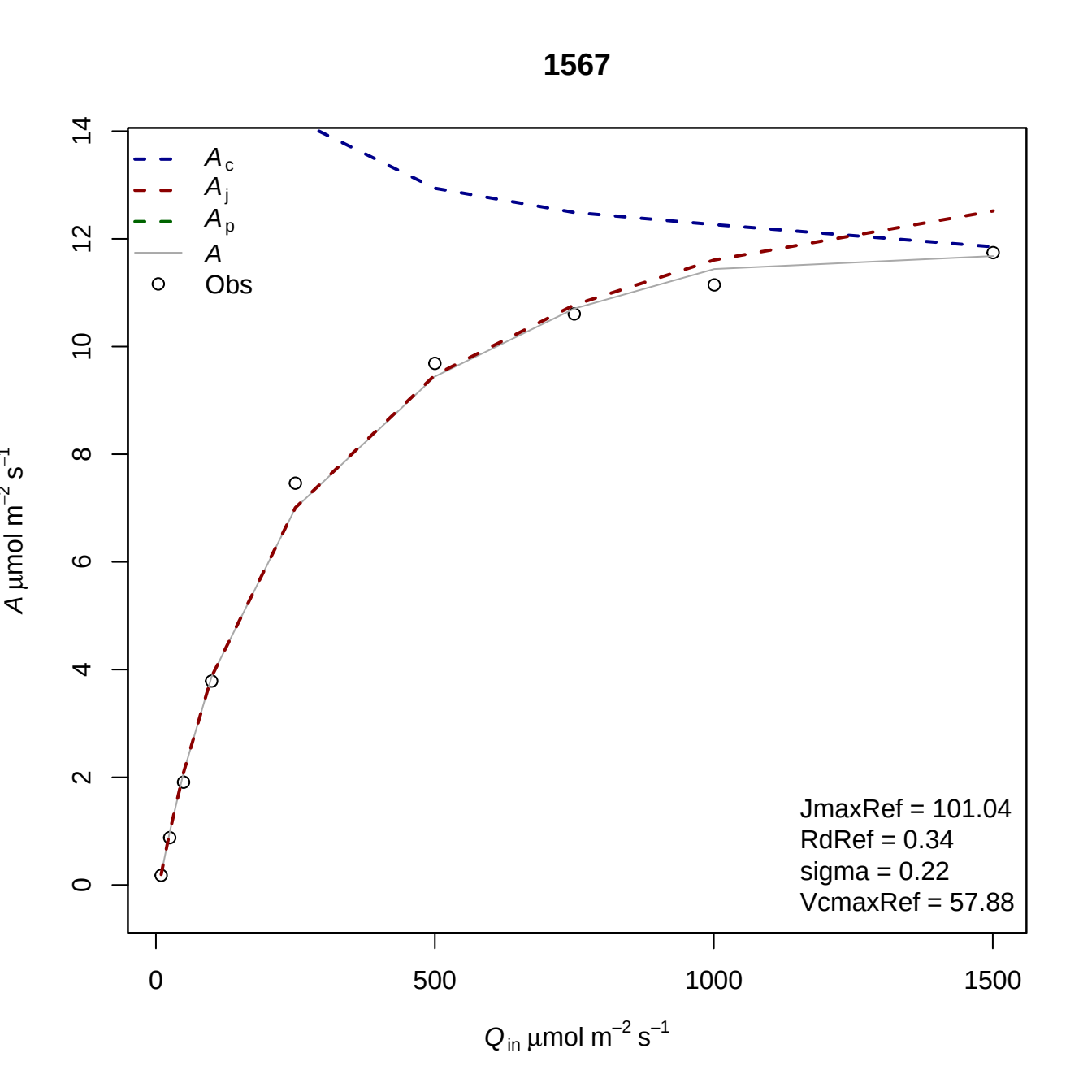

1562
A_c
A_j
A_p
A
Obs
JmaxRef = 22.49
RdRef = 0.85
sigma = 0.09
VcmaxRef = 25101.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500

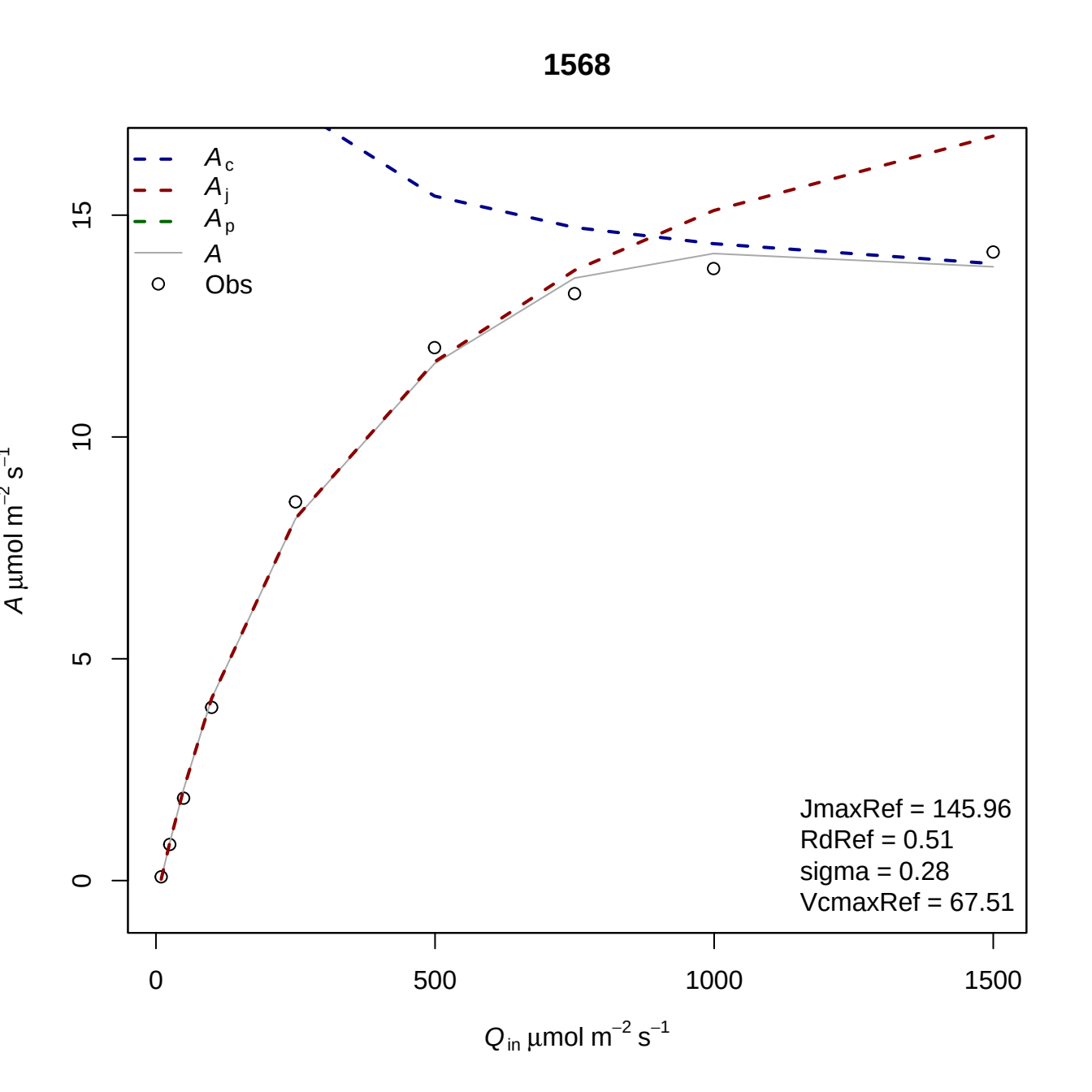

1563
A_c
A_j
A_p
A
Obs
JmaxRef = 86.4
RdRef = 0.48
sigma = 0.21
VcmaxRef = 49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

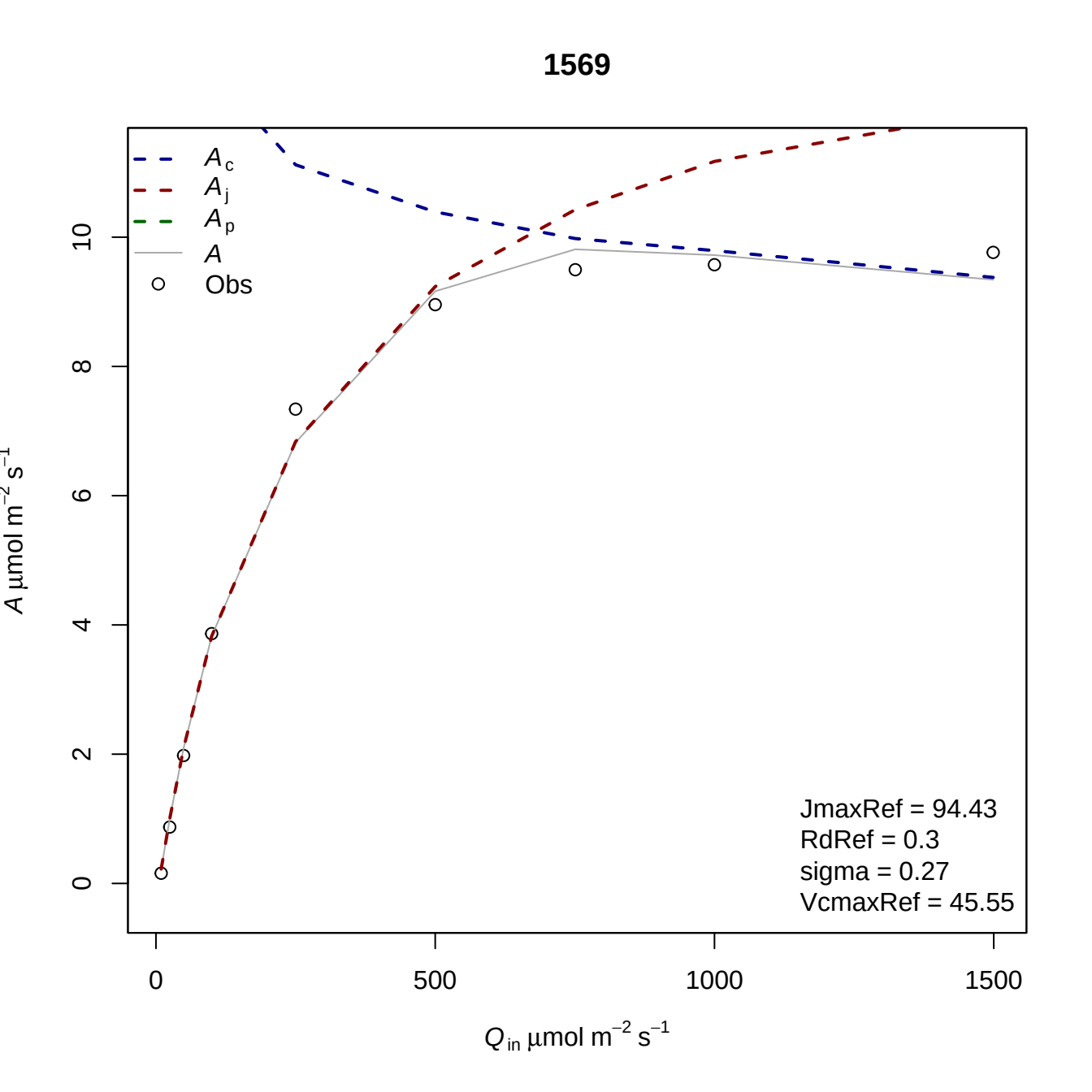

1564
A_c
A_j
A_p
A
Obs
JmaxRef = 39.78
RdRef = 0.74
sigma = 0.09
VcmaxRef = 26.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

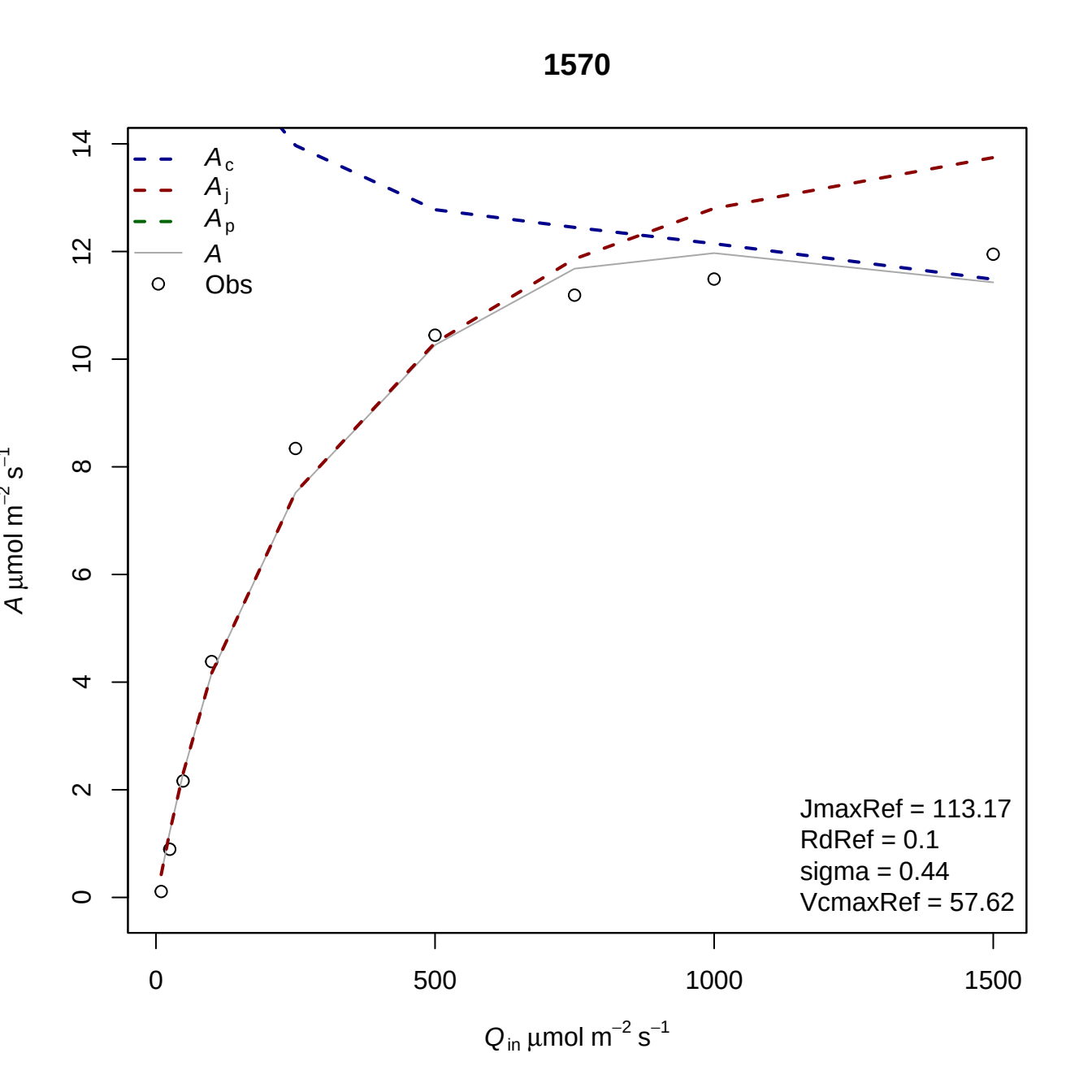

1565
A_c
A_j
A_p
A
Obs
JmaxRef = 32.36
RdRef = 0.79
sigma = 0.08
VcmaxRef = 20.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500

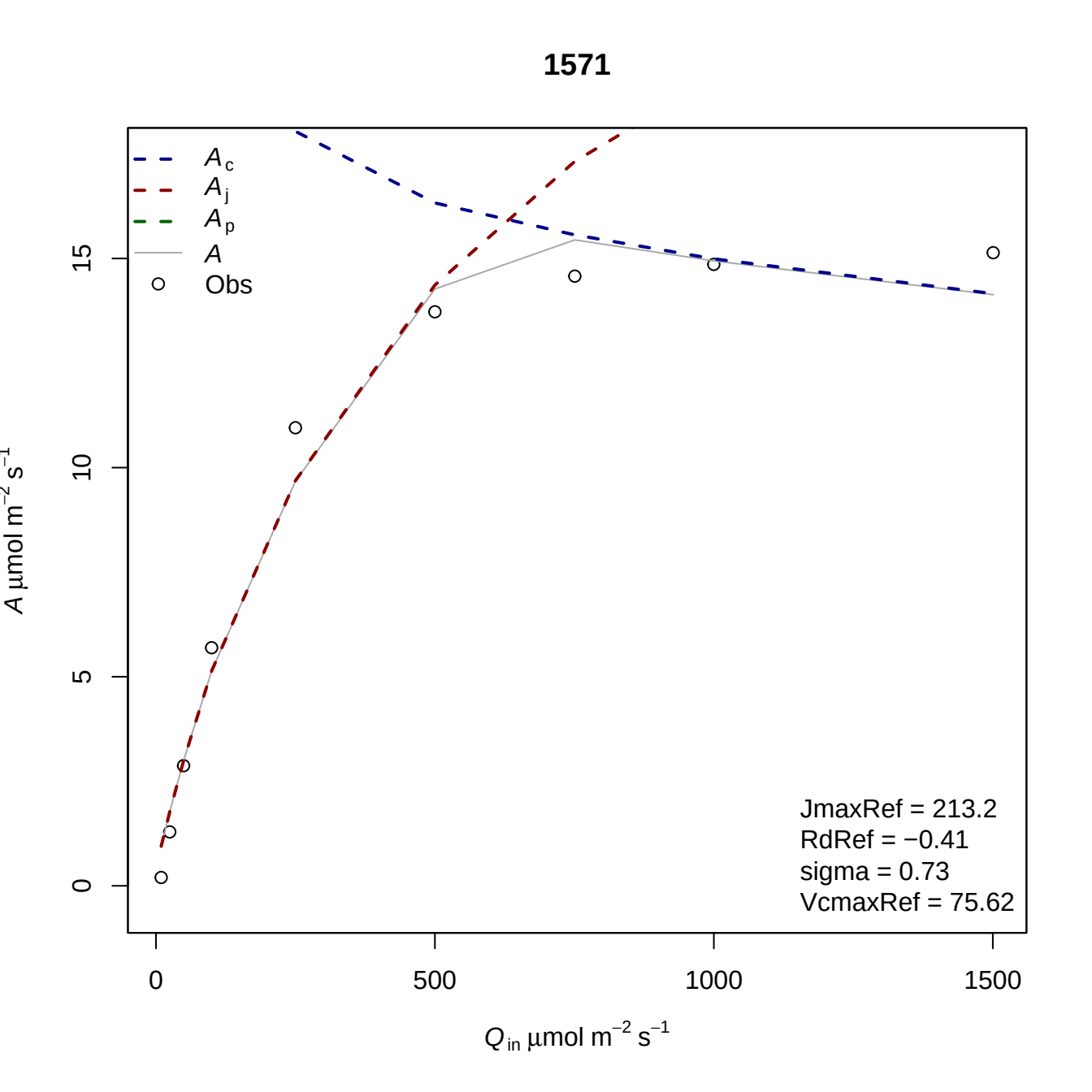

1566
A_c
A_j
A_p
A
Obs
JmaxRef = 91.01
RdRef = 0.31
sigma = 0.24
VcmaxRef = 51.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

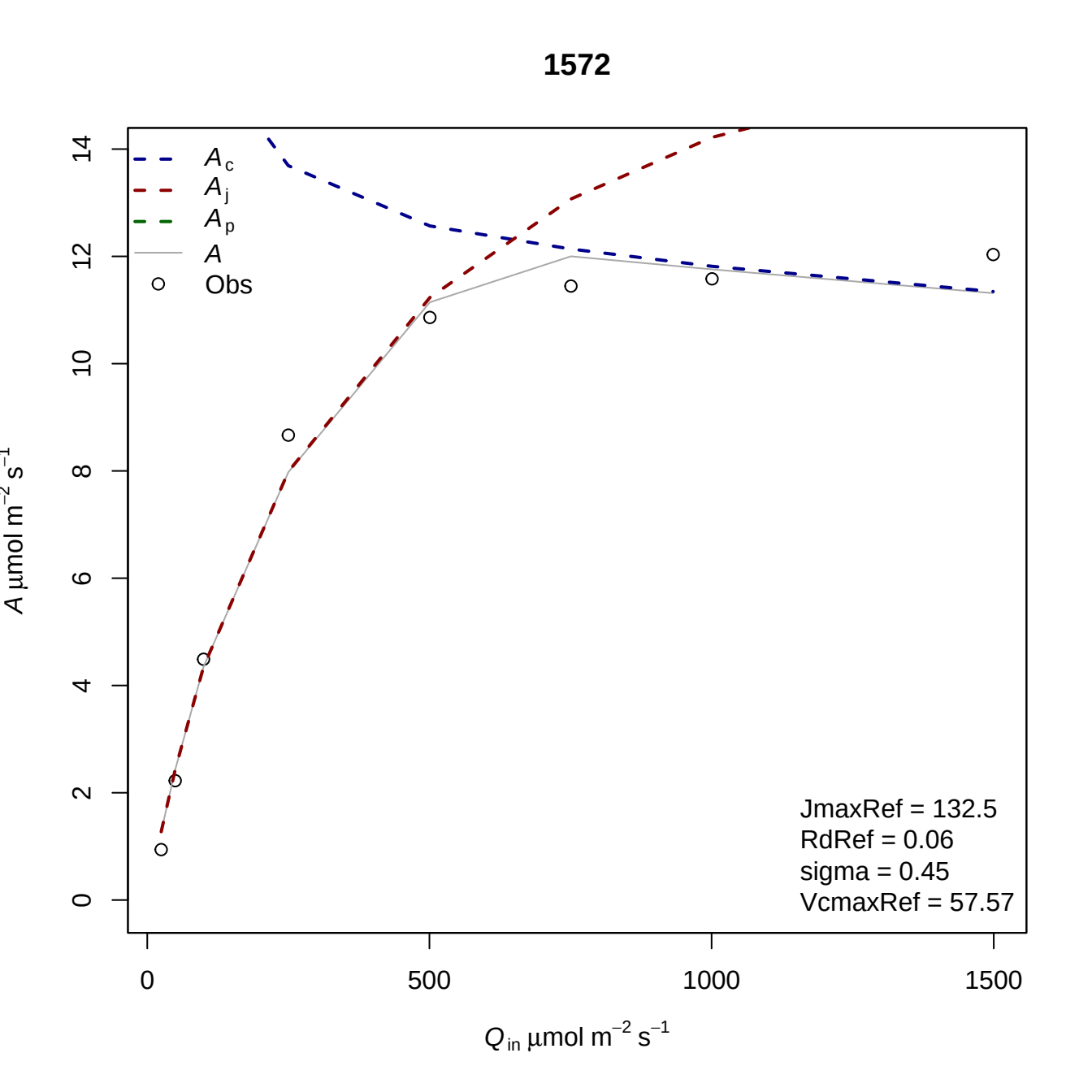

1567
A_c
A_j
A_p
A
Obs
JmaxRef = 101.04
RdRef = 0.34
sigma = 0.22
VcmaxRef = 57.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

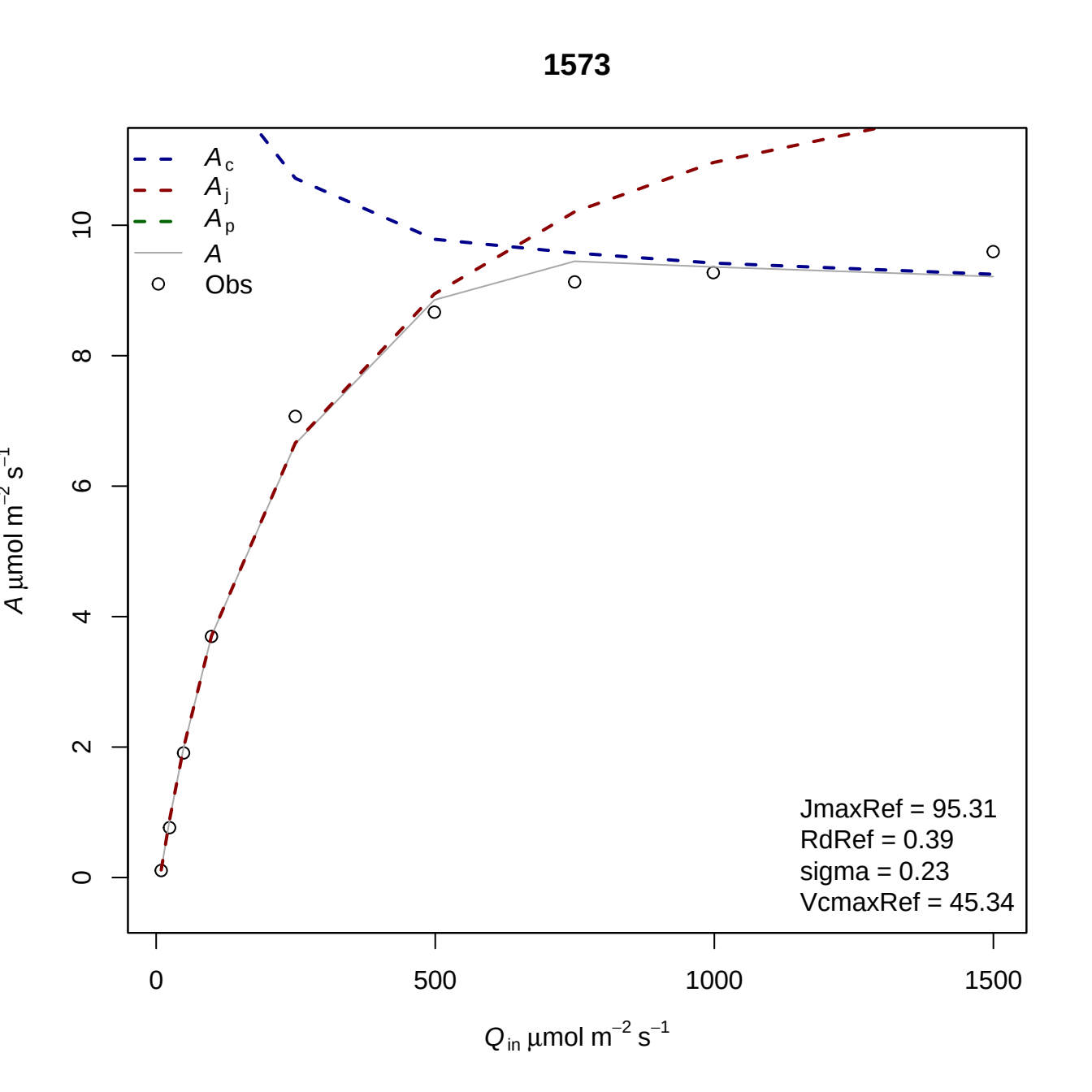

1568
A_c
A_j
A_p
A
Obs
JmaxRef = 145.96
RdRef = 0.51
sigma = 0.28
VcmaxRef = 67.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

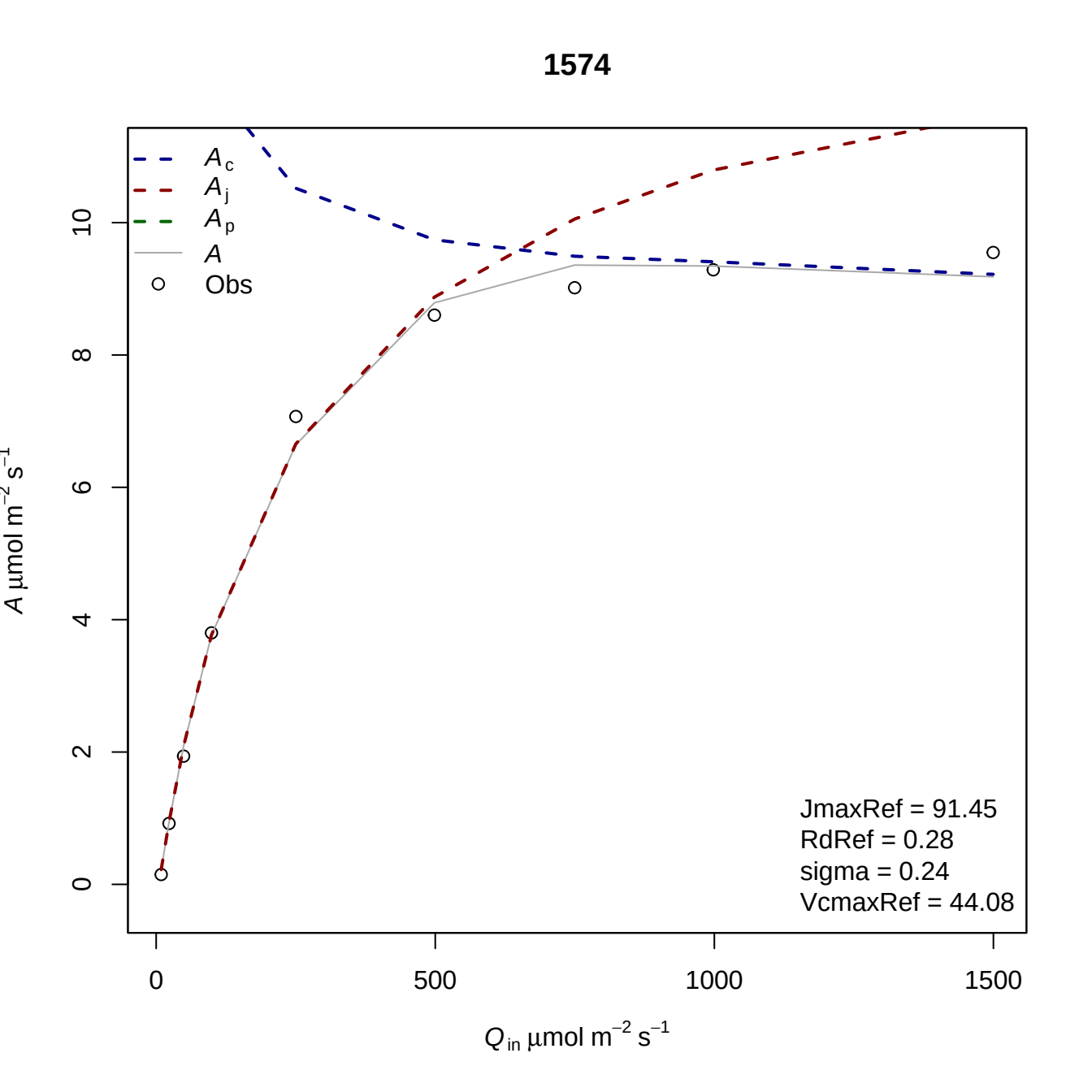

1569
A_c
A_j
A_p
A
Obs
JmaxRef = 94.43
RdRef = 0.3
sigma = 0.27
VcmaxRef = 45.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

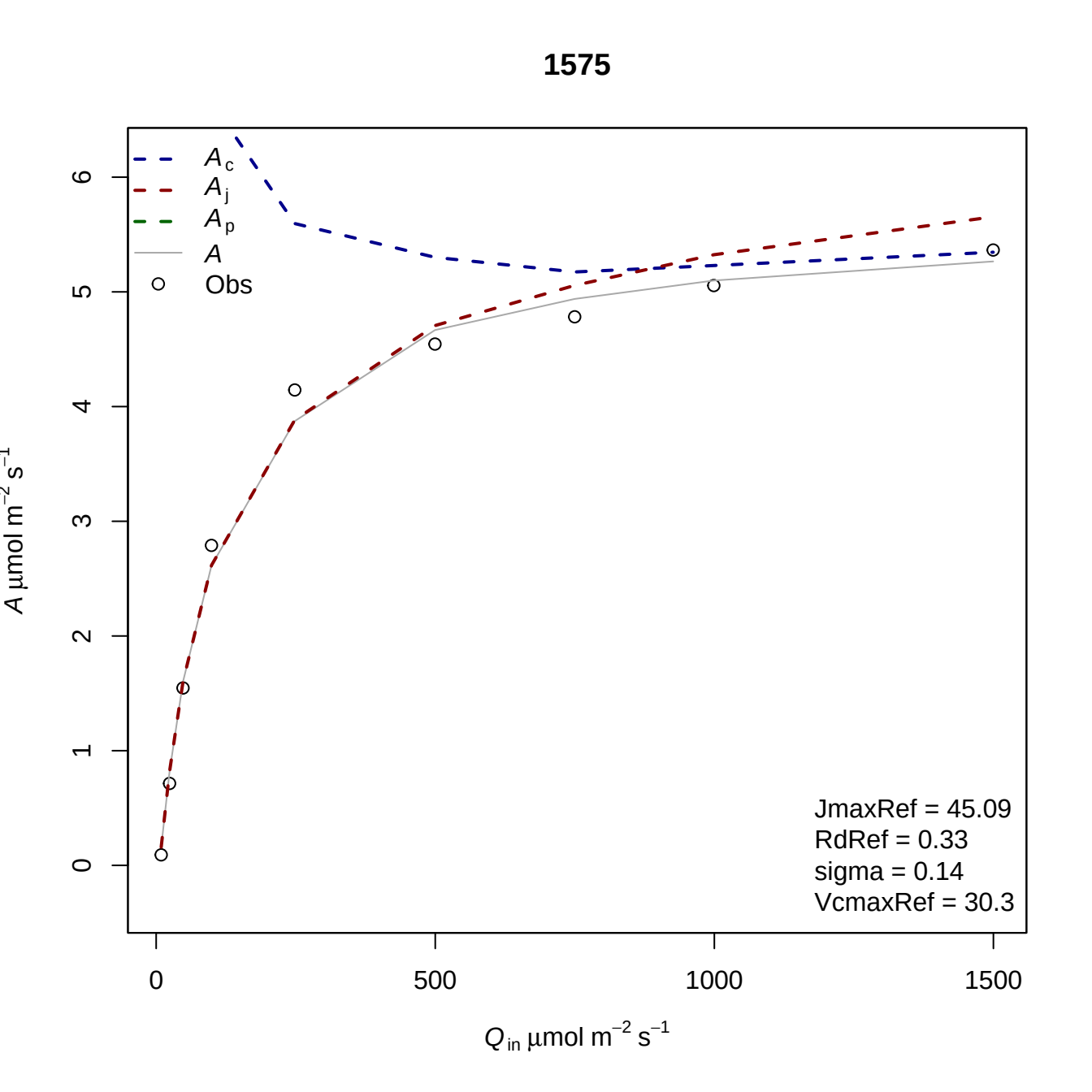

1570
A_c
A_j
A_p
A
Obs
JmaxRef = 113.17
RdRef = 0.1
sigma = 0.44
VcmaxRef = 57.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

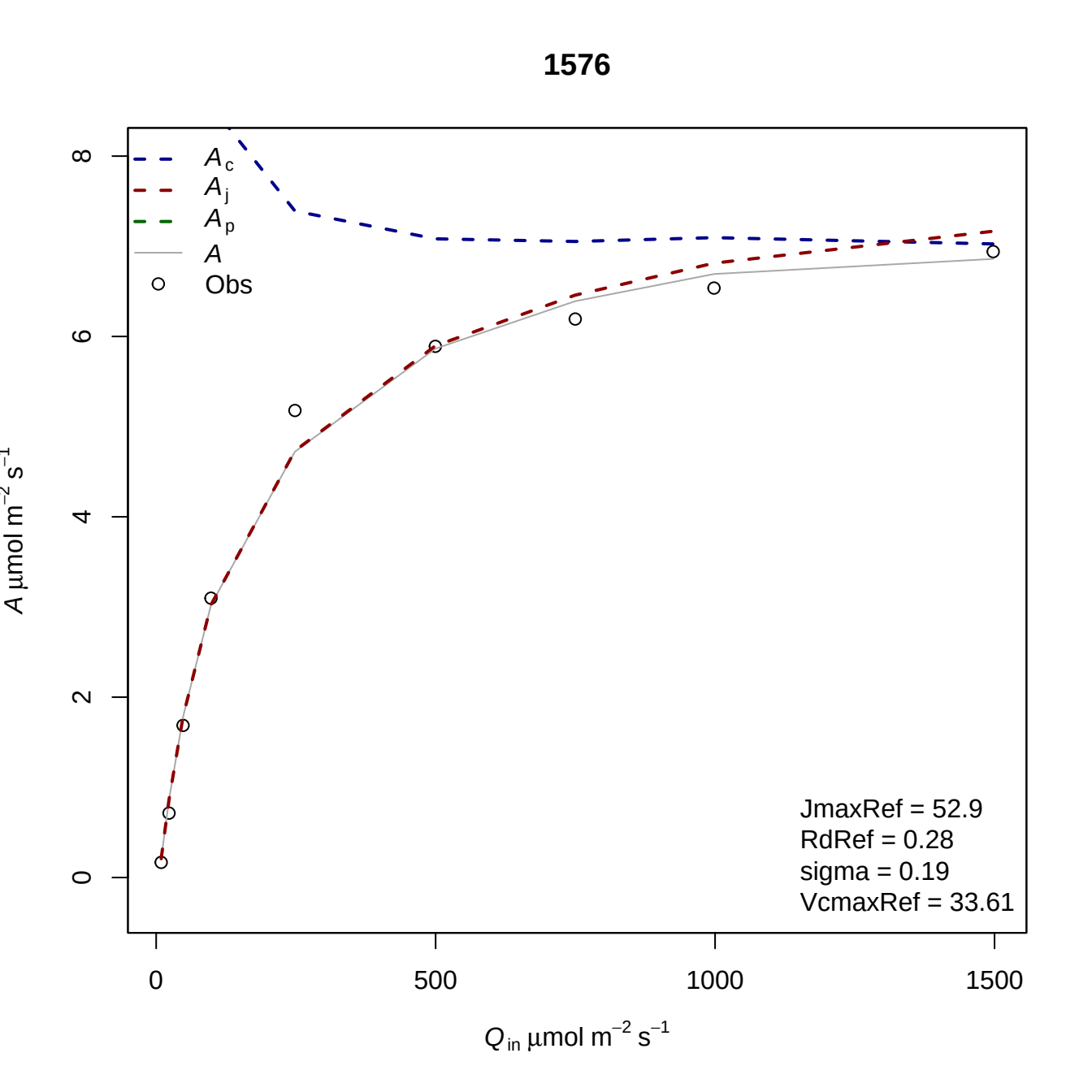

1571
A_c
A_j
A_p
A
Obs
JmaxRef = 213.2
RdRef = −0.41
sigma = 0.73
VcmaxRef = 75.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

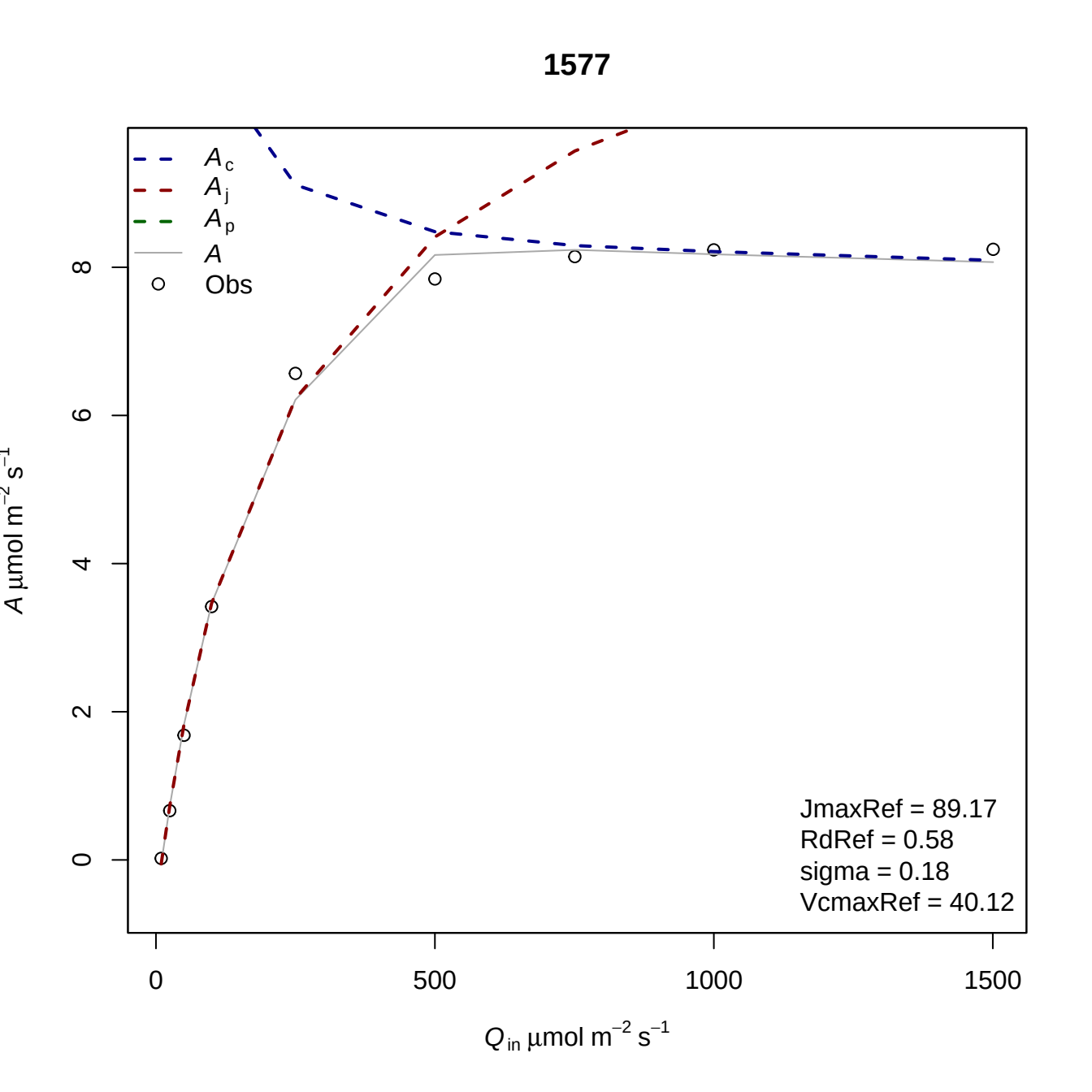

1572
A_c
A_j
A_p
A
Obs
JmaxRef = 132.5
RdRef = 0.06
sigma = 0.45
VcmaxRef = 57.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

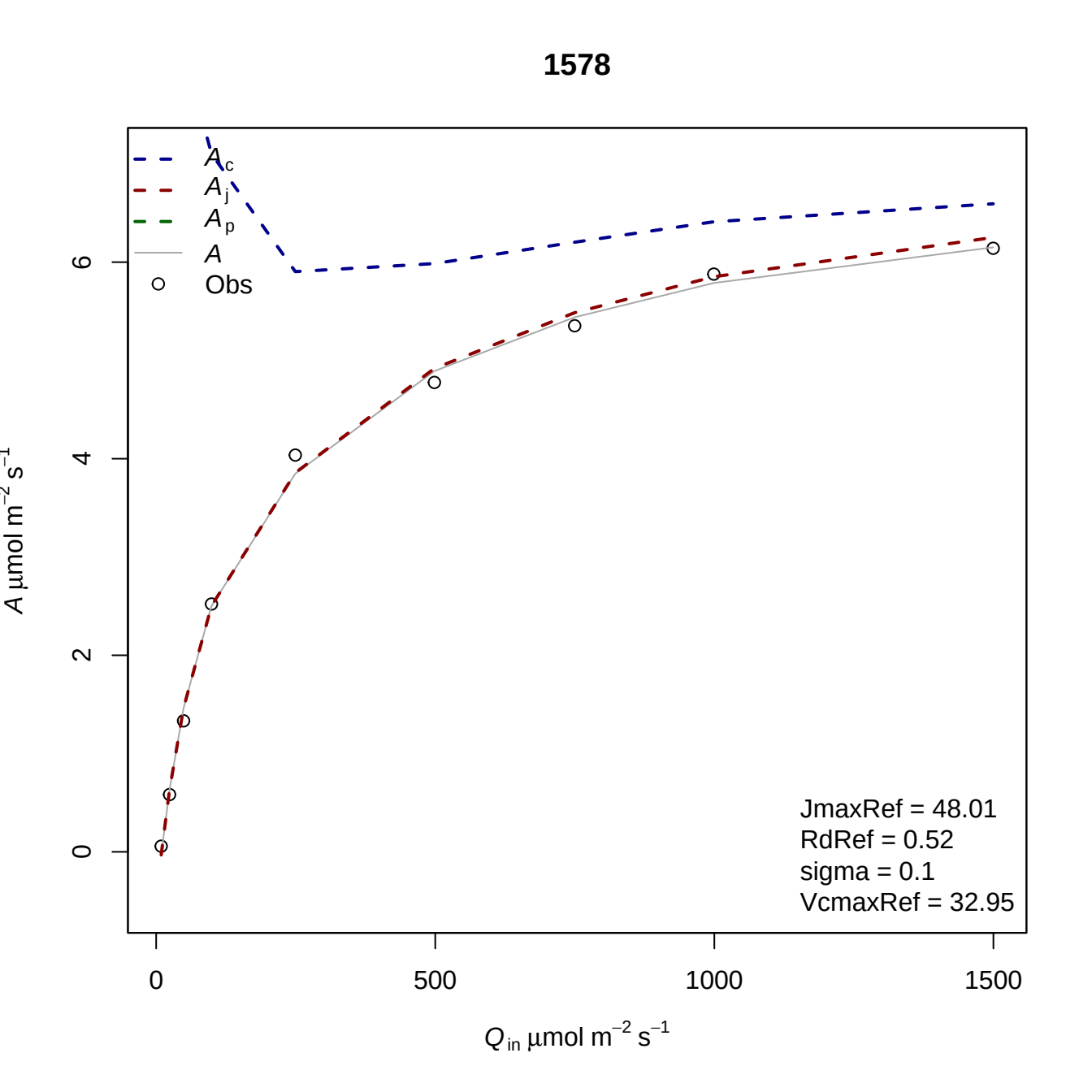

1573
A_c
A_j
A_p
A
Obs
JmaxRef = 95.31
RdRef = 0.39
sigma = 0.23
VcmaxRef = 45.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

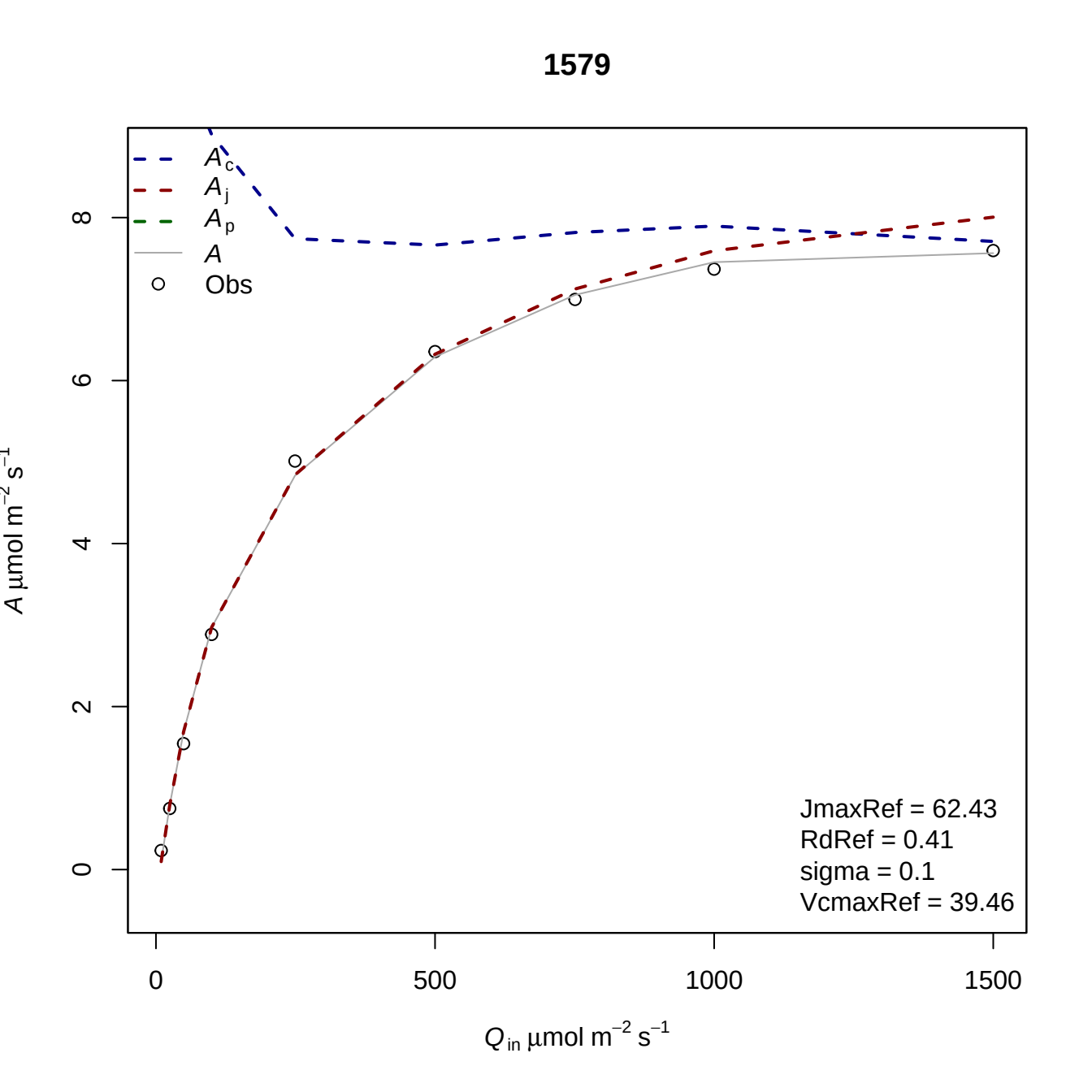

1574
A_c
A_j
A_p
A
Obs
JmaxRef = 91.45
RdRef = 0.28
sigma = 0.24
VcmaxRef = 44.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

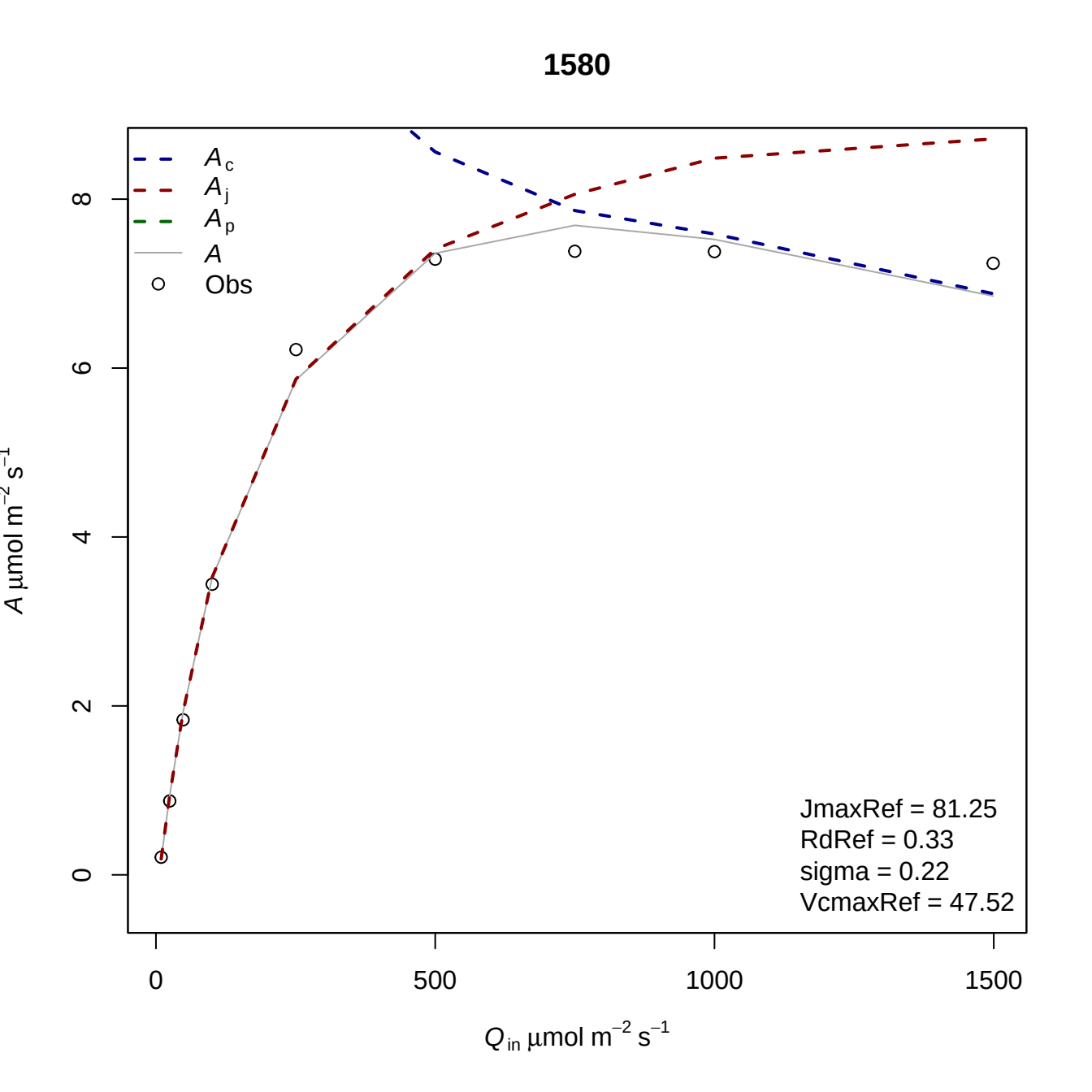

1575
A_c
A_j
A_p
A
Obs
JmaxRef = 45.09
RdRef = 0.33
sigma = 0.14
VcmaxRef = 30.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

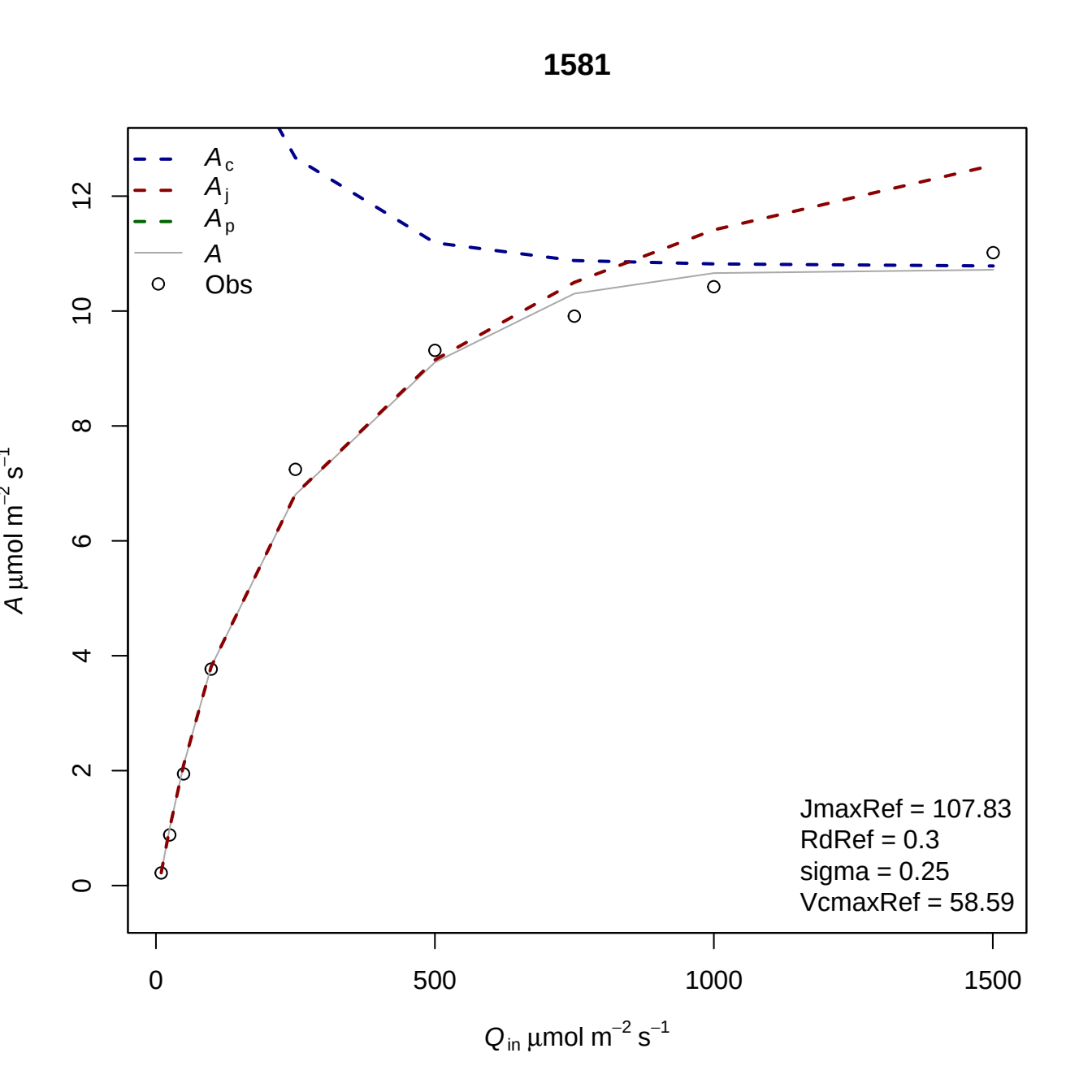

1576
A_c
A_j
A_p
A
Obs
JmaxRef = 52.9
RdRef = 0.28
sigma = 0.19
VcmaxRef = 33.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

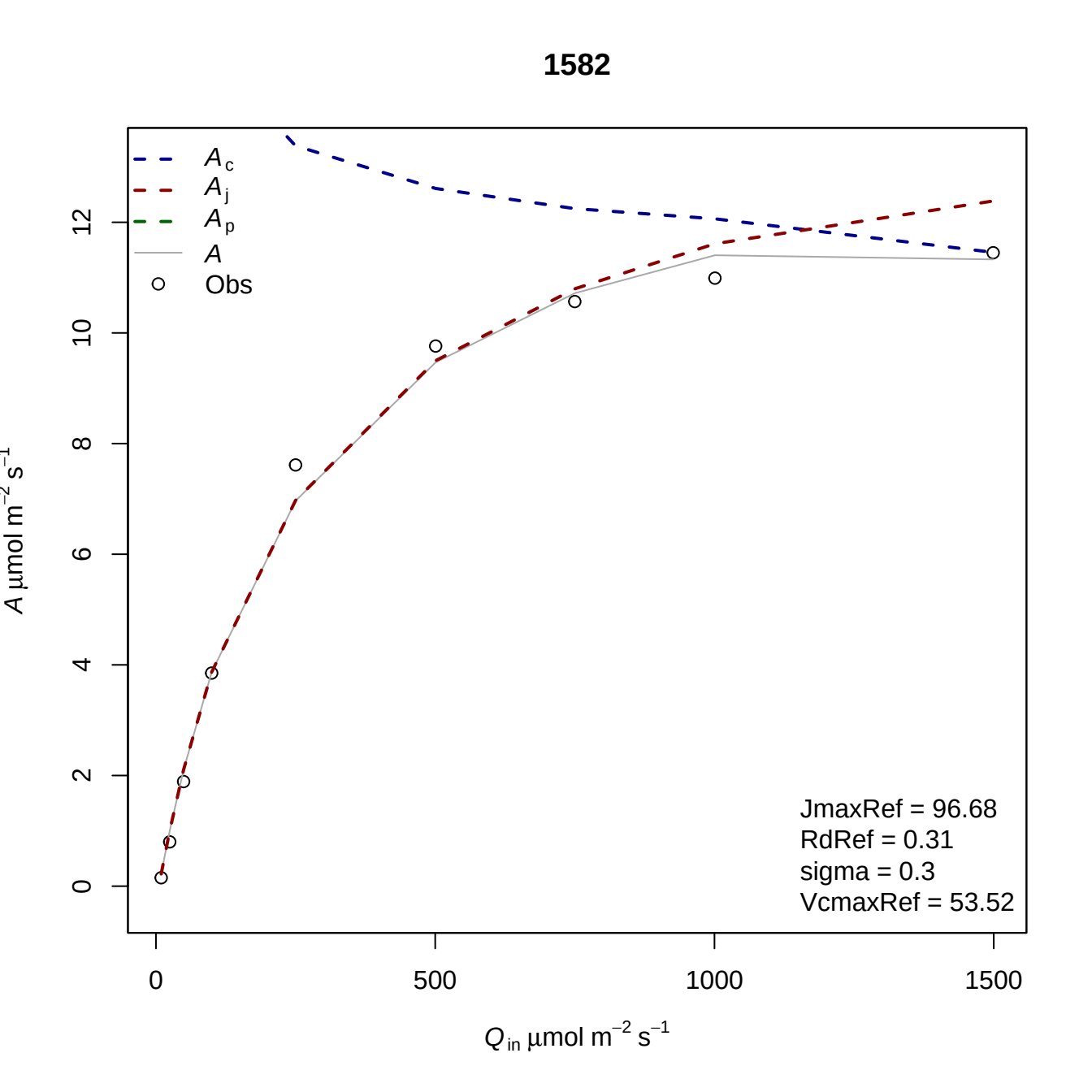

1577
A_c
A_j
A_p
A
Obs
JmaxRef = 89.17
RdRef = 0.58
sigma = 0.18
VcmaxRef = 40.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

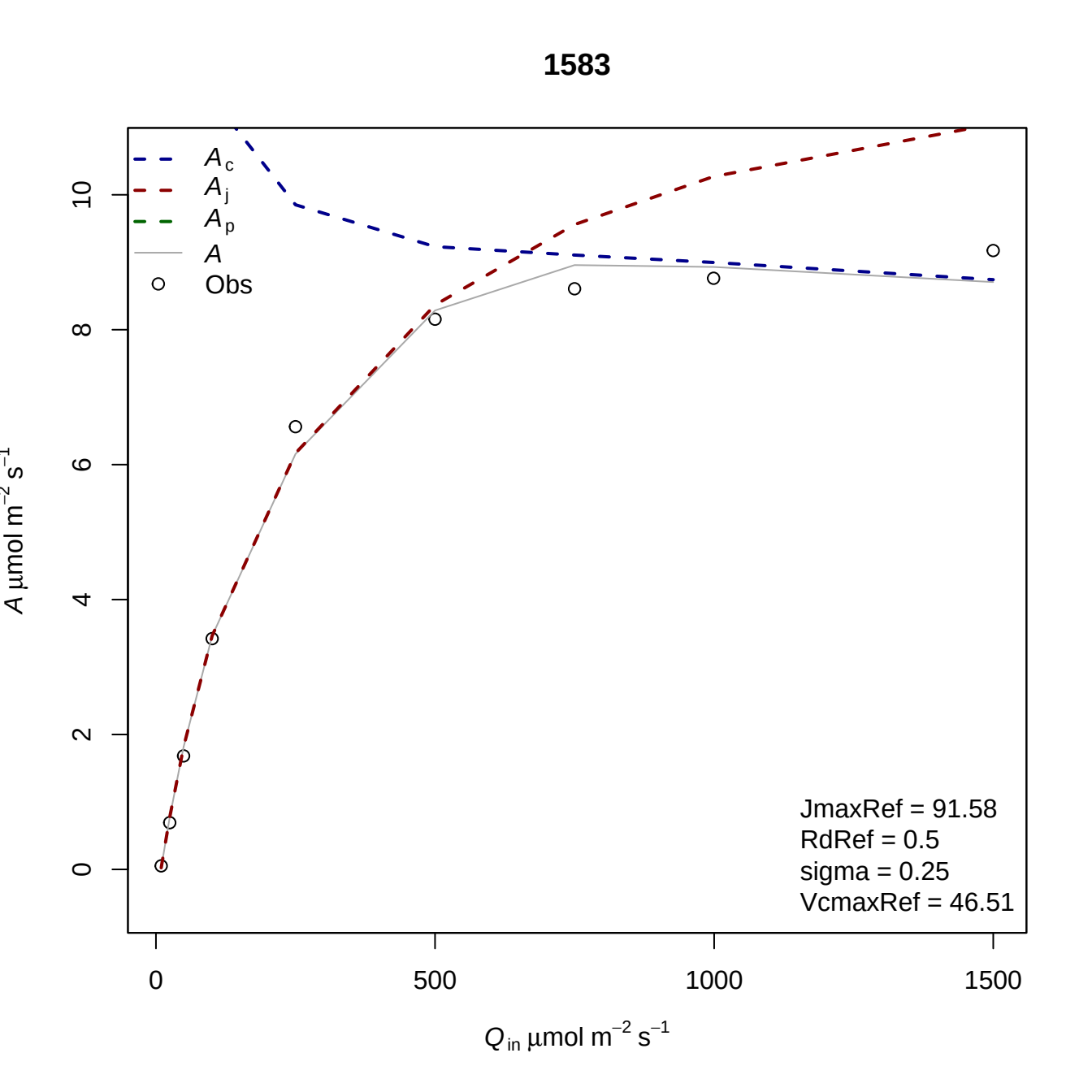

1578
A_c
A_j
A_p
A
Obs
JmaxRef = 48.01
RdRef = 0.52
sigma = 0.1
VcmaxRef = 32.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

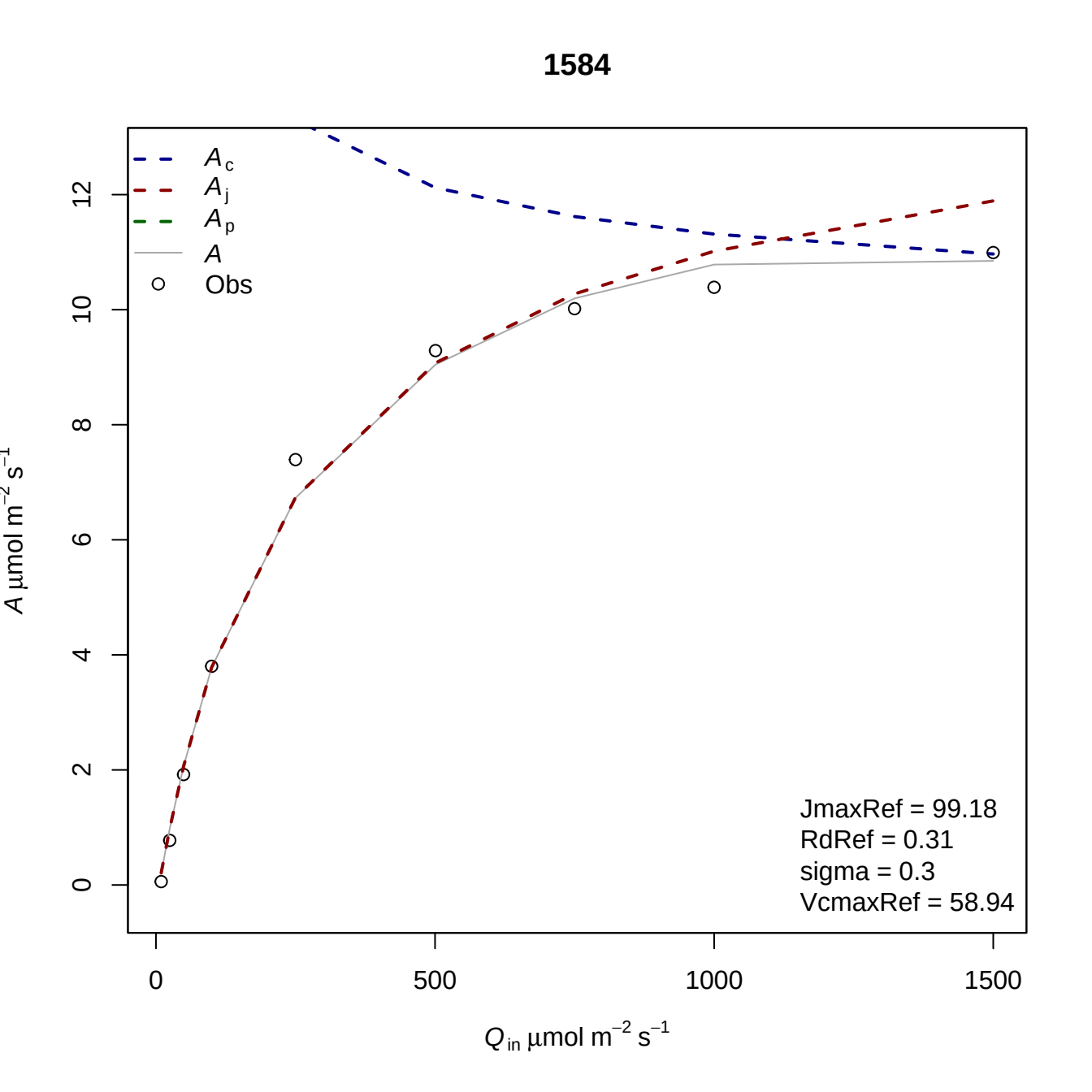

1579
A_c
A_j
A_p
A
Obs
JmaxRef = 62.43
RdRef = 0.41
sigma = 0.1
VcmaxRef = 39.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

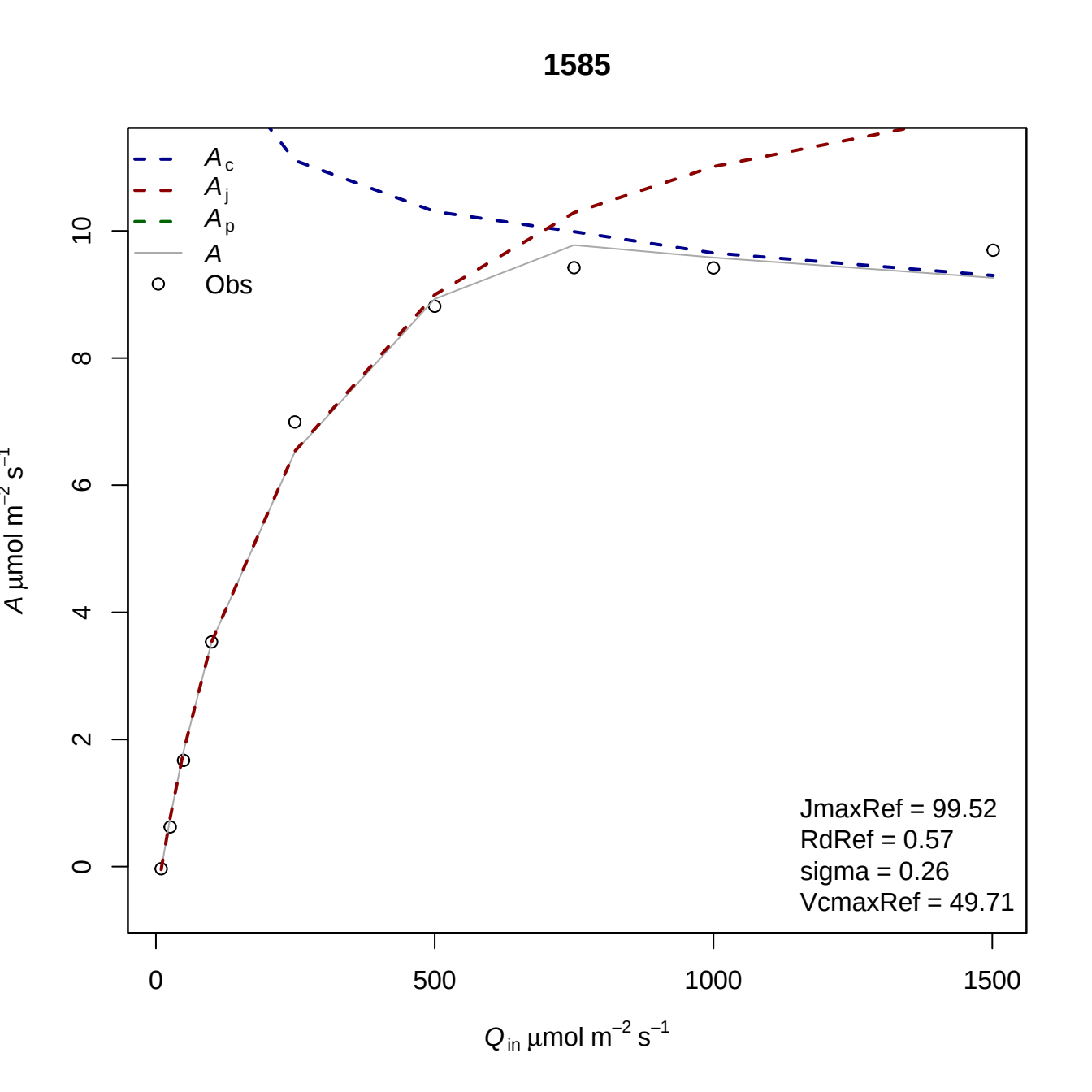

1580
A_c
A_j
A_p
A
Obs
JmaxRef = 81.25
RdRef = 0.33
sigma = 0.22
VcmaxRef = 47.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

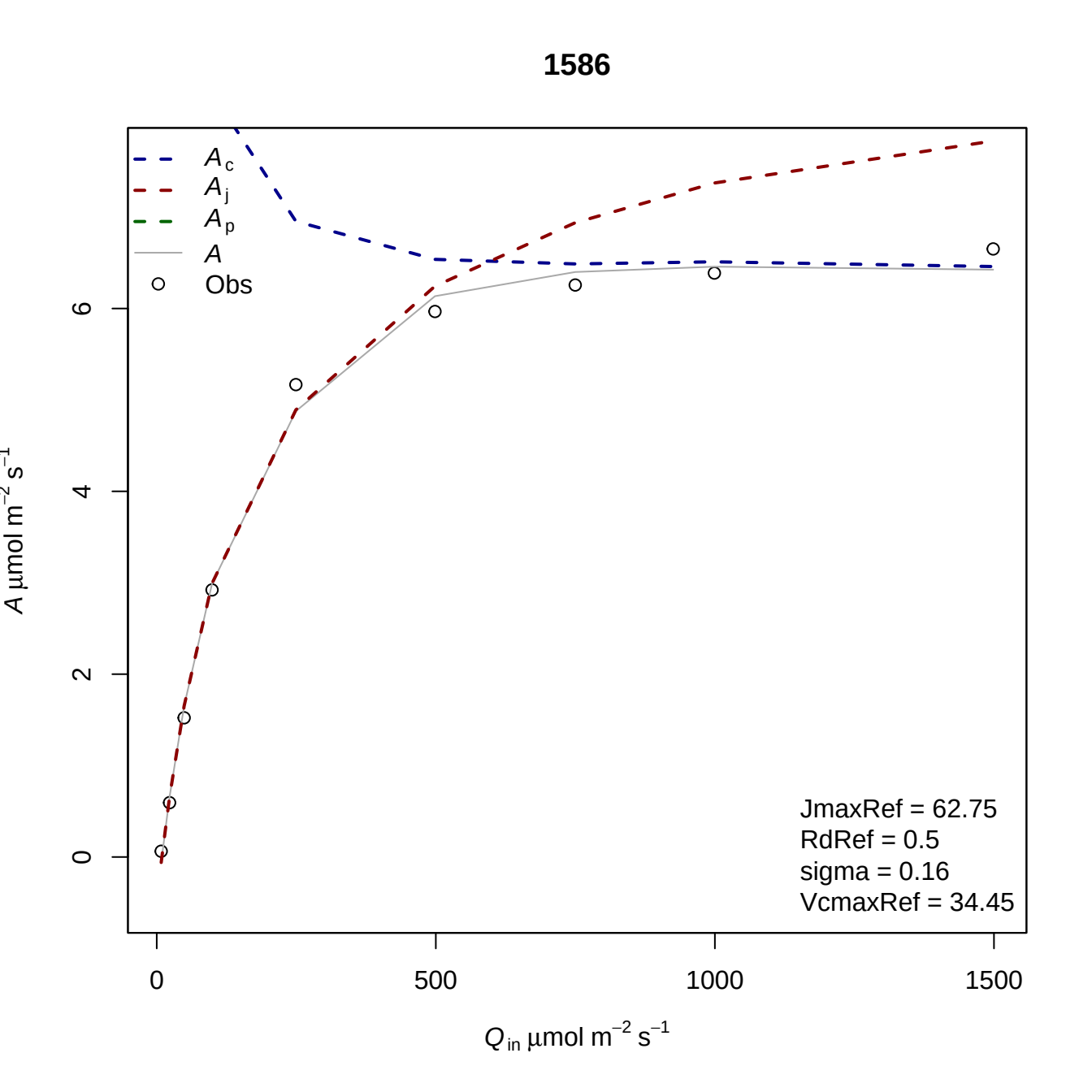

1581
A_c
A_j
A_p
A
Obs
JmaxRef = 107.83
RdRef = 0.3
sigma = 0.25
VcmaxRef = 58.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

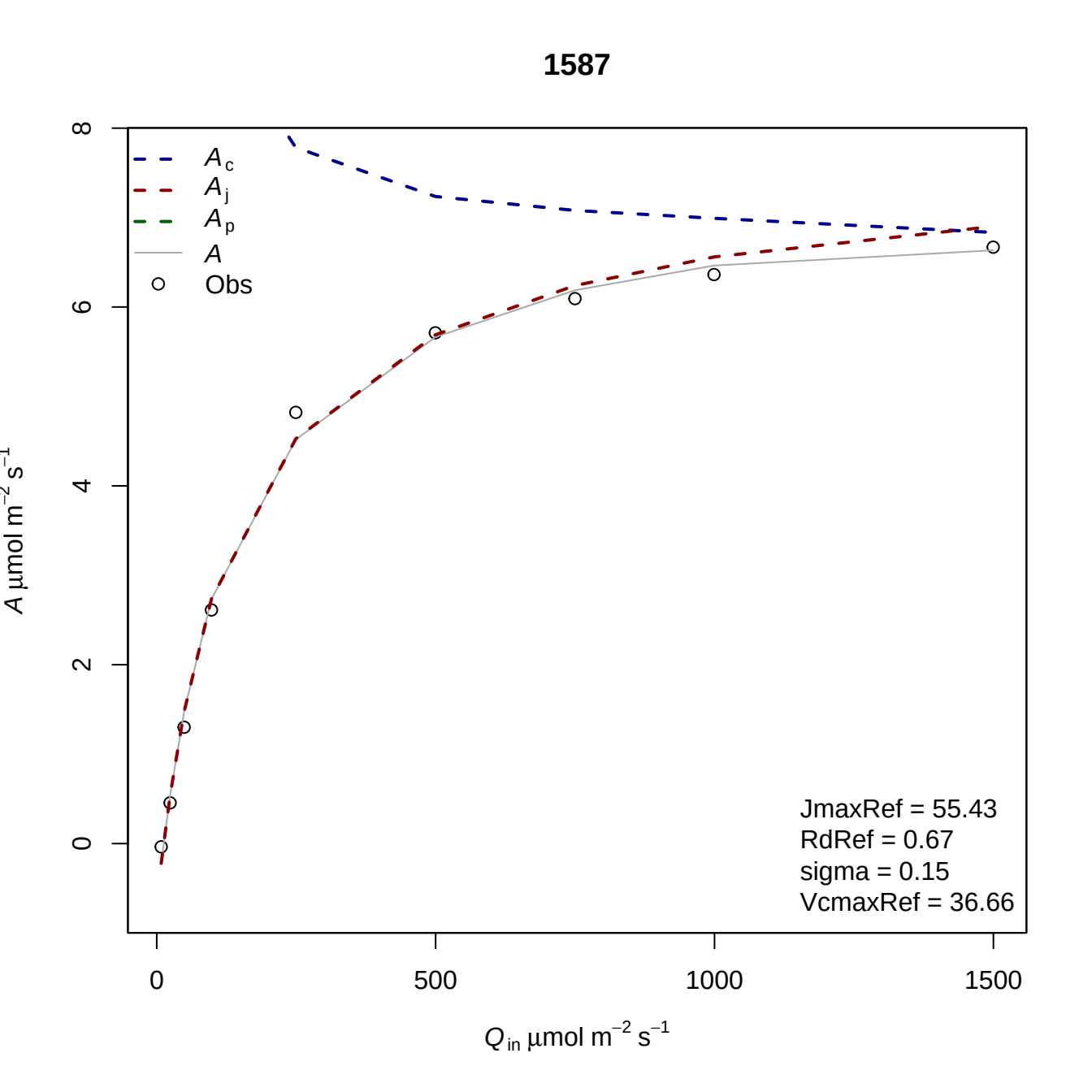

1582
A_c
A_j
A_p
A
Obs
JmaxRef = 96.68
RdRef = 0.31
sigma = 0.3
VcmaxRef = 53.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

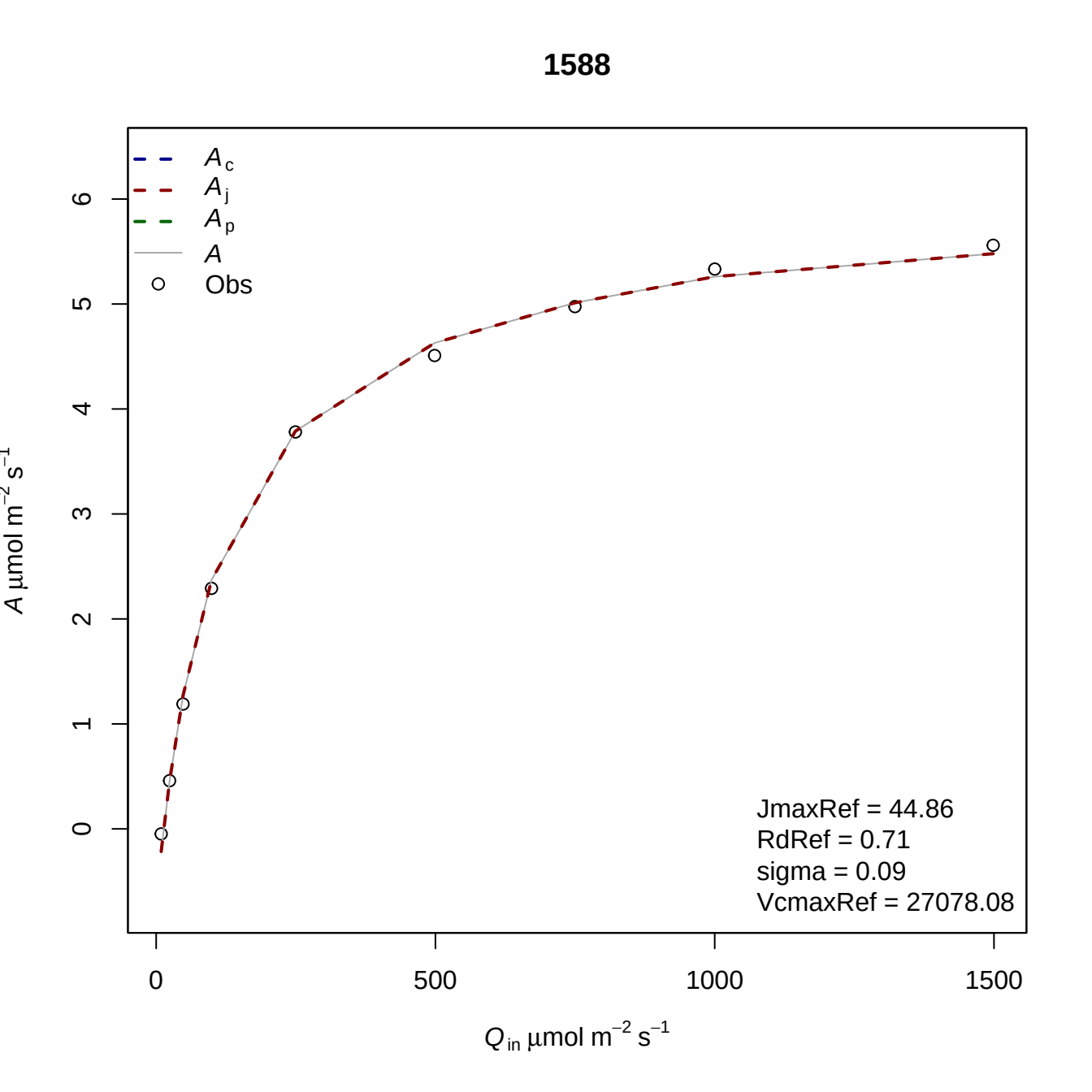

1583
A_c
A_j
A_p
A
Obs
JmaxRef = 91.58
RdRef = 0.5
sigma = 0.25
VcmaxRef = 46.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

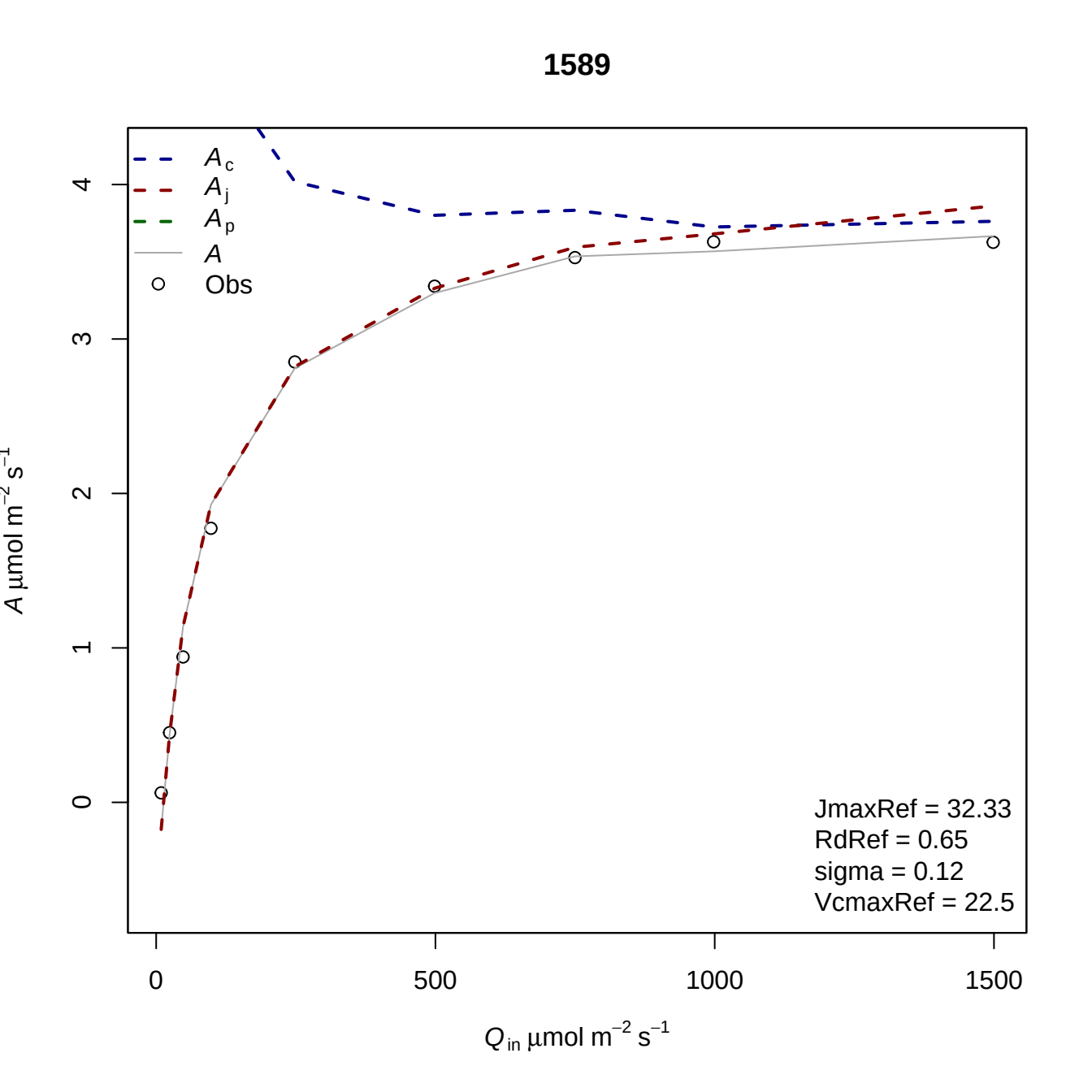

1584
A_c
A_j
A_p
A
Obs
JmaxRef = 99.18
RdRef = 0.31
sigma = 0.3
VcmaxRef = 58.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

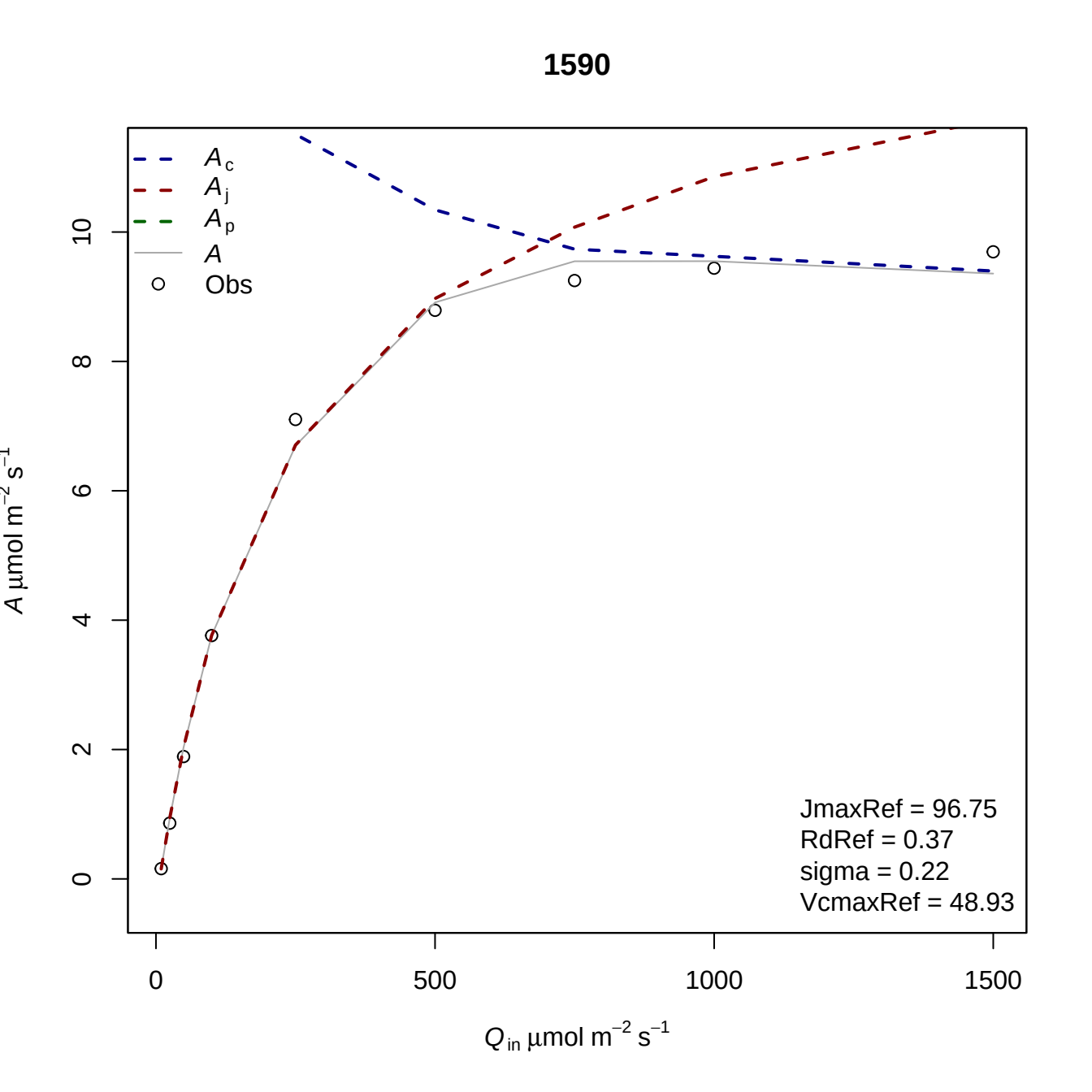

1585
A_c
A_j
A_p
A
Obs
JmaxRef = 99.52
RdRef = 0.57
sigma = 0.26
VcmaxRef = 49.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

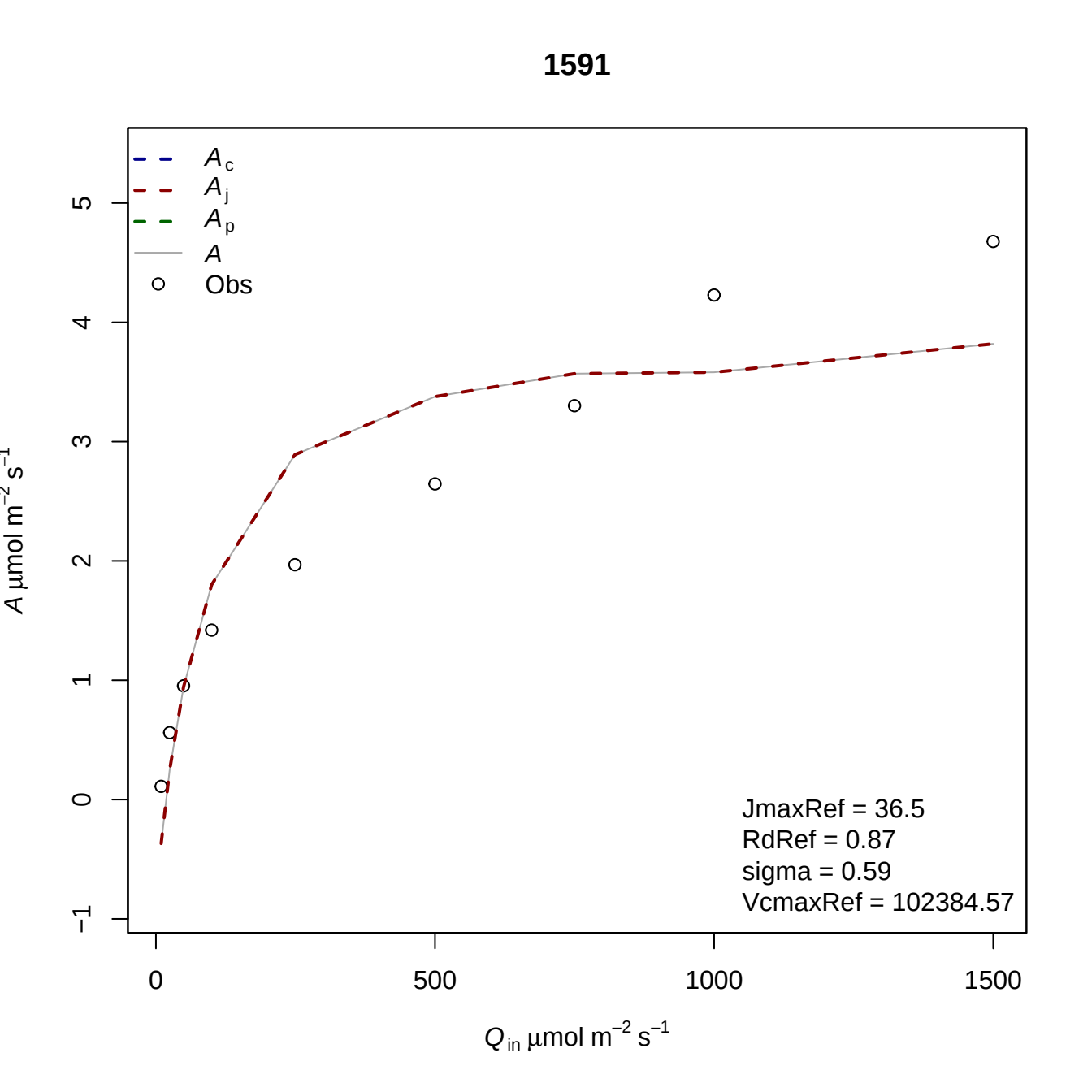

1586
A_c
A_j
A_p
A
Obs
JmaxRef = 62.75
RdRef = 0.5
sigma = 0.16
VcmaxRef = 34.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

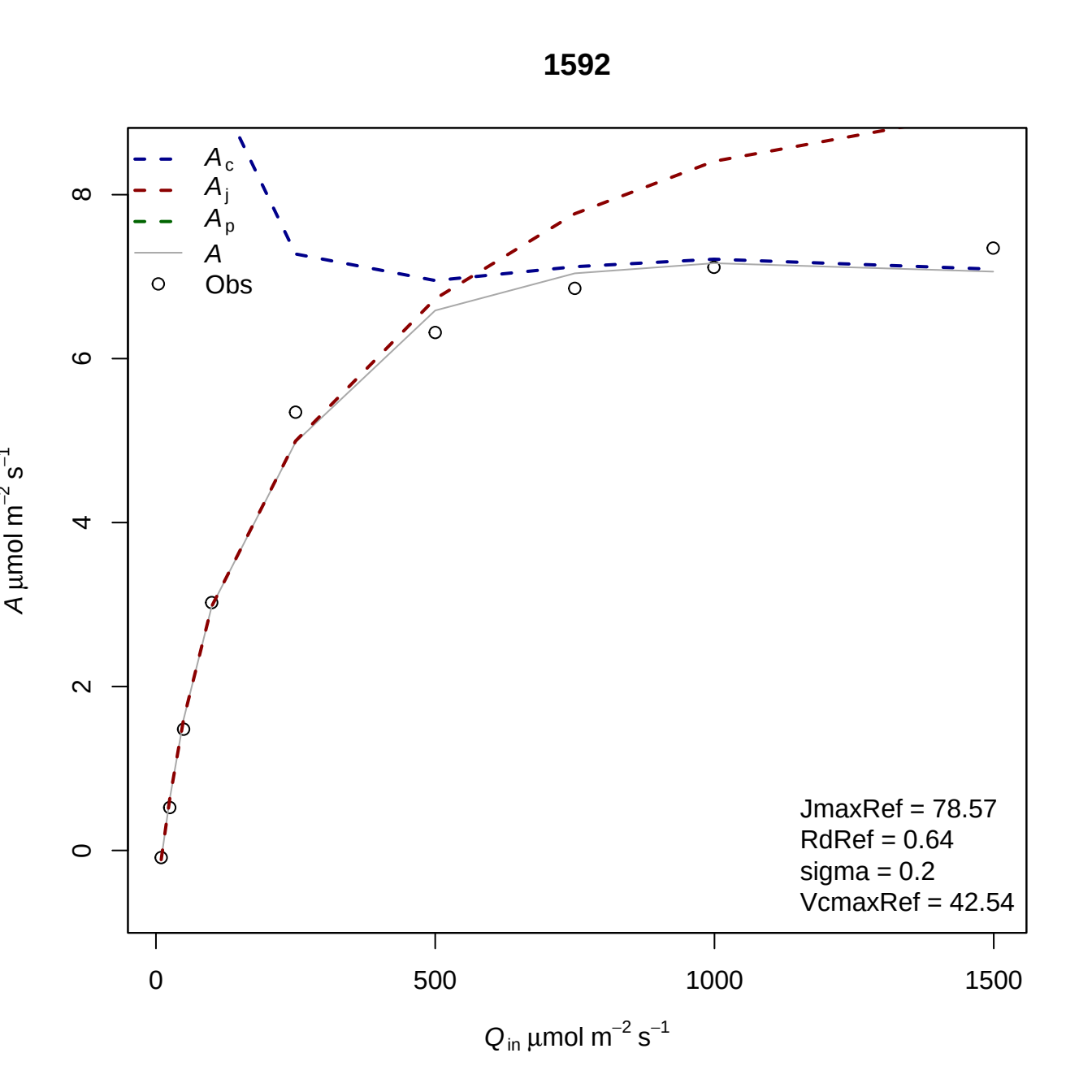

1587
A_c
A_j
A_p
A
Obs
JmaxRef = 55.43
RdRef = 0.67
sigma = 0.15
VcmaxRef = 36.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

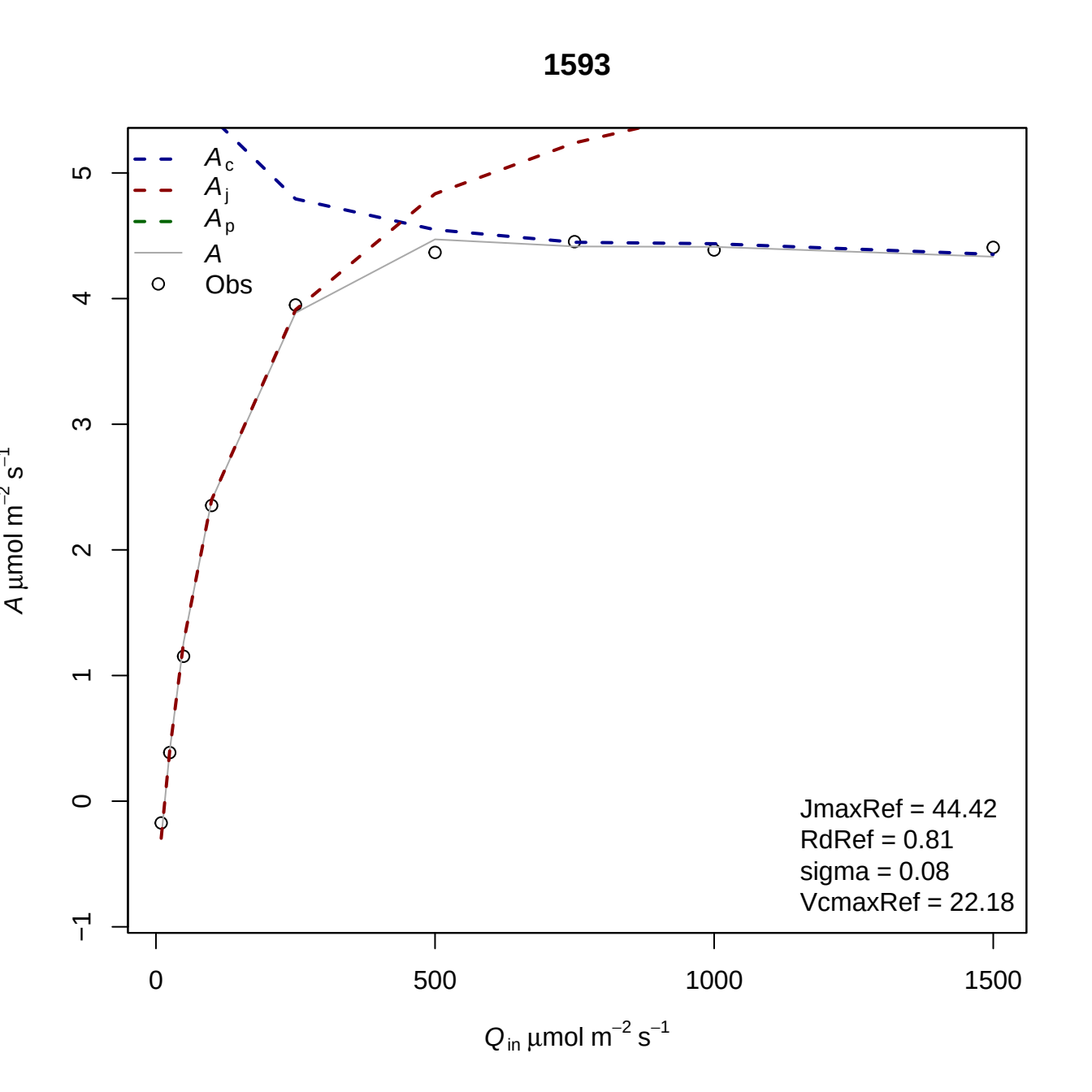

1588
A_c
A_j
A_p
A
Obs
JmaxRef = 44.86
RdRef = 0.71
sigma = 0.09
VcmaxRef = 27078.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

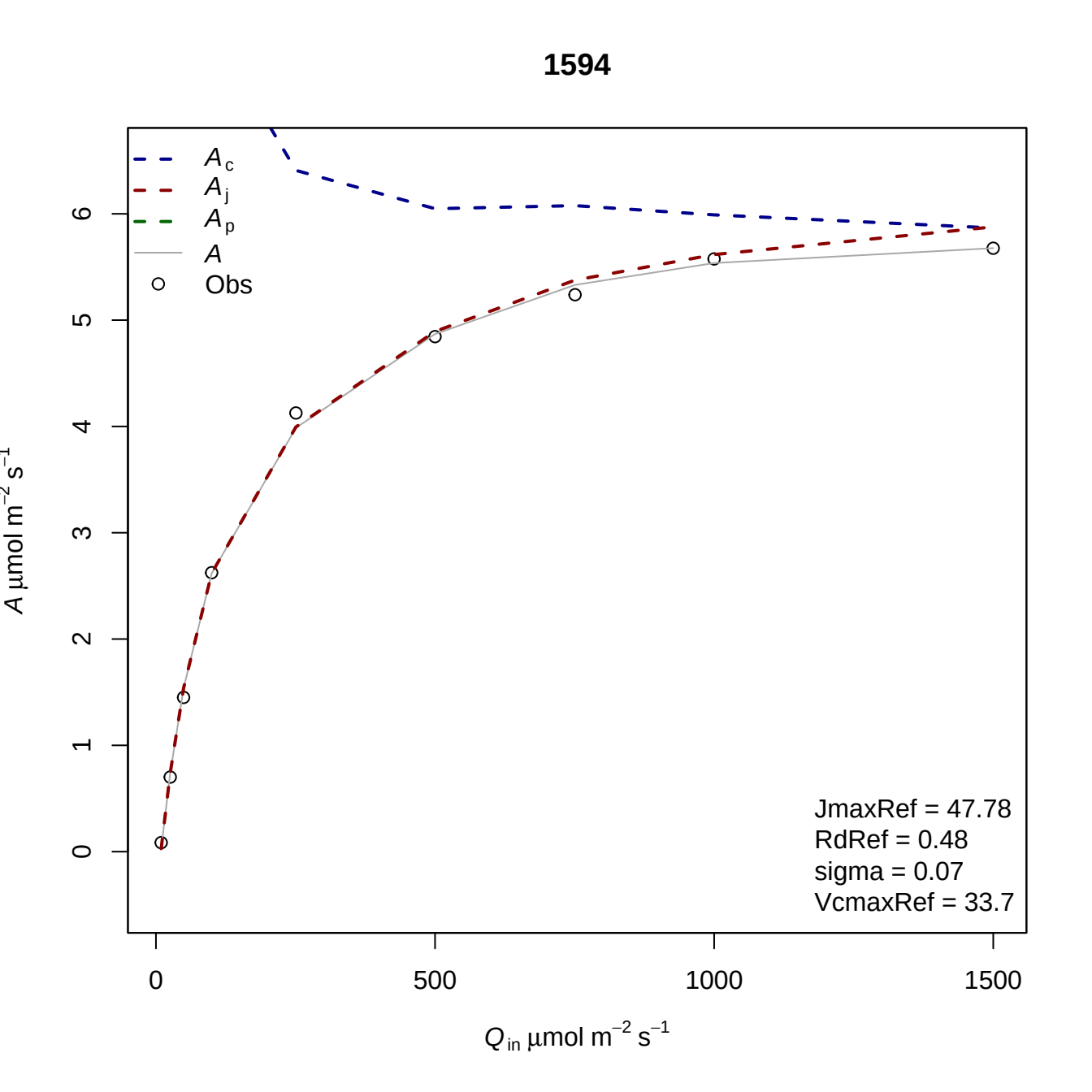

1589
A_c
A_j
A_p
A
Obs
JmaxRef = 32.33
RdRef = 0.65
sigma = 0.12
VcmaxRef = 22.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

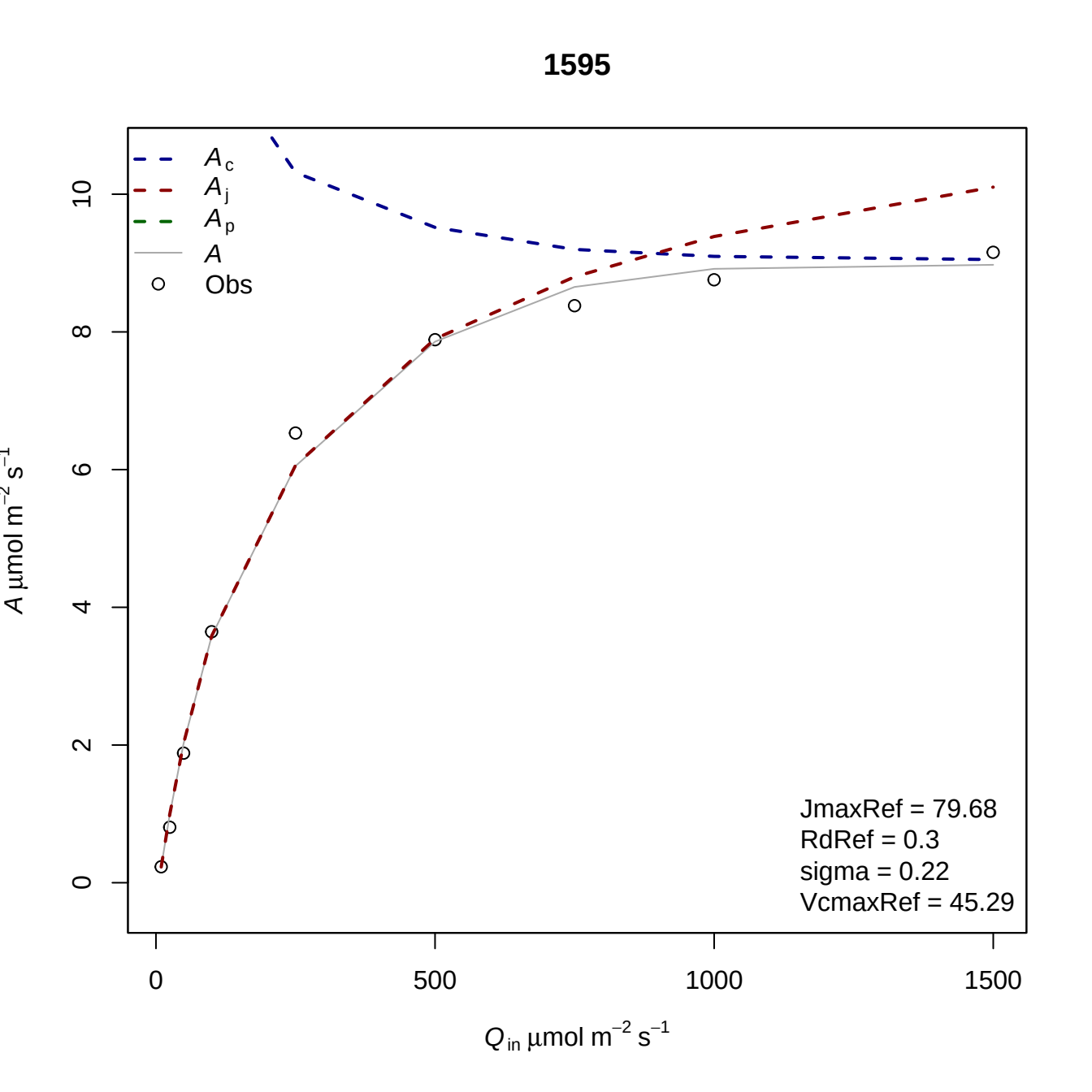

1590
A_c
A_j
A_p
A
Obs
JmaxRef = 96.75
RdRef = 0.37
sigma = 0.22
VcmaxRef = 48.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

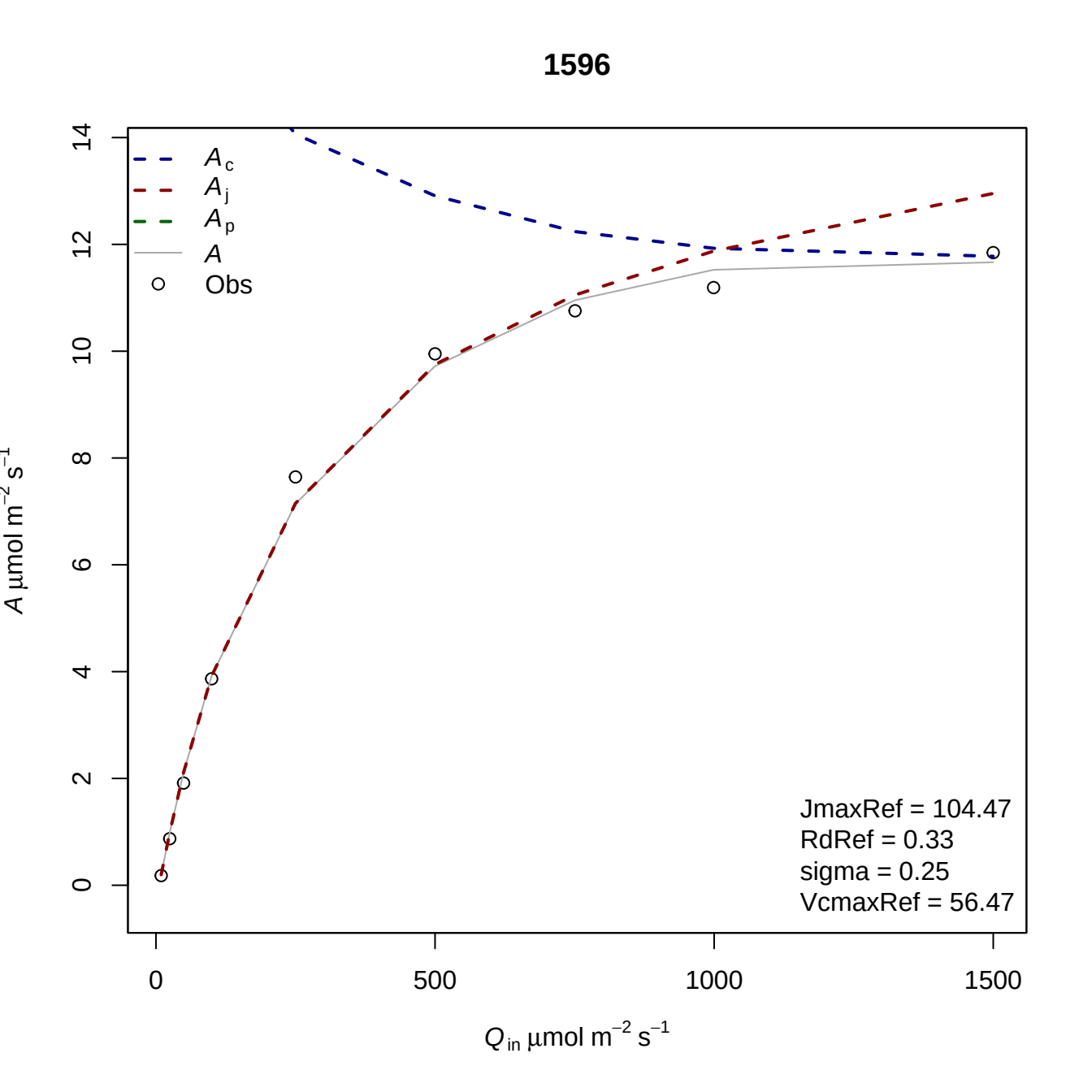

1591
A_c
A_j
A_p
A
Obs
JmaxRef = 36.5
RdRef = 0.87
sigma = 0.59
VcmaxRef = 102384.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

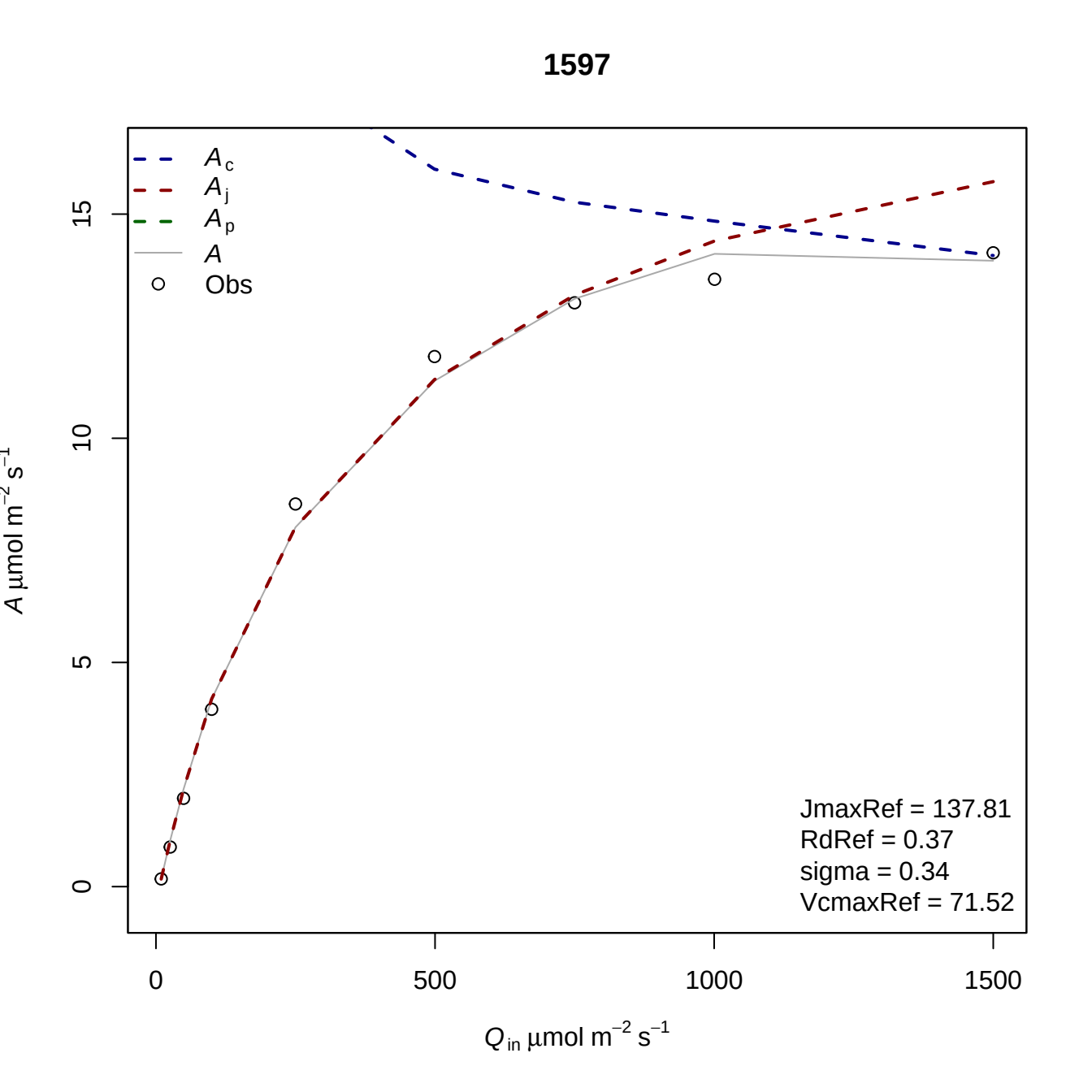

1592
A_c
A_j
A_p
A
Obs
JmaxRef = 78.57
RdRef = 0.64
sigma = 0.2
VcmaxRef = 42.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

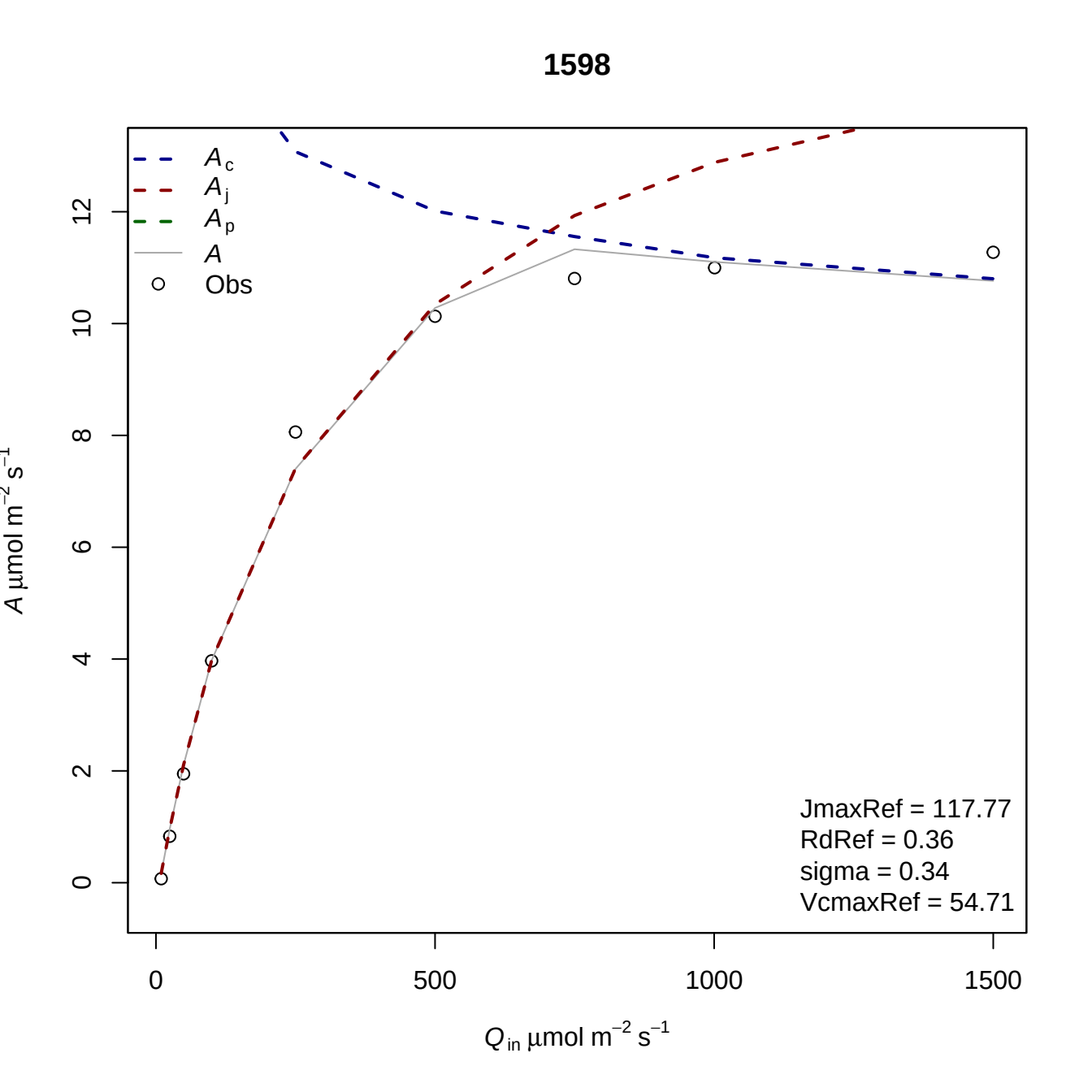

1593
A_c
A_j
A_p
A
Obs
JmaxRef = 44.42
RdRef = 0.81
sigma = 0.08
VcmaxRef = 22.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

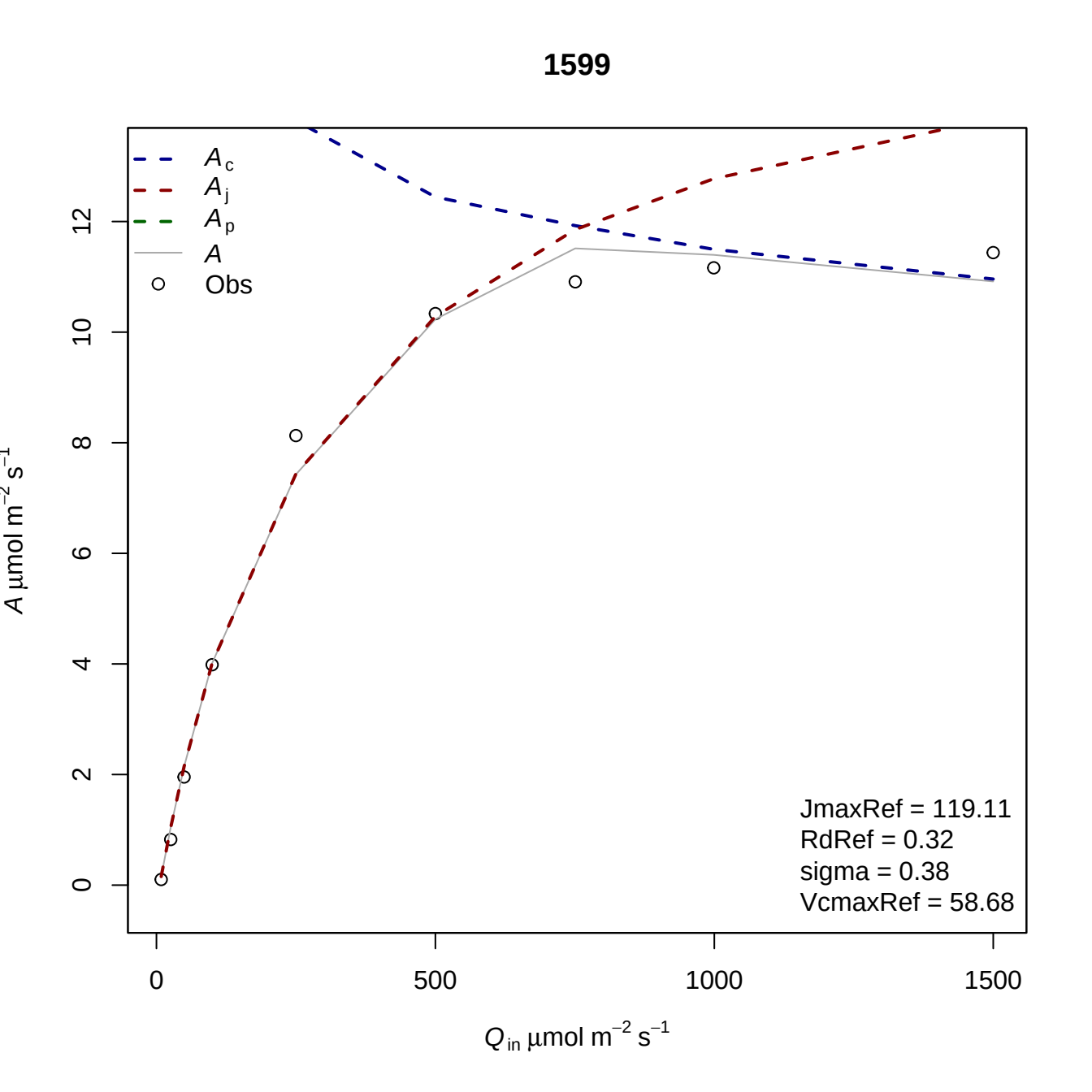

1594
A_c
A_j
A_p
A
Obs
JmaxRef = 47.78
RdRef = 0.48
sigma = 0.07
VcmaxRef = 33.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

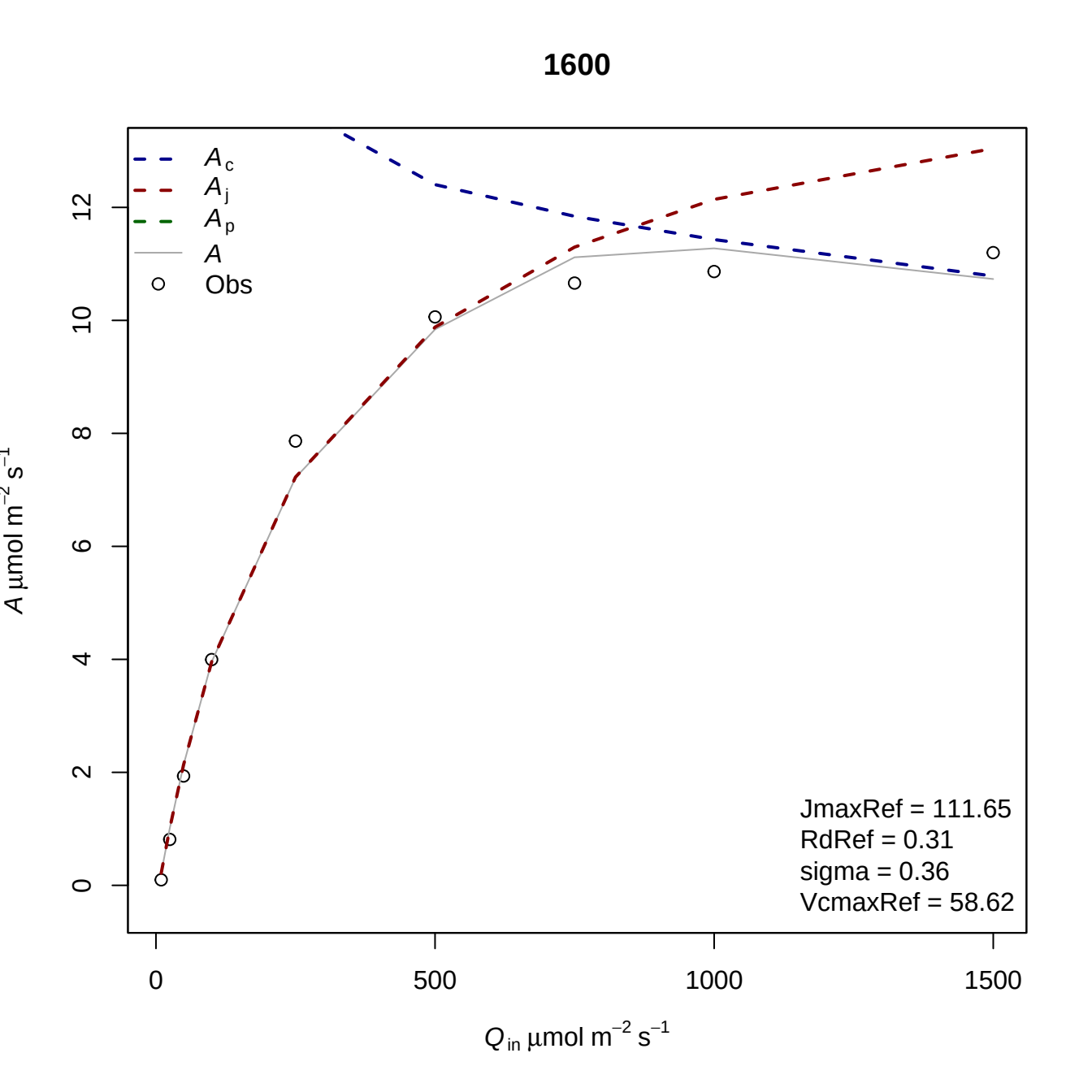

1595
A_c
A_j
A_p
A
Obs
JmaxRef = 79.68
RdRef = 0.3
sigma = 0.22
VcmaxRef = 45.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

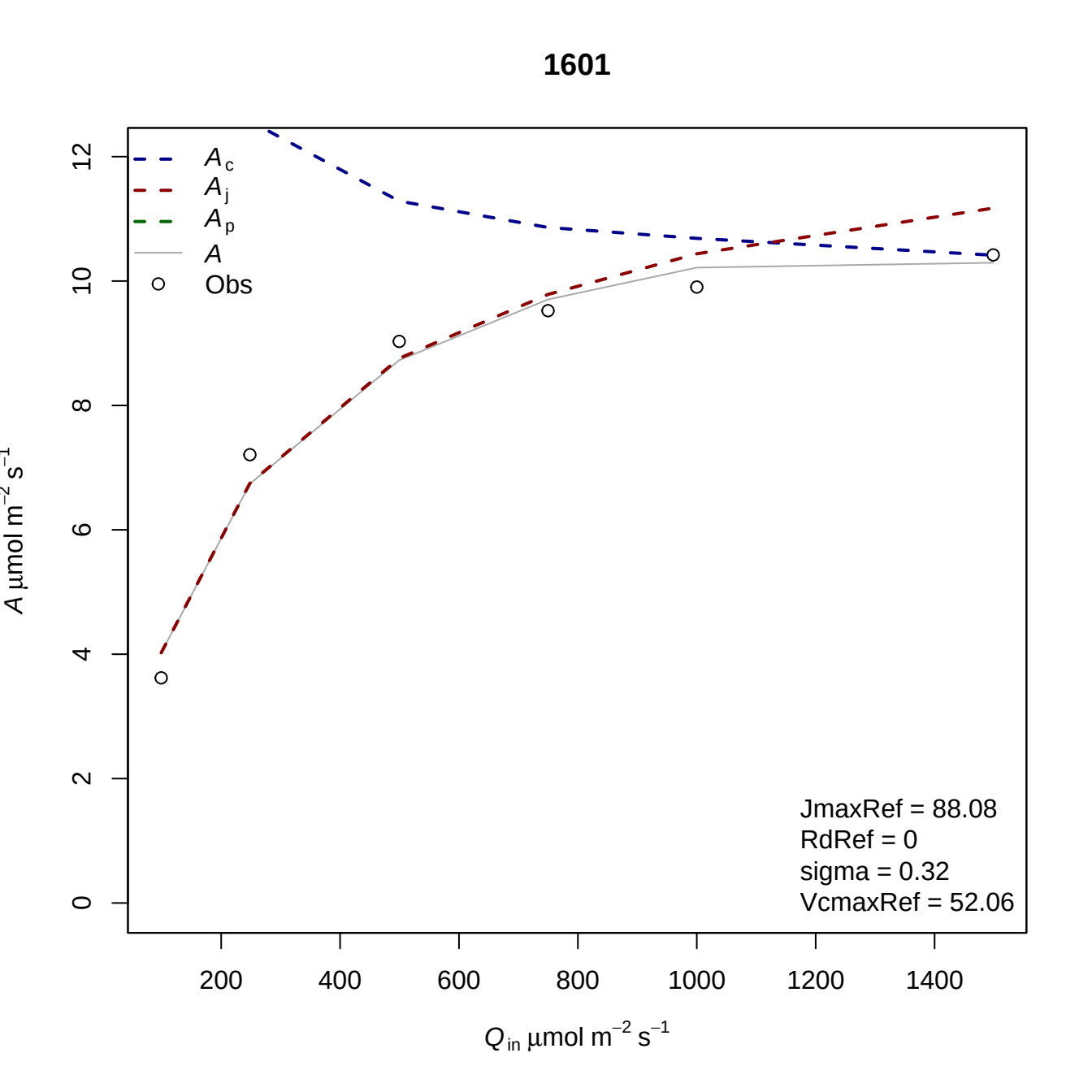

1596
A_c
A_j
A_p
A
Obs
JmaxRef = 104.47
RdRef = 0.33
sigma = 0.25
VcmaxRef = 56.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

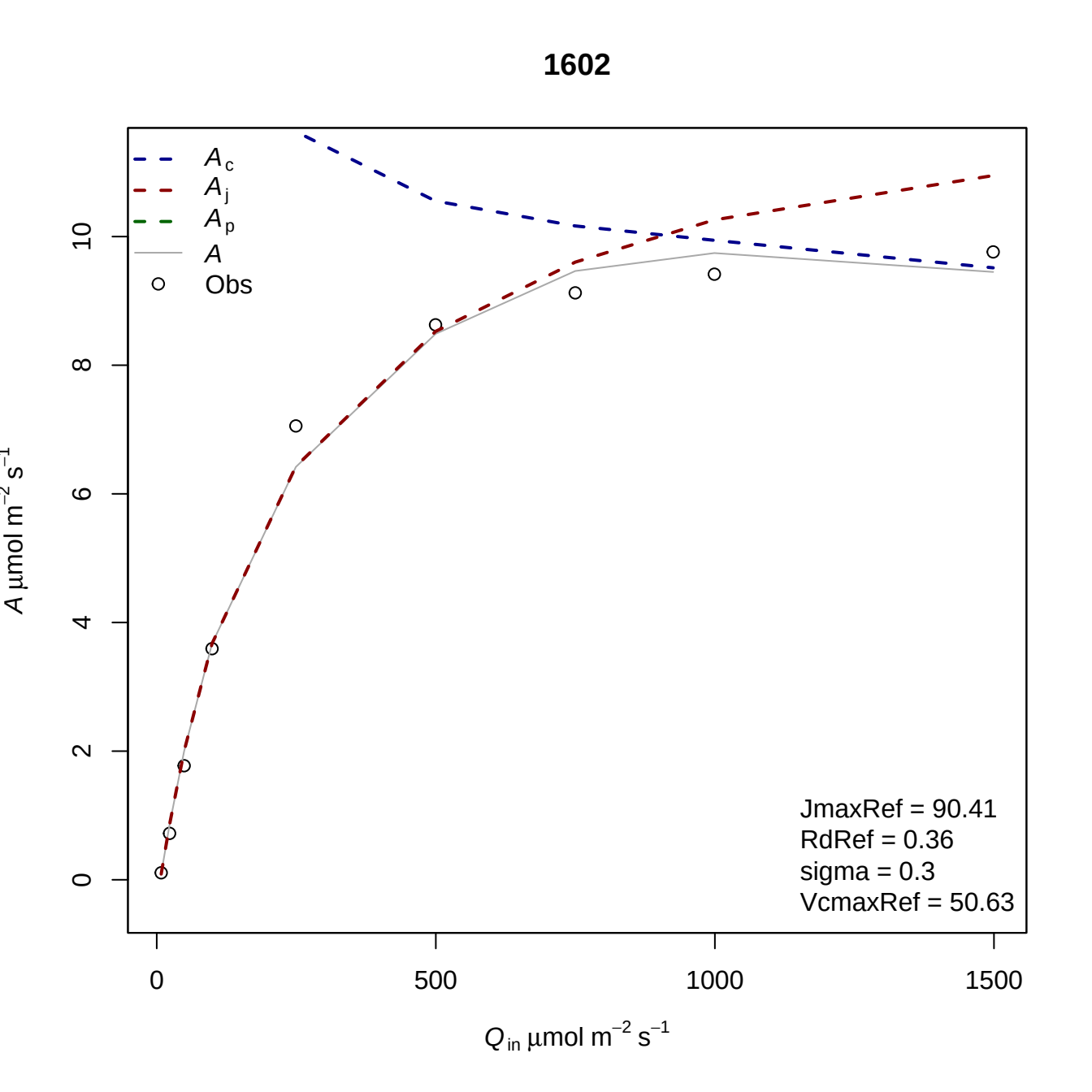

1597
A_c
A_j
A_p
A
Obs
JmaxRef = 137.81
RdRef = 0.37
sigma = 0.34
VcmaxRef = 71.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

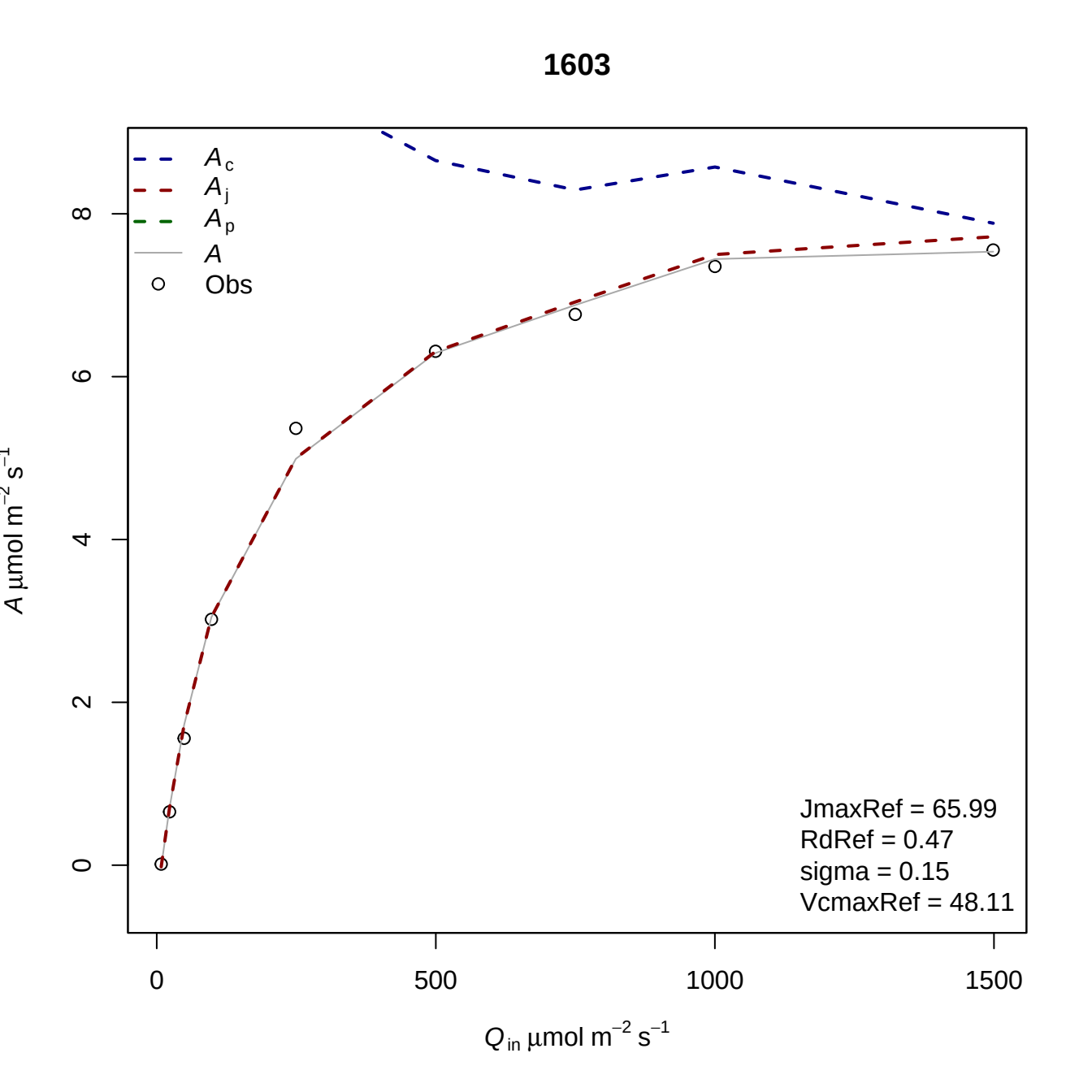

1598
A_c
A_j
A_p
A
Obs
JmaxRef = 117.77
RdRef = 0.36
sigma = 0.34
VcmaxRef = 54.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

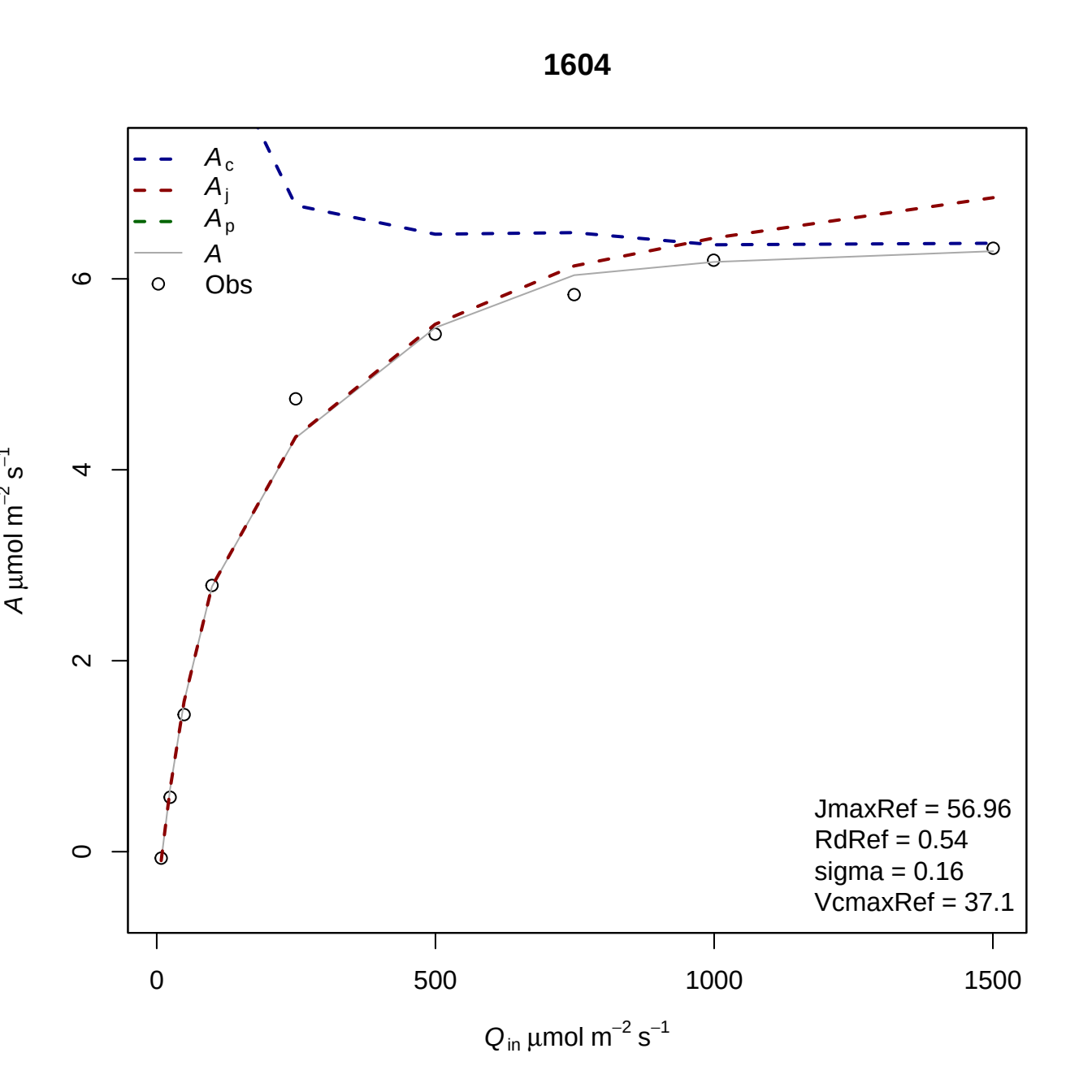

1599
A_c
A_j
A_p
A
Obs
JmaxRef = 119.11
RdRef = 0.32
sigma = 0.38
VcmaxRef = 58.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

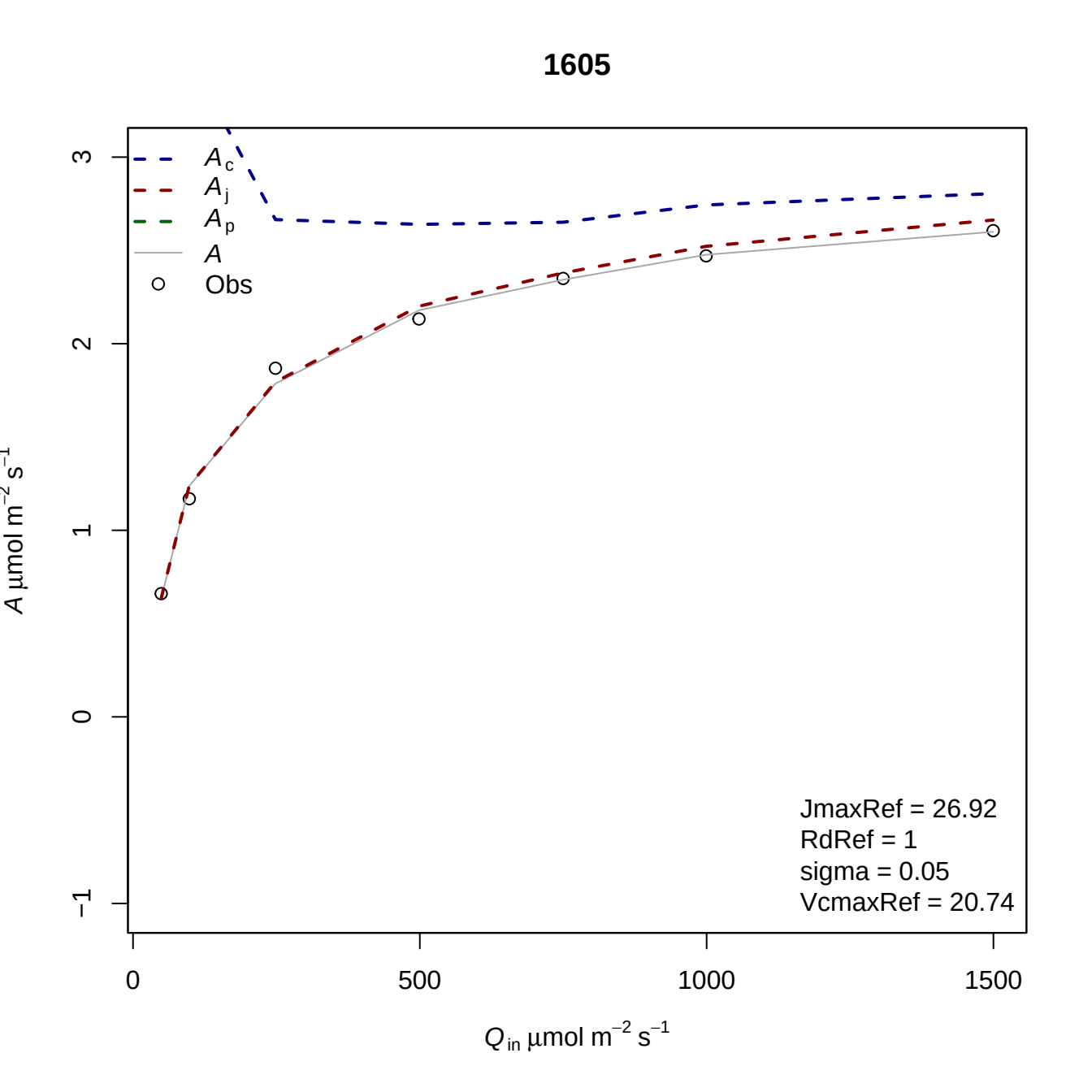

1600
A_c
A_j
A_p
A
Obs
JmaxRef = 111.65
RdRef = 0.31
sigma = 0.36
VcmaxRef = 58.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

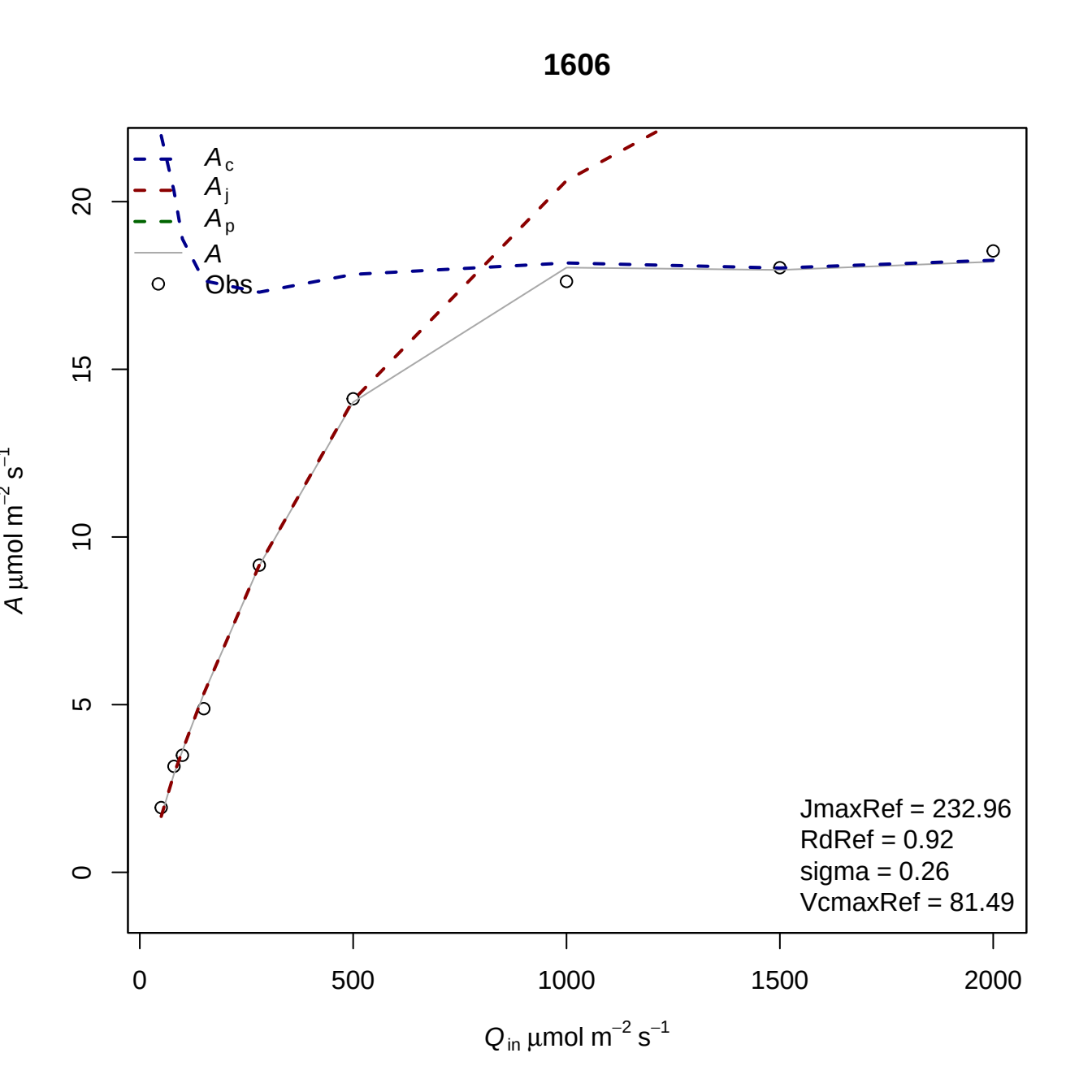

1601
A_c
A_j
A_p
A
Obs
JmaxRef = 88.08
RdRef = 0
sigma = 0.32
VcmaxRef = 52.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
200
400
600
800
1000
1200
1400

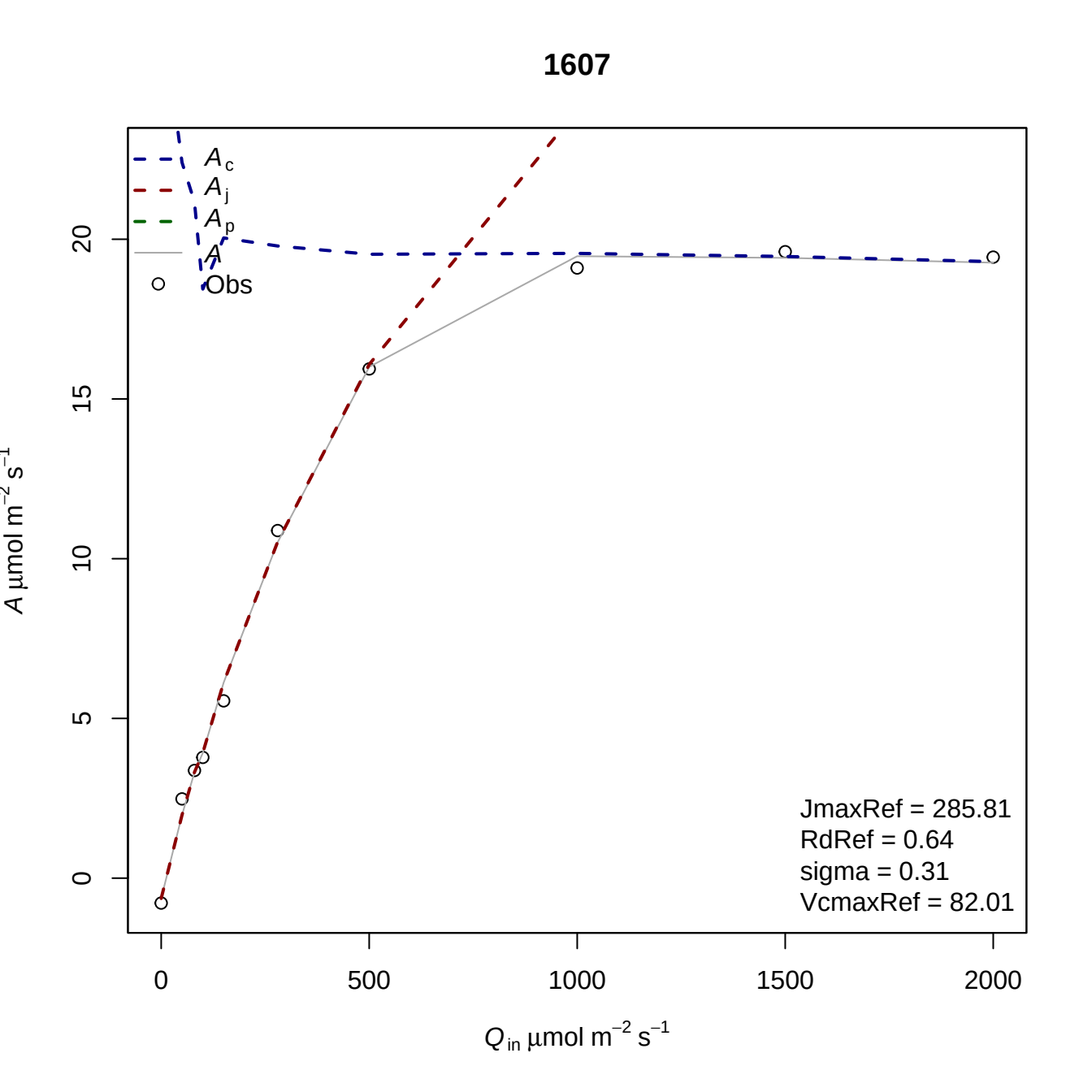

1602
A_c
A_j
A_p
A
Obs
JmaxRef = 90.41
RdRef = 0.36
sigma = 0.3
VcmaxRef = 50.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

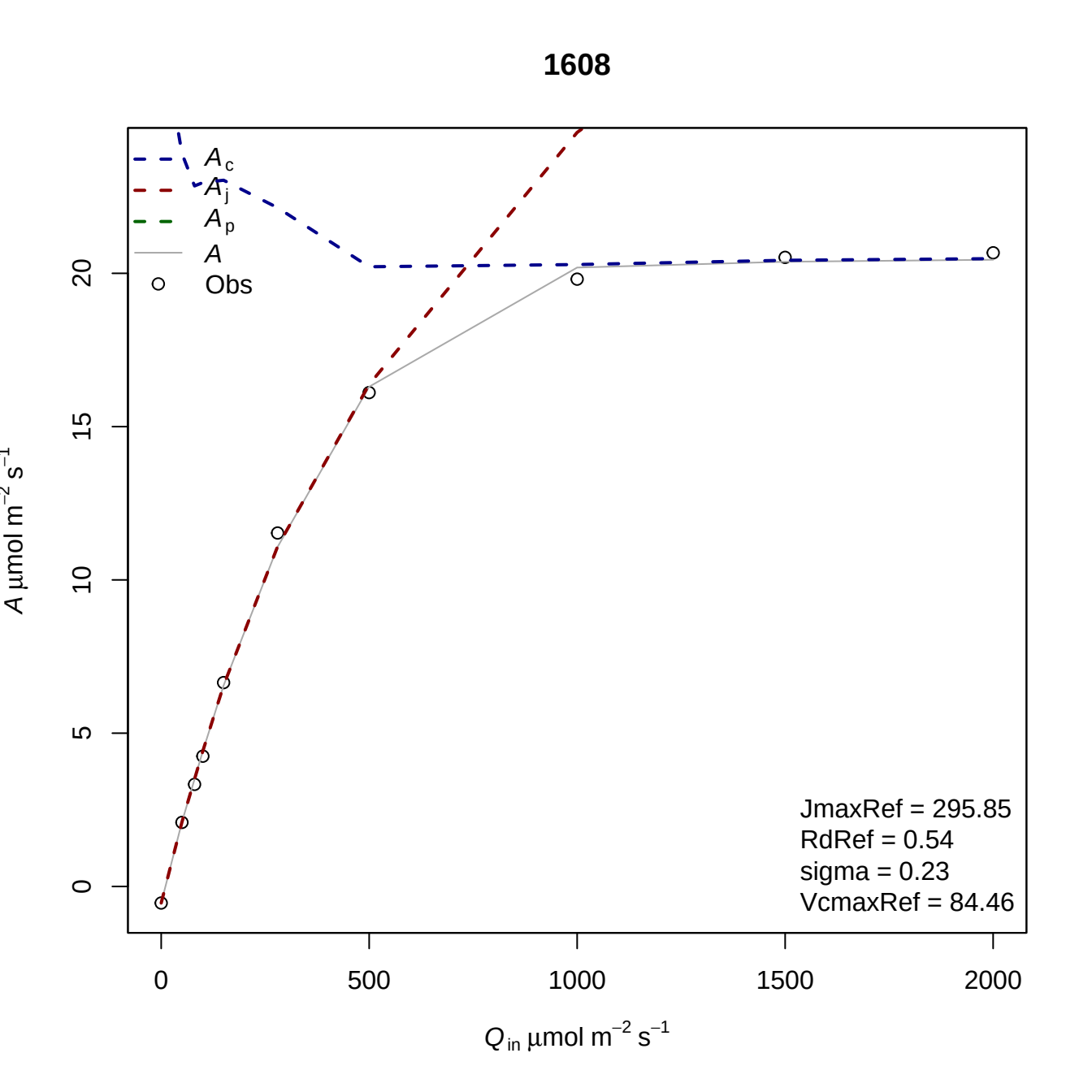

1603
A_c
A_j
A_p
A
Obs
JmaxRef = 65.99
RdRef = 0.47
sigma = 0.15
VcmaxRef = 48.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

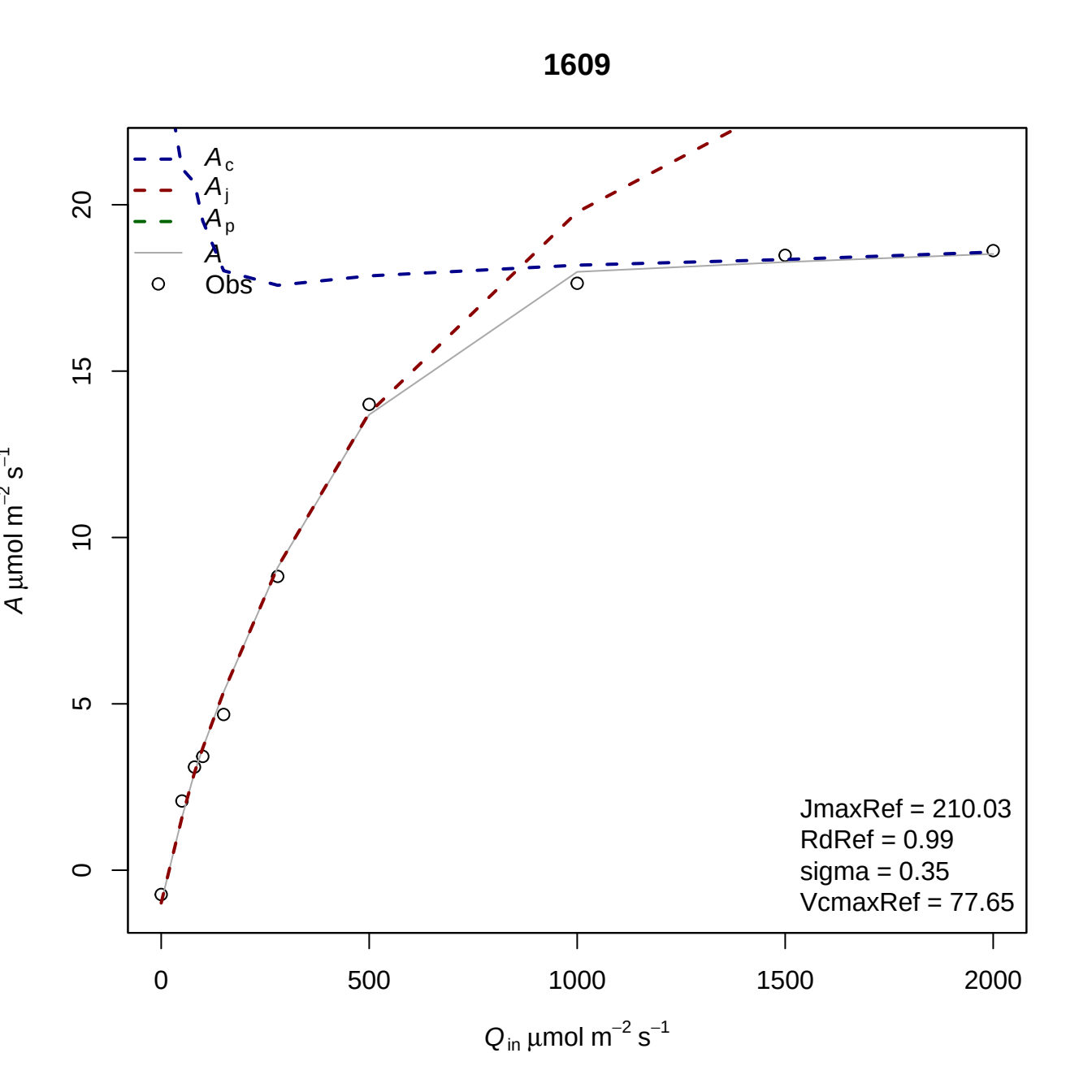

1604
A_c
A_j
A_p
A
Obs
JmaxRef = 56.96
RdRef = 0.54
sigma = 0.16
VcmaxRef = 37.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

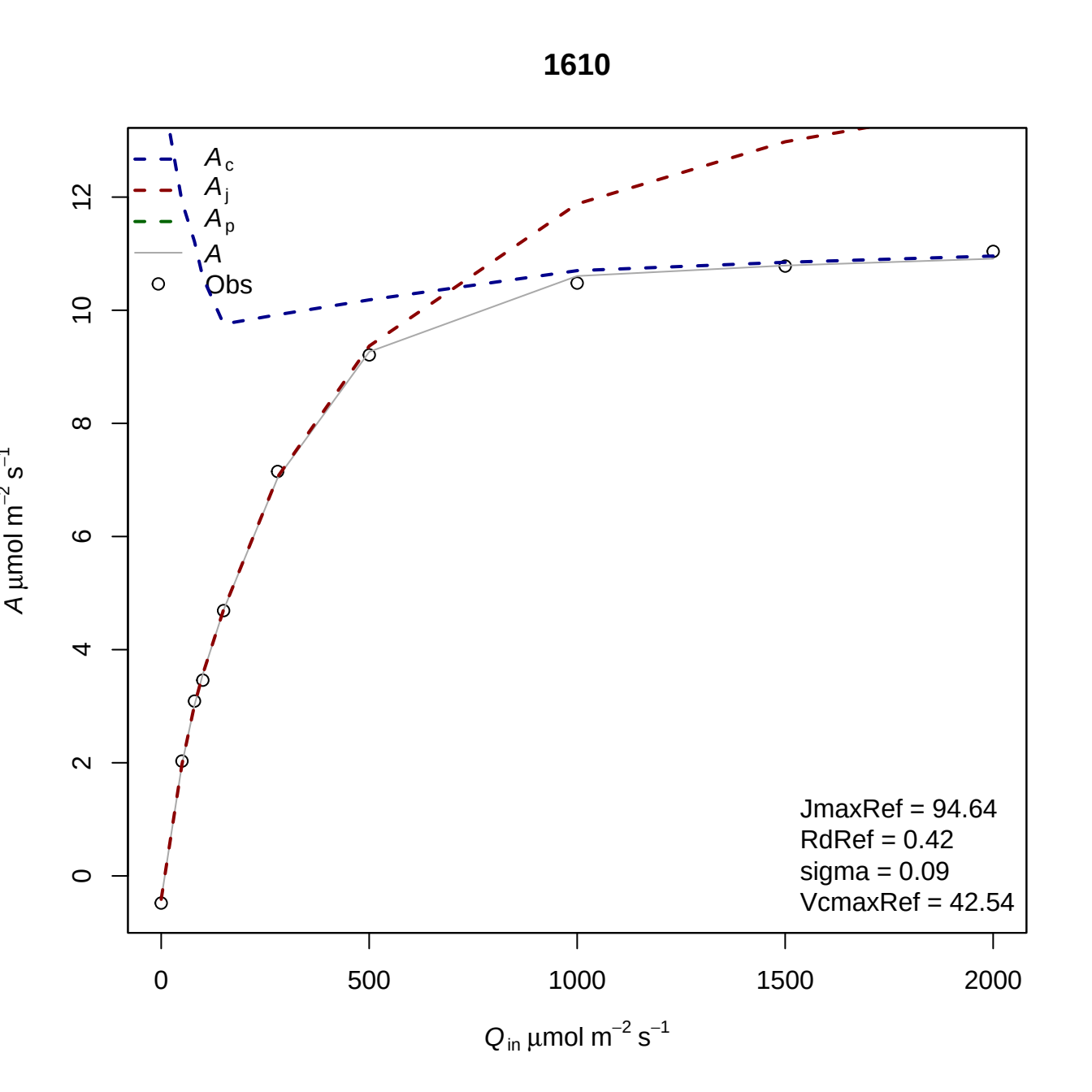

1605
A_c
A_j
A_p
A
Obs
JmaxRef = 26.92
RdRef = 1
sigma = 0.05
VcmaxRef = 20.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
500
1000
1500

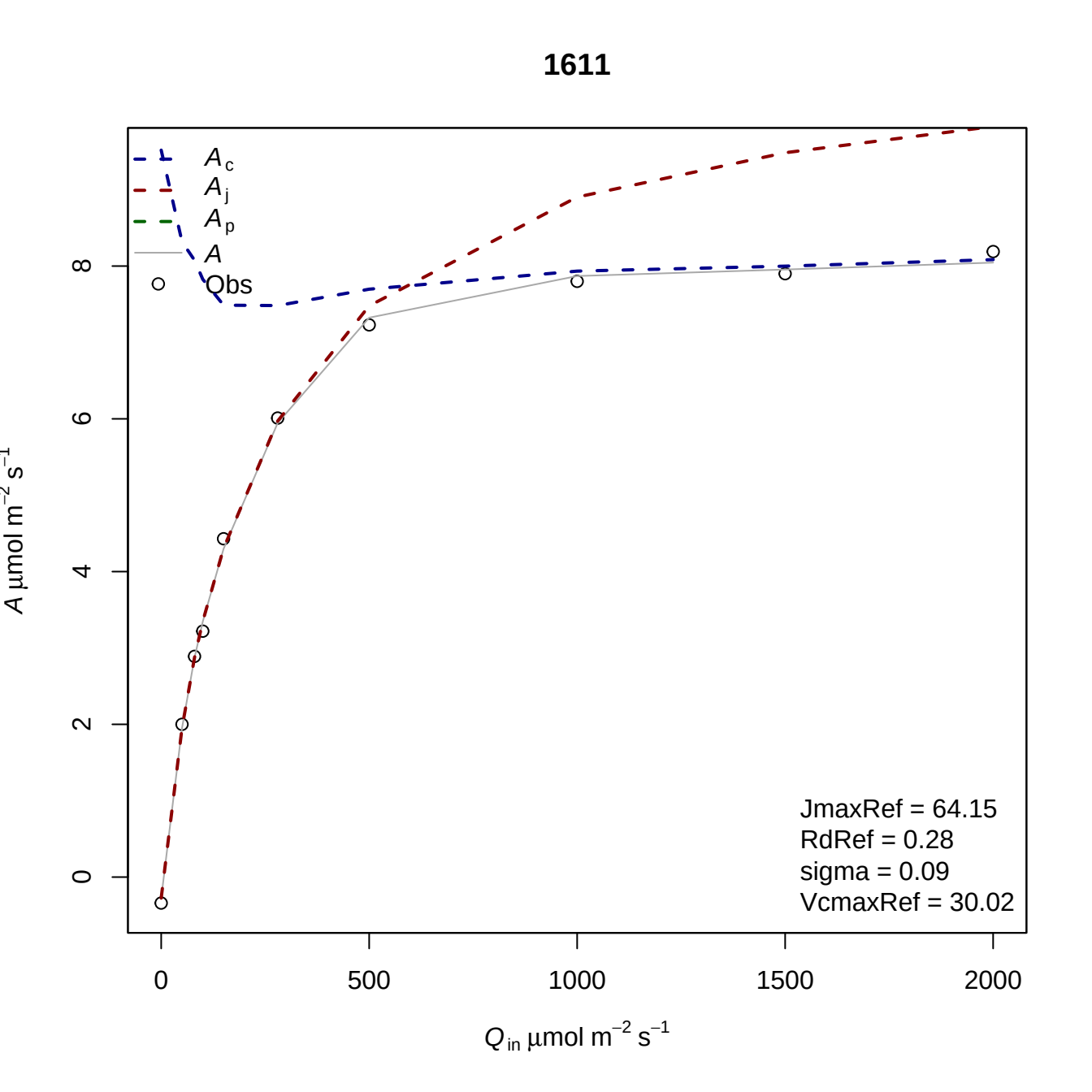

1606
A_c
A_j
A_p
A
Obs
JmaxRef = 232.96
RdRef = 0.92
sigma = 0.26
VcmaxRef = 81.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

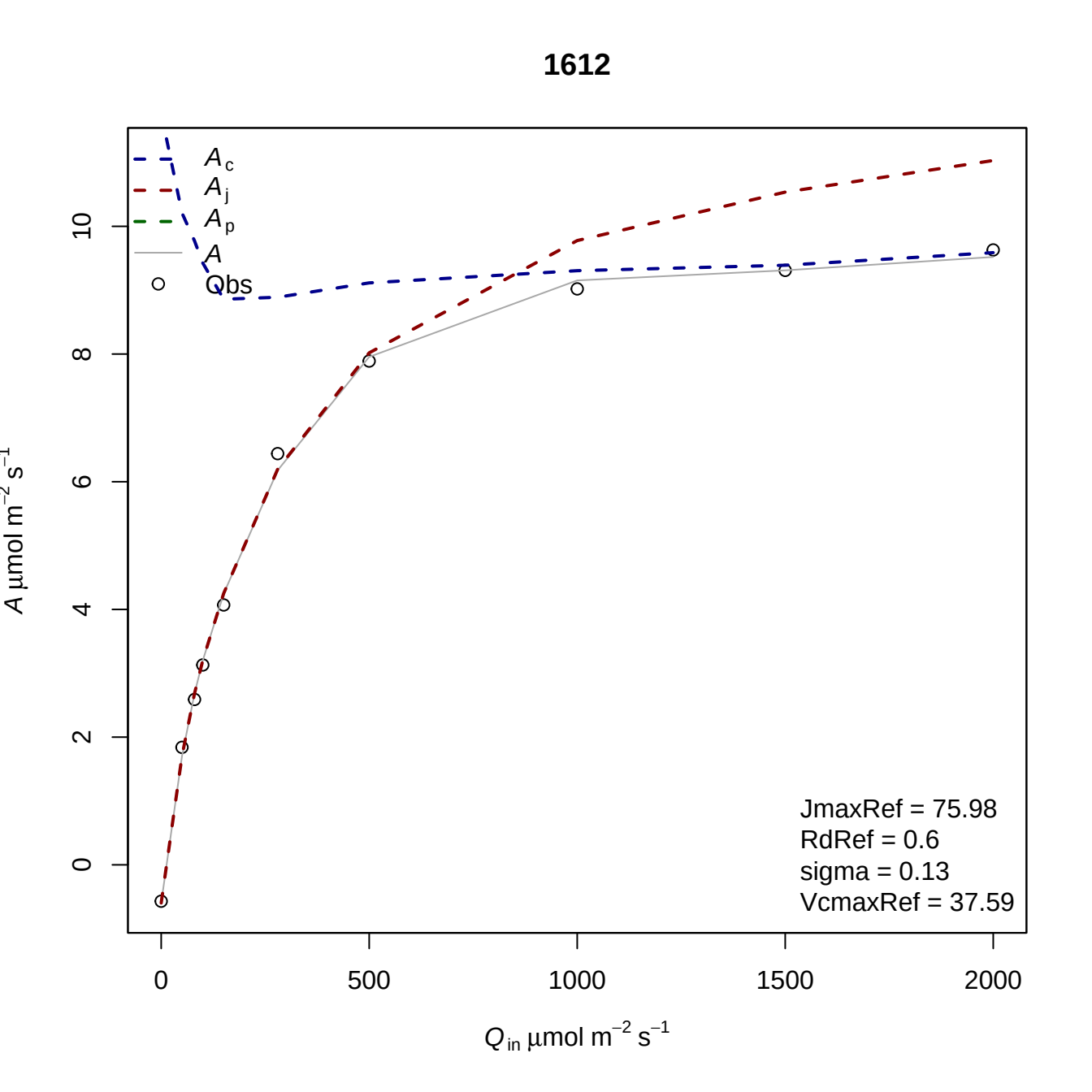

1607
A_c
A_j
A_p
A
Obs
JmaxRef = 285.81
RdRef = 0.64
sigma = 0.31
VcmaxRef = 82.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

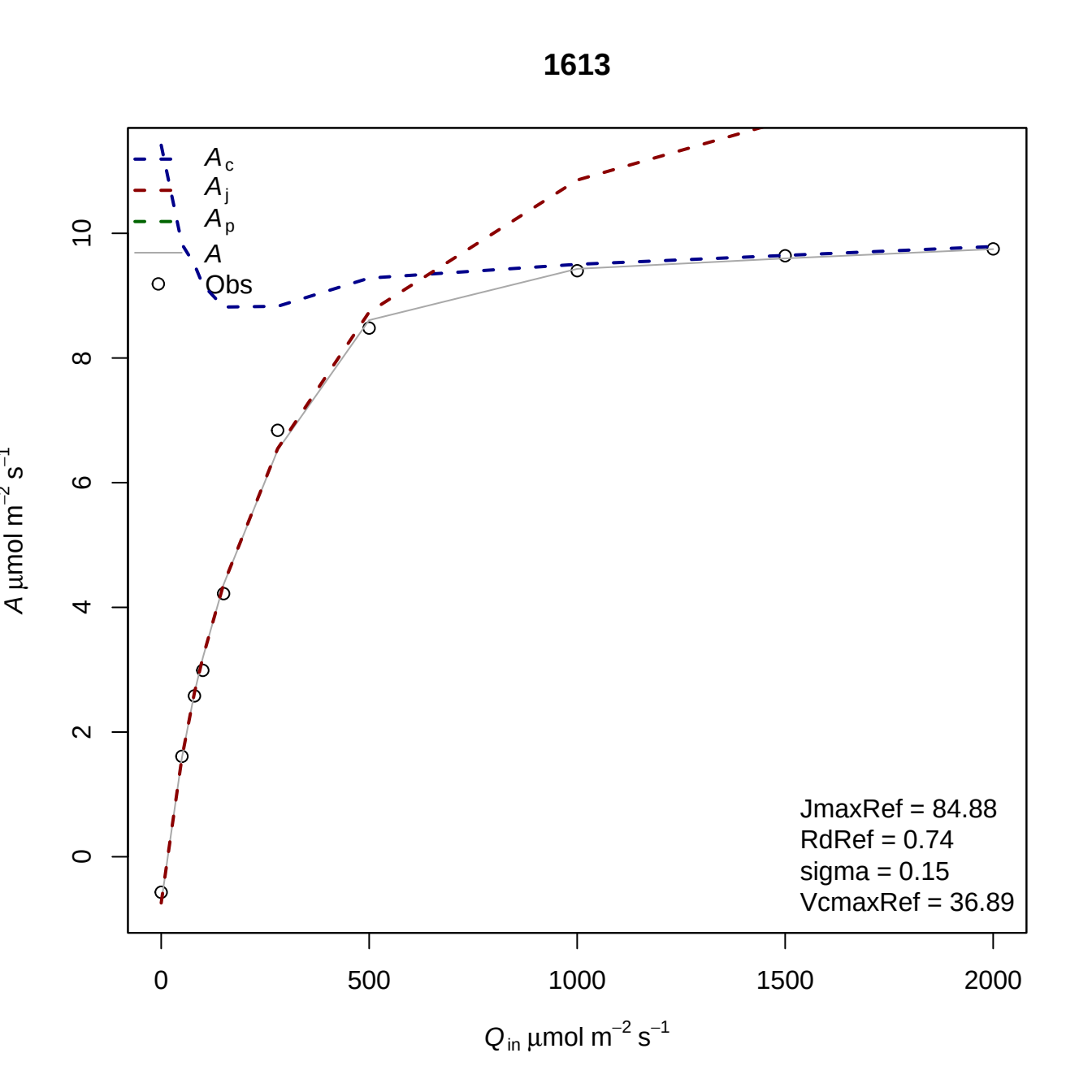

1608
A_c
A_j
A_p
A
Obs
JmaxRef = 295.85
RdRef = 0.54
sigma = 0.23
VcmaxRef = 84.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

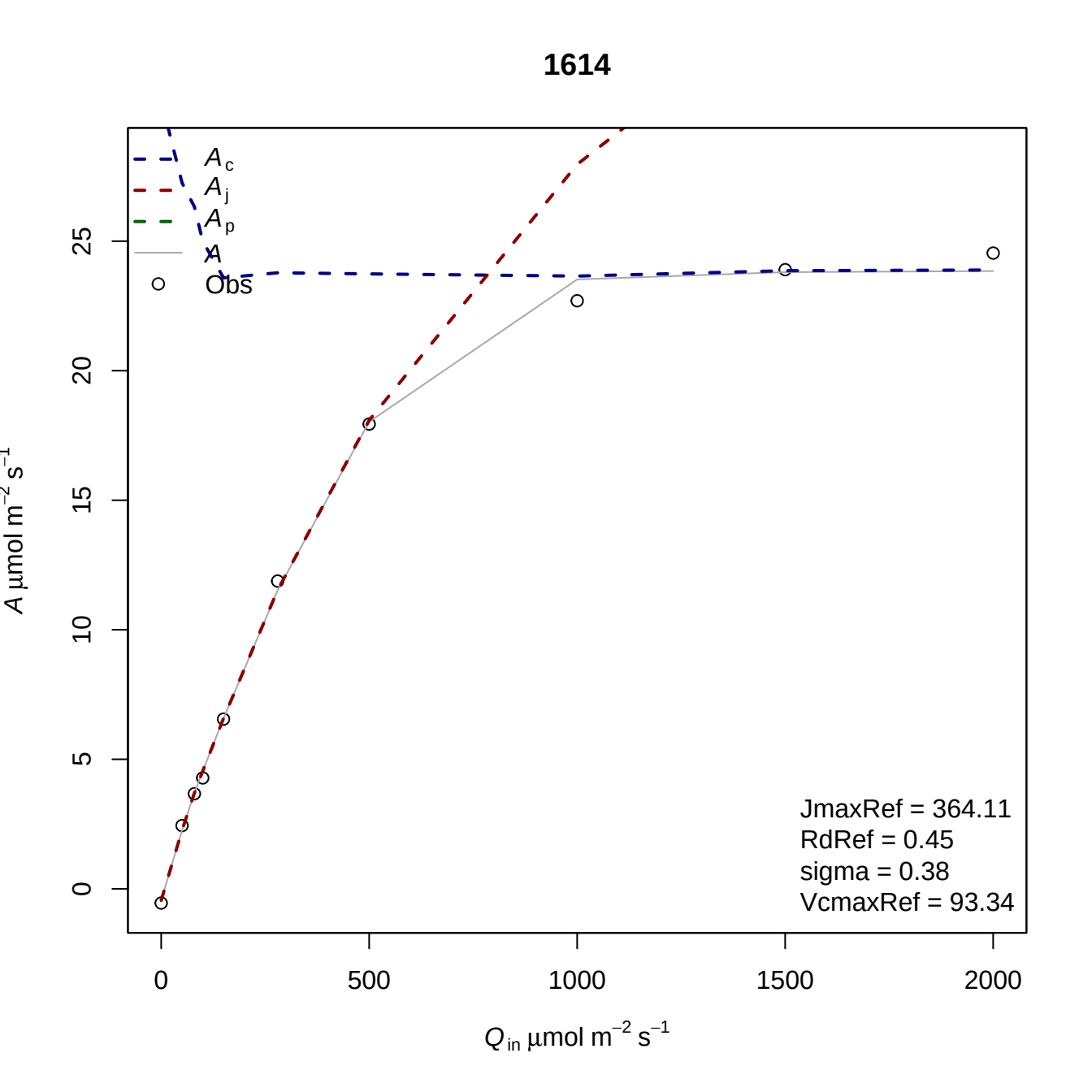

1609
A_c
A_j
A_p
A
Obs
JmaxRef = 210.03
RdRef = 0.99
sigma = 0.35
VcmaxRef = 77.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

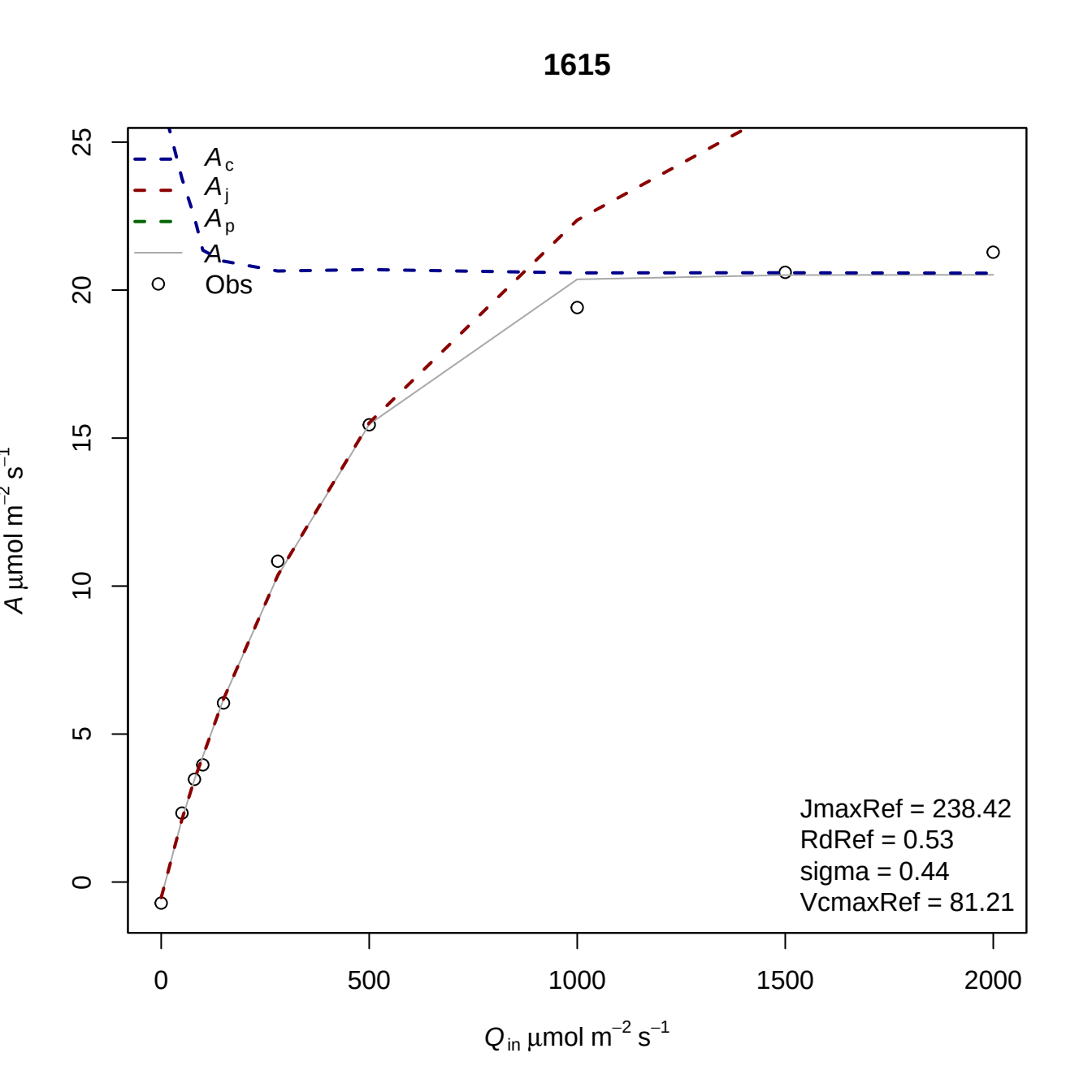

1610
A_c
A_j
A_p
A
Obs
JmaxRef = 94.64
RdRef = 0.42
sigma = 0.09
VcmaxRef = 42.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

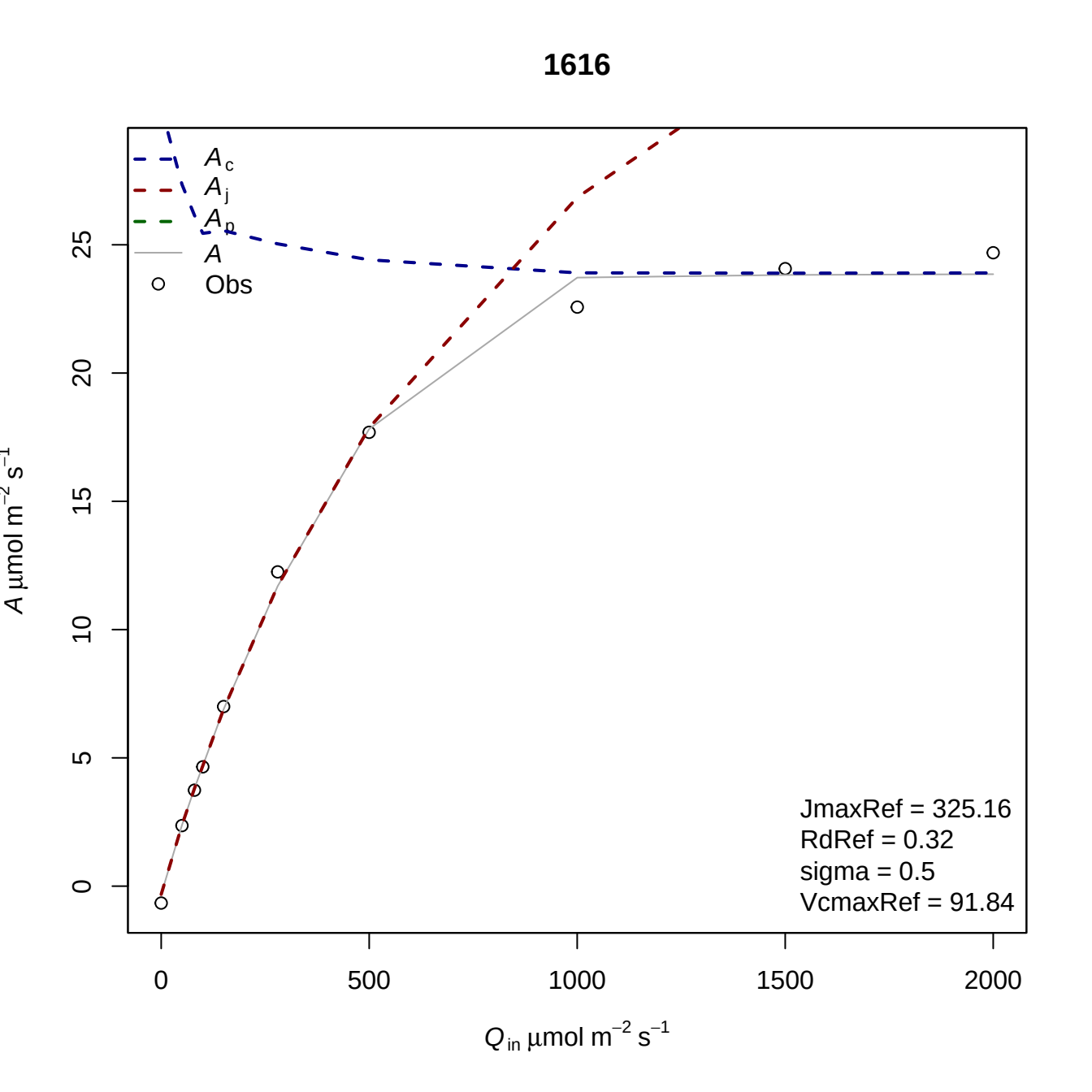

1611
A_c
A_j
A_p
A
Obs
JmaxRef = 64.15
RdRef = 0.28
sigma = 0.09
VcmaxRef = 30.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

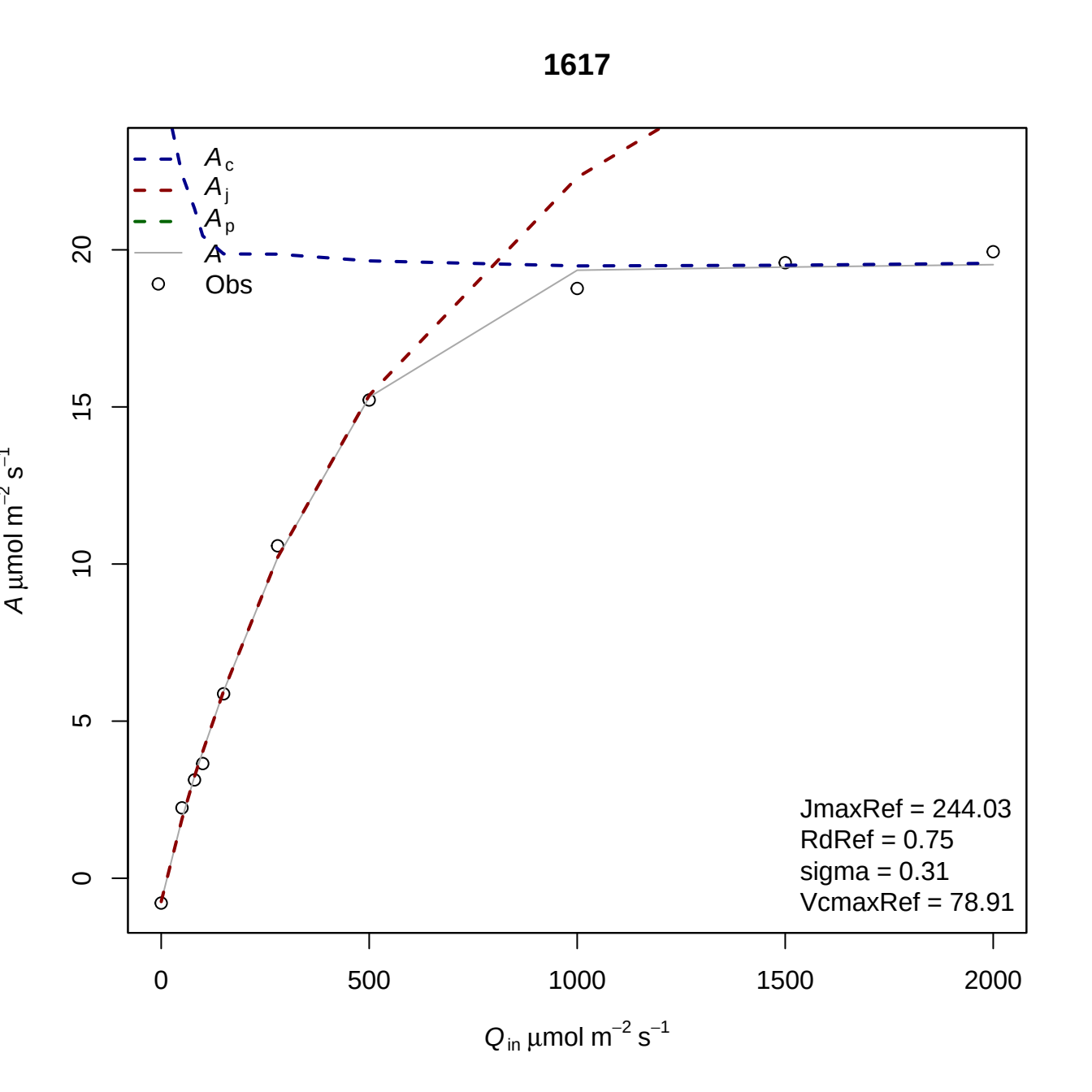

1612
A_c
A_j
A_p
A
Qbs
JmaxRef = 75.98
RdRef = 0.6
sigma = 0.13
VcmaxRef = 37.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

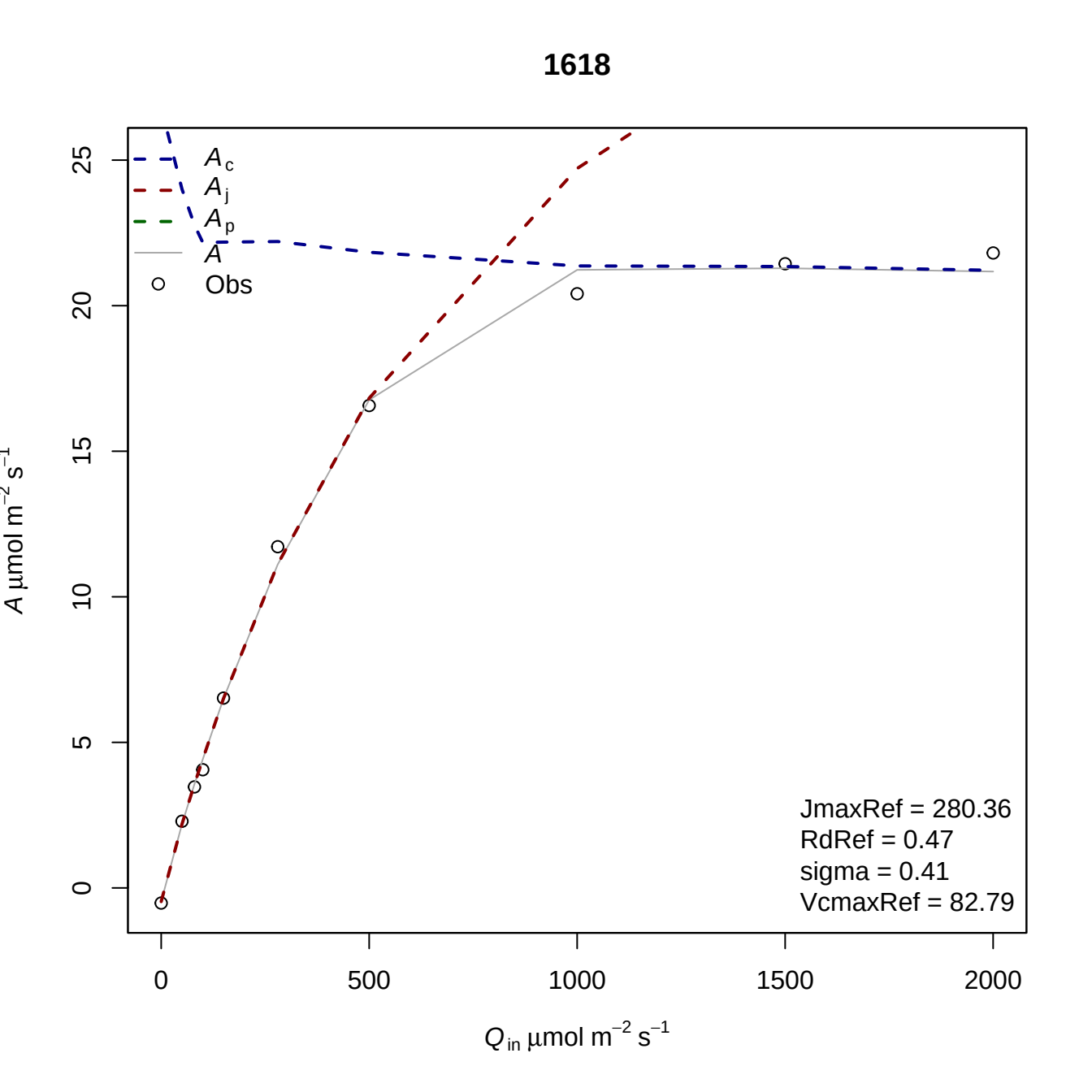

1613
A_c
A_j
A_p
A
Obs
JmaxRef = 84.88
RdRef = 0.74
sigma = 0.15
VcmaxRef = 36.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

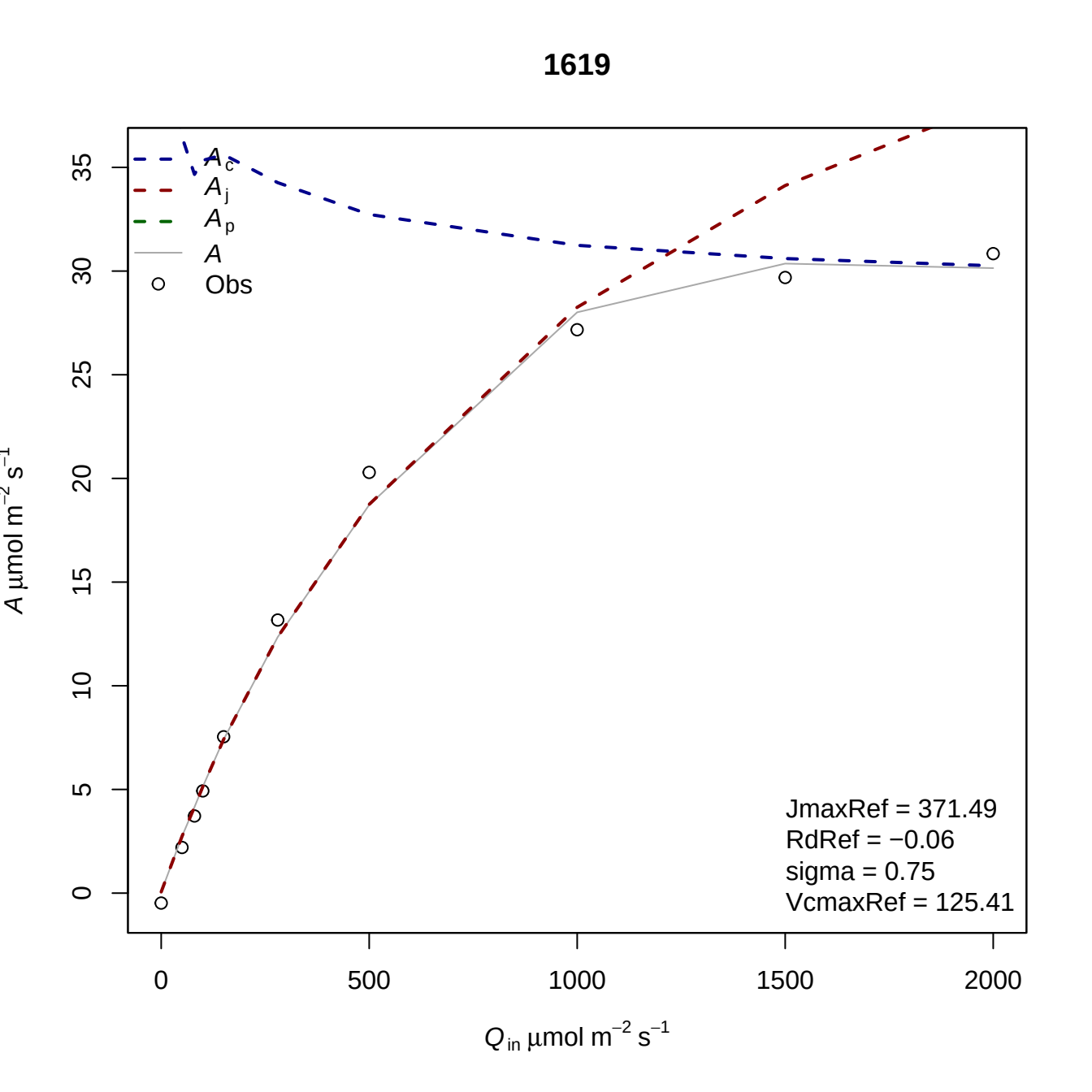

1614
A_c
A_j
A_p
A
Obs
JmaxRef = 364.11
RdRef = 0.45
sigma = 0.38
VcmaxRef = 93.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

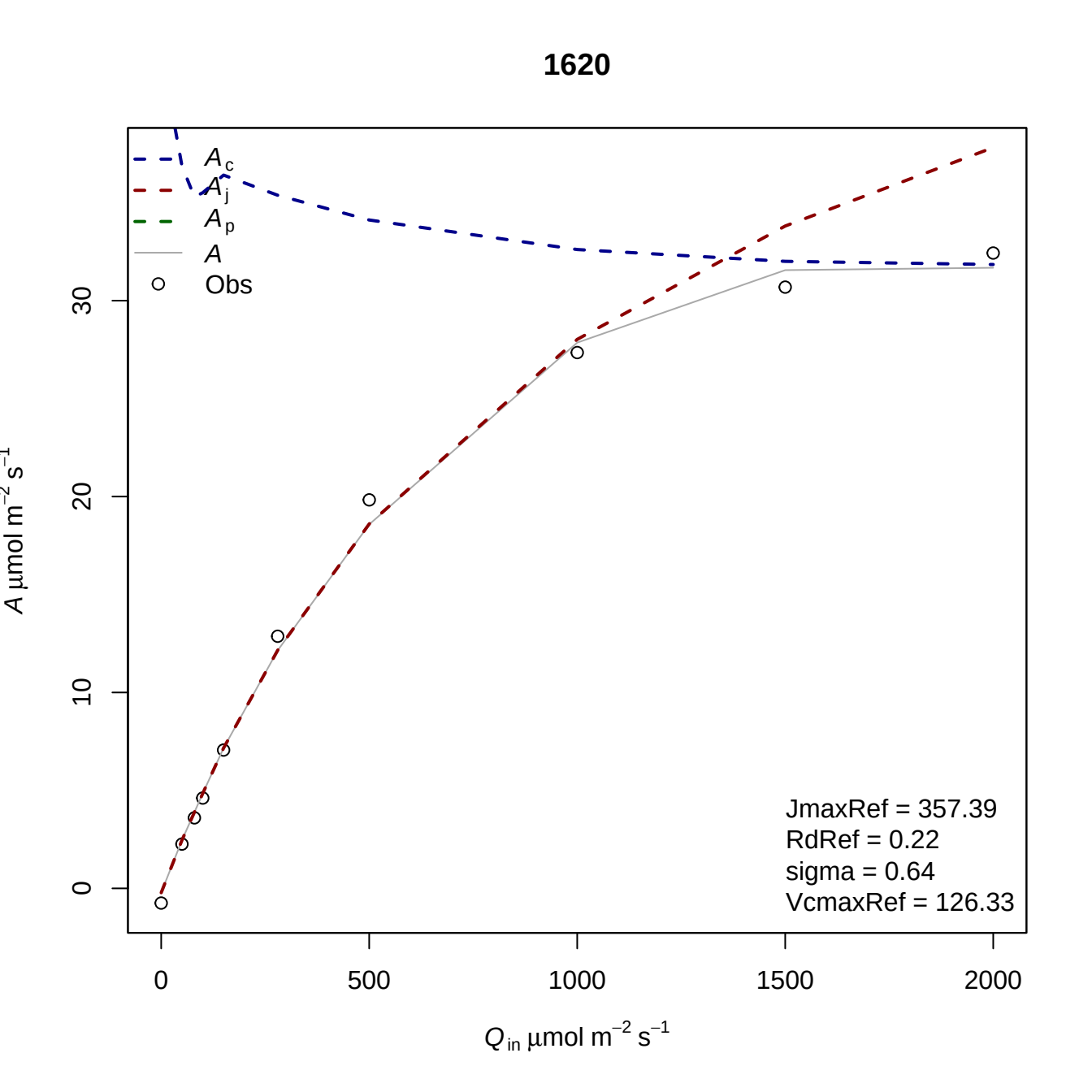

1615
A_c
A_j
A_p
A
Obs
JmaxRef = 238.42
RdRef = 0.53
sigma = 0.44
VcmaxRef = 81.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

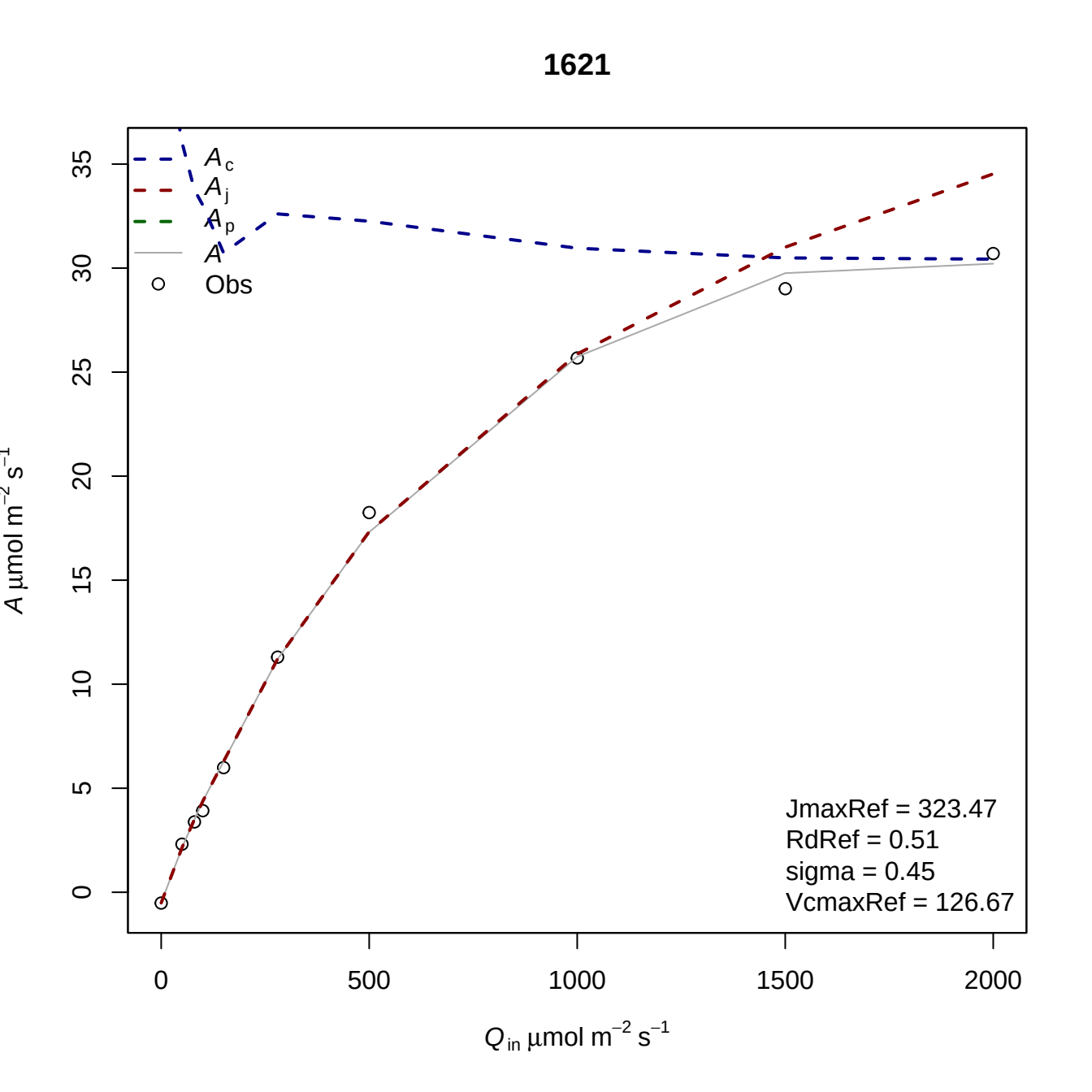

1616
A_c
A_j
A_p
A
Obs
JmaxRef = 325.16
RdRef = 0.32
sigma = 0.5
VcmaxRef = 91.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

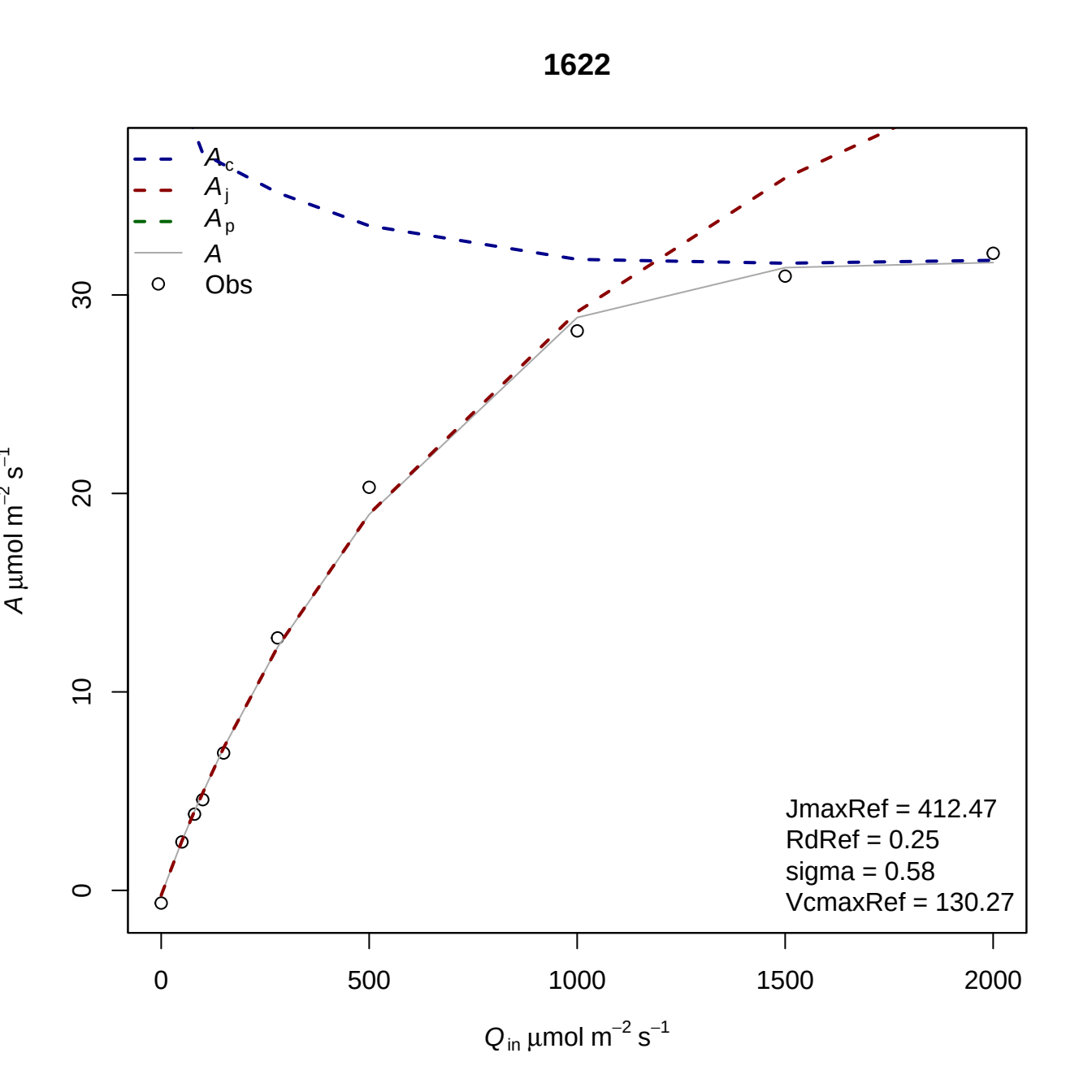

1617
A_c
A_j
A_p
A
Obs
JmaxRef = 244.03
RdRef = 0.75
sigma = 0.31
VcmaxRef = 78.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

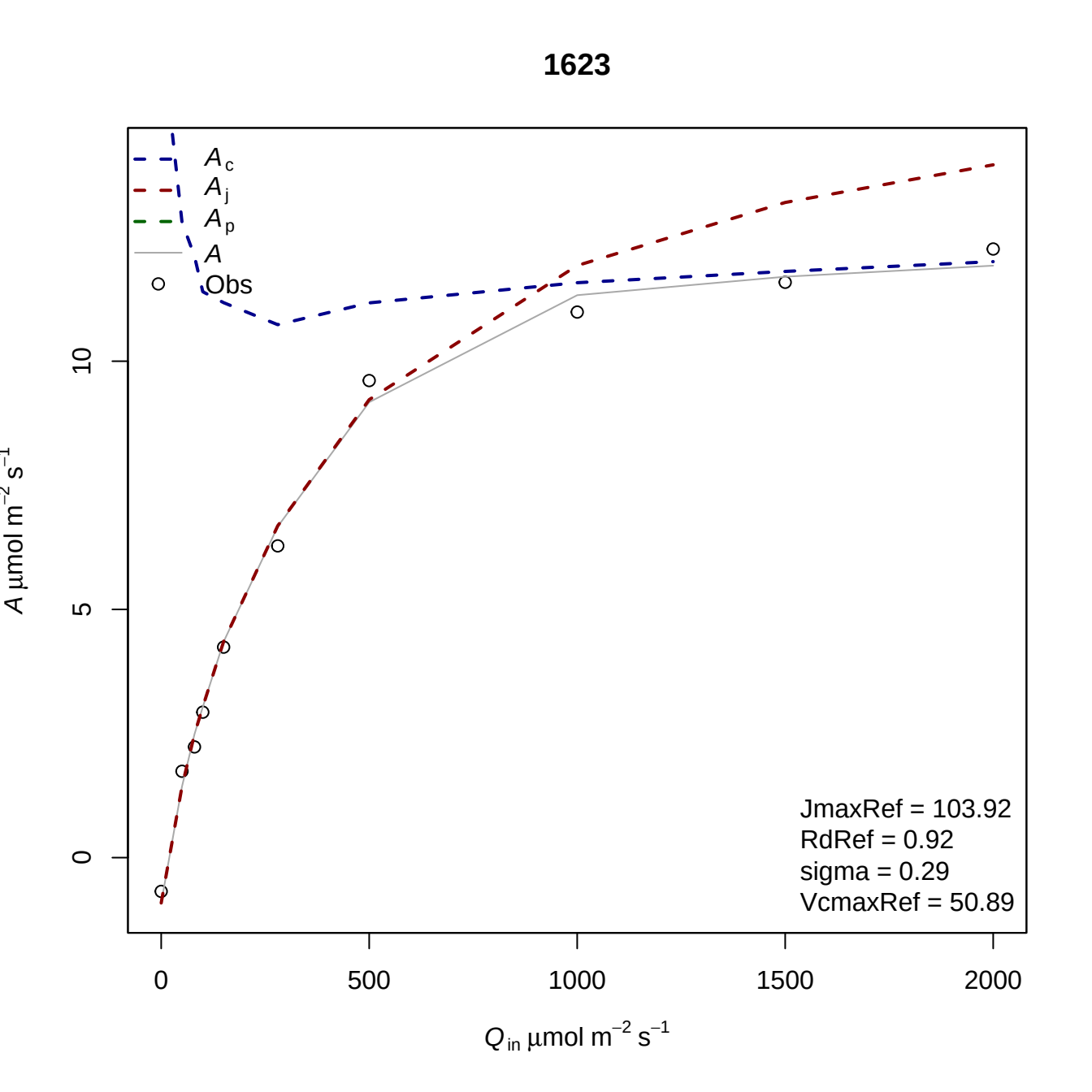

1618
A_c
A_j
A_p
A
Obs
JmaxRef = 280.36
RdRef = 0.47
sigma = 0.41
VcmaxRef = 82.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

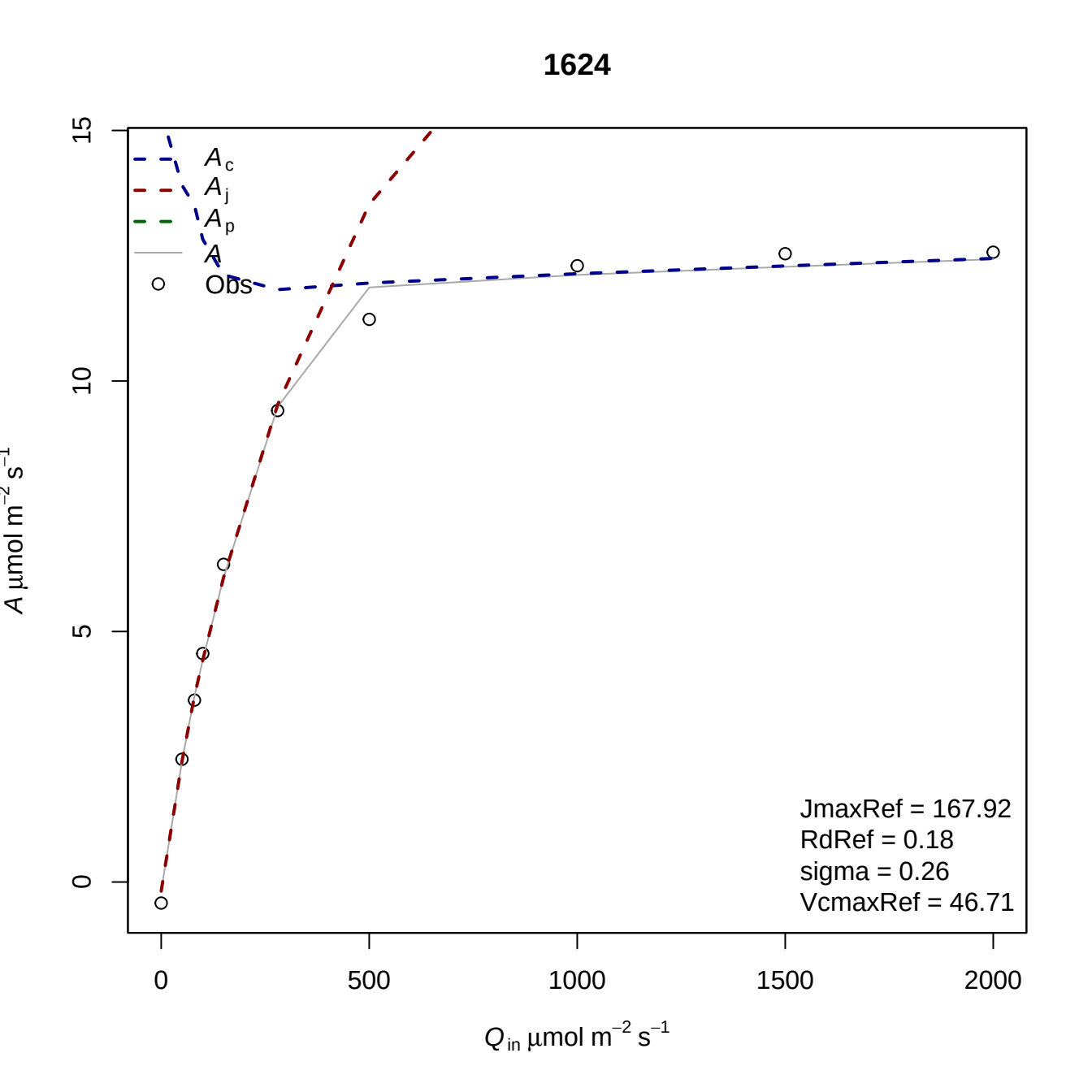

1619
A_c
A_j
A_p
A
Obs
JmaxRef = 371.49
RdRef = −0.06
sigma = 0.75
VcmaxRef = 125.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

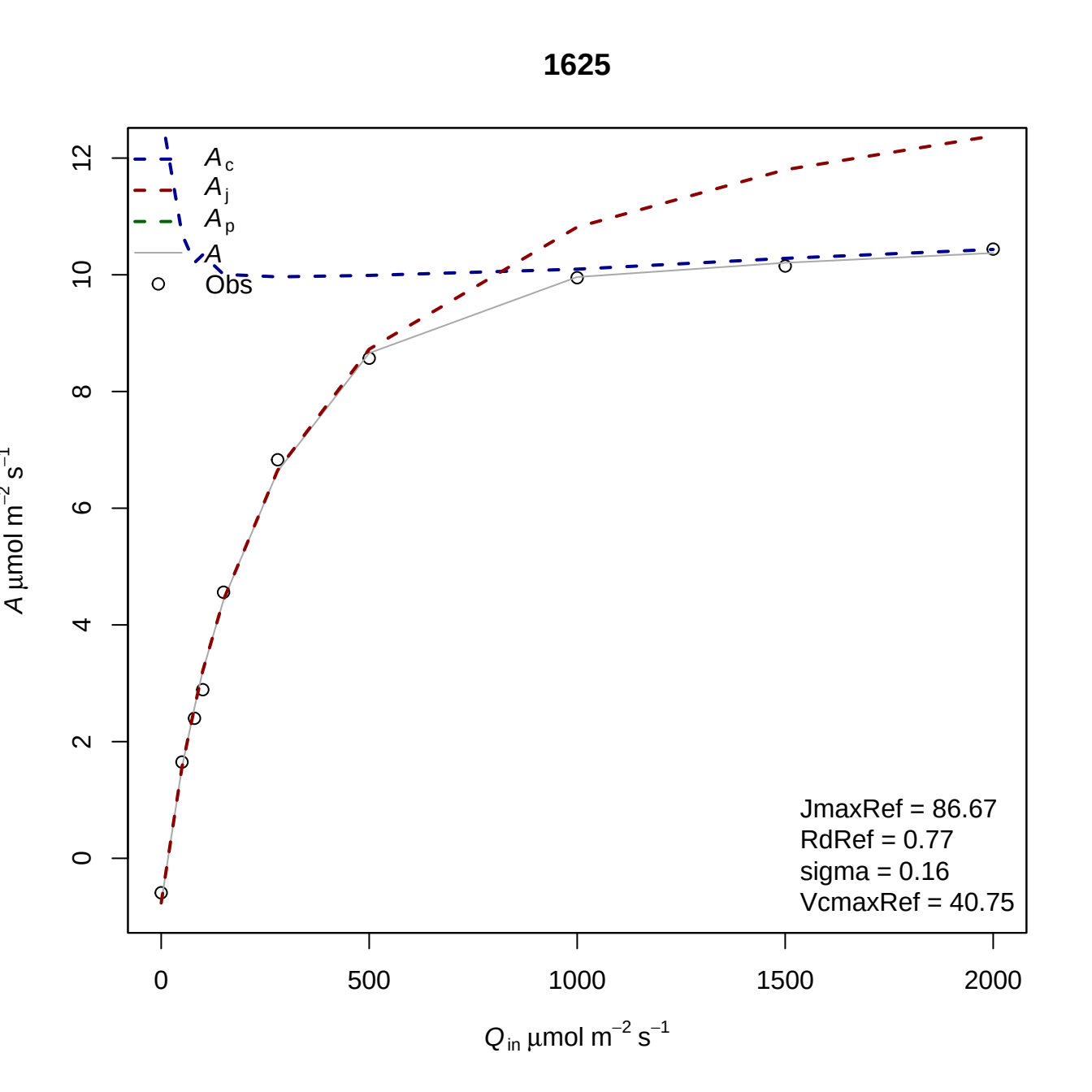

1620
A_c
A_j
A_p
A
Obs
JmaxRef = 357.39
RdRef = 0.22
sigma = 0.64
VcmaxRef = 126.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

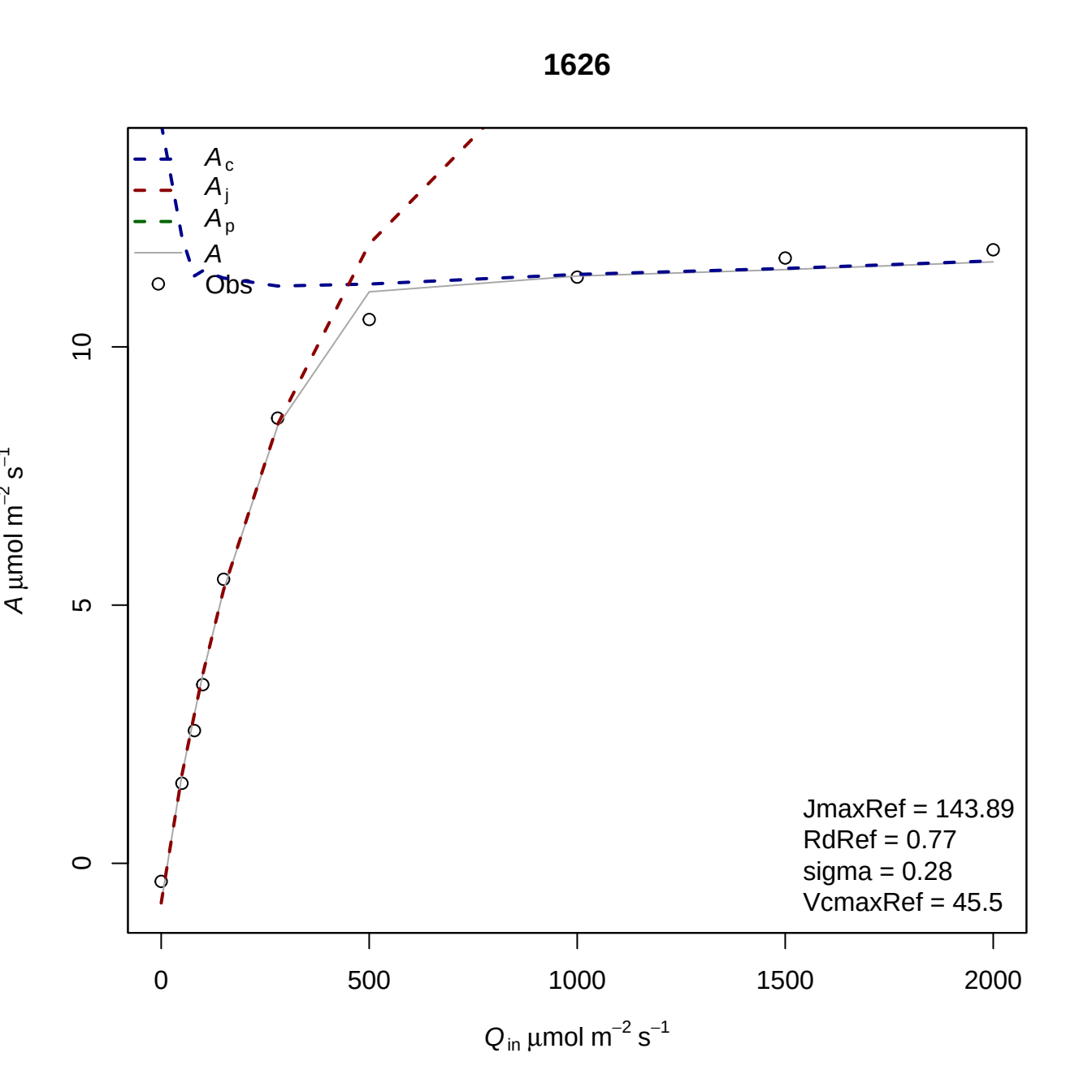

1621
A_c
A_j
A_p
A
Obs
JmaxRef = 323.47
RdRef = 0.51
sigma = 0.45
VcmaxRef = 126.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

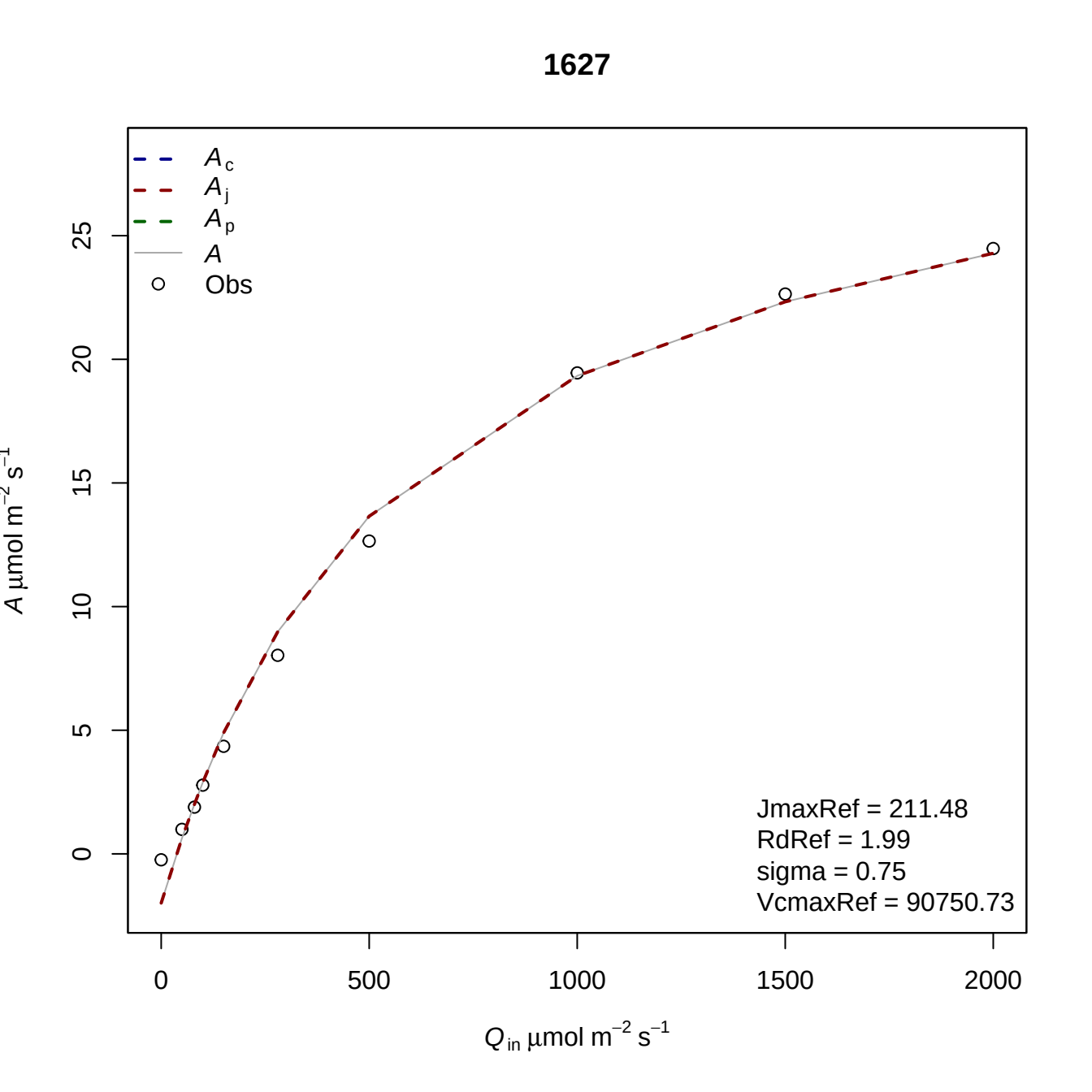

1622
A_c
A_j
A_p
A
Obs
JmaxRef = 412.47
RdRef = 0.25
sigma = 0.58
VcmaxRef = 130.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

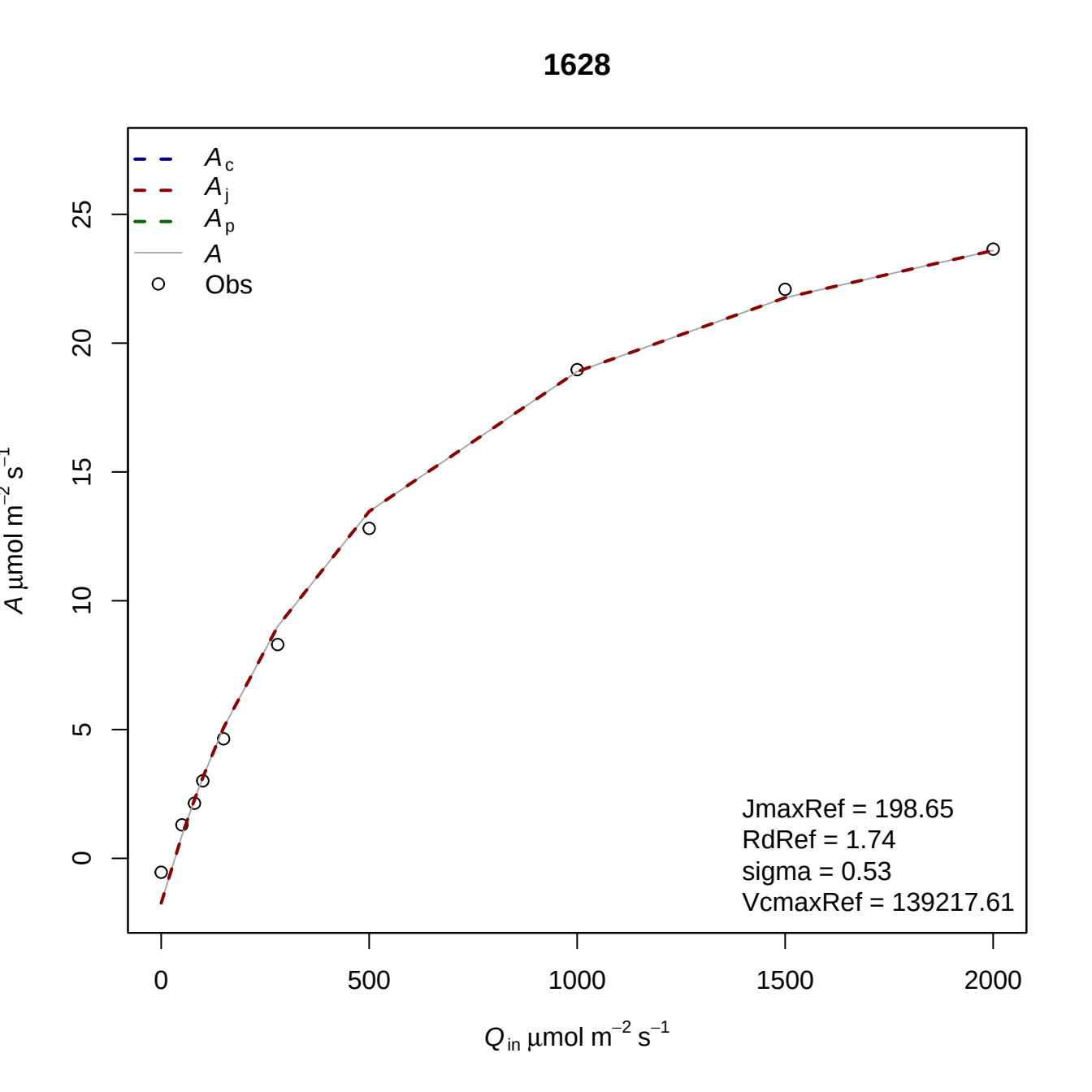

1623
A_c
A_j
A_p
A
Obs
JmaxRef = 103.92
RdRef = 0.92
sigma = 0.29
VcmaxRef = 50.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

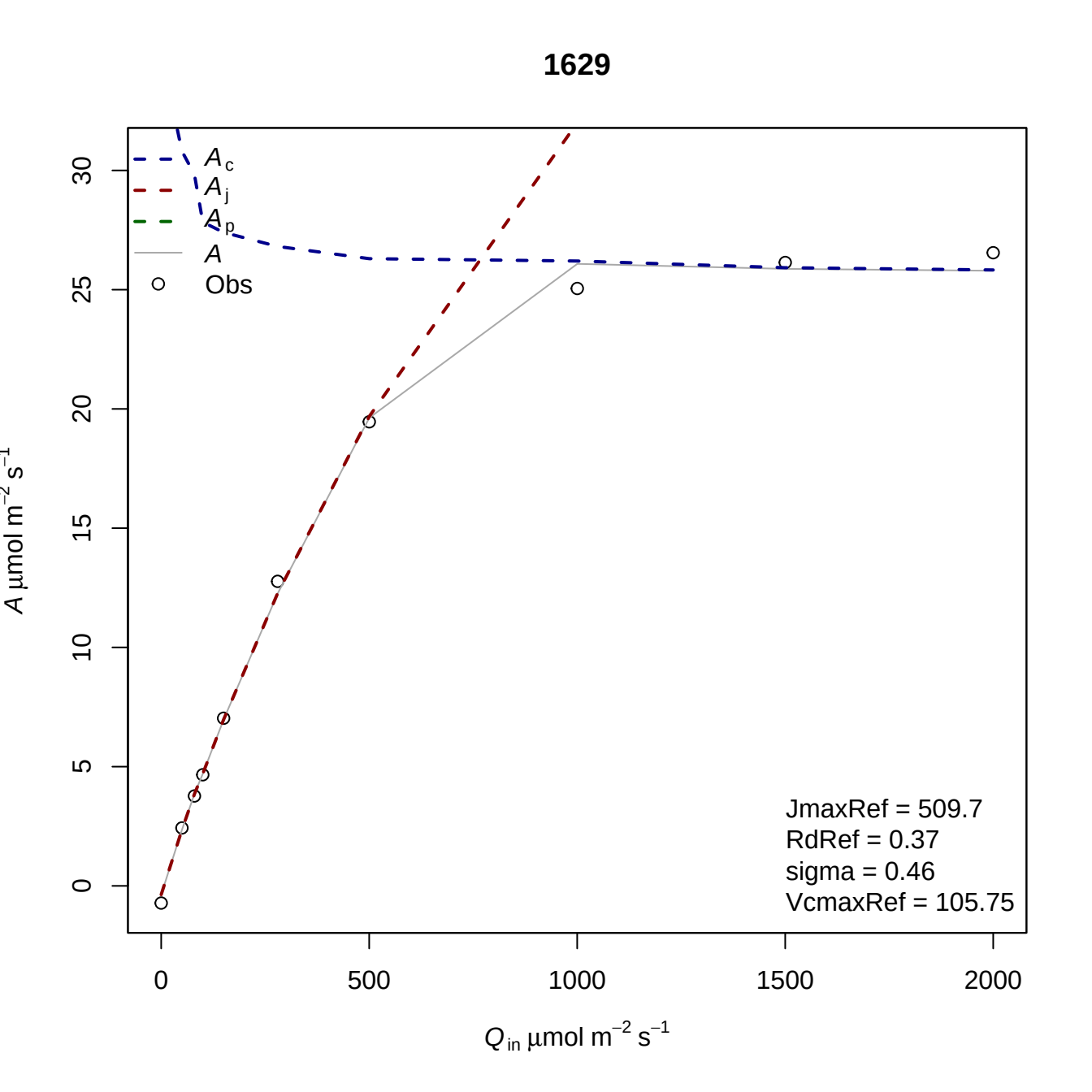

1624
A_c
A_j
A_p
A
Obs
JmaxRef = 167.92
RdRef = 0.18
sigma = 0.26
VcmaxRef = 46.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

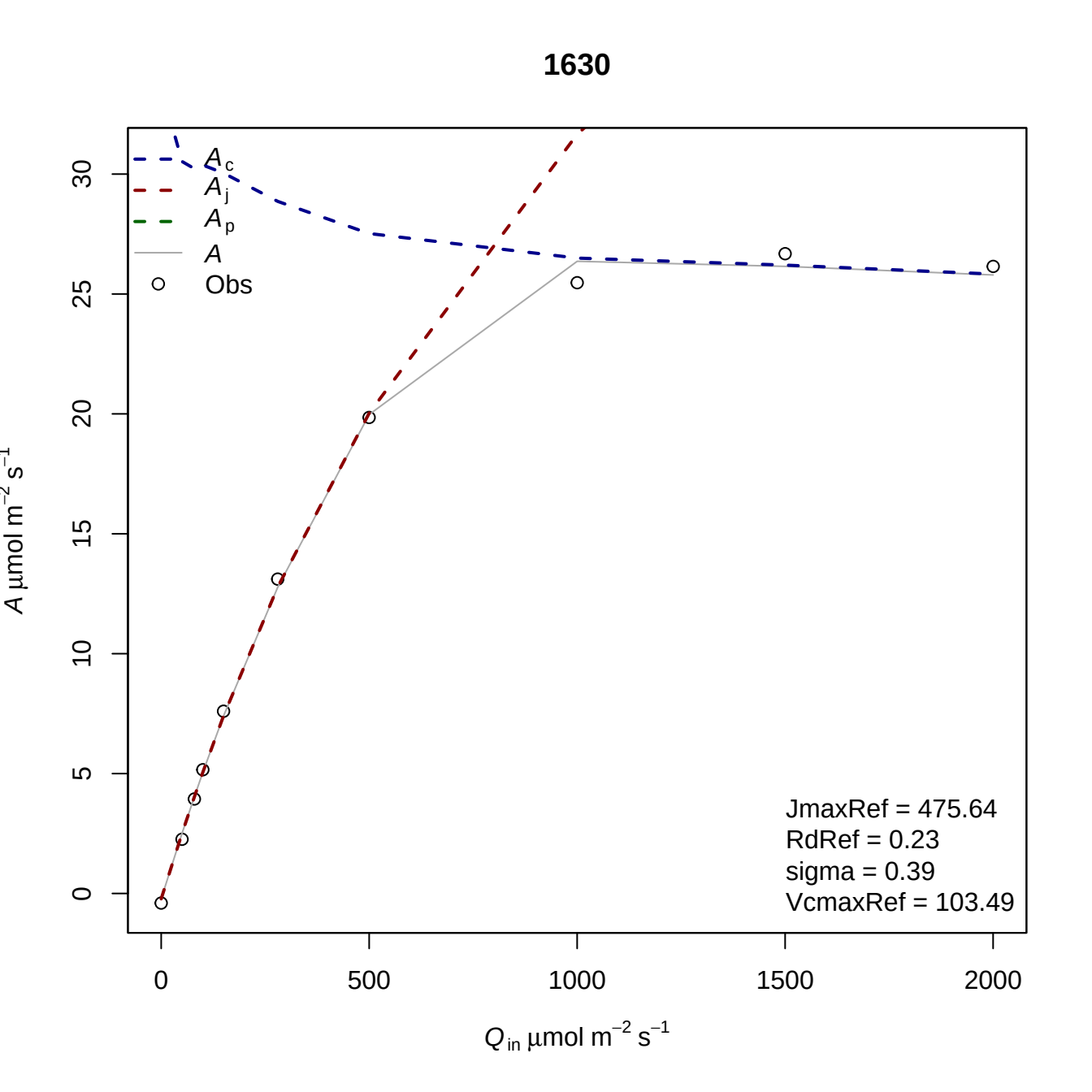

1625
A_c
A_j
A_p
A
Obs
JmaxRef = 86.67
RdRef = 0.77
sigma = 0.16
VcmaxRef = 40.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

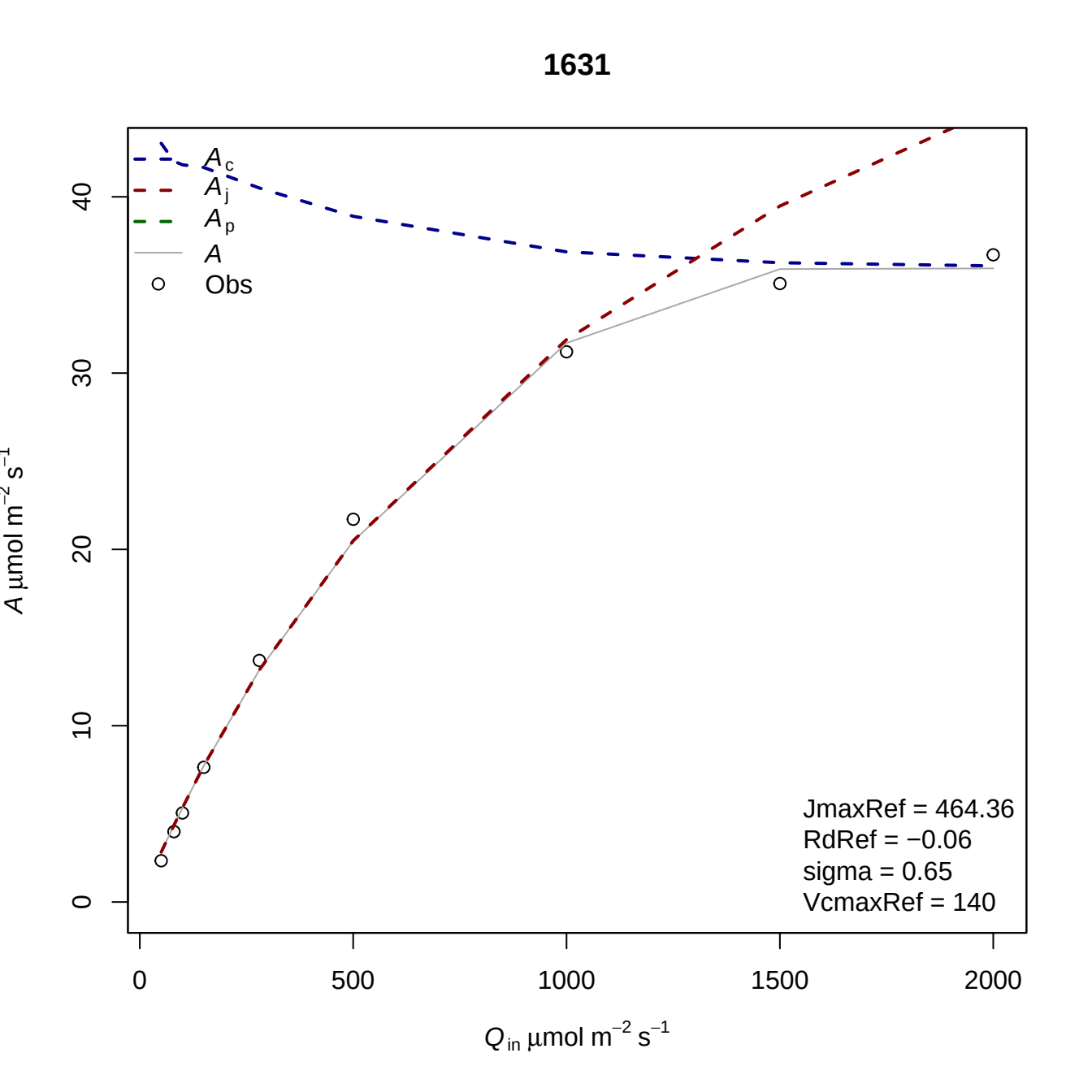

1626
A_c
A_j
A_p
A
Obs
JmaxRef = 143.89
RdRef = 0.77
sigma = 0.28
VcmaxRef = 45.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

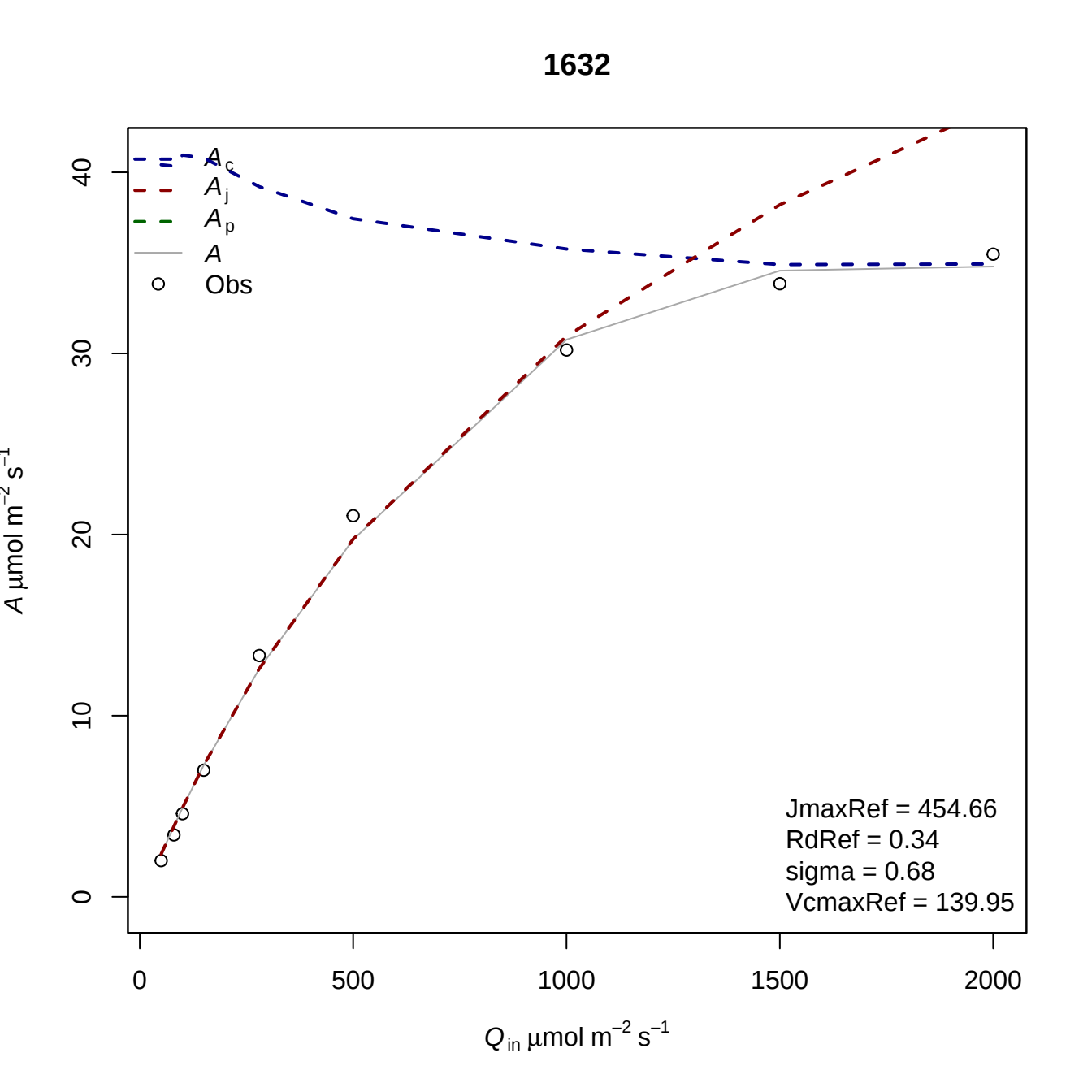

1627
A_c
A_j
A_p
A
Obs
JmaxRef = 211.48
RdRef = 1.99
sigma = 0.75
VcmaxRef = 90750.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

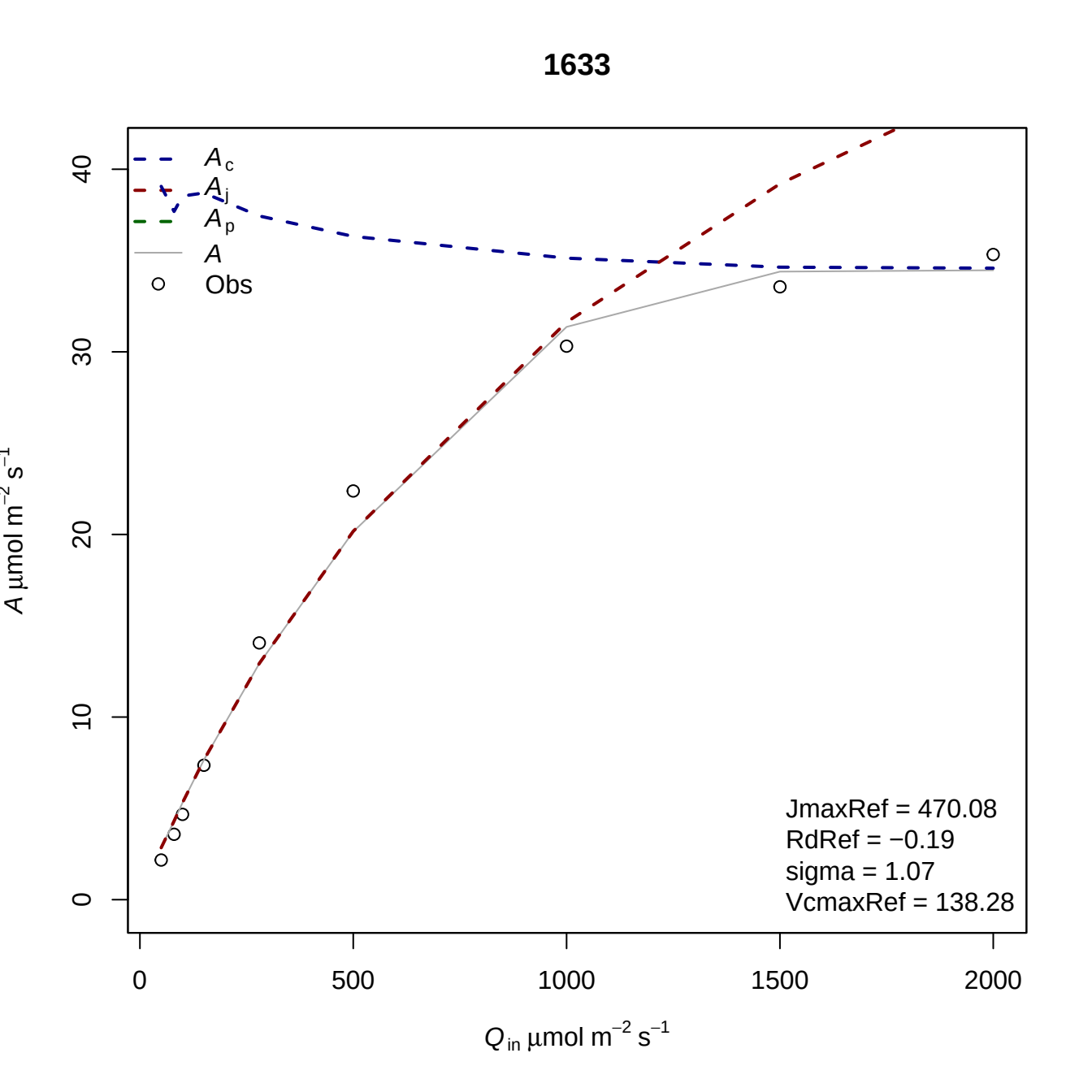

1628
A_c
A_j
A_p
A
Obs
JmaxRef = 198.65
RdRef = 1.74
sigma = 0.53
VcmaxRef = 139217.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

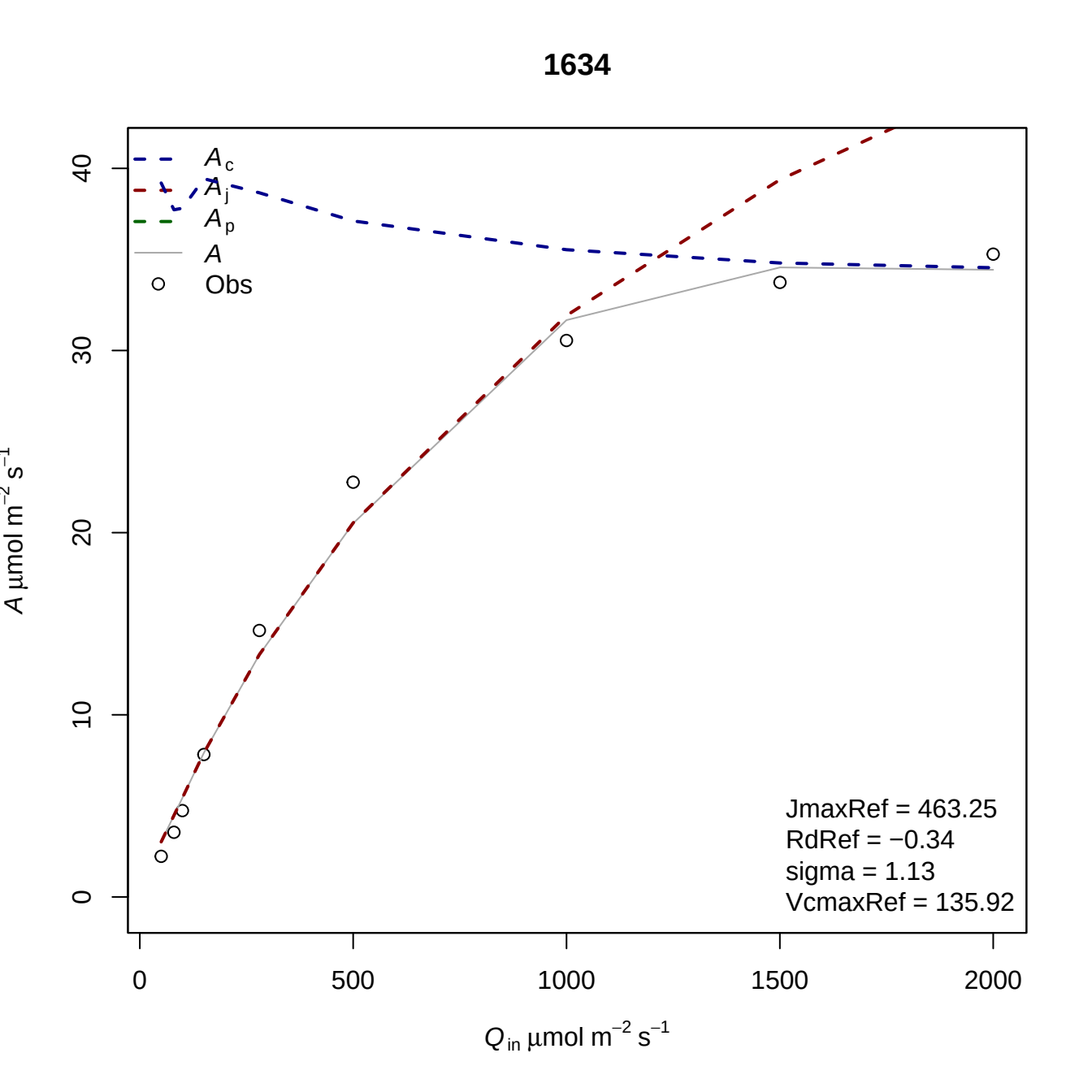

1629
A_c
A_j
A_p
A
Obs
JmaxRef = 509.7
RdRef = 0.37
sigma = 0.46
VcmaxRef = 105.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

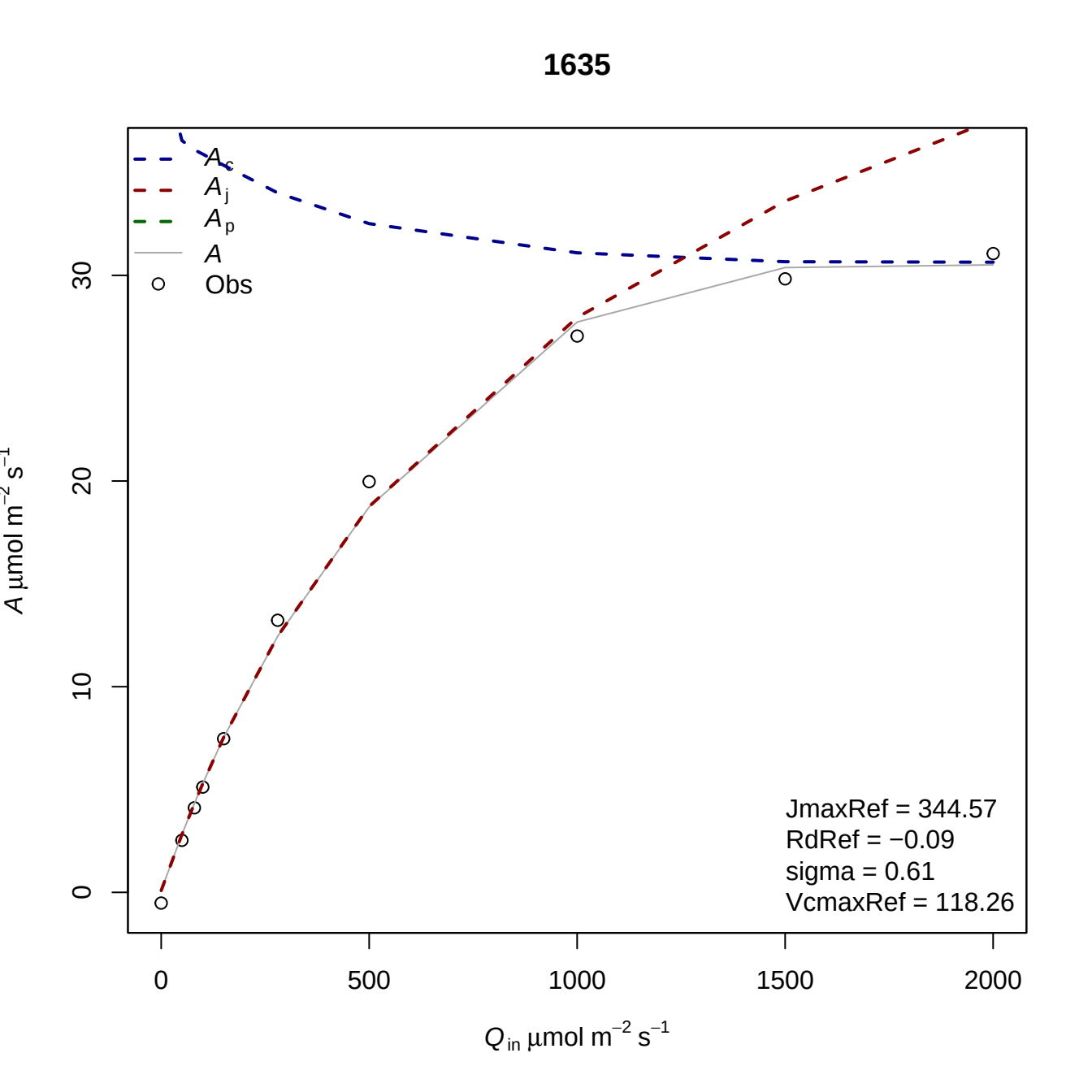

1630
A_c
A_j
A_p
A
Obs
JmaxRef = 475.64
RdRef = 0.23
sigma = 0.39
VcmaxRef = 103.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

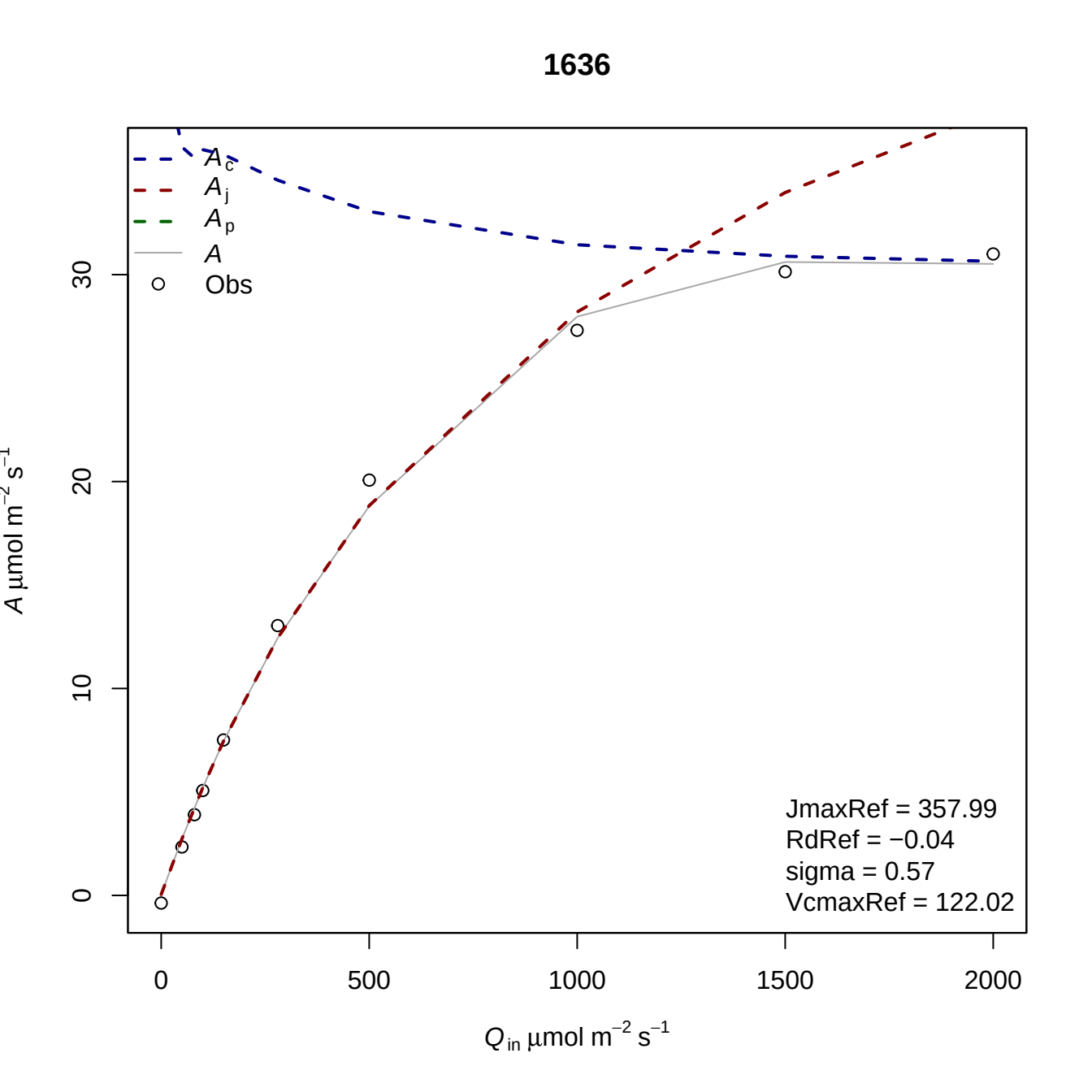

1631
A_c
A_j
A_p
A
Obs
JmaxRef = 464.36
RdRef = −0.06
sigma = 0.65
VcmaxRef = 140
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000

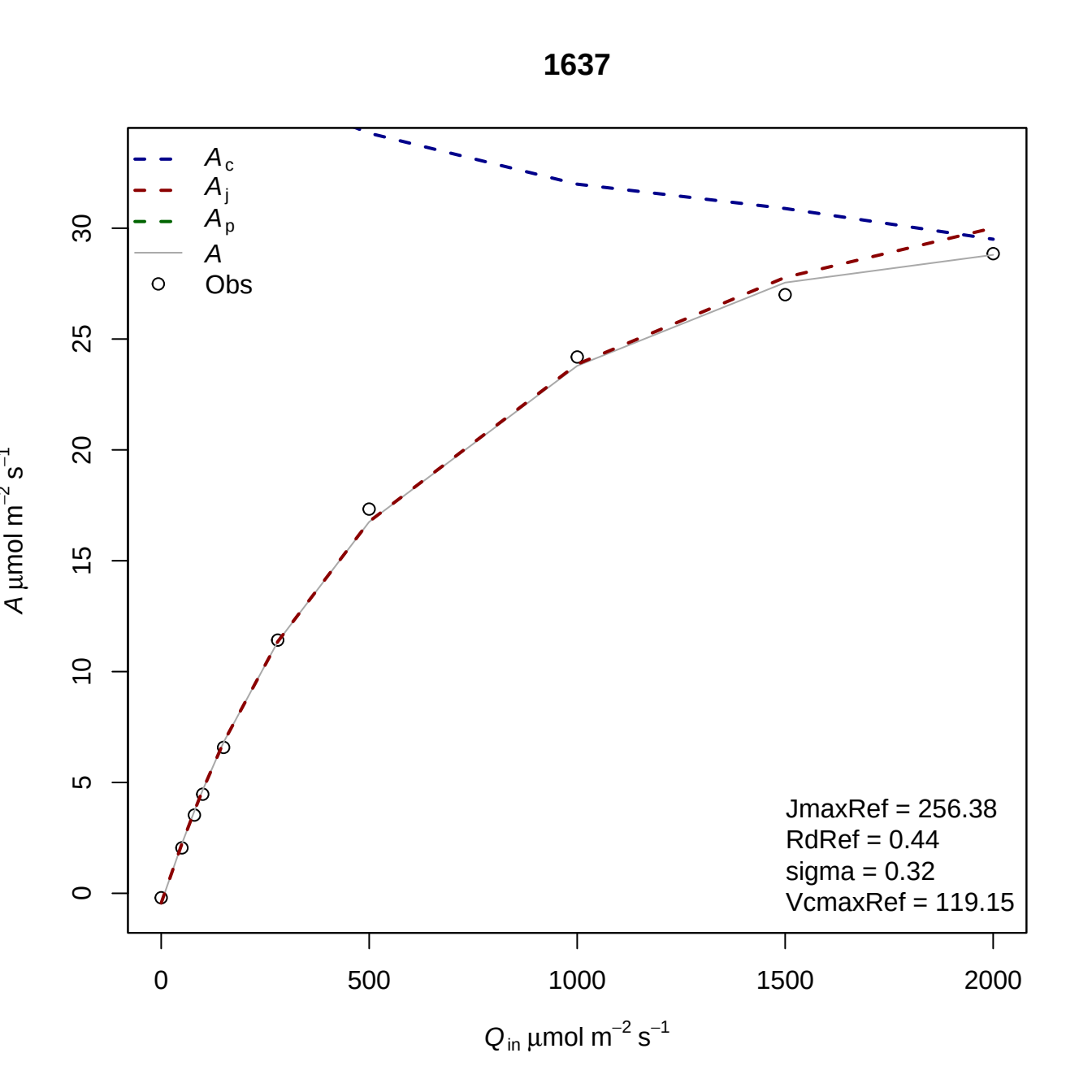

1632
A_c
A_j
A_p
A
Obs
JmaxRef = 454.66
RdRef = 0.34
sigma = 0.68
VcmaxRef = 139.95
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

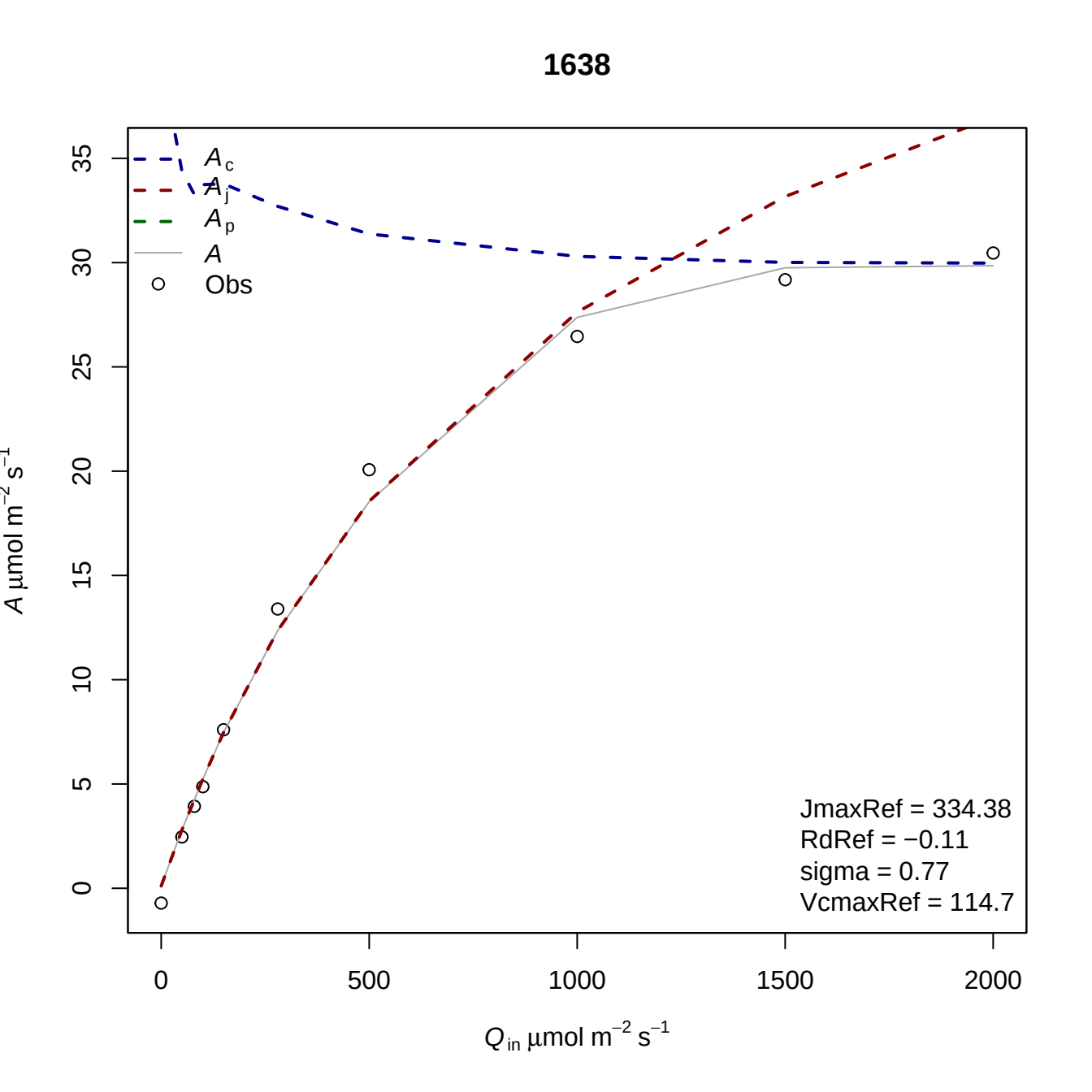

1633
A_c
A_j
A_p
A
Obs
JmaxRef = 470.08
RdRef = −0.19
sigma = 1.07
VcmaxRef = 138.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000

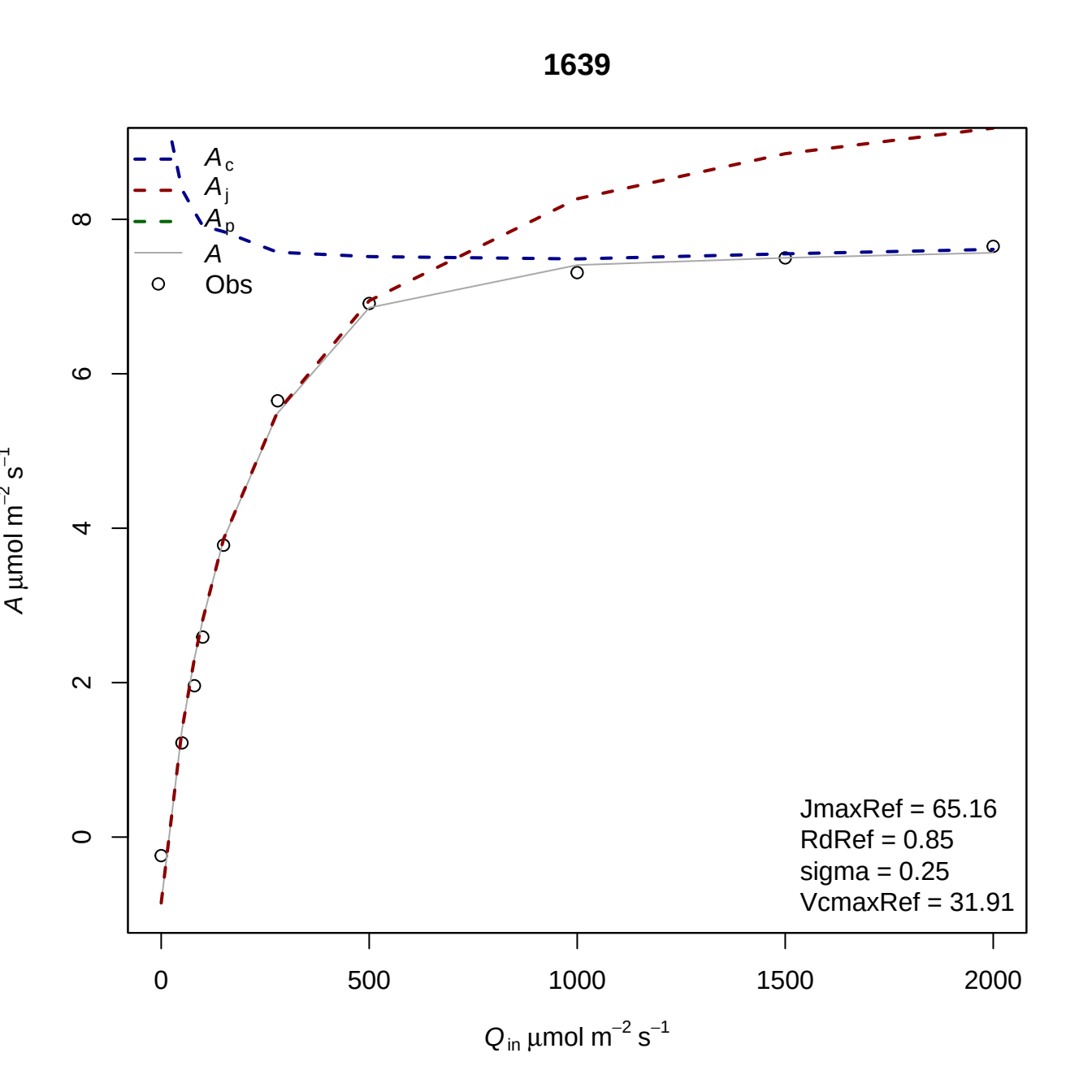

1634
A_c
A_j
A_p
A
Obs
JmaxRef = 463.25
RdRef = −0.34
sigma = 1.13
VcmaxRef = 135.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000

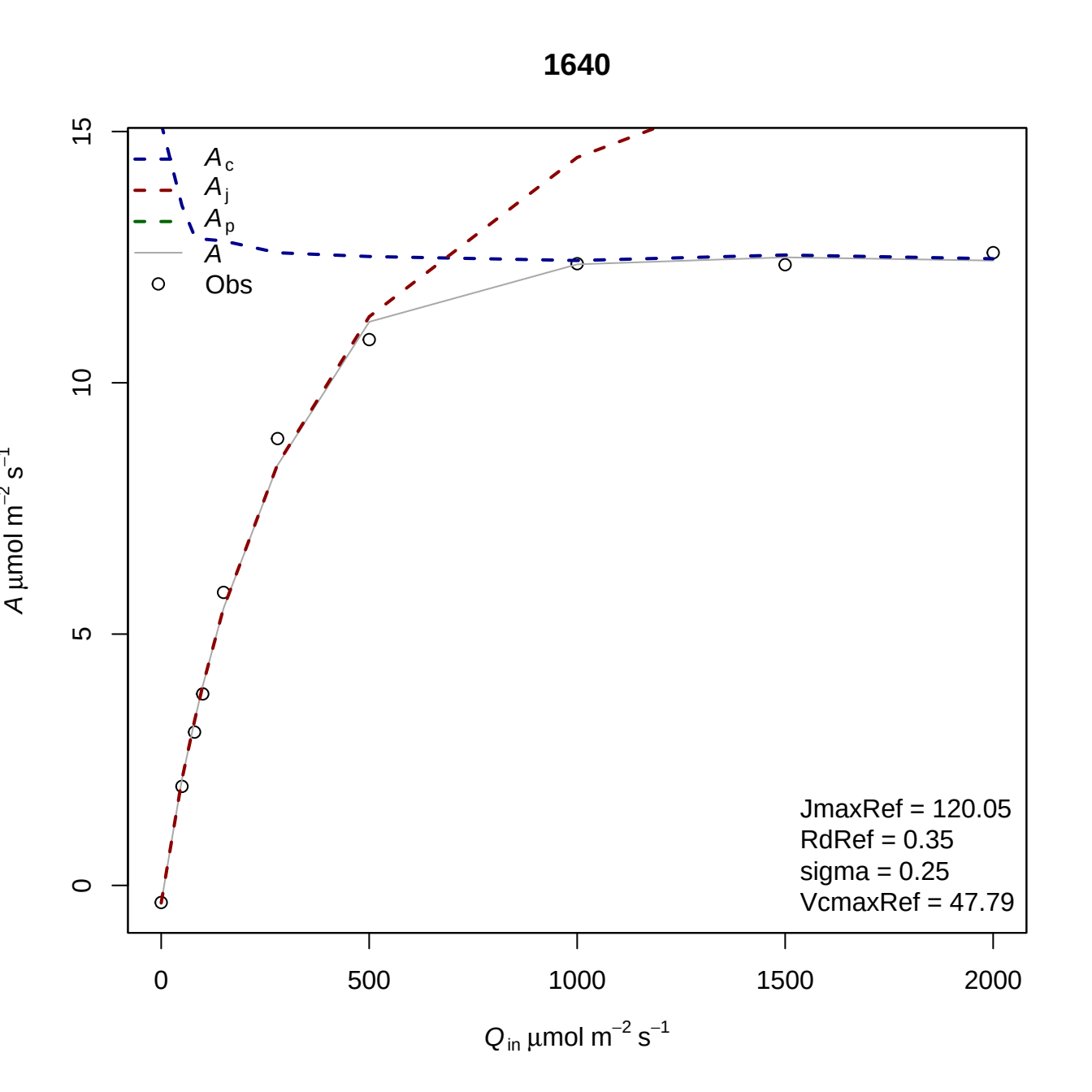

1635
A_c
A_j
A_p
A
Obs
JmaxRef = 344.57
RdRef = −0.09
sigma = 0.61
VcmaxRef = 118.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

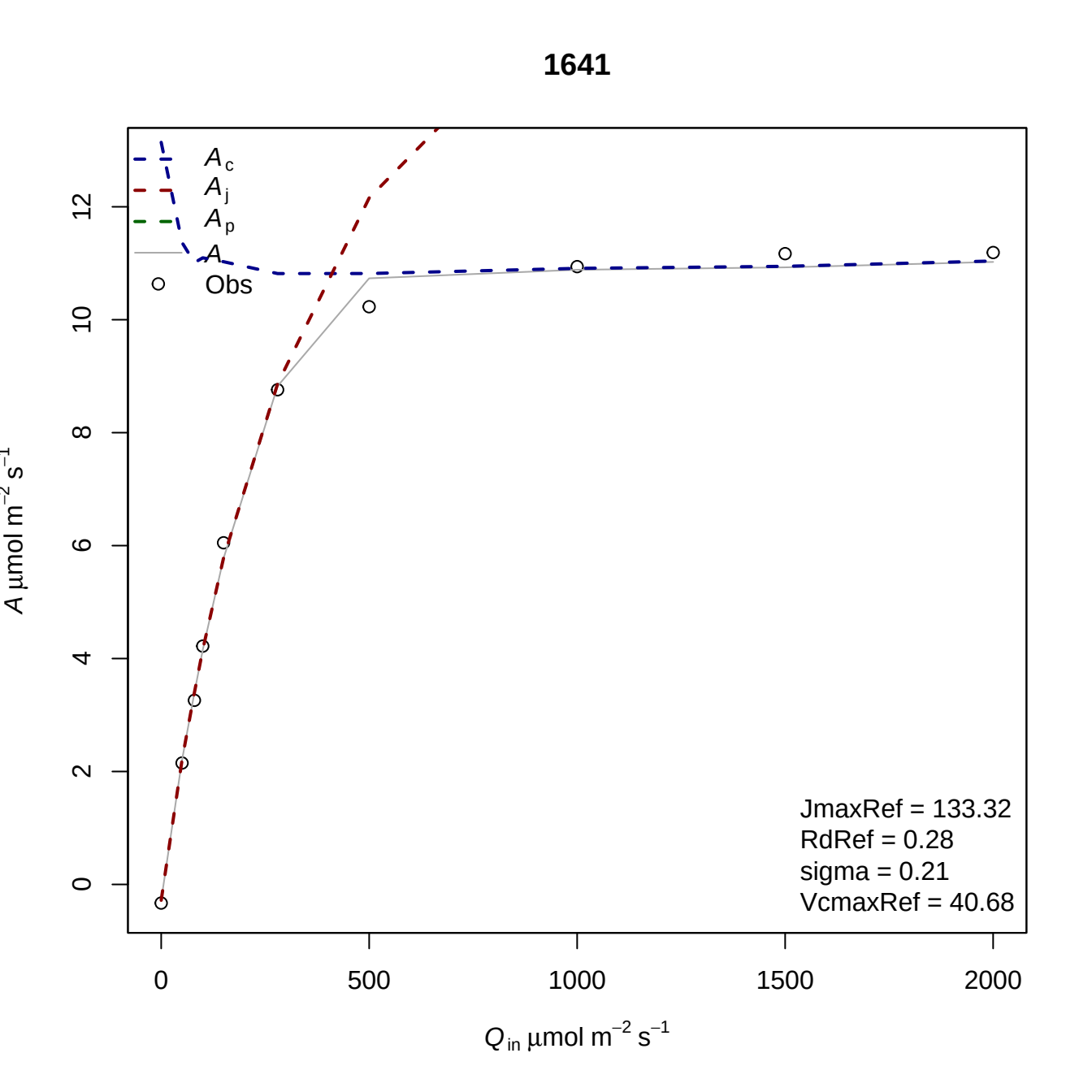

1636
A_c
A_j
A_p
A
Obs
JmaxRef = 357.99
RdRef = −0.04
sigma = 0.57
VcmaxRef = 122.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

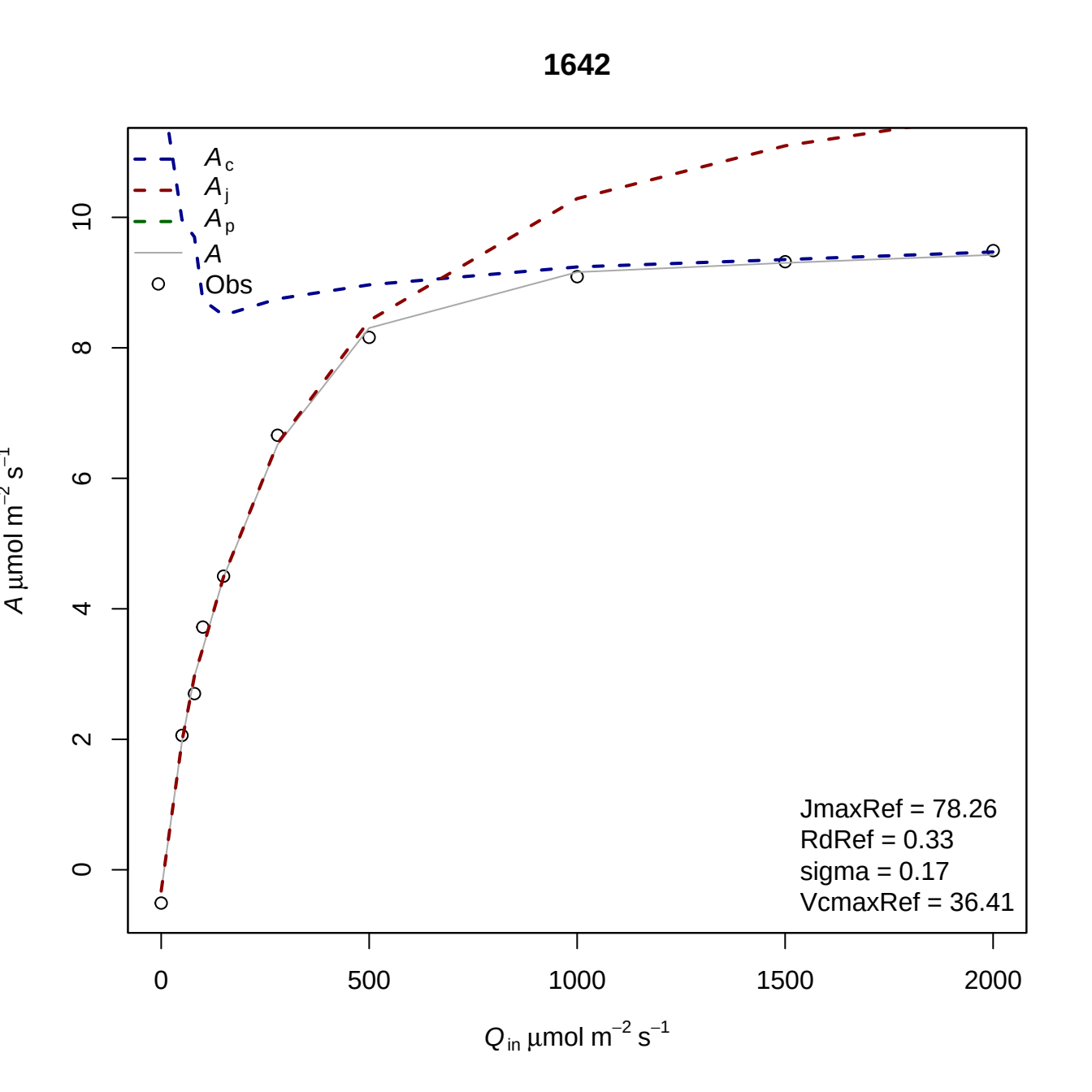

1637
A_c
A_j
A_p
A
Obs
JmaxRef = 256.38
RdRef = 0.44
sigma = 0.32
VcmaxRef = 119.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

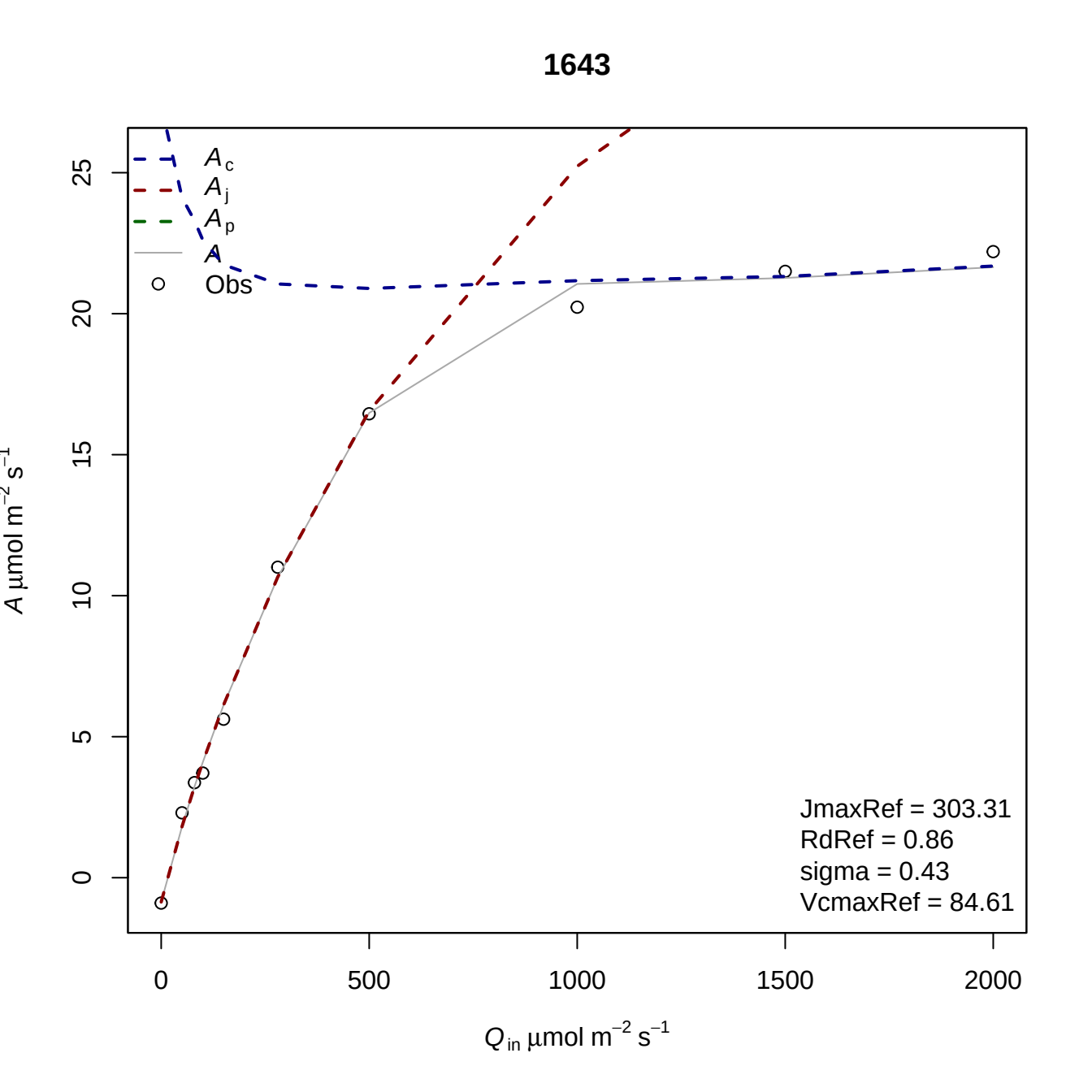

1638
A_c
A_j
A_p
A
Obs
JmaxRef = 334.38
RdRef = −0.11
sigma = 0.77
VcmaxRef = 114.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

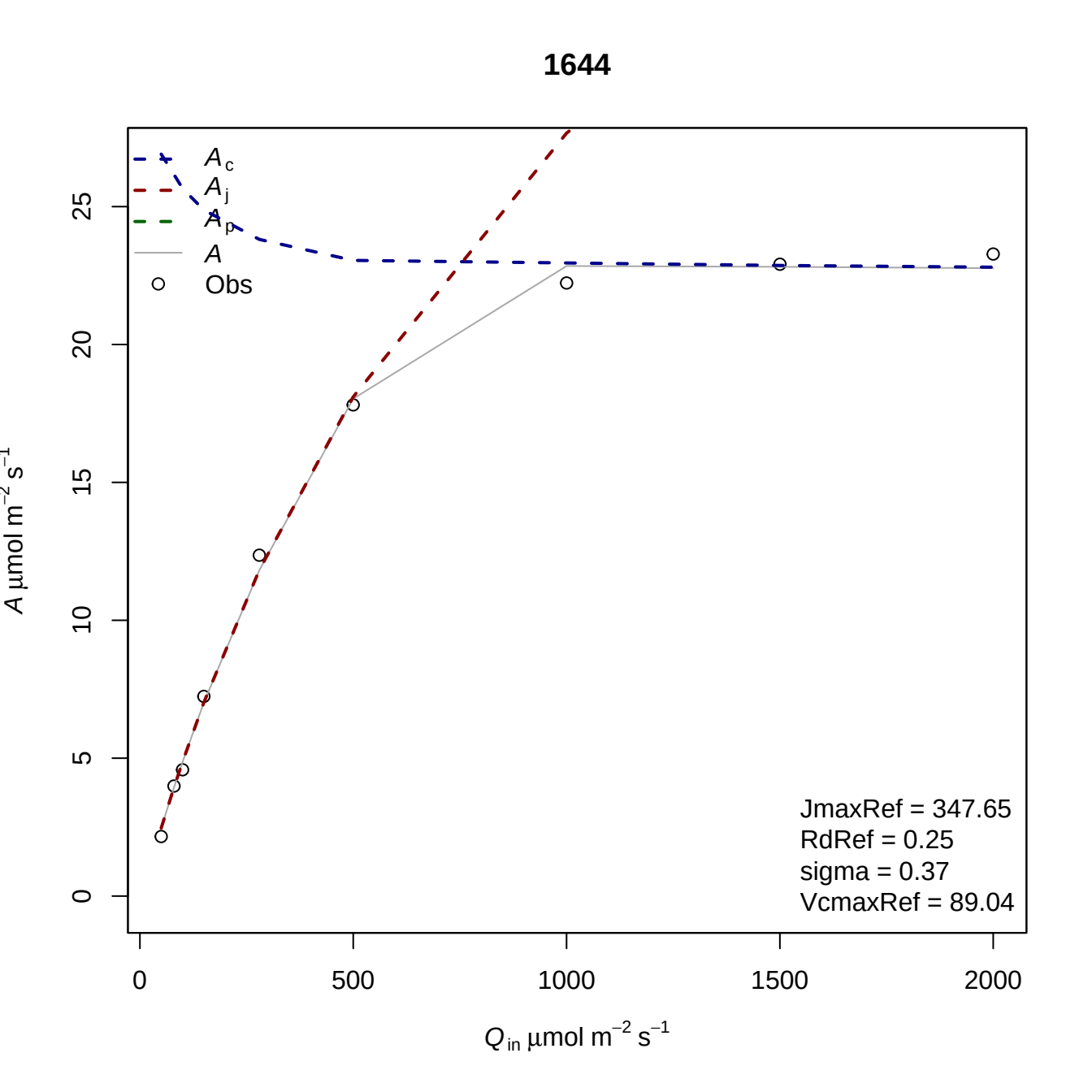

1639
A_c
A_j
A_p
A
Obs
JmaxRef = 65.16
RdRef = 0.85
sigma = 0.25
VcmaxRef = 31.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

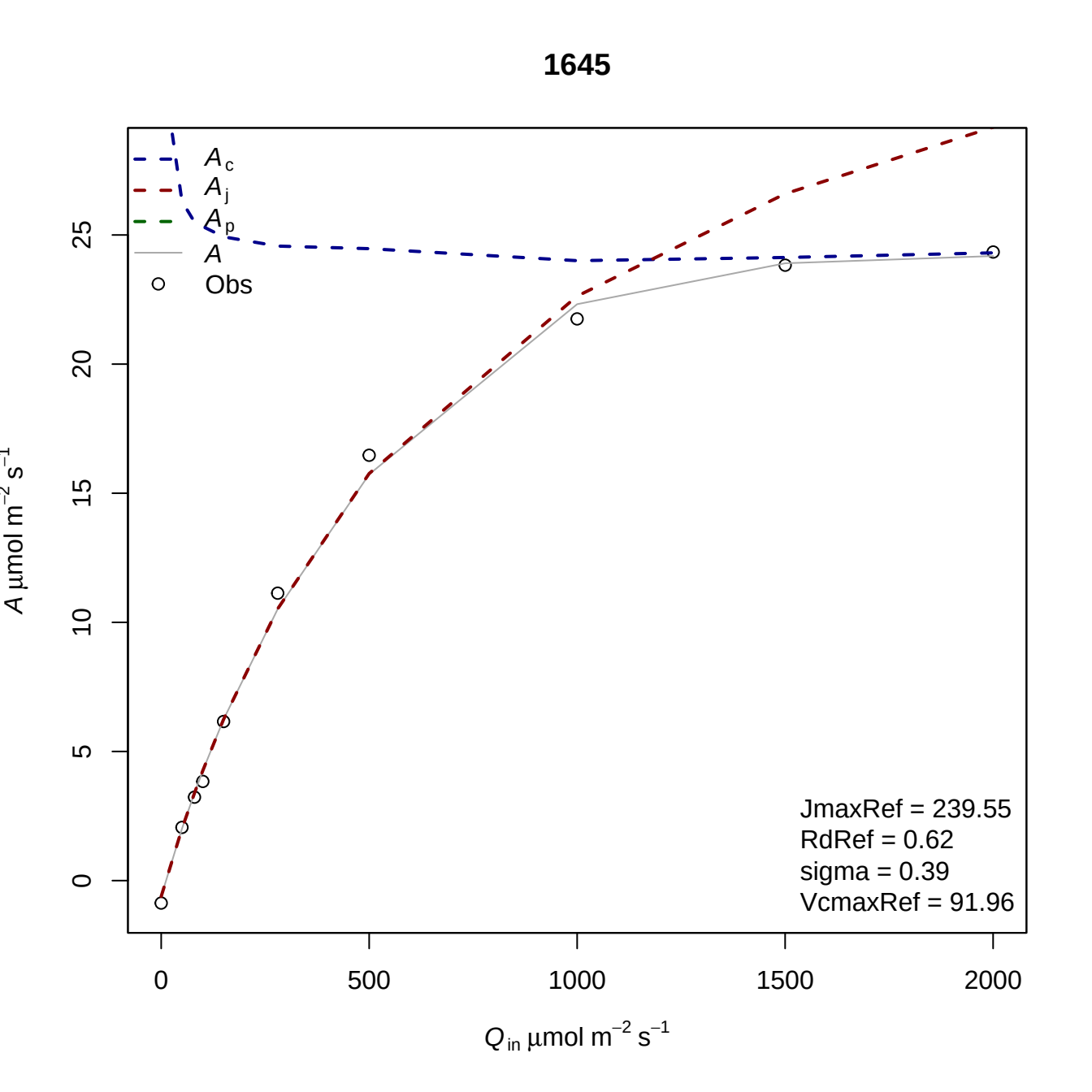

1640
A_c
A_j
A_p
A
Obs
JmaxRef = 120.05
RdRef = 0.35
sigma = 0.25
VcmaxRef = 47.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

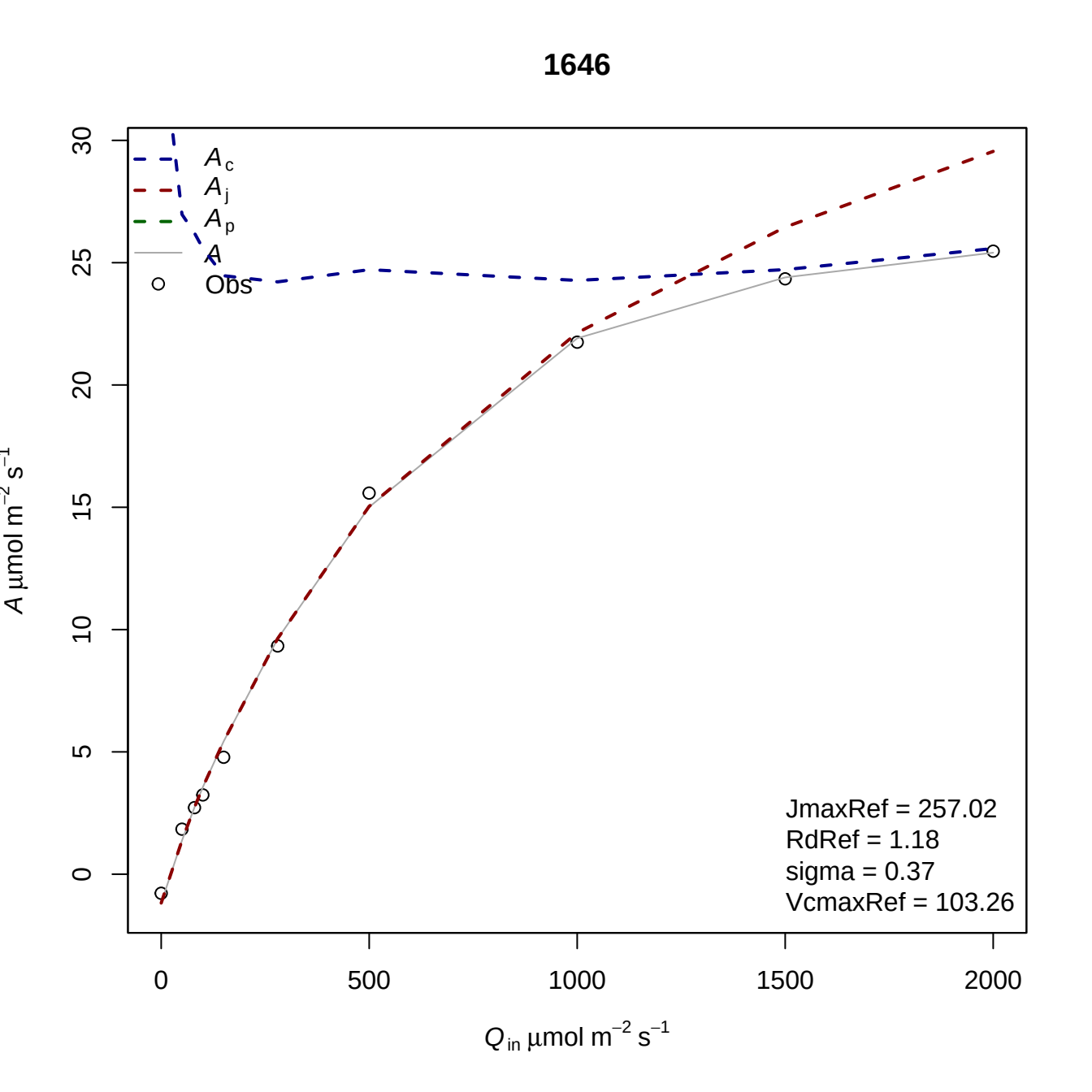

1641
A_c
A_j
A_p
A
Obs
JmaxRef = 133.32
RdRef = 0.28
sigma = 0.21
VcmaxRef = 40.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

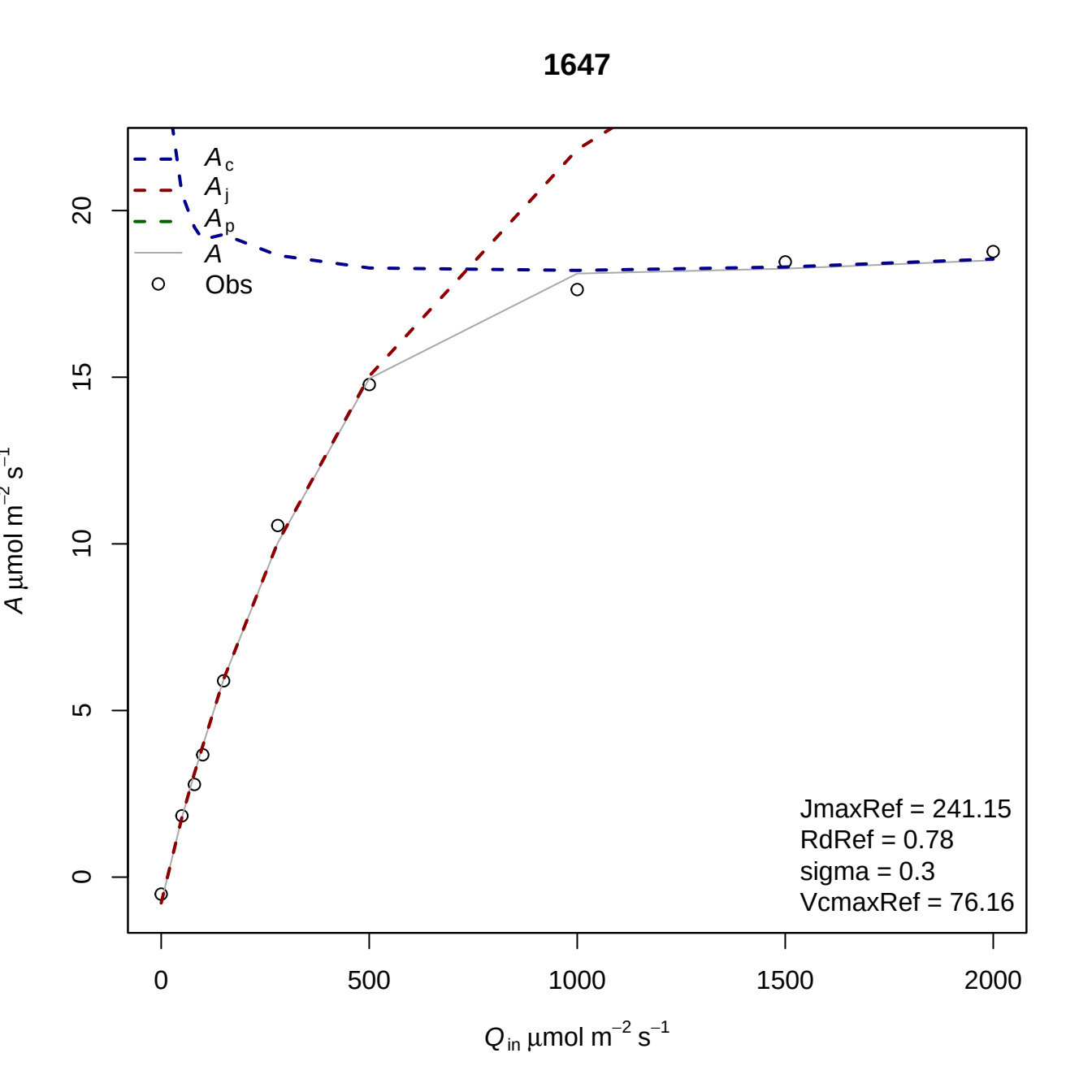

1642
A_c
A_j
A_p
A
Obs
JmaxRef = 78.26
RdRef = 0.33
sigma = 0.17
VcmaxRef = 36.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

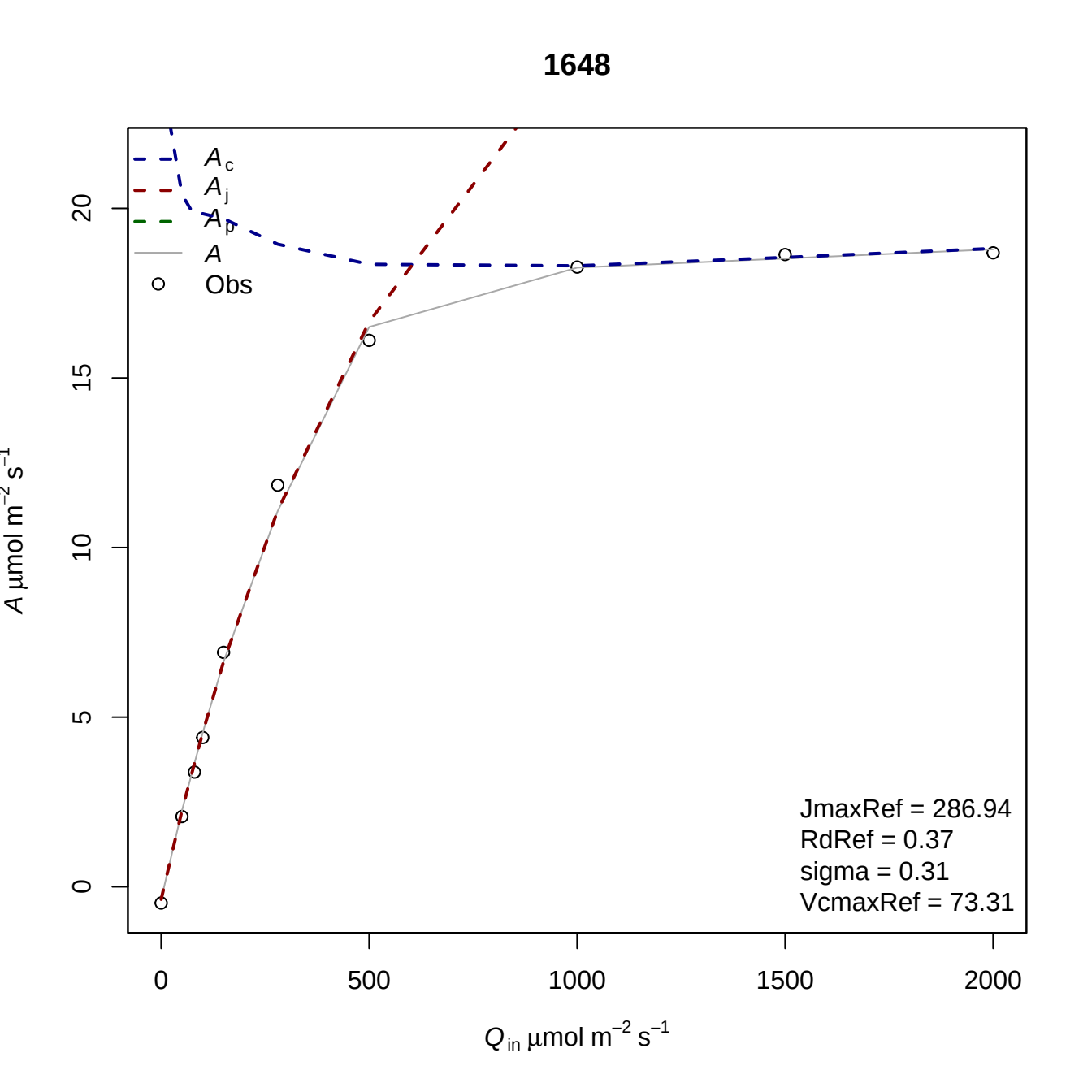

1643
A_c
A_j
A_p
A
Obs
JmaxRef = 303.31
RdRef = 0.86
sigma = 0.43
VcmaxRef = 84.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

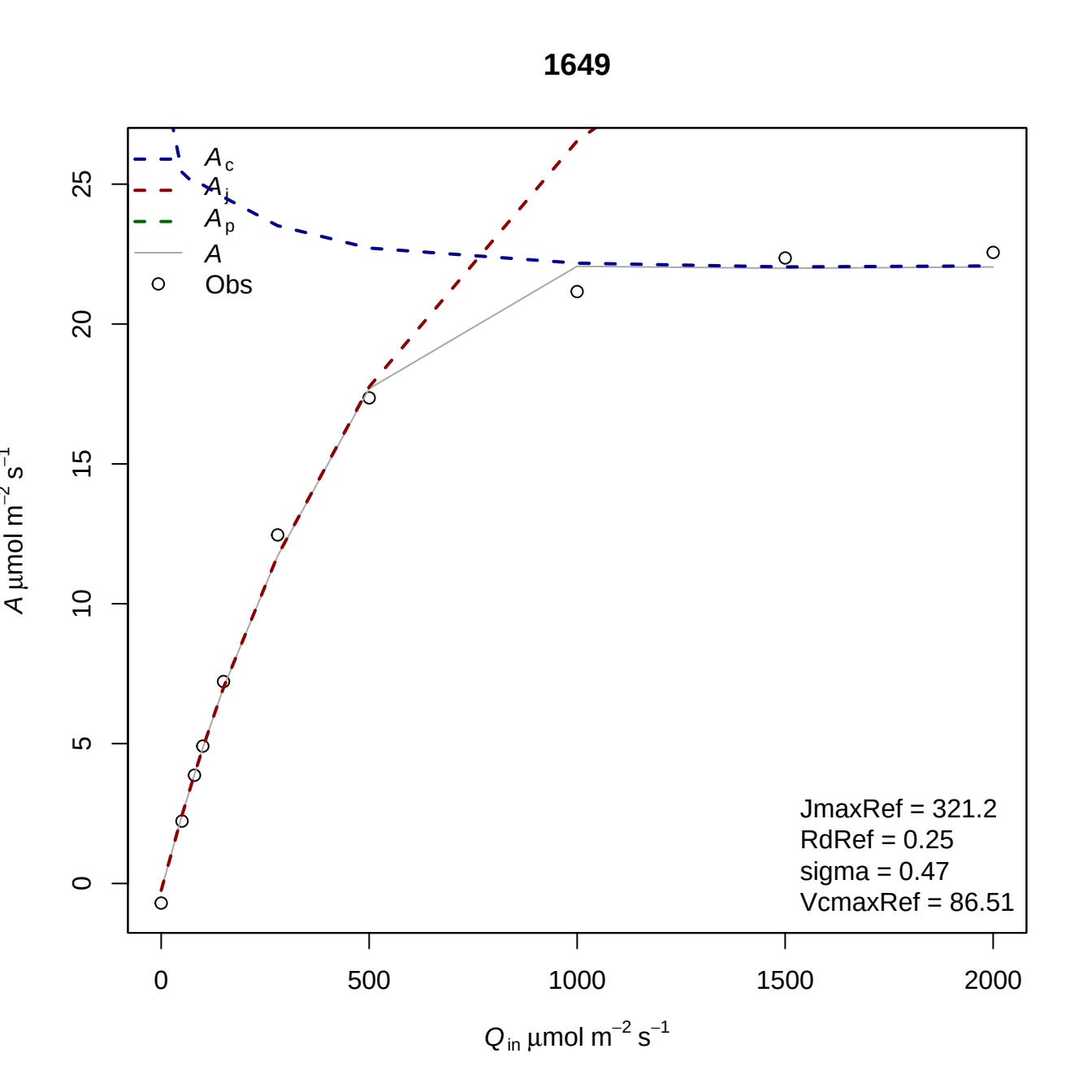

1644
A_c
A_j
A_p
A
Obs
JmaxRef = 347.65
RdRef = 0.25
sigma = 0.37
VcmaxRef = 89.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

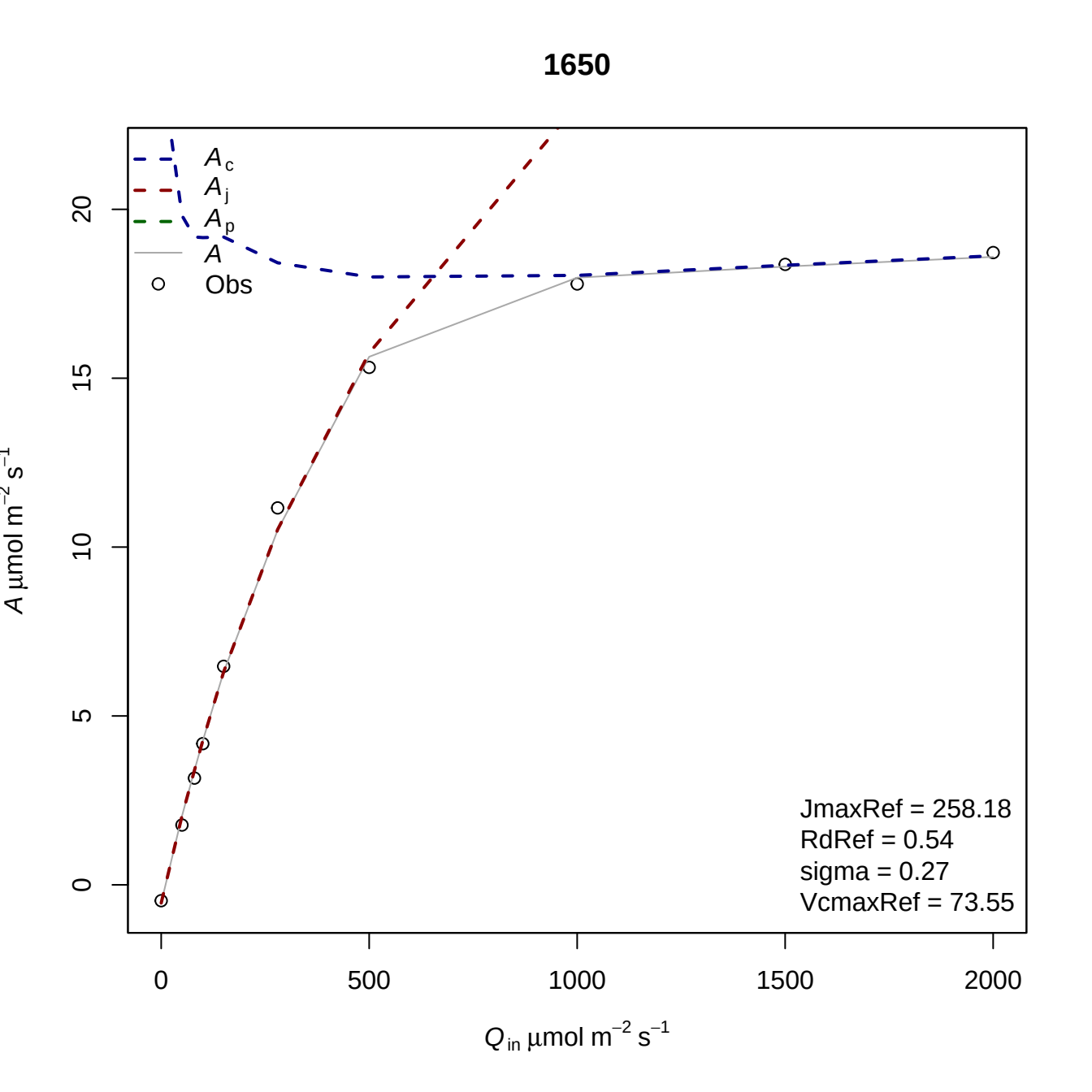

1645
A_c
A_j
A_p
A
Obs
JmaxRef = 239.55
RdRef = 0.62
sigma = 0.39
VcmaxRef = 91.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

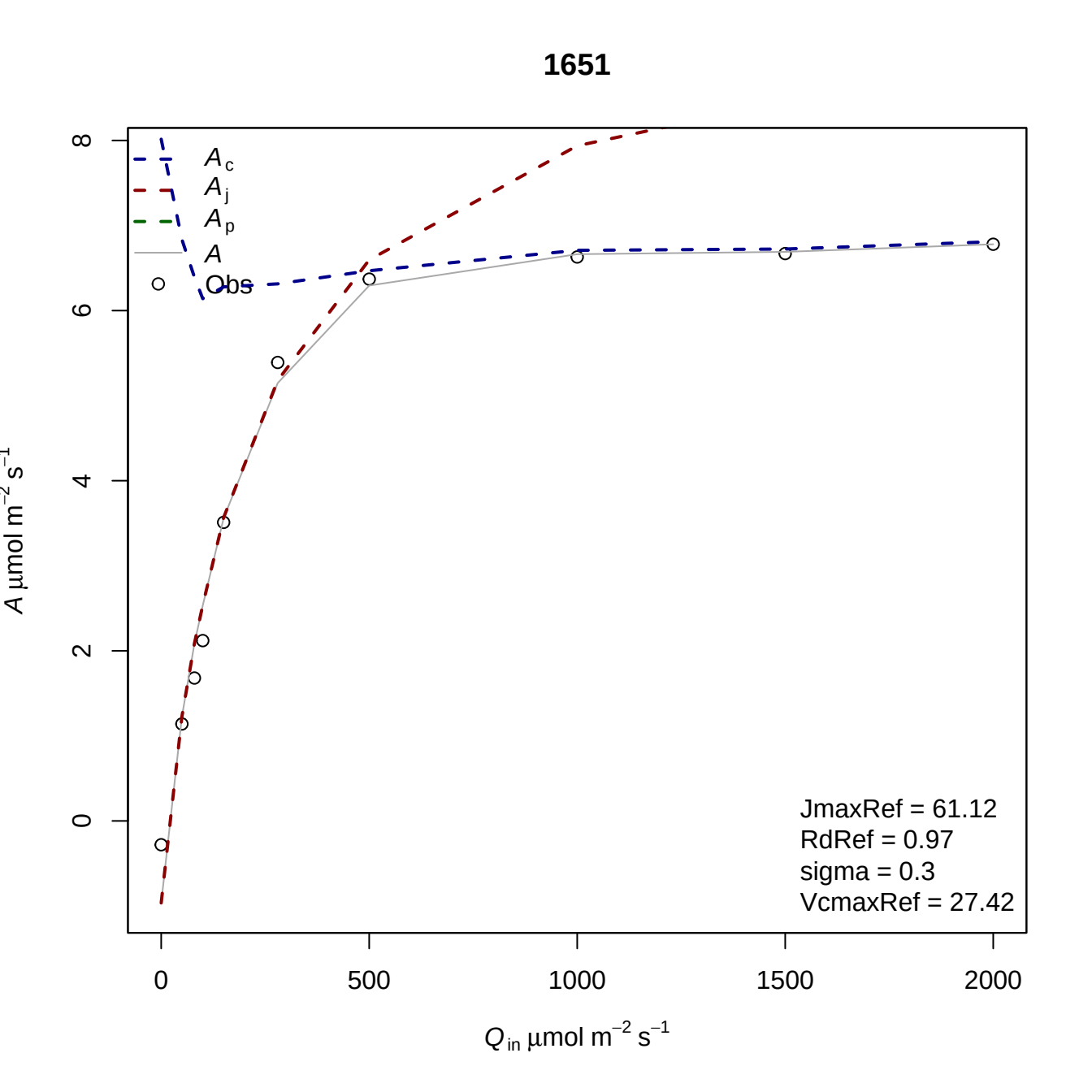

1646
A_c
A_j
A_p
A
Obs
JmaxRef = 257.02
RdRef = 1.18
sigma = 0.37
VcmaxRef = 103.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

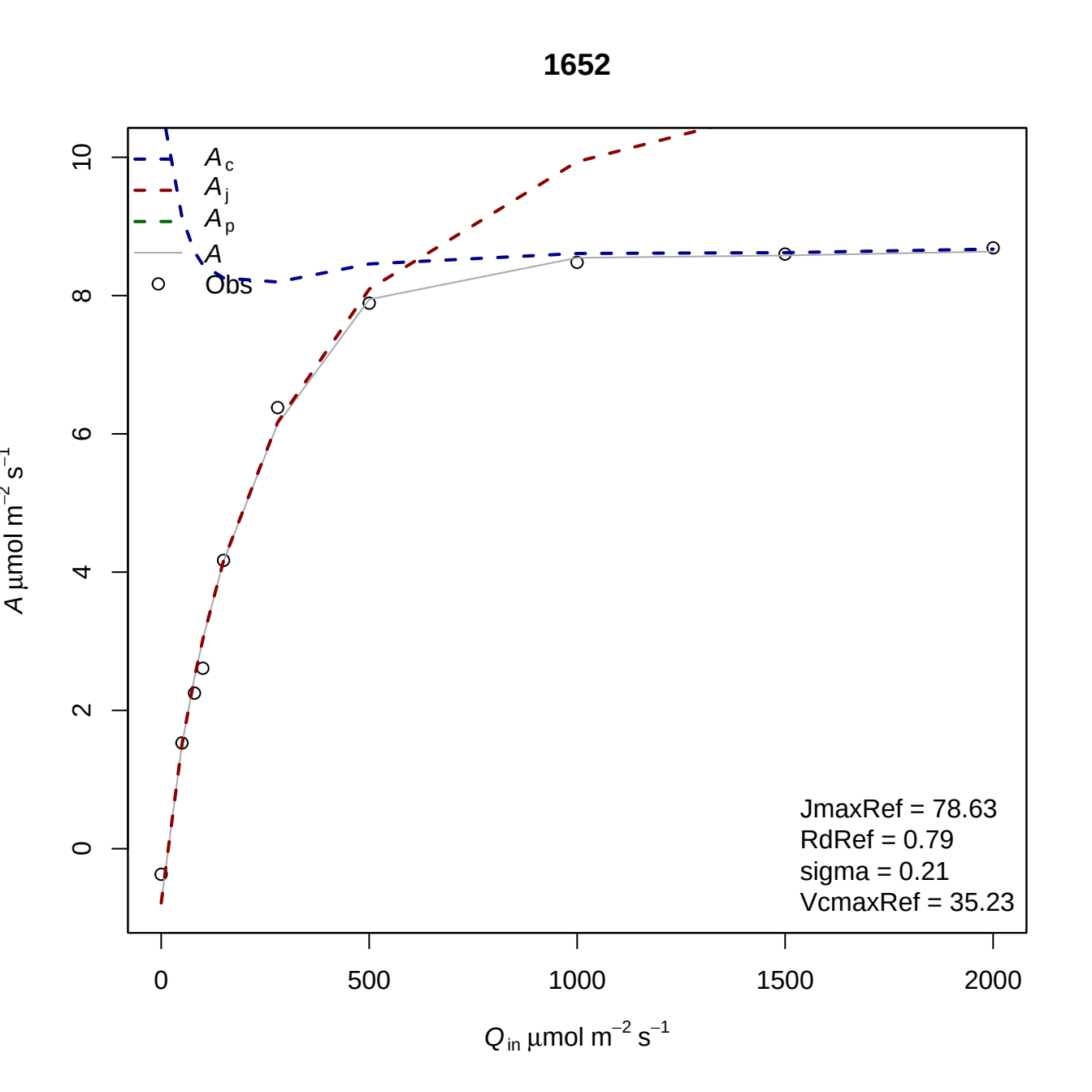

1647
A_c
A_j
A_p
A
Obs
JmaxRef = 241.15
RdRef = 0.78
sigma = 0.3
VcmaxRef = 76.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

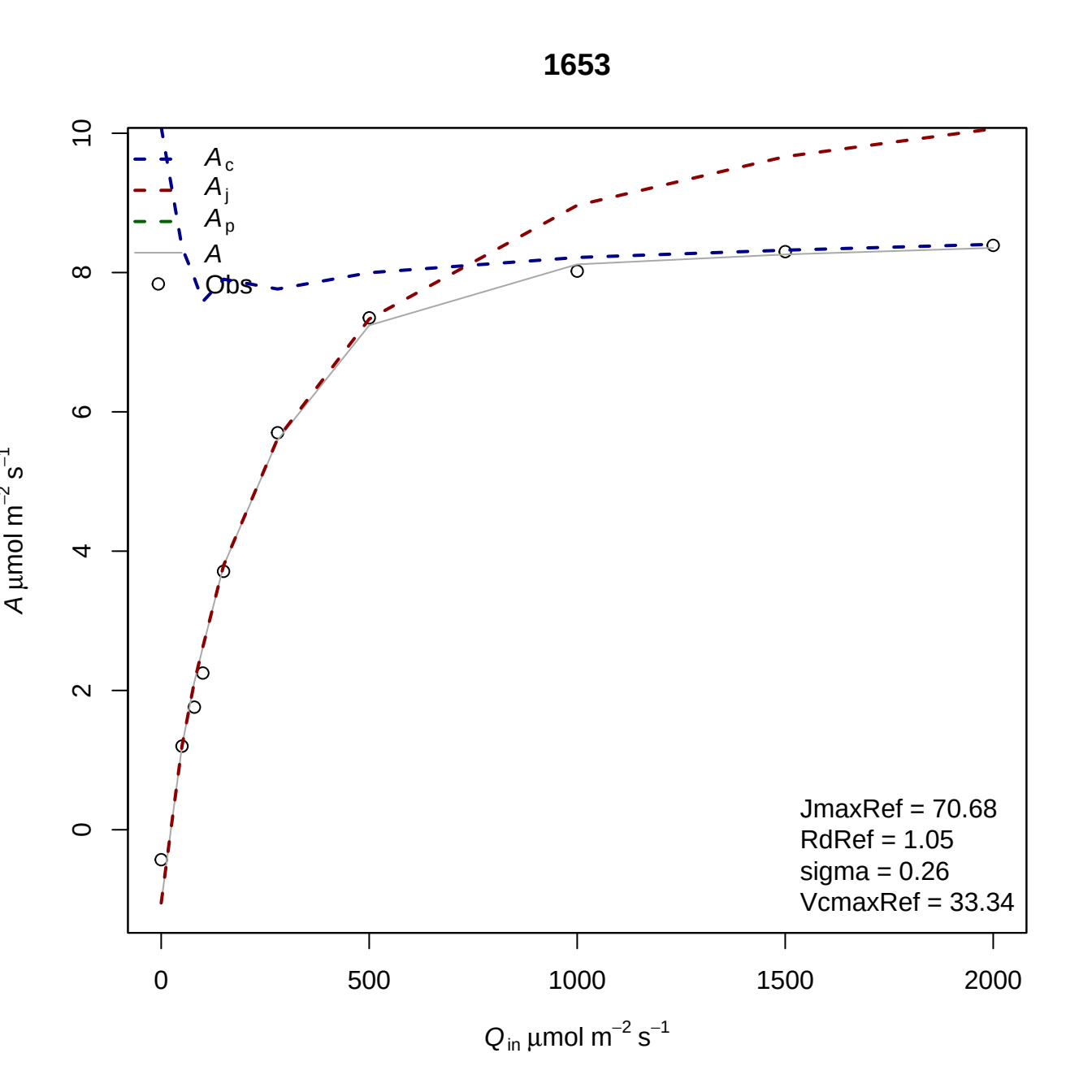

1648
A_c
A_j
A_p
A
Obs
JmaxRef = 286.94
RdRef = 0.37
sigma = 0.31
VcmaxRef = 73.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

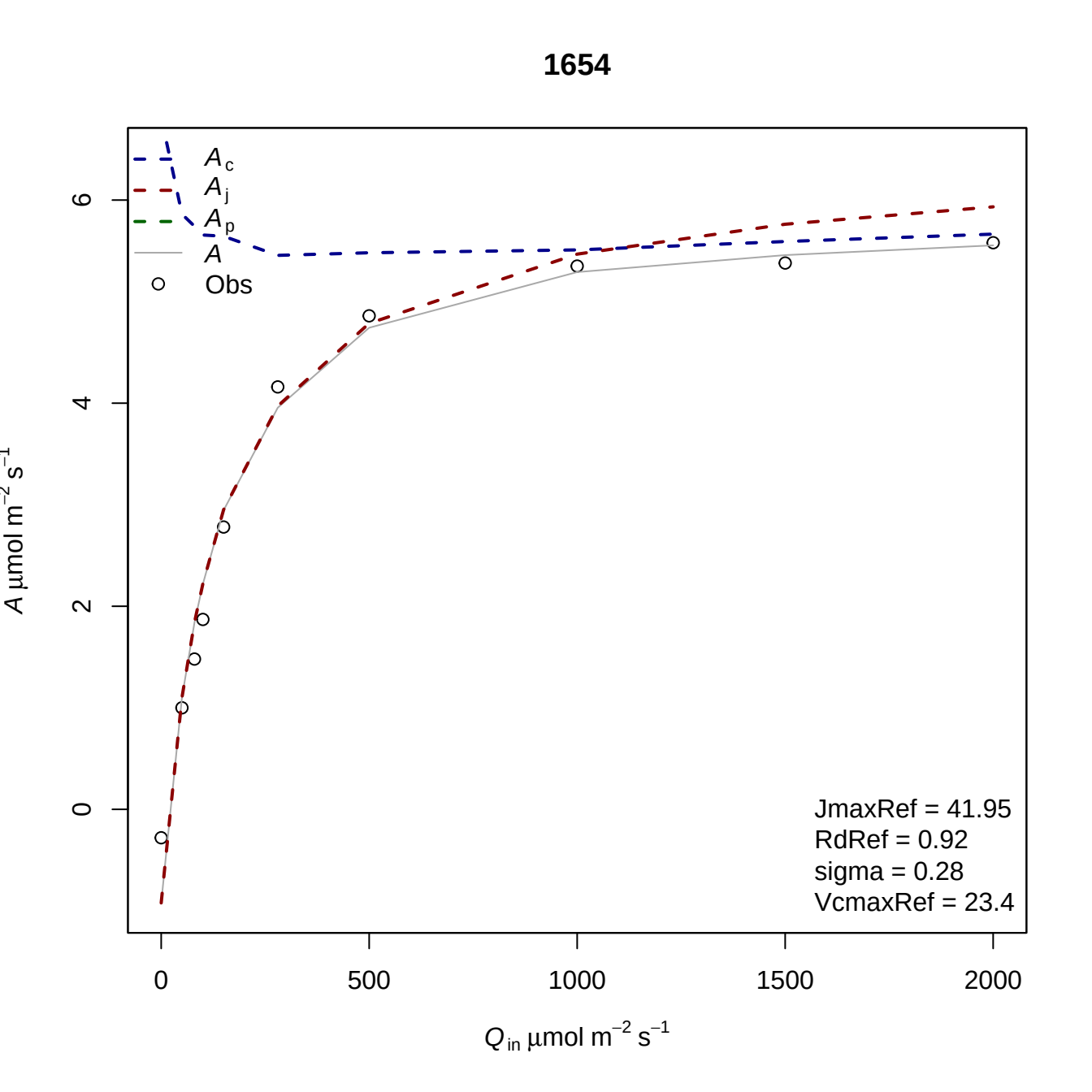

1649
A_c
A_j
A_p
A
Obs
JmaxRef = 321.2
RdRef = 0.25
sigma = 0.47
VcmaxRef = 86.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

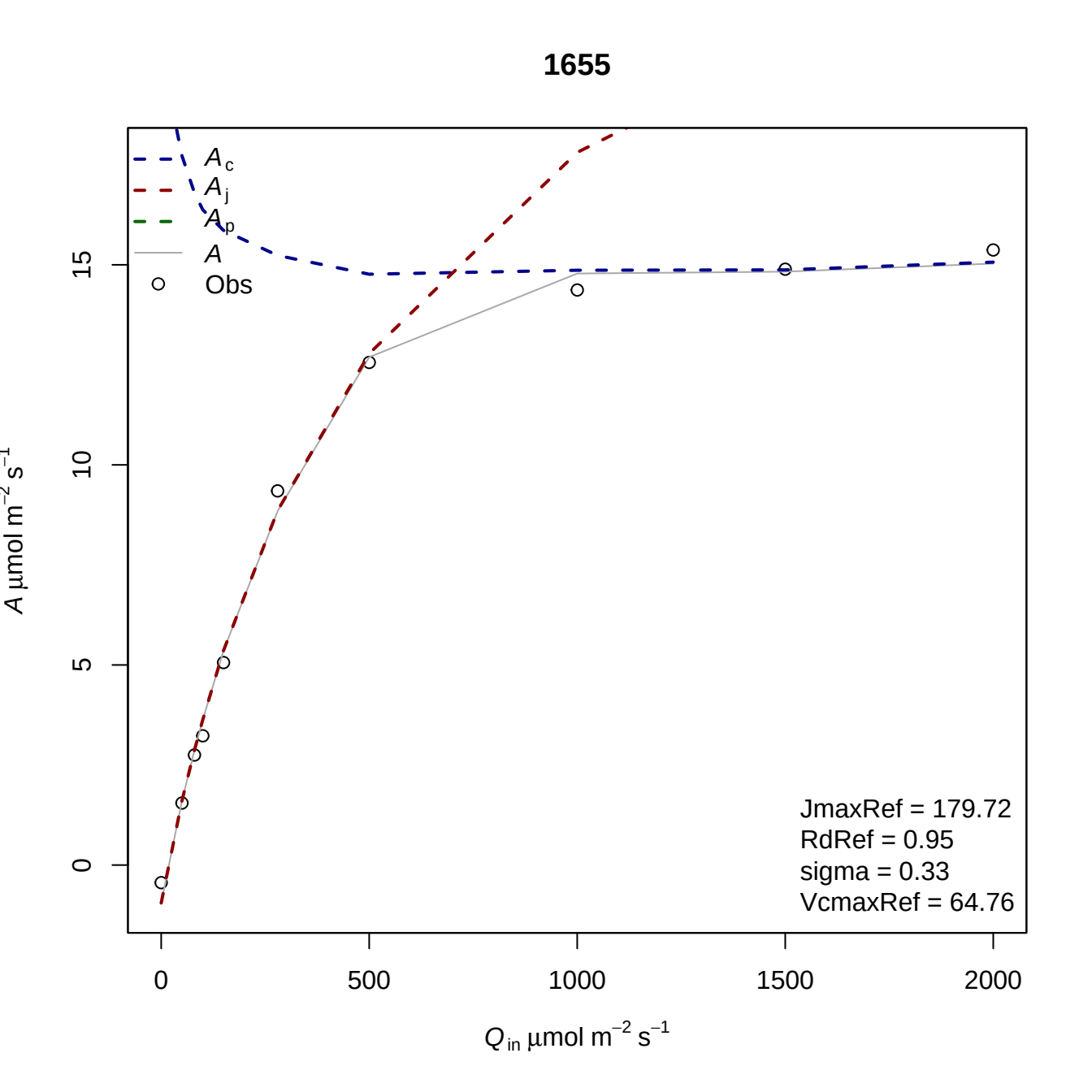

1650
A_c
A_j
A_p
A
Obs
JmaxRef = 258.18
RdRef = 0.54
sigma = 0.27
VcmaxRef = 73.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

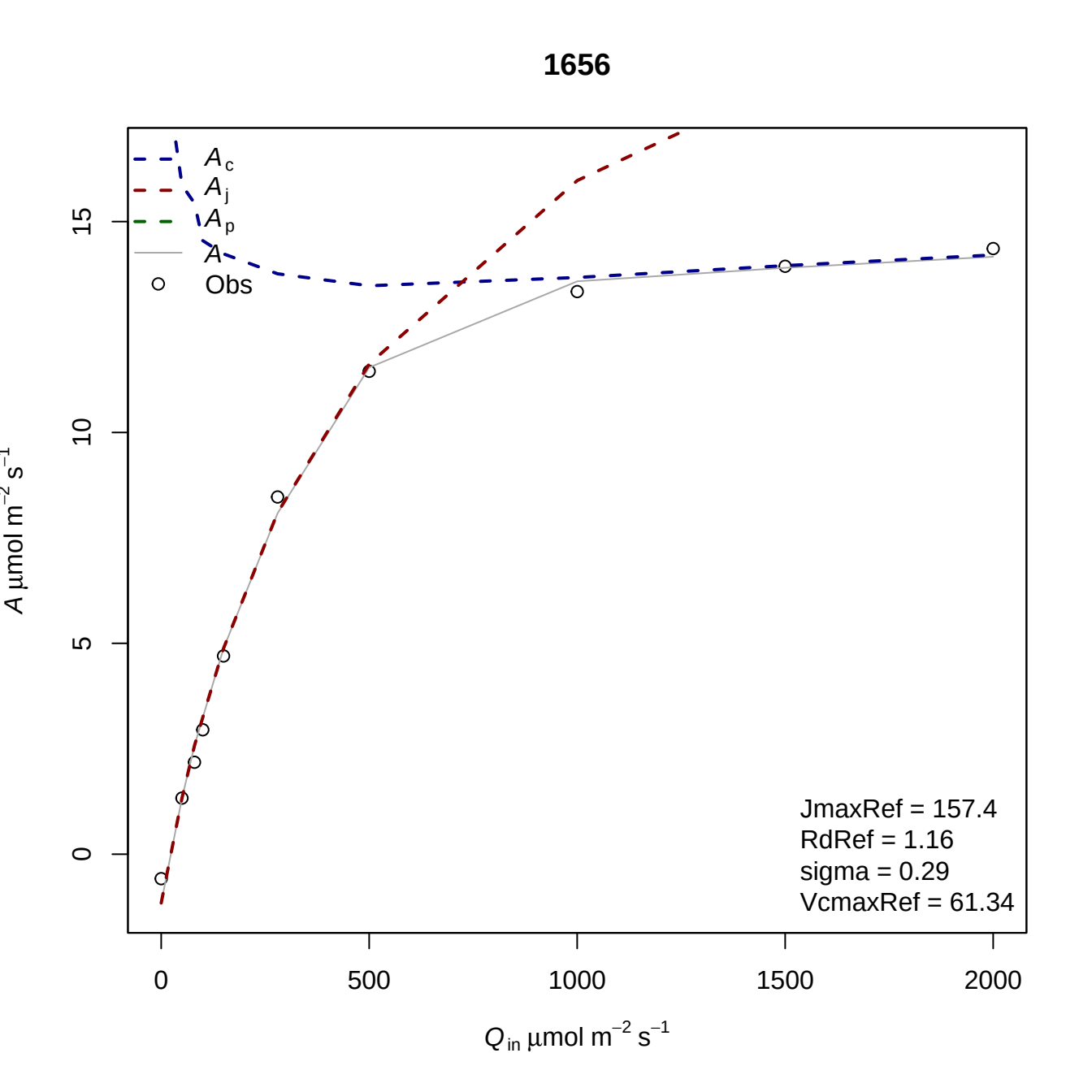

1651
A_c
A_j
A_p
A
Obs
JmaxRef = 61.12
RdRef = 0.97
sigma = 0.3
VcmaxRef = 27.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

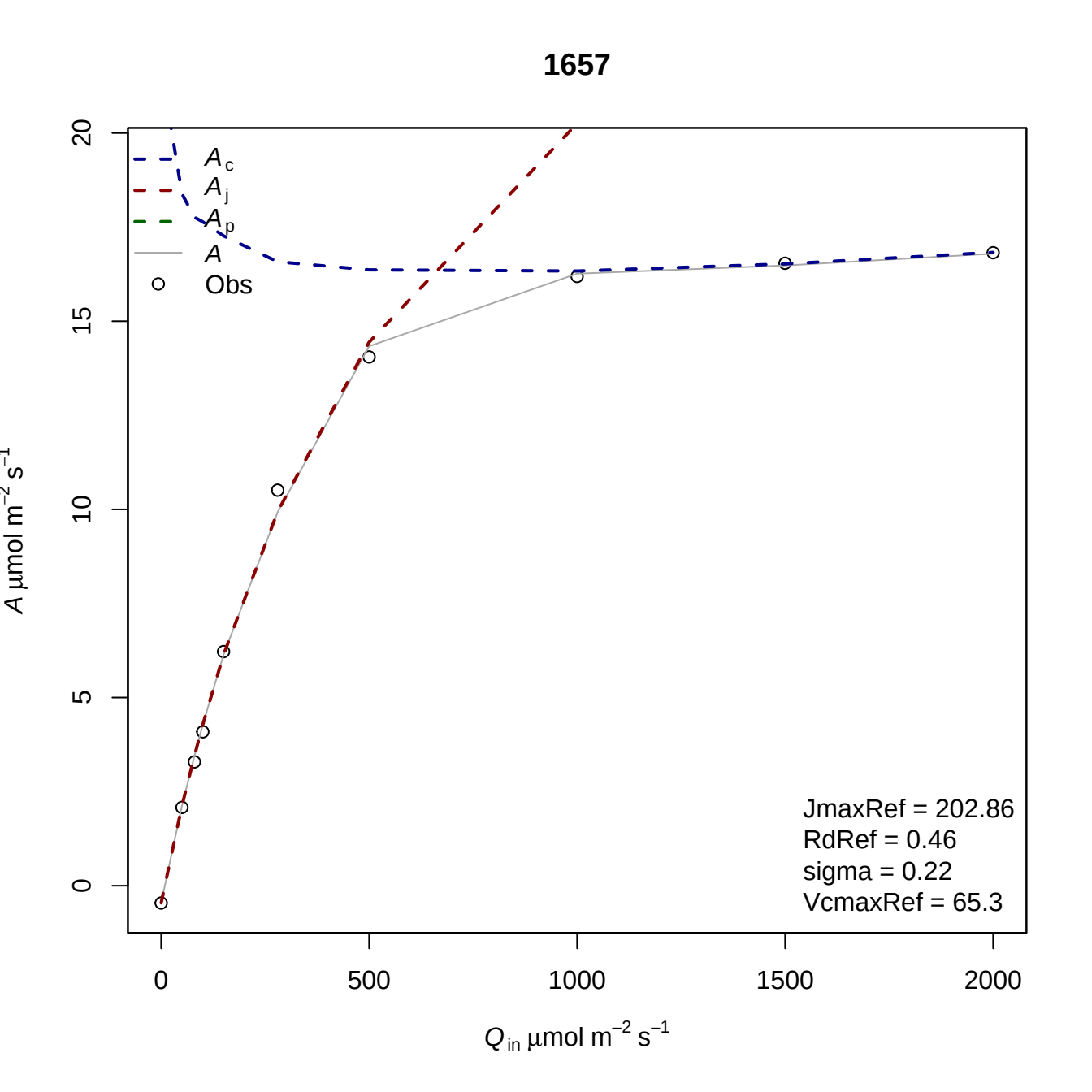

1652
A_c
A_j
A_p
A
Obs
JmaxRef = 78.63
RdRef = 0.79
sigma = 0.21
VcmaxRef = 35.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

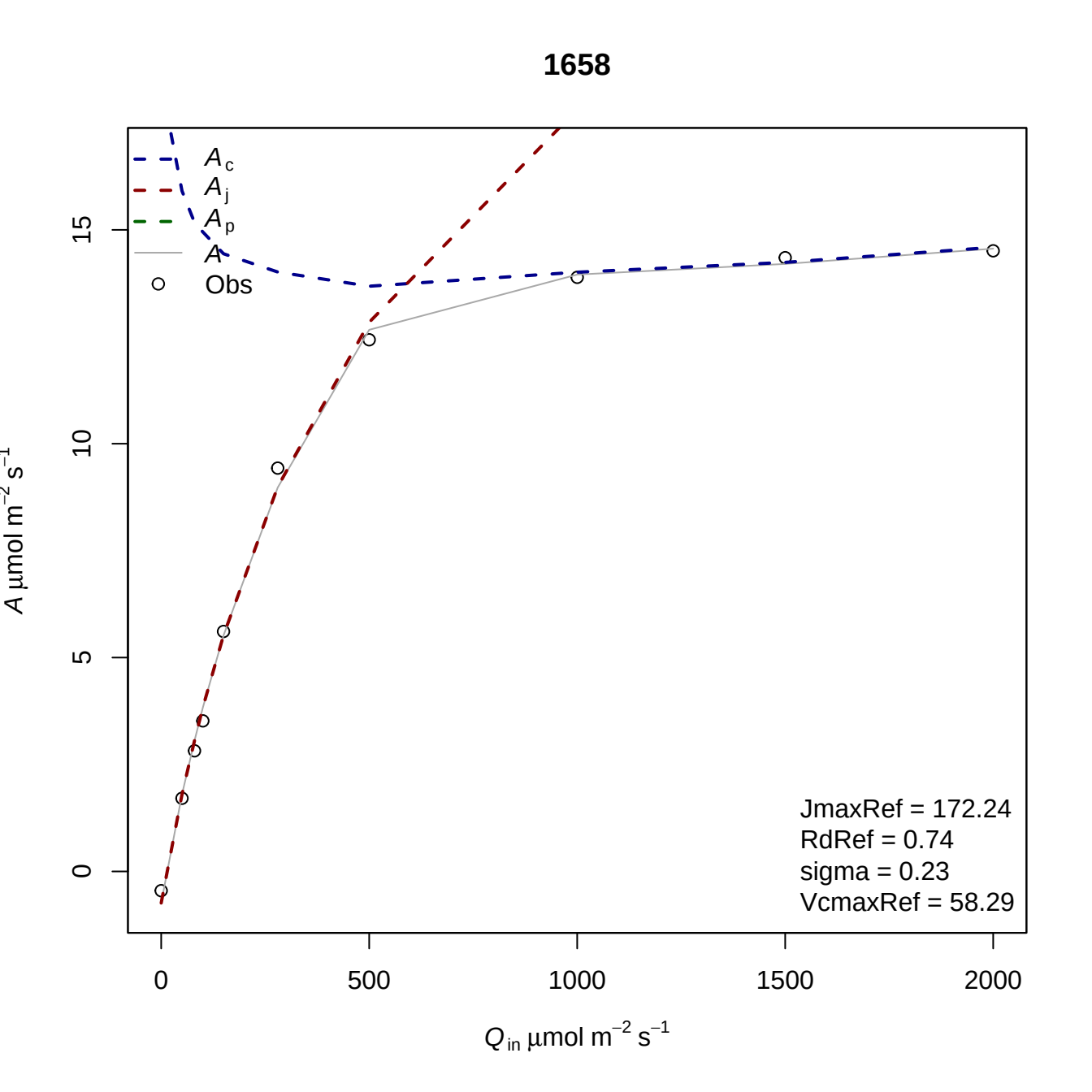

1653
A_c
A_j
A_p
A
Obs
JmaxRef = 70.68
RdRef = 1.05
sigma = 0.26
VcmaxRef = 33.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

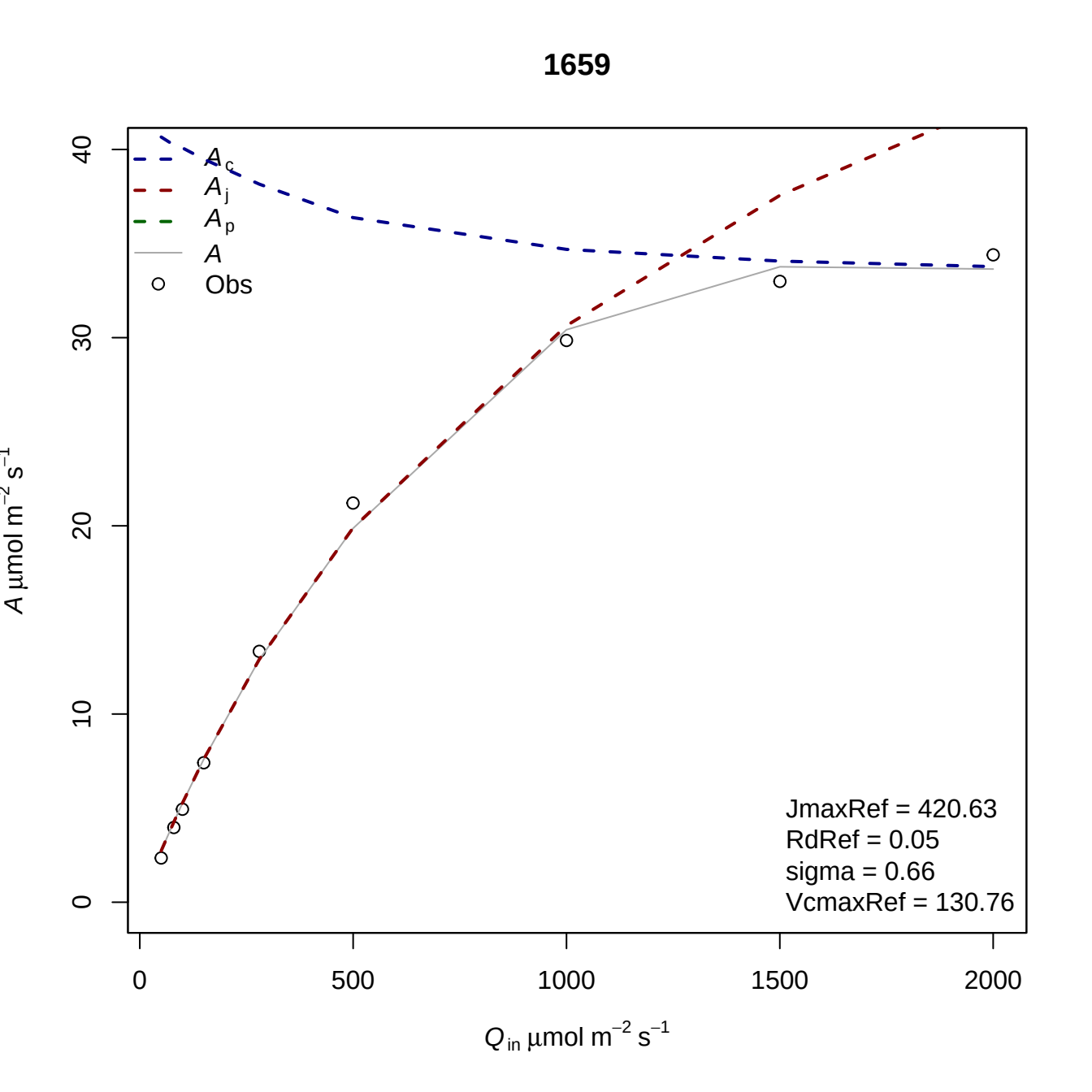

1654
A_c
A_j
A_p
A
Obs
JmaxRef = 41.95
RdRef = 0.92
sigma = 0.28
VcmaxRef = 23.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

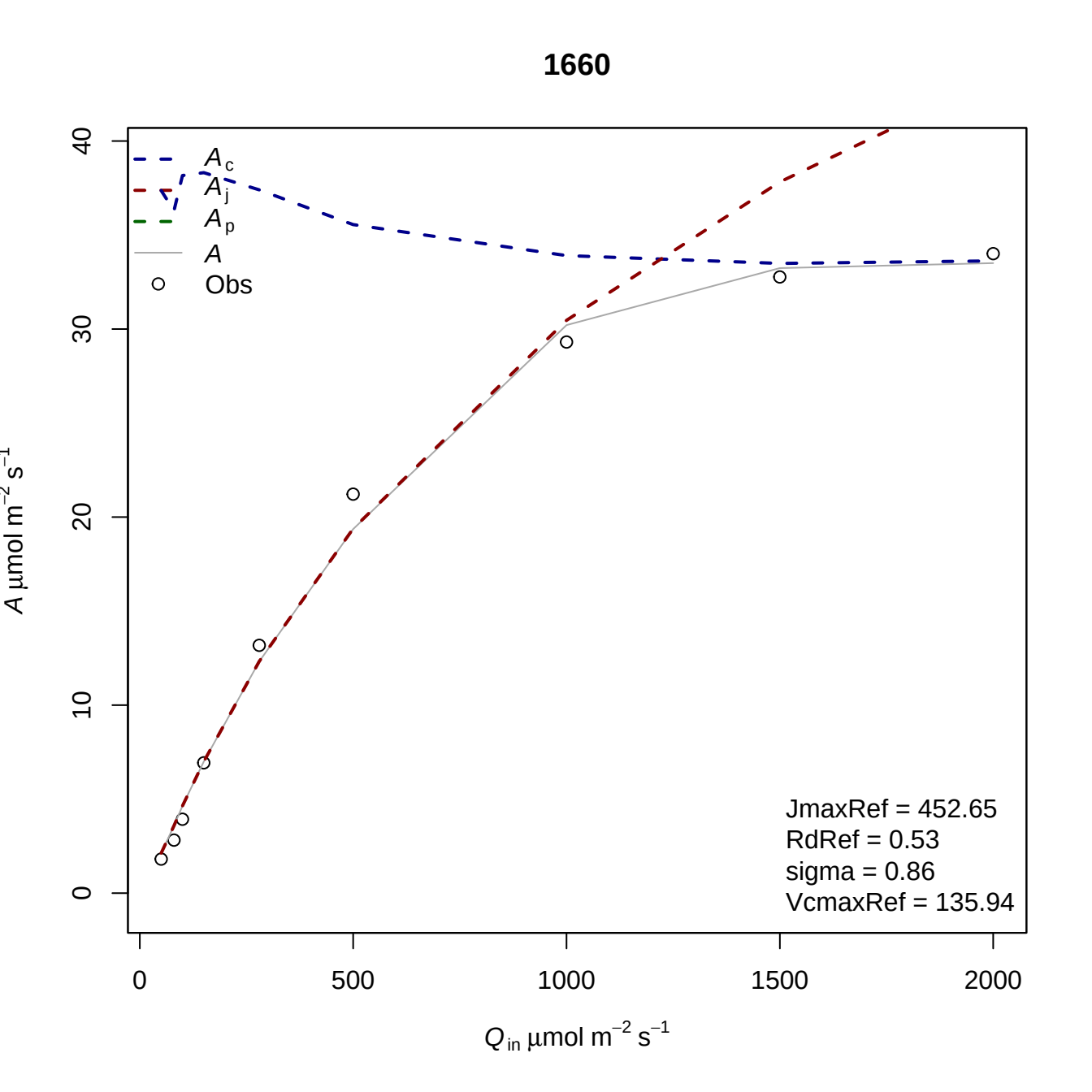

1655
A_c
A_j
A_p
A
Obs
JmaxRef = 179.72
RdRef = 0.95
sigma = 0.33
VcmaxRef = 64.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

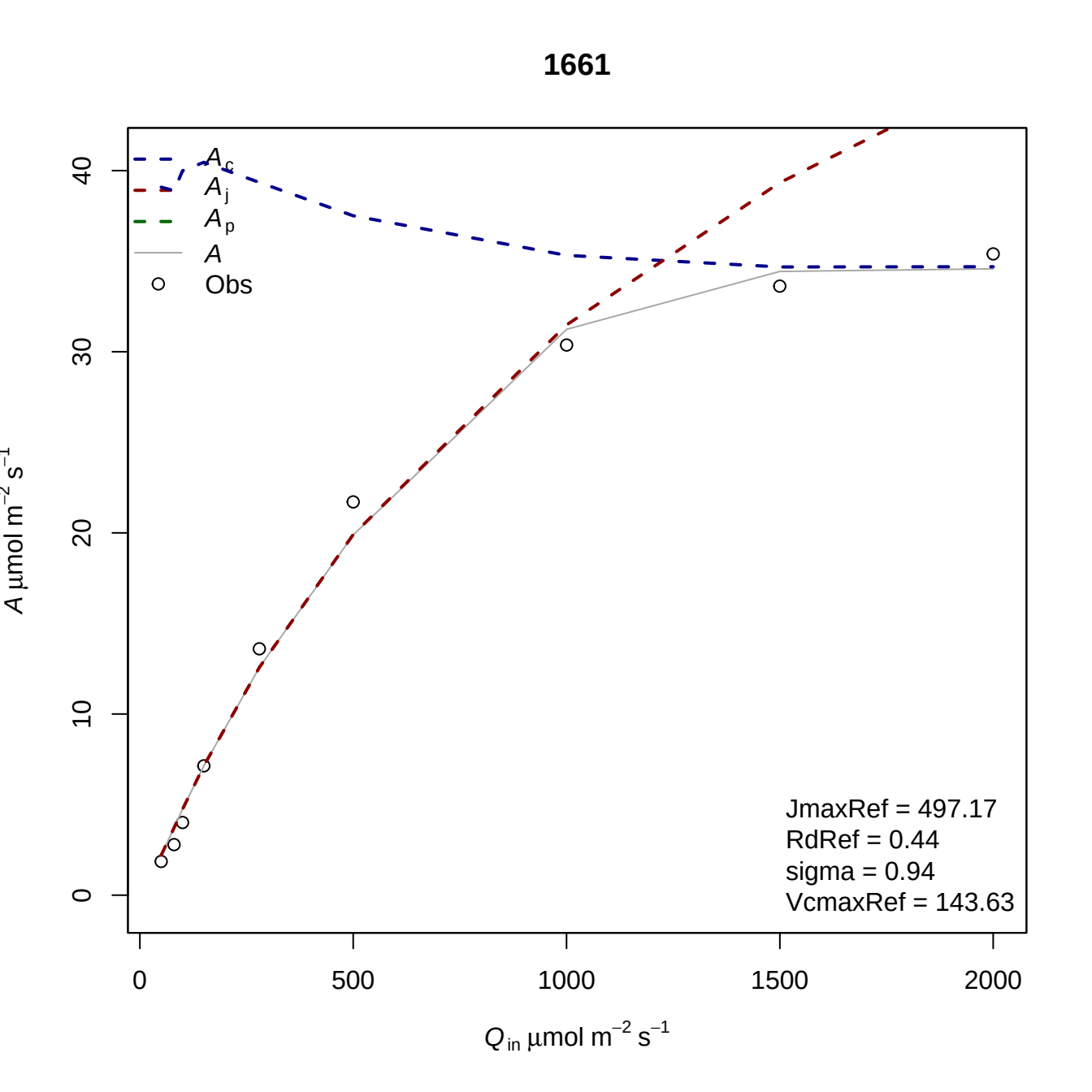

1656
A_c
A_j
A_p
A
Obs
JmaxRef = 157.4
RdRef = 1.16
sigma = 0.29
VcmaxRef = 61.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

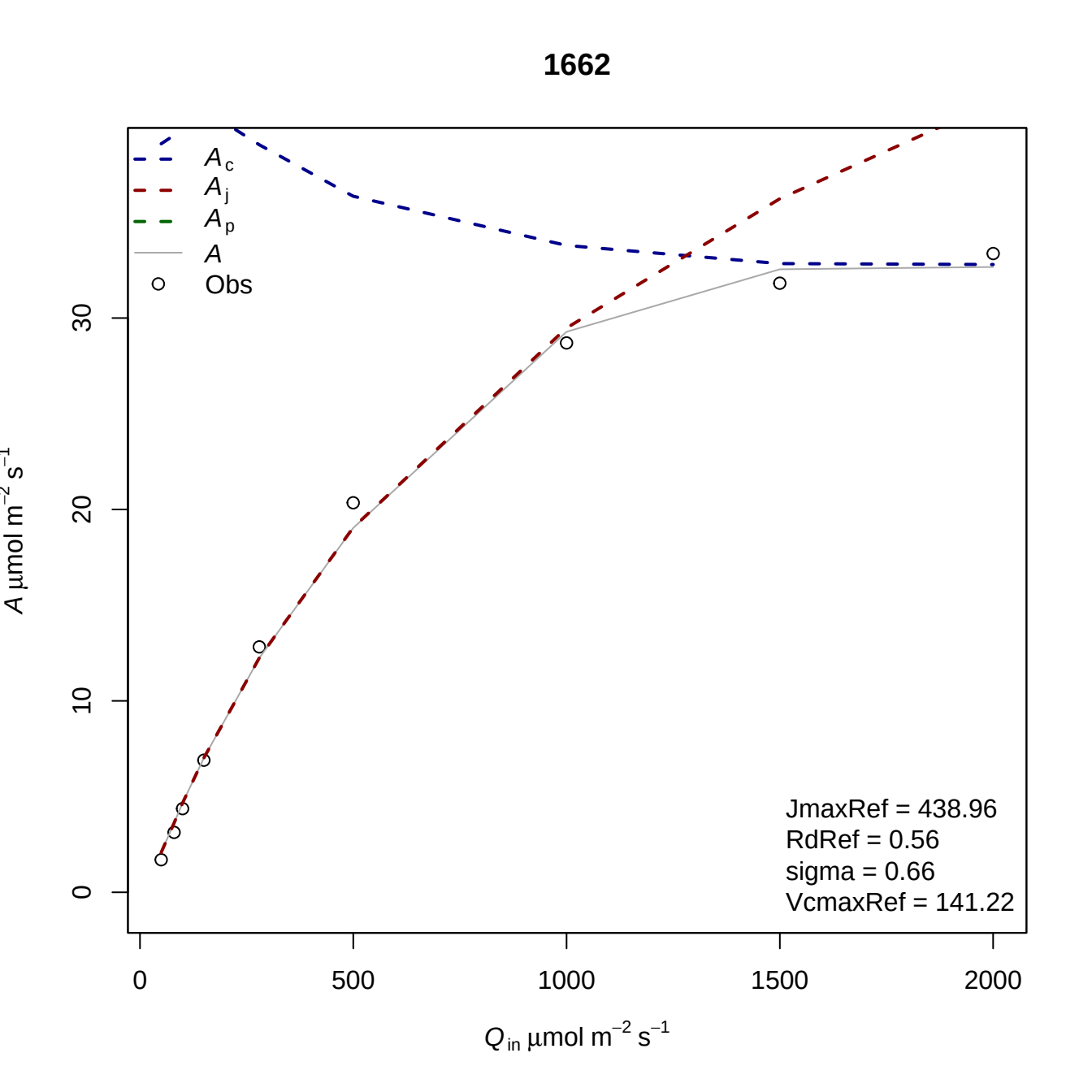

1657
A_c
A_j
A_p
A
Obs
JmaxRef = 202.86
RdRef = 0.46
sigma = 0.22
VcmaxRef = 65.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000


1658
A_c
A_j
A_p
A
Obs
JmaxRef = 172.24
RdRef = 0.74
sigma = 0.23
VcmaxRef = 58.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000


1659
A_c
A_j
A_p
A
Obs
JmaxRef = 420.63
RdRef = 0.05
sigma = 0.66
VcmaxRef = 130.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000


1660
A_c
A_j
A_p
A
Obs
JmaxRef = 452.65
RdRef = 0.53
sigma = 0.86
VcmaxRef = 135.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000


1661
A_c
A_j
A_p
A
Obs
JmaxRef = 497.17
RdRef = 0.44
sigma = 0.94
VcmaxRef = 143.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000


1662
A_c
A_j
A_p
A
Obs
JmaxRef = 438.96
RdRef = 0.56
sigma = 0.66
VcmaxRef = 141.22
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}